# Computing the Riemann Hypothesis

# Computing the Riemann Hypothesis

## Thorin Measures, Pólya Densities and Hitting Times

Nicholas G. Polson
Booth School of Business, University of Chicago

Vadim Sokolov
Department of Systems Engineering and Operations Research
George Mason University

*Computing the Riemann Hypothesis: Thorin Measures, Pólya Densities and Hitting Times*

Copyright © 2026 Nicholas G. Polson and Vadim Sokolov

License: distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), `https://creativecommons.org/licenses/by/4.0/`: you may copy, share and adapt this book for any purpose, provided you credit the authors. The code is released under the MIT License.

Digital Object Identifier (DOI): `10.5281/zenodo.23741450`

Independently published

First edition, October 2026

Cover: the Thorin density of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ at eight resolutions, computed by the authors.

Source, code and data: `https://github.com/VadimSokolov/computing_rh`

Version 1.4.1, typeset by the authors in LaTeX from commit `8af4c0c-dirty` of the source (28 October 2026).

# Preface

The Riemann Hypothesis can be stated as the positivity of a measure. The reciprocal Xi clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is the Laplace transform of a generalised gamma convolution exactly when every nontrivial zero lies on the critical line, and then its Thorin measure is the discrete measure $\sum_{\gamma>0}\delta_{\gamma^2}$ on the squared ordinates. This book asks what computation can say about that positivity condition.

Its answer has three parts. First, the Thorin measure is computable at every positive resolution, with explicit error bounds, from the Pólya kernel alone, with no primes and no list of zeros. Second, the distance between the computed measure and the measure demanded by RH can be measured exactly, and the signature an off line zero would leave can be calibrated. Third, a finite computation could refute RH by certifying a zero off the line, but neither a computation of the zeros up to a finite height nor a finite set of positivity tests on the heat trace can prove it. A last question is how much of the positivity can be proved. Moved from the centre of the critical strip to the basepoint one, the clock has a positive Thorin measure with no hypothesis. Chapters 9 to 11 follow each construction of the book back across the strip as far as it holds unconditionally; below the basepoint one, zero density estimates still show that the zeros to the right of the basepoint, the only possible source of negativity, carry a vanishing fraction of the mass of the density.

The book grew out of the first author's programme on generalised gamma convolutions and the Pólya density, collected in the monograph *On Hilbert's 8th Problem* [424], and out of a joint paper computing the Thorin measure directly [437]. The first author's arXiv paper of the same title is a separate work, which we cite by version: [421] is its version 24, of 2026, and [422] its version 5, of 2017. Chapters 6 and 7 incorporate Part IX of that monograph [424], the first author's work on the reciprocal zeta function, written as the manuscript [430], and Chapters 9 to 11 a paper of the authors on the reciprocal Xi function at the basepoint one and across the critical strip, together with notes on Riesz's criterion and on almost generalised gamma convolutions. Sections 8.5 and 11.3 draw on a manuscript of the first author on the Thorin measures of the Dedekind eta function [427], Chapter 13 on a note of the first author on ferromagnetic random variables, Chapter 14 on a manuscript of the first author on the Pólya density as a maximum order statistic [429], Chapter 15 on a working report of the authors

[438], Chapter 16 on a chapter of the first author on Pólya and the zeros of entire functions, and Chapters 17 to 21 on five further chapters of the first author, on Yoshida's Hermitian forms, on the compact windows of Zhu [526], on three families of generalised gamma convolutions, on deconvolution as a damped Weil criterion, and on Dirichlet $L$ functions and the Linnik constant, and Chapters 22 and 23 on two more chapters of the first author, on friable approximants of $\Xi$ after Miyatake and the cosh ladder and on the work of Haseo Ki read through the Thorin programme, and Chapter 24 on a chapter and two notes of Nick Polson on hitting times and the reciprocal $\xi$ function [238, 251, 250], and Chapter 25 on a chapter of Nick Polson on extreme values, hitting times and a path to RH [231], Chapter 26 on a chapter of Nick Polson on Weil positivity and the Thorin condition [248], Chapter 27 on a note of Nick Polson on the identity of Comtet and Majumdar [229], Chapter 28 on two notes of Nick Polson on the Thorin truncation of $L(s, \chi_4)$ and on the Hermite coefficients of $\Xi$ [245, 237], and Chapter 29 on two notes of Nick Polson on the secant dilates and the Möbius function, Chapter 30 on a manuscript of Nick Polson on the level one measure and the backward Cauchy flow [240], Chapter 31 on his notes on the complexity and acceleration of that flow, Chapter 32 on a manuscript of his on the explicit formula as a deconvolution [230], Chapter 33 on his notes on Cauchy deconvolution, Chapter 34 on his note on the deconvolution from level one to the critical line, and Chapter 35 on his note on the criterion of Balazard, Saias and Yor. The core evaluator runs in ball arithmetic, and Appendix A says where the scripts and data files behind the tables and figures are, and the file `README.md` in `code/` of the repository lists them and what the archive lacks. No proof of RH is claimed.

Chicago and Fairfax, September 2026

# Contents


Preface v

1 Introduction and foundations 1
1.1 The Thorin condition and what this book shows . . . . . 1
1.2 The Xi function, the Pólya density and the clock . . . . . 9
1.3 Generalised gamma convolutions and the Thorin condition . . . . . 17
1.4 Exercises . . . . . 23

2 Evaluating the Xi function 27
2.1 The Pólya route . . . . . 27
2.2 Classical evaluators, Turing's method and the Pólya route . . . . . 33
2.3 A worked computation from first principles . . . . . 37
2.4 Exercises . . . . . 40

3 The Thorin measure at finite resolution 43
3.1 What computation can reach, and what the sine squared lemma adds 43
3.2 Distances to the discrete measure implied by RH . . . . . 47
3.3 What an off line zero would do . . . . . 50
3.4 Counting zeros with the Thorin measure . . . . . 52
3.5 Further readings of the sine squared lemma . . . . . 56
3.6 Exercises . . . . . 60

4 Weak limits, hyperbolic complete monotonicity and a Bosch type theorem 63
4.1 The limit of vanishing resolution . . . . . 63
4.2 Bosch's half Cauchy theorem and HCM of the reciprocal Xi function 65
4.3 A Bosch type theorem for the reciprocal Xi function . . . . . 72
4.4 Exercises . . . . . 90

5 The heat trace, the arithmetic and the limits of finite information 93
5.1 The heat trace and the hierarchy of positivity . . . . . 93
5.2 What finite information cannot decide . . . . . 117
5.3 The Thorin functional from the arithmetic . . . . . 125
5.4 Spacings, irregularity and the cell distance . . . . . 128
5.5 Exercises . . . . . 131

6 The reciprocal zeta function 135
6.1 Thorin type forms of the reciprocal zeta function . . . . . 135
6.2 The gamma factor, the Meixner law and the sine squared lemma . 147

7 Infinite divisibility and truncations of the reciprocal zeta function 165
7.1 Infinite divisibility, necklaces and solved models . . . . . 165
7.2 Truncations, finite conditions and the road to RH . . . . . 188

8 Probability, hitting times and other criteria 211
8.1 The clock as a random variable . . . . . 211
8.2 Brownian boundaries and Williams' structure . . . . . 219
8.3 Keiper Li coefficients from the Pólya density . . . . . 225
8.4 The de Bruijn Newman flow from the Pólya density . . . . . 230
8.5 Beyond zeta: the Dedekind eta kernel, $L(s, \chi_{12})$ and Ramanujan's $\Delta$ 263
8.6 The inverse problem: recovering the atoms by nonnegative deconvolution . . . . . 303

8.7 Pólya's Bessel function: hitting times, a clock and the flow . . . . 306
8.8 Exercises . . . . . 331

9 The basepoint one 335
9.1 The Thorin measure and the Lévy form at the basepoint one . . . 335
9.2 Hitting times at the basepoints . . . . . 367

10 The critical strip and Riesz's criterion 377
10.1 Normal scale mixtures and completeness across the strip . . . . . 377
10.2 Riesz's criterion and the Thorin path . . . . . 399
10.3 The squarefree series and the clock . . . . . 411

11 Infinite divisibility, the eta analogue and almost generalised gamma convolutions 431
11.1 Infinite divisibility and a map of processes . . . . . 431
11.2 Free probability and the Upsilon transform . . . . . 444
11.3 The Dedekind eta analogue at the basepoint one . . . . . 464
11.4 Almost generalised gamma convolutions . . . . . 470
11.5 Pakes, the Lambert law and the Thorin condition . . . . . 487

12 Conclusions and open problems 517
12.1 The Thorin approach in summary, and a path to RH . . . . . 517
12.2 Open problems and lessons for the Thorin programme . . . . . 521

13 Ferromagnetic random variables 529
13.1 Ferromagnetic laws and their clocks . . . . . 529
13.2 Laws on an interval . . . . . 534
13.3 The Mellin transform of a clock . . . . . 540
13.4 The Pólya law . . . . . 544
13.5 Ferromagnetic approximations of the Pólya law . . . . . 548
13.6 The exponential of the Pólya variable . . . . . 563
13.7 Two arithmetic spin chains . . . . . 572
13.8 Open problems . . . . . 579

14 The Pólya density as a maximum order statistic and the Thorin measure of the clock 583
14.1 Notation . . . . . 583
14.2 The Pólya density as a maximum order statistic . . . . . 583
14.3 Khinchin: the zeta distribution inside the tilted maximum . . . . . 585
14.4 The Wald couple of the maximum . . . . . 586
14.5 The clock exists and is infinitely divisible . . . . . 587
14.6 A Stieltjes criterion . . . . . 587
14.7 Is the clock a weak limit of GGC laws? . . . . . 589

14.8 The sine squared lemma and the basepoint family in the $\lambda$ chart . 591
14.9 The maximum representation does not transfer . . . . . 593
14.10Gnedenko: the first zero as a norming constant . . . . . 595
14.11Finding the zeros from the Thorin density in the $z$ chart . . . . . 596
14.12Finite maxima and a limit argument . . . . . 597
14.13Bondesson classes containing GGC . . . . . 603
14.14Is there a new path to the Riemann Hypothesis? . . . . . 605
14.15Open problems . . . . . 608
14.16Numerical checks . . . . . 609

15 The Thorin phase diagram 611
15.1 Which computations see past the verified height . . . . . 611
15.2 Clocks along the flow at every basepoint . . . . . 613
15.3 The Thorin phase diagram . . . . . 619
15.4 What a Thorin computation sees at positive time . . . . . 622
15.5 A certificate for one point of the diagram . . . . . 624
15.6 Where the uniform step could come from . . . . . 629
15.7 Exercises . . . . . 636

16 Pólya and the zeros of entire functions 639
16.1 Pólya's question . . . . . 639
16.2 Notation and the Thorin dictionary . . . . . 640
16.3 Pólya's papers on zeros, one by one . . . . . 641
16.4 The Laguerre Pólya class in the Thorin variable . . . . . 644
16.5 Pólya frequency functions: the two sided criterion . . . . . 645
16.6 The universal factor $\cosh(au)$ and the clock . . . . . 646
16.7 Pólya's $\Xi^*$ as the archimedean model . . . . . 649
16.8 Pólya's Mellin criterion applied to the Riemann kernel . . . . . 651
16.9 Is the $2\sin^2$ lemma useful with regard to Pólya's work? . . . . . 653
16.10What would help show RH . . . . . 654
16.11Progress on the open problems . . . . . 655
16.12Consequences of a quasi Riemann hypothesis at 7/8 . . . . . 667

17 Yoshida's Hermitian forms, Weil positivity and the Thorin programme 679
17.1 Yoshida's paper . . . . . 679
17.2 The ladder is the diagonal of Weil's form . . . . . 681
17.3 Every large rung fails . . . . . 682
17.4 Compact support and the signed Thorin formula . . . . . 684
17.5 Localisation: a modulated heat trace . . . . . 685
17.6 The tent and the sine squared lemma . . . . . 687
17.7 The sine squared lemma off the diagonal . . . . . 691
17.8 Damped forms and sharp coercivity . . . . . 693
17.9 An uncertainty barrier . . . . . 701

17.10 The question of a path to RH . . . 702
17.11 Open problems . . . 703

18 Zhu's compact windows and the Landau Widom horizon 707
18.1 What Zhu proves . . . 707
18.2 The Thorin dictionary . . . 709
18.3 Zhu's test function through a truncated reciprocal Xi transform . . 712
18.4 The Landau Widom horizon . . . 713
18.5 The comb barrier, basepoints and damping . . . 718
18.6 What window certificates reach . . . 722
18.7 Open problems . . . 722

19 Three families of generalised gamma convolutions and the deconvolution obstruction 725
19.1 Three families, and the tilt . . . 725
19.2 The basepoint family: a Poisson semigroup to the right of the zeros 728
19.3 What a single basepoint sees . . . 731
19.4 The heat flow: a stable step that cannot carry real zeros . . . 738

20 Deconvolution as a damped Weil criterion 743
20.1 Weil's distribution and the damped criterion . . . 744
20.2 The ladder . . . 747
20.3 What can be done and what is open . . . 753

21 Thorin representations, Dirichlet $L$ functions and the Linnik constant 755
21.1 Linnik's problem and where zeros enter . . . 755
21.2 The Thorin property at every tilt . . . 758
21.3 The centre clock: moments, transfer function and negative moments 762
21.4 The master theorem route . . . 770
21.5 Dirichlet $L$ functions and number fields . . . 774
21.6 GRH from the Thorin property, and families of characters . . . 784
21.7 Consequences for the Linnik constant and for RH . . . 791
21.8 Open problems . . . 792

22 Friable approximants of $\Xi$ after Miyatake and the cosh ladder 797
22.1 Riemann's representation, the friable approximant and the kink . . 799
22.2 Approximation, the kink constant and no finite proxy . . . 801
22.3 The deficit: positivity and Lorentzian form . . . 804
22.4 Humps, pairs and the count of real zeros . . . 807
22.5 A survey of the breakdown height up to height 21,000 . . . 809
22.6 Where the lost zeros go . . . 810
22.7 Miyatake's theorem and the remainder $R$ . . . 812
22.8 The pole term, the secant and reweighting . . . 814

22.9 The $2\sin^2$ lemma and the Thorin programme . . . . . 815
22.10 The cosh ladder . . . . . 817
22.11 Miyatake's identity, read probabilistically . . . . . 818
22.12 The transition at $b = 2$ . . . . . 820
22.13 The Pólya mixture of $\psi_{\mathrm{M}}$ . . . . . 823
22.14 The ladder and the quasi Riemann abscissa . . . . . 826
22.15 Open problems . . . . . 835

23 Haseo Ki and the Thorin programme 839
23.1 The work of Ki and the Thorin dictionary . . . . . 840
23.2 The heat flow, the barrier and the Gaussian tilt . . . . . 841
23.3 Derivatives and the Brownian exit time . . . . . 843
23.4 Approximate zeta functions and the Euler product . . . . . 847
23.5 Zeros of the derivative of zeta, open problems and notes . . . . . 849

24 Hitting times and the reciprocal xi function 851
24.1 The law and its density . . . . . 853
24.2 Kent's formula . . . . . 856
24.3 Processes behind the reciprocal xi law . . . . . 858
24.4 The candidate families . . . . . 858
24.5 Truncations, total positivity and the HCM route . . . . . 859
24.6 Jensen hitting times . . . . . 866
24.7 Extreme value hitting times . . . . . 871
24.8 The prime side hitting times have a universal limit . . . . . 875
24.9 Thorin rigidity . . . . . 881
24.10 What the prime side limit would have to be . . . . . 882
24.11 Diffusion approximations . . . . . 883
24.12 Curved boundaries: the Roberts and Shortland view . . . . . 884
24.13 Marc Yor's probabilistic Riemann function . . . . . 889
24.14 The Davenport Heilbronn test . . . . . 894
24.15 What the reciprocal Xi function proves . . . . . 895
24.16 The coefficient ladder . . . . . 902
24.17 The angle theorem: what the ladders can and cannot see . . . . . 903
24.18 What a hitting time proof would need . . . . . 908
24.19 Open problems . . . . . 908
24.20 Expansions of the reciprocal Xi function . . . . . 909

25 Extreme values, hitting times and a path to RH 913
25.1 The Pólya density as an extreme value law . . . . . 915
25.2 Smith and Tawn in pictures . . . . . 920
25.3 Hitting times and the Thorin variable . . . . . 921
25.4 B splines, Hadamard products and Thorin . . . . . 923
25.5 An explicit string: the Liouville operator . . . . . 926

25.6 The string test from kernel moments . . . 926
25.7 Is there a path to RH? . . . 928
25.8 Work on the open problems . . . 930
25.9 Second round on the open problems . . . 935
25.10 Devroye's series, the mean excess function and detection . . . 940
25.11 Remaining open problems . . . 946

26 Weil positivity as the Thorin condition 949
26.1 Weil's explicit formula and the heat trace . . . 951
26.2 Weil positivity is the Thorin condition . . . 952
26.3 The archimedean place is a Gumbel Lévy measure . . . 954
26.4 Three regimes . . . 955
26.5 Zeros from primes by Prony's method . . . 958
26.6 The prime point process: first and second order . . . 959
26.7 Does the extreme value approach offer insight? . . . 959
26.8 Windows, primes and rigidity . . . 960
26.9 Certificates, smoothing, energy and moments . . . 964
26.10 The sine squared lemma and a single heat time . . . 971
26.11 Is there a path to RH? . . . 974
26.12 Open problems . . . 975

27 The Comtet Majumdar family and the uniform law 977
27.1 The identity decoded . . . 979
27.2 GGC reciprocals and the Laguerre Pólya class . . . 981
27.3 The shifted Comtet Majumdar family . . . 982
27.4 The uniform law and the $2\sin^2$ lemma . . . 985
27.5 Damping the kernel . . . 990
27.6 The de Bruijn Newman family and what the identity contributes . 993

28 Truncations of the beta lattice and Hermite coefficients 997
28.1 The beta lattice and its truncations . . . 999
28.2 The drift of the truncated zeros . . . 1001
28.3 The side of approach is a Gram parity . . . 1004
28.4 Transient zeros . . . 1007
28.5 Symmetrised truncations . . . 1009
28.6 Where this leaves the route . . . 1011
28.7 Hermite coefficients of $\Xi$ . . . 1012
28.8 The turning point asymptotic . . . 1014
28.9 The first negative coefficient . . . 1016
28.10 Counting zeros from $\Phi$ alone . . . 1017
28.11 The best scale and the run lengths . . . 1019
28.12 The Thorin shadow . . . 1021
28.13 Notes and open problems . . . 1021

29 Completeness of the secant dilates and the Möbius problem 1023
29.1 The secant family and the identity . . . 1025
29.2 Weighted spaces and the Mellin line . . . 1026
29.3 Completeness in one weight is unconditional . . . 1027
29.4 The Möbius route in the strip $\frac{1}{2} < c < 1$ . . . 1029
29.5 Two weights: where the zeros are felt . . . 1030
29.6 The zeta version . . . 1031
29.7 What the quasi Riemann hypothesis gives . . . 1032
29.8 Relation to the Thorin programme . . . 1033
29.9 Numerical check . . . 1033
29.10Necessity of the divergence condition . . . 1034
29.11Wiener, Kac and Salem: the $L^1$ picture . . . 1036
29.12Explicit approximants in one weight . . . 1038
29.13The Meixner family . . . 1041
29.14The Thorin form of the two weight criterion . . . 1044
29.15Numerical checks of the sharpness experiment . . . 1047
29.16Open problems . . . 1048

30 The Thorin measure at level one and the backward Cauchy flow 1049
30.1 The level one measure . . . 1051
30.2 The Fourier side: primes . . . 1053
30.3 The backward flow . . . 1055
30.4 The backward flow as a Bayesian inverse problem . . . 1058
30.5 Generative Bayesian computation for the backward flow . . . 1063
30.6 Is this a path to RH? . . . 1066
30.7 Open problems . . . 1067

31 Complexity and acceleration of the backward flow 1069
31.1 The deconvolution problem, precisely . . . 1073
31.2 Three layers of cost . . . 1076
31.3 EM as majorise minimise . . . 1077
31.4 Acceleration . . . 1083
31.5 Von Neumann's Monte Carlo inversion . . . 1086
31.6 Cauchy as a scale mixture . . . 1089
31.7 What is being computed . . . 1091
31.8 Thorin measures at level 7/8 . . . 1091
31.9 Pictures of the Thorin measures and the decision of positivity . . . 1094
31.10The limits of the computation . . . 1104
31.11Open problems worked . . . 1107
31.12Open problems . . . 1129

32 The explicit formula as a deconvolution 1131
32.1 The deconvolution at level one half . . . 1134

32.2 Sparse recovery . . . 1138
32.3 The computation . . . 1141
32.4 Posterior concentration, and what it cannot do . . . 1142
32.5 The functional equation makes the problem singular . . . 1145
32.6 An adaptive prior, and a ReLU network . . . 1152
32.7 What this does and does not do . . . 1155
32.8 Questions taken up . . . 1156
32.9 Open problems . . . 1162

33 Cauchy deconvolution 1165
33.1 The operator . . . 1167
33.2 Positivity . . . 1169
33.3 Stability . . . 1171
33.4 Statistics . . . 1174
33.5 Atomic sources . . . 1176
33.6 What is known and what is not . . . 1177

34 Cauchy deconvolution from $\mathfrak{U}_1$ to $\mathfrak{U}_{1/2}$ 1179
34.1 What the book already proves . . . 1181
34.2 Two test functions . . . 1183
34.3 A particle filter along the Cauchy flow . . . 1185
34.4 Optimising the filter with generative Bayesian computation . . . 1189
34.5 Concentration of the particle approximation . . . 1194
34.6 From $\mathfrak{U}_1$ to $\mathfrak{U}_{3/4}$ . . . 1200
34.7 Negative mass and converse results . . . 1202
34.8 Zero free regions from positivity . . . 1209
34.9 Nested sampling over heights . . . 1218
34.10 What would close the gap . . . 1221

35 Balazard, Saias and Yor meet Thorin 1225
35.1 Thorin measures along vertical lines . . . 1227
35.2 The Balazard, Saias and Yor criterion at every level and height . . . 1228
35.3 The dictionary: one derivative in $\sigma$ . . . 1229
35.4 The 2 sin-squared lemma, EGGC and Davenport and Heilbronn . . . 1231
35.5 Brownian motion and a probabilistic reading of the criterion . . . 1236
35.6 The Monte Carlo solution . . . 1241
35.7 Two dimensions: the smoothed Riesz identity . . . 1254
35.8 Limits . . . 1264
35.9 Open problems . . . 1265

A Code and reproducibility 1267
A.1 The Pólya evaluators . . . 1267

B A short history of computations on the zeros 1269

C Notation 1271

# List of Figures

1.1 The Thorin density $\rho_\varepsilon(\theta)$ on $10 \le \theta \le 60$, computed from the Pólya density at eight resolutions from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top), on a logarithmic colour scale. As the resolution sharpens, the density concentrates on bright vertical lines at the ordinates of the zeros, all of which lie on the critical line in this window: the Cauchy flow converging to the RH measure $\sum \delta_{\gamma^2}$. . . . . 3

1.2 Left: the Pólya kernel $\Phi(u)$ on the real line. Right: $|\Phi(x+i\eta)|$, on a logarithmic scale, on the lines $\eta = 0.1000$, 0.3927, 0.6854 and 0.7654 inside the strip of analyticity, that is at distances 0.6854, 0.3927, 0.1 and 0.02 below its edge $\eta = \pi/4$ (the tilts of Section 2.1.3). As the line approaches the natural boundary $\eta = \pi/4$ the kernel grows and spreads (compare Table 2.1), and it still decays doubly exponentially in $|x|$. . . . . 12

1.3 Cell distance divided by the number of atoms in the window, for the sinh clock (50 atoms, cells $(k-\frac{1}{2}, k+\frac{1}{2}]$) and for the 79 zeros below height 200 (Voronoi cells of Section 3.2). Both vanish linearly in $\varepsilon$. For the sinh clock the undivided distance stays below $4\varepsilon/\pi$ whatever the number of atoms, so the gap between the curves depends on the window, and the factor of three in the title of the figure holds for these two windows only. . . . . 24

2.1 The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2} + \varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 1.2.3). . 30

2.2 The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed. . . . . 37

2.3 The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of $\gamma_1$, the kernel of $\gamma_2$ and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction. . . . . 39

3.1 Cumulative Thorin mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of $U_{\mathrm{RH}}$, on the whole window (left) and on $28 \le \theta \le 52$ (right). . . . . 49
3.2 The three signatures of Proposition 3.3.1 as functions of $\delta/\varepsilon$: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1. . . . . 52
3.3 Left: absolute error of Selberg's smoothed prime side of $v_{3/4}$, at resolution 0.25, for three cutoffs, and (dotted) with the sum over the first 1700 zeros added. Right: the maximal absolute and relative errors against the cutoff, with $x^{-\varepsilon}$ for comparison. . . . . 59

4.1 Normalised alternating derivatives $(-1)^k\partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\varphi_\alpha(u/v)$ for $1 \le k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{3, 20\}$, computed from the Pólya density, at the centre $\alpha = \frac{1}{2}$ and at the unconditional basepoint $\alpha = 1$. . . . . 66
4.2 Left: the sliver $\mathcal{E}_{T_0}$ of Theorem 4.3.9 for $T_0 = 2$ (dashed) and the discs $D_\gamma$ of six hypothetical zeros $a+ib$ with $a > 2$ and $0 < b < \frac{1}{2}$, with the poles $-\gamma^2$ and $-\bar\gamma^2$ (dots) at the ends of their vertical diameters; the dot at 0 marks the origin. Right: for the toy reciprocal Xi function with the first 200 zeros of $\zeta$ and the pair $60 \pm 0.3i$, the set where $\operatorname{Im}\psi > 0$ (shaded) on a grid of $481 \times 300$ points, which lies inside the upper half of $D_\gamma$ (solid); the cross is the pole $-\bar\gamma^2 = -3599.91 + 36i$. Drawn by `pakes_disc.py`. . . . . 79

5.1 Left: normalised Hankel minors of the heat trace at two times; positivity persists but falls below any fixed precision as $t$ grows. Right: the first six $(-1)^k W^{(k)}(t)$, divided by their values at $t = 0.005$, from 1700 zeros. . . . . 100
5.2 Top: the Williamson measures $U_k$ of the heat trace for $k = 10^4$, $10^5$, $10^6$, computed from 1700 zeros, as densities in $\theta = \sqrt{z}$. Bottom: the Poisson smoothed Thorin densities $\rho_\varepsilon$ with $\varepsilon = 30/(2\sqrt{k})$, so that the widths match at $\theta = 30$. . . . . 108

5.3 Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with $W$ replaced by its sum over the first 1700 zeros. The change is at most $4 \times 10^{-11}$, near $t = 7 \times 10^{-7}$, and falls below $10^{-1000}$ by $t = 10^{-3}$. The sum over 1700 zeros is smaller than $W$, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with $W$ from the explicit formula (1.2.5) the maximum is $3.66 \times 10^{-11}$, at $t = 7.2 \times 10^{-7}$. . . . . 118
5.4 First order cell errors $(\mu_\varepsilon(C_k) - 1)/\varepsilon$ for the 79 cells below height 200, computed from the first 1700 ordinates. . . . . 129
5.5 The 1699 unfolded spacings of the first 1700 zeros against the GUE Wigner surmise and the Poisson law. . . . . 131

6.1 The function $M(\tau^2) = N(\tau) - \theta(\tau)/\pi = 1 + S(\tau)$ against $\tau = \sqrt{t}$, with the zero ordinates marked; under RH it is the distribution function of the Thorin measure of $\varphi$. The dashed curve is the gamma factor contribution $-\theta(\tau)/\pi$ alone. . . . . 143
6.2 The Thorin density of the gamma factor in the variable $\tau = \sqrt{t}$, namely $-\theta'(\tau)/\pi$, together with the first three Poisson kernels $c_n/(\pi(c_n^2 + \tau^2))$ of Theorem 6.2.1. The density is positive for $\tau < \tau_0 \approx 6.29$ and negative beyond, where the flat compensators outweigh the kernels. . . . . 148

7.1 The function $w = W - W_0$ on a logarithmic scale, with its two components: the heat trace $W$ over the zeros and the negative smooth part $-W_0$. The vertical line is $x_* \approx 0.00322$, where $W_0$ changes sign. To the left, $w$ differs from $e^{x/4}$ by less than $2 \times 10^{-13}$; to the right, $-W_0 > 0$, and from $x_1 = 0.004$ on it dominates any contribution of off line zeros to $W$. The faint curve is the asymptote $c/(2\sqrt{\pi x})$. . . . . 167
7.2 The Cardon ratio for the random harmonic law $H_{\kappa^*}$. Left: $\log r(t)$ on the real axis, asymptotically linear with slope $\frac{7}{2}$ (dashed). Centre: $L'(s)$ for $L(s) = \log(r(\sqrt{s})/r(0))$, positive, decreasing and convex as a Stieltjes function must be. Right: the continuation $G(x) = r(-ix)$ on $[0, \pi/4)$, with a simple zero at $x = 0.6388$. . . . . 185
7.3 Left: real parts of the zeros of $F_n$ near $\rho_j/2$, in the variable $u = w + 1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against $n$, converging to $-\operatorname{Re}\kappa(\rho_j)$. . . . . 191

8.1 The law of the clock variable computed from the Pólya density by Talbot inversion (points) and by the spectral expansion with Pólya residues (lines): density, survival function, and the distance of the hazard from $\gamma_1^2$, whose logarithm has slope $-(\gamma_2^2 - \gamma_1^2)$ (dashed). Beyond $t \approx 0.17$ the last curve reflects the double precision of the plotting script, not the hazard. . . . . 214

8.2 Left: the Brownian boundary of the clock computed from the Volterra equation, the Roberts Shortland tangent boundary, and the right tail asymptote with its logarithmic correction, over the three zones. Right: the simulated distribution function of the crossing time minus that of the clock, for both boundaries; the grey band is $\pm 1/\sqrt{n}$, twice the largest standard error of a simulated distribution function. . . . . 223

8.3 The Li test functions $4\sin^2(n \arctan \frac{1}{2\theta})$ for $n = 1, 10, 50, 100, 200$, in the square root chart $\theta = \sqrt{z}$, against the zeros below height 200 (grey vertical lines). Larger $n$ probes higher zeros. . . . . 229

8.4 Left: the prime pieces of the sine squared decomposition at the basepoint $\alpha = 2$, written $\sigma$ in the plot, for $\zeta$, $L(s, \chi_{12})$ and $L(s, \Delta)$, and the pole piece of $\zeta$. For $\zeta$ the prime piece is positive and the pole piece negative; for the two eta based functions there is no pole piece and the prime piece takes both signs. Right: the Thorin densities $\rho_{0.1}$ of $L(s, \chi_{12})$ and $L(s, \Delta)$ on a logarithmic scale, peaking at their zeros (vertical lines). . . . . 267

8.5 Left: the Bondesson function $N_1$ of $M_1$, nonnegative, with its asymptote $\frac{1}{3}\sqrt{3z/\pi}$; each drop is a zero of the transform at $-b_{6k}$. Right: the poles $-2\pi m$, $m \le 60$, and the zeros of $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*}$: the 25 real zeros above $-2\pi \times 61$, one in each gap with $\tau(m+1)\tau(m+2) > 0$, and the five pairs of nonreal zeros, the only ones with $|\operatorname{Re}\theta| \le 2\pi \times 60.5$ and $|\operatorname{Im}\theta| \le 100$. Drawn by `maxexp_figs.py` from `maxexp_psizeros.json`. . . . . 283

8.6 The real zeros of $H_t^{\Delta}$ below height 52 for $-1.1 \le t \le 0.5$, and the seven collisions of Table 8.16; all zeros move down as $t$ grows. Drawn by `maxexp_figs.py` from `maxexp_flow.json`. . . . . 286

8.7 Left: $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha + i\theta, \chi_4) = \pi\rho_{\alpha-1/2}(\theta)$ for $\alpha = 1, 0.75, 0.55$ on a logarithmic scale, with the ordinates of the zeros of $L(s, \chi_4)$ (dashed). Right: at $\alpha = \frac{3}{2}$, the drift and gamma piece $b_\alpha + G_\alpha$, the prime piece $\varpi_\alpha$, of both signs, and their sum $v_\alpha$. Drawn by `dbeta_figs.py`. . . . . 296

8.8 Left: the cumulative Thorin mass $\frac{1}{\pi}\arg\xi(\alpha+i\theta,\chi_4)=\frac{1}{\pi}\int_0^\theta v_\alpha$ for $\alpha=1$, 0.75 and 0.55, against the zero count $N(\theta)$ of $L(s,\chi_4)$. Right: $v_{0.75}(\theta)$, its minimum $b_{0.75}=0.0389784$ at the origin, and the lower bound of Corollary 9.1.16 with $\alpha'=3.15$, which exceeds $b_{0.75}$ for $\theta>2.434$ (`dbeta_minimum.py`). Drawn by `dbeta_figs.py`. . . . . 302

8.9 Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15,55]$ shaded. The zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement $\delta$ of the fifth zero, the $\delta^2$ law through the point at $\delta=0.05$, and the off grid baseline $r_0=0.106$. 305

8.10 Left: the kernel $\Phi$ of (1.2.3), Pólya's asymptote $\Phi^*$ and the kernel $e^{-2\pi\cosh 2u}$ of $P$, each divided by its value at $u=0$. Right: the zero counting functions of $\Xi$, $\Xi^*$ and $P$ on $(0,100]$, against $N_0(T)+\frac{7}{8}$ and $N_0(T)-\frac{1}{4}$, the mean values of the Riemann von Mangoldt formula and of Proposition 8.7.8, with $N_0(T)=\frac{T}{2\pi}\log\frac{T}{2\pi e}$. Drawn by `biane_figs.py` from `biane_polya.json`. . . . . 308

8.11 The clock from $R_0=2\pi e^{2.5}$. Left: $\mathbb{E}e^{-u\tau_{R_0}}$, exact and from the bridge corrected Monte Carlo with two standard errors, on a logarithmic scale; the dotted lines mark the first three Thorin atoms $\nu_1^2=95.4289$, $\nu_2^2=154.965$, $\nu_3^2=215.648$ of $\tau_\infty$. Right: the histogram of the corrected hitting times against the exact densities of $\tau_{R_0}$ and of $\tau_\infty$, by Talbot inversion. Drawn by `biane_figs.py` from `biane_clock.json`. . . . . 322

8.12 The real zeros in $(0,90)$ of $P_t$ (left) and of $\Xi_t$ (right) for $-60\le t\le 0$, on a grid (step 0.5 above $t=-20$, step 1 below). The zeros of $P_t$ drift upward and leave the window one at a time; those of $\Xi_t$ collide in pairs. Drawn by `biane_figs.py` from `biane_flow.json`. . . . . 327

9.1 Left: the Thorin density $\rho_{1/2}(\theta)=v_1(\theta)/\pi$ of $T_1$ on $[0,100]$, with the ordinates of the zeros as thin vertical lines. Right: the cumulative Thorin mass $\int_0^T\rho_{1/2}$ against the zero count $N(T)$. The plot writes $\rho_1$ for $\rho_{1/2}$ and $s$ for $u$. . . . . 339

9.2 Left: the signed canonical measure $x\Pi_1(dx)$ of $\log\xi(1+s)$, made of the gamma density minus the pole, $\big(\frac{e^{-3x}}{1-e^{-2x}}-1\big)dx$, which is negative beyond $x=0.28120$, and the prime atoms $\Lambda(n)/n$ at $\log n$ for $n\le 60$. Right: the bracket $v_1(\theta)=\pi\rho_{1/2}(\theta)$ on $[0,60]$ assembled by the sine squared lemma from the drift $b_1$, the gamma piece $A(\theta)$ and the compensated prime piece $Q(\theta)$; the plot writes $\rho_1$ for $\rho_{1/2}$ and $P(\theta)$ for $Q(\theta)$. Drawn by `figs3.py`. . . . . 346

9.3 Left: absolute error of the compensated formula (9.1.8) at $\theta = \gamma_1$ against the prime cutoff $X$ for $\alpha = 1$, 0.9, 0.75 and 0.6, with the rates $X^{1/2-\alpha}$ (dotted). Right: at the centre, $\alpha = \frac{1}{2}$, the compensated sum at the first ordinate grows like $\log X$ (dashed line), the atom of the RH measure forming. Drawn by `figs4.py`. . . . . 353
9.4 Left: the density of $T_1$ by Talbot inversion of $\varphi_1$ and by the subordination formula (9.2.1), with the tail $0.00652\,x^{-3/2}$. Right: the densities $v_\alpha/\pi$ at the basepoints $\alpha = 0.6$, 0.75 and 1, with the ordinates of the zeros; they sharpen onto the zeros as $\alpha$ decreases, and for $\alpha < 1$ they are Thorin densities of $T_\alpha$ only if no zero lies to the right of the line $\operatorname{Re} s = \alpha$. The plots write $X_1$ for $T_1$, $\rho_\alpha$ for $v_\alpha/\pi$ and $s$ for $u$. Drawn by `figs2.py`. . . . . 370
9.5 Left: the exact Brownian boundary of $T_1$, written $X_1$ in the plot, from the Volterra equation over the three zones of Section 8.2.2.2 (shaded: small time, bulk, right tail), the tangent approximation, which has no solution beyond $t = 0.091$, the flat level $-0.19277$, and the value $\mathbb{E}\,b(T_1) = \sum_\rho \rho^{-1}/\sqrt{2} = 0.0163$. Right: survival of $1.5 \times 10^5$ simulated Brownian paths across the exact boundary against the exact survival function of $T_1$ and the tail $0.01303\,t^{-1/2}$; the Kolmogorov distance is 0.0019. . . . . 373

10.1 Left: the Thorin densities $v_\alpha/\pi$ at $\alpha = 0.6$, 0.75, 1, written $\rho_\alpha$ in the plot, and their Cauchy floors $b_\alpha/\pi$ (dotted). Middle: $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$, tending to one. Right: the Poisson traces $K_1$ and $K_{1/2}$ against $b_1/(\pi y)$. . . . . 384
10.2 Left: the weight $\gamma^n e^{-\gamma y}$ of the condition $(n, y)$ in (10.1.3), normalised to maximum one, for $(n, y) = (100, 1)$ and $(300, 10)$, peaking at $\gamma = n/y$; the marks are the ordinates below 200. Right: the sign factor $\cos(n\varphi - \delta y)$ of a hypothetical zero off the line for $(\gamma, \delta, y) = (50, 0.25, 1)$ and $(14.13, 0.5, 0.5)$, positive below the orders 365 and 52, at which it is first negative. Drawn by `biane_figs.py`. . . . . 391
10.3 Left: $|\eta(\sigma + i\theta)|$, the Fourier transform of Salem's kernel divided by $\Gamma(\sigma + i\theta)$, at $\sigma = \frac{1}{2}$, 0.6, 0.75, 1. Right: the Riesz function $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula (10.1.7) over 60 zeros (`figs_wiener.py`). . . . . 397
10.4 Left: the weights $|\Gamma(1 - \rho/2)/(2\zeta'(\rho))|$ of the Riesz function and $|\Gamma(1-\rho/2+ih)/(2\zeta'(\rho))|$ of the twisted Riesz function with $h = 50$, for the first 300 zeros. Right: $\operatorname{Re} R_{50}(x)/x^{1/4}$ from the Möbius series against the explicit formula (10.2.1) over 300 zeros. The plot writes $T$ for $h$. Data from `twisted.py`; drawn by `figs_strategy.py`. 402

10.5 Left: $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula over 60 zeros, as in the right panel of Figure 10.3. Middle: $1 - 1/\zeta(2k)$ and its Monte Carlo value from the clock (Proposition 10.2.3), against $4^{-k}$. Right: base ten logarithm of the largest term in Riesz's power series and in the clock series of Proposition 10.2.4 for $T = 0.01$, 0.023 and 0.05. . . . . 404
10.6 The normalised error $\Delta(x)/x^{1/4}$ of the squarefree count on $[100, 3000]$ (grey) and the explicit formula (10.3.11) with the first 60 zeros and the trivial terms at $s = -1$, $-3$ and $-5$, divided by $x^{1/4}$ (dark). Drawn by `sqfree_atoms.py`. . . . . 420

11.1 The densities of the centre clock $T$ and of the shifted clock $T_1$, written $X_1$ in the plot, on a logarithmic axis. The density of $T$ is drawn on a grid of spacing 0.0005 on $[0.004, 0.4]$, from the Talbot inversion and the spectral expansion of Section 8.1; that of $T_1$ on the 40 point geometric grid of the Talbot inversion, which is why its curve is piecewise linear. Assembled by `comp.py` and drawn by `figs.py`, both in the folder `sato_bondesson`. . . . . 434
11.2 Four processes of the map. Top left: paths of a three dimensional Bessel process run until they reach the level one; the hitting time has mean $\frac{1}{3}$, and two independent copies make up the variable $S_2$ of Section 9.1.1. Top right: ten units of time of the Riemann Ornstein Uhlenbeck process $F$, written $Y$ in the plot, whose stationary law is the centre clock, with the mean $\mathbb{E}T$, written $\sum \gamma^{-2}$ in the plot. Bottom left: the exact Brownian boundaries of $T_\alpha$ at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 computed under RH beyond $t = 0.03$ (Section 9.2.6), asymptotically linear at the centre and flat above it. Bottom right: Brownian paths crossing the boundary of $T_1$, written $X_1$ in the plot (Section 9.2.4). Drawn by `figs.py` in the folder `basepoint_one/process_map`. . . . . 443
11.3 Left: the distribution function $W(\theta^{-2})$ of the Thorin measure of $\Upsilon(T)$ in the chart $\theta = \sqrt{u}$, against the zero count $N(\theta)$, which under RH is the distribution function of the Thorin measure of $T$ in the same chart. Right: the densities of $T$, of $\Upsilon(T)$ and of $\Lambda_{\mathrm{BP}}(T)$ on $[0, 0.1]$; the last vanishes to the left of $c = 0.0153$. Computed by `wigfree_upsilon.py` and drawn by `wigfree_plots.py`. . . . . 455
11.4 The flow $e^{-(t/2)D^2}p$ for $p$ with the roots $-3$, $-1$, $\frac{1}{2}$, 2 and $1 \pm 0.8i$. Left: the real parts of the six roots for $0 \le t \le 0.4$; the complex pair has one real part until it meets the axis at $t^* = 0.3203$ (dashed), where its two roots split. Right: the imaginary part of the pair, against $\sqrt{0.64 - t}$ for the pair on its own, which reaches the axis at $t = 0.64$. Computed by `wigfree_heat.py` and drawn by `wigfree_plots.py`. . . . . 464

11.5 Left: the Thorin densities at the basepoint one for $L(s, \chi_{12})$ and for $\zeta$ on $[0, 60]$, with the ordinates of the zeros of $L(s, \chi_{12})$ (vertical lines). Right: the cumulative Thorin mass for $L(s, \chi_{12})$ against its zero count $N_{\chi_{12}}(T)$, written $N(T)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`. . . . . 466

11.6 Left: the canonical measure $x\Pi(dx)$ of $\log \xi(1 + s, \chi_{12})$, written with $\Lambda(1+s, \chi_{12})$ in the plot: the positive gamma density $e^{-x}/(1-e^{-2x})$ and the prime atoms $\Lambda_{\chi_{12}}(n)/n$ at $\log n$, of both signs, the positive ones drawn upward and the negative ones downward; there is no pole. Right: $v_1(\theta)$ on $[0, 40]$, written $\pi\rho_1$ in the plot, as the sum of the drift $b_1$, written $b$, the gamma piece $A(\theta)$ and the prime piece $Q(\theta)$, written $P(\theta)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`. . . . . 468

11.7 Left: absolute error at $\theta = \gamma_1 = 3.8046$ for $L(s, \chi_{12})$ against the prime cutoff $X$, for $\alpha = 1$, 0.9, 0.75 and 0.6, from primes alone (solid) and after the zero correction over 297 zeros (dashed). Right: at the centre and the first ordinate the sum grows like $\log X$, the atom of the GRH measure forming. Drawn by `figs.py` of `basepoint_one/eta`. . . . . 470

11.8 Left: negative mass of the density on $[0, 2000]$ against the basepoint $\alpha$ for fifteen hypothetical zeros at real part $\beta$ (`more2.py`). Middle and right: the density $v_{0.6}/\pi$ near the heights 300 and $10^5$, alone and with a hypothetical pair of zeros at $\beta = 0.7$ or 0.9 at the centre of the window (`high.py`). Drawn by `figs_almost.py`. . . . . 480

11.9 Left: the zero density exponents of Theorem 11.4.2 and the region where they give a power saving; the exponent of Guth and Maynard is drawn on $[\frac{7}{10}, 1)$, where it holds by their estimate and, above $\frac{4}{5}$, by Huxley's [279]. Middle and right: the density $v_\alpha/\pi$ at $\alpha = 0.6$ and 0.7 near the hypothetical off line zero at height 292.4 (dotted), with and without the hypothetical zeros, evaluated as in `almost.py` with step 0.005. Drawn by `figs_almost.py`. . . . . 481

12.1 The Thorin path to RH. The sine squared lemma connects the arithmetic to the Thorin density. The programme must carry the GGC property across the strip. . . . . 519

14.1 Histograms of $M_4$ simulated as a maximum of exponentials and as a weighted sum of exponentials. . . . . 585

14.2 The smoothed Thorin density $r_\eta(x)$ in the $(x, \eta)$ plane near $\gamma_5^2 = 1084.7$. Left: zeros on the line. Right: $\gamma_5$ moved off the line; the off line pair is an atom of negative mass at $(x_\rho, b)$. . . . . 589

14.3 The density $u_\alpha$ of (1.2.6), computed from $\operatorname{Re}\xi'/\xi$ on the line $\operatorname{Re} s = \alpha$ with no zero as input, for $\alpha = 1.5, 1, 0.7, 0.55$; for $\alpha = 0.7$ and 0.55 it is the Thorin density of $T_\alpha$ only if no zero lies to the right of the line, as under RH. Dashed lines mark $\gamma_k^2$. The term of a zero on the line has unit mass and is close to a Lorentzian of width $4(\alpha - \frac{1}{2})\gamma_k$ near its peak (Proposition 14.8.1); as $\alpha \downarrow \frac{1}{2}$ these terms converge to unit atoms at $\gamma_k^2$. . . . . 594
14.4 The smoothed Thorin density $r_\eta$ of $T$, $\eta = 3$, computed from $\xi'/\xi$ with no zero as input. Dashed lines mark $\gamma_k^2$. Each spike has height near $1/(\pi\eta)$ and unit mass. . . . . 597

15.1 The Thorin phase diagram. Solid curves are the boundaries of certified GGC regions carried along the flow by Theorem 15.3.2(i), and the dashed curve is that of the region conditional on the half plane of Corollary 15.3.3(d); the dotted curves are the level sets of $t + 2(\alpha - \frac{1}{2})^2$, labelled by the bound on $\Lambda$ that a point on them gives. RH is the point $(0, \frac{1}{2})$. Script `phase_thorin.py`. . . . . 621
15.2 $v_t(\alpha, \theta) = \operatorname{Re}\xi_t'/\xi_t(\alpha + i\theta)$, which is $\pi$ times the Thorin density of the shifted time $t$ clock when $(t, \alpha) \in \mathcal{G}$, on the lines $\operatorname{Re} s = 0.6$ and 0.75, at $t = 0$ (solid) and $t = 0.2$ (dashed), from the Pólya integral. Dotted lines mark the ordinates of the zeros of $\zeta$. Script `phase_thorin.py`. . . . . 623
15.3 Left: $\lambda_{\min}$ of the Weil Gram matrix against the support length $L$ for two resolutions. Right: a pair of zeros at height $g$ moved to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$, signed logarithm of $\lambda_{\min}$ for $w = 0.05$. The pair at $g \approx 17.6$ is detected at every $L$, the pair at $g \approx 54.7$ only at $L = 4$, and the pair at $g \approx 146.7$ not at all within $L \le 4$. Script `phase_weil.py`, output `phase_weil.json`. . . . . 631
15.4 Left: the minimum normalised gap and $\min_k \Delta_k^2 g_k$ along the flow, both monotone on the computed range; the second crosses the Lehmer threshold $\frac{4}{5}$ near $t = 0.17$. Right: $D^2$ for the closest pair below height 200, at ordinates 184.87 and 185.60, against the isolated pair law $D_0^2 + 8t$. . . . . 632
15.5 The bound $\Lambda \le 0.22$ of Polymath15 and the twelve rows of their table of conditional bounds [456] against the barrier $X$ (verification height $X/2$), with the fits $\Lambda = c/\log X$ and $\log X = a/\Lambda + b$. Dashed lines mark $\Lambda = 0.1$ and 0.05. . . . . 634

16.1 Left: the clock $U_{3/4}(\tau)$ of the basepoint law at $\alpha = 3/4$ for $0 \le \tau \le 60$, the zero count $N(\tau)$ (dotted), and the zeros of $\operatorname{Re}\xi(3/4 + i\tau)$, which sit exactly on the half integer levels (grey lines). Right: $\vartheta(t_k^*)/\pi - (k - \frac{3}{2})$ for the 138 zeros of Pólya's $\Xi^*$ below height 300, with the asymptote $-(2\pi^2 - 3)/(\pi t)$ of Proposition 16.7.2. . . . . 652

16.2 Left: $-(\log f_X)''(x)$ on a log scale, from direct quadrature on $[0, 0.95]$ (solid) and from the residue expansion with twelve zeros on $[0.95, 4]$ (dashed); it is positive throughout, which is Theorem 16.11.5. Right: the compensated clock condition $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + ir) - \frac{\xi'}{\xi}(\alpha) \geq 0$ of Theorem 16.11.7 for $\alpha = 1$ (solid), 0.75 (dashed) and 0.6 (dotted), computed from the Hadamard sum over the first 1700 zeros at step 0.02; $D_\alpha$ vanishes at $r = 0$ and is positive after, with a peak of height about $1/(\alpha - \frac{1}{2})$ at each ordinate. Drawn by `polya_fig.py`. . . . . 658

16.3 The function $\operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\tau)$, which is $\pi$ times the Thorin density of Theorem 16.12.1(i) in the ordinate variable, at $\alpha = \frac{7}{8}$ (solid), 1 (dashed) and $\frac{3}{2}$ (dotted) for $0 \leq \tau \leq 60$, computed from $\xi$ with no zeros (`pol_qrh_edge.py`). The zeros in this range lie on the critical line, and each contributes a Lorentzian of width $\alpha - \frac{1}{2}$ and height $1/(\alpha - \frac{1}{2})$, that is $\frac{8}{3}$, 2 and 1; at $\alpha = \frac{7}{8}$ the computed peaks are 2.69 to 2.93. On this range the minimum is at $\tau = 0$, where the value is $\frac{\xi'}{\xi}(\alpha) = 0.0173$, 0.0231 and 0.0461. Positivity at every height for $\alpha > \frac{7}{8}$ is Theorem 16.12.1(ii), under $(\mathrm{Q}_{7/8})$, and not a consequence of the figure. . . . . 669

17.1 The modulated heat trace $\mathcal{G}(u, \frac{x}{2}) + \mathcal{G}(-u, \frac{x}{2})$ against the window centre $u$ at $x = 0.02$ (solid), 0.1 (dashed) and 0.5 (dotted), from the first 1700 zeros; dotted verticals mark the ordinates of the zeros. 687

17.2 The least value of $D$ over windows of length 0.4 on $[0, 60]$, for the actual zeros (solid, the first 1700 zeros with the smooth tail; the least window minimum is 0.0427, in the window centred at $x = 25.5$) and with one hypothetical off line pair added at height $\gamma$ and displacement $\delta$, $(\gamma, \delta) = (100, 0.3)$ (dashed), $(100, 0.2)$ (dash dotted) and $(1000, 0.3)$ (dotted). Vertical dotted lines mark the predicted levels $x_* = 18.1$, 27.2 and 33.5 of Remark 17.6.3. Script `yo_tent.py`. 691

17.3 Left: the excess of the least eigenvalue of the damped Yoshida form over its floor, $\lambda_{\min}(\alpha, a)/(2b_\alpha) - 1$, against the window half length $a$, for $\alpha = 0.55$, 0.6, 0.75, 1 (finite sections of $\max(96, \lceil 24a \rceil)$ modes, logarithmic scales); it tends to zero, roughly like $a^{-2}$. Right: $\log_{10}$ of the finite section value at $\alpha = \frac{1}{2}$ for $N = \max(24, 24a)$ and $N = \max(12, 12a)$ modes, with the level of double precision and the curve $\exp(-10ae^{2a})$. . . . . 700

17.4 Left: the symbol $v_\alpha(\tau) = \operatorname{Re} \xi'/\xi(\alpha + i\tau)$ on $[0, 60]$ for $\alpha = 1$, 0.75, 0.6, 0.55 (logarithmic scale), with $b_\alpha$ as thin horizontal lines. Right: the smallest value of $v_\alpha$ in windows of length 10 on $[0, 2000]$, with the minimum above $\tau = 20$ circled. Above $\tau = 20$ the minima are 0.1901, 0.0968, 0.0389 and 0.0195 for $\alpha = 1$, 0.75, 0.6, 0.55, which are 8.2, 8.4, 8.4 and 8.4 times $b_\alpha$, all at $\tau = 27.67$, between the third and the fourth ordinate (the largest gap above the second); the lowest local minimum other than the one at the origin is lower, $4.90\, b_1$ at $\tau = 17.49$ for $\alpha = 1$ and $4.98\, b_{0.55}$ for $\alpha = 0.55$, in the gap between the first two ordinates. . . . . 700

18.1 Detection thresholds of Table 18.3 and of the displacements $\delta = 0.001$ and 0.003 of the text against the height $\gamma$ of the hypothetical quadruple (logarithmic axis), for $\delta = 0.001$, 0.003, 0.01, 0.05, 0.2, 0.45 (points of equal $\delta$ joined), with the horizon $\frac{1}{2}\log(\gamma/2\pi)$ dashed; the vertical axis is the half length $a$ of the window. Left: criterion (i). Right: criterion (ii), with upper bounds at the heights 1000 and 2000. The points are the thresholds refined by bisection to 0.005, so that they lie up to 0.05 below the entries of the table. Drawn by `code/zh_detect.py` (option `--figure`) from `data/zh_detect.json`. 715

19.1 The Thorin density $\rho_\varepsilon$ across the gap at height 9878 (gap 0.872) with a hypothetical quadruple at distance $\delta = 0.45$ inserted at its midpoint, at four levels $\varepsilon = 0.05$, 0.15, 0.25 and 0.4. It is positive at $\varepsilon = 0.05$ and $\varepsilon = 0.15$ and negative near the midpoint at $\varepsilon = 0.25$ and $\varepsilon = 0.4$. The vertical scale is symmetric logarithmic, linear within $\pm 0.1$. Script `tf_window.py` with the option `--plot`, data `tf_window.json`. . . . . 735

19.2 The visibility region $\varepsilon < \delta < \delta_{\mathrm{det}}(\varepsilon)$ (shaded band) at the midpoint of the 10000th gap. Solid: $\delta_{\mathrm{det}}$ from the actual zeros through $\xi'/\xi$; dashed: the midpoint lattice of Proposition 19.3.2 with the same gap; dotted: $\delta = \varepsilon$, the lower edge of the band. The marked point is $\varepsilon = 0.1895$, $\delta = 0.45$. Script `tf_visibility.py`. . . . . 737

20.1 The converged ladder on a logarithmic scale: $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$ against the window half length $a$ at $a = 0.25$, 0.5, 0.75, 1, 1.5, 2, 3, 4, for $\varepsilon = 0.5$, 0.375, 0.2, 0.1, 0.05 (five marker styles in black and grey), each with the value $2b_{1/2+\varepsilon}$, the limit where the hypotheses of Theorem 20.2.1 hold, as a dashed line labelled $m$, and for $\varepsilon = 0$ at $a = 0.25$ and 0.5 only (heavy line with stars). The values are those of Table 20.1; the legend and the axis write $y$ and $\lambda(a, y)$ for $\varepsilon$ and $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$. Drawn by `code/dc_ladder.py` (part `fig`) from `data/dc_ladder.json`. . . . . 752

22.1 The 31 zeros of $g_{9/2}$ with real part below 160 in the upper half plane of $z$ (circles), the escape law (22.12.1) (curve), and the lines $|\operatorname{Im} z| = \frac{9}{2}$ bounding de Bruijn's strip $|\operatorname{Im} w| \le \frac{9}{4}$ in the scale $w = z/2$ of $g$ (dashed). . . . . 822
22.2 The Thorin path: the first twelve atoms $\gamma_k(b)$ against $\alpha = \frac{1}{2} + b$ at the stored values $\alpha = 1, 0.9, 0.8, 0.7, 0.6, 0.55, 0.5$, from the unconditional rung $\alpha = 1$ to the critical line. . . . . 827

24.1 (a) The inverse first passage boundary $b$ of the reciprocal Xi law on $[0.003, 0.12]$, with the line of slope $-\sqrt{2}\,\gamma_1$ through its last node. (b) The boundary at small times against the asymptote $\log(1/t)/(2\sqrt{2})$ of Proposition 24.12.1, which it approaches only slowly. (c) Normalised boundary changes for off line pairs at the zeros 6, 20, 60, against the shift response $K$ (computed). . . . . 888
24.2 (a) The minimum over $t$ of the Thorin density $N'_\alpha(t) = \pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + it)$ of the base point laws on $[0, 50]$, attained at $t = 0$, for $\alpha = 1, 0.75, 0.6, 0.55$. (b) The values $N_\alpha(50)$ against $N(50) = 10$. Only these summary values are stored, so the curves on $[0, 50]$ are not drawn (computed). . . . . 894

25.1 Left: real and imaginary part of the zero of the Dirichlet polynomial of the Dyck chain near $\gamma_1$ for $n = 100$ to 12800 (dashed: $\gamma_1$, dotted: the real part $\frac{1}{2}$). Right: estimate of the second atom of the hitting time law after deflating the first, against the threshold $x$ (dashed: $\gamma_2$). Drawn from the stored data of `code/ev_g1_tail.py` by `code/ev_figs.py`; the panel of the source with the logistic Markov chain and its exceedance clusters has no stored data and is not drawn. . . . . 922
25.2 Error of the estimate $\sqrt{f(x)/\mathbb{P}(T > x)}$ of $\gamma_1$ from the Laplace inversion of $L$ (Talbot, 100 and 130 nodes) at $x = 0.18, 0.19, 0.20$. Drawn from the stored data of `code/ev_g1_tail.py` by `code/ev_figs.py`; the curve of the tail with the three atom expansion in the source is not drawn, since only these estimates are stored. . . . . 924
25.3 Supremum distance between the mean corrected truncation with knots $\gamma_1^2, \ldots, \gamma_K^2$ and the density of the reciprocal $\Xi$ law, against $K$ (dashed: the peak density 80). The values for $K = 5, 20, 40$ are the stored data of `code/ev_g1_tail.py`, those for $K = 1, 2$ are the printed values above; drawn by `code/ev_figs.py`. The density curves of the source, including the Gumbel limit of the maxima of $n$ exponentials, are not drawn. . . . . 925

35.1 The Davenport and Heilbronn Lévy density. Left: $K_{\mathrm{DH}}$ by the explicit formula and by the sum over zeros, with the off line terms in absolute value. Right: $K_{\mathrm{DH}}(u)e^{\gamma_1 u} - 1$ on a logarithmic scale; the quantity $K_{\mathrm{DH}}(u)e^{\gamma_1 u}$ tends to 1, from above for large $u$ since the first zero is simple and the second lies on the line. Beyond $u = 9$ the right panel shows rounding noise. . . . . 1235

35.2 Vertical curves at $\theta = \frac{1}{2}$ (computed, floating point; `code/bsy_vertical.py`, `data/bsy_vertical.json`, figure by `code/bsy_figs.py`). Left: $P(f > \lambda)$ (black) and $P(f < -\lambda)$ (grey) under the Cauchy law centred at $t_0$, for $\zeta$ at $t_0 = 0$ (solid) and Davenport and Heilbronn at $t_0 = 85.7$ (dashed), $\lambda \le 10$, logarithmic vertical axis. Right: the running balance $\mathbb{E}[\mathrm{clip}(f, -L, L)] - \log|h(1 + it_0)|$, which converges to the defect $D_{1/2}(t_0)$, for $\zeta$ at $t_0 = 0$ and 85.7 and for Davenport and Heilbronn at $t_0 = 85.7$, with the prediction 1.4407 dotted; window $|\tau - t_0| \le 200$, neglected Cauchy mass 0.0016 . . . . . 1245

35.3 Root mean square error against $N$ (from the 400 replications of Table 35.4; code/bsy_yks.py, data/bsy_yks_r400.json), drawn for $\theta = \frac{1}{2}$ and $\frac{7}{8}$ for plain sampling, YKS and YKS with control, with the fitted slopes in the legend. The dotted lines have slope $-2$ (anchored at the first YKS point), the rate of the theorem of Yakowitz, Krimmel and Szidarovszky for integrands in $C^2[0, 1]$, and slope $-\frac{1}{2}$ (anchored at the first plain point). . . . . 1247

35.4 Root mean square error against $N$ for the half disc identity (data of Table 35.5; figure by `code/bsy_figs.py`). The dotted lines have slopes $-\frac{1}{2}$, $-2$ and $-\frac{5}{2}$. . . . . 1252

35.5 The integrand $\log|f|\,\Delta\varphi$ on the box ($121 \times 121$ points, one grey scale for both functions; code/bsy_riesz2d.py). The cross marks the Davenport and Heilbronn zero. The large values are at the left and right edges of the box, where $\Delta\varphi$ is largest and $\log|f|$ is large; the picture does not show the zero, which is why the two look alike. Their integrals are 0 and 0.632. . . . . 1260

35.6 Root mean square error of $I(\varphi)$ against $N$ in two dimensions, 20 replicates, for $\zeta$ (left) and Davenport and Heilbronn (right), with the plain, Voronoi, Delaunay, jittered, grid and (Davenport and Heilbronn) deflated grid rules, the deflating zero being off by $1.4 \times 10^{-4}$. The dotted lines have slopes $-\frac{1}{2}$ and $-\frac{3}{2}$ (code/bsy_riesz2d.py, data/bsy_riesz2d.json, figure by code/bsy_figs.py). . . . . 1261

# List of Tables

1.1 Principal results of Chapters 2 to 5. . . . . 6
1.2 Principal results of Chapters 6 to 8. . . . . 7
1.3 Principal results of Chapters 9 to 11. All are unconditional, and several use verified computations: the verification of RH below height 1000, through Theorem 5.1.1, or below the height $T_0 = 3 \times 10^{12}$, Platt's verification of GRH for small moduli, or computations in ball arithmetic. . . . . 9
1.4 The heat trace from the zeros and from the explicit formula, after [424, Table 4.1]. At $t = 1$ the cancellation between the arithmetic terms reaches about eighty seven digits. . . . . 14
1.5 The bracket $v_\alpha(\theta)$ of the sine squared lemma computed from the canonical measure and from the Pólya density; $\theta = 14.1347$ is the first ordinate $\gamma_1$, rounded. The Pólya values agree with an independent evaluation of $\mathrm{Re}(\xi'/\xi)$ through $\Gamma$ and $\zeta$ to about $10^{-15}$. . 17
1.6 The Bondesson hierarchy on the clock, whose Lévy density is $W(t)/t$. 20
1.7 Diagnostics for the Thorin condition. The pole test, the Thorin density and the zero density scan locate a failure in height; the others detect it only globally. . . . . 21
1.8 GGC Laplace transforms in the variable $u$, their Thorin measures in $z$, and the corresponding laws. . . . . 22

2.1 Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x| \le 6$; the first two rows are the lines $\eta = 0.1$ and $\eta = \pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts. . . . . 31
2.2 The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of $\xi'/\xi$, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height. . . . . 32

2.3 Working precision needed by the untilted and the tilted Pólya route for $D$ target digits. The untilted bits are $(D+\pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D+\delta\theta/\log 10+5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \geq 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \leq 8.8\times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \leq 3.7$, the smallest that suffices. . . 32
2.4 The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached. . . . . 35

3.1 The four pieces of the sine squared decomposition (3.1.1), with the prime sum over $n \leq 2\times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond $2\times 10^6$ ($1.3\times 10^{-3}$ at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece. . . . . 45
3.2 Distances between the Pólya computed Thorin measure $\mu_\varepsilon$ and $U_{\mathrm{RH}}$ on $0 < z \leq 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.2.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \leq 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, $L^1$ in $\theta$ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row. . . . . 48
3.3 Proposition 3.4.2 at $T = 200$ by the Pólya route (computed, not certified). . . . . 56

4.1 The dictionary. In the Riemann column the Mellin transform (Section 1.2 and Theorem 4.3.25) and the reciprocal Xi function are unconditional, the log variable and the Thorin measure presume RH, and the exit time, the hitting time of Kent's theorem (Section 8.1.3), presumes RH and simple zeros. The last entry is the necessary condition of Theorem 4.3.27, which assumes RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$; sufficiency is open (Question 4.3.28). . . . . 69
4.2 Where the lower bound (4.3.4) becomes positive. On a logarithmic grid from $t_e$ to about $10^{14}t_e$, $m_p(t) > 0$ at every grid point $t > t_1$; the ray reaches $t_1$ at $\tau_1 = t_1^{2/p}$. The entries are computed, not proved. 83
4.3 HCM threshold for powers of $W$ in logistic models with frequencies removed, computed as $\beta_* = \pi/h_*$ from the width $h_*$ of the largest strip on which $\mathrm{Im}(-g'/g) \geq 0$. The half Cauchy row is Bosch's theorem, not a computation. . . . . 89

5.1 Normalised leading Hankel minors of the heat trace, $\det[c_{i+j}]_{i,j<N}/\prod_{i<N} c_{2i}$ for the even block and $\det[c_{i+j+1}]_{i,j<N}/\prod_{i<N} c_{2i+1}$ for the odd block, from the first 1700 zeros in sixty digit arithmetic. . 100

5.2 The lowest ordinates from the Stieltjes moments $m_0, \dots, m_{2N-1}$, computed from the Taylor coefficients of $\Xi$ at the centre: the number returned by the Jacobi matrix of order $N$ to within $10^{-6}$, the errors of the first and of the last of them, the last Cholesky pivot of $\mathcal{H}_N$, and the least number of digits, in steps of 10, of the Taylor coefficients and of the arithmetic from which on the same ordinates are returned. . . . . 102

5.3 The Gaussian smoothing of the zeros from the zeros and from the primes, by (5.1.3). $\mathcal{G}(u,t)$ is summed over the zeros below height 2197.26, the rest adding less than $10^{-70000}$; the polar term and $\mathcal{G}(-u,t)$ are below $10^{-8000}$. $P_N$ is $P$ with the sum cut at $n \le N$, and the $P$ column is $P_{10^9}$. The bound on the rest of the prime sum that ignores its phases, from $\psi(x) < 1.03883x$ [465, Theorem 12], is $1.6 \times 10^{-20}$, $2.1 \times 10^{-10}$, $1.0 \times 10^{-8}$ and $1.0 \times 10^{-4}$ at $N = 10^9$ for the four rows; in the first row the difference at $N = 10^9$ is the rounding error of the sums, in extended precision. . . . . 112

5.4 The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, $g$ the midpoint of the pair. $k^*$ is the least order at which $c_k^F(t) < 0$ for some $t > 0$, found by the scan above (computed, not certified); $c_{k^*}^F(t^*) < 0$ is certified in ball arithmetic. The last column is $c_{k^*}^F(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$. . . . . 119

5.5 What finite tests establish and what they cannot exclude. . . . . 121

5.6 The heat trace from the explicit formula and from the zeros. For $t \ge 0.05$ the digits lost are $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10 \approx 0.30 + 86.88\,t$. 128

6.1 Thorin measures of $\varphi$, $G$ and $R$ and of the building blocks used in this chapter and in Chapter 7, with the convention $\log f(s) = -\int \log(1 + s/t)\, U(dt)$ after normalising $f(0) = 1$. Here $b = \frac{1}{2}(\gamma_E + \log \pi)$, $\kappa > 0$ and $c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$. The rows for $\varphi$ and $R$ are Theorem 6.1.10, the row for $G$ is Theorem 6.2.1, the fourth row is Lemma 6.1.7, the fifth and sixth appear in Table 1.8, and the last is Theorem 6.2.12. . . . . 137

7.1 Argument principle counts for Conjecture T: rectangles searched and zeros of $Q_n$ found with $\operatorname{Re} u > \frac{5}{4}$. . . . . 190

7.2 Zeros of the symmetrised truncations $K_n$ in $-4 < \operatorname{Re} w < 4.5$: total found, how many lie on the line, and the number of zeros of $\xi$ in the doubled window for comparison. In the window $195 < \operatorname{Im} w < 215$ the count for $K_8$ exceeds that for $\xi$ by one; see the text. . . . . 194
7.3 Bounds for the torus extremes over the band $1.2 \le x \le 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 7.2.11 holds with these values. The bounds are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 7.2.13); the programs are in `verification_reciprocal_zeta/torus/`. . . . . 198
7.4 As Table 7.3, for the single band variant: bounds for the extremes over the band $1 \le x \le \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values. . . . . 198
7.5 Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none. . . . . 201
7.6 Smallest order $n^*$ at which an inserted off line quadruple at height $\gamma$ and distance $\delta$ makes $\min_t L_n[W](t)$ negative. . . . . 204
7.7 Constants used in Sections 6.1.4 and 7.1.1. . . . . 207

8.1 The saddlepoint density of the clock against the Talbot inversion of its Laplace transform, both from $\xi$ alone. . . . . 215
8.2 Prony extraction of the atoms from $P(T > t)$ on $t \in [0.03, 0.108]$. Identical results for relative noise $10^{-16}$, $10^{-30}$ and $10^{-45}$: the error is model truncation, not noise. . . . . 215
8.3 Monte Carlo estimates of $\Xi(z)$ from Williams' representation with $10^6$ samples. The last column is the sample size at which the standard error equals $|\Xi(z)|$. . . . . 216
8.4 Relative error of the clock transform: partial product over $N$ zeros, $|G/G_N - 1|$, and with the Weyl tail, $|G_N^{\mathrm{W}}/G - 1|$. The two measures of error agree to first order, and $\gg 1$ marks an error above $10^3$. The partial product values agree with [424, Section 18.2.1] where both are given. . . . . 217
8.5 Relative error $|G_{\mathrm{a}}/G - 1|$ of the approximant $G_{\mathrm{a}}$ of $G$ with the Weyl tail and with the moment matched tail. The cutoffs are $\Theta_N^2 = 3.99 \times 10^4$, $2.95 \times 10^5$ and $4.83 \times 10^6$ for $N = 79$, 300 and 1700. . . 218
8.6 Keiper Li coefficients from the Pólya density (all digits shown are correct) and Keiper's leading asymptotic, which becomes accurate only for large $n$. . . . . 226
8.7 The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; $t$ is the de Bruijn Newman time. . . . . 233

8.8 Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \dots, m_{59}(t)$; at $t = 0$ they are $\gamma_1$, $\gamma_2$, $\gamma_3$, $\gamma_5$ and $\gamma_{10}$. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \dots, \theta_{12}(t)$ to within $3.1 \times 10^{-7}$ of the zeros that Newton's method finds on $H_t$, and $\theta_{13}(t)$ only to within $5.2 \times 10^{-4}$. . . . . 234

8.9 The first failing order $n_{\text{fail}}(t) \le 50$ of the Hankel test along the flow on the grid of step 0.01 and, at one time on each step, the least order at which the pair that starts at $(\gamma_k, \gamma_{k+1})$ breaks the test alone (real: the pair has not yet collided; $> 50$: no order up to 50 fails). The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone. . . . . 236

8.10 The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k+1)$ by the zeros $\gamma_k$, $\gamma_{k+1}$ it starts from, and its position $\theta$; the number of real zeros below it; the number of collisions so far among $\gamma_1, \dots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\text{fail}}(t)$ and $n'_{\text{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30. . . 238

8.11 The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time $t_{\text{coll}}$ from the census of Section 8.4.3 (`dbn_collisions.py`), against predictions computed from $H_0$: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1} - \gamma_k)^2/2$, the critical point and Biane forms $\tau_c$ and $\tau_B$, and the frozen predictor of Section 8.4.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of $H_t$. . . . . 241

8.12 Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13, 14)$ and $(19, 20)$, which collide at $s = 1.2853$ and 1.2755, against the bounds of Theorems 8.4.7 and 8.4.9; a bound is shown only where it is finite, for $s < 1/\ell(0)$ (0.9064 and 0.7719) or $s < 1/m(0)$ (1.0334 and 0.9573). Units of $H_t$; `wigbiane_flow.py`. . . . . 249

8.13 Two sided brackets for the collision time of seven pairs, in the units of $H_t$: the lower bound $1/m(0)$ of Corollary 8.4.13, whose hypothesis holds on the grid, $\bar{\delta} < 0$, and the upper bound $s^*$ of Corollary 8.4.15, with $\rho$ the largest value of $Q$ and $\bar{\kappa} = 0$ for these pairs. The correction $\bar{\eta}$ of Theorem 8.4.12 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision. `wigbiane_flow.py`. . . . . 249

8.14 Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [463, Table 1], upper bounds from [112], [310], [456] and [419]. Odlyzko's bound is the one computed in [394, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication. . . . . 252
8.15 The zero density identity for $L(s, \chi_{12})$ at height 100, computed from the eta kernel. The last column is $N_{\chi_{12}}(100) - 2\nu(\frac{1}{2} + \varepsilon, 100)$; the row $\varepsilon = \frac{1}{2}$ gives $N_{\chi_{12}}(100)$. . . . . 265
8.16 The collisions of zeros of $H_t^{\Delta}$ below height 52, in the order in which they occur as $t$ decreases: the pair at $t = 0$, the isolated pair estimate with $D$ the gap of the pair, and the time $t_*$ and height $\theta_*$ of the collision, computed by `dbn_delta.py`. . . . . 286
8.17 The Pólya kernel $\Phi_K$ of $K = \mathbb{Q}(\sqrt{d})$ for the discriminants named in the first author's note: the characters at 2 and 3, the sign of the kernel, the least value of $H_K$ on $[1, \sqrt{|d|}/\pi]$, which decides it by Theorem 8.5.18(iv), with the point where it is attained, and $\zeta_K(\frac{1}{2}) = \zeta(\frac{1}{2})L(\frac{1}{2}, \chi_d)$. Computed by `sqfree_quadratic.py`. . . . . 290
8.18 Positive Pólya kernels $\Phi_K$ among the 6079 fundamental discriminants $-20000 \le d < 0$, by the splitting of 2 and 3 in $K = \mathbb{Q}(\sqrt{d})$ ($\chi_d(p) = 1$ split, 0 ramified, $-1$ inert): the number of positive kernels over the number of discriminants in each class, 2738 positive in all. The empty corner is Theorem 8.5.19. Computed by `sqfree_quadratic.py`. . . . . 290
8.19 The height $t_0(\alpha)$ above which the bound of Proposition 8.5.26 certifies $v_\alpha > 0$, for basepoints below $\alpha_c(\chi_4) = 2.2823$; the last column tends to $-2\gamma_E - \log\frac{8}{\pi} = -2.0891$ (Proposition 8.5.27). Computed by `dbeta_thorin.py`. . . . . 297
8.20 The brackets of $\xi_{\mathbb{Q}(i)}$, $\xi$ and $\xi(\cdot, \chi_4)$ at the basepoint $\frac{3}{2}$; the last row is the difference of the first two (Proposition 8.5.28). Computed by `dbeta_thorin.py`. . . . . 299
8.21 The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15, 55]$: for each zero $\gamma_k$ in the data window, the centre of mass $c_k$ of its cluster, the error of the centre and the total mass of the cluster. . . . . 305
8.22 Nonnegative deconvolution of $\rho_{0.1}$ with the signature $g$ of a displaced fifth zero added to the data. Here $r$ is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$. . . . . 306
8.23 Left: the positive zeros below 60 of $\Xi$, of Pólya's $\Xi^*$ and of $P(\theta) = K_{i\theta/2}(2\pi)$. Right: the numbers of zeros in $(0, T]$ against $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Computed by `biane_polya.py`. . . . . 308
8.24 Index side products for $P$, $\Xi^*$ and $\Xi$ . . . . . 319

8.25 The clock from $R_0 = 2\pi e^{2.5}$: exact moments against 100000 Euler paths with step $4 \times 10^{-6}$, with and without the bridge correction, and the Laplace transform, with the $z$ scores (Monte Carlo − exact)/standard error. The standard error of the mean is $4.75 \times 10^{-5}$; over $0 < u \le 400$ the largest $|z|$ of the corrected transform is 0.72. Computed by `biane_clock.py`. . . . . 322
8.26 Pólya's function and $\Xi$. Apart from the computations of the last row, the left column is a list of theorems; in the right column the entries that concern the zeros are equivalent to RH or open. . . . . 329

9.1 The identity (9.1.1) of Biane, Pitman and Yor, checked with $4 \times 10^5$ draws of $S_2$ by `comp.py`. The sample mean of $S_2$ is 0.6664 against $\mathbb{E}S_2 = \frac{2}{3}$. . . . . 336
9.2 The Thorin density $\rho_{1/2} = v_1/\pi$ of $T_1$ at the basepoint one, computed from $\xi'/\xi$, from the arithmetic form $v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$ with $\zeta'/\zeta$ evaluated exactly, and from the Poisson sum over the first 1700 zeros with $\beta = \frac{1}{2}$. The zero sum misses the zeros above height 2198, whose contribution is a nearly constant $1.6 \times 10^{-4}$. . . . . 338
9.3 The error of the compensated sum (9.1.8) at $\theta = \gamma_1$ with primes up to $X$, its prediction from the explicit formula (9.1.9) over the first 1700 zeros, and what remains after subtracting the prediction. . . 354
9.4 Constants of Theorem 9.1.18. For a comparison line $\alpha' > 1$, $t_1$ is the root of $A_f^{\infty}(\alpha', t) = S_f(\alpha') + (\alpha' - \frac{1}{2})V_f$, found by bisection and rounded up to three decimals, at which the strict inequality is certified in ball arithmetic; the proof needs the convexity margin to be positive. The first comparison line is that of the first author's note; the minimiser of $t_1$ is located on a grid of step 0.005; $\alpha_{\max}$ is the largest $\alpha'$ in steps of 0.05 keeping the margin above 0.02. For $\xi$ the range $\alpha > \alpha_{\max}$ is covered by Proposition 9.1.12. Computed by `dbeta_minimum.py`; the ordinates of $L(s, \chi_4)$ and $L(s, \chi_{12})$ are certified by `dbeta_zeros.py`. . . . . 360
9.5 What the primes show about the clocks $T_\alpha$ . . . . . 365
9.6 Basepoints above the centre: the drift $\frac{\xi'}{\xi}(\alpha)$, the tail constant of $T_\alpha$, and the minimum on $[0, 60]$, attained at the origin, of the density $\rho_{\alpha-1/2} = v_\alpha/\pi$, the Thorin density of $T_\alpha$ when every zero lies to the left of the line $\operatorname{Re} s = \alpha$. At $\alpha = 1$ the minimum at the origin holds at every height (Proposition 9.1.11), as it does at every $\alpha \ge 1$ (Proposition 9.1.12); for $\frac{1}{2} < \alpha < 1$ the minimum at the origin holds on the verified window $|\theta| \le 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)), and at every height for every such $\alpha$ it is equivalent to RH (part (vi) of that theorem). . . . . 375

9.7 Exact Brownian boundaries above the centre: the flat level, the mean boundary value $\mathbb{E}\, b(T_\alpha)$ at the crossing, obtained by integrating the computed boundary against the law of $T_\alpha$, against the Wald type identity, and the computed $\sqrt{100}\, P(T_\alpha > 100)$ against the tail constant of Theorem 9.2.5. Computed by `bnd_alpha.py` for $\alpha = 0.6$ and 0.75 and by `rs1.py` for $\alpha = 1$. . . . . 376

10.1 The Thorin form (10.1.2) against the reciprocal Xi function on the real axis, computed by `comp.py`. . . . . 382
10.2 The EGGC representation in the strip, computed by `strip.py`: the Thorin integral over $\theta \le 200$ plus the Weyl tail, and the product of the factors of Theorem 10.1.3(3) over the 1700 computed zeros plus the Weyl tail. . . . . 384
10.3 The first order at which the term of a hypothetical zero $\frac{1}{2} - \delta + i\gamma$ in (10.1.3) is negative, against Proposition 10.1.7. The last row is a zero at the edge of the strip at the height of the first ordinate. Computed by `biane_traces.py`. . . . . 389
10.4 The Poisson trace $K_{1/2}$ of the zeros of $\zeta$, Pólya's trace $K^P$, their distances from the common main term $M$, which tend to $\frac{7}{8}$ and $-\frac{1}{4}$, and their difference $D$, which tends to $\frac{9}{8}$, against the prediction of Proposition 10.1.11 under RH. Computed by `biane_traces.py`. . . 393
10.5 Best $L^1$ error in approximating a Gaussian of width 0.15 by $N$ translates of $\Phi_\alpha$, computed by `wiener.py`. At the centre the error cannot fall below $|\hat{g}(\gamma_1)| = 0.1056$, and $\hat{h}(\gamma_1)$ vanishes exactly; the computed value is $1.2 \times 10^{-8}$. . . . . 396
10.6 Left: the Riesz coefficients from the clock, Proposition 10.2.3. Right: the Riesz function from the Möbius series, which agrees with the power series to fourteen digits, and from the clock average of Proposition 10.2.4. . . . . 403
10.7 Riesz coefficients as killing probabilities of the basepoint one clock, and the stable factor as a Brownian hitting probability; $E$ is an independent exponential clock of rate $(2k-1)^2$, and the errors are one standard error of $10^6$ draws. . . . . 408
10.8 The sharp counts: the number $Q(x)$ of squarefree integers up to $x$, the normalised error $\Delta(x)/x^{1/4}$, the error $\mathcal{E}_1(x)$ of (10.3.9), which confirms the constant $C$, also multiplied by $x^{3/4}$, and the error $\mathcal{E}_{3/4}(x)$ of (10.3.4) at $\alpha = \frac{3}{4}$ multiplied by $x^{1/4}$. Computed by `sqfree_sieve.py`; the sums of reciprocals are checked in extended precision. . . . . 415

10.9 Root mean square and largest absolute value of the squarefree error $\Delta$ on the ranges $[X, 2X]$, divided by $X^{1/4}$, over real $x$; the largest value on $[10^7, 2 \times 10^7]$ is the left limit at 17199381, and over the integers alone it is 0.982. Computed by `sqfree_sieve.py`. . . . . 423

10.10 The Gamma(2) smoothed count: $Q_2(x) - 6x/\pi^2$, the sum (10.3.13) truncated at $d \le 2 \times 10^5$, the normalised error, the contribution $2\,\mathrm{Re}\sum_{k\le 100} B_{\rho_k} x^{\rho_k/2}$ of the first hundred atoms, the residues at $s = -1, \dots, -6$, whose leading part is $2c_1/x$, and the residual. The constant 1 is the residue $D(0)$ at $s = 0$. Computed by `sqfree_smooth.py` at 45 digits. . . . . 427

10.11 Gamma($\nu$) smoothing above the threshold of Theorem 10.3.18: the limit $-\zeta(-\nu)$ of $q_\nu$ at 0, the least value of $1 - \mathcal{R}_\nu$, which has the sign of $f_\nu$, with the point where it is attained (grid search refined by golden section), and the maximal runs of grid points on which $1 - \mathcal{R}_\nu < 0$. Computed by `sqfree_gammanu.py`. . . . . 430

11.1 The heat trace $W$ and the Lévy density $h = -W'$ of the driving process of the centre clock, from the first 1700 zeros and from the explicit formula (1.2.5) and its derivative with the primes below $2 \times 10^4$. Computed by `weil.py` in the folder `sato_bondesson`. . . 433

11.2 The subexponential principle for $T_1$: the tail of the law is asymptotic to the tail of its Lévy measure. The ratio is that of the second column to the third, and the last column is the common asymptote of (11.1.2). Computed by `comp.py` in the folder `sato_bondesson`. . 435

11.3 What the books of Sato and Bondesson, and the papers around them, give the reciprocal Xi function. Theorem numbers, here and in the text, refer to the first edition of Sato's book [471]. . . . . 441

11.4 The processes by basepoint, with $a = \alpha - \frac{1}{2}$. (RH) marks statements that hold under RH, and the Kent strings need the zeros to be simple as well. The Mellin row gives the size bias of $S_2$ that produces $V_\alpha$ (Section 9.1.1); $\widetilde{T}_\alpha$, $H$ and the sticky clock are those of Theorem 9.2.5 and Corollary 9.2.3, with $\widetilde{T} = \widetilde{T}_1$; and the boundaries $b$ and their Wald defects are those of Sections 9.2.4 and 9.2.6. The boundary row reports computations at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 made under RH beyond $t = 0.03$, and its entry for $\alpha > 1$ is extrapolated, not computed; the flatness is observed, not proved, and above the centre the Wald defect $b_\alpha/\sqrt{2}$ requires the boundary to be bounded at infinity (Lemma 9.2.4). All other entries hold as stated. . . . . 445

11.5 The Laplace exponent of $\Upsilon(T)$, which is minus the free cumulant transform of $\Lambda_{\mathrm{BP}}(T)$ at $-s$: the integral of (11.2.8) from $\xi$ alone, and the sums over the first 300 and 1700 zeros, with no tail, with the tail of order one and with the full smooth tail, as differences from the integral. Computed by `wigfree_upsilon.py`. . . . . 454

11.6 The clock $T$, its Upsilon image and its free counterpart, from the densities of Figure 11.3. The exact means are $m_0 = 0.023104993$; the variances $m_1 = 3.71726 \times 10^{-5}$ for $T$ and $\Lambda_{\mathrm{BP}}(T)$ and $2m_1 = 7.43452 \times 10^{-5}$ for $\Upsilon(T)$; the third central moments $2m_2 = 2.8835 \times 10^{-7}$ for $T$ and $\Lambda_{\mathrm{BP}}(T)$ and $12m_2 = 1.7301 \times 10^{-6}$ for $\Upsilon(T)$. The modes of $T$ and $\Upsilon(T)$ are grid points of spacing 0.0005, that of $\Lambda_{\mathrm{BP}}(T)$ of spacing about $7 \times 10^{-5}$. Computed by `wigfree_upsilon.py`. . . . . 454

11.7 The Jacobi parameters of $\nu$ against the local semicircle model: $\alpha_n$ and $2\beta_n$; half the right end $\frac{1}{2}L_n$ of the local interval from (11.2.13) with the smooth density and $n(L_n) = n$, and the ratios to it; the same with the midpoint rule $n(L_n') = n - \frac{1}{2}$; the cruder rule $\frac{1}{2}\gamma_{2n/\pi}^{-2}$, with $N_0(\gamma_r) = r$; and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac{1}{4}$ in the local model. Computed by `wigfree_jacobi.py`. . . . . 458

11.8 The first order $N$ at which the moments $\mu_k = \mathbb{E}Y^k/(a)_k$ fail the Hankel test for a Gamma($a$) scale mixture, for $Y = |X|$, $X^2$ and $S_2$: the Hamburger test uses $[\mu_{i+j}]_{i,j<N}$ alone, the Stieltjes test also $[\mu_{i+j+1}]_{i,j<N}$. Computed by `wigfree_gamma2.py`. . . . . 462

11.9 The Thorin density $v_1/\pi$ of $\xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$, computed from $\xi'/\xi(\cdot,\chi_{12})$ and from the Poisson sum over the 297 zeros of $L(s,\chi_{12})$ below height 340 with $\beta = \frac{1}{2}$. . . . . 466

11.10 The Lévy form of Proposition 11.3.2 against $\log(\xi(1+s,\chi_{12})/\xi(1,\chi_{12}))$, with primes up to $X$. . . . . 468

11.11 The sine squared formula below the basepoint one for $L(s,\chi_{12})$, with primes up to $X$: the prime piece, the error against $v_\alpha(\theta)$, and the error left after subtracting the explicit formula over the 297 computed zeros; $\gamma_1 = 3.8046276$. . . . . 469

11.12 The two basepoint one examples side by side: the reciprocal Xi function of Chapter 9 and the reciprocal completed $L$ function of $\chi_{12}$ of this section. . . . . 471

11.13 A single hypothetical pair of zeros at real parts $\beta$ and $1-\beta$ inserted into the density $v_{0.6}/\pi$, computed from $\xi'/\xi$ on a window of width 8 at the heights 300 and $10^5$: the minimum of the density and the negative mass $\int \max(-v_{0.6}/\pi, 0)$ on the window. Computed by `high.py`. . . . . 479

11.14Negative and positive mass of the density $v_\alpha/\pi$ on $[0,T]$ with fifteen hypothetical quadruples of zeros at real part 0.75. Computed by `almost.py`. . . . . 479
11.15Quantiles of the normalised Thorin density $\bar{\rho}_\kappa$ at the midpoints between consecutive unfolded ordinates, for the zeros below height 2198 and for CUE eigenvalues. Computed by `sens.py`. . . . . 481
11.16Moments of $\zeta$ on the critical line against the asymptotic of Hardy, Littlewood and Ingham for $k = 1$ and the leading Keating and Snaith terms for $k = 2$ and 3. Computed by `ks.py`. . . . . 483
11.17The ladder functions of $R_t$ along the de Bruijn Newman flow, on the window $0 \le -\log x \le 8$: the first zero $z_1(t)$ of $H_t$, the ratio $a_2/a_1$ of the first two Thorin atoms, the ratios $W^{[2]}/W$ and $W^{[3]}/W$ at $-\log x = 8$, the sets of $-\log x$ on which $W^{[3]}(x;t) < 0$ and $W^{[4]}(x;t) < 0$, to the grid step 0.001, and the largest $n$ with $W^{[1]}, \dots, W^{[n+1]} \ge 0$ on the window: $R_t$ lies in no class above $L_n$, and in $L_n$ if the signs persist for $x < e^{-8}$. On the window $W$, $W^{[1]}$ and $W^{[2]}$ are positive and $W^{[5]}$ takes negative values for every $t$. Numerical for $t > 0$; the row $t = 0$ is Theorem 11.5.8. Computed by `heat_ladder.py`. . . . . 500
11.18The GGC test for the tilted variables $\mathcal{B}_t$ of Section 11.5.5.3: the least order $n \le 12$ at which $\mathcal{K}'_n(\lambda)$ has a negative eigenvalue, and the least eigenvalue of the scaled matrix $\mathcal{K}'_n(2)$ at that order; for $t = 0$, where no order fails, the last column gives $\mathcal{K}'_7(2)$. At $\lambda = 200$ no matrix of order at most 12 fails, for any $t$. Every negative entry of the last column is certified, and so is the entry for $\mathcal{K}'_9(0)$ at $t = -\frac{1}{100}$, $-2.899 \times 10^{-12}$. Computed by `pakes_filter.py` and `pakes_filter_cert.py`. . . . . 508
11.19The GGC test for the variables $\mathcal{B}_a$ of Section 11.5.5.4 and their reciprocals: the sum $S_a$ of Lemma 11.5.17, the least eigenvalues of the scaled matrices $\mathcal{K}'_7(0)$ and $\mathcal{K}'_7(2)$ of $\mathcal{B}_a$, and the verdicts. "Passes": no matrix $\mathcal{K}_n(\lambda)$ or $\mathcal{K}'_n(\lambda)$ with $n \le 12$ has a negative eigenvalue at $\lambda = 0$, $\frac{1}{2}$, 2, 10, 20, 50 and 200. "Fails at $\mathcal{K}'_n(\lambda)$": the least failing order, certified at that $\lambda$. "Zero": the Laplace transform vanishes on $(-\infty, 0)$ (Proposition 11.5.19). GGC for $\xi$ by (11.5.8), for $1/\mathcal{B}_a$ with $\chi_{-3}$ and $\chi_{-4}$ by Proposition 11.5.19. All the functions but $f_{\mathrm{DH}}$ have an Euler product. Computed by `pakes_filter.py` and `pakes_filter_cert.py`. . . . . 511

12.1 The path, step by step. . . . . 518

13.1 Ferromagnetic laws and their clocks. Here $j_{\nu,k}$ are the positive zeros of the Bessel function $J_\nu$, and $\tau_\nu$ is the hitting time of 1 by a Bessel process of dimension $2\nu+2$ started at 0. In the fifth row $\nu_k$ are the zeros of $\nu \mapsto K_{i\nu}(2\pi)$ and $T_P$ is the clock of Theorem 8.7.9. . . . . 532
13.2 Which laws on $[-1,1]$ are ferromagnetic. Every entry is proved in this section or in Section 13.1. . . . . 540
13.3 Even cumulants of the Pólya law, by quadrature of the moments of $\Phi$, from the Taylor coefficients of $\log \xi(\frac{1}{2}+s)$, and from the sums $(-1)^{m+1}\frac{(2m)!}{m}\sum_k \gamma_k^{-2m}$ over the first 1700 ordinates with the smooth count beyond them. The difference $1.3\times10^{-10}$ in $\kappa_2$ is the error of the smooth count, which ignores the oscillation of $S(t)$ above height 2197. Computed by `fm_polya.py`. . . . . 545
13.4 The moment test (13.2.2) for the Pólya law: the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ and the Turán ratios, against the bound $1+\frac{1}{j}$ that every ferromagnetic law satisfies. All 24 computed ratios exceed the bound. Computed by `fm_polya.py`. . . . . 546
13.5 Laws of Theorem 13.5.4 fitted to the Pólya law. The last three rows keep Pólya's kernel $e^{-2\pi\cosh 2t}$ (Remark 13.5.7). The parameters are rounded to seven significant digits from a numerical minimisation (`wf_fit.py`) and then fixed; the distances are upper bounds enclosed in ball arithmetic and rounded up (`wf_tvcert.py`). . . . . 554
13.6 The zero completed laws $f_K$. For $K \le 16$, $f_K$ is a ferromagnetic probability density by Theorem 13.5.12(i) ($s_K \le 2\beta_K$). For every $K \ge 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so $f_K$ is a ferromagnetic probability density by Theorem 13.5.12(ii), and $\min Q_{K,2\beta_K}$ is the value at $y=0$, where it is attained. The threshold $v^*(K)$ of Proposition 13.5.13 lies in a certified interval of width at most $1.2\times10^{-7}$, within $6\times10^{-7}$ of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. Computed by `fm_qcert.py` and `fm_tvcert.py` from the certified zeros and variance, except $v^*(1)=s_1$. . . . . 560
13.7 $-\log|\mathbb{E}e^{iye^{2X}}|^2$ computed four ways: by a direct Fourier integral of the density $y^{1/4}f_0(y)/(2\xi(\frac{1}{2}))$; in the Thorin form with the atoms $\frac{9}{4}$ at the points $\pi m^2$ summed in closed form and the continuous part from the computed $\theta_{1/4}$; in the theta Lévy form $\int 4\sin^2(\frac{yx}{2})k_{1/4}(x)\,dx/x$; and from the atoms alone, which is $-\log|\mathbb{E}e^{iy\frac{\pi}{2}S_{9/4}}|^2$. Computed by `fm_sizebias.py`. . . . . 571
13.8 Knauf's chain on the line $\operatorname{Re}\beta=\frac{3}{2}$, at $\beta=\frac{3}{2}+it$. Each entry is $|Z_N(\beta)-\zeta(\beta-1)/\zeta(\beta)|$ followed, after a slash, by the profile term $|W_N(2-\beta)|/\zeta(2)$; computed. . . . . 575

13.9 Mean distance from $1+\rho_n$, over $n \le 20$, of the zero of Knauf's partition function $Z_N$ and of the zero of the corrected sum $C_N$ in the disc of radius 0.3 about $1+\rho_n$; computed. . . . . 576
13.10 The Farey chain in a field: the computed Lee and Yang threshold $\beta_{\mathrm{LY}}(N)$, the angle $\theta_N$ of the double zeros at the threshold, and the root $\beta^*(N)$ of $2^\beta Z_N^{\mathrm{F}}(\beta,1)=4$, above which Lemma 13.7.5 proves the property; computed. . . . . 578

14.1 The dictionary between the trigonometric side, where everything is classical, and the Riemann side developed in this chapter. The variable $\theta$ is that of Section 1.2; ID and SD abbreviate infinitely divisible and self decomposable, and the Wald couple of the third and fourth rows is that of Theorem 14.4.2 and Remark 14.5.1. . . . 584
14.2 Zeros recovered as atoms of the smoothed Thorin measure, $\eta=3$. The integral of $r_\eta$ over $[50,1700]$ is 6.95, against 7 zeros. . . . . 597
14.3 Zeros of the unsymmetrised finite stage $\xi_N$. For $N=16$ and $N=32$ the zero listed near $\rho_5$ does not track $\rho_5$, since $\rho_5$ lies above the height $4\sqrt{N}$ discussed below. . . . . 600
14.4 Real zeros $t_k$ of the symmetrised stage $\mathrm{Re}\,\xi_N(\frac12+it)$: the column "zeros" is the number of sign changes on $[10,44]$, where $\xi$ has 8 zeros ($10\le\gamma\le44$). At $N=16$ the agreement at $\gamma_5$ is accidental. 600
14.5 Zeros of the symmetric stage $F_N^{\mathrm{sym}}$ in $[-1,2]\times[0,T]$ by the argument principle, against sign changes on the critical line: "terms" is the number of nonzero coefficients $p_m^{(N)}$, $m\ge1$, "height" the height $T$ searched, and "on the line" the number of sign changes out of the number of zeros counted. The windows have height 10. Above $T^*(N)$ the off line zeros arrive in bulk for $N\le32$: for $N=16$ the two windows of $(300,320]$ have 10 of their 13 zeros off the line. The sum $D_N$, whose terms are much larger than the sum itself, is evaluated in double precision, with a bound on its rounding error on the critical line in every window, and in multiprecision in the windows below 100 for $N=96$ and below 130 for $N=128$; above 130 the double precision margin of $N=128$ is lost, so that stage is reported only up to 130. Off line zeros come in pairs $\sigma+it$, $1-\sigma+it$, so an off line pair adds two to the count of its window, and a difference of one between the two counts is an artefact, such as a zero at the window boundary. . . . . 602
14.6 $\min_{0\le t\le60} A_X(\alpha-\frac12,t)$ from primes below $e^X$ (up to $4.4\times10^5$ at $X=13$). The values decrease towards $\xi'/\xi(\alpha)=0.011551$, 0.0046209, 0.0023105 for the three $\alpha$. . . . . 608
14.7 Numerical verification of results of this chapter that are not covered by the book's tables. . . . . 610

15.1 The three routes of the book that reach every height. All other routes are finite information. . . . . 613
15.2 The value at the origin and at the minimum of a selection of the valleys of $v_t(\alpha, \cdot)$ between consecutive zeros. The ordinate $\theta$ is that of the minimum at $t = 0$ and $\alpha = 0.6$; at $t = 0.2$ the minimum lies 0.06 to 0.08 to the left, and at $\alpha = 0.75$ it moves by less than 0.002. Script `phase_thorin.py`. . . . . 622
15.3 Constants for the certificate at $t = 0.2$, in floating point. Script `phase_cert.py`. . . . . 627
15.4 The Weil Gram matrix on hat functions. The last column splits $\lambda_{\min} = v^* G v$ for the minimising unit vector into the pole, prime (with its sign) and archimedean pieces. Script `phase_weil.py`, output `phase_weil.json`. . . . . 630
15.5 The smallest eigenvalue from the arithmetic side against the sum of squares of $\hat{g}_{\min}$ over the first 1500 zeros, with the zeros (by index) that carry the largest shares. Script `phase_weil.py`, output `phase_weil.json`. . . . . 630

16.1 Atoms of the even part law against the rounded clock of the basepoint law. . . . . 648
16.2 $(-1)^j \frac{d^{2+j}}{d\nu^{2+j}} \log K_\nu(2\pi)$. . . . . 653
16.3 Where the half plane $\operatorname{Re} s > \frac{7}{8}$ of [398], which has not been refereed, acts outside this section. . . . . 670
16.4 Heights beyond which Theorem 16.12.7 guarantees the compensated clock inequality under $(\mathrm{Q}_{7/8})$, rounded up, with the half width $d$ of the window and the number $k$ of zeros that the proof asks for in it. "Off line" uses only $\frac{1}{8} \le \beta \le \frac{7}{8}$; "on line" uses $\beta = \frac{1}{2}$, valid for $r + d \le H$. Computed by `polya_qrh.py` with the bound of Lemma 16.12.6. . . . . 674

17.1 The tent value $D(x)$ from the primes (Proposition 17.6.1(c)) and from the zeros, and $V_1(\Delta_x)$ by the closed form (c) and by the residue series (d). The rows $x = \log 2$ and $x = \log 5$ differ by Landau's term, see the text. . . . . 691

17.2 The least eigenvalue $\lambda_{\min}(\alpha, a)$ of the damped Yoshida form on the window $[-a, a]$, from the matrices of the form on the modes $\chi_n$, $|n| \le N$. For $\alpha > \frac{1}{2}$ the entries are converged values, from Richardson extrapolation of $N = 96$, 192, 384, with an uncertainty of at most $3 \times 10^{-7}$ at $a = 0.35$ and $1.1 \times 10^{-8}$ for $a \ge 0.8$ (an independent Legendre basis agrees to $1.5 \times 10^{-7}$ at most). For $\alpha = \frac{1}{2}$ the entries are the finite section values at $N = \max(24, 24a)$, upper bounds computed in extended precision, which are not converged; the entry for $a = 4$ ($N = 96$) was computed with 640 digits. The last row is the floor $2b_\alpha = 2\xi'(\alpha)/\xi(\alpha)$. Computed by `yo_damped.py` with `yo_matrix.py`, data in `yo_damped.json`. . . . . 699

18.1 Least eigenvalues of Weil's form on the window $[-a, a]$ in the even and the odd sector, from the prime side Yoshida matrices in 60 digit arithmetic with cosine and sine modes up to the order $N$ shown (`code/zh_window_eigs.py`). The last column gives Zhu's certified bounds for the even sector. The values at 26 modes and below are not converged, and the odd values at 160 modes are still falling. . . 711

18.2 The best Rayleigh quotient $Q(f_0)/\|f_0\|^2$ of Zhu's product family over the number $K$ of atoms killed and the even exponent $m$, with the optimal $K$ and $m$, the last ordinate $\gamma_K$ killed, the number $N(T_*)$ of ordinates below the resolution height $T_* = 2\pi e^{2a}$ of Section 18.4, and $T_*$ itself. . . . . 713

18.3 Smallest window half length $a$ at which Yoshida's form on $[-a, a]$ detects a hypothetical off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ added to the first 2100 zeros, on a grid of step 0.05 (script `code/zh_detect.py`, record `data/zh_detect.json`). The entries of the lower block are exact on the full matrix at the heights 51.37 and 199.64; at 1000.57 and 1999.99 the entries marked $\le$ are upper bounds, the smaller of the value from a band of modes and the entry of criterion (i) above. At the cell $(1999.99, 0.2)$ the band computation of 150 modes does not detect the quadruple at 2.90 and 2.95 (it does at 3.00, with 2100, 4000 and 8000 zeros and with the margin 90 in place of 45 in the number of modes), so that the entry 2.95 there is the entry of criterion (i); the exact test detects the quadruple in a band of 300 modes at 2.95 and in a band of 600 modes at 2.90. . . . . 716

18.4 The horizon $2\pi e^{2a}$ at the half lengths of some window statements, and the half length $\frac{1}{2}\log(H/2\pi) = 13.45$ at which it equals the height $H = 3 \times 10^{12}$ of the verification of the zeros. The conversion is a heuristic and not a theorem. . . . . 718

18.5 The comb constant $A_a(\alpha)$ of (18.5.1) and the barrier height $T_1(\alpha, a) = 2\pi e^{A_a(\alpha)}$ at three basepoints. The column $\alpha = \frac{1}{2}$ is Zhu's [526], who prints $A_{0.8}(\frac{1}{2}) = 2.9420$ and $T_1 = 119$ at $a = 0.8$, $1.0 \times 10^7$ at $a = 1.6$ and $2.4 \times 10^{11}$ at $a = 2$; the digits here are 2.94197, 119, $1.04 \times 10^7$ and $2.44 \times 10^{11}$. For $\alpha > 1$ the constant is bounded by $-2\zeta'(\alpha)/\zeta(\alpha) = 18.88$, 3.01, 1.14 at $\alpha = 1.1$, 1.5, 2. . . 719

18.6 The crossover $\varepsilon_\times = \lambda/(4p_1)$ of Theorem 18.5.2 for several values of $\lambda = \lambda_{\min}^{\text{even}}(\frac{1}{2}, a)$, with $4p_1 = 0.0924200$. The extrapolated values are three point power law fits to the finite sections of 60 digit matrices (script `code/zh_window_eigs.py`, record `data/zh_window_eigs.json`), not bounds; Zhu's reference values of his Table 1, $9.34 \times 10^{-7}$ and $1.65 \times 10^{-17}$, give $1.01 \times 10^{-5}$ and $1.79 \times 10^{-16}$. . . . . 722

19.1 Mean modulus of $E(N, s)$ of (19.2.2) for $\chi_4$ over $s = \alpha + 10i$ and $s = \alpha + 25i$, with prime powers summed up to $40N$, and the slope of $\log_{10}$ of the mean against $\log_{10} N$ fitted by least squares on the four values of $N$. Four values of $N$ and two values of $b$ illustrate the GRH statement of the proposition and do not verify it. Script `tf_primes.py`. . . . . 731

19.2 The first order visibility distance $\delta_c = \sqrt{2/\mathcal{L}_0}$ at the actual zeros, from $\mathcal{L}_\Xi = -(\log|\Xi|)''$ evaluated with `mpmath` at 30 digits, in the three configurations (i), (ii) and (iii) of the start of this section, at six heights $n_0 = 1, 10, 10^2, \ldots, 10^5$; the height is the ordinate $\gamma_{n_0}$ of the $n_0$th zero. Replacing a zero: the 20 zeros $\gamma_n$, $n_0 < n \le n_0 + 20$, each removed from $\mathcal{L}_0$, with the local gap $(\gamma_{n+1} - \gamma_{n-1})/2$; the mean gap is the mean of the local gap over these 20 zeros. At a midpoint and count preserving: the 20 midpoints of $\gamma_n$ and $\gamma_{n+1}$, $n_0 \le n < n_0 + 20$, with the local gap $\gamma_{n+1} - \gamma_n$, and in the count preserving block the two zeros $\gamma_n$ and $\gamma_{n+1}$ are removed as well. Median and least are over the 20 values of $\delta_c$, and the ratio is the median over the 20 points of $\delta_c$ divided by the local gap. Script `tf_deltac.py`, data `tf_deltac.json`. . . . . 733

19.3 Levels $\varepsilon$ at which $\rho_\varepsilon$ is negative (grid and points as in the text), near a hypothetical quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$ inserted at the midpoint of the gap between the 29th and 30th zeros and between the 10000th and 10001st (configuration (ii)). Script `tf_window.py`, data `tf_window.json`. . . . . 734

19.4 The number $K$ of greedy levels of Corollary 19.3.3, counted without the downward moves, in the three lattice models with $\Delta = 2\pi/\log(h/2\pi)$, against $\frac{1}{4}\log(h/2\pi)$. Script `tf_visibility.py`. . . 738

20.1 The ladder $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$, converged values: the limit of the variational upper bounds on the Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, by power law extrapolation from $N = 384$, 768, 1536 (`code/dc_ladder.py`, parts `grid` for $\varepsilon > 0$ and `y0` for $\varepsilon = 0$, `data/dc_ladder.json`). The uncertainty of an entry is the one defined in the text; it is an error estimate and not a bound. The last row is the limit $2b_{1/2+\varepsilon} = 2\,\xi'/\xi(\frac{1}{2}+\varepsilon)$ of Theorem 20.2.1 where its hypotheses hold (for $\varepsilon < \frac{1}{2}$ they are conditional, and at $\varepsilon = 0$ the limit is 0 under RH), computed by `code/dc_weil.py`, part W2 (`data/dc_weil.json`). At $\varepsilon = 0$ the entries for $a \ge 0.75$ are omitted: in double precision they are below the rounding level of about $10^{-13}$, and the upper bounds from extended precision are given in the text. . . . . 750

20.2 The rung at $\varepsilon = 0$ for $a \le 0.5$: the Legendre bound with $M = 32$, the converged value (Fourier modes to 1536, power law extrapolation) and its uncertainty, the shift of the extrapolation when $N = 1536$ is dropped, an error estimate and not a bound. At $a = \frac{1}{2}\log 2 = 0.3466$, the largest window in which no prime enters, the value is $1.3293 \times 10^{-3}$. Script `code/dc_ladder.py`, part `y0`, data `data/dc_ladder.json`. . . . . 750

20.3 The scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ in units of $10^{-4}$, against the constant $\pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 20.2.1(b): converged $\lambda_{\min}$ (Fourier modes to 1536, power law extrapolation, as in Table 20.1), rounded to three digits. The uncertainty of an entry, the shift of the extrapolation times $a^2$, is at most $4\times10^{-5}$ in these units, an error estimate and not a bound. Script `code/dc_rate.py`, parts `const` and `table`, data `data/dc_rate.json`. . . . . 753

20.4 Bounds on the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$, in units of $10^{-4}$: the lower bound of the proof of Theorem 20.2.1(b) with the single radius $\kappa \le a$ that maximises $\kappa^2(1-\ell_0(\kappa))$ (the radius in brackets), the layer cake lower bound, the converged value (as in Table 20.3, extended to $a = 6$ in `data/dc_rate.json`), and the value of the trial function $\cos(\pi u/2a)$ computed from the prime side, which is an upper bound. Script `code/dc_rate.py`, parts `lower` and `trial`. . . . . 753

21.1 The moments $\mathbb{E}[T^k]$ from the zeros (800 ordinates and the smooth tail from $\gamma_{800}$) and from the function (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`). . . . . 764

21.2 Relative errors of $p_1, p_2, p_3$ for the smooth and the anchored tail with the first $N$ ordinates; the cuts are $T = 1183.1475$ ($S(T) = 0.083$) for $N = 799$, $T = 397.1503$ ($S(T) = 0.244$) for $N = 200$ and $T = 680.8186$ ($S(T) = -0.211$) for $N = 400$ (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`). . . . . 765
21.3 Toeplitz minors $\det[f((r-c)h)]_{r,c<n}$ of the density of $Y$ (`gl_density_pf.py`, 70 digits). . . . . 770
21.4 The minimum of $v_\alpha(b)$ over $0 \le b \le 150$ for $\chi_4$, always at $b = 0$, against the bound $2(\alpha - \frac{1}{2})\, p_1(\chi_4) = (\alpha - \frac{1}{2})V_f$ of Theorem 9.1.18(i), with $V_f = 2p_1(\chi_4) = 0.1560296499$. Computed by `code/gl_sinsq_chi4.py` (`data/gl_sinsq_chi4.json`). . . . . 784
21.5 The power sums $p_j(\chi)$ of the ordinates for the characters modulo 4 and 5: the zero side, the function side, the relative error, and the difference of the two sides times $300^{2j}$. . . . . 786
21.6 The zero $y$ of the mixture $M_\lambda = \lambda A + (1-\lambda)B$ modulo 5 in each of the two intervals between a pole of $A$ and a pole of $B$, for five weights; each interval contains exactly one zero. . . . . 788

22.1 Real zero counts (computed in floating point, not certified). $\tau_P^*$ solves $A(\tau) = D_P(\tau)$. For $P \le 5$, $R_P$ is a full sign change scan; for $P = 7$ it is the hump criterion, with a 215 digit check of every hump in $[481, 501]$, the last surviving pair being 489.7932 and 491.0605. . 809
22.2 The survey (computed). $N = N(\tau_P^*)$, off $= R_P - N$, model is the ramp model's expected offset with the window 200 for $m_Z$. . . . . 810
22.3 Departed zeros of $\Xi_P$ (computed, residuals below $10^{-34}$) against the two forms of the departure law. The six zeros are every second zero of the chain of Remark 22.6.1 for $P = 2$ and a selection for $P = 3$. . . . . 811
22.4 The $2\sin^2$ Lévy density of $\Xi_2$, from its 6 real and 190 traced departed zeros, against that of $\Xi$. $\nu_2 > 0$ on the whole computed range $[0.02, 3]$. . . . . 816
22.5 Critical ladder rung $b_c$ and heat time $\lambda_c$ at which the pair above each Davenport Heilbronn zero becomes real. In every case $\lambda_c L^2 < 1$ (0.30, 0.09, 0.26), so these heights are pre asymptotic for the heat flow. . . . . 833
22.6 The collision of the moved pair $\gamma_1 \pm i\eta$ of $\Xi$: $(\eta^2 - s_c^2)/\eta^4$ at three heights $\eta$, against its limit $d_c = \tau_1^2 - \frac{m_4 - 1}{2}\tau_2$ of Proposition 22.14.8 (computed; the three cosh cells at $\eta = 0.3$ are certified by a Krawczyk step). . . . . 835

23.1 Computed shifts of the first and third positive zeros of $\Xi^{(2n)}$, rescaled by $m_n$, against the Edgeworth prediction $\kappa_n c_k^3/(6m_n^3)$ (computed, `code/ki_tilt.py`, 40 and 50 digits). . . . . 846

24.1 The density of $\mu$ from the residue expansion with the first 120 zeros (computed, 110 digits, agreeing with 140 digits to $10^{-103}$). . . . . 855
24.2 Finite truncation weights $1/G'_N(u_\rho)$ against the residues $w_\gamma = 1.0164 \times 10^4$, $-1.1775 \times 10^6$ and $8.0078 \times 10^9$ at the first, second and fifth zeros. The corrected rows multiply the raw weight by the factor of Proposition 24.5.1(i), with $\sum_{n>N} \gamma_n^{-2}$ taken from the exact total $p_1$; for the first zero and $N = 120$ the factor is $\exp(199.79 \times 0.0027993) = 1.7494$ and the corrected weight agrees with the residue to 0.025 per cent (computed, code/ht_resid.py). . 860
24.3 The lower tail of $\mu$ and the saddle point approximation, as printed in the source (unchecked: not reproduced). The Chernoff exponent $-\Lambda_f(y)$ bounds $\log P(T \le y)$ from above, as it must. . . . . 862
24.4 Windows on which $(\log f)'' < 0$ was verified, given the constants $M_j$ (computed, code/ht_lcert.py). The step count near 0.008 is large because of the cancellation in (24.1.1). . . . . 863
24.5 Relative shift $\ell_{d,n}/\gamma_n^2 - 1$ of the zeros of the Jensen polynomials, against the first order prediction; the last row is $A_n/(2d)$ at $d = 10240$ (computed). . . . . 868
24.6 Distance between the distribution functions of the Jensen law $\mu_d$ and of $\mu$ on the grid of 74 points of $[0.007, 0.08]$ (computed). . . . . 868
24.7 The smoothed zero laws $\hat{\mu}^{[N]}$, with the first $N$ zeros exact and the rest at the Gram midpoints, against $\mu$. The inversion in double precision resolves differences of distribution functions only down to about $2 \cdot 10^{-6}$ for $y \ge 0.011$ (and $1.8 \cdot 10^{-5}$ below), so the supremum for $N = 30$ and $N = 100$ is unchecked; the source printed $2.7 \cdot 10^{-6}$ and $6.8 \cdot 10^{-7}$. . . . . 873
24.8 Theorem 24.8.3 (computed, `code/rx_euler.py`). The upper block compares the two sides of the identity and the lower block gives the zero remainder $R_U$ and the bound of the theorem for the same rows; the residual is below $10^{-30}$ in every row. The rows with $|\operatorname{Re} s| < \frac{1}{2}$ ($s = 0.1$, 0.49 and $0.3 + 3i$) have $\frac{1}{2} + s$ inside the critical strip. At $U = 0.05$ the zero remainder is visible; at $U = 1$ it is below the working precision (30 digits). The bound is loose at small $U$. . . . 880
24.9 The quantity $\log \mathbb{E} e^{-sZ_P}$ for $P = 10^3, \dots, 10^7$, against its limit $(2s)^{3/2}/6$ (computed). . . . . 881
24.10 Boundaries with one off line pair against a GGC control on $[0.0045, 0.12]$; for $\gamma_{60}$ the maximum is over $[0.0045, 0.1]$, since near $t = 0.12$ the solver noise of about $2 \cdot 10^{-7}$ exceeds the signal. "Never" means the density was found positive on a grid of step 0.005 on $(0.0064, 6]$ and at three points below 0.0064, beyond which the tail is governed by $\gamma_1$. All entries computed (`code/ht_curved.py`). . . . . 887

24.11 Normalised boundary change $(b_{\mathrm{toy}} - b_{\mathrm{GGC}})\gamma^4/6\delta^2$ against the shift response $K(t)$, $\delta = 0.3085$ (computed). . . . . 888
24.12 Equivalent forms of RH from the reciprocal Xi law and their unconditional frontier. The Jensen bound is Chassé's, as quoted in [159]. . . . . 896
24.13 Smallest contiguous Toeplitz minors $\det[c_{n+i-j}]_{i,j<k}$ for $k-1 \le n \le 104-k$, normalised by $c_n^k$, for $\xi$ and for the Davenport Heilbronn function. A further 600 random minors of each of the orders 3, 4 and 6 were nonnegative for both functions, none below $-10^{-80}$. . . 905
24.14 First failing order of the coefficient ladder when the $m$th zero is replaced by a pair at angle $\theta$. The bare threshold is the first failing order for the quadratic factor alone, the least $k$ with $\theta > \pi/(k+1)$. The failing index $n$ is 1, 3, 7, 12 and 17 for $m = 1, 3, 6, 11, 16$ (and 6 at $\theta = \pi/10$, $m = 6$), within one of $m$; orders above 40 were not tried. . . . . 906
24.15 First failing order of the density ladder for the same kind of perturbation, as reported in the source note (not reproduced). Order 1 means that the density itself becomes negative. . . . . 907

25.1 Zeros of the extreme value approximations (computed). For the Dyck chain the real parts of the zero near $\gamma_1$ for $n = 800, 1600, 3200, 6400, 12800, 25600$ are $0.2438, 0.3196, 0.3729, 0.4103, 0.4367, 0.4553$, and the extrapolation uses $n = 3200, 6400, 12800$. . . . . 917
25.2 Rouché tests for the folded kernels. The sampled minimum is over 721 points of the quarter circle (computed); the lower bound is over the whole quarter circle (certified in ball arithmetic, and weaker than the sampled minimum because the bound is crude). Every row passes both. . . . . 931
25.3 Partial fraction data for $\beta_\mu$ (computed in ball arithmetic, `code/ev_g1_string.py`). The sign condition of Proposition 25.8.4 is that $m \ge 0$ and all $\sigma_j > 0$; it is a hypothesis of the proposition and does not by itself give a string. . . . . 933
25.4 Absolute errors of $\gamma_k$ recovered by Prony fits of $K$ exponentials to the tail of the hitting time, window starting at threshold $x_0$ (source values, unchecked). . . . . 935
25.5 Rouché tests for $N = 5$ (computed: floating point, sampled on the circle, $E_5$ truncated at $n = 14$). . . . . 936
25.6 The masking of the first nonreal pair in $D_6$ for the two term folded kernel (computed, 250 digits, except the kernel moments row, which is from 120 and 240 digits, and the row marked unchecked, which is the source's value and needs the 19 located zeros). . . . . 939
25.7 Absolute errors of $\gamma_k$ from Prony fits to the tail of the reciprocal $\Xi$ hitting time at 140 digits (computed). . . . . 939

25.8 Calibration on Devroye's two explicit laws, with atoms $\pi^2(k-\frac{1}{2})^2/2$ and $\pi^2k^2/2$, and the reciprocal $\Xi$ law with atoms $\gamma_k^2$. For the two strings $c_k\lambda_k = (-1)^{k-1}\pi(k-\frac{1}{2})$ and $c_k = 2(-1)^{k-1}$ in closed form, so that $x_k = \log\frac{2k+1}{2k-1}/(k\pi^2)$ and $x_k = \log\frac{(k+1)^2}{k^2}/\big((2k+1)\pi^2/2\big)$ (computed, and checked against the product at $K = 4000$ factors). For the reciprocal $\Xi$ law the coefficients come from $\xi'(\rho_k)$ at the first 1700 zeros (computed). . . . . 941

25.9 Long window threshold fits from the Laplace transform alone, as reported by the source (unchecked: the fits were not recomputed; the pair agrees with the zero of $H_{-4}$ found directly, $33.1150940172 \pm 0.150539022\,i$). . . . . 945

26.1 The explicit formula for the heat trace, agreeing to all printed digits (computed). . . . . 952

26.2 Archimedean part, prime part and heat trace (computed). Digits cancelled is $\log_{10}(\mathcal{A}/W)$, which differs from the measure of the caption of Table 5.6. . . . . 956

26.3 The first sign change $t_n$ of $(-1)^n\mathcal{A}^{(n)}$ against $n/\lambda_*$, $\lambda_* = r_*^2 = 39.5620$ (computed). . . . . 957

26.4 Zeros recovered by Prony's method from the heat trace computed from $\Gamma$ and the primes (computed, `code/wt_heat.py`, 110 digits; the source printed errors of $10^{-30}$ for $k \le 3$, which was the floor of its reference, and the digits shown are those of the source). . . . . 958

26.5 End of the positivity window of order $n$ when only the primes in $S$ are kept (computed). The entries $> 0.8$ mean no failure up to the end of the scan. . . . . 962

26.6 $V_\delta$ for the primes, the GUE prediction, the Cramér model and the Poisson level (computed). The source printed $10.19, 7.69, 5.34, 4.06, 3.23, 1.10$ for the primes (all within two standard errors of these) and Cramér values 14.19 to 15.22 with a last entry 10.09, which the recomputation does not reproduce. . . . 963

26.7 Windows for $W'' > 0$ suggested by the bound of Proposition 26.9.2(2) from the primes up to $N_0$ with the Rosser and Schoenfeld tail (computed in floating point on a grid; not certified). 966

26.8 Energy of the normalised prime error on the log scale (computed). The source printed 0.0506 for the first window and 7.54, 9.78, 9.14, 7.15 for a Cramér realisation; the Cramér values are heavy tailed and one realisation does not support a growth rate. . . . . 969

27.1 Displacement $\operatorname{Re}\rho_\delta - \frac{1}{2}$ of the first three zeros of the shifted family, by Newton's method from (27.3.2) at 30 and 45 digits (difference below $5 \times 10^{-25}$; computed). The first row is the real part of the complex slope of (27.3.3), whose imaginary parts $-0.1565$, $+0.0703$, $-0.2961$ show that $\operatorname{Im}\rho_\delta$ moves too. The displacement is linear in $\delta$ only to the digits of the slope for small $\delta$ (at $\delta = 1$ the first entry is 1.53727 against 1.53709). . . . . 984
27.2 The uniform law and the Riemann problem under the $2\sin^2$ lemma. Convergence of the Thorin densities is vague convergence. Without RH the limit at $\alpha = \frac{1}{2}$ records only the zeros on the critical line (zeros at $\frac{1}{2} \pm a + i\gamma$ cancel in the real part), so the limit measure is not a test of RH by itself. . . . . 990

28.1 The drift of the first zero of $M_n$ at $\gamma = 6.0209489$ (computed). The last column is the gamma remainder of Proposition 28.2.4, for which $\delta_2 = -2.5713 - 1.0051i$ is not the limit. . . . . 1003
28.2 The five zeros of $L(s, \chi_4)$ below height 500 with anomalous Gram parity (computed). The partner is the neighbouring zero at the stated gap (the left neighbour for the zero number 291). . . . . 1006
28.3 Zeros of $M_n$ in the left and right half strips. The counts are computed by `code/tr_counts.py` (`data/tr_counts.json`, run on Hopper at two precisions with identical counts). The source note printed 22 for the hard cutoff at $n = 32$, a blank at $n = 64$ and 14 for the mean correction at $n = 32$, $\operatorname{Im} z$ from 0.5 to 20. . . . . 1008
28.4 Symmetrised hard cutoff (computed). . . . . 1010
28.5 The first zeros of $F_8$ and of $L(s, \chi_4)$ (computed). . . . . 1010
28.6 The first negative Hermite coefficient of $\Xi$ (computed). . . . . 1017
28.7 The ratio of $b_m(a)$ to $\sqrt{2a}\, m^{-1/4}\, \Xi(a\sqrt{4m+1})$ (computed). . . . . 1018
28.8 Number of sign changes of $b_m(a)$ for $t_m(a) \le T$ against the zero count $N(T)$ (computed). The coefficient counts of the note (627, 1738, 6947, 7002) were two too large. . . . . 1018
28.9 Mean sign runs of $b_m(4)$ (computed). . . . . 1021

29.1 Weighted errors of the Möbius approximants (computed). The column $c = 0.95$ of the source was $2.9 \times 10^{-1}$, $1.1 \times 10^{-1}$, $1.8 \times 10^{-2}$, $9.8 \times 10^{-3}$, a few percent low. . . . . 1034
29.2 Optimal distance, size of the optimal coefficients and the Möbius error (computed). The source printed $2.4\times10^{15}$, $1.5\times10^{16}$, $2.7\times10^{16}$ at $N = 32$, off by a factor of 5 to 300. . . . . 1039
29.3 The Thorin check at $\theta = 1$ (computed). The first column is the closed forms $\log(\pi/3)$, $\log(3\zeta(3)/\pi)$, $\log(15\zeta(5)/\pi^2)$ evaluated; the source's digits differed in the last place. . . . . 1047

29.4 Distance from $e^{-x}$ to the span of the first $N$ dilates, squares against integers (computed). . . . . 1048

30.1 Spectral peaks of the level one Thorin measure (computed). Every peak lies within $10^{-4}$ of the logarithm of a prime or prime power. The heights, relative to that of $n = 3$, depend on the window and follow the relative value of $\Lambda(n)/n$ to within 0.04. . . . . 1055
30.2 The Fourier backward flow from samples of $u_1$ on $\operatorname{Re} s = 1$ (computed). The error is the largest distance between a zero in $100 < v < 300$ and the nearest peak. At $\Omega = 60$ the sharp cutoff has 213 peaks of height at least a quarter of the maximum, 109 of them at the zeros, and the maximum is +26.6. . . . . 1058
30.3 EM inversion of $u_1$ on $[100, 300]$ (computed). The source printed $4.1 \times 10^{-3}$ in the last row; the recomputation gives $4.05 \times 10^{-3}$, and the grid sensitivity of the residual is of that size. . . . . 1062
30.4 Quartet injection test with controls (computed, `code/lo_em.py inject`); total residual after 10000 EM steps. The last row is the boundary case $\sigma = \theta'$. . . . . 1062

31.1 The mass deficit at $T = 100$, where $M_1(100) = 29.1112$ and $N(100) = 29$ (the author's computation, not rerun). . . . . 1096
31.2 The certificates of Theorem 31.9.6, as recomputed. The whole run takes 29 seconds at 96 bits and 46 at 160 bits on 24 cores of a cluster; the author's run took about ten minutes on one core. . . . 1103

32.1 The matrix pencil at height 200, level one data, taper $\ell = 10$: errors in the position and in the real part of the recovered zeros, by distance from the centre (computed by the author's scripts, not rerun). . . . . 1141

33.1 Stability between levels, computed. The source is a sum of 40 Lorentzians of width $w = 0.05$ on a period 400, the data are at width $a = \frac{1}{2}$ with white noise of size $\varepsilon$ at every frequency, and $P_b * \mu$ is recovered by an oracle spectral cutoff, with $\varepsilon$ from $10^{-12}$ to $10^{-3}$. The fitted slope is that of log of the error in $P_b * \mu$ against $\log \varepsilon$; the predicted slope is the exponent $(b+w)/(a+w)$ of Proposition 33.3.2. Nineteen values of $\varepsilon$ (the error does not depend on the noise draw); script `code/cd_slopes.py`, data `data/cd_slopes.json`. All four fitted slopes are within 0.004 of the prediction. Status: computed, illustration. . . . . 1173

34.1 Particle filter posterior for the zeros of $\zeta$ in the data window $[78, 94]$ after all levels down to $\theta = 0.6$ (computed by the author). The true ordinates are 79.3374, 82.9104, 84.7355, 87.4253, 88.8091 and 92.4919 1186

34.2 The backward Cauchy flow as a filter: the posterior of $\delta^* = \max|\sigma_j - \frac{1}{2}|$ over the zeros of $\zeta$ in $[80, 92]$ as levels are added, with cumulative cost (computed by the author). The bounds measure the displacement of one unpaired zero from the line, which the functional equation forbids. . . . . 1187
34.3 Posterior standard deviation of $\sigma_j$ for the six zeros of $\zeta$ in $[78, 94]$, against the first order Fisher information formula ($\varepsilon = 0.01$, $\Delta v = 0.04$). The predicted row is computed (`code/cf_info.py`); the filter row is the author's run. . . . . 1188
34.4 Contour moments, computed in ball arithmetic at 128 and 192 bits (`code/cf_moments.py`, `data/cf_moments.json`); the two precisions agree, and the integrals are enclosed, so that each $M_0$ is an enclosure containing one integer. The note printed $7 \times 10^{-22}$ and 0.006 for the first two entries, floating point quadrature residuals (the second from zeros at distance 0.05 from the edge); the exact values are 0. The zero of the last row is the Newton root with $K = 1$ and $M_1$ as an interval. . . . . 1190
34.5 Coverage of the GBC posterior quantiles of $\delta^*$ on held out simulations (computed by the author). The middle quantiles undercover, an approximation and calibration error of the quantile network; at an atom exact quantiles overcover. The upper tail is slightly short and is the place to add conformal calibration. . . . . 1191
34.6 Davenport and Heilbronn windows flagged by GBC (computed by the author; the zeros are those of Section 34.2). The four zeros off the line below height 200 are flagged, and located to within 0.1 in the four windows that target them. The window at 180 sees the zero at 176.70 near its edge, outside its target interval $[177, 183]$, and overstates $\delta$: a boundary false alarm, a boundary effect of the simulator. No other window of either function is flagged; for all 35 windows of $\zeta$ the posterior median is 0 and the 95% quantile is at most 0.007. . . . . 1192
34.7 Costs of reported runs (computed by the author). The two runs for $\zeta$ differ in window ($[78, 94]$ with six zeros against $[80, 92]$ with four), in zeros and in data, so their ratio of about 115 in evaluations is a comparison of two different runs and not a matched speedup. For Davenport and Heilbronn the GBC initialised filter succeeds where the plain filter stalls (at tempering exponent $\lambda \approx 0.003$), but the posterior is strongly non Gaussian because the level $\theta = 0.8$ lies within 0.0085 of $\sigma = 0.8085$, so the bridge from the Gaussian proposal still needs many steps. . . . . 1193
34.8 GBC posterior medians of $\delta$. Mean absolute error 0.0096 with the rigid simulator and 0.0034 with the sine kernel simulator. . . . . 1193

34.9 Monte Carlo spread over twelve runs, computed in the source note. Least squares log log slopes of the printed values are $-0.44$, $-0.62$ and $-0.47$, against $-1/2$ from Theorem 34.5.1; with twelve runs per cell each standard deviation has a relative error of about 20%, so the slopes carry about $\pm 0.15$. . . . . 1199

34.10 Negative mass of $u_\theta$ for the Davenport and Heilbronn function on intervals of length 3 centred at its two lowest zeros off the line (computed, `code/cf_xi.py`, `data/cf_xi.json`). The bound is the number of zeros with $\sigma_\rho > \theta$ at the centre. . . . . 1205

34.11 Negative mass of $u_\theta$ on the left half disc around the zeros of the Davenport and Heilbronn function off the line, on a $40 \times 81$ grid, as reported in the source note and not recomputed here. In every row it lies between $r/\pi$ and $2r/\pi$. . . . . 1208

34.12 Zero free constants $1 - \sigma_\rho \geq c/\log \gamma_\rho$ from admissible pairs, in the scheme of Corollary 34.8.6 (computed). The optimal polynomial of degree 4 has $a \approx (1, 1.705, 1.044, 0.425, 0.089)$, and the single level rows to degree 8 are over squares with $b_j \geq 0$ (with only $p \geq 0$ and $a_k \geq 0$ the value grows by about 1% at degree 8); the best level for the derivative family at degree 4 is $x = 0.89$. The rows with derivatives run to $\deg q = 5$ and are optima on discrete constraint grids (computed); at $\deg q = 4$ and 5 the kernels are not certified admissible (at $\deg q = 5$ a dense audit of positive definiteness gives $-1.6 \times 10^{-6}$). The cap of the source note, 0.432 for the degree 4 polynomial, is replaced by the sharper cap of Proposition 34.8.7. The last row is outside the class of Lemma 34.8.5. . . . . 1214

34.13 Ceiling of the scheme of Corollary 34.8.6 over nonnegative positive definite smoothings, by discretised linear programming (computed, `code/cf_lp.py`, `data/cf_lp.json`; not certified). Refining the frequency grid near the origin moves the value for $3 + 4\cos\omega + \cos 2\omega$ from 0.0460 to 0.0450; on the stated uniform grid, changing the number of points or $\lambda_{\max}$ moves the third digit. The rows with the optimised polynomial are lower estimates, since the alternating searches end at local optima (the degree 4 gain over the fixed polynomial is below $10^{-4}$). . . . . 1216

34.14Nested sampling for $\zeta$ on bands of length $10^4$, with $N$ live heights, $M$ moves per replacement and $K$ iterations (computed). In every band the score stays positive. The column min $s$ is the smallest value found by the run, not the band minimum. The scores at the printed heights and the normalised gaps, measured between consecutive zeros found as sign changes of Hardy's $Z$ function, are recomputed with enclosure radius below $6\times10^{-12}$ (`code/cf_ns.py`); the run parameters and $\log X$ are the source's. The smallest values sit in the largest normalised gaps. . . . . 1220

35.1 The dictionary between the two levels. . . . . 1231

35.2 Moments of $Z = \log h_\theta(\theta+i\tau)$ under $P_{1-\theta}(t_0-\cdot)$, computed in floating point (`code/bsy_cm.py`, shared evaluator `code/bsy_zdh.py`): Poisson weighted panels between the zeros with tanh sinh nodes, midpoint cells beyond distance 1000 out to $|\tau-t_0| \le 13000$, and the mass beyond the window ($2.45\times10^{-5}$ at $\theta = \frac{1}{2}$, $1.22\times10^{-5}$ at $\theta = \frac{3}{4}$) estimated from the sample moments of the outer window. Entries are from the finer of two meshes; the larger of the difference between the meshes and half the tail correction is at most $2.1\times10^{-5}$ in every entry (the row at height 5000 is in `data/bsy_cm_5000.json`). For $\zeta$ the equal variance and zero covariance identities of Theorem 35.5.2 hold to that accuracy; for Davenport and Heilbronn (DH), with $h = f$, they fail at both levels. . . . . 1239

35.3 Local defects $D_\theta(t_0)$ (computed, floating point). Davenport and Heilbronn is abbreviated DH. The quadrature columns are the integral over $|\tau - t_0| \le W$ minus $\log|h_\theta(1 + it_0)|$, with the zeros as breakpoints (code/bsy_defects.py): they are realised truncation errors of the window, not accuracies, and change with $W$. Outside the window the Cauchy mass is 0.0053 at $\theta = \frac{1}{2}$, 0.0027 at $\theta = \frac{3}{4}$ and 0.0013 at $\theta = \frac{7}{8}$ for $W = 60$. Monte Carlo (code/bsy_mc.py, computed; the seeds are stored in data/bsy_mc.json) uses $n = 20{,}000$ evaluations for $\zeta$ and 4,000 for DH. Plain Monte Carlo uses exact Cauchy draws, with standard error $\mathrm{sd}/\sqrt{n}$. The vertical sampler takes a pilot mean $c$ from $n/20$ draws, $\epsilon = 0.05$ times the standard deviation of the pilot, an even mixture of Cauchy independence proposals and Gaussian steps of scale $(1-\theta)/2$, and a burn in of $n/20$ steps, with batch means standard errors (50 batches). The Davenport and Heilbronn prediction is the sum of the Green potentials of the zeros found below height 700; for $\theta = \frac{7}{8}$ none has $\beta > \frac{7}{8}$, but $f$ has zeros with $\beta > 1$ [111], so the exact defect is positive. . . . . 1244

35.4 Root mean square error of the Cauchy average of $\log|\zeta(\theta+i\tau)|$, $t_0 = 0$, over 400 replications (computed, floating point; code/bsy_yks.py, data/bsy_yks_r400.json; the sampling error of each entry is about 3.5% and larger for the heavy tailed plain estimator). The control uses $M = 100$ at $\theta = \frac{1}{2}$ and $M = 1000$ at $\theta = \frac{7}{8}$. Gain is the ratio of plain to YKS with control. Fitted slopes over $500 \le N \le 16{,}000$ at $\theta = \frac{1}{2}$ and $\frac{7}{8}$: plain $-0.49$ and $-0.50$, YKS $-0.66$ and $-0.60$, YKS with control $-0.58$ and $-0.52$. . . . . 1246

35.5 Root mean square error at $N = 32{,}768$ for the half disc identity of Theorem 35.6.6, $R = 1$, $t_0 = 85.7$, 200 replications, with the slopes fitted for $N \ge 1024$ in brackets (computed, floating point; `code/bsy_halfdisk.py`, `data/bsy_halfdisk.json`). The target is the right side of the identity; for the Davenport and Heilbronn function (DH) it includes the Green term $G_H(z,\rho)$. . . . . 1252

35.6 Truncated Cauchy average of $\log|\zeta(\theta + i\tau)|$ over $|\tau - t_0| \le T$ at $t_0 = 85.7$, minus $\log|h_\theta(1 + it_0)|$; as in Remark 35.2.2, the zeros above $3 \times 10^{12}$ move the whole line average away from $\log|h_\theta(1+it_0)|$ by less than $10^{-12}$, with no hypothesis. Mean error and spread over 10 random shifts of the grid (computed, floating point; `code/bsy_trunc.py`, `data/bsy_trunc.json`). At $\theta = \frac{1}{2}$ the zeros with $|\gamma - t_0| \le T + 1$ (the zero range convention, counted in the last column) are deflated and their window means added back (the unbiased rule). The shifted grid is unbiased for the window integral, so the mean error is minus the Cauchy tail beyond $T$, up to the spread. In this range $T^2 \cdot |\text{error}|$ stays below 0.27; the bounds of [26] allow slow growth. . . . . 1253

35.7 Root mean square error of the weighted rules at independent uniform nodes on $Q = [-1,1]^d$, 200 replications in one dimension and 30 in two (about 5 percent noise in each root mean square error in one dimension and about 13 percent in two), the largest $N$ in brackets, slopes fitted for $N \ge 4000$ (computed, floating point; `code/bsy_wmc.py`, `data/bsy_wmc.json`). The bump $\prod_i b(x_i/0.9)$ vanishes near $\partial Q$ (case (a) of Theorem 35.7.3); the exponentials do not, and their error is the bias of Remark 35.7.4. The slopes are consistent with the last column, the exponent of Theorem 35.7.3, of Lemma 35.6.8 for the trapezoid row and of the Remark for the last row. . . . . 1259

35.8 Root mean square error of $I(\varphi)$ (computed, floating point, 20 replicates; code/bsy_riesz2d.py, data/bsy_riesz2d.json; the relative standard error of an entry is 16%). Jittered is Delaunay on jittered nodes; deflated grid uses an estimate of the zero that is off by $1.4 \times 10^{-4}$, along $(1+i)/\sqrt{2}$. . . . . 1260

35.9 Deflated shifted grid for Davenport and Heilbronn, root mean square error over 10 shifts, as a function of the error $\varepsilon$ in the deflating zero (computed, floating point; code/bsy_defl_dh.py, data/bsy_defl_dh.json; the relative standard error of an entry is 22%). The last row is the grid without deflation. A synthetic test of the two regimes of Proposition 35.7.5 (code/bsy_defl.py) shows the error falling to $N = 2^{22}$ for every $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, with slope near $-0.6$ while $\varepsilon\sqrt{N} \leq 1$ and near $-1.1$ afterwards; the constant depends on the bump and the box: at $\varepsilon = 10^{-4}$, $N = 262{,}144$ the ratio of the error to $\varepsilon$ is 0.030 here and 0.0050 in the synthetic test, a factor 6. . . . . 1263

B.1 Landmarks in the computation of the zeros of $\zeta$. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [127, Chapters 6 to 8]. . . . . 1270

# Chapter 1

# Introduction and foundations

The book rests on one equivalence, between the Riemann Hypothesis and the positivity of the Thorin measure of a function built from the completed zeta function.

## 1.1 The Thorin condition and what this book shows

Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the completed zeta function, whose zeros are the nontrivial zeros $\rho$ of $\zeta$. The Riemann Hypothesis (RH) asserts that every $\rho$ has real part $\frac{1}{2}$.

This book works with a reformulation. Put

$$G(u) = \frac{\xi(\frac{1}{2} + \sqrt{u})}{\xi(\frac{1}{2})}.$$

Because $\xi(\frac{1}{2} + s)$ is even in $s$, $G$ is an entire function of $u$ with no branch point, of order one half, and Hadamard's theorem gives

$$G(u) = \prod_k \Big(1 + \frac{u}{a_k}\Big), \qquad a_k = -(\rho_k - \tfrac{1}{2})^2,$$

one factor for each pair $\{\rho, 1-\rho\}$ (Section 1.2 derives the product as (1.2.1)). A zero on the line has $a_k = \gamma_k^2 > 0$; a zero off the line has $a_k$ complex. The reciprocal $1/G$ is the *clock*. The classical theory of generalised gamma convolutions (GGC) of Thorin and Bondesson [493, 494, 56] says that a function of the form $\exp\{-\int \log(1+u/z)\,U(dz)\}$ with a positive measure $U$ is the Laplace transform of a positive random variable, and that $U$, the Thorin measure, is unique. The rigidity theorem of [424], which Section 1.2 restates with its source as Proposition 1.2.1, says that the clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is the Laplace transform of a generalised gamma convolution if and only if RH holds, and that its Thorin measure is then

$$U_{\mathrm{RH}} = \sum_{\gamma > 0} \delta_{\gamma^2}, \tag{1.1.1}$$

while the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials $e_\gamma$.

We call the requirement that the Thorin measure of the clock be a positive measure the *Thorin condition*. It has several equivalent faces, each of which is a sign condition. The logarithmic derivative $G'/G(u) = \sum_k (u + a_k)^{-1}$ must be a Stieltjes transform. The heat trace $W(t) = \sum_k e^{-a_k t}$ must be completely monotone. The Thorin density $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2}+\varepsilon+i\theta)$ at every resolution $\varepsilon > 0$ must be nonnegative, which is Lagarias's criterion [320]; for $\varepsilon \geq \frac{1}{2}$, where the basepoint $\frac{1}{2}+\varepsilon$ is at least one, it is positive with no hypothesis (Section 9.1). Weil's explicit formula functional must be positive on positive definite test functions [512]. The clock must be hyperbolically completely monotone (Corollary 4.1.3). All of them are computable at a finite resolution and on a finite window (Section 1.3.3).

### 1.1.1 What computation can and cannot say

Platt and Trudgian verified RH below height $3 \times 10^{12}$ [419]; Appendix B tells the history of such computations, with the landmarks in Table B.1. None of it proves RH, and the reason is structural rather than technological: every computation sees finitely many zeros, and there are functions with the order, type and symmetry of $\xi$ that pass a given finite set of heat trace tests and yet have zeros off the line just above any sufficiently large height $R$ for which RH has been verified up to $R + 50$ (Theorem 5.2.1 [421, Theorem D]).

The Thorin condition gives computation a different role. Instead of locating zeros one at a time, it asks for the sign of a measure, and the measure can be computed at every positive resolution without knowing where a single zero is.

Three ways of evaluating $\xi$ appear throughout. The first is Pólya's integral, $\Xi(\theta) = \int \Phi(u) \cos(\theta u)\, du$, with the Pólya kernel $\Phi$, a positive, doubly exponentially decaying function built from the Jacobi theta function. The integral is exact and needs no primes and no zeros, and since $\Phi$ is analytic in a strip the trapezoid rule applied to it is spectrally accurate; the cost is the cancellation $e^{-\pi\theta/4}$, which the tilted contour of Section 2.1 largely removes. The second is the Riemann Siegel formula and its fast multi point descendant, the Odlyzko Schönhage algorithm [392], the tools of choice at large height. The third is the explicit formula, which evaluates the same objects from the primes and the gamma factor; the sine squared lemma is its Thorin form.

The principle that ties them together is that the Thorin measure at resolution $\varepsilon$ is the Poisson smoothing of the zeros, and simultaneously the sine squared image of the primes. The first description makes it computable from any evaluator of $\xi$ off the line. The second explains why the primes alone cannot compute it near the line. In Selberg's smoothed explicit formula the prime cutoff $x$ plays the part of the resolution: if no zero has real part greater than $\frac{1}{2}$, the error at resolution $\varepsilon$ is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that $k$ correct digits need $x \approx 10^{k/\varepsilon}$, with primes up to $x^2$ (Section 3.5.1). Figure 1.1 shows the first description at work.

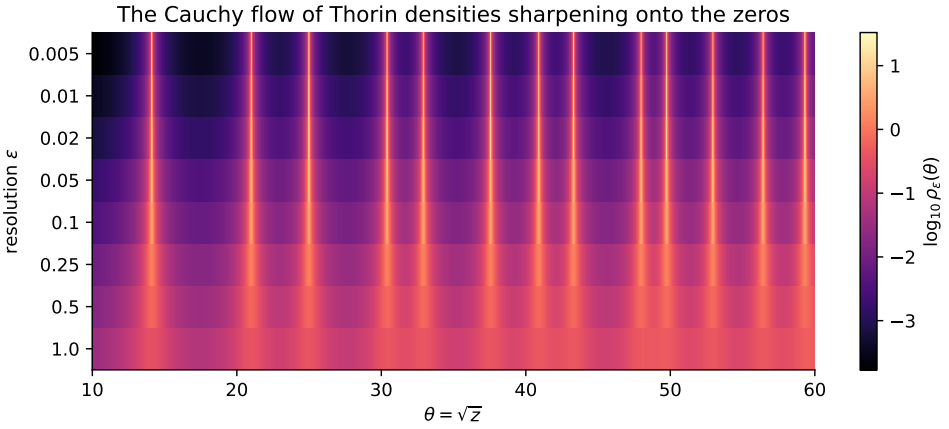



Figure 1.1: The Thorin density $\rho_\varepsilon(\theta)$ on $10 \le \theta \le 60$, computed from the Pólya density at eight resolutions from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top), on a logarithmic colour scale. As the resolution sharpens, the density concentrates on bright vertical lines at the ordinates of the zeros, all of which lie on the critical line in this window: the Cauchy flow converging to the RH measure $\sum \delta_{\gamma^2}$.

### 1.1.2 Plan of the book and its principal results

Two ideas set out in Sections 1.2 and 1.3 carry the rest of the book. The first, stated above, is that the Thorin measure at resolution $\varepsilon$ is the Poisson smoothing of the zeros; as $\varepsilon \downarrow 0$ it converges to the atoms on the line. The second is that the hierarchy of infinitely divisible laws becomes a hierarchy of positivity conditions on one function, the heat trace, of which the lower rungs are theorems and the top rung is RH.

Chapter 2 evaluates $\xi$ and $\xi'$ in the complex plane by the Pólya route. Chapters 3 to 5 are the core of the book. Chapter 3 measures the distance from the computed Thorin measure to $U_{\mathrm{RH}}$ and computes the signature of an off line zero. Chapter 4 treats weak convergence and hyperbolic complete monotonicity. Chapter 5 climbs the hierarchy of positivity on the heat trace.

Chapters 6 and 7 study the reciprocal zeta function in the variable $u$ of the clock. Chapter 8, with the Pólya route as its evaluator, treats the clock as a random variable.

Chapters 9 to 11 move the basepoint. The shifted clock $T_\alpha$ has Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{u})$; at $\alpha = \frac{1}{2}$ it is the clock. These chapters carry both ideas from the basepoint one, the resolution $\varepsilon = \frac{1}{2}$, into the critical strip. Chapter 9 works out the basepoint one and constructs $T_\alpha$ as a hitting time at every basepoint $\alpha > \frac{1}{2}$. Chapter 10 extends the hierarchy to symmetric laws on the line and reads zero free half planes as completeness statements after Wiener, Salem and Riesz.

Chapter 11 reads the hierarchy through the books of Sato and Bondesson, uses zero density estimates to bound how far each $T_\alpha$ can be from a GGC, and closes with the work of Anthony Pakes on size biasing, the Lambert law and ladders of self decomposable laws.

Chapter 12 assembles the results into a path towards RH on which every step is marked proved, computed, announced or open. Chapter 13 reads RH as the statement that the Pólya variable is ferromagnetic, in the sense that its moment generating function vanishes only on the imaginary axis, and builds provably ferromagnetic laws within total variation distance $6.1 \times 10^{-6}$ of it with no hypothesis and $2 \times 10^{-6}$ by a computer assisted proof. Chapter 14 reads the Pólya density as the density of a maximum order statistic of independent exponential variables and follows the clock through that representation. Chapter 15 places the computations of the book on the Thorin phase diagram of the de Bruijn Newman flow, the set of points $(t, \alpha)$ at which the shifted clock of $\xi_t$ is a generalised gamma convolution, and certifies the finite part of one of its points below height 160. Chapter 16 reads Pólya's papers on the zeros of entire functions through the reciprocal Xi law and the basepoint laws, and ends with what the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [398], which has not been refereed, would give. Chapter 17 reads Yoshida's Hermitian forms attached to the explicit formula against the programme. Chapter 18 reads the compact windows of Zhu [526] in the same language. Chapter 19 compares the three one parameter families of generalised gamma convolutions that contain the clock of $\xi$, the Esscher tilt, the basepoint family and the heat flow, and proves the deconvolution theorem. Chapter 20 reads the deconvolution as a damped Weil criterion. Chapter 21 carries the tilt and the Thorin representation to Dedekind zeta functions and Dirichlet $L$ functions and asks what the programme says about the Linnik constant. Chapter 22 studies the friable approximants of $\Xi$, which keep only the integers composed of the primes up to $P$ in Riemann's theta series, as a test case for computational criteria for RH, reads an identity of Miyatake probabilistically and places the bottom of the cosh ladder against the quasi Riemann abscissa. Chapter 23 reads the work of Haseo Ki and his coauthors on the heat flow and on repeated differentiation through the Thorin dictionary and shows that no property that survives a Gaussian tilt of the Pólya density can imply RH. Chapter 24 asks whether RH can be obtained from the limiting behaviour of hitting times, tests the families of approximating laws that can be built without assuming anything about the zeros, proves a universal stable limit for the prime side hitting times and a rigidity theorem for factorisations of the clock law, and studies the density of the clock directly. Chapter 25 reads the Pólya density as a superposition of extreme value laws, builds the hitting times of the approximations that need no hypothesis on the zeros, and asks whether the question of real zeros can be turned into the question of whether such a hitting time belongs to a string of positive mass, with three routes to RH and the place where each one stops. Chapter 26 reads Weil's criterion on Gaussian

test functions as the Thorin condition of the reciprocal $\Xi$ law, with the explicit formula split into an archimedean and a prime part, both of extreme value form. Chapter 27 asks what the identity of Comtet and Majumdar and its cut off regularisation can do for the programme, with a negative answer on the arithmetic side and a positive one for the uniform law. Chapter 28 truncates the lattice of Biane, Pitman and Yor for $L(s, \chi_4)$, ties the side from which the truncated zeros approach the true ones to the Gram parity, and studies the Hermite coefficients of $\Xi$. Chapter 29 answers in weighted $L^2$ spaces the question of Lai and Vere-Jones on the completeness of the dilates of the hyperbolic secant, and says where arithmetic enters. Chapter 30 computes the Thorin measure at level one in closed form and runs the Cauchy semigroup backward, as a positive linear inverse problem solved by the EM algorithm. Chapter 31 asks what that computation costs, how the EM iteration can be accelerated, and what the accelerated iteration computes. Chapter 32 reads Weil's explicit formula as the deconvolution at level one half and recovers zeros from the level one density by the matrix pencil, above the verified range as well. Chapter 33 collects what is known about Cauchy deconvolution as an ill posed problem, proves the facts the book uses, and places the book's results in the literature. Chapter 34 deconvolves from level one to the critical line by particle filters, contour moments and zero free regions from positivity. Chapter 35 proves the Balazard, Saias and Yor criterion at every level and height and solves it by Monte Carlo.

Tables 1.1, 1.2 and 1.3 collect the principal results of Chapters 2 to 11, each with the section where it is developed. A result is marked proved when that section proves it or cites a proof, and computed when it rests on a computation; the section states how far the computation is certified.

| Result | Status | Where |
|---|---|---|
| Pólya route: $\xi$ and $\xi'$ from two fixed polynomials in ball arithmetic, about 60 correct digits at height 200 at 420 bits, 0.7 ms per point | computed | Section 2.1 |
| Tilted contour: 25 digits of $\rho_{0.05}$ at height 5000 against `mpmath`, in 28 ms per point; to leading exponential order the loss to cancellation falls from about $0.341\,\theta$ digits to about $\delta\theta/\log 10$ | proved and computed | Section 2.1 |
| The Lehmer pair near height 7005 resolved by the tilted Pólya route at 34 ms per point | computed | Section 2.2.3.1 |
| $N(1000) = 649$ from the Thorin mass at resolution $\frac{1}{2}$ and the argument along a horizontal segment | computed | Section 2.2.3.2 |
| Raw total variation to $U_{\mathrm{RH}}$ maximal; matched total variation at the floor $3.7 \times 10^{-9}\varepsilon$ of the tail model; cell distance $5.44\,\varepsilon$ | computed | Section 3.2 |
| Mass defect $-(\varepsilon/\pi)\Xi'(\Theta)/\Xi(\Theta)$ to first order; at $\Theta = 200$ the observed slope agrees with it to five digits at $\varepsilon = 0.005$ | proved and computed | Section 3.2 |
| Signatures of an off line pair in total variation and negative mass, exact for an omitted pair and for the negative mass, to leading order for a displaced pair | proved | Section 3.3 |

| Result | Status | Where |
|---|---|---|
| Zero density $\nu(\frac{1}{2}+\varepsilon, 200) = 0$ at eight resolutions, from the argument principle identity of Proposition 3.4.2 | computed | Section 3.4 |
| Prime side of the sine squared lemma: when no zero has real part greater than $\frac{1}{2}$, Selberg's smoothed sum computes $\rho_\varepsilon$ with relative error at most $(x^{-\varepsilon}+x^{-2\varepsilon})/(\varepsilon \log x)$, so with error of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$ at the peaks, the single zero law; tested at $\varepsilon = \frac{1}{4}$ | proved and computed | Section 3.5 |
| The Thorin measures converge vaguely to the measure of the zeros on the line, whether or not RH holds | proved | Section 4.1 |
| The clock is HCM if and only if RH holds, and the shifted clock at any basepoint $\alpha \geq 1$ is HCM unconditionally; the 216 normalised derivatives of a finite HCM test are all positive, certified in ball arithmetic | proved and computed | Section 4.1 |
| Bosch type threshold: $1/G(x^{1/\beta})$ is HCM if and only if $\lvert\beta\rvert \geq 1/(1-\alpha_\xi)$, with $\alpha_\xi$ set by the largest angle of the zeros; unconditionally $1/G(x^p)$ is HCM for $\lvert p\rvert \leq 1-1/(\pi T_0)$, where $T_0 = 3\times 10^{12}$ is the verified height, in particular for $\lvert p\rvert \leq 1 - 1.1\times 10^{-13}$ | proved | Section 4.3 |
| The clock is infinitely divisible and self decomposable with no hypothesis, and $(-1)^k W^{(k)} > 0$ for every $k \leq 1.16\times 10^{25}$, that is, the Williamson measures, inverse gamma smoothings of the Thorin measure, are positive to that order; the two proofs use the verification of RH below heights 1000 and $3\times 10^{12}$ respectively [419] | proved | Section 5.1 |

Table 1.1: Principal results of Chapters 2 to 5.

| Result | Status | Where |
|---|---|---|
| The normalised reciprocal zeta function is infinitely divisible with no hypothesis, and it is not self decomposable, not a GGC and not in the Bondesson class | proved and computed | Section 7.1 |
| Conjecture S, that the symmetrised truncations of the Biane, Pitman and Yor lattice have their zeros in the strip on the line, holds for the sizes $n \leq 24$ and $n = 32, 36$ above explicit heights and numerically below them; for infinitely many $n$ it would imply RH | proved and computed | Section 7.2 |
| Law of the clock by Talbot inversion, with no zero as input; its hazard tends to $\gamma_1^2$, reached to eleven digits at $t = 0.25$ | computed | Section 8.1 |
| Kent approximants with 1700 zeros and a moment matched Weyl tail: relative error $3.5\times 10^{-23}$ at $u = 1$ and $7.0\times 10^{-13}$ at $u = 10^4$ | computed | Section 8.1 |
| The clock is not the first passage time of Brownian motion to an independent random level; the boundary from the Volterra equation, solved in three zones, reproduces its law to Monte Carlo accuracy (Kolmogorov distance 0.0020, against 0.0059 for the tangent boundary) | proved and computed | Section 8.2 |
| Keiper Li coefficients $\lambda_1, \ldots, \lambda_{100}$ from the Pólya density, $\lambda_1$ to 25 digits | computed | Section 8.3 |

| Result | Status | Where |
|---|---|---|
| Closest pair below height 200 collides at de Bruijn Newman time $-0.288$; the Lehmer pair near 7005 at $-7.11 \times 10^{-4}$ | computed | Section 8.4 |
| For $L(s, \chi_{12})$ and $L(s, \Delta)$ the sine squared lemma has no pole piece but a signed prime piece | proved | Section 8.5 |
| The eta kernel of $L(s, \chi_{12})$: GRH below height 100 from the zero density identity and the 67 sign changes | computed | Section 8.5 |
| For an imaginary quadratic field the Pólya kernel of $\zeta_K$ is positive exactly when an explicit theta sum is positive on a finite interval, which gives $L(\frac{1}{2}, \chi_d) > 0$ for $d = -3, -4, -7, -8$; it takes negative values for every $d \equiv 5 \pmod{24}$ | proved | Section 8.5 |
| For the Dirichlet beta function $L(s, \chi_4)$, whose kernel is the cube of the eta function: GRH below height 100; a positive Thorin density at every basepoint $\alpha \geq 1$, which the absolute values of the prime weights certify only for $\alpha > 2.2823$; the density as the difference of those of $\xi_{\mathbb{Q}(i)}$ and $\xi$, and as the derivative of the argument of the completed function on the line | proved and computed | Section 8.5.7 |
| Nonnegative deconvolution of $\rho_{0.1}$ on $[15, 55]$, with the zeros outside $[5, 65]$ as background, recovers every zero in the window to within $10^{-5}$ | computed | Section 8.6 |
| For Pólya's Bessel function $K_{i\theta/2}(2\pi)$ the index side clock is the hitting time of an explicit diffusion, its Thorin atoms are the squared zeros and its trace is completely monotone; its de Bruijn Newman constant is finite, by Newman's theorem, although none of its real zeros below height 90 collide down to $t = -60$ | proved and computed | Section 8.7 |

Table 1.2: Principal results of Chapters 6 to 8.

| Result | Status | Where |
|---|---|---|
| $T_1$, with Laplace transform $\xi(1)/\xi(1+\sqrt{u})$, is a GGC with positive Thorin density, smallest at the origin, by a computation of the zeros below height 2000 in ball arithmetic and explicit zero counts above it, so that $T_1$ is a positive stable one half variable plus an independent GGC; its law has infinite mean and a stable one half tail, and the Lévy form of $\xi(1 + s)$, which converges by the prime number theorem, gives its Thorin density from the primes | proved and computed | Section 9.1 |
| Below the basepoint one, $v_\alpha(\theta)/(\alpha - \frac{1}{2})$ is nonincreasing in $\alpha$ away from the zeros off the line; the Thorin density is smallest at the origin, and only there, for every $\alpha \in (\frac{1}{2}, 7.35]$ on $\lvert\theta\rvert \leq 3 \times 10^{12} - 1.21$, for $L(s, \chi_4)$ and $L(s, \chi_{12})$ at every height for every $\alpha \geq 1$, and at every height for every $\alpha \in (\frac{1}{2}, 1)$ exactly when RH holds; for every $\alpha > \frac{1}{2}$ it is positive on $\lvert\theta\rvert \leq 3 \times 10^{12} - \frac{1}{2}$ | proved and computed | Sections 9.1.6, 11.4 |

| Result | Status | Where |
|---|---|---|
| Every $T_\alpha$, $\alpha > \frac{1}{2}$, exists as the tilted centre clock run through the inverse local time of a sticky Brownian motion, and it is a GGC if and only if no zero has real part greater than $\alpha$ | proved | Section 9.2 |
| For real $u$ and $\alpha \geq \frac{1}{2}$, $\xi(\alpha)/\xi(\alpha + \lvert u\rvert)$ is the characteristic function of an infinitely divisible normal scale mixture; for $\frac{1}{2} < \alpha < 1$ it is an extended GGC, and the translates of every tilted Pólya kernel $\Phi_{\alpha'}$ with $\alpha' > \alpha$ are complete in $L^1$, exactly when no zero has real part greater than $\alpha$ | proved | Section 10.1 |
| The complete monotonicity conditions of the Poisson trace $K_{1/2}$, which together are RH, hold for every order $n < T_0(\pi - y)$ at the point $y$; $K_{1/2}$ and the trace of Pólya's Bessel function differ by $\frac{9}{8} + O(y \log \frac{1}{y})$ as $y \to 0$ | proved and computed | Section 10.1.5 |
| $1/\zeta(2k)$ is, up to an elementary factor, a value of the Laplace transform of the centre clock and of $T_1$; for $\frac{1}{2} \leq \alpha < 1$ the bound $O(x^{\alpha/2+\delta})$ on Riesz's function, for every $\delta > 0$, holds exactly when every $T_{\alpha'}$ with $\alpha' > \alpha$ is a GGC | proved | Section 10.2 |
| The constant of $\sum_{n \leq x} \mu(n)^2/n$ is an elementary term plus $\mathbb{E}V_1 - 2\,\mathbb{E}V_2$, two tilted Pólya means; RH holds exactly when the squarefree count smoothed by an exponential or a Gamma(2) law, or the count in mean square, has error $O(x^{1/4+\varepsilon})$; the error is $\Omega_\pm(x^{1/4})$; Gamma(2) smoothing writes it as an average of the Mertens function under a positive law, and two is the largest shape for which this holds | proved | Section 10.3 |
| The centre clock is driven by a subordinator with Lévy density $-W'$ and is unimodal, as is $T_\alpha$ for $\alpha \geq 1$; the tilted Pólya variable is quasi infinitely divisible for every $\alpha > 1$ but not at $\alpha = 1$ | proved | Section 11.1 |
| For $L(s, \chi_{12})$ the Thorin measure at the basepoint one is positive, and its Lévy form there has no pole to compensate and prime atoms of both signs | proved | Section 11.3 |
| For $\frac{1}{2} < \alpha < 1$, in the density $\rho_{\alpha-1/2}$ the Cauchy kernels of the zeros to the right of $\alpha$ carry at heights $\theta \leq T$ at most $T^{e(\alpha)-1+o(1)}$ times the mass of the kernels of the other zeros, with $e(\alpha) < 1$, by the zero density estimates of Ingham, of Guth and Maynard and of Huxley [284, 166, 279] | proved | Section 11.4 |
| With no hypothesis, the size biased centre clock is the clock plus an independent variable with density $W/\mathbb{E}T$, and RH holds exactly when this variable is a mixture of exponential laws, with weight $\gamma^{-2}/\mathbb{E}T$ at the rate $\gamma^2$, about a fifth at the lowest zero; the Lambert law, with Laplace transform $\mathcal{W}(u)/u$, is a GGC whose Thorin measure is the image of the uniform law on $(0, \pi)$ under $b \mapsto b\, e^{-b \cot b}/\sin b$, by a proof of which only one step fails for the clock, and that step is RH | proved and computed | Sections 11.5.2, 11.5.3 |

| Result | Status | Where |
|---|---|---|
| The centre clock lies in Urbanik's class $L_2$ of multiply self decomposable laws and not in $L_3$; it is not the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter | proved and computed | Section 11.5.4 |
| A property of even kernels that also holds for the damped Pólya kernel $e^{tv^2/4}\Phi(v)$ for some $t < 0$ cannot by itself imply RH, by the theorem of Rodgers and Tao; the GGC property of the variable of Biane, Pitman and Yor fails for the damped kernel at every $t < 0$ tested and, provably, for small $\lvert t\rvert$, so the filter does not exclude it there | proved and computed | Section 11.5.5 |

Table 1.3: Principal results of Chapters 9 to 11. All are unconditional, and several use verified computations: the verification of RH below height 1000, through Theorem 5.1.1, or below the height $T_0 = 3 \times 10^{12}$, Platt's verification of GRH for small moduli, or computations in ball arithmetic.

### 1.1.3 Conventions and how to read this book

The clock is a function of $u = s^2$; earlier papers of the programme used $s^2/2$, which rescales Thorin measures by $z \mapsto z/2$. Thorin measures are measures in the variable $z$, and we write $\theta = \sqrt{z}$ for the chart in which the Thorin measure at the basepoint $\frac{1}{2} + \varepsilon$ has density $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$; $\varepsilon$ is the resolution. The Pólya kernel $\Phi$ is taken in Titchmarsh's normalisation, $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u) \cos(\theta u)\, du$. $N(T)$ counts zeros with $0 < \operatorname{Im}\rho \le T$ and $\nu(\sigma, T)$ those with, in addition, $\operatorname{Re}\rho > \sigma$. Total variation of a signed measure is the total variation norm $\lvert\mu\rvert(E)$ on the stated window. Chapters 6 and 7, and Sections 4.2 and 4.3, write $s$ for $u$, and the Kent approximants of Section 8.1 write $w$; each declares its variables where it starts, and Appendix C lists the notation of the book with the place where each symbol is defined. Numerical values are quoted to the accuracy at which they have been checked.

Readers mainly interested in computation can read Sections 1.2, 2.1 and 2.3 and then go straight to Chapters 3 to 5. Readers interested in the probabilistic side can start with Sections 1.3, 8.1 and 8.2 and continue with Chapters 9 to 11; these chapters build on Sections 1.2 and 1.3 and draw on Section 3.4 for the zero count. Chapters 6 and 7 can be read after Chapter 1 and the statements of Theorems 5.1.1 and 5.1.6 in Section 5.1; they form the most technical part of the book. Readers looking for the conclusions can go directly to Sections 12.1 and 12.2.6. Throughout, statements are marked as proved, computed, announced or open.

## 1.2 The Xi function, the Pólya density and the clock

Most of the material is classical or taken from [424].

### 1.2.1 The completed zeta function and the clock

The completed zeta function $\xi$ of Section 1.1 is entire of order one and maximal type, satisfies $\xi(s) = \xi(1-s)$ and $\overline{\xi(\bar{s})} = \xi(s)$, and is positive on the real axis. On the critical line $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ is real and even. The zeros $\rho = \beta + i\gamma$ of $\xi$ lie in the open strip $0 < \operatorname{Re} s < 1$ and are symmetric under $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar{\rho}$.

Hadamard's theorem applied to $G$, which is entire of order $\frac{1}{2}$, gives, with $u = s^2$,

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \prod_k \Big(1 - \frac{s^2}{(\rho_k - \frac{1}{2})^2}\Big) = \prod_k \Big(1 + \frac{s^2}{a_k}\Big), \tag{1.2.1}$$

with $\sum |a_k|^{-1} < \infty$. Logarithmic differentiation gives the partial fraction expansion

$$\frac{\xi'}{\xi}(s) = \sum_\rho \frac{1}{s-\rho}, \tag{1.2.2}$$

in which each $\rho$ is grouped with $1-\rho$, as the differentiation of (1.2.1) produces, or with $\bar{\rho}$. Grouped either way the series converges absolutely, since each group is $O(\gamma^{-2})$ for fixed $s$ and $\sum_\rho \gamma^{-2} < \infty$; without grouping it converges only when summed symmetrically in $\operatorname{Im}\rho$.

Passing to $u = s^2$ folds each pair $\pm(\rho_k - \frac{1}{2})$ of zeros of $\xi(\frac{1}{2}+s)$ into the single zero $-a_k$ of $G$, so the product has genus zero and needs no exponential factors. Since $\xi$ has no real zeros, $a_k$ is real only when $\rho_k$ lies on the critical line, and then $a_k = \gamma_k^2 > 0$. RH is therefore the statement that every zero of $G$ lies on the negative real axis. The reciprocal of a product of genus zero with negative zeros is a product of gamma Laplace transforms, and this is how generalised gamma convolutions enter.

A random variable $H \geq 0$ is a generalised gamma convolution (GGC) if $\mathbb{E}e^{-uH} = \exp\{-du - \int_{(0,\infty)} \log(1+u/z)\, U(dz)\}$ for $u > 0$, with a drift $d \geq 0$ and a positive measure $U$, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1} U(dz) < \infty$. The drift and the Thorin measure are unique, and the Laplace transform of a GGC extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$.

Unconditionally, $1/G$ is the Laplace transform of a probability law on $[0,\infty)$ [424, Theorem A]; Section 5.1 states this as Theorem 5.1.1 and proves it. A random variable $T$ with $\mathbb{E}e^{-uT} = 1/G(u)$ for $u \geq 0$ is called the clock variable. Its mean and variance are $\mathbb{E}T = \sum_k a_k^{-1} = G'(0) = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ and $\operatorname{Var} T = \sum_k a_k^{-2} = 3.71726 \times 10^{-5}$; these constants, and $\xi(\frac{1}{2}) = 0.4971207781883141$, recur throughout the book.

Proposition 1.2.1 (Rigidity, [424, Proposition 3.4]). *The clock $1/G$ is the Laplace transform of a GGC if and only if RH holds, and then its Thorin measure is $U_{\mathrm{RH}}$ of (1.1.1) and the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials $e_\gamma$.*

If $1/G$ is the Laplace transform of a GGC with drift $d$ and Thorin measure $U$, then $d + \int U(dz)/(u+z) = G'/G(u) = \sum_k (u+a_k)^{-1}$ for $u > 0$. The left side is holomorphic on $\mathbb{C}\setminus(-\infty, 0]$, so every pole $-a_k$ of the right side lies on the negative real axis. Hence every $a_k$ is positive, which is RH, and uniqueness of the Stieltjes representation forces $d = 0$ and $U = \sum_k \delta_{a_k}$. Conversely, under RH, (1.2.1) writes $1/G$ as a product of gamma Laplace transforms.

### 1.2.2 The Pólya kernel and the Pólya density

Riemann's continuation of $\zeta$ writes $\xi$ as a Mellin transform of a combination of $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and its derivatives, and uses the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$, $x > 0$. The substitution $x = e^{2u}$ turns the Mellin transform into a Fourier transform, and in Titchmarsh's normalisation [498, Section 10.1] the result is

$$\begin{aligned} \Phi(u) &= 2\sum_{n\geq 1} \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right) \exp(-\pi n^2 e^{2u}), \\ \Xi(\theta) &= \int_{\mathbb{R}} \Phi(u)\cos(\theta u)\, du. \end{aligned} \tag{1.2.3}$$

Here $u$ is the variable of the kernel, not the argument of the clock. Pólya studied this representation [447], and $\Phi$ is called the Pólya kernel.

The kernel is even and strictly positive, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2}\exp(-\pi e^{2|u|})$ as $|u| \to \infty$. Evenness is the functional equation read through the theta transformation (Exercise 1.2). For $u \geq 0$ every term of (1.2.3) is positive, because $2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2} = \pi n^2 e^{5u/2}(2\pi n^2 e^{2u} - 3)$ and $2\pi n^2 e^{2u} \geq 2\pi > 3$; positivity for $u < 0$ follows by evenness. Evenness decides how $\Phi$ is computed: term $n$ is $n^{-1/2}$ times the first term translated to the left by $\log n$, so for $u \geq 0$ the first term dominates, while for $u < 0$ the terms are individually of order $n^{-1/2}$ and cancel almost completely, and computations therefore evaluate $\Phi(|u|)$.

Each term of (1.2.3) is entire in $u$, and since $\operatorname{Re} e^{2u} = e^{2x}\cos 2y$ for $u = x + iy$, the series converges locally uniformly in the strip $|\operatorname{Im} u| < \pi/4$, so $\Phi$ is analytic there; the boundary lines $\operatorname{Im} u = \pm\pi/4$ are a natural boundary inherited from the theta function [458, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The strip of analyticity makes the trapezoid rule spectrally accurate and permits the tilted contour of Section 2.1.3. Figure 1.2 shows $|\Phi|$ on lines inside the strip.

Since $\Xi(0) = \xi(\frac{1}{2}) = \int_{\mathbb{R}} \Phi$, the Pólya density $p = \Phi/\Xi(0)$ is the law of a symmetric random variable $X$ with all exponential moments, and

$$\frac{\xi(\frac{1}{2} + s)}{\xi(\frac{1}{2})} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad s \in \mathbb{C}. \tag{1.2.4}$$

The clock is $1/\mathbb{E}\cosh(\sqrt{u}\,X)$, and at $u = \theta^2$ it forms with the characteristic function of $X$ a van Dantzig pair, that is, a pair $(\varphi_1, \varphi_2)$ of characteristic functions

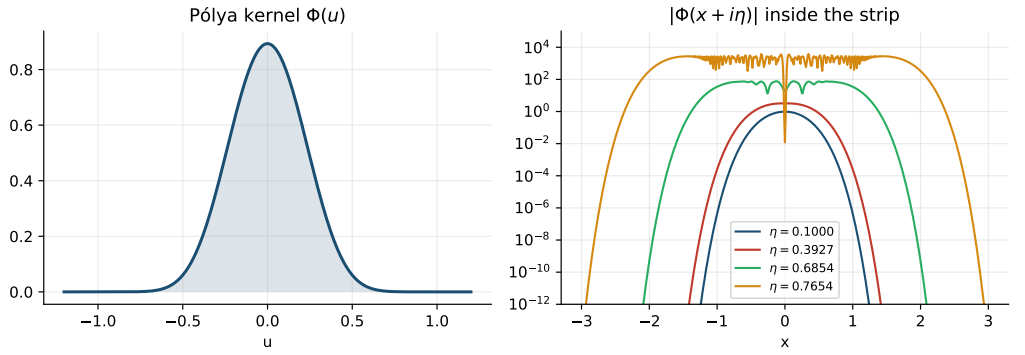



Figure 1.2: Left: the Pólya kernel $\Phi(u)$ on the real line. Right: $|\Phi(x+i\eta)|$, on a logarithmic scale, on the lines $\eta = 0.1000$, 0.3927, 0.6854 and 0.7654 inside the strip of analyticity, that is at distances 0.6854, 0.3927, 0.1 and 0.02 below its edge $\eta = \pi/4$ (the tilts of Section 2.1.3). As the line approaches the natural boundary $\eta = \pi/4$ the kernel grows and spreads (compare Table 2.1), and it still decays doubly exponentially in $|x|$.

with $\varphi_2(\theta) = 1/\varphi_1(i\theta)$; the name goes back to a problem of van Dantzig, which Lukacs studied [366]. Here $\varphi_1(\theta) = \Xi(\theta)/\Xi(0)$ is the characteristic function of $X$, and $1/\varphi_1(i\theta) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta) = \mathbb{E}e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with $Z$ standard normal and independent of $T$ [424, Theorem A]. G. Letac asked whether $\varphi_1$ is the first member of a van Dantzig pair (private communication), and Theorem 5.1.1 shows that it is, unconditionally.

Two classical pairs show the shape to expect. The first is $(\cos\theta, 1/\cosh\theta)$ and the second is $(\sin\theta/\theta, \theta/\sinh\theta)$. In both the first member has only real zeros and the second is the product of the reciprocal factors: $1/\cosh\theta = \prod_{k\geq 1}(1+\theta^2/((k-\frac{1}{2})\pi)^2)^{-1}$ is the characteristic function of $\sum_{k\geq 1} L_k/((k-\frac{1}{2})\pi)$, where the $L_k$ are independent standard Laplace variables, with density $\frac{1}{2}e^{-|x|}$ and characteristic function $1/(1+\theta^2)$, and likewise $\theta/\sinh\theta$ is that of $\sum_{k\geq 1} L_k/(k\pi)$.

The Xi pair has the classical shape exactly when RH holds. If RH holds, $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 1.2.1), so $Z\sqrt{2T}$ has the law of $\sum_{\gamma>0} L_\gamma/\gamma$, the sum that Section 4.2 calls $\log W$. Conversely, if $Z\sqrt{2T}$ has the law of $\sum_k L_k/c_k$ for real $c_k$ with $\sum_k c_k^{-2} < \infty$, then $\xi(\frac{1}{2}+\theta)/\xi(\frac{1}{2}) = \prod_k(1+\theta^2/c_k^2)$ for real $\theta$, hence for complex $\theta$, and every zero of $\xi$ lies on the critical line. Konstantopoulos, Patie and Sarkar state the same equivalence in the language of the Laguerre Pólya class and of Pólya frequency functions [313, Theorem 2.3 and Corollary 2.4].

The Pólya variable also has a probabilistic construction. Let $S_2 = \sum_{k\geq 1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponentials $e_k, e'_k$; it has the law of the sum of two independent hitting times of level 1 by a three dimensional Bessel process started at 0. By the theorem of Biane, Pitman and Yor [49], going

back to Williams [518], $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$. With $Y = \frac{1}{2}\log(\pi S_2/2)$ this reads $\mathbb{E}e^{sY} = 2\xi(s)$, so $X$ is $Y$ tilted by $e^{Y/2}$: $\mathbb{E}f(X) = \mathbb{E}[f(Y)e^{Y/2}]/\mathbb{E}e^{Y/2}$. Tilting $Y$ by $e^{\alpha Y}$ in place of $e^{Y/2}$ gives, at every real basepoint $\alpha$ and with no hypothesis on the zeros, a variable $V_\alpha$ with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$ and $V_{1/2} = X$ (Section 9.1.1).

### 1.2.3 The heat trace: spectral and arithmetic forms

The heat trace of the zeros is

$$W(t) = \sum_k e^{-a_k t} = \tfrac{1}{2}\sum_\rho e^{(\rho-\frac{1}{2})^2 t}, \qquad t > 0,$$

and the series converges absolutely for every $t > 0$ wherever the zeros lie, since $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > \gamma_k^2 - \frac{1}{4}$ for $\rho_k = \beta_k + i\gamma_k$. Integrating $e^{-(u+a_k)t}$ term by term, and using Frullani's integral $\log(1+u/a) = \int_0^\infty (1-e^{-ut})e^{-at}t^{-1}dt$ for $\operatorname{Re} a > 0$, gives, unconditionally for $u > 0$ [424, Theorem 4.3],

$$\frac{G'}{G}(u) = \int_0^\infty e^{-ut} W(t)\,dt, \qquad \log G(u) = \int_0^\infty \bigl(1-e^{-ut}\bigr)\frac{W(t)}{t}\,dt.$$

The second formula shows that $W(t)/t$ is the only possible Lévy density of the clock variable $T$.

The classes of infinitely divisible laws of Section 1.3.2 become sign conditions on $W$ [424, Theorem 4.6]: $T$ is infinitely divisible if and only if $W \geq 0$, self decomposable if and only if $W$ is nonincreasing, and a GGC if and only if $W$ is completely monotone, which by Proposition 1.2.1 happens if and only if RH holds. The first two conditions hold unconditionally, in the strict forms $W > 0$ and $W' < 0$ on $(0,\infty)$ (Theorem 5.1.1, [424, Theorem A]), by a proof that uses the verification of RH below height 1000.

The explicit formula expresses $W$ through the primes. For even test functions $h$ holomorphic and rapidly decreasing in a strip $|\operatorname{Im} r| \leq \frac{1}{2} + c$ with $c > 0$, Weil's explicit formula [512], [288, Theorem 5.12], reads

$$\sum_\rho h(\gamma_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\bigl(\operatorname{Re}\psi(\tfrac{1}{4}+\tfrac{ir}{2}) - \log\pi\bigr)dr \\ - 2\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n),$$

where $\rho = \frac{1}{2}+i\gamma_\rho$, $g(x) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)e^{-irx}dr$ and $\psi = \Gamma'/\Gamma$ is the digamma function. The Gaussian $h(r) = e^{-r^2 t}$ has $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$ and $\sum_\rho h(\gamma_\rho) = 2W(t)$, which gives the arithmetic form [424, Theorem 4.4]

$$W(t) = e^{t/4} + \frac{1}{4\pi}\int_{\mathbb{R}} e^{-r^2 t} m(r)\,dr - \frac{1}{2\sqrt{\pi t}}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}e^{-(\log n)^2/4t}, \\ m(r) = \operatorname{Re}\psi\bigl(\tfrac{1}{4}+\tfrac{ir}{2}\bigr) - \log\pi. \tag{1.2.5}$$

The three terms of (1.2.5) are the polar, archimedean and prime contributions. The polar term $e^{t/4}$ comes from $h(\pm\frac{i}{2})$, that is, from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s=1$ and $s=0$, which the factor $s(s-1)$ of $\xi$ removes. The prime term is the series $\sum \Lambda(n)n^{-1/2}$ smoothed by a Gaussian of standard deviation $\sqrt{2t}$ in $\log n$.

Table 1.4 reproduces the comparison of [424, Table 4.1], and the two sides agree to fifteen digits. The polar and archimedean terms are of order one while $W(t)$ decays like $e^{-\gamma_1^2 t}$ with $\gamma_1^2 = 199.79$. An estimate of $W$ from (1.2.5) at moderate $t$ must therefore survive a cancellation whose depth grows linearly in $t$. Section 5.3.2 extends the comparison to smaller and larger $t$ and gives that depth explicitly.

| $t$ | $W(t)$ from zeros | polar + archimedean − prime | archimedean | prime |
|---|---|---|---|---|
| 0.01 | $0.149698011253780$ | $0.149698011253780$ | $-0.8527967$ | $8.3995\times10^{-6}$ |
| 0.02 | $1.85412917313195\times10^{-2}$ | $1.85412917313195\times10^{-2}$ | $-0.9840612$ | $0.0024101$ |
| 0.05 | $4.58783504857357\times10^{-5}$ | $4.58783504857357\times10^{-5}$ | $-0.9546186$ | $0.0579140$ |
| 0.10 | $2.10479979582235\times10^{-9}$ | $2.10479979582235\times10^{-9}$ | $-0.8625096$ | $0.1628055$ |
| 0.20 | $4.43018218022364\times10^{-18}$ | $4.43018218022364\times10^{-18}$ | $-0.7497538$ | $0.3015173$ |

Table 1.4: The heat trace from the zeros and from the explicit formula, after [424, Table 4.1]. At $t=1$ the cancellation between the arithmetic terms reaches about eighty seven digits.

### 1.2.4 The Thorin density at a shifted basepoint

The formula for the Thorin density in the theorem below defines a density at every basepoint $\alpha > \frac{1}{2}$, whatever the zeros do, and it is the starting point of the computations in Chapters 3 to 5. For a real basepoint $\alpha$ write $v_\alpha(r) = \mathrm{Re}(\xi'/\xi)(\alpha+ir)$, and let $U_\alpha$ be the signed measure on $(0,\infty)$ given by

$$U_\alpha(dz) = \frac{v_\alpha(\sqrt{z})}{2\pi\sqrt{z}}\,dz. \tag{1.2.6}$$

At $\alpha = \frac{1}{2}+\varepsilon$, where $\varepsilon>0$ is the resolution, the chart $\theta=\sqrt{z}$ gives $U_\alpha$ the density

$$\rho_\varepsilon(\theta) = \frac{1}{\pi}\,v_{1/2+\varepsilon}(\theta) = \frac{1}{\pi}\,\mathrm{Re}\,\frac{\xi'}{\xi}\Big(\frac{1}{2}+\varepsilon+i\theta\Big), \tag{1.2.7}$$

so that $U_\alpha(dz) = \rho_\varepsilon(\theta)\,d\theta$. The family $\varepsilon\mapsto\rho_\varepsilon$ is the Cauchy flow of [424, Chapter 8].

Theorem 1.2.2 ([424, Theorem 3.7 and Remark 3.10]). *If $\alpha > \mathrm{Re}\,\rho$ for every zero, in particular for every $\alpha\ge 1$, then $u\mapsto \xi(\alpha)/\xi(\alpha+\sqrt{u})$ is the Laplace transform of a GGC $T_\alpha$ with Thorin density* (1.2.6)*, and $v_\alpha>0$. It is the Laplace transform of a GGC if and only if $\alpha\ge\sup_\rho \mathrm{Re}\,\rho$.*

Until 2026 no basepoint $\alpha < 1$ was known to satisfy the hypothesis of the theorem; the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would give every $\alpha > \frac{7}{8}$ (Section 12.1.2). At every basepoint $\alpha > \frac{1}{2}$ the shifted clock is still the Laplace transform of a probability law, with no hypothesis on the zeros (Theorem 9.2.5); what the zeros decide is whether that law is a GGC. Let $P_a(x) = a/(\pi(a^2 + x^2))$ denote the Poisson kernel, with $P_{-a} = -P_a$. Write $\mu_\varepsilon$ for the signed measure $U_{1/2+\varepsilon}$, and use the same symbol for its image in the chart $\theta = \sqrt{z}$, where it has the density (1.2.7). By Theorem 1.2.2, $\mu_\varepsilon$ is the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac{1}{2} + \varepsilon > \operatorname{Re}\rho$ for every zero, in particular for every $\varepsilon \geq \frac{1}{2}$, and we call it the Thorin measure at resolution $\varepsilon$ in general.

Proposition 1.2.3 (Poisson structure). *For $\varepsilon > 0$ and $\theta \geq 0$ such that $\frac{1}{2} + \varepsilon + i\theta$ is not a zero of $\xi$,*

$$\rho_\varepsilon(\theta) = \sum_{\gamma>0,\ \operatorname{Re}\rho=1/2} \big[P_\varepsilon(\theta-\gamma) + P_\varepsilon(\theta+\gamma)\big] + \sum_{\text{off line } \rho=\frac{1}{2}\pm\delta+i\gamma,\ \gamma>0} \Big(\big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta-\gamma) + \big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta+\gamma)\Big), \quad (1.2.8)$$

*where the second sum counts each off line pair $\{\frac{1}{2}+\delta+i\gamma, \frac{1}{2}-\delta+i\gamma\}$, $0 < \delta < \frac{1}{2}$, once. Equivalently, with $\alpha = \frac{1}{2}+\varepsilon$, $\rho_\varepsilon(\theta)$ is the sum of $P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)$ over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$. In particular, under RH, $\mu_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ restricted to $\theta > 0$, where $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma + \delta_{-\gamma})$.*

*Proof.* Take real parts in (1.2.2) at $s = \alpha + i\theta$ with $\alpha = \frac{1}{2} + \varepsilon$: a zero $\rho = \beta + i\gamma$ contributes $\pi P_{\alpha-\beta}(\theta-\gamma)$ to $v_\alpha(\theta)$, and $\alpha - \beta = \varepsilon \mp \delta$ when $\beta = \frac{1}{2} \pm \delta$. Dividing by $\pi$ and grouping each zero $\rho$ with $\bar\rho$ gives the second form, and collecting the zeros $\rho$, $\bar\rho$, $1-\rho$ and $1-\bar\rho$ gives (1.2.8). The series converges absolutely, and uniformly for $\theta$ in compact sets avoiding the zeros, since each term is $O(\gamma^{-2})$ there and $\sum_\rho \gamma^{-2} < \infty$. □

An off line pair contributes kernels of widths $\varepsilon + \delta$ and $\varepsilon - \delta$, and the second is negative once $\varepsilon < \delta$, that is, once the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ lies to the left of the zero $\frac{1}{2} + \delta + i\gamma$. If every zero lies to the left of the line, every kernel is positive and so is $v_\alpha$; a negative value of $v_\alpha$ therefore proves a zero to the right of $\alpha$.

Remark 1.2.4 (Positivity on one line). The criterion of Theorem 1.2.2 concerns the analytic continuation of the shifted clock, not the sign of $v_\alpha$ on the single line $\operatorname{Re} s = \alpha$. Once a zero $\rho$ lies to the right of the line, the shifted clock ceases to be a GGC, since $\xi(\alpha + \sqrt{u})$ vanishes at $u = (\rho - \alpha)^2 \in \mathbb{C} \setminus (-\infty, 0]$, but the density need not change sign, because the positive kernels of the other zeros can outweigh the negative one. Exercise 1.7 gives an entire function with the order, symmetry and reality of $\xi$ whose density is positive on a line with a zero to its right; for $\xi$ itself it is not known, without assuming RH, whether positivity on one line excludes

zeros to its right. Positivity on every line to the right of $\operatorname{Re} s = \sigma_0$ does exclude zeros there: if $v_\alpha \geq 0$ for every $\alpha > \sigma_0$, then no zero has $\operatorname{Re}\rho > \sigma_0$, because $\operatorname{Re}(\xi'/\xi)(\rho - b) \sim -m/b$ as $b \downarrow 0$ at a zero $\rho$ of multiplicity $m$. With $\sigma_0 = \frac{1}{2}$ this is Lagarias's criterion [320]: RH holds if and only if $v_\alpha \geq 0$ for every $\alpha > \frac{1}{2}$.

Integrated over $\theta > 0$, a line zero contributes mass one, which concentrates at $\theta = \gamma$ as $\varepsilon \downarrow 0$. An off line pair contributes mass two when $\varepsilon > \delta$, and when $\varepsilon < \delta$ a signed measure of mass zero whose total variation tends to zero as $\varepsilon \downarrow 0$. These limits can be summed over the zeros: if $f$ is continuous with support in $[a, b] \subset (0, \infty)$, then for $0 < \varepsilon \leq 1$ each term of (1.2.8) with $\gamma > b + 1$ integrates $f$ to a number of modulus at most $C_f \gamma^{-2}$, where $C_f$ depends only on $f$, because the widths $c$ in (1.2.8) satisfy $|c| < \varepsilon + \frac{1}{2} \leq \frac{3}{2}$, $|P_c(x)| \leq |c|/(\pi x^2)$, and $|\theta \pm \gamma| \geq \gamma - b \geq \gamma/(b+1)$ for $\theta \in [a, b]$; as $\sum_\rho \gamma^{-2} < \infty$, dominated convergence applies to the sum over the zeros. Hence $\mu_\varepsilon \to U_{\text{line}} = \sum_{\text{line zeros}} \delta_{\gamma^2}$ vaguely, whatever the truth of RH [421, Proposition 9.5].

### 1.2.5 The sine squared lemma: the Thorin density from the primes

The sine squared lemma produces the same Thorin density from the Lévy side, where the primes appear as atoms. For $x > 0$ and $s \geq 0$,

$$e^{-sx} + sx - 1 = x \int_0^\infty \log\Big(1 + \frac{s^2}{z}\Big) \frac{\sin^2(x\sqrt{z}/2)}{\pi\sqrt{z}}\, dz. \tag{1.2.9}$$

To prove it, substitute $z = \theta^2$, which turns the right side into $\frac{x}{\pi} \int_0^\infty \log(1 + s^2/\theta^2)(1 - \cos x\theta)\, d\theta$. Both sides vanish at $s = 0$, and their derivatives in $s$ agree because $\int_0^\infty (1 - \cos x\theta)\, 2s(\theta^2 + s^2)^{-1} d\theta = \pi(1 - e^{-sx})$.

The identity (1.2.9) is [421, Lemma 8.1].

Integrating (1.2.9) against a measure turns Lévy representations into Thorin representations. A Lévy representation $\log f(\alpha + s)/f(\alpha) = cs + \int (e^{-sx} + sx - 1) L(dx)$ for $s \geq 0$, with $c$ real and $L$ a real measure on $(0, \infty)$ such that $\int \min(x, x^2)\, |L|(dx) < \infty$, becomes a Thorin representation with density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\, xL(dx)]$, and the bracket equals $\operatorname{Re}(f'/f)(\alpha + i\sqrt{z})$ [421, Theorem 8.2]. For $f = \xi$ at $\alpha > 1$ the input is the signed canonical measure $xL(dx)$, with atoms $\Lambda(n) n^{-\alpha}$ at $\log n$, an archimedean density and a pole density [424, Theorem 8.1].

Table 1.5 compares the two routes: the bracket computed from the canonical measure with the primes truncated at $N = 2 \times 10^6$, and $v_\alpha(\theta)$ computed by the Pólya route of Section 2.1. The difference is the prime tail $\sum_{n > N} \Lambda(n) n^{-\alpha}(1 - \cos(\theta \log n))$, of order $N^{1-\alpha}/(\alpha - 1)$, except at $\alpha = 3$, $\theta = 50$, where it comes from the double precision quadrature of the archimedean integral.

| $\alpha$ | $\theta$ | sine squared, primes $\le 2\times10^6$ | Pólya route | error of prime side |
|---|---|---|---|---|
| 3.0 | 14.1347 | 0.561447238789 | 0.561447238789 | $1.1\times10^{-13}$ |
| 3.0 | 50 | 1.120967014740 | 1.120967015289 | $5.5\times10^{-10}$ |
| 2.0 | 14.1347 | 0.766398694727 | 0.766399169607 | $4.7\times10^{-7}$ |
| 2.0 | 50 | 1.224475714058 | 1.224476216908 | $5.0\times10^{-7}$ |
| 1.5 | 14.1347 | 1.065762504228 | 1.067140080666 | $1.4\times10^{-3}$ |
| 1.5 | 50 | 1.385288214458 | 1.386706213218 | $1.4\times10^{-3}$ |

Table 1.5: The bracket $v_\alpha(\theta)$ of the sine squared lemma computed from the canonical measure and from the Pólya density; $\theta = 14.1347$ is the first ordinate $\gamma_1$, rounded. The Pólya values agree with an independent evaluation of $\mathrm{Re}(\xi'/\xi)$ through $\Gamma$ and $\zeta$ to about $10^{-15}$.

## 1.3 Generalised gamma convolutions and the Thorin condition

The Thorin condition on the clock $1/G$, equivalent to RH by Proposition 1.2.1, is that it be the Laplace transform of a generalised gamma convolution.

### 1.3.1 The definition and its three readings

As in Section 1.2, a random variable $H \ge 0$, or its law, is a generalised gamma convolution (GGC) if

$$\mathbb{E}e^{-uH} = \exp\Big\{ -du - \int_{(0,\infty)} \log\Big(1+\frac{u}{z}\Big)U(dz)\Big\}, \qquad u>0, \tag{1.3.1}$$

with drift $d \ge 0$ and a positive Radon measure $U$ on $(0,\infty)$, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1}U(dz) < \infty$ [493, 494], [56, p. 29]. The drift and the Thorin measure are unique [56, p. 30]. Since $\log(1+u/z)$ is the Laplace exponent of the exponential law with rate $z$, a discrete measure $U = \sum_j c_j\delta_{z_j}$ makes $H$ the sum of $d$ and independent gamma variables with shapes $c_j$ and rates $z_j$. The class is the closure of such sums under weak limits (Thorin's closure theorem, Section 4.1.1).

The first reading is through the Lévy measure. Frullani's integral $\log(1+u/z) = \int_0^\infty (1-e^{-ut})e^{-zt}t^{-1}dt$ turns (1.3.1) into

$$-\log \mathbb{E}e^{-uH} = du + \int_0^\infty \big(1-e^{-ut}\big)\frac{k(t)}{t}\,dt, \qquad k(t) = \int e^{-zt}\,U(dz),$$

so $H$ is infinitely divisible with Lévy density $k(t)/t$, and its canonical function $k$, $t$ times the Lévy density, is completely monotone. Conversely, by Bernstein's theorem, an infinitely divisible law on $[0,\infty)$ with drift $d$ and Lévy density $k(t)/t$, $k$ completely monotone, is a GGC whose Thorin measure is the measure representing $k$ [289, Section 1.2]. For the clock, the unconditional formula for $\log G$ in

Section 1.2 gives $k = W$, the heat trace, so the Thorin condition is the complete monotonicity of $W$.

The second reading is through the Stieltjes transform of the Thorin measure. Differentiating the logarithm of (1.3.1) gives

$$f_H(u) = -\frac{d}{du}\log \mathbb{E}e^{-uH} = d + \int \frac{U(dz)}{u+z},$$

a nonnegative constant plus the Stieltjes transform of a positive measure, holomorphic on $\mathbb{C}\setminus(-\infty,0]$. At distance $\eta>0$ below the cut, $\pi^{-1}\operatorname{Im} f_H(-x-i\eta) = \int P_\eta(x-z)\,U(dz)$ is the Poisson smoothing of $U$, with $P_\eta$ the Poisson kernel of Section 1.2, and letting $\eta\downarrow 0$ recovers $U$ by Stieltjes inversion: for continuity points $a<b$ of $U$,

$$U\big((a,b]\big) = \lim_{\eta\downarrow 0}\frac{1}{\pi}\int_a^b \operatorname{Im} f_H(-x-i\eta)\,dx. \tag{1.3.2}$$

For the clock, $G'/G = -\frac{d}{du}\log(1/G)$ takes the place of $f_H$ whether or not RH holds. The chain rule applied to $G(s^2) = \xi(\frac12+s)/\xi(\frac12)$ gives $(\xi'/\xi)(\frac12+s) = 2s\,(G'/G)(s^2)$. At $s=\varepsilon+i\theta$ we have $\bar s^2 = -x-i\eta$ with $x=\theta^2-\varepsilon^2$ and $\eta=2\varepsilon\theta$, and since $G$ is real on the real axis, (1.2.7) becomes

$$\rho_\varepsilon(\theta) = 2\theta\cdot\frac{1}{\pi}\operatorname{Im}\frac{G'}{G}(-x-i\eta) + \frac{2\varepsilon}{\pi}\operatorname{Re}\frac{G'}{G}(s^2).$$

The first term is the integrand of (1.3.2) at distance $\eta=2\varepsilon\theta$ from the cut, multiplied by the Jacobian $dx/d\theta = 2\theta$; the second carries the factor $\varepsilon$. The Thorin density at resolution $\varepsilon$ is therefore a regularised Stieltjes inversion, taken along the parabola $u=(\varepsilon-i\theta)^2$ rather than along a horizontal line.

Under RH the logarithm gives the counting function directly: for $y>0$ not of the form $\gamma^2$, $U_{\mathrm{RH}}((0,y]) = -\pi^{-1}\operatorname{Im}\log G(-y-i0) = N(\sqrt{y})$, with $\log G$ continued from $(0,\infty)$ through the lower half plane, since the factor $1+u/\gamma^2$ of $G$ has argument $-\pi$ at $u=-y-i0$ when $\gamma^2<y$ and 0 otherwise.

The third reading is as a product. If $U=\sum_j c_j\delta_{z_j}$ is discrete, (1.3.1) is the product $e^{-du}\prod_j(1+u/z_j)^{-c_j}$ of gamma Laplace transforms. Conversely, if $F$ is entire of genus zero with $F(0)=1$ and all its zeros real and negative, then $1/F$ is the Laplace transform of a GGC with $d=0$ and a unit atom at $-u_j$ for each zero $u_j$ of $F$, counted with multiplicity. For the clock $F=G$, the product is Hadamard's (1.2.1), and the Thorin condition says that every zero $-a_k$ of $G$ is real and negative, that is, $a_k>0$.

### 1.3.2 The Bondesson hierarchy and the clock

Between the infinitely divisible laws on $[0,\infty)$ (ID) and the GGCs lie two classes defined by conditions on the Lévy density $k(t)/t$: the self decomposable laws (SD),

for which $k$ is nonincreasing, and the Bondesson class (BO), for which $k(t)/t$ is completely monotone. The GGCs are the laws for which $k$ itself is completely monotone, and inside them lie the laws with a hyperbolically completely monotone density (HCM, defined below). The inclusions are

$$\text{HCM} \subset \text{GGC} \subset \text{SD} \subset \text{ID}, \qquad \text{GGC} \subset \text{BO} \subset \text{ID},$$

and SD and BO are not nested. The Lévy density $e^{-t}(1 + \frac{1}{2t})$ is completely monotone, but $k(t) = e^{-t}(t + \frac{1}{2})$ increases on $(0, \frac{1}{2})$, so the law is in BO and not in SD [421, Section 2]; the Lévy density $t^{-1}\mathbf{1}_{t<1}$ gives a law in SD and not in BO, since a completely monotone function that vanishes on $(1, \infty)$ vanishes identically.

A positive continuous function $\varphi$ on $(0, \infty)$ is hyperbolically completely monotone if for every $u > 0$ the function $\varphi(uv)\varphi(u/v)$ is completely monotone in $w = v + v^{-1}$, and a law is HCM if it has an HCM density. By Bondesson's theorem [56, Theorem 6.1.1], a function $\varphi$ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM, so the Thorin condition becomes a family of sign conditions on real derivatives. An HCM law is a GGC, so the HCM property of a law is stronger than that of its Laplace transform, and strictly so: the GGC of Theorem 4.3.20 has an HCM Laplace transform and a density that is not HCM.

The extended generalised gamma convolutions (EGGC), introduced by Thorin [495], are the weak limits of convolutions of gamma laws and their reflections [56, Section 7.1]. A symmetric law is an EGGC if and only if it is the law of $Z\sqrt{2H}$ with $Z$ standard normal and $H$ an independent GGC [56, Theorem 7.3.1]. Since $\mathbb{E}e^{i\theta Z\sqrt{2T}} = \mathbb{E}e^{-\theta^2 T}$ determines the law of $T$, the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 1.2 is an EGGC if and only if $T$ is a GGC, that is, by Proposition 1.2.1, if and only if RH holds. With no hypothesis $Z\sqrt{2T}$ is self decomposable: subordinating the Lévy measure $W(v)\,dv/v$ of $T$ to the heat kernel gives the Lévy density $k(|x|)/|x|$ with $k(r) = \int_0^\infty e^{-1/4v}(4\pi v)^{-1/2}W(r^2 v)\,dv/v = \sum_j e^{-r\sqrt{a_j}}$ (the first passage identity for each term, with the principal root), and $k$ decreases because $W$ does (Theorem 5.1.1). The $z$ distributions of Barndorff-Nielsen, Kent and Sørensen [31] are the lattice analogue of this family of variance mixtures.

Table 1.6 records what is known about the sign conditions on $W$ that the classes become for the clock. Section 5.2 explains why no finite number of derivative conditions can reach the last row.

Urbanik's classes of multiply self decomposable laws give conditions on $W$ of another kind: with $L_0$ the class of self decomposable laws and $L_n$ that of the self decomposable laws whose driving process has its law in $L_{n-1}$, the clock lies in $L_n$ exactly when $(-t\frac{d}{dt})^j W \geq 0$ for $j \leq n + 1$ (Section 11.5.4), and for $n \geq 1$ this does not follow from the GGC property, since the exponential law is a GGC whose canonical function $e^{-t}$ has $(-t\frac{d}{dt})^2 e^{-t} = -t(1 - t)e^{-t} < 0$ for $t < 1$. For the clock these conditions hold for $n = 2$ and fail for $n = 3$, with no hypothesis

| class or property | condition on $W$ | status for the clock |
|---|---|---|
| ID | $W \geq 0$ | proved, with the zeros verified to height 1000 (Theorem 5.1.1) |
| SD | $W$ nonincreasing | proved, with the zeros verified to height 1000 (Theorem 5.1.1) |
| strict log convexity | $WW'' > W'^2$ and $W'W''' > W''^2$ | proved, with the zeros verified to height 1000 (Section 5.1) |
| derivative signs to order $1.16 \times 10^{25}$ | $(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$ | proved, with the zeros verified to height $3 \times 10^{12}$ (Theorem 5.1.13) |
| complete monotonicity near the origin | $(-1)^n W^{(n)}(t) > 0$ for every $n$ and $t \leq 1.29$ | proved, with the zeros verified to height $10^{12}$ (Theorem 5.1.9) |
| BO | $W(t)/t$ completely monotone | equivalent to RH (Section 8.5.4); open |
| GGC | $W$ completely monotone | equivalent to RH; open |

Table 1.6: The Bondesson hierarchy on the clock, whose Lévy density is $W(t)/t$.

(Theorem 11.5.8), while membership of every $L_n$ would imply RH, since $\bigcap_n L_n$ lies inside the GGCs [44].

### 1.3.3 Diagnostics for the Thorin condition

RH can be stated as a GGC identity on a real interval. The right side of (1.3.1) extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$ (Exercise 1.14), so RH holds if and only if there is a GGC $H$ with $\xi(\frac{1}{2})/\xi(\frac{1}{2}+s) = \mathbb{E}e^{-s^2 H}$ for real $0 < s < 1$ alone [421, Proposition 3.3].

As it stands the identity is of no use to a computation, because analytic continuation from an interval is unstable. A unit Thorin atom at $z_0$ multiplies the transform by $(1 + u/z_0)^{-1}$, which differs from one by less than $1/z_0$ on $(0, 1)$, so transforms that agree to high accuracy on $(0, 1)$ can have very different Thorin measures at large $z$. Riesz's criterion meets the same obstruction: its numbers $1/\zeta(2k)$ are, up to explicit factors, values of the clock at the real points $(2k - \frac{1}{2})^2$, and they bear on the zeros only through the analytic interpolation of $k \mapsto 1/\zeta(2k)$ into the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$ (Section 10.2.6). The information about the zeros enters through the analyticity that the representation forces, and a computation has to supply it in another way: through values of $\xi$ and $\xi'$ at complex arguments, which Chapter 2 computes, or through sign conditions on derivatives of every order. Each characterisation of Sections 1.3.1 and 1.3.2 yields such a test, and Table 1.7 lists them with what a failure proves.

The diagnostics differ in how strongly they weight the higher zeros. Those

| diagnostic | computes | a failure proves |
|---|---|---|
| pole test | poles of $G'/G$ off $(-\infty, 0]$ | not GGC; for the clock, a zero off the line |
| Thorin density at resolution $\varepsilon$ | $\rho_\varepsilon = \pi^{-1}\operatorname{Re}(\xi'/\xi)$ at $\frac{1}{2}+\varepsilon+i\theta$ | a zero with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ (Proposition 1.2.3, [320]) |
| zero density scan | $M_\varepsilon(T)+\tau_\varepsilon(T)$ against $N(T)$ (Proposition 3.4.2) | $\frac{1}{2}[N(T)-M_\varepsilon(T)-\tau_\varepsilon(T)]$ zeros with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ below height $T$ |
| heat trace signs | $(-1)^n W^{(n)}(t)$ | not GGC; finitely many orders are consistent with off line zeros (Section 5.2) |
| Hankel blocks | $\det[(-1)^{i+j}W^{(i+j)}(t)]$ | not GGC (Weil's criterion at one scale, Section 5.1) |
| HCM test | $(-1)^n\partial_w^n[\varphi(uv)\varphi(u/v)]$ | not GGC (Section 4.1) |
| Bernstein test | $(-1)^{n-1}L^{(n)}$, $L=\log G$ | not ID, which Theorem 5.1.1 excludes for the clock |
| Stieltjes moments | Hankel matrices $[m_{i+j}]$ and $[m_{i+j+1}]$, $m_n=\sum_k a_k^{-n-1}$ | not GGC |

Table 1.7: Diagnostics for the Thorin condition. The pole test, the Thorin density and the zero density scan locate a failure in height; the others detect it only globally.

built from $W$ see a zero at height $\gamma$ through $e^{-\gamma^2 t}$: in $(-1)^n W^{(n)}(t)=\sum_j a_j^n e^{-a_j t}$ it enters with weight $(\gamma/\gamma_1)^{2n}e^{-(\gamma^2-\gamma_1^2)t}$ relative to the first zero. These tests certify easily and discriminate poorly. The Bernstein test on $L(u)=\log G(u)$, computed for $u>\frac{1}{4}$, where the Dirichlet series of $\log\zeta(\frac{1}{2}+\sqrt{u})$ converges, is better conditioned [424, Section 18.2.2]: since $(-1)^{n-1}L^{(n)}(u)=(n-1)!\sum_k(u+a_k)^{-n}$, a zero at height $\gamma$ enters with relative weight of order $(\gamma_1/\gamma)^{2n}$. It too sees the higher zeros only through subdominant terms, and for the clock it cannot fail: by the formula for $\log G$ in Section 1.2, $(-1)^{n-1}L^{(n)}(u)=\int_0^\infty t^{n-1}e^{-ut}W(t)\,dt$, which is positive because $W>0$ (Theorem 5.1.1). Read at $u=0$ through (1.2.4), which gives $\log\mathbb{E}e^{hX}=\log G(h^2)$, the same formula makes the even cumulants of the Pólya variable $X$ alternate in sign with no hypothesis: $(-1)^{n-1}\kappa_{2n}(X)=\frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt>0$ for every $n\geq 1$. Under RH this is the sign rule that Newman proved for laws of Lee Yang type [380]. Complete monotonicity of $L'$ is weaker than the Thorin condition, which asks that $L'$ be the Stieltjes transform $\int U(dz)/(u+z)$ of a positive measure $U$.

The Thorin density and the zero density scan built from it are local in the ordinate without a search for zeros: by Proposition 1.2.3, a zero at height $\gamma$ contributes about $1/(\pi\varepsilon)$ to $\rho_\varepsilon$ at $\theta=\gamma$ and about $\varepsilon/(\pi d^2)$ at distance $d\gg\varepsilon$.

### 1.3.4 Explicit Thorin measures and the sinh benchmark

Table 1.8 lists GGCs whose Thorin measures are explicit, in the variable $u$ of the clock. They are models for the clock, test cases for code, and the building blocks of the Vere-Jones representation of Section 6.1 (Theorem 6.1.8 and Table 6.1).

| Laplace transform | Thorin measure | law |
|---|---|---|
| $(1+u/z_0)^{-a}$ | $a\,\delta_{z_0}$ | Gamma$(a, z_0)$ |
| $\pi\sqrt{u}/\sinh(\pi\sqrt{u})$ | $\sum_{k\geq 1}\delta_{k^2}$ | $2K^2$, $K$ Kolmogorov |
| $(\sqrt{u}/\sinh\sqrt{u})^2$ | $2\sum_{k\geq 1}\delta_{\pi^2k^2}$ | $S_2/2$ (Williams) |
| $\operatorname{sech}\sqrt{u}$ | $\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$ | exit time of $\sqrt{2}B$ from $(-1,1)$ |
| $(\operatorname{sech}\sqrt{u})^{1/2}$ | $\frac{1}{2}\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$ | Meixner time |
| $e^{-c\sqrt{u}}$ | $c\,(2\pi\sqrt{z})^{-1}dz$ | positive stable, index $\frac{1}{2}$ |
| $(1+\sqrt{u}/c)^{-1}$ | $c\,\{2\pi\sqrt{z}(z+c^2)\}^{-1}dz$ | $E^2S/c^2$ |
| $e^{x(\alpha-\sqrt{\alpha^2+2u})}$ | $x\{\pi\sqrt{2z-\alpha^2}\}^{-1}\mathbf{1}_{z>\alpha^2/2}dz$ | inverse Gaussian |
| $\exp\{1-x\sqrt{u}-e^{-x\sqrt{u}}\}$ | $x\sin^2(x\sqrt{z}/2)(\pi\sqrt{z})^{-1}dz$ | one compensated jump, by (1.2.9) |
| $\mathcal{W}(u)/u$ | image of $db/\pi$ on $(0,\pi)$ under $r(b)=b\,e^{-b\cot b}/\sin b$ | Lambert law (Theorem 11.5.3) |
| $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ | $\sum_{n\geq 1}\delta_{\nu_n^2}$, $\nu_n$ the zeros of $\nu\mapsto K_{i\nu}(2\pi)$ | hitting time of a diffusion (Theorem 8.7.10) |
| $\xi(\alpha)/\xi(\alpha+\sqrt{u})$, $\alpha\geq 1$ | $v_\alpha(\sqrt{z})(2\pi\sqrt{z})^{-1}dz$ | shifted clock (Theorem 1.2.2) |
| $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ | $\sum_{\gamma>0}\delta_{\gamma^2}$ if and only if RH holds | the clock (Proposition 1.2.1) |

Table 1.8: GGC Laplace transforms in the variable $u$, their Thorin measures in $z$, and the corresponding laws.

In the table, $K$ is the Kolmogorov variable; $S_2$ is as in Section 1.2; $B$ is a standard Brownian motion; the Meixner time $H$ has $\mathbb{E}e^{-uH}=(\operatorname{sech}\sqrt{u})^{1/2}$, so that $Z\sqrt{2H}$, with $Z$ standard normal and independent of $H$, has the Meixner law of Section 6.2, whose time in Proposition 6.2.10 is $2H$; and $S$ is the positive stable variable with $\mathbb{E}e^{-uS}=e^{-\sqrt{u}}$, independent of the standard exponential variable $E$. Rows two to five are the classical hyperbolic laws [49]. At $u=-t^2$ the reciprocals of rows two and four become $\sin(\pi t)/(\pi t)$ and $\cos t$, the characteristic functions of the uniform law on $(-\pi,\pi)$ and of a symmetric sign. Like $\Xi$, both have infinitely many real zeros, so real zeros do not obstruct the Thorin property; the property requires them. For the clock, unlike these models, the positivity of the points $a_k$ is the conclusion to be proved [421, Section 10].

The sinh clock $1/G_0$, with $G_0(u)=\sinh(\pi\sqrt{u})/(\pi\sqrt{u})=\prod_{k\geq 1}(1+u/k^2)$, is the second row of the table. It has the structure that the clock has under RH, a GGC with unit Thorin atoms, but its ordinates are the integers, equally spaced,

and no arithmetic enters. Its Thorin density at resolution $\varepsilon$, the density in $\theta = \sqrt{z}$ of a measure $\mu_\varepsilon^{(0)}$, is

$$\rho_\varepsilon^{(0)}(\theta) = \frac{1}{\pi} \operatorname{Re}\Big[\pi \coth\big(\pi(\varepsilon + i\theta)\big) - \frac{1}{\varepsilon + i\theta}\Big] = \sum_{k \neq 0} P_\varepsilon(\theta - k),$$

exactly the Poisson smoothing of the atoms and their reflections, so the matched total variation of Section 3.2 vanishes identically. The cumulative mass is explicit. As for $\xi$ (Section 3.2), it is $\pi^{-1}$ times the change in the argument of $\sinh(\pi s)/(\pi s)$ along $s = \varepsilon + i\theta$, and at a half integer $\Theta = m + \frac{1}{2}$ the continuous argument of $\sinh(\pi(\varepsilon + i\theta))$ has grown from 0 to $\pi\Theta$ and that of $\varepsilon + i\theta$ to $\frac{\pi}{2} - \arctan(\varepsilon/\Theta)$, so that

$$\mu_\varepsilon^{(0)}\big((0, \Theta^2]\big) = m + \frac{1}{\pi} \arctan \frac{\varepsilon}{\Theta}.$$

The mass defect $\mu_\varepsilon^{(0)}((0, \Theta^2]) - m$ is therefore $\varepsilon/(\pi\Theta) + O(\varepsilon^3)$. On the window $0 < \theta \leq 50.5$, which holds 50 atoms, the defect is $0.00630\,\varepsilon$ to first order and $3.151 \times 10^{-3}$ at $\varepsilon = 0.5$.

The cell distance, the sum over the cells $C$ of $|\mu_\varepsilon^{(0)}(C) - 1|$ as in Section 3.2, is explicit as well. We use the cells $(k - \frac{1}{2}, k + \frac{1}{2}]$, $k = 1, \dots, 50$, and add the mass of the strip $(0, \frac{1}{2}]$, which lies in no cell, as unmatched. For the full lattice $\mathbb{Z}$ each cell would have mass exactly one. The sinh clock lacks the atom at 0, so each of its cells is short by the mass that $P_\varepsilon$ puts there, and the strip carries $\pi^{-1} \arctan 2\varepsilon$; Section 5.4 computes these masses. Summing,

$$d_{\mathrm{cell}}^{(0)}(\varepsilon) = \frac{1}{\pi}\Big[2 \arctan 2\varepsilon - \arctan \frac{\varepsilon}{50.5}\Big] = \frac{4 - 1/50.5}{\pi}\,\varepsilon + O(\varepsilon^3),$$

which is 0.4968, 0.1250, 0.0631, 0.01267 and 0.00633 at $\varepsilon = 0.5$, 0.1, 0.05, 0.01 and 0.005, and $1.267\,\varepsilon$ in the limit. All of it is due to the atom missing at 0.

For the 79 zeros below height 200 the constant is 5.44 (Section 3.2), and it has a different origin. The zeros are irregularly spaced, so the mass that leaks out of a cell is no longer replaced by the mass that leaks in from its neighbours, and every cell contributes; Section 5.4 reads the constant as a statistic of the spacings. Figure 1.3 divides both distances by the number of atoms, which gives $1.267\,\varepsilon/50 = 0.0253\,\varepsilon$ for the sinh clock and $5.44\,\varepsilon/79 = 0.069\,\varepsilon$ for the zeros. Their ratio, nearly three, depends on the window, since for the lattice the distance per atom falls like $1/N$ as the window grows to $N$ atoms. Williams' Thorin measure $2\sum_k \delta_{\pi^2 k^2}$ of Table 1.8 needs no separate benchmark: it is the same lattice rescaled by $\pi$, with atoms of mass two, so its Thorin density at resolution $\varepsilon$ is $\frac{2}{\pi}\rho_{\varepsilon/\pi}^{(0)}(\theta/\pi)$ and its distances follow by scaling from the closed forms for the sinh clock.

## 1.4 Exercises

*On Section 1.2.*

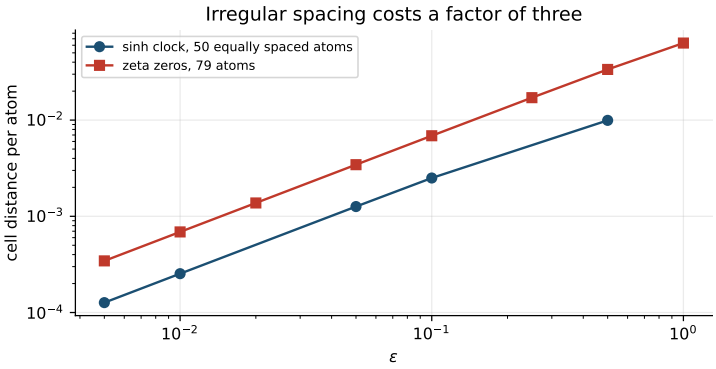



Figure 1.3: Cell distance divided by the number of atoms in the window, for the sinh clock (50 atoms, cells $(k-\frac{1}{2}, k+\frac{1}{2}]$) and for the 79 zeros below height 200 (Voronoi cells of Section 3.2). Both vanish linearly in $\varepsilon$. For the sinh clock the undivided distance stays below $4\varepsilon/\pi$ whatever the number of atoms, so the gap between the curves depends on the window, and the factor of three in the title of the figure holds for these two windows only.

1.1. Show from the Hadamard product (1.2.1) that $\sum_k a_k^{-1} = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$, and compute it to ten digits from the Pólya density by differentiating (1.2.4) twice under the integral.

1.2. Prove that $\Phi$ is even by applying the theta transformation $1+2\omega(1/x) = \sqrt{x}\,(1+2\omega(x))$ to $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and differentiating twice.

1.3. Show that if $X$ has the Pólya density then $\mathbb{E}X^2 = 2\sum_k a_k^{-1}$, and deduce that the Pólya law has variance 0.0462.

1.4. Verify the first row of Table 1.4 by summing $e^{-\gamma^2 t}$ over the first forty zeros, and estimate the error of truncation using $N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e}$.

1.5. Show that an off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ contributes $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, and find the first time at which that contribution is negative.

1.6. Show that at a basepoint $\alpha > 1$ the Thorin density $v_\alpha(\sqrt{z})/(2\pi\sqrt{z})$ is bounded below by a positive multiple of $z^{-1/2}\log(2+z)$ for large $z$. Use Proposition 1.2.3 with the Riemann von Mangoldt formula, or, more simply, $v_\alpha(r) = \mathrm{Re}[\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)]$ at $s = \alpha + ir$ with Stirling's formula.

1.7. Let $\delta = 0.49$, $\gamma_0 = 10.5$ and

$$F(s) = \frac{\sinh(\pi(s-\frac{1}{2}))}{\pi(s-\frac{1}{2})}\Big(1 - \frac{(s-\frac{1}{2})^2}{(\delta+i\gamma_0)^2}\Big)\Big(1 - \frac{(s-\frac{1}{2})^2}{(\delta-i\gamma_0)^2}\Big).$$

Show that $F$ is entire of order one, positive on the real axis and symmetric under $s \mapsto 1-s$, with zeros $\frac{1}{2} \pm ik$ $(k \geq 1)$ and $\frac{1}{2} \pm \delta \pm i\gamma_0$. Show that $\mathrm{Re}(F'/F)(0.6+ir) > 0$ for every real $r$, although $F$ vanishes at $0.99+10.5i$. (Put $\varepsilon = 0.1$. The zeros $\frac{1}{2} \pm \delta + i\gamma_0$ together contribute $2\varepsilon(t^2-\delta^2+\varepsilon^2)/((t^2+(\delta-\varepsilon)^2)(t^2+(\delta+\varepsilon)^2))$ at $t = r-\gamma_0$, and their conjugates the same expression at $t = r+\gamma_0$; it is at least $-2\varepsilon/(\delta^2-\varepsilon^2)$ and is nonnegative for $t^2 \geq \delta^2-\varepsilon^2$; the zeros on the line contribute $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) - \varepsilon/(\varepsilon^2+r^2) > 0$, and $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) \geq \pi\tanh(\pi\varepsilon)$.)

*On Section 1.3.*

1.8. Show that the Gamma$(a,b)$ law, with shape $a$ and rate $b$, has Thorin measure $a\delta_b$, and that the sum of independent Gamma$(a_j,b_j)$ variables has Thorin measure $\sum_j a_j\delta_{b_j}$.

1.9. Derive the Thorin measure of sech $\sqrt{u}$ from the product $\cosh x = \prod_{k\geq 1}(1+4x^2/((2k-1)^2\pi^2))$.

1.10. Prove that $\int_0^\infty \log(1+u/z)\,(2\pi\sqrt{z})^{-1}dz = \sqrt{u}$ (integrate by parts), and deduce the entry for the positive stable law of index $\frac{1}{2}$ in Table 1.8.

1.11. Show that $e^{-t}(1+\frac{1}{2t})$ is completely monotone while $k(t) = e^{-t}(t+\frac{1}{2})$ is not monotone, and conclude that the infinitely divisible law with Lévy density $e^{-t}(1+\frac{1}{2t})$ and no drift is in BO but not in SD.

1.12. For the sinh clock, derive $\mu_\varepsilon^{(0)}((0,\Theta^2]) = m+\pi^{-1}\arctan(\varepsilon/\Theta)$ at $\Theta = m+\frac{1}{2}$ from the periodicity of $\sum_{k\in\mathbb{Z}} P_\varepsilon(\theta-k)$, and check the first order value $0.00630\,\varepsilon$ of the mass defect at $\Theta = 50.5$.

1.13. Show that the HCM property of $\varphi(u) = a/(a+u)$ reduces to complete monotonicity of $a^2/(a^2+auw+u^2)$ in $w$, and deduce that finite products of such functions are HCM.

1.14. Show that the right side of (1.3.1) extends to a holomorphic nonvanishing function on $\mathbb{C}\setminus(-\infty,0]$. Deduce that if an entire function $F$ with $F(0)=1$ satisfies $1/F(u) = \mathbb{E}e^{-uH}$ for $0<u<1$, with $H$ a GGC, then $F$ has no zeros in $\mathbb{C}\setminus(-\infty,0]$.

# Chapter 2

# Evaluating the Xi function

## 2.1 The Pólya route

Pólya's integral (1.2.4) evaluates $\xi$ in the complex plane from the kernel alone, without primes or zeros. In this chapter, $\delta$ always denotes the distance from the line of integration to the edge of the strip of analyticity of $\Phi$, not the displacement of an off line zero.

### 2.1.1 The trapezoid rule on Pólya's integral

Recall the Pólya kernel (1.2.3) in Titchmarsh's normalisation [498, Section 10.1],

$$\Phi(u) = 2\sum_{n\geq 1} \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right) \exp(-\pi n^2 e^{2u}), \tag{2.1.1}$$

even and positive on the real line, analytic in the strip $|\operatorname{Im} u| < \pi/4$ (Section 1.2) and doubly exponentially decaying in every closed substrip (proof of Proposition 2.1.1). With the Pólya variable $X$ of density $p = \Phi/\Xi(0)$, the moment generating function (1.2.4) and its derivative under the integral sign give, for every complex $s$,

$$\frac{\xi(\frac12 + s)}{\xi(\frac12)} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad \frac{\xi'(\frac12 + s)}{\xi(\frac12)} = \mathbb{E}\, X e^{sX}, \qquad X \sim p, \tag{2.1.2}$$

so the Thorin density (1.2.7) is $\pi^{-1}\operatorname{Re}\{\mathbb{E}Xe^{sX}/\mathbb{E}e^{sX}\}$ at $s = \varepsilon + i\theta$. Throughout, log is the natural logarithm.

Analyticity and decay in the strip make the trapezoid rule with nodes $u_j = jh$ converge geometrically in $1/h$ [500, Theorem 5.1] (Exercise 2.1): for $s = \varepsilon + i\theta$,

$$\Big| \int_{\mathbb{R}} \Phi(u) e^{su}\, du - h\sum_{j\in\mathbb{Z}} \Phi(jh) e^{sjh} \Big| \leq \frac{2K_\eta(\varepsilon)\, e^{\eta|\theta|}}{e^{2\pi\eta/h} - 1}, \qquad 0 < \eta < \pi/4,$$

where $K_\eta(\varepsilon) = \int_{\mathbb{R}} |\Phi(x \pm i\eta)|\, e^{\varepsilon|x|}\, dx$, the same for both signs because $\Phi$ is real on the real axis. We write $K_\eta = K_\eta(0)$. Since $|\xi(\frac12 + \varepsilon + i\theta)|$ decays like $e^{-\pi\theta/4}$,

the relative error is at most of order $\exp\{(\eta+\pi/4)\theta - 2\pi\eta/h\}$. With $h = 1/200$ the bound is best for $\eta$ close to $\pi/4$, and at $\theta = 200$ it is about $10^{-289}$; halving $h$ changes the computed values by less than $10^{-64}$.

The limit in practice is the cancellation in (2.1.2): the integrand is of order one and the integral of order $e^{-\pi\theta/4}$, so at $b$ bits the rounding error is of order $2^{-b}e^{\pi\theta/4}$ relative to $|\xi|$, a loss of $\pi\theta/(4\log 10) = 0.341\,\theta$ decimal digits. Discretisation, rounding and the truncation of the $u$ axis at $|u| \le L$ can be budgeted separately. To reach $D$ correct digits at height $\theta$ one needs

$$b \;\ge\; \Big(D + \frac{\pi\theta}{4\log 10}\Big)\log_2 10, \qquad h \;\le\; \frac{2\pi\eta}{(D + 0.682\,\theta)\log 10},$$

where the condition on $h$ uses $\eta + \pi/4 \le \pi/2$. The truncation needs $\pi e^{2L} \ge (D + 0.341\,\theta)\log 10 + |\varepsilon|L$, up to lower order terms.

We take $h = 1/200$ and $L = 3$, beyond which $\Phi$ is below $10^{-500}$, and work at 420 bits (126 decimal digits): at height 200, where $|\xi|$ is about $10^{-63}$, the cancellation consumes about half of them and leaves about 60. Truncating the sum at $|j| \le J = L/h$ and folding it with the evenness of $\Phi$ gives, with $E = e^{sh}$,

$$\xi(\tfrac12 + s) \approx P(E) + P(E^{-1}), \qquad \xi'(\tfrac12 + s) \approx Q(E) - Q(E^{-1}),$$

$$P(x) = \tfrac{h}{2}\Phi_0 + h\sum_{j=1}^{J}\Phi_j x^j, \quad Q(x) = h^2\sum_{j=1}^{J} j\Phi_j x^j,$$

with $\Phi_j = \Phi(jh)$ and $J = 600$. The node values, the only place where the theta series is summed, are computed once for a given step, truncation and precision and serve every resolution and every height; one evaluation of $\xi$ and $\xi'$ then costs four polynomial evaluations in ball arithmetic [290], about 0.7 ms at 420 bits. The ball radii certify the rounding error only, and they overestimate it: at height 200 they certify about 38 digits of $\xi$ and, at $\varepsilon = 0.005$, about 36 of $\mathrm{Re}(\xi'/\xi)$, while comparison with `mpmath` at 90 digits shows that the values themselves carry about 63.

### 2.1.2 Zeros and Thorin masses below height 200

With the parameters of Section 2.1.1, the zeros and the Thorin masses below height $\Theta = 200$ are computed at the eight resolutions $\varepsilon = 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0.01, 0.005$ in five steps.

1. Compute $\Phi_j$, $0 \le j \le J$, from (2.1.1) at 420 bits.

2. For each resolution $\varepsilon$, evaluate $\rho_\varepsilon(\theta_k)$ and $M_\varepsilon(\theta_k) = \pi^{-1}\arg\xi(\frac12 + \varepsilon + i\theta_k)$, with the branch of the argument that vanishes at $\theta = 0$, on the grid $\theta_k = k\varepsilon/8$ ($k\varepsilon/4$ at $\varepsilon = 0.005$), $0 \le \theta_k \le \Theta$. Since $\partial_\theta \arg\xi(\alpha + i\theta) = v_\alpha(\theta)$ with $\alpha = \frac12 + \varepsilon$, $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2])$ is exactly the cumulative Thorin mass.

3. Locate the line zeros as sign changes of $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ on a grid of step 0.05, reading each sign from a ball that excludes zero, and refine them by Newton's method using $\Xi'(\theta) = -\operatorname{Im} \xi'(\frac{1}{2} + i\theta)$.

4. Count all zeros with $0 < \operatorname{Im} \rho \le \Theta$ by the argument principle, $N(\Theta) = \pi^{-1} \Delta \arg \xi(s)$ along $2 \to 2 + i\Theta \to \frac{1}{2} + i\Theta$ [498, Section 9.3].

5. Compare with $U_{\mathrm{RH}}$ in the metrics of Section 3.2.

Steps 3 and 4, which use the same two polynomials, together bound the number of off line zeros. Each sign change of $\Xi$ on the grid encloses at least one zero on the line, so $N(\Theta)$ minus the number of sign changes is an upper bound for the number of off line zeros with $0 < \operatorname{Im} \rho \le \Theta$. When the two numbers agree, every zero below height $\Theta$ is on the line and simple.

In step 2 the argument is accumulated from the increments $\arg(\xi(s_{k+1})/\xi(s_k))$ along $\operatorname{Re} s = \frac{1}{2} + \varepsilon$. The principal value is the true increment only if $\arg \xi$ changes by less than $\pi$ between consecutive points, which the samples cannot show; the grids are chosen so that every observed increment is below $\pi/2$ in modulus, a margin checked after the fact rather than a guarantee, and the largest observed was 0.29. At large height $\xi$ underflows double precision ($|\xi|$ is about $10^{-335}$ at height 1000), so the quotient must be formed in ball arithmetic before the argument is taken.

With $\Theta = 200$ there are 79 sign changes, and at none of the 4001 grid points does the ball for $\Xi$ contain zero. The argument principle gives $N(200) = 79.0000000000004$, with a largest increment of 0.02 radians along the contour, and $\Theta = 200$ lies between $\gamma_{79} = 198.0153$ and $\gamma_{80} = 201.2648$, so it is not an ordinate. The Newton refined roots agree to $5.0 \times 10^{-48}$ with the ordinates computed independently by `mpmath` [376] at 50 digits. The computation thus recovers, with no input but (2.1.1), the classical fact that the 79 zeros below height 200 lie on the line and are simple [15, 419]. It is not yet a proof: the argument increments are summed in double precision from samples, and the analytic error bounds are not added to the radii (Section 12.2.2 lists what a certified version requires). Figure 2.1 shows the computed density.

### 2.1.3 Tilting the contour into the strip

Since $\Phi$ is analytic in $|\operatorname{Im} u| < \pi/4$, the integrals behind (2.1.2) can be taken along any horizontal line inside the strip.

Proposition 2.1.1. *Let $0 < \delta < \pi/4$ and $\eta = \pi/4 - \delta$. For every complex $s$,*

$$\xi\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} \Phi(x + i\eta)\, e^{sx}\, dx,$$
$$\xi'\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} (x + i\eta)\, \Phi(x + i\eta)\, e^{sx}\, dx.$$

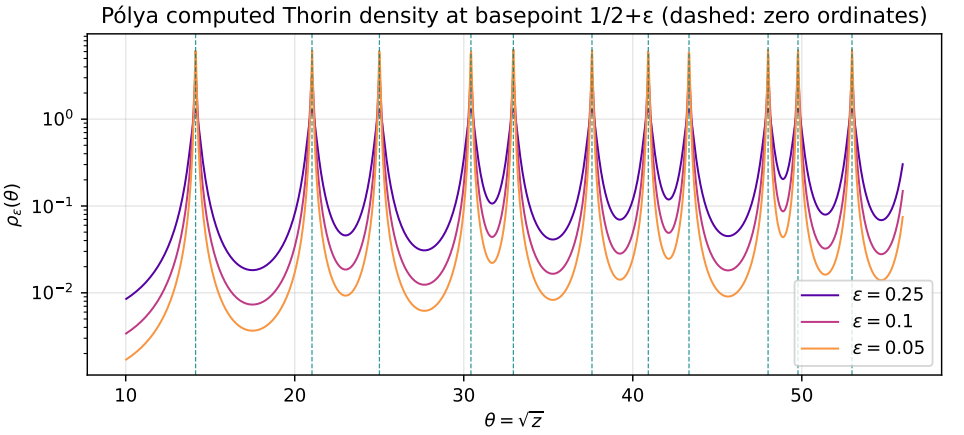



Figure 2.1: The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2}+\varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 1.2.3).

*For real $x$, $|\Phi(x+i\eta)| \le C_\delta$ and $|(x+i\eta)\Phi(x+i\eta)| \le C_\delta$, with $C_\delta = O(\delta^{-5/2})$ as $\delta \to 0$.*

*Proof.* By Section 1.2, $\Phi$ is analytic in $|\operatorname{Im} u| < \pi/4$, and by the identity theorem its evenness $\Phi(u) = \Phi(-u)$ extends from the real line to the strip. For $u = x + iy$ with $|y| \le \eta$ and $x \ge 0$ we have $\operatorname{Re} e^{2u} = e^{2x}\cos 2y \ge e^{2x}\sin 2\delta$, so

$$|\Phi(x+iy)| \le 2\sum_{n\ge 1}\left(2\pi^2 n^4 e^{9x/2} + 3\pi n^2 e^{5x/2}\right)\exp(-\pi n^2 e^{2x}\sin 2\delta), \tag{2.1.3}$$

and $\Phi$ decays doubly exponentially as $x \to +\infty$, uniformly in the closed strip $|y| \le \eta$; by evenness it does so as $x \to -\infty$ as well. Cauchy's theorem therefore moves the line of integration in $\xi(\frac{1}{2}+s) = \int_{\mathbb{R}} \Phi(u)e^{su}\,du$, which is (2.1.2), from $\operatorname{Im} u = 0$ to $\operatorname{Im} u = \eta$, and the second identity follows in the same way from $\xi'(\frac{1}{2}+s) = \int_{\mathbb{R}} u\Phi(u)e^{su}\,du$. Finally, $\sum_{n\ge1} n^k e^{-bn^2} = O(b^{-(k+1)/2})$ uniformly in $b > 0$, so the majorant (2.1.3) is $O(e^{-x/2}\delta^{-5/2})$ for $x \ge 0$, of order $\delta^{-5/2}$ at $x = 0$; with evenness this gives both bounds. □

On the real axis the integrand $\Phi(u)e^{su}$, with $s = \varepsilon + i\theta$, oscillates with frequency $\theta$, and its integral, of order $\theta^{7/4+\varepsilon/2}e^{-\pi\theta/4}|\zeta(\frac{1}{2}+\varepsilon+i\theta)|$ by Stirling's formula, is exponentially smaller than the integrand. On the line $\operatorname{Im} u = \eta$ the factor $e^{i\theta u}$ has constant modulus $e^{-\theta\eta}$ and leaves the integral as the prefactor $e^{i\eta s}$, which carries almost all of the decay of $\xi$; the integral that remains is of order $e^{-\delta\theta}$ up to the same factors, against an integrand of size at most $C_\delta e^{\varepsilon x}$. The cancellation thus

drops from $e^{-\pi\theta/4}$ to $e^{-\delta\theta}$. The price is a larger kernel, by the factor $K_\eta/K_0$ with $K_0 = \xi(\frac{1}{2}) = 0.497$, and a finer step, because the natural boundary at $\operatorname{Im} u = \pi/4$ [458, Chapter 11, Section 1.4, and Chapter 5, Section 3.2] is now only $\delta$ away.

On the tilted line the kernel constant measures the size of the integrand and costs $\log_{10} K_\eta$ digits of precision. By Table 2.1 it grows like $\delta^{-5/2}$, but $\log_{10} K_\eta$ stays under five even at $\delta = 0.01$, which is negligible against the saving $(\pi/4 - \delta)\theta/\log 10$ at any height where tilting is worthwhile.

| $\delta = \pi/4 - \eta$ | $K_\eta$ | $\max_x \lvert\Phi(x + i\eta)\rvert$ |
|---:|---:|---:|
| 0.6854 | 0.554 | 0.981 |
| 0.3927 | 2.52 | 3.24 |
| 0.2 | 19.0 | 15.4 |
| 0.1 | 126 | 76.5 |
| 0.05 | 774 | 415 |
| 0.02 | 7836 | 3938 |
| 0.01 | 43503 | 22207 |

Table 2.1: Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x| \le 6$; the first two rows are the lines $\eta = 0.1$ and $\eta = \pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts.

The trapezoid rule on the tilted line converges geometrically under two conditions (Exercise 2.2). By Poisson summation its error is $\sum_{k\neq 0} I(\theta + 2\pi k/h)$, where $I(t) = \int_{\mathbb{R}} \Phi(x + i\eta) e^{\varepsilon x} e^{itx}\, dx = e^{\eta t - i\eta\varepsilon}\xi(\frac{1}{2} + \varepsilon + it)$. The integrand is analytic in a strip that extends $\delta$ above the line and $\pi/2 - \delta$ below it, and correspondingly $|I(t)|$ is of order $e^{-\delta t}$ as $t \to +\infty$ and $e^{-(\pi/2-\delta)|t|}$ as $t \to -\infty$, up to the factors of Stirling's formula. Relative to $I(\theta)$ the terms with $k \ge 1$ are of order $e^{-2\pi\delta k/h}$, and those with $k \le -1$ are small only when the step resolves the oscillation $e^{i\theta x}$ with a margin. For $D$ correct digits this gives

$$\frac{1}{h} \;\ge\; \frac{D\log 10}{2\pi\delta}, \qquad \frac{2\pi}{h} \;\ge\; \theta + \frac{\delta\theta + D\log 10}{\pi/2 - \delta}, \tag{2.1.4}$$

up to terms of order $\log\theta + \log(1/h)$. The first condition asks for about $D\log 10/(2\pi\delta)$ nodes per unit length, 458 at $D = 25$ and $\delta = 0.02$, that is $h \le 2.2 \times 10^{-3}$. The second is the binding one at large height: at $\delta = 0.02$ and 25 digits it is the stronger from about $\theta = 2800$ on, and it asks for $h \le 1.23 \times 10^{-3}$ at $\theta = 5000$ and $h \le 8.8 \times 10^{-4}$ at $\theta = 7005$.

Taking $\delta$ proportional to $1/\theta$ holds the loss $\delta\theta/\log 10$ fixed, so the working precision grows only through the $\log_{10} K_\eta$ digits of the kernel constant, about $\frac{5}{2}\log_{10}(1/\delta)$, while both conditions (2.1.4) make the number of nodes, and with it the cost per point, grow linearly in $\theta$ up to a logarithmic factor from the truncation.

| $\delta$ | bits | nodes | $\theta$ | error of $\xi'/\xi$ | ms per point | untilted digits needed |
|---|---|---|---|---|---|---|
| 0.10 | 200 | 2001 | 200 | $< 2 \times 10^{-45}$ | 7 | 68 |
| 0.10 | 200 | 2001 | 500 | $9.8 \times 10^{-42}$ | 7 | 171 |
| 0.10 | 200 | 2001 | 1000 | $1.8 \times 10^{-20}$ | 7 | 341 |
| 0.02 | 220 | 8401 | 1000 | $< 6 \times 10^{-45}$ | 28 | 341 |
| 0.02 | 220 | 8401 | 2000 | $< 4 \times 10^{-45}$ | 28 | 682 |
| 0.02 | 220 | 8401 | 5000 | $2.9 \times 10^{-25}$ | 28 | 1705 |

Table 2.2: The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of $\xi'/\xi$, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height.

### 2.1.4 Precision and cost of the tilted route

The tilt makes height 5000 accessible (Table 2.2).

On the tilted line the rounding term of Section 2.1.1 becomes $2^{-b}e^{\delta\theta}$ times the kernel constant, and the step obeys (2.1.4). Table 2.3 compares the working precision of the two routes. The formula of the table for the tilted bits is conservative: it ignores the factor $\theta^{7/4+\varepsilon/2}$ in $|\xi(\frac{1}{2}+\varepsilon+i\theta)|$, worth about six digits at $\theta = 5000$, and it adds five digits for the kernel constant. The run at $\theta = 5000$ in Table 2.2 used 220 bits, fewer than the 244 of Table 2.3, and returned 25 digits; the Lehmer pair of Section 2.2.3.1 was computed with 256 bits at $\delta = 0.01$, where the formula asks for 201.

| height $\theta$ | target digits | bits (untilted) | bits (tilted, $\delta = 0.02$) | nodes (tilted) |
|---|---|---|---|---|
| 100 | 30 | 213 | 119 | 8400 |
| 200 | 30 | 326 | 122 | 8400 |
| 1000 | 30 | 1233 | 145 | 8400 |
| 5000 | 25 | 5748 | 244 | 8400 |
| 7005 | 25 | 8020 | 302 | 8400 |

Table 2.3: Working precision needed by the untilted and the tilted Pólya route for $D$ target digits. The untilted bits are $(D + \pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D + \delta\theta/\log 10 + 5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \geq 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \leq 8.8 \times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \leq 3.7$, the smallest that suffices.

Because the tilt reduces the cancellation to $e^{-\delta\theta}$, ordinary double precision suffices at moderate height. With $\delta = 0.15$ and the node values rounded to double, a vectorised evaluation agrees with the ball arithmetic values of $\rho_{0.05}$ to within an

absolute error of $2.8 \times 10^{-12}$ for $\theta \le 50$, $1.3 \times 10^{-9}$ for $\theta \le 100$ and $7.6 \times 10^{-7}$ for $\theta \le 150$, at 93 $\mu$s per point. The untilted route fails in double precision above $\theta \approx 40$ (Exercise 2.5).

### 2.1.5 Ball arithmetic on the tilted line

Horner's rule, which serves the untilted evaluator well, fails in ball arithmetic on the tilted line. The tilted evaluator writes the trapezoid sum as one polynomial of degree $2J$ in $E = e^{sh}$, with coefficients $h\Phi(x_k + i\eta)$ for $x_k$ running from $-L$ to $L$, times the prefactor $e^{i\eta s}E^{-J}$. With 2001 nodes, $\delta = 0.1$ and 200 bits, the radius of the Horner recursion grew by a factor of about 1.3 per step, and the final ball at $\theta = 100$ had radius $10^{109}$ although its midpoint was correct to many digits. Evaluating each exponential $e^{sx_k}$ directly and summing gives balls of radius about $10^{-55}$ at the same point.

The cause is the wrapping effect of rectangular complex balls. Arb stores a complex number as a real ball and an imaginary ball [290], so a multiplication by $E = e^{\varepsilon h}e^{i\theta h}$ rotates the midpoint but multiplies the radii by $e^{\varepsilon h}(|\cos\theta h| + |\sin\theta h|)$, up to rounding; at $\theta h = 0.4$ this factor is 1.31. In Horner's rule the coefficient of $E^k$ passes through $k$ such multiplications. On the untilted line the effect is contained. The polynomials are folded at $x = 0$, so the coefficient at $x = jh$ passes through at most 600 multiplications, and the coefficients $h\Phi(jh)$ decay like $\exp(-\pi e^{2jh})$, so those that pass through many multiplications are negligible and the radii stay far below $|\xi|$ up to height 200. On the tilted line the coefficients decay only like $\exp(-\pi e^{2x}\sin 2\delta)$, and the coefficient at $x$ passes through $(x+L)/h$ multiplications. With $h = 0.004$ and $L = 4$, which give 2001 nodes, the coefficients near $x = 2$ are still of order $10^{-12}$ at $\delta = 0.1$, and they pass through about 1500 multiplications, which enlarge their radii by a factor of about $10^{176}$.

## 2.2 Classical evaluators, Turing's method and the Pólya route

The classical evaluators on the critical line pay for height differently and, at large height, far less than the Pólya route, tilted or not.

### 2.2.1 The Riemann Siegel formula, Gram points and certified counts

On the critical line $\zeta(\frac{1}{2}+it) = e^{-i\vartheta(t)}Z(t)$, where $\vartheta(t) = \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ is the Riemann Siegel theta function and $Z(t)$ is real for real $t$. Write $m = \lfloor\sqrt{t/2\pi}\rfloor$ and $q = \sqrt{t/2\pi} - m$. For every $K \ge 0$,

$$Z(t) = 2\sum_{n\le m}\frac{\cos(\vartheta(t) - t\log n)}{\sqrt{n}} + (-1)^{m-1}\Big(\frac{t}{2\pi}\Big)^{-1/4}\sum_{j=0}^{K}C_j(q)\Big(\frac{t}{2\pi}\Big)^{-j/2} + R_K(t),$$

where $R_K(t) = O(t^{-(2K+3)/4})$ as $t \to \infty$, $C_0(q) = \cos(2\pi(q^2 - q - \frac{1}{16}))/\cos(2\pi q)$, and the higher $C_j$ are given in [127, Section 7.5] and [143]. This asymptotic expansion was found in Riemann's unpublished notes and published by Siegel [478].

The main sum is the approximate functional equation $\zeta(s) = \sum_{n \le x} n^{-s} + \chi(s) \sum_{n \le y} n^{s-1} + \cdots$, where $\zeta(s) = \chi(s)\zeta(1-s)$ and $2\pi xy = t$ [498, Chapter 4], taken at $x = y = \sqrt{t/2\pi}$ and written for $Z$; the $C_j$ correct for the abrupt truncation at $n = m$. The terms are bounded by 2 and there is no exponential cancellation, so double precision suffices at every height computed in this book; near height $10^{12}$ and above, the phases $\vartheta(t) - t \log n$ must be reduced modulo $2\pi$ in higher precision. Gabcke's thesis gives explicit bounds for the remainders $R_K$ with $K \le 10$ [143], and with them the formula can be used in rigorous computation.

To evaluate $Z$ at many points near height $T$, Odlyzko and Schönhage [392] compute the main sum on a grid by fast Fourier transforms and rational function evaluation, and interpolate between grid points by band limited interpolation. About $T^{1/2}$ values then cost $T^{1/2+o(1)}$ operations, that is $T^{o(1)}$ per point. Odlyzko compared the spacings of the zeros near zero number $10^{12}$ with the Gaussian unitary ensemble [391], and with the algorithm he later took such computations to zero numbers $10^{20}$ [393] and $10^{22}$ [395].

The classical searches for zeros on the line sample $Z$ at the Gram points $g_n$, defined by $\vartheta(g_n) = n\pi$ [158, pp. 301 and 302]. There $Z(g_n)$ usually has the sign of $(-1)^n$, and each Gram interval $[g_{n-1}, g_n)$ usually contains exactly one zero. This is Gram's law, and it fails with positive frequency [502, Corollary 5.4]. Call $g_n$ good when $(-1)^n Z(g_n) > 0$, and call an interval $[g_j, g_{j+k})$ between consecutive good Gram points a Gram block. Rosser's rule asserts that such a block contains exactly $k$ zeros [466], [502, Section 7.3]. It holds far longer than Gram's law, since its first exception is the block that begins at $g_{13\,999\,525}$ [64, Table 3], but it too fails with positive frequency [502, Corollary 7.8]. Counting sign changes of $Z$ gives a lower bound for the number of zeros on the line. The difficulty is to show that none is missing, whether off the line or in a close pair that produces no sign change.

Turing's answer [506] uses the counting function $N(T) = \vartheta(T)/\pi + 1 + S(T)$, where, for $T$ not an ordinate, $\pi S(T) = \arg \zeta(\frac{1}{2} + iT)$, the argument being defined by continuous variation along $2 \to 2 + iT \to \frac{1}{2} + iT$. The function $S$ is small on average: Littlewood proved $\int_0^T S(t)\,dt = O(\log T)$ [498, Theorem 9.9(A)], and Turing gave explicit constants for $|\int_{t_1}^{t_2} S(t)\,dt|$, later sharpened by Trudgian [501]. If the sign changes found near height $T$ were fewer than $N(T)$, then $S$ would be displaced by at least 2 over an interval long enough to violate the integral bound. The count is therefore certified by locating the sign changes in a range slightly beyond $T$ and checking them against Turing's bound for $\int S$. The method never evaluates $\zeta$ off the line. Turing's published argument contained errors, corrected by Lehman [332], and the corrected method underlies every large verification since,

| | Riemann Siegel and Turing | Odlyzko Schönhage | Pólya route |
|---|---|---|---|
| object | sign changes of $Z(t)$ | zeros in blocks, statistics | Thorin density $\rho_\varepsilon$, mass, count |
| where evaluated | critical line | critical line | line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ and contour |
| count certificate | Gram points and $\int S$ | Turing's method | argument principle count equal to the number of sign changes; computed, not yet certified |
| cost per point | $\sqrt{t/2\pi}$ terms, double | amortised, $t^{o(1)}$ | $0.341\,t$ extra digits and step below $\pi/t$ (untilted); $\delta t/\log 10$ digits and step below $2\pi/t$ (tilted) |
| accuracy | asymptotic remainder | asymptotic remainder | spectral; rounding certified by balls |
| reach | $t \approx 1540$ in 1953 | $t \approx 10^{20}$ and beyond [393, 395] | $t \approx 7005$ (Section 2.2.3.1); 25 digits at $t = 5000$ (Table 2.2) |

Table 2.4: The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached.

including Platt and Trudgian's to height $3 \times 10^{12}$ [419].

Before Turing, counts were certified by Backlund's method. Since $\operatorname{Re}\zeta(2+it) \geq 2 - \zeta(2) > 0$ for every $t$, it suffices to check that $\operatorname{Re}\zeta(\sigma + iT) \neq 0$ for $\frac{1}{2} \leq \sigma \leq 2$: then $\arg\zeta$ stays in $(-\frac{\pi}{2}, \frac{\pi}{2})$ along the whole path, so $|S(T)| < \frac{1}{2}$ and $N(T)$ is the integer nearest to $\vartheta(T)/\pi + 1$. Backlund carried this out for $T = 200$ [15] and Hutchinson for $T = 300$ [278, pp. 54 to 57], [501].

### 2.2.2 The Pólya route beside the classical evaluators

The Pólya route evaluates $\xi$ and $\xi'$ on and off the line, at $\frac{1}{2} + \varepsilon + i\theta$, from the Pólya kernel, and so delivers the Thorin density $\rho_\varepsilon$, its mass and the zero count from one evaluator. Its count has the logic of Turing's certificate: when the sign changes of $\Xi$ and the argument principle count $N(T)$ agree, every zero up to height $T$ is on the line and simple (Section 2.1.2).

The methods differ most in how their cost grows with height (Table 2.4). The untilted Pólya route needs about $0.341\,t$ extra digits; and by the budget of Section 2.1.1 its step must be below $\pi/t$, so that its number of nodes grows linearly with $t$ as well. The cause is aliasing. By Poisson summation the trapezoid sum at $s = \varepsilon + it$ equals $\sum_{k\in\mathbb{Z}} \xi(\frac{1}{2} + s - 2\pi ik/h)$, and the term $k = 1$, at height $t - 2\pi/h$, is exponentially larger than $\xi(\frac{1}{2} + s)$ once $t > \pi/h$. The tilted contour of Section 2.1.3 cuts the loss of digits to about $\delta t/\log 10$, but aliasing still requires $2\pi/h > t$, by the second condition in (2.1.4), so there too the number of nodes grows linearly with $t$.

The distances of Section 3.2 do not depend on how $\xi$ is evaluated, and this suggests a hybrid at large height. The approximate functional equation holds off the critical line [498, Chapter 4], so Riemann Siegel type expansions of $\zeta$ [8, Theorems 3.1 and 4.2] and of $\zeta'$, and the Odlyzko Schönhage algorithm for blocks of points, apply at $\frac{1}{2}+\varepsilon+i\theta$ as well. With them as the evaluator and Odlyzko's zeros [396] as the RH reference, one could compute $\rho_\varepsilon$, the matched total variation and the positivity margin at the heights of Odlyzko's computations, from about $10^{11}$ to beyond $10^{20}$, while the tilted Pólya route supplies values with explicitly bounded errors on low windows.

### 2.2.3 Two computations beyond the untilted route

#### *2.2.3.1 A Lehmer pair near height 7005*

Lehmer found two zeros that are unusually close, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ [333, pp. 297 and 298], at distance 0.0377, the classical example of a Lehmer pair; such pairs give lower bounds for the de Bruijn Newman constant [101] (Section 8.4.6). The untilted Pólya route cannot reach this height: it would need about 2400 digits and a step below $4.5\times10^{-4}$. The tilted route with $\delta = 0.01$, 256 bits, 13001 nodes, step $6\times10^{-4}$ and truncation $|x|\le 3.9$ computes $\Xi(\theta)e^{\pi\theta/4}$ at 34 milliseconds per point. On a grid of step 0.005 it finds sign changes in $(7005.060, 7005.065)$ and $(7005.100, 7005.105)$, the two members of the pair (Figure 2.2).

The Thorin density shows what resolution separates them. A single zero on the line produces a Cauchy peak of height $1/(\pi\varepsilon)$, that is 6.4, 15.9 and 31.8 at $\varepsilon = 0.05$, 0.02 and 0.01. The largest values of $\rho_\varepsilon$ on the grid are 11.18, 19.58 and 33.89. At $\varepsilon = 0.05$ the two peaks have merged into one of nearly double height; at $\varepsilon = 0.02$ they are partly separated; at $\varepsilon = 0.01$ each carries its own peak, lifted slightly by its neighbour. The Cauchy kernel predicts this: two kernels with half width at half maximum $\varepsilon$ at distance $g$ have a single maximum when $g \le 2\varepsilon/\sqrt{3}$ and two maxima when $g > 2\varepsilon/\sqrt{3}$. With $g = 0.0377$ the pair is resolved as two peaks for $\varepsilon < \sqrt{3}\,g/2 \approx 0.033$. The criterion concerns two isolated kernels; Exercise 2.11 proves it and asks how much the other zeros move the threshold.

#### *2.2.3.2 The zero count at height 1000*

Proposition 3.4.2 of Section 3.4 gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, where $M_{1/2}(T) = \pi^{-1}\arg\xi(1+iT)$ is the cumulative Thorin mass at resolution $\frac{1}{2}$, that is at the basepoint one of Section 9.1, and $\tau_{1/2}(T)$ is $\pi^{-1}$ times the change of $\arg\xi$ along the segment from $1+iT$ to $\frac{1}{2}+iT$. With the tilted route ($\delta = 0.1$, 200 bits, step 0.004 and truncation $|x|\le 4.2$, that is 2101 nodes), 10000 steps of 0.1 along $\operatorname{Re} s = 1$ and 2000 steps along the segment, we find

$$M_{1/2}(1000) = 648.7562315041018, \qquad \tau_{1/2}(1000) = 0.2437684958980,$$

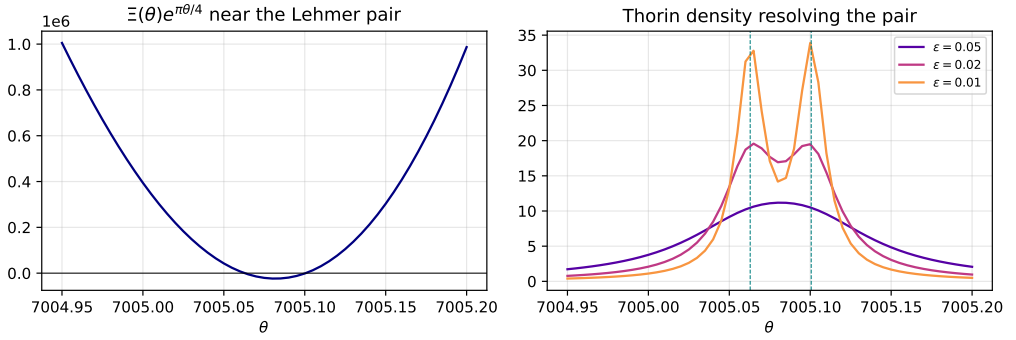



Figure 2.2: The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed.

$$M_{1/2}(1000) + \tau_{1/2}(1000) = 648.9999999999999,$$

so $N(1000) = 649$, with a largest argument increment of 0.43 radians. The computation took 77.5 seconds; `mpmath`'s `nzeros` returns 649 in 0.017 seconds. Where Backlund bounds the argument, this computation accumulates it from samples, and ball arithmetic certifies the rounding in each sampled value but not the absence of winding between samples. The count is therefore computed rather than certified. Certifying it would need, besides the analytic error bounds of Section 2.1, an enclosure of $\xi$ between consecutive sample points $s_k$ and $s_{k+1}$, for instance a check that $\mathrm{Re}\, \xi(s)/\xi(s_k) > 0$ on a ball containing the segment between them.

For counting and locating zeros at height, the Riemann Siegel family with Turing's method is faster by orders of magnitude, and the Pólya route is not a competitor there. It reads the count as an integer from a smooth quantity, the Thorin mass, with no Gram point bookkeeping; it gives the zero density strip by strip (Section 3.4); and it evaluates objects that the classical methods, used on the critical line, do not compute: Thorin densities and their distances to $U_{\mathrm{RH}}$.

## 2.3 A worked computation from first principles

Where Section 2.2 took the tilted route to large height, we return to the untilted route and follow one small computation with it at low height, from the theta series to a Thorin mass and a zero. Every value can be reproduced with `code/polya_flint.py` (Section A.1).

The parameters are those of Section 2.1.1, step $h = 1/200$ and truncation $|u| \le 3$, and the working precision is 240 bits, about 72 decimal digits, the default of the code. As there, $s$ is the displacement from $\frac{1}{2}$: with $E = e^{sh}$ and $\Phi_j = \Phi(jh)$, the evaluator returns $\xi(\frac{1}{2}+s) \approx P(E) + P(E^{-1})$ and $\xi'(\frac{1}{2}+s) \approx Q(E) - Q(E^{-1})$,

from which $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12+\varepsilon+i\theta)$.

*Step one: the kernel at a single point.* At $u=0$ the kernel (1.2.3) is

$$\Phi(0) = 2\sum_{n\ge1}\bigl(2\pi^2n^4-3\pi n^2\bigr)e^{-\pi n^2}.$$

The $n$th term is of order $n^4e^{-\pi n^2}$, and the terms with $n\ge8$ add up to less than $10^{-82}$, so seven terms already give more than eighty digits:

$$\Phi(0) = 0.893393800934246888173969334109418226440848734189 65\ldots.$$

The ball returned by the arithmetic has radius $3\times10^{-71}$.

*Step two: the coefficients.* The truncation leaves 601 nodes $u_j = jh$, $0\le j\le600$. The coefficients of $P$ are $\frac{h}{2}\Phi_0$ and $h\Phi_j$, $1\le j\le600$, and those of $Q$ are $h^2j\Phi_j$; the first is

$$\tfrac{h}{2}\Phi(0) = 0.00223348450233561722043492333527354556610\ldots.$$

The coefficients decay doubly exponentially. By the asymptotic formula of Section 1.2, $\Phi(2.7)\approx4\pi^2e^{12.15}\exp(-\pi e^{5.4})\approx6\times10^{-296}$, so the coefficients beyond $j\approx540$ are below $10^{-297}$.

*Step three: $\xi(\frac12)$ as a trapezoid sum.* At $s=0$, $E=e^{sh}=1$, and $\xi(\frac12)=2P(1)$ is the trapezoid sum of the kernel:

$$\xi(\tfrac12) = 0.49712077818831410991277373968539771980729360955 77\ldots,$$

with ball radius $5\times10^{-71}$. `mpmath`'s evaluation from the definition, $\frac12z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ at $z=\frac12$ in thirty digit arithmetic, gives

$$0.497120778188314109912773739685,$$

agreeing to every digit. By the bound of Section 2.1.1 the quadrature error is below $10^{-400}$, so the seventy digits are set by the working precision, not by the rule.

*Step four: the Thorin density at the first ordinate.* At

$$s = 0.1 + 14.134725141734693790i,$$

that is at resolution $\varepsilon=0.1$ and the first ordinate $\gamma_1$, the evaluator returns $\xi(\frac12+s)$ and $\xi'(\frac12+s)$, and

$$\rho_{0.1}(\gamma_1) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}\Bigl(\frac12+s\Bigr) = 3.18525309376337225967538\ldots.$$

A single Cauchy kernel of width 0.1 contributes exactly $1/(\pi\varepsilon) = 3.18309886183791$ at its centre. By Proposition 1.2.3 the excess, 0.00215423, is the sum of the contributions of every other zero, $\sum_{j\neq 1}[P_{0.1}(\gamma_1-\gamma_j)+P_{0.1}(\gamma_1+\gamma_j)]$, plus the reflected kernel of $\gamma_1$ itself. The nearest neighbour, at distance 6.887, contributes $6.7\times10^{-4}$, the reflected kernel $4\times10^{-5}$, and the many distant zeros, whose contributions decay like the inverse square of the distance, the remaining $1.4\times10^{-3}$. Figure 2.3 shows the same decomposition along the whole stretch between the first two zeros.

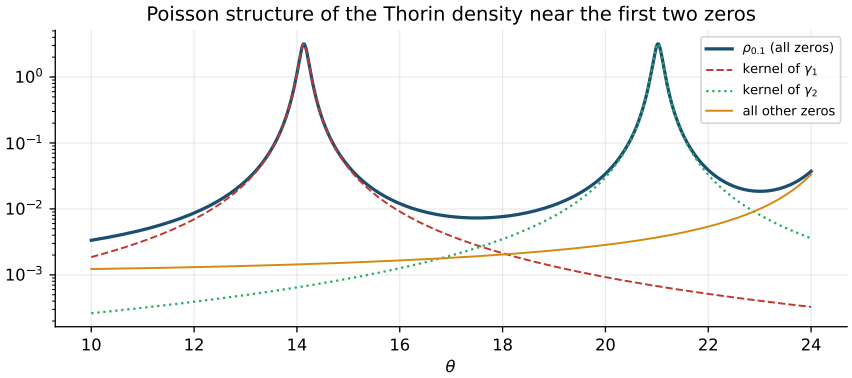



Figure 2.3: The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of $\gamma_1$, the kernel of $\gamma_2$ and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction.

Between zeros the density is small but strictly positive. At $\theta = 17$, between $\gamma_1$ and $\gamma_2 = 21.022$,

$$\rho_{0.1}(17) = 0.0076857707896205949783\ldots,$$

close to the minimum of $\rho_{0.1}$ between the two zeros (Figure 2.3). Lagarias's criterion [320] asks for this positivity at every $\theta$ and every resolution $\varepsilon > 0$.

*Step five: the Thorin mass up to height* 20. The cumulative Thorin mass of Section 2.1.2 is $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$, with the argument continued from the real axis. Stepping along the line $\frac12+0.1+i\theta$ from $\theta=0$ to $\theta=20$ in 1600 steps of $\varepsilon/8 = 0.0125$, as in that section, and accumulating the arguments of the quotients $\xi(\frac12+s_{k+1})/\xi(\frac12+s_k)$, where $s_k = 0.1+i\theta_k$ and $\theta_k = 0.0125\,k$, gives

$$\mu_{0.1}\big((0,400]\big) = M_{0.1}(20) = 1.0509304\ldots,$$

in 0.77 seconds. There is exactly one zero below height 20. By Proposition 3.2.2 the excess 0.0509 over the count agrees to $O(\varepsilon^2)$ with the first order mass defect

$-(\varepsilon/\pi)\Xi'(20)/\Xi(20)$ (Exercise 2.16): it is the mass carried across the edge $\theta = 20$ by the Cauchy tails of the first zero and of the zeros above 20, while what crosses $\theta = 0$ is returned exactly by the reflected kernels. Adding $\tau_{0.1}(20)$, which is $\pi^{-1}$ times the change of $\arg\xi$ along the segment from $\frac{1}{2} + 0.1 + 20i$ to $\frac{1}{2} + 20i$, returns the integer 1, as Proposition 3.4.2 requires.

*Step six: locating the first zero.* The ordinate $\gamma_1$ in step four was only a convenient point, and the density would have revealed it, up to the small shift caused by the other zeros, as the location of its first peak. To locate it from the Pólya route alone, one evaluates $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ on a grid, finds the sign change in $(14.10, 14.15)$, and refines by Newton's method with $\Xi'(\theta) = -\operatorname{Im}\xi'(\frac{1}{2} + i\theta)$; six iterations give $\gamma_1$ to forty digits. Section 2.1.2 refines all 79 zeros below height 200 in the same way, at 420 bits.

## 2.4 Exercises

*On Section 2.1.*

2.1. Let $f$ be analytic in a strip containing $|\operatorname{Im} u| \le \eta$, with $\int_{\mathbb{R}} |f(x \pm i\eta)|\,dx \le M$ and $f(u) \to 0$ uniformly as $|\operatorname{Re} u| \to \infty$ in $|\operatorname{Im} u| \le \eta$. Prove that $|\int_{\mathbb{R}} f(u)\,du - h\sum_j f(jh)| \le 2M/(e^{2\pi\eta/h} - 1)$ by writing the error as the sum of the integrals, from left to right, of $f(u)(e^{-2\pi iu/h} - 1)^{-1}$ along $\operatorname{Im} u = \eta$ and of $f(u)(e^{2\pi iu/h} - 1)^{-1}$ along $\operatorname{Im} u = -\eta$ [500, Theorem 5.1]. Deduce the bound of Section 2.1.1.

2.2. Let $g(x) = \Phi(x + i\eta)e^{\varepsilon x}$ with $\eta = \pi/4 - \delta$ and $\varepsilon \ge 0$, and $I(t) = \int_{\mathbb{R}} g(x)e^{itx}\,dx$. Show by Poisson summation that $h\sum_{j\in\mathbb{Z}} g(jh)e^{i\theta jh} - I(\theta) = \sum_{k\neq 0} I(\theta + 2\pi k/h)$. For $0 < a < \delta$, $0 < b < \pi/2 - \delta$ and $0 < \theta < 2\pi/h$, shift the line of integration up by $a$ when $k \ge 1$ and down by $b$ when $k \le -1$ to bound the error by $Ae^{-a\theta}/(e^{2\pi a/h} - 1) + Be^{b\theta}/(e^{2\pi b/h} - 1)$, where $A$ and $B$ are the suprema of $\int_{\mathbb{R}} |g(x + iy)|\,dx$ over $0 \le y \le a$ and over $-b \le y \le 0$. Derive (2.1.4) from $|I(t)| = e^{\eta t}|\xi(\frac{1}{2} + \varepsilon + it)|$ and Stirling's formula.

2.3. Using the formulas in the caption of Table 2.3, estimate the number of bits the untilted route needs at height 2000 for twenty digits, and compare with the tilted route at $\delta = 0.02$.

2.4. Show that on the tilted line $\operatorname{Im} u = \eta$ the factor $e^{su}$ with $s = \varepsilon + i\theta$ has modulus $e^{\varepsilon x - \theta\eta}$. Explain why the prefactor $e^{i\eta s}$ carries all of the exponential decay of $\xi$ except the factor $e^{-\delta\theta}$, and where that factor comes from.

2.5. Implement the two polynomial evaluator in double precision and find the height at which it first returns the wrong sign of $\Xi$.

2.6. Show that in midpoint radius arithmetic, with a complex number stored as a real and an imaginary ball, multiplication by an exact $w = |w|e^{i\varphi}$ rotates the midpoint and maps real and imaginary radii $r$ to $|w|(|\cos\varphi| + |\sin\varphi|)\,r$, up to rounding. Use this to explain why Horner's rule loses ball accuracy on the tilted line but not on the untilted line.

2.7. Use the tilted route to compute $\Xi(1000)$ and check its sign against $Z(1000)$ from the Riemann Siegel formula, with $Z$ as in Section 2.2 and $\Xi(t) = -\frac{1}{2}\big(t^2 + \frac{1}{4}\big)\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{1}{2}it)|\,Z(t)$.

*On Section 2.2.*

2.8. Compute the Gram points $g_0, \dots, g_{20}$ and check Gram's law for them; find the first Gram interval that contains no zero (it lies beyond $g_{100}$).

2.9. Show that $N(T) = \vartheta(T)/\pi + 1 + S(T)$ follows from the argument principle applied to $\xi$ on a rectangle.

2.10. Explain why a close pair of zeros that falls inside one Gram interval produces no sign change there, and how Turing's method detects that zeros are missing.

2.11. Prove that $P_\varepsilon(x - g/2) + P_\varepsilon(x + g/2)$ has a single maximum if $g \le 2\varepsilon/\sqrt{3}$ and two maxima if $g > 2\varepsilon/\sqrt{3}$, and estimate how the other zeros near height 7005 shift the threshold 0.033 for the Lehmer pair.

2.12. Estimate the cost of counting zeros to height $10^4$ by the Thorin mass at resolution $\frac{1}{2}$ with the tilted route, including the growth of the number of nodes with height, and compare with the Riemann Siegel count.

*On Section 2.3.*

2.13. Compute $\Phi(0)$ with two, three, four and five terms of the series and explain the number of correct digits in each case.

2.14. Repeat step three with $h = 1/5$, $1/10$ and $1/20$ and plot $\log_{10}$ of the error against $1/h$. Show by Poisson summation that the error of the untruncated sum is exactly $2\sum_{k\ge1}\Xi(2\pi k/h)$, and confirm that it falls like $e^{-\pi^2/(2h)}$, the rate set by the strip half width $\pi/4$, up to slowly varying factors.

2.15. Compute $\rho_{0.1}(\gamma_1) - 1/(0.1\pi)$ directly as a sum over the first 1700 zeros with the smooth tail model of Section 3.2, and compare with step four.

2.16. Predict $M_{0.1}(20) - 1$ from the first order formula of Proposition 3.2.2, with $\Xi'(20)/\Xi(20)$ computed by the evaluator of step four, and compare with step five.

# Chapter 3

# The Thorin measure at finite resolution

The Thorin condition is RH (Proposition 1.2.1), and by Lagarias's criterion RH is the positivity of $\rho_\varepsilon$ at every resolution (Remark 1.2.4); the question is what finite computations of $\rho_\varepsilon$ can establish about it. The answer, which Chapters 4 and 5 complete, has two sides. On every window where the zeros are verified, the computed Thorin measure $\mu_\varepsilon$ converges to $U_{\mathrm{RH}}$, the discrete measure demanded by RH, at explicit rates as $\varepsilon \downarrow 0$. Above the computed range every finite computation of the kinds in this book remains consistent with a failure of the Thorin condition, and the step that remains, positivity at every resolution and every height, is not a computational one (Section 5.2).

## 3.1 What computation can reach, and what the sine squared lemma adds

Two facts govern what computation can establish about the Thorin condition and which evaluator it should use (Sections 3.1.1 and 3.1.2). The Pólya density is the evaluator in the rest of this chapter and in Chapters 4 and 5.

### 3.1.1 What a finite computation can measure

Below height 200 the Pólya route finds 79 sign changes of $\Xi$ and an argument principle count $N(200) = 79.0000000000004$ (Section 2.1.2). So, numerically and in agreement with [15, 419], the Thorin functional $U = \sum_k \delta_{a_k}$ of Section 5.3 coincides with $U_{\mathrm{RH}}$ on $(0, 200^2]$. The matched total variation distance of Section 3.2, computed against a model of the zero list, is $3.7 \times 10^{-9}\,\varepsilon$ at every resolution computed, the numerical floor of the comparison.

None of this constrains the zeros above the window. Platt and Trudgian [419] have verified every zero below height $3 \times 10^{12}$, yet the perturbation of $\xi$ in Theorem 5.2.1 [424, Chapter 11] keeps the order, type and symmetry of $\xi(\frac{1}{2} + s)$, puts zeros off the line just above any large enough height $R$ for which RH has been verified up to $R + 50$, and still passes the finite tests of the heat trace, from

Hankel blocks and derivative signs (to order $10^{17}$) to the small time expansion. So a computation from the zeros verified up to a given height, or from finitely many tests of the heat trace, cannot by itself measure a scalar distance to RH, although a finite computation could refute RH by certifying a zero off the line. What it can measure is a margin at a stated resolution and height.

The margin is the minimum of $\rho_\varepsilon$, and positivity on one line is weaker than a zero free half plane (Remark 1.2.4).

Resolution is the first limitation. On $0 \le \theta \le 200$ the computed minimum is $\min \rho_\varepsilon = 0.0147\,\varepsilon$ at each of the eight resolutions of Section 2.1.2, attained at $\theta = 0$, where $\rho_\varepsilon(0) = \pi^{-1}\xi'(\frac{1}{2}+\varepsilon)/\xi(\frac{1}{2}+\varepsilon) = 2\varepsilon\pi^{-1}\sum_k a_k^{-1} + O(\varepsilon^3)$, because $(\xi'/\xi)(\frac{1}{2}+\varepsilon) = \sum_k 2\varepsilon/(\varepsilon^2 + a_k)$ and $\sum_k a_k^{-1} = 0.0231050$ (Section 1.2). So $\inf_\theta \rho_\varepsilon(\theta) \le \rho_\varepsilon(0) \to 0$ as $\varepsilon \downarrow 0$: near the critical line positivity is never won by a fixed margin. On the verified window the origin is in fact the minimum: for every $\varepsilon \in (0, 6.85]$ the minimum of $\rho_\varepsilon$ over $0 \le \theta \le 3\times 10^{12} - 1.21$ is $\rho_\varepsilon(0)$, attained only at the origin, and that this holds at every height for every $\varepsilon \in (0, \frac{1}{2})$ is equivalent to RH (Theorem 9.1.18 and Remark 9.1.19).

Height is the second limitation. Even on the tilted contour of Section 2.1.3 the number of nodes, and with it the cost per point, grows linearly in the height, so the Pólya route is not a competitor for zero verification (Section 2.2.3.2). Its value is accuracy at small $\varepsilon$ on moderate windows, where the Thorin measure is sharp and the prime side is ill conditioned.

### 3.1.2 The sine squared decomposition of the Thorin bracket

The sine squared lemma (1.2.9) converts the Lévy description of $\log \xi$, in which the primes are atoms, into the Thorin description of the reciprocal. At a basepoint $\alpha > 1$, applying the kernel $2\sin^2(x\theta/2) = 1 - \cos(x\theta)$ to the canonical measure of [424, Theorem 8.1] writes the bracket $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ as a sum of four explicit pieces:

$$v_\alpha(\theta) = \underbrace{\frac{\xi'}{\xi}(\alpha)}_{\text{drift}} + \underbrace{\int_0^\infty (1-\cos\theta x)\frac{e^{-\alpha x}}{e^{2x}-1}\,dx}_{\text{archimedean } \ge 0} - \underbrace{\frac{\theta^2}{(\alpha-1)\big((\alpha-1)^2+\theta^2\big)}}_{\text{pole } \ge 0} + \underbrace{\sum_{n\ge 2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big)}_{\text{primes } \ge 0}. \quad (3.1.1)$$

The pole piece is the closed form of $\int_0^\infty (1-\cos\theta x)e^{-(\alpha-1)x}dx$, and the prime piece equals $-\frac{\zeta'}{\zeta}(\alpha) + \mathrm{Re}\,\frac{\zeta'}{\zeta}(\alpha+i\theta)$.

The drift is the linear coefficient $c = \xi'/\xi(\alpha)$ of the Lévy representation of Section 1.2, not the drift $d$ of a GGC. The archimedean piece is the contribution of $\Gamma(s/2)$ together with the factor $s$ of $\frac{1}{2}s(s-1)$; it is smooth in $\theta$ and grows like

| $\alpha$ | $\theta$ | drift | archimedean | pole | primes | $v_\alpha(\theta)$ | $\kappa_\alpha(\theta)$ |
|---|---|---|---|---|---|---|---|
| 3.0 | 2 | 0.1144 | 0.0533 | −0.2500 | 0.1988 | 0.1165 | 3.85 |
| 3.0 | 50 | 0.1144 | 1.2594 | −0.4992 | 0.2464 | 1.1210 | 0.67 |
| 2.0 | 2 | 0.0691 | 0.0859 | −0.8000 | 0.7154 | 0.0704 | 21.5 |
| 2.0 | 50 | 0.0691 | 1.3989 | −0.9996 | 0.7561 | 1.2245 | 1.43 |
| 1.5 | 2 | 0.0461 | 0.1137 | −1.8824 | 1.7683 | 0.0470 | 77.6 |
| 1.5 | 50 | 0.0461 | 1.4863 | −1.9998 | 1.8527 | 1.3867 | 2.78 |

Table 3.1: The four pieces of the sine squared decomposition (3.1.1), with the prime sum over $n \le 2 \times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond $2 \times 10^6$ ($1.3 \times 10^{-3}$ at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece.

$\frac{1}{2}\log\theta$, which accounts for the average density of zeros. The prime piece is a sum of oscillations with frequencies $\log n$ and amplitudes $\Lambda(n)n^{-\alpha}$, one for each prime power. The drift is positive.

Near $\alpha = 1$ the pole piece is almost cancelled by the prime piece (Table 3.1). Both pieces grow like $1/(\alpha - 1)$, the pole explicitly and the primes through the pole of $-\zeta'/\zeta$ at one. At each fixed $\theta \neq 0$ their difference stays bounded as $\alpha \downarrow 1$ because $\zeta$ has no zero on the line $\operatorname{Re} s = 1$, so the analytic input of the prime number theorem appears in the Thorin description as this cancellation, which [424, Section 8.2] identifies with the collapse onto the zeros. Measured by the condition number

$$\kappa_\alpha(\theta) = \frac{|\text{pole}| + |\text{primes}|}{|v_\alpha(\theta)|},$$

the prime side amplifies relative errors by 77.6 at $\alpha = 1.5$ and $\theta = 2$.

The lemma also prices the prime side. By the prime number theorem the truncation tail satisfies, for fixed $\theta \neq 0$,

$$\sum_{n>N} \frac{\Lambda(n)}{n^\alpha}\big(1 - \cos(\theta \log n)\big) \asymp \frac{N^{1-\alpha}}{\alpha - 1},$$

which is $1.4 \times 10^{-3}$ at $\alpha = 1.5$ with $N = 2 \times 10^6$, the size of the errors in Table 1.5; an absolute error of $10^{-10}$ at $\alpha = 1.5$ would need a cutoff $N \approx 4 \times 10^{20}$. For $\alpha \le 1$ the series diverges, and the lemma must be read through the regularised pairing of [424, Chapter 7] and the explicit formula. At the basepoint one the regularisation is explicit: the pole and prime pieces can be compensated against each other, and Section 9.1 converts the resulting Lévy form of $\xi(1 + s)$ term by term into the Thorin density. Near the critical line Selberg's smoothed sum over the primes up to $x^2$ takes the place of the series, and at resolution $\varepsilon$, if no zero lies to the right of the critical line, its error at an ordinate is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that $k$ correct digits need $x \approx 10^{k/\varepsilon}$ (Section 3.5.1). The Pólya route has none

of these costs: its cost per point is polynomial in the height and independent of $\varepsilon$, and at height 200 it reaches $\varepsilon = 0.005$ with about 60 digits (Section 2.1.1).

### 3.1.3 Consequences for computation

The bracket is the derivative of an argument, $v_\alpha(\theta) = \partial_\theta \arg \xi(\alpha + i\theta)$, because $\partial_\theta \log \xi(\alpha + i\theta) = i(\xi'/\xi)(\alpha + i\theta)$. So the cumulative Thorin mass is an argument,

$$\mu_\varepsilon\big((0, \theta^2]\big) = \frac{1}{\pi} \arg \xi\Big(\frac{1}{2} + \varepsilon + i\theta\Big),$$

continued along the line from the real axis, where $\xi > 0$. It is computed without quadrature, by accumulating the arguments of quotients of consecutive values of $\xi$, with steps chosen so that each increment can be read from its principal value, a condition checked after the fact rather than guaranteed (Section 2.1.2). Section 3.2 uses it for masses, cell masses, cumulative distances and the mass defect of Proposition 3.2.2, and Section 3.4 for the zero count.

The lemma is linear with a nonnegative kernel, so a nonnegative Lévy input gives a nonnegative Thorin output, and positivity certificates can be posed as linear programs in the sine squared basis [424, Sections 18.2.8 and 18.2.9]. For $\xi$ the only negative part of the input $xL(dx)$ is the pole density $-e^{-(\alpha-1)x}dx$, so a certificate must show that the positive pieces dominate the pole piece at every $\theta$. For $\alpha > 1$ they do (Theorem 1.2.2), although the proof goes through the zeros and not through the input. For $\frac{1}{2} < \alpha \le 1$, where the lemma is read through the regularised pairing, domination at every basepoint is Lagarias's criterion, that is RH, and Section 3.5.2 states RH as a domination of the same kind at the centre, Weil's criterion on variograms. Beurling generalised prime systems, in which the sign of the input is controlled by construction, are among the open computations of Section 12.2.3.

The lemma also supplies an exact consistency check at $\alpha = \frac{1}{2}$, where the constant parts of the pole and prime pieces diverge and are continued analytically. The constant part $1/(\alpha - 1)$ of the pole piece continues to $-2$ and enters with a minus sign; the constant part $-\zeta'/\zeta(\alpha)$ of the prime piece continues to $-\zeta'/\zeta(\frac{1}{2})$; and the gamma factor contributes the archimedean constant of [424, Section 6.8], $C_\Gamma = \frac{1}{2}\log\pi - \frac{1}{2}\psi(\frac{1}{4}) - 2 = 0.6860917096$, which is minus the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ at $s = \frac{1}{2}$. They sum to

$$C_\Gamma + 2 - \frac{\zeta'}{\zeta}\Big(\frac{1}{2}\Big) = 0,$$

which is equivalent to $\xi'(\frac{1}{2}) = 0$, because $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \zeta'/\zeta(s)$ and $1/s + 1/(s-1)$ vanishes at $s = \frac{1}{2}$.

## 3.2 Distances to the discrete measure implied by RH

On a window where the zeros are known, the Thorin measure $\mu_\varepsilon$ computed by the Pólya route can be compared with $U_{\mathrm{RH}}$ directly (Section 3.1.1), and how close the two are depends on the metric.

Fix $\Theta = 200$ and the window $E = (0, \Theta^2]$ in $z$, equivalently $(0, \Theta]$ in $\theta$. The window contains 79 zeros, all simple and on the line, and $\Theta$ is not an ordinate, since $\gamma_{80} = 201.2648$ (Section 2.1.2). For a signed measure $\nu$ on $E$ write $\|\nu\|_{\mathrm{TV}} = |\nu|(E)$, and write $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2])$ for the cumulative mass.

### 3.2.1 Total variation at matched resolution

A smooth approximation to a sum of point masses never converges to it in total variation.

Proposition 3.2.1. *If $\nu \geq 0$ is absolutely continuous and $U$ is purely atomic with $N$ unit atoms in $E$, then $\|\nu - U\|_{\mathrm{TV}} = \nu(E) + N$.*

*Proof.* Let $A$ be the finite set of atoms. Then $\nu(A) = 0$, so $|\nu - U|(A) = U(A) = N$ and $|\nu - U|(E \setminus A) = \nu(E \setminus A) = \nu(E)$. □

The Thorin measure is positive on $E$, where its density has minimum $0.0147\,\varepsilon$ (Section 3.1.1), so $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}} = \mu_\varepsilon(E) + 79 \approx 158$ for every $\varepsilon$. Total variation becomes informative only after both measures have been brought to the same resolution, or summed over cells.

By Proposition 1.2.3 the object to compare with $\mu_\varepsilon$ is the Poisson smoothing of the RH measure at the same resolution, and we write $d_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * U^{\pm}_{\mathrm{RH}}\|_{\mathrm{TV}}$ for the matched distance. Under RH the two measures coincide and $d_{\mathrm{match}}(\varepsilon) = 0$, so a computation of the matched distance measures its own error unless some zero is off the line.

A computation replaces $U^{\pm}_{\mathrm{RH}}$ by a model $\widetilde{U}$, which we build from the first 1700 ordinates, computed with `mpmath` [376], and from the smooth density $\pi^{-1}\vartheta'(t)\,dt$ of the Riemann Siegel theta function beyond the Gram point $T_* = g_{1699} = 2197.8796$ of Section 2.2.1, at which $\vartheta(T_*)/\pi + 1 = 1700$, each with its reflection. We write $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$ for the matched distance against the model. Since $\gamma_{1700} = 2197.2587 < T_* < \gamma_{1701} = 2198.6875$, we have $N(T_*) = \vartheta(T_*)/\pi + 1$, so the boundary term vanishes when $\int_{T_*}^{\infty} P_\varepsilon(\theta - t)\, d\big(N(t) - \vartheta(t)/\pi - 1\big)$ is integrated by parts, and the error of the tail model is at most of order $\varepsilon/T_*^2$ per unit length. The observed floor is far smaller: $\tilde{d}_{\mathrm{match}}(\varepsilon) = 3.66 \times 10^{-9}\,\varepsilon$ at every resolution from $\varepsilon = 1$ to $\varepsilon = 0.005$ (Table 3.2), that is $1.8 \times 10^{-11}\,\varepsilon$ per unit length. The exact proportionality to $\varepsilon$ identifies the floor as the residual of the tail model; it is an observed value, not a bound for $d_{\mathrm{match}}(\varepsilon)$: such a bound would need certified bounds on the error of the tail model and on the evaluation of $\mu_\varepsilon$. The densities that enter the comparison are stored in double precision.

| $\varepsilon$ | mass | raw TV | matched TV | sup diff | cell TV | $L^1$ in $\theta$ | $W_\varepsilon(0.01)$ |
|---|---|---|---|---|---|---|---|
| 1 | 79.36738 | 158.37 | $3.66 \times 10^{-9}$ | $1.9 \times 10^{-11}$ | 4.992 | 59.23 | 0.29504 |
| 0.5 | 79.19417 | 158.19 | $1.83 \times 10^{-9}$ | $9.6 \times 10^{-12}$ | 2.655 | 41.86 | 0.22362 |
| 0.25 | 79.09887 | 158.10 | $9.14 \times 10^{-10}$ | $4.8 \times 10^{-12}$ | 1.351 | 28.77 | 0.18698 |
| 0.1 | 79.03977 | 158.04 | $3.66 \times 10^{-10}$ | $1.9 \times 10^{-12}$ | 0.5431 | 16.01 | 0.16469 |
| 0.05 | 79.01990 | 158.02 | $1.83 \times 10^{-10}$ | $9.6 \times 10^{-13}$ | 0.2718 | 9.737 | 0.15721 |
| 0.02 | 79.00796 | 158.01 | $7.31 \times 10^{-11}$ | $3.8 \times 10^{-13}$ | 0.1087 | 4.816 | 0.15270 |
| 0.01 | 79.00398 | 158.00 | $3.66 \times 10^{-11}$ | $2.0 \times 10^{-13}$ | 0.05437 | 2.756 | 0.15120 |
| 0.005 | 79.00199 | 158.00 | $1.83 \times 10^{-11}$ | $1.5 \times 10^{-13}$ | 0.02718 | 1.553 | 0.15045 |
| RH | 79 | 0 | 0 | 0 | 0 | 0 | 0.14970 |

Table 3.2: Distances between the Pólya computed Thorin measure $\mu_\varepsilon$ and $U_{\mathrm{RH}}$ on $0 < z \le 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.2.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \le 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, $L^1$ in $\theta$ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row.

### 3.2.2 Cumulative distances and extrapolation in the resolution

The total mass is the simplest quantity to compare, since $U_{\mathrm{RH}}(E) = N(\Theta) = 79$. By Section 3.1.3 it is an argument, $\mu_\varepsilon(E) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\Theta)$, continued from $\arg \xi(\frac{1}{2} + \varepsilon) = 0$.

Proposition 3.2.2. *Suppose $\Theta$ is not an ordinate and every zero with $0 < \operatorname{Im} \rho \le \Theta$ lies on the line. Then*

$$\mu_\varepsilon\big((0, \Theta^2]\big) = N(\Theta) - \frac{\varepsilon}{\pi} \frac{\Xi'(\Theta)}{\Xi(\Theta)} + O(\varepsilon^2), \qquad \varepsilon \downarrow 0.$$

*Proof.* Let $\sigma_1 > 1$. By hypothesis, and because $\xi$ is positive on the real axis, $\xi$ has no zero in the closed rectangle with vertices $\frac{1}{2}+\varepsilon$, $\sigma_1$, $\sigma_1 + i\Theta$ and $\frac{1}{2}+\varepsilon+i\Theta$. So the change of $\arg \xi$ up its left side, which is $\pi\mu_\varepsilon((0, \Theta^2])$, equals the change along the path $\frac{1}{2} + \varepsilon \to \sigma_1 \to \sigma_1 + i\Theta \to \frac{1}{2} + \varepsilon + i\Theta$. The classical formula $\pi N(\Theta) = \Delta \arg \xi$ along $\frac{1}{2} \to \sigma_1 \to \sigma_1 + i\Theta \to \frac{1}{2} + i\Theta$ [498, Section 9.3] differs from it only by the segment from $\frac{1}{2} + \varepsilon + i\Theta$ to $\frac{1}{2} + i\Theta$, since $\arg \xi$ is constant on the real axis. Put $h(\varepsilon) = \xi(\frac{1}{2} + \varepsilon + i\Theta)$. Since $h(0) = \Xi(\Theta) \neq 0$, a branch of $\arg h$ is real analytic near $\varepsilon = 0$, and

$$\pi\mu_\varepsilon\big((0, \Theta^2]\big) = \pi N(\Theta) + \arg h(\varepsilon) - \arg h(0) = \pi N(\Theta) + \varepsilon \operatorname{Im} \frac{\xi'}{\xi}\Big(\frac{1}{2} + i\Theta\Big) + O(\varepsilon^2).$$

From $\frac{d}{d\theta}\Xi(\theta) = i\xi'(\frac{1}{2} + i\theta)$ we get $(\xi'/\xi)(\frac{1}{2} + i\Theta) = -i\,\Xi'(\Theta)/\Xi(\Theta)$, and $\Xi'/\Xi$ is real, so the coefficient of $\varepsilon$ is $-\Xi'(\Theta)/\Xi(\Theta)$. □

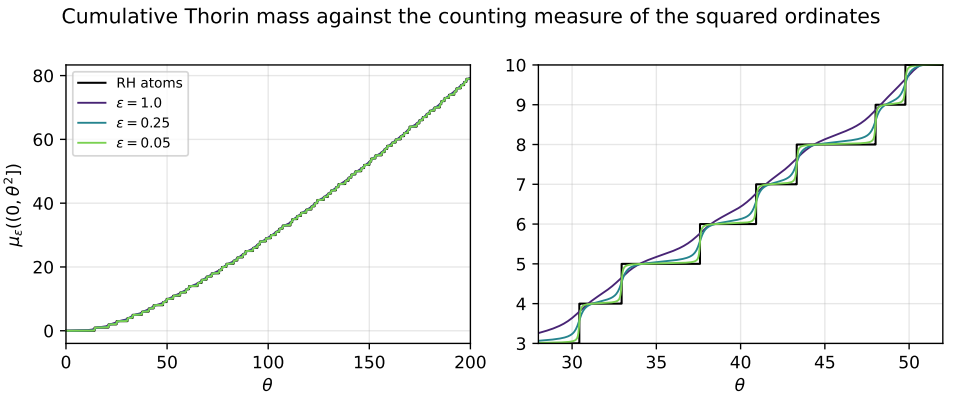



Figure 3.1: Cumulative Thorin mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of $U_{\mathrm{RH}}$, on the whole window (left) and on $28 \le \theta \le 52$ (right).

Remark 3.2.3. The remainder in Proposition 3.2.2 is in fact $O(\varepsilon^3)$: $\xi(s) = \xi(1-s)$ and the reality of $\xi$ on the real axis give $h(-\varepsilon) = \overline{h(\varepsilon)}$, so $\arg h(\varepsilon) - \arg h(0)$ is odd in $\varepsilon$ and its expansion has no term in $\varepsilon^2$.

Choosing $\Theta$ midway between zeros makes the correction small, and adding it recovers $N(\Theta)$ up to $O(\varepsilon^3)$; Proposition 3.4.2 makes the recovery exact. At $\Theta = 200$ the Pólya route gives $\Xi'(200)/\Xi(200) = -1.2506879$, so the proposition predicts $\mu_\varepsilon(E) - 79 = -\varepsilon\,\Xi'(200)/(\pi\,\Xi(200)) + O(\varepsilon^3)$, with $-\Xi'(200)/(\pi\,\Xi(200)) = 0.3981063$; the observed $(\mu_\varepsilon(E) - 79)/\varepsilon$ is 0.398105 at $\varepsilon = 0.005$ and 0.398102 at $\varepsilon = 0.01$. Figure 3.1 compares $M_\varepsilon$ with the counting function of $U_{\mathrm{RH}}$ across the window.

Summing over cells gives a distance that resolves the individual zeros. Let $b_0 = 0$, $b_k = (\gamma_k + \gamma_{k+1})/2$ for $1 \le k \le 78$ and $b_{79} = \Theta$, and let $C_k = (b_{k-1}, b_k]$. The cell masses $\mu_\varepsilon(C_k) = M_\varepsilon(b_k) - M_\varepsilon(b_{k-1})$ are differences of the argument, and the coarse grained distance

$$d_{\mathrm{cell}}(\varepsilon) = \sum_k |\mu_\varepsilon(C_k) - 1|$$

is the total variation distance between the images of $\mu_\varepsilon$ and $U_{\mathrm{RH}}$ under the map sending each cell to its atom. No $b_k$ is an ordinate, so Proposition 3.2.2 applies at both ends of every cell, with $\Xi'(0) = 0$ at $b_0$, and with the remainder of Remark 3.2.3 gives $\mu_\varepsilon(C_k) - 1 = -(\varepsilon/\pi)\big[\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\big] + O(\varepsilon^3)$. Hence $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$, where $\kappa_{\mathrm{cell}} = \pi^{-1}\sum_k |\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})|$, the cell constant of this window, is 5.43665, and every cell mass lies in $[0.99888, 1.00126]$ at $\varepsilon = 0.005$. Section 5.4 reads this constant as a statistic of the irregularity of the spacings.

The $L^1$ distance $\int_0^\Theta |M_\varepsilon(\theta) - N(\theta)|\,d\theta$ between the cumulative mass and the counting function is a transport distance, and because of the Cauchy tails it behaves like $\varepsilon\log(1/\varepsilon)$: it is 1.55 in $\theta$ and 360 in $z$ at $\varepsilon = 0.005$, and Section 3.4.1 derives the rate. The heat trace $W(t) = \int e^{-tz}U_{\mathrm{RH}}(dz)$ is approximated by $W_\varepsilon(t) = \int_0^\infty e^{-t\theta^2}\rho_\varepsilon(\theta)\,d\theta = \int e^{-tz}\mu_\varepsilon(dz)$. At $t = 0.01$ the zeros give $W = 0.1496980112538$ (Table 1.4), while $W_\varepsilon(0.01) = 0.1504500280599$, $0.1512017825455$ and $0.1527045050533$ at $\varepsilon = 0.005$, $0.01$ and $0.02$, again with error linear in $\varepsilon$.

The mass and the heat trace are smooth in $\varepsilon$ at $\varepsilon = 0$, so Richardson extrapolation applies to them: the mass by Proposition 3.2.2, and the heat trace because by Proposition 1.2.3 it is a sum over the zeros of Poisson integrals of the Gaussian $e^{-t\theta^2}$. The combination $2\mu_{0.005}(E) - \mu_{0.01}(E)$ equals $79 + 3.2\times10^{-8}$, against a raw defect of $2.0\times10^{-3}$. For the heat trace the error $7.5\times10^{-4}$ of $W_{0.005}(0.01)$ falls to $2.6\times10^{-7}$ after one Richardson step ($\varepsilon = 0.005$ and $0.01$) and to $1.7\times10^{-10}$ after two (adding $\varepsilon = 0.02$). Without extrapolation the same accuracy would need $\varepsilon \approx 8\times10^{-8}$ for the mass, and $\varepsilon \approx 2\times10^{-6}$ and $10^{-9}$ for the heat trace.

## 3.3 What an off line zero would do

How each distance of Section 3.2 would change if a zero left the line can be computed in closed form, because by Proposition 1.2.3 an off line zero at distance $\delta$ from the line enters the Thorin density, together with its reflection in the line, only through the kernels $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at its ordinate, the second of which is negative once $\varepsilon < \delta$. The three signatures of Proposition 3.3.1 show that an off line zero is most visible at resolutions comparable to its distance from the line, a theme that returns in Section 5.2.

### 3.3.1 Three signatures of an off line zero

Off line zeros come in pairs $\frac12 \pm \delta + i\gamma$ with $0 < \delta < \frac12$, together with their conjugates, and near $\gamma$ a pair contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ to $\rho_\varepsilon$ where two line zeros would contribute $2P_\varepsilon$.

Proposition 3.3.1. *Let $\varepsilon > 0$ and $0 < \delta < \frac12$; when $\delta = \varepsilon$, $P_0$ denotes the unit point mass at the origin.*

*(i) (Displaced zeros, same count.) If two line zeros at height $\gamma$ are replaced by the quadruple $\frac12 \pm \delta \pm i\gamma$, the density near $\gamma$ changes by $2g$, where $g = \frac12(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$ is the change per displaced zero, and*

$$\|g\|_{\mathrm{TV}} = \frac{3\sqrt3}{4\pi}\Big(\frac{\delta}{\varepsilon}\Big)^2 + O\Big(\frac{\delta^4}{\varepsilon^4}\Big), \qquad \delta < \varepsilon.$$

*(ii) (Pair omitted from the list.) If a pair at* $\frac{1}{2} \pm \delta + i\gamma$ *exists but a list of line zeros omits it, the difference is* $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ *and*

$$\|P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\|_{\mathrm{TV}} = \begin{cases} 2, & \delta \le \varepsilon, \\ \dfrac{8}{\pi} \arctan \sqrt{\dfrac{\delta+\varepsilon}{\delta-\varepsilon}} - 2, & \delta > \varepsilon. \end{cases}$$

*(iii) (Negative mass.) For* $\delta > \varepsilon$ *the negative part of* $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$*, the density of one displaced zero, has mass*

$$\frac{2}{\pi} \arctan \sqrt{\frac{\delta+\varepsilon}{\delta-\varepsilon}} - \frac{1}{2} = \frac{\varepsilon}{\pi\delta} + O\Big(\frac{\varepsilon^3}{\delta^3}\Big).$$

*Proof.* (i) Taylor expansion in $\delta$ gives $g = \frac{\delta^2}{2}\partial_a^2 P_a|_{a=\varepsilon} + O(\delta^4)$ [424, Proposition 19.7], with a remainder of $L^1$ norm $O(\delta^4/\varepsilon^4)$. Since $\partial_a^2 P_a(x) = (2a/\pi)(a^2 - 3x^2)/(a^2+x^2)^3$, the substitution $x = \varepsilon v$ gives $\partial_a^2 P_a|_{a=\varepsilon} = 2(1-3v^2)/(\pi\varepsilon^3(1+v^2)^3)$ and $dx = \varepsilon\, dv$. The function $(1-3v^2)/(1+v^2)^3$ is the derivative of $v/(1+v^2)^2$, whose total variation is four times its maximum $9/(16\sqrt{3})$, that is $3\sqrt{3}/4$, so the $L^1$ norm of $\partial_a^2 P_a$ at $a = \varepsilon$ is $(2/(\pi\varepsilon^2)) \cdot 3\sqrt{3}/4 = \frac{3\sqrt{3}}{2\pi\varepsilon^2}$, and multiplying by $\delta^2/2$ gives (i). (ii) For $\delta \le \varepsilon$ both kernels are nonnegative with mass one. For $\delta > \varepsilon$ put $a = \delta + \varepsilon$, $b = \delta - \varepsilon$, so the difference is $P_a - P_b$, which is negative exactly on $|x| < \sqrt{ab}$. Integrating, $\|P_a - P_b\|_{\mathrm{TV}} = \frac{4}{\pi}[\arctan\sqrt{a/b} - \arctan\sqrt{b/a}]$, and $\arctan y + \arctan(1/y) = \pi/2$. (iii) $P_a - P_b$ has total mass zero, so its negative part is half its norm; halve again for the factor $\frac{1}{2}$. □

With $\varepsilon/\delta = \sin\phi$ one has $\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} = \pi/4 + \phi/2$, so for $\delta > \varepsilon$ the norm in (ii) is $(4/\pi)\arcsin(\varepsilon/\delta)$ and the mass in (iii) is $(1/\pi)\arcsin(\varepsilon/\delta)$. Figure 3.2 plots the three signatures.

### 3.3.2 Calibration of the matched distance

Proposition 3.3.1 converts the matched distance of Section 3.2 into detection thresholds. By part (ii), a computed matched distance $\tilde{d}_{\mathrm{match}}(\varepsilon)$ of at most $3.7 \times 10^{-9}\,\varepsilon$ excludes any omitted pair with ordinate in $(0, 200)$: such a pair contributes total variation $(4/\pi)\arcsin(\varepsilon/\delta)$, or 2 if $\delta \le \varepsilon$, less only the part of its kernels outside the window. By part (i), a displaced pair that keeps the count adds $2g$ to a difference whose norm is the floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, so by the triangle inequality the matched distance rises above the floor as soon as $\|g\|_{\mathrm{TV}} > \tilde{d}_{\mathrm{match}}(\varepsilon)$, that is, to leading order, once $\delta$ exceeds

$$\delta_{\min}(\varepsilon) = \varepsilon\Big(\frac{4\pi\,\tilde{d}_{\mathrm{match}}(\varepsilon)}{3\sqrt{3}}\Big)^{1/2}. \tag{3.3.1}$$

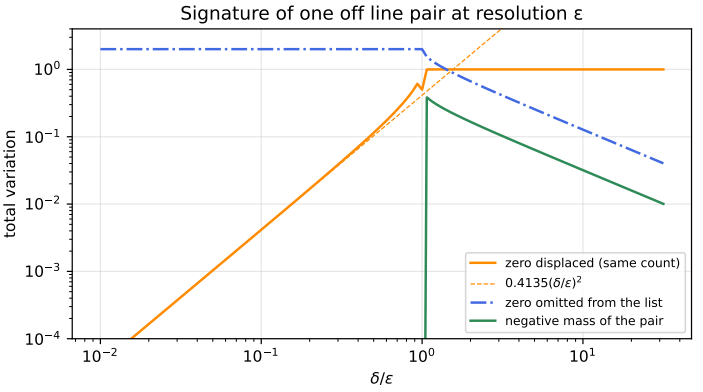



Figure 3.2: The three signatures of Proposition 3.3.1 as functions of $\delta/\varepsilon$: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1.

With the floors of Table 3.2 this threshold is $3.3 \times 10^{-5}$ at $\varepsilon = 0.5$, $1.1 \times 10^{-6}$ at $\varepsilon = 0.05$ and $3.3 \times 10^{-8}$ at $\varepsilon = 0.005$.

The threshold (3.3.1) is the square root law of [424, Proposition 19.7]: the signature grows like the square of the displacement, so the smallest detectable displacement falls like the square root of the achievable distance. The prime side calibration of [424, Section 19.3.6] obeys the same law and reaches $\delta \approx 0.05$ at $\varepsilon = 2$. What changes is the reach: the Pólya route evaluates $\rho_\varepsilon$ at small $\varepsilon$ to about 60 digits below height 200 (Section 2.1.1), so the floor is set by the tail model for the zero list (Section 3.2.1) rather than by the evaluator, whereas the prime side is limited by the truncation of Table 1.5. A positivity test sees less than the matched distance: only the negative part of the whole density, which by part (iii) is at most of order $\varepsilon/\delta$ per displaced zero and which the kernels of the neighbouring zeros can cancel entirely (Section 5.2).

These thresholds concern zeros below height 200 and calibrate the method above the verified window.

## 3.4 Counting zeros with the Thorin measure

Turing's method certifies the count $N(T)$ without evaluating $\zeta$ off the critical line, from the sign changes of $Z$ and an explicit bound on the integral of $S$, where $\pi S(t) = \arg \zeta(\frac{1}{2} + it)$; Backlund's method, which it replaced, bounds the argument of $\zeta$ along a horizontal segment off the line (Section 2.2.1). Without the assumption of Proposition 3.2.2 that the zeros below the window lie on the line,

the argument principle gives at every resolution an exact identity for $N(T)$ and for the number $\nu(\frac{1}{2}+\varepsilon,T)$ of zeros to the right of the line of evaluation, so that off line zeros are counted and not only detected. At $\varepsilon=\frac{1}{2}$ the identity is Backlund's count with the vertical side on $\operatorname{Re} s=1$.

### 3.4.1 The cumulative Thorin mass as a smoothed counting function

Recall that $M_\varepsilon(\theta)=\mu_\varepsilon((0,\theta^2])=\pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$, with the argument continued along the line $\operatorname{Re} s=\frac{1}{2}+\varepsilon$ from $\theta=0$, where $\xi>0$ (Sections 3.1.3 and 3.2). Since $\xi(s)=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the mass splits as

$$M_\varepsilon(\theta)=N_\varepsilon^{\mathrm{sm}}(\theta)+S_\varepsilon(\theta),\qquad S_\varepsilon(\theta)=\frac{1}{\pi}\arg\zeta\Big(\frac{1}{2}+\varepsilon+i\theta\Big),$$

where, with $s=\frac{1}{2}+\varepsilon+i\theta$ and $\theta>0$, the smooth part $N_\varepsilon^{\mathrm{sm}}(\theta)=\pi^{-1}\big[\arg s+\arg(s-1)+\operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{\varepsilon}{2}+\frac{i\theta}{2})-\frac{\theta}{2}\log\pi\big]$ comes from $s(s-1)\pi^{-s/2}\Gamma(s/2)$, with the principal arguments of $s$ and $s-1$, which lie in $(0,\pi)$, and the branch of $\log\Gamma$ that is real on the positive axis. Thus $S_\varepsilon$ is $\pi^{-1}\arg\zeta$ continued along the vertical line from $S_\varepsilon(0+)=-1$ if $0<\varepsilon<\frac{1}{2}$, when $\arg(s-1)\to\pi$ as $\theta\downarrow 0$, and from $S_\varepsilon(0+)=0$ if $\varepsilon>\frac{1}{2}$.

As $\varepsilon\downarrow 0$, $N_\varepsilon^{\mathrm{sm}}\to\vartheta/\pi+1$, where $\vartheta(\theta)=\operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{i\theta}{2})-\frac{\theta}{2}\log\pi$ is the Riemann Siegel theta function, and for $\theta$ not an ordinate Proposition 3.4.2 below gives $M_\varepsilon(\theta)\to N(\theta)-2\nu(\frac{1}{2},\theta)$, where $\nu(\sigma,\theta)$ counts the zeros with real part greater than $\sigma$ and ordinate in $(0,\theta]$ (Section 1.1). When no zero below height $\theta$ lies to the right of the critical line, the limit is therefore the Riemann von Mangoldt formula $N(\theta)=\vartheta(\theta)/\pi+1+S(\theta)$ [498, Section 9.3].

Integrating Proposition 1.2.3 in $\theta$ writes $M_\varepsilon$ as a sum of Cauchy steps. Under RH, for $\theta\geq 0$,

$$M_\varepsilon(\theta)=\sum_{\gamma>0}\big[F_\varepsilon(\theta-\gamma)-F_\varepsilon(-\gamma)+F_\varepsilon(\theta+\gamma)-F_\varepsilon(\gamma)\big],$$

$$F_\varepsilon(x)=\frac{1}{2}+\frac{1}{\pi}\arctan\frac{x}{\varepsilon},$$

with the sum over all positive ordinates. Since $F_\varepsilon(-\gamma)+F_\varepsilon(\gamma)=1$, this says that $M_\varepsilon=P_\varepsilon*\tilde N$, where $\tilde N$ is the odd extension of $N$. As $\vartheta$ is odd, $S_\varepsilon$ is then the Poisson average of the odd extension of $S$ up to an explicit smooth function. If only the zeros below height $T+B$ are known to lie on the line, the display holds on $[0,T]$ up to an error of order $\varepsilon\delta^2/B^3$ for each off line pair above $T+B$ at distance $\delta$ from the line, counted as a double ordinate, because $P_{\varepsilon+\delta}(x)+P_{\varepsilon-\delta}(x)-2P_\varepsilon(x)=-6\varepsilon\delta^2/(\pi x^4)+O(x^{-6})$.

For a single step,

$$\int_{-L}^{L} \bigl|F_\varepsilon(x) - \mathbf{1}_{x\ge 0}\bigr|\,dx = \frac{2}{\pi}\Bigl[L\arctan\frac{\varepsilon}{L} + \frac{\varepsilon}{2}\log\Bigl(1+\frac{L^2}{\varepsilon^2}\Bigr)\Bigr]$$
$$= \frac{2\varepsilon}{\pi}\Bigl(\log\frac{L}{\varepsilon}+1\Bigr) + O\Bigl(\frac{\varepsilon^3}{L^2}\Bigr),$$

which explains the $\varepsilon\log(1/\varepsilon)$ rate of the $L^1$ distance in Table 3.2. Taking half of it on each side of each of the 79 zeros below height 200, with $L$ the half gap on that side, gives 1.61 at $\varepsilon = 0.005$, against the computed 1.55; the difference comes from the tails of the other steps.

### 3.4.2 A Littlewood identity and the zero density

Proposition 3.4.1. *Let $T > 0$. For every $\varepsilon > 0$ such that $\xi$ has no zero on the segment from $\frac12+\varepsilon$ to $\frac12+\varepsilon+iT$,*

$$\pi M_\varepsilon(T) = \frac{\partial}{\partial\varepsilon}\int_0^T \log\Bigl|\xi\Bigl(\frac12+\varepsilon+it\Bigr)\Bigr|\,dt,$$

*and for every $\varepsilon > 0$*

$$\int_0^T \log\Bigl|\xi\Bigl(\frac12+\varepsilon+it\Bigr)\Bigr|\,dt = \int_0^T \log\Bigl|\xi\Bigl(\frac12+it\Bigr)\Bigr|\,dt + \pi\int_0^\varepsilon M_{\varepsilon'}(T)\,d\varepsilon'.$$

*Proof.* Since $\partial_\sigma \log|\xi(\sigma+it)| = \mathrm{Re}(\xi'/\xi)(\sigma+it) = v_\sigma(t)$ and $v_\sigma(t) = \partial_t \arg\xi(\sigma+it)$, we have $M_\varepsilon(T) = \pi^{-1}\int_0^T v_{1/2+\varepsilon}(t)\,dt$, and when no zero lies on the segment, differentiation under the integral gives the first identity. The integral $I(\varepsilon)$ on the left of the second is continuous on $[0,\infty)$, because the logarithmic singularities at the zeros are integrable uniformly in $\varepsilon$, and continuously differentiable except at the finitely many $\varepsilon'$ at which the segment meets a zero. There $I'$ jumps by $2\pi$ for each zero met at an interior point of the segment, since $\int_{-c}^{c} a\,du/(a^2+u^2) \to \pm\pi$ as $a \to 0\pm$. So $I$ is absolutely continuous, and $I(\varepsilon) - I(0) = \int_0^\varepsilon I'(\varepsilon')\,d\varepsilon'$ is the second identity. □

The jump of $2\pi$ in the slope $\pi M_\varepsilon(T)$ when the line crosses a zero is the term $2\nu$ of Proposition 3.4.2 below. Littlewood's lemma [498, Section 9.9] is the same Cauchy Riemann identity with the argument taken along a path through $\mathrm{Re}\,s = 2$; the two arguments differ by $2\pi\nu(\frac12+\varepsilon, T)$, and Exercise 3.14 derives the lemma from the two propositions. Let $\tau_\varepsilon(T) = \pi^{-1}\Delta\arg\xi$ along the horizontal segment from $\frac12+\varepsilon+iT$ to $\frac12+iT$.

Proposition 3.4.2. *Let $\varepsilon > 0$ and $T > 0$, with $T$ not an ordinate and no zero of $\xi$ on the segment from $\frac12+\varepsilon$ to $\frac12+\varepsilon+iT$. Then*

$$M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\,\nu\Bigl(\frac12+\varepsilon, T\Bigr).$$

*The segment condition holds at $\varepsilon = \frac{1}{2}$, and $\nu(1,T) = 0$, which gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, so that $\nu(\frac{1}{2}+\varepsilon,T) = \frac{1}{2}\big[N(T) - M_\varepsilon(T) - \tau_\varepsilon(T)\big]$ for every $\varepsilon$ satisfying the hypotheses.*

*Proof.* Let $\sigma_1 > 1$. Write $\Delta_{\text{right}}$ for the change of $\arg\xi$ along the segment from $\sigma_1$ to $\sigma_1 + iT$, and $\Delta_{\text{top}}(a \to b)$ for its change along the segment from $a + iT$ to $b+iT$. The argument principle on the rectangle $[\frac{1}{2}+\varepsilon, \sigma_1]\times[0,T]$, on whose bottom edge $\xi$ is positive, gives $\Delta_{\text{right}} + \Delta_{\text{top}}(\sigma_1 \to \frac{1}{2}+\varepsilon) - \pi M_\varepsilon(T) = 2\pi\nu(\frac{1}{2}+\varepsilon,T)$. Subtracting this from the classical formula $\pi N(T) = \Delta_{\text{right}} + \Delta_{\text{top}}(\sigma_1 \to \frac{1}{2})$ [498, Section 9.3] leaves $\Delta_{\text{top}}(\frac{1}{2}+\varepsilon \to \frac{1}{2}) + \pi M_\varepsilon(T) = \pi N(T) - 2\pi\nu(\frac{1}{2}+\varepsilon,T)$, and $\Delta_{\text{top}}(\frac{1}{2}+\varepsilon \to \frac{1}{2}) = \pi\tau_\varepsilon(T)$. At $\varepsilon = \frac{1}{2}$ the segment lies on $\operatorname{Re} s = 1$, where $\zeta$ has no zeros, and $\nu(1,T) = 0$ because the Euler product has no zeros in $\operatorname{Re} s > 1$. □

Half the difference of $M_\varepsilon(T) + \tau_\varepsilon(T)$ at two resolutions counts the zeros in the strip between the two lines. The factor two comes from the functional equation: a zero $\frac{1}{2} + \delta + i\gamma$ with $\delta > \varepsilon$ and its partner $\frac{1}{2} - \delta + i\gamma$ add 2 to $N(T)$ and 1 to $\nu(\frac{1}{2}+\varepsilon,T)$, hence nothing to $M_\varepsilon(T) + \tau_\varepsilon(T)$, whereas two zeros on the line at the same ordinate add 2. In the density the pair enters through the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of Proposition 1.2.3, whose masses cancel, as in the proof of Proposition 3.3.1(iii) (Exercise 3.12).

Zero density estimates such as Ingham's [284] bound the number $N(\sigma,T)$ of zeros with real part at least $\sigma$ and ordinate in $(0,T]$, which is at least $\nu(\sigma,T)$, by $T^{e(\sigma)+o(1)}$, with an exponent $e(\sigma) < 1$ for every $\sigma > \frac{1}{2}$, and Section 11.4 turns these bounds, with no hypothesis, into a bound on the mass of the Cauchy kernels of the zeros to the right of the basepoint $\sigma$, the negative part of the split (11.4.1) (Theorem 11.4.2). Proposition 3.4.2 evaluates $\nu$ itself, as an integer, but only up to a finite height.

### 3.4.3 The count and the zero density below height 200

The right side of the identity is an integer, so a computation of the left side with error below one half determines it, provided no winding of the argument is missed. We follow the argument on a grid, summing the principal values of the arguments of quotients of successive values of $\xi$, with steps $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) up the vertical side from $\theta = 0$ and 4000 steps along the horizontal segment. A certificate would need, besides ball arithmetic at the grid points, a bound on the change of the argument between them. We do not supply that bound, and the conclusions below are computed, not certified.

Table 3.3 gives the identity at $T = 200$, evaluated by the Pólya route. Every row returns 79 to within $6 \times 10^{-13}$. The row $\varepsilon = \frac{1}{2}$ gives $N(200) = 79$, and with the 79 sign changes of $\Xi$ on $(0,200)$ (Section 2.1.2) this places every zero below height 200 on the line. Independently of the sign changes, the rows $\varepsilon < \frac{1}{2}$ give

$\nu(\frac{1}{2}+\varepsilon, 200) = 0$ for every $\varepsilon \geq 0.005$, since $\nu$ is nonincreasing in $\sigma$, so that by the functional equation no zero below height 200 lies farther than 0.005 from the line.

| $\varepsilon$ | $M_\varepsilon(200)$ | $\tau_\varepsilon(200)$ | $M_\varepsilon + \tau_\varepsilon = N - 2\nu$ |
|---|---|---|---|
| 1 | 79.36737810665 | −0.36737810665 | 79.00000000000003 |
| 0.5 | 79.19416875933 | −0.19416875933 | 78.99999999999993 |
| 0.25 | 79.09886991193 | −0.09886991193 | 79.00000000000024 |
| 0.1 | 79.03976768776 | −0.03976768776 | 79.00000000000006 |
| 0.05 | 79.01989993142 | −0.01989993142 | 78.99999999999967 |
| 0.02 | 79.00796178162 | −0.00796178162 | 78.99999999999970 |
| 0.01 | 79.00398102016 | −0.00398102016 | 78.99999999999993 |
| 0.005 | 79.00199052625 | −0.00199052625 | 78.99999999999947 |

Table 3.3: Proposition 3.4.2 at $T = 200$ by the Pólya route (computed, not certified).

The identity also gives the mass defect of Proposition 3.2.2 exactly. If $T$ is not an ordinate and every zero with ordinate in $(0, T]$ lies on the line, then $\nu(\frac{1}{2}+\varepsilon, T) = 0$ and $M_\varepsilon(T) = N(T) - \tau_\varepsilon(T)$ for every $\varepsilon > 0$. Comparing with that proposition and Remark 3.2.3 gives $\tau_\varepsilon(T) = (\varepsilon/\pi)\,\Xi'(T)/\Xi(T) + O(\varepsilon^3)$. At $T = 200$ the first order term is $-0.3981063\,\varepsilon$, minus the mass defect found in Section 3.2.2, and for $\varepsilon \leq 0.1$ the column $\tau_\varepsilon(200)$ of Table 3.3 exceeds it by about $0.043\,\varepsilon^3$.

## 3.5 Further readings of the sine squared lemma

This section follows the sine squared lemma (1.2.9) of Section 1.2 in two directions.

### 3.5.1 The prime side near the critical line

For $\alpha > 1$ the integrated lemma is the argument of $\zeta$. In the chart $\theta = \sqrt{z}$ a unit of $xL(dx)$ at $x$ contributes $\pi^{-1}(1-\cos\theta x)$ to the density, and $\pi^{-1}\int_0^\Theta (1-\cos\theta x)\,d\theta = \pi^{-1}(\Theta - \sin(\Theta x)/x)$, so the prime atoms $\Lambda(n)n^{-\alpha}$ at $x = \log n$ return a term linear in $\Theta$ plus $\pi^{-1}$ times

$$-\sum_{n\geq 2} \frac{\Lambda(n)}{n^\alpha}\,\frac{\sin(\Theta \log n)}{\log n} = \arg\zeta(\alpha + i\Theta),$$

on the branch with $\arg\zeta(\alpha) = 0$, since $\log\zeta(s) = \sum_n \Lambda(n)n^{-s}/\log n$ for $\operatorname{Re} s > 1$. Divided by $\pi$, this is $S_{\alpha-1/2}(\Theta)$ in the notation of Section 3.4.1. The series converges absolutely for $\alpha > 1$ and diverges for $\alpha < 1$, so the function $S$ of Turing's method (Section 2.2.1), at $\alpha = \frac{1}{2}$, is reached only through a smoothing.

Selberg [477] truncates the Dirichlet series of $\zeta'/\zeta$ smoothly and accounts for the truncation exactly by a sum over the zeros [498, Section 14.20]. With $\Lambda_x(n) =$

$\Lambda(n)$ for $n \le x$ and $\Lambda_x(n) = \Lambda(n)\log(x^2/n)/\log x$ for $x < n \le x^2$, and for $x > 1$ and $s$ different from 1, from every zero $\rho$ and from every $-2q$,

$$\begin{aligned}\frac{\zeta'}{\zeta}(s) = &-\sum_{n\le x^2}\frac{\Lambda_x(n)}{n^s} + \frac{x^{2(1-s)}-x^{1-s}}{(1-s)^2\log x}\\ &+\frac{1}{\log x}\sum_{q\ge1}\frac{x^{-2q-s}-x^{-2(2q+s)}}{(2q+s)^2} + \frac{1}{\log x}\sum_{\rho}\frac{x^{\rho-s}-x^{2(\rho-s)}}{(s-\rho)^2}.\end{aligned} \tag{3.5.1}$$

At $s = \alpha + i\theta$ with $\alpha = \frac12 + \varepsilon$, a zero $\rho = \beta + i\gamma$ with $\beta < \alpha$ contributes to the last sum a term of modulus about $x^{\beta-\alpha}/(|s-\rho|^2\log x)$. For a line zero at its own ordinate the term is $(x^{-\varepsilon} - x^{-2\varepsilon})/(\varepsilon^2\log x)$, a fraction $x^{-\varepsilon}(1-x^{-\varepsilon})/(\varepsilon\log x)$ of the contribution $1/\varepsilon$ of that zero to $v_\alpha$; we call this the single zero law. The same computation bounds the whole sum. If no zero has real part greater than $\frac12$, then $|x^{\rho-s}| = x^{-\varepsilon}$ and $|x^{2(\rho-s)}| = x^{-2\varepsilon}$ for every zero, and every zero contributes $\mathrm{Re}(1/(s-\rho)) = \varepsilon/|s-\rho|^2$ to $v_\alpha(\theta)$, so that $\sum_\rho |s-\rho|^{-2} = v_\alpha(\theta)/\varepsilon$ by Proposition 1.2.3. Hence the bracket $v_\alpha$ computed from Selberg's smoothed prime sum has error at most

$$\frac{x^{-\varepsilon}+x^{-2\varepsilon}}{\varepsilon\log x}\,v_\alpha(\theta),$$

which is of order $x^{-\varepsilon}/(\varepsilon^2\log x)$ at a peak, where $v_\alpha$ is about $1/\varepsilon$, and $k$ correct digits at an ordinate need $x^\varepsilon\,\varepsilon\log x \approx 10^k$, that is

$$x \approx 10^{k/\varepsilon}$$

to leading order, with primes up to $x^2$. A zero with $\frac12 < \beta < \alpha$ gives a term that decays like $x^{\beta-\alpha}$, more slowly than $x^{-\varepsilon}$, and one with $\beta > \alpha$ a term that grows like $x^{2(\beta-\alpha)}$, since then $x^{2(\rho-s)}$ dominates. With $B = \sup_\rho \mathrm{Re}\,\rho$ the error is $O(x^{B-\alpha}/\log x)$ when $B < \alpha$, and when $B > \alpha$ it reaches $x^{2(B-\alpha)+o(1)}$ along a sequence of cutoffs, so that the prime side diverges as $x \to \infty$ if a zero lies to the right of the line $\mathrm{Re}\,s = \alpha$. Selberg's abscissa $\sigma_{x,t} = \frac12 + 2\max(\beta - \frac12, 2/\log x)$, the maximum over the zeros near height $t$, takes this into account; when none of these zeros lies to the right of $\frac12 + 2/\log x$ it equals $\frac12 + 4/\log x$, where $x^{-\varepsilon} = e^{-4}$ [477]. Section 9.1 meets the same dichotomy without smoothing.

The terms of the zeros to the right of the line $\mathrm{Re}\,s = \alpha$ cannot cancel one another. Suppose that $B > \alpha$, write $E_x(s)$ for the last term of (3.5.1), the sum over the zeros divided by $\log x$, and put $y = \log x$, $\lambda_\rho = \rho - s$ and $R(y) = y\,\mathrm{Re}\,E_{e^y}(s)$. Bounding each term by its modulus gives $|R(y)| \le \sum_\rho |s-\rho|^{-2}\big(e^{(B-\alpha)y} + e^{2(B-\alpha)y}\big)$, so for $\mathrm{Re}\,w > 2(B-\alpha)$ the Laplace transform $\int_0^\infty e^{-wy}R(y)\,dy$ may be taken term by term, and it is

$$\frac12\sum_\rho\Big[\frac{1}{\lambda_\rho^2}\Big(\frac{1}{w-\lambda_\rho} - \frac{1}{w-2\lambda_\rho}\Big) + \frac{1}{\bar\lambda_\rho^2}\Big(\frac{1}{w-\bar\lambda_\rho} - \frac{1}{w-2\bar\lambda_\rho}\Big)\Big].$$

The series converges locally uniformly away from its poles, since $\sum_\rho |\lambda_\rho|^{-2} < \infty$ and $|\lambda_\rho| \to \infty$, and so defines a meromorphic function in the whole plane. A zero $\rho_0$ with $\operatorname{Re}\rho_0 > (B+\alpha)/2$ gives a pole at $2\lambda_{\rho_0}$, to the right of every pole $\lambda_\rho$ and $\bar\lambda_\rho$; each fraction with a pole there contributes $-\frac12\lambda_{\rho_0}^{-2}$ to the residue, which is therefore not zero. If $R(y) = O(e^{\eta y})$, the Laplace integral converges on $\operatorname{Re} w > \eta$ and continues the series holomorphically there, so $\eta \geq 2(\operatorname{Re}\rho_0 - \alpha)$. Letting $\operatorname{Re}\rho_0$ tend to $B$, and using the bound on $|R|$, gives $\limsup_{x\to\infty} \log|\operatorname{Re} E_x(s)|/\log x = 2(B-\alpha)$.

The cutoff interpolates between the zeros and the primes. For $x < \sqrt{2}$ the prime sum is empty, and as $x \downarrow 1$ each remaining term tends to its partial fraction, $1/(1-s)$, $1/(s+2q)$ or $1/(s-\rho)$. The limits cannot be taken inside the sums separately: the sum over $q$ grows like $\frac12\log(1/\log x)$ and the real part of the sum over the zeros falls like $-\frac12\log(1/\log x)$, and only their total, $\zeta'/\zeta(s) - (x^{2(1-s)} - x^{1-s})/((1-s)^2\log x)$, has a limit, $\zeta'/\zeta(s) + 1/(s-1)$. Adding the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ turns this limit into $\xi'/\xi(s)$, whose real part divided by $\pi$ is the density $\rho_\varepsilon(\theta)$, the sum of the kernels of Proposition 1.2.3.

Figure 3.3 tests the formula at $\varepsilon = 0.25$, that is $\alpha = \frac34$, on 501 points of $10 \leq \theta \leq 60$, against the Pólya values of $v_{3/4}$. With the primes alone the maximal relative error is 66%, 20% and 7% for $x = 30$, 300 and 3000, attained where $v_{3/4}$ is small, and the maximal absolute errors 1.14, 0.53 and 0.23, attained at peaks over zeros, follow the single zero law, which predicts 1.15, 0.51 and 0.23; the relative errors stay below the bound $(x^{-\varepsilon} + x^{-2\varepsilon})/(\varepsilon\log x)$, which is 0.72, 0.21 and 0.077. Both errors decrease faster than $x^{-\varepsilon}$. Adding the sum over the first 1700 zeros reduces the error below $1.6\times10^{-7}$, the residual being the zeros above height 2197. At $\varepsilon = 0.005$ ten digits would need $x \approx 10^{2000}$, about $e^{4600}$, to leading order (about $e^{4000}$ by the single zero law), while the cost of one Pólya evaluation does not depend on $\varepsilon$ (Section 3.1.2). So the sine squared lemma identifies the Thorin density, but near the line the primes compute it only at a cost exponential in $1/\varepsilon$.

### 3.5.2 Positivity: Brownian subordination and Weil's criterion

The lemma cannot produce a Brownian first passage time to a random level from a positive input. Brownian motion with variance two per unit time first reaches a level $x > 0$ at a time with Laplace transform $e^{-x\sqrt{u}}$, so its first passage time $T_Y$ to an independent level $Y \geq 0$ has $\mathbb{E}e^{-uT_Y} = \mathbb{E}e^{-Y\sqrt{u}}$. The exponent $\log f(\alpha+s)/f(\alpha) = cs + \int(e^{-sx} + sx - 1)L(dx)$ to which the lemma applies (Section 1.2) would have to equal $-\log\mathbb{E}e^{-sY}$, which is concave in $s$ by Hölder's inequality, whereas $L \geq 0$, $L \neq 0$ makes it strictly convex, with second derivative $\int x^2 e^{-sx}L(dx) > 0$. With $c \geq 0$ the lemma gives a GGC with Thorin density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\,xL(dx)]$, but not a first passage time. Conversely, an infinitely divisible level with $\mathbb{E}e^{-sY} = \exp(-bs - \int(1-e^{-sx})\Pi(dx))$ enters the lemma with $c = b + \int x\,\Pi(dx)$ and the negative input $L = -\Pi$, and $T_Y$ receives the signed density $(2\pi\sqrt{z})^{-1}[b + \int x\cos(x\sqrt{z})\,\Pi(dx)]$. For a Poisson level ($\Pi = \delta_1$,

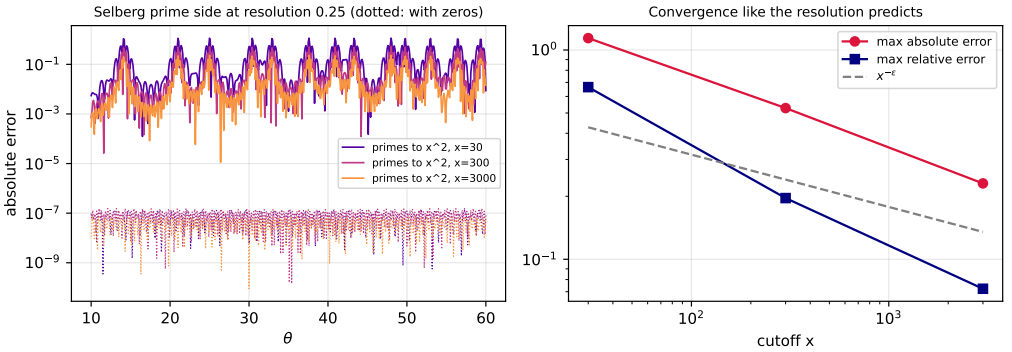



Figure 3.3: Left: absolute error of Selberg's smoothed prime side of $v_{3/4}$, at resolution 0.25, for three cutoffs, and (dotted) with the sum over the first 1700 zeros added. Right: the maximal absolute and relative errors against the cutoff, with $x^{-\varepsilon}$ for comparison.

$b = 0$) this is $\cos(\sqrt{z})/(2\pi\sqrt{z})$, and $T_Y$ is not a GGC: it has an atom $e^{-1}$ at 0. A positive input and a random level coexist only for pure drift, with transform $e^{-c\sqrt{u}}$ and Thorin density $c/(2\pi\sqrt{z})$.

The reciprocal direction does work: for $Y \geq 0$ infinitely divisible with finite mean $m$, $T_m \stackrel{d}{=} T_Y + R$, with $T_m$ the first passage time to the level $m$ and $R$ a GGC independent of $T_Y$, which for the clock is Proposition 9.1.7 and for a Barnes beta law is Proposition 9.1.9.

For $\xi$ there is no random level at any basepoint. By Stirling's formula and the polynomial growth of $\zeta$ in vertical strips, $|\xi(\alpha)/\xi(\alpha + it)|$ is at least $e^{\pi|t|/4}$ divided by a power of $|t|$, while $|\mathbb{E}e^{-itY}| \leq 1$; so $s \mapsto \xi(\alpha)/\xi(\alpha + s)$ is never a Laplace transform in $s$, although for $\alpha \geq 1$ it is one in $u = s^2$ (Theorem 1.2.2). For $\alpha > 1$ the input of $\xi$ is signed, and the positivity of $U_\alpha$ is created by the zeros through the Poisson kernels of Proposition 1.2.3, not inherited from a positive input (Section 3.1.3); compare [424, Remark 3.8]. At $\alpha = \frac{1}{2}$ the obstruction is the drift zero case of Proposition 8.2.1, whose Brownian motion has variance one, and the curved boundaries of Section 8.2 take the place of the straight boundary with a random intercept. The obstruction concerns first passage alone (Section 9.2).

The same kernel puts Weil's criterion in variogram form. Section 5.3 defines the Thorin functional $U$ on the class $\mathcal{T}$ of functions $\varphi(z) = h(\sqrt{z})$ with $h$ even, holomorphic on a strip $|\operatorname{Im} r| \leq \frac{1}{2} + c$ for some $c > 0$, and $O((1 + |r|)^{-2-c})$ there. With $g(x) = \frac{1}{2\pi} \int h(r) e^{-irx}\, dr$,

$$\int_0^\infty \varphi(z)\, \frac{2\sin^2(x\sqrt{z}/2)}{2\pi\sqrt{z}}\, dz = g(0) - g(x), \tag{3.5.2}$$

both sides being $\pi^{-1} \int_0^\infty h(r)(1 - \cos xr)\, dr$. If $h \geq 0$ on the real line, $g$ is positive

definite and $g(0) - g(x) \geq 0$ is a variogram. Weil's criterion [512] states that RH holds if and only if $\langle U, \varphi \rangle \geq 0$ for every such $\varphi$, and by the explicit formula $U$ evaluates $\varphi$ at every zero. Formally, the lemma writes $\langle U, \varphi \rangle$ as the pairing of variograms with the canonical measure of (3.1.1) taken at the centre, the measure $\nu$ of Section 5.3, whose archimedean density and prime atoms are nonnegative and whose only negative input is the pole density $-e^{x/2}dx$. So RH says that on every variogram of a positive definite function the pole mass is dominated by the prime and archimedean mass. Both masses are infinite, and the absolutely convergent form of Section 5.3 makes the statement exact, the divergent constants cancelling through $\xi'(\frac{1}{2}) = 0$ (Section 3.1.3).

The linear programming formulation of [424, Sections 18.2.8 and 18.2.9] works on this form, with the variograms as feasible directions. In a space of test functions of finite dimension the regularised domination is a feasibility problem with dual certificates, which Section 12.2.4 proposes to solve at increasing dimension. Weil's criterion concerns $U$, which sees every zero, whereas the limit $\alpha \downarrow \frac{1}{2}$ of the Thorin measures is $U_{\mathrm{line}}$, which pairs nonnegatively with every such $\varphi$ whatever the zeros do (Section 5.3).

## 3.6 Exercises

*On Section 3.2.*

3.1. Prove Proposition 3.2.1 for a positive measure $\nu$ on a general measurable space that does not charge the atoms of $U$.

3.2. Show that the total variation distance between the unit point masses at $a$ and $b$ is 2 for $a \neq b$, and deduce that total variation cannot compare two atomic measures with slightly displaced atoms.

3.3. Compute the $L^1$ distance between the cumulative functions of a Cauchy kernel of width $\varepsilon$ and a unit step on $[-L, L]$, derive the $\varepsilon \log(1/\varepsilon)$ rate, and compare with the formula for a single step in Section 3.4.1.

3.4. Derive the first order formula of Proposition 3.2.2 for the sinh clock of Section 1.3, where it reads $-(\varepsilon/\pi) f'/f$ with $f(\theta) = \sin(\pi\theta)/(\pi\theta)$, and compare it at $\Theta = m + \frac{1}{2}$ with the exact mass $m + \pi^{-1} \arctan(\varepsilon/\Theta)$ of Section 1.3.4; explain why the difference is of order $\varepsilon^3$.

3.5. Explain why the two point Richardson combination removes the linear term in $\varepsilon$ and the three point combination the quadratic term, and reproduce the heat trace errors $2.6 \times 10^{-7}$ and $1.7 \times 10^{-10}$ from the values of $W_\varepsilon(0.01)$ given in Section 3.2.2.

*On Section 3.3.*

3.6. Verify the expansion $\frac{1}{2}(P_{\varepsilon+\delta}+P_{\varepsilon-\delta})-P_\varepsilon=\frac{\delta^2}{2}\partial_a^2 P_a|_{a=\varepsilon}+O(\delta^4)$, show that the remainder has $L^1$ norm $O(\delta^4/\varepsilon^4)$, and compute the $L^1$ norm of $\partial_a^2 P_a$ at $a=\varepsilon$.

3.7. Show that for $\delta>\varepsilon$ the kernel $P_{\varepsilon+\delta}-P_{\delta-\varepsilon}$ is negative exactly on $|x|<\sqrt{\delta^2-\varepsilon^2}$.

3.8. Compute the depth $2\varepsilon/(\pi(\delta^2-\varepsilon^2))$ of the negative dip of an off line pair at its centre, that is $\varepsilon/(\pi(\delta^2-\varepsilon^2))$ per displaced zero, compare it with the contribution $\varepsilon/(\pi(\varepsilon^2+\delta_j^2))\approx\varepsilon/(\pi\delta_j^2)$ of a line zero at distance $\delta_j$, and derive the local visibility condition of Section 5.2.

3.9. Given the matched floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, derive the smallest displacement $\delta$ of a pair that keeps the count and would be detected, and evaluate it at $\varepsilon=0.05$ using Table 3.2.

*On Section 3.4.*

3.10. Show that $I(\varepsilon)=\int_0^T\log|\xi(\frac{1}{2}+\varepsilon+it)|\,dt$ is continuous for $\varepsilon\geq 0$, that its derivative jumps by $2\pi$ when the line $\operatorname{Re}s=\frac{1}{2}+\varepsilon$ crosses a zero, and deduce the second identity of Proposition 3.4.1.

3.11. Prove Proposition 3.4.2 in detail, paying attention to the orientation of each side of the rectangle.

3.12. Show that the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of an off line pair at distance $\delta>\varepsilon$ from the line have total mass zero, and compute to leading order the mass that they and their mirror images at $-\gamma$ put on a window $(0,T]$ containing the ordinate $\gamma$. Explain how the factor two in Proposition 3.4.2 accounts for the pair.

3.13. Using the first order formula for $\tau_\varepsilon(T)$ in Section 3.4.3, predict $\tau_{0.05}(200)$ from $\Xi'(200)/\Xi(200)=-1.2506879$ and compare with Table 3.3; estimate the coefficient of the $\varepsilon^3$ term from the rows $\varepsilon=0.01$ and $0.02$.

3.14. Show that Littlewood's lemma follows from Proposition 3.4.1 and Proposition 3.4.2 by integrating in $\varepsilon$.

*On Section 3.5.*

3.15. Show that $\frac{1}{\pi}\int_0^\Theta(1-\cos\theta x)\,d\theta=\frac{1}{\pi}(\Theta-\sin(\Theta x)/x)$ and, for $\alpha>1$, derive the Dirichlet series for $\arg\zeta(\alpha+i\Theta)$ from the integrated sine squared lemma.

3.16. Implement Selberg's formula at $s = \frac{3}{4} + i\theta$ with $x = 100$ and check that adding the zero sum over the first hundred zeros reduces the error to the size of the omitted zeros.

3.17. Estimate, from the single zero law, the cutoff $x$ needed for six correct digits of $v_{0.55}(\theta)$ at an ordinate $\theta = \gamma$ from the primes, and compare with the cost of the Pólya route at the same point.

3.18. Show that $g(0) - g(x) \geq 0$ for every real positive definite function $g$, and that $g(0) - g(x)$ is conditionally negative definite.

# Chapter 4

# Weak limits, hyperbolic complete monotonicity and a Bosch type theorem

Sections 4.2 and 4.3 bring in Bosch's theorem on powers of the half Cauchy law. The first explains why his result cannot be transferred as an input, and the second transfers his method instead and proves a threshold theorem for powers of the reciprocal Xi function, in which the angle of the zeros replaces Bosch's drift parameter and RH is the extreme case.

## 4.1 The limit of vanishing resolution

Neither closure under weak limits nor Bondesson's theorem reaches RH.

### 4.1.1 Weak limits of Thorin measures

Theorem 4.1.1 (Thorin [493, 494]; Bondesson [56, Theorem 3.1.5]; see also [424, Theorem 18.9]). *The class of GGC laws is closed under convergence in law.*

Corollary 4.1.2. *RH holds if and only if there are GGC variables $X_N$ with $\mathbb{E}e^{-uX_N} \to 1/G(u)$ for all $u$ in a set with an accumulation point in $(0,\infty)$ [424, Propositions 18.10 and 18.14].*

*Proof.* The transforms are analytic and bounded by one on $\operatorname{Re} u > 0$, so by Vitali's theorem they converge locally uniformly there, and the limit is $1/G$ on $(0,\infty)$. Since $1/G(u) \to 1$ as $u \downarrow 0$, the $X_N$ converge in law to a proper limit with transform $1/G$, which is a GGC by the closure theorem, and Proposition 1.2.1 gives RH. Conversely, under RH the clock variable $T$ is a GGC with transform $1/G$; take $X_N = T$. □

The Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 1.2 is the candidate at hand. Its measures $\mu_\varepsilon$ are the Thorin measures of $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ at $\alpha = \frac{1}{2}+\varepsilon$ whenever these are GGC transforms (Theorem 1.2.2), the transforms converge to $1/G$ pointwise as $\varepsilon \downarrow 0$, and the measures converge vaguely. The flow fails for two reasons.

First, the limit is the wrong measure. As shown after Proposition 1.2.3, $\mu_\varepsilon \to U_{\mathrm{line}}$ vaguely whatever the truth of RH [421, Proposition 9.5], because the contribution of an off line pair tends to zero in total variation (Proposition 3.3.1(ii)); the limit omits the off line zeros. It is the Thorin measure of the clock only under RH, since the Thorin functional $U$ of Section 5.3 counts every zero and $U = U_{\mathrm{line}}$ exactly when RH holds (Section 5.3.1 and [424, Proposition 7.6]). Positivity of the vague limit therefore carries no information.

Second, the closure theorem needs GGC laws along the sequence. The flow supplies a law at every resolution with no hypothesis, that of the shifted clock $T_\alpha$ of Theorem 9.2.5 at $\alpha = \frac{1}{2} + \varepsilon$, but at small $\varepsilon$ it supplies a GGC law only under the hypothesis to be proved: the transform at basepoint $\frac{1}{2} + \varepsilon$ is a GGC transform if and only if no zero lies to the right of the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ (Theorem 1.2.2). Positivity of $\rho_\varepsilon$ at that one $\varepsilon$ is necessary for this but is not known to imply it, since near a cluster of line zeros the negative kernel of an off line pair can be outweighed by the kernels of its neighbours (Exercise 1.7 and Section 5.2.2). Positivity of $\rho_{\varepsilon'}$ for every $\varepsilon' > \varepsilon$ does imply it, as shown in Remark 1.2.4; for $\varepsilon = 0$ this is Lagarias's criterion [320], which is RH. Weak closure adds the endpoint $\varepsilon = 0$ to a family already known to consist of GGCs; it cannot show that the family consists of GGCs [424, Remark 3.10].

On the window $0 \le \theta \le 200$ the Pólya route computes $\min \rho_\varepsilon = 0.0147\,\varepsilon > 0$ at each of the eight resolutions of Table 3.2, down to $\varepsilon = 0.005$ (Section 3.1.1); with the verification of RH to height $3 \times 10^{12}$ this is a theorem, since for every $\varepsilon \le 6.85$ the minimum of $\rho_\varepsilon$ over $[0, 3 \times 10^{12} - 1.21]$ is $\rho_\varepsilon(0)$ (Theorem 9.1.18 and Remark 9.1.19), and Section 12.2.2 lists what would make the Pólya computation itself a certificate.

Weak convergence becomes useful only with approximants whose positivity is proved independently of $\xi$, such as Beurling generalised prime systems whose zeta functions provably have only real ordinates, finite Krein strings, and the partial products over verified zeros completed by the Weyl tail; the last two are the Kent approximants of Section 8.1.3, without and with the Weyl tail. For such approximants the distances of Section 3.2 are the right measure of the approximation. Verification also removes factors from the question: if the first $N$ zeros are verified to lie on the line, then $1/G_N$, with $G_N(u) = \prod_{k \le N}(1 + u/\gamma_k^2)$, is a GGC transform with no hypothesis, and RH holds if and only if the tail factor $G_N/G$ is a GGC transform as well [424, Corollary 18.11].

### 4.1.2 The HCM form of the Thorin condition

Bondesson's theorem (Section 1.3.2; [56, Theorem 6.1.1], see also [289]) and the results of Section 1.2 give the following.

Corollary 4.1.3. *(i) For every $\alpha \ge 1$ the function $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is HCM, unconditionally. (ii) The reciprocal Xi function $\varphi_{1/2}(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$,*

*which is the clock $1/G$, is HCM if and only if RH holds.*

*Proof.* (i) is Theorem 1.2.2 with Bondesson's theorem, and (ii) is Proposition 1.2.1 with Bondesson's theorem. □

By the Hadamard product (1.2.1), with $a_k = -(\rho_k - \frac{1}{2})^2$ and $w = v + 1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v) = \prod_k \frac{a_k^2}{a_k^2 + a_k\,u\,w + u^2} = \prod_k \frac{a_k/u}{w + c_k(u)}, \qquad c_k(u) = \frac{a_k}{u} + \frac{u}{a_k}. \tag{4.1.1}$$

The numbers $c_k(u)$ are local to this subsection, and are not the derivatives $c_k(t)$ of the heat trace in Section 5.1. An off line pair, with values $a$ and $\bar{a}$ of $a_k$, contributes $(|a|^2/u^2)/((w+c)(w+\bar{c}))$ with $c = c(u)$ complex, and $1/((w+c)(w+\bar{c}))$ is the Laplace transform of the signed function $e^{-t\,\mathrm{Re}\,c}\sin(t\,\mathrm{Im}\,c)/\,\mathrm{Im}\,c$. At a single $u$ the remaining factors may smooth this to a positive function; failure is guaranteed only across all $u$, by Bondesson's theorem. At $u = |a|$ both values of $c$ equal $2\cos\arg a$, and the pair contributes $(w + 2\cos\arg a)^{-2}$, which is completely monotone on $[2,\infty)$. Complete monotonicity in $w$ at finitely many $u$ is therefore necessary for HCM but not sufficient.

We test this necessary condition. From 60 digit values of $\xi$ at real arguments, computed by (2.1.2) without cancellation, numerical differentiation gives the normalised derivatives $d_k = (-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\,\varphi_\alpha(u/v)$ for $k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{2.05, 3, 5, 10, 20, 50\}$, at $\alpha = \frac{1}{2}$ and at the control $\alpha = 1$. All 216 values are positive (Figure 4.1 shows $w = 3$ and $w = 20$), and the signs are certified: from the Taylor coefficients of $H_u$ in $w$, computed in ball arithmetic [290] from $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, and so independently of the Pólya route, every $d_k$ is enclosed in a ball of relative radius below $10^{-70}$ that lies in $(0,\infty)$. Under RH, (4.1.1) gives $H_u(w+t)/H_u(w) = \mathbb{E}e^{-tY}$ with $Y = \sum_k e_k/(w + c_k(u))$ and independent standard exponentials $e_k$, so $d_k = \mathbb{E}Y^k$ is of order $d_1^k$. The smallest value, $2.5 \times 10^{-10}$ at $\alpha = \frac{1}{2}$, $u = 1$, $w = 50$, $k = 6$, is $2.54\,d_1^6$ with $d_1 = 0.0215$: it reflects the scale of $Y$, to which many zeros contribute, not closeness to a sign change. The test is consistent with (ii) and cannot decide it (Section 5.2, Table 5.5, and [424, Chapter 11]).

Under RH the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 1.2.1), and if moreover the zeros are simple and satisfy two further hypotheses, its density is not HCM (Theorem 4.3.22); for the Cauchy analogue of the clock variable this holds unconditionally (Theorem 4.3.20).

## 4.2 Bosch's half Cauchy theorem and HCM of the reciprocal Xi function

By Corollary 4.1.3(ii) of Section 4.1 the clock $\varphi = 1/G$ is HCM if and only if RH holds; this is the HCM form of the reciprocal Xi programme of [421]. In

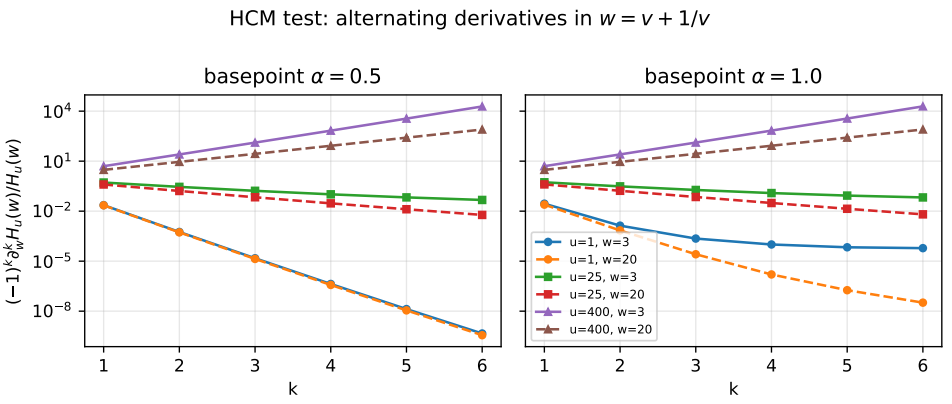



Figure 4.1: Normalised alternating derivatives $(-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\varphi_\alpha(u/v)$ for $1 \le k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{3, 20\}$, computed from the Pólya density, at the centre $\alpha = \frac{1}{2}$ and at the unconditional basepoint $\alpha = 1$.

this section and in Section 4.3 the clock is a function of $s$, the variable $u$ of Section 1.2: $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is the function $\varphi_{1/2}$ of Corollary 4.1.3. A sharp HCM theorem for a nearby, exactly solvable function might transfer to $\varphi$, and the candidate is Bosch's theorem on powers of the half Cauchy law [58].

### 4.2.1 HCM and Bosch's theorem at the half Cauchy point

Bosch's theorem concerns densities, so we recall HCM from Section 1.3.2 in the form that applies to them: a continuous function $f : (0, \infty) \to [0, \infty)$ is HCM if for every $u > 0$ the map $v \mapsto f(uv)f(u/v)$, the pair function of $f$ at $u$, is a completely monotone function of $w = v + v^{-1}$ on $[2, \infty)$, and a positive random variable is HCM if it has an HCM density. In logarithmic coordinates $u = e^a$, $v = e^y$ the condition reads: $y \mapsto f(e^{a+y})f(e^{a-y})$ is completely monotone in $w = 2\cosh y$. We use the following facts, all due to Bondesson; (F1), (F2), (F5) and (F6) are also stated in Section 1.3.2.

(F1) If $X_1, X_2$ are independent and HCM then $X_1X_2$ and $X_1/X_2$ are HCM [56, Theorem 5.1.1].

(F2) If $X$ is HCM then $X^q$ is HCM for $|q| \ge 1$ [56, pp. 68 and 69]. The mechanism is that for $|p| \le 1$ the function $v^p + v^{-p}$ is a Bernstein function of $v + v^{-1}$, so if $h$ is an HCM function then so is $x \mapsto h(x^p)$ for $|p| \le 1$; if $X$ has density $f$, the density of $X^q$ is $|q|^{-1}x^{1/q-1}f(x^{1/q})$, and the power factor is HCM by (F3).

(F3) Pointwise products of HCM functions are HCM, and powers $x^r$, $r \in \mathbb{R}$, are HCM [56, p. 68].

(F4) Pointwise limits of HCM functions, when finite and positive, are HCM [56, p. 68 and Theorem 5.1.3].

(F5) Every HCM law is a GGC [56, Theorem 5.1.2].

(F6) A probability law on $[0,\infty)$ is GGC if and only if its Laplace transform is an HCM function [56, Theorem 6.1.1].

(F7) A function $f$ that is not identically zero is HCM if and only if

$$\begin{aligned} f(x) &= Kx^{r-1}h_1(x)h_2(1/x), \\ h_j(x) &= \exp\Big(-b_j x + \int_{(0,\infty)} \log\frac{1+t}{x+t}\,\Gamma_j(dt)\Big), \end{aligned} \tag{4.2.1}$$

with $K>0$, $r\in\mathbb{R}$, $b_j\ge 0$ and nonnegative measures $\Gamma_j$ satisfying $\int(1+t)^{-1}\Gamma_j(dt)<\infty$ [56, (5.2.3) and Theorem 5.3.1]. In particular every HCM function that is not identically zero extends to an analytic and zero free function on $\mathbb{C}\setminus(-\infty,0]$ [56, p. 68].

THEOREM 4.2.1 (Bosch [58, Theorem 1.2]). *Let $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with $Z_\alpha, \tilde{Z}_\alpha$ independent positive $\alpha$ stable (here $\alpha$ is a stability index, not a basepoint, and $T_\alpha$ is not the variable of Theorem 1.2.2). Then $T_\alpha$ has density $\sin(\pi\alpha)/(\pi\alpha(t^2+2t\cos\pi\alpha+1))$, a half Cauchy law with drift, and the density of $T_\alpha^\beta$ is HCM if and only if $\alpha\le 1/2$ and $|\beta|\ge 1/(1-\alpha)$.*

At $\alpha=1/2$ we have $T_{1/2}\overset{d}{=}|C|$ with density $2/(\pi(1+t^2))$. The density of $|C|^\beta$ is

$$f_\beta(x) = \frac{2}{\pi|\beta|}\,\frac{x^{1/\beta-1}}{1+x^{2/\beta}}, \qquad x>0.$$

With $x=e^{a\pm y}$ and $q=2/|\beta|$ a short computation gives

$$f_\beta(e^{a+y})f_\beta(e^{a-y}) = \frac{4}{\pi^2\beta^2}\,e^{-2a}\,\frac{1}{2\cosh(qa)+2\cosh(qy)}. \tag{4.2.2}$$

So at $\alpha=1/2$ Bosch's theorem is the statement that $1/(c+\cosh(qy))$ is completely monotone in $\cosh y$ for every $c\ge 1$ exactly when $q\le 1$; here $c=\cosh(qa)$, which ranges over $[1,\infty)$.

PROPOSITION 4.2.2. *The half Cauchy density is not HCM.*

*Proof.* Take $\beta=1$, so $q=2$ in (4.2.2). Since $\cosh(2a)+\cosh(2y)=2\sinh^2 a+w^2/2$ with $w=2\cosh y$, the pair function is proportional to $h(w)=1/(c^2+w^2)$ with $c=2|\sinh a|>0$ for $a\ne 0$. For $x>0$,

$$h(2+x) = \int_0^\infty e^{-xt}\,e^{-2t}\,\frac{\sin(ct)}{c}\,dt.$$

If $h$ were completely monotone on $(2,\infty)$, Bernstein's theorem would give $h(2+x)=\int e^{-xt}\mu(dt)$ with $\mu\ge 0$, and uniqueness of the Laplace transform would force $\mu(dt)=e^{-2t}c^{-1}\sin(ct)\,dt$, which is not nonnegative. □

Proposition 4.2.3. *The density of $|C|^\beta$ is HCM for $|\beta|\ge 2$.*

*Proof.* For $q=2/|\beta|\le 1$ the function $\cosh(qy)$ is a Bernstein function of $\cosh y$ by (F2), and $x\mapsto 1/(c+x)$ is completely monotone for $c>0$, so the composition in (4.2.2) is completely monotone. □

At $\alpha=1/2$ sufficiency is therefore elementary; the deep part of Bosch's paper is $\alpha<1/2$.

### 4.2.2 The half Cauchy law and the Riemann Xi function

On the Cauchy side everything is explicit. The logarithm of $|C|$ has density $g_C(y)=1/(\pi\cosh y)$ and characteristic function $\operatorname{sech}(\pi t/2)$. The Weierstrass product

$$\operatorname{sech}\Big(\frac{\pi t}{2}\Big)=\prod_{k\ge 1}\Big(1+\frac{t^2}{(2k-1)^2}\Big)^{-1}$$

says $\log|C|\overset{d}{=}\sum_{k\ge1}L_k/(2k-1)$ with $L_k$ independent standard Laplace variables. The Mellin transform is $\mathbb{E}|C|^z=\sec(\pi z/2)$ for $|\operatorname{Re} z|<1$, the reciprocal of $\cos(\pi z/2)$, a function of the Laguerre Pólya class with real zeros $\pm(2k-1)$; these odd integers are the frequencies of the half Cauchy law. The Cauchy reciprocal Xi function

$$\varphi_C(s)=\mathbb{E}|C|^{i\sqrt{s}}=\operatorname{sech}\Big(\frac{\pi\sqrt{s}}{2}\Big)=\prod_{k\ge1}\Big(1+\frac{s}{(2k-1)^2}\Big)^{-1}$$

is the Laplace transform of the GGC $\sum_k e_k/(2k-1)^2$, with independent standard exponentials $e_k$ and Thorin measure $\sum_k\delta_{(2k-1)^2}$, hence HCM by (F6). The checks below that name no source used mpmath [376] at 20 to 30 digits. Bourgade, Fujita and Yor [60] derive Euler's formulae for $\zeta(2n)$ from products of independent Cauchy variables, whose Mellin transforms are powers of $\sec(\pi z/2)$. Section 10.2 develops their construction.

On the Riemann side the frequencies are the zeros $\gamma$ of $\Xi(z)=\xi(\frac12+iz)$. By (1.2.1), whose product converges absolutely because $\sum|\gamma|^{-2}<\infty$, unconditionally

$$\Xi(z)=\Xi(0)\prod_{\gamma}\Big(1-\frac{z^2}{\gamma^2}\Big),$$

the product over zeros $\gamma$ of $\Xi$ with $\operatorname{Re}\gamma > 0$, counted with multiplicity. No zero is purely imaginary because $\xi(\sigma) > 0$ for real $\sigma$. Hence the clock is

$$\varphi(s) = \frac{\Xi(0)}{\Xi(i\sqrt{s})} = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_{\gamma}\Big(1+\frac{s}{\gamma^2}\Big)^{-1}.$$

Under RH define, in the spirit of Biane, Pitman and Yor [49], $\log W = \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables $L_\gamma$. Then $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1 \approx 14.1347$, the characteristic function of $\log W$ is $\xi(1/2)/\xi(1/2+t)$, and $\varphi(s) = \mathbb{E}W^{i\sqrt{s}}$. The variable $W$, not to be confused with the heat trace $W(t)$, exists without RH: $\log W$ has the law of the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 1.2, where $Z$ is standard normal and independent of the clock variable $T$. Only the series for $\log W$ needs RH. Biane, Pitman and Yor represent $2\xi$ itself, not its reciprocal, as a Mellin transform (Sections 1.2 and 9.1).

| | half Cauchy | Riemann |
|---|---|---|
| entire function | $\cos(\pi z/2)$ | $\Xi(z)/\Xi(0)$ |
| frequencies | $2k-1$ | $\gamma$ |
| log variable | $\sum_k L_k/(2k-1)$ | $\sum_\gamma L_\gamma/\gamma$ |
| Mellin transform | $\sec(\pi z/2)$ | $\Xi(0)/\Xi(z)$ |
| reciprocal Xi | $\operatorname{sech}(\pi\sqrt{s}/2)$ | $\xi(1/2)/\xi(1/2+\sqrt{s})$ |
| Thorin measure | $\sum_k \delta_{(2k-1)^2}$ | $\sum_\gamma \delta_{\gamma^2}$ |
| exit time | Brownian motion on $(-1,1)$ | Krein string with spectrum $\{\gamma^2\}$ |
| density of the power | HCM iff $\lvert\beta\rvert \geq 2$ (Bosch) | HCM only if $\lvert\beta\rvert \geq \gamma_2 - \gamma_1$ |

Table 4.1: The dictionary. In the Riemann column the Mellin transform (Section 1.2 and Theorem 4.3.25) and the reciprocal Xi function are unconditional, the log variable and the Thorin measure presume RH, and the exit time, the hitting time of Kent's theorem (Section 8.1.3), presumes RH and simple zeros. The last entry is the necessary condition of Theorem 4.3.27, which assumes RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$; sufficiency is open (Question 4.3.28).

On the Riemann side the analogue of $g_C$ is the density of $\log W$. The function $t \mapsto \xi(1/2)/\xi(1/2+t)$ is defined for all real $t$ without any hypothesis ($\xi$ has no real zeros) and decays like $\exp(-\frac{1}{2}|t|\log|t|)$, so its Fourier inverse

$$g(y) = \frac{1}{\pi}\int_0^\infty \frac{\xi(1/2)}{\xi(1/2+t)}\cos(ty)\,dt$$

is an even function with $\int g = 1$, entire because the integrand decays faster than any exponential; $g > 0$ unconditionally (Theorem 4.3.25). Under RH, with simple zeros and $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$, the residues at the poles $\pm i\gamma_n$ of the integrand give, for $|y| > u_0$ with the constant $u_0 \geq \pi/4$ of Theorem 4.3.27,

$$g(y) = \sum_n \frac{\gamma_n}{2} A_n e^{-\gamma_n|y|}, \qquad A_n = \prod_{m\neq n}\Big(1-\frac{\gamma_n^2}{\gamma_m^2}\Big)^{-1}, \tag{4.2.3}$$

the analogue of $g_C(y) = (2/\pi)\sum_k (-1)^{k-1} e^{-(2k-1)|y|}$, which converges for every $y \neq 0$ because its coefficients are bounded. The Riemann coefficients $A_1 \approx 50.9$, $A_2 \approx -2664.6$, $A_3 \approx 31420$ (`code/README.md`) alternate in sign, as the Cauchy coefficients do, but grow roughly like $e^{\pi\gamma_n/4}$ (Theorem 4.3.27), so (4.2.3) diverges for $|y| < \pi/4$.

### 4.2.3 The reciprocal Xi function is HCM if and only if RH holds

Bosch's method works with the singularities of a function and with its logarithmic derivative, so we state Corollary 4.1.3(ii) in four forms. Form (iii) is Proposition 1.2.1, and since $\psi = G'/G$, form (iv) is the Stieltjes form of the Thorin condition in Section 1.1. A Stieltjes function is a function $a/s + b + \int_{(0,\infty)} \sigma(dt)/(s+t)$ with $a, b \geq 0$ and $\sigma$ a nonnegative measure satisfying $\int (1+t)^{-1}\sigma(dt) < \infty$ [473, Definition 2.1].

Theorem 4.2.4. *The following are equivalent.*

*(i) The Riemann Hypothesis.*

*(ii)* $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$ *is HCM on* $(0,\infty)$.

*(iii)* $\varphi$ *is the Laplace transform of a GGC, namely* $\sum_\gamma e_\gamma/\gamma^2$.

*(iv)* $\psi(s) = -\varphi'(s)/\varphi(s) = \dfrac{1}{2\sqrt{s}}\dfrac{\xi'}{\xi}\big(\tfrac12 + \sqrt{s}\big)$ *is a Stieltjes function.*

*Proof.* (i) implies (iii): each factor $(1 + s/\gamma^2)^{-1}$ is the Laplace transform of an exponential variable with rate $\gamma^2 > 0$, and the product converges because $\sum \gamma^{-2} < \infty$; the sum $\sum_\gamma e_\gamma/\gamma^2$ converges almost surely. (iii) implies (ii) by (F6).

(ii) implies (i): by (F7), $\varphi$ extends analytically to $\mathbb{C} \setminus (-\infty, 0]$. Suppose a zero $\gamma$ has $\operatorname{Im}\gamma \neq 0$. Since $\operatorname{Re}\gamma > 0$ the point $s_0 = -\gamma^2$ has $\operatorname{Im} s_0 = -2\operatorname{Re}\gamma\,\operatorname{Im}\gamma \neq 0$, and its principal square root is $\pm i\gamma$, so $\xi(1/2 + \sqrt{s_0}) = \Xi(\pm\gamma) = 0$ by evenness. Then $\varphi$ has a pole at a nonreal point of $\mathbb{C} \setminus (-\infty, 0]$, a contradiction.

(i) is equivalent to (iv): under RH, $\psi(s) = \sum_\gamma 1/(s+\gamma^2)$, a Stieltjes function with measure $\sum_\gamma \delta_{\gamma^2}$. Conversely a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and an off line zero produces the nonreal pole $s_0$ of $\psi$ found above. □

Condition (iv) is a companion of Lagarias's criterion [320], and it is the pole test of Table 1.7 in analytic form.

The de Bruijn Newman flow embeds $\varphi$ in a family of the same kind. Let $\Phi_{\mathrm{dBN}}$ and $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be as in Section 8.4, so that $H_0(z) = \frac18\Xi(z/2)$. Put $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$, the function $R_t$ of Theorem 7.2.16, so that $\varphi_0 = \varphi$. The function $H_t$ is even, real on $\mathbb{R}$ and entire of order one, as in the proof of Theorem 7.2.16, so its Hadamard product in $z^2$ converges. It is positive on the imaginary axis, $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, since

$\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u) > 0$; hence $\varphi_t > 0$ on $(0,\infty)$ and no zero of $H_t$ is purely imaginary. With these facts the argument of Theorem 4.2.4 applies verbatim.

Corollary 4.2.5. *$\varphi_t$ is HCM if and only if $t \geq \Lambda$, the de Bruijn Newman constant. Hence $\varphi_t$ is HCM for $t \geq 0.22$ [456], $\varphi_t$ is not HCM for $t < 0$ [463], and RH holds if and only if $\varphi_t$ is HCM for every $t > 0$.*

*Proof.* The zeros of $H_t$ are real if and only if $t \geq \Lambda$ (de Bruijn [112], Newman [381]). If RH holds, $H_0$ has only real zeros, so $\Lambda \leq 0$ and $\varphi_t$ is HCM for every $t > 0$. Conversely, $\varphi_t \to \varphi_0$ pointwise as $t \downarrow 0$, so if every $\varphi_t$ with $t > 0$ is HCM then $\varphi$ is HCM by (F4), and Theorem 4.2.4 gives RH. □

The bound $\Lambda \leq 0.2$ of Platt and Trudgian [419] (Section 8.4) lowers the unconditional threshold from 0.22 to 0.2. The reciprocal Xi function, if HCM at all, is HCM with no room to spare, since the property fails for every $t < 0$. This is the Riemann counterpart of the boundary $|\beta| = 1/(1-\alpha)$ in Bosch's theorem, which is attained; the difference is that Bosch can compute on the boundary because his kernel is explicit.

### 4.2.4 What Bosch's theorem can and cannot supply

Bosch's theorem cannot be the input to a proof of Theorem 4.2.4. For products the reason is elementary.

Proposition 4.2.6. *For any positive numbers $c_n$ with $\sum_n c_n^{-1} < \infty$ the function $\prod_n (1 + s/c_n)^{-1}$ is HCM. If one $c_n$ is replaced by a pair $c, \bar{c}$ with $c$ nonreal, the product is still positive on $(0,\infty)$ but is not HCM.*

*Proof.* The first claim is the argument for (i) implies (ii) in Theorem 4.2.4. For the second, $(1 + s/c)(1 + s/\bar{c}) = |1 + s/c|^2 > 0$ for real $s$, and the product has a pole at the nonreal point $-c$, contradicting (F7). □

Heuristically, every HCM closure operation carries real frequencies to real frequencies and none creates them: on Mellin transforms, products and quotients of independent variables multiply the transforms, and powers rescale the frequencies. Bosch's theorem lives in the class where the frequency set $\{2k-1\}$ is given, so any deduction of Theorem 4.2.4 from it must import the location of the $\gamma$ from elsewhere.

Nor can the Riemann statement be proved factor by factor. Write $\xi(1/2 + \sqrt{s}) = \frac{1}{2}(s - \frac{1}{4})\,\pi^{-(1/2+\sqrt{s})/2}\,\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\,\zeta(\frac{1}{2} + \sqrt{s})$. The reciprocal factors are not individually HCM. The reciprocal of $\pi^{-(1/2+\sqrt{s})/2}$ is $\pi^{1/4}\pi^{\sqrt{s}/2}$, and the pair function of $\pi^{\sqrt{s}/2}$ is $\exp(\frac{1}{2}\log\pi\,\sqrt{u}\,\sqrt{w+2})$, which increases in $w$. The factor $1/\zeta(1/2+\sqrt{s})$ changes sign at $s = 1/4$, since $\zeta < 0$ on $(1/2, 1)$, and HCM functions are positive. The positive combination $1/((s - \frac{1}{4})\zeta(\frac{1}{2} + \sqrt{s}))$ is continuous at $s = 1/4$, and by the argument of Theorem 4.2.4 any HCM statement for it would

again imply RH. It is the function to which the sech expansion of $1/\zeta(\frac{1}{2}+\sqrt{s})$ in Theorem 6.1.5 should be applied.

Bosch's theorem is a density statement, and HCM of a density differs from HCM of a Laplace transform: the density of $|C|$ is not HCM, yet $\varphi_C$ is. On the Cauchy side the two levels are linked only because sech is self dual, its own Fourier transform up to scaling, so that the density $g_C$ of $\log|C|$ and the transform $\varphi_C$ are the same function in different coordinates. On the Riemann side there is no self duality: the density level object is $g$ in (4.2.3), the Fourier inverse of $\Xi(0)/\Xi(i\,\cdot)$, and the function level object is $\varphi$.

Question 4.2.7. The variable $W$, with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, exists unconditionally (Section 4.2.2 and Theorem 4.3.25). For which $\beta$ is the density of $W^\beta$ HCM? On the Cauchy side the answer is $|\beta| \geq 2$.

Question 4.2.8. Does HCM of the density of $W^\beta$ for a single $\beta$ imply RH?

Theorem 4.3.27 answers Question 4.2.7 in part: under RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$, the density of $W^\beta$ can be HCM only if $|\beta| \geq \gamma_2 - \gamma_1 = 6.8873$, and Question 4.3.28 asks whether this is also sufficient. At the Cauchy point the same rule gives $3 - 1 = 2$, Bosch's threshold.

One might hope that an HCM variable has a logarithm with an EGGC law (Section 1.3.2), which would force the singularities of $\mathbb{E}W^z$, that is the zeros of $\Xi$, onto the real axis. This implication is false in general. The generalised inverse Gaussian law with density proportional to $x^{-1}e^{-(x+1/x)}$ is HCM by (F7), yet its Mellin transform at $z = it$, $K_{it}(2)/K_0(2)$, has real zeros in $t$, the first at $t \approx 4.4255$ (by a theorem of Pólya [447], every zero of $t \mapsto K_{it}(2)$ is real). The characteristic function of its logarithm therefore vanishes, and the logarithm is not infinitely divisible. So density level HCM does not by itself give RH.

Two things do transfer. First, the dictionary of Table 4.1 is a solved model in which every quantity is explicit, and it tests any proposed argument: a proof that $\varphi$ is HCM which does not use the values of the frequencies would also prove HCM for $\cos(\pi z/2)$ with frequencies moved off the axis, which is false by Proposition 4.2.6. Second, the boundary phenomenon of Corollary 4.2.5 gives a target: to show that $\varphi_t$ is HCM for all $t > 0$, which by that corollary is equivalent to $\Lambda \leq 0$, that is, by Rodgers and Tao [463], to $\Lambda = 0$.

## 4.3 A Bosch type theorem for the reciprocal Xi function

The observation that closes Section 4.2 becomes a theorem through an elasticity criterion (Theorem 4.3.2). Applied to $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ it gives a Bosch type theorem for $\varphi(x^{1/\beta})$ (Theorem 4.3.5), and the same elasticity identifies the Thorin measure of $\varphi(s^p)$ (Theorem 4.3.12); the positivity of its density is where the Riemann problem now sits.

The notation is that of Section 4.2: the clock $\varphi$ is a function of $s$, and $W$ is the

Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace $W(t)$ of Section 5.1. The numerical checks are made by `code_bosch/verify_bosch_xi_v7.py` in the repository and documented in `code/README.md`; the checks confirm formulas and constants and prove nothing.

### 4.3.1 The elasticity criterion

We use HCM as defined in Section 4.2.1, with the facts (F1) to (F7) listed there, above all Bondesson's representation (4.2.1) [56, (5.2.3) and Theorem 5.3.1]. Throughout, $\Omega = \mathbb{C}\setminus(-\infty, 0]$, $\mathbb{C}^+ = \{\operatorname{Im} x > 0\}$, and a Pick function is an analytic map $\mathbb{C}^+ \to \overline{\mathbb{C}^+}$. For a GGC with Laplace transform $h$, drift $b \geq 0$ and Thorin measure $U$, in the sense of (1.3.1), $-h'/h = b + \int U(dt)/(x+t)$ is a Stieltjes function. We also need two facts about complete Bernstein functions [473, Theorem 6.2]: a function $f$ on $(0,\infty)$ is complete Bernstein if and only if $f \geq 0$ on $(0,\infty)$ and $f$ continues to a Pick function analytic on $\Omega$, and then $f(x)/x$ is a Stieltjes function.

The elasticity criterion below restates (4.2.1) as a Pick function condition, of the kind through which Bosch verifies HCM [58, Section 2.2].

Definition 4.3.1. For $h$ analytic and zero free on $\Omega$ and positive on $(0,\infty)$, let $\log h$ be the branch that is real on $(0,\infty)$ and define the elasticity $\varepsilon_h(x) = -xh'(x)/h(x)$. In logarithmic coordinates $x = e^z$ one has $\varepsilon_h(e^z) = -\frac{d}{dz}\log h(e^z)$.

Theorem 4.3.2. *Let $h > 0$ on $(0,\infty)$. Then $h$ is HCM if and only if $h$ continues to a zero free analytic function on $\Omega$ and $\varepsilon_h$ is a Pick function, that is* $\operatorname{Im}\varepsilon_h(x) \geq 0$ *for* $\operatorname{Im} x > 0$.

*Proof.* Necessity. By (4.2.1), $h$ is zero free and analytic on $\Omega$, and its elasticity is

$$\varepsilon_h(x) = 1 - r + b_1 x + \int \frac{x}{x+t}\,\Gamma_1(dt) - \frac{b_2}{x} - \int \frac{\Gamma_2(dt)}{1+tx}.$$

For $\operatorname{Im} x > 0$ each term has nonnegative imaginary part, for instance $\operatorname{Im}\frac{x}{x+t} = \frac{t\operatorname{Im} x}{|x+t|^2}$.

Sufficiency. Fix $u > 0$. The principal branch $y(w) = \operatorname{arccosh}(w/2)$ maps $\mathbb{C}\setminus(-\infty, 2]$ conformally onto the half strip $S = \{\operatorname{Re} y > 0,\ |\operatorname{Im} y| < \pi\}$, and $w = 2\cosh y$. For $y \in S$ both $ue^y$ and $ue^{-y}$ lie in $\Omega$, so

$$F(w) = -\log h\big(ue^{y(w)}\big) - \log h\big(ue^{-y(w)}\big)$$

is analytic on $\mathbb{C}\setminus(-\infty, 2]$ and real on $(2,\infty)$, and $e^{-F(w)}$ is the pair function. Let $\operatorname{Im} w > 0$, so $y = \varrho + i\theta$ with $\varrho > 0$ and $\theta \in (0,\pi)$. Put $r_1 = ue^{\varrho} > r_2 = ue^{-\varrho}$. Since $\log h(\bar{x}) = \overline{\log h(x)}$,

$$\operatorname{Im} F(w) = -\operatorname{Im}\log h(r_1 e^{i\theta}) + \operatorname{Im}\log h(r_2 e^{i\theta}).$$

Along the ray, $\partial_r \operatorname{Im} \log h(re^{i\theta}) = \operatorname{Im}\big(e^{i\theta}h'/h\big) = -r^{-1}\operatorname{Im}\varepsilon_h(re^{i\theta}) \le 0$, so $\operatorname{Im} F(w) \ge 0$. Thus $F$ is a Pick function that continues analytically across $(2,\infty)$, where it is real. By the Nevanlinna representation $F(w) = a + bw + \int(\frac{1}{t-w} - \frac{t}{1+t^2})\sigma(dt)$ with $b \ge 0$ and $\sigma \ge 0$ carried by $(-\infty, 2]$. Hence for $w > 2$,

$$F'(w) = b + \int_{(-\infty,2]} \frac{\sigma(dt)}{(w-t)^2} = b + \int_{(-\infty,2]} \int_0^\infty \lambda e^{-\lambda(w-t)}\, d\lambda\, \sigma(dt),$$

which is completely monotone. A function whose derivative is completely monotone is a Bernstein function up to an additive constant, and $e^{-F}$ is then completely monotone. So every pair function is completely monotone and $h$ is HCM. □

The two conditions play different roles. Analyticity without zeros is global and rigid: a single zero or pole of $h$ anywhere off the negative axis destroys HCM, however small its effect on the real line. The Pick condition is local and quantitative: it can fail along a ray without any singularity.

### 4.3.2 Bosch's theorem from the criterion

Bosch's variable $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$, $0 < \alpha < 1$, of Section 4.2.1 has the density $\sin(\pi\alpha)/(\pi\alpha(t^2 + 2t\cos\pi\alpha + 1))$ of a half Cauchy law with drift. Let $c = \cos\pi\alpha$ and $q = 1/|\beta|$, half the $q$ of (4.2.2). Since $T_\alpha \overset{d}{=} 1/T_\alpha$ we may take $\beta > 0$. The density of $T_\alpha^\beta$ is proportional to

$$h(x) = \frac{x^{q-1}}{D(x)}, \qquad D(x) = x^{2q} + 2cx^q + 1 = (x^q + e^{i\pi\alpha})(x^q + e^{-i\pi\alpha}).$$

Theorem 4.3.3 (Bosch [58, Theorem 1.2]). *$h$ is HCM if and only if $\alpha \le 1/2$ and $q \le 1 - \alpha$.*

*Proof.* Zeros. For $x \in \Omega$ the value $x^q$ ranges over the sector $|\arg| < q\pi$. The zeros of $D$ satisfy $x^q = e^{\pm i(1-\alpha)\pi}$, so $D$ is zero free on $\Omega$ if and only if $q \le 1 - \alpha$.

Elasticity. For $x = re^{i\theta}$ with $0 < \theta < \pi$ write $x^q = Re^{i\omega}$ with $R = r^q$ and $\omega = q\theta \in (0, q\pi)$. Then $\varepsilon_h = 1 - q + xD'(x)/D(x)$ and $\operatorname{Im}\varepsilon_h = qR\,\partial_R \arg D$. A direct computation of $\operatorname{Im}(\partial_R D \cdot \overline{D})$ gives

$$\operatorname{Im}\varepsilon_h(x) = \frac{2qR\sin\omega\,\big(c(1+R^2) + 2R\cos\omega\big)}{|D(x)|^2}. \tag{4.3.1}$$

If $\alpha \le 1/2$ and $q \le 1 - \alpha$ then $c \ge 0$ and $\cos\omega \ge \cos((1-\alpha)\pi) = -c$, so, using $1 + R^2 \ge 2R$, the bracket is at least $2R(c + \cos\omega) \ge 0$. Theorem 4.3.2 gives HCM. If $\alpha > 1/2$ then $c < 0$ and the bracket tends to $c < 0$ as $R \to 0$, so $\varepsilon_h$ is not Pick. If $q > 1 - \alpha$ then $h$ has poles in $\Omega$. In both cases Theorem 4.3.2 rules out HCM. □

In terms of $\beta$, the theorem says that the density of $T_\alpha^\beta$ is HCM if and only if $\alpha \le 1/2$ and $|\beta| \ge 1/(1-\alpha)$. For the half Cauchy variable, $\alpha = 1/2$, the threshold is $|\beta| \ge 2$; there $c = 0$ and (4.3.1) reduces to $\operatorname{Im}\varepsilon_h = 4qR^2 \sin\omega\cos\omega/|D|^2$, which is nonnegative for every $\omega \in (0, q\pi)$ exactly when $q \le 1/2$. So the half Cauchy threshold is the condition $\omega \le \pi/2$, equivalently that the zeros $x = e^{\pm i\pi/(2q)}$ of $1 + x^{2q}$ stay on or beyond the cut.

### 4.3.3 The Riemann analogue

As in Section 4.2.2, $\gamma$ runs over the zeros of $\Xi(z) = \xi(1/2 + iz)$ with $\operatorname{Re}\gamma > 0$, counted with multiplicity; the nonreal ones come in conjugate pairs $\gamma, \bar\gamma$. By (1.2.1), unconditionally,

$$\varphi(s) = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_\gamma \Big(1 + \frac{s}{\gamma^2}\Big)^{-1}, \qquad \varepsilon_\varphi(s) = \frac{\sqrt{s}}{2}\frac{\xi'}{\xi}\Big(\frac12 + \sqrt{s}\Big) = \sum_\gamma \frac{s}{s+\gamma^2},$$

both absolutely convergent since $\sum |\gamma|^{-2} < \infty$. Write $\gamma = \varrho e^{i\delta}$. Since the zeros of $\xi$ lie in the open critical strip, $|\delta| < \arctan(1/(2\operatorname{Re}\gamma))$, and since every zero has $\operatorname{Re}\gamma > 14$, also $|\delta| < \pi/4$. Define

$$\delta_* = \max_\gamma |\arg\gamma|, \qquad \alpha_\xi = \frac{2\delta_*}{\pi}, \qquad p_* = 1 - \alpha_\xi.$$

The maximum is attained if some zero is off the line, because for every positive angle only finitely many $\gamma$ have $|\arg\gamma|$ larger than it. RH is the statement $\delta_* = 0$.

Lemma 4.3.4. *Let $\gamma = \varrho e^{i\delta}$ with $0 \le \delta < \pi/4$ and $s = re^{i\theta}$ with $0 < \theta < \pi - 2\delta$. Then*

$$\operatorname{Im}\Big(\frac{s}{s+\gamma^2} + \frac{s}{s+\bar\gamma^2}\Big) \ge 0,$$

*with strict inequality. For $\delta = 0$ the single term $\operatorname{Im}\frac{s}{s+\gamma^2} = \frac{\gamma^2 r\sin\theta}{|s+\gamma^2|^2}$ is positive for $0 < \theta < \pi$.*

*Proof.* For $c \in \mathbb{C}$, $\operatorname{Im}\frac{s}{s+c} = \operatorname{Im}\big(-\frac{c}{s+c}\big) = \frac{\operatorname{Im}(\bar c s)}{|s+c|^2}$. With $c = \varrho^2 e^{\pm 2i\delta}$ this is $\varrho^2 r\sin(\theta \mp 2\delta)/|s+c|^2$. The sum is

$$\varrho^2 r\Big(\frac{\sin(\theta-2\delta)}{A_-} + \frac{\sin(\theta+2\delta)}{A_+}\Big), \qquad A_\mp = r^2 + \varrho^4 + 2r\varrho^2\cos(\theta \mp 2\delta).$$

Since $\theta + 2\delta < \pi$ the second sine is positive. If $\theta \ge 2\delta$ both terms are nonnegative. If $\theta < 2\delta$ then $|\theta - 2\delta| < \theta + 2\delta$, so $A_- > A_+$, and $\sin(\theta+2\delta) - \sin(2\delta - \theta) = 2\cos(2\delta)\sin\theta > 0$; the positive term has the larger numerator and the smaller denominator. □

Theorem 4.3.5. *For $p \in \mathbb{R}$ the function $x \mapsto \varphi(x^p)$ is HCM if and only if $|p| \le p_* = 1 - \alpha_\xi$. Equivalently, $\varphi(x^{1/\beta})$ is HCM if and only if $|\beta| \ge 1/(1-\alpha_\xi)$.*

*Proof.* Since $h(x)$ is HCM if and only if $h(1/x)$ is, take $p > 0$. Let $h_p(x) = \varphi(x^p)$, positive on $(0,\infty)$ because $\xi > 0$ on $\mathbb{R}$. Its singularities are the poles $x^p = -\gamma^2 = \varrho^2 e^{\pm i(\pi - 2|\delta|)}$, and $x^p$ ranges over $|\arg| < p\pi$ for $x \in \Omega$. Hence $h_p$ is zero free and analytic on $\Omega$ if and only if $p\pi \le \pi - 2|\delta|$ for every $\gamma$, that is $p \le p_*$. When $p \le p_*$,

$$\varepsilon_{h_p}(x) = p\,\varepsilon_\varphi(x^p) = p\sum_\gamma \frac{x^p}{x^p + \gamma^2},$$

and for $\operatorname{Im} x > 0$ the point $s = x^p$ has $0 < \arg s < p\pi \le \pi - 2\delta_*$. Grouping conjugate pairs, Lemma 4.3.4 makes every term of $\operatorname{Im}\varepsilon_{h_p}$ nonnegative. Theorem 4.3.2 gives HCM. When $p > p_*$ the pole coming from a zero with $|\arg\gamma| = \delta_*$ lies in $\Omega$, and Theorem 4.3.2 rules out HCM. □

Corollary 4.3.6. *The following are equivalent: (i) RH; (ii) $\varphi$ is HCM; (iii) $\varphi(x^p)$ is HCM for every $|p| < 1$; (iv) $\varepsilon_\varphi$ is a Pick function; (v)* $\operatorname{Im}\big[z\,(\xi'/\xi)(\frac12+z)\big] \ge 0$ *for all $z$ with* $\operatorname{Re} z > 0$ *and* $\operatorname{Im} z > 0$.

*Proof.* By Theorem 4.3.5, each of (i), (ii) and (iii) says that $p_* = 1$; the equivalence of (i) and (ii) is also Corollary 4.1.3(ii), restated as Theorem 4.2.4, so only (iii) to (v) are new. (iv) is the elasticity form of (ii) through Theorem 4.3.2. (v) is (iv) with $s = z^2$, which maps the open first quadrant onto $\mathbb{C}^+$. □

Criterion (v) is an angular companion to Lagarias's criterion $\operatorname{Re}(\xi'/\xi)(\frac12+z) > 0$ for $\operatorname{Re} z > 0$ [320].

Corollary 4.3.7 (unconditional). *RH has been verified for all zeros with* $0 < \operatorname{Re}\gamma \le T_0 = 3\times 10^{12}$ *[419]. Hence* $\delta_* < \arctan(1/(2T_0))$ *and*

$$\varphi(x^p) \text{ is HCM for all } |p| \le 1 - \frac{1}{\pi T_0}, \qquad 1 - \frac{1}{\pi T_0} > 1 - 1.1\times 10^{-13}.$$

*Unconditionally,* $\operatorname{Im}\varepsilon_\varphi \ge 0$ *on the sector* $0 < \arg s < \pi - 1/T_0$.

*Proof.* Any off line zero has $\operatorname{Re}\gamma > T_0$ and $|\operatorname{Im}\gamma| < 1/2$, so $|\arg\gamma| < 1/(2T_0)$ and $2\delta_*/\pi < 1/(\pi T_0)$. The sector statement is Lemma 4.3.4 applied termwise. □

Corollary 4.3.7 is a statement about the elasticity $\varepsilon_\varphi(s) = s\psi(s)$, where $\psi = -\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$ is the function of the Stieltjes form (iv) of Theorem 4.2.4.

Lemma 4.3.8. *Let $c = c_1 + ic_2$ with $c_2 > 0$, and let $s = x + iy$ with $y > 0$ and $s \ne c$. Then*

$$\operatorname{Im}\Big(\frac{1}{s-c} + \frac{1}{s-\bar c}\Big) = -\frac{2y\,\big(|s-c_1|^2 - c_2^2\big)}{|s-c|^2\,|s-\bar c|^2},$$

*so the imaginary part is positive on the open disc $|s - c_1| < c_2$, whose vertical diameter is the segment from $\bar c$ to $c$, zero on its boundary and negative outside it.*

*Proof.* With $d = (x - c_1)^2$, $\operatorname{Im} \frac{1}{s-c} = -\frac{y-c_2}{d+(y-c_2)^2}$ and $\operatorname{Im} \frac{1}{s-\bar{c}} = -\frac{y+c_2}{d+(y+c_2)^2}$. Over the common denominator the numerator of the sum is

$$-(y-c_2)\big(d+(y+c_2)^2\big) - (y+c_2)\big(d+(y-c_2)^2\big) = -2y\,(d+y^2-c_2^2),$$

and $d + y^2 = |s - c_1|^2$. □

A nonreal zero $\gamma = a + ib$ of $\Xi$, with $a > 0$ and, after conjugation, $b > 0$, contributes to $\psi$ the pair of terms $1/(s-c)+1/(s-\bar{c})$ with $c = -\bar{\gamma}^2 = b^2-a^2+2iab$. Let

$$D_\gamma = \big\{ s : \ |s - (b^2 - a^2)| \le 2ab \big\}$$

be the closed disc with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$. By Lemma 4.3.8 the pair has positive imaginary part only at interior points of $D_\gamma$, and negative imaginary part at every point of the upper half plane outside $D_\gamma$.

THEOREM 4.3.9 (unconditional). *Let $T_0 = 3 \times 10^{12}$, so that every zero $\gamma$ with $0 < \operatorname{Re}\gamma \le T_0$ is real [419], and let*

$$\mathcal{E}_{T_0} = \Big\{ x + iy : \ x < -T_0^2 + T_0 + \tfrac{1}{4}, \ y^2 < \tfrac{1}{2} - x \Big\}.$$

*Split $\psi = S_{\text{on}} + R_{\text{off}}$, where $S_{\text{on}}(s) = \sum_{\gamma \text{ real}} (s+\gamma^2)^{-1}$ runs over the zeros on the critical line and $R_{\text{off}}$ is the same sum over the nonreal $\gamma$.*

*(i) $S_{\text{on}}$ is a Stieltjes function, with measure $U_{\text{line}} = \sum_{\gamma \text{ real}} \delta_{\gamma^2}$.*

*(ii) $R_{\text{off}}$ is meromorphic on $\mathbb{C}$ and nonnegative on $[0,\infty)$. For every nonreal $\gamma$ the disc $D_\gamma$, and with it the poles $-\gamma^2$ and $-\bar{\gamma}^2$ of $R_{\text{off}}$ on its boundary, lies in $\mathcal{E}_{T_0}$, and $\operatorname{Im} R_{\text{off}} \le 0$ at every point of the upper half plane outside these discs.*

*(iii) $\psi > 0$ on $[0,\infty)$, and $\operatorname{Im}\psi(s) < 0$ at every $s = x + iy$ with $y > 0$ outside $\mathcal{E}_{T_0}$.*

*Proof.* Since $\sum |\gamma|^{-2} < \infty$, the series for $\psi$, $S_{\text{on}}$ and $R_{\text{off}}$ converge absolutely and locally uniformly off the points $-\gamma^2$, so $R_{\text{off}}$ is meromorphic on $\mathbb{C}$, and (i) holds because $\sum_{\gamma \text{ real}} (1+\gamma^2)^{-1} < \infty$. A real $\gamma$ contributes $-y/|s+\gamma^2|^2 < 0$ to $\operatorname{Im}\psi(x+iy)$ for $y > 0$, and $(x+\gamma^2)^{-1} > 0$ to $\psi(x)$ for $x \ge 0$.

A nonreal zero is $\gamma = a + ib$ with $a > T_0$ and, after conjugation, $0 < b < \frac{1}{2}$, because $\xi$ vanishes at $\frac{1}{2} \pm b + ia$. With $c = b^2 - a^2 + 2iab$ as above, its pair of terms equals $2(x - c_1)/((x - c_1)^2 + c_2^2) > 0$ at $x \ge 0$, since $c_1 = b^2 - a^2 < 0$, and its imaginary part is negative at every point of the upper half plane outside $D_\gamma$. It remains to show that $D_\gamma \subset \mathcal{E}_{T_0}$. A point $x + iy$ of $D_\gamma$ has

$$x \le b^2 - a^2 + 2ab = 2b^2 - (a-b)^2 < -T_0^2 + T_0 + \tfrac{1}{4},$$

because the middle term increases in $b$ and, for $a > b$, decreases in $a$, and equals $-T_0^2 + T_0 + \frac{1}{4}$ at $a = T_0$, $b = \frac{1}{2}$; in particular $x < 0$. With $e = x - (b^2 - a^2)$, so that $a^2 = b^2 - x + e$, it also has

$$y^2 \le 4a^2b^2 - e^2 = 8b^4 - 4b^2x - (e - 2b^2)^2 \le 4b^2(2b^2 - x) < 2b^2 - x < \tfrac{1}{2} - x,$$

because $4b^2 < 1$ and $2b^2 - x > 0$. Hence $D_\gamma \subset \mathcal{E}_{T_0}$, and (ii) follows from the signs of the pairs. At a point of the upper half plane outside $\mathcal{E}_{T_0}$ every pair of terms has negative imaginary part, and so has the term of every real $\gamma$; with the signs on $[0, \infty)$, since $\gamma_1$ is real, this gives (iii). □

Both bounds that define $\mathcal{E}_{T_0}$ are sharp for zeros with $a > T_0$ and $0 < b < \frac{1}{2}$. The rightmost point $2b^2 - (a - b)^2$ of $D_\gamma$ tends to $-T_0^2 + T_0 + \frac{1}{4}$ as $b \uparrow \frac{1}{2}$ and $a \downarrow T_0$. For $x < \frac{3}{4} - T_0^2$ the supremum of $y^2$ over the points $x + iy$ of the discs is $\frac{1}{2} - x$: the bound $8b^4 - 4b^2x$ of the proof is attained, at $e = 2b^2$, by the disc with $a^2 = 3b^2 - x$, which has $a > T_0$ once $b$ is close to $\frac{1}{2}$, and it tends to $\frac{1}{2} - x$ as $b \uparrow \frac{1}{2}$. At the tip $x_0 = -T_0^2 + T_0 + \frac{1}{4}$ the height of $\mathcal{E}_{T_0}$ is exactly $T_0 - \frac{1}{2}$, since $\frac{1}{2} - x_0 = (T_0 - \frac{1}{2})^2$, so the ray from the origin to the corner $x_0 + i(T_0 - \frac{1}{2})$ makes the angle $\theta_0 = \arctan \frac{T_0 - 1/2}{T_0^2 - T_0 - 1/4} = 3.33333 \times 10^{-13}$ with the negative axis, which exceeds $1/T_0$ by the factor $1 + 1.7 \times 10^{-13}$. Further out the angle decreases like $|x|^{-1/2}$.

Corollary 4.3.7 gives the sign of the elasticity $s\psi(s)$ outside a sector of opening $1/T_0$ beside the cut, through Lemma 4.3.4; Theorem 4.3.9 gives the sign of $\psi$ itself outside a parabola, through the exact region of Lemma 4.3.8, and the parabola is much thinner. The theorem does not make $\psi$ a Stieltjes function off $\mathcal{E}_{T_0}$, since being a Stieltjes function is a global property; it says that if the Stieltjes form fails, it fails through poles of $\psi$ in $\mathcal{E}_{T_0}$, which come from zeros above the verification height. Like the corollary, it brings no new information about the zeros, and nothing in it limits the number of poles in $\mathcal{E}_{T_0}$ (Section 11.4.7).

Remark 4.3.10 (the right end). The right end of $\mathcal{E}_{T_0}$ cannot be moved to $-(T_0 - \frac{1}{2})^2$, the value the argument would give if the rightmost point of $D_\gamma$ were $-(a-b)^2$: it is $2b^2 - (a - b)^2$, and the extra $2b^2$ reaches into the strip $-(T_0 - \frac{1}{2})^2 \le x < -(T_0 - \frac{1}{2})^2 + \frac{1}{2}$. Whether $\operatorname{Im}\psi$ itself is positive at such points depends on zeros that nobody knows; the termwise argument cannot exclude them.

Figure 4.2 shows the sliver and the discs.

The threshold $p_*$ of Theorem 4.3.5 can be followed along the de Bruijn Newman flow. With $H_t$ and $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$ as in Section 4.2.3, so that $\varphi_0 = \varphi$, let $\delta_*(t)$ be the largest angle $|\arg z|$ over the zeros $z$ of $H_t$ with $\operatorname{Re} z > 0$, and $p_*(t) = 1 - 2\delta_*(t)/\pi$. The proof of Theorem 4.3.5 uses only the product over the zeros, positivity on $(0, \infty)$ and the bound $|\arg z| < \pi/4$ required by Lemma 4.3.4, so whenever the zeros of $H_t$ satisfy that bound, $\varphi_t(x^p)$ is HCM if and only if $|p| \le p_*(t)$. By de Bruijn [112] and Newman [381] (Corollary 4.2.5), $p_*(t) = 1$ exactly for $t \ge \Lambda$; by Rodgers and Tao [463], $p_*(t) < 1$ for every $t < 0$.

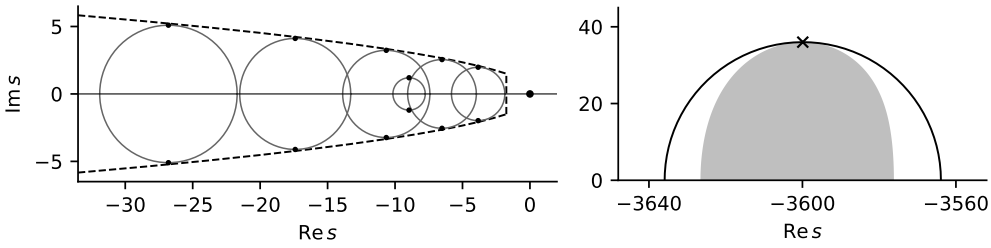



Figure 4.2: Left: the sliver $\mathcal{E}_{T_0}$ of Theorem 4.3.9 for $T_0 = 2$ (dashed) and the discs $D_\gamma$ of six hypothetical zeros $a + ib$ with $a > 2$ and $0 < b < \frac{1}{2}$, with the poles $-\gamma^2$ and $-\bar{\gamma}^2$ (dots) at the ends of their vertical diameters; the dot at 0 marks the origin. Right: for the toy reciprocal Xi function with the first 200 zeros of $\zeta$ and the pair $60 \pm 0.3i$, the set where $\operatorname{Im}\psi > 0$ (shaded) on a grid of $481 \times 300$ points, which lies inside the upper half of $D_\gamma$ (solid); the cross is the pole $-\bar{\gamma}^2 = -3599.91 + 36i$. Drawn by `pakes_disc.py`.

Question 4.3.11. Describe $p_*(t) = 1 - \frac{2}{\pi}\delta_*(t)$ for $t < 0$. By Rodgers and Tao $p_*(t) < 1$ for every $t < 0$. Assume RH; how fast does $1 - p_*(t) \to 0$ as $t \uparrow 0$?

Heuristically, the off line zeros of $H_t$ for small $|t|$ come from nearly coincident pairs of real zeros: by the isolated pair estimate of Section 8.4, a pair at distance $D_0$ collides near $t = -D_0^2/8$ and then has imaginary parts of order $\sqrt{|t|}$, at heights where the gap is of order $\sqrt{|t|}$. Since $1 - p_*(t)$ is governed by $\max \operatorname{Im} z / \operatorname{Re} z$, the answer depends on the lowest height at which such a pair has collided, and so on the small gap statistics of the zeros. Both ingredients of an estimate are missing: de Bruijn's bound on the imaginary parts of the zeros runs only forward in time, from $H_0$ to $H_t$ with $t > 0$, and no lower bound is known for the heights of the pairs that have collided by time $t$.

### 4.3.4 The Thorin measure of $\varphi(s^p)$ and Bosch's variable

By Bondesson's theorem [56, Theorem 6.1.1], a function with value 1 at 0 is HCM exactly when it is the Laplace transform of a GGC, so Theorem 4.3.5 already makes $\varphi(s^p)$ a GGC transform for $0 < p \le p_*$.

Theorem 4.3.12. *Let $0 < p \le p_*$. Then $\varphi(s^p) = \mathbb{E}e^{-sX_p}$ for a GGC variable $X_p$ with no drift. If $p < p_*$, its Thorin measure has the density*

$$u_p(\tau) = \frac{p}{\pi\tau} \operatorname{Im} \varepsilon_\varphi\big(\tau^p e^{ip\pi}\big) = \frac{p}{2\pi\tau} \operatorname{Im}\Big[z\,\frac{\xi'}{\xi}\Big(\frac{1}{2} + z\Big)\Big]_{z=\tau^{p/2}e^{ip\pi/2}} \ge 0, \qquad \tau > 0, \tag{4.3.2}$$

*which in terms of the zeros is*

$$u_p(\tau) = \frac{p}{\pi\tau} \sum_\gamma \operatorname{Im} \frac{\tau^p e^{ip\pi}}{\tau^p e^{ip\pi} + \gamma^2}. \tag{4.3.3}$$

*If $p = p_*$, its Thorin measure is $u_{p_*}(\tau)\, d\tau$ plus a unit atom at $|\gamma|^{2/p_*}$ for every zero $\gamma$ with $\arg\gamma = -\delta_*$, counted with multiplicity, that is, one atom for each conjugate pair of zeros at the extreme angle; when $p_* = 1$ there is an atom at $\gamma^2$ for every zero, and $u_1 = 0$.*

*Proof.* Let $h = h_p$ and $\varepsilon_h(x) = p\,\varepsilon_\varphi(x^p)$. By the proof of Theorem 4.3.5, $\varepsilon_h$ is analytic on $\Omega$ and Pick. On $(0,\infty)$ it is nonnegative: a real zero contributes $x^p/(x^p+\gamma^2) > 0$ and a pair contributes $2\operatorname{Re}\frac{s}{s+\varrho^2 e^{2i\delta}} = \frac{2s(s+\varrho^2\cos 2\delta)}{|s+\varrho^2 e^{2i\delta}|^2} > 0$. Hence $\varepsilon_h$ is complete Bernstein and $\varepsilon_h(x)/x = -(\log h)'(x)$ is a Stieltjes function, $a/x + b + \int U(dt)/(x+t)$. Here $a = \varepsilon_h(0+) = 0$ and $b = \lim_{x\to\infty} \varepsilon_h(x)/x = 0$ because $\varepsilon_\varphi(s) = O(\sqrt{s}\log s)$. Integrating from 0 with $h(0) = 1$ gives $h(x) = \exp(-\int \log(1 + x/t)U(dt))$, a GGC transform. For $p < p_*$ the function $\varepsilon_h$ is continuous up to the upper edge of the cut, and Stieltjes inversion at $x = \tau e^{i\pi}$ gives $U(d\tau) = u_p(\tau)d\tau$ with $u_p(\tau) = \frac{1}{\pi\tau}\operatorname{Im}\varepsilon_h(\tau e^{i\pi})$, which is (4.3.2); (4.3.3) follows from the Hadamard form of $\varepsilon_\varphi$. For $p = p_*$ the same inversion applies away from the points $x_0 = \tau_0 e^{i\pi}$ with $\tau_0 = |\gamma|^{2/p_*}$ and $\arg\gamma = -\delta_*$, at which $x_0^{p_*} = -\gamma^2$. Near such a point the term of $\gamma$ in $\varepsilon_h$ is $p_* x^{p_*}/(x^{p_*} + \gamma^2) = x_0/(x - x_0) + O(1)$, so $\varepsilon_h(x)/x = 1/(x + \tau_0) + O(1)$, and inversion gives a unit atom at $\tau_0$. On the ray itself this term equals $p_*\tau^{p_*}/(\tau^{p_*} - |\gamma|^2)$, which is real, so $u_{p_*}$ does not see the atom. □

Corollary 4.3.13. *Let $L_p = Z_p/\tilde{Z}_p$ be the ratio of independent positive $p$ stable variables, whose density $\frac{\sin p\pi}{\pi}\frac{t^{p-1}}{t^{2p}+2t^p\cos p\pi+1}$ is Lamperti's [323]; its power $L_p^p$ is Bosch's variable $T_\alpha$ with $\alpha = p$. Under RH the Thorin measure of $X_p$ is*

$$U_p = p\sum_\gamma \mathcal{L}\big(\gamma^{2/p} L_p\big),$$

*and $X_p \overset{d}{=} S_p\, X^{1/p}$, where $X = \sum_\gamma e_\gamma/\gamma^2$ is the clock variable of Proposition 1.2.1, with independent standard exponential variables $e_\gamma$, and $S_p$ is a positive $p$ stable variable independent of $X$ with $\mathbb{E}e^{-\lambda S_p} = e^{-\lambda^p}$. At $p = 1/2$, $L_{1/2} \overset{d}{=} C^2$ for a standard Cauchy variable $C$, and $U_{1/2} = \frac{1}{2}\sum_\gamma \mathcal{L}(\gamma^4 C^2)$. As $p \uparrow 1$, $L_p \to 1$ and $U_p \to \sum_\gamma \delta_{\gamma^2}$ vaguely, the Thorin measure of $\varphi$ itself.*

*Proof.* For real $\gamma$ the term of (4.3.3) equals $\frac{p}{\pi}\frac{\gamma^2\tau^{p-1}\sin p\pi}{\tau^{2p}+2\gamma^2\tau^p\cos p\pi+\gamma^4}$; substituting $\tau = \gamma^{2/p}t$ turns it into $p$ times Lamperti's density in $t$. The subordination identity is $\mathbb{E}e^{-sS_pY} = \mathbb{E}e^{-s^pY^p}$ with $Y = X^{1/p}$. □

Bosch's variable is therefore the kernel of the Thorin measure of $\varphi(s^p)$. Formula (4.3.2) makes sense whatever the zeros are, and RH is the statement that it stays nonnegative as $p$ increases to 1 (Question 4.3.16 and the argument after it).

COROLLARY 4.3.14. *RH holds if and only if $X_p$ exists, that is $\varphi(s^p)$ is a GGC transform, for every $p \in (0,1)$. Unconditionally $X_p$ exists for $0 < p \le 1 - 1/(\pi T_0)$, in particular for $0 < p \le 1 - 1.1 \times 10^{-13}$, with the Thorin density* (4.3.2).

*Proof.* If RH fails, $p_* < 1$, and for $p \in (p_*, 1)$ the function $\varphi(s^p)$ has a pole in $\Omega$, which no GGC transform can have, since $\exp(-bs - \int \log(1+s/t)U(dt))$ is analytic and zero free on $\Omega$. The rest is Theorem 4.3.12 with Corollary 4.3.7, which gives $1 - 1/(\pi T_0) < p_*$. □

QUESTION 4.3.15. Construct $X_{1/2}$ from Brownian motion, Bessel processes or the Biane, Pitman and Yor variables [49] without reference to the zeros. A construction that survived $p \uparrow 1$ would prove RH by Corollary 4.3.14.

### 4.3.5 Positivity of the Thorin density from the primes

The method reads the threshold off the zeros: the elasticity has nonnegative imaginary part termwise on the sector $0 < \arg s < \pi - 2\delta_*$, and no information about $\delta_*$ is produced. A proof of RH along these lines must show $u_p \ge 0$ for every $p < 1$ directly from (4.3.2), for instance from the Pólya kernel, from the prime sum of Remark 4.3.17 or from a probabilistic model; this is the Thorin measure form of the reciprocal Xi programme of [421]. It cannot be done factor by factor: only the combination $(s - \frac{1}{4})\zeta(\frac{1}{2} + \sqrt{s})$ can enter an HCM or Pick argument (Section 4.2.4).

QUESTION 4.3.16. Prove $u_p \ge 0$ for every $p \in (0,1)$ directly from (4.3.2), without locating the zeros.

The statement for all $p \in (0,1)$ is equivalent to RH. Under RH it holds by Theorem 4.3.12. Conversely, suppose RH fails, let $\gamma = \varrho e^{i\delta_*}$ be a zero of maximal angle, and let $p \in (p_*, 1)$; formula (4.3.2) still defines $u_p$, except at isolated poles, although it is no longer a Thorin density. At $\tau = \varrho^{2/p}$ the term of $\bar\gamma$ in (4.3.3) equals $-\frac{p}{2\pi\tau} \cot \frac{\pi(p-p_*)}{2}$, which tends to $-\infty$ as $p \downarrow p_*$. The sum of the other terms stays bounded, because the other points $-\gamma'^2$ keep a positive distance from $-\bar\gamma^2$ and the tail of the sum is $O(\sum |\gamma'|^{-2})$. So $u_p$ takes negative values for $p$ slightly above $p_*$.

Along the ray write $\vartheta = p\pi/2 \in (0, \pi/2)$, $t = \tau^{p/2}$, $z = te^{i\vartheta}$ and $w = \frac{1}{2} + z$, so that $u_p(\tau) = \frac{p}{2\pi\tau} \operatorname{Im}[z\, \frac{\xi'}{\xi}(w)]$ by (4.3.2), and split $\xi'/\xi = \Gamma_\xi + \zeta'/\zeta$, where

$$\Gamma_\xi(w) = \frac{1}{w} + \frac{1}{w-1} - \frac{\log \pi}{2} + \frac{1}{2}\psi\Big(\frac{w}{2}\Big)$$

is the logarithmic derivative of the factor $\frac12 w(w-1)\pi^{-w/2}\Gamma(w/2)$ of $\xi$, and $\psi = \Gamma'/\Gamma$ is the digamma function, not the function $\psi$ of Theorem 4.2.4. The ray crosses the critical strip for $t < t_e = 1/(2\cos\vartheta)$ and then enters $\operatorname{Re} w > 1$.

REMARK 4.3.17 (primes). For $t > t_e$, that is $\tau > (2\cos(p\pi/2))^{-2/p}$, we have $\operatorname{Re} w > 1$ and

$$u_p(\tau) = \frac{p}{2\pi\tau}\operatorname{Im}\Big[z\Big(\frac{1}{w} + \frac{1}{w-1} - \frac{\log\pi}{2} + \frac12\psi\Big(\frac{w}{2}\Big)\Big)\Big] - \frac{p}{2\pi\tau}\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\operatorname{Im}\big(z\,n^{-z}\big),$$

with $\Lambda$ the von Mangoldt function.

PROPOSITION 4.3.18. *Let* $\operatorname{Re} w > 1$ *and* $\sigma = \operatorname{Re} w$. *Then*

$$\frac{2\pi\tau}{p}\,u_p(\tau) \;\ge\; m_p(t) := \operatorname{Im}\Big[z\Big(\frac{1}{2w} + \frac{1}{w-1} + \frac12\log\frac{w}{2\pi}\Big)\Big] - \frac{t}{6\sigma^2} - t\Big(-\frac{\zeta'}{\zeta}\Big)(\sigma). \tag{4.3.4}$$

*Every term of* $m_p$ *is explicit, no zero of* $\zeta$ *enters, and* $m_p(t) = \frac{t}{2}\sin\vartheta\log t\,(1+o(1))$ *as* $t\to\infty$. *Consequently for each* $p\in(0,1)$ *there is* $\tau_1(p) < \infty$ *with* $u_p(\tau) > 0$ *for all* $\tau > \tau_1(p)$.

*Proof.* Binet's formula [397, (5.9.13)] gives, for $\operatorname{Re} x > 0$, $\psi(x) = \log x - \frac{1}{2x} - \int_0^\infty b(u)e^{-xu}du$ with $b(u) = \frac{1}{e^u-1} - \frac1u + \frac12 \in [0, u/12]$. Hence $\psi(x) = \log x - \frac{1}{2x} - R$ with $|R| \le \frac{1}{12(\operatorname{Re} x)^2}$. With $x = w/2$ this turns $\Gamma_\xi(w)$ into $\frac{1}{2w} + \frac{1}{w-1} + \frac12\log\frac{w}{2\pi} - \frac{R}{2}$ with $|R| \le \frac{1}{3\sigma^2}$, so $\operatorname{Im}[z\Gamma_\xi(w)]$ is at least the first term of $m_p$ minus $t/(6\sigma^2)$. For the prime part, $\operatorname{Im}[z\,\zeta'/\zeta(w)] = -\sum\Lambda(n)\operatorname{Im}(zn^{-w}) \ge -t\sum\Lambda(n)n^{-\sigma}$. The asymptotic follows from $\operatorname{Im}[z\log z] = t(\sin\vartheta\log t + \vartheta\cos\vartheta)$ and $-\zeta'/\zeta(\sigma)\to 0$ as $\sigma\to\infty$. □

Table 4.2 gives the point beyond which $m_p > 0$, computed on a logarithmic grid of 50 points per decade, with $-\zeta'/\zeta(\sigma)$ evaluated directly. The entry $t_1$ is the last grid point at which $m_p \le 0$, so the sign change lies within one grid step, a factor $10^{1/50}$, above it. These values are computed, not proved: nothing controls $m_p$ between grid points or beyond the last one.

The prime term of $m_p$ can be made elementary, because $-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$ for every $\sigma > 1$. Delange [113, Théorème] proves the sharper bound $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) - 1/(2\sigma^2)$. Since $-\zeta'/\zeta(\sigma)$ enters $m_p$ with the factor $-t < 0$, replacing it by $1/(\sigma-1)$ lowers $m_p$, and (4.3.4) still holds.

REMARK 4.3.19. The ratio $t_1/t_e$ tends to 1. Near $\vartheta = \pi/2$ write $\sigma = 1+x$, so $t = t_e(1+2x)$. The dominant terms of $m_p/t$ are $\frac12\log\frac{t}{2\pi}$ and $-\frac1x$, so positivity begins at $x \approx 2/\log(t_e/2\pi)$, that is $t_1(p) = t_e(p)\big(1 + \frac{4+o(1)}{\log(t_e/2\pi)}\big)$.

So Question 4.3.16 reduces, for each $p$, to the segment $|z| \le t_1(p)$ of the ray, which is asymptotically the part inside the critical strip. That part has length $t_e = 1/(2\cos(p\pi/2)) \approx 1/(\pi(1-p))$, so heights and exponents correspond through $1-p \approx 1/(\pi T)$, as in Corollary 4.3.7.

| $p$ | strip exit $t_e = \frac{1}{2\cos\vartheta}$ | $t_1$ | $t_1/t_e$ | $\tau_1 = t_1^{2/p}$ |
|---|---|---|---|---|
| 0.5 | 0.707 | 4.07 | 5.75 | 274 |
| 0.9 | 3.20 | 11.1 | 3.47 | 210 |
| 0.99 | 31.8 | 69.6 | 2.19 | $5.29 \times 10^3$ |
| 0.999 | 318 | 553 | 1.74 | $3.10 \times 10^5$ |
| 0.9999 | 3183 | 4818 | 1.51 | $2.33 \times 10^7$ |

Table 4.2: Where the lower bound (4.3.4) becomes positive. On a logarithmic grid from $t_e$ to about $10^{14}t_e$, $m_p(t) > 0$ at every grid point $t > t_1$; the ray reaches $t_1$ at $\tau_1 = t_1^{2/p}$. The entries are computed, not proved.

### 4.3.6 The density level and natural boundaries

A variable with an HCM density is a GGC and so has an HCM Laplace transform ((F5) and (F6) of Section 4.2.1). Under RH the clock variable $X = \sum_\gamma e_\gamma/\gamma^2$ is the GGC with transform $\varphi$. Its Cauchy counterpart is $X_C = \sum_k e_k/(2k-1)^2$, with transform $\operatorname{sech}(\pi\sqrt{s}/2)$.

Theorem 4.3.20. *The density of $X_C$ is*

$$f_C(t) = \frac{4}{\pi}\sum_{k\ge1}(-1)^{k-1}(2k-1)e^{-(2k-1)^2t} = \frac{4}{\pi}\,\eta\Big(\frac{4it}{\pi}\Big)^3, \qquad t>0,$$

*with $\eta$ the Dedekind eta function. The line $\operatorname{Re} t = 0$ is a natural boundary of $f_C$. Hence $X_C$ is a GGC whose Laplace transform is HCM, but the density of $X_C$ is not HCM.*

*Proof.* The residue of $\operatorname{sech}(\pi\sqrt{s}/2)$ at $s = -(2k-1)^2$ is $\frac{4}{\pi}(-1)^{k-1}(2k-1)$, which gives the series once the Bromwich contour is moved to the left across the circles $|s| = 4k^2$: for $s = 4k^2e^{i\theta}$, $|\cosh(\pi\sqrt{s}/2)|^2 = \sinh^2(\pi k\cos\frac{\theta}{2}) + \cos^2(\pi k\sin\frac{\theta}{2})$ is at least $\frac14$ for every $\theta$ and at least $\sinh^2(\pi k/2)$ for $|\theta| \le 2\pi/3$, while $|e^{st}| \le e^{-2k^2t}$ for $2\pi/3 \le |\theta| \le \pi$, so for $t > 0$ the integrals over the arcs of these circles to the left of the Bromwich line tend to 0. Jacobi's identity $\eta(z)^3 = \sum_{n\ge0}(-1)^n(2n+1)e^{\pi i(2n+1)^2z/4}$ [174, Theorem 357], at $z = 4it/\pi$, where $e^{\pi i(2k-1)^2z/4} = e^{-(2k-1)^2t}$, gives the eta form. As a modular form $\eta$ has the real axis as natural boundary [458, Chapter 11, Section 1.4, and Chapter 1, Section 4.2], and the real axis in $z$ is the line $\operatorname{Re} t = 0$. An HCM density continues analytically to $\Omega$ by (4.2.1), so $f_C$ is not HCM. □

Question 4.3.21 (density level). Assume RH and that all zeros are simple. Does the density $f_X(t) = \sum_n A_n\gamma_n^2e^{-\gamma_n^2t}$ of $X$, with $A_n$ as in (4.2.3), have $\operatorname{Re} t = 0$ as a natural boundary? If so, $X$ is not HCM.

Pólya's gap theorem for Dirichlet series [445, 451] says that if the exponents $\mu_n$ of a Dirichlet series $\sum_n c_n e^{-\mu_n t}$ have density $d = \lim n/\mu_n$ and satisfy $\inf_n(\mu_{n+1} - \mu_n) > 0$, then every segment of length greater than $2\pi d$ on the line of convergence carries a singularity. When $\limsup \log n/\mu_n = 0$ the abscissa of convergence equals $\limsup_n \log|c_n|/\mu_n$ [367].

THEOREM 4.3.22. *Assume RH and that all zeros are simple, and assume*

*(G)* $\inf_n(\gamma_{n+1}^2 - \gamma_n^2) > 0$;

*(D)* $\log\bigl(1/|\zeta'(\frac12 + i\gamma_n)|\bigr) = o(\gamma_n^2)$.

*Then for* $t > 0$

$$f_X(t) = \sum_n c_n e^{-\gamma_n^2 t}, \qquad c_n = A_n\gamma_n^2 = -\frac{2\gamma_n\,\Xi(0)}{\Xi'(\gamma_n)},$$

*the series has abscissa of convergence* 0, *and* $\operatorname{Re} t = 0$ *is a natural boundary of* $f_X$. *In particular the density of* $X$ *is not HCM.*

*Proof.* Residue expansion. The residue of $\varphi(s) = \Xi(0)/\Xi(i\sqrt{s})$ at $s = -\gamma_n^2$ is $c_n$ as stated. Write $f_X$ as a Bromwich integral and move the contour to the left across the arcs of the circles $|s| = T_k^2$ that lie to the left of the Bromwich line, where, for large $k$, $T_k \in [k, k+1]$ is chosen at distance at least $c/\log k$ from every ordinate, with a small absolute constant $c > 0$; this is possible because $[k-1, k+2]$ contains $O(\log k)$ ordinates [176]. Then $\zeta'/\zeta(\sigma + iy) = O(\log^2 k)$ uniformly in $-1 \le \sigma \le 2$ and $|y - T_k| \le c/(2\log k)$ [498, Theorem 9.6(A) and Section 14.4], and integration from $\sigma = 2$ gives $1/|\zeta(\sigma + iy)| \le \exp(O(\log^2 k))$ there. On the part of such an arc within angle $O(1/T_k)$ of the negative axis, $w = \frac12 + \sqrt{s}$ has $\frac12 \le \operatorname{Re} w \le 2$ and $|\operatorname{Im} w| = \sqrt{T_k^2 - (\operatorname{Re} w - \frac12)^2} = T_k + O(1/T_k)$, so for large $k$ it lies in this band or in its reflection in the real axis, and Stirling's formula gives $1/|\Xi(i\sqrt{s})| \le \exp(O(T_k \log T_k))$, while $|e^{st}| \le e^{-tT_k^2/2}$. On the rest of the arc $\operatorname{Re} w \ge 2$, and Stirling's formula gives $|e^{st}\varphi(s)| \le \exp\bigl(tT_k^2\cos\theta - \frac{T_k}{2}\cos\frac{\theta}{2}\log T_k + O(T_k)\bigr)$ for $s = T_k^2 e^{i\theta}$. For $t > 0$ both bounds tend to zero faster than any power of $T_k$, so the arcs contribute nothing in the limit and $f_X$ equals the residue series.

Abscissa. The exponents $\gamma_n^2$ satisfy $\log n/\gamma_n^2 \to 0$ and have density zero. By Stirling, $|\Xi'(\gamma_n)| = |\xi'(\rho_n)| = \gamma_n^{O(1)} e^{-\pi\gamma_n/4}|\zeta'(\rho_n)|$, so $\log|c_n| = \frac{\pi}{4}\gamma_n + O(\log\gamma_n) + \log(1/|\zeta'(\rho_n)|)$. Since $|\zeta'(\rho_n)| \le \gamma_n^{O(1)}$ always, (D) gives $\log|c_n|/\gamma_n^2 \to 0$, so the abscissa of convergence is 0.

Natural boundary. By (G) and density zero, Pólya's theorem places a singularity in every segment of the imaginary axis. An HCM density continues analytically to $\Omega$, which contains the imaginary axis minus the origin. □

The same argument, with $\mu_k = (2k-1)^2$, gaps $8k$ and $c_k = \frac{4}{\pi}(-1)^{k-1}(2k-1)$, gives a second proof of Theorem 4.3.20 that does not use modularity: by Pólya's gap theorem [445, 451] the line $\operatorname{Re} t = 0$ is a natural boundary of $f_C$.

REMARK 4.3.23 (status of the hypotheses). Hypothesis (D) is far weaker than the conjectured polynomial lower bounds for $|\zeta'(\rho)|$. Hypothesis (G) asks $\gamma_{n+1}-\gamma_n \geq c_0/(\gamma_n+\gamma_{n+1})$ for some $c_0 > 0$, a gap of order $1/\gamma$ against a mean spacing of order $1/\log\gamma$. Under the GUE model the expected number of violations is $\sum_n(\log\gamma_n/\gamma_n)^3 < \infty$, so by the Borel Cantelli heuristic (G) fails at most finitely often, and then holds with a smaller $c_0$ if the zeros are simple.

Numerically, for the first 200 zeros the coefficients $c_n$ computed from $\Xi'(\gamma_n)$ alternate in sign without exception; $\log|c_n|/\gamma_n^2$ equals 0.0462, 0.0144, 0.00524, 0.00322, 0.00195 at $n = 1, 10, 50, 100, 200$, tracking $\pi/(4\gamma_n)$; and the smallest gap $\gamma_{n+1}^2-\gamma_n^2$ is 159, at $n = 4$. The $c_n$ are the residues of the spectral expansion of the clock density in Section 8.1, written $R_k$ there, and Theorem 4.3.22 describes, under its hypotheses, why that expansion cannot be continued past $t = 0$.

For small $p$ the density of $X_p$ escapes this obstruction. The transform $\varphi(s^p)$ is analytic on the wider sector $|\arg s| < (\pi-2\delta_*)/p$ and, by Stirling's formula applied to $\xi(\frac{1}{2}+z)$ for $|\arg z| < \pi/2$, decays along every ray strictly inside it. The inverse Laplace transform then continues the density of $X_p$ to the sector $|\arg t| < (\pi-2\delta_*)/p - \pi/2$, which contains $\Omega$ when $p \leq 2p_*/3$.

QUESTION 4.3.24. Determine $\{p:\ X_p \text{ is HCM}\}$, and the same set for the Cauchy family $X_{C,p}$ with transform $\operatorname{sech}(\pi s^{p/2}/2)$.

Two facts constrain the answer. Under RH, $X_p \overset{d}{=} S_p X^{1/p}$, and positive $p$ stable variables are HCM exactly for $p \leq 1/2$, a conjecture of Bondesson [56, Sections 5.7, 5.8 and 7.2] proved by Bosch and Simon [59, p. 32]. Since HCM is preserved under products of independent variables and under powers of modulus at least one ((F1) and (F2) of Section 4.2.1), HCM of $X$ would give HCM of $X_p$ for $p \leq 1/2$; but Theorem 4.3.22 shows that $X$ is not HCM under (G) and (D), so this route is closed unless those hypotheses fail. The elasticity criterion reduces Question 4.3.24 to the absence of zeros of the density $f$ of $X_p$ in $\Omega$ and the Pick property of $-tf'(t)/f(t)$; by the continuation argument before Question 4.3.24, $f$ is at least analytic on $\Omega$ for small $p$. A numerical scan of this elasticity for $X_{C,1/2}$ would be the first test.

### 4.3.7 THE MELLIN VARIABLE $W$ AND THE FIRST GAP

Question 4.2.7 asks for which $\beta$ the density of $W^\beta$ is HCM; on the Cauchy side the answer is $|\beta| \geq 2$. Let $\chi(t) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+t) = \prod_\gamma(1+t^2/\gamma^2)^{-1}$ and let $g(y) = \frac{1}{\pi}\int_0^\infty \chi(t)\cos(ty)\,dt$ be the function $g$ of Section 4.2.2, which is entire because $\chi(t) = \exp(-\frac{1}{2}|t|\log|t|(1+o(1)))$. By the van Dantzig pair of Section 1.2, recalled in Section 4.2.2, $\chi$ is a characteristic function without RH: the clock variable $T$ exists unconditionally [424, Theorem A], and if $Z$ is a standard normal variable independent of $T$ then $\mathbb{E}e^{itZ\sqrt{2T}} = \mathbb{E}e^{-t^2T} = \chi(t)$. So $g$ is the density of the normal scale mixture $Z\sqrt{2T}$, and $W = \exp(Z\sqrt{2T})$ exists without any

hypothesis. By the elasticity criterion, HCM of $W^\beta$ becomes a sign condition on $-g'/g$ in a strip (Proposition 4.3.26), and since $g$ is entire no natural boundary stands in the way.

#### *4.3.7.1 A lower bound for g*

Theorem 4.3.25 proves the existence of $W$ a second time, independently of [424, Theorem A].

THEOREM 4.3.25. *Unconditionally, by the verification of RH to height $T_0$ [419], $g(y) \geq 0.9998\, g_R(y) > 0$ for all real $y$, where $g_R$ is the density of $\sum_{\gamma \text{ real}} L_\gamma/\gamma$. Hence $\chi$ is a characteristic function and there is a positive random variable $W$ with $\mathbb{E}W^{it} = \chi(t)$ and $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1$.*

*Proof.* Split $\chi = \chi_R \chi_{\text{off}}$ into the products over real zeros and over conjugate pairs of nonreal zeros. For a pair $\gamma = \varrho e^{i\delta} = a + ib$ with $a, b > 0$, the factor $(1 + t^2/\gamma^2)^{-1}(1 + t^2/\bar\gamma^2)^{-1}$ is the Fourier transform of the even kernel

$$k_\gamma(x) = \frac{\varrho}{2\sin 2\delta}\, e^{-a|x|} \sin(b|x| + \delta),$$

of total mass one, positive for $|x| < (\pi - \delta)/b$ and bounded by $\frac{\varrho}{2\sin 2\delta} e^{-a|x|}$. List the pairs as $\gamma^{(1)}, \gamma^{(2)}, \dots$ in order of $\operatorname{Re}\gamma$, write $k_j = k_{\gamma^{(j)}}$, and let $\chi_m$ be the product of the first $m$ pair factors, $\nu_m = k_1 * \cdots * k_m$ and $g_m = g_R * \nu_m$, the function with Fourier transform $\chi_R \chi_m$. We bound $g_m$ uniformly in $m$ and let $m$ grow at the end.

Comparison. $g_R = \ell * \tilde g_R$, where $\ell(z) = \frac{\gamma_1}{2} e^{-\gamma_1|z|}$ is the density of $L_{\gamma_1}/\gamma_1$ and $\tilde g_R$ is the density of the sum over the other real zeros, so $e^{-\gamma_1|x|} g_R(y) \leq g_R(y - x) \leq e^{\gamma_1|x|} g_R(y)$. Writing $\nu_m = \nu_m^+ - \nu_m^-$,

$$g_m(y) \geq g_R(y)\Big( \int e^{-\gamma_1|x|} \nu_m(x)dx - 2\int \cosh(\gamma_1 x)\, \nu_m^-(x)dx \Big) =: g_R(y)(A_m - 2B_m).$$

The term $A_m$. By Parseval, $A_m = \frac{1}{\pi}\int \chi_m(t) \frac{\gamma_1}{\gamma_1^2 + t^2} dt = \mathbb{E}\, \chi_m(\gamma_1 C)$ with $C$ a standard Cauchy variable. For $s \geq 0$ each pair factor is $|1 + s/\gamma^2|^{-2} \geq (1 + s/\varrho^2)^{-2} \geq e^{-2s/\varrho^2}$, so $A_m \geq \mathbb{E}e^{-\kappa C^2} = e^\kappa \operatorname{erfc}(\sqrt\kappa)$ with $\kappa = 2\gamma_1^2 \Sigma_{\text{off}}$ and $\Sigma_{\text{off}} = \sum_{\text{off}} |\gamma|^{-2}$. All nonreal zeros have $\operatorname{Re}\gamma > T_0 = 3 \times 10^{12}$ [419], so with $N(T) \leq T \log T$ for $T \geq T_0$, $\Sigma_{\text{off}} \leq 2\int_{T_0}^\infty N(T) T^{-3} dT \leq 2(\log T_0 + 1)/T_0 < 2 \times 10^{-11}$, $\kappa < 8 \times 10^{-9}$ and $A_m > 0.99989$.

The term $B_m$. For even signed measures $\int \cosh(\gamma_1 x)\, d(|\mu| * |\nu|) = \int \cosh(\gamma_1 x) d|\mu| \cdot \int \cosh(\gamma_1 x) d|\nu|$ exactly, and $(\mu * \nu)^- \leq \mu^+ * \nu^- + \mu^- * \nu^+$. Hence, with $M(\cdot) = \int \cosh(\gamma_1 x)\, d(\cdot)$ and since $M(|k_\gamma|) \geq M(k_\gamma) = |1 - \gamma_1^2/\gamma^2|^{-2} \geq 1$,

$$B_m \leq \sum_{j \leq m} M(k_j^-) \prod_{i \leq m,\ i \neq j} M(|k_i|) \leq SP, \qquad S = \sum_\gamma M(k_\gamma^-), \quad P = \prod_\gamma M(|k_\gamma|),$$

with the sum and the product over all pairs. Here $k_\gamma^-$ lives on $|x| \geq (\pi-\delta)/b$ and $M(k_\gamma^-) \leq \frac{\varrho}{\sin 2\delta\,(a-\gamma_1)} \exp\bigl(-(a-\gamma_1)\frac{\pi-\delta}{b}\bigr)$. As $\varrho/\sin 2\delta = \varrho^3/(2ab)$, this bound is $\frac{\varrho^3}{2a(a-\gamma_1)}\, b^{-1}e^{-K/b}$ with $K = (a-\gamma_1)(\pi-\delta)$. For fixed $a$ it increases with $b$: $\varrho$ increases, $K$ decreases because $\delta = \arctan(b/a)$ increases, and $b \mapsto b^{-1}e^{-K/b}$ increases on $(0, K)$. So, as $b < \frac{1}{2}$, it is at most

$$F(a) = \frac{(a^2+\frac{1}{4})^{3/2}}{a(a-\gamma_1)} \exp\Bigl(-2(a-\gamma_1)\bigl(\pi - \arctan\tfrac{1}{2a}\bigr)\Bigr),$$

and $F(T_0) < \exp(-1.88\times 10^{13})$, while $-(\log F)' > 6$ on $[T_0, \infty)$. Every pair has $a > T_0$, so Stieltjes integration against $N(T) \leq T\log T$ gives

$$S \leq F(T_0)\Bigl(T_0\log T_0 + \frac{\log T_0 + 1}{6} + \frac{1}{36T_0}\Bigr) < \exp(-1.8\times 10^{13}).$$

Moreover $M(|k_\gamma|) = M(k_\gamma) + 2M(k_\gamma^-)$ and $M(k_\gamma) \leq \exp(3\gamma_1^2/\varrho^2)$, so $\log P \leq 3\gamma_1^2\Sigma_{\text{off}} + 2S < 1.2\times 10^{-8}$. So $A_m - 2B_m > 0.9998$ for every $m$.

Limit. For real $t$ each pair factor lies in $(0,1]$, and $\chi_R$ is integrable because it contains the factor $(1+t^2/\gamma_1^2)^{-1}$. By dominated convergence in the Fourier inversion integral, $g_m \to g$ uniformly on $\mathbb{R}$, so $g \geq 0.9998\, g_R$. The comparison also gives $g_m(y) \leq g_R(y)\int e^{\gamma_1|x|}|\nu_m(x)|dx \leq 2P\, g_R(y)$, since $e^{\gamma_1|x|} \leq 2\cosh(\gamma_1 x)$ and $|\nu_m| \leq |k_1| * \cdots * |k_m|$, so $g \leq 2P\, g_R$. Hence $g$ is integrable, its Fourier transform is $\chi$, and $\chi(0) = 1$ makes $g$ a probability density. Finally $\int e^{cy} g_R(y)dy = \prod_{\gamma\ \text{real}}(1 - c^2/\gamma^2)^{-1} < \infty$ for $|c| < \gamma_1$, so $\int e^{c|y|}g(y)dy < \infty$ there, $\mathbb{E}W^z = \int e^{zy}g(y)dy$ is analytic on $|\operatorname{Re} z| < \gamma_1$, and it equals $\Xi(0)/\Xi(z)$ there because the two agree on the imaginary axis. □

The proof uses only that the zeros with $0 < \operatorname{Re}\gamma \leq T_0$ are real. The pair kernel formula and the constants of the proof were also checked numerically (`code/README.md`).

#### *4.3.7.2 Which powers of W are HCM*

For real $\beta \neq 0$ the density of $W^\beta$ is $x^{-1}g(\log x/\beta)/|\beta|$. With $y = \log x$, its elasticity is $1 - \frac{1}{\beta}\frac{g'}{g}(y/\beta)$, and $x \in \mathbb{C}^+$ corresponds to $0 < \operatorname{Im} y < \pi$; for $\beta < 0$ the point $u = y/\beta$ then lies in the lower half of the strip $|\operatorname{Im} u| < \pi/|\beta|$, where the factor $\frac{1}{\beta} < 0$ reverses the sign condition, and $g(\bar u) = \overline{g(u)}$, as $g$ is real on $\mathbb{R}$, moves it to the upper half. Theorem 4.3.2 therefore gives the following.

Proposition 4.3.26. *Let $\beta \in \mathbb{R}\setminus\{0\}$. Then $W^\beta$ is HCM if and only if $g$ has no zeros in the strip $|\operatorname{Im} u| < \pi/|\beta|$ and $\operatorname{Im}\bigl(-g'(u)/g(u)\bigr) \geq 0$ for $0 < \operatorname{Im} u < \pi/|\beta|$.*

As $\operatorname{Im} u \downarrow 0$ the condition requires $(\log g)'' \leq 0$ on $\mathbb{R}$; numerically $(\log g)'' \leq -2.2$ on $[0,1]$. On the Cauchy side $g_C = 1/(\pi\cosh u)$, $-g_C'/g_C = \tanh u$, which

is Pick on $0 < \operatorname{Im} u < \pi/2$ with a pole at $i\pi/2$; Proposition 4.3.26 gives $|\beta| \geq 2$, Bosch's theorem once more.

THEOREM 4.3.27. *Assume RH, simple zeros, and* $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$. *Then* $u_0 = \frac{\pi}{4} + \limsup_n \gamma_n^{-1} \log(1/|\zeta'(\rho_n)|)$ *is finite, and for* $\operatorname{Re} u > u_0$

$$g(u) = -\Xi(0) \sum_n \frac{e^{-\gamma_n u}}{\Xi'(\gamma_n)} = \sum_n b_n e^{-\gamma_n u}, \qquad b_1 > 0 > b_2,$$

*and if* $W^\beta$ *is HCM then* $|\beta| \geq \gamma_2 - \gamma_1 = 6.8873\ldots$.

*Proof.* The expansion is the residue computation for the Fourier integral $g(u) = \frac{1}{2\pi} \int \chi(t) e^{-itu}\, dt$ closed in the lower half plane. We move the line of integration to $\operatorname{Im} t = -T_k$, where $T_k \in [k, k+1]$ are heights at which $1/\zeta(\sigma \pm iT_k) = T_k^{O(1)}$ uniformly in $-1 \leq \sigma \leq 2$ [498, Theorem 9.7]; by Stirling's formula and the functional equation the moved integral is $O(e^{(\pi/4 - \operatorname{Re} u)T_k + O(\log T_k)})$, which tends to 0 for $\operatorname{Re} u > \pi/4$, so there $g(u)$ is the limit of the sums of the residue terms over $\gamma_n < T_k$. By Stirling, $\log|b_n| = \frac{\pi}{4}\gamma_n + \log(1/|\zeta'(\rho_n)|) + O(\log \gamma_n)$, so the series converges absolutely for $\operatorname{Re} u > u_0$, and $u_0 \geq \pi/4$ because $|\zeta'(\rho_n)| \leq \gamma_n^{O(1)}$, so its sum is $g(u)$ there (conjecturally $u_0 = \pi/4$); the signs come from $b_n = \frac{\gamma_n}{2} A_n$. Put $\Delta = \gamma_2 - \gamma_1$. As $\operatorname{Re} u \to \infty$ with $\operatorname{Im} u = v$ fixed,

$$\operatorname{Im}\Big(-\frac{g'}{g}\Big)(x + iv) = \Delta \Big|\frac{b_2}{b_1}\Big| e^{-\Delta x} \sin(\Delta v)\, \big(1 + O(e^{-cx})\big), \qquad c > 0.$$

For $\pi/\Delta < v < 2\pi/\Delta$ this is negative for large $x$. If $|\beta| < \Delta$ the strip $0 < \operatorname{Im} u < \pi/|\beta|$ contains such $v$, and Proposition 4.3.26 fails. □

QUESTION 4.3.28 (first gap conjecture). Is $W^\beta$ HCM exactly when $|\beta| \geq \gamma_2 - \gamma_1$? On the Cauchy side the analogous statement, with frequencies $2k-1$, is Bosch's theorem at $\alpha = 1/2$: $3 - 1 = 2$.

With $b_n$ computed from $\Xi'(\gamma_n)$ ($b_1 = 359.52$, $b_2 = -28007$, $b_3 = 392918$) and $g$ computed by quadrature near the origin, a scan of 4840 points with $0 \leq x \leq 6$ and $0 < v < 0.99\, v_*$, where $v_* = \pi/\Delta = 0.45614$, found no value of $\operatorname{Im}(-g'/g)$ below $-1.3 \times 10^{-15}$, which is rounding; in 40 digit arithmetic the minimum over $1.1 \leq x \leq 9.1$ at $v = 0.999\, v_*$ is $+1.0 \times 10^{-27}$. Just above the threshold the sign flips: the minimum is $-6.5 \times 10^{-8}$ at $v = 1.001\, v_*$, so numerically the threshold is sharp to 0.1 per cent.

Proving sufficiency requires the Pick property of $-g'/g$ in the full strip $0 < \operatorname{Im} u < \pi/\Delta$, which near the origin is a statement about the Pólya type kernel $g$ that does not reduce to the zeros.

#### *4.3.7.3 Solvable models*

To see whether the first gap rule is a coincidence of the half Cauchy case, we test it on a family in which everything is explicit. The logistic variable $\log W$ has characteristic function $\pi t/\sinh(\pi t) = \prod_{k\geq 1}(1+t^2/k^2)^{-1}$. Removing a finite set $R$ of frequencies multiplies the characteristic function by $\prod_{k\in R}(1+t^2/k^2)$, which gives $g = \prod_{k\in R}(1-D^2/k^2)L$ with $D = d/dy$ and $L(y) = \frac{1}{4}\operatorname{sech}^2(y/2)$. For each model we locate numerically the largest strip on which $\operatorname{Im}(-g'/g) \geq 0$; by Theorem 4.3.2, as in Proposition 4.3.26, its width $h_*$ gives the HCM threshold $\beta_* = \pi/h_*$. Table 4.3 lists the results.

| frequencies | gaps | first gap | largest gap | $\beta_* = \pi/h_*$ |
|---|---|---|---|---|
| $1, 2, 3, \ldots$ | $1, 1, 1, \ldots$ | 1 | 1 | 1.000 |
| $1, 3, 5, \ldots$ (half Cauchy) | $2, 2, 2, \ldots$ | 2 | 2 | 2 (Bosch) |
| $1, 3, 4, 5, \ldots$ | $2, 1, 1, \ldots$ | 2 | 2 | 2.000 |
| $1, 3, 5, 6, \ldots$ | $2, 2, 1, \ldots$ | 2 | 2 | 2.000 |
| $1, 3, 4, 6, \ldots$ | $2, 1, 2, 1, \ldots$ | 2 | 2 | 2.000 |
| $1, 4, 5, 6, \ldots$ | $3, 1, 1, \ldots$ | 3 | 3 | 3.000 |
| $1, 4, 5, 7, \ldots$ | $3, 1, 2, 1, \ldots$ | 3 | 3 | 3.000 |
| $1, 4, 7, 8, \ldots$ | $3, 3, 1, \ldots$ | 3 | 3 | 3.000 |
| $1, 5, 6, 7, \ldots$ | $4, 1, 1, \ldots$ | 4 | 4 | 4.000 |
| $1, 2, 4, 5, \ldots$ | $1, 2, 1, \ldots$ | 1 | 2 | 1.269 |
| $1, 2, 3, 6, \ldots$ | $1, 1, 3, 1, \ldots$ | 1 | 3 | 1.317 |
| $1, 2, 5, 6, \ldots$ | $1, 3, 1, \ldots$ | 1 | 3 | 1.483 |

Table 4.3: HCM threshold for powers of $W$ in logistic models with frequencies removed, computed as $\beta_* = \pi/h_*$ from the width $h_*$ of the largest strip on which $\operatorname{Im}(-g'/g) \geq 0$. The half Cauchy row is Bosch's theorem, not a computation.

Whenever the first gap is at least as large as every later gap, the computed threshold equals the first gap, to the resolution of the computation. When some later gap exceeds the first, the threshold lies strictly between the first gap and the largest gap, and depends on where the large gap sits. The asymptotic argument of Theorem 4.3.27 explains the lower bound in all cases. For the Riemann zeros the first gap $\gamma_2 - \gamma_1 = 6.8873$ exceeds every later gap among the first 1000 zeros (the next largest is $\gamma_4 - \gamma_3 = 5.414$). The explicit form of the Riemann von Mangoldt formula of Hasanalizade, Shen and Wong, $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac{7}{8}| \leq 2.5167 + \frac{1}{50T}$ for $e \leq T \leq X_0 = 30\,610\,046\,000$ and $\leq 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}$ for $T > X_0$ [176, (1.7), (2.7) and Corollary 1.4], shows that every interval $(T, T+6.887]$ with $T \geq 1000$ contains an ordinate: the lower bound it gives for the number of ordinates there exceeds 0.52 at $T = 1000$, increases with $T$ while $T + 6.887 \leq X_0$, and exceeds 11 beyond. So every gap that starts above height 1000 is at most 6.887, less than the first, and every gap that starts below it is a gap between two

of the first 650 zeros, since $N(1000) = 649$. The first gap therefore exceeds every later gap. This refines Question 4.3.28.

Question 4.3.29 (first gap conjecture, refined). Let $Y = \sum_n L_n/\lambda_n$ with independent standard Laplace variables $L_n$ and $0 < \lambda_1 < \lambda_2 < \cdots$, and let $W = e^Y$. If the characteristic function of $Y$ decays faster than every exponential, or if the first gap $\lambda_2 - \lambda_1$ is at least every later gap, is the HCM threshold of $W^\beta$ exactly $\lambda_2 - \lambda_1$? In particular, is it $\gamma_2 - \gamma_1$ for the Riemann variable?

### 4.3.8 Where the difficulty lies

Bosch's theorem cannot be the input to a proof that the reciprocal Xi function is HCM, because for a product over the zeros HCM is equivalent to the reality of the zeros (Proposition 4.2.6), and Section 11.1 finds that every GGC criterion in the books of Sato and Bondesson is, at the centre, equivalent to RH. Bosch's method does transfer: at each of three levels it isolates one concrete object on which the question turns.

At the level of transforms the object is the Thorin density $u_p$ of (4.3.2). Its positivity for every $p < 1$ is equivalent to RH; the primes prove it on each ray beyond some point, that point is asymptotically where the ray leaves the critical strip, and what remains is a segment of length about $1/(\pi(1-p))$. The first direction is to prove positivity on that segment from the explicit formula for $\xi'/\xi$ rather than from the zeros; computation alone extends the unconditional range $p \le 1 - 1/(\pi T_0)$ only as fast as the verification height $T_0$ grows.

At the level of the Mellin variable the object is the sign of $\operatorname{Im}(-g'/g)$ in the strip $0 < \operatorname{Im} u < \pi/(\gamma_2 - \gamma_1)$. The second direction is to prove the first gap conjecture, first in the logistic models of Table 4.3, where every quantity is elementary, and then for $W$, whose threshold the scan after Question 4.3.28 already locates to 0.1 per cent. Question 4.2.8 remains open.

At the level of densities the object is the natural boundary. The third direction is to establish it for the clock variable without the hypotheses (G) and (D) of Theorem 4.3.22, which would show what replaces, on the Riemann side, the modularity that produces the natural boundary on the Cauchy side.

Section 5.1 approaches the Thorin functional $\sum_\gamma \delta_{\gamma^2}$ of the clock, its Thorin measure under RH, from the other end of Bondesson's hierarchy, through its Laplace transform, the heat trace $W(t) = \sum_\gamma e^{-\gamma^2 t}$.

## 4.4 Exercises

*On Section 4.1.*

4.1. Show that if $X_N$ are GGCs converging in law to a proper limit, the Thorin measures converge vaguely on $(0, \infty)$, and use the laws Gamma$(n, n)$ of

Section 1.3.1, which converge to the point mass at one, to show that Thorin mass escaping to infinity can reappear in the limit as drift.

4.2. For the sinh clock of Section 1.3.4 with one atom $k^2$ replaced by the conjugate pair $a, \bar{a}$, $a = -(\delta + ik)^2$ with $0 < \delta < \frac{1}{2}$, as in Proposition 1.2.3, construct the Thorin densities at resolution $\varepsilon$ and show that their vague limit is the sinh measure with the atom $k^2$ removed.

4.3. Verify that $\varphi_{1/2}(uv)\varphi_{1/2}(u/v) = \prod_k a_k^2/(a_k^2 + a_k uw + u^2)$ for the clock, and that at $u = |a|$ an off line pair contributes the completely monotone factor $(w + 2\cos\arg a)^{-2}$.

4.4. Compute $(-1)^k \partial_w^k H_u(w)/H_u(w)$ for $k \le 6$ at $u = 1$ and $w = 3$ from the Pólya density and compare with Figure 4.1.

# Chapter 5

# The heat trace, the arithmetic and the limits of finite information

Chapter 4 worked at the top of the Bondesson hierarchy; here we climb it from the bottom. Every rung is a sign condition on the heat trace $W(t)$ of Section 1.2, the pairing of the Thorin functional with $e^{-tz}$ (not the Mellin variable $W$ of Section 4.3). The lower rungs are unconditional theorems, proved with a finite verification of the zeros; the top rung is RH.

## 5.1 The heat trace and the hierarchy of positivity

The rungs of the hierarchy proved unconditionally are infinite divisibility and self decomposability of the clock, log convexity of $W$, positive definite Hankel blocks up to size $3 \times 10^7$, the signs of all derivatives, equivalently the positivity of the Williamson measures, up to order $1.16 \times 10^{25}$, and near the origin the signs of every derivative: $W$ is completely monotone on $(0, 1.29]$. One scheme proves them all, with a reach in size and order set by the height to which RH has been verified; no verification reaches the top, which is where Section 5.2 begins. The results are from [424, Chapters 4 and 5] and [421], except those stated without a citation, which are proved here.

### 5.1.1 The heat trace and the first rungs

By Section 1.2, unconditionally,

$$\log G(u) = \int_0^\infty \bigl(1 - e^{-ut}\bigr) \frac{W(t)}{t}\, dt, \qquad W(t) = \sum_j e^{-a_j t},$$

where $a_j = -(\rho_j - \frac{1}{2})^2$ and the sum runs over one zero of each pair $\{\rho, 1-\rho\}$. So $W(t)/t$ is the only candidate for the Lévy density of the clock, and the classes of Section 1.3 become conditions on $W$: nonnegativity for infinite divisibility, monotonicity for self decomposability, complete monotonicity for the GGC property.

The function $W$ is analytic in $\operatorname{Re} t > 0$, real on the positive axis, and can be differentiated term by term,

$$c_k(t) = (-1)^k W^{(k)}(t) = \sum_j a_j^k e^{-a_j t}.$$

Under RH every $c_k(t)$ is a sum of positive terms. A quadruple of zeros $\frac12 \pm \delta \pm i\gamma$, $0 < \delta < \frac12$, instead contributes the exponents $a = -(\delta + i\gamma)^2 = \gamma^2 - \delta^2 - 2i\delta\gamma$ and $\bar a$, hence the oscillating term $2|a|^k e^{-\operatorname{Re} a\, t}\cos(2\delta\gamma t - k\phi)$, with $|a| = \gamma^2 + \delta^2$ and $\tan\phi = 2\delta\gamma/(\gamma^2 - \delta^2)$. Its amplitude exceeds the contribution $2\gamma^{2k}e^{-\gamma^2 t}$ of the same two zeros placed at $\frac12 + i\gamma$ by the factor $(1+\delta^2/\gamma^2)^k e^{\delta^2 t} \le \exp\{k\delta^2/\gamma^2 + \delta^2 t\}$ [421, Proposition 11.1].

As $t \downarrow 0$ the archimedean term of (1.2.5) dominates. Write $\vartheta$ for the Riemann Siegel theta function. Since $\operatorname{Re}\psi(\frac14 + \frac{iu}{2}) - \log\pi = 2\vartheta'(u)$, the explicit formula gives, with no hypothesis,

$$W(t) = e^{t/4} + \frac{1}{\pi}\int_0^\infty e^{-u^2 t}\vartheta'(u)\,du - \frac{1}{\sqrt{4\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt n}\,e^{-(\log n)^2/4t},$$

and the prime terms are $O(t^{-1/2}e^{-(\log 2)^2/4t})$, smaller than every power of $t$. Mellin's method for the integral [397, Section 2.5] gives, for every $N$ and as $t \downarrow 0$,

$$W(t) = \frac{\log\frac1t - \gamma_E - 4\log 2 - 2\log\pi}{8\sqrt{\pi t}} + \sum_{n=0}^{N} d_n t^n + \sum_{k=1}^{N} b_k t^{k-\frac12} + O\big(t^{N+\frac12}\big),$$

with $d_n = (1 - \frac{E_{2n}}{8})\frac{4^{-n}}{n!}$ and $b_k = \frac{(1-2k)\Gamma(\frac12 - k)}{2\pi}\beta_k$, where $\gamma_E$ is Euler's constant, $E_{2n}$ are the Euler numbers and $\beta_k = \frac{(1-2^{1-2k})|B_{2k}|}{4k(2k-1)}$ are the coefficients of Stirling's series $\vartheta(u) \sim \frac{u}{2}\log\frac{u}{2\pi} - \frac{u}{2} - \frac{\pi}{8} + \sum_k \beta_k u^{1-2k}$. The terms $b_k t^{k-\frac12}$ come from the expansion of $\vartheta'$ at infinity, and the terms $d_n t^n$ from the Taylor series of $e^{-u^2 t}$ through the Hurwitz value $\zeta(-2n, \frac14) = E_{2n}/4^{2n+1}$, together with the coefficients $4^{-n}/n!$ of $e^{t/4}$. After the leading term the series is $\frac78 + \frac{\sqrt t}{48\sqrt\pi} + \frac{9}{32}t - \frac{7t^{3/2}}{2880\sqrt\pi} + \frac{3}{256}t^2 + \frac{31t^{5/2}}{60480\sqrt\pi} + \frac{69}{3072}t^3 + \cdots$, and it diverges; $\frac78 = d_0$ is the constant of the Riemann von Mangoldt formula. The error of the two term expansion (the leading term and $\frac78$) is $b_1\sqrt t + d_1 t + O(t^{3/2})$, which is $6.5\times10^{-4}$, $1.5\times10^{-4}$ and $4.0\times10^{-5}$ at $t = 10^{-3}, 10^{-4}, 10^{-5}$ [421, Corollary 5.5]; with the terms through $t^{5/2}$ the remainder agrees with $\frac{69}{3072}t^3$ to the order of the next term, $t^{7/2}$, in a 40 digit computation for $10^{-6} \le t \le 10^{-3}$, and at $t = 10^{-2}$ it is $-8.377\times10^{-6}$, the prime 2. Consequently $\int_\tau^\infty W(t)\,dt/t = \frac{\log\frac1\tau - \gamma_E - 4\log 2 - 2\log\pi - 2}{4\sqrt{\pi\tau}} + \frac78\log\frac1\tau + O(1)$ as $\tau \downarrow 0$, so the clock has infinitely many jumps. As $t \to \infty$ the first zero dominates, $W(t) = e^{-\gamma_1^2 t}(1 + o(1))$ with $\gamma_1^2 = 199.7904548$. In between, the arithmetic terms cancel to a depth that grows linearly in $t$ (Table 1.4).

THEOREM 5.1.1 ([424, Theorem A]). *Unconditionally, that is, using only the verification of RH below height* 1000 *[419],* $W(t) > 0$ *and* $W'(t) < 0$ *for every* $t > 0$. *Consequently the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density* $W(t)/t$, *and* $(\Xi(\theta)/\Xi(0),\ \xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta))$ *is a van Dantzig pair in the sense of Section 1.2.*

*Proof.* The proof is that of [424, Theorem A], with the constants made explicit. Let $\tau = 1/20000$, a constant of this section only, $\ell = -\frac{1}{2}\log t$, and $s_k(t) = \Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}/(4\pi)$, which is $(-1)^k$ times the $k$th derivative of $s_0(t) = 1/(4\sqrt{\pi t})$. It suffices to show that $c_0$ and $c_1$ are positive, and we do so first for $t \le \tau$, from (1.2.5), and then for $t \ge \tau$, from the zeros.

Let $A$ and $P$ be the archimedean and prime terms of (1.2.5). The series $\psi(z) = -\gamma_E + \sum_{n\ge0}\big(\frac{1}{n+1} - \frac{1}{n+z}\big)$ [397, (5.7.6)] shows that $m$ increases on $[0,\infty)$, from $m(0) = \psi(\frac{1}{4}) - \log\pi = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log\pi = -5.372$. Binet's formula $\psi(z) = \log z - \frac{1}{2z} - 2\int_0^\infty v\,dv/\big((v^2+z^2)(e^{2\pi v}-1)\big)$ [397, (5.9.15)] at $z = \frac{1}{4} + \frac{ir}{2}$, with the integral split at $v = r/4$, gives $|m(r) - \log\frac{r}{2\pi}| \le \frac{9}{8r^2} + 10^{-10} < 0.003$ for $r \ge 20$. Hence $|m| \le 6$ on $[0,20]$ and $\int_0^{20}|m(r) - \log\frac{r}{2\pi}|\,dr \le 120 + 4\pi + 20\log\frac{10}{\pi} - 20 < 136$. Since $\frac{1}{2\pi}\int_0^\infty r^{2k}e^{-r^2t}\log\frac{r}{2\pi}\,dr = s_k(t)(\ell+\alpha_k)$, with $\alpha_0 = -\frac{1}{2}\gamma_E - \log 4\pi = -2.8196$ and $\alpha_1 = \alpha_0 + 1$, it follows that

$$\Big|\frac{(-1)^k A^{(k)}(t)}{s_k(t)} - \ell - \alpha_k\Big| \le \frac{272\cdot 20^{2k}\,t^{k+\frac{1}{2}}}{\Gamma(k+\frac{1}{2})} + 0.003 \qquad (k = 0, 1),$$

and the right side, which increases with $t$, is below 1.089 for $k = 0$ and 0.047 for $k = 1$ at $t = \tau$. For the prime term put $y_n = (\log n)^2/4t$ and $q = (\log 2)/8t$. Both $e^{-y_n}$ and $|y_n - \frac{1}{2}|e^{-y_n}$ are at most $e^{-y_n/2}$, which is at most $n^{-q}$ because $(\log n)^2 \ge (\log 2)\log n$, and $\Lambda(n) \le \log n < \sqrt{n}$; so $|P^{(k)}(t)| \le (2\sqrt{\pi})^{-1}t^{-k-\frac{1}{2}}\sum_{n\ge2} n^{-q} < t^{-k-\frac{1}{2}}2^{-q}$, that is $|P^{(k)}(t)| < t^{-k-\frac{1}{2}}e^{-\kappa/t}$ for $k = 0, 1$ and $t \le \tau$, where $\kappa = (\log 2)^2/8$ is another constant of this section. Since $\kappa/\tau > 1201$, this is below $10^{-500}s_k(t)$. The polar term adds $e^{t/4}$ to $c_0$ and subtracts $\frac{1}{4}e^{t/4} < 10^{-5}s_1(t)$ from $c_1$. So for $t \le \tau$

$$c_0(t) \ge s_0(t)\,(\ell - 3.92), \qquad c_1(t) \ge s_1(t)\,(\ell - 1.87),$$

and both are positive, because $\ell \ge \frac{1}{2}\log 20000 > 4.95$.

For $t \ge \tau$ we use the zeros. Those below height 1000 lie on the line [419] and contribute $\gamma^{2k}e^{-\gamma^2 t} > 0$ to $c_k$, and by the explicit bound for $N(T)$ of [176] one of them lies in $(20, 50]$ and contributes at least $400^k e^{-2500t}$. A zero above height 1000, wherever it lies, has $\operatorname{Re} a \ge \gamma^2 - \frac{1}{4}$ and $|a| \le 2\gamma^2$. The same bound gives $N(r) \le r^2$ for $r \ge 1000$, and integration by parts against $dN$ bounds the total modulus of the contributions of these zeros to $c_0$ and $c_1$ by

$$H_0(t) = e^{-(10^6-\frac{1}{4})t}\big(10^6 + t^{-1}\big), \qquad H_1(t) = 2e^{-(10^6-\frac{1}{4})t}\big(10^{12} + 2\cdot 10^6 t^{-1} + 2t^{-2}\big).$$

At $t = \tau$ these are below $2 \times 10^{-16}$ and $4.1 \times 10^{-10}$, while $e^{-2500\tau} > 0.88$, and $e^{2500t} H_k(t)$ decreases with $t$. So $c_k(t) \geq 400^k e^{-2500t} - H_k(t) > 0$ for $t \geq \tau$ and $k = 0, 1$.

Thus $W > 0$ and $W' < 0$ on $(0, \infty)$. Since $0 < W(t) \leq \sum_j e^{-t \operatorname{Re} a_j}$, the measure $W(t)\, dt/t$ satisfies $\int_0^\infty (1 \wedge t)\, W(t)\, dt/t \leq \sum_j 1/\operatorname{Re} a_j < \infty$, so it is a Lévy measure, and the formula for $\log G$ above makes the clock infinitely divisible with this Lévy measure. A law on the half line whose Lévy density is $W(t)/t$ with $W$ nonincreasing is self decomposable [471, Corollary 15.11]. Finally, $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \theta) = \mathbb{E} e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with $Z$ standard normal and independent of $T$, and with the characteristic function $\Xi(\theta)/\Xi(0)$ of the variable $X$ of Section 1.2 it forms a van Dantzig pair. The program `idsd_check.py` (`code/README.md`) evaluates every numerical inequality of this proof in ball arithmetic. □

Tilting the clock by $e^{-T}$ gives $\xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+s}) = G(1)/G(1+s) = \mathbb{E} e^{-sT^{(1)}}$, with $T^{(1)} \geq 0$ infinitely divisible, so unconditionally $h_k = \xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+k}) = \mathbb{E}(e^{-T^{(1)}})^k$ is a Hausdorff moment sequence.

The proof needs far fewer zeros than those below height 1000. With the same bounds it goes through with $\tau = 1/8300$ and the zeros below height 400. If the archimedean error is bounded by computation instead, from $\int_0^5 r^{2k} |m(r) - \log \frac{r}{2\pi}|\, dr < 0.836$ and 0.404 for $k = 0, 1$ and $r^2 |m(r) - \log \frac{r}{2\pi}| < 0.05$ for $r \geq 5$, it goes through with $\tau = 1/500$ and height 100. If in addition the polar term $e^{t/4}$ is kept in $c_0$, the prime terms with $n \leq 30$ are taken exactly, and the lower bound for $t \geq \tau$ comes from the first zero, in $(14.1347, 14.1348)$, rather than from a zero in $(20, 50]$, it goes through with $\tau = 1/50$ and height 25, below which there are only two zeros. The program `idsd_height.py` evaluates every numerical inequality of the three versions in ball arithmetic.

The same scheme, which we call the two range method, gives more. With the prime term bounded by $100\, t^{-k-\frac{1}{2}} e^{-\kappa/t}$ for $k \leq 3$ and a second zero in $(100, 200]$, it shows that $W$ and $-W'$ are strictly log convex [421, Theorem 7.3], that is $c_0 c_2 > c_1^2$ and $c_1 c_3 > c_2^2$ for every $t > 0$, and that $(-1)^k W^{(k)} > 0$ for $k \leq 3$.

For the first two rungs the explicit formula can be avoided altogether, because where the size of an off line quadruple cannot be bounded its phase fixes its sign, as the first author's note [433] observes. Suppose that the zeros with ordinate in $(0, H]$ lie on the line, with $H \geq 100$, and let $0 \leq k \leq H/8$. A quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ then has $\gamma > H$, and since $0 < \phi < 2\delta/\gamma < 1/\gamma$ its phase satisfies $|2\delta\gamma t - k\phi| < \gamma t + 1$, so its term in $c_k(t)$ is positive unless $\gamma t > \frac{\pi}{2} - 1 > 0.57$. When $\gamma t > 0.57$ the term is at most $2(\gamma^2 + \frac{1}{4})^k e^{-(\gamma^2 - \frac{1}{4})t}$ in modulus, and since $(\gamma^2 + \frac{1}{4})/\gamma_1^2 \leq \gamma^2/99.9$, $\gamma^2 - \frac{1}{4} - \gamma_1^2 \geq \gamma^2/2$ and $\gamma^2 t/2 > 0.285\gamma$, it is at most $2(\gamma^2/99.9)^k e^{-0.285\gamma}$ times the term $\gamma_1^{2k} e^{-\gamma_1^2 t}$ of the first zero. A quadruple has two

zeros of ordinate $\gamma$, so at every $t > 0$

$$c_k(t) \geq \big(1 - S_k(H)\big)\gamma_1^{2k} e^{-\gamma_1^2 t}, \qquad S_k(H) = \sum_{\gamma > H} \Big(\frac{\gamma^2}{99.9}\Big)^k e^{-0.285\gamma},$$

the sum running over the zeros above height $H$, with multiplicity. The explicit bound for $N(T)$ of [176] gives $N(T+1) - N(T) < 2\log T$ for $T \geq 100$, and summing over unit intervals gives $S_0(100) < 10^{-10.5}$ and $S_1(100) < 10^{-8.5}$. Since Hutchinson placed the zeros below height 300 on the line in 1925 [278], this proves Theorem 5.1.1 without the explicit formula. For larger $k$ the factor $(\gamma^2/99.9)^k$ must be absorbed by $e^{-0.285\gamma}$ at $\gamma = H$, so the reachable order, about $0.285H/\log(H^2/99.9)$, stays below the height: the verification below $3 \times 10^{12}$ [419] gives every $k \leq 10^{10}$, with $S_k(H) < 10^{-1.4\times 10^{11}}$. The two range method reaches orders proportional to $H^2$ (Theorems 5.1.6 and 5.1.13). The program `ladder_bound.py` evaluates these bounds in ball arithmetic.

Positivity of $W$ also limits how the clock factorises. By Mase's theorem [371, Theorems 1 and 2], an infinitely divisible law without Gaussian component whose Lévy measure has a density that is positive on the whole half line, or that has mass on both half lines, has an indecomposable factor, hence a factor that is not infinitely divisible, so it lies outside Linnik's class $I_0$ [344] of the infinitely divisible laws all of whose factors are infinitely divisible. Since $W > 0$, this applies unconditionally to the clock variable $T$, to $Z\sqrt{2T}$ and to every nondegenerate GGC [421, Proposition 12.1].

Proposition 5.1.2 (Truncation error). *Let the zeros be simple, so that the $a_j$ are distinct, and let $r_k$ and $r_k^{(K)}$ be the residues at $u = -a_k$, $k \leq K$, of $1/G(u) = \prod_j a_j/(a_j + u)$ and of its product over $j \leq K$. Then $r_k = a_k \prod_{j \neq k} a_j/(a_j - a_k)$ and, when $|a_j| > |a_k|$ for $j > K$,*

$$\log \frac{r_k^{(K)}}{r_k} = \sum_{j>K} \log\Big(1 - \frac{a_k}{a_j}\Big) = -a_k \sum_{j>K} a_j^{-1} - \frac{a_k^2}{2} \sum_{j>K} a_j^{-2} - \cdots .$$

*Under RH $a_j = \gamma_j^2$ and $\sum_{j>K} \gamma_j^{-2} \sim \log\gamma_K/(2\pi\gamma_K)$, so the first term is $-\gamma_k^2 \log\gamma_K/(2\pi\gamma_K)$: the truncated coefficient is close to the residue for $\gamma_k \leq (\gamma_K/\log\gamma_K)^{1/2}$, and not beyond.*

*Proof.* The product converges absolutely, and its pole at $-a_k$ is simple, with residue $a_k$ times the other factors at $u = -a_k$. The ratio of the two residues is the tail product $\prod_{j>K}(a_j - a_k)/a_j$. For the sum, $\sum_{j>K} \gamma_j^{-2} = \int_{\gamma_K}^{\infty} T^{-2}\, dN(T)$ with $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, which is $(\log(\gamma_K/2\pi) + 1)/(2\pi\gamma_K)$ to leading order. □

### 5.1.2 Hankel blocks and derivative signs

Fix $t > 0$ and form the Hankel matrices $[c_{i+j}(t)]$ and $[c_{i+j+1}(t)]$, $i, j \geq 0$.

Theorem 5.1.3 ([421, Theorem 7.1]). *For any fixed $t > 0$ the following are equivalent: both infinite Hankel matrices are positive semidefinite; Weil's quadratic form is positive semidefinite on the span of $r^m e^{-r^2 t/2}$, $m = 0, 1, 2, \ldots$; RH.*

Under RH the derivatives at one time are the moments of the positive measure $e^{-tz}U(dz)$ on the half line, so both matrices are positive semidefinite. Conversely, if both Hankel matrices are positive semidefinite, Stieltjes' theorem makes the $c_k(t)$ the moments of a positive measure $\mu$ on the half line, and since $\sum_k c_k(t)(-s)^k/k! = W(t+s)$ for $|s| < t$, $W(t+s) = \int e^{-sx}\mu(dx)$ there. Both sides are analytic in $\operatorname{Re} s > -t$, the right side because $\int e^{|s|x}\mu(dx) = \sum_k c_k(t)|s|^k/k! = W(t-|s|) < \infty$ for $|s| < t$, so the identity theorem extends the equality to $\operatorname{Re} s > -t$, and $W$ is completely monotone, which is RH. Through the explicit formula, Weil's form on the span of $r^m e^{-r^2 t/2}$ gives the same two matrices, even $m$ the first and odd $m$ the second.

Theorem 5.1.3 tests RH on the real axis. The next one tests it on the vertical line, where $W$ is a sum of oscillating terms.

Theorem 5.1.4. *RH holds if and only if $v \mapsto W(\sigma + iv)$ is bounded on $\mathbb{R}$, for one, equivalently every, $\sigma > 0$.*

*Proof.* Write $s_j = -a_j = (\rho_j - \frac{1}{2})^2 = x_j + iy_j$, $\rho_j = \frac{1}{2} + \delta_j + i\gamma_j$, so that $x_j = \delta_j^2 - \gamma_j^2$, $y_j = 2\delta_j\gamma_j$, $|\delta_j| < \frac{1}{2}$, $|y_j| \leq |\gamma_j|$ and $\gamma_j \neq 0$. Only three facts on the zeros are used: $|\delta_j| < \frac{1}{2}$, $\gamma_j \neq 0$ ($\xi$ has no zeros in $(0,1)$) and $N(T) = O(T \log T)$, so that $\sum_j e^{-c\gamma_j^2} < \infty$ for every $c > 0$ and only finitely many $|\gamma_j|$ lie below any height. Put $F(v) = W(\sigma + iv) = \sum_j c_j e^{is_j v}$, $c_j = e^{s_j \sigma}$, $C = \sum_j |c_j| \leq e^{\sigma/4} \sum_j e^{-\sigma\gamma_j^2} < \infty$. Since $|e^{is_j v}| = e^{-y_j v} \leq e^{|\gamma_j v|}$, the series converges absolutely and locally uniformly in $v$.

Under RH every $s_j = -\gamma_j^2$ is real, so $|F(v)| \leq \sum_j e^{-\sigma\gamma_j^2} = W(\sigma)$ for every $\sigma > 0$. Conversely let $|F| \leq M$ on $\mathbb{R}$ for one $\sigma$, and suppose RH fails. For $\omega \in \mathbb{R}$ and complex $\varepsilon$ with $\operatorname{Re}\varepsilon > 0$ put $L(\varepsilon) = \int_{\mathbb{R}} F(v) e^{-\varepsilon v^2 - i\omega v}\, dv$.

Step 1. On $\operatorname{Re}\varepsilon \geq \eta > 0$ the integrand is entire in $\varepsilon$ and bounded by $Me^{-\eta v^2}$, so $L$ is continuous by dominated convergence and holomorphic by Morera's theorem, with Fubini over triangles justified by the same bound. For real $\varepsilon > 0$, $|L(\varepsilon)| \leq M\sqrt{\pi/\varepsilon}$.

Step 2. For $\varepsilon > 0$ and $b \in \mathbb{C}$, $\int_{\mathbb{R}} e^{-\varepsilon v^2 + bv}\, dv = \sqrt{\pi/\varepsilon}\, e^{b^2/4\varepsilon}$: both sides are entire in $b$ and agree for real $b$ by completing the square. With $b = i(s_j - \omega)$ this gives $\int_{\mathbb{R}} e^{is_j v - \varepsilon v^2 - i\omega v}\, dv = \sqrt{\pi/\varepsilon}\, e^{-(s_j-\omega)^2/4\varepsilon}$. The sum over $j$ may be taken under the integral (Fubini and Tonelli) if $\sum_j |c_j| \int_{\mathbb{R}} e^{-y_j v - \varepsilon v^2}\, dv = \sqrt{\pi/\varepsilon} \sum_j e^{\sigma\delta_j^2 - \sigma\gamma_j^2 + \delta_j^2\gamma_j^2/\varepsilon} < \infty$. For real $\varepsilon \geq 1/(2\sigma)$ we

have $\delta_j^2/\varepsilon \le \sigma/2$, each term is at most $e^{\sigma/4}e^{-\sigma\gamma_j^2/2}$, and the sum is finite. Hence $L(\varepsilon) = \sqrt{\pi/\varepsilon}\,\Psi_\varepsilon(\omega)$ for $\varepsilon \ge 1/(2\sigma)$, where $\Psi_\varepsilon(\omega) = \sum_j c_j e^{-(s_j-\omega)^2/4\varepsilon}$ and the root is principal.

Step 3. The series $\Psi_\varepsilon(\omega)$ converges locally uniformly on $\operatorname{Re}\varepsilon > 0$. Let $K$ be compact there. Then $u = 1/(4\varepsilon)$ stays in a compact set with $\operatorname{Re} u \ge u_0 > 0$ and $|\operatorname{Im} u| \le u_1$. Write $(s_j-\omega)^2 = X + iY$ with $X = (x_j-\omega)^2 - y_j^2$ and $Y = 2(x_j-\omega)y_j$. For $\gamma_j^2 \ge \frac14 + |\omega|$ we have $|x_j-\omega| \ge \gamma_j^2 - \frac14 - |\omega|$, so $X \ge (\gamma_j^2 - \frac14 - |\omega|)^2 - \gamma_j^2$ and $|Y| \le 2|\gamma_j|(\gamma_j^2 + \frac14 + |\omega|)$, of orders $\gamma_j^4$ and $|\gamma_j|^3$. Hence there is $G = G(K,\omega)$ such that $X > 0$ and $u_0X - u_1|Y| \ge 0$ for $|\gamma_j| \ge G$, and then $|e^{-(X+iY)u}| = e^{-(X\operatorname{Re}u - Y\operatorname{Im}u)} \le e^{-(u_0X-u_1|Y|)} \le 1$, so the $j$th term is at most $|c_j|$. Only finitely many $j$ have $|\gamma_j| < G$, and $\sum_j |c_j| = C$, so the M-test applies and $\Psi$ is holomorphic on $\operatorname{Re}\varepsilon > 0$. Now $L$ and $\sqrt{\pi/\varepsilon}\,\Psi_\varepsilon(\omega)$ are holomorphic on the connected half plane and agree on $[1/(2\sigma),\infty)$, so by the identity theorem they agree throughout. With Step 1,

$$|\Psi_\varepsilon(\omega)| \le M \qquad (\varepsilon > 0,\ \omega \in \mathbb{R}). \tag{$*$}$$

Step 4. Put $t = 1/(4\varepsilon)$ and $q_j(\omega) = y_j^2 - (x_j-\omega)^2$. Then $|e^{-(s_j-\omega)^2 t}| = e^{q_j t}$ and its phase is $e^{-i\phi_j t}$, $\phi_j = 2(x_j-\omega)y_j$. For fixed $\omega$ and level $l$ only finitely many $j$ have $q_j(\omega) \ge l$, because $q_j \le \gamma_j^2 - (\gamma_j^2 - \frac14 - |\omega|)^2 \to -\infty$ as $|\gamma_j| \to \infty$.

Step 5. Take a zero off the line, with $s_0 = x_0 + iy_0$, $y_0 = 2\delta\gamma \ne 0$. On the open interval $I_0 = (x_0 - |y_0|, x_0 + |y_0|)$, $q_0 > 0$. For two values $s \ne s'$ in $\{s_j\}$ with $(x,|y|) \ne (x',|y'|)$, the difference $q - q' = y^2 - y'^2 - x^2 + x'^2 + 2\omega(x-x')$ is a nonconstant affine function of $\omega$ if $x \ne x'$ and a nonzero constant if $x = x'$, so it vanishes for at most one $\omega$. Let $E$ be the countable set of these points together with $\{x_j\}$, and fix $\omega \in I_0 \setminus E$. By Step 4, $Q = \max_j q_j(\omega)$ is attained, $Q \ge q_0 > 0$, and by the choice of $\omega$ every maximiser has the same $(x,|y|)$, so $s_j \in \{s^*, \bar s^*\}$ for a fixed $s^* = x^* + iy^*$. Now $Q > 0$ forces $y^* \ne 0$, and $\omega \ne x^*$ forces $\phi := 2(x^*-\omega)y^* \ne 0$. Only finitely many $j$ have $q_j \ge 0$, so $Q' = \max\{0,\, q_j(\omega) : s_j \notin \{s^*, \bar s^*\}\}$ is a maximum over a finite set and $Q' < Q$.

Step 6. Let $m_1 \ge 1$ and $m_2 \ge 0$ be the numbers of $j$ with $s_j = s^*$ and $s_j = \bar s^*$, and $A = e^{s^*\sigma} \ne 0$. The terms with $\bar s^*$ have the same $q$, the phase $e^{+i\phi t}$ and the coefficient $\bar A$. Hence $\Psi_\varepsilon(\omega) = e^{Qt}g(t) + r(t)$ with $g(t) = m_1 A e^{-i\phi t} + m_2 \bar A e^{i\phi t}$ and $|r(t)| \le \sum_j |c_j| e^{Q't} = Ce^{Q't}$ for $t > 0$. The function $g$ is continuous and periodic with period $2\pi/|\phi|$, and not identically zero, since its Fourier coefficient at the frequency $-\phi \ne \phi$ is $m_1 A \ne 0$. So $\kappa = \max|g| > 0$ is attained in every period, at points $t_k \to \infty$, and with $\varepsilon_k = 1/(4t_k)$, using $Q > 0$ for the growth and $Q > Q'$ for the remainder,

$$|\Psi_{\varepsilon_k}(\omega)| \ge \kappa e^{Qt_k} - Ce^{Q't_k} \to \infty,$$

contradicting $(*)$. So RH holds. Since boundedness for one $\sigma$ gives RH, and RH gives it for every $\sigma$, the two readings agree. □

THEOREM 5.1.5 ([421, Theorem 7.4]). *If all zeros below* $T_N = 20000N\sqrt{1+\log N}$ *are on the line, the leading* $N \times N$ *blocks are positive definite at every* $t > 0$. *The verification to height* $3 \times 10^{12}$ *[419] gives this for every* $N \le 3 \times 10^7$.

Table 5.1 shows normalised leading minors recomputed in sixty digit arithmetic from the first 1700 zeros, whose ordinates are stored in double precision; they agree with [421, Numerical observation 7.2] to every digit printed there.

| $t$ | $c_0c_2/c_1^2$ | $N=2$ | $N=3$ | $N=4$ | $N=5$ | block |
|---|---|---|---|---|---|---|
| 0.01 | 1.1346548 | 0.119 | $2.45\times10^{-3}$ | $1.05\times10^{-5}$ | $1.04\times10^{-8}$ | even |
| 0.01 | | 0.182 | $5.87\times10^{-3}$ | $3.30\times10^{-5}$ | $4.57\times10^{-8}$ | odd |
| 0.05 | 1.0000081 | $8.11\times10^{-6}$ | $1.77\times10^{-14}$ | $5.35\times10^{-28}$ | $2.20\times10^{-44}$ | even |
| 0.05 | | $1.79\times10^{-5}$ | $1.22\times10^{-13}$ | $1.71\times10^{-26}$ | $3.81\times10^{-42}$ | odd |

Table 5.1: Normalised leading Hankel minors of the heat trace, $\det[c_{i+j}]_{i,j<N}/\prod_{i<N} c_{2i}$ for the even block and $\det[c_{i+j+1}]_{i,j<N}/\prod_{i<N} c_{2i+1}$ for the odd block, from the first 1700 zeros in sixty digit arithmetic.

At $t = 0.05$ (Figure 5.1, left) the first exponents dominate and the others enter only through $e^{-(a_j-a_1)t}$, so the matrix is nearly of rank one. Computed from the zeros, the minors are still sums of positive terms, by the Cauchy Binet formula $\det[c_{i+j}]_{i,j<N} = \sum_{|S|=N} \prod_{l\in S} e^{-a_l t} \prod_{l<l',\, l,l'\in S} (a_l - a_{l'})^2$ over the sets $S$ of $N$ zeros, so what is lost is conditioning, not sign. The derivative signs (Figure 5.1, right) reach much further, because a sign needs only that the verified terms dominate, not that a small determinant stay positive.

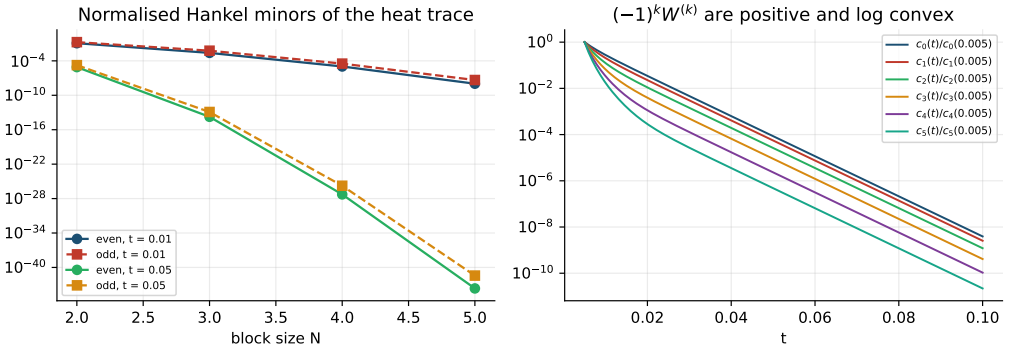



Figure 5.1: Left: normalised Hankel minors of the heat trace at two times; positivity persists but falls below any fixed precision as $t$ grows. Right: the first six $(-1)^k W^{(k)}(t)$, divided by their values at $t = 0.005$, from 1700 zeros.

THEOREM 5.1.6 ([421, Theorem 7.10]). *Unconditionally, that is, using only the verification of RH below height* $10^{12}$ *[419],* $(-1)^k W^{(k)}(t) > 0$ *for every* $t > 0$ *and*

*every integer* $0 \le k \le 10^{17}$.

The proof again splits at $\tau$. Near the origin, Cauchy's estimate on a circle about $t$ of radius close to $t$, such as $(1 - \frac{1}{k+1})t$, on which $\mathrm{Re}(1/z) \ge 1/(2t)$ and $|z| \ge t/(k+1)$, bounds the $k$th derivative of the prime term by a multiple of $(k+1)\Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}e^{-\kappa/t}$. Against $s_k(t)$ this loses a factor of order $k$ and gains $e^{-\kappa/t} \le e^{-1201}$, and for $k \ge 1$ the archimedean term dominates by a margin of more than $2.9\, s_k(t)$, the case $k = 0$ being Theorem 5.1.1. Away from the origin a zero in $(20, 50]$ dominates all zeros above the verified height $H = 10^{12}$, wherever they lie, provided $tH^2 \ge 2(k+1)$ and a logarithmic condition hold; Section 5.1.6 explains why these allow orders up to about $10^{18}$.

The Stieltjes moments of Table 1.7 give a Hankel test at the other end of the heat trace, and the test also returns the zeros [433]. They are the moments of $W$, $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_j a_j^{-n-1}$, and since $\log G(u) = \sum_j \log(1 + u/a_j) = \sum_{n\ge1}(-1)^{n+1}m_{n-1}u^n/n$ they follow from the Taylor coefficients of $G(u) = \Xi(i\sqrt{u})/\Xi(0) = \sum_k M_k u^k/((2k)!\,M_0)$, where $M_k = \int_{\mathbb{R}} v^{2k}\Phi(v)\,dv$; no zero enters. Under RH the $m_n$ are the moments of the positive measure $\nu = \sum_j a_j^{-1}\delta_{1/a_j}$, so every Hankel matrix $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is positive definite. Conversely, if all of them are, Hamburger's theorem makes the $m_n$ the moments of a positive measure, of compact support because $|m_n| \le \sum_j |a_j|^{-n-1}$ grows at most geometrically, and then $\sum_n m_n w^n = \sum_j 1/(a_j - w)$ has all its poles on the real axis: every $a_j$ is real, and positive because $\xi$ has no zeros in $(0, 1)$, which is RH. The shifted matrices $\mathcal{H}'_N = [m_{i+j+1}]_{i,j<N}$ give the same test, and either suffices alone: if every $\mathcal{H}'_N$ is positive definite, Hamburger's theorem applied to $(m_{n+1})$ and $\sum_n m_n w^n = m_0 + w\sum_n m_{n+1}w^n$ give the same conclusion. Positivity is tested by the Cholesky factorisation $\mathcal{H}_N = LL^\top$, and the factor gives more: $J_N = L^{-1}\mathcal{H}'_N L^{-\top}$ is the Jacobi matrix of $\nu$, tridiagonal, and its eigenvalues are the nodes of the $N$ point Gauss rule for $\nu$. They approximate the largest atoms $1/a_j$ first, so their inverse square roots return the lowest ordinates (Table 5.2). With $M_0, \dots, M_{100}$ computed at 1000 digits by the trapezoid rule, every $\mathcal{H}_N$ up to order 50 is positive definite.

This is Gauss quadrature, equivalently Padé approximation of $G'/G$ at the origin, and Table 5.2 shows its two limits. The first is precision. The last Cholesky pivot $\det\mathcal{H}_N/\det\mathcal{H}_{N-1}$ falls from $10^{-61}$ at $N = 10$ to $10^{-398}$ at $N = 50$, and the computation must resolve it. The precision thus grows linearly, by about 17 digits for each further zero. The second is reach. An off line quadruple at height $\gamma$ enters $m_n$ with weight of order $\gamma^{-2n-2}$, against $\gamma_1^{-2n-2}$ for the first zero, so a block of order $N$ sees essentially only the zeros that its Gauss rule resolves; Section 8.4.3 shows this along the de Bruijn Newman flow, where the order 30 test registers the collision of the pair just above the zeros it resolves, and not a collision higher up. The program `jacobi_zeros.py` computes the table.

| $N$ | zeros | error of $\gamma_1$ | last error | $\det \mathcal{H}_N / \det \mathcal{H}_{N-1}$ | digits |
|---|---|---|---|---|---|
| 10 | 2 | $4.4 \times 10^{-18}$ | $1.3 \times 10^{-9}$ | $2.0 \times 10^{-61}$ | 60 |
| 20 | 7 | $1.6 \times 10^{-49}$ | $3.1 \times 10^{-7}$ | $1.2 \times 10^{-138}$ | 140 |
| 30 | 12 | $8.6 \times 10^{-87}$ | $2.4 \times 10^{-7}$ | $1.0 \times 10^{-221}$ | 230 |
| 40 | 17 | $6.1 \times 10^{-128}$ | $5.4 \times 10^{-8}$ | $1.3 \times 10^{-308}$ | 310 |
| 50 | 22 | $5.3 \times 10^{-172}$ | $5.4 \times 10^{-8}$ | $1.4 \times 10^{-398}$ | 400 |

Table 5.2: The lowest ordinates from the Stieltjes moments $m_0, \dots, m_{2N-1}$, computed from the Taylor coefficients of $\Xi$ at the centre: the number returned by the Jacobi matrix of order $N$ to within $10^{-6}$, the errors of the first and of the last of them, the last Cholesky pivot of $\mathcal{H}_N$, and the least number of digits, in steps of 10, of the Taylor coefficients and of the arithmetic from which on the same ordinates are returned.

The Hankel matrices count the zeros off the line as well as test RH. For a real polynomial this is a theorem attributed to Hermite [182]: the Hankel form of the power sums of the roots has as many negative squares as the polynomial has distinct pairs of nonreal roots. Let $\iota_N$ and $\iota'_N$ be the numbers of negative eigenvalues of $\mathcal{H}_N$ and $\mathcal{H}'_N$.

THEOREM 5.1.7. *Let $K \in \{0, 1, 2, \dots\} \cup \{\infty\}$ be the number of distinct zeros $\rho$ of $\xi$ with $\operatorname{Re}\rho > \frac{1}{2}$ and $\operatorname{Im}\rho > 0$. Then $\iota_N$ and $\iota'_N$ are nondecreasing in $N$, $\iota_N \le K$ and $\iota'_N \le K$ for every $N$, and $\iota_N \to K$ and $\iota'_N \to K$ as $N \to \infty$. In particular RH holds if and only if $\iota_N = 0$ for every $N$, and if it fails, $\iota_N \ge 1$ from some $N$ on; the same holds for $\iota'_N$.*

*Proof.* Put $b_k = 1/a_k$, listed with multiplicity, so that $m_n = \sum_k b_k^{n+1}$ and $\sum_k |b_k| < \infty$. For a real vector $x = (x_0, \dots, x_{N-1})$ and $P(w) = \sum_i x_i w^i$,

$$x^\top \mathcal{H}_N x = \sum_k b_k P(b_k)^2, \qquad x^\top \mathcal{H}'_N x = \sum_k b_k^2 P(b_k)^2,$$

the sums converging absolutely. A zero on the line gives $b_k = \gamma_k^{-2} > 0$ and nonnegative terms. A zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\delta > 0$ and $\gamma > 0$, of multiplicity $\mu$, gives $a = -(\delta + i\gamma)^2$ with $\operatorname{Im} a = -2\delta\gamma < 0$, hence $b = 1/a$ with $\operatorname{Im} b > 0$; its mirror image $1 - \bar\rho$ gives $\bar a$ and $\bar b$ with the same multiplicity, and since $P$ has real coefficients the two contribute $2\mu \operatorname{Re}\big(bP(b)^2\big)$ to the first form. With $u = \operatorname{Re} P(b)$ and $v = \operatorname{Im} P(b)$ this is $2\mu\big(\operatorname{Re} b\,(u^2 - v^2) - 2 \operatorname{Im} b\, uv\big)$, a quadratic form in $(u, v)$ whose matrix has determinant $-4\mu^2|b|^2 < 0$, so it has exactly one negative square, and composed with the linear map $x \mapsto (u, v)$ it has at most one. The terms of the zeros on the line add up to a positive semidefinite form, so $\iota_N \le K$. In the second form the weight is $b^2$, and $2\mu \operatorname{Re}\big(b^2 P(b)^2\big)$ is again a form in $(u, v)$ of determinant $-4\mu^2|b|^4 < 0$, so $\iota'_N \le K$. Since $\mathcal{H}_N$ is the leading principal submatrix of $\mathcal{H}_{N+1}$,

Cauchy's interlacing theorem shows that $\iota_N$ does not decrease, and the same holds for $\iota'_N$.

For the lower bound let $\mathcal{B}$ be the closure of $\{b_k\}$. It is a countable compact set, symmetric under conjugation, whose only possible accumulation point is 0, and a countable compact set does not separate the plane, so $\mathbb{C}\setminus\mathcal{B}$ is connected. Fix a finite $m \le K$ and distinct atoms $\omega_1,\dots,\omega_m$ with $\operatorname{Im}\omega_j > 0$, of multiplicities $\mu_j$; they are isolated points of $\mathcal{B}$. On $\mathbb{C}^m = \mathbb{R}^{2m}$ the form $q(y) = \sum_{j\le m} 2\mu_j \operatorname{Re}(\omega_j y_j^2)$ is a sum of $m$ forms in the pairs $(\operatorname{Re} y_j, \operatorname{Im} y_j)$, each with one negative square, so there are an $m$ dimensional real subspace $E$ and $\lambda_0 > 0$ with $q(y) \le -\lambda_0|y|^2$ on $E$. Let $y^{(1)},\dots,y^{(m)}$ be a basis of $E$ and $\lambda_1 > 0$ such that $|\sum_i \alpha_i y^{(i)}|^2 \ge \lambda_1|\alpha|^2$ for $\alpha\in\mathbb{R}^m$. Take open discs $D_j$ about $\omega_j$, and their mirror images $D_j^*$ about $\bar\omega_j$, whose closures are disjoint and meet $\mathcal{B}$ only in their centres, and an open set $O \supset \mathcal{B}\setminus\{\omega_1,\bar\omega_1,\dots,\omega_m,\bar\omega_m\}$, symmetric under conjugation and disjoint from the closed discs, which exists because that set is compact. Let $f_i$ be $y_j^{(i)}$ on $D_j$, $\overline{y_j^{(i)}}$ on $D_j^*$ and 0 on $O$. Each $f_i$ is holomorphic on a neighbourhood of $\mathcal{B}$, so by Runge's theorem there are polynomials $\tilde P_i$ with $|\tilde P_i - f_i| < \varepsilon$ on $\mathcal{B}$, and since $\mathcal{B}$ and $f_i$ are symmetric under conjugation, $P_i(w) = \frac12\big(\tilde P_i(w) + \overline{\tilde P_i(\bar w)}\big)$ has real coefficients and satisfies the same bound. For $\alpha\in\mathbb{R}^m$ and $P = \sum_i \alpha_i P_i$ we have $|P(b_k)| \le \varepsilon\sqrt{m}\,|\alpha|$ at every atom other than the $\omega_j$ and $\bar\omega_j$, and $P(\omega_j) = \sum_i \alpha_i y_j^{(i)} + \eta_j$ with $|\eta_j| \le \varepsilon\sqrt{m}\,|\alpha|$. The atoms $\omega_j$ and $\bar\omega_j$ contribute $\sum_j 2\mu_j \operatorname{Re}\big(\omega_j P(\omega_j)^2\big)$ to the first form, so for $\varepsilon \le 1$ and a constant $C$ that depends on neither $\varepsilon$ nor $\alpha$,

$$\sum_k b_k P(b_k)^2 \le q\Big(\sum_i \alpha_i y^{(i)}\Big) + C\varepsilon|\alpha|^2 \le -(\lambda_0\lambda_1 - C\varepsilon)|\alpha|^2.$$

For $\varepsilon < \lambda_0\lambda_1/C$ the right side is negative for $\alpha \ne 0$, so the coefficient vectors of $P_1,\dots,P_m$ span an $m$ dimensional space on which the form of $\mathcal{H}_N$ is negative definite as soon as $N$ exceeds their degrees, and $\iota_N \ge m$ for those $N$. For $\mathcal{H}'_N$ the same argument applies with $\omega_j^2$ in place of $\omega_j$ in $q$ and the weights $b_k^2$ in the sum. Since $m \le K$ is arbitrary, $\iota_N \to K$ and $\iota'_N \to K$. □

In determinant form the theorem is classical. For a real entire function of order one with zeros $z_j$, Grommer [162] proved that all its zeros are real if and only if the determinants $D_n = \det(s_{i+j})_{i,j<n}$ of the power sums $s_k = \sum_j z_j^{-k-2}$ are positive for every $n \ge 1$, and Chebotarev [505] that if the sequence $D_{n-1}D_n$, $n \ge 1$, with $D_0 = 1$, has exactly $m$ negative terms, the function has $m$ distinct pairs of complex conjugate zeros; Baricz and Štampach state both in this form [29, Theorem 10]. For $\Xi$, whose zeros are the square roots of the $a_k$, $s_{2l} = 2m_l$ and $s_{2l+1} = 0$, so after the even indices are put first $(s_{i+j})_{i,j<n}$ is block diagonal with the blocks $2\mathcal{H}_{\lceil n/2\rceil}$ and $2\mathcal{H}'_{\lfloor n/2\rfloor}$, whose negative indices add up to $\iota_{\lceil n/2\rceil} + \iota'_{\lfloor n/2\rfloor}$; this tends to $2K$, and each zero $\rho$ counted by $K$ gives two pairs of complex conjugate zeros of $\Xi$, as Chebotarev's count requires. Counting by eigenvalues rather than by the signs of determinants, Theorem 5.1.7 needs no determinant to be nonzero,

allows $K = \infty$ and holds for each family separately. Along the de Bruijn Newman flow the proof applies at every time to the zeros of $H_t$, none of which is purely imaginary (Section 8.4.3).

### 5.1.3 Every order near the origin

Let $s_k(t) = \Gamma(k+\frac12)t^{-k-\frac12}/(4\pi)$ as in the proof of Theorem 5.1.1, let $E_k(t)$ be the average of $m$ against the weight $r^{2k}e^{-r^2t}$ on $[0,\infty)$, and let $F_k(x) = {}_1F_1(k+\frac12;\frac12;-x^2)$. Expanding the cosine in its power series gives $\frac{1}{2\pi}\int_{\mathbb{R}} r^{2k}e^{-r^2t}\cos(ry)\,dr = 2s_k(t)F_k(y/2\sqrt{t})$, whose case $k=0$ is the kernel $e^{-y^2/4t}/(2\sqrt{\pi t})$ of the prime term of (1.2.5). Differentiating (1.2.5) $k$ times term by term, which the Gaussian factors in $r$ and in $\log n$ allow, therefore gives, with no hypothesis on the zeros,

$$\frac{c_k(t)}{s_k(t)} = E_k(t) - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}F_k(x_n) + \frac{(-\frac14)^k e^{t/4}}{s_k(t)}, \qquad x_n = \frac{\log n}{2\sqrt{t}}. \tag{5.1.1}$$

In (5.1.1) the archimedean term $E_k(t)$ grows with $k$.

Lemma 5.1.8. (a) *For every integer $k \ge 0$ and every $x \ge 0$, $|F_k(x)| \le B(x)e^{-x^2/2}$, where $B(x) = \max\{2.62,\ (\pi(2ex^2+\frac12))^{1/4}\}$.*

(b) *Let $0 < \varepsilon < 1$, and let $k \ge 1$ and $x \ge 0$ satisfy $e^2x^2 \le 4\varepsilon k$. Then $|F_k(x)| \le C_\varepsilon(k)e^{-x^2/2}$, where $C_\varepsilon(k) = \sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}$.*

*Proof.* Taking $t=1$ and $y=2x$ above, $F_k(x) = \Gamma(k+\frac12)^{-1}\int_{\mathbb{R}} r^{2k}e^{-r^2+2ixr}\,dr$. Put $\mu = x^2/4$. The integrand is entire and decays in every horizontal strip, so the path can be moved to $\operatorname{Im} r = x/2$, where $|r^{2k}| = (\sigma^2+\mu)^k$, $|e^{-r^2}| = e^{-\sigma^2+\mu}$ and $|e^{2ixr}| = e^{-4\mu}$, with $\sigma = \operatorname{Re} r$. Expanding $(\sigma^2+\mu)^k$ by the binomial theorem gives, for every $k$ and $x$,

$$|F_k(x)| \le e^{-3\mu}\sum_{j=0}^{k}\frac{\mu^j}{j!}P_{k,j}, \qquad P_{k,j} = \frac{\Gamma(k+1)}{\Gamma(k+\frac12)}\,\frac{\Gamma(k-j+\frac12)}{\Gamma(k-j+1)}.$$

Kershaw's inequalities $\sqrt{y+\frac14} \le \Gamma(y+1)/\Gamma(y+\frac12) \le \sqrt{y+0.37}$, proved for $y>0$ in [307] and true at $y=0$, where the ratio is $1/\sqrt{\pi}$, give $P_{k,j} \le \sqrt{(k+0.37)/(k-j+\frac14)}$, which is less than $\sqrt{2}$ for $j \le k/2$ and at most $2\sqrt{k+0.37}$ for every $j$. For (a), suppose first that $k \ge 2ex^2$. Since $k/2 \ge 4e\mu$, Chernoff's bound $\sum_{j\ge a}\mu^j/j! \le (e\mu/a)^a$ for the Poisson law, valid for $a > \mu$, gives $\sum_{j\ge k/2}\mu^j/j! \le (2e\mu/k)^{k/2} \le 2^{-k}$, and $2\sqrt{k+0.37}\,2^{-k} \le \sqrt{1.37} < 1.2$ for $k \ge 1$. Hence $|F_k(x)| \le e^{-3\mu}(\sqrt{2}\,e^{\mu}+1.2) \le (\sqrt{2}+1.2)e^{-x^2/2}$, and $\sqrt{2}+1.2 < 2.62$; for $k=0$ the assumption forces $x=0$, where $F_0 = 1$.

Suppose now that $k < 2ex^2$. Kummer's transformation and the Hermite case of ${}_1F_1$ [397, (13.2.39), (13.6.16)] give $F_k(x) = e^{-x^2}\,{}_1F_1(-k;\frac12;x^2) = (-1)^k\frac{k!}{(2k)!}e^{-x^2}H_{2k}(x)$, and the bound $e^{-x^2/2}|H_n(x)| \le (2^n n!)^{1/2}$ [397, (18.14.9)] gives

$$|F_k(x)| \le \Big(\frac{4^k(k!)^2}{(2k)!}\Big)^{1/2} e^{-x^2/2} \le \big(\pi(k+\tfrac12)\big)^{1/4} e^{-x^2/2} < \big(\pi(2ex^2+\tfrac12)\big)^{1/4} e^{-x^2/2},$$

since $4^k(k!)^2/(2k)! = \sqrt{\pi}\,\Gamma(k+1)/\Gamma(k+\frac12) \le \sqrt{\pi(k+\frac12)}$ by Kershaw's upper bound.

For (b), split the sum at $j = \varepsilon k$ instead. For $j \le \varepsilon k$, $P_{k,j} \le \sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}$, and these $j$ contribute at most that times $e^{\mu}$. For $j > \varepsilon k$, $P_{k,j} \le 2\sqrt{k+0.37}$, and since $\varepsilon k \ge e^2\mu > \mu$, Chernoff's bound gives $\sum_{j\ge\varepsilon k}\mu^j/j! \le (e\mu/\varepsilon k)^{\varepsilon k} \le e^{-\varepsilon k}$. Hence $|F_k(x)| \le e^{-3\mu}\big(\sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}\,e^{\mu} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}\big) \le C_\varepsilon(k)e^{-x^2/2}$. □

Theorem 5.1.9. (a) *For every integer $k \ge 1$ and every $t \in (0, 0.019]$, $(-1)^k W^{(k)}(t) > 0$. The proof uses no information on the zeros.*

(b) *Unconditionally, that is, using only the verification of RH below height $10^{12}$ [419], $W$ is completely monotone on $(0, 1.29]$: $(-1)^k W^{(k)}(t) > 0$ for every $t \in (0, 1.29]$ and every integer $k \ge 0$.*

*Proof.* By (5.1.1) and Lemma 5.1.8(a), for every $k \ge 0$ and $t > 0$,

$$\frac{c_k(t)}{s_k(t)} \ge E_k(t) - S_B(t) - Q_k(t),$$

where

$$S_B(t) = 2\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}B(x_n)e^{-(\log n)^2/8t}, \qquad Q_k(t) = \frac{4\pi e^{t/4}t^{k+\frac12}}{4^k\,\Gamma(k+\frac12)}.$$

Above the order $K = 10^{17}$ part (b) of the lemma does better. Take $\varepsilon = 10^{-6}$. The first term of $C_\varepsilon(k)$ decreases in $k$, because $\frac14 < 0.37(1-\varepsilon)$, and the second decreases for $k > 1/(2\varepsilon)$, so $C_\varepsilon(k) \le C = C_\varepsilon(K+1) < 1 + 6\times10^{-7}$ for $k > K$, and $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds $2.3\times10^5$. For $t \ge 0.019$ every prime power $n \le 2^{24}$ has $x_n < 61$. So for $k > K$ and $t \ge 0.019$ the bound holds with $S_B$ replaced by $S_C$, the sum $S_B$ with $B(x_n)$ replaced by $C$ for $n \le 2^{24}$.

The three terms are monotone. Since $m$ increases on $[0,\infty)$, and the weights $r^{2k}e^{-r^2t}$ have likelihood ratios $r^2$ from $k$ to $k+1$ and $e^{r^2(t'-t)}$ from $t'$ to $t < t'$, both increasing in $r$, $E_k(t)$ increases in $k$ and decreases in $t$. $Q_k(t)$ increases in $t$, and for $t \le 2$ it decreases in $k$, the ratio of consecutive terms being $t/(4k+2)$. $B(x)e^{-x^2/2}$ decreases on $(0,\infty)$: the second branch of $B$ increases and is 1.78 at

$x = 2^{-1/2}$, so $B = 2.62$ below that point, and above it the logarithmic derivative of the second branch times $e^{-x^2/2}$ is less than $1/(2x) - x < 0$; since $x_n$ decreases in $t$, $S_B$ and $S_C$ increase in $t$. So for $k \geq k_0$ and $t \leq T \leq 2$ each bound is at least its value at $(k_0, T)$, and it suffices to prove $E_1(0.019) - S_B(0.019) - Q_1(0.019) > 0$, which gives (a), and $E_{K+1}(1.29) - S_C(1.29) - Q_{K+1}(1.29) > 0$, which gives the orders $k > K$ on $[0.019, 1.29]$; with (a) for $t \leq 0.019$ and Theorem 5.1.6 for the orders $k \leq K$, this gives (b).

For the first, lower sums over $[0, 64]$, which the monotonicity of $m$ allows, and the positivity of $m$ beyond 64 give $E_1(0.019) > 0.15924$; the prime powers $n \leq 30000$ and the tail bound below give $S_B(0.019) < 0.13628$; and $Q_1(0.019) < 0.00933$. For the second, the weight $r^{2K+2}e^{-r^2t}$ makes $v = r^2t$ a gamma variable of shape $\lambda = K + \frac{3}{2}$, and Chernoff's bound for the gamma law, $\Pr(v < a) \leq (a/\lambda)^\lambda e^{\lambda - a}$ for $a < \lambda$, gives $\Pr(v < a) < e^{-1000}$ for $a = (1 - 2^{-13})^2\lambda$. As $m$ increases, $E_{K+1}(t) \geq m(r_1) - (m(r_1) - m(0))e^{-1000}$ with $r_1 = \sqrt{a/t}$, which gives $E_{K+1}(1.29) > 17.6066$; the prime powers $n \leq 2^{24}$ and the tail bound below give $S_C(1.29) < 17.5848$; and $Q_{K+1}(1.29) < e^{-1000}$. In both cases the prime powers above the cutoff $n_0$, which is 30000 and $2^{24}$, add less than $2\big(2.62 + 2.08(2\sqrt{t})^{-1/2}\big)n_0^{1-q}/(q-1)$ with $q = \log n_0/8t$, because $x_n \geq x_{n_0} \geq 1$ and $e^{-(\log n)^2/8t} \leq n^{-q}$ for $n > n_0$, $B(x) \leq 2.62 + 2.08\sqrt{x}$ for $x \geq 1$, and $\log n$ and $(\log n)^{3/2}$ are at most $\sqrt{n}$ for $n \geq e^{10}$; this is below $10^{-299}$ and $5 \times 10^{-4}$. The program `cm_orders.py` evaluates every numerical inequality of this proof and of the proof of Lemma 5.1.8 in ball arithmetic. □

Computed in floating point, the bound $E_1 - S_B - Q_1$ of (a) vanishes at $t = 0.0193$; $E_1$ itself vanishes at $t = 0.0261$, so no bound on the size of the prime terms carries (a) much further. Part (b) goes further because $E_k(t) = \frac{1}{2}(\psi(k + \frac{1}{2}) - \log t) - \log 2\pi + O(t/k)$ grows like $\frac{1}{2}\log k$, while at such orders part (b) of the lemma bounds the prime terms at almost no cost: the bound $E_{K+1} - S_C$, computed in the same way, vanishes at $t = 1.2912$, while $E_{K+1} - S_B$ vanishes at $t = 0.578$.

The ray of Section 5.1.5 shows what the proof uses. On the ray $t = k/u^2$, $e^{-f_0}c_k(t)$ is to leading order the Gaussian smoothing $\mathcal{G}(u, t)$ of the zeros (Proposition 5.1.10), $E_k(t)$ is close to $\log(u/2\pi)$, and for large $k$ the factor $F_k(x_n)$ is close to $e^{-(\log n)^2/8t}\cos(u \log n)$. The proof is therefore the bound of Section 12.2.6 that ignores the phases of the primes, made uniform in the order; with $C$ replaced by one, $S_C$ is twice the sum $S$ of that section, up to the prime powers above $2^{24}$. Above the order $10^{17}$ uniformity costs a factor below $1 + 6 \times 10^{-7}$, so 1.2912, the root of $E_{K+1}(t) = 2S(t)$, is as far as this argument can carry (b) with $K = 10^{17}$: it bounds the prime terms by one envelope for all orders above $K$, and the archimedean term by its value at the order $K + 1$. A bound that ignores the phases but treats the orders separately is not covered by this limit. At low orders it costs the factor $B \geq 2.62$ of part (a), and part (a) is not sharp: on a grid of $0.5 \leq x \leq 20$ and $k \leq 1200$ the ratio $|F_k(x)|e^{x^2/2}/B(x)$ is at most 0.44 (computed). The ray also

says where (b) looks. To leading order, complete monotonicity on $(0,T]$ asks for positive Gaussian smoothings at resolutions $1/(2\sqrt{t})$ of at least $1/(2\sqrt{T})$, which is 0.44 for $T = 1.29$, finer than the 0.46 at which the quadruples of Table 5.4, at distance $\frac{1}{4}$ from the line, first make the smoothing negative, at $t = 1.18$. There is no conflict, because the orders above $10^{17}$ at $t \le 1.29$ lie on the rays at heights $u = \sqrt{k/t} > 2.78 \times 10^8$, where the time $t_0(u)$ of Section 12.2.6, below which the phase free bound makes the smoothing positive whatever the zeros, exceeds 1.29. The quadruples lie below height 2000, at heights where $t_0$ is at most 0.511, and there (b) rests on the verified zeros, through Theorem 5.1.6.

### 5.1.4 Williamson measures: the Thorin measure smoothed by gamma laws

Complete monotonicity is the limit of $k$ monotonicity, which asks only that the first $k$ derivatives alternate in sign, and by Williamson's theorem [519] a $k$ monotone function that vanishes at infinity is a positive mixture of the truncated powers $(1 - tz/k)_+^{k-1}$. For the heat trace the mixing measure is explicit,

$$W(t) = \int_0^\infty \Big(1 - \frac{tz}{k}\Big)_+^{k-1} U_k(dz), \qquad U_k(dz) = \frac{k^k}{(k-1)!} z^{-k-1} c_k\Big(\frac{k}{z}\Big) dz,$$

and we call $U_k$ the Williamson measure of order $k$. By Theorem 5.1.6 it is positive for every $k \le 10^{17}$ [421, Corollary 7.13], and by Theorem 5.1.13 for every $k \le 1.16 \times 10^{25}$. RH holds if and only if $U_k \ge 0$ for infinitely many $k$, since then $W$ is $k$ monotone for infinitely many $k$, hence completely monotone.

The summands of $U_k$ are inverse gamma laws, by an elementary identity: for $k \ge 1$ and $c \ge 0$,

$$e^{-c} = \mathbb{E}\Big(1 - \frac{c}{\Gamma_k}\Big)_+^{k-1}, \qquad \Gamma_k \sim \mathrm{Gamma}(k,1), \tag{5.1.2}$$

since $\int_c^\infty (\omega - c)^{k-1} e^{-\omega} d\omega/\Gamma(k) = e^{-c}$. With $c = a_j t$ and $z = k a_j/\Gamma_k$, so that $1 - a_j t/\Gamma_k = 1 - tz/k$, the identity shows that $U_k$ is the sum over the zeros of the densities $(k a_j)^k z^{-k-1} e^{-k a_j/z}/(k-1)!$. For real $a_j$ this is the inverse gamma law of $k a_j/\Gamma_k$. Under RH, therefore, writing $\mathcal{L}(Y)$ for the law of $Y$,

$$U_k = \sum_\gamma \mathcal{L}\Big(\gamma^2 \frac{k}{\Gamma_k}\Big),$$

a sum of probability laws: the Thorin measure with each atom $\gamma^2$ spread over a relative width of about $1/\sqrt{k}$. As $k \to \infty$ the ratio $\Gamma_k/k$ concentrates at one and the atoms reappear, which is the passage from Williamson's theorem to Bernstein's; in statistics, Polson, Scott and Windle used $k$ monotone mixtures of this kind to build samplers for the Bayesian bridge [420, Section 2.2, Theorem 2.1] and noted the same limit [420, Section 3].

The Williamson measures form a second resolution family for the Thorin measure, beside the Poisson family $\rho_\varepsilon$, which is the Cauchy flow of Section 1.2. In the chart $\theta = \sqrt{z}$ the gamma smoothing spreads the atom at $\gamma$ over a width of about $\gamma/(2\sqrt{k})$, proportional to the height, while the Poisson smoothing spreads every atom over the width $\varepsilon$, and the inverse gamma kernel has tails far lighter than the $x^{-2}$ tails of the Cauchy kernel. In Figure 5.2, where the widths match at $\theta = 30$, the Williamson densities fall by many orders of magnitude between neighbouring zeros and the Poisson densities by only one to four.

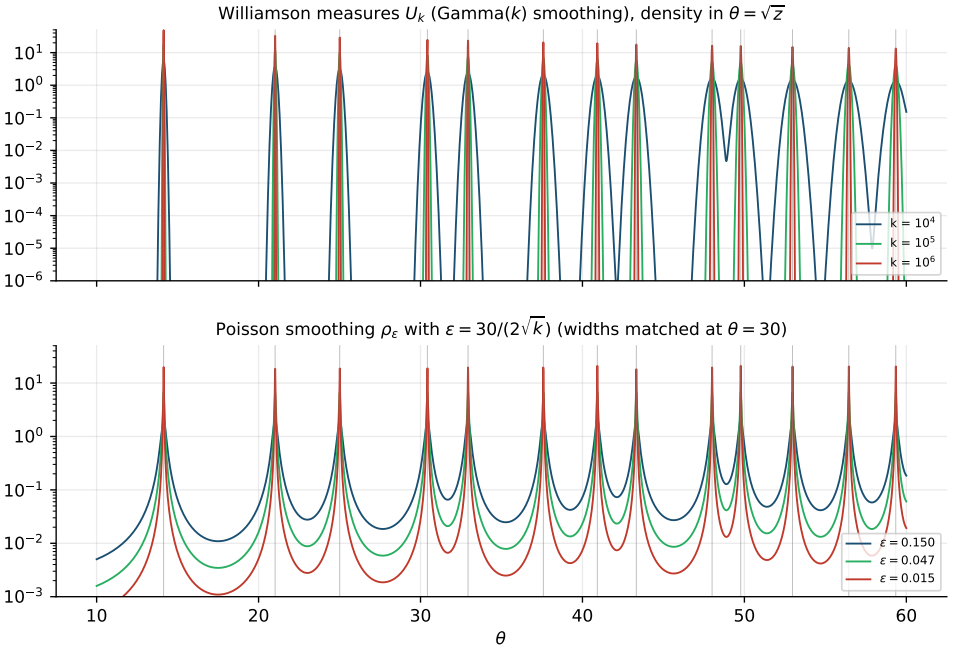



Figure 5.2: Top: the Williamson measures $U_k$ of the heat trace for $k = 10^4$, $10^5$, $10^6$, computed from 1700 zeros, as densities in $\theta = \sqrt{z}$. Bottom: the Poisson smoothed Thorin densities $\rho_\varepsilon$ with $\varepsilon = 30/(2\sqrt{k})$, so that the widths match at $\theta = 30$.

The Poisson smoothing of the zeros is $\pi^{-1}\operatorname{Re}(\xi'/\xi)$ off the line and can be computed from $\xi$ alone at every resolution, which is why $\rho_\varepsilon$ is the central object of Chapter 3 and Section 4.1; the price is the heavy tail, which makes positivity at one resolution a statement about all zeros at once. The gamma smoothing is local, and an off line pair contributes to $U_k$ a signed, oscillating term near its height.

### 5.1.5 Along a ray: the Gaussian smoothing of the zeros

Fix a height $u$ and move along the ray $t = k/u^2$ of the $(k,t)$ plane, on which the weight $x^k e^{-xt}$ of $c_k(t)$ peaks at $x = u^2$ and the density of $U_k$ is taken at $z = u^2$. Write each zero as $\rho = \frac12 + \delta + i\gamma$ and put $\tilde\gamma = \gamma - i\delta$, so that $a = \tilde\gamma^2$; as in every sum of this section, the zero is taken with $\gamma > 0$ from each pair $\{\rho, 1-\rho\}$. The Gaussian smoothing of the zeros at resolution $1/(2\sqrt{t})$ is

$$\mathcal{G}(u,t) = \sum e^{-2t(\tilde\gamma - u)^2}.$$

A zero on the line contributes a Gaussian bump at its ordinate. The zeros $\frac12 \pm \delta + i\gamma$ of a quadruple $\frac12 \pm \delta \pm i\gamma$ have conjugate values of $\tilde\gamma$, so $\mathcal{G}$ is real, and together they contribute $2e^{2t(\delta^2 - (\gamma-u)^2)}\cos(4t\delta(\gamma-u))$, a bump amplified by $e^{2t\delta^2}$ and modulated in $u$ with period $\pi/(2t\delta)$. At $u = 0$, $\mathcal{G}(0,t) = W(2t)$.

Proposition 5.1.10. *Let $u \ge 2$, $t > 0$ and $k = u^2 t$, not necessarily an integer, with $a^k$ the principal power in $c_k(t) = \sum a^k e^{-at}$, and let $f_0 = k(\log u^2 - 1)$. Then*

$$e^{-f_0} c_k(t) = \sum_{|\tilde\gamma - u| \le u/2} e^{-2t(\tilde\gamma - u)^2}\big(1 + \varepsilon_\rho\big) + R, \qquad |\varepsilon_\rho| \le \exp\Big\{\frac{4t}{3u}|\tilde\gamma - u|^3\Big\} - 1,$$

*where, with $\ell_\pm = u \pm \frac12\sqrt{u^2-1}$, $H(r) = u^2 \log\frac{r^2+1/4}{u^2} - r^2 + \frac14 + u^2$ and $\eta = 1 - u^2/(\ell_+^2 + \frac14) > 0$,*

$$|R| \le N(\ell_-) e^{tH(\ell_-)} + \Big(\ell_+^2 + \frac{1}{t\eta}\Big) e^{tH(\ell_+)}.$$

*No hypothesis on the zeros is used.*

*Proof.* Put $\tilde\gamma = u + v$ and $w = v/u$. Since $\operatorname{Re}\tilde\gamma > 0$ and $|\arg\tilde\gamma| < \frac{\pi}{4}$, the principal logarithm gives $e^{-f_0} a^k e^{-at} = e^E$ with

$$E = tu^2\big(2\log(1+w) - 2w - w^2\big) = -2tv^2 + 2tu^2 \sum_{m \ge 3} \frac{(-1)^{m+1} w^m}{m},$$

and for $|w| \le \frac12$ the series has modulus at most $|w|^3/(3(1-|w|)) \le \frac23 |w|^3$. This gives $\varepsilon_\rho = e^{E + 2tv^2} - 1$ and its bound, since $|e^z - 1| \le e^{|z|} - 1$. For a zero with $|v| > u/2$, $|v|^2 = (\gamma - u)^2 + \delta^2$ and $\delta^2 < \frac14$ give $|\gamma - u| > \frac12\sqrt{u^2-1}$, so that $\gamma < \ell_-$ or $\gamma > \ell_+$, and $|e^E| = \exp t\big(u^2 \log(|a|/u^2) - \operatorname{Re} a + u^2\big) \le e^{tH(\gamma)}$, because $|a| = \gamma^2 + \delta^2$ and $\operatorname{Re} a = \gamma^2 - \delta^2$. The function $H$ increases up to $\sqrt{u^2 - \frac14}$, which lies between $\ell_-$ and $\ell_+$, so each of the at most $N(\ell_-)$ zeros below $\ell_-$ contributes at most $e^{tH(\ell_-)}$. For $r \ge \ell_+$, $\log(1+x) \le x$ gives $H(r) \le H(\ell_+) - \eta(r^2 - \ell_+^2)$. The explicit Riemann von Mangoldt formula [176, Corollary 1.2] gives $N(r) \le r^2$ for every $r$, and partial summation then gives $\sum_{\gamma > \ell_+} e^{-q(\gamma^2 - \ell_+^2)} \le \ell_+^2 + 1/q$ for $q > 0$; take $q = t\eta$. □

For $u \geq 10$ the ratios $H(\ell_-)/u^2$ and $H(\ell_+)/u^2$ decrease in $u$, from $-0.616$ and $-0.431$ at $u = 10$, and $\eta > 0.55$, so that $|R| \leq (3u^2 + 2/t)e^{-0.43k}$. What remains is $\varepsilon_\rho$, whose term of order $1/u$ is $\frac{2t}{3u}(\tilde{\gamma} - u)^3$. Along the ray, therefore, $e^{-f_0}c_k(t)$ is $\mathcal{G}(u,t)$ up to an exponentially small remainder and, in each term, a relative error at most $e^{4t|\tilde{\gamma}-u|^3/3u} - 1$. That error is small, of order $t|\tilde{\gamma} - u|^3/u$, for the terms that carry the sum provided $t|\tilde{\gamma} - u|^3$ is small against $u$ for them; for the zeros within a few multiples of $1/\sqrt{t}$ of $u$ it is of order $1/(\sqrt{t}\,u)$. It is not small in general: at a fixed height, as $t$ grows, the term of the nearest zero dominates, and its relative error grows with $t$. For an integer $k$ the density of $U_k$ in the chart $\theta = \sqrt{z}$ is exactly $\sqrt{2t/\pi}\,e^{-\lambda_k}e^{-f_0}c_k(t)$ at $\theta = u$, where $t = k/u^2$ and $\lambda_k \in (\frac{1}{12k+1}, \frac{1}{12k})$ is the remainder in Stirling's formula for $k!$ [460]. The density of $U_k$ at $\theta$ is therefore about $\sqrt{2t/\pi}\,\mathcal{G}(\theta,t)$ with $t = k/\theta^2$: the zeros smoothed by normal densities of standard deviation $1/(2\sqrt{t}) = \theta/(2\sqrt{k})$, which is the width found above.

Proposition 5.1.11. *RH holds if and only if $\mathcal{G}(u,t) \geq 0$ for every real $u$ and every $t > 0$.*

*Proof.* Under RH every term is positive. Suppose that $\frac{1}{2} + d + ic$ is a zero with $d > 0$, and let $|u - c| < \frac{d}{2}$. The term of a zero has modulus $e^{2t\phi}$ with $\phi = \delta^2 - (\gamma - u)^2$. For the zero $\frac{1}{2} + d + ic$, $\phi > \frac{3}{4}d^2$; for a zero with $|\gamma - c| > 1$, $\phi < \frac{1}{4} - \frac{9}{16} < 0$. Group the finitely many zeros with $|\gamma - c| \leq 1$ by their type $(\gamma, |\delta|)$. Two types at the same ordinate have exponents $\phi$ differing by a nonzero constant, and two at different ordinates have exponents differing by a nonconstant affine function of $u$, so outside finitely many points one type has the largest $\phi$. Fix such a $u_*$, different from every ordinate, and let $(\gamma_*, d_*)$ be the maximising type and $\phi_*$ its exponent. Then $\phi_* > \frac{3}{4}d^2$, so $d_* > 0$, and the type consists of $\nu \geq 2$ zeros counted with multiplicity, which contribute $\nu e^{2t\phi_*}\cos(4td_*(\gamma_* - u_*))$. Along $t_j = (2j+1)\pi/(4d_*|\gamma_* - u_*|)$ this is $-\nu e^{2t_j\phi_*}$. The other $n$ zeros with $|\gamma - c| \leq 1$ contribute at most $ne^{2t_j(\phi_* - \Delta)}$, with $\Delta > 0$ the gap to the next exponent, and the zeros with $|\gamma - c| > 1$ at most $32\log(c+2)e^{-5t_j/8}$ once $t_j \geq 1$, since by the explicit Riemann von Mangoldt formula every interval of heights $[y, y+1]$ holds at most $4\log(y+3)$ zeros. Hence

$$e^{-2t_j\phi_*}\mathcal{G}(u_*, t_j) \leq -\nu + ne^{-2t_j\Delta} + 32\log(c+2)e^{-(2\phi_* + 5/8)t_j},$$

which is negative for all large $j$. □

Proposition 5.1.12. *For $\frac{1}{2} < \alpha < 1$ let $\eta = \alpha - \frac{1}{2}$ and $\mathcal{G}_\alpha(u,t) = \operatorname{Re}\sum e^{-2t(\tilde{\gamma} + i\eta - u)^2}$, over the zeros with $\gamma > 0$ as for $\mathcal{G}$, which is $\mathcal{G}$ at $\eta = 0$. Under RH $\mathcal{G}_\alpha(u,t) < 0$ for some $u$ and $t$: there is no Gaussian criterion on the line $\operatorname{Re} s = \alpha$.*

*Proof.* Under RH $\tilde{\gamma} = \gamma$, and the term of $\gamma$ is $e^{2t\eta^2}e^{-2t(\gamma-u)^2}\cos\big(4t\eta(\gamma - u)\big)$. Take $u = \gamma_1 + \pi/(4t\eta)$, where $\gamma_1$ is the lowest ordinate and a simple zero, and let

$g > 0$ be the gap to the next ordinate. The term of $\gamma_1$ is $-e^{2t\eta^2-\pi^2/(8t\eta^2)}$. For $t \geq \pi/(2\eta g)$ every other $\gamma$ has $\gamma - u \geq (\gamma-\gamma_1)/2$, so the other terms total at most $e^{2t\eta^2}\sum_{\gamma>\gamma_1} e^{-t(\gamma-\gamma_1)^2/2} = o(e^{2t\eta^2})$. Hence $\mathcal{G}_\alpha(u,t) = -e^{2t\eta^2}(1+o(1)) < 0$ as $t \to \infty$. □

Proposition 5.1.11 restricts Weil's criterion to the modulated Gaussians. For $f(x) = e^{-x^2/4t-iux}$ the transform $\hat f(r) = \int f(x)e^{irx}dx = \sqrt{4\pi t}\, e^{-t(r-u)^2}$ gives, summed over all zeros $\rho = \frac12 + i\gamma_\rho$,

$$\sum_\rho \hat f(\gamma_\rho)\overline{\hat f(\bar\gamma_\rho)} = 4\pi t\big(\mathcal{G}(u,t) + \mathcal{G}(-u,t)\big),$$

where $\mathcal{G}(-u,t)$ comes from the zeros $1-\rho$ and is at most $e^{t/2}((u+14)^2 + \frac{1}{2t})e^{-2t(u+14)^2}$ in modulus for $u \geq 0$, too little to change the sign in the proof. At $u = 0$ the family gives the first rung, $\mathcal{G}(0,t) = W(2t) > 0$ (Theorem 5.1.1). Since the Gaussian is entire, Weil's formula of Section 1.2 applies to $h(r) = e^{-2t(r-u)^2} + e^{-2t(r+u)^2}$, for which $\sum_\rho h(\gamma_\rho) = 2(\mathcal{G}(u,t) + \mathcal{G}(-u,t))$, and gives $\mathcal{G}$ from the primes:

$$\begin{aligned} \mathcal{G}(u,t) + \mathcal{G}(-u,t) &= 2e^{t/2-2tu^2}\cos 2tu + A(u,t) - P(u,t), \\ A(u,t) &= \frac{1}{4\pi}\int_{\mathbb{R}} \big(e^{-2t(r-u)^2} + e^{-2t(r+u)^2}\big)m(r)\,dr, \\ P(u,t) &= \frac{1}{\sqrt{2\pi t}}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt n}e^{-(\log n)^2/8t}\cos(u\log n), \end{aligned} \tag{5.1.3}$$

with $m$ as in (1.2.5).

Table 5.3 computes both sides. The prime sum converges at every resolution, but more slowly the finer the resolution: since Chebyshev's function $\psi(x) = \sum_{n\leq x}\Lambda(n)$ grows like $x$, the weight $n^{-1/2}e^{-(\log n)^2/8t}$ acts, as a density in $\log n$, like $e^{t/2}$ times a Gaussian with centre $2t$ and standard deviation $2\sqrt t$. At $t = 3$ (last row) the rest of the prime sum beyond $10^9$ is nearly all main term: with the integral $(2\pi t)^{-1/2}\int_N^\infty x^{-1/2}e^{-(\log x)^2/8t}\cos(u\log x)\,dx$ that the prime number theorem predicts for it, the difference at $N = 10^9$ falls to $3\times 10^{-10}$. Where $\mathcal{G}$ is small the two arithmetic terms cancel, the more deeply the finer the resolution, as for $W$ in Table 1.4; these are the points where an off line quadruple would first make $\mathcal{G}$ negative (Section 5.2).

### 5.1.6 What a verification height buys

Chapters 6 and 7 reach further rungs by the same method. Adding to the clock variable an independent positive stable variable of index $\frac12$, that is, multiplying the clock by $e^{-c\sqrt u}$ with $c = \zeta'(\frac12)/\zeta(\frac12) = 2.6860917$, gives the floor $e^{-c\sqrt u}\,\xi(\frac12)/\xi(\frac12 + \sqrt u)$ of Section 6.2, whose canonical function, $y$ times its Lévy density, is $k(y) =$

| $u$ | $t$ | $\mathcal{G}(u,t)$ | $A(u,t)$ | $P(u,t)$ | $\lvert\mathcal{G}-A+P_N\rvert$ | | |
|---|---|---|---|---|---|---|---|
| | | | | | $N=10^4$ | $N=10^6$ | $N=10^9$ |
| 100 | 1 | 0.0960845325 | 0.5519917891 | 0.4559072567 | $2.3\times10^{-6}$ | $1.2\times10^{-10}$ | $9.6\times10^{-19}$ |
| 1977.22 | 1.75 | 1.9835406644 | 0.8672562847 | −1.1162843797 | $1.2\times10^{-3}$ | $1.3\times10^{-7}$ | $2.2\times10^{-13}$ |
| 1000 | 2 | 0.8452691893 | 0.7150931047 | −0.1301760846 | $5.8\times10^{-4}$ | $5.8\times10^{-6}$ | $1.9\times10^{-11}$ |
| 100 | 3 | 0.0003053958 | 0.3186935678 | 0.3183870527 | $2.0\times10^{-3}$ | $5.4\times10^{-4}$ | $1.1\times10^{-6}$ |

Table 5.3: The Gaussian smoothing of the zeros from the zeros and from the primes, by (5.1.3). $\mathcal{G}(u,t)$ is summed over the zeros below height 2197.26, the rest adding less than $10^{-70000}$; the polar term and $\mathcal{G}(-u,t)$ are below $10^{-8000}$. $P_N$ is $P$ with the sum cut at $n\le N$, and the $P$ column is $P_{10^9}$. The bound on the rest of the prime sum that ignores its phases, from $\psi(x)<1.03883x$ [465, Theorem 12], is $1.6\times10^{-20}$, $2.1\times10^{-10}$, $1.0\times10^{-8}$ and $1.0\times10^{-4}$ at $N=10^9$ for the four rows; in the first row the difference at $N=10^9$ is the rounding error of the sums, in extended precision.

$W(y)+c/(2\sqrt{\pi y})$. By Theorem 5.1.1, $k>W>0$, so the floor is infinitely divisible unconditionally; Section 6.2 proves this independently, and proves that the floor is a GGC if and only if RH holds (Theorem 6.2.15). Weil's functional is positive on the cone of test functions completely monotone in $r^2$ (Theorem 7.2.15); this is the first rung integrated against a positive weight, $\sum_\rho \hat h(\gamma_\rho)=2\int_0^\infty W(y)g(y)\,dy$ with $g\ge0$, and for it Section 7.2 proves $W>0$ afresh.

Every rung is reached by the same mechanism: near the origin the archimedean term dominates the primes, and away from it a known positive term, usually a verified low zero, dominates the unverified high zeros. Only the second limits the order, as the proof of Theorem 5.1.6 shows. The first limits the time instead: it gives every order $k\ge1$ up to $t=0.019$, and every order above $10^{17}$ up to $t=1.29$ (Theorem 5.1.9).

THEOREM 5.1.13. *Let $H=3\times10^{12}$, the height to which RH is verified [419]. Then $(-1)^kW^{(k)}(t)>0$ for every $t\ge1.29$ and every integer $k$ with $0\le k\le t(H-23)^2$. Consequently $(-1)^kW^{(k)}(t)>0$ for every $t>0$ and every integer $k$ with $0\le k\le 1.29(H-23)^2$, and $1.29(H-23)^2>1.16\times10^{25}$.*

*Proof.* Fix $t\ge1.29$ and $k$, and put $u=\sqrt{k/t}$, $D=H-u\ge23$ and $f(v)=v^{2k}e^{-tv^2}$. As in Section 5.1.5, write each zero as $\frac12+\delta+i\gamma$ and put $\tilde\gamma=\gamma-i\delta$, so that $c_k(t)=\sum f(\tilde\gamma)$ over one zero of each pair $\{\rho,1-\rho\}$, taken with $\gamma>0$. On $(0,\infty)$, $(\log f)''=-2k/v^2-2t$ is at most $-2t$, and at least $-4t$ on $[u,\infty)$, while $(\log f)'(u)=0$; hence $f(v)\le f(u)e^{-t(v-u)^2}$ for every $v>0$, and $f(v)\ge f(u)e^{-2t(v-u)^2}$ for $v\ge u$. The zeros below height $H$ lie on the line and contribute $f(\gamma)>0$. One of them lies in $[u,u+15]$: for $u\le1000$ because $\gamma_1<15$ and consecutive ordinates below 1015 differ by less than 7, and for $u\ge1000$ because the explicit Riemann von Mangoldt formula of [176, (1.7) and Corollary 1.4]

gives $N(u+15)-N(u)>4$. That zero contributes at least $f(u)e^{-450t}$. A zero above $H$, wherever it lies, has $|\delta|<\frac{1}{2}$ and $|\tilde{\gamma}|\geq\gamma>u$, so its term has modulus $f(|\tilde{\gamma}|)e^{2t\delta^2}\leq e^{t/2}f(u)e^{-t(\gamma-u)^2}$. Partial summation with $N(r)\leq r^2$, and $\int_D^\infty e^{-tv^2}dv\leq e^{-tD^2}/(2tD)$, give

$$\sum_{\gamma>H}e^{-t(\gamma-u)^2}\leq e^{-tD^2}\Big(H^2+\frac{1}{t}+\frac{H}{tD}\Big),$$

and therefore

$$\frac{c_k(t)}{f(u)}\geq e^{-450t}-e^{t/2-tD^2}\Big(H^2+\frac{1}{t}+\frac{H}{tD}\Big)>0,$$

since $t(D^2-450.5)\geq 1.29\cdot 78.5>101$ while $\log(H^2+1+H/23)<57.5$. For the second statement, Theorem 5.1.9(b) gives every order for $t\leq 1.29$, and for $t>1.29$ every $k\leq 1.29(H-23)^2$ satisfies $k\leq t(H-23)^2$. The program `window_orders.py` (`code/README.md`) computes the zeros below height 1015 in ball arithmetic and evaluates every numerical inequality of the proof. □

For the floor, $c/(2\sqrt{\pi y})$ is completely monotone, so Theorem 5.1.13 gives $(-1)^n k^{(n)}>0$ for every $n\leq 1.16\times 10^{25}$; Theorem 6.2.17 proves $n\leq 10^{10}$ independently, with a Cauchy circle of radius $y/2$, which loses a factor $2^n$ and makes its reach grow only linearly in $H$. The same circle applied to $W$ itself, as in a note of Nicholas Polson merged into the book [243], bounds the prime terms of (1.2.5) at order $k$ by a quantity that stays below the archimedean term only for $t\lesssim c_P/k$ up to a constant, where $c_P=\frac{1}{6}(\log 2)^2$ is the constant of the factor $e^{-c_P/t}$ that the circle leaves to the prime 2, while the zeros above $H$ stay below the first zero only for $t\gtrsim 2k\log(H/\gamma_1)/H^2$. Bounding the factor $2^k$ that the circle loses by $e^k$, the two ranges meet at an order of about $H\big(c_P/(2\log(H/\gamma_1))\big)^{1/2}$, which is $1.1754\times 10^{11}$ for $H=3\times 10^{12}$ (heuristic, `pg3_reach.py`); the exact factor $\log 2$ in place of 1 multiplies this by $(\log 2)^{-1/2}$. This describes the circle bound, not a limit on what the derivative signs follow from, and it is far below $1.29(H-23)^2$ in Theorem 5.1.13.

Problem 5.1. At the verified height $H=3\times 10^{12}$, Theorem 5.1.16 gives the signs of the derivatives of $W$ at every $t>0$ of every order $k\leq 1.70(H-23)^2$. Can this range be increased to $k\leq cH^2$ with an explicit $c>1.70$? In the quadruple model of Section 5.2.1, a quadruple just above height $H$ first changes a sign at an order of about $H^2t^*$, with $t^*$ set by the gaps around it, which suggests a scale to investigate (heuristic; the model does not show that such a failure occurs for the zeros of $\zeta$).

### 5.1.7 A compound Poisson defect of the off line zeros

Group the exponents $a_k$ into the real ones, $a_k=\gamma^2>0$, which belong to zeros on the line, and the conjugate pairs $\{a,\bar{a}\}$ of nonreal exponents, one for each off line

quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, for which $a = A - iB$ with $A = \gamma^2 - \delta^2$ and $B = 2\delta\gamma$. Each pair contributes $2e^{-At}\cos(Bt)$ to $W$, and $\cos\theta = 1 - 2\sin^2(\theta/2)$ splits this into a Laplace transform part and a nonnegative remainder.

THEOREM 5.1.14. *Unconditionally, $W(t) = \int_{(0,\infty)} e^{-tz}\, U_+(dz) - R(t)$, where $U_+$ is a Thorin measure, $R \geq 0$, and*

$$U_+ = \sum_{a_k \text{ real}} \delta_{a_k} + 2\sum_{pairs} \delta_A, \qquad R(t) = 4\sum_{pairs} e^{-At}\sin^2\Big(\frac{Bt}{2}\Big).$$

*Let $T_+$ be the GGC with Thorin measure $U_+$, and $Y$ the infinitely divisible variable with Lévy density $R(t)/t$, independent of $T$. Then*

$$T + Y \overset{d}{=} T_+, \qquad \mathbb{E}e^{-uY} = \prod_{pairs} \frac{(1+u/a)(1+u/\bar{a})}{(1+u/A)^2}.$$

*The defect $Y$ is compound Poisson, with rate and jump density*

$$\lambda = \int_0^\infty \frac{R(t)}{t}\, dt = \sum_{pairs} \log\Big(1 + \frac{B^2}{A^2}\Big), \qquad \frac{R(t)}{\lambda t},$$

*and $\lambda < 10^{-12}$, so $\Pr(Y = 0) = e^{-\lambda} > 1 - 10^{-12}$. The following are equivalent: $R \equiv 0$; $Y = 0$; RH. In that case $U_+ = U_{\mathrm{RH}}$ and $T_+ = T$.*

*Proof.* The decomposition is the identity applied pair by pair, and $U_+$ is a Thorin measure because $A \geq \frac{1}{2}|a|$, since $|a| = \gamma^2 + \delta^2$ and $\gamma > 14$, and $\sum_k |a_k|^{-1} < \infty$. By Tonelli and the Frullani type integral $\int_0^\infty e^{-At}\sin^2(Bt/2)\, t^{-1}dt = \frac{1}{4}\log(1+B^2/A^2)$, the function $R(t)/t$ is integrable on $(0,\infty)$ with integral $\lambda$, which is finite by the bound below, and a Lévy measure of finite mass is compound Poisson. For the bound let $H = 3 \times 10^{12}$. Every pair has $\gamma > H$, and $\delta < \frac{1}{2}$ gives $B^2/A^2 = 4\delta^2\gamma^2/(\gamma^2 - \delta^2)^2 < \gamma^2/(\gamma^2 - \frac{1}{4})^2 < (1 + 10^{-24})\gamma^{-2}$. The two zeros $\frac{1}{2} \pm \delta + i\gamma$ of a pair have the same ordinate, so $\lambda < \frac{1}{2}(1+10^{-24})\sum_{\gamma > H} \gamma^{-2}$, the sum running over all zeros of ordinate above $H$. Partial summation with the explicit bound for $N(T)$ of [176, Corollary 1.2], exactly as for $\varepsilon_H$ in the proof of Proposition 14.7.2, gives $\sum_{\gamma > H} \gamma^{-2} < 1.4798 \times 10^{-12}$, hence $\lambda < 7.4 \times 10^{-13}$. The program `xg_defect.py` (`code/README.md`) evaluates this bound in ball arithmetic.

By the formula for $\log G$ at the start of this section and Frullani's integral $\log(1 + u/z) = \int_0^\infty (1 - e^{-ut})e^{-zt}t^{-1}dt$,

$$\log G(u) = \int_0^\infty \big(1 - e^{-ut}\big)\frac{W(t)}{t}\, dt = \int \log\Big(1 + \frac{u}{z}\Big)U_+(dz) - \int_0^\infty \big(1 - e^{-ut}\big)\frac{R(t)}{t}\, dt,$$

that is, $\mathbb{E}e^{-uT}\,\mathbb{E}e^{-uY} = \mathbb{E}e^{-uT_+}$, and the product formula follows from $\mathbb{E}e^{-uT_+} = \prod_{a_k \text{ real}}(1 + u/a_k)^{-1}\prod_{\text{pairs}}(1 + u/A)^{-2}$ and (1.2.1). For the equivalences: every term of $R$ is nonnegative, and a pair contributes a positive term outside the discrete

set $\{t > 0 : Bt \in 2\pi\mathbb{Z}\}$, so $R \equiv 0$ if and only if there is no pair, that is, every $a_k$ is real, which happens only for zeros on the line; and $Y = 0$ if and only if $\lambda = 0$ if and only if $R \equiv 0$. Under RH, $U_+ = \sum_k \delta_{a_k} = U_{\mathrm{RH}}$ and $T_+ = T$ by Proposition 1.2.1. □

Remark 5.1.15. Unconditionally, therefore, the clock plus an independent compound Poisson variable of rate below $10^{-12}$ is a GGC, and the law of the clock is within total variation $10^{-12}$ of that GGC, since $T = T + Y$ on the event $\{Y = 0\}$.

### 5.1.8 The prime side of the ladder

The proofs of Theorems 5.1.9 and 5.1.13 stop at 1.29 and $1.16 \times 10^{25}$ because of the constant $K = 10^{17}$ with which the proof of Theorem 5.1.9 uses Lemma 5.1.8(b): as the discussion after Theorem 5.1.9 says, 1.2912 is the root of $E_{K+1}(t) = 2S(t)$ for that $K$. Theorem 5.1.13 supplies every order up to $1.16 \times 10^{25}$, and the lemma can be used with that order in the place of $10^{17}$. The root of $E_{K+1}(t) = 2S(t)$ then moves to about 1.719 in floating point, and the argument of both theorems runs at $t = 1.70$.

Theorem 5.1.16. *$W$ is completely monotone on $(0, 1.70]$, and $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every integer $k$ with $0 \le k \le 1.70(H-23)^2$, where $H = 3\times10^{12}$; and $1.70(H - 23)^2 > 1.52 \times 10^{25}$.*

*Proof.* Let $K = 1.16 \times 10^{25}$. By Theorem 5.1.13, $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every $k \le K$. For $k > K$ and $t \le 0.019$, Theorem 5.1.9(a) applies. For $k > K$ and $t \in [0.019, 1.70]$ the proof of Theorem 5.1.9(b) applies with $K$ in the place of $10^{17}$ and 1.70 in the place of 1.29, and with the cutoff $n_0 = 10^9$ in the place of $2^{24}$: Lemma 5.1.8(b) with $\varepsilon = 10^{-6}$ gives $C_\varepsilon(k) \le C_\varepsilon(K + 1) < 1 + 6 \times 10^{-7}$ for $k > K$, and the condition $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds $2.5 \times 10^9$, which covers every prime power $n \le 10^9$ at $t \ge 0.019$ (there $x_n < 76$). The monotonicity in $k$ and $t$ is that of the proof, so it suffices to show $E_{K+1}(1.70) - S_C(1.70) - Q_{K+1}(1.70) > 0$, with $S_C$ the sum $S_B$ with $B(x_n)$ replaced by $C$ for $n \le 10^9$. Chernoff's bound for the gamma law with $a = (1 - 2^{-13})^2\lambda$ gives $E_{K+1}(1.70) > 26.7532$; the prime powers $n \le 10^9$ give $S_C(1.70) < 26.2968$, with the tail bound of that proof, which is $2.9 \times 10^{-4}$ at $q = \log n_0/8t = 1.52$; and $Q_{K+1}(1.70) < e^{-1000}$. The margin exceeds 0.4564. So the orders $k > K$ have the right signs on $(0, 1.70]$ and, with the orders $k \le K$, $W$ is completely monotone there.

For $t > 1.70$ and $k \le 1.70(H - 23)^2$ the proof of Theorem 5.1.13 applies with 1.70 in the place of 1.29, since $k \le t(H - 23)^2$ and the only place where the lower bound on $t$ enters is $t(D^2 - 450.5) \ge 1.70 \cdot 78.5 > 133$, against $\log(H^2 + 1 + H/23) < 57.5$. Finally $1.70(H - 23)^2 > 1.52 \times 10^{25}$. The program `cm_orders_k116e25.py` (`code/README.md`) evaluates every numerical inequality of the argument for $k > K$

in ball arithmetic, with the prime powers $n \le 10^9$ (50 851 223 of them): all of them hold. □

In floating point the root of $E_{K+1}(t) = 2S(t)$ for $K = 1.52 \times 10^{25}$ is about 1.724, which is the time $t_0 = 1.72$ of the phase free bound at the verified height (Section 12.2.6); 1.70 is the safe round number.

By Proposition 5.1.10, on the ray $t = k/u^2$ the rung is, up to the errors of that proposition, $\mathcal{G}(u,t)$, and (5.1.3) writes $\mathcal{G}(u,t) + \mathcal{G}(-u,t)$ as a polar term plus $A(u,t) - P(u,t)$, where $A(u,t)$ is close to $\log(u/2\pi)/(2\sqrt{2\pi t})$ and $P(u,t) = (2\pi t)^{-1/2}\Pi(u;t)$, with $\Pi(T;x) = \sum_{n\ge 2}\Lambda(n)n^{-1/2}e^{-(\log n)^2/8x}\cos(T\log n)$ the smoothed prime sum. To leading order, therefore, the rung at height $T$ and scale $x$ is positive when $2\Pi(T;x) < \log(T/2\pi)$, and this is the comparison of the archimedean term with the prime term for which Section 12.2.6 gives the phase free time $t_0(T)$, which agrees to five digits with $x_c(T)$, the root of $M(x) = \log(T/2\pi)$ with $M(x) = 2\sum_{n\ge 2}\Lambda(n)n^{-1/2}e^{-(\log n)^2/8x}$ (twice the sum $S(x)$ of that section). Below $x_c(T)$ it holds whatever the phases are; above it, it is a statement about cancellation among the primes, which Proposition 5.1.11 makes an exact criterion for RH, with $A$ and $\mathcal{G}$ in place of the leading terms.

The comparison can be evaluated at heights far beyond verification, because the cost depends on $T$ only through the precision of $T\log n$ modulo $2\pi$. The table gives $2\Pi(T;x)$ against $\log(T/2\pi)$, with the prime powers $n \le N$, $N = \lfloor e^{\sqrt{8xD}} \rfloor$ for the tolerance $D = 22$ at $x \le 1$, $D = 20$ at $x = 1.7$, $D = 16$ at $x = 2.5$ and $D = 14$ at $x = 3$, and the phases in ball arithmetic at 300 bits. The weight of the omitted prime powers is about $e^{\sqrt{2xD}-D}$ times a factor of order one, not $e^{-D}$; the Prime Number Theorem estimate of the sum of their absolute values is below $1.5 \times 10^{-2}$ in every row. The values are not certified, and the leading term $\log(T/2\pi)$ stands for the archimedean term $A(T,x)$, with a relative error $O(T^{-2})$. The condition holds in every row; the scales $x = 1.7$, 2.5 and 3 at $T = 10^{12}$ are above $x_c(10^{12}) = 1.679$ (computed), where the weights alone no longer decide.

| $T$ | $x$ | $N$ | $2\Pi(T;x)$ | $\log(T/2\pi)$ | $\pi/(4x)$ |
|---|---|---|---|---|---|
| $10^{12}$ | 0.5 | $1.2 \times 10^4$ | −0.187 | 25.79 | 1.57 |
| $10^{12}$ | 1.0 | $5.8 \times 10^5$ | −0.374 | 25.79 | 0.79 |
| $10^{12}$ | 1.7 | $1.5 \times 10^7$ | −0.142 | 25.79 | 0.46 |
| $10^{12}$ | 2.5 | $5.9 \times 10^7$ | 0.630 | 25.79 | 0.31 |
| $10^{12}$ | 3.0 | $9.1 \times 10^7$ | 1.279 | 25.79 | 0.26 |
| $3 \times 10^{12}$ | 1.7 | $1.5 \times 10^7$ | −0.266 | 26.89 | 0.46 |
| $3 \times 10^{12}$ | 3.0 | $9.1 \times 10^7$ | −2.713 | 26.89 | 0.26 |
| $10^{20}$ | 1.0 | $5.8 \times 10^5$ | −0.720 | 44.21 | 0.79 |
| $10^{30}$ | 1.0 | $5.8 \times 10^5$ | −0.077 | 67.24 | 0.79 |
| $10^{50}$ | 1.0 | $5.8 \times 10^5$ | 0.881 | 113.29 | 0.79 |

The last column is the product $\delta\Delta$ of the displacement and the gap of an isolated off line pair at which, in the local model of Proposition 5.2.4 and to

leading order only, a failure of the comparison at the scale $x$ would first be visible, since $t^*_{\mathrm{loc}} = \omega^*/(2\delta\Delta)$ with $\omega^*$ just above $\pi/2$ (Section 5.2.4). The failure sits at one offset of the window in the height, so one value of $T$ excludes nothing and a scan in $T$ over about $\Delta$ is needed. In this reading the prime side resolves only zeros that are well off the line and isolated, and a pair with $\delta\Delta \ll 1$, such as a Lehmer type near miss, needs a scale $x \gg 1$ at which the cutoff $N$ is out of reach. The program `prime_side_check.py` (`code/README.md`) computes the table and the values of $M$ and $x_c$ quoted above.

## 5.2 What finite information cannot decide

Every computation in this book is finite: a finite window, finitely many resolutions, finitely many derivatives or matrix blocks. Theorem 5.2.1 replaces two close zeros of $\xi$ on the line by a quadruple off it [424, Chapter 11] and obtains an entire function whose heat trace passes the finite tests of Section 5.1.

### 5.2.1 A perturbation that passes the finite tests of the heat trace

Theorem 5.2.1 ([421, Theorem D]). (a) *There are ordinates* $1000 < A < B \le 2000$ *of zeros on the critical line, with* $B - A \le 2$*, such that, with* $g = (A+B)/2$ *and* $w = \frac{1}{4} + ig$*,*

$$F(s) = \frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})}\,\frac{(1 - s^2/w^2)(1 - s^2/\bar{w}^2)}{(1 + s^2/A^2)(1 + s^2/B^2)}$$

*is an even entire function with the order, type and reflection symmetry of* $\xi(\frac{1}{2}+s)$ *and with four zeros off the critical line, and the following hold. The reciprocal clock* $1/F(\sqrt{u})$ *is the Laplace transform of an infinitely divisible, self decomposable law, and its heat trace*

$$W_F(t) = W(t) + 2\,\mathrm{Re}\,e^{-at} - e^{-A^2t} - e^{-B^2t}, \qquad a = -w^2 = g^2 - \tfrac{1}{16} - \tfrac{ig}{2},$$

*which is* $t$ *times its Lévy density, is strictly log convex.* $W_F$ *has the same small time expansion as* $W$ *through the constant term, and* $1/F$ *has the same nearest poles as the reciprocal Xi function.*

(b) *Let* $N \ge 1$ *and* $R \ge T_N$*, where* $T_N = 20000N\sqrt{1+\log N}$ *as in Section 5.1, and suppose that every zero of* $\xi$ *with ordinate in* $(0, R+50]$ *lies on the critical line. Then the pair can instead be chosen with* $R < A \le B \le R+50$ *and* $B - A < 2$ *(if* $A = B$*, a multiple zero supplies both), so that* $F$ *keeps every property listed in (a) and the leading* $N \times N$ *Hankel blocks of* $W_F$ *remain positive definite at every* $t > 0$*. If moreover* $R \ge 10^{12}$*, the derivative signs of Theorem 5.1.6 are also preserved:* $(-1)^k W_F^{(k)}(t) > 0$ *for every* $t > 0$ *and every integer* $0 \le k \le 10^{17}$*. For every* $N \le 3 \times 10^7$ *the verification to height* $3 \times 10^{12}$ *[419] establishes the hypothesis on*

*the zeros for* $R = \max(T_N, 10^{12})$*, so that such an* $F$ *with both properties exists unconditionally.*

The construction moves the zeros at heights $A$ and $B$, with their conjugates, to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$. The heat trace changes by the oscillating term $W_F - W$, whose envelope is bounded as in Section 5.1 [421, Proposition 11.1], and at the heights of (b) this change is smaller than the margin by which each of its tests holds. The pair of (b), with the properties of (a) and the Hankel blocks, comes from [421, Theorem 12.5], the derivative signs from [421, Proposition 12.7], which holds for any pair of zeros on the line with $A > 10^{12}$ and $B - A \leq 2$, and the unconditional range from [421, Corollary 12.6], since $T_{3\times 10^7} + 50 < 2.6 \times 10^{12}$. Figure 5.3 shows the change for a pair in the range of (a); the pair it moves, near height 1977.22, is the closest pair among the first 1700 zeros, whether or not the spacings are normalised (Section 5.4.2).

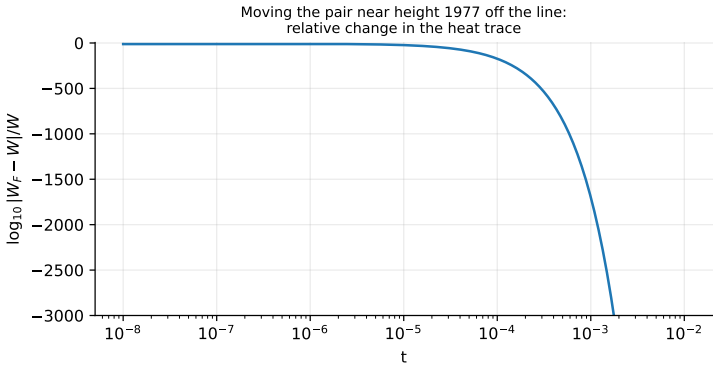



Figure 5.3: Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with $W$ replaced by its sum over the first 1700 zeros. The change is at most $4 \times 10^{-11}$, near $t = 7 \times 10^{-7}$, and falls below $10^{-1000}$ by $t = 10^{-3}$. The sum over 1700 zeros is smaller than $W$, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with $W$ from the explicit formula (1.2.5) the maximum is $3.66 \times 10^{-11}$, at $t = 7.2 \times 10^{-7}$.

Not every finite test is passed. Below any height $T > B$ the argument count of $F$ equals that of $\xi$, since $F$ loses the zeros at heights $A$ and $B$ and gains two at height $g$, but these two are off the line, so $F$ has two zeros fewer on the line, counted with multiplicity; when $A < B$ and both zeros are simple, as for the pair of Figure 5.3, it has two sign changes fewer. Turing's comparison of the two counts, the first test of Table 5.5, then detects the perturbation once it passes height $B$, and the zero density identity at any height $T > g$ returns $\nu(\frac{1}{2} + \varepsilon, T) \geq 1$ for every

$\varepsilon < \frac{1}{4}$, whatever the multiplicities.

The pair of Figure 5.3 lies far below the heights of (b), and at its height the derivative signs of $F$ do fail at a finite order, which can be computed. On the ray $t = k/u^2$ the normalised value $e^{-f_0} c_k^F(t)$ of Proposition 5.1.10, whose proof applies to $F$ because $F$ has at most one zero more than $\xi$ below any height, is a sum of positive terms, one for each zero on the line, the oscillating term of the quadruple and terms, negligible here, from any zeros off the line above $3 \times 10^{12}$, and only the zeros within a few units of $u$ matter. A scan over $u$ and over the orders, at every order up to 500, then on a grid of ratio 1.002 and at every order of the last grid interval, finds that the first order at which $c_k^F(t) < 0$ for some $t > 0$ is $k = 6\,683\,655$, at $t = 1.7113393$. The grid loses nothing, because $c_{k-1}^F(t) = \int_t^\infty c_k^F(s)\,ds$: if $c_k^F \geq 0$ on $(0, \infty)$ then so is $c_{k-1}^F$, and the orders that fail are all those from the first one on. The negative value is certified in ball arithmetic, with explicit bounds for the zeros below $g - 40$, which lie on the line, and for those above $g + 40$, wherever the ones above $3 \times 10^{12}$ lie; the positivity at the lower orders is computed, not certified. Table 5.4 repeats the computation for the closest pair of consecutive zeros in five other windows of height.

| window | zeros moved | $g$ | $\delta$ | $k^*$ | $t^*$ | $c_{k^*}^F(t^*)$, normalised |
|---|---|---|---|---|---|---|
| $(100, 200)$ | 71, 72 | 185.24 | $\frac{1}{4}$ | 40 018 | 1.18413 | $-6.37 \times 10^{-7}$ |
| $(200, 400)$ | 186, 187 | 376.08 | $\frac{1}{4}$ | 219 169 | 1.54072 | $-6.70 \times 10^{-8}$ |
| $(400, 700)$ | 363, 364 | 630.64 | $\frac{1}{4}$ | 624 921 | 1.576635 | $-1.68 \times 10^{-7}$ |
| $(700, 1000)$ | 453, 454 | 750.81 | $\frac{1}{4}$ | 792 746 | 1.410725 | $-9.26 \times 10^{-8}$ |
| $(1000, 1500)$ | 922, 923 | 1329.12 | $\frac{1}{4}$ | 4 433 220 | 2.5121193 | $-3.37 \times 10^{-8}$ |
| $(1500, 2000)$ | 1496, 1497 | 1977.22 | $\frac{1}{4}$ | 6 683 655 | 1.7113393 | $-7.12 \times 10^{-9}$ |
| $(1500, 2000)$ | 1496, 1497 | 1977.22 | 0.1 | 15 722 219 | 4.0257254 | $-6.71 \times 10^{-11}$ |

Table 5.4: The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, $g$ the midpoint of the pair. $k^*$ is the least order at which $c_k^F(t) < 0$ for some $t > 0$, found by the scan above (computed, not certified); $c_{k^*}^F(t^*) < 0$ is certified in ball arithmetic. The last column is $c_{k^*}^F(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$.

The first failing order is $k^* = t^* u_*^2$, where $u_* = \sqrt{k^*/t^*}$ is the height of the failure, between 0.69 and 1.40 from $g$, so that $k^*$ is within 1.6 per cent of $t^* g^2$, with $t^*$ between 1.18 and 2.51 for $\delta = \frac{1}{4}$: it grows roughly with the square of the height. Proposition 5.1.10 explains this. Along the ray, $e^{-f_0} c_k^F(t)$ is, up to relative errors of order $t|\tilde{\gamma} - u|^3/u$ in the terms near $u$, small at these heights and times (Section 5.1.5), the Gaussian smoothing $\mathcal{G}(u, t)$ of the zeros of $F$, in which the quadruple contributes $2e^{2t(\delta^2 - (g-u)^2)} \cos(4t\delta(g - u))$. This term is negative only where $4t\delta|g - u|$ lies in $(\frac{\pi}{2}, \frac{3\pi}{2})$ modulo $2\pi$, hence at distance at least $\pi/(8t\delta)$ from $g$, where its modulus is at most $2e^{2t\delta^2 - \pi^2/(32t\delta^2)}$, while a zero on the line at

distance $d$ from $u$ contributes $e^{-2td^2}$. The first failure time is set by $\delta$ and by the gaps between the line zeros near the moved pair, which its removal widens; the mean spacing $2\pi/\log(g/2\pi)$ falls from 1.86 to 1.09 between the first and last rows, but the gaps near the moved pairs vary more, which is why $t^*$ shows no trend. The first $t$ at which $\min_{|u-g|\le 8} \mathcal{G}(u,t) < 0$, computed in floating point, predicts $k^*$ to 0.21 per cent at height 185 and to 0.08 per cent or better at the other heights, and the minimum lies just beyond the distance $\pi/(8t\delta)$ at which the first negative lobe begins. These failure times of $\mathcal{G}$ within 8 of $g$ are certified in ball arithmetic, with the zeros from Arb, for all seven rows of Table 5.4: $\mathcal{G}(u,t) > 0$ at every centre $u$ within 8 of $g$ for every $t$ up to the failure time rounded down to two decimals, and $\mathcal{G} < 0$ at one centre at the failure time rounded up to four decimals. Far from $g$ the bracket says nothing: near or above $3 \times 10^{12}$, zeros off the line could make $\mathcal{G}$ negative earlier. One time suffices for the first statement, since $\sqrt{2t/\pi}\,\mathcal{G}(\cdot,t)$ at a smaller $t$ is its value at a larger one convolved with a centred normal density: the sum over the zeros below $3 \times 10^{12}$ is positive on the whole line at $t = 1.71$, by an interval cover within 8 of $g$ and by domination beyond, so it has a uniform positive lower bound at every smaller $t$ within 8 of $g$, and the zeros above $3 \times 10^{12}$, wherever they lie, can lower $\mathcal{G}$ there by less than $10^{-2\times 10^{12}}$ (`gsmooth_arb.py`); what they add may be large. Unlike the Hankel condition of Theorem 5.1.3 at one time, positivity of $\mathcal{G}$ at one resolution does not decide whether the zeros are on the line. Moving the pair near 1977.22 to distance 0.1 raises $t^*$ from 1.71 to 4.03: the closer the quadruple to the line, the finer the resolution needed to see it.

The same mechanism puts the first failure for a quadruple just above the verified height far beyond the orders of (b). There $g^2 \approx 10^{25}$, so the first failing order is about $10^{25}t^*$, and $t^*$ depends on the gaps around the quadruple, which a computation at height $3 \times 10^{12}$ would have to supply. The mean spacing there is 0.23, and for $\delta = \frac{1}{4}$ a failure at $t \le 1$ would need a stretch of length 2.9, about twelve mean spacings, free of zeros on the line other than the moved pair. A model with the line zeros spread at their mean density gives $t^* \approx 10$ there, but the same model gives 5.9 at height 1977 against the computed 1.71, because it ignores the gaps, so it gives the order of magnitude only. Theorem 5.1.13 certifies the signs of $\xi$ to every order up to $1.29(H-23)^2 > 1.16 \times 10^{25}$, with $H = 3 \times 10^{12}$, so a quadruple of $\xi$ just above $H$ could make a sign fail only at a time $t^* > 1.29$, at an order of about $t^*H^2$: the orders certified at the time $t^*$ reach that size, and by the model the orders certified at every time fall short of it by a factor of about eight. This is the entry of Table 5.5 on derivative signs.

### 5.2.2 What finite tests establish, and the local visibility condition

Each entry of the middle column of Table 5.5, made rigorous, is an unconditional theorem on its window; the right column is the reason why no accumulation of such entries proves RH.

The second row rests on a local computation. Suppose that $\xi$ has a pair of

| test | establishes | cannot rule out |
|---|---|---|
| sign changes and argument count to height $T$ (Sections 2.1 and 2.2) | all zeros below $T$ on the line | anything above $T$ |
| $\rho_\varepsilon \geq 0$ on $0 < \theta \leq \Theta$ (Section 3.1) | no zero in the window at distance $d > \varepsilon$ from the line with $2/d^2 > \sum_j \delta_j^{-2}$ (to first order in $\varepsilon$) | zeros within $\varepsilon$ of the line; zeros farther from it than about 0.78 of the mean spacing at their height; zeros above the window |
| zero density scan at $T$ (Section 3.4) | $\nu(\frac{1}{2}+\varepsilon, T) = 0$ for the $\varepsilon$ computed | zeros within $\varepsilon$ of the line; zeros above $T$ |
| Hankel blocks of size $N$ (Section 5.1) | Weil positivity on a finite space | off line zeros above $T_N$ |
| derivative signs to order $K$ (Section 5.1) | finite order Williamson positivity | off line zeros above a height of about $\sqrt{K/1.29}$ |
| HCM derivative signs (Section 4.1) | finitely many CM conditions | failure at other $u$ or orders |
| matched total variation at resolution $\varepsilon$ (Section 3.2) | agreement with the Poisson smoothing of the list | zeros outside the window |
| Li coefficients $\lambda_n$, $n \leq N$ (Section 8.3) | finitely many of Li's inequalities | off line zeros $\frac{1}{2} \pm d + i\gamma$ with $Nd/\gamma^2$ small |

Table 5.5: What finite tests establish and what they cannot exclude.

zeros $\frac{1}{2} \pm d + i\gamma$ with $d > \varepsilon$, and that the other zeros near height $\gamma$ lie on the line at the ordinates $\gamma + \delta_j$. A zero $\rho$ contributes $\mathrm{Re}(s-\rho)^{-1}$ to $\pi\rho_\varepsilon(\theta)$, where $s = \frac{1}{2} + \varepsilon + i\theta$, so at the ordinate, ignoring distant zeros and those at negative heights,

$$\pi\rho_\varepsilon(\gamma) = \varepsilon\Big(\sum_j \frac{1}{\delta_j^2 + \varepsilon^2} - \frac{2}{d^2 - \varepsilon^2}\Big).$$

To first order in $\varepsilon$, the off line zero therefore makes $\rho_\varepsilon$ negative at its ordinate if and only if $2/d^2 > \sum_j \delta_j^{-2}$; this is the local visibility condition. For neighbours at the mean spacing $\Delta = 2\pi/\log(\gamma/2\pi)$, that is $\delta_j = j\Delta$ with $j \neq 0$, the sum is $\pi^2/(3\Delta^2)$, and the condition becomes $d < (\sqrt{6}/\pi)\Delta = \sqrt{24}/\log(\gamma/2\pi)$, about 0.78 of the mean spacing. A zero farther from the line leaves $\rho_\varepsilon$ positive at its ordinate at every small resolution and makes it negative there only at resolutions comparable to $d$.

The condition concerns the ordinate itself. For evenly spaced neighbours the neighbour sum is smallest and the pair term most negative there, so the condition also decides whether $\rho_\varepsilon$ is negative near $\gamma$; in general it does not, because away from the ordinate the neighbour sum can fall faster than the pair term grows. With $d = 0.3$, $\varepsilon = 0.005$ and neighbours at $\gamma + 0.2 + m$ and $\gamma - 4 - m$, $m = 0, 1, 2, \ldots$, the sum $\sum_j \delta_j^{-2} = 26.55$ exceeds $2/d^2 = 22.2$ and $\rho_\varepsilon(\gamma) = 0.0069$,

yet $\rho_\varepsilon(\gamma - 0.079) = -0.0060$ (Exercise 5.14). A search for negative values of $\rho_\varepsilon$ must therefore cover every point of the window, not only the ordinates, and every resolution.

A zero on the line at the ordinate of the pair, $\delta_j = 0$, makes the neighbour sum infinite, and the pair is then invisible to first order at every distance. The exact statement is short.

PROPOSITION 5.2.2 (A pair at the height of a line zero). *Let $d > 0$, $\varepsilon > 0$ with $\varepsilon \neq d$, and $u = (\theta - \gamma)^2$. The zeros $\frac{1}{2} + i\gamma$ and $\frac{1}{2} \pm d + i\gamma$ contribute to $\pi\rho_\varepsilon(\theta)$*

$$\frac{\varepsilon\big(3u^2 + 6\varepsilon^2 u + (3\varepsilon^2 - d^2)(\varepsilon^2 - d^2)\big)}{(\varepsilon^2 + u)\big((d-\varepsilon)^2 + u\big)\big((d+\varepsilon)^2 + u\big)},$$

*which is positive at every $\theta$ if and only if $\varepsilon < d/\sqrt{3}$ or $\varepsilon > d$. Hence, if every zero off the line of an entire function with the order, symmetry and reality of $\xi$ shares its ordinate with a zero on the line of at least its multiplicity, and the distances $d$ have a positive infimum $d_0$, its density is positive at every height on every line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ with $0 < \varepsilon < d_0/\sqrt{3}$, although zeros lie to the right of these lines.*

*Proof.* The line zero contributes $\varepsilon/(\varepsilon^2 + u)$ and the pair $2\varepsilon(\varepsilon^2 - d^2 + u)/\big(((\varepsilon - d)^2 + u)((\varepsilon + d)^2 + u)\big)$ (proof of Corollary 11.4.7); over the common denominator the numerator is the one displayed. Its coefficients of $u^2$ and $u$ are positive, so it is positive for every $u \geq 0$ exactly when its value $(3\varepsilon^2 - d^2)(\varepsilon^2 - d^2)$ at $u = 0$ is, that is when $\varepsilon^2 < d^2/3$ or $\varepsilon^2 > d^2$. For the second statement group each pair, and its conjugate at $-\gamma$, with its line zero; every group and every remaining line zero contributes positively, and the sum converges absolutely, as in the proof of Corollary 11.4.7. □

To first order in $\varepsilon$ the contribution is $\varepsilon(3u^2 + d^4)/(u(u + d^2)^2) > 0$ for every $d$: the polynomial $P(x) = x(x^2 + d^2)$ satisfies the Laguerre inequality $P'^2 - PP'' = 3x^4 + d^4 > 0$ on the whole real line without having only real zeros. Exercise 1.7 gives one such function at one line; the proposition gives a band of lines whose width is fixed by the nearest off line pair, so the Hadamard structure and the functional equation alone cannot make positivity on one line detect zeros to its right.

The zero density identity of Proposition 3.4.2 has no blind spot of this kind: one evaluation at resolution $\varepsilon$ returns $\nu(\frac{1}{2} + \varepsilon, T)$, the number of zeros with real part greater than $\frac{1}{2} + \varepsilon$ and ordinate in $(0, T]$, however far from the line they lie. The same count controls the negative part of $\rho_\varepsilon$: the negative kernels in its Poisson form, which come from the zeros to the right of $\frac{1}{2} + \varepsilon$, have total mass on $(0, T]$ at most the number of such zeros below height $2T$ plus a tail that tends to zero as $T \to \infty$ when that number is $O(T^c)$ for some $c < 1$ (Lemma 11.4.1). The zero density estimates of Ingham [284] and of Guth and Maynard [166] give such a bound with no hypothesis and make this mass a vanishing fraction of the positive mass (Theorem 11.4.2).

### 5.2.3 Refuting sufficient routes with one certificate

Not every route to RH through the Pólya density is a positivity question on the clock. Konstantopoulos, Patie and Sarkar [313] introduced a class $\mathcal{D}_P$ of solutions of the van Dantzig problem, the entire functions $\sum_{n\ge0}(-1)^n t^{2n}/(\Psi(1)\cdots\Psi(n))$ in which $\Psi$ is the Laplace exponent of a spectrally negative Lévy process whose largest zero $q$ is at most $\frac12$; then $\Psi(u)=(u-q)\varphi(u)$ with $\varphi$ a Bernstein function [313, Proposition 5.2]. If $q=0$ and $\varphi$ is a Bernstein Pick function whose zeros and poles satisfy a separation condition, every zero of the entire function is real [313, Theorem 4.4], so a representation of $\Xi/\Xi(0)$ of this kind would prove RH; this is the Bernstein Pick route. Let $X$ have the Pólya density and put $M(s)=\mathbb{E}|X|^{2s}$. Since $\Xi(t)/\Xi(0)=\sum_{n\ge0}(-1)^n M(n)t^{2n}/(2n)!$, such a representation needs $\Psi(n)=n\varphi_I(n)$ for every $n\ge1$, where $\varphi_I(s)=4(s-\frac12)M(s-1)/M(s)$ is the canonical interpolant of the moment ratios; that is, $\varphi$ must interpolate $n\varphi_I(n)/(n-q)$ at the integers. The function $M$ is analytic for $\operatorname{Re}s>-\frac12$, and for $-\frac12<\operatorname{Re}s<0$ the classical formula for negative absolute moments gives

$$M(s)=\frac{2\,\Gamma(2s+1)\sin(-\pi s)}{\pi\,\Xi(0)}\int_0^\infty \Xi(\theta)\,\theta^{-2s-1}\,d\theta,$$

whose right side continues $M$ meromorphically to $\operatorname{Re}s<0$.

Theorem 5.2.3 ([421, Theorem E]). *$M$ has exactly one zero in the disc $|s-s_1|<10^{-8}$ with centre $s_1=-8.494182256992559+2.937767068567584i$; the zero is simple, and $M(s-1)$ does not vanish in the disc. Hence $\varphi_I$ has a nonreal pole, and every Bernstein interpolant of the moment ratios agrees with $\varphi_I$ on $(1,\infty)$ and cannot be continued through that pole, so none is a complete Bernstein function. For every $0<q<1$ the sequence $n\varphi_I(n)/(n-q)$ has no Bernstein interpolant either.*

Since the Bernstein Pick functions are the complete Bernstein functions, the theorem closes the Bernstein Pick route, and its last statement excludes $\Xi/\Xi(0)$ from $\mathcal{D}_P$ with $0<q\le\frac12$, the case $q>0$ of the question raised in [313, Section 4.1]. The theorem belongs to the arcsine route of [424, Chapter 12], where its last statement is proved by a beta multiplier obstruction, and at the single remaining value $q=0$ membership in $\mathcal{D}_P$ reduces to a Hausdorff moment criterion.

### 5.2.4 A local model for the first failing order

Once the resolution $1/(2\sqrt{t})$ of the Gaussian smoothing (Section 5.1.5) is below the gaps between the line zeros near the moved pair, $c_k^F(t)$ near the quadruple is governed by the quadruple and its two neighbours on the line, and the first failing order becomes explicit.

Proposition 5.2.4. *Let $F$ be obtained as in Table 5.4 by replacing two consecutive zeros of $\xi$ on the line by the quadruple $\frac12\pm\delta\pm ig$, $g$ their midpoint, let $g-\Delta_-$*

*and $g+\Delta_+$ be the ordinates of the nearest zeros of $F$ on the line below and above $g$, and put $\Delta=\max(\Delta_-,\Delta_+)$. Assume $\delta\ll\Delta\ll g$ and write $\omega=2k\delta\Delta/g^2$, $\kappa=2k\Delta^2/g^2$. Restricted to the quadruple and these two zeros, $c_k^F(t)<0$ for some $t>0$ if and only if, to leading order,*

$$\min_{0\le\eta\le1}\Big[2\cos(\omega\eta)+e^{-\kappa(1-\eta)}\Big]<0,$$

*and the first failing order is*

$$k^*=\frac{\omega^*g^2}{2\delta\Delta},\qquad \omega^*=\frac{\pi}{2}+\frac{\delta}{\Delta}\Big(1+\log\frac{\Delta}{2\delta}\Big)+O\Big(\frac{\delta^2}{\Delta^2}+\frac{\Delta}{g}\Big),$$

*so that $k^*=\frac{\pi}{4}\,g^2/(\delta\Delta)\,(1+o(1))$ as $\delta/\Delta\to0$ and $\Delta/g\to0$. The failure is at the time $t^*=k^*/g^2\,(1+O(\Delta/g))=\omega^*/(2\delta\Delta)\,(1+O(\Delta/g))$, which involves $\delta$ and $\Delta$ only.*

*Proof.* Put $t=k(1+v)/g^2$ and $a=(g-i\delta)^2$. The quadruple contributes $2\operatorname{Re}a^ke^{-at}=2|a|^ke^{-t\operatorname{Re}a}\cos\big(2\delta gt-2k\arctan(\delta/g)\big)$ to $c_k^F(t)$, with $|a|^ke^{-t\operatorname{Re}a}=g^{2k}e^{-tg^2}e^{O(k\delta^2/g^2)}$ and phase $2\delta gk(1+v)/g^2-2k\arctan(\delta/g)=2k\delta v/g+O(k\delta^3/g^3)$. The zero at $g+\Delta_+$ contributes $(g+\Delta_+)^{2k}e^{-t(g+\Delta_+)^2}$, and the logarithm of the ratio of its weight to $g^{2k}e^{-tg^2}$ is $2k\log(1+\Delta_+/g)-t(2g\Delta_++\Delta_+^2)=-(2k\Delta_+/g)(\Delta_+/g+v)+O(k\Delta_+^3/g^3)$. Up to a common factor these are, to this order, the quadruple term and the term of a zero on the line of the Gaussian smoothing of Section 5.1.5, at $u=g(1+v)^{-1/2}$. With $v=-\eta\Delta_+/g$ the phase is $-\omega_+\eta$ and the ratio is $e^{-\kappa_+(1-\eta)}$, the subscripted parameters being built from $\Delta_+$; the other neighbour is reached with $v>0$ and the parameters built from $\Delta_-$. For $\eta>1$ the neighbour dominates exponentially, so the minimum is attained on $[0,1]$, and on the side of the larger gap, where a given phase is reached at a smaller neighbour weight. Since $k\asymp g^2/(\delta\Delta)$ in the relevant range, the neglected terms are a relative error $O(\delta/\Delta)$ in the weight of the quadruple, $O(\delta^2/(g\Delta))$ in its phase and $O(\kappa\Delta/g)$ in the exponent of the neighbour. With $\kappa/\omega=\Delta/\delta\gg1$ the minimum is attained at $\eta=1-\sigma$ with $\omega$ just above $\pi/2$. Writing $\omega=\pi/2+\theta$, the condition is $2(\theta-\omega\sigma)>e^{-\kappa\sigma}$; optimising $\sigma=\kappa^{-1}\log(\kappa/2\omega)$ gives $\theta>(\omega/\kappa)(1+\log(\kappa/2\omega))$, the stated $\omega^*$, with $\omega/\kappa=\delta/\Delta$. Because $e^{-\kappa\sigma}=2\delta/\Delta$ at the optimum, a relative error $O(\delta/\Delta)$ in the weight of the quadruple moves $\theta$ by $O(\delta^2/\Delta^2)$ only, an error $O(\kappa\Delta/g)$ in the exponent of the neighbour by $O(\Delta/g)$, and an error in the phase by that error, $O(\delta^2/(g\Delta))$. Finally $t=k(1+v)/g^2$ with $|v|\le\Delta/g$, which gives the time. □

The table at the end of this section compares $t^*_{\mathrm{loc}}=\omega^*/(2\delta\Delta)$, the time of Proposition 5.2.4 without its remainder, with the certified failure times of Table 5.4, row by row, $\Delta$ being the larger of the gaps from $g$ to the zeros next to the moved pair, read from the zeros; its last column is $100\,(t^*_{\mathrm{loc}}/t^*-1)$. The local model is within 2.8 per cent of the certified time in every row, and the order $\omega^*g^2/(2\delta\Delta)$

within 2.9 per cent of $k^*$. Keeping only the quadruple and its two neighbours, with their exact weights, gives the first failing order of Table 5.4 exactly in four rows and to within 0.03 per cent in the other three (computed, not certified), so the differences of the table belong to the expansion of the proposition and not to the zeros it omits. They are positive because the expansion drops the factor $e^{-2t\delta^2}$ in the weight of the neighbour relative to the quadruple, the amplification $e^{2t\delta^2}$ of the quadruple in Section 5.1.5; with that factor kept, the seven times are within 0.25 per cent of the certified ones. The program `xg_local.py` (`code/README.md`) evaluates the model and the sum over the quadruple and its two neighbours from the zeros.

| $g$ | $\delta$ | $\Delta$ | $t^*_{\mathrm{loc}}$ | $t^*$ | difference, per cent |
|---|---|---|---|---|---|
| 185.24 | $\frac{1}{4}$ | 3.03 | 1.1896 | 1.18413 | 0.46 |
| 376.08 | $\frac{1}{4}$ | 2.36 | 1.5591 | 1.54072 | 1.19 |
| 630.64 | $\frac{1}{4}$ | 2.31 | 1.5941 | 1.576635 | 1.11 |
| 750.81 | $\frac{1}{4}$ | 2.57 | 1.4230 | 1.410725 | 0.87 |
| 1329.12 | $\frac{1}{4}$ | 1.49 | 2.5815 | 2.5121193 | 2.76 |
| 1977.22 | $\frac{1}{4}$ | 2.14 | 1.7341 | 1.7113393 | 1.33 |
| 1977.22 | 0.1 | 2.14 | 4.0339 | 4.0257254 | 0.20 |

## 5.3 The Thorin functional from the arithmetic

The explicit formula gives the Thorin measure from the gamma factor and the primes without reference to any zero [424, Chapter 7].

### 5.3.1 The Thorin functional and the explicit formula

The Thorin measure cannot be introduced as a measure without assuming the conclusion. If $W$ were known to be completely monotone, Bernstein's theorem would supply a unique positive measure $U$ with $W(t) = \int e^{-tz} U(dz)$, but complete monotonicity of $W$ is RH. Nor is $\sum_k \delta_{a_k}$ a measure on $[0,\infty)$ unless every $a_k$ is real and nonnegative, which is RH again; an atom at a nonreal $a_k$ can be paired only with functions defined off the real axis. The remedy is to define $U$ as a linear functional on a class of test functions large enough to determine it and small enough that the arithmetic side converges.

Let $\mathcal{T}$ be the set of functions $\varphi(z) = h(\sqrt{z})$ with $h$ even, holomorphic in $|\operatorname{Im} r| \le \frac{1}{2} + c$ for some $c > 0$ and $O((1+|r|)^{-2-c})$ there, and let $g(x) = \frac{1}{2\pi}\int h(r) e^{-irx} dr$. Then $\varphi$ is holomorphic on the image $\{r^2 : |\operatorname{Im} r| < \frac{1}{2} + c\}$, which contains $-\frac{1}{4}$ and every $a_k$. Define

$$\begin{aligned} \langle U, \varphi \rangle &= \varphi\Big(-\frac{1}{4}\Big) + \frac{1}{4\pi}\int_{\mathbb{R}} \varphi(r^2)\, m(r)\, dr - \sum_{n \ge 2} \frac{\Lambda(n)}{\sqrt{n}}\, g(\log n), \\ m(r) &= \operatorname{Re}\psi\Big(\frac{1}{4} + \frac{ir}{2}\Big) - \log \pi, \end{aligned} \tag{5.3.1}$$

where $\psi = \Gamma'/\Gamma$ and $\Lambda$ is the von Mangoldt function. No zero appears in this definition. Weil's explicit formula [512], [288, Theorem 5.12] identifies the functional: $\langle U, \varphi\rangle = \sum_k \varphi(a_k)$ for every $\varphi \in \mathcal{T}$, the sum running over the exponents $a_k = -(\rho_k - \frac12)^2$ of Section 1.2, one for each pair $\{\rho, 1-\rho\}$, and in particular $\langle U, e^{-t\cdot}\rangle = W(t)$ [421, Proposition 9.4].

The term $\varphi(-\frac14) = h(\frac{i}{2})$ comes from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, which the factor $s(s-1)$ of $\xi$ removes; it is the only evaluation of $\varphi$ off $[0,\infty)$. The integral is the contribution of the gamma factor: with the Riemann Siegel theta function $\vartheta(r) = \operatorname{Im}\log\Gamma(\frac14 + \frac{ir}{2}) - \frac{r}{2}\log\pi$ we have $m = 2\vartheta'$, so the integral is $\pi^{-1}\int_0^\infty h\,d\vartheta$, the integral of $h$ against the smooth part of the zero counting function. The prime sum is the fluctuating part. The half width $\frac12 + c$ of the strip makes $g(x) = O(e^{-(1/2+c)|x|})$, so that the prime sum converges absolutely against the weight $n^{-1/2}$.

The sine squared lemma (1.2.9) rewrites the functional in terms of variograms. By (3.5.2) its kernel carries each $\varphi \in \mathcal{T}$ to $g(0) - g(x)$, a variogram when $h \ge 0$, and the functional takes the absolutely convergent form [421, Proposition 9.2]

$$\langle U, \varphi\rangle = \int_0^\infty \frac{e^{-x/2}}{e^{2x}-1}\big(g(0) - g(x)\big)\,dx - C_\Gamma g(0) + \int_0^\infty e^{x/2} g(x)\,dx - \sum_{n\ge2} \frac{\Lambda(n)}{\sqrt{n}} g(\log n),$$

where $C_\Gamma = \frac12\log\pi - \frac12\psi(\frac14) - 2 = 0.6860917096$ is the archimedean constant of Section 3.1.3. To derive this form, split $\varphi(-\frac14) = \int_0^\infty (e^{x/2} + e^{-x/2}) g(x)\,dx$; the half with $e^{-x/2}$ combines with the gamma integral into the first two terms.

This form regularises a formal pairing. Formally $\langle U, \varphi\rangle = \int_0^\infty \big(g(0) - g(x)\big)\,\nu(dx)$, where $\nu$ is the canonical measure of (3.1.1) at $\alpha = \frac12$,

$$\nu(dx) = \frac{e^{-x/2}\,dx}{e^{2x}-1} - e^{x/2}\,dx + \sum_{n\ge2} \frac{\Lambda(n)}{\sqrt{n}}\,\delta_{\log n}(dx),$$

but the pole and prime parts of this pairing diverge through their $g(0)$ terms. Part by part, the absolutely convergent form exceeds the formal pairing by multiples of $g(0)$: by $-C_\Gamma$ for the gamma factor, by $\int_0^\infty e^{x/2}dx$ for the pole and by $-\sum_n \Lambda(n) n^{-1/2}$ for the primes. Continued analytically as in Section 3.1.3, the last two are $-2$ and $\zeta'(\frac12)/\zeta(\frac12)$, and the three constants cancel by the identity $C_\Gamma + 2 - \zeta'(\frac12)/\zeta(\frac12) = 0$ of that section, which is equivalent to $\xi'(\frac12) = 0$. The only negative part of $\nu$ is the pole density $-e^{x/2}dx$, and Section 3.5.2 reads Weil's criterion from this sign pattern. At the basepoint one the pole density is $-dx$, and no analytic continuation is needed there: the prime atoms compensated by the pole density have a convergent sine squared integral, and its convergence is the prime number theorem (Proposition 9.1.3).

The functional is not in general the limit of the Thorin measures as the base-point approaches the line. As $\alpha \downarrow \frac{1}{2}$ the measures $U_\alpha$ of (1.2.6) converge vaguely to $U_{\text{line}} = \sum \delta_{\gamma^2}$, the sum running over the zeros on the line, whatever the truth of RH (Section 1.2 and [421, Proposition 9.5]), whereas unconditionally

$$U = U_{\text{line}} + \sum_{a_k \notin (0,\infty)} \delta_{a_k} \quad \text{on } \mathcal{T},$$

the sum running over the nonreal exponents. These come in conjugate pairs $(\gamma - id)^2$ and $(\gamma + id)^2$, one pair for each off line pair $\frac{1}{2} \pm d + i\gamma$ of Proposition 1.2.3 (written $\frac{1}{2} \pm \delta + i\gamma$ there), so $U = U_{\text{line}}$ exactly when RH holds [421, Proposition 9.6]. The functional evaluates $\varphi$ at both exponents of an off line pair, while in (1.2.8) the pair contributes kernels $P_{\varepsilon+d} + P_{\varepsilon-d}$ of total mass zero once $\varepsilon < d$, which vanish in the limit.

### 5.3.2 The heat trace from the primes

For $\varphi(z) = e^{-tz}$, that is $h(r) = e^{-tr^2}$ and $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$, the functional is the arithmetic form (1.2.5) of the heat trace, $W = \Pi + A - P$, with the polar term $\Pi = e^{t/4}$, the archimedean term $A$ and the prime term $P$ described in Section 1.2. Table 5.6 extends the comparison of Table 1.4 to smaller and larger $t$, with the three terms in sixty digit arithmetic ($P$ over $n \le 2 \times 10^5$) and the sum over the first 1700 zeros (`data/zeros_1700.txt`); the last column is the number of decimal digits lost to cancellation, $\log_{10}\{(\Pi + |A| + P)/W\}$. The two columns agree to every digit shown.

The cancellation follows a simple law. For $t \ge 0.005$ the archimedean term is negative (Exercise 5.8), so $\Pi + |A| + P = 2e^{t/4} - W$ and the digits lost are $\log_{10}(2e^{t/4}/W - 1)$. Since $W = e^{-\gamma_1^2 t}\big(1 + O(e^{-(\gamma_2^2 - \gamma_1^2)t})\big)$, this is

$$\log_{10} 2 + \frac{(\gamma_1^2 + \frac{1}{4})\, t}{\log 10} \approx 0.30 + 86.88\, t$$

up to an exponentially small error, below $2 \times 10^{-5}$ for $t \ge 0.05$.

The explicit formula holds exactly, and the cancellation costs working precision and nothing more: 26.4 digits at $t = 0.3$, well within sixty digit arithmetic. The zeros give $W$ cheaply at moderate $t$, the primes only near $t = 0$. Section 11.1 finds the same profile for the Lévy density $-W'$ of the subordinator that drives the clock.

The arithmetic side is cheap where $g$ is concentrated near $x = 0$, so that only small $n$ contribute: at small $t$ for the heat kernel, and only at large resolution for the Poisson kernel $h(r) = P_\varepsilon(\theta - r) + P_\varepsilon(\theta + r)$. That kernel is not in $\mathcal{T}$, but for $\varepsilon > \frac{1}{2}$ its transform $g(x) = \pi^{-1} e^{-\varepsilon|x|} \cos \theta x$ makes the prime sum converge absolutely; for smaller $\varepsilon$ the prime side is Selberg's smoothed sum (Section 3.5.1). The spectral side is cheap where the test function is concentrated at the low

| $t$ | polar $\Pi$ | archimedean $A$ | prime $P$ |
|---|---|---|---|
| 0.002 | 1.000500 | 0.782881 | $2.6 \times 10^{-26}$ |
| 0.005 | 1.001251 | −0.464059 | $7.2 \times 10^{-11}$ |
| 0.01 | 1.002503 | −0.852797 | $8.4 \times 10^{-6}$ |
| 0.02 | 1.005013 | −0.984061 | 0.002410 |
| 0.05 | 1.012578 | −0.954619 | 0.057914 |
| 0.1 | 1.025315 | −0.862510 | 0.162806 |
| 0.15 | 1.038212 | −0.797634 | 0.240578 |
| 0.2 | 1.051271 | −0.749754 | 0.301517 |
| 0.3 | 1.077884 | −0.681852 | 0.396032 |

| $t$ | $\Pi + A - P$ | $W$ from zeros | digits lost |
|---|---|---|---|
| 0.002 | 1.78338120537913 | 1.78338120537913 | 0.0 |
| 0.005 | 0.537191964091041 | 0.537191964091041 | 0.4 |
| 0.01 | 0.14969801125378 | 0.14969801125378 | 1.1 |
| 0.02 | 0.0185412917313195 | 0.0185412917313195 | 2.0 |
| 0.05 | $4.58783504857357 \times 10^{-5}$ | $4.58783504857357 \times 10^{-5}$ | 4.6 |
| 0.1 | $2.10479979582235 \times 10^{-9}$ | $2.10479979582235 \times 10^{-9}$ | 9.0 |
| 0.15 | $9.65642094572633 \times 10^{-14}$ | $9.65642094572633 \times 10^{-14}$ | 13.3 |
| 0.2 | $4.43018218022364 \times 10^{-18}$ | $4.43018218022364 \times 10^{-18}$ | 17.7 |
| 0.3 | $9.32464654810622 \times 10^{-27}$ | $9.32464654810622 \times 10^{-27}$ | 26.4 |

Table 5.6: The heat trace from the explicit formula and from the zeros. For $t \geq 0.05$ the digits lost are $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10 \approx 0.30 + 86.88\,t$.

zeros: at large $t$, and at small $\varepsilon$ on a window. In the proofs of the lower rungs in Section 5.1 the two forms meet at $t = 1/20000$, where the inequalities are checked in interval arithmetic. The Thorin condition requires positivity at every scale, and neither side is cheap at every scale. The Pólya route of Section 2.1 is, but only on a window of finite height, and that restriction is the barrier of Section 5.2.

## 5.4 Spacings, irregularity and the cell distance

On a window of finite height the errors of the Pólya route, measured by the cell distance $d_{\mathrm{cell}}(\varepsilon)$ of Section 3.2, are set by the spacings of the zeros, which follow the random matrix prediction already for the first 1700 zeros.

### 5.4.1 The cell distance to first order

We use the cells $C_k = (b_{k-1}, b_k]$ of Section 3.2.2, with $\Theta = 200$ and $K = N(\Theta) = 79$, and the first order formula for $\mu_\varepsilon(C_k) - 1$ derived there, whose remainder is $O(\varepsilon^3)$ (Remark 3.2.3); Exercise 5.10 gives its leading term. The formula needs only that the zeros below $\Theta$ be simple and on the line.

Under RH the formula becomes a sum over the ordinates. By Proposition 1.2.3, $\rho_\varepsilon(\theta) = \sum_j [P_\varepsilon(\theta - \gamma_j) + P_\varepsilon(\theta + \gamma_j)]$ over all positive ordinates, and the kernel

$P_\varepsilon(\cdot-\gamma)$ has distribution function $F_\varepsilon(y-\gamma)$, where $F_\varepsilon(y)=\frac12+\frac1\pi\arctan(y/\varepsilon)=\mathbf{1}_{y\ge0}-\varepsilon/(\pi y)+O(\varepsilon^3/|y|^3)$ as $\varepsilon/|y|\to0$. No ordinate lies on a cell boundary, so

$$\mu_\varepsilon(C_k)-1=-\frac{\varepsilon}{\pi}\sum_j\Big[\frac{1}{b_k-\gamma_j}-\frac{1}{b_{k-1}-\gamma_j}+\frac{1}{b_k+\gamma_j}-\frac{1}{b_{k-1}+\gamma_j}\Big]+O(\varepsilon^3),$$

and by (1.2.1) with $s=i\theta$, $\Xi'/\Xi(\theta)=\sum_j[(\theta-\gamma_j)^{-1}+(\theta+\gamma_j)^{-1}]$ under RH, so the two expansions agree. Summing absolute values,

$$d_{\mathrm{cell}}(\varepsilon)=\kappa_{\mathrm{cell}}\,\varepsilon+O(\varepsilon^3),$$
$$\kappa_{\mathrm{cell}}=\frac1\pi\sum_k\Big|\sum_j\Big[\frac{1}{b_k-\gamma_j}-\frac{1}{b_{k-1}-\gamma_j}+\frac{1}{b_k+\gamma_j}-\frac{1}{b_{k-1}+\gamma_j}\Big]\Big|.$$

Truncated to the first 1700 ordinates, the zero sum gives 5.415, against $\kappa_{\mathrm{cell}}=5.43665$ from the form in $\Xi'/\Xi$ and the observed slope $d_{\mathrm{cell}}(0.005)/0.005=5.43664$. The difference of 0.4 per cent between 5.415 and 5.43665 is the contribution of the zeros above $\gamma_{1700}$. In the truncated formula the cell errors range from $-0.225\,\varepsilon$ to $+0.250\,\varepsilon$ (Figure 5.4), with the two largest positive values at the end cells: at the last because $\Theta$ is not a midpoint between ordinates, and at the first because below $b_1$ there is only $\gamma_1$ and then the reflections $-\gamma_j$, which do not balance the zeros above $b_1$.

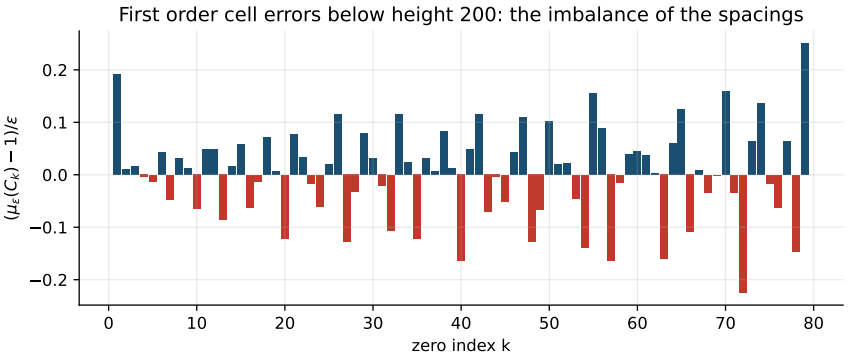



Figure 5.4: First order cell errors $(\mu_\varepsilon(C_k)-1)/\varepsilon$ for the 79 cells below height 200, computed from the first 1700 ordinates.

The atoms $\theta=k\ge1$ of the sinh benchmark of Section 1.3 and their reflections form the lattice $\mathbb{Z}$ without the point 0. For the full lattice every cell $(k-\frac12,k+\frac12]$ would have mass exactly one, by periodicity, so the deficit of each cell is the mass that the missing kernel at 0 would have put in it; the atom free interval $(0,\frac12]$, which the benchmark counts as a cell of mass zero, receives the mass $\frac12$ of the full lattice less that of the missing kernel:

$$\mu_\varepsilon\big((k-\tfrac12,k+\tfrac12]\big)=1-\frac1\pi\arctan\frac{\varepsilon}{\varepsilon^2+k^2-\frac14},\qquad \mu_\varepsilon\big((0,\tfrac12]\big)=\frac1\pi\arctan(2\varepsilon).$$

The interior cells therefore do not balance: each loses $\varepsilon/(\pi(k^2-\frac14))+O(\varepsilon^3)$, while $(0,\frac12]$ gains $2\varepsilon/\pi+O(\varepsilon^3)$. As $\sum_{k\le N}(k^2-\frac14)^{-1}=2-1/(N+\frac12)$, the cell distance of $N$ atoms is $(4-1/(N+\frac12))\,\varepsilon/\pi+O(\varepsilon^3)$, which is $1.26694\,\varepsilon$ for $N=50$, the constant 1.267 of Section 1.3. With the convention used for the zeros, a first cell $(0,\frac32]$, it would be $(\frac43-1/(N+\frac12))/\pi=0.418$. For the zeros no cell is balanced, and the errors in Figure 5.4 do not decay along the window.

The constant vanishes exactly when the sum of $1/(b-x)$ over the ordinates and their reflections vanishes at every cell boundary $b$, as it does when these points are symmetric about every boundary. The lattice $\gamma_j=(j-\frac12)h$, whose reflections complete $(\mathbb{Z}+\frac12)h$, has every cell error zero, whereas the lattice of the benchmark does not. For the zeros every ordinate contributes to a cell error, with a weight that decays like $|C_k|/(\gamma_j-b_k)^2$. The two ordinates nearest an interior boundary are symmetric about it, so the asymmetry comes first from the next gaps on either side, and it is small when those gaps are equal.

### 5.4.2 Spacings of the zeros and the Thorin density

At height $\gamma$ the mean spacing of the zeros is $\Delta=2\pi/\log(\gamma/2\pi)$, as in Section 5.2.2. We unfold by $\tilde\gamma=\vartheta(\gamma)/\pi$. The 1699 normalised spacings $\tilde\gamma_{k+1}-\tilde\gamma_k$ of the first 1700 zeros, which lie below height 2198, have variance 0.146 and minimum 0.089. Figure 5.5 compares their histogram with the Wigner surmise for the Gaussian unitary ensemble, $p_{\mathrm W}(s)=\frac{32}{\pi^2}s^2e^{-4s^2/\pi}$, and with the Poisson law $e^{-s}$ of independent points. The histogram follows the surmise already at this height, including the quadratic repulsion at small spacings that Montgomery's pair correlation conjecture predicts [374, pp. 183 and 184] and that Odlyzko observed at large height [391]. The variance is below the surmise value $3\pi/8-1=0.178$; the spacing distribution approaches the unitary one only slowly with height [374, p. 185], [52, Section 3].

Repulsion affects the Thorin density in two ways. First, it makes very small gaps rare but not absent. The smallest normalised spacing among the first 1700 zeros, 0.089, separates the ordinates 1977.17 and 1977.27, the pair that Section 5.2 moves off the line in Figure 5.3. By the Cauchy kernel criterion of Exercise 2.11, an isolated pair of ordinates at distance $g$ in $\theta$ gives two peaks of $\rho_\varepsilon$ only for $\varepsilon<\sqrt3 g/2$. The resolution needed to separate every pair on a window is therefore set by the smallest gap, and repulsion controls it: if the $N$ normalised spacings were independent, with density $p_{\mathrm W}(s)\propto s^2$ near zero, the smallest would be of order $N^{-1/3}$, which is 0.084 for $N=1699$, close to the observed 0.089.

Second, repulsion sets the scale of the local visibility condition of Section 5.2.2: to first order in $\varepsilon$, an off line zero at distance $d$ from the line makes $\rho_\varepsilon$ negative at its ordinate if and only if $2/d^2$ exceeds $\sum_j\delta_j^{-2}$, where $\delta_j$ are the distances from its ordinate to the other ordinates. The sum is dominated by the nearest neighbours, and repulsion makes it typically of order $\Delta^{-2}$, because $\mathbb{E}s^{-2}$ is finite for the surmise, where it equals $8/\pi$, and infinite for the Poisson law. With

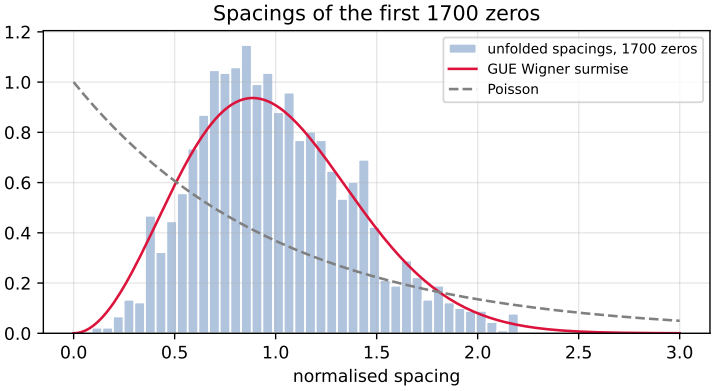



Figure 5.5: The 1699 unfolded spacings of the first 1700 zeros against the GUE Wigner surmise and the Poisson law.

neighbours on a lattice of spacing $\Delta$, the density at the ordinate stays positive at small resolution once $d$ exceeds the threshold $(\sqrt{6}/\pi)\Delta \approx 0.78\,\Delta$ of that section. Since $d < \frac{1}{2}$, a zero hidden in this way can exist only where $\Delta < 0.64$, that is above height about $1.1 \times 10^5$. Hidden here means hidden at the ordinate: beside it the density can be negative even when the condition predicts a positive value there (Exercise 5.14).

The behaviour of the cell constant at large height is open (Section 12.2). In unfolded units the cell errors depend mainly on the nearby ordinates, so the sine kernel process of the unitary ensemble would predict their mean and fluctuations per cell; we have not computed that prediction. Computing the cell errors with Odlyzko's zeros near zero number $10^{12}$ [391], $10^{21}$ or $10^{22}$ [395, 396], at heights about $2.7 \times 10^{11}$, $1.4 \times 10^{20}$ and $1.4 \times 10^{21}$, would compare the Thorin density with the spacing statistics of the zeros at large height. It needs the ordinates, which define the cells, and the hybrid evaluator of Section 2.2.2 for $\Xi'/\Xi$ at the cell boundaries. Section 11.4.5 proposes the same comparison at large height for the positivity margins of the density, its values at the midpoints between consecutive ordinates.

## 5.5 Exercises

*On Section 5.1.*

5.1. Show that $c_0 c_2 - c_1^2 = \frac{1}{2}\sum_{j,l}(a_j - a_l)^2 e^{-(a_j+a_l)t}$ and deduce that log convexity of $W$ holds under RH.

5.2. Recompute the first row of Table 5.1 with the first 100 zeros and explain why no printed digit changes.

5.3. Show that for a real exponent $a$ the summand of the Williamson measure $U_k$ belonging to $a$ is the inverse gamma density of $ka/\Gamma_k$ with $\Gamma_k \sim \mathrm{Gamma}(k,1)$, and compute its variance for $k > 2$.

5.4. Let $\widetilde{W}(t) = e^{-t} + 2e^{-10t}\cos t$, the heat trace of the three exponents 1 and $10 \pm i$. Show that $\widetilde{W} > 0$ and $\widetilde{W}' < 0$ on $(0,\infty)$, using $\cos t > 0$ and $\sin t \geq 0$ for $t \leq 1$ and $2\sqrt{101}\,e^{-9} < 1$ for $t \geq 1$, so that the first two rungs of Theorem 5.1.1 hold. With $\theta = \arctan\frac{1}{10}$, show that $\frac{10}{101} < \theta < \frac{1}{10}$, hence $\frac{2\pi}{3} < 24\theta < \pi$, and deduce that $(-1)^{24}\widetilde{W}^{(24)}(0) = 1 + 2\cdot 101^{12}\cos 24\theta < 0$, so that $\widetilde{W}$ is not completely monotone. Then check numerically that the first order $k$ at which $(-1)^k\widetilde{W}^{(k)}(t) < 0$ for some $t > 0$ is 12, near $t = 2.87$, and compare with the first failures of Table 5.4.

5.5. Show that the term of order $1/u$ of $\varepsilon_\rho$ in Proposition 5.1.10 is $\frac{2t}{3u}(\tilde\gamma - u)^3$. For $u = 2$, $t = 1$ and $k = 4$, show that no zero lies in the disc $|\tilde\gamma - u| \leq u/2$, so that $c_4(1) = e^{f_0}R$ is carried entirely by the remainder, and compare the bound on $R$ with the contribution of the first zero.

5.6. Show that $m(0) > -6$ and, comparing the series for $\psi$ with an integral, that $m(60) > 1$. Deduce that for $t \leq 10^{-6}$ the archimedean term of (1.2.5) is at least $1/(4\sqrt{\pi t}) - 210/\pi > 55$ and the prime term at most $(\pi t)^{-1/2}e^{-(\log 2)^2/4t} < 0.01$. With the argument of the proof of Theorem 5.1.1 for $t \geq \tau$, taken at height $10^4$ and from $t = 10^{-6}$, conclude that $W > 0$ on $(0,\infty)$ follows from the verification of RH below height $10^4$ by estimates that can all be checked by hand.

*On Section 5.3.*

5.7. Verify the arithmetic form of $W(t)$ at $t = 0.01$ with the prime powers up to $10^4$, and explain why so few suffice by showing that only $n = 2$ and $n = 3$ affect the first fifteen digits.

5.8. Show that $m$ increases on $[0,\infty)$ and vanishes at a single point $r_*$, and deduce that $e^{r_*^2 t}A(t)$ decreases in $t$, so that $A$ changes sign at most once; with Table 5.6, conclude that $A < 0$ for $t \geq 0.005$. Then show that for $t \geq 0.05$ the number of digits lost in the table equals $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10$ up to an exponentially small error, and identify the source of each term.

5.9. Show that the regularised constants of the gamma, pole and prime parts of the variogram form sum to zero if and only if $\xi'(\frac{1}{2}) = 0$.

*On Section 5.4.*

5.10. Let $\ell = \Xi'/\Xi$. Show that $\xi(\frac{1}{2} + \varepsilon + ib) = \Xi(b - i\varepsilon)$, and deduce from the Taylor series of $\log \Xi$ about $b$ that $\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi}[\ell(b_k) - \ell(b_{k-1})] + \frac{\varepsilon^3}{6\pi}[\ell''(b_k) - \ell''(b_{k-1})] + O(\varepsilon^5)$. Under RH, check that the term in $\varepsilon^3$ agrees with the expansion of $F_\varepsilon$ to third order.

5.11. Prove the arctangent formulas for the cell masses of the sinh benchmark, and deduce that its cell distance for $N$ atoms is $(4 - 1/(N + \frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$, and $(\frac{4}{3} - 1/(N + \frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$ when the first cell is $(0, \frac{3}{2}]$.

5.12. Show that the Wigner surmise $p_{\mathrm{W}}$ has mean 1 and variance $3\pi/8 - 1$, and that $\mathbb{E}s^{-2} = 8/\pi$ under it, while $\mathbb{E}s^{-2} = \infty$ for the Poisson law.

5.13. Show that $\sum_{m \neq 0} (m\Delta)^{-2} = \pi^2/(3\Delta^2)$, deduce the lattice threshold $(\sqrt{6}/\pi)\Delta$ of the visibility condition, and find the height above which the threshold is less than $\frac{1}{2}$.

5.14. In a local model, put an off line pair at distance $d = 0.3$ from the line at the ordinate $\gamma$ and zeros on the line at $\gamma + 0.2 + m$ and $\gamma - 4 - m$ for $m \geq 0$, and ignore the reflected kernels. At $\varepsilon = 0.005$, show that $2/d^2 < \sum_j \delta_j^{-2}$, so that the visibility condition predicts a positive density at $\gamma$, and check that $\rho_\varepsilon(\gamma) > 0$ while $\rho_\varepsilon(\gamma - 0.079) < 0$.

# Chapter 6

# The reciprocal zeta function

The reciprocal zeta function $1/\zeta(\frac{1}{2}+\sqrt{s})$ is, up to the factor $1-4s$ of the pole and a constant, the reciprocal Xi function times the gamma factor, and unlike $\xi$ it has a Dirichlet series and an Euler product, from which we read its Thorin type measures.

Sections 6.1 and 6.2 compute the Thorin measure of the gamma factor, the Thorin measures of the normalised reciprocal zeta function and of the reciprocal Xi function under RH, and, unconditionally, the formal Thorin measures of these two functions in the Vere-Jones form of Section 6.1.3. After the stable tilt $e^{-c\sqrt{s}}$, with $c=\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, the reciprocal of the gamma factor is a generalised gamma convolution, and multiplying the normalised reciprocal zeta function by it gives the floor $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ of the reciprocal Xi function (Section 6.2.4), in which the zeros are the only possible source of negativity.

The infinite divisibility of the floor, as proved here, rests on the positivity of the canonical function of the normalised reciprocal zeta function, which is Theorem 7.1.1 of Chapter 7. It also follows from the infinite divisibility of the clock (Theorem 5.1.1); what the floor adds is a splitting of its canonical function into an arithmetic part and a completely monotone archimedean part. The floor is a generalised gamma convolution exactly when RH holds, and the derivatives of its canonical function alternate in sign up to order $10^{10}$ by the methods of this chapter and up to order $1.16\times10^{25}$ by the estimates of Section 5.1.

## 6.1 Thorin type forms of the reciprocal zeta function

Under RH the Thorin measure of the normalised reciprocal zeta function is $dM$, where $M-1$ is the argument function $S(\sqrt{t})$. Its heat trace, the canonical function $w$ of Theorem 6.1.11, is explicit in the primes, is an absolutely convergent series of Jacobi theta functions, and gives a criterion for RH (Theorem 6.1.15).

### 6.1.1 The question and the three functions

Write $\sqrt{s}$ for the principal square root on the cut plane $\mathbb{C} \setminus (-\infty, 0]$, so that $\operatorname{Re}\sqrt{s} > 0$ there, and evaluate $\zeta$ at $z = \frac{1}{2} + \sqrt{s}$. A zero $\rho = \frac{1}{2} + i\gamma$ then corresponds to $\sqrt{s} = i\gamma$, that is to $s = -\gamma^2$, a point of the cut, and an atom of a Thorin measure at $t$ produces a singularity at $s = -t$, also on the cut. The question of this chapter and of Chapter 7 is whether the zeros, the primes and the gamma factor assemble into a Thorin measure, and if so which one. The two chapters have their own coordinates, which hold throughout both unless stated otherwise: $s$ is the variable $u$ of Sections 1.2 and 1.3, $t$ is the variable $z$ of their Thorin measures, $x$ is the time of the heat trace $W(x)$, $\tau$ is the ordinate, and $z$ is the argument of $\zeta$ and $\xi$.

The Möbius series and the Euler product converge absolutely to the right of the parabola $x = \frac{1}{4} - y^2$, on

$$H_{1/2} = \{s : \operatorname{Re}\sqrt{s} > \tfrac{1}{2}\} = \{x + iy : \ x > \tfrac{1}{4} - y^2\},$$

and there

$$\frac{1}{\zeta(\frac{1}{2} + \sqrt{s})} = \sum_{n \geq 1} \frac{\mu(n)}{\sqrt{n}}\, e^{-\sqrt{s}\log n} = \prod_p \Big(1 - p^{-1/2} e^{-\sqrt{s}\log p}\Big). \tag{6.1.1}$$

Every term is built from $e^{-\lambda\sqrt{s}}$, which is completely monotone in $s$: it is the Laplace transform of the positive stable law of index $\frac{1}{2}$,

$$e^{-\lambda\sqrt{s}} = \int_0^\infty e^{-sx}\, \frac{\lambda}{2\sqrt{\pi}}\, x^{-3/2} e^{-\lambda^2/4x}\, dx. \tag{6.1.2}$$

The Thorin measure, however, is read from the logarithm. Recall from (1.3.1) that a generalised gamma convolution (GGC) has Laplace transform

$$f(s) = \exp\Big(-as - \int_{(0,\infty)} \log\big(1 + \tfrac{s}{t}\big)\, U(dt)\Big), \tag{6.1.3}$$

with $a \geq 0$ and $U \geq 0$ the Thorin measure; equivalently $-f'/f(s) = a + \int U(dt)/(s+t)$ is a Stieltjes function [56]. We allow $U$ to be signed, with $\int_{(0,1)} |\log t|\, |U|(dt) + \int_{[1,\infty)} t^{-1} |U|(dt) < \infty$ so that the integral converges, and then call (6.1.3) a Thorin type representation.

Three functions of $s$ run through both chapters. The function $1/\zeta(\frac{1}{2} + \sqrt{s})$ vanishes at $s = \frac{1}{4}$, where $\zeta$ has its pole, and equals $1/\zeta(\frac{1}{2}) < 0$ at $s = 0$, so it is not the Laplace transform of a probability law. Both defects are removed by the normalised reciprocal zeta function

$$\varphi(s) = \frac{\zeta(\frac{1}{2})}{(1 - 4s)\, \zeta(\frac{1}{2} + \sqrt{s})}, \qquad \varphi(0) = 1. \tag{6.1.4}$$

The other two are the gamma factor $G(s) = \Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$ and the reciprocal Xi function $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, and $\varphi = RG$ (Proposition 6.1.9). In both chapters $G$ is the gamma factor, not the function $G$ of Section 1.2, whose reciprocal is $R$, except in Section 7.1.3, where $G$ is a Laguerre Pólya function; and $\theta(\tau) = \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{i\tau}{2}) - \frac{\tau}{2}\log\pi$ is the Riemann Siegel theta function, written $\vartheta$ elsewhere in the book (Section 2.2), so that $\theta'(\tau) = \frac{1}{2}\operatorname{Re}\psi(\frac{1}{4} + \frac{i\tau}{2}) - \frac{1}{2}\log\pi$ [127, Section 6.5]. Table 6.1 lists the Thorin measures of the three functions and of the building blocks used below. In it $N(T)$ counts the zeros with $0 < \gamma \le T$, $S(T) = \pi^{-1}\arg\zeta(\frac{1}{2} + iT)$ is the argument function, $c_n = 2n + \frac{1}{2}$, $c = -\theta'(0)$ is the constant of (6.1.6), and $U_\lambda$ is the sawtooth measure of Lemma 6.1.7.

| Function | Thorin measure | Sign |
|---|---|---|
| $\varphi$ | $dM(t)$, $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ (RH) | signed |
| $G$ | $-\dfrac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}dt = \dfrac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\ge0}\Big(\dfrac{c_n}{t + c_n^2} - \dfrac{1}{2n+2}\Big)\Big\}dt$ | signed |
| $R$ | $\sum_{\gamma>0}\delta_{\gamma^2}$ (RH) | positive |
| $2/(1 + e^{-\lambda\sqrt{s}})$ | $-U_\lambda = \sum_k \delta_{((2k+1)\pi/\lambda)^2} - \dfrac{\lambda}{4\pi\sqrt{t}}dt$ | signed |
| $(1 + \sqrt{s}/\kappa)^{-1}$ | $\dfrac{\kappa}{2\pi\sqrt{t}\,(t + \kappa^2)}dt$ | positive |
| $\operatorname{sech}(\lambda\sqrt{s})$ | $\sum_k \delta_{((k+\frac{1}{2})\pi/\lambda)^2}$ | positive |
| $e^{-c\sqrt{s}}/G$ | $\dfrac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}}dt$ | positive |

Table 6.1: Thorin measures of $\varphi$, $G$ and $R$ and of the building blocks used in this chapter and in Chapter 7, with the convention $\log f(s) = -\int \log(1 + s/t)\,U(dt)$ after normalising $f(0) = 1$. Here $b = \frac{1}{2}(\gamma_E + \log\pi)$, $\kappa > 0$ and $c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$. The rows for $\varphi$ and $R$ are Theorem 6.1.10, the row for $G$ is Theorem 6.2.1, the fourth row is Lemma 6.1.7, the fifth and sixth appear in Table 1.8, and the last is Theorem 6.2.12.

### 6.1.2 Stieltjes inversion and the Euler factors

Lemma 6.1.1 recovers the Thorin measure $U$ of a function $F$ on the cut plane from the boundary values of $-F'/F$ on $(-\infty, 0]$ and computes most of the Thorin measures of this chapter and of Chapter 7.

Lemma 6.1.1. *Let $f$ be analytic on $\mathbb{C}\setminus(-\infty,0]$ with $f(\bar{s}) = \overline{f(s)}$, $f(s) \to 0$ as $|s| \to \infty$ uniformly in $\arg s$, $|f(s)| \le C|s|^{-\alpha}$ near 0 for some $\alpha < 1$, and such that $f$ extends continuously from the upper half plane to the points of $(-\infty, 0)$ apart from finitely many points $-t_0$, at each of which $f(s) - r/(s + t_0)$ is bounded near $-t_0$ in $\mathbb{C}\setminus(-\infty,0]$ for some real $r$; $f(-t + i0)$ denotes the values of this extension*

*and* $f(-t-i0)=\overline{f(-t+i0)}$. *Then*

$$f(s)=\int_{(0,\infty)}\frac{m(dt)}{s+t},\qquad m(dt)=-\frac{1}{\pi}\operatorname{Im} f(-t+i0)\,dt,$$

*where the integral is the limit of the integrals over* $(0,T]$ *as* $T\to\infty$, *absolutely convergent when* $\int_{(1,\infty)}t^{-1}|m|(dt)<\infty$, *and where each of the points* $-t_0$, *which we call a simple pole of* $f$, *contributes the atom* $r\,\delta_{t_0}$ *to* $m$.

*Proof.* Fix $s\in\mathbb{C}\backslash(-\infty,0]$ and integrate $f(z)/(z-s)$ over a keyhole contour around $(-\infty,0]$, which by Cauchy's formula gives $2\pi i\,f(s)$; the large circle vanishes by the decay at infinity and the small circle by the bound at the origin. By the continuity of the extension, the integrals along the two sides of the keyhole converge to the integrals of the boundary values as the keyhole closes onto the cut, after subtracting $r/(z+t_0)$ near each pole $-t_0$, where the remainder is bounded. The two sides of the cut combine to $\frac{1}{2\pi i}\{f(-t-i0)-f(-t+i0)\}=-\frac{1}{\pi}\operatorname{Im} f(-t+i0)$. At a simple pole the contour follows small semicircles above and below the cut, which together contribute $r/(s+t_0)$. □

The lemma excludes functions with infinitely many poles on the cut, such as $-R'/R$ under RH, which has a pole at every $-\gamma^2$; these need a direct argument from Hadamard's product, as in the proof of Theorem 6.1.10.

For $f(s)=\lambda e^{-\lambda\sqrt{s}}/(2\sqrt{s})=-\frac{d}{ds}e^{-\lambda\sqrt{s}}$ one has $\sqrt{-t+i0}=i\sqrt{t}$, and the lemma gives the density $\frac{\lambda}{2\pi\sqrt{t}}\cos(\lambda\sqrt{t})$, so that

$$e^{-\lambda\sqrt{s}}=1-\int_0^\infty\log\bigl(1+\tfrac{s}{t}\bigr)\frac{\lambda\cos(\lambda\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

This is not the Thorin measure of $e^{-\lambda\sqrt{s}}$, which is the positive density $\lambda/(2\pi\sqrt{t})$ (Table 1.8), but the Thorin type density that an exponential contributes when it sits in an exponent, as it does in the logarithm of every Euler factor of (6.1.1) and in $\log\zeta(\frac12+\sqrt{s})$.

Proposition 6.1.2. *For* $0<q<1$ *and* $\lambda>0$ *the Euler factor* $E(s)=1-qe^{-\lambda\sqrt{s}}$ *is analytic and zero free on* $\mathbb{C}\setminus(-\infty,0]$, *and*

$$\log\frac{E(s)}{1-q}=-\int_0^\infty\log\bigl(1+\tfrac{s}{t}\bigr)\,u(t)\,dt,$$

$$u(t)=-\frac{q\lambda}{2\pi\sqrt{t}}\,\frac{\cos(\lambda\sqrt{t})-q}{1-2q\cos(\lambda\sqrt{t})+q^2}=-\frac{\lambda}{2\pi\sqrt{t}}\sum_{k\ge1}q^k\cos(k\lambda\sqrt{t}).$$

*Proof.* On $\mathbb{C}\setminus(-\infty,0]$ we have $|e^{\lambda\sqrt{s}}|>1>q$, so $E$ has no zeros. Put $f=-E'/E=-\frac{q\lambda}{2\sqrt{s}}(e^{\lambda\sqrt{s}}-q)^{-1}$, which is $O(|s|^{-1/2})$ at 0 and decays at infinity.

At $\sqrt{s} = i\tau$, $\tau = \sqrt{t}$, we get $f = \frac{iq\lambda}{2\tau}(e^{i\lambda\tau} - q)^{-1}$ and $\operatorname{Im} f = \frac{q\lambda}{2\tau}\,\frac{\cos\lambda\tau - q}{1 - 2q\cos\lambda\tau + q^2}$. Lemma 6.1.1 gives $u$; integrating $-E'/E = \int u(t)\,dt/(s+t)$ from 0 to $s$ gives the logarithm, and the series is the Poisson kernel expansion. □

With $q = p^{-1/2}$ and $\lambda = \log p$, summing $u$ over $p$ formally gives

$$u_\zeta(t) = -\frac{1}{2\pi\sqrt{t}} \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} \cos(\sqrt{t}\log n) \quad \text{“}=\text{”} \quad \frac{1}{2\pi\sqrt{t}} \operatorname{Re}\frac{\zeta'}{\zeta}\big(\tfrac12 + i\sqrt{t}\big), \qquad (6.1.5)$$

a divergent series. Since $\frac{d}{d\tau}\arg\zeta(\frac12 + i\tau) = \operatorname{Re}\frac{\zeta'}{\zeta}(\frac12 + i\tau)$, the right side is formally the density of $dS(\sqrt{t})$, but the series converges only in the sense of distributions. Theorem 6.1.10 makes this precise.

### 6.1.3 The Lai and Vere-Jones identity and sawtooth measures

We trade each exponential for a hyperbolic secant, whose Thorin measure is a lattice of unit atoms, at the price of a Möbius inversion.

Lai and Vere-Jones [322] pose the completeness of the dilates $\operatorname{sech}(\theta x)$, $\theta \ge 0$, as an open problem for the hyperbolic secant law. The Möbius inversion of the alternating exponential series of the secant expands $e^{-x}$ in the dilates $\operatorname{sech}(mx)$. This is Proposition 6.1.3, which we call the Lai and Vere-Jones identity, and the expansions of $1/\zeta$ and $\log\zeta$ built from it form the Vere-Jones representation. Here $\chi_4$ is the nontrivial character modulo 4.

Proposition 6.1.3. *For* $\operatorname{Re} x > 0$,

$$\tfrac12 \operatorname{sech} x = \sum_{m \text{ odd}} \chi_4(m)\, e^{-mx}, \qquad e^{-x} = \tfrac12 \sum_{m\ge 1} \mu(m)\chi_4(m)\, \operatorname{sech}(mx).$$

*Proof.* The first is the geometric series $e^{-x}/(1+e^{-2x})$. Since $\chi_4$ is completely multiplicative, $\sum_m \mu(m)\chi_4(m) \sum_k \chi_4(k) e^{-mkx} = \sum_r e^{-rx} \sum_{mk=r} \mu(m)\chi_4(m)\chi_4(k) = e^{-x}$, the rearrangement being justified by $\sum_{m,k} e^{-mk\operatorname{Re} x} < \infty$ for $\operatorname{Re} x > 0$. □

Lemma 6.1.4. *For* $\lambda > 0$, $\operatorname{sech}(\lambda\sqrt{s}) = \prod_{k\ge 0}\big(1 + s/\tau_k\big)^{-1}$ *with* $\tau_k = \big((k+\frac12)\pi/\lambda\big)^2$, *written* $\tau_k(\lambda)$ *when the dependence on* $\lambda$ *matters. It is the Laplace transform of* $T_\lambda = \sum_k \tau_k^{-1} e_k$ *with* $e_k$ *independent standard exponentials, the exit time of* $\sqrt{2}B$ *from* $(-\lambda, \lambda)$, *and it is a GGC with atomic Thorin measure* $\sum_k \delta_{\tau_k}$; *see [49] and Table 1.8.*

Theorem 6.1.5. *For* $s \in H_{1/2}$,

$$\frac{1}{\zeta(\frac12 + \sqrt{s})} = 1 + \sum_{N\ge 2} A_N \operatorname{sech}(\sqrt{s}\log N) = 1 + \sum_{N\ge 2} A_N\, \mathbb{E}\, e^{-sT_{\log N}},$$

*where* $A_N = \frac12 \mu(m)\chi_4(m)\mu(n)\, n^{-1/2}$ *if* $N = n^m$ *with* $n \ge 2$ *squarefree and* $m$ *odd, and* $A_N = 0$ *otherwise. In particular* $A_N = \mu(N)/(2\sqrt{N})$ *for squarefree* $N$.

*Proof.* Keep the term $n = 1$ of (6.1.1) and apply Proposition 6.1.3 with $x = \sqrt{s}\log n$ to each $n \geq 2$. The pair $(n, m)$ with $n$ squarefree and $n^m = N$ is unique, since all exponents of $N$ must equal $m$. Absolute convergence follows from $\sum_n n^{-1/2}\sum_m n^{-m\,\mathrm{Re}\,\sqrt{s}} < \infty$ for $\mathrm{Re}\,\sqrt{s} > \frac{1}{2}$. □

Each $\mathrm{sech}(\sqrt{s}\log N)$ is a solved model, by which we mean a function whose Thorin measure is explicit. Its atoms $\tau_k(\log N)$ are units at the squares of the odd multiples of $\pi/(2\log N)$, a lattice of spacing $\pi/\log N$ in the variable $\sqrt{t}$; we call them pseudo zeros. The coefficients $A_N$ carry the signs of the Möbius function, so positivity is now a question about the coefficients alone.

Theorem 6.1.5 is additive, but a mixture of GGCs is not in general a GGC, and the Thorin measure is defined through the logarithm. To reach the logarithm we replace sech by $\log(1 + e^{-x})$, whose coefficients have as Dirichlet series Dirichlet's eta function $\eta(1 + v) = (1 - 2^{-v})\zeta(1 + v)$, not the Dedekind eta function of Sections 8.5 and 11.3.

Proposition 6.1.6. *Define* $d_n$ *by* $\sum_n d_n n^{-v} = 1/\eta(1 + v)$, *that is* $d_n = \sum_{2^j m=n} \mu(m)/m$. *Then* $|d_n| \leq 1$ *and for* $\mathrm{Re}\,x > 0$

$$e^{-x} = \sum_{m\geq 1} d_m \log\bigl(1 + e^{-mx}\bigr).$$

*Proof.* Expand $\log(1 + e^{-y}) = \sum_k (-1)^{k+1} e^{-ky}/k$. The coefficients $(-1)^{k+1}/k$ have Dirichlet series $\eta(1 + v)$, so the Dirichlet convolution with $d$ is $\delta_{r,1}$. If $2^\ell \| n$, a squarefree $m = n/2^j$ requires $j \in \{\ell - 1, \ell\}$, so $d_n = \mu(n)/n$ for odd $n$ and $d_n = \mu(k)(2^\ell - 2^{\ell-1})/n = \mu(k)/(2k)$ for $n = 2^\ell k$ with $\ell \geq 1$ and $k$ odd. In particular $|d_n| \leq 1$ and $\sum_n |d_n|/n < \infty$. □

Lemma 6.1.7. *For* $\lambda > 0$ *and* $s \in \mathbb{C} \setminus (-\infty, 0]$,

$$\log\bigl(1 + e^{-\lambda\sqrt{s}}\bigr) = \log 2 - \int \log\bigl(1 + \tfrac{s}{t}\bigr)\, U_\lambda(dt), \qquad U_\lambda(dt) = \frac{\lambda}{4\pi\sqrt{t}}\, dt - \sum_{k\geq 0} \delta_{\sigma_k(\lambda)},$$

*with* $\sigma_k(\lambda) = \bigl((2k + 1)\pi/\lambda\bigr)^2$. *Equivalently* $U_\lambda(dt) = -d_t \Sigma_\lambda(\sqrt{t})$, *where* $\Sigma_\lambda(\tau) = \#\{k \geq 0 : (2k + 1)\pi/\lambda \leq \tau\} - \lambda\tau/2\pi$ *is a sawtooth with* $|\Sigma_\lambda| \leq \frac{1}{2}$.

*Proof.* Write $1 + e^{-y} = 2e^{-y/2}\cosh(y/2)$ with $y = \lambda\sqrt{s}$. Lemma 6.1.4 gives $\log\cosh(\lambda\sqrt{s}/2) = \sum_k \log(1 + s/\sigma_k)$, and $\int_0^\infty \log(1 + s/t)\, t^{-1/2} dt = 2\pi\sqrt{s}$ gives $\lambda\sqrt{s}/2 = \int \log(1 + s/t)\frac{\lambda}{4\pi\sqrt{t}} dt$. □

The reciprocal of each block has Thorin measure $-U_\lambda$: unit atoms at the pseudo zeros, from $\mathrm{sech}(y/2)$, minus the density $\lambda/(4\pi\sqrt{t})$ that counts them on average, from $e^{y/2}$. The function $\Sigma_\lambda$ is a toy model of $S(\tau)$, with $1 + e^{-\lambda\sqrt{s}}$ playing the role of a zeta function all of whose zeros lie on the line.

THEOREM 6.1.8. *Put* $a_n = \Lambda(n)/(\sqrt{n}\log n)$ *and* $B_N = \sum_{n^m=N} d_m a_n$. *For* $s \in H_{1/2}$,

$$\log\zeta\big(\tfrac12+\sqrt{s}\big) = \sum_{N\ge 2} B_N \log\big(1+N^{-\sqrt{s}}\big)$$

*and*

$$\frac{d}{ds}\log\zeta\big(\tfrac12+\sqrt{s}\big) = \sum_{N\ge 2} B_N\Big[\sum_{k\ge 0}\frac{1}{s+\sigma_k(\log N)} - \frac{\log N}{4\sqrt{s}}\Big].$$

*The first values are* $B_p = p^{-1/2}$, $B_4 = \frac14 + \frac{1}{2\sqrt{2}}$, $B_8 = -\frac{1}{6\sqrt{2}}$, $B_9 = \frac16 + \frac{1}{2\sqrt{3}}$.

*Proof.* For $\operatorname{Re} z > 1$, $\log\zeta(z) = \sum_n a_n n^{-(z-1/2)}$. Apply Proposition 6.1.6 with $x = \sqrt{s}\log n$ and regroup by $N = n^m$; since $|d_m| \le 1$ the double series converges absolutely on $H_{1/2}$. Differentiate with Lemma 6.1.7. □

Formally the Thorin measure of $1/\zeta(\frac12+\sqrt{s})$ is $-\sum_N B_N U_{\log N} = \sum_N B_N\, d\Sigma_{\log N}(\sqrt{t})$, a superposition of sawtooth measures. The series does not converge as a series of measures, but its Laplace transforms do; Proposition 6.1.13 and Corollary 6.1.14 make this exact.

### 6.1.4 The normalised reciprocal and its Thorin measure

The factorisation $\varphi = RG$, with $\xi(z) = \frac12 z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ as in Section 1.2, separates the zeros from the archimedean part.

PROPOSITION 6.1.9. *The function* $\varphi$ *is meromorphic on* $\mathbb{C}\setminus(-\infty,0]$ *and*

$$\varphi(s) = R(s)\,G(s), \qquad R(s) = \frac{\xi(\frac12)}{\xi(\frac12+\sqrt{s})}, \qquad G(s) = \frac{\Gamma(\frac14+\frac{\sqrt{s}}{2})\,\pi^{-\sqrt{s}/2}}{\Gamma(\frac14)}.$$

*It is analytic and zero free on* $\mathbb{C}\setminus(-\infty,0]$ *if and only if the Riemann Hypothesis holds. On* $(0,\infty)$ *it is analytic and positive, it is continuous at* 0 *with* $\varphi(0) = 1$, *and* $-\varphi'/\varphi$ *is analytic in a neighbourhood of* $(0,\infty)$.

*Proof.* With $z = \frac12 + \sqrt{s}$ we have $z(z-1) = s - \frac14$, so $(1-4s)\zeta(z) = -8\,\xi(z)\pi^{z/2}/\Gamma(z/2)$; evaluating at $s = 0$ fixes the constant. The gamma factor $G$ is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ since $\operatorname{Re}(\frac14+\frac{\sqrt{s}}{2}) > 0$, so $\varphi$ has no zeros in $\mathbb{C}\setminus(-\infty,0]$, and its poles there are those of $R$, namely the points $\sqrt{s} = \rho - \frac12$ with $\operatorname{Re}\rho > \frac12$. By the functional equation there are none if and only if RH holds. On the real axis $\zeta(\frac12+x)$ is finite and nonzero for $x > 0$, $x \ne \frac12$, and $(1-4s)\zeta(\frac12+\sqrt{s}) \to -4$ as $s \to \frac14$. Since $\zeta(\frac12+x) < 0$ for $0 < x < \frac12$ and $\zeta(\frac12+x) > 0$ for $x > \frac12$, while $1-4s$ changes sign at $s = \frac14$ and $\zeta(\frac12) < 0$, the function $\varphi$ is positive on $(0,\infty)$. □

THEOREM 6.1.10. *Assume RH and let $N(T)$ count the zeros $\frac12+i\gamma$ with $0<\gamma\le T$, with multiplicity, as do the sums over $\gamma$ below. Then for $s\in\mathbb{C}\setminus(-\infty,0]$*

$$\log\varphi(s)=-\int_{(0,\infty)}\log\bigl(1+\tfrac{s}{t}\bigr)\,dM(t),\qquad M(t)=N(\sqrt t)-\frac{\theta(\sqrt t)}{\pi},$$

*that is*

$$dM(t)=\sum_{\gamma>0}\delta_{\gamma^2}-\frac{\theta'(\sqrt t)}{2\pi\sqrt t}\,dt,\qquad -\frac{\varphi'}{\varphi}(s)=\int\frac{dM(t)}{s+t}.$$

*By the Riemann von Mangoldt formula $M(t)=1+S(\sqrt t)$ for $\sqrt t$ not an ordinate, with $S(0^+)=-1$.*

*Proof.* Under RH, Hadamard's product gives $R(s)=\prod_{\gamma>0}(1+s/\gamma^2)^{-1}$, a GGC with Thorin measure $\sum\delta_{\gamma^2}$, and $-R'/R(s)=\sum_{\gamma>0}1/(s+\gamma^2)$. For $G$ put $g=-G'/G=-\frac{1}{4\sqrt s}\{\psi(\frac14+\frac{\sqrt s}{2})-\log\pi\}$, analytic on $\mathbb{C}\setminus(-\infty,0]$, $O(|s|^{-1/2})$ at 0 and $O(|s|^{-1/2}\log|s|)$ at infinity. At $\sqrt s=i\tau$, $g=\frac{i}{4\tau}\{\psi(\frac14+\frac{i\tau}{2})-\log\pi\}$, so $\operatorname{Im}g=\theta'(\tau)/(2\tau)$, and Lemma 6.1.1 gives the density $-\theta'(\sqrt t)/(2\pi\sqrt t)$. Add the two Stieltjes representations and integrate from 0 to $s$; Fubini applies because $|dM|$ integrates $\min(1,1/t)(1+|\log t|)$. Finally $N(T)=\theta(T)/\pi+1+S(T)$ [498, Section 9.3]. □

Under RH the Thorin measure of $\varphi$ has the unit atoms at the squared ordinates, the Thorin measure of $R$, which is a GGC exactly when RH holds (Proposition 1.2.1), and the density $-\theta'(\sqrt t)/(2\pi\sqrt t)$ of the gamma factor, positive for $t<\tau_0^2$ and negative beyond, where $\tau_0\approx6.28984$ is the point at which $\theta$ attains its minimum. Between consecutive ordinates, all of which exceed $\tau_0$, the function $M$ decreases; hence $dM$ is not positive, and $\varphi$ is not a GGC even under RH. Theorem 7.1.3 proves more, unconditionally.

Near the origin $M(t)\sim c\sqrt t/\pi$, with

$$c=-\theta'(0)=\frac{\log\pi-\psi(\frac14)}{2}=\frac{\zeta'(\frac12)}{\zeta(\frac12)}\approx2.68609,\tag{6.1.6}$$

the last equality from $\xi'(\frac12)=0$. The function $M$ rises to $-\theta(\tau_0)/\pi\approx1.12394$ at $t=\tau_0^2$, falls to $\approx0.55025$ just before $\gamma_1^2$, jumps by one, and thereafter is a staircase with mean 1 whose continuous parts decrease (Figure 6.1; Table 7.7 gives these constants to more digits). As $s\to0$, $\varphi(s)=1-c\sqrt s+O(s)$.

### 6.1.5 Lévy density, theta functions and a criterion for RH

Under RH, integrating by parts in Theorem 6.1.10 and using $s/(s+t)=\int_0^\infty(1-e^{-sx})te^{-tx}dx$ turns the Thorin type representation of $\varphi$ into a Lévy Khintchine

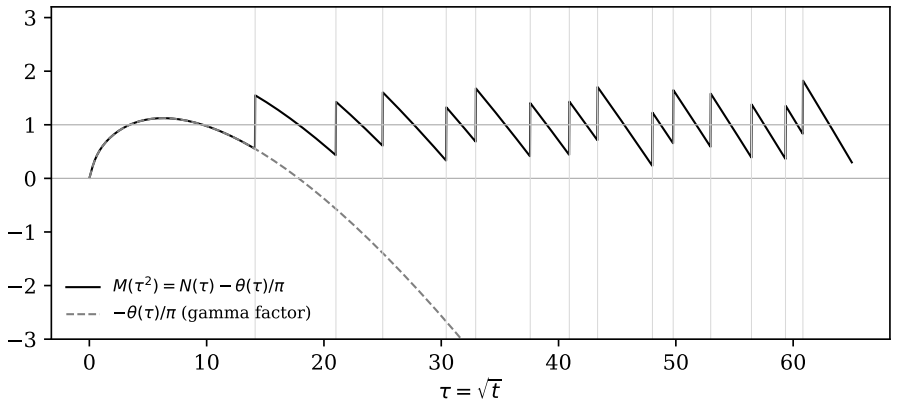



Figure 6.1: The function $M(\tau^2) = N(\tau) - \theta(\tau)/\pi = 1 + S(\tau)$ against $\tau = \sqrt{t}$, with the zero ordinates marked; under RH it is the distribution function of the Thorin measure of $\varphi$. The dashed curve is the gamma factor contribution $-\theta(\tau)/\pi$ alone.

one,

$$\log \varphi(s) = -\int_0^\infty \bigl(1 - e^{-sx}\bigr)\,\frac{w(x)}{x}\,dx,$$
$$w(x) = \int e^{-tx}\,dM(t), \qquad \frac{w(x)}{x} = \int_0^\infty e^{-tx} M(t)\,dt.$$

Theorem 6.1.11. *Define, unconditionally,*

$$w(x) = e^{x/4} - P(x), \qquad P(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x}, \qquad x > 0.$$

*Then* $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)\,dx$ *for* $s > \frac{1}{4}$, *and for every* $x > 0$

$$\begin{aligned} w(x) &= W(x) - W_0(x), \qquad W(x) = \tfrac{1}{2} \sum_\rho e^{x(\rho - \frac{1}{2})^2}, \\ W_0(x) &= \frac{1}{\pi} \int_0^\infty e^{-x\tau^2} \theta'(\tau)\,d\tau, \end{aligned} \tag{6.1.7}$$

*the sum running over all nontrivial zeros with multiplicity. Here* $W$ *is the heat trace of Section 1.2 in the variable* $x$*; under RH,* $W(x) = \sum_{\gamma > 0} e^{-x\gamma^2}$ *and* $w(x) = \int e^{-tx} dM(t)$. *The function* $w$ *satisfies* $w(0^+) = 1$ *and* $w(x) \sim c/(2\sqrt{\pi x})$ *as* $x \to \infty$.

*Proof.* By logarithmic differentiation, $-\varphi'/\varphi = \frac{1}{s - 1/4} + \frac{1}{2\sqrt{s}} \frac{\zeta'}{\zeta}(\frac{1}{2} + \sqrt{s})$. The first term is the transform of $e^{x/4}$; for the second use $\zeta'/\zeta = -\sum \Lambda(n) n^{-z}$ and

$\frac{1}{2\sqrt{s}}e^{-a\sqrt{s}} = \int_0^\infty e^{-sx}(4\pi x)^{-1/2}e^{-a^2/4x}dx$, with Tonelli justified on $s > \frac{1}{4}$. Identity (6.1.7) is Weil's explicit formula for the even test function $h(r) = e^{-xr^2}$, whose Fourier partner is $g(u) = (4\pi x)^{-1/2}e^{-u^2/4x}$: the pole contributes $h(\pm\frac{i}{2}) = e^{x/4}$, the archimedean place contributes $W_0$, and the primes contribute $P$. It is (1.2.5) of Section 1.2 with $t = x$. Under RH, Theorem 6.1.10 gives the same Laplace transform for $\int e^{-tx}dM(t)$ and uniqueness identifies the two. The limit $w(0^+) = 1$ holds because $P(x) = O(x^{-1/2}e^{-(\log 2)^2/4x})$ as $x \to 0$. The behaviour at infinity needs no hypothesis: every term of $W$ has modulus at most $e^{-x(\gamma^2-1/4)}$ with $\gamma > 14$, so $W$ decays exponentially, while $W_0(x) \sim -c/(2\sqrt{\pi x})$ by Lemma 6.1.12 below. □

We call $w$ the canonical function of $\varphi$: when $\varphi$ is infinitely divisible, $w(x)/x$ is its Lévy density, and $\varphi$ is infinitely divisible exactly when $w \geq 0$. The proof of Theorem 7.1.1 uses the prime side $e^{x/4} - P(x)$ for small $x$ and the finitely many verified zeros for large $x$, and it rests on the following integral formula for $W_0$.

Lemma 6.1.12. *For $x > 0$,*

$$W_0(x) = \frac{F(x)}{4\sqrt{\pi x}}, \qquad F(x) = 2\theta'(0) + \int_0^\infty \frac{e^{-u/4}\big(1 - e^{-u^2/16x}\big)}{1 - e^{-u}}\,du.$$

*The function $F$ is strictly decreasing, $F(0^+) = +\infty$ and $F(\infty) = 2\theta'(0) = \psi(\frac{1}{4}) - \log\pi < 0$. Hence $W_0$ has exactly one zero $x_*$, with $W_0 > 0$ on $(0, x_*)$ and $W_0 < 0$ on $(x_*, \infty)$.*

*Proof.* Insert $\psi(z) = \int_0^\infty \big(e^{-u}/u - e^{-zu}/(1-e^{-u})\big)du$ with $z = \frac{1}{4} + \frac{i\tau}{2}$ into $\theta'$, and use $\int_0^\infty e^{-x\tau^2}\cos(\tau u/2)\,d\tau = \frac{1}{2}\sqrt{\pi/x}\,e^{-u^2/16x}$; the interchange is justified after writing the $\psi$ integrand as $e^{-u}/u - e^{-u/4}/(1-e^{-u})$ plus $e^{-u/4}(1-\cos(\tau u/2))/(1-e^{-u})$, both absolutely integrable against $e^{-x\tau^2}$. The terms free of $\tau$ reassemble into $\psi(\frac{1}{4}) - \log\pi$. The integrand of $F$ is positive and decreasing in $x$; as $x \to 0$ it increases to $e^{-u/4}/(1-e^{-u})$, which is not integrable at 0. □

For the sawtooth measure of Lemma 6.1.7 define its heat trace

$$h_\lambda(x) = -\int e^{-tx}\,U_\lambda(dt) = \sum_{k\geq 0} e^{-x\sigma_k(\lambda)} - \frac{\lambda}{4\sqrt{\pi x}} = \frac{\lambda}{2\sqrt{\pi x}}\sum_{j\geq 1}(-1)^j e^{-j^2\lambda^2/4x},$$

the last form by Poisson summation. The first form is a sum over the pseudo zeros minus their smooth count and the second a sum over the multiples $j\lambda$, so this Jacobi theta identity is the explicit formula of the toy zeta function $1 + e^{-\lambda\sqrt{s}}$.

Proposition 6.1.13. *For every $x > 0$, with absolutely convergent sums,*

$$w(x) = e^{x/4} + \sum_{N\geq 2} B_N\, h_{\log N}(x).$$

*Proof.* Substitute the theta form of $h_\lambda$ and regroup the triple sum over $(n, m, j)$ by $r = mj$. The coefficient of $e^{-r^2(\log n)^2/4x}$ is

$$-(4\pi x)^{-1/2} a_n \log n \sum_{mj=r} m\, d_m (-1)^{j+1},$$

and the inner sum is $\delta_{r,1}$ because its Dirichlet series is $\eta(v)^{-1}\cdot\eta(v)$. Since $a_n \log n = \Lambda(n)/\sqrt{n}$ the sum equals $-P(x)$. The bound $|m\, d_m| \le m$ against Gaussian decay gives absolute convergence. □

Proposition 6.1.13 is the Vere-Jones form of Theorem 6.1.11. Unconditionally, $w$ is the Laplace transform, term by term, of the formal series $\delta_{-1/4} + \sum_N B_N\{\sum_k \delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}dt\}$ of sech atom lattices, with all the arithmetic in the weights $B_N$; under RH it is also the Laplace transform of $dM$. The atom at $-\frac{1}{4}$ comes from the pole of $\zeta$ and lies outside $(0,\infty)$, where Thorin measures live, so under RH the sawtooth series must cancel it. Two consequences follow. The first makes the sawtooth expansion of $S$ rigorous after Gaussian averaging.

Corollary 6.1.14. *Assume RH. For every $x > 0$,*

$$\int_0^\infty S(\tau)\, 2\tau e^{-x\tau^2}\, d\tau = \frac{e^{x/4} - 1}{x} + \sum_{N\ge 2} B_N \int_0^\infty \Sigma_{\log N}(\tau)\, 2\tau e^{-x\tau^2}\, d\tau,$$

*and each integral on the right equals $h_{\log N}(x)/x$.*

*Proof.* By Theorem 6.1.11,

$$\int_0^\infty e^{-tx}(1 + S(\sqrt{t}))dt = w(x)/x,$$

and $\int_0^\infty e^{-tx}\Sigma_\lambda(\sqrt{t})\, dt = h_\lambda(x)/x$ by parts since $\Sigma_\lambda(0) = 0$. Subtract $\int e^{-tx}dt = 1/x$ and apply Proposition 6.1.13; substitute $t = \tau^2$. □

The second consequence is a criterion for RH.

Theorem 6.1.15. *The following are equivalent.*

*(i) The Riemann Hypothesis.*

*(ii) There is a real signed Borel measure $m$ on $[0,\infty)$ with $\int |m|(dt)/(1+t) < \infty$ and $w(x) = \int e^{-tx} m(dt)$ for all $x > 0$.*

*In that case $m = dM$.*

*Proof.* If RH holds take $m = dM$; the atoms satisfy $\sum(1+\gamma^2)^{-1} < \infty$ and the density is $O(t^{-1/2}(1+t)^{-1}\log(2+t))$ after division by $1+t$. Conversely, given $m$, put $J(s) = \int m(dt)/(s+t)$, analytic on $\mathbb{C} \setminus (-\infty, 0]$. For $s > \frac{1}{4}$ Fubini gives

$\int_0^\infty e^{-sx}w(x)dx = J(s)$, so $-\varphi'/\varphi = J$ on $(\frac{1}{4},\infty)$ and hence on all of $\mathbb{C}\setminus(-\infty,0]$ by analytic continuation. A zero or pole of $\varphi$ in $\mathbb{C}\setminus(-\infty,0]$ would be a pole of $-\varphi'/\varphi$, so $\varphi$ has none, and Proposition 6.1.9 gives RH. Uniqueness of the Stieltjes transform gives $m = dM$. □

### 6.1.6 Across the critical strip

Fix $\sigma$ and evaluate $\zeta$ on the line $\operatorname{Re} z = \sigma$ through $z = \sigma + \sqrt{s}$. A zero $\rho = \beta + i\gamma$ then sits at $s = (\rho - \sigma)^2$: if $\beta > \sigma$ it gives a pole of $1/\zeta(\sigma + \sqrt{s})$ in $\mathbb{C}\setminus(-\infty,0]$, if $\beta = \sigma$ a point of the cut, and if $\beta < \sigma$ no singularity, only a Poisson kernel of width $\sigma - \beta$ in the Thorin density. On each line one can ask whether a normalised reciprocal of $\zeta$ is infinitely divisible, and whether the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ has the stronger GGC property.

To the right of $\operatorname{Re} z = 1$ the first question has a classical answer, in the variable $\tau$ rather than $s$. The Dirichlet series converges, $\zeta(\sigma+i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau \log N_\sigma}$ for the zeta distribution $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and the Lévy measure of $\log N_\sigma$ is the prime measure $\sum_n \Lambda(n)n^{-\sigma}(\log n)^{-1}\delta_{\log n}$. The Vere-Jones representation writes this measure as a superposition of the alternating lattices $\sum_j (-1)^{j+1}j^{-1}\delta_{jn\log p}$, the signed Lévy measures of fair coins, one for each prime power $p^n$, with weight the necklace polynomial $b_n(p^{-\sigma})$ (Theorem 7.1.8). At $\sigma = 0$ the weight $b_n(1)$ is 1 when $n$ is a power of 2 and 0 otherwise, and the identity becomes Euler's binary product, with every block a solved model of unit multiplicity; for $\sigma > 0$ the weights are fractional and some are negative.

Inside the strip $|\zeta(\sigma+i\tau)|$ is unbounded, so the characteristic function picture breaks down; the Vere-Jones representation survives in the variable $s$. For $0 < \sigma < 1$ put $\varphi_\sigma(s) = \zeta(\sigma)/\{(1 - s/(1-\sigma)^2)\zeta(\sigma + \sqrt{s})\}$, which is $\varphi$ at $\sigma = \frac{1}{2}$. For $s > (1-\sigma)^2$, $-\varphi_\sigma'/\varphi_\sigma$ is the Laplace transform of a function $w_\sigma$, given by the theta series of Proposition 6.1.13 (Section 6.1.5) with $e^{(1-\sigma)^2 x}$ in place of $e^{x/4}$ and the weights $b_n(p^{-\sigma})$ in place of $B_{p^n} = b_n(p^{-1/2})$ (Theorem 7.1.12). When $w_\sigma \geq 0$ this is a Lévy representation, $\log\varphi_\sigma(s) = -\int(1 - e^{-sx})w_\sigma(x)x^{-1}dx$. Theorem 7.1.1 proves $w_{1/2} = w > 0$, and numerically $w_\sigma > 0$ on nine lines from $\sigma = 0.05$ to $\sigma = 0.99$, for $10^{-4} \leq x \leq 10^3$ (Section 7.1); whether infinite divisibility holds on every line of the strip is open (Problem 7.8).

The Thorin property is where the zeros enter: see Proposition 7.1.13 and Theorem 7.1.14 of Section 7.1.2.3.

Chapters 9 to 11 return to the completed functions $R_\sigma$, written there $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, with the basepoint $\alpha$ in place of $\sigma$ and the variable $u$ of the book in place of $s$; they are not the normalised functions $\varphi_\sigma$ of this section.

## 6.2 The gamma factor, the Meixner law and the sine squared lemma

On the critical line $|G|^2$ is proportional to the density of a Meixner hypergeometric law, whose characteristic function is a GGC transform in the square of its variable and whose orthogonal polynomials satisfy a local Riemann Hypothesis. The sine squared lemma turns the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$. It shows that after a stable tilt the reciprocals of $\zeta(\beta+\sqrt{s})$, for $\beta>1$, and of the gamma factor are GGC transforms, and it shows how positivity is lost at the centre.

### 6.2.1 The gamma factor in Vere-Jones form

Recall from Section 6.1.1 the constants $c_n = 2n+\frac12$ for $n\ge 0$ and $b=\frac12(\gamma_E+\log\pi)\approx 0.860973$, where $\gamma_E$ is Euler's constant.

Theorem 6.2.1. *For $s\in\mathbb{C}\setminus(-\infty,0]$,*

$$G(s)=e^{-b\sqrt{s}}\prod_{n\ge0}\frac{e^{\sqrt{s}/(2n+2)}}{1+\sqrt{s}/c_n},$$

*and the Thorin measure of $G$ is*

$$U_G(dt)=\frac{1}{2\pi\sqrt{t}}\Big\{b+\sum_{n\ge0}\Big(\frac{c_n}{t+c_n^2}-\frac{1}{2n+2}\Big)\Big\}dt=-\frac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

*Each factor $(1+\sqrt{s}/\kappa)^{-1}$ is the Laplace transform of a GGC, with positive Thorin density $q_\kappa(t)=\dfrac{\kappa}{2\pi\sqrt{t}\,(t+\kappa^2)}$, a Poisson kernel in $\sqrt{t}$; each factor $e^{a\sqrt{s}}$ with $a>0$ has the negative Thorin density $-a/(2\pi\sqrt{t})$.*

*Proof.* Weierstrass gives $\log\Gamma(z_0+v)-\log\Gamma(z_0)=-\gamma_E v-\log(1+v/z_0)+\sum_{n\ge1}\{v/n-\log(1+v/(n+z_0))\}$. Take $z_0=\frac14$, $v=\sqrt{s}/2$, note $v/(n+\frac14)=\sqrt{s}/c_n$, and shift the index of $v/n$. For $q_\kappa$: $\frac{d}{ds}\log(1+\sqrt{s}/\kappa)=\frac{1}{2\sqrt{s}(\sqrt{s}+\kappa)}$, and Lemma 6.1.1 with $\sqrt{-t+i0}=i\sqrt{t}$ gives the density $\frac{\kappa}{2\pi\sqrt{t}(t+\kappa^2)}$. For $e^{a\sqrt{s}}$ use $\int_0^\infty\log(1+s/t)t^{-1/2}dt=2\pi\sqrt{s}$. To sum the Thorin densities of the factors, write the bracket with index $n$ in $U_G$ as $(\frac{1}{c_n}-\frac{1}{2n+2})-\frac{t}{c_n(t+c_n^2)}$: both parts are nonnegative, and over $n$ the first sums to a constant and the second to $O(\log(2+t))$, so Tonelli's theorem applies to each part against $\log(1+s/t)\frac{dt}{2\pi\sqrt{t}}$ for $s>0$, and analytic continuation extends the sum to $\mathbb{C}\setminus(-\infty,0]$; the same split justifies the termwise integration against $e^{-tx}$ in Proposition 6.2.2. The identification with $\theta'$ is the partial fraction series $\operatorname{Re}\psi(\frac14+\frac{i\tau}{2})=-\gamma_E+\sum_{n\ge0}\{\frac{1}{n+1}-\frac{2c_n}{c_n^2+\tau^2}\}$. □

Formally, the gamma factor is thus a renormalised infinite convolution of GGC laws. Write $X_\varphi$ and $X_R$ for random variables with Laplace transforms $\varphi$ and $R$; both laws exist and are infinitely divisible, by Theorem 7.1.1 and by Theorem 5.1.1 [424, Theorem A]. Write $Y_\kappa$ for the variable with $\mathbb{E}e^{-sY_\kappa} = \kappa/(\kappa+\sqrt{s})$, and $L_a$ for the positive stable variable of index $\frac{1}{2}$ with $\mathbb{E}e^{-sL_a} = e^{-a\sqrt{s}}$. Theorem 6.2.1 and Proposition 6.1.9 then say formally

$$X_\varphi \stackrel{d}{=} X_R + L_b + \sum_{n\ge 0} \big(Y_{c_n} \ominus L_{1/(2n+2)}\big), \tag{6.2.1}$$

where $A \ominus B$ stands for a formal variable whose Laplace transform is the quotient of those of $A$ and $B$. The sum of the $Y_{c_n}$ diverges almost surely because $\sum 1/c_n = \infty$, and the stable deconvolutions are the compensators that make the product of the Laplace transforms converge.

Figure 6.2 shows the Thorin density of $G$ against the first three kernels. It changes sign at $\tau_0$, so the reciprocal $1/G$ has a Thorin density that is negative near the origin.

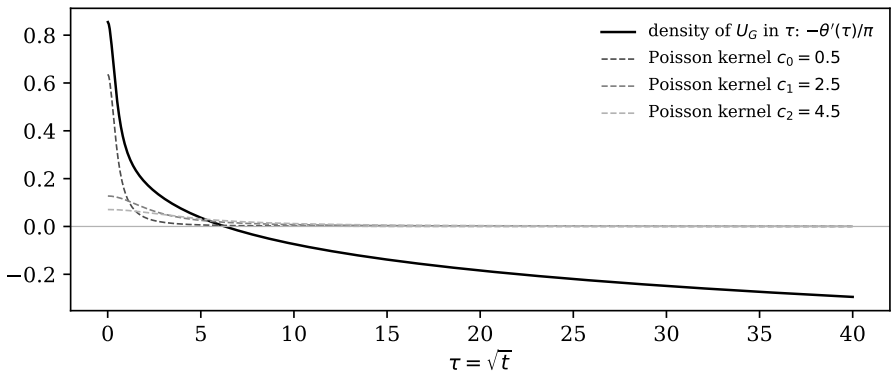



Figure 6.2: The Thorin density of the gamma factor in the variable $\tau = \sqrt{t}$, namely $-\theta'(\tau)/\pi$, together with the first three Poisson kernels $c_n/(\pi(c_n^2+\tau^2))$ of Theorem 6.2.1. The density is positive for $\tau < \tau_0 \approx 6.29$ and negative beyond, where the flat compensators outweigh the kernels.

Proposition 6.2.2. *For $x > 0$,*

$$W_0(x) = -\int e^{-tx}\, U_G(dt) = -\frac{b}{2\sqrt{\pi x}} + \sum_{n\ge 0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2}e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

*Proof.* $\int_0^\infty e^{-tx} q_\kappa(t)\,dt = \frac{\kappa}{\pi}\int_0^\infty \frac{e^{-xv^2}}{v^2+\kappa^2}dv = \frac{1}{2}e^{\kappa^2 x}\operatorname{erfc}(\kappa\sqrt{x})$ and $\int_0^\infty e^{-tx}\frac{dt}{2\pi\sqrt{t}} = \frac{1}{2\sqrt{\pi x}}$. □

THEOREM 6.2.3. *For every $x > 0$, with absolutely convergent sums,*

$$W(x) = e^{x/4} + \sum_{N\geq 2} B_N\, h_{\log N}(x) - \frac{b}{2\sqrt{\pi x}} + \sum_{n\geq 0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2}e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

*Equivalently, $W(x)$ is the Laplace transform, taken term by term, of the formal series*

$$U_R = \delta_{-1/4} + \sum_{N\geq 2} B_N\Big\{\sum_{k\geq 0}\delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}\,dt\Big\} - \frac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\geq 0}\Big(\frac{c_n}{t+c_n^2} - \frac{1}{2n+2}\Big)\Big\}dt,$$

*the formal Thorin measure of $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$. Under RH, $R$ is a GGC, and its Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2} = dN(\sqrt{t})$, where $N(T)$ counts the zeros with $0 < \gamma \leq T$, is a positive measure whose Laplace transform is also $W$.*

*Proof.* $W = w + W_0$ by (6.1.7); substitute Proposition 6.1.13 for $w$ and Proposition 6.2.2 for $W_0$. Term by term, $\delta_{-1/4}$ has Laplace transform $e^{x/4}$, the bracket with index $N$ has transform $h_{\log N}(x)$ by the definition of $h_\lambda$, and the last density is $-U_G$, whose transform is $W_0(x)$ by Proposition 6.2.2. Under RH, Hadamard's product gives $R(s) = \prod_{\gamma>0}(1 + s/\gamma^2)^{-1}$, as in the proof of Theorem 6.1.10, and $W(x) = \sum_{\gamma>0} e^{-x\gamma^2}$ by Theorem 6.1.11. □

In $U_R$ the gamma factor gives negative Poisson kernels of widths $c_n$ in the variable $\sqrt{t}$, with poles at $\sqrt{t} = \pm ic_n$, each compensated by a positive flat density. RH is the statement that the atom at $-\frac{1}{4}$, the sech lattices of the primes and the gamma factor terms combine into unit atoms at the $\gamma^2$ and nothing else. They combine only through Laplace transforms: every partial sum of $U_R$ carries the atom at $-\frac{1}{4}$ and the Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2}$ of $R$ under RH does not, so, like the series after Proposition 6.1.13, $U_R$ does not converge to that measure as a series of measures. By Theorem 6.1.15, $R$ has a Thorin type representation only if RH holds, since adding $U_G$ to its measure would give a measure $m$ as in condition (ii) there.

On the cut the gamma factor has modulus and phase

$$G(-\tau^2 + i0) = \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|}{\Gamma(\frac{1}{4})}\, e^{i\theta(\tau)}.$$

The phase is the Riemann Siegel theta function, and it is $-\pi$ times the distribution function of the Thorin measure: $U_G((0,\tau^2]) = -\theta(\tau)/\pi$, since $\operatorname{Im}\log(1 + s/t) = \pi$

for $t < \tau^2$ at $s = -\tau^2 + i0$. The modulus is the logarithmic potential of the same measure,

$$\log \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|}{\Gamma(\frac{1}{4})} = -\int \log\Big|1 - \frac{\tau^2}{t}\Big| \, U_G(dt), \tag{6.2.2}$$

the integral converging absolutely.

The square of the modulus is proportional to the density of a Meixner hypergeometric law, with parameter $a = \frac{1}{4}$; Lai and Vere-Jones [322] study this family and its case $a = \frac{1}{2}$, the hyperbolic secant law. Since $\int |\Gamma(a+ix)|^2 e^{iux} dx = 2\pi\Gamma(2a)(2\cosh\frac{u}{2})^{-2a}$,

$$p(\tau) = \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2}{2^{3/2}\pi^{3/2}} \quad \text{is a probability density with} \quad \int p(\tau) e^{iv\tau} d\tau = \operatorname{sech}(v)^{1/2}. \tag{6.2.3}$$

In this section $p$ denotes this Meixner density. Since $\Xi(\tau) = -\frac{1}{2}(\tau^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|Z(\tau)$, with Hardy's function $Z(\tau) = e^{i\theta(\tau)}\zeta(\frac{1}{2} + i\tau)$ of Section 2.2, the Meixner density is the archimedean part of $\Xi^2$ on the critical line:

$$\Xi(\tau)^2 = \tfrac{1}{4}\big(\tau^2 + \tfrac{1}{4}\big)^2 \, 2^{3/2}\pi \, p(\tau) \, Z(\tau)^2.$$

### 6.2.2 From Lévy to Thorin: the sine squared lemma

For a function of the form $f(0)/f(\sqrt{s})$ we call $f$ its primal function. The sine squared lemma of [422, Lemma 1] converts the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$. The Vere-Jones representation produces such Lévy measures, sums of atoms at the points $k \log N$, so the lemma applies to them.

Lemma 6.2.4 ([422, Lemma 1]). *Suppose that for some measure $\mu$ on $(0, \infty)$ and some $\alpha$ with $\int_0^\infty (1 \wedge x) e^{-\alpha x} \, \mu(dx) < \infty$,*

$$\log \frac{f(\alpha + s)}{f(\alpha)} - s\frac{f'(\alpha)}{f(\alpha)} = \int_0^\infty \big(e^{-sx} - 1 + sx\big) e^{-\alpha x} \, \frac{\mu(dx)}{x}, \qquad s \geq 0.$$

*Then, for $s \geq 0$,*

$$\log \frac{f(\alpha)}{f(\alpha + s)} + s\frac{f'(\alpha)}{f(\alpha)} = -\int_0^\infty \big(1 - e^{-\frac{1}{2}s^2 t}\big) \frac{\nu_\alpha(t)}{t} \, dt,$$

$$\nu_\alpha(t) = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-tz} \Big\{ \int_0^\infty 2\sin^2\big(x\sqrt{z/2}\big) e^{-\alpha x} \, \mu(dx) \Big\} \frac{dz}{\sqrt{\pi z}}.$$

*The map $\mu \mapsto \nu_\alpha$ is linear and positivity preserving.*

We include the proof by subordination, because its two stages, a Lévy density in $s$ followed by a Thorin density, reappear in Section 6.2.4.

LEMMA 6.2.5 (Lemma 6.2.4 in the scaling $s = u^2$). *Let $\tilde{\mu}$ be a signed measure on $(0,\infty)$ with $\int (1\wedge x)|\tilde{\mu}|(dx) < \infty$, and suppose*

$$\log \frac{f(u)}{f(0)} - u\frac{f'(0)}{f(0)} = \int_0^\infty \big(e^{-ux} - 1 + ux\big)\frac{\tilde{\mu}(dx)}{x}, \qquad u \geq 0.$$

*Then for $s \in \mathbb{C}\setminus(-\infty,0]$*

$$\log \frac{f(0)}{f(\sqrt{s})} + \sqrt{s}\,\frac{f'(0)}{f(0)} = -\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U(t)\,dt,$$
$$U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{t}}{2}\Big)\tilde{\mu}(dx).$$

*If $\tilde{\mu} \geq 0$ then $U \geq 0$, and $s \mapsto f(0)e^{\sqrt{s}f'(0)/f(0)}/f(\sqrt{s})$ is the Laplace transform of a GGC. Since $\sqrt{s} = \int \log(1+s/t)\,\frac{dt}{2\pi\sqrt{t}}$, the Thorin measure of $f(0)/f(\sqrt{s})$ itself is $U(t)\,dt + \frac{f'(0)}{f(0)}\frac{dt}{2\pi\sqrt{t}}$.*

*Proof.* Subordination by the first passage law (6.1.2) gives, for $x, u > 0$,

$$e^{-ux} - 1 + ux = \int_0^\infty \big(1-e^{-u^2y}\big)\big(1-e^{-x^2/4y}\big)\frac{x}{\sqrt{4\pi y^3}}\,dy,$$

because $\int_0^\infty (1-e^{-u^2y})\frac{x}{\sqrt{4\pi y^3}}dy = ux$. Integrating against $\tilde{\mu}(dx)/x$ gives a Lévy form in $s = u^2$ with Lévy density $k(y)/y$, where $k(y) = (4\pi y)^{-1/2}\int(1 - e^{-x^2/4y})\tilde{\mu}(dx)$ is the canonical function. Finally $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos(x\sqrt{t})\}\frac{dt}{\sqrt{\pi t}}$, so $k(y) = \int e^{-yt}U(t)dt$, and Frullani turns $\int(1-e^{-sy})k(y)\frac{dy}{y}$ into $\int \log(1+s/t)U(t)dt$. All kernels in these steps are nonnegative, so by Tonelli's theorem the steps hold with $|\tilde{\mu}|$ in place of $\tilde{\mu}$, and there every integral is finite, since $0 \leq (e^{-ux}-1+ux)/x \leq \min(\frac{u^2x}{2}, u)$ and $\int(1\wedge x)|\tilde{\mu}|(dx) < \infty$; this justifies each interchange for the signed $\tilde{\mu}$. The analytic continuation to $\mathbb{C}\setminus(-\infty,0]$ is immediate. □

Lemma 6.2.5 is (1.2.9) of Section 1.2 integrated against $\tilde{\mu}(dx)/x$; the letters $s$ and $u$ are exchanged between the two sections. We call $\tilde{\mu}$ the primal Lévy measure of $f(0)/f(\sqrt{s})$. Positivity comes from the kernel. The signed cosine density that Lemma 6.1.1 produces for a single exponential in an exponent (Section 6.1.2) is $-\cos = 2\sin^2 - 1$, and Lemma 6.2.5 separates it into the positive part $2\sin^2$ and a flat part $-1$, which is absorbed into the linear tilt $u\,f'(0)/f(0)$, a stable law of index $\frac{1}{2}$.

*6.2.2.1 Zeta to the right of the line, the Vere-Jones blocks and the centre*

PROPOSITION 6.2.6. *For $\beta > 1$ put $\tilde\mu_\beta = \sum_{n\ge2} \Lambda(n) n^{-\beta} \delta_{\log n}$. Then*

$$\frac{\zeta(\beta)}{\zeta(\beta+\sqrt{s})}\, e^{\sqrt{s}\,\zeta'(\beta)/\zeta(\beta)} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{t}\Big) U_\beta(t)\, dt\Big),$$

$$U_\beta(t) = \frac{1}{2\pi\sqrt{t}} \sum_{n\ge2} \frac{\Lambda(n)}{n^\beta}\, 2\sin^2\Big(\frac{\sqrt{t}\log n}{2}\Big) \ge 0,$$

*so the left side is the Laplace transform of a GGC. Adding the tilt, the Thorin measure of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is $-\frac{1}{2\pi\sqrt{t}} \sum \Lambda(n) n^{-\beta} \cos(\sqrt{t}\log n)\, dt$, which is the Euler product density* (6.1.5) *at $\beta$ in place of $\frac12$.*

*Proof.* $\log\zeta(\beta+u) = \sum \Lambda(n) n^{-\beta} e^{-u\log n}/\log n$, so the hypothesis of Lemma 6.2.5 holds with $\tilde\mu_\beta$, which has finite mass for $\beta > 1$. □

The tilt is needed: without it the Thorin density of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is negative near $t = 0$. For $\xi$ no tilt is needed, since $s \mapsto \xi(\alpha)/\xi(\alpha+\sqrt{s})$ is a GGC transform for every $\alpha \ge 1$ (Theorem 1.2.2); at $\alpha = 1$, Section 9.1 obtains its Thorin density from a Lévy form of $\log\xi(1+u)$ by the same lemma.

Apply the lemma to one block $f_\lambda(u) = 1 + e^{-\lambda u}$ of Theorem 6.1.8, shifted by $\alpha > 0$. Since $\log(1+e^{-\lambda(\alpha+u)}) = \sum_k (-1)^{k+1} q^k e^{-k\lambda u}/k$ with $q = e^{-\lambda\alpha}$, the primal Lévy measure is the alternating lattice $\tilde\mu_{\lambda,\alpha} = \lambda \sum_{k\ge1} (-1)^{k+1} q^k \delta_{k\lambda}$.

PROPOSITION 6.2.7. *For $\alpha > 0$ the sine squared transform of $\tilde\mu_{\lambda,\alpha}$ is positive:*

$$U_{\lambda,\alpha}(t) = \frac{\lambda}{2\pi\sqrt{t}}\, \frac{q(1-q)\big(1-\cos\lambda\sqrt{t}\big)}{(1+q)\big(1+2q\cos\lambda\sqrt{t}+q^2\big)} \ge 0,$$

*so $\frac{1+e^{-\lambda\alpha}}{1+e^{-\lambda(\alpha+\sqrt{s})}}\, e^{-\sqrt{s}\lambda q/(1+q)}$ is the Laplace transform of a GGC. As $\alpha \downarrow 0$, $U_{\lambda,\alpha}(t)\, dt \to \sum_{k\ge0} \delta_{\sigma_k(\lambda)}$ vaguely, the tilt density tends to $-\frac{\lambda}{4\pi\sqrt{t}}$, and the Thorin measure of $2/(1+e^{-\lambda\sqrt{s}})$ is recovered as $\sum_k \delta_{\sigma_k(\lambda)} - \frac{\lambda}{4\pi\sqrt{t}} dt = -U_\lambda$, the sawtooth measure of Lemma 6.1.7.*

*Proof.* $\sum_k (-1)^{k+1} q^k = \frac{q}{1+q}$ and $\sum_k (-1)^{k+1} q^k \cos k\omega = \frac{q\cos\omega+q^2}{1+2q\cos\omega+q^2}$; their difference is $\frac{q(1-q)(1-\cos\omega)}{(1+q)(1+2q\cos\omega+q^2)} \ge 0$. As $q \uparrow 1$ this is a Poisson kernel in $\omega = \lambda\sqrt{t}$ concentrating at $\omega = (2k+1)\pi$ with mass $\pi$ per period, which becomes $\sum_k \delta_{\sigma_k(\lambda)}$ after the change of variable. The tilt is $f'(\alpha)/f(\alpha) = -\lambda q/(1+q) \to -\lambda/2$. □

The negative part of the sawtooth measure is entirely the tilt. Summing the blocks with the weights $B_N$ gives $\log\zeta(\frac12+\alpha+\sqrt{s})$, and the sum of Thorin densities $\sum_N B_N U_{\log N,\alpha}$ converges absolutely for $\alpha > \frac12$, where it equals $U_{1/2+\alpha}$ of Proposition 6.2.6. Positivity is then visible termwise in the prime form. It is

not visible blockwise in the Vere-Jones form, because infinitely many weights are negative: $B_{p^3} = \frac{1}{3}(p^{-3/2} - p^{-1/2}) < 0$ for every prime $p$.

At $\beta = \frac{1}{2}$ the prime Lévy measure $\tilde{\mu}_{1/2} = \sum \Lambda(n) n^{-1/2} \delta_{\log n}$ has infinite mass, and both the image and the tilt in Lemma 6.2.5 diverge. The divergences cancel, and the heat trace makes the cancellation exact. Since $\int_0^\infty e^{-xt}\, 2\sin^2(\frac{\sqrt{t}L}{2}) \frac{dt}{2\pi\sqrt{t}} = \frac{1-e^{-L^2/4x}}{2\sqrt{\pi x}}$, the heat trace of the sine squared image is formally

$$\frac{1}{2\sqrt{\pi x}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \big(1 - e^{-(\log n)^2/4x}\big) = \frac{1}{2\sqrt{\pi x}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} - P(x).$$

Replace the divergent constant by its analytic continuation $\sum \Lambda(n) n^{-1/2} \to -\zeta'(\frac{1}{2})/\zeta(\frac{1}{2}) = -c$, and add the tilt $\frac{c}{2\sqrt{\pi x}}$ (the heat trace of $\frac{c}{2\pi\sqrt{t}} dt$) and the heat trace $e^{x/4}$ of the pole atom $\delta_{-1/4}$. The result is $e^{x/4} - P(x) = w(x)$, in agreement with Theorem 6.1.11. So at the centre Lemma 6.2.5 still produces the correct Thorin measure $dM$, but only after a regularisation that subtracts an infinite positive constant from a positive density, and that is where the sign of $dM$ is lost.

### 6.2.2.2 The inverse direction: cosine Lévy forms

The Frullani identity behind Lemma 6.2.5 also runs in the opposite direction, from a Thorin measure in $s$ to a symmetric Lévy measure in $u = \sqrt{s}$: for $\beta > 0$ and real $u$, $\int_0^\infty 2\sin^2(\frac{ux}{2}) e^{-\beta x} \frac{dx}{x} = \frac{1}{2}\log(1 + u^2/\beta^2)$, hence

$$\int \log\Big(1 + \frac{s}{t}\Big) U(dt) = \int_0^\infty 2\sin^2\Big(\frac{ux}{2}\Big) K_U(x)\, \frac{dx}{x}, \qquad K_U(x) = 2\int e^{-\sqrt{t}\,x}\, U(dt). \tag{6.2.4}$$

If $U \geq 0$ then $u \mapsto \exp(-\int \log(1 + u^2/t) U(dt))$ is the characteristic function of a symmetric infinitely divisible law with Lévy density $K_U(|x|)/(2|x|)$. Section 10.1 applies this dictionary to $\xi(\alpha)/\xi(\alpha + |u|)$, which is the characteristic function of a normal scale mixture at every basepoint $\alpha$.

Proposition 6.2.8. *Under* (6.2.4)*:*

*(i) the Meixner law* (6.2.3) *has Lévy density* $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$ *on* $\mathbb{R}$*, and its characteristic function, as a function of* $v^2$*, is a GGC transform with Thorin measure* $\frac{1}{2}\sum_{k\geq 0} \delta_{((k+1/2)\pi)^2}$*, since* $\operatorname{sech}(v)^{1/2} = \prod_k (1 + v^2/((k+\frac{1}{2})\pi)^2)^{-1/2}$*;*

*(ii) the sawtooth measure has* $K_{U_\lambda}(x) = \dfrac{\lambda}{\pi x} - \dfrac{1}{\sinh(\pi x/\lambda)}$*;*

*(iii) under RH, the Thorin measure* $U = \sum_{\gamma>0} \delta_{\gamma^2}$ *of* $R$ *has* $K_U(x) = 2\sum_{\gamma>0} e^{-\gamma x}$*, twice* $e^{-ix/2} V(ix)$ *for Cramér's function* $V(z) = \sum_{\gamma>0} e^{\rho z}$ *[96], and* $\xi(\frac{1}{2})/\xi(\frac{1}{2} + u)$ *is the characteristic function of a sum of independent Laplace variables of scales* $1/\gamma$*.*

PROPOSITION 6.2.9. *For every $x > 0$ the series $\sum_N B_N K_{U_{\log N}}(x)$ diverges.*

*Proof.* $B_N$ vanishes unless $N$ is a prime power. Grouping the powers of a fixed $p$, $\sum_k B_{p^k}/k = \sum_{j,m} d_m\, p^{-j/2}/(j^2 m) = \frac{12}{\pi^2} p^{-1/2} + O(p^{-1})$, using $\sum_m d_m/m = 1/\eta(2) = 12/\pi^2$. Since $K_{U_\lambda}(x) = \frac{\pi x}{6\lambda} + O((x/\lambda)^3)$, the group for $p$ equals $\frac{2x}{\pi}\frac{p^{-1/2}}{\log p}(1 + o(1))$, and $\sum_p p^{-1/2}/\log p = \infty$. The grouping changes the order of summation, but in the order of increasing $N$ the partial sums are dominated by the prime terms $B_p K_{U_{\log p}}(x) \sim \frac{\pi x}{6}\frac{p^{-1/2}}{\log p}$, so the series diverges in that order as well. □

The inverse direction fails to sum over primes for the reason that Lemma 6.2.5 fails at the centre: without Gaussian damping the weights $p^{-1/2}$ are not summable. The Poisson kernel $e^{-\sqrt{t}\,x}$ is the stable $\frac{1}{2}$ average of the heat kernel, $e^{-\sqrt{t}\,x} = \int_0^\infty e^{-ty}\frac{x}{2\sqrt{\pi}} y^{-3/2} e^{-x^2/4y} dy$, and the average extends over $y \gtrsim (\log N)^2$, where the block heat trace $h_{\log N}(y)$ has lost its Gaussian factor $e^{-(\log N)^2/4y}$ and is close to $-\log N/(4\sqrt{\pi y})$. Under the heat kernel alone each block heat trace is a theta function with Gaussian decay in $\log N$, and the series converges (Proposition 6.1.13, Theorem 6.2.3).

### 6.2.3 The Meixner law and its orthogonal polynomials

Let $Y$ have the density $p$ of (6.2.3), so that $\mathbb{E}e^{ivY} = \operatorname{sech}(v)^{1/2}$.

PROPOSITION 6.2.10. *(i) $Y \overset{d}{=} B(T)$, where $B$ is a standard Brownian motion and $T$ is independent with $\mathbb{E}e^{-\lambda T} = \operatorname{sech}(\sqrt{2\lambda})^{1/2}$, the variable $C_{1/2}$ of Biane, Pitman and Yor [49], so that $Y$ is the value at time $\frac{1}{2}$ of the process $\widehat{C}_t = B(C_t)$ of Pitman and Yor [416, (1) and (4)]. The time $T = \sum_{k\ge 0} \frac{8\,\Gamma_k}{(2k+1)^2\pi^2}$, with $\Gamma_k$ independent gamma variables of shape $\frac{1}{2}$ and unit scale, is a GGC with Thorin measure $\frac{1}{2}\sum_k \delta_{(k+1/2)^2\pi^2/2}$.*

*(ii) $Y$ is infinitely divisible with Lévy density $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$, one half of the Lévy density $1/(2|x|\sinh(\pi|x|/2))$ of the process $\widehat{C}$ [416, Table 5]; $Y$ has a Meixner law [475, 161]. As $|\tau| \to \infty$, $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2} e^{-\pi|\tau|/2}$, matching the singularities of $\operatorname{sech}(v)^{1/2}$ at $v = \pm i\pi/2$.*

*(iii) $p$ is not a Pólya frequency function of infinite order.*

*Proof.* (i) $\mathbb{E}e^{ivB(T)} = \mathbb{E}e^{-v^2T/2} = \operatorname{sech}(v)^{1/2}$, and $\operatorname{sech}(\sqrt{2\lambda})^{1/2} = \prod_k (1 + 2\lambda/((k+\frac{1}{2})\pi)^2)^{-1/2}$. (ii) is Proposition 6.2.8(i); the tail follows from Stirling, $|\Gamma(\frac{1}{4} + ix)|^2 \sim 2\pi|x|^{-1/2}e^{-\pi|x|}$ with $x = \tau/2$. (iii) By Schoenberg's theorem [474, Theorem 1] the bilateral Laplace transform of a $\mathrm{PF}_\infty$ density is the reciprocal of an entire function of Laguerre Pólya class, so the reciprocal of its characteristic function is

entire. Here $1/\mathbb{E}e^{ivY} = \cosh(v)^{1/2}$, which has branch points at the simple zeros $v = i(k+\frac{1}{2})\pi$ of cosh and is not entire. □

The time $T$ of (i), which is not the clock variable of Section 1.2, is twice the Meixner time of Table 1.8. The exponent $\frac{1}{2}$ in $\operatorname{sech}(v)^{1/2}$ is the Meixner parameter $2a = \frac{1}{2}$, forced by the factor $\Gamma(z/2)$ of $\xi$ at $z = \frac{1}{2}$. For $2a$ a positive integer the Meixner law is $\mathrm{PF}_\infty$; the zeta function sits at a value where it is not, although its characteristic function is still a GGC transform in $v^2$.

Define the measure $\nu$ on $(0,\infty)$ by

$$\nu(dy) = \frac{e^{y/2}}{\sinh y}\,dy = 2\sum_{k\ge 0} e^{-c_k y}\,dy,$$

with $c_k = 2k + \frac{1}{2}$ as in Theorem 6.2.1.

PROPOSITION 6.2.11. *The function* $\Phi_\Gamma(\lambda) = \psi(\frac{1}{4}+\frac{\lambda}{2}) - \psi(\frac{1}{4}) = \int_0^\infty (1-e^{-\lambda y})\,\nu(dy)$ *is a Bernstein function. Its boundary values on the imaginary axis are*

$$\operatorname{Re}\Phi_\Gamma(i\tau) = 2\big(\theta'(\tau) - \theta'(0)\big) = \int_0^\infty (1-\cos\tau y)\,\nu(dy),$$

$$\operatorname{Im}\Phi_\Gamma(i\tau) = -\frac{d}{d\tau}\log p(\tau) = \int_0^\infty \sin(\tau y)\,\nu(dy).$$

*Consequently* $2\theta(\tau) + \tau\log\pi$ *and* $\log p(\tau)$ *are, up to constants, the imaginary and real parts of* $2\log\Gamma(\frac{1}{4}+\frac{i\tau}{2})$*, the boundary value of a function analytic in the right half plane. So* $2\theta(\tau)+\tau\log\pi$*, which is* $2\pi$ *times the smooth part* $\theta/\pi$ *of the zero counting function plus a linear term, is the harmonic conjugate of the Meixner log density. The smooth heat trace is the heat trace of* $\nu$*:*

$$W_0(x) = \frac{\theta'(0)}{2\sqrt{\pi x}} + \frac{1}{4\sqrt{\pi x}}\int_0^\infty \big(1-e^{-y^2/4x}\big)\,\nu(dy),$$

*which is Lemma 6.1.12 after the substitution* $u = 2y$*; the function* $F$ *of that lemma decreases because* $\nu$ *is positive.*

*Proof.* Substitute $u = 2y$ in $\psi(z) = \int_0^\infty (e^{-u}/u - e^{-zu}/(1-e^{-u}))du$ with $z = \frac{1}{4}+\frac{\lambda}{2}$; note $2e^{-y/2}/(1-e^{-2y}) = e^{y/2}/\sinh y$. Taking $\lambda = i\tau$ gives the boundary values, with $\theta' = \frac{1}{2}\operatorname{Re}\psi(\frac{1}{4}+\frac{i\tau}{2}) - \frac{1}{2}\log\pi$ and $\frac{d}{d\tau}\log|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = -\operatorname{Im}\psi(\frac{1}{4}+\frac{i\tau}{2})$. For $W_0$ integrate $\theta'(\tau) = \theta'(0) + \frac{1}{2}\int(1-\cos\tau y)\nu(dy)$ against $\frac{1}{\pi}e^{-x\tau^2}$. □

THEOREM 6.2.12. *Let* $c = -\theta'(0) = \frac{1}{2}(\log\pi - \psi(\frac{1}{4})) \approx 2.68609$ *be the constant of* (6.1.6)*. Then*

$$\frac{e^{-c\sqrt{s}}}{G(s)} = \frac{\Gamma(\frac{1}{4})\,\pi^{\sqrt{s}/2}\,e^{-c\sqrt{s}}}{\Gamma(\frac{1}{4}+\frac{\sqrt{s}}{2})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U_\Gamma(t)\,dt\Big),$$

*with*

$$U_\Gamma(t) = \frac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}} = \frac{1}{2\pi\sqrt{t}} \sum_{k\geq 0} \frac{t}{c_k\,(c_k^2 + t)} \geq 0.$$

*So $e^{-c\sqrt{s}}/G(s)$ is the Laplace transform of a GGC, and its Thorin distribution function is $(\theta(\sqrt{t}) - \theta'(0)\sqrt{t})/\pi$, a convex increasing function of $\sqrt{t}$.*

*Proof.* With $f(u) = \Gamma(\frac{1}{4} + \frac{u}{2})\pi^{-u/2}$ we have $f'(0)/f(0) = \frac{1}{2}\psi(\frac{1}{4}) - \frac{1}{2}\log\pi = \theta'(0) = -c$, and $\frac{d}{du}[\log f(u) - uf'(0)/f(0)] = \frac{1}{2}\Phi_\Gamma(u) = \frac{1}{2}\int(1 - e^{-uy})\nu(dy)$. Integrating from 0, $\log f(u)/f(0) - uf'(0)/f(0) = \int(e^{-uy} - 1 + uy)\frac{\tilde{\mu}(dy)}{y}$ with $\tilde{\mu} = \frac{1}{2}\nu \geq 0$. Lemma 6.2.5 gives $U_\Gamma(t) = \frac{1}{2\pi\sqrt{t}}\int 2\sin^2(\frac{y\sqrt{t}}{2})\frac{1}{2}\nu(dy) = \frac{1}{2\pi\sqrt{t}}\sum_k \int_0^\infty (1 - \cos y\sqrt{t})e^{-c_k y}dy$, which is the series stated, and Proposition 6.2.11 identifies it with $\theta'$. □

Multiplying $1/G$ by $e^{-c\sqrt{s}}$ adds the Thorin density $\frac{c}{2\pi\sqrt{t}}$, and $U_\Gamma = -U_G + \frac{c}{2\pi\sqrt{t}}$, which is the identity $c - b = \sum_k(\frac{1}{c_k} - \frac{1}{2k+2})$, has no negative part. This is the positive counterpart of Theorem 6.2.1.

### *6.2.3.1 Meixner Pollaczek polynomials and the local Riemann Hypothesis*

Let $\psi_{2m}(x) = H_{2m}(\sqrt{2\pi}\,x)e^{-\pi x^2}$ be the even Hermite functions, eigenfunctions of the Fourier transform $\hat{f}(y) = \int f(x)e^{-2\pi ixy}dx$ with eigenvalue $(-1)^m$. Define the Mellin transform $\mathcal{M}f(s) = \int_0^\infty x^{s-1}f(x)dx$ and $\Gamma_\mathbb{R}(s) = \pi^{-s/2}\Gamma(s/2)$. From here to the end of Section 6.2.3, $s$ is the Mellin variable, the argument of $\zeta$, and not the variable of the clock.

Theorem 6.2.13. *For each $m \geq 0$, $Q_m(s) = \mathcal{M}\psi_{2m}(s)/\Gamma_\mathbb{R}(s)$ is a polynomial of degree $m$ satisfying $Q_m(1-s) = (-1)^m Q_m(s)$, and*

$$Q_m(\tfrac{1}{2} + i\tau) = \kappa_m\, P_m^{(1/4)}\big(\tfrac{\tau}{2}; \tfrac{\pi}{2}\big),$$

*where $P_m^{(\lambda)}(x;\phi)$ are the Meixner Pollaczek polynomials and $\kappa_m \neq 0$. These are the orthogonal polynomials of the Meixner law $p$. Consequently all zeros of $Q_m$ lie on $\operatorname{Re} s = \frac{1}{2}$ and are simple: the Mellin transform of every even Hermite function satisfies the Riemann Hypothesis.*

*Proof.* Expanding $H_{2m}(\sqrt{2\pi}\,x) = \sum_j h_j x^{2j}$ gives

$$\mathcal{M}\psi_{2m}(s) = \frac{1}{2}\Gamma_\mathbb{R}(s)\sum_j h_j \pi^{-j}(s/2)_j$$

with Pochhammer symbols, a polynomial of degree $m$ times $\Gamma_\mathbb{R}(s)$. Tate's local functional equation [492, Theorem 2.4.1 and Section 2.5], $\mathcal{M}f(s)/\Gamma_\mathbb{R}(s) = \mathcal{M}\hat{f}(1-$

$s)/\Gamma_{\mathbb{R}}(1-s)$ for even $f$, applied to the eigenfunction $\psi_{2m}$ gives $Q_m(1-s) = (-1)^m Q_m(s)$, so $i^{-m}Q_m(\frac{1}{2}+i\tau)$ is a real polynomial $R_m(\tau)$. By Mellin Plancherel,

$$\int_0^\infty \psi_{2m}\psi_{2n}\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} Q_m\overline{Q_n}\,|\Gamma_{\mathbb{R}}(\tfrac{1}{2}+i\tau)|^2 d\tau = \sqrt{2}\,i^{m-n}\int_{\mathbb{R}} R_m(\tau)R_n(\tau)\,p(\tau)\,d\tau,$$

since $|\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)|^2 = \pi^{-1/2}|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = 2^{3/2}\pi\,p(\tau)$. The left side vanishes for $m \neq n$, so $(R_m)$ is the orthogonal polynomial sequence of $p$, which with $x = \tau/2$ is the Meixner Pollaczek family with $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$ [312, Section 9.7]. Orthogonal polynomials of a measure with infinite support have real simple zeros. □

The conclusion is the archimedean local Riemann Hypothesis of Bump, Choi, Kurlberg and Vaaler [67], seen through the Meixner law. We make no priority claim for this route; Meixner Pollaczek expansions of $\Xi$ in the literature are reviewed in Section 6.2.3.2.

### *6.2.3.2 Zeta in Meixner coordinates*

Kuznetsov [318, 319] expands $\Xi$ and Dirichlet $L$ functions in Meixner Pollaczek polynomials; Inoue [286] studies the symmetric expansion of the completed zeta function, the multiplicative structure of its coefficients and the zeros of the approximating polynomials; Romik [464] expands $\Xi$ itself in the symmetric Meixner Pollaczek basis $P_n^{(3/4)}(x;\frac{\pi}{2})$, together with the Hermite and continuous Hahn bases, and derives coefficient asymptotics; and Candelpergher [70] gives related polynomial expansions whose polynomials have their roots on the line. The expansions below are included for comparison with the Thorin picture and are not claimed as new.

The map $\mathcal{U}f(\tau) = \mathcal{M}f(\frac{1}{2}+i\tau)/\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)$ is, by the Mellin Plancherel identity in the proof of Theorem 6.2.13, a unitary map from $L^2(0,\infty)$ onto $L^2(\sqrt{2}\,p(\tau)d\tau)$, carrying the even Hermite functions to the Meixner Pollaczek polynomials. Dilations become multipliers, $\mathcal{U}[f(n\,\cdot)] = n^{-1/2-i\tau}\mathcal{U}f$. The function $g(x) = (2\pi^2x^4 - 3\pi x^2)e^{-\pi x^2} = \frac{1}{2}(xD)(xD+1)e^{-\pi x^2}$, with $D = d/dx$, has $\mathcal{U}g = \frac{1}{4}s(s-1)$, a Meixner Pollaczek combination of degree two, and its dilation sum $\Omega$ is, up to the factor $\frac{1}{2}x^{-1/2}$, the Pólya kernel $\Phi$ of Section 1.2 in the variable $x = e^u$:

$$\Omega(x) = \sum_{n\geq 1} g(nx) = \tfrac{1}{2}x^{-1/2}\Phi(\log x), \qquad \mathcal{U}\Omega = \tfrac{1}{4}s(s-1)\zeta(s)\big|_{s=\frac{1}{2}+i\tau}.$$

So in Meixner coordinates $\frac{1}{4}s(s-1)\zeta(s)$ on the critical line is the image of a function in $L^2(0,\infty)$ and has a convergent Meixner Pollaczek expansion

$$\tfrac{1}{4}s(s-1)\zeta(s) = \sum_{m\geq 0} a_{2m}\,Q_m(s), \qquad a_{2m} = \frac{\langle \Omega, \psi_{2m}\rangle}{\langle \psi_{2m}, \psi_{2m}\rangle},$$

in $L^2(p)$. Each $Q_m$ satisfies the local Riemann Hypothesis. We ask whether the property survives in the partial sums.

The coefficients decay faster than geometrically, but the polynomials grow, and the expansion converges slowly as $|\tau|$ grows: near the first zero $\frac{1}{2} + 14.1347i$ the partial sums are still far from zero at $N = 24$. The partial sums $\sum_{m \le N} a_{2m} Q_m(s)$ mix polynomials of both parities under $s \mapsto 1-s$, so they have no reason to vanish on the critical line, and for $N = 12$ and $N = 24$ they do not: all their zeros lie off the line. The asymmetric expansion therefore does not test whether the local Riemann Hypothesis survives the superposition; the symmetric expansion does.

The symmetric choice is the real even function $F(\tau) = (\tau^2 + \frac{1}{4})Z(\tau)$, proportional to $\Xi(\tau)/|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|$, expanded in the even Meixner Pollaczek polynomials $P_{2m}^{(1/4)}(\tau/2; \frac{\pi}{2})$, normalised as in [312, Section 9.7]. Its truncations, the orthogonal projections of $F$ in $L^2(p)$ onto the even polynomials of degree at most $2M$, are real even polynomials. The coefficients alternate in sign from $m = 3$ on, and from $m = 4$ on their modulus decreases slowly, to about 0.01 at $m = 20$. The slow decay reflects the polynomial growth of $F$ against the exponential decay of $p$: the expansion converges in $L^2(p)$, but its truncations approximate $F$ poorly at the ordinates. The real zeros of the truncations do not track the ordinates: the zero nearest 14.13 is 14.50, 14.32, 14.16, 13.73, 13.54, 14.84 for the degrees 38, 40, 42, 50, 60, 80, and there are eight to ten real zeros in $[0, 40]$ against six ordinates. Neither expansion suggests a route from the local Riemann Hypothesis to the global one; such a route must use the arithmetic of the full dilation sum, not only the orthogonality of its archimedean pieces.

The factorisation of $\Xi^2$ in Section 6.2.1 induces a factorisation of the autocorrelation $\Phi * \Phi$ of the Pólya kernel, whose Fourier transform is $\Xi^2$.

PROPOSITION 6.2.14. *The integral* $\mathcal{C}(v) = \int p(\tau)\,|\zeta(\frac{1}{2} + i\tau)|^2 e^{iv\tau}\,d\tau$ *converges absolutely, and*

$$(\Phi * \Phi)(v) = 2^{-3/2}\Big(\frac{d^2}{dv^2} - \frac{1}{4}\Big)^2 \mathcal{C}(v).$$

*Formally* $\mathcal{C} = \operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$, *with*

$$\mathcal{A} = \sum_{m,n \ge 1} \frac{\delta_{\log(m/n)}}{\sqrt{mn}},$$

*where* $\operatorname{sech}(v)^{1/2}$ *is the characteristic function of the Meixner law and* $\mathcal{A}$, *whose Fourier transform is formally* $|\zeta(\frac{1}{2} + i\tau)|^2$, *is the autocorrelation of the measure* $\sum_n n^{-1/2}\delta_{\log n}$. *The measure* $\mathcal{A}$ *is positive and positive definite, and the series for* $\mathcal{C}$ *diverges: truncated at* $m, n \le X$ *it equals* $X \int \operatorname{sech}(v-r)^{1/2} e^{-|r|/2}\,dr + O(X^{1/2})$. *This growth comes from the pole of* $\zeta$, *and the factor* $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ *does not remove it, since* $(\frac{d^2}{dv^2} - \frac{1}{4})e^{-|v|/2} = -\delta_0$.

*Proof.* Since $\Phi$ is even, $\int(\Phi * \Phi)(v)e^{i\tau v}dv = \Xi(\tau)^2 = \frac{1}{4}(\tau^2 + \frac{1}{4})^2\pi^{-1/2}|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2|\zeta(\frac{1}{2} + i\tau)|^2$, and $\pi^{-1/2}|\Gamma|^2 = 2^{3/2}\pi\, p$. Fourier inversion divides by $2\pi$, the factor $(\tau^2 + \frac{1}{4})^2$ becomes $(\frac{d^2}{dv^2} - \frac{1}{4})^2$, and the inverse transform of the even function $p|\zeta|^2$ is $\mathcal{C}/(2\pi)$; the constant is $\frac{1}{4}\,2^{3/2}\pi/(2\pi) = 2^{-3/2}$. The integral $\mathcal{C}$ converges absolutely because $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2}e^{-\pi|\tau|/2}$ (Proposition 6.2.10) and $\zeta(\frac{1}{2} + i\tau)$ grows at most polynomially. Formally $|\zeta|^2 = |\sum n^{-1/2-i\tau}|^2 = \sum_{m,n}(mn)^{-1/2}e^{i\tau\log(m/n)}$, and $\int p(\tau)e^{i\tau(v+r)}d\tau = \operatorname{sech}(v + r)^{1/2}$, which gives $\mathcal{C} = \operatorname{sech}^{1/2} * \mathcal{A}$. For the truncation put $D_X(\tau) = \sum_{n\le X} n^{-1/2-i\tau}$; the truncated series is exactly $\int p(\tau)|D_X(\tau)|^2e^{iv\tau}d\tau$. For $|\tau| \le X$, $D_X(\tau) = \zeta(\frac{1}{2} + i\tau) + X^{1/2-i\tau}/(\frac{1}{2} - i\tau) + O(X^{-1/2})$ [498, Theorem 4.11], and for $|\tau| > X$ the bound $|D_X|^2 \le 4X$ meets the exponential decay of $p$. Since $\int e^{-|r|/2}e^{i\tau r}\,dr = (\frac{1}{4} + \tau^2)^{-1}$, the squared pole term contributes $X\int p(\tau)(\frac{1}{4} + \tau^2)^{-1}e^{iv\tau}d\tau = X\int \operatorname{sech}(v - r)^{1/2}e^{-|r|/2}\,dr$, up to an exponentially small error from $|\tau| > X$; the remaining terms contribute $O(X^{1/2})$ uniformly in $v$. Finally $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ turns $X\int \operatorname{sech}(v - r)^{1/2}e^{-|r|/2}\,dr$ into $-X(\frac{d^2}{dv^2} - \frac{1}{4})\operatorname{sech}(v)^{1/2}$, which is not zero. □

Both factors of $\mathcal{C}$ are positive, the Meixner factor being a GGC transform in $v^2$ and $\mathcal{A}$ a positive measure, but convolution with a positive measure does not preserve the GGC property, so the GGC structure of the Meixner factor does not pass to the product. The lattice sum also needs a counterterm, the pole term of the approximate functional equation, which comes from the same pole of $\zeta$ that produced the atom at $-\frac{1}{4}$ in Proposition 6.1.13. Problem 7.6 of Section 7.2 asks for the regularised form of $\operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$ and whether it has a Thorin or Lee and Yang structure in $v$.

### 6.2.4 An infinitely divisible floor for the reciprocal Xi function

Multiplying $\varphi = RG$ by the GGC transform $e^{-c\sqrt{s}}/G$ of Theorem 6.2.12 gives $e^{-c\sqrt{s}}R$, which we call the floor of the reciprocal Xi function.

Theorem 6.2.15. *Unconditionally, $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density*

$$\frac{W(x) + c/(2\sqrt{\pi x})}{x}, \qquad W(x) = \tfrac{1}{2}\sum_{\rho} e^{x(\rho-\frac{1}{2})^2},$$

*and $W(x) + c/(2\sqrt{\pi x}) > 0$ for all $x > 0$. The following are equivalent: (i) RH; (ii) this law is a GGC; (iii) $W(x) + c/(2\sqrt{\pi x})$ is completely monotone; (iv) $W$ is completely monotone. The proof uses $w > 0$, which is Theorem 7.1.1 of Section 7.1 and rests on the verification of RH to height 50, an explicit bound for $N(T)$ [176] and the enclosures E1 to E3.*

*Proof.* $e^{-c\sqrt{s}}R = \varphi \cdot (e^{-c\sqrt{s}}/G)$ is the product of the transform of an infinitely divisible law (Theorem 7.1.1) and a GGC (Theorem 6.2.12), so it is infinitely divisible, and its Lévy density is the sum $\{w(x) + \int e^{-xt}U_\Gamma(t)dt\}/x$. By Proposition 6.2.2 and Theorem 6.2.12, $\int e^{-xt}U_\Gamma(t)dt = W_0(x) + \frac{c}{2\sqrt{\pi x}}$, and $w + W_0 = W$ by (6.1.7). Positivity holds because $w > 0$ and $U_\Gamma \geq 0$. For the equivalences: GGC means the Lévy density times $x$ is completely monotone, so (ii)⇔(iii). Under RH $W = \sum e^{-x\gamma^2}$ is completely monotone, so (i)⇒(iv)⇒(iii). If (ii) holds, the transform is analytic on $\mathbb{C} \setminus (-\infty, 0]$ and agrees with $e^{-c\sqrt{s}}R$ on $(0, \infty)$, so by uniqueness of analytic continuation $R$ has no poles in $\mathbb{C} \setminus (-\infty, 0]$; since $R$ has no zeros and $G$ is analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, $\varphi = RG$ is analytic and zero free there, and Proposition 6.1.9 gives RH. □

Infinite divisibility alone is not new: by Theorem 5.1.1 [424, Theorem A], $W > 0$ and $W' < 0$ on $(0, \infty)$, so $R$ itself is infinitely divisible and self decomposable, and so is its product with the stable factor $e^{-c\sqrt{s}}$; Section 11.1 draws the consequences of self decomposability. What the floor adds is the splitting of its canonical function $k = W + c/(2\sqrt{\pi x})$ as $k = w + \int e^{-xt}U_\Gamma(t)\,dt$, an arithmetic part plus a completely monotone archimedean part contributed by the Meixner law (Theorem 6.2.12). In law the floor adds to $X_R$ an independent first passage time $L_c$ of $\sqrt{2}B$ to the level $c$. Everything that could fail therefore lies in $w$, and RH is the upgrade of $k$ from positivity, the lowest rung of the ladder of Section 5.1, to complete monotonicity, its top.

### *6.2.4.1 The sine squared lemma at the floor*

As in the proof of Lemma 6.2.5, from here to the end of the section $x$ is the variable of the primal Lévy measure, and the time of the heat trace is written $y$.

The floor is $e^{-c\sqrt{s}}R(s) = f(0)/f(\sqrt{s})$ with primal function $f(u) = e^{cu}\xi(\frac{1}{2} + u)$, and $f'(0)/f(0) = c$ because $\xi'(\frac{1}{2}) = 0$. The tilt contributes the linear term $cu$ to $\log f$, which cancels in $\log f(u)/f(0) - uf'(0)/f(0) = \log \xi(\frac{1}{2} + u)/\xi(\frac{1}{2})$, so the primal Lévy measure of the floor is that of $\xi$ at the centre. Collecting the three factors of $\xi(\frac{1}{2} + u) = \frac{1}{2}(u^2 - \frac{1}{4})\,\pi^{-1/4} f_\Gamma(u)\,\zeta(\frac{1}{2} + u)$, where $f_\Gamma(u) = \pi^{-u/2}\Gamma(\frac{1}{4} + \frac{u}{2})$, gives formally

$$\tilde{\mu}_\xi(dx) = -2\cosh\left(\tfrac{x}{2}\right) dx + \sum_{k\geq 0} e^{-c_k x}\,dx + \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\,\delta_{\log n}(dx). \tag{6.2.5}$$

The first term comes from $u^2 - \frac{1}{4} = (u + \frac{1}{2})(u - \frac{1}{2})$, each linear factor $1 + u/a$ contributing $-e^{-ax}dx$ by Frullani (with $a = -\frac{1}{2}$ by continuation); the second is $\frac{1}{2}\nu$ (Proposition 6.2.11 and the proof of Theorem 6.2.12); the third is the prime measure of Proposition 6.2.6 at $\beta = \frac{1}{2}$. By the explicit formula, as distributions on $x > 0$,

$$\tilde{\mu}_\xi(dx) = -\sum_\rho e^{(\rho - \frac{1}{2})x}\,dx, \qquad \text{which under RH is} \qquad -2\sum_{\gamma > 0} \cos(\gamma x)\,dx. \tag{6.2.6}$$

PROPOSITION 6.2.16. *Apply the two stages of Lemma 6.2.5 to $\tilde{\mu}_\xi$, regularising $\int_0^\infty \cos(\gamma x)dx = 0$ and the growing term of (6.2.5) by analytic continuation.*

*(i) Subordination stage: $(4\pi y)^{-1/2}\int_0^\infty (1-e^{-x^2/4y})\,\tilde{\mu}_\xi(dx) = W(y)$, the heat trace over the zeros. With the tilt, the canonical function of the floor is $k(y) = W(y) + c/(2\sqrt{\pi y})$, as in Theorem 6.2.15, and its positivity is proved.*

*(ii) Bernstein stage: $U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2(\frac{x\sqrt{t}}{2})\,\tilde{\mu}_\xi(dx)$ equals $\sum_{\gamma>0}\delta_{\gamma^2}$ under RH, and the Thorin measure of the floor is $U + \frac{c}{2\pi\sqrt{t}}dt$.*

*The two stages are linked by $W(y) = \int_0^\infty e^{-yt}U(t)\,dt$. The floor is a GGC exactly when $U + \frac{c}{2\pi\sqrt{t}}\,dt \geq 0$, that is, when $k$ is completely monotone; by Theorem 6.2.15 this happens exactly under RH, and then $U = \sum_{\gamma>0}\delta_{\gamma^2} \geq 0$.*

*Proof.* (i) For $\tilde{\mu} = -2\cos(\gamma x)dx$,

$$\int_0^\infty (1-e^{-x^2/4y})(-2\cos\gamma x)\,dx = 2\int_0^\infty e^{-x^2/4y}\cos\gamma x\,dx = \sqrt{4\pi y}\,e^{-\gamma^2 y}.$$

Without RH the same computation with $e^{(\rho-\frac{1}{2})x}$ gives $e^{y(\rho-\frac{1}{2})^2}$, so the result is $W$ in general. (ii) $\int_0^\infty (1-\cos x\sqrt{t})(-2\cos\gamma x)\,dx = \pi\,\delta(\sqrt{t}-\gamma)$, and $\frac{1}{2\pi\sqrt{t}}\pi\,\delta(\sqrt{t}-\gamma) = \delta(t-\gamma^2)$. The link is the identity $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos x\sqrt{t}\}\frac{dt}{\sqrt{\pi t}}$ used in the proof of Lemma 6.2.5. □

The lemma preserves positivity, so if $\tilde{\mu}_\xi$ were a positive Lévy measure the floor would be a GGC and RH would follow. But $\tilde{\mu}_\xi$ is not positive. Its pole part $-2\cosh(x/2)\,dx$ is negative, and at large $x$ it cancels the mean density $e^{x/2}dx$ of the prime atoms, leaving the oscillating sum over zeros (6.2.6). At the centre the lemma turns the question into one about a signed input whose sine squared image is positive if and only if RH holds.

### *6.2.4.2 Complete monotonicity to finite order*

Positivity at stage (i) of Proposition 6.2.16 is Theorem 6.2.15; positivity at stage (ii) is complete monotonicity of $k(y) = W(y) + c/(2\sqrt{\pi y})$, that is $(-1)^n k^{(n)} \geq 0$ for every $n$.

THEOREM 6.2.17. *Let $k(y) = W(y) + c/(2\sqrt{\pi y})$. For every $n \leq 10^{10}$ and every $y > 0$,*

$$(-1)^n\,k^{(n)}(y) > 0.$$

*Consequently, by Williamson's theorem, for each $1 \leq n \leq 10^{10}$ there is a positive measure $\Omega_n$ on $(0,\infty)$ with $k(y) = \int (1-ty)_+^{n-1}\,\Omega_n(dt)$. For $n \geq 1$ the proof uses the verification of RH to height $T_0 = 3\times 10^{12}$ [419], an explicit bound for $N(T)$ [176] and one interval enclosure (E4, Section 7.2.3), and no other information about zeros.*

*Proof.* The case $n = 0$ is the positivity in Theorem 6.2.15. Fix $1 \le n \le 10^{10}$ and put $y_n = 0.05/n$.

Small $y$. Write $k = w + (W_0 + \frac{c}{2\sqrt{\pi y}})$. By Theorem 6.2.12 the bracket is $\int e^{-ty} U_\Gamma(t)\,dt$ with $U_\Gamma \ge 0$, so $(-1)^n$ times its $n$th derivative is $L_n(y) = \int t^n e^{-ty} U_\Gamma(t) dt$, decreasing in $y$. Since $\theta'(\tau) - \theta'(0) \ge \frac{2\tau^2}{\frac14+\tau^2} \ge 1$ for $\tau \ge \frac12$ (the $k = 0$ term of the series in Theorem 6.2.12),

$$L_n(y) = \frac{1}{\pi}\int_0^\infty \tau^{2n} e^{-y\tau^2}\big(\theta'(\tau) - \theta'(0)\big)d\tau \ge \frac{\Gamma(n+\frac12)}{2\pi}\, y^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)}.$$

On the other side $(-1)^n w^{(n)} = (-1)^n 4^{-n} e^{y/4} - (-1)^n P^{(n)}$. $P$ is analytic in $\operatorname{Re} y > 0$, and on the circle $|z - y| = y/2$ one has $|z| \ge y/2$ and $\operatorname{Re}(1/z) \ge 2/(3y)$, so Cauchy's estimate gives

$$\begin{aligned} |P^{(n)}(y)| &\le n!\,(2/y)^n\,(2\pi y)^{-1/2} \sum_m \Lambda(m) m^{-1/2} e^{-(\log m)^2/6y} \\ &\le C_0\, n!\,(2/y)^n\,(2\pi y)^{-1/2} e^{-(\log 2)^2/6y}, \end{aligned}$$

where $C_0 = 0.54993$ bounds $e^{(\log 2)^2/6y}\sum_m \Lambda(m) m^{-1/2} e^{-(\log m)^2/6y}$ for every $y \le 0.05$, each term being increasing in $y$ (enclosure E4). Divide by the leading term of the lower bound for $L_n$ and use Gautschi's inequality $n!/\Gamma(n+\frac12) < \sqrt{n+1}$ [397, (5.6.4)]. For $y \le y_n$ we have $(\log 2)^2/6y \ge 1.6015\,n$, so the ratio is at most

$$r_n = C_0\sqrt{2\pi}\sqrt{n+1}\, e^{(\log 2 - 1.6015)n},$$

which decreases in $n$, since $r_{n+1}/r_n \le \sqrt{3/2}\, e^{\log 2 - 1.6015} < \frac12$, and satisfies $r_n \le r_1 < 0.786$ (enclosure E4). Hence, for $y \le y_n$,

$$(-1)^n k^{(n)}(y) \ge (1 - r_n)\frac{\Gamma(n+\frac12)}{2\pi} y_n^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)} - 4^{-n} e^{y/4}.$$

The first term is at least $0.214\,\Gamma(\frac32)\,20^{3/2}/(2\pi) > 2.6$, since $\Gamma(n+\frac12)(20n)^{n+\frac12}$ increases in $n$, and the two negative terms total less than 0.27. So $(-1)^n k^{(n)} > 0$ on $(0, y_n]$ for every $n \ge 1$.

Large $y$. Here $(-1)^n k^{(n)} = (-1)^n W^{(n)} + \frac{c\,\Gamma(n+\frac12)}{2\pi} y^{-n-\frac12}$. Zeros with $0 < \gamma \le T_0$ lie on the line and contribute $\gamma^{2n} e^{-y\gamma^2} > 0$. Any other zero has $\gamma > T_0$ and contributes at most $(\gamma+1)^{2n} e^{-y(\gamma^2 - \frac14)}$ in modulus. Let $y \ge y_n$. Since $n/y \le 20n^2 < T_0^2$, the function $(\gamma+1)^{2n} e^{-y\gamma^2}$ decreases for $\gamma \ge T_0$, so by $N(t+1) - N(t) \le 2\log(t+1)$, which follows from [176] as in the proof of Theorem 7.1.1, the zeros with $T_0 + j < \gamma \le T_0 + j + 1$ contribute at most $b_j = 2\log(T_0 + j + 1)\,(T_0 + j + 1)^{2n} e^{-y((T_0+j)^2 - \frac14)}$. For $n \le 10^{10}$ we have $b_{j+1}/b_j \le \exp\big(\frac{2n}{T_0} + \frac{1}{T_0 \log T_0} - \frac{2T_0+1}{20n}\big) < 10^{-13}$, so the tail is at most $2b_0 = 4\log(T_0+1)\,(T_0+1)^{2n} e^{-y(T_0^2 - \frac14)}$. Its ratio to

the positive term $\frac{c\,\Gamma(n+\frac{1}{2})}{2\pi} y^{-n-\frac{1}{2}}$ is a constant multiple of $y^{n+\frac{1}{2}} e^{-y(T_0^2-\frac{1}{4})}$, which decreases for $y \geq y_n$ because $T_0^2 - \frac{1}{4} > 20n(n+\frac{1}{2})$. At $y = y_n$, using $\Gamma(n+\frac{1}{2}) \geq \Gamma(\frac{3}{2})$ and $y_n < 1$, the logarithm of the ratio is at most

$$\log \frac{8\pi \log(T_0+1)}{c\,\Gamma(\frac{3}{2})} + 2n \log(T_0+1) - \frac{T_0^2 - \frac{1}{4}}{20n} < -4.4 \times 10^{13},$$

because the left side increases with $n$, and at $n = 10^{10}$ its first term is below 6 while $y_n(T_0^2 - \frac{1}{4}) \approx 4.5 \times 10^{13}$ exceeds $2n \log(T_0+1) < 5.8 \times 10^{11}$ by a wide margin. So the tail is smaller than the positive term.

Since $k(\infty) = 0$, Williamson's theorem [519] gives $\Omega_n$. □

The large $y$ branch covers $y > 0.05/n$, and there the bound of the proof lets the positive term $c\,\Gamma(n+\frac{1}{2}) y^{-n-\frac{1}{2}}/(2\pi)$ contributed by the tilt dominate the zeros above $T_0$ whenever $y_n(T_0^2 - \frac{1}{4}) > 2n \log(T_0+1) + 6$. This sufficient condition holds while $n < 8.8 \times 10^{10}$, and we state the safe round number $10^{10}$.

Section 5.1.6 compares the reach of Theorem 6.2.17 with that of Theorems 5.1.6 and 5.1.13.

# Chapter 7

# Infinite divisibility and truncations of the reciprocal zeta function

Section 7.1 proves with no hypothesis the positivity of the canonical function $w$ of Theorem 6.1.11, which the proof of Theorem 6.2.15 used, so that $\varphi$ itself is infinitely divisible, and classifies $\varphi$ in the Bondesson hierarchy: infinitely divisible, but not self decomposable, not in the Bondesson class and not a generalised gamma convolution.

Section 7.2 studies the truncations of this lattice. One sided truncations fail at zeros where $|S|$ is large, while symmetrised truncations keep their zeros in the strip on the line in every window computed. Conjecture S asserts this for every size; for small sizes it reduces to a finite computation, and for infinitely many sizes it would imply RH. The coordinates of Chapter 6, declared in Section 6.1.1, hold in this chapter too: in the variable $s$ of the clock a zero $\frac{1}{2} + i\gamma$ sits at $s = -\gamma^2$ on the cut.

## 7.1 Infinite divisibility, necklaces and solved models

By Theorem 6.1.11, $-\varphi'/\varphi$ is the Laplace transform of the canonical function $w = e^{x/4} - P = W - W_0$.

### 7.1.1 Infinite divisibility and the class of $\varphi$

Whether $\varphi$ is infinitely divisible is a question about the sign of $w$, since by Theorem 6.1.11 the candidate Lévy density of $\varphi$ is $w(x)/x$. Recall that $w(0^+) = 1$ and $w(x) \sim c/(2\sqrt{\pi x})$ as $x \to \infty$, where $c = -\theta'(0) \approx 2.68609$ is the constant of (6.1.6). The proof uses the verification of RH to height 50 [15, 419], the explicit Riemann von Mangoldt bound of Hasanalizade, Shen and Wong [176], and three interval enclosures, $F(0.004) < -0.111$ for the function $F$ of Lemma 6.1.12, $P(0.004) < 2 \times 10^{-13}$ and $E(0.004) < 0.0011$ for the sum $E$ defined in the proof, computed by `verification_reciprocal_zeta/enclosures.py` (enclosures E1 to E3).

THEOREM 7.1.1. *$w(x) > 0$ for every $x > 0$. Consequently, unconditionally, $\varphi$ is the Laplace transform of an infinitely divisible probability law on $(0,\infty)$ with zero drift and Lévy density $w(x)/x$:*

$$\frac{\zeta(\frac12)}{(1-4s)\,\zeta(\frac12+\sqrt{s})} = \exp\Big(-\int_0^\infty \big(1-e^{-sx}\big)\frac{w(x)}{x}\,dx\Big), \qquad \operatorname{Re} s > 0.$$

*Proof.* Fix $x_1 = 0.004$.

Small $x$. Each term $x^{-1/2}e^{-(\log n)^2/4x}$ increases in $x$ while $(\log n)^2/4x > \frac12$, which holds for $x \le x_1$ and $n \ge 2$. Hence $P(x) \le P(x_1) < 2\times 10^{-13}$ (enclosure E2: the term $n=2$ in interval arithmetic, and $n \ge 3$ through $(\log n)^2/4x_1 \ge \frac{\log 3}{4x_1}\log n$), and $w(x) = e^{x/4} - P(x) > 0$ on $(0, x_1]$.

Large $x$. By Lemma 6.1.12 and $F(x_1) < 0$, for $x \ge x_1$ we have $-W_0(x) \ge |F(x_1)|/(4\sqrt{\pi x})$. Here $F(x_1) < -0.111$ by enclosure E1. In $W$ the zeros with $0 < \gamma \le 50$ lie on the line [15] and contribute $e^{-x\gamma^2} > 0$. Any other zero has $\gamma > 50$ and contributes at most $e^{-x(\gamma^2-1/4)}$ in modulus. The explicit Riemann von Mangoldt formula $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78| \le E(T)$ for $T \ge e$, with $E(T) = 2.5167 + \frac{1}{50T}$ for $T \le X_0 = 30\,610\,046\,000$ and $E(T) = 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T} < 0.3\log T$ for $T > X_0$ [176, (1.7), (2.7) and Corollary 1.4], gives $N(t+1) - N(t) \le 2\log(t+1)$ for $t \ge 50$: the main term grows by at most $\frac{1}{2\pi}\log\frac{t+1}{2\pi}$, and while $t+1 \le X_0$ the two error terms add at most $5.0334 + \frac{1}{25t}$, together less than 5.37 at $t = 50$ against $2\log 51 = 7.86$, with a gap that widens as $t$ grows, since its derivative exceeds $(2 - \frac{1}{2\pi})/(t+1)$; for $t+1 > X_0$ the error terms add at most $0.6\log(t+1)$, and $\frac{1}{2\pi} + 0.6 < 2$. So all such zeros together contribute at most

$$E(x) = \sum_{j\ge 0} 2\log(51+j)\, e^{-x((50+j)^2 - \frac14)}.$$

Enclosure E3 gives $E(x_1) < 0.0011$, against $|F(x_1)|/(4\sqrt{\pi x_1}) > 0.247$. Since $(50+j)^2 - \frac14 > 1/(2x_1)$ for every $j$, $\sqrt{x}\,E(x)$ decreases on $[x_1,\infty)$, so $E(x) < |F(x_1)|/(4\sqrt{\pi x}) \le -W_0(x)$ for all $x \ge x_1$, and $w = W - W_0 > 0$ on $[x_1,\infty)$.

Representation. By Theorem 6.1.11, $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)dx$ for $s > \frac14$. Since $w \ge 0$, Landau's theorem applies [514, Chapter II, Theorem 5b]: if the Laplace transform of a nonnegative function has abscissa of convergence $\sigma_c$, then the transform cannot be continued analytically to a neighbourhood of $\sigma_c$. Here $-\varphi'/\varphi$ is analytic near every point of $(0,\infty)$ by Proposition 6.1.9. Hence $\sigma_c \le 0$, the integral converges for $\operatorname{Re} s > 0$, and equals the continuation of $-\varphi'/\varphi$ there. Integrating over $[a,s]$ for real $0 < a < s$ with Tonelli, and letting $a \downarrow 0$ with $\varphi(a) \to 1$ and monotone convergence, gives the formula for real $s > 0$; then $\int \min(1,x)\, w(x)x^{-1}dx < \infty$ because the left side is finite, and analytic continuation gives the formula for $\operatorname{Re} s > 0$. □

Theorem 7.1.1 is the main unconditional positive result of Chapter 6 and of this chapter, and its proof has a wide margin (Figure 7.1). The only place where

$w$ comes close to a sign change is at infinity, where it decays like $x^{-1/2}$, and there the sign is fixed by $\theta'(0) < 0$.

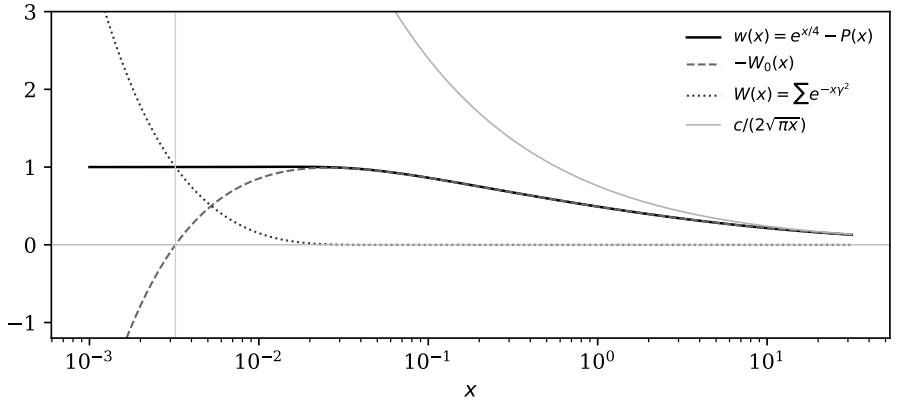



Figure 7.1: The function $w = W - W_0$ on a logarithmic scale, with its two components: the heat trace $W$ over the zeros and the negative smooth part $-W_0$. The vertical line is $x_* \approx 0.00322$, where $W_0$ changes sign. To the left, $w$ differs from $e^{x/4}$ by less than $2 \times 10^{-13}$; to the right, $-W_0 > 0$, and from $x_1 = 0.004$ on it dominates any contribution of off line zeros to $W$. The faint curve is the asymptote $c/(2\sqrt{\pi x})$.

In arithmetic terms, Theorem 7.1.1 is the Gaussian smoothed Chebyshev bound

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x} < \sqrt{4\pi x}\; e^{x/4} \qquad (x > 0).$$

The right side is the same smoothing applied to $dy$ in place of $d\psi(y)$. To see what the primes alone give, write $\psi(y) = y + \Delta(y)$ and put $I(x) = \frac{1}{2} e^{x/4} \operatorname{erfc}(\frac{\sqrt{x}}{2})$.

Proposition 7.1.2. *For $x > 0$,*

$$w(x) = I(x) - Q_\Delta(x), \qquad Q_\Delta(x) = \frac{1}{\sqrt{4\pi x}} \int_{1^-}^{\infty} y^{-1/2} e^{-(\log y)^2/4x}\, d\Delta(y),$$

*with $I(x) \sim (\pi x)^{-1/2}$ and $w(x) \sim \frac{c}{2}(\pi x)^{-1/2}$ as $x \to \infty$. Hence $Q_\Delta(x) \sim -(\frac{c}{2} - 1)(\pi x)^{-1/2}$ with $\frac{c}{2} - 1 \approx 0.343$. On the other hand, any bound $|\Delta(y)| \leq A\sqrt{y} \log^2 y$ for $y \geq 2$, including the one that RH implies, yields only $|Q_\Delta(x)| \leq A\,(x/2 + O(\sqrt{x}))$ after integration by parts.*

*Proof.* With $d\Delta = d\psi - dy$, the main term is $(4\pi x)^{-1/2} \int_1^\infty y^{-1/2} e^{-(\log y)^2/4x} dy = (4\pi x)^{-1/2} \int_0^\infty e^{u/2 - u^2/4x} du = \frac{1}{2} e^{x/4}(1 + \operatorname{erf} \frac{\sqrt{x}}{2})$, and $e^{x/4}$ minus this is $I$. The asymptotics of $w$ are Theorem 6.1.11. For the bound, integrate by parts and

use $\frac{d}{dy}\{y^{-1/2}e^{-(\log y)^2/4x}\} = -y^{-3/2}e^{-(\log y)^2/4x}(\frac{1}{2} + \frac{\log y}{2x})$ and $\int_0^\infty u^2 e^{-u^2/4x}(\frac{1}{2} + \frac{u}{2x})du = \frac{\sqrt{\pi}}{2}(2\sqrt{x})^3 \cdot \frac{1}{4} + 4x$; the boundary term at $1^-$ and the range $1 \le y < 2$ contribute $O(x^{-1/2})$. □

In this Gaussian weighting the error term $\Delta$ of the prime number theorem has a definite negative bias at large $x$, $Q_\Delta(x) \approx -0.343(\pi x)^{-1/2}$, which is $I + W_0$ up to an exponentially small term, and $w > 0$ is the statement that $Q_\Delta(x) < I(x)$ for all $x$. Pointwise bounds on $\Delta$, which oscillates with both signs, cannot give this at large $x$: the margin that keeps $w$ positive there is of order $x^{-1/2}$, while a pointwise bound on $\Delta$ controls $Q_\Delta$ only up to order $x$; the function $D(y) = -A\sqrt{y}\log^2 y$, in place of $\Delta$, gives $Q_D(x) = -A\big(x/2 + 2\sqrt{x/\pi}\big)$ by the computation in the proof, so the bound of Proposition 7.1.2 cannot be improved using the size of $\Delta$ alone. Mean square information on $\Delta(y)/\sqrt{y}$ in the logarithmic scale, of the kind Cramér obtained under RH [97], controls Gaussian smoothed quantities much better, and we make no claim that such information is insufficient.

#### *7.1.1.1 The place of $\varphi$ in the Bondesson hierarchy*

Recall from Section 1.3.2 the classes of laws on $[0,\infty)$ defined through the Lévy density $k(x)/x$: ID, SD ($k$ nonincreasing), BO ($k(x)/x$ completely monotone) and GGC ($k$ completely monotone). They are nested as GGC $\subset$ SD $\subset$ ID and GGC $\subset$ BO $\subset$ ID [56, 489].

THEOREM 7.1.3. *$\varphi \in$ ID but $\varphi \notin$ SD, $\varphi \notin$ BO, and $\varphi$ is not a mixture of exponentials. In particular $\varphi$ is not a generalised gamma convolution.*

*Proof.* ID is Theorem 7.1.1.

SD. If $\varphi$ is not infinitely divisible there is nothing to prove, and if it is then $k = w$ by Theorem 6.1.11. Now $w = e^{x/4} - P$, and $P$ together with every one of its derivatives vanishes at $0^+$, so $w'(0^+) = \frac{1}{4} > 0$ and $w$ strictly increases at the origin.

BO. Membership means that the Lévy density $w(x)/x$ equals $\int e^{-tx}m(dt)$ for some $m \ge 0$. Then $-\varphi'/\varphi(s) = \int_0^\infty xe^{-sx}\int e^{-tx}m(dt)\,dx = \int m(dt)/(s+t)^2$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, so $\varphi$ has no zeros or poles there and RH follows from Proposition 6.1.9. Under RH, Theorem 6.1.11 gives $w(x) = \int e^{-tx}dM(t)$, with $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ as in Theorem 6.1.10, and since $M(0^+) = 0$ an integration by parts gives $w(x)/x = \int e^{-tx}M(t)\,dt$. Uniqueness of Laplace transforms gives $M\,dt = m \ge 0$, that is $S(\tau) \ge -1$ for almost all $\tau$. But by Selberg's central limit theorem [477], $S(\tau) < -1$ for a positive proportion of $\tau \in [T, 2T]$ once $T$ is large, so $M < 0$ on a set of positive measure. Numerically, for $527.71 \le \tau \le 527.90$, just below $\gamma_{289} = 527.9036\ldots$, ball arithmetic gives $N(\tau) = 288$ and $-1.143 < S(\tau) < -1.008$, so $M < 0$ there, and $S(\gamma_{289}^-) = -1.14548\ldots$ (`verification_reciprocal_zeta/certify_k9_k2.py`).

Mixtures of exponentials. The law is a mixture of exponentials exactly when it has a completely monotone density, that is when $\varphi$ is a Stieltjes function, which requires $\operatorname{Im}\varphi(-t+i0)$ to have one sign. With $\zeta(\frac{1}{2}+i\tau) = Z(\tau)e^{-i\theta(\tau)}$ we get $\operatorname{Im}\varphi(-\tau^2+i0) = \zeta(\frac{1}{2})\sin\theta(\tau)/\{(1+4\tau^2)Z(\tau)\}$. On $(0,\gamma_1)$, $Z$ has no zero while $\theta$ decreases through $-\pi$ at $\tau_\pi = 3.43622\ldots$, so $\sin\theta$ changes sign.

GGC. Since GGC is contained in SD and in BO, either clause excludes it. □

The theorem is unconditional.

The failures have two causes, and neither involves a zero off the line. Self decomposability and the GGC property fail because of the pole: the factor $1-4s$ that removes it puts $e^{x/4}$ into $w$, so the canonical function increases near the origin. The Bondesson property fails because the negative Thorin density of the gamma factor meets the fluctuations of the zeros about their smooth count, measured by $S$, which has large negative excursions: the first zero with $S(\gamma^-) < -1$ is $\gamma_{127} = 282.4651$, where $S(\gamma_{127}^-) = -1.0063$, and the most negative value among the first 460 zeros is at $\gamma_{289}$ (the seven such zeros are listed after Theorem 7.2.3, and these values of $S$ are enclosed in ball arithmetic by the program cited in the proof). This failure would occur even if RH were true, and since $S$ is unbounded below, by Selberg and Tsang [477, 504], it recurs infinitely often. Both causes disappear for the completed function $R$, which is infinitely divisible and self decomposable with no hypothesis, by Theorem 5.1.1 [424, Theorem A]. What Theorem 7.1.1 adds there is the floor of Theorem 6.2.15: the canonical function of $e^{-c\sqrt{s}}R(s)$ splits into the arithmetic part $w$ and a completely monotone archimedean part (Section 6.2).

### *7.1.1.2 The density of the law*

Let $X_\varphi$ be, as in Section 6.2, a random variable with $\mathbb{E}e^{-sX_\varphi} = \varphi(s)$. Differentiating the representation of Theorem 7.1.1 gives $-\varphi'(s) = \varphi(s)\int_0^\infty e^{-sx}w(x)\,dx$, which says that the measure $x\Pr(X_\varphi \in dx)$ is the convolution of the law of $X_\varphi$ with $w(x)\,dx$. Since $\varphi(s)\to 0$ as $s\to\infty$, the law has no atom at 0, so it is absolutely continuous, and its density $f$ satisfies Steutel's equation [489, Proposition III.4.16]

$$x\,f(x) = \int_0^x f(x-y)\,w(y)\,dy, \qquad x>0. \tag{7.1.1}$$

The right side is continuous in $x$ by dominated convergence, since $f$ is integrable and $w$ is bounded and continuous on $[0,\infty)$, so $f$ may be taken continuous on $(0,\infty)$. Near the origin, $w(y) = 1+O(y)$ gives $f(x) = \bar f(x)(1+O(x))$ with $\bar f(x) = x^{-1}\int_0^x f$, and $\bar f(x) > 0$ for every $x>0$, since otherwise $s\varphi(s)\to 0$, whereas $s\varphi(s)\to -\zeta(\frac{1}{2})/4 > 0$. As $\bar f' = (f-\bar f)/x$, this gives $(\log\bar f)' = O(1)$ near 0, so $\bar f$ and $f$ have a common positive limit $f(0^+)$, and dominated convergence in $s\varphi(s) = \int_0^\infty e^{-y}f(y/s)\,dy$, $f$ being bounded, gives $f(0^+) = \lim_{s\to\infty} s\varphi(s) = -\zeta(\frac{1}{2})/4 \approx 0.36509$.

At infinity the same equation gives $f(x) \sim w(x)/x \sim \frac{c}{2\sqrt{\pi}} x^{-3/2} \approx 0.75773\, x^{-3/2}$, in three steps. First, $xf(x) \le \sup w$. Second, split the integral in (7.1.1), written as $\int_0^x f(u)w(x-u)\,du$, at $u = x/2$: the part $u \le x/2$ is at most $\sup_{y \ge x/2} w(y) = O(x^{-1/2})$, and the part $u \ge x/2$ is at most $\sup_{[x/2,x]} f \cdot \int_0^{x/2} w = O(x^{-1})\, O(x^{1/2})$, so $f(x) = O(x^{-3/2})$. Third, with this bound the part $u \ge x/2$ is $O(x^{-1}) = o(w(x))$, while the part $u \le x/2$ is $w(x)(1 + o(1))$ by dominated convergence, since $w(x-u)/w(x) \to 1$ for fixed $u$ and stays below 2 for $u \le x/2$ and large $x$. Hence $xf(x) = w(x)(1 + o(1))$. It follows that $\Pr(X_\varphi > x) \sim \frac{c}{\sqrt{\pi}} x^{-1/2}$ and $\mathbb{E}X_\varphi = \infty$, and since $\varphi(s) = 1 - c\sqrt{s} + O(s)$ gives $\varphi(s/n^2)^n \to e^{-c\sqrt{s}}$, the variable $X_\varphi$ lies in the domain of attraction of the positive stable law of index $\frac{1}{2}$ that generates the building blocks (6.1.2).

THEOREM 7.1.4. *$f(0^+) = -\zeta(\frac{1}{2})/4$ and $f'(0^+) = f(0^+)\, w'(0^+) = -\zeta(\frac{1}{2})/16 \approx 0.09127 > 0$. In particular $f$ is not monotone, so $X_\varphi$ is not a mixture of uniform laws on intervals $[0, a]$, and $f$ is not log convex and not completely monotone.*

*Proof.* The value of $f(0^+)$ was found after (7.1.1). Since $P$ and all its derivatives vanish at $0^+$, $w$ is continuously differentiable on $[0, \infty)$ with $w(0^+) = 1$ and $w'(0^+) = \frac{1}{4}$. Write (7.1.1) as $xf(x) = \int_0^x f(u)\, w(x-u)\, du$. Since $f$ is continuous on $(0, \infty)$ and bounded near 0, the right side is differentiable, with derivative $f(x)w(0^+) + \int_0^x f(u)w'(x-u)\,du$, so $f$ is differentiable on $(0, \infty)$ and

$$f'(x) = \frac{1}{x} \int_0^x f(u)\, w'(x-u)\, du \longrightarrow f(0^+)\, w'(0^+) \qquad (x \downarrow 0),$$

because $f$ and $w'$ are continuous at 0. A nonincreasing density is required for a mixture of uniforms by Khintchine's theorem, and log convex or completely monotone densities are nonincreasing. □

PROPOSITION 7.1.5. *$w' > 0$ on $(0, 0.004]$ and $w' < 0$ on $[4.52, \infty)$. On $[0.004, 4.6]$ a grid evaluation of $w'$ (computed, not proved) finds exactly one sign change, at $x_m = 0.015851\ldots$, where $w(x_m) = 1.003409\ldots$.*

*Proof.* On $(0, 0.004]$, $w' = \frac{1}{4}e^{x/4} - P'(x)$ and $0 \le P'(x) \le P'(0.004) < 2 \times 10^{-9}$, since each prime term of $P'$ increases in $x$ there. For $x \ge 0.004$ write $w' = W' - W_0'$. The zeros on the line contribute $-\sum \gamma^2 e^{-x\gamma^2} < 0$ to $W'$. Every zero below height $3 \times 10^{12}$ lies on the line [419], so an off line zero contributes at most $(\gamma^2 + \frac{1}{4})e^{-x(\gamma^2 - 1/4)}$ with $\gamma > 3 \times 10^{12}$. With $N(t+1) - N(t) \le 2\log(t+1)$ from the proof of Theorem 7.1.1, these contributions together are less than $e^{-10^{22}} \gamma_1^2 e^{-x\gamma_1^2}$ for $x \ge 0.004$; hence $W' < 0$. By Proposition 6.2.11, with $\nu(dy) = 2\sum_{k \ge 0} e^{-(2k + \frac{1}{2})y}\, dy$

the measure defined there,

$$-W_0'(x) = \frac{D(x)}{4\sqrt{\pi}\,x^{3/2}}, \qquad D(x) = \theta'(0) + \frac{1}{2}\int_0^\infty g\Big(\frac{y^2}{4x}\Big)\nu(dy),$$
$$g(u) = 1 - (1-2u)e^{-u}.$$

Since $g(0) = 0$, $g'(0) = 3$, $g$ is concave on $[0, \frac{5}{2}]$ and $g \le 1 + 2e^{-3/2} < 3u$ beyond, $g(u) \le 3u$ for all $u \ge 0$. With $\int y^2\nu(dy) = 4\sum_{k\ge 0}(2k+\frac{1}{2})^{-3} < 32.34$ this gives $D(x) \le \theta'(0) + 12.13/x < 0$ for $x \ge 4.52$, and then both parts of $w'$ are negative. □

The computed statement comes from an evaluation of $w'$ on $[0.004, 4.6]$ from the first 300 zeros and the integral for $W_0'$, on a grid of spacing $10^{-4}$ up to 0.05 and $5\times 10^{-3}$ beyond. The prime and zero evaluations of $w'$ agree to $10^{-11}$ at $x = 0.01$.

Numerically $f$ is unimodal as well: it rises from 0.365089 to its maximum 0.366883 at the mode $x = 0.024046$ and then decreases to 0 like $0.75773\,x^{-3/2}$, so its mode lies to the right of the maximum of $w$. Talbot and de Hoog inversion agree to twelve digits at every point used. Unimodality of $f$ remains open (Section 7.2.4). Since $X_\varphi$ is infinitely divisible but not self decomposable, Yamazato's theorem that every self decomposable law is unimodal [521], [471, Theorem 53.1] does not apply. Section 11.1 applies the theorem to the law of $X_R$, which is self decomposable, and to the shifted clocks $T_\alpha$ of Theorem 1.2.2 with $\alpha \ge 1$ (Corollary 11.1.2).

### *7.1.1.3 Closure operations and the harmonic mean*

GGC is closed under independent sums and weak limits, and, by a theorem of Bondesson [57], under independent products, so Theorem 7.1.3 gives at once the following.

Corollary 7.1.6. *$X_\varphi$ is not a sum or a product of independent GGC variables, nor a weak limit of such.*

The reciprocal in $\varphi$ suggests a fourth operation, the harmonic mean. Closure under it would be the statement

(H) if $V_1$ and $V_2$ are independent GGC variables, then so is $2V_1V_2/(V_1+V_2)$,

which would have widened Corollary 7.1.6; it is false.

Proposition 7.1.7. *(H) is false. If $E_1$ and $E_2$ are independent standard exponentials, then $H = 2E_1E_2/(E_1+E_2)$ has density*

$$f_H(h) = e^{-h}\int_0^\infty \Big(1 + \frac{h}{2u}\Big)e^{-u-h^2/4u}\,du, \qquad h > 0,$$

*and $H$ is not a GGC.*

*Proof.* For $h > 0$, $2E_1E_2 > h(E_1 + E_2)$ holds if and only if $E_1, E_2 > \frac{h}{2}$ and $(2E_1 - h)(2E_2 - h) > h^2$: both factors cannot be negative, since $E_1, E_2 < \frac{h}{2}$ forces $|2E_1 - h|, |2E_2 - h| < h$. Substituting $E_1 = \frac{h}{2} + u$, $E_2 = \frac{h}{2} + v$ gives

$$\Pr(H > h) = e^{-h} \iint_{uv > h^2/4} e^{-u-v}\, du\, dv = e^{-h} \int_0^\infty e^{-u-h^2/4u}\, du,$$

so differentiating under the integral, which is justified by the bound $\frac{h}{2u} e^{-u-h^2/4u} \le u^{-1/2} e^{-u} \sup_{y>0} y e^{-y^2}$, locally uniform in $h$, gives $f_H$. In terms of the modified Bessel functions of the second kind the two integrals are $h\,\mathrm{K}_1(h)$ and $h\{\mathrm{K}_0(h) + \mathrm{K}_1(h)\}$, by $\int_0^\infty u^{-\nu-1} e^{-u-h^2/4u}\, du = 2(h/2)^{-\nu}\mathrm{K}_\nu(h)$ [397, (10.32.10)]. Splitting the integral at $u = h$ shows that $f_H$ is bounded with $f_H(0^+) = 1$.

Suppose $H$ were a GGC with drift $a$ and Thorin measure $U$ of total mass $\beta$. Since $f_H > 0$ on $(0, \infty)$ and a GGC satisfies $H \ge a$ almost surely, $a = 0$. Since $f_H$ is bounded, $s\,\mathbb{E}e^{-sH} = \int_0^\infty e^{-y} f_H(y/s)\, dy \to f_H(0^+) = 1$, so $-\log \mathbb{E}e^{-sH} = \log s + o(1)$, while Fatou applied to $\int \log(1 + s/t)\, U(dt)$ gives $\liminf_{s\to\infty}(-\log \mathbb{E}e^{-sH})/\log s \ge \beta$. Hence $\beta \le 1$.

Now let $V$ be any GGC with zero drift and $\beta \le 1$, and $\mathcal{L}(s) = \mathbb{E}e^{-sV}$. For $\operatorname{Im} s > 0$ each $1 + s/t$ lies in the upper half plane, so $\operatorname{Im} \log \mathcal{L}(s) = -\int \arg(1 + s/t)\, U(dt) \in [-\pi\beta, 0) \subseteq [-\pi, 0)$, and $\mathcal{L} > 0$ on $(0, \infty)$: $\mathcal{L}$ is a Stieltjes function [473, Corollary 7.4]. In its representation $\mathcal{L}(s) = b_0 + b_1/s + \int (s + r)^{-1} m(dr)$ [473, Definition 2.1] we have $b_0 = \lim_{s\to\infty} \mathcal{L}(s) = 0$ and $b_1 = 0$, since $\mathcal{L}(0^+) = 1$. So $\mathcal{L}(s) = \int_0^\infty e^{-sx} \big( \int e^{-rx} m(dr) \big) dx$ and the density of $V$ is $\int e^{-rx} m(dr)$, completely monotone and in particular nonincreasing.

But $f_H$ is not nonincreasing: $f_H(h) - 1 \sim h \log(1/h)$ as $h \downarrow 0$, and $f_H$ attains 1.11125 at $h = 0.09765$ against $f_H(0^+) = 1$. So $H$ is not a GGC. □

There is also a structural obstruction. The factorisation of $\varphi$ into $R$ and the gamma factor is a factorisation of Laplace transforms, that is a convolution of laws, while (H) acts on the variables themselves. The gamma factor is not a Laplace transform at all, since it tends to infinity with $s$, and (6.2.1) shows that it enters only through deconvolution by stable laws. The one place where (H) could have acted is the positive part: $R$ is GGC if and only if RH holds, so an unconditional representation $X_R \overset{d}{=} 2V_1V_2/(V_1 + V_2)$ with $V_1, V_2$ independent GGC variables built from the primes and the gamma factor would have proved RH under (H). Proposition 7.1.7 removes the implication, and we know of no candidate representation.

### 7.1.2 The Vere-Jones weights and the Lévy representation across the strip

Throughout this subsection $\sigma$ is the real part of the argument $z$ of $\zeta$.

*7.1.2.1 Necklace polynomials and the sign rule*

Let $M_k(q) = \frac{1}{k}\sum_{d|k}\mu(k/d)\,q^d$ be the necklace polynomial [373].

THEOREM 7.1.8. *$B_N = 0$ unless $N = p^n$ is a prime power, and then*

$$B_{p^n} = b_n\big(p^{-1/2}\big), \qquad b_n(q) = \sum_{i\ge 0,\ 2^i\mid n} M_{n/2^i}(q).$$

*Equivalently, for $|y| < 1$ and $0 < q \le 1$,*

$$\frac{1}{1-qy} = \prod_{n\ge 1}\big(1+y^n\big)^{b_n(q)}, \tag{7.1.2}$$

*so that $\zeta(\frac12+u) = \prod_p\prod_n(1+p^{-nu})^{b_n(p^{-1/2})}$ for $\operatorname{Re} u > \frac12$. At $q = 1$, $b_n(1) = 1$ if $n$ is a power of 2 and 0 otherwise, and (7.1.2) becomes Euler's binary identity $\frac{1}{1-y} = \prod_{j\ge0}(1+y^{2^j})$.*

*Proof.* From the definition of $B_N$ in Theorem 6.1.8, $B_{p^n} = \sum_{jm=n} d_m\,a_{p^j}$ with $a_{p^j} = q^j/j$. Proposition 6.1.6 gives $y^j = \sum_m d_m \log(1+y^{mj})$; multiplying by $q^j/j$ and summing over $j$ gives $\sum_n b_n(q)\log(1+y^n) = \sum_j q^j y^j/j = -\log(1-qy)$, which is (7.1.2). The cyclotomic identity $\frac{1}{1-qy} = \prod_k(1-y^k)^{-M_k(q)}$ and $\log(1+y^n) = \log(1-y^{2n}) - \log(1-y^n)$ give, comparing coefficients of $\log(1-y^n)$, the recursion $b_n = M_n + b_{n/2}$ for even $n$ and $b_n = M_n$ for odd $n$, whose solution is the stated sum. At $q = 1$, $M_k(1) = \frac{1}{k}\sum_{d|k}\mu(k/d) = \delta_{k,1}$. □

The identity (7.1.2) is a 2 adic variant of the cyclotomic identity of Metropolis and Rota [373]; we have not found it stated in this form. Its exponents include $b_2 = \frac{q^2+q}{2}$, $b_3 = \frac{q^3-q}{3}$, $b_4 = \frac{q^4}{4} + \frac{q^2}{4} + \frac{q}{2}$, $b_6 = \frac{q^6+q^3-q^2-q}{6}$, which give $B_4$, $B_8$ and $B_9$ of Theorem 6.1.8 and, at $p = 2$, $B_{16}$ and $B_{64}$.

PROPOSITION 7.1.9. *For every prime $p$, with $q = p^{-1/2}$: $b_{2^a}(q) > 0$ for all $a \ge 1$, and for odd $n > 1$, $b_n(q) \ne 0$ and the sign of $b_n(q)$ is $(-1)^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors of $n$.*

*Proof.* For $n = 2^a$, $b_n = q + \sum_{j=1}^{a}\frac{q^{2^j}-q^{2^{j-1}}}{2^j} = \frac{q}{2} + \frac{q^2}{4} + \cdots + \frac{q^{2^{a-1}}}{2^a} + \frac{q^{2^a}}{2^a} > 0$. For odd $n$, $b_n = M_n$. Put $\ell = \operatorname{rad}(n)$ and $\tilde q = q^{n/\ell} \le q \le 2^{-1/2}$; then $nM_n(q) = \sum_{k|\ell}\mu(\ell/k)\tilde q^k = (-1)^{\omega(n)}\big(\tilde q + \sum_{k|\ell,k>1}\pm\tilde q^k\big)$, and the divisors $k > 1$ of the odd squarefree $\ell$ are distinct odd integers at least 3, so $\sum_{k>1}\tilde q^k < \sum_{m\ \text{odd}\ge3}\tilde q^m = \tilde q\cdot\frac{\tilde q^2}{1-\tilde q^2} \le \tilde q$. □

For even $n$ other than powers of 2 the sign depends on how the 2 adic parts combine. Among $2 \le n \le 60$ there are 42 negative exponents for each of $p = 2, 3, 5, 7$.

The weights $B_N$ and $d_m$ also govern the truncation of the sawtooth expansion of $S$. The sawtooth $\Sigma_\lambda$ of Lemma 6.1.7 has the Fourier series $\Sigma_\lambda(\tau) = \sum_{j\geq 1}(-1)^j \frac{\sin(j\lambda\tau)}{\pi j}$, so the formal series $\sum_N B_N \Sigma_{\log N}(\tau)$ is a double series in $(N, j)$ whose frequencies are $j \log N$.

Proposition 7.1.10. *Truncate the double series by frequency, keeping the terms with $j \log N \leq L$. Then*

$$\sum_{\substack{N,j \\ j\log N\leq L}} B_N\,(-1)^j \frac{\sin(j\tau\log N)}{\pi j} = -\frac{1}{\pi}\sum_{n\leq e^L} \frac{\Lambda(n)}{\sqrt{n}\log n}\sin(\tau\log n),$$

*the classical truncated series for $S(\tau)$. Truncation by $N \leq N_0$ instead is not equivalent: it keeps the harmonics $j \log N > \log N_0$ of each retained sawtooth without the terms of larger $N$ that cancel them.*

*Proof.* With $N = n^m$, the frequency is $r \log n$ with $r = mj$, and the constraint $j \log N \leq L$ is $r \log n \leq L$, which does not separate $m$ from $j$. The coefficient of $\sin(r\tau \log n)$ is $-\frac{a_n}{\pi}\sum_{mj=r} d_m \frac{(-1)^{j+1}}{j}$, and $\sum_{mj=r} d_m (-1)^{j+1} j^{-1}$ has Dirichlet series $\eta(1+v)^{-1}\eta(1+v) = 1$, so it is $\delta_{r,1}$. □

So the sawtooth expansion contains nothing that the classical series lacks once it is truncated consistently, and truncating it by the index $N$ is worse. Numerically, on a uniform grid of step $10^{-3}$ in $\tau \in [10, 50]$ and with the Cesàro weight $1 - \log N/\log N_0$, the root mean square error against $S(\tau)$ is 0.228, 0.223, 0.218 for $N_0 = 10^3, 10^4, 10^5$, against 0.146, 0.128, 0.115 for the classical series with the same weight, the root mean square of $S$ itself being 0.320 (`sawtooth_truncation.py`).

### *7.1.2.2 The cyclotomic form, the prime number theorem and a trace formula*

The cyclotomic identity gives a second factorisation, into the blocks $1 - N^{-u}$:

$$\frac{1}{\zeta(\frac{1}{2}+u)} = \prod_p \prod_{k\geq 1} \bigl(1 - p^{-ku}\bigr)^{M_k(p^{-1/2})}, \qquad \operatorname{Re} u > \tfrac{1}{2}.$$

Each block $1 - e^{-\lambda u} = 2e^{-\lambda u/2}\sinh(\lambda u/2)$ is again a solved model, now with a sinh lattice: with $u = \sqrt{s}$, $\frac{\sinh(\lambda\sqrt{s}/2)}{\lambda\sqrt{s}/2} = \prod_{m\geq 1}\bigl(1 + \frac{s}{(2\pi m/\lambda)^2}\bigr)$, a reciprocal GGC with unit atoms at $(2\pi m/\lambda)^2$. The sinh blocks are the local zeta factors on the line $\sigma = 0$, and unlike the sech blocks they vanish at $u = 0$. Normalising the product at $u \to 0$ therefore requires two identities.

Proposition 7.1.11. *For $0 < q < 1$, summing over $k$ in increasing order,*

$$\sum_{k\geq 1} M_k(q) = 0, \qquad \sum_{k\geq 1} M_k(q)\log k = \log(1-q).$$

*Both follow from the prime number theorem in the forms* $\sum_m \mu(m)/m = 0$ *and* $\sum_m \mu(m)\log m/m = -1$.

*Proof.* $\sum_{k\le K} M_k(q) = \sum_d \frac{q^d}{d}\sum_{m\le K/d}\frac{\mu(m)}{m}$. The inner sums are bounded by 1 in absolute value and tend to 0, so dominated convergence gives 0. Similarly $\sum_{k\le K} M_k(q)\log k = \sum_d \frac{q^d}{d}\sum_{m\le K/d}\frac{\mu(m)\log(dm)}{m} \to \sum_d \frac{q^d}{d}(-1) = \log(1-q)$, using $\sum \mu(m)/m = 0$ for the $\log d$ part. □

Heuristically, the two identities are what make the cyclotomic product of solved models converge at the edge $u \to 0$: $\prod_k (1-y^k)^{-M_k(q)}$ behaves near $y = 1$ like $(1-y)^{-\sum M_k}\prod_k k^{-M_k}$, and the finite limit $\frac{1}{1-q}$ requires both.

In heat trace form the Vere-Jones representation becomes a trace formula. Proposition 6.1.13 and Poisson summation give, for every $x > 0$,

$$\sum_{\gamma>0} e^{-x\gamma^2} - W_0(x) = e^{x/4} + \sum_p \sum_{n\ge1} b_n(p^{-1/2})\Big[\sum_{k\ge0} e^{-x\left(\frac{(2k+1)\pi}{n\log p}\right)^2} - \frac{n\log p}{4\sqrt{\pi x}}\Big], \quad (7.1.3)$$

where the left side is written under RH and is $W(x) - W_0(x)$ in general, and the double series converges absolutely once each bracket is written in its theta form. The left side is the heat trace of the zeros net of its smooth part; the right side is the pole plus the heat traces of the arithmetic progressions $\frac{(2k+1)\pi}{n\log p}$, the pseudo zeros of the blocks, each net of its own mean. So in heat trace the zeros of $\zeta$ are a signed superposition of arithmetic progressions with spacings $2\pi/(n\log p)$. Heuristically, constructive interference of the progressions places the mass at the ordinates $\gamma$, and the negative weights of Proposition 7.1.9 remove the mass from the other points of the progressions.

The trace formula does not give positivity of the Thorin measure of $R$. The obstruction is explicit: the weights $b_n(p^{-1/2})$ are not nonnegative integers, and the passage from the binary product at $q = 1$ to the critical line at $q = p^{-1/2}$ is a deformation of the exponents in which integrality and positivity are lost together. A proof of RH along these lines would have to show that the interference of the progressions in (7.1.3), with exactly these weights, produces a positive measure; Section 7.2.4 poses the deformation from $q = 1$ to $q = p^{-1/2}$ as a problem.

### *7.1.2.3 The Lévy and Thorin representations across the strip*

For $\sigma > 1$ the Vere-Jones representation is a statement about the classical zeta distribution (Section 6.1.6): $\zeta(\sigma + i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau\log N_\sigma}$ where $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and $\log N_\sigma$ is compound Poisson with Lévy measure $\nu_\sigma = \sum_n \frac{\Lambda(n)}{n^\sigma \log n}\delta_{\log n}$. Theorem 7.1.8 decomposes this Lévy measure as

$$\nu_\sigma = \sum_p \sum_{n\ge1} b_n(p^{-\sigma})\,\nu^{\mathrm{alt}}_{n\log p}, \qquad \nu^{\mathrm{alt}}_\lambda = \sum_{k\ge1}\frac{(-1)^{k+1}}{k}\,\delta_{k\lambda},$$

where $\nu_\lambda^{\mathrm{alt}}$ is the formal Lévy measure of the fair coin $\lambda B$, $B \in \{0,1\}$, whose characteristic function $\frac12(1+e^{-i\lambda\tau})$ is the sech block. For $\sigma > 1$ there is also a positive decomposition: by $\frac{1}{1-qy} = \prod_j (1+q^{2^j}y^{2^j})$, the exponent of $p$ in $N_\sigma$ has independent binary digits with $\Pr(\text{digit } j=1) = \frac{p^{-2^j\sigma}}{1+p^{-2^j\sigma}}$. The Vere-Jones form keeps the coins fair and moves $p^{-\sigma}$ into the multiplicities; the binary digit form keeps unit multiplicities and biases the coins.

Inside the critical strip neither form survives as a characteristic function, since $|\zeta(\sigma+i\tau)|$ is unbounded in $\tau$ for $0<\sigma<1$ and the Dirichlet series diverges (Section 6.1.6); the heat kernel supplies the replacement. For $0<\sigma<1$ put

$$\varphi_\sigma(s) = \frac{\zeta(\sigma)}{(1-s/(1-\sigma)^2)\,\zeta(\sigma+\sqrt{s})}, \qquad \varphi_{1/2} = \varphi.$$

THEOREM 7.1.12. *For $0<\sigma<1$ and $s > (1-\sigma)^2$, $-\varphi_\sigma'/\varphi_\sigma(s) = \int_0^\infty e^{-sx} w_\sigma(x)\,dx$, where $w_\sigma$ has the three equivalent forms*

$$\begin{aligned} w_\sigma(x) &= e^{(1-\sigma)^2 x} - \frac{1}{\sqrt{4\pi x}} \sum_{n\ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x} \\ &= e^{(1-\sigma)^2 x} + \sum_p \sum_{n\ge 1} b_n(p^{-\sigma})\, h_{n\log p}(x) \\ &= \tfrac12 e^{(1-\sigma)^2 x} \operatorname{erfc}\big((1-\sigma)\sqrt{x}\big) + \sum_{\gamma>0} \operatorname{Re}\Big[e^{a_\gamma^2 x} \operatorname{erfc}(a_\gamma\sqrt{x})\Big] \\ &\qquad + \frac{\gamma_E + \log\pi}{4\sqrt{\pi x}} + \sum_{k\ge 0}\Big[\tfrac12 e^{a_k^2 x}\operatorname{erfc}(a_k\sqrt{x}) - \frac{1}{4(k+1)\sqrt{\pi x}}\Big], \end{aligned}$$

*with $h_\lambda$ the Jacobi theta function of Proposition 6.1.13, $a_\gamma = \sigma - \frac12 - i\gamma$ (the last form written under RH; in general $\gamma$ runs over $-i(\rho-\frac12)$), and $a_k = 2k+2+\sigma$. As $x\to\infty$, $w_\sigma(x) \sim \frac{\zeta'(\sigma)/\zeta(\sigma)}{2\sqrt{\pi x}}$, with $\zeta'(\sigma)/\zeta(\sigma) > 0$ on $(0,1)$. If $w_\sigma \ge 0$, then $\varphi_\sigma$ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density $w_\sigma(x)/x$:*

$$\log\varphi_\sigma(s) = -\int_0^\infty \big(1-e^{-sx}\big)\frac{w_\sigma(x)}{x}\,dx, \qquad \operatorname{Re} s > 0.$$

*Proof.* The first form is the argument of Theorem 6.1.11 with the pole now at $s=(1-\sigma)^2$. For the second, regroup $\sum_n b_n(q)\, n\log p \sum_j (-1)^j e^{-(jn\log p)^2/4x}$ by $r = jn$: with $nb_n = \sum_{ab=n} a\,d_a q^b$ the coefficient of $e^{-(r\log p)^2/4x}$ is $\sum_{b|r} q^b \sum_{ja=r/b} a\,d_a(-1)^j = -q^r$, since the inner sum is $-\delta_{r/b,1}$ (Proposition 6.1.13); this is the prime $p$ part of the first form. For the third, insert the Hadamard product of $\zeta'/\zeta$ into $-\varphi_\sigma'/\varphi_\sigma = \frac{1}{s-(1-\sigma)^2} + \frac{1}{2\sqrt{s}}\frac{\zeta'}{\zeta}(\sigma+\sqrt{s})$; the pole terms combine to $\frac{1}{2\sqrt{s}(\sqrt{s}+1-\sigma)}$, and each $\frac{1}{\sqrt{s}(\sqrt{s}+a)}$ is the transform of $e^{a^2x}\operatorname{erfc}(a\sqrt{x})$. At $\sigma=\frac12$ this reduces to $w = W - W_0$ with Proposition 6.2.2.

The asymptotic follows from the third form term by term. Uniformly for $|\arg a| \le \frac{3\pi}{4} - \epsilon$ and $|a|\sqrt{x} \ge 1$,

$$\tfrac{1}{2} e^{a^2 x} \operatorname{erfc}(a\sqrt{x}) = \frac{1}{2a\sqrt{\pi x}} + O\big(|a|^{-3} x^{-3/2}\big);$$

every $a$ in the third form has $|\arg a| < \frac{\pi}{2} + \frac{1}{14}$, and $\sum |a|^{-3}$ converges. So $w_\sigma(x) = C_\sigma/(2\sqrt{\pi x}) + O(x^{-3/2})$ with

$$C_\sigma = \frac{1}{1-\sigma} + \sum_\rho \frac{1}{\sigma - \rho} + \frac{\gamma_E + \log \pi}{2} + \sum_{k \ge 0} \Big( \frac{1}{2k+2+\sigma} - \frac{1}{2k+2} \Big),$$

the sum over $\rho$ taken in conjugate pairs, and $C_\sigma = \zeta'(\sigma)/\zeta(\sigma)$ by the Hadamard product of $\xi$. Infinite divisibility follows from $w_\sigma \ge 0$ by Landau's theorem as in Theorem 7.1.1, since $\varphi_\sigma$ is analytic near $(0, \infty)$: $\zeta(\sigma + x) \ne 0$ for $x > 0$ and the pole at $x = 1 - \sigma$ is cancelled. □

Numerically $w_\sigma$ is positive on every line we tried. We evaluated it on 300 points of $x \in [10^{-4}, 10^3]$, from the prime form for $x \le 0.02$ and the zero form beyond, for $\sigma = 0.05, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 0.9, 0.99$. In every case the minimum is attained at the largest $x$, where $2\sqrt{\pi x}\, w_\sigma(x)$ is close to $\zeta'(\sigma)/\zeta(\sigma)$ up to $\sigma = 0.75$ and falls short of it as $\sigma \to 1$, where the pole makes the approach slow and the asymptotic has not been reached at $x = 10^3$. Whether the Lévy density is positive on every vertical line of the strip is open (Problem 7.8).

For the Thorin representation two families have to be distinguished: $\varphi_\sigma$, which carries the gamma factor and the pole, and the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$. For $\sigma > \frac{1}{2}$ the second is, with no hypothesis, the Laplace transform of an infinitely divisible law obtained by tilting and subordinating the law of $X_R$ (Theorem 9.2.5 of Section 9.2 and the proof of Proposition 10.1.1 of Section 10.1, where $R_\sigma$ is written $\varphi_\alpha$). The first never has the Thorin property.

PROPOSITION 7.1.13. *For every $\sigma \in (0,1)$, $\varphi_\sigma$ is not the Laplace transform of a GGC, and if it is infinitely divisible it is not self decomposable.*

*Proof.* The argument is that of Theorem 7.1.3. By Theorem 7.1.12, $-\varphi_\sigma'/\varphi_\sigma(s) = \int_0^\infty e^{-sx} w_\sigma(x)\, dx$ for $s > (1-\sigma)^2$, with

$$w_\sigma = e^{(1-\sigma)^2 x} - P_\sigma, \qquad P_\sigma(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x}.$$

Every derivative of $P_\sigma$ vanishes at $0^+$, so $w_\sigma(x) = 1 + (1-\sigma)^2 x + o(x)$ and $w_\sigma$ strictly increases near the origin. If $\varphi_\sigma$ were a GGC transform, then $-\varphi_\sigma'/\varphi_\sigma = a + \int U(dt)/(s+t)$ with $a \ge 0$ and $U \ge 0$ satisfying the Thorin integrability conditions, so that $k_U(x) = \int e^{-tx} U(dt)$ is integrable near 0 and, by Tonelli,

$\int U(dt)/(s+t) = \int_0^\infty e^{-sx} k_U(x)\,dx$ for $s > (1-\sigma)^2$. Letting $s \to \infty$ gives $a = 0$, and uniqueness of the Laplace transform gives $w_\sigma = k_U$, which is completely monotone and so nonincreasing, a contradiction. If $\varphi_\sigma$ is infinitely divisible, the same uniqueness identifies its Lévy density as $w_\sigma(x)/x$, and self decomposability would require $w_\sigma$ to be nonincreasing. □

As on the critical line, the obstruction is the factor $1 - s/(1-\sigma)^2$ that removes the pole, and it says nothing about the zeros. Beyond it, for $\frac{1}{2} < \sigma < 1$ with $\sup_\rho \operatorname{Re}\rho < \sigma$, the Thorin density $\frac{1}{2\pi\sqrt{t}} \operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + i\sqrt{t})$ given by Lemma 6.1.1 also takes negative values on every line we examined (the minimum of $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\tau)$ over $\tau \in (0,60]$ is $-1.049$, $-0.995$, $-0.867$, $-0.769$, $-0.721$ at $\sigma = 0.55, 0.6, 0.75, 0.9, 0.99$), but the proposition does not need this.

The completed functions $R_\sigma$ carry the zeros. Theorem 7.1.14 is Theorem 1.2.2 of Section 1.2 in the variable of this chapter.

THEOREM 7.1.14. *Let $\sigma \geq \frac{1}{2}$. Then $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is the Laplace transform of a GGC if and only if $\zeta$ has no zero $\rho$ with $\operatorname{Re}\rho > \sigma$. In that case its Thorin measure is the Poisson smoothed zero density*

$$U_\sigma(dt) = \frac{1}{2\pi\sqrt{t}} \sum_\rho \frac{\sigma - \beta_\rho}{(\sigma-\beta_\rho)^2 + (\sqrt{t}-\gamma_\rho)^2}\,dt,$$

*the sum running over all nontrivial zeros $\rho = \beta_\rho + i\gamma_\rho$, counted with multiplicity, of both signs of $\gamma_\rho$, and taken symmetrically; each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ contributes a unit atom at $\gamma_\rho^2$ instead of its kernel. Consequently $R_\sigma$ is a GGC for every $\sigma \geq 1$, unconditionally, and*

$$\Theta := \sup_\rho \operatorname{Re}\rho = \inf\{\sigma \geq \tfrac{1}{2} :\ R_\sigma \text{ is a GGC}\}.$$

*RH is the statement $\Theta = \frac{1}{2}$, that is, that $R_{1/2} = R$ is a GGC.*

*Proof.* A zero with $\operatorname{Re}\rho > \sigma$ gives $\sqrt{s} = \rho - \sigma$ on the principal branch and hence a pole of $R_\sigma$ in $\mathbb{C} \setminus (-\infty, 0]$, so $R_\sigma$ is not a GGC. Suppose there is none. Write the zeros with positive ordinate, counted with multiplicity, as $\rho = \beta + i\gamma$, put $a = \sigma - \beta \geq 0$, and let $\mathcal{P}_a(x) = a/\{\pi(a^2+x^2)\}$, for $a > 0$, be the Cauchy density of scale $a$ (the Poisson kernel $P_a$ of Section 1.2). Hadamard's product [104, Chapter 12], with each $\rho$ paired with $\bar\rho$, gives $\xi(z) = \xi(0)\prod_{\gamma>0}(1 - z/\rho)(1 - z/\bar\rho)$, absolutely convergent since $\sum_\rho |\rho|^{-2} < \infty$, and hence for $s > 0$ and $q = \sqrt{s}$

$$R_\sigma(s) = \prod_{\gamma>0} \frac{a^2+\gamma^2}{(q+a)^2+\gamma^2}.$$

For $a > 0$ let $U_{a,\gamma}(dt) = \{\mathcal{P}_a(\sqrt{t}-\gamma) + \mathcal{P}_a(\sqrt{t}+\gamma)\}\,dt/(2\sqrt{t})$, and let $U_{0,\gamma} = \delta_{\gamma^2}$. The substitution $t = \tau^2$, the identity $1/(s+\tau^2) = (\pi/q)\mathcal{P}_q(\tau)$ and $\mathcal{P}_a * \mathcal{P}_q = \mathcal{P}_{a+q}$

give

$$\int_0^\infty \frac{U_{a,\gamma}(dt)}{s+t} = \frac{q+a}{q\{(q+a)^2+\gamma^2\}} = \frac{d}{ds}\log\{(q+a)^2+\gamma^2\},$$

which also holds for $a = 0$. Integrating over $(0, s]$ with Tonelli, $\int \log(1 + s/t)\, U_{a,\gamma}(dt) = \log\{((q+a)^2+\gamma^2)/(a^2+\gamma^2)\}$, which is nonnegative and at most $(s + 2\sigma\sqrt{s})/\gamma^2$. Since $\sum_{\gamma>0}\gamma^{-2} < \infty$, Tonelli's theorem shows that $U_\sigma = \sum_{\gamma>0} U_{a,\gamma}$ is a positive measure with $\int \log(1 + s/t)\, U_\sigma(dt) = -\log R_\sigma(s) < \infty$, which gives the Thorin integrability conditions, and that $R_\sigma(s) = \exp\{-\int \log(1 + s/t)\, U_\sigma(dt)\}$ for $s > 0$. So $R_\sigma$ is the Laplace transform of a GGC with Thorin measure $U_\sigma$, whose density is the stated sum, grouped by the pairs $\rho, \bar\rho$, and each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ gives the atom $\delta_{\gamma_\rho^2}$. For $\sigma \geq 1$ the hypothesis holds by the Euler product.

For $\sigma > \Theta$ the density also follows from Lemma 6.1.1: then $-R_\sigma'/R_\sigma(s) = \frac{1}{2\sqrt{s}}\frac{\xi'}{\xi}(\sigma + \sqrt{s})$ is $O(|s|^{-1/2})$ near 0, $O(|s|^{-1/2}\log|s|)$ at infinity uniformly on $\mathbb{C} \setminus (-\infty, 0]$ (by the partial fraction formula for $\zeta'/\zeta$ [104, Chapter 15], since every zero lies at distance at least $\sigma - \Theta$ from the half plane $\operatorname{Re} z \geq \sigma$), and continuous up to the cut away from 0, and at $\sqrt{s} = i\tau$ the lemma gives the density $\frac{1}{2\pi\tau}\operatorname{Re}\frac{\xi'}{\xi}(\sigma + i\tau) = \frac{1}{2\pi\tau}\sum_\rho \operatorname{Re}\frac{1}{\sigma+i\tau-\rho}$, summed symmetrically. □

The quasi Riemann Hypothesis $\Theta < 1$, announced in October 2026 with $\Theta \leq \frac{7}{8}$ (Section 12.1.2), is the statement that $R_\sigma$ is a GGC for some $\sigma < 1$; it is step 7 of the path to RH in Section 12.1 (Table 12.1), and short of it Section 11.4 shows with no hypothesis that for $\frac{1}{2} < \sigma < 1$, in the density $\rho_{\sigma-1/2}$ of (1.2.7), the Cauchy kernels of the zeros to the right of $\sigma$, if there are any, carry below height $T$ a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$. Numerically $\min_{0<\tau\leq 70}\operatorname{Re}\frac{\xi'}{\xi}(\sigma + i\tau)$ is $4.6 \times 10^{-5}$, 0.0046, 0.0116, 0.0231 at $\sigma = 0.501, 0.6, 0.75, 1$, attained as $\tau \to 0$.

### 7.1.3 Pólya and Cardon theory for the solved models

Pólya's theory of Fourier transforms with only real zeros, and its extension by Cardon, is built from the objects that the Vere-Jones step produces: cosines, hyperbolic cosines and their products. The solved models are special cases of that theory; $\zeta$ is not, because its multiplicities are neither integers nor nonnegative. Throughout, $\Phi$ is the Pólya kernel and $\Xi(\tau) = \int \Phi(t)\cos(\tau t)\, dt$, as in (1.2.3) with the variable written $t$; $\Phi$ is even, analytic in the strip $|\operatorname{Im} t| < \pi/4$, and has a natural boundary on the lines $\operatorname{Im} t = \pm\pi/4$ [458, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The Pólya variable $X$ of Section 1.2 has density $\Phi/\Xi(0)$ and moment generating function $\mathbb{E}e^{hX} = \xi(\frac{1}{2} + h)/\xi(\frac{1}{2})$, by (1.2.4). In this subsection $G$ denotes a function of the Laguerre Pólya class, not the gamma factor.

*7.1.3.1 Solved models and the Laguerre Pólya class*

On the critical line the block $1 + N^{-\sqrt{s}}$ becomes

$$1 + N^{-i\tau} = 2e^{-i\tau\ell/2}\cos\big(\tfrac{\ell\tau}{2}\big), \qquad \ell = \log N,$$

and $\cos(\ell\tau/2) = \mathbb{E}e^{i\tau(\ell/2)\varepsilon}$ with $\varepsilon = \pm 1$ equally likely. A product of blocks with nonnegative integer multiplicities $m_N$ is, up to the phase, the characteristic function of the Rademacher sum $\sum_N \sum_{j \le m_N} \frac{\log N}{2}\varepsilon_{N,j}$, and all its zeros are real, lying on the lattices $(2k+1)\pi/\log N$. Cardon's theorem on exponential sums [73] extends this: if $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are positive and $G$ is a real entire function of genus 0 or 1 with only real zeros, then

$$\sum_{\varepsilon \in \{\pm 1\}^n} G\Big(i\sum_k a_k\varepsilon_k\Big)\exp\Big(iz\sum_k b_k\varepsilon_k\Big),$$

with the same signs in both sums, has only real zeros.

The function $\zeta$ is not of this form. Formally $\zeta(\frac{1}{2} + i\tau) = \prod_N (1 + N^{-i\tau})^{B_N}$, but the exponents $B_N$ are not integers, and some are negative: by Theorem 7.1.8, at prime cubes $B_{p^3} = \frac{1}{3}p^{-3/2} - \frac{1}{3}p^{-1/2} < 0$.

In the kernel variable the solved models act as multipliers. By Pólya's Hilfssatz II [447, Hilfssatz II], if $a > 0$, $b$ is real and $G$ is a real entire function of genus 0 or 1 with only real zeros, at least one of them, then $e^{ib}G(z+ia) + e^{-ib}G(z-ia)$ has only real zeros. With $G = \Xi$ this says that replacing $\Phi(t)$ by $\Phi(t)\cosh(at)$, a solved model in the variable $t$, cannot create nonreal zeros: if $\Xi$ has only real zeros, so does the transform of the new kernel. For $a \ge \frac{1}{2}$ the conclusion needs no hypothesis: the transform of $\Phi(t)\cosh(at)$ is $\frac{1}{2}(\xi(s+a) + \xi(s-a))$ with $s = \frac{1}{2} + iz$, and Lagarias proved that for $a \ge \frac{1}{2}$ its zeros all lie on the critical line and are simple [321]. Products of such factors converge to Gaussians, $\prod_{k \le n} \cosh(ct/\sqrt{n}) \to e^{c^2t^2/2}$, and this is de Bruijn's route to the Gaussian multipliers $e^{\lambda t^2}$ [112]. In the variable $t$, multiplying $\Phi(t)$ by $e^{\lambda t^2/4}$ is the de Bruijn Newman flow at time $\lambda$ (Section 8.4), the flow of Theorem 7.2.16, so the multiplier $e^{\lambda t^2}$ is the flow at time $4\lambda$. The solved models are therefore the elementary steps of the de Bruijn Newman flow, and the bound $\Lambda \le 0.2$ [419] says that the flow at time 0.2, applied to $\Phi$ or, in the normalisation of Rodgers and Tao, to $\Phi_{\mathrm{dBN}}$ defined before Theorem 7.2.16, already produces a kernel whose transform has only real zeros.

Proposition 7.1.15. *Let $G$ be a real even entire function of order less than 2 with $G(0) \neq 0$. Then $G$ is in the Laguerre Pólya class if and only if $s \mapsto G(0)/G(i\sqrt{s})$ is the Laplace transform of a GGC whose Thorin measure is $\sum_k m_k \delta_{\alpha_k^2}$ with integers $m_k \ge 1$, where $\pm\alpha_k$ are the zeros of $G$ with multiplicities $m_k$.*

*Proof.* By Hadamard, $G(z) = G(0)\prod_k (1 - z^2/\alpha_k^2)^{m_k}$ with no exponential factor, since $G$ is even and of order less than 2. All $\alpha_k$ are real if and only if

$G(0)/G(i\sqrt{s}) = \prod_k (1 + s/\alpha_k^2)^{-m_k}$ is a product of gamma transforms, which is the stated GGC. If some $\alpha_k^2$ is not positive, the product has a zero or a pole off $(-\infty, 0]$, which a GGC transform cannot have. □

The Laguerre Pólya class consists of the locally uniform limits of real polynomials with only real zeros, and a real entire function of order less than 2 belongs to it exactly when its zeros are real. In the language of Proposition 7.1.15, Pólya's multiplier theorem [74, Proposition 1], for even $G$ of order less than 2, reads: if $f$ is even and integrable with $f(t) = O(e^{-|t|^b})$ for some $b > 2$, $\int f(t)e^{izt}dt$ has only real zeros, and $G(0)/G(i\sqrt{s})$ is an integer atom GGC, then $\int G(it)f(t)e^{izt}dt$ has only real zeros. RH is the statement that $R = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is an integer atom GGC. The Thorin measures in the Pólya theory are always integer valued.

### *7.1.3.2 Lee and Yang, and Newman*

Pólya's Hilfssatz II has a physical reading, to which Kac drew attention in his comments on Pólya's 1926 paper [454, pp. 424 to 426]: it is the one spin case of the Lee and Yang circle theorem for the Ising ferromagnet [331]. Cardon's exponential sums make the correspondence explicit. For $\varepsilon \in \{\pm 1\}^n$ the sum

$$\mathcal{Z}_n(z) = \sum_{\varepsilon} G\Big(i \sum_k a_k \varepsilon_k\Big) \exp\Big(iz \sum_k b_k \varepsilon_k\Big)$$

is a partition function. The spins are weighted by $G(i \sum_k a_k \varepsilon_k)$, a function of a magnetisation, and they feel the purely imaginary external fields $izb_k$. For the Gaussian member of the Laguerre Pólya class, $G(x) = e^{-\beta x^2/2}$, which has genus 2 and lies outside Cardon's theorem, the weight is $\exp(\beta(\sum_k a_k \varepsilon_k)^2/2)$, the Curie Weiss ferromagnet with couplings $\beta a_j a_k \geq 0$. The zeros of $\mathcal{Z}_n$ in $z$ are then real by the Lee and Yang theorem [331], since off the real axis the fields $izb_k$ have real parts of one sign. A general even $G$ in the class is a limit of products of such Gaussians and of factors $1 + M^2/\alpha^2$ in the magnetisation $M = \sum_k a_k \varepsilon_k$, and Cardon's theorem is the Lee and Yang theorem for this family of ferromagnetic interactions of the magnetisation.

Newman made the connection with the Riemann Hypothesis precise [381]. In a 1974 paper on generalised Ising systems [379] he had characterised the single spin laws with the Lee and Yang property: a symmetric single spin law $\rho$ with $\int e^{bt^2}\rho(dt) < \infty$ for every $b$ has the Lee and Yang property, that ferromagnetic systems of $\rho$ spins have zeros only on the imaginary field axis, exactly when its moment generating function $\int e^{ht}\rho(dt)$ has only purely imaginary zeros, that is when $\int e^{izt}\rho(dt)$ is in the Laguerre Pólya class. In Thorin language this reads as follows.

PROPOSITION 7.1.16. *Let $\rho$ be a symmetric law with Gaussian moments of all orders. Then $\rho$ has the Lee and Yang property if and only if $s \mapsto 1/\int e^{\sqrt{st}}\rho(dt)$ is*

*the Laplace transform of a GGC with zero drift whose Thorin measure has integer atoms. The atoms are at the squared zeros $\alpha_k^2$ of $\int e^{izt}\rho(dt)$, with the multiplicities of the zeros.*

*Proof.* By Newman's theorem [379] the Lee and Yang property holds exactly when $\int e^{izt}\rho(dt)$ is in the Laguerre Pólya class, and then, by Hadamard, $\int e^{ht}\rho(dt) = e^{bh^2}\prod_k(1+h^2/\alpha_k^2)^{m_k}$ with $b \geq 0$. In fact $b = 0$: for every $\epsilon > 0$, $e^{ht} \leq e^{\epsilon h^2 + t^2/4\epsilon}$ gives $\int e^{ht}\rho(dt) \leq C_\epsilon e^{\epsilon h^2}$ for real $h$, while the right side is at least $e^{bh^2}$. With $h = \sqrt{s}$ the reciprocal is $\prod_k(1+s/\alpha_k^2)^{-m_k}$, a GGC with zero drift and Thorin measure $\sum m_k \delta_{\alpha_k^2}$. Conversely, as in Proposition 7.1.15, a GGC transform with zero drift and integer atoms $m_k$ at $\alpha_k^2$ is $\prod_k(1+s/\alpha_k^2)^{-m_k}$, whose reciprocal at $s = -z^2$ is $\prod_k(1-z^2/\alpha_k^2)^{m_k}$, a function in the Laguerre Pólya class. □

Apply this to the Pólya variable $X$: since $\mathbb{E}e^{hX} = \xi(\frac{1}{2}+h)/\xi(\frac{1}{2})$, we have $1/\mathbb{E}e^{\sqrt{s}X} = R(s)$, the reciprocal Xi function of Section 6.1.1. Three statements therefore coincide: RH; the law of $X$ is a Lee and Yang single spin law; $R$ is a GGC with no drift whose Thorin measure has an atom at each $\gamma^2$, of mass the multiplicity of the zero. The de Bruijn Newman flow at time $\lambda$ multiplies $\Phi(t)$ by $e^{\lambda t^2/4}$, which in Ising terms adds the ferromagnetic self interaction $e^{\lambda t^2/4}$ to the Pólya variable viewed as a spin, and $\Lambda$ is the least $\lambda$ for which the resulting spin is Lee and Yang. Newman's conjecture $\Lambda \geq 0$, proved by Rodgers and Tao [463], says that no antiferromagnetic self interaction can be tolerated.

### *7.1.3.3 A Cardon route*

Cardon's Fourier theorem [74, Theorem 1] gives a constructive sufficient condition. If

$$\Phi(t)\,dt = \Xi(0)\,G(it)\,\Pr\Big(\sum_k a_k\varepsilon_k \in dt\Big), \tag{7.1.4}$$

where the $\varepsilon_k$ are independent fair signs, the $a_k > 0$ are nonincreasing, and $G$ is real, even, entire of order less than 2 with only real zeros, then $\Xi$ has only real zeros. In the Ising reading, (7.1.4) says that the Pólya variable is the magnetisation of a weighted Ising paramagnet with weights $a_k$, tilted by a ferromagnetic interaction $G(iM)$ of its magnetisation $M = \sum_k a_k\varepsilon_k$. Since $|G(it)| \geq |G(0)|$ for even $G$ in the class, the law of $\sum_k a_k\varepsilon_k$ must carry the doubly exponential decay of $\Phi$, and among Rademacher laws the random harmonic series does.

Proposition 7.1.17. *For $\kappa > -1$ let $H_\kappa = \sum_{k\geq 1}\varepsilon_k/(2(k+\kappa))$, with characteristic function $\hat{f}_\kappa(v) = \prod_k \cos(v/(2(k+\kappa)))$ and density $f_\kappa$.*

*(i) As $s \to \infty$, $\log \mathbb{E}e^{sH_\kappa} = \frac{s}{2}\log s + A(\kappa)s + o(s)$ with $A(\kappa) = A(0) - \frac{1}{2}(\psi(1+\kappa)+\gamma_E)$ (here $\psi$ is the digamma function), $A(0) = -0.1485757\ldots$, so $\log\Pr(H_\kappa > u) = -\frac{1}{2}e^{2u-2A(\kappa)-1}(1+o(1))$ as $u \to \infty$. Since $\log\Phi(t) =$*

$-\pi e^{2t} + \frac{9}{2}t + O(1)$ *as* $t \to +\infty$, *the double exponential terms agree exactly when* $A(\kappa) = -\frac{1}{2}(1 + \log 2\pi)$, *that is at* $\kappa^* = 6.61845\ldots$.

*(ii)* $\limsup_{|v|\to\infty} |v|^{-1} \log|\hat{f}_\kappa(v)| = -\pi/4$ *for every* $\kappa$. *Hence* $f_\kappa$ *is analytic in the strip* $|\operatorname{Im} t| < \pi/4$, *which is exactly the strip of analyticity of* $\Phi$.

*Proof.* (i) Put $\ell_\kappa(s) = \log \mathbb{E}e^{sH_\kappa} = \sum_{k\geq 1} \log\cosh \frac{s}{2(k+\kappa)}$. For $\kappa = 0$ put $\lambda = s/2$ and split the sum at $k = \lambda$, using $\log\cosh y = y - \log 2 + \log(1 + e^{-2y})$ for $k \leq \lambda$. Since $\sum_{k\leq\lambda} 1/k = \log\lambda + \gamma_E + O(1/\lambda)$, and since $k \mapsto \log(1 + e^{-2\lambda/k})$ and $k \mapsto \log\cosh(\lambda/k)$ are monotone, so that their sums differ from the corresponding integrals by $O(1)$,

$$\ell_0(s) = \lambda\log\lambda + (\gamma_E - \log 2)\lambda + \lambda\int_1^\infty \frac{\log(1+e^{-2y})}{y^2}\,dy + \lambda\int_0^1 \frac{\log\cosh y}{y^2}\,dy + O(1),$$

which is $\frac{s}{2}\log s + A(0)s + O(1)$ with $A(0) = \frac{1}{2}\big(\gamma_E - 2\log 2 + \int_1^\infty y^{-2}\log(1+e^{-2y})\,dy + \int_0^1 y^{-2}\log\cosh y\,dy\big)$. For general $\kappa$ the mean value theorem gives $s^{-1}(\ell_\kappa - \ell_0)(s) = \sum_k \tanh(s\xi_k)\big(\frac{1}{2(k+\kappa)} - \frac{1}{2k}\big)$ with $\xi_k$ between $\frac{1}{2(k+\kappa)}$ and $\frac{1}{2k}$, and since $\tanh(s\xi_k) \to 1$, dominated convergence gives the limit $\frac{1}{2}\sum_k(\frac{1}{k+\kappa} - \frac{1}{k}) = -\frac{1}{2}(\psi(1+\kappa) + \gamma_E)$. For the tail put $s_u = e^{2u - 2A(\kappa) - 1}$. Chebyshev's exponential inequality at $s = s_u$ gives $\log\Pr(H_\kappa > u) \leq \ell_\kappa(s_u) - s_u u = -\frac{1}{2}s_u(1 + o(1))$. For the lower bound tilt the law of $H_\kappa$ by $e^{sx - \ell_\kappa(s)}$. By convexity of $\ell_\kappa$ the tilted mean is $\ell_\kappa'(s) = \frac{1}{2}\log s + A(\kappa) + \frac{1}{2} + o(1)$, and the tilted variance is $\ell_\kappa''(s) = \sum_k (2(k+\kappa))^{-2}\operatorname{sech}^2\frac{s}{2(k+\kappa)} = O(1/s)$, by comparison with an integral. Given $\delta > 0$, choose $s$ with $\ell_\kappa'(s) = u + \delta/2$; by Chebyshev's inequality the tilted law gives probability at least $\frac{1}{2}$ to $(u, u+\delta]$ for large $u$, so $\log\Pr(H_\kappa > u) \geq \ell_\kappa(s) - s(u+\delta) - \log 2 = -\frac{1}{2}s_u e^{\delta}(1+\delta)(1+o(1))$. Letting $\delta \to 0$ gives the tail.

(ii) Since $\hat{f}_\kappa$ is even we take $v \to +\infty$. First, $\int_0^\infty \log|\cos y|\, y^{-2}dy = -\pi/2$: as $\log|\cos|$ is even and $\pi$ periodic and $\sum_{n\in\mathbb{Z}}(y + n\pi)^{-2} = \csc^2 y$, the integral equals $\int_0^{\pi/2}\log(\cos y)\csc^2 y\,dy$, and an integration by parts against $-\cot y$ turns this into $-\int_0^{\pi/2}\cot y\tan y\,dy = -\pi/2$. For the upper bound fix $M, Y > 0$ and put $g_M = \max(\log|\cos|, -M)$, which is continuous, of bounded variation on $[0, Y]$ and $O(y^2)$ at 0. The points $y_k = v/(2(k+\kappa))$ have spacing at most $2y_k^2/v$, so $v^{-1}\sum_{y_k\leq Y} g_M(y_k) \to \frac{1}{2}\int_0^Y g_M(y)y^{-2}dy$ as a Riemann sum. Since $\log|\cos| \leq g_M \leq 0$, this gives $\limsup_{v\to\infty} v^{-1}\log|\hat{f}_\kappa(v)| \leq \frac{1}{2}\int_0^Y g_M(y)y^{-2}dy$, and letting $Y \to \infty$ and then $M \to \infty$ gives the bound $-\pi/4$. For the lower bound, the same Riemann sum argument, applied to the bounded continuous function $\Psi(y)/y$ with

$\Psi(y) = \int_0^y \log|\cos u|\,du$, gives

$$\int_0^V \log|\hat f_\kappa(v)|\,dv = \sum_k 2(k+\kappa)\Psi\Big(\frac{V}{2(k+\kappa)}\Big) = \frac{V^2}{2}\int_0^\infty \frac{\Psi(y)}{y^3}\,dy + o(V^2) = -\frac{\pi V^2}{8} + o(V^2),$$

since an integration by parts gives $\int_0^\infty \Psi(y)y^{-3}dy = \frac12\int_0^\infty \log|\cos y|\,y^{-2}dy = -\pi/4$. If $\log|\hat f_\kappa(v)| \le -(\pi/4+\epsilon)v$ for all large $v$, the left side would be at most $-(\pi/8+\epsilon/2)V^2 + O(1)$. Finally, $|\hat f_\kappa(v)| \le C_\epsilon e^{-(\pi/4-\epsilon)|v|}$ makes the Fourier integral of $\hat f_\kappa$ converge locally uniformly in $|\operatorname{Im} t| < \pi/4 - \epsilon$, so $f_\kappa$ is analytic there. □

The coincidence of the strips in (ii) has a common source: both densities are Fourier transforms of functions that decay at the exponential rate $\pi/4$ on the real axis. For $f_\kappa$ this is (ii). For $\Phi$, $|\Xi(\tau)| \le e^{-\pi|\tau|/4+o(|\tau|)}$ by Stirling's formula, since $\zeta(\frac12+i\tau)$ grows at most polynomially, and the strip ends at $|\operatorname{Im} t| = \pi/4$ because $\Phi$ is a theta series in $e^{2t}$, which reaches the imaginary axis there.

With both leading orders matched, the question is the ratio

$$r(t) = \frac{\Phi(t)}{\Xi(0) f_{\kappa^*}(t)},$$

which must equal $G(it)$. We computed $f_{\kappa^*}$ on the real axis by a saddle point contour integral of $\mathbb{E}e^{sH_{\kappa^*}}$, and on the imaginary axis by the Fourier integral $f_{\kappa^*}(ix) = \frac1\pi\int_0^\infty \hat f_{\kappa^*}(v)\cosh(vx)\,dv$. The two methods agree at the origin, $f_{\kappa^*}(0) = 2.104533$, to seven digits.

Proposition 7.1.18. *The ratio $r$ satisfies the following necessary conditions for* (7.1.4) *(computed).*

*(i) Growth. $r$ is even, $r(0) = 0.853936$, and $\log r(t)$ becomes linear with slope approaching $\frac72$: the increments over steps of 0.1 fall from 0.3534 to 0.3514 between $t = 1.5$ and $t = 2.2$. The slope is explained exactly: $\Phi$ carries the prefactor $e^{9t/2}$, and the saddle point prefactor $\sqrt{s^*/\pi} \approx e^t$ of $f_{\kappa^*}$ removes one unit. So $G$ would have exponential type $\frac72$, compatible with order one.*

*(ii) Stieltjes signs. With $L(s) = \log r(\sqrt{s}) - \log r(0)$, a Laguerre Pólya $G$ requires $L'(s) = \sum m_k/(s+\alpha_k^2)$, a Stieltjes function. On a grid of 81 points in $s \in [0,4]$ the difference quotients of $L$ give $L' > 0$, $L'' < 0$ and $L''' > 0$ everywhere, and the first order quotients fall from 4.07 on the first grid interval to 0.94 on the last.*

*(iii) The imaginary axis. The continuation $G(x) = r(-ix)$, computed independently from $\Phi(ix)$ and $f_{\kappa^*}(ix)$, is real, agrees with $r$ at second order*

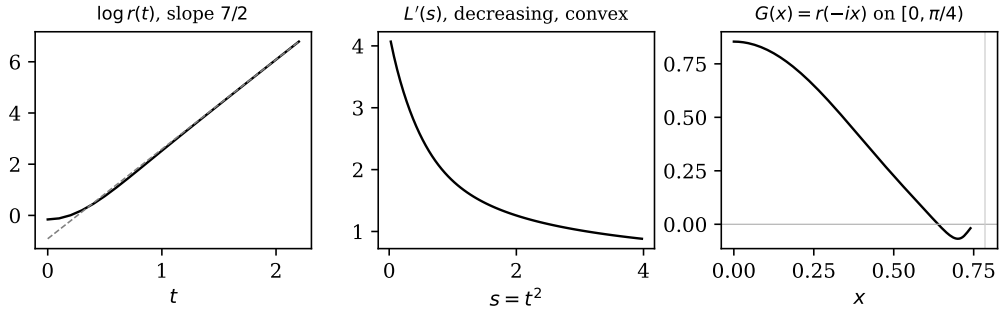



Figure 7.2: The Cardon ratio for the random harmonic law $H_{\kappa^*}$. Left: $\log r(t)$ on the real axis, asymptotically linear with slope $\frac{7}{2}$ (dashed). Centre: $L'(s)$ for $L(s) = \log(r(\sqrt{s})/r(0))$, positive, decreasing and convex as a Stieltjes function must be. Right: the continuation $G(x) = r(-ix)$ on $[0, \pi/4)$, with a simple zero at $x = 0.6388$.

> *(*$\log(G(0.1)/G(0)) = -0.0423$ *against* $\log(r(0.1)/r(0)) = 0.0417$*), decreases from* $0.8539$*, has a simple zero at* $\alpha_1 = 0.6388$*, and satisfies the Laguerre inequality* $G'^2 - GG'' \geq 2.4 > 0$ *on* $[0, 0.735]$*.*

Everything visible inside the strip is consistent with $G$ being a Laguerre Pólya function of exponential type $\frac{7}{2}$. The decisive question cannot be settled inside the strip.

Proposition 7.1.19. *If* (7.1.4) *holds with* $G$ *entire, then*

$$f_{\kappa^*}(t) = \Phi(t)/(\Xi(0)G(it)),$$

*so* $f_{\kappa^*}$ *continues meromorphically exactly as far as* $\Phi$ *does, and the two share their singular behaviour on the lines* $\operatorname{Im} t = \pm\pi/4$*, apart from the zeros of* $G$*. Since* $\Phi$ *has a natural boundary on these lines, coming from the Jacobi theta function at the edge of its modular domain,* $f_{\kappa^*}$ *must have the same natural boundary, with singular behaviour proportional to that of* $\Phi$ *through a factor that continues to an entire function.*

The first sentence of Proposition 7.1.19 is immediate. The second uses the classical fact that the theta function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ has the line $\operatorname{Re} x = 0$ as a natural boundary [458, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], together with $\Phi(t) = 4x^{9/4}\omega''(x) + 6x^{5/4}\omega'(x)$ at $x = e^{2t}$.

So the random harmonic law fits the Pólya kernel inside the strip, and the question is the boundary, where the two functions have different origins: $\Phi$ is a modular theta series and $f_{\kappa^*}$ a Rademacher product whose characteristic function has real zeros at $v = (2j+1)\pi(k+\kappa^*)$, increasingly dense. We see no reason for the Rademacher product to reproduce the modular singularities of $\Phi$, and this

suggests that the right Ising system for the Pólya variable is not the harmonic paramagnet but one whose magnetisation law carries theta function structure.

*7.1.3.4 The Pólya variable as the logarithm of a GGC*

PROPOSITION 7.1.20. *For $0 < q < 1$ the triple product*

$$\sum_{n\in\mathbb{Z}} q^{n^2} e^{inv} = \prod_{m\geq 1} (1-q^{2m})\big(1 + 2q^{2m-1}\cos v + q^{4m-2}\big)$$

*exhibits the discrete Gaussian law $\Pr(J=n) \propto q^{n^2}$ as the law of $\sum_m (B_m - B'_m)$ with independent Bernoulli variables of parameter $q^{2m-1}/(1+q^{2m-1})$. Each factor vanishes at $v = \pi \pm i(2m-1)\log(1/q)$, so the characteristic function has no real zeros and the moment generating function has no purely imaginary zeros: the triple product spins are not Lee and Yang spins, and neither is the discrete Gaussian.*

*Proof.* $1 + 2q^{2m-1}\cos v + q^{4m-2} = (1+q^{2m-1}e^{iv})(1+q^{2m-1}e^{-iv})$, and each factor, normalised, is the characteristic function of a Bernoulli variable or its negative. The zeros are those of $1 + q^{2m-1}e^{\pm iv}$. □

The triple product places the theta function in the field variable. For $\Phi$ the theta function enters multiplicatively, through the modular variable $e^{2t}$; the identity of Biane, Pitman and Yor [49], going back to Williams [518] (Section 1.2), makes this exact.

THEOREM 7.1.21. *Let $S_2 = \sum_{k\geq 1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponentials $e_k, e'_k$, so that $\mathbb{E}e^{-sS_2} = \big(\sqrt{2s}/\sinh\sqrt{2s}\big)^2$ and $S_2$ is a GGC with Thorin measure $2\sum_k \delta_{\pi^2k^2/2}$. Then*

$$2\xi(z) = \mathbb{E}\Big[\Big(\frac{\pi S_2}{2}\Big)^{z/2}\Big].$$

*Consequently, with $Y = \frac{1}{2}\log(\pi S_2/2)$, $2\Xi(\tau) = \mathbb{E}[e^{Y/2}e^{i\tau Y}]$, and the Pólya variable $X$ is $Y$ tilted by $e^{Y/2}$: $\Phi(t)\,dt = \frac{1}{2}e^{t/2}\Pr(Y \in dt)$.*

*Proof.* This is the identity of [49, Section 2]; the density of $S_2$ is a theta series by the partial fraction expansion of $(x/\sinh x)^2$ and Poisson summation, and its Mellin transform is $2\xi$. □

Section 9.1.1 tilts $Y$ to every real basepoint $\sigma$: with no hypothesis, $\xi(\sigma+z)/\xi(\sigma)$ is the moment generating function of $\frac{1}{2}\log(\frac{\pi}{2}S_2)$ under the law of $S_2$ size biased by $S_2^{\sigma/2}$.

Theorem 7.1.21 shows where the modular boundary comes from. It is the quadratic lattice $\pi^2k^2/2$ of the Thorin atoms of $S_2$, through Jacobi's theta function and Poisson summation, that produces the modular natural boundary at $\operatorname{Im} t =$

$\pm\pi/4$: the density of $S_2$ is analytic in $\operatorname{Re} x > 0$ with natural boundary on $\operatorname{Re} x = 0$, and $x = \frac{2}{\pi}e^{2t}$. The Thorin structure thus enters twice: $R$ is the reciprocal moment generating function of the Pólya variable in $\sqrt{s}$, and $S_2$ is a GGC in the usual sense. RH is the statement that every zero of $w \mapsto \mathbb{E}[(\pi S_2/2)^w]$ in the strip $0 < \operatorname{Re} w < \frac{1}{2}$ lies on the line $\operatorname{Re} w = \frac{1}{4}$.

This suggests a multiplicative Lee and Yang property. We say that a positive random variable $V$ has *Mellin zeros on a line* if $\mathbb{E}V^w$, continued, vanishes only on one vertical line. The solved models of the Vere-Jones step have exact multiplicative analogues.

PROPOSITION 7.1.22. *Let $V = \sum_{k\le n} c_k e_k$ with independent standard exponentials $e_k$ and distinct $c_k > 0$. Then $\mathbb{E}V^{u-1} = \Gamma(u)\sum_k A_k c_k^u$ with $A_k = c_k^{n-2}/\prod_{j\neq k}(c_k - c_j)$.*

*(i) For $n = 2$ the zeros are $u = 2\pi i m/\log(c_1/c_2)$, $m \neq 0$: all on the line $\operatorname{Re} u = 0$. The two exponential sum is a multiplicative solved model.*

*(ii) For $n = 3$ with $c = (1, \frac{1}{4}, \frac{1}{9})$, a truncation of the quadratic lattice with single atoms, the zeros in $0 < \operatorname{Im} u < 30$ have real parts 0.488, −1.268, 0.300, 0.387, −1.318, 0.457, 0.136, −1.129, 0.500 (computed): they do not lie on a line.*

*Proof.* The density of $V$ is $\sum_k A_k e^{-x/c_k}$ by partial fractions of $\prod_k (1 + c_k s)^{-1}$, and its Mellin transform is the stated exponential polynomial times $\Gamma(u)$. For $n = 2$ it is proportional to $c_1^u - c_2^u$. For $n = 3$ the zeros are those of $1.5 - 2.4\cdot 4^{-u} + 0.9\cdot 9^{-u}$, which has none with $|\operatorname{Re} u| \ge 8$; the argument principle on $[-8, 8] \times [0.01, 30]$ counts nine, none lies in $0 < \operatorname{Im} u < 0.01$, and Newton's method locates them. □

So the Lee and Yang mechanism survives the passage from additive to multiplicative magnetisations for two components and fails at three.

### 7.1.4 Harmonic means and scale mixtures of the reciprocal Xi function

The reciprocal Xi function $1/G(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, the Laplace transform of the clock $T$ (Section 1.2), is a harmonic mean of transforms. For $x \neq 0$ put $L_x(s) = 1/\cosh(x\sqrt{s})$. From $\cosh z = \prod_{k\ge 1}\left(1 + z^2/(\pi^2(k - \frac{1}{2})^2)\right)$, $L_x$ is a GGC transform with Thorin atoms $\pi^2(k - \frac{1}{2})^2/x^2$; probabilistically, if $C$ is the exit time of Brownian motion from $(-1, 1)$, then $\mathbb{E}e^{-\theta C} = 1/\cosh\sqrt{2\theta}$ and $L_x(s) = \mathbb{E}\exp(-\frac{1}{2}x^2 sC)$.

PROPOSITION 7.1.23. $\mathbb{E}e^{-sT} = 1/\mathbb{E}[1/L_X(s)]$ *with $X$ the Pólya variable: the reciprocal Xi transform is the harmonic mean, over the Pólya density, of Brownian exit time transforms. By Jensen's inequality, $\mathbb{E}e^{-sT} \le \mathbb{E}\exp(-\frac{1}{2}sX^2C)$, with $C$ independent of $X$.*

*Proof.* By (1.2.4), $1/\mathbb{E}e^{-sT} = G(s) = \mathbb{E}\cosh(\sqrt{s}X) = \mathbb{E}[1/L_X(s)]$, and $y \mapsto 1/y$ is convex. □

The arithmetic mean on the right is the Laplace transform of the mixture $\frac{1}{2}X^2C$. A harmonic mean of transforms is not obviously a transform at all. If harmonic means of GGC transforms were GGC, Proposition 7.1.23 would make the clock a GGC, which is RH (Proposition 1.2.1). That fails: a harmonic mean of GGC transforms need not even be a transform.

Example 7.1.24. The harmonic mean of the GGC transforms $(1+s)^{-2}$ and $(1+2s)^{-2}$ is $1/(1+3s+\frac{5}{2}s^2)$, with poles at $-0.6 \pm 0.2i$. Its inverse transform $2e^{-0.6t}\sin(0.2t)$ is negative on $(5\pi, 10\pi)$, so the harmonic mean is not even the Laplace transform of a positive measure.

Example 7.1.25. Every scale mixture $\Gamma_2 Y$ of a Gamma(2) variable $\Gamma_2$ by an independent $Y \geq 0$ is infinitely divisible [316], and it is a GGC if $Y$ is, since products of independent GGC variables are GGC [57]. Without a hypothesis on $Y$ it need not be: let $Y$ equal 1 or $\lambda > 0$, $\lambda \neq 1$, with probability $\frac{1}{2}$ each. The transform $\mathbb{E}(1+sY)^{-2}$ has numerator $(1+\lambda s)^2 + (1+s)^2$, which vanishes at non real $s$. A GGC transform has no zeros off $(-\infty, 0]$, so $\Gamma_2 Y$ is infinitely divisible but not a GGC.

A last negative result excludes Gamma laws as the outer layer of the clock itself, as Theorem 11.2.10 does for $|X|$, $X^2$ and $S_2$.

Theorem 7.1.26. *For no $a > 0$ and no $Y \geq 0$ independent of $\Gamma_a$ does the clock $T$ have the law of $\Gamma_a Y$.*

*Proof.* Suppose it did. Then $Y > 0$ almost surely, since $\mathbb{E}e^{-sT} = 1/G(s) \to 0$ as $s \to \infty$, and $\Gamma_a Y$ has density $t^{a-1}g(t)/\Gamma(a)$ with $g(t) = \mathbb{E}[Y^{-a}e^{-t/Y}]$ completely monotone and not identically zero, so $g \geq g(1) > 0$ on $(0,1]$ and $\mathbb{E}e^{-sT} \geq c_1 s^{-a}$ for large $s$. But $G(s) \geq \cosh(x_0\sqrt{s})\Pr(|X| \geq x_0)$ for every $x_0 > 0$, so $\mathbb{E}e^{-sT} \leq c_2 e^{-x_0\sqrt{s}}$ with $c_2 = 2/\Pr(|X| \geq x_0)$, and the two bounds contradict each other as $s \to \infty$. □

## 7.2 Truncations, finite conditions and the road to RH

Cutting the series for $S_2$ in Theorem 7.1.21 after $n$ terms gives variables whose Mellin transforms are explicit, each a gamma factor times an exponential polynomial, with zeros that converge to the points $\rho/2$; a condition on these zeros, holding for infinitely many $n$, implies RH. The one sided condition fails because the truncated zeros approach the critical line from a side that changes at zeros where $|S(\gamma)|$ is large, with $S(t) = \pi^{-1}\arg\zeta(\frac{1}{2}+it)$ as in Appendix C. Symmetrising by the functional equation removes this mechanism.

### 7.2.1 Truncations of the Biane, Pitman and Yor lattice

In this subsection $w$ is the complex variable of a Mellin transform and $u = w + 1$.

#### 7.2.1.1 Truncated lattices and Conjecture T

Let $e_k, e'_k$ be independent standard exponential variables, and put $\mathcal{S} = \sum_{k\geq 1} k^{-2}(e_k + e'_k) = \frac{\pi^2}{2} S_2$, with $S_2$ the variable of Theorem 7.1.21, and $\mathcal{S}_n = \sum_{k\leq n} k^{-2}(e_k + e'_k)$, a GGC with finitely many Thorin atoms. Put $F(w) = \mathbb{E}\mathcal{S}^{\overline{w}}$ and $F_n(w) = \mathbb{E}\mathcal{S}_n^w$ for $\operatorname{Re} w > -2$. By Theorem 7.1.21,

$$F(w) = 2\pi^w \, \xi(2w),$$

so the zeros of $F$ are the points $w = \rho/2$, and RH says they lie on $\operatorname{Re} w = \frac{1}{4}$. Each $F_n$ is explicit. Define $\alpha_k$ and $\beta_k$, which depend on $n$, by the partial fraction expansion

$$\prod_{j\leq n}\Big(1+\frac{\lambda}{j^2}\Big)^{-2} = \sum_{k\leq n} k^{-2}\Big\{\alpha_k\Big(1+\frac{\lambda}{k^2}\Big)^{-1} + \beta_k\Big(1+\frac{\lambda}{k^2}\Big)^{-2}\Big\},$$

so that $\beta_k = k^2 \prod_{j\leq n,\, j\neq k}(1-k^2/j^2)^{-2}$ and $\alpha_k = -2\beta_k \sum_{j\leq n,\, j\neq k} k^2/(j^2-k^2)$; for $n = 2$ they are $\beta = (\frac{16}{9}, \frac{4}{9})$ and $\alpha = (-\frac{32}{27}, \frac{32}{27})$. The left side is the Laplace transform of $\mathcal{S}_n$, whose density is therefore $\sum_k (\alpha_k + \beta_k k^2 x) e^{-k^2 x}$, and taking moments gives, with $u = w + 1$, $F_n(u-1) = \Gamma(u) Q_n(u)$, where $Q_n(u) = \sum_{k\leq n} k^{-2u}(\alpha_k + \beta_k u)$.

Theorem 7.2.1. *If for infinitely many $n$ the function $F_n$ has no zeros in the half plane $\operatorname{Re} w > \frac{1}{4}$, then the Riemann Hypothesis holds.*

*Proof.* For $\operatorname{Re} w > -2$ the variables $\mathcal{S}_n^w$ are dominated by an integrable variable, since $\mathcal{S}_n \geq e_1 + e'_1$ and $\mathcal{S}_n \leq \mathcal{S}$, so $F_n \to F$ locally uniformly on $\operatorname{Re} w > -2$. If $F$ had a zero $w_0$ with $\operatorname{Re} w_0 > \frac{1}{4}$, Hurwitz's theorem would give zeros of $F_n$ near $w_0$ for all large $n$. Hence every zero $\rho/2$ of $F$ has $\operatorname{Re}\rho \leq \frac{1}{2}$, and the symmetry $\rho \mapsto 1-\rho$ gives $\operatorname{Re}\rho = \frac{1}{2}$. □

Conjecture T. For every $n \geq 1$, $F_n(w) = \mathbb{E}\big[\big(\sum_{k\leq n} k^{-2}(e_k + e'_k)\big)^w\big]$ has no zeros with $\operatorname{Re} w > \frac{1}{4}$.

Conjecture T is a stability statement, of Hermite Biehler or Lee and Yang type [331]. It is false (Corollary 7.2.4), and we state it because the mechanism of its failure also defeats, for large $n$, the Hermite Biehler inequality on which the bounds of Theorems 7.2.9 and 7.2.11 for the symmetrised truncations rest.

At small $n$ the evidence supports it: Table 7.1 finds no zero of $Q_n$ with $\operatorname{Re} u > \frac{5}{4}$ in any rectangle searched. The zeros are counted by the argument principle and located by Newton's method from a grid; the counts and the zeros found agree.

| $n$ | region, in $u = w + 1$ | zeros with $\operatorname{Re} u > \frac{5}{4}$ |
|---|---|---|
| $2, 3, 4, 5, 6, 8, 12, 16, 24, 32$ | $\frac{5}{4} < \operatorname{Re} u < 6,\ 0.3 < \operatorname{Im} u < 40$ | none |
| $8, 16, 32$ | $\frac{5}{4} < \operatorname{Re} u < 6,\ 40 < \operatorname{Im} u < 100$ | none |
| $48, 64$ | $\frac{5}{4} < \operatorname{Re} u < 6,\ 0.3 < \operatorname{Im} u < 30$ | none |
| $16, 32, 64$ | $6 < \operatorname{Re} u < 40,\ 0.3 < \operatorname{Im} u < 40$ | none |

Table 7.1: Argument principle counts for Conjecture T: rectangles searched and zeros of $Q_n$ found with $\operatorname{Re} u > \frac{5}{4}$.

For fixed $n$ and large $|\operatorname{Im} u|$ the zeros move far to the left, because the extreme coefficients $\beta_1$ and $\beta_n$ differ by a factor of order $16^n$, up to a power of $n$. The intermediate heights are therefore where Conjecture T is at risk, and they are the heights that Table 7.1 searches; the failure found in Corollary 7.2.4 lies above them (Remark 7.2.5). An experiment shows that the absence of such zeros at small $n$ is not a consequence of the GGC structure alone: replacing the weights $k^{-2}$ by $k^{-2}(1 + \eta_k)$, with independent $\eta_k$ uniform on $[-0.15, 0.15]$, one of two draws at $n = 16$ gives a zero with $\operatorname{Re} u > \frac{5}{4}$.

*7.2.1.2 How the truncated zeros reach the line, and why Conjecture T fails*

Proposition 7.2.2. *Let $\rho$ be a simple zero of $\xi$ and put*

$$\kappa(\rho) = \frac{\rho\,\xi(\rho - 2)}{2\pi\,\xi'(\rho)} = \frac{\rho\,\xi(3 - \rho)}{2\pi\,\xi'(\rho)}.$$

*There is a constant $C$ such that for all large $n$, $F_n$ has exactly one zero $w_n(\rho)$ in the disc $|w - \frac{\rho}{2} - \frac{\kappa(\rho)}{n}| < C/n^2$; thus $w_n(\rho) = \frac{\rho}{2} + \frac{\kappa(\rho)}{n} + O(n^{-2})$.*

*Proof.* Write $\mathcal{S} = \mathcal{S}_n + \mathcal{T}_n$ with $\mathcal{T}_n = \sum_{k>n} k^{-2}(e_k + e'_k)$, independent of $\mathcal{S}_n$, so that $\mathbb{E}\mathcal{T}_n = 2\sum_{k>n} k^{-2} = \frac{2}{n} + O(n^{-2})$ and $\mathbb{E}\mathcal{T}_n^2 = O(n^{-2})$. For $a > 0$, $b \geq 0$ and $0 < \operatorname{Re} w < 2$, Taylor's formula with integral remainder and $|(a + tb)^{w-2}| = (a + tb)^{\operatorname{Re} w - 2} \leq a^{\operatorname{Re} w - 2}$ give

$$\left|(a + b)^w - a^w - wb\,a^{w-1}\right| \leq \tfrac{1}{2}|w(w - 1)|\,b^2 a^{\operatorname{Re} w - 2},$$
$$\left|(a + b)^{w-1} - a^{w-1}\right| \leq |w - 1|\,b\,a^{\operatorname{Re} w - 2}.$$

Since $\mathcal{S}_n \geq e_1 + e'_1$, a $\Gamma(2)$ variable, $\mathbb{E}\mathcal{S}_n^{\operatorname{Re} w - 2} \leq \Gamma(\operatorname{Re} w)$. Taking expectations with $a = \mathcal{S}_n$ and $b = \mathcal{T}_n$, and using independence,

$$F(w) = F_n(w) + w\,\mathbb{E}\mathcal{T}_n\,F_n(w - 1) + O(n^{-2}), \qquad F_n(w - 1) = F(w - 1) + O(n^{-1}),$$

uniformly on compact subsets of $0 < \operatorname{Re} w < 2$. Hence $F_n(w) = F(w) - \frac{2}{n} wF(w - 1) + O(n^{-2})$ uniformly there. Put $w_0 = \rho/2$, so $0 < \operatorname{Re} w_0 < \frac{1}{2}$, and recall

$F(w_0) = 0$ and $F'(w_0) = 4\pi^{w_0}\xi'(\rho) \neq 0$. For $|w - w_0| = O(1/n)$, Taylor expansion of $F$ and of $wF(w-1)$ about $w_0$ gives

$$F_n(w) = F'(w_0)\Big(w - w_0 - \frac{\kappa(\rho)}{n}\Big) + O(n^{-2}), \qquad \frac{\kappa(\rho)}{n} = \frac{2}{n}\,\frac{w_0 F(w_0 - 1)}{F'(w_0)},$$

and $F(w_0 - 1) = 2\pi^{w_0 - 1}\xi(\rho - 2)$ turns the second identity into the stated $\kappa$. On the circle $|w - w_0 - \kappa/n| = C/n^2$ the linear term has modulus $|F'(w_0)|C/n^2$, which exceeds the $O(n^{-2})$ error once $C$ is larger than its constant divided by $|F'(w_0)|$. Rouché's theorem gives exactly one zero inside for all large $n$. The second form of $\kappa$ follows from $\xi(s) = \xi(1 - s)$. □

If $\operatorname{Re}\kappa(\rho) > 0$ the zero $w_n(\rho)$ lies to the right of the critical line for all large $n$. This happens at zeros where $|S(\gamma)|$ is large, the first being $\gamma_{213}$ (Theorem 7.2.3 and the computation after it), and it is why Conjecture T fails.

For each of the first 100 zeros $\kappa(\rho)$ has negative real part, starting with $\kappa(\rho_1) = -4.762 - 1.944i$, $\kappa(\rho_2) = -8.673 - 0.627i$, $\kappa(\rho_3) = -10.149 - 3.872i$ and $\kappa(\rho_4) = -15.165 + 3.705i$. So these truncated zeros approach the critical line from the left, at distance about $-\operatorname{Re}\kappa(\rho)/n$, which grows with the height. Figure 7.3 shows the approach for the first four zeros, computed by locating the zeros of $F_n$ directly at each $n$ rather than from the expansion, an independent check on Proposition 7.2.2 at low height.

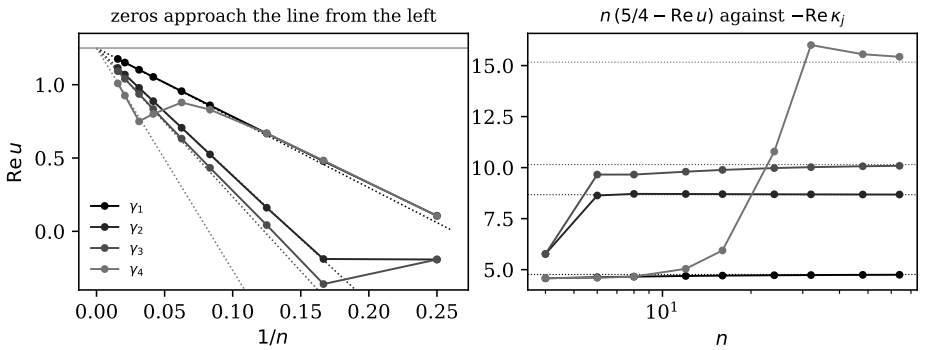



Figure 7.3: Left: real parts of the zeros of $F_n$ near $\rho_j/2$, in the variable $u = w + 1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against $n$, converging to $-\operatorname{Re}\kappa(\rho_j)$.

THEOREM 7.2.3. *Let $\rho = \frac{1}{2} + i\gamma$ be a simple zero on the critical line, let $\theta$ be the Riemann Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ Hardy's function, and*

*put* $a(\gamma) = e^{i\theta(\gamma)}\zeta(\frac{5}{2} + i\gamma)$. *Then*

$$\operatorname{Re}\kappa(\rho) = \frac{(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma) - 4\gamma \operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)},$$
$$\operatorname{Im}\kappa(\rho) = \frac{4\gamma \operatorname{Im} a(\gamma) + (\gamma^2 - \frac{15}{4})\operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)}.$$

*Proof.* On the line $\xi(\frac{1}{2} + it) = \Xi(t)$ is real, so $\xi'(\rho) = -i\,\Xi'(\gamma)$, and $\Xi(t) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ gives $\Xi'(\gamma) = -\frac{1}{2}(\gamma^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|\overline{Z'(\gamma)}$ at a zero. Since $\xi(3-\rho) = \xi(\frac{5}{2} - i\gamma)$ and $\xi$ is real on the real axis, $\xi(3-\rho) = \overline{\xi(\frac{5}{2} + i\gamma)}$, and $\Gamma(\frac{5}{4} + \frac{i\gamma}{2}) = (\frac{1}{4} + \frac{i\gamma}{2})\Gamma(\frac{1}{4} + \frac{i\gamma}{2})$ with $\pi^{-i\gamma/2}\Gamma(\frac{1}{4} + \frac{i\gamma}{2}) = |\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|e^{i\theta(\gamma)}$. Substituting, the moduli of $\Gamma$ cancel, $(\frac{1}{2} + i\gamma)(\frac{1}{4} - \frac{i\gamma}{2}) = \frac{1}{2}(\gamma^2 + \frac{1}{4})$ cancels the remaining factor, and $\kappa = -i(\frac{5}{2} - i\gamma)(\frac{3}{2} - i\gamma)\overline{a(\gamma)}/(4\pi^2 Z'(\gamma))$, whose real and imaginary parts are as stated. □

For large $\gamma$ the term $(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma)$ decides the sign of $\operatorname{Re}\kappa(\rho)$. Since $|\zeta(\frac{5}{2} + i\gamma) - 1| \le \zeta(\frac{5}{2}) - 1 < 0.35$, $a(\gamma)$ is a positive multiple of $e^{i\theta(\gamma)}$ up to a phase below 0.35 in modulus, and up to that phase the sign of $\operatorname{Re}\kappa(\rho)$ is the sign of $\sin\theta(\gamma)/Z'(\gamma)$. Let $S_{\mathrm{mid}}$ be the midpoint of the jump of $S$ at the $n$th zero $\gamma_n$. Then $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S_{\mathrm{mid}})$ and the sign of $Z'(\gamma_n)$ is $(-1)^{n+1}$, so the sign of $\sin\theta(\gamma_n)/Z'(\gamma_n)$ is that of $-\cos(\pi S_{\mathrm{mid}})$: negative when $|S_{\mathrm{mid}}| < \frac{1}{2}$ and positive when $\frac{1}{2} < |S_{\mathrm{mid}}| < \frac{3}{2}$. The phase of $\zeta(\frac{5}{2} + i\gamma)$ moves the threshold $\frac{1}{2}$ by at most about 0.11 either way. For the first 460 zeros the sign of $\operatorname{Re}\kappa$ agrees with the sign of $\operatorname{Im} a(\gamma)/Z'(\gamma)$ in every case, and $\operatorname{Re}\kappa > 0$ occurs three times, at

$$\gamma_{213} = 415.4552, \quad \gamma_{289} = 527.9036, \quad \gamma_{379} = 650.6687,$$

with $\kappa = 233.96 - 2298.91i$, $604.16 + 3074.08i$, $1164.60 - 4296.91i$.

At these three zeros the midpoints of the jumps are 0.598, −0.645 and 0.664. Seven of the first 460 zeros have $S(\gamma^-) < -1$: besides $\gamma_{289}$ these are $\gamma_{127}$, $\gamma_{196}$, $\gamma_{233}$, $\gamma_{368}$, $\gamma_{380}$ and $\gamma_{401}$, with midpoints −0.506, −0.505, −0.537, −0.556, −0.538 and −0.514, and $\operatorname{Re}\kappa$ is negative at every one of them. Three further zeros have midpoints above $\frac{1}{2}$ and $\operatorname{Re}\kappa < 0$: $\gamma_{136}$, $\gamma_{256}$ and $\gamma_{399}$, at 0.507, 0.510 and 0.539. The largest $|S_{\mathrm{mid}}|$ with $\operatorname{Re}\kappa < 0$ is 0.556, at $\gamma_{368}$, and the smallest with $\operatorname{Re}\kappa > 0$ is 0.598, at $\gamma_{213}$, so the threshold on the midpoint lies between about 0.56 and 0.60, inside the range $\frac{1}{2} \pm 0.11$ allowed by the phase of $\zeta(\frac{5}{2} + i\gamma)$.

Corollary 7.2.4. *Conjecture T is false: for all sufficiently large $n$, $F_n$ has a zero with $\operatorname{Re} w > \frac{1}{4}$ near $\rho_{213}/2$.*

*Proof.* The first $1.5 \times 10^9$ zeros, $\rho_{213}$ among them, lie on the critical line and are simple [508], and ball arithmetic applied to the formula of Theorem 7.2.3 gives $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ (Section 7.2.3). By Proposition 7.2.2, for

all large $n$ the function $F_n$ has a zero $w_n$ with $|w_n - \frac{\rho_{213}}{2} - \frac{\kappa(\rho_{213})}{n}| < C/n^2$, so $\operatorname{Re} w_n > \frac{1}{4} + \frac{233.95}{n} - \frac{C}{n^2} > \frac{1}{4}$ once $n > C/233.95$. □

REMARK 7.2.5. The constants in Proposition 7.2.2 depend on $\rho$, and the first order regime requires $n$ large compared with $|\kappa(\rho)|$. For the zero $\rho_{213}$ of Corollary 7.2.4, $|\kappa(\rho_{213})| = 2311$, so the phenomenon begins at $n$ of order $10^3$ or beyond, larger than every $n$ computed in this section.

The failure is intrinsic to the method, not to the particular lattice. Any truncation whose first order error is a multiple of $F(w-1)$ produces the displacement $\kappa(\rho)/n$, and the phase of $\kappa(\rho)$ is governed by $S(\gamma)$. By Selberg's central limit theorem [477] $S$ takes values beyond $\pm\frac{1}{2}$ for a positive proportion of heights, so one expects a positive proportion of zeros to be approached from the right. One sided Hurwitz arguments therefore cannot succeed for truncations of this kind; what is needed is a scheme in which symmetry keeps the zeros on the line. The Dyck chain of Section 25.1 is an analogue in the variable $s = 2w$: its first order error is a multiple of $s\,\xi(s-1)$ where the lattice has $\xi(s-2)$, with a displacement of order $n^{-1/2}$, and the first zero approached from the right is $\rho_{64}$ (the index computed in ball arithmetic; Propositions 25.1.3 and 25.1.4; conditional on Conjecture 25.1.2).

### 7.2.1.3 Symmetrised truncations and Conjecture S

The functional equation supplies that symmetry. Put

$$K_n(w) = \pi^{-w} F_n(w) + \pi^{w-\frac{1}{2}} F_n(\tfrac{1}{2} - w), \qquad Z_n(y) = K_n(\tfrac{1}{4} + iy).$$

PROPOSITION 7.2.6. *$K_n(\frac{1}{2} - w) = K_n(w)$, $K_n(\bar{w}) = \overline{K_n(w)}$, so $Z_n$ is real for real $y$ and the zeros of $K_n$ off the line $\operatorname{Re} w = \frac{1}{4}$ come in quadruples. As $n \to \infty$, $K_n \to 4\,\xi(2w)$ locally uniformly on $-\frac{7}{4} < \operatorname{Re} w < \frac{9}{4}$. Every simple zero $\frac{1}{4} + i\frac{\gamma}{2}$ of $\xi(2w)$ on the line is, for all large $n$, the limit of a zero of $K_n$ lying exactly on the line.*

*Proof.* The symmetries are immediate from the definition and the reality of $F_n$ on the real axis. The limit follows from $F_n \to F = 2\pi^w \xi(2w)$ and $\xi(1-2w) = \xi(2w)$; since $F_n \to F$ on $\operatorname{Re} w > -2$, the convergence holds on $-2 < \operatorname{Re} w < \frac{5}{2}$, which contains the strip of the statement. Near a simple zero $y_0$ of the real function $4\xi(\frac{1}{2} + 2iy)$, $Z_n$ converges uniformly and so changes sign near $y_0$; by Hurwitz, $K_n$ has exactly one zero near $\frac{1}{4} + iy_0$, which must be the real zero of $Z_n$, since an off line zero would bring its mirror image $\frac{1}{2} - \bar{w}$ with it. □

The Hurwitz argument now gives the following.

THEOREM 7.2.7. *If for infinitely many $n$ all zeros of $K_n$ in the strip $0 < \operatorname{Re} w < \frac{1}{2}$ lie on the line $\operatorname{Re} w = \frac{1}{4}$, then RH holds.*

*Proof.* A zero $\rho/2$ of $\xi(2w)$ off the line lies in the strip $0 < \operatorname{Re} w < \frac{1}{2}$, since $0 < \operatorname{Re}\rho < 1$, and by Hurwitz it is the limit of zeros of $K_n$ in the strip, which would be off the line for large $n$. □

Conjecture S. For every $n \geq 1$, all zeros of $K_n$ in $0 < \operatorname{Re} w < \frac{1}{2}$ lie on $\operatorname{Re} w = \frac{1}{4}$.

The symmetrised zeros move along the line. The first order argument of Proposition 7.2.2, applied to $K_n = 2\operatorname{Re}[\pi^{-w}F_n(w)]$ on the line, gives

$$w_n^{\mathrm{sym}}(\rho) = \frac{\rho}{2} + \frac{i\,\operatorname{Im}\kappa(\rho)}{n} + O(n^{-2}) :$$

the symmetrisation removes the real part of the displacement and keeps the imaginary part.

| $n$ | window in $\operatorname{Im} w$ | zeros in $-4 < \operatorname{Re} w < 4.5$: total, on the line | zeros of $\xi$ |
|---|---|---|---|
| 1 | $(0.5, 40)$, $(40, 70)$ | 23, 21; 27, 27 | 21; 27 |
| 2 | $(0.5, 40)$, $(40, 70)$ | 23, 21; 27, 27 | 21; 27 |
| 3 | $(0.5, 40)$, $(40, 70)$ | 23, 21; 27, 27 | 21; 27 |
| 4, 5, 6, 8, 16 | $(0.5, 40)$, $(40, 70)$ | 21, 21; 27, 27 | 21; 27 |
| 32 | $(0.5, 40)$ | 21, 21 | 21 |
| 8 | $(70, 140)$, $(195, 215)$ | 78, 78; 27, 27 | 78; 26 |

Table 7.2: Zeros of the symmetrised truncations $K_n$ in $-4 < \operatorname{Re} w < 4.5$: total found, how many lie on the line, and the number of zeros of $\xi$ in the doubled window for comparison. In the window $195 < \operatorname{Im} w < 215$ the count for $K_8$ exceeds that for $\xi$ by one; see the text.

Table 7.2 counts the zeros of $K_n$ in windows of height. For $n = 1$ the one off line quadruple lies at $\operatorname{Re} w = -2.001, 2.501$, $\operatorname{Im} w = \pm 2.471$, and for $n = 2$ at $\operatorname{Re} w = -2.949, 3.449$, $\operatorname{Im} w = \pm 3.261$; for $n = 3$ it is similar. These quadruples have $|\operatorname{Re} w - \frac{1}{4}| > 2$, so they lie outside the strip $0 < \operatorname{Re} w < \frac{1}{2}$ of Theorem 7.2.7 and do not bear on Conjecture S. For $n \geq 4$ every zero found lies on the line.

The window $195 < \operatorname{Im} w < 215$ matters because it contains $\gamma_{213}/2 = 207.73$, where the unsymmetrised truncations are forced off the line. There the count for $K_8$ exceeds that for $\xi$ by one: the zeros of $\xi$ with $390 < \gamma < 430$ are $\gamma_{196}$ to $\gamma_{221}$, twenty six of them, the nearest excluded ordinates being $\gamma_{195} = 388.85$ and $\gamma_{222} = 430.33$. The difference is not a contradiction: at this height and $n = 8$ the first order displacement along the line, $\operatorname{Im}\kappa(\rho)/n$, is of order 300 for $\rho_{213}$, so $K_8$ is far from its limit and its zeros need not correspond one to one with those of $\xi$. What matters for Conjecture S is that every zero found in the window lies on the line.

The symmetrised truncations are the lattice analogue of the approximations $\zeta_N(s) = \sum_{m\le N} m^{-s} + \chi(s)\sum_{m\le N} m^{s-1}$ derived from the approximate functional equation, with $\chi$ the factor in $\zeta(s) = \chi(s)\zeta(1-s)$. Spira [483, Section 6] and Gonek and Montgomery [155] studied their zeros, and Gonek and Montgomery proved that for $N \le T^{o(1)}$ almost all zeros of $\zeta_N$ with ordinates in $[0,T]$ lie on the critical line and are simple. A proof of Conjecture S for infinitely many $n$ would require controlling the zeros of $Z_n$ uniformly in height, including the range of heights comparable with $n$, where $K_n$ is neither close to $4\xi(2w)$ nor dominated by a single exponential term; this is the same difficulty as for $\zeta_N$.

Proposition 7.2.8. *Write $s = \sigma + i\tau$.*

*(i) All but finitely many zeros of $\zeta_2$ in $0 < \sigma < 1$ lie on the critical line, and the number with $0 < \tau \le T$ is $\vartheta(T)/\pi + O(1)$.*

*(ii) Let $\sigma^*$ be the root of $2^{-\sigma} + 3^{-\sigma} = 1$. For every $\sigma_0 \in (\frac12, \sigma^*)$ there are zeros $\sigma_k + i\tau_k$ of $\zeta_3$ with $\tau_k \to \infty$ and $\sigma_k \to \sigma_0$. In particular $\zeta_3$ has zeros off the critical line at arbitrarily large heights.*

*This follows a note of Nicholas Polson merged into the book [235]; compare [483, 155].*

*Sketch.* (i) With $G(s) = \Gamma_{\mathbb R}(s)(1+2^{-s})$ we have $\Gamma_{\mathbb R}(s)\zeta_2(s) = G(s) + G(1-s)$, which is $2\operatorname{Re} G$ on the critical line. A zero needs $L(\sigma,\tau) = \log|G(\sigma+i\tau)| - \log|G(1-\sigma-i\tau)| = 0$, and by Stirling's formula

$$L(\sigma,\tau) = (\sigma-\tfrac12)\log\frac{|\tau|}{2\pi} + \log|1+2^{-s}| - \log|1+2^{s-1}| + O(|\tau|^{-1}).$$

For $0 < \sigma \le \frac12$ the last two terms are at most $\log 2 - \log(1-2^{-1/2})$, so $L < 0$ unless $\frac12 - \sigma = O(1/\log|\tau|)$; on $[\frac14, \frac12)$ they have bounded $\sigma$ derivative while that of the first term is $\log(|\tau|/2\pi)$, and $L(\frac12,\tau) = 0$, so $L < 0$ there too once $|\tau|$ is large. The symmetry $s \mapsto 1-\bar s$ covers $\sigma > \frac12$. On the line $G(\frac12 + i\tau) = |G|e^{i\Theta(\tau)}$ with $\Theta(\tau) = \vartheta(\tau) + \arg(1+2^{-\frac12 - i\tau})$; the second term has modulus at most $\frac\pi4$ and a bounded derivative, so $\Theta$ is eventually increasing and the count is $\Theta(T)/\pi + O(1)$. (ii) For $\sigma > \frac12$, $|\chi(s)| \asymp (|\tau|/2\pi)^{\frac12-\sigma} \to 0$, so near $\sigma_0 + i\tau$ the function $\zeta_3$ is $1 + 2^{-s} + 3^{-s} + o(1)$. Since $2^{-\sigma_0} + 3^{-\sigma_0} > 1 > |2^{-\sigma_0} - 3^{-\sigma_0}|$ there are unit numbers $u, v$ with $1 + 2^{-\sigma_0}u + 3^{-\sigma_0}v = 0$, and $\sigma_0$ is a simple zero of $w \mapsto 1 + 2^{-w}u + 3^{-w}v$, because a double zero would force $2^{-\sigma_0}u = \log 3/(\log 2 - \log 3)$, of modulus larger than 1. As $\log 2$ and $\log 3$ are linearly independent over $\mathbb Q$, Kronecker's theorem gives $\tau_k \to \infty$ with $(2^{-i\tau_k}, 3^{-i\tau_k}) \to (u, v)$, and Rouché's theorem, applied first to $1 + 2^{-s} + 3^{-s}$ near $\sigma_0 + i\tau_k$ and then to $\zeta_3$, gives the zeros. $\square$

*7.2.1.4 A single term bound for small n*

For small $n$, Conjecture S reduces to a finite computation, because at large heights the gamma factor makes $\log|A_n|$ increase across the strip. Here $A_n(w) = \pi^{-w}F_n(w)$, so that $K_n(w) = A_n(w) + A_n^*(w)$ with $A_n^*(w) = A_n(\frac{1}{2} - w)$ and $|A_n^*(\sigma + iy)| = |A_n(\frac{1}{2} - \sigma + iy)|$. If $\sigma \mapsto \log|A_n(\sigma + iy)|$ is increasing on $(0, \frac{1}{2})$, then $|A_n| > |A_n^*|$ for $\frac{1}{4} < \sigma < \frac{1}{2}$, and $K_n$ has no zeros there at height $y$; by symmetry none to the left of the line either. The derivative is $\operatorname{Re}\frac{A_n'}{A_n} = -\log\pi + \operatorname{Re}\psi(w+1) + \operatorname{Re}\frac{Q_n'}{Q_n}(w+1)$, and for large $y$ the digamma term dominates.

THEOREM 7.2.9. *Let $Q_n(u) = \sum_k k^{-2u}(\alpha_k + \beta_k u)$ as above and, for $y > |\alpha_1|/|\beta_1|$, let $\delta_n(y) = \sum_{k\geq 2} k^{-2}\frac{|\beta_k|y + |\alpha_k|}{|\beta_1|y - |\alpha_1|}$ and*

$$\delta_n'(y) = \sum_{k\geq 2} k^{-2}\frac{2\log k\,(|\beta_k|y + |\alpha_k|) + |\beta_k|}{|\beta_1|y - |\alpha_1|} + \delta_n(y)\,\frac{|\beta_1|}{|\beta_1|y - |\alpha_1|},$$

*which bounds a derivative in the proof and is not the derivative of $\delta_n$. If $\delta_n(y) < 1$ and*

$$\log\frac{y}{\pi} - \frac{1}{y} - \frac{|\beta_1|}{|\beta_1|y - |\alpha_1|} - \frac{\delta_n'(y)}{1 - \delta_n(y)} > 0,$$

*then $K_n$ has no zeros off the line in $0 < \operatorname{Re} w < \frac{1}{2}$ at any height $\geq y$. The condition holds for all $y \geq Y_n$, with $Y_1 = 4.81$, $Y_2 = 5.85$, $Y_3 = 7.11$, $Y_4 = 9.08$, $Y_5 = 12.8$, $Y_6 = 20.8$, $Y_8 = 104.1$ and $Y_{10} = 21232$.*

*Proof.* $|\psi(z) - \log z| \leq 1/|z|$ for $\operatorname{Re} z \geq 1$, so $\operatorname{Re}\psi(w+1) \geq \log y - \frac{1}{y}$ at height $y$. Write $Q_n(u) = (\alpha_1 + \beta_1 u)(1 + \delta(u))$. On $1 \leq \operatorname{Re} u \leq \frac{3}{2}$, $|u| \geq y$, we have $|k^{-2u}| \leq k^{-2}$, so $|\delta(u)| \leq \delta_n(y)$, and differentiating the quotient $\delta(u)$ term by term gives $|\delta'(u)| \leq \delta_n'(y)$; both bounds decrease in $y$. Then $\operatorname{Re}\frac{Q_n'}{Q_n} \geq -\frac{|\beta_1|}{|\alpha_1 + \beta_1 u|} - \frac{|\delta'|}{1 - |\delta|}$, and the condition makes $\operatorname{Re}(A_n'/A_n) > 0$ on the strip at every height $\geq y$. The values of $Y_n$ come from evaluating these bounds with the exact coefficients. □

COROLLARY 7.2.10. *Conjecture S holds for $n = 1, 2, 3, 4, 5, 6, 8$ above the heights $Y_n$ of Theorem 7.2.9, and is verified numerically below them, by a floating point argument principle count and not by an enclosure.*

*Proof.* Above $Y_n$ this is Theorem 7.2.9. Below it, the argument principle counts cover the strip $0 < \operatorname{Re} w < \frac{1}{2}$ for $0 < \operatorname{Im} w < 6, 7, 8$ when $n = 1, 2, 3$, and the windows of Table 7.2 cover $0 < \operatorname{Im} w < 70$ for $4 \leq n \leq 6$ and $0 < \operatorname{Im} w < 140$ for $n = 8$, with $0 < \operatorname{Im} w < 0.6$ checked separately and $K_n > 0$ on the real segment; in each case the total count in the strip equals the number of sign changes of $Z_n$ on the line. □

The single term bound of Theorem 7.2.9 deteriorates quickly: the lattice coefficients $|\beta_k/\beta_1|$ grow with $n$, $Y_{10} = 21232$, and for $n \geq 12$ the bound fails because $\delta_n > 1$.

### 7.2.1.5 Torus bounds and where they stop

Write $u = x + iy$ and $Q_n(u) = u\,P_n(u) + R_n(u)$ with the Dirichlet polynomials $P_n(u) = \sum_{k\leq n} \beta_k k^{-2u}$ and $R_n(u) = \sum_{k\leq n} \alpha_k k^{-2u}$. At height $y$ the values of $P_n$ depend on $y$ only through the prime phases $(p^{-2iy})_{p\leq n}$, which lie on the torus $\mathbb{T}^{\pi(n)}$. Replacing them by independent phases $\omega = (\omega_p)_{p\leq n} \in \mathbb{T}^{\pi(n)}$ gives the torus function

$$P_n(x,\omega) = \sum_{k\leq n} \beta_k k^{-2x} \exp\Big(-i \sum_{p\leq n} v_p(k)\,\omega_p\Big),$$

and in the same way the torus function $P_n'(x,\omega)$ of $P_n'(u) = -\sum_k 2\log k\,\beta_k k^{-2u}$. Their extremes over the torus bound the values at every height. Put $r(x) = \sum_k |\alpha_k| k^{-2x}$ and $r'(x) = \sum_k 2\log k\,|\alpha_k| k^{-2x}$.

THEOREM 7.2.11. *Let $m_n = \min |P_n(x,\omega)|$, $M_n = \max\{-\operatorname{Re}(P_n'/P_n)(x,\omega)\}$ and $\ell_n = \max |P_n'/P_n|(x,\omega)$, over $\omega \in \mathbb{T}^{\pi(n)}$ and the band $1.2 \leq x \leq 1.3$, and let*

$$B_n(y) = \max_{0.3\leq\sigma\leq\frac12,\ \omega} \frac{\begin{array}{r}\log\big(|P_n(\frac32-\sigma,\omega)| + r(\frac32-\sigma)/y\big)\\ -\log\big(|P_n(1+\sigma,\omega)| - r(1+\sigma)/y\big)\end{array}}{2\sigma - \frac12}.$$

*If $m_n > r(1.2)/y$,*

$$\log\frac{y}{\pi} - \frac{2}{y} - M_n - \frac{\ell_n r(1.2) + r'(1.2) + r(1.2)/y}{y\,(m_n - r(1.2)/y)} > 0 \quad \textit{and} \quad \log\frac{y}{\pi} - \frac{1}{y} > B_n(y),$$

*then $K_n$ has no zeros off the line in $0 < \operatorname{Re} w < \frac12$ at height $y$, and the same holds at every larger height.*

*Proof.* For $\frac14 < \sigma \leq 0.3$ both $\sigma$ and $\frac12 - \sigma$ lie in $[0.2, 0.3]$, and the first condition makes $\operatorname{Re}(A_n'/A_n) > 0$ there: $\operatorname{Re}\psi(w+1) \geq \log y - \frac1y$, $\operatorname{Re}\frac1u > 0$, and $\frac{Q_n'}{Q_n} = \frac1u + \frac{P_n' + R_n'/u - R_n/u^2}{P_n + R_n/u}$ differs from $\frac1u + \frac{P_n'}{P_n}$ by at most the stated correction. So $\log|A_n|$ increases on $[\frac12 - \sigma, \sigma]$ and $|A_n| > |A_n^*|$. For $0.3 \leq \sigma \leq \frac12$ compare directly: $\log|\Gamma(1+\sigma+iy)| - \log|\Gamma(\frac32 - \sigma + iy)| \geq (2\sigma - \frac12)(\log y - \frac1y)$ by integrating $\operatorname{Re}\psi$, $|1+\sigma+iy| > |\frac32 - \sigma + iy|$, and $|P_n + R_n/u|$ is bounded below on the right by $|P_n| - r/y$ and above on the left by $|P_n| + r/y$; the second condition then gives $|A_n(\sigma+iy)| > |A_n(\frac12 - \sigma + iy)|$. Both conditions improve as $y$ increases. □

Table 7.3 lists bounds for the torus extremes (Remark 7.2.13) and the smallest integer heights at which the first condition holds with these bounds. For $n = 7$ and

$9 \le n \le 15$ we use a single band variant: if $m_n > r(1)/y$ and the first condition of Theorem 7.2.11 holds with $m_n$, $M_n$, $\ell_n$ the extremes over the band $1 \le x \le \frac{3}{2}$ and with $r(1)$, $r'(1)$ in place of $r(1.2)$, $r'(1.2)$, then $\operatorname{Re}(A_n'/A_n) > 0$ on $0 < \operatorname{Re} w < \frac{1}{2}$ at height $y$, so $\log|A_n|$ increases across the strip and $K_n$ has no zeros off the line there; the condition improves as $y$ increases. The proof is the first part of the proof of Theorem 7.2.11 with the band widened, since $r$ and $r'$ decrease in $x$ and $\operatorname{Re}\psi(w+1) \ge \log y - \frac{1}{y}$ on the whole strip.

| $n$ | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 32 | 36 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $m_n \ge$ | 1.062 | 1.003 | 0.947 | 0.895 | 0.846 | 0.800 | 0.757 | 0.716 | 0.677 | 0.361 | 0.203 |
| $M_n \le$ | 0.862 | 0.895 | 0.926 | 0.956 | 0.985 | 1.012 | 1.039 | 1.064 | 1.089 | 1.259 | 1.331 |
| $\ell_n \le$ | 3.600 | 3.948 | 4.310 | 4.689 | 5.086 | 5.504 | 5.954 | 6.473 | 7.075 | 17.203 | 33.568 |
| $r(1.2) \le$ | 7.522 | 7.776 | 8.009 | 8.223 | 8.420 | 8.600 | 8.834 | 9.062 | 9.280 | 10.675 | 11.313 |
| $r'(1.2) \le$ | 4.648 | 5.019 | 5.361 | 5.675 | 5.961 | 6.222 | 6.650 | 7.078 | 7.489 | 10.181 | 11.577 |
| threshold | 30 | 34 | 37 | 41 | 46 | 50 | 56 | 62 | 69 | 213 | 558 |

Table 7.3: Bounds for the torus extremes over the band $1.2 \le x \le 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 7.2.11 holds with these values. The bounds are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 7.2.13); the programs are in `verification_reciprocal_zeta/torus/`.

| $n$ | 7 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|
| $m_n \ge$ | 1.403 | 1.052 | 0.892 | 0.744 | 0.607 | 0.482 | 0.367 | 0.261 |
| $M_n \le$ | 0.576 | 0.715 | 0.777 | 0.834 | 0.887 | 0.937 | 0.984 | 1.028 |
| $\ell_n \le$ | 1.796 | 3.193 | 4.167 | 5.428 | 7.121 | 9.511 | 13.140 | 19.314 |
| $r(1) \le$ | 4.848 | 5.888 | 6.291 | 6.629 | 6.971 | 7.445 | 7.886 | 8.294 |
| $r'(1) \le$ | 2.014 | 3.096 | 3.520 | 3.869 | 4.282 | 5.032 | 5.746 | 6.418 |
| threshold | 14 | 24 | 31 | 41 | 57 | 84 | 131 | 229 |

Table 7.4: As Table 7.3, for the single band variant: bounds for the extremes over the band $1 \le x \le \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values.

COROLLARY 7.2.12. *Conjecture S holds for $n = 16, \dots, 24, 32, 36$ above the thresholds of Table 7.3 and for $n = 7$ and $9 \le n \le 15$ above those of Table 7.4, and for all these $n$ it is verified numerically below the thresholds, by floating point argument principle counts. With Corollary 7.2.10 this covers every $n \le 24$ and $n = 32, 36$.*

*Proof.* Above the thresholds this is Theorem 7.2.11, with the bounds of Table 7.3 and the second condition of Remark 7.2.13, and its single band variant, with the bounds of Table 7.4. The hypothesis $m_n > r/y$ and the first condition, here and in the variant, hold with the true values wherever they hold with these bounds,

because both improve as $m_n$ increases and as $M_n$, $\ell_n$ and the values of $r$ and $r'$ decrease. Below the thresholds the argument principle count in $0 < \operatorname{Re} w < \frac{1}{2}$ equals the number of sign changes of $Z_n$ on the line, with the following heights and counts:

| $n$ | 7 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|
| height | 15 | 24 | 32 | 43 | 58 | 86 | 132 | 230 | 500 | 35 |
| zeros | 3 | 9 | 14 | 23 | 36 | 64 | 116 | 242 | 649 | 17 |
| $n$ | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 32 | 36 | |
| height | 250 | 43 | 47 | 51 | 57 | 62 | 110 | 215 | 570 | |
| zeros | 269 | 23 | 26 | 30 | 35 | 40 | 90 | 221 | 763 | |

Each height is at least the threshold for the same $n$, with equality for $n = 9$ and $n = 23$, so the counts and the theorem together cover all heights. For $n = 36$ the sampling step on the line is 0.02, cross checked by the argument principle. Every count equals the number of zeros of $\xi$ with ordinate below twice the height, except for $n = 9$, where the ninth zero of $K_9$, at $\operatorname{Im} w = 23.82$, is the image of $\gamma_9/2 = 24.003$ displaced below height 24. □

Remark 7.2.13. The bounds of Tables 7.3 and 7.4 are proved by a branch and bound over the band and the half torus $0 \le \omega_2 \le \pi$, which suffices because $P_n(x, -\omega) = \overline{P_n(x, \omega)}$. On a box with centre $(x_c, \omega_c)$ and half widths $h$ in $x$ and $h_p$ in $\omega_p$, the $k$th term of $P_n$ differs from its value at the centre by at most $\beta_k k^{-2x_c}\{(k^{2h} - 1) + k^{2h} \min(2, \sum_p v_p(k) h_p)\}$, and the resulting enclosures of $P_n$, $P_n'$ and their partial derivatives bound $|P_n|$ from below and $-\operatorname{Re}(P_n'/P_n)$ and $|P_n'/P_n|$ from above on the box; for $M_n$ and $\ell_n$ the proved bound for $m_n$ serves as a lower bound for $|P_n|$. Since $2\sigma - \frac{1}{2} > 0$, the second condition of Theorem 7.2.11 at height $y$ is equivalent to

$$\Big(|P_n(1+\sigma,\omega)| - \frac{r(1+\sigma)}{y}\Big) \exp\Big(\big(2\sigma - \tfrac{1}{2}\big)\Big(\log\frac{y}{\pi} - \frac{1}{y}\Big)\Big) > |P_n(\tfrac{3}{2} - \sigma, \omega)| + \frac{r(\frac{3}{2} - \sigma)}{y}$$

for $0.3 \le \sigma \le \frac{1}{2}$ and all $\omega$, which also gives the positivity of $|P_n(1+\sigma,\omega)| - r(1+\sigma)/y$ that $B_n(y)$ needs. The same branch and bound, applied to the difference of the two sides, proves this at the heights 29, 33, 36, 40, 45, 49, 55, 61, 68, 211, 555 for $n = 16, \dots, 24, 32, 36$, each below the threshold, and so at every larger height. The searches run in double precision and only choose the boxes: every final box, about $1.1 \times 10^8$ in all, is proved again in ball arithmetic with Arb [290], with exact coefficients and on the exact box.

With the band $1.2 \le x \le 1.3$ the method stops between $n = 40$ and $n = 44$. The band minimum $m_n$ falls to 0.20 at $n = 36$, 0.056 at $n = 40$ and 0 at $n = 44$, so at $n = 44$ the hypothesis $m_n > r(1.2)/y$ of Theorem 7.2.11 fails at every height.

This reflects the functions and not only the bound: the torus function $P_n(x,\omega)$ then vanishes at a point of the band, and since by Kronecker's theorem the prime phases at height $y$ come arbitrarily close to every point of the torus, $A_n$ has zeros with real part near $[0.2, 0.3]$ at arbitrarily large heights, where $\log|A_n|$ cannot increase across the strip. On the line $x = \frac{5}{4}$ itself the minimum is still 0.29 at $n = 44$ and 0.19 at $n = 48$, so these torus zeros lie off the line, and a narrower band may postpone the stop. A torus zero with $x > \frac{5}{4}$ would end every comparison of this kind: it gives zeros of $A_n$ with $\operatorname{Re} w > \frac{1}{4}$ at arbitrarily large heights, where $|A_n(w)| > |A_n^*(w)|$ fails. On which side of the line the torus zeros at $n = 44$ lie is not determined here.

The monotonicity used in Theorem 7.2.9 implies the Hermite Biehler inequality $|A_n(w)| > |A_n(\frac{1}{2} - \bar{w})|$ for $\frac{1}{4} < \operatorname{Re} w < \frac{1}{2}$, and the inequality holds far beyond the proven range. On a grid of $\sigma \in \{0.26, 0.3, 0.4, 0.49\}$ and $y \le 100$ it holds for $n = 8, 16, 24$ at every height above 3.6, and it fails only below 3.6, where the strip contains no zeros. For $n = 8, 16$ it also holds for $200 \le y \le 212$, around $\gamma_{213}/2$. It cannot hold for all large $n$: by Corollary 7.2.4, for large $n$ the function $A_n$ has a zero $w$ with $\frac{1}{4} < \operatorname{Re} w < \frac{1}{2}$, where the inequality fails. Quantitatively, near a zero $\rho/2$ the first order margin is governed by the residue of $w\xi(2w-2)/\xi(2w)$, which is a positive multiple of $\kappa(\rho)$, so at zeros with $\operatorname{Re}\kappa(\rho) > 0$ the inequality reverses once $n$ is large compared with $|\kappa(\rho)|$. A proof of Conjecture S for infinitely many $n$ must establish real rootedness without a one sided inequality.

### 7.2.2 What is proved, and how far it reaches

No result of Chapter 6 or of this chapter proves RH. In Table 7.5 every entry in the middle column is a positivity statement about a transform, or about finitely many derivatives. Every entry on the left is a positivity statement about all derivatives, equivalently about a measure. The question is whether the middle column can be pushed to the left. So far it reaches Weil positivity on the Bernstein cone and $(-1)^n W^{(n)} > 0$ up to a finite order, and neither the de Bruijn Newman flow nor a GGC multiplier closes the gap. Chapters 9 to 11 measure the same gap from the right of the centre: $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is a GGC for every $\sigma \ge 1$ with no hypothesis (Theorem 7.1.14; Proposition 9.1.2 computes its Thorin measure at $\sigma = 1$), and for $\frac{1}{2} < \sigma < 1$ zero density estimates show that in the density $\rho_{\sigma-1/2}$ of (1.2.7) the Cauchy kernels of the zeros to the right of $\sigma$ carry, below height $T$, a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$ (Theorem 11.4.2).

#### *7.2.2.1 Weil's criterion in Thorin coordinates*

Proposition 6.2.16 left open which signed primal measures have a positive sine squared image. For $\tilde{\mu}_\xi$ the answer is Weil's criterion.

| Equivalent form of RH | Proved with no unproved hypothesis | Computation the middle column uses |
|---|---|---|
| $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is a GGC, Thorin measure $\sum \delta_{\gamma^2}$ [421, Proposition 3.2] | $e^{-c\sqrt{s}}/G$ is a GGC (Theorem 6.2.12); $e^{\sqrt{s}\zeta'/\zeta(\beta)}\zeta(\beta)/\zeta(\beta + \sqrt{s})$ is a GGC for $\beta > 1$ (Proposition 6.2.6) | none |
| $W(x) = \frac{1}{2}\sum_\rho e^{x(\rho-\frac{1}{2})^2}$ is completely monotone | $W > 0$ (proof of Theorem 7.2.15) and $(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$ (Theorem 5.1.13); $W + c/(2\sqrt{\pi x})$ is positive (Theorem 6.2.15) and $n$ monotone for $n \leq 10^{10}$ (Theorem 6.2.17) | positivity: RH to 50 and enclosures; order $1.16 \times 10^{25}$: RH to $3 \times 10^{12}$ and enclosures; $n$ monotonicity: RH to $3 \times 10^{12}$ |
| $e^{-c\sqrt{s}}R$ is a GGC | it is infinitely divisible (Theorem 6.2.15) | RH to 50 and enclosures |
| $w$ is the Laplace transform of a real measure on $[0, \infty)$ (Theorem 6.1.15) | $w > 0$ (Theorem 7.1.1) | RH to 50 and enclosures |
| the sine squared image of $\tilde{\mu}_\xi$ is positive, equivalently Weil's distribution is positive definite (Proposition 7.2.14) | its subordination stage is positive; Weil positivity holds on the Bernstein cone (Theorem 7.2.15), against every Gaussian (its proof) and against $r^{2n}e^{-xr^2}$ for $n \leq 1.16 \times 10^{25}$ (Theorem 5.1.13); the canonical function $k$ of the floor is $n$ monotone for every $n \leq 10^{10}$ (Theorem 6.2.17), which is strictly weaker than $n$ monotonicity of $W$ | Bernstein cone: RH to 50 and enclosures; order $1.16 \times 10^{25}$: RH to $3 \times 10^{12}$ and enclosures; $n$ monotonicity: RH to $3 \times 10^{12}$ |

Table 7.5: Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none.

PROPOSITION 7.2.14. *Let $D(x) = \sum_\rho e^{(\rho-\frac{1}{2})x}$ be Weil's distribution, whose action on a test function $h$ is $\sum_\rho \hat{h}(\gamma_\rho)$ with $\gamma_\rho = -i(\rho - \frac{1}{2})$. Then $R$ is the Laplace transform of a GGC, with Thorin measure the sine squared image of $\tilde{\mu}_\xi$, if and only if $D$ is positive definite, that is, $\sum_\rho \hat{h}(\gamma_\rho) \geq 0$ for every $h = g * \tilde{g}$ with $g$ smooth and compactly supported. Both conditions are equivalent to RH.*

*Proof.* By Proposition 1.2.1, $R$ is the Laplace transform of a GGC if and only if RH holds, and its Thorin measure is then $\sum_{\gamma>0} \delta_{\gamma^2}$, which is the sine squared image of $\tilde{\mu}_\xi$ by Proposition 6.2.16(ii). Weil's theorem [512, 55] states that RH holds if and only if $\sum_\rho \hat{h}(\gamma_\rho) \geq 0$ for all $h = g * \tilde{g}$ with $g$ smooth and compactly supported, that is, if and only if $D$ is positive definite. □

A formal computation shows how the two formulations correspond. For a positive measure $U$ on $(0,\infty)$ with $\int U(dt)/(1+t) < \infty$, the primal measure $\tilde{\mu}_U(dx) = -2\big(\int \cos(\sqrt{t}\,x)\,U(dt)\big)dx$ on $x > 0$, understood as a distribution, has sine squared image $U$: since $\int_0^\infty (1 - \cos x\sqrt{z})(-2\cos bx)\,dx = \pi\delta(\sqrt{z} - b)$, $\tilde{\mu}_U$ maps to $\int \delta(z-t)U(dt) = U$. The map of Lemma 6.2.5 is injective on the class of that lemma: if $\int_0^\infty \sin^2(x\sqrt{z}/2)\,\nu(dx) = 0$ for every $z > 0$, then $\psi(t) = \int(1 - \cos xt)\,\nu(dx)$ vanishes for all real $t$, and $\psi(t) - \frac{1}{2}\int_{t-1}^{t+1} \psi(s)\,ds = -\int \cos(xt)\big(1 - \frac{\sin x}{x}\big)\nu(dx)$ shows that the finite measure $(1 - \frac{\sin x}{x})\nu(dx)$ has vanishing cosine transform, so $\nu = 0$. Carried over to distributions, these two facts say that a primal measure has a positive sine squared image if and only if $-\tilde{\mu}$ is the restriction to $x > 0$ of an even positive definite distribution on $\mathbb{R}$, that is, by the Bochner Schwartz theorem, the Fourier transform of a positive measure. For $\xi$, $-\tilde{\mu}_\xi$ is the restriction to $x > 0$ of $D$, by (6.2.6); $D$ is even because $\rho \mapsto 1 - \rho$ permutes the zeros, and the explicit formula (6.2.5) is its prime side. The computation is formal, for two reasons: Lemma 6.2.5 is proved only for primal measures with $\int(1 \wedge x)|\tilde{\mu}|(dx) < \infty$, and when RH fails $D$ grows exponentially and is not tempered, so the Bochner Schwartz theorem applies only in Weil's own formulation, with compactly supported test functions. Riesz's criterion has a Thorin form of the same kind (Theorem 10.2.7 of Section 10.2).

The unconditional positivity results of Chapter 6, of this chapter and of Section 5.1 are partial positivity statements of Weil type. The gamma part is a positive definite archimedean piece (Theorem 6.2.12). The heat trace results are Weil positivity along the Gaussian family: with $h$ Gaussian, $\sum_\rho \hat{h}(\gamma_\rho) = 2W(x)$, and $W(x) > 0$ for every $x > 0$ unconditionally (Theorem 5.1.1 [424, Theorem A]). Weil positivity against $r^{2n}e^{-xr^2}$ is $(-1)^n W^{(n)} \geq 0$, which Theorem 5.1.6 [421, Theorem 7.10] proves for every $n \leq 10^{17}$ and Theorem 5.1.13 for every $n \leq 1.16 \times 10^{25}$. Theorem 6.2.17 proves the corresponding statement for the canonical function $k = W + c/(2\sqrt{\pi x})$ of the floor, for $n \leq 10^{10}$, and not for $W$ itself. The known unconditional cases of Weil positivity, due to Yoshida [525, Theorem 1] and to Bombieri [55], concern test functions of small support; the Gaussian family, in

which Theorems 5.1.6, 5.1.13 and 6.2.17 apply, is the opposite regime.

#### 7.2.2.2 The Bernstein cone and the resolution of finite order tests

Proposition 7.2.14 identifies the Thorin positivity of $R$ with Weil positivity for all test functions $\hat{h} = |\hat{g}|^2$. A cone of test functions of another kind, completely monotone in $r^2$, can be settled unconditionally.

THEOREM 7.2.15. *Let* $\hat{h}(r) = \int_0^\infty e^{-yr^2} g(y)\,dy$ *with* $g \geq 0$, $g \not\equiv 0$ *and* $\int_0^1 g(y)\,y^{-1/2}\big(1 + \log(1/y)\big)\,dy + \int_1^\infty g(y)\,dy < \infty$, *so that* $\hat{h}$ *is a completely monotone function of* $r^2$. *Then*

$$\sum_\rho \hat{h}(\gamma_\rho) = 2\int_0^\infty W(y)\,g(y)\,dy > 0,$$

*where* $\gamma_\rho = -i(\rho - \frac{1}{2})$ *and the sum is over all nontrivial zeros. In particular Weil's functional is positive for every test function of Thorin type* $\hat{h}(r) = \prod_k (1 + r^2/t_k)^{-1}$, *a nonempty product with* $t_k > 0$ *and* $\sum_k 1/t_k < \infty$, *for* $\hat{h}(r) = \operatorname{sech}(cr)$ *with* $c > 0$, *and for* $\hat{h}(r) = (1 + r^2/a)^{-m}$ *with* $a > 0$ *and* $m > \frac{1}{2}$. *The proof uses the verification of RH to height* 50 *and the interval enclosures of Theorem 7.1.1 together with the enclosures E5 and E6 of Section 7.2.3.*

*Proof.* For each $y > 0$, $\sum_\rho e^{-y\gamma_\rho^2} = \sum_\rho e^{y(\rho - \frac{1}{2})^2} = 2W(y)$, and $W(y) > 0$ for every $y > 0$. For $y \leq 0.003$ this comes from the prime side: $W = w + W_0$ with $w > 0$ (Theorem 7.1.1) and $W_0 \geq 0$, because the function $F$ of Lemma 6.1.12 satisfies $F(0.003) > 0.0329$ (enclosure E6) and is decreasing. For $y \geq 0.003$ it comes from the zeros below height 50, which lie on the line and contribute $e^{-y\gamma^2} > 0$: the first alone contributes more than $e^{-0.003 \cdot 14.14^2} > 0.548$ at $y = 0.003$, the off line tail above height 50 is below 0.0162 there (enclosure E5, with the bound $N(t+1) - N(t) \leq 2\log(t+1)$ of the proof of Theorem 7.1.1), and the ratio of the tail to the first term decreases as $y$ grows. The interchange of sum and integral is justified by Fubini's theorem: $\sum_\rho |e^{y(\rho - \frac{1}{2})^2}| \leq e^{y/4} \sum_\rho e^{-y(\operatorname{Im}\rho)^2}$, which is $O(y^{-1/2}(1 + \log(1/y)))$ for $y \leq 1$ by the Riemann von Mangoldt formula and bounded for $y \geq 1$, and the integrability condition makes this majorant integrable against $g$. The Thorin type functions are completely monotone in $r^2$ because $\prod(1 + s/t_k)^{-1}$ and $\operatorname{sech}(c\sqrt{s})$ are GGC transforms. The examples satisfy the integrability condition. A nonempty product $\prod_k (1 + s/t_k)^{-1}$ is the Laplace transform of $\sum_k e_k/t_k$, with $e_k$ independent standard exponential variables, and the density $g$ of this sum is at most $t_1$, the maximum of the density of $e_1/t_1$; the function $\operatorname{sech}(c\sqrt{s}) = \prod_{k \geq 1}\big(1 + 4c^2 s/((2k-1)^2\pi^2)\big)^{-1}$ is such a product; and for $(1 + r^2/a)^{-m}$ the function $g(y) = a^m y^{m-1} e^{-ay}/\Gamma(m)$ satisfies the condition because $m > \frac{1}{2}$. □

The Bernstein cone consists of test functions that decrease in $|r|$ and so cannot concentrate near a given height. This is why positivity on it does not imply RH,

and it matches the resolution analysis of Table 7.6: an off line zero at height $\gamma$ and distance $\delta$ from the line is detected only by test functions able to localise at $\gamma$. Theorem 5.1.13 gives $(-1)^n W^{(n)} > 0$ for $n \le 1.16 \times 10^{25}$, which is Weil positivity for $r^{2n} e^{-yr^2}$ and, after integration against $g$ where the integral converges, for the image of the Bernstein cone under multiplication by $r^{2n}$.

Complete monotonicity of $W$ is Bernstein's condition, and its Post Widder form is

$$L_n[W](t) = \frac{(-1)^n}{n!} \Big(\frac{n}{t}\Big)^{n+1} W^{(n)}\Big(\frac{n}{t}\Big) \ge 0 \qquad \text{for all } n, t.$$

$L_n[W]$ is the Thorin measure $\sum \delta_{\gamma^2}$ seen through a kernel of relative width about $n^{-1/2}$, and an off line quadruple $\rho = \frac{1}{2} \pm \delta \pm i\gamma$ contributes $2\,\mathrm{Re}\{(n/t)^{n+1} z^n e^{-zn/t}/n!\}$ with $z = \gamma^2 - \delta^2 - 2i\delta\gamma$. In Table 7.6 the quadruple is inserted midway between two consecutive true ordinates among the first 300 zeros.

| $\gamma$ | $\delta = 0.4$ | $\delta = 0.2$ | $\delta = 0.1$ | $\delta = 0.05$ |
|---|---|---|---|---|
| 27.7 | 724 | 1448 | 2896 | 5792 |
| 60.1 | 11585 | 23170 | 46340 | 92681 |
| 120.1 | 32768 | 65536 | 131072 | 262144 |

Table 7.6: Smallest order $n^*$ at which an inserted off line quadruple at height $\gamma$ and distance $\delta$ makes $\min_t L_n[W](t)$ negative.

In every row the detection order doubles when $\delta$ halves, within the resolution $\sqrt{2}$ of the scan, and across the rows it grows roughly like $\gamma^2$: the ratio $n^*\delta/\gamma^2$ lies between 0.38 and 1.3 in Table 7.6. So an off line zero at height $\gamma$ first becomes visible at order about $\gamma^2/\delta$, and a zero above the verified height $T_0$ of Theorem 6.2.17 at order about $T_0^2/\delta$. Theorem 5.1.13 reaches orders of this size, every order up to $1.29(T_0 - 23)^2$ for $T_0 = 3 \times 10^{12}$ (Section 5.1.6), while the method of Theorem 6.2.17, whose Cauchy estimate loses a factor $2^n$, certifies $n$ monotonicity only up to orders proportional to $T_0$. Finite order tests therefore cannot prove RH: each order sees only a bounded height, and the height it sees tends to infinity only with the order.

### *7.2.2.3 Weak limits and the de Bruijn Newman flow*

The class of GGC laws is closed under convolution and weak limits [56, Section 3.1]. So RH is equivalent to exhibiting $R$ as a limit of Laplace transforms of GGC laws, and positivity then passes to the limit at every order simultaneously. The de Bruijn Newman flow gives a family of candidates. Let $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$, with $\Phi$ the Pólya kernel (1.2.3); $\Phi_{\mathrm{dBN}}$ is the kernel of Section 8.4, in the rescaled variable of Rodgers and Tao [463]. Let $H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du$ be the de Bruijn Newman deformation of $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, with $\Lambda$ defined in this

normalisation, and put

$$R_t(s) = \frac{H_t(0)}{H_t(2i\sqrt{s})}, \qquad R_0 = R.$$

THEOREM 7.2.16. *Let $\Lambda$ be the de Bruijn Newman constant. For every $t \geq \Lambda$, $R_t$ is the Laplace transform of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the zeros of $H_t$. In particular $R_t$ is a GGC for every $t \geq 0.2$, unconditionally. RH holds if and only if $R_t$ is a GGC for arbitrarily small $t > 0$.*

*Proof.* For $t \geq \Lambda$ all zeros of $H_t$ are real by definition of $\Lambda$, and $H_t$ is even and entire of order one with the Hadamard product $H_t(z) = H_t(0)\prod_k(1 - z^2/z_k^2)$, so $R_t(s) = \prod_k(1 + 4s/z_k^2)^{-1}$, an infinite convolution of exponential laws. The bound $\Lambda \leq 0.2$ is [419], improving $\Lambda \leq 0.22$ [456]. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t \to R$ pointwise on $(0, \infty)$; if each $R_t$ is a GGC transform, so is the limit, and Proposition 6.1.9 gives RH. Conversely RH means $\Lambda \leq 0$, so every $R_t$ with $t \geq 0$ is a GGC. By Rodgers and Tao [463], $\Lambda \geq 0$, so RH is equivalent to $\Lambda = 0$. □

Conversely, for $t < \Lambda$, and so for every $t < 0$ [463], $R_t$ is not the Laplace transform of a GGC (Corollary 4.2.5), and Proposition 11.5.12 turns this into a test for the properties of the Pólya kernel that could imply RH.

The zeros $z_k(t)$ move by the backward heat flow, and the Thorin measure of $R_t$ can cease to be positive only by two real zeros colliding and leaving the real line, which is the collision picture of Rodgers and Tao; Section 8.4 computes the flow numerically. What Chapter 6 and this chapter add concerns $t = 0$: any failure there comes from zeros above the verified height. For $t \neq 0$ no prime side is known for the Thorin measure of $R_t$, and changing the test function does not give one. Replacing the Gaussian $e^{-xr^2}$ of Theorem 6.1.11 by a $t$ dependent test function gives a linear statistic $\sum_\rho h(\gamma_\rho)$ of the zeros of $\xi$, whereas the heat trace $\sum_k e^{-xz_k(t)^2/4}$ of the Thorin measure of $R_t$ depends on those zeros nonlinearly: along the equation of motion of the zeros (Section 8.4.1) its derivative in $t$ is a sum over pairs of zeros. No Euler product for $H_t$ is known either.

#### 7.2.2.4 *No GGC multiplier can manufacture positivity*

PROPOSITION 7.2.17. *Let $V$ be the Laplace transform of any GGC. Then $V \cdot R$ is the Laplace transform of a GGC if and only if RH holds. For $\varphi$ only one direction holds for every $V$: if $V \cdot \varphi$ is a GGC transform then RH holds. The converse depends on $V$: it fails for $V \equiv 1$ (Theorem 7.1.3) and holds for $V = e^{-c\sqrt{s}}/G$, for which $V\varphi$ is the floor $e^{-c\sqrt{s}}R$ of Theorem 6.2.15.*

*Proof.* If RH holds, $R$ and $V$ are GGC transforms and so is their product. Conversely, if $VR$ is a GGC transform, then $-(VR)'/(VR)$ and $-V'/V$ are Stieltjes functions, analytic on $\mathbb{C} \setminus (-\infty, 0]$, so $-R'/R$ is analytic there and $R$ has no zeros

or poles in $\mathbb{C} \setminus (-\infty, 0]$. By Proposition 6.1.9 this is RH. The same argument applies to $V \cdot \varphi$, since $\varphi = RG$ with $G$ analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$. For $V = e^{-c\sqrt{s}}/G$, a GGC transform by Theorem 6.2.12, $V\varphi = e^{-c\sqrt{s}}R$, which is a GGC under RH by Theorem 6.2.15. □

Multipliers can remove obstructions that come from normalisation, as the floor does, but not the obstruction that comes from the zeros. Thorin measures add under multiplication, so a multiplier can hide negative Thorin mass. What RH forbids, however, is Thorin mass off the half line, the complex atoms at $-(\rho - \frac{1}{2})^2$ contributed by zeros off the critical line, and no GGC factor can cancel a singularity of $R$ in the cut plane. This is why the floor $e^{-c\sqrt{s}}R$ of Theorem 6.2.15, which uses the best available multiplier $e^{-c\sqrt{s}}/G$, is infinitely divisible but can be shown to be a GGC only by proving RH. Section 11.1 finds the same for the general theory of infinitely divisible laws: each GGC criterion it examines is, for $R$, equivalent to RH (Section 11.1.5).

### 7.2.3 Numerical verification

The computations of Chapter 6 and of this chapter have three levels of certainty. The inputs to Theorems 7.1.1, 6.2.15, 6.2.17 and 7.2.15 are enclosed in outward rounded interval arithmetic by `verification_reciprocal_zeta/enclosures.py` (enclosures E1 to E6, recorded in `verification_reciprocal_zeta/enclosures_report.md`; `code/README.md` lists the scripts). The bounds $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ of Corollary 7.2.4 are enclosed in ball arithmetic with Arb [290] by `verification_reciprocal_zeta/certify_k9_k2.py`, which also encloses the values of $S$ quoted with Theorem 7.1.3. The torus bounds of Tables 7.3 and 7.4 and the second condition of Theorem 7.2.11 are proved by a branch and bound whose final boxes are enclosed in ball arithmetic with Arb (Remark 7.2.13); the programs are in `verification_reciprocal_zeta/torus`. The constants of Table 7.7 are computed in floating point with `mpmath` at 20 to 30 digits, and the argument principle counts of Section 7.2.1 in floating point with 40 or more digits. The torus minima quoted for $n \geq 40$ after Remark 7.2.13, and the threshold at $n = 40$ in Problem 7.2, are estimated by optimisation and are not certified.

The function $w$ at $x = 10^{-3}, 10^{-2}, 0.1, 1, 10$ is 1.00025, 1.00249, 0.86251, 0.49140, 0.21524; its initial increase rules out self decomposability (Theorem 7.1.3). Talbot inversion of $\varphi$ gives the density $f$ of the law with Laplace transform $\varphi$: $f(0.05) = 0.36222$, $f(1) = 0.13508$, $f(16) = 0.0087321$, and $x^{3/2}f(x) = 0.5589, 0.6382, 0.6908, 0.7220, 0.7392$ at $x = 16, 32, 64, 128, 256$, approaching 0.75773.

Three steps remain open, in increasing order of difficulty. First, write the formal Thorin measure of $R_t$ in Vere-Jones form, extending Theorem 6.2.3 with $t$ dependent weights. Second, prove Conjecture S for infinitely many $n$ (Problems 7.1

| Quantity | Value |
|---|---|
| $c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$ | 2.6860917096 |
| $\tau_0$, $\theta(\tau_0)$, $\max M = -\theta(\tau_0)/\pi$ | 6.2898360, −3.5309728, 1.1239436 |
| $M(\gamma_1^{2-})$ | 0.5502528 |
| $\tau_\pi$ with $\theta(\tau_\pi) = -\pi$ | 3.4362182 |
| $x_*$ with $W_0(x_*) = 0$ | 0.0032173906 |
| $F(0.003)$, $F(0.0035)$, $F(0.004)$ | 0.0366850, −0.0442285, −0.1145624 |
| $P(0.004)$ | $1.99 \times 10^{-13}$ |
| $\min S$ over the first 300 ordinates | $S(\gamma_{289}^{-}) = -1.1454808$ |
| $\max S$ over the first 300 ordinates | 1.0975638, at $\gamma = 415.4552$ |

Table 7.7: Constants used in Sections 6.1.4 and 7.1.1.

and 7.2). Third, find a family of positive admissible primal Lévy measures whose sine squared images converge to the image $U$ of $\tilde{\mu}_\xi$ in Proposition 6.2.16.

### 7.2.4 Open problems

Problem 7.1. Conjecture S. Prove that the symmetrised truncations $K_n$ have all their zeros in $0 < \operatorname{Re} w < \frac{1}{2}$ on the line $\operatorname{Re} w = \frac{1}{4}$ for infinitely many $n$; by Theorem 7.2.7 this implies RH. Corollaries 7.2.10 and 7.2.12 give its status for $n \le 24$ and $n = 32, 36$. Since Conjecture T and the Hermite Biehler inequality for $A_n$ fail for large $n$, a proof must establish real rootedness without a one sided inequality, for instance by an interlacing argument in $n$, or by following the zeros of $Z_n$ continuously in a parameter interpolating between $n$ and $n+1$ and excluding collisions of zeros on the line.

Problem 7.2. Beyond the torus method. Corollary 7.2.12 rests on torus bounds proved by a branch and bound up to $n = 36$; at $n = 40$ the threshold, estimated by optimisation, is already near 4900, and with the band $1.2 \le x \le 1.3$ the method ends between $n = 40$ and $n = 44$, where the torus minimum of $P_n$ in the band reaches zero. Extend Theorem 7.2.11 with narrower bands around $x = \frac{5}{4}$, and find the first $n$ at which $P_n$ has torus zeros with $x \ge \frac{5}{4}$.

Problem 7.3. From finite order to all orders. Theorem 6.2.17 proves that $k = W + c/(2\sqrt{\pi y})$ is $n$ monotone for $n \le 10^{10}$, with the bound on $n$ coming only from the height of verified zeros; the small $y$ branch holds for every $n$, on $(0, 0.05/n]$. Can the two regimes be made to overlap for all $n$? The large $y$ branch bounds the zeros above $T_0$ in absolute value, and the proof applies it where $yT_0^2 > 2n \log T_0$. No bound of this kind works where $yT_0^2$ is below about $n$: the peak of $\gamma^{2n} e^{-y\gamma^2}$, at $\gamma^2 = n/y$, then lies above $T_0$, where the zeros are dense enough that their contributions, taken in absolute value, outweigh the positive term $c\,\Gamma(n+\frac{1}{2}) y^{-n-\frac{1}{2}}/(2\pi)$ and the verified zeros together. So this asks for a prime side estimate for $(-1)^n k^{(n)}$

on a range $y \le y_n$ with $y_n$ growing at least linearly in $n$; a range that decreases more slowly than $1/n$ raises the order reached but reaches only finitely many orders. In the proof of Theorem 6.2.17 the obstruction is the factor $2^n$ in the Cauchy estimate for $P^{(n)}$ on the circle of radius $y/2$. A circle of radius close to $y$, as in the proof of Theorem 5.1.6, loses only a factor of order $n$ (Section 5.1.6), but that proof also stops at a finite order, and Proposition 7.1.2 suggests that at large $y$ the obstruction cannot be removed without the explicit formula.

Problem 7.4. Beyond the Bernstein cone. Theorem 7.2.15 proves Weil positivity for test functions completely monotone in $r^2$, under an integrability condition. Find the largest cone of test functions, closed under the operations that preserve the Thorin structure, on which positivity can be proved unconditionally, and determine how far it can localise in height.

Problem 7.5. Deforming the binary product. By Theorem 7.1.8 each Euler factor at $\operatorname{Re} s = \sigma$ is $\prod_n (1 + p^{-nu})^{b_n(p^{-\sigma})}$, with integer multiplicities 0 and 1 at $\sigma = 0$ and necklace multiplicities at $\sigma = \frac{1}{2}$. Follow the pseudo spectral trace formula (7.1.3) along the path $q = p^{-\sigma}$, $0 \le \sigma \le \frac{1}{2}$, and determine where and how the positivity of the interference pattern of the progressions can fail.

Problem 7.6. Regularising the autocorrelation. Proposition 6.2.14 factors $\Phi * \Phi$, formally, into the Meixner characteristic function and the positive lattice measure $\mathcal{A}$. The lattice series diverges: truncated at $m, n \le X$ it grows linearly in $X$, through the square of the pole term of the partial sums of $\zeta$, and the factor $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ does not remove this growth. Find the regularised form of $\operatorname{sech}^{1/2} * \mathcal{A}$ and decide whether it has a Thorin or Lee and Yang structure in $v$.

Problem 7.7. Unimodality. Proposition 7.1.5 gives numerical evidence that $w$ is unimodal. Prove it, and prove that $f$ is unimodal, with mode near 0.0240, for instance from Steutel's equation (7.1.1) and the unimodality of $w$. Yamazato's theorem [521], [471, Theorem 53.1], that every self decomposable law is unimodal, shows with no hypothesis that $X_R$, the variable with Laplace transform $R$, is unimodal (Corollary 11.1.2 of Section 11.1), but it does not apply to $X_\varphi$, whose density is $f$, since $X_\varphi$ is infinitely divisible and not self decomposable (Theorem 7.1.3).

Problem 7.8. Infinite divisibility across the strip. Is $w_\sigma > 0$ for every $\sigma \in (0, 1)$ and every $x > 0$? For $\frac{1}{2} < \sigma < 1$ the method of Theorem 7.1.1 may apply: in the third form of Theorem 7.1.12 the terms of the zeros below the verified height are Poisson kernels of positive width, so the question reduces to the sign of the explicit pole and archimedean terms at large $x$, whose limit is governed by $\zeta'(\sigma)/\zeta(\sigma) > 0$. For the completed functions the answer is known: $R_\sigma$ is infinitely divisible for every $\sigma > \frac{1}{2}$ with no hypothesis (Theorem 9.2.5 and the proof of Proposition 10.1.1). Since $\varphi_\sigma$ differs from $R_\sigma$ by the gamma factor and elementary factors in $\sqrt{s}$, this does not settle the question for $w_\sigma$.

PROBLEM 7.9. Meixner Pollaczek truncations. The symmetric expansion of $(\tau^2 + \frac{1}{4})Z(\tau)$ captures the first ordinate at degree 40 but converges slowly. Is there a weighting of its coefficients, or a Jensen type construction from them, whose polynomials are real rooted and whose zeros converge to the ordinates?

PROBLEM 7.10. Nyman Beurling. Theorem 6.1.5 writes $1/\zeta(\frac{1}{2}+\sqrt{s})$ in the span of $\{\mathrm{sech}(\sqrt{s}\log N)\}$ with Möbius weights on $H_{1/2}$. Relate the completeness question of Lai and Vere-Jones for the dilates $\{\mathrm{sech}(\lambda x)\}$ to the Nyman Beurling criterion [390, 46], which Section 10.1.8 sets beside it, and decide whether the sech expansion admits an $L^2$ version on the boundary $\mathrm{Re}\,\sqrt{s} = \frac{1}{2}$.

PROBLEM 7.11. Product representations of $X_R$. By the closure properties behind Corollary 7.1.6, an unconditional representation of $X_R$ built from independent GGC variables by sums, products and weak limits would make $R$ a GGC and prove RH (Proposition 1.2.1); for $X_\varphi$ the corollary rules out such a representation. The harmonic mean is not available, because Proposition 7.1.7 shows it is not a closure operation for the class. The deconvolution (6.2.1) of Section 6.2.1 gives $X_R$, formally, as $X_\varphi$ minus a stable law and a compensated series of the variables $Y_{c_n}$. Can the compensation be absorbed into a product?

# Chapter 8

# Probability, hitting times and other criteria

The clock $1/G$ is also the Laplace transform of a random variable, whose law the Pólya route of Chapter 2 computes from the same kernel as the Thorin density. This chapter treats the clock as a random variable, through its law, its Kent approximants and its Brownian boundaries, and reads two classical equivalents of RH that the Pólya density computes directly, Li's criterion and the de Bruijn Newman flow, in terms of the Thorin measure.

## 8.1 The clock as a random variable

The clock variable $T$ of Section 1.2, with $\mathbb{E}e^{-uT} = 1/G(u)$, exists unconditionally and is self decomposable (Theorem 5.1.1). Under RH it is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables $e_\gamma$ (Proposition 1.2.1), and if the zeros are also simple, the theorems of Kent and Krein make it the hitting time of a level by a diffusion whose Dirichlet spectrum is $\{\gamma^2\}$ [305], [424, Section 16.3].

### 8.1.1 The law of the clock and its decay rate

The Laplace transform $\mathbb{E}e^{-uT} = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is available from the Pólya route at every complex $u$ other than its poles, untilted for $|\operatorname{Im}\sqrt{u}| < 120$ and tilted above (Section 2.1.3). Talbot's contour inversion [490], as implemented in `mpmath` [376], then gives the density $f$ of $T$ and its survival function $\bar{F}(t) = P(T > t)$ with no reference to the zeros.

Under RH, with simple zeros, the density also has the spectral expansion [305]

$$f(t) = \sum_k R_k e^{-\gamma_k^2 t}, \qquad P(T>t) = \sum_k \frac{R_k}{\gamma_k^2} e^{-\gamma_k^2 t}, \qquad R_k = \frac{2i\gamma_k\,\xi(\frac{1}{2})}{\xi'(\frac{1}{2}+i\gamma_k)},$$

whose coefficients are the residues of the transform at its poles $u = -\gamma_k^2$. The first step of the proof of Theorem 4.3.22 derives it by a contour shift, with

$R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ in the role of $c_n$ there, and the Pólya route supplies both the ordinates, as its Newton roots, and the values of $\xi'$ at them. The residues alternate in sign (Exercise 8.1) and grow like $e^{\pi\gamma_k/4}$ up to powers of $\gamma_k$ and the factor $1/|\zeta'(\rho_k)|$, because Stirling's formula puts the factor $e^{-\pi\gamma_k/4}$ into $|\Xi'(\gamma_k)| = |\xi'(\rho_k)|$; thus $R_1 = 1.0164 \times 10^4$ and $R_2 = -1.1775 \times 10^6$. The series converges for $t > 0$ but cancels heavily at small $t$, so near the origin Talbot's contour is the better tool (Exercise 8.2). Computed with 50 digits from the 79 zeros below 200, the spectral expansion agrees with the Talbot values to 24 digits for $0.010 \le t \le 0.12$. Beyond that range the Talbot values lose relative accuracy as $\bar F(t)$ falls, down to six digits at $t = 0.3$, and the spectral expansion is the accurate one (Figure 8.1).

The primes give a second expansion of $f$, with no hypothesis. Write $\xi = \eta\zeta$ with the gamma factor $\eta(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, and put $\phi_\eta(u) = \xi(\frac12)/\eta(\frac12 + \sqrt{u})$.

Proposition 8.1.1. *Let $c > \frac14$, let $g_1$ be the inverse Laplace transform of $\phi_\eta$ along $\operatorname{Re} u = c$, and let $q_\lambda(x) = \frac{\lambda}{2\sqrt{\pi}} x^{-3/2} e^{-\lambda^2/4x}$ for $x > 0$, both extended by $0$ to $x \le 0$. Then for every $t > 0$*

$$f(t) = \sum_{n\ge1} \frac{\mu(n)}{\sqrt{n}}\, g_n(t), \qquad g_n = g_1 * q_{\log n}, \qquad |g_n(t)| \le \sup_{0<x\le t} |g_1(x)|\ \operatorname{erfc}\Big(\frac{\log n}{2\sqrt{t}}\Big),$$

*so the series converges absolutely, faster than any power of $n$. Moreover $g_1(t) = 2\xi(\frac12)e^{t/4} + O(e^{\varepsilon t})$ as $t \to \infty$ for every $\varepsilon > 0$.*

*Proof.* On $\operatorname{Re} u = c$ we have $\operatorname{Re}\sqrt{u} \ge \sqrt{c} > \frac12$, so the series (6.1.1) for $1/\zeta(\frac12 + \sqrt{u})$ converges absolutely there, with the sum of the absolute values of its terms at most $\zeta(\frac12 + \sqrt{c})$. By Stirling's formula $|\phi_\eta(u)|$ decays faster than any power of $|u|$ on vertical lines in $\operatorname{Re} u > 0$, uniformly on compact ranges of $\operatorname{Re} u$; hence $|g_1(x)| \le C_c e^{cx}$, the Laplace transform of $g_1$ is $\phi_\eta$ on $\operatorname{Re} u > c$, and $f$, whose transform $\phi_\eta(u)/\zeta(\frac12 + \sqrt{u})$ is integrable on the line, is its Bromwich integral along $\operatorname{Re} u = c$. Dominated convergence inverts the series term by term. The term of $n$ is $\phi_\eta(u)e^{-\sqrt{u}\log n}$, and $e^{-\lambda\sqrt{u}}$ is the transform of $q_\lambda$ by (6.1.2), which gives the convolution; the bound follows from $\int_0^t q_\lambda = \operatorname{erfc}(\lambda/2\sqrt{t})$. Finally $\phi_\eta$ is analytic off $(-\infty, 0]$ except for a simple pole at $u = \frac14$, where $\eta(\frac12 + \sqrt{u}) = \frac12(u - \frac14) + O((u-\frac14)^2)$, so the residue is $2\xi(\frac12)$; moving the contour to $\operatorname{Re} u = \varepsilon$ gives the asymptotic of $g_1$. □

So $g_1$ is not a density: it grows like $2\xi(\frac12)e^{t/4}$. The same contour shift gives $g_n(t) = 2\xi(\frac12)n^{-1/2}e^{t/4} + O_n(e^{\varepsilon t})$, so for fixed $N$ the partial sum over $n \le N$ is $2\xi(\frac12)e^{t/4}\sum_{n\le N}\mu(n)/n + O_N(e^{\varepsilon t})$ and grows, while $f$ decays like $e^{-\gamma_1^2 t}$. For large $t$ the series therefore rests on the cancellation $\sum_n \mu(n)/n = 0$, the prime number theorem. For small $t$ it converges very fast. Computed by Talbot inversion at 30

digits (`bn_tail.py`), the terms with $n \le 5$ ($n \le 7$ at $t = 0.04$) agree with the Talbot value of $f$ to at least 20 digits at $t = 0.008, 0.012, 0.016, 0.02, 0.03, 0.04$. Relative to $f$, the term of the prime 2 is $1 \times 10^{-26}$ at $t = 0.008$, $3 \times 10^{-9}$ at $t = 0.02$ and 2.8 percent at $t = 0.04$. The two expansions are complementary. For relative accuracy $10^{-8}$ the spectral expansion needs 50 zeros at $t = 0.008$, 10 at $t = 0.02$ and 2 at $t = 0.06$ (among $1, 2, 5, 10, 20, 50, 120$ zeros), while the Möbius series needs only $n = 1$ up to $t = 0.02$, and $n \le 3$ at $t = 0.04$.

The law is unimodal with no hypothesis (Corollary 11.1.2) and sharply concentrated. Its mean and variance are $\mathbb{E}T = 0.0231050$ and $\operatorname{Var} T = 3.71726 \times 10^{-5}$ (Section 1.2) and its median is 0.0220. At the other end, with no hypothesis, $P(T \le y) \le \exp\{-(1 + o(1))(\log \frac{1}{y})^2/16y\}$ as $y \downarrow 0$, so the density vanishes to infinite order at the origin: Chernoff's bound $P(T \le y) \le e^{uy}/G(u)$, with $\log G(u) \sim \frac{\sqrt{u}}{4} \log u$ by Stirling's formula for $\xi$, is optimised at $\sqrt{u} \approx \log(1/y)/4y$.

PROPOSITION 8.1.2. *RH holds if and only if the density $f$ of $T$ is a Pólya frequency function of infinite order, that is* $\det[f(x_i - y_j)]_{i,j \le r} \ge 0$ *for every $r$ and all $x_1 < \cdots < x_r$, $y_1 < \cdots < y_r$.*

*Proof.* Under RH, $f$ is the density of $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 1.2.1), exponential densities are Pólya frequency functions, and the class is closed under convolution and weak limits. Conversely, by Schoenberg's theorem [474] the bilateral Laplace transform of an integrable Pólya frequency function of infinite order is $1/\Psi$ with $\Psi(u) = Ce^{-cu^2+bu} \prod_k (1 + \lambda_k u)e^{-\lambda_k u}$, $c \ge 0$, $\lambda_k$ real. Here the transform is $1/G$, so $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ has only real zeros. Its zeros are $u = -a_k$ with $a_k = -(\rho - \frac{1}{2})^2$, which is real only if $\rho - \frac{1}{2}$ is real or purely imaginary, and $\xi$ has no real zeros, so every $\rho$ is on the critical line. □

This is the one sided form of the criterion of Proposition 16.5.1 for the two sided density. Under RH with simple zeros, assuming that the spectral expansion converges absolutely for $t > 0$ and may be differentiated termwise, the opposite signs of $R_1$ and $R_2$ make the upper tail of $f$ log concave: $f(t) = R_1 e^{-\gamma_1^2 t}(1 - \varepsilon(t)) + O(e^{-\gamma_3^2 t})$ with $\varepsilon = |R_2/R_1| e^{-\Delta t}$ and $\Delta = \gamma_2^2 - \gamma_1^2$, so $(\log f)''(t) = -\Delta^2 \varepsilon/(1 - \varepsilon)^2 + O(e^{-(\gamma_3^2 - \gamma_1^2)t}) < 0$ for large $t$. On 91 logarithmically spaced points of $[0.006, 0.08]$, which carry all but $6 \times 10^{-6}$ of the mass, a floating point Talbot computation at 25 digits (no interval arithmetic) gives $(\log f)'' < 0$ with maximum $-0.026$, the first two orders of hyperbolic complete monotonicity of $f$. These are necessary conditions for Proposition 8.1.2 tested in the bulk of the law, not a proof.

The hazard $h = f/\bar{F}$ tends to the principal eigenvalue $\gamma_1^2$. Under RH with simple zeros the leading term of the spectral expansion dominates, and $h(t) - \gamma_1^2 = O(e^{-(\gamma_2^2 - \gamma_1^2)t})$ (Exercise 8.3). This is the quasi stationary behaviour of a hitting time; Roberts [461, Section 2] obtains approximations of Brownian boundary hitting laws from approximations of the conditioned diffusion. The Talbot side shows the limit with no zero as input: $h(0.05) = 199.72$, $h(0.1) = 199.7904544$

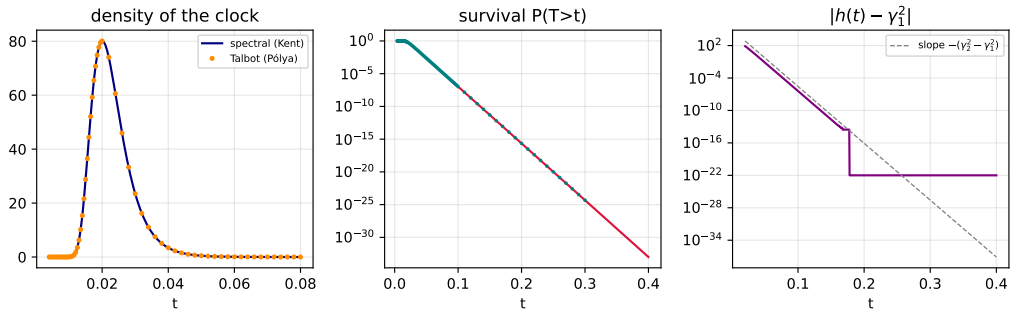



Figure 8.1: The law of the clock variable computed from the Pólya density by Talbot inversion (points) and by the spectral expansion with Pólya residues (lines): density, survival function, and the distance of the hazard from $\gamma_1^2$, whose logarithm has slope $-(\gamma_2^2 - \gamma_1^2)$ (dashed). Beyond $t \approx 0.17$ the last curve reflects the double precision of the plotting script, not the hazard.

and $h(0.25) = 199.79045483$, the last equal to $\gamma_1^2$ to the eleven digits shown. The rate of approach is the spectral gap: a fit of $\log|h(t) - \gamma_1^2|$ on $[0.05, 0.12]$ has slope $-242.12$, against $-(\gamma_2^2 - \gamma_1^2) = -242.14$.

The clock also generates a natural exponential family in the sense of Barndorff-Nielsen. The cumulant generating function $K(\theta) = \log \mathbb{E}e^{\theta T} = \log\{\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{-\theta})\}$ is finite exactly for $\theta < \gamma_1^2$ and $K'(\theta) \to \infty$ as $\theta \uparrow \gamma_1^2$, so the family is regular and steep with natural parameter space $(-\infty, \gamma_1^2)$, whose boundary is the first zero; under RH the tilted laws are $\sum_\gamma e_\gamma/(\gamma^2 - \theta)$. The saddlepoint approximation of Barndorff-Nielsen and Cox [30], $\hat f(y) = \exp\{K(\hat\theta) - \hat\theta y\}/\sqrt{2\pi K''(\hat\theta)}$ with $K'(\hat\theta) = y$, needs $\xi$ only on the real axis $s \ge \frac{1}{2}$ (for $y < \mathbb{E}T$) and on the segment of the critical line from $\frac{1}{2}$ to $\frac{1}{2} + i\gamma_1$ (for $y > \mathbb{E}T$), so it never passes the first zero. It is not normalised (it integrates to 0.98492) and agrees with the Talbot density to about 0.5 percent in the lower tail and within 2.5 percent on $[0.011, 0.035]$ (Table 8.1); in the upper tail the ratio tends to $e/\sqrt{2\pi} = 1.0844$, the error of the approximation for an exponential density, because the tail is $R_1 e^{-\gamma_1^2 y}$. As $y \downarrow 0$ the exponent $K(\hat\theta) - \hat\theta y$ is $-(1 + o(1))(\log\frac{1}{y})^2/16y$, the Chernoff exponent above, but slowly: at $y = 0.006$ the leading term gives $-272$ against $-33.0$ for the exact $\log f$, while the saddlepoint, which keeps all of $\xi$, is within 0.5 percent of $f$. The tail formula of Lugannani and Rice [350], built from the same $\hat\theta$, $K(\hat\theta)$ and $K''(\hat\theta)$, is within 1.4 percent of the Talbot values of $P(T \le y)$ and within 2.4 percent of those of $P(T > y)$ at the sampled $y$ (`bn_tail.py`, part a).

| $y$ | $\hat{\theta}$ | $f(y)$ (Talbot) | $\hat{f}(y)$ | ratio |
|---|---|---|---|---|
| 0.0060 | $-15909.8$ | $5.165 \times 10^{-15}$ | $5.138 \times 10^{-15}$ | 0.9949 |
| 0.0120 | $-1674.1$ | 1.7018 | 1.6953 | 0.9962 |
| 0.0201 | $-114.5$ | 80.1084 | 78.5287 | 0.9803 |
| 0.0239 | 19.4 | 61.4522 | 59.9432 | 0.9754 |
| 0.0401 | 148.8 | 3.3491 | 3.4507 | 1.0303 |
| 0.0800 | 182.8 | 0.0012 | 0.0013 | 1.0762 |

Table 8.1: The saddlepoint density of the clock against the Talbot inversion of its Laplace transform, both from $\xi$ alone.

| terms | $\gamma_1^2$ | $\gamma_2^2$ | $\gamma_3^2$ | $\gamma_4^2$ |
|---|---|---|---|---|
| 2 | 199.86679 | 427.28 | | |
| 3 | 199.79031 | 442.129 | 618.93 | |
| 4 | 199.790455 | 441.9255 | 625.63 | 907.24 |
| true | 199.7904548 | 441.9261506 | 625.5429969 | 925.6730873 |

Table 8.2: Prony extraction of the atoms from $P(T > t)$ on $t \in [0.03, 0.108]$. Identical results for relative noise $10^{-16}$, $10^{-30}$ and $10^{-45}$: the error is model truncation, not noise.

### 8.1.2 The cost of the zeros in time and in samples

We fit sums of $p$ exponentials to $P(T > t)$ by Prony's method, from 40 values at $t = 0.03, 0.032, \ldots, 0.108$ carrying relative noise from $10^{-16}$ down to $10^{-45}$ (Table 8.2). The error of the fitted atoms comes from the neglected terms, whose relative size on a window starting at $t_0$ is $e^{-(\gamma_{p+1}^2 - \gamma_1^2)t_0}$ up to a factor set by the residues. Resolving the $k$th atom therefore needs data accurate to about $(\gamma_k^2 - \gamma_1^2)t_0 / \log 10$ digits, a number that grows without bound in $k$.

Sampling meets the same cancellation in another currency. By the identity $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Section 1.2, with $S_2 = \sum_{k \geq 1} \frac{2}{\pi^2 k^2} \Gamma_k$ for independent Gamma$(2,1)$ variables $\Gamma_k$, the variable $Y = \frac{1}{2}\log(\pi S_2/2)$ satisfies

$$2\,\Xi(z) = \mathbb{E}\big[e^{Y/2}\cos(zY)\big].$$

$\Xi(z)$ is the mean of a random variable with finite variance, since $\mathbb{E}e^Y = 2\xi(1) = 1$, so a sample mean estimates it at every $z$. We draw $n = 10^6$ copies of $S_2$ and compare the estimates with $\Xi(z)$ in Table 8.3, a check of the identity on the critical line; Table 9.1 checks it at real $s$ (Section 9.1.1).

The standard error settles at $3.5 \times 10^{-4}$, the standard deviation of $e^{Y/2}\cos(zY)/2$ over $\sqrt{n}$, while $|\Xi(z)|$ decays like $e^{-\pi z/4}$. The sample size needed to resolve the sign of $\Xi$ therefore grows like $e^{\pi z/2}$. Even the first sign change, at $z = 14.13$, needs about $10^6$ samples: between the first two zeros $|\Xi|$ is at most

| $z$ | Monte Carlo | standard error | $\Xi(z)$ | $n$ with standard error $\lvert\Xi(z)\rvert$ |
|---|---|---|---|---|
| 0 | 0.497061 | $5.4 \times 10^{-5}$ | 0.497121 | $1.2 \times 10^{-2}$ |
| 2 | 0.453136 | $7.3 \times 10^{-5}$ | 0.453100 | $2.6 \times 10^{-2}$ |
| 5 | 0.275888 | $2.4 \times 10^{-4}$ | 0.275550 | 0.77 |
| 8 | 0.103756 | $3.4 \times 10^{-4}$ | 0.103303 | 11 |
| 10 | 0.038401 | $3.5 \times 10^{-4}$ | 0.037968 | 86 |
| 20 | 0.000314 | $3.5 \times 10^{-4}$ | $-3.67 \times 10^{-5}$ | $9.3 \times 10^{7}$ |
| 30 | 0.000289 | $3.5 \times 10^{-4}$ | $-1.50 \times 10^{-8}$ | $5.5 \times 10^{14}$ |

Table 8.3: Monte Carlo estimates of $\Xi(z)$ from Williams' representation with $10^6$ samples. The last column is the sample size at which the standard error equals $\lvert\Xi(z)\rvert$.

$8.0 \times 10^{-4}$ (at $z = 15.59$), and resolving it at two standard errors takes about $8 \times 10^5$ samples. Sampling is the right tool for the law of the Pólya variable itself, for its moments and for the Bayesian inverse problems of [424, Section 19.2], and the wrong tool for the zeros. Devroye's exact samplers for theta type laws [115] make sampling efficient, but the sample size needed for the zeros still grows like $e^{\pi z/2}$.

### 8.1.3 Kent approximants with a Weyl tail

In the rest of this section we write $w$ for the variable $u$ of the clock. Kent's theorem [305] identifies the first hitting time of a level by a diffusion started at a reflecting or entrance end of its interval with the GGC $\prod_n (1 + w/\lambda_n)^{-1}$, where the $\lambda_n$ are the Dirichlet eigenvalues of the interval up to that level. Krein's inverse spectral theory [124] builds a string, and so a generalised diffusion, from any increasing sequence $\lambda_n > 0$ with $\sum 1/\lambda_n < \infty$. The GGCs that arise as such hitting times are therefore exactly those without drift whose Thorin measure consists of unit atoms at distinct points. By Proposition 1.2.1 the clock is one of them exactly when RH holds and the zeros are simple, and its diffusion then has Dirichlet spectrum $\{\gamma^2\}$.

The approximants of the clock that Kent's theorem suggests are the transforms $1/G_N(w)$ of the partial products $G_N(w) = \prod_{k \le N} (1 + w/\gamma_k^2)$, each the transform of the hitting time of a finite string with $N$ point masses. They are admissible in the closure criterion of Corollary 4.1.2, and they are the approximating sequence, under RH, in the proof of [424, Proposition 18.10]. They converge slowly. With $\tau_1(N) = \sum_{k>N} \gamma_k^{-2}$, the relative error $G(w)/G_N(w) - 1$ of the clock transform lies, under RH, between $w\tau_1(N)$ and $e^{w\tau_1(N)} - 1$.

We keep the verified zeros as atoms and replace the others by their average density. By the Riemann von Mangoldt formula $N(t) = \vartheta(t)/\pi + 1 + S(t)$ of Section 2.2, the zeros above the cutoff $\Theta_N$ have density $\vartheta'(\gamma)/\pi \approx \frac{1}{2\pi} \log \frac{\gamma}{2\pi}$,

which is positive for $\gamma > 6.3$. So

$$G_N^{\mathrm{W}}(w) = \prod_{k \le N} \Big(1 + \frac{w}{\gamma_k^2}\Big) \exp\Big\{ \int_{\Theta_N}^{\infty} \log\Big(1 + \frac{w}{\gamma^2}\Big) \frac{\vartheta'(\gamma)}{\pi}\, d\gamma \Big\},$$

with W for Weyl, has reciprocal equal to the Laplace transform of a GGC with $N$ unit atoms and the continuous Thorin density $\vartheta'(\sqrt{z})/(2\pi\sqrt{z})$ on $z > \Theta_N^2$. For $\Theta_N$ we take the Gram point $g_{N-1}$ of Section 2.2.1, where $\vartheta(\Theta_N)/\pi + 1 = N$. For the values of $N$ used here it lies between $\gamma_N$ and $\gamma_{N+1}$, so $S(\Theta_N) = 0$.

Table 8.4 compares the two approximants with the clock computed from $\xi$.

| $N$ | $w = 1$ | $w = 100$ | $w = 1000$ | $w = 10^4$ |
|---|---|---|---|---|
| 1 | $1.8 \times 10^{-2}$ / $1.3 \times 10^{-4}$ | 4.8 / $1.0 \times 10^{-2}$ | $\gg 1$ / $3.3 \times 10^{-2}$ | $\gg 1$ / $4.3 \times 10^{-2}$ |
| 20 | $7.2 \times 10^{-3}$ / $4.3 \times 10^{-7}$ | 1.04 / $4.2 \times 10^{-5}$ | $\gg 1$ / $3.7 \times 10^{-4}$ | $\gg 1$ / $1.5 \times 10^{-3}$ |
| 79 | $3.6 \times 10^{-3}$ / $1.5 \times 10^{-7}$ | 0.43 / $1.5 \times 10^{-5}$ | 34 / $1.5 \times 10^{-4}$ | $\gg 1$ / $1.2 \times 10^{-3}$ |
| 300 | $1.6 \times 10^{-3}$ / $7.4 \times 10^{-9}$ | 0.17 / $7.4 \times 10^{-7}$ | 3.95 / $7.4 \times 10^{-6}$ | $\gg 1$ / $7.2 \times 10^{-5}$ |
| 1700 | $5.0 \times 10^{-4}$ / $2.9 \times 10^{-11}$ | 0.051 / $2.9 \times 10^{-9}$ | 0.64 / $2.9 \times 10^{-8}$ | 142 / $2.9 \times 10^{-7}$ |

Table 8.4: Relative error of the clock transform: partial product over $N$ zeros, $|G/G_N - 1|$, and with the Weyl tail, $|G_N^{\mathrm{W}}/G - 1|$. The two measures of error agree to first order, and $\gg 1$ marks an error above $10^3$. The partial product values agree with [424, Section 18.2.1] where both are given.

At the four points $w = 1, 100, 1000, 10^4$ the table exhibits an explicit positive Thorin measure whose Thorin transform $\int \log(1 + w/z)\, U(dz)$ matches $\log G(w)$ to within $3 \times 10^{-7}$. This is a finite instance of the weak Thorin condition of [424, Section 18.2.3], which asks for such a measure, to within any $\eta > 0$, at every finite set of points, and which is equivalent to RH by the closure criterion of Corollary 4.1.2. A finite instance proves nothing: agreement for $w \le 10^4$ cannot detect off line zeros above the verified height, which change $\log G$ there by far less than the error. The error of the Weyl tail grows linearly in $w$, and to first order it is a weighted average of $S$ beyond the cutoff,

$$\log G(w) - \log G_N^{\mathrm{W}}(w) \approx w\big(\tau_1(N) - \tau_1^{\mathrm{W}}(N)\big) = 2w \int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma^3}\, d\gamma,$$

where $\tau_1^{\mathrm{W}}(N) = \int_{\Theta_N}^{\infty} \gamma^{-2} \vartheta'(\gamma) \pi^{-1} d\gamma$ is the Weyl estimate of $\tau_1(N)$ and the identity is an integration by parts that uses $S(\Theta_N) = 0$.

### 8.1.4 Moment matching and the error at large $w$

The error of the Weyl tail comes from the first order term of $\log(1 + w/\gamma^2) = w/\gamma^2 - w^2/(2\gamma^4) + \cdots$. For $w$ small compared with $\Theta_N^2$ the error of $\log G_N^{\mathrm{W}}$ is approximately $w$ times $\tau_1(N) - \tau_1^{\mathrm{W}}(N)$, and the second order term is $-w^2/2$ times the corresponding difference for $\tau_2(N) = \sum_{k>N} \gamma_k^{-4}$. Both true tail sums

are available without any zero beyond $\gamma_N$, because the power sums $p_m = \sum_k a_k^{-m}$, with $a_k = -(\rho_k - \frac12)^2$ as in Section 1.2 and equal to $\sum_{\gamma>0} \gamma^{-2m}$ under RH, are Taylor coefficients of $\xi$ at the centre: if $\xi(\frac12 + s)/\xi(\frac12) = 1 + c_2 s^2 + c_4 s^4 + \cdots$, then

$$p_1 = \sum_k a_k^{-1} = c_2 = 0.023104993115418970 7889,$$
$$p_2 = \sum_k a_k^{-2} = c_2^2 - 2c_4 = 0.000037172599285269 6862,$$

and $\tau_m(N) = p_m - \sum_{k\le N} \gamma_k^{-2m}$. Here $p_1 = \mathbb{E}T$ and $p_2 = \operatorname{Var} T$ (Section 1.2). We therefore replace the Weyl density $\vartheta'(\gamma)/\pi$ on $[\Theta_N, \infty)$ by $\lambda\,\vartheta'(\gamma)/\pi$ on $[\Theta'_N, \infty)$, and choose the two parameters $\lambda$ and $\Theta'_N$ so that the tail reproduces $\tau_1(N)$ and $\tau_2(N)$ exactly. The tail density remains positive, so the approximant is still a GGC. The adjustments are small: for $N = 79$ the cutoff moves from 199.649 to 199.675 and $\lambda = 1.000059$.

| $N$ | tail | $w=1$ | $w=10^2$ | $w=10^4$ | $w=10^5$ | $w=10^6$ | $w=10^7$ |
|---|---|---|---|---|---|---|---|
| 79 | Weyl | $1.5\times10^{-7}$ | $1.5\times10^{-5}$ | $1.2\times10^{-3}$ | $4.3\times10^{-3}$ | $5.8\times10^{-3}$ | $6.0\times10^{-3}$ |
| | matched | $2.6\times10^{-17}$ | $2.6\times10^{-11}$ | $1.8\times10^{-5}$ | $3.1\times10^{-3}$ | $6.1\times10^{-2}$ | 0.47 |
| 300 | Weyl | $7.4\times10^{-9}$ | $7.4\times10^{-7}$ | $7.2\times10^{-5}$ | $5.5\times10^{-4}$ | $1.7\times10^{-3}$ | $2.1\times10^{-3}$ |
| | matched | $2.3\times10^{-20}$ | $2.3\times10^{-14}$ | $2.2\times10^{-8}$ | $1.5\times10^{-5}$ | $1.8\times10^{-3}$ | $2.8\times10^{-2}$ |
| 1700 | Weyl | $2.9\times10^{-11}$ | $2.9\times10^{-9}$ | $2.9\times10^{-7}$ | $2.8\times10^{-6}$ | $2.4\times10^{-5}$ | $9.3\times10^{-5}$ |
| | matched | $3.5\times10^{-23}$ | $1.1\times10^{-17}$ | $7.0\times10^{-13}$ | $3.6\times10^{-10}$ | $2.5\times10^{-7}$ | $5.0\times10^{-5}$ |

Table 8.5: Relative error $|G_{\mathrm a}/G - 1|$ of the approximant $G_{\mathrm a}$ of $G$ with the Weyl tail and with the moment matched tail. The cutoffs are $\Theta'^2_N = 3.99\times10^4$, $2.95\times10^5$ and $4.83\times10^6$ for $N = 79$, 300 and 1700.

Table 8.5 shows two regimes. For $w$ below $\Theta'^2_N$ moment matching gains ten to twelve digits at $w = 1$ and about two as $w$ approaches $\Theta'^2_N$. Well beyond $\Theta'^2_N$ the gain disappears, and at $w = 10^7$ with $N = 79$ the matched tail is worse than the plain Weyl tail. For $w \gg \Theta'^2_N$ the matched tail differs from the Weyl tail mainly by $(\lambda-1)\int_{\Theta'_N}^\infty \log(1 + w/\gamma^2)\,\vartheta'(\gamma)\pi^{-1}d\gamma$, which grows like $(\lambda-1)\frac{\sqrt{w}}{2}\log\frac{\sqrt{w}}{2\pi}$, while the error of the Weyl tail tends to a constant.

Under RH, since $S(\Theta_N) = 0$ and no ordinate lies in $(\gamma_N, \Theta_N]$, the true tail sum minus its Weyl estimate is a Stieltjes integral against $S$, and an integration by parts gives

$$\log G_N^{\mathrm W}(w) - \log G(w) = -\int_{\Theta_N}^\infty \log\Big(1 + \frac{w}{\gamma^2}\Big)\,dS(\gamma) = -\int_{\Theta_N}^\infty \frac{2w\,S(\gamma)}{\gamma(\gamma^2 + w)}\,d\gamma$$
$$\longrightarrow\ -2\int_{\Theta_N}^\infty \frac{S(\gamma)}{\gamma}\,d\gamma \qquad (w \to \infty).$$

The limit is $-\ell_N$, with $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as in Exercise 8.5. It is finite because $\int_0^t S(u)\,du = O(\log t)$ (Littlewood; Section 2.2.1), and in general it is not zero. The measured errors of the Weyl tail level off at about $6 \times 10^{-3}$ for $N = 79$ and $2 \times 10^{-3}$ for $N = 300$.

The constant is not an obstruction of principle. It depends only on $\gamma_1, \dots, \gamma_N$ and on $\xi$ along the real axis, and Stirling's formula gives $\ell_N$ in closed form (Exercise 8.5). It is also a floor only for the Weyl density. Adding to $\vartheta'/\pi$ a continuous $\omega$ with compact support in $(\Theta_N, \infty)$ and $\int \omega = 0$ changes the limit by $-2\int \omega(\gamma)\log\gamma\,d\gamma$, and two wide bumps of opposite sign placed far apart give this any value while the density stays positive, so a suitable positive tail removes the constant.

## 8.2 Brownian boundaries and Williams' structure

Under RH with simple zeros, Kent's theorem makes the clock variable $T$ the hitting time of a level by a diffusion, but it describes that diffusion only through its spectrum, the set $\{\gamma^2\}$ that is to be found (Section 8.1.3). We look instead for a representation by Brownian motion itself. Unconditionally, $T$ is not a Brownian first passage time to an independent random level for any constant drift (Section 8.2.1). Across a curved boundary it is one: such a boundary exists by a theorem of Ekström and Janson, and we compute it (Section 8.2.2). Williams' representation of the Pólya kernel through Bessel hitting times explains the tilted contour of Section 2.1.3, but it transfers no positivity to the clock (Section 8.2.3).

### 8.2.1 The clock is not a first passage time to a random level

The first passage time of Brownian motion with drift $c \geq 0$ to a level $a > 0$ has the inverse Gaussian law, with transform $\exp\{a(c - \sqrt{c^2 + 2u})\}$. Mixing over an independent random level with law $F$ gives the transforms $\int \exp\{a(c - \sqrt{c^2 + 2u})\}F(da)$, a class of first passage laws.

Proposition 8.2.1. *For no drift $c \geq 0$ and no probability law $F$ on $(0, \infty)$ is the clock variable $T$ equal in law to the first passage time of Brownian motion with drift $c$ to an independent level $A \sim F$.*

*Proof.* On the parabola $u = \frac{1}{2}(-q^2 + 2icq)$, $q \in \mathbb{R}$, we have $c^2 + 2u = (c + iq)^2$, so $\operatorname{Re}\sqrt{c^2 + 2u} = c$ and every mixture transform has modulus at most one there. Suppose that $T$ had such a law, so that the clock transform $1/G(u) = \mathbb{E}e^{-uT}$ and the mixture transform agree for $u \geq 0$. The mixture transform is analytic and bounded by one on the region $D = \{u : \operatorname{Re}\sqrt{c^2 + 2u} > c\}$, and continuous up to its boundary, which is the parabola. The region $D$ is the image of the half plane $\operatorname{Re} v > c$ under $v \mapsto (v^2 - c^2)/2$, so it is connected, and it contains the right half plane. By the identity theorem $1/G$ coincides with the mixture transform

on $D$, so it has no pole in $D$ and is bounded by one there, and by continuity $|1/G| \le 1$ on the parabola away from the zeros of $\xi(\frac12+\sqrt{u})$. For the clock, however, $\sqrt{u} = (c+iq)/\sqrt{2} + O(1/q)$ as $q \to \infty$ (for $c = 0$ the parabola is the negative axis, approached from above, and $\sqrt{u} = i|q|/\sqrt{2}$), so $|\xi(\frac12+\sqrt{u})|$ decays like $\exp(-\pi|q|/(4\sqrt{2}))$ times a power, by Stirling's formula and the polynomial growth of $\zeta$ in vertical strips. Hence $|\xi(\frac12)/\xi(\frac12+\sqrt{u})|$ grows exponentially along the parabola, and the transforms cannot agree. □

The proof covers the nonincreasing lines $a - ct$, $c \ge 0$; an increasing line, with $c < 0$, gives a passage time that is infinite with probability $1 - \mathbb{E}e^{2cA} > 0$, which the clock is not. Expanding $G(u) = 1 + m_1 u + m_2 u^2 + \cdots$ with $m_1 = \mathbb{E}T$ and $m_1^2 - 2m_2 = \operatorname{Var} T$ gives $|G(u)|^2 = 1 - q^2(m_1 - c^2 \operatorname{Var} T) + O(q^3)$ on the parabola, so the clock transform already exceeds one at small $q$ whenever

$$c < c_* = \sqrt{\mathbb{E}T/\operatorname{Var} T} = \sqrt{0.0231050/3.71726 \times 10^{-5}} = 24.93.$$

### 8.2.2 A curved Brownian boundary for the clock

Finding a boundary whose first passage time has a prescribed law is the inverse first passage problem [79, Theorem 1]. By the theorems of Ekström and Janson [129, Theorems 2.3, 8.2 and 9.2], stated in full in Section 9.2.4, the problem has a solution for the clock, which we call the exact boundary. Since the density of the clock is continuous and positive on $(0,\infty)$ (Exercise 8.10), the same theorems make the exact boundary continuous on $(0,\infty)$ and the unique continuous solution $b$ with $\liminf_{t\downarrow 0} b(t) \ge 0$ of the Volterra integral equation (8.2.1) of the first passage density. We call the solution of the discretised Volterra equation the Volterra boundary.

#### *8.2.2.1 The Roberts Shortland tangent boundary*

Roberts and Shortland [462] approximate the hazard of a Brownian first passage across a smooth boundary $b$ by the hazard of the tangent line at the current time, for which the Bachelier Lévy formula is exact:

$$h_{\tan}(t) = \frac{\alpha(2\pi t^3)^{-1/2}\exp\{-(\alpha+\beta t)^2/2t\}}{\mathcal{N}\big((\alpha+\beta t)/\sqrt{t}\big) - e^{-2\alpha\beta}\mathcal{N}\big((\beta t-\alpha)/\sqrt{t}\big)},$$
$$\beta = b'(t),\ \alpha = b(t) - tb'(t),$$

where $\mathcal{N}$ is the standard normal distribution function; we write $\overline{\mathcal{N}} = 1 - \mathcal{N}$ for its tail. We use the formula in reverse. Setting $h_{\tan}(t) = h(t)$, the computed hazard of the clock, defines an ordinary differential equation for $b$, which we integrate by the fourth order Runge Kutta method with step $10^{-4}$ from $t_0 = 0.011$. The starting value $b(t_0) = 0.4023$ is the constant level with the crossing mass of the clock, $2\overline{\mathcal{N}}(b(t_0)/\sqrt{t_0}) = P(T \le t_0) = 1.25 \times 10^{-4}$.

The boundary falls steeply at first, with slope $-87.5$ at $t_0$, and then becomes asymptotically linear with slope tending to $-\sqrt{2}\,\gamma_1 = -19.99$, since an inverse Gaussian hazard tends to $\beta^2/2$.

#### 8.2.2.2 The boundary from the Volterra equation

The Volterra equation of the first passage density determines the exact boundary (Section 8.2.2 and [129, Theorems 9.1 and 9.2]), and solving its discretisation for the clock needs three zones: small times, the bulk and the right tail.

For a continuous boundary $b$ with $b(0+) > 0$, the first passage density $f$ of Brownian motion started at 0 satisfies the Volterra equation of the first kind [412]

$$\begin{aligned} P\big(B_t \ge b(t)\big) &= \int_0^t f(s)\, P\big(B_t \ge b(t) \mid B_s = b(s)\big)\, ds, \\ \text{that is} \quad \overline{\mathcal{N}}\Big(\frac{b(t)}{\sqrt{t}}\Big) &= \int_0^t f(s)\, \overline{\mathcal{N}}\Big(\frac{b(t)-b(s)}{\sqrt{t-s}}\Big) ds, \end{aligned} \tag{8.2.1}$$

with $\overline{\mathcal{N}}$ the standard normal tail. In the inverse problem we know $f$, the density of the clock computed in Section 8.1, and solve for $b$ one time step at a time, since once $b$ is known at the earlier grid points the discretised equation at time $t$ is a single scalar equation for $b(t)$. We discretise with 2900 steps on $[0.009, 0.3]$, midpoint quadrature in $s$, and a scalar root search at each step.

Before $t_0 = 0.009$ the clock has crossed with probability only $P(T \le t_0) = 1.42 \times 10^{-7}$. Its density is super flat: it vanishes to all orders at $t = 0$, like $\exp(-(1+o(1))\log^2(1/t)/(16t))$, as the growth $\log G(u) \sim \frac{1}{4}\sqrt{u}\log u$ at infinity dictates; de Bruijn's exponential Tauberian theorem [50, Theorem 4.12.9] gives this rate for $P(T \le t)$, and the density has the same rate because the clock is unimodal (Corollary 11.1.2). So the exact boundary is unbounded as $t \downarrow 0$: a path above $b(t)$ at time $t$ has crossed by then, so $\overline{\mathcal{N}}(b(t)/\sqrt{t}) \le P(T \le t)$, which forces $b(t) \ge (1+o(1))\log(1/t)/(2\sqrt{2})$. In this first zone we replace the exact boundary by a constant level, fixed by the reflection principle, $2\overline{\mathcal{N}}(b/\sqrt{t_0}) = P(T \le t_0)$, which gives $b(t_0) = 0.4992$. The Volterra boundary then falls from 0.4992 to 0.3908 at $t = 0.011$. The tangent boundary starts there at 0.4023, the constant level with the same crossing mass by time 0.011; a boundary that decreases on $[0, 0.011]$ with that crossing mass must end below this level, so the gap is no defect of the tangent approximation.

In the bulk, between $t = 0.011$ and $t = 0.05$, the Volterra and tangent boundaries agree closely: 0.0721 against 0.0710 at $t = 0.02$, and $-0.5894$ against $-0.5954$ at $t = 0.05$. Here there is no cancellation: both sides of (8.2.1) are of order one.

In the right tail the equation cancels. Solved in the form (8.2.1), the computation fails at $t = 0.0937$, where the clock has survival probability $3.8 \times 10^{-7}$: both sides of the equation are within that amount of $P(T \le t) \approx 1$. In general the absolute error of the quadrature must stay below $P(T > t)$, which is $5 \times 10^{-25}$

at $t = 0.3$, beyond double precision. The cure is to subtract the equation from $P(T \le t) = \int_0^t f$, which gives the complementary form

$$\mathcal{N}\Big(\frac{b(t)}{\sqrt{t}}\Big) - P(T > t) = \int_0^t f(s)\,\mathcal{N}\Big(\frac{b(t) - b(s)}{\sqrt{t-s}}\Big)ds,$$

in which every term is small and computed without cancellation, provided the survival function itself is known to relative accuracy. The spectral expansion of Section 8.1 supplies it to full working precision, even where $P(T > t)$ is of order $10^{-25}$. With the switch made where $P(T > t)$ falls below one half, the solution runs smoothly to $t = 0.3$.

In this zone the boundary is nearly linear with slope $\beta = -\sqrt{2}\gamma_1 = -19.99$. A linear boundary does not suffice. Its crossing time has survival function of order $t^{-3/2}e^{-\beta^2 t/2}$, while that of the clock is asymptotic to a multiple of $e^{-\gamma_1^2 t}$. The power $t^{-3/2}$ is removed by a logarithmic correction: for $b(t) = \beta t + a + \kappa \log t$ the Gaussian factor $e^{-b(t)^2/2t}$ acquires $t^{-\kappa\beta}$, and $\kappa\beta = -\frac{3}{2}$ cancels the power. So the right tail boundary is

$$b(t) \approx \beta t + a + \frac{3}{2|\beta|}\log t, \qquad \beta = -\sqrt{2}\,\gamma_1, \qquad \frac{3}{2|\beta|} = 0.0750.$$

Fitting $a$ on $[0.15, 0.3]$ gives $a = 0.578$ with largest residual 0.0095. The slope of the Volterra boundary over $[0.2, 0.3]$ is $-19.77$. It lies between the slope $-19.99$ without the logarithm and the slope $-19.99 + 0.075/0.25 = -19.69$ of the asymptote at $t = 0.25$.

We simulate $n = 2 \times 10^5$ Brownian paths against both boundaries with step $2 \times 10^{-5}$ and a Brownian bridge crossing correction; sampling noise alone gives a Kolmogorov distance of about $0.87/\sqrt{n} \approx 0.002$. For the Volterra boundary the Kolmogorov distance to the law of the clock is 0.0020, at the level of the Monte Carlo noise, and the mean crossing time is 0.023085 against $\mathbb{E}T = 0.0231050$. For the tangent boundary the distance is 0.0059 and the mean 0.023021 (Figure 8.2). The right tail itself cannot be checked by plain simulation, since only about one path in ten million survives past $t = 0.1$.

Wald's identities constrain the exact boundary as a whole. Since $\mathbb{E}T$ is finite, the first passage time $H$ across it satisfies $\mathbb{E}B_H = 0$ and $\mathbb{E}B_H^2 = \mathbb{E}H$, and since $B_H = b(H)$ and $H$ has the law of the clock, $\int_0^\infty b(t)f(t)\,dt = 0$ and $\int_0^\infty b(t)^2 f(t)\,dt = \mathbb{E}T$.

### 8.2.3 Williams' Gamma(2) structure: the tilt, a benchmark and its limits

Williams [518] and Biane, Pitman and Yor [49] showed that the variable $S_2 = \sum_{n\ge1} \frac{2}{\pi^2 n^2}\Gamma_n$ of Section 8.1.2 is a sum of two independent hitting times of level one by the three dimensional Bessel process started at 0, and that $2\xi(s) =$

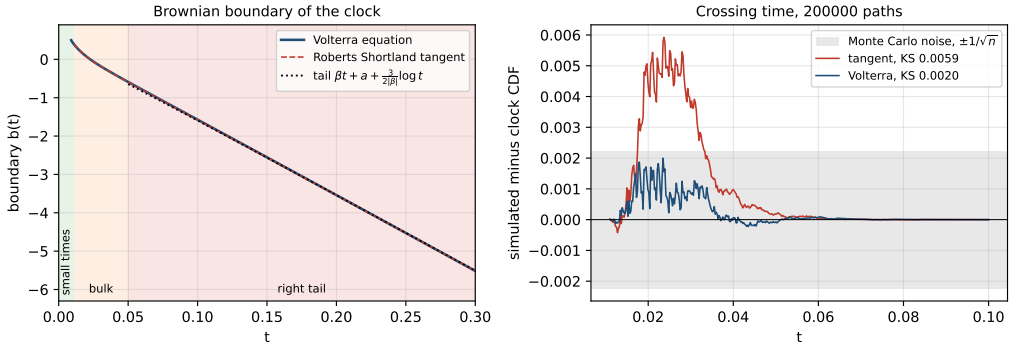



Figure 8.2: Left: the Brownian boundary of the clock computed from the Volterra equation, the Roberts Shortland tangent boundary, and the right tail asymptote with its logarithmic correction, over the three zones. Right: the simulated distribution function of the crossing time minus that of the clock, for both boundaries; the grey band is $\pm 1/\sqrt{n}$, twice the largest standard error of a simulated distribution function.

$\mathbb{E}[(\pi S_2/2)^{s/2}]$, so that the Pólya variable is $Y = \frac{1}{2}\log(\pi S_2/2)$ tilted by $e^{Y/2} = (\pi S_2/2)^{1/4}$ (Section 1.2; [424, Section 16.2.1]). The Thorin measure of $S_2$ has atoms of mass two at the points $\pi^2 n^2/2$, the Gamma(2) shape of a sum of two copies.

First, the tilt. Write $y$ for the variable of the Pólya kernel, so that the value $y$ of $Y$ corresponds to the value $e^{2y}$ of $\pi S_2/2$. Moving the line of integration from the real axis to the line $\operatorname{Im} y = \eta$, as in Proposition 2.1.1, rotates $e^{2y}$ by the angle $2\eta$ into the complex plane. The density of $S_2$, a theta series in its argument, continues analytically to the right half plane, and the boundary of that half plane, the natural boundary of the theta function [458, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], is reached exactly at $\eta = \pi/4$. So the admissible tilts of Proposition 2.1.1 are the rotations of Williams' variable within the half plane of analyticity of its density, and the saving $e^{-\eta\theta}$ is the factor gained by that rotation.

Second, a benchmark. Since $\pi^2 S_2/2 = \sum_n \Gamma_n/n^2$, the Thorin measure of $\pi^2 S_2/2$ is $2\sum_n \delta_{n^2}$, and in the square root variable its Cauchy flow $P_\varepsilon * 2\sum_n \delta_n$ is explicit [424, Section 16.4]. On this measure the distances of Section 3.2 separate the effects of the method from those of the arithmetic, and they need no separate computation: $\pi^2 S_2/2$ is the sum of two independent copies of the sinh clock of Section 1.3.4, so its distances follow from the closed forms of that section (Section 12.2.3).

Third, the limits. Williams' atoms are the Dirichlet eigenvalues $\pi^2 n^2/2$ of an interval, and they live on the Laplace side of the kernel. The zeros of $\xi$ live on the Mellin side of the same variable, and they reappear as unit atoms only on the Laplace side of the reciprocal. The Gamma(2) structure therefore gives positivity

and an exact sampler [115] on the kernel side, but it transfers no positivity to the Thorin measure of the clock. In particular it does not give a positive Gamma(2) scale mixture representation of the Pólya density, a route that [424, Section 12.6] excludes.

Fourth, Maślanka's series. Maślanka [214] expanded $(\sigma-1)\zeta(\sigma) = \sum_{k\ge0} A_k\,(1-\frac{\sigma}{2})_k/k!$, with $A_k = \sum_{j=0}^{k}(-1)^j\binom{k}{j}(2j+1)\zeta(2j+2)$, and Báez-Duarte [17] gave a rigorous proof of this representation. The coefficients are Laguerre moments of Williams' variable [253].

Proposition 8.2.2. *Let $W = \pi^2 S_2/2 = \sum_{n\ge1}\Gamma_n/n^2$, so that $\mathbb{E}e^{-\lambda W} = (\pi\sqrt{\lambda}/\sinh\pi\sqrt{\lambda})^2$, and let $L_k^{(1)}$ be the generalised Laguerre polynomials.*

*(i)* $A_k = \mathbb{E}[W L_k^{(1)}(W)]/(2(k+1))$.

*(ii)* $\sum_{k\ge0} A_k y^k = \big(1-\mathbb{E}e^{-Wy/(1-y)}\big)/(2y)$ *for* $0<|y|<1$.

*(iii) For* $0<\operatorname{Re}\sigma<2$,

$$(\sigma-1)\zeta(\sigma) = \frac{\sin(\pi\sigma/2)}{2\pi}\int_0^\infty r^{-\sigma/2-1}\big(1-\mathbb{E}e^{-rW}\big)\,dr.$$

*For real $0<\sigma<2$ this equals $\mathbb{E}\big[(1-\mathbb{E}[e^{-WR}\mid\mathcal{B}])/(2(1-\mathcal{B}))\big]$, with $\mathcal{B}$ a beta variable of parameters $\frac{\sigma}{2}$ and $1-\frac{\sigma}{2}$, independent of $W$, and $R=(1-\mathcal{B})/\mathcal{B}$; since $\mathbb{E}(1-\mathcal{B})^k = (1-\frac{\sigma}{2})_k/k!$, expanding (ii) at $y=1-\mathcal{B}$ gives Maślanka's series.*

*Proof.* At $s=2j+2$ the identity $2\xi(s) = \mathbb{E}[(W/\pi)^{s/2}]$ reads $(2j+1)\zeta(2j+2) = \mathbb{E}W^{j+1}/(2(j+1)!)$, and $\binom{k}{j}/(j+1)! = \binom{k+1}{k-j}/((k+1)\,j!)$ turns $\sum_j(-1)^j\binom{k}{j}W^{j+1}/(j+1)!$ into $WL_k^{(1)}(W)/(k+1)$, which is (i). For (ii), $\sum_k L_k^{(1)}(w)\,y^{k+1}/(k+1) = \int_0^y(1-t)^{-2}e^{-wt/(1-t)}dt = (1-e^{-wy/(1-y)})/w$, and the bound $|L_k^{(1)}(w)|\le(k+1)e^{w/2}$ for $w\ge0$, with $\mathbb{E}\,We^{W/2}<\infty$ because $\mathbb{E}e^{tW} = \prod_n(1-t/n^2)^{-2}$ is finite for $t<1$, allows the exchange of sum and expectation for $|y|<1$. For (iii) put $z=\sigma/2$. Fubini and $\int_0^\infty r^{-z-1}(1-e^{-rw})\,dr = w^z\Gamma(1-z)/z$ for $0<\operatorname{Re}z<1$, with $\mathbb{E}W^z = 2\pi^z\xi(2z)$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$, give $\pi^z\xi(2z)/\Gamma(z+1) = (2z-1)\zeta(2z)$. The change of variables $r=(1-b)/b$ carries the beta density times $db/(2(1-b))$ to $\frac{\sin\pi z}{2\pi}r^{-z-1}dr$, and the expansion in powers of $1-\mathcal{B}$ is Maślanka's series, proved rigorously in [17]. □

So Maślanka's coefficients determine the law of $W$ and give $\xi$ with no zeros and no primes, but $\xi$ enters through the Mellin transform of $W$, and they transfer no positivity to the Thorin measure of the clock, for the reason given in the third point.

## 8.3 Keiper Li coefficients from the Pólya density

The Pólya density that computes the law of the clock also computes the coefficients of Li's criterion [336], which states that RH holds if and only if

$$\lambda_n = \sum_{\rho} \left[ 1 - \left( 1 - \frac{1}{\rho} \right)^n \right] \geq 0 \qquad \text{for every } n \geq 1,$$

the sum over the zeros taken in conjugate pairs. Keiper [303] introduced the coefficients $\lambda_n$ and computed them before Li proved the criterion, and Bombieri and Lagarias [54] derived an arithmetic expression for them from the explicit formula and related the criterion to Weil's.

### 8.3.1 Taylor coefficients of $\log \xi$ at $s = 1$

Li [336] expresses the coefficients as derivatives at $s = 1$,

$$\lambda_n = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \Big[ s^{n-1} \log \xi(s) \Big]_{s=1}.$$

With $\log \xi(1+w) = \sum_{k \geq 0} a_k w^k$ and $(1+w)^{n-1} = \sum_j \binom{n-1}{j} w^j$, comparing the coefficients of $w^n$ gives

$$\lambda_n = n \sum_{j=0}^{n-1} \binom{n-1}{j} a_{n-j}, \tag{8.3.1}$$

so the computation reduces to the Taylor coefficients of $\log \xi$ at $s = 1$. The zeros nearest to $s = 1$ are $\frac{1}{2} \pm 14.1347i$, so $\log \xi(1+w)$ is analytic in the disc $|w| < R$, where $R = |\rho_1 - 1| = 14.1436$.

The sum (8.3.1) is exact, but its terms can be larger than $\lambda_n$. The Hadamard product $\xi(1+w) = \xi(1) \prod_\rho (1 - w/(\rho - 1))$ gives $m a_m = -\sum_\rho (\rho - 1)^{-m}$ for $m \geq 2$, and for large $m$ the two zeros nearest to $s = 1$ dominate this sum, so that $|m a_m|$ is at most about $2R^{-m}$. Since $n\binom{n-1}{m-1} = m\binom{n}{m}$, the terms of (8.3.1) are $\binom{n}{m} m a_m$ with $m = n - j$, and for large $n$ their absolute sum is at most about $2(1 + 1/R)^n$. Cancellation therefore costs at most about $\log_{10} \big( 2(1 + 1/R)^n / \lambda_n \big)$ digits. This grows like $n \log_{10}(1 + 1/R) \approx 0.03n$ and is about one digit at $n = 100$, where $2(1 + 1/R)^{100} \approx 1.9 \times 10^3$ and $\lambda_{100} = 118.6$.

We compute the $a_k$ by the trapezoid rule for the Cauchy integral on the circle $|w| = 6$,

$$a_k \approx \frac{1}{N r^k} \sum_{j=0}^{N-1} \log \xi\big(1 + r e^{2\pi i j / N}\big) e^{-2\pi i j k / N}, \qquad r = 6,\ N = 512,$$

which converges geometrically. The rule returns $a_k + a_{k+N} r^N + a_{k+2N} r^{2N} + \cdots$, so aliasing changes $a_k$ by a relative amount of about $(r/R)^N \approx 10^{-191}$. Rounding

errors in the sum are amplified in $a_k$ by about $(R/r)^k$, which is $10^{37}$ at $k = 100$ and is absorbed by the working precision.

Each value $\xi(1+w) = \xi(\frac{1}{2} + (\frac{1}{2} + w))$ comes from the Pólya route at 600 bits; the imaginary part of $\frac{1}{2}+w$ never exceeds 6, so the Pólya sum loses almost nothing to cancellation. The principal branch of the logarithm is the analytic one, because $|\arg \xi(1+w)| < 1$ on the circle: in the Hadamard product (1.2.1) with $s = \frac{1}{2} + w$, so that $|s| \le 6.5$, the factor $1 + s^2/\gamma^2$ of the zeros $\frac{1}{2} \pm i\gamma$ has argument at most $\arcsin(6.5^2/\gamma^2) \le 1.01 \cdot 6.5^2/\gamma^2$, and since $\sum_{\gamma>0} \gamma^{-2} = 0.0231$ these arguments add up to less than 0.99.

The first coefficient has the closed form $\lambda_1 = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, where $\gamma_E$ is Euler's constant. The Pólya computation gives

$$\lambda_1 = 0.023095708966121033814 31431025,$$

in agreement with the closed form to all twenty five digits shown. The same number $\lambda_1 = \sum_\rho \rho^{-1} = (\xi'/\xi)(1)$ reappears at the basepoint one (Section 9.1.2): the shifted clock $T_1$ there, with Laplace transform $\xi(1)/\xi(1+\sqrt{u}) = 1 - \lambda_1\sqrt{u} + O(u)$ near $u = 0$, has a tail of order $x^{-1/2}$, and $\lambda_1/\pi$ is its Thorin density at the origin in the chart $\theta = \sqrt{z}$. Table 8.6 lists further values with Keiper's asymptotic $\lambda_n \approx \frac{n}{2}(\log n - 1 - \log 2\pi + \gamma_E)$ [303].

| $n$ | $\lambda_n$ (Pólya) | Keiper asymptotic |
|---|---|---|
| 1 | 0.023095708966121033814 | −1.13 |
| 2 | 0.092345735228046670386 | −1.57 |
| 3 | 0.207638920554324803 79 | −1.74 |
| 5 | 0.575542714461177452 43 | −1.63 |
| 10 | 2.279339363193157743 7 | 0.21 |
| 20 | 8.769276872093215129 6 | 7.35 |
| 50 | 43.531096488374019299 | 41.28 |
| 100 | 118.60377537679132993 | 117.23 |

Table 8.6: Keiper Li coefficients from the Pólya density (all digits shown are correct) and Keiper's leading asymptotic, which becomes accurate only for large $n$.

### 8.3.2 Li's criterion as positivity of the Thorin functional

Li's criterion is a statement about the Thorin functional $U$ of Section 5.3, which has a unit atom at $-(\rho - \frac{1}{2})^2$ for each pair $\{\rho, 1-\rho\}$ of zeros, on the line or off it. For a zero $\rho$ put $r = 1 - 1/\rho$. Then $1 - 1/(1-\rho) = 1/r$, so the pair contributes $2 - r^n - r^{-n}$ to $\lambda_n$; and if $a = -(\rho - \frac{1}{2})^2$, with the square root chosen so that $\rho - \frac{1}{2} = i\sqrt{a}$, then $r = (2\sqrt{a} + i)/(2\sqrt{a} - i)$. This proves the following.

PROPOSITION 8.3.1. *Without any hypothesis on the zeros,*

$$\lambda_n = \sum_{\{\rho,1-\rho\}} \varphi_n\Big(-\big(\rho-\tfrac12\big)^2\Big) = \langle U, \varphi_n\rangle,$$

*where*

$$\varphi_n(z) = 2 - \Big(\frac{2\sqrt{z}+i}{2\sqrt{z}-i}\Big)^n - \Big(\frac{2\sqrt{z}-i}{2\sqrt{z}+i}\Big)^n$$

*is a rational function of $z$, and $\varphi_n(z) = 4\sin^2\big(n\arctan\frac{1}{2\sqrt{z}}\big) \ge 0$ for $z>0$.*

The two fractions are exchanged when $\sqrt{z}$ changes sign, so $\varphi_n$ is a rational function of $z$, with a pole of order $n$ at $z=-\frac14$. At infinity $\varphi_n(z)\sim n^2/z$, so the sum converges absolutely.

For $z>0$ the two fractions are conjugate numbers of modulus one. On the critical line, for $\rho=\frac12+i\gamma$ with $\gamma>0$, the number $1-1/\rho=(\rho-1)/\rho$ has modulus one and argument $\phi_\gamma = 2\arctan\frac{1}{2\gamma}$, and the conjugate zero gives $-\phi_\gamma$. Adding the contributions of $\rho$ and $\bar\rho$,

$$\lambda_n = \sum_{\gamma>0} 2\big(1-\cos(n\phi_\gamma)\big) = \sum_{\gamma>0} 4\sin^2\Big(n\arctan\frac{1}{2\gamma}\Big) \qquad \text{under RH.}$$

Li's criterion thus says that RH holds if and only if $\langle U,\varphi_n\rangle\ge 0$ for every $n$, that is, the Thorin functional is nonnegative on the cone spanned by the $\varphi_n$.

The test functions are Laplace transforms, and this reads Li's coefficients off the heat trace $W$ of Section 5.1 and off the clock tilted at $u=\frac14$ (a note of Nicholas Polson merged into the book [243]). Let $q(y)=y/(1-y)^2$ and define polynomials $\ell_n$ by

$$\sum_{n\ge1}\ell_n(t)\,y^{n-1} = q'(y)\,e^{-tq(y)} = \frac{1+y}{(1-y)^3}\exp\Big(-\frac{ty}{(1-y)^2}\Big),$$

so that $\ell_1=1$, $\ell_2=4-t$, $\ell_3=9-6t+\frac12t^2$, and $\ell_n$ has degree $n-1$ and leading coefficient $(-1)^{n-1}/(n-1)!$. Let $\widetilde T$ be the Esscher tilt of the clock $T$ at $u=\frac14$ (the variable $\widetilde T$ of Theorem 9.2.2), the law $e^{-x/4}P(T\in dx)/\mathbb{E}e^{-T/4}$, with Laplace transform $G(\frac14)/G(\frac14+u)$, and write $\kappa_j(\widetilde T)$ for its cumulants.

PROPOSITION 8.3.2. *Without any hypothesis on the zeros:*

*(i)* $\varphi_n(z)=\int_0^\infty e^{-(z+\frac14)t}\ell_n(t)\,dt$ *for* $\operatorname{Re} z>-\frac14$, *and hence*

$$\lambda_n = \int_0^\infty W(t)\,e^{-t/4}\,\ell_n(t)\,dt;$$

*(ii)* $\kappa_j(\widetilde T)=\int_0^\infty t^{j-1}e^{-t/4}W(t)\,dt=(j-1)!\sum_{\{\rho,1-\rho\}}\big(\rho(1-\rho)\big)^{-j}$ *for* $j\ge1$, *and*

$$\lambda_n = \sum_{j=1}^{n}(-1)^{j+1}\frac{n}{j!}\binom{n+j-1}{2j-1}\kappa_j(\widetilde T), \qquad n\ge1.$$

*Proof.* (i) For $z>0$ put $r=(2\sqrt{z}+i)/(2\sqrt{z}-i)$, so that $\varphi_n(z)=2-r^n-r^{-n}$ and $r+r^{-1}=2-1/(z+\frac14)$. Then $(1-ry)(1-y/r)=(1-y)^2\big(z+\frac14+q(y)\big)/(z+\frac14)$, and for small $|y|$

$$\sum_{n\ge1}\varphi_n(z)\,y^{n-1}=\frac{d}{dy}\log\frac{(1-ry)(1-y/r)}{(1-y)^2}=\frac{q'(y)}{z+\frac14+q(y)}$$
$$=\int_0^\infty e^{-(z+\frac14)t}\,q'(y)e^{-tq(y)}\,dt.$$

Comparing coefficients gives the formula for $\varphi_n(z)$ when $z>0$, and both sides are analytic in $\operatorname{Re}z>-\frac14$. By Proposition 8.3.1, $\lambda_n$ is the sum of $\varphi_n(a)$ over $a=-(\rho-\frac12)^2$, one for each pair $\{\rho,1-\rho\}$. For $\rho=\frac12+\delta+i\gamma$ with $|\delta|<\frac12$ one has $\operatorname{Re}a+\frac14>\gamma^2$, and $\int_0^\infty e^{-ct}|\ell_n(t)|\,dt=O(1/c)$ as $c\to\infty$, so the sum of the integrals converges absolutely, and $W(t)=\sum e^{-at}$ gives the formula for $\lambda_n$. (ii) Since $1+w=1/(1-y)$ for $w=y/(1-y)$, and $[y^n]w^k=\binom{n-1}{n-k}$, (8.3.1) says $\lambda_n=n[y^n]\log\xi\big(1/(1-y)\big)$. With $\sigma=1/(1-y)$ one has $\sigma(\sigma-1)=q(y)$, so $\sigma=\frac12+\big(\frac14+q(y)\big)^{1/2}$ and $\log\xi(\sigma)=\log\xi(\frac12)+\log G\big(\frac14+q(y)\big)$ near $y=0$. By Section 5.1,

$$\log\frac{G(\frac14+v)}{G(\frac14)}=-\log\mathbb{E}e^{-v\widetilde{T}}=\int_0^\infty\big(1-e^{-vt}\big)e^{-t/4}W(t)\,\frac{dt}{t}=\sum_{j\ge1}(-1)^{j+1}\kappa_j(\widetilde{T})\,\frac{v^j}{j!},$$

which gives the first formula for $\kappa_j(\widetilde{T})$; integrating $W(t)=\sum e^{-at}$ term by term, with $a+\frac14=\rho(1-\rho)$, gives the second. Finally $[y^n]q(y)^j=[y^{n-j}](1-y)^{-2j}=\binom{n+j-1}{2j-1}$. □

In terms of the sums $S_j=\sum\big(\rho(1-\rho)\big)^{-j}$, the relation (ii) is that of Voros [360, Section 8.6.2]; the proposition adds that $(j-1)!\,S_j$ are the cumulants of a probability law, the clock tilted to the point $u=\frac14$, which $\sigma=\frac12+\sqrt{u}$ sends to $\sigma=1$. The relation is triangular with diagonal $(-1)^{n+1}/(n-1)!$, so the $\lambda_n$ and the cumulants of $\widetilde{T}$ determine each other; in particular $\mathbb{E}\widetilde{T}=\lambda_1$ and $\operatorname{Var}\widetilde{T}=4\lambda_1-\lambda_2=3.71006364\times10^{-5}$. From 100 independently computed cumulants, (ii) reproduces all 20 digits of Table 8.6 (the rows $n=1,2,3,5,10,20,50,100$), and (i), with $W$ from (1.2.5) for $t<0.004$ and from the zeros above, reproduces $\lambda_1,\dots,\lambda_4$ to 30 digits (computed, `pg3_li.py`).

Under RH, $W(t)=\sum_{\gamma>0}e^{-\gamma^2t}$, and (i) returns the sum of the nonnegative terms $\varphi_n(\gamma^2)$ displayed above. The functional $f\mapsto\int_0^\infty f(t)e^{-t/4}\ell_n(t)\,dt$ is nonnegative on every completely monotone $f$ for which it converges absolutely, by Bernstein's theorem and $\varphi_n\ge0$ on $[0,\infty)$. It is not nonnegative on nonnegative $f$, since $\ell_2<0$ on $(4,\infty)$, nor, in general, on $k$ monotone $f$: $(1-t/20)_+^2$ is 3 monotone, and

$$\int_0^{20}e^{-t/4}\,\ell_3(t)\Big(1-\frac{t}{20}\Big)^2dt=\frac{4-2824e^{-5}}{25}<0.$$

So Li's criterion, read on $W$, asks for the complete monotonicity of Section 5.1, which positivity of $W$ does not give.

The test function $\varphi_n$ is about $n^2/z$ for $\sqrt{z} \gg n$ and oscillates between 0 and 4 for $\sqrt{z} \lesssim n/\pi$, so $\lambda_n$ weighs the zeros below height of order $n$ at full size and the higher zeros by about $n^2/\gamma^2$ (Figure 8.3). Every computed $\lambda_n$ is positive, as it must be: every zero below height $3 \times 10^{12}$ lies on the line [419] and contributes a nonnegative term, while a zero at height $\gamma \gg n$ contributes only about $n^2/\gamma^2$ in absolute value, so the zeros above that height cannot change the sign of $\lambda_n$ for $n \le 100$.

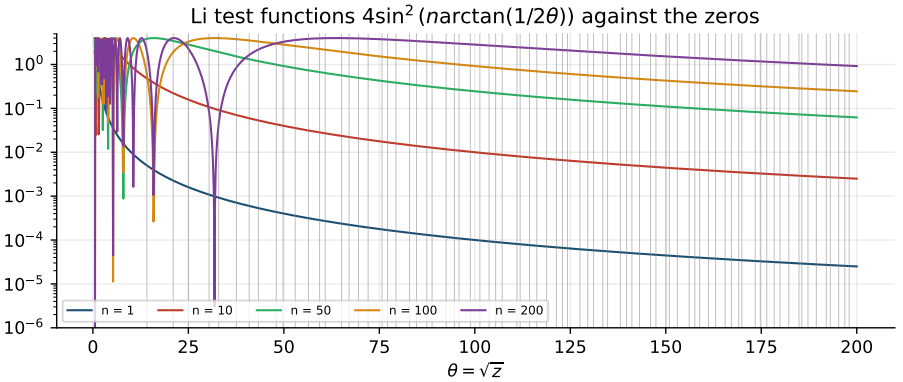



Figure 8.3: The Li test functions $4\sin^2(n \arctan \frac{1}{2\theta})$ for $n = 1, 10, 50, 100, 200$, in the square root chart $\theta = \sqrt{z}$, against the zeros below height 200 (grey vertical lines). Larger $n$ probes higher zeros.

An off line zero makes $|1 - 1/\rho| \neq 1$. For $\rho = \frac{1}{2} - d + i\gamma$ with $d > 0$ one has $|1 - 1/\rho|^2 = 1 + 2d/((\frac{1}{2} - d)^2 + \gamma^2)$, so for large $n$ the term $(1 - 1/\rho)^n$ grows like $e^{nd/\gamma^2}$, and this is how a failure of RH would eventually make $\lambda_n$ negative (Exercise 8.14). Since the exponent is $nd/\gamma^2$, the index at which this happens grows with the height of the zero, so finitely many coefficients, like the finite tests of Section 5.2 (Table 5.5), cannot exclude an off line zero at great height.

Summing $4\sin^2(n \arctan \frac{1}{2\gamma})$ over the first 1700 zeros, computed with `mpmath` [376] as in Section 3.2, and adding the tail above $\gamma_{1700}$ with the smooth zero density $\log(\gamma/2\pi)/2\pi$ gives the Pólya values with relative residuals near $5 \times 10^{-6}$ for $n \le 20$ and $10^{-5}$ at $n = 100$. The absolute residual grows like $n^2$ and comes from the cutoff: at $\gamma_{1700}$ the zero count exceeds its smooth part by about one half, so the smooth tail counts about half a term, $\frac{1}{2}\varphi_n(\gamma_{1700}^2) \approx n^2/(2\gamma_{1700}^2)$, too many.

As a diagnostic the coefficients are poorly localised, so $\lambda_n$ behaves like the heat trace and Bernstein diagnostics of Table 1.7 and unlike the Thorin density, which is local in the ordinate.

Weil's criterion has the same form as Li's, with the functions of the class $\mathcal{T}$ of

Section 5.3 that are nonnegative on $(0,\infty)$ in place of the $\varphi_n$ (Section 3.5.2), but the two cones are different. The pole of $\varphi_n$ lies at $-\frac14$, the point at which (5.3.1) evaluates its test functions, inside the region where $\mathcal{T}$ requires holomorphy, and $\varphi_n$ decays only like $1/z$. So $\varphi_n$ is not in $\mathcal{T}$, $\langle U,\varphi_n\rangle$ is defined through the zeros, and the prime side of $\lambda_n$ is the regularised arithmetic formula of Bombieri and Lagarias [54].

## 8.4 The de Bruijn Newman flow from the Pólya density

The de Bruijn Newman flow reads RH from a deformation of the Pólya kernel. Multiplying the kernel by a Gaussian factor turns $\Xi$ into a family of entire functions $H_t$ that evolve by the backward heat equation. We follow two pairs of zeros backward in time to their collisions and compare the collision times with the isolated pair estimate and with the lower bound of Csordas, Smith and Varga.

### 8.4.1 The flow, the motion of its zeros and its Thorin form

In the normalisation of Rodgers and Tao [463], which Theorem 7.2.16 also uses,

$$H_t(z)=\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du,$$
$$\Phi_{\mathrm{dBN}}(u)=\sum_{n\ge1}\bigl(2\pi^2n^4e^{9u}-3\pi n^2e^{5u}\bigr)\exp(-\pi n^2e^{4u}),$$

so that $H_0(z)=\frac18\Xi(z/2)$. If $v=2u$ denotes the variable of (1.2.3), then $\Phi_{\mathrm{dBN}}(u)=\frac12\Phi(v)$ and

$$8H_t(2\theta)=\int_{\mathbb{R}}\Phi(v)\,e^{tv^2/4}\cos(\theta v)\,dv,$$

so the flow multiplies the kernel of (1.2.3) by $e^{tv^2/4}$, and a zero $z$ of $H_t$ corresponds to the ordinate $\theta=z/2$.

De Bruijn showed that $H_t$ has only real zeros for $t\ge\frac12$ and that the property persists forward in time [112]. Newman showed that there is a constant $\Lambda>-\infty$, at most $\frac12$ by de Bruijn's theorem, such that $H_t$ has only real zeros exactly for $t\ge\Lambda$, and conjectured $\Lambda\ge0$ [381], which Rodgers and Tao proved [463]. Ki, Kim and Lee sharpened de Bruijn's bound to $\Lambda<\frac12$ [310], the Polymath project proved $\Lambda\le0.22$ [456], and Platt and Trudgian improved this to $\Lambda\le0.2$ [419]. De Bruijn's bound comes from a theorem about strips: if the zeros of $H_0$ lie in $|\operatorname{Im}z|\le\Delta$, then $H_t$ has only real zeros for $t\ge\Delta^2/2$ [112, Theorem 13], and the critical strip gives $\Delta=1$. A zero free half plane $\operatorname{Re}s>\alpha$ gives $\Delta=2\alpha-1$, hence $\Lambda\le2(\alpha-\frac12)^2$, which improves 0.2 only if $\alpha<\frac12+\sqrt{0.1}\approx0.816$. RH is the inequality $\Lambda\le0$, so by the theorem of Rodgers and Tao it is equivalent to $\Lambda=0$.

While the zeros are simple they move according to

$$\frac{dz_k}{dt} = 2\sum_{j\neq k}\frac{1}{z_k - z_j},$$

the sum running over all other zeros, positive and negative, with $z_j$ and $-z_j$ taken together. Differentiating $H_t(z_k(t)) = 0$ in $t$ and using the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ gives $dz_k/dt = \partial_z^2 H_t/\partial_z H_t$ at $z_k$, and the Hadamard product of the even function $H_t$ turns this into the display (Exercise 8.15). The equation is the gradient flow of unit charges on the line with logarithmic interaction, so forward in time the zeros repel each other and spread. Backward in time neighbouring zeros approach each other, and a close pair collides and then leaves the real axis as a conjugate pair. A pair at distance $D$ that is isolated from the other zeros obeys $dD/dt \approx 4/D$, so $D^2 \approx D_0^2 + 8t$ with $D_0 = D(0)$, and the pair collides at $t \approx -D_0^2/8$. RH is the statement that no two zeros collide at a positive time. Numerically the flow makes the first zeros more regular: unfolding the first 120 zeros of $H_t$ by $\vartheta(z/2)/\pi$, the variance of the unfolded gaps is 0.1240, 0.1165, 0.1096 and 0.0975 at $t = 0$, 0.05, 0.1 and 0.2, and the smallest gap is 0.3868, 0.4092, 0.4300 and 0.4678, always between the 34th and 35th zeros (computed, `code/gp_flow.py`). This is an observation about these zeros and not a consequence of the repulsion: for the uniform law on $[-1,1]$, whose transform $\sin\theta/\theta$ has the equally spaced zeros $k\pi$, the first order velocity of Remark 13.5.3 at the $k$th zero is $-2/(k\pi)$, so gaps that start equal become unequal. This follows a note of Nicholas Polson merged into the book [242].

At $t = 0$ the velocity has a closed form through Hardy's function $Z$ of Section 2.2 (a note of Nicholas Polson merged into the book [243]).

Proposition 8.4.1. *Let $\gamma_n$ be the $n$th positive zero of $Z$, and suppose it is simple, so that $z_n = 2\gamma_n$ is a simple zero of $H_0$ and moves smoothly with $t$ near $t = 0$. Then*

$$\dot z_n(0) = \frac{1}{2}\Big[\frac{Z''(\gamma_n)}{Z'(\gamma_n)} + \frac{4\gamma_n}{\gamma_n^2 + \frac14} - \operatorname{Im}\psi\big(\tfrac14 + \tfrac{i\gamma_n}{2}\big)\Big],$$

*and the last two terms tend to $-\frac{\pi}{2}$ as $\gamma_n \to \infty$, a drift of $-\frac{\pi}{4}$ in $\dot z_n$.*

*Proof.* Since $H_0(z) = \frac18\Xi(z/2)$, the formula $dz_k/dt = \partial_z^2 H_t/\partial_z H_t$ gives $\dot z_n(0) = \Xi''(\gamma_n)/(2\Xi'(\gamma_n))$. Write $\Xi = gZ$ with $g(\theta) = -\frac12(\theta^2 + \frac14)\pi^{-1/4}|\Gamma(\frac14 + \frac{i\theta}{2})|$, which is $\zeta(\frac12 + i\theta) = e^{-i\vartheta(\theta)}Z(\theta)$ inserted into the definition of $\xi$. At a zero of $Z$, $\Xi' = gZ'$ and $\Xi'' = 2g'Z' + gZ''$, so $\Xi''/\Xi' = Z''/Z' + 2g'/g$, and $2g'/g = 4\theta/(\theta^2 + \frac14) - \operatorname{Im}\psi(\frac14 + \frac{i\theta}{2})$ because $\frac{d}{d\theta}\log|\Gamma(\frac14 + \frac{i\theta}{2})| = -\frac12\operatorname{Im}\psi(\frac14 + \frac{i\theta}{2})$. Finally $\psi(z) = \log z + O(1/z)$ as $z \to \infty$ in $\operatorname{Re} z > 0$ [397, Section 5.11], so $\operatorname{Im}\psi(\frac14 + \frac{i\theta}{2}) \to \frac{\pi}{2}$. □

The rates of Section 8.4.8 are $\dot a_j(0) = \gamma_j \dot z_j(0)$, and the proposition reproduces $\dot a_1(0)$ there. The first 34 zeros move down at $t = 0$, and the 35th, at $\gamma_{35} = 111.874659177$, moves up, with $\dot z_{35}(0) = +0.0595012982$; all 35 signs are certified

in ball arithmetic (`pg3_velocity.py`). Among the first 1500 zeros 286 move up, the velocities average $-0.78280$ against the drift $-\frac{\pi}{4} = -0.7854$, and the largest upward and downward velocities, $+9.16075$ and $-11.21487$, belong to the zeros $n = 1497$ and $n = 1496$, at distance 0.0975025 in $\gamma$ (computed). In the equation of motion a zero at distance $\Delta$ below $z_n$ adds $2/\Delta$ to $\dot z_n$, so the upper zero of a close pair is pushed up, the repulsion that Csordas, Smith and Varga [101] turned into lower bounds for $\Lambda$ (Section 8.4.6); whether a given zero rises depends on all the others, not on one gap.

Let $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$, where $u$ is the variable of the clock (written $s$ in Chapters 6 and 7), not the variable of integration in $H_t$, so that $R_0$ is the clock $1/G$. By Theorem 7.2.16, for every $t \geq \Lambda$, and so for every $t \geq 0.2$, $R_t$ is the Laplace transform of a GGC whose Thorin atoms move with the zeros of $H_t$. Put $\xi_t(s) = 8H_t\big(-2i(s-\frac12)\big)$, so that $\xi_t(\frac12 + i\theta) = 8H_t(2\theta)$ and $\xi_0 = \xi$, and

$$\rho_{\varepsilon,t}(\theta) = \pi^{-1}\,\mathrm{Re}(\xi_t'/\xi_t)\big(\tfrac12 + \varepsilon + i\theta\big).$$

Each zero $z$ of $H_t$ gives the zero $\frac12 + iz/2$ of $\xi_t$, and for every $t$ the function $\rho_{\varepsilon,t}$ is the sum of the Poisson kernels $P_{\frac12+\varepsilon-\beta}(\theta-\gamma)$ over the zeros $\beta+i\gamma$ of $\xi_t$. For $t \geq \Lambda$ every $\beta$ equals $\frac12$ and $\rho_{\varepsilon,t}$ is the Thorin density of $R_t$ at resolution $\varepsilon$. As $t$ decreases through a collision, $\rho_{\varepsilon,t}$ shows the off line signature of Section 3.3: two Poisson peaks merge, and after the collision the pair contributes the kernels $P_{\varepsilon+d(t)} + P_{\varepsilon-d(t)}$ at its ordinate, where $d(t)$, half the imaginary part of the pair in $z$, grows from zero; the second kernel is negative once $d(t) > \varepsilon$. The flow thus produces off line configurations of known geometry, on which the diagnostics of Chapters 3 to 5 can be calibrated; Section 12.2.3 lists this calibration among the open computations.

### 8.4.2 Two collisions computed from the Pólya kernel

The Pólya route evaluates $H_t$ at any $t$ by multiplying the node values $\Phi(v_j)$ by $e^{tv_j^2/4}$ and summing. The factor is entire, so the error analysis of Section 2.1 carries over. Among the 79 zeros below height 200 the closest pair is $\gamma_{71} = 184.8745$, $\gamma_{72} = 185.5988$, at distance 0.7243 in $\theta$, that is $D_0 = 1.4486$ in $z$. We follow the pair by the sign changes of $H_t(2\theta)$ on a grid and refine each zero by Newton's method (Table 8.7).

Bisection on $t$ for the disappearance of the two sign changes gives the collision time

$$t_* = -0.2880,$$

with a grid error of at most $6.1 \times 10^{-5}$. Near a collision the term $4/D$ dominates the motion of the pair, so $D^2 \approx 8(t - t_*)$ and the gap in $\theta$ is about $\sqrt{2(t-t_*)}$; on a grid of step $h$ the sign changes therefore vanish at a time at most $h^2/2$ above $t_*$. The isolated pair estimate is $-D_0^2/8 = -0.262$: the neighbouring zeros, which the

| $t$ | zeros in $\theta = z/2$ | sign changes |
|---|---|---|
| 0.2 | 184.676, 185.590 | 2 |
| 0.1 | 184.771, 185.598 | 2 |
| 0.0 | 184.874, 185.599 | 2 |
| −0.05 | 184.931, 185.595 | 2 |
| −0.1 | 184.992, 185.587 | 2 |
| −0.15 | 185.059, 185.573 | 2 |
| −0.2 | 185.136, 185.550 | 2 |
| −0.3 | none | 0 |

Table 8.7: The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; $t$ is the de Bruijn Newman time.

estimate ignores, delay the collision. Since nonreal zeros persist backward in time, $H_t$ has nonreal zeros for every $t$ below the collision time, so $\Lambda \geq t_* \approx -0.288$.

Lehmer's pair near height 7005, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ (Section 2.2.3.1), is much closer: $D_0 = 2 \times 0.0377 = 0.0754$, and its isolated pair estimate is $-D_0^2/8 = -7.1 \times 10^{-4}$. The tilted Pólya route evaluates $H_t$ at that height at the same cost as $\Xi$. With the parameters of Section 2.2.3.1 we track the two sign changes of the pair on a grid of step 0.0025 in $\theta$ over $[7005.03, 7005.13]$. Both sign changes are present at $t = 0$ and absent at $t = -0.004$, and twelve bisection steps on this interval give $-7.10 \times 10^{-4}$, which by the argument above lies above the collision time by at most $0.0025^2/2 \approx 3 \times 10^{-6}$. Newton's method for the double zero, $H_t(2\theta) = \partial_\theta H_t(2\theta) = 0$ in the unknowns $\theta$ and $t$, started from this value, gives

$$t_* = -7.11 \times 10^{-4}$$

at $\theta = 7005.0817$. Here the isolated pair estimate is accurate: for a pair this close the neighbours hardly matter. The bound $\Lambda \geq t_*$ is much sharper than the one from the pair below height 200.

Pólya's function $K_{i\theta/2}(2\pi)$ behaves differently under the same flow. Down to $t = -60$ none of its real zeros below height 90 collide, and as $t$ decreases from 0 to $-0.5$ they rise almost as a rigid block, at speeds within 2% of the mean speed $\pi/8$ that the two functions share, while the zeros of $\Xi$ rise at speeds between 0.004 and 0.735; yet by Newman's theorem the de Bruijn Newman constant of Pólya's function is finite (Section 8.7.6).

### 8.4.3 The Hankel test along the flow

The Stieltjes moment test of Section 5.1 (Table 5.2) needs only the Taylor coefficients of $\Xi$ at the centre, and along the flow these are moments of the reweighted

kernel [433]. Putting $\theta = i\sqrt{u}$ in the formula for $8H_t(2\theta)$ of Section 8.4.1 gives

$$G_t(u) = \frac{H_t(2i\sqrt{u})}{H_t(0)} = \sum_{k\ge 0} \frac{M_k(t)\,u^k}{(2k)!\,M_0(t)}, \qquad M_k(t) = \int_{\mathbb{R}} v^{2k} e^{tv^2/4}\Phi(v)\,dv,$$

and $G_0$ is the function $G$ of Section 5.1. Write $\pm\theta_j(t)$ for the zeros of $\theta \mapsto H_t(2\theta)$, so that $\theta_j(0) = \gamma_j$ while the zeros are on the line. Since $H_t$ is even and entire of order one, $G_t$ has order $\frac12$ and $G_t(u) = \prod_j (1 + u/a_j(t))$ with $a_j(t) = \theta_j(t)^2$, so the Newton identities turn the $M_k(t)$ into the moments $m_n(t) = \sum_j a_j(t)^{-n-1}$, exactly as at $t = 0$. If every $\theta_j(t)$ is real, the $m_n(t)$ are the moments of the positive measure $\nu_t = \sum_j a_j(t)^{-1}\delta_{1/a_j(t)}$, and every Hankel matrix $\mathcal{H}_N(t) = [m_{i+j}(t)]_{i,j<N}$ is positive definite. The converse is proved as at $t = 0$: if every $\mathcal{H}_N(t)$ is positive definite, every $a_j(t)$ is real, and it is positive because $H_t$ has no zeros on the imaginary axis: $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, the kernel (1.2.3) being strictly positive. So at every time $t$ all the matrices $\mathcal{H}_N(t)$ are positive definite exactly when $H_t$ has only real zeros. The Cholesky factor gives the Jacobi matrix $J_N(t)$, whose eigenvalues return the lowest zeros of $H_t$ as $J_N$ returns the $\gamma_j$ at $t = 0$ (Table 8.8).

| $t$ | $\theta_1(t)$ | $\theta_2(t)$ | $\theta_3(t)$ | $\theta_5(t)$ | $\theta_{10}(t)$ |
|---|---|---|---|---|---|
| 0.2 | 14.07681 | 20.94669 | 24.95221 | 32.88801 | 49.72988 |
| 0 | 14.13473 | 21.02204 | 25.01086 | 32.93506 | 49.77383 |
| −0.4 | 14.25083 | 21.17365 | 25.12774 | 33.02595 | 49.85213 |
| −1.0 | 14.42563 | 21.40341 | 25.30189 | 33.15277 | 49.93350 |

Table 8.8: Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \dots, m_{59}(t)$; at $t = 0$ they are $\gamma_1$, $\gamma_2$, $\gamma_3$, $\gamma_5$ and $\gamma_{10}$. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \dots, \theta_{12}(t)$ to within $3.1 \times 10^{-7}$ of the zeros that Newton's method finds on $H_t$, and $\theta_{13}(t)$ only to within $5.2 \times 10^{-4}$.

Going back from $t = 0$, the matrix $\mathcal{H}_N(t)$ stays positive definite until a time $t_N^*$ at which it ceases to be. Since $\mathcal{H}_N(t)$ is the leading block of $\mathcal{H}_{N+1}(t)$, $t_N^* \le t_{N+1}^*$. Under RH every $t_N^*$ is negative, because $\mathcal{H}_N(0)$ is positive definite, and $t_N^* \to 0$: for each $t < 0$ the function $H_t$ has nonreal zeros by the theorem of Rodgers and Tao, so some $\mathcal{H}_N(t)$ is not positive definite, and then $t_{N'}^* \ge t$ for every $N' \ge N$. With the trapezoid rule of step $1/600$ at $100 + 16N$ digits, the grid $t = 0, -0.05, \dots, -4$ and bisection to $10^{-7}$ give no failure of order 10 down to $t = -4$, and

$$t_{20}^* = -1.8991865, \qquad t_{30}^* = -1.2853067,$$
$$t_{40}^* = -1.2760788, \qquad t_{50}^* = -1.0325048;$$

for $N = 30$ a second discretisation (step $1/550$, 650 digits) gives a bracket that overlaps the first in $[-1.28530669, -1.28530664]$.

Each failure registers a collision. We follow the real zeros of $H_t$ that start at $2\gamma_j$, $j \le 64$, back to $t = -2.05$ by the equation of motion of Section 8.4.1, on the trapezoid sum for $H_t$, and locate each collision as a double zero, $H_t(z) = \partial_z H_t(z) = 0$, by Newton's method in the two unknowns $z$ and $t$; the backward heat equation, which holds exactly for the trapezoid sum, gives the derivatives in $t$. Thirteen of the pairs below index 59 collide in this time range, among them those that start at $(\gamma_k, \gamma_{k+1})$ for $k = 9$, 13, 19, 24, 27 and 34, at

$$t = -1.87791,\ -1.28527,\ -1.27554,\ -1.14387,\ -0.88108,\ -0.38456$$

and the heights $\theta = 49.664$, 60.491, 77.027, 88.466, 95.540 and 111.638; the other seven start at $k = 40$, 43, 45, 48, 51, 53 and 57. To see which collision a failure registers, we continue each double zero into the complex plane, starting from $z_c \pm i\sqrt{2(t_c - t)}$ near the collision, and remove the contribution $2\,\mathrm{Re}\,\theta^{-2n-2}$ of the conjugate pair $\theta = z/2$, $\bar\theta$ from $m_n(t)$. At $t_N^* - \delta$, with $\delta = 0.01$ (for $N = 40$, $\delta = 0.0046$, half the time to the collision of $(\gamma_{13}, \gamma_{14})$), the only one of the thirteen pairs whose removal makes $\mathcal{H}_N(t)$ positive definite again is the one that starts at $(\gamma_9, \gamma_{10})$ for $N = 20$, at $(\gamma_{13}, \gamma_{14})$ for $N = 30$, at $(\gamma_{19}, \gamma_{20})$ for $N = 40$ and at $(\gamma_{27}, \gamma_{28})$ for $N = 50$. The same holds at $t_N^* - 10^{-5}$, except that there removing $(\gamma_{34}, \gamma_{35})$ also restores the order 50 matrix. The failures thus come 0.021, $3.7 \times 10^{-5}$, $5.3 \times 10^{-4}$ and 0.151 after the collisions they register.

A Hankel test of finite order is thus a window on the low part of the spectrum. The order 30 matrix is positive definite at $t = -1.276, -1.278, \ldots, -1.284$ and $-1.2852$, after the collision of $(\gamma_{19}, \gamma_{20})$ at $-1.27554$ and before that of $(\gamma_{13}, \gamma_{14})$ at $-1.28527$: it passes the pair that is too high and too light for its order and registers the lower one. A conjugate pair at height $\theta$ enters $m_n(t)$ with weight of order $\theta^{-2n-2}$, against $\theta_1(t)^{-2n-2}$ for the lowest zero, as at $t = 0$, and increasing the order widens the window.

The whole curve $N \mapsto t_N^*$ up to $N = 50$ follows from one elimination per time. Let $n_{\mathrm{fail}}(t)$ be the index of the first nonpositive pivot of Gaussian elimination on $\mathcal{H}_{50}(t)$, that is, the least $N \le 50$ for which $\mathcal{H}_N(t)$ is not positive definite, so that $t_N^* = \sup\{t < 0 : n_{\mathrm{fail}}(t) \le N\}$. On the grid of step 0.01 it is a staircase [433] (Table 8.9). To see what each step registers, the table gives, at one time on each step, the least order at which each complex pair breaks the test alone, that is, with the contributions of all the other complex pairs removed from the moments. A pair alone breaks the test at a definite order, a higher pair at a higher order, and $n_{\mathrm{fail}}$ is close to the least of these orders. The order at which a pair is seen falls as the pair moves away from the axis: $(\gamma_{27}, \gamma_{28})$, which collides at $-0.881$, breaks order 50 at $t = -1.09$ and order 39 at $-1.29$, and $(\gamma_{34}, \gamma_{35})$, which collides earlier, at $-0.385$, but lies higher, breaks the test only from $t = -1.28$ on, at order 50 and then 47. Removing the pair with the least order moves $n_{\mathrm{fail}}$ up to about the next one: at $t = -1.30$ from 27 to 35, when $(\gamma_{13}, \gamma_{14})$ is removed, at $-1.28$ from 39 to 41, when $(\gamma_{19}, \gamma_{20})$ is removed, and at $-1.09$ from 50 to no failure, when

$(\gamma_{27}, \gamma_{28})$ is removed. With all thirteen pairs removed, no order up to 50 fails at any of these times.

| | | | order broken by $(\gamma_k, \gamma_{k+1})$ alone | | | | |
|---|---|---|---|---|---|---|---|
| $t$ | $n_{\text{fail}}(t)$ | at $t =$ | $k = 13$ | 19 | 24 | 27 | 34 |
| $[-1.03, 0]$ | none | | | | | | |
| $[-1.14, -1.04]$ | 50 | $-1.09$ | real | real | real | 50 | $> 50$ |
| $[-1.18, -1.15]$ | 46 | $-1.16$ | real | real | 44 | 45 | $> 50$ |
| $[-1.27, -1.19]$ | 45 | $-1.23$ | real | real | 44 | 45 | $> 50$ |
| $-1.28$ | 39 | $-1.28$ | real | 38 | 40 | 41 | 50 |
| $-1.29$ | 28 | $-1.29$ | 28 | 35 | 37 | 39 | 47 |
| $-1.30$ | 27 | $-1.30$ | 27 | 35 | 37 | 39 | 47 |

Table 8.9: The first failing order $n_{\text{fail}}(t) \le 50$ of the Hankel test along the flow on the grid of step 0.01 and, at one time on each step, the least order at which the pair that starts at $(\gamma_k, \gamma_{k+1})$ breaks the test alone (real: the pair has not yet collided; $> 50$: no order up to 50 fails). The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone.

A rule suggested in [433] is that the collision of $(\gamma_k, \gamma_{k+1})$ is registered from order about $2k + 2$ on, so that $t_N^* \approx \max\{t_{\text{coll}}(k) : 2k + 2 \le N\}$, where $t_{\text{coll}}(k)$ is the collision time of the pair. It holds for $N = 30$ and 40 ($k = 13$ and 19), roughly for $N = 20$ ($k = 9$, delay 0.021), and fails at $N = 50$: it gives the collision of $(\gamma_{24}, \gamma_{25})$ at $-1.144$, while the order 50 matrix registers $(\gamma_{27}, \gamma_{28})$, for which $2k + 2 = 56$, at $-1.0325$, before $(\gamma_{24}, \gamma_{25})$ has collided. Table 8.9 shows why: a pair beyond the window of the rule enters it after a delay, and $t_N^*$ is the latest of the collision times of the pairs inside the window and of the times at which pairs above it have moved far enough from the axis.

The curve rises to 0 through the collision times of ever closer pairs. Following every real zero that starts below index 215 in the same way (200 digits, step 1/200), 33 of the first 120 gaps close on $[-2.05, 0]$, 28 of them after $t = -1.6$, and 50 of the first 200 gaps close after $t = -1.6$. The latest collisions among the first 200 gaps are those of $(\gamma_{186}, \gamma_{187})$, at distance 0.498, at $t = -0.1310$, of $(\gamma_{159}, \gamma_{160})$ at $-0.1695$ and of $(\gamma_{135}, \gamma_{136})$ at $-0.2015$, and below height 230 those of $(\gamma_{91}, \gamma_{92})$ at $-0.2821$ and of the closest pair below height 200, $(\gamma_{71}, \gamma_{72})$, at $-0.2880$, the value of Section 8.4.2. A collision close to $t = 0$ needs a pair as close as Lehmer's (Section 8.4.2), far beyond the reach of a Hankel test at the centre.

Further back in time the low pairs leave the axis one after another, and Theorem 5.1.7 turns the test into a count. Its proof applies at every time to the zeros of $H_t$, none of which is purely imaginary: the numbers $\iota_N(t)$ and $\iota'_N(t)$ of negative eigenvalues of $\mathcal{H}_N(t)$ and of $\mathcal{H}'_N(t) = [m_{i+j+1}(t)]_{i,j<N}$ do not decrease with $N$, never exceed the number of distinct conjugate pairs $\theta, \bar{\theta}$ of zeros with

$\operatorname{Re}\theta > 0$, and tend to it, so that each negative eigenvalue certifies a pair. The signs of the pivots of one Gaussian elimination give $\iota_1(t), \dots, \iota_{30}(t)$, and $n'_{\text{fail}}(t)$ is to $\mathcal{H}'_N(t)$ what $n_{\text{fail}}(t)$ is to $\mathcal{H}_N(t)$. We follow the real zeros below $\theta = 88$, those that start at $\gamma_1, \dots, \gamma_{24}$, back to $t = -20$ and locate each collision as a double zero as above. Below $\theta = 88$ the collisions found above are those of $(\gamma_9, \gamma_{10})$, $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$, and after them come those of $(\gamma_{22}, \gamma_{23})$, $(\gamma_{15}, \gamma_{16})$, $(\gamma_4, \gamma_5)$, $(\gamma_7, \gamma_8)$, $(\gamma_{17}, \gamma_{18})$ and $(\gamma_2, \gamma_3)$, at

$$t = -2.32414,\ -2.75904,\ -3.95479,\ -4.03382,\ -7.31401,\ -12.49451$$

and the heights $\theta = 84.875$, 67.406, 33.099, 43.403, 72.746 and 27.321. Each pair is followed into the complex plane as above (Table 8.10). Removing the contribution of one pair from the moments shows what the first failure registers: at $t = -2$ and $-3$ the pair that starts at $(\gamma_9, \gamma_{10})$, whose removal moves $n_{\text{fail}}$ from 19 to 21 and from 18 to 21, and from $t = -4$ to $-12$ the pair that starts at $(\gamma_4, \gamma_5)$, whose removal raises $n_{\text{fail}}$ by two or three, from 11 to 14 at $t = -4$; at these times the removal of any other pair that starts below $\theta = 88$ leaves $n_{\text{fail}}$ unchanged. Both pairs are registered soon after their collisions, at the orders 19 and 11, close to $2k + 2 = 20$ and 10. At $t = -2$ the three pairs that start at $(\gamma_9, \gamma_{10})$, $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$ account for every failure up to order 30: with them removed none remains, and alone they break the test at the orders 19, 21 and 30. From $t = -3$ on, pairs that start above $\theta = 88$ enter as well, since with all the pairs that start below $\theta = 88$ removed the test still fails, from order 24 at $t = -3$, 21 at $-4$, 15 at $-8$ and 11 at $-15$ and $-20$. At $t = -15$ and $-20$ the lowest pair is no longer the first to be seen: no single removal raises $n_{\text{fail}}$, and the pair that starts at $(\gamma_4, \gamma_5)$ alone breaks the test as early as the lowest pair alone or earlier, at order 4 against 5 at $t = -20$. The two discretisations of the moments, step 1/300 at 600 digits and step 1/450 at 900 digits, give the same counts and the same removal tests, and at the orders 10 and 30 the eigenvalues of the diagonally scaled matrices give the same counts as the pivots.

The programs are `jacobi_dbn.py` (the moments and the collisions below index 59), `dbn_collisions.py` (the census of the first 200 gaps) and `dbn_hankel_long.py` (the collisions and counts down to $t = -20$).

### 8.4.4 Biane's theorem and Laguerre's quantity

The collision times of Section 8.4.3 can be predicted from $H_0$ alone, before the flow is followed, through a quantity that free probability singles out. The material comes from Section 8 of the first author's note [433], except Corollaries 8.4.13 and 8.4.15 and the halved constant of Remark 8.4.11, which are not in the note; the numbers are recomputed here from $H_t$ itself.

Let $\mu$ be a probability measure on $\mathbb{R}$, $G_\mu(z) = \int \mu(dx)/(z - x)$ its Cauchy transform, and $\sigma_t$ the semicircle law of variance $t > 0$, with density $(2\pi t)^{-1}(4t -$

| $t$ | pair | $\theta$ | real | collided | $n_{\text{fail}}$ | $n'_{\text{fail}}$ | $(\iota_{10}, \iota'_{10})$ | $(\iota_{30}, \iota'_{30})$ |
|---|---|---|---|---|---|---|---|---|
| $-2$ | $(9,10)$ | $49.715 \pm 0.249i$ | 8 | 3 | 19 | 19 | $(0,0)$ | $(2,2)$ |
| $-3$ | $(9,10)$ | $50.139 \pm 0.792i$ | 8 | 5 | 18 | 17 | $(0,0)$ | $(3,3)$ |
| $-4$ | $(4,5)$ | $33.115 \pm 0.151i$ | 3 | 6 | 11 | 10 | $(0,1)$ | $(5,5)$ |
| $-5$ | $(4,5)$ | $33.478 \pm 0.745i$ | 3 | 7 | 9 | 9 | $(1,1)$ | $(6,6)$ |
| $-6$ | $(4,5)$ | $33.841 \pm 1.071i$ | 3 | 7 | 9 | 8 | $(1,1)$ | $(6,6)$ |
| $-8$ | $(4,5)$ | $34.572 \pm 1.592i$ | 3 | 8 | 8 | 8 | $(1,1)$ | $(6,6)$ |
| $-10$ | $(4,5)$ | $35.306 \pm 2.052i$ | 3 | 8 | 8 | 8 | $(1,1)$ | $(6,7)$ |
| $-12$ | $(4,5)$ | $36.043 \pm 2.492i$ | 3 | 8 | 8 | 8 | $(1,1)$ | $(7,7)$ |
| $-15$ | $(2,3)$ | $28.202 \pm 1.166i$ | 1 | 9 | 6 | 5 | $(2,3)$ | $(8,8)$ |
| $-20$ | $(2,3)$ | $29.967 \pm 2.178i$ | 1 | 9 | 5 | 5 | $(3,3)$ | $(8,8)$ |

Table 8.10: The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k+1)$ by the zeros $\gamma_k$, $\gamma_{k+1}$ it starts from, and its position $\theta$; the number of real zeros below it; the number of collisions so far among $\gamma_1, \dots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\text{fail}}(t)$ and $n'_{\text{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30.

$x^2)^{1/2}$ on $[-2\sqrt{t}, 2\sqrt{t}]$. The free convolution $\mu \boxplus \sigma_t$ is the law of $A + S$ for free random variables $A$ and $S$ with laws $\mu$ and $\sigma_t$. Put

$$F_\mu(u) = \int \frac{\mu(dx)}{(u-x)^2} \in (0, \infty], \qquad v_t(u) = \inf\Big\{v \geq 0 : \int \frac{\mu(dx)}{(u-x)^2 + v^2} \leq \frac{1}{t}\Big\},$$
$$\psi_t(u) = u + t \int \frac{(u-x)\,\mu(dx)}{(u-x)^2 + v_t(u)^2}.$$

Theorem 8.4.2 (Biane [47]). *Let $\mu$ be a probability measure on $\mathbb{R}$ and $t > 0$.*

*(a) $\psi_t$ is a homeomorphism of $\mathbb{R}$, and $\mu \boxplus \sigma_t$ has a continuous density $p_t$ with $p_t(\psi_t(u)) = v_t(u)/(\pi t)$ for every real $u$. For $\operatorname{Im} z > 0$, $G_{\mu \boxplus \sigma_t}(z) = G_\mu(\omega_t(z))$, where $\omega_t(z) = z - tG_{\mu \boxplus \sigma_t}(z)$, and $\omega_t$ is the inverse of the map $w \mapsto w + tG_\mu(w)$ on the region $\{u + iv : v > v_t(u)\}$, which that map sends onto the upper half plane.*

*(b) Let $U_t = \{u : F_\mu(u) > 1/t\}$. The interior of the support of $\mu \boxplus \sigma_t$ is $\psi_t(U_t)$, the support is its closure, the density is real analytic on $\psi_t(U_t)$, and the number of components of $U_t$ does not increase with $t$.*

*(c) Let $(a, b)$ be a bounded component of $\mathbb{R} \setminus \operatorname{supp} \mu$ with $\mu(\{a\}) > 0$ and $\mu(\{b\}) > 0$. Then $F_\mu$ is strictly convex on $(a, b)$ and tends to $+\infty$ at both ends; let $u_0$ be its minimiser and*

$$t_0 = \frac{1}{F_\mu(u_0)} = \Big(\min_{a<u<b} \int \frac{\mu(dx)}{(u-x)^2}\Big)^{-1}. \tag{8.4.1}$$

*For $t < t_0$ the set $\{u \in (a,b) : F_\mu(u) \le 1/t\}$ is an interval $[a(t), b(t)]$ with $a(t) < b(t)$, and $\psi_t$ maps $(a(t), b(t))$ onto a component of the complement of the support of $\mu \boxplus \sigma_t$. As $t$ increases to $t_0$, $a(t)$ increases and $b(t)$ decreases to $u_0$, and at $t_0$ the gap closes.*

*Proof.* Part (a) is Lemma 4, Proposition 2 and Corollaries 2 and 3 of [47], where $\sigma_t$ has $R$ transform $tz$ as here; part (b) is Proposition 3, its proof, and Corollary 4 there. For (c), $\mu$ does not charge $(a,b)$, so $F''_\mu(u) = 6\int (u-x)^{-4}\mu(dx) > 0$ there, and the atoms make $F_\mu$ tend to $+\infty$ at $a$ and $b$. The sublevel sets of a strictly convex function are intervals; $\{F_\mu \le 1/t\} \cap (a,b)$ contains $u_0$ in its interior when $1/t > F_\mu(u_0)$, shrinks as $t$ grows, and reduces to $u_0$ at $t = t_0$. Its interior is a component of $\mathbb{R} \setminus \overline{U_t}$, because the points of $(a,b)$ just outside it lie in $U_t$, and $\psi_t$, a homeomorphism, maps it onto a component of the complement of $\overline{\psi_t(U_t)}$, which is the support by (b). This is the discussion following Corollary 5 of [47]. □

The time (8.4.1) is Biane's threshold. For a real entire function $f$ write

$$\mathcal{L}_f(x) = -\frac{d^2}{dx^2}\log|f(x)| = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2},$$

Laguerre's quantity. If $f$ has genus at most one and zeros $z_j$, then $\mathcal{L}_f(x) = \sum_j (x - z_j)^{-2}$. For the normalised counting measure $\mu_p = n^{-1}\sum_j \delta_{z_j}$ of the zeros of a polynomial $p$ of degree $n$ with only real simple zeros, $F_{\mu_p} = \mathcal{L}_p/n$, and by Theorem 8.4.2(c) the gap $(z_k, z_{k+1})$ survives in $\mu_p \boxplus \sigma_t$ until $t_0 = n/\min_{(z_k,z_{k+1})} \mathcal{L}_p$.

For $\Xi$, whose zeros are $\pm\gamma_j$, and for $H_0(z) = \frac{1}{8}\Xi(z/2)$,

$$\mathcal{L}_{H_0}(z) = \tfrac{1}{4}\mathcal{L}_\Xi(z/2), \qquad \mathcal{L}_\Xi(x) = \sum_j \Big[\frac{1}{(x-\gamma_j)^2} + \frac{1}{(x+\gamma_j)^2}\Big] = \sum_j \frac{2(x^2+\gamma_j^2)}{(x^2-\gamma_j^2)^2}.$$

Both are computed from $\Xi$ and its first two derivatives, without the zeros. Since $\Xi(x) = \Xi(0)G(-x^2)$, with $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 1.1,

$$\mathcal{L}_\Xi(x) = 2\frac{G'}{G}(-x^2) - 4x^2\Big(\frac{G'}{G}\Big)'(-x^2) = 2\frac{d}{dx}\Big(x\,\frac{G'}{G}(-x^2)\Big),$$
$$\frac{G'}{G}(u) = \sum_j \frac{1}{u+\gamma_j^2},$$

and $G'/G$ is the Stieltjes transform of the Thorin measure $\sum_j \delta_{\gamma_j^2}$ of the clock $1/G$ (under RH; the identities hold in general with $\gamma_j^2$ replaced by $a_j = -(\rho_j - \frac{1}{2})^2$).

LEMMA 8.4.3. *Let $p$ be a real polynomial of degree $n \ge 2$ with only real simple zeros and critical points $c_1 < \cdots < c_{n-1}$, and let $s$ be real with $s\,\mathcal{L}_p(c_i) < 1$ for every $i$. Then $p + sp''$ has only real simple zeros, one in each of the intervals $(-\infty, c_1)$, $(c_1, c_2)$, …, $(c_{n-1}, \infty)$.*

*Proof.* At a critical point $p'(c_i)=0$, so $p''(c_i)/p(c_i) = (\log|p|)''(c_i) = -\mathcal{L}_p(c_i)$, and $(p+sp'')(c_i) = p(c_i)\big(1-s\mathcal{L}_p(c_i)\big)$ has the sign of $p(c_i)$. There is exactly one critical point between consecutive zeros of $p$, so the signs of $p(c_1),\dots,p(c_{n-1})$ alternate. The polynomial $p+sp''$ has degree $n$ and the sign of $p$ at $\pm\infty$, which is opposite to the sign of $p(c_1)$ at $-\infty$ and to that of $p(c_{n-1})$ at $+\infty$, since $p$ has one zero to the left of $c_1$ and one to the right of $c_{n-1}$. So $p+sp''$ changes sign in each of the $n$ intervals and has $n$ real zeros there. □

Since $e^{sD^2} = \lim_N (1+\frac{s}{N}D^2)^N$, the lemma suggests, without proving it, that the gap $(z_k, z_{k+1})$ of $H_t$ stays open while $|t| < 1/\mathcal{L}_{H_0}(c_k)$, where $c_k$ is the critical point of $H_0$ in the gap. With $z_k = 2\gamma_k$ and $u_k$ the minimiser of $\mathcal{L}_{H_0}$ over the gap, this gives two predictions of the collision time from $H_0$ alone, the critical point form and Biane's form,

$$\tau_c = -\frac{1}{\mathcal{L}_{H_0}(c_k)} = -\frac{4}{\mathcal{L}_{\Xi}(c_k/2)},$$
$$\tau_B = -\frac{1}{m(0)}, \qquad m(0) = \min_{(z_k,z_{k+1})} \mathcal{L}_{H_0} = \mathcal{L}_{H_0}(u_k).$$

For an isolated pair, $\mathcal{L} = 8/D_0^2$ at the midpoint of the gap $D_0 = z_{k+1}-z_k$, and both equal the isolated pair estimate $-D_0^2/8$ of Section 8.4.1, which is the exact collision time of $z^2-a^2$ under $e^{sD^2}$, since $e^{sD^2}(z^2-a^2) = z^2-a^2+2s$. Section 8.4.5 turns both predictions into theorems for functions with only real zeros.

Remark 8.4.4. Biane's theorem and the collisions concern opposite directions of the heat flow. For polynomials of degree $n$ the finite free convolution of Marcus, Spielman and Srivastava [370] is $p \boxplus_n q = \hat{q}(D)p$ when $q = \hat{q}(D)x^n$, so $e^{-aD^2}p = p \boxplus_n e^{-aD^2}x^n$. The polynomial $e^{-aD^2}x^n$ is a rescaled Hermite polynomial [370] and has zeros with mean square $2a(n-1)$, and for large $n$ their distribution is close to the semicircle law $\sigma_{2an}$; in the limit finite free convolution becomes free convolution [370]. So $\mu \mapsto \mu \boxplus \sigma_t$ is the limit of the forward flow $e^{-aD^2}$ with $a \approx t/(2n)$, that is of $H_t$ with $t$ increasing: the zeros repel, and a gap between two groups of zeros fills in as the groups spread, which is what Theorem 8.4.2(c) describes. Zeros collide only in the other direction, under $e^{sD^2}$ with $s = -t > 0$. What carries over is the functional. Biane's threshold for $\mu_p$ is $n/\min\mathcal{L}_p$, and Theorem 8.4.9 below shows that under $e^{sD^2}$ a gap of a function with only real zeros cannot close before $s = 1/\min\mathcal{L}$ while all the zeros stay real. For the pair $\pm a$ the two times are $a^2$ and $a^2/2$, and the factor 2 is the $n$ of the normalisation of $\mu_p$.

Table 8.11 compares the two predictions with the collision times of the census. Every collision of the census, all 58 among the first 200 gaps on $[-2.05, 0]$, comes later in the backward flow than both predictions and the isolated pair estimate: $t_{\mathrm{coll}}/\tau_B \geq 1.083$, $t_{\mathrm{coll}}/\tau_c \geq 1.104$ and $t_{\mathrm{coll}}/(-D_0^2/8) \geq 1.056$, all three for

$(186, 187)$, which collides at $t = -0.1310$. Over the 28 pairs of the table, $t_{\mathrm{coll}}/\tau_B$ lies in $[1.114, 1.719]$, $t_{\mathrm{coll}}/\tau_c$ in $[1.181, 1.951]$ and $t_{\mathrm{coll}}/(-D_0^2/8)$ in $[1.077, 1.518]$, and $|\tau_B|$ falls short of $|t_{\mathrm{coll}}|$ by 25.5% on average. The ratio $t_{\mathrm{coll}}/\tau_B$ is close to 1 when the two zeros dominate their gap, and it is very nearly an increasing function of the share $6Q(0)/m(0)$ of the other zeros in Biane's functional (Section 8.4.5): ordered by this share, the ratios in the table rise with only small inversions, while the gap alone orders them much less well (compare $(114, 115)$ and $(13, 14)$ in the table). Real zeros outside the gap pull the two zeros apart while they fall together. Over all 58 collisions $t_{\mathrm{coll}}/\tau_B$ lies in $[1.083, 2.329]$, and the largest ratios, 2.042 to 2.329, belong to the five collisions with $6Q(0)/m(0) > 0.9$, all at $t < -1.89$.

| pair | $\gamma_{k+1} - \gamma_k$ | $t_{\mathrm{coll}}$ | $-D_0^2/8$ | $\tau_c$ | $\tau_B$ | $t_{\mathrm{coll}}/\tau_B$ | $6Q/m$ | frozen |
|---|---|---|---|---|---|---|---|---|
| $(91, 92)$ | 0.7158 | −0.2821 | −0.2562 | −0.2369 | −0.2455 | 1.149 | 0.250 | −0.2826 |
| $(71, 72)$ | 0.7243 | −0.2880 | −0.2623 | −0.2305 | −0.2517 | 1.144 | 0.242 | −0.2883 |
| $(97, 98)$ | 0.7370 | −0.3040 | −0.2716 | −0.2454 | −0.2587 | 1.175 | 0.286 | −0.3047 |
| $(63, 64)$ | 0.8175 | −0.3686 | −0.3341 | −0.3122 | −0.3199 | 1.152 | 0.256 | −0.3694 |
| $(34, 35)$ | 0.8451 | −0.3846 | −0.3571 | −0.3115 | −0.3452 | 1.114 | 0.200 | −0.3852 |
| $(53, 54)$ | 0.8717 | −0.4231 | −0.3800 | −0.3361 | −0.3624 | 1.167 | 0.277 | −0.4244 |
| $(114, 115)$ | 0.9307 | −0.5380 | −0.4331 | −0.3933 | −0.3997 | 1.346 | 0.462 | −0.5364 |
| $(93, 94)$ | 0.9763 | −0.5533 | −0.4766 | −0.4179 | −0.4471 | 1.237 | 0.371 | −0.5589 |
| $(106, 107)$ | 0.9650 | −0.5691 | −0.4656 | −0.3620 | −0.4306 | 1.321 | 0.451 | −0.5724 |
| $(75, 76)$ | 1.0531 | −0.6572 | −0.5545 | −0.4537 | −0.5166 | 1.272 | 0.410 | −0.6648 |
| $(111, 112)$ | 1.0745 | −0.7102 | −0.5772 | −0.4715 | −0.5310 | 1.338 | 0.481 | −0.7240 |
| $(117, 118)$ | 1.0571 | −0.7653 | −0.5588 | −0.4318 | −0.5029 | 1.522 | 0.599 | −0.7676 |
| $(78, 79)$ | 1.1388 | −0.8235 | −0.6485 | −0.5695 | −0.5930 | 1.389 | 0.513 | −0.8318 |
| $(27, 28)$ | 1.2193 | −0.8811 | −0.7433 | −0.6313 | −0.6937 | 1.270 | 0.400 | −0.8861 |
| $(45, 46)$ | 1.2588 | −0.9600 | −0.7923 | −0.6731 | −0.7325 | 1.311 | 0.452 | −0.9751 |
| $(82, 83)$ | 1.2050 | −0.9734 | −0.7260 | −0.6170 | −0.6533 | 1.490 | 0.601 | −0.9985 |
| $(80, 81)$ | 1.2288 | −1.0044 | −0.7550 | −0.5459 | −0.6808 | 1.475 | 0.589 | −1.0283 |
| $(87, 88)$ | 1.1991 | −1.0343 | −0.7190 | −0.5805 | −0.6390 | 1.619 | 0.667 | −1.0540 |
| $(100, 101)$ | 1.2456 | −1.0435 | −0.7757 | −0.5739 | −0.6963 | 1.499 | 0.614 | −1.0797 |
| $(104, 105)$ | 1.2476 | −1.0485 | −0.7783 | −0.6591 | −0.6983 | 1.502 | 0.617 | −1.0858 |
| $(57, 58)$ | 1.2524 | −1.1004 | −0.7842 | −0.6484 | −0.7031 | 1.565 | 0.621 | −1.0984 |
| $(24, 25)$ | 1.3838 | −1.1439 | −0.9575 | −0.7980 | −0.8898 | 1.286 | 0.424 | −1.1578 |
| $(40, 41)$ | 1.3100 | −1.1519 | −0.8580 | −0.7345 | −0.7771 | 1.482 | 0.566 | −1.1458 |
| $(65, 66)$ | 1.3427 | −1.2447 | −0.9014 | −0.6381 | −0.8066 | 1.543 | 0.631 | −1.2737 |
| $(19, 20)$ | 1.4401 | −1.2755 | −1.0370 | −0.7719 | −0.9573 | 1.332 | 0.461 | −1.2837 |
| $(13, 14)$ | 1.4847 | −1.2853 | −1.1022 | −0.9064 | −1.0334 | 1.244 | 0.374 | −1.2947 |
| $(51, 52)$ | 1.4218 | −1.3190 | −1.0107 | −0.7548 | −0.9126 | 1.445 | 0.582 | −1.3684 |
| $(48, 49)$ | 1.3875 | −1.4611 | −0.9626 | −0.7573 | −0.8497 | 1.719 | 0.703 | −1.4679 |

Table 8.11: The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time $t_{\mathrm{coll}}$ from the census of Section 8.4.3 (`dbn_collisions.py`), against predictions computed from $H_0$: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1} - \gamma_k)^2/2$, the critical point and Biane forms $\tau_c$ and $\tau_B$, and the frozen predictor of Section 8.4.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of $H_t$.

### 8.4.5 Collision brackets from Laguerre's quantity

Write $s = -t$ for the backward time, so that $f_s = H_{-s}$ solves $\partial_s f_s = f_s''$; for a real polynomial $p$, $f_s = e^{sD^2}p = \sum_{k\ge 0} s^k p^{(2k)}/k!$ solves the same equation. In this subsection $f_s$, $s \ge 0$, is a family of real entire functions of genus at most one, smooth in $(z, s)$, with $\partial_s f_s = f_s''$, and $(z_k(s), z_{k+1}(s))$ is a gap between two adjacent simple real zeros, which move continuously until they meet, if they do. On the gap put $q = f_s'/f_s$ and $\mathcal{L} = \mathcal{L}_{f_s} = -q'$. For a point $x$ of the gap write $x_j = 1/(x - z_j)$, where $z_j$ runs over the zeros of $f_s$, and $S_k = \sum_j x_j^k$ for $k \ge 2$, a real and absolutely convergent sum; then

$$\mathcal{L}(x) = S_2, \qquad \mathcal{L}'(x) = -2S_3, \qquad \mathcal{L}''(x) = 6S_4.$$

When all zeros are real, $q$ falls from $+\infty$ to $-\infty$ across the gap and vanishes at a single critical point $c(s)$; we write $\ell(s) = \mathcal{L}(c(s))$. In all cases $\mathcal{L} \to +\infty$ at both ends of the gap, and we write $m(s)$ for its minimum over the gap, attained at a minimiser $u(s)$, where $\mathcal{L}'(u) = 0$, that is $S_3 = 0$.

Lemma 8.4.5. *On the gap $\partial_s \mathcal{L} = \mathcal{L}'' + 2q\mathcal{L}' - 2\mathcal{L}^2$. If all zeros of $f_s$ are real and simple, then $c(s)$ is differentiable with $\dot c = -\mathcal{L}'(c)/\mathcal{L}(c)$, and*

$$\dot\ell = 6S_4 - 2S_2^2 - \frac{4S_3^2}{S_2} \qquad \text{at } x = c(s).$$

*Proof.* With $w = \log f_s$ the heat equation gives $\partial_s w = f_s''/f_s = w'' + w'^2$, hence $\partial_s q = q'' + 2qq'$ and, with $\mathcal{L} = -q'$, $\partial_s \mathcal{L} = -q''' - 2q'^2 - 2qq'' = \mathcal{L}'' - 2\mathcal{L}^2 + 2q\mathcal{L}'$. Since $\partial_z q = -\mathcal{L} < 0$ at $c$, the implicit function theorem applied to $q(c(s), s) = 0$ gives $\dot c = -\partial_s q/\partial_z q = (q'' + 2qq')/\mathcal{L} = -\mathcal{L}'/\mathcal{L}$ at $c$, where $q = 0$. Hence $\dot\ell = \mathcal{L}'\dot c + \partial_s \mathcal{L} = \mathcal{L}'' - 2\mathcal{L}^2 - \mathcal{L}'^2/\mathcal{L}$ at $c$, and $\mathcal{L} = S_2$, $\mathcal{L}' = -2S_3$, $\mathcal{L}'' = 6S_4$. □

Lemma 8.4.6. *For every real square summable sequence $(x_j)$ with power sums $S_k = \sum_j x_j^k$,*

$$E = 3S_2^3 - 6S_2S_4 + 4S_3^2 \ge 0,$$

*and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.*

*Proof.* By homogeneity take $S_2 = 1$. Let $S_k^+ = \sum_{x_j>0} x_j^k$ and $S_k^- = \sum_{x_j<0} |x_j|^k$, and $\alpha = (S_2^+)^{1/2}$, $\beta = (S_2^-)^{1/2}$, so that $\alpha^2 + \beta^2 = 1$, $S_3 = S_3^+ - S_3^-$ and $S_4 = S_4^+ + S_4^-$. Every positive term is at most $\alpha$, so $S_4^+ \le \alpha S_3^+$ and $S_3^+ \le \alpha S_2^+ = \alpha^3$; likewise $S_4^- \le \beta S_3^-$ and $S_3^- \le \beta^3$. Therefore $E \ge g(S_3^+, S_3^-)$ with

$$g(p, n) = 3 - 6(\alpha p + \beta n) + 4(p - n)^2 \qquad \text{on the box } [0, \alpha^3] \times [0, \beta^3].$$

The function $g$ is convex, its Hessian being $\begin{pmatrix} 8 & -8 \\ -8 & 8 \end{pmatrix}$, and its gradient $\big(-6\alpha + 8(p-n), -6\beta - 8(p-n)\big)$, whose components add up to $-6(\alpha+\beta) < 0$, never

vanishes; so its minimum over the box lies on the boundary. On the edge $p = 0$, $g = 3 - 6\beta n + 4n^2 \geq 3 - \frac{9}{4}\beta^2 \geq \frac{3}{4}$, and the edge $n = 0$ is symmetric. At the corner $(\alpha^3, \beta^3)$, using $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$ and $1 - 4\alpha^2\beta^2 = (\alpha^2 - \beta^2)^2$,

$$g = 4(\alpha^3 - \beta^3)^2 - 3(\alpha^2 - \beta^2)^2 = (\alpha - \beta)^2\big[4(\alpha^2 + \alpha\beta + \beta^2)^2 - 3(\alpha + \beta)^2\big],$$

and with $\sigma = \alpha + \beta$, $\alpha\beta = (\sigma^2 - 1)/2$, the bracket is $(\sigma^2+1)^2 - 3\sigma^2 = \sigma^4 - \sigma^2 + 1 > 0$. On the edge $p = \alpha^3$, $g$ is a convex quadratic in $n$ with vertex at $n = \alpha^3 + \frac{3}{4}\beta$. If the vertex lies at or beyond $\beta^3$, the minimum over the edge is at the corner. Otherwise $\beta^3 - \frac{3}{4}\beta > \alpha^3 \geq 0$ forces $\beta^2 > \frac{3}{4}$ and $\alpha < \frac{1}{2}$, and the vertex value is

$$\tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3\beta \geq h(\alpha) = \tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3;$$

$h'(\alpha) = \alpha(\frac{9}{2} - 18\alpha - 24\alpha^2)$, so $h$ increases up to $\alpha \approx 0.198$ and decreases after it, and on $[0, \frac{1}{2}]$ it is at least $\min(h(0), h(\frac{1}{2})) = \min(\frac{3}{4}, \frac{3}{16}) > 0$. The edge $n = \beta^3$ is symmetric. Hence $E \geq 0$. Equality forces the corner with $\alpha = \beta$, and $S_4^+ = \alpha S_3^+$ with $S_3^+ = \alpha^3$, so every positive term equals $\alpha$ and their squares add up to $\alpha^2$: there is exactly one positive term, $\alpha$, and likewise exactly one negative term, $-\alpha$. Conversely $E = 0$ for $\{a, -a\}$. □

THEOREM 8.4.7 (critical point bound). *If all zeros of $f_\sigma$ are real and simple for $0 \leq \sigma < s$, then $\dot{\ell} \leq \ell^2$ on $[0, s)$ and*

$$\ell(\sigma) \leq \frac{\ell(0)}{1 - \sigma\ell(0)} \qquad \big(0 \leq \sigma < \min(s, 1/\ell(0))\big).$$

*In particular, if $p$ is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap $(z_k, z_{k+1})$ at $s = 0$, then $s_1 \geq 1/\mathcal{L}_p(c_k)$. The bound is attained by $p(z) = z^2 - a^2$.*

*Proof.* By Lemma 8.4.5, $\dot{\ell} \leq \ell^2$ reads $6S_4 - 2S_2^2 - 4S_3^2/S_2 \leq S_2^2$, which is $E \geq 0$ divided by $S_2 > 0$, Lemma 8.4.6. Then $\frac{d}{d\sigma}(1/\ell) = -\dot{\ell}/\ell^2 \geq -1$, so $1/\ell(\sigma) \geq 1/\ell(0) - \sigma$, which is the bound. For the polynomial, all zeros are real and simple on $[0, s_1)$, and $\ell \geq (c - z_k)^{-2} + (z_{k+1} - c)^{-2} \geq 8/g^2$, where $g$ is the length of the gap. If $s_1 < 1/\ell(0)$, then $\ell$ stays bounded on $[0, s_1)$, so $g$ stays bounded below and the two zeros cannot meet at $s_1$. For $z^2 - a^2$, $e^{sD^2}p = z^2 - (a^2 - 2s)$, $c = 0$ and $\ell(s) = 2/(a^2 - 2s) = \ell(0)/(1 - s\ell(0))$, with the collision at $s = a^2/2 = 1/\ell(0)$. □

The theorem applies to every real rooted polynomial, in particular to every truncation $\prod_{j \leq J}(1 - z^2/z_j^2)$ of $H_0$. For $H_t$ itself the hypothesis fails at every $t < 0$: by the theorem of Rodgers and Tao [463], $H_t$ has nonreal zeros for every $t < 0$ (Section 8.4.1). The minimum of $\mathcal{L}$ over the gap gives the sharper Biane form; at the minimiser $S_3 = 0$, while $q(u)$ need not vanish.

LEMMA 8.4.8. *For every real square summable sequence with $S_3 = 0$, $2S_4 \le S_2^2$, and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.*

*Proof.* In the notation of the proof of Lemma 8.4.6, without the normalisation, $S_3 = 0$ means $S_3^+ = S_3^- = \mu$, and for a nonzero sequence both $\alpha$ and $\beta$ are then positive. Now $\mu \le \min(\alpha^3, \beta^3)$ and $S_4 \le \alpha S_3^+ + \beta S_3^- = (\alpha + \beta)\mu$. Let $\lambda = \min(\alpha, \beta)$ and $\max(\alpha, \beta) = r\lambda$ with $r \ge 1$. Then $S_4 \le (1 + r)\lambda^4$ and $S_2^2 = (1 + r^2)^2\lambda^4$, and $2(1 + r) \le (1 + r^2)^2$ for $r \ge 1$, with equality only at $r = 1$, since both sides equal 4 there and the right side has derivative $4r(1+r^2) \ge 8 > 2$. Equality in $2S_4 \le S_2^2$ forces $r = 1$, $\mu = \alpha^3$ and $S_4^+ = \alpha S_3^+$, hence one positive term $\alpha$ and one negative term $-\alpha$. □

THEOREM 8.4.9 (Biane form). *If all zeros of $f_\sigma$ are real and simple for $0 \le \sigma < s$, then the upper right Dini derivative of $m$ satisfies $D^+m \le m^2$ on $[0, s)$, and*

$$m(\sigma) \le \frac{m(0)}{1 - \sigma m(0)} \qquad \big(0 \le \sigma < \min(s, 1/m(0))\big).$$

*In particular, if $p$ is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap $(z_k, z_{k+1})$ at $s = 0$, then $s_1 \ge 1/\min_{(z_k, z_{k+1})} \mathcal{L}_p$. This bound on $s_1$ contains that of Theorem 8.4.7, since $m(0) \le \ell(0)$, and it is attained by $z^2 - a^2$.*

*Proof.* Since $\mathcal{L} \to +\infty$ at both ends of the gap, the minimisers are interior and $m$ is continuous. If $u$ is a minimiser at time $\sigma$, then $m(\sigma + h) \le \mathcal{L}(u, \sigma + h)$ for small $h > 0$, so by Lemma 8.4.5, with $\mathcal{L}'(u) = 0$,

$$D^+m(\sigma) \le \partial_\sigma \mathcal{L}(u, \sigma) = \mathcal{L}''(u) - 2\mathcal{L}(u)^2 = 6S_4 - 2S_2^2 \le S_2^2 = m(\sigma)^2,$$

the last step by Lemma 8.4.8, as $S_3 = 0$ at $u$. The lower right Dini derivative of the continuous function $1/m(\sigma) + \sigma$ is therefore nonnegative, so the function does not decrease, and $1/m(\sigma) \ge 1/m(0) - \sigma$. A collision forces $m \ge 8/g^2 \to \infty$, as in the proof of Theorem 8.4.7. For $z^2 - a^2$ the minimiser is the critical point 0, so $m = \ell$. □

Both theorems were also tested numerically. The program `wigbiane_poly.py` draws 2400 polynomials of degree 3 to 12 with only real zeros, 600 in each of four families (zeros uniform on an interval; Gaussian; uniform with one added zero at 2 to 30% of the mean spacing from another, a Lehmer type pair; and a perturbed lattice), follows the zeros of $e^{sD^2}p$ until at most one real zero is left, and locates each collision as a double zero by Newton's method in $(z, s)$ at 40 digits. For the 2398 polynomials it completed (two failed in the location of a collision), the first collision satisfied $s_1\ell(0) \ge 1.0000136$ and $s_1 m(0) \ge 1.0000055$, both for a Lehmer type pair, with medians 1.061 and 1.039.

After the first collision Theorem 8.4.9 no longer applies, yet each of the 4574 later collisions of two zeros adjacent at $s = 0$ satisfied $s_c m(0) \geq 1.0000326$, although the quantity $\delta$ of Corollary 8.4.13 below was positive at some step before 806 of them.

For $H_t$ the hypothesis of Theorem 8.4.9 fails, but the argument survives with a correction. At the minimiser $u$, replace every nonreal pair $a \pm ib$ by the double real point $a$, its shadow.

LEMMA 8.4.10. *(i) For every real square summable sequence,* $S_4 \leq \frac{1}{2}S_2^2 + S_2^{1/2}|S_3|$.

*(ii) If* $w = 1/(d - ib)$ *with* $d \neq 0$ *and* $b$ *real, then* $\left|2\operatorname{Re} w^k - 2d^{-k}\right| \leq k(k+1)\, b^2/|d|^{k+2}$ *for every* $k \geq 1$.

*Proof.* (i) Replacing the sequence by its negative if necessary, let $S_3 = \varepsilon \geq 0$, so $S_3^+ = S_3^- + \varepsilon$ in the notation of the proof of Lemma 8.4.6. Then $S_4 \leq \alpha S_3^+ + \beta S_3^- = (\alpha+\beta)S_3^- + \alpha\varepsilon$, where $S_3^- \leq \beta^3$ and $S_3^- = S_3^+ - \varepsilon \leq \alpha^3$. As in the proof of Lemma 8.4.8, $(\alpha+\beta)\min(\alpha,\beta)^3 \leq \frac{1}{2}(\alpha^2+\beta^2)^2 = \frac{1}{2}S_2^2$, and $\alpha\varepsilon \leq S_2^{1/2}|S_3|$. (ii) With $\epsilon = b/d$ and $\theta = \arctan\epsilon$, $w^k = d^{-k}(1-i\epsilon)^{-k}$ and $\operatorname{Re}(1-i\epsilon)^{-k} = A\cos k\theta$ with $A = (1+\epsilon^2)^{-k/2}$. Now $1 - A\cos k\theta = (1-A) + A(1-\cos k\theta)$, where $0 \leq 1-A \leq \frac{k}{2}\epsilon^2$ and $0 \leq 1-\cos k\theta \leq \frac{1}{2}k^2\theta^2 \leq \frac{1}{2}k^2\epsilon^2$, so $|\operatorname{Re}(1-i\epsilon)^{-k} - 1| \leq \frac{1}{2}(k+k^2)\epsilon^2$; multiply by $2|d|^{-k}$. □

REMARK 8.4.11. The coefficient of the second term in (i) can be halved, and $\frac{1}{2}$ is then best possible, since a single nonzero term gives $S_4 - \frac{1}{2}S_2^2 = \frac{1}{2}S_2^{1/2}|S_3|$. Normalise $S_2 = 1$ and $S_3 = \varepsilon \geq 0$ and put $\mu = S_3^-$, so that $S_4 \leq \alpha(\mu+\varepsilon) + \beta\mu$ with $0 \leq \mu \leq \min(\beta^3, \alpha^3 - \varepsilon)$; the bound increases with $\mu$. For $\varepsilon \leq \alpha^3 - \beta^3$ it is at most $(\alpha+\beta)\beta^3 + \alpha\varepsilon$, which is linear in $\varepsilon$, and for $\varepsilon \geq \max(0, \alpha^3-\beta^3)$ it is at most $\alpha^4 + \alpha^3\beta - \beta\varepsilon$, which decreases. So the bound minus $\frac{1}{2}(1+\varepsilon)$ is largest at $\varepsilon = 0$ or, when $\alpha > \beta$, at $\varepsilon = \alpha^3 - \beta^3$. At $\varepsilon = 0$ the bound is $(\alpha+\beta)\min(\alpha,\beta)^3 \leq \frac{1}{2}$, as in the proof of Lemma 8.4.8. At $\varepsilon = \alpha^3 - \beta^3$ it is $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$, and $\alpha^4 + \beta^4 - \frac{1}{2} \leq \frac{1}{2}(\alpha^3 - \beta^3)$ reads $(\alpha^2-\beta^2)^2 \leq \alpha^3 - \beta^3$, that is $(\alpha-\beta)(\alpha+\beta)^2 \leq 1 + \alpha\beta$; with $\nu = 2\alpha\beta \in [0,1]$ this is $(1-\nu)^{1/2}(1+\nu) \leq 1 + \frac{\nu}{2}$, which after squaring becomes $-\nu^2 - \nu^3 \leq \frac{1}{4}\nu^2$. Hence

$$S_4 \leq \tfrac{1}{2}S_2^2 + \tfrac{1}{2}S_2^{1/2}|S_3|,$$

and the coefficient 72 in Theorem 8.4.12 below may be replaced by 36; we keep the form of [433], which the computations below use. The bound (ii) is sharp as $b/d \to 0$, since $\operatorname{Re}(1-i\epsilon)^{-k} = 1 - \frac{1}{2}k(k+1)\epsilon^2 + O(\epsilon^4)$.

THEOREM 8.4.12 (local bound). *Let* $u$ *be a minimiser of* $\mathcal{L}$ *over the gap at time* $s$, *with* $m = \mathcal{L}(u) > 0$, *and let* $B_p = \sum b^2|u-a|^{-p}$, *summed over the conjugate pairs* $a \pm ib$, $b > 0$, *of nonreal zeros of* $f_s$, *each pair once, be finite for* $p = 4, 5, 6$.

*Then*

$$\mathcal{L}''(u) - 3m^2 \le \eta\, m^2, \qquad \eta = \frac{36\, mB_4 + 108\, B_4^2 + 72\,(m + 6B_4)^{1/2} B_5 + 120\, B_6}{m^2},$$

*and* $D^+ m \le (1+\eta)m^2$. *Hence, if the two zeros bounding the gap meet at* $s_c$, *and* $m > 0$ *and* $\eta \le \bar\eta$ *on* $[0, s_c)$, *then* $s_c \ge 1/\big((1+\bar\eta)m(0)\big)$.

*Proof.* Let $S'_k$ be the power sums of the real sequence in which every nonreal pair $a \pm ib$ is replaced by its shadow, two terms $1/(u-a)$, and let $D_k = S_k - S'_k$. By Lemma 8.4.10(ii) with $d = u - a$, $|D_2| \le 6B_4$, $|D_3| \le 12B_5$ and $|D_4| \le 20B_6$. Since $S_3 = 0$ at $u$, $S'_3 = -D_3$, and Lemma 8.4.10(i) for the shadow sequence gives $S'_4 \le \frac{1}{2} S'^2_2 + S'^{1/2}_2 |D_3|$. With $S_4 = S'_4 + D_4$, $0 \le S'_2 = m - D_2 \le m + 6B_4$ and $S'^2_2 - m^2 = -2mD_2 + D_2^2$,

$$\begin{aligned} 6S_4 - 3m^2 &\le 3(S'^2_2 - m^2) + 6S'^{1/2}_2 |D_3| + 6|D_4| \\ &\le 6m|D_2| + 3D_2^2 + 6S'^{1/2}_2 |D_3| + 6|D_4| \le \eta m^2, \end{aligned}$$

and $\mathcal{L}''(u) = 6S_4$. As in the proof of Theorem 8.4.9, $D^+ m \le \mathcal{L}''(u) - 2m^2 \le (1+\eta)m^2$, so the function $1/m(\sigma) + (1+\bar\eta)\sigma$ does not decrease on $[0, s_c)$. At the collision $m \to \infty$: the share of the two meeting zeros tends to $+\infty$, while that of the other zeros, which stay away from the point where the two meet, remains bounded. No nonreal zero tends to that point: if $z_c$ is a zero of multiplicity $k$ of $f_{s_c}$, then $f_{s_c-\sigma} = e^{-\sigma D^2} f_{s_c}$, so $\sigma^{-k/2} f_{s_c-\sigma}(z_c + \sqrt{\sigma}\, w)$ tends to a nonzero multiple of $H_k(w/\sqrt{2})$ as $\sigma \downarrow 0$, uniformly on compact sets, and by Hurwitz's theorem the $k$ zeros of $f_{s_c-\sigma}$ near $z_c$ lie close to $\sqrt{2\sigma}$ times the zeros of $H_k$, which are real and simple; so they are real, and real zeros only add to $m$. □

Corollary 8.4.13. *In the setting of Theorem 8.4.12 put* $\delta = \mathcal{L}''(u)/m^2 - 3$, *computed from* $f_s$ *and its first four derivatives at* $u$. *Then* $D^+ m \le (1+\delta)m^2$ *and* $\delta \le \eta$, *and* $\delta \le 0$ *when all zeros of* $f_s$ *are real. Hence, if the two zeros bounding the gap meet at* $s_c$, *and* $m > 0$ *and* $\delta \le \bar\delta$ *on* $[0, s_c)$ *with* $\bar\delta > -1$, *then* $s_c \ge 1/\big((1+\bar\delta)m(0)\big)$; *if* $\delta \le 0$ *throughout, the collision cannot come before Biane's time* $1/m(0)$, *exactly as for functions with only real zeros.*

*Proof.* The first display of the proof of Theorem 8.4.9 gives $D^+ m \le \mathcal{L}''(u) - 2m^2 = (1+\delta)m^2$, Theorem 8.4.12 gives $\delta \le \eta$, and Lemma 8.4.8 gives $\delta \le 0$ for real zeros. The rest is the comparison argument of the proof of Theorem 8.4.12. □

Write $m = P + Q$, where $P = (u - z_k)^{-2} + (u - z_{k+1})^{-2}$ is the share of the two zeros bounding the gap and $Q$ the share of all the others, and let $W = \sum |u - z|^{-4}$ over the nonreal zeros $z$ of $f_s$.

THEOREM 8.4.14 (a deadline). *While the two zeros bounding the gap are real, the right derivative $m'(s^+)$ exists and is the least value of $\mathcal{L}''(u) - 2m^2$ over the minimisers $u$, and at every minimiser*

$$\mathcal{L}''(u) - 2m^2 \geq m^2 - 6mQ + 3Q^2 - 6W,$$

*with $Q$ and $W$ taken at $u$. Suppose that $Q \leq \rho$ and $W \leq \bar{W}$ at every minimiser for as long as the two zeros bounding the gap are real, and that $m(0) > r_1$, where $r_{1,2} = 3\rho \pm (9\rho^2 + 6\bar{W})^{1/2}$. Then the gap closes, one of the two zeros bounding it meeting another zero, at a time*

$$s_c \leq s^* = \frac{1}{r_1 - r_2} \log \frac{m(0) - r_2}{m(0) - r_1}.$$

*If all zeros are real, $\bar{W} = 0$ and, for $\rho > 0$, $s^* = (6\rho)^{-1} \log\big(m(0)/(m(0) - 6\rho)\big) = 1/m(0) + 3\rho/m(0)^2 + 12\rho^2/m(0)^3 + \cdots$.*

*Proof.* Near a given time the minimisers lie in a fixed compact subinterval of the gap, and by Danskin's theorem the right derivative of the minimum exists and equals the least value of $\partial_s \mathcal{L}$ over the minimisers, which is $\mathcal{L}''(u) - 2m^2 = 6S_4 - 2m^2$ by Lemma 8.4.5. In $S_4$ the two zeros bounding the gap contribute $x_k^4 + x_{k+1}^4 \geq \frac{1}{2}(x_k^2 + x_{k+1}^2)^2 = \frac{1}{2}P^2$, the other real zeros contribute nonnegative terms, and each nonreal zero $z$ contributes $\mathrm{Re}(u - z)^{-4} \geq -|u - z|^{-4}$. So $\mathcal{L}''(u) - 2m^2 \geq 3P^2 - 6W - 2m^2$, and $P = m - Q$ gives the first inequality. While $m > 0$, $Q \leq \rho$ and $W \leq \bar{W}$, it gives $m'(s^+) \geq m^2 - 6\rho m - 6\bar{W} = (m - r_1)(m - r_2)$, so $m$ stays above the solution $y$ of $y' = (y - r_1)(y - r_2)$, $y(0) = m(0) > r_1 \geq 0$, which increases and tends to $+\infty$ as $s \to s^*$. If the gap were still open on $[0, s^*]$, its two zeros real, simple and apart, then $m$, being at most $\mathcal{L}$ at the midpoint of the gap, would be bounded on $[0, s^*]$, a contradiction. The first collision of either zero need not be with the other: one of them may first meet the zero beyond it, and the theorem then bounds the time of that collision. For a pair that is known to collide, as in the census, $s^*$ bounds the collision time of the pair. □

COROLLARY 8.4.15. *Put $\kappa = \mathcal{L}''(u) - 3P^2$, computed from $f_s$ at $u$ and the two zeros bounding the gap, so that*

$$\kappa = 3\big(x_k^2 - x_{k+1}^2\big)^2 + 6 \sum_{j \neq k, k+1} x_j^4 \geq -6W,$$

*with $\kappa \geq 0$ when all zeros are real. Theorem 8.4.14 holds with $6\bar{W}$ replaced by $\bar{\kappa} = \max\big(0, \sup(-\kappa)\big)$, the supremum over the minimisers at the times before the collision, and this form is never weaker.*

*Proof.* The identity is $6(x_k^4 + x_{k+1}^4) - 3(x_k^2 + x_{k+1}^2)^2 = 3(x_k^2 - x_{k+1}^2)^2$, and the inequality is that of the proof of Theorem 8.4.14. That proof gives $m'(s^+) = \kappa + 3P^2 - 2m^2 \geq m^2 - 6mQ + 3Q^2 - \bar{\kappa}$, and $\bar{\kappa} \leq 6\bar{W}$. □

The two inequalities behind Theorems 8.4.12 and 8.4.14 were tested on random configurations as well. For 29620 configurations of 3 to 11 real zeros and 1 to 7 conjugate pairs, at the critical point $u$ of $\mathcal{L}$ on a random gap, with $\mathcal{L}(u) > 0$, the quantity $6S_4 - 3S_2^2$ never exceeded $0.016\,\eta S_2^2$, and $6S_4 \geq 3P^2 - 6W$ held up to rounding (`wigbiane_ineq.py`). On the polynomials of `wigbiane_poly.py`, with $\rho$ the largest value of $Q$ at the minimisers before the collision, the deadline $s^*$ of Theorem 8.4.14 is defined for 2308 of the 2398 first collisions, with $\bar{W} = 0$, and for 3322 of the 4574 later ones (3345 in the form of Corollary 8.4.15), and no collision came after it: the largest ratio $s_c/s^*$ is $1 - 1.5 \times 10^{-11}$ for the first collisions and $1 - 3.1 \times 10^{-10}$ for the later ones.

Corollaries 8.4.13 and 8.4.15 bracket every collision time by quantities along the flow that need only $H_t$ near the gap, while Theorems 8.4.12 and 8.4.14 need the nonreal zeros. The program `wigbiane_flow.py` (`code/README.md`) computes both kinds for each of the 58 collisions of the census of Section 8.4.3. It evaluates $H_t$ and its first four $z$ derivatives in ball arithmetic at 800 bits, refines the double zero $H_t = \partial_z H_t = 0$ by Newton's method in $(z, t)$, and on a grid of about 140 times $s$ in $[0, |t_{\mathrm{coll}}|)$ it follows the two zeros by Newton's method and computes the minimiser $u$, the critical point $c$ and the quantities $\ell$, $m$, $Q$, $\delta$ and $\kappa$, the last three in ball arithmetic, since near the collision $m$ and $P$ are large and nearly equal. For $\eta$ and $W$ it also follows, by Newton's method in the complex plane, the conjugate pairs born in the census collisions before the time considered (60 pairs, two of them from collisions among the zeros 203 to 206), with their mirror images $-a \pm ib$, and adds a model of the zeros above the census, $z = 2\gamma_j$ for $207 \leq j \leq 2000$, in which every such zero is half of a pair $a \pm ib$ with $b^2 = 4|t|$. Newton's method reproduces the census collision times to $5 \times 10^{-12}$.

First, $\delta < 0$ at every point of the grid on all 58 trajectories; the largest value is $-3.1 \times 10^{-8}$, for $(186, 187)$. So the hypothesis of Corollary 8.4.13 with $\bar{\delta} = 0$ holds on the grid in spite of the nonreal zeros, and its conclusion, the lower bound $1/m(0)$, holds for all 58 collisions of the census, as the direct comparison of Section 8.4.4 shows ($t_{\mathrm{coll}}/\tau_B \geq 1.083$; Table 8.11). Likewise $\ell(s) \leq \ell(0)/(1 - s\ell(0))$ for $s < 1/\ell(0)$ and $m(s) \leq m(0)/(1 - sm(0))$ for $s < 1/m(0)$ at every point of the grid.

Second, $\kappa \geq 0$ at every point of the grid except on the trajectories of $(108, 109)$ and $(153, 154)$, where $\bar{\kappa} = 0.029$ and $0.071$. The upper bound $s^*$ of Corollary 8.4.15, with $\rho$ the largest value of $Q$ on the grid, is defined, $m(0) > r_1$, for 55 of the 58 collisions; it is not for $(153, 154)$, $(178, 179)$ and $(181, 182)$, for which $6\rho/m(0)$ is 0.986, 1.068 and 1.103. Every collision time of the census lies in its bracket $[1/m(0), s^*]$, with $s^* = \infty$ for those three, and Table 8.13 shows the seven pairs of the note [433]. The upper bound is tight when the pair dominates its gap: $s^*/|t_{\mathrm{coll}}|$ ranges from 1.0010, for $(186, 187)$ with $6\rho/m(0) = 0.151$, to 2.021, for $(48, 49)$ with $6\rho/m(0) = 0.965$.

Third, the correction $\eta$ of Theorem 8.4.12, computed from the tracked pairs

and the model above the census, is at most $3.25 \times 10^{-5}$ for the seven pairs of Table 8.13, but it reaches 0.53 for $(153, 154)$ and exceeds $10^{-2}$ for four more collisions, all at $t < -1.89$. With $\bar{\eta}$ in place of $\bar{\delta}$ the lower bound still holds for all 58, the closest case being again $(186, 187)$, whose collision time exceeds the bound by the factor 1.083. For $(34, 35)$, $(71, 72)$, $(91, 92)$ and five other late collisions more than half of $B_4$ at the worst point comes from the model, so $\bar{\eta}$ depends on it, while $\delta$ and $\kappa$ depend on no model. The model is not pessimistic in every respect: the tracked pairs leave the axis faster than an isolated pair would, with $b^2$ up to 3.39 times $2(t_c - t)$.

| $s = -t$ | 0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2 |
|---|---|---|---|---|---|---|---|
| $(13, 14)$: $\ell(s)$ | 1.1033 | 1.2462 | 1.4539 | 1.7833 | 2.3846 | 3.8293 | 12.051 |
| $\ell(0)/(1 - s\ell(0))$ | 1.1033 | 1.4156 | 1.9748 | 3.2638 | 9.3992 | | |
| $m(s)$ | 0.9676 | 1.1099 | 1.3170 | 1.6457 | 2.2461 | 3.6898 | 11.911 |
| $m(0)/(1 - sm(0))$ | 0.9676 | 1.1998 | 1.5787 | 2.3071 | 4.2837 | 29.903 | |
| $(19, 20)$: $\ell(s)$ | 1.2955 | 1.4520 | 1.6768 | 2.0293 | 2.6688 | 4.2153 | 13.848 |
| $\ell(0)/(1 - s\ell(0))$ | 1.2955 | 1.7485 | 2.6889 | 5.8171 | | | |
| $m(s)$ | 1.0446 | 1.1904 | 1.4029 | 1.7413 | 2.3644 | 3.8914 | 13.500 |
| $m(0)/(1 - sm(0))$ | 1.0446 | 1.3205 | 1.7944 | 2.7990 | 6.3583 | | |

Table 8.12: Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13, 14)$ and $(19, 20)$, which collide at $s = 1.2853$ and 1.2755, against the bounds of Theorems 8.4.7 and 8.4.9; a bound is shown only where it is finite, for $s < 1/\ell(0)$ (0.9064 and 0.7719) or $s < 1/m(0)$ (1.0334 and 0.9573). Units of $H_t$; `wigbiane_flow.py`.

| pair | $m(0)$ | $Q(0)$ | $\rho$ | $10^8\bar{\delta}$ | $\bar{\eta}$ | pairs | $1/m(0)$ | $s^*$ | $\lvert t_{\text{coll}}\rvert$ |
|---|---|---|---|---|---|---|---|---|---|
| $(13, 14)$ | 0.9676 | 0.0604 | 0.0610 | $-9.0$ | $2.8 \times 10^{-7}$ | 49 | 1.0334 | 1.2982 | 1.2853 |
| $(19, 20)$ | 1.0446 | 0.0803 | 0.0879 | $-12.8$ | $2.4 \times 10^{-6}$ | 48 | 0.9573 | 1.3325 | 1.2755 |
| $(24, 25)$ | 1.1238 | 0.0794 | 0.0794 | $-10.2$ | $3.2 \times 10^{-5}$ | 44 | 0.8898 | 1.1578 | 1.1439 |
| $(27, 28)$ | 1.4415 | 0.0962 | 0.1012 | $-10.2$ | $4.0 \times 10^{-6}$ | 36 | 0.6937 | 0.9006 | 0.8811 |
| $(34, 35)$ | 2.8970 | 0.0968 | 0.0969 | $-4.3$ | $1.7 \times 10^{-9}$ | 11 | 0.3452 | 0.3853 | 0.3846 |
| $(71, 72)$ | 3.9727 | 0.1605 | 0.1678 | $-5.5$ | $2.0 \times 10^{-9}$ | 4 | 0.2517 | 0.2903 | 0.2880 |
| $(91, 92)$ | 4.0736 | 0.1701 | 0.1757 | $-5.7$ | $3.1 \times 10^{-9}$ | 3 | 0.2455 | 0.2841 | 0.2821 |

Table 8.13: Two sided brackets for the collision time of seven pairs, in the units of $H_t$: the lower bound $1/m(0)$ of Corollary 8.4.13, whose hypothesis holds on the grid, $\bar{\delta} < 0$, and the upper bound $s^*$ of Corollary 8.4.15, with $\rho$ the largest value of $Q$ and $\bar{\kappa} = 0$ for these pairs. The correction $\bar{\eta}$ of Theorem 8.4.12 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision. `wigbiane_flow.py`.

Freezing $Q$ at its value at $s = 0$, that is, taking $\rho = Q(0)$ and $\bar{\kappa} = 0$ in

Corollary 8.4.15, gives a predictor of the collision time from $H_0$ alone,

$$t_{\mathrm{coll}} \approx -\frac{1}{6Q(0)} \log \frac{m(0)}{m(0) - 6Q(0)},$$

the last column of Table 8.11. It is not a bound, since $Q$ grows along the flow ($\rho > Q(0)$ for 41 of the 58 collisions) and the predictor falls on both sides of $t_{\mathrm{coll}}$, but it is accurate when the pair dominates its gap. Over the 28 collisions of Table 8.11 its relative error is at most 0.32% for the six with $6Q(0)/m(0) < 0.3$ and 3.75% in all, for $(51, 52)$; over the 52 collisions of the census with $6Q(0)/m(0) < 0.8$ it is at most 5.75%, for $(130, 131)$. It fails when the other zeros carry most of $m(0)$: the six collisions with $6Q(0)/m(0) \geq 0.8$, all at $t < -1.87$, have errors from 10.4%, for $(68, 69)$, to 46.6%, for $(188, 189)$. To first order in $Q(0)$ the predictor is $1/m(0) + 3Q(0)/m(0)^2$ in absolute value, and if the two zeros sit symmetrically about $u$, so that $P(0) = 8/D_0^2$, it is

$$\frac{D_0^2}{8} + \frac{D_0^4 Q(0)}{32} + O\big(Q(0)^2\big).$$

The bound $\Lambda_k$ of Csordas, Smith and Varga [101] in Section 8.4.6 has the expansion $-\Delta_k^2/8 - \Delta_k^4 g_k/64 - \cdots$ with $\Delta_k = D_0$: the same two terms with $g_k$ in place of $2Q(0)$, and indeed $g_k \approx 2Q(0)$, since $g_k$ adds up, over the other zeros, the inverse squares of their distances to each of the two zeros of the pair. For $(71, 72)$, $g_{71} = 0.33734$ and $2Q(0) = 0.3210$, and the frozen predictor gives $-0.2883$, against $t_{\mathrm{coll}} = -0.2880$ and $\Lambda_{71} = -0.3048$.

Both brackets rest on hypotheses along the flow, on $\delta$ or $\eta$ for the lower bound and on $Q$ and $\kappa$ or $W$ for the upper bound, which the computation checks on the grid of times but which cannot yet be guaranteed in advance.

Problem 8.1. Find a bound for how far the nonreal zeros of $H_t$, $t < 0$, travel, and for how close to a gap between real zeros they come. Such control would make the brackets unconditional statements about $H_t$, computed from $H_0$ alone.

Biane's work meets $\xi$ in two more places. With Pitman and Yor [49], following Williams [518], he wrote $2\xi(s)$ as the Mellin transform of a sum of two Bessel hitting times, the identity behind the Pólya variable of Section 1.2. And in [48] he revisits the paper of Pólya [447], treated in full in Section 8.7.

### 8.4.6 Lehmer pairs and lower bounds for $\Lambda$

Every collision time is a lower bound for $\Lambda$, because nonreal zeros persist backward in time, and Lehmer pairs, pairs with unusually small $D_0$, collide at times close to zero. Csordas, Smith and Varga [101] turned this into a theorem that uses only the zeros of $H_0$. Write $x_j = 2\gamma_j$ for the positive zeros of $H_0$, $x_{-j} = -x_j$,

$\Delta_k = x_{k+1} - x_k$ and

$$g_k = \sum_{j \neq k,k+1}{}' \Big[ \frac{1}{(x_k - x_j)^2} + \frac{1}{(x_{k+1} - x_j)^2} \Big],$$

where the prime restricts the sum to nonzero integers $j$, of either sign. The pair $(x_k, x_{k+1})$ is called a Lehmer pair if $\Delta_k^2 g_k < \frac{4}{5}$, and then [101, Theorem 1]

$$\Lambda \geq \Lambda_k = \frac{\big(1 - \frac{5}{4}\Delta_k^2 g_k\big)^{4/5} - 1}{8 g_k} = -\frac{\Delta_k^2}{8} - \frac{\Delta_k^4 g_k}{64} - \cdots .$$

The leading term is the isolated pair collision time $-D_0^2/8$ with $D_0 = \Delta_k$. The binomial series of $(1-x)^{4/5}$ has only negative coefficients after the constant term, so every further term is negative and decreases in $g_k$, and so does $\Lambda_k$.

For the closest pair below height 200, $\Delta_{71} = 1.44863$ and, from the first 1700 zeros, $g_{71} = 0.33734$, so $\Delta^2 g = 0.708 < \frac{4}{5}$: the pair is a Lehmer pair in this sense, and it gives $\Lambda_{71} = -0.3048$. The omitted zeros above height 2197 increase $g$ by less than $10^{-3}$, which lowers $\Lambda_{71}$ by less than $3 \times 10^{-4}$, so with the tail the bound is $-0.305$. The theorem is proved, but this evaluation of it is a floating point computation with zeros computed by `mpmath` [376]. The bound lies below the computed collision time $-0.288$, as expected. The theorem itself does not order the bound and the collision time: its proof, by contradiction, assumes that $H_t$ has only real zeros for $\Lambda_k \leq t \leq 0$, which fails for every $t < 0$ because $\Lambda \geq 0$.

Table 8.14 lists the published bounds. The last four lower bounds before 2020, among them that of Csordas, Odlyzko, Smith and Varga [100], came from this theorem applied to Lehmer pairs of increasingly high quality, found at heights far greater than 7005. Rodgers and Tao proved $\Lambda \geq 0$ by a different argument, showing that $\Lambda < 0$ would force the zeros into a local equilibrium incompatible with Montgomery's pair correlation estimates [374].

### 8.4.7 The shielding deficit

Let $a < b$ be adjacent simple real zeros of $H_t$ with gap $g = b - a$, and put

$$S_{a,b} = \sum_{\rho \neq a,b} \frac{1}{(a-\rho)(b-\rho)}, \qquad \mathrm{Def} = 2 - g^2 S_{a,b},$$

the sum running over all the other zeros $\rho$ (conjugate pairs, if any, give a real sum). Subtracting the two equations of motion of Section 8.4.1 gives $\dot g = 4/g - 2gS_{a,b}$, that is

$$\frac{d(g^2)}{dt} = 4\,\mathrm{Def}. \tag{8.4.2}$$

For an isolated pair $S_{a,b} = 0$ and $\mathrm{Def} = 2$, which is the law $g^2 = g(0)^2 + 8t$ of Section 8.4.1. For real zeros $S_{a,b} > 0$, and $2xy \leq x^2 + y^2$ gives $S_{a,b} \leq g_k/2$

| bound | authors | year |
|---|---|---|
| $\Lambda > -\infty$ | Newman | 1976 |
| $\Lambda > -50$ | Csordas, Norfolk, Varga | 1988 |
| $\Lambda > -5$ | te Riele | 1991 |
| $\Lambda > -0.385$ | Norfolk, Ruttan, Varga | 1992 |
| $\Lambda > -0.0991$ | Csordas, Ruttan, Varga | 1991 |
| $\Lambda > -4.379 \times 10^{-6}$ | Csordas, Smith, Varga | 1994 |
| $\Lambda > -5.895 \times 10^{-9}$ | Csordas, Odlyzko, Smith, Varga | 1993 |
| $\Lambda \geq -2.63 \times 10^{-9}$ | Odlyzko | 2000 |
| $\Lambda > -1.15 \times 10^{-11}$ | Saouter, Gourdon, Demichel | 2011 |
| $\Lambda \geq 0$ | Rodgers, Tao | 2020 |
| $\Lambda \leq \frac{1}{2}$ | de Bruijn | 1950 |
| $\Lambda < \frac{1}{2}$ | Ki, Kim, Lee | 2009 |
| $\Lambda \leq 0.22$ | Polymath | 2019 |
| $\Lambda \leq 0.2$ | Platt, Trudgian | 2021 |

Table 8.14: Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [463, Table 1], upper bounds from [112], [310], [456] and [419]. Odlyzko's bound is the one computed in [394, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication.

with the $g_k$ of Csordas, Smith and Varga (computed here from the zeros of $H_t$) in Section 8.4.6, so Def $\geq 2 - \Delta_k^2 g_k/2$. A regular lattice of gap $g$ shields exactly: each side contributes $\sum_{m\geq 1} 1/(m(m+1)) = 1$, the factor 2 being what the other zeros must supply. On each side number the exterior gaps outward from the pair, write them $g(1+\varepsilon_m)$ with $\varepsilon_m > -1$, and put $D_0 = 0$, $D_m = \sum_{j\leq m} \varepsilon_j$. The $m$th exterior zero on that side is at distance $g(m + D_m)$ from the nearer end of the pair.

THEOREM 8.4.16 (Newton). *On one side, for every $N \geq 1$,*

$$1 - \sum_{m=1}^{N} \frac{1}{(m+D_m)(m+1+D_m)} = \sum_{m=1}^{N} \frac{\varepsilon_m}{(m+D_{m-1})(m+D_m)} + \frac{1}{N+1+D_N}.$$

*If all zeros are real and the exterior zeros tend to infinity on both sides, then* Def $= \sum_{\text{sides}} \sum_{m\geq 1} \varepsilon_m / \big((m+D_{m-1})(m+D_m)\big)$.

*Proof.* Put $y_m = m + D_{m-1}$ and $z_m = m + D_m$, so $z_m - y_m = \varepsilon_m$, $y_1 = 1$ and $y_{m+1} = z_m + 1$. The $m$th exterior zero contributes $1/\big(z_m(z_m+1)\big)$ to $g^2 S_{a,b}$, because $a - \rho = -g(z_m+1)$ and $b - \rho = -g z_m$. Now $\frac{1}{z_m(z_m+1)} + \frac{z_m - y_m}{y_m z_m} = \frac{1}{y_m} - \frac{1}{z_m+1}$, and summing over $m \leq N$ gives $1/y_1 - 1/(z_N+1) = 1 - 1/(N+1+D_N)$. The last term tends to zero when the zeros tend to infinity; add the two sides. □

COROLLARY 8.4.17. *(a) If all $\varepsilon_m = 0$ then* Def $= 0$ *and $g$ is constant; the lattice of zeros of $e^{\kappa^2 t} \sin \kappa z$ is an example. (b) Keeping $R$ exterior zeros on each side of*

*a regular lattice leaves* $\mathrm{Def} = 2/(R+1)$*, the size of the omitted tail. (c) A single enlarged exterior gap* $g(1+\delta)$ *at position* $j$ *contributes exactly* $\delta/(j(j+\delta))$*. (d) With* Def *as above and* $d_{\nu,j}$ *the exterior gaps on the sides* $\nu = L, R$*,*

$$[\mathrm{Def}]_+ \le g^{-1} \sum_{j\ge 1} j^{-2} \big[(d_{L,j}-g)_+ + (d_{R,j}-g)_+\big],$$

*and by (c) the constant* $j^{-2}$ *is optimal to first order. (e) If the configuration has a smallest gap* $g > 0$ *(a finite configuration or a lattice) and* $g$ *is that gap, then every* $\varepsilon_m \ge 0$*, so* $\mathrm{Def} \ge 0$ *and the smallest gap does not shrink forward in time, with equality only for the lattice.*

*Proof.* (a) to (c) are read off from the theorem; in (b), $2(1-R/(R+1)) = 2/(R+1)$. For (d) let $P_m = \sum_{i\le m}(\varepsilon_i)_+$. Then $z_m \le m + P_m$ and $1/(z(z+1))$ decreases, so by the theorem applied to the sequence $(\varepsilon_i)_+$, one side contributes at most $1 - \sum_m \frac{1}{(m+P_m)(m+P_m+1)} = \sum_m \frac{(\varepsilon_m)_+}{(m+P_{m-1})(m+P_m)} \le \sum_m (\varepsilon_m)_+/m^2$. (e) is the theorem with $\varepsilon_m \ge 0$ and (8.4.2). □

Part (e) says nothing about $H_t$: its mean gap at height $T$ is of order $1/\log T$, so there is no smallest gap. Being minimal among the neighbours is not enough either: take $g = 1$ and exterior zeros at distances 1, 1.01, 1.02 from the pair on both sides, so that the nearest exterior gaps equal $g$ and the next ones are 0.01. Then $g^2 S_{a,b} \ge 1 + \frac{20000}{20301} + \frac{20000}{20604} = \frac{340039}{115039} > 2$, so $\mathrm{Def} < 0$ and this gap shrinks forward. Finally, (8.4.2) integrates to an identity, $4\int \mathrm{Def}\, dt = \Delta(g^2)$, so a bound on the shielding budget that already contains the moving gap is circular; an estimate has to bound Def from the exterior zeros alone.

### 8.4.8 The Thorin atoms of the tilted clock

Multiplying the kernel by $e^{tu^2}$ tilts the clock. For $t \ge \Lambda$ the function $R_t$ of Section 8.4.1 is the Laplace transform of a GGC $T_t$, the tilted clock, with the Thorin atoms $a_j(t) = z_j(t)^2/4 = \theta_j(t)^2$ (Theorem 7.2.16), and $T_0 = T$ is the clock of Section 1.2, which exists unconditionally (Theorem 5.1.1).

Proposition 8.4.18. *For* $t > 0$*, with* $Z$ *a standard normal variable,*

$$H_t(z) = \mathbb{E}\, H_0\big(z + i\sqrt{2t}\, Z\big), \qquad z \in \mathbb{C},$$
$$\frac{1}{R_t(\sigma^2)} = \frac{\mathbb{E}\, \xi\big(\frac12 + \sigma + \sqrt{t/2}\, Z\big)}{\mathbb{E}\, \xi\big(\frac12 + \sqrt{t/2}\, Z\big)}, \qquad \sigma \ge 0.$$

*Proof.* With $c = \sqrt{2t}$ we have

$$\mathbb{E}\cos\big((z + icZ)u\big) = \cos(zu)\, \mathbb{E}\cosh(cuZ) = \cos(zu)\, e^{tu^2},$$

because the part odd in $Z$ has mean zero. Integrating against $\Phi_{\mathrm{dBN}}$ gives the first identity, Fubini applying because $\Phi_{\mathrm{dBN}}(u)$ decays like $\exp(-\pi e^{4u})$; the same computation with $e^{tu^2}\Phi_{\mathrm{dBN}}$ in place of $\Phi_{\mathrm{dBN}}$ gives $H_{t+s}(z) = \mathbb{E}\, H_t\big(z + i\sqrt{2s}\, Z\big)$ for real $t$ and $s > 0$. For the second identity, $8H_0(\pm 2i\sigma) = \xi(\frac{1}{2} + \sigma)$ and $Z$ is symmetric. We checked the first identity at $t = 0.3$ and $z = -4i$ to 40 digits (`xg_tilt.py`, `code/README.md`). □

So for $t > 0$ the tilt is Gaussian convolution, along the real axis of $\sigma$, of the even positive function $\sigma \mapsto \xi(\frac{1}{2} + \sigma)$, with variance $t/2$. Theorem 19.4.1(b) is the smoothing along the real axis of $z$, which lowers $t$.

The Newton identities of Section 8.4.3 give the mean and variance of $T_t$ from the moments $M_k(t)$ of the reweighted kernel,

$$\mathbb{E}T_t = \sum_j \frac{1}{a_j(t)} = \frac{M_1(t)}{2M_0(t)}, \qquad \operatorname{Var} T_t = \sum_j \frac{1}{a_j(t)^2} = \frac{M_1(t)^2}{4M_0(t)^2} - \frac{M_2(t)}{12M_0(t)},$$

and in the variable $a_j$ the motion of the zeros in Section 8.4.1 reads

$$\frac{da_j}{dt} = \frac{1}{2} + 2\sum_{k \neq j} \frac{a_j}{a_j - a_k}, \tag{8.4.3}$$

the term $\frac{1}{2}$ coming from the mirror zero $-z_j$ and the others from the pairs $\pm z_k$. The table gives the first four atoms at $t = 0$ and $t = 0.2$, with $\mathbb{E}T_t$ and $\operatorname{Var} T_t$, the difference quotient $\Delta a_j/\Delta t = (a_j(0.2) - a_j(0))/0.2$ and the rate $\dot a_j(0)$ from (8.4.3).

| | $t = 0$ | $t = 0.2$ | $\Delta a_j/\Delta t$ | $\dot a_j(0)$ |
|---|---|---|---|---|
| $a_1$ | 199.7905 | 198.1565 | −8.170 | −8.193 |
| $a_2$ | 441.9262 | 438.7638 | −15.812 | −15.872 |
| $a_3$ | 625.5430 | 622.6126 | −14.652 | −14.652 |
| $a_4$ | 925.6731 | 919.9043 | −28.844 | −29.069 |
| $\mathbb{E}T_t$ | 0.0231049931 | 0.0232009711 | | |
| $\operatorname{Var} T_t$ | $3.71726 \times 10^{-5}$ | $3.77203 \times 10^{-5}$ | | |

At $t = 0$ the $a_j$ are $\gamma_j^2$, the Thorin atoms of $T$ if RH holds (Proposition 1.2.1), and $\mathbb{E}T_0$ and $\operatorname{Var} T_0$ are the constants of Section 1.2. The rate is the sum (8.4.3) over the first 8000 zeros completed by the tail $-2a_j(\log(\gamma_{8000}/2\pi) + 1)/(2\pi\gamma_{8000})$; a central difference of the zeros of $H_t$ at $t = \pm 0.01$ and $\pm 0.005$, with Richardson extrapolation, reproduces it to $2 \times 10^{-5}$, the order of the error of the tail.

Since $M_k'(t) = \frac{1}{4}M_{k+1}(t)$, the mean of the tilted clock increases along the flow:

$$\frac{d}{dt}\mathbb{E}T_t = \frac{M_0(t)M_2(t) - M_1(t)^2}{8M_0(t)^2} > 0$$

by the Cauchy Schwarz inequality, at every $t$ at which $T_t$ exists (a note of Nicholas Polson merged into the book [243]). The atoms need not all decrease. Take the common coupling $T_t = \sum_j E_j/a_j(t)$ with the same independent standard exponential variables $E_j$ for every $t$, which exists for $t \geq 0.2$ because every zero is real there [419]. The equation of motion gives $\dot z_{64}(0.2) = +0.26897$ and $\dot z_{92}(0.2) = +0.13385$, the only two rising zeros among the first 100, so $a_{64}$ and $a_{92}$ increase (computed, not certified: Gauss Hermite quadrature in double precision, the two velocities unchanged to $3 \times 10^{-17}$ between 60 and 90 nodes). Put $b_j(t) = 1/a_j(t)$ and choose $h > 0$ so small that $c = b_{64}(0.2+h) - b_{64}(0.2) < 0$. Then $T_{0.2+h} - T_{0.2} = cE_{64} + Y$ with $Y = \sum_{j \neq 64} \big(b_j(0.2+h) - b_j(0.2)\big)E_j$, which converges absolutely almost surely, since $|Y| \leq T_{0.2+h} + T_{0.2}$, and is independent of $E_{64}$. Given any value of $Y$, the unbounded variable $E_{64}$ makes $cE_{64} + Y < 0$ with positive probability, so $T_{0.2+h} < T_{0.2}$ with positive probability, and this coupling is not almost surely monotone on any right neighbourhood of $t = 0.2$ (computed). At $t = 0$ the same argument with $\dot z_{35}(0) > 0$ of Proposition 8.4.1 (certified) gives the statement on every right neighbourhood of 0, conditional on RH, which the coupling at $t = 0$ needs. The increase of the mean therefore does not come from monotone atoms.

Problem 8.2. Is the tilted clock stochastically increasing in $t$ on $t \geq \Lambda$ (so, unconditionally, on $t \geq 0.2$), that is, is $P(T_t \leq x)$, whose Laplace transform in $x$ is $R_t(u)/u$, nonincreasing in $t$ for every $x > 0$? The failure of the coupling above does not decide it, since another coupling may be monotone, and an increasing mean is necessary but not sufficient.

For large $t$ the tilt concentrates the kernel. Under the probability density proportional to $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ on $(0,\infty)$, with expectation $\mathbb{E}_t$, we have $1/R_t(s) = \mathbb{E}_t \cosh(2\sqrt{s}\,u)$ and $H_t(z)/H_t(0) = \mathbb{E}_t \cos(zu)$.

Proposition 8.4.19. *As $t \to \infty$ let $u_t$ be the mode of $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$, so that $e^{4u_t} \sim tu_t/(2\pi)$. Then*

$$R_t\big(s/u_t^2\big) \to \frac{1}{\cosh(2\sqrt{s})}, \qquad \frac{T_t}{u_t^2} \Rightarrow 2C, \qquad u_t\, z_j(t) \to \pi\big(j - \tfrac12\big),$$

*where $C$ is the exit time of Brownian motion from $(-1,1)$, whose Laplace transform is $1/\cosh\sqrt{2\theta}$.*

*Proof.* Put $g(u) = tu^2 + \log\Phi_{\mathrm{dBN}}(u) = tu^2 + 9u + \log 2\pi^2 - \pi e^{4u} + o(1)$. The stationary point satisfies $e^{4u_t} = (2tu_t + 9)/(4\pi) + o(1)$ and $g''(u_t) = -8tu_t + O(t)$, so under $\mathbb{E}_t$ the variable $u$ has mean $u_t + o(1)$ and standard deviation of order $(tu_t)^{-1/2} \to 0$, with Gaussian tails in $(u - u_t)\sqrt{tu_t}$. Hence $u/u_t \to 1$ in probability with $\mathbb{E}_t e^{cu/u_t}$ bounded for each $c$, and $\mathbb{E}_t \cosh(2\sqrt{s}\,u/u_t) \to \cosh(2\sqrt{s})$. The Laplace transform $1/\cosh(2\sqrt{s})$ is that of $2C$, and the zeros follow from Hurwitz's theorem applied to $z \mapsto H_t(z/u_t)/H_t(0) \to \cos z$, uniformly on compacts. $\square$

The convergence is fast: at $t = 300$, 1000 and 3000 the mode is $u_t = 0.9602$, 1.3418, 1.6707, and $u_t z_1(t) = 1.57334$, 1.57112, 1.57086 against $\pi/2 = 1.57080$. The path of clocks $T_t$ thus runs from the Brownian exit time $2C$, after the scaling by $u_t^2$, at $t = \infty$ to $T$ at $t = 0$, and consists of GGCs for $t \geq \Lambda$ (Theorem 7.2.16). The limit shape is a fixed shape of the flow: the zeros of $e^{\kappa^2 t}\cos \kappa z$ are the equally spaced points $\pi(j - \frac{1}{2})/\kappa$ at all times [385].

Gaussian smoothing of a reciprocal GGC transform in the variable $\sigma = \sqrt{s}$ is again one, and not only for the tilt.

THEOREM 8.4.20 (Gaussian smoothing of reciprocal GGC transforms). *Let $a_j > 0$ with $\sum_j a_j^{-1} < \infty$, and let $M(\sigma) = \prod_j (1 + \sigma^2/a_j)$, the reciprocal of the Laplace transform, at $s = \sigma^2$, of the GGC with the discrete Thorin measure $\sum_j \delta_{a_j}$ and no drift; $M$ is an even entire function of genus zero in $\sigma^2$. For $c > 0$ put*

$$M_c(\sigma) = \frac{\mathbb{E}\, M(\sigma + cZ)}{\mathbb{E}\, M(cZ)}.$$

*Then $M_c(\sigma) = \prod_j (1 + \sigma^2/a_j(c))$ with real $a_j(c) > 0$, so that $1/M_c(\sqrt{s})$ is the Laplace transform of a GGC $S_\lambda$ with the Thorin measure $\sum_j \delta_{a_j(c)}$, where $\lambda = 2c^2$. The atoms obey* (8.4.3) *with $t$ replaced by $\lambda$ while they are simple, and*

$$\frac{d}{d\lambda}\mathbb{E}S_\lambda = (\mathbb{E}S_\lambda)^2 - \frac{3}{2}\operatorname{Var} S_\lambda.$$

*Proof.* Write $\lambda = 2c^2$, $D = d/d\sigma$ and $u(\lambda, \sigma) = \mathbb{E}M(\sigma + cZ) = e^{\lambda D^2/4} M(\sigma)$, so that $\partial_\lambda u = \frac{1}{4}\partial_\sigma^2 u$, the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ in the variable $\sigma = iz/2$. The function $f(\tau) = M(i\tau) = \prod_j (1 - \tau^2/a_j)$ is an even real entire function of genus at most one with only real zeros, and $u(\lambda, i\tau) = \mathbb{E}f(\tau + icZ) = e^{-\lambda D_\tau^2/4} f(\tau)$ with $D_\tau = d/d\tau$. By Pólya's theorem that $e^{-aD^2}$, $a > 0$, preserves the class of real entire functions of genus at most one with only real zeros [449] (for $H_t$ it is Theorem 19.4.1(c)), $u(\lambda, i\tau)$ has only real zeros in $\tau$. It is even and $u(\lambda, 0) = \mathbb{E}M(cZ) \geq 1$, so $M_c = u/u(\lambda, 0)$ has the product form with real $a_j(c) > 0$. For a single atom, $\mathbb{E}\big(1 + (\sigma + cZ)^2/a\big) = (1 + c^2/a)\big(1 + \sigma^2/(a + c^2)\big)$: the atom moves from $a$ to $a + \lambda/2$, the drift $\frac{1}{2}$ of (8.4.3). In general the atoms obey (8.4.3) by the computation of Section 8.4.1, which uses only the heat equation, the product form above and the simplicity of the zeros.

For the mean write $u = c_0 + c_2\sigma^2 + c_4\sigma^4 + \cdots$, so that $\dot{c}_0 = \frac{1}{2}c_2$ and $\dot{c}_2 = 3c_4$. Since $u/c_0 = \prod_j (1 + \sigma^2/a_j(\lambda)) = 1 + m\sigma^2 + \frac{1}{2}(m^2 - p)\sigma^4 + \cdots$ with $m = \sum_j a_j^{-1}$ and $p = \sum_j a_j^{-2}$, we have $c_2 = mc_0$ and $c_4 = \frac{1}{2}(m^2 - p)c_0$, and

$$\dot{m} = \frac{\dot{c}_2}{c_0} - m\,\frac{\dot{c}_0}{c_0} = \frac{3}{2}(m^2 - p) - \frac{1}{2}m^2 = m^2 - \frac{3}{2}p.$$

Finally $\mathbb{E}S_\lambda = m$ and $\operatorname{Var} S_\lambda = p$. □

Take $M(\sigma) = 1/R_t(\sigma^2)$. The proof of Proposition 8.4.18 gives

$$M_c(\sigma) = \frac{1}{R_{t+2c^2}(\sigma^2)},$$

so for $t \geq \Lambda$ the identity becomes

$$\frac{d}{dt}\mathbb{E}T_t = (\mathbb{E}T_t)^2 - \frac{3}{2}\operatorname{Var} T_t.$$

For the tilt it also follows without the zeros from $dM_k/dt = M_{k+1}/4$ and the formulas for $\mathbb{E}T_t$ and $\operatorname{Var} T_t$ above. At $t = 0$ both sides equal $4.7808 \times 10^{-4}$ and at $t = 0.2$ they equal $4.8170 \times 10^{-4}$.

The clock $T_{0.2}$ is an unconditional GGC (Theorem 7.2.16). By Proposition 8.4.18 its Laplace transform is the reciprocal of the Gaussian smoothing of $\xi(\frac{1}{2} + \sigma)$ with variance 0.1 in $\sigma$, normalised at $\sigma = 0$. It is close to $T$: the means differ by $9.6 \times 10^{-5}$ on 0.0231, and $R_{0.2}(s)/R_0(s) - 1$ is $-9.6 \times 10^{-5}$ at $s = 1$, $-9.3 \times 10^{-4}$ at $s = 10$ and $-7.6 \times 10^{-3}$ at $s = 100$. In this sense $T$ is a GGC up to a smoothing of variance 0.1, and RH is the statement that the variance can be taken to be zero.

### 8.4.9 High order Laguerre expressions along the flow

This subsection and Section 8.4.7 report results of T. Newton [385], rewritten in the notation of the book; the Lambert form of Proposition 8.4.22 is the authors' own. Put $\nu_t(u) = \frac{1}{2}e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ on $\mathbb{R}$, so that $H_t(z) = \int_{\mathbb{R}} \nu_t(u)e^{izu}\,du$, and define the Laguerre expressions $L_{n,t}(x)$ by

$$H_t(x + iy)H_t(x - iy) = \sum_{n\geq 0} L_{n,t}(x)\, y^{2n}.$$

Then $L_{1,t} = H_t'^2 - H_t H_t'' = H_t^2 \mathcal{L}_{H_t}$ with Laguerre's quantity of Section 8.4.4. With $w = u + v$ and $b = u - v$ the odd powers of $b$ cancel under $(u, v) \mapsto (v, u)$, and

$$L_{n,t}(x) = \frac{1}{(2n)!} \iint b^{2n}\, \nu_t(u)\nu_t(v) \cos(xw)\, du\, dv. \tag{8.4.4}$$

Since $H_t$ is even, $H_t(iy) = H_t(0)/R_t(y^2/4)$, so $\sum_n L_{n,t}(0)y^{2n} = H_t(0)^2 R_t(y^2/4)^{-2}$: the $L_{n,t}(0)$ are the Taylor coefficients of the square of the reciprocal tilted clock. By (8.4.4), $L_{n,t}(0) > 0$. Let $\mathbb{P}_{n,t}$ be the probability law on $\mathbb{R}^2$ with density proportional to $b^{2n}\nu_t(u)\nu_t(v)$, with expectation $\mathbb{E}_{n,t}$. Then $L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw)$, and with $k_n = (n + 1)(2n + 1)$ and $k_n(2n)!/(2n + 2)! = \frac{1}{2}$, the central ratio is

$$c_n(t) = \frac{k_n L_{n+1,t}(0)}{L_{n,t}(0)} = \tfrac{1}{2}\,\mathbb{E}_{n,t}[b^2].$$

THEOREM 8.4.21 (Newton). $\lim_{n\to\infty} \sup_{0\le t\le 1/2} \mathbb{E}_{n,t}[w^2] = 0$. *Consequently, for every* $X < \infty$ *there is* $N_X$ *with* $L_{n,t}(x) > 0$ *for all* $n \ge N_X$, $|x| \le X$ *and* $0 \le t \le \frac{1}{2}$.

*Proof.* Let $V = -\log \Phi_{\mathrm{dBN}}$, an even smooth function with $V''(u) \sim 16\pi e^{4u}$ as $u \to \infty$ (the first theta summand dominates). So $V''$ is continuous, bounded below by $-K$ and tends to infinity; Newman's monotonicity of $V''$ [382] is not needed. Given $b = d$, the density of $w$ is proportional to $e^{-Q(w)}$ with $Q(w) = V(\frac{w+d}{2}) + V(\frac{w-d}{2}) - \frac{t}{2}(w^2+d^2)$ and $Q'' = \frac{1}{4}[V''(\frac{w+d}{2}) + V''(\frac{w-d}{2})] - t$. One of the two arguments has modulus at least $|d|/2$, so for $|d| \ge D_0$ and $t \le \frac{1}{2}$ we get $Q'' \ge \kappa(D_0) := \frac{1}{4}[\inf_{|x|\ge D_0/2} V'' - K] - \frac{1}{2}$, which tends to infinity with $D_0$. Since $V$ is even, $Q$ is even, $Q'(0) = 0$ and $wQ'(w) \ge \kappa w^2$; integration by parts gives $\mathbb{E}[wQ'(w) \mid b = d] = 1$, so $\mathbb{E}[w^2 \mid b = d] \le \kappa^{-1}$. For $|d| \le D_0$ the conditional second moment is at most a constant $B(D_0)$ uniform in $t$, because $1 \le e^{t(u^2+v^2)} \le e^{(u^2+v^2)/2}$ and both integrals are continuous in $d$ on a compact set. Finally $\nu_0 \le \nu_t \le \nu_{1/2}$, and comparing the weight $b^{2n}$ on $\{|b| \le D_0\}$ with that on $\{D_0+1 \le |b| \le D_0+2\}$ gives $\mathbb{P}_{n,t}(|b| \le D_0) \le C(D_0)\big(D_0/(D_0+1)\big)^{2n}$ uniformly in $t$. Hence $\mathbb{E}_{n,t}[w^2] \le B(D_0)C(D_0)(\frac{D_0}{D_0+1})^{2n} + \kappa(D_0)^{-1}$; let $n \to \infty$ and then $D_0 \to \infty$. For the second claim, $L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw) \ge 1 - \frac{x^2}{2}\mathbb{E}_{n,t}[w^2]$. □

Newton states that $c_n(t) = (\frac{1}{8} + o(1))(\log n)^2$ uniformly in $t \in [0, \frac{1}{2}]$. The Laplace calculation behind it gives the better centring $\frac{1}{8}\mathcal{W}(n/\pi)^2$, with $\mathcal{W}$ the Lambert function of Section 11.5.3.

PROPOSITION 8.4.22. *Uniformly in* $t \in [0, \frac{1}{2}]$, $c_n(t) = \frac{1}{8}\,\mathcal{W}(n/\pi)^2 + o(\log n)$.

*Sketch.* For large $|b|$ only the first theta summand matters, and $\log \nu_t(u) + \log \nu_t(v) = -2\pi e^{2b}\cosh 2w + O(b^2)$. Integrating out $w$ (a Gaussian of width of order $e^{-b}$) leaves a marginal $-2\pi e^{2b} + O(b^2)$, uniformly in $t$. The law of $b$ has log density $2n\log b - 2\pi e^{2b} + O(b^2)$, maximal where $2b\,e^{2b} = n/\pi$ up to an $O(b^2/n)$ shift, that is at $b_n = \frac{1}{2}\mathcal{W}(n/\pi) + o(1)$, with width $O(\sqrt{b_n/n})$. So $\mathbb{E}_{n,t}[b^2] = b_n^2 + o(\log n)$. □

Since $\mathcal{W}(y) = \log y - \log\log y + o(1)$, the proposition is a statement about the second order term, and the relative error of $\frac{1}{8}(\log n)^2$ is of order $\log\log n/\log n$. The numerical values below, computed from the moment convolution $L_{n,t}(0) = \frac{1}{4}\sum_j M_j M_{n-j}/((2j)!(2n-2j)!)$ with $M_k = \int u^{2k} e^{tu^2}\Phi_{\mathrm{dBN}}(u)\,du$ (`code/newton_central_ratio.py`), suggest that the error is $o(1)$.

| $n$ | $c_n(\frac12)$ | $c_n(0)$ | $\frac18\mathcal{W}(n/\pi)^2$ |
|---|---|---|---|
| $10^2$ | 0.82596 | 0.82413 | 0.80112 |
| $10^3$ | 2.32317 | 2.32216 | 2.31510 |
| $10^4$ | 4.86185 | 4.86152 | 4.85998 |
| $10^5$ | 8.52286 | 8.52278 | 8.52250 |
| $10^6$ | 13.35216 | 13.35215 | 13.35210 |

The difference $c_n(\frac12) - \frac18\mathcal{W}(n/\pi)^2$ is 0.025, 0.0081, 0.0019, 0.00035 and 0.00006, and $c_n(0)$ differs from $c_n(\frac12)$ by $3\times10^{-4}$ at $n=10^4$. At $n=10^5$, $\frac18(\log n)^2 = 16.6$ against $c_n = 8.52$, so an effective version of Theorem 8.4.23 must use the Lambert centring.

THEOREM 8.4.23 (Newton). *For every $X<\infty$, uniformly in $0\le t\le\frac12$ and $|x|\le X$,*

$$\frac{k_n L_{n+1,t}(x)}{L_{n,t}(x)} = c_n(t)\,(1+o(1)) = \tfrac18(\log n)^2(1+o(1)).$$

*Proof.* With $r_n(t,x) = L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw)$, Theorem 8.4.21 gives $r_n\to 1$ uniformly, and $k_nL_{n+1,t}(x)/L_{n,t}(x) = c_n(t)\,r_{n+1}/r_n$; the second equality is Proposition 8.4.22 with $\mathcal{W}(n/\pi)\sim\log n$, used as stated. □

The result has the shape of the theorem of Griffin, Ono, Rolen and Zagier [159] (Problem 16.4): all large orders at each fixed window, for the Laguerre expressions along the whole flow. The index $N_X$ is not effective, and effectivity in $X$ is the open problem here: it would turn the positivity of all $L_{n,t}$ on $|x|\le X$ into a finite computation in $n<N_X$.

### 8.4.10 Forward persistence of real zeros

The motion $\dot z_j = 2\sum_{k\ne j}1/(z_j-z_k)$ of the zeros of $H_t$ (Section 8.4.1) says nothing at a collision, where the sum diverges. The lemma below and the barrier argument after it come from the authors' discussion with a collaborator.

LEMMA 8.4.24. *Let $c$ be a real zero of multiplicity $m\ge1$ of $H_{t^*}$. For $t<t^*$ close to $t^*$, the $m$ zeros of $H_t$ near $c$ are $c+i\sqrt{2(t^*-t)}\,x_k+o(\sqrt{t^*-t})$, $k=1,\dots,m$, with $x_k$ the roots of $\mathrm{He}_m$, so at most one of them is real, and for $t>t^*$ close to $t^*$ all $m$ of them are real and simple. Consequently real zeros of $H_t$ never collide as $t$ increases; a real zero stays real, and a simple real zero is simple except at times when zeros arriving from the complex plane meet it; and nonreal zeros of $H_\tau$ cannot accumulate at a real point as $\tau$ decreases to a limit $\tau_1$.*

*Proof.* Put $t=t^*+s$ and expand $H_t(c+w)=\sum a_{jk}w^js^k$ (a convergent series, since $H_t(z)$ is an entire function of $(z,t)\in\mathbb{C}^2$, by the decay of $\Phi$), and give $w$ weight 1 and $s$ weight 2. The lowest weighted degree part $P$ satisfies $\partial_sP=-\partial_w^2P$, because $\partial_tH_t=-\partial_z^2H_t$ and the two sides have the same weighted degree after the

shift, so $P(w,s) = e^{-s\partial_w^2}P(w,0)$. The part of $P$ with $s^0$ is $a\,w^d$, and $a \neq 0$, since otherwise $P = 0$; as $H_{t^*}(c+w) = a_{m0}w^m + \cdots$ with $a_{m0} \neq 0$, we get $d = m$ and $P = a\,h_m$ with $h_m(w,s) = e^{-s\partial_w^2}w^m$. For $s = \mp\sigma$ the substitution $w = \sqrt{2\sigma}\,x$ turns $h_m$ into $(2\sigma)^{m/2}e^{\pm\partial_x^2/2}x^m$. For $s > 0$ this is $(2s)^{m/2}\mathrm{He}_m(x)$, with $\mathrm{He}_m = e^{-\partial_x^2/2}x^m$ the Hermite polynomial, whose $m$ roots are real and simple. For $s < 0$ put $w = i\sqrt{2|s|}\,x$ instead: $h_m(w,s) = (i\sqrt{2|s|})^m\mathrm{He}_m(x)$, so the zeros of $h_m$ are $i\sqrt{2|s|}$ times the roots of $\mathrm{He}_m$, simple and purely imaginary, with $w = 0$ among them exactly for odd $m$. Now write $w = \sqrt{|s|}\,\omega$ (respectively $\sqrt{s}\,\omega$): every monomial $w^j s^k$ becomes $(\operatorname{sgn} s)^k|s|^{(j+2k)/2}\omega^j$, so $|s|^{-m/2}H_t(c+\sqrt{|s|}\,\omega) \to a\,h_m(\omega,\pm1)$ uniformly on compact sets as $s \to 0$, and the limit has $m$ simple zeros. By Rouché's theorem, in small discs around the zeros of the limit the function has exactly one zero each; by Hurwitz's theorem applied to $H_t(c+w)$ on a fixed circle $|w| = r$, there are exactly $m$ zeros in $|w| < r$, so these are all of them. For $s < 0$ the discs are disjoint, and all lie off the real axis except the one at $\omega = 0$ (odd $m$), which contains exactly one zero; for $s > 0$ the zeros are real because they are simple, near a real simple zero of the limit, and the zero set is symmetric under conjugation. The consequences follow: two real zeros meeting at $c$ at time $t^*$ would give two real zeros near $c$ for $t < t^*$; a real zero stays real by the symmetry (for a multiple one, by the case $t > t^*$), since a simple zero near a real simple zero is unique; and nonreal zeros of $H_{\tau_k}$, $\tau_k \downarrow \tau_1$, with a limit point $c \in \mathbb{R}$ would give nonreal zeros near $c$ just after $\tau_1$, against the case $s > 0$. □

Corollary 8.4.25. *If $\Lambda > 0$, then $H_\Lambda$ has a multiple real zero. Hence RH holds if and only if $H_t$ has only simple zeros for every $t \in (0, 0.2]$. This follows a note of Nicholas Polson merged into the book [242].*

*Proof.* The set of $t$ for which $H_t$ has only real zeros is $[\Lambda,\infty)$, closed by Hurwitz's theorem, so the zeros of $H_\Lambda$ are real. Suppose $\Lambda > 0$ and that they are all simple. By the theorem of the Polymath project that for an absolute constant $C$ and every $0 < t \le \frac12$ every zero $x+iy$ of $H_t$ with $|x| \ge e^{C/t}$ is real and simple [456, Theorem 1.5], there is $X \ge e^{2C/\Lambda}$, with no zero of $H_\Lambda$ near $\pm X$, such that for $t \in [\frac{\Lambda}{2},\Lambda] \subset (0,\frac12]$ every zero of $H_t$ with $|\operatorname{Re} z| > X$ is real. The zeros with $|\operatorname{Re} z| \le X$ lie in a bounded set [112]. As $t \uparrow \Lambda$, Hurwitz's theorem puts exactly one zero of $H_t$ near each zero of $H_\Lambda$ in $[-X,X]$ and no others in the box $[-X,X]\times[-1,1]$ (the strip of de Bruijn holds all zeros), and the zero near a real simple zero is real, since $H_t$ is real on the real line and nonreal zeros come in conjugate pairs. So for $t < \Lambda$ close to $\Lambda$ all zeros of $H_t$ are real, against the definition of $\Lambda$. If RH fails, then $H_0$ has a zero off the real line, so $0 < \Lambda \le 0.2$ [419], and $H_\Lambda$ has a multiple zero. If RH holds, then $\Lambda \le 0$, and a multiple zero of $H_{t^*}$ with $t^* > 0$ would give, by Lemma 8.4.24, zeros of $H_t$ off the real line for $t < t^*$ close to $t^*$, although $t > \Lambda$. □

For a height $X$ let $0 = t_0 < t_1 < \cdots < t_M = 0.2$ and let $Q_i = [u_i, v_i]\times[-1,1]$

be rectangles in the $z$ plane with $u_i < X < v_i$. The hypothesis $B(X)$ is:

(B0) every zero of $H_0$ with $|\operatorname{Re} z| \le u_0$ is real and simple (Platt and Trudgian [419] verified this to $6 \times 10^{12}$ in $z$);

(B1) for each $i$, $H_{t_i}$ has no zero on $\partial Q_i$, and the number of zeros of $H_{t_i}$ in $Q_i$ (counted with multiplicity by the argument principle) is at most the number of sign changes of $H_{t_i}$ observed on $[u_i, v_i]$;

(B2) for each $i$ and every $t \in [t_i, t_{i+1}]$, $|H_t - H_{t_i}| < |H_{t_i}|$ on $\partial Q_i$.

PROPOSITION 8.4.26. *Assume $B(X)$. Then for every $\lambda \in [0, 0.2]$ every zero of $H_\lambda$ with $|\operatorname{Re} z| \le X$ is real and simple.*

*Proof.* Fix $i$. By (B2) and Rouché's theorem, $H_t$ has no zero on $\partial Q_i$ and the same number $N_i$ of zeros in $Q_i$ for all $t \in [t_i, t_{i+1}]$. At $t_i$ a sign change is a real zero of odd multiplicity, so by (B1) the $N_i$ zeros are real and simple. Let $A$ be the set of $t \in [t_i, t_{i+1}]$ at which the $N_i$ zeros in $Q_i$ are real and simple. It contains $t_i$, and it is open in $[t_i, t_{i+1}]$, since a simple real zero persists as a simple zero, real by symmetry. Let $\tau$ be the supremum of the $t$ with $[t_i, t] \subset A$ and take $t'_k \uparrow \tau$ in $A$: the zeros of $H_\tau$ in $Q_i$ are limits of real zeros (Hurwitz's theorem, with no zero on the boundary), hence real, and two of them cannot coincide, since the $N_i$ distinct real zeros of $H_{t'_k}$ near the common point would contradict Lemma 8.4.24; so there are $N_i$ distinct real zeros, all simple, and $\tau \in A$. As $A$ is open, $\tau = t_{i+1}$, so $A = [t_i, t_{i+1}]$, and every zero of $H_t$ in $Q_i$ is real and simple for $t \in [t_i, t_{i+1}]$. The same holds for the mirror image $-Q_i$.

Now let $\lambda \in [0, 0.2]$ and suppose $H_\lambda$ has a nonreal zero $z_0$ with $|\operatorname{Re} z_0| \le X$. By de Bruijn's strip theorem [112, Theorem 13], the zeros of $H_t$ satisfy $|\operatorname{Im} z| \le \sqrt{1 - 2t}$ for $0 \le t \le \frac{1}{2}$ (and $|\operatorname{Im} z| < 1$ at $t = 0$), so no zero meets the lines $\operatorname{Im} z = \pm 1$ and every zero of $H_\tau$ with $|\operatorname{Re} z| \le X$ lies in the compact set $K = \{|\operatorname{Re} z| \le X,\ |\operatorname{Im} z| \le 1\}$. Let $F$ be the set of $\tau \in [0, \lambda]$ at which $H_\tau$ has a nonreal zero in $K$; then $\lambda \in F$. For $\tau \in [t_i, t_{i+1}]$ the first part shows that such a zero has $|\operatorname{Re} z| < u_i < X$ (otherwise it would lie in $Q_i$ or $-Q_i$), and $|\operatorname{Im} z| < 1$, so it is an interior point of $K$ and $F$ is open in $[0, \lambda]$ by Hurwitz's theorem. If $\tau_k \downarrow \tau$ with $\tau_k \in F$, a limit point of the nonreal zeros is a zero of $H_\tau$ in $K$, and it is nonreal by the case $s > 0$ of Lemma 8.4.24. Hence $\tau_0 = \inf\{\tau : [\tau, \lambda] \subset F\}$ lies in $F$, openness forces $\tau_0 = 0$, and $0 \in F$ gives a nonreal zero of $H_0$ with $|\operatorname{Re} z| \le X$; by (B0) it has $|\operatorname{Re} z| > u_0$, so it lies in $\pm Q_0$, against the first part. Hence every zero of $H_\lambda$ with $|\operatorname{Re} z| \le X$ is real. The constants in $B(X)$ do not change if $X$ is replaced by $X' \in (X, \min_i v_i)$, so the same holds for $|\operatorname{Re} z| \le X'$. For simplicity at $\lambda = 0$ use (B0) and the first part. For $\lambda > 0$, a real zero $c$ of multiplicity $m \ge 2$ with $|c| \le X$ would by Lemma 8.4.24 have at least $m - 1 \ge 1$ nonreal zeros of $H_t$ near $c$ for $t < \lambda$ close to $\lambda$, with $|\operatorname{Re} z| \le X'$, which is impossible. □

The hypothesis $B(X)$ has not been verified at any height $X$, and no computation is claimed. The verification intended is the count barrier of the collaborator: evaluate $H_{t_i}$ on the boundary of $Q_i$ and on its real diameter at the grid times, and check (B1) and (B2) there. The status of the problem is recorded in Remark 14.15.1.

Two results of Newton [385] bear on the verification of $B(X)$. The first is the local form of (B2). Let $t_* \geq 0$ and let $a < b$ be adjacent real simple zeros of $H_{t_*}$ with $g_* = b - a$; let $0 < r < g_*/2$ and let $\Omega$ be a conjugation invariant Jordan domain containing the closed discs $D_a$, $D_b$ of radius $r$ about $a$, $b$ and the segment between them, with no other zero of $H_{t_*}$ in $\overline{\Omega}$. Let $\Gamma = \partial D_a \cup \partial D_b \cup \partial\Omega$, $Y = \max_\Gamma |\operatorname{Im} z|$, $m = \min_\Gamma |H_{t_*}|$ and $\psi_Y(\sigma) = \int_0^\infty \Phi_{\mathrm{dBN}}(u) e^{t_* u^2 + Yu}(1 - e^{-\sigma u^2})\, du$.

Proposition 8.4.27 (Newton). *If $\psi_Y(\sigma) < m$, then for $0 \leq \tau \leq \sigma$ the function $H_{t_*-\tau}$ has exactly one zero in each of $D_a$, $D_b$, both real and simple, and no other zero in $\Omega$; so the gap satisfies $g \geq g_* - 2r$ and, by (8.4.2), $4\int \mathrm{Def} \leq g_*^2 - (g_* - 2r)^2$ over the interval.*

*Proof.* Since $|\cos(zu)| \leq e^{|\operatorname{Im} z| u}$, on $\Gamma$ we have $|H_{t_*-\tau} - H_{t_*}| \leq \psi_Y(\tau) \leq \psi_Y(\sigma) < m \leq |H_{t_*}|$. By Rouché's theorem $H_{t_*-\tau}$ has as many zeros in $D_a$, $D_b$ and $\Omega$ as $H_{t_*}$, namely 1, 1 and 2. A single zero in a disc centred on the real axis is real by conjugation symmetry and is simple, and $\Omega$ has no further zero. □

This is the mechanism of (B2) in backward time, with a rectangle replaced by a pair of discs. The second result shows what it cannot do. The decay of $H_t$ along the real axis is that of $\Xi$: for $0 < \delta < \pi/8$, $\Phi_{\mathrm{dBN}}$ is analytic in $|\operatorname{Im} u| \leq \delta$, where $|\exp(-\pi n^2 e^{4(u+i\eta)})| = \exp(-\pi n^2 e^{4u} \cos 4\eta)$ and, $\Phi_{\mathrm{dBN}}$ being even, $\Phi_{\mathrm{dBN}}(u+i\eta) = \Phi_{\mathrm{dBN}}(-u - i\eta)$ decays as fast as $u \to -\infty$, and shifting the contour $u \mapsto u + i\delta$ in $H_t(z) = \int_{\mathbb{R}} \nu_t(u) e^{izu}\, du$ gives, for $\operatorname{Re} z = x > 0$ and $|\operatorname{Im} z| \leq 1$,

$$|H_t(z)| \leq C(\delta, t)\, e^{-\delta x}, \qquad C(\delta, t) = \tfrac{1}{2} \int_{\mathbb{R}} |\Phi_{\mathrm{dBN}}(u + i\delta)|\, e^{t(u^2 - \delta^2) + |u|}\, du < \infty.$$

On the other hand $\psi_Y(\sigma) \geq \psi_0(\sigma)$ for every $Y \geq 0$, and $\psi_0(\sigma) > 0$ does not depend on the height.

Proposition 8.4.28 (Newton, with the mechanism corrected). *Fix $t_* \geq 0$, $\sigma > 0$ and $0 < \delta < \pi/8$. For every real zero $a \geq 1$ of $H_{t_*}$ and every contour $\Gamma$ containing a point $a + i\eta$ with $0 < \eta \leq 1$, we have $\min_\Gamma |H_{t_*}| \leq C(\delta, t_*) e^{-\delta a} < \psi_0(\sigma) \leq \psi_Y(\sigma)$ as soon as $a > \delta^{-1} \log\bigl(C(\delta, t_*)/\psi_0(\sigma)\bigr)$. So no criterion of the form $\psi_Y(\sigma) < m$ certifies a fixed step $\sigma > 0$ at every height, and the step it can certify at height $a$ is $O(e^{-\delta a})$.*

*Proof.* $|H_{t_*}(a + i\eta)| \leq C(\delta, t_*) e^{-\delta a}$ by the display, and $\psi_Y \geq \psi_0 > 0$. For $\sigma \leq 1$, $1 - e^{-x} \geq x/(1+x)$ gives $\psi_0(\sigma) \geq \sigma \int_0^\infty \Phi_{\mathrm{dBN}}(u) e^{t_* u^2} u^2 (1+u^2)^{-1}\, du$, so a certified step is $O(e^{-\delta a})$. □

Newton proved this for pairs with gaps tending to 0, but the obstruction is the decay $e^{-\pi x/8}$ of $H_t$ and not the gaps: it holds at every zero. It concerns bounds through the absolute value of the integrand $\Phi_{\mathrm{dBN}}e^{tu^2}(1-e^{-\sigma u^2})\cos(zu)$; bounds on $H_{t_*-\tau}-H_{t_*}$ that use $H$ itself are not excluded. Jensen's inequality for the convex map $x\mapsto e^{-\sigma x}$ gives $\psi_Y(\sigma)\le A(1-e^{-\sigma M/A})\le 2A\sigma M/(2A+\sigma M)$, with $A$ and $M$ the mass and first moment of the image of $\Phi_{\mathrm{dBN}}(u)e^{t_*u^2+Yu}\,du$ under $x=u^2$ (the second step is $2\operatorname{artanh}(x/2)\ge x$); it improves the linear bound $\sigma M$ only when $\sigma M$ is comparable to $A$, and it bounds backward steps only: the forward profile $\int(e^{\tau x}-1)\,\mu(dx)$ of (B2) is convex in $x$, and Jensen then gives a lower bound.

The practical consequence concerns (B2). On $\partial Q_i$ the minimum of $|H_{t_i}|$ is at most $C(\delta,t_i)e^{-\delta v_i}$, with $v_i>X$, while an absolute value bound for $|H_t-H_{t_i}|$ is at least $\psi_0(t-t_i)$, of order $t-t_i$ by $1-e^{-x}\ge x/(1+x)$. A step certified that way must therefore be $O(e^{-\delta X})$ for every $\delta<\pi/8$, about $10^{-3400}$ at $X=2\times10^4$, and (B2) cannot be checked by it at an appreciable height. The perturbation has to be bounded through $H$: by Taylor's formula in $t$ with $\partial_t^k H_t=(-1)^k H_t^{(2k)}$ and Cauchy estimates from $|H_t|$ on a larger contour at the same height, in ball arithmetic, or by a Riemann–Siegel type approximation with relative error, as in the Polymath project [456].

## 8.5 Beyond zeta: the Dedekind eta kernel, $L(s,\chi_{12})$ and Ramanujan's $\Delta$

The Pólya route uses little that is special to $\zeta$. The zero density identity of Section 3.4 needs a completed function that is positive on the real axis and zero free on $\operatorname{Re}s=1$, and the sine squared decomposition (3.1.1) needs an Euler product. The completed $L$ functions of the real character $\chi_{12}$ modulo 12 and of Ramanujan's cusp form $\Delta$ have these properties, and their kernels, built from the Dedekind eta function, are positive by the product formula. From the eta kernel alone, the zero density identity and a count of sign changes place every zero of $L(s,\chi_{12})$ below height 100 on the critical line, which is the generalised Riemann hypothesis (GRH) for $\chi_{12}$ on that window. The first author's note on the Thorin measures of the Dedekind eta function asks what the eta function says about GRH for $\chi_{12}$ [427, Theorem 12]. Section 11.3 returns to $L(s,\chi_{12})$ at the basepoint one.

### 8.5.1 The eta kernel and GRH for $L(s,\chi_{12})$ below height 100

The character $\chi_{12}$ is the even primitive character modulo 12: $\chi_{12}(n)=1$ for $n\equiv\pm1$, $-1$ for $n\equiv\pm5 \pmod{12}$, and 0 otherwise. Euler's pentagonal number theorem gives

$$\eta(ix)=e^{-\pi x/12}\prod_{n\ge1}\bigl(1-e^{-2\pi nx}\bigr)=\sum_{n\ge1}\chi_{12}(n)\,e^{-\pi n^2x/12},\qquad x>0,$$

so the theta series of $\chi_{12}$ is the Dedekind eta function on the imaginary axis. The completed $L$ function

$$\xi(s,\chi_{12}) = \Big(\frac{12}{\pi}\Big)^{s/2}\Gamma\Big(\frac{s}{2}\Big)L(s,\chi_{12}) = \int_0^\infty \eta(ix)\,x^{s/2}\,\frac{dx}{x}$$

satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ because $\eta(i/x) = \sqrt{x}\,\eta(ix)$, the case $z = ix$ of $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$. With $x = e^{2u}$,

$$\xi\Big(\frac12 + s,\chi_{12}\Big) = \int_{\mathbb{R}} K(u)\,e^{su}\,du, \qquad K(u) = 2e^{u/2}\eta\big(ie^{2u}\big),$$

and the modular transformation makes $K$ even. The product formula makes $\eta(ix) > 0$, so $K$ is positive and $K/\xi(\frac12,\chi_{12})$ is a probability density, the analogue of the Pólya density: if $Y$ has this density, then $\mathbb{E}e^{sY} = \xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ for every complex $s$. Since $\eta(ie^{2u})$ is analytic for $|\operatorname{Im} 2u| < \pi/2$, $K$ is analytic in the same strip $|\operatorname{Im} u| < \pi/4$ as $\Phi$, and like $\Phi$ it decays doubly exponentially, so the trapezoid analysis of Section 2.1 applies unchanged.

The construction extends to every real primitive character $\chi$ modulo $m$, whose root number is always one. For even $\chi$ the kernel is $2e^{u/2}\sum_{n\ge1}\chi(n)e^{-\pi n^2e^{2u}/m}$; for odd $\chi$, whose completed function is $(m/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi)$, it is $2e^{3u/2}\sum_{n\ge1} n\chi(n)e^{-\pi n^2e^{2u}/m}$. Both kernels are even and analytic in $|\operatorname{Im} u| < \pi/4$, but their positivity has to be checked case by case.

The two polynomial evaluator of Section 2.1, run with the eta kernel, agrees with `mpmath`'s Dirichlet $L$ function [376] to sixteen digits. Sign changes of $\xi(\frac12 + i\theta,\chi_{12})$ on a grid of step 0.02, refined by Newton's method, give 67 zeros below height 100, the first of them at

$$3.8046276,\ 6.6922233,\ 8.8905930,\ 11.1883927,\ 12.9661788,\ 15.1814809,\ \ldots.$$

The Thorin density at resolution $\varepsilon$, $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12+\varepsilon+i\theta,\chi_{12})$ as in (1.2.7), is again a sum of Poisson kernels over the zeros, by the Hadamard product (Proposition 1.2.3).

The zero density identity, Proposition 3.4.2, holds for $\xi(s,\chi_{12})$ with the same proof, since $\xi(s,\chi_{12})$ is entire, satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$, is positive on the real axis because $K$ is, and has no zeros with $\operatorname{Re} s \ge 1$. Its hypotheses are also the same. In it $N_{\chi_{12}}(T)$ counts the zeros with $0 < \operatorname{Im}\rho \le T$, and $\nu(\sigma,T)$ those among them with $\operatorname{Re}\rho > \sigma$; $M_\varepsilon(T) = \int_0^T \rho_\varepsilon(\theta)\,d\theta$ is the Thorin mass up to height $T$, and $\tau_\varepsilon(T)$ is the change of argument of $\xi(s,\chi_{12})$ from $\frac12+\varepsilon+iT$ to $\frac12+iT$, divided by $\pi$. Table 8.15 evaluates the identity at $T = 100$.

The row $\varepsilon = \frac12$ gives $N_{\chi_{12}}(100) = 67$, and with the 67 sign changes every zero below height 100 is simple and lies on the critical line. The other rows give $\nu(\frac12+\varepsilon,100) = 0$ without using the sign changes. This is GRH for $\chi_{12}$ below height 100, computed, not certified: the argument is followed on a grid as in Section 3.4,

| $\varepsilon$ | $M_\varepsilon(100)$ | $\tau_\varepsilon(100)$ | $M_\varepsilon+\tau_\varepsilon = N-2\nu$ |
|---|---|---|---|
| 0.5 | 67.604974186 | −0.604974186 | 67.00000000000003 |
| 0.25 | 67.476432836 | −0.476432836 | 67.00000000010355 |
| 0.1 | 67.308171973 | −0.308171973 | 67.00000000014663 |
| 0.05 | 67.185437421 | −0.185437421 | 67.00000000009639 |
| 0.02 | 67.080023774 | −0.080023774 | 67.00000000005207 |

Table 8.15: The zero density identity for $L(s,\chi_{12})$ at height 100, computed from the eta kernel. The last column is $N_{\chi_{12}}(100)-2\nu(\frac12+\varepsilon,100)$; the row $\varepsilon=\frac12$ gives $N_{\chi_{12}}(100)$.

and its increments are summed in double precision. The deviations from 67, at most $1.5\times10^{-10}$, are far below the one half that fixes the integer. GRH for small moduli has long been verified to far greater heights by other methods, by Rumely [467] and later by Platt [417].

### 8.5.2 Ramanujan's $\Delta$ and the cost of the three evaluators

The same construction works for Ramanujan's cusp form $\Delta(z)=\eta(z)^{24}=\sum_{n\ge1}\tau(n)q^n$, $q=e^{2\pi iz}$, of weight 12 and level 1. Its $L$ function, normalised so that the critical line is $\operatorname{Re}s=\frac12$, is $L(s,\Delta)=\sum_{n\ge1}\tau(n)n^{-s-11/2}$, and the completed function $\xi(s,\Delta)=(2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ satisfies $\xi(s,\Delta)=\xi(1-s,\Delta)$. The Mellin transform of $\Delta(iy)$ gives the Pólya type representation

$$\xi\Big(\frac12+s,\Delta\Big)=\int_{\mathbb{R}}K_\Delta(u)\,e^{su}\,du,$$
$$K_\Delta(u)=e^{6u}\,\Delta(ie^u)=e^{6u}e^{-2\pi e^u}\prod_{n\ge1}\big(1-e^{-2\pi n e^u}\big)^{24}.$$

The kernel is even because $\Delta(i/y)=y^{12}\Delta(iy)$, positive by the product formula, and analytic in the strip $|\operatorname{Im}u|<\pi/2$.

We computed $\tau(n)$ for $n\le3000$, the range needed for the prime weights of Section 8.5.3, and evaluated the kernel by its $q$ expansion. At 60 digits the trapezoid evaluator finds the first zeros on the critical line at

$$9.222379,\ 13.907550,\ 17.442777,\ 19.656513,\ 22.336104,\ \ldots,$$

in agreement with the known values [348, modular form 1.12.a.a], and 20 sign changes below height 50, the number of zeros there.

The cost of the three evaluators, for $\Delta$, $\zeta$ and $L(s,\chi_{12})$ at 60 digits (settings, digits lost per unit height and time per point), is compared in `code/README.md`.

### 8.5.3 The sine squared lemma without a pole

The eta based $L$ functions are entire, so two consequences of the pole of $\zeta$ disappear: the negative pole piece of the sine squared decomposition, and the factor that Chapters 6 and 7 need to remove the pole, which by Proposition 7.1.13 alone prevents the normalised reciprocal zeta function from being a GGC. A floor in the sense of Theorem 6.2.15 is still needed for $\chi_{12}$, since its tilt, the factor $e^{-c\sqrt{u}}$ in the variable $u$ of the clock (written $e^{-c\sqrt{s}}$ in Section 6.2), removes the negative part of the Thorin density of the gamma factor, and the gamma factor $(12/\pi)^{s/2}\Gamma(s/2)$ of $\xi(s,\chi_{12})$ is that of $\xi$ times $12^{s/2}$: Theorem 6.2.12 holds for it with $c = 2.6861$ replaced by $c - \frac{1}{2}\log 12 = 1.4436$. The positivity of such a floor would need an analogue of Theorem 7.1.1.

For $f = \chi_{12}$ or $\Delta$ write $\xi(s,f) = L_\infty(s,f)L(s,f)$, with $L_\infty$ the gamma factor, and $-(L'/L)(s,f) = \sum_{n\geq 2}\Lambda_f(n)n^{-s}$ for $\operatorname{Re} s > 1$. At a basepoint $\alpha > 1$ the sine squared lemma gives, as (3.1.1) does for $\zeta$,

$$v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,f) = \underbrace{\frac{\xi'}{\xi}(\alpha,f)}_{\text{drift}} + \underbrace{\operatorname{Re}\frac{L'_\infty}{L_\infty}(\alpha+i\theta,f) - \frac{L'_\infty}{L_\infty}(\alpha,f)}_{\text{archimedean } \geq 0} + \underbrace{\sum_{n\geq 2}\Lambda_f(n)\,n^{-\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big)}_{\text{primes}},$$

where $v_\alpha(\theta)$ is $\pi$ times the Thorin density at resolution $\alpha - \frac{1}{2}$, and the archimedean piece is nonnegative because the digamma function $\psi = \Gamma'/\Gamma$ satisfies $\operatorname{Re}\psi(x+iy) \geq \psi(x)$ for $x > 0$. For $\zeta$, whose gamma factor $\pi^{-s/2}\Gamma(1+\frac{s}{2})$ absorbs the factor $s$ of $\xi$, the remaining factor $s-1$ adds the pole piece $\operatorname{Re}\frac{1}{s-1} - \frac{1}{\alpha-1}$, which is $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ at $s = \alpha + i\theta$, and the identity becomes (3.1.1).

The prime weights are $\Lambda(n) = \log p$ at $n = p^k$ for $\zeta$, all positive; $\Lambda_{\chi_{12}}(p^k) = \chi_{12}(p)^k \log p$ for $\chi_{12}$, of both signs and zero at $p = 2$ and 3; and $\Lambda_\Delta(p^k) = (\alpha_p^k + \bar\alpha_p^k)\log p$ for $\Delta$, where the Satake parameters $\alpha_p$ and $\bar\alpha_p$ are the roots of $X^2 - \tau(p)p^{-11/2}X + 1$. Deligne's theorem $|\tau(p)| \leq 2p^{11/2}$ [114] makes them complex conjugates of modulus one, so the weights of $\Delta$ lie in $[-2\log p, 2\log p]$ and take both signs.

We computed the three decompositions at $\alpha = 2$ for $0 \leq \theta \leq 40$, with prime powers up to 3000, and compared their sum with the left side computed from the kernel. They agree to $5\times 10^{-4}$ for $\zeta$ and $9\times 10^{-6}$ for $\Delta$, the difference being the truncation of the prime sum. For $\chi_{12}$ the largest difference, $2.5\times 10^{-5}$ at $\theta = 40$, is an error of the kernel evaluator, whose truncation at $|u| \leq 2.5$ is too short at that height. Of the 466 prime power weights up to 3000, none is negative for $\zeta$, 222 are negative for $\chi_{12}$ (whose 18 weights at powers of 2 and 3 vanish) and 230 for $\Delta$.

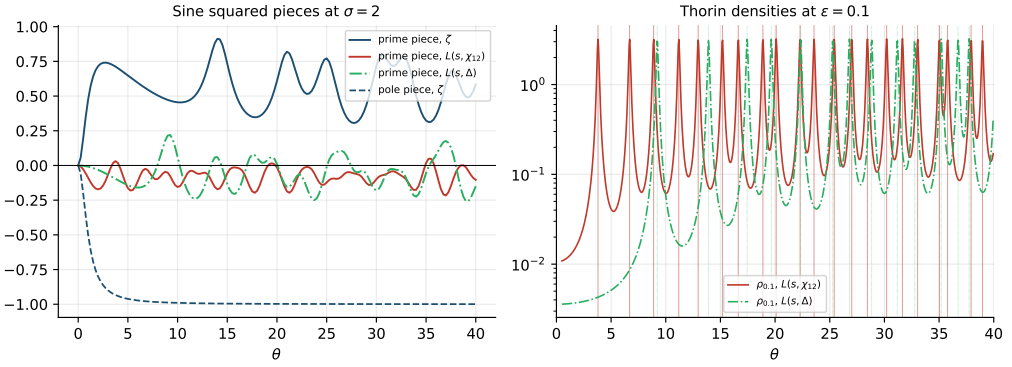



Figure 8.4: Left: the prime pieces of the sine squared decomposition at the basepoint $\alpha = 2$, written $\sigma$ in the plot, for $\zeta$, $L(s, \chi_{12})$ and $L(s, \Delta)$, and the pole piece of $\zeta$. For $\zeta$ the prime piece is positive and the pole piece negative; for the two eta based functions there is no pole piece and the prime piece takes both signs. Right: the Thorin densities $\rho_{0.1}$ of $L(s, \chi_{12})$ and $L(s, \Delta)$ on a logarithmic scale, peaking at their zeros (vertical lines).

On $0 \le \theta \le 40$ the prime piece of $L(s, \chi_{12})$ ranges over $[-0.218, 0.053]$, and that of $L(s, \Delta)$ over $[-0.261, 0.222]$ (Figure 8.4, left). Their left sides are positive, with minima at $\theta = 0$ equal to the drifts, 0.466 and 0.167, so the positivity comes from the drift and the archimedean piece, not from the primes.

For each of the three functions its Riemann hypothesis, RH for $\zeta$ and GRH for the other two, is the statement that $v_\alpha$ stays positive for every $\alpha > \frac{1}{2}$; this is Lagarias's criterion [320], whose proof from the Hadamard product applies unchanged to $\xi(s, \chi_{12})$ and $\xi(s, \Delta)$. Below $\alpha = 1$ the prime piece no longer converges absolutely. For $L(s, \chi_{12})$, Section 11.3.2 shows that the extension of the decomposition down to the centre is GRH for $\chi_{12}$. At resolution 0.1 the Thorin densities of $L(s, \chi_{12})$ and $L(s, \Delta)$ have computed minima 0.011 and 0.004 below height 40 (Figure 8.4, right), positive as GRH predicts.

The distances of Section 3.2, with the zeros of $L(s, \chi_{12})$ or $L(s, \Delta)$ as the reference atoms, Theorem 4.3.5 and the flow of Section 8.4 carry over unchanged, the last two because they use only the Hadamard product, positivity on the real line and the decay of the kernel; we have not computed the first two for these functions, and Section 8.5.5 computes the flow for $\Delta$. Theorem 4.3.5 also needs the bound $|\arg z| < \pi/4$ of Lemma 4.3.4 for the zeros $z$ of $\xi(\frac{1}{2}+iz, f)$ with $\operatorname{Re} z > 0$, and it holds here too: the zeros lie in the open critical strip, so $|\operatorname{Im} z| < \frac{1}{2}$, while the lowest ordinates, 3.80 and 9.22, exceed $\frac{1}{2}$. The arithmetic side, however, offers no termwise positivity, even far to the right of the critical line, so the domination of the pole by positive primes that Sections 3.1 and 3.5 formulate for $\zeta$ has no direct analogue. Far enough to the right the absolute values do suffice: the drift and the

archimedean piece outweigh the prime piece bounded by the absolute values of its weights for $\alpha > 1.8349$ for $L(s, \chi_{12})$ and for $\alpha > 2.2823$ for the Dirichlet beta function $L(s, \chi_4)$, and by Kronecker's theorem no better bound from the sizes of the weights holds at all large heights (Propositions 8.5.26 and 8.5.27). So neither eta based function is theoretically easier than $\zeta$, although $L(s, \chi_{12})$ is the cheapest of the three per zero.

### 8.5.4 Maxima of exponentials

The product formula has a probabilistic reading, which the first author's note [428] develops. The factor $1-e^{-2\pi ny}$ is the distribution function of an exponential variable with rate $2\pi n$, so

$$F(y) = \prod_{n\ge1} \bigl(1-e^{-2\pi ny}\bigr)^{24} = e^{2\pi y}\Delta(iy), \qquad F_1(y) = \prod_{n\ge1} \bigl(1-e^{-2\pi ny}\bigr) = e^{\pi y/12}\eta(iy)$$

are the distribution functions of the maxima $M^*$ and $M_1$ of independent exponential variables.

#### *8.5.4.1 The max representation*

Lemma 8.5.1. *Let $E_{n,j}$, $n \ge 1$, $1 \le j \le 24$, be independent, with $E_{n,j}$ exponential of rate $2\pi n$, and put $M^* = \sup_{n,j} E_{n,j}$ and $M^{(j)} = \sup_n E_{n,j}$. Almost surely these suprema are attained and finite, $\Pr(M^* \le y) = F(y)$ for $y > 0$, and each $M^{(j)}$ has the distribution function $F_1$. So $M^* = \max_{j\le24} M^{(j)}$ is the maximum of 24 independent copies of $M_1$.*

*Proof.* For $y > 0$, $\sum_{n,j} \Pr(E_{n,j} > y) = 24 \sum_n e^{-2\pi ny} < \infty$, so by the Borel Cantelli lemma only finitely many $E_{n,j}$ exceed $y$, almost surely. Taking $y$ rational and below $E_{1,1}$, the supremum is the largest of finitely many values. By independence $\Pr(M^* \le y) = \lim_N \prod_{n\le N}(1 - e^{-2\pi ny})^{24} = F(y)$, and in the same way each $M^{(j)}$ has the distribution function $F_1$; the $M^{(j)}$ are independent because they are built from disjoint sets of the $E_{n,j}$. □

The modular relations $\Delta(i/y) = y^{12}\Delta(iy)$ and $\eta(i/y) = \sqrt{y}\,\eta(iy)$ become self dualities of the two laws,

$$F(y) = y^{-12}e^{2\pi y-2\pi/y}F(1/y), \qquad F_1(y) = y^{-1/2}e^{\pi y/12-\pi/(12y)}F_1(1/y), \quad (8.5.1)$$

so the lower tails are thin: $\Pr(M^* \le y) \sim y^{-12}e^{-2\pi/y}$ and $\Pr(M_1 \le y) \sim y^{-1/2}e^{-\pi/(12y)}$ as $y \downarrow 0$. Since $F(1/y) \le 1$, the first relation also gives $F(y) \le y^{-12}e^{2\pi y-2\pi/y}$ for every $y > 0$, a bound that increases with $y$, because its logarithmic derivative is $(2\pi y^2 - 12y + 2\pi)/y^2$ and $144 < 16\pi^2$.

The density of $M^*$ is Ramanujan's logarithmic derivative. Expanding $1/(1 - e^{-2\pi ny})$ gives $(\log F)'(y) = 24\sum_n 2\pi n\, e^{-2\pi ny}/(1 - e^{-2\pi ny}) =$

$48\pi\sum_{m\ge1}\sigma(m)e^{-2\pi my}$, with $\sigma(m)$ the sum of the divisors of $m$, so that, with Eisenstein's series $E_2(z)=1-24\sum_{m\ge1}\sigma(m)e^{2\pi imz}$,

$$f^*(y)=F(y)\,(\log F)'(y)=2\pi F(y)\big(1-E_2(iy)\big). \tag{8.5.2}$$

Taking the logarithmic derivative of (8.5.1) gives the functional equation of the density,

$$f^*(1/y)=y^{12}e^{2\pi/y-2\pi y}\Big(\big(2\pi y^2-12y+2\pi\big)F(y)-y^2f^*(y)\Big),$$

which in terms of $E_2$ is its quasi modularity, $E_2(i/y)=-y^2E_2(iy)+6y/\pi$ (`maxexp_checks.py`).

THEOREM 8.5.2. *For every complex $s$,*

$$\xi(s,\Delta)=(2\pi)^{-s-\frac{11}{2}}\,\mathbb{E}\,\Gamma\big(s+\tfrac{11}{2},2\pi M^*\big),\qquad L(s,\Delta)=\mathbb{E}\,Q\big(s+\tfrac{11}{2},2\pi M^*\big), \tag{8.5.3}$$

*with $\Gamma(a,x)=\int_x^\infty e^{-v}v^{a-1}dv$ and $Q(a,x)=\Gamma(a,x)/\Gamma(a)$, and both expectations converge absolutely. In particular $L(s,\Delta)=0$ at the points $s=-\frac{11}{2}-k$, $k=0,1,2,\dots$, where $Q(-k,x)=0$.*

*Proof.* Put $a=s+\frac{11}{2}$, so that $\xi(s,\Delta)=\int_0^\infty\Delta(iy)y^{a-1}dy$. Inserting $\Delta(iy)=e^{-2\pi y}\Pr(M^*\le y)=e^{-2\pi y}\mathbb{E}\,\mathbf{1}\{M^*\le y\}$ and integrating in $y$ first gives $\mathbb{E}\int_{M^*}^\infty e^{-2\pi y}y^{a-1}dy=(2\pi)^{-a}\mathbb{E}\,\Gamma(a,2\pi M^*)$. Fubini's theorem applies: with $\sigma=\operatorname{Re}a$, Tonelli's theorem gives $\mathbb{E}\int_{M^*}^\infty e^{-2\pi y}y^{\sigma-1}dy=\int_0^\infty\Delta(iy)y^{\sigma-1}dy$, which is finite for every real $\sigma$, since $\Delta(iy)$ decays exponentially as $y\to\infty$ and like $y^{-12}e^{-2\pi/y}$ as $y\downarrow0$. The same computation gives $\mathbb{E}|\Gamma(a,2\pi M^*)|\le\mathbb{E}\,\Gamma(\sigma,2\pi M^*)<\infty$. Dividing by $(2\pi)^{-a}\Gamma(a)$, with $1/\Gamma$ entire, gives the second identity, in which $Q(a,x)$, for fixed $x>0$, is the entire function $\Gamma(a,x)/\Gamma(a)$ of $a$. At $a=-k$ it vanishes, since $\Gamma(-k,x)$ is finite and $1/\Gamma(-k)=0$. □

COROLLARY 8.5.3 (Gamma mixture and mixed Poisson forms). *Let $G_a$ be a gamma variable of shape $a>0$ and scale one, and $N$ a Poisson process of unit rate, both independent of $M^*$. For real $s>-\frac{11}{2}$, with $a=s+\frac{11}{2}$,*

$$L(s,\Delta)=\Pr\big(G_a>2\pi M^*\big)=\mathbb{E}F\Big(\frac{G_a}{2\pi}\Big)=\mathbb{E}\prod_{n\ge1}\big(1-e^{-nG_a}\big)^{24},$$

*and for every integer $k\ge1$*

$$L\big(k-\tfrac{11}{2},\Delta\big)=\Pr\big(N(2\pi M^*)\le k-1\big)=\mathbb{E}\Big[e^{-2\pi M^*}\sum_{j<k}\frac{(2\pi M^*)^j}{j!}\Big].$$

*Proof.* For real $a>0$, $Q(a,x)=\Pr(G_a>x)$, which gives the first form; conditioning on $G_a$ instead gives $\Pr(M^*<G_a/2\pi)=\mathbb{E}F(G_a/2\pi)$, since the law of $M^*$ has no atoms. For an integer $k$ the $k$th point of $N$ has the law of $G_k$, so $\Pr(G_k>x)=\Pr(N(x)\le k-1)$. □

So $L(-\frac{9}{2},\Delta) = \mathbb{E}e^{-2\pi M^*} = 0.03744128126851554174$ is a value of the Laplace transform of $M^*$, and the central value $L(\frac{1}{2},\Delta) = 0.79212283864603056936$ is the probability that a Poisson clock of rate $2\pi$ rings at most five times before the time $M^*$. Write $P_k = \Pr(N(2\pi M^*) \le k-1) = L(k-\frac{11}{2},\Delta)$. The functional equation $\xi(s,\Delta) = \xi(1-s,\Delta)$ at $s = k-\frac{11}{2}$ becomes the family of identities

$$(2\pi)^{-k}(k-1)!\,P_k = (2\pi)^{k-12}(11-k)!\,P_{12-k}, \qquad 1 \le k \le 11, \tag{8.5.4}$$

both sides being $\xi(k-\frac{11}{2},\Delta)$.

Remark 8.5.4. The identities (8.5.4) follow from the self duality (8.5.1), since $(2\pi)^{-k}(k-1)!\,P_k = \int_0^\infty e^{-2\pi y}F(y)y^{k-1}dy$ and the substitution $y \mapsto 1/y$ turns this integral into the one with $12-k$ in place of $k$. They are five relations, since $k$ and $12-k$ give the same one and $k=6$ gives none, and they do not imply the self duality. Let $g$ be smooth and not identically zero, with compact support in $(0,1)$, such that $g(\log y)$ is orthogonal on $(1,e)$ to the five functions $e^{-2\pi y}F(y)(y^{k-1} - y^{11-k})$, $k \le 5$. For small $\varepsilon$, $\tilde F(y) = F(y)\big(1+\varepsilon g(\log y)\big)$ is again a distribution function, because $F'$ has a positive lower bound and $F$ is below one on $[1,e]$, and it satisfies (8.5.4), because the perturbation changes the difference of the two sides of the $k$th identity by $\varepsilon$ times the integral of $g(\log y)$ against the $k$th function. It is not self dual: on $(1,e)$ self duality would give $\tilde F(y) = y^{-12}e^{2\pi y - 2\pi/y}\tilde F(1/y) = F(y)$, since $g(\log(1/y)) = 0$. With $g$ even and supported in $(-1,1)$ instead, the same construction gives a self dual distribution function other than $F$, so self duality does not determine the law of $M^*$ either. What does determine it is self duality together with the form of $F$ as a power series in $e^{-2\pi y}$. If $F(y) = \sum_{n\ge0} c_n e^{-2\pi n y}$ for $y>0$, with $c_0 = 1$, and $F$ satisfies (8.5.1), then $f(z) = \sum_{n\ge0} c_n e^{2\pi i(n+1)z}$ is holomorphic on the upper half plane, satisfies $f(z+1) = f(z)$, and satisfies $f(-1/z) = z^{12}f(z)$ on the imaginary axis, hence everywhere by analytic continuation. So $f$ is a cusp form of weight 12 for $\mathrm{SL}_2(\mathbb{Z})$, a multiple of $\Delta$ because these cusp forms make up a space of dimension one [288, Chapter 14], and $c_0 = 1$ gives $f = \Delta$. This argument is the simplest case of Hecke's converse theorem [180]. The distribution function of a maximum of independent exponential variables whose rates are integer multiples of $2\pi$ is a power series of this kind, so $M^*$ is the only such maximum with the self duality (8.5.1).

Cutting the integral at $y=1$ and folding $(0,1)$ onto $(1,\infty)$ by modularity gives in the same way the symmetric form

$$\xi(s,\Delta) = A(s) + A(1-s), \qquad A(s) = \int_1^\infty \Delta(iy)\,y^{s+\frac{9}{2}}dy, \tag{8.5.5}$$

$$A(s) = (2\pi)^{-s-\frac{11}{2}}\,\mathbb{E}\,\Gamma\big(s+\tfrac{11}{2}, 2\pi\max(M^*,1)\big) = \sum_{n\ge1}\tau(n)\,\frac{\Gamma(s+\frac{11}{2},2\pi n)}{(2\pi n)^{s+11/2}},$$

where the last form, Hecke's, comes from expanding $\Delta(iy) = \sum_n \tau(n)e^{-2\pi ny}$ instead. Hecke's form is the analogue for $\Delta$ of Riemann's second proof of the functional equation [127, Chapter 1], which cuts the Mellin integral of the theta function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ of Section 1.2 at $x = 1$ in the same way.

PROPOSITION 8.5.5 (Riemann). *For every complex $s \neq 0, 1$,*

$$\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big)\zeta(s) = -\frac{1}{s} - \frac{1}{1-s} + \sum_{n\geq 1}\Big[\frac{\Gamma(\frac{s}{2}, \pi n^2)}{(\pi n^2)^{s/2}} + \frac{\Gamma(\frac{1-s}{2}, \pi n^2)}{(\pi n^2)^{(1-s)/2}}\Big].$$

*Proof.* This is Riemann's second proof of the functional equation, cited above: split the Mellin integral of $\omega$ at $x = 1$, apply the theta transformation on $(0, 1)$, and integrate term by term on $(1, \infty)$. □

The three formulas are incomplete gamma continuations of Mellin integrals, but they randomise different things. In Riemann's and Hecke's the threshold is the fixed point $x = 1$ or $y = 1$, and the rates $\pi n^2$ or $2\pi n$ carry weights: one each for $\zeta$, at the price of the pole terms, and the signed coefficients $\tau(n)$ for $\Delta$. In (8.5.3) there is a single rate $2\pi$, coming from the factor $q = e^{-2\pi y}$ of $\Delta$, and the threshold $M^*$ is random with a positive law; the symmetric form (8.5.5) is Riemann's formula with the sum over $n$ replaced by an expectation over $\max(M^*, 1)$, and with no pole terms.

At $s = \frac{1}{2}$, $\frac{1}{2} + 5i$ and 3 the max representation, the symmetric form and Hecke's formula agree to 25 digits. On the critical line the symmetric form gives $\xi(\frac{1}{2} + i\theta, \Delta) = 2\,\mathrm{Re}\,A(\frac{1}{2} + i\theta)$. It also suggests a Hermite Biehler test, the route that Lagarias studied for $\xi$ [321]: if $|A(s)| > |A(1-s)|$ held throughout $\mathrm{Re}\,s > \frac{1}{2}$, there could be no zeros there. The test fails. On $\mathrm{Re}\,s = 0.6$ the two halves have equal modulus at $\theta = 10.26$, 14.67, 18.04, 20.36 and 22.99, and the dominant half alternates between these ordinates, which separate the zeros 9.22, 13.91, 17.44, 19.66 and 22.34 (`maxlaw.py`, `maxexp_hb.py`). So the cut at $y = 1$ gives no Hermite Biehler function.

The cut can be moved. For every $c > 0$ the same folding gives $\xi(s, \Delta) = A_c(s) + A_{1/c}(1-s)$ with $A_c(s) = \int_c^\infty \Delta(iy)y^{s+9/2}dy$, and $c$ may be chosen separately on each horizontal ray. The freedom is in any case illusory. As $c \to 0$, $A_{1/c}(1-s) \to 0$ and $A_c(s) \to \xi(s, \Delta)$ uniformly on compact sets, so domination for small $c$ only says that $\xi(s, \Delta)$ does not vanish there, and the certificate turns into an evaluation of $\xi$; and at a zero $\frac{1}{2} + i\gamma$ the two halves have equal modulus for every $c$, since their sum vanishes. A Hermite Biehler function that could prove GRH for $\Delta$ would have to come from the structure of the law of $M^*$, not from tuning a cut. The representation is exact and has no signs, but like the kernel it restates the problem without locating the zeros.

*8.5.4.2 Other theta products*

The construction needs a modular form whose expansion on the imaginary axis is a power of $q$ times a product of factors $1-q^a$, $a>0$, with nonnegative exponents, so that the product is a distribution function; eta products with nonnegative exponents qualify.

For $\chi_{12}$ the Mellin integral of Section 8.5.1, with $\eta(ix)=e^{-\pi x/12}F_1(x)$, gives in the same way

$$L(s,\chi_{12})=\mathbb{E}\,Q\Big(\frac{s}{2},\frac{\pi}{12}M_1\Big),\qquad \xi(s,\chi_{12})=A_{12}(s)+A_{12}(1-s),$$

with $A_{12}(s)=\int_1^\infty \eta(ix)x^{s/2-1}dx=(\frac{\pi}{12})^{-s/2}\mathbb{E}\,\Gamma(\frac{s}{2},\frac{\pi}{12}\max(M_1,1))$; the second exponent comes from $\eta(i/x)=\sqrt{x}\,\eta(ix)$. At $s=1$, where $Q(\frac12,x)=\operatorname{erfc}\sqrt{x}$, the class number formula reads $\mathbb{E}\operatorname{erfc}\sqrt{\pi M_1/12}=L(1,\chi_{12})=\log(2+\sqrt3)/\sqrt3$; at $s=2$, where $Q(1,x)=e^{-x}$, $L(2,\chi_{12})=\mathbb{E}e^{-\pi M_1/12}=\pi^2/(6\sqrt3)$ is a value of the Laplace transform of $M_1$, which Section 8.5.4.4 computes in closed form; and at the centre $L(\frac12,\chi_{12})=\Pr(12G_{1/4}>\pi M_1)=0.498557002457815436157$. At $s=0.6+5i$ the max representation, both forms of the symmetric form and `mpmath`'s Dirichlet $L$ function agree to 25 digits, as do the three values above with their closed forms (`maxexp_checks.py`, `maxlaw.py`).

For the elliptic curve of conductor 11 the newform is $f(z)=\eta(z)^2\eta(11z)^2=q\prod_{n\ge1}(1-q^n)^2(1-q^{11n})^2=\sum_{n\ge1}a_nq^n$ [348, modular form 11.2.a.a], so $f(iy)=e^{-2\pi y}F_{11}(y)$, where $F_{11}$ is the distribution function of the maximum $M_{11}$ of independent exponential variables, two of each rate $2\pi n$ and two of each rate $22\pi n$. The Fricke relation $f(i/(11y))=11y^2f(iy)$ makes $f(iy)$ decay faster than any power at both ends, and the proof of Theorem 8.5.2 gives, in the arithmetic normalisation $L(s,E)=\sum_n a_nn^{-s}$ with centre $s=1$,

$$L(s,E)=\mathbb{E}\,Q(s,2\pi M_{11})\quad\text{for every complex } s,\qquad L(1,E)=\mathbb{E}e^{-2\pi M_{11}}.$$

The max representation gives $L(1,E)=0.2538418608559106843378$, equal to all digits to the rapidly convergent series $2\sum_n a_nn^{-1}e^{-2\pi n/\sqrt{11}}$ of root number one.

Jacobi's triple product gives $\vartheta_4(x)=\sum_{n\in\mathbb{Z}}(-1)^ne^{-\pi n^2x}=\prod_{m\ge1}(1-e^{-2\pi mx})(1-e^{-\pi(2m-1)x})^2$, a distribution function with no prefactor: that of the maximum $M_4$ of independent exponential variables, one of each rate $2\pi m$ and two of each rate $\pi(2m-1)$. The partial sums of the alternating series lie between $-1$ and 1, so the series can be integrated term by term against $\theta e^{-\theta y}$, and the partial fraction expansion of $z/\sinh z$ gives

$$\mathbb{E}e^{-\theta M_4}=\sum_{n\in\mathbb{Z}}(-1)^n\frac{\theta}{\theta+\pi n^2}=\frac{\sqrt{\pi\theta}}{\sinh\sqrt{\pi\theta}}=\prod_{n\ge1}\Big(1+\frac{\theta}{\pi n^2}\Big)^{-1},$$

so $M_4\overset{d}{=}\sum_{n\ge1}E_n/(\pi n^2)$ with independent standard exponential variables $E_n$: a GGC with Thorin measure $\sum_n\delta_{\pi n^2}$. Since $\Pr(K\le k)=\sum_n(-1)^ne^{-2n^2k^2}=$

$\vartheta_4(2k^2/\pi)$, $\pi M_4 = 2K^2$ has the Kolmogorov law of Table 1.8. With no prefactor the Mellin transform gives moments instead of incomplete gamma functions: $\mathbb{E}M_4^{s/2} = \frac{s}{2}\int_0^\infty y^{s/2-1}(1-\vartheta_4(y))\,dy$ for $\operatorname{Re} s > 0$, and integrating $1-\vartheta_4(y) = 2\sum_{n\ge1}(-1)^{n+1}e^{-\pi n^2 y}$ term by term, for $\operatorname{Re} s > 1$ and then by analytic continuation, gives

$$(1-2^{1-s})\,\zeta(s) = \frac{\mathbb{E}\big[(2K^2)^{s/2}\big]}{2\,\Gamma(1+\frac{s}{2})}$$

for every complex $s$, both sides being entire, since $\vartheta_4(y) = y^{-1/2}\sum_n e^{-\pi(n+\frac12)^2/y}$ makes every moment of $M_4$ finite. This is a case of the dictionary of Biane, Pitman and Yor [49]. At $s=1$ it gives $\mathbb{E}K = \sqrt{\pi/2}\log 2 = 0.868731160636159141830$, which quadrature of $1-\vartheta_4$ reproduces to $2\times10^{-23}$; at $s=3$ and $\frac12+5i$ the two sides agree to 25 digits.

For $\zeta$ itself the construction stops. The theta function $\vartheta_3(x) = 1+2\omega(x) = \prod_{m\ge1}(1-q^{2m})(1+q^{2m-1})^2$, $q=e^{-\pi x}$, has factors larger than one and is not a distribution function, and the transformation $\vartheta_4(1/x) = \sqrt{x}\,\vartheta_2(x)$ leads to $\vartheta_2(x) = \sum_n e^{-\pi(n+\frac12)^2x}$, which is not one either. So the max structure reaches $\zeta$ only through the factor $1-2^{1-s}$, in moment form. It does not reach it through $\zeta_{\mathbb{Q}(\sqrt3)}(s) = \zeta(s)L(s,\chi_{12})$: the eta law represents the factor $L(s,\chi_{12})$ only, and no product of this kind can represent a Dedekind zeta function, since an eta product with nonnegative exponents decays faster than any power of $y$ at both ends of the imaginary axis, so its Mellin transform is entire, as in the proof of Theorem 8.5.2, while $\zeta_{\mathbb{Q}(\sqrt3)}$ has a pole at $s=1$.

### 8.5.4.3 Gamma shapes and infinite divisibility at real points

The Pólya kernel of $\zeta$ and the kernel $K_\Delta$ of $\Delta$ differ in their gamma shapes. On the logarithmic scale the function $x^a e^{-x}$ of $x = ce^{ku}$ is proportional to the density of $(\log G_a - \log c)/k$, a log gamma density of shape $a$, and its Fourier transform $\int_{\mathbb{R}} x^a e^{-x} e^{i\theta u}du = k^{-1}c^{-i\theta/k}\Gamma(a+i\theta/k)$ has no zeros. With $x = 2\pi e^u$, $K_\Delta(u) = e^{6u}\Delta(ie^u) = (2\pi)^{-6}x^6e^{-x}F(e^u)$ is a single gamma kernel of shape 6 times the distribution function of $\log M^*$, and the gamma kernel's own transform is $(2\pi)^{-6-i\theta}\Gamma(6+i\theta)$, so every zero of $\xi(\frac12+i\theta,\Delta)$ is produced by the multiplier $F(e^u)$. The Pólya kernel (1.2.3) is, with $x_n = \pi n^2 e^{2u}$ and the gamma densities $g_k(x) = x^{k-1}e^{-x}/\Gamma(k)$,

$$\Phi(u) = 2e^{u/2}\sum_{n\ge1}\big(2x_n^2 - 3x_n\big)e^{-x_n} = 2e^{u/2}\sum_{n\ge1}\big(4g_3(x_n) - 3g_2(x_n)\big),$$

a signed mixture of Gamma(2) and Gamma(3) densities over the scales $\pi n^2$. As a function of $x = e^{2u}$ the sum is, up to the factor $x/2$, the density $2\sum_n(2\pi^2n^4x - 3\pi n^2)e^{-\pi n^2x}$ of $\pi S_2/2$ in the representation $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Williams and of Biane, Pitman and Yor (Section 8.2.3). On the logarithmic scale, with the

weight $e^{u/2} = (x_n/\pi n^2)^{1/4}$, the two terms have shapes $\frac{9}{4}$ and $\frac{5}{4}$, while the theta function itself contributes $x_n^{1/4}e^{-x_n}$, of shape $\frac{1}{4}$.

REMARK 8.5.6. Let $D = -x\,d/dx$. Integrating by parts, $\int_0^\infty (Df)(x)x^{s/2-1}dx = \frac{s}{2}\int_0^\infty f(x)x^{s/2-1}dx$ whenever the boundary terms vanish, so the operator $2D(D-\frac{1}{2})$ corresponds to the factor $\frac{1}{2}s(s-1)$ of $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$. It annihilates $-\frac{1}{2}$ and $\frac{1}{2}x^{-1/2}$, the two terms by which $\omega(x)$ and $x^{-1/2}\omega(1/x)$ differ, and it maps $e^{-x}$ to $(2x^2-3x)e^{-x}$. Hence $[2D(D-\frac{1}{2})\omega](x)$, which is also $2D(D-\frac{1}{2})$ applied to $x^{-1/2}\omega(1/x)$, decays faster than any power at both ends,

$$\xi(s) = \int_0^\infty \big[2D(D-\tfrac{1}{2})\omega\big](x)\,x^{s/2-1}dx \quad \text{for every } s,$$
$$\Phi(u) = 2e^{u/2}\big[2D(D-\tfrac{1}{2})\omega\big](e^{2u}),$$

the first for $\operatorname{Re} s > 1$ by the integration by parts and then for every $s$ by analytic continuation, and the second by $x = e^{2u}$. So the gamma shapes of the terms $e^{-\pi n^2 x}$ of $\omega$, one in $x$ and $\frac{1}{4}$ on the logarithmic scale, are raised by one and by two, to the Gamma(2) and Gamma(3) of the Pólya kernel, by the operator that removes the poles of $\zeta$. For $\Delta$, whose $L$ function is entire, no such operator is needed and the kernel keeps the single shape 6; the cancellation between the Gamma(2) and Gamma(3) terms of $\Phi$ is replaced by the positive multiplier $F(e^u)$.

At real points the Gamma(2) shape enters through Kristiansen's theorem [316]: every scale mixture of Gamma(2) laws is infinitely divisible. Since $G_a \overset{d}{=} G_2 B$ for $0 < a < 2$, with $B$ a beta variable of parameters $a$ and $2-a$ independent of $G_2$, the scale mixtures of Gamma($a$) laws with $a \le 2$ are scale mixtures of Gamma(2) laws.

PROPOSITION 8.5.7. *For real $a > 0$ let $V_a = G_a/(2\pi M^*)$, with $G_a$ independent of $M^*$. Then $V_a$ is a scale mixture of Gamma($a$) laws, the mixing law being that of $1/(2\pi M^*)$, and $L(a-\frac{11}{2},\Delta) = \Pr(V_a > 1)$. For $0 < a \le 2$, that is $-\frac{11}{2} < s \le -\frac{7}{2}$, $V_a$ is infinitely divisible. In the same way $L(s,\chi_{12}) = \Pr\big(12G_{s/2}/(\pi M_1) > 1\big)$ for real $s > 0$, where $12G_{s/2}/(\pi M_1)$ is infinitely divisible for $s \le 4$, and is a mixture of exponential laws for $s \le 2$, in particular at the centre $s = \frac{1}{2}$.*

*Proof.* The identities are Corollary 8.5.3 and its analogue for $M_1$. For $a \le 2$, $V_a = G_2 \cdot B/(2\pi M^*)$ is a scale mixture of Gamma(2) laws, infinitely divisible by Kristiansen's theorem. For $s \le 2$, $G_{s/2} \overset{d}{=} G_1 B'$ with $B'$ a beta variable of parameters $\frac{s}{2}$ and $1-\frac{s}{2}$ (or $B' = 1$ at $s = 2$), so $12G_{s/2}/(\pi M_1)$ is a scale mixture of exponential laws, which is infinitely divisible by the theorem of Goldie and Steutel [152, 486]. □

The proposition says nothing about zeros: the representation $\Pr(V_a > 1)$ holds for real $a$ only, since on the critical line the shape $6+i\theta$ is not that of a probability

law; for $\Delta$ the centre $a = 6$ lies outside Kristiansen's range, while for $\chi_{12}$ the centre shape $\frac{1}{4}$ lies inside it. The two Gamma(2) structures are also different in kind: the one in the Pólya kernel is a kernel shape forced by the poles of $\zeta$, and the one in Proposition 8.5.7 is a mixing law.

REMARK 8.5.8 (Gamma($k$) scale mixtures, the Kolmogorov law and a Kristiansen law for $\zeta$). Kristiansen's class also gives a reformulation of RH, but not a route to the GGC property [253]. Let $W(t) = \sum_{\operatorname{Re}\gamma>0} e^{-t\gamma^2}$ be the canonical function of the clock $T$, whose Lévy density is $W(t)/t$. For every integer $k \geq 1$, RH holds if and only if $t^{k-1}W(t)$ is proportional to a scale mixture of Gamma($k$) densities. Indeed RH is equivalent to the complete monotonicity of $W$ (Section 1.2), that is to $W(t) = \int e^{-tz}U(dz)$ with $U \geq 0$ by Bernstein's theorem, and then $t^{k-1}W(t) = \int \Gamma(k)^{-1}z^k t^{k-1}e^{-tz}\,\Gamma(k)z^{-k}U(dz)$; conversely a Gamma($k$) scale mixture divided by $t^{k-1}$ is completely monotone. Under RH the mixing law is discrete: the law with density proportional to $t^{k-1}W(t)$ is that of $G_k/\gamma_J^2$, with $\Pr(J=\gamma)$ proportional to $\gamma^{-2k}$. For $k = 1$ it is a mixture of exponential laws, infinitely divisible by the theorem of Goldie and Steutel [152, 486]; for $k = 2$ it is the Kolmogorov law of $T$, with density $tW(t)/\operatorname{Var} T$, infinitely divisible by Kristiansen's theorem. Neither is a GGC, by the following fact, and Kristiansen's theorem can be applied only once the Gamma(2) mixture is known, which is RH itself.

A discrete mixture of Gamma($k$) laws, with $k$ a positive integer and at least two distinct rates $\lambda_1 < \lambda_2 < \cdots$, finitely many or tending to infinity, is not a GGC. Its Laplace transform $\varphi(z) = \sum_n w_n(1 + z/\lambda_n)^{-k}$ is meromorphic on $\mathbb{C}$, with a pole of order $k$ at each $-\lambda_n$. If it were the transform of a GGC, then $-\varphi'/\varphi = a + \int U(dt)/(z+t)$ with $a \geq 0$ and $U \geq 0$, a function holomorphic off $(-\infty, 0]$ whose poles have positive residues. So $\varphi$ has no zero off the negative axis, and none on it, where $-\varphi'/\varphi$ would have the residue $-m$ at a zero of order $m$; by Stieltjes inversion $U = k\sum_n \delta_{\lambda_n}$, and $\varphi(z) = e^{-az}\prod_n(1 + z/\lambda_n)^{-k} \leq (1 + z/\lambda_1)^{-k}(1 + z/\lambda_2)^{-k}$ for $z > 0$, which contradicts $\varphi(z) \geq w_1(1 + z/\lambda_1)^{-k}$ as $z \to \infty$.

Kristiansen's class carries the completed zeta function itself, with no hypothesis. Let $N$ have the zeta law $\Pr(N = n) = n^{-4}/\zeta(4)$, independent of $G_2$, and put $Q = G_2/(\pi N^2)$. Then $Q$ has the density $\frac{\pi^2}{\zeta(4)}\,x\,\omega(x)$, with $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$, and $\mathbb{E}Q^q = \Gamma(q+2)\pi^{-q}\zeta(2q+4)/\zeta(4)$, so that

$$\pi^{-s/2}\Gamma\left(\tfrac{s}{2}\right)\zeta(s) = \frac{\pi^2}{90}\,\mathbb{E}\big[Q^{s/2-2}\big], \qquad \xi(s) = \frac{\pi^2}{180}\,s(s-1)\,\mathbb{E}\big[Q^{s/2-2}\big], \qquad \operatorname{Re} s > 1,$$

a normalisation of Riemann's integral $\int_0^\infty x^{s/2-1}\omega(x)\,dx$; beyond $\operatorname{Re} s > 1$ the identities hold after continuation, not as expectations. So $Q$ is infinitely divisible by Kristiansen's theorem and, by the fact above with the rates $\pi n^2$, not a GGC. Williams' variable $\pi^2 S_2/2 = \sum_n \Gamma_n/n^2$ is a convolution of Gamma(2) laws of rates $n^2$ and carries the functional equation (Section 8.2.3); $Q$ is a mixture of Gamma(2) laws of rates $\pi n^2$ and carries the Dirichlet series.

*8.5.4.4 The law of $M_1$*

Put $b_n = \pi(n^2-1)/12$; the pentagonal number theorem of Section 8.5.1 reads $F_1(y) = \sum_{n\ge1} \chi_{12}(n) e^{-b_n y}$, so the rates in this expansion are $b_{6k\pm1} = \pi k(3k\pm1)$, $2\pi$ times the generalised pentagonal numbers $k(3k\pm1)/2$.

PROPOSITION 8.5.9. *For $\theta > 0$ let $w = \sqrt{1-12\theta/\pi}$, positive or purely imaginary. Then*

$$\mathbb{E}e^{-\theta M_1} = \frac{\pi(1-w^2)\sin(\pi w/3)}{2\sqrt{3}\, w\cos(\pi w/2)} = \prod_{k\ge1} \frac{1+\theta/b_{6k}}{(1+\theta/b_{6k-1})(1+\theta/b_{6k+1})}.$$

*Proof.* Since $\sum_n \theta/(\theta+b_n) < \infty$, $\theta e^{-\theta y} F_1(y)$ can be integrated term by term, and $\theta + b_n = \pi(n^2-w^2)/12$ gives $\mathbb{E}e^{-\theta M_1} = \sum_{n\ge1} \chi_{12}(n)\theta/(\theta+b_n) = (1-w^2)\sum_{n\ge1}\chi_{12}(n)/(n^2-w^2)$. The function $g(w) = \pi\sin(\pi w/3)/(2\sqrt{3}\, w\cos(\pi w/2))$ is even and meromorphic, its only poles are simple and at the odd integers prime to 3, and its residue at $w=n$ is $-\chi_{12}(n)/(2n)$, as for the series, so their difference is entire. On the circles $|w| = 2k$, $|\cos(\pi w/2)|$ is bounded below, so $g = O(1/k)$ there, and the series is $O(k^{-1}\log k)$; by the maximum modulus principle the difference vanishes. For the product, insert $\sin(\pi w/3) = \frac{\pi w}{3}\prod_{m\ge1}(1-\frac{w^2}{9m^2})$ and $\cos(\pi w/2) = \prod_{n \text{ odd}}(1-\frac{w^2}{n^2})$. The factors with $n = 3m$, $m$ odd, cancel, $1-w^2$ cancels the factor $n=1$, and what remains is a constant times the factors $1-w^2/n^2$ with $n = 6k$ above and $n = 6k\pm1$ below. The value 1 at $\theta = 0$ fixes the constant, and $(1-w^2/n^2)/(1-1/n^2) = 1+\theta/b_n$. □

At $\theta = \pi/12$, where $w = 0$, the closed form gives $\mathbb{E}e^{-\pi M_1/12} = \pi^2/(6\sqrt{3})$, which is $L(2,\chi_{12})$ by Section 8.5.4.2.

COROLLARY 8.5.10 (A Rényi type identity). *$M_1 \overset{d}{=} \sum_{k\ge1} Y_k$ with independent $Y_k$, where $Y_k$ is exponential of rate $b_{6k-1}$ with probability $p_k$ and of rate $b_{6k+1}$ with probability $1-p_k$, and*

$$p_k = \Big(1 - \frac{b_{6k-1}}{b_{6k}}\Big)\frac{b_{6k+1}}{b_{6k+1}-b_{6k-1}} = \frac{(12k-1)(3k+1)}{2(36k^2-1)},$$

*so $p_1 = \frac{22}{35}$, $p_2 = \frac{161}{286}$, and $p_k \downarrow \frac12$.*

*Proof.* For $\alpha < \gamma < \beta$ the partial fractions $\frac{1+\theta/\gamma}{(1+\theta/\alpha)(1+\theta/\beta)} = p\frac{\alpha}{\alpha+\theta} + (1-p)\frac{\beta}{\beta+\theta}$ hold with $p = (1-\alpha/\gamma)\beta/(\beta-\alpha)$ and $1-p = \alpha(\beta-\gamma)/(\gamma(\beta-\alpha))$, both in $(0,1)$; take $\alpha = b_{6k-1} = \pi k(3k-1)$, $\gamma = b_{6k} = \pi(36k^2-1)/12$ and $\beta = b_{6k+1} = \pi k(3k+1)$ in each factor of Proposition 8.5.9, and use the continuity theorem for Laplace transforms. □

By Frullani's integral, as in the first reading of Section 1.3.1, the product gives

$$-\log \mathbb{E}e^{-\theta M_1} = \int_0^\infty \big(1-e^{-\theta t}\big)\frac{k_1(t)}{t}\,dt, \quad k_1(t) = \sum_{k\ge1}\big(e^{-b_{6k-1}t} + e^{-b_{6k+1}t} - e^{-b_{6k}t}\big),$$

so $M_1$ is infinitely divisible, with no drift and Lévy density $k_1(t)/t$; this also follows from Corollary 8.5.10, since mixtures of exponential laws are infinitely divisible [152, 486]. The Lévy density is completely monotone, since $(e^{-b_{6k-1}t}-e^{-b_{6k}t})/t = \int_{b_{6k-1}}^{b_{6k}} e^{-zt}dz$, so $M_1$ lies in the Bondesson class BO of Section 1.3.2. It is not a GGC: its Thorin measure would have to be $U_1 = \sum_k(\delta_{b_{6k-1}}+\delta_{b_{6k+1}}-\delta_{b_{6k}})$, negative at the points $b_{6k}$ where $\chi_{12}$ vanishes between $6k-1$ and $6k+1$. Concretely, the transform vanishes at $\theta=-b_{6k}$, where $w=6k$, so $-(\log \mathbb{E}e^{-\theta M_1})'$ has residue $-1$ there, and if $M_1$ were a GGC the Stieltjes inversion (1.3.2) of this function would give $U(\{b_{6k}\})=-1$. The poles of the transform are at $\theta=-b_n$, $n\equiv\pm1 \pmod 6$, $n\ge 5$, and its zeros at $-b_{6k}$: each zero lies between its own two poles $-b_{6k\pm1}$, but zeros and poles do not alternate, since $-b_7$ and $-b_{11}$, for instance, are consecutive poles. The next lemma reads Lemma 6.2.5 and the cosine dictionary (6.2.4) for a signed Thorin measure.

LEMMA 8.5.11. *Let $X\ge 0$ have Laplace exponent $-\log \mathbb{E}e^{-\theta X} = \int_{(0,\infty)} \log(1+\theta/z)\,U(dz)$ for a signed measure $U$ with $\int \log(1+1/z)\,|U|(dz)<\infty$, and put $N(z)=U((0,z])$, with $N(z)\log(1+u^2/z^2)\to 0$ as $z\to\infty$ for real $u$. Then $-\log \mathbb{E}e^{-\theta X} = \int_0^\infty (1-e^{-\theta t})\nu(t)\,dt$ for $\theta\ge 0$, with the signed function $\nu(t) = t^{-1}\int e^{-zt}U(dz) = \int_0^\infty e^{-zt}N(z)\,dz$, and for real $u$*

$$\begin{aligned} -\log\left|\mathbb{E}e^{iuX}\right|^2 &= \int_0^\infty 4\sin^2\Big(\frac{ut}{2}\Big)\nu(t)\,dt \\ &= \int \log\Big(1+\frac{u^2}{z^2}\Big)U(dz) = \int_0^\infty N(z)\,\frac{2u^2}{z(z^2+u^2)}\,dz. \end{aligned} \tag{8.5.6}$$

*The law of $X$ is a GGC if and only if $U\ge 0$, and it lies in BO, that is, it is infinitely divisible with a completely monotone Lévy density, if and only if $N\ge 0$.*

*Proof.* Frullani's integral $\log(1+\theta/z)=\int_0^\infty (1-e^{-\theta t})e^{-zt}t^{-1}dt$ gives the Lévy form, the exchange of integrals being justified by $\log(1+\theta/z)\le C_\theta\log(1+1/z)$, and integrating $e^{-zt}$ against $dN$ by parts gives the second expression of $\nu$. The first equality of (8.5.6) is twice the real part of the Lévy form at $\theta=-iu$, since $1-\cos ut = 2\sin^2(ut/2)$; the second is the Frullani identity $\int_0^\infty 4\sin^2(ut/2)e^{-zt}t^{-1}dt = \log(1+u^2/z^2)$; the third is an integration by parts, with $\frac{d}{dz}\log(1+u^2/z^2) = -2u^2/(z(z^2+u^2))$, whose boundary terms vanish by the assumptions. The signed function $\nu$ is determined by the law of $X$, since $t\nu(t)$ is the inverse Laplace transform of the derivative of $-\log \mathbb{E}e^{-\theta X}$, and so is the signed measure $U$, by the Stieltjes inversion of $\int U(dz)/(\theta+z)$. If $U\ge 0$ the law is a GGC, and if it is a GGC, its Laplace exponent grows more slowly than $\theta$, so it has no drift, and its Thorin measure is $U$. If $N\ge 0$, $\nu$ is completely monotone and the law is in BO; if the law is in BO, its Lévy density is $\int e^{-zt}\mu(dz)$ with $\mu\ge 0$, it equals $\nu$, and the uniqueness of Laplace transforms gives $\mu(dz)=N(z)\,dz$, so $N\ge 0$, $N$ being right continuous. □

For $M_1$ the Thorin measure is $U_1 = \sum_{n\ge 2} a(n)\delta_{b_n}$ with weights of sine squared form,

$$a(n) = \sin^2\frac{\pi n}{2} + \frac{4}{3}\sin^2\frac{\pi n}{3} - 1 = \mathbf{1}\{2 \nmid n\} - \mathbf{1}\{3 \mid n\} = \frac{1}{6} - \frac{1}{2}\cos \pi n - \frac{2}{3}\cos\frac{2\pi n}{3},$$

equal to 1 for $n \equiv \pm 1 \pmod 6$, to $-1$ for $6 \mid n$, and to 0 otherwise.

Proposition 8.5.12. *The canonical function $k_1(t) = t\nu_1(t)$ of $M_1$ has the theta form*

$$k_1(t) = \tfrac{1}{2}e^{\pi t/12}\Big(1 + \frac{\vartheta_0 - 3\vartheta_{1/2} - 4\vartheta_{1/3}}{\sqrt{3t}}\Big) - 1, \qquad \vartheta_\alpha = \sum_{m\in\mathbb{Z}} e^{-12\pi(m+\alpha)^2/t},$$

*so $\nu_1(t) = (2\sqrt{3})^{-1}t^{-3/2} - (2t)^{-1} + O(t^{-1/2})$ as $t \downarrow 0$ and $\nu_1(t) \sim e^{-2\pi t}/t$ as $t \to \infty$. Its Bondesson function $N_1(z) = U_1((0,z])$ is nonnegative, and $N_1(z) \sim \frac{1}{3}\sqrt{3z/\pi}$ as $z \to \infty$.*

*Proof.* Since $a$ is even, $a(0) = -1$ and $a(1) = 1$,

$$k_1(t) = \sum_{n\ge 2} a(n)e^{-b_n t} = \tfrac{1}{2}e^{\pi t/12}\Big(1 + \sum_{n\in\mathbb{Z}} a(n)e^{-\pi n^2 t/12}\Big) - 1.$$

Poisson summation, $\sum_{n\in\mathbb{Z}} \cos(2\pi n\alpha)e^{-\pi n^2 v} = v^{-1/2}\sum_{m\in\mathbb{Z}} e^{-\pi(m+\alpha)^2/v}$ with $v = t/12$, applied to the three terms of the cosine form of $a(n)$, gives the theta form. As $t \downarrow 0$, $\vartheta_0 = 1 + O(e^{-12\pi/t})$, while $\vartheta_{1/2}$ and $\vartheta_{1/3}$ are $O(e^{-4\pi/(3t)})$, so $k_1(t) = (2\sqrt{3t})^{-1} - \frac{1}{2} + O(\sqrt{t})$; as $t \to \infty$ the first term $e^{-b_5 t} = e^{-2\pi t}$ of the series dominates. Since $b_{6k-1} < b_{6k} < b_{6k+1}$, every negative atom of $U_1$ is preceded by its own positive atom, so $N_1 \ge 0$. Up to $z$ there are about $X/3$ integers $n \equiv \pm 1 \pmod 6$ and $X/6$ multiples of 6 with $b_n \le z$, where $X = \sqrt{12z/\pi + 1}$, so $N_1(z) = X/6 + O(1) \sim \frac{1}{3}\sqrt{3z/\pi}$. □

The Lévy density has the index $\frac{1}{2}$ of a Brownian first passage time, matching $-\log \mathbb{E}e^{-\theta M_1} \sim \sqrt{\pi\theta/3}$, which the lower tail $F_1(y) \sim y^{-1/2}e^{-\pi/(12y)}$ gives; this index comes from modularity. By (8.5.6), the modulus of the characteristic function is

$$\big|\mathbb{E}e^{iuM_1}\big|^{-2} = \prod_{k\ge 1} \frac{(1 + u^2/b_{6k-1}^2)(1 + u^2/b_{6k+1}^2)}{1 + u^2/b_{6k}^2} = \exp\int_0^\infty N_1(z)\,\frac{2u^2\,dz}{z(z^2+u^2)},$$

the exponential of a positive kernel integrated against a nonnegative function. Figure 8.5 (left) shows $N_1$ against its asymptote.

The sine squared form of $a(n)$ splits $U_1$ into two positive measures: $\sin^2(\pi n/2)$ selects the odd $n$, and $1 - \frac{4}{3}\sin^2(\pi n/3)$ the multiples of 3.

THEOREM 8.5.13. *With independent standard exponential variables $E_n$ and $E'_m$, let $S = \sum_{n\ odd,\, n\geq 3} E_n/b_n$ and $Z = \sum_{m\geq 1} E'_m/b_{3m}$, with $Z$ independent of $M_1$. Then $S$ and $Z$ are GGCs,*

$$\mathbb{E}e^{-\theta S} = \frac{\pi}{4}\,\frac{1-w^2}{\cos(\pi w/2)}, \qquad \mathbb{E}e^{-\theta Z} = \frac{\sqrt{3}\,w/2}{\sin(\pi w/3)},$$

*and $M_1 + Z \stackrel{d}{=} S$. Equivalently $U_1 = U_S - U_Z$, with $U_S = \sum_{n\ odd,\, n\geq 3} \delta_{b_n}$ and $U_Z = \sum_{m\geq 1} \delta_{b_{3m}}$, the atoms at $b_{3m}$ with $m$ odd cancelling.*

*Proof.* With $1+\theta/b_n = (n^2-w^2)/(n^2-1)$, the product $\cos(\pi w/2) = \prod_{n\ \mathrm{odd}}(1 - w^2/n^2)$ gives $\prod_{n\ \mathrm{odd},\, n\geq 3}(1+\theta/b_n) = \frac{4}{\pi}\cos(\pi w/2)/(1-w^2)$, the constant $\frac{\pi}{4} = \prod_{n\ \mathrm{odd},\, n\geq 3}(1-n^{-2})$ being the limit at $w=1$; and $\sin(\pi w/3)/(\pi w/3) = \prod_m(1 - w^2/9m^2)$ gives $\prod_m(1+\theta/b_{3m}) = \frac{2}{\sqrt{3}}\sin(\pi w/3)/w$. The quotient of the two transforms is the closed form of Proposition 8.5.9, so $\mathbb{E}e^{-\theta(M_1+Z)} = \mathbb{E}e^{-\theta S}$. □

The rates are shifted eigenvalues of Brownian hitting times [49]. With $\lambda_j = \pi^2(2j-1)^2/8$, the rates of the exit time $C$ of Brownian motion from $(-1,1)$, whose transform is $1/\cosh\sqrt{2\theta} = \prod_j(1+\theta/\lambda_j)^{-1}$, $b_{2j-1} = \frac{2}{3\pi}(\lambda_j - \lambda_1)$, so $S \stackrel{d}{=} \frac{3\pi}{2}\sum_{j\geq 2} E_j/(\lambda_j - \lambda_1)$: the variable $C = \sum_j E_j/\lambda_j$ with its ground mode $j=1$ removed and the others shifted by $\lambda_1$. With $\mu_m = \pi^2 m^2/2$, the rates of the first passage $T$ of the three dimensional Bessel process from 0 to 1, whose transform is $\sqrt{2\theta}/\sinh\sqrt{2\theta}$, $b_{3m} = \frac{3}{2\pi}(\mu_m - \frac{\pi^2}{18})$, so $Z \stackrel{d}{=} \frac{2\pi}{3}T^{(\pi^2/18)}$, where $T^{(c)}$ has the law of $T$ tilted by $e^{cT}$, a proper law since $\frac{\pi^2}{18} < \mu_1$. By Kent's theorem with Krein's inverse spectral theory (Section 8.1.3), $S$ and $Z$, GGCs without drift whose Thorin measures are unit atoms at distinct points, are hitting times of levels by generalised diffusions started at a reflecting or entrance end, so

$$M_1 + \big(\text{a tilted Bessel first passage time}\big) \stackrel{d}{=} \big(\text{the reduced Brownian exit time } S\big),$$

and $M_1$ itself is not a hitting time from such an end, since those are GGCs. Every first hitting time of a one dimensional diffusion lies in BO [306], as $M_1$ does, and from an interior point the transform is a quotient of the kind discussed in Section 11.1.4; whether $M_1$ is the hitting time of some level by some diffusion started at an interior point we do not know.

### 8.5.4.5 The law of $M^*$

The law of $M^*$ has no such product, but its transform $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*} = \theta\int_0^\infty e^{-\theta y}F(y)\,dy$ is meromorphic: since $F(y) = \sum_{n\geq 1}\tau(n)e^{-2\pi(n-1)y}$, for any $c>0$

$$\psi^*(\theta) = \theta\int_0^c e^{-\theta y}F(y)\,dy + \theta\sum_{n\geq 1}\tau(n)\frac{e^{-(\theta+2\pi(n-1))c}}{\theta+2\pi(n-1)},$$

with at most simple poles at $\theta = -2\pi m$, $m \geq 1$, of residue $-2\pi m\,\tau(m+1)$. Just to the right of a pole $\psi^*$ has the sign of the residue and just to the left the opposite sign, so it has a real zero between $-2\pi(m+1)$ and $-2\pi m$ whenever $\tau(m+1)\tau(m+2) > 0$. The first such pair is $\tau(6) = -6048$, $\tau(7) = -16744$, and `maxlaw.py`, with $c = 0.3$ at 40 digits, finds the zero at $\theta = -34.4840085134049704852 = -2\pi \times 5.48830041253$. It is the only sign change of $\psi^*$ on $(-15\pi, 0)$, and the argument principle on the rectangle $|\operatorname{Re}\theta| \leq 15\pi$, $|\operatorname{Im}\theta| \leq 60$, which contains the seven poles $-2\pi, \ldots, -14\pi$, gives winding number $-6$, so it is the only zero there. As for $M_1$, the zero shows that $M^*$ is not a GGC.

The script `maxexp_psizeros.py` certifies the zeros in ball arithmetic, with $c = \frac{3}{100}$, the series to $n = 1500$, its tail bounded by Deligne's bound $|\tau(n)| \leq d(n)n^{11/2} \leq 2n^6$, and the integral bounded by $F(y) \leq y^{-12}e^{2\pi y - 2\pi/y}$, which is increasing.

Proposition 8.5.14. *1. For each of the* 25 *values* $m \leq 60$ *with* $\tau(m+1)\tau(m+2) > 0$, $\psi^*$ *changes sign in an interval of length* $2 \times 10^{-20}$ *inside the gap* $(-2\pi(m+1), -2\pi m)$*; the zeros begin* $\theta/2\pi = -5.48830041$, $-8.66373325$, $-11.4203774$, $-13.2533839$, $-14.7973058$, *and the last is* $-60.7670708$.

*2.* $\psi^*$ *has exactly one zero in the disc of radius* $10^{-7}$ *about each of the ten points*

$$\begin{aligned}
&-155.5940822775511402804 \pm 2.7631188861212609656\,i,\\
&-215.6211277570694988944 \pm 2.9647292148515237348\,i,\\
&-307.4008549057072052743 \pm 3.2283287024583906394\,i,\\
&-322.6135369899940182007 \pm 3.0449254921259967464\,i,\\
&-347.8816009468099010890 \pm 2.7171007701068929577\,i,
\end{aligned}$$

*and it lies within* $10^{-19}$ *of that point.*

*Proof.* For (1), the values of $\psi^*$ at the two ends of each interval are balls of opposite signs. For (2), let $z_0$ be the approximation to the zero computed by Newton's method at 230 bits; the point shown is $z_0$ rounded to nineteen decimals, at distance at most $6.1 \times 10^{-20}$ from it. With $A \geq |\psi^*(z_0)|$, $B \leq |\psi^{*\prime}(z_0)|$ and $C \geq |\psi^{*\prime\prime}|$ on the square of half side $r$ about $z_0$, each computed as a ball, $A + 2Cr^2 < Br/2$ holds for $r = 10^{-6}$ and for $r = 10^{-29}$. On the circle $|\theta - z_0| = r$, inside the square, $|\psi^*(\theta) - \psi^{*\prime}(z_0)(\theta - z_0)| \leq A + Cr^2/2 < Br$, so by Rouché's theorem $\psi^*$ has as many zeros in the disc $|\theta - z_0| < r$ as $\psi^{*\prime}(z_0)(\theta - z_0)$, namely one. The disc of radius $10^{-7}$ about the point shown lies in the disc of radius $10^{-6}$ about $z_0$, and the disc of radius $10^{-29}$ about $z_0$ lies within $10^{-19}$ of the point shown. The zeros come in conjugate pairs because $\psi^*$ is real on the real axis. □

The argument principle on the rectangles $|\operatorname{Re}\theta| \leq 2\pi(k + \frac{1}{2})$, with $|\operatorname{Im}\theta| \leq 30$ for $k = 15$, 25, 40 and 60 and $|\operatorname{Im}\theta| \leq 100$ for $k = 60$, and on a grid of cells, $2\pi$ wide and at most 5 high, covering $-2\pi \times 60.5 \leq \operatorname{Re}\theta \leq 2\pi \times 5.5$, $\frac{1}{2} \leq \operatorname{Im}\theta \leq 30$, finds

exactly these zeros: the winding number on the largest rectangle, which contains the 60 poles $-2\pi,\dots,-120\pi$, is $-26$, so it contains 34 zeros, the 24 real zeros with $m\le 59$ and the ten nonreal zeros of the proposition. This count, unlike the zeros themselves, is computed and not certified: the change of argument is followed along the contours with steps halved until it is below 0.3, in ball arithmetic whose radii stay below a relative $10^{-46}$. Every real zero lies in a gap where the residues force it, every gap with $\tau(m+1)\tau(m+2)<0$ is free of real zeros, and the five nonreal pairs have their real parts in such gaps, those with $m=24$, 34, 48, 51 and 55 ($\operatorname{Re}\theta/2\pi=-24.764$, $-34.317$, $-48.924$, $-51.346$, $-55.367$), at heights between 2.72 and 3.23 (Figure 8.5, right). In a gap where the residues have opposite signs, $\psi^*$ has the same sign at both ends, so its real zeros there come in pairs; the computation finds none, and the five nonreal pairs sit in such gaps, as if a pair of real zeros had collided and left the axis.

COROLLARY 8.5.15. *$M^*$ is not in the Bondesson class BO; in particular it is not a GGC.*

*Proof.* If $M^*$ were in BO, its Lévy density would be $\int e^{-zt}\mu(dz)$ with $\mu\ge 0$, so $-(\log\psi^*)'(\theta)=d+\int\mu(dz)(\theta+z)^{-2}$, with $d\ge 0$ the drift, for $\theta>0$. The right side is holomorphic on $\mathbb{C}\setminus(-\infty,0]$, since $\int\mu(dz)(1+z)^{-2}<\infty$ and $|\theta+z|$ is bounded below by a multiple of $1+z$ for $\theta$ in a compact subset of $\mathbb{C}\setminus(-\infty,0]$. Then the meromorphic function $\psi^{*\prime}/\psi^*$ would have no poles in $\mathbb{C}\setminus(-\infty,0]$, and $\psi^*$ no zeros there, against Proposition 8.5.14(2). □

Nor does $M^*$ have a signed Thorin measure in the sense of Lemma 8.5.11: for a signed measure $U$ on $(0,\infty)$, $-(\log\psi^*)'(\theta)=\int U(dz)(\theta+z)^{-1}$ would again be holomorphic on $\mathbb{C}\setminus(-\infty,0]$. Unit atoms at the points $2\pi m$ with $\tau(m+1)\ne 0$, less unit atoms at the real zeros, account for the poles and the real zeros of $\psi^*$, but no measure on the positive axis accounts for the nonreal zeros. Whether $M^*$ is infinitely divisible remains open. If it is, with Lévy measure $\nu^*$, then $-(\log\psi^*)'(\theta)=d+\int t\,e^{-\theta t}\nu^*(dt)$ is completely monotone on $(0,\infty)$: the cumulants $\kappa_j(\theta)=(-1)^j(\log\psi^*)^{(j)}(\theta)$ of $M^*$ tilted by $e^{-\theta M^*}$ are all positive. At $\theta=0.5$, 2, 8 and 30 the first eight are positive, as the first author's note found. As $\theta\to\infty$ the Laplace method, at the saddle point $y_0=\sqrt{2\pi/\theta}$ of $e^{-\theta y-2\pi/y}$ and with the lower tail $F(y)\sim y^{-12}e^{-2\pi/y}$, gives

$$\psi^*(\theta)\sim 2^{-1/2}(2\pi)^{-21/4}\,\theta^{25/4}\,e^{-2\sqrt{2\pi\theta}},$$

so $-\log\psi^*(\theta)=2\sqrt{2\pi\theta}-\frac{25}{4}\log\theta+\frac{21}{4}\log 2\pi+\frac{1}{2}\log 2+o(1)$. If $M^*$ is infinitely divisible, it has no drift, and Karamata's Tauberian theorem turns the leading term into $\nu^*((t,\infty))\sim 2\sqrt{2}\,t^{-1/2}$ as $t\downarrow 0$.

The first author conjectures that every maximum of independent exponential variables is infinitely divisible [428]; $M_1$, $M_4$, $M^*$ and $M_{11}$ are of this kind. For

finitely many rates $\lambda_1, \dots, \lambda_n$ the transform is rational: with $\lambda_S = \sum_{i \in S} \lambda_i$,

$$\mathbb{E}e^{-\theta M} = \theta \int_0^\infty e^{-\theta y} \prod_{i \le n} (1 - e^{-\lambda_i y})\, dy = \sum_{S \neq \emptyset} (-1)^{|S|+1} \frac{\lambda_S}{\theta + \lambda_S},$$

with simple poles at the values $-\lambda_S$ whose grouped coefficients do not vanish, generically $2^n - 1$ of them, and, since $\mathbb{E}e^{-\theta M} \sim n!\, \lambda_1 \cdots \lambda_n \theta^{-n}$, with $n$ fewer zeros. Writing the poles as $-p$ and the zeros as $-z$, Frullani's integral gives $-\log \mathbb{E}e^{-\theta M} = \int_0^\infty (1 - e^{-\theta t}) k(t) t^{-1} dt$ with $k(t) = \sum_p e^{-pt} - \sum_z e^{-zt}$, as long as every zero has $\operatorname{Re} z > 0$, and by the uniqueness of the Lévy measure the law is infinitely divisible if and only if $k \ge 0$. In particular infinite divisibility forces every zero to have $\operatorname{Re} z$ at least the smallest pole $p_{\min}$, since otherwise $k$ would change sign at large $t$. For two rates $a < b$ the transform is $ab(a+b+2\theta)/((a+\theta)(b+\theta)(a+b+\theta))$ and the canonical function is $e^{-at} + e^{-bt} + e^{-(a+b)t} - e^{-(a+b)t/2}$, positive by the inequality of arithmetic and geometric means; the signed Thorin measure $\delta_a + \delta_b + \delta_{a+b} - \delta_{(a+b)/2}$ has a nonnegative distribution function, so this maximum is in BO and is not a GGC, and $e^{at} k(t) = 1 - x + x^2 + x^{2b/(b-a)} > \frac{3}{4}$ with $x = e^{-(b-a)t/2}$, a bound approached as $b/a \downarrow 1$. In `maxexp_checks.py`, $k$ is evaluated at 181 points of $p_{\min} t \in [10^{-3}, 40]$ for 300 families of random rational rates in $[0.1, 5]$ for each $n = 2, 3, 4, 5$ and 40 families with $n = 6$. In every family $k$ is positive at every point, $e^{p_{\min} t} k(t) \ge 0.53$, and every zero has $\operatorname{Re} z > p_{\min}$. Nonreal zeros are common: they occur in 140, 222, 266 and 40 of the families with 3, 4, 5 and 6 rates, up to 2, 8, 22 and 52 of them. So nonreal zeros, which exclude BO, do not seem to exclude infinite divisibility. The truncations of $M_1$, with the rates $2\pi n$, $n \le N$, have only real zeros for $N = 4, \dots, 10$, and those of $M^*$, with each rate $2\pi n$, $n \le N$, taken 24 times, have 24 real zeros for $N = 2$ and 72 zeros, 52 of them nonreal, for $N = 3$; in each case $k$ is positive on the grid. None of this is a proof.

Question 8.5.16. Is every maximum of independent exponential variables infinitely divisible? In particular, is $M^*$?

For the clock the relaxation from GGC to BO gains nothing. If its Lévy density $W(t)/t$ were completely monotone, equal to $\int e^{-zt} \mu(dz)$ with $\mu \ge 0$, then the formula $(G'/G)(u) = \int_0^\infty e^{-ut} W(t)\, dt$ of Section 1.2 would give $(G'/G)(u) = \int \mu(dz)(u+z)^{-2}$, holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. But $G'/G$ has a pole at every zero $-a_k = (\rho_k - \frac{1}{2})^2$ of $G$, and this point lies in $(-\infty, 0]$ only when $\rho_k$ is on the critical line. So the clock is in BO if and only if RH holds (Table 1.6). The signed Thorin measures of the max laws come from zeros of their Laplace transforms, and the transform $1/G$ of the clock has none. The two laws of this section show the two ways out of the Thorin class: $M_1$ leaves it through real zeros, each preceded by its own pole, and stays in BO, while $M^*$ leaves BO too, through zeros off the real axis, the way a zero of $\xi$ off the critical line would take the clock out of both classes.

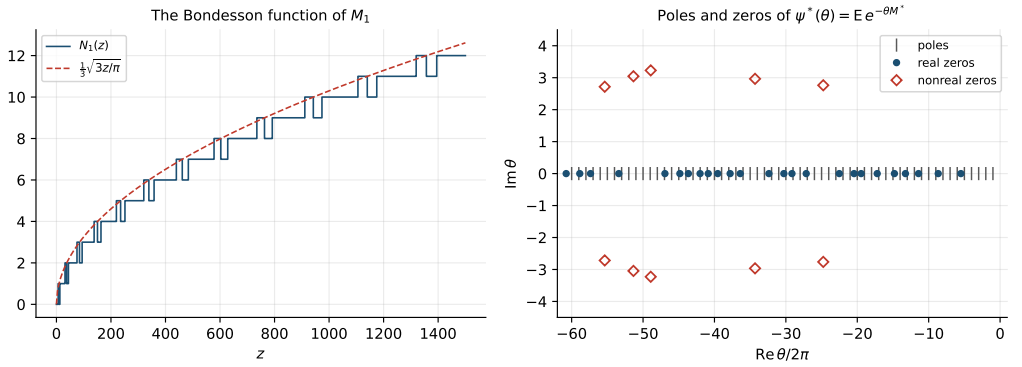



Figure 8.5: Left: the Bondesson function $N_1$ of $M_1$, nonnegative, with its asymptote $\frac{1}{3}\sqrt{3z/\pi}$; each drop is a zero of the transform at $-b_{6k}$. Right: the poles $-2\pi m$, $m \le 60$, and the zeros of $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*}$: the 25 real zeros above $-2\pi \times 61$, one in each gap with $\tau(m+1)\tau(m+2) > 0$, and the five pairs of nonreal zeros, the only ones with $|\operatorname{Re}\theta| \le 2\pi \times 60.5$ and $|\operatorname{Im}\theta| \le 100$. Drawn by `maxexp_figs.py` from `maxexp_psizeros.json`.

### 8.5.5 The de Bruijn Newman flow for $\Delta$

The flow of Section 8.4 needs only an even kernel that decays faster than any Gaussian, so it applies to $K_\Delta$:

$$H_t^\Delta(\theta) = \int_0^\infty e^{tu^2} K_\Delta(u)\cos(\theta u)\,du, \qquad 2H_0^\Delta(\theta) = \xi\big(\tfrac12 + i\theta, \Delta\big).$$

In the variable $\theta$ the times of Section 8.4, whose variable is $z = 2\theta$, are divided by four, and the isolated pair estimate for two zeros at distance $D$ is $-D^2/8$. The zeros of $H_0^\Delta$ lie in the strip $|\operatorname{Im}\theta| < \frac12$, so by de Bruijn's theorem [112] every zero of $H_t^\Delta$ is real for $t \ge \frac18$. Dobner's theorem for the extended Selberg class [119] gives the other side: $\Delta$ has a de Bruijn Newman constant $\Lambda_\Delta$, and $H_t^\Delta$ has nonreal zeros for every $t < 0$. So $0 \le \Lambda_\Delta \le \frac18$, and GRH for $\Delta$ is the statement $\Lambda_\Delta = 0$.

On the critical line the kernel is a probability density up to normalisation, as the eta kernel is in Section 8.5.1: if $V$ has the density $K_\Delta(v)/\xi(\frac12,\Delta)$ on $\mathbb{R}$, then $\mathbb{E}e^{i\theta V} = \xi(\frac12 + i\theta,\Delta)/\xi(\frac12,\Delta)$, and GRH for $\Delta$ is the statement that this characteristic function has only real zeros. In the language of Section 8.5.4, $e^V$ has the density proportional to $y^5\Delta(iy) = y^5e^{-2\pi y}F(y)$, so $V$ is the logarithm of $G_6/2\pi$ conditioned on $G_6/2\pi \ge M^*$, with $G_6$ a gamma variable of shape 6 independent of $M^*$; its law is symmetric by the self duality (8.5.1), with no symmetrisation. The variable $V$ is not infinitely divisible, nor even quasi infinitely divisible, because its characteristic function vanishes at $\theta = 9.2223793999$; this is the argument of Proposition 11.1.3(1) for the Pólya variable. Since $1-\cos x = 2\sin^2(x/2)$ and $K_\Delta$

is even,

$$\xi\big(\tfrac12,\Delta\big)-\xi\big(\tfrac12+i\theta,\Delta\big)=4\int_0^\infty K_\Delta(v)\sin^2\Big(\frac{\theta v}{2}\Big)dv,$$

so the zeros on the critical line are the solutions of $\mathbb{E}[2\sin^2(\theta V/2)]=1$. At the level of logarithms, the factorisation $\xi(s,\Delta)=(2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ splits $\log|\xi(\frac12+i\theta,\Delta)/\xi(\frac12,\Delta)|^2$ into $\log|\Gamma(6+i\theta)/\Gamma(6)|^2$ and $\log|L(\frac12+i\theta,\Delta)/L(\frac12,\Delta)|^2$. The first has the positive sine squared form given by Binet's formula,

$$\log\Big|\frac{\Gamma(6+i\theta)}{\Gamma(6)}\Big|^2=-\int_0^\infty 4\sin^2\Big(\frac{\theta u}{2}\Big)\frac{e^{-6u}}{u(1-e^{-u})}\,du,$$

the Lévy Khintchine form of the infinitely divisible variable $\log G_6$, and the second carries every zero. To the right of the critical strip the second has a sine squared form too: for $\sigma>1$, expanding $\log L(s,f)=\sum_{n\ge2}\Lambda_f(n)n^{-s}/\log n$ gives

$$\log\Big|\frac{L(\sigma+i\theta,f)}{L(\sigma,f)}\Big|^2=-\sum_{n\ge2}\frac{4\Lambda_f(n)}{n^\sigma\log n}\sin^2\Big(\frac{\theta\log n}{2}\Big),$$

for $f=\chi_{12}$ and $\Delta$, and for $\zeta$ with $\Lambda$ in place of $\Lambda_f$. Since $\int_\sigma^\infty n^{-\alpha}d\alpha=n^{-\sigma}/\log n$, the right side is minus twice the integral of the prime piece of the decomposition of Section 8.5.3 over the basepoints $\alpha>\sigma$. For $\zeta$ the weights are positive, which is the infinite divisibility of log of the zeta distribution (Section 6.1.6). For $\Delta$ they are $8\cos(k\vartheta_p)/(kp^{k\sigma})$ at $n=p^k$, where $\alpha_p=e^{i\vartheta_p}$, $0\le\vartheta_p\le\pi$, and they take both signs. The Sato Tate law for $\Delta$, proved by Barnet-Lamb, Geraghty, Harris and Taylor [35], makes the angles $\vartheta_p$ equidistributed for the measure $\frac{2}{\pi}\sin^2\vartheta\,d\vartheta$ on $[0,\pi]$, under which $\cos k\vartheta<0$ with probability $\frac12$ for $k\ne2$ and $\frac12+\frac1\pi$ for $k=2$. So even where the Euler product converges absolutely the sine squared form of $\log|L(\sigma+i\theta,\Delta)|$ has a signed measure, and on the critical line the series diverges: positivity lives in the kernel and in the gamma factor, not in the arithmetic factor.

The kernel factors as $K_\Delta(u)=e^{-2\pi\cosh u}H(u)$, with $H(u)=\sqrt{F(e^u)F(e^{-u})}$ because $K_\Delta(u)^2=K_\Delta(u)K_\Delta(-u)$, and $\log H(u)=-12\sum_m\frac{\sigma(m)}{m}(e^{-2\pi me^u}+e^{-2\pi me^{-u}})$. The expansion comes from $-\log F(y)=24\sum_{m\ge1}\frac{\sigma(m)}{m}e^{-2\pi my}$, so $-\log F$ is completely monotone, the Laplace transform of the positive measure with mass $24\sigma(m)/m$ at each point $2\pi m$; for $\eta$ the masses are $\sigma(m)/m$, and every maximum of independent exponential variables has this property, since $-\log(1-e^{-\lambda y})=\sum_k e^{-k\lambda y}/k$. The transform $K_{i\theta}(2\pi)$ of Pólya's factor has only real zeros [447], the first at $\theta=9.768770083509977867$ (`maxlaw.py`), and the multiplier $H$ moves it to 9.2223793999. Pólya's theorem on universal factors [112, p. 197], which covers the Gaussian multiplier $e^{tu^2}$ of the flow for $t>0$, does not apply to $H$, which is bounded by one and supplies half of the doubly exponential decay of $K_\Delta$. The positivity of the measure of $-\log F$ is automatic and does not by itself decide where the zeros are; whether some multiplier of the same form moves zeros off the line is the following question, and any answer must account for $\Delta$, $\chi_{12}$ and the conductor 11 curve, whose kernels have this form.

QUESTION 8.5.17. For which $a>0$, $c_m\ge 0$ and $\beta_m>0$ does the Fourier transform of $e^{-a\cosh u}\exp\big(-\sum_m c_m(e^{-\beta_m e^u}+e^{-\beta_m e^{-u}})\big)$ have only real zeros?

The kernel $K_\Delta$ is strictly log concave, as the first author's note [432] shows. For $c>0$, $\log(1-e^{-ce^u})$ is the logarithm of the distribution function of $\log E$, with $E$ exponential of rate $c$, and it is strictly concave in $u$: with $x=ce^u$ its second derivative is $x(e^x-1-xe^x)/(e^x-1)^2<0$. So $\log F(e^u)$, a sum of such terms, is strictly concave (in the language of Section 8.5.4, $\log M^*$ has a log concave distribution function), and since $\log K_\Delta(u)=6u-2\pi e^u+\log F(e^u)$ and $K_\Delta$ is even,

$$-\big(\log K_\Delta\big)''(u)>2\pi e^{|u|},\qquad u\in\mathbb{R},$$

where the ratio of the two sides tends to 1 as $|u|\to\infty$. The multiplier $H$ is log concave as well, since $\log H(u)=\frac12\big(\log F(e^u)+\log F(e^{-u})\big)$. In the same way $\log K(u)=\log 2+\frac{u}{2}-\frac{\pi}{12}e^{2u}+\log F_1(e^{2u})$ gives $-(\log K)''(u)>\frac{\pi}{3}e^{2|u|}$ for the kernel of Section 8.5.1. The Pólya kernel $\Phi$ is log concave too [85], though it is not a product of this kind. Log concavity alone does not put the zeros of a transform on a line, or in any strip. The function $(1-|u|)_+^2$ is log concave, and its transform $4(\theta-\sin\theta)/\theta^3$ vanishes at the nonzero roots of $\sin\theta=\theta$, which are nonreal, the first at $7.4977+2.7687i$, with imaginary parts that grow like $\log|\theta|$. Its convolution with $e^{-\cosh u}$ is even, positive, entire and log concave [472], and decays doubly exponentially, like $K_\Delta$; its transform $8(\theta-\sin\theta)K_{i\theta}(1)/\theta^3$ has the same nonreal zeros.

In the language of Section 8.5.4 the flow is a family of Gaussian tilts: for every real $t$, $e^{tv^2}K_\Delta(v)$ is integrable, because $K_\Delta$ decays doubly exponentially, so after normalisation it is the density of the law of $V$ tilted by $e^{tV^2}$, and $H_t^\Delta(\theta)/H_t^\Delta(0)$ is its characteristic function. The question $\Lambda_\Delta=0$ asks whether the smallest tilt whose characteristic function has only real zeros is $t=0$.

With `dbn_delta.py` (Gauss Legendre quadrature at 40 digits) the 21 zeros below height 52, from 9.2224 to 51.1566, stay real for $t=0.05,0.1,\dots,0.5$ and all move down. Backward in time each pair of neighbours is followed through the critical point of $H_t^\Delta$ between them until the critical value changes sign, and each such event is tested by direct sign counts: a collision at $(t_*,\theta_*)$ leaves two sign changes on $[\theta_*-1.2,\theta_*+1.2]$ at $t_*+0.005$ and none at $t_*-0.005$, and the colliding pair is identified by the number of sign changes below $\theta_*-1.2$, since real zeros keep their order. Table 8.16 lists the seven collisions. The numbers of sign changes on $(0.5,52]$ at eleven values of $t$ from 0 to $-1.5$ fall by two at each collision and once by one, between $t=-0.2$ and $-0.3$, where the highest zero, followed back from 51.1566, passes 52 at $t\approx-0.29$.

The script `maxexp_flow.py` recomputes the table independently, at 70 digits and with a different tracking, and finds the same seven collisions, with the same $t_*$ and $\theta_*$ to all digits of the table, and the highest zero passing 52 between $t=-0.29$ and $-0.30$. Figure 8.6 shows the trajectories.

| pair at $t = 0$ | $-D^2/8$ | $t_*$ | $\theta_*$ |
|---|---|---|---|
| 48.3591, 49.2761 | −0.1051 | −0.12622 | 49.18 |
| 36.7415, 37.7539 | −0.1281 | −0.15864 | 37.60 |
| 45.0401, 46.1973 | −0.1674 | −0.24305 | 46.19 |
| 25.2746, 26.8044 | −0.2925 | −0.40713 | 27.26 |
| 40.2190, 41.7305 | −0.2856 | −0.48271 | 42.49 |
| 31.1782, 32.7749 | −0.3187 | −0.48637 | 33.30 |
| 17.4428, 19.6565 | −0.6126 | −0.98116 | 21.09 |

Table 8.16: The collisions of zeros of $H_t^\Delta$ below height 52, in the order in which they occur as $t$ decreases: the pair at $t = 0$, the isolated pair estimate with $D$ the gap of the pair, and the time $t_*$ and height $\theta_*$ of the collision, computed by `dbn_delta.py`.

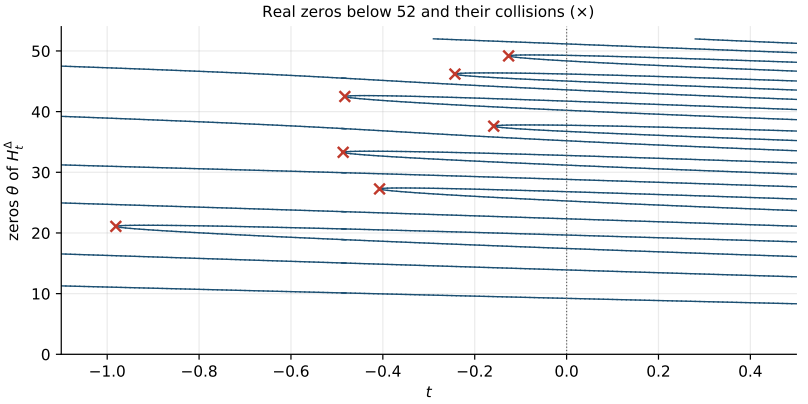



Figure 8.6: The real zeros of $H_t^\Delta$ below height 52 for $-1.1 \le t \le 0.5$, and the seven collisions of Table 8.16; all zeros move down as $t$ grows. Drawn by `maxexp_figs.py` from `maxexp_flow.json`.

The low zeros of $\Delta$ are robust: below height 33, the window of the first author's note, the first pair to leave the line, $(25.27, 26.80)$, does so only at $t = -0.407$. Collisions do not come in order of height, however. The pairs near 37.6 and 49.2 collide at $-0.159$ and $-0.126$, and under GRH for $\Delta$ Dobner's theorem forces collisions in every interval $(t, 0)$, higher up; so no window of heights decides whether $\Lambda_\Delta = 0$ (Dobner's theorem already gives $\Lambda_\Delta \ge 0$), and what a computation can show is how fast the collision times approach 0 with height. The neighbours delay every collision beyond the isolated pair estimate, by factors from 1.20 to 1.69. In the variable $\theta$ the closest pair of $\zeta$ below height 200 collides at $-0.072$, and Lehmer's pair near height 7005 at $-1.8 \times 10^{-4}$ (Section 8.4.2).

### 8.5.6 A Pólya kernel for imaginary quadratic fields

This subsection follows the last part of the first author's note on the squarefree series [431].

Let $d<0$ be a fundamental discriminant, $K=\mathbb{Q}(\sqrt{d})$, $\chi_d$ the Kronecker symbol of $d$, a real primitive odd character modulo $|d|$, $h$ the class number of $K$ and $w$ the number of its roots of unity. Then $\zeta_K(s)=\zeta(s)L(s,\chi_d)=\sum_{n\ge1}a_n n^{-s}$, where $a_n=\sum_{m\mid n}\chi_d(m)\ge0$ is the number of ideals of norm $n$, and we put

$$\Lambda_K(s)=\Big(\frac{\sqrt{|d|}}{2\pi}\Big)^s\Gamma(s)\,\zeta_K(s),\qquad \xi_K(s)=2s(s-1)\Lambda_K(s),$$

the function $\xi_K$ of Section 10.3.3 with $r_1=0$ and $r_2=1$. The duplication formula gives $\xi_K(s)=\frac{2}{\sqrt{|d|}}\,\xi(s)\,\xi(s,\chi_d)$, with $\xi(s,\chi_d)=(|d|/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_d)$ the completed function of Section 8.5.1, so $\xi_K(s)=\xi_K(1-s)$ because the root number of $\chi_d$ is one. Hence $\Lambda_K(s)=\Lambda_K(1-s)$, and $\Lambda_K$ is meromorphic with simple poles at $s=1$ and $s=0$ only, with residues $h/w$ and $-h/w$, by the class number formula $L(1,\chi_d)=2\pi h/(w\sqrt{|d|})$ and the symmetry. Let

$$\vartheta_K(x)=\sum_{n\ge1}a_n\,e^{-2\pi nx/\sqrt{|d|}},\qquad \Psi(y)=y(y-2)\,e^{-y},\qquad H_K(x)=\big(x^2\vartheta_K'(x)\big)',$$

and $\Phi_K(u)=2e^{u/2}H_K(e^u)$.

Theorem 8.5.18. (i) $\vartheta_K(1/x)=x\,\vartheta_K(x)+\frac{h}{w}\,(x-1)$ *for* $x>0$.

(ii) $H_K(x)=\sum_{n\ge1}a_n\Psi(2\pi nx/\sqrt{|d|})$ *and* $H_K(1/x)=x\,H_K(x)$. *The kernel* $\Phi_K$ *is even, analytic in the strip* $|\operatorname{Im}u|<\frac{\pi}{2}$, *and decays doubly exponentially.*

(iii) $\xi_K(\frac12+s)=\int_{\mathbb{R}}\Phi_K(u)\,e^{su}\,du$ *for every* $s\in\mathbb{C}$; *in particular* $\xi_K(\frac12+i\theta)=\int_{\mathbb{R}}\Phi_K(u)\cos(\theta u)\,du$.

(iv) $\Phi_K>0$ *on* $\mathbb{R}$ *if and only if* $H_K>0$ *on* $[1,\sqrt{|d|}/\pi]$. *In particular* $\Phi_K>0$ *when* $|d|<\pi^2$, *that is for* $d=-3,-4,-7,-8$.

(v) *If* $\Phi_K>0$, *then* $\zeta_K(\frac12)<0$ *and* $L(\frac12,\chi_d)>0$. *In particular* $L(\frac12,\chi_d)>0$ *for* $d=-3,-4,-7,-8$.

*Proof.* (i) For $c>1$, Mellin inversion of $\int_0^\infty e^{-2\pi nx/\sqrt{|d|}}x^{s-1}dx=\Gamma(s)\big(\sqrt{|d|}/2\pi n\big)^s$ gives $\vartheta_K(x)=\frac{1}{2\pi i}\int_{(c)}\Lambda_K(s)x^{-s}ds$. Move the line of integration to $\operatorname{Re}s=1-c$, which the exponential decay of $\Gamma$ and the polynomial growth of $\zeta_K$ on vertical strips allow; the residues at $s=1$ and $s=0$ are $\frac{h}{w}x^{-1}$ and $-\frac{h}{w}$. On the new line substitute $s\mapsto1-s$ and use $\Lambda_K(1-s)=\Lambda_K(s)$: the integral becomes $x^{-1}\vartheta_K(1/x)$. So $\vartheta_K(x)=\frac{h}{w}(x^{-1}-1)+x^{-1}\vartheta_K(1/x)$, which is (i).

(ii) Termwise, $\big(x^2\frac{d}{dx}e^{-\kappa x}\big)'=(\kappa^2x^2-2\kappa x)e^{-\kappa x}=\Psi(\kappa x)$. Differentiating (i) gives $\vartheta_K'(1/x)=-x^2\big(\vartheta_K(x)+x\vartheta_K'(x)+\frac{h}{w}\big)$, so $g(x)=x^2\vartheta_K'(x)$ satisfies $g(1/x)=-\big(\vartheta_K(x)+x\vartheta_K'(x)+\frac{h}{w}\big)$, and differentiating again gives $g'(1/x)=x^2\big(2\vartheta_K'(x)+x\vartheta_K''(x)\big)=x\,g'(x)$, that is $H_K(1/x)=xH_K(x)$. Then $\Phi_K(-u)=$

$2e^{-u/2}H_K(e^{-u}) = 2e^{u/2}H_K(e^u) = \Phi_K(u)$. Since $a_n \le n$, the series for $H_K(e^u)$ converges locally uniformly where $\operatorname{Re} e^u > 0$, that is for $|\operatorname{Im} u| < \frac{\pi}{2}$; and $\Phi_K(u) = O\big(e^{5u/2}\exp(-2\pi e^u/\sqrt{|d|})\big)$ as $u \to \infty$, and likewise as $u \to -\infty$ by evenness.

(iii) By (i), $x\vartheta_K(x) \to \frac{h}{w}$ and $x^2\vartheta_K'(x) \to -\frac{h}{w}$ as $x \to 0$, while $\vartheta_K$ and $\vartheta_K'$ decay exponentially at infinity. For $\operatorname{Re} z > 1$ two integrations by parts, whose boundary terms vanish, give $\int_0^\infty H_K(x)x^{z-1}dx = z(z-1)\int_0^\infty \vartheta_K(x)x^{z-1}dx = z(z-1)\Lambda_K(z)$. With $x = e^u$ and $z = \frac12 + s$, $\int_{\mathbb R}\Phi_K(u)e^{su}du = 2\int_0^\infty H_K(x)x^{z-1}dx = \xi_K(z)$. Both sides are entire in $s$, the left by (ii), so the identity holds for every $s$.

(iv) By (ii), $\Phi_K > 0$ exactly when $H_K > 0$ on $(0,\infty)$, and by the symmetry exactly when $H_K > 0$ on $[1,\infty)$. For $x \ge \sqrt{|d|}/\pi$ every argument $2\pi nx/\sqrt{|d|}$ is at least $2n \ge 2$, where $\Psi \ge 0$, and the term $n = 4$ is positive, since $a_4 = 1 + \chi_d(2) + \chi_d(2)^2 \ge 1$; so $H_K > 0$ there. If $|d| < \pi^2$ the interval $[1, \sqrt{|d|}/\pi]$ is empty.

(v) By (iii), $\xi_K(\frac12) = \int_{\mathbb R}\Phi_K > 0$, while

$$\xi_K\big(\tfrac12\big) = -\tfrac12\Lambda_K\big(\tfrac12\big) = -\tfrac12\big(\sqrt{|d|}/2\pi\big)^{1/2}\sqrt{\pi}\,\zeta_K\big(\tfrac12\big).$$

So $\zeta_K(\frac12) < 0$, and $L(\frac12,\chi_d) = \zeta_K(\frac12)/\zeta(\frac12) > 0$ because $\zeta(\frac12) < 0$. □

When $\Phi_K > 0$, the normalised kernel $\Phi_K/\xi_K(\frac12)$ is the density of a symmetric variable $X_K$ with $\mathbb{E}e^{sX_K} = \xi_K(\frac12+s)/\xi_K(\frac12)$, the Pólya variable of $K$, and its tilts $V_{K,\alpha}$, with $\mathbb{E}e^{sV_{K,\alpha}} = \xi_K(\alpha+s)/\xi_K(\alpha)$, have the means $\frac{\xi_K'}{\xi_K}(\alpha)$. The zero part of the constant $C_K$ of Theorem 10.3.7 is then $\mathbb{E}V_{K,1} - 2\,\mathbb{E}V_{K,2}$, as over $\mathbb{Q}$ in Theorem 10.3.5; by (iv) this applies to the Gaussian field $d = -4$ of Section 10.3.3. If moreover the Riemann hypothesis holds for $\zeta_K$, then $1/\mathbb{E}\cosh(\sqrt{u}\,X_K) = \xi_K(\frac12)/\xi_K(\frac12+\sqrt{u})$ is the Laplace transform of a GGC, the clock of $K$ (Proposition 10.3.6).

The kernel $\Phi_K$ is tied to the kernel of $\chi_d$ in Section 8.5.1,

$$K_{\chi_d}(u) = 2e^{3u/2}\sum_{n\ge1} n\chi_d(n)\,e^{-\pi n^2e^{2u}/|d|}, \qquad \xi\big(\tfrac12+s,\chi_d\big) = \int_{\mathbb R} K_{\chi_d}(u)\,e^{su}\,du.$$

Since the two sided Laplace transform turns convolution into multiplication, the identity $\xi_K = \frac{2}{\sqrt{|d|}}\xi\cdot\xi(\cdot,\chi_d)$ and (iii) give $\Phi_K = \frac{2}{\sqrt{|d|}}\,\Phi * K_{\chi_d}$, with $\Phi$ the Pólya kernel (1.2.3). As $\Phi > 0$, the kernel $\Phi_K$ is positive whenever $K_{\chi_d}$ is, and if $\Phi_K$ takes a negative value, so does $K_{\chi_d}$. For $d = -4$ Jacobi's identity $\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8} = q^{1/8}\prod_{n\ge1}(1-q^n)^3$ gives $K_{\chi_{-4}}(u) = 2e^{3u/2}\eta(ie^{2u})^3 > 0$, positive by the product formula as the kernel of $\chi_{12}$ is.

The script `sqfree_quadratic.py` checks the theorem at 30 digits. For $d = -4$, where $h/w = \frac14$, the trapezoid rule with step $1/200$ gives $\int\Phi_K(u)\cos(\theta u)\,du = 0.48753311500024448536$ at $\theta = 0$ and $0.25376667901239784985$ at $\theta = 2.5$, equal to $\xi_K(\frac12+i\theta)$ in every digit shown. The values $\zeta_K(\frac12) = -0.70223711$, $-0.97506623$, $-1.6744215$ and $-1.6070054$ for $d = -3,-4,-7,-8$ are negative, as (v) requires.

The sign of $H_K$ on $[1,\sqrt{|d|}/\pi]$ depends strongly on the first few weights: $\Psi$ is negative on $(0,2)$ and positive beyond, and the weights $a_2 = 1+\chi_d(2)$ and $a_3 = 1+\chi_d(3)$ vanish when 2 and 3 are inert in $K$. The next theorem makes this precise.

THEOREM 8.5.19. *Let $d<0$ be a fundamental discriminant with $\chi_d(2)=\chi_d(3)=-1$, that is $d\equiv 5 \pmod{24}$. Then $H_K(x)<0$ at $x=1.42\sqrt{|d|}/2\pi$, so $\Phi_K$ takes negative values, and so does the kernel $K_{\chi_d}$. There are infinitely many such $d$, for instance $d=-p$ for every prime $p\equiv 19 \pmod{24}$.*

*Proof.* For a fundamental discriminant $d<0$, $\chi_d(2)=-1$ exactly when $d\equiv 5 \pmod 8$, and $\chi_d(3)=-1$ exactly when $d\equiv 2 \pmod 3$; together they give $d\equiv 5 \pmod{24}$. The function $n\mapsto a_n$ is multiplicative, with $a_{p^k}=\frac12(1+(-1)^k)$ when $\chi_d(p)=-1$ and $a_{p^k}\le k+1$ for every $p$. Hence $a_1=1$, $a_2=a_3=0$ and $a_n\le a_n^*$, where $a_n^*=0$ unless 2 and 3 divide $n$ to even powers, and otherwise $a_n^*$ is the number of divisors of $n$ with its factors 2 and 3 removed. At $x=1.42\sqrt{|d|}/2\pi$ the $n$th argument of $\Psi$ in Theorem 8.5.18(ii) is $1.42n$, which exceeds 2 for $n\ge 2$, where $\Psi>0$. So

$$\begin{aligned} H_K(x) &\le \Psi(1.42) + \sum_{n\ge 4} a_n^*\,\Psi(1.42n) \\ &= -0.19907566 + 0.13917904 = -0.05989662 < 0, \end{aligned}$$

the sum taken to $n=2000$ and the rest bounded, with $a_n^*\le 2\sqrt{n}$, by $10^{-1225}$ (`sqfree_quadratic.py`). Then $\Phi_K(\log x)<0$, and $K_{\chi_d}$ takes negative values by the convolution identity above. If $p\equiv 19 \pmod{24}$ is prime, then $p\equiv 3 \pmod 4$, so $-p$ is a fundamental discriminant, and $-p\equiv 5 \pmod{24}$; by Dirichlet's theorem there are infinitely many such primes. □

Table 8.17 shows the discriminants discussed in the first author's note. The four of class number one with 2 and 3 inert, $d=-19,-43,-67,-163$, take the least value of $H_K$ on $[1,\sqrt{|d|}/\pi]$ at the centre $x=1$, the two composite ones with $d\equiv 5 \pmod{24}$ inside the interval, and most of the positive kernels at its right end $\sqrt{|d|}/\pi$, where the term $n=1$ vanishes. All twenty values of $\zeta_K(\frac12)$ computed are negative, so $L(\frac12,\chi_d)>0$ for all of them whether or not the kernel is positive: Theorem 8.5.18(v) is a sufficient condition only.

The script also scans all 6079 fundamental discriminants $-20000\le d<0$. It evaluates $H_K$ in double precision on 3000 geometrically spaced points of $[1,\sqrt{|d|}/\pi]$, refines the least value by golden section search, and would recompute at 30 digits any least value within $10^{-8}$ of zero; there is none. The kernel is positive for 2738 of the discriminants. Table 8.18 sorts them by the splitting of 2 and 3: the proportion of positive kernels falls as either prime goes from split to ramified to inert. By blocks of 2000 in $|d|$ the proportion falls from 337/611 for $|d|\le 2000$ to between 0.41 and 0.45 in each of the five blocks above 10000.

| $d$ | $\chi_d(2)$ | $\chi_d(3)$ | $\Phi_K > 0$ | least $H_K$ on $[1, \sqrt{\lvert d\rvert}/\pi]$ | at $x =$ | $\zeta_K(\frac{1}{2})$ |
|---|---|---|---|---|---|---|
| $-11$ | $-1$ | 1 | yes | 0.14236 | 1.0557 | $-1.4480539$ |
| $-15$ | 1 | 0 | yes | 0.40606 | 1.2328 | $-2.6973176$ |
| $-20$ | 0 | 1 | yes | 0.287033 | 1.4235 | $-2.4529153$ |
| $-23$ | 1 | 1 | yes | 0.4634 | 1.5266 | $-3.5856997$ |
| $-24$ | 0 | 0 | yes | 0.225215 | 1.2409 | $-2.2955534$ |
| $-31$ | 1 | $-1$ | yes | 0.349008 | 1.7723 | $-3.2253855$ |
| $-39$ | 1 | 0 | yes | 0.409694 | 1.9878 | $-4.0547932$ |
| $-47$ | 1 | 1 | yes | 0.46368 | 2.1822 | $-4.8243519$ |
| $-71$ | 1 | 1 | yes | 0.470667 | 2.6821 | $-5.9925928$ |
| $-191$ | 1 | 1 | yes | 0.470667 | 4.3991 | $-8.1483722$ |
| $-19$ | $-1$ | $-1$ | no | $-0.0596605$ | 1 | $-1.1747378$ |
| $-43$ | $-1$ | $-1$ | no | $-0.214707$ | 1 | $-0.76635571$ |
| $-67$ | $-1$ | $-1$ | no | $-0.251274$ | 1 | $-0.54460945$ |
| $-163$ | $-1$ | $-1$ | no | $-0.315818$ | 1 | $-0.10008066$ |
| $-427$ | $-1$ | $-1$ | no | $-0.181949$ | 3.1887 | $-0.20993694$ |
| $-1003$ | $-1$ | $-1$ | no | $-0.228309$ | 3.9479 | $-0.64248774$ |

Table 8.17: The Pólya kernel $\Phi_K$ of $K = \mathbb{Q}(\sqrt{d})$ for the discriminants named in the first author's note: the characters at 2 and 3, the sign of the kernel, the least value of $H_K$ on $[1, \sqrt{\lvert d\rvert}/\pi]$, which decides it by Theorem 8.5.18(iv), with the point where it is attained, and $\zeta_K(\frac{1}{2}) = \zeta(\frac{1}{2})L(\frac{1}{2}, \chi_d)$. Computed by `sqfree_quadratic.py`.

| | $\chi_d(3) = 1$ | $\chi_d(3) = 0$ | $\chi_d(3) = -1$ |
|---|---|---|---|
| $\chi_d(2) = 1$ | 676/767 | 334/507 | 412/757 |
| $\chi_d(2) = 0$ | 489/753 | 178/514 | 131/757 |
| $\chi_d(2) = -1$ | 397/761 | 121/500 | 0/763 |

Table 8.18: Positive Pólya kernels $\Phi_K$ among the 6079 fundamental discriminants $-20000 \le d < 0$, by the splitting of 2 and 3 in $K = \mathbb{Q}(\sqrt{d})$ ($\chi_d(p) = 1$ split, 0 ramified, $-1$ inert): the number of positive kernels over the number of discriminants in each class, 2738 positive in all. The empty corner is Theorem 8.5.19. Computed by `sqfree_quadratic.py`.

Question 8.5.20. Does the proportion of fundamental discriminants $-X \le d < 0$ with $\Phi_K > 0$ tend to a limit as $X \to \infty$, and is the limit positive? Is $\Phi_K$ positive for infinitely many $d$, for instance among the $d$ in which 2 and 3 split, where the proportion up to 20000 is 676/767?

Question 8.5.21. For real quadratic fields and for fields of higher degree the gamma factor $\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}$ replaces the exponentials in $\vartheta_K$ by other kernels, for $\Gamma_{\mathbb{R}}(s)^2$ a Bessel function $K_0$. Is there a criterion for the positivity of the Pólya kernel of $\xi_K$ as simple as Theorem 8.5.18(iv)?

### 8.5.7 The Dirichlet beta function and the eta cube kernel

The $L$ function of the odd character $\chi_4$ modulo 4,

$$L(s,\chi_4)=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)^s}=\prod_p\bigl(1-\chi_4(p)p^{-s}\bigr)^{-1},\qquad \operatorname{Re}s>1,$$

is the Dirichlet beta function, with $L(1,\chi_4)=\frac{\pi}{4}$ and $L(2,\chi_4)$ Catalan's constant. It is the character $\chi_{-4}$ of Section 8.5.6, whose kernel is the cube of the Dedekind eta function and positive by the product formula. This subsection carries the Thorin framework over to it, following the first author's note [425]. The comparison across vertical lines of the note, which applies to $\xi$ and $L(s,\chi_{12})$ as well, is in Section 9.1.6.

#### *8.5.7.1 The kernel, the exit time and the zeros*

Jacobi's identity $\eta(\tau)^3=q^{1/8}\prod_{m\ge1}(1-q^m)^3=\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8}$, with $q=e^{2\pi i\tau}$ (Section 8.5.6), gives at $\tau=ix$ the theta series of $\chi_4$,

$$\eta(ix)^3=\sum_{n\ge1}\chi_4(n)\,n\,e^{-\pi n^2x/4}=e^{-\pi x/4}\prod_{m\ge1}\bigl(1-e^{-2\pi mx}\bigr)^3,\qquad x>0,$$

and the transformation $\eta(i/x)=\sqrt{x}\,\eta(ix)$ gives $\eta(i/x)^3=x^{3/2}\eta(ix)^3$. In the normalisation of Section 8.5.1 for odd characters the completed function is

$$\xi(s,\chi_4)=\Bigl(\frac{4}{\pi}\Bigr)^{(s+1)/2}\Gamma\Bigl(\frac{s+1}{2}\Bigr)L(s,\chi_4)=\int_0^\infty\eta(ix)^3\,x^{(s+1)/2}\,\frac{dx}{x},$$

the integral by termwise integration for $\operatorname{Re}s>1$, since $\int_0^\infty e^{-\pi n^2x/4}x^{(s+1)/2-1}dx=\Gamma(\frac{s+1}{2})(4/\pi n^2)^{(s+1)/2}$, and for every $s$ by analytic continuation, since $\eta(ix)^3$ decays like $e^{-\pi x/4}$ as $x\to\infty$ and, by the transformation, like $x^{-3/2}e^{-\pi/(4x)}$ as $x\to0$. The transformation gives $\xi(s,\chi_4)=\xi(1-s,\chi_4)$, the root number of $\chi_4$ being one, and $\xi(s,\chi_4)$ is entire of order one, with no factor $s(s-1)$ to remove. It is $2/\sqrt{\pi}$ times the function $\Lambda$ of [425], and $\xi(1,\chi_4)=\frac{4}{\pi}L(1,\chi_4)=1$.

Proposition 8.5.22. *The kernel $K_4(u)=2e^{3u/2}\eta(ie^{2u})^3$ satisfies $\xi(\frac12+s,\chi_4)=\int_{\mathbb R}K_4(u)\,e^{su}\,du$ for every complex $s$. It is even, positive and analytic in the strip $|\operatorname{Im}u|<\pi/4$, it has the symmetric product form*

$$K_4(u)=2\exp\Bigl(-\frac{\pi}{4}\cosh2u\Bigr)\prod_{m\ge1}\bigl(1-e^{-2\pi me^{2u}}\bigr)^{3/2}\bigl(1-e^{-2\pi me^{-2u}}\bigr)^{3/2},$$

*so that $K_4(u)=2\exp\bigl(\frac32|u|-\frac{\pi}{4}e^{2|u|}\bigr)\bigl(1+O(e^{-2\pi e^{2|u|}})\bigr)$, and $K_4(0)=2\eta(i)^3=\Gamma(\frac14)^3/(4\pi^{9/4})=0.906767655167731$.*

*Proof.* With $x = e^{2u}$, $x^{(s+1)/2}dx/x = 2e^{(s+1)u}du$, so the Mellin form with $\frac{1}{2}+s$ in place of $s$ is the stated integral. The product formula makes every factor of $K_4(u) = 2e^{3u/2}e^{-\frac{\pi}{4}e^{2u}}\prod_m(1-e^{-2\pi m e^{2u}})^3$ positive. The transformation gives $K_4(-u) = 2e^{-3u/2}\eta(ie^{-2u})^3 = 2e^{-3u/2}e^{3u}\eta(ie^{2u})^3 = K_4(u)$. Hence $K_4(u) = \sqrt{K_4(u)K_4(-u)}$, and in the product $K_4(u)K_4(-u)$ the factors $e^{\pm 3u/2}$ cancel while $e^{-\frac{\pi}{4}(e^{2u}+e^{-2u})} = e^{-\frac{\pi}{2}\cosh 2u}$, which gives the symmetric form. For $u \to +\infty$ the product for $K_4(u)$ itself gives the asymptotic, and evenness gives it for $u \to -\infty$. The function $\eta(ie^{2u})$ is analytic where $\operatorname{Re} e^{2u} > 0$, that is for $|\operatorname{Im} u| < \pi/4$. Finally $\eta(i) = \Gamma(\frac{1}{4})/(2\pi^{3/4})$. □

The evaluator of Section 2.1 applies unchanged to $K_4$. The script `dbeta_thorin.py` computes $\xi(\frac{1}{2}+s,\chi_4)$ from $K_4$ by the trapezoid rule with step $1/64$ on $|u| \le 3$ and agrees with the Dirichlet $L$ function of Arb [290] in all fourteen digits printed at $s = 0$, $\frac{1}{2}$, $0.1+5i$ and $0.1+20i$; for instance $\xi(\frac{1}{2},\chi_4) = 0.98071361405771$, $\xi(0.6+5i,\chi_4) = 0.080871220951695 + 0.011736751015145i$ and $\xi(0.6+20i,\chi_4) = (-2.3183027026127 - 0.20613003173368i)\times 10^{-6}$. Also $L(\frac{1}{2},\chi_4) = 0.667691457189609$.

Remark 8.5.23 (The exit time). By Theorem 4.3.20, $\frac{4}{\pi}\eta(4it/\pi)^3$ is the density of $X_C = \sum_k e_k/(2k-1)^2$, which is $\frac{\pi^2}{8}$ times the exit time $\tau$ of Brownian motion from $(-1,1)$ (Section 4.2.2). With $x = 4t/\pi$, $\eta(ix)^3$ is therefore the density on $(0,\infty)$ of $Z = \frac{4}{\pi}X_C = \frac{\pi}{2}\tau$ [432, Section 3], and

$$\xi(s,\chi_4) = \mathbb{E}\big[Z^{(s-1)/2}\big], \qquad \mathbb{E}e^{-uZ} = \operatorname{sech}\sqrt{\pi u} = \prod_{k\ge 1}\Big(1+\frac{4u}{\pi(2k-1)^2}\Big)^{-1}.$$

So $Z$ is a GGC with Thorin measure $\sum_k \delta_{\pi(2k-1)^2/4}$; its total mass $\xi(1,\chi_4) = 1$ is the value $L(1,\chi_4) = \frac{\pi}{4}$, and its mean $\xi(3,\chi_4) = \frac{4}{\pi}\sum_k(2k-1)^{-2} = \frac{\pi}{2}$ is the value $L(3,\chi_4) = \pi^3/32$. The transformation $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$ says that $1/Z$ has the density $x^{-2}\eta(i/x)^3 = x^{-1/2}\eta(ix)^3$, that is $\mathbb{E}g(1/Z) = \mathbb{E}[Z^{-1/2}g(Z)]$ for $g \ge 0$, and with $g(z) = z^{(1-s)/2}$ this is the functional equation. The primal variable $Y_4$, with density $K_4/\xi(\frac{1}{2},\chi_4)$ and $\mathbb{E}e^{sY_4} = \xi(\frac{1}{2}+s,\chi_4)/\xi(\frac{1}{2},\chi_4)$, is $\frac{1}{2}\log Z$ under the size biasing by $Z^{-1/4}$, since $\mathbb{E}[Z^{-1/4}e^{(s/2)\log Z}]/\mathbb{E}[Z^{-1/4}] = \mathbb{E}[Z^{s/2-1/4}]/\mathbb{E}[Z^{-1/4}]$. So the analogue of the Pólya variable is half the logarithm of a Brownian exit time biased by $Z^{-1/4}$, as for $\zeta$ it is half the logarithm of the variable $\pi S_2/2$ of Williams and of Biane, Pitman and Yor biased by $(\pi S_2/2)^{1/4}$ (Section 8.2.3). In the notation of Section 11.5.5.4, $Z = 1/\mathcal{B}_{\chi_{-4}}$ (Proposition 11.5.19).

The basepoint one has an explicit drift. By the functional equation $b_1 = \frac{\xi'}{\xi}(1,\chi_4) = -\frac{\xi'}{\xi}(0,\chi_4)$. With $L(s,\chi_4) = 4^{-s}\big(\zeta(s,\frac{1}{4}) - \zeta(s,\frac{3}{4})\big)$, $\zeta(0,a) = \frac{1}{2}-a$ and Lerch's formula $\partial_s\zeta(0,a) = \log\Gamma(a) - \frac{1}{2}\log 2\pi$ [397, §25.11], one has $L(0,\chi_4) = \frac{1}{2}$ and $L'(0,\chi_4) = \log\big(\Gamma(\frac{1}{4})/2\Gamma(\frac{3}{4})\big)$, and with $\psi(\frac{1}{2}) = -\gamma_E - 2\log 2$ and $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) =$

$\pi\sqrt{2}$,

$$b_1 = 3\log 2 + \frac{5}{2}\log\pi + \frac{\gamma_E}{2} - 4\log\Gamma\Big(\frac{1}{4}\Big) = 0.0777839899618, \tag{8.5.7}$$

and so, since $b_1 = \frac{1}{2}\log\frac{4}{\pi} - \frac{\gamma_E}{2} + \frac{4}{\pi}L'(1,\chi_4)$, $L'(1,\chi_4) = \frac{\pi}{4}\big(\gamma_E + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac{1}{4})\big) = 0.192901316797$. At the centre $\xi'(\frac{1}{2},\chi_4) = 0$, and the variance of the primal variable is $V_4 = \operatorname{Var} Y_4 = (\log\xi)''(\frac{1}{2},\chi_4) = 0.156029649890$.

Since $K_4 > 0$, $\xi(\frac{1}{2}+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}du > 0$ for every real $s$, so $\xi(s,\chi_4)$ has no real zeros. The script `dbeta_zeros.py` computes the zeros of $L(s,\chi_4)$ below height 100 in ball arithmetic with Arb. The function $\xi(\frac{1}{2}+i\theta,\chi_4)$ is real; its sign is certified on a grid of step $1/32$, and each sign change is refined by bisection to $10^{-12}$. The argument principle along $2 \to 2+100i \to \frac{1}{2}+100i$, on pieces halved until the enclosure of $\xi(s,\chi_4)$ over each piece lies in a disc excluding zero (1633 pieces), gives $N(100) = 50$ for the number of zeros with $0 < \gamma \le 100$, counted with multiplicity, the remaining sides of the rectangle $[-1,2]\times[0,100]$ following by the functional equation and $\xi(\bar{s},\chi_4) = \overline{\xi(s,\chi_4)}$. This equals the number of sign changes, so all fifty zeros are simple and on the critical line. The zero counting function satisfies

$$N(T) = \frac{T}{2\pi}\log\frac{2T}{\pi e} + \frac{1}{8} + \frac{1}{\pi}\arg L\big(\tfrac{1}{2}+iT,\chi_4\big) + \frac{1}{48\pi T} + O(T^{-3}), \tag{8.5.8}$$

with the argument continued from $L(2,\chi_4) > 0$ along $2 \to 2+iT \to \frac{1}{2}+iT$, for $T$ not an ordinate. Indeed, by the argument principle on the rectangle $[-1,2]\times[0,T]$, the functional equation, the reflection $\xi(\bar{s},\chi_4) = \overline{\xi(s,\chi_4)}$ and the positivity of $\xi(s,\chi_4)$ on the real axis, $\pi N(T)$ is the change of $\arg\xi(s,\chi_4)$ along $2 \to 2+iT \to \frac{1}{2}+iT$, and the expansion of the gamma factor by Stirling's series [397, §5.11] in the proof of Proposition 8.5.31 below, at $\alpha = \frac{1}{2}$, gives (8.5.8); compare [104, Chapter 16], where the main term is derived with an error $O(\log T)$. Under GRH for $\chi_4$, $V_4 = \sum_\gamma \gamma^{-2}$ over all the ordinates of both signs; twice the sum over the fifty computed ones is 0.1395621, and the main term $\frac{1}{\pi}(\log\frac{2T}{\pi}+1)/T$ of the rest, from $dN(t) \approx \frac{1}{2\pi}\log\frac{2t}{\pi}\,dt$, is 0.0164 at $T = 100$, together 0.1560.

### *8.5.7.2 The Lévy form and the sine squared decomposition*

The Lévy form of Proposition 11.3.2 for $\chi_{12}$ has an analogue for $\chi_4$ at every basepoint $\alpha \ge 1$.

Proposition 8.5.24. *For $\alpha \ge 1$,*

$$\log\frac{\xi(\alpha+s,\chi_4)}{\xi(\alpha,\chi_4)} = b_\alpha s + \int_0^\infty \big(e^{-sx} - 1 + sx\big)\,\Pi_\alpha(dx), \qquad b_\alpha = \frac{\xi'}{\xi}(\alpha,\chi_4),$$

*with the signed Lévy measure*

$$\Pi_\alpha(dx) = \frac{e^{-\alpha x}}{2x\sinh x}\,dx + \sum_{n\ge 2}\frac{\chi_4(n)\Lambda(n)}{n^\alpha\log n}\,\delta_{\log n}(dx),$$

*whose canonical measure* $x\Pi_\alpha(dx)$ *has the positive density* $e^{-\alpha x}/(2\sinh x)$ *and the atoms* $\chi_4(n)\Lambda(n)n^{-\alpha}$ *at* $\log n$. *For* $\alpha > 1$ *the form holds for* $\operatorname{Re} s > 1-\alpha$, *with absolute convergence. At* $\alpha = 1$ *it holds for* $\operatorname{Re} s > 0$, *the prime sum converging conditionally, as a limit over* $n \le X$.

*Proof.* The factor $(4/\pi)^{(s+1)/2}$ is linear in log and is absorbed by the drift. Binet's formula [397, §5.9], in the integrated form

$$\log\frac{\Gamma(z+v)}{\Gamma(z)} - v\psi(z) = \int_0^\infty (e^{-vx} - 1 + vx)\frac{e^{-zx}}{x(1-e^{-x})}\,dx$$

for $\operatorname{Re} z > 0$ and $\operatorname{Re}(z+v) > 0$, which one checks by differentiating twice in $v$, gives at $z = \frac{\alpha+1}{2}$, $v = \frac{s}{2}$ and after $x \mapsto 2x$ the density $\frac{e^{-(\alpha+1)x}}{x(1-e^{-2x})} = \frac{e^{-\alpha x}}{2x\sinh x}$, for $\operatorname{Re} s > -1-\alpha$. For $\operatorname{Re}(\alpha+s) > 1$ the Euler product gives $\log L(\alpha+s,\chi_4) = \sum_n \chi_4(n)\Lambda(n)n^{-\alpha-s}/\log n$, and subtracting the value and the derivative at $s=0$ gives the atoms; for $\alpha > 1$ the sum converges absolutely for $\operatorname{Re} s > 1-\alpha$, because $|e^{-sx}-1+sx| \le e^{-x\operatorname{Re} s}+1+|s|x$ and $\sum_n \Lambda(n)n^{-\alpha}\bigl(n^{-\operatorname{Re} s}+1+|s|\log n\bigr)/\log n < \infty$ there. At $\alpha = 1$ the linear coefficient is $-\sum_n \chi_4(n)\Lambda(n)/n = \frac{L'}{L}(1,\chi_4)$ and the constant term is $\sum_n \chi_4(n)\Lambda(n)/(n\log n) = \log L(1,\chi_4)$; both series converge by the prime number theorem for arithmetic progressions [104, Chapter 20], as in the proof of Proposition 11.3.2. □

The archimedean density is that of $\xi$ shifted by one basepoint: at the basepoint $\alpha+1$ the gamma density $e^{-(\alpha+2)x}/(1-e^{-2x})$ of $x\Pi_{\alpha+1}$ is the gamma density of $\xi$ at the basepoint $\alpha$ (Section 9.1.4), because the gamma factor $\Gamma(\frac{s+1}{2})$ at $\alpha+1$ and the gamma factor $\Gamma(1+\frac{s}{2})$ of $\xi$, which absorbs the factor $s$, at $\alpha$ have the same argument $\frac{\alpha+2+s}{2}$. There is no pole term, and the atoms carry the sign of $\chi_4(n)$. As for $\chi_{12}$ in Section 11.3.2, the tilted variable $V_\alpha^{(4)}$ with moment generating function $\xi(\alpha+s,\chi_4)/\xi(\alpha,\chi_4)$ is quasi infinitely divisible for $\alpha > 1$, the Euler product writing its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions, and it is not infinitely divisible: by the uniqueness of the Lévy measure of a quasi infinitely divisible law [341], the atom $-3^{-\alpha}$ of $\Pi_\alpha$ at $\log 3$, reflected to $-\log 3$, would have to be positive. At $\alpha = 1$, $V_1^{(4)}$ is not quasi infinitely divisible, since the atoms of each sign have infinite mass, $\sum_{p\equiv 1} 1/p$ and $\sum_{p\equiv 3\,(4)} 1/p$ both diverging, and cancel only conditionally; the argument of Proposition 11.1.3(3) applies.

The sine squared lemma (1.2.9) turns the Lévy form into the Thorin density, the bracket being $\operatorname{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4)$ [421, Theorem 8.2]. With $v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,\chi_4)$ and $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$ as in (9.1.7),

$$v_\alpha(\theta) = b_\alpha + G_\alpha(\theta) + \varpi_\alpha(\theta), \qquad \alpha \ge 1, \tag{8.5.9}$$

where the gamma piece and the prime piece are

$$G_\alpha(\theta) = \int_0^\infty \frac{1-\cos\theta x}{2\sinh x}\, e^{-\alpha x}dx = \frac{1}{2}\Big[\operatorname{Re}\psi\Big(\frac{\alpha+1+i\theta}{2}\Big) - \psi\Big(\frac{\alpha+1}{2}\Big)\Big]$$
$$= \sum_{k\ge 0} q_{\alpha+1+2k}(\theta) \ge 0,$$
$$\varpi_\alpha(\theta) = \sum_{n\ge 2} \frac{\chi_4(n)\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big),$$

the series for $G_\alpha$ from $1/(2\sinh x) = \sum_{k\ge 0} e^{-(2k+1)x}$, and $\varpi_\alpha$ converging conditionally at $\alpha = 1$. So $G_\alpha = \sum_{k\ge 1} q_{(\alpha-1)+2k}$ is the gamma piece of $\xi$ at the basepoint $\alpha - 1$ in (9.1.7), without its pole term $-q_{\alpha-2}$. For $\alpha \ge 1$ the Thorin representation is

$$\log\frac{\xi(\alpha+\sqrt{u},\chi_4)}{\xi(\alpha,\chi_4)} = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\, v_\alpha(\theta)\, d\theta, \qquad u \ge 0, \tag{8.5.10}$$

by Proposition 8.5.25 below.

The first zero of $\xi(\frac{1}{2}+i\theta,\chi_4)$ is $\gamma_1 = 6.0209489$.

Figure 8.7 shows the density and the decomposition. On the left, $v_\alpha$ at $\alpha = 1$, 0.75 and 0.55. Its minimum is $b_\alpha$, at the origin (0.0777840, 0.0389784 and 0.0078013), as Theorem 9.1.18 proves for every $\alpha \in (\frac{1}{2}, 4.70]$ on the verified range and for every $\alpha \ge 1$ at every height. Near each ordinate $v_\alpha$ rises to about $1/(\alpha-\frac{1}{2})$, the height of the Cauchy kernel of a zero on the line, plus the contribution of the other zeros. As $\alpha \downarrow \frac{1}{2}$ the margin at the origin shrinks linearly, $b_\alpha \approx (\alpha-\frac{1}{2})V_4$. On the right, at $\alpha = \frac{3}{2}$, the prime piece $\varpi_\alpha$ is negative on 71% of the grid of step 0.01 on $[0,30]$, with maximum 0.33562 at $\theta = 6.0$, next to $\gamma_1$; the minimum of $v_\alpha$ on the grid is $b_{3/2} = 0.15421515$, at the origin. For $\xi$ the same figure would show a positive prime piece and the negative pole piece (Figure 8.4), while for $\chi_4$ the negative input is spread over the primes $p \equiv 3 \pmod 4$.

Proposition 8.5.25. *For every $\alpha \ge 1$, $u \mapsto \xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$ is the Laplace transform of a GGC with Thorin density $v_\alpha(\theta)/\pi > 0$ in the chart $\theta = \sqrt{z}$, and* (8.5.10) *holds. For $\alpha > \frac{1}{2}$ it is the Laplace transform of a GGC if and only if no zero of $L(s,\chi_4)$ has real part greater than $\alpha$. GRH for $\chi_4$ holds if and only if $v_\alpha \ge 0$ for every $\alpha > \frac{1}{2}$, and the Thorin measures at resolution $\varepsilon = \alpha - \frac{1}{2}$ converge vaguely, as $\varepsilon \downarrow 0$, to the sum of the point masses $\delta_{\gamma^2}$ over the zeros on the critical line, which is the Thorin measure of $\xi(\frac{1}{2},\chi_4)/\xi(\frac{1}{2}+\sqrt{u},\chi_4)$ under GRH.*

*Proof.* The proofs of Theorem 1.2.2, of Lagarias's criterion [320] and of the vague limit stated after Remark 1.2.4 use only the Hadamard product of an entire function of order one whose zeros are symmetric under $\rho \mapsto \bar\rho$ and $\rho \mapsto 1-\rho$, with $\sum_\rho |\rho|^{-2} < \infty$; they apply to $\xi(s,\chi_4)$ [104, Chapter 12], as Section 8.5.3 and Proposition 11.3.1 apply them to $\xi(s,\chi_{12})$. Positivity at $\alpha \ge 1$ needs every zero to

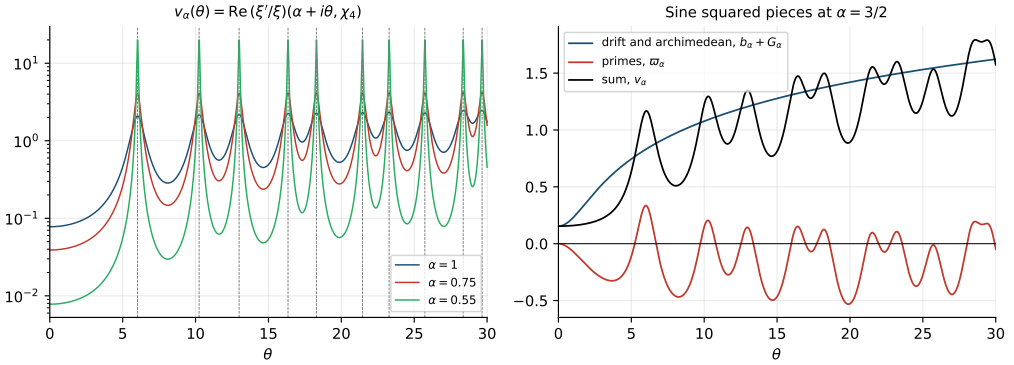



Figure 8.7: Left: $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta, \chi_4) = \pi\rho_{\alpha-1/2}(\theta)$ for $\alpha = 1$, 0.75, 0.55 on a logarithmic scale, with the ordinates of the zeros of $L(s, \chi_4)$ (dashed). Right: at $\alpha = \frac{3}{2}$, the drift and gamma piece $b_\alpha + G_\alpha$, the prime piece $\varpi_\alpha$, of both signs, and their sum $v_\alpha$. Drawn by `dbeta_figs.py`.

have real part less than one, which is the absence of zeros of $L(s, \chi_4)$ on $\mathrm{Re}\, s = 1$ [104, Chapter 14] with the Euler product and the functional equation. □

At the basepoint one the clock $T_1^{(4)}$, with Laplace transform $1/\xi(1 + \sqrt{u}, \chi_4)$, therefore has the expansion $1 - b_1\sqrt{u} + O(u)$ at the origin, and as for $\xi$ in (9.1.3) the Tauberian theorem [136, XIII.5] gives the tail $P(T_1^{(4)} > x) \sim (b_1/\sqrt{\pi})\, x^{-1/2} = 0.0438849\, x^{-1/2}$, with $b_1$ from (8.5.7).

Positivity at $\alpha \geq 1$ uses the zeros, not the Lévy measure. Bounding the prime weights by their absolute values gives positivity only far to the right, and no such bound does better.

Proposition 8.5.26. *For $\alpha > 1$ and real $\theta$,*

$$v_\alpha(\theta) \geq A_4^\infty(\alpha, \theta) - S_4(\alpha),$$
$$A_4^\infty(\alpha, \theta) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\,\mathrm{Re}\,\psi\Big(\frac{\alpha + 1 + i\theta}{2}\Big),$$
$$S_4(\alpha) = \sum_{n \text{ odd}} \frac{\Lambda(n)}{n^\alpha} = -\frac{\zeta'}{\zeta}(\alpha) - \frac{\log 2}{2^\alpha - 1},$$

*and $A_4^\infty(\alpha, \theta)$ increases in $|\theta|$. The bound is positive at every $\theta$ exactly when $\alpha > \alpha_c(\chi_4) = 2.2823106368$, the root of $A_4^\infty(\alpha, 0) = S_4(\alpha)$. For $\chi_{12}$ the same bound, with $A_{12}^\infty(\alpha, \theta) = \frac{1}{2}\log\frac{12}{\pi} + \frac{1}{2}\,\mathrm{Re}\,\psi(\frac{\alpha + i\theta}{2})$ and $S_{12}(\alpha) = S_4(\alpha) - \log 3/(3^\alpha - 1)$, is positive at every $\theta$ exactly when $\alpha > \alpha_c(\chi_{12}) = 1.8348743295$.*

*Proof.* For $\mathrm{Re}\, s > 1$,

$$\frac{\xi'}{\xi}(s, \chi_4) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\psi\Big(\frac{s+1}{2}\Big) + \frac{L'}{L}(s, \chi_4),$$

and $|\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)| \le \sum_n |\chi_4(n)|\Lambda(n)n^{-\alpha} = S_4(\alpha)$. The series $\operatorname{Re}\psi(x+iy)-\psi(x) = \sum_{k\ge0} y^2/\big((x+k)((x+k)^2+y^2)\big)$ shows that $A_4^\infty$ increases in $|\theta|$. Since $A_4^\infty(\alpha,0)$ increases in $\alpha$ and $S_4(\alpha)$ decreases, the root is unique, and `dbeta_thorin.py` locates it by bisection with certified signs. For $\chi_{12}$ the primes 2 and 3 drop out of the sum. □

Proposition 8.5.27. *For $\alpha>1$ and every $T>0$,*

$$\inf_{\theta\ge T}\Big(v_\alpha(\theta)-A_4^\infty(\alpha,\theta)\Big)=\inf_{\theta\ge T}\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)=-S_4(\alpha),$$

*so no lower bound for $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)$ better than $-S_4(\alpha)$ holds at all large heights. The bound of Proposition 8.5.26 certifies $v_\alpha(\theta)>0$ exactly for $|\theta|>t_0(\alpha)$, where $A_4^\infty(\alpha,t_0)=S_4(\alpha)$, for $1<\alpha\le\alpha_c$ (so that $t_0(\alpha_c)=0$), and at every real $\theta$ for $\alpha>\alpha_c$, and*

$$\log t_0(\alpha)=\frac{2}{\alpha-1}-2\gamma_E-\log\frac{8}{\pi}+O(\alpha-1),\qquad \alpha\downarrow1,$$

*with $-2\gamma_E-\log\frac{8}{\pi}=-2.0891$.*

*Proof.* From the Euler product, $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)=-\sum_p \log p\,\operatorname{Re}\frac{w_p}{1-w_p}$ with $w_p=\chi_4(p)p^{-\alpha-i\theta}$, and on the circle $|w|=r<1$ the maximum of $\operatorname{Re}\frac{w}{1-w}=\operatorname{Re}\frac{1}{1-w}-1$ is $\frac{r}{1-r}$, at $w=r$; this gives the lower bound $-S_4(\alpha)$. Since the numbers $\log p$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem [174, Chapter 23] gives, for every $P$ and $\eta>0$, a height $\theta\ge T$ at which $|\chi_4(p)p^{-i\theta}-1|<\eta$ for all odd $p\le P$, and the primes beyond $P$ contribute at most $\sum_{p>P}\log p\,p^{-\alpha}/(1-p^{-\alpha})$, which is small for $P$ large. For the asymptotic, $S_4(\alpha)=\frac{1}{\alpha-1}-\gamma_E-\log2+O(\alpha-1)$ and $A_4^\infty(\alpha,t)=\frac12\log\frac{2t}{\pi}+O(t^{-2})$ as $t\to\infty$, uniformly for $1\le\alpha\le2$, from $\psi(z)=\log z-\frac{1}{2z}+O(|z|^{-2})$; solving $A_4^\infty(\alpha,t_0)=S_4(\alpha)$ gives the formula, since $t_0(\alpha)\to\infty$. □

| $\alpha$ | $S_4(\alpha)$ | $t_0(\alpha)$ | $\log t_0(\alpha)-\frac{2}{\alpha-1}$ |
|---|---|---|---|
| 2 | 0.33891193 | 2.36723 | −1.1383 |
| 1.75 | 0.57950941 | 4.71677 | −1.1155 |
| 1.5 | 1.1261406 | 14.8728 | −1.3005 |
| 1.25 | 2.9637961 | 589.44 | −1.6208 |
| 1.1 | 8.8348984 | $7.41336\times10^{7}$ | −1.8786 |
| 1.05 | 18.784554 | $3.25219\times10^{16}$ | −1.9793 |

Table 8.19: The height $t_0(\alpha)$ above which the bound of Proposition 8.5.26 certifies $v_\alpha>0$, for basepoints below $\alpha_c(\chi_4)=2.2823$; the last column tends to $-2\gamma_E-\log\frac{8}{\pi}=-2.0891$ (Proposition 8.5.27). Computed by `dbeta_thorin.py`.

So for $1 < \alpha < \alpha_c$ the absolute values of the weights reduce the positivity of $v_\alpha$ to a computation on $[0, t_0(\alpha)]$, but $t_0$ grows like $e^{2/(\alpha-1)}$ (Table 8.19). At $\alpha = 1$ the prime piece converges only conditionally, by the prime number theorem for progressions, whose analytic input is the absence of zeros of $L(s, \chi_4)$ on $\operatorname{Re} s = 1$; the positivity of $v_1$, which Proposition 8.5.25 draws from that input, cannot come from the sizes of the weights.

### *8.5.7.3 The Gaussian field*

The pairing of the primes $p \equiv 1$ and $p \equiv 3 \pmod 4$ is their splitting in $K = \mathbb{Q}(i)$. Accordingly $\zeta_K(s) = \zeta(s)L(s, \chi_4)$, and in the normalisation of Section 10.3.3,

$$\xi_K(s) = 2s(s-1)\pi^{-s}\Gamma(s)\zeta_K(s) = \xi(s)\,\xi(s, \chi_4),$$

by the duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$. Write $v_\alpha^K$ and $v_\alpha^\xi$ for the brackets $\operatorname{Re}(\xi_K'/\xi_K)(\alpha + i\theta)$ and $\operatorname{Re}(\xi'/\xi)(\alpha + i\theta)$, and $v_\alpha$ as before for $\chi_4$.

Proposition 8.5.28. *On every vertical line, $v_\alpha = v_\alpha^K - v_\alpha^\xi$. For $\alpha > 1$ the canonical measure of $\xi_K$ at the basepoint $\alpha$ is*

$$x\Pi_\alpha^K(dx) = \Big(\frac{e^{-(\alpha+1)x}}{1-e^{-x}} - e^{-(\alpha-1)x}\Big)dx + \sum_{n\geq 2} \frac{(1+\chi_4(n))\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$

*and the canonical measure of Proposition 8.5.24 is $x\Pi_\alpha = x\Pi_\alpha^K - x\Pi_\alpha^\xi$, with $x\Pi_\alpha^\xi$ that of Section 9.1.4: the pole densities $-e^{-(\alpha-1)x}$ cancel, the gamma densities differ by $e^{-\alpha x}/(2\sinh x)$, and the atoms by $\chi_4(n)\Lambda(n)n^{-\alpha}$.*

*Proof.* Take logarithmic derivatives in $\xi_K = \xi \cdot \xi(\cdot, \chi_4)$. For the measures write $\xi_K(s) = (s-1)\cdot 2\pi^{-s}\Gamma(s+1)\zeta_K(s)$: Binet's formula for $\Gamma(s+1)$ at $\alpha$ gives the density $e^{-(\alpha+1)x}/(x(1-e^{-x}))$, the factor $s-1$ gives $-e^{-(\alpha-1)x}/x$ as for $\xi$, and $-\frac{\zeta_K'}{\zeta_K} = -\frac{\zeta'}{\zeta} - \frac{L'}{L}$ has the coefficients $(1+\chi_4(n))\Lambda(n) \geq 0$. Finally $\frac{e^{-(\alpha+1)x}}{1-e^{-x}} - \frac{e^{-(\alpha+2)x}}{1-e^{-2x}} = e^{-\alpha x}\Big(\frac{1}{e^x-1} - \frac{1}{e^{2x}-1}\Big) = \frac{e^{-\alpha x}e^x}{e^{2x}-1} = \frac{e^{-\alpha x}}{2\sinh x}$. □

Corollary 8.5.29. (i) *For every $\alpha \geq 1$ there are independent GGC variables $H_\alpha^\xi$ and $H_\alpha^{(4)}$, with Laplace transforms $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ and $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, whose sum $H_\alpha^K = H_\alpha^\xi + H_\alpha^{(4)}$ is the GGC with Laplace transform $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ and Thorin density $v_\alpha^K/\pi$.*

(ii) *GRH for $\chi_4$ holds if and only if $v_\alpha^K \geq v_\alpha^\xi$ on every line $\operatorname{Re} s = \alpha > \frac{1}{2}$: the Thorin density of $\xi_K$ dominates that of $\xi$ at every basepoint.*

(iii) *Under RH, GRH for $\chi_4$ holds if and only if, for every $\alpha > \frac{1}{2}$, $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ is the Laplace transform of $H_\alpha^\xi + H$ for some GGC $H$ independent of $H_\alpha^\xi$.*

*Proof.* (i) By Theorem 1.2.2 and Proposition 8.5.25 both factors are GGC transforms for $\alpha \geq 1$, and transforms of independent sums multiply. (ii) By Proposition 8.5.28, $v_\alpha^K \geq v_\alpha^\xi$ is $v_\alpha \geq 0$, and Proposition 8.5.25 applies. (iii) Under RH, $H_\alpha^\xi$ is a GGC for every $\alpha > \frac{1}{2}$ (Theorem 1.2.2). If GRH holds for $\chi_4$, take $H = H_\alpha^{(4)}$. Conversely, the transform of $H$ is then $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, and if it is a GGC transform for every $\alpha > \frac{1}{2}$, Proposition 8.5.25 gives GRH. □

Neither $v_\alpha^K$ nor $v_\alpha^\xi$ is positive term by term: each has the pole piece $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ of (3.1.1). The factorisation is the right language for the pairing of the primes, but it does not by itself lower $\alpha_c$: by Proposition 8.5.27 any argument that uses only the sizes of the weights reaches no further. Table 8.20 shows the three brackets at $\alpha = \frac{3}{2}$.

| $\theta$ | 0 | 2 | 5 | 10 | 20 |
|---|---|---|---|---|---|
| $v_{3/2}^K(\theta)$ | 0.200351 | 0.225243 | 0.685189 | 1.32766 | 1.49863 |
| $v_{3/2}^\xi(\theta)$ | 0.0461359 | 0.0470396 | 0.0526739 | 0.102889 | 0.606114 |
| $v_{3/2}(\theta)$ | 0.154215 | 0.178204 | 0.632515 | 1.22477 | 0.892517 |

Table 8.20: The brackets of $\xi_{\mathbb{Q}(i)}$, $\xi$ and $\xi(\cdot,\chi_4)$ at the basepoint $\frac{3}{2}$; the last row is the difference of the first two (Proposition 8.5.28). Computed by `dbeta_thorin.py`.

The same holds for every real primitive character $\chi_D$, with $D$ a fundamental discriminant and $K = \mathbb{Q}(\sqrt{D})$: $\xi_K$ is a constant multiple of $\xi \cdot \xi(\cdot,\chi_D)$, the constant being $2/\sqrt{|D|}$ for $D < 0$ (Section 8.5.6) and 2 for $D > 0$, since then $\xi_K(s) = s(s-1)D^{s/2}\Gamma_{\mathbb{R}}(s)^2\zeta_K(s)$ and $\xi(s,\chi_D) = D^{s/2}\Gamma_{\mathbb{R}}(s)L(s,\chi_D)$. So $v_\alpha^{(\chi_D)} = v_\alpha^K - v_\alpha^\xi$; the GGC property at every $\alpha \geq 1$, the equivalence of GRH for $\chi_D$ with the domination of the Thorin density of $\xi$ by that of $\xi_K$, and the localisation of Corollary 8.5.32 hold for $\xi(s,\chi_D)$ with the same proofs. For $\chi_{12}$, $K = \mathbb{Q}(\sqrt{3})$, and the numbers are: $\alpha_c(\chi_{12}) = 1.8348743295$ (Proposition 8.5.26); $V_{12} = (\log\xi)''(\frac{1}{2},\chi_{12}) = 0.332923608212$; and at $\alpha = 1$, 0.75 and 0.6 the drift $b_\alpha$ is 0.1650833, 0.0830566 and 0.0332812, the minimum of $v_\alpha$ on a grid of step 0.01 on $[0,60]$, attained at the origin (`dbeta_minimum.py`; Theorem 9.1.18). The script `dbeta_zeros.py` certifies the first five ordinates of $L(s,\chi_{12})$, 3.8046276331, 6.6922233205, 8.8905929587, 11.1883927451 and 12.9661788080, all simple and on the critical line, in agreement with Section 8.5.1; their largest gap, the first, 2.8876, lies below $2/\sqrt{V_{12}} = 3.4662$, and below the value 3.3189 at $a = \frac{1}{2}$ of the threshold of Remark 9.1.20 for $\chi_{12}$.

Question 8.5.30. Is there a construction of $H_\alpha^K = H_\alpha^\xi + H_\alpha^{(4)}$ from the Gaussian primes themselves, two of norm $p$ for each $p \equiv 1$ and one of norm $p^2$ for each $p \equiv 3 \pmod 4$, that persists below the basepoint one? By Proposition 8.5.27 such a construction must see the phases $p^{-i\theta}$, not only the sizes of the weights; below

$\alpha = 1$ its existence is equivalent to GRH for $\chi_4$ under RH, by Corollary 8.5.29(iii). In the language of Section 8.5.7.4, the decomposition is the additivity $\arg \xi_K = \arg \xi + \arg \xi(\cdot, \chi_4)$ on vertical lines, and the domination of Corollary 8.5.29(ii) says that $\arg \xi_K - \arg \xi$ is nondecreasing on every line $\operatorname{Re} s = \alpha > \frac{1}{2}$.

*8.5.7.4 The Thorin measure as an argument*

PROPOSITION 8.5.31. *Let $\alpha > \frac{1}{2}$ be such that no zero of $\xi(s, \chi_4)$ lies on the line $\operatorname{Re} s = \alpha$; this holds for every $\alpha \geq 1$. Then for $\theta > 0$*

$$\frac{1}{\pi} \int_0^\theta v_\alpha(t)\, dt = \frac{1}{\pi} \arg \xi(\alpha + i\theta, \chi_4),$$

*with the argument continuous along the segment from $\alpha$ to $\alpha + i\theta$ and zero at $\alpha$; for $\alpha \geq 1$ the left side is the mass of $(0, \theta^2]$ under the Thorin measure of Proposition 8.5.25. As $\theta \to \infty$, with $a = \frac{\alpha+1}{2}$,*

$$\frac{1}{\pi} \arg \xi(\alpha + i\theta, \chi_4) = \frac{\theta}{2\pi} \log \frac{2\theta}{\pi e} + \frac{\alpha}{4} + \frac{1}{\pi} \arg L(\alpha + i\theta, \chi_4) - \frac{a(a-1) + \frac{1}{6}}{\pi\theta} + O(\theta^{-3}),$$

*with $\arg L$ continuous along the same segment. As $\alpha \downarrow \frac{1}{2}$, for $\theta$ not an ordinate, the left side tends to the number of zeros on the critical line with $0 < \gamma < \theta$, which is $N(\theta)$ under GRH; at $\alpha = \frac{1}{2}$ the expansion is the counting formula* (8.5.8).

*Proof.* Since $\frac{d}{dt} \log \xi(\alpha + it, \chi_4) = i \frac{\xi'}{\xi}(\alpha + it, \chi_4)$, the derivative of $\arg \xi(\alpha + it, \chi_4) = \operatorname{Im} \log \xi(\alpha + it, \chi_4)$ is $v_\alpha(t)$, and $\xi(\alpha, \chi_4) > 0$. The expansion is $\frac{\theta}{2} \log \frac{4}{\pi}$ plus Stirling's series $\log \Gamma(z) = (z - \frac{1}{2}) \log z - z + \frac{1}{2} \log 2\pi + \frac{1}{12z} + O(|z|^{-3})$ [397, §5.11] at $z = a + \frac{i\theta}{2}$: with $y = \frac{\theta}{2}$,

$$\operatorname{Im} \log \Gamma(a + iy) = y \log y - y + \frac{\pi}{2}\Big(a - \frac{1}{2}\Big) - \Big(\frac{a(a-1)}{2} + \frac{1}{12}\Big)\frac{1}{y} + O(y^{-3}),$$

from $\arg z = \frac{\pi}{2} - \frac{a}{y} + O(y^{-3})$ and $\log |z| = \log y + \frac{a^2}{2y^2} + O(y^{-4})$; and $\frac{\theta}{2} \log \frac{4}{\pi} + \frac{\theta}{2} \log \frac{\theta}{2} - \frac{\theta}{2} = \frac{\theta}{2} \log \frac{2\theta}{\pi e}$, $\frac{1}{2}(a - \frac{1}{2}) = \frac{\alpha}{4}$. For the limit, write $v_\alpha$ by the Hadamard product, as in Proposition 1.2.3, as the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$ of $(\alpha - \beta)/((\alpha - \beta)^2 + (t - \gamma)^2)$; each term integrates over $[0, \theta]$ to $\arctan \frac{\theta - \gamma}{\alpha - \beta} + \arctan \frac{\gamma}{\alpha - \beta}$, which for a zero on the line tends to $\pi$ if $0 < \gamma < \theta$ and to 0 if $\gamma > \theta$ or $\gamma < 0$, while the two terms of an off line pair $\frac{1}{2} \pm \delta + i\gamma$ have the limits $\mp\big(\arctan \frac{\theta - \gamma}{\delta} + \arctan \frac{\gamma}{\delta}\big)$, which cancel. The terms are dominated by $\pi$ and by $C\theta/\gamma^2$ for $|\gamma| > 2\theta + 1$, uniformly for $\frac{1}{2} < \alpha \leq 1$, so dominated convergence applies. At $\alpha = \frac{1}{2}$, $a = \frac{3}{4}$ and $a(a-1) + \frac{1}{6} = -\frac{1}{48}$. □

The Thorin measure at resolution $\varepsilon = \alpha - \frac{1}{2}$ is therefore the zero counting function smoothed by the Cauchy kernel of width $\varepsilon$, the analogue for $\chi_4$ of the

cumulative Thorin mass of $\xi$ (Section 3.4.1). At $\alpha = 0.75$ the difference between the left side and the first three terms of the expansion is $-0.0182/\theta$ at $\theta = 50$, 100 and 200, against $-(a(a-1)+\frac{1}{6})/\pi = -11/(192\pi) = -0.018237$ at $a = \frac{7}{8}$ (`dbeta_thorin.py`).

COROLLARY 8.5.32. *GRH for $\chi_4$ holds if and only if, for every $\alpha > \frac{1}{2}$, $\theta \mapsto \arg \xi(\alpha + i\theta, \chi_4)$ is nondecreasing. With no hypothesis, the heights $\theta > 0$ at which it decreases lie in the union of the windows $|\theta - \gamma| < \sqrt{\delta^2 - (\alpha - \frac{1}{2})^2}$, each of length less than 1, over the zeros $\frac{1}{2} + \delta + i\gamma$ with $\delta > \alpha - \frac{1}{2}$, and there are none at heights below $2.5 \times 10^7 - \frac{1}{2}$.*

*Proof.* The derivative is $v_\alpha(\theta)$, and Proposition 8.5.25 gives the equivalence. By the Hadamard product, $v_\alpha$ is a sum of positive Cauchy kernels, one for each zero on the line and two for each off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta \le \alpha - \frac{1}{2}$, plus, for each pair with $\delta > \alpha - \frac{1}{2}$, the difference $\pi\big(P_{\varepsilon+\delta} - P_{\delta-\varepsilon}\big)(\theta \mp \gamma)$ with $\varepsilon = \alpha - \frac{1}{2}$, which is negative exactly where $|\theta \mp \gamma| < \sqrt{\delta^2 - \varepsilon^2}$ (proof of Proposition 3.3.1(ii)); for $\theta > 0$ and $\gamma > 0$ the reflected term at $\theta + \gamma$ is positive, since $\theta + \gamma > \gamma > \delta$. A window has length $2\sqrt{\delta^2 - \varepsilon^2} < 2\delta < 1$. Platt's verification of GRH for the modulus 4 to height $10^8/4$ [417] puts every zero off the line above $2.5 \times 10^7$, and so every window above $2.5 \times 10^7 - \frac{1}{2}$. □

A window holds a decrease only if the negative term of its pair outweighs all the positive terms there, and the decrease is bounded: the negative part of the term $\pi(P_{\varepsilon+\delta} - P_{\delta-\varepsilon})(\theta - \gamma)$ of a pair has integral $4 \arctan \sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - \pi < \pi$, by Proposition 3.3.1(iii), so over any interval the argument falls by less than $\pi$ times the number of pairs whose windows meet it, whatever the other zeros, and the fall caused by a pair tends to zero as $(\alpha - \frac{1}{2})/\delta \to 0$. This answers the quantitative half of a question of the first author's note. No lower bound for the positive terms over a window, such as a mean value estimate for the contribution of the zeros on the line, can exclude this for every pair: at the centre $\theta = \gamma$ of its window the pair contributes $\frac{1}{\varepsilon+\delta} - \frac{1}{\delta-\varepsilon} = -\frac{2\varepsilon}{\delta^2-\varepsilon^2}$, which is unbounded below as $\delta \downarrow \varepsilon$, that is as the zero approaches the line $\operatorname{Re} s = \alpha$ from the right, while by Lemma 11.4.9 a zero far enough to the right of the line cannot produce a decrease against a given positive floor. Outside the strips $|\theta - \gamma| < (1+\sqrt{2})\delta$ around the ordinates of the zeros off the line, Theorem 9.1.15 gives the argument a rate of increase that grows like $\log \theta$ (Corollary 9.1.16). Figure 8.8 shows the smoothing.

### *8.5.7.5 The centre*

For $\zeta$ the positive structure at the centre is the Thorin measure of the clock, not a structure of the Pólya variable itself (Proposition 11.1.3(1)). The same holds for $\chi_4$. The characteristic function $\xi(\frac{1}{2} + i\theta, \chi_4)/\xi(\frac{1}{2}, \chi_4)$ of $Y_4$ is real and changes sign at $\gamma_1 = 6.0209489$ and at each of the other ordinates below height 100; Hardy's

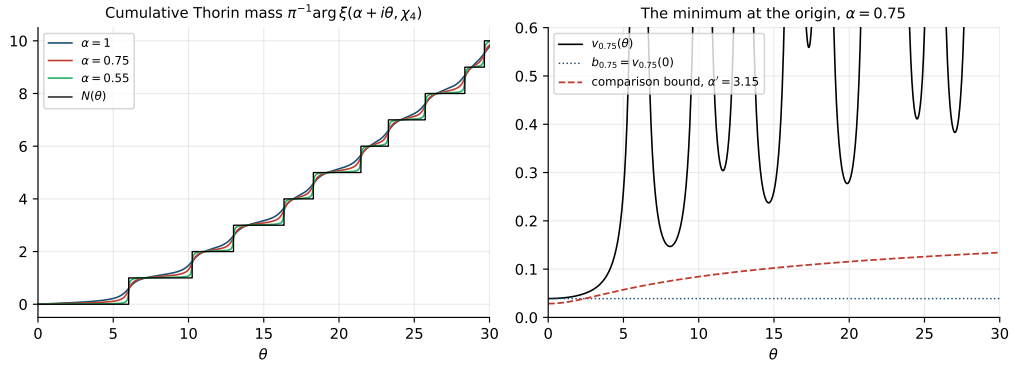



Figure 8.8: Left: the cumulative Thorin mass $\frac{1}{\pi}\arg\xi(\alpha+i\theta,\chi_4) = \frac{1}{\pi}\int_0^\theta v_\alpha$ for $\alpha = 1$, 0.75 and 0.55, against the zero count $N(\theta)$ of $L(s,\chi_4)$. Right: $v_{0.75}(\theta)$, its minimum $b_{0.75} = 0.0389784$ at the origin, and the lower bound of Corollary 9.1.16 with $\alpha' = 3.15$, which exceeds $b_{0.75}$ for $\theta > 2.434$ (`dbeta_minimum.py`). Drawn by `dbeta_figs.py`.

argument for $\zeta$ [498, Chapter 10], applied to the theta series $\eta(ix)^3$ of $\chi_4$, gives infinitely many such sign changes. So $Y_4$ is not quasi infinitely divisible, and in particular not infinitely divisible: a quasi infinitely divisible characteristic function never vanishes. Nor is $\xi(\frac{1}{2}+w,\chi_4)/\xi(\frac{1}{2},\chi_4)$ a positive mixture $\int M(d\lambda)/(w+\lambda)^2$ of Gamma(2) transforms, for a more elementary reason: such a mixture decreases on $w > 0$, while $\xi(\frac{1}{2}+w,\chi_4) = \int_{\mathbb{R}} K_4(u)\cosh(wu)\,du$ increases there. A one sided object remains, the half kernel $K_4(u)\mathbf{1}\{u > 0\}$, and it is not a positive gamma mixture either.

Proposition 8.5.33. *For no $a > 0$ is $u^{1-a}K_4(u)$ completely monotone on $u > 0$; in particular the half kernel is not a positive mixture of Gamma(a) densities $u^{a-1}e^{-\lambda u}$, for any shape $a$. The theta function $\eta(ix)^3$ is not completely monotone on $x > 0$.*

*Proof.* If $g(u) = \int_{[0,\infty)} e^{-\lambda u}M(d\lambda)$ with $M \geq 0$ and $M \neq 0$, choose $L$ with $M([0,L]) > 0$; then $g(u) \geq M([0,L])e^{-Lu}$, so $g$ decays at most exponentially, while $u^{1-a}K_4(u)$ decays like $\exp(-\frac{\pi}{4}e^{2u})$ (Proposition 8.5.22). The Jacobi series $\eta(ix)^3 = \sum_{n\geq 0}(-1)^n(2n+1)e^{-\pi(2n+1)^2x/4}$ is the Laplace transform of the signed measure with the weights $(-1)^n(2n+1)$ at the points $\pi(2n+1)^2/4$, which by the uniqueness of the Laplace transform is its only representing measure, and it is not positive, so Bernstein's theorem excludes complete monotonicity. □

The decay argument applies to the kernel of every completed $L$ function of degree one, all of which decay like $\exp(-ce^{2|u|})$; for $\xi$ it shows that no positive

mixture of Gamma($a$) densities represents the Pólya kernel on the half line, for any shape $a$, as [421, Section 13.6] observes (see also [424, Section 12.6]).

The positive structure is in the reciprocal of the product. With $q = e^{-2\pi x}$,

$$\frac{e^{-\pi x/4}}{\eta(ix)^3} = \prod_{m\ge1} \left(1-q^m\right)^{-3} = \exp\Big(3\sum_{N\ge1} \sigma_{-1}(N)\, e^{-2\pi N x}\Big), \qquad \sigma_{-1}(N) = \sum_{d|N} \frac{1}{d}, \tag{8.5.11}$$

from $-\log(1-q^m) = \sum_k q^{mk}/k$; so the logarithm of the left side is completely monotone, and hence so is the left side. In the language of Section 8.5.4, $\prod_m (1-e^{-2\pi m x})^3$ is the distribution function of the maximum $M_3$ of independent exponential variables, three of each rate $2\pi m$. The representation of Section 8.5.4 also carries over: since $\eta(ix)^3 = e^{-\pi x/4}\Pr(M_3 \le x)$ and $x^{c-1}e^{-\pi x/4}(\pi/4)^c/\Gamma(c)$ is the density of $\frac{4}{\pi}G_c$, with $G_c$ a gamma variable of shape $c$ independent of $M_3$, the Mellin form gives $\xi(s,\chi_4) = (\frac{4}{\pi})^c\Gamma(c)\Pr(\frac{4}{\pi}G_c \ge M_3)$ with $c = \frac{s+1}{2}$, that is

$$L(s,\chi_4) = \Pr\big(4G_{(s+1)/2} \ge \pi M_3\big), \qquad s > -1,$$

which at $s = 1$ is $\mathbb{E}e^{-\pi M_3/4} = \frac{\pi}{4}\int_0^\infty \eta(ix)^3 dx = \frac{\pi}{4}$.

The half kernel has the Laplace transform

$$\int_0^\infty e^{-wu} K_4(u)\, du = A_4\big(\tfrac12 - w\big), \tag{8.5.12}$$

where

$$A_4(s) = \int_1^\infty \eta(ix)^3 x^{(s+1)/2}\frac{dx}{x} = \sum_{n\ge1} \chi_4(n)\, n\, \frac{\Gamma(\frac{s+1}{2}, \frac{\pi n^2}{4})}{(\pi n^2/4)^{(s+1)/2}}$$

is an incomplete Mellin transform of $\eta^3$, and the transformation $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$, applied to the integral over $(0,1)$, gives $\xi(s,\chi_4) = A_4(s) + A_4(1-s)$ for every $s$. This is the analogue of Riemann's formula (Proposition 8.5.5) and of the symmetric form (8.5.5) for $\Delta$, with no pole terms, since $\eta(ix)^3$ has no constant term.

Question 8.5.34. Can the complete monotonicity of (8.5.11), and of its analogue $\prod_m (1-q^m)^{-1}$ for the kernel of $\chi_{12}$, be turned into a positive representation on the reciprocal side, in the spirit of Thorin's construction for the reciprocal of a GGC transform?

## 8.6 The inverse problem: recovering the atoms by nonnegative deconvolution

The inverse problem is to recover the zeros of $\zeta$ from its Thorin density alone. Under RH the density is the Poisson smoothing of a positive measure with an atom at each zero, so the recovery is a deconvolution under a positivity constraint, and

RH on a window becomes a feasibility question. The monograph [424, Section 19.3] poses it as a Bayesian problem with a positivity prior and a posterior predictive check, and notes that the experiment had not been run because $\rho_\varepsilon$ was available only from the primes, at resolutions above one half. The Pólya route of Section 2.1 supplies $\rho_\varepsilon$ at small resolution to about 60 digits below height 200 (Section 2.1.1); the experiment uses it in double precision.

### 8.6.1 Recovering the zeros by nonnegative least squares

Write $P_\varepsilon(x) = \varepsilon/(\pi(\varepsilon^2+x^2))$ and $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$. Under RH, $\rho_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ on $\theta > 0$ (Proposition 1.2.3); since $P_\varepsilon$ has Fourier transform $e^{-\varepsilon|t|}$, recovering $U^{\pm}_{\mathrm{RH}}$ is severely ill posed. An off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta < \varepsilon$ contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at $\gamma$, and $P_{\varepsilon-\delta}$ is sharper than the Poisson smoothing at level $\varepsilon$ of any positive measure, since $P_{\varepsilon-\delta} = P_\varepsilon * \lambda$ would give $\lambda$ the exponentially growing Fourier transform $e^{\delta|t|}$. Positivity of the recovered measure is therefore the test.

Fix a data window $[\theta_a, \theta_b]$ with sample points $\theta_1 < \cdots < \theta_n$, and candidate atom locations $x_1 < \cdots < x_m$ on a grid over a candidate window $[x_a, x_b]$ that contains $[\theta_a, \theta_b]$. With $A_{ij} = P_\varepsilon(\theta_i - x_j) + P_\varepsilon(\theta_i + x_j)$ and data $y_i = \rho_\varepsilon(\theta_i) - b_i$, where $b_i$ is the contribution of the zeros outside the candidate window, the problem is

$$\min_{w \geq 0} \ \|Aw - y\|_2 .$$

Under a Gaussian likelihood with a flat prior on the positive orthant its solution is the maximum a posteriori estimate, and the minimal residual is the discrepancy statistic of the posterior predictive check: up to the error of the grid, it vanishes when the data are the Poisson smoothing of a positive measure. The active set algorithm of Lawson and Hanson [329] solves the problem in finitely many steps.

We took $\varepsilon = 0.1$, the Pólya computed $\rho_{0.1}$ on $[15, 55]$ at 1601 points, and candidates on $[5, 65)$ at spacing 0.02. The background $b_i$ is the Poisson smoothing of the zeros outside $[5, 65]$ among the first 1700 ordinates, the list of Section 3.2 without its smooth tail, which would add about $3 \times 10^{-5}$ to each sample. The nonnegative solution concentrates on clusters, runs of adjacent grid points with weights above $10^{-3}$, one at each zero in the data window (Table 8.21 and Figure 8.9, left): every ordinate is recovered to within $10^{-5}$, with mass one to within half a per cent. The remaining mass, 0.071 of the total 10.094, lies in small clusters at the edges of the data window (0.025), which absorb the tails of the zeros at 14.13 and 56.45 just outside it, and in weights below $10^{-3}$ (0.046). The residual norm is 0.106. It is set by the grid: the true atoms lie between grid points, and a positive combination of two neighbouring Cauchy kernels reproduces an off grid kernel only to second order in the grid spacing.

| $k$ | $\gamma_k$ | $c_k$ | $10^6(c_k-\gamma_k)$ | mass |
|---|---|---|---|---|
| 2 | 21.022040 | 21.022034 | −5.5 | 1.0013 |
| 3 | 25.010858 | 25.010859 | 1.6 | 1.0027 |
| 4 | 30.424876 | 30.424867 | −8.8 | 1.0020 |
| 5 | 32.935062 | 32.935070 | 8.7 | 1.0020 |
| 6 | 37.586178 | 37.586171 | −7.4 | 1.0023 |
| 7 | 40.918719 | 40.918723 | 3.8 | 1.0008 |
| 8 | 43.327073 | 43.327068 | −5.5 | 1.0025 |
| 9 | 48.005151 | 48.005145 | −6.1 | 1.0024 |
| 10 | 49.773832 | 49.773840 | 7.9 | 1.0028 |
| 11 | 52.970321 | 52.970330 | 8.8 | 1.0042 |

Table 8.21: The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15, 55]$: for each zero $\gamma_k$ in the data window, the centre of mass $c_k$ of its cluster, the error of the centre and the total mass of the cluster.

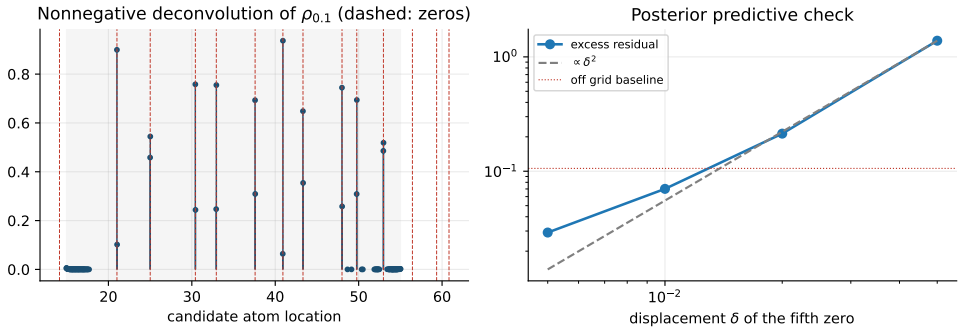



Figure 8.9: Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15, 55]$ shaded. The zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement $\delta$ of the fifth zero, the $\delta^2$ law through the point at $\delta = 0.05$, and the off grid baseline $r_0 = 0.106$.

### 8.6.2 A displaced zero and the threshold of detection

To calibrate the check we displaced the fifth zero, $\gamma_5 = 32.935$, in the sense of Proposition 3.3.1(i): we added to the data the change per displaced zero, $g = \frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$, centred at $\gamma_5$, and refitted. Table 8.22 gives the residuals.

For $\delta \geq 0.02$ the excess residual grows roughly like $\delta^2$, as the signature does (Proposition 3.3.1(i) and [424, Proposition 19.7]): it rises from 0.21 at $\delta = 0.02$ to 1.39 at $\delta = 0.05$, a factor 6.5 for a factor 2.5 in $\delta$, against $2.5^2 = 6.25$. At the smallest displacements it grows more slowly, by a factor 2.4 from $\delta = 0.005$ to 0.01, because there it is dominated by the cross term between the signature and

| $\delta$ | residual $r = \|Aw - y\|$ | excess over baseline | total mass |
|---|---|---|---|
| 0 | 0.1058 | 0 | 10.0942 |
| 0.005 | 0.1098 | 0.029 | 10.0947 |
| 0.01 | 0.1269 | 0.070 | 10.0963 |
| 0.02 | 0.2380 | 0.213 | 10.1031 |
| 0.05 | 1.3891 | 1.385 | 10.1566 |

Table 8.22: Nonnegative deconvolution of $\rho_{0.1}$ with the signature $g$ of a displaced fifth zero added to the data. Here $r$ is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$.

the off grid misfit at $\gamma_5$, which has the same shape. The off grid baseline therefore limits detection: the excess reaches it between $\delta = 0.01$ and 0.02 (Figure 8.9, right). This is the square root law that (3.3.1) states for the matched distance (Section 3.3.2), whatever limits the residual.

For $\delta \geq \varepsilon$ the check is unreliable. At $\delta = \varepsilon$ the zero $\frac{1}{2} + \delta + i\gamma_5$ lies on the line of evaluation, where its kernel is a point mass that the samples miss, and the residual, 0.099, falls below the baseline. For $\delta > \varepsilon$ the atom at $\gamma_5$ leaves the data and is replaced by the signed term $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ of total mass zero, whose negative part, of mass $\varepsilon/(\pi\delta) + O(\varepsilon^3/\delta^3)$ by Proposition 3.3.1(iii), fades as $\delta$ grows. The unit of mass that leaves the data is what the zero density identity of Proposition 3.4.2 counts, and zeros farther from the line than $\varepsilon$ are left to that identity.

As [424, Section 19.3.7] emphasises, this is a model check at finite height and not a route to a proof. What the Pólya route adds is the small resolution, where the signature $(\delta/\varepsilon)^2$ of Proposition 3.3.1(i) is largest for a given $\delta$. The recovery and the calibration are computed, at one resolution and on one window.

The direction of this section is the hard one: Poisson smoothing, which moves the basepoint to the right, preserves positivity, while deconvolution, which moves it back towards the critical line, does not (Section 11.4.7).

## 8.7 Pólya's Bessel function: hitting times, a clock and the flow

Pólya replaced the kernel $\Phi$ of (1.2.3) by its asymptote $\Phi^*$ and proved that the transform $\Xi^*$ has only real zeros [447] (Section 13.5.1). The building block of his proof is the Macdonald function

$$P(\theta) = K_{i\theta/2}(2\pi) = \int_{\mathbb{R}} e^{-2\pi \cosh 2u}\, e^{i\theta u}\, du, \tag{8.7.1}$$

which has only real zeros and whose zero counting function has the main term of the Riemann von Mangoldt formula. Biane [48] showed that $\Xi$ and $P$ are

Mellin transforms of two Brownian hitting times, and that the Thorin measures of these hitting times are spectral measures of the radial Brownian motions on the unitary group $SU(2)$ and on the hyperbolic space $SL_2(\mathbb{C})/SU(2)$. This section follows the first author's note on Biane's paper [426]. For $P$ every statement of the Thorin programme is a theorem with an explicit witness: the zeros are the Dirichlet spectrum of a Liouville operator, the reciprocal $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ is a GGC whose Thorin atoms are the squared zeros, its clock is the hitting time of an explicit diffusion, and its Poisson trace is completely monotone because it is the trace of a semigroup. The model also marks where the analogy stops: by a theorem of Newman, Pólya's function has a finite de Bruijn Newman constant, although its zeros show no sign of colliding in the range we compute.

### 8.7.1 Two hitting times and Biane's dictionary

As $u \to \pm\infty$ only the term $n = 1$ of (1.2.3) matters, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2}\exp(-\pi e^{2|u|})$. Pólya replaced $\Phi$ by the even function with the same asymptotics on both sides,

$$\Phi^*(u) = 8\pi^2 \cosh\tfrac{9u}{2}\, e^{-2\pi\cosh 2u},$$
$$\Xi^*(\theta) = \int_{\mathbb{R}} \Phi^*(u) e^{i\theta u}\, du = 4\pi^2\Big[P\big(\theta - \tfrac{9i}{2}\big) + P\big(\theta + \tfrac{9i}{2}\big)\Big],$$

since the factor $e^{\pm 9u/2}$ shifts the argument of the transform by $\mp\frac{9i}{2}$. This is the form $G(z+ia)+G(z-ia)$ of Pólya's Hilfssatz II (Section 7.1.3) with $G = P$ and $a = \frac{9}{2}$: Pólya proved that $P$, an entire function of order one, has only real zeros, and the Hilfssatz gives the same for $\Xi^*$. In the index of the Bessel function the shift is $\frac{9}{4}$, since $P(\theta \mp \frac{9i}{2}) = K_{i\theta/2\pm 9/4}(2\pi)$, which is the form of $\Xi^*$ in Section 13.5.1. Figure 8.10 compares the three kernels. As a pointwise approximation $\Phi^*$ is poor near the origin, $\Phi^*(0) = 8\pi^2 e^{-2\pi} = 0.147447$ against $\Phi(0) = 0.893394$, and the kernel $e^{-2\pi\cosh 2u}$ of $P$ is narrower still; what the three share is the doubly exponential tail, which governs the zero counting functions. Table 8.23 lists the zeros below height 60 and the counts up to height 1000.

The value $2\pi$ of the argument is forced. Pólya's asymptotics for the zeros of $y \mapsto K_{iy}(2a)$, which Biane derives by stationary phase [48, Section 3.5], give $\frac{T}{\pi}\log\frac{T}{a} - \frac{T}{\pi} + O(1)$ zeros with $0 < y \le T$; for $P$ the zeros are at $\theta = 2y$ with $a = \pi$, and the count in $(0,T]$ becomes $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(1) = N_0(T) + O(1)$. The constant $a = \pi$ comes from the factor $e^{-\pi e^{2u}}$ of the term $n = 1$.

Let $H_a^{(c)}$ be the first passage time of $B_t + ct$ to the level $a > 0$, for a standard Brownian motion $B$ and a drift $c \ge 0$, as in Appendix C. Its density is $a(2\pi\tau^3)^{-1/2}e^{-(a-c\tau)^2/2\tau}$, the inverse Gaussian law of Table 1.8, and its Laplace transform is $e^{a(c-\sqrt{c^2+2u})}$.

Proposition 8.7.1. *Let $H = H_a^{(c)}$ with $a = c = \sqrt{2\pi}$.*

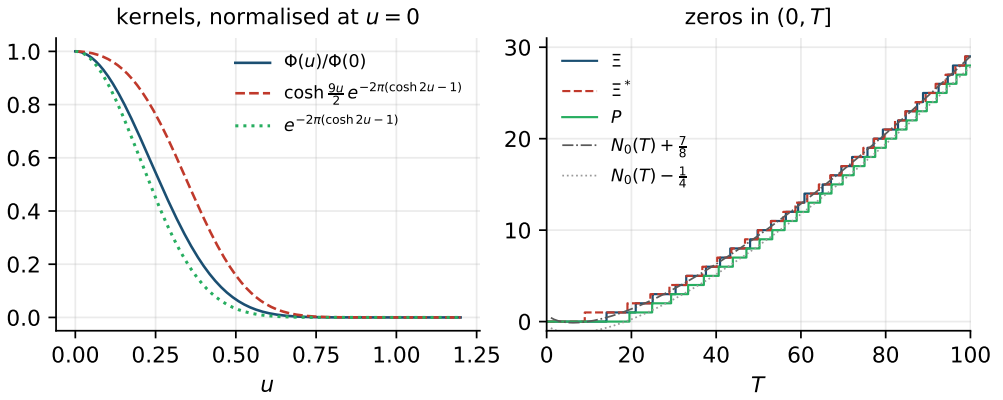



Figure 8.10: Left: the kernel $\Phi$ of (1.2.3), Pólya's asymptote $\Phi^*$ and the kernel $e^{-2\pi\cosh 2u}$ of $P$, each divided by its value at $u = 0$. Right: the zero counting functions of $\Xi$, $\Xi^*$ and $P$ on $(0, 100]$, against $N_0(T) + \frac{7}{8}$ and $N_0(T) - \frac{1}{4}$, the mean values of the Riemann von Mangoldt formula and of Proposition 8.7.8, with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Drawn by `biane_figs.py` from `biane_polya.json`.

| $n$ | $\Xi$ | $\Xi^*$ | $P$ |
|---|---|---|---|
| 1 | 14.135 | 8.993 | 19.538 |
| 2 | 21.022 | 19.065 | 24.897 |
| 3 | 25.011 | 24.475 | 29.370 |
| 4 | 30.425 | 28.975 | 33.383 |
| 5 | 32.935 | 33.007 | 37.099 |
| 6 | 37.586 | 36.737 | 40.600 |
| 7 | 40.919 | 40.249 | 43.936 |
| 8 | 43.327 | 43.595 | 47.140 |
| 9 | 48.005 | 46.807 | 50.234 |
| 10 | 49.774 | 49.908 | 53.236 |
| 11 | 52.970 | 52.916 | 56.158 |
| 12 | 56.446 | 55.843 | 59.009 |
| 13 | 59.347 | 58.700 | |

| $T$ | $\Xi$ | $\Xi^*$ | $P$ | $N_0(T)$ |
|---|---|---|---|---|
| 60 | 13 | 13 | 12 | 11.998 |
| 100 | 29 | 29 | 28 | 28.127 |
| 240 | 102 | 102 | 101 | 100.946 |
| 1000 | 649 | 649 | 647 | 647.741 |

Table 8.23: Left: the positive zeros below 60 of $\Xi$, of Pólya's $\Xi^*$ and of $P(\theta) = K_{i\theta/2}(2\pi)$. Right: the numbers of zeros in $(0, T]$ against $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Computed by `biane_polya.py`.

*(i)* $H$ *has the density* $\tau^{-3/2}e^{2\pi}e^{-\pi(\tau+1/\tau)}$ *on* $(0,\infty)$*, and* $1/H$ *has the law of* $H$ *size biased by* $H$*.*

*(ii)* $\mathbb{E}H^s = 2e^{2\pi}K_{s-\frac12}(2\pi)$ *for every* $s\in\mathbb{C}$*, so that* $\mathbb{E}H^s = \mathbb{E}H^{1-s}$*.*

*(iii)* $P(\theta) = \frac12 e^{-2\pi}\,\mathbb{E}H^{\frac12+\frac{i\theta}{2}}$ *for every* $\theta\in\mathbb{C}$*.*

*(iv)* $U = \frac12\log H$ *has the density* $2e^{2\pi}e^{-u}e^{-2\pi\cosh 2u}$*, and the law of* $U$ *size biased by* $e^U$ *has the density* $e^{-2\pi\cosh 2u}/K_0(2\pi)$*, whose characteristic function is* $P(\theta)/P(0)$*.*

*Proof.* (i) With $a = c = \sqrt{2\pi}$, $(a-c\tau)^2/2\tau = \pi(\tau+1/\tau)-2\pi$ and $a/\sqrt{2\pi}=1$. The density of $1/H$ at $\sigma$ is $\sigma^{-2}$ times that of $H$ at $1/\sigma$, which is $\sigma^{-1/2}e^{2\pi}e^{-\pi(\sigma+1/\sigma)}$, that is $\sigma$ times the density of $H$ at $\sigma$; in particular $\mathbb{E}H = 1$. (ii) The substitution $\tau = e^w$ and $K_\nu(x) = \int_0^\infty e^{-x\cosh w}\cosh(\nu w)\,dw$ [397, (10.32.9)] give $\int_0^\infty \tau^{\nu-1}e^{-\pi(\tau+1/\tau)}\,d\tau = 2K_\nu(2\pi)$, here with $\nu = s-\frac12$; the integral converges for every complex $\nu$, and $K_\nu = K_{-\nu}$ gives the symmetry. (iii) is (ii) at $s = \frac12+\frac{i\theta}{2}$. (iv) The substitution $\tau = e^{2u}$ turns the density of (i) into $2e^{2\pi}e^{-3u}e^{-2\pi\cosh 2u}e^{2u}$; multiplying by $e^u$ and dividing by $\mathbb{E}H^{1/2} = 2e^{2\pi}K_0(2\pi)$ gives the last density, and its characteristic function is $\int e^{-2\pi\cosh 2u}e^{i\theta u}du/K_0(2\pi) = P(\theta)/P(0)$ by (8.7.1). □

The parallel with $\Xi$ is exact. By the identity (9.1.1) of Biane, Pitman and Yor [49], with $V = \frac{\pi}{2}S_2$, $2\xi(s) = \mathbb{E}V^{s/2}$, so $\Xi(\theta) = \frac12\mathbb{E}V^{\frac14+\frac{i\theta}{2}}$, and the kernel is $\Phi(u) = \frac12 e^{u/2}f_Y(u)$ with $Y = \frac12\log V$: the Pólya law of (1.2.4) is the law of $Y$ size biased by $e^{Y/2}$, as the law of (iv) is that of $U$ size biased by $e^U$. In both cases the evenness of the kernel is a reflection symmetry of the hitting time: $\tau\mapsto 1/\tau$ for $H$, which is (i), and the functional equation $\mathbb{E}V^{s/2} = \mathbb{E}V^{(1-s)/2}$ for $V$, which comes from Jacobi's theta transformation.

Both hitting times are GGCs, with the Thorin measures of Table 1.8. For $H_a^{(c)}$ the Thorin measure is $a\,\{\pi\sqrt{2z-c^2}\}^{-1}\mathbf{1}_{z>c^2/2}\,dz$, and the Lévy density, $t^{-1}$ times its Laplace transform, is $a\,e^{-c^2t/2}(2\pi t^3)^{-1/2}$. For the Bessel hitting time, $\mathbb{E}e^{-uJ_a} = a\sqrt{2u}/\sinh(a\sqrt{2u})$ with $J_a \overset{d}{=} a^2J$, so $J_a + J_a' = a^2S_2$ has Thorin measure $2\sum_{n\ge1}\delta_{\pi^2n^2/2a^2}$ and Lévy density $2t^{-1}\sum_n e^{-\pi^2n^2t/2a^2}$. The Thorin measure can also be read from the continuation of $\Psi(u) = -\log\mathbb{E}e^{-uX}$ across the cut: integrating the Stieltjes inversion (1.3.2),

$$U\big((0,z)\big) = \frac{1}{\pi}\operatorname{Im}\Psi(-z+i0)\qquad (z>0 \text{ not an atom}),$$

because each factor $\log(1+u/w)$ with $w<z$ contributes $i\pi$ at $u=-z+i0$. For $H_a^{(c)}$, $\Psi(-z+i0) = a\big(i\sqrt{2z-c^2}-c\big)$ for $z>\frac{c^2}{2}$, so $U((0,z)) = \frac{a}{\pi}\sqrt{2z-c^2}$; for $J_a+J_a'$, $\Psi(-\frac{\lambda^2}{2}+i0) = 2\log\frac{\sin\lambda a}{\lambda a}$, which gains $2\pi i$ each time $\lambda a$ passes a multiple of $\pi$. Biane's dictionary identifies these Thorin measures with spectral

measures. In his normalisation [48, (2.1) to (2.3)] the generator $L_1 = \frac{1}{2}\frac{d^2}{dx^2}$ on $[0,\pi]$ with Dirichlet conditions has the eigenfunctions $\sin(kx)$, $k = 1, 2, \dots$, with eigenvalues $-\frac{k^2}{2}$, and the spectral measure $\sigma_1 = \frac{1}{\pi}\sum_{n\geq 1}\delta_{n^2/2}$; the generator $L_2 = \frac{1}{2}\frac{d^2}{dx^2} + \frac{d}{dx}$ on $[0,\infty)$ with a Dirichlet condition at 0, selfadjoint in $L^2(e^{2x}dx)$, has the generalised eigenfunctions $e^{-x}\sin(kx)$, $k > 0$, with eigenvalues $-\frac{k^2+1}{2}$, and the spectral measure $\sigma_2(d\lambda) = d\lambda/(\pi\sqrt{2\lambda-1})$ on $\lambda > \frac{1}{2}$.

THEOREM 8.7.2 (Biane's dictionary, with corrected constants). *(i) The Thorin measure of $J_\pi + J'_\pi = \pi^2 S_2$ is $2\pi\sigma_1$.*

*(ii) The Thorin measure of $H_1^{(1)}$ is $\sigma_2$.*

*(iii) The radial generator $\frac{1}{2}\frac{d^2}{dx^2} + \cot x\frac{d}{dx}$ of Brownian motion on the unit sphere $S^3 \cong SU(2)$ is the Doob transform of $L_1$ by its ground state $\sin x$, $\sin^{-1}(L_1 + \frac{1}{2})\sin$, and the radial generator $\frac{1}{2}\frac{d^2}{dx^2} + \coth x\frac{d}{dx}$ of Brownian motion on $\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$ is the Doob transform of $L_2$ by the harmonic function $1 - e^{-2x}$.*

*Proof.* (i) With $a = \pi$ the atoms $\pi^2 n^2/2a^2$ are at $n^2/2$, with mass $2 = 2\pi\cdot\frac{1}{\pi}$. (ii) With $a = c = 1$ the density $a\{\pi\sqrt{2z-c^2}\}^{-1}$ on $z > \frac{c^2}{2}$ is that of $\sigma_2$. (iii) For $f$ smooth, $\frac{1}{2}(f\sin)'' = \sin\cdot(\frac{1}{2}f'' + \cot x\, f' - \frac{1}{2}f)$, so $\sin^{-1}(L_1 + \frac{1}{2})(f\sin) = \frac{1}{2}f'' + \cot x\, f'$; and $L_2 h = 0$ for $h = 1 - e^{-2x}$, while $h^{-1}L_2(hf) = \frac{1}{2}f'' + (1 + h'/h)f'$ with $1 + h'/h = 1 + \frac{2e^{-2x}}{1-e^{-2x}} = \coth x$. The eigenfunctions are checked the same way: $L_2(e^{-x}s) = e^{-x}(\frac{1}{2}s'' - \frac{1}{2}s)$. □

For Pólya's function itself the parameters are $a = c = \sqrt{2\pi}$ rather than $a = c = 1$; Brownian scaling gives $H_a^{(c)} \stackrel{d}{=} \frac{a}{c}H_{\sqrt{ac}}^{(\sqrt{ac})}$, so the level and the drift can always be made equal, which is the reciprocal symmetry of Proposition 8.7.1(i), but their common value $\sqrt{ac}$ is $\sqrt{2\pi}$ for $P$ and 1 in the dictionary. Part (iii) corrects the direction of the Doob transforms in the note [426], which has $L_1$ and $L_2$ as the transforms of the radial generators; Biane states them in the direction of (iii) [48, Section 2.4].

REMARK 8.7.3 (Biane's constants). Biane's $K_\mu(x) = \int_0^\infty t^{\mu-1}e^{-\frac{x}{2}(t+t^{-1})}dt$ is twice the standard Macdonald function, and his kernel (3.1) is the $\Phi$ of (1.2.3); with that kernel $\Xi(z) = 2\int_0^\infty \Phi(u)\cos(zu)\,du$ holds for the standard $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, so the factor $\frac{1}{2}$ is missing from his displayed definition of $\xi$, and his Mellin formula $\mathbb{E}W_a^s = 2(2a^2/\pi)^s\xi(2s)$ also uses the standard $\xi$ (both sides are $\frac{2}{3}$ at $s = 1$, $a = 1$). In his Section 4.3 the Lévy measure of $W_a = J_a + J'_a$ should be $2t^{-1}\sum_n e^{-\pi^2 n^2 t/2a^2}dt$, with the factor $t^{-1}$ and the $\frac{1}{2}$ in the exponent, and that of the drifted first passage time should carry the level $x$ as a factor. His Thorin measure (4.1) should have its atoms at $\pi^2 n^2/2a^2$, not at

$n^2/a^2$; the density (4.2) should be $x\,dc/(\pi\sqrt{2c-a^2})$, which for $x=1$ is his density divided by $\sqrt{2}$, and the identity offered for it should read $(\pi t)^{-1/2}e^{-a^2t/2} = \int_{a^2/2}^{\infty} e^{-ct}\,dc/(\pi\sqrt{c-a^2/2})$, with $t$ in place of $t^3$. The measure (2.3) lives on $(\frac{1}{2},\infty)$, not on $[1,\infty)$. In Propositions 2.3 and 2.4 the drift and the harmonic function are oriented for the cone opposite to the one stated: with the cone $\{y_1 \geq \cdots \geq y_n\}$ the drift is $\rho' = (n-1, n-3, \ldots, 1-n)$ and the positive harmonic function is $\prod_{i<j}(1-e^{-2(y_i-y_j)}) = e^{-\langle \rho', y\rangle}\prod_{i<j} 2\sinh(y_i-y_j)$; the Vandermonde function $\prod_{i>j}(x_i-x_j)$ of his Proposition 2.1 is reversed in the same way, and is positive on the cone as $\prod_{i<j}(x_i-x_j)$. The eigenvalue of Proposition 2.6 is discussed in Remark 8.7.5. With these corrections the dictionary of his Section 4.5 is the exact statement of Theorem 8.7.2.

### 8.7.2 The Wick rotation and the curvature family

Biane closes his paper by observing that $\xi$ and Pólya's function are related, through $SU(2)$ and $SL_2(\mathbb{C})/SU(2)$, "in this non obvious way" [48, Section 4.5]. The relation is a Wick rotation, the substitution of $iy$ for a real variable $y$ in a function analytic in $y$. Wick's device [513] rotates the time axis, $t = -i\tau$, which turns the unitary group $e^{-itH}$ of quantum mechanics into the heat semigroup $e^{-\tau H}$. The rotation keeps the spectrum of $H$ and exchanges oscillation and positivity: probability laws, Laplace transforms and completely monotone functions live on the Euclidean side, Fourier transforms and their zeros on the real time side. A Thorin measure is spectral data that can be read on either side. Through Cartan's duality between a symmetric space $G/K$ of noncompact type and its compact dual $U/K$ the substitution $y \mapsto iy$ reverses the sign of the curvature. Biane's paper contains the rotation in four places: in space (the sign of the curvature), in the Laplace variable, in time (heat kernels) and in the Mellin index.

Let $H_n = \{y \in \mathbb{R}^n : \sum_i y_i = 0\}$ and $\mathcal{C} = \{y \in H_n : y_1 \geq \cdots \geq y_n\}$. Biane describes three matrix valued Brownian motions of type A [48, Section 2]. For traceless Hermitian Brownian motion the ordered eigenvalues perform Brownian motion in $\mathcal{C}$ conditioned in Doob's sense never to reach the boundary: the semigroup of Brownian motion killed at $\partial\mathcal{C}$ is the alternating sum $p^0_t(x,y) = \sum_{w\in S_n}\varepsilon(w)p_t(x,wy)$ (the reflection principle), the Vandermonde product $\prod_{i<j}(y_i-y_j)$ is its positive harmonic function vanishing on $\partial\mathcal{C}$, and the Doob transform is Dyson's Brownian motion [125]. For Brownian motion $g_t$ on $SL_n(\mathbb{C})$, the logarithms of the singular values, the radial part of Brownian motion on $SL_n(\mathbb{C})/SU(n)$, perform Brownian motion in $\mathcal{C}$ with generator $\frac{1}{2}\Delta + \langle \nabla \log \tilde{h}, \nabla\rangle$, $\tilde{h}(y) = \prod_{i<j}\sinh(y_i-y_j)$ [292], which is Brownian motion with drift $\rho' = (n-1, n-3, \ldots, 1-n)$, killed at $\partial\mathcal{C}$, transformed by the harmonic function of Remark 8.7.3. For Brownian motion on $SU(n)$ the eigenvalues $e^{i\theta_j}$, with $\sum_j \theta_j = 0$, live in the alcove $\theta_1 \geq \cdots \geq \theta_n$, $\theta_1 - \theta_n \leq 2\pi$, a fundamental domain of the affine Weyl group $S_n \ltimes ((2\pi\mathbb{Z})^n \cap H_n)$; the killed semigroup is the alternating sum over that group, and the eigenvalue process is the Doob transform

by the Dirichlet ground state $\prod_{j>k}(e^{i\theta_j}-e^{i\theta_k})$, which is a constant multiple of $\prod_{j<k}\sin\frac{\theta_j-\theta_k}{2}$ because $\sum_j\theta_j=0$.

The three Doob functions are one function at three values of a curvature parameter. For real $\kappa$ put $s_\kappa(z)=\sin(\sqrt{\kappa}z)/\sqrt{\kappa}$, so that $s_{-1}=\sinh$, $s_0(z)=z$ and $s_1=\sin$, and

$$h_\kappa(y)=\prod_{i<j}s_\kappa(y_i-y_j),\qquad |\rho|^2=\sum_{j=1}^{n}(n+1-2j)^2=\frac{n(n^2-1)}{3}.$$

Proposition 8.7.4. *For every real $\kappa$ and every $n\geq 2$, $\Delta h_\kappa=-\kappa|\rho|^2h_\kappa$ on $H_n$.*

*Proof.* For $\kappa=-1$, write $\prod_{i<j}2\sinh(y_i-y_j)=\prod_{i<j}e^{-(y_i+y_j)}(e^{2y_i}-e^{2y_j})$. The Vandermonde determinant gives $\prod_{i<j}(e^{2y_i}-e^{2y_j})=\det(e^{2(n-j)y_i})$, and on $H_n$ the factor $\prod_{i<j}e^{-(y_i+y_j)}=e^{-(n-1)\sum_i y_i}$ is one, so the exponents $2(n-j)$ may be replaced by $2(n-j)-(n-1)=n+1-2j$: $\prod_{i<j}2\sinh(y_i-y_j)=\sum_{w\in S_n}\varepsilon(w)e^{\langle w\rho',y\rangle}$, the Weyl denominator formula. Each $w\rho'$ lies in $H_n$ and has $|w\rho'|^2=|\rho|^2$, so each exponential is an eigenfunction of the Laplacian of $H_n$ with eigenvalue $|\rho|^2$, and $\Delta h_{-1}=|\rho|^2h_{-1}$. This identity between entire functions of $y\in H_n\otimes\mathbb{C}$ continues analytically. For $\kappa<0$, $h_\kappa(y)$ is a constant times $h_{-1}(\sqrt{-\kappa}\,y)$, and the scaling multiplies the Laplacian by $-\kappa$; for $\kappa>0$, $s_\kappa(z)=\sinh(i\sqrt{\kappa}z)/(i\sqrt{\kappa})$, so $h_\kappa(y)$ is a constant times $h_{-1}(i\sqrt{\kappa}\,y)$, and the substitution multiplies the Laplacian by $(i\sqrt{\kappa})^2=-\kappa$. At $\kappa=0$, $h_0$ is the Vandermonde product, the limit of $h_\kappa$ as $\kappa\to 0$, since $s_\kappa(z)=z-\kappa z^3/6+O(\kappa^2)$, and it is harmonic. □

So Biane's three semigroups are one formula,

$$q_t^{(\kappa)}(x,y)=\frac{h_\kappa(y)}{h_\kappa(x)}\,e^{\kappa|\rho|^2t/2}\,p_t^0(x,y),\qquad \kappa\in\{-1,0,1\},$$

with generator $\frac12\Delta+\langle\nabla\log h_\kappa,\nabla\rangle$, where $p^0$ is the killed kernel of the chamber $\mathcal{C}$ for $\kappa\leq 0$ and of the alcove $\{y_1\geq\cdots\geq y_n,\ y_1-y_n\leq\pi\}$ for $\kappa=1$; for $\kappa=1$ the coordinates are the half angles $y=\theta/2$, in which $h_1$ is the positive Dirichlet ground state of the alcove. The rotation $y\mapsto iy$ sends $h_\kappa$ to a constant times $h_{-\kappa}$, since $s_\kappa(iz)=is_{-\kappa}(z)$: it exchanges the noncompact and the compact Doob functions and reverses the sign of the ground energy, $\frac12\Delta h_\kappa=-\frac{\kappa}{2}|\rho|^2h_\kappa$, which is $+\frac12|\rho|^2$ for $SL_n(\mathbb{C})/SU(n)$ and $-\frac12|\rho|^2=\frac{n-n^3}{6}$ for $SU(n)$ in the half angles.

Remark 8.7.5 (The eigenvalue of Biane's Proposition 2.6). Biane endows the Lie algebra of $SU(n)$ with the form $\langle A,B\rangle=-\operatorname{Tr}(AB)$, in which the angles $\theta_j$ perform standard Brownian motion in $H_n$, and states that the ground state $\prod_{j<k}\sin\frac{\theta_j-\theta_k}{2}$ has the eigenvalue $\frac{n-n^3}{6}$ for $\frac12\Delta$ [48, Proposition 2.6]. By Proposition 8.7.4 in the half angles, the eigenvalue for $\frac12\Delta_\theta$ is $\frac14\cdot\frac{n-n^3}{6}=\frac{n-n^3}{24}$; more generally, for the form $-c\operatorname{Tr}(AB)$ it is $\frac{n-n^3}{24c}$. Biane's value holds for $c=\frac14$, that is when the

half angles are standard Brownian motion. His rank one discussion uses a third normalisation, $c = \frac{1}{2}$, in which $SU(2)$ is the unit sphere $S^3$, the angle $x \in [0, \pi]$ has the generator $\frac{1}{2}\frac{d^2}{dx^2} + \cot x \frac{d}{dx}$ of Theorem 8.7.2(iii), and the ground energy is $\frac{2-8}{12} = -\frac{1}{2}$. `biane_polya.py` confirms $\frac{1}{2}\Delta_\theta h/h = -\frac{1}{4}, -1, -\frac{5}{2}$ for $n = 2, 3, 4$.

In rank one the curvature family is the radial part of Brownian motion on the three dimensional space form of curvature $\kappa$, with generator $\frac{1}{2}\frac{d^2}{dx^2} + \sqrt{\kappa}\cot(\sqrt{\kappa}x)\frac{d}{dx}$: the sphere of radius $\kappa^{-1/2}$ for $\kappa > 0$, the Bessel process of dimension three for $\kappa = 0$, and the hyperbolic space of curvature $\kappa$ for $\kappa < 0$. Let $J_a^{(\kappa)}$ be its hitting time of the radius $a$ from the centre, with $a < \pi/\sqrt{\kappa}$ when $\kappa > 0$; then $J_a^{(0)} = J_a$, whose Laplace transform is classical [84]. For $\kappa < 0$ the generator is that of the Bessel process of dimension three with drift $\sqrt{-\kappa}$ [469, (2.5)], the norm of a three dimensional Brownian motion with drift of that length.

PROPOSITION 8.7.6. *For $u \geq 0$,*

$$\mathbb{E}e^{-uJ_a^{(\kappa)}} = \frac{\sqrt{2u-\kappa}\; s_\kappa(a)}{\sinh\left(a\sqrt{2u-\kappa}\right)},$$

*the Laplace transform of a GGC with Thorin measure $\sum_{n\geq 1}\delta_{\pi^2 n^2/2a^2 - \kappa/2}$. Equivalently, the law of $J_a^{(\kappa)}$ is the exponential tilt $e^{\kappa t/2}\Pr(J_a \in dt)/\mathbb{E}e^{\kappa J_a/2}$ of the law of $J_a$.*

*Proof.* The generator is the Doob transform of $\frac{1}{2}\frac{d^2}{dx^2}$ by $s = s_\kappa$, since $s'' = -\kappa s$ gives $s^{-1}(\frac{1}{2}\frac{d^2}{dx^2} + \frac{\kappa}{2})(sf) = \frac{1}{2}f'' + \frac{s'}{s}f'$ and $s'/s = \sqrt{\kappa}\cot(\sqrt{\kappa}x)$. Hence $f(x) = \sinh(x\sqrt{2u-\kappa})/s_\kappa(x)$ satisfies $\frac{1}{2}f'' + \frac{s'}{s}f' = s^{-1}(\frac{1}{2}g'' + \frac{\kappa}{2}g) = uf$ with $g = \sinh(x\sqrt{2u-\kappa})$, and $f$ is, up to a constant factor, the positive increasing solution that stays bounded at 0, with $f(0+) = \sqrt{2u-\kappa}$. Hence $\mathbb{E}_x e^{-uJ_a^{(\kappa)}} = f(x)/f(a)$ for $0 < x < a$, and letting $x \downarrow 0$ gives $\mathbb{E}e^{-uJ_a^{(\kappa)}} = f(0+)/f(a)$, which is the formula; for $\kappa > 2u$ the square roots are imaginary and the formula is read through $\sinh(ix)/i = \sin x$. As a function of $u$ the denominator divided by $\sqrt{2u-\kappa}$ is entire of order $\frac{1}{2}$, with zeros where $a\sqrt{2u-\kappa} = i\pi n$, that is at $u = -(\pi^2 n^2/2a^2 - \kappa/2)$, all negative under the hypothesis on $a$; Hadamard's product in genus zero gives the Thorin measure. The tilt is the formula at $u - \frac{\kappa}{2}$ for $\kappa = 0$, divided by its value at $u = 0$, since $\sqrt{-\kappa}/\sinh(a\sqrt{-\kappa}) = 1/s_\kappa(a)$. □

Three consequences clarify the dictionary. First, Riemann's clock $J_a + J_a'$ is the flat member $\kappa = 0$, and its link with $SU(2)$ in Theorem 8.7.2 is spectral rather than geometric: the atoms $\pi^2 n^2/2a^2$ are the Dirichlet spectrum of $\frac{1}{2}\frac{d^2}{dx^2}$ on $[0, a]$, which for $a = \pi$ is the spectrum of the radial generator of $SU(2)$ shifted by its ground energy $\frac{1}{2}$. Second, the hyperbolic member $\kappa = -1$ has the discrete atoms $\pi^2 n^2/2a^2 + \frac{1}{2}$; Pólya's continuous measure comes instead from a first passage over a level on the half line, a Dirichlet problem on $[0, \infty)$ with continuous spectrum.

Third, the tilt destroys the reciprocal symmetry of $J_a + J'_a$ that encodes the functional equation. With $W = J_1^{(\kappa)} + J_1'^{(\kappa)}$, the function $(\frac{\pi}{2})^s \mathbb{E}W^s$ equals $2\xi(2s)$ for $\kappa = 0$ and is symmetric under $s \mapsto \frac{1}{2} - s$: it takes the value 0.9963110325 at both $s = 0.1$ and $s = 0.4$. For $\kappa = 1$ the two values are 1.002917197 and 1.023703077. A curvature interpolation between the two clocks must therefore be sought in a class that keeps the symmetry (Question 8.7.20).

Replacing $u$ by a negative value in $\mathbb{E}e^{-uX}$ continues the transform across the origin, and this is the rotation of the Laplace variable used in the inversion formula above. For the curvature family the reciprocal of the transform becomes, for $u < \frac{\kappa}{2}$, the damped sine $\sin(a\sqrt{\kappa - 2u})/(\sqrt{\kappa - 2u}\, s_\kappa(a))$, whose zeros sit at minus the Thorin atoms and slide with $\kappa$; for the drifted clock the reciprocal $e^{a(\sqrt{c^2+2u}-c)}$ becomes $e^{-ac}e^{ia\sqrt{2|u|-c^2}}$ beyond the branch point $u = -\frac{c^2}{2}$, which has constant modulus and no zeros.

The rotation of time relates the heat kernels of the two dual spaces. The radial heat kernels of $\frac{1}{2}\Delta$, relative to the Riemannian volume, are

$$p_t^{\mathbb{H}^3}(r) = \frac{e^{-t/2}}{(2\pi t)^{3/2}} \frac{r}{\sinh r} e^{-r^2/2t},$$
$$p_t^{S^3}(\omega) = \frac{e^{t/2}}{(2\pi t)^{3/2}} \sum_{k\in\mathbb{Z}} \frac{\omega + 2\pi k}{\sin \omega} e^{-(\omega+2\pi k)^2/2t},$$

and the term $k = 0$ of the second is, up to the constant factor $(-1)^{-3/2}$, the first after $r \mapsto i\omega$, $t \mapsto -t$, a genuine Wick rotation of time; the sum over $k$ is the method of images over the affine Weyl group, which makes the kernel periodic.

The fourth rotation acts on the Mellin index. The substitution $s = \frac{1}{2} + i\theta$ turns real Mellin moments into Fourier transforms of a tilted law, and Biane's contour shift $w \mapsto w + \frac{i\pi}{2}$ [48, Section 3.5], with $\cosh(w + \frac{i\pi}{2}) = i \sinh w$, turns the damped integrand $e^{zw - 2a\cosh w}$ into an oscillating one on which stationary phase locates the zeros. For $P$ this is the map $\mu \mapsto i\mu$ on the index of $K_\mu$, and Section 8.7.5 shows that the index is the eigenvalue of a boost, which the rotation turns into an angular momentum. Atoms of a Thorin measure on the Euclidean side become poles after the rotation, and poles require discrete spectrum, which for these radial Brownian motions, with no potential, means a compact space. That is why $\xi$, whose time side Thorin measure is atomic, is attached to $SU(2)$, while $P$, whose time side measure has a density, is attached to $\mathbb{H}^3$.

### 8.7.3 The zeros of $P$ as a Dirichlet spectrum

Put $\psi_\mu(x) = K_\mu(e^x)$. Since $\frac{d}{dx} = r\frac{d}{dr}$ for $r = e^x$, the modified Bessel equation $r^2K'' + rK' - (r^2 + \mu^2)K = 0$ becomes

$$-\psi''_\mu + e^{2x}\psi_\mu = -\mu^2\psi_\mu,$$

a Schrödinger equation with the Liouville potential $e^{2x}$, in which the index enters only through the energy $-\mu^2$. Let $x_0 = \log 2\pi$, and let $A_P = -\frac{d^2}{dx^2} + e^{2x}$ act in $L^2([x_0,\infty))$ with a Dirichlet condition at $x_0$. Biane recalls this spectral interpretation of the zeros of the Macdonald function as well known (Titchmarsh [497]) and notes that Pólya does not mention it [48, Section 3.3]. We write $\nu_1 < \nu_2 < \cdots$ for the positive zeros of $\nu \mapsto K_{i\nu}(2\pi)$, so that $P$ vanishes at $\theta = \pm 2\nu_n$.

PROPOSITION 8.7.7. *(i) $A_P$ is selfadjoint, has compact resolvent, and $A_P \geq 4\pi^2$.*

*(ii) A number $\lambda$ is an eigenvalue of $A_P$ if and only if $\lambda = \nu^2$ with $\nu > 0$ and $K_{i\nu}(2\pi) = 0$, and the eigenfunction is then $K_{i\nu}(e^x)$. In particular every zero of $P$ is real, of absolute value at least $4\pi$, and the zeros of $P$ are the numbers $\pm 2\nu_n$ with $\nu_n^2$ running over the spectrum of $A_P$.*

*(iii) Every eigenvalue of $A_P$ is simple, and every zero of $\mu \mapsto K_\mu(2\pi)$ and of $P$ is simple. More precisely, for $\nu > 0$ and every $y \in \mathbb{R}$,*

$$\psi'_{i\nu}(y)\,\partial_\nu\psi_{i\nu}(y) - \psi_{i\nu}(y)\,\partial_\nu\psi'_{i\nu}(y) = -2\nu\int_y^\infty \psi_{i\nu}(x)^2\,dx < 0. \tag{8.7.2}$$

*Proof.* (i) The endpoint $x_0$ is regular, and the potential is bounded below, so the equation is in the limit point case at $+\infty$ and the Dirichlet realisation is selfadjoint; since $e^{2x} \to \infty$, the resolvent is compact. For $f$ in the form domain, $\langle A_P f, f\rangle = \int_{x_0}^\infty (|f'|^2 + e^{2x}|f|^2)\,dx \geq e^{2x_0}\|f\|^2 = 4\pi^2\|f\|^2$. (ii) For every complex $\mu$, $K_\mu(r) = \sqrt{\pi/2r}\,e^{-r}(1 + O(1/r))$ as $r \to \infty$ [397, (10.40.2)], so $\psi_\mu$ is a nonzero solution in $L^2$ near $+\infty$, and in the limit point case it is the only one up to a constant factor. Hence $-\mu^2$ is an eigenvalue exactly when $\psi_\mu(x_0) = K_\mu(2\pi) = 0$. By (i) such an eigenvalue is real and at least $4\pi^2$, so $\mu = \pm i\nu$ with $\nu \geq 2\pi$; since $P(\theta) = K_{i\theta/2}(2\pi)$, the zeros of $P$ are $\theta = \pm 2\nu$. (iii) Two eigenfunctions with the same eigenvalue both vanish at $x_0$, so they are proportional. For (8.7.2), put $\phi = \partial_\nu\psi_{i\nu}$; differentiating $\psi''_{i\nu} = (e^{2x} - \nu^2)\psi_{i\nu}$ in $\nu$ gives $\phi'' = (e^{2x} - \nu^2)\phi - 2\nu\psi_{i\nu}$, so that $(\psi'_{i\nu}\phi - \psi_{i\nu}\phi')' = 2\nu\psi_{i\nu}^2$. Both functions and their derivatives decay like $\exp(-e^x)$, and integrating from $y$ to $\infty$ gives the identity. The function $\psi_{i\nu}$ is real valued and not identically zero, so the right side is negative. At $y = x_0$ and $\nu = \nu_n$ the second term on the left vanishes, so $\partial_\nu K_{i\nu}(2\pi) \neq 0$ at $\nu_n$, which is the simplicity of the zero. □

The proposition is Pólya's theorem that $P$ has only real zeros, obtained from selfadjointness, and $A_P$ is an explicit Hilbert Pólya operator for $P$. Biane also writes the equation in Dirac form [48, Section 3.4]. With $D = \frac{d}{dx}$ and $\mathcal{A} = D + \frac{1}{2} + e^x$, whose formal adjoint is $\mathcal{A}^\dagger = -D + \frac{1}{2} + e^x$, the system $\mathcal{A}f = \gamma g$, $\mathcal{A}^\dagger g = \gamma f$ gives $\mathcal{A}^\dagger\mathcal{A}f = \gamma^2 f$, and a direct computation gives $\mathcal{A}^\dagger\mathcal{A} = A_P + \frac{1}{4}$. With the Dirichlet condition on $f$ the Dirac eigenvalues are therefore $\gamma = \pm(\nu_n^2 + \frac{1}{4})^{1/2}$,

not $\pm\nu_n$; the shift $\frac14$, which Biane writes as $\gamma^2-\frac14$, is missing from the note [426]. In the variable $u=e^x$ the system reads $(u\frac{d}{du}+\frac12+u)f=\gamma g$, $(-u\frac{d}{du}+\frac12+u)g=\gamma f$, and since $u\frac{d}{du}+\frac12$ is $i$ times the symmetrised dilation operator $\frac12(up+pu)$, $p=-i\frac{d}{du}$, it is the Hamiltonian $xp$ of Berry and Keating [45] perturbed by the potential $u$, as Biane remarks.

The zeros of $P$ obey an exact quantisation rule. Let $I_\mu$ be the modified Bessel function of the first kind, and for real $\nu$ put

$$\vartheta_P(\nu)=\operatorname{Im}\log\Gamma(1+i\nu)-\nu\log\pi-\arg{}_0F_1(;1+i\nu;\pi^2), \tag{8.7.3}$$

with the branches continuous in $\nu$ and equal to zero at $\nu=0$, where ${}_0F_1(;1;\pi^2)=I_0(2\pi)>0$. Write $N_P(T)$ for the number of zeros of $P$ in $(0,T]$ and

$$x(T)=\frac{1}{\pi}\vartheta_P\big(\tfrac{T}{2}\big),\qquad N_0(T)=\frac{T}{2\pi}\log\frac{T}{2\pi e}.$$

Proposition 8.7.8. *(i) For $\nu>0$, $I_{i\nu}(2\pi)\neq0$ and*

$$K_{i\nu}(2\pi)=\frac{\pi\,|I_{i\nu}(2\pi)|\,\sin\vartheta_P(\nu)}{\sinh\pi\nu}.$$

*(ii) $\vartheta_P(\nu)>0$ for $\nu>0$, $\vartheta_P(\nu_n)=n\pi$ for every $n$, and the number of zeros of $\nu\mapsto K_{i\nu}(2\pi)$ in $(0,\nu)$ is $\lceil\vartheta_P(\nu)/\pi\rceil-1$. Equivalently, $N_P(T)=\lfloor x(T)\rfloor$ for every $T>0$ that is not a zero of $P$.*

*(iii) As $\nu\to\infty$,*

$$\vartheta_P(\nu)=\nu\log\frac{\nu}{\pi}-\nu+\frac{\pi}{4}+\frac{\pi^2-\frac{1}{12}}{\nu}+O(\nu^{-3}),$$

*so that $x(T)=N_0(T)+\frac14+\big(2\pi-\frac{1}{6\pi}\big)\frac{1}{T}+O(T^{-3})$ as $T\to\infty$.*

*Proof.* (i) For real $r>0$, $I_{-i\nu}(r)=\overline{I_{i\nu}(r)}$, and $K_\mu=\pi(I_{-\mu}-I_\mu)/(2\sin\mu\pi)$ gives $K_{i\nu}(r)=-\pi\operatorname{Im}I_{i\nu}(r)/\sinh\pi\nu$. The series $I_{i\nu}(r)=(r/2)^{i\nu}\,{}_0F_1(;1+i\nu;r^2/4)/\Gamma(1+i\nu)$ shows that, for $r>0$,

$$\vartheta(\nu,r)=\operatorname{Im}\log\Gamma(1+i\nu)-\nu\log\frac{r}{2}-\arg{}_0F_1\big(;1+i\nu;\tfrac{r^2}{4}\big)$$

satisfies $I_{i\nu}(r)=|I_{i\nu}(r)|e^{-i\vartheta(\nu,r)}$, and $\vartheta_P(\nu)=\vartheta(\nu,2\pi)$; this gives the formula. If $I_{i\nu}(r)$ vanished, so would $I_{-i\nu}(r)$, against the Wronskian $I_\mu I'_{-\mu}-I'_\mu I_{-\mu}=-2\sin(\mu\pi)/(\pi r)$ [397, (10.28.1)], which is not zero for $\mu=i\nu$. So ${}_0F_1$ does not vanish on $[0,\infty)\times(0,\infty)$, and $\vartheta$ is defined there as a continuous function with $\vartheta(0,r)=0$. (ii) The same Wronskian, written for $I=I_{i\nu}$ and $\bar I=I_{-i\nu}$, gives $\operatorname{Im}(I'/I)=\sinh(\pi\nu)/(\pi r|I|^2)$, hence $\partial_r\vartheta(\nu,r)=-\sinh(\pi\nu)/(\pi r|I_{i\nu}(r)|^2)<0$. As $r\to\infty$, $I_{i\nu}(r)\sim e^r/\sqrt{2\pi r}$ and $K_{i\nu}(r)\sim\sqrt{\pi/2r}\,e^{-r}$ uniformly for $\nu$ in compact sets, so $\sin\vartheta(\nu,r)=\sinh(\pi\nu)K_{i\nu}(r)/(\pi|I_{i\nu}(r)|)\to0$ and $\cos\vartheta(\nu,r)>0$ for

large $r$; by continuity in $\nu$ from $\nu = 0$, $\vartheta(\nu, r) \to 0$. So for fixed $\nu > 0$ the function $r \mapsto \vartheta(\nu, r)$ decreases from $\vartheta_P(\nu)$ at $r = 2\pi$ to 0 at $r = \infty$; in particular $\vartheta_P(\nu) > 0$, and $r \mapsto K_{i\nu}(r)$ has exactly $\lceil \vartheta_P(\nu)/\pi \rceil - 1$ zeros in $(2\pi, \infty)$, one at each crossing of a multiple of $\pi$. On the other hand, by (8.7.2) at a zero $y$ of $\psi_{i\nu}$, $\psi'_{i\nu}(y)\, \partial_\nu \psi_{i\nu}(y) < 0$, so every zero of $x \mapsto K_{i\nu}(e^x)$ moves to the right as $\nu$ increases; no zero comes in from $+\infty$, where $K_{i\nu}(r) > 0$ uniformly for $\nu$ in compact sets, and zeros cannot appear in pairs, since they are simple. So the number of zeros in $(x_0, \infty)$ is zero for small $\nu$ and increases by one exactly when $\nu$ passes some $\nu_n$, where a zero enters through $x_0$: it equals the number of $\nu_n < \nu$. Comparing the two counts gives the statement, and $\vartheta_P(\nu_n) \in \pi\mathbb{Z}$ with the count passing from $n-1$ to $n$ gives $\vartheta_P(\nu_n) = n\pi$. (iii) Stirling's series gives $\operatorname{Im} \log \Gamma(1 + i\nu) = \nu \log \nu - \nu + \frac{\pi}{4} - \frac{1}{12\nu} + O(\nu^{-3})$, and with $b = 1 + i\nu$, $\log {}_0F_1(;b;z) = z/b - z^2/(2b^2(b+1)) + O(b^{-3})$, so $\arg {}_0F_1(;1+i\nu;\pi^2) = \operatorname{Im} \frac{\pi^2}{1+i\nu} + O(\nu^{-3}) = -\frac{\pi^2}{\nu} + O(\nu^{-3})$. With $\nu = T/2$, $\frac{1}{\pi}(\nu \log \frac{\nu}{\pi} - \nu) = N_0(T)$. □

Writing $s_P(T) = \{x(T)\} - \frac{1}{2}$ for the sawtooth,

$$N_P(T) = N_0(T) - \frac{1}{4} - s_P(T) + \Big(2\pi - \frac{1}{6\pi}\Big)\frac{1}{T} + O(T^{-3}),$$

against the Riemann von Mangoldt formula $N(T) = \vartheta(T)/\pi + 1 + S(T)$ of Section 2.2, which with $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \frac{1}{48T} + O(T^{-3})$ reads

$$N(T) = N_0(T) + \frac{7}{8} + S(T) + \frac{1}{48\pi T} + O(T^{-3}).$$

The two counting functions share the main term $N_0$, but their mean constant terms differ by $\frac{7}{8} + \frac{1}{4} = \frac{9}{8}$, and the fluctuation of $N_P$ is the bounded sawtooth of a quantisation rule where that of $N$ is $S(T)$. These two facts govern the comparison of the two Poisson traces in Section 10.1.5. Table 8.23 shows the effect: $N(1000) - N_P(1000) = 2$, with $N_0(1000) = 647.741$. The note [426] states that the semiclassical count of $A_P$ is exactly the main term of the Riemann von Mangoldt formula. The count is

$$N_{\mathrm{WKB}}(\nu) = \frac{1}{\pi}\int_{x_0}^{\log \nu} \sqrt{\nu^2 - e^{2x}}\, dx = \frac{1}{\pi}\Big(\nu \operatorname{arccosh}\frac{\nu}{2\pi} - \sqrt{\nu^2 - 4\pi^2}\Big),$$

and its expansion gives $N_{\mathrm{WKB}}(T/2) = N_0(T) + \frac{2\pi}{T} + O(T^{-3})$: it agrees with the main term up to $2\pi/T$, not exactly. Its phase $\varphi(\nu) = \pi N_{\mathrm{WKB}}(\nu) + \frac{\pi}{4}$ is the phase of Debye's expansion of $K_{i\nu}(2\pi)$ for large $\nu$, the term $\frac{\pi}{4}$ being the Maslov correction at the turning point, and by (iii) $\vartheta_P(\nu) - \varphi(\nu) = -\frac{1}{12\nu} + O(\nu^{-3})$.

`biane_polya.py` computes the 1516 zeros $\nu_n \le 1000$ from (8.7.3) with mpmath [376].

### 8.7.4 The index side: a GGC with the squared zeros as Thorin atoms, and its clock

Biane's Thorin measures in Theorem 8.7.2 live on the time side: they describe the hitting time itself, in the energy variable of its Laplace transform. The Thorin programme of this book works on the index side, with the reciprocal $1/\mathbb{E}\cosh(\sqrt{u}\,X)$ of the moment generating function of the Pólya law, (1.2.4), whose Thorin atoms would be the squared ordinates. The same hitting time carries both measures, through the Mellin transforms of Proposition 8.7.1 and (9.1.1).

Theorem 8.7.9. *For $u \geq 0$,*

$$\frac{K_0(2\pi)}{K_{\sqrt{u}}(2\pi)} = \frac{\mathbb{E}H^{1/2}}{\mathbb{E}H^{1/2+\sqrt{u}}} = \prod_{n\geq 1}\Big(1+\frac{u}{\nu_n^2}\Big)^{-1},$$

*with $H = H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$ as in Proposition 8.7.1. This is the Laplace transform of the GGC $T_P = \sum_{n\geq 1} e_n/\nu_n^2$, with independent standard exponentials $e_n$, no drift and Thorin measure $\sum_n \delta_{\nu_n^2}$.*

*Proof.* $K_\mu(2\pi) = \int_0^\infty e^{-2\pi\cosh t}\cosh(\mu t)\,dt$ is entire and even in $\mu$, so $w \mapsto K_{\sqrt{w}}(2\pi)$ is entire. Since $|\cosh(\mu t)| \leq \cosh(t\,\mathrm{Re}\,\mu)$, $|K_\mu(x)| \leq K_{\mathrm{Re}\,\mu}(x)$, and for $\nu > 0$, $K_\nu(x) = \frac{1}{2}\int_0^\infty s^{\nu-1}e^{-\frac{x}{2}(s+1/s)}ds \leq \frac{1}{2}\Gamma(\nu)(2/x)^\nu$. Hence $\log|K_{\sqrt{w}}(2\pi)| = O(|w|^{1/2}\log|w|)$, the function has order at most $\frac{1}{2}$ in $w$, and Hadamard's theorem gives a product of genus zero over its zeros. By Proposition 8.7.7 these are the points $w = -\nu_n^2$, each simple. So $K_{\sqrt{u}}(2\pi) = K_0(2\pi)\prod_n(1+u/\nu_n^2)$, and Proposition 8.7.1(ii) with $s = \frac{1}{2}+\sqrt{u}$ gives the middle expression. Each factor $(1+u/\nu_n^2)^{-1}$ is the Laplace transform of $e_n/\nu_n^2$, and $\sum_n \nu_n^{-2} < \infty$. □

The theorem is an instance of two results of the book. In the language of Section 13.1.3, the law $U$ of Proposition 8.7.1(iv), with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, has the moment generating function $L_U(s) = K_{s/2}(2\pi)/K_0(2\pi)$, which by Proposition 8.7.7 vanishes only at $s = \pm 2i\nu_n$; so $U$ is ferromagnetic, with $c_k = 2\nu_k$ and $b = 0$, and Theorem 13.1.4 gives the clock $T_U = \sum_k e_k/(4\nu_k^2) = T_P/4$, the same variable in the normalisation of the ordinate $\theta$. And Proposition 7.1.15, applied to the even function $P$ of order one, which is in the Laguerre Pólya class by Proposition 8.7.7, gives the GGC $P(0)/P(i\sqrt{s}) = K_0(2\pi)/K_{\sqrt{s}/2}(2\pi)$ with atoms at the squared zeros $4\nu_n^2$. The mean and variance of $T_P$ follow from $K_\mu(x) = k_0(x) + \frac{\mu^2}{2}k_2(x) + \frac{\mu^4}{24}k_4(x) + \cdots$, with $k_{2j}(x) = \int_0^\infty t^{2j}e^{-x\cosh t}dt$:

$$\mathbb{E}T_P = \sum_n \frac{1}{\nu_n^2} = \frac{k_2(2\pi)}{2k_0(2\pi)} = 0.0741239453481,$$

$$\mathrm{Var}\,T_P = \sum_n \frac{1}{\nu_n^4} = 2.25103144196\times 10^{-4},$$

where the sums over the 1516 zeros $\nu_n \le 1000$, with the tail described next, agree with these values.

Table 8.24 compares the closed forms with the products over the zeros, for $P$, for Pólya's $\Xi^*$ and for $\Xi$. The tail beyond the last zero $V$ is $\int_V^\infty \log(1 + u/\nu^2)\, dN(\nu) - \frac{1}{2}\log(1+u/V^2)$, with $dN$ the smooth density of zeros, $\pi^{-1}\vartheta'_P$ from Proposition 8.7.8(iii) for $P$ and $\frac{1}{2\pi}\log\frac{T}{2\pi}$ for $\Xi^*$ and $\Xi$; the subtracted half term accounts for the last zero sitting at the lower end of the integral. The note [426] truncated at the zeros below a fixed height and continued with the plain density from the last zero; its agreement to three or four digits, which the last column reproduces, is limited by that tail, not by the identities, which hold to the accuracy of the full computation. For $\Xi$ the agreement is a check, not a test, since RH is verified far beyond this range.

| function | $u$ | closed form | product | difference | truncated |
|---|---|---|---|---|---|
| $P$ | 0.5 | 0.0370339026997 | $\nu_n \le 1000$ | $-4.6 \times 10^{-11}$ | $1.1 \times 10^{-5}$ |
| | 2 | 0.147801988307 | | $-1.8 \times 10^{-10}$ | $4.5 \times 10^{-5}$ |
| | 5 | 0.367871890035 | | $-4.6 \times 10^{-10}$ | $1.1 \times 10^{-4}$ |
| | 100 | 6.62276702182 | | $-9.1 \times 10^{-9}$ | $2.3 \times 10^{-3}$ |
| $\Xi^*$ | 10 | 0.3044209914 | $\gamma_n^* \le 1000$ | $1.6 \times 10^{-8}$ | $8.8 \times 10^{-5}$ |
| | 60 | 1.661131318 | | $9.4 \times 10^{-8}$ | $5.3 \times 10^{-4}$ |
| | 200 | 4.771881099 | | $3.1 \times 10^{-7}$ | $1.8 \times 10^{-3}$ |
| $\Xi$ | 10 | 0.2292377633 | 1700 zeros | $1.6 \times 10^{-7}$ | $3.4 \times 10^{-5}$ |
| | 60 | 1.328022599 | | $9.8 \times 10^{-7}$ | $2.0 \times 10^{-4}$ |
| | 200 | 4.11056511 | | $3.3 \times 10^{-6}$ | $6.8 \times 10^{-4}$ |

Table 8.24: Index side products. The closed forms are $\log(K_{\sqrt{u}}(2\pi)/K_0(2\pi))$, $\log(\Xi^*(i\sqrt{u})/\Xi^*(0))$ and $\log(\xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2}))$; the next column says which zeros enter the product $\sum \log(1 + u/z^2)$, whose difference from the closed form, after the tail of the text is added, is the fifth column. The last column is the difference for the truncation of the note: the 137 zeros $\nu_n < 150$, the 102 zeros of $\Xi^*$ below 240 and 300 zeros of $\Xi$, each with the plain density from the last zero. Computed by `biane_polya.py`.

Pólya's $\Xi^*$ belongs to the same class. The law with density proportional to $\Phi^*$ is ferromagnetic (Section 13.5.1), so by Theorem 13.1.4, or by Proposition 7.1.15, $\Xi^*(0)/\Xi^*(i\sqrt{u}) = \prod_n (1 + u/\gamma_n^{*2})^{-1}$ is the Laplace transform of a GGC whose Thorin measure $\sum_n \delta_{\gamma_n^{*2}}$ sits at the squares of the positive zeros $\gamma_n^*$ of $\Xi^*$; the measure is positive. In the language of Pólya's Hilfssatz II this is the closure of the class under the cosine tilt: the kernel of $G(\theta + ia) + G(\theta - ia)$ is $2\cosh(au)$ times that of $G$, and Pólya's passage from $P$ to $\Xi^*$ is this closure with $a = \frac{9}{2}$ in the variable $\theta$ (Section 7.1.3); the note gives the shift as $\frac{9}{4}$, which is its value in the index of the Bessel function.

The random variable $T_P$ can be realised pathwise. The index of the

Bessel equation enters as a centrifugal term $\mu^2/2r^2$, so a Feynman Kac weight $\exp(-\frac{u}{2}\int dt/R_t^2)$ along a radial path, the clock of the Hartman Watson law and of Brownian winding, shifts the index from 0 to $\sqrt{u}$.

THEOREM 8.7.10. *Let $R$ be the diffusion on $(0,\infty)$ with generator*

$$\mathcal{G} = \frac{1}{2}\frac{d^2}{dr^2} + \Big(\frac{1}{2r} - \frac{K_1(r)}{K_0(r)}\Big)\frac{d}{dr},$$

*the radial part of planar Brownian motion killed at rate $\frac{1}{2}$ and Doob transformed by $K_0$. For $r > b > 0$ let $T_b$ be the hitting time of $b$ and $\tau_r = \frac{1}{2}\int_0^{T_b} dt/R_t^2$ under $\Pr_r$. Then $T_b < \infty$ almost surely and, for $u \geq 0$,*

$$\mathbb{E}e^{-u\tau_r} = \frac{K_{\sqrt{u}}(r)}{K_0(r)} \cdot \frac{K_0(b)}{K_{\sqrt{u}}(b)}.$$

*As $r \to \infty$, $\tau_r$ converges in law to $\tau_\infty$ with $\mathbb{E}e^{-u\tau_\infty} = K_0(b)/K_{\sqrt{u}}(b)$. For $b = 2\pi$, $\tau_\infty \stackrel{d}{=} T_P$.*

*Proof.* Let $\mathcal{A} = \frac{1}{2}(\frac{d^2}{dr^2} + \frac{1}{r}\frac{d}{dr} - 1)$, the generator of the radial part of planar Brownian motion killed at rate $\frac{1}{2}$. The Bessel equation gives $\mathcal{A}K_\mu = \frac{\mu^2}{2r^2}K_\mu$, in particular $\mathcal{A}K_0 = 0$, and since $K_0' = -K_1$, $\mathcal{G}f = K_0^{-1}\mathcal{A}(K_0 f)$. Hence $v = K_{\sqrt{u}}/K_0$ satisfies $\mathcal{G}v = \frac{u}{2r^2}v$. For real order $K_{\sqrt{u}} \geq K_0$, so $v \geq 1$, and $v(r) \to 1$ as $r \to \infty$ because $K_\mu(r) = \sqrt{\pi/2r}\, e^{-r}(1 + O(1/r))$ uniformly for bounded $\mu$; so $v$ is bounded on $[b,\infty)$. The scale density of $\mathcal{G}$ is $s'(r) = 1/(rK_0(r)^2)$, which is not integrable at $\infty$, so from $r > b$ the diffusion reaches $b$ before any level $M > r$ with probability $(s(M) - s(r))/(s(M) - s(b)) \to 1$, and $T_b < \infty$ almost surely; the drift is $-1 + \frac{1}{8r^2} + O(r^{-3})$ at infinity. By Itô's formula $v(R_{t\wedge T_b})\exp(-\frac{u}{2}\int_0^{t\wedge T_b} ds/R_s^2)$ is a bounded martingale, and optional stopping gives $v(b)\mathbb{E}_r e^{-u\tau_r} = v(r)$. The limit follows from $v(r) \to 1$ and the continuity theorem for Laplace transforms, and the last statement from Theorem 8.7.9. □

In the logarithmic coordinate $X = \log R$, run on the clock $\tau = \frac{1}{2}\int dt/R_t^2$, Itô's formula gives

$$dX_\tau = -2e^{X_\tau}\frac{K_1(e^{X_\tau})}{K_0(e^{X_\tau})}\,d\tau + \sqrt{2}\,dW_\tau,$$

a diffusion with generator $\frac{d^2}{dx^2} + 2\frac{\psi'}{\psi}\frac{d}{dx} = -\psi^{-1}A_P\,\psi$, where $\psi(x) = K_0(e^x)$ is the ground solution of $-\psi'' + e^{2x}\psi = 0$. Its drift is about $-2e^x$, so $+\infty$ is an entrance boundary, and $\tau_\infty$ is its hitting time of $x_0 = \log 2\pi$ from $+\infty$. This is the Kent form of Section 8.1.3: by Kent's theorem [305] the hitting time from an entrance boundary is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues $\lambda_n$ of the generator, here those of $A_P$, which are the $\nu_n^2$. For Pólya's function Theorem 8.7.10 therefore solves the inverse problem that Sections 9.2.1 and 8.1.3 pose for $\xi$: it exhibits

the diffusion, here with the Liouville potential, whose hitting time has its Thorin atoms at the squared zeros, without Krein's inverse spectral theory. For $\Xi$ one wants a half line operator with Dirichlet spectrum $\{\gamma^2/4\}$, the counterpart of $\{\nu_n^2\} = \{\theta_n^2/4\}$ over the zeros $\theta_n = 2\nu_n$ of $P$; the potential $e^{2x}$ on $[\log 2\pi, \infty)$ has the right Weyl law (Proposition 8.7.8), so it is the natural reference to perturb, and the perturbation must account for the difference between $\Phi$ and Pólya's asymptote seen in Figure 8.10 (Question 8.7.17).

COROLLARY 8.7.11. *For real $\lambda$, with $\tau_\infty$ the clock of Theorem 8.7.10 for $b = 2\pi$ and a standard Brownian motion $B$ independent of it,*

$$\frac{K_0(2\pi)}{K_\lambda(2\pi)} = \mathbb{E}e^{i\lambda B_{2\tau_\infty}}.$$

*Proof.* $\mathbb{E}e^{i\lambda B_{2\tau}} = \mathbb{E}e^{-\lambda^2\tau}$ for an independent random time $\tau$; apply Theorem 8.7.10 with $u = \lambda^2$. □

For $\Xi$ the corresponding identity $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\lambda) = \mathbb{E}e^{i\lambda B_{2T}}$ holds with no hypothesis, with the clock $T$ of Theorem 5.1.1, and the variable $B_{2T}$ is the variable $Y_{1/2}$ of Section 10.1.4; what RH adds is that $T$ is the GGC $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 1.2.1). The note [426] states the identity for $\Xi$ under RH and says that for Riemann's function the random time exists if and only if RH holds; the time exists unconditionally, and only its GGC form is equivalent to RH.

REMARK 8.7.12 (The Hartman Watson law). Theorem 8.7.10 is the $K$ counterpart of a formula of Yor [524] for the Hartman Watson law [175]: for planar Brownian motion started at distance $r$ from the origin and conditioned on $|B_t| = \rho$, $\mathbb{E}[\exp(-\frac{u}{2}\int_0^t ds/|B_s|^2)] = I_{\sqrt{u}}(r\rho/t)/I_0(r\rho/t)$. There the endpoint is fixed and the function is $I$; here the path is killed at rate $\frac{1}{2}$, transformed by $K_0$ and stopped at a level, and the function is $K$. Pitman and Yor develop the general theory of such Bessel laws [415].

`biane_clock.py` simulates the clock. With $R_0 = 2\pi e^{2.5} = 76.544867$, so that $X$ starts at $\log 2\pi + 2.5$ as in the note, the exact law of $\tau_{R_0}$ has, from the expansion of $\log v$ in $u$, mean 0.06763390577 and standard deviation 0.01499747823, against 0.07412394535 and 0.01500343775 for $\tau_\infty$. The script runs 100000 paths of the Euler scheme for $X$ with step $\Delta = 4\times10^{-6}$, the step of the note, and records two hitting times on the same paths: the plain one, the first step that ends below $\log 2\pi$, and one with the Brownian bridge correction, under which a step from $x$ to $x'$, both above the level $\ell = \log 2\pi$, crosses it with probability $\exp(-(x-\ell)(x'-\ell)/\Delta)$, the variance of the step being $2\Delta$. The plain scheme overshoots the barrier and is late by about $0.5826\sqrt{2\Delta}/|\beta|$, with $\beta = -13.5317$ the drift at the barrier and $0.5826 = -\zeta(\frac{1}{2})/\sqrt{2\pi}$ the constant of Broadie, Glasserman and Kou [66], that is by $1.218\times10^{-4}$. Table 8.25 shows that the plain mean is 2.9 standard errors late, by $1.39\times10^{-4}$, and that the corrected one agrees with the exact mean. The

exact Laplace transform is computed with Arb and the exact densities by the fixed Talbot contour [490, 1]. Figure 8.11 shows both comparisons. The note's own run, 40000 plain paths with mean $0.067695 \pm 0.000075$, is consistent with the exact mean and with the predicted lateness.

| | mean | standard deviation | $z$ of the mean | quantiles 0.01, 0.5, 0.99 |
|---|---|---|---|---|
| exact | 0.0676339 | 0.0149975 | | |
| plain Euler | 0.0677726 | 0.0150584 | 2.91 | 0.04190, 0.06559, 0.11391 |
| bridge corrected | 0.0676538 | 0.0150221 | 0.42 | 0.04185, 0.06548, 0.11364 |
| $u$ | 25 | 100 | 200 | 400 |
| exact $\mathbb{E}e^{-u\tau_{R_0}}$ | 0.196345 | $2.54238 \times 10^{-3}$ | $1.65093 \times 10^{-5}$ | $2.91684 \times 10^{-9}$ |
| $z$, bridge corrected | $-0.25$ | 0.36 | 0.72 | 0.12 |
| $z$, plain Euler | $-2.81$ | $-1.84$ | $-0.76$ | $-0.48$ |

Table 8.25: The clock from $R_0 = 2\pi e^{2.5}$: exact moments against 100000 Euler paths with step $4 \times 10^{-6}$, with and without the bridge correction, and the Laplace transform, with the $z$ scores (Monte Carlo − exact)/standard error. The standard error of the mean is $4.75 \times 10^{-5}$; over $0 < u \le 400$ the largest $|z|$ of the corrected transform is 0.72. Computed by `biane_clock.py`.

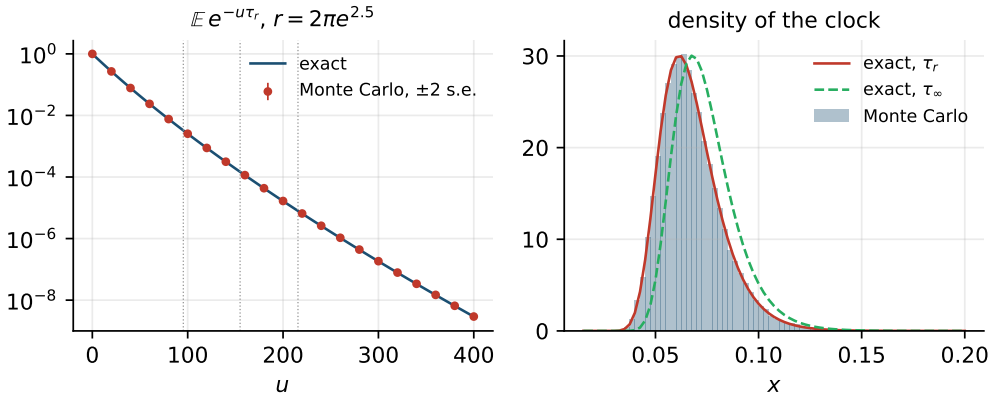



Figure 8.11: The clock from $R_0 = 2\pi e^{2.5}$. Left: $\mathbb{E}e^{-u\tau_{R_0}}$, exact and from the bridge corrected Monte Carlo with two standard errors, on a logarithmic scale; the dotted lines mark the first three Thorin atoms $\nu_1^2 = 95.4289$, $\nu_2^2 = 154.965$, $\nu_3^2 = 215.648$ of $\tau_\infty$. Right: the histogram of the corrected hitting times against the exact densities of $\tau_{R_0}$ and of $\tau_\infty$, by Talbot inversion. Drawn by `biane_figs.py` from `biane_clock.json`.

The cosine dictionary (6.2.4) turns each Thorin measure of this section into a Lévy measure of the reciprocal transform, in the variable conjugate to the square root of the energy.

PROPOSITION 8.7.13. *For real $\lambda$, $a>0$, $c>0$ and $\ell>0$,*

$$2\log\frac{\sinh\lambda a}{\lambda a}=\int_0^\infty(1-\cos\lambda v)\,\frac{4\,dv}{v\,(e^{\pi v/a}-1)},$$

$$\ell\big(\sqrt{c^2+\lambda^2}-c\big)=\int_0^\infty(1-\cos\lambda v)\,\frac{2\ell c}{\pi}K_1(cv)\,\frac{dv}{v},$$

$$\log\frac{K_\lambda(2\pi)}{K_0(2\pi)}=\int_0^\infty 4\sin^2\Big(\frac{\lambda v}{2}\Big)\,\Theta_P(v)\,\frac{dv}{v},\qquad \Theta_P(v)=\sum_{n\ge1}e^{-\nu_n v}=\operatorname{tr}e^{-v\sqrt{A_P}}.$$

*The left sides are $-\log\mathbb{E}e^{-\lambda^2W/2}$ for $W=J_a+J_a'$ and $W=H_\ell^{(c)}$, and $-\log\mathbb{E}e^{-\lambda^2T_P}$.*

*Proof.* By Frullani's integral, $\log(1+\lambda^2/\beta^2)=\int_0^\infty(1-\cos\lambda v)\,2e^{-\beta v}dv/v$. For $W=J_a+J_a'$ the Thorin measure $2\sum_n\delta_{\pi^2n^2/2a^2}$ of Section 8.7.1 gives, at $u=\lambda^2/2$, the rates $\beta=\pi n/a$ with weight 2, and $\sum_n e^{-\pi nv/a}=(e^{\pi v/a}-1)^{-1}$. For $H_\ell^{(c)}$ the density $\ell\{\pi\sqrt{2z-c^2}\}^{-1}$ on $z>\frac{c^2}{2}$ becomes, under $\beta=\sqrt{2z}$, the measure $\ell\beta\,d\beta/(\pi\sqrt{\beta^2-c^2})$ on $(c,\infty)$, and $\int_c^\infty e^{-\beta v}\beta(\beta^2-c^2)^{-1/2}d\beta=cK_1(cv)$. For $T_P$ take Theorem 8.7.9 at $u=\lambda^2$, with rates $\nu_n$, and use $1-\cos=2\sin^2(\cdot/2)$. □

The Bessel clock has the Planck measure, a geometric superposition of the frequencies $\pi n/a$, and the drifted clock has the Macdonald measure, absolutely continuous because its spectrum is. The dictionary acts on Thorin measures, not on Biane's Lévy measures in the time variable, which are heat traces. For Pólya's function the third identity is the exact analogue of the Lévy Khintchine form of $\xi(\frac12)/\xi(\frac12+u)$ in Section 10.1.4, with the Poisson trace $K_{1/2}$ replaced by the wave trace $\Theta_P$ of the selfadjoint operator $A_P$, which is completely monotone with no hypothesis. In the variable $\theta$, where the zeros of $P$ are $2\nu_n$, the trace is $K^P(y)=\Theta_P(2y)=\sum_n e^{-2\nu_n y}$, and $\log(K_{\lambda/2}(2\pi)/K_0(2\pi))=\int_0^\infty 4\sin^2(\lambda y/2)K^P(y)\,dy/y$ has the form of the formula for $\xi$ with $K^P$ in place of $K_{1/2}$; Section 10.1.5 compares the two traces.

### 8.7.5 Rindler modes and a Klein Gordon criterion

Undoing the Wick rotation of the angle in the Euclidean plane of Theorem 8.7.10 gives a Lorentzian picture in which Pólya's function is a standard object of quantum field theory. Put

$$F(x,w)=\exp(-e^x\cosh w),\qquad\text{so that}\qquad K_\mu(e^x)=\frac12\int_{\mathbb{R}}F(x,w)\,e^{\mu w}\,dw.$$

Since $\partial_x^2F=(e^{2x}\cosh^2w-e^x\cosh w)F$ and $\partial_w^2F=(e^{2x}\sinh^2w-e^x\cosh w)F$,

$$\Big(\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial w^2}-e^{2x}\Big)F=0.$$

In the coordinates $X_1 = e^x \cosh w$, $X_0 = e^x \sinh w$ of the Rindler wedge $X_1 > |X_0|$, in which $\rho = e^x$ is the proper distance from the edge of the wedge and $w$ the rapidity, $\partial_{X_1}^2 - \partial_{X_0}^2 = e^{-2x}(\partial_x^2 - \partial_w^2)$, so the equation is $(\partial_{X_1}^2 - \partial_{X_0}^2 - 1)F = 0$ with $F = e^{-X_1}$: the Klein Gordon equation of unit mass in $1+1$ dimensional Minkowski space, solved by a plane wave. The index $\mu$ is the eigenvalue of the boost generator $\partial_w$, and $K_{i\omega}(\rho)$ is the Rindler mode of boost frequency $\omega$ [142]. The zeros of $P(\theta) = K_{i\theta/2}(2\pi)$ are therefore, with $\theta = 2\omega$, the boost frequencies of the normal modes of a unit mass scalar field in the Rindler wedge with a Dirichlet mirror on the hyperbola $\rho = 2\pi$, the configuration of 't Hooft's brick wall model of a black hole horizon [276]. Biane's Dirac system of Section 8.7.3 is the corresponding Dirac field, close to the Rindler space models of Sierra for the Hamiltonian $xp$ [479].

The Wick rotation $w = i\phi$ turns the wedge into the Euclidean plane in polar coordinates $(\rho, \phi)$, the boost into a rotation, the Klein Gordon operator into $\Delta - 1$, and the plane wave $e^{-X_1}$ into $e^{-\rho\cos\phi}$, which is harmonic for $\frac{1}{2}(\Delta - 1)$, the generator of planar Brownian motion killed at rate $\frac{1}{2}$ of Theorem 8.7.10. The boost frequency becomes the angular momentum $\mu$, and the radial operator $\frac{1}{2}(\partial_\rho^2 + \rho^{-1}\partial_\rho - \mu^2/\rho^2 - 1)$ carries the index as the centrifugal term that the Feynman Kac weight of the clock produces.

THEOREM 8.7.14 (A Klein Gordon criterion). *Let $g$ be even, positive and continuous with $\int g(w)e^{c|w|}dw < \infty$ for every $c$, and $G(t) = \int g(w)e^{itw}dw$. Suppose that there are $x_0 \in \mathbb{R}$, a real continuous function $V$ on $[x_0, \infty)$ bounded below, and $F \in C^2([x_0, \infty) \times \mathbb{R})$ such that*

*(a) $(\partial_x^2 - \partial_w^2 - V(x))F = 0$ and $F(x_0, \cdot) = g$;*

*(b) $F$ and its partial derivatives of order at most two are $O(e^{-c|w|})$ for every $c > 0$, uniformly for $x$ in compact subsets of $[x_0, \infty)$, and $\Psi_\mu(x) = \int F(x, w)e^{\mu w}dw$ lies in $L^2((x_0, \infty))$ for every $\mu \in \mathbb{C}$;*

*(c) $\Psi_\mu$ is not identically zero for any $\mu \in \mathbb{C}$.*

*Then every zero of $G$ is real, and its square is an eigenvalue of $-\frac{d^2}{dx^2} + V$ on $[x_0, \infty)$ with a Dirichlet condition at $x_0$.*

*Proof.* By (b), $\Psi_\mu$ is twice differentiable under the integral sign, and two integrations by parts in $w$, whose boundary terms vanish, give $\Psi_\mu'' = \int(\partial_w^2 F + VF)e^{\mu w}dw = (\mu^2 + V)\Psi_\mu$. Since $V$ is bounded below, the operator $-\frac{d^2}{dx^2} + V$ is in the limit point case at $\infty$, and its Dirichlet realisation at $x_0$ is selfadjoint. If $G(t) = 0$, then $\Psi_{it}(x_0) = G(t) = 0$ by (a), and by (b) and (c) $\Psi_{it}$ is an eigenfunction with eigenvalue $-(it)^2 = t^2$, which is therefore real. So $t$ is real or purely imaginary, and $t = -i\mu$ with real $\mu$ is excluded because $G(-i\mu) = \int g(w)e^{\mu w}dw > 0$. □

For Pólya's function the hypotheses hold in the rapidity variable $w = 2u$: with $g(w) = e^{-2\pi\cosh w}$, $G(t) = 2K_{it}(2\pi) = 2P(2t)$, $V(x) = e^{2x}$, $x_0 = \log 2\pi$, $F = \exp(-e^x \cosh w)$ and $\Psi_\mu(x) = 2K_\mu(e^x)$, and the theorem gives Proposition 8.7.7(ii) again. `biane_polya.py` checks the equation for $F$. The criterion says that the reality of the zeros of an even Fourier transform follows from embedding its kernel as the boundary value of a field that satisfies a Klein Gordon equation with a real, time independent potential, which decays fast enough in the rapidity.

For $\Xi$ a Klein Gordon extension of $\Phi$ would prove RH (Question 8.7.18). The scaling matters. For $P$ the eigenvalues of $A_P$ are $\nu_n^2 = \theta_n^2/4$, and the function whose zeros they are, as a function of the spectral parameter $\mu$ at the boundary $x_0$, is the Jost function $\mu \mapsto \Psi_\mu(x_0) = 2K_\mu(2\pi) = 2P(-2i\mu)$. A potential asymptotic to $e^{2x}$ at $+\infty$ has the Weyl law of Proposition 8.7.8, which matches the zeros of $\Xi$ only in the same scaling, $g(w) = \Phi(w/2)$ and $G(t) = 2\Xi(2t)$, and then the Jost function is $\mu \mapsto 2\Xi(-2i\mu) = 2\xi(\frac{1}{2} + 2\mu)$. The note [426] proposes $\mu \mapsto \xi(\frac{1}{2} + \mu)$, which corresponds to the eigenvalues $\gamma^2$, whose density is half that of the Weyl law of $e^{2x}$ (a potential growing like $e^{4x}$ would match them). Inverse spectral theory reconstructs an operator from its eigenvalues and norming constants, a potential in the forms of Gelfand and Levitan or of Marchenko and a string in Krein's form, and under RH and the simplicity of the zeros Krein's theory realises the eigenvalues $\gamma^2/4$ (Section 8.1.3); the content of the question is that the operator be of the form $-\frac{d^2}{dx^2} + V$ and that $V$ admit a Klein Gordon solution whose boundary value is $\Phi(w/2)$, which is a statement about the whole function $\xi$ and not only about its zeros.

### 8.7.6 The de Bruijn Newman flow of $P$ against that of $\Xi$

For real $t$ put

$$P_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4} e^{-2\pi\cosh 2u}\, e^{i\theta u}\, du,$$

$$\Xi_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4}\, \Phi(u)\cos(\theta u)\, du = 8H_t(2\theta),$$

so that $P_0 = P$, $\Xi_0 = \Xi$, and $t$ is the time of the de Bruijn Newman flow in the normalisation of Section 8.4.1. Decreasing $t$ is the direction in which real zeros can be lost: RH is the statement that $\Xi_t$ keeps only real zeros down to $t = 0$, and by the theorem of Rodgers and Tao [463] it has nonreal zeros for every $t < 0$.

Proposition 8.7.15. *There is a constant $\Lambda_P$ with $-\infty < \Lambda_P \le 0$ such that $P_t$ has only real zeros if and only if $t \ge \Lambda_P$.*

*Proof.* Let $S$ be the set of $t$ for which $P_t$ has only real zeros. By Proposition 8.7.7, $0 \in S$. If $t \in S$ and $t' > t$, then $t' \in S$ by de Bruijn's theorem on the Gaussian factor $e^{\lambda u^2}$, $\lambda > 0$ [112, Theorem 13], which applies to the kernel $e^{tu^2/4}e^{-2\pi\cosh 2u}$

because it is $O(e^{-|u|^b})$ for every $b > 2$. If $t_k \in S$ decrease to $t$, then $P_{t_k} \to P_t$ locally uniformly on $\mathbb{C}$ by dominated convergence, with $|e^{t_k u^2/4} e^{i\theta u}| \le e^{t_1 u^2/4} e^{|\operatorname{Im}\theta||u|}$, and $P_t(0) > 0$, so by Hurwitz's theorem every zero of $P_t$ is a limit of real zeros and $t \in S$. Hence $S = [\Lambda_P, \infty)$ with $\Lambda_P \in [-\infty, 0]$. Suppose $\Lambda_P = -\infty$. Then, for the even probability measure $\rho$ with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, the Fourier transform of $e^{\lambda u^2}\rho(du)$ has only real zeros for every real $\lambda$, and by Newman's theorem [381], [383, Theorem 10], $\rho$ is either $\frac{1}{2}(\delta_{u_0} + \delta_{-u_0})$ or has a density $Ku^{2m}\exp(-\alpha u^4 - \beta u^2)\prod_j (1 + u^2/a_j^2)e^{-u^2/a_j^2}$, with $K > 0$, $m \ge 0$ an integer, $a_j > 0$, $\sum_j a_j^{-4} < \infty$, and $\alpha > 0$, or $\alpha = 0$ and $\beta + \sum_j a_j^{-2} > 0$. The inequality $(1 + y)e^{-y} \ge e^{-y^2/2}$ for $y \ge 0$ bounds such a density below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$, while $e^{-2\pi\cosh 2u}$ decays faster than $e^{-Cu^4}$ for every $C$. This contradiction gives $\Lambda_P > -\infty$. □

The proof is Newman's proof that $\Lambda > -\infty$ [381], Theorem 9 of [383], which uses only the fact that the kernel decays faster than $e^{-Cu^4}$ for every $C$; it applies verbatim to $\Phi^*$, and to every even, positive kernel with that decay whose Fourier transform has only real zeros. The note [426] conjectures $\Lambda_P = -\infty$, on the evidence of the computation below; Proposition 8.7.15 shows that the conjecture is false.

`biane_flow.py` computes both transforms by the trapezoid rule with step $1/250$ on $[0, 2]$ at 90 digits. The kernels are analytic in $|\operatorname{Im} u| < \pi/4$ (the kernel of $P_t$ is entire) and below $10^{-68}$ beyond $u = 2$, and at $t = 0$ the sums reproduce Arb's $K_{i\theta/2}(2\pi)$ and mpmath's $\Xi$ at test points up to $\theta = 90$ with relative errors below $1.4 \times 10^{-45}$. The real zeros in $(0, 90)$ are located by sign changes on a grid of step 0.05 and refined, for $t = 0, -0.5, \ldots, -20$ and $t = -21, \ldots, -60$. Figure 8.12 shows the result. The real zeros of $\Xi_t$ in the window fall from 25 at $t = 0$ to 2 at $t = -40$, and they are lost in pairs, neighbouring zeros colliding and leaving the axis: the first collisions in the window, as $t$ decreases, are those of the census of Section 8.4.3, beginning with $(\gamma_{24}, \gamma_{25})$ at $t = -1.144$ and height 88.47 near the top of the window. For $P_t$ nothing of the kind occurs down to $t = -60$: from each time of the grid to the next every real zero increases, and the number in the window falls only as zeros leave it through $\theta = 90$.

Sign changes count only real zeros, so the script also counts all zeros in the box $0.5 \le \operatorname{Re}\theta \le 60.3$, $|\operatorname{Im}\theta| \le 3$ by the argument principle, on a contour refined until the argument changes by less than 0.5 between neighbours. For $P_t$ the box holds 12, 11, 8 and 3 zeros at $t = 0$, $-10$, $-30$ and $-60$, all of them real. For $\Xi_t$ it holds 13, 12 and 12 zeros at $t = 0$, $-2$ and $-4$, of which 13, 10 and 8 are real. Three of the collisions below height 60, those listed in the note [426], are located

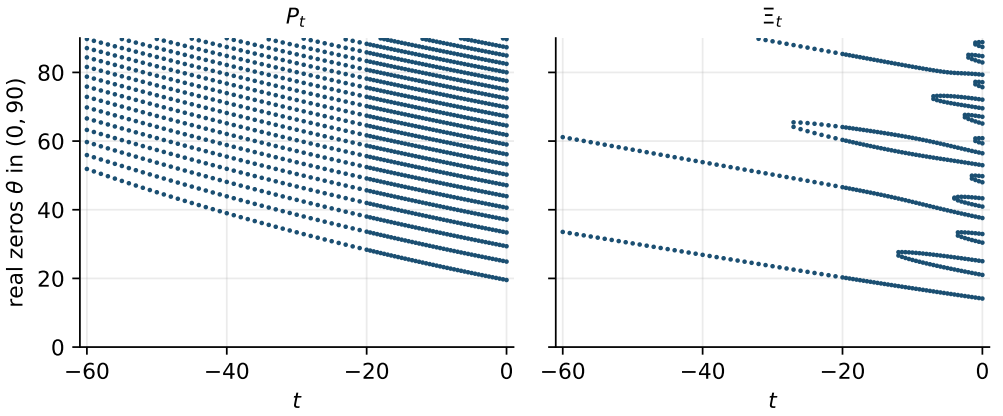



Figure 8.12: The real zeros in $(0, 90)$ of $P_t$ (left) and of $\Xi_t$ (right) for $-60 \le t \le 0$, on a grid (step 0.5 above $t = -20$, step 1 below). The zeros of $P_t$ drift upward and leave the window one at a time; those of $\Xi_t$ collide in pairs. Drawn by `biane_figs.py` from `biane_flow.json`.

by Newton's method for the double zero $\Xi_t = \partial_\theta \Xi_t = 0$ in $(\theta, t)$:

$$
\begin{aligned}
(\gamma_9, \gamma_{10}) &: \; t_c = -1.877909073, \;\; \theta_c = 49.66423465; \\
(\gamma_4, \gamma_5) &: \; t_c = -3.954792524, \;\; \theta_c = 33.09873381; \\
(\gamma_2, \gamma_3) &: \; t_c = -12.49451313, \;\; \theta_c = 27.3212277.
\end{aligned}
$$

The first agrees with the census of Section 8.4.3 ($-1.87791$ at height 49.664); the other two lie below its range $[-2.05, 0]$. The note presents the three as the first collisions below height 60, but the computation, on its grid of step 0.5 in $t$, shows a fourth before $t = -12.5$: the zeros $\gamma_7$ and $\gamma_8$ are at 43.263 and 43.522 at $t = -4$ and have left the axis at $t = -4.5$, and the isolated pair estimate of Section 8.4.1 puts their collision at about $t = -4.03$. In the variable $\theta$ a simple zero moves by

$$
\frac{d\theta_k}{dt} = \frac{\partial_\theta^2 F_t(\theta_k)}{4\, \partial_\theta F_t(\theta_k)} = \frac{1}{2} \sum_{j \neq k} \frac{1}{\theta_k - \theta_j},
$$

for $F_t = P_t$ and for $F_t = \Xi_t$, which is the equation of motion of Section 8.4.1 with $z = 2\theta$. As $t$ decreases every two zeros attract each other; a pair collides when its mutual attraction overcomes the difference of the pulls of all the other zeros on its two members.

The mean motion can be predicted. For $\theta > 4\pi$ the integral $P(\theta)$ is dominated by the conjugate saddle points $u_\pm = \pm \frac{x}{2} + \frac{i\pi}{4}$ of $i\theta u - 2\pi \cosh 2u$, with $x = \operatorname{arccosh}(\theta/4\pi)$, where the exponent is $-\frac{\pi\theta}{4} \pm i(\frac{1}{2}\theta x - 2\pi \sinh x)$. So $P(\theta)$ is $e^{-\pi\theta/4}$ times a slowly varying amplitude times a cosine of phase $\varphi(\theta) = \frac{1}{2}\theta x - 2\pi \sinh x$

plus a constant; since $\varphi'(\theta) = \frac{x}{2}$, the zeros have density $\frac{x}{2\pi} = \frac{1}{2\pi}\log\frac{\theta}{2\pi} + O(\theta^{-2})$. To first order in $t$ the factor $e^{tu^2/4}$ adds to the phase the imaginary part $\frac{\pi t}{16}x$ of $\frac{t}{4}u_+^2$, so the number of zeros below a fixed height changes at the rate $\frac{x}{16}$ per unit of $t$, and each zero moves with

$$\frac{d\theta}{dt} \approx -\frac{x/16}{x/2\pi} = -\frac{\pi}{8},$$

whatever its height. The equation of motion gives the same value: with the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$ on both half lines, $\frac{1}{2}\sum_{j\neq k}(\theta_k - \theta_j)^{-1}$ becomes $\frac{1}{4\pi}$ times the principal value integral of $\log|s|/(\theta_k - s)$, which is $-\frac{\pi^2}{2}$ for $\theta_k > 0$. Over $-0.5 \le t \le 0$ the 24 real zeros of $P$ in the window rise at mean speeds that decrease with the height from 0.3990 (the first zero) to 0.3949 (the last), towards $\pi/8 = 0.3927$, while the 25 zeros of $\Xi$ rise at speeds between 0.0040, for $\gamma_{25}$, held back by $\gamma_{24}$, which is closing on it, and 0.7350, for $\gamma_{19}$, which is closing on $\gamma_{20}$, with mean 0.3647 (`biane_figs.py`). The zeros of $P$ move almost as a rigid block; those of $\Xi$ fluctuate about the same mean motion, and the fast and slow members of the close pairs are the ones that collide.

In the language of Lee and Yang [331] (Section 7.1.3 and Proposition 7.1.16), the single spin law with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$ has the Lee and Yang property, since its moment generating function $K_{h/2}(2\pi)/K_0(2\pi)$ vanishes only at imaginary $h$; it is the ferromagnetic law $U$ of Section 8.7.4. Newman's theorem says that this property survives every Gaussian factor $e^{-bu^2}$, $b > 0$, only for two point laws and for the densities of the proof of Proposition 8.7.15, which include the $\varphi^4$ laws $e^{-au^4-bu^2}$ that Simon and Griffiths realised as limits of Ising ferromagnets [480]. The law with density proportional to $e^{-2\pi\cosh 2u-bu^2}$ has the property exactly when its Fourier transform, a multiple of $P_{-4b}$, has only real zeros, that is when $b \le -\Lambda_P/4$. So the Simon Griffiths type construction for $e^{-2\pi\cosh 2u-bu^2}$ at every real $b$ that the note proposes cannot exist.

### 8.7.7 Pólya's function and Ξ side by side

Table 8.26 follows the comparison in the note [426], corrected in three rows: the index side clock of $\Xi$ exists with no hypothesis, the de Bruijn Newman constant of $P$ is finite, and the computation of the flow describes a window of heights and times, not the whole flow. The Wick rotation of Section 8.7.2 relates the two entries of each of the second, third and fourth rows.

The statements about $\Xi$ in the right column and their relatives, Proposition 1.2.1, Corollary 10.1.4, Proposition 7.1.15 and the GGC form of the clock $T$ discussed after Corollary 8.7.11, are proved by Hadamard factorisation, Bernstein's theorem and integral identities. No property of the primes enters their proofs, and none could, since each is an equivalence between two statements about the zeros of one entire function: they reorganise RH, as Section 12.2 says of the

| | Pólya's $P(\theta) = K_{i\theta/2}(2\pi)$ | Riemann's $\Xi(\theta)$ |
|---|---|---|
| kernel | $e^{-2\pi\cosh 2u}$, (8.7.1) | $\Phi(u)$, (1.2.3) |
| hitting time | $H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$, Brownian motion with drift (Proposition 8.7.1) | $\frac{\pi}{2}S_2$, three dimensional Bessel process, (9.1.1) |
| symmetric space | $\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$ | $SU(2) \cong S^3$ |
| Thorin measure of the hitting time (energy) | continuous: $\sigma_2$ for $H_1^{(1)}$ | discrete: $2\pi\sigma_1$ for $\pi^2 S_2$ (Theorem 8.7.2) |
| index side Thorin measure (angular momentum) | $\sum_n \delta_{\nu_n^2}$, a theorem (Theorem 8.7.9) | $\sum_\gamma \delta_{\gamma^2}$ if and only if RH (Proposition 1.2.1) |
| index side clock | $\tau_\infty$, a hitting time of an explicit diffusion (Theorem 8.7.10) | a random time $T$ (Theorem 5.1.1); no diffusion is known (Question 8.7.17) |
| trace of the sine squared form | $\Theta_P = \operatorname{tr} e^{-v\sqrt{A_P}}$, completely monotone (Proposition 8.7.13) | $K_{1/2}$, completely monotone if and only if RH (Corollary 10.1.4) |
| operator for the zeros | $-\frac{d^2}{dx^2} + e^{2x}$ on $[\log 2\pi, \infty)$ (Proposition 8.7.7) | not known |
| Klein Gordon extension | $\exp(-e^x \cosh w)$ (Theorem 8.7.14) | not known (Question 8.7.18) |
| de Bruijn Newman constant | $-\infty < \Lambda_P \le 0$ (Proposition 8.7.15) | $\Lambda \ge 0$ [463]; RH if and only if $\Lambda = 0$ |
| real zeros in $(0, 90)$ for $-60 \le t \le 0$ | no collision; mean speeds 0.395 to 0.399 on $[-0.5, 0]$ | collisions in pairs from $t = -1.144$; mean speeds 0.004 to 0.735 on $[-0.5, 0]$ |

Table 8.26: Pólya's function and $\Xi$. Apart from the computations of the last row, the left column is a list of theorems; in the right column the entries that concern the zeros are equivalent to RH or open.

programme as a whole, and do not approach it. The proofs of the entries on the left that concern the zeros all go through the selfadjoint operator $A_P$, which is exactly what the right column lacks.

The note [426] proves part (a) of the following proposition; part (b) is the strengthening that Newman's theorem gives.

PROPOSITION 8.7.16. *Let $\mathcal{P}$ be a property of even, positive, integrable kernels such that*

*(i) if $\mathcal{P}(g)$ holds, then $\int g(u)e^{i\theta u}du$ has only real zeros, and*

*(ii) there is $\varepsilon > 0$ such that $\mathcal{P}(g)$ implies $\mathcal{P}(e^{-\varepsilon u^2/4}g)$.*

*(a) If (ii) holds for $g = \Phi$, then $\mathcal{P}(\Phi)$ is false.*

*(b) If (ii) holds for all $g$ with the same $\varepsilon$, then $\mathcal{P}(g)$ is false for every kernel $g$ that is $O(e^{-|u|^b})$ for some $b > 2$ and satisfies $g(u)e^{Cu^4} \to 0$ as $|u| \to \infty$ for every $C$; in particular $\mathcal{P}$ fails for $\Phi$, for $\Phi^*$ and for $e^{-2\pi \cosh 2u}$.*

*Proof.* (a) If $\mathcal{P}(\Phi)$ held, then by (ii) and (i) the transform of $e^{-\varepsilon u^2/4}\Phi$, which is $\Xi_{-\varepsilon}$ of Section 8.7.6, would have only real zeros, so $\Lambda \le -\varepsilon < 0$, against the theorem of Rodgers and Tao [463]. (b) If $\mathcal{P}(g)$ held, then by induction $\mathcal{P}(e^{-k\varepsilon u^2/4}g)$ would hold for $k = 0, 1, 2, \dots$, so the transform of $e^{-k\varepsilon u^2/4}g$ would have only real zeros for each $k$. This kernel is again $O(e^{-|u|^b})$, so by de Bruijn's theorem [112, Theorem 13] the transform of $e^{tu^2/4}g$ would have only real zeros for every $t \ge -k\varepsilon$, and hence for every real $t$. By Newman's theorem, as in the proof of Proposition 8.7.15, $g/\int g$ would be a density of the form given there, which is bounded below by $Ku^{2m}\exp(-C'u^4 - \beta u^2)$ for some $C'$, against the decay of $g$. □

Part (b), which uses only de Bruijn's theorem and Newman's theorem of 1976, shows that the obstruction has nothing to do with the position of $\Phi$ on the boundary of the class of kernels with real zeros: a property that survives a fixed amount of damping for all kernels fails for $P$ as well, although the zeros of $P$ are real for a structural reason, the selfadjointness of $A_P$. So the reading in the note, that Pólya's kernel tolerates every Gaussian damping, that $P$ has slack and $\Xi$ none, is to be replaced by a quantitative one: $P$ has the finite slack $|\Lambda_P|$, whose size is not known, and $\Xi$ has none if RH holds. The note lists among the excluded arguments a ferromagnetic or Lee and Yang construction and a Simon Griffiths representation. The Lee and Yang property of a single spin law is preserved by the ferromagnetic factor $e^{bu^2}$, $b > 0$, which runs the flow forward, but by (b) it is not preserved by a fixed damping except on Newman's class. A Lee and Yang representation of the law with density proportional to $\Phi$, which is equivalent to RH (Proposition 7.1.16), is therefore not excluded by the proposition, and neither is a Simon Griffiths type representation of that law.

Three questions raised by the note are not equivalences and can be studied without assuming what is to be proved. The first two ask for an object, a diffusion or a potential, not for the zeros, and part (b) does not apply to them, because the existence of such an object is not preserved by a fixed Gaussian damping of every kernel: for $P$ it is lost at every $t < \Lambda_P$, where $P_t$ has nonreal zeros. The third is a question about an explicit kernel with no arithmetic in it, and it measures the slack of Pólya's function. A fourth concerns the interpolation between the two clocks of Section 8.7.2.

Question 8.7.17. Is there a diffusion on a half line, with an entrance boundary, whose hitting time of a level from the entrance boundary has the Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + 2\sqrt{u})$, the transform of the clock $T$ of Theorem 5.1.1 scaled by 4, as $\tau_\infty$ has the transform $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ in Theorem 8.7.10? By Kent's theorem [305] such a hitting time is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues $\lambda_n > 0$ of

the generator, so a positive answer implies RH; under RH and the simplicity of the zeros, Krein's inverse spectral theory gives a generalised diffusion with the eigenvalues $\gamma^2/4$ (Section 8.1.3). The question asks for one constructed without RH, for example as a perturbation of the diffusion $X$ of Section 8.7.4, whose potential $e^{2x}$ has the right Weyl law.

Question 8.7.18. Does $g(w) = \Phi(w/2)$ admit a Klein Gordon extension in the sense of Theorem 8.7.14, with a real potential $V$ asymptotic to $e^{2x}$ at $+\infty$? By the theorem a positive answer implies RH. Under RH the question asks whether $\mu \mapsto 2\xi(\frac{1}{2} + 2\mu)$ is, up to normalisation, the Jost function of such a potential whose Klein Gordon solution has the boundary value $g$. A first numerical step is to reconstruct a potential from the first few thousand ordinates and a choice of norming constants by the method of Gelfand and Levitan or of Marchenko, and to test whether it is stable under truncation and close to $e^{2x}$.

Question 8.7.19. What is the de Bruijn Newman constant $\Lambda_P$ of Pólya's function, and that of $\Xi^*$, which is also finite and at most 0 by the proof of Proposition 8.7.15? At what height do the first nonreal zeros of $P_t$ appear as $t$ decreases through $\Lambda_P$, and do they appear, as for $\Xi_t$, through the collision of two real zeros?

Question 8.7.20. Is there a one parameter family of laws joining the drifted first passage time and the sum of two Bessel hitting times, each with a reciprocal symmetry that encodes a functional equation, as $\tau \mapsto 1/\tau$ does for $H$ in Proposition 8.7.1(i) and Jacobi's transformation does for $S_2$, along which the index side Thorin measure can be followed? The curvature family of Proposition 8.7.6 is an Esscher tilt and breaks the symmetry. Positivity of the index side Thorin measure at the Bessel end is RH.

## 8.8 Exercises

*On Section 8.1.*

8.1. Assuming that the zeros are on the critical line and simple, show that the residues of the clock transform at $u = -\gamma_k^2$ are $R_k = 2i\gamma_k \xi(\frac{1}{2})/\xi'(\frac{1}{2} + i\gamma_k)$ and that they alternate in sign. (They are the coefficients $c_n$ of Theorem 4.3.22.)

8.2. Explain why the spectral expansion of the density diverges at $t = 0$ while the Talbot inversion does not, and compare the abscissa of convergence in Theorem 4.3.22.

8.3. Show that the hazard of a finite sum of exponentials with rates $\lambda_1 < \lambda_2 < \cdots$ tends to $\lambda_1$ and that $h(t) - \lambda_1$ decays like $e^{-(\lambda_2 - \lambda_1)t}$.

8.4. Sample Williams' variable with $10^5$ draws and estimate $\Xi(5)$; compare the standard error with Table 8.3.

8.5. Assume RH, let $\Theta_N$ be the cutoff of Section 8.1.3, and assume that $S(\Theta_N) = 0$, as for the values of $N$ used there. Show that $\log G(w) - \log G_N^{\mathrm{W}}(w)$ tends to $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as $w \to \infty$. Then, using Stirling's formula for $\log \xi(\frac12 + x)$ as $x \to +\infty$ and $\int_0^{\infty} \log(1 + x^2/t^2) \log \frac{t}{2\pi}\,dt = \pi x \log \frac{x}{2\pi} - \pi x$, show that

$$\ell_N = 2\sum_{k \le N} \log \gamma_k - \log \xi(\tfrac12) + \tfrac14 \log \tfrac{\pi}{2} - \frac{1}{\pi}\int_0^{\Theta_N} \log t \, \log \frac{t}{2\pi}\,dt + \frac{2}{\pi}\int_{\Theta_N}^{\infty} \Big(\vartheta'(t) - \frac12 \log \frac{t}{2\pi}\Big) \log t \, dt,$$

the terms in $\log x$ cancelling because $\vartheta(\Theta_N) = (N-1)\pi$. So $\ell_N$ is determined by $\gamma_1, \dots, \gamma_N$ and $\xi(\frac12)$.

*On Section 8.2.*

8.6. Show that on the parabola $u = \frac12(-q^2 + 2icq)$ one has $\operatorname{Re}\sqrt{c^2 + 2u} = c$, and hence that every mixture of inverse Gaussian laws with drift $c$ has transform bounded by one there.

8.7. Derive the small $q$ expansion $|G(u)|^2 = 1 - q^2(\mathbb{E}T - c^2 \operatorname{Var} T) + O(q^3)$ on the parabola.

8.8. Show that the hazard of the first passage time of Brownian motion across the line $\alpha + \beta t$, with $\alpha > 0$ and $\beta < 0$, which has the inverse Gaussian law with drift $|\beta|$, tends to $\beta^2/2$ with correction $3/(2t)$.

8.9. Implement the tangent boundary ODE with a different starting time and check that the asymptotic slope is unchanged.

8.10. Show that the density $f$ of the clock is continuous and positive on $(0, \infty)$. (The Laplace transform of $T$ equals $1/G$ on $\operatorname{Re} u \ge 0$, by analytic continuation from $(0, \infty)$ and continuity, so the characteristic function is $1/G(-iy)$; Stirling's formula and the bound $|\zeta(s)| \ge 1/\zeta(\operatorname{Re} s)$ for $\operatorname{Re} s > 1$, applied on the rays $\frac12 + re^{\pm i\pi/4}$, show that it decays faster than any power of $|y|$. Next, $\log G(u) = o(u)$ as $u \to \infty$, so $P(T < t) > 0$ for every $t > 0$; $1/G$ has poles, so the law of $T$ has unbounded support; and the law is unimodal by Corollary 11.1.2, so $f$ itself, which is the derivative of the distribution function at every point, is nondecreasing up to a mode and nonincreasing after it.)

*On Section 8.3.*

8.11. Derive formula (8.3.1) from the derivative representation of $\lambda_n$.

8.12. Show that $\lambda_1 = \sum_\rho \rho^{-1} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, and that under RH $\lambda_1 = \sum_{\gamma>0}(\gamma^2 + \frac{1}{4})^{-1}$.

8.13. Show that under RH $\lambda_n = \sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$, and that the terms are of order $n^2/\gamma^2$ for $\gamma \gg n$.

8.14. For a hypothetical zero $\rho = \frac{1}{2} + d + i\gamma$ with $d > 0$, show that $|1 - 1/(1 - \bar{\rho})| > 1$, and estimate the index $n$ at which its contribution would make $\lambda_n$ negative.

*On Section 8.4.*

8.15. Show that $H_t$ satisfies $\partial_t H_t = -\partial_z^2 H_t$ and derive the equation of motion of a simple zero.

8.16. For an isolated pair at distance $D_0$, solve $dD/dt = 4/D$ and find the collision time.

8.17. Modify the Pólya evaluator to compute $H_t$ and reproduce Table 8.7.

8.18. Explain why the collision time of a pair of zeros gives numerical evidence for, but not a proof of, a lower bound on $\Lambda$.

*On Section 8.5.7.*

8.19. Show that $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$ is even, using $\eta(i/x) = \sqrt{x}\,\eta(ix)$, and deduce the functional equation $\xi(s, \chi_4) = \xi(1 - s, \chi_4)$ from $\xi(\frac{1}{2} + s, \chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}\,du$.

8.20. Show that for real $\sigma > 0$ the series $\sum_{n\ge 0}(-1)^n(2n + 1)^{-\sigma}$ lies between $1 - 3^{-\sigma}$ and 1, and deduce that $L(s, \chi_4)$ has no zeros on $(0, 1)$ without using the kernel.

8.21. For $0 < \varepsilon < \delta$ show that $\pi(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ integrates over the interval $|x| < \sqrt{\delta^2 - \varepsilon^2}$, where it is negative, to $\pi - 4\arctan\sqrt{(\delta + \varepsilon)/(\delta - \varepsilon)}$, which lies in $(-\pi, 0)$ and tends to 0 as $\varepsilon/\delta \to 0$; this is the bound on the fall of the argument after Corollary 8.5.32.

8.22. Show that $g(c, u) = 2c(3u^2 - c^2)/(c^2 + u^2)^3$, the second derivative of $c/(c^2 + u^2)$ in $u$, has $\partial_c g(c, u) = 6(u^4 - 6c^2u^2 + c^4)/(c^2 + u^2)^4$, which is positive exactly when $|c| < (\sqrt{2} - 1)|u|$ or $|c| > (\sqrt{2} + 1)|u|$, and deduce that $g(a + \delta, u) + g(a - \delta, u) > 0$ for $a, \delta > 0$ with $a + \delta < (\sqrt{2} - 1)|u|$, the step for the zeros off the line in the proof of Theorem 9.1.18(ii).

*On Section 8.7.*

8.23. Show that $1/H$ has the law of $H$ size biased by $H$, for $H = H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$ with the density of Proposition 8.7.1(i), and deduce $\mathbb{E}H^s = \mathbb{E}H^{1-s}$ without computing the Mellin transform.

8.24. Show that $F(x,w) = \exp(-e^x \cosh w)$ satisfies $(\partial_x^2 - \partial_w^2 - e^{2x})F = 0$, and that in the coordinates $X_1 = e^x \cosh w$, $X_0 = e^x \sinh w$ one has $\partial_x^2 - \partial_w^2 = e^{2x}(\partial_{X_1}^2 - \partial_{X_0}^2)$, so that $F = e^{-X_1}$ solves the Klein Gordon equation of unit mass.

8.25. Show that $(1+y)e^{-y} \geq e^{-y^2/2}$ for $y \geq 0$, and deduce that each density of Newman's theorem in the proof of Proposition 8.7.15 is bounded below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$.

8.26. Show that the principal value integral of $\log|s|/(1-s)$ over $\mathbb{R}$, taken with symmetric limits at $\pm\infty$, is $-\frac{\pi^2}{2}$, and deduce the mean speed $-\frac{\pi}{8}$ of the zeros under the flow in Section 8.7.6 from the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$.

8.27. For $0 < y < \pi$ show, as claimed in Subsection 10.1.5, that $\gamma^n e^{-\gamma y}$, for $\gamma \leq n/\pi$, is at most $\exp(n(\log\frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its maximum over $\gamma > 0$, and evaluate the exponent for $n = 10^6$ at $y = 1$ and at $y = 3$.

# Chapter 9

# The basepoint one

We now work at the resolution $\varepsilon = \frac{1}{2}$ of the Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 1.2, the basepoint one. The shifted clocks $T_\alpha$, with $\mathbb{E}e^{-uT_\alpha} = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, carry the Thorin programme. At a basepoint $\alpha$ to the right of every zero, $T_\alpha$ is a GGC (Theorem 1.2.2 and Proposition 1.2.3), and from $\alpha = 1$ on its Thorin density is positive with no hypothesis, because no zero lies on the line $\operatorname{Re} s = 1$ or to its right.

We work the basepoint one out completely, as a test bed for the tools of the book. We ask three questions of $T_1$: what its Thorin measure and its law are, whether its Thorin density comes from the sine squared lemma, and whether it is a hitting time to which Kent's theorem applies.

For the Thorin measure and the Lévy form of $\log \xi(1+s)$ alike, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH.

## 9.1 The Thorin measure and the Lévy form at the basepoint one

The scripts of the chapter are in the folders `basepoint_one/xi` and `basepoint_one/process_map` of the archive, except those of Section 9.1.6, which are in `code/`.

### 9.1.1 The Mellin side: Bessel hitting times and size biasing

Let $J$ be the first time a three dimensional Bessel process started at 0 reaches the level one. Its Laplace transform is

$$\mathbb{E}e^{-uJ} = \frac{\sqrt{2u}}{\sinh\sqrt{2u}} = \prod_{n\geq 1}\Big(1 + \frac{2u}{\pi^2 n^2}\Big)^{-1}, \qquad \mathbb{E}J = \frac{1}{3},$$

so $J \stackrel{d}{=} \sum_n \frac{2}{\pi^2 n^2} e_n$ with independent standard exponentials $e_n$. This is a GGC with unit Thorin atoms at the points $\pi^2 n^2/2$, which are the Dirichlet eigenvalues of the Bessel process on $[0,1]$, so $J$ is the simplest instance of Kent's theorem (Section 8.1.3): up to a change of scale the Bessel process is the string diffusion with that spectrum [305]. With $J'$ an independent copy of $J$, the sum $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma$(2,1)$ variables $\Gamma_n$, is Williams' Gamma(2) structure [518] of Section 8.2.3, and Biane, Pitman and Yor [49] proved the identity recalled in Section 1.2 and Theorem 7.1.21,

$$2\xi(s) = \mathbb{E}\Big[\Big(\frac{\pi}{2}S_2\Big)^{s/2}\Big], \qquad s \in \mathbb{C}. \tag{9.1.1}$$

Table 9.1 checks (9.1.1) at real $s$. Table 8.3 checks the same identity on the critical line.

| $s$ | $\mathbb{E}(\pi S_2/2)^{s/2}$, Monte Carlo | standard error | $2\xi(s)$ |
|---|---|---|---|
| 0.5 | 0.99415 | 0.00017 | 0.99424 |
| 1 | 0.99981 | 0.00034 | 1 |
| 2 | 1.04677 | 0.00074 | 1.04720 |
| 3 | 1.14713 | 0.00127 | 1.14788 |

Table 9.1: The identity (9.1.1) of Biane, Pitman and Yor, checked with $4 \times 10^5$ draws of $S_2$ by `comp.py`. The sample mean of $S_2$ is 0.6664 against $\mathbb{E}S_2 = \frac{2}{3}$.

Put $Y = \frac{1}{2}\log(\frac{\pi}{2}S_2)$, as in Section 1.2. By (9.1.1), $2\xi(\alpha + w) = \mathbb{E}[e^{\alpha Y} e^{wY}]$, so for every real $\alpha$

$$\frac{\xi(\alpha + w)}{\xi(\alpha)} = \mathbb{E}e^{wV_\alpha}, \qquad P(V_\alpha \in dy) = \frac{e^{\alpha y} P(Y \in dy)}{\mathbb{E}e^{\alpha Y}}.$$

Since the Pólya variable $X$ is $Y$ tilted by $e^{Y/2}$, the variable $V_\alpha$ has the density $e^{(\alpha - \frac{1}{2})y} p(y)\, \xi(\frac{1}{2})/\xi(\alpha)$, the Esscher tilt of the Pólya density $p$ of (1.2.4); so $V_{1/2} = X$, and we write $V = V_1$ for the tilt at the basepoint one. Equivalently, $V_\alpha$ is $\frac{1}{2}\log(\frac{\pi}{2}S_2)$ under the law of $S_2$ size biased by $S_2^{\alpha/2}$: by $S_2^{1/4}$ at the centre, by $S_2^{1/2}$ at the basepoint one, where the weight $e^Y$ needs no normalisation because $\mathbb{E}e^Y = 2\xi(1) = 1$, and by $S_2^{(1-\varepsilon)/2}$ at $1-\varepsilon$. Every basepoint thus gives a probability law with no hypothesis on this side, and nothing distinguishes $\alpha = 1$ except the weight. The law of $-V_\alpha$, with characteristic function $\xi(\alpha - it)/\xi(\alpha)$, is Nakamura's complete Riemann zeta distribution [377]. He proved that RH holds if and only if, for every $\frac{1}{2} < \alpha < 1$, this characteristic function has a Lévy Khintchine form with a signed measure, with no integrability condition on the measure. The theory of infinite divisibility does distinguish it: $V_\alpha$ is quasi infinitely divisible, though not infinitely divisible, for every $\alpha > 1$, and it is not quasi infinitely divisible at $\alpha = 1$ or at the centre (Proposition 11.1.3; for $\alpha = 1$ see also Section 9.1.4).

Remark 9.1.1 (Ostrovsky's Barnes beta laws). The companion $C_2 = \sum_{k\geq 1} \frac{2}{\pi^2(k-\frac12)^2}(e_k + e'_k)$ of $S_2$, with independent standard exponentials, has $\mathbb{E}e^{-uC_2} = \operatorname{sech}^2\sqrt{2u}$ [49], and its Mellin transform is again $\xi$:

$$\mathbb{E}[C_2^q] = \frac{2^{2q+2}-1}{q+1}\Big(\frac{2}{\pi}\Big)^{q+1}\xi(2q+2),$$

which we checked at $q = -0.3$, $\frac12$, 1, 2 to $10^{-12}$ by quadrature of the Laplace transform. The Lévy density $\frac{1}{4t}\operatorname{sech}^2\frac{t}{4}$ of Proposition 9.1.9 is

$$\frac{1}{4t}\operatorname{sech}^2\frac{t}{4} = \frac{1}{4t}\,\mathbb{E}e^{-t^2C_2/32},$$

a mixture of Gaussians in $t$ whose mixing law has Mellin transform $\xi(2q+2)$. The positivity comes from the gamma side, and the zeros of $\xi$ enter only through the Mellin transform of $C_2$. More generally the Barnes beta laws $\beta_{M,N}(a,b)$ of [351] have Lévy density $e^{-b_0 t}\prod_{j=1}^N(1-e^{-b_j t})/\big(t\prod_{i=1}^M(1-e^{-a_i t})\big)$; for $M = N = 2$, $a = (1,1)$ and $b = (b_0, \frac12, \frac12)$ it is $\frac{e^{-(b_0-\frac12)t}}{4t}\operatorname{sech}^2\frac{t}{4}$, the density above times $e^{-(b_0-\frac12)t}$, and it is again not that of a GGC, since $t$ times it is the Laplace transform of the signed measure $\sum_{j\geq1}(-1)^{j+1}j\,\delta_{b_0-\frac12+\frac{j}{2}}$. The case $M = N = 1$, $a_1 = 1$ [253], is the law of $-\log B$ with $B$ a beta variable of parameters $b_0$ and $b_1$: $t$ times its Lévy density, $e^{-b_0 t}(1-e^{-b_1 t})/(1-e^{-t})$, is the Laplace transform of $\sum_{j\geq0}(\delta_{b_0+j} - \delta_{b_0+b_1+j})$, so $-\log B$ is a GGC if and only if $b_1$ is a positive integer, when it is a sum of independent exponential variables of rates $b_0, \dots, b_0+b_1-1$; for other $b_1$ the negative atoms meet no positive one, and a Laplace transform determines its measure.

### 9.1.2 The Thorin measure and the law of $T_1$

For a real basepoint $\alpha \geq \frac12$ we write $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha+\sqrt{u})$, so that $\varphi_{1/2} = 1/G$ is the clock, and $T_\alpha$ for a random variable with $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha(u)$ for $u \geq 0$, whenever one exists. Theorem 1.2.2 provides it, as a GGC, for $\alpha \geq 1$, and Theorem 9.2.5 in Section 9.2 provides it for every $\alpha > \frac12$. As in Section 1.2, $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha+i\theta)$.

Proposition 9.1.2. *The function $\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$ is the Laplace transform of a GGC $T_1$ with no drift, $\log\varphi_1(u) = -\int_0^\infty \log(1+u/z)\,U_1(dz)$, and in the variable $\theta = \sqrt{z}$ its Thorin measure has density*

$$\rho_{1/2}(\theta) = \frac{1}{\pi}v_1(\theta) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(1+i\theta) = \frac{1}{\pi}\sum_\rho \frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2}, \tag{9.1.2}$$

*the sum running over all nontrivial zeros $\rho = \beta + i\gamma$ in conjugate pairs. Every term is positive, so $U_1$ is a positive measure with no hypothesis on the zeros.*

*Proof.* This is Theorem 1.2.2 at $\alpha = 1$. First, $-\frac{d}{du}\log\varphi_1(u) = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(1+\sqrt{u})$ is analytic on $\mathbb{C}\setminus(-\infty,0]$, because the zeros of $\xi(1+w)$ lie at $w = \rho - 1$ with $\operatorname{Re} w = \beta - 1 < 0$ and so are never reached by the principal branch $w = \sqrt{u}$; Stieltjes inversion (1.3.2) of its jump across the cut, where $\sqrt{u} = \pm i\theta$, gives the density (1.2.6) in $z$, which is (9.1.2) in $\theta$ because $dz = 2\theta\,d\theta$. Second, by (1.2.2) with conjugate pairs grouped, $\operatorname{Re} 1/(1+i\theta-\rho) = (1-\beta)/((1-\beta)^2+(\theta-\gamma)^2) = \pi P_{1-\beta}(\theta-\gamma)$, as in the proof of Proposition 1.2.3, and every term is positive because $\beta < 1$ for all zeros, which is the nonvanishing of $\zeta$ on the line $\operatorname{Re} s = 1$. Third, there is no drift because $\log\xi(1+\sqrt{u})$ grows like $\frac{1}{2}\sqrt{u}\log\sqrt{u}$, more slowly than $u$. □

Under RH every width equals $\frac{1}{2}$, and (1.2.8) at $\varepsilon = \frac{1}{2}$ reads

$$\rho_{1/2}(\theta) = \sum_{\gamma>0}\Big(P_{1/2}(\theta-\gamma) + P_{1/2}(\theta+\gamma)\Big),$$

the RH measure blurred by the Cauchy kernel of width one half. The value at the origin is explicit, $\rho_{1/2}(0) = \frac{1}{\pi}\sum_\rho \frac{1}{\rho} = \frac{1}{\pi}\big(1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi\big) = \frac{0.0230957089661}{\pi} = 0.0073515925$, where $\gamma_E$ is Euler's constant, and it is the minimum of $\rho_{1/2}$, attained only at the origin (Proposition 9.1.11). The cumulative mass $M_{1/2}(T) = \int_0^T \rho_{1/2}(\theta)\,d\theta$ is $\pi^{-1}$ times the argument of $\xi$ along the line $\operatorname{Re} s = 1$, and since no zero lies on or to the right of that line, Proposition 3.4.2 gives the exact unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, in which $\tau_{1/2}(T)$, the change of argument of $\xi$ from $1+iT$ to $\frac{1}{2}+iT$ divided by $\pi$, is $O(\log T)$, like the argument $\arg\zeta(\frac{1}{2}+iT)$ of Section 2.2.1 [498]. Numerically $\int_0^{100}\rho_{1/2} = 29.111$ against $N(100) = 29$, so that $\tau_{1/2}(100) = -0.111$. Table 9.2 lists values of the density, and Figure 9.1 shows it with its mass.

| $\theta$ | $\rho_{1/2}(\theta)$ from $\xi'/\xi$ | from the arithmetic form | from 1700 zeros, $\beta = \frac{1}{2}$ |
|---|---|---|---|
| 0 | 0.0073515925 | 0.0073515925 | 0.0071935 |
| 5 | 0.0084053053 | 0.0084053053 | 0.0082472 |
| 14.134725 | 0.6473699273 | 0.6473699273 | 0.6472119 |
| 20 | 0.1418590769 | 0.1418590769 | 0.1417010 |
| 30 | 0.4070956035 | 0.4070956035 | 0.4069375 |
| 100 | 0.2308349819 | 0.2308349819 | 0.2306766 |

Table 9.2: The Thorin density $\rho_{1/2} = v_1/\pi$ of $T_1$ at the basepoint one, computed from $\xi'/\xi$, from the arithmetic form $v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$ with $\zeta'/\zeta$ evaluated exactly, and from the Poisson sum over the first 1700 zeros with $\beta = \frac{1}{2}$. The zero sum misses the zeros above height 2198, whose contribution is a nearly constant $1.6\times10^{-4}$.

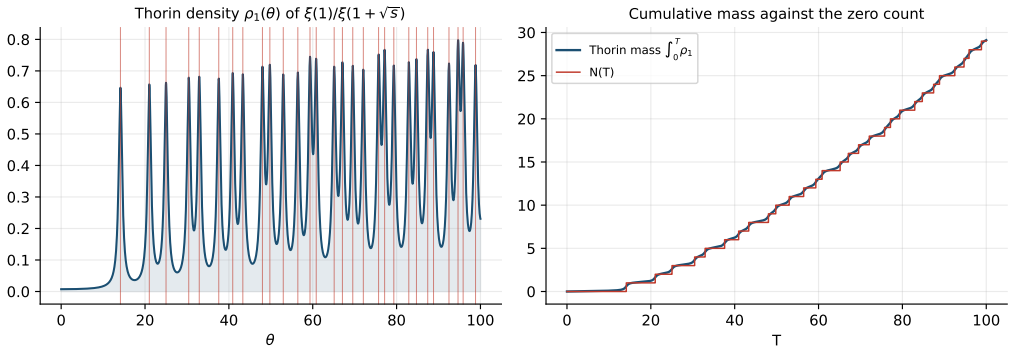



Figure 9.1: Left: the Thorin density $\rho_{1/2}(\theta) = v_1(\theta)/\pi$ of $T_1$ on $[0, 100]$, with the ordinates of the zeros as thin vertical lines. Right: the cumulative Thorin mass $\int_0^T \rho_{1/2}$ against the zero count $N(T)$. The plot writes $\rho_1$ for $\rho_{1/2}$ and $s$ for $u$.

Because $\xi'(1) \neq 0$, the Laplace transform of $T_1$ is not differentiable at the origin:

$$\varphi_1(u) = 1 - \frac{\xi'}{\xi}(1)\sqrt{u} + O(u) = 1 - \Big(\sum_\rho \frac{1}{\rho}\Big)\sqrt{u} + O(u), \qquad u \downarrow 0.$$

By the Tauberian theorem for Laplace transforms [136, XIII.5], $T_1$ has infinite mean and

$$P(T_1 > x) \sim \frac{\sum_\rho \rho^{-1}}{\Gamma(\frac{1}{2})}\, x^{-1/2} = 0.01303\, x^{-1/2}, \qquad x \to \infty. \tag{9.1.3}$$

Equivalently, near the origin the Thorin density in $z$ behaves like $\rho_{1/2}(0)/(2\sqrt{z})$, which makes $\int U_1(dz)/z$ diverge. At the centre, by contrast, $\xi'(\frac{1}{2}) = 0$, the clock has the finite mean $\mathbb{E}T = \sum_k a_k^{-1} = 0.0231050$, which is $\sum_{\gamma>0} \gamma^{-2}$ under RH; at the basepoint one the relevant constant is $\sum_\rho \rho^{-1} = 0.0230957$, and it multiplies $\sqrt{u}$.

### 9.1.3 The sine squared lemma at the basepoint one

At a basepoint $\alpha > 1$ the sine squared lemma (1.2.9), applied to the canonical measure of $\log \xi$, writes $v_\alpha$ as the sum of a drift, an archimedean piece, a pole piece and a prime piece, (3.1.1). The pole piece is the image under the lemma of the measure $e^{-(\alpha-1)x}dx$ and it enters with a minus sign; the prime piece is positive. As $\alpha \downarrow 1$ both diverge, because $\sum \Lambda(n)/n$ diverges (Section 3.1.2). So the lemma cannot be applied to the two pieces separately at the basepoint one. It can be applied to their difference. Throughout Section 9.1, $\psi(y) = \sum_{n \le y} \Lambda(n)$ is Chebyshev's function, whose argument is always the real variable $y \geq 1$ or the

prime cutoff $X$, while at every other argument, as in $\psi(\frac{3}{2})$ or $\psi(\frac{1+i\theta}{2})$, $\psi = \Gamma'/\Gamma$ is the digamma function; from here on $X$ always denotes the cutoff, not the Pólya variable of Section 9.1.1.

PROPOSITION 9.1.3. *Let $\nu_1 = \sum_n \Lambda(n)n^{-1}\delta_{\log n} - \mathbf{1}_{(0,\infty)}(x)\,dx$, the prime measure compensated by Lebesgue measure. For every $\theta \neq 0$,*

$$v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\theta}{2}\Big) + \gamma_E + \lim_{X\to\infty}\int_0^{\log X} 2\sin^2\Big(\frac{\theta x}{2}\Big)\nu_1(dx), \tag{9.1.4}$$

*that is, the limit of $\sum_{n\le X}\Lambda(n)n^{-1}2\sin^2(\frac{1}{2}\theta\log n) - \big(\log X - \sin(\theta\log X)/\theta\big)$. The limit exists by the prime number theorem, and at $\theta = 0$ the formula reduces to $v_1(0) = \sum_\rho \rho^{-1}$.*

*Proof.* On the line $\operatorname{Re} s = 1$ the pole term $1/(s-1) = 1/(i\theta)$ is purely imaginary, so $v_1(\theta) = \operatorname{Re}\frac{1}{1+i\theta} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$. For $\operatorname{Re} s \ge 1$ with $s \neq 1$, partial summation gives

$$-\sum_{n\le X}\frac{\Lambda(n)}{n^s} + \int_1^X y^{-s}dy = -\frac{\psi(X)-X}{X^s} - s\int_1^X (\psi(y)-y)\,y^{-s-1}dy - 1,$$

and the prime number theorem with the de la Vallée Poussin error term, $\psi(y)-y = O(y\exp(-c\sqrt{\log y}))$, makes the boundary term vanish and the integral converge absolutely as $X\to\infty$, uniformly for $\operatorname{Re} s \ge 1$. For $\operatorname{Re} s > 1$ the limit is $\zeta'(s)/\zeta(s) + 1/(s-1)$, and by continuity the same holds on $\operatorname{Re} s = 1$, $s \neq 1$. Taking real parts at $s = 1+i\theta$, where $\operatorname{Re}\int_1^X y^{-1-i\theta}dy = \sin(\theta\log X)/\theta$, gives $\operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta) = \lim_X\big[-\sum_{n\le X}\Lambda(n)n^{-1}\cos(\theta\log n) + \sin(\theta\log X)/\theta\big]$. Writing $-\cos = 2\sin^2 - 1$ and using $\sum_{n\le X}\Lambda(n)/n = \log X - \gamma_E + o(1)$, again a consequence of the prime number theorem, gives (9.1.4), since $\int_0^{\log X} 2\sin^2(\theta x/2)dx = \log X - \sin(\theta\log X)/\theta$. At $\theta = 0$ the bracket vanishes and $1 - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{1}{2}) + \gamma_E = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$. □

So the Thorin density comes from the sine squared lemma, but its positivity does not. Since $\nu_1$ is signed, its sine squared image is not termwise positive. Positivity comes from the zero side, Proposition 9.1.2, through $\beta < 1$, and written arithmetically it is the inequality

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(1+it) < \frac{1}{1+t^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+it}{2}\Big),$$

whose right side grows like $\frac{1}{2}\log(t/2\pi)$: the inequality of Hadamard and de la Vallée Poussin that underlies the proof of the prime number theorem [498].

PROPOSITION 9.1.4 (de la Vallée Poussin in Thorin form). *For $\alpha > 1$ let $c_\alpha = -\frac{\zeta'}{\zeta}(\alpha)$ and let $Z(\alpha,y) = \sum_{n\ge 2}\Lambda(n)n^{-\alpha}(1-\cos(y\log n))$ be the prime part of the Thorin density $g(\alpha,y) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+iy)$, so that $Z(\alpha,y) = c_\alpha + g(\alpha,y) - \operatorname{Re}\mathcal{A}(\alpha+iy)$ with $\mathcal{A}$ as in Theorem 16.12.1(ii).*

*(i)* $0 \le Z(\alpha,y) \le 2c_\alpha$ *and* $4Z(\alpha,y)+Z(\alpha,2y) \le 8c_\alpha$.

*(ii) If* $\beta_0+i\gamma_0$ *is a zero, then* $\dfrac{4}{\alpha-\beta_0} \le 3c_\alpha + 4\operatorname{Re}\mathcal{A}(\alpha+i\gamma_0)+\operatorname{Re}\mathcal{A}(\alpha+2i\gamma_0)$. *With* $\alpha = 1+\frac{1}{4\log\gamma_0}$ *this gives* $1-\beta_0 \ge c/\log\gamma_0$ *for* $\gamma_0 \ge \gamma^*$, *with absolute constants* $c,\gamma^*>0$.

*Proof.* (i) Termwise, $0 \le 1-\cos \le 2$, and $3+4\cos\theta+\cos 2\theta = 2(1+\cos\theta)^2 \ge 0$ gives $3c_\alpha + 4(c_\alpha - Z(\alpha,y)) + (c_\alpha - Z(\alpha,2y)) \ge 0$. (ii) For $\alpha > 1$ every term $(\alpha-\beta)/|\alpha+iy-\rho|^2$ of $g$ is positive, so $g(\alpha,\gamma_0) \ge 1/(\alpha-\beta_0)$ and $g(\alpha,2\gamma_0) \ge 0$; insert these into (i) through $Z = c_\alpha + g - \operatorname{Re}\mathcal{A}$. Then $c_\alpha = \frac{1}{\alpha-1}+O(1)$ and $\operatorname{Re}\mathcal{A}(\alpha+i\tau) = \frac12\log\tau + O(1)$ for $1<\alpha\le 2$, $\tau \ge 2$, and with $x = (1-\beta_0)\log\gamma_0$ the inequality reads $4/(x+\frac14) \le \frac{29}{2}+o(1)$, so $x \ge \frac{8}{29}-\frac14-o(1) = \frac{3}{116}-o(1)$. □

The sine squared lemma is the lower bound $Z \ge 0$ at one frequency. A zero close to $\operatorname{Re} s = 1$ makes $Z(\alpha,\gamma_0)$ large, and what excludes it is the upper bound in (i), which needs the two frequencies $y$ and $2y$. At $\alpha = \frac32$, in floating point over $0 < y \le 100$, $\max_y Z/(2c_\alpha) = 0.716$ and $\max_y\{4Z(\alpha,y)+Z(\alpha,2y)\}/(8c_\alpha) = 0.813$.

The same lemma measures how far it alone reaches. For $\alpha > 1$ let $D(\alpha) = \frac{\xi'}{\xi}(\alpha) - \frac{1}{\alpha-1} = \frac12\psi(1+\frac{\alpha}{2}) - \frac12\log\pi + \frac{\zeta'}{\zeta}(\alpha)$, the logarithmic derivative of $\xi$ without the pole.

Proposition 9.1.5. *The function* $D$ *is increasing and its root is* $\alpha_0 = 5.4962215670\ldots$. *For every* $\alpha \ge \alpha_0$, $u \mapsto \xi(\alpha)/\xi(\alpha+\sqrt{u})$ *is the Laplace transform of a GGC, and the proof uses only the sine squared lemma* (1.2.9), *with no information on the zeros.*

*Proof.* With $\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ and $a = \alpha-1$, write $\xi(\alpha)/\xi(\alpha+s) = \frac{a}{a+s}/G(s)$, where $G(s) = \pi^{-s/2}\Gamma(1+\frac{\alpha+s}{2})\zeta(\alpha+s)/(\Gamma(1+\frac{\alpha}{2})\zeta(\alpha))$. Differentiating $\log\Gamma$ twice, and expanding $\log\zeta$ by the Euler product, gives $\log G(s) = D(\alpha)s + \int_0^\infty (e^{-sx}-1+sx)\,e^{-\alpha x}\big(\frac{dx}{e^{2x}-1} + \sum_n \Lambda(n)\delta_{\log n}\big)\frac{1}{x}$, a positive measure. Put $s = \sqrt{u}$. By (1.2.9), the compensated integral makes $\exp(-\cdot)$ the transform of a GGC with positive Thorin density. The pole factor is $\exp(-\log(1+\sqrt{u}/a))$, and $\log(1+\sqrt{u}/a) = \frac{a}{\pi}\int_0^\infty \log(1+u/\theta^2)\frac{d\theta}{a^2+\theta^2}$ (both sides vanish at $u=0$ and have the derivative $1/(2\sqrt{u}\,(a+\sqrt{u}))$), so it is a GGC factor as well. Finally $e^{-D\sqrt{u}} = \exp(-\frac{D}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta)$ is a GGC transform exactly when $D \ge 0$. The function $D$ increases because $\psi$ increases and $\frac{\zeta'}{\zeta}(\alpha) = -\sum_n \Lambda(n)n^{-\alpha}$ increases, and $D(\alpha) \to \infty$, so $D \ge 0$ exactly for $\alpha \ge \alpha_0$. □

This is weaker than Theorem 1.2.2, which gives the Thorin condition for every $\alpha \ge 1$, and its interest is only the measure it provides. Below $\alpha_0$ the factor $e^{|D|\sqrt{u}}$ is not a Laplace transform, so the factorwise use of the lemma stops there. Keeping the pole and Gamma factors together reaches lower in floating point, as follows, and the condition below that holds only because the three parts combine.

REMARK 9.1.6. In the variable $y = \sqrt{z}$ the Thorin density of $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is $\pi^{-1}g(\alpha, y)$ with $g = P + D + A + Z$, where $P = \frac{\alpha-1}{(\alpha-1)^2+y^2}$ comes from the pole, $A = \frac{1}{2}\big(\operatorname{Re}\psi(1 + \frac{\alpha+iy}{2}) - \psi(1 + \frac{\alpha}{2})\big) \geq 0$ from the Gamma factor, and $Z = \sum_{n\geq 2}\Lambda(n)n^{-\alpha}(1 - \cos(y \log n)) \geq 0$ from the primes. Dropping $Z$ leaves $m(\alpha) = \min_y\{P + A\} + D(\alpha)$, which is $-0.0133$ at $\alpha = 3.5$, 0.0638 at 4, 0.1650 at 5 and 0.2012 at 5.5, and changes sign at $\alpha_1 = 3.5747764\ldots$, with the minimum in $y$ at $y = 3.4506$. Where $m(\alpha) \geq 0$ the Thorin condition holds with only $\Lambda(n) \geq 0$ used, but this is computed in floating point (20 digits) at the sampled points and not proved on $[\alpha_1, \alpha_0)$, where $m$ need not be monotone; a sweep in ball arithmetic is open. No bound on the primes uniform in $y$ goes below $\alpha_1$, since $\inf_y Z(\alpha, y) = 0$ for every $\alpha > 1$: the numbers $\log p$ are independent over $\mathbb{Q}$, so by Kronecker's theorem for any finite set of primes there are arbitrarily large $y$ at which every $\cos(y \log p^k)$ is close to 1, and the tail of $Z$ is uniformly small. The slice version of this obstruction is in Section 12.2.6. For every $\alpha > 1$ a march with step $0.9\, g/L$ and a Lipschitz bound $L(\alpha)$ for $g$ on $[0, Y(\alpha)]$, where $A(\alpha, Y) + D(\alpha) = 0$, checks $g > 0$ from values of $\zeta$ on $\operatorname{Re} s = \alpha$ alone; this is not new, since Theorem 1.2.2 gives $\alpha \geq 1$ from the Hadamard product, and it measures the price of avoiding the zeros: $Y(\alpha)$ grows like $e^{2/(\alpha-1)}$, and $\alpha = 1.3$ took 21 931 evaluations (451 s, floating point, no interval arithmetic).

For $F = \mathbb{Q}(\sqrt{-3})$, with $\xi_F(s) = s(s-1)(\sqrt{3}/2\pi)^s\Gamma(s)\zeta_F(s)$ and $\zeta_F = \zeta\, L(\cdot, \chi_{-3})$, the same splitting has the drift $D_F(\alpha) = \psi(1+\alpha) + \log\frac{\sqrt{3}}{2\pi} + \frac{\zeta_F'}{\zeta_F}(\alpha)$, negative on $(1, \alpha_0^F)$ with root $\alpha_0^F = 3.3098807\ldots$, and the merged threshold, in floating point at sampled $y$ as for $\zeta$, is $\alpha_1^F = 2.38426$. The zeros give the Thorin property of $\xi_F(\alpha)/\xi_F(\alpha + \sqrt{u})$ for every $\alpha \geq 1$ (Theorem 21.5.8(a)), and the half plane $\operatorname{Re} s > \frac{7}{8}$ that [398] claims for every Dirichlet $L$ function, which has not been refereed (Section 12.1.2), would give it for every $\alpha > \frac{7}{8}$.

The lemma also acts on the heat trace. Write $p_1 = \int_0^\infty W(t)\, dt = \frac{1}{2}(\log\xi)''(\frac{1}{2}) = 0.0231049931\ldots$, which is $\sum_{\gamma>0}\gamma^{-2}$ under RH, and $F(v) = \xi(\frac{1}{2} + \sqrt{v})/\xi(\frac{1}{2})$. Theorem 5.1.1 gives $\log F(v) = \int_0^\infty (1 - e^{-vt})W(t)\frac{dt}{t}$ with $W > 0$, hence $p_1 v - \log F(v) = \int_0^\infty (e^{-vt} - 1 + vt)W(t)\frac{dt}{t}$.

PROPOSITION 9.1.7. *Unconditionally, for $s \geq 0$,*

$$\Psi(s) := \frac{\xi(\frac{1}{2} + s^{1/4})}{\xi(\frac{1}{2})}\, e^{-p_1\sqrt{s}} = \exp\Big(-\int_0^\infty \log\Big(1 + \frac{s}{z}\Big)U_*(dz)\Big),$$

*where*

$$U_*(dz) = \frac{dz}{\pi\sqrt{z}}\int_0^\infty \sin^2\Big(\frac{t\sqrt{z}}{2}\Big)W(t)\, dt \geq 0,$$

*so $\Psi$ is the Laplace transform of a GGC with a Thorin measure that is absolutely continuous. Under RH, $\Psi(s) = \prod_{\gamma>0}(1 + \sqrt{s}/\gamma^2)\, e^{-\sqrt{s}/\gamma^2}$.*

*Proof.* Apply (1.2.9) with $x = t$ and $s = v$ inside the last integral above, and Tonelli's theorem (the integrand is positive): $p_1 v - \log F(v) = \int_0^\infty \log(1 + v^2/z)\, U_*(dz)$. Put $v = \sqrt{s}$. Under RH $F(v) = \prod_{\gamma>0}(1 + v/\gamma^2)$. □

REMARK 9.1.8 (The square root is forced). For every real $\alpha$, $q \mapsto \xi(\alpha)/\xi(\alpha+q)$ is not $\int_{[0,\infty)} e^{-qt}\mu(dt)$ for a positive measure $\mu \neq 0$ on any half plane. For real $q \to \infty$, $\xi(\alpha+q)$ grows like $\exp(\frac{q}{2}\log q)$, so the function decays faster than $e^{-qT}$ for every $T$, while $\int e^{-qt}\mu(dt) \geq \mu([0,T])e^{-qT}$; hence $\mu([0,T]) = 0$ for all $T$. (Landau's theorem [514, Chapter II, Theorem 5b] gives it too: the function has no real singularity, since $\xi$ has no real zeros, so the abscissa would be $-\infty$ and the function entire, but it has poles.) No law on $(0,1]$ has Mellin transform $\xi(\alpha)/\xi(\alpha+q)$, and the clock starts at the variable $u = q^2$.

The construction of Proposition 9.1.7 is explicit for a law of Barnes type. Let $Y_{\mathfrak{b}} \geq 0$ be infinitely divisible with Lévy density $\nu(t) = \operatorname{sech}^2(t/4)/(4t)$, so that $\mathbb{E}Y_{\mathfrak{b}} = \frac{1}{4}\int_0^\infty \operatorname{sech}^2(t/4)\, dt = 1$, and let $\mathfrak{b} = e^{-Y_{\mathfrak{b}}}$; written as $e^{-t/2}(1-e^{-t/2})^2/(t(1-e^{-t})^2)$, $\nu$ is the Lévy density of a Barnes beta law of Ostrovsky [351].

PROPOSITION 9.1.9. *(i) For* $\operatorname{Re} q > -\frac{1}{2}$,

$$\mathbb{E}\mathfrak{b}^q = \frac{\mathsf{G}(q+1)^2\mathsf{G}(\frac{1}{2})\mathsf{G}(\frac{3}{2})}{\mathsf{G}(q+\frac{1}{2})\mathsf{G}(q+\frac{3}{2})} = \frac{2^{2q^2}(2\pi)^{-q}\mathsf{G}(1+q)^4}{\mathsf{G}(1+2q)},$$

*with* $\mathsf{G}$ *the Barnes function;* $\mathbb{E}\mathfrak{b} = 2/\pi$, *and* $\mathbb{E}\mathfrak{b}^q \sim Cq^{-1/4}$. *Since* $t\nu(t) = \sum_{k\geq 1}(-1)^{k+1}k\, e^{-kt/2}$, $Y_{\mathfrak{b}}$ *is not a GGC.*

*(ii) For* $s \geq 0$, $\mathbb{E}\mathfrak{b}^{\sqrt{s}} = e^{-\sqrt{s}}/\Psi_{\mathfrak{b}}(s)$, *where* $\Psi_{\mathfrak{b}}(s) = \exp(-\int_0^\infty \log(1+\frac{s}{z})\,\omega_{\mathfrak{b}}(z)\, dz)$ *is the Laplace transform of a GGC with Thorin density*

$$\omega_{\mathfrak{b}}(z) = \frac{1}{2\pi\sqrt{z}} - \frac{1}{\sinh(2\pi\sqrt{z})} > 0.$$

*(iii) For* $-\frac{1}{2} < \operatorname{Re} w < \frac{1}{2}$, $\int_0^\infty z^{w-1}\omega_{\mathfrak{b}}(z)\, dz = -4(2\pi)^{-2w}(1-2^{-2w})\Gamma(2w)\zeta(2w)$, *whose zeros in that strip are the points* $\rho/2$ *and* $\pi ik/\log 2$, $k \neq 0$.

*Proof.* (i) By $\log\Gamma(z) = \int_0^\infty [(z-1)e^{-t} - (e^{-t} - e^{-zt})/(1-e^{-t})]\, dt/t$, both $\log \mathbb{E}\mathfrak{b}^{q+1} - \log \mathbb{E}\mathfrak{b}^q = -\int_0^\infty e^{-qt}(1-e^{-t})\nu(t)\, dt$ and $\log \frac{\Gamma(q+1)^2}{\Gamma(q+\frac{1}{2})\Gamma(q+\frac{3}{2})}$ equal $-\int_0^\infty e^{-qt}\frac{e^{-t/2}(1-e^{-t/2})^2}{t(1-e^{-t})}\, dt$. The first expression for $\mathbb{E}\mathfrak{b}^q$ satisfies the same recursion, by $\mathsf{G}(z+1) = \Gamma(z)\mathsf{G}(z)$, and both sides equal 1 at $q = 0$, so their quotient is 1 periodic on $q > -\frac{1}{2}$. Both sides are $q^{-1/4}(C + o(1))$ as $q \to \infty$: the left because $\nu(t) = \frac{1}{4t} + O(1)$ near 0 and decays exponentially, the right by Stirling's formula for $\mathsf{G}$. So the quotient has a limit, is constant, and equals 1; it extends to $\operatorname{Re} q > -\frac{1}{2}$ by analyticity. The second expression follows from the multiplication formula for $\mathsf{G}$ at $z = q + \frac{1}{2}$. (ii) The

sine squared lemma (1.2.9) with $L(dx) = \nu(x)dx$ gives the Thorin density $\frac{1}{\pi\sqrt{z}}\int_0^\infty \sin^2(\frac{t\sqrt{z}}{2})\frac{\operatorname{sech}^2(t/4)}{4}\,dt$, and $\int_0^\infty \cos(vt)\operatorname{sech}^2(t/4)\,dt = 8\pi v/\sinh(2\pi v)$, from $\int_{\mathbb{R}} e^{i\omega x}\operatorname{sech}^2 x\,dx = \pi\omega/\sinh(\pi\omega/2)$, turns it into $\omega_{\mathfrak{b}}$; positivity is $\sinh x > x$. (iii) With $x = 2\pi\sqrt{z}$ the integral is $2(2\pi)^{-2w}\int_0^\infty x^{2w-1}(\frac{1}{x} - \frac{1}{\sinh x})\,dx$, and $\int_0^\infty x^{s-1}(\frac{1}{\sinh x} - \frac{1}{x})\,dx = 2(1-2^{-s})\Gamma(s)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, the continuation of $\int_0^\infty x^{s-1}\,dx/\sinh x = 2(1-2^{-s})\Gamma(s)\zeta(s)$ past $s = 1$. In the strip $\Gamma(2w)$ has no zeros, its pole at $w = 0$ cancels against $1 - 2^{-2w}$, and $\zeta(2w)$ vanishes only at $2w = \rho$. □

Numerically (25 digits), $\log \mathbb{E}\mathfrak{b} = -0.4515827052894548647$ from the Lévy integral and from both closed forms, and (ii) holds at $s = 10$ and 100 to $10^{-14}$. The GGC $\Psi_{\mathfrak{b}}$ is a ratio of Barnes values and contains no $\zeta$; $\zeta$ enters only through the Mellin transform of its Thorin density, in Riemann's kernel $\frac{1}{x} - \frac{1}{\sinh x}$, which places no constraint on the zeros. The clock is not of this type: here $t\nu(t)$ is bounded near 0, while $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ (Section 5.1).

The lemma runs outward only: from a positive Lévy density in the variable $v$ it gives a Thorin measure in $\sqrt{\ }$ of that variable, here $v = \sqrt{s}$ and the argument $s^{1/4}$ of $\xi$. At the argument $\sqrt{s}$ the Lévy measure of $\log\xi(\frac{1}{2}+v)$ is the signed Weil distribution, so the lemma does not apply, and the Thorin form there is equivalent to RH (Theorem 1.2.2). The continuation of $\Psi$ to $\mathbb{C}\setminus(-\infty,0]$ through the integral is zero free, which says $\xi(\frac{1}{2}+w) \neq 0$ for $|\arg w| < \frac{\pi}{4}$, and this is empty, since every zero has $w = \rho - \frac{1}{2}$ with $|\arg w - \frac{\pi}{2}| \leq \arctan(1/2\gamma_1)$.

### 9.1.4 The Lévy form of $\xi(1+s)$

Proposition 9.1.10 (Lévy form). *For* $\operatorname{Re} s > 0$,

$$\log\frac{\xi(1+s)}{\xi(1)} = b_1 s + \int_0^\infty \big(e^{-sx} - 1 + sx\big)\,\Pi_1(dx), \tag{9.1.5}$$

*with drift* $b_1 = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{3}{2}) + \gamma_E = \sum_\rho \rho^{-1} = 0.0230957089661$ *and the signed Lévy measure*

$$\Pi_1(dx) = \Big(\frac{e^{-3x}}{1-e^{-2x}} - 1\Big)\frac{dx}{x} + \sum_{n\geq 2}\frac{\Lambda(n)}{n\log n}\,\delta_{\log n}(dx),$$

*the integral against the last two pieces being taken as* $\lim_{X\to\infty}\int_0^{\log X}$.

*Proof.* Write $\xi(z) = (z-1)\pi^{-z/2}\Gamma(1+\frac{z}{2})\zeta(z)$ and put $z = 1+s$, so that $\xi(1+s) = \pi^{-(1+s)/2}\Gamma(\frac{3}{2}+\frac{s}{2})\,s\zeta(1+s)$ and $\xi(1) = \frac{1}{2}$. Three classical formulas give the three pieces. The factor $\pi^{-s/2}$ contributes $-\frac{s}{2}\log\pi$. Binet's formula [397, §5.9], $\log\Gamma(a+w) - \log\Gamma(a) = w\psi(a) + \int_0^\infty (e^{-wt} - 1 + wt)\frac{e^{-at}}{t(1-e^{-t})}dt$, with $a = \frac{3}{2}$, $w = \frac{s}{2}$ and $t = 2x$, contributes $\frac{s}{2}\psi(\frac{3}{2})$ and the density $e^{-3x}/(x(1-e^{-2x}))$. For the last

factor, the Euler product gives $\log\zeta(1+s)=\sum_n \frac{\Lambda(n)}{n\log n}e^{-s\log n}$ for $\operatorname{Re} s>0$, and Frullani's integral gives $\log s=\int_0^\infty (e^{-x}-e^{-sx})\frac{dx}{x}$. Their sum is $\log(s\zeta(1+s))$, which is analytic at $s=0$ with value 0 and derivative $\gamma_E$, and subtracting the linear term puts it in the compensated form $\gamma_E s+\int(e^{-sx}-1+sx)\,x^{-1}\nu_1(dx)$, where $x^{-1}\nu_1(dx)=\sum\frac{\Lambda(n)}{n\log n}\delta_{\log n}-\frac{dx}{x}$. The compensated integral converges as $X\to\infty$ because the prime atoms and the measure $dx/x$ cancel at first order, by the prime number theorem in the form $\sum_{n\le X}\Lambda(n)/n=\log X-\gamma_E+o(1)$, and at zeroth order, by the logarithm of Mertens' product theorem [174, Theorem 429], $\sum_{p^k\le X}1/(kp^k)=\log\log X+\gamma_E+o(1)$, which does not need the prime number theorem (the logarithm of the product also contains the terms with $p\le X<p^k$, whose sum is $O(X^{-1/2})$). The drift is $\frac{d}{ds}\log\xi(1+s)|_{s=0}=\frac{\xi'}{\xi}(1)$. □

The variable $V=V_1$ of Section 9.1.1 has $\mathbb{E}e^{sV}=\xi(1+s)/\xi(1)$, so (9.1.5) has the form of a Lévy Khintchine formula for the moment generating function of $V$, with $\Pi_1$, reflected to the negative half line by $x\mapsto -x$, in the place of the Lévy measure. That measure is signed: beyond $x=0.28120$ its absolutely continuous part is negative and the only positive mass is the prime atoms. So the uniqueness theorem for Lévy triplets does not apply to $\Pi_1$ directly. Nor is $\Pi_1$ the signed Lévy measure of a quasi infinitely divisible law in the sense of Lindner, Pan and Sato [341], a law whose characteristic function is the quotient of two infinitely divisible ones: that would require the positive and negative parts of $\Pi_1$ to be Lévy measures, of finite mass away from the origin, while on $(1,\infty)$ the positive part is the prime atoms and the negative part exceeds $0.94\,dx/x$, both of infinite mass, and they cancel only conditionally, which is why (9.1.5) holds only as a compensated limit. Proposition 11.1.3(3) proves that no other signed measure represents the characteristic function of $V$, so $V$ is not quasi infinitely divisible, and hence not infinitely divisible. The left panel of Figure 9.2 shows the canonical measure $x\Pi_1(dx)$.

In the variable $\theta=\sqrt{z}$ the sine squared lemma (1.2.9) reads

$$e^{-sx}+sx-1=\frac{2x}{\pi}\int_0^\infty \log\Big(1+\frac{s^2}{\theta^2}\Big)\sin^2\Big(\frac{x\theta}{2}\Big)d\theta,\qquad \operatorname{Re} s>0,\ x>0,$$

and the drift converts through $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta=\pi s$. Applying both to (9.1.5) and substituting $s=\sqrt{u}$ gives $\log\varphi_1(u)=-\int_0^\infty \log(1+u/\theta^2)\rho_{1/2}(\theta)\,d\theta$ with

$$v_1(\theta)=b_1+\int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)\,x\Pi_1(dx)=b_1+A(\theta)+Q(\theta),\tag{9.1.6}$$

where the drift $b_1$ is the stable part, $A(\theta)=\frac12\big[\operatorname{Re}\psi(\frac{3+i\theta}{2})-\psi(\frac32)\big]$ is the gamma piece and $Q(\theta)=\lim_{X\to\infty}\int_0^{\log X}2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ is the compensated prime piece. This is Proposition 9.1.3 again: the two constant terms $-\frac12\log\pi+\gamma_E$ and $\frac12\psi(\frac32)$

combine into $b_1$, and $\frac{1}{2}\operatorname{Re}\psi(\frac{3+i\theta}{2}) = \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \frac{1}{1+\theta^2}$ accounts for the term $1/(1+\theta^2)$.

The conversion has three readings, each visible in the right panel of Figure 9.2. First, the drift $b_1 = \sum_\rho \rho^{-1}$ becomes the constant Thorin density $b_1/\pi$, which is the Thorin density of the positive stable one half factor $e^{-b_1\sqrt{u}}$. So $\varphi_1$ is $e^{-b_1\sqrt{u}}$ times the transform of a GGC with Thorin density $(A+Q)/\pi$ (Proposition 9.1.11). The stable factor plays the role that the stable tilt plays for the floor of Section 6.2.4. At $u = (2k-1)^2$ the factorisation splits each coefficient $1/\zeta(2k)$ of Riesz's function (10.1.5), up to an explicit factor, into a stable value and a GGC value (Section 10.2.3). Second, the gamma factor contributes $A(\theta) \geq 0$, smooth and growing like $\frac{1}{2}\log(\theta/2)$: the average density of zeros. Third, the primes and the pole contribute $Q(\theta)$, which oscillates with both signs, between $-1.47$ and $1.04$ on $[0, 100]$ and carries the peaks at the ordinates.

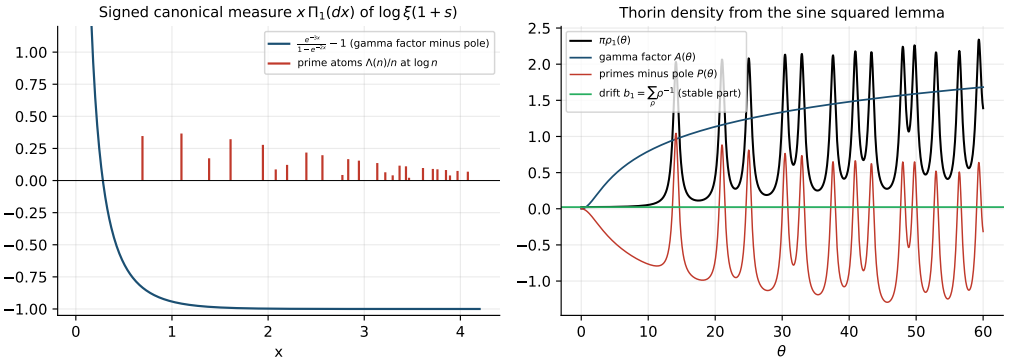



Figure 9.2: Left: the signed canonical measure $x\Pi_1(dx)$ of $\log\xi(1+s)$, made of the gamma density minus the pole, $\big(\frac{e^{-3x}}{1-e^{-2x}} - 1\big)dx$, which is negative beyond $x = 0.28120$, and the prime atoms $\Lambda(n)/n$ at $\log n$ for $n \leq 60$. Right: the bracket $v_1(\theta) = \pi\rho_{1/2}(\theta)$ on $[0, 60]$ assembled by the sine squared lemma from the drift $b_1$, the gamma piece $A(\theta)$ and the compensated prime piece $Q(\theta)$; the plot writes $\rho_1$ for $\rho_{1/2}$ and $P(\theta)$ for $Q(\theta)$. Drawn by `figs3.py`.

The first reading needs $A + Q \geq 0$, that is, $v_1 \geq b_1$, at every height.

Proposition 9.1.11. *With no hypothesis on the zeros, $v_1(\theta) > v_1(0) = b_1$ for every real $\theta \neq 0$; the proof uses a computation of the zeros below height 2000 in ball arithmetic and, above that height, explicit bounds for the number of zeros up to a given height [176, 15]. Consequently $T_1 \overset{d}{=} S + T_1^\circ$, where $S$ and $T_1^\circ$ are independent, $S$ is positive stable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, and $T_1^\circ$ is a GGC with no drift whose Thorin density in the chart $\theta = \sqrt{z}$ is $(v_1 - b_1)/\pi = (A+Q)/\pi$.*

*Proof.* By (9.1.2), $v_1(\theta) = \pi\sum_\rho P_{1-\beta}(\theta - \gamma)$ is a sum of positive terms, with

$\pi P_a(x) = a/(a^2+x^2)$, and $v_1$ is even, because $\xi$ is real on the real axis. We treat three ranges of $\theta \ge 0$.

On $[0, 13.5]$ we use convexity. Every zero has $|\gamma| \ge \gamma_1 = 14.134725$: the certified zeros of the next step, computed in order from the first, show that none has $0 < |\gamma| < \gamma_1$, and none is real, because $\zeta(\sigma) < 0$ for $0 < \sigma < 1$. Since $\pi P_a''(x) = 2a(3x^2-a^2)/(a^2+x^2)^3$, the kernel $P_a$ is strictly convex where $|x| > a/\sqrt{3}$, and for $|\theta| \le 13.5$ every zero has $|\theta - \gamma| \ge \gamma_1 - 13.5 > 0.63 > 1/\sqrt{3} > (1-\beta)/\sqrt{3}$. So $v_1$ is strictly convex on $[-13.5, 13.5]$, and, being even, it has there its unique minimum at $\theta = 0$ and increases on $[0, 13.5]$. By (1.2.2), differentiated twice,

$$v_1''(0) = -\Big(\frac{\xi'}{\xi}\Big)''(1) = \sum_\rho \frac{2(1-\beta)\big(3\gamma^2 - (1-\beta)^2\big)}{\big((1-\beta)^2+\gamma^2\big)^3} > 0.$$

On $[13.5, 2000]$ we use the nearest zero. The script `v1min.py` of the folder `riesz_program/code` computes the first 1800 zeros in ball arithmetic with Arb [290], which isolates them in order by Gram's law and Rosser's rule (Section 2.2.1), each applied only in the range where it is known to hold [64, Section 3], so that none is missed. They reach height 2304.3, all are simple and on the critical line, and consecutive ordinates differ by at most $\gamma_2 - \gamma_1 < 6.8874$. Every $\theta \in [13.5, 2000]$ therefore lies within 3.4437 of an ordinate, half the largest gap, and a zero on the critical line at distance $x$ contributes $\pi P_{1/2}(x) = \frac12/(\frac14 + x^2)$, which exceeds $b_1$ for $x < 4.6259$. All other terms being positive,

$$v_1(\theta) \ge \frac{1/2}{1/4 + 3.4437^2} > 0.041, \qquad 13.5 \le \theta \le 2000.$$

For $\theta \ge 2000$ we count zeros. Write $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2.1). Hasanalizade, Shen and Wong [176, (1.7) and Corollary 1.4] give $|S(T)| \le 2.5167$ for $0 \le T \le X_0 = 30\,610\,046\,000$, from Platt's database of the zeros, and $|S(T)| \le 0.1095 \log T + 0.2042 \log\log T + 3.0305$ for every $T \ge e$, and the smooth part $\vartheta(T)/\pi + 1$ of $N(T)$ differs from $\frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac78$ by at most $\frac{1}{50T}$ for $T \ge \frac57$ [176, (2.7)]. So

$$\Big|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78\Big| \le E(T)$$
$$= \begin{cases} 2.5167 + \frac{1}{50T}, & e \le T \le X_0, \\ 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}, & T > X_0, \end{cases}$$

and $E(T) < 0.3 \log T$ for $T > X_0$, since this holds at $X_0$ and $0.3 - 0.1095 - 0.2042/\log X_0 > 0$. The bound $0.112\log T + 0.278\log\log T + 2.510 + 0.2/T$ of Trudgian [503], which is often quoted, has an incomplete proof [36, 176]. Put $H = 15$. The window $(\theta - H, \theta + H]$ holds $N(\theta+H) - N(\theta-H)$ zeros, counted with multiplicity, and since the main term has derivative $\frac{1}{2\pi}\log\frac{T}{2\pi}$, their number is at least

$$n(\theta) = \frac{H}{\pi}\log\frac{\theta - H}{2\pi} - E(\theta - H) - E(\theta + H).$$

While $\theta + H \le X_0$, $n$ increases, because $E$ decreases there, and $n(2000) > 22.44$; for $\theta+H > X_0$, both values of $E$ are below $0.3\log(\theta+H)$, and $n(\theta) > \frac{H}{\pi}\log\frac{\theta-H}{2\pi} - 0.6\log(\theta+H) > 90.$ So for every $\theta \ge 2000$ the window holds at least 23 zeros. Since $\xi(1-\bar{s}) = \overline{\xi(s)}$, the map $\rho \mapsto 1-\bar{\rho}$ sends each zero to a zero of the same multiplicity; it keeps the ordinate and exchanges $\beta$ and $1-\beta$, so it permutes the zeros of the window, and the sum of $1-\beta$ over them is half their number. Each of them has $0 < 1-\beta < 1$ and $|\theta-\gamma| \le H$, so its term $(1-\beta)/((1-\beta)^2+(\theta-\gamma)^2)$ in $v_1(\theta)$ exceeds $(1-\beta)/(1+H^2)$, and all other terms are positive. Hence

$$v_1(\theta) > \frac{23}{2(1+H^2)} = \frac{23}{452} > 0.05, \qquad \theta \ge 2000,$$

wherever the zeros lie. Backlund's classical constants 0.137, 0.443 and 4.35, valid for $T \ge 200$ [15], give in the same way the increasing bound $E_B(T) = 0.137\log T + 0.443\log\log T + 4.35 + 0.2/T$ in place of $E$ and at least $\frac{H}{\pi}\log\frac{\theta-H}{2\pi} - 2E_B(\theta+H)$ zeros in the window, a number that increases with $\theta$ and exceeds 14.89 at $\theta = 2000$; the window then holds at least 15 zeros, and the bound becomes $15/452 > 0.033$.

It remains to split $\varphi_1$. Since $\frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta = \sqrt{u}$, Proposition 9.1.2 gives

$$\varphi_1(u) = e^{-b_1\sqrt{u}} \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big(v_1(\theta)-b_1\big)\,d\theta\Big\}, \qquad u \ge 0,$$

where the integral is finite because $0 \le v_1 - b_1 \le v_1$ and $\int_0^\infty \log(1+u/\theta^2)\,v_1(\theta)\,d\theta = -\pi\log\varphi_1(u)$ is finite by Theorem 1.2.2 at $\alpha = 1$. The image $U_1^\circ$ of $\pi^{-1}(v_1-b_1)\,d\theta$ under $\theta \mapsto \theta^2 = z$ is a positive measure with $\int \log(1+u/z)\,U_1^\circ(dz) < \infty$, which is equivalent to the integrability condition in (1.3.1), so the second factor is the Laplace transform of a GGC $T_1^\circ$ with no drift. Hence $T_1 \stackrel{d}{=} S + T_1^\circ$, and $v_1 - b_1 = A + Q$ by (9.1.6). □

The stable summand carries the infinite mean and the whole tail: $P(S > t) \sim b_1/\sqrt{\pi t}$ is the tail (9.1.3) of $T_1$, while $T_1^\circ$ has finite mean and infinite variance. The Laplace exponent of $T_1^\circ$ is $\log(\xi(1+\sqrt{u})/\xi(1)) - b_1\sqrt{u} = \frac{1}{2}(\log\xi)''(1)\,u + O(u^{3/2})$ as $u \downarrow 0$, so that

$$\mathbb{E}T_1^\circ = \frac{1}{2}(\log\xi)''(1) = -\frac{1}{2}\sum_\rho \rho^{-2} = 0.0230772,$$

and its Thorin density $(v_1-b_1)/\pi$ vanishes at the origin to exactly second order, since $v_1$ is even and $v_1''(0) > 0$; for instance $v_1(1) - b_1 = 1.12\times10^{-4}$ and $v_1(5) - b_1 = 3.31\times10^{-3}$. Near $z = 0$ the Thorin measure $U_1^\circ$ therefore has a density of exact order $\sqrt{z}$, and the variance $\int z^{-2}\,U_1^\circ(dz)$ is infinite.

For every $\alpha \geq 1$ the same construction gives the Lévy form of $\log(\xi(\alpha+s)/\xi(\alpha))$ with drift $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and

$$x\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)dx + \sum_n \frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$
$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)x\Pi_\alpha(dx),$$

which for $\alpha > 1$ is (3.1.1). For $\alpha > 1$ every piece converges on its own, and the pole contribution is the finite negative term $-\theta^2/((\alpha-1)((\alpha-1)^2+\theta^2))$. Expanding $1/(e^{2x}-1) = \sum_{k\geq 1} e^{-2kx}$ in the archimedean piece of (3.1.1), and writing $q_c(\theta) = \int_0^\infty (1-\cos\theta x)e^{-cx}\,dx = \theta^2/(c(c^2+\theta^2))$, gives

$$v_\alpha(\theta) - b_\alpha = \sum_{k\geq 1} q_{\alpha+2k}(\theta) - q_{\alpha-1}(\theta) + \sum_{n\geq 2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big), \qquad \alpha > 1. \tag{9.1.7}$$

Only the pole term is negative. Near the origin $q_c(\theta) \sim \theta^2/c^3$, and the pole term outweighs the gamma terms when $\alpha < 7.789$, where $\frac{1}{8}\zeta(3, \frac{\alpha}{2}+1) = (\alpha-1)^{-3}$; below that basepoint the arithmetic form needs the primes to keep $v_\alpha - b_\alpha$ positive near the origin.

Proposition 9.1.12. *With no hypothesis on the zeros, $v_\alpha(\theta) > v_\alpha(0) = b_\alpha$ for every $\alpha \geq 1$ and every real $\theta \neq 0$; the proof uses the zeros below height 2000 of Proposition 9.1.11, the explicit bounds of [176] for the number of zeros above that height, and a computation in ball arithmetic. Consequently, for every $\alpha \geq 1$, $T_\alpha \overset{d}{=} S_\alpha + T_\alpha^\circ$, where $S_\alpha$ and $T_\alpha^\circ$ are independent, $S_\alpha$ is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, which is the law of the first passage time of Brownian motion to the level $b_\alpha/\sqrt{2}$, and $T_\alpha^\circ$ is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\mathbb{E}T_\alpha^\circ = \frac{1}{2}(\log\xi)''(\alpha)$.*

*Proof.* At $\alpha = 1$ this is Proposition 9.1.11, so let $\alpha > 1$. Since $v_\alpha$ is even, it suffices to take $\theta > 0$. A zero $\rho = \beta + i\gamma$ contributes $a/(a^2+(\theta-\gamma)^2)$ to $v_\alpha(\theta)$, with $a = \alpha - \beta \in (\alpha-1, \alpha)$, and we put $\theta_0(\alpha) = \gamma_1 - \alpha/\sqrt{3}$. Four steps cover all $\alpha > 1$.

Near the origin, for $\alpha \leq 9$, we use convexity. If $0 < \theta \leq \theta_0(\alpha)$, every zero has $|\theta-\gamma| \geq \gamma_1 - \theta \geq \alpha/\sqrt{3} > a/\sqrt{3}$, so, as in the proof of Proposition 9.1.11, $v_\alpha$ is strictly convex on $[-\theta_0(\alpha), \theta_0(\alpha)]$, and $v_\alpha(\theta) > v_\alpha(0)$ there. Since $\theta_0(9) = 8.94 > 0$, it remains to treat $\theta > \theta_0(\alpha)$ for $\alpha \leq 9$, and every $\theta > 0$ for $\alpha > 9$.

For $1 < \alpha \leq 3$ we use the zeros. Here $\theta_0(\alpha) \geq \theta_0(3) = 12.40$ and $\gamma_1 - 12.40 < 3.4437$, so by the computation in the proof of Proposition 9.1.11 every $\theta \in [\theta_0(\alpha), 2000]$ lies within 3.4437 of the ordinate of a zero on the critical line, whose term exceeds $r(\alpha) = (\alpha - \frac{1}{2})/((\alpha-\frac{1}{2})^2 + 3.4437^2)$. For $\theta \geq 2000$ the window $(\theta-15, \theta+15]$ holds at least 20 zeros, the sum of $a$ over them is $\alpha - \frac{1}{2}$ times their number, and each term exceeds $a/(\alpha^2+225)$, so that $v_\alpha(\theta) > w(\alpha) =$

$20(\alpha-\frac12)/(\alpha^2+225)$. The functions $r$ and $w$ increase on $[1,3]$, and so does $b_\alpha$, whose derivative $\sum_\rho(\gamma^2-a^2)/(a^2+\gamma^2)^2$ is positive for $\alpha<\gamma_1$. It therefore suffices that $r(\alpha_k)>b_{\alpha_{k+1}}$ and $w(\alpha_k)>b_{\alpha_{k+1}}$ at the points $\alpha_k=1+k/256$, and in ball arithmetic the smallest margins are 0.0180 and 0.0209.

For $3\le\alpha\le7$ we use the primes. Adding and subtracting $q_\alpha(\theta)$ in (9.1.7), dropping the prime terms and using $q_c(\theta)<1/c$ gives

$$\begin{aligned} v_\alpha(\theta)-b_\alpha > L(\alpha,\theta) &= \sum_{k\ge0} q_{\alpha+2k}(\theta) - \frac1\alpha - \frac{1}{\alpha-1} \\ &= \frac12\Big[\operatorname{Re}\psi\Big(\frac{\alpha+i\theta}{2}\Big)-\psi\Big(\frac{\alpha}{2}\Big)\Big]-\frac1\alpha-\frac{1}{\alpha-1}, \end{aligned}$$

which increases in $\theta$. Since $|\psi'(z)|\le\psi'(\operatorname{Re}z)$ and $\psi'(x)\le1/x+1/x^2$ for $x>0$, $|\partial L/\partial\alpha|\le M(\alpha)=1/\alpha+3/\alpha^2+1/(\alpha-1)^2$, which decreases in $\alpha$. It therefore suffices that $L(\alpha_k,\theta_0(\alpha_{k+1}))>M(\alpha_k)/256$ at the points $\alpha_k=3+k/256$ of $[3,7]$; the smallest margin is 0.0228, at $\alpha=7$, and the same check succeeds up to $\alpha=7.707$.

For $\alpha\ge7$ the gamma terms suffice. Since $q_c(\theta)$ decreases in $c$, $\sum_{k\ge1}q_{\alpha+2k}(\theta)\ge\frac12\int_{\alpha+2}^\infty q_c(\theta)\,dc=\frac14\log(1+\theta^2/(\alpha+2)^2)$, so with $A=\alpha+2$, $B=\alpha-1$ and $s=\theta^2$,

$$v_\alpha(\theta)-b_\alpha\ge\Phi_\alpha(s)=\frac14\log\Big(1+\frac{s}{A^2}\Big)-\frac{s}{B(B^2+s)}.$$

Here $\Phi_\alpha(0)=0$, and $\Phi'_\alpha(s)=1/(4(A^2+s))-B/(B^2+s)^2$ is nonnegative exactly when $(B^2+s)^2\ge4B(A^2+s)$, the two sides differing by a function that increases in $s$ when $B\ge2$. For $\alpha\ge9$ they are in this order already at $s=0$, because $B^3-4A^2=(\alpha-1)^3-4(\alpha+2)^2$ is 28 at $\alpha=9$ and increases beyond, so $\Phi_\alpha(s)>0$ for every $s>0$, and no zero enters. For $7\le\alpha\le9$ and $\theta>\theta_0(\alpha)\ge\theta_0(9)$, we have $s>s_9=\theta_0(9)^2=79.90$, $6\le B\le8$ and $A\le11$, so $(B^2+s_9)^2\ge(36+s_9)^2>32(121+s_9)\ge4B(A^2+s_9)$; hence $\Phi_\alpha$ increases on $[s_9,\infty)$, where $\Phi_\alpha(s_9)\ge\frac14\log(1+s_9/121)-s_9/(6(36+s_9))>0.0118$.

The checks of the last three steps are carried out in ball arithmetic with Arb [290] (`repro/valpha_min.py`). The factorisation follows from the inequality as in the proof of Proposition 9.1.11, Theorem 1.2.2 giving $\int_0^\infty\log(1+u/\theta^2)\,v_\alpha(\theta)\,d\theta<\infty$, and the mean is the coefficient of $u$ in $\log(\xi(\alpha+\sqrt u)/\xi(\alpha))-b_\alpha\sqrt u=\frac12(\log\xi)''(\alpha)\,u+O(u^{3/2})$. □

As at the basepoint one, $v_\alpha-b_\alpha$ starts as $c(\alpha)\theta^2$, with $c(\alpha)=-\frac12(\xi'/\xi)''(\alpha)>0$, so at every basepoint above one the Thorin density of $T_\alpha^\circ$ vanishes to second order at the origin, $T_\alpha^\circ$ has finite mean and infinite variance, and $S_\alpha$ carries the whole tail $P(T_\alpha>t)\sim b_\alpha/\sqrt{\pi t}$.

More precisely [253], for $\alpha\ge1$ the Lévy density of $T_\alpha^\circ$ is asymptotic to $c(\alpha)\,t^{-5/2}/(4\sqrt\pi)$ as $t\to\infty$, and $P(T_\alpha^\circ>t)\sim c(\alpha)\,t^{-3/2}/(6\sqrt\pi)$.

Indeed $t$ times the Lévy density is $\pi^{-1}\int_0^\infty e^{-t\theta^2}(v_\alpha(\theta)-b_\alpha)\,d\theta$, where $v_\alpha(\theta)-b_\alpha = c(\alpha)\theta^2+O(\theta^4)$ near 0 and $v_\alpha$ grows only logarithmically, so Watson's lemma gives $\pi^{-1}c(\alpha)\sqrt{\pi}/(4t^{3/2})$. The Lévy tail is then asymptotic to $c(\alpha)t^{-3/2}/(6\sqrt{\pi})$, a regularly varying and hence subexponential tail, and for an infinitely divisible law with such a Lévy tail the tail of the law is asymptotic to it [131]. At $\alpha=2$ the ratio of $t$ times the Lévy density to $c(2)/(4\sqrt{\pi}\,t^{3/2})$ is 1.000372, 1.0000929 and 1.0000232 at $t=25$, 100 and 400, with $c(2)=3.25029\times10^{-4}$ (computed, `code/rx_tail.py`).

For $\alpha<1$ the Euler product no longer converges near $\operatorname{Re}s=\alpha$, and the prime atoms $\Lambda(n)n^{-\alpha}$ can be compensated only if the primes are well enough distributed. So the route from the Lévy form through the sine squared lemma computes the Thorin measure unconditionally on $\alpha\ge1$.

### 9.1.5 Below the basepoint one

The construction of Section 9.1.4 can be written down for any basepoint $\alpha=1-\varepsilon$ with $0<\varepsilon<\frac12$, and the question is whether it converges. Formally nothing changes except the pole term, which now grows: with $b_\alpha=\frac{\xi'}{\xi}(\alpha)$,

$$x\Pi_\alpha(dx)=\frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx\;+\;\Big(\sum_n\frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx)-e^{\varepsilon x}\,dx\Big),$$

and the converted Thorin density would be $\pi^{-1}$ times

$$b_\alpha+A_\alpha(\theta)+\lim_{X\to\infty}\Big[\sum_{n\le X}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big)-\int_0^{\log X}2\sin^2\Big(\frac{\theta x}{2}\Big)e^{\varepsilon x}dx\Big],\quad(9.1.8)$$

with $A_\alpha(\theta)=\frac12\big[\operatorname{Re}\psi(\frac{\alpha+2+i\theta}{2})-\psi(\frac{\alpha+2}{2})\big]$. The prime piece and the pole piece now each grow like $X^\varepsilon/\varepsilon$, and whether their difference converges is a question about how well $\psi(y)$ is approximated by $y$.

Proposition 9.1.13. *Let $\Theta=\sup\{\operatorname{Re}\rho\}$ over the nontrivial zeros and let $\frac12<\alpha<1$.*

1. *If $\Theta<\alpha$, the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)}=b_\alpha s+\int_0^\infty(e^{-sx}-1+sx)\Pi_\alpha(dx)$ holds for $\operatorname{Re}s>0$, the compensated limit converging, and the sine squared lemma converts it into (9.1.8), which then equals $v_\alpha(\theta)$ and is positive. The error of the compensated sum with prime cutoff $X$ is $O(X^{\Theta-\alpha+\delta})$ for every $\delta>0$.*

2. *Conversely, if the Lévy form holds for $\operatorname{Re}s>0$, then $\xi(\alpha+s)$ has no zeros with $\operatorname{Re}s>0$, so $\Theta\le\alpha$.*

*Proof.* If $\Theta<\alpha$, the explicit formula gives $\psi(y)-y=O(y^{\Theta+\delta})$ for every $\delta>0$ [498]. Then the Stieltjes integral $\int_1^\infty y^{-w}\,d(\psi(y)-y)$, which equals $-\frac{\zeta'}{\zeta}(w)-\frac{1}{w-1}$

for $\operatorname{Re} w > 1$, converges after one integration by parts for $\operatorname{Re} w > \Theta$ and provides the analytic continuation there, with tail beyond $X$ of order $X^{\Theta - \operatorname{Re} w + \delta}$. Taking $w = \alpha + s$ with $\operatorname{Re} s \geq 0$ and integrating twice in $s$ from $s = 0$ gives the compensated Lévy form, the gamma and polynomial factors being handled exactly as in Proposition 9.1.10. For fixed $\theta$ the sine squared lemma may be applied under the limit, because the compensated measure integrated against $2\sin^2(\theta x/2)$ is a combination of the same convergent Stieltjes integrals at $w = \alpha$ and $w = \alpha \pm i\theta$. Positivity follows from the Poisson form of Proposition 1.2.3, as in Proposition 9.1.2, with widths $\alpha - \beta > 0$. Conversely, suppose the Lévy form holds for $\operatorname{Re} s > 0$. Write $\nu_\alpha$ for the bracket in $x\Pi_\alpha$, so that the compensated part of the right side is the limit of $\int_0^{\log X} (e^{-sx} - 1 + sx)\, x^{-1} \nu_\alpha(dx)$; its part over $[0, 1]$ is an entire function of $s$. For real $s > 1 - \alpha$ the integral $\int_1^\infty e^{-sx} x^{-1} \nu_\alpha(dx)$ converges absolutely, so the convergence of the compensated limit at two such values of $s$ gives that of $\int_1^{\log X} \nu_\alpha(dx)$ and of $\int_1^{\log X} x^{-1} \nu_\alpha(dx)$. The function $y \mapsto \int_1^y \nu_\alpha(dx)$ is therefore bounded on $[1, \infty)$, and one integration by parts against it shows that $\int_1^{\log X} e^{-sx} x^{-1} \nu_\alpha(dx)$ converges uniformly on compact subsets of $\operatorname{Re} s > 0$. So the compensated limit converges locally uniformly, the right side of the Lévy form is analytic in $\operatorname{Re} s > 0$, hence so is $\log \xi(\alpha + s)$, and $\xi(\alpha + s)$ cannot vanish there. □

So the extension to $\xi(1 - \varepsilon + s)$ exists exactly when $\zeta$ has no zeros in the half plane $\operatorname{Re} s > 1 - \varepsilon$, up to the boundary case $\Theta = 1 - \varepsilon$. At the basepoint one the convergence of the compensated sum was the prime number theorem; at the basepoint $1 - \varepsilon$ it is the estimate $\psi(y) - y = O(y^{1 - \varepsilon + \delta})$, which is the zero free half plane $\operatorname{Re} s > 1 - \varepsilon$; and as $\varepsilon \uparrow \frac{1}{2}$ it is $\psi(y) - y = O(y^{1/2 + \delta})$, which is RH. The Lévy form route is an arithmetic restatement of step 7 of the path in Table 12.1, not a way around it.

`below.py` evaluates (9.1.8) with primes up to $10^7$ at $\theta = \gamma_1$ and compares it with $v_\alpha(\gamma_1)$ computed directly. On the range of the computation the primes behave as RH predicts, $\psi(y) - y = O(y^{1/2} \log^2 y)$, so the errors should decay like $X^{1/2 - \alpha}$, and they do (Figure 9.3, left): the decay is fast at $\alpha = 0.9$, slow at $\alpha = 0.6$ and absent at the centre. At $\alpha = \frac{1}{2}$ and $\theta = \gamma_1$ the compensated sum grows like $\log X$ (Figure 9.3, right), which is the atom $\delta_{\gamma_1}$ of the RH measure forming as the cutoff increases; away from the zeros it oscillates without converging. The exponent in $X^{\Theta - \alpha}$ thus measures the largest real part of a zero, but only up to the computable range of $X$, so it can never certify $\Theta$.

Writing the tail of the compensated sum beyond $X$ as a Stieltjes integral against $d(\psi(y) - y)$ and inserting the explicit formula $\psi(y) = y - \sum_\rho y^\rho/\rho - \log 2\pi - \frac{1}{2}\log(1 - y^{-2})$ [104, Chapter 17] gives, for $\Theta < \alpha$ and $X$ not a prime

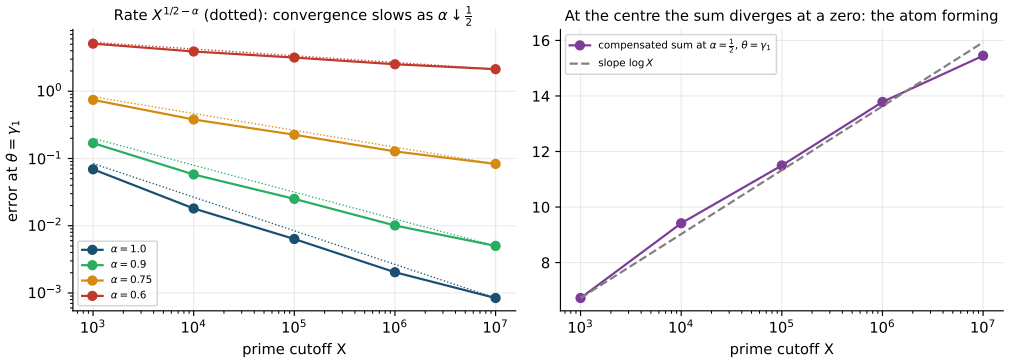



Figure 9.3: Left: absolute error of the compensated formula (9.1.8) at $\theta = \gamma_1$ against the prime cutoff $X$ for $\alpha = 1$, 0.9, 0.75 and 0.6, with the rates $X^{1/2-\alpha}$ (dotted). Right: at the centre, $\alpha = \frac{1}{2}$, the compensated sum at the first ordinate grows like $\log X$ (dashed line), the atom of the RH measure forming. Drawn by `figs4.py`.

power,

$$\text{(formula (9.1.8) at cutoff } X) - v_\alpha(\theta) = \sum_\rho \Big[ -\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)} \Big] + O(X^{-1-\alpha}), \quad (9.1.9)$$

the sum taken over conjugate pairs. Every term has size $X^{\beta-\alpha}$, which makes the rate $X^{\Theta-\alpha}$ visible term by term, and at $\theta = \gamma$ the third term has denominator $\beta - \alpha$, so under RH it approaches $X^{1/2-\alpha}/(1-2\alpha)$ and at the centre it becomes the $\log X$ growth of Figure 9.3. Table 9.3, computed by `errformula.py`, compares the observed error with (9.1.9) truncated to the 1700 zeros below height 2198. The explicit formula accounts for the error to four or five decimal places for $\alpha \geq 0.75$ and to three at $\alpha = 0.6$, and the hybrid of primes up to $X$ plus the first 1700 zeros reduces the error of the primes alone by factors between about 700 and 11000.

This also sharpens the logic of Proposition 9.1.13. A zero with $\beta > \alpha$ would contribute a term of size $X^{\beta-\alpha}$ that grows with the cutoff, so the compensated sum at the basepoint $\alpha$ would diverge, and the size and phase of that growth would locate the zero. The hybrid formula is a zero detector with the same structure as Turing's method (Section 2.2.1): primes up to $X$ and zeros up to height $T$ must balance to within the tail of both.

If instead of compensating one continues the pole contribution analytically, its value at $\alpha = 1-\varepsilon$ is $\operatorname{Re}\frac{1}{-\varepsilon+i\theta} + \frac{1}{\varepsilon} = \frac{\theta^2}{\varepsilon(\varepsilon^2+\theta^2)}$, which is positive, since the pole now lies to the right of the basepoint. The single negative term of the decomposition (3.1.1) above the basepoint one has become positive. Below the line $\operatorname{Re} s = 1$,

| $\alpha$ | $X$ | observed error | predicted by (9.1.9) | error after correction |
|---|---|---|---|---|
| 1 | $10^5$ | $-6.352\times10^{-3}$ | $-6.349\times10^{-3}$ | $-2.9\times10^{-6}$ |
| 1 | $10^7$ | $-8.419\times10^{-4}$ | $-8.431\times10^{-4}$ | $1.2\times10^{-6}$ |
| 0.9 | $10^7$ | $-5.014\times10^{-3}$ | $-5.020\times10^{-3}$ | $6.1\times10^{-6}$ |
| 0.75 | $10^7$ | $-8.297\times10^{-2}$ | $-8.304\times10^{-2}$ | $6.8\times10^{-5}$ |
| 0.6 | $10^5$ | $-3.1668$ | $-3.1665$ | $-2.9\times10^{-4}$ |
| 0.6 | $10^7$ | $-2.1285$ | $-2.1292$ | $7.7\times10^{-4}$ |

Table 9.3: The error of the compensated sum (9.1.8) at $\theta=\gamma_1$ with primes up to $X$, its prediction from the explicit formula (9.1.9) over the first 1700 zeros, and what remains after subtracting the prediction.

therefore, the negativity that RH must control is carried by the primes, not by the pole.

### 9.1.6 The minimum at the origin below the basepoint one

Propositions 9.1.11 and 9.1.12 put the minimum of $v_\alpha$ at the origin for every $\alpha\geq1$, with no hypothesis on the zeros. Below the basepoint one the margin at the origin vanishes linearly, $b_\alpha\approx(\alpha-\frac12)(\log\xi)''(\frac12)$, and Table 9.6 found the minimum at the origin on a window by computation. This subsection proves it on the range of heights where the zeros are known to lie on the critical line, up to $\frac{1+\sqrt2}{2}$ below its top, for every basepoint $\alpha\in(\frac12,7.35]$, and shows that RH is equivalent to it at every height for every $\alpha\in(\frac12,1)$. The tool is a comparison across vertical lines, from the first author's note on the Dirichlet beta function [425]. We state it for $f=\xi$, for $f=\xi(\cdot,\chi_4)$ of Section 8.5.7 and for $f=\xi(\cdot,\chi_{12})$ of Section 8.5.1, and write $v_\alpha(\theta)=\mathrm{Re}(f'/f)(\alpha+i\theta)$, $b_\alpha=v_\alpha(0)=(f'/f)(\alpha)$ and $V_f=(\log f)''(\frac12)$.

Each $f$ is entire of order one, real on the real axis, with $f(s)=f(1-s)$ and no real zeros. Its zeros therefore come in groups: a zero $\frac12+i\gamma$ on the critical line, or a pair $\frac12\pm\delta+i\gamma$ with $0<\delta<\frac12$. At resolution $\varepsilon=\alpha-\frac12$ and offset $u=\theta-\gamma$, a zero on the line contributes $\varepsilon/(\varepsilon^2+u^2)$ to $v_\alpha(\theta)$, and a pair contributes

$$k_\delta(\varepsilon,u)=\frac{\varepsilon-\delta}{(\varepsilon-\delta)^2+u^2}+\frac{\varepsilon+\delta}{(\varepsilon+\delta)^2+u^2}=\frac{2\varepsilon(w+c)}{(w+c)^2+\kappa^2},\tag{9.1.10}$$

where $w=\varepsilon^2$, $c=u^2-\delta^2$ and $\kappa=2\delta|u|$, since the numerator of the sum is $2\varepsilon(\varepsilon^2+u^2-\delta^2)$ and its denominator is $(\varepsilon^2+\delta^2+u^2)^2-4\varepsilon^2\delta^2=(w+c)^2+\kappa^2$. The sum over the groups converges absolutely, since $\sum_\rho|\rho|^{-2}<\infty$. So $v_\alpha(\theta)/(\alpha-\frac12)$ is the sum over the groups of the ratios $1/(w+u^2)$ for the zeros on the line and

$$r_\delta(w,u)=\frac{k_\delta(\varepsilon,u)}{\varepsilon}=2\,\mathrm{Re}\,\frac{1}{w+c+i\kappa}$$

for the pairs, as functions of $w=(\alpha-\frac12)^2$. The ratio of a zero on the line decreases in $w$; that of a pair need not.

LEMMA 9.1.14 (Comparison). *Let $0 < \delta < \frac{1}{2}$ and let $u$ be real. As a function of $w > 0$, the ratio $r_\delta(w, u)$*

*(a) is nonincreasing on $(0, \infty)$ if $|u| \geq (1 + \sqrt{2})\delta$;*

*(b) is nonincreasing on $[2\delta^2, \infty)$ for every $u$, in particular on $w \geq \frac{1}{2}$, that is for $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, whatever $\delta$;*

*(c) if $0 < |u| < (1 + \sqrt{2})\delta$, increases exactly on the interval $\max\big(0, 2\delta^2 - (|u| + \delta)^2\big) < w < 2\delta^2 - (|u| - \delta)^2$, which begins at 0 if and only if $|u| \geq (\sqrt{2} - 1)\delta$; at $u = 0$ it equals $2/(w - \delta^2)$, which has a pole at $w = \delta^2$.*

*In particular the constant $1 + \sqrt{2}$ in (a) cannot be lowered.*

*Proof.* For $u \neq 0$, $\partial_w r_\delta = -2\,\mathrm{Re}(w+c+i\kappa)^{-2} = -2\big((w+c)^2 - \kappa^2\big)/\big((w+c)^2 + \kappa^2\big)^2$, which is positive exactly when $-\kappa < w + c < \kappa$, that is when $2\delta^2 - (|u| + \delta)^2 < w < 2\delta^2 - (|u| - \delta)^2$, since $-\kappa - c = 2\delta^2 - (|u| + \delta)^2$ and $\kappa - c = 2\delta^2 - (|u| - \delta)^2$. The right end is at most $2\delta^2$, which gives (b) for $u \neq 0$; it is positive exactly when $\big||u| - \delta\big| < \sqrt{2}\delta$, that is when $|u| < (1 + \sqrt{2})\delta$, which gives (a) and the first part of (c); and the left end is at most 0 exactly when $|u| + \delta \geq \sqrt{2}\delta$. At $u = 0$, $c = -\delta^2$ and $\kappa = 0$, and $2/(w - \delta^2)$ decreases on $w > \delta^2$, which completes (b). Finally $2\delta^2 < \frac{1}{2}$ because $\delta < \frac{1}{2}$. □

THEOREM 9.1.15 (Comparison across vertical lines). *Let $\frac{1}{2} < \alpha < \alpha'$ and let $\theta$ be real. Suppose that every zero $\frac{1}{2} + \delta + i\gamma$ of $f$ with $\delta > (\alpha - \frac{1}{2})/\sqrt{2}$ satisfies $|\theta - \gamma| \geq (1 + \sqrt{2})\delta$. Then*

$$\frac{v_\alpha(\theta)}{\alpha - \frac{1}{2}} \;\geq\; \frac{v_{\alpha'}(\theta)}{\alpha' - \frac{1}{2}}.$$

*The hypothesis holds at every $\theta$ under RH for $\xi$, or GRH for $\chi_4$ or $\chi_{12}$; at every $\theta$ when $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, with no hypothesis; and at every $\theta$ with $|\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$, where $H_0$ is a height below which every zero of $f$ is known to lie on the critical line: $H_0 = 3 \times 10^{12}$ for $\xi$ [419], and $H_0 = 10^8/q$ for $\xi(\cdot, \chi_q)$, that is $2.5 \times 10^7$ for $\chi_4$ and $10^8/12 > 8.33 \times 10^6$ for $\chi_{12}$ [417].*

*Proof.* A zero on the line has the ratio $1/(w + u^2)$, which decreases in $w$. A pair with $\delta \leq (\alpha - \frac{1}{2})/\sqrt{2}$ has $2\delta^2 \leq (\alpha - \frac{1}{2})^2$, so its ratio is nonincreasing on $[(\alpha - \frac{1}{2})^2, (\alpha' - \frac{1}{2})^2]$ by Lemma 9.1.14(b), and a pair with $|\theta - \gamma| \geq (1 + \sqrt{2})\delta$ by Lemma 9.1.14(a). Summing over the groups, which converges absolutely, gives the inequality. Under RH there are no pairs. If $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, every pair has $\delta < \frac{1}{2} \leq (\alpha - \frac{1}{2})/\sqrt{2}$. If $|\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$, every zero off the line has $|\gamma| > H_0$, so that $|\theta - \gamma| > \frac{1+\sqrt{2}}{2} > (1 + \sqrt{2})\delta$. □

COROLLARY 9.1.16 (Explicit lower bound). *For $\sigma > 1$ and real $\theta$, $v_\sigma(\theta) \geq A_f^\infty(\sigma,\theta) - S_f(\sigma)$, where*

$$\begin{aligned} A_\xi^\infty(\sigma,\theta) &= -\tfrac12 \log\pi + \tfrac12 \operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big), \\ A_{\chi_4}^\infty(\sigma,\theta) &= \tfrac12 \log\tfrac{4}{\pi} + \tfrac12 \operatorname{Re}\psi\big(\tfrac{\sigma+1+i\theta}{2}\big), \\ A_{\chi_{12}}^\infty(\sigma,\theta) &= \tfrac12 \log\tfrac{12}{\pi} + \tfrac12 \operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big), \\ S_f(\sigma) &= \sum_{(n,q)=1} \frac{\Lambda(n)}{n^\sigma}, \end{aligned}$$

*with $q = 1$, 4 and 12 for the three functions, so that $S_\xi = -\zeta'/\zeta$, $S_{\chi_4}(\sigma) = S_\xi(\sigma) - \log 2/(2^\sigma - 1)$ and $S_{\chi_{12}}(\sigma) = S_{\chi_4}(\sigma) - \log 3/(3^\sigma - 1)$; and $A_f^\infty(\sigma,\theta)$ increases in $|\theta|$. Hence, under the hypothesis of Theorem 9.1.15 with $\alpha' > 1$,*

$$\begin{aligned} v_\alpha(\theta) &\geq \frac{\alpha - \frac12}{\alpha' - \frac12}\Big(A_f^\infty(\alpha',\theta) - S_f(\alpha')\Big) \\ &= \frac{\alpha - \frac12}{2\alpha' - 1}\log|\theta| + O(1), \qquad |\theta| \to \infty. \end{aligned}$$

*Proof.* For $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac12 \log\pi + \frac12\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s),$$

where $\operatorname{Re}(\frac1s + \frac{1}{s-1}) \geq 0$ and $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\theta) = -\sum_n \Lambda(n) n^{-\sigma}\cos(\theta\log n) \geq -S_\xi(\sigma)$; for the two $L$ functions the bound is that of Proposition 8.5.26. The monotonicity in $|\theta|$ follows from the series $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\geq 0} y^2/\big((x+k)((x+k)^2+y^2)\big)$, and $\operatorname{Re}\psi(x+iy) = \log|y| + O(1)$ for fixed $x > 0$. Then apply Theorem 9.1.15 between $\alpha$ and $\alpha'$. □

REMARK 9.1.17 (A bound of order $\log\theta$ outside a sparse set). For fixed $\alpha > \frac12$ the hypothesis of Theorem 9.1.15 fails only at heights within $(1+\sqrt2)\delta < \frac{1+\sqrt2}{2}$ of the ordinate of a zero $\frac12 + \delta + i\gamma$ with $\delta > (\alpha - \frac12)/\sqrt2$. For $f = \xi$ the number of such zeros with $0 < \gamma \leq T$ is $N(\sigma_0, T)$ with $\sigma_0 = \frac12 + (\alpha - \frac12)/\sqrt2 > \frac12$, which is $O(T^{3(1-\sigma_0)/(2-\sigma_0)}\log^5 T) = o(T)$ by Ingham's estimate [284] (Section 11.4.1). So Corollary 9.1.16 with $\alpha' = \alpha + 1$ gives $v_\alpha(\theta)/\pi \geq c\log\theta$, for every $c < (\alpha - \frac12)/(\pi(2\alpha+1))$ and all large $\theta$, outside a set of heights of density zero: this is the lower bound of the first of the four questions of Section 11.4.7, for the density $v_\alpha/\pi$ itself. More directly, by the proof of Corollary 11.4.7, $v_\alpha(\theta) \leq 0$ only within $\sqrt{\delta^2 - (\alpha - \frac12)^2} < \frac12$ of the ordinate of a zero $\frac12 + \delta + i\gamma$ with $\frac12 + \delta > \alpha$, so the set of $\theta \in [0,T]$ with $v_\alpha(\theta) \leq 0$ has measure at most $N(\alpha, T + \frac12) \leq T^{e(\alpha)+o(1)}$, with $e(\alpha) < 1$ of Theorem 11.4.2. The exceptional set, however, contains the window of every zero to the right of the line $\operatorname{Re} s = \alpha$, since such a zero has

$\delta > \alpha - \frac{1}{2} > (\alpha - \frac{1}{2})/\sqrt{2}$. So the bound gives no information where Lemma 11.4.9 would need it, and it does not by itself confine a failure of positivity to the thin strips of that question.

THEOREM 9.1.18 (Minimum at the origin). *Let $V_f$, $\gamma_1$, $H_0$ and $\alpha_{\max}$ be as in Table 9.4.*

*(i) For every $\alpha > \frac{1}{2}$, $0 < b_\alpha \le (\alpha - \frac{1}{2})V_f$; the ratio $b_\alpha/(\alpha - \frac{1}{2})$ decreases in $\alpha$ and tends to $V_f$ as $\alpha \downarrow \frac{1}{2}$.*

*(ii) For $\frac{1}{2} < \alpha \le \alpha_{\max}$, $v_\alpha(\theta) > b_\alpha$ for $0 < |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$.*

*(iii) For $\frac{1}{2} + \frac{1}{\sqrt{2}} \le \alpha \le \alpha_{\max}$, $v_\alpha(\theta) > b_\alpha$ for every $\theta \neq 0$.*

*(iv) For $f = \xi(\cdot, \chi_4)$ and $f = \xi(\cdot, \chi_{12})$, $v_\alpha(\theta) > b_\alpha$ for every $\alpha \ge \frac{1}{2} + \frac{1}{\sqrt{2}}$ and every $\theta \neq 0$.*

*(v) For $f = \xi(\cdot, \chi_4)$ and $f = \xi(\cdot, \chi_{12})$, $v_\alpha(\theta) > b_\alpha$ for every $\alpha \ge 1$ and every $\theta \neq 0$.*

*(vi) RH, respectively GRH for $\chi_4$ or for $\chi_{12}$, holds if and only if $v_\alpha(\theta) > b_\alpha$ for every $\alpha \in (\frac{1}{2}, 1)$ and every $\theta \neq 0$.*

*Proof.* (i) The function $f(\frac{1}{2} + s) = \int_{\mathbb{R}} K(u) e^{su}\, du$ is the two sided Laplace transform of a positive even kernel $K$: the Pólya kernel $\Phi$ for $\xi$ (Section 1.2), the kernel $K_4$ of Proposition 8.5.22, and the eta kernel of Section 8.5.1. So $\log f(\frac{1}{2}+s)$ is, up to a constant, the cumulant generating function of a nondegenerate law; it is even and strictly convex in real $s$, and $b_\alpha = (\log f)'(\alpha) = \int_{1/2}^{\alpha} (\log f)''(r)\, dr > 0$. At $\theta = 0$ the offset of a group with ordinate $\gamma$ has $|u| = |\gamma| \ge \gamma_1 > \frac{1+\sqrt{2}}{2} > (1+\sqrt{2})\delta$, so by Lemma 9.1.14(a) the ratio of every group is nonincreasing in $w$, strictly for the zeros on the line, and $b_\alpha/(\alpha - \frac{1}{2})$, their sum, decreases in $\alpha$. As $w \downarrow 0$ the ratios increase to $\gamma^{-2}$ for a zero on the line and to $2\,\mathrm{Re}(|\gamma| + i\delta)^{-2}$ for a pair, and by monotone convergence the sum tends to the sum of these limits, which is $-\sum_\rho (\frac{1}{2} - \rho)^{-2} = (\log f)''(\frac{1}{2}) = V_f$.

(ii) Put $\alpha' = \alpha_{\max}$ and let $t_1$ be its value at $\alpha_{\max}$ in Table 9.4, where $A_f^\infty(\alpha', t_1) - S_f(\alpha') > (\alpha' - \frac{1}{2})V_f$ is certified in ball arithmetic; since $A_f^\infty$ increases in $|\theta|$, the inequality holds for $|\theta| \ge t_1$. For $t_1 \le |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, Corollary 9.1.16 and (i) give $v_\alpha(\theta) > (\alpha - \frac{1}{2})V_f \ge b_\alpha$ (for $\alpha = \alpha'$ the comparison is an identity). For $|\theta| \le t_1$ we show that $v_\alpha$ is strictly convex on $[-t_1, t_1]$; as it is even, it then exceeds $v_\alpha(0) = b_\alpha$ at $0 < |\theta| \le t_1$. The term $\varphi(\theta - \gamma)$, $\varphi(x) = a/(a^2 + x^2)$ with $a = \alpha - \frac{1}{2}$, of a zero on the line is strictly convex where $|\theta - \gamma| > a/\sqrt{3}$, since $\varphi''(x) = 2a(3x^2 - a^2)/(a^2 + x^2)^3$. Below $H_0$ all zeros lie on the line and have $|\gamma| \ge \gamma_1$, and for $|\theta| \le t_1$, $|\theta - \gamma| \ge \gamma_1 - t_1 > (\alpha' - \frac{1}{2})/\sqrt{3} \ge a/\sqrt{3}$ by the convexity margin of the table. Every zero above $H_0$ has $|\theta - \gamma| > H_0 - t_1 > 10^6$. For a

pair there, the second derivative in $\theta$ of $k_\delta(a,\theta-\gamma)$ is $g(a+\delta,u)+g(a-\delta,u)$ with $g(c,u)=2c(3u^2-c^2)/(c^2+u^2)^3$, the second derivative of $c/(c^2+u^2)$ in $u$; since $\partial_c g(c,u)=6(u^4-6c^2u^2+c^4)/(c^2+u^2)^4>0$ for $|c|<(\sqrt{2}-1)|u|$, and $|a\pm\delta|<\alpha_{\max}<(\sqrt{2}-1)10^6$, the odd function $g(\cdot,u)$ increases on an interval containing $\pm(a\pm\delta)$, so that $g(a+\delta,u)>g(\delta-a,u)=-g(a-\delta,u)$. The sum of the second derivatives converges uniformly on $[-t_1,t_1]$, so $v_\alpha''>0$ there.

(iii) For $\alpha\ge\frac12+\frac{1}{\sqrt2}$ the hypothesis of Theorem 9.1.15 holds at every $\theta$, so the bound of (ii) holds for every $|\theta|\ge t_1$, and the convexity argument for $|\theta|\le t_1$ is unchanged.

(iv) It remains to treat $\alpha>\alpha_{\max}$, and there we use the primes. For $\chi_4$, $v_\alpha(\theta)-b_\alpha=G_\alpha(\theta)+\varpi_\alpha(\theta)$ by (8.5.9), with $G_\alpha(\theta)=\frac12[\operatorname{Re}\psi(\frac{\alpha+1+i\theta}{2})-\psi(\frac{\alpha+1}{2})]$; for $\chi_{12}$ the same holds with $G_\alpha(\theta)=\frac12[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2})-\psi(\frac{\alpha}{2})]$ and the weights $\chi_{12}(n)\Lambda(n)$ (Section 8.5.3). Dropping the positive weights and bounding $1-\cos x$ by 2 and by $x^2/2$ gives $\varpi_\alpha(\theta)\ge-R_\alpha(\theta)$ with $R_\alpha(\theta)\le\min\bigl(2S^-(\alpha),\frac12\theta^2M_2(\alpha)\bigr)$, where

$$S^-(\alpha)=\sum_{\chi(n)=-1}\frac{\Lambda(n)}{n^\alpha}=\frac12\Bigl(S_f(\alpha)+\frac{L'}{L}(\alpha)\Bigr),$$
$$M_2(\alpha)=\sum_{\chi(n)=-1}\frac{\Lambda(n)\log^2 n}{n^\alpha}=\frac12\Bigl(S_f''(\alpha)+\Bigl(\frac{L'}{L}\Bigr)''(\alpha)\Bigr).$$

The function $G_\alpha(\theta)$ is a sum of terms $q_c(\theta)=\theta^2/(c(c^2+\theta^2))$, so it increases in $|\theta|$ and decreases in $\alpha$, and $G_\alpha(\theta)/\theta^2$ decreases in $|\theta|$; and $S^-$ and $M_2$ decrease in $\alpha$. So on a cell $s_0\le\alpha\le s_1$ it suffices that $G_{s_1}(T)>\max\bigl(2S^-(s_0),\frac12T^2M_2(s_0)\bigr)$ for one $T>0$: then $G_\alpha(\theta)\ge G_{s_1}(T)>2S^-(\alpha)$ for $|\theta|\ge T$, and $G_\alpha(\theta)\ge\theta^2G_{s_1}(T)/T^2>\frac12\theta^2M_2(\alpha)$ for $0<|\theta|\le T$. The script `dbeta_minimum.py` checks this in ball arithmetic on cells of width $\frac{1}{32}$ covering $[\sigma_2,20]$, with $T\in\{0.25,0.5,\dots,4\}$, where $\sigma_2=3.71875$ for $\chi_4$ and 2.8125 for $\chi_{12}$; the cells just below $\sigma_2$ fail. For $\alpha\ge20$ take $T=2$: since $q_c$ decreases in $c$, $G_\alpha(2)\ge\frac12\int_{\alpha+1}^\infty q_c(2)\,dc=\frac14\log(1+4/(\alpha+1)^2)\ge1/((\alpha+1)^2+4)$, and since every $n$ with $\chi(n)=-1$ is at least 3, $S^-(\alpha)$ and $M_2(\alpha)$ are at most $3^{-(\alpha-20)}$ times their values at 20. With $K=\max(2S^-(20),2M_2(20))$ it therefore suffices that $K((\alpha+1)^2+4)3^{-(\alpha-20)}<1$ for $\alpha\ge20$; the left side decreases in $\alpha$, and at $\alpha=20$ it is $3.4\times10^{-7}$ for $\chi_4$ and $3.9\times10^{-11}$ for $\chi_{12}$. As $\alpha_{\max}=4.70>3.71875$ for $\chi_4$ and $3.10>2.8125$ for $\chi_{12}$, (iii) and this argument cover every $\alpha\ge\frac12+\frac{1}{\sqrt2}$.

(v) By (ii) and (iv) it remains to treat $1\le\alpha<\frac12+\frac{1}{\sqrt2}$ and $|\theta|>H_0-\frac{1+\sqrt2}{2}$, and by symmetry $\theta>0$. Let $n$ be the number of zeros with $|\gamma-\theta|\le h$, counted with multiplicity, where $h=15$ for $\chi_4$ and $h=10$ for $\chi_{12}$. Every zero has $0<\beta<1\le\alpha$, so every term of $v_\alpha(\theta)$ is positive, and a zero in the window contributes $a/(a^2+(\theta-\gamma)^2)\ge a/(\alpha^2+h^2)$, with $a=\alpha-\beta\in(0,\alpha)$; the window contains both members of each pair $\frac12\pm\delta+i\gamma$, whose values of $a$ add up to $2(\alpha-\frac12)$, so the values of $a$ over the window add up to $n(\alpha-\frac12)$. Hence $v_\alpha(\theta)>n(\alpha-\frac12)/(\alpha^2+h^2)$,

which is at least $(\alpha - \frac{1}{2})V_f \geq b_\alpha$ as soon as $n \geq V_f(\alpha^2 + h^2)$. Since $\alpha^2 < (\frac{1}{2} + \frac{1}{\sqrt{2}})^2 = 1.4571\ldots$, it suffices that $n > 35.335$ for $\chi_4$ and $n > 33.778$ for $\chi_{12}$. By the explicit counting formula of Bennett, Martin, O'Bryant and Rechnitzer [36, Theorem 1.1], for a primitive character $\chi$ of conductor $q > 1$, $T \geq \frac{5}{7}$ and $\ell = \log(q(T+2)/(2\pi)) > 1.567$, the number $N(T,\chi)$ of zeros with $|\gamma| \leq T$ satisfies

$$\Big| N(T,\chi) - \frac{T}{\pi}\log\frac{qT}{2\pi e} + \frac{\chi(-1)}{4} \Big| \leq E(T) = 0.22737\,\ell + 2\log(1+\ell) - 0.5.$$

As $f$ has no real zeros and its zeros are symmetric about the real axis, $N(\theta + h,\chi) - N(\theta - h,\chi)$ is at most twice $n$, whatever the convention for zeros at the ends of the window, and since $E$ increases in $T$,

$$n \;\geq\; \frac{1}{2\pi}\int_{\theta-h}^{\theta+h} \log\frac{qt}{2\pi}\,dt - E(\theta+h) \;\geq\; \frac{h}{\pi}\log\frac{q(\theta-h)}{2\pi} - E(\theta+h).$$

The right side increases in $\theta$, its derivative being at least $\big(h/\pi - 0.22737 - 2/(1+\ell)\big)/(\theta+h+2)$ with $\ell = \ell(\theta+h) > 16$, which is positive. At $\theta = H_0 - \frac{1+\sqrt{2}}{2}$, where $qH_0 = 10^8$ for both characters, $\log(q(\theta-h)/(2\pi)) = 16.58280\ldots$ and $E(\theta+h) = 9.00428\ldots$, so that $n \geq 79.177 - 9.005 > 70$ for $\chi_4$ and $n \geq 52.785 - 9.005 > 43$ for $\chi_{12}$, well above the 36 and 34 zeros needed. The script `dbeta_window.py` checks these inequalities and the hypotheses of the formula in ball arithmetic.

(vi) Under RH, or GRH, the hypothesis of Theorem 9.1.15 holds at every $\theta$, and the proof of (ii) applies with $H_0 = \infty$ for every $\alpha < 1 < \alpha_{\max}$. Conversely, the condition and (i) give $v_\alpha \geq b_\alpha > 0$ on every line $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha < 1$, and $v_\alpha > 0$ for $\alpha \geq 1$ (Theorem 1.2.2, Proposition 8.5.25, and Proposition 11.3.1 with Section 8.5.3); so $\operatorname{Re}(f'/f)(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$, which is RH, or GRH, by Lagarias's criterion [320]. □

For the two $L$ functions, part (v) gives the factorisation of Proposition 9.1.12 at every basepoint $\alpha \geq 1$: for the clock $T_\alpha$ with Laplace transform $f(\alpha)/f(\alpha+\sqrt{u})$, $T_\alpha \overset{d}{=} S_\alpha + T_\alpha^\circ$, where $S_\alpha$ and $T_\alpha^\circ$ are independent, $S_\alpha$ is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, and $T_\alpha^\circ$ is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\frac{1}{2}(\log f)''(\alpha)$. The proof is that of Proposition 9.1.12: the Thorin representation $\log(f(\alpha+\sqrt{u})/f(\alpha)) = \frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\,v_\alpha(\theta)\,d\theta$ holds at every $\alpha \geq 1$ (Proposition 8.5.25 and its proof, which applies to $\xi(\cdot,\chi_{12})$), and $\frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta = \sqrt{u}$.

Part (vi) sharpens Lagarias's criterion, which asks only for $v_\alpha > 0$: RH is equivalent to the Thorin density on every line $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha < 1$ being smallest at the origin and only there, and by (ii) this holds with no hypothesis for $|\theta| \leq 3 \times 10^{12} - 1.21$. In the language of Proposition 9.1.12, $b_\alpha/\pi$ is the density of the positive stable part of the clock, so (vi) says that RH holds exactly when, for every $\alpha > \frac{1}{2}$, $T_\alpha$ splits as $S_\alpha + T_\alpha^\circ$ with $T_\alpha^\circ$ a GGC with no drift; under RH every $T_\alpha$ is a GGC (Theorem 1.2.2), and the proof of the factorisation is

| | $\xi$ | $\xi(\cdot,\chi_4)$ | $\xi(\cdot,\chi_{12})$ |
|---|---|---|---|
| $V_f=(\log f)''(\frac12)$ | 0.04620998623 | 0.1560296499 | 0.3329236082 |
| first ordinate $\gamma_1$ | 14.1347251 | 6.0209489 | 3.8046276 |
| verified height $H_0$ | $3\times10^{12}$ | $2.5\times10^{7}$ | $10^8/12$ |
| comparison line $\alpha'$ of [425] | 4.75 | 3.15 | 2.30 |
| $t_1$ at $\alpha'$ | 9.236 | 2.439 | 1.685 |
| convexity margin $\gamma_1-t_1-(\alpha'-\frac12)/\sqrt3$ | 2.4450 | 2.0520 | 1.0804 |
| $\alpha'$ minimising $t_1$, and $t_1$ there | 4.770, 9.2353 | 3.320, 2.4176 | 2.395, 1.6717 |
| $\alpha_{\max}$ | 7.35 | 4.70 | 3.10 |
| $t_1$ at $\alpha_{\max}$ | 10.124 | 3.574 | 2.245 |
| convexity margin at $\alpha_{\max}$ | 0.0559 | 0.0221 | 0.0585 |
| cells of the proof of (iv) pass from $\sigma_2=$ | | 3.71875 | 2.8125 |

Table 9.4: Constants of Theorem 9.1.18. For a comparison line $\alpha'>1$, $t_1$ is the root of $A_f^{\infty}(\alpha',t)=S_f(\alpha')+(\alpha'-\frac12)V_f$, found by bisection and rounded up to three decimals, at which the strict inequality is certified in ball arithmetic; the proof needs the convexity margin to be positive. The first comparison line is that of the first author's note; the minimiser of $t_1$ is located on a grid of step 0.005; $\alpha_{\max}$ is the largest $\alpha'$ in steps of 0.05 keeping the margin above 0.02. For $\xi$ the range $\alpha>\alpha_{\max}$ is covered by Proposition 9.1.12. Computed by `dbeta_minimum.py`; the ordinates of $L(s,\chi_4)$ and $L(s,\chi_{12})$ are certified by `dbeta_zeros.py`.

that of Proposition 9.1.12. Part (i) also shows that the stable part is present at every basepoint: in the chart $z=\theta^2$ the density $v_\alpha(\sqrt z)/(2\pi\sqrt z)$ behaves like $b_\alpha/(2\pi\sqrt z)$ as $z\downarrow0$, with $b_\alpha>0$ for every $\alpha>\frac12$. For $\xi$, part (iii) gives a second proof of Proposition 9.1.12 for $\frac12+\frac{1}{\sqrt2}\le\alpha\le7.35$ that uses of the zeros only the first ordinate and the verification height, not the zeros below height 2000; for $1\le\alpha<\frac12+\frac{1}{\sqrt2}$ the comparison reaches only the verified height, and there and for $\alpha>7.35$ Proposition 9.1.12 is still needed. For the two $L$ functions no zero beyond the first is used: the zeros enter only through the verification heights of Platt and the counting formula of Bennett, Martin, O'Bryant and Rechnitzer.

Remark 9.1.19 (The margin at the origin). For $\xi$, parts (i) and (ii) explain the margin of Section 3.1. For every $\varepsilon\in(0,6.85]$ the minimum of $\rho_\varepsilon=v_{1/2+\varepsilon}/\pi$ over $0\le\theta\le3\times10^{12}-1.21$ is $\rho_\varepsilon(0)=b_{1/2+\varepsilon}/\pi$, attained only at the origin, and $\rho_\varepsilon(0)\le\varepsilon V_\xi/\pi$, with $V_\xi/\pi=2\pi^{-1}\sum_k a_k^{-1}=0.0147091$, the constant of the computed minimum $0.0147\,\varepsilon$ (Section 3.1 and Table 9.6). By (i), $\rho_\varepsilon(0)/\varepsilon$ increases to $V_\xi/\pi$ as $\varepsilon\downarrow0$.

Remark 9.1.20 (The nearest zero under GRH). Under RH, or GRH, there is a shorter route to the minimum at the origin on a finite range of heights, which the first author's note gives first for $\chi_4$ [425]. Let $0<a\le\frac12$ and assume that every zero of $f$ lies on the critical line. Then $v_{1/2+a}(\theta)=\sum_\gamma a/(a^2+(\theta-\gamma)^2)$ over all ordinates, and $b_{1/2+a}=\sum_\gamma a/(a^2+\gamma^2)\le aV_f$, since $V_f=\sum_\gamma\gamma^{-2}$. If every gap

between consecutive positive ordinates up to $\gamma_N$ is less than $2\sqrt{1/V_f - a^2}$, then $v_{1/2+a}(\theta) > b_{1/2+a}$ for $0 < |\theta| \le \gamma_N$. Indeed, on $|\theta| < \gamma_1 - a/\sqrt{3}$ the sum is even and strictly convex; within $a/\sqrt{3}$ of an ordinate the nearest term alone is at least $3/(4a) > aV_f$, as $a^2 \le \frac{1}{4} < 3/(4V_f)$ for each of the three functions; and otherwise $\theta$ lies between two consecutive ordinates at distance at most half their gap $g$ from one of them, whose term is at least $a/(a^2 + g^2/4)$, which exceeds $aV_f$ exactly when $g < 2\sqrt{1/V_f - a^2}$. For $\chi_4$ the threshold is 4.963 at $a = \frac{1}{2}$ and tends to $2/\sqrt{V_{\chi_4}} = 5.063$ as $a \downarrow 0$, and the largest gap below height 100 is the first, 4.2228 (Section 8.5.7.1), so under GRH the conclusion holds for $|\theta| \le 98.75$. Under GRH the gaps tend to zero, since the argument of $L(\frac{1}{2} + iT, \chi_4)$ is $O(\log T/\log\log T)$ by Littlewood's argument for $\zeta$ [498, Theorem 14.13], so the gap condition can fail only on a bounded range of heights. Theorem 9.1.18 needs no gaps: with no hypothesis its part (ii) gives the conclusion for $0 < |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, and under RH, or GRH, its part (vi) gives it at every height.

Remark 9.1.21 (Complete monotonicity in the resolution). Under RH, or GRH, the ratio of Theorem 9.1.15 is $v_{1/2+\sqrt{w}}(\theta)/\sqrt{w} = \sum_\gamma 1/(w + (\theta - \gamma)^2)$, the sum over all ordinates, a Stieltjes function of $w > 0$ and in particular completely monotone. With no hypothesis the ratio is then a Stieltjes function at no height, and it fails to be completely monotone near every zero off the line. The ratio of a pair is $r_\delta(w, u) = 2\int_0^\infty e^{-wx}e^{-cx}\cos(\kappa x)\,dx$ for $w > -c$, so the representing function of the whole ratio is $\sum e^{-(\theta-\gamma')^2 x}$, over the zeros $\gamma'$ on the line, plus $2e^{-cx}\cos\kappa x$ for each pair. If one pair has $\kappa \ne 0$ and $c = u^2 - \delta^2$ smaller than $(\theta - \gamma')^2$ for every other zero and than the value of $c$ of every other pair, its term dominates the representing function as $x \to \infty$ and changes sign, so by Bernstein's theorem the ratio is not completely monotone at that height; at $u = 0$ the pair contributes the pole $2/(w - \delta^2)$. So, in answer to a question of the first author's note [425], no constant plays for complete monotonicity the role that $1 + \sqrt{2}$ plays in Lemma 9.1.14: whether the ratio is completely monotone at a height near a zero off the line depends on the neighbouring zeros, not only on the distance to that zero in units of $\delta$.

### 9.1.7 The heat trace and the Chebyshev error

The compensated sums of Sections 9.1.3 and 9.1.5 converge only conditionally. At $\alpha = 1$ an explicit bound for Chebyshev's error $\psi(y) - y$ bounds the error of a truncation at $X$ only by an amount that grows linearly with the height $\theta$. Averaging over the height against a Gaussian changes this. For $\alpha \ge 1$ the heat trace of $T_\alpha$ is the Laplace transform $k_\alpha(t) = \pi^{-1}\int_0^\infty e^{-t\theta^2} v_\alpha(\theta)\,d\theta$ of its Thorin measure in the variable $z = \theta^2$. The average turns the kernel $2\sin^2(x\theta/2)$ of the sine squared lemma (1.2.9) into a kernel that is positive, increasing and bounded, and one integration by parts then turns an explicit bound for $\psi(y) - y$ into an error bound that holds for all $t$ at once.

LEMMA 9.1.22. *For $x \ge 0$ and $t > 0$,*

$$\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}\,2\sin^2\Big(\frac{x\theta}{2}\Big)\,d\theta = \frac{g_t(x)}{2\sqrt{\pi t}}, \qquad g_t(x) = 1 - e^{-x^2/4t}.$$

*Proof.* Write $2\sin^2(x\theta/2) = 1-\cos x\theta$ and use $\int_0^\infty e^{-t\theta^2}\cos x\theta\,d\theta = \frac{1}{2}\sqrt{\pi/t}\,e^{-x^2/4t}$. □

The kernel $g_t$ increases from 0 to 1 on $[0,\infty)$, it satisfies $g_t(x) \le \min(1, x^2/4t)$, and for fixed $x$ it decreases in $t$.

PROPOSITION 9.1.23 (Heat kernel form). *For $\alpha \ge 1$ and $t > 0$,*

$$2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \bar{A}_\alpha(t) + \int_{[1,\infty)} g_t(\log y)\,y^{-\alpha}\,d\big(\psi(y) - y\big), \tag{9.1.11}$$

*where the gamma piece*

$$\bar{A}_\alpha(t) = \int_0^\infty g_t(x)\,\frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx = \sum_{k\ge1}\Big(\frac{1}{c_k} - \sqrt{\pi t}\,e^{c_k^2 t}\operatorname{erfc}\big(c_k\sqrt{t}\big)\Big), \qquad c_k = \alpha+2k,$$

*is positive and the Stieltjes integral converges absolutely after one integration by parts. For $\alpha > 1$ that integral is $\sum_n \Lambda(n)n^{-\alpha}g_t(\log n) - L_\alpha(t)$, with the Lebesgue piece*

$$L_\alpha(t) = \int_0^\infty g_t(x)\,e^{-(\alpha-1)x}\,dx = \frac{1}{\alpha-1} - \sqrt{\pi t}\,e^{(\alpha-1)^2 t}\operatorname{erfc}\big((\alpha-1)\sqrt{t}\big).$$

*Proof.* Let $\alpha > 1$ and write $v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2(x\theta/2)\,x\Pi_\alpha(dx)$, as at the end of Section 9.1.4. Near the origin $|x\Pi_\alpha|$ has density about $1/(2x)$, and away from it the gamma and pole densities and the prime atoms have finite mass, so $\int_0^\infty \min(1,x^2)\,|x\Pi_\alpha|(dx) < \infty$. By Lemma 9.1.22 the integral of $e^{-t\theta^2}2\sin^2(x\theta/2)$ against $d\theta\,|x\Pi_\alpha|(dx)$ is $\frac{\sqrt{\pi}}{2\sqrt{t}}\int g_t\,d|x\Pi_\alpha| < \infty$, so the order of integration may be exchanged, and the lemma gives $2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \int_0^\infty g_t(x)\,x\Pi_\alpha(dx)$. The gamma density gives $\bar{A}_\alpha$, whose series follows from $e^{-(\alpha+2)x}/(1-e^{-2x}) = \sum_{k\ge1}e^{-c_k x}$ and $\int_0^\infty g_t(x)e^{-cx}\,dx = \frac{1}{c} - \sqrt{\pi t}\,e^{c^2t}\operatorname{erfc}(c\sqrt{t})$ for $c > 0$, the case $c = \alpha-1$ being $L_\alpha$; the prime atoms and the pole density give the Stieltjes integral, written in the variable $y = e^x$.

At $\alpha = 1$ the same computation applies to $v_1 = b_1 + A + Q$ of (9.1.6), in which $Q(\theta) = \lim_{X\to\infty} Q_X(\theta)$ with $Q_X(\theta) = \int_{[1,X]} 2\sin^2(\frac{1}{2}\theta\log y)\,y^{-1}d(\psi(y)-y)$. Integration by parts gives

$$Q_X(\theta) = w(X)\big(\psi(X)-X\big) - \int_1^X \big(\psi(y)-y\big)\,w'(y)\,dy, \qquad w(y) = \frac{2\sin^2(\frac{1}{2}\theta\log y)}{y},$$

and since $|w'(y)| \le (2+|\theta|)\,y^{-2}$, while $\int_1^\infty |\psi(y)-y|\,y^{-2}dy < \infty$ by the prime number theorem with the de la Vallée Poussin error term, $|Q_X(\theta)| \le C(1+|\theta|)$ for all $X$. By dominated convergence the limit in $X$ may therefore be taken outside the integral over $\theta$, and for each $X$ Lemma 9.1.22, applied to the finitely many atoms and to Lebesgue measure on $[0,\log X]$, gives $\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}Q_X(\theta)\,d\theta = (2\sqrt{\pi t})^{-1}\int_{[1,X]} g_t(\log y)\,y^{-1}d(\psi(y)-y)$; the gamma piece $A$ gives $\bar{A}_1$ as before. For every $\alpha \ge 1$ the same integration by parts, now with $|\frac{d}{dy}(g_t(\log y)\,y^{-\alpha})| \le (g_t'(\log y)+\alpha)\,y^{-\alpha-1}$ and $g_t'$ bounded, shows that the Stieltjes integral converges absolutely after one integration by parts, the boundary terms vanishing because $g_t(0)=0$ and $y^{-\alpha}(\psi(y)-y)\to 0$. □

For $\alpha > 1$ every prime power enters (9.1.11) with the positive weight $\Lambda(n)n^{-\alpha}g_t(\log n)$, and the other terms are explicit. Keeping the primes up to $X$ and dropping the rest therefore gives a lower bound with no error term,

$$2\sqrt{\pi t}\,k_\alpha(t) \ge b_\alpha + \bar{A}_\alpha(t) + \sum_{n\le X}\frac{\Lambda(n)}{n^\alpha}\,g_t(\log n) - L_\alpha(t), \qquad \alpha>1, \tag{9.1.12}$$

which uses only $\Lambda(n)\ge 0$, the positivity of the coefficients of $-\zeta'/\zeta$. By Table 9.5, the right side is positive for every $t$ once $\alpha \ge 3.191$ with no prime atoms, the primes then entering only through the drift $b_\alpha$, and once $\alpha \ge 1.533$, 1.367 and 1.284 with the primes up to $10^3$, $10^5$ and $10^7$; the thresholds, to four decimals, are 3.1900, 1.5320, 1.3667 and 1.2840. The same argument pointwise in $\theta$, applied to the sine squared form of $v_\alpha$ with no prime atoms, needs $\alpha \ge 3.575$ (the threshold is 3.5748). At $\alpha$ equal to the four decimal value less $5\times 10^{-5}$ the right side is negative at one $t$, or one $\theta$. From the smaller of the four decimal value plus $5\times 10^{-5}$ and the bound in the table up to $\alpha = 10$, cells cover every $\alpha$ and every $t>0$, or $\theta \ge 0$, and the right side is positive on each cell: $\bar{A}_\alpha(t)$, the prime sum and $L_\alpha(t)$ are monotone in $\alpha$ and in $t$, so each is bounded at a corner of the cell or, near the minimum over $t$, by a mean value form in $t$, and $b_\alpha$ by its mean value form in $\alpha$. No finite $X$ reaches $\alpha = 1$: for fixed $t$, $L_\alpha(t)$ increases to $\int_0^\infty g_t(x)\,dx = \infty$ as $\alpha \downarrow 1$, while the other terms of (9.1.12) stay bounded.

At the basepoint one the primes and the pole must be kept together. Splitting the Stieltjes integral at $X$, and writing $\int_0^{\log X} g_t(x)\,dx = \log X - \sqrt{\pi t}\,\mathrm{erf}\,\big(\log X/(2\sqrt{t})\big)$,

$$2\sqrt{\pi t}\,k_1(t) = b_1 + \bar{A}_1(t) + \sum_{n\le X}\frac{\Lambda(n)}{n}\,g_t(\log n) - \Big(\log X - \sqrt{\pi t}\,\mathrm{erf}\,\frac{\log X}{2\sqrt{t}}\Big) + R_X(t), \tag{9.1.13}$$

with $R_X(t) = \int_{(X,\infty)} g_t(\log y)\,y^{-1}d(\psi(y)-y)$. Suppose that $|\psi(y)-y| \le \omega(\log y)\,y$ for $y \ge X$. Integrating by parts, with $|\frac{d}{dy}(g_t(\log y)/y)| \le (g_t'(\log y)+g_t(\log y))\,y^{-2}$,

$g_t \le 1$ and $\int_{\log X}^{\infty} g_t' = 1 - g_t(\log X)$,

$$|R_X(t)| \le g_t(\log X)\,\frac{|\psi(X)-X|}{X} + \bigl(1-g_t(\log X)\bigr)\sup_{x\ge \log X}\omega(x) + \int_{\log X}^{\infty}\omega(x)\,dx \le \eta_X \tag{9.1.14}$$

for every $t>0$, where $\eta_X = \sup_{x\ge\log X}\omega(x) + \int_{\log X}^{\infty}\omega(x)\,dx$, the last step because $|\psi(X)-X| \le \omega(\log X)X$. Explicit forms of the prime number theorem supply $\omega$: $|\psi(y)-y| \le 0.94\sqrt{y}$ for $11 < y \le 10^{19}$ [68]; $|\psi(y)-y| < 1.93378\times 10^{-8}\,y$ for $y \ge 10^{19}$ and $|\psi(y)-y| < 6.95\times 10^{-9}\,y$ for $y \ge e^{46}$ [65]; and $|\psi(y)-y| \le 8.86\,y(\log y)^{1.514}e^{-0.8288\sqrt{\log y}}$ for $\log y \ge 3000$ [291]. The constants alone would not do, since $\omega$ must be integrable, and the decaying bound of Johnston and Yang closes the integral. With $X = 10^7$ they give $\eta_X < 9.2\times 10^{-4}$.

Proposition 9.1.24. *For every $t>0$, $2\sqrt{\pi t}\,k_1(t) > 0.021$. The proof uses the primes up to $10^7$, the bounds for $\psi(y)-y$ quoted above and a computation in ball arithmetic.*

*Proof.* By (9.1.13) and (9.1.14) it suffices that the right side of (9.1.13), with $R_X(t)$ replaced by $-\eta_X$, exceed 0.021 for every $t$. The three terms of it that depend on $t$, the gamma piece, the prime sum and the bracket, all decrease in $t$, because $g_t$ does, so on a cell $[t_0,t_1]$ the value obtained by evaluating the first two at $t_1$ and the bracket at $t_0$ is a lower bound. The script `heat_primes.py` evaluates it in ball arithmetic with Arb [290], on 65137 cells that cover $[10^{-10},10^8]$ and are refined until each bound exceeds 0.021. It bounds $\bar{A}_1$ from below by the terms of its series with $k \le 1000$ and $c_k\sqrt{t} \le 25$, all terms being positive, and the prime sum from below by grouping the prime powers above $10^4$ into blocks of width $5\times 10^{-4}$ in $\log n$, each weighted by $g_t$ at its left end. For $t \le 10^{-10}$ the same monotonicity, with the bracket at most $\log X$, gives the lower bound 2.86, and for $t \ge 10^8$ dropping $\bar{A}_1$ and the prime sum gives 0.0221. □

From the zero side, Proposition 9.1.11 gives $v_1 \ge b_1$, hence $2\sqrt{\pi t}\,k_1(t) > b_1$ for every $t$, where $b_1 = 0.0230957$. So Proposition 9.1.24 is weaker than what the zeros give. Its interest is its input, the primes up to $10^7$ and published bounds for $\psi(y)-y$, with no zero computed here. It does not avoid the zeros altogether, since those bounds are themselves proved from them: from the verification of RH to a finite height, now $3\times 10^{12}$ [419], and from explicit zero free regions and zero density estimates.

Below the basepoint one this method stops, and the reason is Proposition 9.1.13 again. For $\frac12 < \alpha < 1$ the integration by parts bounds the tail by a boundary term plus $\int_{\log X}^{\infty}\omega(x)\,e^{(1-\alpha)x}(g_t'(x)+\alpha g_t(x))\,dx$, and since $g_t(x)\to 1$ as $x\to\infty$ for every $t$, this is finite only if $\int^{\infty}\omega(x)e^{(1-\alpha)x}\,dx < \infty$, that is, only if $\omega$ decays

| input from the primes | positivity | basepoints |
|---|---|---|
| $\Lambda(n) \geq 0$ | $v_\alpha(\theta) > 0$ for every $\theta$ | $\alpha \geq 3.575$ |
| $\Lambda(n) \geq 0$ | $k_\alpha(t) > 0$ for every $t$ | $\alpha \geq 3.191$ |
| primes up to $10^3$ | $k_\alpha(t) > 0$ for every $t$ | $\alpha \geq 1.533$ |
| primes up to $10^5$ | $k_\alpha(t) > 0$ for every $t$ | $\alpha \geq 1.367$ |
| primes up to $10^7$ | $k_\alpha(t) > 0$ for every $t$ | $\alpha \geq 1.284$ |
| primes up to $10^7$ and bounds for $\psi(y) - y$ | $2\sqrt{\pi t}\, k_1(t) > 0.021$ for every $t$ | $\alpha = 1$ |

Table 9.5: What the primes show about the Thorin density and the heat trace of $T_\alpha$, with no information on the zeros beyond what the published bounds for $\psi(y) - y$ contain. The first five rows are proved in ball arithmetic by `heat_thresholds.py` on cells covering every $\alpha$ from the bound shown to 10 and every time, or height; for $\alpha \geq 10$ positivity is immediate: $b_\alpha = \mathbb{E}V_\alpha$ increases with $\alpha$, its derivative being $\operatorname{Var} V_\alpha$, $b_{10} = 0.391$, and the pole term is at most $\frac{1}{\alpha-1} \leq \frac{1}{9}$ in both forms. The last row is Proposition 9.1.24, also in ball arithmetic. Below $\alpha = 1$ this route needs an explicit quasi Riemann hypothesis as input.

exponentially. Every explicit unconditional bound decays more slowly, like $e^{-c\sqrt{x}}$ from the zero free region of de la Vallée Poussin or like $e^{-cx^{3/5}(\log x)^{-1/5}}$ from that of Vinogradov and Korobov, and so do the published bounds above beyond their finite ranges. The half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 (Section 12.1.2) would give $\psi(y) - y = O(y^{7/8+\delta})$ for every $\delta > 0$ [498], enough in principle for every $\alpha > \frac{7}{8}$; its constant has not been made explicit, and an explicit one would let this method pass a little below the basepoint one. A bound with $\int^\infty \omega(x) e^{(1-\alpha)x}\, dx < \infty$ is itself a quasi Riemann hypothesis: it makes $\int_1^\infty y^{-w}\, d(\psi(y) - y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ converge for $\operatorname{Re} w \geq \alpha$ after one integration by parts, so it forces $\Theta \leq \alpha$, and it follows from $\Theta < \alpha$. Along this route the step below the basepoint one therefore needs a quasi Riemann hypothesis as input, and reaching every $\alpha > \frac{1}{2}$ needs RH itself. The obstruction belongs to the route. Whether $k_\alpha > 0$ just below $\alpha = 1$ needs any hypothesis is not settled by it, since $k_\alpha > 0$ is weaker than $v_\alpha \geq 0$ and only zeros with real part above $\alpha$ contribute negatively to $v_\alpha$.

Remark 9.1.25 (The centre). At $\alpha = \frac{1}{2}$ the prime sum and the Lebesgue piece of (9.1.11) diverge separately, but the identity survives with a finite part, with no hypothesis [253]. Let $W(t) = \sum_{\operatorname{Re}\gamma > 0} e^{-t\gamma^2}$ and

$$\mathcal{F}(\tfrac{1}{2}) = 2 - \frac{\zeta'}{\zeta}(\tfrac{1}{2}) = 2 - \tfrac{1}{2}\log\pi + \tfrac{1}{2}\frac{\Gamma'}{\Gamma}(\tfrac{1}{4}) = -0.686091709613,$$

the value at $s = \frac{1}{2}$ of the continuation of $\int_1^\infty y^{-s}\, d(\psi(y) - y) = -\frac{\zeta'}{\zeta}(s) - \frac{1}{s-1}$. Then for every $t > 0$

$$2\sqrt{\pi t}\, W(t) = \bar{A}_{1/2}(t) + \mathcal{F}(\tfrac{1}{2}) - \int_{[1,\infty)} e^{-(\log y)^2/4t}\, y^{-1/2}\, d\big(\psi(y) - y\big),$$

with $\bar{A}_{1/2}$ the gamma piece of Proposition 9.1.23 at $\alpha = \frac{1}{2}$, and the integral converges absolutely. The kernel is $1 - g_t(\log y)$ instead of $g_t(\log y)$ because the constant $\int_{[1,\infty)} y^{-1/2} d(\psi(y) - y)$, which diverges, has been replaced by $\mathcal{F}(\frac{1}{2})$. With a sharp cutoff, $\int_{[1,X]} y^{-1/2} d(\psi(y) - y)$ has no limit as $X \to \infty$, since a limit would give $\psi(X) - X = o(\sqrt{X})$, which the pole of $\frac{\zeta'}{\zeta}$ at $\frac{1}{2} + i\gamma_1$ excludes; its Gaussian means do converge: as $t \to \infty$, $W(t) \to 0$ exponentially and $\bar{A}_{1/2}(t) = O(1/t)$, so the integral above tends to $\mathcal{F}(\frac{1}{2})$. By Theorem 5.1.1 the left side is positive for every $t$, so the Gaussian smoothed Chebyshev error at the centre stays below $\bar{A}_{1/2}(t) + \mathcal{F}(\frac{1}{2})$ at every time, unconditionally.

*Proof.* The arithmetic form (1.2.5) of $W$, multiplied by $2\sqrt{\pi t}$, reads

$$2\sqrt{\pi t}\, W(t) = 2\sqrt{\pi t}\, e^{t/4} - \tfrac{1}{2}\log\pi + \sqrt{\tfrac{t}{\pi}} \int_0^\infty e^{-tr^2} \operatorname{Re} \frac{\Gamma'}{\Gamma}\Big(\frac{1}{4} + \frac{ir}{2}\Big) dr - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4t}.$$

Completing the square in $\int_{\mathbb{R}} e^{x/2 - x^2/4t} dx$ splits the polar term as

$$2\sqrt{\pi t}\, e^{t/4} = \int_{[1,\infty)} e^{-(\log y)^2/4t} y^{-1/2} dy + \sqrt{\pi t}\, e^{t/4} \operatorname{erfc}\big(\tfrac{\sqrt{t}}{2}\big),$$

and the first piece compensates the prime sum. By Lemma 9.1.22,

$$\bar{A}_{1/2}(t) = 2\sqrt{\pi t}\, \pi^{-1} \int_0^\infty e^{-tr^2} \sum_{k \geq 1} q_{\frac{1}{2}+2k}(r)\, dr,$$

and

$$\sum_{k \geq 1} q_{\frac{1}{2}+2k}(r) = \frac{1}{2}\Big[\operatorname{Re} \frac{\Gamma'}{\Gamma}\Big(\frac{1}{4} + \frac{ir}{2}\Big) - \frac{\Gamma'}{\Gamma}\Big(\frac{1}{4}\Big)\Big] - 2 + \frac{1/2}{1/4 + r^2}.$$

With $\int_0^\infty e^{-tr^2} \frac{1/2}{1/4+r^2} dr = \frac{\pi}{2} e^{t/4} \operatorname{erfc}(\frac{\sqrt{t}}{2})$ and $\int_0^\infty e^{-tr^2} dr = \frac{1}{2}\sqrt{\pi/t}$, this gives

$$\bar{A}_{1/2}(t) + \mathcal{F}(\tfrac{1}{2}) = \sqrt{\tfrac{t}{\pi}} \int_0^\infty e^{-tr^2} \operatorname{Re} \frac{\Gamma'}{\Gamma}(\tfrac{1}{4} + \tfrac{ir}{2})\, dr + \sqrt{\pi t}\, e^{t/4} \operatorname{erfc}(\tfrac{\sqrt{t}}{2}) - \tfrac{1}{2}\log\pi,$$

because $-\frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{1}{4}) - 2 + \mathcal{F}(\frac{1}{2}) = -\frac{1}{2}\log\pi$, and the identity follows. The two forms of $\mathcal{F}(\frac{1}{2})$ agree because $\frac{\xi'}{\xi}(\frac{1}{2}) = 0$. For the sharp cutoff: if $I(X) = \int_{[1,X]} y^{-1/2} d(\psi(y) - y)$ converged, partial summation would give $\psi(X) - X = o(\sqrt{X})$, and then $\varepsilon \int_1^\infty y^{-s} d(\psi(y) - y) \to 0$ at $s = \frac{1}{2} + \varepsilon + i\gamma_1$ as $\varepsilon \downarrow 0$, whereas $-\frac{\zeta'}{\zeta}(s) - \frac{1}{s-1}$ times $\varepsilon$ tends to minus the multiplicity of the zero. Finally $\bar{A}_{1/2}(t) \le \int_0^\infty \min(1, \frac{x^2}{4t}) \frac{e^{-5x/2}}{1-e^{-2x}}\, dx = O(1/t)$. □

At $t = 0.01, 0.025, 0.125, 0.5$ and $2$ the two sides agree to $3 \times 10^{-15}$ (computed, `code/rx_centre.py`, primes up to $10^7$ and the zeros of `data/zeros_1700.txt`). For $t \ge 0.5$ the left side is below $10^{-40}$ and the prime sum is in double precision, so the residual there is the rounding of the prime sum. The compensator $\int_{[1,\infty)} e^{-(\log y)^2/4t} y^{-1/2} dy$ and the prime sum both grow like $2\sqrt{\pi t}\, e^{t/4}$, while their difference stays bounded.

## 9.2 Hitting times at the basepoints

The third question, whether $T_1$ is a hitting time, is probabilistic.

We keep the notation of Section 9.1: $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of $T_\alpha$ whenever that variable exists, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$, and $\rho_{1/2} = v_1/\pi$ is the Thorin density of $T_1$ in the chart $\theta = \sqrt{z}$. In addition $B$ is a standard Brownian motion, $H_a^{(c)}$ is the first passage time of $B_t + ct$ to the level $a \ge 0$, and $H_a = H_a^{(0)}$; this $H$ has nothing to do with the de Bruijn Newman flow $H_t$ of Section 8.4.

### 9.2.1 Kent's theorem and Brownian first passage to a random level

The first candidate is the hitting time of a level by a diffusion. By Kent's theorem [305] and Krein's inverse spectral theory [124], recalled in Section 8.1.3, the first hitting times of levels by diffusions started at a reflecting or entrance end are exactly the GGCs without drift whose Thorin measures consist of unit atoms at distinct points, the Dirichlet eigenvalues of the interval up to the level. The Thorin measure of $\varphi_1$ is absolutely continuous, so $T_1$ is not such a hitting time. It is the clock at the centre that has the discrete form: under RH its Thorin measure has an atom at each $\gamma^2$, of mass the multiplicity of the zero, so that it is such a hitting time exactly when the zeros are also simple. Brownian motion offers a second candidate, a first passage to a random level, and Proposition 8.2.1 excludes it for the clock.

Proposition 9.2.1. *For no drift $c \ge 0$ and no random level $L \ge 0$ independent of the Brownian motion is $T_1$ equal in law to the first passage time $H_L^{(c)}$. (For $c < 0$ the first passage time is infinite with positive probability, so equality is impossible.)*

*Proof.* Conditionally on $L$, the Laplace transform of $H_L^{(c)}$ is $\exp\{-L(\sqrt{c^2+2u}-c)\}$, so $\mathbb{E}e^{-uH_L^{(c)}} = \mathbb{E}\exp\{-L(\sqrt{c^2+2u}-c)\}$ extends analytically, and bounded by one, to the region $R_c = \{u : \operatorname{Re}\sqrt{c^2+2u} > c\}$, which contains the right half plane and lies inside $\mathbb{C}\setminus(-\infty,0]$. The function $\varphi_1$ is analytic there as well, by the proof of Proposition 9.1.2, so if the two agreed on $\operatorname{Re} u > 0$ they would agree on $R_c$, and $|\varphi_1| \le 1$ on $R_c$ and on its boundary, the parabola $u_q = \frac{1}{2}(-q^2+2icq)$, $q \in \mathbb{R}$, on which $\sqrt{c^2+2u_q} = c+iq$, as in the proof of Proposition 8.2.1. On that parabola $\operatorname{Re}\sqrt{u_q}$ stays bounded while $\operatorname{Im}\sqrt{u_q} \sim q/\sqrt{2}$, so by Stirling's formula $|\xi(1+\sqrt{u_q})|$ decays exponentially and $|\varphi_1(u_q)| \to \infty$. For $c = 0$ the parabola is the cut itself and $|\varphi_1(-t^2)| = |\xi(1)/\xi(1+it)|$, which equals 13.0, $1.2\times10^4$, $2.0\times10^{10}$ and $1.5\times10^{23}$ at $t = 10, 20, 40, 80$, as computed by `compute.py`. □

A subordinator with a drift as well as a stable one half part escapes this obstruction: on the cut, where $\sqrt{u} = it$, its exponent $u+\sqrt{u} = -t^2+it$ has negative real part, so mixtures of its transforms can grow.

### 9.2.2 Subordination of the centre clock

Let $T$ be the clock variable of Section 1.2, with $\mathbb{E}e^{-uT} = \varphi_{1/2}(u) = 1/G(u)$, a positive random variable that exists with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000); it is the clock of [421], and we call it the centre clock.

Theorem 9.2.2. *Let $\widetilde{T}$ have the tilted law $P(\widetilde{T} \in dt) = e^{-t/4}P(T \in dt)/\mathbb{E}e^{-T/4}$, and let $(\sigma_\ell)_{\ell\ge0}$ be an independent subordinator with Laplace exponent $u+\sqrt{u}$. Then*

$$T_1 \overset{d}{=} \sigma_{\widetilde{T}} \overset{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}},$$

*where $H_a$ is the first passage time of a standard Brownian motion to the level $a$, independent of $\widetilde{T}$.*

*Proof.* For $\operatorname{Re}\sqrt{u} \ge 0$ the number $\frac{1}{2}+\sqrt{u}$ has positive real part and $(\frac{1}{2}+\sqrt{u})^2 = \frac{1}{4}+u+\sqrt{u}$, so $\xi(1+\sqrt{u}) = \xi\big(\frac{1}{2}+\sqrt{\frac{1}{4}+u+\sqrt{u}}\big)$ with the principal root. Hence

$$\begin{aligned}\varphi_1(u) &= \frac{\xi(1)}{\xi(1+\sqrt{u})} = \frac{\varphi_{1/2}(\frac{1}{4}+u+\sqrt{u})}{\varphi_{1/2}(\frac{1}{4})}\\ &= \frac{\mathbb{E}\big[e^{-T/4}e^{-(u+\sqrt{u})T}\big]}{\mathbb{E}e^{-T/4}} = \mathbb{E}e^{-(u+\sqrt{u})\widetilde{T}} = \mathbb{E}e^{-u\sigma_{\widetilde{T}}}.\end{aligned}$$

Finally $\sigma_\ell \overset{d}{=} \ell + H_{\ell/\sqrt{2}}$, because $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$. □

The subordinator $\sigma$ has a classical description: a subordinator with exponent $\kappa_1 u + \kappa_2\sqrt{2u}$ is the inverse local time at zero of a Brownian motion that is sticky at zero, the drift $\kappa_1$ measuring the time spent at zero per unit of local time [287,

Section 5.7, p. 187]; here $\kappa_1 = 1$ and $\kappa_2 = 1/\sqrt{2}$. So $T_1$ is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$.

Corollary 9.2.3. *Assume RH and that the zeros are simple. Then $\widetilde{T}$ is Kent's hitting time for a Krein string whose Dirichlet spectrum is $\{\gamma^2 + \frac{1}{4}\} = \{|\rho|^2\}$, and $T_1$ is the first hitting time of the level $y_*$ by the process $D(\ell_t)$, where $D$ is that string diffusion started at its left end, $y_*$ is the level whose hitting time by $D$ is $\widetilde{T}$, and $\ell_t$ is the local time at zero of an independent sticky Brownian motion $B$. Equivalently, $T_1$ is the first hitting time of the line $\{y = y_*\}$ by the two dimensional Markov process $(D(\ell_t), B_t)$.*

*Proof.* Under RH, $\mathbb{E}e^{-u\widetilde{T}} = \varphi_{1/2}(\frac{1}{4}+u)/\varphi_{1/2}(\frac{1}{4}) = \prod_{\gamma>0}(\gamma^2+\frac{1}{4})/(\gamma^2+\frac{1}{4}+u)$, the product running over the ordinates $\gamma > 0$ repeated according to multiplicity. Since the zeros are simple, this is a GGC without drift with unit atoms at the distinct points $|\rho|^2$, so Kent's theorem and Krein's inverse theory provide the string. The first time $D(\ell_t)$ reaches $y_*$ is the first $t$ with $\ell_t \geq \widetilde{T}$, which is $\sigma_{\widetilde{T}}$. □

The tail (9.1.3) follows from the theorem: $P(H_a > x) \sim a\sqrt{2/(\pi x)}$, so $P(T_1 > x) \sim \mathbb{E}[\widetilde{T}]/\sqrt{\pi x}$, and $\mathbb{E}\widetilde{T} = -\frac{d}{du}\log\varphi_{1/2}(\frac{1}{4}+u)\big|_{u=0} = \frac{\xi'}{\xi}(1) = \sum_\rho \rho^{-1}$.

### 9.2.3 The density of $T_1$ and harmonic measure

Conditioning on $\widetilde{T}$ in Theorem 9.2.2 and using the density of $H_a$ gives the density of $T_1$ as an expectation over the tilted centre clock,

$$f_{T_1}(x) = \mathbb{E}\Big[\frac{\widetilde{T}}{2\sqrt{\pi}}\,(x-\widetilde{T})^{-3/2}\exp\Big(-\frac{\widetilde{T}^2}{4(x-\widetilde{T})}\Big)\,\mathbf{1}\{\widetilde{T} < x\}\Big], \tag{9.2.1}$$

since $H_a$ has density $a(2\pi h^3)^{-1/2}e^{-a^2/2h}$ and $a = \widetilde{T}/\sqrt{2}$. Figure 9.4 compares (9.2.1), averaged over $4 \times 10^5$ draws of $\widetilde{T}$, against numerical inversion of $\varphi_1$ by Talbot's method [490] and shows the whole density. At the points computed the two agree to about 0.1%, except near the mode, at $x = 0.024$, where the Monte Carlo average is within 1%. For large $x$ the density behaves like $\frac{1}{2\sqrt{\pi}}\sum_\rho \rho^{-1}\,x^{-3/2} = 0.00652\,x^{-3/2}$, the derivative of (9.1.3). Near the origin the density is super flat, inherited from the centre clock through $\widetilde{T}$.

The Cauchy kernels of Proposition 9.1.2 and the stable one half subordinator of Theorem 9.2.2 are both laws of planar Brownian motion at the moment it first reaches a line. Let $B = (B^1, B^2)$ be a planar Brownian motion started at a zero $\rho = \beta + i\gamma$, identified with the point $(\beta, \gamma)$, and let $H$ be the first time $B^1$ reaches the line $\operatorname{Re} s = 1$, a distance $1 - \beta$ away, that is, the first passage time $H_{1-\beta}$ of the one dimensional Brownian motion $B^1 - \beta$. Then $B^2_H$ has the Cauchy law with

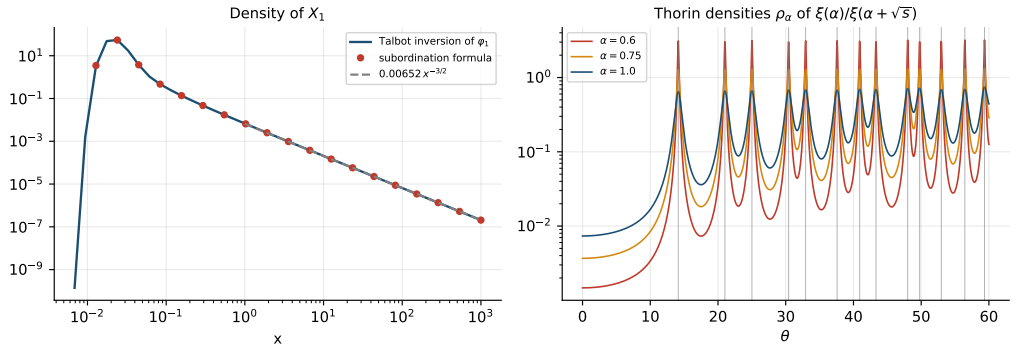



Figure 9.4: Left: the density of $T_1$ by Talbot inversion of $\varphi_1$ and by the subordination formula (9.2.1), with the tail $0.00652\,x^{-3/2}$. Right: the densities $v_\alpha/\pi$ at the basepoints $\alpha = 0.6$, 0.75 and 1, with the ordinates of the zeros; they sharpen onto the zeros as $\alpha$ decreases, and for $\alpha < 1$ they are Thorin densities of $T_\alpha$ only if no zero lies to the right of the line $\operatorname{Re} s = \alpha$. The plots write $X_1$ for $T_1$, $\rho_\alpha$ for $v_\alpha/\pi$ and $s$ for $u$. Drawn by `figs2.py`.

centre $\gamma$ and width $1-\beta$, which is the harmonic measure of the half plane, and $H$ has Laplace transform $e^{-(1-\beta)\sqrt{2u}}$, the stable one half law. So, exactly,

$$\rho_{1/2}(\theta)\,d\theta = \sum_\rho P\big(B_H^2 \in d\theta \mid B_0 = \rho\big).$$

The times $H$ have the same stable one half law, up to scale, as the stable part of the subordinator $\sigma$ in Theorem 9.2.2; the correspondence is of type rather than of scale, since in the theorem the stable part runs per unit of local time $\widetilde{T}$ rather than per unit of distance.

### 9.2.4 The exact Brownian boundary and Wald's identity

The inverse first passage question asks for a boundary $b$ such that the first time $H_b = \inf\{t > 0 : B_t \geq b(t)\}$ at which a standard Brownian motion started at zero crosses $b$ has the law of $T_1$. A deterministic curved boundary always exists. Anulova proved this for two sided symmetric boundaries [7], and Chen, Cheng, Chadam and Saunders for one sided boundaries when the survival function is continuous and positive [79, Theorem 1]. By the theorems of Ekström and Janson [129, Theorems 2.3, 8.2 and 9.2], every law on $(0,\infty)$ is the law of $H_b$ for a lower semicontinuous boundary $b$, which is continuous on every interval where the density of the law is bounded below by a positive constant; when that holds on every compact subinterval of $(0,\infty)$, $b$ is the unique continuous solution of the Volterra equation (8.2.1) with $\liminf_{t\downarrow 0} b(t) \geq 0$. The density (9.2.1) of $T_1$ is continuous and positive on $(0,\infty)$, so the boundary of $T_1$ is continuous there, and

it can be computed from the law of $T_1$ alone through (8.2.1), with $f$ the density of $T_1$, used in the complementary form of Section 8.2.2.2, with $P(T_1 > t)$ in place of $P(T > t)$, once the survival probability falls below one half; `rs1.py` solves it. We computed the distribution function of $T_1$ by Talbot inversion of $\varphi_1(u)/u$ for $t < 0.03$, where $T_1$ is super flat, and by the subordination representation of Theorem 9.2.2 beyond, in the form

$$P(T_1 > t) = \mathbb{E}\Big[\mathbf{1}\{\widetilde{T} \ge t\} + \mathbf{1}\{\widetilde{T} < t\}\Big(2\mathcal{N}\Big(\frac{\widetilde{T}}{\sqrt{2(t-\widetilde{T})}}\Big) - 1\Big)\Big],$$

with $\mathcal{N}$ the standard normal distribution function, as in Section 8.2.2.2. This form is smooth and gives the survival function to relative accuracy even at $t = 100$, where $\sqrt{t}\,P(T_1 > t) = 0.013037$ against the limit 0.013030. As at the centre, the boundary is computed over the three zones of Section 8.2.2.2.

In the small time zone the law of $T_1$ near the origin is that of the tilted centre clock, and the boundary starts from the reflection level 0.505 at $t = 0.009$. In the bulk, from $t = 0.011$ to $t = 0.05$, the boundary falls steeply, to 0.363 at $t = 0.012$ and 0.111 at $t = 0.02$. Here the tangent approximation, taken in its density form, equates the first passage density of the tangent line with the density of $T_1$, where the boundary of Roberts and Shortland [462] in Section 8.2.2.1 equates hazards. Its solution, the boundary $b$ with $(b - tb')(2\pi t^3)^{-1/2}e^{-b^2/2t} = f(t)$ computed by `rs1c.py`, agrees with the exact boundary to $10^{-4}$ at $t = 0.012$ and to $1.5 \times 10^{-3}$ at $t = 0.02$.

The right tail is where the basepoint one differs completely from the centre. At the centre the survival function is exponential and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1$ (Section 8.2.2.2). At the basepoint one the survival function is the stable one half tail $0.01303\,t^{-1/2}$, and a Brownian motion survives that long only below a boundary that stops moving. The exact boundary flattens at the level $b_\infty = -0.19277$: it stays within $10^{-3}$ of this level from $t = 0.056$ on and within $10^{-4}$ from $t = 1.2$ on, up to the end of the computation at $t = 100$; that $b_\infty = \lim_{t\to\infty} b(t)$ is observed, not proved. The tangent approximation fails here: it overshoots downward, to $-0.332$ at $t = 0.05$ against the exact $-0.188$, and has no solution beyond $t = 0.091$. It treats the crossing density at time $t$ as if the path had just started, so a flat boundary at level $a$ would have to produce the density $a\,(2\pi)^{-1/2}t^{-3/2}$, forcing $a = \sqrt{2\pi} \times 0.00652 = 0.0163 > 0$; but by the time the tail is reached the tangent boundary is already negative, and a negative linear boundary can only produce an exponential tail.

The positive number 0.0163 does appear in the exact picture, in a different role. Optional stopping for the martingale $B_{t\wedge H_b}$ gives $\mathbb{E}[b(H_b)\mathbf{1}\{H_b \le t\}] = -\mathbb{E}[B_t\mathbf{1}\{H_b > t\}]$, and once the boundary is flat, a survivor at depth $y$ below it survives a further time $r$ with probability about $y\sqrt{2/(\pi r)}$. The next lemma makes the matching of these two facts exact. For a continuous monotone boundary

that is positive at 0 the limit is Theorem 2 of Novikov [388]; the lemma assumes boundedness in place of monotonicity.

LEMMA 9.2.4. *Let $b$ be continuous on $(0,\infty)$ and bounded on $[1,\infty)$, and let $H_b = \inf\{t > 0 : B_t \geq b(t)\}$ be positive almost surely. Then $b(H_b) = B_{H_b}$ is integrable and $\sqrt{t}\,P(H_b > t) \to \sqrt{2/\pi}\,\mathbb{E}\,b(H_b)$ as $t \to \infty$.*

*Proof.* Since $b$ is bounded above on $[1,\infty)$ and $\limsup_t B_t = \infty$, $H_b$ is finite, and $B_{H_b} = b(H_b)$ by continuity. Choose $u_0 > 0$ with $P(H_b > u_0) > 0$; on that event $b > B$ on $(0,u_0]$, so $b$, which is also continuous on $[u_0,1]$ and bounded on $[1,\infty)$, is bounded below on $(0,\infty)$, say by $-K$. Optional stopping gives $A(s) = \mathbb{E}[b(H_b)\mathbf{1}\{H_b \leq s\}] = -\mathbb{E}[B_s\mathbf{1}\{H_b > s\}]$. Fix $s \geq 1$, let $c_1$ and $c_2$ be the infimum and the supremum of $b$ on $[s,\infty)$, and let $g(y,\tau) = (2\mathcal{N}(y/\sqrt{\tau}) - 1)^+$, the probability that a Brownian motion started at zero stays below the level $y$ up to time $\tau$. By the Markov property at time $s$, on $\{H_b > s\}$ the conditional probability of $\{H_b > t\}$ given the path up to time $s$ lies between $g(c_1 - B_s, t-s)$ and $g(c_2 - B_s, t-s)$, and $c_2 - B_s > 0$ there. Since $\sqrt{t}\,g(y,t-s) \to \sqrt{2/\pi}\,y^+$ as $t \to \infty$, and $\sqrt{t}\,g(y,t-s) \leq 2y^+$ for $t \geq 2s$, dominated convergence and $y^+ \geq y$ give

$$\liminf_{t\to\infty} \sqrt{t}\,P(H_b > t) \geq \sqrt{\tfrac{2}{\pi}}\big(c_1 P(H_b > s) + A(s)\big),$$
$$\limsup_{t\to\infty} \sqrt{t}\,P(H_b > t) \leq \sqrt{\tfrac{2}{\pi}}\big(c_2 P(H_b > s) + A(s)\big).$$

The upper bound at $s = 1$ and the lower bound at every $s$ bound $A(s)$ uniformly, so, since $b(H_b) \geq -K$, monotone convergence makes $b(H_b)$ integrable with $A(s) \to \mathbb{E}\,b(H_b)$. As $s \to \infty$, $c_1$ and $c_2$ stay bounded and $P(H_b > s) \to 0$, so both bounds tend to $\sqrt{2/\pi}\,\mathbb{E}\,b(H_b)$. □

The exact boundary of $T_1$ is continuous; its boundedness on $[1,\infty)$, which the lemma needs, is observed, not proved. Granting it, the lemma and the tail (9.1.3) give

$$\mathbb{E}\,b(T_1) = \lim_{t\to\infty} \mathbb{E}\big[(b_\infty - B_t)\mathbf{1}\{H_b > t\}\big] = \sqrt{\frac{\pi}{2}} \times \frac{\sum_\rho \rho^{-1}}{\sqrt{\pi}} = \frac{1}{\sqrt{2}} \sum_\rho \frac{1}{\rho} = 0.0163311,$$

the first equality by optional stopping, since $P(H_b > t) \to 0$. So, if the boundary is bounded on $[1,\infty)$, Wald's identity $\mathbb{E}B_{H_b} = 0$, which holds for every crossing time of finite mean, fails here, and the defect is exactly $\sum_\rho \rho^{-1}/\sqrt{2}$. Numerically, integrating the computed boundary against the law of $T_1$ gives $\mathbb{E}\,b(T_1) = 0.0163370$. The defect is also the level of the stable summand of $T_1$: since $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$, the variable $S$ of Proposition 9.1.11 has the law of the first passage time $H_{b_1/\sqrt{2}}$ (Section 10.2.4).

We simulated $1.5 \times 10^5$ Brownian paths against the exact boundary on $[0,1]$, with a Brownian bridge crossing correction, using `rs1d.py`. The Kolmogorov

distance between the simulated and the exact survival functions is 0.0019, at the level of the Monte Carlo noise. Figure 9.5, drawn by `figs5.py`, shows the boundary in its three zones and the survival check.

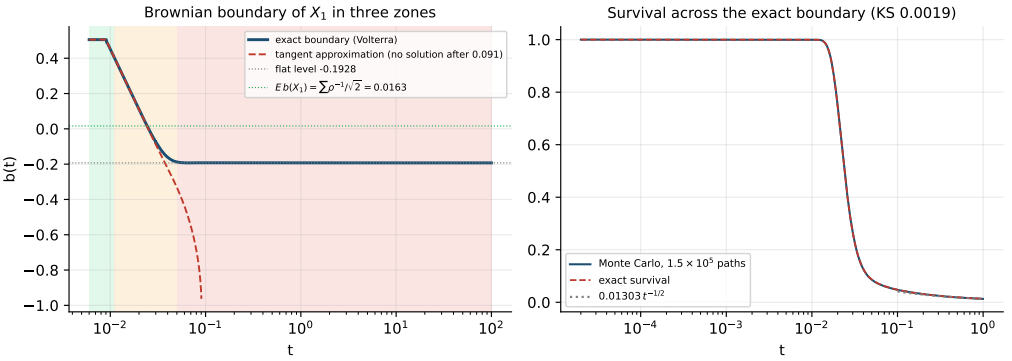



Figure 9.5: Left: the exact Brownian boundary of $T_1$, written $X_1$ in the plot, from the Volterra equation over the three zones of Section 8.2.2.2 (shaded: small time, bulk, right tail), the tangent approximation, which has no solution beyond $t = 0.091$, the flat level $-0.19277$, and the value $\mathbb{E}\,b(T_1) = \sum_\rho \rho^{-1}/\sqrt{2} = 0.0163$. Right: survival of $1.5 \times 10^5$ simulated Brownian paths across the exact boundary against the exact survival function of $T_1$ and the tail $0.01303\,t^{-1/2}$; the Kolmogorov distance is 0.0019.

### 9.2.5 Every basepoint above the centre

Nothing in Theorem 9.2.2 is special to the basepoint one. Since $(\alpha - \frac{1}{2} + \sqrt{u})^2 = (\alpha - \frac{1}{2})^2 + u + 2(\alpha - \frac{1}{2})\sqrt{u}$, the same computation applies at every basepoint to the right of the centre.

Theorem 9.2.5. *For every $\alpha > \frac{1}{2}$, with no hypothesis on the zeros (the existence of the centre clock, Theorem 5.1.1, uses the verification of RH below height 1000), $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha}$, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and $\sigma^{(\alpha)}$ is an independent subordinator with exponent $u + 2(\alpha - \frac{1}{2})\sqrt{u}$, that is $T_\alpha \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$. The law has infinite mean and $P(T_\alpha > x) \sim \frac{\xi'}{\xi}(\alpha)\,(\pi x)^{-1/2}$. The variable $T_\alpha$ is a GGC if and only if no zero has $\beta > \alpha$, in particular for every $\alpha \geq 1$, and when every zero has $\beta < \alpha$ its Thorin density in the chart $\theta = \sqrt{z}$ is $\frac{1}{\pi}\sum_\rho (\alpha - \beta)/((\alpha - \beta)^2 + (\theta - \gamma)^2)$.*

*Proof.* The subordination is the computation of Theorem 9.2.2 with $\frac{1}{4}$ replaced by $(\alpha - \frac{1}{2})^2$ and $\sqrt{u}$ by $2(\alpha - \frac{1}{2})\sqrt{u}$, and $H_a$ has exponent $a\sqrt{2u}$ with $a = \sqrt{2}(\alpha - \frac{1}{2})\ell$. The tail follows as in Section 9.1.2 from $\varphi_\alpha(u) = 1 - \frac{\xi'}{\xi}(\alpha)\sqrt{u} + O(u)$. The Thorin

density and the GGC criterion are the threshold theorem, Theorem 1.2.2: when every zero has $\beta < \alpha$ each Cauchy term is positive, and a zero with $\beta > \alpha$ puts a pole of $\varphi_\alpha$ at the point $(\rho-\alpha)^2$ of $\mathbb{C} \setminus (-\infty, 0]$ (its principal root is $\rho - \alpha$, since $\operatorname{Re}(\rho - \alpha) > 0$), where the Laplace transform of a GGC is holomorphic, whatever the sign of the density near its ordinate (Remark 1.2.4). In the boundary case, when some zero has $\beta = \alpha$ and none has $\beta > \alpha$, every zero has $\beta < \alpha'$ for $\alpha' > \alpha$, so $T_{\alpha'}$ is a GGC; as $\alpha' \downarrow \alpha$, $\varphi_{\alpha'} \to \varphi_\alpha$ pointwise on $[0, \infty)$, so $T_{\alpha'} \to T_\alpha$ in law, and $T_\alpha$ is a GGC by the closure theorem of Section 4.1.1. There the Cauchy kernel of a zero $\alpha + i\gamma$, $\gamma > 0$, of multiplicity $m$ becomes an atom: with its conjugate it contributes the factor $(1+u/\gamma^2)^{-m}$ to $\varphi_\alpha$, a Thorin atom of mass $m$ at $z = \gamma^2$. □

Corollary 9.2.6. *Let $\alpha > \frac{1}{2}$ and $a = \alpha - \frac{1}{2}$. Then $T_\alpha$ is infinitely divisible, with no drift and Lévy measure $W_\alpha(x)\,dx/x$, where*

$$W_\alpha(x) = \frac{ax}{\sqrt{\pi}} \int_0^x (x-s)^{-3/2} e^{-a^2 x s/(x-s)}\, W(s)\, ds$$
$$= \mathbb{E}\Big[(1+4xV)^{-1/2}\, W\Big(\frac{x}{1+4xV}\Big)\Big],$$

*with $V = a^2/(2Z^2)$ and $Z$ standard normal. Hence $W_\alpha > 0$ and $W_\alpha' < 0$ on $(0,\infty)$, so that $T_\alpha$ is self decomposable; and on every interval $(0, T]$ on which $W$ is completely monotone, so is $W_\alpha$.*

*Proof.* By Theorem 9.2.5, $T_\alpha$ is the subordinator $\sigma_s^{(\alpha)} = s + H_{\sqrt{2}as}$ evaluated at the independent time $\widetilde{T}_\alpha$, which is infinitely divisible with no drift and Lévy measure $e^{-a^2 s} W(s)\, ds/s$, and $H_{\sqrt{2}as}$ has the density $(as/\sqrt{\pi}) y^{-3/2} e^{-a^2 s^2/y}$. By Bochner's subordination [471, Theorem 30.1], $T_\alpha$ is infinitely divisible with no drift and Lévy measure $\int_0^\infty P(\sigma_s^{(\alpha)} \in dx)\, e^{-a^2 s} W(s)\, ds/s$, which, since $a^2 s^2/(x-s) + a^2 s = a^2 x s/(x-s)$, is $W_\alpha(x)\,dx/x$ with the first form of $W_\alpha$. The substitution $s = x/(1+4xv)$ turns the integral into an average against the density $\frac{a}{2\sqrt{\pi}} v^{-3/2} e^{-a^2/4v}$ of $V$, which is the second form. There the derivative in $x$ of $(1+4xv)^{-1/2} W(x/(1+4xv))$ is $-2v(1+4xv)^{-3/2} W + (1+4xv)^{-5/2} W'$, negative because $W > 0$ and $W' < 0$ (Theorem 5.1.1), and differentiation under the average is legitimate, because $W^{(n)}(s) = O(s^{-n-1/2} \log(1/s))$ as $s \to 0$ makes every derivative in $x$ of the integrand $O(\log(1+V))$ on compact subsets of $(0,\infty)$, and $\mathbb{E}\log(1+V) < \infty$. Finally, $x \mapsto x/(1+4xv)$ has a completely monotone derivative and maps $(0, T]$ into itself, and $(1+4xv)^{-1/2}$ is completely monotone, so Faà di Bruno's formula and Leibniz's rule carry complete monotonicity of $W$ on $(0, T]$ to every term of the average. □

With Theorem 5.1.9, $W_\alpha$ is therefore completely monotone on $(0, 1.29]$ for every $\alpha > \frac{1}{2}$. When no zero has real part above $\alpha$, in particular for every $\alpha \geq 1$, $T_\alpha$ is a GGC and $W_\alpha$ is the Laplace transform $k_\alpha$ of its Thorin measure (Section 11.1.6);

when some zero has real part above $\alpha$, $T_\alpha$ is not a GGC, so $W_\alpha$ is not completely monotone on $(0,\infty)$, although it is on $(0,1.29]$.

Table 9.6, computed by `improve.py`, and the right panel of Figure 9.4 give the constants and the densities for three basepoints. As $\alpha \downarrow \frac{1}{2}$ the tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$ tends to zero, because $\xi'(\frac{1}{2}) = 0$, and the law acquires the finite mean $\sum_k a_k^{-1}$ of the centre clock; at the same time the subordinator tends to pure drift and the Cauchy widths to $\frac{1}{2} - \beta$, which is where RH is decided.

| $\alpha$ | $\frac{\xi'}{\xi}(\alpha)$ | tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$ | $\min_{[0,60]} \rho_{\alpha-1/2}$ |
|---|---|---|---|
| 0.6 | 0.00462092 | 0.002607 | 0.00147 |
| 0.75 | 0.01155134 | 0.006517 | 0.00368 |
| 1 | 0.02309571 | 0.013030 | 0.00735 |

Table 9.6: Basepoints above the centre: the drift $\frac{\xi'}{\xi}(\alpha)$, the tail constant of $T_\alpha$, and the minimum on $[0,60]$, attained at the origin, of the density $\rho_{\alpha-1/2} = v_\alpha/\pi$, the Thorin density of $T_\alpha$ when every zero lies to the left of the line $\operatorname{Re} s = \alpha$. At $\alpha = 1$ the minimum at the origin holds at every height (Proposition 9.1.11), as it does at every $\alpha \geq 1$ (Proposition 9.1.12); for $\frac{1}{2} < \alpha < 1$ the minimum at the origin holds on the verified window $|\theta| \leq 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)), and at every height for every such $\alpha$ it is equivalent to RH (part (vi) of that theorem).

### 9.2.6 Boundaries across the basepoints

We solved the Volterra equation (8.2.1) [412] at $\alpha = 0.6$ and 0.75, with the law of $T_\alpha$ from Theorem 9.2.5. As at the basepoint one, the distribution function comes from Talbot inversion of $\varphi_\alpha(u)/u$ for $t < 0.03$ and from the subordination identity beyond, here with $3 \times 10^5$ draws of $\widetilde{T}_\alpha$, made under RH as a sum of exponential variables over the first 200 zeros and a gamma variable with the mean and variance of the remaining terms. The bottom left panel of Figure 11.2 in Section 11.1 draws the four boundaries together.

In the small time zone and the bulk the boundaries at the different basepoints are close: they start near the reflection level 0.50, and at $t = 0.02$ they are at 0.072 for the centre and at 0.080, 0.092 and 0.111 for $\alpha = 0.6$, 0.75 and 1. They differ in the right tail. At the centre the survival function is exponential with rate $\gamma_1^2$, and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1 = -19.99$ and the logarithmic correction $\frac{3}{2\sqrt{2}\gamma_1} \log t$; the slope of the computed boundary over $[0.2, 0.3]$ is $-19.77$ (Section 8.2.2.2). Above the centre the exact boundaries flatten: each stays within 0.011 of its level at $t = 100$, the end of the computation, from $t = 0.1$ on, and within $10^{-3}$ from $t = 1$ on. As $\alpha \downarrow \frac{1}{2}$ the flat level moves down, towards the linear boundary of the centre.

The Wald type identity of Section 9.2.4 extends to every basepoint above

the centre. The density of $T_\alpha$ is again continuous and positive on $(0,\infty)$, so its boundary is continuous [129, Theorem 8.2], and as computed it is bounded up to $t = 100$. Granting boundedness at infinity, which here too is observed, not proved, Lemma 9.2.4 and the stable tail $P(T_\alpha > t) \sim b_\alpha(\pi t)^{-1/2}$ of Theorem 9.2.5 give $\mathbb{E}\, b(T_\alpha) = b_\alpha/\sqrt{2}$, so Wald's identity $\mathbb{E}B_{H_b} = 0$ fails by exactly $b_\alpha/\sqrt{2}$. Table 9.7 compares the identity and the tail constant with the computed values; all six pairs agree to within $6 \times 10^{-4}$ in relative terms. The reason given in Section 9.2.4 for the failure of the tangent approximation in the right tail holds at every basepoint above the centre.

| $\alpha$ | flat level | $\mathbb{E}\, b(T_\alpha)$ | $b_\alpha/\sqrt{2}$ | $\sqrt{100}\, P(T_\alpha > 100)$ | $b_\alpha/\sqrt{\pi}$ |
|---|---|---|---|---|---|
| 0.6 | $-0.33384$ | 0.0032687 | 0.0032675 | 0.0026083 | 0.0026071 |
| 0.75 | $-0.25537$ | 0.0081710 | 0.0081680 | 0.0065203 | 0.0065171 |
| 1 | $-0.19277$ | 0.016337 | 0.016331 | 0.013037 | 0.013030 |

Table 9.7: Exact Brownian boundaries above the centre: the flat level, the mean boundary value $\mathbb{E}\, b(T_\alpha)$ at the crossing, obtained by integrating the computed boundary against the law of $T_\alpha$, against the Wald type identity, and the computed $\sqrt{100}\, P(T_\alpha > 100)$ against the tail constant of Theorem 9.2.5. Computed by `bnd_alpha.py` for $\alpha = 0.6$ and 0.75 and by `rs1.py` for $\alpha = 1$.

# Chapter 10

# The critical strip and Riesz's criterion

Chapter 9 read the reciprocal Xi function at a basepoint $\alpha$ as a Laplace transform on the half line, $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ for $u \geq 0$. At the basepoint one its Thorin measure is the Poisson smoothing of the zeros and its Lévy form carries the primes, and by Theorem 9.2.5 it is, for every $\alpha > \frac{1}{2}$ and with no hypothesis, the Laplace transform of a positive random variable $T_\alpha$ obtained by subordinating the centre clock. The same function has two further readings, as the characteristic function of a normal scale mixture and through the completeness of the translates of the tilted Pólya kernels. Both run across the whole critical strip, not only to the right of the zeros, and so show how much of what holds with no hypothesis at the basepoint one survives inside the strip. Section 10.1 develops the two readings in turn and ends with Riesz's criterion, which reads the zero free half planes through growth rather than completeness. Section 10.2 writes that criterion in the language of the clocks, where it becomes the statement that every $T_\alpha$ with $\alpha > \frac{1}{2}$ is a GGC, and shows why none of the resulting representations of its coefficients supplies the growth bound that the criterion needs. Section 10.3 turns to the squarefree integers, whose Dirichlet series $\zeta(s)/\zeta(2s)$ carries the same reciprocal $1/\zeta(2s)$: the constant of $\sum_{n \leq x} \mu(n)^2/n$ is a combination of two tilted Pólya means, and smoothing the count by a gamma law gives criteria for RH of the same kind as Riesz's.

For the scale mixtures and the completeness statements, as for the Thorin measure and the Lévy form in Chapter 9, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH. Chapters 9 to 11 prove no new zero free region.

## 10.1 Normal scale mixtures and completeness across the strip

The zero free half planes of Chapter 9 have two readings, the second being completeness (Section 10.1.6). The first evaluates the Laplace transform $\varphi_\alpha$ at $u^2$. Since $\varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$, the reciprocal of $\xi$ along the real axis is a charac-

teristic function, that of a normal scale mixture, at every basepoint and with no hypothesis (Section 10.1.1); what the zeros decide is the class of the law.

We keep the notation of Chapter 9: $T$ is the centre clock and $T_{1/2} = T$, $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac12)^2 T}$, $H_a$ is the first passage time of a standard Brownian motion to the level $a$, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$. When every zero lies to the left of the line $\mathrm{Re}\, s = \alpha$, the Thorin density of $T_\alpha$ in the chart $\theta = \sqrt{z}$ is $v_\alpha/\pi$, the density $\rho_{\alpha-1/2}$ of (1.2.7) (Theorem 1.2.2); when a zero lies to the right of that line, $T_\alpha$ is not a GGC and $v_\alpha/\pi$ is not a Thorin density for it (Proposition 11.4.3). Throughout, $Z$ is a standard normal variable independent of everything else. The scripts are in the folders `basepoint_one/scale_mixture` and `basepoint_one/wiener` of the archive.

### 10.1.1 The reciprocal on the real axis is a characteristic function

If $X \geq 0$ is independent of $Z$, then $\mathbb{E}e^{iu\sqrt{2X}Z} = \mathbb{E}e^{-u^2 X}$ for real $u$. The centre clock $T$ of [421] is infinitely divisible and self decomposable with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000). For $\alpha > \frac12$, Theorem 9.2.5 writes $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha} \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac12)\widetilde{T}_\alpha}$, where $\sigma^{(\alpha)}$ is an independent subordinator with Laplace exponent $u + 2(\alpha - \frac12)\sqrt{u}$. With $X = T_\alpha$ the evaluation gives the following.

Proposition 10.1.1. *For every $\alpha \geq \frac12$ the function $u \mapsto \varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$ is the characteristic function of $Y_\alpha = \sqrt{2T_\alpha}\, Z$. The law of $Y_\alpha$ is symmetric, unimodal and infinitely divisible with no hypothesis. At the centre it has variance $2\mathbb{E}T = 2\sum_k a_k^{-1} = 0.046210$. For $\alpha > \frac12$ its characteristic function has a corner at the origin, $\xi(\alpha)/\xi(\alpha + |u|) = 1 - b_\alpha|u| + O(u^2)$, so $Y_\alpha$ lies in the domain of attraction of the Cauchy law, with $P(|Y_\alpha| > y) \sim 2b_\alpha/(\pi y)$ as $y \to \infty$.*

*Proof.* The first statement is the evaluation with $X = T_\alpha$. Symmetry and unimodality hold for every normal scale mixture. The tilt $\widetilde{T}_\alpha$ of the infinitely divisible law of $T$ is infinitely divisible, with Lévy measure $e^{-(\alpha-\frac12)^2 t} W(t)\, dt/t$, and a subordinator run for an independent infinitely divisible time is infinitely divisible, because a Bernstein function composed with a Bernstein function is again one [473, Corollary 3.8]. So $T_\alpha$ is infinitely divisible, and $Y_\alpha$, which is a Brownian motion run for the independent time $2T_\alpha$, is infinitely divisible for the same reason. At the centre $\xi'(\frac12) = 0$, so $\xi(\frac12)/\xi(\frac12 + |u|) = 1 - \mathbb{E}T\, u^2 + O(u^4)$, with $\mathbb{E}T = \sum_k a_k^{-1}$ by (1.2.1). For $\alpha > \frac12$ the expansion follows from $\xi(\alpha + |u|) = \xi(\alpha)(1 + b_\alpha|u| + O(u^2))$. The tail comes from that of $T_\alpha$: by Theorem 9.2.5, $P(T_\alpha > t) \sim b_\alpha(\pi t)^{-1/2}$ as $t \to \infty$, so that, since $P(T_\alpha > t) \leq 1$, also $P(T_\alpha > t) \leq Kt^{-1/2}$ for all $t > 0$ and some constant $K$. Conditioning on $Z$, $yP(|Y_\alpha| > y) = \mathbb{E}\big[yP(T_\alpha > y^2/(2Z^2) \mid Z)\big]$, and the integrand is at most $\sqrt{2}K|Z|$ and tends to $\sqrt{2/\pi}\, b_\alpha|Z|$ as $y \to \infty$; by dominated convergence $yP(|Y_\alpha| > y) \to \sqrt{2/\pi}\, b_\alpha \mathbb{E}|Z| = 2b_\alpha/\pi$. □

At the centre $Y_{1/2} = Z\sqrt{2T}$ is the second member of the van Dantzig pair of Section 1.2; it has the same variance as the Pólya variable $X$, the first member, as the relation $\mathbb{E}e^{-\theta^2 T} = 1/\mathbb{E}e^{\theta X}$ requires.

With no hypothesis the law of $Y_{1/2}$ is also self decomposable. For $0 < c < 1$ the self decomposability of $T$ (Theorem 5.1.1) gives $T \stackrel{d}{=} cT' + R_c$ with $T' \stackrel{d}{=} T$ and $R_c \geq 0$ independent, and then $Y_{1/2} \stackrel{d}{=} \sqrt{c}\,Y' + Z'\sqrt{2R_c}$, where $Y' = Z\sqrt{2T'}$ has the law of $Y_{1/2}$ and $Z'$ is standard normal and independent of $(Z, T', R_c)$.

Under RH each factor $(1 + u^2/\gamma^2)^{-1}$ of $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ is a characteristic function, that of $L/\gamma$ for a standard Laplace variable $L$, and a zero off the line would break this. The quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, with $\gamma > 0$ and $0 < \delta < \frac{1}{2}$, contributes to $\xi(\frac{1}{2})/\xi(\frac{1}{2}+|u|)$, by (1.2.1), the factor $g(u) = |1+u^2/(\gamma-i\delta)^2|^{-2}$, which is positive and integrable with $g(0) = 1$, and

$$\frac{1}{2\pi}\int_{\mathbb{R}} g(u)e^{-iux}\,du = \frac{\gamma^2+\delta^2}{4\gamma\delta}\,e^{-\gamma|x|}\big(\gamma\sin\delta|x| + \delta\cos\delta|x|\big).$$

For $x > 0$ the derivative of the right side is $-\frac{(\gamma^2+\delta^2)^2}{4\gamma\delta}e^{-\gamma x}\sin\delta x$, so its least value, attained at $x = \pi/\delta$, is $-e^{-\pi\gamma/\delta}$ times its value $(\gamma^2 + \delta^2)/(4\gamma)$ at $x = 0$. Since $g$ is integrable, it could only be the characteristic function of the law with this density, which is negative there. So $g$ is not a characteristic function, and $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ would still be one, as $W > 0$ ensures, but no longer factor by factor. The defect is tiny. Every ordinate is at least $\gamma_1$ and $\delta < \frac{1}{2}$, so the least value exceeds $-9.5 \times 10^{-39}$, the value of the formula at $\gamma = \gamma_1$ and $\delta = \frac{1}{2}$; and a zero off the line would lie above $T_0$, where $e^{-\pi\gamma/\delta} < e^{-2\pi T_0}$.

Away from the centre neither $V_\alpha$ nor $Y_\alpha$ belongs to a pair. Take $\alpha > \frac{1}{2}$; for $V_\alpha$ the case $\alpha < \frac{1}{2}$ follows, since $V_\alpha \stackrel{d}{=} -V_{1-\alpha}$. The tilted Pólya variable $V_\alpha$ of Section 1.2 has characteristic function $\varphi(\theta) = \xi(\alpha + i\theta)/\xi(\alpha)$, and $1/\varphi(i\theta) = \xi(\alpha)/\xi(\alpha - \theta)$ exceeds 1 for $0 < \theta < 2\alpha - 1$, because $|\alpha - \theta - \frac{1}{2}| < \alpha - \frac{1}{2}$ there and $\xi(\frac{1}{2} + s) = \xi(\frac{1}{2})\mathbb{E}\cosh(sX)$ increases with $|s|$; at $\alpha = 0.75$ and $\theta = 0.25$ it is 1.00145. So it is not a characteristic function. By its Cauchy tail $Y_\alpha$ has no exponential moments, so it cannot be a first member. As a second member its partner $\varphi$ would satisfy $\varphi(i\theta) = \xi(\alpha+|\theta|)/\xi(\alpha)$, and a moment generating function finite on the whole line is analytic, while $\xi(\alpha + |\theta|)/\xi(\alpha) = 1 + b_\alpha|\theta| + O(\theta^2)$ has a corner at $\theta = 0$, since $b_\alpha > 0$ (Proposition 10.1.1).

### 10.1.2 From the Lévy form in $|u|$ to the Thorin form in $u^2$

For $\alpha \geq 1$ the function $\log(\xi(\alpha + s)/\xi(\alpha))$ has, for $\operatorname{Re} s > 0$, the Lévy form (9.1.5) with $b_\alpha$ and $\Pi_\alpha$ in place of $b_1$ and $\Pi_1$, where $x\Pi_\alpha(dx) = \big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\big)dx + \sum_n \Lambda(n)n^{-\alpha}\delta_{\log n}(dx)$ is the signed measure that Section 9.1.4 gives for every

basepoint $\alpha \geq 1$. Putting $s = |u|$ gives

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = e^{-b_\alpha |u|} \exp\Big\{ -\int_0^\infty \big(e^{-|u|x} - 1 + |u|x\big)\Pi_\alpha(dx)\Big\}, \tag{10.1.1}$$

a representation in $|u|$. The first factor is the characteristic function of a Cauchy law of scale $b_\alpha$. The second is not in Lévy Khintchine form, because $e^{-|u|x}$ is a function of $|u|$ rather than of $u$. The sine squared lemma (1.2.9) converts it. In the chart $\theta = \sqrt{z}$ and at $s = |u|$, as in the derivation of (9.1.6), it writes $e^{-|u|x} + |u|x - 1$ as $\frac{2x}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\sin^2(x\theta/2)\,d\theta$ for $x > 0$, and the drift converts through $|u| = \frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\,d\theta$. Both identities follow from $\int_0^\infty \log(1+u^2/\theta^2)\cos(x\theta)\,d\theta = \pi(1-e^{-|u|x})/x$.

By Section 1.3.2, a symmetric law is an extended generalised gamma convolution (EGGC) if and only if it is the law of $Z\sqrt{2T'}$ with $T'$ a GGC independent of $Z$.

THEOREM 10.1.2. *Let $\alpha \geq 1$, or more generally let $\alpha > \frac{1}{2}$ be such that $\operatorname{Re}\rho < \alpha$ for every zero $\rho$. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big) v_\alpha(\theta)\,d\theta\Big\}, \tag{10.1.2}$$

*where*

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big) x\Pi_\alpha(dx) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta) > 0,$$

*the first expression being available for $\alpha \geq 1$, and as the compensated limit* (9.1.8) *(in which $\varepsilon = 1-\alpha$) when $\sup_\rho \operatorname{Re}\rho < \alpha$, so $Y_\alpha$ is a symmetric EGGC with Thorin density $v_\alpha/\pi$ in the chart $\theta$. With no hypothesis on the zeros, for $\frac{1}{2} < \alpha < 1$ the law of $Y_\alpha$ is an EGGC if and only if $\zeta$ has no zeros with real part greater than $\alpha$, and at the centre if and only if RH holds. Under RH,*

$$\frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2}+|u|)} = \prod_{\gamma>0}\Big(1+\frac{u^2}{\gamma^2}\Big)^{-1}, \qquad Y_{1/2} \stackrel{d}{=} \sum_k \frac{L_k}{\gamma_k},$$

*with independent standard Laplace variables $L_k$, of density $\frac{1}{2}e^{-|y|}$.*

*Proof.* For $\alpha \geq 1$, insert the two identities of the lemma into (10.1.1) and exchange the order of integration, which is justified for $\alpha > 1$ by absolute convergence and at $\alpha = 1$ by the compensation argument of Proposition 9.1.10. This gives the first expression for $v_\alpha$, and the second is its identification, (9.1.6) at $\alpha = 1$, which is Proposition 9.1.3 of Section 9.1.3, and (3.1.1) for $\alpha > 1$; for $\frac{1}{2} < \alpha < 1$ with $\sup_\rho \operatorname{Re}\rho < \alpha$ the first expression holds in the compensated form (9.1.8) of Proposition 9.1.13, in which $\varepsilon = 1-\alpha$. In every case (10.1.2) is the Thorin representation of $T_\alpha$ given by Theorem 1.2.2, $\varphi_\alpha(u) = \exp\{-\int \log(1+$

$u/z)\,U_\alpha(dz)\}$ with $U_\alpha$ as in (1.2.6), evaluated at $u^2$ in the chart $z=\theta^2$, and $v_\alpha>0$ because every Cauchy kernel of Proposition 1.2.3 then has positive width $\alpha-\beta$. Since $T_\alpha$ is a GGC, $Y_\alpha=Z\sqrt{2T_\alpha}$ is an EGGC. Conversely, if $Y_\alpha\overset{d}{=}Z\sqrt{2T'}$ with $T'$ a GGC, then $\mathbb{E}e^{-u^2T'}=\mathbb{E}e^{-u^2T_\alpha}$ for all real $u$, so $T'\overset{d}{=}T_\alpha$. Hence $Y_\alpha$ is an EGGC exactly when $T_\alpha$ is a GGC, which by Theorem 1.2.2 happens exactly when $\alpha\ge\sup_\rho\operatorname{Re}\rho$. For $\frac12<\alpha<1$ this is the absence of zeros with real part greater than $\alpha$, and at the centre, by the symmetry $\rho\mapsto1-\rho$, it is RH (Proposition 1.2.1). Under RH the Hadamard product (1.2.1) at $s=|u|$ is the product displayed, and each factor $(1+u^2/\gamma^2)^{-1}$ is the characteristic function of $L/\gamma$, as in Proposition 6.2.8(iii). □

The hypothesis that every zero lies to the left of the line $\operatorname{Re}s=\alpha$ cannot be dropped from (10.1.2). By Proposition 1.2.3, $v_\alpha/\pi$ is, off the ordinates of the zeros on the line $\operatorname{Re}s=\alpha$ and whatever the zeros do, the sum over the zeros with $\gamma>0$ of $P_{\alpha-\beta}(\theta-\gamma)+P_{\alpha-\beta}(\theta+\gamma)$, with $P_{-c}=-P_c$, and by the semigroup property of the Cauchy kernels $\int_0^\infty\frac{2s}{s^2+\theta^2}[P_c(\theta-\gamma)+P_c(\theta+\gamma)]\,d\theta=\frac{2(s+|c|)\operatorname{sgn}c}{(s+|c|)^2+\gamma^2}$ for $s>0$ and real $c\ne0$. Integrating in $s$ from 0 to $|u|$ shows that a zero $\rho=\beta+i\gamma$ with $\gamma>0$ contributes to the right side of (10.1.2) the factor $((\alpha-\beta)^2+\gamma^2)/((|u|+\alpha-\beta)^2+\gamma^2)$ when $\beta<\alpha$, which is its factor in $\xi(\alpha)/\xi(\alpha+|u|)$, but the factor $((|u|+\beta-\alpha)^2+\gamma^2)/((\beta-\alpha)^2+\gamma^2)$ when $\beta>\alpha$. If no zero lies on the line $\operatorname{Re}s=\alpha$, the integral with the signed density therefore converges and equals $\xi(\alpha)/\xi(\alpha+|u|)$ multiplied by

$$\prod_{\beta>\alpha,\ \gamma>0}\frac{\big((|u|+\beta-\alpha)^2+\gamma^2\big)\big((|u|-\beta+\alpha)^2+\gamma^2\big)}{\big((\beta-\alpha)^2+\gamma^2\big)^2},$$

which exceeds one for $u\ne0$ as soon as there is such a zero, since each factor equals $1+u^2(u^2+2\gamma^2-2(\beta-\alpha)^2)/((\beta-\alpha)^2+\gamma^2)^2$ and $0<\beta-\alpha<\frac12<\gamma$. The series of Proposition 1.2.3 may be integrated term by term, because the absolute value of the term of a zero integrates to $\log(((|\alpha-\beta|+|u|)^2+\gamma^2)/((\alpha-\beta)^2+\gamma^2))=O(\gamma^{-2})$. A zero on the line $\operatorname{Re}s=\alpha$ contributes instead an atom of mass one at its ordinate, as every zero does at the centre under RH.

The drift $b_\alpha$ becomes, through $|u|=\frac1\pi\int_0^\infty\log(1+u^2/\theta^2)\,d\theta$, the constant Thorin density $b_\alpha/\pi$, which is also the value $v_\alpha(0)/\pi$. At the basepoint one this constant is the minimum of $v_1/\pi$, attained only at $\theta=0$ (Proposition 9.1.11), and `comp.py` confirms it on $[0,200]$. So $Y_1$ splits as an independent sum of a Cauchy variable of scale $b_1=\sum_\rho\rho^{-1}=0.0230957$ and a symmetric EGGC with Thorin density $(v_1-b_1)/\pi$ (Figure 10.1, left). This is the image at $u^2$ of the factorisation $T_1\overset{d}{=}S+T_1^\circ$ of Proposition 9.1.11 into a positive stable variable of index one half and a GGC with Thorin density $(A+Q)/\pi$: since $\mathbb{E}e^{-u^2S}=e^{-b_1|u|}$, the Cauchy summand is $Z\sqrt{2S}$, and the EGGC summand is $\sqrt{2T_1^\circ}$ times an independent standard normal variable, with characteristic function $e^{b_1|u|}\xi(1)/\xi(1+$

$|u|) = \exp\{-\frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)(v_1-b_1)(\theta)\,d\theta\}$. Table 10.1 checks (10.1.2); at $\alpha = 0.75$ the formula presupposes that no zero has real part 0.75 or more.

| $u$ | Thorin form, $\alpha=1$ | $\xi(1)/\xi(1+\lvert u\rvert)$ | Thorin form, $\alpha=0.75$ | $\xi(0.75)/\xi(0.75+\lvert u\rvert)$ |
|---|---|---|---|---|
| 0.5 | 0.982809 | 0.982838 | 0.988506 | 0.988520 |
| 1 | 0.954904 | 0.954930 | 0.965969 | 0.965980 |
| 2 | 0.871156 | 0.871171 | 0.891317 | 0.891317 |
| 5 | 0.507984 | 0.507962 | 0.536812 | 0.536769 |
| 20 | 0.0003491 | 0.0003487 | 0.0004116 | 0.0004110 |

Table 10.1: The Thorin form (10.1.2) against the reciprocal Xi function on the real axis, computed by `comp.py`.

### 10.1.3 Extended generalised gamma convolutions in the critical strip

Thorin introduced generalised gamma convolutions [493, 494] and extended them to laws on the whole line [495]. Bondesson [56, Section 7.1] calls the resulting class the extended generalised gamma convolutions, and Section 1.3.2 recalls it. A symmetric law without Gaussian part is an EGGC exactly when its characteristic function has the form

$$\exp\Big\{-\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big)m(d\theta)\Big\}$$

with a positive measure $m$ on $(0,\infty)$ that satisfies the integrability conditions of a Thorin measure in the chart $z=\theta^2$; under $z=\theta^2$ it is the Thorin measure of the mixing GGC $T'$ in the representation $Z\sqrt{2T'}$ recalled before Theorem 10.1.2. Whether $T_\alpha$ is a GGC is decided by the location of the zeros and not by the sign of $v_\alpha$ on the single line $\operatorname{Re} s=\alpha$: for $\xi$ it is not known whether positivity on one line excludes zeros to its right (Remark 1.2.4).

Theorem 10.1.3. *Let $\frac12<\alpha<1$ and write $\varepsilon=\alpha-\frac12$.*

1. *With no hypothesis, $\xi(\alpha)/\xi(\alpha+|u|)$ is the characteristic function of the infinitely divisible, symmetric, unimodal law $Y_\alpha$. If every zero has real part less than $\alpha$, then* (10.1.2) *holds with the positive density $v_\alpha(\theta)/\pi = \frac{1}{\pi}\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(\theta-\gamma)^2}$.*
2. *$Y_\alpha$ is an EGGC if and only if no zero has $\beta>\alpha$, that is, if and only if $\zeta$ has no zeros in the half plane $\operatorname{Re} s>\alpha$.*
3. *Let $T_0\ge 2$ be a height up to which all zeros are known to lie on the critical line. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \prod_{0<\gamma\le T_0}\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}\times \mathcal{E}_{\alpha,T_0}(u),$$

*where the finite product is the characteristic function of an EGGC whose Thorin density* $\sum_{0<\gamma\le T_0}[P_\varepsilon(\theta-\gamma)+P_\varepsilon(\theta+\gamma)]$ *is positive, and the remainder satisfies* $|\log \mathcal{E}_{\alpha,T_0}(u)| \le C(|u|+u^2)\frac{\log T_0}{T_0}$ *for an absolute constant* $C$.

*Proof.* (1) is Proposition 10.1.1 and Theorem 10.1.2, the sum being that of Proposition 1.2.3. (2) is part of Theorem 10.1.2: $Y_\alpha$ is an EGGC exactly when $T_\alpha$ is a GGC, and a zero with $\beta > \alpha$ puts a pole of $\varphi_\alpha$ at the point $(\rho-\alpha)^2$ of $\mathbb{C}\setminus(-\infty,0]$, where the Laplace transform of a GGC is holomorphic (Theorem 1.2.2). (3) Group the zeros into those on the line below $T_0$ and the rest. Each pair $\frac12 \pm i\gamma$ contributes the factor $\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}$, whose Thorin density is the pair of Cauchy kernels of width $\varepsilon$ at $\pm\gamma$, by the computation after Theorem 10.1.2, and which is therefore an EGGC characteristic function. Every other zero has $|\gamma| > T_0$, and a pair $\rho, \bar\rho$ with $\gamma > T_0$ contributes $-\log(1+x)$ to $\log \mathcal{E}_{\alpha,T_0}(u)$, with $x = (2(\alpha-\beta)|u|+u^2)/((\alpha-\beta)^2+\gamma^2)$. Since $|\alpha-\beta| < 1$, we have $x \ge -(\alpha-\beta)^2/((\alpha-\beta)^2+\gamma^2) > -\frac12$ and $|x| \le 2(|u|+u^2)/\gamma^2$, so the contribution is at most $4(|u|+u^2)/\gamma^2$ in absolute value. Summing over the zeros with $\gamma > T_0$ gives at most $4(|u|+u^2)\sum_{\gamma>T_0}\gamma^{-2}$, and by partial summation this sum is at most $2\int_{T_0}^\infty N(t)t^{-3}\,dt$. For $t \ge 2$ we have $N(t) \le \frac{t}{2\pi}\log t$: no zero has $0 < \gamma < \gamma_1$, and for $t \ge \gamma_1$ the explicit bound of [176], recalled in the proof of Proposition 9.1.11, gives $N(t) \le \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac78 + E(t)$, where $\frac78 + E(t) < \frac{t}{2\pi}\log(2\pi e)$: up to $X_0$ the left side is at most 3.40 and the right side is at least 6.38, its value at $t=\gamma_1$, and beyond $X_0$ the left side is less than $\frac78 + 0.3\log t$ and the right side exceeds $0.45\,t$. So $\sum_{\gamma>T_0}\gamma^{-2} \le \frac{1}{\pi}\int_{T_0}^\infty t^{-2}\log t\,dt = \frac{1+\log T_0}{\pi T_0} \le \frac{\log T_0}{T_0}$, the last step because $T_0 \ge 2$, and the bound holds with $C = 4$. □

With the verification height $T_0 = 3\times10^{12}$ of Platt and Trudgian [419], which also enters Corollary 4.3.7, the remainder is of relative size at most $8\times10^{-11}$ for $|u| \le 1$, so $Y_\alpha$ is, to that accuracy in characteristic function, an EGGC for every $\alpha > \frac12$, whether or not RH holds; Theorem 11.4.6 gives the same statement for the Laplace transform of $T_\alpha$ with explicit constants. Grouping each zero above $T_0$ with its reflection in the critical line makes the contribution of every off line pair nonnegative below $T_0-\frac12$, so $v_\alpha > 0$ on $[0, T_0-\frac12]$ for every $\alpha > \frac12$ (Corollary 11.4.7). A zero above $T_0$ with $\beta > \alpha$ would stop $T_\alpha$ from being a GGC, and $Y_\alpha$ from being an EGGC, whether or not it made $v_\alpha$ negative near its ordinate (Exercise 1.7), and no finite computation excludes such a zero (Section 3.1.1). What can be said with no hypothesis is in Theorem 11.4.2.

Table 10.2 checks the product over the computed zeros against $\xi(\alpha)/\xi(\alpha+|u|)$ at $\alpha = 0.6$ and 0.75. On $[0,200]$ the Thorin density is positive at both basepoints, and in both cases its minimum is at $\theta = 0$ and equals the Cauchy floor $b_\alpha/\pi$: 0.0014709 at $\alpha = 0.6$ and 0.0036769 at $\alpha = 0.75$ (Figure 10.1, left).

| $\alpha$ | $u$ | Thorin integral (10.1.2) | product over zeros | $\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$ |
|---|---|---|---|---|
| 0.6 | 1 | 0.9726796 | 0.9726812 | 0.9726812 |
| 0.6 | 5 | 0.5549761 | 0.5549190 | 0.5549193 |
| 0.6 | 20 | 0.0004547 | 0.0004539 | 0.0004539 |
| 0.75 | 1 | 0.9659691 | 0.9659798 | 0.9659798 |
| 0.75 | 5 | 0.5368120 | 0.5367687 | 0.5367691 |
| 0.75 | 20 | 0.0004116 | 0.0004110 | 0.0004110 |

Table 10.2: The EGGC representation in the strip, computed by `strip.py`: the Thorin integral over $\theta \le 200$ plus the Weyl tail, and the product of the factors of Theorem 10.1.3(3) over the 1700 computed zeros plus the Weyl tail.

### 10.1.4 The density, the Lévy measure and the Poisson trace

Because $\xi(\alpha)/\xi(\alpha+|u|)$ is integrable, $Y_\alpha$ has the density

$$f_{Y_\alpha}(y) = \frac{1}{\pi}\int_0^\infty \cos(uy)\,\frac{\xi(\alpha)}{\xi(\alpha+u)}\,du,$$

which `dens2.py` computes. At the origin $f_{Y_\alpha}(0) = 1.8994,\ 1.8680,\ 1.8225,\ 1.7507$ for $\alpha = \frac12,\ 0.6,\ 0.75,\ 1$. The corner of the characteristic function at the origin gives the tail $f_{Y_\alpha}(y) \sim b_\alpha/(\pi y^2)$ for $\alpha > \frac12$ (Figure 10.1, middle). At the centre there is no corner, and the density decays like $e^{-\gamma_1|y|}$, the tail of $L_1/\gamma_1$ under RH; unconditionally, the characteristic function continues to the meromorphic function $\xi(\frac12)/\xi(\frac12+u)$ of $u$, whose poles nearest the real axis are $\pm i\gamma_1$.

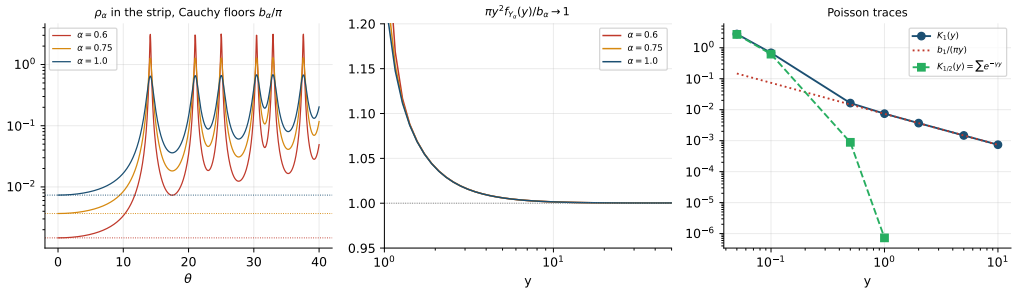



Figure 10.1: Left: the Thorin densities $v_\alpha/\pi$ at $\alpha = 0.6,\ 0.75,\ 1$, written $\rho_\alpha$ in the plot, and their Cauchy floors $b_\alpha/\pi$ (dotted). Middle: $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$, tending to one. Right: the Poisson traces $K_1$ and $K_{1/2}$ against $b_1/(\pi y)$.

Frullani's integral gives $\log(1+u^2/\theta^2) = \int_{\mathbb{R}}(1-\cos uy)\,e^{-\theta|y|}\,\frac{dy}{|y|}$, which is the sine squared lemma read in the other direction, since $1-\cos uy = 2\sin^2(uy/2)$, and which is the cosine dictionary (6.2.4) of Section 6.2. Inserting it in (10.1.2)

puts the characteristic function of $Y_\alpha$ in Lévy Khintchine form in $u$:

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{ -\int_{\mathbb{R}} 2\sin^2\Big(\frac{uy}{2}\Big)\,\frac{K_\alpha(|y|)}{|y|}\,dy\Big\},$$
$$K_\alpha(y) = \frac{1}{\pi}\int_0^\infty e^{-\theta y} v_\alpha(\theta)\,d\theta.$$

So the Lévy density of $Y_\alpha$ is $K_\alpha(|y|)/|y|$, as Section 1.3.2 gives for a symmetric EGGC; in the normalisation of (6.2.4), $K_\alpha = \frac{1}{2}K_{U_\alpha}$. We call $K_\alpha$ the Poisson trace, because at the centre, where under RH the Thorin measure in the chart $\theta$ is $\sum_{\gamma>0}\delta_\gamma$, it is $K_{1/2}(y) = \sum_{\gamma>0} e^{-\gamma y}$, which is $e^{-iy/2}V(iy)$ for Cramér's function $V(z) = \sum_{\gamma>0} e^{\rho z}$ [96] (compare Proposition 6.2.8(iii)): the sum over the zeros of the Poisson kernel in place of the heat kernel of the heat trace $W(t) = \sum_k e^{-a_k t}$, which is $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH. Since $Y_{1/2} = B_{2T}$ for an independent Brownian motion $B$, and $T$ has Lévy density $W(t)/t$ (Theorem 5.1.1), the Lévy density of $Y_{1/2}$ is $\int_0^\infty (4\pi t)^{-1/2} e^{-y^2/4t}\,W(t)\,t^{-1}dt$, so

$$K_{1/2}(y) = \int_0^\infty \frac{y}{\sqrt{4\pi t^3}}\,e^{-y^2/4t}\,W(t)\,dt = \mathbb{E}\,W(H_{y/\sqrt{2}}), \qquad y>0.$$

The Poisson trace is the heat trace averaged over the first passage time of Brownian motion to the level $y/\sqrt{2}$, that is, subordinated by the stable one half first passage kernel, whose Laplace transform $e^{-y\sqrt{a}}$ is the stable one half average of the heat kernel recalled after Proposition 6.2.9. This defines $K_{1/2}$ unconditionally, as $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ with the principal root, and makes it computable from the explicit formula (1.2.5) for $W$.

Corollary 10.1.4. *RH holds if and only if the Poisson trace $K_{1/2}$ is completely monotone, and if and only if the heat trace $W$ is completely monotone. For $\alpha \ge 1$ the Poisson trace $K_\alpha$ is completely monotone with no hypothesis, and $K_\alpha(y) \sim b_\alpha/(\pi y)$ as $y\to\infty$, so that the Lévy density of $Y_\alpha$ is asymptotic to $b_\alpha/(\pi y^2)$, the Lévy density of the Cauchy component.*

*Proof.* For $\alpha \ge 1$, $K_\alpha$ is the Laplace transform of the positive density $v_\alpha/\pi$, hence completely monotone by Bernstein's theorem, and since $v_\alpha$ is continuous with $v_\alpha(0) = b_\alpha$, $yK_\alpha(y) \to b_\alpha/\pi$ as $y\to\infty$. At the centre, $W$ is completely monotone exactly when $T$ is a GGC (Section 1.2), that is exactly when RH holds (Proposition 1.2.1). Under RH, $K_{1/2}(y) = \sum_{\gamma>0} e^{-\gamma y}$ is completely monotone. Conversely, suppose that $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta}\nu(d\theta)$ with a positive measure $\nu$. For $w>0$ the unconditional formula $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ gives

$$\int_0^\infty e^{-wy}\,y\,K_{1/2}(y)\,dy = \sum_k \frac{1}{(w+\sqrt{a_k})^2} = \int_{[0,\infty)} \frac{\nu(d\theta)}{(w+\theta)^2}.$$

The first equality integrates term by term, which is legitimate because $\sqrt{a_k} = \gamma_k - i(\beta_k - \frac{1}{2})$ has real part $\gamma_k$ when $\rho_k = \beta_k + i\gamma_k$ is taken with $\gamma_k > 0$, so the absolute values of the terms integrate to $\sum_k (w+\gamma_k)^{-2} < \infty$. The second is Tonelli's theorem, and it shows that the right side is finite for $w > 0$. The middle expression is meromorphic in $w \in \mathbb{C}$, with a double pole at each point $-\sqrt{a_k}$, while the right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, since $|w+\theta|$ is bounded below by a positive multiple of $1+\theta$ for $w$ in a compact subset of $\mathbb{C} \setminus (-\infty, 0]$. The two sides agree for $w > 0$, hence on $\mathbb{C} \setminus (-\infty, 0]$ with the poles removed, so no pole $-\sqrt{a_k}$ lies in $\mathbb{C} \setminus (-\infty, 0]$. Since $\operatorname{Re}\sqrt{a_k} > 0$, every $\sqrt{a_k}$ is therefore real and positive, that is, every $a_k = -(\rho_k - \frac{1}{2})^2$ is positive, which is RH. □

The traces $K_1$ and $K_{1/2}$ are drawn in Figure 10.1, right.

Propositions 10.1.5 and 10.1.6 follow a note of Nicholas Polson merged into the book [236].

Proposition 10.1.5 (The positive half of the reciprocal law). *Let $A$ be the centred infinitely divisible law without Gaussian part whose Lévy measure is $K_{1/2}(x)\,dx/x$ on $(0,\infty)$, a positive measure by Proposition 35.4.1, and let $A'$ be an independent copy of $A$. Then $Y_{1/2} \stackrel{d}{=} A - A'$, and $\kappa_m(A) = \int_0^\infty x^{m-1} K_{1/2}(x)\,dx$ for $m \geq 2$; in particular* $\operatorname{Var} A = \mathbb{E}T$.

*(i) Under RH, with $\gamma_k$ the positive ordinates counted with multiplicity and $E_k$ independent standard exponential variables, $A$ has the law of $\sum_k (E_k - 1)/\gamma_k$, a series that converges almost surely and in $L^2$; $\mathbb{E}e^{-sA} = 1/\mathcal{P}(s)$ for $s > -\gamma_1$, where $\mathcal{P}(s) = \prod_k (1 + s/\gamma_k) e^{-s/\gamma_k}$ and $\mathcal{P}(s)\mathcal{P}(-s) = \xi(\frac{1}{2} + is)/\xi(\frac{1}{2})$; and $\kappa_m(A) = (m-1)! \sum_k \gamma_k^{-m}$.*

*(ii) If $A$ has the law of $\sum_k (E_k - 1)/c_k$ for positive numbers $c_k$ with $\sum_k c_k^{-2} < \infty$, then RH holds.*

*(iii) For positive $\gamma_k$ with $\sum_k \gamma_k^{-2} < \infty$, the function $u \mapsto \mathcal{P}(\sqrt{u})$ is the Laplace transform of a GGC whose Thorin density in the chart $\theta$ is $\frac{1}{\pi} \sum_k \theta^2 / \big(\gamma_k(\gamma_k^2 + \theta^2)\big)$.*

*For $t \geq \Lambda$, unconditionally for $t \geq 0.2$, the series $A_t = \sum_k (E_k - 1)/\theta_k(t)$ over the zeros $\frac{1}{2} \pm i\theta_k(t)$ of $\xi_t$ satisfies $B_{2T_t} \stackrel{d}{=} A_t - A'_t$, with $T_t$ the tilted clock of Section 8.4.8, and $\mathbb{E}e^{-sA_t} = 1/\mathcal{P}_t(s)$, $\mathcal{P}_t(s)\mathcal{P}_t(-s) = \xi_t(\frac{1}{2} + is)/\xi_t(\frac{1}{2})$, with $\mathcal{P}_t$ the product of (i) over the $\theta_k(t)$.*

*Proof.* By Section 10.1.4, $Y_{1/2}$ is symmetric and infinitely divisible without Gaussian part, with Lévy density $K_{1/2}(|y|)/|y|$. The law $A$ exists because $K_{1/2}(x) = O(x^{-1}\log(1/x))$ at 0 (Proposition 10.1.11) and $K_{1/2}$ decays exponentially at infinity, and $\mathbb{E}e^{iu(A-A')} = \exp\{-\int_0^\infty 4\sin^2(ux/2) K_{1/2}(x)\,dx/x\}$ is the characteristic function of $Y_{1/2}$. The cumulants of $A$ are the moments of its Lévy measure,

and $\int_0^\infty x K_{1/2}(x)\,dx = \sum_k a_k^{-1} = \mathbb{E}T$ by $K_{1/2}(x) = \sum_k e^{-x\sqrt{a_k}}$. (i) Under RH $K_{1/2}(x) = \sum_k e^{-\gamma_k x}$, and by Frullani's integral $(E-1)/\gamma$ has Lévy measure $e^{-\gamma x}\,dx/x$ on $(0,\infty)$. The series has independent centred terms with $\sum_k \gamma_k^{-2} < \infty$, and $\mathbb{E}e^{-s(E-1)/\gamma} = e^{s/\gamma}\gamma/(\gamma+s)$ for $s > -\gamma$; the partial products are bounded uniformly on compact subsets of $(-\gamma_1,\infty)$, which gives uniform integrability and $\mathbb{E}e^{-sA} = 1/\mathcal{P}(s)$. The product $\mathcal{P}(s)\mathcal{P}(-s) = \prod_k (1 - s^2/\gamma_k^2)$ is $\xi(\frac12 + is)/\xi(\frac12)$ by Lemma 15.2.1. (ii) Uniqueness of the Lévy measure gives $K_{1/2}(x) = \sum_k e^{-c_k x}$, which is completely monotone, and Corollary 10.1.4 gives RH. (iii) For $c > 0$, differentiation in $\sqrt{u}$ gives $\log(1 + \sqrt{u}/c) = \int_0^\infty \log(1 + u/\theta^2)\,\frac{c}{\pi(c^2+\theta^2)}\,d\theta$, and $\sqrt{u}/c = \int_0^\infty \log(1+u/\theta^2)\,\frac{d\theta}{\pi c}$. At $c = \gamma_k$ the difference has the positive integrand $\log(1+u/\theta^2)\,\theta^2/\big(\pi\gamma_k(\gamma_k^2+\theta^2)\big)$, and summing over $k$ by Tonelli's theorem,

$$-\log \mathcal{P}(\sqrt{u}) = \int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big)\frac{1}{\pi}\sum_k \frac{\theta^2}{\gamma_k(\gamma_k^2+\theta^2)}\,d\theta,$$

where the left side is finite because $x - \log(1+x) \le x^2/2$ for $x \ge 0$ and $\sum_k \gamma_k^{-2} < \infty$. For the flow, $\mathbb{E}e^{iuB_{2T_t}} = \mathbb{E}e^{-u^2 T_t} = R_t(u^2) = \xi_t(\frac12)/\xi_t(\frac12 + u) = \prod_k (1 + u^2/\theta_k(t)^2)^{-1}$ by Proposition 8.4.18 and Lemma 15.2.1, and the rest is (i) with $\theta_k(t)$ in place of $\gamma_k$. □

So $\operatorname{sd} A = (\mathbb{E}T)^{1/2} = 0.152003$, the excess kurtosis of $A$ is $6\operatorname{Var}T/(\mathbb{E}T)^2 = 0.41779$, and under RH its skewness is $2\sum_k \gamma_k^{-3}/(\mathbb{E}T)^{3/2} = 0.4155$ (computed, `hx_half.py`). For the lattice $\gamma_k = k$ the series is the Gumbel law less Euler's constant (Section 25.4), with skewness $12\sqrt{6}\,\zeta(3)/\pi^3 = 1.1395$ and excess kurtosis $\frac{12}{5}$; the ordinates spread the mass over many comparable rates, and $A$ is much closer to normal.

Proposition 10.1.6 (Primes as poles of the Poisson trace). *With Cramér's function $V$ defined above, whose sum runs over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$, $K_{1/2}(x) = e^{-ix/2}V(ix)$, and the series converges for* $\operatorname{Re} x > 0$. *The function $K_{1/2}$ continues to a meromorphic function on $\mathbb{C}\setminus(-\infty,0]$ (the point $x = 0$ is a logarithmic branch point), whose poles are simple and lie at $x = \pm i\log n$ for the prime powers $n$, with residue $-\Lambda(n)/(2\pi\sqrt{n})$.*

*Proof.* The relation to $V$ is the definition read at $z = ix$, since $\sqrt{a_k} = -i(\rho - \frac12)$ for the zero $\rho$ with $\gamma > 0$ that belongs to $a_k$, and the continuation of $V$ with its simple poles at $z = \pm m\log p$, $p$ prime and $m \ge 1$, is due to Cramér [96]. For the residue fix $n \ge 2$. By Landau's formula [325], $\sum_{0<\gamma\le T} n^\rho = -\frac{T}{2\pi}\Lambda(n) + O(\log T)$, and since $\rho \mapsto 1 - \bar\rho$ permutes the zeros with $\gamma > 0$, $S(T) = \sum_{0<\gamma\le T} n^{-\rho} = n^{-1}\overline{\sum_{0<\gamma\le T} n^\rho} = -cT + O(\log T)$ with $c = \Lambda(n)/(2\pi n)$. For $\varepsilon > 0$,

$$K_{1/2}(\varepsilon + i\log n) = \sqrt{n}\sum_{\gamma>0} e^{i\varepsilon(\rho-\frac12)}n^{-\rho}, \qquad e^{i\varepsilon(\rho-\frac12)} = e^{-\varepsilon\gamma}e^{i\varepsilon(\beta-\frac12)}.$$

Since $|e^{i\varepsilon(\beta-\frac12)}-1|\le\varepsilon/2$, $|n^{-\rho}|\le1$ and $\sum_{\gamma>0}e^{-\varepsilon\gamma}=O(\varepsilon^{-1}\log(1/\varepsilon))$ by the zero count, replacing $e^{i\varepsilon(\beta-\frac12)}$ by 1 changes $\varepsilon K_{1/2}(\varepsilon+i\log n)$ by $O(\varepsilon\log(1/\varepsilon))$. Abel summation against $S$ gives

$$\varepsilon\sum_{\gamma>0}e^{-\varepsilon\gamma}n^{-\rho}=\varepsilon^2\int_0^\infty e^{-\varepsilon T}S(T)\,dT=-c+O\big(\varepsilon\log(1/\varepsilon)\big),$$

hence $\varepsilon K_{1/2}(\varepsilon+i\log n)\to-\sqrt{n}\,c=-\Lambda(n)/(2\pi\sqrt{n})$ as $\varepsilon\to0^+$. This limit is the residue: near $i\log n$ Cramér's continuation has the form $K_{1/2}(x)=r\,(x-i\log n)^{-1}+h(x)$ with $h$ holomorphic, so along the horizontal ray $x=i\log n+\varepsilon$ from the right, on which the series converges, $\varepsilon K_{1/2}(x)=r+\varepsilon h(x)\to r$ (a pole of higher order would make the limit infinite). Since the zeros $\rho$ and $1-\bar\rho$ have conjugate values of $\sqrt{a_k}$, $K_{1/2}(\bar x)=\overline{K_{1/2}(x)}$, so the residue at $-i\log n$ is the conjugate of the real number $-\Lambda(n)/(2\pi\sqrt{n})$, the same value. □

From the 10378 zeros below 10200, $\varepsilon K_{1/2}(\varepsilon+i\log n)$ at $\varepsilon=0.05$, 0.02, 0.01, 0.005, extrapolated linearly to $\varepsilon=0$ from the values at 0.01 and 0.005, agrees with $-\Lambda(n)/(2\pi\sqrt{n})$ to within $7.1\times10^{-4}$ for $n=2,3,4,5,7,8,9$, and for the control $n=6$, where $\Lambda(6)=0$, the extrapolated value is $5.5\times10^{-4}$, of the size of the errors (computed, `hx_cramer.py`). So the Poisson trace carries the zeros as exponentials on the positive axis and the primes as poles on the imaginary axis.

### 10.1.5 The complete monotonicity conditions of the Poisson trace

By Corollary 10.1.4, RH is the complete monotonicity of the Poisson trace $K_{1/2}$. This subsection follows the first author's note on Biane's paper [426]. It compares $K_{1/2}$ with the corresponding trace of Pólya's function $P(\theta)=K_{i\theta/2}(2\pi)$ of Section 8.7, which is completely monotone with no hypothesis.

Write the zeros as $\rho_k=\beta_k+i\gamma_k$ with $\gamma_k>0$, so that $\sqrt{a_k}=\gamma_k+i\delta_k$ with $\delta_k=\frac12-\beta_k\in(-\frac12,\frac12)$, and put $\varphi_k=\arctan(\delta_k/\gamma_k)$. The series $K_{1/2}(y)=\sum_k e^{-y\sqrt{a_k}}$ converges absolutely, with all its derivatives, for $y>0$, and the zeros $\rho_k$ and $1-\bar\rho_k$ have conjugate values of $\sqrt{a_k}$. Hence

$$(-1)^nK_{1/2}^{(n)}(y)=\sum_k\operatorname{Re}\big((\sqrt{a_k})^ne^{-y\sqrt{a_k}}\big)=\sum_k|\sqrt{a_k}|^n\,e^{-\gamma_ky}\cos(n\varphi_k-\delta_ky),\tag{10.1.3}$$

with each zero counted once, with its multiplicity; a zero on the line contributes $\gamma_k^ne^{-\gamma_ky}>0$. So $K_{1/2}$ is completely monotone, and by Corollary 10.1.4 RH holds, exactly when every one of the quantities (10.1.3) is nonnegative, for $n=0,1,2,\dots$ and $y>0$.

Proposition 10.1.7. *The term of a zero with $\delta_k\ne0$ in* (10.1.3) *is positive whenever*

$$n\arctan\frac{|\delta_k|}{\gamma_k}+|\delta_k|\,y<\frac{\pi}{2}.$$

*If $\delta_k > 0$ and $\delta_k y < \frac{\pi}{2}$, the first order $n$ at which the term is negative is the least integer with $n\varphi_k - \delta_k y > \frac{\pi}{2}$, that is the least integer above $(\frac{\pi}{2} + \delta_k y)/\varphi_k$ when this ratio is not an integer. The partner $1 - \bar{\rho}_k$ has the same term.*

*Proof.* $|n\varphi_k - \delta_k y| \le n|\varphi_k| + |\delta_k| y$, and the cosine is positive on $(-\frac{\pi}{2}, \frac{\pi}{2})$. For $\delta_k > 0$ the argument $n\varphi_k - \delta_k y$ increases with $n$ from $-\delta_k y \in (-\frac{\pi}{2}, 0)$, and the cosine is first negative when it passes $\frac{\pi}{2}$. The partner has $-\delta_k$ and $-\varphi_k$, so its argument is the negative of this one. □

`biane_traces.py` finds the first negative order by evaluating the sign factor; Table 10.3 compares it with the formula, and the note's values are the same.

| $\gamma$ | $\delta$ | $y$ | first negative order | $(\frac{\pi}{2} + \delta y)/\varphi$ |
|---|---|---|---|---|
| 50 | 0.25 | 1 | 365 | 364.16 |
| 50 | 0.25 | 0.3 | 330 | 329.16 |
| 200 | 0.1 | 1 | 3342 | 3341.59 |
| 14.13 | 0.5 | 0.5 | 52 | 51.48 |

Table 10.3: The first order at which the term of a hypothetical zero $\frac{1}{2} - \delta + i\gamma$ in (10.1.3) is negative, against Proposition 10.1.7. The last row is a zero at the edge of the strip at the height of the first ordinate. Computed by `biane_traces.py`.

THEOREM 10.1.8. *Suppose that every zero with $0 < \gamma_k \le H$ lies on the critical line and that every zero has $|\delta_k| \le \Delta$, with $0 < \Delta \le \frac{1}{2}$. Then for $0 < y < \pi/(2\Delta)$ and every integer $n$ with $0 \le n < \pi H/(2\Delta)$, every term of* (10.1.3) *is positive, and $(-1)^n K_{1/2}^{(n)}(y) > 0$. Since $0 < \beta_k < 1$, $\Delta = \frac{1}{2}$ is admissible with no hypothesis, and with $H = T_0 = 3 \times 10^{12}$ [419] this covers $0 < y < \pi$ and every $n < \pi T_0 = 9.42 \times 10^{12}$.*

*Proof.* A zero with $\gamma_k \le H$ has $\delta_k = 0$, and its term $\gamma_k^n e^{-\gamma_k y}$ is positive. A zero with $\gamma_k > H$ has $\varphi_k$ of the sign of $\delta_k$, $n|\varphi_k| \le n|\delta_k|/\gamma_k \le n\Delta/H < \frac{\pi}{2}$ and $|\delta_k| y \le \Delta y < \frac{\pi}{2}$. So $n\varphi_k - \delta_k y$, a difference of two numbers of the same sign and of absolute value less than $\frac{\pi}{2}$, lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, and the cosine in (10.1.3) is positive. The series converges absolutely, so its sum is positive. □

Every condition that could fail has $y \ge \pi$ or $n \ge \pi H$, which is above $9.4 \times 10^{12}$; there, complete monotonicity is a characterisation of RH, not a test that a computation could run to a failure.

REMARK 10.1.9. The rectangle uses that $\varphi_k$ and $\delta_k$ have the same sign, which the sum bound of Proposition 10.1.7 discards; the observation is from the first author's note on Ostrovsky's laws and the quasi Riemann hypothesis. Under $(\mathrm{Q}_{7/8})$ of Section 16.12, which rests on the unrefereed half plane of [398], $\Delta = \frac{3}{8}$ and the theorem holds for $0 < y < \frac{4\pi}{3} = 4.1888$ and $n < \frac{4\pi}{3} T_0 = 1.26 \times 10^{13}$. Liu's

claimed half plane $\operatorname{Re} s > 0.874957$ [352], also unrefereed, would give $y < 4.1893$. The half plane at $\frac{7}{8}$ changes the constants by the factor $\frac{4}{3}$ and nothing else.

The weight of the zero $\gamma_k$ in the condition $(n, y)$ is about $\gamma_k^n e^{-\gamma_k y}$, whose logarithm $n \log \gamma - \gamma y$ is largest at

$$\gamma_* = \frac{n}{y},$$

with a width of order $\sqrt{n}/y$. The condition $(n, y)$ is therefore a smoothed probe of the zeros near the height $n/y$. Among the first 300 ordinates the dominant one is, in every case tested, the ordinate nearest to $n/y$, for instance 98.831 for $(n, y) = (100, 1)$ (`biane_traces.py`). Figure 10.2 shows two weight profiles and two sign factors. By Proposition 10.1.7 a zero at height $\gamma$ can make its term negative only when $n \gtrsim \frac{\pi}{2}\gamma/|\delta| \geq \pi\gamma$, that is when $\gamma \lesssim n/\pi$, below the peak at $n/y$ when $y < \pi$; and since $\gamma^n e^{-\gamma y}$ increases up to $n/y$, its value at $\gamma \leq n/\pi$ is at most $\exp(n(\log \frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its value at the peak, which is exponentially small in $n$. Turning this comparison into a proof would need a bound for the number of zeros off the line near each height.

For the whole sum the first failure lies far above the termwise order. We replaced two adjacent ordinates among the first 500 by a pair $\frac{1}{2} \pm \delta + i\bar{\gamma}$ at their midpoint, and located by bisection over integers the order $n^*$ at which the minimum of $(-1)^n K_{1/2}^{(n)}(y)$ over $y \in [0.85, 1.15]\, n/\bar{\gamma}$ first becomes negative; the bisection assumes that the sign is monotone in $n$, and at sampled lower orders no negative value occurs for $y \leq 30$ or $y \leq 5n/\bar{\gamma}$. For $\bar{\gamma} = 54.7$, 125.9, 272.5 and $\delta = 0.1$, 0.375 the ratio $n^*\delta/\bar{\gamma}^2$ lies between 0.63 and 1.19, the law $n^* \approx \gamma^2/\delta$ of the heat trace (Table 7.6), against the termwise order $\pi\bar{\gamma}/(2\delta)$ near $y = 0$.

Proposition 10.1.10. *The following are equivalent:*

*(i) RH;*

*(ii) $K_{1/2}$ is completely monotone on $(0, \infty)$;*

*(iii) for every $x > 0$ the function $y \mapsto K_{1/2}(x + iy)$ is positive definite on $\mathbb{R}$;*

*(iv) for some $x > 0$ the function $y \mapsto K_{1/2}(x + iy)$ is bounded on $[0, \infty)$.*

*Under these conditions $|K_{1/2}(x + iy)| \leq K_{1/2}(x)$.*

*Proof.* The series $K_{1/2}(z) = \sum_k e^{-z\sqrt{a_k}}$ converges absolutely and locally uniformly for $\operatorname{Re} z > 0$, since $|e^{-z\sqrt{a_k}}| = e^{-\gamma_k \operatorname{Re} z + \delta_k \operatorname{Im} z} \leq e^{-\gamma_k \operatorname{Re} z + |\operatorname{Im} z|/2}$, and defines an analytic function there. (i) implies (ii) by Corollary 10.1.4. If (ii) holds, Bernstein's theorem gives $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta}\nu(d\theta)$ with a positive measure $\nu$, the integral converges for $\operatorname{Re} z > 0$ and agrees there with $K_{1/2}(z)$ by analytic continuation, and $K_{1/2}(x + iy) = \int e^{-iy\theta} e^{-x\theta}\nu(d\theta)$ is the Fourier transform of the finite positive

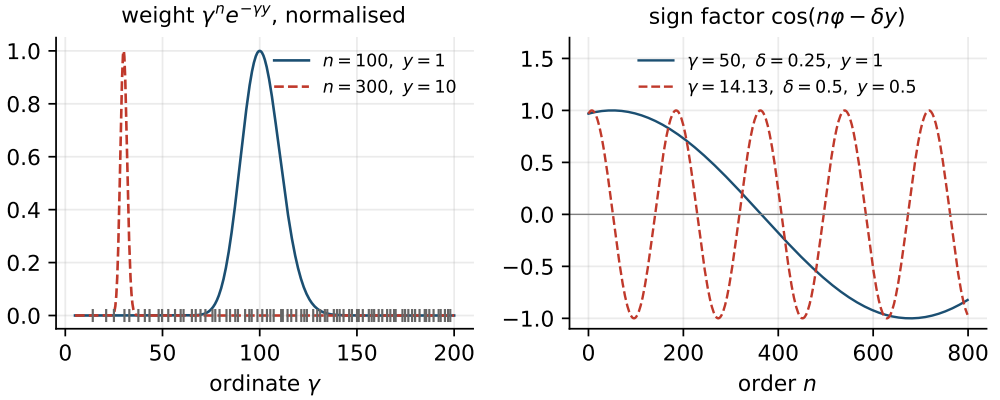



Figure 10.2: Left: the weight $\gamma^n e^{-\gamma y}$ of the condition $(n, y)$ in (10.1.3), normalised to maximum one, for $(n, y) = (100, 1)$ and $(300, 10)$, peaking at $\gamma = n/y$; the marks are the ordinates below 200. Right: the sign factor $\cos(n\varphi - \delta y)$ of a hypothetical zero off the line for $(\gamma, \delta, y) = (50, 0.25, 1)$ and $(14.13, 0.5, 0.5)$, positive below the orders 365 and 52, at which it is first negative. Drawn by `biane_figs.py`.

measure $e^{-x\theta}\nu(d\theta)$, which gives (iii) and the bound. A positive definite function $f$ satisfies $|f(y)| \le f(0)$, so (iii) implies (iv). Suppose (iv) holds at $x$. For $\operatorname{Re} w > \frac{1}{2}$ the bound above allows term by term integration, and

$$\int_0^\infty e^{-wy} K_{1/2}(x + iy)\, dy = \sum_k \frac{e^{-x\sqrt{a_k}}}{w + i\sqrt{a_k}}.$$

The right side is a meromorphic function of $w \in \mathbb{C}$, since $\sum_k e^{-x\gamma_k} < \infty$, with a pole at each point $-i\sqrt{a_k} = \delta_k - i\gamma_k$, of residue $e^{-x\sqrt{a_k}}$ times the multiplicity, which is not zero. Since $K_{1/2}(x + iy)$ is bounded, the left side is analytic in $\operatorname{Re} w > 0$, so no pole has $\delta_k > 0$. The zeros $\rho_k$ and $1 - \bar\rho_k$ have opposite $\delta_k$, so every $\delta_k$ is 0, which is RH. □

Written out, (iii) says that for every $x > 0$, all real $y_1, \dots, y_m$ and all complex $c_1, \dots, c_m$,

$$\sum_{j,l} c_j \bar c_l\, K_{1/2}\big(x + i(y_j - y_l)\big) = \sum_k e^{-x\gamma_k} \Big| \sum_j c_j e^{-i\gamma_k y_j} \Big|^2 \ge 0,$$

the equality holding under RH.

One would like to turn each condition into a statement about primes with the explicit formula of Section 1.2, the display before (1.2.5). The test function of the condition $(n, y)$ is $h(r) = |r|^n e^{-y|r|}$, with

$$g(x) = \frac{1}{2\pi} \int_{\mathbb{R}} h(r) e^{-irx}\, dr = \frac{n!}{\pi} \operatorname{Re} \frac{1}{(y + ix)^{n+1}},$$

which decays only like a power of $|x|$ and has one sign for large $|x|$. The prime side $-2\sum_{m\ge2}\Lambda(m)m^{-1/2}g(\log m)$ then diverges, since $\sum_p p^{-1/2}(\log p)^{-j}=\infty$ for every $j$; the formula holds for $g$ decaying faster than $e^{-|x|/2}$, which goes with $h$ analytic in a strip $|\operatorname{Im} r|<\frac12+\epsilon$, and this $h$ is not even analytic at $r=0$. Smoothing in the variable of the zeros removes the divergence without changing the problem: the Gaussian weight $e^{-r^2t}$ in place of $e^{-y|r|}$ gives the heat trace $W(t)$, whose explicit formula (1.2.5) converges absolutely, and whose complete monotonicity is again equivalent to RH (Corollary 10.1.4). The note [426] identifies this smoothing with the de Bruijn Newman flow, but the Gaussian weight acts on the test function, while the flow multiplies the kernel $\Phi$ by $e^{tu^2/4}$ and moves the zeros.

By Proposition 8.7.13,

$$\log\frac{K_{\lambda/2}(2\pi)}{K_0(2\pi)}=\int_0^\infty 4\sin^2\Big(\frac{\lambda y}{2}\Big)K^P(y)\,\frac{dy}{y},$$
$$K^P(y)=\sum_n e^{-2\nu_n y}=\operatorname{tr}e^{-2y\sqrt{A_P}},$$

the sum over the zeros $\theta_n=2\nu_n$ of $P$, so $K^P$ is the counterpart of $K_{1/2}$ in the formula for $\xi(\frac12)/\xi(\frac12+u)$, and it is completely monotone because $A_P$ is selfadjoint and positive.

Proposition 10.1.11. *Let* $M(y)=\frac{1}{2\pi y}\big(\log\frac{1}{2\pi y}-\gamma_E\big)$ *and* $c=2\pi-\frac{1}{6\pi}-\frac{1}{48\pi}=6.2235022035$. *As* $y\to0+$:

*(i)* $K_{1/2}(y)=M(y)+\frac78+O(y\log\frac1y)$, *and under RH* $K_{1/2}(y)=M(y)+\frac78+\frac{1}{48\pi}y\log\frac1y+o(y\log\frac1y)$;

*(ii)* $K^P(y)=M(y)-\frac14+\big(2\pi-\frac{1}{6\pi}\big)y\log\frac1y+O(y)$;

*(iii)* $D(y)=K_{1/2}(y)-K^P(y)=\frac98+O(y\log\frac1y)$, *and under RH* $D(y)=\frac98-c\,y\log\frac1y+o(y\log\frac1y)$.

*Proof.* (i) The real part of the term $e^{-y\sqrt{a_k}}$ is $e^{-\gamma_k y}\cos(\delta_k y)=e^{-\gamma_k y}+O(y^2e^{-\gamma_k y})$, and the imaginary parts cancel in pairs, so $K_{1/2}(y)=\sum_k e^{-\gamma_k y}+O(y^2\sum_k e^{-\gamma_k y})$, and the sum will turn out to be $O(y^{-1}\log\frac1y)$. By Abel summation $\sum_k e^{-\gamma_k y}=y\int_0^\infty e^{-Ty}N(T)\,dT$, with $N(T)=\vartheta(T)/\pi+1+S(T)$ [498, Section 9.3]. The Riemann Siegel function satisfies $\vartheta(T)/\pi=N_0(T)-\frac18+\frac{1}{48\pi T}+O(T^{-3})$ with $N_0(T)=\frac{T}{2\pi}\log\frac{T}{2\pi e}$, so $N=N_0+\frac78+S+E$ with $E(T)=\frac{1}{48\pi T}+O(T^{-3})$ for $T\ge1$ and $E$ bounded on $(0,1]$. Now $\int_0^\infty Te^{-Ty}\log T\,dT=y^{-2}(1-\gamma_E-\log y)$ gives $y\int_0^\infty e^{-Ty}N_0(T)\,dT=M(y)$; the constant gives $\frac78$; $y\int_1^\infty e^{-Ty}\frac{dT}{48\pi T}=\frac{1}{48\pi}y\log\frac1y+O(y)$, and the rest of $E$ gives $O(y)$. With $S_1(T)=\int_0^T S(t)\,dt$, an integration by parts gives $y\int_0^\infty e^{-Ty}S(T)\,dT=y^2\int_0^\infty e^{-Ty}S_1(T)\,dT$, which is $O(y\log\frac1y)$

since $S_1(T) = O(\log T)$ [498, Theorem 9.9(A)]. Under RH there are no terms $O(y^2 e^{-\gamma_k y})$, and $S_1(T) = O(\log T/(\log\log T)^2)$ [498, Theorem 14.13] makes the last integral $o(y \log \frac{1}{y})$. (ii) By Proposition 8.7.8, the number of zeros $\theta_n \le T$ is $\lfloor x(T) \rfloor$, with $x(T) = N_0(T) + \frac{1}{4} + (2\pi - \frac{1}{6\pi})T^{-1} + O(T^{-3})$ for $T \ge 1$. Write $\lfloor x \rfloor = x - \frac{1}{2} + (\frac{1}{2} - \{x\})$. The terms $N_0$, $\frac{1}{4} - \frac{1}{2}$ and $(2\pi - \frac{1}{6\pi})T^{-1}$ give $M(y)$, $-\frac{1}{4}$ and $(2\pi - \frac{1}{6\pi}) y \log \frac{1}{y} + O(y)$ as in (i). The sawtooth $\frac{1}{2} - \{x(T)\}$ has integral over each interval on which $x$ increases by one equal to $O(T^{-1}(\log T)^{-3})$, because such an interval has length about $2\pi/\log T$ and $x'(T) = \frac{1}{2\pi}\log\frac{T}{2\pi} + O(T^{-2})$ varies over it by $O(T^{-1}(\log T)^{-2})$ relative to its size, so its integral from 0 to $T$ is bounded, and another integration by parts makes its contribution $O(y)$. (iii) is (i) minus (ii). □

`biane_traces.py` tests the proposition. It takes the first 30000 ordinates from Arb, up to 25755.348, cuts at $G = 25754.905$, midway between the last two, and adds the tail $\int_G^\infty e^{-Ty}\vartheta'(T)\,dT/\pi - e^{-Gy}S(G)$, with $S(G) = -0.569$ exact; the neglected term $y\int_G^\infty e^{-Ty}S(T)\,dT$ equals $y^2\int_G^\infty e^{-Ty}\int_G^T S(t)\,dt\,dT$, and Trudgian's bound $|\int_{t_1}^{t_2} S(t)\,dt| \le 2.067 + 0.059\log t_2$ for $t_2 > t_1 > 168\pi$ [501], with Jensen's inequality for the logarithm, bounds it by $ye^{-Gy}(0.059\log(G + 1/y) + 2.067)$; the script records twice this bound, $4.1\times10^{-5}$ and $4.3\times10^{-5}$ at $y = 10^{-4}$ and $10^{-5}$, so the entries of Table 10.4 at $y = 10^{-4}$ and $10^{-5}$ may be off by up to two units in the fifth decimal. The zeros of $P$ come from the phase condition of Proposition 8.7.8, solved by Newton's method for all 30132 zeros to just above the cut, with the tail beyond $G$ added. The factor $e^{-Gy}$ that weights the tails is 0.076 at $y = 10^{-4}$ and 0.77 at $y = 10^{-5}$, so the last row of Table 10.4 rests mostly on the smooth tails. The constant $\frac{9}{8}$ appears clearly, and $D$ approaches it along $\frac{9}{8} - cy\log\frac{1}{y}$.

| $y$ | $K_{1/2}(y)$ | $K^P(y)$ | $K_{1/2} - M$ | $K^P - M$ | $D(y)$ | $\frac{9}{8} - cy\log\frac{1}{y}$ |
|---|---|---|---|---|---|---|
| 0.05 | 2.686015 | 2.230883 | 0.837782 | 0.382650 | 0.455132 | 0.192803 |
| 0.02 | 12.772471 | 12.033884 | 0.860279 | 0.121691 | 0.738587 | 0.638070 |
| 0.01 | 35.723862 | 34.836495 | 0.867696 | −0.019670 | 0.887366 | 0.838397 |
| 0.005 | 92.647265 | 91.663043 | 0.871374 | −0.112848 | 0.984222 | 0.960130 |
| 0.003 | 180.932794 | 179.901902 | 0.872836 | −0.158057 | 1.030892 | 1.016540 |
| 0.001 | 715.903738 | 714.816981 | 0.874286 | −0.212471 | 1.086757 | 1.082010 |
| $10^{-4}$ | 10815.847446 | 10814.727704 | 0.874931 | −0.244811 | 1.119742 | 1.119268 |
| $10^{-5}$ | 144797.380092 | 144796.255761 | 0.874994 | −0.249338 | 1.124332 | 1.124283 |

Table 10.4: The Poisson trace $K_{1/2}$ of the zeros of $\zeta$, Pólya's trace $K^P$, their distances from the common main term $M$, which tend to $\frac{7}{8}$ and $-\frac{1}{4}$, and their difference $D$, which tends to $\frac{9}{8}$, against the prediction of Proposition 10.1.11 under RH. Computed by `biane_traces.py`.

The note [426] states that $K_{1/2}(y) = M(y) + O(\log\frac{1}{y})$, which is true but loses the constant, and it reports $D(y)/\log\frac{1}{y} = 0.195,\ 0.194,\ 0.193,\ 0.193$ at

$y = 0.003, 0.005, 0.01, 0.02$, from which it concludes that $D$ grows like $0.19 \log \frac{1}{y}$. The accurate values of $D/\log\frac{1}{y}$ in this range, 0.177, 0.186, 0.193, 0.189, are also close to 0.19, but that is a coincidence of the range: by Proposition 10.1.11, $D$ tends to $\frac{9}{8}$, and $D/\log\frac{1}{y}$ is 0.098 at $y = 10^{-5}$.

$K^P$ is completely monotone for a structural reason, as the trace of the semigroup $e^{-2y\sqrt{A_P}}$. By Corollary 10.1.4, RH says that $K_{1/2} = K^P + D$ is completely monotone as well, where $D$ is bounded, tends to $\frac{9}{8}$ while both traces grow like $\frac{1}{2\pi y}\log\frac{1}{y}$, and carries the whole difference between $\Phi$ and Pólya's kernel, the arithmetic included. Complete monotonicity constrains the derivatives of every order, and no bound on the size of $D$ can decide it.

### 10.1.6 Wiener's theorem and the basepoint kernels

The zero free half planes have a second reading, as completeness statements, through Wiener's closure theorem [515], for which Kac [294] and Korevaar [314] gave short alternative treatments: the translates $f(\cdot - x)$, $x \in \mathbb{R}$, of a function $f \in L^1(\mathbb{R})$ span a dense subspace of $L^1(\mathbb{R})$ if and only if its Fourier transform $\hat{f}(\theta) = \int f(u)e^{i\theta u}du$ has no real zeros.

Let $\Phi$ be Pólya's kernel (1.2.3), so that $\xi(\frac{1}{2}+s) = \int_{\mathbb{R}} \Phi(u)e^{su}du$ for every complex $s$ by (1.2.4), and for $\alpha \geq \frac{1}{2}$ put

$$\Phi_\alpha(u) = e^{(\alpha-\frac{1}{2})u}\,\Phi(u), \qquad \hat{\Phi}_\alpha(\theta) = \int_{\mathbb{R}} \Phi_\alpha(u)e^{i\theta u}du = \xi(\alpha + i\theta). \tag{10.1.4}$$

Up to the factor $\xi(\alpha)$, $\Phi_\alpha$ is the density of the tilted Pólya variable $V_\alpha$ of Section 9.1.1, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and it is integrable for every $\alpha$ because $\Phi$ decays doubly exponentially. The real zeros of $\hat{\Phi}_\alpha$ are exactly the ordinates of the zeros of $\zeta$ on the line $\operatorname{Re} s = \alpha$, so Wiener's theorem gives the following.

Proposition 10.1.12. *1. The translates of $\Phi_\alpha$ are dense in $L^1(\mathbb{R})$ if and only if $\zeta$ has no zeros on the line $\operatorname{Re} s = \alpha$.*

*2. At the centre they are not dense. For every finite combination $h = \sum_k c_k \Phi(\cdot - x_k)$ and every $g \in L^1(\mathbb{R})$,*

$$\|g - h\|_1 \;\geq\; \sup_\gamma |\hat{g}(\gamma)|,$$

*the supremum over the ordinates of the zeros on the critical line.*

*3. At the basepoint one they are dense, with no hypothesis, because $\xi(1+i\theta) \neq 0$.*

*4. For $\alpha_0 \in [\frac{1}{2}, 1)$, $\zeta$ has no zeros in the half plane $\operatorname{Re} s > \alpha_0$ if and only if the translates of $\Phi_\alpha$ are dense in $L^1(\mathbb{R})$ for every $\alpha > \alpha_0$. In particular RH holds if and only if they are dense for every $\alpha > \frac{1}{2}$.*

5. *If the translates of $\Phi_\alpha$ are dense, approximation near the zeros is expensive: if $h = \sum_k c_k \Phi_\alpha(\cdot - x_k)$ and $\|g-h\|_1 \le \delta$, then $\sum_k |c_k| \ \ge \ (|\hat g(\gamma)| - \delta)/|\xi(\alpha + i\gamma)|$ for every ordinate $\gamma$.*

*Proof.* (1), (3) and (4) are Wiener's theorem applied to $\hat\Phi_\alpha(\theta) = \xi(\alpha + i\theta)$, together with the nonvanishing of $\zeta$ on the line $\operatorname{Re} s = 1$. For (2) and (5), $\hat h(\theta) = \hat\Phi_\alpha(\theta) \sum_k c_k e^{i\theta x_k}$, so at the centre $\hat h(\gamma) = 0$ at every ordinate of a zero on the line, while $|\hat g(\theta) - \hat h(\theta)| \le \|g - h\|_1$ for every $\theta$; for (5), $|\hat h(\gamma)| \le |\xi(\alpha + i\gamma)| \sum_k |c_k|$. $\square$

Wiener's proof of the prime number theorem runs through statement (3), and the proposition shows the whole family interpolating between that theorem at $\alpha = 1$ and RH at $\alpha = \frac12$. It is a restatement rather than a new criterion. By statement (4), completeness at every $\alpha' > \alpha$ is the condition $\alpha \ge \sup_\rho \operatorname{Re}\rho$ under which $T_\alpha$ is a GGC (Theorem 1.2.2) and $Y_\alpha$ an EGGC (Theorem 10.1.2), and, by Lagarias's argument [320], it is also the positivity of $v_{\alpha'}$ for every $\alpha' > \alpha$ (Remark 1.2.4). What the completeness form adds is an explicit, computable obstruction at the centre, statement (2), and an explicit cost of approximation above it, statement (5): near the centre $|\xi(\alpha + i\gamma)| \approx (\alpha - \frac12)|\xi'(\frac12 + i\gamma)|$ at the ordinate $\gamma$ of a simple zero, so the coefficients needed to reproduce $\hat g$ at an ordinate grow like $1/(\alpha - \frac12)$.

We tested the dichotomy with `wiener.py`. We approximated the Gaussian $g(u) = (2\pi\kappa^2)^{-1/2} e^{-u^2/2\kappa^2}$ with $\kappa = 0.15$, for which $\hat g(\gamma_1) = e^{-\kappa^2\gamma_1^2/2} = 0.10565$, by combinations of $N$ equally spaced translates of $\Phi_\alpha$ on $[-1.5, 1.5]$, minimising the $L^1$ error on $[-3, 3]$ exactly by linear programming. Table 10.5 shows the results. At the centre every approximant has $\hat h(\gamma_1) = 0$ up to rounding, the computed values staying below $1.2 \times 10^{-8}$ while the coefficient sums grow to $4 \times 10^9$, and the best error stalls at 0.1345, above the barrier 0.1056 of statement (2), however many translates are used. Above the centre the error falls below the value at which the centre stalls and keeps decreasing, to below the barrier at $\alpha = 0.75$ and $\alpha = 1$. The remaining error is limited by conditioning rather than by completeness. With 80 translates the coefficient sums reach $3\times10^7$ to $2\times10^8$ above the centre, as statement (5) and the rapid decay of $\hat\Phi_\alpha$ require, and double precision linear programming cannot go much further. The closer $\alpha$ is to $\frac12$, the larger the coefficients and the slower the approach, which is the completeness counterpart of the Cauchy kernels sharpening onto the zeros.

### 10.1.7 Salem's criterion and its Pólya analogue

Wiener's theorem has a dual form: for $k \in L^1(\mathbb{R})$, the convolution equation $\int k(x-y) f(y)\, dy = 0$ for all $x$ has a bounded solution $f \ne 0$ if and only if $\hat k$ has a real zero. Salem [468] used it to give one of the classical equivalents of RH: RH holds

| $N$ | $\alpha=\frac12$ | $\alpha=0.6$ | $\alpha=0.75$ | $\alpha=1$ |
|---|---|---|---|---|
| 10 | 0.4705 | 0.4698 | 0.4664 | 0.4540 |
| 20 | 0.1345 | 0.1221 | 0.1060 | 0.0852 |
| 40 | 0.1345 | 0.1152 | 0.0907 | 0.0614 |
| 80 | 0.1365 | 0.1139 | 0.0878 | 0.0570 |
| $\lvert\hat h(\gamma_1)\rvert$ at $N=80$ | 0 | 0.0167 | 0.0377 | 0.0635 |

Table 10.5: Best $L^1$ error in approximating a Gaussian of width 0.15 by $N$ translates of $\Phi_\alpha$, computed by `wiener.py`. At the centre the error cannot fall below $|\hat g(\gamma_1)|=0.1056$, and $\hat h(\gamma_1)$ vanishes exactly; the computed value is $1.2\times10^{-8}$.

if and only if, for every $\frac12<\sigma<1$, the equation

$$\int_{\mathbb R}\frac{e^{-\sigma y}f(y)}{e^{e^{x-y}}+1}\,dy=0\qquad(x\in\mathbb R)$$

has no bounded solution other than $f=0$. Multiplied by $e^{\sigma x}$, it is the convolution equation with the kernel $k_\sigma(u)=e^{\sigma u}/(e^{e^u}+1)$, the Fermi Dirac function $1/(e^w+1)$ in the logarithmic variable $w=e^u$, and that substitution gives its Fourier transform,

$$\hat k_\sigma(\theta)=\int_{\mathbb R}\frac{e^{\sigma u}e^{i\theta u}}{e^{e^u}+1}\,du=\int_0^\infty\frac{w^{\sigma+i\theta-1}}{e^w+1}\,dw=\Gamma(\sigma+i\theta)\,\eta(\sigma+i\theta),$$
$$\eta(s)=(1-2^{1-s})\zeta(s),$$

where $\eta$ is Dirichlet's alternating zeta function, not the Dedekind eta function of Section 11.3. So for $\frac12<\sigma<1$ the real zeros of $\hat k_\sigma$ are exactly the ordinates of the zeros of $\zeta$ on the line $\operatorname{Re}s=\sigma$, and Salem's criterion is Proposition 10.1.12(4) with the Fermi Dirac kernel in place of the tilted Pólya kernel (10.1.4).

Corollary 10.1.13. *RH holds if and only if, for every $\frac12<\alpha<1$, the equation $\int_{\mathbb R}\Phi_\alpha(x-y)f(y)\,dy=0$ for all $x$ has no bounded solution $f\neq0$. At the centre the equation has the explicit bounded solutions $f(y)=\cos(\gamma y)$, one for each ordinate $\gamma$ of a zero on the critical line, because $\Xi(\gamma)=0$ there; at the basepoint one it has none.*

*Proof.* By the dual form of Wiener's theorem with $k=\Phi_\alpha$, a bounded solution $f\neq0$ exists exactly when $\hat\Phi_\alpha(\theta)=\xi(\alpha+i\theta)$ has a real zero, that is, when $\zeta$ has a zero on the line $\operatorname{Re}s=\alpha$. That no such zero exists for any $\alpha\in(\frac12,1)$ means that no zero has $\frac12<\operatorname{Re}\rho<1$, which, since no zero lies on the line $\operatorname{Re}s=1$, is RH by the symmetry $\rho\mapsto1-\rho$. At the centre $\Phi$ is even, so $\int_{\mathbb R}\Phi(x-y)\cos(\gamma y)\,dy=\Xi(\gamma)\cos(\gamma x)$, which vanishes when $\frac12+i\gamma$ is a zero; at the basepoint one $\xi(1+i\theta)\neq0$ for every real $\theta$. □

The factor $1-2^{1-s}$ in $\eta$ vanishes at $s = 1 + 2\pi ik/\log 2$ with $k \neq 0$, so Salem's kernel has real Fourier zeros on the line $\sigma = 1$, at $\theta = 9.0647,\ 18.1294,\ 27.1942, \ldots$, and its translates are not complete there, whereas the Pólya family is complete at $\alpha = 1$ because $\xi(1 + i\theta) \neq 0$. The left panel of Figure 10.3, computed by `salem_riesz.py`, shows $|\eta(\sigma + i\theta)|$. At $\sigma = \frac{1}{2}$ it dips to zero at every ordinate, and at $\sigma = 0.6$ and 0.75 the dips are shallow. The depth of the dips plays for Salem's kernel the role that $|\xi(\alpha + i\gamma)|$ plays in the cost of approximation of Proposition 10.1.12(5), and it is of order $\sigma - \frac{1}{2}$ near each ordinate.

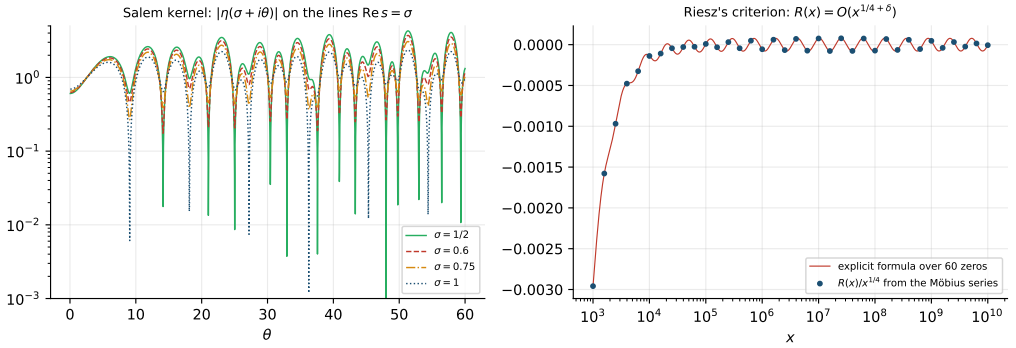



Figure 10.3: Left: $|\eta(\sigma + i\theta)|$, the Fourier transform of Salem's kernel divided by $\Gamma(\sigma + i\theta)$, at $\sigma = \frac{1}{2}$, 0.6, 0.75, 1. Right: the Riesz function $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula (10.1.7) over 60 zeros (`figs_wiener.py`).

### 10.1.8 Riesz's criterion and multiplicative completeness

Riesz [459] found the first criterion of this type, in terms of growth rather than completeness. With the Möbius function $\mu(n)$, which is unrelated to the Thorin measure $\mu_\varepsilon$ of Section 1.2, put

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!\,\zeta(2k)} = x \sum_{n\geq 1} \frac{\mu(n)}{n^2}\, e^{-x/n^2}, \tag{10.1.5}$$

the second form following from $1/\zeta(2k) = \sum_n \mu(n) n^{-2k}$. Riesz proved that RH holds if and only if $R(x) = O(x^{1/4+\delta})$ for every $\delta > 0$. The Mellin transform explains why: for $\frac{1}{2} < \mathrm{Re}\, s < 1$,

$$\int_0^\infty R(x)\, x^{-s-1} dx = \frac{\Gamma(1-s)}{\zeta(2s)}, \tag{10.1.6}$$

whose poles in the strip $0 < \operatorname{Re} s < \frac{1}{2}$ sit at $s = \rho/2$, and shifting the contour gives, when the zeros are simple, the explicit formula

$$R(x) = \sum_{\rho} \frac{\Gamma(1-\rho/2)}{2\zeta'(\rho)}\, x^{\rho/2} + \sum_{k\geq 1} \frac{\Gamma(1+k)}{2\zeta'(-2k)}\, x^{-k}, \tag{10.1.7}$$

in which the sum over the zeros is the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence of heights $T_\nu \to \infty$. The same argument gives the basepoint version: for $\frac{1}{2} \leq \alpha < 1$, $\zeta$ has no zeros in $\operatorname{Re} s > \alpha$ if and only if $R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. Proposition 10.2.1 carries out the proof in detail, for a twisted form of $R$: the zero free half plane bounds $1/\zeta(\sigma + it)$ by a power of $|t|$ on $\sigma \geq \alpha + 2\delta$, by the Borel Carathéodory lemma as in [498, Section 14.2] with the line $\operatorname{Re} s = \alpha$ in place of the critical line.

In the language of this book, $R(x)/x = \sum_n \mu(n) n^{-2} e^{-x/n^2}$ is the Laplace transform of the signed Möbius measure $\sum_n \mu(n) n^{-2} \delta_{1/n^2}$. It is a heat trace on the arithmetic side, with the Möbius function in place of the zeros, and it is the same Möbius series that underlies the representation of the reciprocal zeta function in Section 6.1 (Theorem 6.1.5). The coefficients $1/\zeta(2k)$ of the power series of $R$ are, up to elementary factors, values of the Laplace transforms of the centre clock $T$ and of $T_1$ (Section 10.2.2).

We computed $R(x)$ with `salem_riesz.py` from the Möbius series with $n \leq N = 10^7$, approximating

$$R(x) \approx x\Big[\sum_{n\leq N} \mu(n) n^{-2}\big(e^{-x/n^2} - 1\big) + \frac{6}{\pi^2}\Big]$$

to avoid the slowly converging tail, for $10 \leq x \leq 10^{12}$. Beyond $x \approx 10^{10}$ the truncation at $N = 10^7$ limits the accuracy: the omitted terms are about $-x^2 \sum_{n>N} \mu(n) n^{-4} \approx 9 \times 10^{-26} x^2$, that is $3 \times 10^{-8}$ in $R(x)/x^{1/4}$ at $x = 10^{10}$ and $9 \times 10^{-5}$ at $x = 10^{12}$. For $10^4 \leq x \leq 10^{10}$ the ratio $R(x)/x^{1/4}$ oscillates with amplitude about $1.1 \times 10^{-4}$, and for $10^5 \leq x \leq 10^{10}$ with amplitude $7.8 \times 10^{-5}$, which is the amplitude $|\Gamma(1-\rho_1/2)/\zeta'(\rho_1)|$ of the term of the first zero. The larger swing near $x = 10^4$ comes from the term $\Gamma(2)x^{-1}/(2\zeta'(-2)) = -16.42\, x^{-1}$ of the first trivial zero, which contributes $-1.6 \times 10^{-4}$ to the ratio at $x = 10^4$. The explicit formula (10.1.7) with the first 60 zeros and the first three trivial terms reproduces the computed values to $3.1 \times 10^{-8}$ on $[10^3, 10^{10}]$, against a signal of size up to $3 \times 10^{-3}$ (`riesz_explicit.py`, output `riesz_explicit.json`; Figure 10.3, right). The largest difference, at $x = 10^{10}$, is the truncation error just described, and on $[10^3, 10^9]$ the difference is at most $6.2 \times 10^{-10}$. Section 10.2.1 twists the Möbius series by $(x/n^2)^{ih}$ so that it responds to the zeros near height $2h$.

Wiener's theorem concerns translates. Its multiplicative counterpart concerns dilates, and there completeness is equivalent to RH itself. Nyman and Beurling [390, 46] proved that RH holds if and only if the constant function 1 lies in the

$L^2(0,1)$ closure of the span of the functions $r_\lambda(x) = \{\lambda/x\} - \lambda\{1/x\}$, $0 < \lambda \le 1$, where $\{\cdot\}$ is the fractional part, and Báez-Duarte [16] showed that $\lambda = 1/n$ suffices. Under $x = e^{-u}$ this is a closure problem for translates in a weighted $L^2$ space on the half line, and the Mellin transform of $r_\lambda$ is $-\zeta(s)(\lambda^s - \lambda)/s$, so the zeros of $\zeta$ in the half plane $\operatorname{Re} s > \frac12$ are exactly what obstructs the closure. The same framework governs the dilates $\operatorname{sech}(nx)$ of the Lai and Vere-Jones identity (Proposition 6.1.3), which lies behind the expansion of Theorem 6.1.5: the Mellin transform of sech is $2\Gamma(s)L(s,\chi_4)$, and completeness of systems of dilates is controlled by where the associated Dirichlet series vanishes, in the sense of Beurling's periodic dilation problem and the work of Hedenmalm, Lindqvist and Seip [181]. How the two questions are related is the Nyman Beurling problem of Section 7.2.4. The Nyman Beurling closure problem is equivalent to RH. For the dilates $\operatorname{sech}(nx)$ the answer is different: completeness in one weighted space $L^2(x^{2c-1}dx)$, $c > 0$, holds unconditionally (Theorem 29.3.2), and only the two weight version is equivalent to the generalised Riemann hypothesis for $L(s,\chi_4)$ (Theorem 29.5.2, Chapter 29). The additive family of Proposition 10.1.12 is the probabilistic one, and it reaches RH only as $\alpha \downarrow \frac12$, in the same way as every other construction of Chapters 9 to 11.

## 10.2 Riesz's criterion and the Thorin path

Section 10.1.8 introduced Riesz's criterion [459] for the Riesz function (10.1.5) and its basepoint form, of which RH is the case $\alpha = \frac12$. Hardy and Littlewood [173] gave a companion criterion with $\zeta(2k+1)$ in place of $\zeta(2k)$ and the exponent $-\frac14$, and the constructions of this section apply to it in the same way.

We develop the criterion in two directions. The first is computational (Section 10.2.1). The second reads the coefficients $1/\zeta(2k)$ through the clocks of Chapter 9. None of this supplies the growth bound (Section 10.2.6).

We keep the notation of Chapter 9 and Section 10.1 for $T = T_{1/2}$, $T_\alpha$, $\varphi_\alpha$, $H_a$, $b_\alpha$ and $v_\alpha$; in particular $b_1 = \sum_\rho \rho^{-1} = 0.0230957$. We write $\widetilde{T} = \widetilde{T}_1$ for the centre clock tilted by $e^{-T/4}$, as in Theorem 9.2.2, $\psi(y) = \sum_{n \le y} \Lambda(n)$ for Chebyshev's function, and

$$\kappa_k = \frac{2k(2k-1)(k-1)!}{\pi^k}, \qquad c_k = \frac{1}{\kappa_k\, \zeta(2k)},$$

so that $\xi(2k) = \frac12 \kappa_k \zeta(2k)$ and $c_k = \xi(1)/\xi(2k)$. The scripts named below are in `riesz_program/code` of the archive, their outputs in `riesz_program/data`.

### 10.2.1 Which zeros the Riesz function sees

On the range computed in Section 10.1.8 the Riesz function behaves exactly as RH predicts, and its fluctuations are accounted for by the known zeros, in fact by very few of them. In the explicit formula (10.1.7) a zero $\rho$ enters $R(x)$ through the term

$r_\rho x^{\rho/2}$ with the weight $r_\rho = \Gamma(1-\rho/2)/(2\zeta'(\rho))$. By Stirling's formula $|\Gamma(\frac{3}{4} - \frac{i}{2}\gamma)|$ decays like $e^{-\pi\gamma/4}$. The weight is $3.9\times10^{-5}$ for the first zero, whose term and that of its conjugate give the amplitude $7.8\times10^{-5}$ found in Section 10.1.8, $2.1\times10^{-17}$ for the tenth, $1.5\times10^{-49}$ for the fiftieth and $2.2\times10^{-81}$ for the hundredth (`sens.py`, output `sens.json`; Figure 10.4, left). An off line zero at height 1000 would enter with a weight near $10^{-340}$, and no feasible range of $x$ would reveal it. Riesz's criterion is equivalent to RH, but as a computational test it examines only the bottom of the spectrum.

The remedy is to change the kernel. For real $h$ put

$$R_h(x) = x \sum_{n\ge1} \frac{\mu(n)}{n^2} \Big(\frac{x}{n^2}\Big)^{ih} e^{-x/n^2}, \qquad x > 0,$$

so that $R_0 = R$.

Proposition 10.2.1. *For every real $h$ and $\frac12 < \operatorname{Re} s < 1$,*

$$\int_0^\infty R_h(x)\, x^{-s-1} dx = \frac{\Gamma(1-s+ih)}{\zeta(2s)}.$$

*For $\frac12 \le \alpha < 1$, $\zeta$ has no zeros in the half plane $\operatorname{Re} s > \alpha$ if and only if $R_h(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. In particular RH holds if and only if $R_h(x) = O(x^{1/4+\delta})$ for every $\delta > 0$.*

*Proof.* Term by term, $\int_0^\infty x^{1+ih} n^{-2-2ih} e^{-x/n^2} x^{-s-1} dx = n^{-2s}\Gamma(1-s+ih)$, and summing against $\mu(n)$ gives $\Gamma(1-s+ih)\sum_n \mu(n) n^{-2s}$; the interchange is justified because, with $\sigma = \operatorname{Re} s$, the integrals of the absolute values sum to $\Gamma(1-\sigma)\zeta(2\sigma) < \infty$. Suppose that $\zeta$ has no zeros in $\operatorname{Re} s > \alpha$. Then $1/\zeta(2s)$ is analytic in $\operatorname{Re} s > \frac{\alpha}{2}$. By the argument of [498, Section 14.2] with the line of real part $\alpha$ in place of the critical line, as in Section 10.1.8, $|1/\zeta(2s)| \le |2\operatorname{Im} s|^{C_\delta}$ on $\operatorname{Re} s \ge \frac{\alpha}{2}+\delta$, $|\operatorname{Im} s| \ge 5$, while $|\Gamma(1-s+ih)|$ decays like $e^{-\pi|\operatorname{Im} s - h|/2}$ in vertical strips. So the Mellin inversion integral $R_h(x) = \frac{1}{2\pi i}\int_{(c)} \Gamma(1-s+ih)\zeta(2s)^{-1} x^s\, ds$, with $\frac12 < c < 1$, can be moved to the line $\operatorname{Re} s = \frac{\alpha}{2} + \delta$, which gives $R_h(x) = O(x^{\alpha/2+\delta})$. Conversely, if $R_h(x) = O(x^{\alpha/2+\delta})$ as $x \to \infty$, then, since also $|R_h(x)| \le \zeta(2)\, x$, the Mellin integral converges absolutely for $\frac{\alpha}{2}+\delta < \operatorname{Re} s < 1$ and continues $\Gamma(1-s+ih)/\zeta(2s)$ analytically there. As $\Gamma$ has no zeros, $\zeta(2s)$ has no zeros with $\alpha + 2\delta < \operatorname{Re} 2s < 2$; since $\delta > 0$ is arbitrary and $\zeta$ has no zeros with $\operatorname{Re} s \ge 1$, it has none with $\operatorname{Re} s > \alpha$. □

Shifting the contour further gives, when the zeros are simple, the twisted form of the explicit formula (10.1.7), with the sum over the zeros again the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence $T_\nu \to \infty$:

$$R_h(x) = \sum_\rho \frac{\Gamma(1-\rho/2+ih)}{2\zeta'(\rho)}\, x^{\rho/2} + \sum_{k\ge1} \frac{\Gamma(1+k+ih)}{2\zeta'(-2k)}\, x^{-k}. \tag{10.2.1}$$

The heights $T_\nu$ are those of [498, Theorem 9.7], on which $1/\zeta(\sigma+iT_\nu) = O(T_\nu^A)$ for $-1 \le \sigma \le 2$; with the functional equation for $\operatorname{Re} 2s \le -1$ this makes the integrals over the horizontal segments $\operatorname{Im} 2s = \pm T_\nu$ tend to 0. On the line $\operatorname{Re} s = -q - \frac{1}{2}$ we write $1/\zeta(2s) = \pi^{1/2-2s}\Gamma(s)/(\Gamma(\frac{1}{2}-s)\zeta(1-2s))$; since $|\Gamma(1-s+ih)\Gamma(s)| \le e^{\pi|h|/2}\pi/\cosh(\pi \operatorname{Im} s)$, $|\Gamma(\frac{1}{2}-s)| \ge q!\,(\pi t/\sinh \pi t)^{1/2}$ with $t = \operatorname{Im} s$, and $|\zeta(1-2s)| \ge 2-\zeta(4)$, the integral over that line is at most $Ce^{\pi|h|/2}\pi^{2q}x^{-q-1/2}/q!$, which tends to 0 as $q \to \infty$.

What the twist changes is the weight. Since $|\Gamma(\frac{3}{4}+i(h-\gamma/2))|$ is largest when $\gamma \approx 2h$ and decays like $e^{-\pi|h-\gamma/2|/2}$ away from it, $R_h$ sees the zeros in a band of width about ten around height $2h$. At $h = 50$ the largest weight among the first 300 zeros is 0.124, at $\gamma = 101.3$, where the untwisted weight is $2.9 \times 10^{-35}$ (Figure 10.4, left). With `twisted.py` we computed $R_{50}(x)$ from the Möbius series with $\mu$ sieved to $10^7$ at forty values of $x$ in $[10^4, 10^{10}]$, and compared it with (10.2.1) over the first 300 zeros and three trivial terms. The two agree to $1.2 \times 10^{-8}$ in $|R_h(x)|/x^{1/4}$, against a signal of size up to 0.27, and the zeros with $|\gamma - 100| < 15$ carry 99.9997% of the modulus of the explicit formula, in the median over the forty values of $x$ (Figure 10.4, right).

An off line zero with real part $\beta$ near height $2h$ would contribute a term of size $x^{\beta/2}$, growing relative to the background $x^{1/4}$ by the factor $x^{(\beta-1/2)/2}$: a factor 3.2 between $x = 10^{10}$ and $x = 10^{20}$ when $\beta = 0.6$.

Remark 10.2.2. Twisted Riesz functions are not competitive with direct zero finding as a verification method. Their value is independence: they are computed from $\mu(n)$ only, with no evaluation of $\zeta$, so agreement between $R_h$ and the explicit formula over computed zeros is a check of the zeros that shares no code or method with their computation.

The cost of $R(x)$ or $R_h(x)$ is dominated by sieving $\mu(n)$ to $N = K\sqrt{x}$. Terms with $n < \sqrt{x}/12$ are below $e^{-144}$ and can be omitted, and the terms with $n > N$ are summed through $\sum_n \mu(n)n^{-2} = 6/\pi^2$, or $1/\zeta(2+2ih)$ for $R_h$: the script computes $R(x)/x$ as $\sum_{\sqrt{x}/12 \le n \le N} \mu(n)n^{-2}e^{-x/n^2} + \big(6/\pi^2 - \sum_{n \le N} \mu(n)n^{-2}\big)$, and the neglected remainder $\sum_{n>N} \mu(n)n^{-2}(e^{-x/n^2}-1) \approx -x\sum_{n>N}\mu(n)n^{-4}$ is of order $xN^{-7/2}$ under square root cancellation in the Möbius tail, that is, of order $K^{-7/2}$ in $R(x)/x^{1/4}$. That cancellation is proved in the range used: the Mertens function $\sum_{n \le t}\mu(n)$ is less than $0.571\sqrt{t}$ in absolute value for $33 \le t \le 10^{16}$ [277, Section 6.1], and partial summation then bounds the remainder in $R(x)/x^{1/4}$ by $1.1 \times 10^{-8}$ for $K = 200$ and $1 \le x \le 10^{12}$. The script `riesz_large.py` implements this with a segmented sieve and extended precision. With $K = 200$ it reproduces the explicit formula over 60 zeros and three trivial terms to about $2 \times 10^{-9}$ in $R(x)/x^{1/4}$ at $x = 10^8$, $10^{10}$, $10^{11}$ and $10^{12}$ (`check_large.py`). Its twisted option agrees as well: at $x = 10^{10}$ with $h = 50$ it gives $R_{50}(x) = 37.441416 - 26.042495\,i$, within $3.1 \times 10^{-9}$ of (10.2.1) over 300 zeros in $|R_h(x)|/x^{1/4}$ (`check_large_twisted.py`).

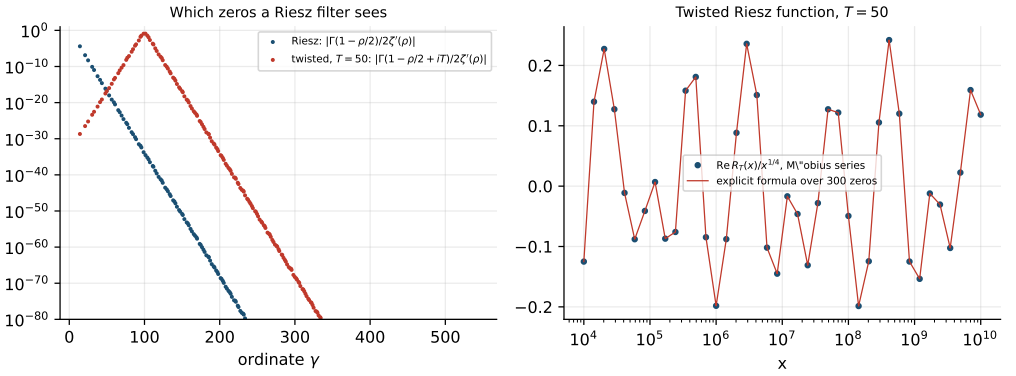



Figure 10.4: Left: the weights $|\Gamma(1-\rho/2)/(2\zeta'(\rho))|$ of the Riesz function and $|\Gamma(1-\rho/2+ih)/(2\zeta'(\rho))|$ of the twisted Riesz function with $h=50$, for the first 300 zeros. Right: $\operatorname{Re} R_{50}(x)/x^{1/4}$ from the Möbius series against the explicit formula (10.2.1) over 300 zeros. The plot writes $T$ for $h$. Data from `twisted.py`; drawn by `figs_strategy.py`.

### 10.2.2 The Riesz coefficients through the centre clock

The Riesz coefficients are values of two Laplace transforms: that of the centre clock $T$ of [421] and that of the basepoint one clock $T_1$ (Proposition 9.1.2).

Proposition 10.2.3. *For every integer $k\ge 1$, with no hypothesis,*

$$\frac{1}{\zeta(2k)}=\frac{\kappa_k}{2\xi(\frac12)}\,\mathbb{E}\,e^{-(2k-\frac12)^2T}=\kappa_k\,\mathbb{E}\,e^{-(2k-1)^2T_1}.$$

*Proof.* From the definition of $\xi$, $\xi(2k)=k(2k-1)\pi^{-k}\Gamma(k)\zeta(2k)=\frac12\kappa_k\zeta(2k)$. Since $2k-\frac12>0$, the Laplace transform of $T$ at $(2k-\frac12)^2$ is $\varphi_{1/2}((2k-\frac12)^2)=\xi(\frac12)/\xi(2k)$; similarly $\mathbb{E}e^{-(2k-1)^2T_1}=\xi(1)/\xi(2k)$, with $\xi(1)=\frac12$. □

Proposition 10.2.4. *For every $x>0$, with no hypothesis,*

$$R(x)=-\frac{1}{\xi(\frac12)}\,\mathbb{E}\Big[\sum_{k\ge1}k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-\frac12)^2T}\Big].$$

*Proof.* Since $\kappa_k/(2(k-1)!)=k(2k-1)\pi^{-k}$, Proposition 10.2.3 turns the term $-(-x)^k/((k-1)!\,\zeta(2k))$ of Riesz's series into $-\xi(\frac12)^{-1}k(2k-1)(-x/\pi)^k\,\mathbb{E}e^{-(2k-\frac12)^2T}$. The exchange of sum and expectation is justified by Fubini's theorem, because

$$\sum_{k\ge1}k(2k-1)\Big(\frac{x}{\pi}\Big)^k\mathbb{E}\,e^{-(2k-\frac12)^2T}=\xi(\tfrac12)\sum_{k\ge1}\frac{x^k}{(k-1)!\,\zeta(2k)}<\infty.$$

□

The terms of the series averaged in Proposition 10.2.4 decay like $e^{-4k^2T}$ rather than like $1/(k-1)!$. Table 10.6, computed by `riesz_thorin.py` (output `riesz_thorin.json`), checks both propositions with $4\times10^5$ draws of $T$, generated under RH as $\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials $e_\gamma$: the first 200 zeros are used exactly, and the remaining terms are replaced by a gamma variable with the same mean and variance, computed from the exact power sums $\sum\gamma^{-2}$ and $\sum\gamma^{-4}$. The coefficients agree within about one standard error, and so do the values of $R$ from the Möbius series, the power series and the clock average.

| $k$ | $1/\zeta(2k)$ | via the clock | standard error |
|---|---|---|---|
| 1 | 0.6079271 | 0.6079115 | 0.0000131 |
| 2 | 0.9239384 | 0.9238153 | 0.0001046 |
| 3 | 0.9829526 | 0.9826581 | 0.0002603 |
| 4 | 0.9959392 | 0.9954609 | 0.0004615 |

| $x$ | $R(x)$, Möbius | clock average | standard error |
|---|---|---|---|
| 0.5 | 0.1252199 | 0.1252287 | 0.0000074 |
| 1 | 0.0439818 | 0.0440007 | 0.0000157 |
| 2 | −0.2677934 | −0.2677918 | 0.0000026 |
| 5 | −0.7705597 | −0.7706050 | 0.0000397 |

Table 10.6: Left: the Riesz coefficients from the clock, Proposition 10.2.3. Right: the Riesz function from the Möbius series, which agrees with the power series to fourteen digits, and from the clock average of Proposition 10.2.4.

The clock average is also far better conditioned than the power series. Riesz's power series is numerically hopeless for large $x$: its largest term is about $e^x\sqrt{x/2\pi}$, that is $10^{4344.5}$ at $x=10^4$, while $R(10^4)$ is of order $10^{-3}$. In the clock average the largest term of the inner series, $\max_k k(2k-1)(x/\pi)^k e^{-(2k-\frac12)^2T}$, is about $10^{81}$ at $x=10^4$ for the typical value $T=0.023$, and its logarithm grows like $(\log x)^2/(16T)$ instead of like $x$ (Figure 10.5, right).

### 10.2.3 The sine squared lemma and a GGC form of Riesz's criterion

The computation of Proposition 10.2.3 works at every basepoint between the centre and one. For $\alpha>\frac12$ the clock $T_\alpha$ exists with no hypothesis (Theorem 9.2.5), and it is a GGC exactly when $\zeta$ has no zeros with $\operatorname{Re}s>\alpha$ (Theorem 1.2.2).

Proposition 10.2.5 (Riesz coefficients as GGC values). *For every $\alpha\in[\frac12,1]$ and every integer $k\ge1$,*

$$\frac{1}{\zeta(2k)}=\frac{\kappa_k}{2\xi(\alpha)}\,\mathbb{E}\,e^{-(2k-\alpha)^2T_\alpha}.$$

*At $\alpha=1$, with no hypothesis, $T_1$ is a GGC with Thorin density $\rho_{1/2}=v_1/\pi$ in*

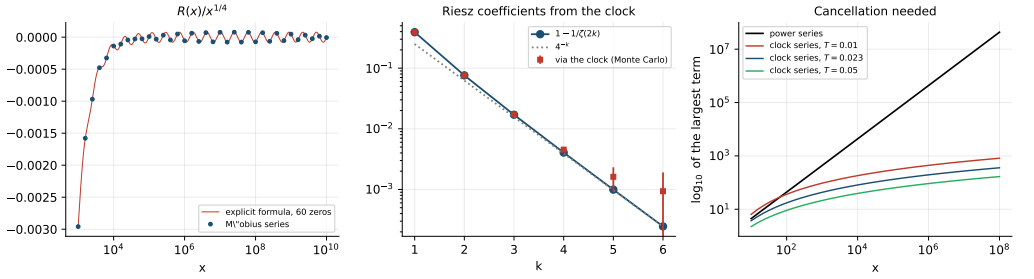



Figure 10.5: Left: $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula over 60 zeros, as in the right panel of Figure 10.3. Middle: $1 - 1/\zeta(2k)$ and its Monte Carlo value from the clock (Proposition 10.2.3), against $4^{-k}$. Right: base ten logarithm of the largest term in Riesz's power series and in the clock series of Proposition 10.2.4 for $T = 0.01$, 0.023 and 0.05.

*the chart* $\theta = \sqrt{z}$, *and*

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ - \int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big)\rho_{1/2}(\theta)\, d\theta \Big\}.$$

*Proof.* Since $2k-\alpha > 0$, the Laplace transform of $T_\alpha$ at $(2k-\alpha)^2$ is $\xi(\alpha)/\xi(2k)$, and $\xi(2k) = \frac{1}{2}\kappa_k\zeta(2k)$. At $\alpha = 1$, $\xi(1) = \frac{1}{2}$, and the Thorin representation $\log\varphi_1(u) = -\int_0^\infty \log(1 + u/\theta^2)\rho_{1/2}(\theta)\, d\theta$ of Proposition 9.1.2, taken at $u = (2k-1)^2$, gives the second formula. □

Remark 10.2.6. All the Riesz coefficients are samples of one GGC Laplace transform: $c_k = \varphi_1((2k-1)^2)$, where $\varphi_1(u) = \mathbb{E}e^{-uT_1}$. As a Laplace transform $\varphi_1$ is completely monotone, and this already makes $\log\varphi_1$ convex in $u$, by Hölder's inequality; so the slopes $(\log c_{k+1} - \log c_k)/8k$ increase with $k$. As the transform of a GGC, $\varphi_1$ is also hyperbolically completely monotone [56, Theorem 6.1.1]: for each $u > 0$ the function $v \mapsto \varphi_1(uv)\varphi_1(u/v)$ is a completely monotone function of $v + 1/v$. In particular it decreases as $v$ moves away from 1, and differentiating $\log\varphi_1(uv) + \log\varphi_1(u/v)$ in $\log v$ shows that $t \mapsto \log\varphi_1(e^t)$ is concave. At the sample points this gives inequalities in the other direction: the slopes $(\log c_{k+1} - \log c_k)/\log\frac{2k+1}{2k-1}$ decrease with $k$, and $c_j c_l \le c_k^2$ whenever $(2j-1)(2l-1) = (2k-1)^2$, for instance $c_1 c_5 \le c_2^2$.

The Thorin density $v_1/\pi$ is built from the primes by the sine squared lemma (1.2.9). Proposition 9.1.10 writes $\log(\xi(1+s)/\xi(1))$ in the Lévy form (9.1.5), with the drift $b_1$ and a signed Lévy measure $\Pi_1$ made of the gamma density, the pole of $\zeta$ and the prime atoms, and the lemma converts it term by term into the Thorin form $v_1 = b_1 + A + Q$ of (9.1.6), with the gamma piece $A$ and the compensated

prime piece $Q$. The Lévy form in $s = 2k-1$ is

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ - b_1(2k-1) - \int_0^\infty \big(e^{-(2k-1)x} - 1 + (2k-1)x\big)\,\Pi_1(dx)\Big\},$$

and in the Thorin form in $u = (2k-1)^2$ the drift splits off a positive stable one half factor:

$$\frac{1}{\zeta(2k)} = \kappa_k\, e^{-b_1(2k-1)}\, \exp\Big\{ - \frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big)\big(A(\theta)+Q(\theta)\big)\,d\theta\Big\}. \tag{10.2.2}$$

This is the factorisation $T_1 \stackrel{d}{=} S + T_1^\circ$ of Proposition 9.1.11 evaluated at $u = (2k-1)^2$: the middle factor is $\mathbb{E}e^{-(2k-1)^2 S}$ for the positive stable variable $S$ of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, and the last factor is $\mathbb{E}e^{-(2k-1)^2 T_1^\circ}$ for the independent GGC $T_1^\circ$ with Thorin density $(A+Q)/\pi = (v_1 - b_1)/\pi$.

Inserting Proposition 10.2.5 at $\alpha = 1$ into the power series gives the Riesz function as a GGC mixture,

$$R(x) = -2\,\mathbb{E}\Big[\sum_{k\ge1} k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-1)^2 T_1}\Big], \tag{10.2.3}$$

where the exchange of sum and expectation is justified as in Proposition 10.2.4, since $\sum_k 2k(2k-1)(x/\pi)^k c_k = \sum_k x^k/((k-1)!\,\zeta(2k)) < \infty$. The variable $T_1$ has two realisations. It is the tilted centre clock run through the inverse local time of a sticky Brownian motion (Theorem 9.2.2 and the paragraph after it), and, like every GGC without drift [289], it is the Wiener gamma integral (11.1.4) against the inverse of the Poisson smoothed zero count $M_{1/2}(\theta) = \pi^{-1}\int_0^\theta v_1$ (Section 11.1.6).

THEOREM 10.2.7. *For $\frac12 \le \alpha < 1$ the following are equivalent:*

1. *$R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$;*
2. *$\zeta$ has no zeros with $\operatorname{Re} s > \alpha$;*
3. *$T_{\alpha'}$ is a GGC for every $\alpha' > \alpha$;*
4. *for every $\alpha' > \alpha$, $v_{\alpha'}(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha' + i\theta) \ge 0$ at every real $\theta$ for which $\alpha' + i\theta$ is not a zero of $\xi$;*
5. *for every $\alpha' > \alpha$ and every real $\theta$ the compensated sine squared sum*

$$\lim_{L\to\infty}\Big[\sum_{\log n \le L} \frac{\Lambda(n)}{n^{\alpha'}}\, 2\sin^2\Big(\frac{\theta\log n}{2}\Big) - \int_0^L 2\sin^2\Big(\frac{\theta x}{2}\Big)e^{(1-\alpha')x}\,dx\Big]$$

*converges.*

*In particular RH holds if and only if $T_\alpha$ is a GGC for every $\alpha > \frac12$.*

*Proof.* Write $\Theta = \sup \operatorname{Re}\rho$ and $D(y) = \psi(y) - y$. Conditions (1) and (2) are equivalent by the basepoint version of Riesz's criterion (Section 10.1.8, and Proposition 10.2.1 with $h = 0$). Conditions (2) and (3) are equivalent by Theorem 1.2.2: $\varphi_{\alpha'}$ is the Laplace transform of a GGC exactly when $\alpha' \geq \Theta$, and this holds for every $\alpha' > \alpha$ exactly when $\Theta \leq \alpha$, which is (2).

(2) implies (4), because if $\Theta \leq \alpha < \alpha'$ every Cauchy kernel in $v_{\alpha'}(\theta) = \sum_\rho (\alpha' - \beta)/((\alpha' - \beta)^2 + (\theta - \gamma)^2)$ has positive width (Proposition 1.2.3). Conversely, let $\rho_0 = \beta_0 + i\gamma_0$ be a zero of multiplicity $m$ with $\beta_0 > \alpha$. The zeros of $\xi$ are isolated, so only finitely many lie on the segment $\{\sigma + i\gamma_0 : 0 \leq \sigma \leq 1\}$. Let $n_0$ be the number of those other than $\rho_0$, counted with multiplicity, and choose $\eta \in (0, \beta_0 - \alpha)$ with $|\beta - \beta_0| \geq 2\eta$ for each of them. For $\alpha' \in (\beta_0 - \eta, \beta_0)$ the point $\alpha' + i\gamma_0$ is not a zero, and Proposition 1.2.3 gives

$$v_{\alpha'}(\gamma_0) = \frac{m}{\alpha' - \beta_0} + \sum_{\rho \neq \rho_0,\ \gamma = \gamma_0} \frac{1}{\alpha' - \beta} + \sum_{\gamma \neq \gamma_0} \frac{\alpha' - \beta}{(\alpha' - \beta)^2 + (\gamma - \gamma_0)^2}.$$

The middle sum is at most $n_0/\eta$ in absolute value. Since $\alpha'$ and $\beta$ lie in $(0, 1)$, each term of the last sum is at most $(\gamma - \gamma_0)^{-2}$ in absolute value, and $\sum_{\gamma \neq \gamma_0} (\gamma - \gamma_0)^{-2} < \infty$: finitely many zeros have $0 < |\gamma - \gamma_0| < 1$, and for $|\gamma - \gamma_0| \geq 1$ we have $(\gamma - \gamma_0)^{-2} \leq 5/(4 + (\gamma - \gamma_0)^2) \leq 5(2 - \beta)/((2 - \beta)^2 + (\gamma - \gamma_0)^2)$, a series bounded by $5v_2(\gamma_0) < \infty$. Hence $v_{\alpha'}(\gamma_0) \leq m/(\alpha' - \beta_0) + n_0/\eta + \sum_{\gamma \neq \gamma_0} (\gamma - \gamma_0)^{-2} < 0$ for $\alpha'$ close to $\beta_0$, and (4) fails at $(\alpha', \gamma_0)$.

(2) implies (5). This is part 1 of Proposition 9.1.13 at the basepoint $\alpha'$, since $\Theta \leq \alpha < \alpha'$, and the argument is short. Choose $\sigma$ with $\Theta < \sigma < \alpha'$, so that $D(y) = O(y^\sigma)$ [498]. With $f(y) = y^{-\alpha'}\, 2\sin^2(\frac{1}{2}\theta \log y)$ the bracket in (5) is

$$\int_{(1, e^L]} f\, dD = f(e^L) D(e^L) - \int_1^{e^L} D(y) f'(y)\, dy,$$

since $f(1) = 0$. The boundary term tends to 0, and the integral converges absolutely because $|f'(y)| \leq (2\alpha' + |\theta|) y^{-\alpha' - 1}$.

(5) implies (2). Suppose (5), and let $\rho_0$ be a zero of multiplicity $m$ with $\operatorname{Re}\rho_0 > \alpha$. Fix $\alpha' \in (\alpha, \operatorname{Re}\rho_0)$ and $\theta \notin \mathcal{E} = \{\theta : \rho_0 + i\theta \text{ or } \rho_0 - i\theta \text{ is a zero}\}$, a countable set that contains 0, and let $M(t) = \int_{(0,t]} (1 - \cos\theta s)\, dD(e^s)$, a function of locally bounded variation. The bracket in (5) is $J(L) = \int_{(0,L]} e^{-\alpha' t} dM(t)$; by hypothesis it converges as $L \to \infty$, so $J$ is bounded. For $\operatorname{Re} w > \alpha'$, integration by parts gives

$$\int_{(0,L]} e^{-wt}\, dM(t) = e^{-(w - \alpha')L} J(L) + (w - \alpha') \int_0^L e^{-(w - \alpha')t} J(t)\, dt,$$

which converges as $L \to \infty$, locally uniformly in $w$, to an analytic function $F_\theta(w)$ on $\operatorname{Re} w > \alpha'$. For $\operatorname{Re} w > 1$ the integral converges absolutely, and writing $1 -$

$\cos\theta t = 1 - \frac{1}{2}e^{i\theta t} - \frac{1}{2}e^{-i\theta t}$ gives

$$F_\theta(w) = g(w) - \tfrac{1}{2}g(w - i\theta) - \tfrac{1}{2}g(w + i\theta),$$
$$g(w) = \int_{(1,\infty)} y^{-w}\, dD(y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}.$$

The right side, call it $\widetilde{F}_\theta(w)$, is meromorphic in the plane with a discrete set $\mathcal{P}$ of poles. On the connected open set $\Omega = \{\operatorname{Re} w > \alpha'\} \setminus \mathcal{P}$ both $\widetilde{F}_\theta$ and $F_\theta$ are analytic, and they agree on $\operatorname{Re} w > 1$, so $\widetilde{F}_\theta = F_\theta$ on $\Omega$ by the identity theorem. The function $g$ is analytic at 1, and in $\operatorname{Re} w > 0$ its only poles are the zeros of $\zeta$, each simple with residue minus its multiplicity. Since $\theta \notin \mathcal{E}$, $g(w \pm i\theta)$ is analytic at $\rho_0$, so $\widetilde{F}_\theta$ has a pole at $\rho_0$ with residue $-m \neq 0$, and $|F_\theta(w)| \to \infty$ as $w \to \rho_0$ in $\Omega$. This contradicts the continuity of $F_\theta$ at $\rho_0$, where $\operatorname{Re}\rho_0 > \alpha'$. Hence no zero has real part greater than $\alpha$, which is (2). □

### 10.2.4 The coefficients as killing probabilities

For a positive random variable $X$ and an independent exponential variable $E_\lambda$ of rate $\lambda$, $\mathbb{E}e^{-\lambda X} = P(X < E_\lambda)$, the probability that $X$ occurs before a killing at rate $\lambda$. Propositions 10.2.3 and 10.2.5 therefore say that every Riesz coefficient is a killing probability, in two ways and with no hypothesis:

$$\frac{1}{\zeta(2k)} = \kappa_k\, P\big(T_1 < E_{(2k-1)^2}\big) = \frac{\kappa_k}{2\xi(\frac{1}{2})}\, P\big(T < E_{(2k-\frac{1}{2})^2}\big). \tag{10.2.4}$$

By Theorem 9.2.2, $T_1 \overset{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$, and by the paragraph after it $T_1$ is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$. So $c_k = 1/(\kappa_k\zeta(2k))$ is the probability that this hitting happens before an independent killing at rate $(2k-1)^2$.

The factorisation (10.2.2) has the same reading. Since $\mathbb{E}e^{-\lambda H_a} = e^{-a\sqrt{2\lambda}}$, the stable factor is

$$e^{-b_1(2k-1)} = P\big(H_{b_1/\sqrt{2}} < E_{(2k-1)^2}\big),$$

the probability that Brownian motion reaches the level $b_1/\sqrt{2} = 0.016331$ before it is killed. So $S \overset{d}{=} H_{b_1/\sqrt{2}}$ and $T_1 \overset{d}{=} H_{b_1/\sqrt{2}} + T_1^\circ$ with $T_1^\circ$ an independent GGC. Thus $c_k = P(H_{b_1/\sqrt{2}} < E_{(2k-1)^2})\, P(T_1^\circ < E'_{(2k-1)^2})$, with independent exponential clocks $E$ and $E'$ of the same rate. If the exact Brownian boundary $b$ of $T_1$ is bounded at infinity, as Section 9.2.4 observes but does not prove, Lemma 9.2.4 and the tail of $T_1$ make the level $b_1/\sqrt{2} = \sum_\rho \rho^{-1}/\sqrt{2}$ its Wald defect $\mathbb{E}\, b(T_1)$.

Under RH, if the zeros are also simple, more is true. The centre clock is then a GGC with unit atoms at the distinct points $\gamma^2$ (Proposition 1.2.1), hence, by Kent's theorem and Krein's inverse spectral theory (Section 8.1.3), the hitting time of a level by the diffusion of a Krein string with Dirichlet spectrum $\{\gamma^2\}$

(Table 4.1). The second form of (10.2.4) then says that $1/\zeta(2k)$ is proportional to the probability that this Riemann string diffusion reaches its level before it is killed at rate $(2k-\frac{1}{2})^2$. At the basepoint one, Corollary 9.2.3 gives the first form the same reading under the same hypotheses.

The Riesz function becomes an alternating sum of killing probabilities,

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!}\,\kappa_k\, P\big(T_1 < E_{(2k-1)^2}\big),$$

and the criterion asks that these alternating sums grow no faster than $x^{1/4+\delta}$. The killing rates $(2k-\alpha)^2$ that the criterion needs, with $k$ in the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$, are complex, and at a complex rate the Laplace transform is no longer a probability; this is the hitting time form of the obstruction of Section 10.2.6.

We checked these identities with `killing.py` (output `killing.json`), using $10^6$ draws of $T_1 = \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ generated under RH as in Section 9.2.3. Table 10.7 compares the killing probabilities with $c_k$, and the Brownian hitting probability with $e^{-b_1(2k-1)}$; all agree within two standard errors. With the averaged killing probabilities for $k \leq 30$ from a second run of $10^6$ draws (`killing2.py`), the alternating sum gives $R(0.5) = 0.125242$, $R(1) = 0.043990$, $R(2) = -0.267895$ and $R(3) = -0.526639$, against the exact 0.125220, 0.043982, −0.267793 and −0.526356.

| $k$ | $P(T_1 < E)$, drawn | $\mathbb{E}e^{-(2k-1)^2 T_1}$ | $c_k = \frac{1}{\kappa_k \zeta(2k)}$ | $P(H_{b_1/\sqrt{2}} < E)$ | $e^{-b_1(2k-1)}$ |
|---|---|---|---|---|---|
| 1 | $0.95481 \pm 0.00021$ | $0.954928 \pm 0.000112$ | 0.954930 | $0.97699 \pm 0.00015$ | 0.97717 |
| 2 | $0.75965 \pm 0.00043$ | $0.759835 \pm 0.000161$ | 0.759909 | $0.93352 \pm 0.00025$ | 0.93306 |
| 3 | $0.50738 \pm 0.00050$ | $0.507865 \pm 0.000156$ | 0.507962 | $0.89098 \pm 0.00031$ | 0.89094 |
| 4 | $0.28834 \pm 0.00045$ | $0.288646 \pm 0.000122$ | 0.288731 | $0.85059 \pm 0.00036$ | 0.85072 |
| 5 | $0.14179 \pm 0.00035$ | $0.141473 \pm 0.000080$ | 0.141535 | $0.81195 \pm 0.00039$ | 0.81232 |
| 6 | $0.06045 \pm 0.00024$ | $0.060638 \pm 0.000045$ | 0.060679 | $0.77555 \pm 0.00042$ | 0.77565 |

Table 10.7: Riesz coefficients as killing probabilities of the basepoint one clock, and the stable factor as a Brownian hitting probability; $E$ is an independent exponential clock of rate $(2k-1)^2$, and the errors are one standard error of $10^6$ draws.

### 10.2.5 Half Cauchy variables, the sech clock and a duality

Bourgade, Fujita and Yor [60] derived Euler's formulae for $\zeta(2n)$ from products of Cauchy variables, and the construction gives a representation of $\zeta(2n)$ dual to that of $1/\zeta(2n)$. Let $C$ and $C'$ be independent standard Cauchy variables, so that $|C|$ and $|C'|$ are half Cauchy. By Section 4.2.2, $\mathbb{E}|C|^s = \sec(\pi s/2)$ for $|\operatorname{Re} s| < 1$, so $\sec^2(\pi s/2) = \mathbb{E}|CC'|^s$ is the moment generating function of $\log|CC'|$ on $|s| < 1$.

Differentiating the series $\tan z = \sum_{n\geq 1} 2^{2n}(2^{2n}-1)|B_{2n}|z^{2n-1}/(2n)!$ and inserting Euler's formula $\zeta(2n) = (2\pi)^{2n}|B_{2n}|/(2(2n)!)$ gives

$$\sec^2 \frac{\pi s}{2} = \sum_{n\geq 1} \frac{8(2n-1)(1-2^{-2n})\,\zeta(2n)}{\pi^2}\, s^{2n-2},$$

and comparing coefficients with the moment generating function,

$$\zeta(2n) = \frac{\pi^2}{8(2n-1)(1-2^{-2n})}\,\frac{\mathbb{E}\big[(\log|CC'|)^{2n-2}\big]}{(2n-2)!}, \qquad n \geq 1. \tag{10.2.5}$$

Let $\tau$ be the exit time of a standard Brownian motion from $(-1,1)$, so that $\mathbb{E}e^{-\lambda\tau} = 1/\cosh\sqrt{2\lambda}$ (Section 4.2.2), and let $B$ be a Brownian motion independent of $\tau$. Then $\mathbb{E}e^{i\theta B_\tau} = \mathbb{E}e^{-\theta^2\tau/2} = \operatorname{sech}\theta$, and since $\log|C|$ has characteristic function $\operatorname{sech}(\pi t/2)$ (Section 4.2.2),

$$\log|C| \overset{d}{=} \frac{\pi}{2}\, B_\tau.$$

We call $\tau$ the sech clock. By the product $\cosh\sqrt{2\lambda} = \prod_{k\geq 1}\big(1+8\lambda/((2k-1)^2\pi^2)\big)$ it is a GGC with unit Thorin atoms at $(2k-1)^2\pi^2/8$, the GGC $\sum_k e_k/(2k-1)^2$ of Section 4.2.2 multiplied by $8/\pi^2$, and it is Kent's hitting time for the interval: the hitting time of 1 by reflected Brownian motion started at 0, whose eigenvalues on $[0,1]$, with reflection at 0 and killing at 1, are these atoms. So its spectrum is proportional to the very points $(2k-1)^2$ at which the Riesz coefficients sample the Laplace transform of $T_1$. For two independent copies, $\log|CC'| \overset{d}{=} \frac{\pi}{2}B_{\tau+\tau'}$. Combining (10.2.5) with Proposition 10.2.5 gives a duality between the two clocks,

$$\frac{\mathbb{E}\big[(\log|CC'|)^{2k-2}\big]}{(2k-2)!}\;\mathbb{E}\,e^{-(2k-1)^2T_1} = \frac{8(2k-1)(1-2^{-2k})}{\pi^2\kappa_k}, \tag{10.2.6}$$

in which a moment of Brownian motion stopped at a sum of two independent sech clocks, times a killing probability of the basepoint one clock, is an elementary constant.

We checked the moment formula (10.2.5) exactly through the Taylor coefficients of $\sec^2$, where it reproduces $\zeta(2)$, $\zeta(4)$, $\zeta(6)$ and $\zeta(8)$ to ten digits. The duality (10.2.6) holds with the killing probabilities simulated in the second run of Section 10.2.4: the products are 0.95508, 1.87524, 2.06191 and 1.63889 for $k = 1,\dots,4$, against 0.95493, 1.87500, 2.06167 and 1.63851.

The duality is structural rather than quantitative: $\zeta(2k)$ as a moment lives at the integers, while Riesz's criterion needs $1/\zeta(2s)$ inside the strip. The half Cauchy law also enters the book through Bosch's theorem [58] (Section 4.2), whose dictionary, Table 4.1, pairs the odd integers $2k-1$ with the ordinates $\gamma$.

### 10.2.6 Why the representation cannot supply the growth bound

The representations of Sections 10.2.2 to 10.2.5 are exact, and all of them except the Kent string readings of Section 10.2.4 hold with no hypothesis. They do not supply the bound $R(x) = O(x^{1/4+\delta})$, for four reasons.

1. The integer values carry almost no information. The growth of $R$ is fixed by the Mellin transform $\Gamma(1-s)/\zeta(2s)$, that is, by the analytic interpolation of $k \mapsto 1/\zeta(2k)$ to complex $k$ with $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$. At the integers $1/\zeta(2k) = 1 - 4^{-k} + O(9^{-k})$, and these values, which Proposition 10.2.3 represents, determine the interpolation only through its analytic continuation.

2. The representation interpolates only on one side of the line $\operatorname{Re} 2k = \alpha$, and as an expectation only on a cone. For complex $k$ the expectation $\mathbb{E}e^{-(2k-\alpha)^2 T_\alpha}$ is finite exactly when $\operatorname{Re}(2k-\alpha)^2 \geq 0$, because $T_\alpha$ has a tail of order $t^{-1/2}$ (Theorem 9.2.5) and no exponential moments; there it equals $\varphi_\alpha((2k-\alpha)^2) = \xi(\alpha)/\xi(\alpha + \sqrt{(2k-\alpha)^2})$, and the principal square root equals $2k-\alpha$ only when $\operatorname{Re}(2k-\alpha) > 0$. So the representation through $T_\alpha$ returns $\xi(\alpha)/\xi(2k)$ as an expectation on the cone $|\operatorname{Im} 2k| \leq \operatorname{Re} 2k - \alpha$, and on the rest of the half plane $\operatorname{Re} 2k > \alpha$ only through the analytic continuation of $\varphi_\alpha$, which is analytic there exactly when no zero lies to the right of $\alpha$. It is a GGC representation with no hypothesis only for $\alpha \geq 1$, that is, on the half plane $\operatorname{Re} 2k > 1$ of absolute convergence, where the criterion carries no information. To reach $\frac{1}{2} < \operatorname{Re} 2k < 1$ one needs a basepoint $\alpha < 1$, and $T_\alpha$ is a GGC there exactly when $\zeta$ has no zeros in $\operatorname{Re} s > \alpha$.

3. The centre clock's Laplace integral converges only up to its first singularity, which under RH is at $-\gamma_1^2$. In the $k$ plane, with $u = (2k - \frac{1}{2})^2$, the region $\operatorname{Re} u > -\gamma_1^2$ is a neighbourhood of the segment $|\operatorname{Im} 2k| < \gamma_1$ of the critical line $\operatorname{Re} 2k = \frac{1}{2}$, not the strip; beyond it one needs analytic continuation, which is again a statement about the zeros.

4. Positivity does not survive. Each coefficient becomes a positive average, but the series is alternating, and the bound requires, at $x = 10^4$ and $T = 0.023$, the cancellation of terms as large as $10^{81}$ after averaging. No monotonicity or positivity argument controls that cancellation; it is carried by the oscillating terms $x^{\rho/2}$ of the explicit formula.

What would help is what the basepoint version of the criterion (Proposition 10.2.1 with $h = 0$) requires: an analytic, and ideally positive, representation of $1/\zeta(2s)$ on whole vertical lines of the strip $\frac{1}{4} < \operatorname{Re} s < \frac{1}{2}$. A representation of the Thorin kind that is positive there is a GGC at a basepoint below one, that is, a zero free half plane. The Thorin representation of the Riesz coefficients is therefore an exact reformulation, with better cancellation and a probabilistic reading, that reaches the criterion only through the same step that every other

construction of Chapters 9 to 11 requires. At a basepoint $\frac{1}{2} < \alpha < 1$, zero density estimates bound the number of Mellin poles to the right of the line $\operatorname{Re} s = \alpha/2$ up to any given height, as they bound the negative part of $v_\alpha/\pi$ (Theorem 11.4.2), but a single such pole breaks the bound $R(x) = O(x^{\alpha/2+\delta})$ for small $\delta$.

## 10.3 The squarefree series and the clock

Riesz's criterion (Sections 10.1.8 and 10.2) reads RH from a function whose Mellin transform is $\Gamma(1-s)/\zeta(2s)$. The squarefree integers carry the same reciprocal $1/\zeta(2s)$. Their Dirichlet series is

$$D(s) = \sum_{n\geq 1} \frac{\mu(n)^2}{n^s} = \prod_p \left(1 + p^{-s}\right) = \frac{\zeta(s)}{\zeta(2s)}, \qquad \operatorname{Re} s > 1, \tag{10.3.1}$$

whose singularities in the half plane $\operatorname{Re} s > 0$ are the pole at $s = 1$ and the poles at the points $\rho/2$ at which $\zeta(\rho/2) \neq 0$ (all of them under RH), and the number $Q(x)$ of squarefree integers up to $x$ oscillates about $6x/\pi^2$ on the scale $x^{1/4}$ that these poles set when they lie on the line $\operatorname{Re} s = \frac{1}{4}$, that is under RH. This section follows the first author's note [431], which starts from a Math StackExchange thread [372] on the partial sums of $\mu(n)^2 n^{-\alpha}$ over $\mathbb{Q}$ and over number fields. Write $\Delta(x) = Q(x) - 6x/\pi^2$ for the error. The bound $\Delta(x) = O(x^{1/4+\varepsilon})$ is open even under RH, but it is equivalent to RH once the count is smoothed by an exponential or a Gamma(2) law, and in mean square; the order $x^{1/4}$ is attained with no hypothesis; and the bound for the count smoothed by a Gamma($n$) law loses a factor $n^{1/8}$, which still gives $\Delta(x) = O(x^{2/5+\varepsilon})$ under RH (Sections 10.3.5 and 10.3.6). The Pólya kernels of the Dedekind zeta functions of imaginary quadratic fields, whose positivity can fail, are in Section 8.5.6 of Chapter 8.

We keep the notation of Section 1.2 for the Pólya variable $X$, with $\mathbb{E}e^{sX} = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, the clock $T$, $G(u) = \xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2})$, the numbers $a_k = -(\rho_k - \frac{1}{2})^2$, the heat trace $W(t) = \sum_k e^{-a_k t}$ and the measure $U_{\mathrm{RH}}$, and the notation of Section 9.1.1 for the tilted variables $V_\alpha$, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and $b_\alpha = \frac{\xi'}{\xi}(\alpha) = \mathbb{E}V_\alpha$. Throughout, $\mu$ is the Möbius function, $M(t) = \sum_{d\leq t} \mu(d)$ is the Mertens function, $\{y\}$ is the fractional part of $y$, and

$$Q(x) = \sum_{n\leq x} \mu(n)^2, \qquad \Delta(x) = Q(x) - \frac{6}{\pi^2}\, x.$$

### 10.3.1 The thread: Perron's formula and an elementary method

The question asks for the behaviour as $x \to \infty$ of

$$S_\alpha(x) = \sum_{N(\mathfrak{a})\leq x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})^\alpha},$$

the sum running over the integral ideals $\mathfrak{a}$ of a number field $K$, with $\mu$ the Möbius function of ideals and $N$ the norm, and, in a comment, for its expansion at $\alpha = 1$. Over $\mathbb{Q}$, $S_\alpha(x) = \sum_{n\le x} \mu(n)^2 n^{-\alpha}$, and another comment points to Abel summation from the classical count

$$Q(x) = \frac{6}{\pi^2}\, x + O(\sqrt{x}). \tag{10.3.2}$$

The accepted answer applies Perron's formula over $K$, and the other answer carries out the elementary method over $\mathbb{Q}$. Neither evaluates the constant at $\alpha = 1$.

Since $\mu(\mathfrak{p}^k)^2$ is 1 for $k \le 1$ and 0 for $k \ge 2$, the Euler factor of $\sum_{\mathfrak{a}} \mu(\mathfrak{a})^2 N(\mathfrak{a})^{-s}$ at a prime ideal $\mathfrak{p}$ is $1 + N(\mathfrak{p})^{-s} = (1 - N(\mathfrak{p})^{-2s})/(1 - N(\mathfrak{p})^{-s})$, so the series is $\zeta_K(s)/\zeta_K(2s)$, with $\zeta_K$ the Dedekind zeta function of $K$, and

$$S_\alpha(x) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{\zeta_K(\alpha + s)}{\zeta_K(2\alpha + 2s)}\, x^s\, \frac{ds}{s}, \qquad \sigma > \max\{1 - \alpha, 0\}. \tag{10.3.3}$$

The integrand of (10.3.3) has a simple pole at $s = 1 - \alpha$, a pole at $s = 0$, and poles at the points $s = \rho/2 - \alpha$ for the zeros $\rho$ of $\zeta_K$, trivial and nontrivial, at which $\zeta_K(\rho/2) \ne 0$. Let $c_K = \lim_{s\to 1}(s-1)\zeta_K(s) = 2^{r_1}(2\pi)^{r_2} hR/(w\sqrt{|d_K|})$ be the residue of the class number formula.

*The case* $\alpha \ne 1$. The residue at $s = 1 - \alpha$ is $c_K x^{1-\alpha}/((1-\alpha)\zeta_K(2))$, as the accepted answer states, and the residue at $s = 0$, which it lists among the poles but does not compute, is $\zeta_K(\alpha)/\zeta_K(2\alpha)$, the value of the continued series. Over $\mathbb{Q}$, for $\alpha > \frac12$ and $\alpha \ne 1$, the two residues are the main terms of

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\, \frac{x^{1-\alpha}}{1-\alpha} + \frac{\zeta(\alpha)}{\zeta(2\alpha)} + O\big(x^{\frac12 - \alpha}\big). \tag{10.3.4}$$

For a proof, Abel summation and (10.3.2) give, for $\alpha > \frac12$, $\alpha \ne 1$,

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\, \frac{x^{1-\alpha}}{1-\alpha} + \Big(\alpha \int_1^\infty \frac{\Delta(t)}{t^{\alpha+1}}\, dt - \frac{6}{\pi^2}\, \frac{\alpha}{1-\alpha}\Big) + \frac{\Delta(x)}{x^\alpha} - \alpha \int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\, dt. \tag{10.3.5}$$

The last two terms are $O(x^{\frac12 - \alpha})$, and the bracket is $D(\alpha)$, because $D(s) = s\int_1^\infty Q(t) t^{-s-1} dt = \frac{6}{\pi^2}\frac{s}{s-1} + s\int_1^\infty \Delta(t) t^{-s-1} dt$ for $\operatorname{Re} s > 1$ and the right side is analytic in $\operatorname{Re} s > \frac12$ apart from $s = 1$. For $\frac12 < \alpha < 1$ the constant is negative (Corollary 10.3.4). Write $\mathcal{E}_\alpha(x)$ for the left side of (10.3.4) minus its two main terms, so that by (10.3.5)

$$\mathcal{E}_\alpha(x) = \frac{\Delta(x)}{x^\alpha} - \alpha \int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\, dt. \tag{10.3.6}$$

*The case* $\alpha = 1$. The poles at $1 - \alpha$ and at 0 merge into a double pole at $s = 0$ of $\zeta_K(1+s)\zeta_K(2+2s)^{-1} e^{s\log x}/s$. The accepted answer computes the

coefficient of $\log x$ and says that the constant term is much more complicated, at least when $K \neq \mathbb{Q}$. It is explicit. Define the Euler Kronecker constant $\gamma_K$ of $K$ by $\zeta_K(s) = c_K/(s-1) + c_K\gamma_K + O(s-1)$ [280], so that $\gamma_{\mathbb{Q}} = \gamma_E$. With $1/\zeta_K(2+2s) = \zeta_K(2)^{-1}\big(1 - 2s\,\zeta_K'(2)/\zeta_K(2) + O(s^2)\big)$ the residue at $s=0$ is

$$\frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big), \qquad C_K = \gamma_K - 2\,\frac{\zeta_K'(2)}{\zeta_K(2)}. \tag{10.3.7}$$

*The elementary method.* The identity $\mu(n)^2 = \sum_{d^2|n}\mu(d)$, the coefficientwise form of $D(s) = \zeta(s)\cdot\zeta(2s)^{-1}$, gives

$$Q(x) = \sum_{d\le\sqrt{x}} \mu(d)\Big\lfloor \frac{x}{d^2}\Big\rfloor = x\sum_{d\le\sqrt{x}}\frac{\mu(d)}{d^2} - \sum_{d\le\sqrt{x}}\mu(d)\Big\{\frac{x}{d^2}\Big\}, \tag{10.3.8}$$

and since $\sum_{d>\sqrt{x}} d^{-2} = O(x^{-1/2})$ and the last sum has at most $\sqrt{x}$ terms, each in $[0,1)$, this is (10.3.2). The second answer then sets

$$B(x) = \sum_{n\le x}\Big(\mu(n)^2 - \frac{6}{\pi^2}\Big) = O(\sqrt{x}),$$
$$D_0(s) = \sum_{n\ge1}\frac{\mu(n)^2 - 6/\pi^2}{n^s} = D(s) - \frac{6}{\pi^2}\,\zeta(s).$$

By Abel summation, $D_0(s) = \lim_{x\to\infty}\big(B(x)x^{-s} + s\int_1^x B(t)t^{-s-1}dt\big)$ converges for $\operatorname{Re} s > \frac12$, so $\sum_{n\le x}\mu(n)^2/n = \frac{6}{\pi^2}(\log x + \gamma_E) + D_0(1) + o(1)$, and there the answer stops. Since $D_0(s) = \zeta(s)\big(\zeta(2s)^{-1} - \zeta(2)^{-1}\big)$ and $\zeta$ has a simple pole with residue 1 at $s=1$,

$$D_0(1) = \lim_{s\to1}\zeta(s)\Big(\frac{1}{\zeta(2s)} - \frac{1}{\zeta(2)}\Big) = -\frac{2\zeta'(2)}{\zeta(2)^2} = 0.69298946940360442669,$$

in agreement with (10.3.7) at $K=\mathbb{Q}$, where $c_{\mathbb{Q}} = 1$ and $\gamma_{\mathbb{Q}} = \gamma_E$.

Proposition 10.3.1. (i) *Let $K$ be a number field of degree $n \ge 2$. As $x\to\infty$,*

$$\sum_{N(\mathfrak{a})\le x}\frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} = \frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big) + O\big(x^{-1/n}\log x\big).$$

(ii) *Over $\mathbb{Q}$,*

$$\sum_{n\le x}\frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\big(\log x + C\big) + O\big(x^{-1/2}\big),$$
$$C = \gamma_E - 2\,\frac{\zeta'(2)}{\zeta(2)} = 24\log A - \gamma_E - 2\log 2\pi, \tag{10.3.9}$$

*where $A$ is the Glaisher Kinkelin constant; $C = 1.7171376510090598473406241$ and $\frac{6}{\pi^2}C = 1.0438945157119382974$. Moreover $\mathcal{E}_1(x) = \sum_{n\le x}\mu(n)^2/n - \frac{6}{\pi^2}(\log x + C)$ is given by* (10.3.6) *with $\alpha = 1$, and*

$$\int_1^\infty \frac{\Delta(t)}{t^2}\,dt = \frac{6}{\pi^2}\,(C-1) = 0.43596741385791166874.$$

*Proof.* (i) The number of ideals of norm at most $y$ is $c_K y + O(y^{1-1/n})$ [328, Chapter VI], so by Abel summation $L(y) = \sum_{N(\mathfrak{c})\le y} N(\mathfrak{c})^{-1} = c_K(\log y + \gamma_K) + O(y^{-1/n})$ for $y \ge 1$, the constant being the one in the Laurent expansion of $\zeta_K$ at $s = 1$. Since $\mu(\mathfrak{a})^2 = \sum_{\mathfrak{b}^2\mid\mathfrak{a}} \mu(\mathfrak{b})$, as one checks on prime powers,

$$\begin{aligned}\sum_{N(\mathfrak{a})\le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} &= \sum_{N(\mathfrak{b})\le\sqrt{x}} \frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\,L\Big(\frac{x}{N(\mathfrak{b})^2}\Big)\\ &= c_K \sum_{N(\mathfrak{b})\le\sqrt{x}} \frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\big(\log x + \gamma_K - 2\log N(\mathfrak{b})\big)\\ &\quad + O\Big(x^{-1/n}\sum_{N(\mathfrak{b})\le\sqrt{x}} N(\mathfrak{b})^{-2+2/n}\Big).\end{aligned}$$

The last sum is $O(\log x)$, since $-2 + 2/n \le -1$. Completing the sums over $\mathfrak{b}$ costs $O(x^{-1/2}\log x)$, which is $O(x^{-1/n}\log x)$ as $n \ge 2$, since $\sum_{N(\mathfrak{b})>Y} N(\mathfrak{b})^{-2}\log N(\mathfrak{b}) = O(Y^{-1}\log Y)$, and the completed sums are $\sum_{\mathfrak{b}} \mu(\mathfrak{b})N(\mathfrak{b})^{-2} = 1/\zeta_K(2)$ and $\sum_{\mathfrak{b}} \mu(\mathfrak{b})\log N(\mathfrak{b})\,N(\mathfrak{b})^{-2} = \zeta_K'(2)/\zeta_K(2)^2$, which give (10.3.7).

(ii) By Abel summation, $\sum_{n\le x}\mu(n)^2/n = Q(x)/x + \int_1^x Q(t)t^{-2}dt$, and inserting $Q(t) = 6t/\pi^2 + \Delta(t)$ gives

$$\sum_{n\le x}\frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\log x + \frac{6}{\pi^2} + \int_1^\infty \frac{\Delta(t)}{t^2}\,dt + \frac{\Delta(x)}{x} - \int_x^\infty \frac{\Delta(t)}{t^2}\,dt,$$

whose last two terms are $O(x^{-1/2})$ by (10.3.2). So the formula holds with some constant, and the constant is identified through $D$: for $\operatorname{Re} s > 1$, $\int_1^\infty \Delta(t)t^{-s-1}dt = D(s)/s - \frac{6}{\pi^2}\frac{1}{s-1}$, and $D(s) = \frac{6}{\pi^2}\zeta(s) + D_0(s) = \frac{6}{\pi^2}\big(\frac{1}{s-1} + \gamma_E - 2\frac{\zeta'(2)}{\zeta(2)}\big) + O(s-1)$, so that $\int_1^\infty \Delta(t)t^{-2}dt = \frac{6}{\pi^2}(C-1)$ by continuity at $s = 1$. This gives (10.3.9) and the formula for $\mathcal{E}_1$, and the classical evaluation $\zeta'(2)/\zeta(2) = \gamma_E + \log 2\pi - 12\log A$ gives the second form of $C$. □

The sieve `sqfree_sieve.py` gives $Q(2\times 10^8) = 121585426$. Table 10.8 shows that $\mathcal{E}_1(x)$ is of order $x^{-3/4}$ and $\Delta(x)$ of order $x^{1/4}$, the sizes that the poles at $\rho/2 - 1$ and $\rho/2$ predict under RH (Section 10.3.4).

| $x$ | $Q(x)$ | $\Delta(x)/x^{1/4}$ | $\mathcal{E}_1(x)$ | $\mathcal{E}_1(x)\,x^{3/4}$ | $\mathcal{E}_{3/4}(x)\,x^{1/4}$ |
|---|---|---|---|---|---|
| $10^3$ | 608 | 0.0130 | $-9.130\times10^{-4}$ | $-0.162$ | $-0.0291$ |
| $10^4$ | 6083 | 0.3729 | $3.031\times10^{-4}$ | 0.303 | 0.0296 |
| $10^5$ | 60794 | 0.0725 | $-2.308\times10^{-6}$ | $-0.0130$ | $-0.00031$ |
| $10^6$ | 607926 | $-0.0348$ | $-1.665\times10^{-6}$ | $-0.0526$ | $-0.00177$ |
| $10^7$ | 6079291 | 0.3553 | $1.729\times10^{-6}$ | 0.308 | 0.00561 |
| $10^8$ | 60792694 | $-0.1619$ | $-1.737\times10^{-7}$ | $-0.174$ | $-0.00174$ |

Table 10.8: The sharp counts: the number $Q(x)$ of squarefree integers up to $x$, the normalised error $\Delta(x)/x^{1/4}$, the error $\mathcal{E}_1(x)$ of (10.3.9), which confirms the constant $C$, also multiplied by $x^{3/4}$, and the error $\mathcal{E}_{3/4}(x)$ of (10.3.4) at $\alpha=\frac34$ multiplied by $x^{1/4}$. Computed by `sqfree_sieve.py`; the sums of reciprocals are checked in extended precision.

### 10.3.2 The Pólya form of the series and the constant

Lemma 10.3.2. *For all $s$,*

$$D(s)=\Gamma_{\mathrm{sq}}(s)\,\frac{\xi(s)}{\xi(2s)},\qquad \Gamma_{\mathrm{sq}}(s)=\frac{2s-1}{s-1}\,2^s\pi^{-(s+1)/2}\,\Gamma\Big(\frac{s+1}{2}\Big).$$

*Proof.* By the definition of $\xi$, $\zeta(s)/\zeta(2s)=\frac{\xi(s)}{\xi(2s)}\cdot\frac{2(2s-1)}{s-1}\,\pi^{-s/2}\,\frac{\Gamma(s)}{\Gamma(s/2)}$, and the duplication formula $\Gamma(s)=2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ gives the factor $\Gamma_{\mathrm{sq}}$. □

The script `sqfree_constants.py` checks the lemma at $s=3$, $0.3+2.1i$, $\frac34$ and $-1.5$, with relative errors below $5\times10^{-41}$.

Proposition 10.3.3. *For every $s$ with $\xi(2s)\neq0$,*

$$\sum_{n\ge1}\frac{\mu(n)^2}{n^s}=\Gamma_{\mathrm{sq}}(s)\,\frac{\mathbb{E}e^{(s-\frac12)X}}{\mathbb{E}e^{(2s-\frac12)X}},$$

*the left side read as the continuation $D(s)$ where the series diverges.*

*Proof.* Lemma 10.3.2 and (1.2.4), which gives $\xi(s)=\xi(\frac12)\mathbb{E}e^{(s-\frac12)X}$ and $\xi(2s)=\xi(\frac12)\mathbb{E}e^{(2s-\frac12)X}$. □

Corollary 10.3.4. *For real $\alpha\notin\{1,\frac12,-1,-3,-5,\dots\}$ the sign of $\zeta(\alpha)/\zeta(2\alpha)$ is the sign of $\frac{2\alpha-1}{\alpha-1}\Gamma(\frac{\alpha+1}{2})$. It is positive for $\alpha>1$ and for $-1<\alpha<\frac12$, negative for $\frac12<\alpha<1$, and equal to $(-1)^k$ on the interval $(-2k-1,-2k+1)$ for $k\ge1$.*

*Proof.* At real $\alpha$ the ratio of moment generating functions in Proposition 10.3.3 is positive, and so is $2^\alpha\pi^{-(\alpha+1)/2}$. The rational factor is positive except on $(\frac12,1)$, and $\Gamma(\frac{\alpha+1}{2})$ has the sign $(-1)^k$ when $\frac{\alpha+1}{2}\in(-k,-k+1)$, that is when $\alpha\in(-2k-1,-2k+1)$. □

At the even negative integers both $\zeta(\alpha)$ and $\zeta(2\alpha)$ vanish and the quotient keeps its sign through them. At $\alpha = \frac{3}{4}$, inside the interval where the series diverges, $\Gamma_{\mathrm{sq}}(\frac{3}{4}) = -1.34612167295$, $\mathbb{E}e^{X/4} = 1.00144503253351$ and $\mathbb{E}e^{X} = 1.02335500958202$, the last two by the trapezoid rule on the Pólya kernel (1.2.3); the ratio 0.978590052481 multiplies $\Gamma_{\mathrm{sq}}(\frac{3}{4})$ to $-1.31730127858 = \zeta(\frac{3}{4})/\zeta(\frac{3}{2})$.

THEOREM 10.3.5. *The constant of* (10.3.9) *is*

$$C = 2 + \log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\gamma_E + b_1 - 2b_2 = 2 + \log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\gamma_E + \mathbb{E}V_1 - 2\,\mathbb{E}V_2.$$

*With no hypothesis on the zeros,*

$$\begin{aligned} b_1 - 2b_2 &= \frac{G'}{G}\Big(\frac{1}{4}\Big) - 6\,\frac{G'}{G}\Big(\frac{9}{4}\Big) = \sum_k \Big(\frac{1}{\frac{1}{4}+a_k} - \frac{6}{\frac{9}{4}+a_k}\Big) \\ &= \int_0^\infty \big(e^{-t/4} - 6e^{-9t/4}\big)\,W(t)\,dt, \end{aligned} \tag{10.3.10}$$

*the sum running over the pairs* $\{\rho, 1-\rho\}$ *and converging absolutely. Under RH,* $a_k = \gamma_k^2$ *and* $b_1 - 2b_2 = \int \big((z+\frac{1}{4})^{-1} - 6(z+\frac{9}{4})^{-1}\big)\,U_{\mathrm{RH}}(dz)$*, a combination of two values of the Stieltjes transform of the Thorin measure of the clock.*

*Proof.* By the proof of Proposition 10.3.1, $(s-1)D(s) = \frac{6}{\pi^2}\big(1 + C(s-1) + O((s-1)^2)\big)$, so $C$ is the derivative of $\log((s-1)D(s))$ at $s=1$. By Proposition 10.3.3,

$$\begin{aligned} \log\big((s-1)D(s)\big) &= \log(2s-1) + s\log 2 - \tfrac{s+1}{2}\log\pi + \log\Gamma\big(\tfrac{s+1}{2}\big) \\ &\quad + \log \mathbb{E}e^{(s-\frac{1}{2})X} - \log \mathbb{E}e^{(2s-\frac{1}{2})X}. \end{aligned}$$

At $s=1$ the derivatives of the first four terms are $2$, $\log 2$, $-\frac{1}{2}\log\pi$ and $\frac{1}{2}\psi(1) = -\frac{1}{2}\gamma_E$, with $\psi$ the digamma function here. The derivative of $\log \mathbb{E}e^{(s-\frac{1}{2})X}$ at $s=1$ is $\mathbb{E}Xe^{X/2}/\mathbb{E}e^{X/2}$, the mean of $X$ tilted by $e^{X/2}$, which is $\mathbb{E}V_1 = b_1$ (Section 9.1.1); likewise the last term gives $-2\mathbb{E}V_2 = -2b_2$. For the zero form, differentiating $\log G(w^2) = \log\xi(\frac{1}{2}+w) - \log\xi(\frac{1}{2})$ gives $\frac{\xi'}{\xi}(\frac{1}{2}+w) = 2w\frac{G'}{G}(w^2)$, so $b_1 = \frac{G'}{G}(\frac{1}{4})$ and $b_2 = 3\frac{G'}{G}(\frac{9}{4})$. Then $\frac{G'}{G}(u) = \sum_k (u+a_k)^{-1}$ by (1.2.1), and this is $\int_0^\infty e^{-ut}W(t)\,dt$ for $u \ge 0$, by termwise integration, since $\operatorname{Re} a_k \ge \gamma_1^2 - \frac{1}{4} > 0$ and $\sum_k |a_k|^{-1} < \infty$. Under RH, $\frac{G'}{G}(u) = \int (u+z)^{-1} U_{\mathrm{RH}}(dz)$ by Proposition 1.2.1. □

The elementary part is $2 + \log 2 - \frac{1}{2}\log\pi - \frac{1}{2}\gamma_E = 1.832174405184478792$, and $b_1 = 0.023095708966121033814$, which is $1 + \frac{1}{2}\gamma_E - \frac{1}{2}\log 4\pi = \sum_\rho \rho^{-1}$, and $b_2 = 0.069066231530000676225$, by the closed forms of $\xi'/\xi$ and by numerical differentiation, which agree to every digit printed (`sqfree_constants.py`). So $b_1 - 2b_2 = -0.11503675409388031864$, and the elementary part plus the zero part reproduces $C$ to all 25 digits.

The variance identity of the first author's note, $\sum_k a_k^{-1} = \frac{1}{2}\operatorname{Var} X$, with $a_k = \gamma_k^2$ under RH, is the formula $\mathbb{E}T = \sum_k a_k^{-1} = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$ of Section 1.2, and the alternation of the even cumulants of $X$, which the note derives under RH, holds with no hypothesis (Section 13.1.2).

### 10.3.3 Number fields

Let $K$ be a number field with $r_1$ real and $r_2$ pairs of complex embeddings and discriminant $d_K$, and put $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(\frac{s}{2})$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_K(s) = s(s-1)\,|d_K|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_K(s).$$

Then $\xi_K$ is entire of order one, $\xi_K(s) = \xi_K(1-s)$ by Hecke's functional equation [328, Chapter XIII], $\xi_K$ is real on the real axis, and its zeros are the nontrivial zeros of $\zeta_K$, all in the strip $0 < \operatorname{Re} s < 1$; for $K = \mathbb{Q}$, $\xi_{\mathbb{Q}} = 2\xi$. For $K = \mathbb{Q}(i)$, where $\zeta_K(s) = \zeta(s)L(s,\chi_{-4})$, the duplication formula gives $\xi_K(s) = \xi(s)\,\xi(s,\chi_{-4})$ exactly, with $\xi(s,\chi_{-4}) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_{-4})$ the completed function of the odd character modulo 4 in the normalisation of Section 8.5.1.

Proposition 10.3.6. *Suppose $\zeta_K(\frac12) \neq 0$ and let $G_K(u) = \xi_K(\frac12 + \sqrt{u})/\xi_K(\frac12)$. Then $1/G_K$ is the Laplace transform of a GGC if and only if every zero of $\zeta_K$ in the critical strip lies on the line $\operatorname{Re} s = \frac12$, the Riemann hypothesis for $\zeta_K$. Its drift is then $0$ and its Thorin measure is $\sum_{\gamma>0}\delta_{\gamma^2}$, over the ordinates of the zeros counted with multiplicity. A real zero $\beta \in (\frac12, 1)$ of $\zeta_K$ would be a pole of $1/G_K$ at the point $(\beta - \frac12)^2$ of the positive axis.*

*Proof.* Since $\xi_K(\frac12 + w)$ is even and entire of order one, $G_K$ is entire of order $\frac12$, and Hadamard's theorem gives $G_K(u) = \prod_k (1 + u/a_{K,k})$ with $a_{K,k} = -(\rho_k - \frac12)^2$, one factor for each pair $\{\rho, 1-\rho\}$ of zeros of $\xi_K$, and $\sum_k |a_{K,k}|^{-1} < \infty$. If every zero lies on the line, every $a_{K,k} = \gamma_k^2$ is positive and $1/G_K$ is a convergent product of gamma Laplace transforms, a GGC with the stated Thorin measure and no drift. Conversely, the Laplace transform of a GGC is holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, so $G_K$ has no zero there, and its zeros $(\rho - \frac12)^2$ lie in $(-\infty, 0]$; since $\rho \neq \frac12$, this means $\rho - \frac12 \in i\mathbb{R} \setminus \{0\}$, that is $\operatorname{Re}\rho = \frac12$. A real zero $\beta$ gives the zero $(\beta - \frac12)^2 > 0$ of $G_K$. □

The hypothesis $\zeta_K(\frac12) \neq 0$ fixes the normalisation. Armitage found number fields whose Dedekind zeta function vanishes at the centre [11]; there the zero of $\xi_K(\frac12 + w)$ at $w = 0$ has even order $2m$, and the proposition applies to $u^{-m}\xi_K(\frac12 + \sqrt{u})$ normalised at $u = 0$.

Theorem 10.3.7. *Let*

$$A_K = 2 + \tfrac12 \log|d_K| + r_1\big(\log 2 - \tfrac12 \log\pi - \tfrac12\gamma_E\big) + r_2\big(2 - \gamma_E - \log 2\pi\big).$$

*Then the constant of* (10.3.7) *is*

$$C_K = A_K + \frac{\xi_K'}{\xi_K}(1) - 2\,\frac{\xi_K'}{\xi_K}(2) = A_K + \sum_k \Big(\frac{1}{\frac14 + a_{K,k}} - \frac{6}{\frac94 + a_{K,k}}\Big),$$

*the sum running over the pairs $\{\rho, 1-\rho\}$ of zeros of $\xi_K$, with multiplicity, a zero of order $2m$ at $s = \frac12$ counting as $m$ pairs with $a_{K,k} = 0$. Under the Riemann*

*hypothesis for $\zeta_K$, and $\zeta_K(\frac12)\neq 0$, the zero part is $\frac{G_K'}{G_K}(\frac14)-6\frac{G_K'}{G_K}(\frac94)$, the same combination of values of the Stieltjes transform of the Thorin measure of Proposition 10.3.6 as over $\mathbb{Q}$.*

*Proof.* From the definition of $\xi_K$,

$$(s-1)\frac{\zeta_K(s)}{\zeta_K(2s)}=\frac{\xi_K(s)}{\xi_K(2s)}\cdot 2(2s-1)\,|d_K|^{s/2}\,\frac{\Gamma_{\mathbb{R}}(2s)^{r_1}\Gamma_{\mathbb{C}}(2s)^{r_2}}{\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}},$$

and $C_K$ is the logarithmic derivative of the left side at $s=1$, by the Laurent expansions behind (10.3.7). With $\Gamma_{\mathbb{R}}'/\Gamma_{\mathbb{R}}(s)=-\frac12\log\pi+\frac12\psi(\frac{s}{2})$, $\Gamma_{\mathbb{C}}'/\Gamma_{\mathbb{C}}(s)=-\log 2\pi+\psi(s)$, $\psi(1)=-\gamma_E$, $\psi(\frac12)=-\gamma_E-2\log 2$ and $\psi(2)=1-\gamma_E$, the logarithmic derivative of the elementary factor at $s=1$ is $A_K$, and that of $\xi_K(s)/\xi_K(2s)$ is $\frac{\xi_K'}{\xi_K}(1)-2\frac{\xi_K'}{\xi_K}(2)$. The Hadamard product of $\xi_K(\frac12+w)$, which is even in $w$, gives $\frac{\xi_K'}{\xi_K}(\frac12+w)=\sum_k 2w/(w^2+a_{K,k})$, and the values at $w=\frac12$ and $w=\frac32$ give the zero form. □

For the Gaussian field $\mathbb{Q}(i)$, with $r_2=1$ and $d_K=-4$, `sqfree_constants.py` gives $c_K/\zeta_K(2)=3/(2\pi L(2,\chi_{-4}))=0.521269392989129$, with $L(2,\chi_{-4})$ Catalan's constant, and

$$\gamma_K=\gamma_E+\frac{L'}{L}(1,\chi_{-4})=0.822825249678847033,$$
$$\frac{\zeta_K'}{\zeta_K}(2)=-0.48089570872664776864,$$

and $C_K=1.784616667132142570284$, where $\frac{L'}{L}(1,\chi_{-4})=\gamma_E+2\log 2+3\log\pi-4\log\Gamma(\frac14)=0.24560958477731417239$. In Theorem 10.3.7, $A_K=4-\gamma_E-\log\pi=2.2780544492490669653$ and the zero part is the sum of the zero part $b_1-2b_2=-0.11503675409388$ of $\zeta$ and the zero part $\frac{\xi_\chi'}{\xi_\chi}(1)-2\frac{\xi_\chi'}{\xi_\chi}(2)=-0.378401028023044$ of $\xi_\chi=\xi(\cdot,\chi_{-4})$: the zeros of $\zeta$ and of $L(s,\chi_{-4})$ simply add their Thorin atoms, and the two sides of the theorem differ by $4.6\times10^{-41}$. The 289 zeros of $L(s,\chi_{-4})$ below height 400, located in Arb, give $-0.365385920572$ for the $L$ part, and their smooth tail $-0.013015062$ brings the sum within $4.6\times10^{-8}$ of the closed form.

### 10.3.4 The Thorin atoms in the squarefree count

Put $s=\frac14+\frac12\sqrt{u}$, so that $2s=\frac12+\sqrt{u}$. Lemma 10.3.2 and the clock give, for $u\geq 0$ and with no hypothesis,

$$D\Big(\frac14+\frac{\sqrt{u}}{2}\Big)=\Gamma_{\mathrm{sq}}\Big(\frac14+\frac{\sqrt{u}}{2}\Big)\,\frac{\xi(\frac14+\frac12\sqrt{u})}{\xi(\frac12)}\,\mathbb{E}e^{-uT},$$

since $1/G$ is the Laplace transform of the clock $T$ (Section 1.2). The poles of $D$ in the strip $0<\operatorname{Re}s<\frac12$ are therefore among the points where $G$ vanishes, $u=-a_k$,

that is $s = \rho_k/2$, and they are all of these points when $\zeta(\rho_k/2) \neq 0$, as under RH; under RH they are the atoms $\gamma_k^2$ of $U_{\mathrm{RH}}$, placed at $u = -\gamma_k^2$, that is at $s = \frac{1}{4} \pm \frac{i}{2}\gamma_k$.

At a simple zero $\rho$, $\zeta(2s) = 2\zeta'(\rho)(s - \rho/2) + O((s - \rho/2)^2)$, so the residue of $D(s)x^s/s$ at $s = \rho/2$ is $R_\rho x^{\rho/2}$, and formally

$$\Delta(x) \approx 1 + \sum_{\rho} R_\rho\, x^{\rho/2} + \sum_{k=1,3,5,\ldots} r_k\, x^{-k}, \qquad R_\rho = \frac{\zeta(\rho/2)}{\rho\,\zeta'(\rho)}, \qquad r_k = \frac{\zeta(-k)}{-2k\,\zeta'(-2k)}, \tag{10.3.11}$$

where $1 = D(0)$ is the residue at $s = 0$ and $r_k x^{-k}$ is the residue at the trivial pole $s = -k$. For the first zero $R_{\rho_1} = 0.028803 - 0.0961835i$ and $|R_{\rho_1}| = 0.100404$, with $\zeta(\rho_1/2) = 1.03142 + 0.452556i$, and for the next four zeros $|R_\rho| = 0.069179$, 0.0338528, 0.0260772 and 0.0464698; $r_1 = -1.3684327776202059$, $r_3 = 0.23541450677954054$ and $r_5 = -0.020962860062791976$ (`sqfree_constants.py`, `sqfree_atoms.py`). Under RH every term of the sum over $\rho$ has size $x^{1/4}$, which is why $\Delta(x) = O(x^{1/4+\varepsilon})$ is the natural conjecture and why $\Delta(x)/x^{1/4}$ stays of order one in Table 10.8; Corollary 10.3.11 shows that the exponent $\frac{1}{4}$ cannot be lowered. The series over $\rho$ is not expected to converge absolutely, and the bound $O(x^{1/4+\varepsilon})$ is open even under RH: the known conditional exponents are larger than $\frac{1}{4}$, among them $\frac{7}{22}$ of Baker and Pintz [18], since improved, and the obstacle is the control of $1/\zeta$ near the critical line.

Figure 10.6 compares $\Delta(x)$ on $[100, 3000]$ with the right side of (10.3.11) truncated after the first $K$ zeros, both divided by $x^{1/4}$ and sampled on the grid $x = 100 + j/20$. The Pearson correlation of the two curves is 0.828, 0.866, 0.915, 0.927 and 0.945 for $K = 10$, 20, 60, 100 and 200, and the root mean square of their difference falls from 0.149 to 0.0868, against 0.304 for $\Delta(x)/x^{1/4}$ itself; what remains includes the unit jumps of $Q$, which no finite sum of atoms reproduces.

For the sum $\sum_{n \le x} \mu(n)^2/n$ of Proposition 10.3.1 the same atoms appear at $s = \rho/2 - 1$, with residues $\zeta(\rho/2)x^{\rho/2-1}/((\rho - 2)\zeta'(\rho))$ of size $x^{-3/4}$ under RH (of modulus $0.09991\,x^{-3/4}$ for the first zero), inside the error $O(x^{-1/2})$. Likewise the error $\mathcal{E}_\alpha$ of (10.3.4), for $\frac{1}{2} < \alpha < 1$, has atoms of size $x^{1/4-\alpha}$. By (10.3.6), a pointwise bound of that order in either case would follow from $\Delta(x) = O(x^{1/4+\varepsilon})$, and it is open for the same reason.

### 10.3.5 Smoothed counts

Smoothing the count removes the obstacle. For $x > 0$ let

$$Q_1(x) = \sum_{n \ge 1} \mu(n)^2 e^{-n/x} = \sum_{d \ge 1} \frac{\mu(d)}{e^{d^2/x} - 1}, \qquad Q_2(x) = \sum_{n \ge 1} \mu(n)^2 \Big(1 + \frac{2n}{x}\Big) e^{-2n/x},$$

the second form of $Q_1$ by $\mu(n)^2 = \sum_{d^2 \mid n} \mu(d)$. If $G_{1,x}$ is exponential and $G_{2,x}$ has the Gamma(2) law, both with mean $x$, then $P(G_{1,x} \ge n) = e^{-n/x}$ and $P(G_{2,x} \ge n) = (1 + 2n/x)e^{-2n/x}$, so $Q_1(x) = \mathbb{E}Q(G_{1,x})$ and $Q_2(x) = \mathbb{E}Q(G_{2,x})$.

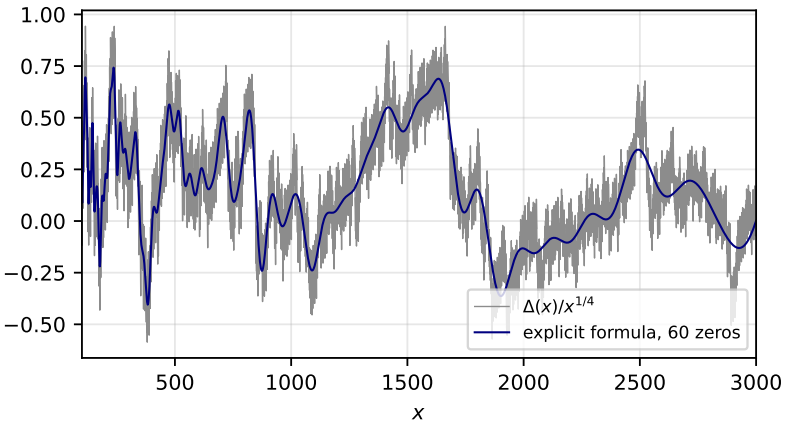



Figure 10.6: The normalised error $\Delta(x)/x^{1/4}$ of the squarefree count on $[100, 3000]$ (grey) and the explicit formula (10.3.11) with the first 60 zeros and the trivial terms at $s = -1$, $-3$ and $-5$, divided by $x^{1/4}$ (dark). Drawn by `sqfree_atoms.py`.

Since $\int_0^\infty e^{-n/x} x^{-s-1} dx = \Gamma(s) n^{-s}$ and $\int_0^\infty (1 + 2n/x) e^{-2n/x} x^{-s-1} dx = (1 + s)\Gamma(s)(2n)^{-s}$, the Mellin transforms are

$$\begin{aligned} \int_0^\infty Q_1(x)\, x^{-s-1} dx &= \Gamma(s) D(s), \\ \int_0^\infty Q_2(x)\, x^{-s-1} dx &= (1+s)\Gamma(s)\, 2^{-s} D(s), \qquad \operatorname{Re} s > 1. \end{aligned} \tag{10.3.12}$$

The kernels $\Gamma(s)$ and $(1 + s)\Gamma(s)2^{-s}$ have no zeros in $\operatorname{Re} s > 0$ and decay exponentially on vertical lines.

Lemma 10.3.8. *If $D(s) = \zeta(s)/\zeta(2s)$ is analytic in the half plane $\operatorname{Re} s > \frac{1}{4}$ apart from $s = 1$, then RH holds.*

*Proof.* Let $\rho$ be a zero with $\beta = \operatorname{Re}\rho > \frac{1}{2}$. Then $\operatorname{Re}(\rho/2) > \frac{1}{4}$, so the pole of $1/\zeta(2s)$ at $s = \rho/2$ must be cancelled, that is $\zeta(\rho/2) = 0$. Since $\frac{1}{4} < \operatorname{Re}(\rho/2) < \frac{1}{2}$, the functional equation and the reflection $s \mapsto \bar{s}$ make $\rho' = 1 - \bar{\rho}/2$ a zero, with $\operatorname{Re}\rho' = 1 - \beta/2 > \frac{1}{2}$ and $\operatorname{Im}\rho' = \operatorname{Im}\rho/2$. Iterating gives zeros $\rho_j$ whose real parts tend to $\frac{2}{3}$, the fixed point of $\beta \mapsto 1 - \beta/2$, and whose imaginary parts are $\operatorname{Im}\rho/2^j$. Since $\zeta < 0$ on $(0, 1)$, $\operatorname{Im}\rho \neq 0$, so the $\rho_j$ are distinct and accumulate at $\frac{2}{3}$, where $\zeta$ is analytic and not identically zero, a contradiction. □

Theorem 10.3.9. (i) *RH holds if and only if $Q_1(x) = \frac{6}{\pi^2} x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$, and if and only if $Q_2(x) = \frac{6}{\pi^2} x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$.*

(ii) *Under RH, if the zeros are simple, then*

$$Q_1(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho A_\rho\, x^{\rho/2} + \frac{c_1 \log x + c_2}{x} + O(x^{-2}),$$
$$Q_2(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho B_\rho\, x^{\rho/2} + \frac{2c_1}{x} + O(x^{-2}),$$
$$A_\rho = \frac{\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)}, \qquad B_\rho = \Big(1+\frac{\rho}{2}\Big)\frac{2^{-\rho/2}\,\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)},$$

*the sums over $\rho$ taken over $|\operatorname{Im}\rho| < 2T_j$ along a sequence $T_j \to \infty$, with*

$$c_1 = -\frac{\zeta(-1)}{2\zeta'(-2)} = -1.36843277762021,$$
$$c_2 = c_1\Big(1 - \gamma_E + \frac{\zeta'(-1)}{\zeta(-1)} - \frac{\zeta''(-2)}{\zeta'(-2)}\Big) = -0.339381226129599.$$

*Proof.* (i) Suppose RH. By Section 14.2 of [498], $1/\zeta(\sigma+it) = O(|t|^\varepsilon)$ uniformly for $\sigma \ge \frac12 + \varepsilon$, and $\zeta(\sigma+it)$ grows at most polynomially in $|t|$ on vertical strips, while $\Gamma(s)$ decays exponentially. Move the line of integration in $Q_1(x) = \frac{1}{2\pi i}\int_{(2)} \Gamma(s)D(s)x^s\,ds$ to $\operatorname{Re} s = \frac14 + \varepsilon$. The only pole crossed is $s=1$, with residue $\Gamma(1)/\zeta(2)\cdot x = 6x/\pi^2$; the point $s=\frac12$ is regular because $\zeta(2s)$ has a pole there. On the new line $|\Gamma(s)D(s)|$ is integrable and the integral is $O(x^{1/4+\varepsilon})$. The same argument with the kernel $(1+s)\Gamma(s)2^{-s}$ treats $Q_2$.

Conversely, suppose $Q_1(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$. For $\operatorname{Re} s > 1$, by (10.3.12),

$$\Gamma(s)D(s) - \frac{6}{\pi^2}\,\frac{1}{s-1} = \int_1^\infty \Big(Q_1(x) - \frac{6}{\pi^2}x\Big)x^{-s-1}dx + \int_0^1 Q_1(x)\,x^{-s-1}dx.$$

The first integral converges and is analytic in $\operatorname{Re} s > \frac14$, by the hypothesis, and the second is entire, since $Q_1(x) = O(e^{-1/x})$ as $x \to 0$. As $\Gamma$ has no zeros, $D$ is analytic in $\operatorname{Re} s > \frac14$ apart from $s=1$, and Lemma 10.3.8 gives RH. The same argument applies to $Q_2$.

(ii) By Theorem 9.7 of [498] there are heights $T_j \in [j, j+1]$ at which $1/\zeta(\sigma \pm 2iT_j) = T_j^{O(1)}$ uniformly for $-1 \le \sigma \le 2$, and on the left of the line $\operatorname{Re} s = -\frac12$ the functional equation bounds $1/\zeta(2s)$ polynomially. Move the line of integration to $\operatorname{Re} s = -\frac52$ along the rectangles of height $T_j$; the exponential decay of $\Gamma$ makes the horizontal sides vanish as $j \to \infty$, and the integral on the new line is $O(x^{-5/2})$. The poles crossed are $s=1$; the points $\rho/2$, with the residues $A_\rho x^{\rho/2}$ and $B_\rho x^{\rho/2}$; $s=0$, where $\Gamma(s)D(s)$ and $(1+s)\Gamma(s)2^{-s}D(s)$ have residue $D(0) = \zeta(0)/\zeta(0) = 1$; $s=-1$; and $s=-2$, a simple pole of both, which gives a multiple of $x^{-2}$. At $s=-1$, $\Gamma$ has a simple pole and $1/\zeta(2s)$ another, from the trivial zero $-2$ of $\zeta$, so

$\Gamma(s)D(s) = c_1(s+1)^{-2} + c_2(s+1)^{-1} + O(1)$, with $c_1$ and $c_2$ from the expansions $\Gamma(s) = -(s+1)^{-1}(1 + (1-\gamma_E)(s+1) + \cdots)$, $\zeta(s) = \zeta(-1)(1 + \frac{\zeta'(-1)}{\zeta(-1)}(s+1) + \cdots)$ and $\zeta(2s) = 2\zeta'(-2)(s+1)(1 + \frac{\zeta''(-2)}{\zeta'(-2)}(s+1) + \cdots)$, and the residue of $\Gamma(s)D(s)x^s$ is $x^{-1}(c_1 \log x + c_2)$. The factor $1+s$ makes the pole of $(1+s)\Gamma(s)2^{-s}D(s)$ at $s=-1$ simple, with residue $2c_1$ times $x^{-1}$. □

At the even negative integers $\Gamma(s)D(s)$ has simple poles, since $\zeta(s)$ vanishes there together with $\zeta(2s)$, and at the odd ones double poles: `sqfree_smooth.py` computes the Laurent coefficients at $s = -1, \dots, -6$ by the trapezoid rule on circles of radius 0.3 and reproduces the closed forms of $c_1$ and $c_2$ within $5 \times 10^{-46}$. For $Q_2$ the double pole at $s=-1$ becomes a simple one, with residue $2c_1$.

The coefficients $A_\rho$ are tiny, $|A_{\rho_1}| = 1.648 \times 10^{-5}$ and $6.29 \times 10^{-82}$ at the hundredth zero, because $|\Gamma(\rho/2)|$ decays like $e^{-\pi\gamma/4}$. `sqfree_smooth.py` checks part (ii) for $Q_1$: the smoothed error is the sum of the atom terms and the trivial terms to the working precision. Table 10.10 in Section 10.3.7 does the same for $Q_2$.

The Perron kernel $x^s/s$ decays only like $1/|t|$, so the atom series for $Q(x)$ has no exponential damping, and bounding its partial sums would need control of $\sum_{|\gamma|\le T} |\zeta(\rho/2)/(\rho\zeta'(\rho))|$, which is out of reach.

### 10.3.6 Mean square, and how much smoothing the sharp count needs

Two further results bracket the sharp problem. The first settles it on average; it is presumably folklore. For $\operatorname{Re} s > 1$ let

$$F(s) = \int_1^\infty \Delta(x)\, x^{-s-1} dx = \frac{D(s)}{s} - \frac{6}{\pi^2}\,\frac{1}{s-1},$$

the second form because $\int_1^\infty Q(x)x^{-s-1}dx = D(s)/s$.

Theorem 10.3.10. *RH holds if and only if $\int_X^{2X} \Delta(x)^2\,dx = O(X^{3/2+\varepsilon})$ for every $\varepsilon > 0$, that is, if and only if the root mean square of $\Delta$ on every range $[X, 2X]$ is $O(X^{1/4+\varepsilon})$.*

*Proof.* Suppose RH. Then $F$ continues analytically to $\operatorname{Re} s > \frac14$: the poles of $D$ at the points $\rho/2$ lie on the line $\operatorname{Re} s = \frac14$, $s = \frac12$ is regular, and the pole at $s=1$ cancels. By Section 14.2 of [498], under RH $\zeta(\sigma+it) = O(|t|^{\max(0,\frac12-\sigma)+\varepsilon})$ and $1/\zeta(2\sigma+2it) = O(|t|^{\varepsilon})$ uniformly for $\sigma \ge \frac14+\varepsilon$ and $|t| \ge 1$, so $F(\sigma+it) = O(|t|^{-3/4+2\varepsilon})$ uniformly there. For $w > 2$ let $I(w) = \int_1^\infty \Delta(x)^2 x^{-w-1}dx$, the Laplace transform in $v = \log x$ of the nonnegative function $\Delta(e^v)^2$. Plancherel's theorem for the function $v \mapsto \Delta(e^v)e^{-wv/2}$ on $(0,\infty)$, whose Fourier transform is $t \mapsto F(\frac{w}{2}+it)$, and the symmetry $F(\bar s) = \overline{F(s)}$ give

$$I(w) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F\Big(\frac{w}{2}+it\Big)\,F\Big(\frac{w}{2}-it\Big)\,dt, \qquad w > 2.$$

The right side is analytic in $w$ on $\operatorname{Re} w > \frac12$, since the integrand is analytic there and $O(|t|^{-3/2+4\varepsilon})$ locally uniformly. So $I$, the Laplace transform of a nonnegative function, continues analytically to a neighbourhood of every point of $(\frac12,\infty)$, and Landau's theorem [514, Chapter II, Theorem 5b] puts its abscissa of convergence at or below $\frac12$. Hence $I(w)<\infty$ for every $w>\frac12$, and $\int_X^{2X}\Delta^2\,dx \le (2X)^{w+1}I(w) = O(X^{3/2+\varepsilon})$ with $w=\frac12+\varepsilon$.

Conversely, by the Cauchy Schwarz inequality the hypothesis gives $\int_X^{2X}|\Delta(x)|x^{-\sigma-1}dx = O(X^{1/4+\varepsilon-\sigma})$, so, summing over $X=2^j$, the integral defining $F$ converges absolutely for $\operatorname{Re} s>\frac14$. Then $D(s)=s\big(F(s)+\frac{6}{\pi^2}\frac{1}{s-1}\big)$ is analytic in $\operatorname{Re} s>\frac14$ apart from $s=1$, and Lemma 10.3.8 gives RH. □

Table 10.9 gives the root mean square of $\Delta$ on dyadic ranges, computed exactly from the sieve: on $[n,n+1)$ the error decreases linearly from $\Delta_n = Q(n)-6n/\pi^2$ to $\Delta_n - 6/\pi^2$, so $\int_n^{n+1}\Delta^2 = \Delta_n^2 - \frac{6}{\pi^2}\Delta_n + \frac{12}{\pi^4}$, and the extremes over real $x$ are taken at integers or as left limits there. Over the eighteen ranges $[X,2X]$ with $X=10^4\cdot 2^j$, $j=0,\dots,13$, and $X=10^4,\dots,10^8$, the root mean square divided by $X^{1/4}$ lies between 0.2337 and 0.3606 and the largest $|\Delta(x)|/X^{1/4}$ between 0.6851 and 1.0361; on $[10^3, 2\times10^8]$ the largest value of $\Delta(x)/x^{1/4}$ is 0.9807, near $x=47523$, and the least $-0.8685$, near $x=80157$.

| $X$ | $\big(\frac1X\int_X^{2X}\Delta^2\big)^{1/2}/X^{1/4}$ | $\max_{[X,2X]}\lvert\Delta(x)\rvert/X^{1/4}$ | near $x=$ |
|---|---|---|---|
| $10^4$ | 0.2701 | 0.8979 | 18471 |
| $10^5$ | 0.2494 | 0.6851 | 128599 |
| $10^6$ | 0.2779 | 0.8382 | 1618611 |
| $10^7$ | 0.3541 | 0.9931 | 17199381 |
| $10^8$ | 0.2812 | 0.8636 | 144151023 |

Table 10.9: Root mean square and largest absolute value of the squarefree error $\Delta$ on the ranges $[X,2X]$, divided by $X^{1/4}$, over real $x$; the largest value on $[10^7, 2\times10^7]$ is the left limit at 17199381, and over the integers alone it is 0.982. Computed by `sqfree_sieve.py`.

Corollary 10.3.11. (i) *Under RH, for every $\varepsilon>0$ the set of $x\in[X,2X]$ with $|\Delta(x)|>X^{1/4+\varepsilon}$ has measure $O(X^{1-\varepsilon})$.*

(ii) *With no hypothesis,*

$$\limsup_{x\to\infty}\frac{\Delta(x)}{x^{1/4}} \ge |R_{\rho_1}|, \qquad \liminf_{x\to\infty}\frac{\Delta(x)}{x^{1/4}} \le -|R_{\rho_1}|,$$

*where $R_{\rho_1}=\zeta(\rho_1/2)/(\rho_1\zeta'(\rho_1))$ and $|R_{\rho_1}|=0.100404$. The same holds for $Q_2(x)-\frac{6}{\pi^2}x-1$ with $|B_{\rho_1}|=9.944\times10^{-5}$ in place of $|R_{\rho_1}|$.*

*Proof.* (i) is Chebyshev's inequality applied to Theorem 10.3.10. For (ii) suppose that $\Delta(x)\le c\,x^{1/4}$ for $x\ge x_0$, with $0<c<|R_{\rho_1}|$. The function $L(s)=$

$\int_{x_0}^{\infty}(cx^{1/4}-\Delta(x))x^{-s-1}dx$ is the Laplace transform in $\log x$ of a nonnegative function, and for $\operatorname{Re} s>1$

$$L(s)=\frac{c\,x_0^{1/4-s}}{s-\frac14}-F(s)+E(s),$$

with $E$ entire. The right side has no singularity on the real half line $(\frac14,\infty)$: a real pole of $D$ there would be a real zero of $\zeta$ in $(\frac12,\infty)$, of which there are none, $s=\frac12$ is regular, and the pole at $s=1$ cancels. By Landau's theorem the integral $L(s)$ converges for $\operatorname{Re} s>\frac14$, so $F$ is analytic there (which already gives RH, by Lemma 10.3.8), and $|L(\sigma+it)|\le L(\sigma)$ for $\sigma>\frac14$. As $\sigma\downarrow\frac14$, $L(\sigma)=c/(\sigma-\frac14)+O(1)$, since $F$ is analytic at $s=\frac14$, where $\zeta(2s)=\zeta(\frac12)\neq0$. At $s=\sigma+\frac{i}{2}\gamma_1$ the first and last terms of the display are $O(1)$, while $F(s)=R_{\rho_1}/(\sigma-\frac14)+O(1)$, because $D(s)/s$ has a simple pole at $\rho_1/2$ with residue $R_{\rho_1}$ (the zero $\rho_1$ is simple and $\zeta(\rho_1/2)\neq0$). Hence $|R_{\rho_1}|\le c$, a contradiction. The lower bound is the same argument applied to $c\,x^{1/4}+\Delta(x)$, and the smoothed case is the same argument with the kernel $(1+s)\Gamma(s)2^{-s}$, whose residue at $\rho_1/2$ is $B_{\rho_1}$. □

The second result measures the cost of sharpening. Let $G_{n,x}$ have the Gamma$(n)$ law with mean $x$, so that $G_{n,x}$ concentrates at $x$ in a window of width $x/\sqrt{n}$; $Q_1(x)=\mathbb{E}Q(G_{1,x})$ and $Q_2(x)=\mathbb{E}Q(G_{2,x})$.

Proposition 10.3.12. *Under RH, uniformly in $n\ge1$,*

$$\mathbb{E}Q(G_{n,x})=\frac{6}{\pi^2}\,x+O_\varepsilon\big(x^{1/4+\varepsilon}\,n^{1/8+\varepsilon}\big).$$

*Proof.* With rate $\lambda=n/x$, $P(G_{n,x}\ge m)=\Gamma(n,\lambda m)/\Gamma(n)$, whose Mellin transform in $\lambda m$ is $\Gamma(s+n)/(s\Gamma(n))$, so $\mathbb{E}Q(G_{n,x})=\frac{1}{2\pi i}\int_{(2)}D(s)\frac{\Gamma(s+n)}{s\Gamma(n)}\big(\frac{x}{n}\big)^s ds$. Move the line to $\operatorname{Re} s=c=\frac14+\varepsilon$, collecting $6x/\pi^2$ at $s=1$. On the new line $|\Gamma(n+c+it)|/\Gamma(n)=O\big(n^c\exp(-\kappa\min(t^2/n,|t|))\big)$ for an absolute $\kappa>0$, since $\Gamma(n+c)/\Gamma(n)=O(n^c)$ and $|\Gamma(n+c+it)/\Gamma(n+c)|^2=\prod_{k\ge0}\big(1+t^2/(n+c+k)^2\big)^{-1}$; the factors with $k<n$ give $e^{-\kappa t^2/n}$ when $|t|\le n$, and those with $k\le|t|$ give $e^{-\kappa|t|}$ when $|t|>n$. Under RH, $|D(c+it)|=O(|t|^{1/4+\varepsilon})$ for $|t|\ge1$ (Section 14.2 of [498] and the functional equation). The remaining integral is therefore $O\big(x^c\int(1+|t|)^{-3/4+\varepsilon}e^{-\kappa\min(t^2/n,|t|)}dt\big)=O(x^cn^{1/8+\varepsilon})$, the integral being dominated by $|t|\lesssim\sqrt{n}$. □

Balancing the window against the loss gives a bound for the sharp count.

Corollary 10.3.13. *Under RH, $\Delta(x)=O(x^{2/5+\varepsilon})$ for every $\varepsilon>0$.*

*Proof.* Since $Q$ is nondecreasing and counts integers, $|Q(y)-Q(x)|\le|y-x|+1$, so $|\mathbb{E}Q(G_{n,x})-Q(x)|\le\mathbb{E}|G_{n,x}-x|+1\le x/\sqrt{n}+1$. With Proposition 10.3.12,

$\Delta(x) = O(x^{1/4+\varepsilon}n^{1/8+\varepsilon} + x/\sqrt{n})$, and $n = \lceil x^{6/5} \rceil$ balances the two terms at $x^{2/5+O(\varepsilon)}$. $\square$

The exponent $\frac{2}{5}$ is weaker than the known conditional exponents. If the loss $n^{1/8}$ were replaced by $n^{\delta}$, the same balance would give $\Delta(x) = O(x^{1-3/(8\delta+4)+\varepsilon})$, which tends to $x^{1/4+\varepsilon}$ as $\delta \to 0$; conversely $\Delta(x) = O(x^{1/4+\varepsilon})$ gives $|\mathbb{E}Q(G_{n,x}) - \frac{6}{\pi^2}x| \le \mathbb{E}|\Delta(G_{n,x})| = O(x^{1/4+\varepsilon})$ uniformly in $n$, by Jensen's inequality. So the sharp conjecture amounts to removing the loss, which needs cancellation in $\int_{|t|\le T} D(\frac{1}{4}+it)x^{it}\,dt/t$ and not only a bound on the size of the integrand.

### 10.3.7 Gamma(2) smoothing and the Mertens function

The elementary method rests on the terms $\mu(d)(\lfloor x/d^2 \rfloor - x/d^2) = -\mu(d)\{x/d^2\}$ of (10.3.8). Following the idea of Steutel and Thiemann of splitting a random variable into its integer and fractional parts [488], we randomise $x$ by a gamma law. Theorem 10.3.18 shows that here shape two is the largest for which the average is taken under a positive law.

Let $Y$ have the Gamma($\nu$) law with rate $\lambda$, let $a > 0$ and $y = \lambda a$. Then $\mathbb{E}\lfloor Y/a \rfloor = \sum_{m\ge1} P(Y \ge ma) = \sum_{m\ge1} \Gamma(\nu, my)/\Gamma(\nu)$ and $\mathbb{E}Y/a = \nu/y$, so the expected fractional part is

$$\mathbb{E}\{Y/a\} = g_\nu(y) = \frac{\nu}{y} - \sum_{m\ge1} \frac{\Gamma(\nu, my)}{\Gamma(\nu)}.$$

For $\nu = 1$, Steutel and Thiemann observed that $\lfloor Y/a \rfloor$ is geometric and $\{Y/a\}$ an independent truncated exponential variable, with $\mathbb{E}\lfloor Y/a \rfloor = 1/(e^y - 1)$ and $g_1(y) = 1/y - 1/(e^y - 1)$.

Lemma 10.3.14. *For $\nu = 2$,*

$$\mathbb{E}\lfloor Y/a \rfloor = \frac{1}{e^y - 1} + \frac{y}{4\sinh^2(y/2)}, \qquad g_2(y) = \frac{2}{y} - \frac{1}{e^y - 1} - \frac{y}{4\sinh^2(y/2)}.$$

*For $0 < |y| < 2\pi$, with the Bernoulli numbers $B_{2k}$,*

$$g_2(y) = \frac{1}{2} + \sum_{k\ge2} \frac{(2k-2)B_{2k}}{(2k)!}\, y^{2k-1} = \frac{1}{2} - \frac{y^3}{360} + \frac{y^5}{7560} - \cdots,$$

*$g_2(y) = 2/y + O(ye^{-y})$ as $y \to \infty$, and for every $\nu > 0$*

$$\int_0^\infty g_\nu(y)\, y^{s-1}\, dy = -\frac{\Gamma(s+\nu)\,\zeta(s)}{s\,\Gamma(\nu)}, \qquad 0 < \operatorname{Re} s < 1,$$

*which is $-(1+s)\Gamma(s)\zeta(s)$ for $\nu = 2$.*

*Proof.* For $\nu = 2$, $P(Y \geq ma) = (1+my)e^{-my}$, and $\sum_m e^{-my} = 1/(e^y-1)$, $\sum_m mye^{-my} = y/(4\sinh^2(y/2))$. For the expansion use $1/(e^y-1) = 1/y - \frac12 + \sum_{k\geq 1} B_{2k} y^{2k-1}/(2k)!$ and $y/(4\sinh^2(y/2)) = -y\frac{d}{dy}(e^y-1)^{-1}$. For the Mellin transform write $g_\nu(y) = \mathbb{E}\{Y/y\}$ with $Y$ of law Gamma($\nu$,1), substitute $u = Y/y$, and use the classical formula $\int_0^\infty \{u\}u^{-s-1}du = -\zeta(s)/s$ for $0 < \operatorname{Re} s < 1$ [498, Section 2.1]: the transform is $\mathbb{E}Y^s \cdot (-\zeta(s)/s)$, with $\mathbb{E}Y^s = \Gamma(s+\nu)/\Gamma(\nu)$. □

The terms of the elementary method now read $\mathbb{E}\big[\mu(d)\big(\lfloor G_{2,x}/d^2\rfloor - G_{2,x}/d^2\big)\big] = -\mu(d)g_2(2d^2/x)$ for $G_{2,x}$ of the Gamma(2) law with mean $x$, and since $0 \leq g_2(y) \leq \min(1, 2/y)$ the sum over all $d$ converges absolutely, with no truncation at $d^2 \leq x$:

$$Q_2(x) = \sum_{n\geq 1} \mu(n)^2 \Big(1 + \frac{2n}{x}\Big) e^{-2n/x} = \frac{6}{\pi^2}\, x - \sum_{d\geq 1} \mu(d)\, g_2\Big(\frac{2d^2}{x}\Big). \tag{10.3.13}$$

Theorem 10.3.15. *Let*

$$f_2(y) = -2g_2'(y) = \frac{4}{y^2} - \frac{y\cosh(y/2)}{2\sinh^3(y/2)}, \qquad y > 0.$$

*Then $f_2 > 0$, $f_2(y) = y^2/60 + O(y^4)$ as $y \to 0$ and $f_2(y) \sim 4/y^2$ as $y \to \infty$, and $f_2$ is the density of a probability law $\mathcal{V}_2$ with $P(\mathcal{V}_2 \geq y) = 2g_2(y)$ and*

$$\mathbb{E}\mathcal{V}_2^s = -2\,\Gamma(s+2)\,\zeta(s), \qquad -3 < \operatorname{Re} s < 1.$$

*For every $x > 0$,*

$$Q_2(x) = \frac{6}{\pi^2}\, x - \frac12\, \mathbb{E}M\Big(\sqrt{x\mathcal{V}_2/2}\Big).$$

*Moreover $f_2$ is twice the difference between the continuous and the integer mixtures of the Gamma(2) densities $\gamma_{2,\lambda}(y) = \lambda^2 y e^{-\lambda y}$ over their rate:*

$$f_2(y) = 2\Big(\int_0^\infty \gamma_{2,\lambda}(y)\, d\lambda - \sum_{m\geq 1} \gamma_{2,m}(y)\Big) = -2\int_0^\infty \Big(\{\lambda\} - \frac12\Big)\, \partial_\lambda \gamma_{2,\lambda}(y)\, d\lambda.$$

*Proof.* With $z = y/2$, $f_2(y) > 0$ is $(\sinh z/z)^3 > \cosh z$, Lazarević's inequality [330]. The expansion at 0 follows from Lemma 10.3.14. The mass is $2(g_2(0+) - g_2(\infty)) = 1$ and the tail is $2g_2(y)$. Integrating by parts in the Mellin formula of Lemma 10.3.14 gives $\mathbb{E}\mathcal{V}_2^s = 2s\int_0^\infty y^{s-1} g_2(y)\, dy = -2s(1+s)\Gamma(s)\zeta(s)$ for $0 < \operatorname{Re} s < 1$, and the behaviour at 0 and at infinity extends the integral and the identity to $-3 < \operatorname{Re} s < 1$. For the representation, $P(\sqrt{x\mathcal{V}_2/2} \geq d) = P(\mathcal{V}_2 \geq 2d^2/x) = 2g_2(2d^2/x)$, so $\frac12 \mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = \sum_d \mu(d) g_2(2d^2/x)$, the interchange being justified by absolute convergence, and (10.3.13) gives the formula. Finally $\int_0^\infty \gamma_{2,\lambda}(y)\, d\lambda = 2/y^2$ and $\sum_m \gamma_{2,m}(y) = y\sum_m m^2 e^{-my} = y\cosh(y/2)/(4\sinh^3(y/2))$, and the last form is the Euler Maclaurin formula $\int_0^\infty h - \sum_{m\geq 1} h(m) = -\int_0^\infty (\{\lambda\} - \frac12) h'(\lambda)\, d\lambda$ for $h = \gamma_{2,\cdot}(y)$, which vanishes at $\lambda = 0$. □

The exponential case is the same with

$$f_1(y) = -2g_1'(y) = \frac{2}{y^2} - \frac{1}{2\sinh^2(y/2)} = 2\Big(\int_0^\infty \lambda e^{-\lambda y}\,d\lambda - \sum_{m\ge 1} m e^{-my}\Big) > 0,$$

positive because $\sinh z > z$, with $f_1(0+) = \frac16$, $\mathbb{E}\mathcal{V}_1^s = -2\Gamma(s+1)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, and

$$Q_1(x) = \frac{6}{\pi^2}\,x - \frac12\,\mathbb{E}M\Big(\sqrt{x\mathcal{V}_1}\Big).$$

Corollary 10.3.16. (i) *With no hypothesis, $Q_\nu(x) = \frac{6}{\pi^2}x + o(x^{1/2})$ for $\nu = 1, 2$.* (ii) *Under RH, $|Q_2(x) - \frac{6}{\pi^2}x| \le \frac12\mathbb{E}|M(\sqrt{x\mathcal{V}_2/2})| = O(x^{1/4+\varepsilon})$, and likewise for $Q_1$; this is a second proof of the forward half of Theorem 10.3.9(i).*

*Proof.* (i) $|M(t)| \le t$ and $M(t) = o(t)$, and $\mathbb{E}\mathcal{V}_\nu^{1/2} < \infty$, so dominated convergence gives $\mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = o(x^{1/2})$, and likewise for $\mathcal{V}_1$. (ii) RH gives $M(t) = O(t^{1/2+\varepsilon})$ [498, Section 14.25], and $\mathbb{E}\mathcal{V}_\nu^{1/4+\varepsilon} < \infty$. □

Table 10.10 checks (10.3.13) and Theorem 10.3.9(ii) for $Q_2$. It computes $Q_2(x) = \sum_d \mu(d)\,\mathbb{E}\lfloor G_{2,x}/d^2\rfloor = \sum_d \mu(d)\big(1/(e^t-1) + t/(4\sinh^2(t/2))\big)$, $t = 2d^2/x$, whose terms with $t > 140$ are below $10^{-55}$, and this agrees with the direct sum over $n$, in double precision, within $10^{-10}$ up to $x = 10^6$. The sum (10.3.13) truncated at $d \le 2\times10^5$, as in the first author's note, loses $2.2\times10^{-4}$ at $x = 10^6$, because $g_2(2d^2/x)$ is close to $x/d^2$ for large $d$ and the tail $x\sum_{d>2\times10^5}\mu(d)/d^2$ is left out.

| $x$ | $Q_2(x) - \frac{6}{\pi^2}x$ | $d \le 2\cdot10^5$ | $\frac{Q_2(x) - \frac{6}{\pi^2}x - 1}{x^{1/4}}$ | 100 atoms | trivial terms | residual |
|---|---|---|---|---|---|---|
| $10^3$ | 0.997161220898 | 0.997161002208 | $-5.048\times10^{-4}$ | $-1.0571\times10^{-4}$ | $-2.73307\times10^{-3}$ | $-1.0\times10^{-23}$ |
| $10^4$ | 1.00094524529 | 1.00094305839 | $9.452\times10^{-5}$ | $1.21889\times10^{-3}$ | $-2.73648\times10^{-4}$ | $-1.3\times10^{-30}$ |
| $10^5$ | 0.996620193442 | 0.99659832449 | $-1.9006\times10^{-4}$ | $-3.35244\times10^{-3}$ | $-2.73683\times10^{-5}$ | $-1.5\times10^{-37}$ |
| $10^6$ | 1.00610290943 | 1.00588421991 | $1.9299\times10^{-4}$ | $6.10565\times10^{-3}$ | $-2.73686\times10^{-6}$ | $-1.8\times10^{-41}$ |
| $10^7$ | 0.992114718377 | | $-1.4022\times10^{-4}$ | $-7.88501\times10^{-3}$ | $-2.73687\times10^{-7}$ | $-3.9\times10^{-42}$ |
| $10^8$ | 1.00440085379 | | $4.4009\times10^{-5}$ | $4.40088\times10^{-3}$ | $-2.73687\times10^{-8}$ | $-2.2\times10^{-38}$ |

Table 10.10: The Gamma(2) smoothed count: $Q_2(x) - 6x/\pi^2$, the sum (10.3.13) truncated at $d \le 2\times10^5$, the normalised error, the contribution $2\operatorname{Re}\sum_{k\le100} B_{\rho_k}x^{\rho_k/2}$ of the first hundred atoms, the residues at $s = -1,\dots,-6$, whose leading part is $2c_1/x$, and the residual. The constant 1 is the residue $D(0)$ at $s = 0$. Computed by `sqfree_smooth.py` at 45 digits.

### 10.3.8 Gamma shapes: two is the threshold

For every real shape $\nu > 0$, $f_\nu = -2g_\nu'$ is a signed density of total mass $2g_\nu(0+) = 1$, and the argument of Theorem 10.3.15 gives, for $Y$ of the Gamma($\nu$) law with

rate $\lambda$,

$$\mathbb{E}Q(Y) = \frac{6}{\pi^2}\,\mathbb{E}Y - \frac{1}{2}\int_0^\infty M\Big(\sqrt{y/\lambda}\Big) f_\nu(y)\,dy,$$
$$\int_0^\infty y^s f_\nu(y)\,dy = -\frac{2\,\Gamma(s+\nu)\,\zeta(s)}{\Gamma(\nu)}, \qquad -\nu < \operatorname{Re} s < 1.$$

Write $\gamma_a(t) = t^{a-1}e^{-t}/\Gamma(a)$ for the Gamma($a$) density, and let

$$\mathcal{R}_\nu(y) = y\sum_{m\ge1}\gamma_{\nu+1}(my) = \frac{y^{\nu+1}}{\Gamma(\nu+1)}\sum_{m\ge1} m^\nu e^{-my},$$
$$q_\nu(y) = \frac{\Gamma(\nu+1)\big(1-\mathcal{R}_\nu(y)\big)}{y^{\nu+1}},$$

so that $\mathcal{R}_\nu(y)$ is the right Riemann sum with step $y$ of the Gamma($\nu+1$) density, whose integral is 1.

Lemma 10.3.17. *For $\nu > 0$ and $y > 0$,*

$$f_\nu(y) = \frac{2\nu\big(1-\mathcal{R}_\nu(y)\big)}{y^2} = \frac{2y^{\nu-1}}{\Gamma(\nu)}\,q_\nu(y),$$

*so $f_\nu \ge 0$ exactly when $\mathcal{R}_\nu \le 1$. For $0 < y < 2\pi$,*

$$q_\nu(y) = -\sum_{k\ge0}\zeta(-\nu-k)\,\frac{(-y)^k}{k!},$$

*and $\mathcal{R}_1(y) = (y/2)^2/\sinh^2(y/2)$, $\mathcal{R}_2(y) = (y/2)^3\cosh(y/2)/\sinh^3(y/2)$.*

*Proof.* With $Y$ of the Gamma($\nu$,1) law and $c = 1/y$, $g_\nu(y) = \mathbb{E}\{cY\} = c\nu - \sum_{m\ge1} P(Y \ge m/c)$, whose derivative in $c$ is $\nu - \sum_m (m/c^2)p_\nu(m/c)$, with $p_\nu = \gamma_\nu$ the density of $Y$. Since $tp_\nu(t) = \nu\gamma_{\nu+1}(t)$, the derivative is $\nu(1-\mathcal{R}_\nu(1/c))$, and returning to $y$ gives $g'_\nu(y) = -\nu(1-\mathcal{R}_\nu(y))/y^2$. The series is the expansion $\sum_m m^\nu e^{-my} = \Gamma(\nu+1)y^{-\nu-1} + \sum_{k\ge0}\zeta(-\nu-k)(-y)^k/k!$ of the polylogarithm, valid for $0 < y < 2\pi$. □

So $\mathcal{R}_1 \le 1$ is $\sinh z \ge z$ and $\mathcal{R}_2 \le 1$ is Lazarević's inequality. Near the origin $f_\nu$ has the sign of $q_\nu(0+) = -\zeta(-\nu)$, and at an even integer $\nu = 2m$, where $\zeta(-2m) = 0$, the sign of $q_{2m}(y) = \zeta(-2m-1)y + O(y^2)$. Since $\zeta$ is positive on $(-4,-2)$, $(-8,-6)$, … and $\zeta(-2m-1) = -B_{2m+2}/(2m+2)$, $f_\nu$ is negative near the origin exactly when $\nu \in (4j+2, 4j+4]$ for some $j \ge 0$.

Theorem 10.3.18. $\{\nu > 0 :\ f_\nu \ge 0 \text{ on } (0,\infty)\} = (0,2]$. *Thus two is the largest shape for which the gamma smoothed squarefree error is a positive average of the Mertens function, and the obstruction is the trivial zero of $\zeta$ at $-2$.*

*Proof.* Let $\nu > 2$ and suppose $f_\nu \geq 0$. Since $f_\nu(y) \sim 2\nu/y^2$ at infinity, $\int_1^\infty y^s f_\nu(y)\,dy$ converges and is analytic for $\operatorname{Re} s < 1$. The function $m(s) = -2\Gamma(s+\nu)\zeta(s)/\Gamma(\nu)$ is analytic in a neighbourhood of $(-\nu, 1)$, so $\int_0^1 y^s f_\nu(y)\,dy = m(s) - \int_1^\infty y^s f_\nu(y)\,dy$, a Laplace transform in $-\log y$ of a nonnegative function, continues analytically along the real interval $(-\nu, 1)$. By Landau's theorem [514, Chapter II, Theorem 5b] it converges on all of $(-\nu, 1)$, and there $m(s) = \int_0^\infty y^s f_\nu(y)\,dy > 0$. But $\Gamma(s+\nu) > 0$ on $(-\nu, 1)$, while $\zeta(s) > 0$ on $(-4, -2)$, which meets $(-\nu, 1)$ because $\nu > 2$, so $m(s) < 0$ there, a contradiction.

Let $0 < \nu \leq 2$. The cases $\nu = 1$ and $\nu = 2$ are the inequalities above. Otherwise write $\nu = k + \mu$ with $k \in \{0, 1\}$ and $0 < \mu < 1$. Bernstein's formula $t^\mu = \frac{\mu}{\Gamma(1-\mu)} \int_0^\infty (1 - e^{-st}) s^{-\mu-1} ds$, applied to $t = mh$ in $\sum_m (mh)^{k+\mu} e^{-mh}$ and to $t$ in $\int_0^\infty t^{k+\mu} e^{-t} dt$, together with $h \sum_m (mh)^k e^{-mhc} = k!\,\mathcal{R}_k(hc)/c^{k+1}$ and $\int_0^\infty t^k e^{-ct} dt = k!/c^{k+1}$ for $c = 1 + s$, gives

$$\Gamma(\nu+1)\big(\mathcal{R}_\nu(h) - 1\big) = \frac{\mu\, h^{k+1}}{\Gamma(1-\mu)} \int_0^\infty \big(q_k(h(1+s)) - q_k(h)\big)\, s^{-\mu-1}\, ds, \qquad h > 0.$$

So $\mathcal{R}_\nu \leq 1$ follows if $q_k$ is nonincreasing. For $k = 0$, $q_0(y) = 1/y - 1/(e^y - 1) = g_1(y)$, which decreases because $f_1 > 0$. For $k = 1$, $q_1(y) = 1/y^2 - 1/(4\sinh^2(y/2)) = f_1(y)/2$, and with $z = y/2$

$$q_1'(y) = \frac{1}{4}\Big(\frac{\cosh z}{\sinh^3 z} - \frac{1}{z^3}\Big) < 0$$

by Lazarević's inequality. □

The script `sqfree_gammanu.py` evaluates $1 - \mathcal{R}_\nu$ for twenty shapes at 40 digits, by the series of Lemma 10.3.17 for $y \leq 2.5$ and by the direct sum beyond, on 400 geometrically spaced points in $[10^{-4}, 10^{0.4}]$ and 799 equally spaced points in $(2.5, 60]$. For the ten shapes $\nu = 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 1.9, 1.99, 2$ no grid point gives a negative value; on the range $[0.02, 15]$ the least values, from $2.68 \times 10^{-3}$ at $\nu = 0.25$ to $6.90 \times 10^{-10}$ at $\nu = 2$, are taken at its left end. Table 10.11 shows the shapes above 2: the negative set starts at the origin for the shapes in $(2, 4]$ and in $(6, 8]$, as the sign rule predicts.

### 10.3.9 The Euler product of the squarefree law

For $\sigma > 1$ let $N_\sigma$ have the law $P(N_\sigma = n) = \mu(n)^2 n^{-\sigma} \zeta(2\sigma)/\zeta(\sigma)$ on the squarefree integers, the squarefree analogue of the zeta distribution of Section 6.1.6. By the Euler product in (10.3.1), $\log N_\sigma = \sum_p B_p \log p$ with independent Bernoulli variables $B_p$ of success probability $p^{-\sigma}/(1 + p^{-\sigma})$, and

$$\mathbb{E} e^{-it \log N_\sigma} = \prod_p \frac{1 + p^{-\sigma-it}}{1 + p^{-\sigma}}, \qquad \log\big(1 + p^{-s}\big) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k}\, p^{-ks}.$$

| $\nu$ | $-\zeta(-\nu)$ | least value of $1-\mathcal{R}_\nu$ | at $y=$ | grid points with $1-\mathcal{R}_\nu<0$ |
|---|---|---|---|---|
| 2.01 | $-3.012\times10^{-4}$ | $-7.310\times10^{-10}$ | 0.0273 | from $10^{-4}$ to 0.0362 |
| 2.1 | $-2.7295\times10^{-3}$ | $-4.941\times10^{-6}$ | 0.265 | from $10^{-4}$ to 0.347 |
| 2.5 | $-8.517\times10^{-3}$ | $-1.0763\times10^{-3}$ | 1.187 | from $10^{-4}$ to 1.512 |
| 3 | $-8.333\times10^{-3}$ | $-7.2304\times10^{-3}$ | 2.144 | from $10^{-4}$ to 2.716 |
| 3.5 | $-4.441\times10^{-3}$ | $-0.019006$ | 2.9745 | from $10^{-4}$ to 3.723 |
| 4 | 0 | $-0.035249$ | 3.726 | from $10^{-4}$ to 4.659 |
| 5 | $3.968\times10^{-3}$ | $-0.076713$ | 5.084 | from 1.716 to 6.386 |
| 6 | 0 | $-0.12519$ | 6.323 | from 3.076 to 7.969 |
| 7 | $-4.167\times10^{-3}$ | $-0.17715$ | 7.4915 | from $10^{-4}$ to 1.234, from 4.227 to 9.409 |
| 8 | 0 | $-0.23064$ | 8.614 | from $10^{-4}$ to 2.269, from 5.307 to 10.85 |

Table 10.11: Gamma($\nu$) smoothing above the threshold of Theorem 10.3.18: the limit $-\zeta(-\nu)$ of $q_\nu$ at 0, the least value of $1-\mathcal{R}_\nu$, which has the sign of $f_\nu$, with the point where it is attained (grid search refined by golden section), and the maximal runs of grid points on which $1-\mathcal{R}_\nu<0$. Computed by `sqfree_gammanu.py`.

So the would be Lévy measure of $\log N_\sigma$ is the signed measure

$$\sum_p \sum_{k\geq 1} \frac{(-1)^{k+1}}{k}\, p^{-k\sigma}\, \delta_{k\log p}$$

of finite total variation, a sum over the primes of alternating lattices like the fair coin lattices of Section 6.1.6, here with the geometric weights $p^{-k\sigma}$ of a biased coin: the law is quasi infinitely divisible, and by the uniqueness of the signed Lévy measure it is not infinitely divisible [341]. Its negative masses, at even $k$, come from the factor $1/\zeta(2s)$, whose weights $-2/k$ at even $k$ outweigh the weights $1/k$ of $\zeta(s)$.

### 10.3.10 Questions

Question 10.3.19. Prove $\Delta(x)=O(x^{1/4+\varepsilon})$ under RH. Equivalent forms above: remove the loss $n^{1/8}$ in Proposition 10.3.12; or show that the exceptional set of Corollary 10.3.11(i), with $X^{1/4+\varepsilon}$ replaced by a constant multiple, is empty for large $X$. By that corollary, the latter would follow from a bound $O(X^{1/4+\varepsilon})$ for the oscillation of $\Delta$ on the subintervals of $[X,2X]$ of length $X^{1-\varepsilon/2}$.

Question 10.3.20. Sharpen the constant $|R_{\rho_1}|=0.1004$ of Corollary 10.3.11(ii) with a Fejér kernel over many zeros, as Ingham did for $M(x)$ [285], and compare it with the observed $\max|\Delta(x)|/x^{1/4}$, about 1 up to $2\times10^8$ (Table 10.9). Is $\limsup|\Delta(x)|/x^{1/4}=\infty$, as the linear independence of the ordinates would suggest?

# Chapter 11

# Infinite divisibility, the eta analogue and almost generalised gamma convolutions

The five sections of this chapter read the laws of Chapters 9 and 10, the shifted clocks $T_\alpha$, the normal scale mixtures $Y_\alpha$ and the tilted Pólya variables $V_\alpha$, in other settings; the question throughout is what can be said below the basepoint one without a zero free half plane, which none of the five sections supplies.

## 11.1 Infinite divisibility and a map of processes

The general theory of infinitely divisible laws is that of two monographs: Sato's book on Lévy processes and infinitely divisible distributions [471] and Bondesson's book on generalised gamma convolutions [56].

None of the results of this section supplies positivity of the Thorin measure uniformly in the height (Section 11.1.5). The laws are also the marginals of processes. Self decomposability makes the centre clock the stationary law of an Ornstein Uhlenbeck process, and the Thorin measure at the basepoint one makes $T_1$ the integral of a gamma process against the inverse of the Poisson smoothed zero count; Section 11.1.6 derives both. Section 11.1.7 adds Brownian motion run on the clocks and places these processes and those of Chapter 9 on one map, sorted by what each needs: nothing, a zero free half plane, or RH.

We use the notation of Chapter 9. The centre clock $T$ has Laplace transform $\varphi_{1/2}(u) = 1/G(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$, and for $\alpha > \frac{1}{2}$ the shifted clock $T_\alpha$ has Laplace transform $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ and the representation $T_\alpha \stackrel{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$ of Theorem 9.2.5, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and $H_a$ is the first passage time of an independent standard Brownian motion to the level $a$. For $\alpha \geq 1$, $T_\alpha$ is a GGC whose Thorin density in the chart $\theta = \sqrt{z}$ is $v_\alpha/\pi$, with $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ (Theorem 1.2.2 and (1.2.7)); at the basepoint one this is the density $\rho_{1/2} = v_1/\pi$ of (9.1.2). We write $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, so that $b_1 = \sum_\rho \rho^{-1} = 0.0230957089661$, and $V_\alpha$ for the tilted Pólya variable of Section 9.1.1, the Esscher tilt of the Pólya density $p$ of (1.2.4) with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha + s)/\xi(\alpha)$, so

that $V_{1/2}$ is the Pólya variable $X$ and $V_1$ is the variable $V$ of Section 9.1.4.

### 11.1.1 Self decomposability: the driving process and unimodality

The Laplace exponent of the centre clock is $\log G(u) = \int_0^\infty (1 - e^{-ut})\, W(t)\, dt/t$, where $W(t) = \sum_k e^{-a_k t}$, with $a_k = -(\rho_k - \frac{1}{2})^2$, is the heat trace of the zeros, equal to $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH and to the arithmetic expression (1.2.5) in any case (Section 1.2). Since $W > 0$ and $W' < 0$ on $(0, \infty)$ with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000), $T$ is infinitely divisible with Lévy measure $W(t)\, dt/t$, and it is self decomposable. A law on $[0, \infty)$ is self decomposable if and only if it is the law of $\int_0^\infty e^{-t}\, dZ_t$ for a subordinator $Z$ with $\mathbb{E}\log(1 + Z_1) < \infty$, its background driving Lévy process [520, 293], [471, Theorem 17.5]. The Lévy measure of a self decomposable law on the half line has the form $k(x)\, dx/x$ with $k$ nonincreasing [471, Corollary 15.11], and the Lévy measure of $Z$ then gives mass $k(x)$ to $(x, \infty)$, so that it has the density $-k'(x)$ when $k$ is differentiable. For the centre clock $k = W$, and self decomposability, which is the unconditional statement that $W$ is nonincreasing, gives the following.

Proposition 11.1.1. *With no hypothesis, $T \stackrel{d}{=} \int_0^\infty e^{-t}\, dZ_t$ for a subordinator $Z$ with Lévy density $h(x) = -W'(x)$, which is $\sum_{\gamma>0} \gamma^2 e^{-\gamma^2 x}$ under RH and is given for every $x > 0$ by the derivative of the explicit formula* (1.2.5) *for $W$. The condition $\mathbb{E}\log(1 + Z_1) < \infty$ holds because $h$ decays exponentially. Moreover RH holds if and only if $h$ is completely monotone.*

*Proof.* The representation is the theorem of Wolfe and of Jurek and Vervaat applied to the self decomposable law of $T$, and the Lévy density of $Z$ can be checked directly. If $\Psi_Z(u) = -\log \mathbb{E}e^{-uZ_1}$, the Laplace exponent of $\int_0^\infty e^{-t}\, dZ_t$ is $\int_0^\infty \Psi_Z(ue^{-t})\, dt = \int_0^u \Psi_Z(w)\, dw/w$, so that $\Psi_Z(u) = u(\log G)'(u)$. By Section 1.2, $(\log G)'(u) = \int_0^\infty e^{-ut} W(t)\, dt$, and since $W(t) = \int_t^\infty h(x)\, dx$ with $h > 0$, Fubini's theorem gives

$$u(\log G)'(u) = \int_0^\infty h(x) \int_0^x ue^{-ut}\, dt\, dx = \int_0^\infty \big(1 - e^{-ux}\big) h(x)\, dx.$$

Here $\int_1^\infty h = W(1)$ and $\int_0^1 xh(x)\, dx = \int_0^1 (W(y) - W(1))\, dy$, which is finite because $W(y)\, dy/y$ is a Lévy measure, so $h$ is the Lévy density of a subordinator without drift. It is not integrable at the origin, since $\int_t^1 h = W(t) - W(1)$ and $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 5.1), so $Z$ makes infinitely many small jumps. As $x \to \infty$, $h(x) \sim \gamma_1^2 e^{-\gamma_1^2 x}$, which gives $\mathbb{E}\log(1 + Z_1) < \infty$ [471, Theorem 25.3]. For the last statement, $h$ is completely monotone exactly when $W$ is, since $W(x) = \int_x^\infty h$, and complete monotonicity of the heat trace is equivalent to RH (Section 1.2 and Table 1.6). □

| $t$ | $W(t)$, zeros | $W(t)$, explicit formula |
|---|---|---|
| 0.005 | 0.537191964091 | 0.537191964091 |
| 0.01 | 0.149698011254 | 0.149698011254 |
| 0.02 | 0.0185412917313 | 0.0185412917313 |
| 0.05 | $4.587835048574 \times 10^{-5}$ | $4.587835048574 \times 10^{-5}$ |
| 0.1 | $2.104799795822 \times 10^{-9}$ | $2.104799798080 \times 10^{-9}$ |
| $t$ | $-W'(t)$, zeros | $-W'(t)$, explicit formula |
| 0.005 | 165.1022811964 | 165.1022811964 |
| 0.01 | 33.72924560562 | 33.72924560562 |
| 0.02 | 3.741068142123 | 3.741068142123 |
| 0.05 | $9.166117863113 \times 10^{-3}$ | $9.166117863113 \times 10^{-3}$ |
| 0.1 | $4.205189085540 \times 10^{-7}$ | $4.205189085735 \times 10^{-7}$ |

Table 11.1: The heat trace $W$ and the Lévy density $h = -W'$ of the driving process of the centre clock, from the first 1700 zeros and from the explicit formula (1.2.5) and its derivative with the primes below $2 \times 10^4$. Computed by `weil.py` in the folder `sato_bondesson`.

Table 11.1 checks the explicit formula and its derivative against the zero sums.

At $t = 0.1$ the terms of the explicit formula cancel to about nine digits (Table 5.6), and `weil.py` takes the weights $\log p$ in double precision; the cancellation magnifies their rounding errors into the differences of $2 \times 10^{-18}$ in $W$ and $2 \times 10^{-17}$ in $h$ that the rows at $t = 0.1$ show. With the weights in multiple precision both explicit columns agree with the zero sums at $t = 0.1$ to every digit shown, as in Table 1.4, so the cancellation costs working precision and nothing more.

Yamazato's theorem states that every self decomposable law on the line is unimodal [521], [471, Theorem 53.1]. The centre clock and every shifted clock $T_\alpha$, $\alpha > \frac{1}{2}$, are self decomposable with no hypothesis (Theorem 5.1.1 and Corollary 9.2.6), so we obtain the following.

Corollary 11.1.2. *The centre clock is unimodal with no hypothesis, and so is $T_\alpha$ for every $\alpha > \frac{1}{2}$.*

The computed densities (Figure 11.1) have modes near 0.020 for the centre clock and 0.024 for $T_1$, the latter located only to the spacing of the geometric grid of the Talbot inversion behind the figure, whose points next to 0.024 are 0.0175 and 0.0328. Yamazato's theorem does not reach the reciprocal zeta law of Section 7.1, which is infinitely divisible but not self decomposable (Theorem 7.1.3), and the unimodality of its density remains open (Section 7.2.4). The density of the centre clock is also $C^\infty$, by Orey's condition [471, Proposition 28.3], since $\int_0^r tW(t)\,dt \geq cr^{3/2}$ by the expansion of $W$ at 0 (Section 5.1).

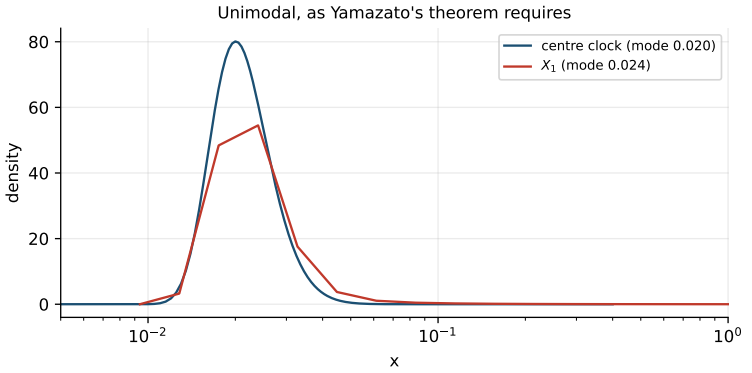



Figure 11.1: The densities of the centre clock $T$ and of the shifted clock $T_1$, written $X_1$ in the plot, on a logarithmic axis. The density of $T$ is drawn on a grid of spacing 0.0005 on $[0.004, 0.4]$, from the Talbot inversion and the spectral expansion of Section 8.1; that of $T_1$ on the 40 point geometric grid of the Talbot inversion, which is why its curve is piecewise linear. Assembled by `comp.py` and drawn by `figs.py`, both in the folder `sato_bondesson`.

### 11.1.2 Moments, the tail and the two ends of the Thorin measure

For a GGC with Thorin measure $U$ the Lévy density is $x^{-1}\int e^{-xz}\,U(dz)$ (Section 1.3.1), so the Lévy measure $\nu_{T_1}$ of $T_1$ and its tail are, in the chart $\theta = \sqrt{z}$,

$$\nu_{T_1}(dx) = \frac{dx}{x}\int_0^\infty e^{-x\theta^2}\rho_{1/2}(\theta)\,d\theta, \qquad \nu_{T_1}((x,\infty)) = \int_0^\infty \rho_{1/2}(\theta)\,E_1(x\theta^2)\,d\theta, \tag{11.1.1}$$

where $E_1(y) = \int_y^\infty e^{-w}\,dw/w$ is the exponential integral. As $x \to \infty$ only $\theta$ near zero contributes, and $\int_0^\infty E_1(w^2)\,dw = \sqrt{\pi}$ gives $\nu_{T_1}((x,\infty)) \sim \rho_{1/2}(0)\sqrt{\pi/x} = b_1(\pi x)^{-1/2}$. The tail is regularly varying of index $-\frac{1}{2}$, so the Lévy measure is subexponential. By the theorem of Embrechts, Goldie and Veraverbeke [131], an infinitely divisible law on $[0,\infty)$ is subexponential if and only if its Lévy measure, restricted to $(1,\infty)$ and normalised, is subexponential, and if and only if the tail of the law is asymptotic to the tail of the Lévy measure. Hence

$$P(T_1 > x) \sim \nu_{T_1}((x,\infty)) \sim \frac{b_1}{\sqrt{\pi x}} = 0.01303\,x^{-1/2}, \qquad x \to \infty. \tag{11.1.2}$$

Theorem 9.2.5 gives the infinite mean and the tail of $T_\alpha$ for every $\alpha > \frac{1}{2}$ by the Tauberian route, and since $T_\alpha$ is infinitely divisible for every $\alpha > \frac{1}{2}$ (Section 11.1.4), the theorem of Embrechts, Goldie and Veraverbeke, read from the tail of the law to that of the Lévy measure, shows that the Lévy tail of $T_\alpha$ is asymptotic to $b_\alpha(\pi x)^{-1/2}$ as well, wherever the zeros lie.

Table 11.2 compares the survival function of $T_1$ from Section 9.2.4 with the Lévy tail computed from (11.1.1) and the Thorin density on $[0, 100]$. In the bulk of the law the two differ widely: at $x = 0.02$, just below the mode, the survival probability is 0.747 while the Lévy tail is 0.101.

| $x$ | $P(T_1 > x)$ | $\nu_{T_1}((x,\infty))$ | ratio | $b_1(\pi x)^{-1/2}$ |
|---|---|---|---|---|
| 0.1 | $4.7639 \times 10^{-2}$ | $4.1558 \times 10^{-2}$ | 1.1463 | $4.1206 \times 10^{-2}$ |
| 0.5 | $1.8907 \times 10^{-2}$ | $1.8458 \times 10^{-2}$ | 1.0243 | $1.8428 \times 10^{-2}$ |
| 1 | $1.3199 \times 10^{-2}$ | $1.3041 \times 10^{-2}$ | 1.0121 | $1.3030 \times 10^{-2}$ |
| 10 | $4.1272 \times 10^{-3}$ | $4.1209 \times 10^{-3}$ | 1.0015 | $4.1206 \times 10^{-3}$ |
| 100 | $1.3037 \times 10^{-3}$ | $1.3030 \times 10^{-3}$ | 1.0005 | $1.3030 \times 10^{-3}$ |

Table 11.2: The subexponential principle for $T_1$: the tail of the law is asymptotic to the tail of its Lévy measure. The ratio is that of the second column to the third, and the last column is the common asymptote of (11.1.2). Computed by `comp.py` in the folder `sato_bondesson`.

Bondesson's results on the two ends of the Thorin measure explain the shape of $T_1$: for a GGC the behaviour of the Thorin measure near zero governs the right tail, and its total mass governs the density at the origin [56]. At the basepoint one the Thorin measure $U_1$ of $T_1$ (Proposition 9.1.2) has $U_1((0,z]) \sim (b_1/\pi)\sqrt{z}$ as $z \downarrow 0$, which gives the expansion $1 - \varphi_1(u) \sim b_1\sqrt{u}$ of Section 9.1. At the other end the total mass is infinite, since $U_1((0, Y^2]) = M_{1/2}(Y)$ differs from the number $N(Y)$ of zeros with ordinates in $(0, Y]$ by the term $\tau_{1/2}(Y) = O(\log Y)$ of Section 9.1.2, and this is why $T_1$, like the centre clock (Section 8.2.2.2), is super flat at the origin: its density vanishes faster than any power. Indeed, for every $z_0 > 0$ the Thorin form gives $\varphi_1(u) \le (1 + u/z_0)^{-U_1((0,z_0])}$, and $U_1((0,z_0]) \to \infty$ as $z_0 \to \infty$, so $\varphi_1(u)$ decays faster than any power of $u$. Markov's inequality $P(T_1 \le x) \le e\,\varphi_1(1/x)$ carries this to the distribution function, and since the density $f_{T_1}$ is nondecreasing up to the mode (Corollary 11.1.2), the bound $x f_{T_1}(x) \le P(T_1 \le 2x)$ for small $x$ carries it to the density.

### 11.1.3 Uniqueness and quasi infinite divisibility

Lindner, Pan and Sato [341] call a law quasi infinitely divisible if its characteristic function is the quotient of two infinitely divisible characteristic functions, equivalently if it has a Lévy Khintchine form with a signed measure $\nu^+ - \nu^-$ whose parts $\nu^+$ and $\nu^-$ are both Lévy measures, so in particular of finite mass away from the origin; the Gaussian coefficient is still nonnegative [341, Lemma 2.7], and the triplet is again unique. Their theory places the tilted Pólya variables $V_\alpha$ precisely. The signed measures in question are those of the Lévy forms of Section 9.1.4: for $\alpha \ge 1$,

$$\log \frac{\xi(\alpha + s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty \left(e^{-sx} - 1 + sx\right) \Pi_\alpha(dx),$$

which at $\alpha = 1$ is (9.1.5), a compensated limit, and which for $\alpha > 1$ converges absolutely when $\operatorname{Re} s > 1 - \alpha$. For $\alpha > 1$, read at $s = it$, it is a Lévy Khintchine form for the characteristic function of $V_\alpha$, with the measure $\Pi_\alpha$ carried to the negative half line by the reflection $x \mapsto -x$. We write $x_* = 0.28120$ for the root of $e^{-3x} = 1 - e^{-2x}$.

PROPOSITION 11.1.3. *1. The Pólya variable $X = V_{1/2}$ is not quasi infinitely divisible, since its characteristic function $\Xi(t)/\Xi(0)$ vanishes at the ordinates.*

*2. For every $\alpha > 1$, $V_\alpha$ is quasi infinitely divisible and not infinitely divisible. Its signed Lévy measure is the reflection to the negative half line of*

$$\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)\frac{dx}{x} + \sum_n \frac{\Lambda(n)}{n^\alpha \log n}\,\delta_{\log n}(dx),$$

*whose atoms have total mass $\log\zeta(\alpha)$ and whose absolutely continuous part is positive on $(0, x_*)$ and negative beyond. The negative part of $\Pi_\alpha$ is therefore carried by $(x_*, \infty)$, and its mass on $(1,\infty)$ is $E_1(\alpha-1) - \sum_{k\ge 0} E_1(\alpha+2+2k)$: it is dominated by the pole term $e^{-(\alpha-1)x}dx/x$, of mass $E_1(\alpha-1)$ on $(1,\infty)$, but is not equal to it.*

*3. At $\alpha = 1$, $V_1$ is not quasi infinitely divisible. The candidate measure $\Pi_1$ of Proposition 9.1.10 has the pole term $-dx/x$ and the prime atoms $\Lambda(n)/(n\log n)$ at $\log n$, each of infinite mass on $(1,\infty)$; since the gamma density $e^{-3x}(1-e^{-2x})^{-1}/x$ is less than $0.06/x$ there, the positive and the negative part of $\Pi_1$ both have infinite mass on $(1,\infty)$, and its Lévy form converges only as a compensated limit, which exists by the prime number theorem.*

Items (1) and (2) are due to Nakamura [377]: (2), with the same measure, is his Theorem 1.4, and (1) is the remark after it. The argument of (1) applied to the moment generating function shows that no $V_\alpha$, $\alpha \in \mathbb{R}$, is infinitely divisible, because $\xi(\alpha + w)/\xi(\alpha)$ is entire with zeros [471, Theorem 25.17].

*Proof.* (1) A quasi infinitely divisible characteristic function is the exponential of the finite exponent of its Lévy Khintchine form, so it never vanishes, while $\mathbb{E}e^{itX} = \Xi(t)/\Xi(0)$ by (1.2.4), which vanishes at $t = 14.134725$ and at every other ordinate of a zero on the critical line.

(2) Let $\alpha > 1$. Every piece of the Lévy form then converges absolutely, and each piece is a Lévy measure: the positive density $e^{-(\alpha+2)x}(1-e^{-2x})^{-1}dx/x$, the pole term and the atoms. Removing the factor $s(s-1)$ from $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ gives, for $\operatorname{Re} s > 1-\alpha$,

$$\frac{\xi(\alpha+s)}{\xi(\alpha)}\cdot\frac{\alpha}{\alpha+s}\cdot\frac{\alpha-1}{\alpha-1+s} = \pi^{-s/2}\,\frac{\Gamma(\frac{\alpha+s}{2})}{\Gamma(\frac{\alpha}{2})}\,\frac{\zeta(\alpha+s)}{\zeta(\alpha)}.$$

Let $e_1$ and $e_2$ be standard exponential variables, $\Gamma_{\alpha/2}$ a gamma variable of shape $\frac{\alpha}{2}$ and unit scale, and $\mathcal{Z}_\alpha$ a variable with the zeta law $P(\mathcal{Z}_\alpha = n) = n^{-\alpha}/\zeta(\alpha)$, all independent of each other and of $V_\alpha$. Since $\mathbb{E}e^{-se_1/\alpha} = \alpha/(\alpha+s)$, $\mathbb{E}\Gamma_{\alpha/2}^{s/2} = \Gamma(\frac{\alpha+s}{2})/\Gamma(\frac{\alpha}{2})$ and $\mathbb{E}\mathcal{Z}_\alpha^{-s} = \zeta(\alpha+s)/\zeta(\alpha)$, the identity says that

$$V_\alpha - \frac{e_1}{\alpha} - \frac{e_2}{\alpha-1} \stackrel{d}{=} \frac{1}{2}\log\Gamma_{\alpha/2} - \frac{1}{2}\log\pi - \log\mathcal{Z}_\alpha.$$

The right side is infinitely divisible: $\frac{1}{2}\log\Gamma_{\alpha/2}$ has, by Binet's formula [397, §5.9] as in the proof of Proposition 9.1.10, the Lévy measure $e^{-\alpha x}(1-e^{-2x})^{-1}dx/x$ reflected to the negative half line, and by the Euler product $-\log\mathcal{Z}_\alpha$ is compound Poisson with Lévy measure $\sum_n \Lambda(n)(n^\alpha \log n)^{-1}\delta_{-\log n}$, of total mass $\log\zeta(\alpha)$. The variable $-e_1/\alpha - e_2/(\alpha-1)$ is infinitely divisible as well, with Lévy measure $(e^{-\alpha x} + e^{-(\alpha-1)x})\,dx/x$ reflected, so the law of $V_\alpha$ convolved with an infinitely divisible law is infinitely divisible, and $V_\alpha$ is quasi infinitely divisible. Its signed Lévy measure is the difference of the two Lévy measures, which is $\Pi_\alpha$ reflected, since $e^{-\alpha x}(1-e^{-2x})^{-1} - e^{-\alpha x} = e^{-(\alpha+2)x}(1-e^{-2x})^{-1}$. The absolutely continuous part of $\Pi_\alpha$ has the density $e^{-(\alpha-1)x}\big(e^{-3x}(1-e^{-2x})^{-1} - 1\big)x^{-1}$, whose sign is that of $e^{-3x} - (1-e^{-2x})$ whatever $\alpha$ is: positive on $(0, x_*)$ and negative beyond, where it is at least $-e^{-(\alpha-1)x}/x$, whose integral over $(1,\infty)$ is $E_1(\alpha-1)$; expanding $(1-e^{-2x})^{-1}$ in powers of $e^{-2x}$ and integrating termwise over $(1,\infty)$ gives the mass of the negative part there. If $V_\alpha$ were infinitely divisible, uniqueness of the triplet [471, Theorem 8.1], applied to the identity in law above, would make its Lévy measure equal to $\Pi_\alpha$ reflected, which is negative on the reflection of $(x_*, \log 2)$, an interval that carries no atom. So $V_\alpha$ is not infinitely divisible.

(3) Suppose that $V_1$ is quasi infinitely divisible, so that $\mathbb{E}e^{itV_1} = \exp\Psi_{V_1}(t)$ with

$$\Psi_{V_1}(t) = ict - \frac{A}{2}t^2 + \int_{\mathbb{R}} \big(e^{ity} - 1 - ity\mathbf{1}\{|y|\le 1\}\big)\,\mu(dy),$$

where $A$ and $c$ are real and $\mu$ is a signed measure on $\mathbb{R}\setminus\{0\}$ with $\int \min(1,y^2)\,|\mu|(dy) < \infty$. As in (1), the characteristic function never vanishes, so $\xi(1+it) \neq 0$ for real $t$; and $\xi(1+s) \neq 0$ for $\operatorname{Re} s > 0$ by the Euler product. So $\ell(s) = \log(\xi(1+s)/\xi(1))$, with $\ell(0) = 0$, is a continuous logarithm on the closed half plane $\operatorname{Re} s \ge 0$, holomorphic inside, and $\Psi_{V_1}(t) = \ell(it)$, since both are continuous logarithms of $\xi(1+it)/\xi(1)$ that vanish at $t=0$. For $r>0$ put $\Delta_r F(t) = F(t+r) + F(t-r) - 2F(t)$. Then

$$\Delta_r\Psi_{V_1}(t) = \int_{\mathbb{R}} e^{ity}\,\lambda_r(dy), \qquad \lambda_r(dy) = -Ar^2\delta_0(dy) + (2\cos ry - 2)\,\mu(dy),$$

and $\lambda_r$ is a finite signed measure because $|2\cos ry - 2| \le \min(4, r^2y^2)$. On the other side, let $\check{\Pi}_1$ be the reflection of $\Pi_1$ to the negative half line. For $\sigma > 0$ the variable $V_{1+\sigma}$ has the characteristic function $\exp(\ell(\sigma+it) - \ell(\sigma))$, and by

(2) its signed Lévy measure is the reflection of $\Pi_{1+\sigma}(dx) = e^{-\sigma x}\Pi_1(dx)$, that is $e^{\sigma y}\check{\Pi}_1(dy)$. The second difference removes the drift and the compensator, so

$$\Delta_r \ell(\sigma + i\,\cdot\,)(t) = \int_{\mathbb{R}} e^{ity}\, m_\sigma(dy), \qquad m_\sigma(dy) = e^{\sigma y}(2\cos ry - 2)\,\check{\Pi}_1(dy),$$

where $m_\sigma$ is a finite signed measure, the atoms contributing at most $4\log\zeta(1+\sigma)$ to its total variation. As $\sigma \downarrow 0$, $m_\sigma$ tends to $m_0 = (2\cos ry - 2)\,\check{\Pi}_1$ in the space of tempered distributions, by dominated convergence, because $|m_0|$ is locally finite and $|m_0|((-R,0)) = O(R)$ as $R \to \infty$: the factor $2\cos ry - 2$ tames $\check{\Pi}_1$ near the origin, and $\Lambda(n) \le \log n$ bounds the atom at $-\log n$ by $1/n$. So the Fourier transforms converge as tempered distributions. On the other hand $\Delta_r \ell(\sigma + it) \to \Delta_r \Psi_{V_1}(t)$ uniformly for $t$ in compact sets, because $\ell$ is uniformly continuous on compact subsets of the closed half plane. Hence the Fourier transforms of $m_0$ and $\lambda_r$ agree on smooth test functions of compact support, which are dense in the Schwartz space, so they agree as tempered distributions, and since the Fourier transform is injective there, $m_0 = \lambda_r$ as measures. The mass at the origin gives $A = 0$. On a compact set $K \subset \mathbb{R} \setminus \{0\}$, $2\cos ry - 2 < 0$ once $r < 2\pi/\max_K |y|$, so $\mu = \check{\Pi}_1$ on $K$, and hence on $\mathbb{R} \setminus \{0\}$. But the atom of $\Pi_1$ at $\log p$ has mass $1/p$, so $|\mu|((-\infty,-1)) \ge \sum_{p\ge 3} 1/p = \infty$, and $|\mu|$ is not a Lévy measure. This contradiction proves (3). □

The proof of (3) uses about the primes only the Euler product, and not the prime number theorem: the nonvanishing of $\zeta$ on the line $\operatorname{Re} s = 1$, the first step towards that theorem, enters only through the hypothesis, which forces it.

Section 9.1.4 reads the conclusion of (3) from the Lévy form itself, and the proof of (3) supplies the uniqueness on which that reading rests. The same argument, with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, shows that the tilted eta variable of Section 11.3.2 is neither quasi infinitely divisible nor infinitely divisible, as that section argues.

### 11.1.4 Bondesson's classes and Thorin Bernstein functions

The class of HCM densities is closed under products and quotients of independent variables and under powers $X^q$ with $|q| \ge 1$ [56, Section 5.1], and together with the characterisation of GGCs by hyperbolic complete monotonicity [56, Theorem 6.1.1] (Section 1.3.2) these are the tools behind the Bosch threshold of Theorem 4.3.5.

Bondesson treats mixtures of exponential laws and their generalised convolutions. This is the natural class for Kent's hitting times started from an interior point, whose Laplace transform is a ratio of two products with interlacing zeros. The law of $T_1$ is not a mixture of exponentials, since a mixture of exponentials has a completely monotone density while that of $T_1$ vanishes at the origin and has an interior mode (Figure 11.1). This is consistent with the conclusion of Section 9.2

that $T_1$ is a subordinated rather than a direct Kent hitting time (Section 9.2.1 and Theorem 9.2.2).

In the language of Schilling, Song and Vondraček [473, Definition 9.10], a law on $[0,\infty)$ is a GGC exactly when its Laplace exponent is a Thorin Bernstein function, equivalently when the derivative of the exponent is a Stieltjes function, the function $f_H$ of Section 1.3.1. The subordination identity of Theorem 9.2.5 reads

$$\begin{aligned} -\log\varphi_\alpha(u) &= \log G\big((\alpha-\tfrac12)^2+f_\alpha(u)\big)-\log G\big((\alpha-\tfrac12)^2\big),\\ f_\alpha(u) &= u+2(\alpha-\tfrac12)\sqrt{u}, \end{aligned} \tag{11.1.3}$$

with $f_\alpha$, the Laplace exponent of the subordinator $\sigma^{(\alpha)}$ of that theorem, a complete Bernstein function. The derivative $\frac{d}{du}[-\log\varphi_\alpha(u)] = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(\alpha+\sqrt{u})$ is a Stieltjes function exactly when no zero lies to the right of $\alpha$, that is when $\alpha \ge \sup_\rho \operatorname{Re}\rho$, which is the criterion of Theorem 1.2.2: a zero with $\operatorname{Re}\rho > \alpha$ makes $\xi(\alpha+\sqrt{u})$ vanish at the point $u=(\rho-\alpha)^2$ of the cut plane $\mathbb{C}\setminus(-\infty,0]$, where a Stieltjes function is holomorphic.

Proposition 11.1.4. *For every $\alpha \ge 1$ the composition $\log G((\alpha-\frac12)^2+f_\alpha) - \log G((\alpha-\frac12)^2)$ is a Thorin Bernstein function with no hypothesis, while $\log G$ itself is a Thorin Bernstein function if and only if RH holds. For $\frac12 < \alpha < 1$ the composition is a Thorin Bernstein function if and only if $\zeta$ has no zeros with real part greater than $\alpha$.*

*Proof.* By (11.1.3) the composition is the Laplace exponent of $T_\alpha$, so it is a Thorin Bernstein function exactly when $T_\alpha$ is a GGC, which by Theorem 1.2.2 happens exactly when no zero has real part greater than $\alpha$. For $\alpha \ge 1$ there is none, since $\zeta$ has no zeros on or to the right of the line $\operatorname{Re} s = 1$. The exponent $\log G$ is a Thorin Bernstein function exactly when $T$ is a GGC, which is RH by Proposition 1.2.1. □

For every $\alpha > \frac12$ the composition is at least a Bernstein function: $u \mapsto \log G((\alpha-\frac12)^2+u) - \log G((\alpha-\frac12)^2)$ is the Laplace exponent of the tilted clock $\widetilde{T}_\alpha$, and a Bernstein function composed with a Bernstein function is again one [473, Corollary 3.8]. So $T_\alpha$ is infinitely divisible for every $\alpha > \frac12$ with no hypothesis, and the location of the zeros decides only whether it is a GGC. The proposition says that composing with the complete Bernstein function $f_\alpha$ creates the Thorin property from a function only known to be Bernstein.

General composition theorems do not settle this. For a Bernstein function $\kappa$ and $a \ge 0$ let $\kappa_a(u) = \kappa((\sqrt{u}+a)^2) - \kappa(a^2)$, so that the composition in (11.1.3) is $\kappa_a$ for $\kappa = \log G$ and $a = \alpha - \frac12$. If $\kappa$ is a Thorin Bernstein function then so is every $\kappa_a$, because the shift turns a drift $cu$ into $c(u+2a\sqrt{u})$ and each factor $(1+u/t)^{-1}$ of the Thorin form into $(a^2+t)/((\sqrt{u}+a)^2+t)$, the Laplace transform of a GGC whose Thorin density in $\theta$ is the pair of Cauchy kernels of width $a$ at $\pm\sqrt{t}$, as in Section 9.2.2. Applied to $\kappa_a$, with $\kappa_{a+a'} = (\kappa_a)_{a'}$, this shows that

the shifts $a$ for which $\kappa_a$ is a Thorin Bernstein function form a half line, and for $\kappa = \log G$ Proposition 11.1.4 identifies it as $[a_*, \infty)$ with $a_* = \sup_\rho \operatorname{Re}\rho - \frac{1}{2}$; RH is the statement $a_* = 0$. For a fixed shift the implication cannot be reversed. The function $\kappa(u) = 2\log(1+u) + \log(1 + (120u + 16u^2)/289)$ is a Bernstein function, since $\kappa'$ is the Laplace transform of $2e^{-x} + 2e^{-15x/4}\cos 2x \geq 0$, and it is not a Thorin Bernstein function, since $\kappa'$ has poles at $-\frac{15}{4} \pm 2i$, where a Stieltjes function is holomorphic; yet

$$\kappa_{1/2}(u) = \log\left(1 + \frac{u}{4}\right) + 2\log\frac{4((\sqrt{u}+\frac{1}{2})^2+1)}{5} + \log\frac{(\sqrt{u}+1)^2+4}{5}$$

is a Thorin Bernstein function, the exponent of an exponential law plus the exponents of GGCs whose Thorin densities are pairs of Cauchy kernels, of width $\frac{1}{2}$ at $\pm 1$ and of width 1 at $\pm 2$; the function $u \mapsto \kappa(u/4a^2)$ gives the same example at any shift $a > 0$.

The Dirichlet mean representation of James, Roynette and Yor [289] writes a GGC whose Thorin measure has finite total mass $c$ as the product $\Gamma_c \mathcal{M}$ of a gamma variable $\Gamma_c$ of shape $c$ and an independent Dirichlet mean $\mathcal{M}$. It applies to the truncated lattice variables $\mathcal{S}_n$ of Section 7.2.1, whose Thorin measures have total mass $2n$, and to the finite sums of exponential variables of Proposition 7.1.22, but not to $T_\alpha$ or, under RH, to $T$, whose Thorin measures have infinite mass.

### 11.1.5 What these results cannot give

Every GGC criterion in these books is, for the reciprocal Xi function at the centre, equivalent to RH: the HCM property of the Laplace transform and the Stieltjes property of the derivative of the exponent (Theorem 4.2.4), the Pick function property (Corollary 4.3.6), and the complete monotonicity of the Lévy density of the driving process (Proposition 11.1.1). The consequences obtained in Sections 11.1.1 to 11.1.4 are mostly unconditional. Table 11.3 collects them with their sources.

They are unconditional for one of two reasons. Some use only the infinite divisibility and self decomposability of the centre clock, that is the inequalities $W > 0$ and $W' < 0$, which hold wherever the zeros above height 1000 lie (Theorem 5.1.1). The others use the region $\operatorname{Re} s \geq 1$, through positivity of the Thorin measure at a basepoint $\alpha \geq 1$, which rests on the nonvanishing of $\zeta$ on the line $\operatorname{Re} s = 1$ (Theorem 1.2.2), or through the Euler product, as in Proposition 11.1.3. So none of the results supplies positivity of the Thorin measure uniformly in the height at a basepoint $\alpha < 1$, in the sense of Theorem 1.2.2 that $T_\alpha$ is a GGC, which is the zero free half plane $\operatorname{Re} s > \alpha$ of step 7 of the path in Table 12.1, and still less at the centre, where it is RH itself. That positivity, uniform in the height, is the one ingredient still missing on the path to RH; the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would supply it for every $\alpha > \frac{7}{8}$, by arithmetic from outside the theory (Section 12.1.2).

| result | source | consequence here | status |
|---|---|---|---|
| background driving Lévy process | Wolfe; Jurek and Vervaat; Sato, Theorem 17.5 | centre clock driven by a subordinator with Lévy density $-W'$; RH if and only if it is completely monotone | proved |
| Yamazato's theorem | Yamazato; Sato, Theorem 53.1 | centre clock and $T_\alpha$, $\alpha > \frac{1}{2}$, unimodal | proved |
| Urbanik's classes $L_n$ | Urbanik; Sato 1980; Berg and Forst | centre clock in $L_2$, not in $L_3$; its driving process and the driving process of that self decomposable, with unimodal marginals | proved |
| moment theorem | Sato, Theorem 25.3 | $\mathbb{E}T_\alpha = \infty$ for $\alpha \geq 1$ | proved |
| subexponential tails | Embrechts, Goldie, Veraverbeke | $P(T_\alpha > x) \sim b_\alpha(\pi x)^{-1/2}$; ratio 1.0005 at $x = 100$ | proved, checked |
| uniqueness, quasi infinite divisibility | Sato, Theorem 8.1; Lindner, Pan, Sato | $V_\alpha$ quasi infinitely divisible for $\alpha > 1$, not at $\alpha = 1$; Pólya variable not quasi infinitely divisible | proved |
| HCM characterisation and closure | Bondesson | Bosch threshold for powers | proved |
| Thorin measure at the two ends | Bondesson | $x^{-1/2}$ tail and super flat origin | proved |
| Thorin Bernstein composition | Schilling, Song, Vondraček | smoothing by $f_\alpha$ creates the Thorin property for $\alpha \geq 1$ | proved |
| Dirichlet means | James, Roynette, Yor | applies to truncations only | remark |

Table 11.3: What the books of Sato and Bondesson, and the papers around them, give the reciprocal Xi function. Theorem numbers, here and in the text, refer to the first edition of Sato's book [471].

### 11.1.6 Wiener gamma integrals and the Riemann Ornstein Uhlenbeck process

A GGC is a superposition of gamma laws, and a gamma process realises the superposition pathwise. Let $(\Gamma_t)_{t\geq 0}$ be a standard gamma process, the subordinator with $\mathbb{E}e^{-u\Gamma_t} = (1+u)^{-t}$. For a measurable $f \geq 0$ the Wiener integral $\int_0^\infty f(t)\, d\Gamma_t$ has Laplace exponent $\int_0^\infty \log(1 + uf(t))\, dt$. Let $U$ be the Thorin measure of a GGC without drift, $M(\theta) = U((0, \theta^2])$ its cumulative mass in the chart $\theta = \sqrt{z}$, and $M^{-1}(t) = \inf\{\theta : M(\theta) > t\}$ the right continuous inverse. The image of Lebesgue measure on $(0, M(\infty))$ under $M^{-1}$ is $dM$, so the substitution $t = M(\theta)$

turns $\int_0^\infty \log(1+u/M^{-1}(t)^2)\,dt$ into $\int \log(1+u/z)\,U(dz)$, and the GGC is the Wiener gamma integral

$$\int_0^\infty \frac{d\Gamma_t}{M^{-1}(t)^2}, \tag{11.1.4}$$

which is the Wiener gamma representation of the class [289]. At the centre, under RH, $M$ is the zero counting function $N$, $M^{-1}(t)=\gamma_k$ for $k-1\le t<k$, the ordinates being counted in increasing order and with multiplicity, and (11.1.4) is $\sum_k(\Gamma_k-\Gamma_{k-1})/\gamma_k^2$, the series $\sum_{\gamma>0} e_\gamma/\gamma^2$ of Proposition 1.2.1. At a basepoint $\alpha$ the count is replaced by the Poisson smoothed count $M_{\alpha-1/2}(\theta)=\pi^{-1}\int_0^\theta v_\alpha$ of Section 3.4, which is $\pi^{-1}$ times the argument of $\xi$ along the line $\operatorname{Re} s=\alpha$. For $\alpha\ge 1$, $v_\alpha>0$ and $T_\alpha$ is a GGC, so with no hypothesis $T_\alpha$ is (11.1.4) with $M=M_{\alpha-1/2}$. For $\frac12<\alpha<1$ the representation holds exactly when $T_\alpha$ is a GGC, that is when no zero has real part greater than $\alpha$ (Theorem 9.2.5), and then $M$ is $M_{\alpha-1/2}$ plus a unit jump at the ordinate of each zero on the line $\operatorname{Re} s=\alpha$ (Theorem 7.1.14). It needs $M_{\alpha-1/2}$ to be nondecreasing, that is $v_\alpha\ge 0$ on the line $\operatorname{Re} s=\alpha$, but for $\xi$ it is not known whether positivity on one line excludes zeros to its right (Remark 1.2.4).

Self decomposability gives a second process. If $Z$ is a subordinator with $\mathbb{E}\log(1+Z_1)<\infty$, the Ornstein Uhlenbeck type process $dF_t=-F_t\,dt+dZ_t$, that is $F_t=e^{-t}F_0+\int_0^t e^{-(t-r)}\,dZ_r$, has a unique stationary law, the law of $\int_0^\infty e^{-t}\,dZ_t$ [471, Theorem 17.5 and Corollary 17.9]. By Proposition 11.1.1 the driving process of the centre clock is the subordinator with Lévy density $h=-W'$, which the explicit formula computes from the primes. So there is, with no hypothesis, an Ornstein Uhlenbeck process $F$ whose stationary law is the centre clock, which we call the Riemann Ornstein Uhlenbeck process. Under RH the Laplace exponent of $Z$ is $\Psi_Z(u)=u(\log G)'(u)=\sum_{\gamma>0} u/(\gamma^2+u)$, so the driver is a superposition of independent compound Poisson processes, one for each zero counted with multiplicity, each with rate one and exponential jumps of mean $\gamma^{-2}$.

The simulation by `ou.py` is exact for the 1700 computed zeros. The zeros above the 1700th enter as a drift of the driver, which adds to the stationary law the mean $\mathbb{E}T-\sum_{k\le 1700}\gamma_k^{-2}$ of their components. The time average is 0.023094 against $\mathbb{E}T=\sum_k a_k^{-1}=0.023105$ (Section 9.1.2). The empirical Laplace transform of the stationary law is $0.79520\pm0.00096$, $0.32828\pm0.00180$ and $0.016404\pm0.00030$ at $u=10$, 50 and 200, with standard errors from 100 batch means, against $\varphi_{1/2}(u)=\xi(\frac12)/\xi(\frac12+\sqrt{u})=0.79514$, 0.32827 and 0.016399. The stationary process at time $\tau\ge0$ is $e^{-\tau}$ times its value at time 0 plus a variable independent of that value, and the centre clock has finite variance, so the autocorrelation at lag $\tau$ is $e^{-\tau}$ with no hypothesis. The estimates at lags 0.5, 1 and 2 are 0.602, 0.352 and 0.119 against 0.607, 0.368 and 0.135; the deficits are within 1.6 standard errors of the estimates, which are about 0.008, 0.010 and 0.013 by batch estimates over 20 blocks of the same run. The top right panel of Figure 11.2 shows ten units of time of the path.

For $\alpha \geq 1$ the shifted clock $T_\alpha$ is a GGC, hence self decomposable, and has its own driver: by (11.1.1), with $v_\alpha/\pi$ in place of $\rho_{1/2}$, its Lévy measure is $k_\alpha(x)\,dx/x$ with $k_\alpha(x) = \pi^{-1}\int_0^\infty e^{-x\theta^2} v_\alpha(\theta)\,d\theta$, and its driver has the completely monotone Lévy density $-k'_\alpha(x) = \pi^{-1}\int_0^\infty \theta^2 e^{-x\theta^2} v_\alpha(\theta)\,d\theta$. For $\frac{1}{2} < \alpha < 1$ the process exists as well, because $T_\alpha$ is self decomposable with no hypothesis (Corollary 9.2.6); its driver has the Lévy density $-W'_\alpha$, which is completely monotone exactly when $T_\alpha$ is a GGC.

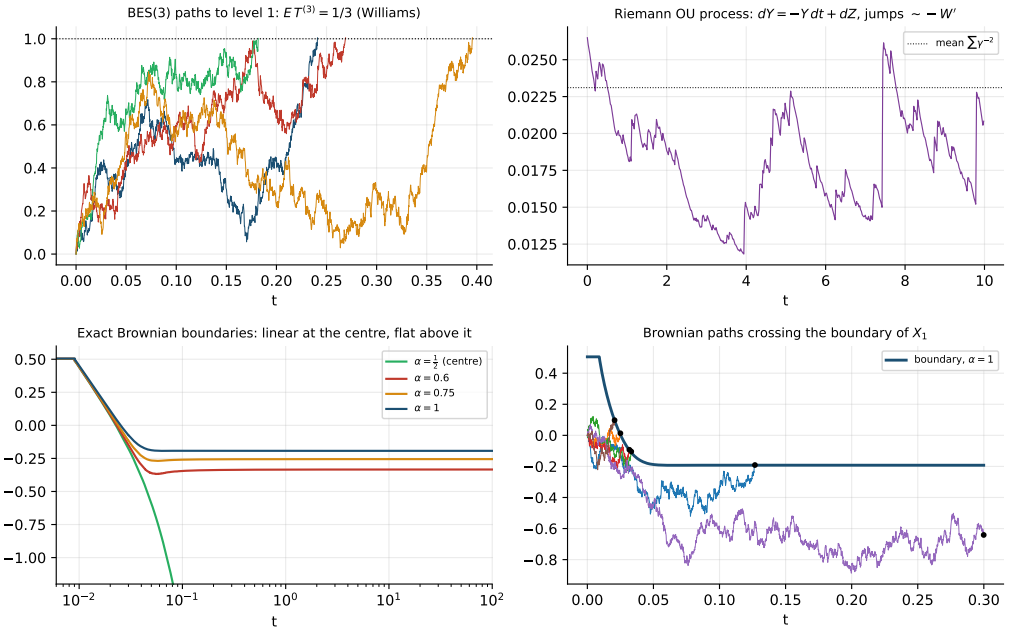



Figure 11.2: Four processes of the map. Top left: paths of a three dimensional Bessel process run until they reach the level one; the hitting time has mean $\frac{1}{3}$, and two independent copies make up the variable $S_2$ of Section 9.1.1. Top right: ten units of time of the Riemann Ornstein Uhlenbeck process $F$, written $Y$ in the plot, whose stationary law is the centre clock, with the mean $\mathbb{E}T$, written $\sum\gamma^{-2}$ in the plot. Bottom left: the exact Brownian boundaries of $T_\alpha$ at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 computed under RH beyond $t = 0.03$ (Section 9.2.6), asymptotically linear at the centre and flat above it. Bottom right: Brownian paths crossing the boundary of $T_1$, written $X_1$ in the plot (Section 9.2.4). Drawn by `figs.py` in the folder `basepoint_one/process_map`.

### 11.1.7 Subordinated Brownian motion and the map of processes

Running an independent standard Brownian motion $B$ on a clock gives a symmetric law with

$$\mathbb{E}e^{iuB_{T_\alpha}} = \mathbb{E}e^{-u^2T_\alpha/2} = \frac{\xi(\alpha)}{\xi(\alpha + |u|/\sqrt{2})}, \qquad u \in \mathbb{R},$$

so $B_{T_\alpha}$ has the law of $Y_\alpha/\sqrt{2}$, the normal scale mixture of Proposition 10.1.1, and what Section 10.1 proves for $Y_\alpha$ carries over with a factor $\sqrt{2}$.

Table 11.4 arranges the processes of Chapter 9 and of this section by basepoint; we write $a = \alpha - \frac{1}{2}$. The Mellin side is the Bessel picture of Williams [518] and of Biane, Pitman and Yor [49] (Sections 1.2 and 9.1.1). The clock side starts from the centre clock, which under RH with simple zeros is Kent's hitting time for a string with spectrum $\{\gamma^2\}$, and passes through the tilt $e^{-a^2T}$ and the sticky Brownian subordination of Theorem 9.2.5 to $T_\alpha$. From $T_\alpha$ it continues to the Wiener gamma integral of Section 11.1.6, to Brownian motion run on the clock, and to the Brownian boundaries of Section 9.2.6.

Three conclusions come out of the map. First, the Mellin side and the clock side differ in kind. The Bessel picture and the size biased variables $V_\alpha$ built from it are unconditional at every basepoint, and on that side the zeros enter only as zeros, of the characteristic function $\Xi(t)/\Xi(0)$ of the Pólya variable and of $\xi(\alpha+s)$ in the Lévy forms of Section 11.1.3; the positivity statements equivalent to RH, the GGC property of the centre clock and the EGGC property of $Y_{1/2}$, live on the clock side, in the reciprocal.

Second, on the clock side the Ornstein Uhlenbeck process at the centre and, at every basepoint above it, the tilted and subordinated clock, Brownian motion run on the clock and the Brownian boundary exist with no hypothesis. The Kent strings need RH at every basepoint, and simple zeros as well, because a Kent hitting time is a GGC whose Thorin measure consists of unit atoms at distinct points (Section 9.2.1). The shifted clock $T_\alpha$ is never a Kent hitting time for $\alpha > \frac{1}{2}$: when it is a GGC, each pair of zeros $\rho, \bar\rho$ with $\beta < \alpha$, such as those on the critical line, contributes to its Thorin measure the pair of Cauchy kernels of width $\alpha - \beta$ at $\pm\gamma$ (Section 11.1.4), so that measure is not a sum of atoms.

Third, the Bessel process is the template. Its hitting time of the level one is Kent's hitting time for a string with spectrum $\{\pi^2 n^2/2\}$, and under RH with simple zeros the centre clock is Kent's hitting time for a string with spectrum $\{\gamma^2\}$.

## 11.2 Free probability and the Upsilon transform

Wigner's semicircle law [517] is the limit law of the free central limit theorem [510, 386]. This section, which follows the first author's note [433], applies the Bercovici Pata bijection [37] and the Upsilon transform [34, 32] to the centre

| | $\alpha = \frac{1}{2}$ | $\frac{1}{2} < \alpha < 1$ | $\alpha = 1$ | $\alpha > 1$ |
|---|---|---|---|---|
| Mellin side | Pólya variable, $S_2^{1/4}$ bias | $S_2^{\alpha/2}$ bias | $S_2^{1/2}$ bias | $S_2^{\alpha/2}$ bias |
| Lévy form of $\log \xi(\alpha + s)$ | none: the characteristic function vanishes at the ordinates | exists if $\sup \operatorname{Re} \rho < \alpha$, only if $\sup \operatorname{Re} \rho \le \alpha$ | compensated limit, by the prime number theorem | absolutely convergent; quasi infinitely divisible |
| clock law | $T$, infinitely divisible and self decomposable | $\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$ | $\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ | $\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$ |
| GGC | if and only if RH | if and only if no zero has $\beta > \alpha$ | yes | yes |
| Thorin measure | $\sum_{\gamma>0} \delta_{\gamma^2}$ (RH) | Cauchy widths $\alpha - \beta$ | Cauchy widths $1 - \beta$ | Cauchy widths $\alpha - \beta$ |
| Kent string | spectrum $\gamma^2$ (RH) | spectrum $\gamma^2 + a^2$ on the sticky clock (RH) | spectrum $\lvert\rho\rvert^2$ on the sticky clock (RH) | spectrum $\gamma^2 + a^2$ on the sticky clock (RH) |
| tail of the clock | exponential, rate $\gamma_1^2$ | $b_\alpha(\pi x)^{-1/2}$ | $0.01303\, x^{-1/2}$ | $b_\alpha(\pi x)^{-1/2}$ |
| Brownian boundary | asymptotically linear, slope $-\sqrt{2}\gamma_1$ | flat: $-0.3338$ ($\alpha = 0.6$), $-0.2554$ ($\alpha = 0.75$) | flat at $-0.19277$ | flat |
| Wald defect $\mathbb{E}\, b(T_\alpha)$ | 0 | $b_\alpha/\sqrt{2}$ | $\sum_\rho \rho^{-1}/\sqrt{2}$ | $b_\alpha/\sqrt{2}$ |
| $B_{T_\alpha}$ | finite variance | Cauchy domain | Cauchy domain, tail $0.0104/y$ | Cauchy domain |
| Ornstein Uhlenbeck driver | jump density $-W'$ | jump density $-W'_\alpha$ | jump density $-k'_1$ | jump density $-k'_\alpha$ |
| symmetric EGGC $Y_\alpha$ | if and only if RH | if and only if no zero has $\beta > \alpha$ | yes | yes |

Table 11.4: The processes by basepoint, with $a = \alpha - \frac{1}{2}$. (RH) marks statements that hold under RH, and the Kent strings need the zeros to be simple as well. The Mellin row gives the size bias of $S_2$ that produces $V_\alpha$ (Section 9.1.1); $\widetilde{T}_\alpha$, $H$ and the sticky clock are those of Theorem 9.2.5 and Corollary 9.2.3, with $\widetilde{T} = \widetilde{T}_1$; and the boundaries $b$ and their Wald defects are those of Sections 9.2.4 and 9.2.6. The boundary row reports computations at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 made under RH beyond $t = 0.03$, and its entry for $\alpha > 1$ is extrapolated, not computed; the flatness is observed, not proved, and above the centre the Wald defect $b_\alpha/\sqrt{2}$ requires the boundary to be bounded at infinity (Lemma 9.2.4). All other entries hold as stated.

clock $T$ of Section 11.1. With no hypothesis, $T$ has an Upsilon image $\Upsilon(T)$, a GGC whose Thorin measure has the distribution function $u \mapsto W(1/u)$, and a free counterpart $\Lambda_{\mathrm{BP}}(T)$, a freely self decomposable law on $[0,\infty)$; and RH holds exactly when $T$ is itself the Upsilon image of an infinitely divisible law (Theorem 11.2.3).

### 11.2.1 The Bercovici Pata bijection and the Upsilon transform

As in Section 8.4.4, $G_\mu(z) = \int \mu(dx)/(z-x)$ is the Cauchy transform of a probability measure $\mu$ on $\mathbb{R}$ and $\sigma_t$ the semicircle law of variance $t$; the subscript distinguishes $G_\mu$ from the function $G$ of the clock. The Cauchy transform maps the upper half plane $\mathbb{C}^+$ into the lower half plane $\mathbb{C}^-$. The function $1/G_\mu$ has a right inverse $(1/G_\mu)^{-1}$ on a region $\Gamma_{\alpha,M} = \{z : |\operatorname{Re} z| < \alpha \operatorname{Im} z,\ \operatorname{Im} z > M\}$, and the Voiculescu transform $\varphi_\mu(z) = (1/G_\mu)^{-1}(z) - z$ linearises free additive convolution: $\varphi_{\mu\boxplus\nu} = \varphi_\mu + \varphi_\nu$ on a common region [510, 42]. Here $\mu \boxplus \nu$ is the law of $X+Y$ for free self adjoint variables $X$ and $Y$ with laws $\mu$ and $\nu$ [386]. The free cumulant transform $\mathcal{C}_\mu(z) = z\varphi_\mu(1/z)$, defined for $z \in \mathbb{C}^-$ with $1/z \in \Gamma_{\alpha,M}$, is additive under $\boxplus$, and $\mu$ is freely infinitely divisible if, for every $n$, it is the $n$ fold free convolution of a law with itself. A freely infinitely divisible law has a Voiculescu transform that extends analytically to all of $\mathbb{C}^+$ [42], so its free cumulant transform is analytic on all of $\mathbb{C}^-$.

A law $\mu$ on $\mathbb{R}$ is infinitely divisible exactly when its cumulant function $C_\mu(u) = \log \int e^{iux}\mu(dx)$ has the Lévy Khintchine form

$$C_\mu(u) = i\eta u - \tfrac{1}{2}Au^2 + \int_{\mathbb{R}} \left(e^{iut} - 1 - iut\,\mathbf{1}_{[-1,1]}(t)\right)\Pi(dt), \qquad u \in \mathbb{R}, \quad (11.2.1)$$

with $\eta \in \mathbb{R}$, $A \geq 0$ and a Lévy measure $\Pi$, that is $\Pi(\{0\}) = 0$ and $\int \min(1,t^2)\,\Pi(dt) < \infty$; the triplet $(\eta, A, \Pi)$ is unique [471, Theorem 8.1]. A law is freely infinitely divisible exactly when

$$\mathcal{C}_\mu(z) = \eta z + Az^2 + \int_{\mathbb{R}} \Big(\frac{1}{1-tz} - 1 - tz\,\mathbf{1}_{[-1,1]}(t)\Big)\,\Pi(dt), \qquad z \in \mathbb{C}^-, \quad (11.2.2)$$

with a triplet of the same kind, again unique [42, 37], [9, Proposition 6].

Definition 11.2.1. The Bercovici Pata bijection $\Lambda_{\mathrm{BP}}$ maps the infinitely divisible law with triplet $(\eta, A, \Pi)$ in (11.2.1) to the freely infinitely divisible law with the same triplet in (11.2.2) [37], [9, Definition 7].

The map is a homomorphism, $\Lambda_{\mathrm{BP}}(\mu * \nu) = \Lambda_{\mathrm{BP}}(\mu) \boxplus \Lambda_{\mathrm{BP}}(\nu)$, and a homeomorphism for weak convergence [37], [9, Section 3.2]. When $\Pi$ has moments of all orders, comparing the expansions of $e^{iut}$ and $1/(1-tz)$ in (11.2.1) and (11.2.2) shows that the free cumulants of $\Lambda_{\mathrm{BP}}(\mu)$ are the classical cumulants of $\mu$: both are $A\mathbf{1}_{\{n=2\}} + \int t^n\,\Pi(dt)$ for $n \geq 2$. The normal law $N(0,t)$, with triplet $(0,t,0)$, goes to the semicircle law $\sigma_t$ [37], [9, Section 3.2 and Remark 8].

Barndorff-Nielsen and Thorbjørnsen [33] proved that $\Lambda_{\mathrm{BP}}$ maps the self decomposable laws onto the freely self decomposable laws, those $\mu$ that for every $c \in (0,1)$ satisfy $\mu = D_c\mu \boxplus \mu_c$ for some law $\mu_c$, $D_c\mu$ being the law of $cX$ when $X$ has law $\mu$; and Hasebe and Thorbjørnsen [177] proved that freely self decomposable laws are unimodal, the free analogue of Yamazato's theorem (Corollary 11.1.2). By [9, Theorem 9] a freely infinitely divisible law $\mu$ is the image of an infinitely divisible law on $[0,\infty)$ if and only if $\mu^{\boxplus t}$ is carried by $[0,\infty)$ for every $t > 0$; these are the free regular laws. If $\mu$ is infinitely divisible on $[0,\infty)$, with drift $d \geq 0$ and Lévy measure $\Pi$ on $(0,\infty)$, $\int \min(1,x)\,\Pi(dx) < \infty$, then (11.2.2) reduces to [9, (7)]

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(z) = dz + \int_{(0,\infty)} \Big(\frac{1}{1-zx} - 1\Big)\,\Pi(dx) = dz + \int_{(0,\infty)} \frac{zx}{1-zx}\,\Pi(dx). \quad (11.2.3)$$

The right side is analytic on $\mathbb{C} \setminus (0,\infty)$, and we use it to extend $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}$ to the negative half line.

The Upsilon transform [34, 32] maps an infinitely divisible law $\mu$ with Lévy process $(X_t)$, $X_1 \sim \mu$, to the law $\Upsilon(\mu)$ of $\int_0^1 \log(1/t)\,dX_t$. It is infinitely divisible, with

$$C_{\Upsilon(\mu)}(u) = \int_0^1 C_\mu\big(u\log(1/t)\big)\,dt = \int_0^\infty C_\mu(ux)\,e^{-x}\,dx. \quad (11.2.4)$$

Barndorff-Nielsen, Maejima and Sato [32] proved that $\Upsilon$ maps the infinitely divisible laws onto the Goldie Steutel Bondesson class (the infinitely divisible laws whose Lévy measure has a completely monotone density in $|x|$ on each half line), and the self decomposable laws onto the Thorin class. On $[0,\infty)$ these are the class BO and the GGCs of Section 1.3.2. Barndorff-Nielsen and Thorbjørnsen [34] found the link with the free side:

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(i\zeta) = C_{\Upsilon(\mu)}(\zeta), \qquad \zeta < 0. \quad (11.2.5)$$

### 11.2.2 The Upsilon transform on the half line

Let $\mu$ be an infinitely divisible law on $[0,\infty)$ with drift $d \geq 0$ and Lévy measure $\Pi$ on $(0,\infty)$,

$$\Psi_\mu(s) = ds + \int_{(0,\infty)} \big(1 - e^{-sx}\big)\,\Pi(dx), \qquad \int \min(1,x)\,\Pi(dx) < \infty.$$

If $\mu$ is self decomposable, $\Pi(dx) = k(x)\,dx/x$ with $k$ nonincreasing, right continuous and $k(\infty) = 0$, the canonical function of $\mu$; if $\mu$ is a GGC with Thorin measure $U$, then $k(x) = \int e^{-tx}\,U(dt)$ (Section 1.3.1). The cumulants of $\mu$, when finite, are $\kappa_1 = d + \int x\,\Pi(dx)$ and $\kappa_n = \int x^n\,\Pi(dx)$ for $n \geq 2$.

Proposition 11.2.2. *Let $\mu$ be as above.*

*(a)* $\Upsilon(\mu)$ *is the infinitely divisible law on* $[0,\infty)$ *with drift* $d$*, Laplace exponent*

$$\Psi_{\Upsilon(\mu)}(s) = \int_0^\infty \Psi_\mu(sx)\, e^{-x}\, dx = ds + \int_{(0,\infty)} \frac{sy}{1+sy}\, \Pi(dy) \tag{11.2.6}$$

*and completely monotone Lévy density* $\ell(z) = \int y^{-1} e^{-z/y}\, \Pi(dy)$*. Every infinitely divisible law on* $[0,\infty)$ *with a completely monotone Lévy density, that is every law in BO, is* $\Upsilon(\mu)$ *for exactly one* $\mu$*. Moreover* $\mathcal{C}_{\Lambda_{\mathrm{BP}(\mu)}}(-s) = -\Psi_{\Upsilon(\mu)}(s)$ *for* $s > 0$*, and* $\kappa_n(\Upsilon(\mu)) = n!\, \kappa_n(\mu)$ *whenever* $\int x^n\, \Pi(dx) < \infty$*.*

*(b)* $\Upsilon(\mu)$ *is a GGC if and only if* $\mu$ *is self decomposable. In that case* $\Upsilon(\mu)$ *has drift* $d$ *and the Thorin measure* $\widetilde{U}$ *with* $\widetilde{U}((0,u]) = k(1/u)$*, the image of the measure* $-dk$ *under* $v \mapsto 1/v$*, and* $\Upsilon$ *is a bijection from the self decomposable laws on* $[0,\infty)$ *onto the GGCs, with inverse given by* $k(v) = \widetilde{U}((0,1/v])$*.*

*(c) If* $\mu$ *is a GGC with Thorin measure* $U$*, then* $\widetilde{U}$ *is the mixture over* $U(dt)$ *of the laws of* $t/E$*, with* $E$ *standard exponential:*

$$\widetilde{U}(du) = u^{-2}(-k')(1/u)\, du, \qquad \Psi_{\Upsilon(\mu)}(s) = ds + \int e^{t/s} E_1(t/s)\, U(dt), \tag{11.2.7}$$

*and* $\int u^p\, \widetilde{U}(du) = \Gamma(1-p) \int t^p\, U(dt)$ *for* $p < 1$*.*

*Proof.* (a) The first equality in (11.2.6) is (11.2.4) for Laplace transforms: for a subordinator, $\mathbb{E} \exp(-s \int_0^1 f(t)\, dX_t) = \exp(-\int_0^1 \Psi_\mu(sf(t))\, dt)$ for every bounded measurable $f \geq 0$, by approximation with step functions and independence of increments, and the case $f = \log(1/t)$ follows by monotone convergence. Since $\int_0^\infty (1 - e^{-sxy}) e^{-x} dx = sy/(1+sy)$, Fubini's theorem gives the second equality, and since

$$\frac{sy}{1+sy} = \int_0^\infty \left(1 - e^{-sz}\right) y^{-1} e^{-z/y}\, dz,$$

it gives $\Psi_{\Upsilon(\mu)}(s) = ds + \int_0^\infty (1 - e^{-sz})\, \ell(z)\, dz$. The function $\ell$ is a mixture of exponentials, hence completely monotone, and it is a Lévy density: $\int_0^\infty \min(1,z)\, y^{-1} e^{-z/y} dz = y(1 - e^{-1/y})$, which lies between $(1-e^{-1}) \min(1,y)$ and $\min(1,y)$, so that $\int \min(1,z)\, \ell(z)\, dz$ is comparable with $\int \min(1,y)\, \Pi(dy) < \infty$. Conversely, let $\lambda$ be infinitely divisible on $[0,\infty)$ with drift $d$ and completely monotone Lévy density $\ell(z) = \int_{[0,\infty)} e^{-zr}\, \sigma(dr)$ (Bernstein's theorem). Integrability of $\ell$ on $(1,\infty)$ forces $\sigma(\{0\}) = 0$. Let $\Pi$ be the image of $r^{-1}\sigma(dr)$ under $r \mapsto 1/r$. Then $\int y^{-1} e^{-z/y}\, \Pi(dy) = \int e^{-zr}\, \sigma(dr) = \ell(z)$, the comparison above shows that $\Pi$ is a Lévy measure with $\int \min(1,y)\, \Pi(dy) < \infty$, and $\lambda = \Upsilon(\mu)$ for the law $\mu$ with drift $d$ and Lévy measure $\Pi$. Uniqueness: $\ell$ determines $\sigma$ by the uniqueness of Laplace transforms, hence $\Pi$, and the drift is read off from $\Psi$ at infinity. By (11.2.3), $\mathcal{C}_{\Lambda_{\mathrm{BP}(\mu)}}(-s) = -ds - \int sx(1+sx)^{-1}\, \Pi(dx) = -\Psi_{\Upsilon(\mu)}(s)$. Finally $\int z^n \ell(z)\, dz = \int \Pi(dy) \int_0^\infty z^n y^{-1} e^{-z/y} dz = n! \int y^n\, \Pi(dy)$, which gives

$\kappa_n(\Upsilon(\mu)) = n!\,\kappa_n(\mu)$ for $n \geq 2$; for $n = 1$ the drift, which is the same for both laws, is added on both sides.

(b) Let $\mu$ be self decomposable with canonical function $k$. With $r = 1/y$,

$$z\,\ell(z) = \int_0^\infty \frac{z}{y^2}\, e^{-z/y}\, k(y)\, dy = \int_0^\infty z\, e^{-zr}\, k(1/r)\, dr = \int_{(0,\infty)} e^{-zr}\, d_r\big[k(1/r)\big],$$

by an integration by parts whose boundary terms vanish: $k(1/r) \to k(\infty) = 0$ as $r \to 0$, and $e^{-zr}k(1/r) \to 0$ as $r \to \infty$ because $yk(y) \leq \int_0^y k \to 0$ as $y \to 0$, $k$ being nonincreasing and integrable near 0. The function $r \mapsto k(1/r)$ is nondecreasing, so $\widetilde{U} = d_r[k(1/r)]$ is a positive measure with $\widetilde{U}((0,u]) = k(1/u)$, and $z\ell(z) = \int e^{-zr}\,\widetilde{U}(dr)$ says that $\Upsilon(\mu)$ is a GGC with Thorin measure $\widetilde{U}$ (Section 1.3.1). Conversely, let $\Upsilon(\mu)$ be a GGC with drift $d$ and Thorin measure $\widetilde{U}$, and put $k(v) = \widetilde{U}((0,1/v])$, a nonincreasing function with $k(\infty) = 0$ (its values at its jumps do not matter). By Fubini's theorem $\int_0^1 k(y)\,dy = \int \min(1,1/r)\,\widetilde{U}(dr)$ and $\int_1^\infty k(y)\,dy/y = \int_{(0,1)} \log(1/r)\,\widetilde{U}(dr)$, both finite by the integrability conditions of (1.3.1) and because $\widetilde{U}$ is finite on compact subsets of $(0,\infty)$. So $k(x)\,dx/x$ is a Lévy measure, the self decomposable law $\mu'$ with drift $d$ and canonical function $k$ exists, and by the first part $\Upsilon(\mu')$ has drift $d$ and Thorin measure $\widetilde{U}$. Hence $\Upsilon(\mu') = \Upsilon(\mu)$, and $\mu = \mu'$ by the uniqueness in (a). The same construction applied to an arbitrary GGC shows that $\Upsilon$ maps the self decomposable laws onto the GGCs, and (a) gives injectivity.

(c) Now $k(x) = \int e^{-tx}\,U(dt)$, so $\widetilde{U}((0,u]) = k(1/u) = \int e^{-t/u}\,U(dt) = \int \Pr(t/E \leq u)\,U(dt)$, and differentiating in $u$, $\widetilde{U}(du) = u^{-2}\int te^{-t/u}\,U(dt)\,du = u^{-2}(-k')(1/u)\,du$. Since $\Psi_\mu(s) = ds + \int \log(1+s/t)\,U(dt)$ and, by an integration by parts, $\int_0^\infty \log(1+sx/t)\,e^{-x}dx = \int_0^\infty e^{-x}(x+t/s)^{-1}dx = e^{t/s}E_1(t/s)$, the formula for $\Psi_{\Upsilon(\mu)}$ follows from (11.2.6) by Fubini's theorem. Finally $\int u^p\,\widetilde{U}(du) = \int \mathbb{E}[(t/E)^p]\,U(dt)$ and $\mathbb{E}E^{-p} = \Gamma(1-p)$ for $p < 1$. □

Part (b) is the half line reading of $\Upsilon(\mathrm{SD}) = $ Thorin class [32], which the note [433] pointed out through (11.2.7). For the standard exponential law, $k(x) = e^{-x}$ and $\Upsilon(\mathrm{Exp}(1))$ has Lévy density $\int_0^\infty y^{-2}e^{-z/y-y}dy = 2z^{-1/2}K_1(2\sqrt{z})$.

### 11.2.3 The Upsilon image and the free counterpart of the clock

We apply Proposition 11.2.2 to the centre clock $T$, whose Laplace transform is $1/G(u)$ with $G(u) = \xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2}) = \prod_k(1+u/a_k)$ (1.2.1). By Section 11.1.1, $T$ is infinitely divisible with no drift and Lévy measure $W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$, and with no hypothesis self decomposable with canonical function $W$ (Theorem 5.1.1); $h = -W'$ is the Lévy density of its driving subordinator (Proposition 11.1.1). The Stieltjes moments of Section 5.1 are $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_k a_k^{-n-1}$, and since $-\log G(u) = \sum_{n\geq 1}(-1)^n m_{n-1}u^n/n$ near $u = 0$, the cumulants of $T$ are $\kappa_n(T) = (n-1)!\,m_{n-1}$.

THEOREM 11.2.3. *(a) With no hypothesis, $\Upsilon(T)$ is a GGC with no drift and Thorin measure $\widetilde{U}(du) = u^{-2}h(1/u)\,du$, so that $\widetilde{U}((0,u]) = W(1/u) > 0$ for every $u > 0$. Its Laplace exponent is*

$$\Psi_{\Upsilon(T)}(s) = \int_0^\infty \log G(sx)\, e^{-x}\, dx = \sum_k e^{a_k/s} E_1(a_k/s), \qquad s > 0, \quad (11.2.8)$$

*with the principal branch of $E_1$; its Lévy density is the completely monotone function $\ell(z) = z^{-1}\int_0^\infty W(z/w)\, e^{-w}\, dw$; its cumulants are $\kappa_n(\Upsilon(T)) = n!\,(n-1)!\,m_{n-1}$; and $\int u^{-n}\,\widetilde{U}(du) = n!\,m_{n-1}$ for $n \ge 1$. Moreover $\mathbb{E}e^{\lambda\Upsilon(T)} = \infty$ for every $\lambda > 0$, while $\mathbb{E}e^{\lambda T} < \infty$ exactly when $\lambda < \gamma_1^2$.*

*(b) With no hypothesis, $\Lambda_{\mathrm{BP}}(T)$ is free regular, freely self decomposable and unimodal. Its free cumulant transform is*

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(z) = \int_0^\infty \frac{z\,W(x)}{1 - zx}\, dx, \qquad z \in \mathbb{C}^-, \quad (11.2.9)$$

*where the right side continues it analytically to $\mathbb{C}\backslash(0,\infty)$, with $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ for $s > 0$; its free cumulants are $(n-1)!\,m_{n-1}$, the cumulants of $T$. Its Cauchy transform satisfies*

$$\Psi_{\Upsilon(T)}\big(-G_{\Lambda_{\mathrm{BP}}(T)}(z)\big) = 1 - z\,G_{\Lambda_{\mathrm{BP}}(T)}(z), \qquad z \in \mathbb{C}^+, \quad (11.2.10)$$

*where $\Psi_{\Upsilon(T)}$ is continued analytically to $\mathbb{C} \setminus (-\infty, 0)$ by the right side of (11.2.6), and its support lies in $[c,\infty)$, where $c = \max_{s>0}(\Psi_{\Upsilon(T)}(s) - 1)/s > 0$.*

*(c) RH holds if and only if $T$ is the Upsilon image of an infinitely divisible law on $[0,\infty)$. In that case $T = \Upsilon(\mu_0)$, where $\mu_0$ is the self decomposable law with no drift and canonical function $N(v^{-1/2})$, that is with Lévy measure $N(x^{-1/2})\,dx/x$; $\mu_0$ is neither a GGC nor in BO. With no hypothesis, $T$ is not the Upsilon image of a law in BO.*

*Proof.* (a) Proposition 11.2.2(b) applies with $k = W$, which is continuously differentiable with $W' = -h$, so $\widetilde{U}((0,u]) = W(1/u)$ and $\widetilde{U}(du) = u^{-2}h(1/u)\,du$; the drift is zero because $T$ has none. The first form of $\Psi_{\Upsilon(T)}$ is (11.2.6) with $\Psi_T = \log G$. For the second, $\operatorname{Re} a_k > 0$ and hence $\operatorname{Re}(1/a_k) > 0$ for every $k$, so $w = sx/a_k$ has $\operatorname{Re} w \ge 0$, and $\log(1+w) = \int_0^1 w\,d\tau/(1+\tau w)$ with $|1+\tau w| \ge 1$ gives $|\log(1+w)| \le |w|$. Since $\sum_k |a_k|^{-1} < \infty$, $\sum_k \int_0^\infty |\log(1 + sx/a_k)|\, e^{-x} dx \le s\sum_k |a_k|^{-1} < \infty$, the series $\log G(sx) = \sum_k \log(1 + sx/a_k)$ can be integrated term by term, and each term gives $e^{a_k/s}E_1(a_k/s)$ by the integration by parts in the proof of Proposition 11.2.2(c), valid for $\operatorname{Re} a_k > 0$ because $e^z E_1(z) = \int_0^\infty e^{-w}(w+z)^{-1}dw$ for $\operatorname{Re} z > 0$. The Lévy density of Proposition 11.2.2(a) is $\ell(z) = \int_0^\infty y^{-2}e^{-z/y}W(y)\,dy$, and the substitution $y = z/w$

gives the stated form. The cumulants follow from $\kappa_n(\Upsilon(T)) = n!\,\kappa_n(T)$, and $\int u^{-n}\widetilde{U}(du) = \int_0^\infty v^n h(v)\,dv = n\int_0^\infty v^{n-1}W(v)\,dv = n!\,m_{n-1}$, by the substitution $v = 1/u$ and an integration by parts whose boundary terms vanish because $W(v) = O(v^{-1/2}\log(1/v))$ at the origin and decays exponentially at infinity. By Sato's moment theorem [471, Theorem 25.17], an infinitely divisible law has $\mathbb{E}e^{\lambda X} < \infty$ exactly when its Lévy measure integrates $e^{\lambda x}$ over $(1,\infty)$. For $\Upsilon(T)$, $z\ell(z) = \int e^{-zr}\widetilde{U}(dr) \geq e^{-z\varepsilon}W(1/\varepsilon)$ for every $\varepsilon > 0$, and with $\varepsilon < \lambda$ the integral $\int_1^\infty e^{\lambda z}\ell(z)\,dz$ diverges. For $T$, every zero other than the first has $\operatorname{Re} a_k \geq \gamma_2^2 - \frac{1}{4} > \gamma_1^2$, so $W(t) = e^{-\gamma_1^2 t}(1+o(1))$ as $t\to\infty$, and $\int_1^\infty e^{\lambda t}W(t)\,dt/t$ converges exactly when $\lambda < \gamma_1^2$.

(b) $T$ is infinitely divisible on $[0,\infty)$ with no drift and Lévy measure $\Pi(dx) = W(x)\,dx/x$, $\int \min(1,x)\,\Pi(dx) < \infty$, so $\Lambda_{\mathrm{BP}}(T)$ is free regular by [9, Theorem 9], freely self decomposable by [33], since $T$ is self decomposable, and unimodal by [177]. Formula (11.2.9) is (11.2.3) with this $\Pi$, and $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ is Proposition 11.2.2(a). The right sides of (11.2.9) and (11.2.6) give $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(w) = -\Psi_{\Upsilon(T)}(-w)$ on all of $\mathbb{C}\setminus(0,\infty)$. Since $\Pi$ has moments of all orders, the free cumulants are the classical cumulants of $T$ (Section 11.2.1). Write $G_\Lambda$ for $G_{\Lambda_{\mathrm{BP}}(T)}$. On a region where $(1/G_\Lambda)^{-1}(1/G_\Lambda(z)) = z$, the definition of $\varphi$ gives $\varphi(1/G_\Lambda(z)) = z - 1/G_\Lambda(z)$, that is $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(G_\Lambda(z)) = zG_\Lambda(z) - 1$; both sides are analytic on $\mathbb{C}^+$, which $G_\Lambda$ maps into $\mathbb{C}^-$, so the identity holds on $\mathbb{C}^+$, and it is (11.2.10). For the support, put $x(s) = (\Psi_{\Upsilon(T)}(s)-1)/s$ for $s>0$. Then $s^2x'(s) = f(s) = s\Psi'_{\Upsilon(T)}(s) - \Psi_{\Upsilon(T)}(s) + 1$, with $f(0+) = 1$, $f'(s) = s\Psi''_{\Upsilon(T)}(s) < 0$, and $f(s)\to-\infty$ as $s\to\infty$, since $s\Psi' - \Psi = -\int(1 - e^{-sz} - sze^{-sz})\ell(z)\,dz$ and $\ell$ is not integrable at the origin. So $x$ increases from $-\infty$ to its maximum $c = x(s^*) = \Psi'_{\Upsilon(T)}(s^*)$ on $(0,s^*]$ and decreases to 0 on $[s^*,\infty)$, because $\Psi_{\Upsilon(T)}(s)/s\to 0$; in particular $c>0$. For $x<0$, which lies outside the support, $G_\Lambda(x) < 0$ and (11.2.10) extends by continuity, so $S = -G_\Lambda(x) > 0$ solves $x(S) = x$, whose only positive solution is $s(x)$, the inverse of $x(\cdot)$ on $(0,s^*)$. This inverse is analytic on a neighbourhood $V$ of $(-\infty,c)$, a union of discs centred on that half line that do not meet $[c,\infty)$, and $-s$ agrees with $G_\Lambda$ on $(-\infty,0)$, hence on the connected set $V\setminus[0,c)$. So $G_\Lambda$ extends analytically across $(-\infty,c)$ with real values there, and by Stieltjes inversion $\Lambda_{\mathrm{BP}}(T)$ gives no mass to $(-\infty,c)$.

(c) By Proposition 11.2.2(a), the Upsilon images of the infinitely divisible laws on $[0,\infty)$ are the laws of the class BO, and the clock is in BO if and only if RH holds (Section 8.5.4 and Table 1.6). Under RH, $W(z)/z = \sum_{\gamma>0}\int_{\gamma^2}^\infty e^{-zr}\,dr = \int_0^\infty e^{-zr}N(\sqrt{r})\,dr$, so in the proof of Proposition 11.2.2(a) $\sigma(dr) = N(\sqrt{r})\,dr$, and the image of $r^{-1}\sigma(dr)$ under $r\mapsto 1/r$ is $N(x^{-1/2})\,dx/x$. Its canonical function $N(v^{-1/2})$ is nonincreasing, so $\mu_0$ is self decomposable, and by Proposition 11.2.2(b) the Thorin measure of $\Upsilon(\mu_0)$ has the distribution function $u\mapsto N(\sqrt{u})$, that is, it is $\sum_{\gamma>0}\delta_{\gamma^2}$, as Proposition 1.2.1 requires. The canonical function and the Lévy density of $\mu_0$ vanish on $(\gamma_1^{-2},\infty)$ without vanishing identically, so neither is

completely monotone: $\mu_0$ is not a GGC and not in BO. If $T = \Upsilon(\mu)$ with $\mu$ in BO, then $T$ is in BO, RH holds, and $\mu = \mu_0$ by the uniqueness in Proposition 11.2.2(a), which is impossible. □

The preimage $\mu_0$ of Theorem 11.2.3(c) has a reading in terms of the Dickman law $D$, the law known from smooth numbers with $\mathbb{E}e^{-sD} = \exp(-\operatorname{Ein}(s))$ and $\operatorname{Ein}(s) = \int_0^s (1-e^{-v})\,dv/v$. Integrating $\operatorname{Ein}(ax) = \int_0^1 (1-e^{-axv})\,dv/v$ against $e^{-x}$ gives $\int_0^\infty \operatorname{Ein}(ax)e^{-x}dx = \log(1+a)$, so $\Upsilon(D/\gamma^2)$ is the exponential law of rate $\gamma^2$, and since $\operatorname{Ein}(s/\gamma^2) = \int_0^{\gamma^{-2}} (1-e^{-sy})\,dy/y$, under RH

$$\mu_0 = \text{law of } \sum_{\gamma>0} D_\gamma/\gamma^2, \qquad D_\gamma \text{ independent Dickman variables,}$$

with Lévy measure $\sum_\gamma \mathbf{1}_{(0,\gamma^{-2})}(y)\,dy/y = N(y^{-1/2})\,dy/y$ and all jumps at most $\gamma_1^{-2}$. With no hypothesis $\psi_0(s) = \sum_k \operatorname{Ein}(s/a_k)$ is entire and $\int_0^\infty \psi_0(sx)e^{-x}dx = \log G(s)$ for $s > 0$: termwise integration is justified because $\operatorname{Re}(sx/a_k) > 0$ and $|\operatorname{Ein}(z)| \le |z|$ for $\operatorname{Re} z \ge 0$, since $|1-e^{-w}| \le |w|$ there. Under RH the density of $\mu_0$, by Fourier inversion because the Laplace transform grows along the negative axis, has mass 0.99993 on $[0.008, 0.045]$ and mean 0.023104, against $p_1 = 0.0231050$.

Proposition 11.2.4. *Let $\mathcal{C}_\xi(z) = \log\{\xi(\frac12)/\xi(\frac12 + \sqrt{-z})\} = \sum_{n\ge1} m_{n-1}z^n/n$ for $|z| < \gamma_1^2$. RH holds if and only if there is a freely infinitely divisible probability measure $\nu_\xi$ whose free cumulant transform is $\mathcal{C}_\xi$. Under RH, $\nu_\xi = \Lambda_{\mathrm{BP}}(\mu_0)$, its free cumulants are $m_{n-1}/n$, its free Lévy measure is $N(y^{-1/2})\,dy/y$, its support is compact and it is freely self decomposable.*

*Proof.* Under RH, (11.2.5) gives $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu_0)}(i\zeta) = C_{\Upsilon(\mu_0)}(\zeta) = C_T(\zeta) = \mathcal{C}_\xi(i\zeta)$, and $\Lambda_{\mathrm{BP}}$ preserves Lévy measures and maps self decomposable laws to freely self decomposable ones. Conversely let $\nu_\xi$ exist. Since $\mathcal{C}_\xi$ is analytic at 0, $\nu_\xi$ has compact support and its free cumulants are $m_{n-1}/n$. For $n \ge 3$ they are the moments $\int t^n \rho(dt)$ of its free Lévy measure $\rho$, and $|m_{n-1}|/n \le \gamma_1^{-2(n-1)} \sum_k |a_k|^{-1}$ because $|a_k| = \gamma^2 + \delta^2 \ge \gamma_1^2$, so $t^2\rho$ lives on $[-\gamma_1^{-2}, \gamma_1^{-2}]$. Then $\sum_{n\ge3} m_{n-1}z^n/n = -\sum_k \{\log(1 - z/a_k) + z/a_k + z^2/2a_k^2\} = \int \frac{z^3t^3}{1-zt}\,\rho(dt)$ is analytic off a subset of the real axis, while the left side has a logarithmic singularity at every $a_k$. So every $a_k$ is real, and RH holds since $\xi$ has no real zeros. □

The Cauchy transform of $\nu_\xi$ solves $\mathcal{C}_\xi(G_{\nu_\xi}(z))/G_{\nu_\xi}(z) + 1/G_{\nu_\xi}(z) = z$, and so the density follows from $\xi$ alone. Solved at 30 digits along $z = x + i\eta$ with $\eta$ down to $10^{-11}$, it is supported on an interval close to $[0.0166, 0.0348]$ with steep edges and its mode is at 0.0193, where it equals 98.6. The trapezoid rule on a grid of spacing 0.00025 gives mass 0.99992, mean 0.0231057 and variance $1.8592 \times 10^{-5}$, against $p_1 = 0.0231050$ and $m_1/2 = 1.8586 \times 10^{-5}$. This is consistent with RH and is no evidence beyond the zeros already verified, since the low cumulants are fixed by the first zeros and the existence in Proposition 11.2.4 concerns all of them

at once. The Upsilon preimage is due to Barndorff-Nielsen, Maejima and Sato [32] and the link between the two sides to Barndorff-Nielsen and Thorbjørnsen [34].

Under RH, $\widetilde{U}$ spreads the atom at $\gamma^2$ into the law of $\gamma^2/E$ (Proposition 11.2.2(c)), and in the chart $\theta = \sqrt{u}$

$$\widetilde{U}((0,\theta^2]) = W(\theta^{-2}) = \sum_{\gamma>0} e^{-\gamma^2/\theta^2} = \mathbb{E}\, N(\theta\sqrt{E}),$$

with $E$ standard exponential, a smoothing of the zero count. With no hypothesis, the asymptotic $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 5.1) and the Riemann von Mangoldt formula give $W(\theta^{-2})/N(\theta) \to \sqrt{\pi}/2 = 0.8862$ as $\theta \to \infty$. Under RH the expectation gives the next term: $\mathbb{E}N_0(\theta\sqrt{E})$, with $N_0(\theta) = (\theta/2\pi)\log(\theta/2\pi e) + \frac{7}{8}$, equals $(\theta/4\sqrt{\pi})\big(\log(\theta/2\pi e) + \frac{1}{2}\psi(\frac{3}{2})\big) + \frac{7}{8}$, where $\psi(\frac{3}{2}) = 2-\gamma_E-2\log 2 = 0.0365$; so the ratio of $W(\theta^{-2})$ to the smooth count $N_0(\theta)$ exceeds $\sqrt{\pi}/2$ by a relative amount of about $\psi(\frac{3}{2})/(2\log(\theta/2\pi e))$.

The free counterpart behaves differently at the left end. The clock and its Upsilon image are infinitely divisible with no drift and infinite Lévy measure, so their supports are $[0,\infty)$ [471, Section 24], and their densities are super flat at the origin; the free law has a hard edge at $c > 0$. Since $\Psi''_{\Upsilon(T)}(s^*) \neq 0$, the solution of $x(s) = x$ leaves the real axis as a square root at $x = c$, so the density $\operatorname{Im} s/\pi$ of $\Lambda_{\mathrm{BP}}(T)$ vanishes like $(x-c)^{1/2}$ there. At the other end, as $z \to \infty$ only the first zero matters in $\ell(z) = z^{-1}\int_0^\infty W(z/w)e^{-w}dw$, the saddle point of $e^{-\gamma_1^2 z/w - w}$ lies at $w = \gamma_1\sqrt{z}$, and

$$\ell(z) \sim 2\gamma_1 z^{-1/2} K_1(2\gamma_1\sqrt{z}) \sim \sqrt{\pi}\,\gamma_1^{1/2} z^{-3/4} e^{-2\gamma_1\sqrt{z}};$$

under RH, $\ell(z) = 2z^{-1/2}\sum_{\gamma>0}\gamma K_1(2\gamma\sqrt{z})$ exactly. The computed density of $\Upsilon(T)$ is 2.552 times $\ell$ at $x = 0.2$ and 1.828 times $\ell$ at $x = 0.4$, so the Lévy density takes over the tail only slowly.

Table 11.5 checks (11.2.8). The integral is computed from $\xi$ alone, by quadrature at 30 digits, and the sums use one of two tails: the tail of order one, $s(\log(\gamma_N/2\pi)+1)/(2\pi\gamma_N)$, which is $s\int_{\gamma_N}^\infty \theta^{-2}dN_0(\theta)$ and the choice of [433], and the full smooth tail $\int_{\gamma_N}^\infty \phi(\theta^2/s)\,dN_0(\theta) + \phi(\gamma_N^2/s)(N_0(\gamma_N) - N)$, where $\phi(z) = e^z E_1(z)$ and $N$ is the number of zeros used. The note's Table 1, the integral and the sums over 300 zeros without and with the tail of order one at $s = 1$, 10 and 100, is reproduced to every digit shown (0.0230681050, 0.0214685871 and 0.0230714840 at $s = 1$). With 1700 zeros and the full tail the difference is $-6.29 \times 10^{-11}s$ at every $s$ up to $10^3$, that is, it comes from the smooth tail of $\sum\gamma^{-2}$ beyond $\gamma_{1700} = 2197.26$, where $\phi(\gamma^2/s) \approx s/\gamma^2$.

Table 11.6 and the right panel of Figure 11.3 compare the three laws with the Lévy measure $W(t)\,dt/t$. The density of $T$ is the fixed Talbot inversion of $1/G$ at 25 digits on the grid 0.002(0.0005)0.15. That of $\Upsilon(T)$ is the fixed Talbot inversion of $e^{-\Psi_{\Upsilon(T)}}$, with $\Psi_{\Upsilon(T)}$ from the 1700 zeros and the tail $u\tau_1 - u^2\tau_2 + 2u^3\tau_3$, $\tau_j$

| | | sum over the zeros minus the integral | | | | |
|---|---|---|---|---|---|---|
| $s$ | $\int_0^\infty \log G(sx)\, e^{-x}\, dx$ | 300 zeros | 300, tail 1 | 300, full tail | 1700, tail 1 | 1700, full tail |
| 1 | 0.0230681049610841 | $-0.0016$ | $3.38 \times 10^{-6}$ | $-1.32 \times 10^{-9}$ | $1.2 \times 10^{-7}$ | $-6.29 \times 10^{-11}$ |
| 10 | 0.227587470315411 | $-0.016$ | $3.39 \times 10^{-5}$ | $-1.32 \times 10^{-8}$ | $1.2 \times 10^{-6}$ | $-6.29 \times 10^{-10}$ |
| 100 | 2.07560251202393 | $-0.16$ | $3.54 \times 10^{-4}$ | $-1.32 \times 10^{-7}$ | $1.23 \times 10^{-5}$ | $-6.29 \times 10^{-9}$ |
| $10^3$ | 15.2403221009594 | $-1.6$ | $4.96 \times 10^{-3}$ | $-1.31 \times 10^{-6}$ | $1.51 \times 10^{-4}$ | $-6.29 \times 10^{-8}$ |
| $10^4$ | 87.4852790298693 | $-15.8$ | 0.187 | $-1.24 \times 10^{-5}$ | $4.28 \times 10^{-3}$ | $-6.27 \times 10^{-7}$ |

Table 11.5: The Laplace exponent of $\Upsilon(T)$, which is minus the free cumulant transform of $\Lambda_{\mathrm{BP}}(T)$ at $-s$: the integral of (11.2.8) from $\xi$ alone, and the sums over the first 300 and 1700 zeros, with no tail, with the tail of order one and with the full smooth tail, as differences from the integral. Computed by `wigfree_upsilon.py`.

the smooth tail of $\sum \gamma^{-2j}$ with its boundary term; it runs in double precision from $x = 0.0125$ on and at 30 digits below, where double precision loses the cancellation. That of $\Lambda_{\mathrm{BP}}(T)$ is $\operatorname{Im} s/\pi$, with $s$ the solution of $\Psi_{\Upsilon(T)}(s) = 1 + xs$ in the upper half plane given by (11.2.10), found by Newton's method on 500 points of $[c, 0.15]$, spaced quadratically near $c$, with residuals below $4.6 \times 10^{-15}$. Here $c = 0.01529224618$ and $s^* = 387.9402$.

| | $T$ | $\Upsilon(T)$ | $\Lambda_{\mathrm{BP}}(T)$ |
|---|---|---|---|
| total mass | 1.00000000 | 1.00000053 | 1.00000021 |
| mean | 0.023104993 | 0.023104991 | 0.023104991 |
| variance | $3.71726 \times 10^{-5}$ | $7.43455 \times 10^{-5}$ | $3.71726 \times 10^{-5}$ |
| third central moment | $2.8835 \times 10^{-7}$ | $1.7301 \times 10^{-6}$ | $2.8835 \times 10^{-7}$ |
| mode | 0.0200 | 0.0185 | 0.017711 |
| density at the mode | 80.1042 | 72.6771 | 91.3878 |
| left end of the support | 0 | 0 | 0.01529224618 |

Table 11.6: The clock $T$, its Upsilon image and its free counterpart, from the densities of Figure 11.3. The exact means are $m_0 = 0.023104993$; the variances $m_1 = 3.71726 \times 10^{-5}$ for $T$ and $\Lambda_{\mathrm{BP}}(T)$ and $2m_1 = 7.43452 \times 10^{-5}$ for $\Upsilon(T)$; the third central moments $2m_2 = 2.8835 \times 10^{-7}$ for $T$ and $\Lambda_{\mathrm{BP}}(T)$ and $12m_2 = 1.7301 \times 10^{-6}$ for $\Upsilon(T)$. The modes of $T$ and $\Upsilon(T)$ are grid points of spacing 0.0005, that of $\Lambda_{\mathrm{BP}}(T)$ of spacing about $7 \times 10^{-5}$. Computed by `wigfree_upsilon.py`.

### 11.2.4 The semicircle as a local model for the Jacobi parameters

Section 5.1 builds from the Stieltjes moments $m_0, \dots, m_{2N-1}$ the Jacobi matrix $J_N$ of the measure $\nu = \sum_j a_j^{-1} \delta_{1/a_j}$, with diagonal entries $\alpha_1, \dots, \alpha_N$ and positive off diagonal entries $\beta_1, \dots, \beta_{N-1}$, the recurrence coefficients of the orthogonal polynomials of $\nu$, computed from the Taylor coefficients of $\Xi$ at the centre alone.

For constant parameters $\alpha_n = \alpha$ and $\beta_n = \beta$ the Jacobi matrix is $\alpha + \beta(S + S^*)$

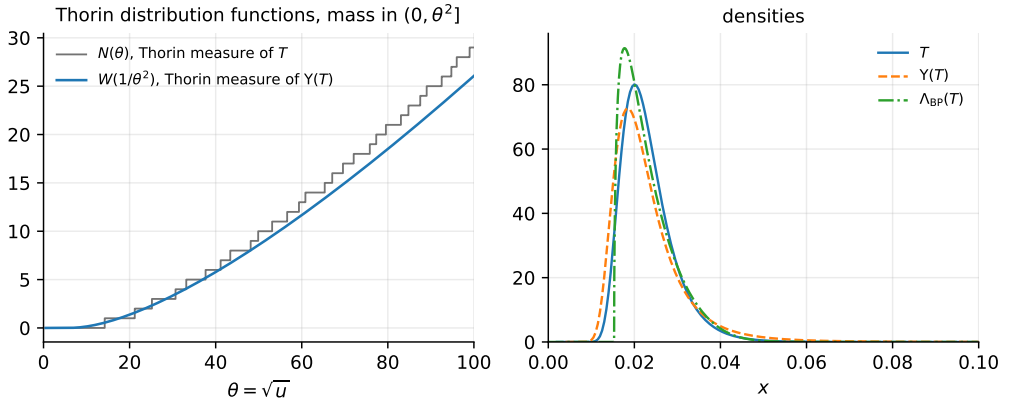



Figure 11.3: Left: the distribution function $W(\theta^{-2})$ of the Thorin measure of $\Upsilon(T)$ in the chart $\theta = \sqrt{u}$, against the zero count $N(\theta)$, which under RH is the distribution function of the Thorin measure of $T$ in the same chart. Right: the densities of $T$, of $\Upsilon(T)$ and of $\Lambda_{\mathrm{BP}}(T)$ on $[0, 0.1]$; the last vanishes to the left of $c = 0.0153$. Computed by `wigfree_upsilon.py` and drawn by `wigfree_plots.py`.

on $\ell^2(\{0, 1, 2, \dots\})$, with $S$ the unilateral shift, and the spectral measure of its first basis vector is the semicircle law on $[\alpha - 2\beta, \alpha + 2\beta]$. The $N \times N$ truncations have the eigenvalues $\alpha + 2\beta\cos(\pi k/(N+1))$, $1 \le k \le N$, whose distribution tends to the arcsine law with density $\big(\pi\sqrt{4\beta^2 - (x-\alpha)^2}\big)^{-1}$ on the same interval, the density of states. If the parameters vary slowly, the eigenvalues are counted by integrating local arcsine laws over the rows. For recurrence coefficients that vary on the scale of the matrix size, $\alpha_{n,N} \to a(n/N)$ and $\beta_{n,N} \to b(n/N)$, this is the theorem of Kuijlaars and Van Assche [317]: the zero distribution of the orthogonal polynomials tends to $\int_0^1 \omega_{[a(t)-2b(t),\,a(t)+2b(t)]}\,dt$, with $\omega_{[A,B]}$ the arcsine law on $[A, B]$. The parameters of $\nu$ decrease to zero with no such scaling, and the count needed below is that of the largest eigenvalues, an edge statement; for $\nu$ the local model is a heuristic. Since $\nu$ has its atoms in $(0, \gamma_1^{-2}]$ and accumulates only at the origin, the local interval $[\alpha_n - 2\beta_n, \alpha_n + 2\beta_n]$ should reach down to zero, which suggests $\alpha_n \approx 2\beta_n$. Write $L(n) = \alpha_n + 2\beta_n$ for its right end, taken to decrease in $n$, and $n(L)$ for the inverse function.

The eigenvalues of $J_N$ above a level $x > 0$ approximate the atoms $1/\gamma_j^2$ above $x$, of which there are $N(x^{-1/2})$. In the local model row $n$ contributes the mass $\frac{2}{\pi}\arccos\sqrt{x/L(n)}$ of the arcsine law on $[0, L(n)]$ above $x$, and nothing when $x \ge L(n)$, so that

$$N\big(x^{-1/2}\big) = \int_0^\infty \frac{2}{\pi}\arccos\sqrt{x/L(n)}_{+}\,dn = \frac{1}{\pi}\int_x^\infty n(L)\,\frac{\sqrt{x}\,dL}{L\sqrt{L-x}}, \qquad (11.2.11)$$

the second form by an integration by parts in $L$. With $u = 1/L$ and $w = 1/x$ it

becomes Abel's equation

$$\int_0^w \frac{f(u)\,du}{\sqrt{w-u}} = \pi N(\sqrt{w}), \qquad f(u) = u^{-1/2}\,n(1/u). \tag{11.2.12}$$

PROPOSITION 11.2.5. *Let $N$ be nondecreasing and right continuous on $[0,\infty)$, with $N = 0$ on $[0,\theta_0)$ for some $\theta_0 > 0$. Then*

$$n(L) = \int_{(0,L^{-1/2})} \frac{dN(T)}{\sqrt{1-LT^2}} \tag{11.2.13}$$

*gives a locally integrable solution $f$ of (11.2.12), with $N(\sqrt{w})$ replaced by its left limit at the jumps of $N$, and it is the only locally integrable solution. If $dN(T) = c\,dT$ on $(0,\infty)$, then $n(L) = \frac{\pi}{2}cL^{-1/2}$, so that $L_n = (\pi c/2n)^2$, which is $\gamma_{2n/\pi}^{-2}$ for the ordinates $\gamma_k = k/c$ of that count.*

*Proof.* For $0 < a < b$ the substitution $u = 1/L$ gives

$$\begin{aligned}\int_a^b \frac{dL}{L\sqrt{(L-a)(b-L)}} &= \int_{1/b}^{1/a} \frac{du}{\sqrt{(1-au)(bu-1)}} \\ &= \frac{1}{\sqrt{ab}} \int_{1/b}^{1/a} \frac{du}{\sqrt{(1/a-u)(u-1/b)}} = \frac{\pi}{\sqrt{ab}}.\end{aligned}$$

Insert (11.2.13) in the right side of (11.2.11), which is equivalent to (11.2.12), and exchange the integrals, all terms being positive. A point $T$ of the support of $dN$ contributes only when $T < x^{-1/2}$, through $L \in (x, T^{-2})$, and with $b = T^{-2}$ and $\sqrt{1-LT^2} = \sqrt{(b-L)/b}$ its contribution is

$$\frac{\sqrt{x}}{\pi} \int_x^b \frac{\sqrt{b}\,dL}{L\sqrt{(L-x)(b-L)}} = 1.$$

So the right side is $N$ of the open interval $(0, x^{-1/2})$, the left limit of $N$ at $x^{-1/2}$. The function $f$ is locally integrable: $n$ vanishes for $L > \theta_0^{-2}$, and for $0 < a < b$, by Fubini's theorem, $\int_a^b n(L)\,dL \le \int_{(0,a^{-1/2})} dN(T) \int_0^{T^{-2}} dL/\sqrt{1-LT^2} = 2\int_{(0,a^{-1/2})} T^{-2}\,dN(T) < \infty$. If $f_1$ and $f_2$ are two locally integrable solutions, $g = f_1 - f_2$ satisfies $\int_0^w g(u)(w-u)^{-1/2}du = 0$ for every $w$; integrating against $(y-w)^{-1/2}$ over $w \in (0,y)$ and using $\int_u^y dw/\sqrt{(y-w)(w-u)} = \pi$ gives $\pi\int_0^y g = 0$ for every $y$, so $g = 0$ almost everywhere. Finally $\int_0^{L^{-1/2}} c\,dT/\sqrt{1-LT^2} = cL^{-1/2}\arcsin 1$. □

The local model therefore predicts $\alpha_n \approx 2\beta_n \approx \frac{1}{2}L_n$, with $n(L_n) = n$ [433]. For the zeros of $\xi$ we use the smooth density $dN_0(T) = (2\pi)^{-1}\log(T/2\pi)\,dT$ on $T > 2\pi$, for which the substitution $T = L^{-1/2}\sin\varphi$ gives

$$n(L) = \frac{1}{2\pi\sqrt{L}} \int_{\varphi_0}^{\pi/2} \log\frac{\sin\varphi}{2\pi\sqrt{L}}\,d\varphi, \qquad \sin\varphi_0 = 2\pi\sqrt{L}.$$

As $L \to 0$, $\int_0^{\pi/2} \log \sin \varphi \, d\varphi = -\frac{\pi}{2} \log 2$ gives $n(L) = (4\sqrt{L})^{-1} \log(1/(4\pi\sqrt{L})) + O(1)$, while the rule of Proposition 11.2.5 for a linear count, applied with the ordinates of the smooth count, $N_0(\gamma_r) = r$ with $N_0(T) = (T/2\pi)\log(T/2\pi e) + \frac{7}{8}$, gives $\frac{1}{2}L_n = \frac{1}{2}\gamma_{2n/\pi}^{-2}$ and corresponds to $n = (4\sqrt{L})^{-1}\log(1/(2\pi e\sqrt{L}))$ to the same order. The two rules differ by a factor that tends to one only logarithmically; this is why the cruder rule $\alpha_n \approx \frac{1}{2}\gamma_{2n/\pi}^{-2}$, which treats the count as linear, is off by a factor near 1.3.

Table 11.7 compares the computed parameters with the two rules. The program `wigfree_jacobi.py` computes $M_0, \dots, M_{100}$ at 1000 digits by the trapezoid rule of step $1/600$ on $[0, 3.6]$, the moments $m_0, \dots, m_{99}$, and the Cholesky factorisation of $\mathcal{H}_{50}$. The matrix $L^{-1}[m_{i+j+1}]L^{-\top}$ is tridiagonal to $1.3 \times 10^{-836}$, and its entries change by at most $2.3 \times 10^{-429}$ relatively when the factorisation runs at 600 digits. The first parameters are irregular, $\alpha_1, \dots, \alpha_5 = (1.61, 3.50, 1.80, 1.22, 1.07) \times 10^{-3}$ and $\beta_1, \dots, \beta_5 = (1.91, 1.15, 0.697, 0.554, 0.498) \times 10^{-3}$, and the model is meant for larger $n$. Over $17 \le n \le 49$ the ratio $\alpha_n / \frac{1}{2}L_n$ has mean 1.033 and standard deviation 0.060 (range 0.940 to 1.178), and $2\beta_n / \frac{1}{2}L_n$ has mean 1.012 and standard deviation 0.060 (range 0.893 to 1.134); the columns $\alpha_n$, $2\beta_n$, $\frac{1}{2}L_n$ and these two ratios reproduce Table 6 of [433] to the digits shown there. Over the same range $\alpha_n/(2\beta_n)$ has mean 1.021 and standard deviation 0.047, and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac{1}{4}$ in the local model, has mean 0.2502 and standard deviation 0.013 (mean 0.256 and standard deviation 0.059 over $1 \le n \le 49$). The cruder rule falls short: $\alpha_n$ and $2\beta_n$ exceed it by the factors 1.286 and 1.26 on average, with standard deviations 0.081 and 0.078.

Two cautions apply. The rule $n(L_n) = n$ and the midpoint rule $n(L_n') = n - \frac{1}{2}$ differ by a shift of $n$ of order one, which the asymptotics cannot see, and they move the mean ratios by two per cent, to 1.011 and 0.991, within the scatter; the data do not decide between them. And the scatter itself, about six per cent, is not explained by this computation: the note attributes it to the discreteness of the zeros, which the smooth density ignores, and that remains to be tested (Problem 12.3). So the local semicircle model predicts the size of the Jacobi parameters to a few per cent over $17 \le n \le 49$ with no fitted constant, but the agreement is not a proof of an asymptotic law.

### 11.2.5 Gamma(2), Laguerre polynomials and finite free convolution

Williams' variable [518] $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma(2) variables $\Gamma_n$, the sum of two independent copies of the hitting time $J = \sum_n \frac{2}{\pi^2 n^2} e_n$ of Section 9.1.1, satisfies $2\xi(s) = \mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$ (9.1.1) [49], and the RH content of this identity lies on its Mellin side (Theorem 7.1.21 and Section 8.2.3). The cumulants of $S_2$ are $\kappa_m = 2(m-1)!\,(2/\pi^2)^m \zeta(2m)$.

| | $10^5\times$ | | | ratio to $\frac12 L_n$ | | $10^5\times$ | ratio to $\frac12 L'_n$ | | $10^5\times$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| $n$ | $\alpha_n$ | $2\beta_n$ | $\frac12 L_n$ | $\alpha_n$ | $2\beta_n$ | $\frac12 L'_n$ | $\alpha_n$ | $2\beta_n$ | $\frac12\gamma^{-2}_{2n/\pi}$ | $\frac{\beta_n^2}{\alpha_n\alpha_{n+1}}$ |
| 5 | 107.3 | 99.54 | 84.84 | 1.27 | 1.17 | 94.11 | 1.14 | 1.06 | 61.8 | 0.2665 |
| 9 | 56.96 | 51.12 | 45.73 | 1.25 | 1.12 | 48.70 | 1.17 | 1.05 | 34.5 | 0.2548 |
| 13 | 30.20 | 32.52 | 30.07 | 1.00 | 1.08 | 31.48 | 0.959 | 1.03 | 23.2 | 0.2611 |
| 17 | 21.71 | 21.51 | 21.83 | 0.995 | 0.986 | 22.64 | 0.959 | 0.950 | 17.0 | 0.2362 |
| 21 | 18.50 | 16.55 | 16.82 | 1.10 | 0.983 | 17.34 | 1.07 | 0.954 | 13.3 | 0.2479 |
| 25 | 15.90 | 14.58 | 13.50 | 1.18 | 1.08 | 13.85 | 1.15 | 1.05 | 10.7 | 0.2588 |
| 29 | 11.25 | 11.45 | 11.16 | 1.01 | 1.03 | 11.41 | 0.986 | 1.00 | 8.93 | 0.2550 |
| 33 | 9.530 | 9.336 | 9.424 | 1.01 | 0.991 | 9.615 | 0.991 | 0.971 | 7.59 | 0.2647 |
| 37 | 8.704 | 8.163 | 8.101 | 1.07 | 1.01 | 8.249 | 1.06 | 0.990 | 6.56 | 0.2463 |
| 41 | 6.998 | 7.053 | 7.062 | 0.991 | 0.999 | 7.179 | 0.975 | 0.982 | 5.74 | 0.2618 |
| 45 | 6.498 | 6.150 | 6.228 | 1.04 | 0.987 | 6.323 | 1.03 | 0.973 | 5.08 | 0.2481 |
| 49 | 5.957 | 6.172 | 5.547 | 1.07 | 1.11 | 5.625 | 1.06 | 1.10 | 4.54 | 0.2738 |

Table 11.7: The Jacobi parameters of $\nu$ against the local semicircle model: $\alpha_n$ and $2\beta_n$; half the right end $\frac12 L_n$ of the local interval from (11.2.13) with the smooth density and $n(L_n) = n$, and the ratios to it; the same with the midpoint rule $n(L'_n) = n - \frac12$; the cruder rule $\frac12\gamma^{-2}_{2n/\pi}$, with $N_0(\gamma_r) = r$; and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac14$ in the local model. Computed by `wigfree_jacobi.py`.

Proposition 11.2.6. *For every complex $s$,*

$$\mathbb{E}\Big[\Big(\frac{\pi}{2}J\Big)^{s/2}\Big] = \frac{2(1-2^{1-s})\,\xi(s)}{s-1} = s\,\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)(1-2^{1-s})\,\zeta(s), \tag{11.2.14}$$

*the value at $s = 1$ being $2\xi(1)\log 2 = \log 2$. Equivalently $J \overset{d}{=} V^{-2}S_2$, with $V$ uniform on $[1,2]$ and independent of $S_2$. The function* (11.2.14) *vanishes at the zeros of $\xi$ and at the points $1 + 2\pi i k/\log 2$, $k \neq 0$, of the line $\operatorname{Re} s = 1$, and, unlike the zero set of $\xi$, its zero set is not symmetric under $s \mapsto 1 - s$.*

*Proof.* This is the multiplicative relation of Biane, Pitman and Yor between $J$, their $\Sigma_1$, and $S_2$, their $\Sigma_2$ [49, Table 1 and (43) to (45)]; we give the computation. For $\operatorname{Re} p < 0$, $\int_0^\infty \lambda^{-p-1}e^{-\lambda x}d\lambda = \Gamma(-p)x^p$ and $\mathbb{E}e^{-\lambda J} = \sqrt{2\lambda}/\sinh\sqrt{2\lambda}$ give, with $r = \sqrt{2\lambda}$,

$$\mathbb{E}J^p = \frac{2^{p+1}}{\Gamma(-p)}\int_0^\infty \frac{r^{-2p}}{\sinh r}\,dr = \frac{2^{p+2}(1-2^{2p-1})\,\Gamma(1-2p)\,\zeta(1-2p)}{\Gamma(-p)},$$

by $1/\sinh r = 2\sum_{k\geq 0} e^{-(2k+1)r}$, which gives $\int_0^\infty r^{z-1}/\sinh r\,dr = 2(1-2^{-z})\Gamma(z)\zeta(z)$ for $\operatorname{Re} z > 1$. Put $p = s/2$ with $\operatorname{Re} s < 0$. The functional equation $\zeta(1-s) = \pi^{1/2-s}\Gamma(\frac{s}{2})\zeta(s)/\Gamma(\frac{1-s}{2})$, the duplication formula $\Gamma(1-s) = 2^{-s}\pi^{-1/2}\Gamma(\frac{1-s}{2})\Gamma(1-\frac{s}{2})$ and $\Gamma(1-\frac{s}{2})/\Gamma(-\frac{s}{2}) = -\frac{s}{2}$ turn this into $\mathbb{E}J^{s/2} = -2^{1-s/2}s(1-2^{s-1})\pi^{-s}\Gamma(\frac{s}{2})\zeta(s)$, and multiplying by $(\pi/2)^{s/2}$ gives

(11.2.14). Both sides are entire, since $J$ has moments of all real orders and the zero of $1-2^{1-s}$ at $s=1$ cancels the pole of $\zeta$, so the identity holds for every $s$. Since $\mathbb{E}V^{-s}=\int_1^2 v^{-s}dv=(1-2^{1-s})/(s-1)$, the right side is $\mathbb{E}V^{-s}\,\mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$. The factor $1-2^{1-s}$ vanishes exactly at $1+2\pi ik/\log 2$; on the line $\operatorname{Re}s=0$, the image of $\operatorname{Re}s=1$ under $s\mapsto 1-s$, it equals $-1$ at the corresponding points $2\pi ik/\log 2$ and $\xi$ has no zeros, so the zero set is not symmetric. □

The elementary factor $(1-2^{1-s})/(s-1)$ adds zeros on the line $\operatorname{Re}s=1$, the first at height $2\pi/\log 2=9.0647$, the zeros of the alternating zeta function that Section 10.1.7 meets.

Proposition 11.2.7. *Let $z_1,z_2,\dots$ be independent standard complex Gaussian vectors in $\mathbb{C}^2$, with $\mathbb{E}z_nz_n^*=I_2$, let $c_n=2/(\pi^2n^2)$ and $W=\sum_n c_nz_nz_n^*$. Then $\operatorname{tr}W\overset{d}{=}S_2$, so that $2\xi(s)=\mathbb{E}(\frac{\pi}{2}\operatorname{tr}W)^{s/2}$, and*

$$\mathbb{E}\det(x-W)=x^2-2m_1x+m_1^2-m_2,\qquad m_k=\sum_n c_n^k,\quad m_1=\tfrac{1}{3},\ m_2=\tfrac{2}{45},$$

*with the real roots $m_1\pm\sqrt{m_2}=0.122515$ and $0.544152$. This is the finite free convolution $\boxplus_2$ of the polynomials $\mathbb{E}\det(x-c_nz_nz_n^*)=x(x-2c_n)$, and the mixed characteristic polynomial of Marcus, Spielman and Srivastava [369] of the matrices $c_nI_2$.*

*Proof.* $|z_n|^2$ is the sum of two independent standard exponential variables, so $\operatorname{tr}W=\sum_n c_n|z_n|^2\overset{d}{=}S_2$. For a $2\times 2$ matrix, $\det(x-W)=x^2-x\operatorname{tr}W+\det W$ and $\det W=\frac{1}{2}((\operatorname{tr}W)^2-\operatorname{tr}W^2)$. Here $\mathbb{E}\operatorname{tr}W=2m_1$ and $\mathbb{E}(\operatorname{tr}W)^2=\operatorname{Var}S_2+(\mathbb{E}S_2)^2=2m_2+4m_1^2$, while $\mathbb{E}\operatorname{tr}W^2=\sum_{n,k}c_nc_k\mathbb{E}|z_n^*z_k|^2=6m_2+2(m_1^2-m_2)$, since $\mathbb{E}|z_n|^4=6$ and $\mathbb{E}|z_n^*z_k|^2=\operatorname{tr}(I_2I_2)=2$ for $n\neq k$. Hence $\mathbb{E}\det W=m_1^2-m_2$. The terms $c_nz_nz_n^*$ are independent and unitarily invariant in law, with $\mathbb{E}\det(x-c_nz_nz_n^*)=x^2-2c_nx$, so by [370, Theorem 1.2] and bilinearity the expected characteristic polynomial of a finite partial sum is the $\boxplus_2$ convolution of these polynomials; with the coefficient formula of $\boxplus_2$, adding $x(x-2c)$ to a polynomial $x^2-Ax+B$ gives $x^2-(A+2c)x+B+Ac$, so the constant term is $2\sum_{n<k}c_nc_k=m_1^2-m_2$ in the limit. □

The reading of the exponent two of $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$ as the size of the matrix is special to the sinh family of [49].

In the normalisation of [370] write a polynomial of degree $d$ as $p(x)=\sum_k(-1)^k\binom{d}{k}e_kx^{d-k}$; the multiplicative convolution $\boxtimes_d$ multiplies the normalised coefficients, $e_k(p\boxtimes_d q)=e_k(p)e_k(q)$ [370, Definition 1.4], and by Szegő's theorem it preserves the property of having only nonnegative roots [370, Theorem 1.6].

Lemma 11.2.8. *Let $a>0$ and $(a)_k=a(a+1)\cdots(a+k-1)$.*

*(i) The reversed Laguerre polynomial* $q_a(x) = x^d L_d^{(a-1)}(1/x)$ *has* $d$ *positive simple roots and normalised coefficients* $e_k = e_0/(a)_k$, $e_0 = \Gamma(d+a)/(d!\,\Gamma(a))$. *Hence* $p \boxtimes_d (q_a/e_0)$ *has the normalised coefficients* $e_k(p)/(a)_k$, *and it has only real roots whenever the roots of* $p$ *are real and of one sign.*

*(ii)* $\{1/(a)_k\}_{k\ge 0}$ *is a multiplier sequence: the function*

$$\sum_k \frac{x^k}{k!\,(a)_k} = \Gamma(a) x^{(1-a)/2} I_{a-1}(2\sqrt{x})$$

*is entire of order* $\frac{1}{2}$ *with only real negative zeros, at* $-j_{a-1,m}^2/4$.

*(iii) A density* $f$ *on* $(0,\infty)$ *is a scale mixture of Gamma*$(a)$ *densities if and only if* $x^{1-a} f(x)$ *is completely monotone; for* $a = 2$, *if and only if* $f(x)/x$ *is.*

*Proof.* (i) By [397, (18.5.12)],

$$x^d L_d^{(a-1)}(1/x) = \sum_k (-1)^k \binom{d+a-1}{d-k} \frac{x^{d-k}}{k!},$$

and $\binom{d+a-1}{d-k}/(k!\binom{d}{k}) = \Gamma(d+a)/(d!\,\Gamma(k+a))$, which is $e_0/(a)_k$. The zeros of $L_d^{(a-1)}$ are positive and simple, those of an orthogonal polynomial for the weight $x^{a-1}e^{-x}$ on $(0,\infty)$ [397, Section 18.2(vi)], and the roots of $q_a$ are their reciprocals. The statement on $\boxtimes_d$ is Szegő's theorem for roots of one sign; for nonpositive roots apply it to $(-1)^d p(-x)$, whose normalised coefficients are $(-1)^k e_k(p)$. (ii) The series is that of the modified Bessel function [397, (10.25.2)]. For $x = -y^2/4 < 0$ it equals $\Gamma(a)(y/2)^{1-a} J_{a-1}(y)$, and for $a-1 > -1$ the zeros of $J_{a-1}(y)/y^{a-1}$ are real [397, Section 10.21(i)]. A real entire function of order less than one with only real zeros is in the Laguerre Pólya class, and by the theorem of Pólya and Schur [455] a sequence whose exponential generating function is in that class with coefficients of one sign is a multiplier sequence. (iii) $f(x) = \int_0^\infty \theta^{-a} x^{a-1} e^{-x/\theta}\, \varpi(d\theta)/\Gamma(a)$ says that $x^{1-a} f(x)$ is the Laplace transform of a positive measure in the variable $1/\theta$; conversely Bernstein's theorem writes a completely monotone $x^{1-a} f(x)$ in this form. □

Let $X$ be the Pólya variable, with density $p = \Phi/\Xi(0)$ for the kernel $\Phi$ of (1.2.3) and moment generating function $\mathbb{E}e^{sX} = G(s^2)$ (1.2.4), and write $G(u) = \sum_k b_k u^k$, so that $b_k = \mathbb{E}X^{2k}/(2k)! = M_k/((2k)!\,M_0)$ with $M_k = \int_{\mathbb{R}} v^{2k}\Phi(v)\,dv$, as in Sections 5.1 and 13.4.1.

Proposition 11.2.9. *For* $a > 0$ *let* $G_a(u) = \sum_k b_k u^k/(a)_k$. *If RH holds, then for every* $a > 0$ *the function* $G_a$ *is in the Laguerre Pólya class with only real negative zeros, and all its Jensen polynomials* $\sum_{i=0}^d \binom{d}{i}(n+i)!\, b_{n+i} x^i/(a)_{n+i}$, $d, n \ge 0$, *have only real negative zeros. Multiplying the coefficients by* $(a)_k$ *instead does not preserve real zeros: it turns* $(1+u)^2 = 1+2u+u^2$ *into* $1+2au+a(a+1)u^2$, *whose discriminant is* $-4a < 0$.

*Proof.* Under RH, $G(u) = \prod_j (1+u/\gamma_j^2)$ with $\sum_j \gamma_j^{-2} < \infty$, so $G$ is in the Laguerre Pólya class, and by Pólya's theorem its Jensen polynomials $J_{d,n}(x) = \sum_i \binom{d}{i}(n+i)!\, b_{n+i} x^i$ of Section 13.4.1 have only real zeros. Since $(a)_{n+i} = (a)_n (a+n)_i$, the Jensen polynomial of $G_a$ of degree $d$ and shift $n$ is $1/(a)_n$ times the image of $J_{d,n}$ under the multiplier sequence $\{1/(a+n)_i\}_i$ of Lemma 11.2.8(ii), so it has only real zeros, negative because its coefficients are positive. For $n = 0$, $\binom{d}{i}\, i!\, d^{-i} \to 1$ gives $\sum_i \binom{d}{i}\, i!\, b_i x^i/((a)_i d^i) \to G_a(x)$ uniformly on compact sets as $d \to \infty$, so by Hurwitz's theorem $G_a$ has only real negative zeros; since $(a)_k \geq a(k-1)!$, $G_a$ has order at most $\frac{1}{3}$, and it is in the Laguerre Pólya class. The example is a direct computation. □

In terms of Lemma 11.2.8(i), dividing the normalised coefficients by $(a)_k$ is the multiplicative convolution with $q_a/e_0$, so peeling a Gamma($a$) layer, the operation that would pass from the moments of a Gamma($a$) scale mixture to those of its mixing law, keeps zeros real, while the inverse operation can destroy them. The note calls the peeling a free Poisson operation. The example does not show that no argument could pass from real zeros of some $G_a$ to real zeros of $G$, as the note puts it.

Lemma 11.2.8(iii) makes the Gamma(2) mixture route of [424, Section 12.6], which Section 8.2.3 recalls as excluded, a statement about complete monotonicity.

THEOREM 11.2.10. *For no $a > 0$ is $|X|$, $X^2$ or $S_2$ a scale mixture of Gamma($a$) laws.*

*Proof.* By Lemma 11.2.8(iii) a mixture would make $g(x) = x^{1-a} f(x)$ completely monotone, with $f$ the density. A completely monotone $g \not\equiv 0$ is positive and log convex, so $g(x) \geq g(1) \exp\big((x-1) g'(1)/g(1)\big)$ for $x \geq 1$: it decays at most exponentially. For $v \geq 0$, dropping the negative terms of (1.2.3) and using $e^{2v} \geq 1$ gives

$$\Phi(v) \leq 4\pi^2 e^{9v/2} \exp(-\pi e^{2v}) \sum_{n \geq 1} n^4 e^{-\pi(n^2-1)},$$

a constant multiple of $e^{9v/2} \exp(-\pi e^{2v})$. The densities of $|X|$ and $X^2$ are $2\Phi(v)/\Xi(0)$ and $\Phi(\sqrt{w})/(\Xi(0)\sqrt{w})$, so $v^{1-a}\Phi(v)$ and $w^{1/2-a}\Phi(\sqrt{w})$ would have to be completely monotone, and both decay faster than any exponential. For $S_2$ the obstruction is at the origin. By Chebyshev's inequality in exponential form, for $\lambda > 0$, $\Pr(S_2 \leq x) \leq e^{\lambda x} \mathbb{E} e^{-\lambda S_2} = e^{\lambda x} \big(\sqrt{2\lambda}/\sinh\sqrt{2\lambda}\big)^2$, and $\lambda = 2/x^2$ gives

$$\Pr(S_2 \leq x) \leq \frac{16\, e^{-2/x}}{x^2 (1 - e^{-4/x})^2} \leq 17\, x^{-2} e^{-2/x}, \qquad 0 < x \leq 1.$$

A mixture would give, $g$ being nonincreasing and $g(1) > 0$, $f(x) \geq g(1) x^{a-1}$ and $\Pr(S_2 \leq x) \geq g(1) x^a/a$ for $0 < x \leq 1$, which is impossible as $x \downarrow 0$. □

The Hankel test sees the failure at low order. If $Y = \Theta\,\Gamma$ with $\Gamma$ a Gamma$(a)$ variable independent of $\Theta \geq 0$, then $\mathbb{E}Y^k = (a)_k \mathbb{E}\Theta^k$, so $\mu_k = \mathbb{E}Y^k/(a)_k$ must be a Stieltjes moment sequence, and both $[\mu_{i+j}]_{i,j<N}$ and $[\mu_{i+j+1}]_{i,j<N}$ must be positive semidefinite for every $N$. Table 11.8 gives the least order $N$ at which each fails to be positive definite, by Cholesky factorisation at 400 digits, with the same orders at 250. The moments of $X^2$ are $M_k/M_0$; those of $|X|$ come from a tanh sinh rule, since odd powers of $|v|$ are not smooth at the origin; those of $S_2$ follow from its cumulants. The table of [433] is the Hamburger row, the first matrix alone.

| $a$ | $\frac{1}{2}$ | 1 | $\frac{3}{2}$ | 2 | $\frac{5}{2}$ | 3 | 4 | 6 | 8 | 12 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\lvert X\rvert$, Stieltjes | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 4 | 4 | 5 |
| $\lvert X\rvert$, Hamburger | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 4 | 4 | 5 | 6 |
| $X^2$, Stieltjes | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 6 | 7 | 9 |
| $X^2$, Hamburger | 2 | 3 | 4 | 4 | 4 | 5 | 5 | 6 | 6 | 7 | 9 |
| $S_2$, both | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 4 | 6 | 12 |

Table 11.8: The first order $N$ at which the moments $\mu_k = \mathbb{E}Y^k/(a)_k$ fail the Hankel test for a Gamma$(a)$ scale mixture, for $Y = |X|$, $X^2$ and $S_2$: the Hamburger test uses $[\mu_{i+j}]_{i,j<N}$ alone, the Stieltjes test also $[\mu_{i+j+1}]_{i,j<N}$. Computed by `wigfree_gamma2.py`.

The theorem also settles which tool applies. Kristiansen proved Steutel's conjecture that scale mixtures of Gamma(2) laws are infinitely divisible [316]; since $x^{\alpha-2}e^{-x}$ is completely monotone for $\alpha \leq 2$, Lemma 11.2.8(iii) makes Gamma$(\alpha)$ itself a scale mixture of Gamma(2) laws, and the theorem covers scale mixtures of Gamma$(\alpha)$ laws for every $\alpha \leq 2$. By Theorem 11.2.10 it cannot be applied to $|X|$, $X^2$ or $S_2$. The Gamma(2) structure that is true is Williams' convolution, a GGC and hence infinitely divisible directly.

### 11.2.6 The heat flow and finite free convolution

Remark 8.4.4 identifies the forward heat flow on polynomials with finite free convolution by a Hermite polynomial and, as the degree grows, with free convolution by a semicircle law. The time is the variance of the semicircle law. For a real polynomial $p$ of degree $d \geq 2$ write $p_t = e^{-(t/2)D^2}p$, a finite sum, and

$$q_t(x) = e^{-(t/2)D^2}x^d = t^{d/2}H_d\big(x/\sqrt{t}\big), \qquad t > 0,$$

with $q_0 = x^d$ and $H_d = e^{-D^2/2}x^d$ the Hermite polynomial of Remark 8.4.4, whose zeros are real. The time $t$ is twice the time of Section 8.4.1, where $\partial_t H_t = -\partial_z^2 H_t$.

Proposition 11.2.11. *Let $p$ be a real polynomial of degree $d \geq 2$ and $s, t \geq 0$.*

*(i)* $p_t = p \boxplus_d q_t$.

*(ii)* $q_s \boxplus_d q_t = q_{s+t}$.

*(iii)* *If $p$ has only real roots, then so has $p_t$, and the largest root of $p_t$ is at most the largest root of $p$ plus $\sqrt{t}\,h_d$, where $h_d$ is the largest zero of $H_d$.*

*(iv)* *While the roots $z_1(t), \dots, z_d(t)$ of $p_t$ are simple, $\dot z_k = \sum_{j\neq k}(z_k - z_j)^{-1}$.*

*Proof.* (i) In operator form, $p \boxplus_d q = \hat p(D)\hat q(D)x^d$ when $p = \hat p(D)x^d$ and $q = \hat q(D)x^d$ [370, Definition 1.1 and (2)]. For $q = q_t$ the operator $\hat q(D)$ acts on polynomials of degree at most $d$ as $e^{-(t/2)D^2}$, and operators in $D$ commute, so $p \boxplus_d q_t = e^{-(t/2)D^2}\hat p(D)x^d = p_t$. (ii) is (i) for $p = q_s$, since $e^{-(t/2)D^2}e^{-(s/2)D^2} = e^{-((s+t)/2)D^2}$; it is the identity $a^d H_d(x/a) \boxplus_d b^d H_d(x/b) = c^d H_d(x/c)$, $c^2 = a^2 + b^2$, of [370, Section 1.2]. (iii) The roots of $q_t$ are $\sqrt{t}$ times those of $H_d$, and by Walsh's theorem [370, Theorem 1.3] the convolution $p \boxplus_d q$ of two polynomials of degree $d$ with only real roots has only real roots, the largest at most the sum of the largest roots of $p$ and $q$. (iv) Differentiate $p_t(z_k(t)) = 0$ in $t$ and use $\partial_t p_t = -\frac12 p_t''$: $\dot z_k = \frac12 p_t''(z_k)/p_t'(z_k)$. Writing $p_t = (x - z_k)r$ gives $p_t'(z_k) = r(z_k)$ and $p_t''(z_k) = 2r'(z_k)$, so that $\dot z_k = r'(z_k)/r(z_k) = \sum_{j\neq k}(z_k - z_j)^{-1}$. □

For a conjugate pair $x \pm iy$, $y > 0$, the partner contributes $-1/(2y)$ to $\dot y$, and each real root $r$ contributes $\operatorname{Im}(x + iy - r)^{-1} = -y/((x-r)^2 + y^2) < 0$: the pair is drawn to the axis by its partner and pulled further by every real root. On its own, $(x-1)^2 + 0.64$ becomes $(x-1)^2 + 0.64 - t$, which reaches the axis at $t = 0.64$. The example of the note adds four real roots,

$$p(x) = (x+3)(x+1)\big(x - \tfrac12\big)(x-2)\big((x-1)^2 + 0.64\big).$$

The discriminant of $p_t$ in $x$ is a polynomial of degree 15 in $t$ whose only root in $(0, 2)$ is $t^* = 0.3203291143987806$. Before $t^*$ the roots are distinct, so the pair stays off the axis, and since $p_t = e^{-((t-t^*)/2)D^2}p_{t^*}$ for $t > t^*$, part (iii) keeps all roots real afterwards: the pair meets the axis at $t^*$, at the point 1.25743, in about half the time it needs on its own. The program `wigfree_heat.py` computes the discriminant exactly.

The Hermite polynomial is the finite free counterpart of the semicircle law: $H_d$ is the expected characteristic polynomial of a symmetric Gaussian matrix [370, Section 1.2], and the roots of $H_d(\sqrt{d}\,x)$, the roots of $q_{1/d}$, are close to the quantiles $x_k$ of $\sigma_1$ given by $\sigma_1([-2, x_k]) = (k - \frac12)/d$. The largest deviation from the quantiles is at the edge, and its products with $d^{2/3}$, from 0.3906 to 0.4962 for $d = 20$ to 320, grow slowly, as they should at the edge scale $d^{-2/3}$ of the extreme zeros. In the limit $d \to \infty$ finite free convolution becomes free convolution [10], [370, Section 1.2]: if the root distributions of polynomials $p_d$ with only real roots tend to a law $\mu$ in moments, those of $p_d \boxplus_d q_{t/d}$ tend to $\mu \boxplus \sigma_t$.

For $H_t$ the identification is a heuristic. The motion of the zeros is the same, up to the factor two in the time (Section 8.4.1), but $H_t$ has infinitely many zeros,

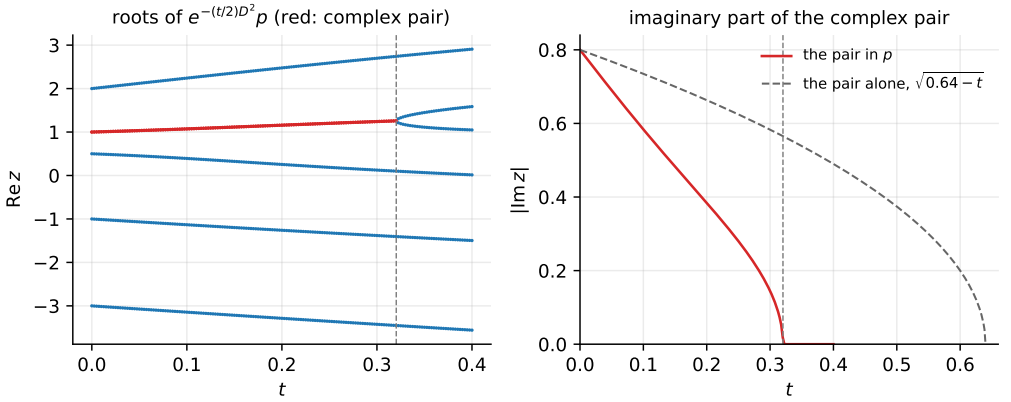



Figure 11.4: The flow $e^{-(t/2)D^2}p$ for $p$ with the roots $-3$, $-1$, $\frac{1}{2}$, $2$ and $1 \pm 0.8i$. Left: the real parts of the six roots for $0 \le t \le 0.4$; the complex pair has one real part until it meets the axis at $t^* = 0.3203$ (dashed), where its two roots split. Right: the imaginary part of the pair, against $\sqrt{0.64 - t}$ for the pair on its own, which reaches the axis at $t = 0.64$. Computed by `wigfree_heat.py` and drawn by `wigfree_plots.py`.

with a density that grows logarithmically, while the limit theorem concerns the distribution of all $d$ roots of a polynomial after scaling; for $H_t$ it would have to be applied to windows of zeros, with the zeros outside a window acting as an external field, and no theorem of this kind is known. De Bruijn's theorem that real zeros of $H_t$ stay real forward in time [112] is the counterpart of (iii), proved directly. In Thorin terms the forward flow is the path of clocks $R_t$ of Section 8.4.1, in the time of $H_t$ (Theorem 7.2.16).

The Jensen polynomials of Section 13.4.1 meet the Hermite polynomials directly. Griffin, Ono, Rolen and Zagier proved that for each degree $d$ the Jensen polynomials $J_{d,n}$, after an affine change of variable and a normalisation depending on $n$, tend to the Hermite polynomial of degree $d$ as $n \to \infty$, and this is how they obtain real zeros for all large shifts [159]. In the language of this subsection the Jensen polynomials of large shift are asymptotically finite free Gaussian; Problem 12.1 asks for an exact finite free structure at every shift.

These limits see the zeros only through their distribution. Moving $o(d)$ of the $d$ roots changes no limit law, so a single nonreal pair is invisible to free convolution limits, and these limits cannot decide RH by themselves [433].

## 11.3 The Dedekind eta analogue at the basepoint one

The $L$ function of the real character $\chi_{12}$ modulo 12, to which Section 8.5 carried the pipeline of the book, has no pole. Its completed function is entire and is a

Mellin transform of the Dedekind eta function, whose product formula makes the kernel positive. We repeat for $L(s,\chi_{12})$ the calculations that Chapter 9 made for $\zeta$ at the basepoint one.

The scripts are in the folder `basepoint_one/eta` of the archive.

### 11.3.1 The completed $L$ function and its Thorin measure

We use the completed function $\xi(s,\chi_{12}) = (12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12})$, its kernel $K$ and the variable $Y$ with density $K/\xi(\frac12,\chi_{12})$ of Section 8.5.1; the function satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ and is entire. We computed its zeros on the critical line by sign changes of $\xi(\frac12+it,\chi_{12})e^{\pi t/4}$ with `zeros12.py`, finding 297 below height 340; Section 8.5.1 lists the first of them. In this section $b_\alpha = (\xi'/\xi)(\alpha,\chi_{12})$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_{12})$, and $\varphi_\alpha$, $T_1$, $\widetilde{T}$, $\Pi$, $A$, $A_\alpha$, $Q$ and $a_k = -(\rho_k-\frac12)^2$ are formed with $\xi(s,\chi_{12})$ and its zeros $\rho=\beta+i\gamma$ in place of $\xi$. So is the tilted eta variable $V_\alpha$, the Esscher tilt of $Y$ by $e^{(\alpha-\frac12)Y}$, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s,\chi_{12})/\xi(\alpha,\chi_{12})$. The constants we need, computed by `base.py`, are

$$L(1,\chi_{12}) = \frac{\log(2+\sqrt3)}{\sqrt3} = 0.760346, \qquad L'(1,\chi_{12}) = 0.362495,$$

$$b_1 = \frac{\xi'}{\xi}(1,\chi_{12}) = \frac12\log\frac{12}{\pi} + \frac12\psi\Big(\frac12\Big) + \frac{L'}{L}(1,\chi_{12}) = \sum_\rho \frac1\rho = 0.1650833123,$$

where $\psi$ is the digamma function. The Taylor coefficients of $\xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ give the power sums $p_1 = \sum_k a_k^{-1} = 0.1664618$ and $p_2 = \sum_k a_k^{-2} = 0.0055985$, which are $\sum_{\gamma>0}\gamma^{-2}$ and $\sum_{\gamma>0}\gamma^{-4}$ under GRH; $p_1$ is the mean of the centre clock.

Proposition 11.3.1. *The function $\varphi_1(u) = \xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$ is the Laplace transform of a GGC with no drift, with Thorin density in $\theta=\sqrt{z}$*

$$\frac1\pi v_1(\theta) = \frac1\pi\,\mathrm{Re}\,\frac{\xi'}{\xi}(1+i\theta,\chi_{12}) = \frac1\pi\sum_\rho \frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2} > 0,$$

*the sum over the nontrivial zeros $\rho=\beta+i\gamma$ of $L(s,\chi_{12})$ in conjugate pairs. Positivity is unconditional because $L(1+it,\chi_{12})\neq0$.*

*Proof.* The proof is that of Proposition 9.1.2, with the Hadamard product of the entire function $\xi(s,\chi_{12})$ in place of that of $\xi$. □

The minimum on $[0,60]$ is at the origin, $v_1(0)/\pi = b_1/\pi = 0.0525477$, and the cumulative mass tracks the zero count: $\int_0^{60} v_1/\pi = 35.81$ against $N_{\chi_{12}}(60) = 36$, where $N_{\chi_{12}}(T)$ counts the zeros with $0<\gamma\le T$ as in Section 8.5.1. Table 11.9, computed by `base.py`, compares direct values with the Poisson sum over the 297 computed zeros; the nearly constant difference of $1.11\times10^{-3}$ is the contribution of the zeros above height 340. Figure 11.5 compares the density with that of $\zeta$: on

this window the zeros of $L(s,\chi_{12})$ are more than twice as dense, and its density is correspondingly higher. As for $\zeta$, $\varphi_1(u) = 1 - b_1\sqrt{u} + O(u)$, so $T_1$ has infinite mean and $P(T_1 > x) \sim b_1(\pi x)^{-1/2} = 0.09314\, x^{-1/2}$.

| $\theta$ | $v_1(\theta)/\pi$ from $\xi'/\xi(\cdot,\chi_{12})$ | from 297 zeros, $\beta = \frac{1}{2}$ |
|---|---|---|
| 0 | 0.0525477 | 0.0514359 |
| 3.8046276 | 0.6889076 | 0.6877958 |
| 5 | 0.1852757 | 0.1841637 |
| 10 | 0.2694467 | 0.2683341 |
| 30 | 0.7445558 | 0.7434362 |
| 60 | 0.7298854 | 0.7287413 |

Table 11.9: The Thorin density $v_1/\pi$ of $\xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$, computed from $\xi'/\xi(\cdot,\chi_{12})$ and from the Poisson sum over the 297 zeros of $L(s,\chi_{12})$ below height 340 with $\beta = \frac{1}{2}$.

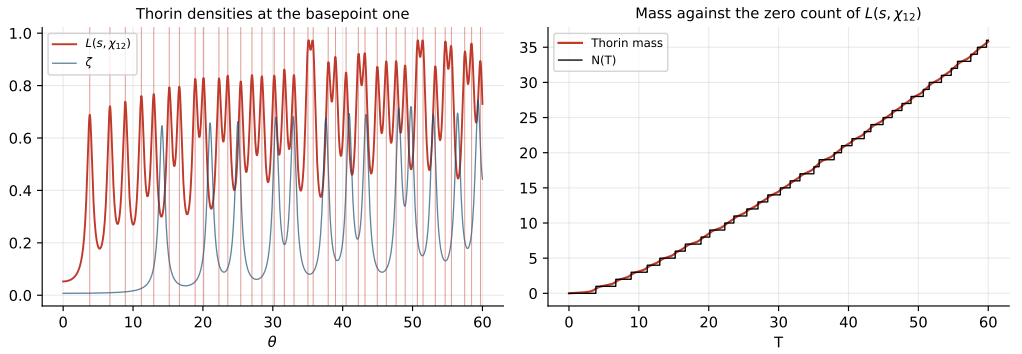



Figure 11.5: Left: the Thorin densities at the basepoint one for $L(s,\chi_{12})$ and for $\zeta$ on $[0,60]$, with the ordinates of the zeros of $L(s,\chi_{12})$ (vertical lines). Right: the cumulative Thorin mass for $L(s,\chi_{12})$ against its zero count $N_{\chi_{12}}(T)$, written $N(T)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`.

### 11.3.2 The Lévy form without a pole, at and below the basepoint one

Proposition 11.3.2. *For* $\operatorname{Re} s > 0$,

$$\log \frac{\xi(1+s,\chi_{12})}{\xi(1,\chi_{12})} = b_1 s + \int_0^\infty (e^{-sx} - 1 + sx)\,\Pi(dx),$$

*where, with* $\Lambda_{\chi_{12}}(n) = \chi_{12}(n)\Lambda(n)$,

$$\Pi(dx) = \frac{e^{-x}}{x(1-e^{-2x})}\,dx + \sum_{n\geq 2} \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\,\delta_{\log n}(dx),$$

*the prime sum converging conditionally, as a limit over* $n \leq X$.

*Proof.* Binet's formula [397, §5.9] for $\log\Gamma(\frac{1}{2}+\frac{s}{2})$, in the integrated form of the proof of Proposition 9.1.10 with $a = \frac{1}{2}$, gives the gamma density; the factor $(12/\pi)^{s/2}$ and the linear terms combine into $b_1$; and for $\operatorname{Re} s > 0$ the Euler product gives

$$\log L(1+s,\chi_{12}) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\,e^{-s\log n}.$$

The linear coefficient is $-\sum_n \Lambda_{\chi_{12}}(n)/n = \frac{L'}{L}(1,\chi_{12})$, which converges by the prime number theorem for arithmetic progressions [104, Chapter 20], in the form $\psi(y,\chi_{12}) = O(y\,e^{-c\sqrt{\log y}})$, where $\psi(y,\chi_{12}) = \sum_{n\leq y}\Lambda_{\chi_{12}}(n)$ is the Chebyshev function of $\chi_{12}$. □

The form differs from that of $\zeta$ in two ways. There is no pole, so no compensating measure: the prime atoms converge on their own at the basepoint one. And the measure is signed for a different reason. For $\zeta$ the gamma density minus the pole term $dx/x$ is negative beyond $x = 0.281$ and the prime atoms are positive; here the gamma density $e^{-x}/(x(1-e^{-2x}))$ is positive everywhere and the atoms carry the sign of $\chi_{12}(n)$ (Figure 11.6, left). Table 11.10, computed by `arith.py`, checks the form.

The tilted eta variable $V_1$, with moment generating function $\xi(1+s,\chi_{12})/\xi(1,\chi_{12})$, is not quasi infinitely divisible. Its positive and its negative atoms each have infinite total mass, since $\sum_{\chi_{12}(p)=1} 1/p$ and $\sum_{\chi_{12}(p)=-1} 1/p$ both diverge by Dirichlet's theorem, and they cancel only conditionally, so the argument of Proposition 11.1.3(3) applies with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, as the discussion after its proof explains. At every basepoint $\alpha > 1$, by contrast, $V_\alpha$ is quasi infinitely divisible: there the atoms of each sign have finite total mass, and the Euler product writes its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions.

The sine squared lemma in the form of Section 9.1.4, with $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta = \pi s$, converts the Lévy form, with $s = \sqrt{u}$, into

$$v_1(\theta) = b_1 + A(\theta) + Q(\theta),$$

with

$$A(\theta) = \tfrac{1}{2}\big[\operatorname{Re}\psi(\tfrac{1+i\theta}{2}) - \psi(\tfrac{1}{2})\big], \qquad Q(\theta) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big).$$

| $s$ | $X$ | $b_1 s$ | gamma piece | primes | total | exact |
|---|---|---|---|---|---|---|
| 0.5 | $10^5$ | 0.08254 | 0.121808 | −0.082565 | 0.121785 | 0.122329 |
| 0.5 | $10^7$ | 0.08254 | 0.121808 | −0.081931 | 0.122419 | 0.122329 |
| 1 | $10^7$ | 0.16508 | 0.409141 | −0.254214 | 0.320011 | 0.320099 |
| 2 | $10^7$ | 0.33017 | 1.270363 | −0.689198 | 0.911332 | 0.911001 |

Table 11.10: The Lévy form of Proposition 11.3.2 against $\log(\xi(1+s,\chi_{12})/\xi(1,\chi_{12}))$, with primes up to $X$.

The drift becomes the stable one half factor $e^{-b_1\sqrt{u}}$, the gamma factor the smooth positive growth $A$, and the primes the oscillating $Q$, now without any divergent partner. With primes to $10^7$ the formula gives 0.84633 at $\theta = 10$ against the exact 0.84649.11.6 (right) shows the three pieces: the prime piece ranges over both signs, and the positivity of the sum comes from the drift and the gamma factor, as Section 8.5.3 found at the basepoint two.

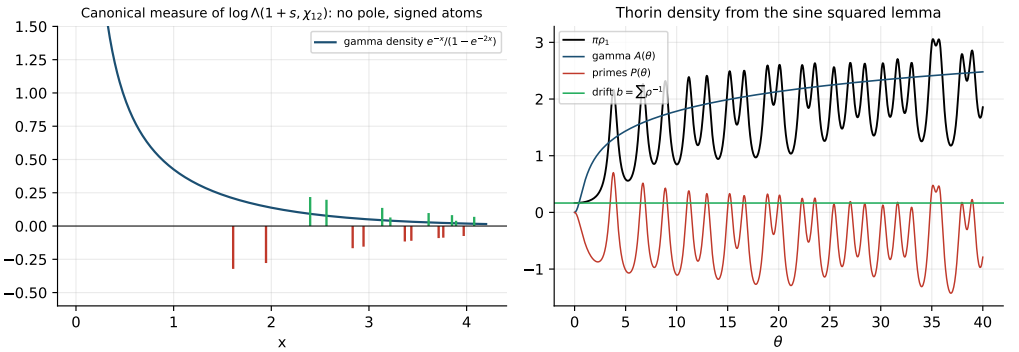



Figure 11.6: Left: the canonical measure $x\Pi(dx)$ of $\log\xi(1+s,\chi_{12})$, written with $\Lambda(1+s,\chi_{12})$ in the plot: the positive gamma density $e^{-x}/(1-e^{-2x})$ and the prime atoms $\Lambda_{\chi_{12}}(n)/n$ at $\log n$, of both signs, the positive ones drawn upward and the negative ones downward; there is no pole. Right: $v_1(\theta)$ on $[0,40]$, written $\pi\rho_1$ in the plot, as the sum of the drift $b_1$, written $b$, the gamma piece $A(\theta)$ and the prime piece $Q(\theta)$, written $P(\theta)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`.

Below the basepoint one, at $\alpha = 1-\varepsilon$, the converted Thorin density is $\pi^{-1}$ times

$$b_\alpha + A_\alpha(\theta) + \lim_{X\to\infty}\sum_{n\le X}\frac{\Lambda_{\chi_{12}}(n)}{n^\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big),$$

with $A_\alpha(\theta) = \frac12[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$. Exactly as for $\zeta$ in Proposition 9.1.13, the limit exists and gives the positive Thorin density $v_\alpha/\pi$ when $L(s,\chi_{12})$ has no zeros with $\operatorname{Re}s > \alpha'$ for some $\alpha' < \alpha$, and conversely its existence for $\operatorname{Re}s > 0$ in the

Lévy form forces the absence of zeros with $\operatorname{Re} s > \alpha$. The error at cutoff $X$ is given by the same explicit formula,

$$\sum_{\rho} \Big[ -\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)} \Big] + O(X^{-\alpha}\log X),$$

because $\psi(y,\chi_{12}) = -\sum_\rho y^\rho/\rho + O(\log y)$ has no main term [104, Chapter 19].

For $\zeta$ the prime piece and the pole piece each grow like $X^{1-\alpha}/(1-\alpha)$ and must cancel; for $L(s,\chi_{12})$ the prime piece stays bounded. Table 11.11, computed by `arith.py`, shows the errors, which decay like $X^{1/2-\alpha}$, and what remains after subtracting the explicit formula over the 297 computed zeros. The correction removes most of the error at $\theta = \gamma_1$; at $\theta = 10$, between zeros, it helps less, because 297 zeros are fewer than the 1700 used for $\zeta$ and the missing zeros above height 340 dominate what remains. At the centre and the first ordinate the sum grows like $\log X$: the atom $\delta_{\gamma_1}$ forming (Figure 11.7).

| $\alpha$ | $\theta$ | $X$ | prime piece | error | error after zero correction |
|---|---|---|---|---|---|
| 1 | $\gamma_1$ | $10^5$ | 0.6964 | $-5.55\times10^{-3}$ | $4.2\times10^{-6}$ |
| 1 | $\gamma_1$ | $10^7$ | 0.7016 | $-4.15\times10^{-4}$ | $2.5\times10^{-5}$ |
| 0.75 | $\gamma_1$ | $10^5$ | 2.0933 | $-2.13\times10^{-1}$ | $1.4\times10^{-4}$ |
| 0.75 | $\gamma_1$ | $10^7$ | 2.2471 | $-5.92\times10^{-2}$ | $1.4\times10^{-3}$ |
| 0.6 | $\gamma_1$ | $10^5$ | 4.8292 | $-3.10$ | $1.1\times10^{-3}$ |
| 0.6 | $\gamma_1$ | $10^7$ | 6.0670 | $-1.86$ | $1.6\times10^{-2}$ |
| 0.6 | 10 | $10^7$ | $-2.4525$ | $-5.68\times10^{-2}$ | $2.6\times10^{-2}$ |

Table 11.11: The sine squared formula below the basepoint one for $L(s,\chi_{12})$, with primes up to $X$: the prime piece, the error against $v_\alpha(\theta)$, and the error left after subtracting the explicit formula over the 297 computed zeros; $\gamma_1 = 3.8046276$.

### 11.3.3 The hitting time and its boundary

We computed the Brownian boundary $b(t)$ whose crossing time has the law of $T_1$ from the Volterra equation (8.2.1), with the distribution function from Talbot inversion for $t < 0.12$ and, under GRH, from the subordination representation of Theorem 9.2.2 beyond; its time scale is about seven times that of $\zeta$, the ratio of the means of the two centre clocks. In the right tail the exact boundary flattens at $-0.2747$, the Brownian form of the stable one half tail.

### 11.3.4 The two examples compared

Table 11.12 sets the two basepoint one examples side by side. Structurally the eta case is the cleaner of the two: below the basepoint one its prime sums stay bounded, so the computation is far better conditioned. Nothing essential becomes

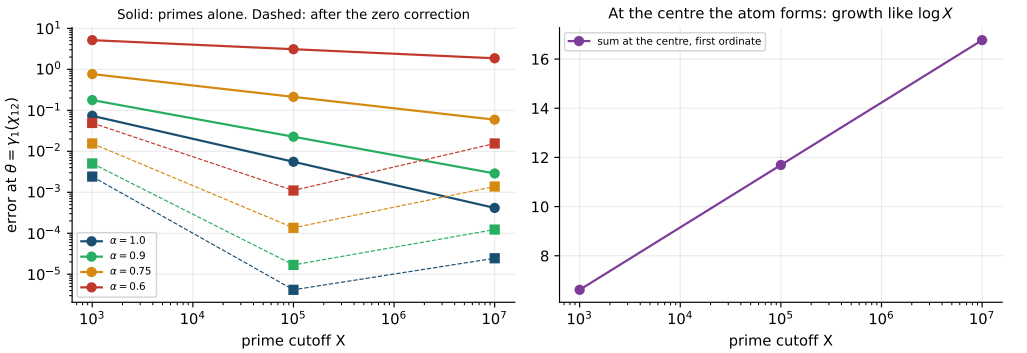



Figure 11.7: Left: absolute error at $\theta = \gamma_1 = 3.8046$ for $L(s, \chi_{12})$ against the prime cutoff $X$, for $\alpha = 1$, 0.9, 0.75 and 0.6, from primes alone (solid) and after the zero correction over 297 zeros (dashed). Right: at the centre and the first ordinate the sum grows like $\log X$, the atom of the GRH measure forming. Drawn by `figs.py` of `basepoint_one/eta`.

easier, however. The negativity that $\zeta$ concentrates in the pole piece is carried here by the signs of $\chi_{12}(p)$; and, under GRH for $\chi_{12}$, the probabilistic picture is the same, with every constant scaled by the denser zeros.

The comparison confirms the two conclusions that fix the shape of the path to RH. First, at the basepoint one the constructions for the reciprocal Xi function are, with two exceptions, unconditional and explicit. The exceptions are the Krein string representation of Corollary 9.2.3, which assumes RH and simple zeros, and the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$: Lemma 9.2.4 gives it for a boundary bounded at infinity, and the boundedness of the computed Brownian boundary is observed, not proved. Second, below the basepoint one each positive statement for $\zeta$ holds, up to the boundary case, exactly when no zero lies to the right of the line $\operatorname{Re} s = \alpha$ (Propositions 9.1.13 and 10.1.12, Theorems 9.2.5 and 10.1.3). The eta example shows that the pole changes how the prime sums converge, not where they stop.

## 11.4 Almost generalised gamma convolutions

Below the basepoint one, as Section 11.3.4 concluded, every positive object built so far from the shifted clock $T_\alpha$, whose Laplace transform is $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha+\sqrt{u})$, is equivalent, up to the boundary case, to the absence of zeros in the half plane $\operatorname{Re} s > \alpha$. These equivalences are restatements rather than reductions. Since no zero free half plane was known before 2026, and none is known to the left of the line $\operatorname{Re} s = \frac{7}{8}$ (Section 12.1.2), the question that remains is quantitative: how negative can the Thorin density be below the basepoint one? Zero density estimates answer it with no hypothesis: with those of Ingham, of Guth and Maynard and of Huxley,

| | $\zeta$ | $L(s,\chi_{12})$ |
|---|---|---|
| kernel | theta series | $\eta(ix)$, a positive product |
| pole | at $s=1$ | none |
| first zero | 14.134725 | 3.804628 |
| drift $b_1=\sum_\rho \rho^{-1}$ | 0.0230957 | 0.1650833 |
| tail constant $b_1/\sqrt{\pi}$ | 0.01303 | 0.09314 |
| mean of the centre clock $\sum_k a_k^{-1}$ | 0.0231050 | 0.1664618 |
| Lévy measure, continuous part | gamma minus pole, negative beyond 0.281 | gamma only, positive |
| Lévy measure, prime atoms | all positive | signs of $\chi_{12}(n)$ |
| compensation at $\alpha=1$ | primes against $dx/x$ (prime number theorem) | none needed |
| prime piece at $\alpha=0.6$, $X=10^7$, $\theta=\gamma_1$ | about 1537, cancelled by the pole | 6.07 |
| flat level of the exact boundary | $-0.19277$ | $-0.27471$ |
| $\mathbb{E}\,b(T_1)=b_1/\sqrt{2}$ | 0.016331 | 0.116732 |
| tangent approximation fails at | $t=0.091$ | $t=1.83$ |

Table 11.12: The two basepoint one examples side by side: the reciprocal Xi function of Chapter 9 and the reciprocal completed $L$ function of $\chi_{12}$ of this section.

every $T_\alpha$ with $\alpha>\frac12$ is a GGC up to a defect carried by the zeros to the right of $\alpha$, whose Cauchy kernels carry a vanishing fraction of the mass of the others, with a power saving; we call this property almost GGC.

With $\xi$ again in place of $\xi(\cdot,\chi_{12})$, we keep the notation of Sections 9.1 and 11.1: $v_\alpha(\theta)=\mathrm{Re}(\xi'/\xi)(\alpha+i\theta)$, and the Thorin density at the basepoint $\alpha$ in the chart $\theta=\sqrt{z}$ is $v_\alpha/\pi$, the density $\rho_{\alpha-1/2}$ of (1.2.7) at the resolution $\alpha-\frac12$; in this section the basepoint, not the resolution, is the index. $P_w(x)=w/(\pi(w^2+x^2))$ is the Poisson kernel of Section 1.2, $P_{-w}=-P_w$, and $P_0$ is the unit point mass, as in Proposition 3.3.1. A zero is $\rho=\beta+i\gamma$; $N(T)$ counts the zeros with $0<\gamma\le T$, and $N(\sigma,T)$ those among them with $\beta\ge\sigma$, so that $N(\sigma,T)\ge\nu(\sigma,T)$ in the notation of Proposition 3.4.2.

### 11.4.1 Mass bookkeeping and the almost GGC theorem

The Poisson form of Proposition 1.2.3 splits according to the side of the line $\operatorname{Re} s = \alpha$ on which each zero lies, $v_\alpha = v_\alpha^+ - v_\alpha^-$, with

$$\begin{aligned} \frac{v_\alpha^+(\theta)}{\pi} &= \sum_{\beta\le\alpha,\ \gamma>0} \big[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\big], \\ \frac{v_\alpha^-(\theta)}{\pi} &= \sum_{\beta>\alpha,\ \gamma>0} \big[P_{\beta-\alpha}(\theta-\gamma) + P_{\beta-\alpha}(\theta+\gamma)\big]. \end{aligned} \tag{11.4.1}$$

Both parts are positive, and a zero on the line $\operatorname{Re} s = \alpha$, if there is one, contributes a unit atom at its ordinate to the positive part. Each pair of kernels has mass one on $(0,\infty)$. A zero with $\beta > \alpha$ has the partner $1-\bar\rho = 1-\beta+i\gamma$, whose real part $1-\beta < \frac12 < \alpha$ puts it in the positive part. Write $m_\alpha^\pm(T) = \pi^{-1}\int_0^T v_\alpha^\pm(\theta)\,d\theta$ for the positive and negative masses on $[0,T]$.

Lemma 11.4.1 (Mass bookkeeping). *Let $\alpha > \frac12$, and suppose that $N(\alpha,t) \le t^{e+o(1)}$ as $t\to\infty$ for some $e<1$. Then*

$$m_\alpha^-(T) \le N(\alpha,2T) + O\big(T^{e-1+o(1)}\big), \qquad m_\alpha^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} + O(T).$$

*Proof.* A zero with $\beta>\alpha$ and $\gamma \le 2T$ contributes at most its total mass one. For a zero with $\gamma > 2T$ put $w = \beta-\alpha$, so that $0<w<\frac12$. Since $\int_0^T P_w(\theta+\gamma)\,d\theta = \int_{-T}^0 P_w(\theta-\gamma)\,d\theta$, its pair of kernels has mass on $[0,T]$ at most

$$\int_{-\infty}^{T} P_w(\theta-\gamma)\,d\theta \le \frac{w}{\pi(\gamma-T)} \le \frac{1}{\pi\gamma},$$

and $\sum_{\beta>\alpha,\ \gamma>2T}\gamma^{-1} = \int_{2T}^\infty t^{-1}dN(\alpha,t) \le \int_{2T}^\infty N(\alpha,t)t^{-2}dt \ll T^{e-1+o(1)}$. The positive part contains the kernels of all zeros with $\beta\le\alpha$, among them every zero on the critical line and every partner $1-\bar\rho$. Its mass on $[0,T]$ is therefore $N(T)$ up to the $N(\alpha,T) = O(T^{e+o(1)})$ zeros to the right of $\alpha$, whose kernels it lacks, and up to the tails of the kernels of the zeros near height $T$, which change it by $O(\log^2 T)$. Since $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, the second claim follows. □

Theorem 11.4.2 (Almost GGC). *For every $\frac12 < \alpha < 1$, with no hypothesis,*

$$\frac{m_\alpha^-(T)}{m_\alpha^+(T)} \le T^{e(\alpha)-1+o(1)}, \qquad e(\alpha) = \begin{cases} \dfrac{3(1-\alpha)}{2-\alpha}, & \frac12 < \alpha < \frac{7}{10}, \\[2mm] \dfrac{15(1-\alpha)}{3+5\alpha}, & \frac{7}{10} \le \alpha < 1, \end{cases}$$

*and $e(\alpha) < 1$ throughout, so the negative part $v_\alpha^-/\pi$ of (11.4.1), the Cauchy kernels of the zeros to the right of $\alpha$, carries on $[0,T]$ a vanishing fraction of the mass of*

*the positive part, with a power saving. For $\alpha \geq 1$ no zero lies to the right of $\alpha$, and $m_\alpha^- = 0$. Under the density hypothesis $e(\alpha) = 2(1-\alpha)$. RH holds if and only if $v_\alpha^-$ vanishes identically for every $\alpha > \frac{1}{2}$.*

*Proof.* Combine Lemma 11.4.1 with three zero density estimates: Ingham's $N(\sigma,T) \ll T^{3(1-\sigma)/(2-\sigma)}\log^5 T$ [284]; Guth and Maynard's $N(\sigma,T) \leq T^{15(1-\sigma)/(3+5\sigma)+o(1)}$ for $\frac{7}{10} \leq \sigma \leq \frac{4}{5}$ [166, Theorem 1.2]; and Huxley's $N(\sigma,T) \leq T^{3(1-\sigma)/(3\sigma-1)+o(1)}$ for $\sigma \geq \frac{3}{4}$ [279], whose exponent is at most $15(1-\sigma)/(3+5\sigma)$ when $\sigma \geq \frac{4}{5}$. Since $3(1-\alpha) < 2-\alpha$ exactly when $\alpha > \frac{1}{2}$, and $15(1-\alpha) < 3+5\alpha$ exactly when $\alpha > \frac{3}{5}$, the exponent is below one for every $\frac{1}{2} < \alpha < 1$. Finally, $v_\alpha^-$ vanishes identically exactly when no zero has $\beta > \alpha$, which is the GGC criterion of Theorem 1.2.2. □

Even the classical estimate of Bohr and Landau [53], $N(\sigma,T) = O(T)$ for fixed $\sigma > \frac{1}{2}$, gives a vanishing fraction $O(1/\log T)$; the zero density estimates improve this to a power of $T$. First, the theorem bounds the decomposition mass $m_\alpha^-(T)$, which overcounts the negative part $\int_0^T \max(-v_\alpha/\pi, 0)\,d\theta$ of the density itself, because the positive kernels of nearby zeros cancel much of each negative kernel. The partner of a zero already cancels part of it: with $\varepsilon = \alpha - \frac{1}{2}$ and $\delta = \beta - \frac{1}{2}$, twice the mass in Proposition 3.3.1(iii) gives the negative part of $P_{\alpha+\beta-1} - P_{\beta-\alpha} = P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ the mass $\frac{4}{\pi}\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - 1 = \frac{2}{\pi}\arcsin\frac{\varepsilon}{\delta}$, less than the mass one that the decomposition counts, and Section 11.4.4 shows the effect of the other zeros. Second, on the verified range the negative mass is essentially zero. All zeros with $0 < \gamma \leq T_0$ lie on the critical line [419], so for $T \leq T_0/2$ the only negative contributions come from zeros above $T_0$, and by the proof of Lemma 11.4.1 their total mass on $[0,T]$ is $O(T_0^{e(\alpha)-1+o(1)})$.

The left panel of Figure 11.9 draws the exponents. The saving $1 - e(\alpha)$ is small near the centre, where Ingham's exponent gives $(2\alpha-1)/(2-\alpha)$. Sharper exponents are known near $\alpha = 1$: the density hypothesis itself holds for $\alpha > \frac{25}{32}$ [62, p. 133], and Huxley's exponent [279] is 0.176 at $\alpha = 0.9$; those of the theorem suffice for what follows.

### 11.4.2 The canonical factorisation and explicit unconditional bounds

Proposition 11.4.3 (Canonical factorisation). *For every $\alpha > \frac{1}{2}$ and $u \geq 0$, $\varphi_\alpha(u) = \varphi_\alpha^+(u)\,D_\alpha(u)$, where*

$$\varphi_\alpha^+(u) = \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big)\frac{v_\alpha^+(\theta)}{\pi}\,d\theta \Big\},$$

$$D_\alpha(u) = \prod_{\beta>\alpha,\ \gamma>0} \frac{(\beta-\alpha)^2+\gamma^2}{(\sqrt{u}-\beta+\alpha)^2+\gamma^2};$$

*here $\varphi_\alpha^+$ is the Laplace transform of a GGC $T_\alpha^+$ with Thorin density $v_\alpha^+/\pi$ in the chart $\theta = \sqrt{z}$, and the product defining the defect $D_\alpha$ runs over the zeros to the right of the line $\operatorname{Re} s = \alpha$ and converges absolutely. The defect is identically one if and only if $\zeta$ has no zeros with $\operatorname{Re} s > \alpha$, and then $T_\alpha \stackrel{d}{=} T_\alpha^+$. Otherwise $D_\alpha(u) \to 0$ as $u \to \infty$, and $T_\alpha^+$ is not the sum of $T_\alpha$ and an independent nonnegative variable.*

*Proof.* Grouping the Hadamard product of $\xi$ in conjugate pairs, as in the proof of Proposition 1.2.3, gives for $u \geq 0$

$$\varphi_\alpha(u) = \prod_{\gamma>0} \frac{(\alpha-\beta)^2+\gamma^2}{(\alpha-\beta+\sqrt{u})^2+\gamma^2},$$

over the zeros with $\gamma > 0$, counted with multiplicity; each factor is $1 + O((u + \sqrt{u})\gamma^{-2})$ and $\sum \gamma^{-2} < \infty$, so the product converges absolutely. For $\beta < \alpha$, with $w = \alpha - \beta > 0$, the factor is $(w^2+\gamma^2)/((w+\sqrt{u})^2+\gamma^2) = \exp\{-\int_0^\infty \log(1 + u/\theta^2)[P_w(\theta-\gamma) + P_w(\theta+\gamma)]\, d\theta\}$, the Laplace transform of a GGC whose Thorin density is the pair of Cauchy kernels of width $w$ at $\pm\gamma$ (the computation after Theorem 10.1.2); for $\beta = \alpha$ it is $\gamma^2/(\gamma^2+u)$, an exponential law, whose Thorin measure is the atom at $\gamma$. The partial products over the zeros with $\beta \leq \alpha$ are therefore Laplace transforms of GGCs, they converge for every $u \geq 0$ to a limit continuous at the origin, and by the closure theorem of Section 4.1.1 the limit is the Laplace transform of a GGC, with Thorin density $v_\alpha^+/\pi$. The remaining factors form $D_\alpha$. If a zero lies to the right of $\alpha$, put $w = \beta - \alpha \in (0, \frac{1}{2})$; its factor tends to zero as $u \to \infty$, and every factor of $D_\alpha$ is at most one for $u \geq 1$, because $(\sqrt{u} - w)^2 \geq w^2$ when $\sqrt{u} \geq 2w$. So $D_\alpha(u) \to 0$. If $T_\alpha^+$ were $T_\alpha + S$ with $S \geq 0$ independent of $T_\alpha$, then $\mathbb{E}e^{-uS} = \varphi_\alpha^+(u)/\varphi_\alpha(u) = 1/D_\alpha(u)$ would be unbounded, which is impossible. □

The proposition also says what the signed density $v_\alpha/\pi$ is. When every zero lies to the left of the line $\operatorname{Re} s = \alpha$ it is the Thorin density of $T_\alpha$ (Theorem 1.2.2); a zero on the line adds to the Thorin measure an atom at its ordinate, the term $P_0$ of (11.4.1), which the function $v_\alpha$ does not show. When a zero lies to the right of the line, $T_\alpha$ is not a GGC and $v_\alpha/\pi$ is not a signed Thorin density for it in any sense: the negative kernel $-P_w$ of such a zero integrates to the factor $((w+\sqrt{u})^2 + \gamma^2)/(w^2+\gamma^2)$, while the zero contributes $(w^2+\gamma^2)/((\sqrt{u}-w)^2+\gamma^2)$ to $\varphi_\alpha$. The bookkeeping of Theorem 11.4.2 applies to the other reformulations in Chapters 9 to 11. In the normal scale mixture of Theorem 10.1.3 it bounds the negative part of the candidate EGGC density $v_\alpha/\pi$; in Wiener's theorem (Proposition 10.1.12) it bounds the proportion of frequencies at which the Fourier transform of the tilted Pólya kernel can vanish; and in Riesz's criterion (Section 10.1.8) it bounds the number of Mellin poles to the right of $\alpha/2$.

The pole factors can be written down. Let $U_\alpha^\circ$ denote the signed measure on $\theta > 0$ with density $v_\alpha/\pi$ and a unit atom at $\gamma$ for every zero $\beta + i\gamma$, $\gamma > 0$, with $\beta = \alpha$ (the term $P_0$ of (11.4.1)), counted with multiplicity.

THEOREM 11.4.4 (Signed Thorin representation). *For every $\alpha \ge \frac12$ and $u \ge 0$,*

$$\varphi_\alpha(u) = \exp\Big\{ -\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big) U_\alpha^\circ(d\theta)\Big\} \prod_{\beta>\alpha,\ \gamma>0} \Big|1-\frac{u}{(\rho-\alpha)^2}\Big|^{-2},$$

*the integral and the product converging absolutely. For a zero to the right of $\alpha$, with $p=(\rho-\alpha)^2=-c+id$, $c=\gamma^2-(\beta-\alpha)^2>0$ and $d=2\gamma(\beta-\alpha)>0$, the factor is*

$$\Big|1-\frac{u}{p}\Big|^{-2} = \frac{|p|^2}{(u+c)^2+d^2} = |p|^2\int_0^\infty e^{-ut}\,\frac{e^{-ct}\sin dt}{d}\,dt,$$

*the Laplace transform of a damped sinusoid, which changes sign, so the factor is not completely monotone.*

*Proof.* Write $f_w(u)=(w^2+\gamma^2)/((w+\sqrt{u})^2+\gamma^2)$. By the Hadamard product, as in the proof of Proposition 11.4.3, $\varphi_\alpha(u)=\prod_{\gamma>0} f_{\alpha-\beta}(u)$ over all zeros with $\gamma>0$, absolutely convergent. For $w\ge 0$ that proof gives $f_w(u)=\exp\{-\int_0^\infty \log(1+u/\theta^2)[P_w(\theta-\gamma)+P_w(\theta+\gamma)]\,d\theta\}$, with the atom at $\gamma$ for $w=0$. For $w=-\omega<0$ the kernels carry the sign $P_{-\omega}=-P_\omega$, so the same integral equals $f_\omega^{-1}$, and a direct computation gives $f_{-\omega}=f_\omega^{-1}\,|p|^2/|p-u|^2$ with $p=(\omega+i\gamma)^2$, since $|p-u|^2=((\sqrt{u}-\omega)^2+\gamma^2)((\sqrt{u}+\omega)^2+\gamma^2)$. Multiplying over the zeros gives the formula, once the integrals are summed under the integral sign: the integral of each pair of kernels against $\log(1+u/\theta^2)$ is $\pm\log f_{|w|}$, which is $O_\alpha((u+\sqrt{u})\gamma^{-2})$, and $\sum\gamma^{-2}<\infty$. Finally $|p-u|^2=(u+c)^2+d^2$, and $c>0$ because $\beta-\alpha<\frac12<\gamma_1$. □

The theorem needs no hypothesis on the zeros: the atoms carry the zeros on the line $\operatorname{Re} s=\alpha$, which is why the case $\alpha=\frac12$ is included. It is the identity behind Proposition 11.4.3, which combines the pole factors with the negative kernels they cancel into $D_\alpha$. At $\alpha=\frac12$ every term is explicit.

COROLLARY 11.4.5 (The critical line). *Unconditionally, $U_{1/2}^\circ=\sum_{\beta=1/2,\ \gamma>0}\delta_\gamma\ge 0$ and*

$$\varphi_{1/2}(u) = \prod_{\beta=1/2,\ \gamma>0}\frac{\gamma^2}{\gamma^2+u}\prod_{\beta>1/2,\ \gamma>0}\Big|1-\frac{u}{(\rho-\frac12)^2}\Big|^{-2}.$$

*Proof.* A zero $\rho$ with $\beta=\frac12+\delta$, $\delta>0$, and its partner $1-\bar\rho$ have the same ordinate, and their kernels $P_{-\delta}+P_\delta$ cancel, so $v_{1/2}=0$ away from the atoms (the term by term form in the proof of Proposition 1.2.3, which holds at $\alpha=\frac12$ away from the ordinates). The atoms come from the zeros with $\beta=\frac12$, each with the factor $f_0=\gamma^2/(\gamma^2+u)$; apply Theorem 11.4.4. □

So on the critical line positivity of $U_{1/2}^\circ$ costs nothing, and RH is the statement that the product of damped sinusoids is empty. For $\frac12<\alpha<\sup_\rho\operatorname{Re}\rho$ the negative kernels of Proposition 11.4.3 are present, and whether they can be

outweighed on the single line $\operatorname{Re} s = \alpha$ is open, except near a zero, where they cannot (Remark 1.2.4).

THEOREM 11.4.6 (Explicit bounds from verification). *Let $T_0 = 3 \times 10^{12}$, below which all zeros lie on the critical line [419], and let $\Sigma_0 = \sum_{\gamma > T_0} \gamma^{-2}$, the sum over all zeros above $T_0$, counted with multiplicity. By the Riemann von Mangoldt formula with the explicit error term of Hasanalizade, Shen and Wong [176], recalled in the proof of Proposition 9.1.11, $\Sigma_0$ differs from $\frac{\log(T_0/2\pi)+1}{2\pi T_0} = 1.4797 \times 10^{-12}$ by less than $2 \times 10^{-24}$, so $\Sigma_0 < 1.48 \times 10^{-12}$. Then for every $\alpha > \frac{1}{2}$ and every $u \geq 0$:*

1. *$|\log D_\alpha(u)| \leq (u + \sqrt{u})\,\Sigma_0$, so that*

$$e^{-1.5\times 10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u) \leq \mathbb{E}e^{-uT_\alpha} \leq e^{1.5\times 10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u);$$

2. *the negative part of the density satisfies $v_\alpha^-(\theta)/\pi \leq 2\Sigma_0/\pi < 9.43 \times 10^{-13}$ for $0 \leq \theta \leq T_0/2$;*

3. *consequently $v_\alpha(\theta) > 0$ for all $0 \leq \theta \leq T_0/2$ whenever $\alpha - \frac{1}{2} > 6 \times 10^{-10}$.*

*Proof.* Every zero to the right of $\alpha$ has $\gamma > T_0$ and $\beta > \frac{1}{2}$. The zeros above $T_0$ off the critical line come in pairs $\beta + i\gamma$, $1 - \beta + i\gamma$ of equal multiplicity, by $\xi(s) = \xi(1-s)$ and $\xi(\bar{s}) = \overline{\xi(s)}$, and at most one member of each pair lies to the right of $\alpha$, so $\sum_{\beta>\alpha,\ \gamma>0} \gamma^{-2} \leq \Sigma_0/2$. (1) For a zero to the right of $\alpha$ put $w = \beta - \alpha \in (0, \frac{1}{2})$ and $x = (u - 2w\sqrt{u})/(w^2+\gamma^2)$, so that its factor in $D_\alpha$ is $1/(1+x)$. Since $u - 2w\sqrt{u} \geq -w^2$, we have $x \geq -w^2/\gamma^2 > -\frac{1}{2}$, and $|x| \leq (u + \sqrt{u})/\gamma^2$; for $x \geq -\frac{1}{2}$, $|\log(1+x)| \leq 2|x|$. Hence the logarithm of its factor is at most $2(u + \sqrt{u})/\gamma^2$ in absolute value, and summing gives $|\log D_\alpha(u)| \leq (u + \sqrt{u})\Sigma_0$; the inequalities for $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha^+(u) D_\alpha(u)$ follow. (2) For $\theta \leq T_0/2$ and $\gamma > T_0$, $P_w(\theta - \gamma) + P_w(\theta + \gamma) \leq \frac{2w}{\pi(\gamma-\theta)^2} \leq \frac{4}{\pi\gamma^2}$, and summing over the zeros to the right of $\alpha$ gives at most $\frac{4}{\pi} \cdot \frac{\Sigma_0}{2}$. (3) Write $a = \alpha - \frac{1}{2}$. If $a \geq \frac{1}{2}$, then $\alpha \geq 1$, no zero lies to the right of $\alpha$, and $v_\alpha = v_\alpha^+ > 0$; so let $6 \times 10^{-10} < a < \frac{1}{2}$. The zeros below $T_0$ lie on the line and contribute to $v_\alpha^+/\pi$ the kernels $P_a(\theta - \gamma)$, and all other kernels of $v_\alpha^+$ are positive, so $v_\alpha^+(\theta)/\pi \geq \frac{a}{\pi(a^2+\gamma_1^2)}$ at every $\theta$ within $\gamma_1$ of an ordinate below $T_0$. Every $\theta \in [0, T_0/2]$ is such a point. On $[0, \gamma_1]$ the ordinate $\gamma_1$ serves. On $[\gamma_1, 2000]$ one of the first 1800 ordinates serves: they are the certified zeros of the proof of Proposition 9.1.11, they reach height 2304.3, and consecutive ones differ by less than $6.8874 < 2\gamma_1$. For $2000 \leq \theta \leq T_0/2$, the count of that proof, with $H = 14$ in place of 15, shows that the window $(\theta - 14, \theta + 14]$ holds at least $n(\theta) = \frac{14}{\pi}\log\frac{\theta-14}{2\pi} - E(\theta-14) - E(\theta+14)$ zeros; while $\theta + 14 \leq X_0$, $n$ increases and $n(2000) > 20$, and for $\theta + 14 > X_0$, $n(\theta) > \frac{14}{\pi}\log\frac{\theta-14}{2\pi} - 0.6\log(\theta+14) > 80$, so the window holds a zero, which lies within $14 < \gamma_1$ of $\theta$ and, since $\theta + 14 < T_0$, on the line. The script `explicit_bounds.py`, with output `explicit_bounds.json`,

recomputes the 1800 zeros and checks these inequalities in ball arithmetic. Finally, since $a < \frac{1}{2}$,

$$\frac{a}{\pi(a^2+\gamma_1^2)} > \frac{a}{\pi(\frac{1}{4}+\gamma_1^2)} > \frac{6\times 10^{-10}}{\pi(\frac{1}{4}+\gamma_1^2)} > 9.5\times 10^{-13},$$

which is more than the bound $2\Sigma_0/\pi < 9.43\times 10^{-13}$ of (2) on $v_\alpha^-/\pi$, so $v_\alpha = v_\alpha^+ - v_\alpha^- > 0$ on $[0, T_0/2]$. □

Theorem 11.4.6 restates the verification of RH to height $3\times 10^{12}$ in Thorin form. Part (3) of Theorem 10.1.3 is the same statement for the characteristic function of the normal scale mixture, with a remainder of relative size $\log T_0/T_0$, and the paragraph after its proof bounds the negative part of $v_\alpha/\pi$ for $\theta \le 10^9$ in the same way as part (2).

Part (3) of Theorem 11.4.6 treats every zero above $T_0$ as if its kernel could be negative wherever the density is evaluated. As the first author's note on the Dirichlet beta function observes [425], grouping each such zero with its reflection in the critical line does better, because the sum of the two kernels of an off line pair is negative only near its ordinate.

Corollary 11.4.7 (Positivity on the verified window). *Let* $T_0 = 3\times 10^{12}$ *and* $V_\xi = (\log\xi)''(\frac{1}{2}) = 0.04620998623$.

1. *For every* $\alpha > \frac{1}{2}$, $v_\alpha(\theta) > 0$ *for* $|\theta| \le T_0 - \frac{1}{2}$.

2. *For every* $\alpha \in (\frac{1}{2}, 7.35]$, $v_\alpha(\theta) > v_\alpha(0)$ *for* $0 < |\theta| \le T_0 - \frac{1+\sqrt{2}}{2}$, *and* $0 < v_\alpha(0) \le (\alpha - \frac{1}{2})V_\xi$.

*Proof.* (1) Put $\varepsilon = \alpha - \frac{1}{2}$. By the Hadamard product, as in Proposition 1.2.3, $v_\alpha(\theta)$ is the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$, counted with multiplicity, of $(\alpha-\beta)/((\alpha-\beta)^2+(\theta-\gamma)^2)$. A zero on the critical line contributes $\pi P_\varepsilon(\theta-\gamma) > 0$. The zeros off the line come in pairs $\frac{1}{2}\pm\delta+i\gamma$ with $0<\delta<\frac{1}{2}$ and equal multiplicities, and a pair contributes $\pi(P_{\varepsilon+\delta}+P_{\varepsilon-\delta})(\theta-\gamma)$, which is nonnegative if $\delta \le \varepsilon$ and, if $\delta > \varepsilon$, negative exactly where $|\theta-\gamma| < \sqrt{\delta^2-\varepsilon^2}$, by the proof of Proposition 3.3.1(ii); and $\sqrt{\delta^2-\varepsilon^2} < \delta < \frac{1}{2}$. Every zero off the line has $|\gamma| > T_0$ [419], so for $|\theta| \le T_0 - \frac{1}{2}$ every pair has $|\theta-\gamma| > \frac{1}{2}$ and contributes a nonnegative term, while the zero $\frac{1}{2}+i\gamma_1$ contributes a positive one.

(2) This is Theorem 9.1.18(i) and (ii) for $\xi$, with $\alpha_{\max} = 7.35$ and $H_0 = T_0$ (Table 9.4). □

Remark 11.4.8. The corollary does not touch the defect of part (1) of Theorem 11.4.6: a zero above $T_0$ to the right of the line $\operatorname{Re} s = \alpha$ stops $T_\alpha$ from being a GGC whatever $v_\alpha$ does on the window (Proposition 11.4.3), and near its own ordinate it can make $v_\alpha$ negative (Section 11.4.3).

The proof of part (1) gives, with no hypothesis, $|\log D_\alpha(u)| \le 2(u + \sqrt{u})\sum_{\beta>\alpha,\ \gamma>0}\gamma^{-2}$. With the verification height and the trivial count the sum is at most $\Sigma_0/2$, with $\Sigma_0 < 1.48\times10^{-12}$. The zero density estimates of Section 11.4.1 [284, 166, 279] improve this to $\sum_{\beta>\alpha}\gamma^{-2} \le \int_{T_0}^{\infty} t^{-2}dN(\alpha,t) \ll T_0^{e(\alpha)-2+o(1)}$, smaller by a further power of $T_0$ since $e(\alpha) < 1$.

### 11.4.3 Where negativity can come from: a cancellation lemma

The counting bound treats every zero to the right of $\alpha$ as if its kernel were entirely negative. Pointwise, a negative kernel is cancelled wherever the rest of the density exceeds it, and the kernel of width $w$ has height at most $1/(\pi w)$.

Lemma 11.4.9 (Cancellation). *Let $\rho = \beta + i\gamma$ be a zero with $\beta > \alpha$, and let $m > 0$ be a lower bound on an interval $I$ for the rest of the density, $v_\alpha/\pi + P_{\beta-\alpha}(\cdot - \gamma)$. If $\beta - \alpha \ge 1/(\pi m)$, the kernel of $\rho$ cannot make $v_\alpha$ negative on $I$.*

*Proof.* On $I$, $v_\alpha/\pi \ge m - \max_x P_{\beta-\alpha}(x) = m - 1/(\pi(\beta-\alpha)) \ge 0$. □

The lemma has a heuristic consequence. At height $\gamma$ the density $v_\alpha/\pi$ averages $\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, the density of the ordinates, and when $\alpha - \frac12$ is large compared with the mean spacing $2\pi/\log\gamma$ it stays close to that average. Taking this average for $m$ in the lemma, a zero to the right of $\alpha$ can then produce negativity only if

$$\beta - \alpha < \frac{2\pi}{\pi\log(\gamma/2\pi)} = \frac{2}{\log(\gamma/2\pi)}.$$

Heuristically, then, positivity of the Thorin density at the basepoint $\alpha$ can be broken at height $\gamma$ only by zeros in the thin strip $\alpha < \beta < \alpha + 2/\log(\gamma/2\pi)$. The local visibility condition of Section 5.2.2, which concerns small resolutions, has the same scale $1/\log\gamma$. The heuristic proves nothing, because a rigorous lower bound for the background near an arbitrary height would itself control $\zeta'/\zeta$ in the strip. It suggests combining zero density estimates for thin strips with lower bounds for $\operatorname{Re}\xi'/\xi$ on average.

We tested the lemma at the heights 300 and $10^5$ with `high.py`. The script computes the background $v_{0.6}/\pi$ from $\xi'/\xi$ on a window of width 8 and adds the kernels of a single hypothetical pair of zeros at $\beta + i\gamma_0$ and $1 - \beta + i\gamma_0$, with $\gamma_0$ the centre of the window (Figure 11.8, middle and right). At height 300 the background has mean 0.51 and minimum 0.069; at height $10^5$ it has mean 1.59, close to the density of ordinates $\frac{1}{2\pi}\log\frac{10^5}{2\pi} = 1.54$, and minimum 0.34. Table 11.13 shows the result. At both heights zeros with $\beta - \alpha \le 0.2$ produce negativity and zeros with $\beta - \alpha \ge 0.3$ are absorbed. At height $10^5$ the heuristic threshold $2/\log(10^5/2\pi) = 0.21$ is close to the observed one, while the rigorous guarantee of Lemma 11.4.9 with the window minimum, $1/(\pi m) = 0.93$, is conservative. For a fixed width the negative dip is deeper at the greater height, because the kernel is

the same while the local minimum of the background near the hypothetical zero differs.

| | height 300 | | height $10^5$ | |
|---|---|---|---|---|
| $\beta$ | min of $v_{0.6}/\pi$ | negative mass | min of $v_{0.6}/\pi$ | negative mass |
| 0.62 | −13.54 | 0.521 | −13.97 | 0.567 |
| 0.65 | −4.16 | 0.299 | −4.59 | 0.365 |
| 0.70 | −1.24 | 0.121 | −1.62 | 0.193 |
| 0.75 | −0.41 | 0.041 | −0.72 | 0.097 |
| 0.80 | −0.10 | 0.008 | −0.32 | 0.041 |
| 0.90 | 0.072 | 0 | 0.023 | 0 |
| 0.95 | 0.075 | 0 | 0.100 | 0 |

Table 11.13: A single hypothetical pair of zeros at real parts $\beta$ and $1-\beta$ inserted into the density $v_{0.6}/\pi$, computed from $\xi'/\xi$ on a window of width 8 at the heights 300 and $10^5$: the minimum of the density and the negative mass $\int \max(-v_{0.6}/\pi, 0)$ on the window. Computed by `high.py`.

### 11.4.4 Hypothetical off line zeros in the computed density

No off line zero is known, so the negative part cannot be observed. To see the mechanism, `almost.py` inserts fifteen hypothetical quadruples of zeros $0.75 \pm i\gamma_j$, $0.25 \pm i\gamma_j$, at the heights $\gamma_j = 20\, j^{5/3}$ below 2000, into the Poisson form of Proposition 1.2.3 together with the first 1700 ordinates, taken on the line; the zeros above height 2198 are omitted. The count of the hypothetical zeros grows like $T^{0.6}$, as a zero density estimate with $e = 0.6$ would allow. Table 11.14 compares the negative mass $\int_0^T \max(-v_\alpha/\pi, 0)\, d\theta$ of the resulting density with its positive mass and with the counting bound $N_{\text{off}}(2T)$ of Lemma 11.4.1, the number of hypothetical zeros with ordinate at most $2T$.

| $\alpha$ | $T$ | negative mass | positive mass | fraction | counting bound $N_{\text{off}}(2T)$ |
|---|---|---|---|---|---|
| 0.6 | 250 | 0.842 | 108.8 | $7.7 \times 10^{-3}$ | 6 |
| 0.6 | 1000 | 1.655 | 650.5 | $2.5 \times 10^{-3}$ | 15 |
| 0.6 | 2000 | 2.278 | 1519.2 | $1.5 \times 10^{-3}$ | 15 |
| 0.7 | 250 | 2.130 | 110.1 | $1.9 \times 10^{-2}$ | 6 |
| 0.7 | 1000 | 4.815 | 653.6 | $7.4 \times 10^{-3}$ | 15 |
| 0.7 | 2000 | 6.938 | 1523.8 | $4.6 \times 10^{-3}$ | 15 |
| 0.8 | any | 0 | | 0 | 0 |

Table 11.14: Negative and positive mass of the density $v_\alpha/\pi$ on $[0, T]$ with fifteen hypothetical quadruples of zeros at real part 0.75. Computed by `almost.py`.

Three features are visible. The negative mass is well below the counting bound, because each negative kernel is partly cancelled by its partner and by the positive

kernels of the neighbouring zeros on the line: at $\alpha = 0.6$ the negative kernel has width 0.15 and height $1/(0.15\pi) = 2.1$, and most of it is cancelled, while at $\alpha = 0.7$ the width is 0.05 and the kernel, of height 6.4, appears as a sharp negative spike that takes the density to $-5.39$ at the height 292.40 (Figure 11.9; the minimum is computed by `almost_dip.py`). The fraction decreases with $T$ at the rate the counting predicts. And at $\alpha = 0.8$, to the right of the hypothetical zeros, the density is positive everywhere, as Theorem 1.2.2 requires: a zero with real part $\beta$ contributes a positive kernel at every basepoint $\alpha > \beta$.

A zero is most visible from basepoints just to its left. The left panel of Figure 11.8, computed by `more2.py`, shows the negative mass on $[0, 2000]$ as a function of $\alpha$ for the same fifteen heights and three real parts $\beta = 0.6$, 0.75 and 0.9. As $\alpha$ increases towards $\beta$ the negative kernels narrow, cancellation fails, and the negative mass climbs towards the number of hypothetical zeros, reaching 7.55 at $\alpha = 0.85$ for $\beta = 0.9$. At $\alpha = \beta$ the zeros lie on the line of evaluation and the negative mass drops to zero. This is the Thorin counterpart of the fact that the translates of the tilted Pólya kernel fail to be complete exactly on the line through the zero (Proposition 10.1.12).

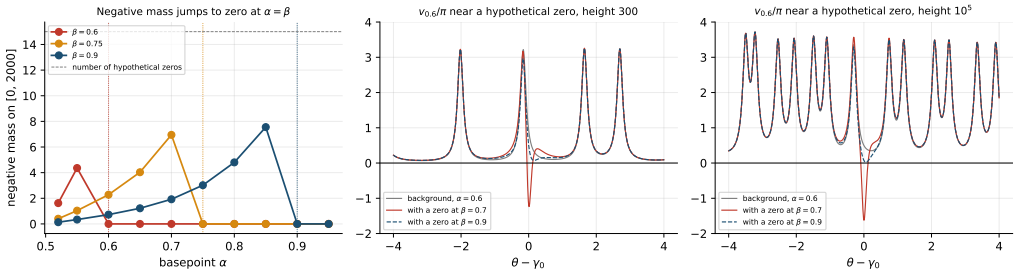



Figure 11.8: Left: negative mass of the density on $[0, 2000]$ against the basepoint $\alpha$ for fifteen hypothetical zeros at real part $\beta$ (`more2.py`). Middle and right: the density $v_{0.6}/\pi$ near the heights 300 and $10^5$, alone and with a hypothetical pair of zeros at $\beta = 0.7$ or 0.9 at the centre of the window (`high.py`). Drawn by `figs_almost.py`.

### 11.4.5 Positivity margins and the circular unitary ensemble

By Lemma 11.4.9, a zero to the right of $\alpha$ at height $\gamma$ can make the density negative only if its distance to the line $\operatorname{Re} s = \alpha$ is below $1/(\pi m)$, where $m$ is the local minimum of the rest of the density. The distribution of the local minima, the positivity margins, therefore determines how much room RH has at each height. The minima occur between consecutive ordinates. After unfolding the ordinates to unit mean spacing, $x_n = \vartheta(\gamma_n)/\pi + 1$ with $\vartheta$ the Riemann Siegel theta function,

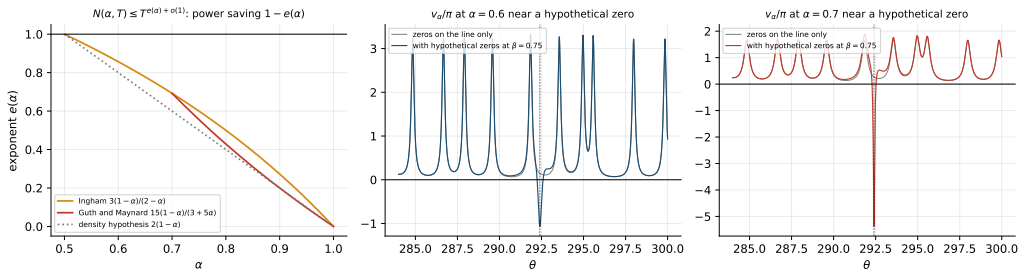



Figure 11.9: Left: the zero density exponents of Theorem 11.4.2 and the region where they give a power saving; the exponent of Guth and Maynard is drawn on $[\frac{7}{10}, 1)$, where it holds by their estimate and, above $\frac{4}{5}$, by Huxley's [279]. Middle and right: the density $v_\alpha/\pi$ at $\alpha = 0.6$ and 0.7 near the hypothetical off line zero at height 292.4 (dotted), with and without the hypothetical zeros, evaluated as in `almost.py` with step 0.005. Drawn by `figs_almost.py`.

as in Section 5.4.2, the density at the resolution of $\kappa$ mean spacings is

$$\bar{\rho}_\kappa(x) = \sum_n P_\kappa(x - x_n) = \sum_n \frac{\kappa}{\pi(\kappa^2 + (x - x_n)^2)},$$

whose mean is one.

The script `sens.py` evaluates $\bar{\rho}_\kappa$ at the 1600 midpoints between consecutive unfolded ordinates below height 2198, leaving out 50 mean spacings at each end, and the same statistic at about 3600 midpoints between the eigenvalues of twelve Haar distributed unitary matrices of size 400, unfolded and extended periodically. The circular unitary ensemble (CUE) of these matrices models the local statistics of the zeros. Table 11.15 shows the result. The medians agree to 1% at every resolution. In the lower tail the zeros have slightly larger margins than the CUE, a finite height effect in the expected direction, since the low zeros are more rigid than the limit of large matrices.

| $\kappa$ | zeta, 0.1% | CUE, 0.1% | zeta, 1% | CUE, 1% | zeta, median | CUE, median |
|---|---|---|---|---|---|---|
| 0.1 | 0.1075 | 0.0919 | 0.1177 | 0.1095 | 0.3349 | 0.3343 |
| 0.25 | 0.2617 | 0.2178 | 0.2849 | 0.2659 | 0.7116 | 0.7092 |
| 0.5 | 0.4803 | 0.4047 | 0.5150 | 0.4645 | 0.9739 | 0.9796 |
| 1 | 0.7477 | 0.6552 | 0.7730 | 0.7313 | 1.0264 | 1.0240 |

Table 11.15: Quantiles of the normalised Thorin density $\bar{\rho}_\kappa$ at the midpoints between consecutive unfolded ordinates, for the zeros below height 2198 and for CUE eigenvalues. Computed by `sens.py`.

The next step is to repeat the comparison with Odlyzko's tables of zeros near the $10^{12}$th zero [391] and near the $10^{21}$st and $10^{22}$nd zeros [395, 396], where the

finite height corrections are smaller, and to fit the lower tail of the margin distribution. The CUE lower tail is governed by large gaps, whose probability decays like a Gaussian in the gap length, so the smallest margin among the first $N$ midpoints should decrease only like a power of $\sqrt{\log N}$. Converted through the cancellation lemma, this predicts the width, as a function of the height, of the strip to the right of each basepoint in which an off line zero could hide without being cancelled. That is the quantitative input that question (1) of Section 11.4.7 needs.

### 11.4.6 Moments of zeta, the Barnes function and zero density

The strength of Theorem 11.4.2 is set by the zero density exponents, which rest on mean value estimates on the critical line, Ingham's on the fourth moment of $\zeta$. Keating and Snaith [301] modelled $\zeta(\frac{1}{2}+it)$ by the characteristic polynomial $Z_N(\theta)=\det(I-Ae^{-i\theta})$ of a Haar distributed unitary matrix $A$ of size $N$, a matrix of the circular unitary ensemble of Section 11.4.5, with $N$ in the role of $\log(T/2\pi)$. By analogy with the exact formula

$$\mathbb{E}|Z_N(\theta)|^{2k}=\prod_{j=1}^{N}\frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2}\sim\frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\,N^{k^2},\qquad N\to\infty,$$

they conjectured that the moments of $\zeta$ grow in the same way, with $\log(T/2\pi)$ in place of $N$ and an arithmetic factor $a_k$ in front (Section 11.4.6.1). Here $\mathsf{G}$ is the Barnes $G$ function, $\mathsf{G}(1+z)=\Gamma(z)\mathsf{G}(z)$ with $\mathsf{G}(1)=1$, written in sans serif to keep $G$ for the clock of Section 1.2. Nikeghbali and Yor [387] gave the random matrix factor a probabilistic meaning in GGC language: the Mellin transform of $|Z_N|$ and the Barnes function are expressed through products and sums of independent gamma, beta and log gamma variables, and the reciprocal of $\mathsf{G}$ has a Lévy Khintchine type representation. The source of these representations is the decomposition of Bourgade, Hughes, Nikeghbali and Yor [61],

$$Z_N\overset{d}{=}\prod_{j=1}^{N}\Big(1+e^{i\omega_j}\sqrt{B_{1,j-1}}\Big),$$

with independent angles $\omega_j$ uniform on $[0,2\pi)$ and independent beta variables $B_{1,j-1}$ with parameters 1 and $j-1$, where $B_{1,0}=1$.

#### *11.4.6.1 The Keating and Snaith moments*

Write $I_k(T)=\frac{1}{T}\int_0^T|\zeta(\frac{1}{2}+it)|^{2k}dt$. The conjecture of Keating and Snaith [301], in the form of Conrey, Farmer, Keating, Rubinstein and Snaith [91], is

$$I_k(T)\sim a_k\,\frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\Big(\log\frac{T}{2\pi}\Big)^{k^2},\qquad a_k=\prod_p\Big(1-\frac{1}{p}\Big)^{k^2}\sum_{m\ge0}\Big(\frac{\Gamma(m+k)}{m!\,\Gamma(k)}\Big)^2p^{-m},$$

the product of an arithmetic factor from the primes and the random matrix factor. It is a theorem for $k = 1$ and $k = 2$ and open for $k \geq 3$. For $k = 1$ Hardy and Littlewood [173] proved the leading term, and Ingham [282] the mean square with its secondary term, $I_1(T) = \log \frac{T}{2\pi} + 2\gamma_E - 1 + o(1)$; for $k = 2$ Ingham [282] proved the asymptotic with the leading coefficient $\frac{1}{2\pi^2}$. The local factor of $a_k$ is $(1 - \frac{1}{p})^{k^2} {}_2F_1(k, k; 1; \frac{1}{p})$. Since ${}_2F_1(1, 1; 1; x) = (1 - x)^{-1}$ and ${}_2F_1(2, 2; 1; x) = (1 + x)(1 - x)^{-3}$, it equals 1 for $k = 1$ and $1 - p^{-2}$ for $k = 2$, so that $a_1 = 1$ and $a_2 = \prod_p (1 - p^{-2}) = 6/\pi^2$ exactly. The Euler product over the primes below $2 \times 10^5$ gives $a_3 = 0.0493218$ (`repro/ks_ak.py`); since the local factor for $k = 3$ is $1 - 9p^{-2} + O(p^{-3})$, the omitted primes lower it by a relative $9 \sum_{p > 2\times 10^5} p^{-2} \approx 4 \times 10^{-6}$, to about 0.0493217. With $\mathsf{G}(3)^2/\mathsf{G}(5) = \frac{1}{12}$ and $\mathsf{G}(4)^2/\mathsf{G}(7) = \frac{1}{8640}$ the leading coefficients are 1, $0.0506606 = \frac{1}{2\pi^2}$, which is Ingham's constant, and $5.709 \times 10^{-6}$, the value $42a_3/9!$ of [91].

Table 11.16, computed by `ks.py`, compares the moments with the leading terms. For $k = 1$ the full asymptotic is already accurate at $T = 1000$. For $k = 2$ and $k = 3$ the leading term is a small part of the moment at these heights, because the lower order terms of the polynomial in $\log T$ predicted in [91] dominate.

| $T$ | $I_1(T)$ | $\log \frac{T}{2\pi} + 2\gamma_E - 1$ | $I_2(T)$ | leading, $k = 2$ | $I_3(T)$ | leading, $k = 3$ |
|---|---|---|---|---|---|---|
| 125 | 3.062 | 3.145 | 28.0 | 4.05 | 396 | 0.11 |
| 250 | 3.869 | 3.838 | 53.1 | 9.33 | 1136 | 0.71 |
| 500 | 4.550 | 4.531 | 84.8 | 18.6 | 2632 | 3.36 |
| 1000 | 5.211 | 5.224 | 127.2 | 33.5 | 5309 | 12.6 |

Table 11.16: Moments of $\zeta$ on the critical line against the asymptotic of Hardy, Littlewood and Ingham for $k = 1$ and the leading Keating and Snaith terms for $k = 2$ and 3. Computed by `ks.py`.

### *11.4.6.2 The moments as exponential moments of the Thorin density*

The densities $v_\sigma$ are the derivatives of $\log|\xi|$ across the basepoints: $\frac{\partial}{\partial\sigma} \log|\xi(\sigma + it)| = \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it) = v_\sigma(t)$ wherever $\xi(\sigma + it) \neq 0$. Integrating from the critical line to the basepoint one, for every $t$ such that $\xi$ has no zero on the segment from $\frac{1}{2} + it$ to $1 + it$, which excludes only countably many $t$,

$$\log|\xi(\tfrac{1}{2} + it)| = \log|\xi(1 + it)| - \int_{1/2}^{1} v_a(t)\, da. \tag{11.4.2}$$

The moments concern $\zeta$, not $\xi$. With the gamma factor $h(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, so that $\xi = h\zeta$, the identity becomes

$$|\zeta(\tfrac{1}{2} + it)|^{2k} = |\zeta(1 + it)|^{2k} \Big|\frac{h(1 + it)}{h(\frac{1}{2} + it)}\Big|^{2k} \exp\Big\{ - 2k \int_{1/2}^{1} v_a(t)\, da\Big\},$$

and Stirling's formula gives $|h(1+it)/h(\frac{1}{2}+it)| \sim (t/2\pi)^{1/4}$ as $t \to \infty$. So the Keating and Snaith moments are exponential moments, over $t$ uniform on $[0,T]$, of the density $v_a(t)$ integrated across the basepoints from $\frac{1}{2}$ to 1, with the weight $|\zeta(1+it)|^{2k}|h(1+it)/h(\frac{1}{2}+it)|^{2k}$, which involves no zero of $\zeta$. In the same terms, RH, which is the positivity of $v_a$ for every $a > \frac{1}{2}$ (Lagarias's criterion, Remark 1.2.4), is the statement that $|\xi(\sigma+it)|$ increases with $\sigma \geq \frac{1}{2}$ for every $t$, a reformulation due to Sondow and Dumitrescu [482].

In the language of (11.4.2) the conjecture of Keating and Snaith is a large deviation statement for the integrated Thorin density. Selberg's central limit theorem [477] says that for $t$ uniform on $[T,2T]$ the variable $\log|\zeta(\frac{1}{2}+it)|$ is approximately Gaussian with variance $\frac{1}{2}\log\log T$. Gaussian exponential moments would give $(\log T)^{k^2}$, and the constant $a_k\mathsf{G}(1+k)^2/\mathsf{G}(1+2k)$ is the correction to them. On $[500,1000]$ the sample variance of $\log|\zeta(\frac{1}{2}+it)|$ is 1.53, against $\frac{1}{2}\log\log 750 = 0.95$. The difference is a constant term, not slow convergence: the random matrix model predicts the variance $\frac{1}{2}\log\log\frac{t}{2\pi} + c + o(1)$ with $c = \frac{1}{2}\big(\gamma_E + 1 - \sum_p \sum_{m\geq 2}(\frac{1}{m} - \frac{1}{m^2})p^{-m}\big) = 0.70$ [301], a conjecture, and this gives 1.48 on $[500,1000]$, close to the sample value.

### *11.4.6.3 What the moments give*

Ingham's exponent $3(1-\sigma)/(2-\sigma)$, which gives Theorem 11.4.2 its saving $(2\alpha-1)/(2-\alpha)$ for $\frac{1}{2} < \alpha < \frac{7}{10}$, rests on the fourth moment, $k=2$, proved by Ingham [282]. Bounds for all moments, that is the Lindelöf hypothesis in the form $\int_0^T |\zeta(\frac{1}{2}+it)|^{2k}dt \ll T^{1+\varepsilon}$ for every $k$, imply the density hypothesis $N(\sigma,T) \ll T^{2(1-\sigma)+\varepsilon}$, which would give the saving $2\alpha-1$, larger than that of Theorem 11.4.2 for $\frac{1}{2} < \alpha < \frac{9}{10}$. The Keating and Snaith asymptotics would give this and much more.

The GGC structure of Nikeghbali and Yor belongs to the random matrix model, not to $\zeta$: it explains the conjecture without proving it, and only the second and fourth moments of $\zeta$ are known asymptotically. And moments, even the Lindelöf hypothesis, bound the number of zeros off the line but do not exclude them.

## 11.4.7 From almost to exactly

Theorem 11.4.2 cannot be upgraded to exact positivity by improving the exponents, or by the moment bounds of Section 11.4.6. Even the density hypothesis, or the Lindelöf hypothesis which implies it, leaves room for a sparse set of zeros off the line, and a single off line zero $\beta+i\gamma$ stops $T_\alpha$ from being a GGC for every $\alpha$ in $[\frac{1}{2},\beta)$ (Theorem 1.2.2) and makes $v_\alpha$ negative near $\gamma$ for every $\alpha$ close enough to $\beta$ (Section 11.4.3). The step from "almost" to "exactly" is a step from counting zeros to excluding them.

Four facts about the programme say that this step needs input from outside it. The smoothing runs the wrong way: when no zero lies to the right of $\alpha$ the density at a basepoint $\alpha' > \alpha$ is the Poisson smoothing of the density at $\alpha$ by the kernel of width $\alpha' - \alpha$, and positivity propagates to the right; RH asks to propagate it to the left, which is deconvolution, and deconvolution does not preserve positivity. Arithmetic domination requires a zero free region: below the basepoint one the compensated Lévy form, which the sine squared lemma converts into the density, holds if $\psi(x) - x = O(x^{\alpha'})$ for some $\alpha' < \alpha$, and only if $\psi(x) - x = O(x^{\alpha+\delta})$ for every $\delta > 0$, where $\psi$ is Chebyshev's function (Proposition 9.1.13). Either condition is a zero free half plane, none was known for any $\alpha < 1$ before the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026, which came from outside the programme (Section 12.1.2), and removing the pole does not help: for $L(s, \chi_{12})$ the prime sums need no compensation, and the boundary is the same (Section 11.3.2). There is no margin: along the de Bruijn Newman flow the reciprocal is a GGC for $t \geq 0.2$ [419] (Theorem 7.2.16 and Section 8.4), RH is the statement at $t = 0$, and $\Lambda \geq 0$ [463], so a proof must establish a closed condition at the boundary itself. And stronger positivity conditions can be false: Conrey and Li showed that a positivity condition in de Branges' approach to RH fails [90], so any positivity statement stronger than RH should be tested numerically before it is pursued (Section 12.2.4).

In the Stieltjes form (iv) of Theorem 4.2.4 the difference between counting zeros and excluding them is explicit. Split $G'/G = S_{\mathrm{on}} + R_{\mathrm{off}}$, where $S_{\mathrm{on}}(u) = \sum (u + \gamma^2)^{-1}$ runs over the zeros $\frac{1}{2} + i\gamma$ on the line with $\gamma > 0$ and is a Stieltjes function with measure $U_{\mathrm{line}}$, and $R_{\mathrm{off}}(u) = \sum (u - (\rho - \frac{1}{2})^2)^{-1}$ runs over the zeros $\rho = \beta + i\gamma$ off the line with $\gamma > 0$. By Theorem 4.3.9, all the poles of $R_{\mathrm{off}}$ lie in a parabolic sliver $\mathcal{E}_{T_0}$ beside the negative axis, which begins at $-T_0^2 + T_0 + \frac{1}{4}$, and in the upper half plane outside it $\operatorname{Im} R_{\mathrm{off}} \leq 0$ and $\operatorname{Im}(G'/G) < 0$. The size of $R_{\mathrm{off}}$ away from the sliver needs no count of zeros: if $|u| \leq T_0^2/2$, a disc that misses $\mathcal{E}_{T_0}$, then every term has $|u - (\rho - \frac{1}{2})^2| \geq |\rho - \frac{1}{2}|^2 - |u| > \gamma^2/2$, since $\gamma > T_0$, so $|R_{\mathrm{off}}(u)| \leq 2\Sigma_0 < 3 \times 10^{-12}$ by Theorem 11.4.6.

The number of terms of $R_{\mathrm{off}}$, on the other hand, is not known to be small. Zero density estimates bound the number of those with $|\beta - \frac{1}{2}| \geq \sigma - \frac{1}{2}$ and $\gamma \leq T$ by $T^{e(\sigma)+o(1)}$ with $e(\sigma) < 1$, for each fixed $\sigma > \frac{1}{2}$ (Theorem 11.4.2), which is $o(T)$; but zeros off the line may approach the line as $\gamma$ grows, and for all of them together only a proportion is known. At most $(1 - \kappa + o(1))N(T)$ of the zeros up to height $T$ lie off the line, with $\kappa > \frac{5}{12}$ in the last published record [457] and $\kappa > 0.6725$, counting only simple zeros on the line, in two preprints of 2026 [3, 324]; that is not $o(T)$, nor even $o(N(T))$. No count, however strong, touches the Thorin condition, which requires $R_{\mathrm{off}} \equiv 0$, because a single pair puts a pole of $G'/G$ off the cut. Section 11.5.5 gives a test for arguments that claim to exclude such poles: since the de Bruijn Newman flow has nonreal zeros at every $t < 0$ [463], no property that the kernel $e^{tv^2/4}\Phi(v)$ keeps for some $t < 0$ can exclude them, so a proof must

use one that fails for every $t < 0$, and the Euler product and the explicit formula are the natural candidates.

The forms that the new input could take are those of Section 12.1. The first new theorem any of them would give is the GGC property of $T_\alpha$ at a single basepoint $\alpha < \frac{7}{8}$, that is $v_\alpha^- \equiv 0$, a zero free half plane beyond the one announced in 2026 (Section 12.1.2); positivity of $v_\alpha$ on the single line $\operatorname{Re} s = \alpha$ is not known to suffice (Remark 1.2.4).

Sections 11.4.2 and 11.4.3 make the remaining problem concrete. The growth of the defect is not part of it. For $u \geq 1$ every factor of $D_\alpha$ is at most one, and a factor with $\gamma \leq \sqrt{u}$ is at least $\gamma^2/(u+\gamma^2) \geq \gamma^2/(2u) \geq 1/u$, since $\gamma \geq \gamma_1 > \sqrt{2}$. Bounding the factors with $\gamma > \sqrt{u}$ as in the proof of Theorem 11.4.6 therefore gives, for $u \geq 1$,

$$\begin{aligned} |\log D_\alpha(u)| &\leq N(\alpha, \sqrt{u}) \log u + 2(u+\sqrt{u}) \sum_{\beta > \alpha,\ \gamma > \sqrt{u}} \gamma^{-2} \\ &\leq u^{e(\alpha)/2 + o(1)}, \qquad u \to \infty, \end{aligned}$$

by the exponents of Theorem 11.4.2. So bounds on the growth of the defect follow from zero density estimates alone, and what remains open is whether it is identically one. Four questions would each be a genuine step.

1. Cancellation on average: show that for fixed $\alpha > \frac{1}{2}$ the background density $v_\alpha/\pi$ is bounded below by $c \log \gamma$ outside a set of heights of density zero. With Lemma 11.4.9 this would confine any failure of positivity to zeros in the strips $\alpha < \beta < \alpha + O(1/\log \gamma)$. Remark 9.1.17 proves the lower bound for $v_\alpha/\pi$ itself, but its exceptional set contains a neighbourhood of the ordinate of every zero to the right of $\alpha$, which is where Lemma 11.4.9 would need it.

2. Thin strip density: bound the number of zeros with $\alpha < \beta < \alpha + 2/\log \gamma$ and $\gamma \leq T$ by $o(T^{1-\delta})$ for some $\delta > 0$, uniformly in $\alpha$. Combined with (1) this would give a pointwise almost positivity theorem.

3. Explicit defect: bound $\log D_\alpha(u)$ explicitly, as described at the end of Section 11.4.2.

4. Positivity at one basepoint: prove $D_\alpha \equiv 1$, that is $v_\alpha^- \equiv 0$, for a single $\alpha < \frac{7}{8}$, which is a zero free half plane beyond the one announced in 2026 (Section 12.1.2).

What the Thorin framework contributes is a precise target, the GGC property of one explicit clock $T_\alpha$ at one basepoint below $\frac{7}{8}$, together with computational tools that test candidate arguments against it: certified densities on windows, positivity margins calibrated against random matrix theory, and the explicit defect. Chapter 12 takes stock of the whole path, beginning with Section 12.1.

## 11.5 Pakes, the Lambert law and the Thorin condition

Anthony Pakes has worked on the objects that surround the Thorin condition: size biasing and the classical moment problem, the Laplace transforms of positive infinitely divisible laws, the Lambert function, and ladders of self decomposable laws. This section follows the first author's account of that work and of what it gives for the clock [434], verified, corrected where needed, and written in the notation of the book. In Pakes's language the question of Section 1.3 is whether $\log G$ is a Thorin Bernstein function, that is, a Bernstein function whose Lévy measure has the form $k(t)\,dt/t$ with $k$ completely monotone [473, Chapter 8], or equivalently the Laplace exponent of a GGC. By Theorem 4.2.4 this holds if and only if $(\log G)' = G'/G$ is a Stieltjes function, and if and only if RH holds.

What the section adds is a dictionary between Pakes's language and that of the book, a few unconditional facts about the clock, the solved Lambert problem as a template whose single failing step for the clock is RH itself, a list of routes that are closed, and one property, the GGC property of the variable of Biane, Pitman and Yor, that passes the filter of Section 11.5.5 wherever it has been tested, but fails for most of the real Dirichlet $L$ functions tested there, so that it does not detect the Euler product.

### 11.5.1 Two strands of Pakes's work

The first strand is the classical moment problem. For a random variable $X \geq 0$ with $0 < \mathbb{E}X < \infty$, the size biased variable $\widehat{X}$, length biased in Pakes's word, has the law $x\,P(X \in dx)/\mathbb{E}X$. Pakes and Khattree [409] characterise laws by in law equations linking $\widehat{X}$ and $X$, such as $X \stackrel{d}{=} \mathsf{U}\widehat{X}$ with $\mathsf{U}$ uniform on $(0,1)$ and independent of $\widehat{X}$, which holds exactly for the exponential laws. Pakes [400] studies size biasing for the laws with the moments of the log normal law, and in [401] he proves converse forms of the Carleman and Krein criteria and introduces size biasing and the stationary excess operation of order $r$; Berg [43] showed, correcting a remark in [401], that a law can be determinate in the sense of Stieltjes while its size biased and stationary excess versions of order one are not. Pakes, Hung and Wu [408] give criteria for the determination of a law by its moments, and Pakes [402] describes the structure of the Stieltjes classes of moment equivalent laws.

Size biasing meets infinite divisibility in Steutel's theorem: $X$ is infinitely divisible if and only if $\widehat{X} \stackrel{d}{=} X + A$ for some $A \geq 0$ independent of $X$, and the law of $A$ is then unique [487], [471, Theorem 51.1], [489]; in this form the theorem is [12, Theorem 11.1], and Pakes takes it for positive laws from Pakes, Sapatinas and Fosam [410]. In Laplace transforms, with $\Psi(u) = -\log \mathbb{E}e^{-uX}$, it reads $\mathbb{E}e^{-u\widehat{X}} = \mathbb{E}e^{-uX}\,\Psi'(u)/\Psi'(0)$, so that $A$ has the Laplace transform $\Psi'(u)/\Psi'(0)$. In [404] Pakes assumes the decomposition and asks when $A$ or $\widehat{X}$ is again infinitely

divisible; he shows that size biasing up to the fourth order preserves infinite divisibility for the Lambert law and for a compound Poisson law, and conjectures that it does so at every order.

For the laws of this book the moment problem is determinate. The moment generating function $\xi(\frac{1}{2}+z)/\xi(\frac{1}{2})$ of the Pólya density is entire (1.2.4). The clock has $\mathbb{E}e^{sT} < \infty$ exactly for $s < \gamma_1^2 = 199.7904548\ldots$: every zero has $|\rho - \frac{1}{2}| \geq |\operatorname{Im}\rho| \geq \gamma_1$, so $1/G$ is analytic in the disc $|u| < \gamma_1^2$ and has a pole at $-\gamma_1^2$, and by Landau's theorem [514, Chapter II, Theorem 5b] the abscissa of convergence is $-\gamma_1^2$. Under RH the measure $\nu = \sum_k a_k^{-1}\delta_{1/a_k}$ of the Stieltjes moments $m_n$ is supported in $[0, \gamma_1^{-2}]$, so its moment problem is a determinate Hausdorff problem (Section 5.1), while the Thorin measure $U_{\mathrm{RH}}$ itself has unbounded support and infinite mass.

The second strand concerns the Laplace transforms of positive infinitely divisible laws, and the Lambert function $\mathcal{W}$ (Section 11.5.3). Pakes [403] and, independently, Kalugin and Jeffrey [295] proved that $\mathcal{W}$ is a Bernstein function, from a formula for the derivatives of $\mathcal{W}$ that involves a sequence of polynomials [407, (8)]. So $e^{-\mathcal{W}(u)} = \mathcal{W}(u)/u$ is the Laplace transform of an infinitely divisible law, the Lambert law. In the same paper, by Pakes's own account [407, Section 2], a complex variable argument shows that $\mathcal{W}'$ is a Stieltjes function, so that $\mathcal{W}$ is a Thorin Bernstein function and the Lambert law is a GGC. Kalugin, Jeffrey, Corless and Borwein [297] obtained Stieltjes representations independently and showed that $\mathcal{W}$ is a complete Bernstein function; the Thorin representation of $\mathcal{W}$ is the theorem proved below. Pakes gave two further proofs that $\mathcal{W}$ is Thorin Bernstein: from Nuttall's integral for the coefficients $n^n/n!$ of $\mathcal{W}'$, which yields an explicit quantile function of the Thorin measure [405], and from continuous state branching processes, one of whose martingale limits has the cumulant function $\mathcal{W}$ [406], which also gives the self decomposability of the Lambert law.

In [407] Pakes identifies the background driving Lévy process of the Lambert law: it is compound Poisson with unit rate and jump density $\int ze^{-xz}\,U_{\mathcal{W}}(dz)$, where $U_{\mathcal{W}}$ is the Thorin measure of $\mathcal{W}$. With $T_{\mathcal{W}}$ a variable with the Lambert law and $J$ a jump of the driving process, this gives the in law relations $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}}+J)$ and $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, with $\mathsf{U}$ uniform on $(0,1)$, $\mathcal{S}$ the stationary excess operator and the factors on the right independent; and if both relations hold for a variable $X$ and some $J \geq 0$ of finite mean $\mu > 0$, then $X \stackrel{d}{=} \mu T_{\mathcal{W}}$ [407, Theorem 1] (Remark 11.5.5). He embeds the Lambert law in a family of GGCs whose Laplace transforms solve Lambert's trinomial equation [407, Theorem 6], which gives one more proof that $\mathcal{W}$ is Thorin Bernstein [407, Corollary 2], and he introduces BDLP ladders: from an infinitely divisible $X_0 \geq 0$ with $\mathbb{E}(\log(1+X_0))^n < \infty$ for every $n$, the law of $X_{n+1}$ is the self decomposable law driven by the Lévy process of $X_n$ [407, Section 5]. In his indexing $X_1$ is self decomposable; this is a unit shift from Urbanik's classes $L_n$ used in Section 11.5.4, in which $L_0$ is the class of self decomposable laws [507, 470, 44]. Along a ladder he finds the size biased

decomposition $\widehat{X}_{n+1} \stackrel{d}{=} X_{n+1} + A_n$, with $A_n$ of density proportional to the tail of the Lévy measure of $X_n$ [407, (53)], and this is the form in which Section 11.5.2 applies Steutel's theorem to the clock.

### 11.5.2 The Thorin question in Pakes's form

The Thorin question asks whether $\log G$ is a Thorin Bernstein function, that is, whether $(\log G)' = G'/G$ is a Stieltjes function (Theorem 4.2.4); then the Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, with no drift (Proposition 1.2.1), and Stieltjes inversion (1.3.2) recovers it from the boundary values of $G'/G$ on the cut, or, in integrated form, from those of $\log G$ (Section 1.3.1). With no hypothesis, $\log G$ is a Bernstein function with Lévy measure $W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$ is the heat trace (Theorem 5.1.1); under RH, $W(t) = \int e^{-tz}\, U_{\mathrm{RH}}(dz)$, the counterpart for the clock of the formula that gives the canonical function of the Lambert law as the Laplace transform of the density of its Thorin measure (Corollary 11.5.4; [407, (12)]).

The Bernstein property is weaker than the Thorin Bernstein property, but it is not blind to the zeros off the line. For $\log G$ it says that $W(t) \geq 0$ for every $t > 0$, and that would exclude, for instance, a zero $\rho = \frac{1}{2} + \delta + i\gamma$ off the line below all the other zeros: if $\gamma^2 - \delta^2$ were smaller than $\operatorname{Re} a_k$ for every other $a_k$, which is $\gamma_k^2$ for a zero on the line, the four zeros $\rho$, $1-\rho$, $\bar\rho$, $1-\bar\rho$ would contribute $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, which dominates as $t \to \infty$ and changes sign. The property holds with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000), so it cannot decide RH.

Size biasing gives the Thorin question a form with one variable. The size biased clock $\widehat{T}$ has the law $t\,P(T \in dt)/\mathbb{E}T$, and $\mathbb{E}T = m_0 = \sum_k a_k^{-1} = 0.0231049931\ldots$, the first Stieltjes moment of Section 5.1.

Proposition 11.5.1. *With no hypothesis, $\widehat{T} \stackrel{d}{=} T + A$, with $A$ independent of $T$ and of density $W(t)/m_0$ on $(0,\infty)$. RH holds if and only if $A$ is a mixture of exponential laws, and then $A \stackrel{d}{=} E/V$, with $E$ standard exponential and independent of $V$, and $V$ of law $m_0^{-1}\sum_{\gamma>0}\gamma^{-2}\delta_{\gamma^2}$, the ordinates counted with multiplicity.*

*Proof.* The Laplace transform of $\widehat{T}$ is

$$\frac{\mathbb{E}[Te^{-uT}]}{m_0} = -\frac{1}{m_0}\,\frac{d}{du}\,\frac{1}{G(u)} = \frac{1}{m_0}\,\frac{1}{G(u)}\,\frac{G'}{G}(u).$$

By Section 1.2, $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ for $u > 0$ with no hypothesis, and $W > 0$ (Theorem 5.1.1); letting $u \downarrow 0$, monotone convergence gives $\int_0^\infty W(t)\,dt = (G'/G)(0) = G'(0) = \sum_k a_k^{-1} = m_0$. So $W/m_0$ is a probability density with Laplace transform $(G'/G)(u)/m_0$, the Laplace transform of $\widehat{T}$ is the product of those of $T$ and of a variable $A$ with this density, and by the uniqueness of Laplace

transforms $\widehat{T} \stackrel{d}{=} T + A$ with $A$ independent of $T$. A variable $A > 0$ is a mixture of exponential laws, $A \stackrel{d}{=} E/V'$ with $V' > 0$ independent of $E$, exactly when it has a completely monotone density: the density of $E/V'$ is $\mathbb{E}[V'e^{-V't}]$, and conversely, by Bernstein's theorem, a completely monotone probability density is $\int_{[0,\infty)} e^{-vt}\,\sigma(dv)$ with $\int v^{-1}\sigma(dv) = 1$, which forces $\sigma(\{0\}) = 0$, and this is the density of $E/V'$ with $P(V' \in dv) = v^{-1}\sigma(dv)$. The density $W/m_0$ is completely monotone exactly when RH holds (Section 1.2 and Table 1.6), and then $W(t) = \sum_{\gamma>0} e^{-\gamma^2 t} = m_0\mathbb{E}[Ve^{-Vt}]$ with $V$ as stated. □

This is Steutel's decomposition of Section 11.5.1 with $\Psi = \log G$, and it is the identity that Pakes derives along BDLP ladders [407, (53)], here for the clock and the subordinator of Proposition 11.1.1 that drives it, whose Lévy density is $h = -W'$. Pakes's variable $A_n$ has the density $x \mapsto \int_x^\infty \omega_n(y)\,dy$, the tail of the Lévy density $\omega_n$ of the driving law, normalised by the mean $\int_0^\infty x\,\omega_n(x)\,dx$, and for the clock the tail is $\int_t^\infty h(x)\,dx = W(t)$ and the mean is $\int_0^\infty xh(x)\,dx = \int_0^\infty W(t)\,dt = m_0$. The variable $A$ has mean $\mathbb{E}A = m_1/m_0 = \operatorname{Var} T/\mathbb{E}T = 1.60885567 \times 10^{-3}$, from the Taylor coefficients of $\xi$ at $\frac{1}{2}$, with no zero entering. Under RH, $A$ is infinitely divisible, as a mixture of exponential laws [152, 486], and so is $\widehat{T} = T + A$; so under RH both parts of the question that Pakes asks in [404] have a positive answer for the clock. The weights $P(V = \gamma^2) = \gamma^{-2}/m_0$ of the first six ordinates are 0.21663, 0.09794, 0.06919, 0.04676, 0.03990 and 0.03064: the first six carry 0.50105, about a half. The first 1700 ordinates carry 0.978508 (`pakes_sizebias.py`).

Pakes's first strand, the moment problem, enters through the Taylor coefficients at the origin. For $|u| < \gamma_1^2$ and for $|u| < 1/e$ respectively,

$$(\log G)'(u) = \sum_{n\ge 0} m_n(-u)^n, \qquad \mathcal{W}'(u) = \sum_{n\ge 1} \frac{n^n}{n!}(-u)^{n-1},$$

with the Stieltjes moments $m_n = \sum_k a_k^{-n-1}$ of Section 5.1; the second series is the derivative of the Lagrange series of $\mathcal{W}$ [94, (3.1)]. Under RH, $m_{n-1} = \int z^{-n}\,U_{\mathrm{RH}}(dz)$ for $n \ge 1$, just as $n^n/n! = \int z^{-n}\,U_{\mathcal{W}}(dz)$ (Corollary 11.5.6): both coefficient sequences are then moment sequences of positive measures. For the clock the positive definiteness of every $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is equivalent to RH (Section 5.1); the number of negative eigenvalues of $\mathcal{H}_N$ does not decrease with $N$ and tends to the number of distinct zeros $\rho$ with $\operatorname{Re}\rho > \frac{1}{2}$ and $\operatorname{Im}\rho > 0$ (Theorem 5.1.7).

Li's criterion is a positivity condition on the same zeros, read on the Thorin functional through the test functions $\varphi_n$ of Section 8.3.2. It sees a zero off the line more slowly: the term of $\rho = \frac{1}{2} - d + i\gamma$, with $0 < d < \frac{1}{2}$, grows like $e^{nd/\gamma^2}$, so it can dominate only from $n$ of order $\gamma^2/d$, which exceeds $2\gamma^2$, while in the data of Section 8.4.3 the Hankel test often registers a pair at an order close to

$2k+2$, set by its index $k$, at a cost in precision that grows linearly with the order (Section 5.1).

### 11.5.3 The Lambert law, a Thorin problem solved

Let $\mathcal{W}$ be the principal branch of Lambert's function, the solution of $\mathcal{W}(u)e^{\mathcal{W}(u)} = u$ that is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real and increasing on $(-1/e, \infty)$, with $\mathcal{W}(0) = 0$, $\mathcal{W}(-1/e) = -1$ and $|\operatorname{Im} \mathcal{W}| < \pi$. Its Lagrange series $\mathcal{W}(u) = \sum_{n\geq 1}(-n)^{n-1}u^n/n!$ has radius $1/e$, and $\mathcal{W}(u) = \log u - \log\log u + o(1)$ as $u \to \infty$ [94, (3.1), (4.19)]. We write $T_{\mathcal{W}}$ for a variable with the Lambert law. The proof that it is a GGC is a template that fails for the clock at exactly one step.

Proposition 11.5.2 ([94, Section 4, (4.4)], [296, (10), (11)]). *For $0 < b < \pi$ put $r(b) = b\, e^{-b\cot b}/\sin b$. Then $r$ increases from $1/e$ to $\infty$ on $(0, \pi)$, and the boundary values of $\mathcal{W}$ on the cut, from the upper half plane, are $\mathcal{W}(-r(b)+i0) = -b\cot b + ib$.*

*Proof.* Put $w = \alpha + ib$ with $0 < b < \pi$ in $we^w = -x$, $x > 0$. The imaginary part gives $e^\alpha(\alpha \sin b + b\cos b) = 0$, so $\alpha = -b\cot b$, and then the real part gives $e^\alpha(\alpha\cos b - b\sin b) = -e^{-b\cot b}\, b/\sin b$, so that $x = r(b)$. Next, $(\log r)'(b) = 1/b - 2\cot b + b/\sin^2 b$ and $b\sin^2 b\,(\log r)'(b) = (b - \sin b\cos b)^2 + \sin^4 b > 0$, while $r(0+) = 1/e$ and $r(\pi-) = \infty$; so for each $x > 1/e$ there is exactly one point $w$ of the curve $\{-b\cot b + ib : 0 < b < \pi\}$ with $we^w = -x$, namely the point with $r(b) = x$. This curve bounds the range of the principal branch in the upper half plane [94, Section 4], and $\mathcal{W}$ is continuous on the cut from above, with values on it; so $\mathcal{W}(-r(b) + i0) = -b\cot b + ib$. □

Kalugin, Jeffrey and Corless obtain the next theorem from the Stieltjes representation of $\mathcal{W}'$ [296, Theorem 3.3] and the characterisation of Thorin Bernstein functions in [473, Chapter 8].

Theorem 11.5.3 (Kalugin, Jeffrey and Corless [296, Theorem 6.2], [298]). *Let $U_{\mathcal{W}}$ be the image of the uniform law $db/\pi$ on $(0, \pi)$ under $r$, a probability measure on $(1/e, \infty)$. Then for $u \in \mathbb{C} \setminus (-\infty, -1/e]$*

$$\mathcal{W}(u) = \int \log\Big(1 + \frac{u}{z}\Big)\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \log\Big(1 + \frac{u}{r(b)}\Big)\, db.$$

*Hence $\mathcal{W}$ is a Thorin Bernstein function, and $T_{\mathcal{W}}$ is a GGC with no drift and Thorin measure $U_{\mathcal{W}}$, of total mass one, whose distribution function is $U_{\mathcal{W}}((1/e, y]) = \pi^{-1}\operatorname{Im}\mathcal{W}(-y + i0)$ for $y > 1/e$.*

*Proof.* Let $\Psi(u)$ be the integral. Since $U_{\mathcal{W}}$ is carried by $(1/e, \infty)$, $\Psi$ is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real on $(-1/e, \infty)$, and $\Psi(\bar u) = \overline{\Psi(u)}$; so is $\mathcal{W}$. Let $x > 1/e$, and let $b(x) \in (0, \pi)$ be the point with $r(b(x)) = x$. As $\eta \downarrow 0$, $1 + (-x + i\eta)/z$ tends to a negative number from the upper half plane when $z < x$ and to a positive number when $z > x$, and $U_{\mathcal{W}}$ has no atoms, so by dominated convergence $\operatorname{Im}\Psi(-x + i0) =$

$\pi\, U_{\mathcal{W}}((1/e, x)) = b(x)$; by Proposition 11.5.2 also $\operatorname{Im} \mathcal{W}(-x + i0) = b(x)$, which is the formula for the distribution function. Near $1/e$, $r(b) - 1/e \sim b^2/(2e)$, so $U_{\mathcal{W}}$ has a density asymptotic to $(e/2)^{1/2}\pi^{-1}(z - 1/e)^{-1/2}$ there, and a bounded continuous density on $[1/e + c, \infty)$ for every $c > 0$. Hence $\Psi$ extends continuously to the closed upper half plane, and so does $\mathcal{W}$, with $\mathcal{W}(-1/e) = -1$. The difference $D = \mathcal{W} - \Psi$ is therefore continuous on the closed upper half plane and real on the real axis, where on $(-\infty, -1/e)$ the imaginary parts of the two boundary values agree. The Schwarz reflection principle continues $D$ across $(-\infty, -1/e)$, the continuation coincides with $D$ in the lower half plane because $D(\bar{u}) = \overline{D(u)}$, and the singularity at $-1/e$, where $D$ is bounded, is removable. So $D$ is entire, and $D(0) = 0$.

It remains to bound $D$, and one sided bounds on real parts suffice. First, $\operatorname{Re} \Psi(u) = \int \log|1 + u/z|\, U_{\mathcal{W}}(dz) \le \log(1 + e|u|)$, since $z > 1/e$. Next let $|u| = R \ge e$ and $w = \mathcal{W}(u)$, so that $|w|e^{\operatorname{Re} w} = R$ and $|\operatorname{Im} w| < \pi$. If $|w| > \log R + \pi$, then $|\operatorname{Re} w| > |w| - \pi > \log R$. This is impossible when $\operatorname{Re} w \ge 0$, since then $\operatorname{Re} w = \log R - \log|w| < \log R$; and when $\operatorname{Re} w < 0$ it gives $\log|w| - \log R > |w| - \pi$, that is $|w| - \log|w| < \pi - \log R \le \pi - 1$, which fails for $|w| > 1 + \pi$, since $x - \log x$ increases for $x > 1$ and exceeds $\pi - 1$ at $x = 1 + \pi$. Hence $|w| \le \log R + \pi$ and $\operatorname{Re} \mathcal{W}(u) = \log R - \log|w| \ge \log R - \log(\log R + \pi)$. By continuity both bounds hold on the whole circle, its point on the cut included, so

$$\max_{|u|=R} \operatorname{Re}\big(-D(u)\big) \le \log(1 + eR) - \log R + \log(\log R + \pi), \qquad R \ge e.$$

The Borel Carathéodory lemma for the entire function $-D$, which vanishes at the origin, gives $\max_{|u|=R/2} |D(u)| \le 2 \max_{|u|=R} \operatorname{Re}(-D(u))$, which is $O(\log\log R)$ by this bound. By Cauchy's estimates every Taylor coefficient of $D$ of positive order vanishes, so $D \equiv D(0) = 0$. □

Corollary 11.5.4 (first part from [296, Theorem 6.5] and [298]). *The canonical function of $T_{\mathcal{W}}$ is*

$$k_{\mathcal{W}}(x) = \int e^{-xz}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi} \int_0^\pi e^{-x\, r(b)}\, db,$$

*so that $\mathcal{W}(u) = \int_0^\infty (1 - e^{-ux})\, k_{\mathcal{W}}(x)\, dx/x$ for $u \ge 0$. The symmetric law of $Z\sqrt{2T_{\mathcal{W}}}$, with $Z$ standard normal and independent of $T_{\mathcal{W}}$, has characteristic function $e^{-\mathcal{W}(w^2)}$, is an EGGC (Section 1.3.2), and has Lévy density $k^{\mathrm{s}}_{\mathcal{W}}(|x|)/|x|$ with*

$$k^{\mathrm{s}}_{\mathcal{W}}(x) = \int e^{-x\sqrt{z}}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi} \int_0^\pi e^{-x\sqrt{r(b)}}\, db,$$

*that is, $\mathcal{W}(w^2) = \int_0^\infty 4\sin^2(wx/2)\, k^{\mathrm{s}}_{\mathcal{W}}(x)\, dx/x$ for real $w$.*

*Proof.* Insert Frullani's integral $\log(1 + u/z) = \int_0^\infty (1 - e^{-ux}) e^{-zx} x^{-1} dx$ in Theorem 11.5.3 and exchange the integrals by Tonelli's theorem. For the symmetric

law, $\mathbb{E}e^{iwZ\sqrt{2T_{\mathcal{W}}}} = \mathbb{E}e^{-w^2 T_{\mathcal{W}}}$, and the cosine dictionary (6.2.4) with $U = U_{\mathcal{W}}$, for which $K_U = 2k^{\mathrm{s}}_{\mathcal{W}}$, gives the Lévy density $K_U(|x|)/(2|x|)$. □

Remark 11.5.5 (Self decomposability, after Pakes). Since $k_{\mathcal{W}}$ is nonincreasing, $T_{\mathcal{W}}$ is self decomposable, and as in Proposition 11.1.1 its driving subordinator $\mathsf{Z}$ has the Lévy density $-k'_{\mathcal{W}}(x) = \int z e^{-xz}\, U_{\mathcal{W}}(dz)$, of total mass $k_{\mathcal{W}}(0+) = 1$ and with mean $\int_0^\infty k_{\mathcal{W}}(x)\, dx = \int z^{-1}\, U_{\mathcal{W}}(dz) = 1$ (Corollary 11.5.6). So the driving process is compound Poisson with unit rate and jumps $J$ of mean one; its Laplace exponent is $u\mathcal{W}'(u) = \mathcal{W}(u)/(1+\mathcal{W}(u))$, and $\mathbb{E}e^{-uJ} = 1/(1+\mathcal{W}(u))$ [407, (5), (6), (18)]. Stopping $T_{\mathcal{W}} = \int_0^\infty e^{-t}\, d\mathsf{Z}_t$ at the first jump of $\mathsf{Z}$, at an exponential time $\tau$, gives the perpetuity $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$, with $\mathsf{U} = e^{-\tau}$ uniform on $(0,1)$ and the three variables independent. The stationary excess $\mathcal{S}J$ has the density $P(J > x)$, since $\mathbb{E}J = 1$, and the Laplace transform $(1 - \mathbb{E}e^{-uJ})/u = \mathcal{W}'(u)$; since $\mathbb{E}e^{-u\mathsf{U}Y} = u^{-1}\int_0^u \mathbb{E}e^{-sY} ds$ for $Y \geq 0$ independent of $\mathsf{U}$, also $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, and $\mathcal{S}J \stackrel{d}{=} J + T_{\mathcal{W}}$.

Corollary 11.5.6 ([296, Corollary 3.4], [297]; Nuttall's integral [389, 63]). *For $n \geq 0$,*

$$\int z^{-n}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \Big(\frac{\sin b\; e^{b\cot b}}{b}\Big)^n db = \frac{n^n}{n!} \qquad (0^0 = 1),$$

*so $(n^n/n!)_{n\geq 0}$ is the moment sequence of the image of $U_{\mathcal{W}}$ under $z \mapsto 1/z$, a probability measure on $(0, e)$. Nuttall conjectured, and Bouwkamp proved, the identity with $\nu^\nu/\Gamma(\nu+1)$ for every real $\nu \geq 0$.*

*Proof.* Differentiating Theorem 11.5.3 gives

$$\mathcal{W}'(u) = \int \frac{U_{\mathcal{W}}(dz)}{u+z} = \sum_{n\geq 1}(-u)^{n-1}\int z^{-n}\, U_{\mathcal{W}}(dz), \qquad |u| < 1/e,$$

the series converging uniformly in $z$ since $z > 1/e$. Compare with the derivative of the Lagrange series $\mathcal{W}(u) = \sum_{n\geq 1}(-n)^{n-1}u^n/n!$, whose coefficients are $(-n)^{n-1}/(n-1)! = (-1)^{n-1}n^n/n!$. For $n = 0$ the integral is the total mass, one. □

The program `pakes_lambert.py` checks the section.

What transfers to the clock. The proof of Theorem 11.5.3 has four steps: the boundary values on the cut, the positivity of the boundary measure, the identification by reflection and a growth bound, and, for the symmetric law, the cosine dictionary (6.2.4). For the clock the first step is the argument count. Continued along the upper side of the negative axis, past the zeros of $G$ on it by small half circles in the upper half plane, $\operatorname{Im}\log G(-x + i0) = \pi N_{\mathrm{line}}(\sqrt{x})$, where $N_{\mathrm{line}}(t)$

counts with multiplicity the zeros on the critical line with ordinate in $(0,t)$, because the four zeros $\rho$, $1-\rho$, $\bar\rho$, $1-\bar\rho$ of a zero off the line contribute to $G$ the factor $(1+u/a)(1+u/\bar a)=|1+u/a|^2>0$ at real $u$; so $N_{\mathrm{line}}(\sqrt{x})$, the distribution function of $U_{\mathrm{line}}$ (Proposition 1.2.3), is read from the boundary values of $\log G$ with no hypothesis, and it is $N(\sqrt{x})$ under RH. The second step is then automatic, since a counting function is nondecreasing. The fourth step gives the Poisson trace $K_{1/2}(y)=\sum_k e^{-y\sqrt{a_k}}$, which is $\sum_{\gamma>0}e^{-\gamma y}$ under RH, as the Lévy side of $Y_{1/2}=Z\sqrt{2T}$ (Section 10.1.4), while the heat trace $W$ is the canonical function of $T$ itself (Section 1.3.1); the two traces are linked with no hypothesis by subordination. The third step is the obstruction: $\mathcal{W}$ is analytic off $(-\infty,-1/e]$ by construction, whereas $\log G$ is analytic off $(-\infty,0]$ if and only if RH holds (Proposition 1.2.1 and the argument after it).

### 11.5.4 Multiply self decomposable laws and branching

The centre clock is self decomposable with no hypothesis (Theorem 5.1.1), so it is the integral $\int_0^\infty e^{-t}\,dZ_t$ of its driving subordinator $Z$ (Proposition 11.1.1). The law of $Z_1$ may be self decomposable in its turn, with a driving process of its own, and so on. Read upwards, from the last driving process to the clock, such a chain is Pakes's BDLP ladder [407, Section 5], and how far it goes places a law in one of Urbanik's classes $L_n$. This subsection shows that for the centre clock the chain has exactly three steps, with no hypothesis on the zeros (Theorem 11.5.8), and that the clock is not the martingale limit of a supercritical continuous state branching process, the construction behind one of Pakes's proofs for the Lambert law (Theorem 11.5.11). Neither says anything about RH.

#### *11.5.4.1 Urbanik's classes and the ladder functions*

A law $\mu$ on $\mathbb{R}$ is self decomposable if for every $c\in(0,1)$ there is a law $\mu_c$, its $c$ component, with $\widehat\mu(z)=\widehat\mu(cz)\,\widehat\mu_c(z)$. Urbanik [507] let $L_0$ be the class of self decomposable laws, $L_n$ for $n\ge1$ the class of laws in $L_{n-1}$ all of whose components lie in $L_{n-1}$, and $L_\infty=\bigcap_n L_n$. Sato proved that $L_\infty$ is the smallest class of laws that contains the stable laws and is closed under convolution and weak limits [470, Theorem 3.5]. In the language of Section 11.1.1, a law lies in $L_n$, $n\ge1$, exactly when it is self decomposable and the law at time one of its driving process lies in $L_{n-1}$ [281].

For laws on $[0,\infty)$ Berg and Forst put the classes in analytic form. Let $f$ be the Laplace exponent of an infinitely divisible law $\mu$ on $[0,\infty)$, so that $\int e^{-ux}\,\mu(dx)=e^{-f(u)}$, and let $(Sf)(u)=uf'(u)$. Then $\mu\in L_n$ if and only if $Sf, S^2f,\dots,S^{n+1}f$ are Bernstein functions [44, Theorem 1.3].

For the centre clock $f=\log G$, and $S$ acts on the canonical function $W$ through

the ladder functions

$$W^{[0]} = W, \qquad W^{[j+1]}(t) = -t\,\frac{d}{dt}W^{[j]}(t), \qquad \text{so that} \quad W^{[j]} = \Big(-t\frac{d}{dt}\Big)^j W.$$

For one exponential, $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)\,e^{-at}$, where $\beta_0 = 1$ and $\beta_{j+1}(u) = u\beta_j(u) - u\beta_j'(u)$:

$$\beta_1(u) = u, \quad \beta_2(u) = u^2 - u, \quad \beta_3(u) = u^3 - 3u^2 + u, \quad \beta_4(u) = u^4 - 6u^3 + 7u^2 - u.$$

In general $\beta_j(u) = \sum_{i=1}^{j}(-1)^{j-i}S(j,i)\,u^i = (-1)^j B_j(-u)$, with $S(j,i)$ the Stirling numbers of the second kind and $B_j$ the Touchard polynomials, and since $\beta_j(u)e^{-u} = (-u\frac{d}{du})^j e^{-u}$ and $-u\frac{d}{du}$ multiplies Mellin transforms by $s$,

$$\int_0^\infty \beta_j(u)\,e^{-u}u^{s-1}\,du = s^j\,\Gamma(s) \qquad (s > 0). \tag{11.5.1}$$

The positive roots of $\beta_3$ are $(3 \mp \sqrt{5})/2 = 0.381966$ and $2.618034$, and those of $\beta_4$ are $0.165757$, $1.343380$ and $4.490864$, so that $\beta_4 < 0$ on $(0, 0.165757) \cup (1.343380, 4.490864)$. Since $W(t) = \sum_k e^{-a_k t}$ with $\operatorname{Re} a_k \geq \gamma_k^2 - \frac{1}{4} > 0$ for every zero (Section 11.1.1),

$$W^{[j]}(t) = \sum_k \beta_j(a_k t)\,e^{-a_k t}, \tag{11.5.2}$$

the series converging absolutely and locally uniformly on $(0,\infty)$.

Lemma 11.5.7. *For every $j \geq 0$ and $u \geq 0$,*

$$\Big(u\frac{d}{du}\Big)^j \log G(u) = \int_0^\infty \big(1 - e^{-ut}\big)\,W^{[j]}(t)\,\frac{dt}{t}. \tag{11.5.3}$$

*For $n \geq 0$ the centre clock lies in $L_n$ if and only if $W^{[n+1]} \geq 0$ on $(0,\infty)$, and then $W^{[1]}, \ldots, W^{[n]}$ are nonnegative too. In that case the driving processes $Z^{[1]} = Z$, $Z^{[2]}$, …, $Z^{[n+1]}$ exist, each the driving process of the law at time one of the one before; $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G(u)$ and Lévy density $W^{[j]}(t)/t$; and for $1 \leq j \leq n$ the law of $Z_s^{[j]}$ is self decomposable for every $s > 0$.*

*Proof.* By (11.5.2), $W^{[j]}(t) = O(t^j e^{-\gamma_1^2 t})$ as $t \to \infty$, and by the explicit formula (11.5.4) below, $W^{[j]}(t) = O(t^{-1/2}\log(1/t))$ as $t \downarrow 0$. The case $j = 0$ of (11.5.3) is the Lévy Khintchine form of $\log G$ (Section 11.1.1). If it holds for $j$, then, since $ue^{-ut} = \frac{d}{dt}(1 - e^{-ut})$,

$$\begin{aligned} u\frac{d}{du}\int_0^\infty \big(1 - e^{-ut}\big)W^{[j]}(t)\,\frac{dt}{t} &= \int_0^\infty ue^{-ut}\,W^{[j]}(t)\,dt \\ &= \int_0^\infty \big(1 - e^{-ut}\big)\,W^{[j+1]}(t)\,\frac{dt}{t}, \end{aligned}$$

the boundary terms $(1-e^{-ut})W^{[j]}(t)$ vanishing at both ends by the two bounds. A function of the form (11.5.3) is a Bernstein function exactly when $W^{[j]} \geq 0$: if $W^{[j]} \geq 0$ it is a Lévy Khintchine form, and conversely, if it is a Bernstein function, its derivative $\int_0^\infty e^{-ut} W^{[j]}(t)\,dt$ is completely monotone, hence the Laplace transform of a nonnegative measure, which by the uniqueness of Laplace transforms is $W^{[j]}(t)\,dt$. So by the theorem of Berg and Forst the clock lies in $L_n$ exactly when $W^{[1]}, \ldots, W^{[n+1]}$ are nonnegative. If $W^{[n+1]} = -t\,(W^{[n]})' \geq 0$, then $W^{[n]}$ is nonincreasing, and since it tends to 0 at infinity it is nonnegative; descending in the same way, so are $W^{[n-1]}, \ldots, W^{[1]}$. For the driving processes, the Laplace exponent of $\int_0^\infty e^{-t}\,dY_t$ is $\int_0^u \Psi_Y(w)\,dw/w$ when the subordinator $Y$ has Laplace exponent $\Psi_Y$ (proof of Proposition 11.1.1), so a self decomposable law with Laplace exponent $f$ has a driving process with Laplace exponent $uf'(u)$. Starting from $f = \log G$ and using (11.5.3), $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G$ and Lévy density $W^{[j]}(t)/t$ for as long as the laws at time one are self decomposable, and the law of $Z_s^{[j]}$, with Lévy density $sW^{[j]}(t)/t$, is self decomposable exactly when $W^{[j]}$ is nonincreasing [471, Corollary 15.11], that is, when $W^{[j+1]} \geq 0$. □

### *11.5.4.2 The ladder index of the centre clock*

THEOREM 11.5.8. *Unconditionally, that is, using only the verification of RH below height* 2197.3 *[419], $W^{[3]}(t) > 0$ for every $t > 0$, and $W^{[4]}(e^{-4.15}) < 0$. Consequently $W$, $W^{[1]}$ and $W^{[2]}$ are positive on $(0,\infty)$; the centre clock lies in $L_2$ and not in $L_3$; its driving process $Z = Z^{[1]}$ and the driving process $Z^{[2]}$ of $Z$ are self decomposable at every time, and the driving process $Z^{[3]}$ of $Z^{[2]}$ is self decomposable at no time.*

*Proof.* By Lemma 11.5.7 it suffices to prove the two inequalities; the lower ladder functions are then positive, not only nonnegative, since $W^{[j]}$ strictly decreases to 0 when $W^{[j+1]} > 0$.

Large $t$. Let $\tau_1 = (3+\sqrt{5})/(2\gamma_1^2) = 0.0131039$, so that $\gamma_1^2\tau_1 = 2.618034$ is the largest root of $\beta_3$; beyond it $\beta_3$ is positive and increasing. The zeros with ordinates below $\Gamma = 2197.3$ lie on the line [419]; there are $N(\Gamma) = 1700$ of them, the last at 2197.2587, and `bdlp_ladder.py` encloses each one in ball arithmetic. For $t \geq \tau_1$ each of them contributes $\beta_3(\gamma^2 t)e^{-\gamma^2 t} \geq 0$ to $W^{[3]}(t)$, and the second contributes at least $\beta_3(\gamma_2^2\tau_1)e^{-\gamma_2^2 t} > 99.38\,e^{-\gamma_2^2 t}$, since $\gamma_2^2\tau_1 = 5.7910$. A zero with ordinate $\gamma > \Gamma$, wherever it lies, has $\operatorname{Re} a \geq \gamma^2 - \frac{1}{4}$ and $|a| \leq \gamma^2 + \frac{1}{4}$, so its term has modulus at most $f(\gamma) = B_3((\gamma^2+\frac{1}{4})t)\,e^{-(\gamma^2-\frac{1}{4})t}$, where the Touchard polynomial $B_j$ has the moduli of the coefficients of $\beta_j$. Since $vB_j'(v) \leq jB_j(v)$, the logarithm of $B_j(v)e^{-v}$ decreases at a rate of at least $\frac{1}{2}$ for $v \geq 2j$, so for $(\Gamma^2+\frac{1}{4})t \geq 6$ the function $f$ decreases and $f(\gamma) \leq f(\Gamma)e^{-(\gamma^2-\Gamma^2)t/2}$. Let

$$N^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + 0.1038\log T + 0.2573\log\log T + 9.3675,$$

an upper bound for $N(T)$ for $T \ge e$ [176, Corollary 1.2], whose derivative is at most $\frac{1}{2\pi}\log\frac{T}{2\pi} + \frac{0.4}{T}$ for $T \ge \Gamma$. Partial summation against $N \le N^+$, with $\gamma^2 - \Gamma^2 \ge 2\Gamma(\gamma - \Gamma)$ and $\log\frac{\gamma}{2\pi} \le \log\frac{\Gamma}{2\pi} + \frac{\gamma-\Gamma}{\Gamma}$, bounds the sum of these moduli by

$$\varepsilon(t) = f(\Gamma)\Big(N^+(\Gamma) - 1700 + \frac{L}{\Gamma t} + \frac{1}{2\pi\Gamma^3 t^2}\Big), \qquad L = \frac{1}{2\pi}\log\frac{\Gamma}{2\pi} + \frac{0.4}{\Gamma}.$$

The logarithmic derivative of $e^{\gamma_2^2 t}\varepsilon(t)$ is at most $3/t - (\Gamma^2 - \frac14 - \gamma_2^2) < 0$ for $t \ge \tau_1$, and $e^{\gamma_2^2\tau_1}\varepsilon(\tau_1) < 10^{-27458}$. So $W^{[3]}(t) \ge e^{-\gamma_2^2 t}\big(99.38 - e^{\gamma_2^2 t}\varepsilon(t)\big) > 0$ for $t \ge \tau_1$.

Small $t$. Applying $(-t\frac{d}{dt})^j$ term by term to (1.2.5), which the Gaussian factors in $r$ and in $\log n$ allow, gives with no hypothesis

$$W^{[j]}(t) = \beta_j\big(-\tfrac{t}{4}\big)e^{t/4} + A_j(t) - P_j(t), \tag{11.5.4}$$

where

$$A_j(t) = \frac{1}{2\pi}\int_0^\infty \beta_j(r^2 t)\, e^{-r^2 t}\, m(r)\, dr,$$
$$P_j(t) = \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, e^{-v_n} q_j(v_n), \qquad v_n = \frac{(\log n)^2}{4t},$$

and $q_0 = 1$, $q_{j+1}(v) = (\frac12 - v)q_j(v) + vq_j'(v)$, because $-t\frac{d}{dt}$ maps $t^{-1/2}$ to $\frac12 t^{-1/2}$ and $v_n$ to $v_n$; thus

$$q_3(v) = -v^3 + \tfrac92 v^2 - \tfrac{13}{4}v + \tfrac18, \qquad q_4(v) = v^4 - 8v^3 + \tfrac{29}{2}v^2 - 5v + \tfrac{1}{16}.$$

In the notation of the proof of Theorem 5.1.1, $\frac{1}{2\pi}\int_0^\infty e^{-r^2 t}\log\frac{r}{2\pi}\,dr = s_0(t)(\ell+\alpha_0)$, and since $-t\frac{d}{dt}$ multiplies $s_0$ by $\frac12$ and maps $\ell$ to $\frac12$, replacing $m(r)$ by $\log\frac{r}{2\pi}$ in $A_j$ gives

$$\bar A_j(t) = 2^{-j} s_0(t)\,(\ell + \alpha_0 + j),$$

which also follows from (11.5.1) and its derivative in $s$ at $s = \frac12$. For the difference we split the integral at $r = 5$. For $r \le 5$ we use $|\beta_j(u)|e^{-u} \le u$, valid for $j = 2, 3, 4$, and the bound $\int_0^5 r^2|m(r) - \log\frac{r}{2\pi}|\,dr < 0.404$ stated after the proof of Theorem 5.1.1; for $r \ge 5$ we use the bound $r^2|m(r) - \log\frac{r}{2\pi}| < 0.05$ stated there, and substitute $u = r^2 t$. This gives

$$|A_j(t) - \bar A_j(t)| \le E_j(t) = \frac{1}{2\pi}\Big(0.404\, t + 0.025\sqrt{t}\, J_j\Big),$$

where $J_j = \int_0^\infty |\beta_j(u)|\, e^{-u} u^{-3/2}\, du$ satisfies $J_3 < 1.3313$ and $J_4 < 1.8295$. Let $\tau_2 = 1/75$. For $t \le \tau_2$ the main term $\bar A_3(t)$ decreases in $t$, because $-t\frac{d}{dt}\bar A_3 = \bar A_4$ and $\ell + \alpha_0 + 4 > 0$; the modulus $|\beta_3(-\frac{t}{4})|e^{t/4}$ of the polar term and $E_3(t)$ increase; and so does the modulus of each term of $P_3(t)$, because $v_n \ge v_2 > 9$,

the polynomials $q_3$ and $q_4$ have no root in $[9,\infty)$, so that $q_3 < 0 < q_4$ there, and $-t\frac{d}{dt}$ maps the $n$th term of $P_3$ to the $n$th term of $P_4$. At $t = \tau_2$ the four quantities are $\bar{A}_3(\tau_2) = 0.357155$, $0.003378$, $E_3(\tau_2) < 0.001469$, and at most $0.057882$ for the sum of the moduli of the terms of $P_3$, in which the terms with $n > 200$ are bounded through $e^{-(\log n)^2/8t} \le n^{-\sigma}$, $\sigma = (\log 200)/8t$, as in the proof of Theorem 5.1.1. Hence $W^{[3]}(t) > 0.2944$ for $t \le \tau_2$, and since $\tau_2 > \tau_1$, $W^{[3]} > 0$ on $(0,\infty)$.

Rung four. Let $t_0 = e^{-4.15}$. Then $\gamma_1^2 t_0 = 3.1496$ lies in the interval $(1.343380, 4.490864)$ on which $\beta_4 < 0$. The first five zeros contribute $-0.97604$, $0.62189$, $0.22811$, $0.01295$ and $0.00216$ to $W^{[4]}(t_0)$, and all zeros together, with the tail bound above, give $W^{[4]}(t_0) = -0.11089729616255 \pm 10^{-23}$ in ball arithmetic. The explicit formula confirms the sign without the zeros: at $t_0$ the polar term is $0.004066$, $\bar{A}_4(t_0) = 0.228563$, $P_4(t_0) = 0.344026$ and $E_4(t_0) < 0.001928$, so that $W^{[4]}(t_0) < 0.004066 + 0.228563 + 0.001928 - 0.344026 < -0.109$. □

The ladder breaks where the first zero dominates. At $t_0$ its term, $\beta_4(3.1496)e^{-3.1496} = -0.97604$, is only partly cancelled by the others, which leave $W^{[4]}(t_0) = -0.11090$: the second zero, for which $\gamma_2^2 t_0$ already exceeds the largest root $4.490864$ of $\beta_4$, contributes $0.62189$, and the zeros above it are too far out. On the arithmetic side the same failure is the prime 2: its term in $P_4(t_0)$, $0.343448$, outweighs the archimedean main term $\bar{A}_4(t_0) = 0.228563$. On $-\log t \in [0,10]$ a scan in double precision on a grid of step $0.001$, with the sign at the ends of every interval certified from the zeros, finds no negative value of $W$, $W^{[1]}$, $W^{[2]}$ or $W^{[3]}$, and finds $W^{[4]} < 0$ exactly for $-\log t \in (3.946, 4.335)$, $W^{[5]} < 0$ on $(3.443, 4.126)$ and $W^{[6]} < 0$ on $(3.146, 3.773) \cup (4.403, 5.051)$. Towards the origin the ladder functions approach fixed fractions of $W$: by (11.5.4), $W^{[j]}(t) = \bar{A}_j(t) + O(1)$ as $t \downarrow 0$, so that $W^{[2]}/W \to \frac{1}{4}$ and $W^{[3]}/W \to \frac{1}{8}$. The two ratios fall from $5.73$ and $5.95$ at $-\log t = 4.30$ to $0.614$ and $0.404$ at $8$ and to $0.470$ and $0.291$ at $10$, decreasing throughout $[4.30, 10]$. The least values of $W^{[2]}$ and $W^{[3]}$ on $4.30 \le -\log t \le 8$ are $0.396547$, at $4.30$, and $0.410976$, at $4.335$.

Corollary 11.5.9. *With no hypothesis, for every $s > 0$ the laws of $Z_s$ and of $Z_s^{[2]}$ are self decomposable, hence unimodal, and the law of $Z_s$ lies in $L_1$.*

*Proof.* The first statement is Theorem 11.5.8 with Lemma 11.5.7, and Yamazato's theorem [521], [471, Theorem 53.1] gives unimodality. The Laplace exponent of $Z_s$ is $s\,u(\log G)'(u)$, so by (11.5.3) its images under $S$ and $S^2$ correspond to $sW^{[2]}$ and $sW^{[3]}$, which are positive, and the theorem of Berg and Forst places $Z_s$ in $L_1$. □

Remark 11.5.10. The perpetuity $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$ of Remark 11.5.5 comes from a driving process that is compound Poisson, and the argument there applies to any such process: if the driving process $Y$ of a self decomposable law is compound Poisson with rate $c$ and jumps distributed as $J$, stopping $X = \int_0^\infty e^{-t}\,dY_t$ at the

first jump gives $X \stackrel{d}{=} \mathsf{U}^{1/c}(X+J)$, with $\mathsf{U}$ uniform on $(0,1)$ and the three variables independent. The centre clock satisfies no relation of this form, for any $c>0$ and any $J \geq 0$. Such a $J$ would have $\mathbb{E}\log(1+J) < \infty$, since $\log(1+\mathsf{U}^{1/c}J) \geq \log(1+J) + c^{-1}\log \mathsf{U}$ and $\mathbb{E}\log(1+\mathsf{U}^{1/c}J) \leq \mathbb{E}\log(1+T) \leq \mathbb{E}T = m_0$, and iterating the relation would write $T$ as $\int_0^\infty e^{-t}\,dY_t$ with $Y$ compound Poisson; but the driving process of a self decomposable law is unique in law, its Laplace exponent being $u(\log G)'(u)$, and that of the clock has the infinite Lévy measure $-W'(x)\,dx$ (Proposition 11.1.1).

The ladder of this subsection is not the derivative ladder of Theorems 5.1.1, 5.1.6, 5.1.9 and 5.1.13 and Table 1.6, which climbs the signs $c_k = (-1)^k W^{(k)} > 0$. Its rungs are alternating combinations of the same $c_k$, with Stirling numbers as coefficients:

$$W^{[j]}(t) = \sum_{i=1}^{j} (-1)^{j-i} S(j,i)\, t^i c_i(t),$$

$$W^{[3]} = t^3 c_3 - 3t^2 c_2 + t c_1, \qquad W^{[4]} = t^4 c_4 - 6t^3 c_3 + 7t^2 c_2 - t c_1.$$

Of the BDLP ladder RH gives only the first rung, since a GGC is self decomposable and a gamma law, a GGC, lies outside $L_1$. In the other direction, membership of the clock in $L_\infty$ would imply RH, by [44, Section 4], but no finite rung does: for every $n$ the canonical function $(\log \frac{1}{t})_+^n$, which vanishes on $[1,\infty)$ and so is not completely monotone, defines a law in $L_n$ that is not a GGC, its ladder functions being $n!/(n-j)!\,(\log \frac{1}{t})_+^{n-j}$ for $j \leq n$ and the measure $n!\,\delta_1(dt)$ for $j = n+1$.

### *11.5.4.3 The ladder along the de Bruijn Newman flow*

Theorem 7.2.16 places the centre clock in a family of laws. For every $t \geq \Lambda$, and so for every $t \geq 0.2$ with no hypothesis, the function $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ of Section 8.4.1 is the Laplace transform of a GGC whose Thorin measure has the atoms $a_k(t) = (z_k(t)/2)^2$, where $0 < z_1(t) < z_2(t) < \cdots$ are the zeros of $H_t$, and $R_0 = 1/G$. Since $t$ now denotes the time of the flow, we write $x$ for the variable of the canonical function $W(x;t) = \sum_k e^{-a_k(t)x}$ of $R_t$, whose ladder functions are

$$W^{[j]}(x;t) = \Big(-x\frac{\partial}{\partial x}\Big)^j W(x;t) = \sum_k \beta_j\big(a_k(t)x\big)\, e^{-a_k(t)x}.$$

Lemma 11.5.7 holds for these laws with the same proof, the bound for small $x$ now coming from the number of zeros of $H_t$, so the law with Laplace transform $R_t$ lies in $L_n$ exactly when $W^{[n+1]}(\cdot\,;t) \geq 0$.

The program `heat_ladder.py` computes $H_t$ by Gauss Legendre quadrature in ball arithmetic, its zeros on $(0,600]$ by a sign scan of step 0.2 and bisection, and the ladder functions $W^{[j]}(x;t)$, $j \leq 5$, from these zeros on the fixed window

$0 \le -\log x \le 8$, on a grid of step 0.001, for nine values of $t$ (Table 11.17). Nothing needs computing for $-\log x \le 0$: the largest roots of $\beta_2, \dots, \beta_5$ are 1, 2.618034, 4.490864 and 6.510134, and $a_1(t)x \ge a_1(t) \ge 12.47$ in every row. The computation is numerical for $t > 0$. The quadrature is checked against a rule with 1024 nodes, with relative differences below $10^{-46}$ at $z = 100$, 300 and 590, but its error is not bounded, and the zeros above $z = 600$ are left out; an estimate from the density $\frac{1}{4\pi}\log\frac{z}{4\pi}$ of the zeros of $H_0$ puts their share of $W^{[3]}$ at $-\log x = 8$ below $2 \times 10^{-9}$, and their share of $W^{[5]}$ below $10^{-5}$. At $t = 0$ the computed zeros are the numbers $2\gamma_k$ to double precision, and there are $N(300) = 138$ of them, the number that Arb's count of the zeros of $\zeta$ gives; $H_0$ agrees with $\Xi(z/2)/8$, computed from $\zeta$, to $10^{-46}$.

| $t$ | $z_1(t)$ | $a_2/a_1$ | $W^{[2]}/W$ | $W^{[3]}/W$ | $W^{[3]} < 0$ | $W^{[4]} < 0$ | $n$ |
|---|---|---|---|---|---|---|---|
| 0 | 28.2695 | 2.2119 | 0.6144 | 0.4038 | none | $(3.946, 4.335)$ | 2 |
| 0.2 | 28.1536 | 2.2142 | 0.6131 | 0.4029 | none | $(3.935, 4.333)$ | 2 |
| 1 | 27.6920 | 2.2239 | 0.6078 | 0.3993 | none | $(3.893, 4.321)$ | 2 |
| 4 | 25.9874 | 2.2678 | 0.5887 | 0.3864 | none | $(3.737, 4.263)$ | 2 |
| 10 | 22.7127 | 2.3915 | 0.5545 | 0.3630 | none | $(3.426, 4.090)$ | 2 |
| 20 | 17.7111 | 2.7188 | 0.5069 | 0.3301 | none | $(2.884, 3.675)$ | 2 |
| 30 | 13.3727 | 3.2530 | 0.4687 | 0.3030 | $(2.983, 3.343)$ | $(2.303, 3.153)$ | 1 |
| 40 | 9.7966 | 4.0960 | 0.4374 | 0.2807 | $(2.236, 3.014)$ | $(1.676, 2.593)$ | 1 |
| 50 | 7.0629 | 5.3383 | 0.4115 | 0.2620 | $(1.563, 2.539)$ | $(1.022, 2.035)$ | 1 |

Table 11.17: The ladder functions of $R_t$ along the de Bruijn Newman flow, on the window $0 \le -\log x \le 8$: the first zero $z_1(t)$ of $H_t$, the ratio $a_2/a_1$ of the first two Thorin atoms, the ratios $W^{[2]}/W$ and $W^{[3]}/W$ at $-\log x = 8$, the sets of $-\log x$ on which $W^{[3]}(x;t) < 0$ and $W^{[4]}(x;t) < 0$, to the grid step 0.001, and the largest $n$ with $W^{[1]}, \dots, W^{[n+1]} \ge 0$ on the window: $R_t$ lies in no class above $L_n$, and in $L_n$ if the signs persist for $x < e^{-8}$. On the window $W$, $W^{[1]}$ and $W^{[2]}$ are positive and $W^{[5]}$ takes negative values for every $t$. Numerical for $t > 0$; the row $t = 0$ is Theorem 11.5.8. Computed by `heat_ladder.py`.

Along the flow the zeros move towards the origin and apart, as the table shows, and the number of zeros in $(0, 600]$ rises from 138 to 149. The set on which $W^{[4]} < 0$ follows the first atom to smaller values of $-\log x$. It lies where $a_1(t)x$ is in the interval $(1.343380, 4.490864)$ on which $\beta_4 < 0$, less the part on which the other atoms compensate, and at $t = 50$, where it covers most of that interval, the first atom alone almost decides the sign. Up to $t = 20$ the third ladder function stays nonnegative on the window, and $R_t$ keeps the index two of the clock, as far as the window shows. Between $t = 27.8$ and $t = 27.9$ it turns negative. The program finds $W^{[3]} \ge 0$ on the window at $t = 21, \dots, 27$ and $t = 27.1, \dots, 27.8$, which gives $W^{[2]} \ge 0$ and $W^{[1]} \ge 0$ there as well, by the descent in the proof of Lemma 11.5.7 applied on $[e^{-8}, \infty)$; at $t = 27.9$ it finds $W^{[3]} < 0$ for $3.230 < -\log x < 3.307$, with least value $-0.0015$, and at $t = 30$, 40 and 50 on the intervals of the table.

So, by this computation, for these $t$ the law with Laplace transform $R_t$, a GGC with no hypothesis, is not in $L_2$, and its index on the window drops to one. The mechanism is that of Theorem 11.5.8, one rung lower: the term of the first atom in $W^{[3]}$ is negative where $a_1(t)x$ lies in $(0.381966, 2.618034)$, and once the first two atoms are far enough apart the second no longer compensates it; at the threshold $a_2/a_1$ is between 3.113 and 3.119, against 2.212 for the clock. At $t = 50$ the ratio $W^{[2]}/W$, which for $t \le 40$ decreases across the whole window, has an interior minimum, 0.276 at $-\log x = 2.682$: the same mechanism has begun to act on the second rung, which a single atom already violates.

Below the window the ratios $W^{[2]}/W$ and $W^{[3]}/W$ tend to $\frac{1}{4}$ and $\frac{1}{8}$, at $t = 0$ by (11.5.4) and for every $t \ge 0.2$ by the count of the zeros of $H_t$. For fixed $t > 0$ the number $N_t(T)$ of zeros of $H_t$ in $(0, T]$ is $\Psi(T) + \frac{t}{16}\log T + O(1)$, with $\Psi(T) = \frac{T}{4\pi}\log\frac{T}{4\pi} - \frac{T}{4\pi}$: Ki, Kim and Lee proved a formula of this kind for every $t > 0$ [310, Theorem 1.4], as recalled in [463, Section 3], and the Polymath project proved it in this form for $0 < t \le \frac{1}{2}$, with the error $O(1)$ for $T \ge e^{C/t}$ and $O(\log T)$ for $T \ge 2$, the constants being absolute but not numerical [456, Theorem 1.5]. An error $O(\log T)$ would do; a bound with numerical constants is Problem 15.6(b). Partial summation against $d\Psi$, with the substitution $u = z^2x/4$ and (11.5.1) at $s = \frac{1}{2}$, gives

$$W^{[j]}(x;t) = \frac{\log(1/x)}{2^{j+3}\sqrt{\pi x}}\big(1 + o(1)\big) \qquad (x \downarrow 0),$$

which at $t = 0$ is the leading part of $\bar{A}_j$. The approach is slow, as the table shows, so for $t > 0$ the signs below $x = e^{-8}$ are plausible but not proved. Either way the index is set by the positions and the spacing of the first zeros: it changes along the flow where every law is a GGC with no hypothesis, and it carries no information about RH.

### *11.5.4.4 No branching process realisation*

One of Pakes's proofs that $\mathcal{W}$ is a Thorin Bernstein function obtains the Lambert law as the limit of a branching process [406] (Section 11.5.1). Let $Y$ be a spectrally positive Lévy process with Laplace exponent $\Psi_Y(u) = -\log \mathbb{E}e^{-uY_1}$, with the sign of Proposition 11.1.1, and with finite Malthusian parameter $\Psi_Y'(0) \in (0, \infty)$. Lamperti's time change turns $Y$ into a continuous state branching process $(\mathsf{B}_t)_{t\ge0}$; started at $\mathsf{B}_0 = z > 0$ it has $\mathbb{E}\mathsf{B}_t = ze^{\Psi_Y'(0)t}$, and the nonnegative martingale $e^{-\Psi_Y'(0)t}\mathsf{B}_t$ converges almost surely to a limit $\mathsf{M}$. By the branching property $\mathbb{E}e^{-u\mathsf{M}} = e^{-z\omega(u)}$ for a function $\omega$ that does not depend on $z$. Conditioning on $\mathsf{B}_s$ gives $\omega(u) = v_s\big(\omega(ue^{-\Psi_Y'(0)s})\big)$, where $e^{-zv_s(\theta)} = \mathbb{E}e^{-\theta\mathsf{B}_s}$ and $\partial_s v_s = \Psi_Y(v_s)$, and differentiating at $s = 0$ gives

$$\Psi_Y'(0)\, u\, \omega'(u) = \Psi_Y\big(\omega(u)\big), \tag{11.5.5}$$

which is [407, (68)]. For the Lambert law, $Y$ is the compound Poisson process with unit rate and standard exponential jumps, $\Psi_Y(u) = u/(1+u)$ and $\Psi_Y'(0) = 1$, and

$\omega = \mathcal{W}$ solves (11.5.5), since differentiating $\mathcal{W}e^{\mathcal{W}} = u$ gives $u\mathcal{W}' = \mathcal{W}/(1+\mathcal{W})$ [407, Section 7].

For the centre clock, or $\lambda T$ with $\lambda > 0$, to be such a limit we need $z\omega(u) = \log G(\lambda u)$. Then (11.5.5) determines $\Psi_Y$ on the range $[0,\infty)$ of $\omega$:

$$\Psi_Y(w) = \frac{\Psi_Y'(0)}{z}\, g(zw), \qquad g = \Psi_Z \circ H = \frac{H}{H'},$$

where $H$ is the inverse function of $\log G$ and $\Psi_Z(u) = u(\log G)'(u)$ is the Laplace exponent of the driving process of Proposition 11.1.1. The function $g$ is analytic at 0, with $g(w) = \sum_{k\ge1} g_k w^k$ and $g_1 = 1$. For every spectrally positive $Y$ the function $-\Psi_Y''(u) = \sigma^2 + \int_0^\infty x^2 e^{-ux}\,\Pi(dx)$, with $\sigma^2$ the Gaussian coefficient and $\Pi$ the Lévy measure of $Y$, is completely monotone. Since $\Psi_Y^{(k)}(0) = \Psi_Y'(0)\, z^{k-1} k!\, g_k$, this forces

$$(-1)^{k-1} g_k \ge 0 \qquad (k \ge 2). \tag{11.5.6}$$

The coefficients $g_k$ need no hypothesis. The Taylor series of $\xi$ at $\frac12$, computed in ball arithmetic, gives that of $\log G(u) = \log\xi(\frac12 + \sqrt{u}) - \log\xi(\frac12) = \sum_{n\ge1}(-1)^{n+1} m_{n-1} u^n/n$, with the Stieltjes moments $m_n = \sum_k a_k^{-n-1}$ of Section 5.1; reverting the series gives $H$, and composing gives $g = \Psi_Z \circ H$. The program `branch_coeffs.py` does this at 3200 bits for $k \le 60$, checks it against the second route $g = H/H'$, which agrees for $k \le 59$, and checks $m_1, \dots, m_{11}$ against the sums over the first 1700 zeros. Every sign for $k \le 60$ is certified.

THEOREM 11.5.11. *With no hypothesis, $(-1)^{15} g_{16} < 0$: in ball arithmetic $g_{16} = 1.49524474945112 \times 10^{-15} \pm 2 \times 10^{-30}$. Consequently no positive multiple of the centre clock is the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter, from any initial state.*

*Proof.* Such a limit would satisfy (11.5.6), which the sign of $g_{16}$ violates. □

The signs keep failing for a structural reason, Pringsheim's theorem, the power series form of Landau's theorem [514, Chapter II, Theorem 5b]: if $(-1)^{k-1} g_k \ge 0$ for all large $k$ and the series $\sum g_k w^k$ has a finite radius of convergence $R$, then its sum is singular at $w = -R$. But $g$ is analytic on a neighbourhood of the negative axis. With no hypothesis, $(\log G)'(u) = \sum_k 1/(u + a_k) > 0$ for $-\gamma_1^2 < u \le 0$: a zero on the line contributes $1/(u+\gamma^2) > 0$, and the conjugate atoms $a$, $\bar a$ of an off line zero together contribute $2\,\mathrm{Re}(u+a)/|u+a|^2$, with $\mathrm{Re}(u+a) \ge \gamma^2 - \frac14 - \gamma_1^2 > 0$. So $\log G$ increases from $-\infty$ to 0 on $(-\gamma_1^2, 0]$, its inverse $H$ is analytic on a neighbourhood of $(-\infty, 0]$, and so is $g = H/H'$. Hence (11.5.6) fails for infinitely many $k$ unless $g$ is entire.

Numerically the radius is finite, $R = 7.1565$, and the singularities that fix it lie off the axis. The function $\Xi$ has a critical point $t^* = 15.5857085898$ between $\gamma_1$ and $\gamma_2$, certified by bisection on $\Xi'$, and at $u^* = -t^{*2} = -242.914312$ the function $G(u) = \Xi(\sqrt{-u})/\Xi(0)$ has $G'(u^*) = 0$ and $G(u^*) = -0.00161241048$. Continued

from 0 around the zero $-\gamma_1^2$ through the upper half plane, $\log G$ reaches $u^*$ with the value

$$\lambda^* = \log|G(u^*)| + i\pi = -6.43002502663 + i\pi,$$

of modulus $|\lambda^*| = 7.156453$ and argument $\arg\lambda^* = 2.687121$, and since $(\log G)'(u^*) = 0$, the inverse $H$ has a square root branch point at $\lambda^*$, and by symmetry at $\bar\lambda^*$. Continuing $H$ numerically along the segment from 0 to $\lambda^*$ in 3000 steps keeps $|G| = e^{\operatorname{Re}\lambda} \geq 0.0016$ on the path, away from the zeros of $G$, and brings the preimage to within 2.86 of $u^*$, at a distance proportional to the square root of the distance from $\lambda^*$ over the last steps, as at a simple critical point; so $\lambda^*$ and $\bar\lambda^*$ are singularities of the principal branch of $g$ (a numerical continuation, not a certificate). The root test agrees: $|g_k|^{1/k}$ is 0.1269, 0.1274, 0.1296, 0.1288 and 0.1317 at $k = 20$, 30, 40, 50 and 60, approaching $1/|\lambda^*| = 0.13973$ slowly, as the factor $k^{-3/2}$ of a square root singularity predicts. The conjugate pair makes $(-1)^{k-1}g_k$ oscillate in $k$ with period $2\pi/(\pi - \arg\lambda^*) = 13.8253$.

The next critical points of $\Xi$ do not matter. The second, $t^{**} = 22.0979772804$ between $\gamma_2$ and $\gamma_3$, has $G(-t^{**2}) = 1.56051 \times 10^{-5} > 0$ and $\log G(-t^{**2}) = -11.0679$, and the third, 26.2722473569 between $\gamma_3$ and $\gamma_4$, has $G = -1.34670 \times 10^{-6}$. So the branching obstruction, like the end of the ladder, comes from the bottom of the spectrum: the singularity that forces the changes of sign is created by the critical point of $\Xi$ between $\gamma_1$ and $\gamma_2$.

### 11.5.5 The criticality filter

Along the de Bruijn Newman flow the function $R_t$ of Theorem 7.2.16 is the Laplace transform of a GGC exactly when $t \geq \Lambda$ (Corollary 4.2.5), and $\Lambda \geq 0$ by the theorem of Rodgers and Tao [463]. So the Pólya kernel sits at the edge of a family of kernels $e^{tv^2/4}\Phi(v)$, $t < 0$, as close to it as one wishes, whose transforms have nonreal zeros and for which the Thorin condition fails. This subsection turns that fact into a test for the properties of the kernel that might imply RH. It applies the test to two representations of the Pólya law as a mixture, and to the property that the first author's account singles out, the GGC property of the variable of Biane, Pitman and Yor, which passes the test at every $t < 0$ tested but does not detect the Euler product. Every inequality below that is called certified is checked in ball arithmetic [290] by `pakes_filter_cert.py`.

#### *11.5.5.1 The filter*

For an even, positive, integrable function $g$ on $\mathbb{R}$ put

$$\widehat{g}(\theta) = \int_{\mathbb{R}} g(v)\cos(\theta v)\,dv, \qquad R_g(s) = \frac{\widehat{g}(0)}{\widehat{g}(i\sqrt{s})} = \frac{\int_{\mathbb{R}} g(v)\,dv}{\int_{\mathbb{R}} g(v)\cosh(\sqrt{s}\,v)\,dv}, \quad (11.5.7)$$

for the $s \geq 0$ at which the last integral converges. For the damped Pólya kernel $\Phi_t(v) = e^{tv^2/4}\Phi(v)$, with $\Phi$ as in (1.2.3), the transform $\widehat{\Phi_t}$ is the function $\Xi_t$

of Section 8.7.6, and $R_{\Phi_t}$ is the function $R_t$ of Theorem 7.2.16, written $\varphi_t$ in Corollary 4.2.5; at $t = 0$ it is the clock $1/G$.

PROPOSITION 11.5.12. *Let $\mathcal{P}$ be a property of even, positive, integrable functions on $\mathbb{R}$ that $\Phi_t$ has for some $t < 0$. Then $\mathcal{P}$ implies neither that $\widehat{g}$ has only real zeros nor that $R_g$ is the Laplace transform of a GGC, for the functions $g$ with $\mathcal{P}$. At every $t \le 0$ the kernel $\Phi_t$*

*(i) is positive and even, and strictly decreasing on $(0, \infty)$;*

*(ii) is log concave;*

*(iii) satisfies $\log \Phi_t(v) = -\pi e^{2|v|} + O(v^2)$ as $|v| \to \infty$;*

*(iv) extends analytically to the strip $|\operatorname{Im} v| < \pi/4$.*

*So no combination of (i) to (iv) implies that $\widehat{g}$ has only real zeros, or that $R_g$ is the Laplace transform of a GGC.*

*Proof.* Let $t < 0$ be such that $\Phi_t$ has $\mathcal{P}$. The zeros of $H_t$, and so those of $\Xi_t(\theta) = 8H_t(2\theta)$, are all real if and only if $t \ge \Lambda$ (de Bruijn [112], Newman [381]), and $\Lambda \ge 0$ [463], so $\Xi_t$ has a zero $z \notin \mathbb{R}$. It is not on the imaginary axis either, since $\Xi_t(iy) = \int \Phi_t(v) \cosh(yv)\, dv > 0$ for real $y$. Hence $s_0 = -z^2$ has $\operatorname{Im} s_0 = -2 \operatorname{Re} z \operatorname{Im} z \ne 0$. As $\Xi_t$ is even and entire, $s \mapsto \Xi_t(i\sqrt{s})$ is entire, and it vanishes at $s_0$, so $R_t$ is meromorphic on $\mathbb{C}$ with a pole at $s_0 \notin (-\infty, 0]$. By (F7) of Section 4.2.1 an HCM function extends analytically to $\mathbb{C} \setminus (-\infty, 0]$, and the extension would agree with $R_t$ off its poles; so $R_t$ is not HCM, and by (F6) it is not the Laplace transform of a GGC.

For (i), $\Phi$ is positive and even, and $\Phi_t'(v) = e^{tv^2/4}\big(\Phi'(v) + \frac{t}{2} v\Phi(v)\big) < 0$ for $v > 0$, since $\Phi' < 0$ on $(0, \infty)$ [99, Theorem A] and $tv\Phi(v) \le 0$. For (ii), $(\log \Phi_t)'' = (\log \Phi)'' + \frac{t}{2} \le (\log \Phi)'' \le 0$, since $\Phi$ is log concave [85]. Properties (iii) and (iv) hold for $\Phi$ by (1.2.3), and the factor $e^{tv^2/4}$ is entire and adds $tv^2/4$ to the logarithm. □

A property that is to imply RH through the kernel alone must fail for $\Phi_t$ at every $t < 0$; we then say that it *passes the filter*. Passing is necessary and far from sufficient: it says only that the theorem of Rodgers and Tao does not rule the property out. For the properties (i), (ii) and (iv) the conclusion needs nothing so deep: the convolution of $(1 - |u|)_+^2$ with $e^{-\cosh u}$ in Section 8.5.5 is even, positive, entire, log concave and doubly exponentially decaying, and its transform has nonreal zeros. What the filter adds is a counterexample that has every property of the list and differs from $\Phi$ only by the factor $e^{tv^2/4}$, with $t$ as close to 0 as one wishes.

The filter is the counterpart, for properties of the kernel, of the perturbation test of Section 12.2.6. There the function $F$ of Theorem 5.2.1 keeps the zeros

of $\xi$ below a height $R$, the order, the type and the functional equation, and has four zeros off the line above $R$; here $\Phi_t$ keeps every property of the kernel that a Gaussian factor $e^{tv^2/4}$ with $t<0$ preserves, and has nonreal zeros by the theorem of Rodgers and Tao, which does not say where. For $t>0$ the zeros of $H_t$ obey a formula of Riemann and von Mangoldt type, $N_t(T)=\Psi(T)+\frac{t}{16}\log T+O(1)$ (Section 11.5.4.3); for $t<0$, the case the filter needs, we know of no count in the literature: Rodgers and Tao prove one for $\Lambda<t\le 0$, with the error $O(\log^2 T)$ [463, Section 3], as a step in showing that this range of $t$ is empty. A proof of RH must therefore use, at some step, a property that $\Phi_t$ lacks for every $t<0$. The Euler product and the explicit formula are the natural candidates, as they are for $F$ in Section 12.2.6, but this is a reading and not a theorem: no Euler product for $H_t$ with $t\neq 0$ is known (see the discussion after Theorem 7.2.16), and none is known not to exist. The Lambert proof of Section 11.5.3 is untouched by the filter, since it concerns a single function, whose analyticity off the cut is forced by the equation $\mathcal{W}e^{\mathcal{W}}=u$, and no family of comparison functions sharing its properties is in sight.

### *11.5.5.2 Mixtures of arcsine laws*

For $t\le 0$ let $X_t$ be the variable with density $p_t=\Phi_t/\Xi_t(0)$, so that $X_0$ is the Pólya variable $X$. Since $p_t$ is even and decreasing on $(0,\infty)$ by Proposition 11.5.12(i), Khinchine's theorem writes $X_t=UQ_t$ with $U$ uniform on $[-1,1]$ and $Q_t\ge 0$ independent of $U$, of density $-2rp_t'(r)$; at $t=0$ this is the mixture of uniform laws of Section 13.4.2.

Proposition 11.5.13. *For every $t\le 0$ the function*

$$h_t(\rho)=-2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}},\qquad \rho>0,$$

*is a positive probability density, and*

$$p_t(v)=\frac{1}{\pi}\int_{|v|}^\infty \frac{h_t(\rho)\,d\rho}{\sqrt{\rho^2-v^2}},\qquad v\neq 0.$$

*Equivalently, $X_t\overset{d}{=}\varrho_t\cos\Theta$ with $\Theta$ uniform on $[0,2\pi)$ and $\varrho_t$ of density $h_t$ independent of $\Theta$: $X_t$ is a mixture of arcsine laws, and it is the first coordinate of the rotation invariant vector $\varrho_t(\cos\Theta,\sin\Theta)$ in $\mathbb{R}^2$, as well as of a rotation invariant vector in $\mathbb{R}^3$.*

*Proof.* By Archimedes' theorem the coordinates of a uniform point of the unit sphere in $\mathbb{R}^3$ are uniform on $[-1,1]$. Writing the point as $(B\cos\Theta,B\sin\Theta,V)$, with $V$ uniform on $[-1,1]$, $B=\sqrt{1-V^2}$ and $\Theta$ uniform on $[0,2\pi)$ independent of $V$, the first coordinate gives $U\overset{d}{=}B\cos\Theta$, where $B$ has density $b/\sqrt{1-b^2}$ on

$(0,1)$. Hence $X_t = UQ_t \overset{d}{=} (BQ_t)\cos\Theta$, and $Q_t$ times the uniform point is a rotation invariant vector in $\mathbb{R}^3$ with first coordinate $X_t$. The density of $\varrho_t = BQ_t$ is

$$\int_\rho^\infty \big(-2rp_t'(r)\big)\,\frac{\rho/r}{\sqrt{1-\rho^2/r^2}}\,\frac{dr}{r} = -2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}} = h_t(\rho),$$

positive since $p_t' < 0$ on $(0,\infty)$. Given $\varrho_t = \rho$, the variable $\rho\cos\Theta$ has the arcsine density $1/(\pi\sqrt{\rho^2-v^2})$ on $(-\rho,\rho)$, which gives the formula for $p_t$. □

At $t = 0$ the proposition is the arcsine route of [424, Chapter 12] recalled in Section 5.2, which writes $X$ as $2\sqrt{I}$ times the cosine of an independent angle; since $\mathbb{E}|\cos\Theta|^s \neq 0$ for $s > -1$, the Mellin transform of $|X|$ determines the law of the radius, so $\varrho_0 \overset{d}{=} 2\sqrt{I}$, and $h_0$ is the density of the variable $I$ of that route after the change of variable. Both representations, the uniform mixture and the arcsine mixture, hold at every $t \le 0$, so by Proposition 11.5.12 neither can decide RH, or the Thorin condition, by itself: whatever is to be proved about $Q_0$ or $\varrho_0$ must use something that $Q_t$ or $\varrho_t$ lacks for $t < 0$. The scale mixtures of Gamma laws, the other mixture route of [424, Section 12.6], are excluded already at $t = 0$, for every shape (Theorem 11.2.10).

### *11.5.5.3 The variable of Biane, Pitman and Yor under damping*

Let $\mathcal{B} = \frac{\pi}{2}S_2$, where $S_2 = J + J'$ is the sum of two independent hitting times of 1 by a three dimensional Bessel process started at 0, as in (9.1.1) and Theorem 7.1.21. Then

$$\mathbb{E}e^{-\lambda\mathcal{B}} = \Big(\frac{\sqrt{\pi\lambda}}{\sinh\sqrt{\pi\lambda}}\Big)^2 = \prod_{m\ge1}\Big(1+\frac{\lambda}{\pi m^2}\Big)^{-2}, \qquad \mathbb{E}\mathcal{B}^{s/2} = 2\xi(s), \tag{11.5.8}$$

so $\mathcal{B}$ is a GGC with Thorin measure $2\sum_m \delta_{\pi m^2}$, and its density is

$$f(b) = \sum_{m\ge1}\big(4\pi^2m^4b - 6\pi m^2\big)e^{-\pi m^2 b} = b^{-5/4}\,\Phi\big(\tfrac12\log b\big), \qquad b > 0.$$

More generally, for an even positive kernel $g$ with $\int g(v)e^{|v|/2}\,dv < \infty$ let $\mathcal{B}_g$ have the density proportional to $b^{-5/4}g(\frac12\log b)$; the Pólya variable of $g$, of density $g/\widehat{g}(0)$, is $\frac12\log\mathcal{B}_g$ tilted by $\mathcal{B}_g^{1/4}$, and $\mathcal{B}_\Phi = \mathcal{B}$ (Theorem 7.1.21). For the damped kernels write $\mathcal{B}_t = \mathcal{B}_{\Phi_t}$: its density is proportional to $e^{t(\log b)^2/16}f(b)$, finite for every real $t$ because $f(b)$ decays like $e^{-\pi b}$ as $b\to\infty$ and, up to a power of $b$, like $e^{-\pi/b}$ as $b\downarrow 0$, and $\mathbb{E}\mathcal{B}_t^{s/2} = \Xi_t(i(\frac12 - s))/\Xi_t(\frac{i}{2})$. Damping the kernel thus tilts $\mathcal{B}$ by a lognormal factor, and "$\mathcal{B}_g$ is a GGC" is a property of the kernel $g$ that $\Phi$ has. The filter asks whether $\Phi_t$ lacks it for every $t < 0$.

Two tests are available for a positive variable $\mathcal{V}$ with $\ell(\lambda) = \mathbb{E}e^{-\lambda\mathcal{V}}$. Let $\kappa_k(\lambda) = (-1)^k\frac{d^k}{d\lambda^k}\log\ell(\lambda)$, $k \ge 1$, be the cumulants of the law $e^{-\lambda b}\,P(\mathcal{V}\in$

$db)/\ell(\lambda)$. The variable is infinitely divisible if and only if $-\log \ell$ is a Bernstein function, that is, if and only if $\kappa_k(\lambda) \geq 0$ for all $k \geq 1$ and $\lambda > 0$ [473]. If it is a GGC, with $-\log \ell(\lambda) = d\lambda + \int \log(1 + \lambda/z)\, U(dz)$, then for $k \geq 0$

$$c_k(\lambda) = \frac{\kappa_{k+1}(\lambda)}{k!} = d\,\delta_{k0} + \int \frac{U(dz)}{(\lambda + z)^{k+1}} = \int x^k\, \sigma_\lambda(dx), \tag{11.5.9}$$

where $\sigma_\lambda$ is $d\,\delta_0$ plus the image of $(\lambda + z)^{-1}U(dz)$ under $z \mapsto (\lambda + z)^{-1}$. Hence the Hankel matrices $\mathcal{K}_n(\lambda) = [c_{i+j}(\lambda)]_{i,j=0}^{n-1}$ and $\mathcal{K}'_n(\lambda) = [c_{i+j+1}(\lambda)]_{i,j=0}^{n-1}$ are positive semidefinite for every $n$ and $\lambda \geq 0$, since $\sum_{i,j} y_i y_j c_{i+j+r}(\lambda) = \int x^r \big(\sum_i y_i x^i\big)^2 \sigma_\lambda(dx)$ for $r = 0, 1$, and a negative eigenvalue of one of them proves that $\mathcal{V}$ is not a GGC. The smallest case is already useful.

LEMMA 11.5.14. *If $\mathcal{V}$ is a GGC with a finite fourth moment, its cumulants $\kappa_k = \kappa_k(0)$ satisfy $2\kappa_2\kappa_4 \geq 3\kappa_3^2$, with equality for the gamma laws.*

*Proof.* With $c_k = \kappa_{k+1}/k!$ the inequality reads $c_1 c_3 \geq c_2^2$, which is $\det \mathcal{K}'_2(0) \geq 0$. For a gamma law $U$ is a single atom $\alpha\delta_z$ and $c_k = \alpha z^{-k-1}$, so equality holds. □

The program `pakes_filter.py` computes the moments

$$\int_{\mathbb{R}} e^{2kv}\, e^{-\lambda e^{2v}}\, e^{tv^2/4}\, e^{-v/2}\, \Phi(v)\, dv,$$

which are those of $\mathcal{B}_t$ under the law of the cumulants $\kappa_k(\lambda)$ up to normalisation, by the trapezoid rule on $[-5, 5]$. The program forms $\kappa_1, \ldots, \kappa_{26}$, at $\lambda = 0$, $\frac{1}{2}$, 2, 10, 20, 50 and 200, and the matrices $\mathcal{K}_n(\lambda)$ and $\mathcal{K}'_n(\lambda)$ of orders $n \leq 12$. Dividing $c_{i+j+r}$ by $c^{i+j+r}$, with $c = c_1/c_0$, is a congruence that keeps the inertia, and the eigenvalues quoted below are those of the matrices so scaled. The program `pakes_filter_cert.py` certifies the failures: it encloses the moments in balls (Arb, 256 bits, each tail bounded by $2^{-3000}$), computes the cumulants and the scaled matrix $M$ in ball arithmetic, and encloses $y^{\mathsf{T}} M y$ for a rational vector $y$ close to the eigenvector of the least eigenvalue; an enclosure inside $(-\infty, 0)$ proves that $M$ is not positive semidefinite. In every certificate below the cumulants used are certified positive and $y^{\mathsf{T}} M y$ agrees with the computed least eigenvalue to all the digits quoted.

Infinite divisibility shows nothing. In all 301 cases computed, the tilts $\mathcal{B}_t$ for the twenty values of $t$ of the next paragraph, the reciprocals $1/\mathcal{B}_t$ for $t = -1$, 0, $\frac{1}{5}$, and the ten functions other than $\xi$ of Table 11.19 below, on both sides, each at the seven values of $\lambda$, none of $\kappa_1, \ldots, \kappa_{26}$ is negative.

The GGC test separates $t = 0$ from every other value. It was run for $t = -20$, $-10$, $-5$, $-2$, $-1$, $-\frac{1}{2}$, $-\frac{1}{5}$, $-\frac{1}{10}$, $-\frac{1}{20}$, $-\frac{1}{50}$, $-\frac{1}{100}$, 0, $\frac{1}{100}$, $\frac{1}{50}$, $\frac{1}{20}$, $\frac{1}{10}$, $\frac{1}{5}$, $\frac{1}{2}$, 1 and 5. At $t = 0$ no matrix of order at most 12 has a negative eigenvalue, as the theorem requires, though the matrices are nearly singular. For every $t \neq 0$ some $\mathcal{K}'_n(\lambda)$ with $n \leq 12$ has a negative eigenvalue at each $\lambda \leq 50$. Table 11.18 shows eleven of the twenty values; the other nine give the same picture.

| $t$ | $\lambda=0$ | $\frac{1}{2}$ | 2 | 10 | 20 | 50 | eigenvalue at $\lambda=2$ |
|---|---|---|---|---|---|---|---|
| $-20$ | 8 | 9 | 8 | 8 | 7 | 9 | $-1.582\times10^{-7}$ |
| $-5$ | 8 | 8 | 7 | 7 | 8 | 10 | $-8.095\times10^{-8}$ |
| $-1$ | 8 | 8 | 7 | 7 | 8 | 11 | $-1.533\times10^{-8}$ |
| $-\frac{1}{2}$ | 8 | 8 | 7 | 7 | 9 | 11 | $-4.725\times10^{-9}$ |
| $-\frac{1}{10}$ | 8 | 8 | 8 | 8 | 9 | 12 | $-1.800\times10^{-9}$ |
| $-\frac{1}{100}$ | 9 | 9 | 8 | 8 | 10 | 12 | $-4.119\times10^{-11}$ |
| 0 | none | none | none | none | none | none | $1.247\times10^{-13}$ |
| $\frac{1}{100}$ | 4 | 4 | 5 | 6 | 7 | 10 | $-2.357\times10^{-5}$ |
| $\frac{1}{5}$ | 4 | 4 | 4 | 5 | 7 | 9 | $-1.101\times10^{-4}$ |
| 1 | 3 | 3 | 4 | 5 | 6 | 8 | $-2.102\times10^{-3}$ |
| 5 | 3 | 3 | 3 | 5 | 6 | 8 | $-2.141\times10^{-3}$ |

Table 11.18: The GGC test for the tilted variables $\mathcal{B}_t$ of Section 11.5.5.3: the least order $n\le12$ at which $\mathcal{K}_n'(\lambda)$ has a negative eigenvalue, and the least eigenvalue of the scaled matrix $\mathcal{K}_n'(2)$ at that order; for $t=0$, where no order fails, the last column gives $\mathcal{K}_7'(2)$. At $\lambda=200$ no matrix of order at most 12 fails, for any $t$. Every negative entry of the last column is certified, and so is the entry for $\mathcal{K}_9'(0)$ at $t=-\frac{1}{100}$, $-2.899\times10^{-12}$. Computed by `pakes_filter.py` and `pakes_filter_cert.py`.

Proposition 11.5.15. *(i) $\mathcal{B}_t$ is not a GGC for $t=-20$, $-5$, $-1$, $-\frac{1}{2}$, $-\frac{1}{10}$, $-\frac{1}{100}$, $\frac{1}{100}$, $\frac{1}{5}$, 1 and 5.*

*(ii) There is $t_0>0$ such that $\mathcal{B}_t$ is not a GGC for $0<|t|<t_0$.*

*Proof.* (i) For each of these $t$ the matrix $\mathcal{K}_n'(2)$ at the order $n$ of Table 11.18 is certified not to be positive semidefinite.

(ii) Put $\ell_t(\lambda)=\mathbb{E}e^{-\lambda\mathcal{B}_t}$ and split $\int_0^\infty e^{-\lambda b}e^{t(\log b)^2/16}f(b)\,db$, the numerator of $\ell_t$, at $b=1$. The part on $(0,1]$ is entire in $(t,\lambda)\in\mathbb{C}^2$, since $|e^{t(\log b)^2/16}|=e^{O((\log b)^2)}$ is negligible against $f(b)=O(b^{-7/2}e^{-\pi/b})$ as $b\downarrow0$, and the denominator, the value at $\lambda=0$, is analytic in $t$ and positive for real $t$. On $[1,\infty)$ write $f$ as the sum of its terms. Fix $M\ge1$ and $x$ with $\pi M^2<x<\pi(M+1)^2$. The terms with $m>M$ give a function analytic in $t\in\mathbb{C}$ and $\operatorname{Re}\lambda>-\pi(M+1)^2$. For $m\le M$ turn the path $[1,\infty)$ into the ray $b=1+re^{-3\pi i/4}$, $r\ge0$, which avoids $(-\infty,0]$ and along which $(\log b)^2$ grows like $(\log r)^2$. Writing $\lambda+\pi m^2=u+iv$, the integrand decays along every ray $b=1+re^{i\alpha}$ with $-\frac{3\pi}{4}\le\alpha\le0$ when $v>u>0$, so there the rotated integral equals the original one; and it converges, and is analytic in $(t,\lambda)$, on the half plane $v>u$, which contains $\lambda=-x$. So $\ell_t$ continues analytically, jointly in $(t,\lambda)$ near $(0,-x)$, from the upper half plane across the negative axis at $-x$.

Suppose that $\mathcal{B}_t$ is a GGC, with Thorin measure $U_t$. Then $-\log\ell_t(\lambda)=d_t\lambda+\int\log(1+\lambda/z)\,U_t(dz)$ on $\mathbb{C}\setminus(-\infty,0]$, and as $\varepsilon\downarrow0$ its imaginary part at $\lambda=-y+i\varepsilon$ tends to $\pi\big(U_t((0,y))+\frac{1}{2}U_t(\{y\})\big)$. For $t$ near 0, $\ell_t$ has no zero near

$-x$, since $\ell_0(-x) = \pi x/\sin^2\sqrt{\pi x} > 0$ by (11.5.8). So the limit is a real analytic function of $y$ near $x$, $U_t$ has no atom there, and $U_t((0,y)) = -\frac{1}{\pi}\arg \ell_t(-y+i0)$, with the argument continued from 0 on $(0,\infty)$ through the upper half plane, is nondecreasing in $y$; at $t=0$ it equals $2M$. Write $\ell_t = \ell_0 + t\big(\dot\ell - \dot\ell(0)\,\ell_0\big) + O(t^2)$, with $\dot\ell(\lambda) = \frac{1}{16}\int_0^\infty e^{-\lambda b}(\log b)^2 f(b)\,db$ continued in the same way; by joint analyticity the remainder is $O(t^2)$ together with its derivative in $\lambda$, uniformly near $-x$. Since $\ell_0$ and $\dot\ell(0)$ are real there,

$$U_t((0,y)) = 2M - \frac{t}{\pi}\mathcal{J}(y) + O(t^2), \qquad \mathcal{J}(y) = \frac{\operatorname{Im}\dot\ell(-y+i0)}{\ell_0(-y)},$$

with the same expansion for the derivatives in $y$. In $\dot\ell(-y+i0)$ the part on $(0,1]$ and the terms with $m > M$ are real, and for $m \le M$ the integral over the rotated ray differs by a real number from $\int_0^\infty e^{-\sigma b}(\log b)^2(4\pi^2m^4b - 6\pi m^2)\,db$ with $\sigma = -s_m + i0$, $s_m = y - \pi m^2$, which is a combination of $\partial_a^2\big[\Gamma(a)\sigma^{-a}\big]$ at $a=2$ and $a=1$, with $\log\sigma = \log s_m + i\pi$. This gives

$$\operatorname{Im}\dot\ell(-y+i0) = \frac{\pi}{8}\sum_{m\le M}\Big(\frac{4\pi^2m^4(\log s_m - 1 + \gamma_E)}{s_m^2} + \frac{6\pi m^2(\log s_m + \gamma_E)}{s_m}\Big),$$

where $\gamma_E$ is Euler's constant. At $y = 1.015\pi$, where $M=1$, the certificate program gives $\mathcal{J}'(y) = 8.52630079173 \pm 3.4\times10^{-12}$, and at $y = 2.47\pi$ it gives $\mathcal{J}'(y) = -0.0671813667518 \pm 10^{-14}$. So for small $t>0$ the derivative of $U_t((0,y))$ at $y = 1.015\pi$ is $-\frac{t}{\pi}\mathcal{J}'(1.015\pi) + O(t^2) < 0$, and for small $t<0$ the derivative at $y = 2.47\pi$ is negative; both contradict monotonicity. □

The program `pakes_characters.py` scans $\mathcal{J}'$ at 399 points in each of the first four gaps $(\pi M^2, \pi(M+1)^2)$: in each gap $\mathcal{J}'$ is positive on a left part and negative on a right part, for $y/\pi$ in $[1.8925, 3.9925]$, $[5.5125, 8.9875]$, $[11.1, 15.9825]$ and $[18.6775, 24.9775]$, so every one of these gaps excludes both signs of $t$ on its own.

Remark 11.5.16. The variable $\mathcal{B}$ is a GGC, but its density $f$ is not HCM. Indeed $f$ is analytic in $\operatorname{Re} b > 0$ with the natural boundary $\operatorname{Re} b = 0$ (the discussion after Theorem 7.1.21), while an HCM function extends analytically to $\mathbb{C}\setminus(-\infty,0]$ by (F7).

The GGC property of $\mathcal{B}_g$ passes the filter wherever it has been tested, and it is the only property in this subsection that does; whether $\mathcal{B}_t$ fails to be a GGC for every $t<0$ is open (Problem 11.2). It is not a property of the zeros alone: for $t \ge 0.2$ the zeros of $\Xi_t$ are all real [419] and $R_t$ is the Laplace transform of a GGC (Theorem 7.2.16), yet $\mathcal{B}_t$ is not a GGC at $t = \frac{1}{5}$, 1 and 5.

### *11.5.5.4 Theta kernels, characters and the function of Davenport and Heilbronn*

The variable $\mathcal{B}_g$ can be formed for other Dirichlet series with a functional equation. Let $q \ge 3$, $p \in \{0,1\}$, and let $a = (a_n)_{n\ge1}$ be a real sequence of period $q$ with

$a_1 = 1$ and $a_{q-n} = (-1)^p a_n$ whose theta series

$$\vartheta_a(x) = \sum_{n\ge1} n^p a_n e^{-\pi n^2 x/q}, \qquad x > 0,$$

satisfies $\vartheta_a(1/x) = x^{p+1/2}\vartheta_a(x)$. Then

$$\xi(s,a) = \int_0^\infty \vartheta_a(x)\, x^{(s+p)/2}\, \frac{dx}{x} = \Big(\frac{q}{\pi}\Big)^{(s+p)/2}\Gamma\Big(\frac{s+p}{2}\Big)\sum_{n\ge1}\frac{a_n}{n^s}$$

is entire with $\xi(s,a) = \xi(1-s,a)$, and $K_a(v) = 2e^{(p+1/2)v}\vartheta_a(e^{2v})$ is an even kernel with $\xi(\frac12 + s, a) = \int_{\mathbb{R}} K_a(v)e^{sv}\,dv$. The real primitive characters $\chi_d$, $d$ a fundamental discriminant, are of this kind, with $q = |d|$ and $p = 1$ for $d < 0$; for $\chi_{-4}$, written $\chi_4$ in Section 8.5.7, $\vartheta_a(x) = \eta(ix)^3$ and $K_a$ is the kernel $K_4$ of Proposition 8.5.22. When $\vartheta_a > 0$, the variable $\mathcal{B}_a = \mathcal{B}_{K_a}$ has the density $b^{p/2-1}\vartheta_a(b)/\xi(0,a)$ and $\mathbb{E}\mathcal{B}_a^{s/2} = \xi(s,a)/\xi(0,a)$, and by the functional equation its reciprocal has the density $x^{(p-1)/2}\vartheta_a(x)/\xi(0,a)$. We test both. For $\chi_{-4}$ the reciprocal $1/\mathcal{B}_a$ is a GGC: it is $\frac{\pi}{2}$ times the exit time of Remark 8.5.23. Whether $\mathcal{B}_{\chi_{-4}}$ itself is a GGC is open, and HCM cannot decide it: the density of $1/\mathcal{B}_{\chi_{-4}}$ is not HCM (Theorem 4.3.20), so by (F2) neither is that of $\mathcal{B}_{\chi_{-4}}$.

Lemma 11.5.17. *Let* $S_a = \sum_{n\ge2} n^p|a_n|e^{-\pi(n^2-1)/q}$. *If* $S_a < 1$, *then* $\vartheta_a > 0$ *on* $(0,\infty)$.

*Proof.* For $x \ge 1$, $\vartheta_a(x) = e^{-\pi x/q}\big(1 + \sum_{n\ge2} n^p a_n e^{-\pi(n^2-1)x/q}\big) \ge e^{-\pi x/q}(1 - S_a) > 0$, and for $0 < x < 1$, $\vartheta_a(x) = x^{-p-1/2}\vartheta_a(1/x) > 0$. □

The certificate program encloses $S_a$ in ball arithmetic: it is at most 0.659 for the nine characters and the function of Davenport and Heilbronn below, so these kernels are positive. Positivity can fail: for $d = -19$ and $d = -40$ the program `pakes_filter.py` finds $e^{\pi/q}\vartheta_a(1) = -0.567$ and $-0.421$. For $d = -19 \equiv 5 \pmod{24}$ this is a case of Theorem 8.5.19, and kernels of this kind go back to Pólya's examples recalled in [49, Section 6.1].

The function of Davenport and Heilbronn [111] is $f_{\mathrm{DH}}(s) = \sum_n a_n n^{-s}$ with $a$ of period 5 and $(a_1,\dots,a_5) = (1, \varkappa, -\varkappa, -1, 0)$, where

$$\varkappa = \frac{\sqrt{10 - 2\sqrt5} - 2}{\sqrt5 - 1} = 0.2840790438404122960283.$$

With $\chi$ the character modulo 5 with $\chi(2) = i$, $f_{\mathrm{DH}} = \frac{1-i\varkappa}{2}L(s,\chi) + \frac{1+i\varkappa}{2}L(s,\bar\chi)$, and $\varkappa$ is chosen so that $(1+i\varkappa)/(1-i\varkappa)$ is the root number

$$\varepsilon(\chi) = 0.85065080835203993218 + 0.52573111211913360603\,i;$$

then $p = 1$ and $\vartheta_a(1/x) = x^{3/2}\vartheta_a(x)$. The coefficients are not multiplicative, since $a_6 = 1$ while $a_2 a_3 = -\varkappa^2$, so $f_{\mathrm{DH}}$ has no Euler product, and it has infinitely many zeros in $\operatorname{Re} s > 1$ [498, Chapter 10]. Zeros off the critical line inside the strip have been computed by Spira [484] and by Balanzario and Sánchez-Ortiz [28]; the program confirms two of them,

$$0.808517182456637 + 85.6993484853776\,i,$$
$$0.650830080609737 + 114.163342730757\,i,$$

where $|f_{\mathrm{DH}}| < 10^{-25}$. Modulo 5 there is no other function of the class built from $\chi$ and $\bar\chi$: a combination $\alpha L(s,\chi) + \beta L(s,\bar\chi)$ with $a_1 = 1$ whose completed function is $w$ times its reflection must have $w = \pm 1$ and $\beta = w\,\varepsilon(\chi)\alpha$, and the choice $w = -1$ gives the coefficients $(1, -1/\varkappa, 1/\varkappa, -1, 0)$, whose theta series satisfies $\vartheta(1/x) = -x^{3/2}\vartheta(x)$, so that it vanishes at $x = 1$ and takes both signs.

Table 11.19 gives the test for $\xi$, for nine real characters and for $f_{\mathrm{DH}}$. For the characters and for $f_{\mathrm{DH}}$ the moments are computed as for $\mathcal{B}_t$, from the kernels $K_a$, and at $\lambda = 0$ also from the values of $L$ functions alone: $\mathbb{E}\mathcal{B}_a^k = \xi(2k,a)/\xi(0,a)$ and $\mathbb{E}\mathcal{B}_a^{-k} = \xi(2k+1,a)/\xi(0,a)$, with $\sum_n a_n n^{-s} = q^{-s}\sum_{r=1}^{q} a_r \zeta(s, r/q)$ through the Hurwitz zeta function. The two routes agree, and the failures marked in the table are certified, by the first route for $\lambda > 0$ and by the second at $\lambda = 0$.

| function | $S_a$ | $\mathcal{K}_7'(0)$ | $\mathcal{K}_7'(2)$ | $\mathcal{B}_a$ | $1/\mathcal{B}_a$ |
|---|---|---|---|---|---|
| $\xi$ | | $6.2\times10^{-16}$ | $1.2\times10^{-13}$ | GGC | fails at $\mathcal{K}_3'(0)$ |
| $\chi_{-3}$ | 0.086 | $2.4\times10^{-8}$ | $4.9\times10^{-8}$ | passes | GGC |
| $\chi_{-4}$ | 0.006 | $1.2\times10^{-8}$ | $4.6\times10^{-8}$ | passes | GGC |
| $\chi_5$ | 0.158 | $7.8\times10^{-8}$ | $7.1\times10^{-8}$ | passes | fails at $\mathcal{K}_{10}'(0)$ |
| $\chi_8$ | 0.043 | $4.8\times10^{-8}$ | $3.8\times10^{-8}$ | passes | fails at $\mathcal{K}_7'(2)$ |
| $\chi_{12}$ | 0.002 | $2.1\times10^{-8}$ | $4.2\times10^{-8}$ | passes | fails at $\mathcal{K}_9'(10)$ |
| $\chi_{24}$ | 0.045 | $7.5\times10^{-9}$ | $2.0\times10^{-8}$ | fails at $\mathcal{K}_{11}'(2)$ | fails at $\mathcal{K}_6'(0)$ |
| $\chi_{-7}$ | 0.608 | $-0.149$ | $-0.221$ | fails at $\mathcal{K}_3'(0)$ | fails, zero |
| $\chi_{-8}$ | 0.130 | $-0.057$ | $-0.477$ | fails at $\mathcal{K}_2'(0)$ | ID, fails, zero |
| $\chi_{13}$ | 0.659 | $-0.0023$ | $-0.256$ | fails at $\mathcal{K}_3'(0)$ | fails at $\mathcal{K}_3'(0)$ |
| $f_{\mathrm{DH}}$ | 0.092 | $-1.2\times10^{-5}$ | $-1.4\times10^{-4}$ | fails at $\mathcal{K}_6'(0)$ | fails, zero |

Table 11.19: The GGC test for the variables $\mathcal{B}_a$ of Section 11.5.5.4 and their reciprocals: the sum $S_a$ of Lemma 11.5.17, the least eigenvalues of the scaled matrices $\mathcal{K}_7'(0)$ and $\mathcal{K}_7'(2)$ of $\mathcal{B}_a$, and the verdicts. "Passes": no matrix $\mathcal{K}_n(\lambda)$ or $\mathcal{K}_n'(\lambda)$ with $n \le 12$ has a negative eigenvalue at $\lambda = 0$, $\frac12$, 2, 10, 20, 50 and 200. "Fails at $\mathcal{K}_n'(\lambda)$": the least failing order, certified at that $\lambda$. "Zero": the Laplace transform vanishes on $(-\infty, 0)$ (Proposition 11.5.19). GGC for $\xi$ by (11.5.8), for $1/\mathcal{B}_a$ with $\chi_{-3}$ and $\chi_{-4}$ by Proposition 11.5.19. All the functions but $f_{\mathrm{DH}}$ have an Euler product. Computed by `pakes_filter.py` and `pakes_filter_cert.py`.

Proposition 11.5.18. *Neither $\mathcal{B}_{\chi_{-8}}$ nor $1/\mathcal{B}_{\chi_{-8}}$ is a GGC.*

*Proof.* From the values of $L(s,\chi_{-8})$ at $s=0,2,4,6,8$ and at $s=3,5,7,9$, the certificate program gives

$$3\kappa_3^2-2\kappa_2\kappa_4=7.55465911093\pm4\times10^{-12}\qquad\text{and}\qquad 78.8854168657\pm2\times10^{-11}$$

for $\mathcal{B}_{\chi_{-8}}$, where $(\kappa_2,\kappa_3,\kappa_4)=(3.0767,15.997,123.54)$, and for its reciprocal. Both are positive, against Lemma 11.5.14. □

The same four numbers settle $1/\mathcal{B}_{\chi_{-7}}$ (27.8926758865) and $1/\mathcal{B}_a$ for $f_{\mathrm{DH}}$ (0.738104955084). So $L(s,\chi_{-8})$, which has an Euler product and whose zeros lie on the critical line to height $10^8/8$ at least [417], fails the test already at order two, on both sides.

Proposition 11.5.19. *Let $p=1$ and $\vartheta_a>0$, and put $w=\sqrt{\pi q\lambda}$ and $\Sigma_a(w)=\sum_{1\le n<q/2}a_n\sinh\frac{(q-2n)w}{q}$. Then*

$$\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx=\frac{\sqrt{q}\,\Sigma_a(w)}{\sinh w},\qquad \mathbb{E}e^{-\lambda/\mathcal{B}_a}=\frac{\sqrt{q}\,\Sigma_a(w)}{\xi(0,a)\sinh w},\tag{11.5.10}$$

*and $\xi(0,a)=q^{-1/2}\sum_{1\le n<q/2}(q-2n)a_n$. In particular:*

*(i) for $\chi_{-4}$, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=\operatorname{sech}\sqrt{\pi\lambda}$, and $1/\mathcal{B}_a$, which is $\frac{\pi}{2}$ times the exit time of Brownian motion from $(-1,1)$, is a GGC;*

*(ii) for $\chi_{-3}$, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=3\sinh(w/3)/\sinh w=\prod_{3\nmid m}\bigl(1+3\lambda/\pi m^2\bigr)^{-1}$, so $1/\mathcal{B}_a$ is a GGC with Thorin measure $\sum_{3\nmid m}\delta_{\pi m^2/3}$, and it is $\frac{3\pi}{2}$ times the hitting time of 1 by a three dimensional Bessel process started at $\frac13$;*

*(iii) for $\chi_{-8}$, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=\cosh(w/4)/\cosh(w/2)=\prod_{k\ge1}(1+\lambda/4\alpha_k)/(1+\lambda/\alpha_k)$ with $\alpha_k=\frac{\pi}{8}(2k-1)^2$, so $1/\mathcal{B}_a$ is infinitely divisible, with Lévy density $\sum_k(e^{-\alpha_k x}-e^{-4\alpha_k x})/x>0$, and it is $\pi$ times the hitting time of 1 by a reflected Brownian motion started at $\frac12$; it is not a GGC, because its Laplace transform vanishes at $\lambda=-\pi/2$;*

*(iv) if $a=\chi_d$ with $d<0$ and $\chi_d(2)=1$, that is $d\equiv1\pmod8$, the Laplace transform of $1/\mathcal{B}_a$ vanishes at a point of $(-4\pi/q,-\pi/q)$, so $1/\mathcal{B}_a$ is not a GGC; for $\chi_{-7}$ the point is $\lambda=-1.033610573$. The same holds for $f_{\mathrm{DH}}$, at $\lambda=-2.046575671$.*

*Proof.* By the functional equation $\vartheta_a(x)=x^{-3/2}\sum_n na_ne^{-\pi n^2/qx}$, and

$$\int_0^\infty e^{-\lambda x-c/x}x^{-3/2}\,dx=\sqrt{\pi/c}\,e^{-2\sqrt{c\lambda}}$$

with $c=\pi n^2/q$ gives $\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx=\sqrt{q}\sum_{n\ge1}a_ne^{-2nw/q}$ for $\lambda>0$. Summing over a period and pairing $n$ with $q-n$, where $a_{q-n}=-a_n$, gives the first formula

of (11.5.10), which is [49, (96)] in our normalisation; at $\lambda = 0$ it gives $\int_0^\infty \vartheta_a = \xi(1,a) = \xi(0,a)$, and since $1/\mathcal{B}_a$ has the density $\vartheta_a/\xi(0,a)$ the second formula follows. (i) and (ii) follow from the product formulas for cosh and sinh, and [49, (92)] identifies the transforms in (ii) and (iii) with those of the hitting times. In (iii) Frullani's integral $\log(1+\lambda/c) = \int_0^\infty (1-e^{-\lambda x})e^{-cx}\,dx/x$ gives the Lévy density. A GGC transform $\ell$ cannot vanish on $(-\infty,0)$ where it extends meromorphically: for $0<\varepsilon\le 1$, $|\ell(-x+i\varepsilon)| \ge e^{dx-C}$ with $C = \int \log(1+(x+1)/z)\,U(dz) < \infty$, because $|1+(-x+i\varepsilon)/z| \le 1+(x+1)/z$. In (iii) the zero is that of $\cosh(w/4)$ at $w = 2\pi i$. In (iv) put $w = iy$; then $\Sigma_a(iy)/\sinh(iy) = \nu(y)/\sin y$ with $\nu(y) = \sum_{1\le n<q/2} a_n \sin\frac{(q-2n)y}{q}$, and since $a_{q-n} = -a_n$, $\nu(\pi) = \frac12\sum_{n=1}^{q-1} a_n \sin\frac{2\pi n}{q}$ and $\nu(2\pi) = -\frac12\sum_{n=1}^{q-1} a_n \sin\frac{4\pi n}{q}$. For an odd real primitive character the Gauss sum is $i\sqrt{q}$, so $\nu(\pi) = \frac{\sqrt{q}}{2}$ and $\nu(2\pi) = -\chi_d(2)\frac{\sqrt{q}}{2}$. If $\chi_d(2) = 1$, $\nu$ vanishes at some $y_0 \in (\pi, 2\pi)$, where $\sin y_0 \ne 0$, and the transform vanishes at $\lambda = -y_0^2/\pi q$. For $f_{\mathrm{DH}}$, $\nu(\pi) = \frac{\sqrt5}{2}$ and $\nu(2\pi) = -0.317610026505$. □

A survey extends the table. For each of the 26 fundamental discriminants $d$ with $3 \le |d| \le 40$, the program `pakes_filter.py` tests the positivity of $\vartheta_{\chi_d}$ and, where it holds, runs the test at 80 digits to order 10 at $\lambda = 0$, $\frac12$, 2, 10 and 20, on both sides. Of the 24 positive kernels, $\mathcal{B}_{\chi_d}$ passes only for $d = -4$, $-3$, 5, 8, 12 and 24 (the last fails at order 11, beyond the survey), and $1/\mathcal{B}_{\chi_d}$ only for $d = -4$ and $-3$. The 18 failures of $\mathcal{B}_{\chi_d}$ occur at order 2 or 3 for $d = -39$, $-35$, $-31$, $-24$, $-23$, $-20$, $-15$, $-11$, $-8$, $-7$, 13, 21, 29 and 37, and at orders 6 to 8 for $d = 17$, 28, 33 and 40; the failures of $1/\mathcal{B}_{\chi_d}$ for $d = -7$, $-15$, $-23$, $-31$ and $-39$ are cases of Proposition 11.5.19(iv).

Combinations of $L$ functions do not change the picture. Modulo 7 the odd characters are $\chi_{-7}$ and a complex pair $\chi$, $\bar\chi$, with $\chi(3) = e^{\pi i/3}$, and for real $r$

$$f_r = \big(1 - 2r\cos\tfrac{\phi}{2}\big)L(s,\chi_{-7}) + re^{-i\phi/2}L(s,\chi) + re^{i\phi/2}L(s,\bar\chi), \qquad e^{i\phi} = \varepsilon(\chi),$$

has real coefficients, $a_1 = 1$ and $\vartheta(1/x) = x^{3/2}\vartheta(x)$. For 17 values of $r$ between $-3$ and 3 the kernel is positive for $-3 \le r \le 2$ and not for $r = 3$, and for each of the 16 positive kernels both $\mathcal{B}$ and $1/\mathcal{B}$ fail at $\lambda = 0$, at order 2 or 3. None of the characters whose variable passes has a complex character of the same conductor and parity to combine with, so the smallest family through a passing character would need a larger modulus.

The pattern is not that of the Euler product. All nine $L$ functions of Table 11.19 have one, and GRH has been verified for them to height at least $10^8/q$ [417]; yet $\mathcal{B}_{\chi_d}$ fails for $\chi_{24}$, $\chi_{-7}$, $\chi_{-8}$ and $\chi_{13}$, and for 18 of the 24 positive kernels of the survey. It fails for $f_{\mathrm{DH}}$, which has zeros off the line, but it fails in the same way for $\chi_{-7}$ and $\chi_{13}$, whose zeros are on the line as far as they have been computed. At $\lambda = 0$ the test of order $n$ reads only the moments $\xi(2k,a)/\xi(0,a)$ with $1 \le k \le 2n$, whose numerators are values of Dirichlet series where they converge

absolutely, and for $\lambda > 0$ it reads Laplace transforms of the theta series: it sees the shape of the theta transform and not the arithmetic of the coefficients. Where the GGC property is proved, it comes from a product formula for that transform, $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$ for $\xi$ and the products of Proposition 11.5.19(i) and (ii) for the reciprocals of $\chi_{-4}$ and $\chi_{-3}$. The computations find no function with $\mathcal{B}_a$ a GGC and a zero off the line.

### 11.5.6 What survives, and open problems

Apart from the Lambert template of Section 11.5.3, the results of this section, and those of the other chapters that it uses, fall into three classes. The first consists of equivalences with RH: the Thorin Bernstein property of $\log G$ (Theorem 4.2.4), the mixture of exponential laws in the size biased clock (Proposition 11.5.1), the vanishing of the Hankel inertia indices (Theorem 5.1.7), and the GGC property of $R_t$ at $t = 0$ (Theorem 7.2.16 and Proposition 1.2.1). The second consists of facts that hold with no hypothesis: the clock is infinitely divisible and self decomposable (Theorem 5.1.1), it lies in $L_2$ and not in $L_3$ (Theorem 11.5.8), and its driving subordinator has self decomposable, hence unimodal, laws (Corollary 11.5.9); the points of the upper half plane where $\mathrm{Im}(G'/G)$ could fail to be negative lie in a thin parabola beyond about $-T_0^2$ (Theorem 4.3.9); the clock is not the martingale limit of a supercritical continuous state branching process (Theorem 11.5.11); $R_t$ is a GGC for $t \geq 0.2$ [419] and for no $t < 0$ (Corollary 4.2.5); and $\mathcal{B}_t$ is not a GGC for $t$ near 0, $t \neq 0$ (Proposition 11.5.15). Being unconditional does not by itself prevent a statement from implying RH, since a proof of RH would be unconditional too. What prevents it is that none of these facts distinguishes $\xi$ from functions with zeros off the line: the statements about the flow at $t \neq 0$ are removed by the filter of Section 11.5.5, infinite divisibility and self decomposability hold as well for the comparison function $F$ of Section 12.2.6, the ladder index and the branching obstruction are set by the first zeros (Section 11.5.4), and the localisation uses only the zeros below the verification height. The third class consists of the routes that are closed. The Hankel route is closed for practical purposes, on the heuristic of Table 7.6. Properties of the kernel and its representations as mixtures are removed by the filter (Propositions 11.5.12 and 11.5.13), and the GGC property of the variable of Biane, Pitman and Yor does not detect the Euler product (Section 11.5.5.4).

#### *11.5.6.1 Open problems*

Problem 11.1. Find a natural class of functions $\xi(s, a)$ with positive kernels, containing $\xi$, in which the GGC property of $\mathcal{B}_a$ implies that every zero of $\xi(s, a)$ lies on the critical line, or a function in such a class whose variable $\mathcal{B}_a$ is a GGC and which has a zero off the line. Since $\mathcal{B}$ is a GGC, the first answer for a class containing $\xi$ proves RH. No candidate for the second is known: the function of

Davenport and Heilbronn, the family $f_r$ modulo 7 at the 16 tested values of $r$ with a positive kernel and the tilts $\mathcal{B}_t$ at the eleven tested values of $t < 0$ all fail the test. The question is related to Question 8.7.20 and to the structural reason for $R$ to be a GGC that the discussion of Table 7.6 asks for.

PROBLEM 11.2. Is $\mathcal{B}_t$ a GGC for no $t \neq 0$? For $t < 0$ this would be a property of the kernel that passes the filter at every $t$. Proposition 11.5.15 proves it for $0 < |t| < t_0$, through the first order change of the Thorin measure in the gaps between its atoms, and at the ten values of part (i).

PROBLEM 11.3. Which of the variables $\mathcal{B}_{\chi_d}$ for $d = -4$, $-3$, 5, 8 and 12, which pass the test to order 12, are GGCs? (For $d = 24$ the test fails at order 11.) On the reciprocal side, $1/\mathcal{B}_{\chi_d}$ is a GGC for $d = -3$ and $-4$ (Proposition 11.5.19); for $d = -4$ it is the exit time variable of Remark 8.5.23, whose density is not HCM (Theorem 4.3.20), so the question for $\mathcal{B}_{\chi_{-4}}$ cannot be settled through HCM. For even characters the reciprocal transform is the series $\frac{2}{\xi(0,a)} \sum_n a_n K_0\big(2n\sqrt{\pi\lambda/q}\big)$ of Bessel functions, and no product formula for it is known to us.

PROBLEM 11.4. With no hypothesis, the Li coefficients of Section 8.3.2 are

$$\lambda_n = \int_0^\infty k_n(t) W(t)\, dt = m_0\, \mathbb{E} k_n(A),$$

where $W$ is the heat trace, $A$ is the variable of Proposition 11.5.1, and $k_n$ is the function with $\int_0^\infty e^{-zt} k_n(t)\, dt = \varphi_n(z)$. Since $\varphi_n$ is a polynomial of degree $n$ in $(z+\frac{1}{4})^{-1}$ with no constant term, $k_n(t) = e^{-t/4} p_n(t)$ with $p_n$ a polynomial of degree $n-1$; for instance $k_1(t) = e^{-t/4}$ and $k_2(t) = (4-t)e^{-t/4}$. Which unconditional properties of $W$, or of $A$, force $\mathbb{E} k_n(A) \geq 0$ for every $n$, which is RH by Li's criterion [336, 54]? Positivity of $W$ (Theorem 5.1.1) gives $n = 1$; complete monotonicity of $W$, which is RH, gives every $n$, since $\varphi_n \geq 0$ on $(0, \infty)$.

PROBLEM 11.5. Describe the ladder index of $R_t$ along the de Bruijn Newman flow. The computation of Section 11.5.4.3 finds the index two of the clock up to $t = 27.8$ and one from $t = 27.9$ on, on its window. Is the index nonincreasing in $t$, and is it one for all large $t$? The answer carries no information about RH, but it would show how the spacing of the first zeros sets the index.

# Chapter 12

# Conclusions and open problems

The results of Chapters 1 to 11 can be placed on a single path, which runs along the basepoint from the region where positivity of the Thorin measure is a theorem to the centre, where it is RH.

Chapter 13 follows as a complement on the side of statistical mechanics, where RH is the statement that the Pólya law is ferromagnetic.

## 12.1 The Thorin approach in summary, and a path to RH

The path is a programme with its gap identified, not a proof.

### 12.1.1 The path along the basepoint

Figure 12.1 and Table 12.1 set out the path.

Section 11.3.4 draws the two conclusions that fix the shape of the path. At the basepoint one the constructions attached to the reciprocal Xi function are unconditional and explicit (Sections 9.1, 9.2 and 10.1, Table 11.3), with two exceptions: the Krein string representation of Corollary 9.2.3 assumes RH and simple zeros, and Lemma 9.2.4 gives the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$ only for a boundary bounded at infinity, while the boundedness of the computed boundary is observed, not proved. Below the basepoint one each construction holds, up to a boundary case, exactly when no zero lies to the right of $\alpha$ (Theorems 9.2.5 and 10.1.3, Propositions 9.1.13 and 10.1.12), and the theory of Sato and Bondesson reaches nothing there (Section 11.1.5).

### 12.1.2 The open steps

By Theorem 1.2.2, that is [424, Theorem 3.7 and Remark 3.10], step 7 is exactly a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < 1$, a quasi Riemann hypothesis. Until 2026 no such half plane was known. In October 2026 OpenAI released a proof, produced with one of its AI models, that $\zeta(s) \neq 0$ for $\operatorname{Re} s > \frac{7}{8}$, and the same for every Dirichlet $L$ function [398], together with a formal proof of both statements in the

| step | statement | status |
|---|---|---|
| 1 | The clock is infinitely divisible and self decomposable; the heat trace is positive and decreasing | proved (Theorem 5.1.1, which is Theorem A of [424]), with the verification of RH below height 1000 as input |
| 2 | The heat trace is log convex, its leading Hankel blocks of size up to $3 \times 10^7$ are positive definite, derivative signs hold to order $1.16 \times 10^{25}$ | proved, with verification height as input |
| 3 | The reciprocal at basepoint $\alpha \geq 1$ is a GGC with Poisson Thorin density, and at $\alpha = 1$ its Lévy form, hitting time, scale mixture and completeness are explicit; for every $\alpha > \frac{1}{2}$ the shifted clock $T_\alpha$ exists and is an almost GGC | proved (Theorems 1.2.2, 9.2.5 and 11.4.2; Chapters 9 to 11), with the verification of RH below height 1000 as input to Theorem 9.2.5 |
| 4 | The normalised reciprocal zeta function is infinitely divisible; the canonical function of the floor of the reciprocal Xi function is an arithmetic part plus a completely monotone archimedean part, and RH is the complete monotonicity of the sum | proved (Chapters 6 and 7) |
| 5 | The Thorin density is computable at every resolution, with an explicit bound for each source of error, and is positive at every sample point of every window computed; the runs do not yet add these bounds to the ball radii (Section 12.2.2) | computed (Chapters 2 to 5) |
| 6 | Zero density $\nu(\frac{1}{2}+\varepsilon, 200) = 0$ for every computed $\varepsilon$, down to $\varepsilon = 0.005$ | computed (Section 3.4) |
| 7 | For some $\alpha < 1$ the shifted clock $T_\alpha$ is a GGC, equivalently, the half plane $\operatorname{Re} s > \alpha$ is zero free (Theorem 1.2.2); then for every $\alpha' > \alpha$ the measure $U_{\alpha'}$ is the Thorin measure of $T_{\alpha'}$ and is positive at all heights | announced for $\alpha = \frac{7}{8}$ in October 2026, with a formal proof in Lean, not yet refereed (Section 12.1.2); open for every $\alpha < \frac{7}{8}$ |
| 8 | Positivity of $U_\alpha$ for every $\alpha > \frac{1}{2}$ (Lagarias), equivalently the Thorin condition | open; equivalent to RH |

Table 12.1: The path, step by step.

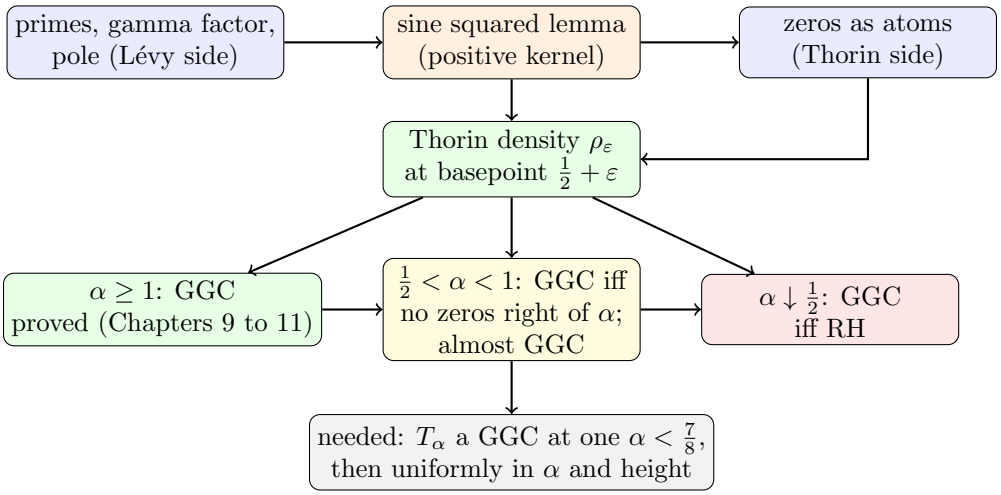



Figure 12.1: The Thorin path to RH. The sine squared lemma connects the arithmetic to the Thorin density. The programme must carry the GGC property across the strip.

Lean proof assistant and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [399]. The argument proves a power saving for smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$, and it uses nothing from the Thorin picture. It has not yet been refereed. We rebuilt the formal proof from the public repository (commit `adc7f12`, Lean 4.34.1, Mathlib `d13f23b`): the Lean kernel accepts both proofs, which use only the axioms `propext`, `Classical.choice` and `Quot.sound`, and the comparator of the Lean Focused Research Organization, run without its sandbox, confirms that the theorems proved are the published challenge statements. If it stands, step 7 holds with $\alpha = \frac{7}{8}$, and the constructions of this book follow. By Theorem 1.2.2 the shifted clock $T_\alpha$ is a GGC for every $\alpha \geq \frac{7}{8}$, with Thorin density (1.2.6) and $v_\alpha > 0$ for $\alpha > \frac{7}{8}$, while at $\alpha = \frac{7}{8}$ each zero $\frac{7}{8} + i\gamma$ with $\gamma > 0$, which the announcement does not exclude, would add an atom at $\gamma^2$ (Theorem 7.1.14); for $\alpha > \frac{7}{8}$ the clock $T_\alpha$ is a Brownian first passage time plus an independent GGC with no drift, because the first two steps of the proof of Proposition 9.1.12 go through for $\frac{7}{8} < \alpha < 1$, where every zero has $0 < \alpha - \beta < \alpha$, with smallest margins 0.0137 and 0.0157 in ball arithmetic (`valpha_min.py`); by Proposition 9.1.13 the compensated Lévy form converges for $\frac{7}{8} < \alpha < 1$, with error $O(X^{7/8-\alpha+\delta})$ at prime cutoff $X$ for every $\delta > 0$, and the same holds for $L(s, \chi_{12})$ (Section 11.3.2); by Theorem 10.2.7 Riesz's function satisfies $R(x) = O(x^{7/16+\delta})$ for every $\delta > 0$; and by de Bruijn's theorem $\Lambda \leq \frac{9}{32}$ (Section 8.4), weaker than the bound 0.2, while for the cosine tilts of Section 13.5.2 his theorem for cosine factors gives $\Omega \leq \frac{3}{8}$, against $\frac{1}{2}$ with no hypothesis. Below $\frac{7}{8}$ nothing changes. Positivity of the Thorin density on the single line $\operatorname{Re} s = \alpha$ is a weaker statement

than a zero free half plane (Remark 1.2.4). So the first theorem the Thorin path needs beyond the announcement, a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < \frac{7}{8}$, is an open problem in its own right; the Lévy form, the scale mixture, completeness and Riesz's criterion (Sections 9.1, 10.1 and 10.2) restate it and do not get round it. The Thorin formulation offers two specific tools for it. The sine squared form, compensated as in Proposition 9.1.13, writes the required positivity as domination of the pole piece by the prime and archimedean pieces at basepoint $\alpha$, a statement about a single explicit function of $\theta$ for each $\alpha$; and the Poisson form shows that a zero to the right of $\alpha$ appears as a Cauchy kernel of negative width, whose signature Section 3.3 calibrates exactly.

Step 8 carries the GGC property from any fixed $\alpha$ to every $\alpha > \frac{1}{2}$. Weak limits cannot do this on their own (Section 4.1), because the limit of the Cauchy flow is positive whether or not RH holds. Three mechanisms could do it, each consistent with everything computed in this book:

1. a positive approximating family whose positivity is proved independently, such as zeta functions of Beurling generalised prime systems that provably have only real ordinates, finite Krein strings, or the partial products over verified zeros completed by the Weyl tail; once the first $N$ zeros are verified, RH is equivalent to the GGC property of the tail factor left when their factors are removed from the clock [424, Corollary 18.11];

2. an operator whose spectrum is the set of ordinates, where positivity would follow from self adjointness, the Hilbert Pólya idea in Thorin form (Section 8.1.3);

3. a direct inequality for the variogram pairing that controls the pole by the primes and the gamma factor at every scale, the Weil form of step 8.

Spin chains give finite approximations of another kind: the partition functions of Knauf's chain are Dirichlet polynomials that converge to $\zeta(\beta-1)/\zeta(\beta)$ for $\operatorname{Re}\beta > 2$, and by Remark 13.7.4 the convergence of their corrected form on a strip in $1 < \operatorname{Re}\beta < 2$ would carry their zeros to the zeros $1+\rho$ of $\zeta(\beta-1)$ in it, a consequence that does not decide RH, while its convergence on $\operatorname{Re}\beta > \frac{1}{2}$ without the point 2, where the zeros of $\zeta(\beta)$ are poles of the limit, would imply it (Section 13.7).

### 12.1.3 What computation contributes

Computation cannot supply the uniformity in the height that steps 7 and 8 require: neither finitely many heat trace tests (Theorem 5.2.1) nor positivity at a fixed resolution (Section 5.2.2) can exclude an off line zero above the verified height. What it can do is developed in Section 12.2.1. It calibrates the size of the positivity margins, which tells a proof how much room it has: the Thorin density at resolution

$\varepsilon$ has minimum $0.0147\,\varepsilon$ on the window below height 200 (Section 3.1), a margin that vanishes linearly and must be controlled uniformly. That this margin is the value at the origin is a theorem on the whole verified window, and at every height for every $\varepsilon < \frac{1}{2}$ it is equivalent to RH (Theorem 9.1.18).

Section 3.5 gives the variogram form of the domination inequality, on which the linear programming formulation of [424, Sections 18.2.8 and 18.2.9] works, the most concrete target for step 8. Both are posed at the centre, where the variogram form is Weil's criterion and the linear program asks whether the clock $1/G$ is a GGC, so each is equivalent to RH. Section 4.1 and [424, Section 18.2.3] give the closure criterion in its weakest form, the target for approximating families in step 8. Section 11.4 closes with four questions between almost and exact positivity (Section 11.4.7), the last of which is step 7 below $\frac{7}{8}$, the GGC property at a single basepoint $\alpha < \frac{7}{8}$.

## 12.2 Open problems and lessons for the Thorin programme

The programme of computations below is aimed at the gap that Section 12.1 located, positivity uniform in the height, and the lessons that follow it are drawn for anyone working on the Thorin route to RH, including the first author's own programme [424]. The open problems on the reciprocal zeta function are stated in Section 7.2.4, those on the basepoint one and the critical strip in Section 11.4.7, and those of Pakes's framework in Section 11.5.6.1.

### 12.2.1 What computation can and cannot do on the path

Computation cannot supply uniformity in height (Section 12.1.3); what it can do falls under four aims. It can extend and certify: prove positivity of the Thorin density on explicit windows, at every resolution, with interval arithmetic. It can test intermediate conjectures: the four questions of Section 11.4.7 all have numerically testable consequences. It can calibrate: measure how much room positivity has, and compare with random matrix predictions, so that a proof knows which inequalities are sharp. And it can cross check: compute the same quantities from the primes, through the criteria of Riesz and Salem and the compensated Lévy form, independently of evaluations of $\zeta$.

Five work packages follow: twisted Riesz functions as an arithmetic filter bank, the cross check (Section 10.2.1); certified Thorin densities at height (Section 12.2.2); the positivity margins against the circular unitary ensemble (Section 11.4.5); explicit tables of the almost GGC defect (Section 11.4.2); and testing stronger conjectures before pursuing them (Section 12.2.4).

### 12.2.2 Certified Thorin densities at height

At resolution $\varepsilon = \alpha - \frac{1}{2}$ the Thorin density $\rho_\varepsilon(\theta) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\theta)$ is a smooth function of $\theta$. Its positivity on a window of ordinates is what the absence of zeros with $\beta > \alpha$ predicts there; since an off line zero need not make the density negative (Section 5.2.2), a certificate for the window should also count such zeros. Two computations do this. The first evaluates $\rho_\varepsilon$ on a grid with rigorous error bounds, using the ball arithmetic Riemann Siegel and Euler Maclaurin evaluators of Arb [290], exposed in python flint, at moderate height and the Pólya route where it is faster, and bounds the variation between grid points by the derivative. The second counts the zeros to the right of the line $\operatorname{Re} s = \alpha$ through the zero density identity $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac{1}{2} + \varepsilon, T)$ of Proposition 3.4.2. It is the count of Section 3.4, which at $\varepsilon = \frac{1}{2}$ is Backlund's method. Section 3.4.3 followed the argument in steps of $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) without a bound on its change between grid points, so its counts are computed, not certified; with such a bound the count would certify a whole window. The deliverables are certified positivity of $\rho_\varepsilon$ for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ on windows of length $10^4$ at heights $10^6$, $10^8$ and $10^{10}$, together with the minimum density on each window, which feeds the margin statistics of Section 11.4.5. Below the verified height $3 \times 10^{12}$ the positivity itself needs no new computation: an off line pair makes the density negative only within $\frac{1}{2}$ of its ordinate, so $\rho_\varepsilon > 0$ on $[0, 3 \times 10^{12} - \frac{1}{2}]$ for every $\varepsilon > 0$ (Corollary 11.4.7); there the certificates test the evaluators against the verification and measure the margins.

Below height 200 the certificate is within reach now. The ball arithmetic of Section 2.1 certifies the rounding only. Adding the discretisation bound of Section 2.1.1 and of Proposition 2.1.1, bounds for the truncations of the theta series and of the $u$ axis, a bound on the change of the argument between grid points, and an explicit bound on the remainder of Proposition 3.2.2, which is $O(\varepsilon^3)$ (Remark 3.2.3), turns positivity of $\rho_\varepsilon$ on $0 < z \le 200^2$ at each computed $\varepsilon$, the integer identity of Proposition 3.4.2 and the HCM derivative signs of Section 4.1.2 into theorems on that window.

Above that window the evaluator changes. The tilted contour gives 25 digits at height 5000 and resolves the Lehmer pair near height 7005 (Sections 2.1.3 and 2.2.3.1), but its number of nodes grows linearly with the height. Chirp transform batching of the Pólya route, and the hybrid of Section 2.2.2, a Riemann Siegel or Odlyzko Schönhage evaluator at $\frac{1}{2} + \varepsilon + i\theta$ with Odlyzko's zeros as the RH reference, would give the positivity margin, the matched total variation and the zero density scan near the $10^{12}$th [391], $10^{21}$st and $10^{22}$nd zeros [395, 396]. The same zeros would measure the cell constant of Section 5.4 at large height, whose behaviour is open (Section 5.4.2), and allow a comparison of the cell errors with the prediction of the sine kernel process of the unitary ensemble.

### 12.2.3 Calibration against known configurations

The diagnostics of Chapters 3 to 5 are calibrated when they are run on configurations whose zeros are known exactly, and three such computations remain. Section 8.6 ran the posterior predictive check of [424, Section 19.3.7] at $\varepsilon = 0.1$ on one window and calibrated it with a displaced zero. Three extensions are open: smaller resolutions, where the signature of a displaced zero is largest and the Pólya route supplies $\rho_\varepsilon$ to about 60 digits below height 200 (Section 2.1.1), whereas the deconvolution used double precision; continuous atom positions, which would remove the baseline of the grid; and larger heights. The square root law (3.3.1) of Section 3.3.2, which follows from Proposition 3.3.1, sets what the check can detect.

The de Bruijn Newman flow of Section 8.4 manufactures off line pairs in a controlled way. Running these Thorin diagnostics along the flow through the collision of the Lehmer pair near 7005 would calibrate the zero density scan, the matched total variation and the negative mass against configurations that are known exactly.

Generalised prime systems of Beurling whose zeta functions provably have only real ordinates give positive Thorin measures by construction. Measuring how fast their clock transforms approach $1/G$, in the distances of Section 3.2 and along a sequence with an accumulation point as the closure criterion of Corollary 4.1.2 [424, Propositions 18.10 and 18.14] allows, is the computational side of that criterion; the Kent approximants of Section 8.1.3 are the first instance.

Two computations of Chapter 8 remain at their edges. The Brownian boundary of Section 8.2 reproduces the law of the clock in the bulk, to 0.0020 in Kolmogorov distance for the boundary of the Volterra equation and to 0.0059 for the tangent approximation of Roberts and Shortland (Section 8.2.2.2), but the zone of large times $t$ is checked by no simulation, since only about one path in ten million survives past $t = 0.1$. Importance sampling with an exponentially tilted drift, or multilevel splitting, would check that zone and the logarithmic correction of the asymptote of the boundary directly. The Cauchy integral method of Section 8.3 extends to $n$ in the thousands, at a cost of about $0.03n$ digits lost to cancellation. Comparing the Pólya values of the Keiper Li coefficients with the sums $\sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$ over the zeros to large $n$ would test how the Li functional sees the zeros near height $n$.

### 12.2.4 Testing stronger conjectures before pursuing them

Conrey and Li showed that a positivity condition in de Branges' approach to RH is false even though RH may be true [90]. Any intermediate statement proposed on the Thorin path should therefore be tested numerically first. The candidates are a lower bound $\rho_{\alpha-1/2}(\theta) \geq c(\alpha) \log \theta$ outside a sparse set, the first of the four questions of Section 11.4.7; monotonicity or convexity of the margin in $\alpha$; positivity of the Thorin density for $\alpha$ slightly below one, uniformly on large windows;

hyperbolic complete monotonicity of the Laplace transform $\varphi_\alpha$ of $T_\alpha$ below one; and variogram inequalities at single scales. The first two are partly settled: the lower bound holds for $\rho_{\alpha-1/2}$ itself outside a set of heights of density zero, although not near the zeros where Lemma 11.4.9 would need it (Remark 9.1.17), and the margin divided by $\alpha - \frac{1}{2}$ decreases in $\alpha$, with the ratio $\rho_{\alpha-1/2}(\theta)/(\alpha - \frac{1}{2})$ nonincreasing in $\alpha$ at every height outside the windows of the zeros off the line (Theorems 9.1.15 and 9.1.18). Each can be evaluated on the computed windows and at the heights of Odlyzko's tables, and any statement that fails there should be discarded.

The variogram inequalities can be tested in finite dimension. The reformulation of Section 3.5 poses finite dimensional feasibility problems: is the regularised pole mass dominated by the prime and archimedean mass on the cone of variograms of positive definite functions of a given dimension? Solving them with explicit dual certificates at increasing dimension would measure how the margin behaves.

The cross checks test the zeros themselves. Agreement of the twisted Riesz functions (Proposition 10.2.1) with the explicit formula over the computed zeros is a check that shares no code or method with the computation of the zeros (Remark 10.2.2).

### 12.2.5 Open problems from free probability

The note [433] closes with eight problems, and five of them are settled or taken up in the book. They are settled by Theorem 5.1.1 through Theorem 11.2.3, by Propositions 11.2.7 and 11.2.9 with Theorem 11.2.10, or taken up in Sections 8.4.3 and 8.4.5. is the following.

Problem 12.1. Jensen polynomials as finite free products. The multiplicative convolution $\boxtimes_d$ multiplies normalised coefficients and keeps roots real and of one sign (Lemma 11.2.8(i)), and the normalised coefficients of the reversed Jensen polynomial $x^d J_{d,n}(1/x)$ are, up to the signs $(-1)^i$, the numbers $(n+i)!\, b_{n+i}$, $0 \le i \le d$. A factorisation $(n+i)!\, b_{n+i} = \alpha_i \beta_i$ into the normalised coefficients of two polynomials with only real roots of one sign would therefore give real zeros of $J_{d,n}$. Find such factorisations for every shift $n$ at a fixed degree $d$, or show that none exists with factors of a given kind. The obvious split, a Gaussian multiplier sequence $q^{i^2}$, $0 < q \le 1$, times a residual close to the coefficients of $(1+x)^d$, fails, because a root of multiplicity $d$ splits into nonreal roots under a generic perturbation; and by Proposition 11.2.9 the factor $1/(a)_i$ can be multiplied in but not divided out. A factorisation has to put real structure into both factors [433].

Problem 12.2. Laguerre and Meixner Pollaczek expansions. Theorem 11.2.10 excludes scale mixtures of gamma laws for $|X|$ and $X^2$, and Proposition 11.2.9 shows that removing a gamma layer keeps zeros real while adding one does not. Find a variable in which the Pólya density, or $\Xi$ itself, has an expansion in Laguerre or Meixner Pollaczek polynomials with positive coefficients and truncations

with only real zeros [433]. The symmetric Meixner Pollaczek expansions of Section 6.2.3.2 converge slowly, and a real rooted construction from their coefficients is a problem of Section 7.2.4.

Problem 12.3. Jacobi parameters. Under RH, where $\nu$ is a positive measure and all its Jacobi parameters exist, prove that $\alpha_n \sim 2\beta_n \sim \frac{1}{2}L_n$, with $L_n$ given by (11.2.13) for the smooth count: a spectral form of the Riemann von Mangoldt formula. Table 11.7 supports it to a few per cent for $n \leq 49$, but the local model of Section 11.2.4 is a heuristic, outside the scaling of [317]. Decide between the rules $n(L_n) = n$ and $n(L'_n) = n - \frac{1}{2}$, which the data do not separate, and explain the scatter of the ratios, about six per cent, through the individual gaps between the zeros, as [433] proposes.

Problem 12.4. The order at which a collision is seen. Section 8.4.3 computes the failure times $t_N^*$ of the Hankel test along the flow to $N = 50$ and explains their steps by collisions; the rule of [433] that the collision of $(\gamma_k, \gamma_{k+1})$ is registered from order $2k + 2$ on fails at $N = 50$ (Table 8.9). Find the rule that gives the order at which a pair is registered, as a function of the index $k$ of the pair and of its distance from the axis, and follow the staircase of $t_N^*$ to orders of a few hundred, towards the limit 0 that it has under RH.

Problem 12.5. Unconditional collision brackets. Section 8.4.5 brackets the collision times by quantities along the flow (Corollaries 8.4.13 and 8.4.15) and through the nonreal zeros (Theorems 8.4.12 and 8.4.14), and checks the hypotheses on a grid of times for the 58 collisions of the census. Find the a priori control of the nonreal zeros of $H_t$, $t < 0$, asked for in Problem 8.1, which would make the brackets statements about $H_t$ computed from $H_0$ alone.

Problem 12.6. Moving the ladder without new zeros. Theorems 5.1.9 and 5.1.13 prove the derivative signs $(-1)^k W^{(k)} > 0$ on ranges that grow with the height to which RH is verified (Section 5.1.6), and complete monotonicity of $W$ on $(0, \infty)$ is RH. Find a mechanism that extends these ranges with no new information on the zeros, for instance a functional equation for the law of $T$ under $\Upsilon$, as [433] suggests. Theorem 11.2.3(c) restricts the form such an equation can take: $T$ is not the Upsilon image of a law in BO, and under RH it is the image of the law $\mu_0$, which is neither a GGC nor in BO.

### 12.2.6 Lessons from the computations

Each lesson is a statement that a computation in the book established or refuted.

Equivalent statements are not equally computable. The Thorin condition has at least eight equivalent faces: positivity of the Thorin measure, complete monotonicity of the heat trace, Stieltjes form of $G'/G$, Hankel positivity at one time, HCM of the clock, Lagarias positivity, Weil positivity and Li positivity. Mathematically they are the same statement. Computationally they differ by many

orders of magnitude in conditioning. The heat trace at moderate time loses about $87t$ digits to cancellation (Section 5.3); the normalised minors of the even Hankel block at $t = 0.05$ fall to $10^{-44}$ by size five (Section 5.1); the Keiper Li coefficients weight a zero at height $\gamma$ by $n^2/\gamma^2$ until $n$ reaches $\gamma$ (Section 8.3); the Thorin density at resolution $\varepsilon$ is local in the ordinate and sees each zero at full size $1/\varepsilon$ (Section 3.2). A computational attack should choose its face by conditioning, and the local face is the Thorin density.

The primes compute the wrong end: the sine squared lemma is the right way to understand the structure and the wrong way to compute near the line, where $k$ digits at resolution $\varepsilon$ need primes up to about $10^{2k/\varepsilon}$ even when no zero lies to the right of the line (Section 3.5.1). Any argument that proposes to establish positivity near the line by bounding prime sums must overcome that exponential, and Section 7.1 shows the same obstruction in another form (Proposition 7.1.2).

The heat trace meets the same obstruction in its own terms. By Proposition 5.1.11, RH is the positivity of the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros at every height $u$ and every $t > 0$, and (5.1.3) writes the symmetric smoothing $\mathcal{G}(u,t)+\mathcal{G}(-u,t)$ as a polar term plus an archimedean term $A(u,t)$ minus a prime sum $P(u,t)$. Bounding $|\cos(u\log n)|$ by one bounds $P(u,t)$ by $S(t)/\sqrt{2\pi t}$, where $S(t) = \sum_{n\geq 2}\Lambda(n)n^{-1/2}e^{-(\log n)^2/8t}$ is, up to a factor, the prime term of (1.2.5) at time $2t$, and so proves $\mathcal{G}(u,t)+\mathcal{G}(-u,t) > 0$ for $t$ below the time $t_0(u)$ at which the bound meets the archimedean and polar terms, which agrees to five digits with the root of $2S(t) = \log(u/2\pi)$ (computed by `phasefree.py`): $t_0$ is 0.45 at height $10^3$, 1.72 at the verified height $3\times 10^{12}$ and 3.64 at $10^{50}$, to the digits shown (proved in ball arithmetic by `phasefree_arb.py`), and it grows only doubly logarithmically in $u$, since $S(t)$ is asymptotic to $\sqrt{8\pi t}\,e^{t/2}$. These resolutions are too coarse to see the quadruples of Table 5.4: at their heights $t_0$ lies between 0.311 and 0.511, while within 8 of each moved pair they make $\mathcal{G}$ negative only for $t > 1.18$ (both proved in ball arithmetic). Nor can any bound on $P$ that holds uniformly in the height do better, since by Dirichlet's theorem on simultaneous approximation the phases $u\log n$ of any finite set of prime powers come arbitrarily close to multiples of $2\pi$ at arbitrarily large $u$, where $P(u,t)$ therefore comes arbitrarily close to $S(t)/\sqrt{2\pi t}$. Nor does positivity at a single resolution give a zero free strip: at fixed $t$ the bound $S(t)/\sqrt{2\pi t}$ on $P$ proves $\mathcal{G}(u,t)+\mathcal{G}(-u,t) > 0$ at every height $u$ with $t_0(u) > t$, that is above about $2\pi e^{2S(t)}$, which is $8.1\times 10^{212}$ at $t = 6$, and $\mathcal{G}(u,6) > 0$ itself for every $u \geq 10^{213}$ (proved in ball arithmetic, with the bound on $|\mathcal{G}(-u,6)|$ above), whatever the zeros are, so a strip needs $t$ to grow with the height. A proof along the heat trace must use the phases, uniformly in the height.

Limits do not create positivity: positivity must be proved at every resolution, not in the limit. A proposal that reaches the Thorin condition as a limit must say where the positivity of its approximants comes from.

The perturbation theorem gives a test that a proposed proof of RH must pass. Replace $\xi$ by the function $F$ of Theorem 5.2.1, with $R$ above every height that the

proof uses explicitly. A step that holds for $F$ as well as for $\xi$ cannot be the step that excludes zeros off the line, because $F$ has four. The steps that survive include the Riemann von Mangoldt formula with an explicit error, which $F$ satisfies with the error enlarged by one on $[A, B)$, the only heights where the two counts differ; the verified zeros below $R$; the small time expansion of the heat trace through the constant term, since $W_F - W = \left(\frac{1}{2}(B-A)^2 + \frac{1}{8}\right)t + O(t^2)$; infinite divisibility, self decomposability and log convexity; the Hankel blocks of size $N$ when $R \geq T_N$ and the derivative signs up to order $10^{17}$ when $R \geq 10^{12}$; and the order, type and functional equation. What separates $\xi$ from $F$ is arithmetic, the Euler product and the explicit formula with the prime weights $\Lambda(n)$, together with local counts of the zeros on the line in windows that contain the height $g$ of the quadruple (Section 5.2.1). A proof through the derivative signs can do without them only below the order at which the signs of $F$ first fail, which is about $t^* R^2$, with $t^*$ between 1.18 and 4.03 in the cases computed below height 2000 and, by the model of Section 5.2.1, of order one or more at larger heights, far beyond $10^{17}$. The function $F$ exists unconditionally when $R + 50 \leq 3 \times 10^{12}$; for larger $R$ the test still applies, because a correct proof of RH makes the hypothesis of Theorem 5.2.1(b) true for every $R$. The criticality filter of Section 11.5.5 is the companion test for properties of the Pólya kernel: a step that holds for $e^{tv^2/4}\Phi(v)$ with some $t < 0$, whose transform has nonreal zeros by the theorem of Rodgers and Tao, cannot be the step that excludes zeros off the line.

# Chapter 13

# Ferromagnetic random variables

A symmetric random variable is ferromagnetic if its moment generating function vanishes only on the imaginary axis. By (1.2.4) the moment generating function of the Pólya variable $X$ is $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, so RH is the statement that one explicit law, the Pólya law with density $p = \Phi/\Xi(0)$, is ferromagnetic.

With no hypothesis, the Pólya law lies within total variation distance $6.1\times10^{-6}$ of a law that is ferromagnetic by an explicit construction, and within $2 \times 10^{-6}$ of one whose positivity, and hence ferromagnetism, rests on a computer assisted proof in ball arithmetic (Section 13.5).

The scripts are in the folder `ferro` (`code/README.md`). We use the units of (1.2.3) throughout, in which the zeros of the characteristic function of $X$ are the ordinates $\gamma$ and $\operatorname{Var} X = 2\mathbb{E}T = 0.0462099862$. The normalisation of Rodgers and Tao in Section 8.4 doubles the variable of the kernel, and statements quoted from the literature have been converted. As in Section 1.2, $Z$ is a standard normal variable, the $e_k$ are standard exponential variables, the $L_k$ are standard Laplace variables, $\varepsilon$ and the $\varepsilon_k$ are fair random signs, $U$ is uniform on $[-1,1]$, and all of them are independent of each other and of everything else.

## 13.1 Ferromagnetic laws and their clocks

### 13.1.1 Definition and background

Definition 13.1.1. A real random variable $X$ is *ferromagnetic* if it is symmetric, $\mathbb{E}e^{b_0X^2} < \infty$ for some $b_0 > 0$, and its moment generating function $L_X(z) = \mathbb{E}e^{zX}$, which is then entire, vanishes only on the imaginary axis. Equivalently, every zero of the characteristic function $\varphi_X(\theta) = \mathbb{E}e^{i\theta X} = L_X(i\theta)$ is real.

These are Newman's laws of type $\mathcal{L}$ [380].

The name comes from statistical mechanics. Lee and Yang proved that the partition function of an Ising ferromagnet, with spins $\pm1$ and a common complex field, vanishes only when the field is purely imaginary [331], and Newman showed that a symmetric single spin law with Gaussian moments of all orders has the same

property in every ferromagnetic system exactly when its own moment generating function has only imaginary zeros [379]; Lieb and Sokal gave another proof and extended the theorem to spins with several components [339]. Section 7.1.3 recalls these results, and Proposition 7.1.16 restates Newman's theorem in the language of Thorin measures. Two smaller classes of single spin laws carry the correlation inequalities of the Ising model. The Griffiths Simon class consists of the weak limits of suitably scaled sums of ferromagnetically coupled Ising spins; it contains the laws with density proportional to $e^{-ax^4-bx^2}$, $a > 0$, and its laws inherit the Lee and Yang property and the inequalities of Griffiths, Kelly and Sherman and of Griffiths, Hurst and Sherman [480, 160]. The class of Ellis, Monroe and Newman consists of the laws with density $e^{-V(x)}$, $V$ even and $V'$ convex on $[0,\infty)$, and their weak limits; the Griffiths, Hurst and Sherman inequality holds for ferromagnets with such single spin laws [130].

### 13.1.2 The Hadamard form and the cumulants

Theorem 13.1.2 (Newman [380]). *Let $X$ be symmetric with $\mathbb{E}e^{b_0X^2} < \infty$ for some $b_0 > 0$. Then $X$ is ferromagnetic if and only if*

$$L_X(s) = e^{bs^2} \prod_k \Big(1 + \frac{s^2}{c_k^2}\Big), \qquad b \ge 0, \quad c_k > 0, \quad \sum_k c_k^{-2} < \infty, \tag{13.1.1}$$

*with each $c_k$ repeated according to the multiplicity of the zero $ic_k$. If $\mathbb{E}e^{bX^2} < \infty$ for every $b > 0$, then $b = 0$.*

*Proof.* Since $|L_X(z)| \le \mathbb{E}e^{|\operatorname{Re} z||X|} \le Ce^{|z|^2/4b_0}$, the function $\varphi_X$ is entire of order at most 2; it is even and real on $\mathbb{R}$, and $\varphi_X(0) = 1$. Suppose that its zeros are real, at $\pm c_k$ with $c_k > 0$. Hadamard's theorem, with the primary factors of genus 2 paired over $\pm c_k$, gives $\varphi_X(\theta) = e^{-b'\theta^2} \prod_k (1 - \theta^2/c_k^2) e^{\theta^2/c_k^2}$ with $b'$ real and $\sum_k c_k^{-4} < \infty$, so that, for real $s$,

$$\frac{\log L_X(s)}{s^2} = b' + \sum_k \frac{1}{c_k^2} \frac{\log(1+x_k) - x_k}{x_k}, \qquad x_k = \frac{s^2}{c_k^2}.$$

The function $(\log(1+x)-x)/x$ decreases from 0 to $-1$ on $(0,\infty)$, so by monotone convergence the sum tends to $-\sum_k c_k^{-2}$ as $s \to \infty$. If this sum diverged, $L_X(s)$ would tend to 0, which is impossible because $L_X(s) = \mathbb{E}\cosh(sX) \ge 1$. Hence $\sum_k c_k^{-2} < \infty$, the factors $e^{\theta^2/c_k^2}$ can be collected into the exponential, and (13.1.1) holds with $b = b' - \sum_k c_k^{-2}$. Then $\log L_X(s) = bs^2 + o(s^2)$, since $\log(1+x)/x$ is bounded by 1 and tends to 0, and $L_X \ge 1$ forces $b \ge 0$. The converse is clear, since the right side of (13.1.1) vanishes only at $\pm ic_k$. If $\mathbb{E}e^{bX^2} < \infty$ for every $b$, then $L_X(s) \le C_\epsilon e^{\epsilon s^2}$ for every $\epsilon > 0$, so $b = 0$, as in the proof of Proposition 7.1.16. □

COROLLARY 13.1.3 (Newman [380]). *If $X$ is ferromagnetic, with* (13.1.1), *then* $\operatorname{Var} X = 2b + 2\sum_k c_k^{-2}$, *and for* $m \geq 2$

$$\frac{\kappa_{2m}(X)}{(2m)!} = \frac{(-1)^{m+1}}{m} \sum_k c_k^{-2m}. \tag{13.1.2}$$

*So the even cumulants alternate in sign:* $\kappa_4 \leq 0$, $\kappa_6 \geq 0$, *and so on. Moreover* $\mathbb{E}e^{sX} \leq e^{s^2 \operatorname{Var} X/2}$ *for real* $s$, *and* $\mathbb{E}X^4 \leq 3(\operatorname{Var} X)^2$.

*Proof.* The first two statements are the expansion of $\log L_X(s) = bs^2 + \sum_k \log(1 + s^2/c_k^2)$. The bound on $\mathbb{E}e^{sX}$ follows from $\log(1+x) \leq x$, and the bound on $\mathbb{E}X^4 = \kappa_4 + 3\kappa_2^2$ from $\kappa_4 \leq 0$. □

For the Pólya law the alternation holds with no hypothesis: by the formula for $\log G$ in Section 5.1, $(-1)^{n-1}\kappa_{2n}(X) = \frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt > 0$ for every $n$ (Section 1.3), and under RH this is (13.1.2) with $c_k = \gamma_k$ and $b = 0$.

### 13.1.3 THE CLOCK OF A FERROMAGNETIC LAW

THEOREM 13.1.4. *Let $X$ be ferromagnetic, with* (13.1.1), *and put*

$$T_X = b + \sum_k \frac{e_k}{c_k^2}, \qquad Y_X = Z\sqrt{2T_X},$$

*the series converging almost surely and in mean.*

*(i)* $\mathbb{E}e^{-uT_X} = 1/L_X(\sqrt{u})$ *for* $u \geq 0$, *and $T_X$ is a GGC with drift $b$ and Thorin measure* $\sum_k \delta_{c_k^2}$, *one unit atom for each pair of zeros* $\pm ic_k$.

*(ii)* $\mathbb{E}e^{i\theta Y_X} = 1/L_X(\theta)$ *for real* $\theta$, *so that* $(\varphi_X, \varphi_{Y_X})$ *is a van Dantzig pair, and* $Y_X \overset{d}{=} \sqrt{2b}\,Z + \sum_k L_k/c_k$.

*(iii) The moment generating function of $Y_X$ is* $1/\varphi_X(t)$ *for* $|t| < c_1$ *and infinite for* $|t| \geq c_1$, *unless $X$ is Gaussian.*

*(iv) Unless $X$ is degenerate, $Y_X$ has a Pólya frequency density of infinite order.*

*(v)* $\kappa_m(T_X) = (-1)^{m+1}m!\,\kappa_{2m}(X)/(2m)!$ *for* $m \geq 1$; *in particular* $\mathbb{E}T_X = \frac{1}{2}\operatorname{Var} X$ *and* $\operatorname{Var} T_X = -\kappa_4(X)/12$.

*Proof.* (i) $\mathbb{E}e^{-ue_k/c_k^2} = (1 + u/c_k^2)^{-1}$; multiply over $k$ and compare with (13.1.1) at $s = \sqrt{u}$. (ii) Conditionally on $T_X$, $Y_X$ is centred normal with variance $2T_X$, so $\mathbb{E}e^{i\theta Y_X} = \mathbb{E}e^{-\theta^2 T_X} = 1/L_X(\theta)$, which is also the characteristic function of $\sqrt{2b}\,Z + \sum_k L_k/c_k$, since $\mathbb{E}e^{i\theta L} = (1+\theta^2)^{-1}$. (iii) $\mathbb{E}e^{tY_X} = \mathbb{E}e^{t^2 T_X} = e^{bt^2}\prod_k(1 - t^2/c_k^2)^{-1}$ while $t^2 < c_1^2$, and $\mathbb{E}e^{t^2 e_1/c_1^2} = \infty$ for $t^2 \geq c_1^2$. (iv) The Laplace density $\frac{1}{2}e^{-|x|}$ and the normal densities are Pólya frequency functions of infinite

order, and the class is closed under convolution and under the limits involved here. Equivalently, the bilateral Laplace transform of $Y_X$ is the reciprocal of $\varphi_X(s) = e^{-bs^2}\prod_k(1 - s^2/c_k^2)$, which has Schoenberg's form [474]. (v) For $u$ near 0, $\log \mathbb{E}e^{-uT_X} = -\log L_X(\sqrt{u}) = -\sum_m \kappa_{2m}(X)u^m/(2m)!$, and comparing coefficients with $\sum_m \kappa_m(T_X)(-u)^m/m!$ gives the formula. □

Part (i) is Proposition 7.1.16 with the Gaussian factor allowed. The series of exponential variables in (i), with its atomic Thorin measure, also appears in [423]. For the Pólya law under RH, $b = 0$, $c_k = \gamma_k$ and $T_X$ is the clock $T$ of Proposition 1.2.1, and $(\varphi_X, \varphi_{Y_X})$ is the van Dantzig pair of Section 1.2, with $Y_X = Z\sqrt{2T}$ the variable $Y_{1/2}$ of Proposition 10.1.1. Without RH the clock $T$ still exists (Theorem 5.1.1), and so does the pair; what fails without RH is the product form of the clock. Konstantopoulos, Patie and Sarkar study van Dantzig pairs of this kind through the Laguerre Pólya class and Pólya frequency functions, and connect them with the Lee and Yang property [313]. The correspondence runs one way only: every ferromagnetic law gives a Pólya frequency law, but $\varphi(s) = 1 - s^2$, the reciprocal of the bilateral Laplace transform of the Laplace law, is not a characteristic function.

Part (v) holds for the Pólya law with no hypothesis, with the clock $T$ of Theorem 5.1.1 in place of $T_X$: the identity $\log \mathbb{E}e^{-uT} = -\log G(u) = -\log \mathbb{E}e^{\sqrt{u}X}$ is all the proof uses. With the cumulants of Table 13.3 it gives $\mathbb{E}T = 0.0231049931$ and $\operatorname{Var} T = -\kappa_4/12 = 3.71725993 \times 10^{-5}$, the constants of Section 1.2.

### 13.1.4 Examples

Table 13.1 lists the basic ferromagnetic laws with their clocks. The variables are those of Biane, Pitman and Yor [49]: $C$ is the exit time of Brownian motion from $(-1, 1)$, with $\mathbb{E}e^{-uC} = 1/\cosh\sqrt{2u}$, and $J$ is the hitting time of 1 by a Bessel process of dimension 3 started at 0, with $\mathbb{E}e^{-uJ} = \sqrt{2u}/\sinh\sqrt{2u}$, as in Section 9.1.1. The first two rows are the classical van Dantzig pairs of Section 1.2.

| law of $X$ | $L_X(s)$ | $c_k$ | clock $T_X$ | $Y_X$ |
|---|---|---|---|---|
| fair sign $\varepsilon$ | $\cosh s$ | $(k - \frac{1}{2})\pi$ | $C/2$ | hyperbolic secant |
| uniform on $[-1, 1]$ | $\sinh s/s$ | $k\pi$ | $J/2$ | logistic, scale $1/\pi$ |
| $c_\nu(1 - x^2)^{\nu-1/2}$, $\nu > -\frac{1}{2}$ | $\Gamma(\nu + 1)(2/s)^\nu I_\nu(s)$ | $j_{\nu,k}$ | $\frac{1}{2}\tau_\nu$ | Bessel type |
| $N(0, \sigma^2)$ | $e^{\sigma^2 s^2/2}$ | none | $\sigma^2/2$ | $N(0, \sigma^2)$ |
| $e^{-2\pi\cosh 2x}/K_0(2\pi)$ | $K_{s/2}(2\pi)/K_0(2\pi)$ | $2\nu_k$ | $T_P/4$ | $Z\sqrt{T_P/2}$ |
| Pólya, under RH | $\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ | $\gamma_k$ | $T$ | $Z\sqrt{2T}$ |

Table 13.1: Ferromagnetic laws and their clocks. Here $j_{\nu,k}$ are the positive zeros of the Bessel function $J_\nu$, and $\tau_\nu$ is the hitting time of 1 by a Bessel process of dimension $2\nu + 2$ started at 0. In the fifth row $\nu_k$ are the zeros of $\nu \mapsto K_{i\nu}(2\pi)$ and $T_P$ is the clock of Theorem 8.7.9.

The third row rests on Kent's theorem for Bessel processes [304]: for the Bessel process of dimension $2\nu+2$ started at 0, the hitting time of 1 satisfies $\mathbb{E}e^{-u\tau_\nu} = (\sqrt{2u})^\nu/(2^\nu\Gamma(\nu+1)I_\nu(\sqrt{2u}))$, so that $\mathbb{E}e^{-u\tau_\nu/2} = 1/L_X(\sqrt{u})$ and $T_X = \frac{1}{2}\tau_\nu = \sum_k e_k/j_{\nu,k}^2$. The zeros $j_{\nu,k}$ are real for $\nu > -1$, so every symmetric beta law $c_\nu(1-x^2)^{\nu-1/2}$ with $\nu > -\frac{1}{2}$ is ferromagnetic. The family contains the arcsine law ($\nu = 0$), the uniform law ($\nu = \frac{1}{2}$) and the semicircle law ($\nu = 1$), and it tends to the law of $\varepsilon$ as $\nu \downarrow -\frac{1}{2}$, where $\tau_{-1/2} = C$.

### 13.1.5 Closure

Products of functions of the form (13.1.1) have the same form, so sums of independent ferromagnetic variables are ferromagnetic. The class is closed under convergence in law, and this is what the Gaussian factor in Definition 13.1.1 is for: the laws of $n^{-1/2}\sum_{k\le n}\varepsilon_k$ have Gaussian moments of all orders, and their limit $N(0,1)$ does not.

Theorem 13.1.5. *Let $X_n$ be ferromagnetic and suppose that $X_n \to X$ in law. Then $X$ is ferromagnetic,* $\operatorname{Var} X_n \to \operatorname{Var} X$*, and $L_{X_n} \to L_X$ locally uniformly on* $\mathbb{C}$.

*Proof.* Let $\sigma_n^2 = \operatorname{Var} X_n$. By Corollary 13.1.3, $\mathbb{E}X_n^4 \le 3\sigma_n^4$, so the Paley Zygmund inequality applied to $X_n^2$ gives $P(X_n^2 > \sigma_n^2/4) \ge (\frac{3}{4})^2\sigma_n^4/\mathbb{E}X_n^4 \ge \frac{3}{16}$. A sequence that converges in law is tight, so the $\sigma_n$ are bounded, say by $\sigma$. By Corollary 13.1.3 again, $|L_{X_n}(z)| \le L_{X_n}(\operatorname{Re} z) \le e^{\sigma^2(\operatorname{Re} z)^2/2}$, so the $L_{X_n}$ form a locally bounded, hence normal, family of entire functions, and for each real $s$ the variables $e^{sX_n}$ are uniformly integrable, since their second moments are bounded. Hence $\mathbb{E}e^{sX_n} \to \mathbb{E}e^{sX}$ for real $s$, which gives $\mathbb{E}e^{sX} \le e^{\sigma^2 s^2/2}$ and, averaging over $s = \sqrt{2b}Z$, $\mathbb{E}e^{bX^2} < \infty$ for $b < 1/2\sigma^2$. So $L_X$ is entire, every locally uniform limit point of the $L_{X_n}$ agrees with it on $\mathbb{R}$, and by Vitali's theorem $L_{X_n} \to L_X$ locally uniformly. Each $L_{X_n}$ has no zero in the half plane $\operatorname{Re} z > 0$, and $L_X(0) = 1$, so by Hurwitz's theorem $L_X$ has no zero there either; by symmetry it has none in $\operatorname{Re} z < 0$. The fourth moments are bounded, so the $X_n^2$ are uniformly integrable and the variances converge. □

Newman and Wu proved the closure of this class under convergence in law [384].

Proposition 13.1.6. *Let $X$ be ferromagnetic with law $\rho$, and let $\mathbb{E}e^{b_0X^2} < \infty$.*

*(i) For every $a > 0$ the law $\cosh(ax)\,\rho(dx)/\mathbb{E}\cosh(aX)$ is ferromagnetic.*

*(ii) For every $0 < \lambda < b_0$ the law $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$ is ferromagnetic.*

*Proof.* (i) The moment generating function of the tilted law is proportional to $L_X(z+a) + L_X(z-a)$. For $w$ with $\operatorname{Re} w > 0$ and $a > 0$ we have $|w+a| >$

$|w-a|$. Applied to $w = z \pm ic_k$ this gives $|c_k^2 + (z+a)^2| > |c_k^2 + (z-a)^2|$ when $\operatorname{Re} z > 0$, and $|e^{b(z+a)^2}| = e^{4ab \operatorname{Re} z} |e^{b(z-a)^2}|$. So for $\operatorname{Re} z > 0$ every factor of (13.1.1) has larger modulus at $z+a$ than at $z-a$, and $L_X(z+a) + L_X(z-a) \neq 0$ unless $L_X$ is constant, in which case the sum is 2. The sum is even in $z$, so it has no zeros with $\operatorname{Re} z < 0$ either, and the tilted law satisfies the moment condition because $\cosh(ax) \leq e^{a|x|}$. (ii) By (i), applied $n$ times, the law with density proportional to $\cosh(x\sqrt{2\lambda/n})^n$ with respect to $\rho$ is ferromagnetic. Since $\cosh y \leq e^{y^2/2}$, these densities are dominated by $e^{\lambda x^2}$, which is $\rho$ integrable, and they converge to $e^{\lambda x^2}$, so the laws converge in total variation to $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$. Theorem 13.1.5 applies. □

Part (i) is the one spin case of the Lee and Yang theorem, and for the Pólya law, whose transform has genus 1, it is Pólya's Hilfssatz II of Section 7.1.3. For the Pólya law, part (ii) is the forward half of de Bruijn's theorem [112]: multiplying $p$ by $e^{\lambda x^2/4}$ is the de Bruijn Newman flow at time $\lambda$ (Section 8.4).

Remark 13.1.7. Even polynomial factors with imaginary zeros also preserve the class. If $X$ is ferromagnetic with law $\rho$ and $r > 0$, then $(x^2+r^2)\rho(dx)/\mathbb{E}(X^2+r^2)$ is ferromagnetic. It is symmetric with a Gaussian moment, and its characteristic function is proportional to $r^2\varphi_X - \varphi_X'' = (r-D)(r+D)\varphi_X$, $D = d/d\theta$; by (13.1.1) $\varphi_X(\theta) = L_X(i\theta)$ is in the Laguerre Pólya class [335], and each factor $r \pm D$ preserves that class by Laguerre's theorem, as in the proof of Proposition 13.5.17. Products of such factors are the polynomial case of the multipliers $\psi(it)$ of Proposition 13.5.8. This remark follows a note of Nicholas Polson merged into the book [235].

Corollary 13.1.8. *RH holds if and only if the Pólya law is a limit in law of ferromagnetic laws.*

*Proof.* If RH holds, the Pólya law is ferromagnetic by (1.2.4), and it is its own limit. Conversely, a limit in law of ferromagnetic laws is ferromagnetic by Theorem 13.1.5, and then every zero of $\Xi$ is real. □

Let $\delta$ be the infimum of the total variation distances $\sup_A |P(X \in A) - P(Y \in A)|$ from the Pólya variable $X$ to ferromagnetic variables $Y$; when $Y$ has a density $f$ the distance is $\mathrm{TV}(f,p) = \frac{1}{2}\int |f-p|$. If RH holds, $\delta = 0$; conversely, ferromagnetic laws that converge to the Pólya law in total variation converge in law, so $\delta = 0$ implies RH.

## 13.2 Laws on an interval

The Pólya law is symmetric and unimodal, and every such law is a mixture of uniform laws. This section asks which mixtures are ferromagnetic, for laws on an interval. Mixing towards the centre can destroy the property, as an atom at

the origin shows; a concave shoulder guarantees it, and a convex one rules it out unless the density is linear on its support, a scaled triangle; and mixing towards the edges is always safe. The Pólya law falls between the two shoulder cases, which is one way to see why its zeros are hard.

### 13.2.1 Mixtures of uniform laws

By Khinchine's theorem [136], a random variable $X$ is symmetric and unimodal about 0 if and only if $X \stackrel{d}{=} UR$ with $R \geq 0$ independent of $U$. Then $L_X(s) = \mathbb{E}[\sinh(sR)/(sR)]$, and if $X$ has a density $p$ that is absolutely continuous on $(0,\infty)$ and tends to 0 at infinity, $R$ has the density $-2rp'(r)$. Williamson's theorem refines the representation [519]: a density on $(0,\infty)$ is $k$ monotone exactly when it is a scale mixture of $k(1-x)_+^{k-1}$, so the symmetric $k$ monotone densities are the scale mixtures of

$$f_k(x) = \tfrac{k}{2}(1-|x|)_+^{k-1},$$

of which $f_1$ is the uniform density and $f_2$ the triangle. The uniform law itself is ferromagnetic, with $L_U(s) = \sinh s/s$, and Viète's product

$$\frac{\sinh s}{s} = \prod_{k\geq 1} \cosh \frac{s}{2^k}, \qquad U \stackrel{d}{=} \sum_{k\geq 1} 2^{-k}\varepsilon_k,$$

shows why: the uniform variable is a sum of independent fair signs, and the property is that of a single spin. So is every sum $\sum_k c_k U_k$ of independent scaled uniform variables. Mixtures behave differently.

Theorem 13.2.1. *For every $0 < q < 1$ the law $(1-q)\,U + q\,\delta_0$ is symmetric and unimodal but not ferromagnetic. Consequently, for every $0 < q < 1$ and every small enough $a > 0$, the mixture of the uniform laws on $[-1,1]$ and on $[-a,a]$ with weights $1-q$ and $q$ is not ferromagnetic.*

*Proof.* The moment generating function $L(s) = (1-q)\sinh s/s + q$ vanishes exactly where $\sinh s/s = -c$, with $c = q/(1-q) > 0$. There is no real zero, since $\sinh s/s \geq 1$ on $\mathbb{R}$, and only finitely many imaginary ones, since $s = iy$ needs $\sin y/y = -c$ and $|\sin y/y| \leq 1/|y|$. But $L$ has infinitely many zeros. It is entire of order 1, so with finitely many zeros it would be $e^{\alpha s}P(s)$ with a polynomial $P$; evenness forces $\alpha = 0$, and $L$ is not a polynomial, since it grows like $e^s/s$ on the positive axis. So all but finitely many zeros lie off the imaginary axis. The second statement follows from Theorem 13.1.5, since the mixtures converge in law to $(1-q)U + q\delta_0$ as $a \downarrow 0$. □

Both components of the mixture are ferromagnetic, so the class is not closed under mixing, even of unimodal laws with a common mode. For $c > c_*$, where $c_* = -\min_y \sin y/y = 0.2172336\ldots$, the minimum being at the first positive root $y = 4.4934095\ldots$ of $\tan y = y$, there are no imaginary zeros at all.

PROPOSITION 13.2.2. *For $0 \le q \le 1$ the law $q\,\delta_0 + \frac{1-q}{2}(\delta_{-1}+\delta_1)$ is ferromagnetic if and only if $q \le \frac{1}{2}$ or $q = 1$.*

*Proof.* For $q = 1$ the law is $\delta_0$, and $L = 1$ has no zeros. For $q < 1$ the zeros of $L(s) = (1-q)\cosh s + q$ solve $\cosh s = -c$ with $c = q/(1-q)$. If $c \le 1$, write $-c = \cos\theta_0$ with $\theta_0 \in [\frac{\pi}{2}, \pi]$; the solutions $s = \pm i\theta_0 + 2\pi i k$ are imaginary. If $c > 1$ the solutions are $s = \pm \operatorname{arccosh} c + i\pi(2k+1)$, which are not; for $q = 0.6$ the first is $0.9624237 + i\pi$. □

For $k = 2$ the representation is explicit and is a statement about convexity. If $p$ is a symmetric density, twice continuously differentiable on $(0,\infty)$, with $r^2 p'(r) \to 0$ as $r \to \infty$ and $\int_0^\infty (r + r^2)|p''(r)|\,dr < \infty$, two integrations by parts give, for $x \ne 0$ (and at $x = 0$ with $p(0) = \lim_{x\downarrow 0} p(x)$),

$$p(x) = \int_0^\infty \frac{1}{r}\Big(1 - \frac{|x|}{r}\Big)_+ r^2 p''(r)\,dr, \qquad \mathbb{E}e^{i\theta X} = \frac{4}{\theta^2}\int_0^\infty p''(r)\sin^2\frac{\theta r}{2}\,dr \quad (\theta \ne 0),$$

so $X$ is a scale mixture of $f_2$ for the signed measure $r^2 p''(r)\,dr$, which has mass one and is the only finite signed measure on $(0,\infty)$ with this property (the second distributional derivative of $r^{-1}(1 - x/r)_+$ is $r^{-2}\delta_r$ on $(0,\infty)$); it is a probability law exactly when $p$ is convex on $(0,\infty)$. This is Williamson's representation for $k = 2$ [519], in the form used by Polson, Scott and Windle [420, Section 2.2].

PROPOSITION 13.2.3. *(i) If $X$ is a scale mixture of $f_2$ for a probability law $\nu$ on $(0,\infty)$, then $\mathbb{E}e^{i\theta X} \ge 0$ for real $\theta$, and for $\theta \ne 0$ it vanishes exactly when $\nu$ is carried by $\{2\pi j/|\theta| : j \ge 1\}$. In particular, if $\nu$ has a nonzero absolutely continuous part, $\mathbb{E}e^{i\theta X}$ has no real zeros.*

*(ii) The Pólya law is not a scale mixture of $f_2$ for any probability law: with $y_n = \pi n^2 e^{2u}$ and $R(y) = 32y^4 - 224y^3 + 330y^2 - 75y$, the kernel (1.2.3) satisfies $\Phi''(u) = \frac{1}{2}e^{u/2}\sum_{n\ge 1} R(y_n)e^{-y_n}$, $\Phi''(0) < 0$, and $\Phi''(u) > 0$ for $u \ge \frac{1}{2}\log(7/\pi)$.*

*(iii) The kernel $\Phi$ has exactly one positive inflection point $u^*$: $\Phi'' < 0$ on $[0,u^*)$, $\Phi'' > 0$ on $(u^*,\infty)$, and $\Phi'''(u^*) > 0$.*

*Proof.* (i) $\mathbb{E}e^{i\theta X} = \int_0^\infty \hat{f}_2(r\theta)\,\nu(dr)$ with $\hat{f}_2(\theta) = 4\sin^2(\theta/2)/\theta^2$, which is nonnegative and, for $\theta \ne 0$, vanishes exactly on $2\pi\mathbb{Z}\setminus\{0\}$. A nonnegative integral vanishes exactly when its integrand vanishes $\nu$ almost everywhere, and a countable set is null for an absolutely continuous measure. (ii) Each $(1 - x/r)_+$ is convex on $(0,\infty)$, so a mixture of them with a probability law is convex there, and it suffices to show $\Phi''(0) < 0$. The formula for $\Phi''$ is termwise differentiation of (1.2.3), written in $y_n$. At $u = 0$ the term $n = 1$ of the sum is below $-34$, since $R(\pi) < -800$ and $e^{-\pi} > 0.043$, while $0 < R(\pi n^2) < 32\pi^4 n^8$ for $n \ge 2$ and $32\pi^4\sum_{n\ge 2} n^8 e^{-\pi n^2} < 3$ (the three inequalities are certified in ball arithmetic,

`pg3_fejer.py`). For $y \ge 7$, $R(y) = y\{32y^2(y-7) + 330y - 75\} > 0$, so every term is positive when $u \ge \frac{1}{2}\log(7/\pi)$.

(iii) Put $y = \pi e^{2u}$ and $D(y) = \sum_{n\ge 2} R(n^2 y) e^{-(n^2-1)y}$, so that $\Phi''(u) = \frac{1}{2} e^{u/2-y} F(y)$ with $F = R + D$; it suffices to show that $F$ has exactly one zero on $[\pi, \infty)$ and that it is simple. For $y \ge \pi$ and $n \ge 2$, $0 < R(n^2 y) < 32 n^8 y^4$ and $y^4 e^{-(n^2-1)y}$ decreases, so $0 < D(y) < 32\pi^4 e^{\pi} \sum_{n\ge 2} n^8 e^{-\pi n^2} < 3e^{\pi} < 72$. Since $R''(y) = 384y^2 - 1344y + 660 > 0$ for $y \ge 3$, $R$ is convex on $[\pi, \frac{9}{2}]$ and lies below the larger of its endpoint values, $R(\pi) < -800$ and $R(\frac{9}{2}) = -945$, so $F < -728$ there. For $y \ge \frac{9}{2}$, $R'(y) \ge R'(\frac{9}{2}) = 951$, and termwise differentiation gives $|D'(y)| \le 64 y^4 \sum_{n\ge 2} n^{10} e^{-(n^2-1)y} < 40$ (the sum is at its largest at $y = \frac{9}{2}$, where its first term dominates). Hence $F' > 911$ on $[\frac{9}{2}, \infty)$, $F(\frac{9}{2}) < 0$ and $F(7) > 0$, so $F$ has exactly one zero, which lies in $(\frac{9}{2}, 7)$ and is simple. The prefactor is positive, which gives every sign claimed in (iii). □

The location of $u^*$ is certified in ball arithmetic, $u^* = 0.23726528718742524911$ (`pg3_fejer.py`), so the Pólya density is concave on $|u| \le u^*$ and convex outside. Its mixing measure $r^2 p''(r)\, dr$ has mass $-0.05106$ on $(0, u^*)$, negative because $p'' < 0$ there, and $1.05106$ on $(u^*, \infty)$ (computed; a note of Nicholas Polson merged into the book [243]). Part (i) gives a second reason why this measure is signed: $\Xi$ vanishes at $\gamma_1$, and a probability mixing law with a nonzero absolutely continuous part would leave no real zero. At every real zero of $\Xi$ the contributions of $(0, u^*)$ and $(u^*, \infty)$ to the integral above cancel, but that cancellation is only the Fourier identity $\int_0^\infty p''(r) \sin^2(\theta r/2)\, dr = \theta^2 \hat{p}(\theta)/4$ at a zero of $\hat{p}$, so it says nothing beyond the zero itself. The heat trace behaves the other way: by Theorem 5.1.13 it is 2 monotone unconditionally, so $W(t) = \int (1 - tz/2)_+ \, U_2(dz)$ with $U_2 \ge 0$ (Section 5.1.4), a positive mixture of triangles, and RH is the same positivity at every order.

### 13.2.2 The shoulder theorem

Let $p$ be a symmetric density on $[-1, 1]$ that is absolutely continuous and nonincreasing on $[0, 1]$, with $p(1) = 0$, and put $g = -p' \ge 0$ on $(0, 1)$. Integrating $\varphi_X(z) = 2\int_0^1 p(y)\cos(zy)\, dy$ by parts turns the cosine transform of the density into a sine transform of its slope:

$$z\, \varphi_X(z) = 2 \int_0^1 g(y) \sin(zy)\, dy. \tag{13.2.1}$$

Pólya proved that if $f$ is positive, integrable and nondecreasing on $(0, 1)$, then $\int_0^1 f(t)\cos(zt)\, dt$ and $\int_0^1 f(t)\sin(zt)\, dt$ have only real zeros [442].

Theorem 13.2.4. *Let $p$ be a symmetric density on $[-1, 1]$, absolutely continuous and nonincreasing on $[0, 1]$, with $p(1) = 0$, and let $[-c, c]$, $0 < c \le 1$, be its support.*

*(i) If $p$ is concave on $[0,1]$, then $X$ is ferromagnetic.*

*(ii) If $p$ is convex on $[0,1]$ and not linear on $[0,c]$, then $X$ is not ferromagnetic. Moreover $\varphi_X \geq 0$ on $\mathbb{R}$, and its real zeros, if any, are double and lie in a lattice $(2\pi/s_0)\mathbb{Z}$ with $0 < s_0 < c$.*

*Proof.* (i) If $p$ is concave, then $c = 1$ and $g$ is nondecreasing on $(0,1)$. If $g > 0$ on $(0,1)$, Pólya's theorem in its sine form and (13.2.1) show that $\varphi_X$ has only real zeros. Otherwise $p$ is constant near 0; then $(p + \epsilon(1-|x|))/(1+\epsilon)$ is concave with a positive slope, so it is ferromagnetic, and Theorem 13.1.5 applies as $\epsilon \downarrow 0$.

(ii) If $p$ is convex, $g$ is nonincreasing on $(0,c)$, so $g(y) = g(c-) + \mu((y,c))$ with a positive measure $\mu$ on $(0,c)$, which is not zero because $p$ is not linear. Since $\int_0^s \sin(zy)\,dy = 2\sin^2(zs/2)/z$, Fubini's theorem in (13.2.1) gives, for real $z \neq 0$,

$$\frac{z^2}{4}\,\varphi_X(z) = g(c-)\sin^2\frac{zc}{2} + \int_{(0,c)} \sin^2\frac{zs}{2}\,\mu(ds) \;\geq 0.$$

A real zero $z_0$ needs $z_0 s \in 2\pi\mathbb{Z}$ for every $s$ in the support of $\mu$, so with any such $s_0 < c$ the real zeros lie in $(2\pi/s_0)\mathbb{Z}$. At a real zero the right side has vanishing first derivative and second derivative $\frac{1}{2}g(c-)c^2 + \frac{1}{2}\int s^2\mu(ds) > 0$, because every cosine $\cos(z_0 s)$ that enters equals 1; so the zero is double. Suppose now that every zero of $\varphi_X$ were real. Since $\varphi_X$ is even and entire of exponential type, Hadamard's theorem would give $\varphi_X(z) = \prod_{m\in M}(1 - z^2/z_m^2)^2$ for a set $M$ of positive integers, with $z_m = 2\pi m/s_0$, and hence

$$\varphi_X(iy) \leq \prod_{m\geq 1}\Big(1 + \frac{y^2 s_0^2}{4\pi^2 m^2}\Big)^2 = \Big(\frac{\sinh(ys_0/2)}{ys_0/2}\Big)^2 \leq e^{s_0|y|}.$$

But $p > 0$ on $[0,c)$, so for $0 < \epsilon < c - s_0$ and $y > 0$, $\varphi_X(iy) = L_X(y) \geq \int_{c-\epsilon}^{c} e^{yx}p(x)\,dx \geq C_\epsilon e^{(c-\epsilon)y}$ with $C_\epsilon > 0$, which contradicts the bound as $y \to \infty$. □

Corollary 13.2.5. *(i) The law with density proportional to $1-|x|^b$ on $[-1,1]$, $b > 0$, is ferromagnetic if and only if $b \geq 1$. In Khinchine's form, $UR$ with $R$ distributed as $\mathrm{Beta}(a,1)$ is ferromagnetic if and only if $a \geq 2$.*

*(ii) The symmetric $k$ monotone kernel $f_k$ is ferromagnetic if and only if $k \leq 2$.*

*(iii) The laws with density proportional to $(1-x^2)^m$, $0 < m \leq 1$, which include the semicircle law, are ferromagnetic.*

*(iv) The product $U_1U_2$ of two independent uniform variables, with density $\frac{1}{2}\log(1/|x|)$, is not ferromagnetic.*

*Proof.* (i) $1-x^b$ is concave on $[0,1]$ for $b \geq 1$ and convex and not linear for $b < 1$; the mixing variable has density $-2rp'(r) \propto r^b$, that of Beta$(b+1,1)$. This settles $a = b+1 > 1$. For $0 < a \leq 1$ the density of $UR$ is unbounded at 0 and the theorem does not apply, but Proposition 13.2.7 below excludes these laws: with $\mu_j = a/(a+2j)$, its case $j=1$ requires $a \geq \sqrt{10}-2 > 1$. (ii) $f_1$ is uniform and $f_2$ the triangle; for $k \geq 3$, $(1-x)^{k-1}$ is convex and not linear. (iii) $\frac{d^2}{dx^2}(1-x^2)^m = -2m(1-x^2)^{m-2}(1-(2m-1)x^2) \leq 0$ on $[0,1]$ when $m \leq 1$. (iv) The density is unbounded at the origin, so Theorem 13.2.4 does not apply, but directly $\varphi(z) = \int_0^1 \sin(zu)/(zu)\,du = \mathrm{Si}(z)/z$, and $\mathrm{Si}(z) > 0$ for $z > 0$. So $\varphi$ has no real zero, while it has infinitely many zeros, by the argument in the proof of Theorem 13.2.1. □

### 13.2.3 Mixing towards the edges

Theorem 13.2.6. *Every symmetric density on $[-1,1]$ that is nondecreasing in $|x|$ is ferromagnetic. Equivalently, every mixture of the uniform laws on the shells $\{a \leq |x| \leq 1\}$, $0 \leq a < 1$, is ferromagnetic.*

*Proof.* Here $\varphi_X(z) = 2\int_0^1 p(t)\cos(zt)\,dt$ with $p$ nondecreasing on $(0,1)$. If $p > 0$ on $(0,1)$, Pólya's theorem in its cosine form [442] applies; otherwise add $\epsilon$ times the uniform density, normalise, and let $\epsilon \downarrow 0$ (Theorem 13.1.5). A density nondecreasing in $|x|$ is a mixture of the shell densities, and conversely. □

### 13.2.4 A moment test

Ferromagnetic laws satisfy inequalities between consecutive moments. If $X$ is ferromagnetic, $L_X(\sqrt{x}) = \sum_j b_j x^j$, with $b_j = \mathbb{E}X^{2j}/(2j)!$, equals $e^{bx}\prod_k(1 + x/c_k^2)$, an entire function of $x$ in the Laguerre Pólya class with only negative zeros. Its coefficients satisfy the Turán inequalities $(j!\,b_j)^2 \geq (j-1)!\,b_{j-1}(j+1)!\,b_{j+1}$, the limit as $n, K \to \infty$ of Newton's inequalities for the polynomials $(1 + bx/n)^n \prod_{k \leq K}(1 + x/c_k^2)$, whose zeros are real; that is

$$b_j^2 \geq \Big(1 + \frac{1}{j}\Big) b_{j-1} b_{j+1}, \qquad j \geq 1. \tag{13.2.2}$$

For a mixture $X = UR$ the moments factor as $\mathbb{E}X^{2j} = \mu_j/(2j+1)$ with $\mu_j = \mathbb{E}R^{2j}$.

Proposition 13.2.7. *If $UR$ is ferromagnetic and $R$ is not 0 almost surely, then for every $j \geq 1$*

$$\frac{2j+1}{2j+3} \leq \frac{\mu_j^2}{\mu_{j-1}\mu_{j+1}} \leq 1.$$

*Proof.* The moments $\mu_j$ are finite, since $\mathbb{E}(UR)^{2j} = \mu_j/(2j+1)$ and $L_{UR}$ is entire, and positive, since $P(R > 0) > 0$. Substitute $b_j = \mu_j/(2j+1)!$ in (13.2.2): the

factor is $(1+\frac{1}{j})\frac{(2j+1)!^2}{(2j-1)!(2j+3)!} = \frac{2j+1}{2j+3}$. The upper bound is the Cauchy Schwarz inequality and holds for every $R$. □

The test is necessary and weak. For $R$ distributed as $\mathrm{Beta}(a,1)$, $\mu_j = a/(a+2j)$; at $j=1$ the condition is $a^2+4a-6 \geq 0$, that is $a \geq \sqrt{10}-2 = 1.1622777$, and for $j \geq 2$ it holds for every $a > 0$. Corollary 13.2.5(i) gives the true threshold $a \geq 2$.

### 13.2.5 Summary

Table 13.2 collects the answers. The Pólya law is unbounded, so none of them applies to it.

| law on $[-1,1]$ | ferromagnetic |
|---|---|
| uniform, and sums $\sum_k c_k U_k$ | yes |
| symmetric beta $c_\nu(1-x^2)^{\nu-1/2}$, $\nu > -\frac{1}{2}$ | yes |
| concave shoulder | yes |
| nondecreasing in $\lvert x\rvert$ (shell mixtures) | yes |
| convex shoulder, not linear on the support | no |
| $1-\lvert x\rvert^b$ | if and only if $b \geq 1$ |
| $k$ monotone kernel $f_k$ | if and only if $k \leq 2$ |
| uniform with an atom at 0, weight in $(0,1)$ | no |
| product of two uniform variables | no |
| spin one, $q\delta_0 + \frac{1-q}{2}(\delta_{-1}+\delta_1)$ | if and only if $q \leq \frac{1}{2}$ or $q = 1$ |

Table 13.2: Which laws on $[-1,1]$ are ferromagnetic. Every entry is proved in this section or in Section 13.1.

## 13.3 The Mellin transform of a clock

Let $X$ be ferromagnetic with $b = 0$ in (13.1.1) and infinitely many zeros, so that $L_X(s) = \prod_k (1+s^2/c_k^2)$, and let $T_X = \sum_k e_k/c_k^2$ be its clock (Theorem 13.1.4). For every $m$, $T_X$ is at least $c_m^{-2}$ times a sum of $m$ independent standard exponential variables, and $\mathbb{E}e^{tT_X} < \infty$ for $t < c_1^2$, so $T_X$ has moments of every real order and $\mathbb{E}T_X^{-w}$ is entire. Put

$$\mathcal{D}_X(w) = \Gamma(w)\,\mathbb{E}T_X^{-w} = \int_0^\infty u^{w-1}\Big(\frac{1}{L_X(\sqrt{u})} - 1\Big)\,du, \qquad -1 < \operatorname{Re} w < 0,$$

the Mellin transform of the Laplace transform $1/L_X(\sqrt{u})$ of $T_X$, a meromorphic function whose poles are among those of $\Gamma$. For the uniform pair $\pi U + \pi U'$, with $U'$ an independent copy of $U$, the moment generating function is $(\sinh \pi s/\pi s)^2$, the zeros are $c_k = k$, each counted twice, and the clock is $\sum_k (e_k + e_k')/k^2$, a GGC with Thorin measure $2\sum_k \delta_{k^2}$. The Gamma(2) variables of Williams' structure

(Section 8.2.3) are the sums $e_k + e'_k$ over the two spins, and the clock is $\frac{\pi^2}{2} S_2$ in the notation of (9.1.1). By (9.1.1) with $s = -2w$, and the functional equation,

$$\mathcal{D}_{\pi U + \pi U'}(w) = 2\pi^{-w}\,\Gamma(w)\,\xi(1+2w). \tag{13.3.1}$$

So the nonreal zeros of $\mathcal{D}_{\pi U+\pi U'}$ are the points $\frac{\rho-1}{2}$, and RH is the statement that they lie on the line $\operatorname{Re} w = -\frac{1}{4}$. We write $z = 2w+1$, the variable in which they are the zeros of $\xi$.

PROPOSITION 13.3.1. *Let $X$ be as above, with simple zeros $\pm i c_k$, $0 < c_1 < c_2 < \cdots$, and put*

$$A_k = -\frac{2}{c_k \varphi'_X(c_k)}, \qquad B_k = \frac{2\big(c_k \varphi''_X(c_k) - \varphi'_X(c_k)\big)}{c_k^2\, \varphi'_X(c_k)^3}.$$

*(i) Near $u = -c_k^2$, $1/L_X(\sqrt{u}) = A_k(1+u/c_k^2)^{-1} + O(1)$ and $1/L_X(\sqrt{u})^2 = A_k^2(1+u/c_k^2)^{-2} + B_k(1+u/c_k^2)^{-1} + O(1)$.*

*(ii) If $1/L_X(\sqrt{u}) = 1 + \sum_k A_k\big((1+u/c_k^2)^{-1} - 1\big)$ for $u \ge 0$ and $\sum_k |A_k| c_k^{2\sigma} < \infty$ for some $\sigma \in (-1,0)$, then*

$$\mathcal{D}_X(w) = \frac{\pi}{\sin \pi w} \sum_k A_k c_k^{2w}, \qquad -1 < \operatorname{Re} w \le \sigma.$$

*(iii) If $1/L_X(\sqrt{u})^2 = 1 + \sum_k \big[A_k^2\big((1+u/c_k^2)^{-2} - 1\big) + B_k\big((1+u/c_k^2)^{-1} - 1\big)\big]$ for $u \ge 0$ and $\sum_k (A_k^2 + |B_k|) c_k^{2\sigma} < \infty$ for some $\sigma \in (-1,0)$, then, with $X'$ an independent copy of $X$,*

$$\mathcal{D}_{X+X'}(w) = \frac{\pi}{\sin \pi w} \sum_k \big[(1-w)A_k^2 + B_k\big] c_k^{2w}, \qquad -1 < \operatorname{Re} w \le \sigma.$$

*Proof.* (i) Put $\theta = \sqrt{-u}$, so that $1/L_X(\sqrt{u}) = 1/\varphi_X(\theta)$ and $1 + u/c_k^2 = (c_k - \theta)(c_k+\theta)/c_k^2$, and expand $\varphi_X$ to second order at its simple zero $c_k$. (ii) and (iii). For $c > 0$ and $-1 < \operatorname{Re} w < 0$, the beta integral gives $\int_0^\infty u^{w-1}\big((1+u/c)^{-1} - 1\big)\,du = c^w \pi / \sin \pi w$ and $\int_0^\infty u^{w-1}\big((1+u/c)^{-2} - 1\big)\,du = c^w \Gamma(w)\Gamma(2-w) = c^w (1-w)\pi / \sin \pi w$. Since $|(1+u/c)^{-j} - 1| \le \min(2u/c, 1)$ for $u \ge 0$ and $j = 1, 2$, and only finitely many $c_k$ are below 1, the hypothesis makes the series of the integrals of the absolute values converge on every line $\operatorname{Re} w = \sigma' \le \sigma$, so sum and integral can be exchanged. □

The expansion in (ii) has the form of the partial fraction expansion that defines Krein's class, recalled before Proposition 24.2.2, for the function $u \mapsto L_X(\sqrt{u})$, whose zeros are the points $-c_k^2$, with the coefficients $A_k c_k^2$. For $X = \pi U$, $\varphi_X(\theta) =$

$\sin\pi\theta/(\pi\theta)$, so $A_k = -2(-1)^k$ and $B_k = -6$ for every $k$, and the two expansions are the classical partial fractions

$$\frac{\pi\sqrt{u}}{\sinh\pi\sqrt{u}} = 1 + 2u\sum_{k\ge1}\frac{(-1)^k}{u+k^2}, \qquad \Big(\frac{\pi\sqrt{u}}{\sinh\pi\sqrt{u}}\Big)^2 = 1 + \sum_{k\ge1}\frac{2u(u-k^2)}{(u+k^2)^2},$$

the second from $\pi^2/\sinh^2\pi x = \sum_{k\in\mathbb{Z}}(x-ik)^{-2}$. The conditions on the weights hold for every $\sigma < -\frac12$. So $\mathcal{D}_{\pi U}(w) = (2\pi/\sin\pi w)\,\eta(-2w)$, with $\eta(s) = \sum_k (-1)^{k-1}k^{-s}$, and the weights of the pair, $(1-w)A_k^2 + B_k = -(4w+2)$, do not depend on $k$:

$$\mathcal{D}_{\pi U+\pi U'}(w) = -\frac{2\pi(2w+1)}{\sin\pi w}\,\zeta(-2w), \qquad -1 < \operatorname{Re} w < -\tfrac12. \tag{13.3.2}$$

Status: proved. By the functional equation of $\zeta$ this is (13.3.1), and read the other way it proves the functional equation of $\zeta$: for $\operatorname{Re} w > 0$, $\mathcal{D}_{\pi U+\pi U'}(w) = 2\int_0^\infty x^{2w-1}(\pi x/\sinh\pi x)^2\,dx$, and the expansion $\sinh^{-2} y = 4\sum_{m\ge0}(m+1)e^{-(2m+2)y}$ gives $2(2\pi)^{-2w}\Gamma(2w+2)\zeta(2w+1)$, so the two expressions are one meromorphic function. This is the argument of Riemann's first proof of the functional equation [127, Chapter 1], a sum over residues: one integration by parts turns the integral into $4\pi(2w+1)\int_0^\infty x^{2w}/(e^{2\pi x}-1)\,dx$, and the poles $\pm ik$, $k\ge1$, of $(\pi x/\sinh\pi x)^2$, over which (13.3.2) sums, are the zeros of the moment generating function of the pair, its Lee and Yang zeros.

The hypotheses of (ii) fail for smooth densities. If $X$ has a density $q$ for which $tq(t)$ is smooth with all derivatives integrable, then $\varphi_X'(\theta) = O(\theta^{-N})$ for every $N$, so $|A_k|$ grows faster than every power of $c_k$ and $\sum_k A_k c_k^{2w}$ diverges for every $w$. This covers the laws of Theorem 13.5.4; for $m \equiv 1$, $|A_k|$ grows like $e^{\pi c_k/2\ell}$ up to powers of $c_k$ and of $\log c_k$, by the asymptotic formula for $K_{i\nu}(a)$ as $\nu\to\infty$, as the residues of the Pólya law grow like $\gamma^{-3/4}e^{\pi\gamma/4}/|\zeta'(\rho)|$ in the proof of Proposition 24.2.2. The zeros of $\mathcal{D}$ do not need the expansion. Let $W$ have a law of Theorem 13.5.4, with an independent copy $W'$, and put $X_\delta = \pi U + \delta W$, a ferromagnetic law with a smooth density of this kind. Since $0 \le 1-1/L_W(\delta\sqrt{u})^2 \le \min(1, 2\delta^2 u\,\mathbb{E}T_W)$, dominated convergence gives $\mathcal{D}_{X_\delta+X_\delta'} \to \mathcal{D}_{\pi U+\pi U'}$ uniformly on compact subsets of $-1 < \operatorname{Re} w < 0$ as $\delta\to0$, and by Hurwitz's theorem, for each $k$ with $\rho_k$ simple, $\mathcal{D}_{X_\delta+X_\delta'}$ has a zero near $\frac{\rho_k-1}{2}$ once $\delta$ is small enough (proved).

Proposition 13.3.2. *Let $X_k = \pi(U_1+\cdots+U_k)$ with independent copies $U_i$ of $U$. For $\operatorname{Re} w > 0$,*

$$\mathcal{D}_{X_k}(w) = 2(2\pi)^{-2w}\,\Gamma(2w+k)\sum_{m\ge0}\binom{m+k-1}{k-1}\Big(m+\frac{k}{2}\Big)^{-2w-k}.$$

*For $k = 2r$ the series is $\sum_{i=0}^{r-1} a_{r,i}\,\zeta(2w+1+2i)$, where $a_{r,i}$ is the coefficient of $j^{2r-1-2i}$ in $j\prod_{l=1}^{r-1}(j^2-l^2)/(2r-1)!$, and for $k = 2r+1$ it is $\sum_{i=0}^{r} b_{r,i}\,(2^{2w+1+2i}-$*

$1)\,\zeta(2w+1+2i)$, *where* $b_{r,i}$ *is the coefficient of* $h^{2r-2i}$ *in* $\prod_{l=1}^{r}\big(h^2-(l-\frac12)^2\big)/(2r)!$. *The first and the last of these coefficients are not* 0, *so for* $k\ge 3$ *the series has at least two terms. In particular*

$$\mathcal{D}_{X_1}(w)=4\pi^{-2w}\big(1-2^{-2w-1}\big)\Gamma(2w+1)\,\zeta(2w+1),$$
$$\mathcal{D}_{X_4}(w)=\tfrac13(2\pi)^{-2w}\,\Gamma(2w+4)\big[\zeta(2w+1)-\zeta(2w+3)\big],$$

*and* $\mathcal{D}_{X_6}(w)=\frac{1}{60}(2\pi)^{-2w}\,\Gamma(2w+6)\big[\zeta(2w+1)-5\zeta(2w+3)+4\zeta(2w+5)\big]$, *while* $\mathcal{D}_{X_2}$ *is* (13.3.1).

*Proof.* For $\operatorname{Re} w>0$,

$$\mathcal{D}_{X_k}(w)=\int_0^\infty u^{w-1}L_{X_k}(\sqrt u)^{-1}\,du=2\int_0^\infty x^{2w-1}\Big(\frac{\pi x}{\sinh\pi x}\Big)^k\,dx.$$

Expand $\sinh^{-k}y=2^k\sum_{m\ge0}\binom{m+k-1}{k-1}e^{-(2m+k)y}$ and integrate term by term. The binomial coefficient is $\prod_{l=1}^{k-1}(m+l)/(k-1)!$, which is the stated polynomial in $j=m+r$ for $k=2r$ and in $h=m+r+\frac12$ for $k=2r+1$. It vanishes at $j=1,\dots,r-1$ and at $h=\frac12,\dots,r-\frac12$, so the sums run over all positive integers $j$ and all positive half integers $h$, and $\sum_{m\ge0}(m+\frac12)^{-v}=(2^v-1)\zeta(v)$. □

For $k=1$ the nonreal zeros of $\mathcal{D}_{X_1}$ are those of $\zeta(z)$ and the points $z=2\pi im/\log 2$, $m\ne0$, so they lie on the two lines $\operatorname{Re}z=\frac12$ (under RH) and $\operatorname{Re}z=0$; for $k=2$ they are the zeros of $\xi(z)$. For $k=4$ and $k=6$ they do not lie on one vertical line: $\zeta(z)-\zeta(z+2)$ vanishes at $z=1.76891972665+23.3078297413i$ and $z=0.650238826497+31.7124838244i$, and $\zeta(z)-5\zeta(z+2)+4\zeta(z+4)$ at $z=3.18634599215+31.979112615i$ and $z=1.51363208762+41.215007487i$ (computed in ball arithmetic, each zero alone in a square of half side $7.5\cdot10^{-9}$ by the argument principle, `tlz_k.py`).

Laws whose characteristic function is that of $\pi U$ divided by a polynomial behave in the same way. If $\varphi_X(\theta)=\varphi_{\pi U}(\theta)/Q(\theta^2)$ for a polynomial $Q$ with $Q(0)=1$, then $1/L_X(\sqrt u)^2=Q(-u)^2/L_{\pi U}(\sqrt u)^2$, and integrating term by term for $\operatorname{Re}w>0$ gives $\mathcal{D}_{X+X'}(w)=\sum_j q_j\,\mathcal{D}_{\pi U+\pi U'}(w+j)$, where $Q(-u)^2=\sum_j q_ju^j$, an identity of meromorphic functions (proved).

For the density $(1+\cos t)/2\pi$ on $[-\pi,\pi]$, which is $C^1$ and not $C^2$ at $\pm\pi$, $Q(x)=1-x$, the zeros are $c_k=k+1$, $k\ge1$, the law is ferromagnetic, and

$$\mathcal{D}_{X+X'}(w)=\mathcal{D}_{\pi U+\pi U'}(w)+2\,\mathcal{D}_{\pi U+\pi U'}(w+1)+\mathcal{D}_{\pi U+\pi U'}(w+2),$$

a combination of $\zeta(-2w)$, $\zeta(-2w-2)$ and $\zeta(-2w-4)$ by (13.3.2). Its weights $A_k=2(-1)^{k+1}(k^2+2k)$ grow like $k^2$, so (ii) and (iii) do not apply. In the box $-1<\operatorname{Re}w<1$, $\frac12<\operatorname{Im}w<30$, which holds the 13 zeros $\frac{\rho_k-1}{2}$, $\gamma_k<60$, of $\mathcal{D}_{\pi U+\pi U'}$, it has no zero, and in $-3<\operatorname{Re}w<-2$, $\frac12<\operatorname{Im}w<30$ it has 6, at $-2.443554+15.837821i$, $-2.976747+19.238797i$, $-2.479242+21.119434i$,

$-2.273308 + 24.426843i$, $-2.862245 + 26.206760i$ and $-2.756588 + 28.553832i$, so not on one vertical line (computed in ball arithmetic, `tlz_onecos.py`).

The line is not stable inside the laws of Theorem 13.2.6, scaled by $\pi$. For $\varepsilon \geq 0$ let $X_\varepsilon$ have the density $(1 + \varepsilon t^2)/(2\pi + \frac{2}{3}\pi^3\varepsilon)$ on $[-\pi, \pi]$, which is nondecreasing in $|t|$, so that $X_\varepsilon$ and the pair $X_\varepsilon + X'_\varepsilon$ are ferromagnetic. At $\varepsilon = 0$ the zero $w_k(0) = \frac{\rho_k - 1}{2}$ of $\mathcal{D}_{\pi U + \pi U'}$ is simple whenever $\rho_k$ is, as for $k = 1, 2$, and then by the implicit function theorem it moves with

$$\frac{dw_k}{d\varepsilon}(0) = -\frac{\dot{\mathcal{D}}(w_k(0))}{\mathcal{D}'_{\pi U + \pi U'}(w_k(0))}, \qquad \dot{\mathcal{D}}(w) = -2\int_0^\infty u^{w-1}\,\frac{\dot{L}(\sqrt{u})}{L_{\pi U}(\sqrt{u})^3}\,du,$$

where $\dot{L}(s) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\big(t^2 - \frac{\pi^2}{3}\big)e^{st}\,dt$ is the derivative of $L_{X_\varepsilon}(s)$ at $\varepsilon = 0$ and $\dot{\mathcal{D}}$ that of $\mathcal{D}_{X_\varepsilon + X'_\varepsilon}$.

The computed values are $\frac{dw_1}{d\varepsilon}(0) = 5.89145 - 13.09321i$ and $\frac{dw_2}{d\varepsilon}(0) = 1.56442 - 16.25014i$, with real parts different from 0 and from each other (computed by quadrature with three rules that agree, `tlz_deform.py`). So for small $\varepsilon > 0$ the first two zeros leave the line $\operatorname{Re} w = -\frac{1}{4}$ at different rates, and the nonreal zeros of $\mathcal{D}_{X_\varepsilon + X'_\varepsilon}$ do not lie on one vertical line although every $X_\varepsilon$ is ferromagnetic. By this computation no theorem whose hypotheses hold for all symmetric densities on an interval that are nondecreasing in $|t|$ can put the zeros of $\mathcal{D}$ of the pair on one line (computed).

This section follows notes of Nicholas Polson merged into the book [246, 241].

## 13.4 The Pólya law

The Pólya variable $X$ has the density $p = \Phi/\Xi(0)$, with $\Xi(0) = \xi(\frac{1}{2}) = 0.4971207782$, and $L_X(s) = \xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ is an entire function of order 1. Its density decays like $\exp(-\pi e^{2|t|})$, so $X$ has Gaussian moments of all orders, and RH is the statement that $X$ is ferromagnetic; by Theorem 13.1.2, with $b = 0$, it is the product formula $L_X(s) = \prod_k (1 + s^2/\gamma_k^2)$ over the positive ordinates. The law is symmetric and unimodal, since $\Phi'(t) < 0$ for $t > 0$ [99, Theorem A].

### 13.4.1 Necessary conditions

If $X$ is ferromagnetic, Corollary 13.1.3 with $c_k = \gamma_k$ gives $\kappa_2 = 2\sum_k \gamma_k^{-2}$, $\kappa_4 = -12\sum_k \gamma_k^{-4}$ and $\kappa_6 = 240\sum_k \gamma_k^{-6}$. Table 13.3 computes the cumulants. The agreement of the columns is not evidence for RH. The Hadamard product $\xi(\frac{1}{2} + s)/\xi(\frac{1}{2}) = \prod_k (1 + s^2/\gamma_k^2)$, with one factor for each pair of zeros $\frac{1}{2} \pm i\gamma_k$, holds with no hypothesis if $\gamma_k$ is allowed to be complex, and it converges absolutely because $\sum_k |\gamma_k|^{-2} < \infty$; it gives the same formulas for the cumulants.

The moment test of Section 13.2.4 is sharper. For the Pólya law the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ are the Taylor coefficients of $\Xi$, up to sign and the

| | moments of $\Phi$ | Taylor coefficients of $\log\xi$ | zero sums |
|---|---|---|---|
| $\kappa_2$ | 0.0462099862308 | 0.0462099862308 | 0.046209986105 |
| $\kappa_4$ | $-4.46071191423\times10^{-4}$ | $-4.46071191423\times10^{-4}$ | $-4.46071191423\times10^{-4}$ |
| $\kappa_6$ | $3.46017435362\times10^{-5}$ | $3.46017435362\times10^{-5}$ | $3.46017435362\times10^{-5}$ |

Table 13.3: Even cumulants of the Pólya law, by quadrature of the moments of $\Phi$, from the Taylor coefficients of $\log\xi(\frac12+s)$, and from the sums $(-1)^{m+1}\frac{(2m)!}{m}\sum_k\gamma_k^{-2m}$ over the first 1700 ordinates with the smooth count beyond them. The difference $1.3\times10^{-10}$ in $\kappa_2$ is the error of the smooth count, which ignores the oscillation of $S(t)$ above height 2197. Computed by `fm_polya.py`.

factor $\Xi(0)$:

$$\frac{\Xi(t)}{\Xi(0)}=\sum_{j\ge0}(-1)^jb_jt^{2j}.$$

In this form (13.2.2), $b_j^2\ge(1+\frac1j)b_{j-1}b_{j+1}$, is the system of Turán inequalities for the moments $\int_0^\infty t^{2j}\Phi(t)\,dt$, which Csordas, Norfolk and Varga proved for every $j\ge1$, answering a question of Pólya [99]. Table 13.4 shows the margins. They decrease slowly; the Gaussian law, for which $b_j=(\sigma^2/2)^j/j!$, satisfies every inequality with equality.

The Turán inequalities do not characterise the class. The law with mass $\frac35$ at 0 and $\frac15$ at $\pm\sqrt{5/2}$ has $b_0=1$ and $b_j=\frac25(\frac52)^j/(2j)!$ for $j\ge1$, so $b_1^2=\frac14>\frac{5}{24}=2b_0b_2$, and $b_j^2/(b_{j-1}b_{j+1})=\frac{(j+1)(2j+1)}{j(2j-1)}>1+\frac1j$ for $j\ge2$: every inequality (13.2.2) holds. Yet its characteristic function $\frac35+\frac25\cos(\sqrt{5/2}\,\theta)$ has no real zero. The cumulants show it at once: by (13.1.1) a ferromagnetic law has $(-1)^{n+1}\kappa_{2n}=\frac{(2n)!}{n}\sum_k c_k^{-2n}\ge0$ for $n\ge2$, while here $\kappa_6=-\frac54$. In the numbers $p_n=(-1)^{n+1}n\kappa_{2n}/(2n)!$, which for a ferromagnetic law are $\sum_k c_k^{-2n}$ when $n\ge2$ and $b+\sum_k c_k^{-2}$ when $n=1$, the Hankel determinant $p_1p_3-p_2^2$ of $p_1,p_2,p_3$ is $-\frac{5}{1152}$, whereas for a ferromagnetic law it is $bp_3+\sum_{j<k}c_j^{-2}c_k^{-2}(c_j^{-2}-c_k^{-2})^2\ge0$.

The Turán inequalities are the first of a hierarchy. For $d,n\ge0$ let

$$J_{d,n}(x)=\sum_{i=0}^{d}\binom{d}{i}(n+i)!\,b_{n+i}\,x^i,$$

the Jensen polynomial of degree $d$ and shift $n$ of the sequence $j!\,b_j$. Pólya proved that RH holds if and only if every $J_{d,n}$ has only real zeros, and for $d=2$ this is (13.2.2) with $j=n+1$. Griffin, Ono, Rolen and Zagier proved that for each $d$ all but finitely many of the $J_{d,n}$ have only real zeros, and that all of them do when $d\le8$ [159]. Each degree is a necessary condition for the Pólya law to be ferromagnetic, and none of them is known to fail. With $F(x)=\sum_{j\ge0}b_jx^j$, so that $F(-t^2)=\varphi_X(t)$, the polynomials $J_{d,n}$ with $n$ fixed are the Jensen polynomials

| $j$ | $b_j$ | $b_j^2/(b_{j-1}b_{j+1})$ | $1+1/j$ |
|---|---|---|---|
| 1 | $2.3104993\times10^{-2}$ | 2.1496879 | 2 |
| 2 | $2.4833405\times10^{-4}$ | 1.5941155 | 1.5 |
| 3 | $1.6743526\times10^{-6}$ | 1.4057378 | 1.3333333 |
| 4 | $8.0306974\times10^{-9}$ | 1.3100631 | 1.25 |
| 5 | $2.9401358\times10^{-11}$ | 1.2518550 | 1.2 |
| 10 | $1.1310848\times10^{-24}$ | 1.1322525 | 1.1 |
| 15 | $1.7510030\times10^{-39}$ | 1.0907154 | 1.0666667 |
| 20 | $2.9919818\times10^{-55}$ | 1.0693835 | 1.05 |
| 24 | $2.1176410\times10^{-68}$ | 1.0585214 | 1.0416667 |

Table 13.4: The moment test (13.2.2) for the Pólya law: the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ and the Turán ratios, against the bound $1+\frac{1}{j}$ that every ferromagnetic law satisfies. All 24 computed ratios exceed the bound. Computed by `fm_polya.py`.

of the derivative $F^{(n)}$, and all of them have only real zeros exactly when $F^{(n)}$ is in the Laguerre Pólya class. For $n \geq 1$ that says nothing about the zeros of $\varphi_X$. For the spin one law of Table 13.2 with $q > \frac{1}{2}$, $F(x) = q + (1-q)\cosh\sqrt{x}$ and $F'(x) = (1-q)\sinh\sqrt{x}/(2\sqrt{x})$, whose zeros $-k^2\pi^2$ are real, so every $J_{d,n}$ with $n \geq 1$ has only real zeros, while $\varphi_X(t) = q + (1-q)\cos t$ has no real zero at all. In the other direction, if $\xi$ had only finitely many zeros off the critical line, then $F$, which has order $\frac{1}{2}$, would have only finitely many nonreal zeros, and $F^{(n)}$ would be in the Laguerre Pólya class for every large $n$, by the theorem of Craven, Csordas and Smith on real entire functions of order less than two [98].

A necessary condition of a different kind comes from the correlation inequalities of Section 13.1.1. Write $\Phi = e^{-F}$. Newman proved that $F'$ is convex on $[0,\infty)$ [382]; in his normalisation $\Xi(x/2) = 4\int e^{ixt - V(t)}\,dt$, and $V(t) = F(2t) + \log 2$. So the Pólya law belongs to the class of Ellis, Monroe and Newman, and every ferromagnet whose single spins are independent copies of $X$ satisfies the Griffiths, Hurst and Sherman inequality [130].

### 13.4.2 A mixture of uniform laws

Since $p$ is symmetric and unimodal, Khinchine's theorem of Section 13.2.1 writes $X \overset{d}{=} UR$, with $R$ independent of $U$ and with the density $h(r) = -2rp'(r) = -2r\Phi'(r)/\Xi(0)$ on $(0,\infty)$. Since $\mathbb{E}|X|^m = \mathbb{E}R^m/(m+1)$, the mixing variable has $\mathbb{E}R = 2\mathbb{E}|X| = 0.3464543599$ and $\mathbb{E}R^2 = 3\operatorname{Var}X = 0.1386299587$. Integrating by parts as in (13.2.1), now on the half line,

$$z\,\varphi_X(z) = 2\int_0^\infty \big(-p'(u)\big)\sin(zu)\,du,$$

so RH says that the sine transform of the positive function $-\Phi'$ on $(0,\infty)$ has only real zeros. The shape of this slope is that of neither part of Theorem 13.2.4. Since $p'' = p\,(F'^2 - F'')$, the density is concave near the origin, where $p''(0) = -F''(0)p(0) = -33.65$, and convex for large $u$, where $F'^2/F'' \to \infty$. On a bounded support, a slope that rises gives only real zeros, by Pólya's theorem, and one that falls, and is not constant on the support, gives zeros off the real line, by Theorem 13.2.4(ii); the slope of the Pólya law does both. The damped densities, proportional to $e^{tu^2/4}\Phi(u)$ with $t < 0$, are symmetric and unimodal too, hence mixtures of uniform laws, and they are mixtures of arcsine laws as well (Proposition 11.5.13), while their transforms have nonreal zeros, so neither mixture can decide RH by itself (Section 11.5.5).

The mixture does not discretise well. Replacing $R$ by $J$ equally weighted atoms at its quantiles $r_j$ of order $(j - \frac{1}{2})/J$ gives the characteristic function $\varphi_J(z) = J^{-1}\sum_j \sin(r_j z)/(r_j z)$. For $J = 5$ it has 17 zeros in the rectangle $(0.25, 100) \times (-30, 30)$, of which only 7 are real, and for $J = 20$ only 3 of 19 are real, while $\varphi_X$ has 29 zeros in $(0, 100)$, all real (computed by `fm_polya.py`). As Theorem 13.2.1 suggests, mixing in the $R$ variable destroys the property easily.

### 13.4.3 No component with finitely many zeros

A sum of independent ferromagnetic variables is ferromagnetic (Section 13.1.5), and one might hope to reach the Pólya law, or to approximate it, as such a sum of simple pieces. The growth of $L_X$ rules out most pieces.

Theorem 13.4.1. *Let $X$ be a symmetric random variable with $\mathbb{E}e^{sX} < \infty$ for every real $s$ and $\log \mathbb{E}e^{sX} = o(s^2)$ as $s \to \infty$, and let $X \stackrel{d}{=} D + R$ with $D$ and $R$ independent. Then $\log \mathbb{E}e^{sD} = o(s^2)$ as $|s| \to \infty$. In particular, if $D$ is ferromagnetic and not 0, then $b = 0$ in its Hadamard form* (13.1.1)*, and its characteristic function has infinitely many zeros. The Pólya law satisfies the hypothesis, so it has no Gaussian component, and it is not a finite or convergent infinite sum of independent ferromagnetic variables of which one is not 0 and has finitely many zeros.*

*Proof.* By independence $\mathbb{E}e^{sX} = \mathbb{E}e^{sD}\,\mathbb{E}e^{sR}$, and both factors are finite because the product is. Jensen's inequality gives $\mathbb{E}e^{sR} \geq e^{s\mathbb{E}R}$, and $\mathbb{E}R = -\mathbb{E}D$ because $\mathbb{E}X = 0$. So $\log \mathbb{E}e^{sD} \leq \log \mathbb{E}e^{sX} + s\mathbb{E}D$, and the right side is $o(s^2)$ as $s \to \pm\infty$, by the symmetry of $X$. If $D$ is ferromagnetic, $\mathbb{E}e^{sD} = e^{bs^2}\prod_k (1 + s^2/c_k^2) \geq e^{bs^2}$, so $b = 0$. If moreover there are finitely many zeros, then $\varphi_D(t) = \prod_{k\leq K}(1 - t^2/c_k^2)$ is a polynomial bounded on the real line, hence constant, so $K = 0$ and $D = 0$. For the Pólya law, $\log \mathbb{E}e^{sX} = \log \xi(\frac{1}{2} + s) - \log \xi(\frac{1}{2}) = O(s \log s)$, since $\xi$ has order 1. A summand of an almost surely convergent sum of independent variables is an independent component of the sum. □

The laws of Section 13.5 that keep the first $K$ zeros of $\varphi_X$ exactly carry a

Gaussian factor, so none of them is a component of the Pólya law. If RH holds, any decomposition of $X$ into independent ferromagnetic pieces splits the ordinates into infinite sets, each the zero set of a characteristic function $\prod_k (1 - t^2/c_k^2)$ with no Gaussian factor, like those of the uniform law, the fair sign and the symmetric beta laws in Table 13.1.

The simplest such splitting would be a Viète product $L_X(s) = \prod_j \cosh(a_j s)$, which would write $X = \sum_j a_j \varepsilon_j$ as a sum of independent scaled fair signs, a member of the Griffiths Simon class with uncoupled spins. The ordinates would then split into arithmetic progressions, and three consecutive terms of a progression satisfy $\gamma - 2\gamma' + \gamma'' = 0$. The hypothesis that the positive ordinates are linearly independent over the rationals, which Ingham used in his work on the Mertens conjecture [285], forbids such a relation. Under that hypothesis the Pólya law is not a sum of independent scaled fair signs, and if it belongs to the Griffiths Simon class at all, the spins must be coupled. Section 13.8 returns to this question.

## 13.5 Ferromagnetic approximations of the Pólya law

By Corollary 13.1.8 and the remark after it, RH holds exactly when the total variation distance $\delta$ from the Pólya law to the class of ferromagnetic laws is 0. This section bounds $\delta$ from above with no hypothesis, by the distances to laws that are proved to be ferromagnetic. The distances are enclosed in ball arithmetic, with explicit bounds for the discretisation and the tails, so the bounds on $\delta$ they give are proved.

### 13.5.1 Pólya's approximations

The Gaussian law $N(0, \operatorname{Var} X)$ is ferromagnetic and lies at total variation distance $1.48 \times 10^{-2}$ from the Pólya law. Pólya kept the leading part of the first term of $\Phi$ and made it even:

$$\Phi^*(t) = 8\pi^2 \cosh \tfrac{9t}{2}\, e^{-2\pi \cosh 2t},$$

$$\int \Phi^*(t) e^{i\theta t}\, dt = 4\pi^2 \Big[ K_{\frac{9}{4} + \frac{i\theta}{2}}(2\pi) + K_{\frac{9}{4} - \frac{i\theta}{2}}(2\pi) \Big],$$

by $\int e^{-a \cosh u + \nu u}\, du = 2K_\nu(a)$, and he proved that this transform has only real zeros [447]. So the law with density proportional to $\Phi^*$ is ferromagnetic. It is the law of $\frac{\varepsilon}{2} \log G$, where $G$ is generalised inverse Gaussian with density proportional to $g^{5/4} e^{-\pi(g + 1/g)}$, since the substitution $g = e^{2t}$ turns $e^{9t/2} e^{-2\pi \cosh 2t}$ into that density; so it can be sampled exactly. Keeping the whole first term gives the positive kernel

$$\Psi(t) = \Big( 8\pi^2 \cosh \tfrac{9t}{2} - 12\pi \cosh \tfrac{5t}{2} \Big) e^{-2\pi \cosh 2t},$$

and the same theorem of Pólya covers it.

PROPOSITION 13.5.1. *The law with density proportional to* $\Psi$ *is ferromagnetic.*

*Proof.* Write $\Psi(t) = \psi(it)f(t)$ with $f(t) = e^{-2\pi\cosh 2t}$ and $\psi(z) = 8\pi^2\cos\frac{9z}{2} - 12\pi\cos\frac{5z}{2}$. In the variable $w = z/2$, $\psi$ is the trigonometric polynomial $8\pi^2\cos 9w - 12\pi\cos 5w$ of degree 9. At $w_j = j\pi/9$, $0 \le j < 18$, the first term is $8\pi^2(-1)^j$ and the second is at most $12\pi < 8\pi^2$ in absolute value, so $\psi$ changes sign at least 18 times in a period, and since a trigonometric polynomial of degree 9 has at most 18 zeros in a period strip, all its zeros are real. So $\psi$ is a real entire function of order 1 with only real zeros. The transform of $f$ is $K_{i\theta/2}(2\pi)$, which has only real zeros, by Pólya's theorem on $\int e^{-\lambda\cosh t}e^{izt}dt$ [112, (1.2)], and $f$ decays faster than $e^{-|t|^b}$ for every $b$. Pólya's theorem on universal factors [112, p. 197], [74, Proposition 1] then says that the transform of $\psi(it)f(t)$ has only real zeros. The density decays like $f$, so it has Gaussian moments of all orders. □

As densities both approximations are poor: their total variation distances to $p$ are 0.099 and 0.161, their variances 0.0624 and 0.0726 against $\operatorname{Var} X = 0.0462$, and their values at the origin 1.404 and 1.163 against $p(0) = 1.797$, all enclosed in ball arithmetic (`fm_tvcert.py`). Their transforms have 14 zeros each in $(0.25, 62.5)$, all real, and an argument principle count in $(0.25, 62.5) \times (-2, 2)$ finds no others, the same number as the ordinates below 62.5. The positions differ: the first five are 8.99, 19.07, 24.47, 28.97, 33.01 for $\Phi^*$ and 7.60, 18.72, 24.28, 28.83, 32.89 for $\Psi$, against 14.13, 21.02, 25.01, 30.42, 32.94 (computed by `fm_polya.py`).

### 13.5.2 TILTS THAT SHRINK THE STRIP

Every zero $\gamma$ of $\varphi_X = \Xi/\Xi(0)$ satisfies $|\operatorname{Im}\gamma| < \frac{1}{2}$, because $0 < \operatorname{Re}\rho < 1$. Multipliers of the density move the zeros towards the real axis, and de Bruijn measured by how much. For $\lambda_1, \dots, \lambda_N > 0$ and $\mu \ge 0$ put

$$m(t) = e^{\mu t^2}\prod_{k\le N}\cosh(\lambda_k t), \qquad c(m) = 2\mu + \sum_{k\le N}\lambda_k^2.$$

Then $m(t) = \mathbb{E}e^{tY} = \mathbb{E}\cosh(tY)$ for $Y = \sqrt{2\mu}\,Z + \sum_k \lambda_k\varepsilon_k$, and $c(m) = \operatorname{Var} Y$.

PROPOSITION 13.5.2. *The zeros of* $\int \Phi(t)m(t)e^{izt}\,dt$ *lie in the strip* $|\operatorname{Im} z| \le (\max(\frac{1}{4} - c(m), 0))^{1/2}$. *The same holds for* $N = \infty$ *when* $\sum_k \lambda_k^2 < \infty$. *In particular, with no hypothesis, the law with density proportional to* $m\,p$ *is ferromagnetic whenever* $\operatorname{Var} Y \ge \frac{1}{4}$; *for example* $\cosh(at)\,p$ *for* $a \ge \frac{1}{2}$, $\frac{\sinh(ct)}{ct}\,p$ *for* $c \ge \frac{\sqrt{3}}{2}$, *and* $e^{\mu t^2}p$ *for* $\mu \ge \frac{1}{8}$. *If no zero of* $\zeta$ *has real part greater than* $\frac{7}{8}$, *as announced in October 2026 (Section 12.1.2), all of this holds with* $\frac{9}{64}$ *in place of* $\frac{1}{4}$; *then* $\cosh(at)\,p$ *is ferromagnetic for* $a \ge \frac{3}{8}$ *and* $\frac{\sinh(ct)}{ct}\,p$ *for* $c \ge \frac{3\sqrt{3}}{8}$.

*Proof.* For finite $N$ this is de Bruijn's Theorem 13, for the Gaussian factor $e^{\frac{1}{2}(2\mu)t^2}$, followed by his Theorem 12, for the product of cosines [112]; both apply to an even

kernel $F$ with $F(t) = O(e^{-|t|^b})$ for some $b > 2$, which $\Phi$ and $\Phi e^{\mu t^2}$ are, and both take as input a closed strip containing the zeros, here $|\operatorname{Im} z| \le \frac{1}{2}$. Under the hypothesis of the last sentence every zero has $\frac{1}{8} \le \operatorname{Re}\rho \le \frac{7}{8}$, by $\xi(s) = \xi(1-s)$, so the zeros of $\varphi_X$ lie in $|\operatorname{Im} z| \le \frac{3}{8}$, and the same two theorems give the strip with $\frac{9}{64}$ in place of $\frac{1}{4}$. For $N = \infty$ the partial products $m_N$ have $c(m_N) \uparrow c(m)$, and their transforms converge to that of $\Phi m$ locally uniformly in $z$, by dominated convergence with $m_N \le m$. By Hurwitz's theorem the zeros of the limit lie in the intersection of the strips. The tilted densities decay like $\Phi$, so they have Gaussian moments of all orders. The three examples have $\operatorname{Var} Y = a^2$, $c^2/3$ (by Viète's product $\frac{\sinh ct}{ct} = \prod_k \cosh \frac{ct}{2^k}$) and $2\mu$. □

The single cosine is elementary, and the proof shows where the number $\frac{1}{2}$ comes from. The characteristic function of $\cosh(at)p(t)$ is proportional to $\varphi_X(z+ia) + \varphi_X(z-ia)$. Take $z = x + iy$ with $y > 0$ and a pair of zeros $\beta \pm i\eta$ of $\varphi_X$, with $c = (x-\beta)^2$. De Bruijn's identity [112, Lemma 1]

$$\big(c + (y+a-\eta)^2\big)\big(c + (y+a+\eta)^2\big) - \big(c + (y-a-\eta)^2\big)\big(c + (y-a+\eta)^2\big) \\ = 8ay\,(c + y^2 + a^2 - \eta^2)$$

shows that the pair contributes a factor larger than 1 to $|\varphi_X(z+ia)|^2/|\varphi_X(z-ia)|^2$ as soon as $a > |\eta|$. With $a \ge \frac{1}{2}$ this holds for every zero, by the Hadamard product, so the two terms cannot cancel above the real axis, nor, by symmetry, below it. In the variable $s = \frac{1}{2} + iz$ the transform is $\frac{1}{2}(\xi(s+a) + \xi(s-a))$, and Lagarias proved, in addition, that its zeros are simple [321]. For the de Bruijn Newman flow the multiplier is $e^{\lambda t^2/4}$, with $c(m) = \lambda/2$; the proposition gives de Bruijn's bound $\Lambda \le \frac{1}{2}$, and the bound $\Lambda \le 0.2$ of Platt and Trudgian [419] makes $e^{\lambda t^2/4}p$ ferromagnetic for every $\lambda \ge 0.2$, again with no hypothesis.

The three multipliers of the proposition with $\operatorname{Var} Y = \frac{1}{4}$ give total variation distances $2.731 \times 10^{-3}$, $2.739 \times 10^{-3}$ and $2.751 \times 10^{-3}$ from $p$, in the order listed. The flow at $\lambda = 0.2$, with $c(m) = 0.1$, gives $1.097 \times 10^{-3}$. For comparison, $\cosh(\sqrt{0.1}\,t)\,p$, with the same contraction, gives $1.094 \times 10^{-3}$, but whether it is ferromagnetic is not known: the bound of Platt and Trudgian is a statement about the Gaussian flow, and Proposition 13.5.2 needs $\operatorname{Var} Y \ge \frac{1}{4}$, or $\frac{9}{64}$ if the announced half plane stands, both above 0.1 (the distances are enclosed in ball arithmetic by `fm_tvcert.py`).

The cosine tilts define a constant in the spirit of $\Lambda$. Let $\Omega$ be the infimum of the $a_0 \ge 0$ such that $\cosh(at)\,p$ is ferromagnetic for every $a \ge a_0$. Proposition 13.5.2 gives $\Omega \le \frac{1}{2}$, and $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is free of zeros. If RH holds, Proposition 13.1.6(i) gives $\Omega = 0$. Conversely, if $\Omega = 0$, the tilted laws converge to $p$ in total variation as $a \downarrow 0$, and $p$ is ferromagnetic by Theorem 13.1.5. So RH holds if and only if $\Omega = 0$. Unlike the flow, the operators $\varphi \mapsto \varphi(\cdot + ia) + \varphi(\cdot - ia)$ do not form a semigroup, and we do not know whether the set of good values of $a$ is an interval.

Remark 13.5.3. To first order the multipliers of this section, and the factor $1 + t^2/r^2$ of Remark 13.1.7, act on a simple zero in the same way. Let $\varphi(\theta) = L_X(i\theta) = e^{-b\theta^2}\prod_k(1-\theta^2/c_k^2)$ for a ferromagnetic $X$, as in (13.1.1), and let $c_j$ be a simple zero. The factors $\cosh(\lambda t)$, $e^{\mu t^2}$ and $1 + t^2/r^2$ replace $\varphi$, up to a constant, by $\varphi - \kappa\varphi'' + O(\kappa^2)$ locally uniformly, with $\kappa = \frac{\lambda^2}{2}$, $\mu$ and $r^{-2}$ respectively, so the zero moves to $c_j + \kappa\varphi''(c_j)/\varphi'(c_j) + O(\kappa^2)$, where

$$\frac{\varphi''(c_j)}{\varphi'(c_j)} = \frac{1}{c_j} + 4c_j\sum_{k\neq j}\frac{1}{c_j^2 - c_k^2} - 4bc_j.$$

This is the velocity of the de Bruijn Newman flow of Section 8.4.1, in the variable $\theta = z/2$ and the time $\mu = t/4$: to first order each multiplier runs the flow for time $\kappa$. The three differ at the next order, where the coefficients of $\varphi''''$ are $\frac{\kappa^2}{6}$, $\frac{\kappa^2}{2}$ and 0. They are not the only operations that preserve the class; dilations do as well. This remark follows a note of Nicholas Polson merged into the book [235].

### 13.5.3 Multipliers on Pólya's kernel

The multipliers $m(t) = e^{\mu t^2}\prod_{k\leq N}\cosh(\lambda_k t)$ of Section 13.5.2 need a hypothesis when they act on $p$, unless $c(m) \geq \frac{1}{4}$ (Proposition 13.5.2), but none when they act on Pólya's kernel $e^{-2\pi\cosh 2t}$ of Section 13.5.1, and the kernel itself can be varied. This section follows notes of Nicholas Polson merged into the book [247, 234, 235, 226]. As in Section 13.5.2, $Y = \sqrt{2\mu}\,Z + \sum_{k\leq N}\lambda_k\varepsilon_k$, so that $m(t) = \mathbb{E}e^{tY}$.

Theorem 13.5.4. *Let $a, \ell > 0$, $\mu \geq 0$ and $\lambda_1, \dots, \lambda_N \geq 0$, and let $q$ be the probability density proportional to $m(t)e^{-a\cosh\ell t}$.*

*(i) The law $q$ is ferromagnetic, it has Gaussian moments of all orders, and $b = 0$ in* (13.1.1).

*(ii) $\int_{\mathbb{R}} m(t)e^{-a\cosh\ell t}\,dt = \frac{2}{\ell}\,\mathbb{E}K_{Y/\ell}(a)$, and $L_q(s) = \mathbb{E}K_{(s+Y)/\ell}(a)/\mathbb{E}K_{Y/\ell}(a)$ for complex $s$. For $\mu = 0$ the expectations are averages over the $2^N$ values of $Y$.*

*(iii) The clock of $q$ in Theorem 13.1.4 has $\mathbb{E}e^{-uT_q} = \mathbb{E}K_{Y/\ell}(a)/\mathbb{E}K_{(\sqrt{u}+Y)/\ell}(a)$ for $u \geq 0$. For $m \equiv 1$, $T_q = \ell^{-2}\sum_j e_j/\nu_j(a)^2$, where $0 < \nu_1(a) \leq \nu_2(a) \leq \dots$ are the zeros of $\nu \mapsto K_{i\nu}(a)$ on $(0,\infty)$, listed with multiplicity.*

For $a = 2\pi$, $\ell = 2$ and $m \equiv 1$, part (iii) is the fifth row of Table 13.1. For $a = 2\pi$, $\ell = 2$ and $m(t) = \cosh\frac{9t}{2}$, $q$ is Pólya's $\Phi^*$, normalised, and (ii) is Proposition 16.7.1. With one field and no tilt the transform is a multiple of $\operatorname{Re}K_{(\lambda_1+i\theta)/\ell}(a)$, a shifted Bessel kernel, as in Theorem 25.8.3 for integer $\lambda_1/\ell$.

*Proof.* (i) For $m \equiv 1$, the substitution $v = \ell t$ and $\int_{\mathbb{R}} e^{-a\cosh v + \nu v}\,dv = 2K_\nu(a)$ give the characteristic function $K_{i\theta/\ell}(a)/K_0(a)$. Pólya proved that $z \mapsto K_{iz}(a)$ has only real zeros [447]; we recall a short proof. If $K_{i\zeta}(a) = 0$, then $f(x) = K_{i\zeta}(ae^x)$

satisfies $-f''+a^2e^{2x}f=\zeta^2 f$ on $[0,\infty)$, by Bessel's equation, with $f(0)=0$, and $f$ and $f'$ decay like $\exp(-ae^x)$ times a power of $e^x$. Integration by parts against $\bar f$ gives

$$\zeta^2\int_0^\infty |f|^2\,dx=\int_0^\infty \bigl(|f'|^2+a^2e^{2x}|f|^2\bigr)\,dx>0,$$

since $f\neq 0$, so $\zeta$ is real. Every density in the construction is at most a constant times $\exp(\mu t^2+\sum_k\lambda_k|t|-\frac{a}{2}e^{\ell|t|})$, so it has Gaussian moments of all orders. Proposition 13.1.6(i), applied once for each field, and Proposition 13.1.6(ii), with any $b_0>\mu$, give the first statement, and $b=0$ by the last statement of Theorem 13.1.2. (ii) For real $s$, Tonelli's theorem gives $\int e^{st}m(t)e^{-a\cosh\ell t}\,dt=\mathbb{E}\int e^{(s+Y)t}e^{-a\cosh\ell t}\,dt=\frac{2}{\ell}\mathbb{E}K_{(s+Y)/\ell}(a)$, which is finite because the left side is; both sides are entire in $s$, since $|K_\nu(a)|\le K_{\operatorname{Re}\nu}(a)$. (iii) is Theorem 13.1.4(i) with $b=0$ and (ii) at $s=\sqrt{u}$. For $m\equiv 1$, $L_q(s)=K_{s/\ell}(a)/K_0(a)$ vanishes exactly at $s=\pm i\ell\nu_j(a)$, so $c_j=\ell\nu_j(a)$. □

Proposition 13.5.5. *For real $v$ let $G_v$ have the generalised inverse Gaussian density $g^{v-1}e^{-\frac{a}{2}(g+1/g)}/(2K_v(a))$ on $(0,\infty)$. In Theorem 13.5.4, let $V$ have the law of $Y/\ell$ weighted by $K_V(a)$, that is $\mathbb{E}h(V)=\mathbb{E}[h(Y/\ell)K_{Y/\ell}(a)]/\mathbb{E}K_{Y/\ell}(a)$, and given $V$ let $G_V$ have the density of index $V$. Then $q$ is the law of $\ell^{-1}\log G_V$, and also of $\varepsilon\,\ell^{-1}\log G_{|V|}$, with $\varepsilon$ independent of $V$ and $G_{|V|}$. For $\mu=0$, $V$ takes the values $\ell^{-1}\sum_k\eta_k\lambda_k$, $\eta\in\{\pm1\}^N$, with probabilities proportional to $K_{\ell^{-1}\sum_k\eta_k\lambda_k}(a)$.*

*Proof.* The substitution $g=e^{\ell t}$ turns $e^{vt}e^{-a\cosh\ell t}\,dt$ into $\ell^{-1}g^{v/\ell-1}e^{-\frac{a}{2}(g+1/g)}\,dg$, and $m(t)=\mathbb{E}e^{tY}$. So $e^{\ell X}$, for $X$ with density $q$, has density proportional to $\mathbb{E}\bigl[g^{Y/\ell-1}e^{-\frac{a}{2}(g+1/g)}\bigr]=\mathbb{E}\bigl[2K_{Y/\ell}(a)f_{Y/\ell}(g)\bigr]$, where $f_v$ is the density of $G_v$; this is the mixture stated. The law of $V$ is symmetric, because that of $Y$ is and $K_{-v}=K_v$, and $1/G_v$ has the law of $G_{-v}$, which gives the second form. □

With one field the two signs have equal weights, and $a=2\pi$, $\ell=2$, $\lambda_1=\frac{9}{2}$ give the description of $\Phi^*$ in Section 13.5.1. With more fields the weights differ: for $\lambda_1=\lambda_2>0$ the index 0 and the indices $\pm2\lambda_1/\ell$ have total weights in the ratio $K_0(a):K_{2\lambda_1/\ell}(a)$, and $K_v(a)=\int_0^\infty e^{-a\cosh u}\cosh(vu)\,du$ increases in $v>0$. For $\mu=0$ every law $q$ can be sampled exactly, by drawing $V$ and then $G_V$.

Remark 13.5.6. The kernel is a product of two Gumbel distribution functions, and this builds $q$ from independent variables. Let $W_1,W_2$ be independent with $\mathbb{P}(W_i\le x)=e^{-e^{-x}}$, put $c=\log\frac{a}{2}$, $A=(W_1+c)/\ell$ and $B=-(W_2+c)/\ell$. Then

$$\mathbb{P}(A\le t\le B)=e^{-\frac{a}{2}e^{-\ell t}}\,e^{-\frac{a}{2}e^{\ell t}}=e^{-a\cosh\ell t},$$

which for $a=2\pi$ and $\ell=2$ is the two sided Gumbel form of Pólya's kernel in the list after Theorem 25.1.1 (there in the variable $u=t/2$). By Fubini's theorem $\mathbb{E}(B-A)^+=\int_{\mathbb{R}}e^{-a\cosh\ell t}\,dt=\frac{2}{\ell}K_0(a)$, where $(B-A)^+=\ell^{-1}\log^+\bigl(4e^{-W_1-W_2}/a^2\bigr)$. Weight the pair $(A,B)$ by $(B-A)^+/\mathbb{E}(B-A)^+$

and, given the pair, let $U$ be uniform on $[A,B]$; then $U$ has density $\mathbb{P}(A \le t \le B)/\mathbb{E}(B-A)^+$, which is $q$ for $m \equiv 1$. For general $m$ take $Y$ independent of $(W_1,W_2)$, weight by $\mathbf{1}_{\{A<B\}}\int_A^B e^{Ys}\,ds$, and give $U$ the density proportional to $e^{Ys}$ on $[A,B]$; the same computation gives the density proportional to $m(t)e^{-a\cosh \ell t}$. The length bias cannot be dropped: an interval equal to $[0,1]$ or $[0,2]$ with probability $\frac12$ each puts a uniform point in $[0,1]$ with probability $\frac34$, whereas the density proportional to $\mathbb{P}(t \in [A,B])$ gives $[0,1]$ the mass $\frac23$. In the same way the logistic law of scale $1/\pi$ in the second row of Table 13.1 is the law of $(W_1-W_2)/\pi$, because $\Gamma(1+i\theta)\Gamma(1-i\theta) = \pi\theta/\sinh \pi\theta$.

Remark 13.5.7. The proof of Proposition 8.7.8 holds for $K_{i\nu}(r)$ at every $r>0$, and its Stirling step gives $\vartheta(\nu,r) = \nu\log\frac{2\nu}{er} + \frac{\pi}{4} + O(1/\nu)$. So for $m\equiv 1$ the number of zeros of $\varphi_q(\theta) = K_{i\theta/\ell}(a)/K_0(a)$ in $(0,\theta]$ is

$$\frac{\theta}{\pi\ell}\log\frac{2\theta}{ea\ell} + O(1).$$

The zeros of $\varphi_X$ with real part in $(0,\theta]$ are those with $0 < \operatorname{Im}\rho \le \theta$, counted with no hypothesis by the Riemann von Mangoldt formula $\frac{\theta}{2\pi}\log\frac{\theta}{2\pi e} + O(\log\theta)$, and the two main terms agree exactly when $\ell = 2$ and $a = 2\pi$: among the kernels $e^{-a\cosh \ell t}$, Pólya's is the only one whose zero count has the main term of the Riemann von Mangoldt formula (compare Section 8.7). Fields keep the main term. If $\mu = 0$, a further field $\lambda$ replaces the transform $F$ by $\operatorname{Re} F(\theta - i\lambda)$, up to a constant; as in the paragraph before Proposition 16.7.2, $\arg F(\theta - i\lambda)$ increases with $\theta$, the zeros on $(0,\infty)$ are where it crosses $(\frac12+\mathbb{Z})\pi$, and an integration by parts against the counting function of $F$ shows that the count changes by $O(1)$. So for $m(t) = \prod_{k\le N}\cosh(\lambda_k t)$ the count is again $\frac{\theta}{\pi\ell}\log\frac{2\theta}{ea\ell} + O(1)$. A factor $1+t^2/r^2$ (Remark 13.1.7) also changes it by $O(1)$, because the zeros of $\varphi \mp \varphi'/r$ interlace with those of $\varphi$, by Rolle's theorem applied to $e^{\mp r\theta}\varphi$. On Pólya's kernel, then, $N(\theta) - N_q(\theta) = S(\theta) + O(1)$, where $N(\theta)$ counts the zeros of $\xi$ with $0 < \operatorname{Im}\rho \le \theta$ and $N_q(\theta)$ the zeros of $\varphi_q$ in $(0,\theta]$; Selberg's central limit theorem [477] makes $S$ unbounded above and below, so $\sup_\theta |N(\theta) - N_q(\theta)| = \infty$ for every such $q$. A Gaussian factor changes the main term: for $\mu > 0$ the count is

$$\frac{\theta}{\pi\ell}\log\frac{2\theta}{ea\ell} + \frac{\mu}{\ell^2}\log\theta + O(1),$$

on Pólya's kernel $\vartheta(\theta)/\pi + \frac{\mu}{4}\log\theta + O(1)$, the analogue of the term $\frac{t}{16}\log T$ of Proposition 15.2.7(ii) with $\mu = \frac{t}{4}$. For the proof write $e^{\mu t^2} = \mathbb{E}e^{\sqrt{2\mu}Zt}$ with $Z$ standard normal, so that $\varphi_q(\theta)$ is proportional to $\mathbb{E}\,F(\theta - i\sqrt{2\mu}Z)$, with $F$ the transform for $\mu = 0$. Stirling's formula for $K_{(c+i\theta)/\ell}(a)$, uniform for $|c| \le \theta^{1/3}$, and a saddle point at $\sqrt{2\mu}Z \approx \frac{2\mu}{\ell}\log\theta$ give $\varphi_q(\theta) = A(\theta)\big(\cos\Theta(\theta) + o(1)\big)$ in a strip about the real axis, with $A>0$ and $\Theta(\theta) = \frac{\theta}{\ell}\log\frac{2\theta}{ea\ell} + \frac{\pi\mu}{\ell^2}\log\theta + O(1)$; the tails $|c| > \theta^{1/3}$ are controlled by the bound $|\Gamma(\sigma+i\tau)| \ll |\sigma+i\tau|^{\sigma-1/2}e^{\sigma-\pi|\tau|/2}$ for $\sigma \ge 0$. As the zeros are real (Theorem 13.5.4), the count is $\Theta(\theta)/\pi + O(1)$.

Table 13.5 lists members fitted to $p$ by minimising $\mathrm{TV}(q,p)$ numerically. The parameters are exact decimals, so each row is one explicit ferromagnetic law, and its distance is enclosed in ball arithmetic as in Section 13.5.6.

| $m(t)$ | $a$ | $\ell$ | $\mathrm{TV}(q,p)$ |
|---|---|---|---|
| $1$ | $2.838657$ | $2.574646$ | $< 3.2 \times 10^{-4}$ |
| $\cosh(\lambda t)$, $\lambda = 2.146184$ | $5.483952$ | $2.062866$ | $< 6.1 \times 10^{-6}$ |
| $\cosh(\lambda t)$, $\lambda = 2.618518$ | $2\pi$ | $2$ | $< 2.0 \times 10^{-3}$ |
| $\cosh^2(\lambda t)$, $\lambda = 1.792548$ | $2\pi$ | $2$ | $< 2.1 \times 10^{-4}$ |
| $e^{\mu t^2}\cosh(\lambda t)$, $\lambda = 2.125355$, $\mu = 0.9417395$ | $2\pi$ | $2$ | $< 8.0 \times 10^{-6}$ |

Table 13.5: Laws of Theorem 13.5.4 fitted to the Pólya law. The last three rows keep Pólya's kernel $e^{-2\pi\cosh 2t}$ (Remark 13.5.7). The parameters are rounded to seven significant digits from a numerical minimisation (`wf_fit.py`) and then fixed; the distances are upper bounds enclosed in ball arithmetic and rounded up (`wf_tvcert.py`).

The second row is the closest law the search found: a second free field did not lower the distance, nor did a tilt with a free kernel (`wf_fit.py`). It is ferromagnetic by Theorem 13.5.4, with no positivity proof, and its distance improves the bound $1.266 \times 10^{-4}$ that the zero completed law $f_{16}$ of Section 13.5.4 gives in Proposition 13.5.15, though not the bound $1.932 \times 10^{-6}$ of $f_{150}$.

A small distance does not pin the zeros. For every $\theta$, $|\varphi_q(\theta) - \varphi_X(\theta)| \le 2\,\mathrm{TV}(q,p)$, but for the second row, numerically, $|\varphi_X(\theta)| < 2\,\mathrm{TV}(q,p)$ on $[23,100]$ (computed, not certified), so there the inequality does not determine the sign of $\varphi_q$ and the distance does not control its zeros (`wf_zeros.py`). The first zero of $\varphi_q$ for the second row is $14.133924$, against $\gamma_1 = 14.134725$, and over the first twenty zeros the mean and largest distances to $\gamma_1, \dots, \gamma_{20}$ are $0.449$ and $1.145$; for the fourth row, on Pólya's kernel, they are $0.501$ and $1.284$. The zeros are located by certified sign changes in ball arithmetic; that there are no further zeros in the strip $|\operatorname{Im}\theta| < 2$ up to the twenty first is computed, by the change of phase on a grid, and not certified. The clock of the second row has $\mathbb{E}T_q = 0.0231055$ and $\operatorname{Var} T_q = 3.71451 \times 10^{-5}$, against $\mathbb{E}T = 0.0231049931$ and $\operatorname{Var} T = 3.71725993 \times 10^{-5}$ (`wf_clock.py`).

A sequence of such laws on one kernel cannot approach $p$, with or without RH.

Proposition 13.5.8. *Let $a_n \to a$ and $\ell_n \to \ell$ with $a, \ell > 0$, and let $q_n$ be the probability density proportional to $m_n(t)e^{-a_n\cosh \ell_n t}$, where $m_n(t) = \psi_n(it)$ for an even function $\psi_n$ of the Laguerre Pólya class with $\psi_n(0) > 0$. Then $q_n$ does not converge in law to $p$. The multipliers of Theorem 13.5.4 are of this form, and so are $e^{\mu t^2}\prod_{k\ge 1}\cosh(\lambda_k t)$ with $\sum_k \lambda_k^2 < \infty$ and the multiplier $\psi(it)$ of $\Psi$ in Proposition 13.5.1.*

*Proof.* By the Laguerre Pólya theorem [335], $\psi_n(z) = \psi_n(0)e^{-\alpha_n z^2}\prod_j (1 - z^2/z_{n,j}^2)$

with $\alpha_n \geq 0$ and $z_{n,j} > 0$, so $m_n(t) = \psi_n(0)e^{\alpha_n t^2}\prod_j(1 + t^2/z_{n,j}^2)$ is even and nondecreasing in $|t|$, and $|m_n(w)| \leq m_n(|w|)$ for complex $w$, factor by factor. Write $q_n = g_n k_n$ with $k_n(t) = e^{-a_n \cosh \ell_n t}$ and $g_n$ a multiple of $m_n$. Since $k_n$ decreases in $|t|$, for $U \geq 0$

$$1 \geq \int_U^{U+1} q_n(t)\,dt \geq g_n(U)\,k_n(U+1),$$

and $k_n(U+1) \to e^{-a\cosh \ell(U+1)} > 0$, so the entire functions $g_n$ are bounded on every disc uniformly in $n$ and form a normal family. If $q_n \to p$ in law, a subsequence of $(g_n)$ converges locally uniformly on $\mathbb{C}$ to an entire function $g$, so $q_n \to g(t)e^{-a\cosh \ell t}$ locally uniformly on $\mathbb{R}$ along it, and $\int_\alpha^\beta p = \int_\alpha^\beta g(t)e^{-a\cosh \ell t}\,dt$ for all $\alpha < \beta$. Then $\Phi = \Xi(0)\,g(t)e^{-a\cosh \ell t}$ on $\mathbb{R}$ would extend to an entire function, whereas $\Phi$ has a natural boundary on the lines $\operatorname{Im} t = \pm\pi/4$ (Section 7.1.3). □

In particular no choice of fields on one kernel, finitely or infinitely many, gives the Pólya law; this answers the question of the notes [247, 234, 235]. A sequence of laws of Theorem 13.5.4 converging in law to $p$ would prove RH, by Theorem 13.1.5; by the proposition its kernel parameters $(a_n, \ell_n)$ could have no limit point in $(0,\infty)^2$, and whether such a sequence exists is not known.

Spins with these laws can be coupled.

Corollary 13.5.9. *Let $q_1, \dots, q_N$ be laws of Theorem 13.5.4, with parameters $a_i, \ell_i$, and let $J = (J_{ij})$ be symmetric with $J_{ij} \geq 0$ and $J_{ii} = 0$. Let $(X_1, \dots, X_N)$ have density proportional to $\exp\big(\sum_{i<j} J_{ij}t_it_j\big)\prod_i q_i(t_i)$ on $\mathbb{R}^N$. Then $M = \sum_i c_iX_i$ is ferromagnetic for all $c_1, \dots, c_N \geq 0$, not all 0.*

*Proof.* Each $q_i(t)$ is at most a constant times $\exp(-\frac{a_i}{4}e^{\ell_i|t|})$, so the density is integrable and symmetric under $t \mapsto -t$, and $\mathbb{E}e^{bM^2} < \infty$ for every $b$. By Theorem 13.5.4(i) each $q_i$ has Gaussian moments of all orders and a moment generating function with only imaginary zeros, so by Newman's theorem [379, 339], in its form for fields $h_i$ with $\operatorname{Re} h_i > 0$ for every $i$, $Z(h) = \mathbb{E}\exp(\sum_i h_iX_i)$ has no zero with $\operatorname{Re} h_i > 0$ for every $i$. For $\varepsilon > 0$ the function $z \mapsto Z(zc + \varepsilon\mathbf{1})$ therefore has no zero in $\operatorname{Re} z > 0$; letting $\varepsilon \downarrow 0$, Hurwitz's theorem and $Z(0) = 1$ show that $L_M(z) = Z(zc)$ has none there, and by symmetry none in $\operatorname{Re} z < 0$. □

For two spins with $J_{12} = J$ and $c_1 = c_2 = 1$, expanding $e^{Jt_1t_2}$ gives the transform of $X_1 + X_2$, up to a constant,

$$Z_J(\theta) = \sum_{n\geq 0} \frac{(-J)^n}{n!}\,\varphi_1^{(n)}(\theta)\,\varphi_2^{(n)}(\theta),$$

with $\varphi_i$ the characteristic function of $q_i$. Whether such coupled laws can approach $p$ is not known; Proposition 13.5.8 does not apply to them, since the law of $X_1+X_2$ is not a multiplier on one kernel.

At $J = 0$ the zero counts add: with every $\mu_i = 0$, the transform $\prod_i \varphi_i(\theta)$ of $X_1 + \cdots + X_N$ has $\sum_i \frac{\theta}{\pi \ell_i} \log \frac{2\theta}{e a_i \ell_i} + O(1)$ zeros in $(0, \theta]$, by Remark 13.5.7, and this has the main terms of the Riemann von Mangoldt formula exactly when

$$\sum_i \frac{1}{\ell_i} = \frac{1}{2}, \qquad \sum_i \frac{\log(a_i \ell_i)}{\ell_i} = \frac{1}{2} \log 4\pi;$$

for $N = 1$ this is Pólya's kernel, and for $\ell_i = 2N$ it asks $\prod_i a_i = (2\pi/N)^N$. Coupling moves the count by an unbounded amount. For two spins and $J > 0$, Jensen's inequality under the product of the tilted laws proportional to $q_i(t)e^{-yt}$ gives $\log Z_J(iy) - \log Z_0(iy) \geq J\, m_1(y) m_2(y) + O(1)$, where $-m_i(y)$ is the tilted mean and $m_i(y) \sim \ell_i^{-1} \log y$ by Stirling's formula. Both laws are ferromagnetic with $b = 0$ in (13.1.1) (Theorem 13.1.2), so $\log \big(Z(iy)/Z(0)\big) = \int_0^\infty \log(1 + y^2/r^2)\, dN(r)$ for their counting functions, and an $N_J - N_0$ bounded above would make the difference at most $O(\log y)$. So $N_J(\theta) - N_0(\theta)$ is unbounded above. A saddle point computation suggests $N_J(\theta) - N_0(\theta) = \big(\sum_{i<j} J_{ij}/(\ell_i \ell_j)\big) \log \theta + O(1)$ for $X_1 + \cdots + X_N$ (heuristic).

For $N = 2$ and $\ell_1 = \ell_2 = 4$ the condition displayed above is $a_1 a_2 = \pi^2$, met by two copies of the law $q$ with $(a, \ell) = (\pi, 4)$ and $m \equiv 1$, for which $\varphi_q(\theta) = K_{i\theta/4}(\pi)/K_0(\pi)$. The Mellin transform $\mathcal{D}$ of the clock of this pair (Section 13.3), and that of a single copy, have no zero in the box $-0.95 < \operatorname{Re} w < -0.05$, $\frac{1}{2} < \operatorname{Im} w < 11$, in which $\mathcal{D}_{\pi U + \pi U'}$ has the two zeros $\frac{\rho_1 - 1}{2}$ and $\frac{\rho_2 - 1}{2}$ (computed, `tlz_bessel.py`; the argument principle on the boundaries of the boxes $\frac{1}{2} < \operatorname{Im} w < \frac{17}{2}$ and $\frac{17}{2} < \operatorname{Im} w < 11$, sampled, not a ball proof, with the uniform pair computed by the same quadrature as a control, which gives winding 1 in each box). The zeros $\pm i c_k$ of this pair have the main term of the Riemann von Mangoldt formula, and those of the uniform pair the count $2\theta$ (Remark 13.5.7). This paragraph follows a note of Nicholas Polson merged into the book [246].

Proposition 13.5.10. *For two spins as above let $J \downarrow 0$.*

*(i) If $\alpha$ is a simple zero of $\varphi_1$ and $\varphi_2(\alpha) \neq 0$, then $Z_J$ has a zero $\alpha + J\varphi_2'(\alpha)/\varphi_2(\alpha) + O(J^2)$.*

*(ii) If $\alpha$ is a simple zero of both $\varphi_1$ and $\varphi_2$, then $Z_J$ has the two real zeros*

$$\alpha \pm \sqrt{J} + \frac{J}{4}\Big(\frac{\varphi_1''}{\varphi_1'} + \frac{\varphi_2''}{\varphi_2'}\Big)(\alpha) + O(J^{3/2}),$$

*at distance $2\sqrt{J} + O(J^{3/2})$.*

*Proof.* The expansion gives $Z_J = \varphi_1\varphi_2 - J\varphi_1'\varphi_2' + O(J^2)$ locally uniformly, and (i) is the implicit function theorem at the simple zero $\alpha$ of $\varphi_1\varphi_2$. For (ii) put $\theta = \alpha + \sqrt{J}\, v$, $A = \varphi_1'(\alpha)\varphi_2'(\alpha) \neq 0$ and $B = (\varphi_1'\varphi_2'' + \varphi_1''\varphi_2')(\alpha)$; then $Z_J/(AJ) = v^2 - 1 + \sqrt{J}\, \frac{B}{2A}(v^3 - 2v) + O(J)$, and the implicit function theorem at $v = \pm 1$ gives

$v = \pm 1 + \sqrt{J}\,\frac{B}{4A} + O(J)$. Both zeros are real because $X_1 + X_2$ is ferromagnetic (Corollary 13.5.9). □

Numerically (`bs_coupled.py`), for the spins $(a, \ell) = (2\pi, 1)$ and $(2\pi, 1.3)$ with $m \equiv 1$, the transform of $X_1 + X_2$ has 39, 39 and 40 zeros in $(0, 45)$ for $J = 0$, $\frac{1}{8}$ and $\frac{1}{2}$. At $J = \frac{1}{8}$ Proposition 13.5.10(i) predicts the zeros to within 0.084 for the 17 zeros that lie at distance more than $\frac{1}{2}$ from the zeros of the other spin, and to within 0.017 for the 4 beyond 1; the prediction is poor near a near coincidence, at the zeros 43.66 and 43.69 of the two spins, which is the case of a shared zero split by (ii). For two spins $(2\pi, 1)$ the double zeros of $Z_0$ split into 24 pairs in $(0, 45)$, and their separations lie between 0.458 and 0.534 (mean 0.486) at $J = \frac{3}{40}$, against $2\sqrt{J} = \sqrt{3/10} = 0.548$, and between 0.580 and 0.919 (mean 0.688) at $J = \frac{1}{4}$, against $2\sqrt{J} = 1$. The separation is below $2\sqrt{J}$ and decreases with the height (computed).

PROPOSITION 13.5.11. *For $n \geq 1$ let $r_n(t) = e^{|t|/2}\exp(-\pi n^2 e^{2|t|})$, and consider the laws with densities proportional to $r_n$ and to $\sum_{m\geq 1} r_m$: the components of the theta series in Riemann's formula and their mixture (Remark 22.3.2). Each of these laws is symmetric, has Gaussian moments of all orders and a characteristic function positive on $\mathbb{R}$, and none of them is ferromagnetic: each characteristic function has infinitely many zeros, and none of them is real. Status: proved.*

*Proof.* Let $X$ have one of these laws and characteristic function $\varphi$. As $\sum_{m\geq 1} e^{-\pi m^2 x} \leq e^{-\pi x}/(1-e^{-3\pi})$ for $x \geq 1$, each density is at most a constant times $e^{|t|/2}\exp(-\pi e^{2|t|})$, which gives the moments, and $\varphi > 0$ on $\mathbb{R}$ by Remark 22.3.2. The same bound gives $\mathbb{E}e^{b|X|} \leq C\int_0^\infty e^{(b+\frac{1}{2})t}e^{-\pi e^{2t}}\,dt \leq C\pi^{-b/2-1/4}\,\Gamma(\frac{b}{2}+\frac{1}{4})$ for $b \geq 0$, so $\varphi$ is entire and, since $|\varphi(z)| \leq \mathbb{E}e^{|\operatorname{Im} z||X|}$, of order at most one by Stirling's formula. If $\varphi$ had finitely many zeros, Hadamard's theorem would give $\varphi(z) = P(z)e^{az}$ with a polynomial $P$, of degree $d$ say, and a constant $a$, so that $|\varphi(iy)| \leq C'(1+|y|)^d e^{|a||y|}$. But $\varphi(iy) = \mathbb{E}\cosh(yX) \geq \frac{1}{2}e^{A|y|}\,\mathbb{P}(|X| > A)$ for every $A > 0$, and $\mathbb{P}(|X| > A) > 0$ because the density is positive; with $A > |a|$ this is a contradiction for large $|y|$. So $\varphi$ has infinitely many zeros, none of them real, and $X$ is not ferromagnetic (Definition 13.1.1). □

The components are laws of $\pm\frac{1}{2}\log X$ with $X$ of gamma type truncated to $[1, \infty)$, whose densities have a corner at the origin; as the cusped tilts of $p$ in Theorem 13.5.16 are not, by the same mechanism. The product of two Gumbel distribution functions in Remark 13.5.6 gives instead the smooth kernel $e^{-a\cosh \ell t}$ and the ferromagnetic laws of Theorem 13.5.4. Since $\varphi > 0$ on $\mathbb{R}$, the real zeros of $\Xi$ are zeros of no component: by Remark 22.3.2 they are the points $\theta$ where the positive function $(\theta^2 + \frac{1}{4})W(\theta)$, with $W$ the function of Remark 22.3.2, equals 1. For $n = 1$ the characteristic function has 5 zeros in the box $0 \leq \operatorname{Re}\theta \leq 40$, $0 \leq \operatorname{Im}\theta \leq 30$, the two of least modulus being $13.0885{+}10.8707i$ and $20.1987{+}15.0786i$

(computed; argument principle, `px_cbox.py`). This proposition follows a note of Nicholas Polson merged into the book [244].

### 13.5.4 Laws that keep the first zeros

Let $0 < \gamma_1 \le \gamma_2 \le \dots$ be the ordinates of the zeros on the critical line, with multiplicity, and put $s_K = \sum_{k\le K} \gamma_k^{-2}$ and $\beta_K = \frac{1}{2}\operatorname{Var} X - s_K$. The idea is to keep the first $K$ factors of the Hadamard product exactly and to replace the others by the Gaussian factor with the same effect on the variance:

$$\varphi_K(\theta) = e^{-\beta_K \theta^2} \prod_{k\le K} \Big(1 - \frac{\theta^2}{\gamma_k^2}\Big).$$

A symmetric Maxwell variable $M$, with density $x^2 e^{-x^2/2}/\sqrt{2\pi}$, has $\mathbb{E}e^{i\theta M} = (1-\theta^2)e^{-\theta^2/2}$ and $\operatorname{Var} M = 3$; it is ferromagnetic, with $L_M(s) = (1+s^2)e^{s^2/2}$. Let $M_1, M_2, \dots$ be independent copies.

Theorem 13.5.12. *(i) $\beta_K > 0$ for every $K$. If $3s_K \le \operatorname{Var} X$, then $\varphi_K$ is the characteristic function of*

$$X_K = \big(\operatorname{Var} X - 3s_K\big)^{1/2} Z + \sum_{k\le K} \frac{M_k}{\gamma_k},$$

*whose law $f_K$ is ferromagnetic, with $\operatorname{Var} X_K = \operatorname{Var} X$ and $\kappa_4(X_K) = -12\sum_{k\le K}\gamma_k^{-4}$. The condition holds for $K \le 16$ and fails for $K = 17$.*

*(ii) For every $K$, if the inverse Fourier transform $f_K$ of $\varphi_K$ is nonnegative, then $f_K$ is a ferromagnetic probability density.*

*(iii) RH holds if and only if $\varphi_K \to \varphi_X$ pointwise as $K \to \infty$. Under RH, $f_K \to p$ in law along every sequence of $K$ for which $f_K \ge 0$.*

*Proof.* With no hypothesis, $\frac{1}{2}\operatorname{Var} X = \sum_k \gamma_k^{-2}$, where the sum runs over one zero of each pair $\pm\gamma$ and complex $\gamma$ are allowed (Section 13.4.1). A zero off the line contributes, with its conjugate, $2\operatorname{Re}\gamma^{-2} > 0$, since $|\operatorname{Re}\gamma| > 14 > |\operatorname{Im}\gamma|$. So $\beta_K$ is the sum of $\gamma_k^{-2}$ over the line zeros with $k > K$ and of these positive terms, and it is positive.

(i) $M_k/\gamma_k$ has characteristic function $(1-\theta^2/\gamma_k^2)e^{-\theta^2/2\gamma_k^2}$, so the sum over $k \le K$ accounts for every factor of $\varphi_K$ except $e^{-(\beta_K - s_K/2)\theta^2}$, a Gaussian factor of variance $2\beta_K - s_K = \operatorname{Var} X - 3s_K$. A sum of independent ferromagnetic variables is ferromagnetic. The cumulants come from $\log\varphi_K$. Since $s_K$ increases with $K$, the last statement is $3s_{16} < \operatorname{Var} X < 3s_{17}$. In ball arithmetic, with the certified zeros of Arb [290] and $\operatorname{Var} X = (\log\xi)''(\frac{1}{2})$ from the Taylor series of $\log\xi$, $3s_{16} = 0.0456803$, $\operatorname{Var} X = 0.0462100$ and $3s_{17} = 0.0463006$ (`fm_cutoff.py`).

(ii) $f_K$ has total mass $\varphi_K(0) = 1$, and its moment generating function $e^{\beta_K s^2}\prod_{k\le K}(1 + s^2/\gamma_k^2)$ vanishes only on the imaginary axis. By Proposition 13.5.13 below, $f_K$ is a Gaussian density times a polynomial, so it has Gaussian moments.

(iii) As $K \to \infty$, $\beta_K \downarrow \beta_\infty$, the sum of the terms $2\,\mathrm{Re}\,\gamma^{-2}$ over the zeros off the line, and $\varphi_K \to \varphi_\infty = e^{-\beta_\infty \theta^2}\prod_k(1-\theta^2/\gamma_k^2)$, a product over the line zeros, whose zeros are all real. If RH holds, $\beta_\infty = 0$ and $\varphi_\infty = \varphi_X$ by the Hadamard product. If RH fails, $\varphi_X$ vanishes at a nonreal point and $\varphi_\infty$ does not, so $\varphi_\infty \neq \varphi_X$. Along a sequence with $f_K \ge 0$ the $\varphi_K$ are characteristic functions, and Lévy's continuity theorem gives convergence in law. □

The law $f_{16}$ is proved ferromagnetic with no hypothesis, and its distance from the Pólya law is $\mathrm{TV}(f_{16}, p) = 1.27 \times 10^{-4}$, ten times smaller than that of the tilt by the flow at $\lambda = 0.2$. Beyond $K = 16$ the Maxwell construction runs out of Gaussian variance, and positivity must be checked directly. The exact condition comes from a closed form, in which $\mathrm{He}_n$ is the Hermite polynomial with $\mathrm{He}_n(y) = \mathbb{E}(y + iZ)^n$, $g_v$ is the $N(0, v)$ density, and $e_j$ is the $j$th elementary symmetric function of $\gamma_1^{-2}, \dots, \gamma_K^{-2}$.

Proposition 13.5.13. *For $v > 0$ the function $e^{-v\theta^2/2}\prod_{k\le K}(1 - \theta^2/\gamma_k^2)$ is the Fourier transform of*

$$f_{K,v}(x) = g_v(x)\,Q_{K,v}\big(x/\sqrt{v}\big), \qquad Q_{K,v}(y) = \sum_{j=0}^{K} \frac{e_j}{v^j}\,\mathrm{He}_{2j}(y),$$

*and $f_K = f_{K,2\beta_K}$. There is a threshold $v^*(K)$ such that $f_{K,v} \ge 0$ if and only if $v \ge v^*(K)$, and $v^*(K) \le s_K$. Moreover $Q_{K,v}(y) > 0$ for $|y| \ge \sqrt{8K+2}$, so the sign of $f_{K,v}$ is decided on $|x| < \sqrt{(8K+2)v}$.*

*Proof.* Differentiating $g_v$ gives $g_v^{(n)}(x) = (-1)^n v^{-n/2}\mathrm{He}_n(x/\sqrt{v})g_v(x)$, and the transform of $g_v^{(2j)}$ is $(-1)^j\theta^{2j}e^{-v\theta^2/2}$. Expanding $\prod_k(1-\theta^2/\gamma_k^2) = \sum_j(-1)^j e_j\theta^{2j}$ gives the formula. For $v < v'$, $f_{K,v'} = f_{K,v} * g_{v'-v}$, so positivity persists as $v$ increases, and the set of good $v$ is closed. At $v = s_K$ the transform is that of $\sum_{k\le K} M_k/\gamma_k$, so $v^*(K) \le s_K$. The zeros of $\mathrm{He}_n$ lie in $|y| < \sqrt{4n+2}$, by a Sturm comparison for the Hermite equation, so for $|y| \ge \sqrt{8K+2}$ every $\mathrm{He}_{2j}$ with $j \le K$ is positive, and so are the coefficients $e_j v^{-j}$. □

For $K = 1$, $Q_{1,v}(y) = 1 + (y^2-1)/(v\gamma_1^2)$, so $v^*(1) = \gamma_1^{-2} = s_1$, the Maxwell value. For larger $K$ the threshold is far smaller, and Table 13.6 brackets it for $K \le 800$. The Hermite sum cancels catastrophically, so the computation runs in ball arithmetic [290] at thousands of bits (8210 at $K = 800$), from the certified ordinates $\gamma_1, \dots, \gamma_{800}$ and $\mathrm{Var}\,X = (\log\xi)''(\frac{1}{2})$. The recurrence $\mathrm{He}_{n+1} = y\mathrm{He}_n - n\mathrm{He}_{n-1}$ gives $Q_{K,v}$ in the monomial basis. On an interval $[c-r, c+r]$ its Taylor

coefficients $T_m$ at $c$ come exactly from one polynomial product, and $Q_{K,v}(c+h) \geq T_0 - \sum_{m\geq 1} |T_m| r^m$ for $|h| \leq r$, with no remainder; intervals on which this bound is not positive are bisected. A cover of $[0, \sqrt{8K+2}]$ by such intervals, with Proposition 13.5.13 beyond it and the evenness of $Q_{K,v}$, proves $Q_{K,v} > 0$ on the line. In this way $f_K > 0$ is proved for every $1 \leq K \leq 200$ and for $K = 300$, 400, 600 and 800. For all these $K$ the minimum of $Q = Q_{K,2\beta_K}$ is attained only at $y = 0$: near 0 because $Q(y) - Q(0) \geq y^2\big(b_2 - \sum_{m\geq 2} |b_{2m}| y^{2m-2}\big) > 0$ for $0 < |y| \leq 2^{-5}$, the $b_n$ being the coefficients of $Q$, and elsewhere by covers on which the lower bound exceeds $Q(0)$. For the column $v^*(K)$, a complete cover at $v = t_+ \cdot 2\beta_K$ gives $v^*(K) \leq t_+ \cdot 2\beta_K$, and a certified negative value of $Q_{K,v}$ at $v = t_- \cdot 2\beta_K$ gives $v^*(K) > t_- \cdot 2\beta_K$, for dyadic rationals $t_- < t_+$ at most $2^{-18}$ apart.

| $K$ | $2\beta_K$ | $s_K$ | $v^*(K)$ | $v^*(K)/2\beta_K$ | $\min Q_{K,2\beta_K}$ | $\mathrm{TV}(f_K, p)$ |
|---|---|---|---|---|---|---|
| 0 | 0.046210 | 0 | 0 | 0 | 1 | $1.48 \times 10^{-2}$ |
| 1 | 0.036199 | 0.005005 | 0.005005 | 0.138 | 0.862 | $3.81 \times 10^{-3}$ |
| 2 | 0.031674 | 0.007268 | 0.004955 | 0.156 | 0.804 | $2.09 \times 10^{-3}$ |
| 3 | 0.028477 | 0.008867 | 0.005416 | 0.190 | 0.762 | $1.30 \times 10^{-3}$ |
| 5 | 0.024472 | 0.010869 | 0.005527 | 0.226 | 0.705 | $6.96 \times 10^{-4}$ |
| 10 | 0.019121 | 0.013544 | 0.005441 | 0.285 | 0.623 | $2.64 \times 10^{-4}$ |
| 16 | 0.015756 | 0.015227 | 0.005222 | 0.331 | 0.566 | $1.27 \times 10^{-4}$ |
| 17 | 0.015343 | 0.015434 | 0.005190 | 0.338 | 0.558 | $1.14 \times 10^{-4}$ |
| 20 | 0.014273 | 0.015969 | 0.005065 | 0.355 | 0.538 | $8.75 \times 10^{-5}$ |
| 50 | 0.009108 | 0.018551 | 0.004146 | 0.455 | 0.430 | $1.70 \times 10^{-5}$ |
| 100 | 0.006222 | 0.019994 | 0.003303 | 0.531 | 0.355 | $4.42 \times 10^{-6}$ |
| 150 | 0.004907 | 0.020652 | 0.002816 | 0.574 | 0.316 | $1.93 \times 10^{-6}$ |
| 200 | 0.004122 | 0.021044 | 0.002487 | 0.603 | 0.289 | |
| 300 | 0.003199 | 0.021505 | 0.002059 | 0.644 | 0.255 | |
| 400 | 0.002659 | 0.021775 | 0.001784 | 0.671 | 0.232 | |
| 600 | 0.002036 | 0.022087 | 0.001441 | 0.708 | 0.203 | |
| 800 | 0.001677 | 0.022267 | 0.001229 | 0.733 | 0.184 | |

Table 13.6: The zero completed laws $f_K$. For $K \leq 16$, $f_K$ is a ferromagnetic probability density by Theorem 13.5.12(i) ($s_K \leq 2\beta_K$). For every $K \geq 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so $f_K$ is a ferromagnetic probability density by Theorem 13.5.12(ii), and $\min Q_{K,2\beta_K}$ is the value at $y = 0$, where it is attained. The threshold $v^*(K)$ of Proposition 13.5.13 lies in a certified interval of width at most $1.2 \times 10^{-7}$, within $6 \times 10^{-7}$ of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. Computed by `fm_qcert.py` and `fm_tvcert.py` from the certified zeros and variance, except $v^*(1) = s_1$.

The question is the trend of the ratio $v^*(K)/2\beta_K$. While it stays below 1, $f_K$ is a probability density. It increases slowly, to 0.733 at $K = 800$, and its slope in $\log K$ falls from 0.110 between $K = 30$ and 50 to 0.086 between $K = 600$ and 800.

A straight line in $\log K$ through the last two points reaches 1 near $K = 1.8 \times 10^4$, while the power law $1 - v^*(K)/2\beta_K \approx 1.65K^{-0.27}$, fitted for $K \geq 100$, never does. The data do not decide between them. Either outcome is consistent with RH, since Theorem 13.5.12(iii) asks only for the convergence of $\varphi_K$, not for positivity.

### 13.5.5 The Maxwell series

For $K \leq 16$, Theorem 13.5.12(i) reads as a law the factorisation

$$\varphi_K(\theta) = e^{-(\operatorname{Var} X - 3s_K)\theta^2/2}\, \mathbb{E}e^{i\theta \sum_{k \leq K} M_k/\gamma_k}.$$

It holds for every $K$, though from $K = 17$ on its Gaussian factor has negative variance, and letting $K \to \infty$ in it, with that factor moved to the other side, gives an identity in law. Let $\mathcal{M} = \sum_k M_k/\gamma_k$, summed over the zeros on the critical line. The series converges almost surely, because its terms are independent and centred with summable variances $3\gamma_k^{-2}$.

Theorem 13.5.14. *Let $Z$ be independent of $X$. Then RH holds if and only if*

$$X + \left(\tfrac{1}{2} \operatorname{Var} X\right)^{1/2} Z \stackrel{d}{=} \mathcal{M}.$$

*Proof.* $\mathbb{E}e^{i\theta\mathcal{M}} = \prod_k (1 - \theta^2/\gamma_k^2) e^{-\theta^2/2\gamma_k^2}$, an entire function of $\theta$ with only real zeros. Under RH, $\sum_k \gamma_k^{-2} = \frac{1}{2} \operatorname{Var} X$, and the product is $\varphi_X(\theta) e^{-\operatorname{Var} X\, \theta^2/4}$, the characteristic function of the left side. If RH fails, the characteristic function of the left side vanishes at a nonreal point, and that of $\mathcal{M}$ does not. □

A simulation checks the identity numerically (computed by `fm_maxwell.py`, with $10^7$ draws). Like Table 13.3, this checks the computation, not RH.

### 13.5.6 How close

Proposition 13.5.15. *With no hypothesis, the distance $\delta$ of Section 13.1.5 is at most $\mathrm{TV}(f_{150}, p) < 1.932 \times 10^{-6}$, since $f_{150}$ is a ferromagnetic law by Theorem 13.5.12(ii) and the positivity proof of Section 13.5.4; without that computer assisted proof, $f_{16}$ gives $\delta < 1.266 \times 10^{-4}$, and the second law of Table 13.5, ferromagnetic by Theorem 13.5.4, gives $\delta < 6.024 \times 10^{-6}$.*

Every density here is even, and $\mathrm{TV}(q, p) = \int_0^\infty |q - p|$ is enclosed in ball arithmetic. On $[0, 3]$, cut into intervals $[c - r, c + r]$ with $r = 2^{-12}$, the integral of $|d|$, $d = q - p$, differs from the exact integral of $|d(c) + d'(c)h|$ over $|h| \leq r$ by at most $|d''(c)|r^3/3 + M_3 r^4/12$, where $M_3$ bounds $|d'''|$ by interval evaluation (for the laws $f_K$, by $\frac{1}{\pi} \int_0^\infty \theta^3 |\varphi_K(\theta)|\, d\theta$); beyond 3 each density is bounded by an explicit function of known logarithmic decay rate. The laws $f_K$ are built from the certified zeros and variance of Section 13.5.4, so the distances are those of the true laws (`fm_tvcert.py`). For the laws of Table 13.5 the normalising constants are Arb's

Bessel functions $K_v(a)$ at real order, by Theorem 13.5.4(ii), or integrals enclosed by Arb's quadrature when $\mu > 0$, and $M_3$ comes from interval evaluation of the closed form (`wf_tvcert.py`).

### 13.5.7 A cusped weight never works

The multipliers of Proposition 13.5.2 are entire. A weight with a cusp at the origin behaves differently: no tilt of $p$ by $e^{-a|t|}$, $a > 0$, is ferromagnetic, however small $a$ is. We write the tilts in the variable $u = t/2$ of the de Bruijn Newman flow (Section 8.4). An even weight $w(u)$ gives the multiplier $m(t) = w(t/2)$ in the notation of Proposition 13.5.2, and $8\int_0^\infty w(u)\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du = \int_{\mathbb{R}} \Phi(t)m(t)\cos(\theta t)\,dt$ with $\theta = z/2$, so the zeros of the first function are twice those of the transform of the tilt $m\,p$. Thus $e^{-\mu|u|}$ is $e^{-a|t|}$ with $a = \mu/2$, and $\cosh(\mu u)$ is $\cosh(at)$.

Theorem 13.5.16. *Let $\mu > 0$ and $H^{(\mu)}(z) = \int_0^\infty e^{-\mu u}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$. With $C$ a standard Cauchy variable,*

$$H^{(\mu)}(z) = \mathbb{E}\,H_0(z + \mu C),$$
$$z^2 H^{(\mu)}(z) \to \mu\,\Phi_{\mathrm{dBN}}(0) = 0.4467\mu \quad (z \to \infty,\ z \in \mathbb{R}).$$

*Hence $H^{(\mu)}$ is positive beyond some height, has finitely many real zeros and infinitely many non real ones, the law with density proportional to $e^{-\mu|t|/2}p(t)$ is not ferromagnetic, and the tilted clock $R^{(\mu)}(s) = H^{(\mu)}(0)/H^{(\mu)}(2i\sqrt{s})$ is not the Laplace transform of a GGC, for any $\mu > 0$.*

*Proof.* Since $e^{-\mu|u|} = \mathbb{E}\cos(\mu C u)$ and $\mathbb{E}\sin(\mu C u) = 0$, Fubini gives $H^{(\mu)}(z) = \mathbb{E}H_0(z + \mu C) = \int H_0(y)P_\mu(z - y)\,dy$ with $P_\mu(y) = \mu/(\pi(\mu^2 + y^2))$. As $H_0(y) = \frac{1}{8}\Xi(y/2)$ decays exponentially and $\int_{\mathbb{R}} H_0(y)\,dy = \pi\Phi_{\mathrm{dBN}}(0)$ by Fourier inversion at $u = 0$, dominated convergence, with $z^2 P_\mu(z - y) \le 2(\mu + y^2/\mu)/\pi$, gives $z^2 H^{(\mu)}(z) \to \mu\Phi_{\mathrm{dBN}}(0)$. The function $H^{(\mu)}$ is even and entire of order one; with finitely many zeros it would be a polynomial times $e^{\alpha z}$, hence a polynomial by evenness, contradicting $H^{(\mu)}(z) \to 0$. So it has infinitely many zeros, almost all non real. It is also positive on the imaginary axis, $H^{(\mu)}(iy) = \int_0^\infty e^{-\mu u}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, so the argument of Theorem 4.2.4 applies to $R^{(\mu)}$ verbatim, as it does to $R_t$ in Corollary 4.2.5, and $R^{(\mu)}$ is the Laplace transform of a GGC only if all zeros of $H^{(\mu)}$ are real. By the relation between the variables above, the characteristic function of the tilt $e^{-\mu|t|/2}p(t)$ is $H^{(\mu)}(2\theta)/H^{(\mu)}(0)$, whose non real zeros show that the tilt is not ferromagnetic. □

Numerically the real zeros are lost at once (`xg_cauchy.py`). For $\mu = 1$ the function $H^{(1)}$ has no sign change on $[0, 80]$, where $H_0$ has six, and the Laguerre inequality $(H')^2 - HH'' \ge 0$, satisfied by every real rooted even entire function of order one, fails for $24 \le z \le 38$ and for $43 \le z \le 80$. Even for $\mu = 0.01$ only two sign changes survive on $[0, 400]$, against the six of $H_0$ on $[0, 80]$. The mechanism

is the cusp of $e^{-\mu|u|}$ at $u=0$, which adds to $H^{(\mu)}$ a positive term decaying like $z^{-2}$ and swamps the exponentially decaying oscillation that carries the zeros of $\Xi$. The table gives $H^{(\mu)}(z)$ against the tail value $\mu\Phi_{\mathrm{dBN}}(0)/z^2$ of the theorem.

| $\mu$ | $z$ | $H^{(\mu)}(z)$ | $\mu\Phi_{\mathrm{dBN}}(0)/z^2$ |
|---|---|---|---|
| 1 | 80 | $7.237\times10^{-5}$ | $6.980\times10^{-5}$ |
| 1 | 400 | $2.796\times10^{-6}$ | $2.792\times10^{-6}$ |
| 3 | 400 | $8.387\times10^{-6}$ | $8.376\times10^{-6}$ |
| 0.01 | 400 | $2.796\times10^{-8}$ | $2.792\times10^{-8}$ |

A weight that is smooth at the origin but decays only like $u^{-2}$ is the other natural candidate [253]. For $a>0$ let

$$H^{[a]}(z)=\int_0^\infty \frac{\Phi_{\mathrm{dBN}}(u)}{1+a^2u^2}\cos(zu)\,du.$$

Proposition 13.5.17. *$H^{[a]}(z)=\mathbb{E}\,H_0(z+aL)$, with $L$ a standard Laplace variable of density $\frac12 e^{-|y|}$; $H^{[a]}=\int_0^\infty e^{-v}H_{-va^2}\,dv$, an exponential mixture of backward heat flows; and $H_0=(1-a^2\frac{d^2}{dz^2})H^{[a]}$. Consequently, if $H^{[a]}$ has only real zeros for one $a>0$, then RH holds.*

*Proof.* $\mathbb{E}\cos(aLu)=(1+a^2u^2)^{-1}=\int_0^\infty e^{-v}e^{-va^2u^2}dv$ and $\mathbb{E}\sin(aLu)=0$, so Fubini gives the first two forms, with $H_t(z)=\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ as in Section 8.4; the third follows from $-\frac{d^2}{dz^2}\cos(zu)=u^2\cos(zu)$. The function $H^{[a]}$ is even, real on $\mathbb{R}$ and entire of order one, since its kernel decays like $\Phi_{\mathrm{dBN}}$; if its zeros are real it is therefore in the Laguerre Pólya class, and so is $(1-a\frac{d}{dz})(1+a\frac{d}{dz})H^{[a]}=H_0$, because $1\pm a\frac{d}{dz}$ preserve the class (Laguerre's theorem; see [335]). So $H_0$ has only real zeros. □

The weight $(1+a^2u^2)^{-1}$ has poles at $u=\pm i/a$, so it is not one of the entire multipliers of Proposition 13.5.2 and not a universal factor in Pólya's sense; RH is not known to imply that $H^{[a]}$ has real zeros, and whether it has them for some $a>0$ is open.

## 13.6 The exponential of the Pólya variable

The positive variable $e^{2X}$ lives on the Mellin side of Section 9.1.1: it is the variable $\frac{\pi}{2}S_2$ of Biane, Pitman and Yor, size biased by its power $\frac14$. Along the family $e^{\beta X}$ the picture is complete except on one interval: $e^{\beta X}$ is not infinitely divisible for $0<|\beta|<2$, it is self decomposable and not a GGC at $|\beta|=2$, and it has a hyperbolically completely monotone density, so it is a GGC, for $|\beta|\geq 5.5791422$, a bound certified in ball arithmetic. None of this involves a hypothesis on the zeros. GGC, HCM and self decomposable laws are as in Section 1.3.2.

### 13.6.1 Size biased laws and their phase

Write $B_0 = \frac{\pi}{2} S_2 = \sum_{m\ge1} \Gamma_m/(\pi m^2)$, with independent Gamma$(2,1)$ variables $\Gamma_m$ as in Section 9.1.1. It is a GGC with Thorin atoms of mass 2 at the points $\pi m^2$ and Laplace transform $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$, and by (9.1.1) $\mathbb{E}B_0^\sigma = 2\xi(2\sigma)$ for every complex $\sigma$. With Riemann's function $\omega(x) = \sum_{n\ge1} e^{-\pi n^2 x}$ of Section 1.2, its density is

$$f_0(y) = \sum_{m\ge1} \big(4\pi^2 m^4 y - 6\pi m^2\big) e^{-\pi m^2 y} = 4y\,\omega''(y) + 6\,\omega'(y)\,:$$

two integrations by parts give $\int_0^\infty y^\sigma (4y\omega'' + 6\omega')\,dy = 2\sigma(2\sigma-1)\pi^{-\sigma}\Gamma(\sigma)\zeta(2\sigma) = 2\xi(2\sigma)$. For real $q$ let $B_q$ be $B_0$ size biased by $B_0^q$, with density $y^q f_0(y)/(2\xi(2q))$, so that

$$\mathbb{E}B_q^\sigma = \frac{\xi(2\sigma+2q)}{\xi(2q)}, \qquad e^{2X} \stackrel{d}{=} B_{1/4},$$

the second because $\mathbb{E}e^{2\sigma X} = \xi(\frac12 + 2\sigma)/\xi(\frac12)$. In the notation of Section 9.1.1, where $V_\alpha$ is the Esscher tilt of the Pólya density at the basepoint $\alpha$, $B_q = e^{2V_{2q}}$; so the $B_q$ with $0 < q < 1$ correspond to the basepoints $0 < \alpha < 2$, and $e^{2X} = e^{2V_{1/2}}$ to the centre. Since $X$ is symmetric, $e^{-\beta X} \stackrel{d}{=} e^{\beta X}$, and for $\beta > 0$, $e^{\beta X} = B_{1/4}^{\beta/2}$.

Proposition 13.6.1. *For $0 < |\beta| < 2$ the variable $e^{\beta X}$ is not infinitely divisible.*

*Proof.* Take $\beta > 0$. The first term of $\Phi$ dominates for large $t$, so $\log P(X > x) = -\pi e^{2x} + O(x)$ and, with $Y = e^{\beta X}$, $\log P(Y > y) = -\pi y^{2/\beta} + O(\log y)$. Since $2/\beta > 1$, $\mathbb{E}\exp(cY\log Y) < \infty$ for every $c > 0$. By Sato's theorem on the tails of infinitely divisible laws [471, Theorem 26.1], an infinitely divisible law with this property has no Lévy measure, so it is Gaussian or degenerate, and $Y$ is neither. □

The other results of the section come from the continuation of the Laplace transform

$$\Lambda_q(\lambda) = \mathbb{E}e^{-\lambda B_q} = \frac{1}{2\xi(2q)} \int_0^\infty y^q f_0(y)\, e^{-\lambda y}\, dy,$$

which is finite for $\lambda > -\pi$. Let $0 < q < 1$, the range of the results below, fix $M \ge 1$ and let $R_M$ be the tail $\sum_{m>M}$ of the series for $f_0$. It is bounded near 0, because $f_0(y) = y^{-5/2} f_0(1/y)$ by the evenness of $\Phi$, so that $f_0(y) \to 0$ as $y \downarrow 0$, and it is $O(y\, e^{-\pi(M+1)^2 y})$ at infinity. Integrating the first $M$ terms gives, for $\operatorname{Re}\lambda > -\pi$,

$$2\xi(2q)\,\Lambda_q(\lambda) = \sum_{m\le M} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(\pi m^2+\lambda)^{q+2}} - \frac{6\pi m^2 \Gamma(q+1)}{(\pi m^2+\lambda)^{q+1}}\Big] + \int_0^\infty y^q R_M(y)\, e^{-\lambda y}\, dy, \tag{13.6.1}$$

with principal powers. The last integral is analytic for $\operatorname{Re}\lambda > -\pi(M+1)^2$ and real on the real axis. So (13.6.1) continues $\Lambda_q$ analytically to the plane cut along

$(-\infty,-\pi]$, and $\Lambda_q$ has boundary values $\Lambda_q(-r+i0)$ that are continuous in $r \neq \pi m^2$, while near each $-\pi m^2$ it behaves like a multiple of $(\pi m^2+\lambda)^{-q-2}$.

The boundary values carry the Thorin measure. If $G$ is a GGC with drift $b$ and Thorin measure $\mu$, its Laplace transform $H(\lambda) = \exp\{-b\lambda - \int \log(1+\lambda/t)\,\mu(dt)\}$ continues analytically to $\mathbb{C} \setminus (-\infty, 0]$, and at a point $r$ that is not an atom of $\mu$ and where $H(-r+i0)$ is finite and not zero, the argument of $H(-r+i0)$ is $-\pi\mu((0,r))$, modulo $2\pi$, as the proof of the lemma below shows. We call a continuous determination of $-\frac{1}{\pi}\arg H(-r+i0)$ on an interval a phase of $H$ there. Fourati used the argument of such boundary values to show that certain powers of positive stable variables are not GGC [140].

LEMMA 13.6.2. *Let $G$ be a GGC with drift $b$, Thorin measure $\mu$ and Laplace transform $H$, and let $I \subset (0,\infty)$ be an open interval such that for each $r \in I$, $H(-r+i\epsilon)$ tends as $\epsilon \downarrow 0$ to a nonzero limit $H(-r+i0)$, continuous in $r$. Then $\mu$ has no atoms in $I$, and $\mu((0,r)) - \phi(r)$ is constant on $I$ for every phase $\phi$ of $H$ on $I$.*

*Proof.* Let $L(\lambda) = -b\lambda - \int \log(1+\lambda/t)\,\mu(dt)$, so that $H = e^L$. For $r \in I$ and $0 < \epsilon \le 1$, $\operatorname{Im}\log(1+(-r+i\epsilon)/t) = \arg(t-r+i\epsilon)$ lies in $(0,\pi)$, tends to $\pi$, $\frac{\pi}{2}$ or 0 according as $t < r$, $t = r$ or $t > r$, and is at most $2\epsilon/t$ for $t \ge 2r$. Since $\mu((0,2r)) < \infty$ and $\int_{[2r,\infty)} \mu(dt)/t < \infty$, dominated convergence gives $\operatorname{Im} L(-r+i\epsilon) \to -\pi\big(\mu((0,r)) + \frac{1}{2}\mu(\{r\})\big)$. If $\mu(\{r\}) = a > 0$ then, since $|t-r+i\epsilon| \le t+r+1$,

$$\operatorname{Re} L(-r+i\epsilon) \;\ge\; br - \int_{t\neq r} \log\Big(1+\frac{r+1}{t}\Big)\,\mu(dt) - a\log\frac{\epsilon}{r} \;\longrightarrow\; \infty,$$

so $|H(-r+i\epsilon)| \to \infty$, which is excluded. Hence $\mu$ has no atoms in $I$, and $e^{i\operatorname{Im} L(-r+i\epsilon)}$, the argument factor of $H(-r+i\epsilon)$, tends to that of $H(-r+i0)$. So $\mu((0,r)) - \phi(r)$ is an even integer for each $r \in I$, and being continuous on $I$ it is constant. □

The same lower bound for $\operatorname{Re} L$ shows that the boundary values of a GGC transform never vanish.

THEOREM 13.6.3. *For $0 < q < 1$ the variable $B_q$ is not a GGC. In particular $e^{2X}$ is not a GGC.*

*Proof.* Let $r > \pi$, $r \neq \pi m^2$, and take $M$ with $\pi(M+1)^2 > r$ in (13.6.1). At $\lambda = -r+i0$ the terms with $\pi m^2 > r$ and the last integral are real, and $(\pi m^2 - r + i0)^{-p} = (r-\pi m^2)^{-p} e^{-i\pi p}$ for $\pi m^2 < r$, so

$$2\xi(2q)\,\operatorname{Im}\Lambda_q(-r+i0) = -\sin(\pi q) \sum_{\pi m^2 < r} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(r-\pi m^2)^{q+2}} + \frac{6\pi m^2 \Gamma(q+1)}{(r-\pi m^2)^{q+1}}\Big] < 0. \tag{13.6.2}$$

Suppose that $B_q$ is a GGC, and apply Lemma 13.6.2 on $I = (\pi, 4\pi)$. By (13.6.2) a phase $\phi$ of $\Lambda_q$ on $I$ takes its values in $(2k, 2k+1)$ for a single integer $k$. As $r \downarrow \pi$ the first term of (13.6.1) dominates, so that $\Lambda_q(-r + i0) \sim c\,(r-\pi)^{-q-2} e^{-i\pi(q+2)}$ with $c > 0$, and $\phi(r) \to 2k + q$. As $r \uparrow 4\pi$ the term with $m = 2$ dominates and is positive, and $\phi(r) \to 2k$. So $\phi$ falls by $q$ across $I$, while $\mu((0,r)) = \phi(r) + \text{const}$ is nondecreasing, a contradiction. □

Size biasing a GGC by a negative power gives a GGC: Fourati's Lemma 6.3 [140] states this for every real exponent, and its proof, which multiplies the variable by one plus an independent gamma variable of shape $\beta > 0$ and lets the rate of the gamma variable tend to 0, covers the size bias by $x^{-\beta}$, once slips in the printed normalising constants are corrected. For positive exponents the statement fails. By Theorem 13.6.3 it fails for every exponent in $(0,1)$, and it fails at 1 as well: the size bias of order one of the GGC $e_1 + \frac{1}{2}e_2$ has Laplace transform $\frac{4}{3}(2\lambda+3)(1+\lambda)^{-2}(2+\lambda)^{-2}$, which vanishes at the point $-\frac{3}{2}$ of the cut. With the negative orders, $B_q$ is a GGC for every $q \le 0$ and for no $q \in (0,1)$.

### 13.6.2 The size bias principle

A gamma variable of shape $t$ size biased by $y^q$ is a gamma variable of shape $t+q$ with the same rate. For a sum of gamma variables with different rates the same happens at every rate, up to an independent GGC that lives between the rates. The proof uses divided differences. For a polynomial $Q$ with real zeros $s_1 \le \cdots \le s_n$, listed with multiplicity, and a function $h$ analytic near $[s_1, s_n]$, put

$$h[Q] = \frac{1}{2\pi i} \oint \frac{h(\zeta)}{Q(\zeta)}\, d\zeta = \sum_{\zeta_0} \mathrm{Res}_{\zeta=\zeta_0} \frac{h(\zeta)}{Q(\zeta)},$$

the contour enclosing the zeros of $Q$ inside the domain of $h$ and the sum running over the distinct zeros $\zeta_0$ of $Q$, so that $h[Q]$ is the divided difference of $h$ at $s_1, \dots, s_n$ when $Q$ is monic. By the formula of Hermite and Genocchi, $h[Q]$ is the integral of $h^{(n-1)}(\sum_j w_j s_j)$ over the simplex of weights $w_j \ge 0$, $\sum_j w_j = 1$, with its uniform measure of total mass $1/(n-1)!$; so $h[Q]$ has the sign of $h^{(n-1)}$ when that sign is constant on $[s_1, s_n]$.

Lemma 13.6.4. *Let $Q(\zeta) = \prod_{n=1}^{K} (\zeta - \tau_n)^{t_n}$ with $\tau_1 < \cdots < \tau_K$ and integers $t_n \ge 1$, let $1 \le k < K$ and $\tau_k < x < \tau_{k+1}$, and write $Q = Q_- Q_+$ with $Q_-(\zeta) = \prod_{n \le k} (\zeta - \tau_n)^{t_n}$, of degree $N_-$, and $Q_+$ of degree $N_+$. For $p > 0$ put*

$$D_+(x) = h_+[Q_+], \qquad h_+(\zeta) = \frac{(\zeta - x)^{-p}}{Q_-(\zeta)},$$

$$D_-(x) = h_-[Q_-], \qquad h_-(\zeta) = \frac{(x - \zeta)^{-p}}{Q_+(\zeta)}.$$

*Then* $(-1)^{N_+-1}D_+(x) > 0$ *and* $(-1)^{N_+}D_-(x) > 0$, *and as functions of* $x \in (\tau_k, \tau_{k+1})$, $|D_+|$ *increases and* $|D_-|$ *decreases.*

*Proof.* On $(x, \infty)$ the function $h_+$ is a product of the completely monotone functions $(\zeta - x)^{-p}$ and $(\zeta - \tau_n)^{-1}$, $n \le k$, so $(-1)^j h_+^{(j)} > 0$ there for every $j$, and the formula of Hermite and Genocchi with $j = N_+ - 1$ gives the first sign. On $(-\infty, x)$ the function $(-1)^{N_+}h_-(\zeta) = (x-\zeta)^{-p}\prod_{n>k}(\tau_n - \zeta)^{-t_n}$ is a product of functions all of whose derivatives are positive, which gives the second. Finally $\partial_x(\zeta - x)^{-p} = p(\zeta - x)^{-p-1}$ and $\partial_x(x-\zeta)^{-p} = -p(x-\zeta)^{-p-1}$, so $D'_+$ and $D'_-$ are $p$ and $-p$ times the same expressions with $p+1$ in place of $p$: $D'_+$ has the sign of $D_+$, and $D'_-$ the sign opposite to that of $D_-$. □

THEOREM 13.6.5 (size bias principle). *Let* $Y = \sum_{n=1}^K \Gamma_n/\tau_n$ *with* $0 < \tau_1 < \cdots < \tau_K$ *and independent gamma variables* $\Gamma_n$ *with integer shapes* $t_n \ge 1$ *and unit scale, let* $0 < q < 1$, *and let* $Y_q$ *be* $Y$ *size biased by* $Y^q$. *Then*

$$Y_q + G \stackrel{d}{=} \sum_{n=1}^K \frac{\Gamma'_n}{\tau_n},$$

*where the* $\Gamma'_n$ *are independent gamma variables with shapes* $t_n + q$ *and unit scale, and* $G$, *independent of* $Y_q$, *is a GGC without drift whose Thorin measure has a continuous distribution function* $\theta_G$ *that vanishes on* $(0, \tau_1]$, *increases by exactly* $q$ *on each interval* $[\tau_k, \tau_{k+1}]$ *and is constant on* $[\tau_K, \infty)$.

*Proof.* Let $N = \sum_n t_n$ and $Q(\zeta) = \prod_n(\zeta - \tau_n)^{t_n}$. The density of $Y$ is $c_0(-1)^{N-1}(e^{-\zeta y})[Q]$ with $c_0 = \prod_n \tau_n^{t_n}$, as taking Laplace transforms under the contour integral shows: $(1/(\zeta+\lambda))[Q] = -1/Q(-\lambda)$, by the residue at $\zeta = -\lambda$. Multiplying by $y^q$ before integrating gives, for $\lambda \ge 0$,

$$\Lambda(\lambda) = \mathbb{E}e^{-\lambda Y_q} = c\,(-1)^{N-1}g_\lambda[Q], \qquad g_\lambda(\zeta) = (\zeta+\lambda)^{-1-q},$$

with a constant $c > 0$, and since $g_\lambda^{(N-1)} = (-1)^{N-1}(1+q)\cdots(N-1+q)\,(\zeta+\lambda)^{-N-q}$, the formula of Hermite and Genocchi gives

$$\Lambda(\lambda) = \frac{\mathbb{E}(\bar{s}+\lambda)^{-N-q}}{\mathbb{E}\bar{s}^{\,-N-q}}, \qquad \bar{s} = \sum_{j=1}^N w_j s_j,$$

where $s_1, \ldots, s_N$ are the rates listed with multiplicity and $(w_j)$ is uniform on the simplex. This continues $\Lambda$ analytically to $\mathbb{C} \setminus (-\infty, -\tau_1]$, where the residue form $c(-1)^{N-1}g_\lambda[Q]$ remains valid, and shows that $|\Lambda(\lambda)|$ is of exact order $|\lambda|^{-N-q}$ as $|\lambda| \to \infty$, uniformly up to the cut.

Boundary values. Let $\tau_k < r < \tau_{k+1}$. At points $\zeta < r$ the derivatives of $(\zeta - r + i0)^{-1-q}$ are $e^{-i\pi(1+q)}$ times those of $(r-\zeta)^{-1-q}$. Splitting the residues

of $g_\lambda/Q$ into those at $\tau_n$ with $n > k$ and those with $n \le k$ gives, with $D_\pm$ of Lemma 13.6.4 at $p = 1+q$ and the signs found there,

$$\Lambda(-r+i0) = c\,(-1)^{N-1}\big(D_+(r) + e^{-i\pi(1+q)}D_-(r)\big) = c\,(-1)^{N_-}z(r),$$
$$z(r) = |D_+(r)| + e^{-i\pi q}|D_-(r)|.$$

The point $z(r)$ is the sum of two nonzero vectors at the angle $\pi q < \pi$, so its argument lies in $(-\pi q, 0)$. The ratio $|D_-|/|D_+|$ decreases, from $\infty$ as $r \downarrow \tau_k$, where $D_-$ has the singular term $(r-\tau_k)^{-t_k-q}$, to 0 as $r \uparrow \tau_{k+1}$; so $\arg z(r)$ increases from $-\pi q$ to 0. For $r > \tau_K$ the same computation with $Q_+ = 1$ gives $\Lambda(-r+i0) = c(-1)^{N-1}e^{-i\pi(1+q)}D_-(r)$ with $D_- > 0$.

The factor $G$. Put $\theta_G = 0$ on $(0,\tau_1]$, $\theta_G(r) = kq + \frac{1}{\pi}\arg z(r)$ for $\tau_k < r < \tau_{k+1}$, and $\theta_G = (K-1)q$ on $[\tau_K, \infty)$. This function is continuous and nondecreasing. Near $\tau_k$, $|D_+/D_-|$ is $(r-\tau_k)^{t_k+q}$ times a function that is continuously differentiable up to $\tau_k$, and similarly near $\tau_{k+1}$; since $t_k+q > 1$, $\theta_G$ is continuously differentiable, with derivative 0 at the points $\tau_n$. Let $G$ be the GGC without drift and with Thorin measure $d\theta_G$, with Laplace transform $H_G(\lambda) = \exp\{-\int \log(1+\lambda/t)\,d\theta_G(t)\}$, and let $\Lambda_+(\lambda) = \prod_n (1+\lambda/\tau_n)^{-t_n-q}$, the Laplace transform of the right side. The function $J = \Lambda H_G/\Lambda_+$ is analytic on $\mathbb{C}\setminus(-\infty,-\tau_1]$ and positive on $(-\tau_1,\infty)$. For $\tau_k < r < \tau_{k+1}$ the arguments of $\Lambda$, $H_G$ and $1/\Lambda_+$ at $-r+i0$ are, modulo $2\pi$,

$$\pi N_- + \arg z(r), \qquad -\pi kq - \arg z(r), \qquad \pi(N_- + kq),$$

and for $r > \tau_K$ they are $\pi N - \pi q$, $-\pi(K-1)q$ and $\pi(N+Kq)$. In both cases they add up to a multiple of $2\pi$, so $J$ has real boundary values from above, by symmetry the same boundary values from below, and it continues analytically across the cut between the points $-\tau_n$. Near $-\tau_n$, $\Lambda = O(|\lambda+\tau_n|^{-t_n-q})$ by the residue form, and $1/\Lambda_+$ cancels this, while $H_G$ is bounded because $\theta_G$ has a bounded density; so these singularities are removable, and $J$ is entire. As $|\lambda| \to \infty$, $|\Lambda|$, $|1/\Lambda_+|$ and $|H_G|$ are of the exact orders $|\lambda|^{-N-q}$, $|\lambda|^{N+Kq}$ and $|\lambda|^{-(K-1)q}$, so $J$ is bounded. By Liouville's theorem $J \equiv J(0) = 1$, that is $\Lambda H_G = \Lambda_+$, which is the identity in law. □

THEOREM 13.6.6. *For $0 < q < 1$,*

$$B_q + G_q \;\stackrel{d}{=}\; \frac{\pi}{2}S_{2+q},$$

*where $G_q$ is independent of $B_q$ and is a GGC without drift whose Thorin measure has a continuous distribution function $\theta_q$ with $\theta_q = 0$ on $(0,\pi]$ and $\theta_q(\pi m^2) = (m-1)q$ for every $m \ge 1$. In particular $e^{2X} + G_{1/4} \stackrel{d}{=} \frac{\pi}{2}S_{9/4}$.*

*Proof.* Let $B^{(M)} = \sum_{m\le M}\Gamma_m/(\pi m^2)$ and let $B_q^{(M)}$ be its size bias of order $q$. By Theorem 13.6.5, $B_q^{(M)} + G_M \stackrel{d}{=} \sum_{m\le M}\Gamma'_m/(\pi m^2)$, with shapes $2+q$ on the

right. As $M \to \infty$, $\mathbb{E}[(B^{(M)})^q e^{-\lambda B^{(M)}}] \to \mathbb{E}[B_0^q e^{-\lambda B_0}]$ for $\lambda \geq 0$ by dominated convergence, since $B^{(M)} \leq B_0$; so $B_q^{(M)} \to B_q$ in law, and the right sides converge in law to $\frac{\pi}{2} S_{2+q}$, whose Laplace transform is $\Lambda_+(\lambda) = (\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^{2+q}$. Hence the Laplace transforms of the $G_M$ converge to $H = \Lambda_+/\Lambda_q$, which is continuous at 0. So $G_M$ converges in law to a variable $G_q$ with Laplace transform $H$, $G_q$ is a GGC by the closure theorem of Section 4.1.1, and $B_q + G_q \stackrel{d}{=} \frac{\pi}{2} S_{2+q}$. A GGC with drift $b$ is at least $b$, while $\frac{\pi}{2} S_{2+q}$ is smaller than any given positive number with positive probability, so $G_q$ has no drift.

It remains to find the Thorin measure $\mu$ of $G_q$. By (13.6.1), near each $-\pi m^2$ both $\Lambda_q$ and $\Lambda_+$ are $(\pi m^2 + \lambda)^{-2-q}$ times functions with continuous nonzero boundary values, and between these points $\Lambda_q(-r + i0) \neq 0$ by (13.6.2). So $H$ has continuous nonzero boundary values on all of $(0, \infty)$, and by Lemma 13.6.2 $\mu$ has no atoms and $\mu((0, r)) = \phi(r)$, where $\phi$ is the phase of $H$ that vanishes on $(0, \pi)$, where $H > 0$. On $(\pi m^2, \pi(m+1)^2)$ the argument of $\Lambda_+(-r + i0)$ is $-\pi(2+q)m$, and $\Lambda_q(-r+i0)$ lies in the lower half plane by (13.6.2), with argument tending to $-\pi(2+q)$ modulo $2\pi$ at the left end and to 0 at the right end, as in the proof of Theorem 13.6.3. So, modulo 2, $\phi$ takes its values in $(mq - 1, mq)$ on this interval and tends to $(m-1)q$ at its left end and to $mq$ at its right end. Since $\phi$ is continuous and $\phi(\pi) = 0$, induction on $m$ gives $\phi(\pi m^2) = (m-1)q$. □

The theorem gives $B_q$ a signed Thorin measure, $(2+q)\sum_m \delta_{\pi m^2} - d\theta_q$, with a negative continuous part of mass $q$ on each gap, which is what Theorem 13.6.3 detects.

### 13.6.3 Self decomposability and the theta Lévy density

Theorem 13.6.7. *For $0 < q < 1$ the variable $B_q$ is self decomposable, with no drift and Lévy density $k_q(x)/x$, where*

$$k_q(x) = (2+q)\,\omega(x) - \int_0^\infty e^{-tx}\, d\theta_q(t), \qquad 2\,\omega(x) \leq k_q(x) \leq (2+q)\,\omega(x).$$

*In particular $e^{2X}$ is self decomposable, hence infinitely divisible and unimodal, its Lévy density lies between $2\omega(x)/x$ and $\frac{9}{4}\omega(x)/x$, and it is not a GGC.*

*Proof.* By Theorem 13.6.6, $-\log\Lambda_q = -\log\Lambda_+ + \log H$, and Frullani's integral $\log(1 + \lambda/t) = \int_0^\infty (1 - e^{-\lambda x}) e^{-tx}\, dx/x$, applied to the atoms of $\frac{\pi}{2} S_{2+q}$ and to the Thorin measure of $G_q$, gives

$$-\log\Lambda_q(\lambda) = \int_0^\infty (1 - e^{-\lambda x})\, k_q(x)\, \frac{dx}{x}.$$

On $[\pi m^2, \pi(m+1)^2]$ the measure $d\theta_q$ has mass $q$ and $e^{-tx} \leq e^{-\pi m^2 x}$, so $0 \leq \int e^{-tx}\, d\theta_q(t) \leq q\,\omega(x)$, which gives the bounds. In particular $k_q > 0$, so $B_q$ is

infinitely divisible with Lévy density $k_q(x)/x$, and it is self decomposable if $k_q$ is nonincreasing (Section 1.3.2), that is, if

$$\int_0^\infty te^{-tx}\,d\theta_q(t) \;\le\; (2+q)\sum_{m\ge1}\pi m^2e^{-\pi m^2x}, \qquad x>0.$$

Bound the contribution of $[\pi m^2,\pi(m{+}1)^2]$ to the left side by $q\,s_m(x)$, with $s_m(x)=\max\{te^{-tx}:\pi m^2\le t\le\pi(m+1)^2\}$. For $m\ge2$, $s_m(x)\le\pi(m+1)^2e^{-\pi m^2x}\le\frac94\pi m^2e^{-\pi m^2x}$, and $\frac94q<2+q$, which leaves $(2-\frac54q)\pi m^2e^{-\pi m^2x}$ of the $m$th term on the right. It remains to show $q\,s_1(x)\le(2+q)\pi e^{-\pi x}+(2-\frac54q)4\pi e^{-4\pi x}$. If $x\ge1/\pi$, then $s_1(x)=\pi e^{-\pi x}$. If $x\le1/(4\pi)$, then $s_1(x)=4\pi e^{-4\pi x}$, and $4q\le4(2-\frac54q)+(2+q)$ because $8q\le10$. Otherwise $s_1(x)=1/(ex)$, and $(2+q)\pi x\,e^{1-\pi x}\ge(2+q)\frac14e^{3/4}>1>q$, because $ue^{1-u}$ increases on $(\frac14,1)$. Self decomposable laws are unimodal [521], and $e^{2X}=B_{1/4}$ is not a GGC by Theorem 13.6.3. □

Near 0, $\omega(x)\sim1/(2\sqrt{x})$, and $0\le q\sqrt{t/\pi}-\theta_q(t)\le2q$, since $\theta_q=0$ on $(0,\pi]$ and, on $[\pi m^2,\pi(m+1)^2]$, the first lies between $mq$ and $(m+1)q$ and the second between $(m-1)q$ and $mq$; so the subtracted fraction $1-k_q/((2+q)\omega)$ tends to $q/(2+q)$ as $x\to0$.

The script `fm_sizebias.py` checks the three theorems numerically. It continues $\Lambda_q$ by (13.6.1) with $M=5$, at $30+r/4$ digits, on 1800 points of $r\in[0.05,90]$, for $q=0.1$, 0.25, 0.5, 0.75 and 0.9. In every case the phase of $\Lambda_q$ decreases along the grid on each of the four gaps $(\pi m^2,\pi(m{+}1)^2)$, $m\le4$, so that $\theta_q$ increases, and at the last grid point below $\pi(m+1)^2$, $\theta_q$ is within $10^{-7}$ of $mq$, as Theorem 13.6.6 requires. Extending $\theta_q$ beyond the scan by $q(\sqrt{t/\pi}-1)$, the value it takes at the points $\pi m^2$, the function $k_q$ is positive and decreasing on 300 points of $[10^{-3},3]$, and the subtracted fraction at $x=10^{-3}$ is 0.046, 0.108, 0.194, 0.264 and 0.300, against the limits 0.048, 0.111, 0.200, 0.273 and 0.310.

The factorization can also be tested away from the cut, through the characteristic function on the real line. For a positive infinitely divisible variable $B$ with no drift and Lévy density $k(x)/x$, $-\log|\mathbb{E}e^{iyB}|^2=\int_0^\infty4\sin^2(\frac{yx}{2})\,k(x)\,dx/x$, and Frullani's integral $\int_0^\infty4\sin^2(\frac{yx}{2})e^{-tx}\,dx/x=\log(1+y^2/t^2)$ turns this into the Thorin form $\int\log(1+y^2/t^2)$ against the Thorin measure, here the signed one. For $B_q$ the atoms sum in closed form, by $\prod_m(1+u^4/m^4)=(\cosh\sqrt2\pi u-\cos\sqrt2\pi u)/(2\pi^2u^2)$:

$$-\log\left|\mathbb{E}e^{iyB_q}\right|^2=(2+q)\log\frac{\cosh\sqrt{2\pi y}-\cos\sqrt{2\pi y}}{2\pi y}-\int_0^\infty\log\left(1+\frac{y^2}{t^2}\right)d\theta_q(t).$$

Table 13.7 compares, for $e^{2X}$, the direct Fourier integral of the density with the Thorin form and the theta Lévy form, both built from the computed $\theta_{1/4}$. The two forms agree with each other to a relative $2\times10^{-7}$, and with the direct integral to a relative $1.2\times10^{-5}$. As a last check, $\mathbb{E}G_{1/4}=\mathbb{E}\frac\pi2S_{9/4}-\mathbb{E}e^{2X}=\frac{3\pi}{8}-\xi(\frac52)/\xi(\frac12)=0.0815947$, and $\int\theta_{1/4}(t)\,t^{-2}dt=0.0815351$ from the computed Thorin measure;

| $y$ | direct | Thorin form | theta Lévy form | atoms alone |
|---|---|---|---|---|
| 0.5 | 0.0588838 | 0.0588840 | 0.0588840 | 0.0609722 |
| 1 | 0.227615 | 0.227616 | 0.227616 | 0.235869 |
| 2 | 0.808286 | 0.808290 | 0.808290 | 0.839868 |
| 4 | 2.347416 | 2.347431 | 2.347431 | 2.457076 |
| 8 | 5.261493 | 5.261552 | 5.261553 | 5.575941 |

Table 13.7: $-\log|\mathbb{E}e^{iye^{2X}}|^2$ computed four ways: by a direct Fourier integral of the density $y^{1/4}f_0(y)/(2\xi(\frac{1}{2}))$; in the Thorin form with the atoms $\frac{9}{4}$ at the points $\pi m^2$ summed in closed form and the continuous part from the computed $\theta_{1/4}$; in the theta Lévy form $\int 4\sin^2(\frac{yx}{2})k_{1/4}(x)\,dx/x$; and from the atoms alone, which is $-\log|\mathbb{E}e^{iy\frac{\pi}{2}S_{9/4}}|^2$. Computed by `fm_sizebias.py`.

the extension beyond $t = 90$ differs from $\theta_{1/4}$ by at most $q$, which can move the integral by at most $q/90 < 0.003$.

On the dual side the roles are reversed. Section 10.1.4 puts the characteristic function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ of $Y_{1/2}$ in the same form, with no hypothesis: $\log(\xi(\frac{1}{2}+u)/\xi(\frac{1}{2})) = 2\int_0^\infty(1-\cos ux)\,K_{1/2}(x)\,dx/x$ for real $u$. So the logarithm of the moment generating function of $X$ has the form of $-\log|\mathbb{E}e^{iyB_q}|^2$, with the Poisson trace $K_{1/2}$, a sum over the zeros, in place of the theta kernel $k_q$. The Poisson trace is completely monotone exactly when RH holds (Corollary 10.1.4), while $k_{1/4}$ is positive and decreasing with no hypothesis.

### 13.6.4 Hyperbolically completely monotone powers

The kernel $\Phi$ is analytic in the strip $|\operatorname{Im} t| < \pi/4$, where $\operatorname{Re} e^{2t} > 0$, and since it is real and even, $F(v) = \Phi(iv)$ is real for real $v$. We have $F(0) = 0.8933938$; $F$ has a local maximum $F(v_m) = 7.0315837$ at $v_m = 0.5630962646$, and it vanishes at $v = 0.6388301$, which is the zero $\alpha_1$ of the continuation in Proposition 7.1.18(iii).

Proposition 13.6.8. *The variable $e^{\beta X}$ has an HCM density, and so is a GGC, for $|\beta| \geq 5.5791422$, and it does not have an HCM density for $|\beta| \leq 5.5791395$. The proof of the first statement is computer assisted.*

*Proof.* Let $\beta > 0$. The density $p(\beta^{-1}\log y)/(\beta y)$ of $e^{\beta X}$ has elasticity $1-\beta^{-1}(\Phi'/\Phi)(\beta^{-1}\log y)$, and the upper half plane in $y$ corresponds to the strip $0 < \operatorname{Im} t < \pi/\beta$. By Theorem 4.3.2, as in Proposition 4.3.26, $e^{\beta X}$ has an HCM density if and only if $\Phi$ continues to a zero free analytic function on $|\operatorname{Im} t| < \pi/\beta$ and

$$h(x,v) = \operatorname{Im}\Big(-\frac{\Phi'}{\Phi}\Big)(x+iv) \geq 0 \qquad \text{for } 0 < v < \pi/\beta,$$

and HCM laws are GGC by (F5) of Section 4.2.1. On the imaginary axis $h(0,v) = F'(v)/F(v)$. Ball arithmetic [290] encloses $v_m$ in an interval of width $2\times 10^{-17}$

with certified signs $F' > 0$ at its left end and $F > 0 > F'$ at its right end $v_b$. If $\pi/\beta > v_b$, the strip contains $iv_b$, where $\Phi$ is analytic and $h < 0$, so $e^{\beta X}$ is not HCM. Since $\pi/v_b > 5.5791395$, this proves the second statement.

For the first, let $v_* = 0.563096$, so that $\pi/v_* < 5.5791422$, and let $\varphi_1$ be the term $n = 1$ of $\Phi$. Three estimates, in ball arithmetic with rigorous bounds for the tail of the series, show that $h \geq 0$ on $0 < v \leq v_*$. (Z) For $\operatorname{Re} t \geq 0$ and $|\operatorname{Im} t| \leq 0.5634$, $|\Phi/\varphi_1 - 1| \leq 0.6015$, so $\Phi \neq 0$ there, and also for $\operatorname{Re} t \leq 0$, because $\Phi(-\bar{t}) = \overline{\Phi(t)}$. (T) On the segment $v = v_*$, $0 \leq x \leq 2$, adaptive bisection into 2896 intervals encloses $h$ in positive balls. The smallest lower bound, $1.8 \times 10^{-8}$, occurs next to $x = 0$, where $h(0, v)$ tends to 0 as $v \uparrow v_m$. Since $h(-x, v) = h(x, v)$, $h > 0$ on the whole segment $|x| \leq 2$. (S) For $x \geq 2$ and $0 < v \leq v_*$, the first term gives $\operatorname{Im}(-\varphi_1'/\varphi_1) \geq 2\pi e^{2x} \sin 2v \geq 8e^4 v$, and Cauchy's estimate on discs of radius 0.1 bounds the imaginary part of the derivative of $\log(\Phi/\varphi_1)$ by $10^{-40}v$; so $h > 0$ there, and by symmetry also for $x \leq -2$. By (Z), $h$ is harmonic on the strip $|\operatorname{Im} t| < 0.5634$. It vanishes on the real axis and, by (T) and (S), is positive on the top and the sides of the rectangle $[-2, 2] \times [0, v_*]$, so $h \geq 0$ inside by the minimum principle, and with (S), $h \geq 0$ on $0 < v \leq v_*$. So $e^{\beta X}$ is HCM for $\beta \geq \pi/v_*$. □

The certificate stops $3 \times 10^{-6}$ short of $\pi/v_m = 5.5791396$, because $h$ vanishes at $iv_m$ and the strip of height $v_m$ cannot be covered by balls with positive lower bounds; presumably the threshold is exactly $\pi/v_m$. The computation is `fm_hcm.py`, which works in the variable $t/2$ of Rodgers and Tao [463], where the heights are halved and the thresholds doubled.

The HCM threshold, the least $|\beta|$ for which $e^{\beta X}$ has an HCM density (these powers form a closed half line, by (F2) and (F4) of Section 4.2.1), lies between 5.5791395 and $\pi/v_*$, and for $2 < |\beta|$ below it infinite divisibility and the GGC property are open (Section 13.8). The bracket comes from the kernel, not from the zeros, and involves no hypothesis. It is the Pólya side counterpart of Theorem 4.3.27, where, under RH and two further hypotheses on the zeros, the HCM powers of the Mellin variable $W$ are limited by the first gap $\gamma_2 - \gamma_1$. For the clock, whose Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is the moment generating function of $X$ inverted at $\sqrt{u}$, the GGC property is RH itself (Proposition 1.2.1).

## 13.7 Two arithmetic spin chains

In this section $\beta = \sigma + it$ is a complex inverse temperature, $s(n)$ is Stern's diatomic sequence, given by $s(1) = 1$, $s(2n) = s(n)$ and $s(2n + 1) = s(n) + s(n + 1)$ [274], and $\mu$ and $\varphi$ are the Möbius and Euler functions ($\varphi$ is Euler's function here, not the characteristic function $\varphi_X$ of Definition 13.1.1). Knauf's number theoretical spin chain [203, 204] has, at length $N$, $2^N$ configurations, whose energies are the

numbers $\log s(k)$, $2^N \le k < 2^{N+1}$. Its partition function is therefore

$$Z_N(\beta) = \sum_{k=2^N}^{2^{N+1}-1} s(k)^{-\beta} = \sum_{m\ge 1} c_N(m)\, m^{-\beta}, \tag{13.7.1}$$

where $c_N(m)$ is the number of times $m$ occurs in the row $s(2^N), \dots, s(2^{N+1}-1)$. Knauf proved that the chain is ferromagnetic, in the sense that minus its energy, expanded in products of the spins, has nonnegative coefficients apart from the constant term, and that its interactions are many body [203]; Contucci and Knauf proved that it has a phase transition at $\beta = 2$ [93]. Section 13.7.3 turns to a second chain, of Kleban and Özlük, in a magnetic field. This section follows a note of Nicholas Polson merged into the book [233].

### 13.7.1 Knauf's chain and the Stern row

Proposition 13.7.1. *(i) $c_N(1) = 1$, and for $m \ge 2$, $c_N(m)$ is the number of coprime pairs $(p, q)$ of positive integers with $p + q = m$ for which $p/q$ has depth at most $N-1$ in the Calkin Wilf tree. Hence $c_N(m)$ does not decrease with $N$, $c_N(m) = \varphi(m)$ for $m \le N+1$, $c_N(m) = 0$ for $m > F_{N+2}$, where $F_n$ are the Fibonacci numbers, $F_1 = F_2 = 1$, and $c_N(m) \to \varphi(m)$ as $N \to \infty$.*

*(ii) $Z_N(\beta) \to \zeta(\beta-1)/\zeta(\beta)$ as $N \to \infty$ for $\operatorname{Re}\beta > 2$, increasingly for real $\beta > 2$.*

*(iii) $Z_N(-1) = \sum_m m\, c_N(m) = 3^N$.*

Part (ii) is due to Knauf [203]. The proof below goes through the enumeration of the positive rationals by Calkin and Wilf [69].

*Proof.* The Calkin Wilf tree has the root $1/1$, and the children of $p/q$ are $p/(p+q)$ and $(p+q)/q$. Read level by level it lists the fractions $s(k)/s(k+1)$, $k \ge 1$, those at depth $j$ being those with $2^j \le k < 2^{j+1}$, and every positive rational occurs in it exactly once, in lowest terms [69]. Since $s(2k) = s(k)$ and $s(2k+1) = s(k) + s(k+1)$, row $N+1$ consists of row $N$, at its even places, and of the sums $p+q$ over the fractions $p/q$ at depth $N$, at its odd places. Row 0 is $\{1\}$, so by induction row $N$ consists of 1 and of one entry $p+q$ for each fraction $p/q$ at depth at most $N-1$. The coprime pairs with $p+q = m \ge 2$ are the pairs $(p, m-p)$ with $p$ prime to $m$; there are $\varphi(m)$ of them, each at one depth. Passing from $p/q$ to a child adds $p$ or $q$ to $p+q$, so a fraction at depth $j$ has $j+2 \le p+q \le F_{j+3}$, the upper bound by induction together with $\max(p,q) \le F_{j+2}$. Hence every pair with $p+q = m \le N+1$ has depth at most $m-2 \le N-1$, and no fraction at depth at most $N-1$ has $p+q > F_{N+2}$. This proves (i). Part (ii) follows from (i) by monotone convergence for real $\beta$ and by dominated convergence for complex $\beta$, since $\sum_m \varphi(m) m^{-\beta} = \zeta(\beta-1)/\zeta(\beta)$ converges absolutely for $\operatorname{Re}\beta > 2$. For (iii), let $S_N$ be the sum of row $N$. Then $S_{N+1} = S_N + \sum_{k=2^N}^{2^{N+1}-1} \big(s(k)+s(k+1)\big) = 3S_N$, because $s(2^N) = s(2^{N+1}) = 1$, and $S_0 = 1$. □

The finite statements of (i) and (iii) were also checked with exact integers for every $N \le 34$, and the depth count of (i) against Euclid's algorithm for $m \le 800$ (computed; `fc_stern.py`). The continuation of $\zeta(\beta-1)/\zeta(\beta)$ vanishes at the points $1+\rho$, for the nontrivial zeros $\rho$ of $\zeta$, at $\beta = 1$ and at $\beta = 1-2k$, $k \ge 1$. The series $\sum_m \varphi(m) m^{-\beta}$ diverges on the strip $1 < \operatorname{Re}\beta < 2$, which contains the points $1+\rho$, but the finite sums $Z_N$ have zeros close to them (Section 13.7.2). As in statistical mechanics, zeros of a partition function in the complex inverse temperature are called Fisher zeros.

### 13.7.2 The Perron identity

Let $w_N(x) = c_N(m)/\varphi(m)$ for $m \le x < m+1$, $m \ge 1$, and $w_N(x) = 1$ for $0 < x < 1$. By Proposition 13.7.1(i) this depth profile equals 1 on $(0, N+2)$ and 0 on $[F_{N+2}+1, \infty)$, so its Mellin transform $W_N(z) = \int_0^\infty w_N(x) x^{z-1}\,dx$, defined for $\operatorname{Re} z > 0$, is $1/z$ plus an entire function.

Proposition 13.7.2. *For* $\operatorname{Re}\beta > 1$, $\beta \ne 2$,

$$Z_N(\beta) = \frac{\zeta(\beta-1)}{\zeta(\beta)} + \frac{W_N(2-\beta)}{\zeta(2)} + R_N(\beta), \qquad R_N(\beta) = \sum_{d\ge 1} \mu(d) d^{-\beta} r_d(\beta),$$

*where* $r_d(\beta) = \sum_{n\ge1} n^{1-\beta} w_N(dn) - \zeta(\beta-1) - d^{\beta-2} W_N(2-\beta)$, *and the series converges absolutely.* $R_N$ *extends analytically to* $\beta = 2$, *and* $C_N(\beta) = Z_N(\beta) - W_N(2-\beta)/\zeta(2)$ *is meromorphic on the plane with a single pole, at* $\beta = 2$, *which is simple with residue* $1/\zeta(2)$.

*Proof.* Since $\varphi(m) = \sum_{d\mid m} \mu(d)\, m/d$ and $c_N(m) = \varphi(m) w_N(m)$ vanishes for $m > F_{N+2}$,

$$Z_N(\beta) = \sum_m \varphi(m) w_N(m) m^{-\beta} = \sum_{d\ge1} \mu(d) d^{-\beta} \sum_{n\ge1} n^{1-\beta} w_N(dn),$$

a finite sum. For $\operatorname{Re}\beta > 1$ the series $\sum_d \mu(d) d^{-\beta} = 1/\zeta(\beta)$ and $\sum_d \mu(d) d^{-2} = 1/\zeta(2)$ converge absolutely, and $r_d(\beta) = -\zeta(\beta-1) - d^{\beta-2} W_N(2-\beta)$ for $d > F_{N+2}$, so the series for $R_N$ converges absolutely, and inserting the definition of $r_d$ gives the identity. $Z_N$ is entire, and $W_N(2-\beta)$ is analytic except for a simple pole at $\beta = 2$ with residue $-1$, which gives the statement on $C_N$. The main term $\zeta(\beta-1)/\zeta(\beta)$ has a simple pole at $\beta = 2$ with residue $1/\zeta(2)$, so the poles of the two main terms cancel and $R_N$ extends analytically to $\beta = 2$. □

The proposition holds for every $N$, and its content is the form of $R_N$. For $1 < \operatorname{Re}\beta < 2$, $d^{\beta-2} W_N(2-\beta) = \int_0^\infty w_N(dx) x^{1-\beta}\,dx$, so $r_d(\beta)$ is the error made in replacing the sum $\sum_n n^{1-\beta} w_N(dn)$ by the corresponding integral, less the value $\zeta(\beta-1)$ of the same difference for $w_N \equiv 1$ (the limit of the truncated differences): an Euler Maclaurin remainder. Heuristically $R_N$ is small when the depth profile

| $t$ | $N = 16$ | $N = 24$ | $N = 32$ | $\lvert\zeta(\beta-1)/\zeta(\beta)\rvert$ |
|---|---|---|---|---|
| 0.3 | 15.0197 / 14.9781 | 43.7060 / 43.7165 | 126.5819 / 126.5812 | 0.5617 |
| 1 | 4.7475 / 4.7440 | 10.6377 / 10.6446 | 23.8043 / 23.7874 | 0.5873 |
| 2 | 1.0789 / 1.0509 | 1.4826 / 1.5005 | 2.0082 / 2.0524 | 0.6557 |
| 5 | 0.0938 / 0.1117 | 0.0446 / 0.0652 | 0.0376 / 0.0503 | 0.9003 |
| 10 | 0.0660 / 0.0812 | 0.0762 / 0.0805 | 0.0415 / 0.0462 | 1.2084 |

Table 13.8: Knauf's chain on the line $\operatorname{Re}\beta = \frac{3}{2}$, at $\beta = \frac{3}{2} + it$. Each entry is $|Z_N(\beta) - \zeta(\beta-1)/\zeta(\beta)|$ followed, after a slash, by the profile term $|W_N(2-\beta)|/\zeta(2)$; computed.

is regular on the scale of the integers. The two main terms are the residues that a smooth cutoff produces (Remark 13.7.4(i)).

On the line $\operatorname{Re}\beta = \frac{3}{2}$ the profile term $W_N(2-\beta)/\zeta(2)$ grows with $N$ at small heights and accounts there for almost all of $Z_N - \zeta(\beta-1)/\zeta(\beta)$ (Table 13.8): at the tabulated points with $t \le 2$ the two moduli agree to within 3 percent. The profile term falls below $|\zeta(\beta-1)/\zeta(\beta)|$ at a height $t_N$ between 2.1 and 2.5 for $10 \le N \le 24$ (first crossing on a grid of step 0.02). At the first twenty points $1 + \rho_n$ the mean of $|R_N|$ is 0.081 at $N = 12$ and 0.010 at $N = 34$, with maxima 0.22 and 0.026. A power fit over $12 \le N \le 34$ gives the exponent $-1.82$, but the data fix no law: the exponent is $-1.69$ over $12 \le N \le 24$ and $-2.23$ over $24 \le N \le 34$. These values are computed in double precision from the exact counts $c_N(m)$ (`fc_perron.py`); the double precision sums agree with ball arithmetic at 128 bits to $2 \times 10^{-11}$ (seven values of $N$ up to 26).

For every even $N$ from 10 to 34 and every $n \le 20$, the disc of radius 0.3 about $1 + \rho_n$ contains exactly one zero of $Z_N$ and exactly one zero of the corrected sum $C_N$ of Proposition 13.7.2, found by Newton's method from $1 + \rho_n$ (the winding number on the circle, sampled; computed, not a ball proof); the largest distance of any of these zeros from $1 + \rho_n$ is 0.243. Table 13.9 gives the mean distance of these zeros from $1 + \rho_n$. Removing the profile term does not bring the zeros consistently closer: the mean distance is smaller for $C_N$ only at $N = 14$ and at every computed $N$ from 24 on, and larger at the other $N \le 22$. At $N = 34$ the correction brings 12 of the twenty zeros closer, 14 zeros of $C_N$ lie within 0.005 of $1 + \rho_n$ against 9 of $Z_N$, and the largest distances are 0.011 for $C_N$ and 0.021 for $Z_N$. At the points $1 + \rho_n$, where $\zeta(\beta-1)/\zeta(\beta)$ vanishes, the means of $|Z_N|$ and of the profile term at $N = 34$ are 0.016 and 0.021, against the mean 0.010 of $|R_N|$ given above: there the profile term and $R_N$ are of the same size and partly cancel, and removing the profile term lowers the mean error from 0.016 to 0.010. So the profile term is the systematic error of $Z_N$ at small heights, where it dominates, and at $N = 34$ it is part of the error at the points $1 + \rho_n$ too (computed, `fc_fisher.py`; the means at $N = 34$ and the windings for $N = 30$ to 34, `fc_extra.py`).

| $N$ | 10 | 14 | 18 | 22 | 24 | 26 | 30 | 34 |
|---|---|---|---|---|---|---|---|---|
| $Z_N$ | 0.0676 | 0.0304 | 0.0148 | 0.0107 | 0.0114 | 0.0094 | 0.0076 | 0.0081 |
| $C_N$ | 0.0693 | 0.0280 | 0.0186 | 0.0121 | 0.0103 | 0.0092 | 0.0065 | 0.0047 |

Table 13.9: Mean distance from $1+\rho_n$, over $n \le 20$, of the zero of Knauf's partition function $Z_N$ and of the zero of the corrected sum $C_N$ in the disc of radius 0.3 about $1+\rho_n$; computed.

Conjecture 13.7.3. *There is an open strip $\sigma_1 < \operatorname{Re}\beta < \sigma_2$, with $1 \le \sigma_1 < \frac{3}{2} < \sigma_2 \le 2$, on which $R_N \to 0$ locally uniformly as $N \to \infty$.*

If the conjecture holds, then $C_N = \zeta(\beta-1)/\zeta(\beta) + R_N$ converges to $\zeta(\beta-1)/\zeta(\beta)$ locally uniformly on the strip, and by Hurwitz's theorem every point $1+\rho$ in the strip is the limit of exactly as many zeros of $C_N$, counted with multiplicity, as its multiplicity as a zero of $\zeta(\beta-1)$, and every limit point of zeros of $C_N$ in the strip is such a point (conditional). The computations support the conjecture only at the twenty points $1+\rho_n$ of the line $\operatorname{Re}\beta = \frac{3}{2}$; at $\beta = \frac{3}{2}+2i$ the value of $|R_N|$ lies between 0.021 and 0.051 for every $N$ from 10 to 34, with no decrease (computed, `fc_perron.py`).

Remark 13.7.4. (i) For a smooth cutoff the analogue of Conjecture 13.7.3 is a theorem, and its main terms do not depend on the zeros of $\zeta$. Let $\Sigma_Y(\beta) = \sum_m \varphi(m) e^{-m^2/Y^2} m^{-\beta}$, whose cutoff $e^{-x^2}$ has the Mellin transform $W(z) = \frac{1}{2}\Gamma(\frac{z}{2})$. For every $\delta > 0$, as $Y \to \infty$,

$$\Sigma_Y(\beta) - \frac{Y^{2-\beta} W(2-\beta)}{\zeta(2)} = \frac{\zeta(\beta-1)}{\zeta(\beta)} + O\big(Y^{1+\delta-\operatorname{Re}\beta}\big),$$

uniformly on compact subsets of the strip $1+\delta < \operatorname{Re}\beta < 3$ without the point 2. Indeed, $\Sigma_Y(\beta) = \frac{1}{2\pi i}\int_{(c)} \frac{\zeta(\beta+z-1)}{\zeta(\beta+z)} Y^z W(z)\,dz$ for $c > \max(0, 2-\operatorname{Re}\beta)$. Moving the line of integration to $\operatorname{Re}(\beta+z) = 1+\delta$ crosses the poles at $z=0$ and $z = 2-\beta$, with residues $\zeta(\beta-1)/\zeta(\beta)$ and $Y^{2-\beta}W(2-\beta)/\zeta(2)$ (no other pole of $W$ is crossed for $\operatorname{Re}\beta < 3$); on the new line $|1/\zeta(\beta+z)| \le \zeta(1+\delta)$, $\zeta(\beta+z-1)$ grows at most polynomially in $|\operatorname{Im} z|$, and $W(z)$ decays exponentially.

(ii) Hence the consequence of Conjecture 13.7.3 does not decide RH. If it holds, the zeros of $C_N$ in the strip converge to the points $1+\rho$ in it whether or not $\operatorname{Re}\rho = \frac{1}{2}$, and by (i) the two terms removed from $Z_N$ are those that a regular cutoff produces whatever the zeros. The conjecture is a statement about the regularity of the depth profile.

(iii) A statement that would imply RH concerns the zeros of $\zeta(\beta)$, which are poles of $\zeta(\beta-1)/\zeta(\beta)$: if $C_N$ converges locally uniformly on the half plane $\operatorname{Re}\beta > \frac{1}{2}$ with the point 2 removed, then RH holds. The limit is analytic there, by Weierstrass's theorem, and it equals $\zeta(\beta-1)/\zeta(\beta)$ on $\operatorname{Re}\beta > 2$, by Proposition 13.7.1(ii) and because $W_N(2-\beta) \to 0$ there: for $\operatorname{Re} z < 0$, $W_N(z) =$

$1/z + \int_1^\infty w_N(x) x^{z-1}\,dx \to 1/z - 1/z$ by dominated convergence, as $w_N \uparrow 1$. So $\zeta(\beta-1)/\zeta(\beta)$ would continue analytically to the half plane without the point 2, while a zero $\rho$ of $\zeta$ with $\operatorname{Re}\rho > \frac12$ is a pole of it, because $\zeta(\rho-1) \neq 0$, as $-\frac12 < \operatorname{Re}(\rho-1) < 0$. The hypothesis of this implication, the convergence of $C_N$ on the half plane, is open. For the smooth cutoff of (i), whose term $Y^{2-\beta}W(2-\beta)$ has further poles at $\beta = 4, 6, \dots$, the corresponding convergence on the strip $\frac12 < \operatorname{Re}\beta < \frac52$ without the point 2 holds if and only if RH holds: under RH the line in (i) can be moved to $\operatorname{Re}(\beta+z) = \frac12 + \delta$, where $1/\zeta(\beta+z) = O(|\operatorname{Im} z|^\delta)$ [498, Section 14.2]. That version therefore restates RH, as the convergence of $\sum_n \mu(n) n^{-s}$ on $\operatorname{Re} s > \frac12$ does [498, Section 14.25].

### 13.7.3 The Farey chain in a field

Kleban and Özlük introduced a second chain [200]. Its configurations are the words $\varpi_1 \cdots \varpi_N$ in the letters $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $B = A^{\mathsf T}$, and the energy of a word is $\log \operatorname{tr}(\varpi_1 \cdots \varpi_N)$. Its free energy equals that of Knauf's chain [137], so it has the same transition at $\beta = 2$, and it is fully magnetised for $\beta > 2$ [92]. Conjugation by $\Pi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ exchanges $A$ and $B$, so the energy does not change when the two letters are exchanged, which is the flip of all spins; Knauf's chain has a single ground state, $s(k) = 1$ only at $k = 2^N$, and no such symmetry. In a field $x = e^h$ the partition function of the Farey chain is

$$Z_N^{\mathrm F}(\beta, x) = \sum_{\varpi_1, \dots, \varpi_N \in \{A,B\}} \bigl(\operatorname{tr} \varpi_1 \cdots \varpi_N\bigr)^{-\beta} x^{n_B},$$

where $n_B$ is the number of letters $B$ in the word. It is a polynomial of degree $N$ in $x$ with positive coefficients, palindromic by the symmetry. Under the Gibbs law at zero field the centred magnetisation $n_B - \frac{N}{2}$ is bounded and symmetric, with moment generating function $e^{-Nz/2} Z_N^{\mathrm F}(\beta, e^z)/Z_N^{\mathrm F}(\beta, 1)$, so it is ferromagnetic in the sense of Definition 13.1.1 exactly when every zero of $x \mapsto Z_N^{\mathrm F}(\beta, x)$ lies on the unit circle. We then say that the chain has the Lee and Yang property at $(\beta, N)$.

Lemma 13.7.5. *Let $P(x) = 1 + x^N + Q(x)$ with $Q(x) = \sum_{k=1}^{N-1} a_k x^k$ and $a_k = a_{N-k} \geq 0$. If $Q(1) < 2$, the zeros of $P$ are simple and lie on the unit circle.*

*Proof.* For real $\theta$, $g(\theta) = e^{-iN\theta/2} P(e^{i\theta}) = 2\cos\frac{N\theta}{2} + \sum_k a_k \cos\bigl((k - \frac{N}{2})\theta\bigr)$ is real, and the sum has modulus at most $Q(1) < 2$. So $g(2\pi j/N)$ has the sign of $(-1)^j$ for $j = 0, 1, \dots, N$, and $g$ has at least $N$ zeros in $(0, 2\pi)$. They give $N$ distinct zeros $e^{i\theta}$ of $P$, which has degree $N$. □

For the Farey chain the words $A^N$ and $B^N$ have trace 2. Every other word contains both letters, so it is a matrix with positive integer entries and determinant 1, whose diagonal entries $a$ and $d$ satisfy $ad = 1 + bc \geq 2$, and its trace is at least 3.

| $N$ | 10 | 14 | 18 | 22 | 26 | 30 | 32 |
|---|---|---|---|---|---|---|---|
| $\beta_{\mathrm{LY}}(N)$ | 1.2424 | 1.4885 | 1.6201 | 1.7005 | 1.7543 | 1.7926 | 1.8079 |
| $\theta_N$ | 0.9952 | 0.6802 | 0.5130 | 0.4105 | 0.3414 | 0.2918 | 0.2719 |
| $N\theta_N$ | 9.952 | 9.522 | 9.235 | 9.030 | 8.876 | 8.754 | 8.702 |
| $\beta^*(N)$ | 2.1056 | 2.0924 | 2.0809 | 2.0719 | 2.0648 | 2.0592 | 2.0567 |

Table 13.10: The Farey chain in a field: the computed Lee and Yang threshold $\beta_{\mathrm{LY}}(N)$, the angle $\theta_N$ of the double zeros at the threshold, and the root $\beta^*(N)$ of $2^\beta Z_N^{\mathrm{F}}(\beta,1)=4$, above which Lemma 13.7.5 proves the property; computed.

Hence $Z_N^{\mathrm{F}}(\beta,x)=2^{-\beta}\big(1+x^N+Q_{N,\beta}(x)\big)$, where $Q_{N,\beta}(1)=\sum(2/\operatorname{tr}\varpi_1\cdots\varpi_N)^\beta$ over the other words decreases strictly in $\beta$, and Lemma 13.7.5 gives the Lee and Yang property at $(\beta,N)$ for every $\beta>\beta^*(N)$, where $\beta^*(N)$ is the root of $2^\beta Z_N^{\mathrm{F}}(\beta,1)=4$.

The computation finds, for the even $N$ from 10 to 36, a threshold $\beta_{\mathrm{LY}}(N)$ such that the Lee and Yang property holds at every computed $\beta\ge\beta_{\mathrm{LY}}(N)$ and fails at every computed $\beta$ with $0<\beta<\beta_{\mathrm{LY}}(N)$ (grid of step 0.1 on $0.2\le\beta\le4$, one transition at every $N$, then bisection; for $N\le32$ the bracket of width $3\times10^{-11}$ is certified by a Sturm count in ball arithmetic at 400 bits, for $N=34,36$ it is a double precision bisection; `fc_ly.py`, and `fc_extra.py` for $\beta^*(N)$); at $\beta=0$ it holds trivially, since $Z_N^{\mathrm{F}}(0,x)=(1+x)^N$. At the threshold four zeros meet on the circle as two double zeros $e^{\pm i\theta_N}$, and just below it they leave the circle as a quartet $e^{\pm u\pm i\theta}$. The threshold rises with $N$, from 1.2424 at $N=10$ to 1.8329 at $N=36$ (computed; `fc_ly.py`), and stays below $\beta^*(N)$, which falls with $N$ (Table 13.10, which stops at $N=32$; computed).

Conjecture 13.7.6. *For every $\beta>2$ the Farey chain has the Lee and Yang property at $(\beta,N)$ for all large $N$, and for every $\beta$ with $0<\beta<2$ it fails to have it for all large $N$.*

The value 2 comes from the transition [93, 137], not from the data: free fits of $\beta_{\mathrm{LY}}(N)$ in powers of $1/N$ over $10\le N\le22$ extrapolate to 2.086 (linear) and 2.046 (quadratic), and over $20\le N\le36$ to 2.044 and 2.017 (computed), all above 2. For every $N$ let $\beta_{\mathrm{LY}}(N)$ be the infimum of the $\beta_0>0$ such that the property holds at $(\beta,N)$ for all $\beta>\beta_0$; it is the computed threshold for the even $N$ above. By Lemma 13.7.5, $\beta_{\mathrm{LY}}(N)\le\beta^*(N)$, so a limit of $\beta_{\mathrm{LY}}(N)$ is at most $\liminf_N\beta^*(N)$.

The theorems of Lee and Yang, of Newman and of Lieb and Sokal recalled in Section 13.1.1 concern pair interactions. The interactions of both chains are many body [203, 200], so these theorems do not apply to them; for the Farey chain, Lemma 13.7.5 gives the Lee and Yang property above $\beta^*(N)$, and the computation finds it down to $\beta_{\mathrm{LY}}(N)$, below the transition, and not for $0<\beta<\beta_{\mathrm{LY}}(N)$. The zeros of this section are of two kinds: the Fisher zeros of Section 13.7.2 are zeros of Knauf's chain in the inverse temperature, and those of this subsection are zeros

of the Farey chain in the field, at real $\beta$.

## 13.8 Open problems

Problems 13.7 and 13.8 concern the variable $e^{2X}$ of Section 13.6, where no hypothesis on the zeros enters, and the last two concern the spin chains of Section 13.7.

Problem 13.1. Mixed shoulders. Theorem 13.2.4 decides the symmetric laws on $[-1,1]$ whose density is concave on $[0,1]$ and those whose density is convex there and not linear on its support: by (13.2.1) the question is whether the sine transform of the slope $g=-p'$ has only real zeros, and the theorem answers it when $g$ is monotone. Decide the slopes that first rise and then fall. One class is settled. If $g$ is symmetric about $\frac12$, then $p(0)=\int_0^1 g=1$, because $1=2\int_0^1 p=2\int_0^1 xg(x)\,dx=\int_0^1 g$ by the symmetry of $g$, so $u\mapsto g(\frac12+u)$ is a probability density on $[-\frac12,\frac12]$; and $X\stackrel{d}{=}\frac12 U+B$, with $B$ independent of $U$ and distributed with that density, since the density of $\frac12U+B$ has slope $g$ on $(0,1)$. As $\sin(z/2)/(z/2)$ has only real zeros, $X$ is then ferromagnetic exactly when $B$ is. For the beta slopes $g(y)\propto y^a(1-y)^b$, $a,b\ge0$, this decides the diagonal $a=b$, where $B$ is a symmetric beta law and $X$ is ferromagnetic (Section 13.1.4), and Theorem 13.2.4 decides the edges: $X$ is ferromagnetic for $b=0$, where $p$ is concave, and not for $a=0<b$, where $p$ is convex and not linear. The chapter does not decide the rest of the quadrant. On the half line the slope of the Pólya law rises and then falls (Section 13.4.2), and RH is the statement that its sine transform has only real zeros.

Problem 13.2. Lines of zeros of the Mellin transform of a clock. For which ferromagnetic laws $X$ does $\mathcal{D}_X$ of Section 13.3 have nonreal zeros, all of them on one vertical line? The uniform pair $\pi U+\pi U'$ has this property if and only if RH holds, by (13.3.1). The sums of one, four and six independent copies of $\pi U$ do not have it (Proposition 13.3.2; for four and six computed, Section 13.3), and neither do the pairs $X_\varepsilon+X'_\varepsilon$ of Section 13.3 for small $\varepsilon>0$ (computed), so no theorem whose hypotheses hold for every symmetric density on an interval that is nondecreasing in $|t|$ can give it. Is the uniform pair, up to scale, the only pair $X+X'$ with $X$ in the class of Theorem 13.2.6 that has it? A first step is the first order motion of the zeros along the densities $\frac{1}{2\pi}+\varepsilon g$ on $[-\pi,\pi]$, which is linear in $g$: is there an even $g\neq0$ with $\int g=0$ along which every zero $\frac{\rho_k-1}{2}$ keeps its real part to first order? This problem follows notes of Nicholas Polson merged into the book [246, 241].

Problem 13.3. The Griffiths Simon class. Newman's theorem places the Pólya law in the class of Ellis, Monroe and Newman (Section 13.4.1). Does it belong to the Griffiths Simon class of Section 13.1.1? The laws of that class are ferromagnetic, so a positive answer would imply RH, and under Ingham's hypothesis the

spins of any approximating system would have to be coupled (Section 13.4.3).

PROBLEM 13.4. Splitting the zeros. Assume RH, so that $\varphi_X(t) = \prod_\gamma (1 - t^2/\gamma^2)$ over the positive ordinates, with no Gaussian factor by Theorem 13.4.1. Can the ordinates be split into two nonempty sets $A$ and $A'$ such that $\prod_{\gamma \in A}(1 - t^2/\gamma^2)$ and $\prod_{\gamma \in A'}(1 - t^2/\gamma^2)$ are both characteristic functions? This is the same as writing $X = D + R$ with $D$ and $R$ independent, ferromagnetic and not 0, and by Theorem 13.4.1 both sets are then infinite. Under Ingham's hypothesis no three ordinates are in arithmetic progression, so no piece is a scaled fair sign or a scaled uniform variable, whose zeros form arithmetic progressions.

PROBLEM 13.5. Positivity of the zero completed laws. The laws $f_K$ of Section 13.5.4 are proved to be ferromagnetic probability densities for $K \le 16$ by the Maxwell construction, and for every $K \le 200$ and for $K = 300$, 400, 600 and 800 by a computer assisted proof in ball arithmetic. Compute the threshold $v^*(K)$ near $K = 5000$, where the linear and the power law fits of the ratio $v^*(K)/2\beta_K$ part; the cancellation there calls for a working precision of tens of thousands of bits. Is $v^*(K) < 2\beta_K$ for every $K$? If so, every $f_K$ is a ferromagnetic probability density by Theorem 13.5.12(ii), and with no hypothesis their limit in law, with characteristic function $e^{-\beta_\infty \theta^2} \prod_k (1 - \theta^2/\gamma_k^2)$ over the zeros on the critical line, is a ferromagnetic law by Theorem 13.1.5; by Theorem 13.5.12(iii) it is the Pólya law exactly when RH holds.

PROBLEM 13.6. The cosine constant. By Section 13.5.2, $\Omega \le \frac{1}{2}$ with no hypothesis, $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is free of zeros (Section 12.1.2), and RH holds if and only if $\Omega = 0$. Is the set of $a \ge 0$ for which $\cosh(at)\, p$ is ferromagnetic an interval? Improve these bounds by computation, as computations with the zeros have improved de Bruijn's bound $\Lambda \le \frac{1}{2}$ on the de Bruijn Newman constant, to $\Lambda \le 0.2$ [419]. A first case is $a = \sqrt{0.1}$, below both bounds: the law $\cosh(\sqrt{0.1}\, t)\, p$ is as close to $p$ as the flow at $\lambda = 0.2$, which is ferromagnetic, and deciding it is a computation with the zeros of $\frac{1}{2}(\xi(s+a) + \xi(s-a))$, of the kind that gave the bound 0.2 for the flow.

PROBLEM 13.7. The size bias principle for real shapes. Theorem 13.6.5 is proved for integer shapes. Does it hold for all real shapes $t_n > 0$? For $K = 1$ it is the gamma identity, with $G = 0$. Integer shapes enter the proof twice, through the divided difference formula for the density of $Y$ and through Lemma 13.6.4, and together they show that on each gap $(\tau_k, \tau_{k+1})$ the phase of the Laplace transform of $Y_q$ is nonincreasing and falls by exactly $q$. Is this still true for real shapes? With it, the rest of the proof goes through, provided the estimates of the transform near the points $-\tau_n$ and at infinity can be checked as well.

PROBLEM 13.8. The powers of $e^X$. For $2 < |\beta|$ below the HCM threshold, which lies between 5.5791395 and $\pi/0.563096 = 5.57914219\ldots$ (Proposition 13.6.8), is $e^{\beta X}$ infinitely divisible, self decomposable or a GGC? Below the threshold it has no

HCM density, which leaves the GGC property open, and Theorem 13.6.7 gives self decomposability only at $|\beta| = 2$. Is the HCM threshold exactly $\pi/v_m = 5.5791396$? That requires $h \geq 0$ on the open strip $0 < v < v_m$, where $h(0, v) \to 0$ as $v \uparrow v_m$, so a certificate must replace the balls with positive lower bounds near $iv_m$ by a local argument.

Problem 13.9. The depth profile of Knauf's chain. Prove Conjecture 13.7.3, or find the strips on which it holds. By Proposition 13.7.2, $R_N$ measures, heuristically, how far the profile $w_N = c_N/\varphi$ of Section 13.7.2 is from being regular on the scale of the integers. The fraction $a/m$ is a child of $a/(m-a)$ in the Calkin Wilf tree, and each step down the tree adds one to the sum of the partial quotients, the root $1/1 = [1]$ having sum 1; so by Proposition 13.7.1(i), for $m \geq 2$, $c_N(m)$ is the number of $a$ prime to $m$, $1 \leq a < m$, for which $a/m = [0; b_1, \dots, b_k]$ has $b_1 + \cdots + b_k \leq N + 1$. Yao and Knuth proved that the mean of $b_1 + \cdots + b_k$ over $1 \leq a \leq m$ is asymptotic to $\frac{6}{\pi^2} \log^2 m$ [522]; over the $a$ prime to $m$, which are those counted here, a quadratic fit in $\log m$ to the computed means at $m = 10^5, \dots, 10^8$ has leading coefficient 0.6073 (computed; `fc_stern.py`, the mode `--yk`). The consequence of the conjecture does not decide RH (Remark 13.7.4), and the convergence in Remark 13.7.4(iii) would imply it.

Problem 13.10. The Lee and Yang threshold of the Farey chain. Prove Conjecture 13.7.6. By Lemma 13.7.5 its first half follows if, for every inverse temperature $\beta > 2$, the words other than $A^N$ and $B^N$ carry less weight than these two for all large $N$, that is $\sum (2/\operatorname{tr} \varpi_1 \cdots \varpi_N)^\beta < 2$ over the other words. The computed $\beta^*(N)$ falls from 2.1056 to 2.0525 for $10 \leq N \leq 36$, and for $2 < \beta \leq \beta^*(N)$ the lemma does not decide. The second half asks that for every $\beta$ with $0 < \beta < 2$ some zeros leave the circle once $N$ is large; at the computed thresholds the zeros that leave it first are at the angle $\theta_N$, with $N\theta_N$ between 8.61 and 9.95 (computed; `fc_ly.py`, and Table 13.10 for $N \leq 32$).

# Chapter 14

# The Pólya density as a maximum order statistic and the Thorin measure of the clock

This chapter follows the first author's manuscript [429], which starts from one observation: the law behind the Pólya kernel of the Riemann Xi function is built from the maximum $M_4$ of independent exponential random variables of Section 8.5.4.2, in the sense that $e^{2Y}$ is a sum of two independent copies of $M_4$ (Proposition 14.2.1). Everything else follows from asking which parts of that observation survive the passage to the Thorin measure of the clock.

The Thorin programme of this book asks whether the clock, with Laplace transform $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{\lambda})$, is a generalised gamma convolution. The unconditional statements about the clock one level below the GGC property are those of Section 5.1, and this chapter uses them as they stand.

*What is not claimed.* Nothing here proves RH. Every statement equivalent to RH is labelled as such.

*The dictionary.* Table 14.1 sets the trigonometric side against the Riemann side.

## 14.1 Notation

The Laplace variable of this chapter is $\lambda$, the variable $u$ of Section 1.2, so that the Laplace transform of the clock is $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{\lambda})$, written $\varphi_{1/2}$ when the basepoint family is in view.

## 14.2 The Pólya density as a maximum order statistic

Section 8.5.4.2 shows that $\vartheta_4(t) = \sum_{n\in\mathbb{Z}}(-1)^n e^{-\pi n^2 t}$ is the distribution function of the maximum $M_4$ of independent exponential variables, one of each rate $\pi n$

| | Trigonometric side | Riemann side |
|---|---|---|
| entire function | $\sin(\sqrt{\pi}\theta)/(\sqrt{\pi}\theta)$ | $\Xi(\theta)/\Xi(0)$ |
| real zeros | $\theta = \pm\sqrt{\pi}\, n$ | $\theta = \pm\gamma_k$ (all real iff RH) |
| space component | $\sqrt{\pi/2}\, v$, with $v$ uniform on $(-1,1)$ | $X/\sqrt{2}$, with $X$ the Pólya variable |
| time component | $M_4$ | $T$, the clock |
| Laplace transform | $\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda}$ | $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda})$ |
| exists as a law | yes | yes (Theorem 5.1.1) |
| ID and SD | yes | yes (Theorem 5.1.1) |
| $x$ times the Lévy density | $\sum_{n\geq 1} e^{-\pi n^2 x}$ | $W(x) = \sum_k e^{-a_k x}$ |
| GGC, Thorin measure | $\sum_{n\geq 1} \delta_{\pi n^2}$ | $\sum_k \delta_{\gamma_k^2}$ if and only if RH (Proposition 1.2.1) |
| maximum of exponentials | yes, with rates $\pi n$ and $\pi(2m-1)$ | no (Proposition 14.9.1) |
| Gnedenko norming | $a_n = 1/\pi$ | $a_n = 1/\gamma_1^2$ |

Table 14.1: The dictionary between the trigonometric side, where everything is classical, and the Riemann side developed in this chapter. The variable $\theta$ is that of Section 1.2; ID and SD abbreviate infinitely divisible and self decomposable, and the Wald couple of the third and fourth rows is that of Theorem 14.4.2 and Remark 14.5.1.

for $n \geq 1$ and one of each rate $\pi(2m-1)$ for $m \geq 1$: by independence and Jacobi's triple product, $\Pr(M_4 \leq t) = \prod_{n\geq 1}(1-e^{-\pi n t})\prod_{m\geq 1}(1-e^{-\pi(2m-1)t}) = \vartheta_4(t)$. Since the Kolmogorov distribution function is $k \mapsto \vartheta_4(2k^2/\pi)$, $\pi M_4$ is twice the square of a variable with the Kolmogorov law. The same section has $\mathbb{E}e^{-\lambda M_4} = \sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda} = \prod_{n\geq 1}(1+\lambda/(\pi n^2))^{-1}$, so that $M_4 \stackrel{d}{=} \sum_{n\geq 1} E_n/(\pi n^2)$ with independent standard exponential variables $E_n$, a GGC with Thorin measure $\sum_n \delta_{\pi n^2}$; and $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\,\xi(s)/(s-1)$ for every complex $s$, with the right side continued at $s=1$ by $\log 2$. The Mellin transform is a case of the dictionary of Biane, Pitman and Yor [49]. By the Mellin transform, inside the open critical strip the zeros of $s \mapsto \mathbb{E}M_4^{s/2}$ are exactly the nontrivial zeros of $\zeta$, since $1-2^{1-s}$ vanishes only on $\operatorname{Re} s = 1$. The same law is thus a maximum of exponentials with linear rates $\pi n$ and a sum of exponentials with quadratic rates $\pi n^2$; this coincidence is the Jacobi triple product, and Section 14.9 shows that it has no counterpart for the clock.

Up to a constant factor, $e^Y$ is the maximum of a normalised Brownian excursion [49].

Proposition 14.2.1. *Let $M_4$ and $M_4'$ be independent, and let $Y$ be the variable of Section 1.2, $\mathbb{E}e^{sY} = 2\xi(s)$. Then:*

*(i)* $e^{2Y} = \frac{\pi}{2} S_2 \stackrel{d}{=} M_4 + M_4'$.

*(ii) If $f$ is the density of $e^Y = \sqrt{M_4 + M_4'}$, then the Pólya kernel* (1.2.3) *is* $\Phi(u) = \frac{1}{2}\, e^{3u/2} f(e^u)$, *and*

$$f(y) = 4 \sum_{n \geq 1} \bigl(2\pi^2 n^4 y^3 - 3\pi n^2 y\bigr) e^{-\pi n^2 y^2}.$$

*Proof.* (i) By definition $e^{2Y} = \frac{\pi}{2} S_2$, and $\frac{\pi}{2} S_2 = \sum_{k \geq 1} (e_k + e_k')/(\pi k^2)$ by the definition of $S_2$. Since $M_4 \stackrel{d}{=} \sum_{n \geq 1} E_n/(\pi n^2)$, a sum of two independent copies of $M_4$ has the same law.

(ii) Let $\mu$ be the law of $Y$. By Section 1.2, $2\Xi(\theta) = 2\xi(\frac{1}{2} + i\theta) = \mathbb{E} e^{(\frac{1}{2} + i\theta)Y} = \int_{\mathbb{R}} e^{u/2} e^{i\theta u} \mu(du)$, and the measure $e^{u/2}\mu(du)$ is finite because $\mathbb{E} e^{Y/2} = 2\xi(\frac{1}{2})$. On the other hand $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u) e^{i\theta u}\, du$ by (1.2.3), since $\Phi$ is even. The finite measures $\frac{1}{2} e^{u/2}\mu(du)$ and $\Phi(u)\, du$ have the same Fourier transform, so they are equal: $\mu(du) = 2e^{-u/2}\Phi(u)\, du$. The density of $e^Y$ is $f(y) = y^{-1} \cdot 2y^{-1/2}\Phi(\log y)$, which is the first claim, and substituting (1.2.3) gives the series. □

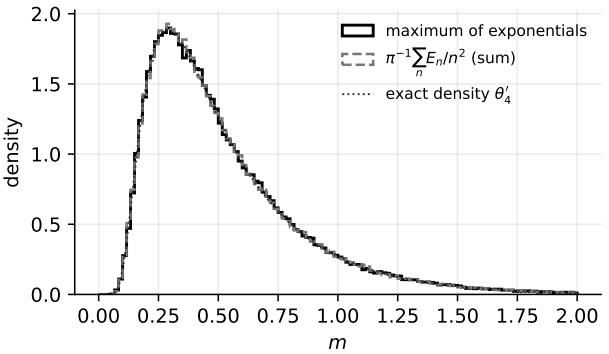



Figure 14.1: Histograms of $M_4$ simulated as a maximum of exponentials and as a weighted sum of exponentials.

## 14.3 Khinchin: the zeta distribution inside the tilted maximum

Khinchin [308] proved that for $\sigma > 1$ the function $\zeta(\sigma + it)/\zeta(\sigma)$ is an infinitely divisible characteristic function; Gnedenko and Kolmogorov [151] reproduce it as a worked example, and Lin and Hu [340] name the law the Riemann zeta distribution. If $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, the zeta distribution of Sections 6.1.6 and 7.1.2.3, then $\mathbb{E} N_\sigma^{-it} = \zeta(\sigma + it)/\zeta(\sigma)$, and the Euler product gives $N_\sigma = \prod_p p^{G_p}$ with independent geometric variables $G_p$, $\Pr(G_p = k) = (1 - p^{-\sigma})p^{-k\sigma}$. Neither Khinchin nor Gnedenko worked on the zeros or on $\Xi$; the work that continues their line stays at $\sigma > 1$ [340, 165, 378]. The completed ratio $\xi(\sigma + it)/\xi(\sigma)$ is the

characteristic function of the variable $V_\sigma$ of Section 9.1.1, whose reflection $-V_\sigma$ is Nakamura's complete Riemann zeta distribution [377], and $V_\sigma$ is quasi infinitely divisible but not infinitely divisible for $\sigma > 1$ (Proposition 11.1.3). The Pólya density, on the other hand, is not infinitely divisible, because its characteristic function $\Xi(\theta)/\Xi(0)$ has real zeros.

THEOREM 14.3.1. *Fix $\sigma > 1$ and let $\widetilde{M}_4$ have the tilted law $\Pr(\widetilde{M}_4 \in dm) \propto m^{\sigma/2} \Pr(M_4 \in dm)$. Let $K$, $N_\sigma$ and $\Gamma_{1+\sigma/2}$ be independent of each other and of $\widetilde{M}_4$, with $\Pr(K = k) = (1-a)a^k$ for $k \geq 0$ and $a = 2^{1-\sigma}$, with $N_\sigma = \prod_p p^{G_p}$ zeta distributed with parameter $\sigma$, and with $\Gamma_{1+\sigma/2}$ gamma distributed with shape $1 + \sigma/2$ and scale 1. Then*

$$\widetilde{M}_4\, 4^{-K} \;\stackrel{d}{=}\; \frac{\Gamma_{1+\sigma/2}}{\pi N_\sigma^2} \;\stackrel{d}{=}\; \frac{\Gamma_{1+\sigma/2}}{\pi} \prod_p p^{-2G_p}.$$

*Proof.* By Section 8.5.4.2, $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\,\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ for every complex $s$, because $2\xi(s)/(s-1) = 2\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$. With $s = \sigma + it$, $\mathbb{E}\widetilde{M}_4^{it/2}$ is the ratio of this at $s$ and at $\sigma$, that is

$$\mathbb{E}\widetilde{M}_4^{it/2} = \frac{1 - ae^{-it\log 2}}{1-a} \cdot \pi^{-it/2} \cdot \frac{\Gamma(1+\frac{\sigma+it}{2})}{\Gamma(1+\frac{\sigma}{2})} \cdot \frac{\zeta(\sigma+it)}{\zeta(\sigma)}.$$

The first factor is the reciprocal of $\mathbb{E}e^{-itK\log 2} = \mathbb{E}(4^{-K})^{it/2}$. The remaining factors are $\pi^{-it/2}$, $\mathbb{E}\Gamma_{1+\sigma/2}^{it/2}$ and $\mathbb{E}N_\sigma^{-it}$. Hence $\mathbb{E}(\widetilde{M}_4\, 4^{-K})^{it/2} = \mathbb{E}\big(\Gamma_{1+\sigma/2}/(\pi N_\sigma^2)\big)^{it/2}$ for every real $t$, which is the first identity in law; the second is the Euler product. □

## 14.4 The Wald couple of the maximum

DEFINITION 14.4.1. A pair $(\tau, \varsigma)$ of independent variables with $\tau \geq 0$ is a Wald couple if $\mathbb{E}e^{s\varsigma} < \infty$ and $\mathbb{E}e^{s\varsigma}\,\mathbb{E}e^{-s^2\tau/2} = 1$ for all real $s$. Equivalently, with $Z$ standard normal and independent of $\tau$, the characteristic functions of $\varsigma$ and of $\sqrt{\tau}Z$ form a van Dantzig pair in the sense of Section 1.2: $\mathbb{E}e^{-\theta^2\tau/2} = 1/\mathbb{E}e^{-\theta\varsigma}$ for real $\theta$.

The name comes from Wald's identity $\mathbb{E}\exp(sB_\tau - s^2\tau/2) = 1$: if a stopping time $\tau$ of Brownian motion $B$ has $B_\tau$ independent of $\tau$ and the stopped martingales $\exp(sB_{t\wedge\tau} - s^2(t\wedge\tau)/2)$ are uniformly integrable, then $(\tau, B_\tau)$ is a Wald couple.

THEOREM 14.4.2. *Let $\upsilon, \upsilon_1, \upsilon_2$ be uniform on $(-1,1)$, with $\upsilon_1$ and $\upsilon_2$ independent, let $M_4'$ be an independent copy of $M_4$, and let $c = \sqrt{\pi/2}$. Then $(M_4, c\upsilon)$ and $(M_4 + M_4', c(\upsilon_1 + \upsilon_2))$ are Wald couples. The associated van Dantzig pairs are $(\sin ct/(ct),\ ct/\sinh ct)$ and their squares, and $\sqrt{M_4}\,Z$ has the logistic law with characteristic function $ct/\sinh ct$.*

*Proof.* $\mathbb{E}e^{-s^2M_4/2} = cs/\sinh(cs)$ by Section 8.5.4.2, with $\lambda = s^2/2$, and $\mathbb{E}e^{scv} = \sinh(cs)/(cs)$. Squaring gives the second couple. □

The pair $(\sin\theta/\theta, \theta/\sinh\theta)$ is the classical van Dantzig pair of Section 1.2 [366], at $\theta = ct$, and the language of Wald couples is that of the first author's [423]. The logistic statement is the representation of the logistic law as a normal scale mixture with Kolmogorov mixing of Andrews and Mallows [6] and Stefanski [485], read through the maximum.

## 14.5 The clock exists and is infinitely divisible

Theorem 5.1.1 proves unconditionally, using only the verification of RH below height 1000, that $W > 0$ and $W' < 0$ on $(0, \infty)$, so that the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density $W(x)/x$; the law is unimodal by Yamazato's theorem [521] (Section 11.1), and its mean is $\mathbb{E}T = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ (Section 1.2). The ladder of signs $(-1)^k W^{(k)} > 0$ is Theorem 5.1.6 for $0 \le k \le 10^{17}$, Theorem 5.1.9 for every order on $(0, 1.29]$ and Theorem 5.1.13 for every $t > 0$ and every $k \le 1.16 \times 10^{25}$.

Remark 14.5.1. The clock gives a Wald couple on the Riemann side: $(T, X/\sqrt{2})$ is a Wald couple. Indeed $T$ exists by Theorem 5.1.1, $\mathbb{E}e^{sX/\sqrt{2}} = \xi(\frac{1}{2} + s/\sqrt{2})/\xi(\frac{1}{2})$ and $\mathbb{E}e^{-s^2T/2} = \xi(\frac{1}{2})/\xi(\frac{1}{2} + |s|/\sqrt{2})$, and $\xi(\frac{1}{2} + w)$ is even in $w$. This is the van Dantzig pair $(\Xi(\theta)/\Xi(0),\ \xi(\frac{1}{2})/\xi(\frac{1}{2} + \theta))$ of Theorem 5.1.1 read at $\theta = s/\sqrt{2}$.

## 14.6 A Stieltjes criterion

Section 1.3 reads the GGC property through the Stieltjes transform of the Thorin measure. For the clock, $G'/G = -\frac{d}{d\lambda}\log(1/G)$ takes the place of the function $f_H$ there, whether or not RH holds, and by Section 1.2 $(G'/G)(\lambda) = \sum_k (\lambda + a_k)^{-1} = \int_0^\infty e^{-\lambda x} W(x)\, dx$. The quantity of this section is

$$r_\eta(x) = \frac{1}{\pi}\ \mathrm{Im}\,\frac{G'}{G}(-x - i\eta), \qquad x \in \mathbb{R},\ \eta > 0,$$

defined wherever $-x - i\eta$ is not a pole of $G'/G$. If the clock is a GGC with Thorin measure $U$, then $r_\eta$ is the Poisson smoothing of $U$ at distance $\eta$ below the cut, as in (1.3.2).

Theorem 14.6.1. *The following are equivalent.*

*(a) The Riemann Hypothesis.*

*(b) $T$ is a GGC; then $T \overset{d}{=} \sum_{\gamma>0} e_\gamma/\gamma^2$ and its Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, the ordinates counted with multiplicity.*

*(c)* $G'/G$ *is a Stieltjes function.*

*(d)* $r_\eta(x) \geq 0$ *for all* $x \in \mathbb{R}$ *and* $\eta > 0$ *for which* $-x - i\eta$ *is not a pole of* $G'/G$.

*(e)* $W$ *is completely monotone.*

*In that case* $r_\eta$ *is the Poisson smoothing of* $U_{\mathrm{RH}}$ *at scale* $\eta$, *and* $U_{\mathrm{RH}}((0,x]) = N(\sqrt{x})$.

Items (a), (b), (c) and (e) are in the book: (a) if and only if (b) is Proposition 1.2.1, (b) implies (c) by the Stieltjes form of the GGC property in Section 1.3, and (b) if and only if (e) is the characterisation of a GGC by the complete monotonicity of $k$, $x$ times its Lévy density [56], with $k = W$ for the clock (Sections 1.2 and 1.3).

*Proof.* It remains to prove that (c) implies (d) and that (d) implies (a). (c) implies (d): a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$ and maps the upper half plane into the closed lower half plane, and $(G'/G)(\bar\lambda) = \overline{(G'/G)(\lambda)}$, so $\mathrm{Im}(G'/G)(-x - i\eta) \geq 0$ for $\eta > 0$. (d) implies (a): if RH fails, then some zero has $\beta \neq \frac12$, and since the zeros are symmetric under $\rho \mapsto \bar\rho$ and $\rho \mapsto 1-\rho$ there is a zero $\rho = \beta + i\gamma$ with $\gamma > 0$ and $\beta > \frac12$; Lemma 14.6.2 gives $r_\eta(x_\rho) \to -\infty$ as $\eta \uparrow b$, so $r_\eta(x_\rho) < 0$ for $\eta < b$ close to $b$ and (d) fails. The last statement is (1.3.2) and the counting formula of Section 1.3 for $U_{\mathrm{RH}}$. □

Lemma 14.6.2 (Detection). *Let* $\rho = \beta + i\gamma$ *be a zero of multiplicity* $m$ *with* $\beta > \frac12$ *and* $\gamma > 0$, *put* $\delta = \beta - \frac12$, *and write* $a_\rho = x_\rho - ib$ *with* $x_\rho = \gamma^2 - \delta^2$ *and* $b = 2\delta\gamma$, *so that* $\lambda_\rho = -a_\rho = -x_\rho + ib$. *Then* $r_\eta(x_\rho) \to -\infty$ *as* $\eta \uparrow b$. *Moreover, for* $0 < \eta < b$ *such that* $-x_\rho - i\eta$ *is not a pole of* $G'/G$,

$$\pi r_\eta(x_\rho) = \eta\Big(A_\eta - \frac{2m}{b^2 - \eta^2}\Big),$$

*where*

$$A_\eta = \frac{1}{\eta} \sum_{\lambda_{\rho'} \neq \lambda_\rho, \bar\lambda_\rho} \mathrm{Im}\,\big(-x_\rho - i\eta - \lambda_{\rho'}\big)^{-1},$$

*the sum running over the zeros* $\rho'$ *with* $\mathrm{Im}\,\rho' > 0$, *counted with multiplicity, whose* $\lambda_{\rho'}$ *is neither* $\lambda_\rho$ *nor* $\bar\lambda_\rho$. *Hence* $r_\eta(x_\rho) < 0$ *at every* $\eta \in (0,b)$ *with* $A_\eta < 2m/b^2$. *If all the other zeros lie on the line,* $A_\eta \to A_0 = \sum_{\gamma'} (\gamma'^2 - x_\rho)^{-2}$ *as* $\eta \downarrow 0$.

*Proof.* At $x = x_\rho$ we have $-x_\rho - i\eta - \lambda_\rho = -i(\eta + b)$ and $-x_\rho - i\eta - \bar\lambda_\rho = -i(\eta - b)$. Since $\mathrm{Im}(-iy)^{-1} = 1/y$, each of the $m$ copies of the pair $\lambda_\rho, \bar\lambda_\rho$ contributes $1/(\eta + b) + 1/(\eta - b) = -2\eta/(b^2 - \eta^2)$, which gives the display. This tends to $-\infty$ as $\eta \uparrow b$, while all other terms stay bounded, because the other poles of $G'/G$ lie at a positive distance from $-x_\rho - ib$. If $A_\eta < 2m/b^2$ then $A_\eta < 2m/(b^2 - \eta^2)$ for every $\eta \in (0,b)$. For a zero $\rho'$ on the line, $\lambda_{\rho'} = -\gamma'^2$ and $\mathrm{Im}(-x_\rho - i\eta + \gamma'^2)^{-1} = \eta/((\gamma'^2 - x_\rho)^2 + \eta^2)$, which gives the limit $A_0$. □

For $x \neq x_\rho$ the same computation shows that the $m$ copies of the pair contribute $m\,[P_{b+\eta}(x-x_\rho)-P_{b-\eta}(x-x_\rho)]$ to $r_\eta(x)$ for $0<\eta<b$, with $P_h$ the Poisson kernel of width $h$: a positive kernel of width $b+\eta$ and a negative kernel of width $b-\eta$, which becomes an atom of mass $-m$ at $x_\rho$ as $\eta \uparrow b$. A zero with $b<\sqrt{x_\rho}$ is therefore seen by a search over the window $\{0<\eta<\sqrt{x},\ x\le X\}$ as soon as $x_\rho \le X$. The condition $b<\sqrt{x_\rho}$ is $\delta<\gamma/\sqrt{4\gamma^2+1}$, that is $\beta<1-1/(16\gamma^2)$ up to terms of higher order, and it follows from the zero free region $\beta<1-c/\log\gamma$ for $\gamma$ large [498, Theorem 3.8]; it does not follow from $b<\gamma$. A search that finds $r_\eta \ge 0$ on the whole window therefore certifies that no such zero has $x_\rho \le X$, while a search on a grid is evidence only. Figure 14.2 tests the detection on a model in which $G'/G$ is built from the first 60 zeros: with all zeros on the line, $r_\eta$ is positive on the grid of the figure; after moving $\gamma_5$ to $\beta=0.6$ (so $b=6.59$) a negative region appears at $x\approx 1084.7$ for $\eta$ up to 6.5, and $r_\eta(x_\rho)\to-\infty$ as $\eta\uparrow b$, as the lemma predicts.

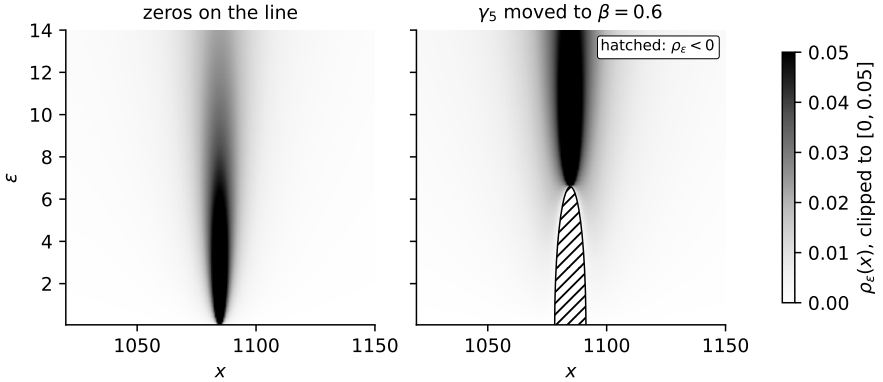



Figure 14.2: The smoothed Thorin density $r_\eta(x)$ in the $(x,\eta)$ plane near $\gamma_5^2 = 1084.7$. Left: zeros on the line. Right: $\gamma_5$ moved off the line; the off line pair is an atom of negative mass at $(x_\rho, b)$.

## 14.7 Is the clock a weak limit of GGC laws?

The class of GGC laws is closed under weak limits [56, Theorem 3.1.5] (Section 4.1.1), and with Proposition 1.2.1 this gives a sharp answer: $T$ is a weak limit of GGC laws if and only if RH holds (Corollary 4.1.2).

The de Bruijn Newman flow of Section 8.4 is the second family of candidates, and Theorem 7.2.16 places RH at the end of a path of GGC laws. Let $\Phi_{\mathrm{dBN}}(u)=\frac{1}{2}\Phi(2u)$ be the kernel in the rescaled variable of Rodgers and Tao [463], as in Theorem 7.2.16, let $H_t(z)=\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be the de Bruijn Newman deformation of $H_0(z)=\frac{1}{8}\xi(\frac{1}{2}+\frac{iz}{2})$, with $\Lambda$ the de Bruijn Newman constant in this normalisation, and let $V_t$ be the variable with density proportional to $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$,

the Gaussian tilt of the Pólya variable $V_0 = X/2$, so that $\mathbb{E}e^{sV_t} = H_t(is)/H_t(0)$ for real $s$. Whenever a law $T_t$ makes $(T_t, V_t)$ a Wald couple (Definition 14.4.1), $\mathbb{E}e^{-\lambda T_t} = H_t(0)/H_t(i\sqrt{2\lambda}) = R_t(\lambda/2)$, where $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ is the transform of the flow clock of Section 8.4 and $R_0 = 1/G$. Thus $T_t$ is half the flow clock, and $T_0 = T/2$; at $t = 0$ the couple $(T/2, X/2)$ is the one of Remark 14.5.1 rescaled by $\sqrt{2}$.

By Theorem 7.2.16, for every $t \geq \Lambda$ the transform $R_t$ is that of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the positive zeros of the even function $H_t$ (each pair $\pm z$ gives one atom). Hence for $t \geq \Lambda$ the law $T_t$ exists and is a GGC with Thorin measure $\sum_k \delta_{z_k(t)^2/2}$, and since $\Lambda \leq 0.2$ [419, 456] this holds unconditionally for $t \geq 0.2$. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t(\lambda/2) \to 1/G(\lambda/2)$ pointwise, which is the transform of $T/2$, and $T_t \to T/2$ in law along any sequence $t_n \downarrow 0$ for which $T_{t_n}$ exists. With the bound $\Lambda \geq 0$ of Rodgers and Tao [463], Theorem 7.2.16 gives that RH holds if and only if $T_t$ is a GGC for every $t > 0$, if and only if $\Lambda = 0$.

Two cautions. Unconditionally $T_t$ is known to exist only for $t \geq 0.2$ and at $t = 0$ (Theorem 5.1.1), so the convergence $T_t \to T/2$ is a statement about the range of $t$ on which $T_t$ exists. Under RH this is every $t \geq 0$; without RH, whether $T_t$ exists as a probability law for $0 < t < 0.2$ is open (Problem 14.3 in Section 14.15). And the Thorin atoms $z_k(t)^2/2$ of $T_t$ move with the zeros of $H_t$ only on the range $t \geq \Lambda$ where $T_t$ is a GGC; Table 8.8 gives the zeros of $H_t(2\theta)$, in the variable $\theta = z/2$.

Definition 14.7.1 (Laplace distance). For laws $P$ and $Q$ on $[0, \infty)$ with Laplace transforms $L_P$ and $L_Q$, the Laplace distance is $d_L(P, Q) = \sup_{\lambda \geq 0} |L_P(\lambda) - L_Q(\lambda)|$.

Proposition 14.7.2. *Let $H = 3 \times 10^{12}$, let $N = N(H)$ be the number of zeros $\rho_k$ with $0 < \gamma_k \leq H$, and let $T^{[N]} = \sum_{k \leq N} e_k/\gamma_k^2$ with independent standard exponential variables $e_k$, the GGC with transform $1/G_N$ of Section 4.1.1. With $a_k$ as in Section 1.2, put $\varepsilon_H = \sum_{k>N} 1/|a_k|$ and, for $K \leq N$, $S_K = \sup_{\lambda \geq 0} \lambda/G_K(\lambda)$. Then*

$$d_L\big(T, T^{[N]}\big) \leq S_K\, \varepsilon_H, \qquad \varepsilon_H < 1.4798 \times 10^{-12}, \qquad S_1 = \gamma_1^2, \qquad S_{1700} < 16.85, \tag{14.7.1}$$

*so that $d_L(T, T^{[N]}) < 2.5 \times 10^{-11}$, while the first factor alone gives $\gamma_1^2 \varepsilon_H < 2.96 \times 10^{-10}$. The proof uses only the verification of RH up to height $H$ [419], and makes no assumption on the zeros above $H$.*

*Proof.* By (1.2.1), $1/G = \prod_k f_k$ with $f_k(\lambda) = (1 + \lambda/a_k)^{-1}$, and $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > 0$ for every zero, so for $\lambda \geq 0$ we have $|1 + \lambda/a_k| \geq \operatorname{Re}(1 + \lambda/a_k) \geq 1$, hence $|f_k| \leq 1$ and $|f_k - 1| = |\lambda/a_k|\,|f_k| \leq \lambda/|a_k|$. For complex numbers of modulus at most one, $|\prod_j f_j - 1| \leq \sum_j |f_j - 1|$. The zeros with $k \leq N$ lie on the line, so $\prod_{k \leq N} f_k = 1/G_N$ is the transform of $T^{[N]}$, and its factors lie in $(0, 1]$; leaving out all but the first $K$ of them can only increase the product, so $1/G_N \leq 1/G_K$.

Therefore

$$\Big|\frac{1}{G(\lambda)}-\frac{1}{G_N(\lambda)}\Big|=\frac{1}{G_N(\lambda)}\Big|\prod_{k>N}f_k(\lambda)-1\Big|\le\frac{\lambda}{G_K(\lambda)}\,\varepsilon_H\le S_K\,\varepsilon_H.$$

For $K=1$, $\lambda/G_1(\lambda)=\lambda/(1+\lambda/\gamma_1^2)$ increases to $\gamma_1^2$, a supremum that is not attained. For $K\ge 2$, $\frac{d}{d\lambda}\log(\lambda/G_K(\lambda))=-\phi_K(\lambda)/\lambda$ with $\phi_K(\lambda)=\sum_{k\le K}\lambda/(\lambda+\gamma_k^2)-1$, which increases strictly from $-1$ to $K-1$, so $\lambda/G_K$ increases up to the root $\lambda^*$ of $\phi_K$ and decreases after it. For $K=1700$ the root is $\lambda^*=47.362$, and evaluating $\phi_{1700}$ and $\lambda/G_{1700}$ in ball arithmetic with the first 1700 zeros, on both sides of $\lambda^*$, gives $S_{1700}<16.8497$ (code/maxwald_weak.py).

For $\varepsilon_H$, $|a_k|=|\rho_k-\frac12|^2\ge\gamma_k^2$, and partial summation gives $\varepsilon_H\le\sum_{\gamma_k>H}\gamma_k^{-2}=2\int_H^\infty(N(t)-N(H))\,t^{-3}\,dt$. Write $N(t)=N_0(t)+E(t)$ with $N_0(t)=\frac{t}{2\pi}\log\frac{t}{2\pi e}$; by [176, Corollary 1.2], $|E(t)|\le 0.1038\log t+0.2573\log\log t+9.3675$ for $t\ge e$. The main term integrates to exactly $(\log\frac{H}{2\pi}+1)/(2\pi H)=1.47970\times10^{-12}$, and the bound for $E$, used at $t$ and at $H$, adds less than $10^{-23}$. □

## 14.8 The sine squared lemma and the basepoint family in the $\lambda$ chart

The sine squared lemma (1.2.9) [421, Lemma 8.1] turns the Lévy measure of $\log\xi$ at a basepoint $\alpha$ into the Thorin measure of the reciprocal at the same basepoint. For $\alpha>1$ its input is the signed canonical measure of [424, Theorem 8.1]; for $\frac12<\alpha\le1$ there is no such Lévy form, and the family is defined by the Hadamard product, not by the lemma. The family is that of the shifted clocks $T_\alpha$ with transform $\varphi_\alpha(\lambda)=\xi(\alpha)/\xi(\alpha+\sqrt{\lambda})$, $T_{1/2}=T$. For every $\alpha>\frac12$ put $u_\alpha(\lambda)=\mathrm{Re}(\xi'/\xi)(\alpha+i\sqrt{\lambda})/(2\pi\sqrt{\lambda})$, the Thorin density of $T_\alpha$ when every zero has $\beta<\alpha$ (Theorem 1.2.2) and in every case the density of $U_\alpha$ of (1.2.6) in the chart $\lambda=\theta^2$, a sum over the zeros of Poisson kernels (Proposition 1.2.3), and for $\alpha>1$ it has the arithmetic form (3.1.1) with $\theta=\sqrt{\lambda}$ (drift, archimedean, pole and prime pieces), which Table 1.5 compares with the Pólya route. Since $\varphi_\alpha\to1/G$ pointwise as $\alpha\downarrow\frac12$ and $T_\alpha$ exists for every $\alpha>\frac12$, $T_\alpha\to T$ in law, with no hypothesis (Section 4.1.1).

In the chart $\theta$ a zero $\rho=\beta+i\gamma$ with $\gamma>0$ contributes, paired with $\bar\rho$, the sum $P_\delta(\theta-\gamma)+P_\delta(\theta+\gamma)$ of two Poisson kernels with $\delta=\alpha-\beta$ (Proposition 1.2.3; $P_{-a}=-P_a$ as there). In the chart $\lambda=\theta^2$ the map $\theta\mapsto\theta^2$ folds the kernel centred at $-\gamma$ onto the one centred at $\gamma$ and brings in the Jacobian $1/(2\sqrt{\lambda})$.

Proposition 14.8.1. *Let $\alpha>\frac12$, let $\rho=\beta+i\gamma$ be a zero with $\gamma>0$, and put $\delta=\alpha-\beta\ne0$ and $c=\alpha-\rho=\delta-i\gamma$. The term of $\rho$ in $u_\alpha=\sum_{\gamma>0}u_\rho$, where the zeros $\rho$ and $\bar\rho$ are paired, is*

$$u_\rho(\lambda)=\frac{P_\delta(\sqrt{\lambda}-\gamma)+P_\delta(\sqrt{\lambda}+\gamma)}{2\sqrt{\lambda}},\qquad\lambda>0,\tag{14.8.1}$$

*and it satisfies the following.*

*(a)* $u_\rho(\lambda) = \kappa_\rho(\lambda)\, C_\rho(\lambda) = \delta\,(\lambda + |c|^2)/(\pi\sqrt{\lambda}\,|c^2+\lambda|^2)$, *where*

$$\kappa_\rho(\lambda) = \frac{\lambda + \gamma^2 + \delta^2}{2\gamma\sqrt{\lambda}},$$
$$C_\rho(\lambda) = \frac{1}{\pi}\,\frac{2\delta\gamma}{(\lambda - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2}.$$

*For $\delta > 0$, $C_\rho$ is the Cauchy density centred at $\gamma^2 - \delta^2$ with half width $2\delta\gamma$.*

*(b)* $\int_0^\infty u_\rho(\lambda)\, d\lambda = \operatorname{sign}\delta$, *so for $\delta > 0$ the term has unit mass.*

*(c)* $\kappa_\rho(\lambda) \ge \sqrt{1 + \delta^2/\gamma^2} > 1$ *for every* $\lambda > 0$, $\kappa_\rho(\gamma^2 - \delta^2) = (1 - \delta^2/\gamma^2)^{-1/2}$ *when* $|\delta| < \gamma$, *and for* $K \ge 1$ *and* $|\delta| \le \gamma/(2K)$,

$$|\kappa_\rho(\lambda) - 1| \le \Big(\frac{K^2}{2} + 1\Big)\frac{\delta^2}{\gamma^2} \qquad \text{for } |\lambda - \gamma^2| \le K|\delta|\gamma.$$

*Hence near its peak the term is a Lorentzian of full width $4|\delta|\gamma$ at half height, with the sign of $\delta$, up to a relative error $O(\delta^2/\gamma^2)$; away from the peak it is not: $u_\rho(\lambda) \sim \delta/(\pi(\gamma^2+\delta^2)\sqrt{\lambda})$ as $\lambda \downarrow 0$ and $u_\rho(\lambda) \sim \delta/(\pi\lambda^{3/2})$ as $\lambda \to \infty$, while $C_\rho$ is bounded at 0 and $O(\lambda^{-2})$ at infinity.*

*Proof.* Write $\theta = \sqrt{\lambda}$ and $A = \lambda + \gamma^2 + \delta^2$. Since $(\theta \mp \gamma)^2 + \delta^2 = A \mp 2\gamma\theta$,

$$P_\delta(\theta - \gamma) + P_\delta(\theta + \gamma) = \frac{\delta}{\pi}\Big[\frac{1}{A - 2\gamma\theta} + \frac{1}{A + 2\gamma\theta}\Big] = \frac{\delta}{\pi}\,\frac{2A}{A^2 - 4\gamma^2\lambda},$$

and $A^2 - 4\gamma^2\lambda = (\lambda - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2$, since both sides equal $\lambda^2 + 2(\delta^2 - \gamma^2)\lambda + (\gamma^2 + \delta^2)^2$. For $c = \delta - i\gamma$ the right side is also $|c^2 + \lambda|^2$, because $c^2 + \lambda = (\lambda + \delta^2 - \gamma^2) - 2i\delta\gamma$, and $|c|^2 = \gamma^2 + \delta^2$. Dividing by $2\theta$ gives $u_\rho = \delta A/(\pi\sqrt{\lambda}\,|c^2+\lambda|^2)$, which is the second form in (a), and writing $\delta A/\sqrt{\lambda} = \frac{A}{2\gamma\sqrt{\lambda}} \cdot 2\delta\gamma$ gives $\kappa_\rho C_\rho$. For (b) substitute $\lambda = \theta^2$: $\int_0^\infty u_\rho\, d\lambda = \int_0^\infty [P_\delta(\theta-\gamma) + P_\delta(\theta+\gamma)]\, d\theta = \int_{-\gamma}^\infty P_\delta + \int_\gamma^\infty P_\delta$, which is 1 for $\delta > 0$ because $P_\delta$ is an even probability density, and $-1$ for $\delta < 0$ because $P_{-a} = -P_a$. For (c), $\lambda + (\gamma^2 + \delta^2) \ge 2\sqrt{\lambda}\sqrt{\gamma^2 + \delta^2}$ gives the lower bound, and $A = 2\gamma^2$ at $\lambda = \gamma^2 - \delta^2$ gives the value there. For the window put $\lambda = \gamma^2(1 + \varepsilon)$, so that $|\varepsilon| \le K|\delta|/\gamma \le \frac{1}{2}$ and

$$\kappa_\rho(\lambda) = \frac{1 + \varepsilon/2}{\sqrt{1+\varepsilon}} + \frac{\delta^2}{2\gamma^2}\,\frac{1}{\sqrt{1+\varepsilon}}.$$

The first term is 1 at $\varepsilon = 0$ and has derivative $\varepsilon/(4(1+\varepsilon)^{3/2})$, so it differs from 1 by at most $\varepsilon^2/2$ for $|\varepsilon| \le \frac{1}{2}$, and the second term is at most $\delta^2/\gamma^2$. Take $\delta > 0$, the case $\delta < 0$ being the same with a change of sign. The half height points of

$C_\rho$ are at $\gamma^2 - \delta^2 \pm 2\delta\gamma$, inside the window with $K = 3$ (for $\delta \le \gamma/6$), where $|(\log C_\rho)'| = 1/(2\delta\gamma)$; a relative perturbation $\eta$ of the density therefore moves them by $O(\eta\,\delta\gamma)$, which gives the width $4\delta\gamma\,(1 + O(\delta^2/\gamma^2))$. The behaviour at 0 and at infinity is read off (a). □

Remark 14.8.2. The leading correction to the width is the factor $1 + \eta$ with $\eta = \delta^2/(2\gamma^2)$: with $x = (\lambda - \gamma^2 + \delta^2)/(2\delta\gamma)$ the expansion above gives $\kappa_\rho = 1 + \eta(1 + x^2) + O(\delta^3/\gamma^3)$, so near the peak $u_\rho \approx C_\rho + \eta/(2\pi\delta\gamma)$, with a relative error $O(\delta^3/\gamma^3)$, is a Lorentzian on a flat background of relative height $\eta$, whose half height points are at $x = \pm(1 + \eta)$ to first order.

Numerically the three expressions for $u_\alpha$ agree, up to their truncations: the line form $\mathrm{Re}(\xi'/\xi)(\alpha + i\sqrt{\lambda})/(2\pi\sqrt{\lambda})$ computed through the Pólya density, the zero form $\sum_\rho u_\rho$ of Proposition 14.8.1, and the prime form (3.1.1). At $\alpha = 2$ the three forms agree to 4 to 5 significant digits at $\lambda = 100, 250, 500, 1000$; at $\alpha = 1.5$ the prime form with $n \le N = 2 \times 10^5$ agrees to 1 to 2 digits, the gap being the truncated prime tail $\sum_{n>N} \Lambda(n) n^{-\alpha} \approx N^{1-\alpha}/(\alpha - 1)$ as for Table 1.5. The minimum of $u_\alpha$ on $[1, 2500]$ is positive for $\alpha = 0.55, 0.6, 0.75, 1, 1.5$ and lies near $\lambda = 59$ for each of them, which on a window where every zero is known to lie on the line checks the computation and does not test RH.

## 14.9 The maximum representation does not transfer

The distribution function of $T$ can be computed by residues, as $\vartheta_4$ is for $M_4$ (Section 8.5.4). Along a vertical line $\mathrm{Re}\,\lambda = -L$ that contains no pole of $1/G$, the point $\frac{1}{2} + \sqrt{\lambda}$ runs off to the right as $|\,\mathrm{Im}\,\lambda| \to \infty$, where $|\xi|$ grows, so $1/(\lambda G(\lambda))$ is integrable on the line and the Bromwich contour for $F_T(t) = \Pr(T \le t)$, whose transform is $1/(\lambda G(\lambda))$, can be shifted from $\mathrm{Re}\,\lambda > 0$ to $\mathrm{Re}\,\lambda = -L$ past the finitely many poles in between, because $\mathrm{Re}(-a_k) \le \frac{1}{4} - \gamma_k^2$. Write $R_k$ for the residue of $1/G$ at $-a_k$, which for a simple zero $\rho_k = \frac{1}{2} + i\gamma_k$ is $R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ (Section 8.1.1). When the poles with $\mathrm{Re}\,a_k < L$ are simple,

$$\Pr(T > t) = \sum_{\mathrm{Re}\,a_k < L} \frac{R_k}{a_k}\, e^{-a_k t} + O(e^{-Lt}), \qquad t \to \infty. \tag{14.9.1}$$

Only this finite shift is used below: the full residue series needs contours between the poles on which $1/|\xi|$ is controlled, and under RH with simple zeros it is the spectral expansion of Section 8.1.1.

The coefficients $R_k/a_k$ alternate in sign, because $\Xi'$ changes sign at consecutive simple zeros, and they grow like $e^{\pi\gamma_k/4}$ up to a power of $\gamma_k$ and the factor $1/|\zeta'(\rho_k)|$ (Section 8.1.1). The factor $e^{-\gamma_k^2 t}$ still wins for every $t > 0$.

A law $F$ on $(0, \infty)$ that is the distribution function of a maximum of independent exponential variables $E_i/r_i$ with deterministic rates $r_i$, where $\sum_i e^{-r_i t} < \infty$

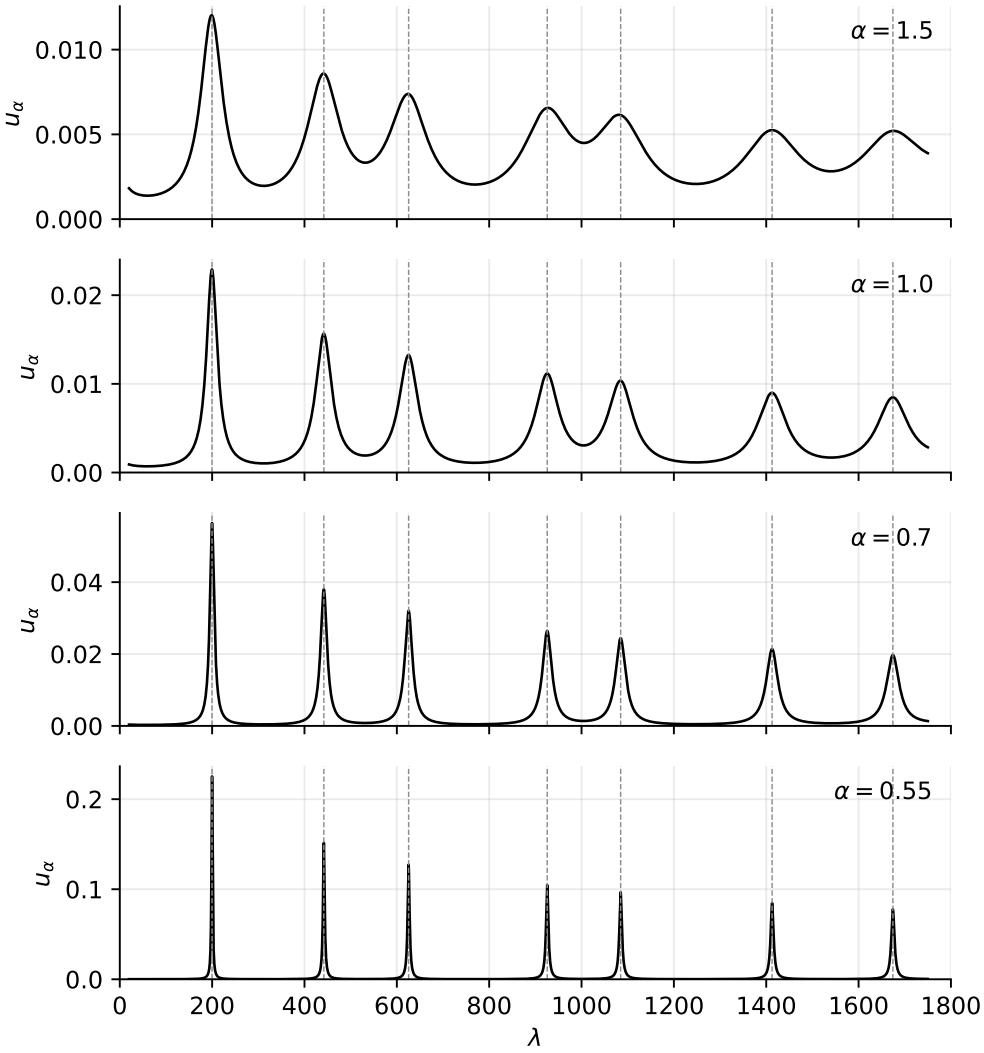



Figure 14.3: The density $u_\alpha$ of (1.2.6), computed from $\operatorname{Re}\,\xi'/\xi$ on the line $\operatorname{Re} s = \alpha$ with no zero as input, for $\alpha = 1.5, 1, 0.7, 0.55$; for $\alpha = 0.7$ and 0.55 it is the Thorin density of $T_\alpha$ only if no zero lies to the right of the line, as under RH. Dashed lines mark $\gamma_k^2$. The term of a zero on the line has unit mass and is close to a Lorentzian of width $4(\alpha - \frac{1}{2})\gamma_k$ near its peak (Proposition 14.8.1); as $\alpha \downarrow \frac{1}{2}$ these terms converge to unit atoms at $\gamma_k^2$.

for $t > 0$, has $-\log F(t) = \sum_i \sum_{m\ge1} e^{-mr_i t}/m$, a sum of exponentials with positive coefficients. The same holds, with $-\log F(t) = \int e^{-xt}\,\nu(dx)$, for the maximum of a Poisson cloud of exponential clocks with a rate measure $\nu$, that is, of the clocks $E_i/x_i$ attached to the points $x_i$ of a Poisson process of intensity $\nu$, since the number of clocks above $t$ is Poisson with mean $\int e^{-xt}\nu(dx)$.

PROPOSITION 14.9.1. *$-\log F_T$ is not completely monotone. Hence $T$ is not distributed as a maximum of independent exponential variables, for any choice of rates, nor as the maximum of a Poisson cloud of exponential clocks.*

*Proof.* Put $q_k = R_k/a_k$. For $k \le 3$ the zero $\rho_k$ lies on the line, so the $a_k = -(\rho_k - \frac12)^2$ of Section 1.2 equals $\gamma_k^2$. The zeros with $\operatorname{Re} a < 3a_1 = 599.37$ are $\rho_1$ and $\rho_2$ only: $a_1 = 199.79$, $a_2 = 441.93$ and $a_3 = 625.54$, the zeros below height $H = 3\times10^{12}$ lie on the line [419], and a zero above $H$ has $\operatorname{Re} a \ge \gamma^2 - \frac14 > H^2 - \frac14$ wherever it lies. The zeros below height 200 are simple (Section 3.1.1). Equation (14.9.1) with $L = 3a_1$ gives $\Pr(T > t) = q_1 e^{-a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t})$, and since $-\log(1-y) = y + \frac12 y^2 + O(y^3)$ and $2a_1 < a_2 < 3a_1 < a_1 + a_2$,

$$-\log F_T(t) = q_1 e^{-a_1 t} + \tfrac12 q_1^2 e^{-2a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t}).$$

Suppose that $-\log F_T(t) = \int_{[0,\infty)} e^{-xt}\nu(dx)$ for $t > 0$ with $\nu \ge 0$ (Bernstein's theorem). First $\nu([0,a_1)) = 0$: for $x_0 < a_1$, $\nu([0,x_0])e^{-x_0 t} \le -\log F_T(t) = O(e^{-a_1 t})$. Then $e^{a_1 t}(-\log F_T(t)) = \nu(\{a_1\}) + \int_{(a_1,\infty)} e^{-(x-a_1)t}\nu(dx)$, whose last term tends to 0 by dominated convergence, so $\nu(\{a_1\}) = q_1$, and $f_1(t) = \int_{(a_1,\infty)} e^{-xt}\nu(dx) = \frac12 q_1^2 e^{-2a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t})$. The same two steps give $\nu((a_1, 2a_1)) = 0$ and $\nu(\{2a_1\}) = \frac12 q_1^2$, and then $f_2(t) = \int_{(2a_1,\infty)} e^{-xt}\nu(dx) \ge 0$ satisfies $e^{a_2 t} f_2(t) \to q_2$ because $a_2 < 3a_1$. So $q_2 \ge 0$. But $R_2 = -2\gamma_2 \Xi(0)/\Xi'(\gamma_2)$ and $\Xi'(\gamma_2) > 0$, since $\Xi$ is positive on $[0,\gamma_1)$, negative on $(\gamma_1,\gamma_2)$ and changes sign at the simple zero $\gamma_2$; so $q_2 = R_2/a_2 < 0$, a contradiction. □

For $M_4$ the coefficient at $a_2 = 4\pi$ also has the wrong sign: $\Pr(M_4 > t) = \sum_{n\ge1} 2(-1)^{n+1} e^{-\pi n^2 t}$ has $q_1 = 2$ and $q_2 = -2$. But $a_2 = 4\pi = 4a_1$ coincides with a term of the expansion in $q_1$, and the coefficient of $e^{-4\pi t}$ in $-\log\vartheta_4$ becomes $q_2 + \frac14 q_1^4 = -2 + 4 = 2 > 0$. Commensurability of the exponents $\pi n^2$ rescues complete monotonicity (the Jacobi triple product writes $-\log\vartheta_4$ as a sum of exponentials with positive coefficients), while $\gamma_2^2/\gamma_1^2 = 2.2119$ is not an integer, so nothing rescues it for $T$.

## 14.10 Gnedenko: the first zero as a norming constant

Gnedenko's theorem [150] classifies the limits of normalised maxima of independent identically distributed variables, and a law with a tail $\Pr(X > t) \sim Ce^{-at}$ lies in the Gumbel domain with norming constant $1/a$ and centring $\log(Cn)/a$.

The proposition reads the norming constant of $T$ as the first atom $\gamma_1^2$ of its Thorin measure, with the constant given by the first residue.

PROPOSITION 14.10.1. *(i)* $\Pr(M_4 > t) \sim 2e^{-\pi t}$, *so for independent copies* $M_4^{(j)}$, $j \le n$,

$$\Pr\left(\pi \max_{j\le n} M_4^{(j)} - \log(2n) \le x\right) \to \exp(-e^{-x}).$$

*(ii)* $\Pr(T > t) \sim (R_1/\gamma_1^2)\, e^{-\gamma_1^2 t}$, *so for independent copies* $T^{(j)}$, $j \le n$, *of* $T$,

$$\Pr\left(\gamma_1^2 \max_{j\le n} T^{(j)} - \log \frac{R_1 n}{\gamma_1^2} \le x\right) \to \exp(-e^{-x}),$$

$$\max_{j\le n} T^{(j)} \Big/ \log n \to \frac{1}{\gamma_1^2} \quad \textit{in probability.}$$

*Proof.* (i) is $1 - \vartheta_4(t) = 2e^{-\pi t} + O(e^{-4\pi t})$. For (ii), $T$ exists by Theorem 5.1.1, and (14.9.1) with $a_1 < L < a_2$ gives $\Pr(T > t) = (R_1/a_1)e^{-a_1 t} + O(e^{-Lt})$, because the simple pole at $-\gamma_1^2$ is the rightmost singularity of $1/(\lambda G(\lambda))$ after 0, every other pole having $\operatorname{Re}\lambda \le -\gamma_2^2$. In both cases, if $\Pr(X > t) \sim Ce^{-at}$ and $y_n = (\log(Cn) + x)/a$, then $n \Pr(X > y_n) \to e^{-x}$ and $\Pr(\max_{j\le n} X^{(j)} \le y_n) = (1 - \Pr(X > y_n))^n \to \exp(-e^{-x})$. The last statement follows because $a \max_{j\le n} X^{(j)} - \log n$ is then tight. □

This gives a way to find $\gamma_1$ that uses no zeros at all: invert the Laplace transform $(1 - 1/G(\lambda))/\lambda$ numerically and read off the exponential rate of the tail. It is the experiment of Section 8.1.1, where the hazard of $T$ tends to $\gamma_1^2$ by Talbot inversion with no zero as input. The local logarithmic slopes of $\Pr(T > t)$ on $[0.04, 0.05]$, $[0.05, 0.06]$, $[0.06, 0.07]$ and $[0.07, 0.08]$ are 199.496, 199.764, 199.788 and 199.790, converging to $\gamma_1^2 = 199.7905$ at the rate $e^{-(\gamma_2^2 - \gamma_1^2)t}$ predicted by the second term of (14.9.1), and the intercept $e^{\gamma_1^2 t} \Pr(T > t)$ at $t = 0.08$ gives $R_1/\gamma_1^2 = 50.87103$ against the residue value 50.87104.

## 14.11 FINDING THE ZEROS FROM THE THORIN DENSITY IN THE $z$ CHART

Theorem 14.6.1 gives a procedure in the chart $z = \lambda$: compute the smoothed Thorin density $r_\eta$ of that theorem on a grid from $\xi'/\xi$, with no zero used, locate its local maxima, and take square roots. Under RH $r_\eta = P_\eta * U_{\mathrm{RH}}$ is the Poisson smoothing of the Thorin measure $U_{\mathrm{RH}} = \sum_k \delta_{\gamma_k^2}$ at distance $\eta$ below the cut (Theorem 14.6.1 and Section 1.3), so each atom appears as a spike of unit mass and height near $1/(\pi\eta)$ at $x \approx \gamma_k^2$. The book carries out the same recovery in the chart $\theta = \sqrt{z}$ (Sections 3.1.1 and 8.6). The table below is a cross check in the other chart, at $\eta = 3$.

The peak of $r_\eta$ near $\gamma_k^2$ is displaced from it by the pull of the neighbouring spikes, to leading order by $\eta^4 \sum_{j\ne k} (\gamma_j^2 - \gamma_k^2)^{-3}$, which is of modulus below $2 \times 10^{-5}$

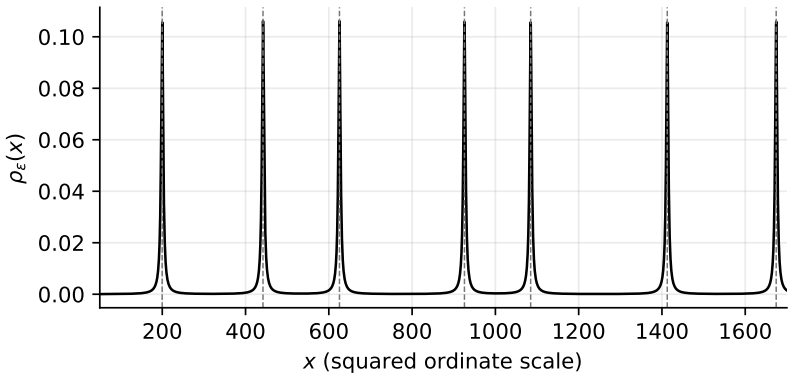



Figure 14.4: The smoothed Thorin density $r_\eta$ of $T$, $\eta = 3$, computed from $\xi'/\xi$ with no zero as input. Dashed lines mark $\gamma_k^2$. Each spike has height near $1/(\pi\eta)$ and unit mass.

| $k$ | peak of $r_\eta$ | $\sqrt{\text{peak}}$ | $\gamma_k$ |
|---|---|---|---|
| 1 | 199.790462 | 14.1347254 | 14.1347251 |
| 2 | 441.926159 | 21.0220398 | 21.0220396 |
| 3 | 625.542987 | 25.0108574 | 25.0108576 |
| 4 | 925.673105 | 30.4248764 | 30.4248761 |
| 5 | 1084.718263 | 32.9350613 | 32.9350616 |
| 6 | 1412.720791 | 37.5861782 | 37.5861782 |
| 7 | 1674.341571 | 40.9187191 | 40.9187190 |

Table 14.2: Zeros recovered as atoms of the smoothed Thorin measure, $\eta = 3$. The integral of $r_\eta$ over $[50, 1700]$ is 6.95, against 7 zeros.

in the table. On the window $[50, 1700]$ with $\eta = 3$ the minimum of $r_\eta$ is $5.6\times10^{-5} > 0$, attained at the endpoint $x = 50$. This is a consistency check of the computation with Theorem 14.6.1(d) and Lemma 14.6.2, not a test of RH: the window contains only the squares of the zeros below height $\sqrt{1700} = 41.2$, which are known to lie on the line.

## 14.12 Finite maxima and a limit argument

The maximum interpretation of Section 14.2 suggests a direct strategy: truncate the maximum at $N$ terms, show that the finite object has no zeros off the critical line, and pass to the limit. For the sum representation instead of the maximum the book has the truncations of the Biane, Pitman and Yor lattice (Subsection 7.2.1).

### 14.12.1 Finite maxima

For $N \geq 1$ let

$$M_4^{(N)} = \max\Big\{ \frac{E_n}{\pi n} \ (n \leq N), \quad \frac{E'_m}{\pi(2m-1)} \ (m \leq N) \Big\},$$

$$\mathcal{P}_N(q) = \prod_{n \leq N} (1 - q^n) \prod_{m \leq N} (1 - q^{2m-1}) = \sum_{m \geq 0} p_m^{(N)} q^m,$$

where $(E_n)_{n \geq 1}$ and $(E'_m)_{m \geq 1}$ are independent standard exponential variables, and $M_4 = \lim_N M_4^{(N)}$ is the maximum of Subsection 8.5.4, written with the rates $\pi n$ and $\pi(2m-1)$. Then $\Pr(M_4^{(N)} \leq t) = \mathcal{P}_N(e^{-\pi t})$ exactly, a polynomial in $e^{-\pi t}$ with integer coefficients. Expanding the product and integrating term by term, for $\operatorname{Re} s > 0$,

$$\mathbb{E}\big(M_4^{(N)}\big)^{s/2} = -\Gamma(1 + \tfrac{s}{2})\, \pi^{-s/2} D_N(s), \qquad D_N(s) = \sum_{m \geq 1} p_m^{(N)}\, m^{-s/2},$$

a finite Dirichlet polynomial $D_N$ with integer coefficients. Since the factors left out of $\mathcal{P}_N$ are $1 - q^a$ with $a > N$, $p_m^{(N)}$ equals the coefficient of $\vartheta_4$ for $m \leq N$: it is $2(-1)^n$ at $m = n^2$ and 0 otherwise. So the first block of $D_N$ is exactly minus the partial sum $2\sum_{n^2 \leq N} (-1)^{n+1} n^{-s}$ of $2(1 - 2^{1-s})\zeta(s)$, followed by a tail of integer coefficients at the frequencies $\frac{1}{2}\log m$ with $N < m \leq N(3N+1)/2$, the degree of $\mathcal{P}_N$.

By Subsection 8.5.4, $(1 - 2^{1-s})\zeta(s) = \mathbb{E}[(2K^2)^{s/2}]/(2\Gamma(1 + \frac{s}{2}))$ for every complex $s$, with $\pi M_4 = 2K^2$ and $K$ the Kolmogorov variable, and since $2\xi(s)/(s-1) = 2\pi^{-s/2}\Gamma(1 + \frac{s}{2})\zeta(s)$ this reads

$$\mathbb{E} M_4^{s/2} = \frac{2(1 - 2^{1-s})\, \xi(s)}{s - 1} \qquad \text{for every complex } s.$$

Define the normalised finite stage and its symmetrisation

$$\xi_N(s) = \frac{(s-1)\, \mathbb{E}(M_4^{(N)})^{s/2}}{2(1 - 2^{1-s})}, \qquad \xi_N^{\mathrm{sym}}(s) = \tfrac{1}{2}\big(\xi_N(s) + \xi_N(1-s)\big),$$

so that the same expression with $M_4$ in place of $M_4^{(N)}$ is $\xi$. In the open strip $0 < \operatorname{Re} s < 1$ the factor $1 - 2^{1-s}$ does not vanish and $\mathbb{E}(M_4^{(N)})^{s/2}$ is holomorphic (it is holomorphic for $\operatorname{Re} s > -4N$, since $\Pr(M_4^{(N)} \leq t) \sim C t^{2N}$ as $t \to 0$), so $\xi_N$ and $\xi_N^{\mathrm{sym}}$ are holomorphic there. The function $\xi_N^{\mathrm{sym}}$ is invariant under $s \mapsto 1 - s$ and, because $\xi_N$ is real on the real axis, $\xi_N^{\mathrm{sym}}(\frac{1}{2} + it) = \operatorname{Re}\, \xi_N(\frac{1}{2} + it)$ is real for real $t$.

THEOREM 14.12.1. *(i)* $M_4^{(N)} \uparrow M_4$ *almost surely, and* $\Pr(M_4^{(N)} \neq M_4) \leq C\sqrt{N}\, e^{-\pi\sqrt{N}/2}$.

*(ii) On every compact* $K$ *in the open strip* $0 < \operatorname{Re} s < 1$,

$$\sup_K |\xi_N - \xi| \leq C_K\, N^{1/4} e^{-\pi\sqrt{N}/4}, \qquad \sup_K |\xi_N^{\mathrm{sym}} - \xi| \leq C_K\, N^{1/4} e^{-\pi\sqrt{N}/4}.$$

*Proof.* (i) The variables of the stages are the same for all $N$, so adding variables can only raise the maximum, and $M_4^{(N)} \neq M_4$ requires a left out variable to exceed $M_4^{(N)}$. For any $\tau > 0$,

$$\Pr(M_4^{(N)} \neq M_4) \leq \Pr(M_4^{(N)} \leq \tau) + \sum_{n>N} e^{-\pi n\tau} + \sum_{m>N} e^{-\pi(2m-1)\tau}.$$

The last two sums are at most $2e^{-\pi N\tau}/(1-e^{-\pi\tau})$. For the first, $\Pr(M_4^{(N)} \leq \tau) = \vartheta_4(\tau)/\Pi_N(\tau)$, where $\Pi_N(\tau) = \prod_{n>N}(1-e^{-\pi n\tau}) \prod_{m>N}(1-e^{-\pi(2m-1)\tau})$ is the product of the left out factors. Since $1 - x \geq e^{-x/(1-x)}$, $\Pi_N(\tau) \geq \exp\bigl(-2e^{-\pi N\tau}/((1-e^{-\pi\tau})(1-e^{-\pi N\tau}))\bigr)$, while Jacobi's transformation $\vartheta_4(\tau) = \tau^{-1/2} \sum_n e^{-\pi(n+1/2)^2/\tau}$ gives $\vartheta_4(\tau) \leq 3\tau^{-1/2} e^{-\pi/(4\tau)}$ for $\tau \leq 1$. The choice $\tau = 1/(2\sqrt{N})$ balances $\pi/(4\tau)$ against $\pi N\tau$, both equal to $\pi\sqrt{N}/2$, and gives the bound.

(ii) For $\sigma = \operatorname{Re} s > 0$, $M_4^{(N)} \leq M_4$ gives $|\mathbb{E}M_4^{s/2} - \mathbb{E}(M_4^{(N)})^{s/2}| \leq 2\mathbb{E}[M_4^{\sigma/2}\mathbf{1}\{M_4 \neq M_4^{(N)}\}] \leq 2(\mathbb{E}M_4^{\sigma})^{1/2} \Pr(M_4^{(N)} \neq M_4)^{1/2}$ by Cauchy and Schwarz. The prefactor $(s-1)/(2(1-2^{1-s}))$ is bounded on $K$ because $1 - 2^{1-s}$ vanishes only on $\operatorname{Re} s = 1$. The symmetrised bound follows since $K$ and $1 - K$ both lie in the strip and $\xi(1-s) = \xi(s)$. □

The exponent in (ii) is not sharp. A heuristic count of the event $M_4^{(N)} \neq M_4$, in which the $j$th left out variable wins with probability $\mathbb{E}e^{-\pi j M_4} = \pi\sqrt{j}/\sinh(\pi\sqrt{j})$, of order $\sqrt{j}\, e^{-\pi\sqrt{j}}$, gives a rate of order $Ne^{-\pi\sqrt{N}}$.

COROLLARY 14.12.2 (Hurwitz). *(a) Every zero of* $\xi$ *in the open strip is a limit of zeros of* $\xi_N$*, and of zeros of* $\xi_N^{\mathrm{sym}}$*, with multiplicity; every limit of zeros of* $\xi_N$ *inside the open strip is a zero of* $\xi$*.*

*(b) If* $\rho = \frac{1}{2} + i\gamma$ *is a simple zero of* $\xi$*, then for all large* $N$*,* $\xi_N^{\mathrm{sym}}$ *has exactly one zero in a small disc about* $\rho$*, and that zero lies exactly on the critical line.*

*(c) Suppose that there are* $N_j \to \infty$ *and heights* $h_j \to \infty$ *such that* $\xi_{N_j}^{\mathrm{sym}}$ *has no zeros in* $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < h_j\}$ *for every* $j$*. Then the Riemann Hypothesis holds. In particular this is so if there are* $c > 0$ *and* $N_0$ *such that for every* $N \geq N_0$*,* $\xi_N^{\mathrm{sym}}$ *has no zeros in* $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < c\sqrt{N}\}$*.*

*Proof.* (a) is Hurwitz's theorem with Theorem 14.12.1(ii). For (b), Hurwitz gives exactly one zero $z_N$ of $\xi_N^{\mathrm{sym}}$ in a disc $D$ centred at $\rho$ for large $N$. The symmetries $s \mapsto 1 - s$ and $s \mapsto \bar{s}$ of $\xi_N^{\mathrm{sym}}$ make its zero set invariant under $s \mapsto 1 - \bar{s}$, which

maps $D$ to itself; uniqueness forces $1-\bar{z}_N = z_N$, that is $\operatorname{Re} z_N = \frac{1}{2}$. For (c), fix $\Gamma$ and let $U = \{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < \Gamma\}$. For $j$ so large that $h_j > \Gamma$ the functions $\xi^{\mathrm{sym}}_{N_j}$ have no zeros in $U$, and they converge to $\xi \not\equiv 0$ uniformly on compact subsets of $U$, so by Hurwitz $\xi$ has no zeros in $U$. Letting $\Gamma \to \infty$ and using $\xi(s) = \xi(1-s)$ gives RH. □

Part (b) is pointwise in $\rho$ and gives no height up to which the stages reproduce all the zeros of $\xi$, since that would need an error bound uniform up to the height together with lower bounds for $|\xi|$ on circles around every zero below it, that is, bounds for $1/\zeta'(\rho)$, and none is proved here.

### 14.12.2 What the finite stages look like

The unsymmetrised stage $\xi_N$ is not zero free off the line at any of the $N$ from 4 to 128 that we tested, so the naive version of the strategy fails. Its zeros near $\rho_k$ approach $\rho_k$ as $N$ grows, and more slowly the larger $\gamma_k$ is (Table 14.3), and it carries spurious zeros elsewhere, such as the zero of $\xi_{16}$ at $1.055 + 17.984i$, just to the right of the strip.

| $N$ | zero near $\rho_1$ | zero near $\rho_2$ | zero near $\rho_5$ |
|---|---|---|---|
| 16 | $0.51577 + 14.09915i$ | $0.95703 + 20.91901i$ | $1.56809 + 34.68128i$ |
| 32 | $0.49948 + 14.13483i$ | $0.48159 + 21.02790i$ | $1.02797 + 32.22350i$ |
| 64 | $0.50000052 + 14.1347256i$ | $0.50003418 + 21.0220191i$ | $0.51254 + 32.92764i$ |

Table 14.3: Zeros of the unsymmetrised finite stage $\xi_N$. For $N = 16$ and $N = 32$ the zero listed near $\rho_5$ does not track $\rho_5$, since $\rho_5$ lies above the height $4\sqrt{N}$ discussed below.

| $N$ | zeros | $\lvert t_1 - \gamma_1\rvert$ | $\lvert t_2 - \gamma_2\rvert$ | $\lvert t_3 - \gamma_3\rvert$ | $\lvert t_5 - \gamma_5\rvert$ |
|---|---|---|---|---|---|
| 16 | 7 | $3.6 \times 10^{-2}$ | $9.9 \times 10^{-2}$ | $1.7 \times 10^{-1}$ | $1.3 \times 10^{-3}$ |
| 32 | 10 | $1.0 \times 10^{-4}$ | $5.9 \times 10^{-3}$ | $6.8 \times 10^{-2}$ | 1.1 |
| 64 | 8 | $4.1 \times 10^{-7}$ | $2.1 \times 10^{-5}$ | $3.8 \times 10^{-4}$ | $7.5 \times 10^{-3}$ |

Table 14.4: Real zeros $t_k$ of the symmetrised stage $\operatorname{Re} \xi_N(\frac{1}{2} + it)$: the column "zeros" is the number of sign changes on $[10, 44]$, where $\xi$ has 8 zeros ($10 \le \gamma \le 44$). At $N = 16$ the agreement at $\gamma_5$ is accidental.

From $N = 32$ to $N = 64$ the error at $\gamma_1$ falls by a factor of about 250, against $e^{\pi(8-\sqrt{32})} = 1574$ for the exponential factor of the heuristic rate (about 800 with its prefactor $N$) and $e^{\pi(8-\sqrt{32})/4} = 6.3$ for that of the proven one; this is a rate of about $e^{-0.75\pi\sqrt{N}}$ (about $e^{-0.60\pi\sqrt{N}}$ between $N = 8$ and $N = 16$). The supremum of $|\xi_N - \xi|$ over the grid $\{0.3, 0.5, 0.7\} + i\{0, 2, \dots, 10\}$ decays like $e^{-c\pi\sqrt{N}}$ with

$c \approx 0.93$ between $N = 32$ and $N = 64$ and $c \approx 0.80$ between $N = 8$ and $N = 16$, and that of $|\xi_N^{\mathrm{sym}} - \xi|$ with $c \approx 0.81$ and $c \approx 0.80$, all between the exponent $\frac{1}{4}$ of the proof and the exponent 1 of the heuristic. Between consecutive stages the exponents for $\xi_N^{\mathrm{sym}}$ and for the error at $\gamma_1$ oscillate, between 0.73 and 1.10 and between 0.49 and 1.37, so the exponent 1 of the heuristic is not a bound. Since $|\xi(\frac{1}{2} + i\gamma)|$ decays like $e^{-\pi\gamma/4}$, the heuristic predicts that zeros up to height $\gamma$ are captured once $\pi\sqrt{N}$ exceeds $\pi\gamma/4$ by a margin, that is $N \gtrsim \gamma^2/16$. The tables agree: the symmetrised stage $N = 64$ captures every zero up to height 33 to within $10^{-2}$ (and $\xi_{64}$ itself to within $1.5 \times 10^{-2}$), and the accuracy degrades beyond 37. This suggests that the stage of index $N$ tracks the zeros of $\xi$ up to a height of about $4\sqrt{N}$; this tracking height is not the zero free height of Corollary 14.12.2(c), which may grow with $N$ in any way.

### 14.12.3 The symmetric stages $F_N^{\mathrm{sym}}$ and how far they stay on the line

The normalisation $\xi_N$ divides by $1 - 2^{1-s}$, which has zeros on $\operatorname{Re} s = 1$ and, after symmetrisation, on $\operatorname{Re} s = 0$. For counting zeros it is better to keep the factor. Put

$$F_N(s) = \frac{s-1}{2}\,\mathbb{E}\big(M_4^{(N)}\big)^{s/2} = -\frac{s-1}{2}\,\Gamma\big(1+\tfrac{s}{2}\big)\pi^{-s/2}D_N(s),$$

$$F_N^{\mathrm{sym}}(s) = \tfrac{1}{2}\big(F_N(s) + F_N(1-s)\big),$$

so that $F_N = (1 - 2^{1-s})\xi_N$. These functions are holomorphic on $-4N < \operatorname{Re} s < 4N + 1$, but they are not entire: $F_N$ has a pole at $s = -4N$ and $F_N^{\mathrm{sym}}$ also at $s = 4N + 1$, far from every region used below. As $N \to \infty$, $F_N \to (1 - 2^{1-s})\xi(s)$ and $F_N^{\mathrm{sym}} \to \mathfrak{q}(s)\xi(s)$, with $\mathfrak{q}(s) = 1 - 2^{-s} - 2^{s-1}$. The zeros of $\mathfrak{q}$ are simple and are the solutions of $2^s = 1 \pm i$, all on $\operatorname{Re} s = \frac{1}{2}$, at $t = (2\pi k \pm \pi/4)/\log 2$.

Theorem 14.12.1 and Corollary 14.12.2 hold for $F_N^{\mathrm{sym}}$ with $\mathfrak{q}\xi$ in place of $\xi$, on compact subsets of the open strip, since $1 - 2^{1-s}$ is bounded there. Part (c) becomes: if there are $N_j \to \infty$ and $h_j \to \infty$ such that $F_{N_j}^{\mathrm{sym}}$ has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < h_j\}$, then RH holds, by the same argument, since a zero of $\xi$ in that region is a zero of $\mathfrak{q}\xi$. We call (c) for $F_N^{\mathrm{sym}}$ the $F_N^{\mathrm{sym}}$ form of Corollary 14.12.2(c): it is a different statement from (c) for $\xi_N^{\mathrm{sym}}$, since $F_N^{\mathrm{sym}}$ is not $\xi_N^{\mathrm{sym}}$ times a fixed factor.

Zeros of $F_N^{\mathrm{sym}}$ in a rectangle $[-1, 2] \times [t_1, t_2]$ can be counted by the argument principle, and zeros on the line are sign changes of the real function $F_N^{\mathrm{sym}}(\frac{1}{2} + it)$. When the two counts agree, every zero in the rectangle is on the line. Write $T^*(N)$ for the height below which the counts agree in every window of height 10 and above which off line zeros appear in the rectangle. Since the rectangle contains the strip, $F_N^{\mathrm{sym}}$ has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < T^*(N)\}$, which is the hypothesis of the $F_N^{\mathrm{sym}}$ form of Corollary 14.12.2(c) at the height $T^*(N)$. Table 14.5 reports the counts.

| $N$ | terms | height | on the line | first off line window | $T^*(N)$ |
|---|---|---|---|---|---|
| 8 | 72 | 400 | 70 of 192 | $(110, 120]$ | $\approx 110$ |
| 16 | 350 | 1000 | 146 of 510 | $(250, 260]$ | $\approx 250$ |
| 32 | 1478 | 800 | 312 of 360 | $(530, 540]$ | $\approx 530$ |
| 48 | 3372 | 1000 | 494 of 496 | $(870, 880]$ | $\approx 870$ |
| 64 | 6034 | 1500 | 791 of 797 | $(1060, 1070]$ | $\approx 1060$ |
| 96 | 13670 | 2500 | 1463 of 1465 | $(1590, 1600]$ | $\approx 1590$ |
| 128 | 24354 | 130 | 64 of 66 | $(60, 70]$ | $\approx 60$ |

Table 14.5: Zeros of the symmetric stage $F_N^{\mathrm{sym}}$ in $[-1, 2] \times [0, T]$ by the argument principle, against sign changes on the critical line: "terms" is the number of nonzero coefficients $p_m^{(N)}$, $m \geq 1$, "height" the height $T$ searched, and "on the line" the number of sign changes out of the number of zeros counted. The windows have height 10. Above $T^*(N)$ the off line zeros arrive in bulk for $N \leq 32$: for $N = 16$ the two windows of $(300, 320]$ have 10 of their 13 zeros off the line. The sum $D_N$, whose terms are much larger than the sum itself, is evaluated in double precision, with a bound on its rounding error on the critical line in every window, and in multiprecision in the windows below 100 for $N = 96$ and below 130 for $N = 128$; above 130 the double precision margin of $N = 128$ is lost, so that stage is reported only up to 130. Off line zeros come in pairs $\sigma + it$, $1 - \sigma + it$, so an off line pair adds two to the count of its window, and a difference of one between the two counts is an artefact, such as a zero at the window boundary.

The first off line pairs of $F_N^{\mathrm{sym}}$ are $1.5643 + 111.7906i$ for $N = 8$, $1.1950 + 254.3196i$ for $N = 16$, $1.2079 + 537.2287i$ for $N = 32$, $1.0434 + 870.3807i$ for $N = 48$, $1.1025 + 1063.9243i$ for $N = 64$, $1.1177 + 1593.1234i$ for $N = 96$ and $1.5311 + 60.7314i$ for $N = 128$, each with its mirror image under $s \mapsto 1 - \bar{s}$. The first off line zero inside the strip appears for $N = 8$ in $(120, 130]$, for $N = 16$ in $(330, 340]$ and for $N = 32$ in $(550, 560]$, and none was found there for $N = 48$, 64, 96 and 128 up to the heights 1000, 1500, 2500 and 130 searched, so the height up to which $F_N^{\mathrm{sym}}$ has no off line zeros in the strip is at least $T^*(N)$ and may be larger. The counts depend on the rectangle: for $N = 16$ the rectangle $[-8, 9] \times [240, 250]$ contains 7 zeros where $[-1, 2] \times [240, 250]$ contains 3, all on the line, so zeros of $F_N^{\mathrm{sym}}$ far from the strip exist below $T^*(N)$, and $T^*(N)$ refers to $[-1, 2]$.

Three things follow. First, the symmetric stages are not real rooted: off line zeros exist in $[-1, 2] \times [0, T]$ for every $N$ we tested, but only above a height $T^*(N)$, and the first ones found lie outside the critical strip. Second, up to $N = 96$ the height $T^*(N)$ grows roughly linearly, while the height up to which the stage approximates $\mathfrak{q}\xi$ well enough to capture its zeros is only about $4\sqrt{N}$ (Table 14.4 is for $\xi_N^{\mathrm{sym}}$; the same height is expected for $F_N^{\mathrm{sym}}$, whose limit $\mathfrak{q}\xi$ has the same zeros besides those of $\mathfrak{q}$). At $N = 128$ the growth breaks: $T^*(128)/128 = 0.47$, although the strip has no off line zero of that stage below 130, the height to which

it was counted. Below $4\sqrt{N}$ the zeros of $F_N^{\mathrm{sym}}$ are the zeros of $\mathfrak{q}\xi$ moved along the line; between $4\sqrt{N}$ and $T^*(N)$ the stage has its own zeros, fewer than $\mathfrak{q}\xi$ has, but still all on the line; above $T^*(N)$ it has zeros off the line. The linear growth is what a heuristic based on the approximate functional equation suggests for a symmetrised Dirichlet polynomial whose longest frequency is $\frac{1}{2}\log m_{\max}$ with $m_{\max} = N(3N+1)/2$: the two halves of $F_N^{\mathrm{sym}}$ can balance off the line only when $(t/2\pi)^{\sigma-1/2}$ is matched by a ratio of the Dirichlet polynomial at $\sigma$ and $1-\sigma$, which first becomes possible at $t$ of order $N$; the pair of $N = 128$, far from the strip, falls outside this heuristic. Third, the hypothesis of the $F_N^{\mathrm{sym}}$ form of Corollary 14.12.2(c) holds up to height $T^*(N)$ for the $N$ tested, but finitely many $N$ cannot verify an asymptotic hypothesis, and (c) needs no particular height: it suffices that $T^*(N_j) \to \infty$ along some sequence $N_j \to \infty$. The finite $N$ question becomes the growth of $T^*(N)$: the conjecture $T^*(N) \geq \kappa N$ for some $\kappa > 0$ and all $N$ would imply RH, as would any $T^*(N_j) \to \infty$, in particular the far weaker $T^*(N) \geq c\sqrt{N}$.

We do not have a mechanism for $T^*(N)$. The classical argument that puts zeros of a symmetrised Dirichlet polynomial on the line, dominance of the first term of $D_N$, that is $|D_N(s) - p_1| < |p_1|$ on $\mathrm{Re}\, s \geq \frac{1}{2}$ (with $|p_1| = 2$), fails here because $\sum_m |p_m^{(N)}| m^{-1/4}$ is large: about 50 for $N = 8$ and $2.4 \times 10^9$ for $N = 64$. The observed $T^*(N)$ is nevertheless what the balance heuristic predicts, so the right statement to prove may be about the ratio $|D_N(\sigma + it)/D_N(1 - \sigma + it)|$ rather than about $D_N$ alone.

### 14.12.4 Comparison with the symmetrised lattice truncations

Subsection 7.2.1 carries out the same strategy for the symmetrised lattice truncations $K_n$, with Proposition 7.2.6 and Theorem 7.2.7 as the analogues of Corollary 14.12.2(b) and (c); the off line zeros of $F_N^{\mathrm{sym}}$ above $T^*(N)$ do not bear on Conjecture S of that subsection, which concerns $K_n$.

## 14.13 Bondesson classes containing GGC

The classes BO, GGC and SD (Subsection 1.3.2) have finite order versions. This section establishes, for the clock $T$, the finite order versions up to the orders $1.16 \times 10^{25}$ on the Lévy side and $9 \times 10^{12}$ on the transform side from finite verification, and shows that every infinite order version is equivalent to RH.

Definition 14.13.1. A function $f$ on an interval is $k$ monotone if $(-1)^j f^{(j)} \geq 0$ for $0 \leq j \leq k$, and completely monotone if it is $k$ monotone for every $k$. For $k \geq 1$ let $\mathrm{GGC}_k$ be the laws on $[0,\infty)$ with a Lévy density $\ell$ such that $x\ell(x)$ is $k$ monotone, and $\mathrm{BO}_k$ those with $\ell$ itself $k$ monotone. Thus $\mathrm{GGC}_1 = \mathrm{SD}$ is the self decomposable class, $\bigcap_k \mathrm{GGC}_k = \mathrm{GGC}$ and $\bigcap_k \mathrm{BO}_k = \mathrm{BO}$. A positive function $\varphi$ on $(0,\infty)$ is HCM of order $k$ if for every $u > 0$ the function $w \mapsto \varphi(uv)\varphi(u/v)$,

$w = v + 1/v \geq 2$, is $k$ monotone on $[2,\infty)$; HCM means HCM of every order (Subsection 4.1.2).

The finite order versions are introduced here. They are not Urbanik's nested classes, and other finite order notions in the literature (Williamson's $k$ monotone functions, Bondesson's classes of finite order for densities) use other conventions. By the Leibniz identity $(-1)^j(fg)^{(j)} = \sum_i \binom{j}{i}[(-1)^i f^{(i)}][(-1)^{j-i} g^{(j-i)}]$ a product of two nonnegative $k$ monotone functions is $k$ monotone; as $x^{-1}$ is completely monotone and $\ell(x) = x\ell(x)\cdot x^{-1}$, this gives $\mathrm{GGC}_k \subset \mathrm{BO}_k$ for every $k$.

THEOREM 14.13.2. *Let $T$ be the clock, with Lévy density $W(x)/x$ and Laplace transform $\varphi_{1/2}(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda}) = 1/G(\lambda)$, and let $H = 3\times 10^{12}$. Statements (i) and (ii) are unconditional: they use the verification of RH to height $H$ [419].*

*(i) $T \in \mathrm{GGC}_k$, and hence $T \in \mathrm{BO}_k$, for every $k \leq 1.16\times 10^{25}$.*

*(ii) $\varphi_{1/2}$ is HCM of order $k$ for every $k \leq 9\times 10^{12}$.*

*(iii) The following are equivalent: RH; $T$ is a GGC; $T \in \mathrm{BO}$; $\varphi_{1/2}$ is HCM; $\varphi_{1/2}$ is HCM of every order.*

*Proof.* (i) By Theorem 5.1.1 the clock $T$ exists and is infinitely divisible with Lévy density $\ell(x) = W(x)/x$, and Theorem 5.1.13 gives $(-1)^j W^{(j)}(x) > 0$ for every $x > 0$ and every integer $0 \leq j \leq 1.29(H-23)^2$, which exceeds $1.16\times 10^{25}$. Since $x\ell(x) = W(x)$, this says $T \in \mathrm{GGC}_k$ for every $k \leq 1.16\times 10^{25}$, and the Leibniz step above gives $T \in \mathrm{BO}_k$.

(ii) Zeros are counted with multiplicity, and $\rho = \frac{1}{2}+b+i\gamma$ with $\gamma > 0$ runs over the zeros of positive ordinate. Put $a_\rho = -(\rho-\frac{1}{2})^2 = \gamma^2 - b^2 - 2ib\gamma$, as in Section 1.2, so that $\operatorname{Re} a_\rho > 0$ and $\sum_\rho |a_\rho|^{-1} < \infty$, and by the Hadamard product (1.2.1), $\varphi_{1/2}(\lambda) = \prod_\rho (1+\lambda/a_\rho)^{-1}$, absolutely convergent. Since $(1+uv/a)(1+u/(va)) = 1 + uw/a + u^2/a^2$ with $w = v + 1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v) = \prod_{b=0,\ \gamma>0} \Big(1 + \frac{uw}{a_\rho} + \frac{u^2}{a_\rho^2}\Big)^{-1} \times \prod_{b>0,\ \gamma>0} \Big|1 + \frac{uw}{a_\rho} + \frac{u^2}{a_\rho^2}\Big|^{-2}, \qquad w = v + \frac{1}{v},$$

where in the second product $\rho$ stands for the pair $\{\rho, 1-\bar\rho\}$ of zeros with $\gamma > 0$ and $b \neq 0$, whose values $a_\rho$ and $a_{1-\bar\rho} = \bar a_\rho$ are conjugate, so that their two factors multiply to a squared modulus. Put $c_\rho(u) = a_\rho/u + u/a_\rho$, so that $1 + uw/a + u^2/a^2 = (u/a)(w + c)$. For a zero on the line $a_\rho = \gamma^2$ and $c_\rho(u) \geq 2$ is real, and the factor $(a_\rho/u)/(w + c_\rho(u))$ is completely monotone in $w$ on $[2,\infty)$. For a pair the factor is $(|a_\rho|^2/u^2)/((w+c)(w+\bar c))$ with $c = c_\rho(u)$ complex. Write $a_\rho = re^{-i\phi}$ with $0 < \phi < \pi/2$. Then $\tan\phi = 2b\gamma/(\gamma^2 - b^2) \leq \gamma/(\gamma^2 - \frac{1}{4}) < 1.01/\gamma$

for $\gamma \geq 10$, because $0 < b < \frac{1}{2}$. Moreover $\operatorname{Re} c = (r/u + u/r)\cos\phi$ and $|\operatorname{Im} c| = |r/u - u/r|\sin\phi \leq (r/u + u/r)\sin\phi$, so for every $w \geq 0$ the angle $\vartheta = |\arg(w+c)|$ satisfies $\tan\vartheta = |\operatorname{Im} c|/(w + \operatorname{Re} c) \leq \tan\phi$, that is $\vartheta \leq \phi$. Partial fractions give $1/((w+c)(w+\bar{c})) = -\operatorname{Im}[(w+c)^{-1}]/\operatorname{Im} c$, whence

$$(-1)^j \frac{d^j}{dw^j} \frac{1}{(w+c)(w+\bar{c})} = \frac{j!\,\sin((j+1)\vartheta)}{|\operatorname{Im} c|\,|w+c|^{j+1}},$$

which is positive as long as $(j+1)\vartheta < \pi$, hence for all $j < \pi/\phi - 1$ (when $\operatorname{Im} c = 0$, that is $u = r$, the factor is $(w+c)^{-2}$, whose signed derivatives $(j+1)!\,(w+c)^{-j-2}$ are positive). By the verification of RH to height $H$ [419] every zero with $\gamma \leq H$ is on the line, so every pair has $\gamma > H$ and $\phi < 1.01/H$, and its factor is $k$ monotone for every $k \leq \pi H/1.01 - 1$, which is $9.33 \times 10^{12}$ and exceeds $9 \times 10^{12}$. Products of nonnegative $k$ monotone functions are $k$ monotone by the Leibniz identity, and the partial products converge, with all their $w$ derivatives, locally uniformly on $\operatorname{Re} w > 0$: the factors have no zeros there, since the poles $w = -c$ lie in $\operatorname{Re} w < 0$, and $\sum_\rho |uw/a_\rho + u^2/a_\rho^2| < \infty$ locally uniformly. So the limit is $k$ monotone.

(iii) RH holds if and only if $T$ is a GGC by Proposition 1.2.1, and if and only if $\varphi_{1/2}$ is HCM by Corollary 4.1.3(ii), that is by Bondesson's theorem [56, Theorem 6.1.1] (a function $\varphi$ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM; see also [289]). A GGC is in BO by the Leibniz step. Conversely suppose that $\ell(x) = W(x)/x$ is completely monotone, $\ell(x) = \int_{[0,\infty)} e^{-xy} m(dy)$ with $m \geq 0$. Taking Laplace transforms of $W(x) = x\ell(x)$ for $\operatorname{Re}\lambda > 0$,

$$\int_0^\infty e^{-\lambda x} W(x)\,dx = \int_{[0,\infty)} \frac{m(dy)}{(\lambda+y)^2} = \sum_\rho \frac{1}{\lambda + a_\rho} = \frac{G'}{G}(\lambda).$$

The middle expression is analytic on $\mathbb{C} \setminus (-\infty, 0]$, while $G'/G$ has a pole at every $-a_\rho$, and $-a_\rho \notin (-\infty, 0]$ exactly when $a_\rho \notin (0, \infty)$, that is when $\rho$ is off the line. So BO forces RH. □

For a model pair factor with a zero at $\beta = 0.6$, $\gamma = 100$, the first order at which the closed form fails is 1571 at $u = 1$, $w = 2$, and 1570 as $u \to 0$, against the bound $\pi\gamma/(2(\beta - \frac{1}{2})) = 1570.8$.

## 14.14 Is there a new path to the Riemann Hypothesis?

The honest answer is no.

### 14.14.1 Three reformulations and what each costs

The derivative ladder: RH holds if and only if $W$ is completely monotone (Theorem 14.6.1(e)), and order $k$ and height $H$ trade quadratically (Subsection 5.1.6), so climbing the ladder to infinity is verifying RH to infinite height; the ladder

is a reformulation, not a path. The basepoint family: the basepoint clock $T_\alpha$ (Section 14.8) is GGC for every $\alpha > \frac{1}{2}$ if and only if RH (Theorem 10.2.7), and this is Lagarias's criterion [320] (Remark 1.2.4), so the Thorin route at basepoints and the Lagarias route are one route. The finite maxima: Corollary 14.12.2(c) reduces RH to zero freeness of the symmetrised stages $\xi_N^{\mathrm{sym}}$ off the line up to heights $h_j \to \infty$, for infinitely many $N$, a hypothesis that is not weaker than RH (whether RH implies it is open). Unsymmetrised sections of $\zeta$ are known to have zeros off the line and even to the right of $\operatorname{Re} s = 1$ [499, 375], which is why the symmetrisation is essential; for the symmetrised sections there are results of Spira and of Gonek and Montgomery [483, 155] (the paragraph after Table 7.2); we know of no result of this kind for the symmetrised stages of the maximum.

### 14.14.2 The arithmetic form below $\alpha = 1$, and how far it sees

The sine squared prime form of Section 1.2 (the paragraph after (1.2.9), with Table 1.5) writes $v_\alpha(\theta) = \operatorname{Re} \xi'/\xi(\alpha + i\theta)$ for $\alpha > 1$ through the signed canonical measure with atoms $\Lambda(n)n^{-\alpha}$ at $\log n$, an archimedean density and a pole density: the prime and gamma terms are nonnegative and the pole terms negative, so the positivity of $v_\alpha$ for $\alpha > 1$, which is Lagarias positivity there, comes from the absence of zeros with $\beta \geq 1$ and not from the terms one by one. Theorem 10.2.7 already continues it below $\alpha = 1$ by compensation: its last condition, that the compensated sine squared sum converges for every $\alpha' > \alpha$, is equivalent to zero freeness above $\alpha$. A smoothed form of the continuation, with explicit pole terms, is the following.

Proposition 14.14.1. *Let $\alpha = \frac{1}{2} + \delta$ with $\delta > 0$, $t \in \mathbb{R}$ and $X > 0$. Put $g_X(x) = e^{-\delta|x|}\cos(tx)\,(1 - |x|/X)_+$ and $h_X(r) = \int_{\mathbb{R}} g_X(x)e^{irx}dx$, so that $h_X \geq 0$ on $\mathbb{R}$: it is $(2\pi)^{-1}$ times the convolution of the two sided Poisson kernel $h_\infty(r) = \delta/(\delta^2 + (t-r)^2) + \delta/(\delta^2 + (t+r)^2)$ with the Fejér kernel $4\sin^2(Xr/2)/(Xr^2)$. Define*

$$A_X(\delta,t) = \frac{1}{2}\Big[h_X(\tfrac{i}{2}) + h_X(-\tfrac{i}{2}) - \log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h_X(r)\,\operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\,dr - 2\sum_{n<e^X}\frac{\Lambda(n)}{n^\alpha}\cos(t\log n)\Big(1 - \frac{\log n}{X}\Big)\Big],$$

*where $\psi$ is the digamma function and $h_X(\frac{i}{2}) + h_X(-\frac{i}{2}) = 2\operatorname{Re}[J(-\delta + \frac{1}{2} + it) + J(-\delta - \frac{1}{2} + it)]$ with $J(c) = -1/c + (e^{cX} - 1)/(Xc^2)$. Then:*

*(a) $A_X(\delta,t) = \frac{1}{2}\sum_\rho h_X(\gamma_\rho)$, with $\gamma_\rho = (\rho - \frac{1}{2})/i$ and the sum over all nontrivial zeros, unconditionally.*

*(b) Under RH, $A_X(\delta,t) \geq 0$ for every $X$, $\delta$ and $t$. In particular a single triple $(\delta,t,X)$ with $A_X(\delta,t) < 0$ would disprove RH.*

*(c) If every zero has $|\beta - \frac12| < \delta$ (under RH, or unconditionally when $\delta \ge \frac12$), then $A_X(\delta,t) \to \operatorname{Re}\,\xi'/\xi(\alpha+it)$ as $X\to\infty$.*

*Proof.* (a) This is Weil's explicit formula $\sum_\rho h(\gamma_\rho) = h(\frac{i}{2}) + h(-\frac{i}{2}) - g(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb R} h(r)\operatorname{Re}\psi(\frac14 + \frac{ir}{2})\,dr - 2\sum_n \Lambda(n)n^{-1/2}g(\log n)$, in the form of [288], which evaluates the distribution $D$ of Proposition 7.2.14 on a test function $g$ with Fourier transform $h$, applied to the even test function $g_X$, which is continuous, compactly supported and piecewise smooth, with Fourier transform $h_X$, so that $g_X(0) = 1$ and $n^{-1/2}g_X(\log n) = n^{-\alpha}\cos(t\log n)(1-\log n/X)_+$; the sum over zeros converges absolutely because $h_X(r) = O(r^{-2})$ uniformly in $|\operatorname{Im} r| \le \frac12$. Here $J(c) = \int_0^X e^{cx}(1-x/X)\,dx$, and $h_X(r) = 2\int_0^X e^{-\delta x}\cos(tx)\cos(rx)(1-x/X)\,dx = \frac12\sum_{s_1,s_2=\pm1} J(-\delta + i(s_1t+s_2r))$. At $r = \pm\frac{i}{2}$ two of the four terms are the complex conjugates of the other two, so $h_X(\pm\frac{i}{2}) = \operatorname{Re}[J(-\delta+\frac12+it) + J(-\delta-\frac12+it)]$ and the two pole terms add up to twice that, which is the stated value.

(b) Under RH every $\gamma_\rho$ is real and $h_X \ge 0$ on $\mathbb R$, so every term of (a) is nonnegative.

(c) For a zero $\rho = \frac12 + b + i\gamma$ the four exponentials of $h_X(\gamma_\rho) = h_X(\gamma - ib)$ have $\operatorname{Re} c = -\delta \pm b$, which is negative when $|b| < \delta$. Then $J(c) \to -1/c$ as $X\to\infty$, and $|J(c) + 1/c| \le 2/(X|c|^2)$, so $h_X(\gamma_\rho) \to h_\infty(\gamma_\rho) = \delta/(\delta^2 + (t-\gamma_\rho)^2) + \delta/(\delta^2 + (t+\gamma_\rho)^2)$, with $|h_X(\gamma_\rho)| \le C(1+|\gamma_\rho|)^{-2}$ uniformly in $X \ge 1$, where $C$ depends on $\delta$, $t$ and the finitely many zeros with $||\gamma_\rho| - t| \le 1$, because the four terms $-1/c$ combine in pairs to $O(\gamma^{-2})$ and $|c| \ge 1$ for the other zeros. By dominated convergence $A_X \to \frac12\sum_\rho h_\infty(\gamma_\rho)$. Finally $h_\infty(\gamma_\rho)$ is the sum of the four terms $\frac12/(\alpha \pm it - \rho)$ and $\frac12/(\rho - (1-\alpha) \pm it)$, and since the zeros are symmetric under $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar\rho$, $\frac12\sum_\rho h_\infty(\gamma_\rho) = \operatorname{Re}\sum_\rho (\alpha + it - \rho)^{-1} = \operatorname{Re}\,\xi'/\xi(\alpha+it)$ by the symmetric Hadamard product. □

Remark 14.14.2. The hypothesis of (c) is needed. If a zero has $b = \beta - \frac12 > \delta$, the terms of $h_X(\gamma - ib)$ with $\operatorname{Re} c = b - \delta > 0$ contain $e^{(b-\delta)X}/(Xc^2)$, which grows with $X$, and $h_\infty(\gamma - ib)$ is only the analytic continuation of a divergent integral. At $t = \gamma$ the four zeros $\frac12 \pm b \pm i\gamma$ contribute $e^{(b-\delta)X}/(X(b-\delta)^2)\,(1 + O((b-\delta)^2/\gamma^2)) + O(1)$ to $A_X(\delta,\gamma)$, which is positive and unbounded in $X$. No converse is claimed: whether an off line zero forces $A_X(\delta,t) < 0$ somewhere for large $X$ is not shown here.

The proposition is Lagarias's criterion restricted to a family, and it is also a form of Weil's positivity criterion (compare Proposition 7.2.14) for the positive definite test functions $g_X$, which have corners. Table 14.6 gives the minimum over $t \in [0, 60]$, positive at every $X$ and every $\alpha$ tested.

The positivity in the table is not evidence of anything new, and it is important to say why. The contribution of zeros above height $H$ to $A_X$ is at most about $(e^{X/2}/X + \delta)\log H/H$, because $|h_X(\gamma_\rho)| \lesssim e^{|\operatorname{Im}\gamma_\rho| X}/(X\gamma^2)$ and $|\operatorname{Im}\gamma_\rho| \le \frac12$. With $H = 3\times 10^{12}$ and $X = 13$ that is below $10^{-8}$, about $4.9\times 10^{-10}$, far under the minima in the table. So the verified zeros force $A_X \ge 0$ for $t$ in the range of the

| $X$ | $\alpha = 0.75$ | $\alpha = 0.6$ | $\alpha = 0.55$ |
|---|---|---|---|
| 2 | 0.03275 | 0.02495 | 0.02230 |
| 4 | 0.02208 | 0.01432 | 0.01160 |
| 8 | 0.01707 | 0.00954 | 0.00681 |
| 13 | 0.01510 | 0.00812 | 0.00576 |

Table 14.6: $\min_{0\le t\le 60} A_X(\alpha - \frac{1}{2}, t)$ from primes below $e^X$ (up to $4.4 \times 10^5$ at $X = 13$). The values decrease towards $\xi'/\xi(\alpha) = 0.011551$, 0.0046209, 0.0023105 for the three $\alpha$.

table as long as this bound stays below the minima, that is up to $X$ of about 46 to 50, where it reaches the values from 0.0058 down to the limit 0.0023; the bound itself stays small up to about $2\log H \approx 57$. The arithmetic test at length $e^X$ sees zeros up to a height of order $e^{X/2}$, the usual duality between primes and zeros, and to see past the present verification one would need primes up to about $10^{25}$. As a computation the arithmetic form is strictly weaker than zero verification. As a target for proof it is sharper than before: RH needs $A_X \ge 0$ uniformly in $X$ (whether this suffices is open, Remark 14.14.2), with the prime sum, the gamma integral and the pole terms each explicit, and the whole difficulty is that the pole terms, of size $e^{(1/2-\delta)X}/X$, are cancelled by the prime sum only through the prime number theorem with an error term, which is to say through the zeros.

## 14.15 Open problems

Problem 14.1. Prove the hypothesis of the $F_N^{\mathrm{sym}}$ form of Corollary 14.12.2(c): find heights $h_N \to \infty$ such that for infinitely many $N$ the symmetric stage $F_N^{\mathrm{sym}}$ has no zeros with $\frac{1}{2} < \operatorname{Re} s < 1$ and $|\operatorname{Im} s| < h_N$, for instance $h_N = c\sqrt{N}$ with some $c > 0$. Subsection 14.12.3 finds, for the $N$ tested, that every zero of the stages with $-1 \le \operatorname{Re} s \le 2$ and $0 < \operatorname{Im} s \le T^*(N)$ lies on the line, with $T^*(N)$ of order $N$ for $N \le 96$, far more than needed (at $N = 128$ a pair outside the strip comes down to height 60.7, while the strip has none below 130), and the sharper conjecture is $T^*(N) \ge \kappa N$. Either statement is about explicit functions built from the Dirichlet polynomials $D_N$ with integer coefficients, and either implies RH. A mechanism for $T^*(N)$, presumably through the ratio $|D_N(\sigma+it)/D_N(1-\sigma+it)|$, is the first thing to find. The corresponding statement for the symmetrised lattice truncations $K_n$ is Conjecture S of Subsection 7.2.1 (Theorem 7.2.7).

Problem 14.2. Theorem 10.2.7 continues the sine squared prime form below $\alpha = 1$ by compensation, and shows that the compensated sum converges for every $\alpha' > \alpha$ exactly when $\zeta$ has no zero with $\operatorname{Re} s > \alpha$; Proposition 14.14.1 gives a smoothed form of the continuation with explicit pole terms, which is not positive term by term. Is there a continuation of the prime form below $\alpha = 1$ that is positive term by term? The prime sum diverges there, but smoothed versions

of it, with $n^{-\alpha}$ replaced by a weight that cuts off at $n \approx e^{1/\varepsilon}$, remain positive termwise, and the question is whether the loss from the cutoff can be controlled by $(\xi'/\xi)(\alpha)$, the gamma sum and the pole terms. A positive answer on $\alpha \in (\alpha_0, 1]$ would prove the zero free region $\beta \le \alpha_0$.

Problem 14.3. For $0 < t < 0.2$, does the law $T_t$ of Section 14.7, half of the flow clock of Section 8.4, with transform $R_t(\lambda/2)$, exist as a probability law, and is it infinitely divisible? Unconditionally it is known to be a GGC only for $t \ge 0.2$, and to exist only for $t \ge 0.2$ and at $t = 0$, where $T_0 = T/2$ (Theorem 5.1.1); under RH it exists and is a GGC for every $t \ge 0$. This is the analogue along the de Bruijn Newman flow of the existence of $T$ (Theorem 5.1.1); it would need an explicit formula for the heat trace of the flow, which the Gaussian tilt destroys.

Remark 14.15.1. No published result gives an explicit height below which the zeros of $H_t$ are real for $0 < t < 0.2$: [456] and [419] give $\Lambda \le 0.2$ through a barrier at $|\operatorname{Im} z| \ge y_0 > 0$, and [310] show that all but finitely many zeros are real, without a height. Real zeros do not collide forward in time (a multiple real zero splits into real zeros, as in Lemma 8.4.24, after the statement of [112] cited in Section 8.4), so the verification of RH to $3 \times 10^{12}$ passes to $H_t$ except for zeros entering from above, which a box barrier touching the real axis would exclude. That route has not been carried out. The forward statement is Lemma 8.4.24, and the barrier hypothesis and its consequence are in Proposition 8.4.26. With a certified height $H'$, an explicit count of the zeros of $H_t$ and the low zeros of $H_t$ located, the certificate of Section 5.1.8 would give about $0.76H'$ rungs, and rungs 0 and 1 (infinite divisibility and self decomposability of $T_t$ for every $t \in [0, 0.2]$) would need only an explicit zero density bound for $H_t$ and a small computation. Proposition 15.2.7 reduces an explicit count of the zeros of $H_t$ to two inequalities for the heat average of $\zeta$, which are not proved here (Problem 15.6).

Problem 14.4. Is there an analogue of Theorem 14.3.1 at $\sigma = \frac{1}{2}$, that is, a factorisation of the tilted law of $T$ or of the Pólya variable through a law built from the primes, with the explicit formula replacing the Euler product?

## 14.16 Numerical checks

The numerical results of this chapter are recomputed by the programs `maxwald_*.py` of the repository (`code/README.md`). The zero counts of Table 14.5 are the most delicate, and the other checks use multiprecision arithmetic, except the grids of Figure 14.2, the grid values and the gamma factors of the finite stages, and the sums and integrals of Table 14.6, which are in double precision; those of Table 14.6 are checked against the sum over the zeros, to $5 \times 10^{-7}$. Table 14.7 lists the checks of the results kept in this chapter.

| result | test points | agreement |
|---|---|---|
| kernel identity of Proposition 14.2.1 | six values of $u$ | 24 digits |
| $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\xi(s)/(s-1)$ | $s = 0.5,\ 2,\ 3+2i$ | 8 to 30 digits |
| Theorem 14.3.1, Mellin form | $\sigma = 1.5,\ 2,\ 3.5,\ t = 1,\ 5,\ 20$ | 30 digits |
| low coefficients of $\mathcal{P}_N$ equal those of $\vartheta_4$ | $N = 4, 8, 16$ | exact |
| zero of $\xi_{64}$ near $\rho_1$ | $N = 64$ | $\lvert \operatorname{Re} - \frac{1}{2}\rvert = 5.2 \times 10^{-7}$ |
| zero counts of $F_N^{\mathrm{sym}}$ in $[-1,2] \times [0,T]$ against sign changes on the line (Table 14.5) | $N = 8,\ 16,\ 32,\ 48,\ 64,\ 96,\ 128$ | windows of height 10 |
| unit mass of the term of one zero, Proposition 14.8.1(b) | $\alpha = 0.55$ | 10 digits |
| width at half height of the term of $\rho_1$, Proposition 14.8.1 | $\alpha = 0.55$ | 6 digits, up to the factor $1 + \delta^2/(2\gamma_1^2)$ |
| $\gamma_1^2$ by Laplace inversion, Proposition 14.10.1 | $t \in [0.04, 0.08]$ | 199.790 |

Table 14.7: Numerical verification of results of this chapter that are not covered by the book's tables.

# Chapter 15

# The Thorin phase diagram

This chapter sets out the most promising computational path to RH that the book has found, and says exactly where it stops.

The sine squared lemma and the Thorin side of Chapter 3 offer no computational alternative to the de Bruijn Newman flow, only the right language for it. The Thorin side is the object to compute, prime free and well defined at every basepoint, and the flow is the only known source of an analytic tail below the basepoint $\frac{7}{8}$. What separates the theorems $\Lambda \leq 0.22$ of Polymath15 [456] and $\Lambda \leq 0.2$ of Platt and Trudgian [419] from RH is not anything a computed density shows; it is the existence of the analytic tail at $t = 0.2$ and its absence at $t = 0$. Sections 15.2, 15.5 and the two experiments of Section 15.6 were first written as a working report by the authors [438], which this chapter supersedes. We use the normalisation of Section 8.4: $H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du$, $\xi_t(s) = 8H_t(-2i(s - \frac{1}{2}))$, a zero $x + iy$ of $H_t$ is the zero $\frac{1}{2} - \frac{y}{2} + \frac{ix}{2}$ of $\xi_t$, and $P_a(x) = a/(\pi(a^2 + x^2))$ is the Cauchy density of scale $a > 0$.

## 15.1 Which computations see past the verified height

Every route to RH in this book replaces the hypothesis by an infinite family of statements, each of which a finite computation can certify. The question that separates the routes is what happens above the height the computation reaches. If nothing does, the route is finite information in the sense of Section 5.2: by Theorem 5.2.1 a perturbation with the same zeros below that height and zeros off the line above it passes the same test, so the test certifies no more than a verification of the zeros up to a height determined by its size.

Definition 15.1.1. A route *reaches every height* if there are a finite computation $C$ and an analytic statement $A$ valid above a barrier $X$ such that $C$ and $A$ together imply a statement about all zeros of $\xi$, or of a fixed approximant or deformation of $\xi$, that is not implied by the zeros below $X$ alone.

### 15.1.1 The finite information routes

Most of the book's tests are finite information, and the book measures exactly how much height each one buys. The heat trace has its derivative signs to order $1.16 \times 10^{25}$ from the verification to $3 \times 10^{12}$ (Theorem 5.1.13), and the order is a height in disguise: an off line zero at height $\gamma$ first breaks the alternation at an order of about $t\gamma^2$, with $t$ the parameter of the heat trace and not the time of the flow. The Bosch function $\varphi(x^p)$ of Section 4.3 is HCM for $|p| \leq 1 - 1/(\pi T_0)$, and $1 - p \approx 1/(\pi T)$ is again a height (Corollary 4.3.7). Li coefficients see an off line zero at height $\gamma$ and distance $d$ from the line only from an order of about $\gamma^2/d$, while the Hankel test of Section 8.4.3 often registers a pair at an order close to $2k + 2$, set by its index $k$, at a cost in precision that grows linearly with the order (Section 11.5). The Cauchy flow $\rho_\varepsilon$ is positive on the verified window for every $\varepsilon$, but its minimum is attained at the origin and tends to zero with $\varepsilon$ (Theorem 9.1.18 and Remark 9.1.19), so positivity is never won by a fixed margin, and positivity on one line is not known to exclude zeros to its right (Remark 1.2.4). The ferromagnetic distance of Section 13.5 is below $1.932 \times 10^{-6}$, and it is zero exactly under RH, but the approximating laws keep only the first $K$ zeros, so it sees nothing above them. The Weil Gram matrices and the arithmetic forms of Section 15.6.1 below have the same property, with a resolution in height set by the support of the test functions.

### 15.1.2 The three routes that reach every height

Three routes in the book reach every height in the sense of Definition 15.1.1.

*The Euler product: basepoints $\alpha \geq 1$.* At the basepoint one, the Thorin density of the shifted clock is positive at every height, and the zeros below 2000 in ball arithmetic and explicit bounds for $S(T)$ [176, 15] certify that it is smallest only at the origin (Propositions 9.1.11 and 9.1.12). Above the barrier, the analytic input is the explicit bound for $S(T)$, and positivity itself comes from the Euler product, through the absence of zeros in $\operatorname{Re} s \geq 1$. Below $\alpha = 1$ the same structure gives a statement at every height only together with a zero free half plane (Proposition 9.1.13); the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [398] would extend the route to $\alpha \geq \frac{7}{8}$ (Section 12.1.2), and below $\frac{7}{8}$ nothing is known.

*The functional equation: symmetrised truncations.* The symmetrised truncations of the Biane, Pitman and Yor lattice of Section 7.2 have all their zeros in the strip on the line, at every height, for $n \leq 24$ and $n = 32, 36$, proved above explicit thresholds and computed below them (Theorem 7.2.7 and Corollaries 7.2.10 and 7.2.12). Above the threshold the analytic input is the symmetry of the functional equation. RH would follow from infinitely many $n$ (Problem 7.1), and no argument uniform in $n$ is known.

*The heat flow: positive time.* At time $t > 0$ and for a suitable $y_0 > 0$, the function $H_t$ has no zeros with $\operatorname{Im} z \geq y_0$ above a barrier, by an analytic argument, the effective Riemann Siegel approximation of Polymath15 [456]; a computation of winding numbers keeps zeros from entering through a strip of width one at the barrier for the times up to $t$, and the verification of RH covers the zeros below it. Together they give an upper bound for $\Lambda$, since then $\Lambda \leq t + y_0^2/2$ by de Bruijn's strip theorem [112, Theorem 13] (Polymath15 take $t = y_0 = 0.2$): $\Lambda \leq 0.22$ [456] and, from the verification to $3 \times 10^{12}$, $\Lambda \leq 0.2$ [419].

| route | analytic input above the barrier | what is certified at every height | what is missing |
|---|---|---|---|
| basepoint $\alpha \geq 1$ | Euler product, explicit bounds for $S(T)$ | the shifted clock is a GGC, its density minimal at the origin | any basepoint below $\frac{7}{8}$ |
| symmetrised truncations | functional equation symmetry | zeros of $K_n$ in the strip on the line, $n \leq 24$ and $n = 32, 36$ (floating point counts below the thresholds) | infinitely many $n$ |
| heat flow | Riemann Siegel at $t > 0$ | $\Lambda \leq 0.2$; GGC at points $(t, \alpha)$ | a tail at $t = 0$ |

Table 15.1: The three routes of the book that reach every height. All other routes are finite information.

The first and the third routes are the two coordinates of the phase diagram of Section 15.3. The second route has no coordinate on the diagram and stands apart; Problem 15.3 asks whether it can be given one.

## 15.2 Clocks along the flow at every basepoint

This section extends Theorem 1.2.2, which decides when the shifted clock $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is a generalised gamma convolution, and Theorem 7.2.16 with Corollary 4.2.5, which decide it for the centre clock $R_t$ of the flow, to every time $t \geq 0$ and every basepoint $\alpha \geq \frac{1}{2}$.

Lemma 15.2.1 (Structure of $\xi_t$). *For every $t \geq 0$: (i) $\xi_t$ is entire of order at most one, real on the real axis, and $\xi_t(1 - s) = \xi_t(s)$; (ii) $\xi_t(\frac{1}{2} + x) > 0$ for real $x$; (iii) with $w = s - \frac{1}{2}$ and the zeros written $\rho = \frac{1}{2} + \tilde{\rho}$,*

$$\xi_t(\tfrac{1}{2} + w) = \xi_t(\tfrac{1}{2}) \prod \Big(1 - \frac{w^2}{\tilde{\rho}^2}\Big),$$

*the product running over one representative of each pair $\{\tilde{\rho}, -\tilde{\rho}\}$, with multiplicity, and $\sum_\rho |\tilde{\rho}|^{-2} < \infty$.*

*Proof.* Since $\Phi_{\mathrm{dBN}}(u) \le Ce^{9u-\pi e^{4u}}$ for $u \ge 0$ and $|\cos(zu)| \le e^{|z|u}$,

$$|H_t(z)| \le C\int_0^\infty e^{tu^2+|z|u+9u-\pi e^{4u}}\,du \le \exp(C'|z|\log|z|)$$

for large $|z|$; so $H_t$ and $\xi_t$ are entire of order at most one. Evenness of $H_t$ gives the reflection symmetry, and reality of the integrand on the imaginary axis gives $\xi_t$ real on the real axis, with

$$\xi_t(\tfrac12+x) = 8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(2xu)\,du > 0,$$

since $\Phi_{\mathrm{dBN}} > 0$. Hadamard's theorem for order at most one gives $H_t(z) = e^{a+bz}\prod_k E_1(z/z_k)$ over the zeros, none of which is zero since $H_t(0) > 0$. Pairing $z_k$ with $-z_k$, a zero by evenness, gives $E_1(z/z_k)E_1(-z/z_k) = 1 - z^2/z_k^2$, and $\sum|z_k|^{-2} < \infty$ since the convergence exponent is at most one. Evenness forces $b = 0$, and the substitution $z = -2iw$, $\tilde\rho = iz_k/2$ gives (iii). □

Lemma 15.2.2 (One conjugate pair). *Let $d, \gamma \in \mathbb{R}$, not both zero, and $\varphi(u) = (d^2+\gamma^2)/((\sqrt{u}+d)^2+\gamma^2)$ for $u \ge 0$.*

*(a) If $d > 0$, then $\varphi$ is the Laplace transform of a GGC whose Thorin measure, in the chart $\theta = \sqrt{z}$, is $[P_d(\theta-\gamma) + P_d(\theta+\gamma)]\,d\theta$ on $\theta > 0$ (compare Proposition 1.2.3):*

$$-\log\varphi(u) = \int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big[P_d(\theta-\gamma)+P_d(\theta+\gamma)\big]\,d\theta.$$

*(b) If $d = 0$, then $\varphi(u) = (1+u/\gamma^2)^{-1}$, an exponential law, with Thorin measure $\delta_{\gamma^2}$ in the variable $z$, that is, $\delta_\gamma$ in the chart $\theta = \sqrt{z}$.*

*(c) If $d < 0$, then $\varphi$ has a pole at $u_0 = (-d+i\gamma)^2 \in \mathbb{C}\setminus(-\infty,0]$ and is not the Laplace transform of a GGC.*

*Proof.* (a) Both sides vanish at $u = 0$, so it suffices to compare derivatives in $u > 0$. On the left, with $F(u) = (\sqrt{u}+d)^2+\gamma^2$, $F'/F = (\sqrt{u}+d)/(\sqrt{u}\,F)$. On the right, differentiating under the integral and using the symmetry $\theta \mapsto -\theta$ of the second kernel,

$$\begin{aligned}\int_{-\infty}^\infty \frac{P_d(\theta-\gamma)}{\theta^2+u}\,d\theta &= \frac{\pi}{\sqrt{u}}\int_{-\infty}^\infty P_d(\theta-\gamma)P_{\sqrt{u}}(\theta)\,d\theta\\ &= \frac{\pi}{\sqrt{u}}P_{d+\sqrt{u}}(\gamma) = \frac{d+\sqrt{u}}{\sqrt{u}\,((d+\sqrt{u})^2+\gamma^2)},\end{aligned}$$

by the semigroup property $P_a * P_b = P_{a+b}$. The integral converges, since $\log(1+u/\theta^2)$ is integrable at 0 and $O(\theta^{-2})$ at infinity, and the measure is positive since

$d > 0$. (b) is immediate. (c) The equation $\sqrt{u} + d - i\gamma = 0$ has the solution $\sqrt{u} = -d + i\gamma$, whose real part is positive, so it is the principal root and $u_0$ lies off $(-\infty, 0]$ (if $\gamma = 0$ then $u_0 = d^2 > 0$); the Laplace transform of a GGC is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$ [56]. □

THEOREM 15.2.3 (Threshold along the flow). *Let $t \geq 0$ and $\alpha \geq \frac{1}{2}$. The shifted time $t$ clock*

$$\varphi_{t,\alpha}(u) = \frac{\xi_t(\alpha)}{\xi_t(\alpha + \sqrt{u})}$$

*is the Laplace transform of a GGC if and only if $\alpha \geq \sup_\rho \operatorname{Re}\rho$ over the zeros $\rho$ of $\xi_t$. In that case its Thorin measure, in the chart $\theta = \sqrt{z}$, is*

$$\sum_{\substack{\rho=\beta+i\gamma \\ \gamma>0,\ \beta<\alpha}} \left[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\right] d\theta \ + \sum_{\substack{\rho=\alpha+i\gamma \\ \gamma>0}} m_\rho\, \delta_\gamma,$$

*where the first sum counts each zero with its multiplicity and $m_\rho$ is the multiplicity of $\rho$; this is the Poisson form of Proposition 1.2.3 for $\xi_t$.*

*Proof.* By Lemma 15.2.1, with $a = \alpha - \frac{1}{2} \geq 0$,

$$\varphi_{t,\alpha}(u) = \prod \frac{1 - a^2/\tilde{\rho}^2}{1 - (a + \sqrt{u})^2/\tilde{\rho}^2} = \prod_\rho \Big(1 - \frac{\sqrt{u}}{\rho - \alpha}\Big)^{-1},$$

where the last product runs over all zeros and is grouped so that each finite partial product contains, with every zero, its partner under $\tilde{\rho} \mapsto -\tilde{\rho}$; the denominators do not vanish since $\xi_t(\alpha) > 0$. Because $\xi_t$ is real on the real axis, the zeros are also closed under conjugation, and regrouping into conjugate pairs is legitimate: both groupings give absolutely convergent products, the conjugate pairing because $(1-\sqrt{u}/(\rho-\alpha))(1-\sqrt{u}/(\bar{\rho}-\alpha)) = 1+(2d\sqrt{u}+u)/(d^2+\gamma^2)$ with $d = \alpha-\beta$ bounded (the zeros of $H_t$ lie in $|\operatorname{Im} z| \leq 1$ by de Bruijn's strip theorem [112, Theorem 13], as in Section 8.4.1; and $0 \leq d \leq 2\alpha - 1$ directly when $\alpha \geq \sup \operatorname{Re}\rho$, the only case in which the pairing is used, by the symmetry $\rho \mapsto 1 - \rho$) and $\sum |\rho - \alpha|^{-2} < \infty$, and the partial products over $|\gamma| \leq T$ agree. Hence

$$\varphi_{t,\alpha}(u) = \prod_{\{\rho,\bar{\rho}\}} \frac{d^2 + \gamma^2}{(\sqrt{u} + d)^2 + \gamma^2},$$

a product of the functions of Lemma 15.2.2, one for each conjugate pair, every pair being genuine because $\xi_t$ has no real zeros by Lemma 15.2.1(ii).

If $\alpha \geq \sup \operatorname{Re}\rho$, every factor is a GGC transform by Lemma 15.2.2(a) and (b). Finite products of GGC transforms are GGC transforms with Thorin measures adding, the partial products converge pointwise to $\varphi_{t,\alpha}$, which is continuous with $\varphi_{t,\alpha}(0) = 1$, so by the continuity theorem and the closure of the class under

weak limits ([56, Theorem 3.1.5], Section 4.1.1) the limit is a GGC transform. Its Thorin measure is the full sum $\mu$ displayed in the theorem, and there is no drift, by the following monotone convergence argument. Write $\mu_T$ for the sum of the kernels and atoms over the zeros with $0 < \gamma \le T$, so that $\mu_T \uparrow \mu$ as $T \to \infty$, and $\varphi_T$ for the partial product over the same zeros. By Lemma 15.2.2, $-\log \varphi_T(u) = \int \log(1 + u/\theta^2)\, \mu_T(d\theta)$ for $u \ge 0$, and the integrand is nonnegative, so by monotone convergence $-\log \varphi_T(u)$ increases to $\int \log(1 + u/\theta^2)\, \mu(d\theta)$. Since $\varphi_T(u) \to \varphi_{t,\alpha}(u) > 0$, this gives $-\log \varphi_{t,\alpha}(u) = \int \log(1 + u/\theta^2)\, \mu(d\theta) < \infty$ for every $u \ge 0$. If some zero has $\beta > \alpha$, then $\xi_t(\alpha + \sqrt{u_0}) = 0$ at $u_0 = (\rho - \alpha)^2$, whose principal root is $\rho - \alpha$ because $\operatorname{Re}(\rho - \alpha) > 0$, so $\varphi_{t,\alpha}$ has a pole in $\mathbb{C} \setminus (-\infty, 0]$ and is not a GGC transform. □

Remark 15.2.4. The proof uses nothing about $\xi_t$ beyond Lemma 15.2.1, so it applies to any even real entire function of order at most one with a genus zero product in $w^2$, positive on the real axis, in particular to the perturbation of Theorem 5.2.1, whose shifted clocks are therefore not GGC transforms at any basepoint below the real part of the displaced zeros. The Thorin density of $\varphi_{t,\alpha}$ in the chart $\theta$ is $\pi^{-1} \operatorname{Re}(\xi_t'/\xi_t)(\alpha + i\theta)$ away from atoms, which is $\rho_{\alpha - 1/2, t}(\theta)$ of Section 8.4.1. The shifted time $t$ clock is a function of $u$ only: where the threshold condition fails the theorem claims no probability law behind it, and for $\alpha = \frac{1}{2}$ and $0 < t < 0.2$ it is open, without RH, whether $R_t$ is the transform of a probability law at all (Problem 14.3); the basepoint family at $t = 0$ and the flow at $\alpha = \frac{1}{2}$ are the two one parameter families of laws compared in Section 14.7.

### 15.2.1 The minimum at the origin along the flow

For $t \ge 0$ and $\alpha > \frac{1}{2}$ put $v_t(\alpha, \theta) = \operatorname{Re}(\xi_t'/\xi_t)(\alpha + i\theta)$, $b_t(\alpha) = v_t(\alpha, 0)$ and $V_t = (\log \xi_t)''(\frac{1}{2})$. For $\alpha \ge \sup \operatorname{Re} \rho$, $v_t(\alpha, \cdot)/\pi$ is the Thorin density of the shifted clock $\varphi_{t,\alpha}$ in the chart $\theta$ (Remark 15.2.4), and since $\int_0^\infty \log(1 + u/\theta^2)\, d\theta = \pi\sqrt{u}$,

$$\tilde{\varphi}_{t,\alpha}(u) = \varphi_{t,\alpha}(u)\, e^{b_t(\alpha)\sqrt{u}} = \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big) \frac{v_t(\alpha, \theta) - b_t(\alpha)}{\pi}\, d\theta \Big\}.$$

So $\tilde{\varphi}_{t,\alpha}$, the shifted clock with the constant part of its Thorin density removed, is the Laplace transform of a GGC exactly when $v_t(\alpha, \theta) \ge b_t(\alpha)$ for every $\theta$, that is when the minimum of $v_t(\alpha, \cdot)$ is at the origin. At $t = 0$ this is the question of Theorem 9.1.18. For $t \ge \Lambda$, unconditionally for $t \ge 0.2$, the zeros of $\xi_t$ are $\frac{1}{2} \pm i\theta_k(t)$ with $0 < \theta_1(t) \le \theta_2(t) \le \cdots$, and $V_t = 2 \sum_k \theta_k(t)^{-2} = 2\mathbb{E}T_t$ (Section 8.4.8).

Proposition 15.2.5 (The minimum at the origin along the flow). *Let $t \ge \Lambda$, $\alpha > \frac{1}{2}$, $a = \alpha - \frac{1}{2}$, $\theta_k = \theta_k(t)$, $v = v_t(\alpha, \cdot)$ and $b = b_t(\alpha)$.*

*(i) $b \le a V_t$.*

*(ii) If $a^2 \le \frac{3}{5}\theta_1^2$, then $v(\theta) \ge b$ for $|\theta| \le \theta_1$. If $a \le \frac{1}{2}$ and $V_t < 4$, then $v(\theta) > b$ at every $\theta$ within $D_t = (1/V_t - \frac{1}{4})^{1/2}$ of some $\theta_k$.*

*(iii)* *Let $a \le \frac12$ and $V_t < 4$. If every $\theta \in [\theta_1, T_*]$ is within $D_t$ of some $\theta_k$, and for every $\theta > T_*$ some $w > 0$ has $\#\{k : |\theta_k - \theta| \le w\} > V_t(w^2 + \frac14)$, then $v \ge b$.*

*(iv)* *Let $\alpha' > \frac12$ and $a' = \alpha' - \frac12$ with $a'^2 \le \frac35\theta_1^2$. If $v_t(\alpha', \theta) \ge a' V_t$ for every $\theta \ge \theta_1$, then $v_t(\alpha, \cdot) \ge b_t(\alpha)$ for every $\alpha \in (\frac12, \alpha']$.*

*(v)* *As $\theta \to 0$, $(v(\theta) - b)/\theta^2 \to \sum_k 2a(3\theta_k^2 - a^2)/(a^2 + \theta_k^2)^3 = 6a \operatorname{Var} T_t + O(a^3)$.*

*Under the hypotheses of (iii), respectively (iv), $\tilde\varphi_{t,\alpha}$ is the Laplace transform of a GGC for every $\alpha \in [\frac12, 1]$, respectively $[\frac12, \alpha']$, the case $\alpha = \frac12$ being Theorem 15.2.3; there $b_t(\frac12) = \operatorname{Re}(\xi_t'/\xi_t)(\frac12) = 0$ by the symmetry $\xi_t(1-s) = \xi_t(s)$, so that $\tilde\varphi_{t,1/2} = \varphi_{t,1/2}$.*

*Proof.* By Lemma 15.2.1, with $q(u) = a/(a^2+u^2)$, $v(\theta) = \sum_k [q(\theta-\theta_k) + q(\theta+\theta_k)]$ and $b = \sum_k 2q(\theta_k)$. (i) $2q(\theta_k) \le 2a\theta_k^{-2}$. (ii) For $0 \le \theta \le \theta_k$ put $p_k(\theta) = q(\theta_k - \theta) + q(\theta_k + \theta)$. Since $u \mapsto u/(a^2+u^2)^2$ decreases on $u \ge a/\sqrt3$, $p_k$ increases where $\theta_k - \theta \ge a/\sqrt3$, and elsewhere $p_k \ge q(a/\sqrt3) = \frac{3}{4a} \ge 2q(\theta_k)$, because $\theta_k^2 \ge \frac53 a^2$. So $p_k(\theta) \ge p_k(0)$ for $|\theta| \le \theta_1$, and summing gives the first claim. If $|\theta - \theta_k| = d$, then $v(\theta) \ge q(d) > aV_t \ge b$ as soon as $a^2 + d^2 < 1/V_t$, which holds for $d < D_t$ when $a \le \frac12$. (iii) Since $V_t \ge 2\theta_1^{-2}$, $\frac35\theta_1^2 > \frac14 \ge a^2$, so (ii) gives $v \ge b$ on $[0, T_*]$; above $T_*$ the more than $V_t(w^2+\frac14)$ zeros of the window give $v(\theta) > aV_t(w^2+\frac14)/(a^2+w^2) \ge aV_t \ge b$. (iv) Each $q(u)/a = 1/(a^2+u^2)$ decreases in $a$, so for $a \le a'$ and $|\theta| \ge \theta_1$, $v_t(\alpha,\theta) \ge \frac{a}{a'} v_t(\alpha', \theta) \ge aV_t \ge b_t(\alpha)$, as in Theorem 9.1.15; for $|\theta| \le \theta_1$ apply (ii), since $a^2 \le a'^2 \le \frac35\theta_1^2$. (v) $v(\theta) - b = \sum_k [p_k(\theta) - p_k(0)]$, $p_k''(0) = 2q''(\theta_k)$ with $q''(u) = 2a(3u^2 - a^2)/(a^2+u^2)^3$, the series of second derivatives converges uniformly near 0, and $\sum_k \theta_k^{-4} = \operatorname{Var} T_t$ (Section 8.4.8). The last assertion is the identity displayed before the proposition. □

At $t = 0.2$, with $\operatorname{Var} T_{0.2} = 3.77203 \times 10^{-5}$ from the table of Section 8.4.8, the coefficient in (v) is $2.2632 \times 10^{-4} a$ to first order in $a$, against $2.2304 \times 10^{-4} a$ at $t = 0$ under RH: the flow raises the curvature of $v$ at its minimum slightly. From the same table, $V_{0.2} = 0.0464019$, $D_{0.2} = 4.6153$ and $\theta_1(0.2) = 14.0768$.

Lemma 15.2.6 (The heat average as a shifted Gaussian average). *Write $\xi(s) = g(s)\zeta(s)$ with $g(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, and $\ell = g'/g$. For $t \ge 0$ and $\operatorname{Im} s > 0$,*

$$\xi_t(s) = g(s)\, e^{t\ell(s)^2/4}\, \zeta_t(s), \qquad \zeta_t(s) = \mathbb{E}\, \Psi\Big(s, \frac{t}{2}\ell(s) + \sqrt{\tfrac{t}{2}}\, Z\Big),$$
$$\Psi(s,h) = \frac{\xi(s+h)}{g(s)\, e^{h\ell(s)}},$$

*with $Z$ standard normal. Moreover $\Psi(s,h) = \zeta(s+h)e^{r(s,h)}$ with $r(s,h) = h^2 \int_0^1 (1-u)\,\ell'(s+uh)\,du$, and if $Y = \min\big(\operatorname{Im} s, \operatorname{Im} s + \frac{t}{2}\operatorname{Im}\ell(s)\big) > 0$, then $|r(s,h)| \le |h|^2\big(\frac{1}{4Y} + \frac{7}{4Y^2}\big)$ for every $h \in \frac{t}{2}\ell(s) + \mathbb{R}$.*

*Proof.* By Proposition 8.4.18, since $\xi_t(\frac12+i\theta)=8H_t(2\theta)$ and $Z$ is symmetric, $\xi_t(s)=\mathbb{E}\,\xi(s+\tau Z)$ with $\tau=\sqrt{t/2}$. With $\phi_{\mathrm N}$ the normal density, $u\mapsto\xi(s+\tau u)\phi_{\mathrm N}(u)$ is entire, and because $\xi$ has order one the Gaussian factor lets the line of integration move to $u=v+\tau\ell(s)$. Since $\phi_{\mathrm N}(v+\tau\ell)=\phi_{\mathrm N}(v)e^{-\tau\ell v-t\ell^2/4}$, this gives $\xi_t(s)=e^{-t\ell^2/4}\mathbb{E}[\xi(s+h)e^{-\tau\ell Z}]$ with $h=\frac t2\ell+\tau Z$, and $\xi(s+h)=g(s)e^{h\ell}\Psi(s,h)$ with $h\ell-\tau\ell Z=\frac t2\ell^2$ gives the identity. Off the real axis $g$ has no zeros or poles, so $\log g(s+h)-\log g(s)-h\ell(s)=r(s,h)$ along the segment from $s$ to $s+h$, by Taylor's formula. For $h\in\frac t2\ell(s)+\mathbb{R}$ the imaginary part of $s+uh$ is linear in $u$, hence at least $Y$ on the segment. Euler summation gives $\psi'(z)=\frac1z+\frac{1}{2z^2}-2\int_0^\infty(\{x\}-\frac12)(z+x)^{-3}\,dx$, whose last term is at most $2/(\operatorname{Im}z)^2$ in absolute value; with $\ell'(v)=-(v-1)^{-2}+\frac14\psi'(\frac v2+1)$, which follows from $\psi'(z)=z^{-2}+\psi'(z+1)$, this gives $|\ell'(v)|\le\frac{1}{2y}+\frac{7}{2y^2}$ for $\operatorname{Im}v=y>0$, and the bound on $r$ follows. □

Proposition 15.2.7 (The zero count of $\xi_t$). *Let $t\ge0$ and let $N_t(T)$ be the number of zeros $\rho$ of $\xi_t$ with $0<\operatorname{Im}\rho\le T$, counted with multiplicity whether or not they lie on the critical line. For $T>0$ not an ordinate let $\pi S_t(T)$ be the change of a continuous argument of $\zeta_t=\xi_t/(g\,e^{t\ell^2/4})$ along the segments from 2 to $2+iT$ and from $2+iT$ to $\frac12+iT$, starting from $\arg\zeta_t(2)=0$, and at ordinates take right limits.*

*(i)* $N_t(T)=1+\vartheta(T)/\pi+\frac{t}{4\pi}\operatorname{Im}\ell(\frac12+iT)^2+S_t(T)$, *and* $S_0=S$.

*(ii) The main term is* $\frac{T}{2\pi}\log\frac{T}{2\pi e}+\frac78+\frac{t}{16}\log\frac{T}{2\pi}+O\big((1+t\log T)/T\big)$; *for* $0<t\le\frac12$ *a count of this form, with error terms, is [456, Theorem 1.5].*

*(iii) Let* $T>\frac52$. *If* $\operatorname{Re}\zeta_t(2+iy)>0$ *for* $0\le y\le T$, *and* $M(\omega)\ge\max\big(|\zeta_t(2+iT+\frac52e^{i\omega})|,|\zeta_t(2+iT+\frac52e^{-i\omega})|\big)$ *for* $0\le\omega\le2\pi$, *then*

$$|S_t(T)|\le\frac12+\frac{1}{\log\frac53}\Big(\frac{1}{2\pi}\int_0^{2\pi}\log M(\omega)\,d\omega-\log\operatorname{Re}\zeta_t(2+iT)\Big).$$

*Proof.* (i) By Lemma 15.2.1, $\xi_t$ is positive on the real axis and $\xi_t(1-s)=\xi_t(s)=\overline{\xi_t(\bar s)}$; by de Bruijn's strip theorem, as in the proof of Theorem 15.3.2, its zeros satisfy $(\operatorname{Re}s-\frac12)^2\le\max(\frac14-\frac t2,0)$, so for $t>0$ they lie in $0<\operatorname{Re}s<1$, as they do for $t=0$ because $\zeta$ has no zeros on $\operatorname{Re}s=1$. The argument principle on the rectangle with corners $-1$, 2, $2+iT$, $-1+iT$ and the two symmetries give $\pi N_t(T)$ as the change of $\arg\xi_t$ along the two segments. There $\xi_t=g\,e^{t\ell^2/4}\zeta_t$ with each factor continuous and nonzero, and $g(2)=1/\pi$ and $\ell(2)$ are real. The argument of $g$ changes by $\vartheta(T)+\pi$: $s(s-1)=(s-\frac12)^2-\frac14$ stays in the closed upper half plane and ends on the negative axis, and $\pi^{-s/2}\Gamma(\frac s2)$ has no zeros in $\operatorname{Re}s>0$. The factor $e^{t\ell^2/4}$ contributes $\frac t4\operatorname{Im}\ell(\frac12+iT)^2$. (ii) Stirling's series gives $\operatorname{Re}\ell(\frac12+iT)=\frac12\log\frac{T}{2\pi}+O(T^{-2})$, $\operatorname{Im}\ell(\frac12+iT)=\frac\pi4-\frac{7}{4T}+O(T^{-3})$ and $\vartheta(T)=\frac T2\log\frac{T}{2\pi e}-\frac\pi8+O(T^{-1})$. (iii) This is Backlund's argument [15]. Along the

vertical segment $|\arg\zeta_t| < \frac{\pi}{2}$, and along the horizontal one the argument moves by at most $\pi$ between consecutive zeros of $\operatorname{Re}\zeta_t(\sigma+iT)$, so $|S_t(T)| \le m+\frac12$ with $m$ the number of these zeros on $[\frac12, 2]$. They are zeros of $f(z) = \frac12[\zeta_t(z+iT)+\overline{\zeta_t(\bar z+iT)}]$, which is analytic on $|z-2| \le \frac52$, has $f(2) = \operatorname{Re}\zeta_t(2+iT)$ and $|f| \le M$ on the circle, and they lie in $|z-2| \le \frac32$; Jensen's formula on the circle of radius $\frac52$ gives $m\log\frac53 \le \frac{1}{2\pi}\int_0^{2\pi}\log M(\omega)\,d\omega - \log f(2)$. □

The count includes zeros off the line, so (iii) applies below $\Lambda$ as well. Ki, Kim and Lee [310, Theorem 1.4] give $S_t(T) = O_t(1)$ for $t > 0$, and the Polymath project [456, Theorem 1.5] gives $S_t(T) = O(\log T)$ uniformly in $0 < t \le \frac12$, with constants that are not numerical; an explicit bound needs the two hypotheses of (iii) with an explicit $M$ (Problem 15.6).

Corollary 15.2.8 (Conditional; the finite range computed). *Suppose that $v_{0.2}(10,\theta) \ge \frac{19}{2}V_{0.2}$ for every $\theta \ge \theta_1(0.2)$. Then $\tilde\varphi_{0.2,\alpha}$ is the Laplace transform of a GGC for every $\alpha \in [\frac12, 10]$. For $\theta_1(0.2) \le \theta \le 10^4$ the hypothesis is computed, not certified: on a grid of step $0.05$, with a Gauss Hermite quadrature whose error is not bounded, the least margin is $v_{0.2}(10,\theta) - \frac{19}{2}V_{0.2} = 0.1163$, at $\theta = 14.0768$ (`hx_far.py`).*

*Proof.* $\Lambda \le 0.2$ [419], and Proposition 15.2.5(iv) applies with $\alpha' = 10$, since $(\frac{19}{2})^2 \le \frac35\theta_1(0.2)^2$. This needs only $\theta_1(0.2) \ge (\operatorname{Var}T_{0.2})^{-1/4} = 12.76$, which follows from $\operatorname{Var}T_{0.2} = \sum_k \theta_k(0.2)^{-4} \ge \theta_1(0.2)^{-4}$ and gives $\frac35\theta_1(0.2)^2 \ge 97.69 \ge 90.25$. □

Remark 15.2.9. At $t = 0$ the hypothesis of the corollary, with $\xi$ in place of $\xi_{0.2}$, is a statement about the Gamma factor alone, up to an error below $10^{-3}$, since $|\operatorname{Re}(\zeta'/\zeta)(10+i\theta)| \le \sum_n \Lambda(n)n^{-10} < 10^{-3}$; but there the comparison in (iv) needs the zeros on the line. Along the flow the zeros are on the line and the Euler product is lost: by Lemma 15.2.6, $v_t(\alpha',\theta) = \operatorname{Re}\big[\ell + \frac{t}{2}\ell\ell' + \zeta_t'/\zeta_t\big](\alpha'+i\theta)$, and what is missing is a bound for $\zeta_t'/\zeta_t$ on the line $\operatorname{Re}s = 10$ that is uniform in the height (Problem 15.6(a)). The route of the note [236] is (iii): it needs the gaps between consecutive zeros of $\xi_{0.2}$ below a height $T_*$ to be below $2D_{0.2} = 9.2306$, and an explicit count above $T_*$ (Problem 15.6(b)). On our scan of $\xi_{0.2}(\frac12 + i\theta)$ through Lemma 15.2.6, for $0 \le \theta \le 4200$, the largest gap between consecutive sign changes is 6.87, at $\theta = 14.0768$, and the number of sign changes below $T$ differs from the main term of Proposition 15.2.7 by less than 0.65 at $T = 500, 1000, \dots, 4000$ (computed, `hx_scan.py`).

## 15.3 The Thorin phase diagram

Theorem 15.2.3 gives every time $t$ its own threshold basepoint, $\alpha^*(t) = \max(\frac12, \sup\operatorname{Re}\rho)$ over the zeros of $\xi_t$.

Definition 15.3.1. The *GGC region* $\mathcal{G}$ is the set of points $(t,\alpha)$ with $t \geq 0$ and $\alpha \geq \frac{1}{2}$ at which the shifted clock $\varphi_{t,\alpha}$ is the Laplace transform of a GGC, where at $\alpha = \frac{1}{2}$ the clock is $R_t$ of Section 8.4.

By Theorem 15.2.3, $(t,\alpha) \in \mathcal{G}$ exactly when every zero $x+iy$ of $H_t$ has $|y| \leq 2(\alpha - \frac{1}{2})$; at $\alpha = \frac{1}{2}$ this is the statement that the zeros are real, which is the GGC property of $R_t$. The slice $t = 0$ of $\mathcal{G}$ concerns the basepoint family of Section 14.8, a GGC for $\alpha \geq \sup \operatorname{Re}\rho$ and unconditionally for $\alpha \geq 1$, and the slice $\alpha = \frac{1}{2}$ the flow family $R_t$ of Section 8.4, known to be a GGC unconditionally only for $t \geq 0.2$ (Problem 14.3); the diagram joins these two one parameter families. The region is closed upward in $\alpha$, since the threshold condition is monotone.

Theorem 15.3.2 (de Bruijn's strip theorem in Thorin form). *(i) If $(t_0,\alpha_0) \in \mathcal{G}$, then $(t,\alpha) \in \mathcal{G}$ for every $t \geq t_0$ and every $\alpha \geq \frac{1}{2}$ with*

$$(\alpha - \tfrac{1}{2})^2 + \tfrac{t}{2} \;\geq\; (\alpha_0 - \tfrac{1}{2})^2 + \tfrac{t_0}{2}.$$

*(ii)* $\Lambda = \min\{\, t + 2(\alpha - \frac{1}{2})^2 : (t,\alpha) \in \mathcal{G} \,\}$.

*(iii) RH holds if and only if $(0,\frac{1}{2}) \in \mathcal{G}$, and if and only if $(t, \frac{1}{2} + \sqrt{\delta - t}/\sqrt{2}) \in \mathcal{G}$ for every $\delta > 0$ and some $t \in [0,\delta]$.*

*Proof.* De Bruijn's theorem on strips [112, Theorem 13], in the form used by Polymath15 [456], says that if the zeros of $H_{t_0}$ lie in $|\operatorname{Im} z| \leq \Delta$, then the zeros of $H_t$ lie in $(\operatorname{Im} z)^2 \leq \max(\Delta^2 - 2(t-t_0), 0)$ for $t \geq t_0$. With $\Delta = 2(\alpha_0 - \frac{1}{2})$ the zeros of $H_t$ then satisfy $|\operatorname{Im} z| \leq 2(\alpha - \frac{1}{2})$ whenever $4(\alpha - \frac{1}{2})^2 \geq 4(\alpha_0 - \frac{1}{2})^2 - 2(t - t_0)$, which is (i). For (ii), a point $(t,\alpha) \in \mathcal{G}$ puts the zeros of $H_t$ in $|\operatorname{Im} z| \leq 2(\alpha - \frac{1}{2})$, so by (i) those of $H_{t'}$ are real for $t' = t + 2(\alpha - \frac{1}{2})^2$, and $\Lambda \leq t + 2(\alpha - \frac{1}{2})^2$. Conversely $\Lambda \geq 0$ [463], the zeros of $H_\Lambda$ are real, and so $(\Lambda, \frac{1}{2}) \in \mathcal{G}$, where the quantity equals $\Lambda$. (iii) The first equivalence is (ii) with $\Lambda \geq 0$. For the second, the points named lie on the level set $t + 2(\alpha - \frac{1}{2})^2 = \delta$, so their membership gives $\Lambda \leq \delta$ for every $\delta > 0$, and conversely RH puts $(0,\frac{1}{2})$ in $\mathcal{G}$, from which (i) gives every point of every level set. □

Corollary 15.3.3 (The known region). *The following points lie in $\mathcal{G}$, and with them every point that Theorem 15.3.2(i) carries them to; the bounds on $\Lambda$ in (a) to (c) are those of Table 8.14.*

*(a) $(0,1)$, by the critical strip; it gives de Bruijn's $\Lambda \leq \frac{1}{2}$.*

*(b) $(0.2, 0.6)$, by the certificate of Polymath15 [456]; it gives $\Lambda \leq 0.22$.*

*(c) A point on the level set $t + 2(\alpha - \frac{1}{2})^2 = 0.2$, by the certificate of Polymath15 with the verification of Platt and Trudgian [419]; it gives $\Lambda \leq 0.2$.*

*(d) Conditionally on the half plane announced in October 2026 [398], $(0, \frac{7}{8})$; it gives $\Lambda \leq \frac{9}{32}$.*

*Proof.* (a) is the absence of zeros of $\zeta$ in $\operatorname{Re} s \geq 1$ and the functional equation. (b) is the statement proved in [456]: $H_{0.2}$ has no zeros with $\operatorname{Im} z > 0.2$, which is $(0.2, 0.6) \in \mathcal{G}$. (c) is [419], which takes a row of the table of [456] whose hypothesis their verification satisfies. (d) is immediate from [398]. The bounds on $\Lambda$ are Theorem 15.3.2(ii). □

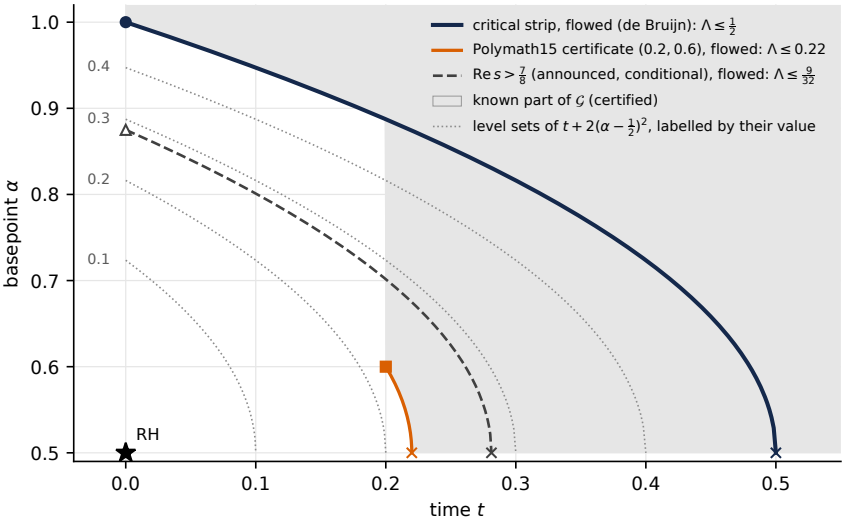



Figure 15.1: The Thorin phase diagram. Solid curves are the boundaries of certified GGC regions carried along the flow by Theorem 15.3.2(i), and the dashed curve is that of the region conditional on the half plane of Corollary 15.3.3(d); the dotted curves are the level sets of $t + 2(\alpha - \frac{1}{2})^2$, labelled by the bound on $\Lambda$ that a point on them gives. RH is the point $(0, \frac{1}{2})$. Script `phase_thorin.py`.

Figure 15.1 shows the region. A half plane at time zero improves the current bound only below $\alpha = \frac{1}{2} + \sqrt{0.1} \approx 0.816$, as Section 8.4 notes.

Remark 15.3.4 (Two semigroups). The two coordinates of the diagram carry two monotonicities, and they run in compatible directions. In $\alpha$ the Cauchy densities form a semigroup, so when no zero of $\xi_t$ lies to the right of $\alpha$ the Thorin density at a basepoint $\alpha' > \alpha$ is the Poisson smoothing of the one at $\alpha$, and positivity propagates to the right; going left is deconvolution, which does not preserve positivity (Section 11.4.7). In $t$ the heat flow carries real zeros to real zeros and narrows strips, so membership propagates forward in time. RH asks for the corner where both coordinates are smallest, and neither semigroup can be run backward.

## 15.4 What a Thorin computation sees at positive time

The Pólya route computes $\xi_t$ at the cost of $\xi$, since the flow only reweights the kernel (Section 8.4.1), and, when $(t,\alpha)\in\mathcal{G}$, the Thorin density of the shifted time $t$ clock is $\pi^{-1}v_t(\alpha,\theta)$ with

$$v_t(\alpha,\theta)=\operatorname{Re}\frac{\xi_t'}{\xi_t}(\alpha+i\theta),$$
$$\xi_t(s)=8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(2wu)\,du,$$
$$\xi_t'(s)=8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\,2u\sinh(2wu)\,du,$$

$w=s-\frac12$, by Remark 15.2.4; where membership in $\mathcal{G}$ is not known, as at $t=0$ with $\alpha=0.6$, the function $v_t/\pi$ is still defined. We computed $v_t(\alpha,\theta)$ at the two times $t=0$ and $t=0.2$ and the two basepoints $\alpha=0.6$ and 0.75, for $0\le\theta\le100$ in steps of 0.25, with no primes and no list of zeros. The integrals were evaluated by the trapezoid rule of Section 2.1.1 with step $1/160$ on $[0,1.6]$ (the tail beyond 1.6 is below $10^{-812}$) at $\lfloor 0.36\,\theta\rfloor+35$ digits, the term $0.36\,\theta$ absorbing the cancellation $|\xi_t(\alpha+i\theta)|\approx e^{-\pi\theta/4}$ that the Pólya route always carries (Section 2.1). A second discretisation agrees with the first in $v_t$ at all 1604 points of the four grids to a relative $2\times10^{-35}$; at $t=0$ the integrals, evaluated with 10 more digits, agree with the closed forms of $\xi$ and $\xi'/\xi$ through $\Gamma$, $\psi$, $\zeta$ and $\zeta'$ to at least 43 digits at both basepoints and at $\theta=0$, 17.5, 50 and 100, and at $t=0.2$ the integral for $\xi_t'$ agrees with a central difference of $\xi_t$ to 30 digits at two points.

| | $\alpha=0.6$ | | | $\alpha=0.75$ | | |
|---|---|---|---|---|---|---|
| $\theta$ | $t=0$ | $t=0.2$ | change | $t=0$ | $t=0.2$ | change |
| 0 | 0.004621 | 0.004640 | +0.4% | 0.011551 | 0.011599 | +0.4% |
| 17.50 | 0.0230 | 0.0231 | +0.4% | 0.0573 | 0.0575 | +0.4% |
| 23.01 | 0.0582 | 0.0578 | −0.7% | 0.1439 | 0.1429 | −0.7% |
| 27.67 | 0.0389 | 0.0393 | +0.9% | 0.0968 | 0.0976 | +0.9% |
| 35.27 | 0.0520 | 0.0526 | +1.1% | 0.1291 | 0.1304 | +1.1% |
| 42.12 | 0.1539 | 0.1496 | −2.8% | 0.3729 | 0.3627 | −2.7% |
| 45.63 | 0.0569 | 0.0579 | +1.8% | 0.1412 | 0.1438 | +1.8% |
| 95.26 | 0.5625 | 0.4829 | −14.2% | 1.2474 | 1.0906 | −12.6% |

Table 15.2: The value at the origin and at the minimum of a selection of the valleys of $v_t(\alpha,\cdot)$ between consecutive zeros. The ordinate $\theta$ is that of the minimum at $t=0$ and $\alpha=0.6$; at $t=0.2$ the minimum lies 0.06 to 0.08 to the left, and at $\alpha=0.75$ it moves by less than 0.002. Script `phase_thorin.py`.

Proposition 15.4.1 (Computed). *On $0\le\theta\le100$:*

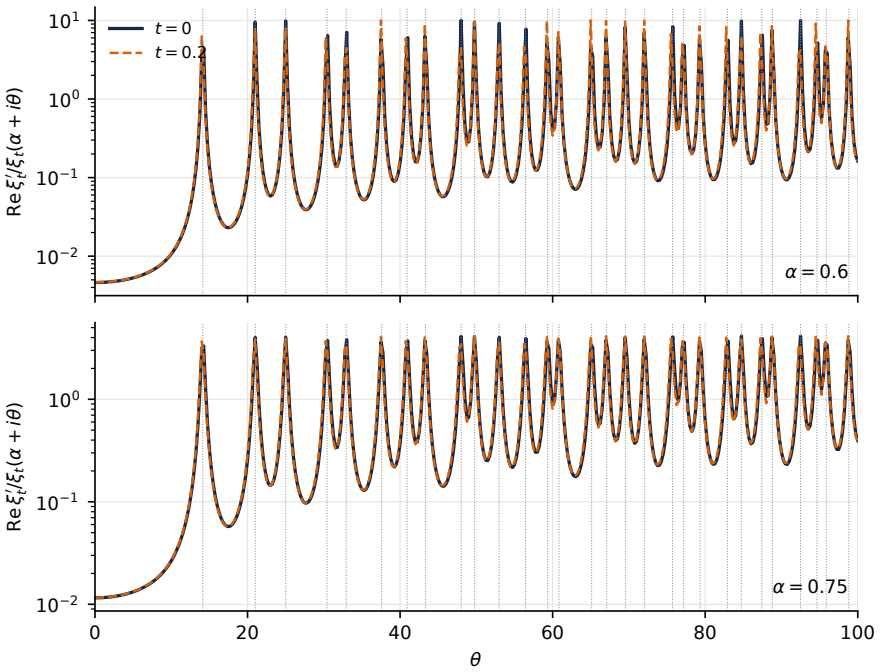



Figure 15.2: $v_t(\alpha, \theta) = \mathrm{Re}\ \xi_t'/\xi_t(\alpha + i\theta)$, which is $\pi$ times the Thorin density of the shifted time $t$ clock when $(t, \alpha) \in \mathcal{G}$, on the lines $\mathrm{Re}\, s = 0.6$ and $0.75$, at $t = 0$ (solid) and $t = 0.2$ (dashed), from the Pólya integral. Dotted lines mark the ordinates of the zeros of $\zeta$. Script `phase_thorin.py`.

*(i)* *At both times and both basepoints the minimum of $v_t(\alpha, \cdot)$ is at the origin, $v_t(\alpha, 0) = \xi_t'/\xi_t(\alpha)$, in agreement with Theorem 9.1.18 at $t = 0$, and $v_t(\alpha, \cdot)$ is positive.*

*(ii)* *Between consecutive zeros the valleys are set by the gaps relative to the kernel width $\alpha - \frac{1}{2}$, and they scale roughly in proportion to it: at $t = 0$ and $\alpha = 0.6$ they range from $0.023$ near $\theta = 17.5$ to $0.56$ near $\theta = 95$, and at $\alpha = 0.75$ they are $2.2$ to $2.5$ times larger.*

*(iii)* *The $28$ valleys above $\theta = 10$, each taken at its minimum, change between $t = 0$ and $t = 0.2$ by a median of $2.3\%$ and at most $14.2\%$ at $\alpha = 0.6$ ($2.2\%$ and $12.6\%$ at $\alpha = 0.75$), the largest change being that of the valley near $\theta = 95$; the zeros below $60$ move by $0.044$ to $0.118$, with mean $0.075$, the first from $14.135$ to $14.077$ (Section 15.5.2).*

Part (ii) corrects a natural expectation. Averaged over a window wider than the spacing of the zeros, $v_t(\alpha, \theta)$ is close to $\frac{1}{2}\log(\theta/2\pi)$, so one might expect the certificate to enjoy a margin of that size. At $t = 0$ Remark 9.1.17 proves a margin

that does grow with the height, $v_\alpha(\theta)/\pi \geq c \log\theta$ for every $c < (\alpha - \frac{1}{2})/(\pi(2\alpha+1))$ and all large $\theta$ outside a set of density zero, but its constant is proportional to $\alpha - \frac{1}{2}$: at $\alpha = 0.6$ it is below $0.015 \log\theta$, against the average $(2\pi)^{-1}\log(\theta/2\pi)$ of $v_\alpha/\pi$, and its exceptional set contains the window of every zero to the right of $\operatorname{Re} s = \alpha$. Below height 100 the computed margin is smaller still, set by the largest gaps between consecutive zeros, because the kernel width $\alpha - \frac{1}{2}$ is small against the mean spacing $2\pi/\log(\theta/2\pi)$.

On the windows computed here, below height 100, the clock of $\xi_{0.2}$, which is a GGC at the basepoint 0.6 by Corollary 15.3.3(b), and the clock of $\xi$, which is a GGC at the basepoint 0.6 only if there is no zero with $\beta > 0.6$ anywhere, are described by the functions $v_{0.2}/\pi$ and $v_0/\pi$, whose valleys differ by a median of 2.3% and at most 14.2%, and both are positive. Nothing in the computed density distinguishes the theorem from the conjecture. The Thorin side therefore does not offer a cheaper finite computation than the flow; it offers the same finite computation in the language of this book, and the whole question is where the tail comes from. The same comparison at $t = 0.2$ and $t = 0$ above the barrier of Polymath15, at heights near $3 \times 10^{10}$, would require a Riemann Siegel type evaluator of $H_{0.2}$ such as the effective approximation of [456] (Section 2.2 gives the one for $\zeta$); Problem 15.1 asks for it.

## 15.5 A certificate for one point of the diagram

This section writes out the finite part of the statement $(0.2, 0.6) \in \mathcal{G}$ as a certified computation with explicit constants, and runs it in ball arithmetic below height 160. The statement is that $\xi_{0.2}$ has no zero with real part above 0.6. Given a barrier $X$ (in the variable of $H_{0.2}$, so that it sits at height $X/2$ for $\xi_{0.2}$; see Theorem 15.5.6) beyond which this is known [456], it is the vanishing of the winding number of $\xi_{0.2}$ around the rectangle $Q = [0.6, 0.9] \times [0, X/2]$. Throughout, $t = 0.2$.

### 15.5.1 The lemmas

Lemma 15.5.1 (Bottom edge). *$\xi_t(\sigma) > 0$ for every real $\sigma$ and every $t \geq 0$.*

*Proof.* Lemma 15.2.1(ii). □

Lemma 15.5.2 (Right edge). *For $0 \leq t \leq \frac{1}{2}$ every zero of $\xi_t$ satisfies $|\beta - \frac{1}{2}| \leq \frac{1}{2}\sqrt{1-2t}$. In particular for $t = 0.2$ every zero has $\beta \leq 0.8873$, so $\xi_{0.2}$ has no zero on $\operatorname{Re} s = 0.9$.*

*Proof.* De Bruijn's theorem on strips, as in the proof of Theorem 15.3.2, with $\Delta = 1$ from the critical strip and $\beta - \frac{1}{2} = \frac{1}{2}\operatorname{Im} z$ up to sign. □

LEMMA 15.5.3 (Tilted bound). *Let $0 < \eta < \pi/8$, $t \geq 0$ and $s = \frac{1}{2} + x + i\theta$ with $\theta \geq 0$. Then*

$$|\xi_t(s)| \leq K_\eta(t,x)\, e^{-2\eta\theta}, \qquad K_\eta(t,x) = 4\int_{-\infty}^{\infty} \left|e^{t(u+i\eta)^2}\right| |\Phi_{\mathrm{dBN}}(u+i\eta)|\, e^{2xu}\, du,$$

*and $K_\eta(t,x)$ is finite and even in $x$. Write $\bar K_\eta(t,R) = \max_{|x|\leq R} K_\eta(t,x)$.*

*Proof.* Since $\Phi_{\mathrm{dBN}}$ is even, $\xi_t(s) = 4\int_{\mathbb{R}} e^{tu^2}\Phi_{\mathrm{dBN}}(u)e^{2(x+i\theta)u}\, du$. The kernel extends analytically to $|\operatorname{Im} u| < \pi/8$ with the same doubly exponential decay, which is Proposition 2.1.1 in the variable $v = 2u$, so the contour may be moved to $\operatorname{Im} u = \eta$, and $|e^{2i\theta(u+i\eta)}| = e^{-2\eta\theta}$. □

LEMMA 15.5.4 (Derivative bound). *With the notation of Lemma 15.5.3, for $r > 0$, $|\xi_t'(s)| \leq r^{-1}\bar K_\eta(t, |x|+r)\, e^{-2\eta(\theta-r)}$, and the factor $e^{-2\eta(\theta-r)}$ may be replaced by 1 when $\theta \leq r$.*

*Proof.* Cauchy's estimate on the disc of radius $r$ about $s$. A point $z$ of the disc has $|\operatorname{Re} z - \frac{1}{2}| \leq |x| + r$ and $\operatorname{Im} z \geq \theta - r$. If $\operatorname{Im} z \geq 0$, Lemma 15.5.3 bounds $|\xi_t(z)|$ by $\bar K_\eta(t,|x|+r)e^{-2\eta \operatorname{Im} z}$, which is at most $\bar K_\eta(t,|x|+r)e^{-2\eta(\theta-r)}$ and at most $\bar K_\eta(t,|x|+r)$. If $\operatorname{Im} z < 0$, which occurs only when $\theta < r$, then $\xi_t(\bar z) = \overline{\xi_t(z)}$ gives the bound $\bar K_\eta(t,|x|+r)$ for $|\xi_t(z)|$. So on the disc $|\xi_t| \leq \bar K_\eta(t,|x|+r)e^{-2\eta(\theta-r)}$, and $|\xi_t| \leq \bar K_\eta(t,|x|+r)$ when $\theta \leq r$. □

LEMMA 15.5.5 (Grid). *Let $s_0, s_1$ be points of a segment on which $\xi_t$ has no zero, and suppose $|s_1 - s_0| \cdot \sup_{[s_0,s_1]} |\xi_t'| < |\xi_t(s_0)|$. Then the change of $\arg \xi_t$ along the segment equals the principal value $\operatorname{Arg}(\xi_t(s_1)/\xi_t(s_0)) \in (-\pi, \pi)$.*

*Proof.* The image of the segment lies in the open disc of radius $|\xi_t(s_0)|$ about $\xi_t(s_0)$, which does not contain 0, and on that disc a continuous branch of the argument varies by less than $\pi$ from its value at the centre. □

THEOREM 15.5.6 (The certificate). *Let $X$ be a barrier for $(0.2, 0.6)$ in the variable of $H_{0.2}$, as Polymath15 provide one [456] (no script of this book checks it), that is, a number such that $H_{0.2}$ has no zero $z$ with $\operatorname{Re} z \geq X$ and $\operatorname{Im} z > 0.2$, so that $\xi_{0.2}$ has no zero with $\operatorname{Re} s > 0.6$ and $\operatorname{Im} s \geq X/2$. Let $s_0, \ldots, s_M$ be points of the left, top and right edges of $Q$, in order along $\partial Q$ from one bottom corner, $s_0$, to the other, $s_M$, and including the two top corners, and let $B_j$ be ball enclosures of $\xi_{0.2}(s_j)$ with radii $\varepsilon_j$. Suppose that for every $j$ there are $r > 0$ and $\eta \in (0, \frac{\pi}{8})$ such that*

$$|s_{j+1} - s_j|\, \frac{\bar K_\eta(0.2,\, 0.4 + r)}{r}\, e^{-2\eta(\theta_j - r)} + \varepsilon_j + \varepsilon_{j+1} < \min |B_j|,$$
$$\theta_j = \min(\operatorname{Im} s_j, \operatorname{Im} s_{j+1}),$$

*where the factor* $e^{-2\eta(\theta_j - r)}$ *may be replaced by* 1 *when* $\theta_j \le r$*, and that* $\sum_j \operatorname{Arg}(B_{j+1}/B_j)$*, evaluated in ball arithmetic, lies within* $(-\pi, \pi)$*. Then* $\xi_{0.2}$ *has no zero in* $Q$*, so* $(0.2, 0.6) \in \mathcal{G}$ *and* $\Lambda \le 0.22$*.*

*Proof.* By Lemma 15.5.1, $\xi_{0.2}$ is real and positive on the bottom edge, so $\arg \xi_{0.2}$ does not change along it, and by Lemma 15.5.2 the right edge is zero free. The hypothesis gives $|\xi_{0.2}(s_j)| \ge \min |B_j|$ at the grid points, and Lemma 15.5.4 with $|x| \le 0.4$ and $\operatorname{Im} z \ge \theta_j$ on the segment from $s_j$ to $s_{j+1}$ bounds the derivative between them, so Lemma 15.5.5 applies to every step, and the winding number is the integer nearest to $\frac{1}{2\pi} \sum_j \operatorname{Arg}(B_{j+1}/B_j)$, which is 0. So $\xi_{0.2}$ has no zero in $Q$; it has none with $\operatorname{Re} s > 0.9$ by Lemma 15.5.2, none with $\operatorname{Re} s > 0.6$ and $\operatorname{Im} s \ge X/2$ by the barrier, and its zeros in $\operatorname{Im} s < 0$ are the conjugates of those in $\operatorname{Im} s > 0$. Hence it has no zero with $\operatorname{Re} s > 0.6$, which is $(0.2, 0.6) \in \mathcal{G}$, and the last clause is Theorem 15.3.2. □

The number of grid points is of order $(X/2) \sup_j K_\eta e^{2\eta r}/(r\,|\xi_t(s_j)| e^{2\eta\theta_j})$. The decay of $\xi_t$, of order $e^{-\pi\theta/4}$ away from its zeros, cancels against the decay $e^{-2\eta\theta}$ of the derivative bound only at $\eta = \pi/8$, which Lemma 15.5.3 excludes, and $K_\eta$ grows without bound as $\eta \uparrow \pi/8$ (Table 15.3). With a tilt $\eta < \pi/8$ the count grows like $e^{(\pi/4 - 2\eta)\theta}$. We have no proof that the count is $O(X \log X)$, or polynomial in $X$, when the tilt is chosen at each height. In the runs below the left edge needs on average about 45 steps per unit height up to height 60, 73 up to 120 and 94 up to 160 (Theorem 15.5.8).

Proposition 15.5.7 (The certificate in Thorin form). *Let* $t \ge 0$ *and* $\frac{1}{2} < \alpha < \sigma_1$*, and suppose* $v_t(\sigma, \theta) > 0$ *for every* $(\sigma, \theta) \in [\alpha, \sigma_1] \times [0, T]$ *at which* $\xi_t(\sigma + i\theta) \ne 0$*. Then* $\xi_t$ *has no zero* $\beta + i\gamma$ *with* $\alpha < \beta \le \sigma_1$ *and* $0 < \gamma < T$*.*

*Proof.* Near a zero $\rho$ of multiplicity $m$, $\xi_t'/\xi_t(s) = m/(s-\rho) + O(1)$, so at $s = \rho - \delta$ with small $\delta > 0$ the real part is $-m/\delta + O(1) < 0$. If $\alpha < \beta \le \sigma_1$, these points lie in the rectangle for small $\delta$. This is the argument of Remark 1.2.4, restricted to a rectangle of finite height. □

Positivity on the single line $\operatorname{Re} s = \alpha$ is not enough (Remark 1.2.4), which is why the proposition asks for the whole rectangle. It is the pointwise counterpart of Theorem 15.5.6: the same evaluations of $\xi_t$ and $\xi_t'$ certify it, with a two dimensional grid and a bound on the derivative of $v_t$ in place of the argument increments, and the margin is the one measured in Proposition 15.4.1.

### 15.5.2 Constants and the march

Table 15.3 gives the constants of Lemma 15.5.3 at $t = 0.2$ and the values of $\xi_{0.2}$ on the real axis that fix the scale.

A floating point march of Lemma 15.5.5 around $Q_{60} = [0.6, 0.9] \times [0, 60]$, with the Pólya route evaluator (the trapezoid rule of Section 2.1.1 with the weight $e^{tu^2}$)

| $\eta$ | decay $e^{-2\eta\theta}$ | $K_\eta(0.2, 0.4)$ | $K_\eta(0.2, 0.9)$ |
|---|---|---|---|
| 0.30 | 0.60 | 23.67 | 25.31 |
| 0.35 | 0.70 | 199.2 | 231.1 |
| 0.38 | 0.76 | 4847 | 6776 |

| $\sigma$ | $\xi_{0.2}(\sigma)$ |
|---|---|
| 0.5 | 0.49827 |
| 0.6 | 0.49839 |
| 0.9 | 0.50013 |
| 1.0 | 0.50117 |
| 1.5 | 0.50996 |

Table 15.3: Constants for the certificate at $t = 0.2$, in floating point. Script `phase_cert.py`.

at 40 digits (working precision 72 digits) and an adaptive step keeping $|\xi_t(s_{j+1}) - \xi_t(s_j)| < \frac{1}{2}|\xi_t(s_j)|$, returned the winding $-2.9 \times 10^{-72}$ in 910 evaluations. The same evaluator located the zeros of $\xi_{0.2}$ on the critical line below 60,

$$14.077,\ 20.947,\ 24.952,\ 30.330,\ 32.888,\ 37.503,$$
$$40.833,\ 43.282,\ 47.887,\ 49.730,\ 52.905,\ 56.358,\ 59.237,$$

thirteen zeros, as for $\xi$ below 60, each moved slightly by the flow (five of them are in Table 8.8). For the control function that moves the pair at 30.330 and 32.888 to the quadruple $\frac{1}{2} \pm \frac{1}{4} + 31.609i$ and its conjugates, as in Theorem 5.2.1 (there for $\xi$ and ordinates $1000 < A < B \le 2000$), the computed winding was 1 to within $10^{-35}$ in 905 evaluations: exactly one of the four new zeros, $0.75 + 31.609i$, lies in $Q_{60}$.

The finite part of Theorem 15.5.6 was then run in ball arithmetic, adding every truncation error to the radius of the enclosure. The evaluator is the same trapezoid rule, with $w = s - \frac{1}{2} = x + i\theta$ and $u_j = jh$: $\xi_t(s) = 8h \sum_{j=0}^{J} \omega_j e^{tu_j^2} \Phi_{\mathrm{dBN}}(u_j) \cosh(2wu_j) + E$, where $Jh \ge 3$, $\omega_0 = \frac{1}{2}$, $\omega_j = 1$ for $j \ge 1$, and $|E| \le E_{\mathrm{cut}} + E_{\mathrm{trap}}$. For $u \ge 3$ the kernel satisfies $|\Phi_{\mathrm{dBN}}(u)| \le 60\, e^{9u} e^{-\pi e^{4u}}$, so that for $|x| \le 1$ the summand is at most $8h\, G(u_j)$ with $G(u) = 60 \exp(tu^2 + 11u - \pi e^{4u})$, which decreases with $-G'/G \ge \lambda = 4\pi e^{4Jh} - 2tJh - 11$ for $u \ge Jh$, and the nodes beyond $Jh$ contribute at most $E_{\mathrm{cut}} = 8G(Jh)/\lambda < 10^{-229000}$. The error $E_{\mathrm{trap}}$ of the rule is bounded by Poisson summation on the strip $|\operatorname{Im} u| < a$, $a = 0.30 < \pi/8$, in which the integrand of Lemma 15.5.3 is analytic: $E_{\mathrm{trap}} \le 2K_{0.30}(0.2, x) \cosh(2a\theta)/(e^{2\pi a/h} - 1)$, with $K_{0.30}(0.2, x) \le 23.72$ for $|x| \le 0.4$, bounded in the way described below, and $h$ is taken so that this is below $5 \times 10^{-4} e^{-\pi T/4}$ at the top of the rectangle of height $T$: $h = 0.0199$, $0.0106$ and $0.008$ at $T = 60$, $120$ and $160$, at the working precisions 219, 288 and 334 bits. The series for the kernel is summed until its tail

is below $2^{-\mathrm{prec}-30}$, the tail after $N$ terms being enclosed by

$$\sum_{n>N} \left|(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}\right| \\ \le |e^{9u}| \Big(2\pi^2 + \frac{3\pi}{\kappa}\Big)\Big(\frac{4}{\pi\kappa}\Big)^2 e^{-2}\, \frac{e^{-\pi(N+1)^2\kappa/2}}{1 - e^{-\pi(N+1)\kappa}}, \qquad \kappa = \mathrm{Re}(e^{4u}),$$

using $|e^{5u}| \le |e^{9u}|/\kappa$, $n^4 e^{-\pi n^2\kappa/2} \le (4/\pi\kappa)^2 e^{-2}$ and $n^2 \ge (N+1)^2 + 2(N+1)(n - N-1)$ for the geometric sum, for every $N$ and every $\kappa > 0$. At a real node $\kappa = e^{4u}$; at a complex $u$ $\kappa$ is a lower bound of $\mathrm{Re}(e^{4u})$ over the input ball. The constants $\bar{K}_\eta(0.2, 0.4 + r)$ were bounded above by interval subdivision of $[0, 3]$ into $3 \cdot 2^{14}$ cells, plus the tail $u \ge 3$, below $e^{-25000}$ since $\mathrm{Re}\, e^{4(u+i\eta)} \ge e^{12}\cos(4\eta) > 8000$ there, for the three tilts $\eta = 0.30$, 0.35 and 0.38 and the three radii $r = \frac{1}{4}$, $\frac{3}{8}$ and $\frac{1}{2}$: 24.37, 24.82 and 25.36, then 212.33, 221.11 and 231.71, then 5633.6, 6170.8 and 6841.2, each rounded up. The ordinates of the grid points are exact dyadic numbers, and the abscissae 0.6 and 0.9 are balls of radius $2^{-\mathrm{prec}}$ enclosing them, so that the radius of $\xi_t(s_j)$ is that of the evaluator. At each step the tilt $\eta$ and the radius $r$ giving the smallest derivative bound are used, the step is $0.8\,|\xi_t(s_j)|_{\mathrm{low}}$ divided by that bound, capped at 0.5, and the inequality of Theorem 15.5.6, with the exact length of the step and its exact lowest height, is then verified in ball arithmetic, the step being halved when it fails.

THEOREM 15.5.8. *$\xi_{0.2}$ has no zero $s$ with $0.6 \le \mathrm{Re}\, s \le 0.9$ and $0 \le \mathrm{Im}\, s \le 160$.*

*Proof.* For $T = 60$, 120 and 160 the script walks the boundary of $Q_T = [0.6, 0.9] \times [0, T]$ counterclockwise, up the right edge, along the top and down the left edge, with the steps of Theorem 15.5.6. The bottom edge is not walked, since $\xi_{0.2}$ is real and positive on it (Lemma 15.5.1). The run took 4483 steps for $T = 60$ (1737 up the right edge, 31 along the top and 2715 down the left edge), 13,679 for $T = 120$ (4955, 11 and 8713) and 23,219 for $T = 160$ (8142, 20 and 15,057), and returned the winding number enclosures $[\pm 3.94 \times 10^{-5}]$, $[\pm 1.24 \times 10^{-6}]$ and $[\pm 3.47 \times 10^{-7}]$. In every run the inequality of Theorem 15.5.6 was verified in ball arithmetic at every step and no step failed, and the sum of the arguments lies within $\pi$ of 0, so by Lemmas 15.5.1 to 15.5.5 the winding number is 0 and $\xi_{0.2}$ has no zero in $Q_T$, which for $T = 160$ is the theorem. Script `phase_rig.py`, data `phase_rig.json`. □

With Lemma 15.5.2 this says that every zero of $\xi_{0.2}$ below height 160 has $|\beta - \frac{1}{2}| < 0.1$; as a control, the same script run around $[0.4, 0.6] \times [0, 60]$, which contains the thirteen zeros on the line, returns the winding number enclosure $[13.0000 \pm 5.04 \times 10^{-5}]$ in 4841 steps. The barrier of Polymath15 is near $X = 6 \times 10^{10}$, height $3 \times 10^{10}$ for $\xi_{0.2}$, and reaching it in this framework requires the evaluator of the programme below.

## 15.6 Where the uniform step could come from

Section 12.1.2 names three mechanisms that could supply the missing uniform step: a positive approximating family whose positivity is proved independently, a Hilbert Pólya operator, or a direct Weil inequality.

### 15.6.1 The Weil Gram matrix on compactly supported test functions

For $h$ smooth enough, even or not (each term depends on $h$ only through its even part $h_e$, defined below), with $\hat h(r) = \int h(x)e^{irx}\,dx$, the explicit formula reads

$$\sum_\rho \hat h(\gamma_\rho) = \hat h(\tfrac{i}{2}) + \hat h(-\tfrac{i}{2}) - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\big(h(\log n) + h(-\log n)\big) + \frac{1}{2\pi}\int_{\mathbb{R}} \hat h(r)\ \mathrm{Re}\,\psi\big(\tfrac14 + \tfrac{ir}{2}\big)\,dr - h(0)\log\pi,$$

with $\rho = \frac12 + i\gamma_\rho$ and $\psi = \Gamma'/\Gamma$. For $h = g * \tilde g$ with $\tilde g(x) = \overline{g(-x)}$ and $g$ supported in $[-L/2, L/2]$, the left side is $\sum_\rho |\hat g(\gamma_\rho)|^2 \ge 0$ under RH, which is Weil's criterion [512], $h$ is supported in $[-L, L]$, and the prime sum runs over prime powers $n \le e^L$. The archimedean term was evaluated in $x$ space,

$$\frac{1}{2\pi}\int \hat h(r)\ \mathrm{Re}\,\psi\big(\tfrac14 + \tfrac{ir}{2}\big)\,dr = \int_0^\infty \Big[\frac{h(0)e^{-2x}}{x} - \frac{2h_e(x)e^{-x/2}}{1-e^{-2x}}\Big]dx,$$
$$h_e(x) = \tfrac12\big(h(x) + h(-x)\big),$$

which is far better conditioned than the $r$ integral for compactly supported $h$: truncated at $r = 400$, the $r$ integral carries an error of $6\times10^{-6}$ in the archimedean term of the diagonal entry at $w = 0.1$, more than two thousand times the smallest eigenvalue $2.5\times10^{-9}$ of Table 15.4. With hat functions $g_j(x) = \max(0, 1 - |x - c_j|/w)$ at nodes spaced $w$ apart inside $[-L/2, L/2]$, $N = L/w - 1$ functions in all, $h_{jk} = g_j * \tilde g_k$ is an explicit piecewise cubic, and the Gram matrix $G_{jk} = W(h_{jk})$ is Hermitian, with $W(g*\tilde g) \ge 0$ on the span exactly when $G$ is positive semidefinite.

The experiment says three things (Tables 15.4 and 15.5, Figure 15.3). First, $\lambda_{\min} > 0$ throughout and it falls roughly geometrically in $L$, by a factor between about 2.5 and 20 per unit of $L$ up to $L = 6$, as it must, since the nodes are nested (for the values of $L$ used), so that the matrix for a shorter support is a principal submatrix of the matrix for a longer one, and the smallest eigenvalue of the Weil form on $[-L/2, L/2]$ tends to zero as $L$ grows: a long enough support can make $\hat g$ small at every low ordinate. Second, the two sides of Weil's identity agree to a relative $2\times10^{-4}$ at every $(w, L)$, down to $\lambda_{\min} = 2.5\times10^{-9}$; and the margin is carried by a handful of zeros, zero 41 at height 124.3 carrying 32% at $w = 0.1$, $L = 6$. Third, at $w = 0.05$ a pair displaced at height $g \approx 17.6$ makes $\lambda_{\min}$ negative

| $w$ | $L$ | $N$ | $\lambda_{\min}$ | $\lambda_{\max}$ | pole, −prime, archimedean |
|---|---|---|---|---|---|
| 0.25 | 1 | 3 | $4.22 \times 10^{-4}$ | 0.034 | 0.3468, −0.0141, −0.3322 |
| 0.25 | 2 | 7 | $2.22 \times 10^{-5}$ | 0.050 | 0.8410, −0.2472, −0.5938 |
| 0.25 | 4 | 15 | $4.70 \times 10^{-7}$ | 0.096 | 1.6494, −0.8550, −0.7944 |
| 0.25 | 6 | 23 | $3.56 \times 10^{-8}$ | 0.144 | 2.7255, −1.8067, −0.9187 |
| 0.1 | 1 | 9 | $1.84 \times 10^{-5}$ | 0.083 | 0.1287, −0.0019, −0.1268 |
| 0.1 | 2 | 19 | $1.27 \times 10^{-6}$ | 0.149 | 0.2900, −0.0676, −0.2224 |
| 0.1 | 4 | 39 | $5.70 \times 10^{-8}$ | 0.290 | 0.7281, −0.3981, −0.3300 |
| 0.1 | 6 | 59 | $2.48 \times 10^{-9}$ | 0.431 | 1.1050, −0.7353, −0.3697 |

Table 15.4: The Weil Gram matrix on hat functions. The last column splits $\lambda_{\min} = v^{*}Gv$ for the minimising unit vector into the pole, prime (with its sign) and archimedean pieces. Script `phase_weil.py`, output `phase_weil.json`.

| $w$ | $L$ | $\lambda_{\min}$ (arithmetic) | $\sum_{\rho} \lvert\hat{g}(\gamma_{\rho})\rvert^2$ (zeros) | zeros carrying most of the sum |
|---|---|---|---|---|
| 0.25 | 1 | $4.2184 \times 10^{-4}$ | $4.2184 \times 10^{-4}$ | #2 (56%), #4 (20%), #5 (11%) |
| 0.25 | 3 | $1.8238 \times 10^{-6}$ | $1.8237 \times 10^{-6}$ | #9 (44%), #11 (24%), #20 (5%) |
| 0.25 | 6 | $3.5562 \times 10^{-8}$ | $3.5560 \times 10^{-8}$ | #10 (40%), #20 (13%), #30 (9%) |
| 0.1 | 2 | $1.2691 \times 10^{-6}$ | $1.2691 \times 10^{-6}$ | #13 (37%), #15 (15%), #16 (15%) |
| 0.1 | 4 | $5.7008 \times 10^{-8}$ | $5.7004 \times 10^{-8}$ | #14 (36%), #13 (23%), #16 (7%) |
| 0.1 | 6 | $2.4846 \times 10^{-9}$ | $2.4841 \times 10^{-9}$ | #41 (32%), #15 (17%), #14 (8%) |

Table 15.5: The smallest eigenvalue from the arithmetic side against the sum of squares of $\hat{g}_{\min}$ over the first 1500 zeros, with the zeros (by index) that carry the largest shares. Script `phase_weil.py`, output `phase_weil.json`.

at every $L \le 4$, one at $g \approx 54.7$ only at $L = 4$ and one at $g \approx 146.7$ at no $L \le 4$, an instance of the finite information principle of Section 15.1.1. How fine $w$ must be to see a pair at height $g$ is not fixed by one resolution: at $w = 0.2$ the pair at $g \approx 17.6$ is already seen at $L = 2$, with $wg \approx 3.5$.

The sign structure is the reverse of the Lévy picture of Section 9.1, where the pole of $\zeta$ is a negative input to $\log \xi$; in the Weil form the pole, on the minimising vectors of Table 15.4, contributes positively. A certificate for the Weil form would therefore have to show, for every $L$, that the three terms together are nonnegative on the whole span, although the pole term alone is not (for real odd $g$ it is $-2G(\frac{1}{2})^2$, with $G(a) = \int g(x)e^{ax}\,dx$), and on the minimising vectors the pole term outweighs the prime and archimedean terms only by a margin that tends to zero. No uniformly positive quadratic form can do that. What could is a factorisation $W(g * \tilde{g}) = \|Ag\|^2$ with $A$ built from the arithmetic and exact positivity, the shape of the Hilbert Pólya mechanism, and Problem 15.4 asks for the first experiment in that direction.

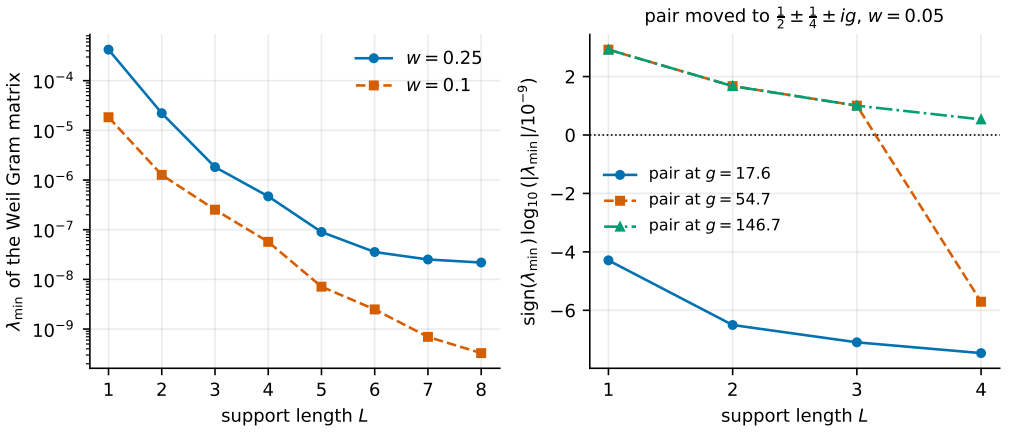



Figure 15.3: Left: $\lambda_{\min}$ of the Weil Gram matrix against the support length $L$ for two resolutions. Right: a pair of zeros at height $g$ moved to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$, signed logarithm of $\lambda_{\min}$ for $w = 0.05$. The pair at $g \approx 17.6$ is detected at every $L$, the pair at $g \approx 54.7$ only at $L = 4$, and the pair at $g \approx 146.7$ not at all within $L \leq 4$. Script `phase_weil.py`, output `phase_weil.json`.

### 15.6.2 The zero gas along the flow

In the variable $z = 2\gamma$ the zeros of $H_t$ move, while simple, by $dz_k/dt = 2\sum_{j \neq k}(z_k - z_j)^{-1}$ (Section 8.4.1, with five of the zeros at four times in Table 8.8). We integrated this system for the first 600 zeros, heights up to 939.02, with a frozen tail for the omitted zeros (the zeros 601 to 1700 held at their positions at $t = 0$ and a continuum of density $\frac{1}{4\pi}\log(z/4\pi)$ beyond them, whose drift on $z_k$ is $-4z_k \sum_{j>600} z_j^{-2}$ to leading order), backward from $t = 0$ until the first gap fell below 0.02 and forward to $t = 0.3$. Along the trajectory we recorded the minimum normalised gap $s_{\min}(t) = \min_k (z_{k+1} - z_k)\frac{1}{4\pi}\log(z_{k+1}/4\pi)$ and the Csordas, Smith and Varga quantity $\min_k \Delta_k^2 g_k$ over the first 400 pairs, with $\Delta_k = z_{k+1} - z_k$ and $g_k = \sum_{j \neq k,k+1}[(z_k - z_j)^{-2} + (z_{k+1} - z_j)^{-2}]$, where $j$ runs over all the other zeros $\pm z_j$ of the model, the frozen tail and the continuum included; a pair with $\Delta_k^2 g_k < \frac{4}{5}$ is a Lehmer pair in the sense of [101] and Section 8.4.6.

Proposition 15.6.1 (Computed; the upper end of (iii) certified). *(i) The backward integration stops at $t = -0.0493$, when the pair at ordinates 750.656 and 750.966, the 453rd and 454th zeros, collides; the isolated pair estimate is $-D_0^2/8 = -0.048$. (ii) At the 171 times sampled in $[-0.049, 0.30]$ both $s_{\min}(t)$ and $\min_k \Delta_k^2 g_k$ are strictly increasing, from 0.0076 to 0.502 and from 0.035 to 0.977; at $t = 0$ the minimiser of $\Delta_k^2 g_k$ is the pair at 564.16 and 564.51, with value 0.238. (iii) For the pair near 185, the pair $(\gamma_{71}, \gamma_{72})$ of Section 8.4.2, certified signs of $\xi_t(\frac{1}{2} + i\theta)$ in Arb at 640 bits (the trapezoid sum of Section 2.1.1 with an enclosure of its dis-*

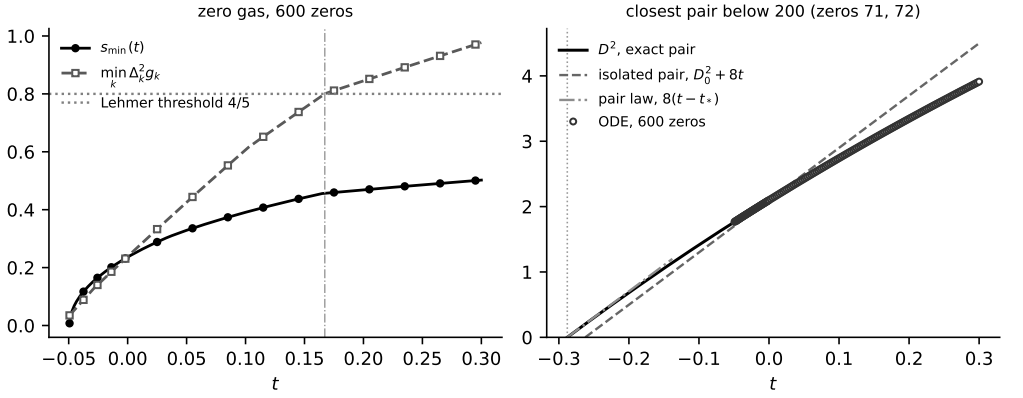



Figure 15.4: Left: the minimum normalised gap and $\min_k \Delta_k^2 g_k$ along the flow, both monotone on the computed range; the second crosses the Lehmer threshold $\frac{4}{5}$ near $t = 0.17$. Right: $D^2$ for the closest pair below height 200, at ordinates 184.87 and 185.60, against the isolated pair law $D_0^2 + 8t$.

*cretisation error) at* 131 *points of* $[184.6, 185.9]$*, with bisection in* $t$ *on the number of sign changes, show that the pair is still real at* $t = -0.287979$*, and with the pair law below place the collision in* $[-0.28803, -0.287979]$*, in agreement with the value* $-0.2880$ *of Section 8.4 and inside the bracket of Section 8.4.5. Newton's method on the same sum puts the double zero at* $z_* = 370.78087$*,* $t_* = -0.287979837009$*; a Krawczyk test certifies a real double zero of the trapezoid sum itself in a box of half width* $2^{-400}$ *about this point (a statement about the sum, not about* $H_t$*), and at* $t = t_* + 1.2 \times 10^{-12}$ *the signs* $+, -, +$ *of* $H_t$ *at* $z_* - 4 \times 10^{-6}$*,* $z_*$ *and* $z_* + 4 \times 10^{-6}$ *show that the pair is still real, which certifies that the collision lies below* $-0.287979837008$*: the upper end of the bracket is certified and the lower end is not. The extrapolation of the integrated* $D^2$ *was* 7% *early and the isolated pair estimate* $-0.262$ *was* 9% *late. Script* `phase_gas.py` *for (i) and (ii), output* `phase_gas.json`*; script* `phase_collide.py` *for (iii), output* `phase_collide.json`*.*

The bracket in (iii) uses that a pair at distance $D$ in $z$ obeys $D^2 \approx 8(t - t_*)$ near the collision (Section 8.4.2), so sign changes on a grid of step 0.01 in $\theta$ vanish at most $0.01^2/2 = 5 \times 10^{-5}$ above $t_*$. At $t = -0.287981$ the grid finds none, which does not exclude two real zeros less than 0.01 apart, and the lower end rests on the pair law, an asymptotic statement and not an enclosure.

The question is whether a gap functional, monotone as in (ii), can carry a positive lower bound from $t > 0$ down to $t = 0$ uniformly in the height, and the data suggest that gap functionals cannot. Both are minimised at Lehmer pairs, whose normalised gaps are expected, on the GUE conjecture, to be arbitrarily small at large height, so their infimum over all heights would be zero at $t = 0$. A

Lyapunov functional for RH must be global, insensitive to individual close pairs, in the manner of the pair correlation that Rodgers and Tao used to prove the reverse inequality [463]; we did not find a candidate. Two rigorous companions of the data are in Sections 8.4.7 and 8.4.10: an exact formula for the drift of a gap, and a proof that no bound through the absolute value of the integrand gives a step uniform in the height [385].

### 15.6.3 The cost of the flow route

The published points of the phase diagram give the cost of descending along it (Table 8.14 lists the bounds on $\Lambda$). Polymath15 proved $\Lambda \le 0.22$ with the barrier $X \approx 6\times 10^{10}$, and Table 1 of their paper lists, for each bound from $\Lambda \le 0.21$ down to $\Lambda \le 0.1$, the barrier $X$ that the same method needs, with RH verified to height $X/2$ [456]: $X \approx 2\times 10^{12}$ for 0.21, $5\times 10^{12}$ for 0.2, $2\times 10^{13}$ for 0.19, $7\times 10^{15}$ for 0.15 and $9\times 10^{21}$ for 0.1. Platt and Trudgian obtained $\Lambda \le 0.2$ from the second row, which by their account needs verification to $2.51\times 10^{12}$, and the third row, 0.19, requires verification slightly above $10^{13}$ [419]. The product $\Lambda \log X$ is 5.46 at 0.22 and falls along the table from 5.95 at 0.21 to 5.06 at 0.1.

Proposition 15.6.2 (Computed). *The least squares fit of $\Lambda = c/\log X$ in $1/\log X$ to the bound $\Lambda \le 0.22$ and the twelve rows of the table gives $c = 5.59$, with residuals below 0.013 in $\Lambda$, which along the table are errors of up to a factor of about 200 in $X$; on this law a bound $\Lambda \le \delta$ requires a barrier $X$ of order $e^{c/\delta}$, that is verification to height $X/2$: $X = 10^{48.6}$ for $\delta = 0.05$ and $10^{242.9}$ for $\delta = 0.01$. Since $\Lambda \log X$ falls along the table, the law $\log X = a/\Lambda + b$ fits its twelve rows better, with $a = 4.24$, $b = 8.28$ and residuals below 0.003 in $\Lambda$, within a factor 1.6 in $X$, and gives $X = 10^{40.4}$ for $\delta = 0.05$ and $10^{187.7}$ for $\delta = 0.01$. Script `phase_scaling.py`, output `phase_scaling.json`.*

Beyond the table the two laws should be read as orders of magnitude. Their form agrees with the heuristic that the flow at time $t$ makes the zeros near height $x$ real once $t \log x$ exceeds a constant, which is how the Riemann Siegel type approximation of $H_t$ enters [456]. Along the table, halving the bound from 0.2 to 0.1 raises the required barrier from $5\times 10^{12}$ to $9\times 10^{21}$, to about its power 1.7, and the power is 2 at every bound on the law $c/\log X$ and tends to 2 as the bound decreases on the law $\log X = a/\Lambda + b$. By the table itself the bound $\Lambda \le 0.15$ needs verification to a height of about $3.5\times 10^{15}$, some 1200 times the record $3\times 10^{12}$ [419], and $\Lambda \le 0.1$ about $4.5\times 10^{21}$; no conceivable computation reaches 0.05. The flow route produces a theorem at every stage, a monotone sequence of upper bounds for $\Lambda$, which is 0 exactly when RH holds [463], as no other route does, and it cannot reach the corner of the diagram by computation.

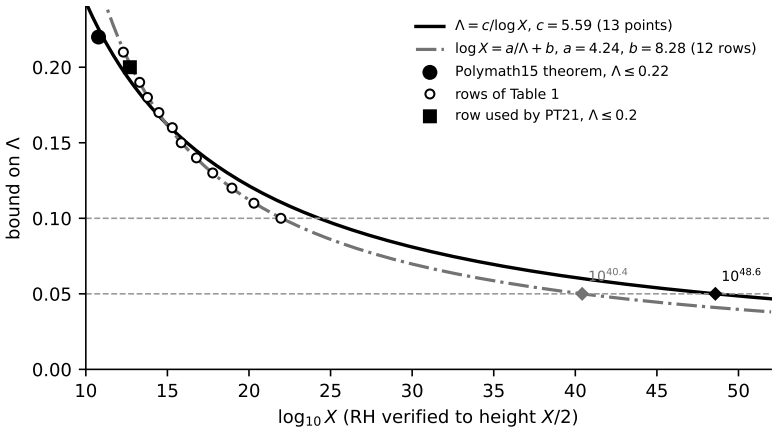



Figure 15.5: The bound $\Lambda \le 0.22$ of Polymath15 and the twelve rows of their table of conditional bounds [456] against the barrier $X$ (verification height $X/2$), with the fits $\Lambda = c/\log X$ and $\log X = a/\Lambda + b$. Dashed lines mark $\Lambda = 0.1$ and 0.05.

### 15.6.4 A programme

The phase diagram suggests five steps, of which the first three are engineering.

1. An evaluator: a Riemann Siegel type form of $\xi_t$ and $\xi_t'$, such as the effective approximation of $H_t$ in [456], in ball arithmetic, at a cost of about $\sqrt{\theta/2\pi}$ terms per point (Table 2.4), replacing the Pólya integral, whose required precision grows like $0.341\,\theta$ digits (Section 2.1.1); see Section 12.2.2 for $t = 0$. Lemma 15.2.6 gives another evaluator: at $s = \frac{1}{2} + i\theta$, $\xi_t(s)$ is $g(s)e^{t\ell(s)^2/4}$ times a Gaussian average of $\zeta e^r$ centred at $s + \frac{t}{2}\ell(s)$, close to $\frac{1}{2} + \frac{t}{4}\log\frac{\theta}{2\pi} + i(\theta + \frac{\pi t}{8})$, with variance $\frac{t}{2}$ in the real direction, which Gauss Hermite quadrature computes from ordinary evaluations of $\zeta$ and $\Gamma$; at $t = 0.2$ the relative differences between the rules with 40 and 60 nodes are $6.8\times10^{-56}$ at $\theta = 1000.3$ and $2.4 \times 10^{-47}$ at $\theta = 4000.3$ (`hx_shift.py`), and an error bound for the quadrature is open.

2. Certificates in Thorin form: Theorem 15.5.6 or Proposition 15.5.7 at a target $(t_0, \alpha_0)$ up to the barrier.

3. The tail in the same framework: the barrier and the analytic region beyond it [456], so that the points of Corollary 15.3.3(b) and (c) are certified end to end and stated as GGC theorems.

4. A measurement: the optimal $(t_0, \alpha_0)$ on each level set as a function of the verification height, and the constant $c$ of Proposition 15.6.2 as a function of the height.

5. The structural step: a tail at $t = 0$ below the basepoint $\frac{7}{8}$. Every certified point is a theorem, and by Proposition 15.4.1 no computed window distinguishes the corner from them; the step is the existence of the tail, and it must come from one of the mechanisms of Section 12.1.2.

### 15.6.5 Open problems

Problem 15.1. Compute $v_t(\alpha, \theta)$ at $t = 0$ and $t = 0.2$ near the Polymath barrier, heights near $3 \times 10^{10}$, with a Riemann Siegel type evaluator of $H_{0.2}$ such as the effective approximation of [456] (Section 2.2 gives the one for $\zeta$), and measure how the difference between the functions $v_{0.2}/\pi$ and $v_0/\pi$ grows with the height. Proposition 15.4.1(iii) measures it below 100. For $t = 0$ see work package 2 of Section 12.2.2 and Section 12.1.3.

Problem 15.2. Describe the boundary $\alpha^*(t)$ of $\mathcal{G}$ for $0 \le t \le 0.2$ as far as computation and the known zero free regions allow. Is $\alpha^*$ continuous, and is $(\alpha^*(t) - \frac{1}{2})^2 + \frac{t}{2}$ strictly decreasing in $t$ where $\alpha^*(t) > \frac{1}{2}$? Theorem 15.3.2(i) shows that it is nonincreasing there. Problem 14.3 asks the neighbouring question for the flow clock $T_t$, $0 < t < 0.2$.

Problem 15.3. Give the symmetrised truncations of Section 7.2 a coordinate on the phase diagram: for some $(t, \alpha)$ outside the region of Corollary 15.3.3, is there a family of explicit truncations of $\xi_t$, built from finitely many terms of the Pólya integral or of a Dirichlet series and making no use of the zeros, as the symmetrised lattice truncations $K_n$ of Subsection 7.2.1 are, that converges to $\xi_t$ on compact sets, together with heights $h_n \to \infty$ such that for infinitely many $n$ the truncation has no zero with $\operatorname{Re} s > \alpha$ and $|\operatorname{Im} s| < h_n$? Hurwitz's theorem then makes $(t, \alpha)$ a new point of $\mathcal{G}$, as in Corollary 14.12.2(c); at $t = 0$ and $\alpha = \frac{1}{2}$ the question for the $K_n$ is Conjecture S (Theorem 7.2.7), and for the stages of Chapter 14 it is Problem 14.1.

Problem 15.4. Compute the Cholesky factor of the Weil Gram matrix of Section 15.6.1 for decreasing $w$ and increasing $L$, and decide whether its entries converge to an object built from the arithmetic, the first test of a factorisation $W(g * \tilde{g}) = \|Ag\|^2$.

Problem 15.5. Find a global functional of the zeros of $H_t$, insensitive to individual close pairs, that is monotone along the flow and whose positivity at $t = 0$ excludes collisions at positive time. Biane's gap functional of Section 8.4.4 is local, the minimum over one gap of Laguerre's quantity, and Problems 8.1 and 12.5 ask for the control of the nonreal zeros that would make its brackets unconditional; the functional asked for here should instead be one quantity for all the zeros.

Problem 15.6. (a) Prove that $v_{0.2}(10, \theta) \ge \frac{19}{2} V_{0.2}$ for every $\theta \ge \theta_1(0.2)$, which by Corollary 15.2.8 makes $\tilde{\varphi}_{0.2,\alpha}$ a GGC transform for every $\alpha \in [\frac{1}{2}, 10]$. By

Lemma 15.2.6 and Cauchy's estimate it suffices to bound $|\zeta_{0.2}-1|$ on the discs of radius 5 about $10+i\theta$ by a small constant uniform in $\theta$; on these discs the shifted Gaussian average of the lemma reaches $\operatorname{Re} s \le 2$ with probability below $10^{-20}$ at every height, and at the four heights sampled the largest of 256 sampled values of $|\zeta_{0.2}-1|$ on each boundary circle is at most 0.0326 (computed, not a bound on the discs). (b) For $0 < t \le 0.2$ prove an explicit bound $|S_t(T)| \le c_1 \log T + c_2$ for $T \ge T_1$, for instance by proving the two hypotheses of Proposition 15.2.7(iii) with an explicit $M$, from Lemma 15.2.6, Rademacher's convexity bound for $-\frac{1}{2} \le \operatorname{Re} s \le \frac{3}{2}$ [260] and the functional equation, with its Gamma factor kept, for $\operatorname{Re} s < -\frac{1}{2}$. The minimum of $\operatorname{Re}\zeta_{0.2}(2+iy)$ over $0 \le y \le 200$ is 0.69006, near $y = 14.015$ (computed, `hx_edge.py`). At $t = 0$ much sharper explicit bounds are known [176].

### Notes

The phase diagram theorem is de Bruijn's strip theorem [112, Theorem 13] read through Theorem 15.2.3; its case $t = 0$ is in Section 8.4. Theorem 15.2.3, the certificate of Section 15.5 with Theorem 15.5.8, and the experiments of Sections 15.6.1 and 15.6.2 are from the authors' working report [438]. The bounds on $\Lambda$ are those of de Bruijn [112], Ki, Kim and Lee [310], Polymath15 [456] and Platt and Trudgian [419], and the lower bound is Rodgers and Tao [463]. The half plane $\operatorname{Re} s > \frac{7}{8}$ is the announcement [398], discussed in Section 12.1.2. The description of the Thorin measure as one object with a prime face and a zero face, joined by the sine squared lemma, is developed for automorphic $L$ functions in [439]. Section 15.2.1, except part (iv) of Proposition 15.2.5 and Corollary 15.2.8, and part (b) of Problem 15.6 are from a note of Nicholas Polson merged into the book [236].

## 15.7 Exercises

*On Sections 15.2 and 15.3.*

15.1. Prove the semigroup identity $P_a * P_b = P_{a+b}$ from the characteristic function $e^{-a|\xi|}$ of $P_a$, and deduce the identity $\int_{\mathbb{R}} P_d(\theta-\gamma)(\theta^2+u)^{-1}d\theta = \pi P_{d+\sqrt{u}}(\gamma)/\sqrt{u}$ used in Lemma 15.2.2.

15.2. Show that the Thorin density of Lemma 15.2.2(a), in the chart $z = \theta^2$, has total mass one, and that it converges vaguely to $\delta_{\gamma^2}$ as $d \downarrow 0$.

15.3. Show that the level sets of $t + 2(\alpha - \frac{1}{2})^2$ are the parabolas along which Theorem 15.3.2(i) carries the boundary of a certified region, and that the parabola through $(t_0, \alpha_0)$ meets the axis $\alpha = \frac{1}{2}$ at $t_0 + 2(\alpha_0 - \frac{1}{2})^2$.

15.4. Using Theorem 15.3.2(ii) and Corollary 15.3.3(d), find the $\alpha$ for which a zero free half plane $\operatorname{Re} s > \alpha$ at time zero would improve the bound $\Lambda \le 0.2$,

and the smallest $t$ for which the flowed half plane $\operatorname{Re} s > \frac{7}{8}$ gives a point of $\mathcal{G}$ at the basepoint 0.6.

15.5. Let $F$ be the perturbation of Theorem 5.2.1, with a quadruple of zeros at $\frac{1}{2} \pm \frac{1}{4} \pm ig$. Assuming RH, show that the shifted clock of $F$ at the basepoint $\alpha$ is a GGC exactly for $\alpha \geq \frac{3}{4}$, and compute its Thorin density near $\theta = g$ for $\alpha = 0.8$.

*On Sections 15.4 and 15.5.*

15.6. Assuming RH, show that the average of $v_0(\alpha, \theta)$ over a window of length $H$ about $\theta$ is $\frac{1}{2}\log(\theta/2\pi) + O(\log\theta/H)$ when $H$ is large compared with $\alpha - \frac{1}{2}$, and explain why the minimum over the window can be much smaller than the average when $\alpha - \frac{1}{2}$ is small compared with the mean spacing.

15.7. Show that $v_t(\alpha, 0) = \xi_t'/\xi_t(\alpha)$ is positive for every $t \geq 0$ and $\alpha > \frac{1}{2}$, using Lemma 15.2.1(ii), and compute it at $t = 0.2$, $\alpha = 0.6$ from the Pólya integral.

15.8. Prove Lemma 15.5.5 and show that it remains true with $\leq$ in place of $<$, the change of argument then lying in $(-\frac{\pi}{2}, \frac{\pi}{2})$.

15.9. Explain why the number of grid points in Theorem 15.5.6 grows like $e^{(\pi/4 - 2\eta)\theta}$ for a tilt $\eta < \pi/8$.

15.10. Show that positivity of $v_t$ on the boundary of a rectangle does not by itself exclude a zero inside it, and explain why Proposition 15.5.7 asks for the whole rectangle.

*On Section 15.6.*

15.11. Show that for $g$ supported in $[-L/2, L/2]$ the function $h = g * \tilde{g}$ is supported in $[-L, L]$, and that the prime sum in Weil's explicit formula runs over $n \leq e^L$.

15.12. Derive the $x$ space form of the archimedean term of Section 15.6.1 from the integral representation $\psi(z) = \int_0^\infty (e^{-t}/t - e^{-zt}/(1 - e^{-t}))\, dt$.

15.13. For an isolated pair of zeros of $H_t$ at distance $D$, show from the equation of motion that $D^2 \approx D_0^2 + 8t$ (compare Exercises 8.15 and 8.16). Explain why two certified sign changes of $\xi_t(\frac{1}{2} + i\theta)$ on a grid of step 0.01 show that the pair of Proposition 15.6.1(iii) is still real at $t$, while their absence does not show that it has left the line, and why the lower end of the bracket there rests on the law $D^2 \approx 8(t - t_*)$.

15.14. Fit $\Lambda = c/\log X$ to the thirteen points of Proposition 15.6.2 by least squares in $1/\log X$ and check the value $c = 5.59$; then fit $\Lambda = c'\log\log X/\log X$ to the same points and compare the barrier each law gives for $\Lambda = 0.1$ with the last row of the table, $X \approx 9 \times 10^{21}$.

# Chapter 16

# Pólya and the zeros of entire functions

This chapter reads Pólya's work through the reciprocal Xi law $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$, its Thorin measure, the basepoint laws $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ and the $2\sin^2$ lemma. The statement that carries the chapter is Proposition 16.6.3: the atoms of the even part laws produced by Pólya's universal factor $\cosh(au)$ sit exactly where the argument clock of the basepoint law crosses half integers.

*What is not claimed.* Nothing here proves RH. Section 16.12 is conditional on a half plane that has not been refereed. A number is computed in floating point at the precision stated, and a certificate that says so is a floating point certificate with explicit analytic bounds, not an interval arithmetic proof.

## 16.1 Pólya's question

Pólya spent a decade, from 1918 to 1927, on a single question: when does the Fourier transform of a positive even kernel have only real zeros? The motivation was Riemann's formula

$$\begin{aligned} \Xi(t) &= \xi\big(\tfrac{1}{2}+it\big) = 2\int_0^\infty \Phi(u)\cos(tu)\,du, \\ \Phi(u) &= 2\sum_{n\geq 1}\Big(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\Big)\,e^{-\pi n^2 e^{2u}}, \end{aligned} \tag{16.1.1}$$

in which the Riemann Hypothesis is the statement that $\Xi$ has only real zeros. The Thorin programme of this book reads the same function from the other side, through the reciprocal $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$, which is unconditionally the Laplace transform of a self decomposable law $\mu$ on $[0,\infty)$ (Theorem 5.1.1), and is the Laplace transform of a generalised gamma convolution (GGC) exactly when RH holds (Proposition 1.2.1). Pólya works on the kernel $\Phi$, multiplying it by factors that preserve real zeros, while the Thorin programme works on the logarithm of the transform, asking for positivity of a measure.

This chapter asks three questions. Which of Pólya's results have not yet been brought into the Thorin programme? Which of them would help to show RH? Is the $2\sin^2$ lemma useful with regard to his work?

A remark on sources. Pólya's papers on zeros are collected in Volume II of his Collected Papers, edited by Boas [454]. We did not have the full text of every paper discussed below. The Acta paper of 1926 [447] was read in part, and what is said of it comes from its first page and from the careful secondary accounts of Newman and Wu [383], Lagarias [321], Gasper [146] and Haglund [168]; every formula used below for Pólya's $\Xi^*$ was rederived and checked numerically. Whether a result is already part of the programme is judged by comparison with the earlier chapters, the first author's paper [421] (version 24 of arXiv 1708.02653) and monograph [424].

## 16.2 Notation and the Thorin dictionary

We use the Titchmarsh normalisation (16.1.1). Since $\Phi$ is even and positive,

$$\xi\big(\tfrac12+w\big)=2\int_0^\infty \Phi(u)\cosh(wu)\,du=\sum_{k\ge0}c_k w^{2k},\qquad c_k=\frac{2}{(2k)!}\int_0^\infty \Phi(u)u^{2k}\,du>0. \tag{16.2.1}$$

Put $F(s)=\xi(1/2+\sqrt{s})/\xi(1/2)=\sum_{k\ge0}a_k s^k$ with $a_k=c_k/c_0>0$. Then $F$ is entire of order one half with $F(s)=\prod_\rho(1-s/s_\rho)$, $s_\rho=(\rho-1/2)^2$, the product taken over zeros in the upper half plane (one factor for each pair $\rho,1-\rho$), and $\varphi=1/F$ is the Laplace transform of the reciprocal Xi law $\mu$; we write $Y\sim\mu$. Its exponent $\psi=\log F$ satisfies $\psi'(s)=\int_0^\infty e^{-sx}W(x)\,dx$ with the heat trace $W(x)=\sum_{\gamma>0}e^{(\rho-1/2)^2x}$, and the dictionary of Section 5.1 reads: $W\ge0$ (infinite divisibility, unconditional), $-W'\ge0$ (self decomposability, unconditional), $W$ completely monotone (GGC, equivalent to RH). Under RH the Thorin measure is $U=\sum_{\gamma>0}\delta_{\gamma^2}$ and $Y\overset{d}{=}\sum_\gamma E_\gamma/\gamma^2$.

At a basepoint $\alpha$ the law $\mu^{(\alpha)}$ has Laplace transform $\xi(\alpha)/\xi(\alpha+\sqrt{s})$; it is a GGC if and only if $\alpha\ge\sigma_{\max}=\sup\operatorname{Re}\rho$, unconditionally for $\alpha\ge1$ (Theorem 1.2.2), and its Thorin distribution in the ordinate variable is the clock

$$U_\alpha(\tau)=\frac{1}{\pi}\,\Delta_0^\tau\arg\xi(\alpha+i\tau),\qquad \frac{d}{d\tau}U_\alpha(\tau)=\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau). \tag{16.2.2}$$

With $\tau_\alpha$ the inverse of the clock and $g$ a standard gamma subordinator, $\mu^{(\alpha)}$ is the law of $\int_0^\infty dg_v/\tau_\alpha(v)^2$.

The $2\sin^2$ lemma (Lemma 6.2.4) rests on the identity, for $x,w>0$ and $u=$

$w^2/2$,

$$e^{-wx}+wx-1=\int_0^\infty \log\Big(1+\frac{u}{z}\Big)\,\varpi_x(z)\,dz,$$
$$\varpi_x(z)=\frac{x}{\pi\sqrt{2z}}\,2\sin^2\Big(x\sqrt{z/2}\Big)\ge 0. \tag{16.2.3}$$

Consequently, if $\log f(\alpha+w)/f(\alpha)-wf'(\alpha)/f(\alpha)=\int(e^{-wx}-1+wx)e^{-\alpha x}\mu_f(dx)/x$ with $\mu_f\ge 0$, the compensated reciprocal $e^{wf'(\alpha)/f(\alpha)}f(\alpha)/f(\alpha+w)$, $w=\sqrt{2u}$, is a GGC in $u$ with Thorin density $\int\varpi_x(z)e^{-\alpha x}\mu_f(dx)/x$. Differentiating twice in $w$, the hypothesis $\mu_f\ge 0$ is the statement that $(\log f)''$ is completely monotone on $(\alpha,\infty)$.

*Notation.* The function $F$ is the function $G$ of Section 1.2, and $Y\sim\mu$ is the clock variable $T$, so that $\varphi=1/F$ is its Laplace transform. The coefficients $a_k$ are the scaled moments $b_k$ of Section 13.2.4 (Table 13.4), not the exponents $a_j$ of Section 5.1. The word clock in this chapter means the argument clock $U_\alpha$, the cumulative Thorin mass $M_\varepsilon(\theta)=\pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$ of Section 3.4 with $\varepsilon=\alpha-\frac12$ and $\theta=\tau$, and not the Laplace transform $1/G$ of the earlier chapters. We write $b_\alpha=\frac{\xi'}{\xi}(\alpha)$ for the drift of Section 9.1. Finally $X=Z\sqrt{2Y}$ is the variable $Y_{1/2}$ of Section 10.1, with density $f_X$, and not the Pólya variable $X$ of Section 1.2, whose density is $\Phi/\Xi(0)$.

## 16.3 Pólya's papers on zeros, one by one

### P1. Approximation by polynomials with real zeros (1913)

*Über Annäherung durch Polynome mit lauter reellen Wurzeln*, Rend. Circ. Mat. Palermo 36 [440]. Completing Laguerre, Pólya characterised the entire functions that are locally uniform limits of real polynomials with only real zeros: they are exactly

$$f(z)=cz^me^{-\lambda z^2+\kappa z}\prod_k\Big(1-\frac{z}{b_k}\Big)e^{z/b_k},$$
$$m\in\{0,1,2,\dots\},\ \lambda\ge 0,\ c,\kappa\in\mathbb{R},\ b_k\in\mathbb{R}\setminus\{0\},\ \sum_k b_k^{-2}<\infty, \tag{16.3.1}$$

the Laguerre Pólya class $\mathcal{LP}$. The functions of type I, $cz^me^{\sigma z}\prod_k(1+z/b_k)$ with $c\ge 0$, $\sigma\ge 0$, $b_k>0$ and $\sum_k b_k^{-1}<\infty$, are exactly the locally uniform limits of polynomials with only negative zeros and positive coefficients. RH is the statement $\Xi\in\mathcal{LP}$.

P2. Pólya and Schur, two kinds of factor sequences (1914)

*Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen*, J. reine angew. Math. 144 [455]. A sequence $(g_k)$ is a multiplier sequence if $\sum p_k x^k$ with only real zeros implies $\sum g_k p_k x^k$ with only real zeros. Pólya and Schur proved that a positive sequence is a multiplier sequence if and only if $\sum g_k z^k/k!$ is in $\mathcal{LP}$ of type I, if and only if every Jensen polynomial $\sum_{k=0}^{d} \binom{d}{k} g_k x^k$ has only real zeros.

P3. Genus zero and one (1915)

*Algebraische Untersuchungen über ganze Funktionen vom Geschlechte Null und Eins*, J. reine angew. Math. 145 [441]. Operations on entire functions of genus at most one that preserve reality of zeros. The statement used later is the shift lemma, quoted as Hilfssatz II of the 1926 paper P8 in Section 7.1.3: if $G$ is real, of genus 0 or 1, with only real zeros, at least one of them, then $G(z+ia)+G(z-ia)$ has only real zeros for real $a$.

P4. Zeros of certain entire functions (1918)

*Über die Nullstellen gewisser ganzer Funktionen*, Math. Z. 2 [442]. The continuous Eneström Kakeya theorem: if $f$ is positive and nondecreasing on $(0,1)$, then $\int_0^1 f(t)\cos(zt)\,dt$ and $\int_0^1 f(t)\sin(zt)\,dt$ have only real zeros. The Riemann kernel is positive but decreasing on $(0,\infty)$, so the theorem does not apply.

P5. Number theoretic remarks (1919)

*Verschiedene Bemerkungen zur Zahlentheorie*, Jahresber. DMV 28 [443]. Among other things, the Pólya conjecture that the summatory Liouville function $L(x) = \sum_{n\le x} \lambda(n)$ satisfies $L(x) \le 0$ for $x \ge 2$. It implies RH and is false: Haselgrove (1958) [178] disproved it and Tanaka (1980) [491] found the first sign change at $x = 906{,}150{,}257$.

P6. Zeros of successive derivatives (1922)

*Über die Nullstellen sukzessiver Derivierten*, Math. Z. 12 [444]. Pólya introduced the final set of a function, the set of limit points of the zeros of $f^{(n)}$ as $n \to \infty$, and determined it in several classes. The statements about real entire functions that we use are recorded under P13.

P7. A Fourier integral (1923)

*On the zeros of an integral function represented by Fourier's integral*, Messenger of Math. 52 [446]. A short note on the zeros of cosine transforms of positive kernels, a precursor of the 1926 and 1927 papers.

P8. The integral representation of the Riemann xi function (1926)

*Bemerkung über die Integraldarstellung der Riemannschen $\xi$ Funktion*, Acta Math. 48 [447]. As $u \to +\infty$ the kernel is dominated by its $n = 1$ term, and Pólya replaced it by the even function

$$\Phi^*(u) = 4\pi^2 \left(e^{9u/2} + e^{-9u/2}\right) e^{-\pi(e^{2u}+e^{-2u})} = 8\pi^2 \cosh\left(\tfrac{9u}{2}\right) e^{-2\pi \cosh 2u}, \quad (16.3.2)$$

and proved that $\Xi^*(t) = 2\int_0^\infty \Phi^*(u)\cos(tu)\,du$ has only real zeros, with a zero count asymptotic to that of $\Xi$. The proof combines the reality of the zeros of $z \mapsto K_{iz}(a)$, $a > 0$, with the shift lemma P3. A refinement keeping the second exponential term is also real rooted.

P9. Zeros of certain trigonometric integrals (1926)

J. London Math. Soc. 1 [448]. Following Hardy's proof of infinitely many zeros on the critical line, Pólya explained why infinitely many real zeros is much easier than no complex zeros, and gave a general criterion for infinitely many real zeros. In Thorin language the infinitude of real zeros is the statement that the on line part $\varphi_{\text{on}} = \prod_{\text{on line}}(1 + s/\gamma^2)^{-1}$, a GGC whatever the zero set, has a Thorin measure with infinitely many atoms.

P10. Trigonometric integrals with only real zeros (1927)

*Über trigonometrische Integrale mit nur reellen Nullstellen*, J. reine angew. Math. 158 [449]. The central paper. A real analytic $\phi$ is a universal factor if $\int \mathcal{F}(t)e^{izt}dt$ with only real zeros implies the same for $\int \phi(t)\mathcal{F}(t)e^{izt}dt$, for every admissible kernel $\mathcal{F}$, that is, every real kernel that decays fast enough for these integrals to define entire functions of $z$. Pólya proved that $\phi$ is a universal factor if and only if $z \mapsto \phi(iz)$ belongs to $\mathcal{LP}$. Hence $e^{\lambda t^2}$ ($\lambda > 0$) and $\cosh(bt)$ ($b \in \mathbb{R}$) are universal factors. He also proved a Mellin criterion: if $\mathcal{F}$ is admissible, holomorphic near the origin, and $H(z) = \int_0^\infty \mathcal{F}(t)t^{z-1}dt$ has no zeros in $\mathbb{C} \setminus (-\infty, 0]$, then $\int \mathcal{F}(t^{2q})e^{izt}dt$ has only real zeros for every positive integer $q$. Consequences: $\int e^{-t^{2q}}e^{izt}dt$ and $\int \cosh(bt)e^{-a\cosh t}e^{izt}dt$ with $a > 0$ and $b \in \mathbb{R}$ have only real zeros. The factor $e^{\lambda t^2}$ is the starting point of de Bruijn (1950) and Newman (1976), hence of the de Bruijn Newman constant $\Lambda$ (Table 8.14).

P11. Jensen's investigations (1927)

*Über die algebraisch funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Kgl. Danske Vid. Selsk. Math. Fys. Medd. 7 [450]. Pólya edited and proved Jensen's criteria: a real entire function of genus at most one is in $\mathcal{LP}$ if and only if all its Jensen polynomials have only real zeros, together with Jensen's

integral criteria. This is the source of the Jensen polynomial approach to RH (Section 13.4.1); Farmer (2022) [133] argued that this direction cannot reach RH.

P12. Fourier's work on transcendental equations (1930)

Quart. J. Math. Oxford 1 [452]. The Fourier Pólya conjectures on the number of nonreal zeros of successive derivatives.

P13. Derivatives and analytic character (1943)

Bull. AMS 49 [453]. The Pólya Wiman conjecture: if $f$ is real entire of order below two with finitely many nonreal zeros, then $f^{(n)}$ has only real zeros for all large $n$; proved by Craven, Csordas and Smith (1987) [98] and Kim (1990) [311], with the Fourier Pólya conjecture settled by Ki and Kim (2000) [309]. For $\Xi$, whose zeros lie in a strip, the theorem of Ki and Kim gives the local statement that for every $R$ the zeros of $\Xi^{(n)}$ in $|z| < R$ are real for all large $n$; Conrey showed that the proportion of real zeros of $\Xi^{(n)}$ tends to one [89]. Global real rootedness of $\Xi^{(n)}$ for some $n$ is not known.

P14. The Hilbert Pólya suggestion

In a 1982 letter to Odlyzko, Pólya recalled suggesting to Landau, around 1914, that the zeros might be the eigenvalues of a self adjoint operator. The Krein string route of Corollary 9.2.3, which assumes RH and simple zeros, is a probabilistic form of this.

## 16.4 The Laguerre Pólya class in the Thorin variable

Proposition 16.4.1. *The following are equivalent.*

*(i) RH.*

*(ii)* $F(s) = \xi(1/2 + \sqrt{s})/\xi(1/2)$ *belongs to the Laguerre Pólya class of type I, that is, it is a locally uniform limit of polynomials with only negative zeros.*

*(iii) The sequence* $g_k = k!\, a_k$*, where* $F(s) = \sum a_k s^k$*, is a Pólya Schur multiplier sequence.*

*(iv) For every* $d$ *the Jensen polynomial* $J_d(x) = \sum_{k=0}^{d} \binom{d}{k} g_k x^k$ *has only real, negative zeros.*

*(v) The reciprocal Xi law is a GGC.*

*Proof.* Under RH every $s_\rho = -\gamma^2$ is negative and $F(s) = \prod_\gamma (1 + s/\gamma^2)$ with $\sum \gamma^{-2} < \infty$, which is of type I; conversely a function of type I has only real

nonpositive zeros, so every $s_\rho \le 0$, and $s_\rho = (\rho - 1/2)^2 \le 0$ forces $\rho - 1/2 \in i\mathbb{R}$. The equivalence of (ii), (iii) and (iv) is the theorem of Pólya and Schur, since $\sum g_k s^k/k! = F(s)$. The equivalence with (v) is Proposition 1.2.1. □

The ladder (iv) is the coefficient counterpart of the monotonicity ladder $T_k = (-1)^k W^{(k)} \ge 0$ of Theorem 5.1.6. Its first rung, for $F$ and for every derivative of $F$, is the Turán inequality $g_k^2 \ge g_{k-1} g_{k+1}$ for $k \ge 1$, the condition that the Jensen polynomials $J_{2,n}$, $n \ge 0$, are real rooted; for $F$ alone it is the case $k = 1$. With $m_k = (2k)!\, c_k = 2\int_0^\infty \Phi(u) u^{2k}\, du$ it reads $m_k^2 \ge \frac{2k-1}{2k+1} m_{k-1} m_{k+1}$, which is the Turán inequality proved unconditionally by Csordas, Norfolk and Varga (1986) [99]; in the scaled moments $a_k$, which are the $b_j$ of Section 13.2.4, it is (13.2.2), whose margins are in Table 13.4.

*Numerics.* Computing the $c_k$ from (16.2.1) by quadrature on the kernel reproduces $\Xi(0) = \xi(1/2) = 0.497120778188314$ to fifteen digits and gives $a_1 = 0.0231049931$ and the values $a_2, a_3, a_4$ of Table 13.4. The Turán ratios $g_k^2/(g_{k-1} g_{k+1})$ decrease slowly from 1.0748 at $k = 1$ and stay above 1.0109 for all $k \le 40$. The Jensen polynomials $J_d$ of every degree $2 \le d \le 30$, computed with 120 digits, have only real negative zeros.

Remark 16.4.2. The ladder (iv) at fixed degree $d$ is a finite set of polynomial inequalities in $a_1, \dots, a_d$, and so is checkable to any degree; but for each fixed $d$ it is implied by information about finitely many zeros, which is why it cannot prove RH by itself.

## 16.5 Pólya frequency functions: the two sided criterion

Schoenberg (1951) [474] proved that an integrable function $f$ on $\mathbb{R}$ is a Pólya frequency function of every order ($\mathrm{PF}_\infty$) if and only if its bilateral Laplace transform is $1/\Psi(s)$ on a strip around the origin, with $\Psi$ a function of the form (16.3.1) with $m = 0$ and $\Psi(0) > 0$ that is not of the form $Ce^{\kappa s}$. The name honours Pólya, and the theorem is the two sided counterpart of the Thorin theory.

Proposition 16.5.1. *Let $Y \sim \mu$ be the reciprocal Xi law and $Z$ a standard normal variable independent of $Y$. Unconditionally, $X = Z\sqrt{2Y}$ has a smooth symmetric density $f_X$ with characteristic function and bilateral Laplace transform*

$$\mathbb{E}e^{itX} = \frac{\xi(1/2)}{\xi(1/2+t)}, \qquad \mathbb{E}e^{zX} = \frac{\Xi(0)}{\Xi(z)} \quad (|\operatorname{Re} z| < \gamma_1). \tag{16.5.1}$$

*Moreover RH holds if and only if $f_X$ is $\mathrm{PF}_\infty$.*

This is the equivalence of Section 1.2, between RH and the law of $Z\sqrt{2Y}$ as a sum of Laplace variables, read through Schoenberg's theorem; the same equivalence is stated for the Laguerre Pólya class in [313, Theorem 2.3 and Corollary 2.4], and its forward half is item (iv) of Theorem 13.1.4.

*Proof.* $\mathbb{E}e^{itX} = \mathbb{E}e^{-t^2Y} = \varphi(t^2) = \xi(1/2)/\xi(1/2+t)$, which is integrable in $t$ since $\xi(1/2+t)$ grows like $\exp(\frac{t}{2}\log t)$, so $f_X$ exists and is smooth. For real $z$, $\mathbb{E}e^{zX} = \mathbb{E}e^{z^2Y} = \varphi(-z^2)$, and $\varphi(-z^2) = \xi(1/2)/\xi(1/2+iz) = \Xi(0)/\Xi(z)$ by evenness; the Laplace transform of the positive measure $\mu$ converges up to its first singularity on the negative real axis (Landau's theorem), which is at $-\gamma_1^2$, because a zero off the line gives a nonreal $s_\rho$ and the zero of lowest ordinate is on the line. Under RH, $\Xi(z)/\Xi(0) = \prod_\gamma(1 - z^2/\gamma^2) = \prod_\gamma(1 - z/\gamma)e^{z/\gamma}(1 + z/\gamma)e^{-z/\gamma}$ is of Schoenberg's form, so $f_X$ is $\mathrm{PF}_\infty$. Conversely if $f_X$ is $\mathrm{PF}_\infty$ then $\Xi(z)/\Xi(0)$ agrees on a strip with an entire function having only real zeros, hence everywhere, and RH follows. □

The Thorin atoms $\gamma^2$ become Schoenberg's $\pm 1/\gamma$. Under RH, $X \overset{d}{=} \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables $L_\gamma$, because a Laplace variable is $Z\sqrt{2E}$. The PF ladder $\mathrm{PF}_2 \supset \mathrm{PF}_3 \supset \cdots \supset \mathrm{PF}_\infty$ is a third ladder, after $T_k$ and the Jensen degrees. $\mathrm{PF}_2$ is log concavity of $f_X$.

*Numerics.* Fourier inversion of (16.5.1) gives $f_X$. On $[0, 1.6]$ with step 0.02 every second difference of $\log f_X$ is negative (the largest is $-2.6 \times 10^{-5}$), so $\mathrm{PF}_2$ holds on the grid; all 3000 random pairs of ordered triples $x_1 < x_2 < x_3$, $y_1 < y_2 < y_3$ in $[-0.8, 0.8]$ have a positive normalised determinant $\det[f_X(x_i - y_j)]_{i,j\le 3} / \prod_i f_X(x_i - y_i)$, the smallest being $6.7 \times 10^{-9}$.

Problem 16.1. Prove unconditionally, in exact or interval arithmetic, that $f_X$ is log concave, that is $\mathrm{PF}_2$ (Theorem 16.11.5 gives a floating point certificate), and then that it is $\mathrm{PF}_n$ for $n \ge 3$ (Section 16.11). The case of $\mathrm{PF}_2$ would be a two sided analogue of the unconditional self decomposability of $\mu$, one rung up the Schoenberg ladder.

## 16.6 The universal factor $\cosh(au)$ and the clock

Multiplying the kernel by $e^{au}$ is an Esscher tilt of the Pólya density and moves the basepoint: $\int_{-\infty}^{\infty} \Phi(u)e^{au}\cosh(wu)\,du = \frac{1}{2}\left(\xi(\frac{1}{2}+a+w) + \xi(\frac{1}{2}+a-w)\right)$. The tilt alone destroys evenness. Pólya's universal factor $\cosh(au)$ is the even average of the two tilts. Define, for $a \ge 0$ and $\alpha = 1/2 + a$,

$$A_\alpha(w) = 2\int_0^\infty \Phi(u)\cosh(au)\cosh(wu)\,du = \tfrac{1}{2}\left(\xi(\alpha+w) + \xi(\alpha-w)\right), \quad (16.6.1)$$

the even part in $w$ of the basepoint function $w \mapsto \xi(\alpha + w)$. For real $t$, $A_\alpha(it) = \mathrm{Re}\,\xi(\alpha + it)$, and in the variable $s = 1/2 + w$ this is Lagarias's function $A_h(s) = \frac{1}{2}(\xi(s+h) + \xi(s-h))$ with $h = a$.

Theorem 16.6.1 (Pólya, de Bruijn, Lagarias). *(i) Under RH, $A_\alpha$ has only purely imaginary zeros for every $\alpha > 1/2$.*

*(ii) If every zero of $\Xi$ satisfies $|\operatorname{Im} t| \le \Delta$, then every zero of $t \mapsto A_\alpha(it)$ satisfies $|\operatorname{Im} t| \le \max(\Delta^2 - a^2, 0)^{1/2}$. In particular, unconditionally, $A_\alpha$ has only purely imaginary zeros for $\alpha \ge 1$, and they are simple.*

*(iii) If $A_{\alpha_n}$ has only imaginary zeros along a sequence $\alpha_n \downarrow 1/2$, then RH holds.*

*Proof.* (i) $\cosh(au)$ is a universal factor since $\cos(az) \in \mathcal{LP}$. (ii) is de Bruijn's strip theorem for the factor cosh, and $\Delta \le 1/2$ because $\zeta$ has no zeros with $\operatorname{Re} s > 1$; simplicity is Lagarias's theorem [321]. (iii) $A_\alpha \to \xi(1/2 + w)$ locally uniformly as $\alpha \downarrow 1/2$, and Hurwitz's theorem applies. □

Corollary 16.6.2 (Even part basepoint laws). *Whenever $A_\alpha$ has only imaginary zeros $\pm i\tau_k$, the function $\varphi_\alpha^P(s) = \xi(\alpha)/A_\alpha(\sqrt{s})$ is the Laplace transform of a GGC law $\mu_\alpha^P$ with Thorin measure $\sum_k \delta_{\tau_k^2}$, so $\mu_\alpha^P$ is the law of $\sum_k E_k/\tau_k^2$. This holds unconditionally for $\alpha \ge 1$, for all $\alpha > 1/2$ under RH, and RH follows if it holds along a sequence $\alpha_n \downarrow 1/2$.*

*Proof.* $A_\alpha(\sqrt{s})$ is entire of order one half in $s$ with positive Taylor coefficients by (16.6.1), value $\xi(\alpha) > 0$ at $s = 0$, and zeros $s = -\tau_k^2$; the genus zero product gives $\varphi_\alpha^P(s) = \prod_k (1 + s/\tau_k^2)^{-1}$. □

Proposition 16.6.3 (Half integer crossings). *Let $\alpha$ be such that $\xi$ has no zeros in $\operatorname{Re} s \ge \alpha$ (this holds for $\alpha > \sigma_{\max}$, and unconditionally for every $\alpha \ge 1$, since $\zeta$ has no zeros on $\operatorname{Re} s = 1$). Then the clock $U_\alpha$ is continuous and strictly increasing from $U_\alpha(0) = 0$, and the positive zeros of $t \mapsto \operatorname{Re} \xi(\alpha + it)$ are exactly*

$$\tau_k = \tau_\alpha\big(k - \tfrac{1}{2}\big), \qquad k = 1, 2, \dots, \tag{16.6.2}$$

*where $\tau_\alpha$ is the inverse of the clock. Consequently the Thorin distribution of the even part law is the rounding of the Thorin distribution of the basepoint law,*

$$U_\alpha^P\big((0, T^2]\big) = \big\lfloor U_\alpha(T) + \tfrac{1}{2} \big\rfloor, \tag{16.6.3}$$

*and in law $\mu_\alpha^P = \mathcal{L}\big(\sum_k E_k/\tau_\alpha(k - \frac{1}{2})^2\big)$ while $\mu^{(\alpha)} = \mathcal{L}\big(\int_0^\infty dg_v/\tau_\alpha(v)^2\big)$: the even part law is the midpoint rule for the gamma clock integral.*

*Proof.* Since no zero has $\beta_\rho \ge \alpha$, the Hadamard sum $\operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau) = \sum_\rho \frac{\alpha - \beta_\rho}{|\alpha + i\tau - \rho|^2}$ has every term positive, so the clock is strictly increasing, and $\xi(\alpha + i\tau) \ne 0$. Write $\xi(\alpha + i\tau) = |\xi(\alpha + i\tau)| e^{i\pi U_\alpha(\tau)}$, which is legitimate since $\xi(\alpha) > 0$. Then $\operatorname{Re} \xi(\alpha + i\tau) = |\xi(\alpha + i\tau)| \cos(\pi U_\alpha(\tau))$ vanishes exactly when $U_\alpha(\tau) \in \mathbb{Z} + \frac{1}{2}$, and a strictly increasing clock starting at 0 and tending to infinity (by Stirling, $U_\alpha(\tau) \sim \frac{\tau}{2\pi} \log \tau$) crosses each level $k - \frac{1}{2}$ exactly once, transversally. This gives (16.6.2) and simple zeros. Every zero $\rho$ has $|\beta_\rho - \frac{1}{2}| < \alpha - \frac{1}{2}$, by the hypothesis and the functional equation, so Theorem 16.6.1(ii) with $\Delta = \alpha - \frac{1}{2}$ shows that $A_\alpha$ has no zeros off the imaginary axis, and the corollary on even part laws applies;

counting levels below $U_\alpha(T)$ then gives (16.6.3). The two laws follow from the gamma process representation: the increments of $g$ over $[k-1,k]$ are independent standard exponentials, and replacing $\tau_\alpha(v)$ on $[k-1,k]$ by its midpoint value gives the sum. □

Under the hypothesis of Proposition 16.6.3, which holds unconditionally for $h \geq \frac{1}{2}$ and under RH for every $h > 0$, the proposition explains why the zeros of $A_h$ are asymptotically equally spaced, as in Lagarias's theorem [321]: they are level crossings of a smooth increasing clock whose derivative $\pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + i\tau)$ is a Cauchy smoothing of the zero density, with fluctuations of lower order than its mean $\frac{1}{2\pi} \log \frac{\tau}{2\pi}$.

*Numerics.* Table 16.1 lists, for four basepoints, the number of zeros of $\operatorname{Re} \xi(\alpha+it)$ on $(0, 100]$, the clock at 100, its rounding, and the largest distance of the clock from a half integer at the computed zeros (on a grid of step 0.01 with linear interpolation). As $\alpha \downarrow 1/2$ the atoms converge to the ordinates.

| $\alpha$ | zeros on $(0, 100]$ | $U_\alpha(100)$ | $\lfloor U_\alpha(100) + \frac{1}{2} \rfloor$ | max distance to $\mathbb{Z} + \frac{1}{2}$ |
|---|---|---|---|---|
| 1.00 | 29 | 29.1112 | 29 | $2.7 \times 10^{-5}$ |
| 0.75 | 29 | 29.0548 | 29 | $5.6 \times 10^{-5}$ |
| 0.60 | 29 | 29.0218 | 29 | $1.7 \times 10^{-4}$ |
| 0.52 | 29 | 29.0044 | 29 | $1.6 \times 10^{-3}$ |

Table 16.1: Atoms of the even part law against the rounded clock of the basepoint law.

Remark 16.6.4 (Is the Pólya family easier?). Real rootedness of $A_\alpha$ is a counting condition: it can be certified to a given height by sign changes of $\operatorname{Re} \xi(\alpha + it)$ together with an argument principle count, exactly as in Turing's method. So the Pólya family is not easier in principle, but it converts the analytic positivity of $\operatorname{Re} \xi'/\xi$ into a discrete, verifiable statement about atoms. The zero trajectories $\alpha \mapsto \tau_k(\alpha)$ satisfy $d\tau_k/d\alpha = \operatorname{Re} \xi'(s)/\operatorname{Im} \xi'(s)$ at $s = \alpha + i\tau_k$. Under RH no two trajectories collide for $\alpha > \frac{1}{2}$, because the zeros are simple (Proposition 16.6.3); conversely a nonimaginary zero of $A_\alpha$ can appear only through a collision of two imaginary zeros, provided no zeros enter from infinity as $\alpha$ decreases, which needs uniform control of the zero counts in $\alpha$. In this heuristic sense RH is the statement that no two trajectories collide before $\alpha = \frac{1}{2}$, a discrete analogue of the de Bruijn Newman dynamics.

## 16.7 Pólya's $\Xi^*$ as the archimedean model

Proposition 16.7.1. *With* (16.3.2),

$$\Xi^*(t) = 4\pi^2\left[K_{\frac{9}{4}+\frac{it}{2}}(2\pi) + K_{\frac{9}{4}-\frac{it}{2}}(2\pi)\right] = 8\pi^2 \operatorname{Re} K_{\frac{9}{4}+\frac{it}{2}}(2\pi). \tag{16.7.1}$$

*Proof.* Substitute $v = 2u$ in $2\int_0^\infty \Phi^*(u)\cos(tu)\,du$, write $\cosh(9v/4)\cos(tv/2)$ as the average of $\cos((t/2 \mp 9i/4)v)$, and use $\int_0^\infty e^{-a\cosh v}\cos(\omega v)\,dv = K_{i\omega}(a)$ and the evenness of $K_\nu$ in $\nu$. □

This is the identity of Section 8.7, where $P(\theta) = K_{i\theta/2}(2\pi)$ and $\Xi^*(\theta) = 4\pi^2[P(\theta - \frac{9i}{2}) + P(\theta + \frac{9i}{2})]$.

Formula (16.7.1) says that $\Xi^*$ is exactly the construction (16.6.1) applied to the Bessel function: it is the even part, at basepoint $9/4$ in the order variable, of $\nu \mapsto K_\nu(2\pi)$. Pólya's theorem that $K_{iz}(a)$ has only real zeros is the Riemann Hypothesis for this function: its zeros in $\nu$ lie on $\operatorname{Re}\nu = 0$. The basepoint $9/4$ lies strictly to the right of that line, so by the argument of Theorem 16.6.1 and Proposition 16.6.3 the even part has only real zeros (the argument needs $\operatorname{Re}\frac{d}{d\nu}\log K_\nu(2\pi) > 0$ on $\operatorname{Re}\nu > 0$, and this holds because $\nu \mapsto K_\nu(2\pi)$ is even, of order one, with only the zeros $\pm iz_n$, $z_n > 0$, so that $K_\nu(2\pi) = K_0(2\pi)\prod_n(1+\nu^2/z_n^2)$ and each term $\frac{d}{d\nu}\log(1+\nu^2/z_n^2) = (\nu+iz_n)^{-1} + (\nu - iz_n)^{-1}$ has positive real part there), located where the Bessel clock $\pi^{-1}\arg K_{9/4+it/2}(2\pi)$ crosses half integers.

Proposition 16.7.2 (Half Gram points). *Let $t_k^*$ be the positive zeros of $\Xi^*$ in increasing order and $\vartheta$ the Riemann Siegel theta function. Then*

$$\frac{\vartheta(t_k^*)}{\pi} = k - \frac{3}{2} - \frac{2\pi^2 - 3}{\pi\, t_k^*} + O\big((t_k^*)^{-2}\big). \tag{16.7.2}$$

*Hence, with $N^*(T) = \#\{k : t_k^* \le T\}$, $N^*(T) = \lfloor \vartheta(T)/\pi + \frac{3}{2}\rfloor$ for all large $T$ outside the intervals $[t_k^*, t_k^* + O(1/(t_k^*\log t_k^*)))$, on which $N^*(T)$ is one larger; in particular $N^*(T) - \vartheta(T)/\pi$ is bounded, whereas $N(T) = \vartheta(T)/\pi + 1 + S(T)$.*

*Sketch.* For $\nu = \frac{9}{4} + \frac{it}{2}$, the ascending series of $I_{-\nu}$ and the reflection formula give $K_\nu(2\pi) = \frac{1}{2}\Gamma(\nu)\pi^{-\nu}\big(1 + \frac{\pi^2}{1-\nu} + O(t^{-2})\big)$, the companion term $\frac{1}{2}\Gamma(-\nu)\pi^{\nu}$ being smaller by a factor of order $t^{-9/2}$. With $s = \frac{1}{2} + it$ we have $\Gamma(\nu) = (\frac{s}{2}+1)\frac{s}{2}\,\Gamma(\frac{s}{2})$ and $\Gamma(\frac{s}{2})\pi^{-it/2} = |\cdot|e^{i\vartheta(t)}$, so the phase of $K_\nu(2\pi)$ is $\vartheta(t) + \arg s + \arg(s+2) + \operatorname{Im}\frac{\pi^2}{1-\nu} + O(t^{-2})$. Here $\arg s + \arg(s+2) = \pi - \frac{3}{t} + O(t^{-3})$ and $\operatorname{Im}\frac{\pi^2}{1-\nu} = \frac{2\pi^2 t}{t^2+25/4} = \frac{2\pi^2}{t} + O(t^{-3})$, so the phase is $\vartheta(t) + \pi + \frac{2\pi^2-3}{t} + O(t^{-2})$. The zeros of the real part are where this phase lies in $\frac{\pi}{2} + \pi\mathbb{Z}$, which is (16.7.2); the indexing is fixed by the first zero. The integer can also be fixed without computation, by letting the argument $2\pi$ of $K$ decrease to 0 as in the proof of Proposition 16.7.3(ii); at basepoint $\frac{9}{4}$ the ratio of $K_\nu(a)$ to $\frac{1}{2}\Gamma(\nu)(a/2)^{-\nu}$ is then $1 + O(a^2)$ uniformly in $t$, the companion term contributing $O(a^{9/2})$. □

Compare the Thorin form of the zero count, $U((0,t]) = \vartheta(\sqrt{t})/\pi + 1 + S(\sqrt{t})$ under RH. Pólya's $\Xi^*$ replaces the arithmetic fluctuation $S$ by a bounded rounding error: by Proposition 16.7.2 its Thorin distribution $N^*(\sqrt{t})$ is the rounding of the smooth archimedean distribution $\vartheta(\sqrt{t})/\pi + 1$. This is the distribution that the $2\sin^2$ lemma computes for the gamma factor in Theorem 6.2.12, up to an affine function of $\sqrt{t}$: that theorem gives $(\vartheta(\sqrt{t}) - \vartheta'(0)\sqrt{t})/\pi$ for the compensated factor $e^{-c\sqrt{s}}/G(s)$, with $c = -\vartheta'(0)$. In this sense $\Xi^*$ is the zeta function with $\zeta$ replaced by its first Dirichlet term and the result symmetrised, and its reciprocal $\Xi^*(0)/\Xi^*(\sqrt{-s})$ is, by Pólya's theorem, an unconditional GGC with Thorin measure $\sum_k \delta_{(t_k^*)^2}$.

*Numerics.* $\Xi^*$ has 29 zeros in $(0, 100]$, as does $\Xi$, but they are displaced: the first eight are 8.9928, 19.0654, 24.4748, 28.9750, 33.0069, 36.7367, 40.2491, 43.5947. There are 138 zeros up to height 300; at the last, $t^*_{138} = 299.627$, the deviation from $k - \frac{3}{2}$ times $t_k^*$ is $-5.329$, against the constant $-(2\pi^2 - 3)/\pi = -5.328$ of (16.7.2) (Figure 16.1, right).

*Basepoint $\frac{1}{4}$.* At basepoint $\frac{1}{4}$ the zeros of the even part sit at half Gram points with no shift of index, and their displacement from the zeros of $\Xi$ is governed by $S$. This follows a note of Nicholas Polson merged into the book [235]. Let

$$\Xi^\circ(t) = \operatorname{Re} K_{\frac{1}{4}+\frac{it}{2}}(2\pi), \qquad \psi(t) = \arg K_{\frac{1}{4}+\frac{it}{2}}(2\pi), \quad \psi(0) = 0,$$

so that $\Xi^\circ$ is, up to a constant, the transform of the law proportional to $\cosh(\frac{x}{2})e^{-2\pi\cosh 2x}$, the member of Theorem 13.5.4 with $a = 2\pi$, $\ell = 2$ and $\mu = 0$ and one field $\lambda_1 = \frac{1}{2}$.

PROPOSITION 16.7.3. *(i) The zeros of $\Xi^\circ$ are real and simple, $\psi$ is strictly increasing on $[0,\infty)$, and the positive zeros $t_1^\circ < t_2^\circ < \cdots$ are the solutions of $\psi(t_n^\circ) = (n - \frac{1}{2})\pi$. So $N^\circ(T) = \#\{n : t_n^\circ \le T\} = \lfloor \psi(T)/\pi + \frac{1}{2} \rfloor$.*

*(ii) $\psi(t) = \vartheta(t) + O(t^{-1/2})$. Hence $\vartheta(t_n^\circ)/\pi = n - \frac{1}{2} + O((t_n^\circ)^{-1/2})$, and $t_n^\circ - g_n = O\big((t_n^\circ)^{-1/2}/\log t_n^\circ\big)$ for the half Gram points $\vartheta(g_n) = (n - \frac{1}{2})\pi$.*

*(iii) Let $0 < \gamma_1 \le \gamma_2 \le \cdots$ be the ordinates of all zeros of $\xi$ in the upper half plane, with multiplicity and whatever their real parts. Suppose that exactly one zero of $\xi$, counted with multiplicity, has ordinate $\gamma_{n+1}$, and let $S(\gamma_{n+1})$ be the mean of the one sided limits of $S$ there. Then $\vartheta(\gamma_{n+1}) = (n-\frac{1}{2})\pi - \pi S(\gamma_{n+1})$, and*

$$t_n^\circ - \gamma_{n+1} = \frac{\pi S(\gamma_{n+1})}{\vartheta'(\gamma_{n+1})} + O\Big(\frac{\gamma_{n+1}^{-1/2}}{\log \gamma_{n+1}}\Big)$$

*with an absolute implied constant.*

*Sketch.* (i) The zeros are real by Theorem 13.5.4. As in the paragraph before Proposition 16.7.2, $K_\nu(2\pi) = K_0(2\pi)\prod_n(1+\nu^2/z_n^2)$, and at $\nu = \frac14 + \frac{it}{2}$ the factor of index $n$ contributes

$$\frac{1/8}{\frac{1}{16}+(\frac{t}{2}+z_n)^2} + \frac{1/8}{\frac{1}{16}+(\frac{t}{2}-z_n)^2} > 0$$

to $\psi'(t)$. Since $\Xi^\circ(t) = |K_{\frac14+\frac{it}{2}}(2\pi)|\cos\psi(t)$ and the modulus does not vanish for real $t$, the zeros on $(0,\infty)$ are where $\psi$ crosses $(\frac12+\mathbb{Z})\pi$, each once and simply. (ii) For $\nu = \frac14 + \frac{it}{2}$ the ascending series and the reflection formula give, as in the sketch of Proposition 16.7.2, $K_\nu(2\pi) = \frac12\Gamma(\nu)\pi^{-\nu}R(t)$ with $R(t) = 1 + O(t^{-1/2})$: the companion term $\frac12\Gamma(-\nu)\pi^{\nu}$ is now smaller only by a factor of order $t^{-1/2}$. As $\arg\Gamma(\frac14+\frac{it}{2}) - \frac{t}{2}\log\pi = \vartheta(t)$, $\psi - \vartheta$ is a continuous argument of $R$, zero at $t=0$, and it tends to a multiple of $2\pi$. The multiple is 0. Replace $2\pi$ by $a \in (0, 2\pi]$ in the argument of $K$ and in $(a/2)^{-\nu}$: the ratio $R_a(t)$ is continuous and nonzero, positive at $t=0$, equal to $1+O(t^{-1/2})$ uniformly for $a$ in compact sets, and within $\frac12$ of 1 for every $t$ once $a$ is small, so the limit of its argument is a multiple of $2\pi$ that depends continuously on $a$ and vanishes for small $a$. The statement on $g_n$ follows with $\vartheta'(t) = \frac12\log\frac{t}{2\pi} + O(t^{-2})$. (iii) Away from ordinates $N(T) = \vartheta(T)/\pi + 1 + S(T)$, and $N$ jumps from $n$ to $n+1$ at $\gamma_{n+1}$; averaging the one sided limits gives the identity, which needs no hypothesis on the other zeros. Subtracting it from (ii) gives $\vartheta(t_n^\circ) - \vartheta(\gamma_{n+1}) = \pi S(\gamma_{n+1}) + O(\gamma_{n+1}^{-1/2})$, and the mean value theorem with $\vartheta''(t) = O(1/t)$ and $S(T) = O(\log T)$ gives the displacement. □

Numerically, $\Xi^\circ$ has 28 zeros in $(0, 100]$ (the 29th is $t_{29}^\circ = 101.0946$), the first at $t_1^\circ = 19.332507$, and $\vartheta(t_n^\circ)/\pi - (n-\frac12)$ increases from $-0.2434$ at $n=1$ to $-0.0155$ at $n=28$ (computed; the existence of the 28 zeros is certified by sign changes in ball arithmetic, `code/gp_skeleton.py`). For $n \le 28$ the displacement $t_n^\circ - \gamma_{n+1}$ has mean absolute value 0.513 and correlation 0.889 with $\pi S(\gamma_{n+1})/\vartheta'(\gamma_{n+1})$ (computed). Each $t_n^\circ$ is paired with $\gamma_{n+1}$, and $\gamma_1$ has no partner.

## 16.8 Pólya's Mellin criterion applied to the Riemann kernel

Take $q=1$ and $\mathcal{F}(v) = \Phi(\sqrt{v})$ in P10, so that $\mathcal{F}(t^2) = \Phi(t)$ and the conclusion would be RH. The criterion asks that

$$M(z) = \int_0^\infty \Phi(u)\,u^{2z-1}\,du = \tfrac12\int_0^\infty \mathcal{F}(v)v^{z-1}dv \tag{16.8.1}$$

have no zeros in $\mathbb{C}\setminus(-\infty,0]$. The integral converges for $\operatorname{Re} z > 0$, and $M(k+\frac12) = \frac12(2k)!\,c_k > 0$ and $2M(\frac12) = \xi(\frac12)$. Near the origin $\Phi(u) = \sum_{j\ge0} p_j u^{2j}$, with radius of convergence $\pi/4$, $p_0 = \Phi(0) = 0.893394$ and $p_1 = -8.36525$; integrating termwise over $(0,\delta)$, $\delta < \pi/4$, gives $\int_0^\delta \Phi(u)u^{2z-1}du = \sum_j p_j\delta^{2z+2j}/(2z+$

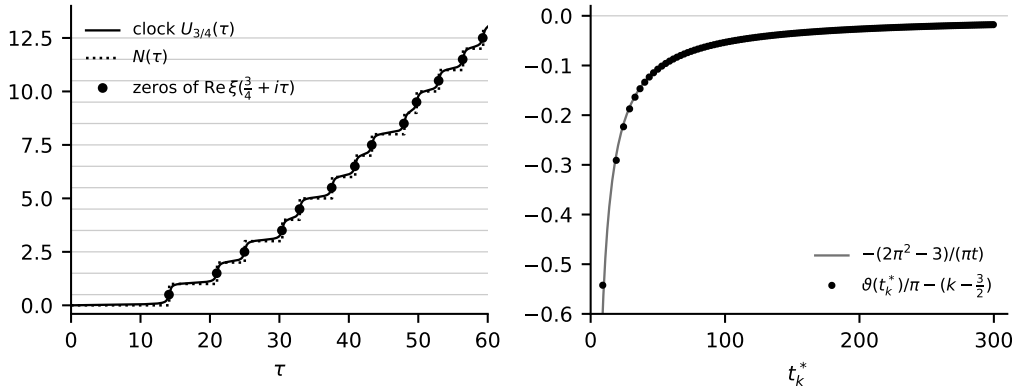



Figure 16.1: Left: the clock $U_{3/4}(\tau)$ of the basepoint law at $\alpha = 3/4$ for $0 \le \tau \le 60$, the zero count $N(\tau)$ (dotted), and the zeros of $\mathrm{Re}\,\xi(3/4 + i\tau)$, which sit exactly on the half integer levels (grey lines). Right: $\vartheta(t_k^*)/\pi - (k - \frac{3}{2})$ for the 138 zeros of Pólya's $\Xi^*$ below height 300, with the asymptote $-(2\pi^2 - 3)/(\pi t)$ of Proposition 16.7.2.

$2j)$, which continues $M$ to a meromorphic function with simple poles at $z = 0, -1, -2, \ldots$, and this continuation is the function that the criterion refers to.

The criterion does not apply to $\Phi$. If $X_\Phi$ has the Pólya density $\Phi/\Xi(0)$ (the variable $X$ of Theorem 5.2.3, and not the $X$ of this chapter), then $\mathbb{E}|X_\Phi|^{2s} = 2M(s + \frac{1}{2})/\Xi(0)$ with the $M$ of (16.8.1), so that theorem says that $M$ has a simple zero at $z_1 = s_1 + \frac{1}{2} = -7.99418226 + 2.93776707\,i$, which lies in $\mathbb{C} \setminus (-\infty, 0]$. We checked this independently (`polya_mellin_z1.py`, at 60 digits, with two quadrature layouts): $|M| = 8.0 \times 10^{-16}$ at the point $s_1 + \frac{1}{2}$ printed in that theorem, and on the circle $|z - z_1| = \frac{1}{2}$ we have $|M| \ge 0.224$ and the winding number of $M$ is 1, so that $z_1$ is the only zero in the disc. So the hypothesis of the criterion fails for $q = 1$. For $q \ge 2$ the function $\mathcal{F}(t^{2q})$ has no $t^2$ term whenever $\mathcal{F}$ is holomorphic near the origin, whereas $\Phi(t) = 0.893394 - 8.36525\,t^2 + \cdots$, so there is no factorisation $\Phi(t) = \mathcal{F}(t^{2q})$ with $q \ge 2$. The Mellin route of the 1927 paper is therefore closed for $\Phi$ itself, by the zero that closes the Bernstein Pick route in Theorem 5.2.3.

The zero $z_1$ is the only obstruction that we found. An argument principle scan of the continuation (`polya_mellin.py`, floating point) finds no zero with $0.05 \le |\mathrm{Im}\,z| \le 14.05$ in $-3.95 \le \mathrm{Re}\,z \le 8.05$. A plain Gauss Legendre rule for (16.8.1) produces spurious zeros with $\mathrm{Re}\,z < \frac{1}{2}$, because the integrand has an integrable singularity $u^{2\,\mathrm{Re}\,z-1}$ at the origin that oscillates like $\cos(2\,\mathrm{Im}\,z \log u)$; the termwise integration on $(0, \frac{1}{4})$ removes the difficulty.

## 16.9 Is the $2\sin^2$ lemma useful with regard to Pólya's work?

### 16.9.1 Where it helps

The lemma is applied at a basepoint $\alpha$ through the weight $e^{-\alpha x}$ in its hypothesis, which on the kernel side is the tilt $e^{au}$, the non even half of Pólya's universal factor $\cosh(au)$; it computes the continuous Thorin densities of which Pólya's approximations are the roundings, up to the affine term of its compensation (Theorem 6.2.12, Proposition 16.7.2); and positivity of the Thorin density at a basepoint makes the clock increasing along $\operatorname{Re} s = \alpha$ and places the real zeros of $\operatorname{Re}\xi(\alpha + it)$ at its half integer crossings (Proposition 16.6.3).

### 16.9.2 Where it cannot help

*The Bessel test.* The Bessel model $f(\nu) = K_\nu(2\pi)$ is even in $\nu$, of order one, with zeros $\pm iz_n$ on $\operatorname{Re}\nu = 0$ (Section 16.7) and $\sum z_n^{-2} < \infty$, so the product formula and the basepoint theorem (Theorem 1.2.2) apply verbatim, as in Theorem 8.7.10, which is the case $\alpha = 0$: $K_\alpha(2\pi)/K_{\alpha+\sqrt{s}}(2\pi)$ is a GGC for every $\alpha \geq 0$, with Lorentzian Thorin density when $\alpha > 0$. The hypothesis of the lemma at basepoint $\alpha$ is that $(\log f)''$ is completely monotone on $(\alpha, \infty)$. Table 16.2 shows the signs of $(-1)^j d^{2+j} \log K_\nu(2\pi)/d\nu^{2+j}$: the third entry is negative at every $\nu \leq 5$ and a later entry fails at $\nu = 8$. So the primal measure of the lemma is signed at every basepoint tested, although the reciprocal law is a GGC at every basepoint. The lemma's success for $\xi$ at $\alpha > 1$ is a cancellation, as Chapter 9 already observes.

| $\nu$ | $j=0$ | $j=1$ | $j=2$ | $j=3$ | $j=4$ | $j=5$ |
|---|---|---|---|---|---|---|
| 0.25 | 0.1482 | $6.7\times10^{-4}$ | $-2.7\times10^{-3}$ | $-9.8\times10^{-5}$ | $3.9\times10^{-4}$ | $3.6\times10^{-5}$ |
| 1 | 0.1469 | $2.6\times10^{-3}$ | $-2.5\times10^{-3}$ | $-3.7\times10^{-4}$ | $3.2\times10^{-4}$ | $1.3\times10^{-4}$ |
| 3 | 0.1373 | $6.6\times10^{-3}$ | $-1.3\times10^{-3}$ | $-6.8\times10^{-4}$ | $-1.4\times10^{-5}$ | $1.5\times10^{-4}$ |
| 5 | 0.1223 | $8.0\times10^{-3}$ | $-1.5\times10^{-4}$ | $-4.6\times10^{-4}$ | $-1.6\times10^{-4}$ | $6.9\times10^{-6}$ |
| 8 | 0.0991 | $7.1\times10^{-3}$ | $5.6\times10^{-4}$ | $-6.9\times10^{-5}$ | $-7.7\times10^{-5}$ | $-3.5\times10^{-5}$ |

Table 16.2: $(-1)^j \frac{d^{2+j}}{d\nu^{2+j}} \log K_\nu(2\pi)$.

*Kernel multipliers.* Pólya's universal factors $e^{\lambda u^2}$ and $\cosh(au)$ multiply the kernel, but the lemma acts on $\log f$ and has no composition rule that carries a lemma statement at one value of $\lambda$ or $a$ to another, which is why the heat flow row of Table 15.1 stands apart from the basepoint row.

### 16.9.3 Verdict

The lemma is not a mechanism for producing real zeros: in the Bessel model, where Pólya's theorem gives real zeros and every basepoint law is a GGC, the positivity the lemma needs is false. Any route to RH through the lemma must

therefore control a cancellation between signed primal measures, and Pólya's own tools (universal factors, difference equations for $K_\nu$, Hermite Biehler) are the natural candidates for that control.

## 16.10 What would help show RH

None of Pólya's theorems crosses into the critical strip. Each stops at the same place as the basepoint programme.

1. The cosh factor and de Bruijn's strip theorem give real zeros of the even part for $\alpha \geq 1/2 + \Delta$, the same threshold $\sigma_{\max}$ as the basepoint laws. Proposition 16.6.3 identifies the two families, so no gain in the strip.

2. The heat flow factor $e^{\lambda u^2}$ reaches $\Lambda \leq 0.2$ (Platt and Trudgian, sharpening the bound 0.22 of Polymath 15), inside the strip in the Gaussian sense, and $\Lambda \geq 0$ by Rodgers and Tao; the remaining gap is the whole problem.

3. The Pólya Schur and Schoenberg ladders are equivalent to RH; their first rungs hold: the Turán inequality for the first is a theorem, and log concavity for the second is settled by a floating point certificate in Section 16.11.

4. The Mellin criterion does not apply to $\Phi$ (Section 16.8): for $q = 1$ the Mellin transform has the zero of Theorem 5.2.3, and for $q \geq 2$ there is no factorisation $\Phi(t) = \mathcal{F}(t^{2q})$. A different criterion, applied to a different factorisation of the Pólya density, such as its signed mixture of Gamma(2) and Gamma(3) densities (Section 8.5.4.3) or its representation through maxima (Chapter 14), would be needed.

5. The Bessel model is a laboratory. Since its analogue of RH is a theorem of Pólya, any proposed mechanism can be tested on it before it is tried on $\xi$; the $2\sin^2$ positivity mechanism already fails this test.

Problem 16.2. Find a sign condition on the primal measure of the lemma that holds for the Bessel model at every basepoint $\alpha > 0$, is preserved by the Euler product, that is, by adding the primal measures of the Euler factors, and fails for the Davenport Heilbronn function.

Problem 16.3. Prove real rootedness of $A_\alpha = \frac{1}{2}(\xi(\alpha + w) + \xi(\alpha - w))$ for one fixed $\alpha \in (\frac{1}{2}, 1)$ without proving the quasi Riemann hypothesis at $\alpha$, by a Pólya type argument that uses the Euler product (the odd part $B_\alpha$ need not be real rooted).

Problem 16.4. Determine whether the derivative laws $\xi^{(2n)}(\frac{1}{2})/\xi^{(2n)}(\frac{1}{2} + \sqrt{s})$, or their counterparts $F^{(n)}(0)/F^{(n)}(s)$ with derivatives taken in the Thorin variable $s$, are GGCs for some $n$. For the first family, Ki and Kim give a local statement and

Conrey the proportion of zeros on the line, for large $n$; for the second, Griffin, Ono, Rolen and Zagier give the first $d$ rungs of the Pólya Schur ladder for $n \geq N(d)$ (Section 16.11). For neither family is a global statement known for any $n$, and one for a single $n$ would be the first unconditional GGC in its family.

Problem 16.5. Under RH, which is what makes the midpoint rule of Proposition 16.6.3 available for every $\alpha > \frac{1}{2}$, prove that it converges, $\mu_\alpha^P - \mu^{(\alpha)} \to 0$ weakly as $\alpha \downarrow \frac{1}{2}$, with an explicit rate, so that the atomic and continuous descents can be run together.

## 16.11 Progress on the open problems

### 16.11.1 Problem 16.5: the even part and basepoint laws differ by an explicit factor

Theorem 16.11.1. *For every $\alpha > 1/2$ and real $s \geq 0$ put $R_\alpha(s) = \xi(\alpha - \sqrt{s})/\xi(\alpha + \sqrt{s})$. Then, unconditionally, $R_\alpha(0) = 1$, $0 < R_\alpha(s) < 1$ for $s > 0$, and*

$$\begin{aligned} \frac{\varphi_\alpha^P(s)}{\varphi^{(\alpha)}(s)} &= \frac{2}{1 + R_\alpha(s)}, \\ \psi_\alpha^P(s) - \psi_\alpha(s) &= \log \frac{1 + R_\alpha(s)}{2}, \end{aligned} \tag{16.11.1}$$

*where $\psi = -\log \varphi$ denotes the Laplace exponent, and for $s > 0$ the difference of the exponents lies in $(-\log 2,\, 0)$. Hence $\varphi^{(\alpha)} \leq \varphi_\alpha^P \leq 2\varphi^{(\alpha)}$ on $[0, \infty)$, and as $\alpha = \frac{1}{2} + a \downarrow \frac{1}{2}$,*

$$\psi_\alpha^P(s) - \psi_\alpha(s) = -a\, \frac{\xi'}{\xi}\big(\tfrac{1}{2} + \sqrt{s}\big) + O(a^2) \tag{16.11.2}$$

*locally uniformly in $s$. In particular $\varphi_\alpha^P \to \varphi$ and $\varphi^{(\alpha)} \to \varphi$ pointwise, so whenever the two families are laws they converge weakly to the reciprocal Xi law.*

*Proof.* By definition $A_\alpha(\sqrt{s}) = \frac{1}{2}\xi(\alpha + \sqrt{s})\,(1 + R_\alpha(s))$, which gives (16.11.1). For real $\sigma$, $\xi(\sigma) > 0$, $\xi$ is symmetric about $\frac{1}{2}$ and increasing on $[\frac{1}{2}, \infty)$, because $\xi(\frac{1}{2} + w) = \sum c_k w^{2k}$ with $c_k > 0$. Since $\xi(\alpha - \sqrt{s}) = \xi(1 - \alpha + \sqrt{s})$ and $|\alpha + \sqrt{s} - \frac{1}{2}| > |\frac{1}{2} - \alpha + \sqrt{s}|$ when $\alpha > \frac{1}{2}$ and $s > 0$, we get $0 < R_\alpha < 1$ for $s > 0$, while $R_\alpha(0) = 1$. For (16.11.2), $\log R_\alpha(s) = \log \xi(\frac{1}{2} + \sqrt{s} - a) - \log \xi(\frac{1}{2} + \sqrt{s} + a) = -2a\frac{\xi'}{\xi}(\frac{1}{2} + \sqrt{s}) + O(a^3)$, and $\log \frac{1+R}{2} = \frac{1}{2}\log R + O((\log R)^2)$. □

Combined with Proposition 16.6.3, Theorem 16.11.1 evaluates the error of the midpoint rule exactly. For every $\alpha$ as in Proposition 16.6.3, that is for $\alpha > \sigma_{\max}$ and unconditionally for $\alpha \geq 1$, and every $s \geq 0$,

$$\sum_{k \geq 1} \log\!\left(1 + \frac{s}{\tau_\alpha (k - \frac{1}{2})^2}\right) = \int_0^\infty \log\!\left(1 + \frac{s}{\tau_\alpha(v)^2}\right) dv + \log \frac{1 + R_\alpha(s)}{2}. \tag{16.11.3}$$

The defect of the midpoint sum is a ratio of two values of $\xi$ on the real axis, and no modulus of continuity of the clock is needed. Numerically, for $\alpha = 1, 0.75, 0.6, 0.52$ the defect divided by $a = \alpha - \frac{1}{2}$ varies by 1.2% at $s = 1$ and by 10% at $s = 100$, and approaches $-\frac{\xi'}{\xi}(\frac{1}{2} + \sqrt{s}) = -0.0461, -0.408$, as (16.11.2) predicts.

### 16.11.2 Problem 16.1: $f_X$ is log concave

Write $N = f_X f_X'' - (f_X')^2$, so that $\mathrm{PF}_2$ is $N < 0$, and recall $\varphi(t^2) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+t) > 0$ for real $t$.

Lemma 16.11.2 (Origin). *$f_X'(0) = 0$ and $f_X''(0) = -\frac{1}{\pi}\int_0^\infty t^2\varphi(t^2)\,dt < 0$, so $f_X$ is log concave on a neighbourhood of 0, unconditionally.*

Lemma 16.11.3 (Residue expansion). *Let $H = 3 \times 10^{12}$ be the height up to which the zeros of $\xi$ are verified to be simple and on the critical line [419], let $0 < T < H$ be different from every ordinate, and put $\eta_k = -\Xi(0)/\Xi'(\gamma_k)$. Then, unconditionally, for $x > 0$,*

$$f_X(x) = \sum_{\gamma_k < T} \eta_k\, e^{-\gamma_k x} + E_T(x), \qquad \bigl|E_T^{(j)}(x)\bigr| \le C_j(T)\, e^{-Tx},$$
$$C_j(T) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{|u - iT|^j\, \xi(\frac{1}{2})}{|\xi(\frac{1}{2} + u - iT)|}\, du. \tag{16.11.4}$$

*The residues alternate in sign, $\eta_1 > 0 > \eta_2$, and so on.*

*Proof.* $f_X(x) = \frac{1}{2\pi}\int \varphi(t^2)e^{-itx}dt$. Move the line of integration to $\operatorname{Im} t = -T$. The poles crossed are the zeros of $\xi(\frac{1}{2} + t)$ with $-T < \operatorname{Im} t < 0$, which below $H$ are $t = -i\gamma_k$, simple; the vertical sides vanish because $|\xi(\sigma + i\tau)|$ grows like a gamma function as $|\sigma| \to \infty$, uniformly for bounded $\tau$. The residue at $-i\gamma_k$ is computed from $\xi'(\frac{1}{2} - i\gamma) = i\,\Xi'(\gamma)$. The signs alternate because $\Xi$ changes sign at each simple real zero, starting from $\Xi(0) > 0$. □

Numerically $\log|\eta_k| - \pi\gamma_k/4$ equals $-5.2, -8.0, -8.8, -9.7, -11.1$ for $k = 1, 10, 20, 40, 60$, so $|\eta_k|$ grows like $e^{\pi\gamma_k/4}$ up to a power of $\gamma_k$, and the expansion is useful only for $x > \pi/4$.

Lemma 16.11.4 (Two exponentials). *If $f = Ae^{-ax} + Be^{-bx}$, then $ff'' - (f')^2 = AB\,(b-a)^2 e^{-(a+b)x}$. With $A = \eta_1 > 0 > B = \eta_2$ this is negative.*

The proof is a one line expansion. With $T = 23$, between $\gamma_2$ and $\gamma_3$, Lemmas 16.11.3 and 16.11.4 give $Ne^{(\gamma_1+\gamma_2)x} = \eta_1\eta_2(\gamma_2 - \gamma_1)^2 + \varepsilon(x)$, where $|\varepsilon(x)|$ is bounded by a sum of terms each decreasing in $x$, built from $C_0, C_1, C_2$ and $|\eta_1|, |\eta_2|$. Numerically $\eta_1 = 359.524$, $\eta_2 = -28007.2$, $\eta_1\eta_2(\gamma_2 - \gamma_1)^2 = -4.776 \times 10^8$, and $C_0, C_1, C_2 = 5.99 \times 10^4,\ 1.38 \times 10^6,\ 3.19 \times 10^7$.

THEOREM 16.11.5 ($\mathrm{PF}_2$, floating point certificate). *The density $f_X$ of $X = Z\sqrt{2Y}$ is log concave on $\mathbb{R}$. The certificate has three parts, for $x \geq 0$ (the density is even).*

*(A) On $[0, 0.95]$, direct quadrature of $f^{(j)}(x) = \frac{1}{\pi}\int_0^\infty \varphi(t^2)\frac{d^j}{dx^j}\cos(tx)\,dt$ at the left end $x_i$ of each of 770 cells, with $\sup|N'|$ on a cell of length $h$ bounded by $B(h)$ through $|f^{(j)}(y)| \leq |f^{(j)}(x_i)| + h\,m_{j+1}$ and $m_j = \frac{1}{\pi}\int_0^\infty t^j\varphi(t^2)dt$ ($m_0, \ldots, m_4 = 1.899,\ 7.321,\ 44.94,\ 355.8,\ 3361$), proves $N < 0$. The cells are chosen from the left, each as long as possible with $h\,B(h) \leq 0.99\,|N(x_i)|$, so that $N \leq N(x_i) + h\,B(h) < 0$ on the cell; any factor in $(0, 1)$ in place of 0.99 works (814 cells at 0.9, 1165 at 0.5).*

*(B) On $[0.95, 2.1]$, the expansion (16.11.4) with the first twelve zeros and $T = \frac{1}{2}(\gamma_{12} + \gamma_{13}) = 57.90$ gives $Ne^{(\gamma_1+\gamma_2)x}$ as an explicit exponential sum plus a remainder with decreasing majorant; 26 equal cells certify negativity (22 equal cells are the least that do, and two adaptive cells suffice). At $x = 0.95$ the sum is $-2.18 \times 10^8$ and the remainder at most $7.03 \times 10^6$.*

*(C) For $x \geq 2.1$ the two term form with $T = 23$ has $|\varepsilon(x)| \leq 4.687 \times 10^8 < 4.776 \times 10^8$: the majorant decreases and equals $4.687 \times 10^8$ at $x = 2.1$.*

Figure 16.2 (left) shows $-(\log f_X)''$. It is 23.66 at the origin and then decays like $\frac{|\eta_2|}{\eta_1}(\gamma_2 - \gamma_1)^2 e^{-(\gamma_2-\gamma_1)x}$, which is $4.0 \times 10^{-9}$ at $x = 4$: the log concavity of the tail is exponentially thin, and it is held up by the sign of $\eta_2$ alone.

*Numerics.* The certificate is reproduced by `code/polya_density.py` (`code/README.md`) in mpmath at 80 digits. The values $f_X^{(j)}$ on $[0, 1.6]$ come from composite Gauss Legendre quadrature and beyond 1.6 from the residue expansion; the two agree to a relative $2 \times 10^{-20}$ at $x = 1.6$ with fourteen zeros and to $8 \times 10^{-42}$ with forty (for the derivatives of orders one to three, to $2 \times 10^{-18}$ and $6 \times 10^{-39}$). The quadrature errors are many orders of magnitude below the margins of Parts (A) and (B). Part (C) is thin: the majorant falls below $4.776 \times 10^8$ from $x = 2.0904$ on, so that at $x = 2.1$ the margin is 1.9% of the bound, and a relative error of 2% in $C_0, C_1, C_2$ would break it; the constants are stable to many digits under two quadrature layouts. The only arithmetic input is that the first thirteen zeros are simple and on the line.

The mechanism of Lemma 16.11.4 extends, as a sketch, to every order $n$ with $\gamma_n < H$ in the far tail. If $x_1 < \cdots < x_n$, $y_1 < \cdots < y_n$ and every $x_i - y_j$ is large, the Cauchy Binet expansion of $\det[f_X(x_i - y_j)]$ is dominated by the term from $\gamma_1, \ldots, \gamma_n$, whose sign is $\operatorname{sgn}(\eta_1 \cdots \eta_n)\,(-1)^{n(n-1)/2} = (-1)^{\lfloor n/2 \rfloor + n(n-1)/2} = +1$. So the $\mathrm{PF}_n$ minors with $\gamma_n < H$ are positive far in the tail, unconditionally, provided the remainder $E_T$ of (16.11.4), which is not a sum of exponentials, can be dominated inside the determinant; Proposition 16.11.15 does this for coalescing points.

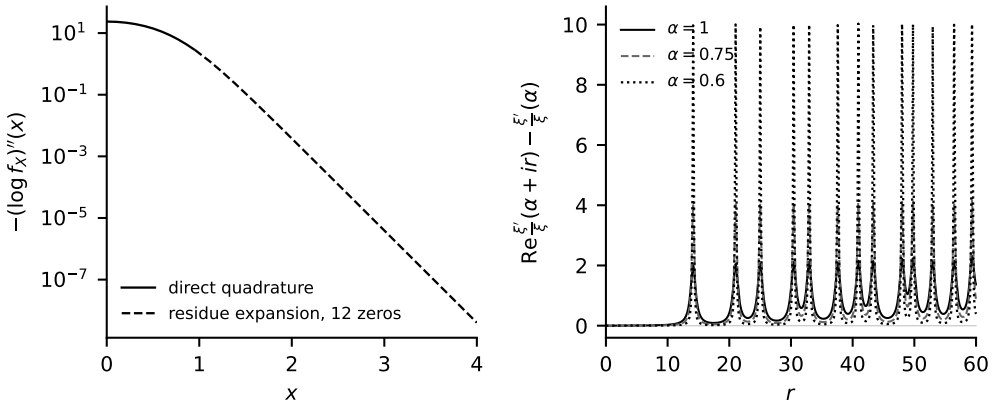



Figure 16.2: Left: $-(\log f_X)''(x)$ on a log scale, from direct quadrature on $[0, 0.95]$ (solid) and from the residue expansion with twelve zeros on $[0.95, 4]$ (dashed); it is positive throughout, which is Theorem 16.11.5. Right: the compensated clock condition $D_\alpha(r) = \mathrm{Re}\,\frac{\xi'}{\xi}(\alpha + ir) - \frac{\xi'}{\xi}(\alpha) \geq 0$ of Theorem 16.11.7 for $\alpha = 1$ (solid), 0.75 (dashed) and 0.6 (dotted), computed from the Hadamard sum over the first 1700 zeros at step 0.02; $D_\alpha$ vanishes at $r = 0$ and is positive after, with a peak of height about $1/(\alpha - \frac{1}{2})$ at each ordinate. Drawn by `polya_fig.py`.

THEOREM 16.11.6 ($f_X$ is uniformly close to a $\mathrm{PF}_\infty$ density). *Let $X_{\mathrm{on}} = \sum L_\gamma/\gamma$, the sum over the zeros on the critical line with independent standard Laplace variables. Unconditionally $X_{\mathrm{on}}$ has a $PF_\infty$ density $f_{\mathrm{on}}$, and for $j = 0, 1, 2$*

$$\sup_x \left| f_X^{(j)}(x) - f_{\mathrm{on}}^{(j)}(x) \right| \leq \varepsilon_j := \frac{1}{\pi} \int_0^\infty t^j \, \Pi_K(t) \left( 2S\, t^2 + S^2 t^4 \right) dt, \tag{16.11.5}$$

*where $\Pi_K(t) = \prod_{k \leq K} (1 + t^2/\gamma_k^2)^{-1}$ for any $K$ with $\gamma_K < H$, and $S = \sum_{\gamma > H} \gamma^{-2} \leq (\log \frac{H}{2\pi} + 2)/(2\pi H) = 1.533 \times 10^{-12}$ for $H = 3 \times 10^{12}$. With $K = 300$, $\varepsilon_0 \leq 1.57 \times 10^{-10}$, $\varepsilon_1 \leq 1.3 \times 10^{-9}$ and $\varepsilon_2 \leq 1.3 \times 10^{-8}$. Consequently every $n \times n$ minor $\det[f_X(x_i - y_j)]$ with ordered $x$ and $y$ is at least $-n \cdot n!\,(2\varepsilon_0 + f_{\mathrm{on}}(0))^{n-1} \varepsilon_0$, which is $-1.1 \times 10^{-8}$ for $n = 3$ and $-1.2 \times 10^{-7}$ for $n = 4$, since $f_{\mathrm{on}}(0) \leq \frac{1}{\pi} \int_0^\infty \Pi_K = 1.976$.*

*Proof.* The characteristic function of $X_{\mathrm{on}}$ is $\varphi_{\mathrm{on}}(t^2) = \prod_{\text{on line}} (1 + t^2/\gamma^2)^{-1}$, the reciprocal of a function of type (16.3.1), so $f_{\mathrm{on}}$ is $\mathrm{PF}_\infty$ by Schoenberg's theorem whatever the zeros off the line do. Write $\varphi(t^2) = \varphi_{\mathrm{on}}(t^2) R(t^2)$, where $R$ collects the off line zeros. Each conjugate pair $s_\rho, \bar{s}_\rho$ with $s_\rho = -a^2$, $a = \gamma - i\delta$, contributes $1/P_\rho(t^2)$ with $P_\rho(v) = 1 + 2v\,\mathrm{Re}(a^{-2}) + v^2 |a|^{-4}$. Since $\gamma > H > \frac{1}{2} > |\delta|$, $\mathrm{Re}\, a^{-2} = (\gamma^2 - \delta^2)/|a|^4 > 0$, so $P_\rho \geq 1$ on $[0, \infty)$, $0 < R \leq 1$ and $1 - R \leq \sum_\rho (P_\rho - 1) \leq 2St^2 + S^2 t^4$. Fourier inversion gives $|f_X^{(j)} - f_{\mathrm{on}}^{(j)}| \leq \frac{1}{\pi} \int_0^\infty t^j \varphi_{\mathrm{on}}(t^2)(1 - R(t^2))\, dt$, and $\varphi_{\mathrm{on}} \leq \Pi_K$ because the first $K$ zeros are on the line. The tail sum $S$ follows

by partial summation from the zero counting formula with an explicit error term [176]; Proposition 14.7.2 gives the sharper value $1.4798\times10^{-12}$ for the same sum, and the constants of the theorem were computed with the larger one. For the minors, expand the determinant multilinearly in the differences $f_X - f_{\mathrm{on}}$ and use $\det[f_{\mathrm{on}}(x_i-y_j)]\ge 0$ and $|f_{\mathrm{on}}|\le f_{\mathrm{on}}(0)$. □

RH is the statement that the uniform distance of $f_X$ to a $\mathrm{PF}_\infty$ density is zero. The bound on the minors grows quickly with $n$ (it exceeds 1 at $n=10$), so for large orders the theorem gives no information, and a failure of $\mathrm{PF}_3$ would be invisible at the level $10^{-8}$. Numerically there is none: over 6050 Toeplitz type minors for each of $n=3$ and $n=4$, with spacings $0.02d$ and $0.02(d+1)$, $1\le d\le 25$, and offsets up to 1.2, all are positive and the smallest normalised minor $\det[f_X(x_i-y_j)]/\prod_i f_X(x_i-y_i)$ is $1.6\times10^{-10}$ for $n=3$ and $3.5\times10^{-19}$ for $n=4$. Both occur at the smallest spacing; most of the smallness comes from nearly coinciding points, since for $n=3$ the smallest value over the offsets grows with $d$ from $1.6\times10^{-10}$ at $d=1$ to $6.0\times10^{-2}$ at $d=25$.

### 16.11.3 Problem 16.2: the exact form of the $2\sin^2$ lemma

Stieltjes inversion gives the Thorin density itself, with no sign assumption on the primal measure.

Theorem 16.11.7. *Let $f$ be a real entire function with no zeros in $\operatorname{Re} z>\alpha-\delta$ for some $\delta>0$ and with $\frac{f'}{f}(z)=O(\log|z|)$ uniformly on $\operatorname{Re} z\ge\alpha$. This holds for $\xi$ with $\alpha>\sigma_{\max}$, with $\delta=\alpha-\sigma_{\max}$, the bound following from the Hadamard product because $O(\log T)$ zeros have ordinate in $[T,T+1]$, and for $\nu\mapsto K_\nu(2\pi)$ with $\alpha>0$, with $\delta=\alpha$, the zeros being purely imaginary with spacing about $\pi/\log y$ at height $y$. Let $\psi(u)=\log f(\alpha+w)-\log f(\alpha)-w\frac{f'}{f}(\alpha)$ with $w=\sqrt{2u}$ the principal root. Then $\psi'$ is holomorphic on $\mathbb{C}\setminus(-\infty,0]$, $\psi'(u)=O(|u|^{-1/2}\log|u|)$, and*

$$\begin{aligned}\psi'(u)&=\int_0^\infty\frac{\omega_\alpha(z)}{z+u}\,dz,\\ \omega_\alpha(z)&=\frac{1}{\pi\sqrt{2z}}\left[\operatorname{Re}\frac{f'}{f}\Big(\alpha+i\sqrt{2z}\Big)-\frac{f'}{f}(\alpha)\right].\end{aligned}\tag{16.11.6}$$

*Hence the compensated reciprocal $e^{wf'(\alpha)/f(\alpha)}f(\alpha)/f(\alpha+w)$ is the Laplace transform in $u$ of a GGC if and only if*

$$\operatorname{Re}\frac{f'}{f}(\alpha+ir)\ge\frac{f'}{f}(\alpha)\qquad\text{for all } r>0,\tag{16.11.7}$$

*that is, if and only if the clock at basepoint $\alpha$ runs at least as fast at every height as at height zero.*

*Proof.* For $u \notin (-\infty, 0]$ the root $w$ has positive real part, so $\alpha + w$ stays in the zero free half plane and $\psi'(u) = \big[\frac{f'}{f}(\alpha + w) - \frac{f'}{f}(\alpha)\big]/w$ is holomorphic there. By the hypothesis on $\frac{f'}{f}$ the numerator of $\psi'$ is $O(\log|w|)$, whence the bound on $\psi'$; the zero free strip of width $\delta$ makes $\psi'$ continuous up to the cut. Cauchy's formula on a keyhole contour around the cut, with the bound to discard the large circle and the integrability of $\omega_\alpha$ at 0 (where $\omega_\alpha(z) = O(\sqrt{z})$ by the evenness of $r \mapsto \operatorname{Re}\frac{f'}{f}(\alpha+ir)$), gives the Stieltjes representation with density $-\frac{1}{\pi}\operatorname{Im}\psi'(-z+i0)$. At $u = -z+i0$ the root is $w = ir$, $r = \sqrt{2z}$, and $-\frac{1}{\pi}\operatorname{Im}\psi' = \frac{1}{\pi r}\operatorname{Re}\big[\frac{f'}{f}(\alpha+ir) - \frac{f'}{f}(\alpha)\big]$, which is $\omega_\alpha$. A Stieltjes function with zero constant term is the derivative of a Thorin Bernstein function exactly when its representing density is nonnegative, which is (16.11.7). Finally, if $\log f$ has the representation of the lemma with $\nu(dx) = e^{-\alpha x}\mu_f(dx)$, then $\frac{f'}{f}(\alpha + w) - \frac{f'}{f}(\alpha) = \int(1 - e^{-wx})\nu(dx)$ and $\omega_\alpha(z) = \frac{1}{\pi r}\int 2\sin^2(rx/2)\,\nu(dx)$, which is the $2\sin^2$ lemma. □

The right sign condition on the primal measure is not $\nu \geq 0$ but

$$\int \big(1 - \cos(rx)\big)\,\nu(dx) \geq 0 \qquad \text{for all } r, \tag{16.11.8}$$

that the cosine transform of $\nu$ is maximal at the origin, read in this way because $\nu$ has infinite mass for $\xi$ and for the Bessel model. Its local form at $r = 0$ is $-(\log f)'''(\alpha) \geq 0$, the $j = 1$ column of Table 16.2, which is positive at every basepoint; the failures in that table are in higher columns, which (16.11.8) does not require. Numerically, (16.11.7) holds for the Bessel model at $\alpha = 0.25, 1, 3, 8$ on $0 < r \leq 60$ (grid of step 0.02), and for $\xi$ at $\alpha = 1, 0.75, 0.6, 0.55$ on $0 < r \leq 100$ (grid of step 0.01). That the condition holds for the Bessel model at every $\alpha > 0$ rests on these numerics, and that it fails for the Davenport Heilbronn function (Problem 16.2) is not discussed here. For $\xi$, condition (16.11.7) reads $\operatorname{Re}\frac{\xi'}{\xi}(\alpha + ir) \geq b_\alpha = \frac{\xi'}{\xi}(\alpha)$, the compensated companion of Lagarias's criterion $\operatorname{Re}\frac{\xi'}{\xi} > 0$; in the variable $u = w^2/2$ the Thorin densities of the basepoint law and of the compensated reciprocal differ by the stable term $b_\alpha/(\pi\sqrt{2z})$ of Proposition 9.1.12, where the stable factor is $e^{-b_\alpha\sqrt{u}}$ in the variable $u$ of that proposition. To the right of 1 the primes enter exactly as the lemma predicts. Write $\frac{\xi'}{\xi} = \mathcal{A} + \frac{\zeta'}{\zeta}$ with the archimedean part $\mathcal{A}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})$.

Proposition 16.11.8 (Primes and the compensated clock). *For $\alpha > 1$ and real $r$, write $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + ir) - \frac{\xi'}{\xi}(\alpha)$, which is $v_\alpha(r) - b_\alpha$ in the notation of Section 9.1. Then*

$$\begin{aligned} D_\alpha(r) = {} & \big[\operatorname{Re}\mathcal{A}(\alpha + ir) - \mathcal{A}(\alpha)\big] \\ & + \sum_{n \geq 2} \frac{\Lambda(n)}{n^\alpha}\, 2\sin^2\Big(\frac{r\log n}{2}\Big). \end{aligned} \tag{16.11.9}$$

*The prime sum is nonnegative; it is the* $2\sin^2$ *transform of the positive measure* $\sum \Lambda(n)n^{-\alpha}\delta_{\log n}$. *The archimedean bracket equals* $\sum_{k\ge1} q_{\alpha+2k}(r) - q_{\alpha-1}(r)$ *with* $q_c(r) = r^2/(c(c^2+r^2))$, *so that only the pole term is negative, and it is nonnegative for every* $r$ *if and only if* $\zeta(3, 1+\frac{\alpha}{2}) \ge 8(\alpha-1)^{-3}$, *that is, if and only if* $\alpha \ge \alpha_1 = 7.78918\ldots$; *the compensated condition then holds from the identity alone. For* $1 < \alpha < \alpha_1$ *the bracket dips below zero (its minimum is* $-4.83$ *at* $\alpha = 1.2$, $r = 1.57$; $-0.74$ *at* $\alpha = 2$; $-0.011$ *at* $\alpha = 6$*), and the primes must make up the deficit. They do, for every* $\alpha \ge 1$, *by Proposition 9.1.12, whose proof combines the zeros below height* 2000, *the bounds of [176] for the zeros above, and a computation in ball arithmetic;* (16.11.9) *is* (9.1.7).

*Proof.* For $\alpha > 1$, $\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+ir) - \frac{\zeta'}{\zeta}(\alpha) = \sum \Lambda(n)n^{-\alpha}(1-\cos(r\log n))$ by the Dirichlet series, and $1-\cos x = 2\sin^2(x/2)$. For the bracket, $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\ge0} y^2/((x+k)((x+k)^2+y^2))$ for the digamma function $\psi$, so with $x = \alpha/2$, $y = r/2$ the term $\frac12\Gamma'/\Gamma(s/2)$ contributes $\sum_{k\ge0} q_{\alpha+2k}(r)$; the term $k=0$ cancels the contribution $-q_\alpha(r)$ of $1/s$, and $1/(s-1)$ contributes $-q_{\alpha-1}(r)$. Each ratio $((\alpha-1)^2+r^2)/((\alpha+2k)^2+r^2)$ increases with $r$, so the bracket $r^2\big[\sum_{k\ge1} \frac{1}{(\alpha+2k)((\alpha+2k)^2+r^2)} - \frac{1}{(\alpha-1)((\alpha-1)^2+r^2)}\big]$ is nonnegative for every $r$ if and only if it is nonnegative as $r\to0$, that is, if and only if $\sum_{k\ge1}(\alpha+2k)^{-3} \ge (\alpha-1)^{-3}$, which is $\zeta(3,1+\frac{\alpha}{2}) \ge 8(\alpha-1)^{-3}$. The quantity $(\alpha-1)^3\sum_{k\ge1}(\alpha+2k)^{-3} = \sum_{k\ge1}\big(\frac{\alpha-1}{\alpha+2k}\big)^3$ increases with $\alpha$, so the condition holds exactly for $\alpha \ge \alpha_1$, where this quantity equals 1. □

As $\alpha\downarrow1$ the deficit of the bracket comes from the pole term, $\operatorname{Re}\frac{1}{\alpha-1+ir} - \frac{1}{\alpha-1} \sim -\frac{1}{\alpha-1}$ for fixed $r\neq0$, and the prime sum compensates it because $-\frac{\zeta'}{\zeta}(\alpha) \sim \frac{1}{\alpha-1}$. Grouping $\frac{1}{s-1}$ with $\frac{\zeta'}{\zeta}(s)$, which is holomorphic on $\operatorname{Re}s\ge1$ by the prime number theorem, both pieces stay finite at $\alpha=1$. Below 1 the Dirichlet series is gone and (16.11.9) has no direct analogue; this is the wall that Proposition 9.1.12 meets at $\alpha=1$.

Remark 16.11.9 (The pole needs the gamma factor). The gamma terms of the archimedean bracket of (16.11.9) cannot be dropped while its pole term is kept [253]. Let $f(s) = (s-1)\zeta(s)$, a real entire function with no zeros in $\operatorname{Re}s\ge1$. For every $\alpha>1$ the compensated reciprocal $e^{w\frac{f'}{f}(\alpha)}f(\alpha)/f(\alpha+w)$ is not the Laplace transform in $u=w^2/2$ of a GGC. By Theorem 16.11.7, whose hypotheses $f$ satisfies, and the uniqueness of the Stieltjes representation, it would be one only if

$$D^f_\alpha(r) = \operatorname{Re}\frac{f'}{f}(\alpha+ir) - \frac{f'}{f}(\alpha) = \sum_{n\ge2}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{r\log n}{2}\Big) - q_{\alpha-1}(r)$$

were nonnegative for every $r$. But the mean of $D^f_\alpha$ over $[0,R]$ tends to $\sum_n \Lambda(n)n^{-\alpha} - \frac{1}{\alpha-1} = -\frac{f'}{f}(\alpha)$ as $R\to\infty$, because the prime series converges ab-

solutely, the mean of $\cos(r\log n)$ over $[0,R]$ is at most $1/(R\log 2)$ in absolute value, and $q_{\alpha-1}(r)\to\frac{1}{\alpha-1}$; and $\frac{f'}{f}(\alpha)>0$. Indeed $\Gamma(\alpha)\zeta(\alpha)=\int_0^\infty x^{\alpha-1}(e^x-1)^{-1}dx$ gives $f(\alpha)=\mathbb{E}h(G_{\alpha-1})$, with $h(x)=x/(1-e^{-x})=x+x/(e^x-1)$ and $G_{\alpha-1}$ a Gamma variable of shape $\alpha-1$; since $h'\ge\frac12$ and $G_{\alpha-1+\varepsilon}$ is $G_{\alpha-1}$ plus an independent Gamma variable of mean $\varepsilon$, $f'(\alpha)\ge\frac12$. So $D^f_\alpha$ is negative somewhere. The primes are bounded on average while the pole term tends to $-\frac{1}{\alpha-1}$; in $\xi$ the gamma terms $\sum_{k\ge1}q_{\alpha+2k}(r)$, which grow like $\frac12\log r$, pay for it, and one prime at a time is harmless (Remark 24.8.2).

Proposition 16.11.10. *If* $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)\ge b_\alpha$ *for all real* $r$ *and every* $\alpha>\alpha_0\ge\frac12$, *then no zero of* $\zeta$ *has real part greater than* $\alpha_0$. *For* $\alpha_0=\frac12$ *it implies RH.*

*Proof.* $b_\alpha=\frac{\xi'}{\xi}(\alpha)>0$ for $\alpha>\frac12$, because $\xi$ is increasing on $(\frac12,\infty)$. The hypothesis therefore gives $\operatorname{Re}\frac{\xi'}{\xi}>0$ on the half plane $\operatorname{Re}s>\alpha_0$. Directly, near a zero $\rho$ with $\operatorname{Re}\rho>\alpha_0$ and multiplicity $m\ge1$ the function $\frac{\xi'}{\xi}(s)-m/(s-\rho)$ is holomorphic, hence bounded, so $\operatorname{Re}\frac{\xi'}{\xi}(\rho-\varepsilon)=-m/\varepsilon+O(1)$ as $\varepsilon\downarrow0$, while $b_{\operatorname{Re}\rho-\varepsilon}$ stays bounded; the hypothesis at $\alpha=\operatorname{Re}\rho-\varepsilon>\alpha_0$ and $r=\operatorname{Im}\rho$ fails for small $\varepsilon>0$, so there is no such zero; this is condition (iv) of Theorem 10.2.7. □

Under RH the compensated condition holds at every $\alpha>\frac12$, and it holds unconditionally for $\alpha\ge1$ (Theorem 9.1.18 and Proposition 9.1.12), so Proposition 16.11.10 has a converse for $\alpha_0=\frac12$. The delicate range is below the first zero: there the numerics show $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)$ rising from its value at $r=0$ like $r^2$ (Figure 16.2, right). The refined problem is to prove $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)\ge b_\alpha$ at one $\alpha<1$ for all heights. By Proposition 16.11.10, doing so for all $\alpha$ above some $\alpha_0<1$ implies a quasi Riemann hypothesis at $\alpha_0$, and by Corollary 16.12.10 it is equivalent to one, whereas on a single line it is not known to imply anything about the zeros to its right (compare Remark 1.2.4).

### 16.11.4 Problem 16.3: local reality and certificates to height 300

Proposition 16.11.11 (Local reality, unconditional). *Let* $K$ *be such that the first* $K$ *zeros of* $\xi$ *in the upper half plane, ordered by ordinate, are simple and on the line, with ordinates* $\gamma_1<\cdots<\gamma_K$, *and fix* $\Gamma\in(\gamma_K,\gamma_{K+1})$. *There is* $\alpha_K>\frac12$ *such that for* $\frac12<\alpha<\alpha_K$ *the zeros of* $A_\alpha$ *in* $|w|<\Gamma$ *are exactly* $\pm i\tau_k(\alpha)$, $k\le K$, *simple and purely imaginary, with* $\tau_k(\alpha)\to\gamma_k$.

*Proof.* $A_\alpha\to\xi(\frac12+w)$ uniformly on $|w|\le\Gamma$, so by Hurwitz's theorem $A_\alpha$ has exactly one zero in a small disc around each $\pm i\gamma_k$ and none elsewhere in $|w|<\Gamma$. $A_\alpha$ is even and real on the real axis, so its zero set is invariant under $w\mapsto-\bar w$; a unique zero in a disc centred on the imaginary axis is therefore on that axis. □

This is the even part analogue of the local theorem of Ki and Kim for the derivatives of $\xi$ (item P13 of Section 16.3). For $\alpha = 0.6$ and 0.75 the argument principle on the rectangle $|\operatorname{Re} w| \le 1$, $0 \le \operatorname{Im} w \le 100$, which contains every zero of $A_\alpha$ up to that height by de Bruijn's strip theorem, counts 29.000 zeros, and Table 16.1 lists 29 sign changes of $\operatorname{Re}\xi(\alpha+it)$. So every zero of $A_{0.6}$ and $A_{0.75}$ up to height 100 is purely imaginary. At $\alpha = 0.55$ the same count to height 300 gives 138.000 zeros against 138 sign changes, so every zero of $A_{0.55}$ up to height 300 is purely imaginary as well. Together with Proposition 16.6.3 this is a Turing type method for Thorin atoms in the strip: it verifies the even part laws to a given height without any statement about $\operatorname{Re}\xi'/\xi$.

### 16.11.5 Problem 16.4: derivatives in the Thorin variable are the shifted Jensen polynomials

Since $\xi(\frac12+z) = \frac18(4z^2-1)\widehat{\zeta}(\frac12+z)$ with $\widehat{\zeta}(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, the sequence of Griffin, Ono, Rolen and Zagier [159], defined by $(4z^2-1)\widehat{\zeta}(\frac12+z) = \sum_n \gamma_{\mathrm{GORZ}}(n) z^{2n}/n!$, is $\gamma_{\mathrm{GORZ}}(n) = 8\xi(\frac12)\,n!\,a_n$, a constant multiple of our $g_n = n!\,a_n$. Their shifted Jensen polynomials are therefore, up to a constant factor, the polynomials $J_{d,n}(x) = \sum_{j=0}^{d}\binom{d}{j} g_{n+j}x^j$ of Section 13.2.4, the Jensen polynomials of the $n$th derivative $F^{(n)}(s)$ in the Thorin variable. By Pólya and Schur, the law with Laplace transform $F^{(n)}(0)/F^{(n)}(s)$ is a GGC if and only if $J_{d,n}$ is real rooted for every $d$.

Corollary 16.11.12. *For every $d$ there is $N(d)$ such that for all $n \ge N(d)$ the derivative laws $F^{(n)}(0)/F^{(n)}(s)$ satisfy the first $d$ rungs of the Pólya Schur ladder [159]. Under RH they are GGCs for every $n$, by Laguerre's theorem that the Laguerre Pólya class is closed under differentiation.*

The first case is classical. Since $F'(s) = \xi'(\frac12+\sqrt{s})/(2\sqrt{s}\,\xi(\frac12))$, the zeros of $F'$ are the squares $s = w^2$ of the zeros $w \ne 0$ of $\xi'(\frac12+w)$, so the law $F'(0)/F'(s)$ is a GGC if and only if every zero of $\xi'$ other than $\frac12$ lies on the critical line. That is the Riemann Hypothesis for $\xi'$, which RH implies and which is not known to imply RH; Conrey's theorem [89] gives a positive proportion of its zeros on the line. In general $F^{(n)}(w^2) = (D^n\xi)(\frac12+w)/\xi(\frac12)$ with $D = \frac{1}{2w}\frac{d}{dw}$, so the derivative laws in the Thorin variable correspond to powers of this Bessel type operator rather than of $\frac{d}{dw}$. Numerically $\xi'(\frac12+w)$ has 29 zeros with $1 < \operatorname{Im} w < 100$ in the box $|\operatorname{Re} w| \le 0.6$, all on the imaginary axis.

Numerically every $J_{d,n}$ with $2 \le d \le 15$ and $0 \le n \le 25$, 364 polynomials in all, has only real negative zeros, computed at 150 digits from the coefficients of Section 16.4. The refined problem is a uniform statement: one $n$ for which all rungs hold. As in Section 13.2.4, and following Farmer [133], large $n$ carries little information about RH, so the useful direction is the opposite one, small $n$ with all $d$, which is RH itself at $n = 0$.

### 16.11.6 Problem 16.2 again: the compensated condition at almost all heights in the strip

Write $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + ir) - \frac{\xi'}{\xi}(\alpha)$ and let $N(\sigma, T)$ count zeros with $\beta > \sigma$ and $0 < \gamma \le T$.

Lemma 16.11.13. *For $\alpha > \frac12$, unconditionally, $0 < b_\alpha = \frac{\xi'}{\xi}(\alpha) \le 2a_1(\alpha - \frac12) < 0.04621\,(\alpha - \frac12)$.*

*Proof.* With $a = \alpha - \frac12$, $b_\alpha = \frac{d}{da}\log F(a^2) = 2a\,\psi'(a^2)$, where $\psi = \log F$. By Theorem 5.1.1 $\psi$ is a Bernstein function, so $\psi'$ is completely monotone and in particular $\psi'(a^2) \le \psi'(0) = a_1 = 0.0231050$. This is part (i) of Theorem 9.1.18, where $V_f = 2a_1 = 0.04620998623$ (Table 9.4) and the bound is proved for every $\alpha > \frac12$. □

Theorem 16.11.14 (Almost all heights). *Fix $\frac12 < \alpha \le 1$ and put $\sigma = \frac12(\alpha + \frac12)$. There is $T_0$ such that for $T \ge T_0$ the exceptional set $E_\alpha(T) = \{r \in [T, 2T] : D_\alpha(r) \le 0\}$ has measure*

$$|E_\alpha(T)| \le C\Big(1 + \frac{1}{\alpha - \frac12}\Big)\big(N(\sigma, 3T) + 1\big) \tag{16.11.10}$$

*with an absolute constant $C$. By Ingham's density theorem [284]*

$$|E_\alpha(T)| \ll_\alpha T^{3(1-\sigma)/(2-\sigma)}(\log T)^5 = o(T),$$

*and under the density hypothesis $|E_\alpha(T)| \ll T^{2(1-\sigma)+\epsilon}$. In particular the compensated clock condition holds at every basepoint in the strip on a set of heights of full density.*

*Proof.* By the Hadamard product, summed symmetrically, $\operatorname{Re}\frac{\xi'}{\xi}(s) = \sum_\rho \operatorname{Re}\frac{1}{s-\rho}$ over all nontrivial zeros, so with $c_\rho = \alpha - \beta_\rho$,

$$D_\alpha(r) = \sum_\rho \left[\frac{c_\rho}{c_\rho^2 + (r - \gamma_\rho)^2} - \frac{c_\rho}{c_\rho^2 + \gamma_\rho^2}\right],$$

the sum running over zeros of both signs of $\gamma$. Call a zero harmful if $\beta > \alpha$, so $c < 0$, and good if $|\beta - \frac12| \le \sigma - \frac12$, so $c \in [c_{\min}, c_{\max}]$ with $c_{\min} = \alpha - \sigma = \frac12(\alpha - \frac12)$ and $c_{\max} = \alpha - 1 + \sigma \le \frac34$. Every zero that is not harmful contributes a nonnegative first term. Hence

$$D_\alpha(r) \ge L(r) - b_\alpha - N_h(r),$$
$$N_h(r) = \sum_{\text{harmful}} \frac{|c|}{c^2 + (r - \gamma)^2},$$

where $L(r)$ is the first term of any one good zero. The second terms of all zeros, harmful ones included, sum to $b_\alpha$, so no separate bound for the harmful second terms is needed.

The local term. By Littlewood's theorem [345] that gaps between consecutive ordinates tend to zero, for $r \geq T_0$ the window $[r-1, r+1]$ contains an ordinate. If one such zero is good, $L(r) \geq \frac{c}{c^2+1} \geq \kappa := \frac{c_{\min}}{c_{\min}^2+1} \geq \frac{4}{5} c_{\min}$, since $c \mapsto c/(c^2+1)$ increases on $(0,1)$. Otherwise $r$ lies within 1 of a zero that is not good, that is one with $\beta > \sigma$ or its mirror with $\beta < 1-\sigma$; the measure of such $r$ in $[T, 2T]$ is at most $4N(\sigma, 2T+1)$.

The harmful sum. Since $\int_{\mathbb{R}} \frac{|c|}{c^2+(r-\gamma)^2} dr = \pi$, the harmful zeros with $\frac{T}{2} \leq \gamma \leq 3T$ contribute at most $2\pi N(\alpha, 3T) \leq 2\pi N(\sigma, 3T)$ to $\int_T^{2T} N_h$, and the remaining harmful zeros contribute $o(1)$ by the trivial bound $|c|/(r-\gamma)^2$ and Ingham's bound for $N(\alpha, t)$. By Markov's inequality, $N_h(r) > \kappa/2$ on a set of measure at most $\frac{2}{\kappa}(2\pi N(\sigma, 3T) + o(1))$.

Outside these two sets, Lemma 16.11.13 gives $b_\alpha \leq 0.04621(\alpha - \frac{1}{2}) = 0.09242\, c_{\min}$, hence $D_\alpha(r) \geq \kappa - \frac{\kappa}{2} - 0.0925\, c_{\min} \geq (0.4 - 0.0925) c_{\min} > 0$. Adding the two exceptional measures and using $\kappa \geq \frac{4}{5} c_{\min}$ gives (16.11.10). □

Theorem 16.11.14 has the same content as Remark 9.1.17, which obtains the comparison from Theorem 9.1.15 with $\sigma_0 = \frac{1}{2} + (\alpha - \frac{1}{2})/\sqrt{2}$ in place of $\sigma$ and a lower bound growing like $\log \theta$; the proof here is direct, and Theorem 11.4.2 gives a statement of the same kind for the mass of the negative part of the density.

To the right of the strip the inequality holds on the grid numerically (Section 16.12), and at all heights for every $\alpha \geq 1$ by Proposition 9.1.12. Below 1 it holds at all heights, on every line to the right of $\alpha_0$, under a quasi Riemann hypothesis at $\alpha_0$ (Theorem 16.12.8).

### 16.11.7 Problem 16.1 again: coalescing points and the $\mathrm{PF}_3$, $\mathrm{PF}_4$ conditions

Theorem 16.11.6 controls $\mathrm{PF}_n$ minors up to an additive error of $1.1 \times 10^{-8}$ for $n = 3$ and $1.2 \times 10^{-7}$ for $n = 4$, which is enough wherever the minors of $f_{\mathrm{on}}$ are larger than that, but not where they vanish. They vanish in two places: at coalescing points and far in the tail. When all $x_i$ and all $y_j$ coalesce at points whose difference is $x$, the minor divided by the two Vandermonde products tends to a constant multiple of the Hankel determinant $\mathcal{H}_n(x) = \det[f_X^{(i+j)}(x)]_{i,j<n}$, and the $\mathrm{PF}_n$ condition becomes $(-1)^{n(n-1)/2}\mathcal{H}_n(x) \geq 0$; for $n = 2$ this is log concavity.

Proposition 16.11.15. *In the tail the coalescing conditions hold, unconditionally, for every $n$ with $\gamma_n < H$: for an exponential sum $f = \sum_k \eta_k e^{-\gamma_k x}$,*

$$\mathcal{H}_n(x) = \sum_{|S|=n} \Big( \prod_{k \in S} \eta_k \Big) \prod_{k<l \in S} (\gamma_l - \gamma_k)^2 \, e^{-x \sum_{k \in S} \gamma_k},$$

*and the term of $S = \{1, \dots, n\}$ dominates as $x \to \infty$, with sign $(-1)^{\lfloor n/2 \rfloor} = (-1)^{n(n-1)/2}$ by the alternation of the residues.*

*Proof.* $[f^{(i+j)}(x)]_{i,j<n} = V\,\mathrm{diag}(\eta_k e^{-\gamma_k x})\,V^{\top}$ with $V = [(-\gamma_k)^i]_{i<n,\,k}$, so the Cauchy Binet formula gives the sum over $n$ element subsets $S$, each weighted by $\det(V_S)^2 = \prod_{k<l\in S}(\gamma_l-\gamma_k)^2$. The residue expansion of Lemma 16.11.3 with $T$ between $\gamma_n$ and $\min(\gamma_{n+1}, H)$ shows that the remaining terms, including those that contain the remainder $E_T$ and its derivatives, are smaller by a factor of order $e^{-(T-\gamma_n)x}$. □

Numerically, combining quadrature on $[0, 1.6]$ with the residue expansion with fourteen zeros beyond, $-\mathcal{H}_3/f_X^3$ and $\mathcal{H}_4/f_X^4$ are positive at every point of a grid of step 0.02 on $[0,4]$; their smallest values, $1.05\times10^{-21}$ and $2.08\times10^{-39}$, occur at $x=4$, where Proposition 16.11.15 predicts exponential decay. So $\mathrm{PF}_3$ and $\mathrm{PF}_4$ hold at coalescing points on the grid and, by Proposition 16.11.15, as $x\to\infty$, and the scan of separated minors in Section 16.11 finds none that is negative. A complete proof of $\mathrm{PF}_3$ would need three things that are not proved here: an explicit threshold in $x$ in Proposition 16.11.15, a bound for nearly coalescing points, and a bound for separated points in the tail.

### 16.11.8 Problem 16.3 again: almost all zeros of the even and odd parts are on the line

Let $B_\alpha(w) = \frac{1}{2i}(\xi(\alpha+w)-\xi(\alpha-w))$ be the odd part, so that $A_\alpha(it) = \mathrm{Re}\,\xi(\alpha+it)$ and $B_\alpha(it) = \mathrm{Im}\,\xi(\alpha+it)$, and let $N_A(T)$, $N_B(T)$ count the zeros with $0<\mathrm{Im}\,w\le T$ and $N_A^0(T)$, $N_B^0(T)$ those on the imaginary axis.

Theorem 16.11.16 (Zero accounting for the even and odd parts). *Let $\frac12<\alpha\le1$ be such that no zero of $\xi$ lies on the line $\mathrm{Re}\,s=\alpha$, which excludes at most countably many $\alpha$. Then, with no other hypothesis, as $T\to\infty$ through values that are not ordinates of zeros of $\xi$, $A_\alpha$ or $B_\alpha$,*

$$\begin{aligned} N_A(T) &= U_\alpha(T) + 2N(\alpha,T) + O(\log T),\\ N_A^0(T) &\ge \left\lfloor U_\alpha(T)+\tfrac12\right\rfloor, \end{aligned} \tag{16.11.11}$$

*and the same holds for $B_\alpha$ with $\lfloor U_\alpha(T)\rfloor$ in place of $\lfloor U_\alpha(T)+\frac12\rfloor$. Consequently*

$$\begin{aligned} N_A(T)-N_A^0(T) &\le 2N(\alpha,T)+O(\log T),\\ N_B(T)-N_B^0(T) &\le 2N(\alpha,T)+O(\log T), \end{aligned} \tag{16.11.12}$$

*so by Ingham's density theorem all but $O(T^{3(1-\alpha)/(2-\alpha)}\log^5 T)$ of the zeros of $A_\alpha$ and of $B_\alpha$ up to height $T$ are purely imaginary, a proportion tending to one.*

*Proof.* For $\mathrm{Re}\,w\ge\alpha$ both $\alpha+w$ and $\alpha-w$, reflected by the functional equation to $1-\alpha+w$, have real part at least 1, where $\log|\xi(\sigma+it)|$ is strictly increasing in $\sigma$ because every zero has $\beta<1$; as $\alpha+\mathrm{Re}\,w>1-\alpha+\mathrm{Re}\,w$, $|\xi(\alpha+w)|>|\xi(\alpha-w)|$ and $A_\alpha(w)\neq0$. So all zeros lie in $|\mathrm{Re}\,w|<\alpha$. Apply the argument principle to $A_\alpha$

on the rectangle $|\operatorname{Re} w| \le X$, $0 \le \operatorname{Im} w \le T$ with $X = 2$. On the real axis $A_\alpha > 0$. On the right side $A_\alpha = \frac{1}{2}\xi(\alpha + w)(1 + q)$ with $|q| < 1$, so its argument changes by that of $\xi$ along $\operatorname{Re} s = \alpha + X$ up to $O(1)$, and the left side contributes the same by the symmetries $A_\alpha(-w) = A_\alpha(w)$ and $A_\alpha(\bar{w}) = \overline{A_\alpha(w)}$. The argument principle for $\xi$ on $[\alpha, \alpha + X] \times [0, T]$ converts the change along $\operatorname{Re} s = \alpha + X$ into $\pi U_\alpha(T) + 2\pi N(\alpha, T)$ plus a horizontal variation at height $T$. The two horizontal variations, of $\xi$ and of $A_\alpha$, are $O(\log T)$ by Backlund's argument [15] with Jensen's formula, since $\log|\xi|$ and $\log|A_\alpha|$ vary by $O(\log T)$ on discs of bounded radius at height $T$, by Stirling's formula and polynomial bounds for $\zeta$. This gives the first identity in (16.11.11). For the second, $A_\alpha(it) = |\xi(\alpha + it)| \cos(\pi U_\alpha(t))$ with $U_\alpha$ continuous from $U_\alpha(0) = 0$, so every level $k - \frac{1}{2} \le U_\alpha(T)$ is crossed at least once. The odd part is treated in the same way with sin, except that $B_\alpha$ is odd, with a simple zero at $w = 0$ on the bottom edge, so the contour is indented above 0, which changes the count by $\frac{1}{2}$, inside $O(\log T)$, and $B_\alpha(\bar{w}) = -\overline{B_\alpha(w)}$ replaces $A_\alpha(\bar{w}) = \overline{A_\alpha(w)}$. Subtracting gives (16.11.12). □

Theorem 16.11.16 sharpens the half integer crossing picture into a balance sheet. Numerically, at $\alpha = 0.55$ the clock reaches $U_{0.55}(300) = 137.921$; $\operatorname{Re}\xi(0.55 + it)$ has $138 = \lfloor 137.921 + \frac{1}{2} \rfloor$ zeros on $(0, 300]$ and $\operatorname{Im}\xi(0.55 + it)$ has $137 = \lfloor 137.921 \rfloor$, and the argument principle counts 138.000 zeros of $A_{0.55}$ and 137.0 zeros of $B_{0.55}$ in the box, so the balance is exact there with no off axis zeros.

The odd part family also clarifies Problem 16.4. Since $B_{\frac{1}{2}+a}(w)/a \to -i\,\xi'(\frac{1}{2} + w)$ locally uniformly as $a \downarrow 0$, the derivative $\xi'$ is the endpoint of the odd part family, just as $\xi$ is the endpoint of the even part family. Under RH every $B_\alpha$ has only imaginary zeros, and Hurwitz's theorem then gives the Riemann Hypothesis for $\xi'$. Theorem 16.11.16 cannot be pushed to $a \to 0$ directly, because the known bounds for $N(\alpha, T)$ grow to a positive proportion of all zeros as $\alpha \downarrow \frac{1}{2}$, and Hurwitz's theorem needs every zero of $B_\alpha$ on the axis, not almost every one; this is why the Riemann Hypothesis for $\xi'$ remains open.

## 16.12 Consequences of a quasi Riemann hypothesis at 7/8

### 16.12.1 The claim and its status

On 6 October 2026 OpenAI released a manuscript dated 30 September 2026 [398], with a formal proof in Lean, that every Dirichlet $L$ function, $\zeta(s)$ included, has no zeros in the half plane $\operatorname{Re} s > \frac{7}{8}$, and a companion manuscript dated 5 October 2026 [399] with a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$. The work has not been refereed; Section 12.1.2 describes it. Everything in this section is conditional on

($\mathrm{Q}_{7/8}$) $\zeta(s) \ne 0$ for $\operatorname{Re} s > \frac{7}{8}$, so that $\sigma_{\max} \le \frac{7}{8}$ and every zero lies in $\frac{1}{8} \le \beta \le \frac{7}{8}$.

Write $\Delta = \sigma_{\max} - \frac{1}{2} \le \frac{3}{8}$ for the half width of the strip containing the zeros of $\Xi$.

### 16.12.2 What moves from 1 to 7/8

Theorem 16.12.1 (Consequences of $\mathrm{Q}_{7/8}$). *Assume* ($\mathrm{Q}_{7/8}$).

*(i) Basepoint laws. For every $\alpha \ge \frac{7}{8}$ the law $\mu^{(\alpha)}$ with Laplace transform $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ is a GGC. For $\alpha > \frac{7}{8}$ its Thorin measure is absolutely continuous, with density $\pi^{-1}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$ in the ordinate variable, a sum of Lorentzians of widths at least $\alpha - \frac{7}{8}$; it has a stable tail $\mu^{(\alpha)}((y,\infty)) \sim b_\alpha/\sqrt{\pi y}$.*

*(ii) The clock and Lagarias positivity. For $\alpha > \frac{7}{8}$ the clock $U_\alpha$ is strictly increasing and $\operatorname{Re}\frac{\xi'}{\xi} > 0$ on $\operatorname{Re} s > \frac{7}{8}$. In terms of $\zeta$ this is the one sided bound $-\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+i\tau) < \operatorname{Re}\mathcal{A}(\alpha+i\tau) = \frac{1}{2}\log\frac{\tau}{2\pi} + O(\tau^{-2})$, with $\mathcal{A}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})$, at every height $\tau$ and for every $\alpha > \frac{7}{8}$.*

*(iii) Even and odd parts. For $\alpha \ge \frac{7}{8}$ the functions $A_\alpha$ and $B_\alpha$ have only purely imaginary zeros, simple for $\alpha > \frac{7}{8}$. Proposition 16.6.3, the exact midpoint formula* (16.11.3) *and, for $\alpha \le 1$, the accounting theorem with $N(\alpha, T) = 0$ hold for every $\alpha > \frac{7}{8}$, and the even part laws $\mu_\alpha^P$ are GGCs there.*

*(iv) Heat flow. The de Bruijn Newman constant satisfies $\Lambda \le 2\Delta^2 \le \frac{9}{32} = 0.28125$ by de Bruijn's theorem alone.*

*(v) Densities. $N(\sigma, T) = 0$ for $\sigma > \frac{7}{8}$, so in Theorem 16.11.14 the exceptional set at a basepoint $\alpha \le \frac{7}{8}$ is generated only by zeros with $\frac{1}{2}(\alpha+\frac{1}{2}) < \beta \le \frac{7}{8}$.*

*Proof.* (i) By Theorem 1.2.2 and Theorem 7.1.14, $\mu^{(\alpha)}$ is a GGC when no zero has real part greater than $\alpha$, and its Thorin measure is the Poisson smoothed zero density, in which a zero of real part $\beta < \alpha$ contributes a Lorentzian of width $\alpha - \beta \ge \alpha - \frac{7}{8}$; a zero produces an atom only when $\beta = \alpha$, which is why absolute continuity is claimed for $\alpha > \frac{7}{8}$ only. For the tail, $1 - \xi(\alpha)/\xi(\alpha+\sqrt{s}) \sim b_\alpha\sqrt{s}$ as $s \downarrow 0$, and Karamata's Tauberian theorem for the survival function, with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, gives $\mu^{(\alpha)}((y,\infty)) \sim b_\alpha/\sqrt{\pi y}$. (ii) For $\alpha > \sigma_{\max}$ every term $\frac{\alpha-\beta}{|\alpha+i\tau-\rho|^2}$ of the Hadamard sum is positive, so $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau) > 0$ and the clock, whose derivative is $\pi^{-1}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$, runs forward. Since $\frac{\xi'}{\xi} = \mathcal{A} + \frac{\zeta'}{\zeta}$, positivity is the stated bound, and $\operatorname{Re}\mathcal{A}(\alpha+i\tau) = \frac{1}{2}\log\frac{\tau}{2\pi} + O(\tau^{-2})$ by Stirling's formula for $\frac{\Gamma'}{\Gamma}$. (iii) For $A_\alpha$ this is Theorem 16.6.1(ii) with $\Delta \le \frac{3}{8}$ and $a = \alpha - \frac{1}{2} \ge \frac{3}{8}$. For $B_\alpha$ it is the Hermite Biehler property of $z \mapsto \xi(\alpha - iz)$, which holds whenever $\sigma_{\max} \le \alpha$ (Remark 16.6.4; the pairing of $\rho$ with $1-\bar\rho$ proves it). Simplicity follows from the strict monotonicity of the clock in (ii), since the zeros of $A_\alpha(it)$ are transversal

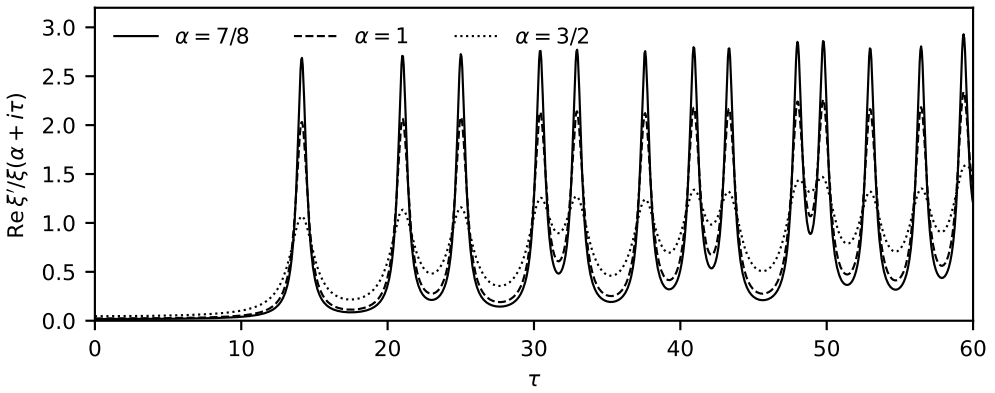



Figure 16.3: The function $\mathrm{Re}\,\frac{\xi'}{\xi}(\alpha+i\tau)$, which is $\pi$ times the Thorin density of Theorem 16.12.1(i) in the ordinate variable, at $\alpha=\frac{7}{8}$ (solid), 1 (dashed) and $\frac{3}{2}$ (dotted) for $0\le\tau\le 60$, computed from $\xi$ with no zeros (`pol_qrh_edge.py`). The zeros in this range lie on the critical line, and each contributes a Lorentzian of width $\alpha-\frac{1}{2}$ and height $1/(\alpha-\frac{1}{2})$, that is $\frac{8}{3}$, 2 and 1; at $\alpha=\frac{7}{8}$ the computed peaks are 2.69 to 2.93. On this range the minimum is at $\tau=0$, where the value is $\frac{\xi'}{\xi}(\alpha)=0.0173$, 0.0231 and 0.0461. Positivity at every height for $\alpha>\frac{7}{8}$ is Theorem 16.12.1(ii), under $(\mathrm{Q}_{7/8})$, and not a consequence of the figure.

crossings of half integer levels. The remaining statements are the propositions cited with $\alpha>\sigma_{\max}$. (iv) In the normalisation $H_0(z)=\frac{1}{8}\Xi(\frac{z}{2})$ the zeros of $H_0$ lie in $|\operatorname{Im} z|\le 2\Delta$, and de Bruijn's theorem [112] makes $H_t$ real rooted for $t\ge\frac{1}{2}(2\Delta)^2$. (v) is immediate. □

The bound $\Lambda\le\frac{9}{32}$ is weaker than the bound 0.2 of Platt and Trudgian [419], which sharpens the bound 0.22 of the Polymath project [456] (Table 8.14), obtained by their barrier method and extensive computation; a smaller a priori $\Delta$ might lower those bounds, but that requires redoing the computation. And the stable tails of (i) show what the GGC laws inside the strip look like: the laws at basepoints between $\frac{7}{8}$ and 1 have heavy tails of nearly the same weight as the law at 1, and only the law at $\frac{1}{2}$ has a finite mean. At the edge the density of (i) can be computed from $\xi$ with no zeros: on $0\le\tau\le 60$, $\mathrm{Re}\,\frac{\xi'}{\xi}(\frac{7}{8}+i\tau)$ is smallest at $\tau=0$, where it equals $\frac{\xi'}{\xi}(\frac{7}{8})=0.0173248$, and near each ordinate it has a peak of height at least about $\frac{8}{3}$ and width $\frac{3}{4}$, narrower than the smallest gap 1.77 between ordinates there, so the Thorin measure at $\alpha=\frac{7}{8}$ already separates the zeros (Figure 16.3).

Elsewhere in the book the half plane acts as Table 16.3 lists; every entry is conditional on [398] and is proved where it is cited, except the heuristic count in the last row.

| Chapter | What $(\mathrm{Q}_{7/8})$ gives there |
|---|---|
| 9 | the compensated Lévy form for $\frac{7}{8} < \alpha < 1$, with error $O(X^{7/8-\alpha+\delta})$ (Proposition 9.1.13) |
| 10 | $R(x) = O(x^{7/16+\delta})$ (Theorem 10.2.7); the range $0 < y < \frac{4\pi}{3}$ of Remark 10.1.9 |
| 12 | step 7 of the path, with $\alpha = \frac{7}{8}$ (Section 12.1.2) |
| 13 | $\cosh(at)\,p$ ferromagnetic for $a \geq \frac{3}{8}$; $\Omega \leq \frac{3}{8}$ (Section 13.5.2) |
| 15, 19 | $\Lambda \leq \frac{9}{32}$, weaker than 0.2 (Corollary 15.3.3) |
| 18 | the bound on $\zeta'/\zeta$ behind $v_\alpha > 0$ is Theorem 16.12.1(ii) (Section 18.5) |
| 20 | equivalent to the damped form being nonnegative; $\lambda_{\min}(\frac{7}{8}, a) \geq 0$ (Theorem 20.1.2) |
| 21 | with the half plane for every Dirichlet $L$ function, Theorem 21.6.5 for every $q \geq 3$; the heuristic count of Section 21.1.2 does not lower $K$ |

Table 16.3: Where the half plane $\operatorname{Re} s > \frac{7}{8}$ of [398], which has not been refereed, acts outside this section.

Proposition 16.12.2 (Poles of the reciprocal under $\mathrm{Q}_{7/8}$). *Assume* $(\mathrm{Q}_{7/8})$. *Every pole* $(\rho-\frac{1}{2})^2$ *of* $1/\xi(\frac{1}{2}+\sqrt{w})$ *lies in the closed region* $(\operatorname{Im} w)^2 \leq 4\Delta^2(\Delta^2 - \operatorname{Re} w) \leq \frac{9}{16}(\frac{9}{64} - \operatorname{Re} w)$, *bounded by a parabola with focus* 0 *and vertex* $\Delta^2 \leq \frac{9}{64}$, *against the vertex* $\frac{1}{4}$ *with no hypothesis. Theorem 4.3.9 holds with the sliver* $\mathcal{E}_{T_0}$ *replaced by the thinner set*

$$\mathcal{E}'_{T_0} = \Big\{ x+iy : \; x < -T_0^2 + \tfrac{3}{4}T_0 + \tfrac{9}{64}, \; y^2 \leq \tfrac{9}{16}\big(\tfrac{9}{32} - x\big) \Big\},$$

*whose half width at the abscissa* $-X^2$ *is about* $\frac{3}{4}X$, *against about* $X$ *for* $\mathcal{E}_{T_0}$.

*Proof.* In the notation of the proof of Theorem 4.3.9 a zero is $\gamma = a + ib$ with $0 \leq b \leq \Delta \leq \frac{3}{8}$, and its pole $w = -\gamma^2$ has $\operatorname{Re} w = b^2 - a^2$ and $(\operatorname{Im} w)^2 = 4a^2b^2 = 4b^2(b^2 - \operatorname{Re} w)$, which increases in $b$. In that proof, $b \leq \frac{3}{8}$ in place of $b < \frac{1}{2}$ gives $x \leq 2b^2 - (a-b)^2 < -T_0^2 + \frac{3}{4}T_0 + \frac{9}{64}$ and $y^2 \leq 4b^2(2b^2 - x) \leq \frac{9}{16}(\frac{9}{32} - x)$ on $D_\gamma$, and the rest is unchanged. □

### 16.12.3 The Gaussian smoothed Möbius sum

The proof of [398] bounds smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$ (Section 12.1.2), and its last step is a continuation: a bound on the growth of such a sum makes the reciprocal $L$ function analytic in a half plane. Liu [352] claims an optimisation of its exponents to the half plane $\operatorname{Re} s > 0.874957\ldots$, and in a second preprint [353], the half plane $\operatorname{Re} s > B_\flat = 0.8748866470\ldots$, with $B_\flat$ the root near 0.8749 of $133488B^4 - 376596B^3 + 384534B^2 - 167541B + 26231$, for every Hecke $L$ function of finite order over every number field $K$, obtained over $K(\zeta_{12})$ and carried down to $K$. Xu and Sang [364], assuming the analytic inputs of [398], give $8748867/10^7 \approx 0.8748867$,

which agrees with $B_\flat$ to six digits. All of these claims rest on the argument of [398]; none is refereed, and the Lean proof covers $\frac{7}{8}$ only. We checked only that $B_\flat$ is a root of the quartic. Nothing below uses them: $B_\flat$ lies $1.13\times10^{-4}$ below $\frac{7}{8}$ and would change the bound of Theorem 16.12.1(iv) from 0.28125 to $2(B_\flat-\frac12)^2=0.28108$. For $\zeta$ alone, with a Gaussian weight in the Mellin variable, the sum and its explicit formula can be written down and computed, and nothing in this subsection depends on $(\mathrm{Q}_{7/8})$. For $x>0$ put

$$M_{\mathrm g}(x)=\frac{1}{2\pi i}\int_{(2)}x^{v-2/3}\,\frac{e^{(v-5/6)^2}}{\zeta(v)}\,dv=\frac{x^{-2/3}}{2\sqrt{\pi}}\sum_{n\ge1}\mu(n)\Big(\frac{x}{n}\Big)^{5/6}\exp\Big(-\frac14\log^2\frac{x}{n}\Big),\tag{16.12.1}$$

where the second form follows from $1/\zeta(v)=\sum_n\mu(n)n^{-v}$ and

$$\frac{1}{2\pi i}\int_{(c)}e^{vw}e^{(v-c)^2}dv=\frac{e^{cw-w^2/4}}{2\sqrt{\pi}},$$

a Möbius sum smoothed by a Gaussian of standard deviation $\sqrt2$ in $\log n$ around $\log x$.

Proposition 16.12.3. *Let $\sigma_{\max}=\sup\operatorname{Re}\rho$.*

*(i) For every $K\ge1$ there are heights $T_m\in[m,m+1]$ such that for every $x>0$*

$$M_{\mathrm g}(x)=\lim_{m\to\infty}\sum_{|\gamma|<T_m}\operatorname*{Res}_{v=\rho}\frac{x^{v-2/3}e^{(v-5/6)^2}}{\zeta(v)}+\sum_{k=1}^{K}\frac{e^{(2k+5/6)^2}}{\zeta'(-2k)}\,x^{-2k-2/3}+I_K(x),$$

*with*

$$I_K(x)=\frac{1}{2\pi i}\int_{(-2K-1)}x^{v-2/3}e^{(v-5/6)^2}\zeta(v)^{-1}dv=O(x^{-2K-5/3}).$$

*At a simple zero the residue is $e^{(\rho-5/6)^2}x^{\rho-2/3}/\zeta'(\rho)$. As $K\to\infty$ the trivial zero series diverges: its $k$th term has modulus*

$$\frac{2(2\pi)^{2k}e^{(2k+5/6)^2}x^{-2k-2/3}}{(2k)!\,\zeta(2k+1)}.$$

*(ii) $\sigma_{\max}\le\sigma$ if and only if $M_{\mathrm g}(x)=O(x^{\sigma-2/3+\varepsilon})$ as $x\to\infty$ for every $\varepsilon>0$. RH is the exponent $-\frac16$ and $(\mathrm{Q}_{7/8})$ the exponent $\frac{5}{24}$.*

*(iii) $M_{\mathrm g}(x)=c_1x^{-8/3}+E(x)+O(x^{-11/3})$, with $c_1=e^{(17/6)^2}/\zeta'(-2)=-1.00659\times10^5$ and $E$ the sum over the zeros in (i), and $E(x)$ is not $O(x^{\sigma-2/3})$ for any $\sigma<\sigma_{\max}$. So $c_1x^{-8/3}$ is the leading term of $M_{\mathrm g}$ on a finite range of $x$ only, not its asymptotic.*

*Proof.* (i) Move the line of integration from $\operatorname{Re} v = 2$ to $\operatorname{Re} v = -2K-1$ inside the rectangles $|\operatorname{Im} v| < T_m$. On the horizontal sides $|e^{(v-5/6)^2}| = e^{(\operatorname{Re} v-5/6)^2 - T_m^2}$, and $T_m \in [m, m+1]$ can be chosen with $|\zeta(\sigma \pm iT_m)|^{-1} \le e^{C(\log T_m)^2}$ for $-2K-1 \le \sigma \le 2$ ([498, Chapter 9] for $-1 \le \sigma \le 2$, the functional equation to the left of $-1$), so the horizontal integrals tend to zero. The residues at $v = -2k$ use $\zeta'(-2k) = (-1)^k (2k)!\,\zeta(2k+1)/(2(2\pi)^{2k})$; at $v = 1$ the integrand vanishes. (ii) If $M_{\mathrm{g}}(x) = O(x^{\sigma - 2/3 + \varepsilon})$, then $\int_0^\infty M_{\mathrm{g}}(x) x^{2/3 - v}\, dx/x$ converges absolutely for $\operatorname{Re} v > \sigma + \varepsilon$, since near $x = 0$ the series in (16.12.1) gives $M_{\mathrm{g}}(x) x^{2/3} = O(x^{5/6} e^{-\log^2 x/4})$. By Mellin inversion it equals $e^{(v-5/6)^2}/\zeta(v)$ for $\operatorname{Re} v > 1$, hence for $\operatorname{Re} v > \sigma + \varepsilon$ by continuation, so $\zeta$ has no zero there. Conversely, if $\sigma_{\max} \le \sigma$, then $1/\zeta(v) = O(|\operatorname{Im} v|^{\varepsilon})$ on $\operatorname{Re} v \ge \sigma + \varepsilon$, $|\operatorname{Im} v| \ge 1$, by the Borel and Carathéodory argument of [498, Section 14.2], which uses only the absence of zeros to the right of $\sigma$; moving the line to $\operatorname{Re} v = \sigma + \varepsilon$ gives the bound. (iii) Take $K = 1$ in (i). If $E(x) = O(x^{\sigma - 2/3})$ with $\sigma < \sigma_{\max}$, then $M_{\mathrm{g}}(x) = O(x^{\sigma' - 2/3})$ with $\sigma' = \max(\sigma, -2) < \sigma_{\max}$, and (ii) gives $\sigma_{\max} \le \sigma'$, a contradiction. □

Computed three ways, from the Möbius sum over $n \le 4 \times 10^7$ and from the integral on $\operatorname{Re} v = 2$ and on $\operatorname{Re} v = 0.6$, between which no zero lies below height $3 \times 10^{12}$ [419], far beyond the reach of the Gaussian factor, $M_{\mathrm{g}}(10^3) = -5.53094 \times 10^{-4}$; the ratio $M_{\mathrm{g}}(x)/(c_1 x^{-8/3})$ is 0.5495, 0.9470, 0.99856 and 0.999983 at $x = 10^3$, $10^4$, $10^5$ and $10^6$ (`qrh.py`; the Möbius sum is accurate only while its window $n \lesssim x e^8$ fits below $4 \times 10^7$). The first pair of zeros has coefficient $|e^{(\rho_1 - 5/6)^2}/\zeta'(\rho_1)| = 2.404 \times 10^{-87}$ and overtakes $c_1 x^{-8/3}$ only at $x \approx 10^{36.5}$. Below that the sum is numerically its trivial zero term, and whether its exponent is $-\frac{1}{6}$ or larger cannot be seen at any computable $x$. The continuation step does not need to see it: it reads the exponent, which the rightmost zero sets however small its coefficient. Since $\sigma_{\max} = \frac{1}{2}$ is RH, reading an exponent is no obstruction in itself; what stops a method of this kind is the size of the exponents it can prove.

$M_{\mathrm{g}}$ is the Mellin counterpart of the density $f$ of the clock $T$ (Section 8.1.1). Both invert a reciprocal, $e^{(v-5/6)^2}/\zeta(v)$ on a vertical line and $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ on a Bromwich line, and both have two expansions. On the prime side each is a Möbius series with no hypothesis, (16.12.1) here and Proposition 8.1.1 there. On the zero side each is a residue series, Proposition 16.12.3(i) here and, under RH with simple zeros, $f(t) = \sum_k R_k e^{-\gamma_k^2 t}$ there. They differ in the archimedean factor. The Gaussian is chosen, and it puts the factor $e^{-\gamma^2}$ into the weight of each zero. The clock carries the gamma factor of $\xi$, which puts $e^{-\pi\gamma_k/4}$ into $|\xi'(\rho_k)|$, so that the residues $R_k$ grow like $e^{\pi\gamma_k/4}$ up to powers of $\gamma_k$ and the factor $1/|\zeta'(\rho_k)|$. That is why the spectral expansion of $P(T > t)$ at $t = 0.008$ is far off with 20 zeros and within $10^{-8}$ with 50 (`bn_tail.py`, part b), while $M_{\mathrm{g}}$ shows no zero below $x = 10^{36}$.

Proposition 16.12.4 (Mertens function under $\mathrm{Q}_{7/8}$). *Assume* ($\mathrm{Q}_{7/8}$). *Then* $M(x) = \sum_{n \le x} \mu(n) = O(x^{7/8+\varepsilon})$ *for every* $\varepsilon > 0$, *and* $\sum_n \mu(n) n^{-s}$ *converges*

*to* $1/\zeta(s)$ *on* $\operatorname{Re} s > \frac{7}{8}$. *So the Möbius series of* $1/\zeta(\frac{1}{2}+\sqrt{s})$ *converges on* $H_{3/8} = \{\operatorname{Re}\sqrt{s} > \frac{3}{8}\} = \{u+iy:\ u > \frac{9}{64} - \frac{16}{9}y^2\}$, *which contains the region* $H_{1/2}$ *of absolute convergence of Section 6.1.*

*Proof.* With no zero in $\operatorname{Re} s > \frac{7}{8}$, $\log\zeta$ is analytic there, and the argument of [498, Section 14.2], with $\frac{7}{8}$ in place of $\frac{1}{2}$, gives $1/\zeta(s) = O(|t|^{\varepsilon})$ uniformly on $\operatorname{Re} s \geq \frac{7}{8}+\varepsilon$, $|t| \geq 2$. Perron's formula with the contour on $\operatorname{Re} s = \frac{7}{8}+\varepsilon$, as in [498, Section 14.25], gives the bound on $M$. Partial summation makes the series converge locally uniformly on $\operatorname{Re} s > \frac{7}{8}$, and its sum, analytic there, equals $1/\zeta$ on $\operatorname{Re} s > 1$. With $\sqrt{s} = c+id$, $s = c^2-d^2+2icd$, and $c = \frac{3}{8}$ traces $x = \frac{9}{64} - \frac{16}{9}y^2$. □

This is the unsmoothed form of Proposition 16.12.3(ii), whose exponent $\frac{5}{24}$ is $\frac{7}{8} - \frac{2}{3}$.

Remark 16.12.5 (Least prime in a progression). Assume, as [398] claims, that no Dirichlet $L$ function vanishes in $\operatorname{Re} s > \frac{7}{8}$, so that every nontrivial zero has $\frac{1}{8} \leq \beta \leq \frac{7}{8}$ and $|\rho| \geq \frac{1}{8}$. The explicit formula for $\psi(x,\chi)$ [104, Chapters 19 and 20] with $T = x$, and $\sum_{|\gamma|\leq x} |\rho|^{-1} = O(\log^2(qx))$, give $\psi(x;q,a) = x/\varphi(q) + O(x^{7/8}\log^2(qx))$ uniformly in $(a,q) = 1$, so the least prime $p \equiv a \pmod q$ is $O(q^{8+\varepsilon})$ (7.9928 with the unrefereed $B_\flat$). This is weaker than the unconditional exponent 5.2 of Xylouris [365], and it is not the Linnik constant $K$ of Chapter 21.

### 16.12.4 The compensated clock condition at every height

Lemma 16.12.6 (Window count). *For* $d > 0$ *and* $r > d+e$, *the number of zeros, counted with multiplicity, with ordinates in* $[r-d, r+d]$ *is at least*

$$\nu(r,d) = \frac{\vartheta(r+d) - \vartheta(r-d)}{\pi} - s(r+d) - s(r-d),$$

*where* $\vartheta$ *is the Riemann Siegel theta function,* $s(t) = 2.5167$ *for* $e \leq t \leq X_0 = 30\,610\,046\,000$, *and* $s(t) = 0.1095\log t + 0.2042\log\log t + 3.0305$ *for* $t > X_0$.

*Proof.* $N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t)$ (Section 2.2.1), and $|S(t)| \leq s(t)$ for $t \geq e$ by (1.7) and Corollary 1.4 of Hasanalizade, Shen and Wong [176], as in the proof of Proposition 9.1.11. The bound of Trudgian [503], which is often quoted, has an incomplete proof [36, 176] and is not used. □

Theorem 16.12.7 (Compensated clock under $\mathrm{Q}_{7/8}$, counting form). *Assume* $(\mathrm{Q}_{7/8})$ *and let* $\alpha > \frac{7}{8}$. *Put* $c_{\min} = \alpha - \frac{7}{8}$, $c_{\max} = \alpha - \frac{1}{8}$, $f(c,x) = c/(c^2+x^2)$ *and* $m(d) = \min\{f(c_{\min},d), f(c_{\max},d)\}$. *If for some* $d > 0$ *and some* $T_0 > d+e$ *one has* $\nu(r,d)\,m(d) > b_\alpha$ *for all* $r \geq T_0$, *then*

$$D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) - \frac{\xi'}{\xi}(\alpha) > 0 \qquad \text{for all } r \geq T_0.$$

*For* $r+d \leq H = 3\times10^{12}$ *the conclusion also holds if* $m(d)$ *is replaced by* $f(\alpha-\frac{1}{2},d)$.

*Proof.* By the Hadamard product, $D_\alpha(r) = \sum_\rho f(\alpha - \beta_\rho, r - \gamma_\rho) - b_\alpha$, summed symmetrically over all zeros $\rho = \beta_\rho + i\gamma_\rho$ counted with multiplicity. Under $(\mathrm{Q}_{7/8})$ every width $c_\rho = \alpha - \beta_\rho$ lies in $[c_{\min}, c_{\max}]$, because $\frac{1}{8} \le \beta_\rho \le \frac{7}{8}$, so every term is positive. For $|r - \gamma_\rho| \le d$ we have $f(c_\rho, r - \gamma_\rho) \ge f(c_\rho, d)$, and $f(\cdot, d)$ increases on $(0, d)$ and decreases on $(d, \infty)$, so its minimum on an interval is attained at an endpoint and $f(c_\rho, d) \ge m(d)$. Lemma 16.12.6 counts the terms, whence $D_\alpha(r) \ge \nu(r, d)\, m(d) - b_\alpha$. For $r + d \le H$ every zero with ordinate in $[r - d, r + d]$ lies on the critical line by the verification of [419], so $c_\rho = \alpha - \frac{1}{2}$ and $f(c_\rho, d) = f(\alpha - \frac{1}{2}, d)$. $\square$

Table 16.4 gives the heights obtained. For each $\alpha$ and each half width $d$ put $k = \lfloor b_\alpha / m(d) \rfloor + 1$; if the window holds at least $k$ zeros then $k\, m(d) > b_\alpha$, and by Lemma 16.12.6 it does as soon as $\nu(r, d) \ge k$. The table lists, for the $d$ that makes the last height smallest, the height $T_0$ beyond which $\nu(r, d) \ge k$. Under $(\mathrm{Q}_{7/8})$ the off line threshold is finite for every $\alpha > \frac{7}{8}$, because the window count grows like $(\frac{d}{\pi} - 0.219) \log r$ for $d > 0.69$ while $m(d)$ is fixed. For $\alpha$ close to $\frac{7}{8}$, however, it exceeds the verified height $H$, and then the off line range $r \ge T_0$ and the on line range $r + d \le H$ leave the heights between $H$ and $T_0$ uncovered. The two ranges overlap from $\alpha_* = 0.8888$ on: at $\alpha = 0.8888$ the count gives $T_0 = 2.30 \times 10^{12}$ with $d = 3.5$ and sixteen zeros, whereas at $\alpha = 0.8887$ it gives $3.19 \times 10^{12} > H$, and the smallest $\alpha$ for which the off line threshold $T_0$ is at most $H$ is 0.88875. For the eight basepoints of the table the ranges overlap, and the count covers every height beyond the on line threshold. The scan in $\alpha$ is made by the script on a grid; we do not prove that the thresholds are monotone in $\alpha$, and Theorem 16.12.8 makes this unnecessary.

| $\alpha$ | $b_\alpha$ | $T_0$ off line | $(d, k)$ | $T_0$ on line | $(d, k)$ |
|---|---|---|---|---|---|
| 0.900 | 0.01848 | $1.9 \times 10^6$ | $(3, 7)$ | 147 | $(13, 8)$ |
| 0.925 | 0.01963 | $5.2 \times 10^4$ | $(3.5, 5)$ | 147 | $(13, 8)$ |
| 0.950 | 0.02079 | $1.3 \times 10^4$ | $(5, 7)$ | 147 | $(13, 8)$ |
| 1.000 | 0.02310 | $3.5 \times 10^3$ | $(6.5, 8)$ | 147 | $(13, 8)$ |
| 1.500 | 0.04614 | 324 | $(8, 5)$ | 147 | $(13, 8)$ |
| 2.000 | 0.06907 | 231 | $(10.5, 7)$ | 147 | $(13, 8)$ |
| 5.000 | 0.20166 | 191 | $(12, 8)$ | 167 | $(12.5, 8)$ |
| 7.800 | 0.31322 | 238 | $(13, 10)$ | 214 | $(12.5, 9)$ |

Table 16.4: Heights beyond which Theorem 16.12.7 guarantees the compensated clock inequality under $(\mathrm{Q}_{7/8})$, rounded up, with the half width $d$ of the window and the number $k$ of zeros that the proof asks for in it. "Off line" uses only $\frac{1}{8} \le \beta \le \frac{7}{8}$; "on line" uses $\beta = \frac{1}{2}$, valid for $r + d \le H$. Computed by `polya_qrh.py` with the bound of Lemma 16.12.6.

The count loses information in one place. A zero at distance $c_{\min} = \alpha - \frac{7}{8}$ from the line $\operatorname{Re} s = \alpha$ contributes only about $c_{\min}/d^2$ to $D_\alpha$, which tends to 0 with $\alpha - \frac{7}{8}$, whereas $b_\alpha$ stays near 0.0173; this is why the method stops at $\alpha_* = 0.8888$

rather than at $\frac{7}{8}$, and why any extension of the verified height only lowers $\alpha_*$ towards $\frac{7}{8}$.

THEOREM 16.12.8 (Compensated clock under a quasi Riemann hypothesis). *Let $\frac{1}{2} \le \alpha_0 < 1$ and assume that $\zeta$ has no zero with $\operatorname{Re} s > \alpha_0$. Then $D_\alpha(r) > 0$ for every $\alpha$ with $\alpha_0 < \alpha \le 1$ and every real $r \neq 0$. Together with Proposition 9.1.12, which has no hypothesis, $D_\alpha(r) > 0$ for every $\alpha > \alpha_0$ and every real $r \neq 0$.*

*Proof.* Fix $\alpha$ and put $a = \alpha - \frac{1}{2}$ and $c_{\max} = \alpha + \alpha_0 - 1$, which is less than 1 because $\alpha \le 1$ and $\alpha_0 < 1$. As $D_\alpha$ is even we take $r > 0$. By the Hadamard product, as in the proof of Theorem 16.12.7, $D_\alpha(r) = \sum_\rho f(\alpha - \beta_\rho, r - \gamma_\rho) - b_\alpha$ with $f(c,x) = c/(c^2 + x^2)$. By hypothesis and the functional equation, $1 - \alpha_0 \le \beta_\rho \le \alpha_0 < \alpha$ for every zero, so every width $c_\rho = \alpha - \beta_\rho$ lies in $(0, c_{\max}]$ and every term is positive.

Window estimate. Fix $d > 0$ and let $W$ be the multiset of zeros with ordinate in $[r-d, r+d]$, with $n$ elements. The map $\rho \mapsto 1 - \bar\rho$ preserves the ordinate and the multiplicity and fixes exactly the zeros on the critical line, so $W$ consists of zeros on the line, of width $a$, and of pairs $\{\rho, 1-\bar\rho\}$ of the same ordinate whose widths $c, c'$ satisfy $c + c' = 2a$ however far from the line the zero lies. For $0 < c \le c_{\max}$ and $|x| \le d$ we have $c^2 + x^2 \le c_{\max}^2 + d^2$, hence $f(c,x) \ge c/(c_{\max}^2 + d^2)$; summing over $W$, in which the widths add up to $na$,

$$D_\alpha(r) \;\ge\; \frac{na}{c_{\max}^2 + d^2} - b_\alpha \;\ge\; a\Big(\frac{n}{c_{\max}^2 + d^2} - 0.04621\Big),$$

by Lemma 16.11.13. With $d = 4$ and $c_{\max} < 1$ we have $c_{\max}^2 + d^2 < 17$ and $0.04621 \cdot 17 = 0.7856 < 1$, so $n \ge 1$ suffices: if the window $[r-4, r+4]$ contains the ordinate of at least one zero, then $D_\alpha(r) > a\big(\frac{1}{17} - 0.04621\big) > 0.0126\, a$.

It remains to show that the window contains an ordinate, or that $D_\alpha(r) > 0$ for another reason, at every $r > 0$. There are three ranges.

(a) $0 < r \le \theta_0 = \gamma_1 - \alpha/\sqrt{3}$, where $\theta_0 \ge \gamma_1 - 1/\sqrt{3} = 13.557$. Every zero has $|r - \gamma_\rho| \ge \gamma_1 - r \ge \alpha/\sqrt{3} > c_\rho/\sqrt{3}$, because $c_\rho < \alpha$, and $x \mapsto f(c,x)$ is strictly convex where $|x| > c/\sqrt{3}$, since $\partial_x^2 f = 2c(3x^2 - c^2)/(c^2 + x^2)^3$. So $D_\alpha$ is strictly convex on $[-\theta_0, \theta_0]$ (the series of second derivatives converges uniformly there), even, and zero at the origin, hence positive for $0 < |r| \le \theta_0$; this is the argument of Proposition 9.1.11.

(b) $\theta_0 < r \le \gamma_{1700} + 4 = 2201.26$. The first 1700 ordinates, computed in ball arithmetic and stored in `data/zeros_1700.txt`, have consecutive differences at most 6.8873 (the largest, between $\gamma_1 = 14.1347$ and $\gamma_2 = 21.0220$; the next is 5.4140), so the windows $[\gamma_k - 4, \gamma_k + 4]$ cover $[\gamma_1 - 4, \gamma_{1700} + 4]$, and $\gamma_1 - 4 = 10.13 < \theta_0$. Every $r$ of the range lies in one of them, that is, its window contains an ordinate.

(c) $r \ge 2200$. By Lemma 16.12.6 with $d = 4$ the window holds at least $\nu(r,4)$ zeros, and we show $\nu(r,4) \ge 2.42$. Put $I(r) = (\vartheta(r+4) - \vartheta(r-4))/\pi$. Since $\vartheta''(t) = \frac{1}{4}\sum_{n \ge 0}(n + \frac{1}{4})t\,\big((n+\frac{1}{4})^2 + \frac{t^2}{4}\big)^{-2} > 0$, $I$ increases, and on $[2200, X_0 - 4]$

we have $\nu(r,4) = I(r) - 2\cdot 2.5167 \geq \nu(2200,4) = 7.4591 - 5.0334 = 2.4257$. For $r \geq X_0 - 4$ Stirling's formula gives $I(r) = \frac{4}{\pi}\log\frac{r}{2\pi} + O(r^{-2})$ with an error below $10^{-9}$, and $s$ is nondecreasing and, at $t = r+4 \geq X_0$, at most $s_{\log}(t) = 0.1095\log t + 0.2042\log\log t + 3.0305$, so $\nu(r,4) \geq \frac{4}{\pi}\log\frac{r}{2\pi} - 2s_{\log}(r+4) - 10^{-9}$. The derivative of the right side is at least $\big(\frac{4}{\pi} - 2\cdot 0.1095 - \frac{2\cdot 0.2042}{\log X_0}\big)/(r+4) > 1/(r+4)$, so it increases, and at $r = X_0 - 4$ it equals $28.4018 - 12.6490 - 10^{-9} > 15.75$. Hence $\nu(r,4) \geq 2.42 > 0$ for all $r \geq 2200$, and the window contains a zero. Ranges (b) and (c) overlap in $[2200, 2201.26]$. □

For $\alpha_0 = \frac{1}{2}$ the theorem is the direction RH ⇒ compensated positivity of Theorem 9.1.18(vi), and the window argument is that of the proof of part (v) of Theorem 9.1.18.

Remark 16.12.9 (One block for each zero). Under the hypothesis of Theorem 16.12.8 every width $c = \alpha - \beta_\rho$ is positive, and the Hadamard product splits the compensated reciprocal into one factor for each pair $\rho, \bar\rho$ with $\gamma = \gamma_\rho > 0$ [253]:

$$\Phi_{\alpha,\rho}\Big(\frac{w^2}{2}\Big) = \frac{c^2+\gamma^2}{(c+w)^2+\gamma^2}\, e^{2cw/(c^2+\gamma^2)}.$$

By Frullani's integral $-\log\Phi_{\alpha,\rho}(w^2/2) = \int_0^\infty (e^{-wy} + wy - 1)\,\varrho_\rho(dy)$, with the damped cosine wave $\varrho_\rho(dy) = -2y^{-1}e^{-cy}\cos(\gamma y)\,dy$, a signed measure with $\int (y \wedge y^2)\,|\varrho_\rho|(dy) < \infty$. The $2\sin^2$ lemma, applied to its positive and negative parts, gives the density $\pi^{-1}[f(c, r-\gamma) + f(c, r+\gamma) - 2f(c,\gamma)]$ in the ordinate $r$, a recentred pair of Cauchy bumps. Summed over the zeros these are $\pi^{-1}D_\alpha(r)$, as in the proof of Theorem 16.12.8, the constants $2f(c_\rho, \gamma_\rho)$ adding up to $b_\alpha$. No single block is a GGC, nor even a Laplace transform, since $\Phi_{\alpha,\rho}(w^2/2) \to \infty$ as $w \to \infty$: positivity is a property of the sum, in which the bumps of the zeros near height $r$ outweigh the accumulated constants. As $c \downarrow 0$ the block becomes $(1 + w^2/\gamma^2)^{-1}$, an exponential law in $u = w^2/2$ whose Thorin measure is the unit atom at $r = \gamma$; this is Proposition 6.2.16, zero by zero.

Corollary 16.12.10. *For $\frac{1}{2} \leq \alpha_0 < 1$ the following are equivalent: (a) $\zeta$ has no zero with $\operatorname{Re} s > \alpha_0$; (b) $D_\alpha(r) \geq 0$ for every $\alpha \in (\alpha_0, 1)$ and every real $r$ at which $D_\alpha$ is defined. When they hold, $D_\alpha(r) > 0$ for every $\alpha \in (\alpha_0, 1)$ and every real $r \neq 0$.*

*Proof.* (a) implies (b) by Theorem 16.12.8, and (b) implies (a) by Proposition 16.11.10, since $D_\alpha \geq 0$ for $\alpha \geq 1$ by Proposition 9.1.12. □

For $\alpha_0 = \frac{1}{2}$ this is part (vi) of Theorem 9.1.18.

The basepoints at which the compensated reciprocal is a GGC form a closed half line [253]. Let $\Theta = \sup\operatorname{Re}\rho$, the $\sigma_{\max}$ of Theorem 16.11.7. For $\alpha < \Theta$ a zero with $\operatorname{Re}\rho > \alpha$ gives the transform a pole at $u = (\rho - \alpha)^2/2$, off the negative axis, so it is not a GGC, whose transform is holomorphic off $(-\infty, 0]$ [56]. For $\alpha > \Theta$

it is one: by Theorem 16.12.8 with $\alpha_0 = \Theta$ and Theorem 16.11.7 with $\delta = \alpha - \Theta$ when $\alpha < 1$, and by Proposition 9.1.12 when $\alpha \geq 1$. At $\alpha = \Theta$, as $\alpha \downarrow \Theta$ the transforms $e^{\sqrt{2u}\, b_\alpha}\xi(\alpha)/\xi(\alpha + \sqrt{2u})$ converge for every $u \geq 0$ to the transform at $\Theta$, which is continuous at $u = 0$; so the laws converge weakly and the limit is a GGC, the class being closed under weak limits. So the set is exactly $[\Theta, \infty)$; for $\Theta = \frac{1}{2}$ this is RH. A zero on the line $\operatorname{Re} s = \Theta$ gives the Thorin measure at $\Theta$ an atom at its ordinate, of mass its multiplicity, the vague limit of its Cauchy kernel of width $\alpha - \Theta$; if $\Theta$ is not attained there is no atom.

COROLLARY 16.12.11. *Assume* $(\mathrm{Q}_{7/8})$ *and let* $\alpha > \frac{7}{8}$. *Then the compensated reciprocal* $e^{wb_\alpha}\xi(\alpha)/\xi(\alpha + w)$ *is the Laplace transform in* $u = w^2/2$ *of a GGC, with the Thorin density* (16.11.6) *of the* $2\sin^2$ *lemma, positive at every height. For* $\alpha \geq 1$ *this holds with no hypothesis, by the factorisation of Proposition 9.1.12.*

*Proof.* The hypotheses of Theorem 16.11.7 hold for $\xi$ with $\delta = \alpha - \frac{7}{8}$, and its condition (16.11.7) is $D_\alpha \geq 0$, which holds by Theorem 16.12.8 for $\frac{7}{8} < \alpha \leq 1$ and by Proposition 9.1.12 for $\alpha \geq 1$. □

So under the claimed theorem the conclusion of the $2\sin^2$ lemma, positivity of the Thorin density of the compensated reciprocal, holds on every line to the right of $\frac{7}{8}$, inside the critical strip for $\frac{7}{8} < \alpha < 1$, even though the hypothesis of the lemma, positivity of the primal measure, cannot even be stated there, because the Euler product diverges.

*Numerics.* The script `polya_qrhpair.py` (`code/README.md`) computes the constants of the proof of Theorem 16.12.8 for $\alpha_0 = \frac{7}{8}$ and writes them to `data/polya_qrhpair.json`. The margin $a/(c_{\max}^2 + 16) - b_\alpha$ for one zero in the window, with $b_\alpha$ computed by mpmath at sixty values of $\alpha$ in $(\frac{7}{8}, 1]$, is at least 0.0053, with the minimum at $\alpha = 0.8751$, and the ratio $b_\alpha/a$ is at most 0.04620; the inequality $f(c,x) + f(2a - c, x) \geq 2a/(c_{\max}^2 + d^2)$ for a pair holds on grids of $\alpha$, $c$ and $x$ with smallest ratio 1.0000047. The quantity $\nu(r, 4)$ was recomputed with the Stirling series for $\vartheta$ and with mpmath at ten heights up to $10^{16}$, agreeing to $7.1 \times 10^{-15}$; it never exceeds the number of ordinates in $[r - 4, r + 4]$ for $r$ from 40 to 2190.

The direct computation of $D_\alpha$ agrees. The script `polya_primes.py` finds $D_\alpha(r) > 0$ on $(0, 600]$ at step 0.01 (60 000 heights) at $\alpha = 0.9$, 0.95, 1, 1.001, 2, 4 and 7.8, and on $(0, 200]$ at the fifteen equally spaced basepoints from 1 to 7.8 and at twelve further ones (1.001, 1.01, 1.1, 1.2, 1.5, 2, 3, 4, 5, 6, 7 and 7.789), with the smallest value of $D_\alpha$ at the first height of the grid, where $D_\alpha$ grows like $r^2$ and $D_\alpha(r)/r^2$ is $1.1 \times 10^{-4}$ at $\alpha = 1$ and $9.0 \times 10^{-4}$ at $\alpha = 7.8$. At heights above 2 the minimum over $(2, 600]$ is $3.7 \times 10^{-4}$, $4.2 \times 10^{-4}$ and $4.6 \times 10^{-4}$ at $\alpha = 0.9$, 0.95 and 1, attained just above $r = 2$. These computations use no hypothesis on the zeros, and for $\alpha < 1$ they are floating point evidence, not a proof.

### 16.12.5 What it does not give, and what to look for in the proof

The claimed theorem is a fixed half plane. The routes of this chapter and of the book need more.

1. Bondesson's closure gives RH from GGC basepoint laws along a sequence $\alpha_n \downarrow \frac{1}{2}$ (Corollary 4.1.2). ($\mathrm{Q}_{7/8}$) supplies them for $\alpha \geq \frac{7}{8}$ only; the open strip is $\frac{1}{2} < \alpha < \frac{7}{8}$.

2. The even part laws and the accounting theorem need no zero to the right of $\alpha$; below $\frac{7}{8}$ the term $2N(\alpha, T)$ of Theorem 16.11.16 remains and is controlled only by density estimates.

The structural point is Corollary 16.12.10: a quasi Riemann hypothesis at $\alpha_0$ is equivalent to the positivity of the compensated Thorin densities of the $2\sin^2$ lemma on every line to the right of $\alpha_0$. The proof announced in [398] does not go through the Thorin picture (Section 12.1.2), but the equivalence says that whatever it proves has a Thorin reading. If it contains an explicit lower bound for $\operatorname{Re}\frac{\zeta'}{\zeta}$ or for a mollified Dirichlet polynomial on $\operatorname{Re} s > \frac{7}{8}$, then a bound at least as strong as $-\operatorname{Re}\mathcal{A}$ is, by part (ii) of Theorem 16.12.1, positivity of the clock, a Thorin statement in disguise, and the natural next step towards $\alpha_0 < \frac{7}{8}$ is to transfer it directly to the clock: the identity (16.11.9) at $\alpha < 1$, with the prime sum replaced by whatever the proof controls.

# Chapter 17

# Yoshida's Hermitian forms, Weil positivity and the Thorin programme

In 1992 Hiroyuki Yoshida studied Weil's explicit formula as a Hermitian form on functions supported in a window $[-a, a]$ [525]. As the first author reads the paper, Yoshida proved the form positive definite for $a \leq \frac{1}{2}\log 2$, the range in which no prime power enters, and reduced the Riemann Hypothesis to its nondegeneracy on a completed space for every $a$ (Section 17.1). This chapter reads Yoshida against the Thorin programme of this book (Section 1.1) and the sine squared lemma (1.2.9), and works the open problems that come out of the comparison.

Yoshida's functional $\mathcal{Y}(F) = \sum_\rho \Phi(\rho)$ is Weil's distribution, and the heat trace $W$ of the reciprocal Xi law is half its value on a Gaussian. The sine squared lemma is the hinge: its Brownian level $x$ is Yoshida's support length $2a$. There is an uncertainty barrier for one coefficient: Proposition 17.9.1 bounds the off line part of the tent coefficient $D(x) = x\langle \chi_0, \chi_0 \rangle$ at $a = x/2$ by an explicit $\varepsilon_H(x)$; against the lower bound for the on line part that the first 1700 zeros give (a double precision computation), this keeps $D$ positive at levels below about 47 (Proposition 17.9.2) whatever the zeros above the verified height $3 \times 10^{12}$ [419] do, so that no computation of $D$ at those levels can detect an off line zero.

*What is not claimed.* Nothing here proves RH. A number is computed in floating point at the precision stated, or in ball arithmetic where the text says so. The barrier concerns the tent coefficient only. The account of Yoshida's paper in Section 17.1 is the first author's reading of it, used here as context; it was not checked against the paper.

## 17.1 Yoshida's paper

For $F \in C_c^\infty(\mathbb{R})$ put

$$\Phi(s) = \int_{\mathbb{R}} F(y) e^{(s-1/2)y}\, dy, \qquad \widehat{F}(t) = \Phi(\tfrac{1}{2} + it) = \int_{\mathbb{R}} F(y) e^{ity}\, dy.$$

The letter $\Phi$ is local to this chapter and is not the Pólya kernel of Section 1.2. As there, write each nontrivial zero as $\rho = \frac{1}{2} + i\gamma_\rho$, that is $\gamma_\rho = -i(\rho - \frac{1}{2})$. The number $\gamma_\rho$ is complex for a zero off the line: if $\rho = \frac{1}{2} + \delta + i\gamma$ with $\gamma$ and $\delta$ real, then $\gamma_\rho = \gamma - i\delta$, whereas $\gamma = \operatorname{Im}\rho$ is the ordinate and $\beta_\rho = \operatorname{Re}\rho$ the real part. Then $\Phi(\rho) = \widehat{F}(\gamma_\rho)$, the entire extension of $\widehat{F}$, and Weil's explicit formula reads

$$\begin{aligned} \mathcal{Y}(F) := \sum_\rho \widehat{F}(\gamma_\rho) = {} & \int_{\mathbb{R}} F(y)\big(e^{y/2} + e^{-y/2}\big)dy - (\log\pi)F(0) \\ & - \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\big(F(\log n) + F(-\log n)\big) + V_1(F), \end{aligned} \tag{17.1.1}$$

summed over all nontrivial zeros with multiplicity, with the archimedean term in two forms,

$$\begin{aligned} V_1(F) &= \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{F}(t)\operatorname{Re}\psi\Big(\frac{1}{4} + \frac{it}{2}\Big)dt \\ &= -\tfrac{1}{2}\operatorname{Pf}\Big(\frac{1}{|y|}\Big)(F) - (\gamma_E + \log 2)F(0) - \int_{\mathbb{R}}\Big(\frac{e^{|y|/2}}{2\sinh|y|} - \frac{1}{2|y|}\Big)F(y)\,dy, \end{aligned} \tag{17.1.2}$$

where $\psi = \Gamma'/\Gamma$ and $\operatorname{Pf}(1/|y|)(F) = \lim_{\epsilon\downarrow 0}\big(\int_{|y|>\epsilon} F(y)\,|y|^{-1}dy + 2F(0)\log\epsilon\big)$ is the finite part of $1/|y|$, the integral cut at $|y| = 1$. Each term of (17.1.1) depends on $F$ only through its even part, because the zeros are symmetric under $\rho \mapsto 1-\rho$, which negates $\gamma_\rho$. The formula holds for the Gaussians used below, whose transforms are entire and decrease rapidly in every strip (the class of Section 1.2), and, in Weil's original form [512, 55], for continuous, piecewise $C^1$ functions of compact support, the class of the tent and of the products $\chi_n * \widetilde{\chi_m}$ of Sections 17.6 and 17.7.

Remark 17.1.1 (The archimedean term in the real variable). The second line of (17.1.2) is the explicit formula of Section 1.2 with the archimedean integral carried to the variable $y$ and its divergence at $y = 0$ cut at $|y| = 1$; the forms of Theorem 20.1.1 and Section 15.6.1 differ from it only in where $\gamma_E$ is placed.

The Hermitian form is $\langle\varphi_1, \varphi_2\rangle = \mathcal{Y}(\varphi_1 * \widetilde{\varphi}_2)$ with $\widetilde{\varphi}(y) = \overline{\varphi(-y)}$. Since $\widehat{\varphi * \widetilde{\varphi}} = |\widehat{\varphi}|^2$ on $\mathbb{R}$, Weil's criterion [512, 55] says that RH holds if and only if $\langle\varphi,\varphi\rangle \geq 0$ for all $\varphi \in C_c^\infty(\mathbb{R})$ (Proposition 7.2.14 and Section 3.5.2). Let $C(a)$ be the space of the functions of $C_c^\infty(\mathbb{R})$ supported in $[-a,a]$, let $\chi_n(y) = (2a)^{-1/2}e^{\pi i n y/a}1_{[-a,a]}(y)$ for $n \in \mathbb{Z}$, an orthonormal basis of $L^2[-a,a]$, and let $K(a)$ be the space of the restrictions to $[-a,a]$ of smooth $2a$ periodic functions, that is the series $\sum c_n\chi_n$ with rapidly decreasing coefficients. Yoshida's results, as the first author reads the paper, are the following.

(Y1) Positivity on odd $\varphi$ alone is equivalent to RH; on even $\varphi$ alone, to RH up to

real zeros. The proof is a localisation lemma that concentrates a test vector on one zero.

(Y2) The prime sum in $\langle \varphi_1, \varphi_2 \rangle$ for $\varphi_1, \varphi_2 \in C(a)$ runs over $\log n \le 2a$ only. The form is positive definite on $C(a)$ for small $a$, and on the complement of the low Fourier modes for any $a$.

(Y3) The form is nondegenerate on $C(a)$ for every $a$, unconditionally.

(Y4) There are closed formulas for the coefficients $\langle \chi_n, \chi_m \rangle$ on the basis of $K(a)$.

(Y5) Theorem 1: the form is positive definite on $K(a)$ for $a = \frac{1}{2}\log 2$.

(Y6) Theorem 2: RH holds if and only if the form is nondegenerate on a completion of $K(a)$ for every $a$. If RH fails there is a first $a_0$ beyond which positivity is lost, and continuity in $a$ forces degeneracy there.

(Y7) For $\mathbb{Q}$ the form is nondegenerate on $K(a)$ for every $a$.

The statements (Y1) to (Y7) were not checked against the paper, and the proofs of Sections 17.2, 17.3, 17.4 and 17.5 do not use them.

Throughout, $W(x) = \sum_{\gamma>0} e^{(\rho-1/2)^2 x}$ is the heat trace of Section 1.2, the sum over one zero from each pair $\{\rho, 1-\rho\}$; $T_k = (-1)^k W^{(k)}$ is the ladder, the function written $c_k$ in Section 5.1; $U$ is the Thorin functional of Section 5.3.1, which under RH is the measure $\sum_{\gamma>0} \delta_{\gamma^2}$; and $p_1 = \sum_{\gamma>0} \gamma_\rho^{-2} = \frac{1}{2}(\log \xi)''(\frac{1}{2}) = 0.0231049931$, so that $2p_1$ is the number $V_\xi$ of Remark 9.1.19. The sine squared lemma, in the chart of this chapter, is

$$k_x(w) := e^{-wx} + wx - 1 = \int_0^\infty \log\Big(1 + \frac{u}{z}\Big)\varpi_x(z)\,dz, \qquad u = \frac{w^2}{2},$$
$$\varpi_x(z) = \frac{x}{\pi\sqrt{2z}}\, 2\sin^2\Big(x\sqrt{\tfrac{z}{2}}\Big). \tag{17.1.3}$$

It is (1.2.9) after the substitution $z \mapsto 2z$, and (16.2.3) of Chapter 16.

## 17.2 The ladder is the diagonal of Weil's form

Let $h_x(y) = (4\pi x)^{-1/2} e^{-y^2/(4x)}$, so $\widehat{h_x}(t) = e^{-t^2 x}$ and $h_x = h_{x/2} * \widetilde{h_{x/2}}$.

Proposition 17.2.1. *For $k \ge 0$ and $x > 0$ let $g_{k,x} = (i\,d/dy)^k h_{x/2}$, so that $\widehat{g_{k,x}}(t) = t^k e^{-t^2 x/2}$. Then*

$$\langle g_{k,x}, g_{k,x} \rangle = \sum_\rho \gamma_\rho^{2k} e^{-\gamma_\rho^2 x} = 2\,T_k(x), \qquad W(x) = \tfrac{1}{2}\langle h_{x/2}, h_{x/2} \rangle.$$

*Proof.* $F = g_{k,x} * \widetilde{g_{k,x}}$ has $\widehat{F}(t) = t^{2k}e^{-t^2x}$, so $\Phi(\rho) = \gamma_\rho^{2k}e^{-\gamma_\rho^2 x}$. The summand is even in $\gamma_\rho$ and $\rho \mapsto 1-\rho$ negates $\gamma_\rho$, so the sum over all zeros is twice the sum over the zeros with $\gamma > 0$, which is $2T_k(x)$. □

Dividing the prime side of $\langle h_{x/2}, h_{x/2}\rangle$ by two gives the explicit formula (1.2.5) for $W$. Both sides give $W(0.003) = 1.09522012145284$ (the first 1700 zeros against the primes, agreeing to thirty digits; script `yo_explicit.py`). Since $W$ is completely monotone exactly under RH (Proposition 1.2.1), Proposition 17.2.1 says that RH is positivity of Weil's form on the diagonal of a single two parameter family $(k, x)$, with no cross terms.

## 17.3 Every large rung fails

The rungs $T_k(x) = \sum_{\gamma>0} w_\rho^k e^{-w_\rho x}$, with $w_\rho = \gamma_\rho^2$, satisfy $T'_{k-1} = -T_k$, and $T_{k-1}(x) \to 0$ as $x \to \infty$ because every exponent has real part $\gamma^2 - \delta^2 > \gamma_1^2 - \frac{1}{4} > 0$, where $\gamma_1 = 14.1347\ldots$ is the least ordinate. Hence

$$T_{k-1}(x) = \int_x^\infty T_k(s)\, ds, \qquad k \geq 1,\ x > 0. \tag{17.3.1}$$

So $T_k \geq 0$ on $(0,\infty)$ forces $T_{k-1} \geq 0$ there: the orders for which $T_k \geq 0$ form an initial segment. This is the argument of Section 5.1.4, where it is stated for the Williamson measures $U_k$, which are positive exactly when $T_k \geq 0$ on $(0,\infty)$.

Theorem 17.3.1. *(i) If $T_k \geq 0$ on $(0,\infty)$ then $T_j \geq 0$ on $(0,\infty)$ for every $j \leq k$, so the set of such orders is an initial segment of $\mathbb{N}$, and it is all of $\mathbb{N}$ if and only if RH holds. (ii) If RH fails there is $K_0$ such that for every $k \geq K_0$ there is $x > 0$ with $T_k(x) < 0$. Consequently, for any infinite set $\mathcal{K}$ of orders, RH holds if and only if $T_k \geq 0$ on $(0,\infty)$ for every $k \in \mathcal{K}$. In particular RH is equivalent to $\langle g_{k,x}, g_{k,x}\rangle \geq 0$ for all odd $k$ and all $x > 0$, a positivity statement on odd Hermite Gaussians only.*

*Proof.* (i) The first claim is (17.3.1). All $T_k$ are nonnegative on $(0,\infty)$ if and only if $W$ is completely monotone, which holds if and only if RH holds (Proposition 1.2.1). (ii) If RH fails, some $T_{k_0}$ takes a negative value, and then by (i) so does every $T_k$ with $k \geq k_0$; let $K_0$ be the least such $k_0$. If $T_k \geq 0$ for every $k$ in an infinite set $\mathcal{K}$, then by (i) it holds for every $k$, since $\mathcal{K}$ is unbounded, and RH holds; the converse is clear. The last statement is Proposition 17.2.1 for odd $k$. □

For infinite sets $\mathcal{K}$ this is the equivalence of Section 5.1.4 between RH and $U_k \geq 0$ for infinitely many $k$. What is added here is the reading of $T_k$ as the diagonal of Yoshida's form (Proposition 17.2.1), which turns the odd orders into the Hermite counterpart of Yoshida's parity reduction (Y1), and the location of the failure in the next proposition, which is not needed for the theorem. With

$y = u^2$ and $x = k/y$ the computation is the ray $t = k/u^2$ of Proposition 5.1.10: $e^{-k(\log y - 1)} T_k(k/y)$ is the quantity $e^{-f_0} c_k(t)$ there.

PROPOSITION 17.3.2 (Where the rungs fail). *Suppose RH fails. For a zero* $\rho = \frac{1}{2} + \delta + i\gamma$ *with* $\gamma > 0$ *write* $w_\rho = \gamma_\rho^2 = r_\rho e^{i\theta_\rho}$, *so that* $\theta_\rho = -2\arctan(\delta/\gamma)$ *and* $\sec\theta_\rho = (\gamma^2 + \delta^2)/(\gamma^2 - \delta^2)$. *Then* $S = \max \log\sec\theta_\rho > 0$ *is attained, by finitely many zeros. Let* $w_* = r_* e^{i\theta_*}$ *be one maximiser, of multiplicity* $m$, *and* $y_* = r_* \cos\theta_* = \gamma_*^2 - \delta_*^2$. *There are* $\eta > 0$ *and* $K_1$ *and, for every* $k \geq K_1$, *a point* $y_k$ *with* $|y_k - y_*| = O(1/k)$ *such that*

$$e^{-k(\log y_k - 1)} T_k(k/y_k) = -2m\, e^{kS}\big(1 + O(1/k)\big) + O\big(e^{k(S-\eta)}\big) < 0.$$

*Proof.* Since $|\delta| < \frac{1}{2} < \gamma$, $\operatorname{Re} w_\rho = \gamma^2 - \delta^2 > 0$. The zeros $\rho$ and $1 - \overline{\rho}$ give conjugate values of $w$, so $T_k$ is real. Evaluate at $x = k/y$ and multiply by the positive factor $e^{-k(\log y - 1)}$:

$$e^{-k(\log y - 1)} T_k(k/y) = \sum_{\gamma > 0} e^{k g(w_\rho, y)}, \qquad g(w, y) = \log\frac{w}{y} - \frac{w}{y} + 1.$$

For fixed $w = re^{i\theta}$, $\operatorname{Re} g(w, y) = \log(r/y) - r\cos\theta/y + 1$ has a unique maximum in $y$, equal to $\log\sec\theta$, at $y = r\cos\theta$. On line zeros $\theta = 0$ and $\operatorname{Re} g \leq 0$. Suppose an off line zero exists. Since $|\theta_\rho| < 1/\gamma \to 0$, only finitely many zeros have $|\theta_\rho|$ above any given positive bound, so $S = \max\log\sec\theta_\rho$ is attained, by finitely many zeros, and $S > 0$. Every zero other than $w_*$ and $\overline{w_*}$ has, for $y$ in a small neighbourhood $J$ of $y_*$, $\operatorname{Re} g(w_\rho, y) \leq S - \eta$ for some $\eta > 0$: on line zeros because $\operatorname{Re} g \leq 0$, for zeros with $|\theta_\rho| < |\theta_*|$ because only finitely many have $\sec\theta_\rho$ near the maximum, and for the other maximisers because $y_*$ is not their optimal point. Since $e^{ku} \leq e^u$ for $u \leq 0$ and $k \geq 1$, the remainder satisfies

$$\Big|\sum_{\rho \neq w_*, \overline{w_*}} e^{k g(w_\rho, y)}\Big| \leq e^{k(S-\eta)} \sum_\rho e^{\operatorname{Re} g(w_\rho, y) - S + \eta} \leq C\, e^{k(S-\eta)}, \qquad y \in J,$$

with $C$ independent of $k$, because $\sum_\rho (r_\rho/y) e^{1 - r_\rho \cos\theta_\rho / y}$ converges uniformly on $J$. The main term is $2m\, e^{k \operatorname{Re} g(w_*, y)} \cos\big(k \operatorname{Im} g(w_*, y)\big)$ with $\operatorname{Im} g(w_*, y) = \theta_* - r_* \sin\theta_*/y$, whose derivative in $y$ is $r_* \sin\theta_*/y^2 \neq 0$. Choose $y_k$ with $k \operatorname{Im} g(w_*, y_k) \equiv \pi \pmod{2\pi}$ and $|y_k - y_*| = O(1/k)$. Then $\operatorname{Re} g(w_*, y_k) = S - O(1/k^2)$, the main term is $-2m\, e^{kS}(1 + O(1/k))$, and it dominates the remainder for $k \geq K_1$. Hence $T_k(k/y_k) < 0$. □

The proposition gives no explicit $K_1$, and so no explicit bound for $K_0$, because $C$ depends on the zeros, and two constraints enter. The first is the phase condition: $y_k$ lies at distance about $\pi y_*/(k\tan\theta_*)$ from $y_*$, which costs a factor $\exp(-\pi^2/(2k\tan^2\theta_*))$ in the main term, so the asymptotics start only once $k\theta_*^2$ is of order $\pi$. The second is the competition of the main term $e^{kS}$, with $S = \log\sec\theta_*$ and $\sec\theta_* - 1 = 2\delta_*^2/(\gamma_*^2 - \delta_*^2)$, with the nearest true zeros.

Remark 17.3.3 (A stress test). Add a hypothetical pair at $\gamma = 60.5$, $\delta = 0.45$ to the true zeros (only those between heights 40 and 85 matter at these orders: every excluded term is below $e^{-0.23k}$ and, relative to the main terms, below $10^{-420}$ for $k \geq 3349$), so that $\sec\theta_* - 1 = 1.1065 \times 10^{-4}$ and $y_* = 3660.05$, and test the sign of $T_k(k/y)$ on 3001 points of $|y/y_* - 1| \leq 0.06$. The first failing order is $K_0 = 6349$. Every one of the 3000 orders in $[6349, 9348]$ fails, every one of 395 sampled orders $6349 + 997j$ up to 399167 fails, and none of the orders 1 to 6348 fails on this grid. For the order 6348 the sign is positive at every point of two further grids (200001 points near $y_*$, with both local minima of the scale free margin polished, where the margin is at least $2.9 \times 10^{-4}$, and 18421 points of $y \in [1, 10^8]$ with all 1700 zeros), while at $y = 3584.65$, that is $x = 1.771$, arithmetic with 60 digits gives $T_{6348}(6348/y) = +1.289 \times 10^{19803}$ and $T_{6349}(6349/y) = -5.700 \times 10^{19805}$. By Theorem 17.3.1(i) positivity of $T_{6348}$ gives that of every lower order, so 6349 is the first failing order of this configuration; all of this is computed, not proved (script `yo_ladder.py`, `code/README.md`). For this pair $K_0(\sec\theta_* - 1) = 0.7025$.

The first failing order found on a grid is an upper bound for $K_0$. For the pair $(\gamma, \delta) = (30.5, 0.45)$ the grid of the stress test gives 1064, while a grid of $\pm 30$ percent gives 396, the negative lobe lying 27 percent from $y_*$; the other thirteen pairs of the next sentence give the same order on both grids. Over fourteen pairs, with $\gamma$ from 30.5 to 250.5 and $\delta$ from 0.02 to 0.45 (`data/yo_ladder.json`), $K_0(\sec\theta_* - 1)$ lies between 0.03 and 0.82, and at fixed $\gamma$ it is proportional to $\delta$: the logarithmic slope of $K_0(\sec\theta_* - 1)$ against $\delta$ is 1.04 for $\gamma = 60.5$ and 1.05 for $\gamma = 120.5$, and the ratio $K_0(\sec\theta_* - 1)/\delta$ lies between 0.97 and 2.01 for thirteen of the fourteen pairs. So $K_0$ grows with $\gamma^2$ and, at fixed $\gamma$ over the range tested, like $1/\delta$, not like $(\sec\theta_* - 1)^{-1} \approx \gamma^2/(2\delta^2)$: for $\gamma = 60.5$ and $\delta = 0.02$ the second quantity is $4.6 \times 10^6$ and the computed order is 128610. This is an observation from computation, not a theorem; a scale argument (the pair is a Gaussian envelope times a cosine in the variable $\sqrt{k}\log(y/y_*)$, and its negative lobe has to beat the true zeros that sit near it) suggests that $K_0$ is of order $\gamma^2/(\delta d)$, where $d$ is the distance from $\gamma$ to the nearest true ordinate, and that $\gamma^2/\delta^2$ appears only when $\delta$ is of the size of $d$.

## 17.4 Compact support, the signed Thorin formula, and the first windows

Section 5.3.1 defines the Thorin functional $U$ on the class $\mathcal{T}$ of the functions $\varphi(t) = h(\sqrt{t})$ by (5.3.1), and identifies it with $\varphi \mapsto \sum_k \varphi(a_k)$ through Weil's formula, with no hypothesis on the zeros. In the variable $t = r^2$ the same functional

is the formal signed measure

$$U^{\pm}(dt) = \delta_{-1/4}(dt) + \frac{\operatorname{Re}\psi(\frac{1}{4}+\frac{i\sqrt{t}}{2}) - \log\pi}{4\pi\sqrt{t}}\,dt - \frac{1}{2\pi\sqrt{t}}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\cos(\sqrt{t}\log n)\,dt, \tag{17.4.1}$$

that is $\langle U,\varphi\rangle = \int \varphi\, dU^{\pm}$: the atom is the first term of (5.3.1), the density is its integral term with $t = r^2$, and the third term is the prime sum. The prime series in (17.4.1) is not a tempered distribution in the ordinate $r = \sqrt{t}$. Let $Z$ be the space of the functions $\widehat{F}$ with $F \in C_c^\infty(\mathbb{R})$, and $Z'$ its dual.

Proposition 17.4.1. *In the ordinate $r = \sqrt{t}$ the prime series of* (17.4.1) *defines an element of $Z'$, and for every even $F \in C_c^\infty(\mathbb{R})$ with support in $[-L,L]$,*

$$\mathcal{Y}(F) = 2\int \widehat{F}(\sqrt{t})\,U^{\pm}(dt), \qquad \text{with } \widehat{F}(\sqrt{-1/4}) = \widehat{F}(i/2);$$

*only the prime powers $n \leq e^L$ contribute.*

*Proof.* Pairing $\cos(r\log n)$ with $\widehat{F}(r)$ gives $\pi\big(F(\log n) + F(-\log n)\big)$, which vanishes for $\log n > L$; hence the series is a finite sum on every element of $Z$ and defines an element of $Z'$. With $t = r^2$ the three parts of (17.4.1) give, after the factor 2, which is the passage from one zero of each pair $\{\rho, 1-\rho\}$ in (5.3.1) to all zeros in (17.1.1), the pole term $\widehat{F}(i/2) + \widehat{F}(-i/2)$, the term $V_1(F) - (\log\pi)F(0)$, and the prime sum of (17.1.1). The function $\varphi(t) = \widehat{F}(\sqrt{t})$ lies in $\mathcal{T}$, because $\widehat{F}$ is even, entire and $O((1+|r|)^{-N})$ in every strip. □

For $\varphi \in K(a)$ with $a = \frac{1}{2}\log 2$, $F = \varphi * \widetilde{\varphi}$ is continuous, vanishes at $\pm\log 2$, and the only prime power level in its support is that endpoint. On Yoshida's window, then, the form is the pole and archimedean part alone: his Theorem 1, as the first author reads it, uses only the gamma factor, the pole and the fact that Dirichlet series begin at $n = 2$. Connes and Consani reproved this window from the semilocal trace formula [87]. Beyond it the primes enter: the window of half length 0.8, of autocorrelation support 1.6, contains the prime powers 2, 3 and 4, and Zhu [526] has certified positivity of the form on it (Chapter 18). Windows are indexed by their half length $a$ in the ladder of Section 20.2, each of them a finite problem in the prime powers up to $e^{2a}$.

## 17.5 Localisation: a modulated heat trace

Yoshida's localisation lemma (Y1) concentrates a test vector on one zero. Its Gaussian form is already in the book as the Gaussian smoothing $\mathcal{G}(u,t) = \sum_{\gamma>0} e^{-2t(\gamma_\rho - u)^2}$ of the zeros of Proposition 5.1.10, the sum over one zero of each pair $\{\rho, 1-\rho\}$ (there $\widetilde{\gamma}$ is the $\gamma_\rho$ of this chapter). For real $u$ and $x > 0$ let $g$ be the vector with $\widehat{g}(t) = e^{-(t-u)^2x/2}$, so that $g * \widetilde{g} = h_x e^{-iuy}$ and $\widehat{g * \widetilde{g}}(t) = e^{-(t-u)^2x}$.

PROPOSITION 17.5.1. *(i)* $\langle g,g\rangle = \mathcal{Y}(h_x e^{-iuy}) = \sum_\rho e^{-(\gamma_\rho - u)^2 x} = \mathcal{G}(u,\frac{x}{2}) + \mathcal{G}(-u,\frac{x}{2})$, *which is* $2W(x)$ *at* $u=0$. *(ii) Unconditionally,*

$$\mathcal{G}(u,\tfrac{x}{2}) + \mathcal{G}(-u,\tfrac{x}{2}) = 2e^{-(u^2-1/4)x}\cos(ux) - \frac{\log\pi}{\sqrt{4\pi x}} + \frac{1}{2\pi}\int_{\mathbb{R}} e^{-(t-u)^2 x}\,\operatorname{Re}\psi\Big(\frac{1}{4}+\frac{it}{2}\Big)dt - \frac{1}{\sqrt{\pi x}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\cos(u\log n)\,e^{-(\log n)^2/(4x)}.$$

*(iii) Under RH it equals* $\int \big[e^{-(\sqrt{t}-u)^2x} + e^{-(\sqrt{t}+u)^2x}\big]U(dt)$. *(iv) RH holds if and only if it is nonnegative for all real* $u$ *and all* $x>0$.

*Proof.* (i) The transform $e^{-(t-u)^2x}$ is entire, so (17.1.1) applies; the zero $1-\rho$ of the pair $\{\rho,1-\rho\}$ has $\gamma_{1-\rho} = -\gamma_\rho$ and gives $\mathcal{G}(-u,\frac{x}{2})$, and at $u=0$ the sum is $2W(x)$ by Proposition 17.2.1. (ii) Apply (17.1.1) to $F = h_x e^{-iuy}$: the pole term is $\widehat{F}(i/2)+\widehat{F}(-i/2) = e^{-(u+i/2)^2x} + e^{-(u-i/2)^2x} = 2e^{-(u^2-1/4)x}\cos(ux)$, the term $-(\log\pi)F(0)$ is $-(\log\pi)/\sqrt{4\pi x}$, $V_1(F)$ is the integral, and the prime sum is $-\sum\Lambda(n)n^{-1/2}h_x(\log n)\big(e^{-iu\log n}+e^{iu\log n}\big)$. This is (5.1.3) at $t=x/2$, term by term: its polar term $2e^{t/2-2tu^2}\cos 2tu$, its integral $A(u,t)$ (with $m(r) = \operatorname{Re}\psi(\frac14+\frac{ir}{2}) - \log\pi$) and its prime term $P(u,t)$ are the four terms above, because $2t=x$. (iii) Group $\pm\gamma$ in $\sum_{\gamma>0}\big[e^{-x(\gamma-u)^2}+e^{-x(\gamma+u)^2}\big]$ and use $U = \sum_{\gamma>0}\delta_{\gamma^2}$. (iv) This is Proposition 5.1.11, in the form of the remark after its proof: the term $\mathcal{G}(-u,t)$, which comes from the zeros $1-\rho$, is too small there to change the sign. □

The formula (ii) agrees with (5.1.3); Table 5.3 computes the same identity. A vector modulated to a centre $u$ within $\frac{\delta}{2}$ of the ordinate $\gamma$ of an off line zero $\rho = \frac12+\delta+i\gamma$, with finitely many exceptional centres, makes the form negative at arbitrarily large $x$ (Proposition 5.1.11), and $u$ is free: modulation reaches a zero above any verified height, which is why the barrier of Proposition 17.9.1 is stated for the tent coefficient only.

The ladder reaches the height $u$ only at order $k \approx u^2x$, because the weight $r^{2k}e^{-r^2x}$ of $T_k(x)$ peaks at $r=\sqrt{k/x}$ and the power $r^{2k}$ is its only way to move the window; Theorem 5.1.13 states the consequence, that verification to a height $H$ reaches the orders up to about the square of $H$. Modulation moves the window at no cost. Figure 17.1 shows the modulated heat trace resolving isolated zeros into bumps of height one at $x=0.5$; nearby zeros overlap, and zeros less than 2 apart merge into one bump (computed, script `yo_ladder.py`).




Figure 17.1: The modulated heat trace $\mathcal{G}(u, \frac{x}{2}) + \mathcal{G}(-u, \frac{x}{2})$ against the window centre $u$ at $x = 0.02$ (solid), 0.1 (dashed) and 0.5 (dotted), from the first 1700 zeros; dotted verticals mark the ordinates of the zeros.

## 17.6 The tent and the sine squared lemma

Yoshida's basic vector is $\chi_0 = (2a)^{-1/2} 1_{[-a,a]}$, and $\chi_0 * \widetilde{\chi_0} = (2a)^{-1}(2a - |y|)_+$. Put $x = 2a$,

$$\Delta_x(y) = (x - |y|)_+, \qquad D(x) = \mathcal{Y}(\Delta_x) = x\,\langle \chi_0, \chi_0 \rangle,$$

where $\chi_0$ is taken for $a = x/2$. So $\Delta_x$ is the tent of height $x$ on $[-x, x]$, and $D(x)$ is the single coefficient $\langle \chi_0, \chi_0 \rangle$ of Yoshida's matrix multiplied by the length $x$ of the window.

Proposition 17.6.1. *Let $k_x(w) = e^{-wx} + wx - 1$ be the compensated exponential on the left of the sine squared lemma* (1.2.9)*, and let $\gamma_\rho = -i(\rho - \frac{1}{2})$, which is complex for an off line zero.*

*(a) For complex $w$ and real $t$,*

$$\int_{\mathbb{R}} \Delta_x(y) e^{wy}\, dy = \frac{k_x(w) + k_x(-w)}{w^2} = \frac{4\sinh^2(wx/2)}{w^2},$$

$$\widehat{\Delta_x}(t) = \frac{4\sin^2(xt/2)}{t^2} \geq 0.$$

*(b) Unconditionally $D(x) = \sum_\rho 4\sin^2(\gamma_\rho x/2)/\gamma_\rho^2$, the sum over all nontrivial zeros. Under RH, with the Thorin measure $U = \sum_{\gamma>0} \delta_{\gamma^2}$ of Proposition 1.2.1,*

$$D(x) = 8 \int \frac{\sin^2(x\sqrt{t}/2)}{t}\, U(dt)$$

$$= \frac{8\pi}{x} \int \frac{x \sin^2(x\sqrt{t}/2)}{\pi\sqrt{t}}\, \frac{U(dt)}{\sqrt{t}},$$

*so $D$ samples the Thorin density of the sine squared lemma* (1.2.9) *at level $x$ on the Thorin atoms $t = \gamma^2$, with the weight $t^{-1/2}$.*

*(c) Unconditionally,*

$$D(x) = 32\sinh^2\frac{x}{4} - x\log\pi - 2\sum_{n\le e^x}\frac{\Lambda(n)}{\sqrt{n}}\log\frac{e^x}{n} + V_1(\Delta_x),$$

$$V_1(\Delta_x) = x - x\log x - (\gamma_E + \log 2)x \\ - 2\int_0^x \Big(\frac{e^{y/2}}{2\sinh y} - \frac{1}{2y}\Big)(x-y)\,dy,$$

*and the prime term is a Riesz mean of order one of $\Lambda(n)n^{-1/2}$.*

*(d) With $\psi$ the digamma function, $V_1(\Delta_x)$ is also the residue series*

$$V_1(\Delta_x) = x\,\psi(\tfrac14) + \tfrac12\psi'(\tfrac14) - 2\sum_{k\ge0}\frac{e^{-(2k+1/2)x}}{(2k+\frac12)^2}.$$

*Proof.* (a) Taylor's formula with integral remainder gives $k_x(w) = w^2\int_0^x (x-y)e^{-wy}dy$, so $\int \Delta_x e^{wy}dy = (k_x(w)+k_x(-w))/w^2$, and $k_x(w)+k_x(-w) = e^{wx} + e^{-wx} - 2 = 4\sinh^2(wx/2)$; put $w = it$.

(b) Evaluate $\widehat{\Delta_x}$ at $\gamma_\rho$ and sum over the zeros; the series converges absolutely, since $|\sin^2(x\gamma_\rho/2)| \le \cosh^2(x/4)$ for $|\operatorname{Im}\gamma_\rho| < \frac12$ and $\sum|\gamma_\rho|^{-2} < \infty$. Under RH group $\pm\gamma$, which gives $D(x) = 2\sum_{\gamma>0} 4\sin^2(\gamma x/2)/\gamma^2$.

(c) Insert $\Delta_x$ in (17.1.1). By (a) at $w = \pm\frac12$ the first term is $2\cdot 4\sinh^2(x/4)/(1/4) = 32\sinh^2(x/4)$, the second is $-(\log\pi)\Delta_x(0) = -x\log\pi$, and $\Delta_x(\log n) + \Delta_x(-\log n) = 2(x - \log n)_+$. For $V_1$ use the second form of (17.1.2), with $\operatorname{Pf}(1/|y|)(\Delta_x) = 2(x\log x - x)$ for all $x > 0$, because $\int_{\epsilon<|y|<x}(x-|y|)\,dy/|y| = 2x\log(x/\epsilon) - 2x + 2\epsilon$. To see that (17.1.1) holds for the tent, apply it to $\Delta_x * h_\eta$, with $h_\eta(y) = (4\pi\eta)^{-1/2}e^{-y^2/(4\eta)}$, whose transform $\widehat{\Delta_x}(t)e^{-\eta t^2}$ decays rapidly in every strip, and let $\eta\downarrow 0$. The zero sum converges by dominated convergence, since $|e^{-\eta\gamma_\rho^2}| \le e^{\eta/4}$ and the bound of (b) holds; the Fourier form of $V_1$ converges because $\widehat{\Delta_x}(t) = O(t^{-2})$; and the prime sum converges because $\Delta_x * h_\eta \to \Delta_x$ uniformly with Gaussian tails.

(d) By the Fourier form of (17.1.2), $V_1(\Delta_x) = \frac{1}{2\pi}\int \widehat{\Delta_x}(t)\psi(\frac14 + \frac{it}{2})\,dt$, because the imaginary part of $\psi(\frac14+\frac{it}{2})$ is odd in $t$ and $\widehat{\Delta_x}$ is even. With $w = it$ and (a) this is

$$V_1(\Delta_x) = \frac{1}{2\pi i}\int_{\operatorname{Re} w=0}\frac{e^{wx}+e^{-wx}-2}{w^2}\,\psi\big(\tfrac14+\tfrac{w}{2}\big)\,dw.$$

The integrand is regular at $w = 0$, so the line may be moved to $\operatorname{Re} w = -\frac14$. Split the numerator into its three terms, and close the contour to the left for $e^{wx}$

and to the right for $e^{-wx}$ and for $-2$; the arcs, which pass between the poles of $\psi$, contribute nothing in the limit, because the integrands are $O(|w|^{-2}\log|w|)$ on them. On the left, $\psi(\frac14+\frac{w}{2})$ has simple poles at $w=-(2k+\frac12)$ with residue $-2$, which gives $-2\sum_k e^{-(2k+1/2)x}/(2k+\frac12)^2$. On the right the double pole at $w=0$ gives, with the clockwise orientation, $x\psi(\frac14)-\frac12\psi'(\frac14)$ from $e^{-wx}$ and $\psi'(\frac14)$ from $-2$. □

The kernel $K(w,x)=\int\Delta_x(y)e^{wy}dy=(k_x(w)+k_x(-w))/w^2$ of (a) has two readings. The sine squared lemma fixes $w$ and sums over the level $x$ against the canonical measure of Section 5.3, built from $\Lambda(n)n^{-1/2}$ at $x=\log n$, the gamma factor and the pole; Yoshida fixes the level $x$ and sums over the zeros.

*The tent in the variogram form.* Put $\varphi_x(z)=\widehat{\Delta_x}(\sqrt z)=4\sin^2(x\sqrt z/2)/z$. By (b) and (c), $D(x)=2\langle U,\varphi_x\rangle$, where $\langle U,\cdot\rangle$ is the Thorin functional of Section 5.3.1, and (c) is twice (5.3.1) for $\varphi_x$. The tent lies just outside the class $\mathcal T$ of that section, because $\widehat{\Delta_x}(r)=O(r^{-2})$, and the limiting argument in the proof of (c) takes the place of the decay condition. In the variogram form of Weil's criterion (Section 3.5.2) the tent is the function $g=\Delta_x$ of (3.5.2): its transform $h=\widehat{\Delta_x}$ is nonnegative, and its variogram is $g(0)-g(u)=\min(u,x)$ for $u\ge0$,

$$\int_0^\infty\varphi_x(z)\,\frac{2\sin^2(u\sqrt z/2)}{2\pi\sqrt z}\,dz=\min(u,x),$$

since both sides are $\pi^{-1}\int_0^\infty\widehat{\Delta_x}(r)(1-\cos ur)\,dr$, and (3.5.2) needs only $h\in L^1$. Theorem 17.6.2 below is therefore Weil's criterion restricted to the one parameter family of variograms $\min(u,x)$, $x>0$, of tents.

THEOREM 17.6.2. *RH holds if and only if $D(x)\ge0$ for all $x>0$, if and only if $D$ is bounded below. Equivalently, RH holds if and only if the single coefficient $\langle\chi_0,\chi_0\rangle$ is nonnegative for every $a>0$.*

*Proof.* Under RH every term in (b) is nonnegative. Conversely put $\omega(x)=\sum_\rho e^{(\rho-1/2)x}/(\rho-\frac12)^2$, which converges absolutely. Using $\rho\mapsto1-\rho$ and $(\rho-\frac12)^2=-\gamma_\rho^2$ we get $D=2\omega+4p_1$, where $p_1=\sum_k a_k^{-1}=\frac12\sum_\rho\gamma_\rho^{-2}$ is the mean of the clock (Section 1.2). For $\operatorname{Re}s>\frac12$,

$$\int_0^\infty\omega(x)e^{-sx}dx=R(s):=\sum_\rho\frac{1}{(\rho-\frac12)^2\,(s-\rho+\frac12)},$$

which is meromorphic on $\mathbb C$ with poles only at the points $\rho-\frac12$, and with the nonzero residues $m/(\rho-\frac12)^2$, $m$ the multiplicity. If $D\ge-C$, then $f=\omega+2p_1+C/2\ge0$ has the Laplace transform $R(s)+(2p_1+C/2)/s$. By Landau's theorem [514, Chapter II, Theorem 5b] this transform is singular at the real point of its abscissa of convergence $\sigma_c$. No $\rho-\frac12$ is real, because $\xi$ has no real zero, so the

only real singularity is 0, $\sigma_c \le 0$, and the transform is holomorphic in $\operatorname{Re} s > 0$; hence no zero has $\operatorname{Re}(\rho - \frac{1}{2}) > 0$, and by the symmetry $\rho \mapsto 1-\rho$ none has $\operatorname{Re}(\rho - \frac{1}{2}) < 0$. □

The equivalence is classical in substance (Landau's theorem applied to a Riesz mean, as in Ingham [283]); its interest is its place in Yoshida's matrix and in the lemma. Yoshida's Theorem 1 gives $D > 0$ on $(0, \log 2]$ (Section 17.1), where no prime power contributes to (c), and $\langle \chi_0, \chi_0 \rangle = D(\log 2)/\log 2 = 0.18338$ at $a = \frac{1}{2}\log 2$. Computed from the prime side (c), which is explicit between consecutive prime powers, on a grid of step $10^{-4}$, $D(x) \ge 0.0550$ on $[0.05, 16]$, with the minimum at $x = 8.04$, and the mean of $D$ on $[2, 16]$ is 0.092409 against $4p_1 = 0.092420$. The difference $-1.07 \times 10^{-5}$ is the mean of the oscillating part $2\omega$: it equals $-\frac{4}{14}\sum_{\gamma>0}(\sin 16\gamma - \sin 2\gamma)/\gamma^3$, which the first 1700 zeros give to the same digits (script `yo_tent.py`, `code/README.md`).

Remark 17.6.3 (Detection scale). A hypothetical off line zero $\rho = \frac{1}{2} + \delta + i\gamma$ with its three companions adds to $D$ the term $4\operatorname{Re}\big[4\sin^2\big((\gamma - i\delta)x/2\big)/(\gamma - i\delta)^2\big]$, whose amplitude is about $4e^{\delta x}/\gamma^2$ with oscillating sign. Against the mean level $4p_1$ of $D$ this makes $D$ first turn negative near $x_* \approx (2\log\gamma + \log p_1)/\delta$. For $(\gamma, \delta) = (100, 0.3)$, $(100, 0.2)$, $(1000, 0.3)$ the first negative values of $D$ plus the pair term occur at $x = 17.3$, 25.4, 32.4, against the predictions 18.1, 27.2, 33.5 (Figure 17.2); the predictions are upper estimates. On a grid of step 0.01 the third value is 32.83 and not 32.4, because at $\gamma = 1000$ the pair term has period $2\pi/\gamma = 0.0063$, shorter than the step. A negative value of the single coefficient $D(2a)$ shows that the form is not positive on $C(a)$, so Yoshida's threshold $a_0$ of Section 17.1 is at most about $x_*/2$, logarithmic in the height. This is a bound and not an estimate, because the tent is one vector of the window: a vector that vanishes at the on line ordinates near $\gamma$ sees the pair earlier, and Table 18.3 of Chapter 18 gives, for the full form on a window, detection at half lengths far below $x_*/2$ in the cases computed there (a heuristic experiment of that chapter, with its own cutoff).

Table 17.1 compares the two sides of Proposition 17.6.1. The prime side is (c) with the exact $\Lambda(n)$. The zero side is $\sum_{\gamma>0} 8\sin^2(\gamma x/2)/\gamma^2$ over the first 1500 zeros (to height 1980.91) plus the mean tail $2(\log(T/2\pi)+1)/(\pi T)$ at $T = \gamma_{1500}$. The two sides agree to $3.6 \times 10^{-6}$ at every level of the table except $x = \log 2$ and $x = \log 5$, where they differ by $1.6 \times 10^{-4}$ and $2.3 \times 10^{-4}$. These are the corners of the prime side at $x = \log n$, where its slope drops by $2\Lambda(n)/\sqrt{n}$, which a zero sum cut at $T$ and completed by a mean tail cannot follow: by Landau's formula [325] for $\sum_{0<\gamma\le T} n^{i\gamma}$ the zeros above $T$ contribute $2\Lambda(n)/(\pi\sqrt{n}\,T)$ at $x = \log n$, which is $1.575 \times 10^{-4}$ and $2.313 \times 10^{-4}$ at $T = \gamma_{1500}$, and with it the two corner rows agree to $1.3 \times 10^{-6}$. The last two columns agree to $10^{-29}$ in every row (eight decimals are printed). The values come from the script `yo_explicit.py` (`code/README.md`).

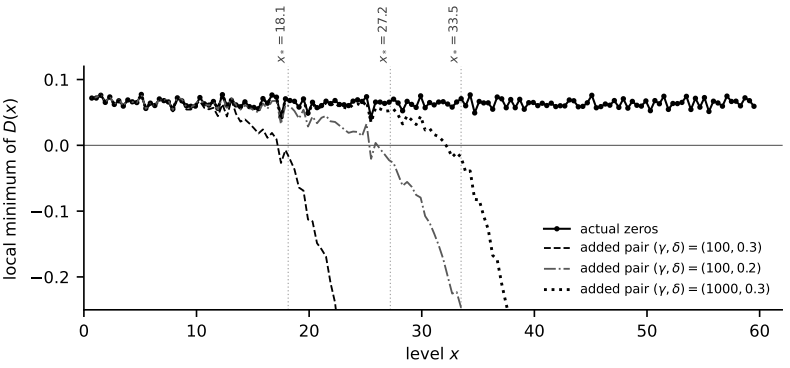



Figure 17.2: The least value of $D$ over windows of length 0.4 on $[0, 60]$, for the actual zeros (solid, the first 1700 zeros with the smooth tail; the least window minimum is 0.0427, in the window centred at $x = 25.5$) and with one hypothetical off line pair added at height $\gamma$ and displacement $\delta$, $(\gamma, \delta) = (100, 0.3)$ (dashed), $(100, 0.2)$ (dash dotted) and $(1000, 0.3)$ (dotted). Vertical dotted lines mark the predicted levels $x_* = 18.1$, 27.2 and 33.5 of Remark 17.6.3. Script `yo_tent.py`.

| $x$ | prime side | zero side | $V_1(\Delta_x)$ by (c) | by (d) |
|---|---|---|---|---|
| 0.2 | 0.1089996 | 0.1089981 | 0.25787887 | 0.25787887 |
| 0.5 | 0.0803252 | 0.0803270 | 0.15008051 | 0.15008051 |
| $\log 2$ | 0.1271112 | 0.1269547 | −0.04998528 | −0.04998528 |
| 1.0 | 0.0880146 | 0.0880147 | −0.50847597 | −0.50847597 |
| 1.5 | 0.0898189 | 0.0898186 | −1.52909231 | −1.52909231 |
| $\log 5$ | 0.1343264 | 0.1340964 | −1.78866457 | −1.78866457 |
| 2.0 | 0.1066822 | 0.1066850 | −2.80144646 | −2.80144646 |
| 3.0 | 0.0794186 | 0.0794176 | −5.86891443 | −5.86891443 |
| 4.0 | 0.0702819 | 0.0702855 | −9.39384635 | −9.39384635 |
| 6.0 | 0.0962464 | 0.0962437 | −17.16435327 | −17.16435327 |

Table 17.1: The tent value $D(x)$ from the primes (Proposition 17.6.1(c)) and from the zeros, and $V_1(\Delta_x)$ by the closed form (c) and by the residue series (d). The rows $x = \log 2$ and $x = \log 5$ differ by Landau's term, see the text.

## 17.7 The sine squared lemma off the diagonal: Yoshida's matrix is a Löwner matrix

Recall Yoshida's basis $\chi_n = (2a)^{-1/2} e^{\pi i n y/a} 1_{[-a,a]}$ of $K(a)$ (Section 17.1, [525]), and write $\nu_n = \pi n/a$. The Fourier transform of the basis vector is

$$\widehat{\chi_n}(t) = (2a)^{-1/2} \int_{-a}^{a} e^{i(t+\nu_n)y}\, dy = (2a)^{-1/2}\, \frac{2(-1)^n \sin(at)}{t + \nu_n},$$

because $\sin(a(t+\nu_n)) = (-1)^n \sin(at)$. Every basis vector carries the same factor $\sin(at)$; only the pole moves. The Löwner matrix of a function $g$ on the nodes $\nu_n$ has the entries $(g(\nu_n) - g(\nu_m))/(\nu_n - \nu_m)$ off the diagonal and $g'(\nu_n)$ on it, and a Pick function is a function holomorphic on the upper half plane with nonnegative imaginary part there.

THEOREM 17.7.1. *(i) Unconditionally,*

$$\langle \chi_n, \chi_m \rangle = (-1)^{n+m} \frac{2}{a} \sum_\rho \frac{\sin^2(a\gamma_\rho)}{(\gamma_\rho + \nu_n)(\gamma_\rho + \nu_m)},$$

*an absolutely convergent series.*

*(ii) Let $g_a(\nu) = \sum_\rho \sin^2(a\gamma_\rho)/(\gamma_\rho - \nu)$, the sum over all zeros taken in symmetric pairs; unconditionally it is an odd meromorphic function of $\nu$. Then for $n \neq m$*

$$\langle \chi_n, \chi_m \rangle = (-1)^{n+m} \frac{2}{a} \frac{g_a(\nu_n) - g_a(\nu_m)}{\nu_n - \nu_m}, \qquad \langle \chi_n, \chi_n \rangle = \frac{2}{a} g_a'(\nu_n),$$

*so Yoshida's matrix is, up to the sign change* $\operatorname{diag}((-1)^n)$, *$\frac{2}{a}$ times the Löwner matrix of $g_a$ on the lattice $\frac{\pi}{a}\mathbb{Z}$.*

*(iii) Under RH, with the weighted Thorin measure $U_a^\sharp(dt) = 2\sin^2(a\sqrt{t})\, U(dt) \geq 0$ and its Stieltjes transform $\mathcal{S}_a(s) = \int U_a^\sharp(dt)/(t+s)$,*

$$g_a(\nu) = \nu\, \mathcal{S}_a(-\nu^2),$$
$$\langle \chi_n, \chi_m \rangle = (-1)^{n+m} \frac{2}{a} \int \frac{2\sin^2(a\sqrt{t})\,(t + \nu_n\nu_m)}{(t-\nu_n^2)(t-\nu_m^2)}\, U(dt),$$

*and $g_a$ is an odd Pick function. By the elementary half of Löwner's theorem every finite Yoshida matrix is then positive semidefinite, for every $a$.*

*(iv) For each fixed $a > 0$, RH holds if and only if $g_a$ is a Pick function.*

*Proof.* (i) $F = \chi_n * \widetilde{\chi_m}$ has $\widehat{F}(t) = \widehat{\chi_n}(t)\overline{\widehat{\chi_m}(\bar{t})} = (-1)^{n+m}\frac{2}{a}\sin^2(at)/((t+\nu_n)(t+\nu_m))$, entire because $\sin^2(at)$ vanishes at $-\nu_n$ and $-\nu_m$. It is $O(t^{-2})$ in every strip, and $F$ is continuous, piecewise smooth and compactly supported, so (17.1.1) applies to it as it does to the tent (proof of Proposition 17.6.1(c)). Sum $\widehat{F}(\gamma_\rho)$.

(ii) Replace $\gamma_\rho$ by $-\gamma_\rho$ (the zero set is symmetric), then use $\frac{1}{(\gamma-\nu_n)(\gamma-\nu_m)} = \frac{1}{\nu_n-\nu_m}\big(\frac{1}{\gamma-\nu_n} - \frac{1}{\gamma-\nu_m}\big)$; the diagonal is the limit. The series of $g_a$ converges in symmetric pairs, since the pair $\pm\gamma_\rho$ contributes $2\nu\sin^2(a\gamma_\rho)/(\gamma_\rho^2 - \nu^2) = O(|\nu|\,|\gamma_\rho|^{-2})$, and $g_a(-\nu) = -g_a(\nu)$ because the zero set is symmetric. A zero with $\gamma_\rho = \nu_n$ contributes nothing at the lattice, since $\sin^2(a\gamma_\rho) = 0$.

(iii) Under RH group $\pm\gamma$: $\sin^2(a\gamma)\big(\frac{1}{\gamma-\nu}-\frac{1}{\gamma+\nu}\big)=\frac{2\nu\sin^2(a\gamma)}{\gamma^2-\nu^2}$, and with $t=\gamma^2$ this is $\nu\int U_a^\sharp(dt)/(t-\nu^2)$. In the same way the pair $\pm\gamma$ in (i) gives $2\sin^2(a\gamma)(\gamma^2+\nu_n\nu_m)/((\gamma^2-\nu_n^2)(\gamma^2-\nu_m^2))$, which is the second formula. Each atom $\nu/(t-\nu^2)$ has imaginary part $\operatorname{Im}\nu\,(t+|\nu|^2)/|t-\nu^2|^2>0$ for $\operatorname{Im}\nu>0$, so $g_a$ is a Pick function. For a Pick function $g$ and real nodes $\nu_1,\dots,\nu_N$, the matrix $[(g(\nu_i)-g(\nu_j))/(\nu_i-\nu_j)]$ is positive semidefinite, which is Löwner's easy direction [349, 120].

(iv) If RH fails, an off line pair gives the nonreal ordinates $\pm\gamma\pm i\delta$, two of them in the upper half plane, where $g_a$ has a pole with residue $-m\sin^2(a\gamma_\rho)$, $m$ the multiplicity. This residue is nonzero because sin vanishes only on the real axis, and a Pick function is holomorphic in the upper half plane. The converse is (iii). □

The mechanism of the theorem is that of the rigidity of the clock, Proposition 1.2.1: there $G'/G$ is the Stieltjes transform of $U$, holomorphic off the negative real axis exactly when every zero is on the line, and $\mathcal{S}_a$ is the same transform with the atoms weighted by $2\sin^2(a\gamma)$, sampled on the cut at the squared modal frequencies $\nu_n^2$ and differenced.

Numerically, the prime side of $\langle\chi_n,\chi_m\rangle$, computed from (17.1.1) and (17.1.2) by Gauss Legendre quadrature, gives the coefficient $\langle\chi_0,\chi_0\rangle=0.18338$ of Yoshida's matrix at $a=\frac12\log 2$, and at $a=1$ it matches part (i) summed over the first 1500 zeros, entry by entry, up to the zero tail. The tail of the smooth count from $T=1982.0$ accounts for the difference up to $9\times10^{-7}$ (`yo_explicit.py`, `code/README.md`).

## 17.8 Basepoints: damped forms, multipliers and sharp coercivity

The Thorin programme of this book leaves the critical line for a basepoint $\alpha>\frac12$, where the Thorin density of the shifted clock is $\rho_{\alpha-1/2}=\pi^{-1}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$ (equation (1.2.7)). On Yoshida's side the same move damps the autocorrelation $\varphi*\widetilde{\varphi}$ by $e^{-(\alpha-1/2)|y|}$.

### 17.8.1 The multiplier identity

For a basepoint $\alpha\ge\frac12$ put $\varepsilon=\alpha-\frac12$ and define the damped Yoshida form

$$\langle\varphi_1,\varphi_2\rangle_\alpha=\mathcal{Y}\big(e^{-\varepsilon|y|}\,\varphi_1*\widetilde{\varphi_2}\big),$$

where $\mathcal{Y}$ is the functional of (17.1.1) and $\widetilde{\varphi}(y)=\overline{\varphi(-y)}$; at $\alpha=\frac12$ it is the form of Section 17.1, and $C(a)\subset K(a)$ are the spaces of that section. On the prime side the damping replaces $\Lambda(n)n^{-1/2}$ by $\Lambda(n)n^{-\alpha}$, and in the pole term $e^{y/2}+e^{-y/2}$ becomes $e^{(1-\alpha)|y|}+e^{-\alpha|y|}$; on $K(a)$ the prime sum is still finite, since only $\log n\le 2a$ occurs. Let $\sigma_{\max}=\sup_\rho\operatorname{Re}\rho\in[\frac12,1]$, let $v_\alpha(\tau)=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$

and $b_\alpha = v_\alpha(0) = \xi'(\alpha)/\xi(\alpha)$, so that $\rho_\varepsilon = \pi^{-1} v_{1/2+\varepsilon}$ is the density (1.2.7), and write $P_w(\tau) = w/(\pi(w^2+\tau^2))$ for the Poisson kernel of width $w$, with $P_{-w} = -P_w$, as in Proposition 1.2.3.

PROPOSITION 17.8.1. *Let $\alpha \geq \sigma_{\max}$, and suppose that no zero has real part $\alpha$. Then $\rho_\varepsilon$ is real analytic and positive, $(1+\tau^2)^{-1}\rho_\varepsilon$ is integrable over $\mathbb{R}$, and for every $a > 0$ and every $\varphi \in K(a)$*

$$\langle \varphi, \varphi \rangle_\alpha = \int_{\mathbb{R}} |\widehat{\varphi}(\tau)|^2 \, \rho_\varepsilon(\tau)\, d\tau = \frac{1}{\pi} \int_{\mathbb{R}} |\widehat{\varphi}(\tau)|^2 \, v_\alpha(\tau)\, d\tau.$$

*Proof.* For $w > 0$ the kernel $P_w$ has Fourier transform $\int P_w(\tau) e^{i\tau y}\, d\tau = e^{-w|y|}$, so for a bounded $F$ with compact support and $\gamma \in \mathbb{R}$, $\int \widehat{F}(\tau) P_w(\tau - \gamma)\, d\tau = \int F(y) e^{-w|y|} e^{i\gamma y}\, dy$. Let $F = \varphi * \widetilde{\varphi}$, which is continuous, piecewise smooth and supported in $[-2a, 2a]$, so that (17.1.1) applies to $e^{-\varepsilon|y|}F$ and

$$\mathcal{Y}(e^{-\varepsilon|y|}F) = \sum_\rho \int F(y) e^{-\varepsilon|y|} e^{i\gamma_\rho y}\, dy,$$

the series converging absolutely. Group the zeros in fours $\{\rho, \bar\rho, 1-\rho, 1-\bar\rho\}$, counted with multiplicity; for a zero on the critical line the group is $\{\rho, \bar\rho\}$. For the four zeros $\frac12 \pm \delta \pm i\gamma$ with $\gamma > 0$ and $\delta > 0$ we have $e^{i\gamma_\rho y} = e^{\pm i\gamma y} e^{\pm \delta y}$, so their contributions add up to

$$\begin{aligned} &\int F(y) e^{-\varepsilon|y|} \big(e^{i\gamma y} + e^{-i\gamma y}\big)\big(e^{\delta y} + e^{-\delta y}\big) dy \\ &\quad = \int F(y) \big(e^{-(\varepsilon-\delta)|y|} + e^{-(\varepsilon+\delta)|y|}\big)\big(e^{i\gamma y} + e^{-i\gamma y}\big) dy, \end{aligned}$$

because $e^{\delta y} + e^{-\delta y}$ is even. The widths $\varepsilon - \delta = \alpha - \beta$ and $\varepsilon + \delta = \alpha - (1-\beta)$, with $\beta = \frac12 + \delta$, are the differences between $\alpha$ and the real parts of the zeros of the group, and they are positive, since every zero has real part less than $\alpha$. By the identity for $P_w$ the group contributes $\int \widehat{F}(\tau) \sum P_{\alpha-\beta_{\rho'}}(\tau - \operatorname{Im}\rho')\, d\tau$, the sum over its four members $\rho'$ with real part $\beta_{\rho'}$; the same holds for a zero on the line, with the single width $\varepsilon$. Hence $\mathcal{Y}(e^{-\varepsilon|y|}F) = \sum_\rho \int \widehat{F}(\tau) P_{\alpha-\beta_\rho}(\tau - \operatorname{Im}\rho)\, d\tau$ over all zeros, where $\beta_\rho = \operatorname{Re}\rho$. Now $\widehat{F} = |\widehat{\varphi}|^2 \geq 0$ on $\mathbb{R}$, so every term is nonnegative and the sum and the integral may be interchanged. By Proposition 1.2.3, which is Hadamard's product, $\sum_\rho P_{\alpha-\beta_\rho}(\tau - \operatorname{Im}\rho) = \pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau) = \rho_\varepsilon(\tau)$, a sum of positive kernels; it is real analytic because no zero lies on the line $\operatorname{Re} s = \alpha$. Finally $\int P_w(\tau - t)(1+\tau^2)^{-1} d\tau = (1+w)/((1+w)^2 + t^2) \leq (1+\alpha)/(1+t^2)$ for $0 < w \leq \alpha$, so by Tonelli $\int (1+\tau^2)^{-1} \rho_\varepsilon\, d\tau \leq (1+\alpha) \sum_\rho (1 + (\operatorname{Im}\rho)^2)^{-1} < \infty$, because the number of zeros with $|\operatorname{Im}\rho| \leq T$ is $O(T \log T)$. The proof used only that $F = \varphi * \widetilde{\varphi}$ is continuous, piecewise $C^1$ and supported in $[-2a, 2a]$, so the identity holds as well for every bounded piecewise $C^1$ function $\varphi$ supported in $[-a, a]$. □

For the Fejér taper $\varphi = e^{ity}\chi_0$ on the window $a = X/2$ one has $\varphi * \widetilde{\varphi} = e^{ity}(1-|y|/X)_+$, and since $\mathcal{Y}(F)$ depends only on the even part of $F$, $\langle\varphi,\varphi\rangle_\alpha = \mathcal{Y}\big(e^{-\varepsilon|y|}\cos(ty)(1-|y|/X)_+\big) = 2A_X(\varepsilon,t)$ in the notation of Proposition 14.14.1, whose part (c), $A_X(\varepsilon,t) \to v_\alpha(t)$ as $X\to\infty$ when $\alpha > \sigma_{\max}$, so that every zero has $|\beta - \frac12| < \varepsilon$, is the concentration argument in the proof of Theorem 17.8.5(i) below for this family of test functions.

Remark 17.8.2. For $\alpha < \sigma_{\max}$ the two sides of the identity differ: a zero with $\beta > \alpha$ contributes the growing factor $e^{(\beta-\alpha)|y|}$ to the damped Weil form, but the negative kernel $P_{\alpha-\beta} = -P_{\beta-\alpha}$ to the symbol. The script `yo_damped.py` checks the identity at $\alpha = 1$, $a = 0.8$ and $\varphi = \chi_0$ (data in `yo_damped.json`, `code/README.md`).

Theorem 17.8.3. *The damped forms have the following properties.*

*(i) For $\alpha \ge \sigma_{\max}$ the damped form is positive semidefinite on $C_c^\infty(\mathbb{R})$; unconditionally this holds for every $\alpha \ge 1$.*

*(ii) RH holds if and only if the damped form is positive semidefinite for every $\alpha > \frac12$.*

*(iii) If the damped form at $\alpha$ is positive semidefinite on $C(a)$, so is the form at every $\alpha' > \alpha$; thus $a_0(\alpha) = \sup\{a : \langle\cdot,\cdot\rangle_\alpha \ge 0 \text{ on } C(a)\}$ is nondecreasing in $\alpha$.*

*Proof.* (i) If no zero has real part $\alpha$ this is Proposition 17.8.1, because $\rho_\varepsilon \ge 0$ and every $\varphi \in C_c^\infty(\mathbb{R})$ lies in some $K(a)$. If a zero has real part $\alpha$, then $\alpha = \sigma_{\max}$ and $\langle\varphi,\varphi\rangle_\alpha = \lim_{\alpha'\downarrow\alpha}\langle\varphi,\varphi\rangle_{\alpha'} \ge 0$, since for fixed $\varphi$ each part of the form is continuous in $\varepsilon$: the pole term and the finite prime sum obviously, and $V_1(e^{-\varepsilon|y|}F)$ because $\widehat{e^{-\varepsilon|y|}F} = P_\varepsilon * \widehat{F}$ with $|\widehat{F}(t)| \le C(1+t^2)^{-1}$, which gives a bound $C'(1+t^2)^{-1}$ uniform in $\varepsilon \le 1$, while $\operatorname{Re}\psi(\frac14 + \frac{it}{2}) = O(\log(2+|t|))$, and dominated convergence applies to the Fourier form of $V_1$ in (17.1.2). Unconditionally $\sigma_{\max} \le 1$ and no zero has real part 1, so every $\alpha \ge 1$ is covered.

(ii) Under RH, $\sigma_{\max} = \frac12$ and (i) applies to every $\alpha > \frac12$. Conversely, positivity for a sequence $\alpha_n \downarrow \frac12$ gives $\langle\varphi,\varphi\rangle \ge 0$ for every $\varphi \in C_c^\infty(\mathbb{R})$ by the same continuity, which is Weil's criterion (Section 17.1).

(iii) Write $e^{-\varepsilon'|y|} = \int P_{\varepsilon'}(\lambda)e^{i\lambda y}\,d\lambda$ with $\varepsilon' = \alpha' - \alpha$. Since $(\varphi_\lambda * \widetilde{\varphi_\lambda})(y) = e^{i\lambda y}(\varphi * \widetilde{\varphi})(y)$ for $\varphi_\lambda = e^{i\lambda y}\varphi \in C(a)$, we get $\langle\varphi,\varphi\rangle_{\alpha'} = \int P_{\varepsilon'}(\lambda)\langle\varphi_\lambda,\varphi_\lambda\rangle_\alpha\,d\lambda$ (Fubini, as in (i)), which is nonnegative when the form at $\alpha$ is positive semidefinite on $C(a)$. □

Remark 17.8.4 (One basepoint). Part (ii) uses every $\alpha > \frac12$. The converse of (i) holds at a single basepoint: if $\frac12 \le \alpha \le 1$ and $\langle\varphi,\varphi\rangle_\alpha \ge 0$ for every $\varphi \in C_c^\infty(\mathbb{R})$, then no zero has real part greater than $\alpha$. This is Theorem 20.1.2 of Chapter 20. With (i), the damped form at $\alpha \in [\frac12, 1]$ is positive semidefinite if and only if

$\sigma_{\max} \le \alpha$. A statement about the symbol on one line is weaker: positivity of $v_\alpha$ on the line $\operatorname{Re} s = \alpha$ is not known to exclude zeros to its right (Remark 1.2.4).

### 17.8.2 Sharp coercivity

For $a > 0$ let

$$\lambda_{\min}(\alpha, a) = \inf\Big\{ \frac{\langle \varphi, \varphi \rangle_\alpha}{\|\varphi\|^2} : \ 0 \ne \varphi \in K(a) \Big\},$$

the least eigenvalue of the damped form on the window $[-a, a]$, where $\|\cdot\|$ is the norm of $L^2(\mathbb{R})$ and the test functions are complex valued. The matrix of the form on the basis $\chi_n$, $|n| \le N$, is the form restricted to a subspace of $K(a)$, so its least eigenvalue is an upper bound for $\lambda_{\min}(\alpha, a)$: these are the finite sections of Table 17.2. Under the hypotheses of Proposition 17.8.1 the infimum over the smaller space $C(a)$ is the same. Indeed, let $\varphi \in K(a)$ and let $\psi_\delta \in C_c^\infty(-a, a)$ satisfy $0 \le \psi_\delta \le 1$ and $\psi_\delta = 1$ on $[-a + \delta, a - \delta]$, with $\psi_\delta$ monotone on each of the two end intervals. Then $\eta_\delta = \varphi(1 - \psi_\delta)$ lies in $K(a)$, has total variation bounded independently of $\delta$ and $L^1$ norm at most $2\delta\|\varphi\|_\infty$, so that $|\widehat{\eta_\delta}(\tau)| \le C \min(\delta, |\tau|^{-1})$, and $\langle \eta_\delta, \eta_\delta \rangle_\alpha = \int |\widehat{\eta_\delta}|^2 \rho_\varepsilon \to 0$ as $\delta \downarrow 0$ by dominated convergence, since $(1 + \tau^2)^{-1} \rho_\varepsilon$ is integrable. The form being positive semidefinite on $K(a)$, the Rayleigh quotients of $\varphi\psi_\delta \in C(a)$ tend to that of $\varphi$. The argument applies verbatim to the functions supported in $[-a, a]$ that are smooth on $[-a, a]$, to which Proposition 17.8.1 extends, so the infimum is the same over that larger class; it contains the Legendre polynomials on the window used in Chapter 20, whose finite sections are therefore upper bounds as well. Put $m_\alpha = \inf_{\tau \in \mathbb{R}} v_\alpha(\tau)$.

Theorem 17.8.5. *Let $V_\xi = (\log \xi)''(\frac{1}{2})$ and $p_1 = \frac{1}{2} V_\xi = 0.0231049931\ldots$, which is $\sum_{\gamma > 0} \gamma^{-2}$ under RH, and, under RH, let $p_2 = \sum_{\gamma > 0} \gamma^{-4} = 3.71726\ldots \times 10^{-5}$.*

*(i) If $\alpha \ge \sigma_{\max}$ and no zero has real part $\alpha$, then $\lambda_{\min}(\alpha, a) \ge 2m_\alpha$ for every $a > 0$, $\lambda_{\min}(\alpha, a)$ is nonincreasing in $a$, and $\lambda_{\min}(\alpha, a) \to 2m_\alpha$ as $a \to \infty$.*

*(ii) Under RH, $m_\alpha = b_\alpha = \xi'(\alpha)/\xi(\alpha)$ for every $\alpha > \frac{1}{2}$, and $v_\alpha(\tau) > b_\alpha$ for $\tau \ne 0$: the minimum of $\operatorname{Re} \xi'/\xi$ on each vertical line lies on the real axis, and only there. For $\alpha \ge 1$ this holds with no hypothesis (Proposition 9.1.12, and Proposition 9.1.11 for $\alpha = 1$); for $\frac{1}{2} < \alpha < 1$ it holds with no hypothesis for $0 < |\tau| \le 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)); and RH is equivalent to it on every line $\frac{1}{2} < \alpha < 1$ (Theorem 9.1.18(vi), which sharpens Lagarias's criterion [320]).*

*(iii) Hence RH holds if and only if*

$$\langle \varphi, \varphi \rangle_\alpha \ \ge \ 2b_\alpha \, \|\varphi\|^2 \qquad \text{for all } \alpha \in (\tfrac{1}{2}, 1],\ a > 0,\ \varphi \in K(a),$$

*and under RH the constant is sharp: $\lambda_{\min}(\alpha, a) \downarrow 2b_\alpha$ as $a \to \infty$.*

(iv) *For every* $\alpha > \frac12$, *with no hypothesis,* $2b_\alpha \le 4p_1(\alpha-\frac12)$; *under RH also* $4p_1(\alpha-\frac12) - 4p_2(\alpha-\frac12)^3 \le 2b_\alpha$. *The coercivity constant vanishes linearly at* $\alpha = \frac12$.

*Proof.* (i) By Proposition 17.8.1 and Plancherel, $\int|\widehat{\varphi}|^2 = 2\pi\|\varphi\|^2$, we have, for $\varphi \in K(a)$,

$$\langle\varphi,\varphi\rangle_\alpha = \pi^{-1}\int|\widehat{\varphi}|^2 v_\alpha \ge \pi^{-1}m_\alpha\int|\widehat{\varphi}|^2 = 2m_\alpha\|\varphi\|^2.$$

Monotonicity holds because $\lambda_{\min}(\alpha,a)$ is also the infimum over $C(a)$ and $C(a)\subset C(a')$ for $a<a'$. For the limit, fix $\tau_0$ and $\psi\in C_c^\infty(-1,1)$ with $\|\psi\|=1$, and let $\varphi_a(y) = a^{-1/2}e^{-i\tau_0 y}\psi(y/a)\in C(a)$. Then $\|\varphi_a\|=1$, $|\widehat{\varphi_a}(\tau)|^2 = a\,|\widehat{\psi}(a(\tau-\tau_0))|^2$, and $|\widehat{\varphi_a}|^2/2\pi$ is a probability density that concentrates at $\tau_0$ as $a\to\infty$. The function $v_\alpha$ is continuous and $(1+\tau^2)^{-1}v_\alpha$ is integrable (Proposition 17.8.1), and $\widehat{\psi}$ is a Schwartz function, so $\langle\varphi_a,\varphi_a\rangle_\alpha = \pi^{-1}\int|\widehat{\varphi_a}|^2 v_\alpha \to 2v_\alpha(\tau_0)$. Hence $\limsup_a \lambda_{\min}(\alpha,a)\le 2v_\alpha(\tau_0)$ for every $\tau_0$, that is, $\le 2m_\alpha$.

(ii) Under RH the whole of (ii) is in the book, at the places cited; for $\alpha\ge1$ no hypothesis is needed.

(iii) Under RH, (i) and (ii) give the inequality and its sharpness. Conversely, $b_\alpha>0$ for every $\alpha>\frac12$ with no hypothesis (Theorem 9.1.18(i)), so the inequality gives positive semidefiniteness for $\alpha\in(\frac12,1]$, Theorem 17.8.3(i) covers $\alpha>1$, and part (ii) of that theorem gives RH.

(iv) The upper bound is Theorem 9.1.18(i), $b_\alpha \le (\alpha-\frac12)V_\xi$, with $V_\xi = 2p_1$. Under RH, $b_\alpha = \sum_{\gamma>0} 2\varepsilon/(\varepsilon^2+\gamma^2)$ and $\gamma^{-2} - \varepsilon^2\gamma^{-4} \le (\varepsilon^2+\gamma^2)^{-1} \le \gamma^{-2}$. □

Remark 17.8.6. At a single $\alpha\in(\frac12,1)$ with no zero on the line, the inequality $\langle\varphi,\varphi\rangle_\alpha \ge 2b_\alpha\|\varphi\|^2$ for all $a$ and $\varphi$ holds if and only if $\sigma_{\max}\le\alpha$ and $v_\alpha\ge b_\alpha$ on the line: if it holds, the form is positive semidefinite (as $b_\alpha>0$), so $\sigma_{\max}\le\alpha$ by Remark 17.8.4, and then (i) gives $2m_\alpha\ge 2b_\alpha$, while $m_\alpha\le v_\alpha(0)=b_\alpha$; the converse is (i).

Lemma 17.8.7. *Consecutive ordinates of the zeros of* $\zeta$ *differ by at most* 6.8874. *Hence every real* $\tau$ *with* $|\tau|\ge\gamma_1$ *lies within* 3.4437 *of an ordinate.*

*Proof.* The lemma needs no hypothesis on the zeros. The first 1800 zeros of Proposition 9.1.11, computed in ball arithmetic, reach height 2304.3: consecutive ordinates among them differ by at most $\gamma_2-\gamma_1 = 6.88731\ldots$, and the later gaps by at most 5.4141 (between $\gamma_3=25.0109$ and $\gamma_4=30.4249$). It remains to show that $\gamma'-\gamma\le h := 6.8874$ for consecutive ordinates $\gamma<\gamma'$ with $\gamma\ge 2304.3$. Take $T$ with $\gamma<T<\gamma'$ and write $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2.1). The second derivative $\vartheta''(T) = \frac{T}{4}\sum_{k\ge0}(k+\frac14)\big((k+\frac14)^2+\frac{T^2}{4}\big)^{-2}$ is positive, so $\vartheta$ is convex, $\vartheta(T+h)-\vartheta(T)\ge h\vartheta'(T)$, and

$$N(T+h)-N(T) \;\ge\; \frac{h\,\vartheta'(T)}{\pi} - |S(T)| - |S(T+h)|.$$

We use the bounds of [176] as they are quoted in the proof of Proposition 9.1.11: $|S(t)| \le 2.5167$ for $0 \le t \le X_0 = 30\,610\,046\,000$, and $|S(t)| \le B(t) = 0.1095 \log t + 0.2042 \log\log t + 3.0305$ for $t \ge e$, with $B$ increasing. For $2304.3 \le T \le X_0 - h$ the first bound applies to both terms, and $\vartheta'(T) \ge \vartheta'(2304.3) = 2.9523$ gives $N(T+h) - N(T) \ge 6.4725 - 5.0334 > 1.4$. For $T > X_0 - h$ both $|S(T)|$ and $|S(T+h)|$ are at most $B(T+h)$, because $B(X_0) = 6.32 > 2.5167$, so $N(T+h) - N(T) \ge D(T) = h\vartheta'(T)/\pi - 2B(T+h)$. At $T = X_0 - h$, $h\vartheta'(T)/\pi = 24.452$ and $2B(X_0) = 12.649$, so $D(X_0 - h) = 11.80 > 0$; and $D$ increases, because a comparison of the sum for $\vartheta''$ with an integral gives $\vartheta''(T) \ge \frac{1}{2T} - \frac{1}{T^2}$, while $2T\,B'(T+h) \le 0.219 + 0.4084/\log T$, so that $T\,D'(T) \ge \frac{h}{2\pi}(1 - \frac{2}{T}) - 0.219 - \frac{0.4084}{\log T} > 0.8$ for $T \ge X_0 - h$. In every case $N(T+h) - N(T) > 0$, so some ordinate lies in $(T, T+h]$, and letting $T \downarrow \gamma$ gives $\gamma' - \gamma \le h$. Finally, a real $\tau$ with $|\tau| \ge \gamma_1$ lies between two consecutive ordinates, hence within half a gap of one of them. □

Corollary 17.8.8. *For every $a > 0$ and every $\varphi \in K(a)$, with no hypothesis on the zeros,*

$$\langle \varphi, \varphi \rangle_1 \;\ge\; 2b_1 \, \|\varphi\|^2 = \bigl(2 + \gamma_E - \log 4\pi\bigr) \|\varphi\|^2 = 0.0461914\ldots \, \|\varphi\|^2,$$

*and no larger constant works for every $a$: $\lambda_{\min}(1, a) \downarrow 2 + \gamma_E - \log 4\pi$ as $a \to \infty$.*

*Proof.* At $\alpha = 1$ no zero has real part 1 and $\sigma_{\max} \le 1$, so Theorem 17.8.5(i) applies, and $m_1 = b_1$ by Proposition 9.1.11. Finally $b_1 = \xi'(1)/\xi(1) = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, twice which is $2 + \gamma_E - \log 4\pi = 0.0461914179\ldots$. □

Remark 17.8.9 (A Laguerre type inequality). Laguerre's quantity is

$$\mathcal{L}_\Xi(\tau) = -(\log \Xi)''(\tau) = \sum_\rho (\tau - \gamma_\rho)^{-2}, \qquad \Xi(\tau) = \xi(\tfrac{1}{2} + i\tau),$$

and it is the limit of $v_{1/2+\varepsilon}(\tau)/\varepsilon$ as $\varepsilon \downarrow 0$, taken termwise: a zero on the line gives $(\tau - \gamma)^{-2}$, and a pair $\frac{1}{2} \pm \delta + i\gamma$ gives $2\,\mathrm{Re}(\tau - \gamma + i\delta)^{-2}$. Dividing $v_\alpha(\tau) \ge b_\alpha$ by $\varepsilon$, part (ii) of Theorem 17.8.5 becomes, as $\varepsilon \downarrow 0$,

$$\mathcal{L}_\Xi(\tau) \;\ge\; \mathcal{L}_\Xi(0) = 2p_1,$$

a sharpening of Laguerre's inequality $\Xi'^2 - \Xi\Xi'' \ge 0$, which says $\mathcal{L}_\Xi \ge 0$. The sharpened inequality holds under RH for every $\tau$, and with no hypothesis for $|\tau| \le 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)). Numerically (`yo_symbol.py`) $\mathcal{L}_\Xi(0) = 0.0462100$ is the global minimum of $\mathcal{L}_\Xi$ on $[0, 1800]$, and the next local minimum, 0.2301087 at $\tau = 17.498$ between the first two ordinates, is 4.98 times larger.

Table 17.2 lists $\lambda_{\min}(\alpha, a)$ for five windows and the four levels $\alpha = 0.55$, 0.6, 0.75, 1, with the floor $2b_\alpha$ of Theorem 17.8.5 in the last row, and the finite section values at $\alpha = \frac{1}{2}$. The matrices of the form are assembled from the prime side of (17.1.1) with the damping.

| $a$ | $\alpha=\frac{1}{2}$ | $\alpha=0.55$ | $\alpha=0.6$ | $\alpha=0.75$ | $\alpha=1$ |
|---|---|---|---|---|---|
| 0.35 | $1.20\times10^{-3}$ | 0.0066141 | 0.0120197 | 0.0281496 | 0.0547526 |
| 0.8 | $2.28\times10^{-17}$ | 0.0047355 | 0.0094659 | 0.0236477 | 0.0472506 |
| 1.5 | $2.43\times10^{-64}$ | 0.0046489 | 0.0092974 | 0.0232400 | 0.0464620 |
| 2.5 | $1.95\times10^{-145}$ | 0.0046305 | 0.0092608 | 0.0231497 | 0.0462846 |
| 4.0 | $1.59\times10^{-286}$ | 0.0046246 | 0.0092490 | 0.0231206 | 0.0462270 |
| $2b_\alpha$ | 0 | 0.0046210 | 0.0092418 | 0.0231027 | 0.0461914 |

Table 17.2: The least eigenvalue $\lambda_{\min}(\alpha,a)$ of the damped Yoshida form on the window $[-a,a]$, from the matrices of the form on the modes $\chi_n$, $|n|\le N$. For $\alpha>\frac{1}{2}$ the entries are converged values, from Richardson extrapolation of $N=96$, 192, 384, with an uncertainty of at most $3\times10^{-7}$ at $a=0.35$ and $1.1\times10^{-8}$ for $a\ge0.8$ (an independent Legendre basis agrees to $1.5\times10^{-7}$ at most). For $\alpha=\frac{1}{2}$ the entries are the finite section values at $N=\max(24,24a)$, upper bounds computed in extended precision, which are not converged; the entry for $a=4$ ($N=96$) was computed with 640 digits. The last row is the floor $2b_\alpha=2\xi'(\alpha)/\xi(\alpha)$. Computed by `yo_damped.py` with `yo_matrix.py`, data in `yo_damped.json`.

Every entry for $\alpha>\frac{1}{2}$ exceeds the floor and decreases with $a$, as Theorem 17.8.5 says (under RH for $\alpha<1$). The excess over the floor falls roughly like $a^{-2}$, as in Theorem 20.2.1(b) of Chapter 20.

The column $\alpha=\frac{1}{2}$ is of another kind. The entries depend strongly on $N$ (at $a=0.8$ the value is $6.9\times10^{-17}$ at $N=12$, $2.3\times10^{-17}$ at $N=24$ and $1.9\times10^{-17}$ at $N=48$). Two other tables of the book compute the same eigenvalue differently: Table 18.1 of Chapter 18 at $\alpha=\frac{1}{2}$, to 60 digits, for even and for odd test functions separately, and Table 20.1 of Chapter 20, the rungs at more levels $\alpha>\frac{1}{2}$, as converged values.

Figure 17.4 shows the symbol on four lines.

At $\alpha=\frac{1}{2}$ Theorem 17.8.5 does not apply. Under RH the symbol there is the atomic measure on the ordinates, the vague limit of $\rho_\varepsilon$ as $\varepsilon\downarrow0$; it vanishes between the atoms, which is why $\lambda_{\min}(\frac{1}{2},a)\to0$, and under RH the decay is faster than any power of $a$ (Theorem 20.2.1(c) of Chapter 20). The speed of the collapse in Table 17.2 is a numerical observation. The finite section values at $\alpha=\frac{1}{2}$ follow $\exp(-2c\,ae^{2a})$ with $c=-\log\lambda/(2ae^{2a})=4.12$, 4.77, 5.10, 5.01, 4.86 at $a=0.25$, 0.35, 0.5, 0.65, 0.8 (so $4.1\le c\le5.1$, at $N=48$). Zhu proves, under RH and for all large $a$, only the bound $\exp(-ae^a)$ for the least eigenvalue over real even test functions (Theorem 1.3 of [526], Chapter 18): it has $e^a$ where the observed decay has $e^{2a}$. For $a\ge1.5$ the finite sections saturate and $c$ falls, so those values say nothing about the infimum. A heuristic for the form of the decay: $\widehat{\varphi}$ is an entire function of exponential type $a$ and has at most about $2aH/\pi$ real zeros in $[-H,H]$, against about $(H/\pi)\log(H/2\pi)$ ordinates there, so a vector of type $a$ can vanish at every ordinate up to a height of about $2\pi e^{2a}$ and no further, which

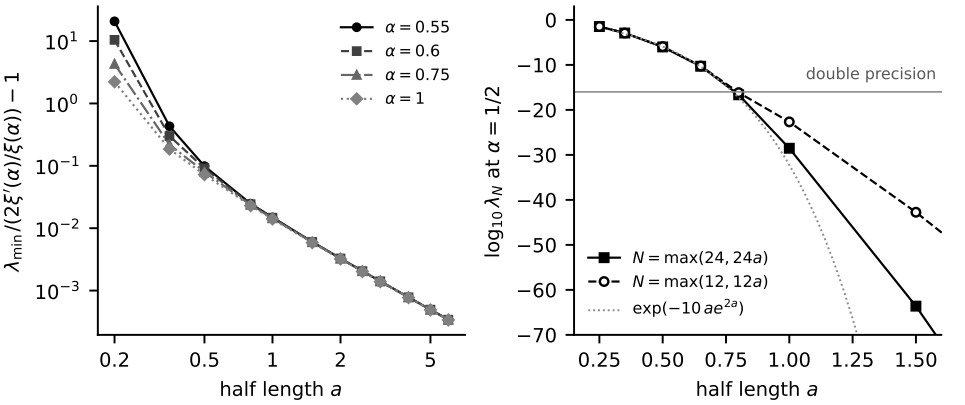



Figure 17.3: Left: the excess of the least eigenvalue of the damped Yoshida form over its floor, $\lambda_{\min}(\alpha, a)/(2b_\alpha) - 1$, against the window half length $a$, for $\alpha = 0.55$, 0.6, 0.75, 1 (finite sections of $\max(96, \lceil 24a \rceil)$ modes, logarithmic scales); it tends to zero, roughly like $a^{-2}$. Right: $\log_{10}$ of the finite section value at $\alpha = \frac{1}{2}$ for $N = \max(24, 24a)$ and $N = \max(12, 12a)$ modes, with the level of double precision and the curve $\exp(-10ae^{2a})$.

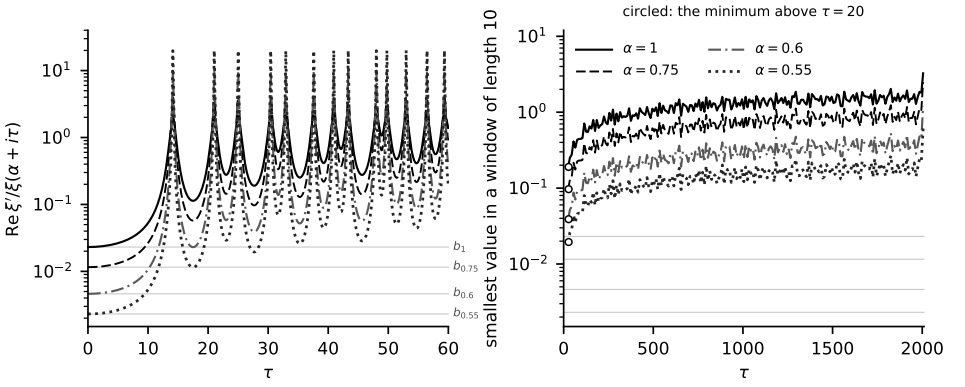



Figure 17.4: Left: the symbol $v_\alpha(\tau) = \mathrm{Re}\,\xi'/\xi(\alpha + i\tau)$ on $[0, 60]$ for $\alpha = 1$, 0.75, 0.6, 0.55 (logarithmic scale), with $b_\alpha$ as thin horizontal lines. Right: the smallest value of $v_\alpha$ in windows of length 10 on $[0, 2000]$, with the minimum above $\tau = 20$ circled. Above $\tau = 20$ the minima are 0.1901, 0.0968, 0.0389 and 0.0195 for $\alpha = 1$, 0.75, 0.6, 0.55, which are 8.2, 8.4, 8.4 and 8.4 times $b_\alpha$, all at $\tau = 27.67$, between the third and the fourth ordinate (the largest gap above the second); the lowest local minimum other than the one at the origin is lower, $4.90\,b_1$ at $\tau = 17.49$ for $\alpha = 1$ and $4.98\,b_{0.55}$ for $\alpha = 0.55$, in the gap between the first two ordinates.

suggests a decay like $\exp(-c\,ae^{2a})$. This is not a proof.

## 17.9 An uncertainty barrier

Every prime side computation is limited in one of two ways. Compact support in the level variable keeps the prime sum finite but limits the level to $\log n \le X$. A finite basis limits the frequency, the ordinate $|\tau| \le T$. For the tent coefficient $D$ of Proposition 17.6.1 the limit in level runs into zero verification, which is what the next proposition makes precise.

Proposition 17.9.1. *Suppose all zeros with $0 < \gamma \le H$ lie on the critical line, and let $D_{\mathrm{on}} \ge 0$ be the part of $D$ from on line zeros. Then for every $x > 0$*

$$|D(x) - D_{\mathrm{on}}(x)| \le \varepsilon_H(x) := 8\cosh^2\frac{x}{4}\sum_{\gamma>H}\frac{1}{\gamma^2},$$

*the sum being over all zeros above $H$, counted with multiplicity. With $H = 3\times10^{12}$ (Platt and Trudgian [419], the height $H_0$ of Theorem 9.1.18) the sum differs from $(\log(H/2\pi)+1)/(2\pi H) = 1.4797\times10^{-12}$ by less than $2\times10^{-24}$ (Theorem 11.4.6), so that $\varepsilon_H(x) < 1.184\times10^{-11}\cosh^2(x/4)$.*

*Proof.* Off line zeros have $|\gamma| > H$ and $|\delta| < \frac12$, so $|\sin(\gamma_\rho x/2)|^2 \le \cosh^2(\delta x/2) \le \cosh^2(x/4)$ and $|\gamma_\rho|^2 \ge \gamma^2$. Each off line zero therefore contributes at most $4\cosh^2(x/4)/\gamma^2$ to $D$, and summing over all such zeros, two (a zero and its conjugate) for each ordinate $\gamma > H$ of the upper half plane, gives the bound. □

The proposition is of use only against a lower bound for $D_{\mathrm{on}}$. Every term of $D_{\mathrm{on}}$ is nonnegative, so the first $N$ zeros, which are on the line, give one: $D_{\mathrm{on}}(x) \ge L_N(x) := \sum_{n\le N} 8\sin^2(\gamma_n x/2)/\gamma_n^2$. For the $N = 1700$ certified zeros of the book the least value of $L_N$ on $[0.05, 47.1]$ is 0.040704, at $x = 25.383$, and 0.04070 is a lower bound for it computed by branch and bound (`yo_tent.py`, `code/README.md`); on $[0.05, 16]$ the least value is 0.0531.

Proposition 17.9.2 (Computed positivity of the tent coefficient). *For the $N = 1700$ certified zeros of the book let $L_N(x) = \sum_{n\le N} 8\sin^2(\gamma_n x/2)/\gamma_n^2$, and let $H = 3\times10^{12}$. Since all zeros with $0 < \gamma \le H$ lie on the critical line [419], $D(x) \ge L_N(x) - \varepsilon_H(x)$ for every $x > 0$. Moreover $L_N(x) > \varepsilon_H(x)$ on $[0.05, 47.48]$, so $D > 0$ on $[0.05, 47.48]$ from the verification of RH up to $H$ and the first 1700 zeros alone, with no prime sum. The first inequality is proved. The second is computed in double precision and not in ball arithmetic: for $x < 46.688$ it follows from the floor 0.04070 of $L_N$ on $[0.05, 47.1]$, found by branch and bound, and for $46.688 \le x \le 47.48$ from a grid of step $5\times10^{-4}$ and the Lipschitz constant 10.85 of $L_N$.*

The level 47.10 at which $\varepsilon_H$ reaches 0.05 ($e^{47.10} = 2.85 \times 10^{20}$) is arithmetically right, but it is the barrier only if $D_{\mathrm{on}}$ stays above 0.05, which it does not; with the floor 0.0407 the bound reaches it at $x = 46.69$ ($e^{46.69} = 1.9 \times 10^{20}$), and pointwise at $x = 47.48$ ($e^{47.48} = 4.2 \times 10^{20}$). The barrier is at about 47 either way, because $\varepsilon_H$ grows exponentially. A prime side computation of $D$ at a level below 47.48, which needs all prime powers up to $e^x$, can therefore only confirm a positive sign that the zeros already give; it cannot produce a negative value of $D$, which is how an off line zero would show. To see one through $D$ a computation must reach levels beyond 47.48, with all prime powers up to $4 \times 10^{20}$, or heights beyond $H$, and the second is the zero verification itself.

In frequency, Proposition 17.8.1 makes the same point for a basepoint $\alpha \geq \sigma_{\max}$: a test vector with spectrum concentrated in $|\tau| \leq T < H$ sees only the symbol there, and a zero $\beta + i\gamma$ with $\gamma > H$ contributes to the Hadamard sum for the symbol the term $(\alpha - \beta)/((\alpha - \beta)^2 + (\tau - \gamma)^2)$, of modulus at most $|\alpha - \beta|/(\gamma - T)^2$.

What the barrier does not say. Proposition 17.9.1 is proved for the single coefficient $D$ and for no other vector. A normalised function supported in the window $[-a, a]$ and modulated to a height above $H$ (Proposition 17.5.1) sees an off line zero there with a weight that does not decay with the height, and $\varepsilon_H$, which carries the factor $\gamma^{-2}$ of the tent, does not bound the contribution of the zeros above $H$ to the form of such a vector. Nor could a perturbation bound of size $10^{-11}$ settle the sign of a windowed form at $\alpha = \frac{1}{2}$, whose least eigenvalue is below $10^{-15}$ from $a = 0.8$ on (Table 17.2). The tent is one vector of the window, so the level $x_*$ of Remark 17.6.3 bounds Yoshida's threshold only from above, $a_0 \leq x_*/2$, and the window experiment of Chapter 18 (Table 18.3) finds a hypothetical off line pair at half lengths far below $x_*/2$.

## 17.10 The question of a path to RH

The chapter does not give a path to RH. Every criterion proved here has the same structure. On the RH side it is a positivity statement: $D \geq 0$, $T_k \geq 0$ for infinitely many $k$, the modulated heat trace of Proposition 17.5.1 nonnegative, a Pick function, a coercive damped form. On the other side it is the presence of a pole off the real axis, or a negative Cauchy kernel, at an off line ordinate. The barrier of Section 17.9 then says, for the tent coefficient $D$, that a computation at levels below 47 cannot go beyond what zero verification already knows; for the other quantities no such statement is proved there. A proof needs an inequality valid at all levels or all heights at once, and it must use the Euler product: the function $f_{\mathrm{DH}}$ of Davenport and Heilbronn [111] (Section 11.5.5.4) shares the functional equation and has zeros off the line, so it violates the analogue of every criterion in this chapter.

What the chapter does change is the shape of the missing inequality. Theo-

rem 17.8.5 puts it in its sharpest form:

$$\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)\ \geq\ \frac{\xi'}{\xi}(\alpha)\qquad\text{for all }\tau,\ \alpha\in(\tfrac12,1],\tag{17.10.1}$$

which we write $v_\alpha(\tau)\geq b_\alpha$ as in Theorem 9.1.18. Taken as a family over $\alpha\in(\frac12,1]$, and not line by line, this is equivalent to the coercivity of every damped Yoshida form with the constant it has on functions of low frequency. Three facts about it:

1. It is true with no hypothesis for every $\alpha\geq 1$ and every $\tau$ (Propositions 9.1.11 and 9.1.12), and for $\frac12<\alpha\leq\alpha_{\max}=7.35$ and $0<|\tau|\leq H-\frac{1+\sqrt2}{2}$ (Theorem 9.1.18(ii)); for $\frac12<\alpha<1$ and all $\tau$ it is equivalent to RH (Theorem 9.1.18(vi)). On the line $\alpha=1$ the minimum is at $\tau=0$, the next lowest local minimum of $v_1$, at $\tau=17.49$, is 4.90 times $b_1$, and on $[20,2000]$ the least value is $8.23\,b_1$, at $\tau=27.67$ (computed by `yo_symbol.py`, `code/README.md`).

2. If it holds, even in the weak form $v_{\alpha'}(\tau)>0$, on every line $\operatorname{Re}s=\alpha'$ with $\alpha<\alpha'\leq 1$, then $\sigma_{\max}\leq\alpha$, since a zero $\rho_0=\beta_0+i\gamma_0$ with $\beta_0>\alpha$ sends $v_{\alpha'}(\gamma_0)$ to $-\infty$ as $\alpha'\uparrow\beta_0$; on all of $(\frac12,1]$ it is RH. On a single line it is not known to exclude a zero to its right (Remark 1.2.4). Its $\alpha\downarrow\frac12$ limit, under RH, is the sharpened Laguerre inequality $-(\log\Xi)''(\tau)\geq 2p_1$.

3. Its failure mode is local. A zero with $\beta>\alpha$ would create a negative Cauchy kernel of height $1/(\pi(\beta-\alpha))$ at its ordinate, competing against a symbol of size $\frac{1}{2\pi}\log\tau$ there. So the inequality at $\alpha$ asks for control of $-\operatorname{Re}\zeta'/\zeta(\alpha+i\tau)$ by $\frac12\log(\tau/2\pi)$, a one sided bound on $\zeta'/\zeta$; the Fejér tapered form of the question is Proposition 14.14.1. No method known today proves it at any fixed $\alpha<1$ for all $\tau$; the zero free half plane $\operatorname{Re}s>\frac78$ announced in October 2026 (Section 12.1.2) has not been refereed and is not counted.

## 17.11 Open problems

What remains:

Problem 17.1 (The symbol inequality below the line $\alpha=1$). The inequality $v_\alpha(\tau)\geq b_\alpha$ is a theorem of the book for every $\alpha\geq 1$ and every $\tau$ (Propositions 9.1.11 and 9.1.12), and for $\frac12<\alpha\leq\alpha_{\max}$ up to height $3\times10^{12}-1.21$ (Theorem 9.1.18(ii)); for $\frac12<\alpha<1$ and all $\tau$ it is equivalent to RH (Theorem 9.1.18(vi)). What is open is the region $\alpha<1$ above the verified height. Determine the part of it in which $v_\alpha(\tau)>b_\alpha$ can be proved with no hypothesis and with explicit constants; the Vinogradov Korobov region $\alpha\geq 1-c(\log|\tau|)^{-2/3}(\log\log|\tau|)^{-1/3}$ [509, 315] is the natural target. A proof on every line of a strip $\alpha_0<\alpha\leq 1$ and for all $\tau$ would give $\sigma_{\max}\leq\alpha_0$ (Section 17.10), which is why the target is a region that narrows with the height.

Problem 17.2 (A prime side proof at $\alpha = 1$). The book proves $v_1(\tau) \geq b_1$ through the zeros (Proposition 9.1.11), by the Cauchy kernels of the ordinates nearest to $\tau$ and a bound on the gaps between them (compare Lemma 17.8.7). At $\alpha = 1$ the statement can be written entirely in primes. Since $\xi'/\xi(1) = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$ and $\operatorname{Re}\frac{1}{i\tau} = 0$, it reads

$$\sum_{n\geq 2} \frac{\Lambda(n)}{n}\cos(\tau \log n) \;\leq\; \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\tau}{2}\Big) - \frac{1}{2}\log\pi + \frac{1}{1+\tau^2} - \frac{\xi'}{\xi}(1) \qquad (\tau \neq 0),$$

the left side being the conditionally convergent series for $-\operatorname{Re}\zeta'/\zeta(1+i\tau)$. Find a proof using only Mertens type information, in the spirit of the Fejér taper of Proposition 14.14.1; the right side grows like $\frac{1}{2}\log(\tau/2\pi)$ and the left side is $O(\log\log\tau)$ under RH.

Problem 17.3 (Löwner at a single window). By Theorem 17.7.1, positivity on $K(a)$ is positivity of the Löwner matrices of $g_a$ on the lattice $\frac{\pi}{a}\mathbb{Z}$. Löwner's hard theorem [349, 120] recovers the Pick property from Löwner matrices on all finite subsets of an interval. What does positivity on the lattice alone, for one fixed $a$, imply about the location of the zeros? A positive answer that grew with $a$ would quantify Yoshida's threshold $a_0$.

Problem 17.4 (The constants of the ladder and of the window). Determine the order of $K_0$ for the ladder and of $a_0$ for Yoshida's window, for a single off line pair, and compare with the displacement $\sigma_{\max} - \frac{1}{2}$ and with the de Bruijn Newman constant $\Lambda \leq 2(\sigma_{\max} - \frac{1}{2})^2$ (de Bruijn [112], in the normalisation of Rodgers and Tao [463] used in Section 8.4). For the ladder the computations of Section 17.3 give $K_0(\sec\theta_* - 1) \approx 0.70$ for the pair of the stress test (Remark 17.3.3) and, over fourteen pairs, a first failing order that grows like $\gamma^2$ and, at fixed $\gamma$, like $1/\delta$ and not like the heuristic $\gamma^2/\delta^2$; its dependence on $\delta$ and on the spacing of the true ordinates is what is to be determined. For the window only $a_0 \leq x_*/2$ is known (Remark 17.6.3), and the window experiment of Chapter 18 (Table 18.3) finds the threshold far below it.

Problem 17.5 (Zero density and the barrier). Proposition 17.9.1 allows every zero above $H$ to have $\delta$ arbitrarily close to $\frac{1}{2}$. Zero density estimates make zeros with large $\delta$ rare. Sharpen the barrier level using an explicit $N(\sigma, T)$ bound, and determine whether any prime side computation of $D$ at feasible levels can, even in principle, exclude off line zeros beyond the verified height. For windows no barrier is proved: decide whether a window of half length $a$ below about 23.7 (level 47.48) can see an off line zero above $H$, for which Section 18.4 of Chapter 18 gives numerical evidence and a heuristic horizon, not a proof.

## Notes

The forms of this chapter are Yoshida's [525], and the explicit formula is Weil's [512]. Lagarias's criterion [320] asks only for $v_\alpha > 0$; the minimum at the origin of Theorem 9.1.18(vi) sharpens it. In integrated form, under RH, the minimum at the origin says that $\sigma \mapsto |\xi(\sigma+i\tau)|/\xi(\sigma)$ is nondecreasing on $(\frac{1}{2}, \infty)$, a normalised version of the reformulation of Sondow and Dumitrescu [482] that $|\xi(\sigma + i\tau)|$ increases in $\sigma$. Theorem 17.6.2 is classical in substance.

# Chapter 18

# Zhu's compact windows and the Landau Widom horizon

Xuefeng Zhu's preprint on Weil positivity in compact windows [526] continues Yoshida's programme of Chapter 17 by certified computation.

*What is not claimed.* The account of Zhu's preprint is the first author's reading of version 2 of 2 September 2026, checked statement by statement: his theorems are quoted with the status he gives them and are not reproved, and his certificates are not rechecked. A number is computed in floating point at the precision stated; the least eigenvalues of Table 18.1 are consistent with Zhu's certified intervals and carry no interval certificate of their own. The passage from a statement about a window to a statement about the height of the zeros (the horizon of Section 18.4) is Zhu's heuristic and is marked as one wherever it is used. The detection experiment of that section rests on a hypothetical configuration of zeros, a quadruple added to the actual zeros of $\xi$; it is a computation about that configuration and not a statement about $\xi$.

## 18.1 What Zhu proves

Zhu [526] writes $L$ for the half length of the window. Here it is $a$, as in Chapter 17, where $L$ is the support of an autocorrelation, so that his $\lambda^*(L)$, $A_L$ and $T_1(L)$ appear below as $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)$, $A_a$ and $T_1(a)$. For real even continuous $f$ supported in $[-a,a]$, with $F=\widehat{f}$, $\widehat{f}(t)=\int_{\mathbb{R}} f(y)e^{ity}\,dy$, Zhu writes Weil's quadratic form as

$$Q(f)=2F\big(\tfrac{i}{2}\big)^2+\frac{1}{2\pi}\int_{\mathbb{R}}|F(t)|^2\,\Psi_a(t)\,dt,\qquad \Psi_a(t)=\operatorname{Re}\psi\Big(\frac{1}{4}+\frac{it}{2}\Big)-\log\pi-\mathcal{P}_a(t), \tag{18.1.1}$$

where $\mathcal{P}_a(t)=\sum_{\log n<2a}2\Lambda(n)n^{-1/2}\cos(t\log n)$ is the prime comb. He sets $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)=\inf Q(f)/\|f\|^2$ over the real even $f\neq 0$ supported in $[-a,a]$. His results, with the status he gives them, are the following.

(Z1) One stroke reduction (Theorem 1.1, unconditional). Let $A_a = \sup_t \mathcal{P}_a(t) = \sum_{\log n<2a} 2\Lambda(n)n^{-1/2}$: the supremum is attained at $t = 0$, and by Weyl equidistribution of the phases $t \log p$ values arbitrarily close to it occur at arbitrarily large $t$ (his Lemma 3.2). The envelope $\operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2}) - \log\pi \geq \log\frac{t}{2\pi} - \frac{1}{t}$ for $t \geq \frac{15}{4}$ (Lemma 3.1) gives $\Psi_a(t) \geq \beta^* > 0$ for $t \geq T_\sharp$, where $\beta^* = \log\frac{T_\sharp}{2\pi} - \frac{1}{T_\sharp} - A_a$ and $\beta^* > 0$ holds for every $T_\sharp$ above a threshold that exceeds $T_1 := 2\pi e^{A_a}$ by the factor $1 + O(T_1^{-1})$ (120.1 against 119.1 at $a = 0.8$). Positivity on the window then reduces to the positive semidefiniteness of a finite matrix in an even Legendre basis, with explicit tails.

(Z2) Certificate (Theorem 1.2, a computer assisted theorem). At $a = 0.8$, with $T_\sharp = 200$, 200 even Legendre modes (orders up to 398), working precision 50 digits and a verified Cholesky factorisation, $Q(f) \geq 8.9\times10^{-18}\|f\|^2$ for every real even $f$ supported in $[-0.8, 0.8]$. The autocorrelation then has support $[-1.6, 1.6]$, which is 2.3 times the range $2a \leq \log 2$ that Zhu attributes to Yoshida [525] and, by a different method, to Connes and Consani [87]. His variational upper bound, computed on the geometric side in interval arithmetic, is $\lambda_{\min}^{\text{even}}(\frac{1}{2}, 0.8) \leq 2.27 \times 10^{-17}$, which encloses the infimum within a factor 2.6.

(Z3) Under RH, $\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) \leq \exp(-ae^a)$ for $a$ large (Theorem 1.3, conditional on RH), by an explicit test function: a canonical product over the ordinates up to $T = 2\pi e^a$ times a power of the sinc function. The construction works at half saturation, and with $T = 2\pi e^{2\theta a}$ it closes only for $\theta \lesssim 0.65$ (his Remark 13.6).

(Z4) Decay law (Conjecture 12.1, fitted to computed bounds). The certified variational upper bounds for $a$ from 0.5 to 2, unconditional and computed on the geometric side, fall from $10^{-6}$ to $3.2 \times 10^{-283}$ and follow $-\log\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) = C\,N(T_*)/\log N(T_*)\,(1 + o(1))$, where $T_* = 2\pi e^{2a}$, $N(T)$ is the number of ordinates up to $T$ and $C = 20.13\ldots \approx 2\pi^2$ (the fit of his equation (5)). The constant is fitted and not derived: Zhu calibrates the law on $a \leq 1.2$ and tests it on $a = 1.4$ to 2, where it underpredicts by about 5 percent with no trend. That the law describes the infimum itself, with $2\pi^2$ in place of the fitted $C$, is his Conjecture 12.1.

(Z5) Barrier (Theorem 1.4, unconditional). Since $A_a = (4 + o(1))e^a$ by the prime number theorem, any application of (Z1) needs $T_\sharp > T_1(a) = 2\pi e^{A_a}$, about $1.0 \times 10^7$ at $a = 1.6$ and $2.4 \times 10^{11}$ at $a = 2$ ($A_{1.6} = 14.323$, $A_2 = 24.383$), so that the size of the matrix grows doubly exponentially in $a$; and since $\sup_t \mathcal{P}_a(t) = A_a$ exactly, no pointwise bound on the comb lowers $T_1$. To improve on pointwise envelopes one would have to prove that the phases $t \log p$ cannot align, which Zhu calls an arithmetic statement about simultaneous Diophantine approximation of $\{\log p\}$ (his Remark 1.6).

(Z6) Parity (Lemma 6.1, Theorem 6.2 and Corollary 6.3). For real $f = f_e + f_o$, $Q(f) = Q(f_e) + Q(f_o)$, with the pole term entering as $+2c^2$ on the even part and $-2s^2$ on the odd part, where $c = \int f_e(x)\cosh\frac{x}{2}\,dx$ and $s = \int f_o(x)\sinh\frac{x}{2}\,dx$. At $a = 0.8$ the even ground state is simple, with the certified bounds $8.9\times10^{-18} \le \lambda_1^{\text{even}} \le 2.523\times10^{-16}$ (the upper bound $2.27\times10^{-17}$ of (Z2) is the sharper), $\lambda_2^{\text{even}} \ge 2.085\times10^{-12}$ and $8.206\times10^{-15} \le \lambda_1^{\text{odd}} \le 2.347\times10^{-14}$, where $\lambda_1^{\text{even}} \le \lambda_2^{\text{even}}$ are the first two min max values of $Q(f)/\|f\|^2$ over real even $f$ supported in $[-0.8, 0.8]$ and $\lambda_1^{\text{odd}}$ is the infimum over real odd $f$. Hence $Q(f) \ge 8.9\times10^{-18}\|f\|^2$ for complex $f$ as well. Zhu connects the simple even ground state with one of the two spectral steps that the programme of Connes, Consani and Moscovici [88] isolates as missing.

Zhu also retracts a claim of an earlier draft at autocorrelation support 2.38 ($a = 1.19$), keeping a computation that is consistent with positivity there and no certificate (his Section 7), and he warns that evaluating the geometric side of $Q$ in double precision produces spurious negative eigenvalues (his Remark 2.1 and Section 10).

For complex test functions the least eigenvalue of Chapter 17 is $\lambda_{\min}(\frac12, a)$, the infimum of $\langle\varphi,\varphi\rangle/\|\varphi\|^2$ over $0 \ne \varphi \in C(a)$ (Section 17.8.2). Write $\lambda_{\min}^{\text{odd}}(\frac12, a)$ for the infimum over real odd test functions. The form has no cross terms between the real and the imaginary part of $\varphi$, nor between its even and its odd part (Zhu's Lemma 6.1), so that

$$\lambda_{\min}(\tfrac12, a) = \min\bigl(\lambda_{\min}^{\text{even}}(\tfrac12, a),\, \lambda_{\min}^{\text{odd}}(\tfrac12, a)\bigr),$$

and the two sides $\lambda_{\min}(\frac12, a)$ and $\lambda_{\min}^{\text{even}}(\frac12, a)$ are equal when the ground state is even, which Zhu's Theorem 6.2 shows at $a = 0.8$. Yoshida's parity reduction (Y1) of Section 17.1 concerns the same two sectors. The splitting holds for the damped forms of Section 17.8.1 at every $\alpha$ as well, because the damping $e^{-\varepsilon|y|}$ is even.

## 18.2 The Thorin dictionary

The functional $\mathcal{Y}$, the form $\langle\varphi_1,\varphi_2\rangle = \mathcal{Y}(\varphi_1 * \widetilde{\varphi}_2)$ and the spaces $C(a)$ and $K(a)$ are those of Section 17.1. Call a real even $f$ supported in $[-a, a]$ admissible if the explicit formula applies to $f * f$, as it does for every $f \in C_c^\infty(\mathbb{R})$. Let $PW_a$ be the Paley Wiener space of the entire functions of exponential type at most $a$ that are square integrable on $\mathbb{R}$, and $PW_a^{\text{even}}$ its real even members, which contain the transforms $F = \widehat{f}$ of the admissible $f$. The measure $U = \sum_{\gamma>0}\delta_{\gamma^2}$ is the Thorin measure of the clock under RH (Section 1.2), each ordinate counted with its multiplicity.

Proposition 18.2.1. *For admissible $f$, with $F = \widehat{f}$, Zhu's form is Yoshida's form*

*on the window $C(a)$: $Q(f)=\langle f,f\rangle=\mathcal{Y}(f*f)$. Under RH*

$$Q(f)=\sum_{\rho}|F(\gamma_\rho)|^2=2\int_{(0,\infty)}|F(\sqrt{t})|^2\,U(dt),$$

$$\lambda^{\mathrm{even}}_{\min}(\tfrac12,a)=\inf_f\frac{2\int_{(0,\infty)}|F(\sqrt{t})|^2\,U(dt)}{\frac{1}{2\pi}\int_{\mathbb{R}}|F|^2},$$

*so that $\lambda^{\mathrm{even}}_{\min}(\frac12,a)$ is the lower sampling constant of the Thorin measure $U$ for the transforms $F\in PW^{\mathrm{even}}_a$ of the admissible $f$.*

*Proof.* The autocorrelation $g=f*f$ is even, supported in $[-2a,2a]$ and has $\widehat{g}=F^2=|F|^2\geq 0$ on $\mathbb{R}$. By Proposition 17.4.1 (there $L=2a$; the proposition is stated for smooth $f$, and for an admissible $f$ the identity is the term by term explicit formula that follows), with no hypothesis on the zeros, $\langle f,f\rangle=\mathcal{Y}(g)=2\int\widehat{g}(\sqrt{t})\,U^{\pm}(dt)$, and only the prime powers $n\leq e^{2a}$ enter; under RH the functional $\varphi\mapsto\int\varphi\,dU^{\pm}$ is the sum of $\varphi$ over the atoms of $U$ (Section 5.3.1). Term by term, the explicit formula (17.1.1) for $g$ has the pole term $\widehat{g}(\frac{i}{2})+\widehat{g}(-\frac{i}{2})=2F(\frac{i}{2})^2$. Its prime term $-\sum_{n\geq2}\Lambda(n)n^{-1/2}\big(g(\log n)+g(-\log n)\big)$ equals $-\frac{1}{2\pi}\int|F|^2\mathcal{P}_a$, because $\widehat{g}=|F|^2$ and $g$ vanishes beyond $2a$. Its archimedean term is $V_1(g)-g(0)\log\pi=\frac{1}{2\pi}\int|F|^2\big(\operatorname{Re}\psi(\frac14+\frac{it}{2})-\log\pi\big)dt$, by the first line of (17.1.2) and $g(0)=\frac{1}{2\pi}\int|F|^2$. The three terms add up to (18.1.1). Under RH every $\gamma_\rho$ is real, and the zero side of (17.1.1) is $\sum_\rho\widehat{g}(\gamma_\rho)=\sum_\rho F(\gamma_\rho)^2=2\sum_{\gamma>0}F(\gamma)^2$, because $F$ is even and the ordinates come in pairs $\pm\gamma$; this is $2\int F(\sqrt{t})^2\,U(dt)$. Finally $\|f\|^2=\frac{1}{2\pi}\int|F|^2$ by Plancherel. □

Two consequences follow. First, under RH the positivity of the infimum for every $a$ is a sampling statement, and it is Zhu's Proposition 2.3. Littlewood's theorem on the gaps between consecutive ordinates [498, Section 9.12], which holds without RH although Zhu states it as a consequence of RH, says that the gaps tend to zero, so under RH the set $\{\pm\gamma\}$ contains, above some height, a separated subset $\Lambda$ of lower uniform density larger than $a/\pi$ (a maximal subset of points at least $\pi/(4a)$ apart), and Beurling's sampling theorem [476, Chapters 2 and 3] gives $c>0$ with $\sum_{\lambda\in\Lambda}|F(\lambda)|^2\geq c\int_{\mathbb{R}}|F|^2$ for all $F\in PW_a$. Hence $Q(f)\geq 2\pi c\|f\|^2$ and $\lambda^{\mathrm{even}}_{\min}(\frac12,a)>0$, with a constant that is not explicit (Zhu's Remark 2.4). Without RH the argument gives nothing (Zhu's Remark 2.4 counts the gap bound as a second use of RH and says that unconditionally the gaps are only known to be bounded). Second, in this picture the smallness of the infimum is the poor conditioning of sampling by a set that is sparse at low frequency: there is no ordinate in $(-14.13,14.13)$.

Zhu's parity lemma (Z6) has a Thorin reading. The pole is the atom of mass one of the signed Thorin measure $U^{\pm}$ at $t=-\frac14$ (equation (17.4.1)), that is at the imaginary ordinates $\pm\frac{i}{2}$, and it is paired with $F(\frac{i}{2})F(-\frac{i}{2})$. For even $f$ this is

| $a$ | modes $N$ | $\lambda_{\min}^{\text{even}}$ | $\lambda_{\min}^{\text{odd}}$ | Zhu, even |
|---|---|---|---|---|
| 0.5 | 8 | $1.336 \times 10^{-6}$ | $2.51 \times 10^{-4}$ | |
| 0.5 | 14 | $1.082 \times 10^{-6}$ | $2.39 \times 10^{-4}$ | |
| 0.5 | 20 | $1.005 \times 10^{-6}$ | $2.30 \times 10^{-4}$ | |
| 0.5 | 26 | $9.743 \times 10^{-7}$ | $2.24 \times 10^{-4}$ | $\leq 1.05 \times 10^{-6}$ |
| 0.5 | 160 | $9.344 \times 10^{-7}$ | $1.997 \times 10^{-4}$ | |
| 0.8 | 8 | $5.49 \times 10^{-15}$ | $2.67 \times 10^{-12}$ | |
| 0.8 | 14 | $4.127 \times 10^{-17}$ | $3.86 \times 10^{-14}$ | |
| 0.8 | 20 | $2.651 \times 10^{-17}$ | $2.333 \times 10^{-14}$ | |
| 0.8 | 26 | $2.144 \times 10^{-17}$ | $2.157 \times 10^{-14}$ | $[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$ |
| 0.8 | 160 | $1.673 \times 10^{-17}$ | $1.644 \times 10^{-14}$ | |

Table 18.1: Least eigenvalues of Weil's form on the window $[-a, a]$ in the even and the odd sector, from the prime side Yoshida matrices in 60 digit arithmetic with cosine and sine modes up to the order $N$ shown (`code/zh_window_eigs.py`). The last column gives Zhu's certified bounds for the even sector. The values at 26 modes and below are not converged, and the odd values at 160 modes are still falling.

$c^2 \geq 0$ and for odd $f$ it is $-s^2 \leq 0$ (Proposition 17.4.1): it helps the even sector and hurts the odd one. That the even ground state is nevertheless the smaller is explained by Zhu (Section 6.3) through the constraint $F_o(0) = 0$ on the odd sector, which denies odd test functions the zero free interval $[0, \gamma_1)$ in which even minimisers park most of their mass: the low frequency gap, and not the pole, controls the conditioning. At $a = 0.8$ the odd sector lies about three orders of magnitude above the even one and at $a = 0.5$ two orders (Table 18.1), and Zhu reports that the ratio grows almost geometrically in $a$ over $[0.5, 1.2]$, as $10^{2.07a+1.32}$.

The least eigenvalues of the two sectors are computed from the prime side matrices of Yoshida's form. The even sector has the basis $(2a)^{-1/2}$ and $a^{-1/2}\cos(\pi jy/a)$, the odd sector the basis $a^{-1/2}\sin(\pi jy/a)$, for $1 \leq j \leq N$, and $N$ is the number of modes in Table 18.1.

The values of Table 18.1 are consistent with Zhu's numbers, which come from another basis and another code: Fourier modes on $[-a, a]$ against his Legendre polynomials, and quadrature of the explicit formula in the variable $y$ against his spherical Bessel integrals. At $a = 0.8$ both sectors lie inside his certified intervals at 26 modes. At 160 modes the even values $9.344 \times 10^{-7}$ and $1.673 \times 10^{-17}$ lie 0.04 and 1.4 percent above the reference floors $9.34 \times 10^{-7}$ and $1.65 \times 10^{-17}$ of Zhu's Table 1 (a converged exact assembly, which he gives as reference values and not as certificates), and the odd values $1.997 \times 10^{-4}$ and $1.644 \times 10^{-14}$ lie 2.9 and 4.7 percent above his $1.94 \times 10^{-4}$ and $1.57 \times 10^{-14}$ and are still falling. At $a = 0.8$ the second even eigenvalue is $8.42 \times 10^{-12}$ at 160 modes (Zhu: $8.38 \times 10^{-12}$ in his

Table 1, and the certified bound $\lambda_2^{\mathrm{even}} \geq 2.085 \times 10^{-12}$), about $5 \times 10^5$ times the first, in agreement with the simple ground state of (Z6). Raising the precision to 90 digits, or the number of quadrature nodes by half, changes the least eigenvalues by a relative $2 \times 10^{-58}$ at most (checked at 26 and 40 modes), so every value is an upper bound for the infimum over its sector.

As a check of the normalisation of Proposition 18.2.1, take the unit bottom eigenvector $f$ of the matrices of 26 and 40 modes at $a = 0.5$ and 0.8, in both sectors. The partial sums $2\sum_{\gamma \leq \gamma_K} F(\gamma)^2$ over the first $K$ ordinates of `data/zeros_1700.txt` increase with $K$ and stay below the eigenvalue, and with the mean tail $2\int_{\gamma_K}^{\infty} F(t)^2 \, \frac{1}{2\pi} \log \frac{t}{2\pi} \, dt$ of the missing ordinates they reproduce it to within 0.16 percent for $K = 300$, 400 and 1700 in all eight cases.

Three other tables of the book list the same eigenvalue at other levels: Table 17.2 of Chapter 17 for the windows $a = 0.35$ to 4 (its entry $2.28 \times 10^{-17}$ at $a = 0.8$, from 24 modes, lies between our values at 20 and 26 modes), and Tables 20.1 and 20.2 of Chapter 20. Table 18.1 is at $\alpha = \frac{1}{2}$ only, with the even and the odd sector apart, in 60 digits, and it shows the convergence in the number of modes without extrapolating; it reaches $a = 0.8$.

## 18.3 Zhu's test function through a truncated reciprocal Xi transform

Zhu's proof of (Z3) takes $\widehat{f_0}(r) = B(r)h(r)$, with $B(r) = \prod_{0<\gamma_j \leq T}(1 - r^2/\gamma_j^2)$ a finite product over the ordinates up to a truncation height $T$, and a window kernel $h$ of type $a$. In the notation of Section 1.2, $G(u) = \prod_{\gamma>0}(1 + u/\gamma^2)$ is the reciprocal of the Laplace transform of the clock, which under RH is $\sum_{\gamma>0} e_\gamma/\gamma^2$. Truncate the product, $G_T(u) = \prod_{0<\gamma_j \leq T}(1 + u/\gamma_j^2)$. Then $1/G_T$ is the Laplace transform of the finite gamma convolution $\sum_{\gamma_j \leq T} e_{\gamma_j}/\gamma_j^2$ and

$$B(r) = G_T(-r^2),$$

the reciprocal of that Laplace transform continued to the cut $s = -r^2$. Zhu's test function is the window kernel divided by the truncated reciprocal Xi transform on the cut. In the variable $t = r^2$ it vanishes at the Thorin atoms $\gamma_j^2 \leq T^2$ (and at the zeros of $h$), and under RH

$$Q(f_0) = 2\int_{t>T^2} h(\sqrt{t})^2 \, G_T(-t)^2 \, U(dt).$$

The optimisation is a trade: each atom killed costs a factor $r^2/\gamma_j^2$ of growth, which $h$ must absorb with its limited type.

We evaluated this family numerically (script `zh_product_bound.py`), with $h(r) = \big(\sin(ar/m)/(ar/m)\big)^m$, which has type $a$, optimising over the number $K$ of atoms killed and over the even exponent $m \geq 2K+2$. Table 18.2 is a computation

| $a$ | best bound | $K$ | $m$ | $\gamma_K$ | $N(T_*)$ | $T_* = 2\pi e^{2a}$ |
|---|---|---|---|---|---|---|
| 0.5 | $3.88 \times 10^{-4}$ | 2 | 6 | 21.0 | 1 | 17.1 |
| 0.8 | $4.03 \times 10^{-9}$ | 6 | 18 | 37.6 | 4 | 31.1 |
| 1.0 | $7.04 \times 10^{-14}$ | 9 | 26 | 48.0 | 8 | 46.4 |

Table 18.2: The best Rayleigh quotient $Q(f_0)/\|f_0\|^2$ of Zhu's product family over the number $K$ of atoms killed and the even exponent $m$, with the optimal $K$ and $m$, the last ordinate $\gamma_K$ killed, the number $N(T_*)$ of ordinates below the resolution height $T_* = 2\pi e^{2a}$ of Section 18.4, and $T_*$ itself.

of this chapter and not of Zhu. The quotient $Q(f_0)/\|f_0\|^2$ is evaluated on the zero side under RH, with the 1700 ordinates of `data/zeros_1700.txt` and an explicit Riemann von Mangoldt tail [176], and on the geometric side by Zhu's formula (18.1.1), which needs no hypothesis. The two agree to 13 digits or more at $a = 0.8$ and 1.0 and to $3.5 \times 10^{-7}$ relative at $a = 0.5$, where the tail beyond 1700 ordinates limits the comparison, so every entry is an unconditional upper bound for $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2}, a)$.

The product construction is crude. At $a = 0.8$ it gives $4.03 \times 10^{-9}$ against about $1.7 \times 10^{-17}$ for the window infimum (the entry at 160 modes of Table 18.1, an upper bound), and a general element of $PW_a$ does much better. The number of atoms worth killing grows with $a$ and exceeds the number $N(T_*)$ of ordinates below the horizon $T_*$ of the next section: in each row the last atom killed lies above $T_*$ and below $eT_*$. When $K$ is restricted to $N(T_*)$ the bound is worse by the factors 1.27, 1.40 and 1.05. At $a = 0.35$ and $a = 1.2$ (not tabulated) the optimum is $K = 0$ and $K = 16$, equal to $N(T_*)$ in both cases. So atoms above the horizon are worth killing at the three tabulated half lengths, and three rows with $a \le 1$ cannot settle the asymptotics (Zhu's Remark 13.6 in (Z3)).

## 18.4 The Landau Widom horizon

Zhu's resolution height comes from a comparison of densities (his Section 9). By Jensen's formula a real entire function of exponential type $a$ has real zeros of average density at most $a/\pi$ on long intervals $[-R, R]$ about the origin, which Zhu takes as a bound for the local density, while the ordinates have density $\nu(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$ at height $t$. The two meet at

$$\frac{a}{\pi} = \frac{1}{2\pi} \log \frac{T}{2\pi} \iff T = T_*(a) := 2\pi e^{2a}, \tag{18.4.1}$$

which we call the Landau Widom horizon, or the horizon. Zhu's argument is that below $T_*$ a test function of type $a$ can in principle interpolate the ordinates, placing a zero of $F$ on each of them, and that above $T_*$ the ordinates outnumber

the available oscillations, so that $|F|^2$ must leak onto them. A count of degrees of freedom gives the same height up to a factor. An even function of $PW_a$ concentrated on the frequencies $[0,T]$ has about $aT/\pi$ degrees of freedom, since the time frequency rectangle $[-a,a]\times[-T,T]$ has about $2aT/\pi$ [481, 326], halved for evenness, while $N(T)\approx\frac{T}{2\pi}\log\frac{T}{2\pi e}$ ordinates lie below $T$. Equating the two counts gives $2\pi e^{2a+1}=eT_*$.

Zhu's decay law (Z4) uses $T_*$. He models the cost of killing $N(T_*)$ ordinates at saturated budget as an excursion of order $N(T_*)$ eigenvalues beyond the Shannon number of the window, at the plunge rate $\lambda_{K+j}\approx\exp(-\pi^2 j/\log K)$ of the eigenvalues of the concentration operator of time and band limiting [327]. This gives $-\log\lambda_{\min}^{\mathrm{even}}(\frac12,a)\asymp\pi^2N(T_*)/\log N(T_*)$ up to a constant, and the fit supplies the constant (his Section 9 and Conjecture 12.1). It is his model, fitted and not derived.

The passage from a window statement to a statement about the height of the zeros is a heuristic and not a theorem. Nothing in this book proves that positivity of Weil's form on every window of half length $a$ is implied by, or implies, the verification of the zeros up to a height of order $T_*(a)$, and Problem 18.1 asks for the theorem that the heuristic would need.

A quadruple of off line zeros $\frac12\pm\delta\pm i\gamma$ contributes $4\operatorname{Re}F(\gamma-i\delta)^2$ to $Q(f)$ for real even $f$, and a negative value of $Q$ needs a test function that nearly vanishes at the on line ordinates near $\gamma$ while keeping its energy near $\gamma\pm i\delta$. Heuristically this is possible only when the window has more degrees of freedom near $\gamma$ than the ordinates have, $a/\pi>\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, which predicts a detection threshold

$$a_{\mathrm{det}}(\gamma,\delta)\approx\tfrac12\log\frac{\gamma}{2\pi}+O(1),$$

with a weak dependence on $\delta$ for small $\delta$. Once the test function vanishes at the neighbours and at $\gamma$ itself, the contribution of the quadruple is $4\operatorname{Re}F(\gamma-i\delta)^2=-4\delta^2F'(\gamma)^2+O(\delta^4)$, which is negative for small $\delta>0$, and it must exceed the residual of the on line zeros.

We tested this on a hypothetical configuration (scripts `zh_detect.py` and `zh_zeros.py`). Add to the on line zeros one quadruple $\frac12\pm\delta\pm i\gamma$, where $\gamma$ is the midpoint of the gap between two consecutive ordinates (numbers 10 and 11, 79 and 80, 649 and 650, 1517 and 1518). The matrix is the zero side formula of Theorem 17.7.1 for Yoshida's form on $C(a)$, with the first 2100 ordinates (the last at 2620.28) and the modes $|n|\le\lceil a(\gamma+45)/\pi\rceil+2$, whose frequencies $\pi n/a$ reach 45 beyond the height of the quadruple. The on line zeros contribute a positive semidefinite Gram matrix and the quadruple a real symmetric matrix of rank at most four, so any negativity comes from the quadruple. Two criteria are used. In criterion (i) the entry is the smallest $a$ on a grid of step 0.05 at which the least eigenvalue of the float64 matrix is below $-10^{-11}$. In criterion (ii) it is the smallest $a$ at which the exact least eigenvalue is negative, which is decided in ball arithmetic: in each parity sector the quadruple is a perturbation of rank two of

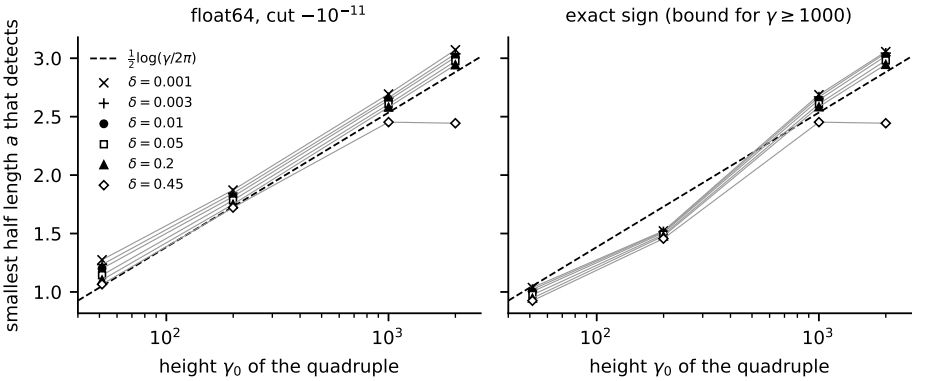



Figure 18.1: Detection thresholds of Table 18.3 and of the displacements $\delta = 0.001$ and 0.003 of the text against the height $\gamma$ of the hypothetical quadruple (logarithmic axis), for $\delta = 0.001$, 0.003, 0.01, 0.05, 0.2, 0.45 (points of equal $\delta$ joined), with the horizon $\frac{1}{2}\log(\gamma/2\pi)$ dashed; the vertical axis is the half length $a$ of the window. Left: criterion (i). Right: criterion (ii), with upper bounds at the heights 1000 and 2000. The points are the thresholds refined by bisection to 0.005, so that they lie up to 0.05 below the entries of the table. Drawn by `code/zh_detect.py` (option `--figure`) from `data/zh_detect.json`.

a positive definite matrix, so the sign is decided by a two by two determinant. At the heights 1000.57 and 1999.99 criterion (ii) is only an upper bound, from the principal submatrix on the modes with $|\pi n/a - \gamma| \leq 150$: by interlacing a detection there is a detection for the full matrix, and a failure to detect proves nothing. The entry of the table is the smaller of this band value and the entry of criterion (i).

The rounding error of the float64 eigenvalue is bounded a priori, from the rounding of the Gram matrix and a textbook bound for the eigensolver, by $4 \times 10^{-12}$ to $5 \times 10^{-12}$ at height 51, $9 \times 10^{-12}$ at 200, about $2.6 \times 10^{-11}$ at 1000 and $4 \times 10^{-11}$ at 2000. The cut $-10^{-11}$ of criterion (i) is therefore safe in double precision at the heights 51 and 200, whereas at 1000 and 2000 the bound exceeds the cut, so that there a detection is certified only when the float64 eigenvalue lies below minus the bound. At the printed value of $a$ every one of the sixteen events is certified: in fourteen cells by the float64 eigenvalue and this bound, in the other two (at $(51.37, 0.01)$ and $(199.64, 0.2)$) by the exact test of the cut. The grid point below the printed value is certified as a non detection at the heights 51 and 200 only. Each entry of criterion (i) is an upper bound, for the model of the table, for the sign change in exact arithmetic.

For $\delta \leq 0.2$ every threshold of criterion (i) lies 0.02 to 0.17 above $\frac{1}{2}\log(\gamma/2\pi)$. For $\delta = 0.45$ it lies above it by 0.05 and 0.02 at the heights 51 and 200 and below it

| $\gamma$ | $\frac{1}{2}\log\frac{\gamma}{2\pi}$ | $\delta = 0.01$ | $\delta = 0.05$ | $\delta = 0.2$ | $\delta = 0.45$ |
|---|---|---|---|---|---|
| Criterion (i): the float64 least eigenvalue falls below $-10^{-11}$ | | | | | |
| 51.37 | 1.051 | 1.20 | 1.15 | 1.15 | 1.10 |
| 199.64 | 1.729 | 1.85 | 1.80 | 1.75 | 1.75 |
| 1000.57 | 2.535 | 2.65 | 2.65 | 2.60 | 2.50 |
| 1999.99 | 2.882 | 3.05 | 3.00 | 2.95 | 2.45 |
| Criterion (ii): the exact least eigenvalue is negative | | | | | |
| 51.37 | 1.051 | 1.05 | 1.00 | 0.95 | 0.95 |
| 199.64 | 1.729 | 1.55 | 1.50 | 1.50 | 1.50 |
| 1000.57 | 2.535 | $\le 2.65$ | $\le 2.65$ | $\le 2.60$ | $\le 2.50$ |
| 1999.99 | 2.882 | $\le 3.05$ | $\le 3.00$ | $\le 2.95$ | $\le 2.45$ |

Table 18.3: Smallest window half length $a$ at which Yoshida's form on $[-a, a]$ detects a hypothetical off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ added to the first 2100 zeros, on a grid of step 0.05 (script `code/zh_detect.py`, record `data/zh_detect.json`). The entries of the lower block are exact on the full matrix at the heights 51.37 and 199.64; at 1000.57 and 1999.99 the entries marked $\le$ are upper bounds, the smaller of the value from a band of modes and the entry of criterion (i) above. At the cell $(1999.99, 0.2)$ the band computation of 150 modes does not detect the quadruple at 2.90 and 2.95 (it does at 3.00, with 2100, 4000 and 8000 zeros and with the margin 90 in place of 45 in the number of modes), so that the entry 2.95 there is the entry of criterion (i); the exact test detects the quadruple in a band of 300 modes at 2.95 and in a band of 600 modes at 2.90.

by 0.04 and 0.43 at the heights 1000 and 2000 (Figure 18.1). Refined by bisection to 0.005 the thresholds of criterion (ii) are 1.003, 0.975, 0.947, 0.925 at $\gamma = 51.37$ and 1.506, 1.491, 1.478, 1.456 at $\gamma = 199.64$, for $\delta = 0.01, 0.05, 0.2, 0.45$, which lie below the horizon by 0.05 to 0.13 and by 0.22 to 0.27. So the horizon is neither an upper bound nor a lower bound for the detection threshold in these data. The table is set by the cut and by the model as much as by the horizon: near the cut the negative eigenvalue grows by a factor $2 \times 10^2$ to $2 \times 10^4$ per step 0.05 in $a$, so that any cut between $10^{-16}$ and $10^{-9}$ moves the thresholds by less than 0.15, but double precision cannot see the sign change itself, because between the exact sign change and the cut the float64 eigenvalue is rounding noise (between $-2 \times 10^{-15}$ and $-5 \times 10^{-15}$ for every $a$ from 1.3 to 1.7 at height 200).

The dependence on $\delta$ is weak for small $\delta$. At $\delta = 0.001$ the thresholds of criterion (i) are 1.30, 1.90, 2.70, 3.10 at the four heights, and at $\delta = 0.003$ they are 1.25, 1.85, 2.70, 3.05. From $\delta = 0.001$ to 0.2 they fall by 0.10 to 0.15, and they are nonincreasing in $\delta$ in every row. So a displacement of 0.01 is detected almost as early as one of 0.2, but a displacement of 0.45 is detected at height 2000 much earlier than the horizon suggests. The weak dependence is a property of

these computations and not a consequence of the expansion above. What lets a large displacement be seen early is the growth of a test function off the real axis: for $F \in PW_a$ bounded on $\mathbb{R}$, the Phragmén Lindelöf bound gives $|F(\gamma - i\delta)|^2 \le e^{2a\delta} \sup_{\mathbb{R}} |F|^2$, which is 9.1 at $a = 2.45$, $\delta = 0.45$. It is an upper bound and not a measured gain, and the data are consistent with the reading that at height 2000 the term of the quadruple, amplified by up to that factor, can beat on line neighbours spaced about 0.9 apart (the gap that contains the quadruple is 0.889, the mean spacing 1.09) before the window resolves them individually. The experiment does not test this reading.

Three features of the model limit what the table says. First, the configuration is not count preserving. The variant that removes the two on line zeros flanking the gap is detected earlier, by 0.15 to 0.30 in each of the eight cells computed ($\delta = 0.01$ and 0.2 at the four heights), and every one of its thresholds lies at or below the horizon, by 0.03 to 0.24. The thresholds of the table are therefore those of a configuration with two extra zeros and not a statement about a configuration with the true zero count. Second, the truncation to 2100 ordinates can only advance detection, because the omitted zeros add positive semidefinite terms. In the eight cells that were also run with 4000 and with 8000 ordinates the float64 thresholds are unchanged, but the rounding bound grows with the number of zeros, and the entry $(1999.99, 0.2)$ is no longer certified at either size (the certified value is 3.00), nor is $(1000.57, 0.01)$ at 8000. Third, with the margin 90 in place of 45 in the number of modes the entry $(1999.99, 0.01)$ is 3.00 and not 3.05.

Compare the tent coefficient $D(x) = x\langle \chi_0, \chi_0 \rangle$ of Chapter 17 at the level $x = 2a$. By Remark 17.6.3 it turns negative near $x_* \approx (2\log\gamma + \log p_1)/\delta$, with $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231049931$. The tent is one vector of the window $C(a)$, so $x_*/2$ is only an upper bound for the half length at which the window detects the quadruple. For $\delta = 0.2$ the level at which the quadruple can first push $D$ negative for some phase, computed from the first 8000 ordinates, gives $x_*/2 = 8.68$, 16.2, 24.2, 27.8 at the four heights, against the thresholds 1.15, 1.75, 2.60, 2.95 of Table 18.3. Over the 16 cells of the table the window thresholds are 4.0 to 177 times smaller than $x_*/2$, the ratio being roughly $2/\delta$ (144 to 177 at $\delta = 0.01$, 29 to 37 at 0.05, 7.5 to 9.4 at 0.2 and 4.0 to 5.1 at 0.45). The formula for $x_*$ overestimates the computed level by 2 to 22 percent. The window data do not support an order $\delta^{-1}\log\gamma$ for the window threshold itself. For the height $3 \times 10^{12}$ the formula gives $x_*/2 = 26.85/\delta$, that is 59.7 at $\delta = 0.45$ and 134 at $\delta = 0.2$, against the half length 13.45 of the horizon, with the same ratio $2/\delta$. Proposition 17.9.1 (level 47) is proved for $D$ only, and Table 18.3 shows that the window is far more sensitive than the coefficient. The level 47 is therefore not the level at which windows begin to see beyond the verification of the zeros, and the conversion from half lengths to levels, here and in Chapter 17, is a heuristic.

The horizon turns a window statement into a height statement only heuristically, and only for small displacements. Large displacements are seen below the

| window statement | half length $a$ | height $2\pi e^{2a}$ |
|---|---|---|
| Yoshida's Theorem 1, as Zhu reports it | 0.347 | 12.6 |
| Zhu's certificate (Z2) | 0.8 | 31.1 |
| practical limit of envelope certificates, Zhu's Remark 1.6 | $\approx 1.6$ | 154 |
| verification of the zeros to $3 \times 10^{12}$ [419] | $\approx 13.4$ | $3 \times 10^{12}$ |
| RH | $\infty$ | $\infty$ |

Table 18.4: The horizon $2\pi e^{2a}$ at the half lengths of some window statements, and the half length $\frac{1}{2}\log(H/2\pi) = 13.45$ at which it equals the height $H = 3 \times 10^{12}$ of the verification of the zeros. The conversion is a heuristic and not a theorem.

horizon at the heights 1000 and 2000 (Table 18.3), and if the margin below the horizon keeps growing beyond the range tested, a window of half length near 13 could see a large displacement far above $3 \times 10^{12}$. Table 18.4 lists the heights $2\pi e^{2a}$ of some window statements.

Yoshida's window lies below the first ordinate: its height $4\pi = 12.57$ is less than $\gamma_1 = 14.13$. Zhu's certificate reaches about the fourth ordinate ($\gamma_4 = 30.42 < 31.12 < \gamma_5 = 32.94$). Under the heuristic, a statement about these windows speaks about the zeros below the height shown, which the verification of the zeros covers far beyond.

## 18.5 The comb barrier, basepoints and damping

Damping to a basepoint $\alpha \geq \frac{1}{2}$ (Section 17.8) replaces $\Lambda(n)n^{-1/2}$ by $\Lambda(n)n^{-\alpha}$ in the comb. Put

$$A_a(\alpha) = \sum_{\log n < 2a} \frac{2\Lambda(n)}{n^{\alpha}}, \qquad T_1(\alpha, a) = 2\pi e^{A_a(\alpha)}, \tag{18.5.1}$$

so that $A_a(\frac{1}{2}) = A_a$ and $T_1(\frac{1}{2}, a) = T_1$. For $\alpha > \frac{1}{2}$ the height $T_1(\alpha, a)$ is a definition by analogy and not the cutoff of a theorem: Zhu's Theorem 1.1 is a statement about the undamped form, no damped analogue is proved here, and for $\frac{1}{2} < \alpha < 1$ the pole part of the damped form, with kernel $2e^{-(\alpha-1/2)|y|}\cosh(y/2)$, grows like $e^{(1-\alpha)|y|}$ and is not a positive rank one term. By the prime number theorem, as $a \to \infty$,

$$\begin{gathered} A_a(\alpha) \sim \frac{2e^{2(1-\alpha)a}}{1-\alpha} \quad (\tfrac{1}{2} \leq \alpha < 1), \qquad A_a(1) = 4a - 2\gamma_E + o(1), \\ A_a(\alpha) \text{ increases to } -2\frac{\zeta'}{\zeta}(\alpha) \quad (\alpha > 1). \end{gathered} \tag{18.5.2}$$

Table 18.5 lists the values, computed by `zh_comb.py` (`code/README.md`).

| | $\alpha=\frac12$ | | $\alpha=0.75$ | | $\alpha=1$ | |
|---|---|---|---|---|---|---|
| $a$ | $A_a(\alpha)$ | $\log_{10}T_1$ | $A_a(\alpha)$ | $\log_{10}T_1$ | $A_a(\alpha)$ | $\log_{10}T_1$ |
| 0.35 | 0.980 | 1.22 | 0.824 | 1.16 | 0.693 | 1.10 |
| 0.80 | 2.942 | 2.08 | 2.278 | 1.79 | 1.772 | 1.57 |
| 1.60 | 14.323 | 7.02 | 8.497 | 4.49 | 5.222 | 3.07 |
| 2.00 | 24.383 | 11.39 | 12.566 | 6.26 | 6.874 | 3.78 |
| 3.00 | 75.190 | 33.45 | 26.647 | 12.37 | 10.856 | 5.51 |
| 4.00 | 213.548 | 93.54 | 49.930 | 22.48 | 14.855 | 7.25 |
| 5.00 | 588.024 | 256.17 | 88.183 | 39.10 | 18.844 | 8.98 |

Table 18.5: The comb constant $A_a(\alpha)$ of (18.5.1) and the barrier height $T_1(\alpha,a)=2\pi e^{A_a(\alpha)}$ at three basepoints. The column $\alpha=\frac12$ is Zhu's [526], who prints $A_{0.8}(\frac12)=2.9420$ and $T_1=119$ at $a=0.8$, $1.0\times10^7$ at $a=1.6$ and $2.4\times10^{11}$ at $a=2$; the digits here are 2.94197, 119, $1.04\times10^7$ and $2.44\times10^{11}$. For $\alpha>1$ the constant is bounded by $-2\zeta'(\alpha)/\zeta(\alpha)=18.88$, 3.01, 1.14 at $\alpha=1.1$, 1.5, 2.

Corollary 18.5.1. *Let $\alpha\ge1$, $a>0$ and $f\in C(a)$, and write $F=\widehat{f}$ and $b_\alpha=\xi'(\alpha)/\xi(\alpha)$. With no hypothesis on the zeros,*

$$\langle f,f\rangle_\alpha=\frac{1}{\pi}\int_{\mathbb{R}}|F(t)|^2\,\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)\,dt\;\ge\;2b_\alpha\|f\|^2,\qquad b_\alpha=\inf_{t\in\mathbb{R}}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)>0,$$

*and the constant $2b_\alpha$ does not depend on $a$.*

*First proof, $\alpha\ge1$.* Let $\sigma_{\max}=\sup_\rho\operatorname{Re}\rho$ be the largest real part of a zero (Section 17.8.1). Then $\sigma_{\max}\le1\le\alpha$ and no zero has real part $\alpha$, so Proposition 17.8.1 gives the identity, and Theorem 17.8.5(i) gives the inequality with $m_\alpha=\inf_t\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$. By part (ii) of that theorem, which for $\alpha\ge1$ needs no hypothesis (Propositions 9.1.11 and 9.1.12), $m_\alpha=b_\alpha$, and $b_\alpha>0$ because the density $v_\alpha=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\cdot)$ is positive (Theorem 1.2.2). □

*Second proof, $\alpha>1$.* Let $g=f*\widetilde{f}$, which is supported in $[-2a,2a]$ and has $\widehat{g}=|F|^2$. By (17.1.1) with the damping $e^{-(\alpha-1/2)|y|}$, the form is the sum of a pole part, an archimedean part and the prime part $-\sum_{n\ge2}\Lambda(n)n^{-\alpha}\big(g(\log n)+g(-\log n)\big)$. Since $\int_{\mathbb{R}}|F(t)|^2\cos(t\log n)\,dt=\pi\big(g(\log n)+g(-\log n)\big)$, which vanishes for $\log n>2a$, and since for $\alpha>1$ the full comb $\sum_{n\ge2}2\Lambda(n)n^{-\alpha}\cos(t\log n)$ converges absolutely and uniformly in $t$, the prime part equals $-\frac{1}{2\pi}\int|F|^2\sum_{n\ge2}2\Lambda(n)n^{-\alpha}\cos(t\log n)\,dt$ with the full comb, inserted term by term. The pole part is $\int g(y)\big(e^{-(\alpha-1)|y|}+e^{-\alpha|y|}\big)dy$, which is $\frac{1}{2\pi}\int|F|^2\,2\operatorname{Re}\big(\frac1s+\frac{1}{s-1}\big)dt$ at $s=\alpha+it$, because $e^{-w|y|}$ has Fourier transform $2w/(w^2+t^2)$. The archimedean part is the Poisson smoothing by $P_{\alpha-1/2}$ of the multiplier $\operatorname{Re}\psi(\frac14+\frac{it}{2})-\log\pi$, and this is $\operatorname{Re}\psi(\frac s2)-\log\pi$ at $s=\alpha+it$, because $z\mapsto\psi(\frac14+\frac{iz}{2})$ is holomorphic for $\operatorname{Im}z<\frac12$.

Now $\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)$ and $-\frac{\zeta'}{\zeta}(s) = \sum\Lambda(n)n^{-s}$ for $\operatorname{Re} s > 1$, so the three multipliers add up to $2\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$, and $\langle f,f\rangle_\alpha = \frac{1}{2\pi}\int|F|^2\,2\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)\,dt$. This symbol is positive (Theorem 1.2.2) and tends to infinity like $\frac{1}{2}\log|t|$, because $\operatorname{Re}\frac{\zeta'}{\zeta}$ is bounded on the line $\operatorname{Re} s=\alpha$; its infimum is therefore positive and equals $2b_\alpha$ by Theorem 17.8.5(ii), and Plancherel, $\int|F|^2 = 2\pi\|f\|^2$, gives the bound. □

For $\alpha \geq 1$ no certificate is needed: the damped form is positive on every window (Theorem 17.8.3(i)), although the envelope height $T_1(1,a)$ still grows like $e^{4a}$ (Table 18.5). The line $\operatorname{Re} s = 1$ is the same boundary as in the basepoint picture: the shifted clock $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is a GGC with Thorin density $v_\alpha > 0$ for every $\alpha \geq 1$ with no hypothesis, and for $\alpha < 1$ exactly when $\alpha \geq \sigma_{\max}$ (Theorem 1.2.2). The one sided bound on $\zeta'/\zeta$ that corresponds to $v_\alpha > 0$ is the bound of Theorem 16.12.1(ii), which for $\alpha < 1$ holds under the hypothesis of that theorem.

For $\frac{1}{2} \leq \alpha < 1$ the full comb diverges, since $\sum\Lambda(n)n^{-\alpha}$ diverges, and the second proof is not available. The symbol $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$ still exists by analytic continuation, and the damped form is its multiplier whenever $\alpha \geq \sigma_{\max}$ and no zero has real part $\alpha$ (Proposition 17.8.1). Positivity of the symbol on this one line follows from the absence of zeros with real part at least $\alpha$ and is not known to imply it (Remark 1.2.4), whereas positivity of the damped form on every window is equivalent to the absence of zeros with real part greater than $\alpha$ (Theorem 20.1.2, Remark 17.8.4). What the comb would have to supply there is the one sided bound for $-\operatorname{Re}\zeta'/\zeta(\alpha+it)$ of Section 17.10, that is cancellation in $\sum\Lambda(n)n^{-\alpha-it}$ beyond what absolute convergence gives, which is Problem 17.1 of Chapter 17.

### 18.5.1 Damping removes the collapse

Write $\lambda_{\min}(\alpha,a)$ for the least eigenvalue of the damped form of Section 17.8 on the window $[-a,a]$, over complex test functions in $C(a)$ (Section 17.8.2), and $\lambda_{\min}^{\text{even}}(\alpha,a)$ for the infimum of $\langle f,f\rangle_\alpha/\|f\|^2$ over real even $f \in C(a)$. By the splitting of Section 18.1, $\lambda_{\min}(\alpha,a) \leq \lambda_{\min}^{\text{even}}(\alpha,a)$.

Let $p_1 = \frac{1}{2}(\log\xi)''(\frac{1}{2}) = 0.0231049931\ldots$, which is $\sum_{\gamma>0}\gamma^{-2}$ under RH, and $p_2 = 3.7172599\ldots\times10^{-5}$, which is $\sum_{\gamma>0}\gamma^{-4}$ under RH, as in Theorem 17.8.5.

Theorem 18.5.2 (Two parameter bound). *Let $a > 0$ and $\alpha > \frac{1}{2}$, and write $b_\alpha = \xi'(\alpha)/\xi(\alpha)$.*

*(i) With no hypothesis, $\lambda_{\min}(\alpha,a) \geq \lambda_{\min}(\frac{1}{2},a)$.*

*(ii) If RH holds, or if $\alpha \geq 1$, then $\lambda_{\min}(\alpha,a) \geq 2b_\alpha$. Here $2b_\alpha \leq 4p_1(\alpha-\frac{1}{2})$ with no hypothesis, and under RH $2b_\alpha \geq 4p_1(\alpha-\frac{1}{2}) - 4p_2(\alpha-\frac{1}{2})^3$.*

*Hence, under RH,* $\lambda_{\min}(\alpha,a) \geq \max\bigl(\lambda_{\min}(\frac12,a),\, 2b_\alpha\bigr)$ *for every* $\alpha > \frac12$ *and every* $a > 0$.

*Proof.* (i) Put $\varepsilon = \alpha - \frac12$ and recall $e^{-\varepsilon|y|} = \int_{\mathbb{R}} P_\varepsilon(\lambda) e^{i\lambda y} d\lambda$ with the Poisson kernel $P_\varepsilon(\lambda) = \varepsilon/(\pi(\varepsilon^2+\lambda^2))$. For $f \in C(a)$ put $f_\lambda(y) = e^{i\lambda y} f(y) \in C(a)$, so that $f_\lambda * \widetilde{f_\lambda} = e^{i\lambda y}\,(f * \widetilde{f})$. The interchange of $\mathcal{Y}$ and the integral over $\lambda$ is that of the proof of Theorem 17.8.3(iii), taken from the basepoint $\frac12$, and gives

$$\begin{aligned}\langle f,f\rangle_\alpha = \mathcal{Y}\bigl(e^{-\varepsilon|y|} f * \widetilde{f}\bigr) &= \int_{\mathbb{R}} P_\varepsilon(\lambda)\langle f_\lambda, f_\lambda\rangle_{1/2}\, d\lambda \\ &\geq \lambda_{\min}(\tfrac12,a)\int_{\mathbb{R}} P_\varepsilon(\lambda)\|f_\lambda\|^2 d\lambda = \lambda_{\min}(\tfrac12,a)\|f\|^2,\end{aligned}$$

because $\|f_\lambda\| = \|f\|$ and $\int P_\varepsilon = 1$.

(ii) Under RH, $\sigma_{\max} = \frac12$ and no zero has real part $\alpha > \frac12$; for $\alpha \geq 1$, $\sigma_{\max} \leq 1 \leq \alpha$ and no zero has real part $\alpha$. In both cases Theorem 17.8.5(i) gives $\lambda_{\min}(\alpha,a) \geq 2m_\alpha$, and part (ii) of that theorem gives $m_\alpha = b_\alpha$, under RH and with no hypothesis for $\alpha \geq 1$. The bounds on $2b_\alpha$ are Theorem 17.8.5(iv). $\square$

At $a = 0.8$ the first bound is certified: Corollary 6.3 of [526] gives $\lambda_{\min}(\frac12, 0.8) \geq 8.9 \times 10^{-18}$ for complex test functions, so by (i) $\langle f,f\rangle_\alpha \geq 8.9\times10^{-18}\|f\|^2$ for every $\alpha \geq \frac12$ and every $f \in C(0.8)$, with no hypothesis. The second bound is not a multiple of $4p_1(\alpha-\frac12)$ but only comparable to it: $2b_\alpha$ lies between $4p_1(\alpha-\frac12) - 4p_2(\alpha-\frac12)^3$ and $4p_1(\alpha-\frac12)$, so under RH a bound $\lambda_{\min}(\alpha,a) \geq 4p_1(\alpha-\frac12)$ would fail for large $a$, where $\lambda_{\min}(\alpha,a) \to 2b_\alpha$ (Theorem 17.8.5(i) and (ii)). Unconditionally the second bound is available for $\alpha \geq 1$ only (Corollary 18.5.1); at a single $\alpha < 1$ with no zero on the line the inequality $\lambda_{\min}(\alpha,a) \geq 2b_\alpha$ for all $a$ is equivalent to $\sigma_{\max} \leq \alpha$ together with $v_\alpha \geq b_\alpha$ on the line (Remark 17.8.6).

The theorem has a numerical counterpart for the windows of Table 18.1. The matrices of that table, with the damping, give upper bounds for $\lambda_{\min}^{\mathrm{even}}(\alpha,a)$ (computed at 60 digits, for 26 and 40 modes, `code/README.md`). In all 44 values, for $a = 0.5$ and $0.8$ and $\alpha - \frac12$ between $10^{-17}$ and $10^{-3}$, the finite section value lies above both bounds of the theorem: the least ratio to the floor $2b_\alpha$ is 1.030 at $a = 0.8$, $\alpha - \frac12 = 10^{-4}$ and 1.121 at $a = 0.5$, $\alpha - \frac12 = 10^{-3}$, and the least ratio to the undamped value of the same size is 1.011 at $a = 0.5$, $\alpha - \frac12 = 10^{-7}$ and 1.047 at $a = 0.8$, $\alpha - \frac12 = 10^{-17}$.

The damped value is the sum of the two terms and not their maximum. For $\alpha - \frac12 = \varepsilon$ the difference $\lambda_{\min}^{\mathrm{even}}(\frac12+\varepsilon,a) - \lambda_{\min}^{\mathrm{even}}(\frac12,a)$ equals $s \cdot 4p_1\varepsilon$ with $s = 1.111$ at $a = 0.5$ for $\varepsilon$ from $10^{-7}$ to $10^{-4}$ and $s = 1.082$ at $a = 0.8$ for $\varepsilon$ from $10^{-17}$ to $10^{-14}$, the same at 26 and 40 modes to the digits quoted. For larger $\varepsilon$ at $a = 0.8$ the factor $s$ falls, to 1.067, 1.043, 1.040 and 1.030 at $\varepsilon = 10^{-10}$, $10^{-8}$, $10^{-6}$ and $10^{-4}$, so that from $\varepsilon = 10^{-8}$ on the finite section value lies within 4.5 percent of the floor $2b_\alpha$; the fall begins at $\varepsilon$ of order $10^{-10}$, where $\varepsilon$ times the slope is of the size of the gap of about $10^{-11}$ to the second even eigenvalue.

Far above the crossover the least eigenvalue is close to the floor, as the converged entries of Table 17.2 show.

The two terms of the theorem are equal, to first order in the damping, at the crossover $\varepsilon_\times = \lambda^{\text{even}}_{\min}(\frac{1}{2}, a)/(4p_1)$, at which the least eigenvalue is about twice $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$ (the damping at which it exactly doubles is 0.90 times $\varepsilon_\times$ at $a = 0.5$ and 0.92 times $\varepsilon_\times$ at $a = 0.8$, for 26 modes). Its value depends on which number is used for $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$, and Table 18.6 gives the choices. The 26 mode values are not converged.

| $a$ | value used for $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$ | $\lambda$ | $\lambda/(4p_1)$ |
|---|---|---|---|
| 0.5 | Zhu, certified upper bound | $1.05 \times 10^{-6}$ | $1.14 \times 10^{-5}$ |
| 0.5 | 26 modes, Table 18.1 | $9.743 \times 10^{-7}$ | $1.05 \times 10^{-5}$ |
| 0.5 | extrapolated from 256 to 400 modes | $9.33 \times 10^{-7}$ | $1.01 \times 10^{-5}$ |
| 0.8 | Zhu, certified lower bound | $8.9 \times 10^{-18}$ | $9.6 \times 10^{-17}$ |
| 0.8 | extrapolated from 256 to 400 modes | $1.63 \times 10^{-17}$ | $1.77 \times 10^{-16}$ |
| 0.8 | 26 modes, Table 18.1 | $2.144 \times 10^{-17}$ | $2.32 \times 10^{-16}$ |
| 0.8 | Zhu, certified upper bound | $2.27 \times 10^{-17}$ | $2.46 \times 10^{-16}$ |

Table 18.6: The crossover $\varepsilon_\times = \lambda/(4p_1)$ of Theorem 18.5.2 for several values of $\lambda = \lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$, with $4p_1 = 0.0924200$. The extrapolated values are three point power law fits to the finite sections of 60 digit matrices (script `code/zh_window_eigs.py`, record `data/zh_window_eigs.json`), not bounds; Zhu's reference values of his Table 1, $9.34 \times 10^{-7}$ and $1.65 \times 10^{-17}$, give $1.01 \times 10^{-5}$ and $1.79 \times 10^{-16}$.

The explanation of the collapse is the one given at the end of Section 17.8. A damping of the size of the crossover already doubles the finite section values (computed); for larger damping the least eigenvalue is, under RH, at least $2b_\alpha$, a margin of order $\alpha - \frac{1}{2}$ in place of $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$. Whether certificates for the damped forms say more or less about the zeros than the certificates at $\alpha = \frac{1}{2}$ is not settled here (Problem 18.3).

## 18.6 What window certificates reach

A window of half length $a$ resolves the zeros up to the horizon $2\pi e^{2a}$ (a heuristic, Section 18.4, after Section 9 of [526]), and whether the verification of the zeros up to a height $H$ implies positivity on the window of half length $\frac{1}{2}\log(H/2\pi) - C$ is Problem 18.1; for large displacements it may fail.

## 18.7 Open problems

Problem 18.1 (The horizon as a theorem). Determine the largest half length $a(H, \delta_0)$ such that, if all zeros with $0 < \gamma \le H$ lie on the line and every zero above $H$ has $|\beta - \frac{1}{2}| \le \delta_0$, Weil's form is positive on the window $[-a, a]$ with $a = a(H, \delta_0)$.

Is $a(H,\delta_0) \geq \frac{1}{2}\log\frac{H}{2\pi} - C(\delta_0)$, with a constant $C(\delta_0)$ that depends only on $\delta_0$? Status: open. The heuristic of Section 18.4 suggests the bound for small $\delta_0$. A window gains up to a factor $e^{2a\delta}$ on a zero at distance $\delta$ from the line, and in Table 18.3 the largest displacement is detected below the horizon at the largest heights, so a constant $C$ that does not depend on $\delta_0$ may not exist. A proof would need a quantitative form of the lower sampling inequality of Proposition 18.2.1 for the ordinates, with constants that depend only on the local density of the zeros. Compare Proposition 17.9.1, which is proved for the tent coefficient only.

Problem 18.2 (Decay law from the Thorin side). Derive the exponent of Zhu's decay law, $-\log\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a) = C\,N(T_*)/\log N(T_*)\,(1+o(1))$ with $T_* = 2\pi e^{2a}$ and the fitted constant $C = 20.13\ldots$ of his equation (5) (his Conjecture 12.1 [526] states the law with $C = 2\pi^2$), from Zhu's test function $\widehat{f} = h\,G_T(-r^2)$ of Section 18.3, optimising $T$ and $h$ jointly, and explain the constant, which Zhu finds close to $2\pi^2 = 19.74\ldots$, through the Landau Widom plunge. Status: open. Zhu fits the law to certified upper bounds for $a \leq 1.2$, tests it on $a = 1.4$ to 2, and states that the constant is fitted and not derived. The only proved decay beyond the bound $O(a^{-N})$ for every $N$ of Theorem 20.2.1(c) of Chapter 20 is his bound $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a) \leq \exp(-ae^a)$ for large $a$, under RH (Theorem 1.3), which kills the zeros only up to $2\pi e^a$, half the exponent of $T_*$, and is weaker than the law.

Problem 18.3 (Certified damped windows). Extend the certification of Zhu's Theorem 1.2 to the damped forms at $\alpha \in (\frac{1}{2},1)$, where under RH the margin is at least $2\xi'(\alpha)/\xi(\alpha)$, of order $\alpha - \frac{1}{2}$, in place of $\lambda_{\min}(\frac{1}{2},a)$ (Theorem 18.5.2), and measure how far $a$ can be pushed. Status: open. No damped analogue of Zhu's Theorem 1.1 is proved here, and below $\alpha = 1$ the damped pole part is not a positive rank one term (Section 18.5); for $\alpha \geq 1$ no certificate is needed (Corollary 18.5.1). The question for the rungs of Chapter 20 is Problem 20.1.

Problem 18.4 (Comb cancellation on average). Zhu's barrier concerns the supremum of the comb $\mathcal{P}_a(t) = \sum_{\log n<2a} 2\Lambda(n)n^{-1/2}\cos(t\log n)$. The form sees the comb only through $\int |F|^2\mathcal{P}_a$ with $F$ in the Paley Wiener space $PW_a$ of Proposition 18.2.1, an average over a frequency band of width about $1/a$. Determine the supremum of $\int_{|t|\geq T}|F|^2\mathcal{P}_a \big/ \int_{|t|\geq T}|F|^2$ over $F \in PW_a$, as a function of $T$ and $a$. Status: open. Zhu's Lemma 3.2 shows that the pointwise supremum of $\mathcal{P}_a$ is $A_a$, which bounds the ratio from above. The average over the whole line is smaller than $A_a$ for every $F$, because $g = f * \widetilde{f}$ vanishes at $\pm 2a$ and damps the terms with $\log n$ near $2a$; how much smaller the restricted ratio is, uniformly in $T$, is not known, and a bound of smaller order than $A_a = (4+o(1))e^a$ would be a statement about short sums of $\Lambda(n)n^{-1/2-it}$, the object of the Fejér tapered sums of Proposition 14.14.1.

## Notes

The theorems of Zhu quoted in this chapter are from [526] (version 2 of 2 September 2026), which the chapter reads in the language of the book and does not reprove; they and his heuristics (the horizon and the plunge rate of his Section 9, the parity of the ground state) are credited where they are stated. Yoshida's form is [525], Weil's criterion is [512, 55], and the verification of the zeros to $3 \times 10^{12}$ is [419]. The computations of the chapter are its own: the table of least eigenvalues in 60 digit arithmetic (Table 18.1), the optimisation of Zhu's product test function (Section 18.3), the comb table (Table 18.5), the detection experiment (Table 18.3 and Figure 18.1) and the damping numbers (Table 18.6). `code/README.md` lists the scripts.

# Chapter 19

# Three families of generalised gamma convolutions and the deconvolution obstruction

Three one parameter families of generalised gamma convolutions (GGCs) contain the clock of Riemann's $\xi$ as a boundary case: the Esscher tilt, the basepoint family, which smooths the zero counting measure by Cauchy kernels, and the de Bruijn Newman heat flow; in each the Thorin property propagates away from the centre, and RH asks for it at the centre.

*What is not claimed.* Nothing here proves RH, and the word obstruction in the title names what the theorems on the three families have in common, not a theorem of its own. A number of the chapter's own is computed in floating point at the precision stated. Where a sentence reads a theorem or an identity as saying what a route needs, it is a heuristic and is marked as one.

## 19.1 Three families, and the tilt

Let $\xi$ be Riemann's function, with zeros $\rho = \beta + i\gamma$, and $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$. The centre clock is the variable $T \geq 0$ with

$$\mathbb{E}\big[e^{-uT}\big] = \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{u})}, \qquad u \geq 0,$$

which exists with no hypothesis (Theorem 5.1.1) and is a GGC exactly under RH, with Thorin measure $\sum_{\gamma>0} \delta_{\gamma^2}$ (Proposition 1.2.1). Heights are written $h$ in this chapter. A basepoint $\alpha = \frac{1}{2} + \varepsilon$ is also called the level $\varepsilon$, the resolution of Section 1.2. Put

$$\sigma_{\max} = \sup_{\rho} \operatorname{Re} \rho \in [\tfrac{1}{2}, 1],$$

so that RH is $\sigma_{\max} = \frac{1}{2}$. Since $\beta < 1$ for every zero, every level $\varepsilon \geq \frac{1}{2}$ lies strictly to the right of every zero, $\varepsilon > \beta - \frac{1}{2}$, whether or not $\sigma_{\max} < 1$; the latter is the

quasi Riemann hypothesis, and the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [398] (Section 12.1.2), not yet refereed, would give $\sigma_{\max} \le \frac{7}{8}$.

The book meets the clock inside three one parameter families, each containing it as the boundary case $p = 0$:

(A) the *tilt* $u_0 \ge 0$, the Esscher transform $\xi(\frac{1}{2} + \sqrt{u_0})/\xi(\frac{1}{2} + \sqrt{u_0 + u})$ (Theorem 9.2.2, Corollary 9.2.3 and Proposition 8.3.2 at $u_0 = \frac{1}{4}$);

(B) the *basepoint* $\alpha = \frac{1}{2} + \varepsilon$, the shifted clock $\xi(\alpha)/\xi(\alpha + \sqrt{u})$, whose Thorin density is a sum of Cauchy kernels over the zeros (Theorem 1.2.2 and Proposition 1.2.3);

(C) the *heat time* $t \ge 0$, the clock of the de Bruijn Newman function $H_t$ (Section 8.4.1 and Theorem 7.2.16).

This chapter adds the converse half of the Poisson semigroup and the deconvolution form of RH (Theorems 19.2.1(b) and 19.2.2), the measured exponents of Table 19.1 for the smoothed prime sums, and, in Section 19.3, the visibility curve at every level with its closed forms in three lattice models and its level count.

Proposition 19.1.1 (Invertible families do not localise the difficulty). *Let $\{S_p\}_{p \ge 0}$ be maps on a set $\mathcal{D}$ of Laplace transforms, with $S_0$ the identity. If each $S_p$ is injective on $\mathcal{D}$, maps GGC Laplace transforms in $\mathcal{D}$ to GGC Laplace transforms, and has an inverse on $S_p(\mathcal{D})$ that does the same, then for every $\varphi \in \mathcal{D}$ and every $p$, $S_p\varphi$ is a GGC Laplace transform exactly when $\varphi$ is.*

*Proof.* If $\varphi$ is a GGC transform so is $S_p\varphi$, and if $S_p\varphi$ is one so is $S_p^{-1}S_p\varphi = \varphi$. □

In such a family the Thorin property is equally hard at every parameter: knowing it at some $p > 0$ would already prove RH. The tilt is a family of this kind, and no unconditional point is known in it. The basepoint and heat families are not of this kind: the property is known at $\varepsilon \ge \frac{1}{2}$ and at $t \ge 0.2$, and their inverses do not preserve it. The two fail in opposite ways. The inverse of the Cauchy smoothing is an unbounded operator, with Fourier multiplier $e^{a|x|}$, so the step towards RH is an ill posed computation. The inverse of the heat flow is bounded, a Gaussian average, so the step towards RH is a stable computation; but by Corollary 19.4.2 a property that Gaussian averaging preserves and that implies real zeros holds for no $H_t$.

### 19.1.1 The tilt: invertible and uninformative

Write $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 1.2, an entire function of $u$, and for $u_0 \ge 0$

$$\widetilde{\varphi}_{u_0}(u) = \frac{G(u_0)}{G(u_0 + u)}, \qquad u \ge 0,$$

the Laplace transform of the clock tilted by $e^{-u_0 T}$, that is of the variable $\widetilde{T}_{u_0}$ with law $e^{-u_0 t}P(T \in dt)/\mathbb{E}e^{-u_0 T}$ (the $\widetilde{T}$ of Theorem 9.2.2 for $u_0 = \frac{1}{4}$). Each pair $\{\rho, 1-\rho\}$ of zeros gives the number $a_\rho = -(\rho - \frac{1}{2})^2 = \gamma_\rho^2 = \gamma^2 - (\beta - \frac{1}{2})^2 - 2i\gamma(\beta - \frac{1}{2})$, the $a_k$ of (1.2.1), where $\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ is the complex ordinate of $\rho$, and Hadamard's product (1.2.1) gives $G(u) = \prod(1 + u/a_\rho)$, absolutely convergent since $\sum |a_\rho|^{-1} < \infty$. On the line $a_\rho = \gamma^2$.

THEOREM 19.1.2 (The tilt family). *(a) For $u_1 \geq 0$,*

$$\widetilde{\varphi}_{u_0+u_1}(u) = \frac{\widetilde{\varphi}_{u_0}(u_1 + u)}{\widetilde{\varphi}_{u_0}(u_1)},$$

*so the forward tilt is the Esscher transform by $e^{-u_1 \widetilde{T}_{u_0}}$, and it maps GGCs to GGCs, moving the Thorin measure up by $u_1$. Backward, if $\widetilde{\varphi}_{u_0+u_1}$ is a GGC transform with Thorin measure in $(u_1, \infty)$, then*

$$\widetilde{\varphi}_{u_0}(u) = \frac{\widetilde{\varphi}_{u_0+u_1}(u - u_1)}{\widetilde{\varphi}_{u_0+u_1}(-u_1)}$$

*is a GGC transform, with the Thorin measure moved down by $u_1$; the denominator is $G(u_0 + u_1)/G(u_0)$, finite and positive.*

*(b) $\widetilde{\varphi}_{u_0}$ is the Laplace transform of a GGC exactly when every $u_0 + a_\rho$ is real and positive. In particular* $\mathrm{Im}(u_0 + a_\rho) = -2\gamma(\beta - \frac{1}{2})$ *does not depend on $u_0$.*

*(c) For $\xi$, the Thorin property at one tilt is equivalent to the property at every tilt and to RH.*

*Proof.* (a) The identity is the definition. If $\varphi(u) = \exp(-du - \int \log(1 + u/z)\, U(dz))$, then $\varphi(u_1 + u)/\varphi(u_1) = \exp(-du - \int \log(1 + u/(z + u_1))\, U(dz))$, a GGC with the same drift and the Thorin measure moved up by $u_1$. Backward, if $U$ lies in $(u_1, \infty)$ and $\varphi(-u_1)$ is finite, as it is here, then $\varphi(u - u_1)/\varphi(-u_1) = \exp(-du - \int \log(1 + u/(z - u_1))\, U(dz))$, a GGC with the measure moved down. (b) If every $u_0 + a_\rho$ is real and positive, $\widetilde{\varphi}_{u_0}(u) = \prod(u_0 + a_\rho)/(u_0 + a_\rho + u)$ is the transform of $\sum_\rho e_\rho/(u_0 + a_\rho)$ with independent unit exponentials $e_\rho$, a GGC. Conversely a GGC transform extends to an analytic zero free function on $\mathbb{C} \setminus (-\infty, 0]$, while $\widetilde{\varphi}_{u_0}$ has poles at $u = -(u_0 + a_\rho)$, so each of these lies in $(-\infty, 0)$. (c) The zeta function has no real zeros in $(0, 1)$, so $\gamma \neq 0$ for every zero, and $\xi(\frac{1}{2}) \neq 0$. By (b) the property at any $u_0$ says that every $a_\rho$ is real, that is $\beta = \frac{1}{2}$; conversely, under RH every $a_\rho = \gamma^2 > 0$, and (b) gives the property at every $u_0 \geq 0$. □

Theorem 21.2.3 states the equivalence for the Dedekind zeta function of every number field and every $u_0 \geq 0$, and Theorem 19.1.2 is its case of $\mathbb{Q}$.

In the chart $\theta = \sqrt{z}$ of the Thorin measure the tilt is the deformation $\theta \mapsto \sqrt{\theta^2 + u_0}$ of the atoms, not a smoothing. What the tilt can move is real zeros: for

the Dedekind zeta function $\zeta_K$ of a quadratic field $K$ with $\zeta_K(\frac12)\neq 0$, a real zero $\beta\in(\frac12,1)$ of $\zeta_K$, which is a real zero of $L(s,\chi)$ for the quadratic character $\chi$ of $K$, gives a real $a_\rho=-(\beta-\frac12)^2<0$, admissible exactly when $u_0>(\beta-\frac12)^2$.

## 19.2 The basepoint family: a Poisson semigroup to the right of the zeros

For real $\varepsilon$ and $\theta$ such that $\frac12+\varepsilon+i\theta$ is not a zero of $\xi$, let

$$\rho_\varepsilon(\theta)=\frac{1}{\pi}\,\mathrm{Re}\,\frac{\xi'}{\xi}\big(\tfrac12+\varepsilon+i\theta\big)$$

be the density of (1.2.7), now for every real $\varepsilon$ and not only for $\varepsilon>0$, and let $P_a(x)=a/(\pi(a^2+x^2))$ for real $a\neq 0$ be the Poisson kernel of Section 1.2, with $P_a*P_b=P_{a+b}$ for $a,b>0$. For $\varepsilon>0$ the shifted clock at the basepoint $\frac12+\varepsilon$ has Thorin density $\rho_\varepsilon$ in the chart $\theta=\sqrt{z}$ whenever it is a GGC and no zero lies on the line $\mathrm{Re}\,s=\frac12+\varepsilon$ (Theorem 1.2.2). The function $(\theta,\varepsilon)\mapsto\rho_\varepsilon(\theta)$ is harmonic off the zeros, odd in $\varepsilon$ by the functional equation, and Hadamard's product gives the absolutely convergent expansion

$$\rho_\varepsilon(\theta)=\sum_\rho P_{\varepsilon-(\beta-\frac12)}(\theta-\gamma),\qquad \varepsilon\neq\beta-\tfrac12, \tag{19.2.1}$$

over all zeros with multiplicity (Proposition 1.2.3). A zero with $\beta-\frac12<\varepsilon$ contributes a positive Cauchy density and one with $\beta-\frac12>\varepsilon$ a negative one.

Theorem 19.2.1 (Where the Poisson semigroup holds). *(a) If $\varepsilon_0>\beta-\frac12$ for every zero, then $\rho_{\varepsilon_0+a}=P_a*\rho_{\varepsilon_0}$ for every $a>0$, and $\rho_\varepsilon>0$ for every $\varepsilon\geq\varepsilon_0$. This holds for every $\varepsilon_0\geq\frac12$, with no hypothesis.*

*(b) If $\varepsilon_0\geq 0$ lies at positive distance from the set $\{\beta-\frac12\}$ and some zero has $\beta-\frac12>\varepsilon_0$, then $\rho_{\varepsilon_0+a}\neq P_a*\rho_{\varepsilon_0}$ for some $a>0$.*

*Proof.* (a) Every term of (19.2.1) at $\varepsilon_0$ is a Cauchy density $P_c(\theta-\gamma)$ with $c=\varepsilon_0-(\beta-\frac12)>0$, and $P_a*P_c=P_{a+c}$. The terms are positive, so the convolution passes through the sum by monotone convergence. For $\varepsilon_0\geq\frac12$, $c\geq 1-\beta>0$. (b) Now $|c|\geq c_0>0$ for every term, and $|c|<1$ because $0\leq\varepsilon_0<\frac12$, so $|P_c(\theta-\gamma)|\leq\min\big((\pi c_0)^{-1},(\pi(\theta-\gamma)^2)^{-1}\big)$, and by the Riemann von Mangoldt formula the sum is $O(\log(2+|\theta|))$. Hence $w(\theta,a)=(P_a*\rho_{\varepsilon_0})(\theta)$ is defined and harmonic in $a>0$. If $w(\cdot,a)=\rho_{\varepsilon_0+a}$ for every $a>0$, then off the zeros $\rho_\varepsilon(\theta)$ would agree with a function harmonic on the whole half plane $\{\varepsilon>\varepsilon_0\}$, and in particular bounded near every point $(\gamma,\beta-\frac12)$ with $\beta-\frac12>\varepsilon_0$. But near a zero, $\xi'/\xi(s)=m_\rho/(s-\rho)+O(1)$, so $\rho_\varepsilon(\theta)$ is unbounded there. □

Part (a) is the semigroup of Remark 15.3.4. Part (b) is its converse (a rightmost zero need not exist).

THEOREM 19.2.2 (Deconvolution form of RH). *Let $\varepsilon_0 > \beta - \frac{1}{2}$ for every zero (for instance $\varepsilon_0 \geq \frac{1}{2}$), and let $0 \leq \varepsilon_1 < \varepsilon_0$.*

*(a) There is a positive measure $\nu$ on $\mathbb{R}$ with $\int (1+x^2)^{-1}\nu(dx) < \infty$ and $\rho_{\varepsilon_0} = P_{\varepsilon_0-\varepsilon_1} * \nu$ if and only if $\beta - \frac{1}{2} \leq \varepsilon_1$ for every zero. When it exists, $\nu$ is unique: it is $\rho_{\varepsilon_1}\, d\theta$ if $\beta - \frac{1}{2} < \varepsilon_1$ for every zero, and otherwise the vague limit of $\rho_\varepsilon\, d\theta$ as $\varepsilon \downarrow \varepsilon_1$, which carries point masses $m_\rho \delta_\gamma$ at the zeros with $\beta - \frac{1}{2} = \varepsilon_1$.*

*(b) In particular RH holds if and only if $\rho_{\varepsilon_0} = P_{\varepsilon_0} * \mu$ for a positive measure $\mu$ (with $\int (1+x^2)^{-1}\mu(dx) < \infty$, which is automatic since $P_{\varepsilon_0} * \mu = \rho_{\varepsilon_0}$ is finite), and then $\mu = \sum_\gamma \delta_\gamma$, the sum over the ordinates of all zeros, of both signs, with multiplicity.*

*Proof.* If every zero has $\beta - \frac{1}{2} < \varepsilon_1$, Theorem 19.2.1(a) gives $\nu = \rho_{\varepsilon_1}\, d\theta$. If every zero has $\beta - \frac{1}{2} \leq \varepsilon_1$ with equality for some, write $\rho_{\varepsilon_0} = P_{\varepsilon_0-\varepsilon} * \rho_\varepsilon$ for $\varepsilon_1 < \varepsilon < \varepsilon_0$ and let $\varepsilon \downarrow \varepsilon_1$: by (19.2.1) the terms of the zeros with $\beta - \frac{1}{2} < \varepsilon_1$ converge to Cauchy densities and those with $\beta - \frac{1}{2} = \varepsilon_1$ to point masses at their ordinates, and the sum over the zeros passes to the limit by the domination of the discussion after Remark 1.2.4, so $\rho_\varepsilon\, d\theta$ converges vaguely to a positive measure $\nu$, the sum of these limits. Then $P_{\varepsilon_0-\varepsilon_1} * \nu = \rho_{\varepsilon_0}$ term by term, because $P_{\varepsilon_0-\varepsilon_1} * P_{\varepsilon_1-(\beta-\frac{1}{2})} = P_{\varepsilon_0-(\beta-\frac{1}{2})}$ and $P_{\varepsilon_0-\varepsilon_1} * \delta_\gamma = P_{\varepsilon_0-(\beta-\frac{1}{2})}(\cdot - \gamma)$ when $\beta - \frac{1}{2} = \varepsilon_1$, all terms being positive; in particular $\int (1+x^2)^{-1}\nu(dx) < \infty$. Conversely, suppose $\nu$ exists and some zero has $\beta - \frac{1}{2} > \varepsilon_1$. The Poisson integral $u(\theta,\varepsilon) = (P_{\varepsilon-\varepsilon_1} * \nu)(\theta)$ is harmonic on $\{\varepsilon > \varepsilon_1\}$, and for $\varepsilon > \varepsilon_0$ it equals $P_{\varepsilon-\varepsilon_0} * \rho_{\varepsilon_0} = \rho_\varepsilon(\theta)$ by Theorem 19.2.1(a). The set $\{\varepsilon > \varepsilon_1\}$ minus the zeros is connected and $u(\theta,\varepsilon) - \rho_\varepsilon(\theta)$ is harmonic on it and vanishes on the open set $\{\varepsilon > \varepsilon_0\}$, so $u = \rho_\varepsilon$ there. Then $\rho_\varepsilon(\theta)$ is bounded near a zero with $\beta - \frac{1}{2} > \varepsilon_1$, which contradicts the pole of $\xi'/\xi$. For uniqueness, let $\nu$ and $\nu'$ both qualify and put $\sigma = \nu - \nu'$, a signed measure with $\int (1+x^2)^{-1}|\sigma|(dx) < \infty$. For $\varepsilon > \varepsilon_0$ the Poisson integrals of $\nu$ and $\nu'$ at height $\varepsilon - \varepsilon_1$ both equal $P_{\varepsilon-\varepsilon_0} * \rho_{\varepsilon_0}$, so the harmonic function $(P_{\varepsilon-\varepsilon_1} * \sigma)(\theta)$ on $\{\varepsilon > \varepsilon_1\}$ vanishes for $\varepsilon > \varepsilon_0$, hence everywhere by real analyticity, and letting $\varepsilon \downarrow \varepsilon_1$ gives $\sigma = 0$, since the Poisson integral of a signed measure $\sigma$ with $\int (1+x^2)^{-1}|\sigma|(dx) < \infty$ converges vaguely to $\sigma$. Part (b) is the case $\varepsilon_1 = 0$: $\beta - \frac{1}{2} \leq 0$ for every zero means $\beta = \frac{1}{2}$ for every zero by the symmetry $\rho \mapsto 1 - \rho$, and (19.2.1) at $\beta = \frac{1}{2}$ gives $\mu$. □

The *if* direction of (b), and the atoms at a boundary line in (a), are Theorem 1.2.2, Proposition 1.2.3 and Theorem 15.2.3; what this chapter adds is the *only if* direction and the uniqueness. Theorem 19.2.2 turns the obstruction of Section 11.4.7 into an exact statement: $\sigma_{\max} - \frac{1}{2}$ is the depth to which the Thorin density at a basepoint beyond the zeros can be deconvolved into a positive measure. Two remarks qualify it. First, positivity of $\rho_{\varepsilon_1}$ is necessary for deconvolvability to $\varepsilon_1$ but not sufficient: below $\sigma_{\max} - \frac{1}{2}$ the function $\rho_{\varepsilon_1}$ may well be positive (Remark 1.2.4 and Section 19.3), and still $P_{\varepsilon_0-\varepsilon_1} * \rho_{\varepsilon_1} \neq \rho_{\varepsilon_0}$. Second,

the deconvolution is unbounded. On the Fourier side $\widehat{P_a}(x) = e^{-a|x|}$, and in the explicit formula the frequency $x = \log n$ carries the weight $\Lambda(n)n^{-\alpha}$ at basepoint $\alpha$. Deconvolving by $P_a$ multiplies that weight by $n^a$, so the prime sums converge absolutely only for $\alpha > 1$, and below the basepoint one they need cancellation, which is a zero free region (Proposition 9.1.13).

PROPOSITION 19.2.3 (The smoothed prime sums at a basepoint). *Let $\chi$ be a primitive nonprincipal character, $N > 0$, and $s = \alpha + ib$ such that $s - k$ is not a zero of $L(s,\chi)$ for any integer $k \geq 0$. Put*

$$E(N,s) = \sum_{n\geq 1} \chi(n)\Lambda(n)n^{-s}e^{-n/N} + \frac{L'}{L}(s,\chi),$$

*the difference between the smoothed prime sum and $-L'/L(s,\chi)$. Then*

$$\begin{aligned} E(N,s) = & -\sum_{\rho} m_\rho\, \Gamma(\rho - s)N^{\rho - s} \\ & + \sum_{k\geq 1} \frac{(-1)^k}{k!}\Big(-\frac{L'}{L}\Big)(s-k,\chi)N^{-k}, \end{aligned} \tag{19.2.2}$$

*where $\rho$ runs over all zeros of $L(s,\chi)$, trivial ones included, with multiplicity $m_\rho$. Hence, for fixed $s$, $E(N,s) = O(N^{\sigma_{\max}(\chi)-\alpha}) + O(N^{-1})$ as $N \to \infty$, where $\sigma_{\max}(\chi)$ is the supremum of $\beta$ over the nontrivial zeros $\beta + i\gamma$ of $L(s,\chi)$. Under GRH the sum over the nontrivial zeros is $N^{1/2-\alpha}F_s(\log N)$, where $F_s(x) = -\sum m_\rho \Gamma(\rho - s)e^{i(\gamma-b)x}$, summed over the nontrivial zeros $\rho = \frac{1}{2} + i\gamma$, is an almost periodic function none of whose coefficients vanishes; so $F_s(\log N)$ does not tend to zero, and for $\alpha < \frac{3}{2}$ the error $E(N,s)$ is not $o(N^{1/2-\alpha})$.*

*Proof.* This is the explicit formula for the smoothed prime sum (compare [104, Chapter 19]); the derivation is given for the order of the error. Insert $e^{-x} = \frac{1}{2\pi i}\int_{(c)} \Gamma(w)x^{-w}dw$ with $c > \max(0, 1-\alpha)$ and sum: the smoothed prime sum is $\frac{1}{2\pi i}\int_{(c)}(-L'/L)(s+w,\chi)\Gamma(w)N^w\,dw$. Move the contour to $\operatorname{Re} w = -R$ with $R \to \infty$ along a sequence avoiding the poles; $\Gamma(w)$ decays like $e^{-\pi|\operatorname{Im} w|/2}$ and $L'/L$ grows at most polynomially on suitable horizontal segments and vertical lines [104, Chapters 16 and 19], so the integrals vanish in the limit. The poles are simple by the hypothesis on $s$. The residues are $(-L'/L)(s,\chi)$ at $w = 0$, which is moved to the left side, $-m_\rho\Gamma(\rho - s)N^{\rho-s}$ at $w = \rho - s$, since $-L'/L$ has residue $-m_\rho$ at $\rho$, and $(-1)^k(k!)^{-1}(-L'/L)(s-k,\chi)N^{-k}$ at $w = -k$. The sum over $\rho$ converges absolutely and is dominated by the zeros with $|\gamma - b|$ small, because $|\Gamma(\rho - s)|$ decays exponentially in $|\gamma - b|$; the nontrivial zeros contribute $O(N^{\sigma_{\max}(\chi)-\alpha})$, the trivial zeros $O(N^{-\alpha})$, and $O(N^{-1-\alpha})$ for odd $\chi$, and the last sum is $O(N^{-1})$. Under GRH each nontrivial term is $|\Gamma(\rho - s)|N^{1/2-\alpha}$ times a phase, $\Gamma$ has no zeros, and the series defining $F_s$ converges absolutely, so $F_s$ is

almost periodic in the sense of Bohr with the nonzero coefficients $-m_\rho\Gamma(\rho - s)$, and its mean square is the sum of the squares of their moduli, which is positive. $\square$

So recovering the Thorin density at basepoint $\alpha$ to accuracy $\eta$ from the primes with this smoothing needs primes up to about $N \approx \eta^{-1/(\alpha-\sigma_{\max}(\chi))}$, which is unbounded as $\alpha \downarrow \sigma_{\max}(\chi)$. This is a heuristic reading of the proposition as the deconvolution of Theorem 19.2.2 paid for in primes: the identity is for the weight $e^{-n/N}$, and it does not bound every method that uses the primes up to $N$. Proposition 9.1.13 gives the same exponent for the compensated prime sums of $\xi$ (the book writes $\Theta$ for $\sigma_{\max}$ there), and Section 11.3.2 treats $L(s, \chi_{12})$, where no pole needs compensating. Table 19.1 shows the exponent for $\chi_4$, for which GRH is verified to height $2.5 \times 10^7$ [417]: the measured exponent matches $\frac{1}{2} - \alpha$ to within 0.006 for $0.6 \le \alpha \le 1.2$, while at $\alpha = 1.5$ the $O(N^{-1})$ term is of the same size as the zero sum, between 1.1 and 2.3 times it.

| $\alpha$ | $N = 10^3$ | $10^4$ | $10^5$ | $2.5 \times 10^5$ | slope | $\frac{1}{2} - \alpha$ |
|---|---|---|---|---|---|---|
| 1.5 | $3.83 \times 10^{-3}$ | $3.97 \times 10^{-4}$ | $4.72 \times 10^{-5}$ | $2.00 \times 10^{-5}$ | $-0.949$ | $-1.00$ |
| 1.2 | $1.56 \times 10^{-2}$ | $3.03 \times 10^{-3}$ | $6.24 \times 10^{-4}$ | $3.36 \times 10^{-4}$ | $-0.695$ | $-0.70$ |
| 1.0 | $5.86 \times 10^{-2}$ | $1.83 \times 10^{-2}$ | $5.73 \times 10^{-3}$ | $3.78 \times 10^{-3}$ | $-0.499$ | $-0.50$ |
| 0.9 | $1.21 \times 10^{-1}$ | $4.78 \times 10^{-2}$ | $1.87 \times 10^{-2}$ | $1.35 \times 10^{-2}$ | $-0.400$ | $-0.40$ |
| 0.8 | $2.60 \times 10^{-1}$ | $1.29 \times 10^{-1}$ | $6.36 \times 10^{-2}$ | $4.95 \times 10^{-2}$ | $-0.302$ | $-0.30$ |
| 0.7 | $5.66 \times 10^{-1}$ | $3.57 \times 10^{-1}$ | $2.22 \times 10^{-1}$ | $1.86 \times 10^{-1}$ | $-0.203$ | $-0.20$ |
| 0.6 | $1.23 \times 10^{0}$ | $9.78 \times 10^{-1}$ | $7.69 \times 10^{-1}$ | $6.93 \times 10^{-1}$ | $-0.103$ | $-0.10$ |

Table 19.1: Mean modulus of $E(N, s)$ of (19.2.2) for $\chi_4$ over $s = \alpha + 10i$ and $s = \alpha + 25i$, with prime powers summed up to $40N$, and the slope of $\log_{10}$ of the mean against $\log_{10} N$ fitted by least squares on the four values of $N$. Four values of $N$ and two values of $b$ illustrate the GRH statement of the proposition and do not verify it. Script `tf_primes.py`.

The identity (19.2.2) itself was checked in the 56 cases of the table: $E(N, s)$ and the right side, built from the zeros of $L(s, \chi_4)$ of height below 62, the trivial zeros $-1, -3, \dots, -15$ and the terms $m \le 9$, agree to $5.4 \times 10^{-14}$ in absolute value (script `tf_primes.py`).

The book's route through basepoints, positivity at every height for $\alpha \ge 1$ (Section 15.1.2), stops at $\alpha = 1$, the edge of the known zero free half plane; the cost exponent and the zero free half plane are two faces of one fact (Proposition 9.1.13), not cause and effect.

## 19.3 What a single basepoint sees

Theorem 19.2.2, like Lagarias's criterion [320] (Remark 1.2.4), needs every level. A test at one level $\varepsilon$ sees less, and Section 5.2.2, with its Table 5.5 and Exercise 5.14, sets out what it sees in a lattice model. Two facts frame it. A zero at distance

$\delta = \beta - \frac{1}{2} > 0$ from the line contributes positively at every level $\varepsilon > \delta$, so those levels cannot see it. As $\varepsilon \uparrow \delta$ its contribution at $\theta = \gamma$ is $-1/(\pi(\delta - \varepsilon)) \to -\infty$, so the levels just below $\delta$ always see it. Near the line the test reduces to the sign of Laguerre's quantity $\mathcal{L}_\Xi = -(\log|\Xi|)''$.

A quadruple of zeros $\frac{1}{2} \pm \delta \pm i\gamma^*$ with $\delta > 0$ is added by hand to quantities computed from the actual zeros, to $\xi'/\xi$ and to $\mathcal{L}_\Xi$, in the way that Section 11.4.4 inserts quadruples into the computed density, and nothing below says that $\xi$ has such zeros. The tables, the figures and Corollary 19.3.3 describe what a test would see if it did, and are statements about a model. A quadruple puts two zeros in the upper half plane, and it is placed among the actual zeros in one of three ways, which differ in the number of zeros up to a given height:

(i) the quadruple replaces one on line zero, at the ordinate $\gamma^*$ of that zero, as in the lattice of Section 5.2.2; the count of zeros up to a given height above $\gamma^*$ grows by one;

(ii) the quadruple is inserted at the midpoint $\gamma^*$ of a gap between consecutive on line zeros; the count grows by two;

(iii) the quadruple replaces the two on line zeros that bound a gap and sits at the midpoint $\gamma^*$ of the gap, as in the perturbation of Theorem 5.2.1; the count is unchanged.

Proposition 19.3.1 (The test near the line). *Let $\theta$ be real and not the ordinate of a zero on the line, and let $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$, defined by the same formula for every real $\varepsilon$. Then*

$$\rho_\varepsilon(\theta) = \frac{\varepsilon}{\pi}\,\mathcal{L}_\Xi(\theta) + O(\varepsilon^3) \quad (\varepsilon \to 0),$$

*where*

$$\mathcal{L}_\Xi(\theta) = -\frac{d^2}{d\theta^2}\log|\Xi(\theta)| = \sum_\rho \frac{1}{(\theta - \gamma_\rho)^2},$$

*$\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ are the zeros of $\Xi$, and the sum is real, since the zeros of $\Xi$ come in conjugate pairs. Under RH, $\mathcal{L}_\Xi(\theta) = \sum_\gamma (\theta - \gamma)^{-2} > 0$. A quadruple of zeros $\frac{1}{2} \pm \delta \pm i\gamma^*$ contributes*

$$2\operatorname{Re}\frac{1}{(\theta - \gamma^* + i\delta)^2} + 2\operatorname{Re}\frac{1}{(\theta + \gamma^* + i\delta)^2},$$

*which equals $-2/\delta^2 + O(\gamma^{*-2})$ at $\theta = \gamma^*$.*

*Proof.* As a function of $(\theta, \varepsilon)$, $\rho_\varepsilon(\theta)$ is harmonic off the zeros and odd in $\varepsilon$ by the functional equation, so near $(\theta, 0)$ it equals $\varepsilon\,\partial_\varepsilon \rho_\varepsilon(\theta)|_{\varepsilon=0} + O(\varepsilon^3)$. With $F = \xi'/\xi$ and $\Xi'(\theta) = iF(\frac{1}{2} + i\theta)\Xi(\theta)$, differentiating gives $F'(\frac{1}{2} + i\theta) = -(\Xi'/\Xi)'(\theta)$, which

is real, and $\partial_\varepsilon \rho_\varepsilon = \pi^{-1} \operatorname{Re} F'$. Hadamard's product for $\Xi$, with the zeros $\pm\gamma_\rho$ paired, gives $\Xi'/\Xi = \sum (\theta - \gamma_\rho)^{-1}$, and differentiating once more gives the sum. For the quadruple the zeros of $\Xi$ are $\gamma^* \pm i\delta$ and $-\gamma^* \pm i\delta$, and each conjugate pair contributes $2\operatorname{Re}(\theta \mp \gamma^* + i\delta)^{-2}$; at $\theta = \gamma^*$ the first is $2\operatorname{Re}(i\delta)^{-2} = -2/\delta^2$. □

At first order the test at $\gamma^*$ sees the quadruple exactly when

$$\delta < \delta_c(\gamma^*) = \sqrt{2/\mathcal{L}_0(\gamma^*)},$$

with $\mathcal{L}_0(\gamma^*)$ the contribution of the other zeros to $\mathcal{L}_\Xi(\gamma^*)$; this is the local visibility condition of Section 5.2.2, written with Laguerre's quantity. In (i) $\mathcal{L}_0$ omits the replaced zero; with neighbours at the mean spacing $\Delta = 2\pi/\log(h/2\pi)$ this gives $\mathcal{L}_0 = \pi^2/(3\Delta^2)$ and $\delta_c = (\sqrt{6}/\pi)\Delta = 0.780\,\Delta$. In (ii), midway between two on line zeros, $\mathcal{L}_0 = \pi^2/\Delta^2$ and $\delta_c = (\sqrt{2}/\pi)\Delta = 0.450\,\Delta$. In (iii) the two zeros at $\gamma^* \pm \frac{1}{2}\Delta$ are removed from the lattice of (ii), $\mathcal{L}_0 = (\pi^2 - 8)/\Delta^2$ and $\delta_c = \sqrt{2/(\pi^2 - 8)}\,\Delta = 1.034\,\Delta$. A distance $\delta_c = c\Delta$ is below $\frac{1}{2}$ exactly when $h > 2\pi e^{4\pi c}$, that is above $2\pi e^{4\sqrt{6}} = 1.131 \times 10^5$ in (i), above $2\pi e^{4\sqrt{2}} = 1799$ in (ii) and above $2.8 \times 10^6$ in (iii). Table 19.2 replaces the lattice by the actual zeros.

| | | (i) replacing a zero | | | (ii) at a midpoint | | | (iii) count preserving | | |
|---|---|---|---|---|---|---|---|---|---|---|
| height | mean gap | median | least | ratio | median | least | ratio | median | least | ratio |
| 14 | 3.177 | 2.065 | 1.503 | 0.716 | 1.339 | 0.692 | 0.450 | 2.987 | 2.364 | 1.035 |
| 50 | 2.557 | 1.824 | 1.335 | 0.719 | 1.128 | 0.589 | 0.447 | 2.551 | 2.187 | 1.000 |
| 237 | 1.679 | 1.201 | 0.951 | 0.733 | 0.674 | 0.447 | 0.454 | 1.691 | 1.508 | 1.093 |
| 1419 | 1.178 | 0.799 | 0.655 | 0.711 | 0.469 | 0.297 | 0.452 | 1.167 | 0.975 | 1.062 |
| 9878 | 0.848 | 0.579 | 0.419 | 0.701 | 0.375 | 0.164 | 0.444 | 0.856 | 0.715 | 0.973 |
| 74921 | 0.661 | 0.446 | 0.235 | 0.662 | 0.296 | 0.088 | 0.437 | 0.651 | 0.527 | 0.902 |

Table 19.2: The first order visibility distance $\delta_c = \sqrt{2/\mathcal{L}_0}$ at the actual zeros, from $\mathcal{L}_\Xi = -(\log|\Xi|)''$ evaluated with `mpmath` at 30 digits, in the three configurations (i), (ii) and (iii) of the start of this section, at six heights $n_0 = 1, 10, 10^2, \ldots, 10^5$; the height is the ordinate $\gamma_{n_0}$ of the $n_0$th zero. Replacing a zero: the 20 zeros $\gamma_n$, $n_0 < n \le n_0 + 20$, each removed from $\mathcal{L}_0$, with the local gap $(\gamma_{n+1} - \gamma_{n-1})/2$; the mean gap is the mean of the local gap over these 20 zeros. At a midpoint and count preserving: the 20 midpoints of $\gamma_n$ and $\gamma_{n+1}$, $n_0 \le n < n_0 + 20$, with the local gap $\gamma_{n+1} - \gamma_n$, and in the count preserving block the two zeros $\gamma_n$ and $\gamma_{n+1}$ are removed as well. Median and least are over the 20 values of $\delta_c$, and the ratio is the median over the 20 points of $\delta_c$ divided by the local gap. Script `tf_deltac.py`, data `tf_deltac.json`.

The ratio $\delta_c$/gap is stable across heights in each configuration: about 0.45 at midpoints, as in the lattice; somewhat below the lattice value 0.78 when a zero is replaced; around the lattice value 1.034 when the count is preserved. The median of $\delta_c$ falls below $\frac{1}{2}$, the largest possible distance of a zero from the line, already by

height 1419 at midpoints and by height 74921 when replacing a zero, below the lattice estimate $1.1 \times 10^5$ above. Above these heights an off line zero at distance between $\delta_c$ and $\frac{1}{2}$ passes the first order test at its own ordinate wherever $\delta_c < \frac{1}{2}$, which is so at half or more of the sampled points once the median is below $\frac{1}{2}$, that is, the test does not see it.

Table 19.3 and Figure 19.1 show what happens beyond $\delta_c$. We insert a hypothetical quadruple at the midpoint $\gamma^*$ of a gap, configuration (ii), add its Cauchy terms exactly to $\rho_\varepsilon$ computed from $\xi'/\xi$, and record the levels $\varepsilon \in (0, \delta)$, on the grid $\varepsilon = \delta k/40$, $k = 1, \dots, 39$, with $0.999\,\delta$, at which $\rho_\varepsilon$ is negative at some of 41 equally spaced points of the gap. At height 9878, where $\delta_c = 0.381$, the quadruple with $\delta = 0.45$ is seen only for $0.19125 \le \varepsilon < 0.45$; at every level of the grid below 0.19, the first order test included, $\rho_\varepsilon$ stays positive at the 41 points of the gap. The level at which the sign changes at the ordinate is $\varepsilon = 0.1895$ (Proposition 19.3.2 below), and the minimum over the gap, which lies about 0.004 to the left of the midpoint, changes sign at $\varepsilon = 0.1893$; the grid level 0.19125 is the first above both.

| $\gamma^*$ | $\delta_c$ | $\delta = 0.05$ | $\delta = 0.2$ | $\delta = 0.3$ | $\delta = 0.45$ |
|---|---|---|---|---|---|
| 100.07 | 1.109 | all $\varepsilon < \delta$ | all $\varepsilon < \delta$ | all $\varepsilon < \delta$ | all $\varepsilon < \delta$ |
| 9878.22 | 0.381 | all $\varepsilon < \delta$ | all $\varepsilon < \delta$ | all $\varepsilon < \delta$ | $0.19125 \le \varepsilon < 0.45$ |

Table 19.3: Levels $\varepsilon$ at which $\rho_\varepsilon$ is negative (grid and points as in the text), near a hypothetical quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$ inserted at the midpoint of the gap between the 29th and 30th zeros and between the 10000th and 10001st (configuration (ii)). Script `tf_window.py`, data `tf_window.json`.

### 19.3.1 The visibility curve

Proposition 19.3.2 (The visibility curve). *Let $\rho_\varepsilon^0$ be the Thorin density of all zeros other than the two zeros $\frac{1}{2} \pm \delta + i\gamma^*$ of a quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$, so that the terms of the quadruple's images at $-\gamma^*$ are included in it (and $\delta_{\mathrm{det}}$ below depends on $\delta$ through them, by $10^{-9}$ in the computations of this section), and suppose $\rho_\varepsilon^0(\gamma^*) > 0$. For $0 < \varepsilon < \delta$, $\rho_\varepsilon(\gamma^*) < 0$ if and only if*

$$\delta < \delta_{\mathrm{det}}(\varepsilon), \qquad \delta_{\mathrm{det}}(\varepsilon)^2 = \varepsilon^2 + \frac{2\varepsilon}{\pi \rho_\varepsilon^0(\gamma^*)}.$$

*In the lattice models the other zeros are those of the lattice alone, and the images at $-\gamma^*$ and the distant zeros are neglected, as in Section 5.2.2. Let the other zeros lie on the line at the points $\gamma^* \pm (k + \frac{1}{2})\Delta$ with $k \ge 0$ in configuration (ii), $\gamma^* \pm k\Delta$*

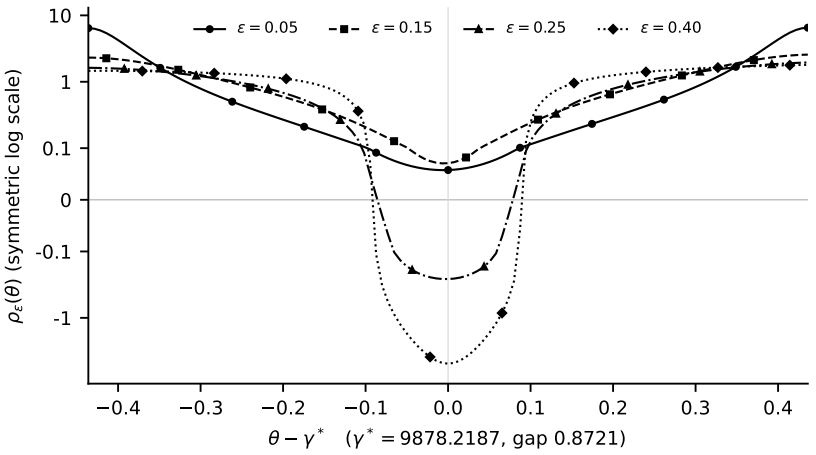



Figure 19.1: The Thorin density $\rho_\varepsilon$ across the gap at height 9878 (gap 0.872) with a hypothetical quadruple at distance $\delta = 0.45$ inserted at its midpoint, at four levels $\varepsilon = 0.05$, 0.15, 0.25 and 0.4. It is positive at $\varepsilon = 0.05$ and $\varepsilon = 0.15$ and negative near the midpoint at $\varepsilon = 0.25$ and $\varepsilon = 0.4$. The vertical scale is symmetric logarithmic, linear within $\pm 0.1$. Script `tf_window.py` with the option `--plot`, data `tf_window.json`.

*with $k \geq 1$ in (i), and $\gamma^* \pm (k + \frac{1}{2})\Delta$ with $k \geq 1$ in (iii). Then*

$$\delta_{\mathrm{det}}(\varepsilon)^2 = \varepsilon^2 + \begin{cases} \dfrac{2\varepsilon\Delta}{\pi \tanh(\pi\varepsilon/\Delta)} & \text{in (ii),} \\[2ex] \dfrac{2\varepsilon}{\frac{\pi}{\Delta}\coth\frac{\pi\varepsilon}{\Delta} - \frac{1}{\varepsilon}} & \text{in (i),} \\[2ex] \dfrac{2\varepsilon}{\frac{\pi}{\Delta}\tanh\frac{\pi\varepsilon}{\Delta} - \frac{2\varepsilon}{\varepsilon^2+\Delta^2/4}} & \text{in (iii).} \end{cases}$$

*As $\varepsilon \to 0$ these tend to $\delta_c^2 = 2\Delta^2/\pi^2$, $6\Delta^2/\pi^2$ and $2\Delta^2/(\pi^2 - 8)$, and for $\varepsilon \geq \Delta$ all three satisfy $\delta_{\mathrm{det}}(\varepsilon)^2 - \varepsilon^2 = 2\varepsilon\Delta/\pi + O(\Delta^2)$, that is $\delta_{\mathrm{det}}(\varepsilon) - \varepsilon = (\Delta/\pi)\big(1 + O(\Delta/\varepsilon)\big)$. In all three, $\delta_{\mathrm{det}}$ is increasing.*

*Proof.* The two zeros at height $\gamma^*$ contribute $\pi^{-1}\big((\varepsilon - \delta)^{-1} + (\varepsilon + \delta)^{-1}\big) = -2\varepsilon/(\pi(\delta^2 - \varepsilon^2))$ at $\theta = \gamma^*$, so $\rho_\varepsilon(\gamma^*) < 0$ exactly when $\pi\rho_\varepsilon^0(\gamma^*) < 2\varepsilon/(\delta^2 - \varepsilon^2)$, which rearranges to the stated inequality. A zero on the line at distance $\delta_j$ from $\gamma^*$ contributes $\varepsilon/(\pi(\varepsilon^2 + \delta_j^2))$ to $\rho_\varepsilon^0(\gamma^*)$ (Proposition 1.2.3), so in the lattice models $\pi\rho_\varepsilon^0(\gamma^*) = \varepsilon\sum_j(\varepsilon^2 + \delta_j^2)^{-1}$, which is positive. The sum decreases with $\varepsilon$, and $\delta_{\mathrm{det}}(\varepsilon)^2 = \varepsilon^2 + 2\big/\sum_j(\varepsilon^2 + \delta_j^2)^{-1}$ increases. For the lattices, $\sum_{k\in\mathbb{Z}} \varepsilon/(\varepsilon^2 + (k + \frac{1}{2})^2\Delta^2) = (\pi/\Delta)\tanh(\pi\varepsilon/\Delta)$ and

$\sum_{k\neq 0}\varepsilon/(\varepsilon^2+k^2\Delta^2)=(\pi/\Delta)\coth(\pi\varepsilon/\Delta)-1/\varepsilon$, by the partial fractions of tanh and coth, and removing the two zeros at $\gamma^*\pm\frac12\Delta$ subtracts $2\varepsilon/(\varepsilon^2+\Delta^2/4)$ from the first. The limits follow from $\tanh x=x+O(x^3)$ and $\coth x=1/x+x/3+O(x^3)$, which give $\pi\rho_\varepsilon^0(\gamma^*)/\varepsilon\to\pi^2/\Delta^2$, $\pi^2/(3\Delta^2)$ and $(\pi^2-8)/\Delta^2$. For $\varepsilon\geq\Delta$, $\tanh(\pi\varepsilon/\Delta)$ and $\coth(\pi\varepsilon/\Delta)$ equal 1 up to $O(e^{-2\pi\varepsilon/\Delta})$, so that $\pi\rho_\varepsilon^0(\gamma^*)=\pi/\Delta+O(1/\varepsilon)$ in all three, whence $\delta_{\det}^2-\varepsilon^2=2\varepsilon\Delta/\pi+O(\Delta^2)$; dividing by $\delta_{\det}+\varepsilon=2\varepsilon+O(\Delta)$ gives $\delta_{\det}-\varepsilon=(\Delta/\pi)(1+O(\Delta/\varepsilon))$. □

The first display of the proposition is the formula for $\pi\rho_\varepsilon(\gamma)$ at the ordinate in Section 5.2.2, solved for the distance, with all the other zeros and not only the neighbours in $\rho_\varepsilon^0$. So a level $\varepsilon$ sees, at the ordinate, exactly the zeros at distances in $(\varepsilon,\delta_{\det}(\varepsilon))$: a band that starts at width $\delta_c$ near the line and has width close to $\Delta/\pi=2/\log(h/2\pi)$ well above it, the scale $1/\log\gamma$ of Section 5.2.2 with its constant. The width $\delta_{\det}(\varepsilon)-\varepsilon$ is $\Delta/\pi+c\,\Delta^2/\varepsilon+\cdots$ with $c=\frac{1}{2\pi^2}$, $-\frac{1}{2\pi^2}$ and $\frac{3}{2\pi^2}$ in (i), (ii) and (iii), so that for large $\varepsilon$ it is above $\Delta/\pi$ in (i) and (iii) and below it in (ii); in (i) and (iii) it falls monotonically to $\Delta/\pi$ (computed), while in (ii) it falls from $0.450\,\Delta$ to a least value $0.272\,\Delta$ at $\varepsilon=0.581\,\Delta$ and then rises to $\Delta/\pi$ from below (computed by `tf_visibility.py`). The curve explains Table 19.3. At the midpoint of the 10000th gap, $\pi\rho_\varepsilon^0(\gamma^*)$ computed from $\xi'/\xi$ gives $\delta_{\det}(0.19)=0.4503$, and solving $\delta_{\det}(\varepsilon)=0.45$ gives $\varepsilon=0.1895$. Figure 19.2 draws the region for the actual zeros and for the lattice with the same gap, which lies within 0.015 of it.

Corollary 19.3.3 (How many levels, in the lattice model). *Take any of the three lattice models, with gap $\Delta=2\pi/\log(h/2\pi)$, and put $\varepsilon_1=\delta_c$ and $\varepsilon_{k+1}=\delta_{\det}(\varepsilon_k)$. Let $K$ be the least $k$ with $\varepsilon_{k+1}>\frac12$, so that $K=0$ if $\delta_c>\frac12$.*

*(a) The sign of $\mathcal{L}_\Xi(\gamma^*)$, together with positivity of $\rho_\varepsilon(\gamma^*)$ at the levels $\varepsilon_1,\dots,\varepsilon_K$, each moved down slightly so that consecutive bands overlap, detects a single off line quadruple at every distance in $(0,\frac12]$.*

*(b) No test made at the ordinate, that is the sign of $\mathcal{L}_\Xi(\gamma^*)$ together with the sign of $\rho_\varepsilon(\gamma^*)$ at finitely many levels, that detects a single off line quadruple at every distance in $(0,\frac12]$ uses fewer than $K$ levels.*

*(c) $K=\frac14\log(h/2\pi)+O(\log\log h)$.*

*Proof.* By Proposition 19.3.1 the first order test detects the distances in $(0,\delta_c)$, and by Proposition 19.3.2 the level $\varepsilon$ detects exactly those in $(\varepsilon,\delta_{\det}(\varepsilon))$, since a zero at distance $\delta\leq\varepsilon$ contributes positively and $\rho_\varepsilon^0(\gamma^*)>0$. Write $f=\delta_{\det}$, so that $\varepsilon_{k+1}=f(\varepsilon_k)$; $f$ is continuous and increasing with $f(\varepsilon)>\varepsilon$.

(a) The bands $(0,\varepsilon_1)$ and $(\varepsilon_k,\varepsilon_{k+1})$, $1\leq k\leq K$, cover $(0,\varepsilon_{K+1})$ except the points $\varepsilon_1,\dots,\varepsilon_K$. Moving the levels down by suitably small amounts, which is possible because $f$ is continuous and $\varepsilon_{K+1}=f(\varepsilon_K)>\frac12$, makes consecutive bands overlap and still reach beyond $\frac12$.

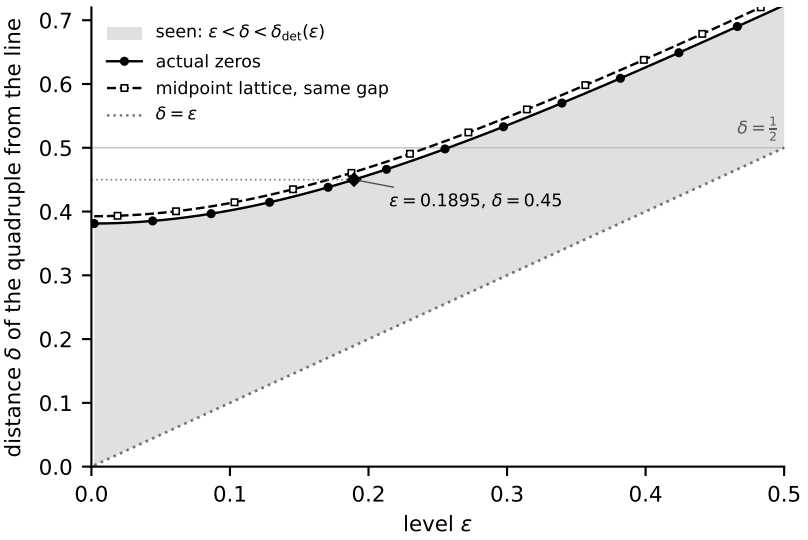



Figure 19.2: The visibility region $\varepsilon < \delta < \delta_{\det}(\varepsilon)$ (shaded band) at the midpoint of the 10000th gap. Solid: $\delta_{\det}$ from the actual zeros through $\xi'/\xi$; dashed: the midpoint lattice of Proposition 19.3.2 with the same gap; dotted: $\delta = \varepsilon$, the lower edge of the band. The marked point is $\varepsilon = 0.1895$, $\delta = 0.45$. Script `tf_visibility.py`.

(b) We claim that if $j \geq 1$ bands $(\varepsilon', f(\varepsilon'))$ cover $[\ell, x)$ with $\ell > 0$, then $x < f^{(j)}(\ell)$, the $j$th iterate. A band containing $\ell$ has $\varepsilon' < \ell$, so its right end is $r = f(\varepsilon') < f(\ell)$. For $j = 1$ this gives $x \leq r < f(\ell)$. For $j > 1$, either $x \leq r < f(\ell) \leq f^{(j)}(\ell)$, or the other $j-1$ bands cover $[r, x)$ and the claim for $j-1$ gives $x < f^{(j-1)}(r) < f^{(j-1)}(f(\ell)) = f^{(j)}(\ell)$. Now suppose that $m$ levels, with the first order test, detect every distance in $(0, \frac{1}{2}]$. The point $\delta_c$ is not in $(0, \delta_c)$, so unless $\delta_c > \frac{1}{2}$ (and then $K = 0$) some bands cover $[\delta_c, \frac{1}{2}]$, and the claim with $\ell = \delta_c$ and $x = \frac{1}{2}$ gives $\frac{1}{2} < f^{(m)}(\delta_c) = \varepsilon_{m+1}$. By the definition of $K$, $m \geq K$.

(c) The bands advance by $a(\varepsilon) = \delta_{\det}(\varepsilon) - \varepsilon = \big(\delta_{\det}(\varepsilon)^2 - \varepsilon^2\big)/\big(\delta_{\det}(\varepsilon) + \varepsilon\big)$, which by the formulas of Proposition 19.3.2 is $\frac{\Delta}{\pi}\big(1 + O(\Delta/\varepsilon)\big)$ for $\varepsilon \geq \Delta$ and, for $\varepsilon \leq \Delta$, lies between two positive constants times $\Delta$, since $a(\varepsilon)/\Delta$ is a positive continuous function of $\varepsilon/\Delta$. The number of steps from $\delta_c$ to $\frac{1}{2}$ is therefore $\int_\Delta^{1/2} \frac{d\varepsilon}{a(\varepsilon)} + O(\log(1/\Delta)) = \frac{\pi}{\Delta}\big(\frac{1}{2} + O(\Delta \log(1/\Delta))\big)$, and $\pi/(2\Delta) = \frac{1}{4}\log(h/2\pi)$, while $\log(1/\Delta) = O(\log\log h)$. □

Table 19.4 gives the numbers of levels. The count preserving lattice has the larger $\delta_c = 1.034\,\Delta$, which stays above $\frac{1}{2}$ up to $h = 2.8 \times 10^6$, so it uses fewer levels, and at $10^5$ the $\delta_c = 0.506$ of the replacement lattice already covers every distance. At the height $3 \times 10^{12}$ to which RH has been verified [419] the three

| $h$ | $\frac{1}{4}\log(h/2\pi)$ | (i) replacing | (ii) midpoint | (iii) count preserving |
|---|---|---|---|---|
| $10^5$ | 2.4 | 0 | 2 | 0 |
| $10^8$ | 4.1 | 2 | 4 | 1 |
| $3\times10^{12}$ | 6.7 | 4 | 6 | 3 |
| $10^{22}$ | 12.2 | 9 | 12 | 7 |

Table 19.4: The number $K$ of greedy levels of Corollary 19.3.3, counted without the downward moves, in the three lattice models with $\Delta = 2\pi/\log(h/2\pi)$, against $\frac{1}{4}\log(h/2\pi)$. Script `tf_visibility.py`.

models use between three and six levels. The corollary is a statement about a model with a single quadruple and regularly spaced neighbours, and about tests at the ordinate: a test that looks for negative values anywhere in the window can see more (Section 5.2.2, Exercise 5.14), and for it no lower bound is proved here. Turning it into a certificate for the actual zeros is Problem 19.1.

## 19.4 The heat flow: a stable step that cannot carry real zeros

With the de Bruijn Newman kernel $\Phi_{\mathrm{dBN}}$ of Section 8.4.1,

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du, \qquad H_0(z) = \tfrac{1}{8}\,\Xi(z/2),$$

the function $H_t$ is defined for every real $t$, since the kernel decays like $e^{9u}\exp(-\pi e^{4u})$, and the zeros of $H_0$ are the points $2\gamma_\rho$, with imaginary parts $-2(\beta-\frac{1}{2})$. The de Bruijn Newman constant $\Lambda$ satisfies $0 \le \Lambda \le 0.2$ [463, 419].

Theorem 19.4.1 (The heat family). *(a) $\partial_t H_t = -\partial_z^2 H_t$, so $H_t = e^{-tD^2}H_0$ with $D = d/dz$.*

*(b) For real $t$, $s>0$ and complex $z$, with $Z$ a standard normal variable,*

$$H_{t-s}(z) = \mathbb{E}\big[H_t\big(z+\sqrt{2s}\,Z\big)\big].$$

*(c) Forward in $t$ real zeros persist, and if the zeros of $H_{t_0}$ lie in $|\operatorname{Im} z| \le r$, then those of $H_t$ lie in $(\operatorname{Im} z)^2 \le \max(r^2 - 2(t-t_0), 0)$ for $t \ge t_0$ [112, Theorem 13].*

*(d) Backward in $t$ real zeros are not preserved: $f(z) = z^2 - c$, $c>0$, has real zeros, and $\mathbb{E}f(z+\sqrt{2s}Z) = z^2 - c + 2s$ has none once $s > c/2$. For $H$ itself, $H_t$ has nonreal zeros for every $t<0$ [463].*

*Proof.* (a) Differentiate under the integral: $\partial_t$ brings down $u^2$ and $\partial_z^2$ brings down $-u^2$. (b) $\mathbb{E}\cos((z+\sigma Z)u) = \cos(zu)e^{-\sigma^2u^2/2}$ for complex $z$, and with $\sigma^2 = 2s$ the

factor $e^{-su^2}$ turns the kernel $e^{tu^2}\Phi_{\mathrm{dBN}}$ into $e^{(t-s)u^2}\Phi_{\mathrm{dBN}}$; Fubini applies because $\Phi_{\mathrm{dBN}}$ decays like $e^{9u}\exp(-\pi e^{4u})$, for every real $t$. (c) is de Bruijn's theorem, in the form used by Polymath15 [456] and in the proof of Theorem 15.3.2. (d) is the computation shown and the theorem of Rodgers and Tao. □

COROLLARY 19.4.2 (No averaging stable property carries real zeros). *Let $\mathcal{P}$ be a property of entire functions that is preserved by $f \mapsto \mathbb{E}f(\cdot + \sqrt{2s}Z)$ for some $s > 0$. If $\mathcal{P}$ implies that all zeros are real, then $H_t$ has $\mathcal{P}$ for no real $t$.*

*Proof.* If $H_{t_0}$ has $\mathcal{P}$, then by (b), applied $k$ times, so does $H_{t_0-ks}$ for every integer $k \geq 1$, and for $t_0 - ks < 0$ it has only real zeros, against (d). The proof uses only that some $H_t$ has a nonreal zero, which is Newman's theorem $\Lambda > -\infty$ [381]. □

The corollary is the heat flow form of the criticality filter of Section 11.5.5: Proposition 11.5.12 shows that no property of the kernel $\Phi_t$ that holds at some $t < 0$ can force real zeros. The step towards RH, from $t = 0.2$ down to 0, is an average of $H_{0.2}$ against a Gaussian of variance 0.4: a contraction, stable to compute.

A Gauss Legendre rule with 200 nodes in $u$ on $[0, 1.5]$, beyond which $\Phi_{\mathrm{dBN}} < 10^{-540}$, and a Gauss Hermite rule with 80 nodes reproduce $H_0(z)$ from $H_{0.2}$ at $z = 0$, 10 and 28.27 to a relative error below $10^{-35}$ in 40 digit arithmetic. The step away from RH, $e^{-tD^2}$, preserves real zeros, as the operator of the Laguerre Pólya theory, and is a deconvolution in $z$. In the basepoint family the roles are reversed: the positivity preserving step is a convolution and the step towards RH an unbounded deconvolution.

The two informative families are tied by the conserved quantity of Theorem 15.3.2: a basepoint offset $\varepsilon$ is worth heat time $2\varepsilon^2$, and $\Lambda \leq 2(\sigma_{\max} - \frac{1}{2})^2$. The critical strip, $\sigma_{\max} \leq 1$, gives de Bruijn's $\Lambda \leq \frac{1}{2}$; the half plane announced in October 2026 [398], not yet refereed, would give $\sigma_{\max} \leq \frac{7}{8}$ and, conditionally on it, $\Lambda \leq \frac{9}{32}$, weaker than the known 0.2; and $\Lambda \leq 0.2$ corresponds to the basepoint $\frac{1}{2} + \sqrt{0.1} = 0.816$ (Corollary 15.3.3). In the language of Theorem 19.2.2, whose *if* direction holds for $\xi_t$ by the product of Lemma 15.2.1, the bound $\Lambda \leq 0.2$ of Platt and Trudgian, obtained through the Polymath15 framework, says that the Thorin density of $\xi_{0.2}$ deconvolves to a positive measure all the way to the line; RH asks for the same at $t = 0$, and by Corollary 19.4.2 no property preserved by the Gaussian average joining the two times can carry it.

*What would get past the obstruction.* This paragraph is a heuristic reading of the theorems above. In the basepoint family a proof needs a bound, uniform in the height, for the deconvolution of $\rho_{\varepsilon_0}$; from the primes that means cancellation in $\sum \Lambda(n)n^{-\alpha-ib}$ below $\alpha = 1$, which is a zero free region by Proposition 9.1.13, and the half plane announced in October 2026 [398], not yet refereed, would move the end of the range to $\alpha = \frac{7}{8}$. In the heat family a proof needs a property of $H_0$ that

implies real zeros and that fails for every $H_t$ with $t < 0$, since those have nonreal zeros (Theorem 19.4.1(d)): a property that sees the boundary $t = \Lambda$ itself. In the tilt family nothing can be gained. And since $\Lambda \geq 0$, any such statement must hold at the boundary with no slack (Remark 15.3.4).

### 19.4.1 Open problems

Problem 19.1 (How many levels, for the actual zeros). Corollary 19.3.3 counts levels in a model with one quadruple among regularly spaced zeros, and only at its ordinate. Turn it into a certificate: find levels $\varepsilon_1 < \cdots < \varepsilon_K$ such that positivity of every $\rho_{\varepsilon_k}$ on $[0, h]$, together with positivity of $\mathcal{L}_\Xi$, excludes all off line zeros up to height $h$, allowing several off line zeros, irregular spacing, and minima off the ordinate (Exercise 5.14). Is $K$ still of order $\log h$? Such a certificate is finite information in the sense of Section 5.2 and Theorem 5.2.1, so its interest is as a different verification of RH up to height $h$, from $\xi$ on lines to the right of the critical line rather than from sign changes on it.

Problem 19.2 (Stability with an a priori bound). On a strip $\frac{1}{2}+\varepsilon_1 \leq \operatorname{Re} s \leq \frac{1}{2}+\varepsilon_0$ free of zeros, Hadamard's three lines theorem bounds the error of an approximation $g$ to $\xi'/\xi$ at level $\varepsilon$ by $\eta^\lambda M^{1-\lambda}$, $\lambda = (\varepsilon - \varepsilon_1)/(\varepsilon_0 - \varepsilon_1)$, where $\eta$ is the error at $\varepsilon_0$ and $M$ a bound at $\varepsilon_1$. The bound $M$ presupposes the zero free strip. Is there a formulation in which the a priori bound comes from the Thorin structure itself, for instance from positivity of $\rho_\varepsilon$ at the levels already certified?

Problem 19.3 (Seeing the boundary). By Theorem 19.4.1(d), a property of $H_0$ that implies real zeros fails for $H_t$ at every $t < 0$. Which of the positivity statements of the book, for instance the GGC property of the clock, whose Laplace transform is $R_0$, the case $t = 0$ of $R_t(s) = H_t(0)/H_t(2i\sqrt{s})$ in Theorem 7.2.16, or the Pólya frequency property of the density of $Z\sqrt{2T}$ (Proposition 16.5.1, where the clock is written $Y$), can be shown to fail for $t < 0$ by a mechanism visible at finite height, as Lehmer pairs make $\Lambda$ close to 0 visible [101]?

Problem 19.4 (The window below $\delta$). For a quadruple at distance $\delta > \delta_c$ the level test detects it on a set of levels contained in $(0, \delta)$, an interval $(\varepsilon_c, \delta)$ in our examples. At the ordinate $\varepsilon_c$ is the inverse of the visibility curve of Proposition 19.3.2. Describe the distribution of $\delta_{\mathrm{det}}$, and of the onset $\varepsilon_c(\delta, \gamma^*)$, over heights in terms of the local zero statistics, and compare it with the prediction of the sine kernel; Table 19.2 gives its lower end, $\delta_c$.

### Notes

The ordinate formula for $\pi\rho_\varepsilon(\gamma)$, the first order visibility condition and the lattice threshold $(\sqrt{6}/\pi)\Delta$ of Section 19.3 are those of Section 5.2.2, and inserting hypothetical quadruples into the density computed from the zeros is the device

of Section 11.4.4; the closed forms of Proposition 19.3.2, the level count of Corollary 19.3.3 and the computations at the actual zeros are in this chapter. The bound $\Lambda \leq 0.2$ is that of Platt and Trudgian [419], and the visibility of a Lehmer pair through $\Lambda$ is the work of Csordas, Smith and Varga [101].

# Chapter 20

# Deconvolution as a damped Weil criterion

*What is not claimed.* The half plane $\operatorname{Re} s > \frac{7}{8}$ has not been refereed. The eigenvalues of the ladder are variational upper bounds, the converged values are extrapolations of such bounds with an error estimate, and none of them is a certificate.

*The question.* Let $\xi$ have zeros $\rho = \beta + i\gamma$, let $\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ be the zeros of $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$, and put $\sigma_{\max} = \sup_\rho \operatorname{Re}\rho \in [\frac{1}{2}, 1]$. We say that a level $\varepsilon > 0$ is *to the right of the zeros* if $\beta < \frac{1}{2} + \varepsilon$ for every zero. Every $\varepsilon \geq \frac{1}{2}$ is, since $\beta < 1$, whether or not $\sigma_{\max} < 1$; the latter is the quasi Riemann hypothesis. For real $\alpha$ write $v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\theta)$ and $b_\alpha = v_\alpha(0) = \frac{\xi'}{\xi}(\alpha)$, so that $\rho_\varepsilon = \pi^{-1} v_{1/2+\varepsilon}$ is the density (1.2.7): for $\varepsilon > 0$ the Poisson smoothing of the zeros (Proposition 1.2.3) and, when $\varepsilon$ is to the right of the zeros, the Thorin density of the shifted clock at basepoint $\frac{1}{2}+\varepsilon$ (Theorem 1.2.2). Write $P_a(x) = a/(\pi(a^2+x^2))$ for the Cauchy density of scale $a > 0$. Theorem 19.2.2 of Chapter 19 says that, for $\varepsilon_0$ to the right of the zeros and $0 \leq \varepsilon < \varepsilon_0$, there is a positive measure $\nu$ with $\int (1+\theta^2)^{-1}\nu(d\theta) < \infty$ and $\rho_{\varepsilon_0} = P_{\varepsilon_0-\varepsilon} * \nu$ if and only if no zero has $\beta > \frac{1}{2}+\varepsilon$. The proof runs a Poisson integral of $\nu$ down from $\varepsilon_0$ and uses harmonic continuation: a positive $\nu$ would make $\rho_\varepsilon(\theta)$ harmonic in $(\theta, \varepsilon)$ near a zero with $\beta - \frac{1}{2} > \varepsilon$, where $\xi'/\xi$ has a pole.

Asking whether the deconvolution *can be done* has two meanings. One is constructive: can $P_{\varepsilon_0}^{-1}\rho_{\varepsilon_0}$ be produced, from data we can compute, without assuming where the zeros are? The other is the theorem itself: can it be shown to be a positive measure? We answer the first completely (Section 20.1.1) and reduce the second to a two parameter family of finite problems (Sections 20.1.2 and 20.2), whose behaviour we partly prove and compute.

## 20.1 Weil's distribution and the damped criterion

### 20.1.1 The deconvolved object exists: it is Weil's distribution

With $\hat f(t)=\int f(x)e^{itx}dx$ the Fourier transform of $P_a$ is $e^{-a|x|}$, so deconvolution by $P_a$ multiplies Fourier transforms by $e^{a|x|}$. Let $\mathcal B$ be the space of even functions $\varphi(\theta)=\frac{1}{2\pi}\int\hat\varphi(x)e^{-i\theta x}dx$ with $\hat\varphi$ of compact support, Lipschitz, and with $\hat\varphi'$ of bounded variation. Then $\varphi$ is entire of exponential type and $\varphi(\theta+is)=O(\theta^{-2})$ uniformly for $|s|\le 1$, since $\hat\varphi(x)e^{sx}$ has the same properties. For $a\ge 0$ let $P_a^{-1}\varphi\in\mathcal B$ have Fourier transform $\hat\varphi(x)e^{a|x|}$; the corner of $e^{a|x|}$ at 0 keeps it in $\mathcal B$.

Theorem 20.1.1 (The deconvolved functional). *Let $\varepsilon_0$ be to the right of the zeros. For $\varphi\in\mathcal B$ put $D(\varphi)=\int\rho_{\varepsilon_0}(\theta)\,(P_{\varepsilon_0}^{-1}\varphi)(\theta)\,d\theta$. Then $D$ does not depend on $\varepsilon_0$, and with $g(x)=\hat\varphi(x)/2\pi$, so that $\varphi(\theta)=\int g(x)e^{i\theta x}dx=\hat g(\theta)$,*

$$D(\varphi)=\sum_\rho\varphi(\gamma_\rho)=\hat g\big(\tfrac{i}{2}\big)+\hat g\big(-\tfrac{i}{2}\big)-(\gamma_E+\log\pi)\,g(0)\\ +\int_0^\infty\frac{2\big(e^{-2x}g(0)-e^{-x/2}g(x)\big)}{1-e^{-2x}}\,dx\\ -2\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt n}\,g(\log n),$$

*where the sum over $\rho$ runs over all zeros with multiplicity, in conjugate pairs, and converges absolutely. Through the $x$ form of its archimedean term, the Thorin functional $U$ of Section 5.3.1 extends to $\mathcal B$, and then $D(\varphi)=2\langle U,\varphi(\sqrt{\cdot})\rangle$.*

*Proof.* By the expansion of $\rho_{\varepsilon_0}$ over the zeros (Proposition 1.2.3, summed over all zeros $\rho=\beta+i\gamma$, with $\gamma$ of both signs), $D(\varphi)=\sum_\rho\int P_{c_\rho}(\theta-\gamma)\,(P_{\varepsilon_0}^{-1}\varphi)(\theta)\,d\theta$ with $c_\rho=\varepsilon_0-(\beta-\frac12)>0$; the interchange is justified because $\rho_{\varepsilon_0}=O(\log(2+|\theta|))$ and $P_{\varepsilon_0}^{-1}\varphi=O(\theta^{-2})$. By Parseval each term is $\frac{1}{2\pi}\int\hat\varphi(x)e^{(\varepsilon_0-c_\rho)|x|}e^{-i\gamma x}\,dx$. A zero $\rho$ off the line and its partner $1-\bar\rho$ have the same $\gamma$ and opposite $\beta-\frac12=\pm\delta$, so the pair gives $\frac{1}{2\pi}\int\hat\varphi(x)e^{-i\gamma x}\,2\cosh(\delta x)\,dx=\varphi(\gamma-i\delta)+\varphi(\gamma+i\delta)$, because $e^{\delta|x|}+e^{-\delta|x|}=e^{\delta x}+e^{-\delta x}$; a zero on the line is its own partner and gives the single term $\varphi(\gamma)$. This is $\sum_\rho\varphi(\gamma_\rho)$, absolutely convergent by the uniform $O(\theta^{-2})$ bound, and it does not involve $\varepsilon_0$. The second expression is the explicit formula of Weil [512], [288, Theorem 5.12], with the archimedean term written in the variable $x$ by $\operatorname{Re}\psi(\frac14+\frac{it}{2})=-\gamma_E+\int_0^\infty\frac{e^{-u}-e^{-u/4}\cos(tu/2)}{1-e^{-u}}\,du$; this integral converges because $g$ is Lipschitz at 0, and the formula extends from the usual class to $\mathcal B$ by approximation: mollifying $g$ gives functions of the usual class supported in a fixed interval, with uniformly bounded Lipschitz constants, and both sides are continuous along such sequences, the zero side by the uniform $O(\theta^{-2})$ bound. The comparison with (5.3.1), whose functional sums once over each pair $\{\rho,1-\rho\}$, gives the factor 2. □

For $g$ even, Lipschitz and of compact support, with $g'$ of bounded variation, so that $\hat{g} \in \mathcal{B}$, write $\langle D, g\rangle = D(\hat{g})$, and for $g$ not even apply it to the even part of $g$. By the theorem $\langle D, g\rangle = \sum_\rho \hat{g}(\gamma_\rho)$, which is the action on $g$ of Weil's distribution $D(x) = \sum_\rho e^{(\rho-\frac{1}{2})x}$ of Proposition 7.2.14.

The distribution $P_{\varepsilon_0}^{-1}\rho_{\varepsilon_0}$ on $\mathcal{B}$, which exists with no hypothesis, is a positive measure exactly when RH holds: under RH it is $\sum_\gamma \delta_\gamma$, and if $D(\varphi) = \int \varphi\, d\mu$ on $\mathcal{B}$ with $\mu \geq 0$, then $\int \rho_{\varepsilon_0}\psi = D(P_{\varepsilon_0} * \psi) = \int (P_{\varepsilon_0} * \psi)\, d\mu$ for every $\psi \in \mathcal{B}$, which gives $\int (1+\theta^2)^{-1} d\mu < \infty$ (take $\psi \geq 0$ with $P_{\varepsilon_0} * \psi \geq c(1+\theta^2)^{-1}$) and $\rho_{\varepsilon_0} = P_{\varepsilon_0} * \mu$, so that RH follows from Theorem 19.2.2. On the functions $\varphi = |\hat{h}|^2$ the statement $D(\varphi) \geq 0$ is Weil's criterion [512], in the form the book uses in Section 3.5.2. We checked the second expression against $2\sum_{\gamma>0} \hat{g}(\gamma)$ for the Gaussian $g(x) = e^{-(x/c)^2}$ (`code/dc_weil.py`): at $c = \frac{1}{2}$ both sides are $6.69239857682 \times 10^{-6}$ and they differ by less than $10^{-69}$ at seventy digits of working precision.

### 20.1.2 The damped Weil criterion

Partial deconvolution, to depth $\varepsilon$ instead of to the line, corresponds to damping. For bounded $h$ of bounded variation and compact support put $\tilde{h}(x) = \overline{h(-x)}$, so that $h * \tilde{h}$ is Lipschitz and Hermitian, and

$$Q_\varepsilon(h) = \big\langle D,\, e^{-\varepsilon|x|}\,(h * \tilde{h})\big\rangle.$$

The Fourier transform of $e^{-\varepsilon|x|}$ is $2\pi P_\varepsilon$ and that of $h * \tilde{h}$ is $|\hat{h}|^2$, so

$$\widehat{e^{-\varepsilon|\cdot|}(h * \tilde{h})} = P_\varepsilon * |\hat{h}|^2,$$

a nonnegative function. Real $h$ suffice in what follows, since $Q_\varepsilon(h) = Q_\varepsilon(\operatorname{Re} h) + Q_\varepsilon(\operatorname{Im} h)$.

Theorem 20.1.2 (Damped Weil criterion). *Let $0 \leq \varepsilon \leq \frac{1}{2}$. Then $Q_\varepsilon(h) \geq 0$ for every $h \in C_c^\infty(\mathbb{R})$ if and only if no nontrivial zero of $\zeta$ has $\beta > \frac{1}{2} + \varepsilon$. When $\varepsilon$ is to the right of the zeros,*

$$Q_\varepsilon(h) = \int_{\mathbb{R}} |\hat{h}(\theta)|^2\, \rho_\varepsilon(\theta)\, d\theta \tag{20.1.1}$$

*for every bounded $h$ of bounded variation and compact support. In particular: at $\varepsilon = 0$ this is Weil's criterion for RH; at $\varepsilon = \frac{1}{2}$ it holds with no hypothesis; and the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [398], which has not been refereed, is equivalent to $Q_{3/8} \geq 0$.*

*Proof.* Formula (20.1.1) follows from Theorem 20.1.1. Let $h$ be real, bounded, of bounded variation and of compact support. Then $F = h * \tilde{h}$ is Lipschitz and of compact support, and its derivative $h * d\tilde{h}$, the convolution of a bounded

function of bounded variation with a finite measure, is of bounded variation; so $e^{-\varepsilon|x|}F$ is Lipschitz, of compact support, with a derivative of bounded variation, and $\varphi = P_\varepsilon * |\hat{h}|^2$, the Fourier transform of $e^{-\varepsilon|x|}F$, and $|\hat{h}|^2$, that of $F$, lie in $\mathcal{B}$. As $P_\varepsilon^{-1}\varphi = |\hat{h}|^2$, the first identity of Theorem 20.1.1 with $\varepsilon_0 = \varepsilon$ gives $Q_\varepsilon(h) = D(\varphi) = \int |\hat{h}|^2 \rho_\varepsilon \, d\theta$; for complex $h$ add the two parts, the cross terms of $|\hat{h}|^2$ being odd and $\rho_\varepsilon$ even. For bounded piecewise $C^1$ functions $h$ this is Proposition 17.8.1 of Chapter 17 at $\alpha = \frac{1}{2} + \varepsilon$, where the form $\langle \varphi, \varphi \rangle_\alpha$ is $Q_\varepsilon(\varphi)$. The *if* direction is Theorem 17.8.3(i): if no zero has $\beta > \frac{1}{2} + \varepsilon$, then either $\varepsilon$ is to the right of the zeros and $\rho_\varepsilon > 0$, so that $Q_\varepsilon \geq 0$ by (20.1.1), or some zero has $\beta = \frac{1}{2} + \varepsilon$, and then $Q_\varepsilon(h) = \lim_{\varepsilon' \downarrow \varepsilon} Q_{\varepsilon'}(h) \geq 0$, the limit holding because $Q_\varepsilon(h)$ is a finite combination of integrals continuous in $\varepsilon$.

Conversely, suppose $Q_\varepsilon \geq 0$. The map $T_\varepsilon : g \mapsto \langle D, e^{-\varepsilon|x|} g \rangle$ is a distribution of order at most one, the corner of $e^{-\varepsilon|x|}$ at 0 being harmless in the archimedean integral, and $Q_\varepsilon(h) = T_\varepsilon(h * \tilde{h}) \geq 0$ says that it is positive definite. By the Bochner Schwartz theorem [147, Chapter II] it is tempered and $T_\varepsilon(g) = \int \hat{g} \, d\nu_\varepsilon$ for a positive tempered measure $\nu_\varepsilon$ and every Schwartz $g$. Fix $\varepsilon_0 > \varepsilon$ with $\varepsilon_0 \geq \frac{1}{2}$, put $c = \varepsilon_0 - \varepsilon$, and let $k_\eta(x) = e^{-c\sqrt{x^2+\eta^2}}$, smooth, positive definite and of exponential decay, with $\hat{k}_\eta \geq 0$, $\hat{k}_\eta \to 2\pi P_c$ pointwise and $\hat{k}_\eta \leq C(1+\theta^2)^{-1}$ uniformly in $\eta \leq 1$. For $h \in C_c^\infty$ and $g = h * \tilde{h}$, $T_\varepsilon(k_\eta g) = \frac{1}{2\pi} \int (\hat{k}_\eta * |\hat{h}|^2) \, d\nu_\varepsilon$, and as $\eta \to 0$ the left side tends to $T_\varepsilon(e^{-c|x|} g) = Q_{\varepsilon_0}(h) = \int |\hat{h}|^2 \rho_{\varepsilon_0}$, because $k_\eta g \to e^{-c|x|} g$ uniformly, with supports in a fixed interval and uniformly bounded Lipschitz constants, which is enough for dominated convergence in the archimedean integral and in the other terms of $\langle D, \cdot \rangle$. By Fatou, $\int (P_c * |\hat{h}|^2) \, d\nu_\varepsilon \leq \int |\hat{h}|^2 \rho_{\varepsilon_0} < \infty$; choosing $h$ with $|\hat{h}|^2 > 0$ near 0 gives $P_c * |\hat{h}|^2 \geq c'(1+\theta^2)^{-1}$, so $\int (1+\theta^2)^{-1} d\nu_\varepsilon < \infty$, and dominated convergence then gives $\int (P_c * |\hat{h}|^2) \, d\nu_\varepsilon = \int |\hat{h}|^2 \rho_{\varepsilon_0}$, that is $\int |\hat{h}|^2 \, (P_c * \nu_\varepsilon) = \int |\hat{h}|^2 \rho_{\varepsilon_0}$. Polarising, the same holds with $|\hat{h}|^2$ replaced by $\hat{h}_1 \overline{\hat{h}_2}$, and these products span a dense set, so $\rho_{\varepsilon_0} = P_{\varepsilon_0 - \varepsilon} * \nu_\varepsilon$. Theorem 19.2.2 gives $\beta \leq \frac{1}{2} + \varepsilon$ for every zero. □

The theorem is the Fourier dual of the deconvolution theorem; at $\varepsilon = 0$ it is a proof of Weil's criterion along the lines of the harmonic continuation of Theorem 19.2.2. The damping also explains why $\varepsilon = \frac{1}{2}$ is free. The primes enter with weight $\Lambda(n) n^{-1/2-\varepsilon}$, absolutely summable exactly for $\varepsilon > \frac{1}{2}$: the damped criterion at $\varepsilon = \frac{1}{2}$ sits at the edge of the Euler product, the basepoint one of Chapter 9.

## 20.2 The ladder

Restrict the support of $h$. For $a > 0$ and $\varepsilon \geq 0$ let

$$\lambda_{\min}\big(\tfrac12+\varepsilon, a\big) = \inf\Big\{ \frac{Q_\varepsilon(h)}{\|h\|_2^2} : h \text{ bounded, real, of bounded variation, supported in } [-a,a],\ h \neq 0 \Big\}.$$

This is the least eigenvalue $\lambda_{\min}(\alpha, a)$, $\alpha = \frac12 + \varepsilon$, of Chapter 17, with the same normalisation. The function $e^{-\varepsilon|x|}(h * \tilde h)$ is supported in $[-2a, 2a]$, so $Q_\varepsilon$ involves only the prime powers $n \leq e^{2a}$: each $\lambda_{\min}(\frac12+\varepsilon, a)$ is the bottom of the spectrum of an explicit form. RH is $\lambda_{\min}(\frac12, a) \geq 0$ for every $a$. Yoshida [525, Theorem 1] proved positivity of Weil's form for $a \leq \frac12 \log 2$, the range in which no prime enters, Bombieri [55] treated other test functions of small support, and Connes and Consani reproved Yoshida's range from the semilocal trace formula [87]. Zhu certifies the lower bound $8.9 \times 10^{-18}$ for $\lambda_{\min}(\frac12, a)$ at $a = 0.8$ (Corollary 6.3 of [526]), and reports an empirical law for the decay of $\lambda_{\min}(\frac12, a)$ [526] (Chapter 18), and Connes, Consani and Moscovici find that the Fourier transform of the lowest eigenvector of a truncated form reproduces the low zeros to high accuracy [88]. The last two papers are cited as preprints.

Theorem 20.2.1 (The ladder at depth $\varepsilon > 0$). *Let $\varepsilon > 0$ be to the right of the zeros, and suppose that $v_{1/2+\varepsilon}(\theta) > v_{1/2+\varepsilon}(0)$ for $\theta \neq 0$, with $v''_{1/2+\varepsilon}(0) > 0$ and $\liminf_{|\theta|\to\infty} v_{1/2+\varepsilon}(\theta) > v_{1/2+\varepsilon}(0)$.*

*(a) $\lambda_{\min}(\frac12+\varepsilon, a)$ is nonincreasing in $a$, and $\lambda_{\min}(\frac12+\varepsilon, a) \geq 2b_{1/2+\varepsilon} > 0$ for every $a$.*

*(b) $\lambda_{\min}(\frac12+\varepsilon, a) \to 2b_{1/2+\varepsilon}$ as $a \to \infty$, and there are constants $c, a_0 > 0$ with*

$$\frac{c}{a^2} \leq \lambda_{\min}\big(\tfrac12+\varepsilon, a\big) - 2b_{1/2+\varepsilon} \leq \frac{\pi^2 v''_{1/2+\varepsilon}(0)/4 + o(1)}{a^2} \qquad (a \geq a_0).$$

*(c) At $\varepsilon = 0$, with none of the hypotheses above, under RH, $\lambda_{\min}(\frac12, a) \geq 0$ and $\lambda_{\min}(\frac12, a) = O(a^{-N})$ for every $N$, so $\lambda_{\min}(\frac12, a) \to 0$; if RH fails, $\lambda_{\min}(\frac12, a) < 0$ for all large $a$.*

*Proof.* (a) is Theorem 17.8.5(i) of Chapter 17 at $\alpha = \frac12 + \varepsilon$, whose $m_\alpha = \inf_\theta v_\alpha(\theta)$ is $b_\alpha$ here, the minimum being at the origin; and $b_{1/2+\varepsilon} = \sum_\rho \mathrm{Re}(\frac12 + \varepsilon - \rho)^{-1} > 0$ because every term is positive. We use the first step of that proof: by (20.1.1) and Parseval, $\|\hat h\|_2^2 = 2\pi\|h\|_2^2$, so $Q_\varepsilon(h)/\|h\|_2^2 = 2\langle v_{1/2+\varepsilon}\rangle_h$, the average of $v_{1/2+\varepsilon}$ against the probability density $|\hat h|^2/\|\hat h\|_2^2$, which is at least $2\min v_{1/2+\varepsilon} = 2b_{1/2+\varepsilon}$.

(b) Write $v = v_{1/2+\varepsilon}$. Upper bound: take $h(u) = \cos(\pi u/2a)$ on $[-a, a]$. Then $\langle\theta^2\rangle_h = \|h'\|_2^2/\|h\|_2^2 = \pi^2/4a^2$, and the normalised density satisfies $|\hat h(\theta)|^2/\|\hat h\|_2^2 =$

$O(a^{-3}\theta^{-4})$ for $|\theta| \geq 1/a$. Write $v(\theta) = v(0) + \frac{1}{2}v''(0)\theta^2 + R(\theta)$, with $|R(\theta)| \leq C\theta^4$ for $|\theta| \leq \eta$ and $|R(\theta)| \leq C\theta^2 \log(2 + |\theta|)$ for all $\theta$. Splitting at $|\theta| = \eta$ gives $\langle R \rangle_h = O(\eta a^{-3}) + O(a^{-3}\eta^{-1})$, and with $\eta = a^{-1/2}$ this is $o(a^{-2})$. Hence $\lambda_{\min}(\frac{1}{2} + \varepsilon, a) \leq 2v(0) + v''(0)\pi^2/4a^2 + o(a^{-2})$. Lower bound: by the hypotheses there is $c_1 > 0$ with $v(\theta) - v(0) \geq c_1 \min(\theta^2, 1)$. For $h$ supported in $[-a, a]$ the fraction of $\|\hat{h}\|_2^2$ carried by $|\theta| \leq \kappa/a$ is at most the largest eigenvalue $\ell_0(\kappa)$ of the time and band limiting operator with product $\kappa$, which is less than 1 for every finite $\kappa$ and tends to 0 with $\kappa$ [481]. So $\langle \min(\theta^2, 1) \rangle_h \geq (\kappa/a)^2(1 - \ell_0(\kappa))$ once $\kappa \leq a$, which is the lower bound with $c = 2c_1\kappa^2(1 - \ell_0(\kappa))$. Convergence $\lambda_{\min}(\frac{1}{2} + \varepsilon, a) \to 2b_{1/2+\varepsilon}$ follows from the two bounds.

(c) Under RH, $Q_0(h) = \sum_\gamma |\hat{h}(\gamma)|^2 \geq 0$ by Theorem 20.1.1. Fix a real $h_1 \in C_c^\infty[-1, 1]$, $h_1 \neq 0$, and put $h_a(u) = h_1(u/a)$. Then $\hat{h}_a(\theta) = a\hat{h}_1(a\theta)$ with $|\hat{h}_1(t)| \leq C_N|t|^{-N}$, so $Q_0(h_a)/\|h_a\|_2^2 \leq C'_N a^{1-2N} \sum_\gamma \gamma^{-2N}$, every $|\gamma|$ being larger than 14. If RH fails, Weil's criterion gives $h \in C_c^\infty$, which may be taken real, with $Q_0(h) < 0$; it is supported in some $[-a_0, a_0]$, and by monotonicity $\lambda_{\min}(\frac{1}{2}, a) \leq \lambda_{\min}(\frac{1}{2}, a_0) < 0$ for $a \geq a_0$. □

At $\varepsilon = \frac{1}{2}$ the hypotheses of the theorem hold with no assumption on the zeros: the level is to the right of the zeros because $\zeta$ has no zeros on $\operatorname{Re} s = 1$, and Proposition 9.1.11 gives the minimum at the origin, with $v_1''(0) > 0$, and in its proof $v_1 > 0.05 > b_1$ for $|\theta| \geq 2000$, which is the third condition. For $0 < \varepsilon < \frac{1}{2}$ the level is to the right of the zeros exactly when the closed half plane $\operatorname{Re} s \geq \frac{1}{2} + \varepsilon$ is free of zeros, which the announced half plane gives only for $\varepsilon > \frac{3}{8}$; the minimum at the origin is certified for $0 < |\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$, $H_0 = 3 \times 10^{12}$, by Theorem 9.1.18(ii) (Table 9.4), and its holding on every line with $\frac{1}{2} < \alpha < 1$ jointly is equivalent to RH (part (vi) there). The third condition is not known for $0 < \varepsilon < \frac{1}{2}$ without RH; under RH it holds, because $v_{1/2+\varepsilon}$ is then a sum of Poisson kernels of the zeros and tends to infinity.

Remark 20.2.2. At $\varepsilon = \frac{1}{2}$ the hypotheses of Theorem 20.2.1(a) and (b) hold, as just said, and $2b_1 = 2\frac{\xi'}{\xi}(1) = 2\big(1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi\big) = 2 + \gamma_E - \log 4\pi = 0.0461914\ldots$. So for every $a > 0$ and every bounded real $h$ of bounded variation supported in $[-a, a]$,

$$Q_{1/2}(h) \;\geq\; \big(2 + \gamma_E - \log 4\pi\big)\,\|h\|_2^2 = 0.0461914\ldots\,\|h\|_2^2,$$

and by (20.1.1) this is Proposition 9.1.11 read through Plancherel, and Theorem 17.8.5(i) at $\alpha = 1$, which is Corollary 17.8.8 of Chapter 17; what the ladder adds is the rate in $a$ of Theorem 20.2.1(b).

### 20.2.1 Numerical results

We computed $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$ variationally, with $h$ in the span of the first $M$ Legendre polynomials $q_0, \ldots, q_{M-1}$ on $[-a, a]$, normalised in $L^2$: the correlations $q_j * \tilde{q}_k$ by

Gauss Legendre quadrature, which is exact for polynomials; the archimedean and pole integrals by Gauss Legendre quadrature on $[0, 2a]$, plus the exact tail of the $g(0)$ term; and the primes exactly. The script `dc_ladder.py` (`code/README.md`) repeats this for $M = 24$ to 128 and computes the same eigenvalue on a second basis, the normalised Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, for $N = 24$ to 1536, from the matrices of the module `yo_matrix.py` of Chapter 17. Every computed eigenvalue is an upper bound for $\lambda_{\min}$, since a finite basis is a variational class, and needs no hypothesis on the zeros, because $Q_\varepsilon$ is an explicit finite form; the hypotheses of Theorem 20.2.1 enter only in what the numbers are compared with. Rounding is not the limit: double precision agrees with thirty digit arithmetic to $1.4 \times 10^{-14}$ at most, and 1.5 times the quadrature nodes changes the eigenvalue by at most $2 \times 10^{-15}$. The limit is the size of the basis.

Between $M = 24$ and $M = 32$ the Legendre bounds change by at most $2.6 \times 10^{-6}$, but they are not converged at that level: for $\varepsilon > 0$ the $M = 32$ values lie above the converged values defined next by $0.6 \times 10^{-7}$ to $1.8 \times 10^{-6}$, so the change between two sizes understates the error. The Fourier sections are nested, and their eigenvalues converge like a power, $\lambda_N - \lambda_\infty \propto N^{-p}$ with $p$ between 1.18 and 1.69 over the whole grid. The *converged value* is the power law extrapolation from the three largest sizes $N = 384$, 768, 1536, and its uncertainty is the shift of that value when the largest size is dropped (sizes 192, 384, 768): at most $2.7 \times 10^{-8}$ at $a = 0.25$, $3.0 \times 10^{-9}$ at $a = 0.5$ and $1.7 \times 10^{-9}$ for $a \ge 0.75$. This is an error estimate and not a bound. Two checks support it. The Legendre bound with $M = 128$ lies above the converged value in all 40 entries with $\varepsilon > 0$, by $1.8 \times 10^{-8}$ to $1.5 \times 10^{-7}$, as an upper bound must. And the converged values of Table 17.2 of Chapter 17, obtained there by Richardson extrapolation of the sizes $N = 96$, 192, 384, agree with the values computed here for the same cells to $2.0 \times 10^{-8}$ at most over its twenty entries with $\alpha > \frac{1}{2}$ and to $4.1 \times 10^{-9}$ for $a \ge 0.8$.

Table 20.1 shows the converged values for $\varepsilon > 0$ and, at $\varepsilon = 0$, for $a = 0.25$ and 0.5, and Table 20.2 the rung at $\varepsilon = 0$ for $0.25 \le a \le 0.5$. Table 17.2 of Chapter 17 and Table 18.1 of Chapter 18 list eigenvalues of the same kind for other windows and levels; the level 0.375 of Table 20.1 is that of the announced half plane. Figure 20.1 plots the eigenvalue on a logarithmic scale.

For every level in Table 20.1 the converged values decrease to $2b_{1/2+\varepsilon}$ and never cross it (nor does any computed upper bound, in either basis), as Theorem 20.2.1(a) predicts when its hypotheses hold; at $\varepsilon = \frac{1}{2}$ this is Remark 20.2.2, and at $\varepsilon = \frac{3}{8}$ the announced half plane gives $\lambda_{\min}(\frac{7}{8}, a) \ge 0$ (Theorem 20.1.2), while $\lambda_{\min} \ge 2b_{7/8}$ needs in addition the minimum of $v_{7/8}$ at the origin, which Theorem 9.1.18(ii) certifies only up to height $3 \times 10^{12}$. The margin is linear in $\varepsilon$. With no hypothesis $2b_{1/2+\varepsilon} \le 2V_\xi\, \varepsilon$, where $V_\xi = (\log \xi)''(\frac{1}{2}) = 0.04620998623$ is the constant of Theorem 9.1.18(i), and $2b_{1/2+\varepsilon}/\varepsilon \to 2V_\xi = 0.0924200$ as $\varepsilon \downarrow 0$; under RH $V_\xi = 2p_1$ with $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231050$.

The rate of approach changes character at the line. Table 20.3 compares the

| $a$ | $\varepsilon = 0.5$ | 0.375 | 0.2 | 0.1 | 0.05 | 0 |
|---|---|---|---|---|---|---|
| 0.25 | 0.0894856 | 0.0756372 | 0.0560458 | 0.0447429 | 0.0390618 | 0.0333609 |
| 0.5 | 0.0495266 | 0.0372606 | 0.0200130 | 0.0100860 | 0.0050779 | $9.33 \times 10^{-7}$ |
| 0.75 | 0.0474318 | 0.0355980 | 0.0190070 | 0.0095123 | 0.0047593 | |
| 1 | 0.0468396 | 0.0351417 | 0.0187506 | 0.0093779 | 0.0046899 | |
| 1.5 | 0.0464620 | 0.0348541 | 0.0185931 | 0.0092974 | 0.0046489 | |
| 2 | 0.0463394 | 0.0347613 | 0.0185428 | 0.0092720 | 0.0046361 | |
| 3 | 0.0462555 | 0.0346979 | 0.0185087 | 0.0092548 | 0.0046275 | |
| 4 | 0.0462270 | 0.0346764 | 0.0184971 | 0.0092490 | 0.0046246 | |
| $2b_{1/2+\varepsilon}$ | 0.0461914 | 0.0346497 | 0.0184828 | 0.0092418 | 0.0046210 | 0 |

Table 20.1: The ladder $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$, converged values: the limit of the variational upper bounds on the Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, by power law extrapolation from $N = 384$, 768, 1536 (`code/dc_ladder.py`, parts `grid` for $\varepsilon > 0$ and `y0` for $\varepsilon = 0$, `data/dc_ladder.json`). The uncertainty of an entry is the one defined in the text; it is an error estimate and not a bound. The last row is the limit $2b_{1/2+\varepsilon} = 2\,\xi'/\xi(\frac{1}{2}+\varepsilon)$ of Theorem 20.2.1 where its hypotheses hold (for $\varepsilon < \frac{1}{2}$ they are conditional, and at $\varepsilon = 0$ the limit is 0 under RH), computed by `code/dc_weil.py`, part W2 (`data/dc_weil.json`). At $\varepsilon = 0$ the entries for $a \ge 0.75$ are omitted: in double precision they are below the rounding level of about $10^{-13}$, and the upper bounds from extended precision are given in the text.

| $a$ | $M = 32$ | converged | uncertainty |
|---|---|---|---|
| 0.25 | $3.3363 \times 10^{-2}$ | $3.3361 \times 10^{-2}$ | $2.7 \times 10^{-8}$ |
| 0.3 | $7.5722 \times 10^{-3}$ | $7.5715 \times 10^{-3}$ | $1.4 \times 10^{-8}$ |
| $\frac{1}{2}\log 2$ | $1.3294 \times 10^{-3}$ | $1.3293 \times 10^{-3}$ | $3.4 \times 10^{-9}$ |
| 0.35 | $1.1938 \times 10^{-3}$ | $1.1935 \times 10^{-3}$ | $1.3 \times 10^{-9}$ |
| 0.4 | $1.8173 \times 10^{-4}$ | $1.8126 \times 10^{-4}$ | $1.1 \times 10^{-9}$ |
| 0.45 | $1.6232 \times 10^{-5}$ | $1.6149 \times 10^{-5}$ | $1.7 \times 10^{-10}$ |
| 0.5 | $9.4039 \times 10^{-7}$ | $9.3317 \times 10^{-7}$ | $1.4 \times 10^{-11}$ |

Table 20.2: The rung at $\varepsilon = 0$ for $a \le 0.5$: the Legendre bound with $M = 32$, the converged value (Fourier modes to 1536, power law extrapolation) and its uncertainty, the shift of the extrapolation when $N = 1536$ is dropped, an error estimate and not a bound. At $a = \frac{1}{2}\log 2 = 0.3466$, the largest window in which no prime enters, the value is $1.3293 \times 10^{-3}$. Script `code/dc_ladder.py`, part `y0`, data `data/dc_ladder.json`.

scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ of the converged values with the upper constant $\bar{c}_\varepsilon = \pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 20.2.1(b): the ratio falls towards 1 as $a$ grows. An upper bound for $\lambda_{\min}$ that approaches the constant from above says nothing about whether the true limit is that constant or a smaller one, and the

converged values are extrapolations, so we fit them: a least squares fit on $2 \le a \le 7$ (prime powers to $e^{14}$ at $a = 7$) of $a^2(\lambda_{\min} - 2b_{1/2+\varepsilon}) = \bar{c}_\varepsilon(C_0 + C_1/a + C_2/a^2)$ gives $C_0 = 1.0002$, 1.0003, 1.0005 at $\varepsilon = \frac{1}{2}$, 0.2, 0.1, with an uncertainty from the errors of the data of 0.00004, 0.0001, 0.0002, and over the choice of the window ($a \ge 1.5$, 2 or 3) and of the degree (2 or 3) $C_0$ ranges over 0.9999 to 1.0004, 0.9995 to 1.0010 and 0.9991 to 1.0016. The data therefore support the limit $\bar{c}_\varepsilon$ to within 0.04 to 0.2 percent, with a positive correction $C_1 = 0.139$, 0.159, 0.166 and $C_2 = 0.037$, 0.045, 0.051 and residuals below $1.1 \times 10^{-4}\,\bar{c}_\varepsilon$; the term in $a^{-2}$ is needed, a fit by $C_0 + C_1/a$ alone being rejected by its residuals. This is numerical evidence and not a proof (Problem 20.2). It also indicates, numerically, that $\bar{c}_\varepsilon$ is not a bound at finite $a$: the scaled excess exceeds it at every computed $a$ ($1.024\,\bar{c}_\varepsilon$ at $a = 6$, $\varepsilon = \frac{1}{2}$).

At $\varepsilon = 0$ and $a = 0.35$ the $1.20 \times 10^{-3}$ of Table 17.2 is the finite section with $N = 24$, an upper bound for the value $1.19 \times 10^{-3}$ of Table 20.2. From $a = 0.75$ on double precision returns only rounding noise, below about $10^{-13}$, and extended precision gives the upper bounds $\lambda_{\min}(\frac{1}{2}, a) < 4.4 \times 10^{-15}$, $1.92 \times 10^{-17}$ and $7.3 \times 10^{-30}$ at $a = 0.75$, 0.8 and 1 (Fourier modes $|n| \le 48$, 48 and 60, at 60, 60 and 80 digits). They fall steeply with the size of the section and depend on the basis (the Legendre bounds with $M = 24$ are $2.7 \times 10^{-13}$, $2.7 \times 10^{-14}$ and $2.1 \times 10^{-16}$, and with $M = 96$ they are $3.7 \times 10^{-15}$, $1.66 \times 10^{-17}$ and $6.3 \times 10^{-30}$, each of these three below the Fourier bound above), so they are upper bounds and not the order of the eigenvalue. Zhu's certified lower bound at $a = 0.8$ is $8.9 \times 10^{-18}$ (Corollary 6.3 of [526], Chapter 18), and the bounds $1.92 \times 10^{-17}$ and $1.66 \times 10^{-17}$ lie above it, the second within a factor 1.9.

The decay of the rung at $\varepsilon = 0$ is stated four ways in this book: Section 15.6.1 observes that the least eigenvalue on hat functions falls roughly geometrically in the support length $L = 2a$, up to $L = 8$; Chapter 17 observes that the finite sections follow $\exp(-2c\,ae^{2a})$ with $4.1 \le c \le 5.1$ for $0.25 \le a \le 0.8$, and does not prove it; Zhu proves, under RH and for large $a$, that the least eigenvalue over real even test functions, and so $\lambda_{\min}(\frac{1}{2}, a)$, is at most $\exp(-ae^a)$ (Theorem 1.3 of [526], Chapter 18, a preprint); and Theorem 20.2.1(c) proves, under RH, that the rung is $O(a^{-N})$ for every $N$. The four do not conflict: each of these falls is faster than any power of $a$, and Theorem 20.2.1(c) is the weakest of the conditional bounds and the only one whose proof is in the text.

The proof of Theorem 20.2.1(b) can be run with numbers, and Table 20.4 does so (`dc_rate.py`, parts `lower` and `trial`). Write $\psi(\theta) = v_{1/2+\varepsilon}(\theta) - v_{1/2+\varepsilon}(0)$. The infimum $c_1$ of $\psi(\theta)/\min(\theta^2, 1)$ is the limit $v''_{1/2+\varepsilon}(0)/2$ at the origin, on a grid of 600 points to height 40 and with the assumption that $\psi$ stays above its minimum on $[20, 40]$ beyond 40, and $\max_\kappa \kappa^2(1 - \ell_0(\kappa)) = 0.5348$ at $\kappa = 1.54$, where $\ell_0(1) = 0.5726$ and $\ell_0(2) = 0.8806$. So the lower constant of the theorem is $c = 2c_1 \cdot 0.5348 = 0.2167\,\bar{c}_\varepsilon$ for $a \ge 1.54$. Using at every radius at once the bound $P(|\theta| \le r) \le \ell_0(ar)$ for the probability density $|\hat{h}|^2/\|\hat{h}\|_2^2$ gives the layer

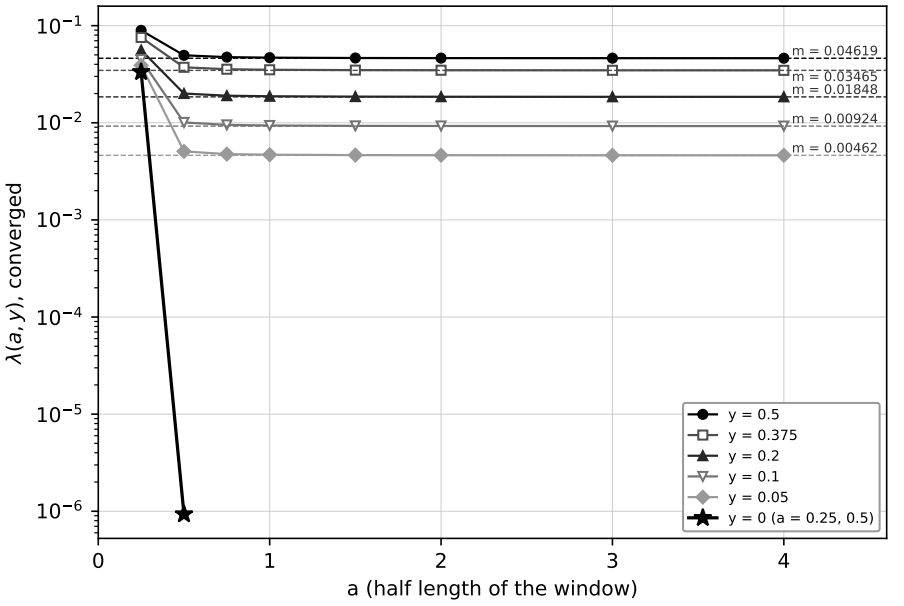



Figure 20.1: The converged ladder on a logarithmic scale: $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ against the window half length $a$ at $a = 0.25$, 0.5, 0.75, 1, 1.5, 2, 3, 4, for $\varepsilon = 0.5$, 0.375, 0.2, 0.1, 0.05 (five marker styles in black and grey), each with the value $2b_{1/2+\varepsilon}$, the limit where the hypotheses of Theorem 20.2.1 hold, as a dashed line labelled $m$, and for $\varepsilon = 0$ at $a = 0.25$ and 0.5 only (heavy line with stars). The values are those of Table 20.1; the legend and the axis write $y$ and $\lambda(a, y)$ for $\varepsilon$ and $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$. Drawn by `code/dc_ladder.py` (part `fig`) from `data/dc_ladder.json`.

cake bound $\lambda_{\min} - 2b_{1/2+\varepsilon} \geq 2\int_0^\infty \tilde{\psi}'(r)\,(1-\ell_0(ar))\,dr$, with $\tilde{\psi}$ the nondecreasing minorant of $\psi$ in $|\theta|$; it is $0.666\,\bar{c}_\varepsilon$ at $a = 4$, $\varepsilon = \frac{1}{2}$ ($3.66 \times 10^{-4}$), and tends to $8I_1/\pi^2\,\bar{c}_\varepsilon = 0.683\,\bar{c}_\varepsilon$ as $a \to \infty$, with $I_1 = \int_0^\infty \kappa\,(1-\ell_0(\kappa))\,d\kappa = 0.8425$. These lower bounds are floating point computations on a grid, numerical and not certificates, and for $\varepsilon < \frac{1}{2}$ the identity $Q_\varepsilon(h)/\|h\|_2^2 = 2\langle v_{1/2+\varepsilon}\rangle_h$ behind them presupposes that the level is to the right of the zeros.

The cosine trial function of the proof gives upper bounds, computed from the prime side (Table 20.4, last column). At $\varepsilon = \frac{1}{2}$ the ratio of its scaled excess to $\bar{c}_\varepsilon$ is 1.52, 1.37, 1.17 and 1.11 at $a = 1$, 2, 4 and 6, and $a$ times its remainder over $\bar{c}_\varepsilon$ is 0.65 to 0.68 for $3 \leq a \leq 8$, a positive correction of order $a^{-1}$. At $\varepsilon = 0.2$ and 0.1 the remainder oscillates in $a$ ($a$ times it over $\bar{c}_\varepsilon$ lies between 1.4 and 2.2 at $\varepsilon = 0.2$ and between 2.3 and 4.8 at $\varepsilon = 0.1$ for $1.5 \leq a \leq 8$), because the tail of $|\hat{h}|^2$, of order $\cos^2(a\theta)/\theta^4$, meets the peaks of $v_{1/2+\varepsilon}$ at the zeros, of height $1/\varepsilon$, and the cosine is a poor trial function for small $\varepsilon$.

| $\varepsilon$ | $\pi^2 v''_{1/2+\varepsilon}(0)/4$ | $a=1$ | $a=2$ | $a=3$ | $a=4$ |
|---|---|---|---|---|---|
| 0.5 | 5.485 | 6.48 | 5.92 | 5.76 | 5.69 |
| 0.2 | 2.200 | 2.68 | 2.40 | 2.33 | 2.29 |
| 0.1 | 1.100 | 1.36 | 1.21 | 1.17 | 1.15 |

Table 20.3: The scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ in units of $10^{-4}$, against the constant $\pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 20.2.1(b): converged $\lambda_{\min}$ (Fourier modes to 1536, power law extrapolation, as in Table 20.1), rounded to three digits. The uncertainty of an entry, the shift of the extrapolation times $a^2$, is at most $4\times10^{-5}$ in these units, an error estimate and not a bound. Script `code/dc_rate.py`, parts `const` and `table`, data `data/dc_rate.json`.

| $\varepsilon$ | $a$ | lower ($\kappa$) | lower (layer cake) | converged | upper (trial) |
|---|---|---|---|---|---|
| 0.5 | 1 | 0.950 (1.00) | 3.834 | 6.482 | 8.353 |
| 0.5 | 2 | 1.189 (1.54) | 3.692 | 5.919 | 7.493 |
| 0.5 | 4 | 1.189 (1.54) | 3.656 | 5.690 | 6.413 |
| 0.5 | 6 | 1.189 (1.54) | 3.661 | 5.619 | 6.095 |
| 0.2 | 1 | 0.381 (1.00) | 1.538 | 2.678 | 4.291 |
| 0.2 | 2 | 0.477 (1.54) | 1.481 | 2.401 | 4.584 |
| 0.2 | 4 | 0.477 (1.54) | 1.466 | 2.295 | 3.258 |
| 0.2 | 6 | 0.477 (1.54) | 1.468 | 2.262 | 2.857 |
| 0.1 | 1 | 0.191 (1.00) | 0.769 | 1.361 | 2.813 |
| 0.1 | 2 | 0.239 (1.54) | 0.741 | 1.206 | 3.745 |
| 0.1 | 4 | 0.239 (1.54) | 0.733 | 1.150 | 2.331 |
| 0.1 | 6 | 0.239 (1.54) | 0.734 | 1.133 | 1.863 |

Table 20.4: Bounds on the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$, in units of $10^{-4}$: the lower bound of the proof of Theorem 20.2.1(b) with the single radius $\kappa \le a$ that maximises $\kappa^2(1-\ell_0(\kappa))$ (the radius in brackets), the layer cake lower bound, the converged value (as in Table 20.3, extended to $a=6$ in `data/dc_rate.json`), and the value of the trial function $\cos(\pi u/2a)$ computed from the prime side, which is an upper bound. Script `code/dc_rate.py`, parts `lower` and `trial`.

## 20.3 What can be done and what is open

### 20.3.1 What can be done

Partly, and the parts can be stated exactly: the deconvolution of the Thorin density always exists, as Weil's distribution computed from the primes with no hypothesis (Theorem 20.1.1), and its positivity to depth $\varepsilon$ is a zero free half plane (Theorem 20.1.2), known for $\varepsilon = \frac{1}{2}$ and announced for $\varepsilon = \frac{3}{8}$, each rung $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \ge 0$ being a finite problem in the primes up to $e^{2a}$ whose limit, under the

hypotheses of Theorem 20.2.1, is a margin $2b_{1/2+\varepsilon}$ of about $0.092\,\varepsilon$, approached at a rate of order $a^{-2}$.

A rung is finite information in the sense of Section 5.2.2 (Theorem 5.2.1 is the rigorous form of this for the heat trace), so heuristically no finite set of rungs sees every height, and at $\varepsilon = 0$ the margin is 0 and, under RH, the approach to it is faster than any power of $a$ (Theorem 20.2.1(c)): this is the deconvolution obstruction, consistent with the theorem of Rodgers and Tao that RH, if true, holds with no slack [463].

### 20.3.2 Open problems

Problem 20.1 (Certified rungs). Certify $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) > 0$ in interval arithmetic for $\varepsilon \in \{0.05, 0.1, 0.2, 0.375\}$ and $a = 4$, which by monotonicity in $a$ covers every $a \le 4$, and determine how far in $a$ a certificate can be pushed at fixed $\varepsilon$. Each certificate is unconditional, and at $\varepsilon = \frac{3}{8}$ each agrees with what the announced half plane gives, namely $\lambda_{\min} \ge 0$ (Theorem 20.1.2), and can be checked independently of it.

Problem 20.2 (The exact rate). Prove that, under the hypotheses of Theorem 20.2.1, $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon}) \to \pi^2 v''_{1/2+\varepsilon}(0)/4$, and find the next term. The converged values of Table 20.3 exceed the constant by a relative amount that falls like $a^{-1}$: a least squares fit on $2 \le a \le 7$ gives $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon}) = \bar{c}_\varepsilon(C_0 + C_1/a + C_2/a^2)$, with $\bar{c}_\varepsilon = \pi^2 v''_{1/2+\varepsilon}(0)/4$, $C_0 = 1.0002,\ 1.0003,\ 1.0005$, $C_1 = 0.139,\ 0.159,\ 0.166$ and $C_2 = 0.037,\ 0.045,\ 0.051$ at $\varepsilon = \frac{1}{2},\ 0.2,\ 0.1$, and residuals below $1.1 \times 10^{-4}\,\bar{c}_\varepsilon$. So, numerically, the correction is positive and $\bar{c}_\varepsilon$ is not a bound at finite $a$ (the scaled excess is $1.024\,\bar{c}_\varepsilon$ at $a = 6$, $\varepsilon = \frac{1}{2}$); the fit is numerical evidence for the limit, to within 0.04 to 0.2 percent, and for the order $a^{-1}$ of the correction, and neither is proved.

Problem 20.3 (The transition at the line). Describe $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ jointly as $a \to \infty$ and $\varepsilon \to 0$. For fixed $\varepsilon > 0$, under the hypotheses of Theorem 20.2.1, the limit is $2b_{1/2+\varepsilon}$, which is at most $2V_\xi\,\varepsilon$ and asymptotic to it as $\varepsilon \downarrow 0$, and the rate is algebraic; at $\varepsilon = 0$, under RH, the decay is faster than any power. On which curves $\varepsilon = \varepsilon(a) \to 0$ does $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ stay comparable to $2b_{1/2+\varepsilon}$, and does the crossover occur at $\varepsilon \approx$ the mean spacing of the zeros at height $e^{2a}$?

Problem 20.4 (Uniformity from structure). Find a structural reason, valid for all $a$, why $Q_\varepsilon \ge 0$ on $L^2[-a, a]$ for some fixed $\varepsilon < \frac{1}{2}$, using properties of the archimedean and pole terms that dominate the primes up to $e^{2a}$ uniformly. By Theorem 20.1.2 any such argument proves a zero free half plane $\operatorname{Re} s > \frac{1}{2} + \varepsilon$.

# Chapter 21

# Thorin representations, Dirichlet $L$ functions and the Linnik constant

For a number field $F$, the reciprocal of the completed Dedekind zeta function, tilted by $u_0 \geq 0$ and so read at the point $\frac{1}{2} + \sqrt{u_0}$, is a generalised gamma convolution (GGC) exactly when every nontrivial zero lies on the critical line or is real and at distance less than $\sqrt{u_0}$ from the centre, with no zero at the centre when $u_0 = 0$, and its Thorin measure is then the zero counting measure shifted by $u_0$ (Theorem 21.2.3). For cyclotomic fields the Thorin property at the centre, for every modulus, is equivalent to GRH for all Dirichlet $L$ functions together with $L(\frac{1}{2}, \chi) \neq 0$ for every primitive character $\chi$, so it implies the GRH value $K = 7$ of the Linnik constant (Corollary 21.7.1); this is weaker than the known conditional result, because the nonvanishing at the centre is an extra hypothesis, and with the present methods it gives nothing beyond what GRH gives (Section 21.1.2).

*What is not claimed.* Nothing here proves RH, and nothing here lowers the Linnik constant. A statement that rests on RH, on GRH or on the zero free half plane of October 2026, which has not been refereed, says so. A number not attributed to a source is computed in floating point at the precision stated.

## 21.1 Linnik's problem and where zeros enter

### 21.1.1 The theorem and the records

Linnik [342, 343] proved that there is an absolute constant $K$ such that every sufficiently large even integer $N$ can be written

$$N = p_1 + p_2 + 2^{k_1} + \cdots + 2^{k_K}, \qquad p_1, p_2 \text{ prime}, \quad k_i \in \mathbb{Z}_{\geq 0}. \tag{21.1.1}$$

The 1951 paper assumes GRH and the 1953 paper does not; neither gives a value of $K$. Gallagher [144] recast the argument so that explicit constants became possible, and the values of $K$ that followed were lowered in a series of papers [346, 347, 337, 511, 338], from 54000 (Liu, Liu and Wang) to 1906 (Li), and under

GRH to 200 (Liu, Liu and Wang), until Heath-Brown and Puchta [179] reached the values in the first line of the table below.

The best values on record are the following.

| Authors | $K$ unconditionally | $K$ under GRH |
|---|---|---|
| Heath-Brown and Puchta [179] | 13 | 7 |
| Pintz and Ruzsa [413] | | 7 |
| Pintz and Ruzsa [414] | 8 | |

Platt and Trudgian [418] sharpened the numerical constants in the method of [179].

### 21.1.2 Where $K$ is determined

Fix a small constant $\varpi > 0$, write $e(x) = \exp(2\pi i x)$ and $L = \lfloor \log_2(N/2K) \rfloor$, and put

$$\mathcal{S}(\alpha) = \sum_{\varpi N < p \le N} e(p\alpha), \qquad \mathcal{D}(\alpha) = \sum_{1 \le \nu \le L} e(2^\nu \alpha).$$

The number of representations (21.1.1) with both primes in $(\varpi N, N]$ and the exponents restricted to $1 \le k_i \le L$, so that the powers of two add up to at most $N/2$, is

$$R(N) = \int_0^1 \mathcal{S}(\alpha)^2 \, \mathcal{D}(\alpha)^K \, e(-N\alpha)\, d\alpha. \tag{21.1.2}$$

Let $\mathfrak{M} \subset [0,1]$ be the set of major arcs, the neighbourhoods of rationals with small denominator, and $\mathfrak{m} = [0,1] \setminus \mathfrak{M}$ the minor arcs.

We follow the argument of Heath-Brown and Puchta [179], whose constants we quote with their pinpoints; the proof has five steps.

(1) *Major arcs.* Gallagher's method, with the major arcs $q \le P$, $|\alpha - a/q| \le P/(qN)$ and $P \le N^{45/154 - 4\varpi}$, gives on $\mathfrak{M}$ a main term of at least

$$2.7895\,(1 - 3\varpi)\, C_0\, N (\log 2)^{-2} L^{K-2}$$

(Lemma 7 of [179], stated there for $K \ge 9$), where $C_0 = \prod_{p>2} \big(1 - (p-1)^{-2}\big) = 0.6601\ldots$ is the twin prime constant, and 2.7895 is $2\big(1 + \frac{1}{3}(1 - 2^{-6}) + \frac{1}{15}(1 - 4^{-6})\big) = 2.78955\ldots$ rounded down, the contribution of the divisors 1, 3 and 5 to the singular series of Goldbach's problem averaged over the powers of two (equation (25) there). No hypothesis on zeros is used, and no explicit numerical zero free region. [179] control the error terms, character sums over primes, by the zero density estimates of Huxley and Jutila, $\sum_{r \le R} N(\sigma, r, T) \ll (R^2 T)^{\kappa(\sigma)(1-\sigma)}$ with $\kappa(\sigma) = \frac{3}{2-\sigma} + \varpi$ for $\frac{1}{2} \le \sigma \le \frac{3}{4}$ and $\frac{12}{5} + \varpi$ for $\frac{3}{4} \le \sigma \le 1$, which also fix the restriction on $P$, and, for the few characters with a zero near 1 and for the principal character,

by the zero free regions of Iwaniec (for nonreal characters, with an implied constant that is not explicit) and of Vinogradov. Zeros close to $\sigma = 1$ play no role.

(2) *Minor arc mean square.* $\int_0^1 |\mathcal{S}(\alpha)\mathcal{D}(\alpha)|^2 d\alpha$ counts the solutions of $p_1 - p_2 = 2^\mu - 2^\nu$ with $\mu, \nu \le L$. Chen's sieve bound for the number of prime differences, $r(n) \le C_0 C_1 h(n) N (\log N)^{-2}$ with $C_1 = 7.8342$ (Theorem 3 of Chen's 1978 paper on Goldbach's problem and the sieve methods, as used in [179]), bounds the integral over $[0,1]$ from above, and a lower bound for the part over $\mathfrak{M}$ is subtracted, which gives $\int_{\mathfrak{m}} |\mathcal{S}\mathcal{D}|^2 d\alpha \le (13.968 + o(1))\, C_0 N (\log 2)^{-2}$ (Lemmas 8 to 10 of [179]).

(3) *Minor arc sup bound for the primes.* $\mathcal{S}(\alpha) \ll N^{\theta + o(1)}$ on $\mathfrak{m}$, with $\theta = 263/308$ unconditionally, from Theorem 3.1 of Vaughan's book on the Hardy Littlewood method with $P = N^{45/154 - 4\varpi}$, and $\theta = 3/4$ under GRH, from Lemma 12 of Baker and Harman (Section 6 of [179]).

(4) *Large deviations for the powers of two.* The set $\mathcal{A}_\lambda = \{\alpha : |\mathcal{D}(\alpha)| \ge \lambda L\}$ has measure $\ll N^{-E(\lambda)}$, with $E(0.863665) > \frac{109}{154}$ and $E(0.722428) > \frac{1}{2}$ (Lemma 1 of [179], proved there by a martingale argument suggested by Keith Ball). The contribution of $\mathfrak{m} \cap \mathcal{A}_\lambda$ is $\ll N^{-E(\lambda) + 2\theta + o(1)} L^K$, negligible once $E(\lambda) > 2\theta - 1$, that is $\frac{109}{154}$ unconditionally and $\frac{1}{2}$ under GRH.

(5) *Comparison.* Off $\mathcal{A}_\lambda$, $|\mathcal{D}| < \lambda L$, so the contribution of $\mathfrak{m} \setminus \mathcal{A}_\lambda$ is at most $(\lambda L)^{K-2} \cdot 13.968\, C_0 N (\log 2)^{-2}$, and the count is positive, for $N$ large and $\varpi$ small, when $13.968\, \lambda^{K-2} < 2.7895$ (end of Section 6 of [179]). This needs $K - 2 > 10.99$ at $\lambda = 0.863665$ and $K - 2 > 4.95$ at $\lambda = 0.722428$, which gives $K = 13$ unconditionally and $K = 7$ under GRH.

In this summary a hypothesis on zeros enters only at step (3): GRH lowers $\theta$ to $\frac{3}{4}$ and hence lowers $\lambda$. In the scheme of [179] the remaining gains come from the constants of steps (1), (2) and (4), which do not depend on a hypothesis on zeros; Elsholtz is reported there to have obtained $K = 12$ unconditionally, in unpublished work, by improving the constants of the first two steps. The refinements of Pintz and Ruzsa [413, 414] differ from this argument in a number of respects, as [179] remark, and we do not describe them. In the scheme of [179] an input that delivered GRH would be worth the passage from $K = 13$ to $K = 7$, and against the records it is the passage from $K = 8$ to $K = 7$.

*The zero free half plane of October 2026.* In October 2026 OpenAI released a proof, produced with one of its AI models, that $\zeta(s)$ and every Dirichlet $L$ function have no zeros in $\operatorname{Re} s > \frac{7}{8}$ [398], and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [399]. The claims have not been refereed (Section 12.1.2). A zero free region can act in step (3), through the error term of the exponential sum, and in

step (1), through the zero density estimates. For step (3), the exponent $\frac{3}{4}$ under GRH comes from the lemma of Baker and Harman used in [179]: for $\alpha = a/q + \beta$ with $|\beta| \le N^{-1/2}$, GRH gives $\mathcal{S}(\alpha) \ll (\log N)\{N/q + N^{1/2}q^{1/2} + N(q|\beta|)^{1/2}\}$. By Dirichlet's theorem every $\alpha$ has such an approximation with $q \le N^{1/2}$ and $|\beta| \le 1/(qN^{1/2})$, which makes each of the three terms at most $N^{3/4}$ once $q > N^{1/4}$, and $q \le N^{1/4}$ does not occur on the minor arcs because $P = N^{45/154-4\varpi} \ge N^{1/4}$. In the second term GRH enters as the factor $N^{1/2}$, the size of the error term in the explicit formula for $\psi(x,\chi)$. Replacing GRH by a zero free half plane $\operatorname{Re} s > \theta_0$ for all Dirichlet $L$ functions would replace that factor by $N^{\theta_0}$, and the same choice of approximation, with $q \le N^{1-\theta_0}$, would give $\theta = \frac{1}{2}(1+\theta_0)$. This is a heuristic count: the lemma is proved under GRH and its analogue for a half plane is not proved here. It beats the unconditional exponent $\frac{263}{308}$ only if $\theta_0 < \frac{109}{154} = 0.7078$. The half plane $\operatorname{Re} s > \frac{7}{8}$ gives $\frac{15}{16}$, worse than the unconditional exponent, so used this way it does not lower $K$. For step (1), the constant 2.7895 is arithmetic, and zeros close to $\sigma = 1$ play no role in [179], so excluding exceptional zeros is not what a half plane would add. Zeros enter the major arcs through the zero density estimates, which fix the restriction $P \le N^{45/154-4\varpi}$ by way of the bound $(\frac{45}{77}\kappa(\sigma) - 1)(1-\sigma) \le \frac{31}{77}(1-\sigma) \le \frac{31}{154}$ for the exponent of the estimate (Section 3 there), taken at the left end $\sigma = \frac{1}{2}$; a half plane $\operatorname{Re} s > \theta_0$ removes only the zeros to the right of $\theta_0$, so it does not relax this restriction. Whether a finer combination of the half plane with Vaughan's identity does better, on the major or the minor arcs, is Problem 21.9. In the Thorin language of this chapter, the half plane would give the Thorin property of the shifted clock at every basepoint $\alpha > \frac{7}{8}$ for $\xi_F$ with $F$ abelian over $\mathbb{Q}$ (quadratic and cyclotomic fields in particular, by (21.5.4)) and for every real character (Theorem 21.5.8; for $\xi$ this is the remark after Theorem 1.2.2). Liu [353] claims the half plane $\operatorname{Re} s > B_\flat = 0.87489$ for every Hecke $L$ function of finite order over every number field (Section 16.12.3); if that unrefereed and unformalised claim stands, the same holds at every basepoint $\alpha > B_\flat$ for $\xi_F$ with $F$ any number field, abelian or not.

## 21.2 The Thorin property at every tilt

### 21.2.1 Conventions

The centre clock is the book's $T$ (Section 1.2), the variables $e_\gamma$ are independent standard exponentials as in the book, and $Y = \sqrt{2T}\,Z$, with $Z$ a standard normal variable independent of $T$, is the variable $Y_{1/2}$ of Section 10.1. The letter $T$ also stands for a height, but only in $N(T)$ and $S(T)$, as in the book, and as the cut height of the anchored tail of Section 21.3.3. The transfer function from the real axis to the critical line is written $\mathfrak{T}(t)$ (Proposition 21.3.6), and $\mathfrak{B}(z)$ is its Blaschke product in $z = t^2$, so that $\mathfrak{T}(t) = \mathfrak{B}(t^2)$ (Proposition 21.3.8). Completed Dirichlet $L$ functions are written $\xi(s,\chi)$, as in the book, and $\Lambda$ is the von Mangoldt function. The tilt $u_0$ and the basepoint $\alpha$ are different parameters: the clock tilted by $u_0$

reads $\xi$ at $\frac{1}{2}+\sqrt{u_0+u}$, the shifted clock at basepoint $\alpha$ reads it at $\alpha+\sqrt{u}$, and the two agree only at the centre. In Section 21.1.2 only, $\alpha$ is also the circle method variable, and $N$, $L$, $\lambda$, $\theta$, $c$ and $\varpi$ have their circle method meanings there.

### 21.2.2 Generalised gamma convolutions

A probability law on $[0,\infty)$ is a generalised gamma convolution (GGC) if its Laplace transform has the form

$$\varphi(s)=\exp\Big(-bs-\int_{(0,\infty)}\log\Big(1+\frac{s}{t}\Big)\,U(dt)\Big),\qquad s\ge 0, \tag{21.2.1}$$

with $b\ge 0$ and a nonnegative Thorin measure $U$ satisfying $\int_{(0,1]}|\log t|\,U(dt)<\infty$ and $\int_{(1,\infty)}t^{-1}U(dt)<\infty$; see Bondesson [56] and equation (1.3.1), where the drift is $d$ and the Thorin variable $z$. We use three standard facts.

(G1) The right side of (21.2.1) extends to an analytic, zero free function on $\mathbb{C}\setminus(-\infty,-t_0]$, where $t_0=\inf\operatorname{supp}U$.

(G2) If $t_j>0$ and $\sum_j 1/t_j<\infty$, then $\prod_j t_j/(t_j+s)$ is the Laplace transform of $\sum_j e_j/t_j$ with $e_j$ independent unit exponentials, a GGC with $b=0$ and $U=\sum_j\delta_{t_j}$.

(G3) If $\inf\operatorname{supp}U>c$ with $c>0$, then $\varphi(-c)<\infty$ and $\varphi(s-c)/\varphi(-c)$ is again of the form (21.2.1), with Thorin measure $U(\cdot+c)$. This is closure under exponential tilting.

The class is also closed under independent sums, positive scaling and weak limits (equation (1.3.1)), and, by a theorem of Bondesson [57], under products of independent GGC variables. It is not closed under products with arbitrary independent positive variables (an exponential times an independent two point variable is a mixture of two exponentials with rates $a\neq b$, whose Thorin measure would be the signed measure $\delta_a+\delta_b-\delta_c$ for a rate $c$ between $a$ and $b$), under harmonic means (Proposition 7.1.7), or under mixtures (Proposition 21.6.3).

### 21.2.3 Completed Dedekind zeta functions

Let $F$ be a number field of degree $n=r_1+2r_2$ and discriminant $d_F$. Put $\Gamma_{\mathbb{R}}(s)=\pi^{-s/2}\Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s)=2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_F(s)=s(s-1)\,|d_F|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\,\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_F(s). \tag{21.2.2}$$

Then $\xi_F$ is entire of order one, real on the real axis, and $\xi_F(s)=\xi_F(1-s)$. For $F=\mathbb{Q}$ it is $2\xi(s)$, with Riemann's $\xi(s)=\frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$. Its zeros are the nontrivial zeros of $\zeta_F$, written $\rho=\frac{1}{2}+w_\rho$ with $|\operatorname{Re}w_\rho|<\frac{1}{2}$.

LEMMA 21.2.1 (Evenness). *$E_F(w) := \xi_F(\frac{1}{2}+w)$ is even and entire, so $E_F(w) = G_F(w^2)$ with $G_F$ entire. The zeros of $G_F$ are the points $z_\rho = w_\rho^2$, one for each pair $\{\rho, 1-\rho\}$.*

*Proof.* The functional equation gives $E_F(-w) = \xi_F(\frac{1}{2}-w) = \xi_F(\frac{1}{2}+w) = E_F(w)$, and an even entire function is an entire function of $w^2$. □

LEMMA 21.2.2 (Product formula). *Let $Z_F$ be the multiset of the values $z_\rho \neq 0$, one for each pair of zeros, and let $2m_F$ be the order of vanishing of $\xi_F$ at $\frac{1}{2}$. Then, with absolute convergence since $\sum_{z\in Z_F} |z|^{-1} < \infty$,*

$$G_F(u) = c_F\, u^{m_F} \prod_{z\in Z_F} \Big(1 - \frac{u}{z}\Big), \qquad c_F \neq 0. \tag{21.2.3}$$

*Proof.* $G_F$ is entire of order $\frac{1}{2}$ because $E_F$ has order one, so Hadamard's theorem gives the product with no exponential factor. Convergence of $\sum |z_\rho|^{-1} = \sum |\rho - \frac{1}{2}|^{-2}$ follows from the Riemann von Mangoldt formula $N_F(T) \asymp T\log T$. □

The function $G_F$ is not normalised, unlike the function $G$ of Section 1.2: for $F = \mathbb{Q}$ one has $G_{\mathbb{Q}} = 2\xi(\frac{1}{2})\, G$.

Under GRH every $z \in Z_F$ equals $-\gamma^2$ for an ordinate $\gamma > 0$, while a real zero $\beta \in (\frac{1}{2}, 1)$, which GRH excludes, would give $z = (\beta - \frac{1}{2})^2 \in (0, \frac{1}{4})$.

### 21.2.4 The equivalence theorem

For $u_0 \geq 0$ with $\xi_F(\frac{1}{2} + \sqrt{u_0}) \neq 0$, tilt the centre clock by $u_0$:

$$\varphi_{F,u_0}(s) := \frac{\xi_F(\frac{1}{2}+\sqrt{u_0})}{\xi_F(\frac{1}{2}+\sqrt{u_0+s})} = \frac{G_F(u_0)}{G_F(u_0+s)}, \qquad s \geq 0. \tag{21.2.4}$$

By Lemma 21.2.1 the branch of the square root does not matter. For $F = \mathbb{Q}$ this is the function $\widetilde{\varphi}_{u_0}$ of Section 19.1.1.

THEOREM 21.2.3 (Thorin property of the tilted clock). *Let $u_0 \geq 0$. The following are equivalent.*

*(i) $\xi_F(\frac{1}{2}+\sqrt{u_0}) \neq 0$ and $\varphi_{F,u_0}$ is the Laplace transform of a GGC.*

*(ii) Every $z_\rho$ is real with $z_\rho < u_0$, and $m_F = 0$ if $u_0 = 0$. Equivalently, every nontrivial zero of $\zeta_F$ lies on the critical line or is a real zero $\beta$ with $|\beta - \frac{1}{2}| < \sqrt{u_0}$, and $\zeta_F(\frac{1}{2}) \neq 0$ when $u_0 = 0$.*

*When they hold, $b = 0$ and the Thorin measure is*

$$U_{F,u_0} = \sum_{\gamma>0} \delta_{u_0+\gamma^2} + m_F\, \delta_{u_0} + \sum_{\beta} \delta_{u_0-(\beta-\frac{1}{2})^2}, \tag{21.2.5}$$

*where $\frac{1}{2}+i\gamma$ runs over critical line zeros with $\gamma > 0$ and $\beta$ over real zeros in $(\frac{1}{2}, 1)$, all with multiplicity.*

*Proof.* (i) implies (ii). By (G1) the Laplace transform extends to an analytic zero free function on $\mathbb{C} \setminus (-\infty, 0]$, and it agrees with $G_F(u_0)/G_F(u_0+s)$, which is meromorphic, on $s > 0$. Hence $G_F(u_0+s)$ has no zeros off $(-\infty, 0]$, so every zero $z$ of $G_F$ satisfies $z - u_0 \le 0$, and $z \ne u_0$ because $G_F(u_0) \ne 0$. Imaginary $w_\rho$ is a critical line zero, real $w_\rho$ a real zero with $(\beta - \frac{1}{2})^2 < u_0$. At $u_0 = 0$ a central zero would make $G_F(0) = 0$.

(ii) implies (i). Under (ii), Lemma 21.2.2 gives

$$\varphi_{F,u_0}(s) = \Big(\frac{u_0}{u_0+s}\Big)^{m_F} \prod_{z \in Z_F} \frac{u_0 - z}{u_0 - z + s},$$

where every rate $u_0 - z$ is positive, equal to $u_0 + \gamma^2$ or to $u_0 - (\beta - \frac{1}{2})^2$, and the reciprocals are summable. Fact (G2) gives the GGC with Thorin measure (21.2.5). □

For a general field at $u_0 = 0$ the theorem is Proposition 10.3.6, which assumes $\zeta_F(\frac{1}{2}) \ne 0$. That assumption is not automatic, since Armitage found fields with $\zeta_F(\frac{1}{2}) = 0$ [11], and here a central zero appears in the conclusion as the atom $m_F \delta_{u_0}$.

COROLLARY 21.2.4 (The tilts). *(a) At $u_0 = 0$: $\xi_F(\frac{1}{2})/\xi_F(\frac{1}{2} + \sqrt{s})$ is a GGC Laplace transform if and only if all nontrivial zeros of $\zeta_F$ lie on the critical line and $\zeta_F(\frac{1}{2}) \ne 0$.*

*(b) For every $u_0 \ge \frac{1}{4}$, in particular at $u_0 = \frac{1}{4}$ and $u_0 = 1$, where $\xi_F$ is read from 1 and from $\frac{3}{2}$: $\varphi_{F,u_0}$ is a GGC Laplace transform if and only if every nontrivial zero lies on the critical line or on the real axis.*

*(c) Define the Thorin threshold $u_F^* = \max\{(\beta - \frac{1}{2})^2\}$ over real zeros $\beta \in [\frac{1}{2}, 1)$, with $u_F^* = 0$ if there are none. If all nonreal zeros are on the critical line, the Thorin property holds at tilt $u_0$ exactly for $u_0 > u_F^*$, and also at $u_0 = 0$ when there are no real zeros at all.*

*(d) When the property holds at two tilts $u_1 < u_0$, the measures are shifts of one another, $U_{F,u_0} = U_{F,u_1}(\cdot - (u_0 - u_1))$, and $\varphi_{F,u_1}(s) = \varphi_{F,u_0}(s - (u_0 - u_1))/\varphi_{F,u_0}(-(u_0 - u_1))$, the tilt (G3).*

*Proof.* (a) and (b) are Theorem 21.2.3 at $u_0 = 0$ and for $u_0 \ge \frac{1}{4}$, since every real zero in $(0,1)$ has $(\beta - \frac{1}{2})^2 < \frac{1}{4}$. (c) restates the condition $z_\rho < u_0$ for the real values $z_\rho$. (d) follows from (21.2.5) and (21.2.4); for $\xi$ it is the forward and backward tilt of Theorem 19.1.2(a). □

REMARK 21.2.5 (The tilt and Landau's theorem). Suppose the Thorin property holds at the tilt 1. Descending from tilt 1 to tilt 0 is the tilt (G3) with $c = 1$, admissible exactly when $\varphi_{F,1}(-1) < \infty$. Since $\varphi_{F,1}$ is the Laplace transform of a

positive measure, Landau's theorem [514, Ch. II, Theorem 5b] puts its abscissa of convergence at a real singularity. So $\varphi_{F,1}(-1) < \infty$ if and only if $G_F$ has no zero in $[0,1]$, that is, $\xi_F$ has no real zero in $[\frac{1}{2}, \frac{3}{2}]$. For $\xi$ there is no real zero, so under RH the descent to tilt 0 is possible, and it is Theorem 19.1.2(a) (Section 19.1.1). What is specific to number fields is that a central zero or a real zero in $(\frac{1}{2}, 1)$ blocks the descent to tilt 0.

Remark 21.2.6 (The Thorin measure is the zero counting measure). For $F = \mathbb{Q}$, $\zeta(\frac{1}{2}) \neq 0$ and $\zeta(\sigma) < 0$ on $(0,1)$, so there are no central or real zeros. By Corollary 21.2.4, the Thorin property of $\xi$ at any tilt $u_0$ is equivalent to RH, with Thorin measure $\sum_{\gamma>0} \delta_{u_0+\gamma^2}$; at tilt $u_0 = 1$ the smallest atom is $1+\gamma_1^2 = 200.7905$.

## 21.3 The centre clock: moments, transfer function and negative moments

### 21.3.1 The zeta function at $\sigma > 1$

Lemma 21.3.1. *For $\sigma > 1$ and $\mathrm{Re}(s) \geq 0$,*

$$\log \frac{\zeta(\sigma + s)}{\zeta(\sigma)} = \int_0^\infty (e^{-sx} - 1)\, \mu^\sigma(dx), \tag{21.3.1}$$

*where $\mu^\sigma = \sum_{p,\, r \geq 1} r^{-1} p^{-r\sigma}\, \delta_{r \log p}$, the sum over primes $p$ and integers $r \geq 1$, has total mass $\log \zeta(\sigma)$.*

*Proof.* Take logarithms of the Euler product, expand $\log(1-z)^{-1} = \sum_{r \geq 1} z^r / r$, subtract $\log \zeta(\sigma)$ and write $p^{-rs} = e^{-rs \log p}$. □

Lemma 21.3.2. *Let $\mathsf{N}$ follow the Riemann zeta distribution $\mathrm{RZ}(\sigma)$, $\Pr(\mathsf{N} = n) = n^{-\sigma}/\zeta(\sigma)$, and $J^\sigma := \log \mathsf{N}$. Then $J^\sigma$ is compound Poisson with Lévy measure $\mu^\sigma$ and $\mathbb{E}[e^{itJ^\sigma}] = \zeta(\sigma - it)/\zeta(\sigma)$.*

*Proof.* $\mathbb{E}[e^{itJ^\sigma}] = \zeta(\sigma)^{-1} \sum_n n^{-\sigma} n^{it}$, and (21.3.1) at $s = -it$ identifies the exponent. □

The Riemann zeta distribution and the infinite divisibility of $\log \mathsf{N}$ go back to Khinchin [308]; see Lin and Hu [340], Theorem 14.3.1 and Section 6.1.6.

Theorem 21.3.3 (Wiener spectral identity). *For $\sigma > 1$ and integer $k \geq 1$,*

$$\lim_{X \to \infty} \frac{1}{2X} \int_{-X}^{X} \left|\zeta(\sigma + it)\right|^{2k} dt = \sum_{n \geq 1} \frac{d_k(n)^2}{n^{2\sigma}}, \tag{21.3.2}$$

*where $d_k(n)$ counts ordered factorisations $n = n_1 \cdots n_k$.*

*Proof.* By Lemma 21.3.2, $|\zeta(\sigma+it)|^{2k} = \zeta(\sigma)^{2k}\mathbb{E}[e^{itV}]$ with $V = \sum_{i\le k} J_i - \sum_{i\le k} J_i'$ for independent copies of $J^\sigma$. Wiener's theorem on the atom of a measure at 0 [516] gives the limit $\Pr(V=0) = \Pr(\mathsf{N}_1\cdots\mathsf{N}_k = \mathsf{N}_1'\cdots\mathsf{N}_k')$. The product $\mathsf{N}_1\cdots\mathsf{N}_k$ has law $d_k(m)m^{-\sigma}/\zeta(\sigma)^k$, so the collision probability is $\zeta(\sigma)^{-2k}\sum_m d_k(m)^2 m^{-2\sigma}$. □

Corollary 21.3.4. *The right side of* (21.3.2) *is* $\zeta(2\sigma)$ *for* $k=1$, $\zeta(2\sigma)^4/\zeta(4\sigma)$ *for* $k=2$ *(through Ramanujan's identity), and in general the Euler product*

$$\prod_p \sum_{r\ge 0} \binom{r+k-1}{k-1}^2 p^{-2r\sigma}.$$

These are the classical mean value theorems of Carlson ($k=1$) and Ingham ($k=2$), which hold for $\sigma > \frac12$ (see [498, Chapter VII]); for $k\ge 3$ they are known only on a smaller range of $\sigma$, and on the whole range $\sigma > \frac12$ they follow from the Lindelöf hypothesis [498, Chapter XIII].

### 21.3.2 The Riemann case in detail

Take $F=\mathbb{Q}$ and $u_0 = 0$, and assume RH. By Corollary 21.2.4,

$$\frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u})} = \mathbb{E}\big[e^{-uT}\big], \qquad T \overset{d}{=} \sum_{\gamma>0}\frac{e_\gamma}{\gamma^2}, \qquad u\ge 0, \tag{21.3.3}$$

with independent standard exponential variables $e_\gamma$. The measure $\sum_{\gamma>0}\delta_{\gamma^2}$ is the measure $U_{\mathrm{RH}}$ of Proposition 1.2.1, which Chapters 3 to 5 compute at finite resolution. Without RH the clock still exists, as a self decomposable variable (Theorem 5.1.1), with the same Laplace transform, and the moment identities, the Mellin transform and the interpolation below hold for it with $\gamma^2$ replaced by $a_k = -(\rho_k - \frac12)^2$, the numbers of the product $G(u) = \prod_k (1+u/a_k)$ of equation (1.2.1).

#### *21.3.2.1 The normal scale mixture and the moments*

Define $Y := \sqrt{2T}\,Z$ with $Z\sim N(0,1)$ independent of $T$. Conditioning on $T$,

$$\mathbb{E}\big[e^{itY}\big] = \mathbb{E}\big[e^{-t^2T}\big] = \frac{\xi(\frac12)}{\xi(\frac12+|t|)}, \tag{21.3.4}$$

which evaluates $\xi$ on the real axis, where it is positive, even and super exponentially growing. This is the variable $Z\sqrt{2T}$ of Section 1.2. It is the variable $X = Z\sqrt{2Y}$ of Proposition 16.5.1, where Chapter 16 writes $Y$ for the clock. The moments of $T$ are the Taylor coefficients of (21.3.3): with $p_j := \sum_{\gamma>0}\gamma^{-2j}$,

$$\mathbb{E}[T^k] = k!\,h_k\big(\{\gamma^{-2}\}\big), \qquad \mathbb{E}[Y^{2k}] = (2k-1)!!\,2^k\,\mathbb{E}[T^k], \tag{21.3.5}$$

where $h_k$ is the complete homogeneous symmetric polynomialpolynomial. In particular $\mathbb{E}[T] = p_1$ and $\mathrm{Var}(T) = p_2$.

*21.3.2.2 The Hadamard product and the transfer function*

The same factorisation gives the critical line value

$$\frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2})} = \prod_{\gamma>0}\Big(1-\frac{t^2}{\gamma^2}\Big), \tag{21.3.6}$$

which alternates in sign; read in $u = t^2$ it is the Laplace transform of no measure, while in $t$ it is the characteristic function of the Pólya density (Section 1.2). Multiplying (21.3.4) by (21.3.6) gives

$$\mathbb{E}[e^{itY}]\cdot\frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2})} = \frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2}+|t|)} = \prod_{\gamma>0}\frac{\gamma^2-t^2}{\gamma^2+t^2}, \tag{21.3.7}$$

with zeros at $\pm\gamma$ and, continued in $t$, poles at $\pm i\gamma$, where the Laplace transform of $T$ stops being analytic. The left side is the characteristic function of the sum of $Y$ and an independent variable with the Pólya density, so without any hypothesis the middle term is a characteristic function; the product on the right is its form under RH. The critical line moments $\int_0^X |\zeta(\frac{1}{2}+it)|^{2k}dt$ are governed by (21.3.6), not by $Y$; the transfer function links the two and has a closed form (Proposition 21.3.6).

## 21.3.3 Closed forms and their verification

*21.3.3.1 The moment identity*

We computed $\mathbb{E}[T^k]$ two ways. The zero side uses (21.3.5), with $p_j$ summed over the first 800 ordinates plus the Riemann von Mangoldt tail $\int_{\gamma_{800}}^{\infty} t^{-2j}\,\frac{1}{2\pi}\log\frac{t}{2\pi}\,dt$, started at the ordinate $\gamma_{800} = 1183.7128$. The function side uses the Taylor expansion of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+w)$ in powers of $w^2$ at $w=0$, from the Taylor coefficients of $\xi(\frac{1}{2}+s)$ at $s=0$. Table 21.1 gives the result. The residual is mostly the error

Table 21.1: The moments $\mathbb{E}[T^k]$ from the zeros (800 ordinates and the smooth tail from $\gamma_{800}$) and from the function (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`).

| $k$ | zero side | function side | relative error |
|---|---|---|---|
| 1 | 0.0231054 | 0.0231050 | $1.9\times10^{-5}$ |
| 2 | $5.71033\times10^{-4}$ | $5.71013\times10^{-4}$ | $3.5\times10^{-5}$ |
| 3 | $1.52001\times10^{-5}$ | $1.51994\times10^{-5}$ | $4.9\times10^{-5}$ |
| 4 | $4.38851\times10^{-7}$ | $4.38824\times10^{-7}$ | $6.0\times10^{-5}$ |

of starting the smooth tail at an ordinate: with the first 800 ordinates and the tail started at the midpoint of $\gamma_{800}$ and $\gamma_{801}$ the relative errors for $k = 1,\dots,4$ are $3.1\times10^{-7}$, $5.8\times10^{-7}$, $8.1\times10^{-7}$ and $1.0\times10^{-6}$, and with the anchored

tail below they are $8.8\times10^{-10}$, $1.6\times10^{-9}$, $2.3\times10^{-9}$ and $2.8\times10^{-9}$. The function side computes $\sum_k a_k^{-j}$ with no hypothesis (the cumulants of $T$ from $G(u)=\prod_k(1+u/a_k)$, Section 11.2.3), so the agreement checks the zero list and the tail. One missing zero at $\gamma_{800}$ would change $\mathbb{E}[T]$ by $3.1\times10^{-5}$ in relative terms, the order of the residuals in Table 21.1, so the table alone cannot show that none is missing; the anchored residual quoted above is smaller than that change by a factor of $3.5\times10^{4}$, and with it the agreement shows that no zero below the cut is missing from the list. It does not test that the zeros lie on the line: moving a pair of zeros at height $\gamma_{800}$ off the line by 0.25 changes $p_1$ by $8.3\times10^{-12}$ in relative terms, below every residual. Nor is it a check on the Thorin property itself, which no finite computation can establish (Theorem 5.2.1).

*Anchoring the tail.* Most of the residual comes from the fluctuation $S(T)$ of the zero count at the cut. Write $N(t)=N_0(t)+1+S(t)$ with $N_0=\vartheta/\pi$. Integration by parts gives, exactly,

$$\sum_{\gamma>T}\gamma^{-2j}=\int_T^\infty t^{-2j}\,\frac{\vartheta'(t)}{\pi}\,dt+\big(N_0(T)+1-N(T)\big)T^{-2j}\\ +2j\int_T^\infty S(t)\,t^{-2j-1}\,dt. \tag{21.3.8}$$

The smooth tail keeps only the first term, with the error $-S(T)T^{-2j}$ of the second, of order $T^{-2j}\log T$ at most. Since $\int_T^X S(t)\,dt=O(\log X)$ [498, Section 9.9], the third term is $O(jT^{-2j-1}\log T)$, so the anchored tail, which keeps the first two terms, typically gains over the smooth tail a factor of the order of the height of the cut. Table 21.2 gives the relative errors of $p_1,p_2,p_3$ against the values of the function side, for the first $N$ ordinates and the cut at the midpoint of $\gamma_N$ and $\gamma_{N+1}$, so that $N(T)=N$. The gain is typical, not guaranteed: the smooth error

Table 21.2: Relative errors of $p_1,p_2,p_3$ for the smooth and the anchored tail with the first $N$ ordinates; the cuts are $T=1183.1475$ ($S(T)=0.083$) for $N=799$, $T=397.1503$ ($S(T)=0.244$) for $N=200$ and $T=680.8186$ ($S(T)=-0.211$) for $N=400$ (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`).

| $N$ | tail | $p_1$ | $p_2$ | $p_3$ |
|---|---|---|---|---|
| 799 | smooth | $2.6\times10^{-6}$ | $1.1\times10^{-9}$ | $2.1\times10^{-13}$ |
| | anchored | $6.3\times10^{-9}$ | $5.6\times10^{-12}$ | $1.6\times10^{-15}$ |
| 200 | smooth | $6.7\times10^{-5}$ | $2.6\times10^{-7}$ | $4.3\times10^{-10}$ |
| | anchored | $6.8\times10^{-8}$ | $5.1\times10^{-10}$ | $1.2\times10^{-12}$ |
| 400 | smooth | $2.0\times10^{-5}$ | $2.6\times10^{-8}$ | $1.4\times10^{-11}$ |
| | anchored | $1.4\times10^{-7}$ | $3.7\times10^{-10}$ | $3.1\times10^{-13}$ |

is $|S(T)|T^{-2j}$ to 0.1%, so the gain is large only when $|S(T)|$ is not small. Over

the 1650 cuts $N = 50$ to 1699 the ratio of the smooth to the anchored error of $p_1$ has median 354, about a third of the height of the cut, and deciles 42 and 2700, and it is below 10 at 3.2% of the cuts. This is the first half of Problem 21.3.

*21.3.3.2 The tilt by one quarter and a classical constant*

The classical evaluation

$$\sum_{\rho} \frac{1}{\rho} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi = 0.0230957089661\ldots,$$

with $\gamma_E$ Euler's constant, which under RH is $\sum_{\gamma>0}(\gamma^2+\frac14)^{-1}$, differs from $\mathbb{E}[T] = 0.0231049931\ldots$ by about $9.3\times10^{-6}$. The shift by $\frac14$ is the tilt $u_0 = \frac14$. The book identifies the classical constant as the mean of this tilted clock in Section 9.2 (after Corollary 9.2.3), and as $\pi$ times the Thorin density of the shifted clock at basepoint one at the origin (Proposition 9.1.2); what we add is the expansion (21.3.11) in the cumulants.

Proposition 21.3.5 (Esscher tilt). *Assume* (21.3.3) *and let* $u_0 > 0$. *The tilted law* $\Pr_{u_0}(dh) = e^{-u_0 h}\Pr(T \in dh)/\mathbb{E}[e^{-u_0 T}]$ *is the GGC with Thorin measure* $U_{\mathbb{Q},u_0} = \sum_{\gamma>0}\delta_{u_0+\gamma^2}$ *of Theorem 21.2.3, which is the forward tilt of Theorem 19.1.2(a), and*

$$\mathbb{E}\big[e^{-u_0 T}\big] = \frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u_0})}, \tag{21.3.9}$$

$$\mathbb{E}_{u_0}[T] = \sum_{\gamma>0}\frac{1}{\gamma^2+u_0} = \frac{1}{2\sqrt{u_0}}\frac{\xi'}{\xi}\big(\tfrac12+\sqrt{u_0}\big). \tag{21.3.10}$$

*At* $u_0 = \frac14$, *since* $\xi(1) = \frac12$,

$$\mathbb{E}\big[e^{-T/4}\big] = 2\,\xi(\tfrac12) = 0.99424155637\ldots,$$
$$\mathbb{E}_{1/4}[T] = \frac{\xi'(1)}{\xi(1)} = 1 + \frac{\gamma_E}{2} - \frac12\log 4\pi.$$

*Proof.* $\mathbb{E}_{u_0}[e^{-vT}] = \mathbb{E}[e^{-(u_0+v)T}]/\mathbb{E}[e^{-u_0T}] = \prod_{\gamma>0}(\gamma^2+u_0)/(\gamma^2+u_0+v)$, which is $\varphi_{\mathbb{Q},u_0}(v)$. Equation (21.3.9) is (21.3.3) at $u = u_0$, and $\mathbb{E}_{u_0}[T] = -\frac{d}{du}\log\mathbb{E}[e^{-uT}]$ at $u_0$, computed either from the product or from $\log\xi(\frac12+\sqrt{u})$. At $u_0 = \frac14$ the evaluation point is $\frac12+\sqrt{u_0} = 1$, where $\xi(1) = \xi(0) = \frac12$ and $\xi'(1)/\xi(1) = \sum_\rho \rho^{-1}$. □

From the first 800 zeros with the anchored tail, $\prod_{\gamma>0}\gamma^2/(\gamma^2+\frac14) = 0.99424155637$ and $\sum_{\gamma>0}(\gamma^2+\frac14)^{-1} = 0.02309570899$, within $5\times10^{-12}$ and

$2 \times 10^{-11}$ of $2\xi(\frac{1}{2})$ and of the constant. Expanding $(\gamma^2 + \frac{1}{4})^{-1}$ in powers of $\gamma^{-2}$, valid since $\gamma_1 > \frac{1}{2}$,

$$1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi \;=\; \sum_{j\ge 1} (-1)^{j+1} 4^{1-j}\, p_j, \tag{21.3.11}$$

ties it to the whole cumulant sequence of $T$. With the $p_j$ of the function side the series agrees with the constant to a relative $2 \times 10^{-30}$ after ten terms, the size of the first omitted term.

### *21.3.3.3 The transfer function*

Proposition 21.3.6. *Assume RH. For all real $t$,*

$$\mathfrak{T}(t) \;:=\; \frac{\xi(\frac{1}{2} + it)}{\xi(\frac{1}{2} + |t|)} \;=\; \prod_{\gamma > 0} \frac{\gamma^2 - t^2}{\gamma^2 + t^2}, \tag{21.3.12}$$

*and $\mathfrak{T}$ is real, even, bounded by 1 in modulus, and vanishes exactly at $t = \pm\gamma$. For $|t| < \gamma_1$,*

$$\log \mathfrak{T}(t) \;=\; -2 \sum_{\gamma > 0} \operatorname{artanh}\Big(\frac{t^2}{\gamma^2}\Big) \;=\; -2 \sum_{m \ge 1} \frac{p_{2m-1}}{2m-1}\, t^{4m-2}, \tag{21.3.13}$$

*which involves only the odd index power sums $p_1, p_3, p_5, \dots$.*

*Proof.* Lemma 21.2.2 gives $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2}) = \prod_{\gamma>0}(1 + s^2/\gamma^2)$ for all $s \in \mathbb{C}$. Setting $s = it$ and $s = |t|$ and dividing gives (21.3.12), each factor real with modulus at most 1. For (21.3.13), $\log \frac{\gamma^2 - t^2}{\gamma^2 + t^2} = -2\operatorname{artanh}(t^2/\gamma^2)$; expand and interchange the absolutely convergent double sum. □

Proposition 21.3.7. *Assume RH. The odd index power sums determine the zeros: the series $\sum_{m\ge1} p_{2m-1} z^{2m-1}$ converges for $|z| < \gamma_1^2$ and continues to the meromorphic function*

$$\sum_{m \ge 1} p_{2m-1} z^{2m-1} = \frac{1}{2} \sum_{\gamma > 0} \Big( \frac{1}{1 - z/\gamma^2} - \frac{1}{1 + z/\gamma^2} \Big),$$

*whose poles are the points $\pm\gamma^2$, each with residue $-\frac{1}{2}\gamma^2$ times the multiplicity of $\gamma$. Hence $\{p_{2m-1}\}$ determines $\{\gamma\}$ with multiplicities, and with it every $p_j$.*

*Proof.* For $|z| < \gamma_1^2$, $\sum_m (z/\gamma^2)^{2m-1} = (z/\gamma^2)/(1 - z^2/\gamma^4)$, which is half the bracket; summing over $\gamma$ and interchanging is justified by $\sum \gamma^{-2} < \infty$. The sum over $\gamma$ converges locally uniformly off $\{\pm\gamma^2\}$, since the bracket is $O(|z|\gamma^{-2})$ for $\gamma^2 > 2|z|$, so it defines a meromorphic function with the stated poles. A meromorphic function is determined by its Taylor coefficients at 0, and its poles and residues give back the multiset $\{\gamma^2\}$. □

Numerically, $\mathfrak{T}(2) = 0.8312$, $\mathfrak{T}(5) = 0.3145$ and $\mathfrak{T}(13) = 1.7242 \times 10^{-4}$, the last close to the first zero $\gamma_1 \approx 14.13$. The Hadamard product over the first 1699 ordinates with the anchored tail reproduces these three values, computed from $\xi$ at the two points $\frac{1}{2} + it$ and $\frac{1}{2} + t$, to relative errors $3.8 \times 10^{-10}$, $2.4 \times 10^{-9}$ and $1.6 \times 10^{-8}$.

PROPOSITION 21.3.8. *Let* $\mathfrak{B}(z) = \prod_{\gamma>0}(\gamma^2 - z)/(\gamma^2 + z)$, *so* $\mathfrak{T}(t) = \mathfrak{B}(t^2)$. *Under RH,* $\mathfrak{B}$ *is a Blaschke product for the right half plane: it converges there,* $|\mathfrak{B}| < 1$ *for* $\mathrm{Re}(z) > 0$, $|\mathfrak{B}| = 1$ *for* $\mathrm{Re}(z) = 0$, *its zeros are the points* $\gamma^2$, *and it has no singular inner factor.*

*Proof.* Each factor $b_\gamma(z) = (\gamma^2 - z)/(\gamma^2 + z)$ is a Cayley map taking $\mathrm{Re}(z) > 0$ into the unit disc with $|b_\gamma| = 1$ on $\mathrm{Re}(z) = 0$. Since $1 - b_\gamma(z) = 2z/(\gamma^2 + z)$ and $\sum \gamma^{-2} < \infty$, the product converges locally uniformly away from $\{-\gamma^2\}$; this is the half plane Blaschke condition. A convergent product of Blaschke factors is a Blaschke product, so there is no singular inner factor, and by Lemma 21.2.2 its value is $\mathfrak{B}(z) = \xi(\frac{1}{2} + i\sqrt{z})/\xi(\frac{1}{2} + \sqrt{z})$, the product formula of the lemma having no exponential factor. □

REMARK 21.3.9 (Without RH). Let $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2}) = \prod_k (1 + u/a_k)$ be the function of Section 1.2 (equation (1.2.1)), with $a_k = -(\rho_k - \frac{1}{2})^2$ and one factor for each pair $\{\rho_k, 1 - \rho_k\}$ of zeros. Then $\mathfrak{T}(t) = G(-t^2)/G(t^2)$ for real $t$ with no hypothesis, and $\mathfrak{B}(z) := G(-z)/G(z) = \prod_k (a_k - z)/(a_k + z)$ is still a Blaschke product for the right half plane, with zeros at the points $a_k = -(\rho_k - \frac{1}{2})^2$. They lie in the open right half plane, since $\mathrm{Re}\, a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > 0$ for $\rho_k = \beta_k + i\gamma_k$ ($|\beta_k - \frac{1}{2}| < \frac{1}{2}$ and $|\gamma_k| \geq \gamma_1$). The product converges because $1 - (a_k - z)/(a_k + z) = 2z/(a_k + z)$ and $\sum_k |a_k|^{-1} < \infty$ (Lemma 21.2.2). RH is the statement that these zeros are real, hence equal to the $\gamma^2$, not that $\mathfrak{B}$ is inner. In particular $|\mathfrak{B}(iy)| = 1$ is an identity for every real entire $G$, and the reality of $\mathfrak{T}(t)$ for real $t$ is the functional equation; a numerical check of either tests rounding only.

### *21.3.3.4 Negative moments as a Mellin transform*

PROPOSITION 21.3.10. *For* $\mathrm{Re}(s) > 0$, *with no hypothesis on the zeros,*

$$\begin{aligned} \mathbb{E}\big[T^{-s}\big] &= \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \, \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{u})} \, du \\ &= \frac{2\,\xi(\frac{1}{2})}{\Gamma(s)} \int_0^\infty \frac{w^{2s-1}}{\xi(\frac{1}{2} + w)} \, dw. \end{aligned} \tag{21.3.14}$$

*Proof.* Since $T > 0$ almost surely, $T^{-s} = \Gamma(s)^{-1} \int_0^\infty u^{s-1} e^{-uT} \, du$. Take expectations, with Fubini justified by positivity, substitute the Laplace transform (21.3.3),

which holds without RH, and put $u = w^2$. The integral converges at 0 for $\mathrm{Re}(s) > 0$ and at $\infty$ for every $s$. □

In a computation at 30 digits the two forms agree to $10^{-18}$ at $s = \frac{1}{2}$ and to $1.4 \times 10^{-27}$ at $s = \frac{3}{4}$; at $s = \frac{1}{4}$ the integrand $u^{-3/4}$ of the first form is singular at the origin, and the value below is that of the second form. The values are

$$\mathbb{E}[T^{-1/2}] = 6.7332723127566\ldots,$$
$$\mathbb{E}[T^{-1/4}] = 2.590049154\ldots, \qquad \mathbb{E}[T^{-3/4}] = 17.567694990\ldots.$$

The value $\mathbb{E}[T^{-1/2}]$ is a spectral average weighted toward the low lying zeros; Problem 21.4 compares it with random matrix theory.

PROPOSITION 21.3.11. *With no hypothesis on the zeros, $s \mapsto \mathbb{E}[T^{-s}]$ is an entire function, equal for $\mathrm{Re}\, s > 0$ to the Mellin integral* (21.3.14)*, and at the nonpositive integers it takes the values of the positive moments:*

$$\mathbb{E}[T^{-s}]\big|_{s=-n} \;=\; \mathbb{E}[T^n], \qquad n = 0, 1, 2, \ldots \tag{21.3.15}$$

*Proof.* Since $T > 0$ almost surely and $T$ has all positive and all negative moments, $\mathbb{E}|T^{-s}| = \mathbb{E}[T^{-\mathrm{Re}\, s}] < \infty$ for every $s$, so $\mathbb{E}[T^{-s}]$ is entire as an expectation and (21.3.15) holds by definition. What the residues add is that the continuation of the Mellin formula is this function. Write $M(s) = \int_0^\infty u^{s-1}\varphi(u)\,du$ with $\varphi(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u}) = \sum_{n\ge 0} c_n u^n$ near 0, $c_n = (-1)^n \mathbb{E}[T^n]/n!$. Subtracting Taylor polynomials continues $M$ to a meromorphic function with simple poles at $s = -n$ and $\mathrm{Res}_{s=-n} M = c_n$. Since $1/\Gamma$ is entire with simple zeros exactly there, $M/\Gamma$ is entire and

$$\mathbb{E}[T^{-s}]\big|_{s=-n} = \frac{\mathrm{Res}_{s=-n} M(s)}{\mathrm{Res}_{s=-n} \Gamma(s)} = \frac{(-1)^n \mathbb{E}[T^n]/n!}{(-1)^n/n!} = \mathbb{E}[T^n].$$

□

The entire function $\mathbb{E}[T^{-s}]$ is not of exponential type, because the radius of convergence of $\sum_n \mathbb{E}[T^n]\lambda^n/n!$ is $\gamma_1^2$ (Theorem 11.2.3(a)), so $\mathbb{E}[T^n]$ is not bounded by any geometric sequence, whereas the values of a function of exponential type along the integers grow at most geometrically. Its modulus is nevertheless bounded on every vertical line by $|\mathbb{E}[T^{-s}]| \le \mathbb{E}[T^{-\mathrm{Re}\, s}]$. We claim no uniqueness theorem of Carlson type.

### *21.3.3.5 Total positivity of the density of Y*

The density of $Y = \sqrt{2T}\, Z$ is

$$f(y) = \frac{1}{\pi} \int_0^\infty \cos(ty)\, \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + t)}\, dt,$$

computed from $\xi$ on the real axis alone. By Proposition 16.5.1, $f$ is unconditionally a smooth symmetric density, and RH holds if and only if $f$ is a Pólya frequency function of infinite order. One direction is Theorem 13.1.4(iv); the equivalence in the language of the Laguerre Pólya class is in [313, Theorem 2.3 and Corollary 2.4].

The low orders are decided by the low zeros (Chapter 16). Order two holds without RH, that is, $f$ is log concave (Theorem 16.11.5, a floating point certificate), and, given the verification of RH to height $3\times10^{12}$ [419], $f$ lies within $1.6\times10^{-10}$ of a Pólya frequency function of every order (Theorem 16.11.6), so that every minor of order 3 and 4 is at least $-1.1\times10^{-8}$ and $-1.2\times10^{-7}$. What follows tests $f$ at the orders two to eight on other grids. The computation is by `gl_density_pf.py` at 70 digits, with the trapezoid rule in $t$; $f(2)$ agrees to $3\times10^{-62}$ with the residue series $\sum_k r_k e^{-\gamma_k y}$, $r_k = -\xi(\frac12)/\Xi'(\gamma_k)$, over the first 60 zeros.

At the origin $f(0) = 1.8994210510239\ldots$, which equals $\mathbb{E}[T^{-1/2}]/(2\sqrt{\pi})$ with $\mathbb{E}[T^{-1/2}] = 6.7332723127566\ldots$ (Proposition 21.3.10 at $s=\frac12$). On $[0,2]$ with step 0.05, $ff''-f'^2<0$ at all 41 grid points; the largest value, $-1.37\times10^{-22}$, is at $y=2$, where $f(2) = 1.89857001286\times10^{-10}$. The 121 minors $\det[f(x_r-y_c)]_{r,c<3}$ of order three with $x_r=(i+r)/10$ and $y_c=(j+c)/10$, $0\le i,j\le10$, depend only on $i-j$ and are all positive; the smallest, $6.8275\times10^{-16}$ at $|i-j|=10$, is far smaller than the $1.1\times10^{-8}$ by which Theorem 16.11.6 allows a minor of order three to be negative. The Toeplitz minors $\det[f((r-c)h)]_{r,c<n}$ are positive for $n=3$ to 8 at $h=0.05$, 0.2 and 0.4 (Table 21.3), the smallest being $1.4971\times10^{-21}$. These are

Table 21.3: Toeplitz minors $\det[f((r-c)h)]_{r,c<n}$ of the density of $Y$ (`gl_density_pf.py`, 70 digits).

| $n$ | $h=0.05$ | $h=0.2$ | $h=0.4$ |
|---|---|---|---|
| 3 | $2.7009\times10^{-3}$ | 2.1714 | 6.4789 |
| 4 | $6.4860\times10^{-6}$ | 2.0489 | 11.961 |
| 5 | $3.8720\times10^{-9}$ | 1.9020 | 22.083 |
| 6 | $7.2088\times10^{-13}$ | 1.7560 | 40.770 |
| 7 | $5.0043\times10^{-17}$ | 1.6184 | 75.271 |
| 8 | $1.4971\times10^{-21}$ | 1.4906 | 138.97 |

finite order tests: a negative minor would refute RH, and positive minors cannot prove it (Problem 21.10).

## 21.4 The master theorem route

By Theorem 21.2.3 and Remark 21.2.6, the Thorin property of the centre clock is equivalent to RH, so a proof of it that avoids the zeros (Problem 21.1) cannot take RH as an input. The route of [421] expands $\xi(\frac12)/\xi(\frac12+\sqrt{u})$ term by term and

passes to an integral representation of the GGC through the Ramanujan master theorem. We isolate what the extension of Chaudhry and Qadir [78] settles in that step and what it leaves open.

### 21.4.1 The growth hypothesis

Chaudhry and Qadir [78, §4] enlarge Hardy's class to $Q_\infty(\kappa,\delta)$ (their $\varepsilon$ is called $\kappa$ here, because $\varepsilon=\alpha-\frac12$ in Section 21.5): meromorphic $F(s;0)$, analytic for $\operatorname{Re}(s)<\delta$, not identically zero at the nonpositive integers, with

$$|\Gamma(s)F(s;0)|\le C\,|\tau|^{\kappa(\sigma-\delta)},\qquad s=\sigma+i\tau,\ |\tau|\to\infty, \tag{21.4.1}$$

uniformly for $\sigma_1\le\sigma\le\sigma_2<\delta$. Their Theorem 4.1 grants the master theorem to the class, and their Examples 5.5 and 5.6 place $\zeta(s)/[\Gamma(1-s)\zeta(1-s)]$ and $\zeta(1-s)/[\Gamma(s)\zeta(s)]$ in $Q_\infty(1,\frac12)$. The function to which the route applies the theorem is $F(s;0)=\xi(\frac12)/\xi(\frac12+s)$.

Lemma 21.4.1. *Assume RH, and let $0<\sigma_1\le\sigma_2<\frac12$. Then $F(s;0)=\xi(\frac12)/\xi(\frac12+s)$ satisfies*

$$|\Gamma(s)\,\xi(\tfrac12)/\xi(\tfrac12+s)|\le C_\eta\,e^{-\pi|\tau|/4}\,|\tau|^{\sigma/2-9/4+\eta},\qquad |\tau|\ge1,$$

*uniformly for $\sigma_1\le\sigma\le\sigma_2$, for every $\eta>0$. In particular the growth condition* (21.4.1) *of $Q_\infty(1,\frac12)$ holds on such strips, with exponential room. By the functional equation the same argument gives, for $-\sigma_2\le\sigma\le-\sigma_1$, the exponent $\frac32\sigma-\frac94+\eta$, which is smaller.*

*Proof.* Write $\xi(\frac12+s)=\frac12 w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)$ with $w=\frac12+s$, so $\frac12<\operatorname{Re}w<1$. Stirling's formula $|\Gamma(x+iy)|\asymp|y|^{x-1/2}e^{-\pi|y|/2}$ gives $|\Gamma(w/2)|\asymp|\tau|^{\sigma/2-1/4}e^{-\pi|\tau|/4}$, and $|w(w-1)|\asymp\tau^2$. Under RH, $1/\zeta(w)=O(|\tau|^{\eta})$ uniformly for $\operatorname{Re}w\ge\frac12+\sigma_1$ [498, Section 14.2]. So $|1/\xi(\frac12+s)|\ll e^{\pi|\tau|/4}|\tau|^{-\sigma/2-7/4+\eta}$, and multiplying by $|\Gamma(s)|\asymp|\tau|^{\sigma-1/2}e^{-\pi|\tau|/2}$ gives the bound. For $-\sigma_2\le\sigma\le-\sigma_1$ use $F(s;0)=F(-s;0)$, which is the functional equation $\xi(\frac12+s)=\xi(\frac12-s)$, and $|\Gamma(s)/\Gamma(-s)|\asymp|\tau|^{2\sigma}$: the bound at $-s$ has the exponent $-\sigma/2-\frac94+\eta$, and the factor $|\tau|^{2\sigma}$ turns it into $\frac32\sigma-\frac94+\eta$. □

The strip must avoid $\sigma=0$, where $F(s;0)$ has its poles at $s=i\gamma$. And without RH no polynomial bound for $1/\zeta$ is known on any fixed vertical strip $\sigma_0<\operatorname{Re}w<1$ of the critical strip, so unconditionally the growth condition is only known where $\operatorname{Re}(\frac12+s)\ge1$, outside the range of the class.

### 21.4.2 The absolute convergence hypothesis

With $g$ the density of the clock $T$ (it exists, since $T$ is self decomposable and not degenerate), the interchange of sum and integral requires

$$\int_0^\infty g(y)\,\mathfrak{R}(z/y)\,dy < \infty, \qquad \mathfrak{R}(w) := \sum_{k\ge 1} b_k w^k, \quad b_k := \frac{1}{\Gamma(k)}\,\frac{\xi(\frac12)}{\xi(\frac12+k)}. \tag{21.4.2}$$

Lemma 21.4.2. *With no hypothesis on the zeros, $\mathfrak{R}$ is entire of order $\frac23$, and* (21.4.2) *holds for every $z>0$.*

*Proof.* Stirling applied to $\Gamma(k/2+1/4)$ gives $\log\xi(\frac12+k) = \frac{k}{2}\log k + O(k)$, so the coefficients satisfy $\log(1/b_k) = \frac{3k}{2}\log k + O(k)$ and $\mathfrak{R}$ has order $\frac23$. In particular $\mathfrak{R}(w) \le e^{w^{3/4}}$ for $w \ge w_0$, so $\mathfrak{R}(z/y)$ grows at most like $e^{(z/y)^{3/4}}$ as $y \to 0^+$, and convergence is a condition on the left tail of $g$. Stirling's formula for $\xi(\frac12+x)$ gives $\log \mathbb{E}[e^{-uT}] = -\log\big(\xi(\frac12+\sqrt{u})/\xi(\frac12)\big) \sim -\frac14\sqrt{u}\log u$ as $u\to\infty$, and the exponential Tauberian theorem of Kasahara [299] converts this into $-\log\Pr(T\le y) \sim \frac{1}{16}\,y^{-1}\log^2(1/y)$ as $y\to 0^+$. The clock is self decomposable (Theorem 5.1.1) with the infinite Lévy measure $W(x)\,dx/x$, so it has no atom, and it is unimodal [521] (Section 11.1.1), with a mode $m>0$ and a density $g$ that is nondecreasing on $(0,m)$. Hence $g(y) \le y^{-1}\Pr(T\le 2y)$ for $2y\le m$, that is, $g(y)\le y^{-1}\exp\{-\frac{1}{32}\,y^{-1}\log^2(1/y)\,(1+o(1))\}$, which beats $e^{(z/y)^{3/4}}$ since $1>\frac34$. For $y\ge y_0$ the argument $z/y$ stays in $[0, z/y_0]$, where $\mathfrak{R}$ is bounded, and $\int g = 1$. □

### 21.4.3 What remains open

Two obstructions remain.

(i) *The poles at the zeros.* The class requires analyticity on $\operatorname{Re} s < \frac12$, and the residue argument of Chaudhry and Qadir assumes that the only poles are those of $\Gamma(s)$. But $\xi(\frac12)/\xi(\frac12+s)$ has further poles at $s=\rho-\frac12$, which lie on $\operatorname{Re} s = 0$ under RH and in the strip $|\operatorname{Re} s| < \frac12$ in general, so it is not in $Q_\infty(1,\frac12)$ with or without RH. A contour argument that handles them is available only once their location is known, and assuming it is assuming RH.

(ii) *The $\sqrt{u}$ distortion.* The theorem of Chaudhry and Qadir is stated on vertical strips in the variable $s$, while the Laplace variable of the clock is $u=s^2$, and the squaring map carries the strip $|\operatorname{Re} s|<1$ onto the parabolic region $\{|\operatorname{Im} u| < 2\sqrt{1-\operatorname{Re} u}\}$. Whether the master theorem survives this change of variable needs a separate argument; Proposition 21.4.3 gives it in the shifted variable $w=u-1$, where it holds and is empty.

### 21.4.4 The master theorem in other variables

Hardy's class consists of the $\varphi$ analytic on $\operatorname{Re} w \ge -\delta$ $(0<\delta<1)$ with $|\varphi(v+i\tau)| \le Ce^{Pv+A|\tau|}$, $A<\pi$; for them $\int_0^\infty x^{w-1}\sum_{k\ge0}(-x)^k\varphi(k)/k!\,dx = \Gamma(w)\varphi(-w)$ on $0<\operatorname{Re} w<\delta$.

Proposition 21.4.3. *The function $\varphi(w)=\xi(\frac32)/\xi(\frac12+\sqrt{1+w})$, with the principal root, is analytic and bounded on $\operatorname{Re} w\ge-\delta$. Hence it is in Hardy's class with $P=0$ and any $A$, with no hypothesis on the zeros. The image of $\operatorname{Re} w\ge-\delta$ under $w\mapsto\frac12+\sqrt{1+w}$ lies in $\{\sigma+i\tau:\sigma-\frac12\ge\sqrt{1-\delta+\tau^2}\}$, which contains no zero, so the identity carries no information on them.*

*Proof.* Let $z=x+iy=\sqrt{1+w}$. Then $x^2-y^2=\operatorname{Re}(1+w)\ge1-\delta>0$, which gives the region $(\sigma-\frac12=x,\ \tau=y)$ and shows that $|\arg z|<\frac\pi4$ and $|z|^2\ge1-\delta$. A zero $\rho=\beta+i\gamma$ with $\frac12+z=\rho$ would need $|\gamma|<\beta-\frac12<\frac12$, but $|\gamma|>14$, so $\varphi$ is analytic. For large $|z|$, $\operatorname{Re}(\frac12+z)\ge\frac12+|z|/\sqrt2>1$ and $\zeta(\frac12+z)\to1$, so Stirling gives $\log|\xi(\frac12+z)|=\frac12|z|(\cos\theta\log|z|-\theta\sin\theta)(1+o(1))$ with $\theta=\arg z$, and $\cos\theta\ge1/\sqrt2$. Thus $\xi(\frac12+z)$ grows faster than any $e^{c|z|}$ and $\varphi$ is bounded. □

By contrast, the Thorin property is a statement on the cut $w\in(-\infty,-1]$, which maps onto the critical line. This does not contradict Lemma 21.4.1, which concerns the variable $s$ and needs RH. The other two variables fail for exact reasons.

*The variable $\alpha$.* Fix $s>0$ and $\delta<\min(1,\sqrt s)$. The function $a\mapsto\xi(1+a)/\xi(1+a+\sqrt s)$ is analytic on $\operatorname{Re} a\ge-\delta$, since $\operatorname{Re}(1+a+\sqrt s)>1$ there, and of polynomial growth by Stirling and the convexity bound for $\zeta$. The master theorem evaluates $\xi(\alpha)/\xi(\alpha+\sqrt s)$ at $\alpha=1-w\in(1-\delta,1)$, but only while $\alpha+\sqrt s>1$, in the region of the Euler product. Letting $s\to0$, which the Thorin property at $\alpha$ requires, needs $\delta\to0$ unless there is a zero free strip $\beta\le1-\delta$, that is $\Theta\le1-\delta$ for $\Theta=\sup\beta$, which is not known for any $\delta>0$. So the route is exactly as strong as the hypothesis on $\Theta$ it needs.

*The Thorin variable.* Unconditionally $\xi(\frac12)/\xi(\frac12+\sqrt u)=\exp\{-\int_0^\infty(1-e^{-ut})W(t)\,dt/t\}$ with $W(t)=\sum_{\operatorname{Im}\rho>0}e^{-\lambda_\rho t}$, $\lambda_\rho=-(\rho-\frac12)^2$, as in Theorem 5.1.1. The master theorem needs $W$ to be given by a Taylor series at 0. But $W(t)\sim\log(1/t)/(8\sqrt{\pi t})$ (Section 5.1), so $W$ has none: its Mellin transform is $\Gamma(w)Z(w)$ with $Z(w)=\sum_{\operatorname{Im}\rho>0}\lambda_\rho^{-w}$, convergent for $\operatorname{Re} w>\frac12$, and the singular term gives the double pole $\int_0^1 t^{w-3/2}\log(1/t)\,dt=(w-\frac12)^{-2}$, whereas a Taylor series would give only simple poles at $w=0,-1,\dots$.

### 21.4.5 Why the square root stays

Let $L$ be the class of the functions $s\mapsto\mathbb{E}e^{-sX}$, $X$ a real random variable, finite on an interval.

THEOREM 21.4.4. *(a) For every real $\alpha$, $s \mapsto \xi(\alpha)/\xi(\alpha+s)$ is not in $L$, whether or not RH holds.*

*(b) For every real $\alpha$, $t \mapsto \xi(\alpha+it)/\xi(\alpha)$ is, unconditionally, the characteristic function of the density $\Phi(u)e^{(\alpha-\frac{1}{2})u}/\xi(\alpha)$, with $\Phi$ of (1.2.3), but it is never infinitely divisible, hence never an EGGC. At $\alpha=\frac{1}{2}$ it is the member $\Xi(t)/\Xi(0)$ of the van Dantzig pair of Theorem 5.1.1.*

*(c) For every real $\alpha$, $t \mapsto \xi(\alpha)/\xi(\alpha+it)$ is unbounded, so it is not a characteristic function.*

*Proof.* (a) By Stirling, $\log\{\xi(\alpha)/\xi(\alpha+s)\} = -\frac{1}{2}s\log s\,(1+o(1))$, while, if $\mathbb{E}e^{-sX}$ agreed with it on an interval, it would be finite and agree with it on all of $\mathbb{R}$, since the transform of a positive measure is singular at the real end points of its interval of convergence [514] and $\xi(\alpha)/\xi(\alpha+s)$ is analytic near the real axis; then $\mathbb{E}e^{-sX} \geq \Pr(X \leq M)e^{-sM}$ decays at most exponentially as $s \to \infty$. (b) $\Phi$ is even and positive, since $2\pi n^2 e^{2u} > 3$ for $u \geq 0$, and $\xi(\frac{1}{2}+iz) = \int_{\mathbb{R}} \Phi(u)e^{izu}du$ continues to $z = t - i(\alpha-\frac{1}{2})$, which gives the density, integrable because $\Phi(u) = O(e^{\frac{9}{2}|u|-\pi e^{2|u|}})$. The function is entire in $t$ and vanishes at $t = \gamma - i(\beta-\alpha)$ for every zero $\rho$, whereas an infinitely divisible law with $\mathbb{E}e^{cX} < \infty$ for every real $c$ has $\mathbb{E}e^{zX} = e^{\psi(z)}$ for every complex $z$, with $\psi$ its Lévy Khintchine exponent [471], so its characteristic function has no zero in $\mathbb{C}$. (c) $|\xi(\alpha+it)| \to 0$ faster than any power. □

The EGGC reading of the reciprocal that survives is therefore the symmetric one, $\xi(\alpha)/\xi(\alpha+|t|) = \mathbb{E}e^{-t^2H}$, with $H$ a GGC when $\alpha \geq \sup_\rho \operatorname{Re}\rho$ (Theorem 1.2.2, Section 1.3.2), which contains the square root. Without it, the structure that exists lives on the $\xi$ side and stops at $\alpha = 1$. For $\alpha > 1$, with $G \sim \text{Gamma}(1+\frac{\alpha}{2}, 1)$ and $N$ independent, $\Pr(N=n) = n^{-\alpha}/\zeta(\alpha)$, the identity $\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ gives $\xi(\alpha+s)/\xi(\alpha) = (1+\frac{s}{\alpha-1})\mathbb{E}e^{-sL}$ with $L = \frac{1}{2}\log\pi - \frac{1}{2}\log G + \log N$ infinitely divisible, and $\xi(\alpha)/\xi(\alpha+s)$ is the quotient of a gamma transform by an infinitely divisible one, whose Lévy measure enters with its sign reversed, $1/\mathbb{E}N^{-s} = \zeta(\alpha)\sum_n \mu(n)n^{-\alpha-s}$ being Möbius signed.

## 21.5 Dirichlet $L$ functions and number fields

This section carries the construction of Section 21.2 from $\xi$ to Dirichlet $L$ functions and to Dedekind zeta functions. The numbers are computed by the scripts `gl_dirichlet.py` and `gl_sinsq_chi4.py` (`code/README.md`).

Let $\chi$ be a primitive character modulo $q > 1$ of parity $a \in \{0,1\}$, $\chi(-1) =$

$(-1)^a$, and

$$\xi(s,\chi) = \Big(\frac{q}{\pi}\Big)^{(s+a)/2}\Gamma\Big(\frac{s+a}{2}\Big)L(s,\chi),$$
$$\xi(s,\chi) = W_\chi\,\xi(1-s,\bar\chi), \qquad |W_\chi| = 1,$$

an entire function of order one [104].

### 21.5.1 The positivity obstacle

For $\mathrm{Re}(s) > 1$,

$$\log L(s,\chi) = \sum_{n\ge 2} \frac{\chi(n)\Lambda(n)}{\log n}\, n^{-s}, \tag{21.5.1}$$

so the analogue of $\mu^\sigma$ (Lemma 21.3.1) carries the weights $\chi(p)^r$: complex for complex $\chi$, signed for real nonprincipal $\chi$. Then $L(\sigma+it,\chi)/L(\sigma,\chi)$ is the characteristic function of no probability law: by uniqueness its only candidate has the weights $\chi(n)n^{-\sigma}/L(\sigma,\chi)$ at $-\log n$, and these are not all nonnegative.

On the critical line the Thorin construction needs no Euler product at all, only evenness and the Hadamard product of the completed function.

### 21.5.2 Real characters

For a real primitive character, $\bar\chi = \chi$ and $W_\chi = 1$, so $\xi(s,\chi) = \xi(1-s,\chi)$ and $\xi(\cdot,\chi)$ is real on the real axis.

Proposition 21.5.1. *Let $\chi$ be a real primitive character, and assume GRH for $L(s,\chi)$ and $L(\frac12,\chi)\neq 0$. Write the zeros as $\frac12 \pm i\gamma$ with $\gamma>0$, counted with multiplicity. Then $T_\chi \overset{d}{=} \sum_{\gamma>0} e_\gamma/\gamma^2$ is a GGC with*

$$\frac{\xi(\frac12,\chi)}{\xi(\frac12+\sqrt{u},\chi)} = \mathbb{E}\big[e^{-uT_\chi}\big], \qquad u\ge 0, \tag{21.5.2}$$

*with cumulants $(j-1)!\,p_j(\chi)$, $p_j(\chi) = \sum_{\gamma>0}\gamma^{-2j}$. Propositions 21.3.5 to 21.3.11 hold with $\xi$ replaced by $\xi(\cdot,\chi)$, with GRH for $L(s,\chi)$ and $L(\frac12,\chi)\neq 0$ in the place of RH and $\xi(\frac12)\neq 0$, and with the values at $u_0 = \frac14$, which use $\xi(1) = \frac12$, replaced by those below; in particular*

$$\sum_{\gamma>0}\frac{1}{\gamma^2+\frac14} = \frac{\xi'}{\xi}(1,\chi) = \frac12\log\frac{q}{\pi} + \frac12\,\psi\Big(\frac{1+a}{2}\Big) + \frac{L'}{L}(1,\chi), \tag{21.5.3}$$

$$\mathbb{E}\big[e^{-T_\chi/4}\big] = \frac{\xi(\frac12,\chi)}{\xi(1,\chi)}.$$

*Proof.* $\xi(\frac12+s,\chi)$ is even, entire of order one and nonzero at $s=0$, so the argument of Lemmas 21.2.1 and 21.2.2 gives $\xi(\frac12+s,\chi)/\xi(\frac12,\chi)=\prod_\gamma(1+s^2/\gamma^2)$, and (G2) gives (21.5.2). The later propositions used only this product, evenness and growth on the real axis. For (21.5.3), take the logarithmic derivative of the product at $s=\frac12$ and of the definition of $\xi(\cdot,\chi)$ at $s=1$. □

The hypothesis $L(\frac12,\chi)\neq0$ is Chowla's conjecture for real characters, still open; a central zero would put an atom at $u_0$ in the tilted Thorin measure (Theorem 21.2.3) and block the untilted clock (Corollary 21.2.4(a)). Lemma 21.4.1 holds for $\xi(\cdot,\chi)$ under GRH, with the same decay $e^{-\pi|\tau|/4}$ and the power $|\tau|^{\sigma/2-a/2-1/4+\eta}$ in place of $|\tau|^{\sigma/2-9/4+\eta}$, so the master theorem analysis of Section 21.4 carries over with the same two obstructions.

### 21.5.3 Complex characters

For complex $\chi$ the functional equation links $\xi(s,\chi)$ to a different function, $\xi(1-s,\bar\chi)$. So $\xi(\frac12+s,\chi)$ is not even, its ordinates are not symmetric under $\gamma\mapsto-\gamma$, and the pairing behind (21.5.2) fails. Rotating by the root number makes the function real on the critical line, but not even, and it stays complex on the real axis, where a Laplace transform must be positive.

Proposition 21.5.2. *Let $\chi$ be a complex primitive character and $\xi_{\chi\bar\chi}(s):=\xi(s,\chi)\xi(s,\bar\chi)$. Then $\xi_{\chi\bar\chi}(s)=\xi_{\chi\bar\chi}(1-s)$ and $\xi_{\chi\bar\chi}(\sigma)=|\xi(\sigma,\chi)|^2$ for real $\sigma$. If GRH holds for $L(s,\chi)$ and $L(\frac12,\chi)\neq0$, then*

$$\frac{\xi_{\chi\bar\chi}(\frac12)}{\xi_{\chi\bar\chi}(\frac12+\sqrt{u})}=\mathbb{E}\big[e^{-uT_{\chi\bar\chi}}\big],\qquad T_{\chi\bar\chi}\overset{d}{=}\sum_{\gamma\in\Gamma_\chi}\frac{e_\gamma}{\gamma^2},$$

*where $\Gamma_\chi$ is the multiset of all ordinates of zeros of $L(s,\chi)$, of both signs.*

*Proof.* The root numbers multiply to $W_\chi W_{\bar\chi}=|W_\chi|^2=1$, and $\xi(\sigma,\bar\chi)=\overline{\xi(\sigma,\chi)}$ for real $\sigma$. The zeros of $L(s,\bar\chi)$ are the conjugates $\frac12-i\gamma$, so $\xi_{\chi\bar\chi}$ has zeros at $\frac12\pm i\gamma$ for every $\gamma\in\Gamma_\chi$, and the even product argument applies: each ordinate of $L(s,\chi)$ gives one factor $1+u/\gamma^2$, its partner $\frac12-i\gamma$ among the zeros of $L(s,\bar\chi)$ being the other zero of the pair, so no ordinate is counted twice. □

### 21.5.4 The Dedekind route

Let $\chi$ be a real primitive character of conductor $q$ and $F=\mathbb{Q}(\sqrt{D})$, with $D=\chi(-1)q$ a fundamental discriminant. Then $\zeta_F(s)=\zeta(s)L(s,\chi)$, and $F$ is real quadratic, $(r_1,r_2)=(2,0)$, for even $\chi$ and imaginary quadratic, $(r_1,r_2)=(0,1)$, for odd $\chi$. More generally, for $q\geq3$ the cyclotomic field $F_q=\mathbb{Q}(\zeta_q)$ is totally complex of degree $\varphi(q)$ and

$$\zeta_{F_q}(s)=\prod_{\chi \bmod q}L(s,\chi^*),\tag{21.5.4}$$

with $\chi^*$ the primitive character inducing $\chi$. Every primitive character of conductor dividing $q$ occurs, complex ones together with their conjugates. And Dedekind zeta functions have nonnegative coefficients,

$$\log \zeta_F(s) = \sum_{n\geq 2} \frac{\Lambda_F(n)}{\log n}\, n^{-s}, \qquad \Lambda_F(n) = \sum_{N\mathfrak{p}^r = n} \log N\mathfrak{p} \;\geq 0,$$

which is why one passes to fields.

Proposition 21.5.3. *For a quadratic field, $\xi_F(s) = 2\,\xi(s)\xi(s,\chi)$ when $\chi$ is even and $\xi_F(s) = 2q^{-1/2}\xi(s)\xi(s,\chi)$ when $\chi$ is odd. Consequently, at any tilt $u_0 \geq 0$ at which $\xi$ and $\xi(\cdot,\chi)$ both have the Thorin property in the sense of Theorem 21.2.3, so does $\xi_F$, its tilted clock is the independent sum of the tilted clocks of $\xi$ and of $\xi(\cdot,\chi)$, and*

$$U_{F,u_0} = U_{\mathbb{Q},u_0} + U_{\chi,u_0},$$

*where $U_{\chi,u_0}$ is the Thorin measure of the tilted clock of $\xi(\cdot,\chi)$, defined as $U_{F,u_0}$ is in Theorem 21.2.3; at $u_0 = 0$ this reads $T_F \overset{d}{=} T_\zeta + T_\chi$, with $T_\zeta = T$ the centre clock of $\xi$. For a cyclotomic field, under the hypotheses of Propositions 21.5.1 and 21.5.2, $T_{F_q}$ is the independent sum of $T_\zeta$, of $T_\chi$ over the real primitive characters of conductor $f \mid q$ with $f > 1$, and of $T_{\chi\bar\chi}$ over the conjugate pairs of complex primitive characters of conductor dividing $q$. In particular the cumulants add: $p_j(F) = p_j(\zeta) + p_j(\chi)$ for a quadratic field.*

For $\mathbb{Q}(i)$ at basepoints $\alpha \geq 1$, where no hypothesis is needed, this is Proposition 8.5.28 and Corollary 8.5.29, with the constant $2 \cdot 4^{-1/2} = 1$.

*Proof.* For even $\chi$ the gamma factors agree, $\Gamma_{\mathbb{R}}(s)^2$ on both sides, and $|d_F| = q$. For odd $\chi$ the duplication formula gives $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$, while $\xi(s,\chi) = q^{(s+1)/2}\Gamma_{\mathbb{R}}(s+1)L(s,\chi)$ carries an extra $q^{1/2}$. The constants cancel in the ratios (21.2.4), so the Laplace transforms multiply and the variables add. The cyclotomic case follows from (21.5.4) in the same way: the factors there are primitive, half of the characters modulo $q \geq 3$ are odd, so the gamma factors multiply to $\Gamma_{\mathbb{C}}(s)^{\varphi(q)/2}$, and the discriminant is the product of the conductors. □

### 21.5.5 The Lévy decomposition of $\xi_F$, with signs

For $\alpha > 1$ and $s \geq 0$ put $h_s(x) = e^{-sx} - 1 + sx \geq 0$. Binet's formula gives

$$\log \frac{\Gamma_{\mathbb{R}}(\alpha+s)}{\Gamma_{\mathbb{R}}(\alpha)} - s\frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}(\alpha) = \int_0^\infty h_s(x)\frac{e^{-\alpha x}}{1-e^{-2x}}\frac{dx}{x},$$
$$\log \frac{\Gamma_{\mathbb{C}}(\alpha+s)}{\Gamma_{\mathbb{C}}(\alpha)} - s\frac{\Gamma'_{\mathbb{C}}}{\Gamma_{\mathbb{C}}}(\alpha) = \int_0^\infty h_s(x)\frac{e^{-\alpha x}}{1-e^{-x}}\frac{dx}{x},$$

a linear factor gives $\log\frac{z+s}{z}-\frac{s}{z}=-\int_0^\infty h_s(x)e^{-zx}\frac{dx}{x}$ for $z>0$, and

$$\log\frac{\zeta_F(\alpha+s)}{\zeta_F(\alpha)}-s\frac{\zeta_F'}{\zeta_F}(\alpha)=\sum_{n\geq 2}\frac{\Lambda_F(n)}{\log n}\,n^{-\alpha}h_s(\log n).$$

Proposition 21.5.4. *For $\alpha>1$ and $s\geq 0$,*

$$\log\frac{\xi_F(\alpha+s)}{\xi_F(\alpha)}-s\,\frac{\xi_F'}{\xi_F}(\alpha)=\int_0^\infty h_s(x)\,e^{-\alpha x}\,\frac{\nu_F(dx)}{x} \tag{21.5.5}$$

*with the signed measure*

$$\nu_F(dx)=\Big(\frac{r_1}{1-e^{-2x}}+\frac{r_2}{1-e^{-x}}-1-e^x\Big)dx+\sum_{n\geq 2}\Lambda_F(n)\,\delta_{\log n}(dx). \tag{21.5.6}$$

*Proof.* Add the decompositions above. The density $-(1+e^x)$ comes from the factors $\alpha+s$ and $\alpha-1+s$ of $s(s-1)$, both with positive $z$ since $\alpha>1$, and $|d_F|^{s/2}$ is linear in $s$ and only changes the drift. □

For $F=\mathbb{Q}$, where $\xi_{\mathbb{Q}}=2\xi$, this is Proposition 9.1.10; for $\chi_4$ and $\chi_{12}$ it is Propositions 8.5.24 and 11.3.2.

Remark 21.5.5 (Where positivity fails, and why it is the same question). The polynomial factor contributes the density $-(1+e^x)$, which dominates the positive continuous part for large $x$. By the prime ideal theorem $\Lambda_F$ has mean value one, so on average the atoms $\sum\Lambda_F(n)\delta_{\log n}$ cancel $-e^x\,dx$, and by the explicit formula what remains is a sum over the zeros of $\zeta_F$. Positivity of any smoothed $\nu_F$ is therefore again a statement about zeros: the Dedekind route does not give the Thorin property for free.

### 21.5.6 The sine squared lemma: from primes to the Thorin measure

Proposition 21.5.4 writes $\log\xi_F$ against a signed Lévy measure. The sine squared lemma [421, Lemma 8.1], first in [422, Lemma 1], converts any such decomposition into a Thorin representation of the reciprocal [421, Theorem 8.2]; in the book it is equation (1.2.9) of Section 1.2, applied to $\xi$ in Proposition 9.1.3 and to $\xi(\cdot,\chi_4)$ after the Lévy form of Proposition 8.5.24 (Section 8.5.3). In the chart $z=b^2$ the kernel identity (1.2.9) reads, for $w,x>0$,

$$e^{-wx}-1+wx=\frac{x}{\pi}\int_0^\infty\log\Big(1+\frac{w^2}{b^2}\Big)\,2\sin^2\Big(\frac{bx}{2}\Big)\,db, \tag{21.5.7}$$

and the book gives its proof. For a function $f$ below write $v_\alpha(b)=\operatorname{Re}\frac{f'}{f}(\alpha+ib)$. For $f=\xi$ this is the function $v_\alpha$ of Section 1.2, $v_\alpha/\pi$ is the density $\rho_{\alpha-1/2}$ of (1.2.7), and the measure $U_\alpha(dt)=v_\alpha(\sqrt{t})\,dt/(2\pi\sqrt{t})$ is that of (1.2.6).

PROPOSITION 21.5.6 (The lemma on the prime side). *Let $f$ be $\xi_F$, or $\xi(\cdot,\chi)$ for a real primitive $\chi$, and let $\alpha > 1$. Then*

$$\frac{f(\alpha)}{f(\alpha+\sqrt{u})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{u}{b^2}\Big)\,\frac{v_\alpha(b)}{\pi}\,db\Big), \qquad u \ge 0, \tag{21.5.8}$$

*where*

$$v_\alpha(b) = \frac{f'}{f}(\alpha) + \int_0^\infty 2\sin^2\Big(\frac{bx}{2}\Big)e^{-\alpha x}\,\nu(dx) = \operatorname{Re}\frac{f'}{f}(\alpha+ib), \tag{21.5.9}$$

*and $\nu$ is the Lévy measure of $f$: $\nu_F$ of* (21.5.6) *for $\xi_F$, and for $\xi(\cdot,\chi)$*

$$\nu_\chi(dx) = \sum_{n\ge2}\chi(n)\Lambda(n)\,\delta_{\log n}(dx) + \frac{e^{(1-a)x}}{2\sinh x}\,dx,$$

*that is $(2\sinh x)^{-1}dx$ for odd $\chi$ and $(1-e^{-2x})^{-1}dx$ for even $\chi$, from the gamma factor $\Gamma_{\mathbb{R}}(s+a)$.*

*Proof.* For $\xi_F$ the decomposition

$$\log\frac{f(\alpha+w)}{f(\alpha)} - w\,\frac{f'}{f}(\alpha) = \int_0^\infty h_w(x)\,e^{-\alpha x}\,\frac{\nu(dx)}{x}$$

is Proposition 21.5.4, and for $\xi(\cdot,\chi)$ it follows in the same way from Binet's formula for $\Gamma_{\mathbb{R}}(s+a)$ and (21.5.1). Insert (21.5.7) and interchange the integrals; this is justified for $\alpha > 1$ because the kernels are nonnegative and $\int \min(1,x)\,e^{-\alpha x}|\nu|(dx) < \infty$ (near 0 the density of $|\nu|$ is of order $1/x$, and at infinity $e^{-\alpha x}|\nu|(dx)$ is dominated by $e^{-(\alpha-1)x}dx$ and by the prime sum $\sum\Lambda_F(n)n^{-\alpha}$). The linear term is absorbed using $\int_0^\infty \log(1+w^2/b^2)\,db = \pi w$, which gives the first form of $v_\alpha$ with $u = w^2$. For the second, differentiate the decomposition in $w$: $\int(1-e^{-wx})e^{-\alpha x}\nu(dx) = (f'/f)(\alpha+w) - (f'/f)(\alpha)$. The left side is analytic for $\operatorname{Re} w > 1-\alpha$ by the same bound, the right side is meromorphic with poles at the zeros, and they agree for $w > 0$, so both continue to $w = ib$; taking real parts gives $\int 2\sin^2(bx/2)e^{-\alpha x}\nu(dx) = \operatorname{Re}(f'/f)(\alpha+ib) - (f'/f)(\alpha)$. □

For a real character the arithmetic part of $v_\alpha(b)$ is

$$-\sum_{n\ge2}\chi(n)\Lambda(n)\,n^{-\alpha}\cos(b\log n),$$

absolutely convergent for $\alpha > 1$. At $\alpha = 1$ it converges conditionally for nonprincipal $\chi$, and for $\xi_F$ the prime sum diverges and converges only together with the density $-e^x$ of (21.5.6), as in Proposition 9.1.3.

LEMMA 21.5.7. *For $c \in \mathbb{C}$ with $\operatorname{Re} c > 0$ and $w > 0$,*

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \log\Big(1+\frac{w^2}{b^2}\Big)\,\operatorname{Re}\frac{1}{c+ib}\,db = 2\log\Big|1+\frac{w}{c}\Big|.$$

*For $\operatorname{Re} c < 0$ the right side is replaced by $-2\log|1-w/c|$.*

*Proof.* Write $\log(1+w^2/b^2) = \int_0^w 2w'(b^2+w'^2)^{-1}dw'$. For $w' > 0$ the function $b \mapsto (c+ib)^{-1}(b^2+w'^2)^{-1}$ is rational and $O(|b|^{-3})$, with poles at $ic$, $iw'$ and $-iw'$. If $\operatorname{Re} c > 0$ the pole $ic$ lies in the upper half plane, and closing the contour in the lower half plane picks up only $-iw'$, so that $\int_{-\infty}^{\infty} db/((c+ib)(b^2+w'^2)) = -2\pi i/((c+w')(-2iw')) = \pi/(w'(c+w'))$. Since $|\operatorname{Re}(c+ib)^{-1}| \le |\operatorname{Re} c|^{-1}$ and $\int_{-\infty}^{\infty} 2w'(b^2+w'^2)^{-1}db = 2\pi$, Fubini applies, and taking real parts,

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{c+ib}\,db = \operatorname{Re}\int_0^w \frac{2\,dw'}{c+w'} = 2\log\Big|\frac{c+w}{c}\Big|.$$

If $\operatorname{Re} c < 0$ the same computation, closing in the upper half plane, gives $-2\log|(c-w)/c|$. □

Zeros $\rho = \beta + i\gamma$ are counted with multiplicity. Say that $f$ has the *Thorin property at the basepoint* $\alpha > \frac12$ if $f(\alpha) \neq 0$ and $u \mapsto f(\alpha)/f(\alpha+\sqrt{u})$ is the Laplace transform of a GGC.

THEOREM 21.5.8 (The basepoint family of the lemma). *Let $f$ be as in Proposition 21.5.6, with zeros $\rho = \beta + i\gamma$.*

*(a) If every zero has $\beta < \alpha$, then* (21.5.8) *holds with*

$$\frac{v_\alpha(b)}{\pi} = \frac{1}{\pi}\sum_\rho \operatorname{Re}\frac{1}{\alpha+ib-\rho} = \frac{1}{\pi}\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(b-\gamma)^2} > 0,$$

*so $f$ has the Thorin property at the basepoint $\alpha$, with no drift and Thorin measure $U_\alpha(dt) = v_\alpha(\sqrt{t})\,dt/(2\pi\sqrt{t})$. Since every zero has $\beta < 1$, this holds unconditionally for every $\alpha \ge 1$.*

*(b) If some zero has $\beta > \alpha$ and $f(\alpha) \neq 0$, then $f$ does not have the Thorin property at the basepoint $\alpha$. Hence $f$ has it at every basepoint $\alpha > \frac12$ if and only if every zero lies on the critical line.*

*(c) Under GRH, and if $f(\frac12) \neq 0$, with $\varepsilon = \alpha - \frac12$,*

$$\frac{v_\alpha(b)}{\pi} = \sum_{\gamma>0}\big(P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)\big), \qquad P_\varepsilon(x) = \frac{\varepsilon}{\pi(\varepsilon^2+x^2)},$$

*the Poisson smoothing at height $\varepsilon$ of the zero counting measure. As $\alpha \downarrow \frac12$, $(v_\alpha(b)/\pi)\,db \to \sum_{\gamma>0}\delta_\gamma$ vaguely on $[0,\infty)$, and $U_\alpha$ tends to the untilted Thorin measure $\sum_\gamma \delta_{\gamma^2}$ of Theorem 21.2.3.*

*Proof.* (a) By Hadamard's factorisation $f(s) = e^{A+Bs}\prod_\rho (1-s/\rho)e^{s/\rho}$ and the functional equation, $\operatorname{Re}(f'/f)(s) = \sum_\rho \operatorname{Re}(s-\rho)^{-1}$, absolutely convergent, for all $s$ off the zeros, and $\operatorname{Re} B = -\sum_\rho \operatorname{Re}\rho^{-1}$ [104, Ch. 12]. Each term of the sum is positive when $\operatorname{Re} s > \beta$, which gives the formula for $v_\alpha$ and its positivity. For a zero $\rho$ put $\omega_\rho = \alpha - \rho$, so that $\operatorname{Re}\omega_\rho = \alpha - \beta > 0$ and the term of $\rho$ in $v_\alpha(b)$ is $\operatorname{Re}(\omega_\rho + ib)^{-1}$, and for $w > 0$ put

$$I_\rho = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{\omega_\rho + ib}\,db \;\ge 0.$$

Since $\operatorname{Re}(\omega_{\bar\rho}+ib)^{-1} = \operatorname{Re}(\omega_\rho - ib)^{-1}$, the substitution $b \mapsto -b$ and Lemma 21.5.7 give $I_\rho + I_{\bar\rho} = 2\log|1+w/\omega_\rho|$, and $|1+w/\omega_{\bar\rho}| = |1+w/\omega_\rho|$. The function $f$ is real on the real axis, so $\rho \mapsto \bar\rho$ permutes the zeros, and since all terms are nonnegative we may sum:

$$\int_0^\infty \log\Big(1+\frac{w^2}{b^2}\Big)\frac{v_\alpha(b)}{\pi}\,db = \sum_\rho I_\rho = \sum_\rho \log\Big|1+\frac{w}{\omega_\rho}\Big|.$$

On the other side the Hadamard product gives $f(\alpha+w)/f(\alpha) = e^{Bw}\prod_\rho(1+w/\omega_\rho)e^{w/\rho}$, an absolutely convergent product, and the left side is positive because $f$ is real and has no zero on $[\alpha,\infty)$. Taking logarithms and using $\operatorname{Re} B = -\sum_\rho \operatorname{Re}\rho^{-1}$, we get $\log f(\alpha+w)/f(\alpha) = \sum_\rho \log|1+w/\omega_\rho|$, the series converging absolutely. With $u = w^2$ this is (21.5.8) with density $v_\alpha/\pi$. In the variable $t = b^2$ the exponent is $\int_0^\infty \log(1+u/t)\,U_\alpha(dt)$ and it is finite; this is equivalent to the integrability conditions on a Thorin measure stated with (21.2.1), so $U_\alpha$ is a Thorin measure with no drift.

(b) Let $\rho$ be a zero with $\beta > \alpha$ and put $u_\rho = (\rho-\alpha)^2$. Since $\operatorname{Re}(\rho-\alpha) = \beta-\alpha > 0$, the principal square root of $u_\rho$ is $\rho-\alpha$, and $u_\rho \notin (-\infty,0]$, because otherwise $\rho-\alpha$ would be purely imaginary. The function $f(\alpha+\sqrt{u})$ is analytic on $\mathbb{C}\setminus(-\infty,0]$ and vanishes at $u_\rho$, so $f(\alpha)/f(\alpha+\sqrt{u})$ has a pole there. If $f$ had the Thorin property at $\alpha$, this function would coincide on $u>0$, hence on $\mathbb{C}\setminus(-\infty,0]$, with a GGC transform, which is analytic there by (G1): a contradiction. If every zero lies on the critical line, then $\beta = \frac12 < \alpha$ for every $\alpha > \frac12$ and (a) gives the Thorin property at every basepoint $\alpha > \frac12$. If some zero is off the line, the functional equation gives one with $\beta > \frac12$, and the property fails at every $\alpha \in (\frac12,\beta)$, by the argument above where $f(\alpha)\ne 0$ and by definition where $f(\alpha) = 0$.

(c) Under GRH every zero is $\frac12 \pm i\gamma$ with $\gamma > 0$, since $f(\frac12) \ne 0$, and (a) applies for every $\alpha > \frac12$; with $\alpha-\beta = \varepsilon$ the two zeros $\frac12 \pm i\gamma$ contribute $P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)$. Let $h$ be continuous with compact support in $[0,R]$ and let $0 < \varepsilon \le 1$. For $\gamma \le R+1$ the integral of $h$ against $P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)$ over $[0,\infty)$ tends to $h(\gamma)$, because $\gamma > 0$ is a point of continuity of $h$ extended by zero and $\int_0^\infty P_\varepsilon(b+\gamma)\,db \le \varepsilon/(\pi\gamma)$. For $\gamma > R+1$ and $b \in [0,R]$ one has $|b\mp\gamma| \ge \gamma - R \ge \gamma/(R+1)$, so the integral is at most $2R\|h\|_\infty(R+1)^2\varepsilon/(\pi\gamma^2)$. The zeros with $\gamma > R+1$ therefore contribute

at most a constant times $\varepsilon \sum_\gamma \gamma^{-2}$, which tends to 0, and the finitely many others give $\sum_{\gamma>0} h(\gamma)$ in the limit, so $\int h\,(v_\alpha/\pi)\,db \to \sum_{\gamma>0} h(\gamma)$. Under $b \mapsto b^2$ this is the convergence of $U_\alpha$ to $\sum_\gamma \delta_{\gamma^2}$. □

If the half plane of [398], which has not been refereed (Section 12.1.2), stands, then every zero of every Dirichlet $L$ function has $\beta \le \frac{7}{8}$, and part (a) gives the Thorin property of $\xi(\cdot,\chi)$ for every real primitive character $\chi$, and of $\xi_F$ for every quadratic field $F$, at every basepoint $\alpha > \frac{7}{8}$, with Thorin density $v_\alpha/\pi$ a sum of Lorentzians of widths at least $\alpha - \frac{7}{8}$. This is Theorem 16.12.1(i) carried from $\zeta$ to the family, and it lowers the unconditional end $\alpha \ge 1$ of part (a) to $\alpha > \frac{7}{8}$ under that hypothesis. At $\alpha = \frac{7}{8}$ itself part (a) does not apply, since the half plane does not exclude a zero on the line $\operatorname{Re} s = \frac{7}{8}$.

For $\xi$ these statements are in the book: part (a) is the first sentence of Theorem 1.2.2; parts (a) and (b) together, with the borderline case in which a zero on the line $\operatorname{Re} s = \alpha$ contributes an atom, are Theorem 7.1.14, and Theorem 10.2.7 treats the basepoints $\frac{1}{2} \le \alpha < 1$; and the Poisson form of part (c) is Proposition 1.2.3. For $\chi_4$ the corresponding result is Proposition 8.5.25, and for $\chi_{12}$ at the basepoint one Proposition 11.3.1. In part (c) the vague convergence holds without RH, as shown after Remark 1.2.4 and in [421, Proposition 9.5]: the limit is then the counting measure of the ordinates of the zeros on the line, the pairs off the line contributing signed mass whose total variation tends to zero. Without $f(\frac{1}{2}) \ne 0$ a central zero of order $2m$ adds the term $2m\,P_\varepsilon(b)$ to $v_\alpha/\pi$, of mass $m$ on $(0,\infty)$, which concentrates at $b = 0$ as $\varepsilon \downarrow 0$; the limit is then not a Thorin measure (Corollary 21.2.4(a)).

The tilt family and the basepoint family meet at the centre. Theorem 21.2.3 tilts by $u_0$, reads $f$ at $\frac{1}{2} + \sqrt{u_0 + u}$, and shifts each atom from $\gamma^2$ to $u_0 + \gamma^2$. The lemma's family reads $f$ at $\alpha + \sqrt{u}$ and smooths each atom by a Poisson kernel of width $\alpha - \frac{1}{2}$. One consequence differs. At the centre $f(\frac{1}{2} + w)$ is even in $w$ and $v_{1/2}(0) = 0$; for $\alpha > \frac{1}{2}$ the function $f(\alpha + w)$ is not even, and under the hypothesis of part (a) $v_\alpha(0) = (f'/f)(\alpha) > 0$ is a sum of positive terms. The Thorin measure then has a component like that of a stable law of index $\frac{1}{2}$, its density behaving like $v_\alpha(0)/(2\pi\sqrt{t})$ as $t \downarrow 0$, and the shifted clock $T_\alpha$ has $\mathbb{E}[T_\alpha] = \infty$. At every basepoint $\alpha \ge 1$ the book proves the exact split into a positive stable variable of index $\frac{1}{2}$ and a GGC, for $\xi$ in Propositions 9.1.11 and 9.1.12 and for $\chi_4$ and $\chi_{12}$ in Theorem 9.1.18(v) with the paragraph after it. The moment dictionary of Section 21.3.3 lives only at the centre.

Remark 21.5.9 (What the lemma buys, and what it cannot). Reaching the centre means running the Poisson smoothing backwards from $\alpha$ to $\frac{1}{2}$. In the variable $x = \log n$ the smoothing multiplies the weight of $n$ by $n^{-(\alpha - 1/2)}$, so undoing it replaces $n^{-\alpha}$ by the divergent $n^{-1/2}$: the backward step is the explicit formula, and it is ill posed. Chapter 19 measures its cost. By Theorem 19.2.2 the Thorin density at a basepoint beyond the zeros is the Poisson smoothing of a positive

measure down to the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon_1$ exactly when no zero has $\beta - \frac{1}{2} > \varepsilon_1$. Under GRH, by Proposition 19.2.3 and Table 19.1 (for $\chi_4$) the number of primes needed for accuracy $\delta$ is of order $\delta^{-1/(\alpha-1/2)}$, unbounded as $\alpha \downarrow \frac{1}{2}$. Section 20.1.1 of Chapter 20 shows that the deconvolved object is Weil's distribution of the explicit formula. Positivity on a single line is not known to exclude zeros to its right (Remark 1.2.4 and Section 12.1.2). For $\xi$, Proposition 9.1.12 proves that the minimum of $v_\alpha$ on the line is at $b = 0$ for every $\alpha \geq 1$, and for $\chi_4$ and $\chi_{12}$ Theorem 9.1.18(v) does; the theorem also certifies it for every $\alpha \in (\frac{1}{2}, \alpha_{\max}]$ up to the height to which GRH is verified (Table 9.4), and shows that the statement on the whole lines for all $\alpha \in (\frac{1}{2}, 1)$ together is equivalent to GRH. Our computations for $\chi_4$ below agree. A real zero $\beta > \alpha$ adds the term $-(\beta-\alpha)/((\beta-\alpha)^2 + b^2)$ to $v_\alpha(b)$, whose modulus is largest at $b = 0$, so the origin is where a real exceptional zero acts first: a Siegel zero is a failure of the lemma's positivity at the origin. The classical zero free region arguments live on the Lévy side of the lemma, which restates their conclusions as positivity of a Thorin density. That is a translation, and this chapter obtains no new estimate from it, for Linnik's constant $K$ or for the least prime in a progression.

*Numerical check for $\chi_4$.* The script `code/gl_sinsq_chi4.py` checks the lemma for $\chi_4$ at 30 digits. At $\alpha = 2$, with the prime powers up to $N$ and the integral over $b$ done term by term (each prime term is the kernel identity), the integral in (21.5.8) differs from $\log \xi(\alpha + w, \chi_4)/\xi(\alpha, \chi_4)$ by $-4.4 \times 10^{-8}$, $-1.2 \times 10^{-9}$ and $6.2 \times 10^{-11}$ for $N = 10^4$, $10^5$ and $10^6$ at $w = 1$ and $w = 3$; this difference is the tail of the prime sum beyond $N$. The prime side of (21.5.9), over the prime powers up to $2 \times 10^6$, agrees with $v_\alpha(b)/\pi$ computed directly from $\xi'/\xi$ to $9.4 \times 10^{-11}$ at $\alpha = 2$ and to $1.9 \times 10^{-7}$ at $\alpha = \frac{3}{2}$. The zero side, the sum of part (a) over the 203 zeros of $L(s, \chi_4)$ below height 300 found by the script, with the tail beyond height 300 taken from the smooth zero density, differs from the direct value by $1.1 \times 10^{-6}$ to $1.7 \times 10^{-6}$ at $\alpha = \frac{3}{2}$ and 2 for $0 \leq b \leq 50$; once the tail is anchored to the exact zero count at the cut, as in (21.3.8), the difference falls to between $9.7 \times 10^{-9}$ and $1.7 \times 10^{-8}$, far inside the enclosure of $1.6 \times 10^{-5}$ to $2.7 \times 10^{-5}$ that the explicit bound for the zero count used in the proof of Theorem 9.1.18(v) guarantees.

On the lines $\alpha = 1, \frac{3}{4}, \frac{3}{5}, \frac{11}{20}$ the minimum of $v_\alpha(b)$ over $0 \leq b \leq 150$, on grids of step 0.1 and 0.005, is at $b = 0$ (Table 21.4), where $v_\alpha(0) = b_\alpha = (\xi'/\xi)(\alpha, \chi_4)$.

These checks test different things. For $\alpha = 2$ and $\frac{3}{2}$ every zero of $L(s, \chi_4)$ has real part below $1 < \alpha$, so Proposition 21.5.6 and part (a) of Theorem 21.5.8 hold with no hypothesis, and the comparisons above are identities up to truncation: they test the quadrature, the truncation of the prime sums, the zero list and its tail, and the code, not the theorem and not GRH. The grid of the table is a finite test of the function $v_\alpha$, consistent with GRH; part (ii) of Theorem 9.1.18 turns it into a theorem for $0 < |b| \leq 2.5 \times 10^7 - 1.21$, using the verification of GRH for $\chi_4$ to height $2.5 \times 10^7$ [417], and for $\alpha < 1$ the function $v_\alpha/\pi$ is the Thorin density of $\xi(\alpha, \chi_4)/\xi(\alpha + \sqrt{u}, \chi_4)$ only if no zero lies to the right of $\alpha$, which is known below

| $\alpha$ | $v_\alpha(0) = b_\alpha$ | $2(\alpha - \frac{1}{2})\, p_1(\chi_4)$ | ratio |
|---|---|---|---|
| 1 | 0.0777839900 | 0.0780148250 | 0.99704 |
| $\frac{3}{4}$ | 0.0389784293 | 0.0390074125 | 0.99926 |
| $\frac{3}{5}$ | 0.0156011077 | 0.0156029650 | 0.99988 |
| $\frac{11}{20}$ | 0.0078012503 | 0.0078014825 | 0.99997 |

Table 21.4: The minimum of $v_\alpha(b)$ over $0 \le b \le 150$ for $\chi_4$, always at $b = 0$, against the bound $2(\alpha - \frac{1}{2})\, p_1(\chi_4) = (\alpha - \frac{1}{2})V_f$ of Theorem 9.1.18(i), with $V_f = 2p_1(\chi_4) = 0.1560296499$. Computed by `code/gl_sinsq_chi4.py` (`data/gl_sinsq_chi4.json`).

that height and open above it. The lemma's output is not computable from the primes alone in practice for $\alpha \le 1$: the plain prime sum at $b = 0$ over the prime powers up to $2 \times 10^6$ differs from $v_\alpha(0)$ by 0.17, 12, 260 and 1100 percent of the value at $\alpha = 1, \frac{3}{4}, \frac{3}{5}, \frac{11}{20}$, so the minima of the table come from the zeros and from the direct values of $\xi'/\xi$, not from the primes.

Finally, with the smoothing run backwards by a smooth cutoff, the peaks of $-\sum_{n \le N} \chi_4(n)\Lambda(n)n^{-1/2}\cos(b\log n)\,\cos^2\left(\frac{\pi}{2}\frac{\log n}{\log N}\right)$, $N = 2 \times 10^6$, lie within 0.003 of all fourteen zeros of $L(s, \chi_4)$ below height 40. Up to height 45 the seventeen zeros are matched by seventeen peaks and there is no other peak above half the largest value. This tests the explicit formula with the smoothing run backwards, and it locates the ordinates from the primes; it does not test the lemma.

## 21.6 GRH from the Thorin property, and families of characters

Theorem 21.6.1. *Let $\chi$ be a real primitive character with quadratic field $F$.*

*(a) If $\varphi_{F,1}$ is the Laplace transform of a GGC, then RH holds and every non-trivial zero of $L(s, \chi)$ lies on the critical line or is real.*

*(b) If $\varphi_{F,0}$ is the Laplace transform of a GGC, equivalently $\varphi_{F,1}$ is one with Thorin measure in $(1, \infty)$, then $L(s, \chi)$ satisfies GRH and $L(\frac{1}{2}, \chi) \ne 0$.*

*If $\varphi_{F_q,0}$ is a GGC Laplace transform for every $q \ge 3$, with $F_q = \mathbb{Q}(\zeta_q)$, then GRH holds for all Dirichlet L functions, and $L(\frac{1}{2}, \chi) \ne 0$ for every primitive character $\chi$.*

*Proof.* Apply Corollary 21.2.4 to $\zeta_F = \zeta\, L(s, \chi)$, part (b) of the corollary for the tilt $u_0 = 1$ and part (a) for $u_0 = 0$; for $\zeta$ itself real zeros in $(0, 1)$ are excluded because $\zeta(\sigma) < 0$ there, and $\zeta(\frac{1}{2}) \ne 0$. The last statement uses equation (21.5.4): the zeros of $\zeta_{F_q}$ are those of the functions $L(s, \chi)$ for the primitive characters $\chi$ of conductor dividing $q$, and every primitive character of conductor $f \ge 3$ occurs

for $q = f$. Part (a) of the corollary then puts all of them on the critical line and excludes $L(\frac{1}{2}, \chi) = 0$. □

REMARK 21.6.2 (The Siegel zero is a threshold crossing). A real zero $\beta$ contributes the atom $u_0 - (\beta - \frac{1}{2})^2$, positive while $u_0 > (\beta - \frac{1}{2})^2$ and crossing zero as the tilt descends to $(\beta - \frac{1}{2})^2$. A Siegel zero near 1 pushes the threshold $u_F^*$ of Corollary 21.2.4(c) towards $\frac{1}{4}$, leaving only tilts above a value near $\frac{1}{4}$.

### 21.6.1 Numerical verification for the characters modulo 4 and 5

We take $\chi_4$ (odd, so $a = 1$) and the real character $\chi_5$ modulo 5, the Legendre symbol (even, so $a = 0$). The numbers are computed by `gl_dirichlet.py`, in ball arithmetic where a count or a sign is certified and otherwise in mpmath, except those of the last paragraph, by `gl_zeta_moments.py` (`code/README.md`). Write $N(T, \chi)$ for the number of zeros of $\xi(s, \chi)$ with $0 < \operatorname{Im} s < T$, $\theta_\chi(T)/\pi = \frac{T}{2\pi} \log \frac{qT}{2\pi e} + \frac{a}{4} - \frac{1}{8}$ for the smooth count and $S(T, \chi) = N(T, \chi) - \theta_\chi(T)/\pi$ for its fluctuation.

*Zeros.* Sign changes of the real function $\xi(\frac{1}{2} + it, \chi)$ on $0 < t < 300$ give the first ordinates 6.0209489 and 6.6484533. The counts, 203 and 214, are certified: the argument principle in ball arithmetic gives $N(300, \chi) = 203$ and 214, equal to the numbers of sign changes, so no zero below 300 was missed and all are simple and on the line. The smooth counts $\theta_\chi(T)/\pi$ at $T = 300$ are 203.15 and 213.56, so $S(300, \chi_4) = -0.153$ and $S(300, \chi_5) = +0.443$.

*No real or central zeros.* On a grid of step 0.005 in $[\frac{1}{2}, 1)$ the smallest values of $L(\sigma, \chi)$ are the central ones, $L(\frac{1}{2}, \chi_4) = 0.66769$ and $L(\frac{1}{2}, \chi_5) = 0.23175$, and ball arithmetic certifies $L(\sigma, \chi) > 0$ on $[\frac{1}{2}, 2]$, with 32 pieces and smallest lower bound 0.033 for $\chi_4$ and 101 pieces and 0.0031 for $\chi_5$. Since $\xi(\sigma, \chi) = \xi(1 - \sigma, \chi)$ and the gamma factor is positive, $\xi(\sigma, \chi) > 0$ on $[0, 1]$, so neither function has a real zero in the critical strip, and neither has a central zero. So both Thorin thresholds are 0, and the untilted clock is available under GRH. For $\chi_4$ the book also certifies the absence of real zeros with a positive kernel (Section 8.5.7.1).

*Power sums.* Let $p_j(\chi) = \sum_{\gamma > 0} \gamma^{-2j}$ be the power sums of the ordinates. The zero side is

$$\sum_{0 < \gamma < X} \gamma^{-2j} + \int_X^\infty t^{-2j} \, \frac{1}{2\pi} \log \frac{qt}{2\pi} \, dt, \qquad X = 300,$$

the sum over the ordinates below $X$ plus the tail started at $X$. The function side is read from the Taylor coefficients of $\log \xi(\frac{1}{2} + s, \chi)$, in which $s^{2j}$ has the coefficient $(-1)^{j+1} p_j / j$; it uses no zeros. Table 21.5 gives the two sides.

With $N(t, \chi) = \theta_\chi(t)/\pi + S(t, \chi)$, integration by parts gives the error of a tail started at the height $X$ as $S(X, \chi) X^{-2j} - 2j \int_X^\infty S(t, \chi) \, t^{-2j-1} \, dt$, up to the difference between the density of the tail and $\theta_\chi'/\pi$, which is of order $X^{-2j-1}$. So the error falls with $j$ like $X^{-2j}$, and it is not of the size $X^{1-2j} \log X$ of the tail

Table 21.5: The power sums $p_j(\chi)$ of the ordinates for the characters modulo 4 and 5: the zero side, the function side, the relative error, and the difference of the two sides times $300^{2j}$.

| $q$ | $j$ | zero side | function side | relative error | difference $\times 300^{2j}$ |
|---|---|---|---|---|---|
| 4 | 1 | 0.0780131 | 0.0780148 | $2.2 \times 10^{-5}$ | $-0.1512$ |
| | 2 | $9.288458 \times 10^{-4}$ | $9.288458 \times 10^{-4}$ | $2.0 \times 10^{-8}$ | $-0.1498$ |
| | 3 | $2.216962 \times 10^{-5}$ | $2.216962 \times 10^{-5}$ | $9.2 \times 10^{-12}$ | $-0.1484$ |
| 5 | 1 | 0.0784620 | 0.0784571 | $6.3 \times 10^{-5}$ | 0.4423 |
| | 2 | $7.178001 \times 10^{-4}$ | $7.178001 \times 10^{-4}$ | $7.6 \times 10^{-8}$ | 0.4414 |
| | 3 | $1.316152 \times 10^{-5}$ | $1.316152 \times 10^{-5}$ | $4.6 \times 10^{-11}$ | 0.4405 |

itself. In Table 21.5 the last column is within 3 percent of $S(300, \chi_4) = -0.1526$ and $S(300, \chi_5) = +0.4431$ for every $j$.

*Transfer function and products.* The transfer function $\mathfrak{T}_\chi(t)$, the ratio $\xi(\frac12 + it, \chi)/\xi(\frac12 + t, \chi)$, has the values $\mathfrak{T}_{\chi_4}(2) = 0.53522$ and $\mathfrak{T}_{\chi_4}(5) = 0.014573$. The Hadamard product $\prod_{\gamma>0}(1 + s^2/\gamma^2)$ at $s = 3$, over the 203 and 214 ordinates with the tail started at 300, is 1.9524380 and 1.9736682, against $\xi(\frac72, \chi)/\xi(\frac12, \chi) =$ 1.9524675 and 1.9735809 for $\chi_4$ and $\chi_5$. The relative differences, $-1.5 \times 10^{-5}$ and $+4.4 \times 10^{-5}$, are the size $S(300, \chi)\, s^2/300^2$ of the fluctuation term of the tail. A match at this level does not by itself show that the product has no exponential factor; that is Lemma 21.2.2, whose argument covers $\xi(s, \chi)$ for a real character, because $\xi(\frac12 + s, \chi)$ is even in $s$ (Proposition 21.5.1).

*Negative moments and the tilt.* The negative moments, computed as the Mellin integral of equation (21.3.14) for $\xi(s, \chi)$, are

$$\mathbb{E}\big[T_{\chi_4}^{-1/2}\big] = 3.7574677\ldots, \qquad \mathbb{E}\big[T_{\chi_5}^{-1/2}\big] = 3.7097072\ldots,$$

smaller than 6.7333 for $\zeta$ because the first zeros are lower. In equation (21.5.3), $(\xi'/\xi)(1, \chi_4) = 0.0777840$ and $(\xi'/\xi)(1, \chi_5) = 0.0782785$. The difference from $p_1(\chi)$ is $2.30835 \times 10^{-4}$ for $\chi_4$ and $1.78632 \times 10^{-4}$ for $\chi_5$; the series of equation (21.3.11) through $j = 12$ agrees to $3 \times 10^{-28}$. Since $\xi(1, \chi_4) = \frac{4}{\pi} \cdot \frac{\pi}{4} = 1$, $\mathbb{E}[e^{-T\chi_4/4}] = \xi(\frac12, \chi_4) = 0.9807136$, and the product over the zeros with the tail gives 0.9807140, a relative difference of $4.2 \times 10^{-7}$.

*The tilt $u_0 = 1$.* For $\zeta$, the product $\prod_\gamma(1 + \gamma^2)/(1 + \gamma^2 + s)$ over the first 400 ordinates with the Riemann von Mangoldt tail started at $\gamma_{400}$ matches $\xi(\frac32)/\xi(\frac12 + \sqrt{1 + s})$ to the tail error: the zero side is 0.9332865 against 0.9332901 at $s = 3$, a relative difference of $3.8 \times 10^{-6}$, and 0.3287603 against 0.3287814 at $s = 50$, a relative difference of $6.4 \times 10^{-5}$. With the anchored tail of equation (21.3.8) the differences are $9.5 \times 10^{-9}$ and $1.6 \times 10^{-7}$.

### 21.6.2 The family: sum and mixture

The cyclotomic field settles one family question. By Proposition 21.5.3, the independent sum of the character variables of conductor dividing $q$ is the Thorin variable of $\mathbb{Q}(\zeta_q)$, a GGC whenever the Thorin property holds there. Averages are another matter. Let $\mathcal{F}_q$ be the real primitive characters modulo $q$ together with one representative of each conjugate pair of complex ones, and let $T^{(q)}$ be the variable of a uniformly chosen member, under GRH for the members and with their central values nonzero, so that their clocks exist (Propositions 21.5.1 and 21.5.2). Its moments are the averages

$$\mathbb{E}\big[(T^{(q)})^k\big] = \frac{1}{|\mathcal{F}_q|}\sum_{\mathcal{F}_q}\mathbb{E}[T_\bullet^k],$$

but the next proposition shows that $T^{(q)}$ is in general not a GGC. Here the Thorin measures of the members are sums of unit point masses, repeated according to multiplicity, so their Laplace transforms $\varphi_i(s) = \prod_t t/(t+s)$ are meromorphic on $\mathbb{C}$ with poles at the points $s = -t$.

Proposition 21.6.3 (Mixtures of clocks). *Let $\varphi_1, \varphi_2$ be Laplace transforms of GGCs with zero drift and Thorin measures $U_1, U_2$ that are sums of unit point masses, repeated according to multiplicity, let $0 < \lambda < 1$, and put $\varphi = \lambda\varphi_1 + (1-\lambda)\varphi_2$.*

*(a) If $\varphi(-x) = 0$ for some $x > 0$ that is not an atom of $U_1$ or $U_2$, then $\varphi$ is not the Laplace transform of a GGC.*

*(b) Let $t_1 < t_2$ be consecutive points of $\operatorname{supp} U_1 \cup \operatorname{supp} U_2$ with $t_1$ an atom of only one of the measures and $t_2$ an atom of only the other. If $U_1 + U_2$ has an odd number of atoms in $(0, t_1]$, counted with multiplicity, then $\varphi$ has a zero in $(-t_2, -t_1)$, so $\varphi$ is not the Laplace transform of a GGC, for any $\lambda$.*

*Proof.* (a) Suppose $\varphi = g$ on $(0,\infty)$ with $g(s) = \exp(-ds - \int \log(1 + s/t)\, U(dt))$. Both are analytic on $\mathbb{C} \setminus (-\infty, 0]$, so they agree there. For $|s| \le x+1$ off the negative axis, $|1 + s/t| \le 1 + (x+1)/t$, so $|g(s)| \ge \exp(-d(x+1) - \int \log(1 + (x+1)/t)\, U(dt)) > 0$, the integral being finite by the conditions on a Thorin measure. Letting $s \to -x$ from the upper half plane, $\varphi(s) \to \varphi(-x) = 0$ because $-x$ is not a pole, a contradiction. (b) For $y > 0$ off the atoms, $\varphi_i(-y) = \prod_t t/(t-y)$ has the sign $(-1)^{N_i(y)}$, with $N_i(y)$ the number of atoms of $U_i$ in $(0, y)$. Say $t_1$ is an atom of $U_1$ and $t_2$ of $U_2$. As $y \downarrow t_1$, $|\varphi_1(-y)| \to \infty$ while $\varphi_2(-y)$ stays bounded, so $\varphi(-y)$ has the sign $(-1)^{n_1}$, where $n_i$ is the number of atoms of $U_i$ in $(0, t_1]$. As $y \uparrow t_2$, $\varphi_2$ dominates and the sign is $(-1)^{n_2}$, since $U_2$ has no atom in $(t_1, t_2)$. These signs differ when $n_1 + n_2$ is odd, and the intermediate value theorem gives the zero; (a) applies. □

Part (b) concerns two clocks. If no atom is common to the two measures, its hypothesis fails only when every maximal run of consecutive atoms of one measure, except the last run, has an even number of atoms (counted with multiplicity), because the numbers of atoms at the changes from one measure to the other are the partial sums of the run lengths; the odd configuration of (b) is the case in which some such partial sum is odd.

Modulo 5 the family has two members, $\chi_5$ and the conjugate pair $\{\psi, \bar{\psi}\}$ with $\psi(2) = i$, so that $\lambda = \frac{1}{2}$ is the uniform average. Under GRH and with $L(\frac{1}{2}, \chi_5)L(\frac{1}{2}, \psi) \neq 0$ the clocks of the two members are the Thorin variables of Propositions 21.5.1 and 21.5.2, the second for the product $\xi_{\psi\bar{\psi}}(s) = \xi(s, \psi)\,\xi(s, \bar{\psi})$; numerically $L(\frac{1}{2}, \chi_5) = 0.23175$ and $L(\frac{1}{2}, \psi) = 0.76375 + 0.21696\,i$, with enclosures that exclude 0 (the part `central` of `gl_mixture_mod5.py`, at 128 bits). In the variable $y = \sqrt{-s}$ the atoms are the ordinates: for the pair 4.1329037, 6.1835782, 8.4572292, 9.4429311 (the absolute values of the ordinates of $L(s, \psi)$), and for $\chi_5$ 6.6484533, 9.8314444. These are all the atoms below 9.84: the argument principle certifies exactly 3 zeros of $\xi(\frac{1}{2} + it, \chi_5)$ and 5 zeros of $L(s, \psi)$, two with positive and three with negative ordinate, below height 12. Three atoms lie in $(0, 6.6484533]$, and the next atom belongs to the pair, so by (b) the mixture vanishes between $y = 6.6484533$ and $y = 8.4572292$ for every $\lambda$; the same happens between 9.4429311 and 9.8314444, with five atoms below.

Numerically, on the negative axis the mixture is $M_\lambda(y) = \lambda A(y) + (1-\lambda)B(y)$, with $A(y) = \xi(\frac{1}{2}, \chi_5)/\xi(\frac{1}{2} + iy, \chi_5)$ and $B(y) = \xi_{\psi\bar{\psi}}(\frac{1}{2})/\xi_{\psi\bar{\psi}}(\frac{1}{2} + iy)$. We computed it in ball arithmetic at 128 bits for $\lambda = 0.01$, 0.1, 0.5, 0.9 and 0.99. In all ten cases the signs at the two ends of the interval are opposite, and each interval contains exactly one zero (the argument principle on a thin rectangle about the interval gives the count); Table 21.6 lists the zeros.

Table 21.6: The zero $y$ of the mixture $M_\lambda = \lambda A + (1-\lambda)B$ modulo 5 in each of the two intervals between a pole of $A$ and a pole of $B$, for five weights; each interval contains exactly one zero.

| $\lambda$ | zero in (6.6484533, 8.4572292) | zero in (9.4429311, 9.8314444) |
|---|---|---|
| 0.01 | 6.64874689 | 9.83141547 |
| 0.1 | 6.65169528 | 9.83112598 |
| 0.5 | 6.67865792 | 9.82859297 |
| 0.9 | 6.98112355 | 9.80692785 |
| 0.99 | 8.03540132 | 9.65957988 |

Proposition 21.6.4 (The family modulo 5). *Under GRH for $L(s, \chi_5)$ and $L(s, \psi)$, for every $\lambda \in (0, 1)$ the function $\lambda\,\xi(\frac{1}{2}, \chi_5)/\xi(\frac{1}{2} + \sqrt{u}, \chi_5) + (1-\lambda)\,\xi_{\psi\bar{\psi}}(\frac{1}{2})/\xi_{\psi\bar{\psi}}(\frac{1}{2} + \sqrt{u})$ of $u \geq 0$ is not the Laplace transform of a GGC. In particular the variable $T^{(5)}$, the case $\lambda = \frac{1}{2}$, is not a GGC.*

*Proof.* The central values are nonzero, as computed, so under GRH the clocks of the two members are the Thorin variables of Propositions 21.5.1 and 21.5.2, whose Thorin measures are sums of unit point masses at the squared ordinates. The certified atoms above put the two members in the odd configuration of Proposition 21.6.3(b). The location of the zeros uses no hypothesis; the hypothesis enters only through the identification of the Thorin measures of the two members. □

### 21.6.3 Landau Siegel zeros and the twisted heat trace

Let $\chi$ be a primitive real character of conductor $q \geq 3$. Siegel's theorem gives $1-\beta \gg_\varepsilon q^{-\varepsilon}$ for a real zero $\beta$ of $L(s,\chi)$, with an ineffective constant, and the classical effective bound is only $1-\beta \gg q^{-1/2}\log^{-2} q$ [104].

Theorem 21.6.5 (Banks, Kerr; unrefereed). *There are absolute constants $q_0$ and $\delta_0 > 0$ such that for every primitive real character $\chi$ of conductor $q \geq q_0$, every real zero $\beta \in (0,1)$ of $L(s,\chi)$ satisfies $(1-\beta)\log q \geq \delta_0$. Banks gives $q_0 = 10^{10}$ and $\delta_0 = 3.0 \times 10^{-3}$.*

The preprints [20, 302] are dated October 2026 and have not been refereed (Section 21.1). Banks describes the proof as an arithmetic comparison of the archimedean and the $p$ adic size of an interpolation determinant in the field $\mathbb{Q}(\sqrt{2}, \sqrt{\chi(-1)q})$, and Kerr gives a simplification; we have not checked either. The half plane $\operatorname{Re} s > \frac{7}{8}$ for every Dirichlet $L$ function reported in Section 21.1 [398], if it stands, would give $1-\beta \geq \frac{1}{8}$ for every real zero and so imply Theorem 21.6.5 for every $q \geq 3$. Banks refers to a proof by OpenAI; whether that is the release of [398] we cannot say.

*The twisted heat trace.* Let $z_\rho = (\rho - \frac{1}{2})^2$, the variable of this chapter (the atom of a zero is $u_0 - z_\rho$, Theorem 21.2.3), so that $z_\rho = -\gamma^2$ for $\rho = \frac{1}{2} + i\gamma$ and $z_\rho = (\beta - \frac{1}{2})^2 > 0$ for a real zero $\beta$. As in Section 5.1, with the zeros counted with multiplicity and $t > 0$ the heat time, the twisted heat trace is

$$W(t;\chi) = \sum_{\operatorname{Im}\rho > 0} e^{t z_\rho} + \tfrac{1}{2} \sum_{\rho \text{ real}} e^{t z_\rho}.$$

This symbol $W(t;\chi)$ is not the root number $W_\chi$ of Section 21.5, which equals 1 for real $\chi$. If $L(\frac{1}{2},\chi) \neq 0$ and $|\gamma| > |\beta - \frac{1}{2}|$ for every zero $\beta + i\gamma$ (so that $\operatorname{Re}(-z_\rho) > 0$ and the integral converges), then for $u \geq 0$

$$\log \frac{\xi(\frac{1}{2} + \sqrt{u}, \chi)}{\xi(\frac{1}{2}, \chi)} = \int_0^\infty (1 - e^{-ut}) \, \frac{W(t;\chi)}{t} \, dt,$$

and the ratio $\xi(\frac{1}{2},\chi)/\xi(\frac{1}{2} + \sqrt{u},\chi)$ is the Laplace transform of a GGC if and only if $W(\cdot\,;\chi)$ is completely monotone, which as in Section 5.1 is the ladder

$(-1)^k W^{(k)}(t;\chi) > 0$. The Gaussian explicit formula, with $a \in \{0,1\}$ the parity of $\chi$ and $\psi$ the digamma function, reads

$$W(t;\chi) = \frac{\log(q/\pi)}{4\sqrt{\pi t}} + \frac{1}{4\pi}\int_{-\infty}^{\infty} e^{-tr^2}\,\mathrm{Re}\,\psi\Big(\frac{1+2a}{4}+\frac{ir}{2}\Big)dr \\ - \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)\chi(n)}{\sqrt{n}}\,e^{-(\log n)^2/4t}. \tag{21.6.1}$$

Compared with the formula for $\xi$ in equation (1.2.5) the polar term $e^{t/4}$ is absent, because $L(s,\chi)$ is entire, $-\log\pi$ in $m(r)$ becomes $-\log(\pi/q)$, and the prime sum carries the factor $\chi(n)$.

*A real zero as a failure of rung one.* A real zero $\beta \in (\frac12, 1)$ and its mirror $1-\beta$ contribute together the single term $e^{ct}$ with $c = (\beta - \frac12)^2 = \frac14 - \beta(1-\beta)$, which grows with $t$. This is the defect that a complex off line pair cannot model in Theorem 5.1.14, whose defect needs a nonzero imaginary part.

Proposition 21.6.6. *Let $\beta \in (\frac12, 1)$ be the largest real zero of $L(s,\chi)$, suppose every nonreal zero of $L(s,\chi)$ lies on the critical line, and put $c = (\beta - \frac12)^2$. Then $W(t;\chi) \to \infty$ and $-W'(t;\chi) < 0$ for all $t > t_1 = \max\{1, c^{-1}\log(C'\log q/c)\}$, where $C'$ is an absolute constant. If moreover $q \ge q_0$, put $\eta = \delta_0/\log q$; then $c \le \frac14 - \eta(1-\eta)$, so by Theorem 21.6.5 the growth rate $c$ of such a term is at least $\eta(1-\eta)$ below the rate $\frac14$ of the pole term $e^{t/4}$ of $\zeta$.*

*Proof.* Under the hypotheses,

$$W(t;\chi) = \sum_{\gamma>0} e^{-t\gamma^2} + me^{ct} + (\text{terms } e^{c't} \text{ from the other real zeros, } c' \le c),$$

and $-W'(t;\chi) \le \sum_{\gamma>0}\gamma^2 e^{-t\gamma^2} - ce^{ct}$. By the bound $N(T,\chi) \ll (T+1)\log(q(T+2))$ for the zero counting function, the sum is at most $C'\log q$ at $t=1$ and decreases in $t$, so $-W' < 0$ as soon as $ce^{ct} > C'\log q$, which holds for $t \ge t_1$; and $W \ge e^{ct} \to \infty$. For the last statement, the identity $c = \frac14 - \beta(1-\beta)$ and $\beta \le 1-\eta$ (Theorem 21.6.5), with $\beta(1-\beta)$ decreasing on $(\frac12,1)$, give $\beta(1-\beta) \ge \eta(1-\eta)$. □

Thus a real zero with $\beta > 1-\eta$, which Theorem 21.6.5 excludes for $q \ge q_0$, would, under the hypotheses of Proposition 21.6.6, be seen on rung one at the scale $t_1$. The difficulty is uniformity in $q$: for a given $\chi$ of small conductor the verification of GRH by Platt [417] covers a real zero, and for general $q$ equation (21.6.1) is the only handle. For complex pairs the book's detection of a pair at height $a$ is Proposition 5.1.10, and nothing in Theorem 21.6.5 touches complex zeros.

*The pole lock, a heuristic.* The prime sum of equation (21.6.1) suggests why a zero near 1 is tied to the primes with $\chi(p) = -1$. Suppose, as a model and not as a claim that such a character exists, that $\chi(p) = -1$ for every prime $p$. The prime term of equation (21.6.1) is then

$$\frac{1}{2\sqrt{\pi t}} \sum_{p} \frac{\log p}{\sqrt{p}}\, e^{-(\log p)^2/4t} + O(1) \;\approx\; \frac{1}{2\sqrt{\pi t}} \int_0^{\infty} e^{u/2 - u^2/4t}\, du = e^{t/4}\, \Phi\big(\sqrt{t/2}\big),$$

where $\Phi$ is the standard normal distribution function and the $O(1)$, which comes from the prime powers (mainly the squares, where $\chi(p^2) = 1$) and tends to $-\frac{1}{4}$ as $t \to \infty$, is not negligible for $t \le 3$. As $t \to \infty$ this is $e^{t/4}(1 + o(1))$: on the prime side, such a character produces the growth that the pole of $\zeta$ produces in equation (1.2.5). A real zero $\beta$ is the same picture with the rate reduced from $\frac{1}{4}$ to $(\beta - \frac{1}{2})^2$, and the reduction $\beta(1-\beta)$ is what Theorem 21.6.5 bounds below. Numerically (the script `gl_pole_lock.py`, primes up to $6 \times 10^6$) the ratio of the prime sum to $e^{t/4}$ is 0.402, 0.510, 0.645, 0.735 This is a heuristic about a hypothetical character.

*What does not transfer.* The proof of Theorem 21.6.5 as Banks describes it is not a positivity statement and not a complete monotonicity statement, and it concerns the single point $s = 1$ through the pole of $\zeta_K(s) = \zeta(s)L(s,\chi)$ of the quadratic field $K = \mathbb{Q}(\sqrt{\chi(-1)q})$ of $\chi$. The Thorin measure is a statement about the whole zero set in the variable $z_\rho$, and we see no way to recast the determinant argument as a rung of the ladder.

## 21.7 Consequences for the Linnik constant and for RH

### 21.7.1 The Linnik constant

Corollary 21.7.1. *Suppose that for every $q \ge 3$ the function $\varphi_{\mathbb{Q}(\zeta_q),0}$ is the Laplace transform of a GGC. Then every sufficiently large even integer is a sum of two primes and at most seven powers of two.*

*Proof.* Theorem 21.6.1 gives GRH for all Dirichlet $L$ functions, and under GRH every sufficiently large even integer is a sum of two primes and at most seven powers of two, by the theorem of Heath-Brown and Puchta [179] (see also Pintz and Ruzsa [413]). □

The corollary is weaker than the known conditional result. By Corollary 21.2.4(a) and equation (21.5.4), its hypothesis is GRH for every Dirichlet $L$ function together with $L(\frac{1}{2}, \chi) \neq 0$ for every primitive character $\chi$. It contains GRH and adds a statement that is not known to follow from it, because the nonvanishing at the centre is open (for real characters it is Chowla's conjecture), whereas $K = 7$ already follows from GRH alone. A direct proof of the

Thorin property for all cyclotomic fields would prove GRH, and with it $K = 7$ unconditionally, one better than the record $K = 8$ of Pintz and Ruzsa [414].

For $K$ itself the Thorin property gives what GRH gives. Nothing in the proof of Corollary 21.7.1 uses the nonvanishing at the centre, and the inputs that decide $K$ below seven, the large deviation estimate for the sum $\mathcal{D}(\alpha)$ over the powers of two (step (4) of Section 21.1.2), the sieve constant for prime differences (step (2)) and the refinements of Pintz and Ruzsa, are additive questions about the powers of two and not statements about zeros. With the present methods the framework therefore does not reach below seven.

A weaker hypothesis may suffice for seven. The Thorin property at tilt $u_0 = 1$ for cyclotomic fields (Corollary 21.2.4(b) with equation (21.5.4)) is GRH with possible real zeros. Where could a real zero hurt? Write $\mathcal{S}(a/q+\eta)$ as a combination of twisted prime sums over the characters modulo $q$. A real zero $\beta$ of $L(s,\chi^*)$, with $\chi^*$ of conductor $q^* \mid q$, contributes a term of size about $q^{1/2}\varphi(q)^{-1}N^{\beta}(1+|\eta|N)^{-1}$, which exceeds $N^{3/4}$ only if $q < N^{2\beta-3/2}$ and $|\eta| < N^{\beta-7/4}$. Those arcs lie close to the major arcs, and moving them there, where a real zero changes the main term by a factor of the shape $1-\chi^*(\cdot)N^{\beta-1}$ that Gallagher's method [144] already handles for one exceptional zero, would reduce the question to major arc estimates with real zeros (Problem 21.2). If it works, real zeros, and with them Siegel zeros, become irrelevant to $K = 7$.

### 21.7.2 What would help towards RH

The three smoothings of the centre clock are compared in Chapter 19.

A proof has to establish a closed positivity condition at the boundary, from input that sees the arithmetic beyond positivity. Zero free regions of the kind announced in October 2026 [398] would be such input, if confirmed, and each one would move the unconditional end of the basepoint family inwards (Section 21.1.2); averages over families of $L$ functions, where the low lying zeros are conjectured to obey the symmetry laws of Katz and Sarnak [300], are another (Problem 21.4). Section 19.4 says what would get past each of the three obstructions of Chapter 19.

## 21.8 Open problems

Problem 21.1 (The Thorin property without the zeros). Establish condition (i) of Theorem 21.2.3 for $\mathbb{Q}$, or for cyclotomic fields, without first knowing (ii). In the master theorem route, settle the parabolic distortion of Section 21.4.3, uniformly in the field, and find a substitute for the residue bookkeeping at the zeros. *Status.* Open. For $\mathbb{Q}$ at $u_0 = 0$ the statement to be proved is RH, so a growth bound that is itself proved under RH, such as Lemma 21.4.1, cannot be an input to its proof.

PROBLEM 21.2 (Seven with real zeros). Show that $K = 7$ follows from the Thorin property at tilt $u_0 = 1$ for cyclotomic fields, that is from GRH with possible real zeros. Section 21.7 locates the arcs where a real zero can exceed the bound $N^{3/4}$; what remains is to move them into the major arcs and to redo the main term with several real zeros, checking that the constant 2.7895 survives. In the paper of Heath-Brown and Puchta [179] the major arc estimate that gives this constant (their Lemma 7) is stated for $K \geq 9$, with zero density estimates in place of GRH, and the paper does not say how it is used for $K = 7$; the computation of the constant itself holds for $K \geq 7$. *Status.* Open. Only the location of the arcs is done, and it is a heuristic estimate.

PROBLEM 21.3 (The moment identity at scale). Anchoring the tail to the exact zero count, equation (21.3.8), already brings the check of $p_1$ from $2.6\times10^{-6}$ to $6.3\times10^{-9}$ with 799 zeros. Carry it to large height with Odlyzko's tables [396], where the remaining term of (21.3.8), the integral of $S(t)\,t^{-2j-1}$ over the tail, sets the limit and calls for an explicit formula for $\int S$, and to $L(s,\chi)$ with the zeros of Platt [417]. *Status.* Partly done. The script `gl_zeta_moments.py` (`code/README.md`) reproduces the anchoring with the first 1700 ordinates at the cuts $N = 200$ and $N = 799$ and at every cut $N = 50, \dots, 1699$, at 100 digits (`data/gl_zeta_moments.json`). For $p_1$ the anchored tail typically gains about a third of the height of the cut over the smooth tail (the first and ninth deciles of the gain are 42 and 2700), but it gains nothing where $|S|$ is small at the cut: the gain is below a factor 10 at 3.2% of the cuts. Large heights and the characters are open.

PROBLEM 21.4 (The negative moment as a family statistic). The value $\mathbb{E}[T^{-1/2}] = 6.7333$ (and 3.7575, 3.7097 for $T_{\chi_4}$, $T_{\chi_5}$) is computed from $\xi$ alone: $\mathbb{E}[T^{-1/2}] = \Gamma(\frac12)^{-1}\int_0^\infty u^{-1/2}\prod_\gamma(1+u/\gamma^2)^{-1}\,du$, the product being over the ordinates $\gamma > 0$. It is mostly a smooth statistic. Let $E_m$ be this integral with the first $m$ ordinates exact and the factors of the others replaced by the smooth model $\exp\big(-\frac{1}{\pi}\int_{t_m}^\infty \log(1+u/t^2)\,\vartheta'(t)\,dt - a_m\log(1+u/t_m^2)\big)$, where $\vartheta$ is the Riemann Siegel theta function, $t_m$ is the midpoint of $\gamma_m$ and $\gamma_{m+1}$, and $a_m = N_0(t_m) + 1 - m$, with $N_0 = \vartheta/\pi$, is the mass of the atom at $t_m$ that the anchored tail (21.3.8) adds. For $m = 0$ the cut is $t_0 = 9.6669$, the point where $N_0 + 1 = 0$, and there is no atom. The relative error $E_m/\mathbb{E}[T^{-1/2}] - 1$ is $-7.0\times10^{-3}$ for $m = 0$, $-2.5\times10^{-3}$ for $m = 1$, $1.0\times10^{-4}$ for $m = 2$, $-3.9\times10^{-4}$ for $m = 5$ and $1.4\times10^{-5}$ for $m = 20$. The atom matters: cutting at the midpoint without it gives $-5.5\times10^{-3}$, $-1.2\times10^{-3}$, $-1.4\times10^{-3}$ and $-6.5\times10^{-4}$ for $m = 1$, 2, 5, 20, and cutting at the point where $N_0 + 1 = m$ without it gives $-2.5\times10^{-3}$, $9.1\times10^{-5}$, $-4.0\times10^{-4}$ and $1.0\times10^{-5}$. The elasticity $(\gamma_k/E)\,\partial E/\partial\gamma_k$ of $E = \mathbb{E}[T^{-1/2}]$ with respect to one ordinate, the others fixed in the integral above, is 0.1799 for $\gamma_1$ and falls to 0.0319 for $\gamma_6$ (over all ordinates the elasticities sum to 1, by scaling). A single function therefore says little, and the natural comparison with random matrices is in families. Compute the average of $\mathbb{E}[T_\chi^{-1/2}]$ over real primitive characters of conductor in $[Q, 2Q]$ and

compare it with the prediction of the symplectic ensemble. Since $\mathbb{E}[T^{-1/2}]$ is not a linear statistic of the zeros, the one level density of Katz and Sarnak [300] alone does not give this prediction, and all the correlation functions enter. The circular unitary comparisons of Section 11.4.5 and the moments of Keating and Snaith [301] (Section 11.4.6.1) are the models for a single function. *Status.* Partly done: the single function is computed, with the model above, by `gl_density_pf.py`; the family average is open.

Problem 21.5 (The family mixture). Let $T^{(q)}$ be the variable of a uniformly chosen member of $\mathcal{F}_q$ (Section 21.6.2, under GRH for the members and with their central values nonzero). Is $T^{(q)}$ infinitely divisible, and does any $q$ with two or more members avoid the odd configuration of Proposition 21.6.3(b), which is stated for two clocks, for every pair of its members? *Status.* Open. That $T^{(q)}$ is not a GGC is settled for $q = 5$ and every weight, under GRH for the two members (their values at $\frac{1}{2}$ are nonzero numerically; Proposition 21.6.4), and holds, under the same hypotheses, for every $q$ whose family has exactly two members, when these are in the odd configuration (Proposition 21.6.3(b)). It does not decide whether $T^{(5)}$ is infinitely divisible.

Problem 21.6 (Sign definite explicit formulae). Under the hypothesis of Corollary 21.7.1 the explicit formula for $\psi(x; q, 1)$, which up to the factor $\varphi(q)$ counts the prime ideals of $\mathbb{Q}(\zeta_q)$, becomes an integral against the nonnegative measure $U_{F_q,0}$, and Theorem 21.5.8 supplies the positive family $\pi^{-1}\operatorname{Re}(\xi'_{F_q}/\xi_{F_q})(\alpha + ib)$ joining it to the primes. For $a \not\equiv 1 \pmod q$ the characters enter the formula for $\psi(x; q, a)$ with the weights $\bar\chi(a)$, which are complex or negative, and no sign is available. These are zero counting measures and their Poisson smoothings, so any gain must come from structure beyond counting, such as monotonicity or log concavity of the associated GGC densities [56, Ch. 4], transported to bounds on $L'/L$ near the line. Is there a sign definite explicit formula for $\psi(x; q, a)$ with $a \not\equiv 1 \pmod q$, in terms of such measures? *Status.* Open.

Problem 21.7 (Certified backward continuation). The Thorin density $\pi^{-1}\operatorname{Re}(f'/f)(\alpha + ib)$ of Theorem 21.5.8, which is $\rho_{\alpha-1/2}(b)$ for $f = \xi$, is computable from the primes, with rigorous error bounds, for $\alpha > 1$, and its positivity on every line $\operatorname{Re} s = \alpha'$ with $\alpha' > \alpha$ is a zero free half plane $\operatorname{Re} s > \alpha$. The book certifies the minimum at the origin for $\xi$, $\chi_4$ and $\chi_{12}$ for $\alpha < 1$ up to a large height, from the zeros (Theorem 9.1.18). How far into the strip can positivity be certified from the primes alone, using smoothed prime sums of the kind that located the zeros of $L(s, \chi_4)$ in Section 21.5.6, before the ill posedness of the backward step takes over? Chapter 19 measures the cost of the step: by Theorem 19.2.2 it is a deconvolution, which is unbounded for the basepoint family, and Proposition 19.2.3 counts the primes it needs, up to about $\eta^{-1/(\alpha-\sigma_{\max})}$ for accuracy $\eta$; for the heat flow the corresponding step is stable but cannot carry real zeros (Section 19.4). Section 20.1.1 of Chapter 20 reads the same step as

a deconvolution of the explicit formula. *Status.* Partly done. For $\chi_4$ the prime side (21.5.9) of $\pi$ times the density at $b = 0$, summed over the prime powers up to $2 \times 10^6$ without smoothing, differs from $(\xi'/\xi)(\alpha, \chi_4)$ by $1.3 \times 10^{-4}$, $4.8 \times 10^{-3}$, $4.1 \times 10^{-2}$ and $8.3 \times 10^{-2}$ at $\alpha = 1$, $\frac{3}{4}$, $\frac{3}{5}$ and $\frac{11}{20}$, that is by 0.2%, 12%, 260% and 1100% of the value, so that for $\alpha \leq 1$ the minima computed in Section 21.5.6 come from the zeros and not from the primes. What smoothed prime sums can certify below $\alpha = 1$ is open.

Problem 21.8 (The arithmetic Lévy measure and the minor arcs). For $F = \mathbb{Q}$ the atoms of $\nu_F$ in (21.5.6) at the primes $p$ have mass $\log p$, the weights that enter $\mathcal{S}(\alpha)$; the prime powers of (21.5.6) do not enter $\mathcal{S}(\alpha)$. Can $\mathcal{S}(\alpha)$ on the minor arcs be written as an expectation against the GGC laws above? Such a representation would be a different route to the sup bound in step (3); nothing in the chapter shows that it exists. *Status.* Open.

Problem 21.9 (Zero free half planes in the circle method). Assume $L(s, \chi) \neq 0$ for $\operatorname{Re} s > \theta_0$ and every Dirichlet character. What is the best minor arc exponent $\theta(\theta_0)$ obtainable by combining the explicit formula with Vaughan's identity? The one line balance of Section 21.1.2 gives $\frac{1}{2}(1+\theta_0)$, which is better than the unconditional exponent $\frac{263}{308}$ of step (3) exactly when $\theta_0 < \frac{109}{154}$, so it does nothing at $\theta_0 = \frac{7}{8}$; it is not shown to be optimal. Does a hybrid bound, using the half plane only for the characters of small conductor, improve $\lambda$ in step (4)? And does the half plane, which excludes exceptional zeros of large conductor, raise the constant 2.7895 of step (1)? The paper of Heath-Brown and Puchta [179] suggests that it does not by itself: they state that zeros close to $\sigma = 1$ play no role and that no explicit numerical zero free region is needed, the constant is a sum over the divisors $d \leq 5$ (the improvement of Elsholtz to 2.96169 that they report takes $d \leq 21$ and a better sieve constant), and the restriction $q \leq P = N^{45/154 - 4\varpi}$ of their major arcs comes from zero density estimates, whose critical zeros have $\sigma \leq \frac{3}{4}$ and are not removed by a half plane at $\frac{7}{8}$. That restriction is not sharp for those estimates: the exponent $(\frac{45}{77}\kappa(\sigma) - 1)(1 - \sigma)$ of their Section 3 has its maximum 0.1050 at $\sigma = 0.676$, against the bound $\frac{31}{154} = 0.2013$ that they use. *Status.* Open.

Problem 21.10 (Finite order total positivity). RH holds if and only if the density of $Y = \sqrt{2T}\, Z$ is a Pólya frequency function of infinite order (Proposition 16.5.1, where the variable is called $X$; Section 21.7.2). For this one density the question whether a finite order already implies RH has a trivial answer. The orders decrease, so if RH holds every such implication is true, and if RH fails the property fails at some finite order, where the implication is vacuous. The question has content for a class of functions. Let $\mathcal{E}$ be the class of even entire functions $F$ with $F(0) = 1$ and the order, type and reflection symmetry of $\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ for which $1/F$ is the characteristic function of a probability density $f_F$; the density of $Y$ is $f_F$ for $F = \xi(\frac{1}{2} + \cdot)/\xi(\frac{1}{2})$ (Proposition 16.5.1). Is there a finite $k$ such that, for every $F \in \mathcal{E}$, the property that $f_F$ is a Pólya frequency function of order $k$ implies that

every zero of $F$ lies on the imaginary axis? *Status.* Open. Chapter 16 shows that the low orders are decided by the low zeros: order two holds without RH (Theorem 16.11.5), and the density lies within $1.6 \times 10^{-10}$ of a Pólya frequency function of every order (Theorem 16.11.6), so that, given the verification of RH to height $3 \times 10^{12}$ [419], every minor of order 3 and 4 is at least $-1.1 \times 10^{-8}$ and $-1.2 \times 10^{-7}$. Problem 16.1 asks for proofs of the low orders, not for the converse asked here. Section 21.3.3.5 reports the computations of this chapter. The perturbation of Theorem 5.2.1, made at a height that is large compared with $k$, is the natural candidate for a negative answer in the class $\mathcal{E}$; whether it keeps the order $k$ has not been computed here.

## Notes

The Thorin representation and the van Dantzig pair are from [421, 423]; the GGC background is Bondesson's monograph [56]; the master theorem class is from Chaudhry and Qadir [78], and the exponential Tauberian theorem from Kasahara [299]. Completed Dirichlet $L$ functions and Dedekind zeta functions are in Davenport [104], mean values of $\zeta$ in Titchmarsh [498]. The account of the circle method in Section 21.1.2 is a summary of the paper of Heath-Brown and Puchta [179].

# Chapter 22

# Friable approximants of $\Xi$ after Miyatake and the cosh ladder

Keep in Riemann's theta series only the integers composed of the primes up to $P$, and the result is a function $\Xi_P$ that differs from $\Xi$ by an exponentially small amount and yet has infinitely many zeros off the real line. This chapter studies that function, the friable approximant, as a test case for any computational criterion for the Riemann Hypothesis. It asks how many real zeros $\Xi_P$ shares with $\Xi$, why no finite set of primes reaches beyond a height of about $4q^2$, where $q$ is the first prime left out (a heuristic formula, checked by computation), where the remaining zeros go, and what the $2\sin^2$ lemma and the Thorin condition can and cannot detect. The last sections read an identity of Miyatake, who worked with the primes up to $P$ inside Pólya's symmetrised kernel, probabilistically (Lévy's area, a Pólya Gamma clock, a logistic partition of unity), and place the bottom of the cosh ladder against the quasi Riemann abscissa.

Riemann's $\xi$ function is $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, and $\Xi(t) = \xi(\frac{1}{2}+it)$ is even, entire and real on $\mathbb{R}$ (Chapter 1, Section 1.2); RH says that all zeros of $\Xi$ are real. Miyatake [217, 216] approximated $\xi$ using only the integers composed of primes $p \le P$, so that the Dirichlet series becomes the finite Euler product $\zeta_P(s) = \prod_{p\le P}(1-p^{-s})^{-1} = \sum_{n\in\mathcal{F}(P)} n^{-s}$, with $\mathcal{F}(P)$ the set of $P$ friable integers. He worked inside Pólya's symmetrised kernel rather than Riemann's (Section 22.7). Here Riemann's own kernel is truncated instead, and the friable approximant $\Xi_P$ of Section 22.1 is the object of study. The incomplete gamma form of truncated theta series, which is the form used for all computations below, is also the subject of Haglund [168].

The chapter is organised around one function, the deficit $D_P = \Xi_P - \Xi$, and one geometric object, the negative hump of $\Xi$. Every number printed below was computed by the scripts listed for this chapter in the code archive (file code/README.md), in floating point at the precision stated unless a certified method is named. The results are these.

1. $|\Xi_P - \Xi| \le |t^2 + \frac{1}{4}|\,\delta_P$ in $|\operatorname{Im} t| \le \frac{3}{2}$ with $\delta_P \approx e^{-\pi q^2}/(\pi q^2)$, and $|\Xi_P - \Xi| \le$

$(1+\delta(a_q))c_P$ on $\mathbb{R}$ with $c_P \approx 4\pi q^2 e^{-\pi q^2}$: the first omitted prime sets the accuracy (Theorems 22.2.1 and 22.3.4, proved).

2. The truncated kernel has a kink of exact size $c_P > 0$, so $\Xi_P \to c_P$ on $\mathbb{R}$; no reweighting of the theta series with $\sum_n |w_n| n^{-\sigma} < \infty$ for every $\sigma > 0$ (this includes the friable sets) has infinitely many real zeros, and the same holds for any set of omitted integers (Theorems 22.2.3 and 22.2.4, proved).

3. $D_P > 0$ on $\mathbb{R}$ and is, with an explicit uniform error, a sum of Lorentzians (Theorems 22.3.1 and 22.3.4, proved).

4. Real zeros of $\Xi_P$ live in negative humps of $\Xi$, two per surviving hump, and are lost in pairs by collision; the count is twice the number of humps deeper than $D_P$, at every height below $3 \cdot 10^{12}$ where the humps are short enough (Theorems 22.4.1 and 22.4.2, proved given the verification of RH to that height [419]). For $P = 7$ the computed count is 264.

5. $\Xi_P$ has more zeros than $\Xi$: heuristically $N_P(X) = N(X) + y_P(X)/4 + O(1)$, compared with argument principle counts for $P = 2, 3$, which fall below it by 4.86 at $X = 600$ for $P = 2$ (Remark 22.6.1, heuristic and computed).

6. The $2\sin^2$ Lévy density of $\Xi_2$ agrees with that of $\Xi$ to seven digits for $x \le 0.1$ (computed), and on the computed range it is positive although $\Xi_2$ has zeros off the line; the support of the Thorin measure can (Section 22.9).

7. Multiplying Riemann's kernel by $\cosh(bt)$ gives, unconditionally for $b \ge \frac{1}{2}$ (Lagarias, Theorem 16.6.1), reciprocals that are Laplace transforms of generalised gamma convolutions, and RH is equivalent to this persisting as $b \downarrow 0$ (Section 22.10).

8. Dividing instead by $\cosh(bt)$, as Miyatake does with $b = \frac{9}{2}$, is smoothing by Lévy's stochastic area, equivalently the de Bruijn Newman flow at a Pólya Gamma random negative time; at $b = 2$ the transform changes character, and for $b > 2$ with $C_b \neq 0$ it is never in the Laguerre Pólya class (Sections 22.11 and 22.12).

9. The bottom of the cosh ladder is bounded by the quasi Riemann abscissa, and in paired form (real and imaginary parts of $\xi$ on vertical lines, a Thorin pair) it lies exactly at it (Section 22.14).

## 22.1 Riemann's representation, the friable approximant and the kink

### 22.1.1 Riemann's representation and the kink identity

Let $\psi(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and $\Theta_3 = 1 + 2\psi$, so that $\Theta_3(1/x) = \sqrt{x}\,\Theta_3(x)$. Riemann's formula

$$\xi(s) = \frac{1}{2} + \frac{s(s-1)}{2}\int_1^\infty \psi(x)\big(x^{s/2-1} + x^{-(1+s)/2}\big)\,dx$$

becomes, with $s = \frac{1}{2} + it$ and $x = e^{2u}$,

$$\begin{aligned}\Xi(t) &= \frac{1}{2} - 2\left(t^2 + \tfrac{1}{4}\right)\int_0^\infty \psi(e^{2u})e^{u/2}\cos(tu)\,du \\ &= \left(t^2 + \tfrac{1}{4}\right)\int_0^\infty k(u)\cos(tu)\,du, \end{aligned} \tag{22.1.1}$$

where $k(u) = e^{-u/2} - 2e^{u/2}\psi(e^{2u})$; the pole term $\frac{1}{2}$ has been absorbed using

$$\int_0^\infty e^{-u/2}\cos(tu)\,du = \frac{1/2}{t^2 + \frac{1}{4}}.$$

Since $k(u) = 2\cosh(u/2) - e^{u/2}\Theta_3(e^{2u})$ and the theta functional equation makes $e^{u/2}\Theta_3(e^{2u})$ even, $k$ extends to an even real analytic function on $\mathbb{R}$. This folded kernel is the new ingredient; the classical kernel is the one of Equation (1.2.3), $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(tu)\,du$ with $\Phi$ positive, even and analytic exactly in the strip $|\operatorname{Im} u| < \pi/4$ (Titchmarsh [498, (2.16.1)]). The two are related by

$$\Phi(u) = 2\sum_{n\geq 1}\big(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\big)e^{-\pi n^2 e^{2u}} = \tfrac{1}{2}\big(\tfrac{1}{4}k(u) - k''(u)\big), \qquad u \geq 0,$$

which follows from two integrations by parts in (22.1.1): since $k'(0) = 0$ (Lemma 22.1.1 below), $\int_0^\infty (\frac{1}{4}k - k'')\cos(tu)\,du = (t^2 + \frac{1}{4})\int_0^\infty k\cos(tu)\,du = \Xi(t)$, and the terms $e^{-u/2}$ cancel in $\frac{1}{4}k - k''$.

Lemma 22.1.1 (Kink identity). $\frac{1}{2} + \psi(1) + 4\psi'(1) = 0$; *equivalently* $k'(0) = 0$.

*Proof.* Differentiate $\Theta_3(1/x) = \sqrt{x}\,\Theta_3(x)$ at $x = 1$ to get $-\Theta_3'(1) = \frac{1}{2}\Theta_3(1) + \Theta_3'(1)$, that is $\Theta_3(1) + 4\Theta_3'(1) = 0$, and insert $\Theta_3 = 1 + 2\psi$. Then $k'(0) = -\frac{1}{2} - \psi(1) - 4\psi'(1) = 0$. □

The folded pole part $e^{-|u|/2}$ has a corner at the origin and the theta part has exactly the opposite corner. Their cancellation is the smoothness of $k$ at 0, equivalent to the evenness of the analytic continuation of $\Phi$; it is a consequence of the functional equation, and Theorem 22.2.4 shows that no reweighting of the theta series with $\sum_n |w_n| n^{-\sigma} < \infty$ for every $\sigma > 0$ keeps it. In terms of the maximum $M$ of a standard Brownian excursion, $k(u) = e^{-u/2}\,\mathbb{E}\min\big(\sqrt{2/\pi}\,M, e^u\big)$ for real $u$, and Lemma 22.1.1 says $\mathbb{P}(M > \sqrt{\pi/2}) = \frac{1}{2} - \psi(1)$ (Remark 24.13.2).

### 22.1.2 The friable approximant

Definition 22.1.2. For a prime $P$ let $\psi_P(x) = \sum_{n\in\mathcal{F}(P)} e^{-\pi n^2 x}$ and

$$\Xi_P(t) = \frac{1}{2} - 2\left(t^2+\tfrac{1}{4}\right)\int_0^\infty \psi_P(e^{2u})e^{u/2}\cos(tu)\,du, \qquad \xi_P(\tfrac{1}{2}+it) = \Xi_P(t).$$

Let $q = q(P)$ be the least prime above $P$.

$\Xi_P$ is even, entire of order one, real on $\mathbb{R}$, and $\xi_P(s) = \xi_P(1-s)$ by construction. The Mellin transform $\int_0^\infty \psi_P(x)x^{s/2-1}dx$ of $\psi_P$, taken at $s/2$, is $\pi^{-s/2}\Gamma(s/2)\zeta_P(s)$ for $\operatorname{Re} s > 0$, which is the sense in which $\Xi_P$ is the Euler product truncation of Riemann's exact kernel.

Proposition 22.1.3 (Incomplete gamma form).

$$\xi_P(s) = \frac{1}{2} + \frac{s(s-1)}{2}\sum_{n\in\mathcal{F}(P)}\Big[(\pi n^2)^{-s/2}\,\Gamma\big(\tfrac{s}{2},\pi n^2\big) + (\pi n^2)^{-(1-s)/2}\,\Gamma\big(\tfrac{1-s}{2},\pi n^2\big)\Big].$$

*Proof.* With $x = e^{2u}$, Definition 22.1.2 is Riemann's formula with $\psi_P$ in place of $\psi$. The series converges absolutely and uniformly for $x \geq 1$, so it can be integrated term by term, and $\int_1^\infty e^{-\pi n^2 x}x^{z-1}dx = (\pi n^2)^{-z}\Gamma(z,\pi n^2)$ for $z = s/2$ and, since $x^{-(1+s)/2} = x^{(1-s)/2-1}$, for $z = (1-s)/2$. □

All computations in this chapter use this form; with all integers it reproduces $\xi(\frac{1}{2}+it)$ to working precision (computed: relative difference below $10^{-26}$ for $t \leq 50$ at 40 digits).

#### *22.1.2.1 The literal Euler product and its poles*

One could instead symmetrise the finite Euler product directly, setting $F_P(s) = E_P(s) + E_P(1-s)$ with $E_P(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta_P(s)$ and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. Since $\zeta_P$ has poles at $s = 2\pi ik/\log p$, $F_P$ has poles on $\operatorname{Re} s = 0$ and $\operatorname{Re} s = 1$.

Proposition 22.1.4. *For $0 < \operatorname{Re} s < 1$, $\xi_P(s) = \frac{1}{2} + F_P(s) - \frac{s(s-1)}{2}[\mathcal{I}_P(s) + \mathcal{I}_P(1-s)]$ with $\mathcal{I}_P(s) = \sum_{n\in\mathcal{F}(P)}(\pi n^2)^{-s/2}\gamma(\frac{s}{2},\pi n^2)$. The poles of $F_P$ cancel against those of $\mathcal{I}_P$, so the theta form is the pole free regularisation of the symmetrised Euler product.*

*Proof.* For $0 < \operatorname{Re} s < 1$ both $\operatorname{Re} s > 0$ and $\operatorname{Re}(1-s) > 0$, so the Dirichlet series $\sum_{n\in\mathcal{F}(P)}(\pi n^2)^{-s/2}\Gamma(\frac{s}{2}) = \Gamma_{\mathbb{R}}(s)\zeta_P(s)$ and its reflection converge absolutely, and $\Gamma = \gamma + \Gamma(\cdot,\pi n^2)$ termwise in Proposition 22.1.3 gives the identity. The statement about poles concerns the meromorphic continuations: $\xi_P$ is entire, $F_P$ has the poles above, and on $\operatorname{Re} s = 0$ and $\operatorname{Re} s = 1$ the function $\mathcal{I}_P$ is its continuation, not the series, which diverges there. □

The unregularised $F_P$ pays for its poles with spurious zeros near them. For $P = 2$, near the poles $1+2\pi ik/\log 2$, the computed zeros of $F_2$ include $1.5151+9.1951i$, $1.3184 + 18.0602i$, $1.2268 + 27.3286i$, $1.1815 + 45.4005i$, $0.7885 + 72.7840i$ (inside the critical strip, with its mirror at 0.2115) and $1.1200 + 108.8198i$ (computed, residual below $10^{-26}$; this is a selection and not a list of all the zeros of $F_2$).

Proposition 22.1.5. *Let $P \geq 2$ and let $p \leq P$ be prime. For infinitely many integers $k$, $F_P$ has a zero $s_k$ with $|s_k - 1 - \frac{2\pi ik}{\log p}| = O(|k|^{-1/2})$, and for $P = 2$ every $k$ with $|k|$ large enough qualifies. So $\operatorname{Re} s_k \to 1$, and the zeros $1 - \bar{s}_k$ have real parts tending to 0. This follows a note of Nicholas Polson merged into the book [235].*

*Sketch.* At $\sigma_k = 1 + \frac{2\pi ik}{\log p}$ the term $E_P(1 - s)$ has a simple pole, with residue $R_k$, while $E_P(s)$ is analytic and nonzero, so by Rouché's theorem $F_P$ has a zero within $2|R_k/E_P(\sigma_k)|$ of $\sigma_k - R_k/E_P(\sigma_k)$ once this ratio is small. The factors $s(s - 1)$ in $R_k$ and in $E_P(\sigma_k)$ have equal moduli, Stirling's formula gives $|\Gamma_{\mathbb{R}}(1 - \sigma_k)/\Gamma_{\mathbb{R}}(\sigma_k)| \asymp |k|^{-1/2}$, the Euler factors of $\zeta_P(\sigma_k)$ lie on $\operatorname{Re} s = 1$ and are bounded above and below, and the Euler factors $(1 - q^{-(1-s)})^{-1}$ of the residue, $q \leq P$, $q \neq p$, are bounded when every $q^{2\pi ik/\log p}$ stays away from 1. As 1 and the numbers $\log q/\log p$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem gives infinitely many such $k$; for $P = 2$ there is no such $q$. The regular part of $E_P(1-s)$ and the variation of $E_P(s)$ are $O(|k|^{-1/2}\log|k|\,|E_P(\sigma_k)|)$ on the disc, by the same estimates and $\Gamma'/\Gamma(s) = O(\log|s|)$. Then $|R_k/E_P(\sigma_k)| = O(|k|^{-1/2})$. The second statement is the symmetry $F_P(s) = F_P(1 - s) = \overline{F_P(\bar{s})}$. □

## 22.2 Approximation, the kink constant and no finite proxy

Theorem 22.2.1. *Let $\delta_P = \sum_{m\notin\mathcal{F}(P)} e^{-\pi m^2}/(\pi m^2)$. For $|\operatorname{Im} t| \leq \frac{3}{2}$,*

$$|\Xi_P(t) - \Xi(t)| \leq \left|t^2 + \tfrac{1}{4}\right| \delta_P, \qquad \delta_P \leq \frac{e^{-\pi q^2}}{\pi q^2(1 - e^{-2\pi q})}.$$

*Proof.* Every $m < q$ is $P$ friable, so every excluded $m$ satisfies $m \geq q$, and $\Xi_P - \Xi = 2(t^2+\frac{1}{4})\sum_{m\notin\mathcal{F}(P)} \int_0^\infty e^{-\pi m^2 e^{2u}} e^{u/2}\cos(tu)\,du$. For $t = x+iy$ with $|y| \leq \frac{3}{2}$ we have $|\cos(tu)|e^{u/2} \leq e^{3u/2}e^{u/2} = e^{2u}$, and $\int_0^\infty e^{-\alpha e^{2u}} e^{2u} du = e^{-\alpha}/(2\alpha)$; with $\alpha = \pi m^2$ and the factor 2 this gives the first bound. For the second, $m^2 \geq q^2 + 2q(m - q)$ gives $e^{-\pi m^2}/(\pi m^2) \leq e^{-\pi q^2} e^{-2\pi q(m-q)}/(\pi q^2)$, and the geometric series in $m - q$ is summed. □

Computed values: $\delta_P = 1.85875 \cdot 10^{-14}$, $9.8969 \cdot 10^{-37}$, $9.0827 \cdot 10^{-70}$, $2.1406 \cdot 10^{-168}$ for $P = 2, 3, 5, 7$; on a grid of $|\operatorname{Im} t| \leq \frac{3}{2}$ the ratio $|\Xi_P - \Xi|/(|t^2 + \frac{1}{4}|\delta_P)$ is at most 0.9756, 0.9907, 0.9952, 0.9980, attained at $t = -\frac{3}{2}i$.

COROLLARY 22.2.2 (Certified tracking). *If $\gamma$ is a simple zero of $\Xi$ and $|\Xi| > |t^2+\frac14|\delta_P$ on $|t-\gamma| = r \le \frac32$, then $\Xi_P$ has exactly one zero in the disc, it is real, and $\gamma_P - \gamma = -(\Xi_P - \Xi)(\gamma)/\Xi'(\gamma)\,(1+O(\eta))$ with $\eta$ the ratio of $\max|\Xi_P-\Xi|$ on the disc to $\min|\Xi|$ on the circle.*

*Proof.* On the circle $|\Xi_P - \Xi| \le |t^2+\frac14|\delta_P < |\Xi|$ by Theorem 22.2.1, so Rouché's theorem gives $\Xi_P$ the same number of zeros in the disc as $\Xi$, namely one. The disc is symmetric about the real axis and $\Xi_P$ is real on $\mathbb{R}$, so the conjugate of the zero is also a zero in the disc, and uniqueness makes it real. The first order formula is the Taylor expansion of $\Xi_P = \Xi + D_P$ at $\gamma$, whose remainder is controlled on the disc by Cauchy's estimates. □

For $P=2$ the predicted shifts $-D_2(\gamma_k)/\Xi'(\gamma_k)$ are $+2.487\times10^{-9}$, $-4.027\times10^{-7}$, $+7.648\times10^{-6}$, $-5.539\times10^{-4}$, $+3.695\times10^{-3}$ at the first five zeros, and the computed shifts are $+2.487\times10^{-9}$, $-4.027\times10^{-7}$, $+7.648\times10^{-6}$, $-5.536\times10^{-4}$, $+3.702\times10^{-3}$ (computed at 40 digits); the first three agree to the digits shown, and the fourth and fifth agree to three digits, the relative difference at the fifth zero being $2\cdot10^{-3}$ (second order effects). The shifts grow like $e^{\pi\gamma/4}$, one decade every $4\ln10/\pi = 2.93$ units of height.

THEOREM 22.2.3 (Kink constant). *Folding $\Xi_P$ as in* (22.1.1) *with kernel $k_P(u) = e^{-u/2} - 2e^{u/2}\psi_P(e^{2u})$, put $c_P = -k_P'(0)$. Then*

$$c_P = \sum_{m\notin\mathcal{F}(P)} (4\pi m^2-1)e^{-\pi m^2} = 4\pi q^2 e^{-\pi q^2}(1+O(q^{-2})) > 0,$$

*and $\Xi_P(t) = c_P + O_P(t^{-2})$ for real $t\to\infty$. Hence $\Xi_P$ has finitely many real zeros.*

*Proof.* Differentiating, $k_P'(0) = -\frac12 - \psi_P(1) - 4\psi_P'(1)$; subtracting Lemma 22.1.1 leaves $\sum_{m\notin\mathcal{F}(P)}(1-4\pi m^2)e^{-\pi m^2}$. Since $k_P$ is smooth on $[0,\infty)$ with every derivative $O(e^{-u/2})$, two integrations by parts give $\int_0^\infty k_P(u)\cos(tu)\,du = -k_P'(0)t^{-2} + O(t^{-4})$, and multiplying by $t^2+\frac14$ gives the limit. Since $c_P>0$, $\Xi_P$ is positive for large real $t$, so its real zeros lie in a bounded set. □

Computed: $c_2 = 5.89126\cdot10^{-11}$ against the leading term $5.944\cdot10^{-11}$, and $c_3 = 2.43420\cdot10^{-32}$ against $2.442\cdot10^{-32}$. For $P=2$ the ratio $\Xi_2(t)/c_2$ is 0.450, 0.775, 0.934, 0.983, 0.9956 at $t = 50, 100, 200, 400, 800$, and $t^2(\Xi_2 - c_2) \to -1.65\cdot10^{-7}$, the predicted $c_P/4 + k_P'''(0)$. The limit regime is visible only above the breakdown height $\tau_P^*$ of Section 22.4: for $P=3$ the function $\Xi_3$ is still dominated by $\Xi$ at $t=50$ and $t=100$.

THEOREM 22.2.4 (No summable proxy). *For any real weights $w \ne 0$ with $\sum_n |w_n| n^{-\sigma} < \infty$ for every $\sigma > 0$ (this includes the weights $1_{\mathcal{F}(P)}$, for which the*

*sum is $\zeta_P(\sigma)$, and so $\Xi_P$), put $\psi_w = \sum w_n e^{-\pi n^2 x}$ and*

$$\Xi_w(t) = \frac{1}{2} - 2\left(t^2 + \tfrac{1}{4}\right) \int_0^\infty \psi_w(e^{2u}) e^{u/2} \cos(tu)\, du.$$

*Then $\Xi_w$ is even, entire of order at most one, and has finitely many real zeros and infinitely many nonreal zeros.*

*Proof.* Let $k_w(u) = e^{-u/2} - 2e^{u/2}\psi_w(e^{2u})$. The series $\psi_w(x)$ converges absolutely and locally uniformly on $\operatorname{Re} x > 0$ with $|\psi_w(x)| = O_\sigma(x^{-\sigma/2})$ as $x \downarrow 0$ for every $\sigma > 0$ (since $\sup_n n^\sigma e^{-\pi n^2 x} = O_\sigma(x^{-\sigma/2})$), and $e^{2u}$ has positive real part for $|\operatorname{Im} u| < \pi/4$, so $k_w$ is analytic in that strip and real analytic on $\mathbb{R}$; all its derivatives are $O(e^{-u/2})$ at $+\infty$. Repeated integration by parts gives $\int_0^\infty k_w(u)\cos(tu)\, du = \sum_j (-1)^{j+1} k_w^{(2j+1)}(0) t^{-2j-2} + \dots$, and hence $\Xi_w(t) \sim (-1)^{j_0+1} k_w^{(2j_0+1)}(0)\, t^{-2j_0}$ as $t \to \infty$ on $\mathbb{R}$, for the least $j_0$ with $k_w^{(2j_0+1)}(0) \neq 0$. Such a $j_0$ exists. Otherwise the odd part of $k_w$, which is real analytic on $\mathbb{R}$, would vanish to infinite order at 0, hence identically; then $k_w(-u) = k_w(u) \to 0$ as $u \to \infty$, and this gives $\psi_w(x) \sim \frac{1}{2} x^{-1/2}$ as $x \downarrow 0$, impossible since $|\psi_w(x)| = O_\sigma(x^{-\sigma/2})$ with $\sigma < 1$. So $\Xi_w$ has a nonzero power law limit or decay on $\mathbb{R}$ and finitely many real zeros. With $s = \frac{|y|}{2} + \frac{1}{4}$, the substitution $v = e^{2u}$ gives the bound $|\Xi_w(x+iy)| \leq \frac{1}{2} + |t^2 + \frac{1}{4}|\, \pi^{-s}\Gamma(s) \sum_n |w_n| n^{-1/2}$, from $\sum_n |w_n|(\pi n^2)^{-s} \leq \pi^{-s} \sum_n |w_n| n^{-1/2}$ for $s \geq \frac{1}{4}$, for $t = x + iy$, so the order is at most one. Suppose $\Xi_w$ had finitely many zeros. By Hadamard's theorem it would be $P(t)e^{at}$ with $P$ a polynomial; evenness forces $a = 0$, and boundedness on $\mathbb{R}$ (from the asymptotics) makes $P$ constant. The constant is $\Xi_w(i/2) = \frac{1}{2}$, since $t^2 + \frac{1}{4} = 0$ there. Then the cosine transform of $\psi_w(e^{2u})e^{u/2}$ vanishes for all real $t$, so $\psi_w = 0$ on $[1, \infty)$; if $n$ is the least index with $w_n \neq 0$, multiplying $\sum_{m \geq n} w_m e^{-\pi m^2 x} = 0$ by $e^{\pi n^2 x}$ and letting $x \to \infty$ gives $w_n = 0$, a contradiction. Hence $\Xi_w$ has infinitely many zeros, and so infinitely many nonreal ones. □

The proof of Theorem 22.2.3 never uses friability. For every nonempty set $E$ of omitted integers, the function built from the series over the integers outside $E$ has $c_E = \sum_{m \in E} (4\pi m^2 - 1) e^{-\pi m^2} > 0$, so it tends to $c_E$ on $\mathbb{R}$ and has finitely many real zeros. Hardy's theorem [169] that $\Xi$ has infinitely many real zeros therefore uses every integer in the theta series: the primes hide in the smoothness of the kernel at the origin. (Theorem 22.2.4 is the statement for summable weights; the statement for omitted integers is the one just given, and the weights equal to 1 except at one integer are outside its hypothesis, since $\sum n^{-\sigma}$ diverges for $\sigma \leq 1$.)

For $E = \{n > N\}$, that is for the weights $w_n = \mathbf{1}_{n \leq N}$ of Theorem 22.2.4, the constant is $c_E = 1.71806 \cdot 10^{-4}$ for $N = 1$ and $c_E = 5.89126 \cdot 10^{-11}$ for $N = 2$, where $\Xi_w$ and $c_E$ agree with $\Xi_2$ and $c_2$ to the digits shown, the integers $n \leq 2$ and the 2 friable integers first differing at $n = 4$. In $0 < t \leq 60$ the function $\Xi_w$ has 2 real zeros for $N = 1$, at 14.1659 and 19.5639, and 6 for $N = 2$, at 14.1347, 21.0220,

25.0109, 30.4243, 32.9388 and 37.4954 (as for $\Xi_2$, $R_2 = 6$ in Section 22.4), against the 13 zeros of $\Xi$ there. Its nonreal zeros with $0 < \operatorname{Re} t \le 60$ and $0 < \operatorname{Im} t \le 12$ are $22.915 + 4.666i$ and $27.930 + 9.028i$ for $N = 1$, and $41.190 + 1.462i$, $45.418 + 4.421i$, $49.549 + 7.364i$ and $53.720 + 10.074i$ for $N = 2$ (computed; argument principle, `px_xiw.py`). These functions are not the folded approximations of Section 25.1: by the identity $\Xi_P^{\mathrm{M}} = \Xi_P - c_P$ of Section 22.7, that is two integrations by parts with $k'_w(0) = -c_E$, the transform of the folded sum of the first $N$ terms of (1.2.3) is $\Xi_w - c_E$, so that $8H_N^F(z) = \Xi_w(\frac{z}{2}) - c_E$ in the notation there, and its zeros are the solutions of $\Xi_w = c_E$. The computation follows a note of Nicholas Polson merged into the book [233].

## 22.3 The deficit: positivity and Lorentzian form

For real $t$, the termwise form of the definition gives

$$D_P(t) = \big(t^2 + \tfrac14\big) \sum_{m \notin \mathcal{F}(P)} C_{a_m}(t), \qquad C_a(t) = \int_0^\infty f_a(w) \cos\big(\tfrac{tw}{2}\big)\, dw, \quad (22.3.1)$$

where $w = 2u$, $f_a(w) = e^{-ae^w} e^{w/4}$ and $a_m = \pi m^2 \ge 9\pi$. Put $b = a - \frac14$ and $\mathcal{L}_a(t) = e^{-a} b/(b^2 + t^2/4)$, the cosine transform of $e^{-a} e^{-bw}$.

Theorem 22.3.1. *$D_P(t) > 0$ for all real $t$, and $D_P > D_{P'}$ when $P < P'$. Hence $\Xi_P > \Xi_{P'} > \Xi$ on $\mathbb{R}$.*

*Proof.* For $a \ge \frac32$, $f_a' = f_a(\frac14 - ae^w) < 0$ and $f_a'' = f_a(x^2 - \frac32 x + \frac{1}{16}) > 0$ with $x = ae^w \ge a$, so $f_a$ is integrable, strictly convex and decreasing to 0. With $K(w) = (1 - \cos\lambda w)/\lambda$, $K' = \sin\lambda w$, two integrations by parts give Pólya's formula $\int_0^\infty f_a(w)\cos(\lambda w)\, dw = \lambda^{-2} \int_0^\infty f_a''(w)(1 - \cos\lambda w)\, dw$ for $\lambda > 0$, which is strictly positive because $f_a'' > 0$ (and the case $\lambda = 0$ is $\int f_a > 0$). So $C_a > 0$. Monotonicity in $P$ holds term by term, since $\mathcal{F}(P) \subset \mathcal{F}(P')$. □

Remark 22.3.2. The proof of Theorem 22.3.1 uses only $a \ge \frac32$, so $C_{\pi n^2}(t) > 0$ for every $n \ge 1$ and every real $t$, with $C_a$ as in (22.3.1), extended to complex $t$ by the same integral. Put

$$W(t) = 4 \int_0^\infty \psi(e^{2u}) e^{u/2} \cos(tu)\, du = 2 \sum_{n \ge 1} C_{\pi n^2}(t),$$

so that $2\Xi(t) = 1 - (t^2 + \frac14) W(t)$ for every complex $t$, by (22.1.1). Then $W > 0$ on $\mathbb{R}$, the zeros of $\Xi$ in $\mathbb{C}$ are exactly the solutions of $(t^2 + \frac14) W(t) = 1$, and RH says that every solution is real. So the transform of no term of the theta series vanishes on the real line, and the zeros of $\Xi$ come from the cancellation between the pole term and the whole series. In terms of laws, $C_{\pi n^2}(t)/C_{\pi n^2}(0)$ is the characteristic function of the symmetric density proportional to $e^{|u|/2} \exp(-\pi n^2 e^{2|u|})$, the law

of $\pm\frac12\log X$ with an independent fair sign, where $X$ has the density proportional to $x^{-3/4}e^{-\pi n^2 x}$ on $[1,\infty)$. Hence $W(t)/W(0)$ is the characteristic function of the mixture of these laws with the weights $2C_{\pi n^2}(0)/W(0)$, whose density is

$$p_V(u)=\frac{2e^{|u|/2}\psi(e^{2|u|})}{W(0)}=\frac{e^{-|u|/2}-k(u)}{W(0)},$$
$$W(0)=2\sum_{n\ge1}(\pi n^2)^{-1/4}\,\Gamma(\tfrac14,\pi n^2)=0.0230338$$

(computed, `px_const.py`). None of these laws is ferromagnetic (Proposition 13.5.11). Next, $k$ is bounded, $(\frac14-\frac{d^2}{du^2})e^{-|u|/2}=\delta_0$ in the sense of distributions, and $\Phi=\frac12(\frac14k-k'')$ holds on all of $\mathbb{R}$, both sides being even. So $k-2e^{-|\cdot|/2}*\Phi$ is bounded and annihilated by $\frac14-\frac{d^2}{du^2}$, whose null solutions $Ae^{u/2}+Be^{-u/2}$ are unbounded unless $A=B=0$, and

$$k(u)=2\int_{\mathbb{R}}e^{-|u-w|/2}\,\Phi(w)\,dw.$$

With the Laplace density $\varkappa(u)=\frac14e^{-|u|/2}$, and since $\Phi/\xi(\frac12)$ is the density of the Pólya law of (1.2.4), this reads

$$\varkappa=(1-\upsilon)\,\varkappa*\frac{\Phi}{\xi(\frac12)}+\upsilon\,p_V,\qquad \upsilon=\frac{W(0)}{4}=1-2\xi(\tfrac12)=0.00575844$$

(computed, `px_const.py`): the Laplace law is the mixture, with weights $1-\upsilon$ and $\upsilon$, of the Pólya law plus an independent Laplace variable and of the law $p_V$. Status: proved. This remark follows notes of Nicholas Polson merged into the book [244, 232].

Lemma 22.3.3. *If $F(0)=F'(0)=0$ and $F,F',F''$ are integrable on $[0,\infty)$, then the cosine and sine transforms of $F$ at $\lambda>0$ are at most* $\min(\|F\|_1,\|F''\|_1/\lambda^2)$ *in absolute value, and the sine bound needs only $F(0)=0$. Hence $|\cdot|(\beta^2+\lambda^2)\le\beta^2\|F\|_1+\|F''\|_1$ for every $\beta>0$.*

*Proof.* Integrability of $F$ and $F'$ makes $F\to0$ at infinity, and integrability of $F'$ and $F''$ makes $F'\to0$. Two integrations by parts give $\int_0^\infty F\cos\lambda w\,dw=-\lambda^{-1}\int_0^\infty F'\sin\lambda w\,dw=-\lambda^{-2}\int_0^\infty F''\cos\lambda w\,dw$, where the boundary terms vanish because $F(0)=F'(0)=0$. For the sine transform, $\int_0^\infty F\sin\lambda w\,dw=\lambda^{-1}\int_0^\infty F'\cos\lambda w\,dw=-\lambda^{-2}\int_0^\infty F''\sin\lambda w\,dw$, which uses only $F(0)=0$. The bound by $\|F\|_1$ is trivial, and $\min(A,B/\lambda^2)(\beta^2+\lambda^2)\le\beta^2A+B$. □

Theorem 22.3.4 (Uniform Lorentzian bound). *For $a\ge9\pi$ and all real $t$, $|C_a(t)-\mathcal{L}_a(t)|\le\delta(a)\mathcal{L}_a(t)$ with*

$$\delta(a)=\frac1b\left[\frac{2ab^2}{(b-1)^3}+\frac{2ab}{(b-1)^2}+\frac{2a^2}{(b-2)^3}+\frac{a}{b-1}\right]\le\frac{5.61}{a}.$$

*Proof.* Write $f_a = e^{-a}e^{-bw}e^{-a\varphi}$ with $\varphi = e^w - 1 - w \ge 0$. Then $C_a - \mathcal{L}_a$ is the cosine transform of $h = e^{-a}e^{-bw}g$ with $g = e^{-a\varphi} - 1$, and $h(0) = h'(0) = 0$. The bounds $|g| \le \frac{a}{2}w^2e^w$, $|g'| \le awe^w$, $|g''| \le a^2w^2e^{2w} + ae^w$ in $h'' = e^{-a}e^{-bw}(b^2g - 2bg' + g'')$ give $e^a\|h\|_1 \le a/(b-1)^3$ and $e^a\|h''\|_1 \le ab^2/(b-1)^3 + 2ab/(b-1)^2 + 2a^2/(b-2)^3 + a/(b-1)$. Lemma 22.3.3 with $\lambda = t/2$ and $\beta = b$, together with $\mathcal{L}_a(t)(b^2 + t^2/4) = e^{-a}b$, gives the claim. □

The last inequality follows from monotonicity, which is elementary. With $b = a - \frac{1}{4}$ and $a \ge 9\pi$ (so $b > 2$), every term of $\delta(a)$, $a\delta(a)$, $r_1(a)$ and $r_2(a)$ (the last two in Theorem 22.3.5) is a positive constant times a product of factors $a/(a-c)$, $(a-c_1)/(a-c_2)$ with $c_2 > c_1$, and $1/(a-c)$, where $c \ge 0$ and $a > c$; for example $2ab^2/(b(b-1)^3) = 2\,(a/(b-1))(b/(b-1))(1/(b-1))$ and $a^2/(b-2)^2 = (a/(b-2))^2$. Each such factor is positive and nonincreasing in $a$, so $\delta$, $a\delta$, $1/(1-\delta)$ (as $\delta(9\pi) = 0.198 < 1$), $r_1$ and $r_2$ are nonincreasing on $[9\pi, \infty)$. As an illustration, $a\delta(a)$ equals 5.6067 at $a = 9\pi$, 5.204, 5.103, 5.041, 5.0016 and 5.00002 at $a = 25\pi$, $49\pi$, $121\pi$, $10^4$, $10^6$, with limit 5 (the leading terms are $2 + 2 + 0 + 1$ over $b$), and the bound $\delta(a) \le 5.61/a$ is the value at $a = 9\pi$ propagated by the decrease of $a\delta(a)$. By quadrature the true error at $a = 9\pi$ is 0.034 (computed, 0.03365), and the bound gives $\delta(9\pi) = 0.198$.

Summing, $D_P \approx (t^2 + \frac{1}{4})\sum_{m \notin \mathcal{F}(P)} \mathcal{L}_{a_m}(t)$. Since $(t^2 + \frac{1}{4})\mathcal{L}_a(t) = e^{-a}b\,(t^2 + \frac{1}{4})/(b^2 + t^2/4) \le 4e^{-a}b$, Theorem 22.3.4 gives $D_P \le (1 + \delta(a_q))c_P$ on $\mathbb{R}$, so $\Xi_P$ differs from $\Xi$ on $\mathbb{R}$ by at most $(1 + \delta(a_q))c_P \approx 4\pi q^2 e^{-\pi q^2}$, and the deficit rises to its limit $c_P$ in the form of the Lorentzian $\mathcal{L}_{a_q}$, whose half width is $2b_q \approx 2\pi q^2$. Near the breakdown height $4q^2$ the deficit is only $D_P(4q^2)/c_P \approx 4/(\pi^2 + 4) = 0.2884$ of its limit; the measured values are 0.2929, 0.2901, 0.2893, 0.2888 for $P = 2, 3, 5, 7$ (computed).

THEOREM 22.3.5 (Curvature). *For every prime $P$ and $t \ge 14$, $(\log D_P)''(t) \ge -2/t^2 - r_2(a_q) - r_1(a_q)^2 \ge -0.0239$, where, with $\delta = \delta(a)$,*

$$r_1(a) = \frac{1}{2(1-\delta)b}\Big(1 + \frac{a^2}{(b-2)^2} + \frac{a}{(b-1)^2} + \frac{2a}{b-1}\Big),$$
$$r_2(a) = \frac{1}{4(1-\delta)b}\Big(\frac{4}{b} + \frac{2a^2}{(b-2)^3} + \frac{2a}{(b-1)^3} + \frac{4a}{(b-1)^2}\Big)$$

*bound $|C_a'|/C_a$ and $|C_a''|/C_a$ uniformly in $t$.*

*Proof.* Apply Lemma 22.3.3 to $f_a w$ (sine) and $f_a w^2$ (cosine), with $f_a \le e^{-a}e^{-bw}$, $|f_a'| \le f_a a e^w$, $|f_a''| \le f_a(a^2e^{2w} + ae^w)$, and divide by $C_a \ge (1-\delta)\mathcal{L}_a$. Positivity of every $C_{a_m}$ transfers the bounds to the sum $S = \sum_m C_{a_m}$, namely $|S'| \le r_1 S$ and $|S''| \le r_2 S$, because $r_1, r_2$ do not increase with $a$ on $[a_q, \infty) \subset [9\pi, \infty)$ (every term of their formulas is of the form treated after Theorem 22.3.4; for illustration $r_1 = 0.0960$, 0.0281, 0.0136 and $r_2 = 0.00435$, 0.00045, 0.00011 at $a = 9\pi$, $25\pi$, $49\pi$, computed). Then $(\log D_P)'' = (\log(t^2 + \frac{1}{4}))'' + S''/S - (S'/S)^2 \ge -2/t^2 - r_2 - r_1^2$,

because $(\log(t^2+\frac14))'' = 2(\frac14 - t^2)/(t^2+\frac14)^2 > -2/t^2$. At $a = 9\pi$ one finds $\delta = 0.19830$, $r_1 = 0.095954$, $r_2 = 0.0043527$ and $2/14^2 + r_2 + r_1^2 = 0.023764 \le 0.0239$ (computed). □

The value $-0.0239$ is slightly looser than the computed $-0.02376$. The computed minima of $(\log D_P)''$ over $t \ge 14$ are $-0.010642$, $-0.010240$, $-0.010186$, $-0.010169$ for $P = 2, 3, 5, 7$, all at $t = 14$, well above the bound.

## 22.4 Humps, pairs and the count of real zeros

Since $\Xi(0) > 0$, call the maximal intervals $I_j \subset (0,\infty)$ on which $\Xi < 0$ the negative humps, numbered from the left, with length $\ell_j$ and depth $d_j = \max_{I_j} |\Xi|$; the endpoints of $I_j$ are zeros of $\Xi$ (when the zeros are simple, $I_j = (\gamma_{2j-1}, \gamma_{2j})$ for the positive zeros $\gamma_1 \le \gamma_2 \le \dots$). Let $N(T)$ be the number of zeros $\gamma \in (0,T]$ counted with multiplicity. On $\mathbb{R}$, $\Xi = -AZ$ with $Z$ Hardy's function (Section 2.2) and $A(t) = \frac12(t^2+\frac14)\pi^{-1/4}|\Gamma(\frac14 + \frac{it}{2})| \sim (\pi/2)^{1/4} t^{7/4} e^{-\pi t/4}$ (the archimedean factor; the asymptotic follows from Stirling's formula). The first hump begins at $\gamma_1 = 14.13\ldots > 14$, so Theorem 22.3.5 applies on every hump.

Theorem 22.4.1.

*(1) Every real zero of $\Xi_P$ lies in a negative hump; $\Xi_P(\gamma_k) = D_P(\gamma_k) > 0$ at the zeros $\gamma_k$ of $\Xi$ (so no zero of $\Xi_P$ is a zero of $\Xi$).*

*(2) Each hump holds an even number of zeros of $\Xi_P$, counted with multiplicity, and the number $R_P$ of positive real zeros is even.*

*(3) Assume that every zero of $\Xi$ whose real part lies within $\frac12$ of $I_j$ is real (by [419] this holds for every hump below $3 \cdot 10^{12}$), and that $\ell_j < 18.29$. Then $I_j$ holds no zero, one double zero or two simple zeros of $\Xi_P$, and two simple zeros precisely when $\max_{I_j}(|\Xi| - D_P) > 0$.*

*(4) Surviving humps are nested in $P$: if $I_j$ carries zeros for $\Xi_P$, it does for $\Xi_{P'}$, $P' > P$.*

*Proof.* (1) Where $\Xi \ge 0$ we have $\Xi_P = \Xi + D_P > 0$. (2) $\Xi_P$ is positive at both ends of each hump, and $\Xi_P(0) > 0$, $\Xi_P(t) \to c_P > 0$. (3) By the Hadamard product (Equation (1.2.1)), $\Xi$ is of order one and even, so $(\log|\Xi|)''(t) = -\sum_w \mathrm{Re}(t-w)^{-2}$ over all zeros $w$ of $\Xi$ (both signs). A real zero contributes $-(t-w)^{-2} \le 0$. A nonreal zero $w = \gamma + i\delta'$ has $0 < |\delta'| < \frac12$ (the zeros lie in the strip $|\operatorname{Im} w| < \frac12$), and contributes $(\delta'^2 - x^2)/(x^2+\delta'^2)^2$ with $x = t - \gamma$, which is negative as soon as $|x| \ge \frac12$. For $t \in I_j$ the zeros with $|\operatorname{Re} w - t| < \frac12$ are real by hypothesis and all other zeros contribute at most 0. Keeping only the two endpoints $\alpha < \beta$ of $I_j$, $(\log|\Xi|)'' \le -(t-\alpha)^{-2} - (\beta - t)^{-2} \le -8/\ell_j^2$ on $I_j$, the minimum of the right side being at the midpoint. When $8/\ell_j^2 > 0.0239$, that is $\ell_j < 18.29$, Theorem 22.3.5

makes $g = \log|\Xi| - \log D_P$ strictly concave on $I_j$, and $g \to -\infty$ at both ends. So $g$ has no zero (if $\max g < 0$), one double zero (if $\max g = 0$) or two simple zeros (if $\max g > 0$). A zero of $g$ is a zero of $\Xi_P$ because $\Xi = -|\Xi|$ on $I_j$, and the multiplicities agree: $\Xi_P' = -|\Xi| g'$ where $|\Xi| = D_P$. (4) follows from Theorem 22.3.1. □

The hypothesis of (3) is local: it uses only the zeros near the hump. The humps used in this chapter lie below 21,400 and $\ell_j < 18.29$ for them, since the gaps between consecutive zeros there are of the order of $2\pi/\log(t/2\pi)$ (computed from the tabulated zeros), so for those humps (3) is an unconditional theorem given [419].

The two zeros of a surviving hump sit near its ends and are pushed inward, which explains the alternating signs of the first order shifts. Real zeros leave the axis only in pairs. To see this one deforms continuously, for example $\Xi + sD_P$ for $0 \le s \le 1$; a pair of real zeros of $\Xi + sD_P$ collides where $\log|\Xi| - \log(sD_P)$ attains its maximum zero, near but not at the extremum $t_j$ of the hump. Near the collision $\Xi_P \approx D_P(t_j) - d_j + \frac{1}{2}\Xi''(t_j)(t-t_j)^2$, so the pair is $t_j \pm \sqrt{2(d_j - D_P)/\Xi''(t_j)}$ before and $t_j \pm i\sqrt{2(D_P - d_j)/\Xi''(t_j)}$ after (heuristic).

THEOREM 22.4.2 (Count). *Under the hypotheses of Theorem 22.4.1(3) for every hump meeting* $(0, \tau_P^+)$,

$$R_P = 2\#\{j : \max_{I_j}(|\Xi| - D_P) > 0\},$$

*where the count is over all humps. Let* $\tau_P^+$ *solve* $D_P(\tau) = 0.63\,A(\tau)\,\tau^{1/6}\log\tau$*; the solution is unique for* $\tau \ge 14$*, and* $\Xi_P > 0$ *on* $(\tau_P^+, \infty)$*. Then* $R_P \le 2\lceil N(\tau_P^+)/2\rceil$.

*Proof.* The formula is part (3) summed over the humps meeting $(0, \tau_P^+)$; beyond $\tau_P^+$ there is no zero, as follows. Put $\mathcal{R}(t) = D_P(t)/(0.63A(t)t^{1/6}\log t)$. Since $D_P = (t^2 + \frac{1}{4})S$ and $|S'| \le r_1(a_q)S$ (Theorem 22.3.5), $(\log D_P)' \ge 2t/(t^2 + \frac{1}{4}) - r_1(a_q)$. Also $(\log A)' = 2t/(t^2 + \frac{1}{4}) - \frac{1}{2}\operatorname{Im}\frac{\Gamma'}{\Gamma}(\frac{1}{4} + \frac{it}{2})$. The terms $2t/(t^2 + \frac{1}{4})$ cancel, and with $\operatorname{Im}\frac{\Gamma'}{\Gamma}(\frac{1}{4} + iy) = y\sum_{n\ge 0}\big((n + \frac{1}{4})^2 + y^2\big)^{-1} \ge \frac{\pi}{2} - \frac{1}{4y}$ (compare the sum with an integral) we get, for $t \ge 14$ (so $y = t/2 \ge 7$),

$$(\log\mathcal{R})' \ge \tfrac{1}{2}\Big(\tfrac{\pi}{2} - \tfrac{1}{2t}\Big) - r_1(a_q) - \frac{1}{6t} - \frac{1}{t\log t} > 0.6,$$

using $r_1(a_q) \le r_1(9\pi) = 0.0960$. With $\mathcal{R}(14) < 1$ (computed: $D_2(14) < 10^{-10}$, and $0.63A(14)14^{1/6}\log 14$ is of order $10^{-3}$) and $\mathcal{R} \to \infty$, the equation $\mathcal{R} = 1$ has exactly one solution $\tau_P^+$ for $t \ge 14$ and $\mathcal{R} > 1$ beyond it. By Hiary's estimate $|Z(t)| = |\zeta(\frac{1}{2} + it)| \le 0.63t^{1/6}\log t$ [172], $|\Xi| = A|Z| < D_P$ for $t > \tau_P^+$, so $\Xi_P > 0$ there. Only humps meeting $(0, \tau_P^+)$ count, and there are at most $\lceil N(\tau_P^+)/2\rceil$ of them, each carrying at most two zeros. □

| $P$ | $q$ | $\tau_P^*$ | $N(\tau_P^*)$ | $\tau_P^+$ | $N(\tau_P^+)$ | $R_P$ |
|---|---|---|---|---|---|---|
| 2 | 3 | 39.73 | 6 | 41.63 | 7 | 6 |
| 3 | 5 | 104.68 | 31 | 107.06 | 32 | 32 |
| 5 | 7 | 201.32 | 80 | 203.99 | 81 | 80 |
| 7 | 11 | 490.18 | 263 | 493.23 | 264 | 264 |

Table 22.1: Real zero counts (computed in floating point, not certified). $\tau_P^*$ solves $A(\tau) = D_P(\tau)$. For $P \le 5$, $R_P$ is a full sign change scan; for $P = 7$ it is the hump criterion, with a 215 digit check of every hump in $[481, 501]$, the last surviving pair being 489.7932 and 491.0605.

The monotonicity of $r_1$ used here is the proved one in the proof of Theorem 22.3.5, so the uniqueness of $\tau_P^+$ and the positivity beyond it are proved.

The primes $2, 3, 5, 7$ therefore reproduce the first 264 zeros of $\Xi$ in the sense that the computed zeros of $\Xi_7$ pair with the first 264 zeros $\gamma_k$ of $\Xi$, the largest displacement being 0.338 at $k = 264$ (computed); for $P = 3$ the largest displacement is 0.764, for $P = 5$ it is 0.208 and for $P = 2$ it is 0.091. The counts $R_P$ of Table 22.1 are computed, and become theorems once a ball arithmetic certificate of the sign of $\max_{I_j}(\log|\Xi| - \log D_P)$ on every hump below $\tau_P^+$ is supplied. The breakdown height has the heuristic form

$$\tau_P^* \approx 4q^2 + \frac{6}{\pi}\log q + \frac{4}{\pi}\log\frac{\pi^2+4}{4} + 0.03 \approx 4q^2 + \frac{6}{\pi}\log q + 1.61,$$

where $\frac{4}{\pi}\log\frac{\pi^2+4}{4} = 1.583$ comes from the Lorentzian and Stirling estimates and the remaining 0.03 is a fitted constant. The formula gives 39.71, 104.68, 201.33, 490.19 for $q = 3, 5, 7, 11$, against the computed 39.73, 104.68, 201.32, 490.18, and it reproduces all twenty values of Table 22.2 to within 0.05 (computed).

## 22.5 A survey of the breakdown height up to height 21,000

For every prime $P \le 71$ the hump criterion of Theorem 22.4.2 (computed, with $Z$ evaluated by the Riemann Siegel formula and the zeros located by sign changes on a grid of step 0.02, repeated at step 0.016 with no change in any count) gives $R_P - N(\tau_P^*) \in \{-1, 0, 1, 2, 3\}$, with mean 0.90 and standard error 0.28 (sample standard deviation divided by $\sqrt{20}$) over the twenty primes (Table 22.2); the heights reach 21,326, and no hump was marginal. The offset can be compared with a heuristic ramp model. Near $\tau_P^*$, $\log A$ falls at rate $\pi/4$ while $\log D_P$ is nearly flat, so hump $j$ survives when $\log\max_{I_j}|Z| > \frac{\pi}{4}(t_j - \tau_P^*)$. Negative humps have density $\log(t/2\pi)/4\pi$ (computed: 0.6475 against 0.6469 in a window of 400 around $\tau_P^*$ at $P = 71$) and carry two zeros each, so the model offset is $(2/\pi^2)\log(\tau_P^*/2\pi)\, m_Z(\tau_P^*)$, with $m_Z(T)$ the mean of $\log\max|Z|$ over the negative humps within 200 of $T$. The model gives the right order of magnitude and no more: its mean over the survey is 0.447 against the observed 0.90, and the rows do not follow it closely.

| $P$ | $\tau_P^*$ | $N$ | $R_P$ | off | model | $P$ | $\tau_P^*$ | $N$ | $R_P$ | off | model |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 39.7 | 6 | 6 | 0 | 0.16 | 31 | 5484 | 5039 | 5040 | 1 | 0.42 |
| 3 | 104.7 | 31 | 32 | 1 | 0.23 | 37 | 6733 | 6405 | 6408 | 3 | 0.53 |
| 5 | 201.3 | 80 | 80 | 0 | 0.29 | 41 | 7405 | 7157 | 7160 | 3 | 0.60 |
| 7 | 490.2 | 263 | 264 | 1 | 0.37 | 43 | 8845 | 8799 | 8802 | 3 | 0.58 |
| 11 | 682.5 | 401 | 400 | −1 | 0.37 | 47 | 11245 | 11616 | 11616 | 0 | 0.54 |
| 13 | 1163 | 782 | 782 | 0 | 0.37 | 53 | 13933 | 14867 | 14868 | 1 | 0.44 |
| 17 | 1451 | 1027 | 1028 | 1 | 0.47 | 59 | 14893 | 16050 | 16052 | 2 | 0.52 |
| 19 | 2124 | 1631 | 1632 | 1 | 0.43 | 61 | 17966 | 19897 | 19898 | 1 | 0.54 |
| 23 | 3372 | 2838 | 2840 | 2 | 0.41 | 67 | 20174 | 22715 | 22714 | −1 | 0.56 |
| 29 | 3852 | 3323 | 3324 | 1 | 0.48 | 71 | 21326 | 24201 | 24200 | −1 | 0.64 |

Table 22.2: The survey (computed). $N = N(\tau_P^*)$, off $= R_P - N$, model is the ramp model's expected offset with the window 200 for $m_Z$.

Conjecture 22.5.1.

$$R_P = N(\tau_P^*) + \frac{2}{\pi^2}\log\frac{\tau_P^*}{2\pi}\, m_Z(\tau_P^*) + O\big((\log q)^{1/2}(\log\log q)^{1/2}\big),$$

*with $m_Z(T)$ the mean of $\log\max|Z|$ over negative humps near $T$.*

The status of Conjecture 22.5.1 is open. The twenty offsets of Table 22.2 are all in $\{-1,\dots,3\}$ and are consistent with a bounded or slowly growing offset, but they differ from the model by a factor of about two on average, the error term is not motivated by the ramp model, and a growth like $\log q$ cannot be tested up to height 21,326.

## 22.6 Where the lost zeros go

Beyond $\tau_P^*$ the zeros of $\Xi_P$ solve $\Xi = -D_P$. Stirling's formula and the functional equation give $|\Xi(x+iy)| \approx A(x)(x/2\pi)^{|y|/2}$, and matching moduli gives the departure law (heuristic)

$$|y_P(x)| \approx \frac{2\log(D_P(x)/A(x))}{\log(x/2\pi)} \sim \frac{\pi}{2}\,\frac{x-\tau_P^*}{\log(x/2\pi)}, \tag{22.6.1}$$

where the second form follows from $\log(D_P/A) \approx \frac{\pi}{4}(x-\tau_P^*)$. Using $D_P(x)$ rather than its limit $c_P$ matters: near $\tau_P^*$, $D_P$ is only about $0.3c_P$ (computed: 0.336 at $P=2$ and 0.310 at $P=3$), and with $D_P$ the law fits the computed zeros to within 0.11 (Table 22.3; the largest difference is 0.104). Since $|y_P|$ soon exceeds $\frac{1}{2}$, the departed zeros leave the critical strip altogether.

Remark 22.6.1 (Zero count, heuristic). With $N_P(X)$ the number of zeros of $\Xi_P$ in $0 < \operatorname{Re} t < X$ (both half planes), and $y_P(X)$ the departure law (22.6.1), the heuristic is

$$N_P(X) = N(X) + \tfrac{1}{4}y_P(X) + O(1).$$

| $P$ | $\operatorname{Re} z$ | $\lvert\operatorname{Im} z\rvert$ | law with $c_P$ | law (22.6.1) |
|---|---|---|---|---|
| 2 | 45.418 | 4.421 | 5.38 | 4.46 |
| 2 | 57.879 | 12.612 | 13.23 | 12.65 |
| 2 | 70.174 | 19.406 | 19.90 | 19.51 |
| 3 | 111.384 | 3.663 | 4.40 | 3.64 |
| 3 | 128.300 | 12.233 | 12.84 | 12.24 |
| 3 | 148.311 | 21.542 | 22.03 | 21.56 |

Table 22.3: Departed zeros of $\Xi_P$ (computed, residuals below $10^{-34}$) against the two forms of the departure law. The six zeros are every second zero of the chain of Remark 22.6.1 for $P = 2$ and a selection for $P = 3$.

The $O(1)$ is not derived, and this statement is a heuristic, not a proposition.

*Derivation (heuristic).* Apply the argument principle to $\Xi_P$ and to $\Xi$ on $[0, X] \times [-Y, Y]$ with $Y \gg y_P(X)$. On the horizontal sides $|\Xi| \gg |D_P|$, and on the left side both functions are real and positive. On the right side $\Xi_P \approx D_P$ for $|y| < y_P$, where $\arg D_P$ is nearly constant, while $\partial_y \arg \Xi = \operatorname{Re}(\Xi'/\Xi) \approx -\pi/4 + y/(2X)$. The odd term integrates to zero, so the two counts differ by $\frac{\pi}{2} y_P$ radians, that is $y_P/4$ zeros, up to $O(1)$ from the transitions. □

The evidence is a computation, not a proof. Argument principle counts, evaluated at 20 to 40 digits and equal to an integer within $10^{-15}$, give $N_P(X) - N(X)$ equal to 2, 3, 4, 5 for $P = 2$ at $X = 51.5$, 60, 68, 80, against $y_P/4$ equal to 2.17, 3.47, 4.59, 6.12 ($N_P = 12$, 16, 20, 26 against $N = 10$, 13, 16, 21). For $P = 3$ at $X = 120$, 140, 150 they give 2, 4, 6 against 2.03, 4.45, 5.58. Every count was repeated with a second contour height and precision (20 and 28 digits), with the same integer. The difference $N_P - N - y_P/4$ is $-0.17$, $-0.47$, $-0.59$, $-1.12$ for $P = 2$, and $-4.86$ at $X = 600$ ($N_P = 384$ at two precisions, $N = 341$, $y_P/4 = 47.86$), so the computed counts do not support a bounded $O(1)$. Tracing the departed zeros of $\Xi_2$ one by one gives, for $P = 2$, a single smooth chain running from $41.19 + 1.46i$ through $45.42 + 4.42i$ out to $601.90 + 191.23i$. It has 190 zeros up to 601.90 and 189 with real part below 600. Each of these zeros was verified individually (residual test), and the chain was traced in one continuous job to 601.90. The argument principle on the whole region $\operatorname{Re} t < 600$ gives $N_P = 384 = 6 + 2 \cdot 189$, the six real zeros and the chain with its conjugates. The chain count (189 departed zeros below 600) and the difference $N_P - N = 43$ are different quantities, and 43 is compared with $y_P/4 = 47.86$ above. So for $P = 2$ the lost humps release their zeros in pairs, and these, together with about $y_P/4$ new zeros, populate one curve (observed for $P = 2$, and in the counts for $P = 3$). Its density in the upper half plane is, heuristically,

$$\frac{1}{4\pi} \log \frac{x}{2\pi} + \frac{\pi}{16 \log(x/2\pi)},$$

larger than the density $\frac{1}{4\pi}\log\frac{x}{2\pi}$ of lost humps by the factor $1+\pi^2/(4\log^2(x/2\pi))$, which is algebra: $(\pi/(16L))/(L/(4\pi)) = \pi^2/(4L^2)$ with $L=\log(x/2\pi)$. Integrated from 41.19 to 600 this density gives 197.4 zeros, against the 189 of the chain (four per cent more), and the spacings of the chain, 4.13, 3.72 and 3.39 near $x=45$, 100 and 150, exceed the reciprocal density there, 3.90, 3.43 and 3.18, by 6 to 9 per cent (over the whole chain the spacings lie between 2.537 and 4.228).

## 22.7 Miyatake's theorem and the remainder $R$

Write $\Xi$ in the form used by Miyatake [217], as a combination of two shifted Laplace transforms,

$$\Xi(z) = 2\pi^2\Big\{G_{\mathrm M}\big(\tfrac12 iz-\tfrac94\big)+G_{\mathrm M}\big(\tfrac12 iz+\tfrac94\big)\Big\},\qquad G_{\mathrm M}(w)=\int_{-\infty}^{\infty}\psi_{\mathrm M}(t)e^{wt}\,dt. \tag{22.7.1}$$

Let $\Phi$ be the kernel of (1.2.3), so that $\Xi(z)=\int_{\mathbb R}\Phi(u)e^{izu}\,du$, which equals $2\int_0^\infty\Phi(u)\cos(zu)\,du$, and

$$\Phi(t)=2\pi e^{5t/2}\sum_{n\ge1}(2\pi n^2e^{2t}-3)\,n^2e^{-\pi n^2e^{2t}}.$$

The identity (22.7.1) holds exactly when

$$8\pi^2\cosh(9t/2)\,\psi_{\mathrm M}(2t)=\Phi(t). \tag{22.7.2}$$

Indeed, (22.7.2) gives $\Xi(z)=8\pi^2\int\cosh(9t/2)\psi_{\mathrm M}(2t)e^{izt}\,dt$, and the substitution $s=2t$ turns this into $4\pi^2\int\cosh(9s/4)\psi_{\mathrm M}(s)e^{izs/2}\,ds = 2\pi^2\{G_{\mathrm M}(\frac12 iz+\frac94)+G_{\mathrm M}(\frac12 iz-\frac94)\}$. Condition (22.7.2) forces

$$\psi_{\mathrm M}(t)=\sum_{n\ge1}n^4e^{-2\pi n^2\cosh t}\big(1-R_{\mathrm M}(n^2\pi,t)\big), \tag{22.7.3}$$

where

$$1-R_{\mathrm M}(x,t)=\frac{1-\frac{3}{2x}e^{-t}}{1+e^{-9t/2}}\,e^{xe^{-t}}\qquad(t>0),$$

extended evenly. Each term of $\Phi(s/2)/(8\pi^2\cosh(9s/4))$ equals $n^4(1-\frac{3}{2x}e^{-s})e^{-xe^s}/(1+e^{-9s/2})$ with $x=\pi n^2$, and $e^{-xe^s}=e^{-2x\cosh s}e^{xe^{-s}}$, which proves (22.7.3). The sum $\psi_{\mathrm M}$ is even, a single term is not, which is the corner discussed below. A numerical check of (22.7.1) and (22.7.2) agrees to thirteen digits at $z=0$, at $z=5$ and at the first zero of $\Xi$. The remainder $R_{\mathrm M}$ undoes Pólya's three simplifications: $e^{xe^{-t}}$ desymmetrises the exponential, $1/(1+e^{-9t/2})$ removes the universal factor, and $1-\frac{3}{2x}e^{-t}$ restores the second theta term. Each term of (22.7.3) has a corner at $t=0$, with one sided logarithmic slopes $+0.0222$, $-10.18$, $-25.97$ for $n=1,2,3$ (the slope of term $n$ is $3/(2x-3)+\frac94-x$ with

$x = \pi n^2$); only the full sum is smooth, which is Lemma 22.1.1 in Miyatake's coordinates. The ratio $Q = \psi_{\mathrm{M}}/e^{-2\pi\cosh t}$ has $Q(0) = 6.0590692$ and tends to 1 as $t \to \infty$ (the factor of the term $n = 1$ tends to 1); numerically it decreases on $[0, 6]$, and $Q(0) - 1 = 5.059069$.

Dropping the factors $1 - R_{\mathrm{M}}$ and summing over the $P$ friable integers $n$ (those with every prime factor at most $P$) gives

$$\xi^{(P)}(z) = 4\pi^2 \sum_{n \text{ friable}} n^4 \Big\{ K_{iz/2-9/4}(2\pi n^2) + K_{iz/2+9/4}(2\pi n^2) \Big\},$$

the function that Miyatake studies in paper B [217]. His Theorem I states that $\xi^{(P)}$ has all but finitely many zeros real, with $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(1)$ of them in $(0, T)$; the proof uses Watson type steepest descent asymptotics for $K_z(a)$, a lower bound $|\zeta_P| \geq \sigma > 0$ for the finite Euler product $\zeta_P$ on a strip $\frac{1}{2} - \varepsilon \leq x \leq A$, the argument principle and Rouché's theorem. Hejhal [170] proved a theorem of this kind for truncated symmetrised kernels. For $P = 2$ we find exactly 79 real zeros of $\xi^{(2)}$ in $(0, 200)$, as many as $\Xi$; the first is 8.9929, against 14.1347 for $\Xi$, and the last is 198.07. The lower bound $\sigma$ on $|\zeta_P|$ tends to 0 as $P \to \infty$, so the method does not reach the full sum.

The remainder $R_{\mathrm{M}}$ separates two families. Keeping it with the friable integers gives $\Xi_P^{\mathrm{M}} = \Xi_P - c_P$ (two integrations by parts with $k_P'(0) = -c_P$), extremely close to $\Xi$ and, like $\Xi_P$, with finitely many real zeros (Theorem 22.2.3 for $\Xi_P$). Dropping it gives functions of the type of $\xi^{(P)}$ and of Pólya's approximant, with almost all zeros real but at a distance from the zeros of $\Xi$. For $P = 2$ the zeros of $\Xi_2$ differ from the first five zeros of $\Xi$ by $2.5 \times 10^{-9}$, $4.0 \times 10^{-7}$, $7.7 \times 10^{-6}$, $5.536 \times 10^{-4}$ and $3.702 \times 10^{-3}$, while those of $\xi^{(2)}$ are off by several units already at the first zero. The Pólya shift route to RH through (22.7.1) would need the transform $g(x) = G_{\mathrm{M}}(ix)$ to lie in the Laguerre Pólya class. It does not. Since $\psi_{\mathrm{M}}(t) = \Phi(t/2)/(8\pi^2\cosh(9t/4))$, $\psi_{\mathrm{M}}$ has a pole at $t = 2\pi i/9$, inside the strip where $\Phi(t/2)$ is analytic; hence $g(x) = Ce^{-2\pi x/9}(1 + o(1))$ with $C = 0.0899917116445 > 0$ (Theorem 22.12.1, which proves that $g$ is not Laguerre Pólya, Corollary 22.12.3), and numerically $g$ has no zeros in $(0, 26) \times (-4, 4)$ and none on $(0, 60]$. The pole terms cancel exactly in the shifted sum (22.7.1). In the variable $\cosh t$ this pole is a simple pole at $\cos\frac{2\pi}{9}$, the exponent of the exponential part of Section 22.13 (Proposition 22.13.1). Moreover $Q$ decreases numerically, whereas every universal factor $\mathbb{E}\cosh(tX)$ in the sense of Cardon [74] increases in $|t|$.

Three further facts, all computed except the second half of the third, sharpen this picture. First, interpolating with factors $1 - \lambda R_{\mathrm{M}}$ gives $\Xi_\lambda = (1-\lambda)\xi^{(2)} + \lambda\Xi_P^{\mathrm{M}}$, where $\Xi_P^{\mathrm{M}}$ keeps $R_{\mathrm{M}}$ and has 7 real zeros in $(0, 200)$, the last at 39.53. The real part $X$ of the first nonreal zero of $\Xi_\lambda$ (the height to which the real zeros survive) satisfies $h(\lambda) \approx h(1) + \frac{4}{\pi}\log\frac{1}{\lambda}$: for $\lambda = 10^{-k}$, $k = 0, \ldots, 8$, one finds $h = 44, 45, 50, 51, 56, 57, 62, 63, 67$ on a grid of step 1, with a least squares slope of 2.93 per decade against $\frac{4}{\pi}\log 10 = 2.93$. The survivors are zeros of the Pólya type

function, not of $\Xi$. Second, dividing Riemann's kernel $\Phi(u)$ by $\cosh(bu)$ creates a pole inside its strip of analyticity when $b > 2$, and for $b \le 2$ the transform acquires nonreal zeros at a height that recedes as $b \to 0$, in every case computed: the largest admissible $b$ lies in $(1, 1.25)$ up to height 40, in $(0.5, 0.75)$ up to 100 and in $(0.25, 0.5)$ up to 160. This is the computed analogue, for cosh factors, of the theorem of Rodgers and Tao [463] that the de Bruijn Newman constant is nonnegative (Sections 22.10 and 22.11; the threshold $b = 2$ is explained in Section 22.12). Third, de Bruijn's strip theorem [112] reduces Miyatake's Theorem I to the zeros of $g^{(P)}(w) = 2\sum_{n \text{ friable}} n^4 K_{iw}(2\pi n^2)$: $\xi^{(P)}(z) = 2\pi^2\{g^{(P)}(z/2 + 9i/4) + g^{(P)}(z/2 - 9i/4)\}$, so if the zeros of $g^{(P)}$ lie in $|\operatorname{Im} w| \le \frac{9}{4}$, every zero of $\xi^{(P)}$ is real. For $P = 2$ the ten nonreal zeros of $g^{(2)}$ in the upper half plane with real part below 60 have imaginary parts rising to 1.964 (at real part 58.5), and they appear to approach 2, because the governing friable Dirichlet polynomial is geometric with zeros on $\operatorname{Re} s = 0$. For the full sum the condition becomes that partial sums of $\zeta$ have zeros only in $\operatorname{Re} s \le \frac{1}{2}$, which is false [375]; and Riemann's own $g$ has, computationally, zeros on a cone starting at $8.96 + 5.20i$ (Section 22.12).

## 22.8 The pole term, the secant and reweighting

The pole term $\frac{1}{2}/(t^2 + \frac{1}{4})$ is the first term of $\frac{\pi}{2}\operatorname{sech}(\pi t) = \sum_k (-1)^k (k + \frac{1}{2})/((k + \frac{1}{2})^2 + t^2)$, the transform of $\frac{1}{2}\operatorname{sech}(u/2) = \sum_k (-1)^k e^{-(2k+1)|u|/2}$. That kernel is smooth because its kinks cancel in Abel sum, $\sum_k (-1)^k (2k + 1) = 0$.

Proposition 22.8.1 (Kink budget). *Replace the pole kernel $e^{-u/2}$ in $k_P$ by an even kernel $\kappa$, smooth on $[0, \infty)$ with derivatives $O(e^{-u/2})$. Then $\lim_{t\to\infty} \Xi_P^\kappa(t) = c_P - \frac{1}{2} - \kappa'(0+)$ along $\mathbb{R}$.*

*Proof.* The kernel $k_P^\kappa = k_P + \kappa - e^{-u/2}$ is smooth on $[0, \infty)$ with every derivative $O(e^{-u/2})$, and $\Xi_P^\kappa(t) = (t^2 + \frac{1}{4}) \int_0^\infty k_P^\kappa(u) \cos(tu)\, du$. Two integrations by parts give $\int_0^\infty k \cos(tu)\, du = -k'(0+)t^{-2} - t^{-2} \int_0^\infty k'' \cos(tu)\, du$, and the last integral tends to 0 by the Riemann Lebesgue lemma, so $(t^2 + \frac{1}{4}) \int_0^\infty k \cos(tu)\, du \to -k'(0+)$. For $k = k_P$ the limit is $c_P = -k_P'(0)$ (Theorem 22.2.3); for $k = k_P^\kappa$ it is $-k_P'(0) - \kappa'(0+) - \frac{1}{2} = c_P - \frac{1}{2} - \kappa'(0+)$, since $(e^{-u/2})'(0) = -\frac{1}{2}$. □

So the secant completion, with $\kappa'(0+) = 0$, lowers the limit from $c_P$ to $c_P - \frac{1}{2}$, which is negative as soon as $c_P < \frac{1}{2}$, unless the slope $-\frac{1}{2}$ is restored elsewhere. Miyatake's paper A [216] is on the consistent side: there the pole term becomes $\frac{1}{4}\operatorname{sech}(u/4)$, with transform $\pi \operatorname{sech}(2\pi z)$ in the variable $z$ of $\Xi(z) = \int \Phi(u) e^{izu}\, du$, while every theta term is symmetrised, so the whole kernel is smooth.

Proposition 22.8.2 (Reweighting). *For weights $w$ put $D_w = \Xi_w - \Xi = (t^2 + \frac{1}{4}) \sum_n (1 - w_n) C_{\pi n^2}$. If $0 \le w_n \le w_n' \le 1$ for every $n$ then $D_w \ge D_{w'} \ge 0$, with $D_{w'} > 0$ unless every $w_n' = 1$. So among weights in $[0, 1]$ that vanish on a given*

*set, the indicator of its complement has the smallest deficit; the friable truncation is of this kind.*

*Proof.* Every $C_{\pi n^2}$ is positive on $\mathbb{R}$ (proof of Theorem 22.3.1), so $1-w_n \geq 1-w'_n \geq 0$ termwise gives $D_w \geq D_{w'} \geq 0$, with strict inequality for $D_{w'}$ as soon as one $w'_n < 1$. □

Doing better requires some $w_n > 1$; then $D_w$ can change sign, and each sign change plants a spurious real zero. For $P = 2$, matching the deficit at the heights $\frac{40}{3}$, $\frac{80}{3}$, 40 with weights on $n \in \{1\}, \{1,2\}, \{1,2,4\}$ gives 2, 8, 9 real zeros in $(0, 80)$, each time with one spurious zero (computed). The weights grow like $e^{\pi n^2}$ ($w_4 - 1 \approx 3.7 \times 10^9$), because each correction is a Lorentzian of height $e^{-\pi n^2}$ and width $2\pi n^2$ trying to imitate one of width $2\pi q^2$, where $q$ is the least integer outside the friable set.

## 22.9 The $2\sin^2$ lemma and the Thorin programme

By Hadamard, $\Xi_P(t)/\Xi_P(0) = \prod_{\operatorname{Re} t_0 > 0}(1 - t^2/t_0^2)$, the product over the zeros $t_0$ with positive real part, which is complete when $\Xi_P$ has no zero on $i\mathbb{R}$ (Proposition 22.9.1). On the imaginary axis the sine squared lemma in Frullani form (the factor by factor form of (1.2.9)),

$$\log(t_0 + iy) + \log(t_0 - iy) - 2\log t_0 = 4\int_0^\infty \sin^2\Big(\frac{xy}{2}\Big)e^{-t_0 x}\,\frac{dx}{x} \qquad (\operatorname{Re} t_0 > 0),$$

with principal logarithms (for real $t_0$ the left side is $\log(1 + y^2/t_0^2)$; for complex $t_0$ it can differ from the principal $\log(1 + y^2/t_0^2)$ by $2\pi ik$), applies factor by factor:

$$\log\frac{\Xi_P(iy)}{\Xi_P(0)} = \int_0^\infty 2\sin^2\Big(\frac{xy}{2}\Big)\nu_P(x)\,dx, \tag{22.9.1}$$

where, with $\gamma_j^{(P)}$ the $R_P$ positive real zeros and $x_0 + iy_0$ the departed zeros in the first quadrant,

$$\nu_P(x) = \frac{2}{x}\Big[\sum_{j \leq R_P} e^{-\gamma_j^{(P)} x} + 2\sum_{\text{departed}} e^{-x_0 x}\cos(y_0 x)\Big].$$

For $\Xi$ under RH, $\nu(x) = \frac{2}{x}\sum_\gamma e^{-\gamma x} > 0$; this positive density is the starting point of the Thorin representation of the reciprocal $\Xi$ function [421]. In $\nu_P$ the departed zeros enter through oscillating terms.

Proposition 22.9.1. $\sup_{y\in\mathbb{R}} |\Xi_P(iy)/\Xi(iy) - 1|$ *is at most* $6.8 \times 10^{-13}$ *for* $P = 2$ *and at most* $3.4 \times 10^{-35}$ *for* $P = 3$. *In particular* $\Xi_P$ *has no zero on* $i\mathbb{R}$, *and the* $2\sin^2$ *exponents of* $\Xi_P$ *and* $\Xi$ *agree uniformly to that accuracy.*

*Proof.* Theorem 22.2.1 is stated only for $|\operatorname{Im} t| \le \frac{3}{2}$, so its proof is redone on $i\mathbb{R}$. There $\cos(iyu) = \cosh(yu)$, and by the formula in that proof $\Xi_P(iy) - \Xi(iy) = 2(y^2 - \frac14)\sum_{m\notin\mathcal{F}(P)} \int_0^\infty e^{-\pi m^2 e^{2u}} e^{u/2}\cosh(yu)\,du$ (with the sign of $t^2 + \frac14 = \frac14 - y^2$ absorbed in the modulus), whose integrand is positive. Using $\cosh(yu) \le e^{|y|u}$ and the substitution $v = e^{2u}$ gives $|\Xi_P(iy) - \Xi(iy)| \le |y^2 - \frac14| \sum_{m\notin\mathcal{F}(P)} (\pi m^2)^{-s}\Gamma(s, \pi m^2)$ with $s = \frac12(|y| + \frac12)$, for every real $y$, and $\Xi(iy) = \xi(\frac12 + y) > 0$. The supremum of the quotient of this bound by $\xi(\frac12 + y)$ was computed at 30 digits on a grid of step 0.1 in $[0, 40]$: it is $6.75 \times 10^{-13}$ near $y = 7.0$ for $P = 2$ ($q = 3$) and $3.37 \times 10^{-35}$ near $y = 6.9$ for $P = 3$ ($q = 5$), and it has fallen below $10^{-20}$ at $y = 40$ (the tail decays because $\xi(\frac12 + y)$ grows faster than $|y^2 - \frac14|\Gamma(s)$). Both suprema are less than 1, so $\Xi_P(iy) \neq 0$ (the supremum over the grid is computed, the bound itself is proved). □

The true suprema of the ratio, computed at 40 digits on a grid, are $6.05 \times 10^{-13}$ at $y = 6.85$ for $P = 2$ and $3.24 \times 10^{-35}$ at $y = 6.8$ for $P = 3$.

| $x$ | $\frac{x}{2}\nu_2(x)$ | $\frac{x}{2}\nu(x)$ | departed part |
|---|---|---|---|
| 0.03 | 6.6432 | 6.6432 | 3.8860 |
| 0.05 | 2.6860 | 2.6860 | 0.9924 |
| 0.1 | 0.62104 | 0.62104 | 0.06521 |
| 0.5 | $8.8375 \times 10^{-4}$ | $8.8375 \times 10^{-4}$ | $1.50 \times 10^{-9}$ |

Table 22.4: The $2\sin^2$ Lévy density of $\Xi_2$, from its 6 real and 190 traced departed zeros, against that of $\Xi$. $\nu_2 > 0$ on the whole computed range $[0.02, 3]$.

The departed zeros sit up to 0.34 radians off the real axis (the maximum, 0.343, is at $174.36 + 62.25i$). The six real zeros of $\Xi_2$ differ from the first six zeros of $\Xi$, mainly the sixth (37.495 against 37.586), and the real zeros of $\Xi_2$ together with the departed zeros reproduce $\nu$ to seven digits for $x \le 0.1$ (the difference of $\frac{x}{2}\nu_2$ and $\frac{x}{2}\nu$ is $3.4 \times 10^{-7}$ at $x = 0.03$, a bound set by cutting the chain of departed zeros at height 602, and at most $2 \times 10^{-12}$ at $x = 0.05$, 0.1, 0.5). The departed part alone reproduces the tail of $\nu$ beyond the zeros of $\Xi_2$ to three digits at $x = 0.03$ and to one digit at $x = 0.5$. On the computed range, $\nu_2$ is positive with minimum $2.6 \times 10^{-19}$ at $x = 3$.

Remark 22.9.2. Positivity of the $2\sin^2$ Lévy density does not force the zeros of an even entire function of order one to be real. For $f(t) = \cos t\,(1 - t^2/(10 + i)^2)(1 - t^2/(10 - i)^2)$ the factor by factor form gives

$$\nu(x) = \frac{2}{x}\Big[\sum_{k\ge0} e^{-(k+\frac12)\pi x} + 2e^{-10x}\cos x\Big] > 0 \qquad (x > 0),$$

since $\sum_k e^{-(k+\frac12)\pi x} > e^{-\pi x/2} + e^{-3\pi x/2} \ge 2e^{-\pi x} \ge 2e^{-10x} \ge -2e^{-10x}\cos x$. On the computed range $[0.02, 3]$ the friable approximant $\Xi_2$ behaves the same way:

$\nu_2 > 0$ there although $\Xi_2$ has infinitely many nonreal zeros. Whether $\nu_P \geq 0$ on all of $(0,\infty)$ is open (Problem 22.2).

What does see the departed zeros is the Thorin structure in $s = y^2$. We have

$$\log \frac{\Xi_P(i\sqrt{s})}{\Xi_P(0)} = \sum_{\operatorname{Re} t_0 > 0} \log\Big(1 + \frac{s}{t_0^2}\Big),$$

which is $\int \log(1 + s/u)\, U_P(du)$ with $U_P \geq 0$ on $(0,\infty)$ only if every $t_0^2 > 0$. Equivalently, $\nu_P(x) = \frac{2}{x}\int e^{-x\sqrt{u}}\, U_P(du)$ must come from a positive measure supported on the positive axis. The departed zeros place mass of $U_P$ at the complex points $x_0^2 - y_0^2 + 2ix_0y_0$. For $1/\xi_P(\frac12 + \sqrt{s})$ this means a pole at $s = -t_0^2 \notin (-\infty, 0]$, so by Theorem 22.2.4, for every finite $P$ and every $w \neq 0$ with $\sum_n |w_n| n^{-\sigma} < \infty$ for all $\sigma > 0$ (the weights $1_{\mathcal{F}(P)}$ included), $1/\xi_w(\frac12 + \sqrt{s})$ is not the Laplace transform of a generalised gamma convolution. By Remark 22.9.2 the sign of the Lévy density alone is not enough: a route to RH through the $2\sin^2$ form must use the support of the Thorin measure.

## 22.10 The cosh ladder

The universal factor experiments of Section 22.7 point to a constructive use of the same idea. For real $b$ let

$$F_b(z) = \int_{-\infty}^{\infty} \Phi(u)\cosh(bu) e^{izu}\, du = \tfrac12\big\{\Xi(z + ib) + \Xi(z - ib)\big\}, \tag{22.10.1}$$

which is entire for every $b$ because $\Phi$ decays double exponentially, even, real on $\mathbb{R}$, of order one, and satisfies $F_b(0) = 2\int_0^\infty \Phi\cosh(bu)\, du > 0$. With $\alpha = \frac12 + b$ and the variable $z$ real, $F_b(z) = \operatorname{Re}\xi(\alpha + iz)$; in the notation of Section 16.6 it is $A_\alpha(iz)$.

Theorem 22.10.1 (de Bruijn, Lagarias). *For every $b \geq \frac12$ all zeros of $F_b$ are real, and $s \mapsto F_b(0)/F_b(i\sqrt{s})$ is the Laplace transform of a generalised gamma convolution with Thorin measure $\sum_j \delta_{\gamma_j(b)^2}$, where $\pm\gamma_j(b)$ are the zeros of $F_b$. This holds unconditionally.*

The statement is Theorem 16.6.1(ii) and Corollary 16.6.2 of Section 16.6 (the simplicity of the zeros is Lagarias's theorem [321]); we keep the proof, which is short.

*Proof.* The zeros of $\Xi$ lie in $|\operatorname{Im} z| < \frac12$, the critical strip. By de Bruijn's strip theorem [112], all zeros of $\Xi(z+ib)+\Xi(z-ib)$ then lie in $|\operatorname{Im} z| \leq (\max(\frac14 - b^2, 0))^{1/2}$, so they are real for $b \geq \frac12$. Since $F_b$ is even of order one, Hadamard gives $F_b(z) = F_b(0)\prod_j (1 - z^2/\gamma_j(b)^2)$ with $\sum_j \gamma_j(b)^{-2} < \infty$. Hence $F_b(0)/F_b(i\sqrt{s}) = \prod_j (1 + s/\gamma_j(b)^2)^{-1}$, the Laplace transform of $\sum_j e_j/\gamma_j(b)^2$ with independent standard exponentials $e_j$. $\square$

Let $B_{\cosh} = \inf\{b \geq 0 : \ F_b \text{ has only real zeros}\}$. Then $B_{\cosh} \leq \frac{1}{2}$ unconditionally, and $B_{\cosh} = 0$ if and only if RH holds: RH gives $B_{\cosh} = 0$ because $F_0 = \Xi$, and conversely if $F_{b_n}$ is real rooted along a sequence $b_n \downarrow 0$, which exists because $B_{\cosh}$ is an infimum, Hurwitz's theorem passes real rootedness to the limit $\Xi$. The set of real rooted $b$ need not be an interval, and the argument does not use that. So RH is the statement that the ladder of Theorem 22.10.1 reaches the bottom rung. The constant $\Omega$ of Chapter 13 is defined through ferromagnetic cosine tilts for every $a \geq a_0$; both constants are at most $\frac{1}{2}$ and both vanish under RH. Two remarks fix the place of the ladder in the Thorin programme.

The first concerns the space below the bottom. Dividing by $\cosh(bu)$ instead of multiplying, the transform $g_b$ of Section 22.11 has nonreal zeros at a height that recedes as $b \to 0$, in every case computed (Section 22.7); for Gaussian factors the corresponding statement is the theorem of Rodgers and Tao [463]. If RH holds, $\Lambda = 0$ by their theorem, so along the de Bruijn Newman flow the reciprocal $\Xi$ sits at the edge of the generalised gamma convolution class.

The second concerns computation. One might hope to lower the unconditional bound $B_{\cosh} \leq \frac{1}{2}$ in the spirit of the Polymath bound [456] for the de Bruijn Newman constant. Section 22.14 proves $B_{\cosh} \leq \sigma_{\max} - \frac{1}{2}$, with $\sigma_{\max}$ the supremum of the real parts of the zeros, and that the paired version of the ladder bottoms out exactly at $\sigma_{\max} - \frac{1}{2}$; it conjectures that lowering $B_{\cosh}$ is, at large height, a quasi Riemann hypothesis (Conjecture 22.14.2). Any zero free half plane $\operatorname{Re} s > \sigma_0$ gives $B_{\cosh} \leq \sigma_0 - \frac{1}{2}$; the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 and recorded in Chapter 13 would give $B_{\cosh} \leq \frac{3}{8}$.

## 22.11 Miyatake's identity, read probabilistically

Throughout this and the next two sections $\Phi$ is the kernel of Section 22.7, with $\Xi(z) = \int_{\mathbb{R}} \Phi(u) e^{izu}\, du$; it is even, positive, analytic in $|\operatorname{Im} u| < \pi/4$ and decays like $\exp(-\pi e^{2|u|})$. For $b > 0$ put

$$g_b(z) = \int_{\mathbb{R}} \Phi(u) \operatorname{sech}(bu)\, e^{izu}\, du,$$

the transform of $\Phi/\cosh(bu)$ studied in Section 22.7. It is entire of order at most one, even and real on $\mathbb{R}$, and Miyatake's transform of $\psi_{\mathrm{M}}$ is $g(x) = G_{\mathrm{M}}(ix) = g_{9/2}(2x)/(4\pi^2)$.

Proposition 22.11.1 (Shift identity). *For every $b > 0$ and $z \in \mathbb{C}$, $\Xi(z) = \frac{1}{2}\{g_b(z + ib) + g_b(z - ib)\}$, that is $\Xi = \cos(b\partial) g_b$ with $\partial = d/dz$. Miyatake's identity* (22.7.1) *is the case $b = \frac{9}{2}$.*

*Proof.* Write $\Phi = \Phi \operatorname{sech}(bu) \cdot \frac{1}{2}(e^{bu} + e^{-bu})$; the integrals converge for all $z$. $\square$

So $g_b = \sec(b\partial)\Xi$ is $\Xi$ pushed backwards through the universal factor $\cos(b\partial)$, which is why $Q$ decreases while universal factors increase. Three classical objects describe the inverse map. Let $\mathcal{H}_\lambda(z) = \int_{\mathbb{R}} \Phi(u) e^{\lambda u^2} e^{izu}\, du$ be the de Bruijn Newman family in the normalisation of this chapter; in that of Section 8.4.1, $H_t(z) = \frac{1}{8}\mathcal{H}_{t/4}(z/2)$.

THEOREM 22.11.2 (Lévy area, Pólya Gamma clock, Thorin atoms). *Let* $\mathcal{A} = \frac{1}{2}\int_0^1 (W^1 dW^2 - W^2 dW^1)$ *be Lévy's stochastic area of a planar Brownian motion* $(W^1, W^2)$*, with* $\mathbb{E}e^{i\lambda\mathcal{A}} = \operatorname{sech}(\lambda/2)$*, and let* $\varpi \sim \mathrm{PG}(1,0)$ *be a Pólya Gamma variable, with* $\mathbb{E}e^{-s\varpi} = \operatorname{sech}\sqrt{s/2}$*. Then for* $x \in \mathbb{R}$

$$g_b(x) = \mathbb{E}\,\Xi(x + 2b\mathcal{A}) = \int_{\mathbb{R}} \Xi(x-y)\frac{1}{2b}\operatorname{sech}\frac{\pi y}{2b}\, dy = \mathbb{E}_\varpi\big[\mathcal{H}_{-2b^2\varpi}(x)\big],$$

*and* $2b\mathcal{A} \overset{d}{=} 2b\sqrt{\varpi}\,\mathcal{N} \overset{d}{=} \sum_{k\ge1} \frac{2b}{(2k-1)\pi} L_k$ *with* $\mathcal{N}$ *standard normal and* $L_k$ *independent standard Laplace variables. Hence* $\Xi \mapsto g_b$ *is convolution with an extended generalised gamma convolution whose Thorin measure has atoms at* $\pm(2k-1)\pi/(2b)$*, the odd multiples of* $\pi/9$ *for Miyatake.*

*Proof.* $\mathbb{E}e^{iu\,2b\mathcal{A}} = \operatorname{sech}(bu) = \mathbb{E}e^{-2b^2\varpi u^2} = \prod_k (1 + 4b^2u^2/((2k-1)\pi)^2)^{-1}$, the last equality being the product $\cosh x = \prod_k (1 + 4x^2/((2k-1)\pi)^2)$ inverted; insert each form into the definition of $g_b$ and use Fubini. □

So Miyatake's $\psi_{\mathrm{M}}$ lives on the negative side of the de Bruijn Newman flow, at a random time with mean $-b^2/2$ (as $\mathbb{E}\varpi = \frac{1}{4}$). The real zero counts of $g_b$ and of $\mathcal{H}_{-b^2/2}$ in $(0, 250)$ track each other: 94 and 96 at $b = 0.5$, 78 and 72 at 0.75, 51 and 51 at 1, 29 and 31 at 1.5, against 108 for $\Xi$ (computed).

In the Mellin variable the identity is a partition of unity. Let $\Upsilon > 0$ have $\log\Upsilon$ with density $2\Phi(u)e^{-u/2}$, so that $\mathbb{E}\Upsilon^s = 2\xi(s)$; this is the square root of the variable $V$ of Biane, Pitman and Yor [49], with $2\xi(s) = \mathbb{E}V^{s/2}$, realised through Brownian excursion and Bessel(3) functionals, and $\xi(s) = \xi(1-s)$ says that $\Upsilon^{-1}$ has the size biased law of $\Upsilon$.

PROPOSITION 22.11.3 (Logistic partition). *For* $c > 0$, $2\xi(s) = \mathbb{E}[\Upsilon^s/(1+\Upsilon^c)] + \mathbb{E}[\Upsilon^{s+c}/(1+\Upsilon^c)]$*; with* $c = 2b$ *and* $s = \frac{1}{2} + iz$ *the two terms are* $g_b(z+ib)$ *and* $g_b(z-ib)$*. The logistic weight is the Pólya Gamma mixture* $\Upsilon^c/(1+\Upsilon^c) = \frac{1}{2}\Upsilon^{c/2}\mathbb{E}_\varpi \exp(-\frac{1}{2}\varpi c^2(\log\Upsilon)^2)$ *[258].*

*Proof.* The first identity is $1/(1+\Upsilon^c) + \Upsilon^c/(1+\Upsilon^c) = 1$ under $\mathbb{E}\Upsilon^s = 2\xi(s)$. With $s = \frac{1}{2} + iz$ and $c = 2b$ the first term is $\int 2\Phi(u)e^{izu}/(1+e^{2bu})\, du = \int \Phi(u)\operatorname{sech}(bu)e^{-bu}e^{izu}\, du = g_b(z+ib)$, and the second is $g_b(z-ib)$ likewise. The mixture is the case $a = b = 1$ of the identity of Polson, Scott and Windle, $\operatorname{sech}(cL/2) = \mathbb{E}e^{-\varpi c^2 L^2/2}$ with $L = \log\Upsilon$. □

Miyatake's identity is therefore the Pólya Gamma augmentation of a Bernoulli likelihood with log odds $9\log\Upsilon$, the constant 9 being the one that removes the growth factor $e^{9u/2}$ of the leading term of $\Phi$. In the same language each term $n^4e^{-2\pi n^2\cosh t}$ of $\psi_{\mathrm{M}}$ is a generalised inverse Gaussian density in $r=e^t$, and Pólya's approximant is $8\pi^2K_{9/4}(2\pi)\,\mathbb{E}\cos(\frac{z}{2}\log\varrho)$ with $\varrho\sim\mathrm{GIG}(\frac{9}{4},2\pi,2\pi)$, so that $\mathbb{E}\varrho^{iy}=K_{9/4+iy}(2\pi)/K_{9/4}(2\pi)$; the law of $\varrho$ is hyperbolically completely monotone [56].

## 22.12 The transition at $b=2$

Theorem 22.12.1 (Exponential tail). *Let $b>2$, $\pi/(2b)<\tau<\min(\pi/4,3\pi/(2b))$ and $Y>0$. Then, uniformly for $\operatorname{Re}z\ge0$ and $|\operatorname{Im}z|\le Y$,*

$$g_b(z)=C_be^{-\pi z/(2b)}+O(e^{-\tau\operatorname{Re}z}),\qquad C_b=\frac{2\pi}{b}\,\Phi\Big(\frac{i\pi}{2b}\Big).$$

*For $b<2$ there is no residue and $g_b(x)=O(e^{-\tau x})$ for every $\tau<\pi/4$.*

*Proof.* Shift the contour from $\mathbb{R}$ to $\operatorname{Im}u=\tau$. This is allowed because $\Phi(u+i\sigma)$ decays doubly exponentially in $\operatorname{Re}u$ for $0\le\sigma\le\tau<\pi/4$ (the factor $\exp(-\pi e^{2|\operatorname{Re}u|}\cos2\sigma)$ with $\cos2\sigma>0$) and $\Phi$ is analytic in $|\operatorname{Im}u|<\pi/4$. The only pole of $\operatorname{sech}(bu)$ between the two lines is $u_0=i\pi/(2b)$, since $\tau<3\pi/(2b)$, and it lies between them since $\tau>\pi/(2b)$. The residue of $\operatorname{sech}(bu)$ at $u_0$ is $1/(b\sinh(bu_0))=1/(ib)$, so $2\pi i$ times the residue of the integrand is $\frac{2\pi}{b}\Phi(u_0)e^{izu_0}=\frac{2\pi}{b}\Phi(\frac{i\pi}{2b})e^{-\pi z/(2b)}$; here $\Phi(iy)$ is real. On the line $\operatorname{Im}u=\tau$ we have $|e^{izu}|\le e^{-\tau\operatorname{Re}z+Y|\operatorname{Re}u|}$, $\operatorname{sech}(bu)$ is bounded because $\tau$ is not an odd multiple of $\pi/(2b)$, and $\Phi(v+i\tau)e^{Y|v|}$ is integrable in $v$; this gives the error term. For $b<2$ the pole $i\pi/(2b)$ lies outside the strip $|\operatorname{Im}u|<\pi/4$ of $\Phi$, so the shift crosses no pole and gives $O(e^{-\tau x})$ for every $\tau<\pi/4$. □

The pole of $\operatorname{sech}(bu)$ enters the strip of $\Phi$ exactly when the Lévy area kernel $e^{-\pi|y|/(2b)}$ outlasts the decay $|y|^{7/4}e^{-\pi|y|/4}$ of $\Xi$, so the two thresholds of Section 22.7 coincide. For Miyatake, $C_{9/2}=\frac{4\pi}{9}\Phi(i\pi/9)=3.5527303732$, and in the scale of $g$ the constant is $C_{9/2}/(4\pi^2)=0.0899917116445$, matched by $g(x)e^{2\pi x/9}$ at $x=40,60,80$. The sign is rigorous: on the imaginary axis the $n$th term of $\Phi(i\pi/9)$ is at most $2(2\pi^2+3\pi)n^4e^{-2.4066n^2}$ (the exponent is $\pi\cos(2\pi/9)=2.4065996$), which sums to less than $2\times10^{-6}$ over $n\ge3$, while the first two terms have real parts 2.50683 and 0.03762.

Theorem 22.12.2 (Cusp asymptotics). *With $\delta=\pi(b-2)/(4b)$, as $b\downarrow2$,*

$$C_b=-\frac{\pi^3}{32b}(2\delta)^{-1/2}\delta^{-4}e^{-\frac{\pi}{8}\cot\delta}\big(1+O(b-2)\big).$$

*Proof.* Use theta functions with nome $e^{i\pi\tau}$, so that $\vartheta_4(\tau)=\vartheta_3(\tau+1)$ and $\vartheta_4(\tau)=(-i\tau)^{-1/2}\vartheta_2(-1/\tau)$. With Jacobi's $\omega(x)=\sum_{n\ge1}e^{-\pi n^2x}$ of (1.2.3),

$\Phi(u) = 2e^{u/2}(2x^2\omega''(x) + 3x\omega'(x))$ with $x = e^{2u}$. Put $u = i\pi/(2b) = i(\frac{\pi}{4} - \delta)$. Then $x = ie^{-2i\delta}$ is near the cusp $x = i$. With $\varepsilon = 1 + ix = 2i\sin\delta\, e^{-i\delta}$ and $w = -i\varepsilon = 2\sin\delta\, e^{-i\delta}$ we have $1 + 2\omega(x) = \vartheta_3(ix) = \vartheta_4(\varepsilon) = w^{-1/2}\vartheta_2(-1/\varepsilon)$ and $1/\varepsilon = \frac{1}{2} - \frac{i}{2}\cot\delta$ exactly. The term $n = 0$ of $\vartheta_2(-1/\varepsilon) = 2\sum_{n\ge 0} e^{-i\pi(n+\frac{1}{2})^2/\varepsilon}$ is $2e^{-i\pi/8}e^{-(\pi/8)\cot\delta}$, and the derivative of its exponent with respect to $x$ is $-\pi/(4\varepsilon^2)$. Hence the leading part of $\omega''$ is $w^{-1/2}e^{-i\pi/8}e^{-(\pi/8)\cot\delta}\pi^2/(16\varepsilon^4)$, and that of $\Phi(u)$ is $2e^{u/2}\cdot 2x^2\omega''$. The phases are $e^{i\pi/8}e^{-i\delta/2}$ (from $e^{u/2}$), $e^{-4i\delta}$ (from $x^2$ up to the sign $-1$, as $x^2 = -e^{-4i\delta}$), $e^{i\delta/2}$ (from $w^{-1/2}$), $e^{4i\delta}$ (from $\varepsilon^{-4}$) and $e^{-i\pi/8}$; they multiply to exactly 1, so the leading term is real and negative, equal to $-\frac{\pi^2}{64}(2\delta)^{-1/2}\delta^{-4}e^{-(\pi/8)\cot\delta}(1 + O(\delta))$ (moduli: $4\cdot|w|^{-1/2}\cdot\frac{\pi^2}{16}|\varepsilon|^{-4}$ with $|w| = 2\sin\delta$, $|\varepsilon| = 2\sin\delta$). Multiplying by $2\pi/b$ gives the formula. The relative corrections are of the following orders: the second derivative of the exponent and the derivative of $w^{-1/2}$ give $O(\delta)$; the term $3x\omega'$ is relatively $O(\delta^2)$; the terms $n \ge 1$ of $\vartheta_2$ are relatively $O(e^{-\pi\cot\delta})$. Since $\delta = \pi(b-2)/(4b)$, $O(\delta) = O(b-2)$. Here $\delta$ is the distance of the pole $i\pi/(2b)$ from the edge $i\pi/4$ of the strip of $\Phi$. □

The exact modular series (the $\vartheta_2$ side) agrees with the asymptotics to ratios 0.9409, 0.9851, 0.9970, 0.9997 at $b = 2.02$, 2.005, 2.001, 2.0001 (where $C_b \approx -1.58\times 10^{-4324}$); the first ratio is reproduced by the direct series. By Theorem 22.12.2, $C_b < 0$ for $b$ close to 2, and $C_{9/2} > 0$, so $C_b$ has at least one zero in $(2, \frac{9}{2})$. A scan of $(2, 2.5)$ with step $10^{-5}$ and of $[2.47, 20]$ with step $10^{-3}$ finds exactly one sign change, at $b_0 = 2.4588641657$, with $C_b < 0$ on $(2, b_0)$ and $C_b > 0$ beyond; this uniqueness is computed on a grid and not certified (Problem 22.11).

COROLLARY 22.12.3. *For $b > 2$ with $C_b \ne 0$, $g_b$ has finitely many real zeros, infinitely many nonreal zeros, and is not in the Laguerre Pólya class. In particular Miyatake's $g$, for which $b = \frac{9}{2}$ and $C_{9/2} > 0$, is not.*

*Proof.* Finitely many real zeros follows from Theorem 22.12.1, since $g_b(x) = C_b e^{-\pi x/(2b)}(1 + o(1))$ has no sign change for large $|x|$ ($g_b$ is even). A Laguerre Pólya function has only real zeros, so if $g_b$ were in the class it would have finitely many zeros altogether; being of order at most one it would then be $P(z)e^{\alpha z}$ with $P$ a polynomial, evenness forces $e^{2\alpha z} = P(-z)/P(z)$, hence $\alpha = 0$, and a nonzero polynomial cannot tend to 0 on $\mathbb{R}$, while $g_b(0) > 0$ and $g_b(x) \to 0$. The same Hadamard argument shows that $g_b$ cannot have finitely many zeros, so it has infinitely many, all but finitely many nonreal. □

The nonreal zeros escape, heuristically, as follows. Writing $\Xi \approx E(\frac{1}{2} + iz) + E(\frac{1}{2} - iz)$ with $E(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$, the operator $\sec(b\partial)$ damps each term by about $2(x/2\pi)^{-b/2}$, and balancing against $C_b e^{-\pi z/(2b)}$ gives, with $L = \log(x/2\pi)$,

$$|\operatorname{Im} z| \approx b - \tfrac{1}{2} + \frac{2}{L}\Big[\frac{\pi(b-2)}{4b}x - \tfrac{3}{2}\log x + \log C_b - \tfrac{1}{2}\log 4\pi\Big]. \tag{22.12.1}$$

For $b = \frac{9}{2}$ this matches all 31 zeros of $g_{9/2}$ with real part below 160 (from 17.92 to 158.58) to within 0.14 to 0.27, about 0.15 beyond the first four; the first is $17.9168 + 10.3988i$, which in the scale of $g$ is the vertex $8.9584 + 5.1994i$ of the cone of Section 22.7. By (22.12.1) the cone opens like $x/\log x$, so de Bruijn's strip $|\operatorname{Im} w| \le \frac{9}{4}$ fails, already at the first zero (computed), and by a margin that grows along the computed zeros (Figure 22.1). The formula (22.12.1) is a heuristic (Problem 22.10).

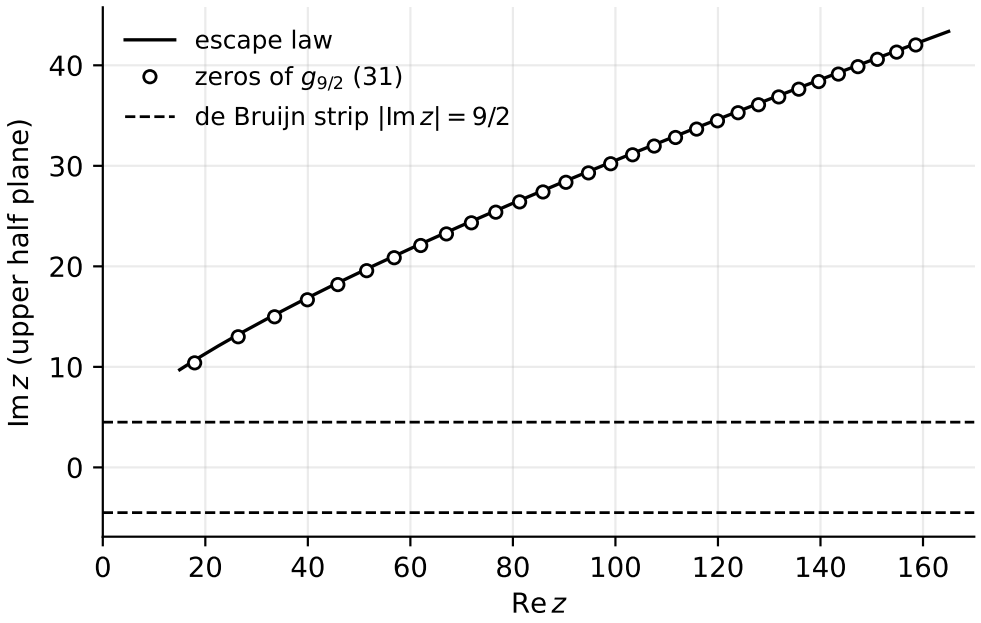



Figure 22.1: The 31 zeros of $g_{9/2}$ with real part below 160 in the upper half plane of $z$ (circles), the escape law (22.12.1) (curve), and the lines $|\operatorname{Im} z| = \frac{9}{2}$ bounding de Bruijn's strip $|\operatorname{Im} w| \le \frac{9}{4}$ in the scale $w = z/2$ of $g$ (dashed).

For $b < 2$ the root mean square de Bruijn radius of the Pólya Gamma clock, $(\mathbb{E}\, 4b^2\varpi)^{1/2} = b$, equals the strip width that de Bruijn's theorem needs for the shift identity. The strip nevertheless fails: below height 250 the largest imaginary parts of zeros of $g_b$ are 0.367, 1.258, 2.153 for $b = 0.5$, 1, 1.5. Concentrating the clock does not help. Let $\varpi_h = \mathrm{PG}(h, 0)/h$, so that $\mathbb{E}e^{-s\varpi_h} = \cosh(\sqrt{s/2h})^{-h}$ with mean $\frac{1}{4}$; the corresponding weight at radius $b$ is $\operatorname{sech}(bu/\sqrt{h})^h$, and $h \to \infty$ is the deterministic flow $\mathcal{H}_{-b^2/2}$. At radius 1 the clocks PG(1), PG(4), PG(16) and the deterministic flow $\mathcal{H}_{-1/2}$ give 28 nonreal pairs in $[0, 250] \times [-3, 3]$ for each (only 27 of the pairs of PG(4) have imaginary part above 0.125) and largest imaginary parts 1.258, 1.331, 1.322, 1.325. The obstruction is the backward direction of the flow, for which de Bruijn's radius has no counterpart.

## 22.13 The Pólya mixture of $\psi_{\mathrm{M}}$

This section follows a note of Nicholas Polson merged into the book [435]. Write $c = \cosh t$. The map $t \mapsto \cosh t$ takes the strip $|\operatorname{Im} t| < \frac{\pi}{2}$, modulo $t \mapsto -t$, one to one onto the half plane $\operatorname{Re} c > 0$, and $\psi_{\mathrm{M}}(t) = \Phi(t/2)/(8\pi^2 \cosh(9t/4))$ is even, so $\psi_{\mathrm{M}}$ is a function of $c$ there. Without their factors $1 - R_{\mathrm{M}}$ the terms of (22.7.3) are $n^4 e^{-2\pi n^2 c}$, Laplace transforms in $c$ of atoms at $a = 2\pi n^2$, and by (22.7.2) they contribute to $\Xi$ the function $\xi^{(P)}$ of Section 22.7 with every $n$ admitted, which we write $\xi^{(\infty)}$. Each of its terms $8\pi^2 n^4 \operatorname{Re} K_{\frac{9}{4}+\frac{it}{2}}(2\pi n^2)$, the transform of $8\pi^2 n^4 \cosh(9u/2) e^{-2\pi n^2 \cosh 2u}$, has only real zeros: $\cosh(9u/2)$ is a universal factor [449], and $K_{\frac{it}{2}}(2\pi n^2)$, the transform of $e^{-2\pi n^2 \cosh 2u}$, has only real zeros [447]; for $n = 1$ this is $\Xi^*$ of Proposition 16.7.1.

Proposition 22.13.1 (The polar part). *As a function of $c = \cosh t$, $\psi_{\mathrm{M}}$ is analytic on $\operatorname{Re} c > 0$ except for a simple pole at $c_0 = \cos \frac{2\pi}{9}$, with residue*

$$r_0 = \frac{\sin \frac{2\pi}{9}}{18\pi^2}\, \Phi\Big(\frac{i\pi}{9}\Big) = \frac{\sin \frac{2\pi}{9}}{8\pi^3}\, C_{9/2} > 0.$$

*Hence, with $\psi_{\mathrm{E}}(t) = r_0\, e^{-2\pi(\cosh t - c_0)}/(\cosh t - c_0) = r_0 \int_{2\pi}^{\infty} e^{c_0 a} e^{-a \cosh t}\, da$,*

$$\psi_{\mathrm{M}}(t) = \sum_{n \ge 1} n^4 e^{-2\pi n^2 \cosh t} + \psi_{\mathrm{E}}(t) + \psi_{\mathrm{A}}(t), \tag{22.13.1}$$

*where $\psi_{\mathrm{A}}$ is analytic in $c$ on all of $\operatorname{Re} c > 0$.*

*Proof.* $\Phi(t/2)$ is analytic for $|\operatorname{Im} t| < \frac{\pi}{2}$ (Section 22.11), and an even analytic function of $t$ is an analytic function of $c$ near $c = 1$, the only critical value of cosh in the strip. In the strip the zeros of $\cosh(9t/4)$ are $t = \pm 2\pi i/9$; they are simple, both map to $c_0$, and $dc/dt = \sinh t = i \sin \frac{2\pi}{9} \neq 0$ at $t = 2\pi i/9$. The residue in $c$ is therefore $\Phi(i\pi/9)\, i \sin \frac{2\pi}{9}/(8\pi^2 \cdot \frac{9}{4} i)$, which is the formula, and $C_{9/2} = \frac{4\pi}{9}\Phi(i\pi/9) > 0$ by Section 22.12. The atoms are entire in $c$, and $\psi_{\mathrm{E}}$ has the same simple pole with the same residue and no other singularity, so $\psi_{\mathrm{A}}$ is analytic on $\operatorname{Re} c > 0$. The integral form is $\int_{2\pi}^{\infty} e^{-a(c-c_0)}\, da = e^{-2\pi(c-c_0)}/(c - c_0)$ for $\operatorname{Re} c > c_0$. □

So $r_0 = 0.009206406367$, computed in interval arithmetic, all digits shown certified (the script is `code/pg8_constants.py`), and the exponent $c_0$ of the exponential part depends only on the factor $\cosh(9t/4)$ of (22.7.2) and on the map $c = \cosh t$; $\Phi$ enters only through $r_0$. The lower limit $2\pi$, the position of the first atom, is a choice: any $a_0 > 0$ gives the same pole.

PROPOSITION 22.13.2 (The exponent is forced). *Let* $0 < \theta < \frac{\pi}{2}$, $a_0 > 0$, *and*

$$\begin{aligned}\mathcal{E}_\theta(z) &= \int_{\mathbb{R}} \frac{\cosh(9u/2)\, e^{-a_0(\cosh 2u - \cos\theta)}}{\cosh 2u - \cos\theta}\, e^{izu}\, du \\ &= \int_{a_0}^{\infty} e^{a\cos\theta}\, \tfrac{1}{2}\Big\{K_{\frac{9}{4}+\frac{iz}{2}}(a) + K_{\frac{9}{4}-\frac{iz}{2}}(a)\Big\}\, da,\end{aligned}$$

*the exponential mixture with exponent* $\cos\theta$ *of the Pólya functions* $\frac{1}{2}\{K_{\frac{9}{4}+\frac{iz}{2}}(a) + K_{\frac{9}{4}-\frac{iz}{2}}(a)\}$ *of Proposition 16.7.1. For* $\theta/2 < \tau < \frac{\pi}{4}$ *and* $Y > 0$, *uniformly for* $\operatorname{Re} z \geq 0$ *and* $|\operatorname{Im} z| \leq Y$,

$$\mathcal{E}_\theta(z) = \frac{\pi\cos(9\theta/4)}{\sin\theta}\, e^{-\theta z/2} + O(e^{-\tau \operatorname{Re} z}).$$

*If* $\theta \neq \frac{2\pi}{9}$, *then* $\mathcal{E}_\theta$ *has finitely many real zeros and infinitely many nonreal zeros, and it is not in the Laguerre Pólya class.*

*Proof.* Shift the contour from $\mathbb{R}$ to $\operatorname{Im} u = \tau$ as in the proof of Theorem 22.12.1: on $\operatorname{Im} u = \sigma$, $0 \leq \sigma \leq \tau < \frac{\pi}{4}$, the factor $e^{-a_0\cosh 2u}$ decays doubly exponentially, and the only pole between the two lines is $u_0 = i\theta/2$, where $\cosh 2u_0 = \cos\theta$; the next poles, $\pm i\theta/2 + i\pi\mathbb{Z}$, lie above $\frac{\pi}{4}$. $2\pi i$ times the residue at $u_0$ is $2\pi i\cos(9\theta/4)e^{-\theta z/2}/(2i\sin\theta)$. For $0 < \theta < \frac{\pi}{2}$, $\cos(9\theta/4) = 0$ only at $\theta = \frac{2\pi}{9}$. The kernel is positive, so $\mathcal{E}_\theta$ is even, real, of order at most one, positive at 0 and tends to 0 on $\mathbb{R}$, and the proof of Corollary 22.12.3 gives the rest. The Bessel form follows from $e^{-a_0(c-\cos\theta)}/(c-\cos\theta) = \int_{a_0}^{\infty} e^{-a(c-\cos\theta)}\, da$ for $c \geq 1$ and $\int_{\mathbb{R}} \cosh(9u/2)e^{-a\cosh 2u}e^{izu}\, du = \frac{1}{2}\{K_{\frac{9}{4}+\frac{iz}{2}}(a) + K_{\frac{9}{4}-\frac{iz}{2}}(a)\}$, which is $\int_{\mathbb{R}} e^{\nu v - a\cosh v}\, dv = 2K_\nu(a)$ after $v = 2u$. □

So among the exponential mixtures with exponent in $(0,1)$ only $c_0 = \cos\frac{2\pi}{9}$ can have only real zeros, and by Proposition 22.13.1 this is the exponent of the polar part of $\psi_{\mathrm{M}}$: with $a_0 = 2\pi$, $\Xi^{\mathrm{E}} = 8\pi^2 r_0\, \mathcal{E}_{2\pi/9}$ is the transform of $8\pi^2\cosh(9u/2)\psi_{\mathrm{E}}(2u)$, whose apparent pole at $u = i\pi/9$ is cancelled by the zero of $\cosh(9u/2)$, and by (22.7.2) and (22.13.1), $\Xi = \xi^{(\infty)} + \Xi^{\mathrm{E}} + \Xi^{\mathrm{A}}$ with $\Xi^{\mathrm{A}}$ the transform of $8\pi^2\cosh(9u/2)\psi_{\mathrm{A}}(2u)$.

PROPOSITION 22.13.3 (A Meixner convolution of a Pólya type function). *Let* $c_1 = \cos\frac{\pi}{9}$, *so that* $c_0 = 2c_1^2 - 1$, *and*

$$\mathcal{U}(u) = \frac{\cosh(9u/2)}{2(\cosh u - c_1)} = \cosh\tfrac{7u}{2} + 2c_1\cosh\tfrac{5u}{2} + (1+2c_0)\cosh\tfrac{3u}{2} + (1+2c_1)\cosh\tfrac{u}{2}, \tag{22.13.2}$$

*so that* $\cosh(9u/2)/(\cosh 2u - c_0) = \mathcal{U}(u)/(\cosh u + c_1)$.

*(i)* $\mathcal{U}$ *is a universal factor, and* $\Xi^{\mathcal{U}}(z) = \int_{\mathbb{R}} \mathcal{U}(u)e^{-2\pi\cosh 2u}e^{izu}\, du = \sum_j m_j\cos(\frac{b_j}{2}\partial)P$, *with* $(b_j, m_j) = (7,1), (5,2c_1), (3,1+2c_0), (1,1+2c_1)$ *and* $P$ *of* (8.7.1), *is in the Laguerre Pólya class.*

*(ii) For $0 \le \varphi < \pi$ the function $q_\varphi(x) = \frac{(1+\cos\varphi)\sinh(\varphi x)}{\sin\varphi\,\sinh(\pi x)}$, with $q_0(x) = 2x/\sinh(\pi x)$, is a probability density with $\int_{\mathbb{R}} q_\varphi(x)e^{iux}\,dx = \frac{1+\cos\varphi}{\cosh u+\cos\varphi}$ and variance $1/(1+\cos\varphi)$, and it is a Pólya frequency function. Moreover $\frac{1+\cos\varphi}{\cosh u+\cos\varphi} = \mathbb{E}e^{-\varsigma_\varphi u^2}$ with*

$$\varsigma_\varphi = \sum_{k\ge 0}\Big(\frac{e_k}{((2k+1)\pi-\varphi)^2} + \frac{e'_k}{((2k+1)\pi+\varphi)^2}\Big), \qquad \mathbb{E}\varsigma_\varphi = \frac{1}{2(1+\cos\varphi)},$$

*$e_k, e'_k$ independent standard exponential variables, so the transform of $k(u)(1+\cos\varphi)/(\cosh u+\cos\varphi)$ is $q_\varphi * \hat{k}$ for every kernel $k$ for which these integrals converge; for $k=\Phi$ it is $\mathbb{E}\,\mathcal{H}_{-\varsigma_\varphi}$, a de Bruijn Newman flow at a random negative time, and $\varphi=\frac{\pi}{2}$ is $g_b$ of Theorem 22.11.2 with $b=1$.*

*(iii)* $\Xi^{\mathrm{E}} = \dfrac{8\pi^2 r_0 e^{2\pi c_0}}{1+c_1}\; q_{\pi/9} * \Xi^{\mathcal{U}}$.

*Proof.* The identity: $\cosh 2u - c_0 = 2(\cosh u - c_1)(\cosh u + c_1)$, and multiplying the four term sum by $2(\cosh u - c_1)$ with $2\cosh u\cosh\frac{bu}{2} = \cosh\frac{(b+2)u}{2} + \cosh\frac{(b-2)u}{2}$ leaves $\cosh\frac{9u}{2}$, the other coefficients vanishing by $2+2c_0 = 4c_1^2$ and $4c_1c_0 = 1+2c_1$, the latter being $4c_1^3 - 3c_1 = \cos\frac{\pi}{3} = \frac{1}{2}$. (i) $\mathcal{U}(iw) = \cos\frac{9w}{2}/(2(\cos w - c_1))$, and the zeros $\pm\frac{\pi}{9} + 2\pi\mathbb{Z}$ of $\cos w - c_1$ are simple zeros of $\cos\frac{9w}{2}$, so $\mathcal{U}(iw)$ is an even real entire function of exponential type whose zeros, the remaining zeros of $\cos\frac{9w}{2}$, are real; it is in the Laguerre Pólya class, and $\mathcal{U}$ is a universal factor by Pólya's theorem [449]. $P$ has only real zeros [447], hence so has $\Xi^{\mathcal{U}}$, which is even of order one, so it is in the class. The shift form is $\int_{\mathbb{R}} k(u)\cosh(bu)e^{izu}\,du = \frac{1}{2}\{\hat{k}(z+ib) + \hat{k}(z-ib)\}$, as in (22.10.1), for $k(u) = e^{-2\pi\cosh 2u}$, whose transform is $P$, and $b = \frac{b_j}{2}$. (ii) For $0<\varphi<\pi$ the residue theorem on the rectangle of height $2\pi$, with poles at $i(\pi\mp\varphi)$, gives $\int_{\mathbb{R}} e^{ivx}\,dx/(\cosh x+\cos\varphi) = 2\pi\sinh(\varphi v)/(\sin\varphi\sinh(\pi v))$; Fourier inversion gives the pair, $\varphi\to 0$ gives $q_0$, the value 1 at $u=0$ the mass, and the coefficient of $u^2$ the variance. The bilateral Laplace transform $\int q_\varphi(x)e^{sx}\,dx = (1+\cos\varphi)/(\cos s+\cos\varphi)$, $|\operatorname{Re} s| < \pi-\varphi$, is the reciprocal of an even function of order one with the real zeros $\pm((2k+1)\pi\mp\varphi)$, so $q_\varphi$ is a Pólya frequency function by Schoenberg's theorem [474] (Theorem 25.10.3). The same zeros give $(\cosh u+\cos\varphi)/(1+\cos\varphi) = \prod_{k\ge0}(1+u^2/((2k+1)\pi-\varphi)^2)(1+u^2/((2k+1)\pi+\varphi)^2)$, and inverting each factor as in the proof of Theorem 22.11.2 gives $\mathbb{E}e^{-\varsigma_\varphi u^2}$, with the mean read off the coefficient of $u^2$. The convolution form is Fubini's theorem. (iii) Multiply kernels, using (22.13.2) and $1/(\cosh u + c_1) = (1+c_1)^{-1}\int q_{\pi/9}(x)e^{iux}\,dx$. □

For $0<\varphi<\pi$, $q_\varphi$ is the law of $(\log V - \log V')/(2\pi)$, where $V$ and $V'$ are independent with the inverted beta density $v^{a-1}/(B(a,1-a)(1+v))$ on $(0,\infty)$ and $a = \frac{1}{2} + \frac{\varphi}{2\pi}$: by Euler's beta integral and the reflection formula,

$\mathbb{E}e^{iu\log V/(2\pi)} = \cos\frac{\varphi}{2}/\cos\frac{\varphi+iu}{2}$, and the product of this with its complex conjugate is $(1+\cos\varphi)/(\cosh u+\cos\varphi)$. Each $\log V/(2\pi)$ is a Meixner variable [161], the logarithm of the local variance in the shrinkage priors of Polson and Scott [254]; for $\varphi=\frac{\pi}{9}$, $a=\frac{5}{9}$.

Three computations bear on $\Xi^{\mathrm{E}}$ and the pieces of (22.13.1) (computed, `code/pg8_E.py`, `code/pg8_meixner.py`, `code/pg8_main.py`). First, $\Xi^{\mathrm{E}}$ has 452 zeros in $[0.25,750]\times[-3,3]$, all real, with least gap 1.314 at height about 749, against 442.9 for $\frac{T}{2\pi}\log\frac{T}{2\pi e}$ evaluated between 50 and 750, where $\Xi^{\mathrm{E}}$ has 443 zeros. Second, the same convolution does not keep the zeros of $\Xi$ real: in $(0,250]$, $q_\varphi * \Xi$ has 80 real zeros for $\varphi=\frac{\pi}{9}$, 80 for $\varphi=0$ and 72 for $\varphi=\frac{\pi}{4}$, against 108 for $\Xi$ and 51 for $g_b$ with $b=1$, the case $\varphi=\frac{\pi}{2}$ (Section 22.11); for $\varphi=\frac{\pi}{9}$ the zeros in $[0.5,250]\times[-3,3]$ are real below height 95.5, and the first nonreal one is $95.520+0.2361\,i$. For $0\le\varphi\le\frac{\pi}{2}$, $(\cosh u+\cos\varphi)/(1+\cos\varphi)=\mathbb{E}\cosh(uX)$ with $\Pr(X=0)=\cos\varphi/(1+\cos\varphi)$ and $\Pr(X=\pm1)=1/(2(1+\cos\varphi))$, so these are instances of Problem 22.6, answered in the expected direction below height 250. Third, $\xi^{(\infty)}+\Xi^{\mathrm{E}}$ has 229 real zeros in $(50,500]$ and 261 zeros in $[50,500]\times[-3,3]$, that is 16 conjugate pairs of nonreal zeros, the first at $188.502+0.228\,i$, while every term of $\xi^{(\infty)}$ has only real zeros: real zeros of the pieces of (22.13.1) do not pass to their sum. The function $\xi^{(\infty)}$ alone has the same two counts, so $\Xi^{\mathrm{E}}$ moves the zeros slightly and changes neither count. The counts come from the argument principle at 160 bits with two quadratures agreeing to $10^{-36}$, and interval Newton encloses the two first nonreal zeros named above, $95.5202007665+0.2361064242\,i$ and $188.5020229515+0.2278349245\,i$, each as a simple zero up to the quadrature allowance. Heuristically the random negative time of Proposition 22.13.3(ii) pulls the close pairs of zeros of $\Xi$ off the line, as for $g_b$, while the zeros of $\Xi^{\mathcal{U}}$, like those of Pólya's $\Xi^*$, are close to evenly spaced.

## 22.14 The ladder and the quasi Riemann abscissa

The ladder $F_b$ of Section 22.10 is the opposite symmetrisation: $F_b=\cos(b\partial)\Xi$ while $g_b=\sec(b\partial)\Xi$, and in the Mellin variable $F_b(z)=\frac14\mathbb{E}[\Upsilon^s(\Upsilon^b+\Upsilon^{-b})]$ with $s=\frac12+iz$, the Rademacher Esscher weight in place of the logistic coin. On $\mathbb{R}$, $F_b(x)=\operatorname{Re}\xi(\alpha+ix)$ with $\alpha=\frac12+b$. Theorem 22.10.1 is due to de Bruijn and Lagarias, who also showed that on RH all rungs $b>0$ are real rooted (Theorem 16.6.1). We now locate the bottom rung $B_{\cosh}$. Let $\sigma_{\max}=\sup\{\operatorname{Re}\rho\}$ over the nontrivial zeros of $\zeta$, and let $\gamma_k(b)$ be as in Theorem 22.10.1.

Proposition 22.14.1. $B_{\cosh}\le\sigma_{\max}-\frac12$.

*Proof.* The zeros of $\Xi$ satisfy $|\operatorname{Im}t|\le\Delta=\sigma_{\max}-\frac12$. De Bruijn's strip theorem with $\Delta$ in place of $\frac12$ in the proof of Theorem 22.10.1 shows that the zeros of $F_b$ satisfy $|\operatorname{Im}z|\le(\max(\Delta^2-b^2,0))^{1/2}$, so $F_b$ is real rooted for $b\ge\Delta$ (Theorem 16.6.1(ii)). □

Numerically, below height 100 the number of real zeros of $\operatorname{Re}\xi(\alpha+it)$ is $29 = N(100)$ for every $\alpha \in \{0.5, 0.55, 0.6, 0.7, 0.8, 1, 1.25, 1.5\}$ tested, and the first twelve Thorin atoms $\gamma_k(b)$, which move continuously where the zeros are simple, decrease strictly in $b$ on the grid $\alpha = 0.50, 0.51, \dots, 1.00$ (Figure 22.2); the smallest gap $\gamma_{k+1} - \gamma_k$ is 1.77.

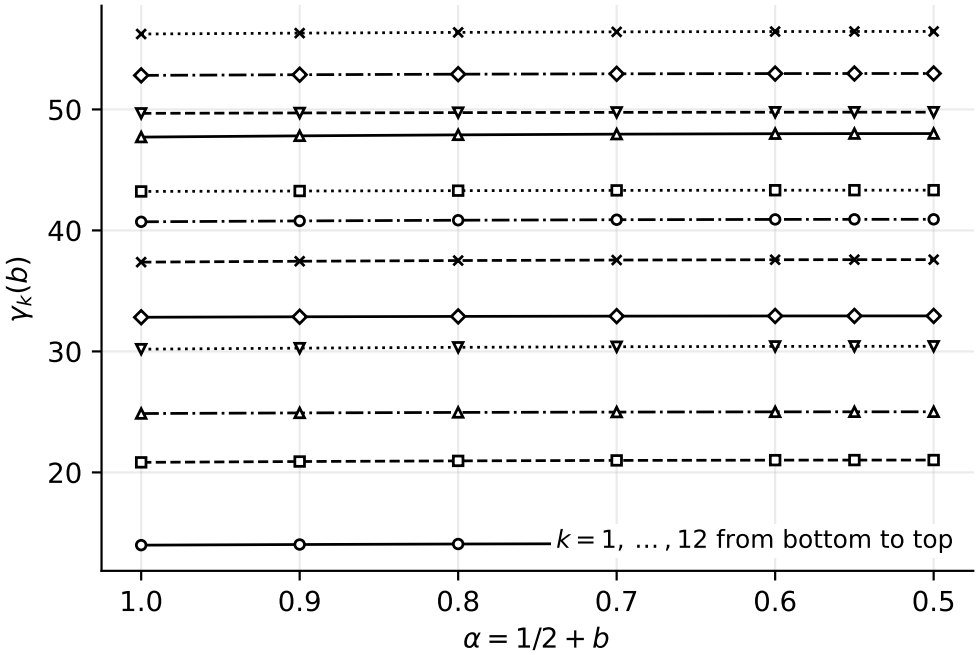



Figure 22.2: The Thorin path: the first twelve atoms $\gamma_k(b)$ against $\alpha = \frac{1}{2} + b$ at the stored values $\alpha = 1, 0.9, 0.8, 0.7, 0.6, 0.55, 0.5$, from the unconditional rung $\alpha = 1$ to the critical line.

Conversely, at height $x$ and $z = x + iy$ with $y > 0$, Stirling and the functional equation give

$$\frac{|\Xi(z+ib)|}{|\Xi(z-ib)|} \approx \Big(\frac{x}{2\pi}\Big)^{y} \frac{|\zeta(\alpha+y+ix)|}{|\zeta(\alpha-y+ix)|},$$

so the two terms of $F_b$ can cancel off the axis only near zeros of $\zeta$ with $\beta \approx \alpha + y$, and, heuristically, Rouché near a simple zero with $\beta > \alpha$ produces a zero of $F_b$ near $\gamma + i(\beta - \alpha)$. The cosh shift moves the effective abscissa by the fixed amount $b$; the heat flow, in the Rodgers Tao approximation, moves it by $\lambda \log(x/2\pi)$, which clears every zero at large height and is why Ki, Kim and Lee [310] have only finitely many nonreal zeros for every $\lambda > 0$ and the Polymath bound [456] is reachable.

Conjecture 22.14.2. *Let* $\sigma_{\max}^{\infty} = \limsup_{\gamma\to\infty}\beta$. *Then* $\sigma_{\max}^{\infty} - \frac{1}{2} \le B_{\cosh} \le \sigma_{\max} - \frac{1}{2}$. *The upper bound is Proposition 22.14.1; the lower bound is conjectured. Under the conjecture, lowering the unconditional bound* $B_{\cosh} \le \frac{1}{2}$ *is equivalent to a quasi Riemann hypothesis whenever* $\sigma_{\max}$ *is approached at large height.*

The mechanism can be tested on a function with genuine off line zeros. Let $\chi$ be the character modulo 5 with $\chi(2) = i$ and $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1) = 0.2840790438$. The Davenport Heilbronn function [111] is $f(s) = \frac{1-i\kappa}{2}L(s,\chi) + \frac{1+i\kappa}{2}L(s,\bar{\chi})$, completed as $\Lambda(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$; it satisfies $\Lambda(s) = \Lambda(1-s)$ (to $2.4\times10^{-29}$ at 30 digits) and is real on the critical line, and its first zero off the line is $0.80851718246 + 85.69934848538i$. The ladder built from $\Lambda$ has a nonreal pair above this zero for $\alpha$ up to about 0.798, slightly below $\beta$. The imaginary part of the pair is 0.2893, 0.2244, 0.1019, 0.0685, 0.0425 at $\alpha = 0.6$, 0.7, 0.78, 0.79, 0.795, and its square falls almost linearly near the collision (the slopes of the square are $-0.33$, $-0.50$, $-0.57$, $-0.57$ between consecutive values). The pair collides on the axis at $\alpha \approx 0.798$ (the double root is at $\alpha = 0.798095$, $x = 85.6255$), and is real for $\alpha \ge 0.80$, with the total count of zeros in the box $[83, 88.5]\times[-0.9, 0.9]$ conserved at 4. The small early collision is a finite height effect, since $(x/2\pi)^y$ is only $13.6^y$ at height 85; it is the reason for $\sigma_{\max}^{\infty}$ rather than $\sigma_{\max}$ in the lower bound.

### 22.14.1 An exact paired form

The ladder uses only the real part of $\xi$ on the vertical line. Pairing it with the imaginary part gives an exact statement. For $\alpha > \frac{1}{2}$ put $E_\alpha(z) = \xi(\alpha - iz)$, $A_\alpha = \frac{1}{2}(E_\alpha + E_\alpha^{\#})$ and $B_\alpha = \frac{i}{2}(E_\alpha - E_\alpha^{\#})$, with $E^{\#}(z) = \overline{E(\bar{z})}$, so that $A_\alpha(t) = \operatorname{Re}\xi(\alpha + it) = F_{\alpha-1/2}(t)$ and $B_\alpha(t) = \operatorname{Im}\xi(\alpha + it)$ for real $t$, and $E_\alpha = A_\alpha - iB_\alpha$. Here $A_\alpha$ is the even part of Section 16.6 evaluated at $iz$.

Theorem 22.14.3. *For* $\alpha > \frac{1}{2}$ *the following are equivalent: (i)* $\zeta(s) \ne 0$ *for* $\operatorname{Re} s \ge \alpha$; *(ii)* $E_\alpha$ *is Hermite Biehler, that is* $|E_\alpha(z)| > |E_\alpha^{\#}(z)|$ *for* $\operatorname{Im} z > 0$; *(iii)* $\operatorname{Re}\xi(\alpha + it)$ *and* $\operatorname{Im}\xi(\alpha + it)$, *as entire functions of* $t$, *have only real simple zeros, and these strictly interlace. Hence the paired bottom rung, the infimum of the* $\alpha - \frac{1}{2}$ *for which these hold, is exactly* $\sigma_{\max} - \frac{1}{2}$, *and* RH *holds if and only if (iii) holds for every* $\alpha > \frac{1}{2}$.

*Proof.* (i)⇒(ii): for $z = x + iy$ with $y > 0$ we have $E_\alpha(z) = \xi(\alpha + y - ix)$ and $E_\alpha^{\#}(z) = \xi(\alpha - y + ix)$. Let $\rho = \beta + i\gamma$ run over the zeros of $\xi$, a set symmetric under $\rho \mapsto \bar{\rho}$. By the Hadamard product, pairing $\rho$ with $\bar{\rho}$,

$$\frac{|E_\alpha(z)|^2}{|E_\alpha^{\#}(z)|^2} = \prod_{\rho} \frac{(\alpha + y - \beta)^2 + (x - \gamma)^2}{(\alpha - y - \beta)^2 + (x - \gamma)^2} = \prod_{\rho}\Big(1 + \frac{4y(\alpha - \beta)}{|\alpha - y + ix - \rho|^2}\Big),$$

and every factor exceeds 1 when $\beta < \alpha$. The product converges absolutely because $\sum\gamma^{-2} < \infty$, and $E_\alpha$ has no real zeros since no $\beta$ equals $\alpha$. Hence $|E_\alpha| > |E_\alpha^{\#}|$

on the upper half plane. (ii)⇒(iii) is the Hermite Biehler theorem in de Branges's form: if $E = A - iB$ is Hermite Biehler, then $A$ and $B$ are real entire with only real simple zeros that strictly interlace. (iii)⇒(i): $A_\alpha$ and $B_\alpha$ are real entire of order one with only real zeros, hence of genus at most one; $A_\alpha$ is even and $B_\alpha$ is odd, so their Hadamard products carry no factor $e^{az}$: $A_\alpha(z) = \xi(\alpha)\prod_k(1 - z^2/a_k^2)$ and $B_\alpha(z) = \xi'(\alpha)z\prod_k(1 - z^2/b_k^2)$ (without the parity the step fails, since interlacing alone allows an extra factor $e^{az}$). For strictly interlacing zeros the quotient of these products is, up to sign, a Pick function by Chebotarev's theorem [505]. Since $B_\alpha/A_\alpha \sim (\xi'(\alpha)/\xi(\alpha))z$ at 0, with $\xi(\alpha) > 0$ and $\xi'(\alpha) > 0$ ($\xi'(\alpha) > 0$ holds unconditionally for $\alpha > \frac12$, because $\xi'/\xi(\alpha) = \sum_\rho(\alpha - \rho)^{-1}$ is real and each pair $\rho$, $1 - \rho$ contributes $\operatorname{Re}\left[(\alpha - \rho)^{-1} + (\alpha - 1 + \rho)^{-1}\right] = (2\alpha - 1)(xy + \gamma^2)/((x^2 + \gamma^2)(y^2 + \gamma^2)) > 0$, with $x = \alpha - \beta$, $y = \alpha - 1 + \beta$ and $xy > -\frac14 > -\gamma^2$), the sign is +: $B_\alpha/A_\alpha$ is a Pick function and $\operatorname{Im}(B_\alpha/A_\alpha) > 0$ on the upper half plane. The identity $|E_\alpha^\#|^2 - |E_\alpha|^2 = -4|A_\alpha|^2\operatorname{Im}(B_\alpha/A_\alpha)$ then gives $|E_\alpha| > |E_\alpha^\#| \geq 0$ on the upper half plane, and $E_\alpha \neq 0$ on $\mathbb{R}$ because the zeros of $A_\alpha$ and $B_\alpha$ are disjoint. So $E_\alpha(z) = \xi(\alpha - iz)$ has no zeros in the closed upper half plane; since $E_\alpha(z) = 0$ at $z = -\gamma + i(\beta - \alpha)$, this says $\beta < \alpha$ for every zero. (The other sign is excluded by Hardy: $E_\alpha^\#(z) = \xi(\alpha + iz)$ vanishes at $\gamma + i(\alpha - \frac12)$ for every zero on the line, so it is not Hermite Biehler.) The last two statements follow. □

In the language of de Branges spaces this is a statement about the function $E_\alpha$; its Thorin form is the following. Since $A_\alpha$ is even with $A_\alpha(0) = \xi(\alpha) > 0$ and $B_\alpha$ is odd with $B_\alpha'(0) = \xi'(\alpha) > 0$,

$$\frac{2\xi(\alpha)}{\xi(\alpha + \sqrt{s}) + \xi(\alpha - \sqrt{s})} \quad \text{and} \quad \frac{2\sqrt{s}\,\xi'(\alpha)}{\xi(\alpha + \sqrt{s}) - \xi(\alpha - \sqrt{s})}$$

(these are $\xi(\alpha)/A_\alpha(i\sqrt{s})$ and $B_\alpha'(0)/(B_\alpha(z)/z)$ at $z = i\sqrt{s}$) are Laplace transforms of generalised gamma convolutions with strictly interlacing Thorin atoms exactly when $\zeta$ has no zeros with $\operatorname{Re} s \geq \alpha$. Call two such generalised gamma convolutions, whose Thorin atoms $a_k^2$ and $b_k^2$ strictly interlace, a Thorin pair: the even and odd parts of $\xi$ about $\alpha$ form a Thorin pair precisely to the right of every zero. For $\zeta$ the interlacing holds on $[0, 100]$ at $\alpha = 0.55$, 0.6, 0.7, 1 (and 0.8, 1.25, 1.5), with 29 positive zeros of each function and smallest distance between a zero of $A_\alpha$ and one of $B_\alpha$ equal to 0.50, 0.51, 0.54, 0.69 for $\alpha = 0.55$, 0.6, 0.7, 1. For Davenport Heilbronn the criterion is sharp where the ladder alone is not: at $\alpha = 0.800$ and 0.805 the real part already has only real zeros on $(83, 88.5)$, yet the zeros of the real (A) and imaginary (B) parts come in the order ABAABBAB, while at $\alpha = 0.82$ and 0.85 the order is ABABABAB.

### 22.14.2 Density one by phase counting

Let $\theta_\alpha(t) = \arg\xi(\alpha + it)$; for $\alpha > \sigma_{\max}$ this is $\pi$ times the clock $U_\alpha$ of Proposition 16.6.3. Since $\operatorname{Re}(\xi'/\xi)(s) = \sum_\rho \operatorname{Re}(s - \rho)^{-1}$ (Hadamard's product; the same

identity is used in the proof of Proposition 16.6.3),

$$\theta'_\alpha(t) = \sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(t-\gamma)^2},$$

a sum of Cauchy bumps of mass $+\pi$ for $\beta<\alpha$ and $-\pi$ for $\beta>\alpha$. Write $N^\pm(T)$ for the number of zeros with $|\gamma|\le T$ (so $N^\pm = 2N$ with the usual $N$ counting $0<\gamma\le T$) and $N^\pm_{>\alpha}(T)$ for those with $\beta>\alpha$.

THEOREM 22.14.4. *Let $\alpha>\frac12$ with $\beta\ne\alpha$ for every zero. Then $\theta_\alpha(T)-\theta_\alpha(-T) = \pi(N^\pm(T)-2N^\pm_{>\alpha}(T))+O(\log T)$. Hence $\operatorname{Re}\xi(\alpha+it)$ and $\operatorname{Im}\xi(\alpha+it)$ each have at least $N^\pm(T)-2N^\pm_{>\alpha}(T)-O(\log T)$ real zeros in $[-T,T]$. If moreover $F_b$, $b=\alpha-\frac12$, satisfies the Riemann von Mangoldt formula with error $O(\log T)$, its nonreal zeros with $|\operatorname{Re} z|\le T$ number at most $2N^\pm_{>\alpha}(T)+O(\log T)$.*

*Proof.* Integrating $\theta'_\alpha$ over $[-T,T]$, a zero $\rho$ with $|\gamma|\le T$ contributes $\operatorname{sgn}(\alpha-\beta)\,(\arctan\frac{T-\gamma}{|\alpha-\beta|}+\arctan\frac{T+\gamma}{|\alpha-\beta|})$, which is $\pm\pi$ up to a remainder of size $|\alpha-\beta|/(T-|\gamma|)$, and a zero with $|\gamma|>T$ contributes a remainder of size $|\alpha-\beta|/(|\gamma|-T)$, of the opposite sign. Zeros within distance 1 of $\pm T$ number $O(\log T)$ and contribute $O(\log T)$ in all. The remaining remainders sum to $O(\log^2 T)$ if estimated separately; pair $\rho$ with $1-\rho$, which has the same ordinate and combined weight $(\alpha-\beta)+(\alpha-1+\beta)=2\alpha-1$, independent of $\beta$. Up to $O(\log T)$ the remainders are then $(2\alpha-1)\sum_\gamma \operatorname{sgn}(|\gamma|-T)\big(\frac{1}{||\gamma|-T|}-\frac{1}{|\gamma|+T}\big)$, and partial summation with $N^\pm(t)=\frac{t}{\pi}\log\frac{t}{2\pi e}+O(\log t)$ shows that the contributions from below and from above $T$ cancel to $O(\log T)$. This proves the first formula. A continuous phase $\theta_\alpha$ with total change $\Delta$ crosses the lattice $\pi\mathbb{Z}$ (zeros of $\operatorname{Im}\xi(\alpha+it)$) and the lattice $\pi\mathbb{Z}+\frac{\pi}{2}$ (zeros of $\operatorname{Re}\xi(\alpha+it)$) at least $|\Delta|/\pi-1$ times each, with $\Delta=\theta_\alpha(T)-\theta_\alpha(-T)$. Under the Riemann von Mangoldt formula $F_b$ has $N^\pm(T)+O(\log T)$ zeros with $|\operatorname{Re} z|\le T$, of which at least $N^\pm(T)-2N^\pm_{>\alpha}(T)-O(\log T)$ are real. □

With Ingham's estimate [284], $N^\pm_{>\alpha}(T)=O(T^{3(1-\alpha)/(2-\alpha)}\log^5 T)=o(T)$ for every $\alpha=\frac12+b>\frac12$, since $3(1-\alpha)/(2-\alpha)<1$. So, unconditionally, $\operatorname{Re}\xi(\alpha+it)$ has at least $N^\pm(T)-o(T)$ real zeros in $[-T,T]$, against $N^\pm(T)\sim\frac{T}{\pi}\log\frac{T}{2\pi}$ (this is the lower bound of the theorem). If the rung $F_b$ satisfies the Riemann von Mangoldt formula with error $O(\log T)$, which is open (Problem 22.8), almost all zeros of $\frac12(\xi(s+b)+\xi(s-b))$ lie on the critical line, and as a count the number of conjugate pairs of nonreal zeros up to height $T$ is at most $N^\pm_{>\alpha}(T)+O(\log T)$. For the Davenport Heilbronn function, whose zeros off the line below height 200 have $\beta=0.8085,\ 0.6508,\ 0.5744,\ 0.7243$ (computed in $0.52\le\operatorname{Re}s\le1.5$, $0.1\le\operatorname{Im}s\le200$, at heights 85.70, 114.16, 166.48, 176.70), the real part of $\Lambda(\alpha+it)$ has 130, 130, 126, 124 real zeros on $(0.5,200)$ at $\alpha=0.95,\ 0.85,\ 0.7,\ 0.6$, where 0, 0, 2, 3 of those zeros lie to the right of $\alpha$: the deficits 4 and 6 are two real zeros per zero to the right, as the count predicts. A direct count of the nonreal zeros of $F_b$ was not run, so that the statement that each such zero costs one conjugate pair is established

at the level of these counts only. For $\zeta$ itself, the error $\theta_\alpha(T)-\theta_\alpha(-T)-\pi N^{\pm}(T)$ at $T=20,40,60,80,100$ stays below 1.6 for $\alpha=0.6,\ 0.75,\ 1$ (computed).

### 22.14.3 Interlacing across rungs

This subsection follows a note of Nicholas Polson merged into the book [436]. Theorem 22.14.3 interlaces $A_\alpha$ with $B_\alpha$ on one rung; for $\alpha>1$ the Euler product compares different rungs. Normalise $\theta_\alpha(0)=0$, write $M(\sigma)=-\zeta'(\sigma)/\zeta(\sigma)=\sum_n \Lambda(n)n^{-\sigma}$ for $\sigma>1$, so that $|(\zeta'/\zeta)(\sigma+it)|\le M(\sigma)$, and take $\arg\zeta(\sigma+it)=\operatorname{Im}\log\zeta(\sigma+it)$ with $\log\zeta(s)=\sum_n\Lambda(n)n^{-s}/\log n$.

Proposition 22.14.5. *Let $1<\alpha<\alpha'$, $h=\alpha'-\alpha$, $D=\theta_{\alpha'}-\theta_\alpha$,*

$$\mathcal{I}=\int_\alpha^{\alpha'} M(\sigma)\,d\sigma=\log\frac{\zeta(\alpha)}{\zeta(\alpha')},\qquad J=\sum_p\big(\arctan p^{-\alpha}-\arctan p^{-\alpha'}\big)\le\mathcal{I}.$$

*Then $D(t)=\frac{h\pi}{4}+\arg\zeta(\alpha'+it)-\arg\zeta(\alpha+it)+O(1/t)$, so $|D(t)-\frac{h\pi}{4}|\le\mathcal{I}+O(1/t)$.*

*(i) If $[\frac{h\pi}{4}-\mathcal{I},\frac{h\pi}{4}+\mathcal{I}]\subset(k\pi,(k+1)\pi)$ for an integer $k$, the zeros of $A_\alpha$ and $A_{\alpha'}$ alternate for large $t$, and so do those of $B_\alpha$ and $B_{\alpha'}$.*

*(ii) If $[\frac{h\pi}{4}-\mathcal{I},\frac{h\pi}{4}+\mathcal{I}]\subset(k\pi-\frac{\pi}{2},k\pi+\frac{\pi}{2})$ for an integer $k$, the zeros of $A_\alpha$ and $B_{\alpha'}$ alternate for large $t$, and so do those of $B_\alpha$ and $A_{\alpha'}$.*

*(iii) If a multiple of $\pi$ lies in $(\frac{h\pi}{4}-J,\frac{h\pi}{4}+J)$, the zeros of $A_\alpha$ and $A_{\alpha'}$ fail to alternate at arbitrarily large $t$; if an odd multiple of $\frac{\pi}{2}$ does, so do those of $A_\alpha$ and $B_{\alpha'}$.*

*Proof.* For $\alpha>1$, $\xi(\alpha+it)\neq0$ and $\theta'_\alpha>0$ by the display before Theorem 22.14.4, so $A_\alpha=|\xi|\cos\theta_\alpha$ and $B_\alpha=|\xi|\sin\theta_\alpha$ vanish exactly where $\theta_\alpha$ crosses $\frac{\pi}{2}+\pi\mathbb{Z}$ and $\pi\mathbb{Z}$. $D(t)$ is the integral of $\operatorname{Im}(\xi'/\xi)(\sigma+it)$ over $\alpha\le\sigma\le\alpha'$; by Stirling's formula the factor $\frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$ contributes $\frac{h\pi}{4}+O(1/t)$, and $|\int_\alpha^{\alpha'}\operatorname{Im}(\zeta'/\zeta)(\sigma+it)\,d\sigma|\le\mathcal{I}$. Also $J\le\mathcal{I}$ termwise: $\arctan x-\arctan y\le\log\frac{1-y}{1-x}$ for $0\le y\le x<1$, since $\frac{1}{1+u^2}\le\frac{1}{1-u}$, and with $x=p^{-\alpha}$, $y=p^{-\alpha'}$ the right sides sum to $\mathcal{I}$. Let $n_1$ and $n_2$ count the zeros in $(0,t]$ of the two functions compared, with phases $\phi_1=\theta_\alpha$ and $\phi_2=\theta_{\alpha'}$ in (i), $\phi_2=\theta_{\alpha'}-\frac{\pi}{2}$ in (ii). Under (i) or (ii), $\phi_2-\phi_1$ stays in one interval $(k'\pi,(k'+1)\pi)$ for $t\ge t_0$, so $n_2-n_1$ takes only two adjacent values there, which is alternation. For (iii), $\arg\zeta(\alpha'+it)-\arg\zeta(\alpha+it)=\sum_p\{\arg(1-p^{-\alpha}w_p)-\arg(1-p^{-\alpha'}w_p)\}$ with $w_p=p^{-it}$, a sum that converges absolutely and uniformly. By Kronecker's theorem [174, Chapter 23] for the numbers $\log p$ it comes within any $\varepsilon>0$ of its value $J$ at $w_p=-i$ for all $p$, and of its value $-J$ at $w_p=i$, at arbitrarily large $t$, and $D$ is uniformly continuous, since $|D'|\le M(\alpha)+M(\alpha')+o(1)$. So at arbitrarily large $t$ there are intervals of a fixed length on which $\phi_2-\phi_1$ lies below the critical multiple of $\pi$ by $\varepsilon$, followed by intervals on which it lies above it by $\varepsilon$.

Since $\phi_1$ advances by about $\frac{1}{2}\log\frac{t}{2\pi}$ per unit of $t$, $n_2 - n_1$ takes some value $m-1$ on the first and $m+1$ on the second; between the last visit to $m-1$ and the first visit to $m+1$ come two zeros of one function with no zero of the other between them. The pairs $(B_\alpha, B_{\alpha'})$ and $(B_\alpha, A_{\alpha'})$ are treated in the same way, with the phases $\theta_\alpha - \frac{\pi}{2}$ and $\theta_{\alpha'} - \frac{\pi}{2}$ or $\theta_{\alpha'}$. □

For $\alpha = 5$, $\alpha' = 11$, $\frac{h\pi}{4} = \frac{3\pi}{2}$ and $\mathcal{I} = 0.035768 < \frac{\pi}{2}$, so $A_5$ and $A_{11}$ alternate for large $t$ by (i), while $A_5$ and $B_{11}$ fail to alternate at arbitrarily large $t$ by (iii), which applies because $\frac{h\pi}{4} = \frac{3\pi}{2}$ is itself an odd multiple of $\frac{\pi}{2}$ and $J > 0$; no failure occurs below $t = 1000$ (computed). For $\alpha = 2$, $\alpha' = 3.9$, $\frac{h\pi}{4} = \frac{\pi}{2} - \frac{\pi}{40}$ and $J \geq \arctan\frac{1}{4} - \arctan 2^{-3.9} = 0.17809 > \frac{\pi}{40}$, so $A_2$ and $B_{3.9}$ fail to alternate at arbitrarily large $t$; in $(0, 100]$ they have 29 and 29 zeros, which alternate up to height 71.518, and the first failure is the zero 74.163 of $B_{3.9}$, the second of two consecutive zeros of $B_{3.9}$ with no zero of $A_2$ between them (computed on a grid, not certified; `code/pg10_pairs.py`). For $\alpha = 2$, (i) applies for $0 < h < 3.4004$ with $k = 0$ and for $4.6198 < h < 7.3682$ with $k = 1$, and (ii) for $0 < h < 1.5161$ with $k = 0$ and for $2.5695 < h < 5.3744$ with $k = 1$ (certified, `code/pg10_window.py`). The lower condition of (i) with $k = 0$, $\mathcal{I} < \frac{h\pi}{4}$, holds for every $h$ once $\alpha \geq \sigma_1 = 1.80732$, where $M(\sigma_1) = \frac{\pi}{4}$.

PROPOSITION 22.14.6. *(i) For $\alpha > 1$, $A_\alpha$ has $N(T) - S(T) + O_\alpha(1)$ zeros in $(0, T]$, where $N(T) = \frac{\vartheta(T)}{\pi} + 1 + S(T)$ counts the zeros of $\Xi$ with $0 < \gamma \leq T$. Hence the zeros of $A_\alpha$ do not alternate with the real zeros of $\Xi$ for large $t$.*

*(ii) For $a > 0$, $\operatorname{Re} K_{\frac{9}{4}+\frac{it}{2}}(a)$ has $\frac{T}{2\pi}(\log\frac{T}{a} - 1) + O_a(1)$ zeros in $(0, T]$. Hence two of these functions with different $a$ do not alternate for large $t$, and none alternates with the real zeros of $\Xi$ for large $t$ if $a \leq 2\pi$, or, under the Riemann hypothesis, if $a > 2\pi$.*

*Proof.* Zero sets that alternate for $t \geq t_0$ have counting functions that differ by $O(1)$. (i) By the proof of Proposition 22.14.5, $\theta_\alpha(t) = \vartheta(t) + \pi + (\alpha - \frac{1}{2})\frac{\pi}{4} + \arg\zeta(\alpha + it) + O(1/t)$ is increasing, $|\arg\zeta(\alpha + it)| \leq \log\zeta(\alpha)$, and $A_\alpha$ vanishes where $\theta_\alpha$ crosses $\frac{\pi}{2} + \pi\mathbb{Z}$; this gives the count. Alternation with the real zeros of $\Xi$, at most $N(T)$ in number, would give $S(T) \geq -O(1)$, but $S$ is unbounded in both directions [477]. (ii) For $\nu = \frac{9}{4} + \frac{it}{2}$, $K_\nu(a) = \frac{\pi}{2}(I_{-\nu}(a) - I_\nu(a))/\sin\nu\pi = \frac{1}{2}\Gamma(\nu)(\frac{a}{2})^{-\nu}(1 + O_a(1/t))$, so, as in the proof of Proposition 16.7.2, the zeros for large $t$ are the crossings of $\frac{\pi}{2} + \pi\mathbb{Z}$ by an increasing function $\arg\Gamma(\nu) - \frac{t}{2}\log\frac{a}{2} + O_a(1/t) = \frac{t}{2}\log\frac{t}{a} - \frac{t}{2} + \frac{7\pi}{8} + O_a(1/t)$. The counts of two members differ by $\frac{T}{2\pi}\log\frac{a'}{a} + O(1)$. Against $\Xi$: for $a < 2\pi$ the count exceeds $N(T)$ by $\frac{T}{2\pi}\log\frac{2\pi}{a} + O(\log T)$; for $a = 2\pi$ it is $N(T) - S(T) + O(1)$ and the argument of (i) applies; for $a > 2\pi$ it falls short of $N(T)$ by $\frac{T}{2\pi}\log\frac{a}{2\pi} + O(\log T)$, which excludes alternation when all zeros of $\Xi$ are real. □

For $a = 2\pi$ the function is $\Xi^*/(8\pi^2)$ of Proposition 16.7.1, whose zeros follow the half Gram points of Proposition 16.7.2.

### 22.14.4 Universal factors as vertical averages

Every universal factor acts as a vertical average, $\int_{\mathbb{R}} \Phi(u)\mathbb{E}\cosh(uX)e^{izu}\,du = \mathbb{E}\,\Xi(z+iX)$: the ladder is $X = \pm b$ and the de Bruijn Newman flow is $X = \sqrt{2\lambda}\mathcal{N}$ with $\mathcal{N}$ standard normal. Since $|\Xi(x+iu)|$ grows like $(x/2\pi)^{|u|/2}$, the average is dominated near the Esscher tilt location

$$u^*(L) = \arg\max_u \left[\tfrac{L}{2}u + \log p_X(u)\right], \qquad L = \log\frac{x}{2\pi},$$

which is $\lambda L$ for the Gaussian, Rodgers and Tao's effective abscissa, and $\operatorname{ess\,sup} X$ for bounded $X$. Below the asymptotic regime $\lambda L^2 \gg 1$ both flows obey the local model of a pair at $\pm i\eta$, $\eta = \beta - \frac{1}{2}$: on $z^2+\eta^2$, $\cos(b\partial)$ gives $z^2+\eta^2-b^2$ and $e^{-\lambda\partial^2}$ gives $z^2+\eta^2-2\lambda$, so the model predicts $b_c = \eta$ and $\lambda_c = \eta^2/2$. The heat flow of $\Lambda$ is the flow by $e^{\lambda u^2}$ of its kernel $\varphi(u) = 2(5/\pi)^{-1/2}e^{3u/2}\sum_{n\ge1} n\,(a\chi(n) + \bar{a}\bar{\chi}(n))\exp(-\pi n^2 e^{2u}/5)$, $a = \frac{1-i\kappa}{2}$, which is even and gives $\Lambda(\frac{1}{2}+ix)$ by its Fourier transform. Table 22.5 compares the model with three Davenport Heilbronn zeros.

| zero | $\eta$ | $L$ | $b_c$ | $b_c/\eta$ | $\lambda_c$ | $\lambda_c/(\eta^2/2)$ |
|---|---|---|---|---|---|---|
| $0.8085 + 85.699i$ | 0.3085 | 2.613 | 0.2981 | 0.966 | 0.0441 | 0.927 |
| $0.6508 + 114.163i$ | 0.1508 | 2.900 | 0.1504 | 0.997 | 0.0111 | 0.978 |
| $0.7243 + 176.702i$ | 0.2243 | 3.337 | 0.2190 | 0.977 | 0.0237 | 0.941 |

Table 22.5: Critical ladder rung $b_c$ and heat time $\lambda_c$ at which the pair above each Davenport Heilbronn zero becomes real. In every case $\lambda_c L^2 < 1$ (0.30, 0.09, 0.26), so these heights are pre asymptotic for the heat flow.

For $\zeta$, where an off line zero would lie above $3\times10^{12}$ [419] ($L \approx 27$), the heat flow would absorb a zero at $\lambda \approx \eta/L$, far below the local value $\eta^2/2$ once $\eta > 2/L$ (with $L = 26.89$, $2/L = 0.074$), while the ladder needs $b \approx \eta$. In this local model the cosh ladder, unlike the de Bruijn Newman constant, admits no Polymath style improvement short of a quasi Riemann hypothesis.

Remark 22.14.7 (Mixtures of rungs). For symmetric $X$, the vertical average $\mathbb{E}\,\Xi(z+iX) = \mathbb{E}F_{|X|}(z)$ is a positive mixture of rungs, and when $\mathbb{E}\cosh(uX) = \prod_k \cosh(\lambda_k u)$, so that $\mathbb{E}X^2 = \sum_k \lambda_k^2$, Proposition 13.5.2, de Bruijn's theorem, puts its zeros in $|\operatorname{Im} z| \le (\max(\frac{1}{4} - \mathbb{E}X^2, 0))^{1/2}$. Following [436], each of the following therefore has only real zeros, unconditionally: $\frac{1}{2}(\Xi + F_b)$ for $b \ge 2^{-1/2}$, from $X = \frac{b}{2}(\varepsilon_1+\varepsilon_2)$ with independent signs $\varepsilon_k$; $\frac{3}{4}F_b + \frac{1}{4}F_{3b}$ for $b \ge \frac{1}{2\sqrt{3}}$, from $X = b(\varepsilon_1+\varepsilon_2+\varepsilon_3)$; $\frac{1}{2}(F_{b_1} + F_{b_2})$ for $b_1^2 + b_2^2 \ge \frac{1}{2}$, from $\frac{1}{2}(\cosh b_1 u + \cosh b_2 u) = \cosh\frac{(b_1+b_2)u}{2}\cosh\frac{(b_2-b_1)u}{2}$; and $\frac{1}{b_2-b_1}\int_{b_1}^{b_2} F_b\,db$ for $b_1^2 + b_1b_2 + b_2^2 \ge \frac{3}{4}$, from $\frac{\sinh x}{x} =$

$\prod_{k\ge1}\cosh(2^{-k}x)$. Conversely $\Xi$ is no mixture $\int F_b\,\mu(db)$ with $\mu\ge0$ other than $\mu=\delta_0$: on the kernel side $\int\cosh(bu)\,\mu(db)\equiv1$ forces $\mu=\delta_0$. It is instead the hyperbolic secant smoothing of every rung, $\Xi=p_b*F_b$ with $p_b(x)=\frac{1}{2b}\operatorname{sech}\frac{\pi x}{2b}$, whose characteristic function is $\operatorname{sech}(bu)$; the same smoothing takes $\Xi$ to $g_b$ (Theorem 22.11.2). Beyond products of cosh the variance does not decide. For $\Pr(X=0)=\frac14$ and $\Pr(X=\pm2/\sqrt3)=\frac38$, $\mathbb{E}X^2=1$ and $\mathbb{E}\cosh(uX)=\frac14+\frac34\cosh\frac{2u}{\sqrt3}$ is a universal factor, by Pólya's theorem on universal factors [112, p. 197], since $\frac14+\frac34\cos\frac{2w}{\sqrt3}$ has only real zeros; yet for $p(z)=(z^2+1)^3$, with zeros at height 1,

$$\mathbb{E}\,p(z+iX)=z^6-12z^4+5z^2+\tfrac29$$

has two zeros on the imaginary axis, while the two laws of variance one in the local model above give $\cos(\partial)p=z^4(z^2-12)$ and $e^{-\partial^2/2}p=z^6-12z^4+30z^2-8$, both with only real zeros.

The local model above predicts the collision of a pair at $\pm i\eta$ from the variance alone. To the next order the shape of the law enters; the note [436] tests the model numerically, and the following statement makes the test precise.

Proposition 22.14.8 (One pair). *Let $G$ be a real entire function of order less than two, $a\in\mathbb{R}$ with $G(a)\ne0$, $\tau_j=G^{(j)}(a)/G(a)$ and $F_\eta(z)=((z-a)^2+\eta^2)G(z)$. Let $\ell$ be symmetric, bounded or standard normal, with $\mathbb{E}\ell^2=1$ and $m_4=\mathbb{E}\ell^4$, and put $d_c=\tau_1^2-\frac{m_4-1}{2}\tau_2$. For every $K>|d_c|$ there are $r>0$ and $\eta_0>0$ such that for $0<\eta<\eta_0$ and $|s^2-\eta^2|\le K\eta^4$ the function $\mathbb{E}F_\eta(z+is\ell)$ has exactly two zeros in $|z-a|<r$, and there is $s_c$ with*

$$s_c^2=\eta^2-d_c\,\eta^4+O(\eta^6)$$

*such that the two zeros are a nonreal conjugate pair for $s<s_c$, a double real zero at $a+\tau_1\eta^2+O(\eta^4)$ for $s=s_c$, and real and distinct for $s>s_c$.*

*Proof.* Put $z=a+\eta^2x$ and $s^2=\eta^2-\eta^4d$. Since $\ell$ is symmetric, $\mathbb{E}F_\eta(z+is\ell)$ is an analytic function of $z$ and $s^2$, and expanding $G(z+is\ell)=G(a)\sum_j\tau_j(z-a+is\ell)^j/j!$ with $\mathbb{E}\ell^2=1$ and $\mathbb{E}\ell^4=m_4$ gives

$$\frac{\mathbb{E}F_\eta(z+is\ell)}{G(a)\,\eta^4}=(x-\tau_1)^2+d-d_c+\eta^2R(x,d,\eta),$$

with $R$ analytic and bounded for bounded $x$, $d$ and small $\eta$; the expectation converges because $G$ has order less than two. As $\eta\to0$ the function tends to $(z-a)^2G(z)$ uniformly near $a$, so by Hurwitz's theorem it has exactly two zeros in a fixed disc about $a$, and they lie at bounded $x$, by Rouché's theorem against $(x-\tau_1)^2$ on $|x-\tau_1|=C$. At $\eta=0$ the system $P=\partial_xP=0$ for the right side has the nondegenerate solution $x=\tau_1$, $d=d_c$, which the implicit function theorem continues to small $\eta$. The function is real on $\mathbb{R}$, so the two zeros are real or

conjugate according to the sign of the discriminant; $(x_+ - x_-)^2 = 4(d_c - d) + O(\eta^2)$ is analytic in $d$ with $d$-derivative $-4 + O(\eta^2)$, so its only zero is $d_c(\eta) = d_c + O(\eta^2)$ (this is where $K > |d_c|$ is used), and its sign is that of $d_c(\eta) - d$. □

With $\ell = (\varepsilon_1 + \cdots + \varepsilon_n)/\sqrt{n}$, the ladder of $n$ independent cosh steps, $m_4 = 3 - \frac{2}{n}$, that is 1, 2 and $\frac{7}{3}$ for one, two and three steps, and $m_4 = 3$ for the Gaussian, the de Bruijn Newman flow. So the variance decides the collision to first order, as in the local model, and the law enters at order $\eta^4$ through its kurtosis. Table 22.6 tests the expansion on $\Xi$ itself, with its first zero moved to $\gamma_1 \pm i\eta$ and its mirror to $-\gamma_1 \pm i\eta$, so that $a = \gamma_1$ and $G(z) = \Xi(z)((z+\gamma_1)^2 + \eta^2)/(z^2 - \gamma_1^2)$, which is $\Xi(z)(z+\gamma_1)/(z-\gamma_1)$ times $1 + O(\eta^2)$ near $\gamma_1$; this factor changes the expansion in the proof of Proposition 22.14.8 only at order $\eta^6$, so the proposition holds with the $\tau_j$ of $\Xi(z)(z+\gamma_1)/(z-\gamma_1)$, $\tau_1 = -0.54426$ and $\tau_2 = 0.22728$ (computed, `code/pg10_collision.py`).

| law of $\ell$ | $m_4$ | $d_c$ | $\eta = 0.3$ | $\eta = 0.15$ | $\eta = 0.075$ |
|---|---|---|---|---|---|
| one cosh step | 1 | 0.2962 | 0.2827 | 0.2927 | 0.2953 |
| two cosh steps | 2 | 0.1826 | 0.1777 | 0.1813 | 0.1823 |
| three cosh steps | $\frac{7}{3}$ | 0.1447 | 0.1417 | 0.1439 | 0.1445 |
| Gaussian | 3 | 0.0689 | 0.0684 | 0.0688 | 0.0689 |

Table 22.6: The collision of the moved pair $\gamma_1 \pm i\eta$ of $\Xi$: $(\eta^2 - s_c^2)/\eta^4$ at three heights $\eta$, against its limit $d_c = \tau_1^2 - \frac{m_4 - 1}{2}\tau_2$ of Proposition 22.14.8 (computed; the three cosh cells at $\eta = 0.3$ are certified by a Krawczyk step).

At height $T$ the coefficients $\tau_1$ and $\tau_2$ are typically of size $L$ and $L^2$, $L = \log \frac{T}{2\pi}$, so the expansion describes pairs with $\eta L$ small, the regime $\eta < 2/L$ of the paragraph after Table 22.5, and it says nothing uniform in the height (heuristic).

## 22.15 Open problems

The computations of this chapter leave the following questions, from the friable approximants of Section 22.7 to the ladder of Section 22.14. The scripts that produce the numbers of the chapter are listed in `code/README.md` of the repository of the book.

Problem 22.1. Decide whether the heuristic of Remark 22.6.1 holds with a bounded error.

Problem 22.2. Assuming RH for $\Xi$, decide whether $\nu_P \geq 0$ on all of $(0, \infty)$ for every $P$. The angle $y_0/x_0 \sim (\pi/2)/\log x_0$ of the departed chain tends to zero, which suggests yes; the computed range is $[0.02, 3]$.

Problem 22.3. Prove Conjecture 22.5.1, or at least $|R_P - N(\tau_P^*)| = O(\log q)$ under RH and a lower bound for the peaks of Hardy's function $Z$ on negative humps.

Problem 22.4. Find a computable Thorin criterion that detects the departed zeros, for instance the failure of positivity of Hankel or moment matrices built from $\nu_P$ beyond a certain order, and use the friable approximants to calibrate it.

Problem 22.5. Prove that all zeros of $g^{(2)}$ lie in $|\operatorname{Im} w| < 2$, and deduce from de Bruijn's theorem that every zero of Miyatake's $\xi^{(2)}$ is real.

Problem 22.6. Prove that no factorisation $\Phi = U\chi$ with $U = \mathbb{E}\cosh(tX)$ a nontrivial universal factor has $\hat{\chi}$ in the Laguerre Pólya class, which would extend the theorem of Rodgers and Tao from Gaussian factors to Cardon's factors.

Problem 22.7. Make the local mechanism of Section 22.14 rigorous at all large heights, proving $B_{\cosh} \geq \sigma_{\max}^{\infty} - \frac{1}{2}$ in Conjecture 22.14.2; Theorem 22.14.3 already settles the paired version. Proposition 22.14.8 proves the mechanism for one pair as $\eta \to 0$ with the other zeros fixed; what is missing is uniformity in the height and in the other zeros.

Problem 22.8. Prove the Riemann von Mangoldt formula for the rungs $F_b$ with error $O(\log T)$, which makes the density one statement after Theorem 22.14.4 unconditional.

Problem 22.9. Prove a Ki, Kim and Lee theorem for the vertical average $\mathbb{E}\,\Xi(z + iX)$ whenever $X$ is unbounded, and show that for bounded $X$ the nonreal zeros at large height sit above zeros of $\zeta$ with $\beta > \frac{1}{2} + \operatorname{ess\,sup} X$.

Problem 22.10. Make the escape law (22.12.1) rigorous for $b > 2$, $C_b \neq 0$.

Problem 22.11. Certify, by interval arithmetic, that $C_b$ has exactly one zero $b_0$ on $(2, \infty)$: an interval Newton enclosure of $b_0 = 2.4588641657$, the sign of $\Phi(i\pi/(2b))$ on $(2, 20]$ by subdivision with a rigorous tail for the series over $n$, Theorem 22.12.2 with an explicit error near $b = 2$, and a proof that $\Phi(iy) > 0$ for $0 < y \leq \pi/40$ to cover $b > 20$.

Problem 22.12. Decide whether $\mathcal{E}_{2\pi/9}$ of Proposition 22.13.2, equivalently $q_{\pi/9} * \Xi^{\mathcal{U}}$ of Proposition 22.13.3, is in the Laguerre Pólya class, for $a_0 = 2\pi$ or for every $a_0 > 0$. That $q_{\pi/9}$ is a Pólya frequency function does not suffice: convolution with a symmetric probability density of variance $v$ sends $z^2 - r^2$ to $z^2 - r^2 + v$, which has nonreal zeros for $r^2 < v$, and the same convolution gives nonreal zeros for $\Xi$ (Section 22.13). A proof has to use more of $\Xi^{\mathcal{U}}$, for instance the spacing of its zeros.

*Notes.* The identity (22.7.1) and the function $\xi^{(P)}$ with its Theorem I are Miyatake's [217], and the symmetrised kernel with a secant pole term is that of his

paper A [216]; Hejhal [170] treats truncated symmetrised kernels. The ladder is that of de Bruijn and Lagarias, the shift identity and the Brownian and Mellin readings of Section 22.11 use the Biane Pitman Yor identity [49] and the Pólya Gamma mixture of Polson, Scott and Windle [258], the paired form uses Chebotarev's theorem [505], and the counting in Section 22.14.2 uses Ingham's density estimate [284]. The statements about Miyatake's two papers rest on their description by the authors of the chapter and have not been checked against the papers. Section 22.13 follows a note of Polson [435]; there the universal factor theorem is Pólya's [449], the reality of the zeros of $K_{iz}(a)$ is Pólya's [447], the Pólya frequency criterion is Schoenberg's [474], and the Meixner reading of $q_{\varphi}$ uses [161, 254]. Section 22.14.3, Remark 22.14.7 and the experiment behind Table 22.6 follow a note of Polson [436]; the strip bound for products of cosh is de Bruijn's [112], and the unboundedness of $S(T)$ in both directions is Selberg's [477].

# Chapter 23

# Haseo Ki and the Thorin programme

Haseo Ki and his coauthors have studied the objects this book computes with: Fourier transforms of rapidly decaying positive kernels, their zeros under the heat flow and under repeated differentiation, and zeta functions that share the functional equation of $\zeta$ but not its Euler product. The de Bruijn Newman flow of Section 8.4 and the Thorin condition of Chapter 1 meet in this work, and this chapter reads it through the Thorin dictionary: a zero of an even entire function is an atom of the Thorin measure of its reciprocal, and the question whether the reciprocal is a generalised gamma convolution (GGC) is the question whether the zeros are real.

Four results come out. The theorem of Ki, Kim and Lee on the de Bruijn Newman flow says that after any positive time the reciprocal of the flowed function is a GGC transform up to a finite rational factor, and de Bruijn's strip theorem confines the atoms of that factor to confocal parabolas that shrink with time (Section 23.2). Read together with the theorem of Rodgers and Tao, the flow gives an obstruction: no property that survives a Gaussian tilt of the Pólya density can imply RH (Proposition 23.2.3). Ki's theorem on repeated differentiation is a statement about moment tilts of the Pólya density, and it implies that under RH the reciprocals of the derivatives are GGC transforms whose clocks converge in law to the exit time of Brownian motion from an interval (Section 23.3); a conjecture, supported by computation, gives the rate at which the atoms converge. Ki's approximation theorems for Epstein type zeta functions and the Davenport Heilbronn functions indicate that a Thorin argument for RH must use the Euler product (Section 23.4). Ki's work on the zeros of $\zeta'$ concerns the spacing of the zeros (Section 5.4) and gives no route to positivity; his uniqueness theorems in the extended Selberg class and his work in descriptive set theory are not used.

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it, and heuristic, conditional or open as stated. Results of Ki and his coauthors are cited, not reproved.

## 23.1 The work of Ki and the Thorin dictionary

### 23.1.1 The strands of the work

The work of Ki and his coauthors falls into four strands that matter here. The first concerns the heat flow on Fourier transforms of rapidly decaying kernels: Ki and Kim extend a result of Newman on the zeros of Fourier transforms [197] and answer a question of de Bruijn [198], and Ki, Kim and Lee prove that for every positive time the flowed function has all but finitely many of its zeros real and simple [310]. The second concerns repeated differentiation: Ki and Kim settle the Fourier Pólya conjecture [309], and Ki studies the Riemann $\Xi$ function under repeated differentiation [193]. The third concerns approximate zeta functions, which keep the functional equation of a zeta function and lose its Euler product: approximations of the Epstein zeta function [191], modified Epstein zeta functions [192], approximations of the Ramanujan $\Xi$ function [194], Weng's zeta functions [196] and, with Komori and Suzuki, Weng zeta functions for Chevalley groups [199]. The fourth concerns the zeros of $\zeta'$ [195, 134, 154] and is taken up in Section 23.5.

### 23.1.2 Zeros as Thorin atoms

The Pólya kernel $\Phi$ and the function $\Xi(\theta) = \xi(\frac{1}{2} + i\theta) = \int_{\mathbb{R}} \Phi(u)\cos(\theta u)\,du$ are those of (1.2.3), and $X$ is the Pólya variable with density $\Phi/\Xi(0)$, so that $\Xi(\theta)/\Xi(0) = \mathbb{E}[e^{i\theta X}]$. With $s = -\theta^2$ this reads $\Xi(i\sqrt{s})/\Xi(0) = \mathbb{E}[\cosh(\sqrt{s}\,X)] = G(s)$, the function of Section 1.2, because $\xi(\frac{1}{2} - \sqrt{s}) = \xi(\frac{1}{2} + \sqrt{s})$. Ki's results concern real entire functions of this kind, and the following lemma carries the Thorin reading over to all of them.

Lemma 23.1.1. *(a) Let $F$ be an even entire function, real on $\mathbb{R}$, of order less than two, with $F(0) \neq 0$. Let $\pm w_k$ be its zeros, with multiplicity (one sign chosen for each pair, and $w_k \neq 0$), and put $\lambda_k = w_k^2$. Then $s \mapsto F(i\sqrt{s})$ is entire of order less than one and*

$$\frac{F(0)}{F(i\sqrt{s})} = \prod_k \Big(1 + \frac{s}{\lambda_k}\Big)^{-1},$$

*locally uniformly off the points $-\lambda_k$. If every $w_k$ is real, this is the Laplace transform of $T = \sum_k \lambda_k^{-1} e_k$, with $e_k$ independent standard exponential variables: a GGC with zero drift and discrete Thorin measure $\sum_k \delta_{\lambda_k}$. If some $w_k$ is not real, then $\lambda_k \notin (0,\infty)$, and $F(0)/F(i\sqrt{s})$ is not the Laplace transform of any GGC.*

*(b) Let $K$ be even and positive, with $K(u) = O(\exp(-|u|^p))$ for some $p > 2$, and let $F(z) = \int_{\mathbb{R}} K(u)e^{izu}\,du$. Then $F$ satisfies the hypotheses of (a).*

*Proof.* (a) The function $f(s) = F(i\sqrt{s})$ is entire because $F$ is even, and it has the same order as $F$ divided by two, so its order is less than one. Hadamard's theorem gives $f(s) = f(0)\prod_k (1 + s/\lambda_k)$ with convergent product, because an entire function of order less than one has genus zero. The product is the Laplace transform of $T$ when all $\lambda_k > 0$, since $\mathbb{E}e^{-se_k/\lambda_k} = (1 + s/\lambda_k)^{-1}$. A zero $w_k \notin \mathbb{R}$ has $w_k^2 \notin (0,\infty)$. A GGC transform $\exp\{-\int \log(1 + s/t)\,\mu(dt)\}$ with $\mu \geq 0$ on $(0,\infty)$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, while $1/f$ has a pole at $-\lambda_k \notin (-\infty, 0]$, of the order of the multiplicity of the zero, which no other factor cancels.

(b) $F(0) = \int K > 0$. The bound on $K$ gives $|F(x + iy)| \leq C \int e^{|y||u|-|u|^p} du \leq C' \exp(c_p |y|^{p/(p-1)})$, so $F$ has order at most $p/(p-1) < 2$, and $F$ is even and real. □

For $F = \Xi$ the zeros $\pm\theta$ are $\pm\gamma$ with $\gamma > 0$, and Lemma 23.1.1(a) gives $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s}) = \prod_{\gamma>0}(1 + s/\gamma^2)^{-1}$, the clock of Section 1.2. RH is the statement that its Thorin measure $\sum_{\gamma>0} \delta_{\gamma^2}$ is a positive measure on $(0,\infty)$. Special cases of the lemma, and the pole argument in its proof, appear in Propositions 1.2.1, 7.1.16 and 11.5.12; the general form is needed for the approximations of Section 23.4. The atoms of a hypothetical counterexample are confined by a simple map.

Lemma 23.1.2 (Strips become confocal parabolas). *For $\Delta > 0$ the map $w \mapsto w^2$ sends the strip $\{|\operatorname{Im} w| \leq \Delta\}$ onto the closed region to the right of the parabola $\mathcal{P}_\Delta = \{x + iy :\ x = y^2/(4\Delta^2) - \Delta^2\}$. Every $\mathcal{P}_\Delta$ has its focus at the origin, the parabolas are nested, and they collapse onto $[0,\infty)$ as $\Delta \downarrow 0$.*

*Proof.* The line $w = a + ib$, $a \in \mathbb{R}$, $0 < b \leq \Delta$, goes to $w^2 = (a^2 - b^2) + 2iab$, that is, to the parabola $x = y^2/(4b^2) - b^2$. For fixed $y \neq 0$ the right side decreases from $+\infty$ to $y^2/(4\Delta^2) - \Delta^2$ as $b$ increases from 0 to $\Delta$, so the images of the lines with $0 < b \leq \Delta$ fill the points with $x \geq y^2/(4\Delta^2) - \Delta^2$, $y \neq 0$. The real axis ($b = 0$) goes to $[0,\infty)$, and the points $x \in [-\Delta^2, 0)$ on the axis $y = 0$ are the images of $w = i\sqrt{-x}$. The lines with $b \leq \Delta$ lie in the strip and the same computation shows that the image has no point left of $\mathcal{P}_\Delta$. The focus of $x = y^2/(4\Delta^2) - \Delta^2$ is at $x = -\Delta^2 + \Delta^2 = 0$. For $\Delta' < \Delta$ the parabola $\mathcal{P}_{\Delta'}$ lies to the right of $\mathcal{P}_\Delta$ at every height $y \neq 0$, and for fixed $y \neq 0$ its abscissa tends to $+\infty$ as $\Delta' \downarrow 0$. □

## 23.2 The heat flow, the barrier and the Gaussian tilt

For real $t$ let $\Xi_t$ be the transform of the damped kernel $e^{tu^2/4}\Phi(u)$ of Section 8.7.6, so that $\Xi_t(\theta) = \int_{\mathbb{R}} \Phi(u) e^{tu^2/4} \cos(\theta u)\, du$ and $\Xi_0 = \Xi$. With $D = d/d\theta$ we have $D^2 \cos(\theta u) = -u^2 \cos(\theta u)$, hence $\Xi_t = e^{-(t/4)D^2}\Xi$: the backward heat operator is the Gaussian tilt of the Pólya density, and $\Xi_t(\theta) = 8H_t(2\theta)$ with $H_t$ the function of Section 8.4.1, so that the constants of that section carry over without change. The zeros of $\Xi$ lie in $|\operatorname{Im}\theta| < \frac{1}{2}$ and the zeros of $\Xi_t$ in $(\operatorname{Im}\theta)^2 \leq \max(\frac{1}{4} - \frac{t}{2}, 0)$ by de Bruijn's theorem [112, Theorem 13] (Theorem 19.4.1(c) with $r = 1$, since the zeros

of $H_0$ lie in $|\operatorname{Im} z| \le 1$); in particular $\Xi_t$ has only real zeros for $t \ge \frac{1}{2}$. Newman showed that there is a constant $\Lambda$ such that $\Xi_t$ has only real zeros exactly for $t \ge \Lambda$ [381], and Rodgers and Tao proved $\Lambda \ge 0$ [463]; RH is $\Lambda \le 0$. By Lemma 23.1.1(b), applied to $K = \Phi e^{tu^2/4}$, the atoms of the Thorin measure of $1/\Xi_t$ are the squares of the zeros of $\Xi_t$ in the upper half plane and on the positive axis.

### 23.2.1 The theorem of Ki, Kim and Lee in Thorin form

Ki, Kim and Lee proved that for every $t > 0$ the function $\Xi_t$ has infinitely many zeros, all but finitely many of them real and simple [310]; this gives $\Lambda < \frac{1}{2}$ and no height beyond which the zeros are real (Remark 14.15.1 records the same statement). Lemma 23.1.1 turns it into a statement on the Thorin measure.

Proposition 23.2.1 (Eventually Thorin). *For every $t > 0$ there are a finite, conjugation invariant multiset $E_t \subset \mathbb{C} \setminus (0, \infty)$ and a GGC Laplace transform $L_t$, with discrete Thorin measure on $(0, \infty)$ whose atoms are simple beyond a finite level, such that*

$$\frac{\Xi_t(0)}{\Xi_t(i\sqrt{s})} = L_t(s) \prod_{\eta \in E_t} \Big(1 + \frac{s}{\eta}\Big)^{-1}.$$

*$\Xi_t$ has only real zeros exactly when $E_t = \emptyset$.*

*Proof.* The kernel $\Phi e^{tu^2/4}$ is even and positive and decays like $\exp(-\pi e^{2u}/2)$, so Lemma 23.1.1(b) applies to it for every $p$. Split the zeros of $\Xi_t$ into the real ones and the finitely many nonreal ones, which by the theorem of Ki, Kim and Lee form a conjugation invariant set. The nonreal zeros $w$ give the multiset $E_t = \{w^2\}$, which misses $(0, \infty)$, and the real zeros give $L_t$ by Lemma 23.1.1(a). Only finitely many real zeros are multiple. □

For $t > 0$ the tail of the Thorin measure of the reciprocal is therefore positive, and only a finite rational factor has to be settled by computation. The case $t = 0$ of the identity would be the clock of $\xi$ with $E_0$ the multiset of squared off line zeros, and RH is the statement that $E_0$ is empty. The barrier of the Polymath project is a separate device, for the zeros that enter the strip from large heights (Chapter 15); the Polymath project proved $\Lambda \le 0.22$ [456] with an effective version of the theorem and a barrier, and Platt and Trudgian improved this to $\Lambda \le 0.2$ [419].

De Bruijn's strip theorem says where $E_t$ can be; Ki and Kim [198] proved the form of it that allows finitely many exceptional zeros: if for every $\varepsilon > 0$ all but finitely many zeros of a real entire function $f$ of genus at most one lie in $|\operatorname{Im} z| \le \Delta + \varepsilon$, then for every $\varepsilon > 0$ all but finitely many zeros of $e^{-\lambda D^2} f$ lie in $|\operatorname{Im} z| \le \sqrt{\max(\Delta^2 - 2\lambda, 0)} + \varepsilon$. That statement is about all but finitely many zeros and does not locate the finitely many exceptions, so by itself it adds nothing to Proposition 23.2.1. What does locate $E_t$ is de Bruijn's theorem for all zeros.

Corollary 23.2.2 (Shrinking parabolas). *For $0 < t < \frac{1}{2}$ every Thorin atom of $\Xi_t$, in particular every $\eta \in E_t$, lies in the closed region to the right of $\mathcal{P}_{\Delta(t)}$, where $\Delta(t)^2 = \frac{1}{4} - \frac{t}{2}$. The parabolas share the focus 0 and collapse onto $[0, \infty)$ at $t = \frac{1}{2}$, where de Bruijn's theorem takes over.*

*Proof.* By Theorem 19.4.1(c) with $r = 1$, all zeros of $\Xi_t$ lie in $(\operatorname{Im}\theta)^2 \le \frac{1}{4} - \frac{t}{2} = \Delta(t)^2$, and an atom is the square of a zero. Lemma 23.1.2 gives the region. □

### 23.2.2 The Gaussian tilt obstruction

Read through Lemma 23.1.1, the theorem of Rodgers and Tao says that for every $t < 0$ the Thorin measure of $1/\Xi_t$ has atoms off $(0, \infty)$ (Proposition 23.2.3). This is a filter on proof strategies.

Proposition 23.2.3 (No slack in the Gaussian direction). *Let $\mathcal{K}$ be a class of positive even kernels closed under $K \mapsto Ke^{-bu^2}$ for every $b > 0$. If every $K \in \mathcal{K}$ has a Fourier transform with only real zeros, then $\Phi \notin \mathcal{K}$.*

*Proof.* If $\Phi \in \mathcal{K}$, then $\Phi e^{-bu^2} \in \mathcal{K}$ for every $b > 0$, so $\Xi_t$ has only real zeros for every $t < 0$, that is, $\Lambda < 0$, contradicting $\Lambda \ge 0$ [463]. □

The statement is Proposition 8.7.16(a) and Proposition 11.5.12 in other words, and the remark that follows is what the proposition adds. Multiplication by $e^{-bu^2}$ adds $-2b$ to $(\log K)''$ and the linear function $-bv$ to $\log K(\sqrt{v})$. Every condition that bounds $(\log K)''$ from above is therefore preserved, including log concavity of $K(u)$ and of $K(\sqrt{v})$. No such condition can imply RH together with any further hypotheses that the tilt also preserves. For log concavity this is Proposition 11.5.12(ii); see also [385]. The same holds for any form of the sine squared lemma (eq. (1.2.9)) whose positivity hypothesis on the Lévy measure survives the corresponding tilt. A Thorin argument for RH must therefore use a property that $\Phi$ has and that $\Phi e^{-bu^2}$ loses for every $b > 0$.

Remark 23.2.4 (Newman's class). Newman [379] characterised the Lee and Yang single spin laws as those whose Fourier transform is in the Laguerre Pólya class (Proposition 7.1.16); the class is preserved by $e^{bt^2}$, $b > 0$. The law with density $\Phi/\Xi(0)$ belongs to it if and only if RH holds, and by Rodgers and Tao $\Phi e^{-bu^2}$ does not belong to it for any $b > 0$, so under RH the Pólya law lies on its boundary.

## 23.3 Derivatives and the Brownian exit time

Ki proved [193] that there are real constants $A_n$ and $\tau_n \to 0$ with

$$A_n \, \Xi^{(2n)}(\tau_n \theta) \to \cos\theta \tag{23.3.1}$$

locally uniformly on $\mathbb{C}$, where $\Xi$ is the function of Section 23.1 and $\Xi^{(2n)}$ its $2n$th derivative. The theorem is stated for the function $\xi(\frac{1}{2} + i\theta)$, which is the present

$\Xi$, so no rescaling of the variable is needed. Ki also treats the Xi functions of automorphic $L$ functions, and Gunns and Hughes extend the result to the Selberg class [164]. This section reads (23.3.1) as a statement about moment tilts of the Pólya density.

### 23.3.1 Moment tilts

Since $\Xi^{(2n)}(\theta) = (-1)^n \int_{\mathbb{R}} u^{2n}\Phi(u)\cos(\theta u)\,du$, the ratio $\Xi^{(2n)}(\theta)/\Xi^{(2n)}(0)$ is the characteristic function of $X_n$, the symmetric variable with density proportional to $u^{2n}\Phi(u)$. Write $\ell_n(u) = 2n\log u + \log\Phi(u)$ for the logarithm of the tilted density on $u > 0$. Because $\log\Phi(u) = \frac{9}{2}u - \pi e^{2u} + O(1)$ as $u \to +\infty$, the tilt concentrates $|X_n|$ near the mode $u_n$, the root of $2\pi e^{2u} = 2n/u + \frac{9}{2}$, so that $u_n \sim \frac{1}{2}\log n$; the convergence is slow (computed: $u_n = 3.4186$ at $n = 10^4$, against $\frac{1}{2}\log n = 4.605$). Setting $\theta = 0$ in (23.3.1) gives $A_n\Xi^{(2n)}(0) \to 1$, so (23.3.1) is equivalent to

$$\mathbb{E}\big[e^{i\theta\tau_n X_n}\big] \to \cos\theta \quad \text{locally uniformly on } \mathbb{C}: \tag{23.3.2}$$

$\tau_n X_n$ converges in law to a symmetric sign, and all exponential moments converge. Put $m_n = \mathbb{E}|X_n|$. The family $|\tau_n X_n|$ is uniformly integrable, because its exponential moments are bounded on compact sets by (23.3.2), so $\tau_n m_n \to 1$, and since the locally uniform convergence in (23.3.2) is stable under replacing $\tau_n$ by a factor tending to one, one may take $\tau_n = 1/m_n$.

Proposition 23.3.1 (Brownian exit time limit). *Let*

$$\psi_n(s) = \frac{\Xi^{(2n)}(0)}{\Xi^{(2n)}(i\sqrt{2s}/m_n)} = \frac{1}{\mathbb{E}[\cosh(\sqrt{2s}\,X_n/m_n)]}.$$

*Then $\psi_n(s) \to 1/\cosh\sqrt{2s}$ locally uniformly off the poles of the limit. The limit is $\mathbb{E}[e^{-sT}]$ for $T$ the exit time of standard Brownian motion from $(-1,1)$, a GGC with Thorin measure*

$$\sum_{k\ge 1} \delta_{c_k^2/2}, \qquad c_k = \tfrac{(2k-1)\pi}{2}.$$

*For each $K$ there is $n_K$ such that for $n \ge n_K$ the $K$ smallest Thorin atoms of $\psi_n$ are positive and simple and converge to $c_k^2/2$. If RH holds, each $\psi_n$ is the Laplace transform of a GGC $T_n$, and $T_n \to T$ in law.*

*Proof.* Evaluate the locally uniform limit (23.3.2), with $\tau_n = 1/m_n$, at $\theta = i\sqrt{2s}$. The formula $\mathbb{E}e^{-sT} = 1/\cosh\sqrt{2s}$ is classical, and $\cosh\sqrt{2s} = \prod_k (1 + s/(c_k^2/2))$. The statement about atoms is Hurwitz's theorem applied to the entire functions $s \mapsto 1/\psi_n(s)$: the zeros of $\cosh\sqrt{2s}$ are simple and real, so for large $n$ there is exactly one zero of $1/\psi_n$ in a small disc around each of them, and since $1/\psi_n$ is real on the real axis the conjugate zero lies in the same disc, so the zero is real. Under RH, $\Xi$ lies in the Laguerre Pólya class, which is closed under differentiation,

so $\Xi^{(2n)}$ has only real zeros; the kernel $u^{2n}\Phi(u)$ is even and positive with the decay required in Lemma 23.1.1(b), and Lemma 23.1.1(a) applies to each $\psi_n$. Convergence of Laplace transforms on $[0,\infty)$ to a transform that is continuous at 0 gives convergence in law. □

By Kent's theorem (Section 9.2, [305]) a GGC with discrete Thorin measure is the hitting time of a diffusion governed by a Krein string, and the Riemann string is the unknown of the inverse spectral problem. Under RH, Proposition 23.3.1 says that the hitting times of the strings of the scaled high derivatives converge in law to the exit time of Brownian motion from an interval, as the clocks of the de Bruijn Newman flow do (Proposition 8.4.19). Convergence of the strings themselves would need the continuity of Krein's inverse spectral correspondence, which is not used here. The statement about the $K$ smallest atoms is also implied by the local theorem of Ki and Kim recalled in Section 16.3.

### 23.3.2 The rate of convergence of the atoms

Write $|X_n| = m_n + V_n$, with $\mathbb{E}V_n = 0$, $\sigma_n^2 = \mathbb{E}V_n^2$ and $\kappa_n = \mathbb{E}V_n^3$ (the third central moment, which is the third cumulant). The following exact equation for the zeros leads to a conjecture for the rate. Let $\mathcal{A}_n(\theta) = \mathbb{E}\cos(\theta V_n)$ and $\mathcal{B}_n(\theta) = \mathbb{E}\sin(\theta V_n)$.

Lemma 23.3.2. *A real $\theta$ is a zero of $\Xi^{(2n)}$ if and only if*

$$\cos(\theta m_n)\,\mathcal{A}_n(\theta) = \sin(\theta m_n)\,\mathcal{B}_n(\theta).$$

*In particular, where $\sin(\theta m_n) \neq 0$ and $\mathcal{A}_n(\theta) \neq 0$, the zeros satisfy $\cot(\theta m_n) = \mathcal{B}_n(\theta)/\mathcal{A}_n(\theta)$.*

*Proof.* By symmetry of $X_n$, $\mathbb{E}\cos(\theta X_n) = \mathbb{E}\cos(\theta(m_n + V_n)) = \cos(\theta m_n)\mathcal{A}_n(\theta) - \sin(\theta m_n)\mathcal{B}_n(\theta)$. □

Now $\mathcal{A}_n(\theta) = 1 - \frac{1}{2}\theta^2\sigma_n^2 + \dots$ is even in the fluctuation and $\mathcal{B}_n(\theta) = -\frac{1}{6}\theta^3\kappa_n + \dots$ is odd. Near $\theta m_n = c_k$ we have $\cot(c_k + \delta) = -\tan\delta \approx -\delta$, so the shift of the $k$th zero is $\delta \approx -\mathcal{B}_n/\mathcal{A}_n \approx \kappa_n\theta^3/6$. The variance cancels to leading order: broadening the tilted law symmetrically does not move the zeros, and to first order only the third central moment does. This computation is heuristic. It suggests the following statement, which is supported by the numbers below and is not proved.

Remark 23.3.3 (Conjecture: the Edgeworth rate). As $n \to \infty$, for each fixed $k$ the $k$th positive zero $z_{k,n}$ of $\Xi^{(2n)}$ satisfies

$$z_{k,n}\,m_n = c_k + \frac{\kappa_n\,c_k^3}{6\,m_n^3}\big(1 + o(1)\big),$$

and, with $\tau_n = 1/m_n$,

$$\sup_{0\le s\le S}\Big|\psi_n(s) - \frac{1}{\cosh\sqrt{2s}}\Big| = \Big(\sup_{x\le\sqrt{2S}}\frac{x^2}{2\cosh x}\Big)\frac{\sigma_n^2}{m_n^2}\big(1 + o(1)\big).$$

For $S \geq 2.133$ the constant is $\max_x x^2/(2\cosh x) = 0.53222$, attained at $x = 2.06534$ (computed: the maximiser solves $\tanh x = 2/x$).

The second formula comes from $\mathbb{E}\cosh(x|X_n|/m_n) = \cosh x\,(1+x^2\sigma_n^2/(2m_n^2))+ O(\kappa_n/m_n^3)$. A saddle point expansion of $\ell_n$ at $u_n$ gives the orders of the cumulants (heuristic). At the mode, $\ell_n''(u_n) \approx -2n/u_n^2 - 4n/u_n - 9$ and $\ell_n'''(u_n) \approx 4n/u_n^3 - 8n/u_n - 18$, so that

$$\sigma_n^2 \approx -\frac{1}{\ell_n''(u_n)}, \qquad \kappa_n \approx \ell_n'''(u_n)\,\sigma_n^6,$$

and to leading order $\sigma_n^2 \approx u_n/(4n)$ and $\kappa_n \approx -u_n^2/(8n^2)$. The low atoms then move by $O\big(1/(n^2\log n)\big)$ while the Laplace transform converges at rate $O\big(1/(n\log n)\big)$. The leading forms are poor at the values of $n$ computed: at $n = 10^4$ they exceed the computed $\sigma_n^2$ by 15 percent and $|\kappa_n|$ by 57 percent, and the logarithmic factors are not visible, so the numbers below use the full saddle forms.

### 23.3.3 Numerical evidence

The script `code/ki_tilt.py` computes the tilted law by the trapezoid rule on a window around $u_n$, with $\log\Phi$ used to avoid underflow, in mpmath at 40 and 50 digits with two windows and steps for each $n$, its cumulants, the first six zeros of $\int u^{2n}\Phi(u)\cos(\theta u)\,du$, and the Laplace transform error on $0 \leq s \leq 20$. Its output for the normalisation of this book is `data/ki_tilt_book.json` ($n = 10, 10^2, 10^4$) and, for the chapter source's normalisation $\Phi_{\mathrm{dBN}}$ of Section 8.4.1, `data/ki_tilt_chapter.json` ($n = 5$ and $n = 10, 10^2, \dots, 10^6$); the two runs differ by the scaling $u \mapsto 2u$, which doubles $m_n$ and $\sigma_n$ and multiplies $\kappa_n$ by 8 and leaves $z_{k,n}m_n$, $\sigma_n/m_n$, $\kappa_n/m_n^3$ and the Laplace error unchanged. The rows for $n = 10^3$ and $10^6$ of Table 23.1 are those of the second file with $m_n$ and $\sigma_n$ doubled. The two runs for each $n$ agree to $4 \times 10^{-11}$ relative at worst ($n = 5$) and to $10^{-26}$ otherwise.

| $n$ | $m_n$ | $\sigma_n$ | $z_{1,n}m_n - c_1$ | Edgeworth | $z_{3,n}m_n - c_3$ | Edgeworth |
|---|---|---|---|---|---|---|
| 10 | 0.7832 | 0.1019 | $-4.01 \times 10^{-5}$ | $-4.00 \times 10^{-5}$ | $-5.28 \times 10^{-3}$ | $-5.00 \times 10^{-3}$ |
| 100 | 1.5319 | 0.0530 | $-1.70 \times 10^{-6}$ | $-1.70 \times 10^{-6}$ | $-2.12 \times 10^{-4}$ | $-2.12 \times 10^{-4}$ |
| $10^3$ | 2.4382 | 0.0224 | $-1.72 \times 10^{-8}$ | $-1.72 \times 10^{-8}$ | $-2.15 \times 10^{-6}$ | $-2.15 \times 10^{-6}$ |
| $10^4$ | 3.4185 | 0.0086 | $-1.50 \times 10^{-10}$ | $-1.50 \times 10^{-10}$ | $-1.87 \times 10^{-8}$ | $-1.87 \times 10^{-8}$ |
| $10^6$ | 5.4844 | 0.0011 | $-1.11 \times 10^{-14}$ | $-1.11 \times 10^{-14}$ | $-1.39 \times 10^{-12}$ | $-1.39 \times 10^{-12}$ |

Table 23.1: Computed shifts of the first and third positive zeros of $\Xi^{(2n)}$, rescaled by $m_n$, against the Edgeworth prediction $\kappa_n c_k^3/(6m_n^3)$ (computed, `code/ki_tilt.py`, 40 and 50 digits).

For $100 \leq n \leq 10^4$ the prediction agrees with the computed shift to within 0.02 percent for every $k \leq 6$ (at worst 0.018 percent, at $n = 100$). At $n = 10$ the worst $k$ gives 23 percent, and for small $n$ the higher zeros are not in the regime

$c_k^3|\kappa_n| \ll m_n^3$ where the expansion applies. At $n = 10^6$ the shift of the first zero is $-1.11 \times 10^{-14}$ in the mpmath runs and agrees with the prediction. The ratio of the sup of the Laplace transform error on $0 \le s \le 20$ to $\sigma_n^2/m_n^2$ rises from 0.514 at $n = 5$ through 0.5307 at $n = 10^2$ and 0.53220 at $n = 10^4$ to 0.53221 at $n = 10^6$, against the predicted 0.53222. At $n = 10^4$ the quadrature gives $\sigma_n^2 = 7.452 \times 10^{-5}$ and $\kappa_n = -9.27 \times 10^{-9}$, and the saddle point formulas displayed above agree with these values.

Two consequences matter for computation. The low Thorin atoms of $\Xi^{(2n)}$ are rigid: by $n = 100$ the first six rescaled zeros $z_{k,n}m_n$ lie within $3 \times 10^{-3}$ of the Brownian values $c_k$ (computed: at most $2.26 \times 10^{-3}$), and by $n = 1000$ within $3 \times 10^{-5}$ ($2.29 \times 10^{-5}$), and the deviation is governed, to first order, by the third central moment $\kappa_n$, a functional of the moments of $\Phi$. Because the variance drops out of this first order term, any model of $\Phi$ that reproduces the tilted mean and the third central moment reproduces the low atoms of the high derivatives to first order in $\kappa_n/m_n^3$; the variance enters at second order through $\mathcal{A}_n$.

Remark 23.3.4 (Scale mixtures survive the tilt). If $\Phi(u) = \int \sigma^{-1} g(u/\sigma)\, F(d\sigma)$ is a positive scale mixture, as in the uniform and arcsine mixtures of Sections 13.4.2 and 11.5.5.2, then

$$u^{2n}\Phi(u) = \int \sigma^{2n}\, \sigma^{-1} \big[(u/\sigma)^{2n} g(u/\sigma)\big] F(d\sigma)$$

is again a positive scale mixture, of the tilted kernel $u^{2n}g(u)$ with mixing measure reweighted by $\sigma^{2n}$. Concentration of the rescaled mixture needs concentration of every factor after scaling, since a product of independent positive factors concentrates only if each factor does; Remark 23.3.3 would then say how the third central moment of the reweighted mixture controls the atoms.

## 23.4 Approximate zeta functions and the Euler product

### 23.4.1 Ki's approximation theorems

Ki's 2005 paper truncates the Chowla Selberg expansion of an Epstein zeta function [191]. The partial sums keep the functional equation and have no Euler product, and Ki proves that in every strip containing the critical line all but finitely many of their zeros are simple and on the line. Later papers do the same for modified Epstein zeta functions [192] and for approximations of the Ramanujan $\Xi$ function attached to $\Delta$ [194]. Lemma 23.1.1(a) turns each of these into a Thorin statement once the approximation $F$ is even, real on the real axis, of order less than two in the variable, has $F(0) \neq 0$, and all but finitely many of all its zeros, not only those in a fixed strip, are real; positivity of a kernel is not needed for this step, and where the approximation is not known to be a Fourier transform of a positive kernel Lemma 23.1.1(b) does not apply and these hypotheses must

be checked. The conclusion is then that the reciprocal $F(0)/F(i\sqrt{s})$ is eventually Thorin in the sense of Proposition 23.2.1: a GGC transform up to a finite rational factor. The truncations and approximations $\zeta_N$ of Section 7.2 and the friable approximant of Chapter 22 belong to the same family, and Ki's method, an argument principle count set against a count of real zeros by sign changes, is a template for proving eventual Thorin statements about them.

### 23.4.2 The Davenport and Heilbronn obstruction

Davenport and Heilbronn [111] showed that the Epstein zeta function of a positive definite binary quadratic form of class number greater than one has zeros off the critical line (in the half plane $\operatorname{Re} s > 1$). Such a function has a functional equation of Riemann type, and its completed form has the inversion $t \mapsto 1/t$ of a theta function. Its reciprocal Xi function is not a GGC, by Lemma 23.1.1(a).

Remark 23.4.1 (What a Thorin proof must use, heuristic). Any property shared by $\xi$ and the Davenport Heilbronn functions cannot imply RH. The functional equation and the inversion of the theta function are shared. If the Pólya type kernel of such an Epstein zeta function is positive and has the same doubly exponential decay as $\Phi$, which is not established here, then positivity and decay of the kernel are shared as well. An argument that derives a positive Thorin measure for $1/\Xi$ from these properties alone would then apply equally to those functions and would be invalid. This indicates that a correct argument must use the Euler product, or another property that separates $\zeta$ from them. It is a heuristic principle, not a theorem.

The principle supports placing the Euler product Lévy representation, which writes $\log(1/\zeta)$ through the primes (Section 9.1.4, Section 5.3) and feeds that Lévy measure to the sine squared lemma, at the centre of the programme. Proposition 23.2.3 and Remark 23.4.1 point the same way: the Gaussian tilt keeps the positivity, evenness and decay of the kernel but not the theta inversion, while the passage to Epstein functions keeps the functional equation and the theta inversion, and positivity and decay only under the unproved hypothesis of the remark. An argument blind to the Euler product cannot work.

### 23.4.3 Weng zeta functions as a calibration set

Weng's high rank zeta functions are built from shifted completed zeta values with a functional equation of Riemann type; they have no Euler product. Ki [196], and Ki, Komori and Suzuki [199] for Chevalley groups, proved that their nontrivial zeros lie on the critical line. If that holds, the reciprocals of the corresponding Xi functions have positive Thorin measure by Lemma 23.1.1(a), and they are a test set for the numerical pipeline of this book (Chapter 3), not for an argument that uses the Euler product: compute the Thorin measure through the sine squared

lemma, confirm positivity and measure the margin. Because they have no Euler product, they are a different test from the one that Remark 23.4.1 describes.

## 23.5 Zeros of the derivative of zeta, open problems and notes

Ki showed, on RH, that $\liminf(\beta' - \frac{1}{2})\log\gamma' \neq 0$ over the zeros $\beta' + i\gamma'$ of $\zeta'$ holds exactly when $\zeta'/\zeta(s) = 1/(s-\rho) + O(\log t)$ near the line, with $\rho$ the nearest zero [195]. Farmer and Ki tie zeros of $\zeta'$ near the line to small gaps between zeros of $\zeta$ and to Landau Siegel zeros [134], and Gonek and Ki study the pair correlation of the zeros of the real and imaginary parts of $\zeta$ [154]. In Thorin language $\zeta'/\zeta$ is the derivative of the Lévy exponent, so Ki's criterion is a local regularity condition on the Lévy measure near the critical line, and in the Farmer and Ki dichotomy small gaps appear as nearly coalescing Thorin atoms. These results concern the spacing of the zeros (Section 5.4); they give no new route to positivity.

For the programme the chapter yields the following, in order of use. Proposition 23.2.3 (proved) says that no property of $\Phi$ that survives a Gaussian tilt can carry a proof of RH. Remark 23.4.1 (heuristic) says that an argument blind to the Euler product cannot work, and Ki's approximation theorems and the Weng functions supply test cases for the numerical pipeline. Proposition 23.3.1 (proved under RH, with Ki's theorem as input) and Remark 23.3.3 (conjecture, computed) say that under RH the scaled clocks of the high derivatives converge to the Brownian exit time and conjecturally their atoms move, to first order, only through the third central moment. Proposition 23.2.1 and Corollary 23.2.2 (proved, from the theorem of Ki, Kim and Lee and de Bruijn's strip theorem) put the theorem of Ki, Kim and Lee and the confocal parabola picture in Thorin form.

Problem 23.1 (Prove the Edgeworth rate). Prove the conjecture of Remark 23.3.3, with explicit error terms, by a saddle point analysis of $\ell_n$ and Lemma 23.3.2. The local theorem of Ki and Kim (Section 16.3) gives that for every $R$ the zeros of $\Xi^{(n)}$ in $|\theta| < R$ are real for large $n$, so that the number $K_n$ of atoms of $\psi_n$ known to be real tends to infinity; determine how large $K_n$ may be unconditionally. The expansion is governed by the condition $c_{K_n}^3|\kappa_n|/m_n^3 \to 0$, that is $K_n = o\big((n^2\log n)^{1/3}\big)$; reality of the atoms is a separate question.

Problem 23.2 (Descending the derivative ladder). Find an explicit $n_0$ and heights $T(n) \to \infty$ such that all zeros of $\Xi^{(2n)}$ below $T(n)$ are real for $n \geq n_0$. This is the effective form of the local theorem of Ki and Kim [309] and of Problem 16.4. It would give a sequence of certified positive Thorin measures computed from the moments of $\Phi$ alone, and a derivative analogue of the barrier.

Problem 23.3 (A property that breaks the obstruction). Identify a property of $\Phi$ used by the sine squared lemma that is destroyed by $\Phi \mapsto \Phi e^{-bu^2}$ for every $b > 0$ and that fails for the Pólya kernels of Epstein zeta functions of class number greater than one.

Problem 23.4 (Weng calibration). Compute the Thorin measures of reciprocal Weng Xi functions of rank two and three through the sine squared lemma and compare their margin of positivity with that of $\sum_{\gamma>0} \delta_{\gamma^2}$.

Problem 23.5 (Eventual Thorin for the friable approximant). Adapt Ki's argument principle and sign change method to prove that the reciprocal of the friable approximant of Chapter 22 is eventually Thorin, with an explicit bound on the exceptional set.

*Notes.* The theorems on the heat flow and on repeated differentiation are due to Ki and his coauthors as cited, to de Bruijn, Newman and to Rodgers and Tao; Proposition 23.2.3 is the theorem of Rodgers and Tao read through the Thorin dictionary, in the form of Proposition 8.7.16(a), and Remark 23.4.1 rests on the Davenport Heilbronn functions. The computations of Section 23.3.3 are those of `code/ki_tilt.py` and `code/ki_check.py` (data in `data/ki_tilt_book.json`, `data/ki_tilt_chapter.json` and `data/ki_check.log`), which compute the tilted laws in mpmath and take about two seconds for each $n$.

# Chapter 24

# Hitting times and the reciprocal xi function

The reciprocal of the normalised $\xi$ function is the Laplace transform of a probability law, the law of the clock $T$ of Section 1.2. Under the Riemann Hypothesis that law is the hitting time of a Krein string, the explosion time of a pure birth chain, and its density is Kent's eigenfunction expansion over the zeros. This chapter asks whether the argument can be run backwards: whether RH can be obtained from the limiting behaviour of hitting times. It shows where the content of such an argument has to lie. It tests the families of approximating laws that can be built without assuming anything about the zeros (Jensen laws, Gram smoothed zeros, truncated Euler products, boundaries of Brownian motion), proves a universal stable limit for the prime side hitting times and a rigidity theorem for factorisations of the clock law, and studies the density of the clock directly: its residue expansion, its log concavity, and the failure of hyperbolic complete monotonicity. No step proves RH.

The chapter develops the source chapter on hitting times and two notes, one on a residue expansion of the reciprocal $\xi$ function and one on the Euler product as a Lévy measure, all by Nick Polson [238, 251, 250]. Its numbers were recomputed for this book by scripts written from the mathematics alone (the scripts of the source were not used); every number printed below is one of those recomputations unless it is marked unchecked, and where the recomputation differed from the source the recomputed value is printed. Results that no independent computation reached are said to be unchecked. The notation is the book's: the entire function is $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ of Section 1.2, the clock is $T$ with law $\mu$ and Laplace transform $1/G$, the canonical function is the heat trace $W$ (Chapter 5), and the kernel of de Bruijn and Newman is $\Phi_{\mathrm{dBN}}$ of Section 8.4, not the kernel $\Phi$ of (1.2.3).

Several parts of the sources are already in the book and are cited, not restated: the self decomposability of $\mu$ and the equivalence of the GGC property with RH (Theorem 5.1.1 and Proposition 1.2.1), the spectral expansion of the density and the Kent approximants (Section 8.1), the criterion by Pólya frequency functions (Proposition 8.1.2), the Jensen ladder (Proposition 16.4.1), the closure of the GGC

class under weak limits (Corollary 4.1.2), the sine squared lemma (Lemma 6.2.5) and the monotonicity ladder of the heat trace (Section 5.1).

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it (code/README.md lists the scripts `ht_*.py` of this chapter), conditional, heuristic or open as stated. The results are these.

1. With no hypothesis on the zeros the density of $\mu$ is the limit of grouped residue sums, and its tail is $w_1 e^{-\gamma_1^2 y} + w_2 e^{-\gamma_2^2 y} + O(e^{-c_3 y})$ (Theorem 24.1.1, proved, and Proposition 24.1.3, proved). Absolute convergence of the residue series needs a lower bound for $|\xi'(\rho)|$ and is conditional (Theorem 24.1.2). Under RH the partial fraction series of $1/G$ does not converge absolutely, because the weights $|w_\gamma|$ grow like $e^{\pi\gamma/4}$ (Proposition 24.2.2, proved).

2. Truncating the Hadamard product gives GGC laws exactly while the zeros used lie on the line, and $\mu_N$ is a GGC for every $N$ up to the zeros below $3 \cdot 10^{12}$ (Proposition 24.5.1, proved given the verification of RH to that height [419]). The density is not HCM without RH, which is proved (HCM implies GGC, and GGC implies RH); the sum of the first 120 residue terms vanishes at a point $z_*$ off the positive axis (computed), which would prove that the density is not HCM if the omitted terms are controlled by an explicit remainder bound on a sector containing $z_*$ (Theorem 24.5.7), and non HCM is proved for every finite sum of two or more exponentials (same theorem).

3. The density of $\mu$ is log concave in the tail (Theorem 24.5.2, proved), on $[0.008, \infty)$ (Theorem 24.5.5, computed, with one numerical input not enclosed) and near the origin conditionally on a saddle point approximation (Proposition 24.5.3, conditional).

4. The Jensen polynomials of $G$ give GGC laws $\mu_d$ for every $d \le 2 \cdot 10^{17}$ (Proposition 24.6.1, proved given a theorem of Chasse that was not read), converging to $\mu$ at the rate $2.8/d$ in distribution function (computed, on a grid of $[0.007, 0.08]$). The resolution law $d^* = \gamma^2/(2t_c)$ for an off line pair is heuristic, and it reproduces the directly tracked Davenport and Heilbronn value to 0.1% (computed) (Section 24.6).

5. Every truncation of $\mu$ with real knots (zeros on the line, or under RH) is a B spline transform and a linear functional of the Gumbel extremal process (Proposition 24.7.1 and Theorem 24.7.2, proved). The smoothed zero laws are within 1.3% of $\mu$ in distribution function with no zero and within $2.5 \cdot 10^{-5}$ with ten zeros (computed). The Pólya density is not $\mathrm{PF}_5$ (computed).

6. The truncated Euler hitting times satisfy $T_{\mathbb{E}\log N_P} \stackrel{d}{=} Y_P + T_{\log N_P}$ with $Y_P$ a GGC, and after centring and scaling they converge to a spectrally positive

stable law of index $\frac{3}{2}$ that does not depend on the zeros (Proposition 24.8.1 and Theorem 24.8.4, proved; the rate is computed).

7. Factorisations $\mu = \mu_1 * \mu_2$ into two GGC laws are rigid (Theorem 24.9.1, proved). What this and the stable limit say about a hitting time proof built from the primes is a heuristic and is stated as one (Remark 24.10.1).

8. Hitting times of diffusions started at an entrance or reflecting end are GGC (Kent, proved); the identification of the limiting diffusion is open (Section 24.11).

The later sections treat curved boundaries and the inverse first passage problem (Section 24.12), Yor's probabilistic Riemann function (Section 24.13), the Davenport and Heilbronn test (Section 24.14), the frontier of what the reciprocal $\xi$ law proves (Section 24.15), the ladder of Taylor coefficients, what a hitting time proof would need, and open problems.

## 24.1 The law and its density

Let $u_\rho = (\rho - \frac{1}{2})^2$ for each zero $\rho = \frac{1}{2} + \delta + i\gamma$ of $\zeta$ in the upper half plane, so that $G(u) = \prod_{\gamma>0}(1 - u/u_\rho)$ is an entire function of order one half, and under RH $u_\rho = -\gamma^2$. With no hypothesis the function $1/G$ is the Laplace transform of a self decomposable law $\mu$ on $[0, \infty)$, the law of the clock $T$, whose canonical function is $W(x) = \sum_{\gamma>0} e^{u_\rho x}$; the law $\mu$ is a generalised gamma convolution (GGC) if and only if RH holds, and then its Thorin measure is $\sum_{\gamma>0} \delta_{\gamma^2}$, so that $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables $e_\gamma$ (Theorem 5.1.1 and Proposition 1.2.1). The mean of $T$ is $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231049931$ (Section 1.2).

The spectral expansion of the density $f$ of $\mu$ over the zeros is in Section 8.1.1, with the residues $R_k$ computed there. Here the residues are written $w_\gamma = 1/G'(-\gamma^2) = R_k$, and the point is what holds without any hypothesis on the zeros.

### 24.1.1 The partial fraction series diverges, the density series converges

If the zeros are simple, $1/G$ has the poles $u_\rho$ with residues $1/G'(u_\rho)$, and a Mittag Leffler expansion would read $1/G(u) = \sum_\rho 1/(G'(u_\rho)(u - u_\rho))$. It does not converge. Under RH, $G'(-\gamma^2) = \xi'(\frac{1}{2} + i\gamma)/(2i\gamma\,\xi(\frac{1}{2}))$, and the gamma factor of $\xi$ gives $|w_\gamma| \asymp \gamma^{-3/4} e^{\pi\gamma/4}/|\zeta'(\rho)|$, so the weights alternate in sign and are enormous: $w_{\gamma_1} = 1.0164 \times 10^4$, $w_{\gamma_2} = -1.1775 \times 10^6$, $w_{\gamma_5} = 8.0078 \times 10^9$, and $w_{\gamma_{120}} = -3.909 \times 10^{89}$ at $\gamma_{120} = 269.970$, where $\log|w_{\gamma_{120}}| = 206.3$ against $\pi\gamma/4 = 212.0$ (computed, code/ht_residues.py and code/ht_resid.py). Neither

$\sum w_\gamma/(u+\gamma^2)$ nor $\sum w_\gamma/\gamma^2$ converges. Multiplying by $e^{uy}$ before summing cures this, because $e^{-\gamma^2 y}$ beats $e^{\pi\gamma/4}$ for every $y>0$.

THEOREM 24.1.1 (Grouped residue expansion). *There are radii $R_n\to\infty$ with $R_n\in[n,2n]$ such that for every $y>0$*

$$f(y)=\lim_{n\to\infty}\sum_{|u_\rho|<R_n}\operatorname*{Res}_{u=u_\rho}\frac{e^{uy}}{G(u)}, \tag{24.1.1}$$

*with no hypothesis on the zeros. A simple zero contributes $e^{u_\rho y}/G'(u_\rho)$, a zero of multiplicity $m$ contributes $e^{u_\rho y}P_\rho(y)$ with $P_\rho$ a polynomial of degree $m-1$, and a pair of zeros $\frac12\pm\delta+i\gamma$ off the line has conjugate values $u_\rho$ and contributes $2\operatorname{Re}\big(e^{u_\rho y}/G'(u_\rho)\big)$. For $y$ in a compact subset of $(0,\infty)$ the error after the $n$th group is at most $R_n\exp(-\frac12 R_n y+C\sqrt{R_n}\log R_n)+\exp(-c\sqrt{R_n}\log R_n)$ for constants $c,C>0$.*

*Sketch.* Write $f(y)=(2\pi i)^{-1}\int_{1-i\infty}^{1+i\infty}e^{uy}G(u)^{-1}\,\mathrm{d}u$, which converges absolutely because $\log|G(1+it)|$ grows like $c\sqrt{|t|}\log|t|$ with $c>0$. Close the contour by an arc of the circle $|u|=R$. The function $G$ has order one half and $\log\max_{|u|=r}|G|\sim\frac14\sqrt{r}\log r$, so the minimum modulus theorem for functions of finite order [335] produces $R_n\in[n,2n]$ with $\log|G(u)|\ge -C\sqrt{R_n}\log R_n$ on the whole circle $|u|=R_n$. On the part of the arc in $\operatorname{Re}u\le -R_n/2$ the factor $|e^{uy}|\le e^{-R_n y/2}$ dominates. On the rest of the arc $\sqrt{u}$ has argument at most $\pi/3$ and $\operatorname{Re}(\frac12+\sqrt{u})\ge\frac12\sqrt{R_n}$, so $\zeta(\frac12+\sqrt{u})$ is bounded away from zero, Stirling's formula gives $\log|G(u)|\ge c\sqrt{R_n}\log R_n$, and $|e^{uy}|\le e^{y}$. Both contributions tend to zero, and the residue theorem gives (24.1.1). □

The expansion therefore holds for every $y>0$ as it stands, multiple zeros and off line zeros included, provided the residues are summed in the order of $|u_\rho|$ and grouped between the radii $R_n$. Only absolute convergence needs a hypothesis.

THEOREM 24.1.2 (Residue expansion). *Assume that the zeros of $\zeta$ are simple and that $|\xi'(\rho)|\ge e^{-C|\gamma|}$ for some constant $C$ and all zeros. Then the series*

$$f(y)=\sum_{\gamma>0}\frac{e^{u_\rho y}}{G'(u_\rho)},\qquad y>0, \tag{24.1.2}$$

*converges absolutely and locally uniformly. Under RH it reads $f(y)=\sum_{\gamma>0}w_\gamma e^{-\gamma^2 y}$ with $w_\gamma=1/G'(-\gamma^2)=\gamma^2\prod_{\gamma'\neq\gamma}(1-\gamma^2/\gamma'^2)^{-1}$.*

*Proof.* The hypothesis gives $|1/G'(u_\rho)|\le 2|\gamma||\xi(\frac12)|e^{C|\gamma|}$, against $|e^{u_\rho y}|\le e^{-(\gamma^2-1/4)y}$. The product form of $w_\gamma$ is the derivative of the Hadamard product at $-\gamma^2$. □

The hypothesis is conditional: it is a lower bound for $|\zeta'(\rho)|$ times the gamma factor, expected to hold with room to spare (Gonek's conjectures predict polynomial bounds for $1/|\zeta'(\rho)|$ on the line), and it is not known.

The weights of (24.1.2) are exactly those of the classical formula for the density of a sum of independent exponentials with rates $\lambda_k$, so under RH Theorem 24.1.2 says that $\mu$ is the law of $\sum_\gamma e_\gamma/\gamma^2$, written out as a density. The series cannot be integrated term by term, since $\sum w_\gamma/\gamma^2$ diverges: the mass $\int f = 1$ and the mean $\int yf = p_1$ are properties of the sum, not of the terms. With the first 120 zeros the series reproduces the density at $y \geq 0.008$ to the digits of Table 24.1, its integral over $[0.008, \infty)$ is 0.9999999995 (so $5 \times 10^{-10}$ of the mass lies below 0.008, where the series is not used), and its first moment over $[0.008, \infty)$ is 0.0231049931114 against $p_1 = 0.0231049931154$, although the zeros used have $\sum \gamma^{-2} = 0.0203057$: the residues carry the whole function $G$, not just the zeros summed (computed, code/ht_resid.py). The positivity of $f$, which holds unconditionally because $\mu$ is a probability law, is a massive cancellation: near $y = 0.008$ the largest single term exceeds $f(y)$ by a factor of about $e^{28.5}$ (the term $-7.83 \times 10^6$ at $\gamma = 49.77$ against $f(0.008) = 3.42 \times 10^{-6}$, computed).

| $y$ | 0.010 | 0.015 | 0.020 | 0.025 | 0.030 | 0.040 | 0.050 | 0.060 | 0.100 |
|---|---|---|---|---|---|---|---|---|---|
| $f(y)$ | 0.0244 | 28.765 | 80.104 | 53.175 | 23.433 | 3.4137 | 0.4660 | 0.0632 | $2.14 \times 10^{-5}$ |

Table 24.1: The density of $\mu$ from the residue expansion with the first 120 zeros (computed, 110 digits, agreeing with 140 digits to $10^{-103}$).

Proposition 24.1.3 (Tail of the density). *Let $\gamma_1 < \gamma_2 < \gamma_3$ be the first three ordinates and $c_3 = \frac{1}{2}(\gamma_2^2 + \gamma_3^2)$. With no hypothesis,*

$$f(y) = w_1 e^{-\gamma_1^2 y} + w_2 e^{-\gamma_2^2 y} + O\big(e^{-c_3 y}\big), \qquad y \to \infty,$$

*with the same bound for each derivative, where $w_1 = 1.01635 \times 10^4 > 0$ and $w_2 = -1.17754 \times 10^6 < 0$.*

*Proof.* Shift the Bromwich line to $\operatorname{Re} u = -c_3$. Only the poles $-\gamma_1^2$ and $-\gamma_2^2$ are crossed: a zero with $u_\rho$ to the right of $-c_3$ has $\gamma^2 - \delta^2 < c_3$, so $\gamma < 23.2$, and every such zero is known to lie on the line and to be simple. On $\operatorname{Re} u = -c_3$ the function $|G|$ is bounded below and grows like $e^{c\sqrt{|t|}\log|t|}$, so the shifted integral is $O(e^{-c_3 y})$, and differentiating under the integral gives the derivatives. □

### 24.1.2 Density and canonical function

The density (24.1.1) and the canonical function $W(x) = \sum e^{u_\rho x}$ are sums over the same exponentials, with weights $1/G'(u_\rho)$ and 1 respectively. They are tied by

Steutel's equation for infinitely divisible laws on $[0,\infty)$,

$$y\,f(y) = \int_0^y f(y-x)W(x)\,\mathrm{d}x, \tag{24.1.3}$$

which is the Laplace transform of the identity $-\varphi' = \varphi\,G'/G$ for $\varphi = 1/G$, with $G'/G = \int e^{-ux}W(x)\,\mathrm{d}x$. The GGC property is a statement about $W$ (complete monotonicity, whose weights are all one); the residue expansion moves the question to $f$, whose weights alternate.

## 24.2 Kent's formula

Kent [305] showed that the first passage time of a generalised diffusion, equivalently of a Krein string, from its entrance point to a level is a sum $\sum_n e_n/\lambda_n$ of independent exponentials, where the $\lambda_n$ are the Dirichlet eigenvalues of the string, and that its density has the eigenfunction expansion

$$f(t) = \sum_n \lambda_n \prod_{m\neq n} \frac{\lambda_m}{\lambda_m - \lambda_n}\, e^{-\lambda_n t}. \tag{24.2.1}$$

Writing $G(u) = \prod(1 + u/\lambda_m)$, the weight is $1/G'(-\lambda_n)$, so (24.2.1) is the residue expansion. The theorem is used in the book in Section 8.1.3 (Kent approximants) and in Corollary 9.2.3.

Proposition 24.2.1 (Kent's formula and the residue expansion). *If RH holds and the zeros are simple,* (24.1.2) *is Kent's expansion* (24.2.1) *with* $\lambda_n = \gamma_n^2$, *and* $f$ *is the density of the hitting time of a Krein string with Dirichlet spectrum* $\{\gamma_n^2\}$. *Without these hypotheses the grouped expansion* (24.1.1) *is the formal continuation of Kent's formula to complex eigenvalues: an off line pair contributes* $2\,\mathrm{Re}\left(e^{u_\rho y}/G'(u_\rho)\right)$, *a multiple zero contributes a polynomial times an exponential, and no string has such a spectrum, so the formula survives while its probabilistic meaning does not.*

*Proof.* The weights agree by the product formula of Theorem 24.1.2. The existence of the string for an increasing sequence $\lambda_n$ with $\sum 1/\lambda_n < \infty$ is Krein's inverse spectral theorem [124], as in Section 8.1.3. The spectrum of a string is real, positive and simple. □

### 24.2.1 A check: Brownian exit from an interval

The exit time of standard Brownian motion from $(-1,1)$, started at 0, has Laplace transform $1/\cosh\sqrt{2u}$, eigenvalues $\lambda_k = (k+\tfrac12)^2\pi^2/2$ and residues $(-1)^k\pi(k+\tfrac12)$, which gives the theta series $f(t) = \sum_k (-1)^k\pi(k+\tfrac12)e^{-(k+\frac12)^2\pi^2 t/2}$. Numerically it has mass 1 and mean 1 (to 40 digits, the theta series and

its dual series agreeing to $4\times10^{-61}$), and its Laplace transform at $u=1$ equals $1/\cosh\sqrt{2}=0.459098131085425$ to fifteen digits. The truncated Kent products $\lambda_k\prod_{m<N,m\neq k}\lambda_m/(\lambda_m-\lambda_k)$ converge to the residues, slowly for large $k$ (computed, code/ht_kent.py):

| $k$ | $N=50$ | $N=400$ | residue |
|---|---|---|---|
| 0 | 1.5630 | 1.5698 | 1.5708 |
| 1 | −4.5050 | −4.6860 | −4.7124 |
| 5 | −9.4241 | −16.0202 | −17.2788 |

### 24.2.2 What is special about $\xi$

For the exit time the weights grow like $\sqrt{\lambda_k}$: the function $\cosh\sqrt{2u}$ has exponential type in $\sqrt{u}$ and lies in Krein's class, so that $\sum|w_k|/\lambda_k^2<\infty$ and the partial fraction series of $1/\cosh\sqrt{2u}$ converges, although $\sum w_k/\lambda_k$ converges only conditionally. For $\xi$ the weights grow like $e^{\pi\sqrt{\lambda}/4}$ up to the factor $1/|\zeta'(\rho)|$. Krein's class $K$ consists of the real entire functions $E$ with simple real zeros $a_k$ whose reciprocal has the expansion

$$\frac{1}{E(z)}=c+\sum_k c_k\Big(\frac{1}{z-a_k}+\frac{1}{a_k}\Big),\qquad \sum_k\frac{|c_k|}{a_k^2}<\infty,$$

and Krein's theorem says that every function in $K$ has exponential type and lies in the Cartwright class [209].

Proposition 24.2.2 (Not in Krein's class). *The function $\xi(\frac12+it)$ does not belong to Krein's class, whether or not RH holds. Moreover, under RH the partial fraction series of $1/G$ over the points $u_\rho=-\gamma_n^2$ does not converge absolutely.*

*Proof.* Along the imaginary direction of $t$, $\log|\xi(\frac12+it)|$ grows like $\frac12|t|\log|t|$, so $\xi(\frac12+it)$ has order one and infinite type, hence is not of exponential type, and Krein's theorem excludes it from $K$. For $1/G$ the argument is different, since $G$ has exponential type zero. Under RH absolute convergence needs $\sum|w_\gamma|/\gamma^4<\infty$, but $|w_\gamma|\asymp\gamma^{-3/4}e^{\pi\gamma/4}/|\zeta'(\rho)|$ and $|\zeta'(\frac12+i\gamma)|\ll\gamma^{1/4}\log^2\gamma$ by the convexity bound, so the terms tend to infinity. □

The residue expansion of the density is the natural substitute for the partial fraction expansion that Krein's theorem forbids: multiplying by $e^{uy}$ is the regularisation that a function of infinite type requires. Under RH the truncations $G_N$ have finitely many zeros and are trivially in $K$; the divergence appears only in the limit, as the weights $1/G_N'(u_\rho)$ grow to the residues (Table 24.2). The theory of Ostrovskii on the class $K$, and the work of Kamynin, Ostrovskii and Vishnyakova on zeros of entire functions that are characteristic functions, concern the class to which $\xi(\frac12+it)=\mathbb{E}\cos(tV)$ belongs; we state no result from it.

## 24.3 Processes behind the reciprocal xi law

Under RH there are two processes whose hitting times have law $\mu$.

1. The Krein string of Proposition 24.2.1 (for simple zeros), a generalised diffusion whose time to reach the end has law $\sum e_\gamma/\gamma^2$.

2. A pure birth chain that leaves its $n$th state at rate $\gamma_n^2$ (with multiplicities counted as repeated rates). Since $\sum \gamma^{-2} < \infty$ the chain explodes in finite time, and its explosion time has law $\mu$.

Conversely, any process with nonnegative speed measure, or any family of GGC laws, that produces $1/G$ proves RH. This is the probabilistic form of the Hilbert and Pólya idea; its operator theoretic counterpart, de Branges spaces and canonical systems attached to $\zeta$ and to Dirichlet $L$ functions, has been studied by Lagarias [207].

Proposition 24.3.1 (Reduction to approximating hitting times). *RH holds if and only if there are GGC laws, for instance hitting times of strings, converging weakly to $\mu$.*

*Proof.* Under RH the laws of $\sum_{n\le N} e_n/\gamma_n^2$ converge to $\mu$. Conversely, a weak limit of GGC laws is a GGC (Corollary 4.1.2, Bondesson's continuity theorem [56]), and $\mu$ is a GGC only under RH. □

The limit therefore costs nothing. The entire content of a hitting time proof is the construction of approximating processes whose speed measures remain nonnegative.

## 24.4 The candidate families

**De Bruijn and Newman.** Replacing Riemann's kernel by $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ gives laws $\mu_t$ that are GGC exactly when $t \ge \Lambda$ (Corollary 4.2.5), and hitting times of strings when moreover the zeros are simple, which is so for $t > \Lambda$. Rodgers and Tao proved $\Lambda \ge 0$ [463], and $\Lambda \le 0.2$ is known [419]. RH is $\Lambda = 0$: the strings would have to exist for every $t > 0$ and fail for every $t < 0$, so any positivity preserving evolution in $t$ would have to be sharp at $t = 0$.

**Basepoint.** For $\alpha \ge 1$ the law $\mu^{(\alpha)}$ with Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is a GGC with Lorentzian Thorin density $\pi^{-1}\operatorname{Re}\xi'/\xi(\alpha + i\tau)$, the hitting time of a diffusion with continuous spectrum, and as $\alpha \downarrow \frac{1}{2}$ the spectrum sharpens onto the atoms $\gamma^2$. Survival of these laws down to $\alpha$ is the quasi Riemann hypothesis at $\alpha$ (Theorem 7.1.14, Chapter 9, Section 16.12).

**Truncated Hadamard products.** These are hitting times exactly when the zeros used lie on the line (Proposition 24.5.1); the construction is circular.

**Riemann's integral.** Since $G(u) = \mathbb{E}\cosh(\sqrt{u}\,V)$ with $V$ symmetric and density proportional to $\Phi_{\mathrm{dBN}}(|v|/2)$, the function $\varphi = 1/G$ is the harmonic mean, over the random half width $V/\sqrt{2}$, of the transforms $\mathrm{sech}(\sqrt{u}\,V)$ of Brownian exit times. Each is a GGC, and a harmonic mean of GGC transforms need not be one. Jensen's inequality gives $\varphi(u) \le \mathbb{E}\,\mathrm{sech}(\sqrt{u}\,V)$, a scale mixture of GGC transforms (the exit time from $(-a,a)$ has Thorin atoms at $(k+\frac{1}{2})^2\pi^2/(2a^2)$), and the reciprocal of a mixture is not a mixture of reciprocals, so this route gives bounds and not the GGC property.

**Tilted balance.** For $\lambda > \frac{1}{4}$, $Y_\lambda + T_{\log N}$ equals an explicit archimedean law, where $Y_\lambda$ is the Esscher tilt of $\mu$ and $T_{\log N}$ the first passage of drifted Brownian motion to the random level $\log N$, with $N$ zeta distributed [421], and the limit $\lambda \downarrow \frac{1}{4}$ meets the pole of $\zeta$.

**Truncated Euler products.** For every $\alpha > 0$ and every $P$ there is a GGC $Y_P$ with $T_{\mathbb{E}\log N_P} \overset{d}{=} Y_P + T_{\log N_P}$ (Proposition 24.8.1). It is the one family built from the primes that exists unconditionally at $\alpha = \frac{1}{2}$, and it escapes as $P \to \infty$.

## 24.5 Truncations, total positivity and the HCM route

### 24.5.1 Bondesson's finite approximations

Bondesson's construction of GGC laws starts from finite convolutions of exponentials and passes to weak limits [56]. For $N \ge 1$ order the zeros by height, never splitting a pair $\frac{1}{2} \pm \delta + i\gamma$, and put

$$G_N(u) = \prod_{n \le N} \Big(1 - \frac{u}{u_{\rho_n}}\Big), \qquad \varphi_N = 1/G_N.$$

Then $\varphi_N$ is the Laplace transform of a signed measure $\mu_N$ of total mass one, a probability law when the first $N$ zeros lie on the line, with density the finite residue sum $f_N(y) = \sum_{n \le N} e^{u_{\rho_n} y}/G_N'(u_{\rho_n})$ (for simple zeros).

Proposition 24.5.1 (Truncations). *(i) $\varphi_N \to \varphi$ pointwise on $[0,\infty)$, and for each fixed zero the finite weight converges to the residue, $1/G_N'(u_\rho) \to 1/G'(u_\rho)$, with correction factor $\prod_{n>N}(1 - u_\rho/u_{\rho_n})^{-1} \approx \exp\big(+u_\rho \sum_{n>N} u_{\rho_n}^{-1}\big)$, which under RH equals $\exp(\gamma^2 \sum_{n>N} \gamma_n^{-2}) > 1$.*

*(ii) $\mu_N$ is a GGC if and only if the first $N$ zeros lie on the line; in particular it is a GGC, unconditionally, for every $N \le N_0(H)$, the number of zeros below the verified height $H = 3 \cdot 10^{12}$ [419].*

*(iii) RH holds if and only if every $\mu_N$ is a GGC, and then $\mu$ is a GGC by Bondesson's continuity theorem (Corollary 4.1.2).*

*(iv) Under RH, for $u \geq 0$, $0 \leq \log G(u) - \log G_N(u) \leq u\,\tau_1(N)$ with $\tau_1(N) = \sum_{n>N} \gamma_n^{-2} \approx (\log(T_N/2\pi) + 1)/(2\pi T_N)$, where $T_N$ is the height of the $N$th zero.*

*Proof.* (i) is the convergence of the Hadamard product and of its derivative at $u_\rho$; the sign follows from $\prod(1 - x_n)^{-1} \approx \exp(\sum x_n)$ with $x_n = u_\rho/u_{\rho_n}$, which is positive under RH. (ii) A finite product $\prod(1-u/u_n)^{-1}$ is a GGC Laplace transform exactly when every $u_n$ is real and negative, and $u_\rho = (\delta + i\gamma)^2$ is real and negative exactly when $\delta = 0$; when a pair is not split, $G_N$ is real. (iii) If some zero is off the line, $\mu_N$ fails to be a GGC for all large $N$; conversely use (ii) and Bondesson's theorem. (iv) Under RH $\log G(u) - \log G_N(u) = \sum_{n>N} \log(1 + u/\gamma_n^2) \leq u\tau_1(N)$ for $u \geq 0$. □

The proposition shows both the reach and the limit of the method. The approximants are explicit GGC laws, unconditionally, up to a height of $3 \cdot 10^{12}$ (the relative error of the clock transform is also bounded in Section 8.1.3). But Bondesson's closure theorem needs all $N$, and (iii) only restates RH: the method converts RH into the statement that no truncation ever leaves the GGC class, and gives no mechanism to prevent it.

| | $N = 10$ | $N = 30$ | $N = 60$ | $N = 120$ |
|---|---|---|---|---|
| first zero, raw | 1476 | 3118 | 4447 | 5808 |
| first zero, corrected | 9970 | 10132 | 10155 | 10161 |
| second zero, corrected | $-1.068 \times 10^6$ | $-1.160 \times 10^6$ | $-1.173 \times 10^6$ | $-1.176 \times 10^6$ |
| fifth zero, corrected | $4.15 \times 10^9$ | $7.28 \times 10^9$ | $7.80 \times 10^9$ | $7.96 \times 10^9$ |

Table 24.2: Finite truncation weights $1/G_N'(u_\rho)$ against the residues $w_\gamma = 1.0164 \times 10^4$, $-1.1775 \times 10^6$ and $8.0078 \times 10^9$ at the first, second and fifth zeros. The corrected rows multiply the raw weight by the factor of Proposition 24.5.1(i), with $\sum_{n>N} \gamma_n^{-2}$ taken from the exact total $p_1$; for the first zero and $N = 120$ the factor is $\exp(199.79 \times 0.0027993) = 1.7494$ and the corrected weight agrees with the residue to 0.025 per cent (computed, code/ht_resid.py).

### 24.5.2 Pólya frequency functions

By Proposition 8.1.2, RH holds if and only if the density $f$ of $\mu$ is a Pólya frequency function of infinite order, so that RH sits at the top of a ladder $\mathrm{PF}_2 \Leftarrow \mathrm{PF}_3 \Leftarrow \cdots \Leftarrow \mathrm{PF}_\infty \iff$ RH of determinant inequalities on the explicit function (24.1.1), alongside the ladder of derivatives of $W$ (Section 5.1). The lowest rung, $\mathrm{PF}_2$, is log concavity of $f$. It is not settled by self decomposability, which gives unimodality

(Corollary 11.1.2) but not log concavity. If an off line zero exists, Schoenberg's theorem [474] says some rung fails, and the heuristic of the Gaussian window argument suggests that it fails only at an order that grows with the height of that zero.

### 24.5.3 Log concavity

The rung $\mathrm{PF}_2$ splits into three ranges of $y$.

Theorem 24.5.2 (Log concavity in the tail). *There is $Y$ such that $f$ is strictly log concave on $[Y,\infty)$, with no hypothesis.*

*Proof.* By Proposition 24.1.3, $f(y) = w_1 e^{-\gamma_1^2 y}(1+\varepsilon(y)+\eta(y))$ with $\varepsilon(y) = (w_2/w_1)e^{-\Delta y}$, $\Delta = \gamma_2^2 - \gamma_1^2 = 242.1$, and $\eta$, together with its first two derivatives, of size $O(e^{-(c_3-\gamma_1^2)y})$. Then $(\log f)'' = \Delta^2\varepsilon(1+o(1)) + O(e^{-(c_3-\gamma_1^2)y})$. Since $w_2/w_1 < 0$ the first term is negative, and since $c_3 - \gamma_1^2 = 333.9 > \Delta$ it dominates the second for large $y$. □

The sign of $w_2/w_1$ is the whole point: the second residue is negative, and this negative correction bends $\log f$ downwards. The same mechanism operates at every pair of consecutive rates in a sum of exponentials. On the numerical grid of step 0.001 in $[0.008, 0.2]$ one finds $(\log f)'' < 0$ with minimum $-2.59 \times 10^6$ at $y = 0.008$ and value $-6.3\times10^{-15}$ at $y = 0.2$ (computed, code/ht_resid.py; the value $-4.8 \times 10^{-5}$ near $y = 0.106$ is the two term tail value, not a grid computation). That this holds on all of $[0.008,\infty)$ is Theorem 24.5.5, a computation based on an explicit remainder bound that needs no hypothesis on the zeros but one numerical input that is not enclosed, so its status is computed.

Near the origin the density is governed by the archimedean growth of $\xi$. Write $\psi = \log G$ and $\Lambda_f(y) = \sup_{u>0}(\psi(u) - uy)$. Chernoff's bound gives $\log P(T \le y) \le -\Lambda_f(y)$ (Section 8.1), and the saddle point approximation

$$\log f(y) \approx -\Lambda_f(y) - \tfrac{1}{2}\log\big(2\pi|\psi''(u_y)|\big), \qquad \psi'(u_y) = y, \tag{24.5.1}$$

gives $(\log f)'' \approx -1/|\psi''(u_y)|$, which is large and negative: since $\psi(u) \sim \frac{1}{4}\sqrt{u}\log u$, one finds $\sqrt{u_y} \approx \log(1/y)/4y$ and $\Lambda_f''(y)$ of order $\log^2(1/y)/y^3$.

Proposition 24.5.3 (Log concavity near the origin). *If the saddle point approximation* (24.5.1) *holds as $y \to 0$ together with two derivatives, then $f$ is strictly log concave on $(0, y_0]$ for some $y_0 > 0$.*

The hypothesis is the usual one for densities of infinitely divisible laws with regularly varying exponent. Making it rigorous with derivatives requires control of $\psi$ in a sector around the positive axis, which the growth of $\xi$ to the right of the critical line supplies but which is not written out here, so the status is conditional. Table 24.3 shows the approximation against $\log f$ at five points as reported in the source; these values were not reproduced and are unchecked. What remains of log

concavity, unlike RH, is a problem about a fixed analytic function on a bounded range, and it is expected to be true.

| $y$ | $\log f(y)$ | saddle (24.5.1) | $\log P(T \le y)$ | $-\Lambda_f(y)$ | $\Lambda_f''(y)$ |
|---|---|---|---|---|---|
| 0.008 | $-12.5846$ | $-12.5857$ | $-21.40$ | $-19.06$ | $2.62 \times 10^6$ |
| 0.010 | $-3.7121$ | $-3.7144$ | $-11.82$ | $-9.75$ | $1.09 \times 10^6$ |
| 0.012 | 0.5749 | 0.5710 | $-6.91$ | $-5.09$ | $5.19 \times 10^5$ |
| 0.015 | 3.3592 | 3.3511 | $-3.30$ | $-1.84$ | $2.01 \times 10^5$ |
| 0.020 | 4.3833 | 4.3636 | $-1.07$ | $-0.16$ | $5.38 \times 10^4$ |

Table 24.3: The lower tail of $\mu$ and the saddle point approximation, as printed in the source (unchecked: not reproduced). The Chernoff exponent $-\Lambda_f(y)$ bounds $\log P(T \le y)$ from above, as it must.

### 24.5.4 A certificate for log concavity on $[0.008, \infty)$

The grouped expansion has an explicit error once the Bromwich line is moved to a vertical line that passes between two consecutive poles. Let $N = 120$ and $c = \frac{1}{2}(\gamma_{120}^2 + \gamma_{121}^2) = 73296.53$, with $\gamma_{120} = 269.970$ and $\gamma_{121} = 271.494$ (computed).

Proposition 24.5.4 (Explicit truncation). *With no hypothesis, for every $y > 0$ and $j \ge 0$,*

$$\Big| f^{(j)}(y) - \sum_{n \le 120} w_n (-\gamma_n^2)^j e^{-\gamma_n^2 y} \Big| \le M_j e^{-cy}, \qquad M_j = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|-c+it|^j}{|G(-c+it)|}\, \mathrm{d}t.$$

*Numerically $\log M_0 = 206.26$ and $M_{j+1}/M_j = 1.0000185\, c$ for $j = 0, 1, 2$ (computed).*

*Proof.* Shift the Bromwich line to $\operatorname{Re} u = -c$. The poles crossed are those with $\operatorname{Re} u_\rho = \delta^2 - \gamma^2 > -c$, which forces $\gamma < \sqrt{c + \frac{1}{4}} < 270.8$; these are the first 120 zeros, which are simple and on the line. On $\operatorname{Re} u = -c$ the function $1/|G|$ is integrable, because $\operatorname{Re} \sqrt{u}$ grows like $|t|^{1/2}$ and with it $|\xi(\frac{1}{2} + \sqrt{u})|$. Differentiating under the integral gives the derivatives. □

The bound uses nothing about zeros above height 270.8, and it is minute: with $\log M_j = 206.26, 217.46, 228.67, 239.87, 251.07$ for $j \le 4$, $M_j e^{-cy} \le e^{-330}$ for $y \ge 0.008$. The constants $M_j$, $j \le 8$, were enclosed by ball arithmetic with the integral cut at $|t| = 2^{20}$ and the zeros and the derivative of $\zeta$ taken from Arb; $M_0 = 3.78781207977 \times 10^{89}$ is enclosed to a relative $1.6 \times 10^{-12}$. The integrand at $|t| = 2^{20}$ is $10^{-572}$ of its peak, but the tail beyond is not bounded analytically, and the aliasing of the circle quadrature for $\zeta'$ was checked at two sizes and not certified. So the $M_j$ are computed, and not proved.

Two devices make log concavity checkable. The first is an exact identity for sums of exponentials: if $f = \sum_k w_k e^{-\lambda_k y}$ then

$$ff'' - f'^2 = \frac{1}{2}\sum_{j,k} w_j w_k (\lambda_j - \lambda_k)^2 e^{-(\lambda_j+\lambda_k)y}. \tag{24.5.2}$$

Every pair term decays like $e^{-(\lambda_j+\lambda_k)y}$, and the slowest is the pair $(1,2)$, whose coefficient $w_1 w_2(\lambda_2 - \lambda_1)^2$ is negative. Once that term dominates the sum of the absolute values of the others, which happens at some $y$ and then persists, $g = ff'' - f'^2$ is negative for good. With the truncation bound added, the ratio of the other terms to the leading one is 2.83 at $y = 0.02$, 1.39 at 0.022, 0.622 at 0.025, 0.033 at 0.04 and $2.2 \times 10^{-5}$ at 0.08 (computed), and in ball arithmetic it is below one and decreasing on the grid $0.025, 0.030, \ldots, 0.3$, with the enclosure $0.621855669841 \pm 2 \times 10^{-13}$ at $y = 0.025$. Hence $g < 0$ on $[0.025, \infty)$, given the $M_j$.

The second device is a Taylor model for the window below 0.025. At a point $y_i$ compute $g^{(m)}(y_i)$ for $m < K = 6$ from (24.5.2) or by Leibniz's rule, and bound $|g^{(K)}|$ on $[y_i, y_i + h]$ by replacing every $f^{(j)}$ with $A_j(y_i) = \sum_{n \le 120} |w_n| \lambda_n^j e^{-\lambda_n y_i} + M_j e^{-c y_i}$, which dominates $|f^{(j)}|$ on the whole step because each term decreases in $y$. If the Taylor polynomial of $g$ plus the remainder is negative on $[y_i, y_i + h]$ after adding the truncation errors, then $g < 0$ there; take the largest such $h$ and move on. In ball arithmetic at 400 bits, with the range $[0.008, 0.08]$ split into seven sub windows (five of them in $[0.008, 0.012]$), every sub window was covered. Table 24.4 lists the windows; the step counts depend on the rule that chooses $h$, which the source does not give, so those of the check (a greedy doubling rule) are printed and the source's 25856 steps (its parts sum to 25859), 73 and 74 are not reproduced.

| window | Taylor steps | method |
|---|---|---|
| $[0.008, 0.012]$ | 29984 | Taylor model, $K = 6$, 400 bits |
| $[0.012, 0.020]$ | 88 | Taylor model, $K = 6$, 400 bits |
| $[0.020, 0.080]$ | 90 | Taylor model, $K = 6$, 400 bits |
| $[0.025, \infty)$ | none | pair identity (24.5.2) with Proposition 24.5.4 |

Table 24.4: Windows on which $(\log f)'' < 0$ was verified, given the constants $M_j$ (computed, code/ht_lcert.py). The step count near 0.008 is large because of the cancellation in (24.1.1).

THEOREM 24.5.5 (Log concavity, computed). *Assume the values of the constants $M_j$, $j \le 8$, of Proposition 24.5.4, which are computed by quadrature. Then, with no hypothesis on the zeros, the density $f$ of $\mu$ is strictly log concave on $[Y_*, \infty)$ with $Y_* = 0.008$.*

Its status is computed until the $M_j$ are enclosed with a bound on the tail of the integral. The range it leaves open, $(0, Y_*)$, carries mass $P(T < Y_*) = 5 \times 10^{-10}$ (computed), and there the saddle point approximation (24.5.1) would give log concavity with $\Lambda_f''(y)$ of order $10^6$ and growing (Proposition 24.5.3, conditional). Closing it rigorously needs either more zeros, so that $c$ grows and the window can be pushed down (at $y = 0.005$ the 120 term sum with its truncation balls still encloses $g = -6.5 \times 10^{-40}$ away from zero, with ball radius $7.6 \times 10^{-49}$ against a cancellation of $6.3 \times 10^{35}$ in $f$, but no Taylor model was run below 0.008), or a rigorous saddle point bound. Neither involves the location of the zeros. Log concavity of $f$ is the strong unimodality of $\mu$, by the theorem of Ibragimov [184].

### 24.5.5 Off line zeros are invisible to the low rungs

The pair of zeros $\frac{1}{2} \pm \delta + i\gamma$ contributes $2|G'(u_\rho)|^{-1} e^{(\delta^2-\gamma^2)y} \cos(2\delta\gamma y - \arg G'(u_\rho))$ to the density, while $u_\rho$ makes the angle $\theta_\rho = 2\arctan(\delta/\gamma)$ with the negative axis. Over the range $y \lesssim \gamma^{-2}$ in which it is not exponentially negligible its phase moves by only about $2\delta/\gamma$, and Schoenberg's theorem requires exactly that $u_\rho$ stay near the negative axis. A determinant of order $k$ resolves the density to roughly $k$ derivatives, so the first failing rung is expected, heuristically, to be of order $\gamma/\delta$ at best, and larger once the lower zeros are allowed to smooth the pair. For a zero above the verified height this is at least of order $10^{12}$. The question which rung fails first is therefore answered, heuristically, by: none that can be computed. In an experiment reported in the source, nine toy densities were built from the residue formula with the first $m-1$ zeros on the line and the $m$th replaced by an off line pair (at $\gamma_m = 30.4$, 43.3, 67.1 and $\delta = 0.45$, 0.25, 0.1); all were positive, log concave on $(0, 0.2]$ and had nonnegative Toeplitz determinants of orders three and four. This experiment was not reproduced here and is unchecked. Numerical confirmation of low rungs, for either ladder, carries no information about zeros above the verified height.

### 24.5.6 The HCM route

Bondesson calls a density $f$ on $(0, \infty)$ hyperbolically completely monotone (HCM) if, for each $u > 0$, $f(uv)f(u/v)$ is a completely monotone function of $v + 1/v$. HCM densities are GGC densities, and the class is closed under products, quotients and powers $|q| \geq 1$ of independent variables [56] (Chapter 4). HCM is a sufficient condition for the GGC property, not a necessary one, so it gives a route to RH in one direction only.

Proposition 24.5.6 (HCM route). *If the density $f$ of $\mu$ is HCM, then $\mu$ is a GGC and RH holds.*

The $k$th rung of this route is $(-1)^k \partial_w^k [f(uv)f(u/v)] \geq 0$ with $w = v + 1/v$. Bondesson's representation $f(x) = Cx^{\beta-1} h_1(x) h_2(1/x)$, with $h_1, h_2$ Laplace trans-

forms of GGC laws, gives a consequence that is easy to test: an HCM density extends to a holomorphic function on $\mathbb{C} \setminus (-\infty, 0]$ with no zeros there, since $h_1(x)$ and $h_2(1/x)$ have the form $\exp(-ax - \int \log(1 + x/t)U(\mathrm{d}t))$ and are holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$.

THEOREM 24.5.7 (No HCM). *(i) No density of a sum of $n \geq 2$ independent exponential variables with distinct rates is HCM (proved).*

*(ii) With the first* 120 *terms of* (24.1.2)*, the continuation of $f$ to $\operatorname{Re} z > 0$ vanishes at*

$$z_* = 0.013831874754506 + 0.020001801712034\, i$$

*(computed). If the omitted terms are controlled by an explicit remainder bound on a sector containing $z_*$, this proves that $f$ is not HCM.*

*Proof.* (i) The density is an exponential polynomial $\sum_{k \leq n} c_k e^{-\lambda_k z}$ with distinct real $\lambda_k$ and nonzero $c_k$. It has at most $n - 1$ real zeros, by the Descartes rule for exponential sums, and infinitely many zeros altogether, since an entire function of order one with finitely many zeros is a polynomial times an exponential, which a sum of two or more exponentials with distinct rates is not. So it has nonreal zeros, which lie in $\mathbb{C} \setminus (-\infty, 0]$, and no HCM density has any. For $e^{-x} - e^{-2x}$ they are the points $2\pi i k$, $k \neq 0$.

(ii) Newton's method on the 120 term sum at 50 digits gives the stated $z_*$, floating point and not enclosed; the largest term of the normalised sum $f(z)e^{\gamma_1^2 z}/w_1$ at $z_*$ has modulus 5.4, and the normalised sum is $1.9 \times 10^{-12}$ at the printed fifteen digits (the root to forty digits was not tested). The argument principle on the box $0.012 \leq \operatorname{Re} z \leq 0.15$, $0 < \operatorname{Im} z \leq 0.05$ gives winding number 5 for the 120 term sum, and 1 on a box of half width $10^{-3}$ around $z_*$ (floating point, code/ht_resid.py). □

Part (ii) is a computation on the 120 term sum and not a proof, for four reasons. The series (24.1.2) presupposes RH and simple zeros, and its convergence for $\operatorname{Re} z > 0$ needs the hypothesis of Theorem 24.1.2. The termwise estimate for the omitted zeros uses residues that are not controlled. The remainder bound of Proposition 24.5.4 does not apply: it is for real $y$ only, because on $\operatorname{Re} u = -c$ the factor $|e^{uz}| = e^{-c \operatorname{Re} z - t \operatorname{Im} z}$ grows with $|t|$ while $1/|G(-c + it)|$ decays only like $\exp(-c'\sqrt{|t|}\log|t|)$, so the integral diverges for every nonreal $z$. And the argument principle needs that bound on the boundary of the box. A proof is within reach and needs no hypothesis on the zeros: bend the contour of the Bromwich integral at $-c$ into the two rays $\arg(u + c) = \pm(\pi/2 + \eta)$ with $\eta$ a little above $\arg z_* = 55.3°$, where the poles are real up to $|u| = 9 \times 10^{24}$ and within angle $1/\gamma$ of the negative axis above, the rays cross none of them, and $1/|G|$ decays on them, which gives an explicit remainder bound on a sector containing $z_*$. If RH fails then $f$ is not HCM at all, because HCM densities are GGC, so only the case

of RH needs the computation. Under RH, simple zeros and two hypotheses on the zeros, Theorem 4.3.22 proves that the density is not HCM.

For the two exponential density the failure of the HCM ladder appears only at high order. With $w = 2.1$, the normalised derivative $(-1)^k \partial_w^k [f(uv)f(u/v)]/f(uv)f(u/v)$ of $f = e^{-x} - e^{-2x}$ at $u = 5$ is positive for $k \le 14$ and equals $-1.21 \times 10^9$ at $k = 15$ (stable between 300 and 450 digits). At $u = 6$ and $u = 7$ the first negative order is 14 and 18, and for $u = 3, 4, 8, 10$ none appears up to order 40 (computed, code/ht_hcm2exp.py). Low order evidence for an HCM property of $f$ would therefore have been misleading. Since the computed zero indicates that the HCM class is not available even under RH, Proposition 24.5.6, while true, is vacuous for this law as far as the computed zero shows: the HCM route to RH appears closed, provided the bound of the bent contour is supplied.

## 24.6 Jensen hitting times

Proposition 24.3.1 asks for GGC laws converging to $\mu$. Truncated Hadamard products supply them but are circular, because they are built from the zeros. A family built from data that are known unconditionally comes from the Taylor coefficients of $G$, and it is the probabilistic form of the Jensen and Pólya criterion (Proposition 16.4.1).

### 24.6.1 The family

Write $G(u) = \sum_{k \ge 0} c_k u^k$. The coefficients are positive and computable without any hypothesis: $G(u) = \mathbb{E} \cosh(\sqrt{u}\, V)$ with $V$ the symmetric variable of Section 24.4 (Proposition 24.13.1 identifies it with the logarithm of a size biased range of a Brownian bridge), so that

$$c_k = \frac{\mathbb{E} V^{2k}}{(2k)!},$$

the even moments of $V$ divided by $(2k)!$. For $d \ge 1$ put

$$J_d(u) = \sum_{k=0}^{d} \binom{d}{k} \frac{k!}{d^k}\, c_k u^k = \sum_{k=0}^{d} c_k \prod_{j<k} \Big(1 - \frac{j}{d}\Big) u^k.$$

Up to the scaling $X = u/d$ and a constant this is the Jensen polynomial of degree $d$ and shift 0 of Griffin, Ono, Rolen and Zagier [159], whose sequence $g(n)$ (not to be confused with the ordinates) is defined by $(-1+4z^2)\Lambda(\frac{1}{2}+z) = \sum g(n) z^{2n}/n!$; since $(-1+4z^2)\Lambda(\frac{1}{2}+z) = 8\xi(\frac{1}{2}+z)$, their $g(k)$ is a constant multiple of $k!\, c_k$.

Proposition 24.6.1. *1. $J_d \to G$ uniformly on compact sets, without any hypothesis.*

2. *If $J_d$ has only real zeros, they are negative, $J_d(u) = \prod_{j\le d}(1 + u/\ell_{d,j})$, and $1/J_d$ is the Laplace transform of the GGC law $\mu_d$ of $\sum_{j\le d} e_j/\ell_{d,j}$: the time a pure birth chain with rates $\ell_{d,1}, \dots, \ell_{d,d}$ takes to pass through $d$ states.*

3. *RH holds if and only if $\mu_d$ exists for every $d$, and then $\mu_d \Rightarrow \mu$.*

4. *$\mu_d$ exists for every $d \le 2 \cdot 10^{17}$, unconditionally (proved, quoted through [159]).*

5. *Every $\mu_d$ has the same mean $c_1 = p_1$ as $\mu$, and its variance exceeds that of $\mu$ by $2c_2/d$.*

*Proof.* For (1), the factors $\prod_{j<k}(1-j/d)$ lie in $[0,1]$ and tend to 1, and $\sum c_k|u|^k < \infty$. For (2), a polynomial with positive coefficients has no zeros in $[0,\infty)$, and a product of linear factors with negative zeros is the transform of a sum of independent exponentials. For (3), if every $J_d$ is hyperbolic then $G$, as a locally uniform limit of polynomials with only negative zeros, has only negative zeros by Hurwitz's theorem, which is RH; Bondesson's continuity theorem (Corollary 4.1.2) gives the same conclusion directly from $\mu_d \Rightarrow \mu$. Conversely under RH $G$ is in the Laguerre and Pólya class, and the Jensen polynomials of such a function are hyperbolic (Proposition 16.4.1, [159]). Part (4) is the theorem of Chasse on the Jensen polynomials of this sequence, as quoted in a footnote of [159]. Part (5) follows from the factors 1 and $1 - 1/d$ at $k = 1, 2$: the cumulants are $\kappa_1 = c_1$ and $\kappa_2 = c_1^2 - 2c_2(1 - 1/d)$, while $\operatorname{Var}\mu = c_1^2 - 2c_2 = \sum \gamma^{-4}$. □

So the Jensen laws are what Proposition 24.3.1 asks for, built from Brownian moments rather than from zeros, and they exist up to degree $2 \cdot 10^{17}$ by the cited theorem. Their existence for every $d$ is RH. The questions are how fast they converge, and what the existence of $\mu_d$ at a given degree says about the zeros.

### 24.6.2 Rate of convergence

Since $\prod_{j<k}(1-j/d) = 1-k(k-1)/(2d)+O(k^4/d^2)$ and $\sum_k k(k-1)c_k u^k = u^2G''(u)$,

$$J_d(u) = G(u) - \frac{u^2 G''(u)}{2d} + O(d^{-2}).$$

At a simple zero $u_n = -\gamma_n^2$ this moves the zero by

$$\frac{\ell_{d,n}}{\gamma_n^2} - 1 = \frac{A_n}{2d} + O(d^{-2}), \qquad A_n = \frac{u_n G''(u_n)}{G'(u_n)}.$$

The coefficients $c_k$, $k < 600$, were computed from the moments of Pólya's kernel by a trapezoid rule at 260 digits; two step sizes agree to a relative $2 \times 10^{-205}$ for every $k < 600$, and $c_1 = 0.02310499311541$ and $c_2 = 2.48334053 \times 10^{-4}$

agree with $(\log\xi)''(\frac12)/2$ and with $(c_1^2-\sum\gamma^{-4})/2$ to twenty and twelve digits (computed, code/ht_jensen.py). The zeros of $J_d$ near $-\gamma_n^2$ were found by Newton's method at full and half precision, with relative changes below $10^{-110}$. Table 24.5 compares the relative shifts with $A_n/(2d)$; here $A_1,\dots,A_6 = -8.69, -16.37, -15.15, -29.57, -15.83, -32.02$ (computed, from the zeros plus a smooth tail they agree to five digits).

| $d$ | $n=1$ | $n=2$ | $n=3$ | $n=4$ | $n=5$ | $n=6$ |
|---|---|---|---|---|---|---|
| 80 | −0.0513 | −0.0905 | −0.0871 | −0.140 | −0.104 | −0.153 |
| 640 | −0.00674 | −0.0126 | −0.0117 | −0.0221 | −0.0127 | −0.0241 |
| 10240 | −0.000424 | −0.000799 | −0.000739 | −0.00144 | −0.000774 | −0.00156 |
| $A_n/2d$ | −0.000424 | −0.000799 | −0.000740 | −0.00144 | −0.000773 | −0.00156 |

Table 24.5: Relative shift $\ell_{d,n}/\gamma_n^2-1$ of the zeros of the Jensen polynomials, against the first order prediction; the last row is $A_n/(2d)$ at $d=10240$ (computed).

The laws converge at the same rate. Inverting $1/(uJ_d(u))$ by Talbot's method (at two contour sizes, agreeing to $3\times10^{-17}$) and comparing with the residue expansion of $\mu$, the largest difference of distribution functions over the grid $y = 0.007+0.001\,i$, $i=0,\dots,73$, is given in Table 24.6; the maximum sits at $y\approx0.016$ to 0.017, and $d\cdot\sup|F_d-F|$ increases towards a limit of about 2.76 (computed, code/ht_jlaws.py). On a coarser grid of 18 equispaced points of this interval the supremum is smaller by up to fourteen per cent, so the grid has to be named.

| $d$ | 10 | 40 | 160 | 640 | 2560 |
|---|---|---|---|---|---|
| $\sup\lvert F_d-F\rvert$ | 0.155 | 0.0560 | 0.0163 | 0.00425 | 0.00107 |
| $d\cdot\sup\lvert F_d-F\rvert$ | 1.55 | 2.24 | 2.60 | 2.72 | 2.75 |

Table 24.6: Distance between the distribution functions of the Jensen law $\mu_d$ and of $\mu$ on the grid of 74 points of $[0.007, 0.08]$ (computed).

### 24.6.3 What a degree can see

The first order formula also says what the existence of $\mu_d$ certifies. Near a zero at $u$ the correction $-u^2G''/(2d)$ acts like a heat flow of duration proportional to $u^2/d$. For a quadratic model $G\approx(u-a)^2+b^2$ of an off line pair with $a=\delta^2-\gamma^2$ and $b=2\delta\gamma$, the correction is $-u^2/d$, and the zeros of $J_d$ are real exactly when $b^2<|u|^2/d$. An off line pair at height $\gamma$ and displacement $\delta$ is therefore invisible to the Jensen polynomials below the degree

$$d^*(\gamma,\delta)\approx\frac{|u|^2}{b^2}\approx\frac{\gamma^2}{4\delta^2}$$

(heuristic: the quadratic model ignores the other zeros).

This was tested on Davenport and Heilbronn. With 600 coefficients of $G_{DH}$ computed in the same way from the positive kernel of Proposition 24.13.3, the zero of $J_d^{DH}$ starting at $u_\rho = (0.3085 + 85.6993i)^2 = -7344.28 + 52.88i$ was followed as $1/d$ increases from 0 (computed at 240 and 280 digits, code/ht_dh.py). Its imaginary part squared decreases linearly in $1/d$, with slope $-6.27 \times 10^7$ at $1/d = 0$ against $|u_\rho|^2 = 5.39 \times 10^7$ in the quadratic model, and the pair is real below and complex above the double root $d^* = 20784.9$ of $J_d$ (both coefficient sets give 20784.92): at $d = 20619$ it is real and at $d = 20833$ it has imaginary part 2.44. The model value is $\gamma^2/(4\delta^2) = 19290$. So the Jensen polynomials of Davenport and Heilbronn are not all hyperbolic, as they cannot be, but the first failure caused by this zero is near degree $2 \cdot 10^4$, far beyond the degrees at which the coefficient ladder of Table 24.13 was tested.

For $\xi$ the same scaling turns Chasse's theorem around. An off line zero at height $\gamma$ would first make $\mu_d$ fail near $d \approx \gamma^2/(4\delta^2)$. Since every zero below $3 \cdot 10^{12}$ is on the line and $\delta < \frac{1}{2}$, the heuristic predicts that the Jensen laws exist up to degree of order $10^{25}$, well beyond the cited $2 \cdot 10^{17}$, which corresponds through the same scaling to heights near $\sqrt{2 \cdot 10^{17}} \approx 4.5 \cdot 10^8$. Conversely, certifying $\mu_d$ for all $d \le D$ says about the zeros what a verification up to height about $\sqrt{D}$ says, at a cost that grows like $D \approx \gamma^2$ rather than the $\gamma^{1+o(1)}$ of direct zero verification. This is a heuristic: the other zeros change $d^*$ by about 8% for Davenport and Heilbronn (the ratio 20785/19290 is 1.0775). The unconditional ladder of Section 5.1 trades height against order in a similar way.

### 24.6.4 A Hermite identity and the de Bruijn flow

Write $u = -z^2$ and use $c_k = \mathbb{E}V^{2k}/(2k)!$. Since $(2k)! = 4^k k!(\frac{1}{2})_k$ and $\prod_{j<k}(1 - j/d)\,(-1)^k = (-d)_k/d^k$,

$$J_d(-z^2) = \mathbb{E}\, C_d(zV),$$
$$C_d(x) = {}_1F_1\Big(-d; \tfrac{1}{2}; \frac{x^2}{4d}\Big) = (-1)^d \frac{d!}{(2d)!} H_{2d}\Big(\frac{x}{2\sqrt{d}}\Big),$$

with $H_{2d}$ the Hermite polynomial (proved). So the Jensen polynomial is the expectation, over the Brownian variable $V$, of a fixed even polynomial with $2d$ real zeros; for each $d$ it plays the role that cos plays for $G$ itself, since $G(-z^2) = \mathbb{E}\cos(zV)$. The identity holds to forty digits at the five pairs $(d, z) = (5, 0.7), (10, 2), (20, 3.5), (40, 5), (100, 7)$ (computed, code/ht_jensen.py). The Plancherel and Rotach asymptotics of $H_{2d}$ give, for bounded $x$,

$$C_d(x) = e^{x^2/(8d)} \cos\Big(x\sqrt{1 + \tfrac{1}{4d}}\Big) + O(x^3/d^2)$$

with an additive error (heuristic: the error term is quoted from the leading term of Szegő's expansion and was not proved here), which is of the size seen at $x = 7$: the

two sides differ by $5.5\cdot10^{-2}$, $3.1\cdot10^{-4}$, $2.9\cdot10^{-6}$ for $d=10,100,1000$ (computed). With $V=2u$ in Pólya's variable this is

$$J_d(-z^2)\approx\frac{8}{\Xi(0)}H_t\big(2z(1+\tfrac{1}{8d})\big),\qquad t=\frac{z^2}{2d},$$
$$H_t(x)=\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(xu)\,\mathrm{d}u,$$

the de Bruijn and Newman function at the height dependent time $t=z^2/(2d)$ (Section 8.4). Near height $\gamma$ the Jensen truncation of degree $d$ therefore resembles the de Bruijn flow run for time $\gamma^2/(2d)$, to first order. Heuristically this gives the resolution law: an off line pair at height $\gamma$ which the de Bruijn flow merges at time $t_c$ is seen by $J_d$ when

$$d>d^*=\frac{\gamma^2}{2t_c}.$$

For an isolated pair $t_c\approx2\delta^2$, which recovers $\gamma^2/(4\delta^2)$. For Davenport and Heilbronn the flow $H_t$ of the kernel of $G_{DH}$ was computed: the double root of $H_t$ near $z=85.6175$ occurs at $t_c=0.176507$ (steps $h=1/250$ and $1/300$ give 0.17650714 and 0.17650712; the step $h=1/200$ gave 0.17639 and is not converged; computed, code/ht_dh.py, data/ht_dh_flow.log), hence $d^*=85.699^2/(2\cdot0.176507)=20804.8$, against the directly tracked 20784.9: agreement to 0.1% (computed). The approximation 19290 does not give this agreement; the value 20805 comes from the exact $t_c$.

This also answers the converse half of Problem 24.6 at the heuristic level. If RH holds up to height $T$ and $\Lambda\le\Lambda_0$, then near every height $\gamma\ge T$ the Jensen truncation of degree $d\le T^2/(2\Lambda_0)$ is the de Bruijn flow at time at least $\Lambda_0$, where all zeros are real, while below $T$ the zeros are real to begin with. So one expects $J_d$ hyperbolic for

$$d\le D(T)=\frac{T^2}{2\Lambda_0}.$$

With de Bruijn's $\Lambda_0=\frac12$ [112] and the height $5.45\cdot10^8$ reached by van de Lune, te Riele and Winter [508] this gives $D\approx3.0\cdot10^{17}$, the order of Chasse's $2\cdot10^{17}$ (it has not been checked that this is how Chasse's bound was obtained). With $T=3\cdot10^{12}$ and $\Lambda_0=0.2$ [419] it gives $D\approx2.25\cdot10^{25}$. Making this rigorous needs a uniform error bound in the Plancherel and Rotach step and control of the height dependence of $t$, which is the remaining content of Problem 24.6.

### 24.6.5 Assessment

The Jensen laws are a concrete, unconditional and non circular family of hitting times converging to $\mu$, at the rate $2.8/d$ in distribution function on the grid above. They realise Proposition 24.3.1 up to degree $2\cdot10^{17}$, and their existence at every degree is RH. But each degree sees only an off line pair of displacement $\delta$ at

heights up to about $2\delta\sqrt{d}$, an insensitivity of the same $\delta^2$ type as that found for boundaries in Proposition 24.12.2 (with $\gamma^{-2}$ here and $\gamma^{-4}$ there). Approximating hitting times built this way do not escape the need to control the zeros at every height; they convert it into a sequence of finite positivity problems whose size grows like the square of the height.

## 24.7 Extreme value hitting times

The simplest hitting times with an infinite spectrum are extreme values. If $e_1, \dots, e_n$ are independent standard exponentials, Rényi's representation gives

$$\max_{k \le n} e_k \stackrel{d}{=} \sum_{k=1}^{n} \frac{e_k}{k},$$

the passage time of a pure birth chain with rates $1, 2, \dots, n$, and $\max_k e_k - \log n$ tends to the Gumbel law, whose bilateral Laplace transform is $\Gamma(1+s)$ [261] (compare Chapter 14, Proposition 14.2.1 and Corollary 8.5.10). Since $1/\Gamma(1+s) = e^{\gamma_E s}\prod_k (1+s/k)e^{-s/k}$ is in the Laguerre and Pólya class, the Gumbel density $e^{-x-e^{-x}}$ is a Pólya frequency function of infinite order in Schoenberg's sense [474], and its Hadamard truncations are the centred maxima of $n$ exponentials. Curry and Schoenberg [103] showed that Pólya frequency functions are limits of B splines and used this extreme value example as an illustration. For $\mu$ the same structure holds, with the integers replaced by the $\gamma^2$.

### 24.7.1 B splines and Dirichlet means

Proposition 24.7.1. *Let $0 < \lambda_1 < \cdots < \lambda_N$ and let $f_N$ be the density of $\sum_{k \le N} e_k/\lambda_k$. Then*

$$f_N(y) = \Big(\prod_{k \le N} \lambda_k\Big) \frac{y^{N-1}}{(N-1)!} \int M(x; \lambda_1, \dots, \lambda_N) e^{-xy}\, \mathrm{d}x,$$

*where $M(\cdot; \lambda_1, \dots, \lambda_N)$ is the Curry and Schoenberg B spline with knots $\lambda_k$, normalised to integral one. Equivalently $M$ is the density of the Dirichlet mean $D_N = \sum_k W_k \lambda_k$ with $(W_k)$ uniform on the simplex, and the formula is the case of Dirichlet parameter $N$ of the density formula of James, Roynette and Yor [289, Theorem 2.1].*

*Proof.* By partial fractions $f_N(y) = (\prod \lambda_k)(-1)^{N-1}[\lambda_1, \dots, \lambda_N]e^{-\lambda y}$, a divided difference in $\lambda$. The Hermite and Genocchi formula $[\lambda_1, \dots, \lambda_N]g = \mathbb{E}\, g^{(N-1)}(D_N)/(N-1)!$, with $g(\lambda) = e^{-\lambda y}$ and $g^{(N-1)} = (-y)^{N-1}e^{-\lambda y}$, gives the result, and the density of $D_N$ is the B spline [103]. □

For the knots $1, \dots, n$ the B spline is the cardinal one and the formula returns $ne^{-y}(1-e^{-y})^{n-1}$, the density of the maximum. For the knots $\gamma_1^2, \dots, \gamma_N^2$ it returns the Kent density of the truncation $\mu_N$: the reciprocal $\xi$ law is, under RH, the limit of $y^{N-1}$ times Laplace transforms of B splines whose knots are the squared zeros. The positivity of these densities is the positivity of B splines with real knots, and it is the reality of the knots that RH asserts. As $N \to \infty$ the Dirichlet mean $D_N$ concentrates near $\frac{1}{N}\sum \gamma_k^2$ and the representation degenerates, as noted in Section 24.13.

### 24.7.2 The extremal process

The infinite version uses the extremal process of the Gumbel law. Let $\Gamma_1 < \Gamma_2 < \cdots$ be the points of a unit Poisson process on $(0, \infty)$ and $G_k = -\log \Gamma_k$; then $G_1 > G_2 > \cdots$ are the points of the Poisson process with intensity $e^{-x}\,dx$, the limit of the top order statistics of $n$ exponentials minus $\log n$, and $G_1$ is standard Gumbel. (These $G_k$ are the extremal points, and have nothing to do with the function $G$.)

Theorem 24.7.2. *For any $0 < \lambda_1 \le \lambda_2 \le \cdots$ with $\sum 1/\lambda_k < \infty$, and $b_k = k/\lambda_k$,*

$$\sum_k \frac{e_k}{\lambda_k} \overset{d}{=} \sum_{k \ge 1} b_k\,(G_k - G_{k+1}),$$

*and if moreover $k \log k/\lambda_k \to 0$, the series may be rearranged:*

$$\sum_{k \ge 1} b_k\,(G_k - G_{k+1}) = b_1 G_1 + \sum_{k \ge 2} (b_k - b_{k-1}) G_k.$$

*In particular, under RH, $\mu$ is the law of $\sum_k k\gamma_k^{-2}(G_k - G_{k+1})$, a linear functional of the Gumbel extremal process. For finite $N$ the truncation $\mu_N$ is the L statistic $\sum_{k \le N} b_k(e_{(N-k+1)} - e_{(N-k)})$ of $N$ exponentials, with $e_{(0)} = 0$.*

*Proof.* The ratios $\Gamma_k/\Gamma_{k+1}$ are independent with laws $\mathrm{Beta}(k, 1)$, so $G_k - G_{k+1} = \log(\Gamma_{k+1}/\Gamma_k)$ are independent exponentials of rates $k$. Then $b_k(G_k - G_{k+1}) \overset{d}{=} e_k/\lambda_k$, which gives the first statement. The partial sums of the rearranged series equal those of the first series plus $b_K G_{K+1}$, and $G_{K+1} \sim -\log K$ almost surely, so the rearrangement converges when $b_K \log K \to 0$; for $\lambda_k = \gamma_k^2$ this holds since $b_k \asymp (\log k)^2/k$. Without the condition on $\lambda_k$ the second form can fail: the condition $\sum 1/\lambda_k < \infty$ with $\lambda_k$ increasing gives only $k/\lambda_k \to 0$. The finite statement is Rényi's representation of the spacings. □

The weights $b_k = k/\gamma_k^2$ are 0.0050, 0.0045, 0.0048, 0.0043, 0.0046 for $k \le 5$ (computed), they are not monotone, and for large $k$ they behave like $(\log k)^2/(4\pi^2 k)$. The top of the extremal process, the largest values, determines the bulk of $\mu$, and the deep points, which accumulate at $-\infty$ like $-\log k$,

determine its lower tail. RH is the statement that these weights are attached to real atoms; the representation itself holds for any positive spectrum.

### 24.7.3 Smoothed zeros

The extremal form suggests replacing the zeros by their expected positions. Let $\hat{t}_k$ solve $\vartheta(\hat{t}_k) = (k - \frac{3}{2})\pi$, with $\vartheta$ the Riemann and Siegel theta function, so that $\hat{t}_k$ is the midpoint of the Gram interval in which Gram's law puts $\gamma_k$; then $\hat{t}_1, \hat{t}_2, \hat{t}_3 = 14.52, 20.65, 25.49$ against $14.13, 21.02, 25.01$ (computed). The law

$$\hat{\mu} = \mathcal{L}\Big( \sum_k \frac{e_k}{\hat{t}_k^2} \Big) = \mathcal{L}\Big( \sum_k k \hat{t}_k^{-2} (G_k - G_{k+1}) \Big)$$

is an explicit GGC hitting time defined without reference to any zero. It has the same counting function of Thorin atoms as $\mu$ to leading order, hence the same small time behaviour $-\log \hat{\mu}(0,t] \sim (\log(1/t))^2/(16t)$ (Section 8.1). More generally let $\hat{\mu}^{[N]}$ use the true $\gamma_k$ for $k \le N$ and $\hat{t}_k$ beyond. Since the first $1.2 \cdot 10^{13}$ zeros are on the line, $\hat{\mu}^{[N]}$ is a GGC unconditionally for every $N$ up to that size. Table 24.7 compares them with $\mu$, using $4 \cdot 10^5$ smoothed zeros plus an integral for the rest, Talbot inversion in double precision, and the residue expansion of $\mu$ on the grid $y = 0.007 + 0.001\, i$ of Section 24.6 (computed, code/ht_ev.py).

| $N$ | $\sup \lvert F_N - F \rvert$ | mean error |
|---|---|---|
| 0 | $1.25 \cdot 10^{-2}$ | $-2.2 \cdot 10^{-4}$ |
| 1 | $3.2 \cdot 10^{-3}$ | $4.6 \cdot 10^{-5}$ |
| 2 | $2.7 \cdot 10^{-3}$ | $-3.6 \cdot 10^{-5}$ |
| 5 | $8.6 \cdot 10^{-4}$ | $1.1 \cdot 10^{-5}$ |
| 10 | $2.5 \cdot 10^{-5}$ | $-3.2 \cdot 10^{-7}$ |
| 30 | not resolved | $3.4 \cdot 10^{-8}$ |
| 100 | not resolved | $-8.4 \cdot 10^{-9}$ |

Table 24.7: The smoothed zero laws $\hat{\mu}^{[N]}$, with the first $N$ zeros exact and the rest at the Gram midpoints, against $\mu$. The inversion in double precision resolves differences of distribution functions only down to about $2 \cdot 10^{-6}$ for $y \ge 0.011$ (and $1.8 \cdot 10^{-5}$ below), so the supremum for $N = 30$ and $N = 100$ is unchecked; the source printed $2.7 \cdot 10^{-6}$ and $6.8 \cdot 10^{-7}$.

The smoothed law alone is within 1.3% of $\mu$ in distribution function, and inserting the first ten zeros brings the error to $2.5 \cdot 10^{-5}$, forty times better than the Jensen law of degree 2560 ($1.07 \cdot 10^{-3}$). The error is governed by $\sum_{k>N} |\gamma_k^{-2} - \hat{t}_k^{-2}|$, that is by the fluctuation $S(t)$ of the zero counting function, and the extremal representation shows where it enters: through the weights on the points $G_k$ with $k > N$ (heuristic). These laws are as unconditional as the zero verification they use, and their convergence to $\mu$ as $N \to \infty$ is again RH. What they add is speed:

the smooth part carries the archimedean information exactly, and only the fluctuation $S(t)$ needs the zeros. Section 8.1.3 has the Kent approximants with a Weyl tail above a Gram point, which are the continuous counterpart; the two tables are not the same experiment.

### 24.7.4 The Pólya density

Curry and Schoenberg's theory applies to Pólya frequency functions, those whose bilateral Laplace transform is the reciprocal of a Laguerre and Pólya function. The Pólya density $p(u) = 4\Phi_{\mathrm{dBN}}(|u|)/\Xi(0)$ of $V/2$ has bilateral transform $\xi(\frac{1}{2} + s/2)/\xi(\frac{1}{2}) = G(s^2/4)$. It is not a Pólya frequency function of infinite order: an entire bilateral transform does not exclude that (the Gaussian density is one, with transform $e^{cs^2}$), but a transform that is entire and has zeros is not the reciprocal of an entire function. This was also confirmed directly. With $p$ normalised by $p(0) = 1$, the Toeplitz determinants $\det[p((i-j)h)]_{i,j\le k}$ are nonnegative for $k \le 4$ at every tested step, but for $k = 5$ they are $-4.0\cdot10^{-11}$ and $-1.4\cdot10^{-7}$ at $h = 0.03$ and $h = 0.05$ (also $-2.1\cdot10^{-20}$ at 0.01 and $-1.8\cdot10^{-14}$ at 0.02; the same to eighty digits at 90 and 130 digit precision) and positive for $h \ge 0.07$ among the tested steps (computed, code/ht_ev.py). So $p$ appears log concave and to be $\mathrm{PF}_4$, but it is not $\mathrm{PF}_5$; the $k = 6$ minor is negative at $h = 0.05$ and 0.07 ($-2.3\cdot10^{-9}$ and $-5.2\cdot10^{-6}$). The object to which Curry and Schoenberg apply is the dual density $f$ of $\mu$, whose transform $1/G = 1/\mathbb{E}\cosh(\sqrt{u}\,V)$ is the reciprocal of the transform of $p$ in the variable $\sqrt{u}$; $f$ is $\mathrm{PF}_\infty$ if and only if RH holds (Proposition 8.1.2). The extreme value structure is on the Pólya side as well, in a different form: Biane, Pitman and Yor's variable [49] is the range of a Brownian bridge, and by the Jacobi triple product the alternating theta function

$$\sum_{n\in\mathbb{Z}}(-1)^n q^{n^2} = \prod_{m\ge1}(1-q^{2m})(1-q^{2m-1})^2, \qquad q = e^{-x},$$

is the distribution function of the maximum of independent exponentials of rates $2m$ and, twice, $2m-1$. With $x = 2a^2$ this is Kolmogorov's law of $\sup|b|$. Its Mellin transform carries the factor $1 - 2^{1-s}$, whose zeros lie on $\mathrm{Re}\,s = 1$; heuristically this is a reason, unrelated to RH, why a reciprocal law built from this maximum is not a GGC, and only the nonalternating theta function, which is not a product of distribution functions, gives $\xi$ alone.

### 24.7.5 Assessment

The extreme value view gives three things: the B spline and Dirichlet mean form of every truncation (Proposition 24.7.1), a representation of $\mu$ as a linear functional of the Gumbel extremal process (Theorem 24.7.2), and the smoothed zero laws, the fastest converging approximating hitting times found here, at the cost of using the verified zeros. None of them is non circular in the way the Jensen laws are,

and the Jensen laws converge slowly. The trade off between the two families is the same as before: the information about the location of the zeros has to be put in, either through the zeros themselves or through a positivity condition at every degree.

## 24.8 The prime side hitting times have a universal limit

### 24.8.1 Truncated Euler products

For $P \geq 2$ let $\zeta_P(z) = \prod_{p\leq P}(1-p^{-z})^{-1}$ and let $N_P$ be the random integer with $\Pr(N_P = n) = n^{-\alpha}/\zeta_P(\alpha)$ for $n$ composed of primes up to $P$. Then $L = \log N_P = \sum_{p\leq P}(\log p)G_p$ with independent geometric variables $G_p$, $\Pr(G_p = k) = (1-p^{-\alpha})p^{-k\alpha}$, so $L$ is infinitely divisible with finite mean $\mathbb{E}L = B_{\alpha,P} = \sum_{p\leq P}\log p\, p^{-\alpha}/(1-p^{-\alpha})$, and

$$-\log \mathbb{E}e^{-wL} = \sum_{p\leq P}\sum_{k\geq 1}\frac{p^{-k\alpha}}{k}\big(1-e^{-wk\log p}\big), \qquad \operatorname{Re} w \geq 0.$$

Let $T_a$ denote the first passage time of standard Brownian motion to the level $a$, so that $\mathbb{E}e^{-uT_a} = e^{-a\sqrt{2u}}$.

Proposition 24.8.1 (Truncated Euler products and first passage to the mean level). *For every $\alpha > 0$ and every $P$, the function*

$$\Phi_{\alpha,P}(u) = \frac{e^{-B_{\alpha,P}\sqrt{2u}}}{\mathbb{E}e^{-\sqrt{2u}\,L}} = \frac{\zeta_P(\alpha)}{\zeta_P(\alpha+\sqrt{2u})}\,e^{-B_{\alpha,P}\sqrt{2u}}$$

*is the Laplace transform of a GGC law $Y_P$ with zero drift, whose Thorin measure has density in the ordinate $\tau$ (with atom positions $\tau^2/2$)*

$$\frac{1}{\pi}\sum_{p\leq P}\sum_{k\geq 1}\log p\, p^{-k\alpha}\big(1-\cos(k\tau\log p)\big) \;\geq 0.$$

*In particular this holds on the critical line $\alpha = \frac{1}{2}$, unconditionally. If $Y_P$ is independent of $N_P$ and of the Brownian motion, then*

$$T_{\mathbb{E}\log N_P} \overset{d}{=} Y_P + T_{\log N_P}.$$

*Proof.* Put $\tilde\mu = \sum_{p\leq P}\sum_{k\geq 1}\log p\, p^{-k\alpha}\delta_{k\log p}$, a positive measure with $\int(1\wedge x)\tilde\mu(\mathrm{d}x) < \infty$ because the sum is finite over $p$ and converges over $k$. With $f(w) = \mathbb{E}e^{-wL}$ the formula above reads $\log f(w)/f(0) - wf'(0)/f(0) = \int(e^{-wx}-1+wx)\tilde\mu(\mathrm{d}x)/x$, since $\tilde\mu(\mathrm{d}x)/x$ puts the mass $p^{-k\alpha}/k$ at $k\log p$ and $-f'(0)/f(0) = \mathbb{E}L$. The sine squared lemma (Lemma 6.2.5) therefore gives $\Phi_{\alpha,P}(u) = \exp\big(-\int_0^\infty \log(1+2u/\tau^2)U(\mathrm{d}\tau)\big)$ with $U(\mathrm{d}\tau) =$

$\pi^{-1}\int(1-\cos\tau x)\tilde\mu(\mathrm{d}x)\,\mathrm{d}\tau$, which is the stated density and is nonnegative termwise; there is no drift because $-\log\Phi_{\alpha,P}(u)=O(\sqrt{u})$. For the second statement, $\mathbb{E}e^{-uT_{\log N_P}}=\mathbb{E}e^{-\sqrt{2u}L}$ and $\mathbb{E}e^{-uT_{\mathbb{E}L}}=e^{-\sqrt{2u}\mathbb{E}L}$, so $\mathbb{E}e^{-uT_{\mathbb{E}L}}=\Phi_{\alpha,P}(u)\,\mathbb{E}e^{-uT_{\log N_P}}$. □

The zeros of $1/\zeta_P(\alpha+w)$ lie on the line $\operatorname{Re}w=-\alpha$ and cause no difficulty: the principal branch $w=\sqrt{2u}$ maps $\mathbb{C}\setminus(-\infty,0]$ into the half plane $\operatorname{Re}w>0$, where $\zeta_P(\alpha+w)$ has neither zeros nor poles, as a GGC Laplace transform requires. The mechanism of Proposition 24.8.1 is a Jensen gap: for any infinitely divisible $L\ge0$ with finite mean the same argument gives $T_{\mathbb{E}L}\overset{d}{=}Y+T_L$ with $Y$ a GGC independent of $L$, a sharpened form of Jensen's inequality $\mathbb{E}e^{-wL}\ge e^{-w\mathbb{E}L}$; nothing arithmetic is used beyond the infinite divisibility of $\log N_P$. The proposition is the statement of [250] for truncated products, proved here from the sine squared lemma of the book; the corresponding statement at $\alpha>1$ for the full product $\zeta$ is not used in this chapter (Remark 24.8.2). As a check, the source reports at $\alpha=\frac12$, $P=13$ and $u=0.7$ the Thorin integral 5.92746438 against $-\log\Phi_{1/2,13}(0.7)=5.92746429$; this was not recomputed.

Remark 24.8.2 (One prime blocks: Möbius, Liouville and the Pólya Gamma law). The following is from [253]. Fix a prime $p$ and write $\ell=\log p$, $q=p^{-\alpha}$ with $\alpha>0$, $w=\sqrt{2u}$ and $\vartheta=\tau\ell$. For a single prime the density of Proposition 24.8.1 sums in closed form,

$$\frac{1}{\pi}\sum_{k\ge1}\ell\,q^k\bigl(1-\cos k\vartheta\bigr)=\frac{1}{\pi}\,\frac{\ell\,q(1+q)(1-\cos\vartheta)}{(1-q)(1-2q\cos\vartheta+q^2)},$$

which vanishes at $\tau=2\pi k/\ell$, the ordinates of the zeros of $1-p^{-s}$ on $\operatorname{Re}s=0$, that is of the poles of the Euler factor $e_p(s)=(1-p^{-s})^{-1}$: it is the constant $\frac{\ell}{2\pi}\coth\frac{\alpha\ell}{2}$ minus the periodic row of Cauchy kernels $\pi^{-1}\alpha/(\alpha^2+(\tau-2\pi k/\ell)^2)$ centred there, so as $\alpha\downarrow0$ its plateau rises like $1/(\pi\alpha)$. The Liouville factor $1/(1+p^{-s})$, the Euler factor of $\sum_n\lambda(n)n^{-s}=\zeta(2s)/\zeta(s)$, gives the odd lattice. Put

$$\Phi^{+}_{p,\alpha}(u)=\frac{1+q}{1+q\,p^{-w}}\exp\Bigl(-\frac{w\ell q}{1+q}\Bigr).$$

For every $\alpha>0$ this is the Laplace transform of a GGC with zero drift, with Thorin density in the ordinate $\tau$

$$\frac{1}{\pi}\,\frac{\ell\,q(1-q)(1-\cos\vartheta)}{(1+q)(1+2q\cos\vartheta+q^2)}\ \ge0,$$

whose bumps sit at $\tau=\pi(2k-1)/\ell$, the ordinates of the zeros of $1+p^{-s}$ on $\operatorname{Re}s=0$. Indeed $-\log\Phi^{+}_{p,\alpha}(u)=\sum_{k\ge1}(-1)^{k+1}\frac{q^k}{k}(e^{-wk\ell}-1+wk\ell)$, so the proof of Proposition 24.8.1 applies to the signed measure $\sum_k(-1)^{k+1}\ell q^k\delta_{k\ell}$, whose

total variation is that of the Möbius case, and the series $\frac{1}{\pi}\sum_k (-1)^{k+1}\ell q^k(1-\cos k\vartheta)$ sums to the displayed density, which is nonnegative although its terms are not. As $\alpha \downarrow 0$, $\Phi^+_{p,\alpha}(u) \to 1/\cosh(\ell\sqrt{u/2})$, the Laplace transform of $\ell^2\omega$ with $\omega$ a Pólya Gamma variable PG(1, 0) [258], and the Thorin densities converge vaguely to $\sum_{k\geq 1}\delta_{\pi(2k-1)/\ell}$, the Thorin measure of that law (compare Theorem 22.11.2). In the Möbius case the compensator $\ell q/(1-q)$ diverges as $\alpha \downarrow 0$, because $e_p$ has a pole at $s = 0$. The even lattice appears instead through $g(s) = (1-p^{-s})/(s\ell) = p^{-s/2}\sinh(s\ell/2)/(s\ell/2)$, the factor $1-p^{-s}$ with its zero at $s = 0$ removed, whose zeros $2\pi ik/\ell$, $k \neq 0$, all lie on $\operatorname{Re} s = 0$: at $\alpha = 0$ its compensated reciprocal $e^{wg'(0)/g(0)}g(0)/g(w)$ equals $(w\ell/2)/\sinh(w\ell/2)$, the Laplace transform of $(\ell/2\pi)^2 H$, with $H$ the logistic mixing law $H(1, 0)$ of [31], $\mathbb{E}e^{-uH} = \pi\sqrt{2u}/\sinh(\pi\sqrt{2u})$, whose Thorin atoms are the even lattice $\tau = 2\pi k/\ell$, $k \geq 1$.

For $\alpha > 1$ the products over all primes converge and give the GGC transforms

$$\frac{\zeta(\alpha)}{\zeta(\alpha+w)}\,e^{w\frac{\zeta'}{\zeta}(\alpha)} \qquad \text{and} \qquad \frac{\zeta(2\alpha+2w)\,\zeta(\alpha)}{\zeta(\alpha+w)\,\zeta(2\alpha)}\,e^{wG_\lambda(\alpha)},$$

$$G_\lambda(s) = \frac{\zeta'}{\zeta}(s) - 2\frac{\zeta'}{\zeta}(2s),$$

with Thorin densities $\pi^{-1}[\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+i\tau) - \frac{\zeta'}{\zeta}(\alpha)]$ and $\pi^{-1}[\operatorname{Re} G_\lambda(\alpha+i\tau) - G_\lambda(\alpha)]$, the sums of the one prime densities. The exponents carry the sign + because $\sum_p \ell q/(1-q) = -\frac{\zeta'}{\zeta}(\alpha)$ and $\sum_p \ell q/(1+q) = -G_\lambda(\alpha)$. The logarithms of the finite products converge for every $u$, since $|e^{-x} - 1 + x| \leq x$ and $\sum_p \ell q/(1-q) < \infty$, the limits are continuous at $u = 0$, and the class of GGC laws is closed under weak limits [56]. For $0 < \alpha < 1$ neither continued function is the Laplace transform of a probability law (the second is defined only for $\alpha \neq \frac{1}{2}$, where $\zeta(2\alpha)$ is infinite), whatever the zeros of $\zeta$ do: the first vanishes at $u = (1-\alpha)^2/2$, where $\alpha + w = 1$; the second vanishes there when $\frac{1}{2} < \alpha < 1$, and has a pole at $u = (\frac{1}{2} - \alpha)^2/2$ when $0 < \alpha < \frac{1}{2}$.

The difficulty is the limit $P \to \infty$. For $\alpha \leq 1$ the compensator $B_{\alpha,P}$ diverges, like $P^{1-\alpha}/(1-\alpha)$ when $\alpha < 1$ and like $\log P$ when $\alpha = 1$, and it does so faster than $\log\zeta_P(\alpha+w)$ for any $w > 0$. Hence $-\log\Phi_{\alpha,P}(u) \to \infty$ for every $u > 0$. The laws escape to infinity and have no weak limit, so Bondesson's closure theorem has nothing to act on.

### 24.8.2 A Brownian truncation of the Euler product centred at one half

In the strip the primes can be truncated not by size but in Brownian time [253]. Let $W(t) = \sum_{\operatorname{Re}\gamma>0} e^{-t\gamma^2}$ be the heat trace over the zeros, counted with multiplicity, and let $T_x$ be the first passage time of standard Brownian motion to the level $x > 0$, as above.

THEOREM 24.8.3 (Brownian truncated Euler product). *Let $U>0$ and let $s\in\mathbb{C}$ with $\operatorname{Re}(s^2)>-\gamma_1^2$, a region that contains $\mathbb{R}$ and the band $|\operatorname{Im} s|<\gamma_1$, and in which $\operatorname{Re}(\gamma^2+s^2)>0$ for every zero. Put*

$$\begin{aligned} B_U(x,s)&=\mathbb{E}\big[1-e^{-s^2T_x/2};\,T_x\le U\big]\\ &=\operatorname{erfc}\frac{x}{\sqrt{2U}}-\frac12\Big[e^{-xs}\operatorname{erfc}\frac{x-sU}{\sqrt{2U}}+e^{xs}\operatorname{erfc}\frac{x+sU}{\sqrt{2U}}\Big],\end{aligned}$$

$$k_\infty(u)=e^{u/8}-\frac{\log\pi}{2\sqrt{2\pi u}}+\frac{1}{2\pi}\int_0^\infty e^{-ur^2/2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac14+\frac{ir}{2}\Big)dr,$$

$$A_U(s)=\int_0^U\big(1-e^{-s^2u/2}\big)k_\infty(u)\,\frac{du}{u},$$

$$R_U(s)=\sum_{\operatorname{Re}\gamma>0}\Big[E_1\Big(\frac{\gamma^2U}{2}\Big)-E_1\Big(\frac{(\gamma^2+s^2)U}{2}\Big)\Big],\qquad E_1(z)=\int_1^\infty e^{-zv}\,\frac{dv}{v}.$$

*Then, with no hypothesis on the zeros,*

$$\frac{\xi(\frac12)}{\xi(\frac12+s)}=e^{-A_U(s)}\prod_p E_{p,U}(s)\,e^{-R_U(s)},$$

$$E_{p,U}(s)=\exp\Big(\sum_{m\ge1}\frac{p^{-m/2}}{m}\,B_U(m\log p,s)\Big),$$

*the product converging absolutely, and*

$$|R_U(s)|\le\frac{2}{U}\,e^{-m_sU/2}\sum_{\operatorname{Re}\gamma>0}\Big(\frac{1}{\operatorname{Re}\gamma^2}+\frac{1}{\operatorname{Re}(\gamma^2+s^2)}\Big),$$

$$m_s=\gamma_1^2+\min\big(0,\operatorname{Re}(s^2)\big)>0.$$

*If moreover $\operatorname{Re}s>0$ and $\operatorname{Re}(s^2)\ge0$, then $E_{p,U}(s)\to(1-p^{-1/2-s})/(1-p^{-1/2})$ as $U\to\infty$, the Euler factor of $1/\zeta(\frac12+s)$ normalised at the centre.*

*Proof.* For $\operatorname{Re}a,\operatorname{Re}b>0$, Frullani's integral gives $\log(b/a)=\int_0^\infty(e^{-au}-e^{-bu})\frac{du}{u}$ with the principal logarithm. With $a=\gamma^2/2$, $b=(\gamma^2+s^2)/2$ and the paired Hadamard product $\xi(\frac12+s)/\xi(\frac12)=\prod_{\operatorname{Re}\gamma>0}(1+s^2/\gamma^2)$,

$$\log\frac{\xi(\frac12+s)}{\xi(\frac12)}=\int_0^\infty\big(1-e^{-s^2u/2}\big)W\Big(\frac{u}{2}\Big)\frac{du}{u},$$

the sum over the zeros and the integral being exchangeable because the integrals of the absolute values are $O(|s|^2/\operatorname{Re}\gamma^2)$. Every zero has $\operatorname{Re}\gamma^2\ge\gamma_1^2$: on the line trivially, and off it because its ordinate exceeds $3\times10^{12}$ [419] while $|\operatorname{Im}\gamma|<\frac12$. Split the integral at $U$. On $(U,\infty)$, $\int_U^\infty(e^{-au}-e^{-bu})\frac{du}{u}=E_1(aU)-E_1(bU)$ gives

$R_U(s)$, and $|E_1(z)| \le e^{-\operatorname{Re} z}/\operatorname{Re} z$ for $\operatorname{Re} z > 0$, with $\operatorname{Re}\gamma^2 \ge \gamma_1^2$ and $\operatorname{Re}(\gamma^2+s^2) \ge m_s$, gives the bound. On $(0,U]$ insert the arithmetic form (1.2.5) of the heat trace,

$$W\Big(\frac{u}{2}\Big) = k_\infty(u) - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\,\frac{e^{-(\log n)^2/2u}}{\sqrt{2\pi u}}.$$

Near $u = 0$, $k_\infty(u) = O(u^{-1/2}\log(1/u))$ while $1 - e^{-s^2u/2} = O(u)$, so the archimedean part converges and gives $A_U(s)$. In the prime part $e^{-x^2/2u}/(u\sqrt{2\pi u}) = f_x(u)/x$, where $f_x(u) = x\,e^{-x^2/2u}/\sqrt{2\pi u^3}$ is the density of $T_x$; since $|1 - e^{-s^2u/2}| \le 1 + e^{|s|^2U/2}$ on $[0,U]$ and $\Pr(T_x \le U) = \operatorname{erfc}(x/\sqrt{2U}) \le e^{-x^2/2U}$, the double sum and integral converge absolutely, so they may be exchanged, and the prime power $n = p^m$ contributes $-\frac{p^{-m/2}}{m}B_U(m\log p, s)$. The closed form of $B_U$ is the reflection principle, $\mathbb{E}[e^{-\lambda T_x}; T_x \le U] = \frac12[e^{-x\sqrt{2\lambda}}\operatorname{erfc}(\frac{x}{\sqrt{2U}} - \sqrt{\lambda U}) + e^{x\sqrt{2\lambda}}\operatorname{erfc}(\frac{x}{\sqrt{2U}} + \sqrt{\lambda U})]$ with $\sqrt{2\lambda} = s$, an even entire function of $s$. Exponentiating gives the identity. If $\operatorname{Re} s > 0$ and $\operatorname{Re}(s^2) \ge 0$, then $|e^{-s^2T_x/2}| \le 1$, dominated convergence gives $B_U(x,s) \to 1 - \mathbb{E}e^{-s^2T_x/2} = 1 - e^{-xs}$, and $\sum_m \frac{p^{-m/2}}{m}(1-p^{-ms}) = \log\frac{1-p^{-1/2-s}}{1-p^{-1/2}}$. $\square$

When $\operatorname{Re}(s^2) < 0$ the truncation cannot be removed: the integrand of $B_U(x,s)$ then grows exponentially in $u$, and $B_U(x,s)$ is unbounded as $U \to \infty$. The structure, a smoothed Euler product times a smoothed product over the zeros, is that of the hybrid Euler Hadamard product of Gonek, Hughes and Keating [149]; here the smoothing kernel is the first passage density and the identity is exact for every $U$. Table 24.8 checks it, with primes up to $10^3$ for $U \le 0.25$ and $10^6$ for $U = 1$ and the first 1700 zeros (`data/zeros_1700.txt`) for $R_U$, the rest bounded by the theorem (computed, `code/rx_euler.py`).

### 24.8.3 The universal limit

The escape of the truncated Euler family of Proposition 24.8.1 can be undone by centring and scaling. At $\alpha = \frac12$ write $L_P = \log N_P = \sum_{p\le P}(\log p)G_p$; its cumulants satisfy $\kappa_j(P) = \sum_{p\le P}(\log p)^j \sum_{m\ge1} m^{j-1}p^{-m/2} \asymp \sqrt{P}(\log P)^{j-1}$. The Laplace exponent of $Y_P$ is $-\log\mathbb{E}e^{-uY_P} = K_P(\sqrt{2u})$ with

$$K_P(w) = \log\mathbb{E}e^{-w(L_P - \mathbb{E}L_P)} = \sum_{p\le P}\sum_{m\ge1}\frac{p^{-m/2}}{m}\big(e^{-wm\log p} - 1 + wm\log p\big)$$
$$= \sum_{j\ge2}\frac{(-1)^j\kappa_j w^j}{j!} \quad (|w| < \tfrac12;\ \text{the proof does not use the series}).$$

In particular $Y_P$ has mean $\kappa_2(P) = \operatorname{Var} L_P$ and a tail of index $\frac32$ governed by $\kappa_3(P)$.

| $s$ | $U$ | $\log \frac{\xi(1/2+s)}{\xi(1/2)}$ | $A_U - \sum_p \log E_{p,U}$ |
|---|---|---|---|
| 0.1 | 0.05 | $2.31048072572 \times 10^{-4}$ | $2.30708738670 \times 10^{-4}$ |
| 0.1 | 1 | $2.31048072572 \times 10^{-4}$ | $2.31048072572 \times 10^{-4}$ |
| 0.49 | 0.05 | $5.54643804846 \times 10^{-3}$ | $5.53831869645 \times 10^{-3}$ |
| 0.49 | 1 | $5.54643804846 \times 10^{-3}$ | $5.54643804846 \times 10^{-3}$ |
| 2 | 1 | $9.21256255914 \times 10^{-2}$ | $9.21256255914 \times 10^{-2}$ |
| $0.3+3i$ | 0.25 | $-0.207311444533 + 0.0422063284094i$ | $-0.207311444532 + 0.0422063284090i$ |

| $s$ | $U$ | $R_U$ | bound |
|---|---|---|---|
| 0.1 | 0.05 | $3.39 \times 10^{-7}$ | $1.25 \times 10^{-2}$ |
| 0.1 | 1 | $2.06 \times 10^{-48}$ | $3.83 \times 10^{-45}$ |
| 0.49 | 0.05 | $8.12 \times 10^{-6}$ | $1.25 \times 10^{-2}$ |
| 0.49 | 1 | $4.68 \times 10^{-47}$ | $3.83 \times 10^{-45}$ |
| 2 | 1 | $3.55 \times 10^{-46}$ | $3.81 \times 10^{-45}$ |
| $0.3+3i$ | 0.25 | $(-1.15 + 0.406i) \times 10^{-12}$ | $1.62 \times 10^{-11}$ |

Table 24.8: Theorem 24.8.3 (computed, `code/rx_euler.py`). The upper block compares the two sides of the identity and the lower block gives the zero remainder $R_U$ and the bound of the theorem for the same rows; the residual is below $10^{-30}$ in every row. The rows with $|\operatorname{Re} s| < \frac{1}{2}$ ($s = 0.1$, 0.49 and $0.3 + 3i$) have $\frac{1}{2} + s$ inside the critical strip. At $U = 0.05$ the zero remainder is visible; at $U = 1$ it is below the working precision (30 digits). The bound is loose at small $U$.

THEOREM 24.8.4 (Universal limit of the Euler hitting times). *As $P \to \infty$,*

$$Z_P = \frac{Y_P - \kappa_2(P)}{\kappa_3(P)^{2/3}} \implies Z, \qquad \log \mathbb{E} e^{-sZ} = \frac{(2s)^{3/2}}{6}, \quad \operatorname{Re} s \geq 0,$$

*a spectrally positive stable law of index $\frac{3}{2}$. More precisely, for $\operatorname{Re} s \geq 0$,*

$$\left| \log \mathbb{E} e^{-sZ_P} - \frac{(2s)^{3/2}}{6} \right| \leq \frac{|s|^2}{6} \rho_4(P), \qquad \rho_j(P) = \frac{\kappa_j(P)}{\kappa_3(P)^{j/3}},$$

*and $\rho_4(P) \to 0$.*

*Proof.* Put $c = \kappa_3^{2/3}$ and $w = \sqrt{2s/c}$ with the principal root, so $\operatorname{Re} w \geq 0$. Then $\log \mathbb{E} e^{-sZ_P} = s\kappa_2/c - K_P(w)$. For $\operatorname{Re} x \geq 0$ one has $|e^{-x} - 1 + x - x^2/2 + x^3/6| \leq |x|^4/24$, since $|e^{-\xi}| \leq 1$ on the segment from 0 to $x$. Apply this with $x = wm \log p$ in the second form of $K_P$: the terms of degree two and three sum to $\kappa_2 w^2/2 - \kappa_3 w^3/6$, and the rest is at most $\kappa_4 |w|^4/24$. Now $\kappa_2 w^2/2 = \kappa_2 s/c$ cancels the centring, $\kappa_3 w^3/6 = \kappa_3 (2s/c)^{3/2}/6 = (2s)^{3/2}/6$ because $c^{3/2} = \kappa_3$, and $\kappa_4 |w|^4/24 = |s|^2 \rho_4/6$. Finally $\rho_4 \asymp P^{-1/6} (\log P)^{1/3} \to 0$. At $\operatorname{Re} s = 0$ this gives the characteristic functions, so Lévy's continuity theorem gives weak convergence; the limit $\exp((2s)^{3/2}/6)$ has a Laplace exponent $Cs^{3/2}$ with $C > 0$, that of a spectrally positive stable law of index $\frac{3}{2}$. □

Exact computation from the truncated Euler product confirms the limit, with the slow rate $P^{-1/6}$ that the cumulant estimate predicts (computed, code/ht_euler32.py; primes up to $10^7$):

| $P$ | $10^3$ | $10^4$ | $10^5$ | $10^6$ | $10^7$ | limit |
|---|---|---|---|---|---|---|
| $s=0.5$ | 0.1466 | 0.1523 | 0.1563 | 0.1591 | 0.1612 | 0.1667 |
| $s=1$ | 0.3946 | 0.4158 | 0.4307 | 0.4416 | 0.4498 | 0.4714 |
| $s=2$ | 1.0443 | 1.1200 | 1.1752 | 1.2168 | 1.2484 | 1.3333 |

Table 24.9: The quantity $\log \mathbb{E}e^{-sZ_P}$ for $P = 10^3, \dots, 10^7$, against its limit $(2s)^{3/2}/6$ (computed).

The rate is explicit. Continuing the expansion one term further,

$$\log \mathbb{E}e^{-sZ_P} = \frac{(2s)^{3/2}}{6} - \rho_4(P)\frac{(2s)^2}{24} + \rho_5(P)\frac{(2s)^{5/2}}{120} - \cdots,$$

and $\rho_4(P) \sim 2^{-1/3}P^{-1/6}(\log P)^{1/3}$. The exact values of $\rho_4(P)$ are 0.54, 0.37, 0.26, 0.19, 0.13 at $P = 10^3, \dots, 10^7$, while the asymptotic expression gives 0.48, 0.36, 0.26, 0.19, 0.14 (computed). With the $\rho_4$ and $\rho_5$ terms the expansion reproduces every entry of Table 24.9 to within $1.8 \cdot 10^{-2}$ at $P = 10^3$ and $2.2 \cdot 10^{-4}$ at $P = 10^7$; at $P = 10^7$ and $s = 1$ it gives 0.44986 against the exact 0.44983 (computed). The bound of the theorem gives the rate for the first half of Problem 24.1 in the sense of a bound on the Laplace exponent, and the expansion is a computed description of the next terms.

The limit depends only on the central limit structure of $\log N_P$. It is the same for every Euler product with the same prime weights, whatever its zeros, and being centred and supported on the whole line it is not even a hitting time. The renormalisation that rescues the Euler family erases exactly the information that RH concerns. This is the prime side counterpart of the Davenport and Heilbronn test of Section 24.14.

## 24.9 Thorin rigidity

A second obstruction is structural.

Theorem 24.9.1 (Thorin rigidity). *If $\mu = \mu_1 * \mu_2$ with $\mu_1$ and $\mu_2$ GGC laws, then RH holds and*

$$U_1 = \sum_{\gamma>0} a_\gamma \delta_{\gamma^2}, \qquad U_2 = \sum_{\gamma>0} (m_\gamma - a_\gamma)\delta_{\gamma^2}, \qquad 0 \le a_\gamma \le m_\gamma,$$

*where $m_\gamma$ is the multiplicity of the zero (so $m_\gamma = 1$ for simple zeros), and $\mu_1$ is the law of $\sum_\gamma g_\gamma/\gamma^2$ with independent gamma variables $g_\gamma$ of shapes $a_\gamma$. No GGC law with drift or with a nonzero absolutely continuous Thorin part is a GGC factor of $\mu$ with a GGC cofactor.*

*Proof.* Drifts and Thorin measures add under convolution of GGC laws and are determined by the Laplace transform. So $\mu$ is a GGC, which is RH, with drift zero and Thorin measure $\sum m_\gamma \delta_{\gamma^2}$; the nonnegative measures $U_1, U_2$ summing to it are carried by the points $\gamma^2$ with masses in $[0, m_\gamma]$. □

The hypothesis that the cofactor is also a GGC cannot be dropped. Assume RH with simple zeros and take $\gamma_1^2 < a < \gamma_2^2$. The function $(1 + u/a)/((1 + u/\gamma_1^2)(1 + u/\gamma_2^2))$ is a mixture of two exponential transforms with weights $(1 - \gamma_1^2/a)/(1 - \gamma_1^2/\gamma_2^2)$ and $(1 - \gamma_2^2/a)/(1 - \gamma_2^2/\gamma_1^2)$, both positive with sum one, so the exponential law of rate $a$ is an exact factor of $\mu$, with a probability cofactor that is the convolution of this mixture with the remaining exponentials. The same holds for a GGC whose Thorin density is supported inside $(\gamma_1^2, \gamma_2^2)$ with total mass at most one: each cofactor term is a mixture of exponentials, hence infinitely divisible [152, 486], so the exact factor exists with a cofactor that is a probability law (proved). Hence only factorisations into two GGC laws are rigid.

Every prime built law met so far is a GGC with an absolutely continuous Thorin measure: the Euler laws for $\alpha > 1$, the truncated Euler laws $Y_P$, and the basepoint laws $\mu^{(\alpha)}$ for $\alpha \geq 1$. None can be an exact factor of $\mu$ with a GGC cofactor. A proof through such pieces must therefore be a limit in which absolutely continuous Thorin measures converge vaguely to $\sum \delta_{\gamma^2}$. The basepoint descent is a limit of this kind; the Euler family, by Theorem 24.8.4, is not, after the one centring and scaling treated there.

## 24.10 What the prime side limit would have to be

Remove the compensator $B_{1/2,P}/\pi$ from the Thorin density of $Y_P$ at $\alpha = \frac{1}{2}$ (it is the Thorin density of the stable factor $e^{-B\sqrt{2u}}$) and add the archimedean density $\pi^{-1} A_{1/2}(\tau)$, where $A_{1/2}(\tau) = \operatorname{Re} A(\frac{1}{2} + i\tau)$ and $A(z) = \frac{1}{z} + \frac{1}{z-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{z}{2})$ is the archimedean part of $\xi'/\xi$. What remains is the formal Thorin density $\pi^{-1}[A_{1/2}(\tau) + \operatorname{Re}\zeta_P'/\zeta_P(\frac{1}{2} + i\tau)]$. Against test functions whose Fourier transforms decay fast enough the prime sums converge, and the explicit formula identifies the limit. In the Gaussian case it is the identity (Weil's explicit formula with the test function $e^{-xr^2}$, proved)

$$\sum_{\gamma>0} e^{u_\rho x} = e^{x/4} + (\text{archimedean terms}) \\ - \frac{1}{2\sqrt{\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/(4x)},$$

whose left side is $W(x)$ and in which the pole of $\zeta$ appears as $e^{x/4}$. The zeros emerge from the primes as atoms, but only weakly and only after the pole term is added; at no finite stage is the prime side density positive, already at $\tau =$

0, where $A_{1/2}(0) = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{1}{4}) = -2.686 < 0$ and $\zeta_P'/\zeta_P(\frac{1}{2}) < 0$. The positivity a prime side hitting time proof needs is positivity of this limit against every admissible test function, which is a Weil type criterion at the centre of the strip (compare Chapters 17 and 20), the same inequality that is stated at $\alpha = 1$ on windows of one mean zero spacing in [421], now required at $\alpha = \frac{1}{2}$ where the pole term has exponential size.

Remark 24.10.1 (Shape of a hitting time proof, heuristic). A proof of RH through hitting times built from the primes would have to produce GGC laws whose Thorin measures converge vaguely to $\sum \delta_{\gamma^2}$, which is what weak convergence to $\mu$ means by the continuity theorem. By Theorem 24.9.1 they cannot be exact factors of $\mu$ with GGC cofactors. The truncated Euler family does not do it after centring and scaling (Theorem 24.8.4), but that excludes one renormalisation of one family only. And the positivity of the limit would be a Weil type criterion at the centre, by the heuristic of the preceding paragraph. Only the statement about exact factors is a theorem; the others are heuristic.

## 24.11 Diffusion approximations

Diffusion approximations in the sense of Roberts, Gelman and Gilks [262], in which high dimensional Markov chains, suitably sped up, converge to a Langevin diffusion, offer half of what is needed. (The boundary crossing approach of Roberts and Shortland is a different tool and is treated in Section 24.12.) The hitting time of a level by a one dimensional diffusion started at an entrance or reflecting end is a GGC, by Kent's theorem [305] (Section 8.1.3); from an interior point it need not be. For reflected Brownian motion on $[0,\infty)$ started at $x \in (0,1)$ the time to reach 1 has transform $\cosh(x\sqrt{2u})/\cosh(\sqrt{2u})$, and $-\varphi'/\varphi = \sum_k 1/(u+\lambda_k) - \sum_k 1/(u+\mu_k)$ with $\lambda_k = (k+\frac{1}{2})^2\pi^2/2$ and $\mu_k = \lambda_k/x^2 > \lambda_k$, so the Thorin measure would have negative atoms. Hence any family of processes converging weakly to a one dimensional diffusion started at an entrance or reflecting end, with hitting times converging in law, has a GGC limit, so positivity comes for free from the diffusion limit; and RH together with the simplicity of the zeros is equivalent to $\mu$ being the hitting time of a level by a one dimensional generalised diffusion started at an entrance end.

The other half is identification. The limiting diffusion must have speed measure with spectrum exactly $\{\gamma^2\}$, and diffusion limits of this type are universal: the limit depends only on averaged quantities such as a Fisher information constant. That is the same washing out as in Theorem 24.8.4; a diffusion scaling of the Euler family is the cruder Gaussian scaling, under which $Y_P/\operatorname{Var} L_P \to 1$ in probability (the fluctuation scale $\kappa_3^{2/3}$ is $o(\kappa_2)$) and every trace of the zeros disappears. A diffusion approximation would supply positivity only if the chains before the limit already carried the arithmetic at the scale the limit sees, and by Theorem 24.9.1 this must happen through Thorin measures concentrating onto the $\gamma^2$, not through

exact factors with GGC cofactors. Its natural role is as a tool: if Markov chains built from the primes approximated the Krein string operator, for example through discretisations of the canonical systems of [207], a weak convergence theorem of Roberts type is what would carry nonnegativity to the limit.

## 24.12 Curved boundaries: the Roberts and Shortland view

Roberts and Shortland [462] compute the law of the first time a Brownian motion crosses a curved, time dependent boundary, the problem behind barrier options with time dependent coefficients. Their question runs from boundary to law. Ours runs the other way: for a law $\nu$ on $(0,\infty)$, find $b$ such that

$$\tau_b = \inf\{t > 0 : W_t \geq b(t)\}, \qquad W_0 = 0,$$

has law $\nu$. This is the inverse first passage problem. Under mild conditions on $\nu$ it has a unique solution, in a suitable viscosity sense [7, 79]. So $\mu$ has a boundary whether or not RH holds, and the existence of the representation says nothing. The question is whether the boundary has a shape that sees the zeros. The boundary of $\mu$ is the one computed in Section 8.2 and in Sections 9.2 and 11.3; here we add the enclosure at small times, the validation of the solver, and the effect of moving a zero. Three statements result: the two ends of the boundary are determined by quantities that do not depend on RH (Proposition 24.12.1 and the slope at infinity); the boundary in between is computed to about $2 \cdot 10^{-4}$; and moving a zero off the line changes it, to first order and heuristically, by an amount proportional to $\delta^2/\gamma^4$ times a profile that does not depend on the zero (Remark 24.12.3).

### 24.12.1 The small time end

Proposition 24.12.1. *Let $b$ be the boundary of $\mu$, write $F(t) = \mu(0,t]$, and let $m(t) = \inf_{0<s\leq t} b(s)$. For every $t > 0$,*

$$\sqrt{t}\,\bar{\Phi}_{\mathrm{N}}^{-1}\big(F(t)\big) \leq b(t), \qquad m(t) \leq \sqrt{t}\,\bar{\Phi}_{\mathrm{N}}^{-1}\big(\tfrac{1}{2}F(t)\big),$$

*where $\Phi_{\mathrm{N}}$ is the standard normal distribution function and $\bar{\Phi}_{\mathrm{N}} = 1-\Phi_{\mathrm{N}}$. Moreover $-\log F(t) \sim (\log(1/t))^2/(16t)$ as $t \downarrow 0$, so*

$$\liminf_{t\downarrow 0} \frac{b(t)}{\log(1/t)} \geq \frac{1}{2\sqrt{2}}, \qquad \limsup_{t\downarrow 0} \frac{m(t)}{\log(1/t)} \leq \frac{1}{2\sqrt{2}},$$

*and if $b$ is nonincreasing near $0$ then $b(t) \sim \log(1/t)/(2\sqrt{2})$. None of this depends on RH. Status: proved.*

*Proof.* If $W_t \geq b(t)$ then $\tau_b \leq t$, so $\bar{\Phi}_{\mathrm{N}}(b(t)/\sqrt{t}) \leq F(t)$, which is the first inequality. If $\tau_b \leq t$ then $\sup_{s\leq t} W_s \geq m(t)$, and by the reflection principle

$F(t) \le 2\bar{\Phi}_{\mathrm{N}}(m(t)/\sqrt{t})$, which is the second. For the asymptotics, Stirling's formula and $\zeta(\sigma) \to 1$ give $-\log \mathbb{E}e^{-s\tau} = \log \xi(\frac{1}{2} + \sqrt{s}) - \log \xi(\frac{1}{2}) \sim \frac{1}{4}\sqrt{s}\log s$ as $s \to \infty$ through the reals, unconditionally. This is regularly varying of index $\frac{1}{2}$, and Kasahara's exponential Tauberian theorem [299], [50, Section 4.12] converts it into the Legendre transform asymptotics: the supremum of $\frac{1}{4}\sqrt{s}\log s - st$ is attained at $\sqrt{s} \sim \log(1/t)/(4t)$ and equals $(\log(1/t))^2/(16t)$ to leading order. Since $\bar{\Phi}_{\mathrm{N}}^{-1}(p) \sim \sqrt{2\log(1/p)}$ as $p \downarrow 0$, both bounds become $\sqrt{2t \cdot (\log(1/t))^2/(16t)} = \log(1/t)/(2\sqrt{2})$. When $b$ is nonincreasing, $m = b$ and the two bounds meet. $\square$

The proposition is also a check on the numerics. For a nonincreasing boundary the two bounds enclose $b$, and they are explicit in $F$. Evaluating the Legendre transform numerically (computed), the ratio $b(t)/\log(1/t)$ is 0.269, 0.328, 0.342, 0.348 at $t = 10^{-4}, 10^{-8}, 10^{-12}, 10^{-16}$, approaching $1/(2\sqrt{2}) = 0.354$ slowly, as the log log corrections suggest. Every finite sum of exponentials has a polynomial lower tail and hence $b(0+) = 0$. The logarithmic blow up is the trace of infinitely many zeros, through the growth of $\xi$ along the real axis, and it is the same wherever the zeros lie.

### 24.12.2 Computing the boundary

Conditioning on the crossing time and using the strong Markov property gives, for every $t > 0$,

$$\bar{\Phi}_{\mathrm{N}}\Big(\frac{b(t)}{\sqrt{t}}\Big) = \int_0^t \bar{\Phi}_{\mathrm{N}}\Big(\frac{b(t) - b(s)}{\sqrt{t-s}}\Big)\,\nu(\mathrm{d}s),$$

a nonlinear Volterra equation of the first kind for $b$ [412]. We solve it forward with a midpoint rule in $s$. At each step the right side minus the left side need not be monotone in $b(t)$, and a global root search occasionally jumps to a spurious root; bracketing the root next to the previous value removes this. The distribution function of $\mu$ comes from a contour integral of $\xi(\frac{1}{2})/(s\,\xi(\frac{1}{2} + \sqrt{s}))$ along a vertical line through the saddle point below $t = 0.0064$ and from the residue expansion with 120 zeros above it; the two agree to a relative difference of $10^{-33}$ where they overlap. The code is `code/ht_curved.py`. We validated the solver in three ways (all computed).

1. On the linear boundary $b(t) = 1 - t/2$, whose crossing law is known in closed form, it recovers $b$ to $10^{-6}$ for $t \ge 0.05$; the start of the recursion carries a transient of a few $10^{-3}$ that does not reach the range of interest.

2. Halving the step from $4 \cdot 10^{-5}$ to $2 \cdot 10^{-5}$ changes the boundary of $\mu$ by at most $1.5 \cdot 10^{-4}$ on $[0.0045, 0.12]$, at most $8 \cdot 10^{-5}$ for $t \ge 0.006$, and at most $2.2 \cdot 10^{-5}$ for $t \ge 0.01$.

3. The computed boundary is decreasing, and at every grid point it lies inside the enclosure of Proposition 24.12.1, whose width is 0.004 at $t = 0.0045$ and

0.015 at $t = 0.01$. Figure 24.1(b) compares the boundary at small times with the asymptote $\log(1/t)/(2\sqrt{2})$ of Proposition 24.12.1.

The interval $[0.0045, 0.12]$ carries all but about $2 \cdot 10^{-9}$ of the mass of $\mu$, whose mean is 0.0231.

### 24.12.3 The large time end and the shape

The tail $\mu(t,\infty) \sim c\, e^{-\gamma_1^2 t}$ forces $b'(t) \to -\sqrt{2}\,\gamma_1 = -19.99$, the slope of a Brownian motion whose drift gives hitting hazard $\gamma_1^2$. Matching the survival function of a Brownian motion with drift $c = \sqrt{2}\,\gamma_1$ below a slowly rising level $g(t)$, namely $\mathrm{const}\cdot g\, t^{-3/2} e^{cg - c^2 t/2}$, against $c' e^{-\gamma_1^2 t}$ gives $cg + \log g = \frac{3}{2}\log t + O(1)$, so (heuristic: the matching is quasi static and has no error control)

$$b(t) = -\sqrt{2}\,\gamma_1 t + \frac{3}{2\sqrt{2}\,\gamma_1}\log t - \frac{1}{\sqrt{2}\,\gamma_1}\log\log(\gamma_1^2 t) + O(1),$$
$$t\big(b'(t) + c\big) \to \frac{3}{2c}.$$

We checked this in two ways (computed). For the exponential law, $t(b' + \sqrt{2})$ is 0.80, 0.88, 0.92, 0.94 at $t = 20, 50, 100, 150$, against $3/(2\sqrt{2}) - 1/(\sqrt{2}\log t) = 0.82$, 0.88, 0.91, 0.92. For $\mu$ we continued the solution to $t = 2$ with the complementary form of the integral equation, $\Phi_{\mathrm{N}}(b(t)/\sqrt{t}) - \mu(t,\infty) = \int_0^t \Phi_{\mathrm{N}}((b(t) - b(s))/\sqrt{t-s})\,\mu(\mathrm{d}s)$, which keeps full relative accuracy when the survival function is as small as $e^{-400}$. Then $t(b' + c)$ is 0.0645, 0.0665, 0.0677, 0.0688 at $t = 0.8, 1.2, 1.6, 2$, against 0.0652, 0.0659, 0.0664, 0.0667 from the formula, with $3/(2c) = 0.0750$ as the limit. The limit is not reached on the range computed, and the slow approach seen on $[0.03, 0.12]$, where the slope is still $-19.6$ at the right end, is consistent with the $\log t$ correction rather than with a numerical artefact. This end sees only the first zero, which is on the line.

Between the two ends the shape gives no criterion either. The exponential law, the simplest GGC, has a boundary that starts at 0, rises to a maximum of 0.664 at $t = 0.25$, and then falls with slope tending to $-\sqrt{2}$; the Gamma law of shape 2 behaves the same way. The boundary of $\mu$ falls monotonically from $+\infty$. GGC boundaries thus come in shapes that no simple convexity or monotonicity condition separates.

### 24.12.4 Moving a zero off the line

To see what a zero off the line does, we replace one factor $1 + s/\gamma_m^2$ of the Hadamard product by the pair $(1 + s/\lambda)(1 + s/\bar{\lambda})$ with $-\lambda = (\delta + i\gamma_m)^2$, the image of the quartet $\frac{1}{2} \pm \delta \pm i\gamma_m$. As a GGC control we use the split real pair $(\gamma_m \pm 0.001)^2$, so that both laws have the same number of factors. The value $\delta = 0.3085$ is the displacement of the Davenport Heilbronn zero of Section 24.14.

| zero | $\delta$ | $\max \lvert b_{\mathrm{toy}} - b_{\mathrm{GGC}} \rvert$ | density negative from | a law |
|---|---|---|---|---|
| $\gamma_1$ | 0.05 | $2.25 \cdot 10^{-4}$ | $t = 2.25$ | no |
| $\gamma_1$ | 0.15 | $2.0 \cdot 10^{-3}$ | $t = 0.77$ | no |
| $\gamma_1$ | 0.3085 | $8.8 \cdot 10^{-3}$ | $t = 0.39$ | no |
| $\gamma_6$ | 0.3085 | $8.2 \cdot 10^{-6}$ | never | yes |
| $\gamma_{20}$ | 0.3085 | $6.3 \cdot 10^{-7}$ | never | yes |
| $\gamma_{60}$ | 0.3085 | $5.5 \cdot 10^{-8}$ | never | yes |

Table 24.10: Boundaries with one off line pair against a GGC control on $[0.0045, 0.12]$; for $\gamma_{60}$ the maximum is over $[0.0045, 0.1]$, since near $t = 0.12$ the solver noise of about $2 \cdot 10^{-7}$ exceeds the signal. "Never" means the density was found positive on a grid of step 0.005 on $(0.0064, 6]$ and at three points below 0.0064, beyond which the tail is governed by $\gamma_1$. All entries computed (`code/ht_curved.py`).

When the off line pair is the lowest zero, the pole closest to the origin is complex, the density oscillates in its tail and becomes negative; at $\delta = 0.3085$ this happens at $t = 0.39$, where the remaining mass is about $e^{-78}$. Beyond that point no boundary exists, and before it the boundary moves by less than $10^{-2}$. For $\xi$ this case cannot occur, because the lowest zero is on the line. When the pair sits higher the tail is still governed by $\gamma_1$, and the toy law is, on the evidence of the computed density, a probability law that is not a GGC (it has complex poles, Theorem 24.9.1). The size of the change then follows a simple rule. Two parts of it are proved.

Proposition 24.12.2. *Moving the pair at height $\gamma$ off the line by $\delta$ multiplies the Laplace transform, relative to the control, by $1+6\delta^2 s/\gamma^4+O(\delta^2 s^2/\gamma^6)$ for $|s| \ll \gamma^2$. To first order this is the shift $\tau \mapsto \tau - 6\delta^2/\gamma^4$ of the mean. Status: proved.*

*Proof.* The mean of a GGC with atoms $\lambda_j$ is $\sum 1/\lambda_j$. For the toy pair this contribution is $2\,\mathrm{Re}(1/\lambda) = 2(\gamma^2-\delta^2)/(\gamma^2+\delta^2)^2 = 2/\gamma^2 - 6\delta^2/\gamma^4 + O(\delta^4/\gamma^6)$, and for the control it is $2/\gamma^2 + O(10^{-6}/\gamma^4)$. Higher cumulants change by $O(\delta^2/\gamma^{2k+2})$, and the ratio of the two Laplace transforms is $\big((1 + s/\lambda)(1 + s/\bar{\lambda})\big)^{-1}$ over the control factor, which expands as stated. □

The step from the law to the boundary is not proved, and we state it as an observation.

Remark 24.12.3 (Boundary response, heuristic). If the map from a law to its boundary is linear to first order and the change of the law is, to leading order, a pure shift, then

$$b_{\mathrm{toy}}(t) - b_{\mathrm{GGC}}(t) \approx \frac{6\delta^2}{\gamma^4}\, K(t),$$

where $K$ is the linear response of the boundary to a shift of the law, a profile that

does not depend on $\gamma$ or $\delta$. Linearity, the universality of $K$ and the size of the neglected terms are assumed, not proved. Status: heuristic, checked numerically (Table 24.11).

We computed $K$ directly by solving for the law of $\tau - \Delta$ with $\Delta = 2 \cdot 10^{-6}$; it is $-72$, $-29$, $-13$, $-11$ at $t = 0.01, 0.02, 0.05, 0.08$, and it is linear in $\Delta$ to four digits (the values at $\Delta = 2 \cdot 10^{-6}$ and $4 \cdot 10^{-6}$ agree to $10^{-2}$). Table 24.11 compares it with the normalised differences $(b_{\mathrm{toy}} - b_{\mathrm{GGC}})\gamma^4/6\delta^2$, and Figure 24.1(c) shows the collapse. At $t = 0.02$ the three heights agree with $K$ to 3%, 2% and 0.5%. At small times the agreement sets in only once $t$ exceeds a multiple of $1/\gamma$, where the relevant $s$ are below $\gamma^2$.

| | $t = 0.01$ | $t = 0.02$ | $t = 0.05$ | $t = 0.08$ |
|---|---|---|---|---|
| $\gamma_6$ | $-11.2$ | $-28.3$ | $-17.0$ | $-14.4$ |
| $\gamma_{20}$ | $-36.5$ | $-29.8$ | $-13.7$ | $-11.7$ |
| $\gamma_{60}$ | $-60.6$ | $-29.5$ | $-13.1$ | $-11.2$ |
| $K(t)$ | $-71.9$ | $-29.3$ | $-12.9$ | $-11.0$ |

Table 24.11: Normalised boundary change $(b_{\mathrm{toy}} - b_{\mathrm{GGC}})\gamma^4/6\delta^2$ against the shift response $K(t)$, $\delta = 0.3085$ (computed).

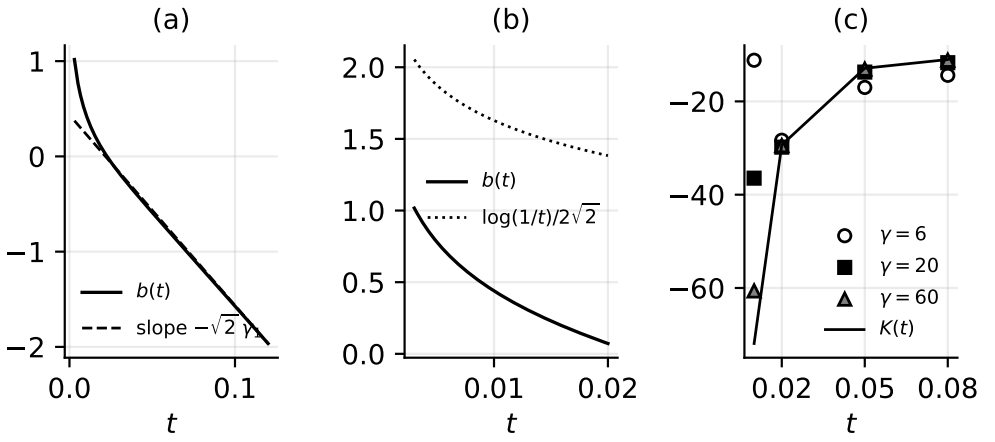



Figure 24.1: (a) The inverse first passage boundary $b$ of the reciprocal Xi law on $[0.003, 0.12]$, with the line of slope $-\sqrt{2}\,\gamma_1$ through its last node. (b) The boundary at small times against the asymptote $\log(1/t)/(2\sqrt{2})$ of Proposition 24.12.1, which it approaches only slowly. (c) Normalised boundary changes for off line pairs at the zeros 6, 20, 60, against the shift response $K$ (computed).

For $\xi$ the consequence is quantitative, and it is first order and heuristic (Remark 24.12.3). Any zero off the line lies above height $3 \cdot 10^{12}$ [419] and has $\delta < \frac{1}{2}$,

so its shift is at most $6 \cdot \frac{1}{4}/(3 \cdot 10^{12})^4 < 2 \cdot 10^{-50}$. The computed response satisfies $|K| \le 215$ on $[0.0045, 0.12]$ (its computed maximum is 195.5 at $t = 0.0045$). For one off line quartet the boundary of $\mu$ on the range carrying all but $2 \cdot 10^{-9}$ of the mass would therefore change, to first order, by less than $10^{-47}$. RH failing allows every zero above $3 \cdot 10^{12}$ to be off the line, and then the first order shifts add up to $\frac{3}{2} \sum_{\gamma > T} \gamma^{-4} \approx 8 \cdot 10^{-38}$ at most, so the change is less than $10^{-34}$ (the computed value of the product with 215 is $1.7 \cdot 10^{-35}$; here $\sum_{\gamma > T} \gamma^{-4} \ll T^{-3} \log T$). The change is larger only at times of order $\log \gamma / \gamma$, where the saddle point $s$ reaches $\gamma^2$; by Proposition 24.12.1 the mass there is about $\exp(-\frac{1}{4} \gamma \log \gamma)$.

### 24.12.5 Assessment

The Roberts and Shortland view is a faithful and cheap encoding of $\mu$: the boundary is computable to about $2 \cdot 10^{-4}$ on the range that carries the mass, and Proposition 24.12.1 gives an explicit enclosure at small times. But its two ends are fixed by the growth of $\xi$ and by the first zero, and a violation of RH would move it in between, to first order (heuristic, Remark 24.12.3), by about $6\delta^2 \gamma^{-4} K(t)$, which is the same first order shift that any other representation of $\mu$ would show. A shape test could see RH only if it detected the GGC property itself, through a characterisation of the boundaries whose crossing laws are GGCs. Kent's theorem [305] makes the hitting times of levels by one dimensional diffusions GGCs, but we do not know which curved boundaries for Brownian motion reduce to that case, and we record the question as Problem 24.5. Without such a theorem the method is a computational tool, useful for the prime side laws of Theorem 24.8.4 and the Davenport Heilbronn comparison, with no positivity of its own.

## 24.13 Marc Yor's probabilistic Riemann function

Marc Yor and his coauthors built a substantial probabilistic account of $\xi$. Four of their papers bear on this chapter: Biane, Pitman and Yor on Brownian laws whose Mellin transform is $\xi$ [49]; Pitman and Yor on the hyperbolic laws [416]; James, Roynette and Yor on GGCs, Thorin measures and Dirichlet means [289]; and Nikeghbali and Yor on the Barnes $G$ function [387]. We take them in turn and ask of each what it adds to the reciprocal Xi law $\mu$. Section 8.7 and (9.1.1) of the basepoint chapter already use the first of these papers; this section adds the harmonic mean form, the Davenport Heilbronn counterpart and the Wiener and Gamma coupling.

### 24.13.1 The Biane, Pitman and Yor variable

Let $b$ be a standard Brownian bridge and $Y = \sqrt{2/\pi}\,(\max_u b_u - \min_u b_u)$. Biane, Pitman and Yor prove [49, (4), (5)]

$$\mathbb{E}\, Y^s = 2\xi(s) \qquad (s \in \mathbb{C}).$$

The same law appears in several Brownian guises: $Y^2 \stackrel{d}{=} \frac{\pi}{2}\Sigma_2$ with $\Sigma_2 = \frac{2}{\pi^2}\sum_{n\geq 1}(\varepsilon_n + \hat{\varepsilon}_n)/n^2$, whose Laplace transform is $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$; $\Sigma_2$ is the sum of two independent hitting times of 1 by a three dimensional Bessel process; and $\frac{\pi}{2}\sqrt{\Sigma_2}$ is the maximum of a standard Brownian excursion [49, (57)]. The functional equation $\xi(s) = \xi(1-s)$ becomes the size bias identity $\mathbb{E}g(1/Y) = \mathbb{E}\,Y g(Y)$ [49, (20)]. The reciprocal of $\xi$ follows at once.

PROPOSITION 24.13.1. *Let $\mathbb{P}^*$ have density $Y^{1/2}/\mathbb{E}Y^{1/2}$ with respect to $\mathbb{P}$, and let $U = \log Y$. Under $\mathbb{P}^*$ the variable $U$ is symmetric and has the law of the Pólya variable of* (1.2.4)*, with density $\Phi/\Xi(0)$, where $\Xi(z) = \xi(\frac{1}{2}+iz)$ and $\Phi$ is the kernel of Riemann and Pólya of* (1.2.3)*; equivalently $U/2$ has density $2\Phi(2u)/\Xi(0)$. Moreover*

$$\mathbb{E}\,e^{-s\tau} = \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2}+\sqrt{s})} = \frac{1}{\mathbb{E}^*\cosh(\sqrt{s}\,U)} = \frac{1}{\mathbb{E}^*\big[\mathbb{E}[e^{-sT}\,|\,U]^{-1}\big]},$$

*where, given $U$, $T$ is the exit time of a standard Brownian motion from $(-|U|/\sqrt{2}, |U|/\sqrt{2})$. Status: proved.*

*Proof.* From $\mathbb{E}Y^s = 2\xi(s)$, $\mathbb{E}^*e^{wU} = \mathbb{E}\,Y^{1/2+w}/\mathbb{E}\,Y^{1/2} = \xi(\frac{1}{2}+w)/\xi(\frac{1}{2})$. Taking $g(y) = y^{-1/2}f(\log y)$ in the size bias identity gives $\mathbb{E}\,Y^{1/2}f(-\log Y) = \mathbb{E}\,Y^{1/2}f(\log Y)$, so $U$ is symmetric under $\mathbb{P}^*$ and $\mathbb{E}^*e^{wU} = \mathbb{E}^*\cosh(wU)$. Putting $w = iz$ gives $\mathbb{E}^*e^{izU} = \Xi(z)/\Xi(0) = \int_{\mathbb{R}}\Phi(u)e^{izu}\mathrm{d}u/\Xi(0)$, which identifies the density of $U$ (this is (1.2.4)). Finally $\mathbb{E}e^{-sT} = 1/\cosh(a\sqrt{2s})$ for exit from $(-a,a)$, and $a = |U|/\sqrt{2}$. □

So the harmonic mean route of Section 24.4 has an explicit Brownian mixing variable: the logarithm of the normalised range of a Brownian bridge, size biased by its square root. Each conditional exit time is a GGC, and $\mu$ is a GGC exactly when the harmonic mean over $U$ preserves that property, which is RH. Proposition 24.13.1 puts the problem in Brownian language without changing it.

REMARK 24.13.2. Let $M$ be the maximum of a standard Brownian excursion. Its law is explicit [83, 190]:

$$\mathbb{P}(M \leq m) = \sum_{n\in\mathbb{Z}}(1-4n^2m^2)e^{-2n^2m^2} \qquad (m>0),$$

and $Y \stackrel{d}{=} \sqrt{2/\pi}\,M$, because the range of a Brownian bridge has the law of $M$, by the path transformation of Vervaat [357]. With $\psi(x) = \sum_{n\geq 1}e^{-\pi n^2 x}$ as in Section 22.1.1, and for $y>0$,

$$\begin{aligned}\mathbb{P}(Y>y) &= 2\sum_{n\geq 1}(2\pi n^2y^2-1)e^{-\pi n^2y^2},\\ \mathbb{E}(Y-y)^+ &= 2y\,\psi(y^2),\\ f_Y(y) &= 2\Phi(\log y)\,y^{-3/2},\end{aligned}$$

where $f_Y$ is the density of $Y$ and $\Phi$ the kernel of (1.2.3). The first is the formula above at $m = \sqrt{\pi/2}\,y$. It equals $-\frac{d}{dy}\big(2y\psi(y^2)\big)$, and both sides of the second tend to 0 as $y \to \infty$, which gives the second. The third is $-\frac{d}{dy}$ of the first, compared term by term with (1.2.3) at $u = \log y$. Hence $\mathbb{E}Y^s = 2\int_{\mathbb{R}} \Phi(u) e^{(s-\frac12)u}\,\mathrm{d}u = 2\xi(s)$ for every complex $s$: the identity of Biane, Pitman and Yor follows from the law of $M$, term by term. With $y = e^u$ and $\mathbb{E}Y = 2\xi(1) = 1$, the second line gives the folded kernel of (22.1.1) as

$$k(u) = e^{-u/2} - 2e^{u/2}\psi(e^{2u}) = e^{-u/2}\,\mathbb{E}\min(Y, e^u), \qquad u \in \mathbb{R}.$$

Its evenness is the size bias identity with $g(y) = \min(1, e^u/y)$, and the kink identity $k'(0) = 0$ of Lemma 22.1.1 is its case $g = \mathbf{1}_{(0,1)}$, since $k'(0) = \frac12\big(\mathbb{P}(Y > 1) - \mathbb{E}[Y; Y < 1]\big)$. As $\mathbb{E}[Y; Y > 1] - \mathbb{P}(Y > 1) = \mathbb{E}(Y-1)^+ = 2\psi(1)$, the kink identity reads

$$\mathbb{P}\big(M > \sqrt{\pi/2}\big) = \mathbb{P}(Y > 1) = \tfrac12 - \psi(1) = 0.456783$$

(computed, `px_const.py`). Status: proved. This remark follows a note of Nicholas Polson merged into the book [232].

### 24.13.2 Davenport and Heilbronn have the same structure

The Biane, Pitman and Yor structure rests on two facts: the theta function behind $\xi$ is positive, and it is self dual. The Davenport Heilbronn function shares both.

Proposition 24.13.3. *Let $a_n$ be the Davenport Heilbronn coefficients, periodic modulo 5 with $(a_0, \dots, a_4) = (0, 1, \kappa, -\kappa, -1)$ and $\kappa = 0.2840790438$, and let $\theta_{DH}(x) = \sum_{n\ge1} a_n n\, e^{-\pi n^2 x/5}$. Then $\theta_{DH}(x) > 0$ for every $x > 0$. Consequently there is a probability law on $(0,\infty)$ whose Mellin transform is a constant multiple of $\xi_{DH}$, a symmetric $U_{DH}$ with $\mathbb{E}^* e^{wU_{DH}} = \xi_{DH}(\frac12 + w)/\xi_{DH}(\frac12)$, and a positive Pólya kernel $\Phi_{DH}$. Status: proved.*

*Proof.* For $x \ge 1$, using $|a_n| \le 1$,

$$\theta_{DH}(x) \ge e^{-\pi x/5}\Big(1 - \sum_{n\ge2} n\, e^{-\pi(n^2-1)x/5}\Big) \ge e^{-\pi x/5}(1 - 0.324) > 0.$$

The functional equation of $f_{DH}$ is equivalent to the self duality $\theta_{DH}(1/x) = x^{3/2}\theta_{DH}(x)$, which we also checked numerically (to a relative $10^{-40}$ at 37 points), and it carries positivity to $0 < x < 1$. Mellin transformation gives $\int_0^\infty \theta_{DH}(x) x^{(s+1)/2}\,\mathrm{d}x/x = \sqrt{5/\pi}\,\xi_{DH}(s)$ (checked to $2 \cdot 10^{-41}$ at $s = \frac12$ and $s = 0.8 + i$), so $\theta_{DH}(x)x^{1/2}/x$, normalised, is a probability density with Mellin transform $\xi_{DH}(s)/\xi_{DH}(0)$ in the variable $s/2$. The rest follows as in Proposition 24.13.1. □

The harmonic mean $1/\mathbb{E}^*\cosh(\sqrt{s}\,U_{DH})$ has the complex pole $s = (0.3085 + 85.699i)^2$ coming from the off line zero, so it is not a GGC (Theorem 24.9.1). The positivity of a self dual kernel, the size bias symmetry and the harmonic mean representation therefore all hold for a function that violates RH. What Davenport and Heilbronn may lack is the Brownian meaning: we do not know whether $\theta_{DH}$ is the theta function of any natural Brownian or Bessel functional. That is the only place in the Biane, Pitman and Yor picture where arithmetic could enter, and we record it as Problem 24.7.

### 24.13.3 The hyperbolic laws as solvable models

Pitman and Yor [416] study the laws $C_1$, $S_1$, $T_1$ with Laplace transforms $1/\cosh\sqrt{2s}$, $\sqrt{2s}/\sinh\sqrt{2s}$ and $\tanh\sqrt{2s}/\sqrt{2s}$. The first is the exit time of Brownian motion from $(-1,1)$, and

$$\frac{1}{\cosh\sqrt{2s}} = \prod_{n\geq 1}\Big(1 + \frac{2s}{\pi^2(n-\frac{1}{2})^2}\Big)^{-1}$$

is the exact model of $\mu$ under RH, with $\pi^2(n-\frac{1}{2})^2/2$ in place of $\gamma_n^2$: a GGC whose Thorin measure is a sum of unit atoms, of infinite total mass [289, Section 1.6]. The law $T_1$ shows how a GGC fails. Its transform has zeros and poles interlacing on the negative axis, so its Thorin measure has atoms of both signs, and it is not a GGC [289]. For $\mu$ an off line zero produces instead a pair of complex atoms. Both failures are visible only in the canonical function, not in positivity of the law.

### 24.13.4 The Wiener and Gamma representation

James, Roynette and Yor [289, Proposition 1.1] show that a positive variable is a GGC if and only if it is a Wiener and Gamma integral $\int_0^\infty h(u)\,\mathrm{d}\gamma_u$ against a standard gamma process, where the Thorin measure is the image of Lebesgue measure under $u \mapsto 1/h(u)$, and $h$ can be taken nonincreasing and is then unique. For $\mu$ under RH this gives $h(u) = 1/\gamma^2_{\lceil u\rceil}$ and $\mu = \sum \varepsilon_n/\gamma_n^2$.

The base point laws $\mu^{(\alpha)}$ with transform $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ fit the same frame, but only when $\alpha \geq \sup_\rho \operatorname{Re}\rho$, which is where $\mu^{(\alpha)}$ is a GGC (Theorem 1.2.2; below that line the Hadamard product has zeros in the slit plane and no Thorin representation exists, see Proposition 11.4.3). For such $\alpha$ the Hadamard product gives

$$\log\frac{\xi(\alpha+\sqrt{s})}{\xi(\alpha)} = \int_0^\infty \log\Big(1+\frac{s}{t^2}\Big)\,\mathrm{d}N_\alpha(t),$$
$$N'_\alpha(t) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it) = \frac{1}{\pi}\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(t-\gamma)^2},$$

so the Thorin measure in the variable $t$ is $\mathrm{d}N_\alpha$, a Poisson smoothing of the zero counting function at width $\alpha-\beta$. Then $\mu^{(\alpha)}$ is the Wiener and Gamma integral of $h_\alpha(u)=N_\alpha^{-1}(u)^{-2}$. The density $N'_\alpha$ is also defined for $\alpha<\sup_\rho \operatorname{Re}\rho$, and its positivity on every line $\operatorname{Re}s=\alpha>\frac12$ is Lagarias's criterion (Remark 1.2.4), but on a single such line positivity of the density does not make $\mu^{(\alpha)}$ a GGC.

Proposition 24.13.4. *On one gamma process $\gamma$, set $X_\alpha=\int_0^\infty h_\alpha(u)\,\mathrm{d}\gamma_u$. Then $X_\alpha$ has law $\mu^{(\alpha)}$ for every $\alpha\ge\sup_\rho\operatorname{Re}\rho$, in particular for every $\alpha\ge1$. RH holds if and only if $N_\alpha$ is nondecreasing for every $\alpha>\frac12$, and then $N_\alpha\to N$ pointwise at continuity points and $X_\alpha\to X_{1/2}=\sum\varepsilon_n/\gamma_n^2$ in probability as $\alpha\downarrow\frac12$. Status: proved for the representation and the equivalence, the convergence under RH is heuristic (the proof below is a sketch).*

*Sketch.* The representation is Theorem 1.2.2 together with the Wiener and Gamma characterisation. The equivalence is Lagarias's criterion: each zero with $\beta<\alpha$ contributes a positive Poisson kernel, and a zero with $\beta>\alpha$ contributes a negative kernel of height $1/(\pi(\beta-\alpha))$ at $t=\gamma$, which dominates all other terms as $\alpha\uparrow\beta$, so $N_\alpha$ fails to be monotone for some $\alpha$ close to $\beta$. Under RH the kernels tend to unit atoms at the $\gamma_n$. The part of $\mathrm{d}N_\alpha$ on $(0,\gamma_1/2)$ has mass $O(\alpha-\frac12)$ and contributes a variable whose transform is $\exp(-O(\alpha-\frac12)\sqrt{s})$, which tends to 1; on the rest the kernels $h_\alpha$ converge in $L^1(\mathrm{d}u)$ to $h$ (this needs $N_\alpha-N=O(\log t)$ on the tail, which we have not written out), and the Wiener and Gamma integrals converge in probability. □

Figure 24.2 shows, against $\alpha=1,0.75,0.6,0.55$, the minimum of $N'_\alpha$ on $[0,50]$ and the value $N_\alpha(50)$. The densities are positive, with minima 0.0074, 0.0037, 0.0015, 0.0007 (all at $t=0$), and $N_\alpha(50)=9.72$, 9.78, 9.88, 9.94 approaches $N(50)=10$ (computed). The coupling gives a pathwise picture of the base point descent under RH: all the laws $\mu^{(\alpha)}$ are functionals of one gamma process, and an off line zero would appear as a point where $N_\alpha$ stops being monotone, at the moment $\alpha$ passes $\beta$. Like Proposition 24.13.1, this reformulates RH and does not prove it.

Dirichlet means, the other main theme of [289], need a Thorin measure of finite total mass $\theta$, in which case the GGC is $\gamma_\theta\cdot D_\theta$ with $D_\theta$ a Dirichlet mean [289, (31) to (35)]. The Thorin measures of $\mu$ and of every $\mu^{(\alpha)}$ with $\alpha\ge\sup_\rho\operatorname{Re}\rho$ have infinite mass. For the truncation $\mu_N=\sum_{n\le N}\varepsilon_n/\gamma_n^2$ the representation holds with $\theta=N$, but $D_N$ tends to $\frac1N\sum_{n\le N}\gamma_n^{-2}\to0$ [289, Proposition 1.7], so it degenerates in the limit and carries no information about $\mu$.

### 24.13.5 Moments and the Barnes function

Nikeghbali and Yor [387] express the Barnes $G$ function, and through it the Keating and Snaith moments of characteristic polynomials of random unitary matrices,

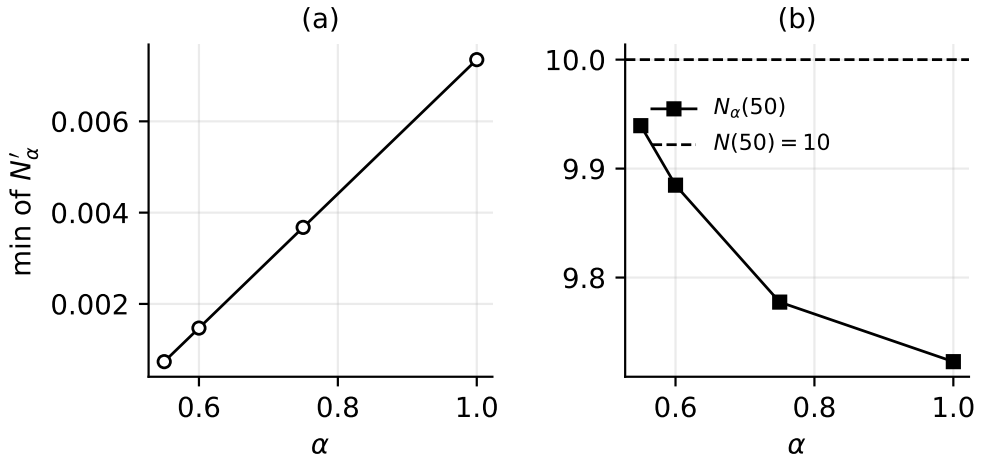



Figure 24.2: (a) The minimum over $t$ of the Thorin density $N'_\alpha(t) = \pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + it)$ of the base point laws on $[0, 50]$, attained at $t = 0$, for $\alpha = 1, 0.75, 0.6, 0.55$. (b) The values $N_\alpha(50)$ against $N(50) = 10$. Only these summary values are stored, so the curves on $[0, 50]$ are not drawn (computed).

by sums and products of GGC variables (see Section 11.4.6). This links GGCs to the conjectured moments of $\zeta$ on the critical line, but moments of $|\zeta(\frac{1}{2} + it)|$ do not locate zeros, and we see no route from it to the positivity of the Thorin measure of $\mu$.

### 24.13.6 Assessment

Yor's work supplies three things this programme can use: a Brownian meaning for Pólya's kernel and hence for the harmonic mean form of $\mu$ (Proposition 24.13.1), exactly solvable models of $\mu$ and of its failure modes in the hyperbolic laws, and a pathwise coupling of the base point laws through the Wiener and Gamma representation (Proposition 24.13.4). It does not supply the arithmetic. Proposition 24.13.3 shows that the positivity and symmetry behind the Biane, Pitman and Yor variable hold for Davenport and Heilbronn too, so they cannot imply RH; the Brownian interpretation of $\theta$ is the one ingredient that might separate the two.

## 24.14 The Davenport Heilbronn test

Every proposed argument must fail for the Davenport Heilbronn function $f_{DH}$ [111], which has the functional equation $\xi_{DH}(s) = \xi_{DH}(1 - s)$ with $\xi_{DH}(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f_{DH}(s)$, no Euler product, and zeros off the line; the zero at $0.808517182456637 + 85.6993484853776\,i$ is classical [484] (see Section 22.14

for the function and its zeros). We confirmed it by the argument principle (one zero in the box $[0.78, 0.84] \times [85.67, 85.73]$, at two step sizes). Its reciprocal law passes every positivity test we can compute: the Taylor coefficients of $\xi_{DH}(\frac{1}{2} + \sqrt{s})/\xi_{DH}(\frac{1}{2})$ have all contiguous Toeplitz minors positive up to order 10 over the first 100 indices (computed; the check ran to order 14 and index 119 at two precisions, which agree to 80 digits, not an interval proof), indistinguishable from those of $\xi$, as the angle theorem predicts (Section 24.17). For hitting times the test rules out the de Bruijn and Newman route and the harmonic mean route unless the Euler product enters them, and it rules out any argument that uses only an archimedean factor of this growth and the functional equation.

## 24.15 What the reciprocal Xi function proves

Every route in this chapter ends in an equivalence with RH whose unconditional part stops exactly where classical knowledge stops. This section records the frontier for each, adds one quantitative statement about the canonical function, and identifies the single inequality at which arithmetic would have to enter.

### 24.15.1 Partial complete monotonicity

The law $\mu$ is a GGC if and only if its canonical function $\Theta(x) = \sum_{\gamma>0} e^{s_\rho x}$ is completely monotone, that is $(-1)^n \Theta^{(n)}(x) \geq 0$ for all $n$ and $x$. Each zero on the line contributes $\gamma^{2n} e^{-\gamma^2 x} > 0$; an off line pair contributes an oscillating term. The verified zeros give a quantitative piece of this condition.

Theorem 24.15.1. *Let $T = 3 \cdot 10^{12}$, so that every zero with $0 < \gamma \leq T$ is on the line [419]. For every $x \geq 1/T$ and every integer*

$$0 \leq n \leq \frac{(T^2 - \gamma_1^2 - 1)\, x - 5}{\log(T^2/\gamma_1^2)},$$

*we have $(-1)^n \Theta^{(n)}(x) > 0$. Numerically the bound is $n \leq 1.7 \cdot 10^{23} x$. RH holds if and only if the bound can be replaced by $n < \infty$ for every $x > 0$. Status: proved, with the verification height $T$ as input; the constants are computed (`code/ht_misc.py`).*

*Proof.* The zeros with $\gamma \leq T$ contribute $\sum \gamma^{2n} e^{-\gamma^2 x} \geq \gamma_1^{2n} e^{-\gamma_1^2 x}$. A zero with $\gamma > T$ has $s_\rho = (\delta + i\gamma)^2$ with $|\delta| < \frac{1}{2}$, so its contribution has modulus at most $g(\gamma) = (\gamma^2 + \frac{1}{4})^n e^{-(\gamma^2 - \frac{1}{4})x}$. Under the bound on $n$ we have $n \leq T^2 x/52$, so $g$ is decreasing for $\gamma \geq T$ and $g(T + j + 1)/g(T + j) \leq e^{-1.9Tx} \leq e^{-1.9}$. There are at most $2\log(T + j + 1)$ zeros with $T + j < \gamma \leq T + j + 1$, by an explicit bound for the argument of $\zeta$ [176], so the tail is at most $4\log(T + 1) g(T)/(1 - e^{-1.9}) < 136\, g(T) < e^5 g(T)$. The bound on $n$ gives $n \log((T^2 + \frac{1}{4})/\gamma_1^2) + 5 < (T^2 - \frac{1}{4} - \gamma_1^2)x$, which makes this smaller than $\gamma_1^{2n} e^{-\gamma_1^2 x}$. □

At scale $x$ the $n$th Bernstein condition weighs the zeros by $\gamma^{2n}e^{-\gamma^2 x}$, which peaks at $\gamma^2 = n/x$. The theorem says that the conditions which only look below height $T$ hold, and no more: it is zero verification written as complete monotonicity, with the same height to order trade off as the Jensen degree ($d \approx \gamma^2$, a heuristic of Section 24.6) and the first order boundary response ($\delta^2/\gamma^4$, Remark 24.12.3).

### 24.15.2 The frontier

Table 24.12 lists the equivalent forms of RH that the reciprocal Xi law provides, the classical statement each is equivalent to, and how far each is known unconditionally.

| Reciprocal Xi form | Classical form | Known unconditionally |
|---|---|---|
| $\mu$ is a GGC | RH | Theorem 24.15.1: Bernstein order $1.7 \cdot 10^{23}x$ |
| density of $\mu$ is $\mathrm{PF}_\infty$ | RH (Schoenberg [474]) | log concave on $[0.008, \infty)$ (computed, Section 24.20) |
| $\mu^{(\alpha)}$ GGC for all $\alpha > \frac{1}{2}$ | $\operatorname{Re} \xi'/\xi > 0$ on $\operatorname{Re} s > \frac{1}{2}$ (Lagarias [320]) | $\alpha \geq 1$ only |
| $\mu^{(\alpha)}$ GGC at one $\alpha < 1$ | quasi RH at $\alpha$ | announced for $\alpha \geq \frac{7}{8}$ [398], open below |
| de Bruijn laws $\mu_t$ GGC | $t \geq \Lambda$ | $t \geq 0.2$ [419] |
| Jensen laws $\mu_d$ exist | Jensen and Pólya criterion | $d \leq 2 \cdot 10^{17}$ [159] |
| hybrid laws $\tilde{\mu}^{[N]}$ GGC | zeros up to $\gamma_N$ on the line | $N \leq 1.2 \cdot 10^{13}$ |
| GGC limit of prime built laws | Weil type positivity at the centre | the scaling of Theorem 24.8.4 erases the zeros; Theorem 24.15.4 for $\theta < \beta < 2 - \theta$ under QRH($\theta$) |
| boundary, Brownian, extremal representations | none | exist for every law |

Table 24.12: Equivalent forms of RH from the reciprocal Xi law and their unconditional frontier. The Jensen bound is Chassé's, as quoted in [159].

Two entries carry the diagnosis. First, the base point row: Lagarias proved that RH is equivalent to $\operatorname{Re} \xi'/\xi(s) > 0$ throughout $\operatorname{Re} s > \frac{1}{2}$ [320], and the Thorin density of $\mu^{(\alpha)}$ is exactly $\pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + it)$ whenever $\mu^{(\alpha)}$ is a GGC (Section 24.13, Theorem 1.2.2). So the base point descent is Lagarias' criterion, and pushing it below $\alpha = 1$ at a single value is the quasi Riemann hypothesis there. That is the announced half plane $\operatorname{Re} s > \frac{7}{8}$ of [398] (Section 12.1.2), not yet refereed, and it is open for every $\alpha < \frac{7}{8}$; before the announcement the zero free regions of de la Vallée Poussin and of Vinogradov and Korobov, which shrink to $\operatorname{Re} s = 1$ as the height grows, were the best known. Second, the last row: the probabilistic representations exist for every law and carry no information. Every unconditional

entry in the right column except the base point, quasi Riemann hypothesis and prime built rows also holds for Davenport and Heilbronn below the height of their first off line zero, apart from the log concavity row, which has been computed for $\mu$ only and is not claimed for them. The base point and prime built rows rest on the zero free half plane $\operatorname{Re} s \geq 1$ and on the prime sum, which come from the Euler product and fail for Davenport and Heilbronn, whose function has zeros with real part greater than 1 [111]. So the entries that survive for Davenport and Heilbronn cannot be extended to RH without the Euler product, and the entries that use the Euler product are the ones that fail for it.

### 24.15.3 The quasi Riemann hypothesis and the prime sum

Write QRH($\alpha$) for the statement that every zero has real part at most $\alpha$, where $\frac{1}{2} \leq \alpha < 1$. It was announced for $\alpha = \frac{7}{8}$ in October 2026, with a formal proof in Lean that has not yet been refereed [398], and it is open for every $\alpha < \frac{7}{8}$. It is exactly a statement about the prime sum, and exactly a statement about the base point hitting times.

Proposition 24.15.2. *For $\frac{1}{2} \leq \alpha < 1$ the following are equivalent.*

1. QRH($\alpha$).
2. $\psi(x) = x + O(x^{\alpha+\varepsilon})$ *for every* $\varepsilon > 0$.
3. *The Dirichlet series* $D(s) = \sum_{n \geq 1} (\Lambda(n) - 1) n^{-s}$ *converges for* $\operatorname{Re} s > \alpha$, *so that* $-\zeta'/\zeta = \zeta + D$ *there.*
4. $\operatorname{Re}\, \xi'/\xi(s) > 0$ *for* $\operatorname{Re} s > \alpha$.
5. *The base point laws* $\mu^{(\beta)}$ *are GGCs, that is diffusion hitting times, for every* $\beta > \alpha$.

*Status: proved (the equivalence of (4) and (5) with (1) is Theorem 1.2.2 and Remark 1.2.4).*

*Proof.* (1) and (2) are equivalent by the explicit formula. (2) gives $\sum_{n \leq x} (\Lambda(n) - 1) = \psi(x) - \lfloor x \rfloor = O(x^{\alpha+\varepsilon})$, and partial summation gives (3). Under (3), $-\zeta'/\zeta = \zeta + D$ is analytic in $\operatorname{Re} s > \alpha$ apart from the pole at 1, so $\zeta$ has no zeros there, which is (1). Under (1), the zeros pair as $\rho$ and $1 - \bar{\rho}$ with real parts in $[1 - \alpha, \alpha]$, so each term $\operatorname{Re} 1/(s - \rho)$ of $\operatorname{Re}\, \xi'/\xi(s)$ is positive when $\operatorname{Re} s > \alpha$, which is (4). By Theorem 1.2.2, $\mu^{(\beta)}$ is a GGC if and only if $\beta \geq \sup_\rho \operatorname{Re} \rho$, which gives (1) if and only if (5); and (4) is Lagarias' criterion on the half plane, Remark 1.2.4, whose last step shows that positivity of $\operatorname{Re}\, \xi'/\xi$ on every line to the right of $\alpha$ excludes zeros there. □

So the quasi Riemann hypothesis does control the prime sum below 1: under QRH($\alpha$), a smoothed explicit formula expresses the Thorin density of $\mu^{(\beta)}$ at height $t$, for $\beta > \alpha$, as the archimedean term plus a prime sum $\operatorname{Re}\sum_n \Lambda(n)n^{-\beta-it}w(n/X)$ minus a smooth main term in $X^{1-\beta-it}$, with an error $O(X^{\alpha-\beta}\log t)$ from the zeros (compare Proposition 9.1.13). The base point hitting times are then built from the primes up to $X$ to within that error. Two facts stop this from being used to go further.

First, positivity cannot be carried downward in $\beta$. In the variable $w = \sqrt{s}$, moving the base point is a translation, and if $w \mapsto 1/\xi(\beta + w)$ were the Laplace transform of a positive measure in $w$, a translation would be an exponential tilt of that measure and positivity would pass from $\beta$ to $\beta - \Delta$ for free. It never is (for the one sided case see Remark 9.1.8; the lemma adds the two sided case).

LEMMA 24.15.3. *For no real $\beta$ is $w \mapsto 1/\xi(\beta + w)$ the one or two sided Laplace transform of a positive measure. Status: proved.*

*Proof.* By Landau's theorem the abscissa of convergence of such a transform is a singular point on the real axis [514]. The function $1/\xi(\beta + w)$ is meromorphic with poles only at $w = \rho - \beta$, none of them real, since $\xi(\sigma) > 0$ for real $\sigma$. A finite abscissa would therefore have to be a regular point, and an infinite one would make the transform entire, contradicting the poles. □

This is why every positive representation in this chapter lives in the variable $s = w^2$, where the poles $-\gamma^2$ of the transform are real and negative, and why no translation argument in $\beta$ is available. Second, any step from QRH($\alpha$) to QRH($\alpha - \varepsilon$) is new information about zeros in the strip $\alpha - \varepsilon < \operatorname{Re} s \le \alpha$. Davenport and Heilbronn satisfy all the structural properties used here and have zeros in such strips, so the information cannot come from structure; it must come from the Euler product, through the prime sum above.

### 24.15.4 Prime built hitting times under the quasi Riemann hypothesis

Suppose QRH($\theta$) holds for some $\theta < 1$; the case $\theta = \frac{7}{8}$ is the one announced in [398], so the statements below are conditional on that announcement for $\theta = \frac{7}{8}$. Then the base point laws above $\theta$ can be built from the primes, and this is the correct renormalisation of the Euler hitting times of Theorem 24.8.4. Let $w(u) = (1-u)_+^3$, with Mellin transform $W(z) = 3!\,\Gamma(z)/\Gamma(z+4) = 6/(z(z+1)(z+2)(z+3))$, and for $\beta > \theta$ define

$$d_X(t) = \frac{1}{\pi}\Big( \operatorname{Re} A(\beta + it) - \operatorname{Re}\sum_{n \le X} \frac{\Lambda(n)}{n^{\beta+it}}\Big(1 - \frac{n}{X}\Big)^3 + \operatorname{Re}\, X^{1-\beta-it} W(1-\beta-it)\Big),$$

which uses only the primes up to $X$ and the gamma function. Shifting the contour in $\frac{1}{2\pi i}\int(-\zeta'/\zeta)(s+z)X^zW(z)\,\mathrm{d}z$ gives the explicit formula

$$\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\beta+it)=d_X(t)-\frac{1}{\pi}\operatorname{Re}\sum_{\rho}X^{\rho-\beta-it}W(\rho-\beta-it)\\ +O\big(X^{-1}\log(|t|+2)\big).$$

The poles of $W$ at $z=-1,-2,-3$, with residues $-3$, $3$, $-1$, produce the remainder, which is of order $X^{-1}$ and not the $X^{-2-\beta}$ of the first trivial zero alone. The formula is used only for $\theta<\beta<2-\theta$ under $\mathrm{QRH}(\theta)$: for $2-\theta\le\beta\le1+\theta$ a zero with $\beta_\rho=\beta-1$ meets a pole of $W$ (for $\beta>1+\theta$ nothing collides and the $X^{-1}$ remainder dominates), and the remainder $-3X^{-1}(-\zeta'/\zeta)(s-1)$ has poles in the critical strip, so the bound is not uniform.

THEOREM 24.15.4. *Assume* $\mathrm{QRH}(\theta)$ *and let* $\theta<\beta<2-\theta$. *Then* $|d_X(t)-\pi^{-1}\operatorname{Re}\xi'/\xi(\beta+it)|\ll_\beta X^{\theta-\beta}\log(t+2)$ *uniformly in* $t$, *and there is* $X_0(\beta,\theta)$ *such that for* $X\ge X_0$ *the function* $d_X$ *is nonnegative. The GGC laws* $\mu_X^{(\beta)}$ *with Thorin density* $d_X$ *in the variable* $z=t^2$ *are then hitting times built from the primes up to* $X$, *and* $\mu_X^{(\beta)}\Rightarrow\mu^{(\beta)}$ *as* $X\to\infty$. *Conversely, if for some* $\beta$ *and every* $t$ *the values* $d_X(t)$ *are nonnegative for all large* $X$, *then* $\mathrm{QRH}(\beta)$ *holds. Status: proved, conditional on* $\mathrm{QRH}(\theta)$ *for the first part. The unconditional range* $1<\beta<2$ *is not covered, see the discussion after the proof.*

*Proof.* Under $\mathrm{QRH}(\theta)$, $|X^{\rho-\beta-it}|\le X^{\theta-\beta}$ for every zero, and $|W(\rho-\beta-it)|\ll(1+|\gamma-t|)^{-4}$, so the zero sum is $\ll X^{\theta-\beta}\log(t+2)$, and the remainder $O(X^{-1}\log(|t|+2))$ is no larger because $\beta<2-\theta\le1+\theta$. By Proposition 24.15.2 the exact density is positive, and it is continuous, so it has a positive minimum on any compact set. For large $t$, $\pi^{-1}\operatorname{Re}\xi'/\xi(\beta+it)=\frac{1}{2\pi}\log(t/2\pi)-\pi^{-1}\operatorname{Re}(-\zeta'/\zeta)(\beta+it)+O(1/t)$ and $\zeta'/\zeta(\beta+it)=o(\log t)$ under $\mathrm{QRH}(\theta)$, so the exact density is at least $c\log t$ for large $t$, and the error is a factor $X^{\theta-\beta}$ smaller, which gives $X_0$ (no appeal to the smoothed counting function of zeros is needed). The Laplace exponents converge because $d_X\to\pi^{-1}\operatorname{Re}\xi'/\xi$ uniformly against $\log(1+s/t^2)$.

For the converse we cannot use the pointwise limit, since a zero with $\beta_\rho>\beta$ makes the term $X^{\rho-\beta-it}W(\rho-\beta-it)$ grow like $X^{\beta_\rho-\beta}$. Fix $t$ that is not an ordinate of a zero and not half the difference of two ordinates, and put $s=\beta+it$. For $\operatorname{Re}u$ large, $\int_1^\infty\sum_{n\le X}\Lambda(n)n^{-s}(1-n/X)^3X^{-u-1}\mathrm{d}X=(-\zeta'/\zeta)(s+u)W(u)$, because $\int_0^1(1-v)^3v^{u-1}\mathrm{d}v=W(u)$. So the Mellin transform $M(u)=\int_1^\infty\pi d_X(t)X^{-u-1}\mathrm{d}X$ is a combination of $\operatorname{Re}A(s)/u$, of $(\zeta'/\zeta)(s+u)W(u)$ and its conjugate counterpart at $\bar s$, and of $W(1-s)/(u+s-1)$ with its conjugate; the pole at $u=1-s$ of $\zeta'/\zeta(s+u)$ cancels against the main term. It is meromorphic in $\operatorname{Re}u>0$ with poles only at $u=\rho-s$ and $u=\rho-\bar s$, none of them real, and every zero with $\beta_\rho>\beta$ gives a pole with $\operatorname{Re}u=\beta_\rho-\beta>0$ (the choice of $t$ keeps the poles from cancelling). If $d_X(t)\ge0$ for $X\ge X_0$, the transform of the tail $X\ge X_0$ has a

real singularity at its abscissa of convergence by Landau's theorem [514], as in Lemma 24.15.3, while $M$ is regular at every real $u > 0$; so the abscissa is at most 0, $M$ is analytic in $\operatorname{Re} u > 0$, and no zero has real part above $\beta$. Since $t$ ranges over a set of full measure, this is QRH($\beta$). □

So under QRH($\frac{7}{8}$) the reciprocal Xi programme has what the heuristic Remark 24.10.1 asks for, at every base point $\frac{7}{8} < \beta < \frac{9}{8}$: GGC hitting times built from the primes and the gamma function, converging to the base point law. The renormalisation that makes this work is the subtraction of the smooth main term $X^{1-s}W(1-s)$, the contribution of the smoothed prime number theorem, rather than the central limit scaling of Theorem 24.8.4, which erases the zeros. Whether the same holds unconditionally for $1 < \beta < 2$ is left open: the two singular terms at $\beta_\rho = \beta - 1$ may cancel in the sum, but the proof above does not show it. What QRH($\frac{7}{8}$) does not give is anything at $\beta \le \frac{7}{8}$, and in particular at $\frac{1}{2}$: there the error bound $X^{\theta-\beta}\log(t+2)$ of Theorem 24.15.4 does not decay.

We computed $d_X$ with $X = 10^2, \dots, 10^5$ on $0 < t < 2000$. The error against the exact density is 0.34, 0.11, 0.038, 0.011 at $\beta = 1$, and 0.90, 0.44, 0.22, 0.10 at $\beta = 0.8$ (computed, `code/ht_primebuilt.py`): it decays like $X^{1/2-\beta}$, the rate set by the actual zeros, which lie on the line as far as they have been computed. The densities are nonnegative on the window for $X \ge 10^3$ at $\beta = 1, 0.9, 0.8, 0.7$, with minimum close to the exact minimum (0.0073 against 0.0074 at $\beta = 1$). At $\beta = 0.6$ they are still negative at $X = 10^5$, with minimum $-0.16$ near $t = 1164$, against an exact minimum of 0.0015. The obstacle is visible: when the kernel width $\beta - \frac{1}{2}$ is smaller than the zero spacing, the exact density between consecutive zeros is tiny, and the error must be pushed below it. Near the centre this costs a power of $X$ that grows without bound as $\beta \downarrow \frac{1}{2}$.

Other consequences of QRH($\frac{7}{8}$) in this chapter's terms are weaker than what is already known. De Bruijn's theorem turns zeros in the strip $|\operatorname{Re}\rho - \frac{1}{2}| \le \frac{3}{8}$ into $\Lambda \le 2(\frac{3}{8})^2 = \frac{9}{32} = 0.28$, which is worse than the proven 0.2 [419], so it improves neither the de Bruijn laws nor the Jensen degree heuristic $D(T) = T^2/(2\Lambda_0)$ of Section 24.6. It removes the defect $D_\beta$ of Proposition 11.4.3 for $\beta > \frac{7}{8}$, and it gives $\psi(x) = x + O(x^{7/8+\varepsilon})$, a power saving in the prime number theorem that follows from the announcement [398] if it stands.

### 24.15.5 Zero density as an almost GGC statement

What is known unconditionally is QRH on average: zero density estimates. Section 11.4.6 and Theorem 11.4.2 already turn them into a statement about base point hitting times, and we do not repeat it. Fix $\beta > \frac{1}{2}$ and split the Thorin density $\pi^{-1}\operatorname{Re}\,\xi'/\xi(\beta+it)$ into the Poisson kernels of the zeros with $\operatorname{Re}\rho < \beta$, which are positive, and those with $\operatorname{Re}\rho > \beta$, which are negative, with total masses $m_\beta^+(T)$ and $m_\beta^-(T)$ on $[0,T]$. The book's theorem bounds $m_\beta^-(T)/m_\beta^+(T) \le T^{e(\beta)-1+o(1)}$ with $e(\beta) < 1$ from Ingham's density estimate [284] and from Guth

and Maynard [166, Theorem 1.2]. Selberg's estimate $N(\sigma,T)\ll T^{1-(\sigma-1/2)/4}\log T$ [477] gives the weaker relative size $T^{-(\beta-1/2)/4}$ for every $\beta>\frac12$, and $N(\sigma,T)\ll T^{30(1-\sigma)/13+o(1)}$ gives $T^{30(1-\beta)/13-1+o(1)}$, which is the stronger of the two once $\beta>0.575$. Both are beaten by Ingham's exponent on $(\frac12,\frac{7}{10})$, and Guth and Maynard's $15(1-\sigma)/(3+5\sigma)$ is sharper on $[\frac{7}{10},\frac45]$. The zero count satisfies $\nu_-[0,T]\ll N(\beta,2T)+1$ in the notation above, with $\nu_-$ the negative part of the Thorin measure, because a kernel centred at height $\gamma>2T$ puts mass $O(1/(\gamma-T))$ times its width into $[0,T]$, which sums to $O(1)$ over the sparse set of zeros to the right of $\beta$.

What this does not give is a factor. The negative kernels are not the Thorin density of a second hitting time $G_-$ with $\mu^{(\beta)}*G_-=G_+$: the negative Poisson kernel of a zero to the right of $\beta$ is the Thorin density of the reflected zero, so the product of the two exponentials is analytic in the slit plane, while $\xi(\beta)/\xi(\beta+\sqrt{s})$ has poles there. The correct statement is the canonical factorisation of Proposition 11.4.3: $\mu^{(\beta)}$ has Laplace transform $\varphi_\beta^+D_\beta$, where $\varphi_\beta^+$ is the Laplace transform of the GGC $T_\beta^+$ whose Thorin density is the positive part above and $D_\beta$ is the defect over the zeros to the right of $\beta$, and the book shows that $T_\beta^+$ is not the sum of a variable with law $\mu^{(\beta)}$ and an independent nonnegative variable. Density estimates say that the defect $D_\beta$ is sparse. Proving that it is empty is $\mathrm{QRH}(\beta)$, and no density estimate, however strong, can do it, because density estimates allow a sparse set of exceptions.

### 24.15.6 Where arithmetic would enter

Writing $\xi'/\xi=A+\zeta'/\zeta$ with $A$ the archimedean part, the base point Thorin density is

$$\frac{1}{\pi}\Big(\operatorname{Re}A(\alpha+it)-\operatorname{Re}\sum_{n\ge2}\frac{\Lambda(n)}{n^{\alpha+it}}\Big),$$

$$\operatorname{Re}A(\alpha+it)=\tfrac12\log\frac{t}{2\pi}+O(t^{-1}),$$

with the Dirichlet series read as $-\zeta'/\zeta(\alpha+it)$ when $\alpha\le1$. For $\alpha>1$ the series converges absolutely and is bounded by $-\zeta'/\zeta(\alpha)$, so positivity for large $t$ is automatic. That the base point laws are GGCs for $\alpha\ge1$ comes from a different fact, which is also arithmetic: every term $\operatorname{Re}1/(s-\rho)$ of $\operatorname{Re}\xi'/\xi$ is positive because no zero has real part $\ge1$ (Theorem 1.2.2). For $\alpha<1$ positivity at every $t$ requires the prime sum to stay below $\frac12\log(t/2\pi)$ in real part, uniformly in $t$, with no absolute convergence available. That uniform inequality, a positivity statement about $\sum\Lambda(n)n^{-\alpha-it}$ in the critical strip, is the arithmetic input that the routes of this chapter would need, in the form of Lagarias' criterion. Davenport and Heilbronn fail it, because their function has no prime sum: its logarithmic derivative has no Euler product expansion at all.

*Notes.* The sections on curved boundaries, Yor's function, the Davenport Heilbronn test and the frontier are from the chapter of Nick Polson on hitting times and the reciprocal xi function; the statements of Biane, Pitman and Yor, of Pitman and Yor, of James, Roynette and Yor, of Roberts and Shortland and of Kent are those of the cited papers. The factorisation of the base point law used in the discussion of zero density, and the status of QRH($\frac{7}{8}$), are those of Sections 11.4 and 12.1; the corrections to the identity with $G_-$, to the explicit formula and to the converse of Theorem 24.15.4 were made in the merge.

## 24.16 The coefficient ladder

The density of the reciprocal Xi law has a Pólya frequency ladder (Section 24.6 and Chapter 16), and so does the canonical function $\Theta$. This section adds a third object with such a ladder, the Taylor coefficients of $F(s) = \xi(\frac{1}{2} + \sqrt{s})/\xi(\frac{1}{2})$, and the next one shows what the three ladders can and cannot see. The results and computations of both sections come from notes contributed by Nick Polson [251]; the scripts that produced the numbers are listed in `code/README.md`.

Riemann's integral gives, with the de Bruijn Newman kernel $\Phi_{\mathrm{dBN}}$ of Section 8.4 (so that $\xi(\frac{1}{2} + w) = 8\int_0^\infty \Phi_{\mathrm{dBN}}(u)\cosh(2wu)\,\mathrm{d}u$),

$$F(s) = \sum_{n\ge 0} c_n s^n, \qquad c_n = \frac{\mathbb{E}V^{2n}}{(2n)!} = \frac{8}{\xi(\frac{1}{2})}\int_0^\infty \Phi_{\mathrm{dBN}}(u)\frac{(2u)^{2n}}{(2n)!}\,\mathrm{d}u > 0,$$

where $V$ is symmetric with the Pólya density $\Phi/\Xi(0)$ of (1.2.3), so that $V/2$ has density proportional to $\Phi_{\mathrm{dBN}}$. Then $c_0 = 1$ and, computed at 110 digits from the first 120 zeros and Riemann's integral, $c_1 = \sum_{\mathrm{Im}\,\rho>0}(-s_\rho)^{-1} = 0.023104993$ (with $s_\rho = (\rho - \frac{1}{2})^2$), $c_2 = 2.4833405 \times 10^{-4}$ and $c_3 = 1.6743526 \times 10^{-6}$ (status: computed). As a check on all the coefficients, $\sum_n c_n = F(1) = \xi(\frac{3}{2})/\xi(\frac{1}{2}) = 1.02335500958$, which the computed series reproduces to the printed digits.

Theorem 24.16.1 (Coefficient ladder). *(i) RH holds if and only if $(c_n)$ is a Pólya frequency sequence, that is, every minor of the Toeplitz matrix $(c_{i-j})$ is nonnegative.*

*(ii) Unconditionally, $(c_n)$ is PF$_2$: $c_n^2 \ge c_{n-1}c_{n+1}$ for all $n \ge 1$.*

*Proof.* (i) By the theorem of Aissen, Schoenberg and Whitney and of Edrei [13, 126], a sequence with $c_0 = 1$ is PF$_\infty$ if and only if its generating function is $e^{bs}\prod_i(1 + \alpha_i s)/\prod_i(1 - \beta_i s)$ with $b, \alpha_i, \beta_i \ge 0$ and $\sum_i(\alpha_i + \beta_i) < \infty$. Since $F$ is entire of order one half, so that it has no poles and is a product over its zeros, this says exactly that every zero of $F$ is real and negative, and the zeros of $F$ are the numbers $(\rho - \frac{1}{2})^2$, so that this is RH.

(ii) Write $b_n = (2n)!\,c_n$. Csordas, Norfolk and Varga proved the Turán inequalities $b_n^2 > \frac{2n-1}{2n+1}b_{n-1}b_{n+1}$ [99], which is the reduction already used in Section 16.4

(the normalisations differ by the factor $4^n$, which cancels). Then

$$\frac{c_n^2}{c_{n-1}c_{n+1}} = \frac{b_n^2}{b_{n-1}b_{n+1}} \cdot \frac{(2n-2)!\,(2n+2)!}{(2n)!^2} > \frac{2n-1}{2n+1} \cdot \frac{(2n+2)(2n+1)}{2n(2n-1)} = \frac{n+1}{n} > 1.$$

□

Part (ii) restates in the Toeplitz form the Turán paragraph of Section 16.4; part (i) is the statement not made elsewhere in the book, and it places the coefficients on the same footing as the other two objects. The three ladders are

$$\Theta \text{ completely monotone} \iff f \text{ PF}_\infty \iff (c_n) \text{ PF}_\infty \iff \text{RH},$$

with unconditional low rungs for each: $T_k \geq 0$ for $k \leq H^2/80$ for $\Theta$ [421, Section 6], in another form Theorem 24.15.1; log concavity of $f$ on $[Y_*, \infty)$ for the density (Theorem 24.5.5, a computer assisted theorem whose enclosed inputs are listed there); and $\text{PF}_2$ of $(c_n)$ by Theorem 24.16.1.

The consecutive Toeplitz minors $\det[c_{n+i-j}]_{i,j<k}$, normalised by $c_n^k$, were computed for $k = 2, \ldots, 7$ and $k - 1 \leq n \leq 35 - k$ (status: computed). They are all positive, and they decrease with $n$, so the smallest sit at the largest $n$ allowed; with this range they are $4.16 \times 10^{-2}$, $1.52 \times 10^{-4}$, $7.71 \times 10^{-8}$, $7.66 \times 10^{-12}$, $1.98 \times 10^{-16}$ and $1.70 \times 10^{-21}$. A different range gives different digits (for $n \leq 36-k$ the same minima are $4.05 \times 10^{-2}$, $1.4 \times 10^{-4}$, $6.5 \times 10^{-8}$, $5.8 \times 10^{-12}$, $1.3 \times 10^{-16}$ and $9.1 \times 10^{-22}$), so the range and the normalisation are part of the statement. Among 399 random minors of orders 3, 4 and 5 (133 of each), none was below $-10^{-40}$. The higher order Turán inequalities of Dimitrov and Lucas [116] and the hyperbolicity of Jensen polynomials of Griffin, Ono, Rolen and Zagier [159] are rungs of a closely related ladder on the same coefficients. Of the three ladders, finite rungs of the coefficient ladder are the ones that can be proved, as Theorem 24.17.2 shows next; finite rungs of the density ladder can be certified by computation; and the ladder of $\Theta$ is the one that connects to the Thorin measure.

## 24.17 The angle theorem: what the ladders can and cannot see

The coefficient ladder has a multiplicative structure that the other two lack. It yields an unconditional theorem, a test against a function that violates RH, and a clear statement of what climbing the ladders can give.

### 24.17.1 The angle theorem

For a zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\gamma > 0$, let $s_\rho = (\delta + i\gamma)^2$ and let $\theta_\rho \in [0, \pi)$ be the angle between $s_\rho$ and the negative real axis. Since $\tan \theta_\rho = 2|\delta|\gamma/(\gamma^2 - \delta^2)$,

$$\theta_\rho = 2 \arctan \frac{|\delta|}{\gamma}, \tag{24.17.1}$$

and RH is the statement that every $\theta_\rho$ vanishes.

Lemma 24.17.1 (Three term sequences). *For $r > 0$ and $0 \le \theta < \pi$, the sequence $(1, 2r^{-1}\cos\theta, r^{-2})$, the coefficient sequence of $(1 + s/re^{i\theta})(1 + s/re^{-i\theta})$, is $PF_k$ if and only if $\theta \le \pi/(k+1)$.*

*Proof.* Scale $s$ so that $r = 1$. The contiguous Toeplitz minor of order $m$ is the tridiagonal determinant $D_m$ with $D_m = 2\cos\theta\, D_{m-1} - D_{m-2}$, $D_0 = 1$, $D_1 = 2\cos\theta$, so $D_m = U_m(\cos\theta) = \sin((m+1)\theta)/\sin\theta$. These are positive for $m \le k$ exactly when $(k+1)\theta \le \pi$, which gives necessity. For sufficiency, every minor of a Toeplitz matrix of bandwidth two is a product of contiguous minors and of powers of the end coefficients, which are positive [188]. □

Theorem 24.17.2 (Angle theorem). *If every zero of $\zeta$ satisfies $\theta_\rho \le \pi/(k+1)$, then the Taylor coefficients $(c_n)$ of $F$ form a $PF_k$ sequence.*

*Proof.* Since $\sum |s_\rho|^{-1} < \infty$, $F$ is the product of the factors $1 + s/\gamma^2$ over the zeros on the line and of the real quadratic factors $(1 - s/s_\rho)(1 - s/\overline{s_\rho})$ over the pairs off the line, with multiplicity. The first have $\mathrm{PF}_\infty$ coefficient sequences $(1, \gamma^{-2})$; the second have $\mathrm{PF}_k$ sequences by Lemma 24.17.1, because $1 - s/s_\rho = 1 + s/(|s_\rho| e^{\pm i\theta_\rho})$. Multiplying generating functions convolves coefficient sequences, which multiplies their Toeplitz matrices, and by the Cauchy Binet formula a product of $\mathrm{PF}_k$ matrices is $\mathrm{PF}_k$. The partial products converge coefficientwise, and coefficientwise limits preserve the sign of every minor. □

Corollary 24.17.3 (An unconditional rung count). *Unconditionally, $(c_n)$ is $PF_k$ for every $k \le \lfloor \pi H \rfloor - 1 = 9424777960768$, where $H = 3 \cdot 10^{12}$ is the height to which all zeros are known to lie on the line [419]. More generally, $(c_n)$ is $PF_K$ as soon as every zero with $\gamma \le (K+1)/\pi$ lies on the line.*

*Proof.* Any zero has $|\delta| < \frac{1}{2}$, so by (24.17.1) $\theta_\rho < 2\arctan(1/(2\gamma)) < 1/\gamma$. Zeros up to the verified height have $\theta_\rho = 0$; zeros above it have $\theta_\rho < 1/H \le \pi/(k+1)$ for the stated $k$. In the second statement, a zero off the line has $\gamma > (K+1)/\pi$, so $\theta_\rho < 1/\gamma < \pi/(K+1)$. Apply Theorem 24.17.2. □

The bound in the proof is the exact angle (24.17.1), not the cruder $1/(\gamma - 1)$, which would fall three short of the count in the statement. The corollary contains part (ii) of Theorem 24.16.1 as the case $k = 2$, although the Turán inequalities of

Csordas, Norfolk and Varga are sharper than $\mathrm{PF}_2$, and it explains the numerics of Section 24.16: no computable rung of the coefficient ladder can fail. The same angle of the zeros, there of the poles of the factors, drives the unconditional range of the power closure of Theorem 4.3.5 and Corollary 4.3.7, and the two should be read side by side.

### 24.17.2 The Davenport Heilbronn test

The Davenport Heilbronn function $f_{DH}(s) = \sum a_n n^{-s}$ [111], with $a_n$ periodic modulo 5 with values $(1, \kappa, -\kappa, -1, 0)$ and $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1)$, has the completed form $\xi_{DH}(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f_{DH}(s)$ with $\xi_{DH}(s) = \xi_{DH}(1-s)$, which we checked to 60 digits. It has no Euler product and it violates RH. It has the functional equation of $\xi$ and an archimedean term of the same growth, $\frac{1}{2}\log\tau + O(1)$ at height $\tau$, but not the same archimedean factor: its term is $\frac{1}{2}\log(5/\pi) + \frac{1}{2}\psi(\frac{z+1}{2})$ in place of that of $\xi$. The classical off line zero is $0.8085171824566374 + 85.69934848537759\,i$ [484] (computed again here by root finding at 60 digits).

We computed the first 110 Taylor coefficients of $F_{DH}(s) = \xi_{DH}(\frac{1}{2} + \sqrt{s})/\xi_{DH}(\frac{1}{2})$ by a Cauchy integral on the circle of radius 1000 with 1024 nodes at 220 digits. For $\xi$ itself they agree with the coefficients from Riemann's integral to $3.9 \times 10^{-141}$ for $n \le 109$, and a second radius (700) agrees with the first to $4.8 \times 10^{-126}$ for $\xi$ and $2.7 \times 10^{-158}$ for $F_{DH}$ (status: computed).

| order $k$ | 2 | 3 | 4 | 5 | 6 | 10 |
|---|---|---|---|---|---|---|
| $\xi$: smallest minor | $1.47 \times 10^{-2}$ | $6.52 \times 10^{-6}$ | $1.32 \times 10^{-10}$ | $1.66 \times 10^{-16}$ | $1.65 \times 10^{-23}$ | $5.87 \times 10^{-61}$ |
| DH: smallest minor | $1.58 \times 10^{-2}$ | $7.98 \times 10^{-6}$ | $1.98 \times 10^{-10}$ | $3.26 \times 10^{-16}$ | $4.56 \times 10^{-23}$ | $1.25 \times 10^{-59}$ |

Table 24.13: Smallest contiguous Toeplitz minors $\det[c_{n+i-j}]_{i,j<k}$ for $k - 1 \le n \le 104 - k$, normalised by $c_n^k$, for $\xi$ and for the Davenport Heilbronn function. A further 600 random minors of each of the orders 3, 4 and 6 were nonnegative for both functions, none below $-10^{-80}$.

The two rows are indistinguishable. The off line zero at height 85.7 has $\theta = 0.0072$ by (24.17.1). If it has the largest angle among the zeros of $f_{DH}$, Theorem 24.17.2 shows that $(c_n^{DH})$ is $\mathrm{PF}_k$ for every $k \le 435$, because $\pi/\theta - 1 = 435.3$; the condition is not verified here, and zeros of $f_{DH}$ in $\mathrm{Re}\, s > 1$ at low height, which Davenport and Heilbronn showed to exist for such functions, would have larger angles. By the theorem of Aissen, Schoenberg, Whitney and Edrei, $(c_n^{DH})$ is not $\mathrm{PF}_\infty$, since $F_{DH}$ has a zero off the negative axis. So, under that condition, the coefficient ladder cannot tell $\xi$ from a function that violates RH until an order of several hundred, and by Corollary 24.17.3 it cannot tell a hypothetical off line zero of $\zeta$ above height $H$ from RH before order $9.4 \times 10^{12}$.

### 24.17.3 How sharp is the angle theorem?

Theorem 24.17.2 says that an off line pair at angle $\theta$ cannot break the coefficient ladder below order about $\pi/\theta$. To see whether it breaks near that order, we took the first 30 zeros of $\xi$, replaced the $m$th by a pair at angle $\theta$ from the negative axis with the same modulus $\gamma_m^2$, and found the first order $k$ and index $n$ at which a contiguous Toeplitz minor of the polynomial's coefficients is negative, at 800 bits with the zeros from interval arithmetic (status: computed).

| | $m=1$ | $m=3$ | $m=6$ | $m=11$ | $m=16$ |
|---|---|---|---|---|---|
| $\theta=\pi/4$ (bare threshold 4) | 6 | 12 | 17 | 21 | 24 |
| $\theta=\pi/6$ (bare threshold 6) | 8 | 15 | 24 | 31 | 35 |
| $\theta=\pi/10$ (bare threshold 10) | 13 | 20 | 30 | none to 40 | none to 40 |

Table 24.14: First failing order of the coefficient ladder when the $m$th zero is replaced by a pair at angle $\theta$. The bare threshold is the first failing order for the quadratic factor alone, the least $k$ with $\theta > \pi/(k+1)$. The failing index $n$ is 1, 3, 7, 12 and 17 for $m = 1, 3, 6, 11, 16$ (and 6 at $\theta = \pi/10$, $m = 6$), within one of $m$; orders above 40 were not tried.

Two things stand out. When the pair is the lowest zero, the ladder fails only two or three orders beyond the bare threshold, so the angle theorem is nearly sharp. When the pair sits higher, the zeros below it smooth it, and the failing order grows: across the table it is $c\,(\pi/\theta)\sqrt{m}$ with $c$ between about 1.2 and 1.7, and the failure appears at index $n \approx m$. This is an observation on the computed entries, not a theorem (Problem 24.8). If the scaling persists, which is heuristic, an off line zero at height $\gamma$, position $m \approx N(\gamma)$ and displacement $\delta$ first breaks the coefficient ladder at order about $(\pi\gamma/2|\delta|)\sqrt{N(\gamma)}$, of order $\gamma^{3/2}\sqrt{\log\gamma}/|\delta|$. That lies between the angle bound $\gamma/|\delta|$ and the order about $\gamma^2\log\gamma/|\delta|$ at which the ladder of $\Theta$ would be expected to fail. For the Davenport Heilbronn zero at height 85.7, which is about the 44th zero of that function (the main term of the zero count is 43.95 at $T = 85.7$, and 44.8 with the constant $\frac{7}{8}$), the scaling predicts a first failure between about 3500 and 4900 ($c$ from 1.2 to 1.7 times 2894) at index near 44. We cannot compute that far, since minors of that order need thousands of digits.

### 24.17.4 The density ladder compared

The same experiment on the density used the first 20 zeros and Toeplitz determinants $\det[f(x_0 + (i-j)h)]$ on a grid of base points $x_0$ and spacings $h$, with the arguments computed in exact multiprecision. Table 24.15 is printed as the source note reports it. It has not been reproduced: the note does not define the 79 base points and 5 spacings, and no script of the book recomputes it. The control, in which the pair is replaced by two real zeros $10^{-6}$ apart, passes every order to 6 in

the same report.

| | $m = 1$ | $m = 3$ | $m = 6$ |
|---|---|---|---|
| $\theta = \pi/4$ | 1 | 3 | 5 |
| $\theta = \pi/10$ | none to 7 | 3 | 6 |
| $\theta = \pi/40$ | none to 7 | none to 7 | 6 |
| $\theta = \pi/100,\ \pi/400$ | none to 8 | none to 8 | none to 8 ($m = 6, 11$) |

Table 24.15: First failing order of the density ladder for the same kind of perturbation, as reported in the source note (not reproduced). Order 1 means that the density itself becomes negative.

If the table is right, then at large angles the density ladder reacts much sooner than the coefficient ladder, at order 3 to 6 where the coefficient ladder needs 12 to 30 (Table 24.14). A plausible reason is that the density sees the pair as an oscillation in time, while the coefficients see it only through a smoothed generating function (heuristic). At angles of order 1/100 and below, both ladders are blind at every order computed. Such angles are still far larger than any that an off line zero of $\zeta$ above the verified height can have, since $\theta_\rho < 1/H < 3.4 \times 10^{-13}$ there by the proof of Corollary 24.17.3. The interval arithmetic repetition of the log concavity theorem, and the open window below $Y_*$, are in Theorem 24.5.5 and the discussion after it.

### 24.17.5 Is there a path to RH?

For the coefficient ladder the angle theorem makes the answer precise. Rung $K$ is implied by the location of the zeros up to height about $K/\pi$, and every off line zero is invisible below order about $\pi\gamma/(2|\delta|)$. The ladder of $\Theta$ behaves in the same way with order about $\gamma^2$, up to a logarithm, in place of $\gamma$ [421, Section 6]. The density ladder is, by the computations above if they are confirmed, equally blind at realistic angles, although it reacts sooner when the angle is large. So each ladder is a reformulation of RH in which a finite rung is implied by a finite verification of zeros, and the evidence is that a finite rung is also no stronger than one. A proof of RH by climbing a ladder rung by rung would therefore need an argument that controls every rung at once.

The Davenport Heilbronn test sharpens this. The function $f_{DH}$ has the functional equation and an archimedean term of the growth of $\xi$, it violates RH, and its low rungs hold. So an argument that applies verbatim to $f_{DH}$ would prove RH for a function for which it is false. Such an argument must use a property that $f_{DH}$ lacks, and the Euler product is the evident one. Chapter 9 shows where the Euler product gives positivity for free, to the right of 1, where its first moment converges, and Proposition 9.1.13 extends this to the compensated form below 1.

Carrying the positivity into the strip is the hard step, and nothing in this chapter does it.

## 24.18 What a hitting time proof would need

Taken together, the results of this chapter reduce a hitting time proof of RH to one of three constructions, none of which is known.

1. A construction from the primes at basepoint $\alpha = \frac{1}{2}$, finer than central limit scaling, of GGC laws whose Thorin measures converge vaguely to $\sum \delta_{\gamma^2}$. The heuristic Remark 24.10.1 reads its positivity as a Weil type criterion. Equivalently, one would carry the arithmetic positivity of the Euler product at basepoint $\alpha > 1$ into the critical strip. Section 12.1 records what is known about the zero free half planes that this would require: the announced half plane $\operatorname{Re} s > \frac{7}{8}$ is not enough, and the step needed beyond it is itself open.

2. A positivity preserving evolution of the Krein string mass in the de Bruijn and Newman parameter or in the basepoint, sharp at $t = 0$ or $\alpha = \frac{1}{2}$.

3. A proof that the harmonic mean of Brownian exit time transforms given by Riemann's integral is a GGC.

By the Davenport Heilbronn argument of Section 24.17.5, a proof of the second or third kind that does not use the Euler product cannot be expected to exist, which leaves the first. The statement is a heuristic one: it excludes arguments that apply verbatim to $f_{DH}$, not every argument that uses its functional equation.

## 24.19 Open problems

Problem 24.1. The rate in Theorem 24.8.4 is explicit, through the cumulant expansion that follows its table. Determine whether any conditioning of $Y_P$ on $N_P$, or any coupling with the archimedean factor, has a limit that retains the zeros.

Problem 24.2. Compute the Krein string mass functions of the de Bruijn and Newman laws $\mu_t$ for $t \geq 0.2$ from their spectral data, and study their evolution as $t$ decreases. Is there a monotone quantity of the string that would force $\Lambda = 0$?

Problem 24.3. Is the harmonic mean $1/\mathbb{E}\cosh(\sqrt{s}V)$ of Brownian exit time transforms a GGC for any explicit class of symmetric $V$ other than those already known to give real zeros through Pólya's theorems?

Problem 24.4. Build Markov chains from the primes whose generators approximate a canonical system of de Branges type for $\xi$, and prove a diffusion limit theorem for them in the style of Roberts, Gelman and Gilks.

Problem 24.5. Characterise the boundaries $b$ for which the Brownian crossing time $\tau_b$ is a GGC. A characterisation that is closed under vague limits of Thorin measures, checked on the computed boundary of $\mu$, would be equivalent to RH; a necessary condition alone would only test it. If the first order response of Section 24.12 is a guide (it is heuristic, Remark 24.12.3), any such condition must be sensitive to changes of order $\delta^2\gamma^{-4}$ in the middle of the boundary, or must act at times of order $\log\gamma/\gamma$.

Problem 24.6. Prove the resolution law of Section 24.6: if $F$ has a non real zero $s_\rho$ with $\operatorname{Im} s_\rho = \omega$ and all other zeros at distance at least $\eta$, show that $J_d$ has a non real zero for every $d \geq C|s_\rho|^2/\omega^2$, with $C$ depending on $\eta$ and the local zero density. Conversely, is there an explicit $D(T)$ such that RH up to height $T$ implies that $J_d$ is hyperbolic for every $d \leq D(T)$?

Problem 24.7. Is the Davenport Heilbronn theta function $\theta_{DH}$ the theta function of a natural Brownian or Bessel functional, in the sense in which $\theta$ is that of the range of a Brownian bridge? A property shared by all such functionals and failed by $\theta_{DH}$ would be a genuinely arithmetic input in probabilistic form.

Problem 24.8. Prove the scaling of Table 24.14: that an off line pair at angle $\theta$ and position $m$ first breaks the coefficient ladder at order of size $(\pi/\theta)\sqrt{m}$, at index near $m$. A local central limit theorem for the coefficients of the product of the $m$ factors below the pair is the natural tool.

Problem 24.9. Explain why the density ladder, if Table 24.15 is confirmed, reacts at low order to pairs at large angles, and find its scaling in $\theta$ and $m$. The question is how fast the advantage over the coefficient ladder disappears as $\theta \to 0$.

Problem 24.10. Enclose the constants $M_j$ of Proposition 24.5.4 rigorously, with a bound on the tail of the integral, completing the computer proof of Theorem 24.5.5, and close the window $(0, Y_*)$ by a rigorous saddle point bound.

## 24.20 Expansions of the reciprocal Xi function

For reference, here are the expansions of $1/F(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, of its law $\mu$, or of the associated canonical function, with the region of validity, the status and the hypotheses of each. Most are stated and proved elsewhere in the book, and only a pointer is given for those.

### Over the zeros

**Hadamard product.** $1/F(s) = \prod_{\operatorname{Im}\rho>0}(1 - s/s_\rho)^{-1}$, converging absolutely because $\sum |s_\rho|^{-1} < \infty$ (proved); it exhibits $\mu$ as a GGC exactly under RH (Section 1.2).

**Basepoint product and Lorentzian Thorin density.** Unconditional identities, with a positive Thorin density for $\alpha \geq 1$ (proved; Chapter 9).

**Residue expansion.** $f(y) = \sum_\rho e^{s_\rho y}/F'(s_\rho)$, unconditional when grouped between suitable circles, with an explicit truncation bound (Theorems 24.1.2 and 24.1.1, Proposition 24.5.4; proved, the constants $M_j$ computed).

**Kent's formula.** The same series read as the eigenfunction expansion of a Krein string hitting time (Proposition 24.2.1); conditional on RH and simple zeros.

**Mittag Leffler series.** $1/F(s) = \sum_\rho 1/(F'(s_\rho)(s - s_\rho))$ diverges, because $\xi$ is not in Krein's class (Proposition 24.2.2; proved).

**Canonical function over zeros.** $\Theta(x) = \sum_{\gamma>0} e^{s_\rho x}$; completely monotone exactly under RH (Section 1.2).

Over the primes

**Euler product as a Lévy measure.** Valid for $\alpha > 1$, where it gives a GGC after compensation; in the strip the compensator is infinite (proved; Chapter 9 and [250]).

**Truncated Euler products.** A GGC at every $\alpha > 0$, but the laws escape to infinity as $P \to \infty$ (Theorem 24.8.4).

**Möbius series.** A signed mixture of first passage times, converging for $\operatorname{Re} z > 1$ (Proposition 8.1.1); inside the strip its convergence is tied to RH.

**Explicit formula for $\Theta$.** Unconditional for every $x > 0$ (Section 1.2).

From Riemann's integral and theta functions

**Cosh transform.** $F(s) = \mathbb{E}\cosh(\sqrt{s}\,V)$, hence $1/F(s) \leq \mathbb{E}\operatorname{sech}(\sqrt{s}V)$, a scale mixture of Brownian exit time laws; unconditional and valid for real $s \geq 0$ (Theorem 16.6.1).

**Deformations.** Pólya's approximations and the de Bruijn Newman flow (Section 8.4); the deformed reciprocals are GGC transforms for $t \geq \Lambda$.

**Theta function and Brownian representations.** $\xi$ as the Mellin transform of a Jacobi theta law [49], giving the reciprocal through moment and Mellin identities rather than termwise series.

Power series

**Taylor series of $1/F$.** $1/F(s) = \sum(-1)^n m_n(\mu)s^n/n!$, with $m_n$ the moments of $\mu$ and radius of convergence $\gamma_1^2 = 199.79$, set by the first pole (computed: 199.79045).

**Cumulant series.** $\log F(s) = \sum_{n\geq 1}(-1)^{n+1}\kappa_n s^n/n!$ with $\kappa_n = (n-1)!\sum_{\mathrm{Im}\,\rho>0}(-s_\rho)^{-n}$, which is $(n-1)!\sum_{\gamma>0}\gamma^{-2n}$ under RH (proved).

**Taylor coefficients of $F$.** $F(s) = \sum c_n s^n$; $\mathrm{PF}_\infty$ exactly under RH and $\mathrm{PF}_k$ unconditionally for $k \leq \lfloor \pi H \rfloor - 1$ (Theorem 24.16.1 and Corollary 24.17.3), with the Turán inequalities [99], the higher order Turán inequalities [116] and the hyperbolicity of Jensen polynomials [159].

Asymptotic expansions

**Upper tail.** $f(y) = w_1 e^{-\gamma_1^2 y} + w_2 e^{-\gamma_2^2 y} + O(e^{-c_3 y})$ with $c_3 = \frac{1}{2}(\gamma_2^2 + \gamma_3^2)$ (Proposition 24.1.3; proved).

**Lower tail.** $\log f(y) = -\Lambda(y) - \frac{1}{2}\log(2\pi|\psi''(s_y)|) + o(1)$ as $y \to 0$, with $\Lambda$ the Legendre transform of $\psi = \log F$ (Table 24.3); heuristic, since a rigorous version with derivatives is open.

Of these, only the residue expansion in its grouped form is explicit term by term and valid throughout the critical strip without hypotheses. The prime side expansions converge only to the right of 1, or give $\Theta$ rather than the density; the Hadamard product, the Lorentzian sums and Kent's formula give positive objects only when the zeros cooperate.

# Chapter 25

# Extreme values, hitting times and a path to RH

The Pólya density whose cosine transform is the Riemann $\Xi$ function is an exact superposition of extreme value laws, and the reciprocal $\xi$ function is the Laplace transform of a sum of independent exponentials whose rates are the squared ordinates $\gamma_k^2$. Kent's theorem reads such a sum as a hitting time of a one dimensional diffusion, and Krein's inverse spectral theorem says that every strictly increasing sequence of positive rates arises this way. This chapter puts the two halves together. Extreme value theory supplies explicit approximations to the Pólya density, each approximation defines a candidate hitting time, and the question of real and simple zeros becomes the question of whether that hitting time belongs to a string with positive mass. The chapter asks whether this gives a path to RH. It does not prove RH. It gives a precise target, three routes to it with the place where each one stops, a family of approximations (the folded Gumbel kernels) whose zeros can be tracked rigorously to a stated height, and a set of diagnostics, from Hankel determinants of the kernel moments to threshold fits of the hitting time tail, whose power and limits are measured.

The chapter develops a source chapter on extreme values, hitting times and a path to RH by Nick Polson [231], together with the computations done for this book. Its numbers were recomputed here by scripts written from the mathematics alone (the scripts of the source were not used; the scripts of this chapter are `code/ev_g1_*.py`, `code/ev_evt_lib.py` and the other files `code/ev_*.py`, listed in `code/README.md`). Every number printed below is one of those recomputations unless it is marked unchecked, and where the recomputation differed from the source the recomputed value is printed. Statements of papers that were not read are marked unchecked in the text. The notation is the book's, with the following warnings. The kernel $\Phi$ of this chapter is the kernel $\Phi_{\mathrm{dBN}}$ of Section 8.4 (the normalisation of Rodgers and Tao, with $e^{9u}$ and $e^{4u}$), not the kernel $\Phi$ of (1.2.3); with it $H(z) = \int_0^\infty \Phi(u)\cos(zu)\,du = \frac{1}{8}\Xi(z/2)$, so the zeros of $H$ are at $z = \pm 2\gamma_k$ and the variable of the zeros is $x = z/2$. The eigenvalues of a string are $\lambda_k$, the time of the de Bruijn and Newman flow is $t$ as in Section 8.4 and $\Lambda$ is its constant,

and Euler's constant is $\gamma_E$.

Several parts of the source are already in the book and are cited, not restated: Kent's theorem and the reduction of RH to a string with simple spectrum (Proposition 24.2.1, Proposition 24.3.1, Section 8.1.3), the B spline form of the truncated hitting time and its relation to the Gumbel law (Section 24.7, Proposition 24.7.1), the hazard rate and the Prony extraction of the atoms (Sections 8.1.1 and 8.1.2), the Liouville operator and the explicit diffusion for Pólya's kernel (Proposition 8.7.7 and Theorem 8.7.10), the Stieltjes criterion for the logarithmic derivative (Theorem 14.6.1), the incomplete gamma form of the truncated theta series, the kink at the origin and the tracking of zeros by Rouché's theorem (Proposition 22.1.3, Theorem 22.2.3, Corollary 22.2.2) and the de Bruijn and Newman flow (Section 8.4). What is new here is the decomposition of the Pólya density into Gumbel pieces, the sharp cutoff (folded) family and its height rule, the Hankel string test, the shifted Bessel strings, the detection of the first nonreal pair and the diagnostics of the hitting time tail.

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it, certified when it was computed in ball arithmetic and the sentence says what the certificate covers, conditional, heuristic or open as stated. A computation in ordinary floating point or on a sampled set of points is called computed and never certified. The results are these.

1. Each term of the theta series for $\Phi$ is an explicit combination of two Gumbel kernels of fractional order, and the Mellin transform of the term is $n^{-s}\xi(s)/(4\zeta(s))$ (Theorem 25.1.1, proved). This gives three families of approximations: the folded Gumbel kernels, Pólya's two sided kernel and the Dyck chain. Their real zeros, to the heights tabulated, are computed (Table 25.1). The height $T_N \approx 4(N+1)^2$ at which the folded family fails is a heuristic balance of the kink against the size of $H$.

2. The zeros of $\Xi$ are all real and simple if and only if the reciprocal $\xi$ law is the hitting time of a string (Corollary 25.3.1, proved, with the hypotheses of simplicity and of the class of $\widehat{H}$ that the source left out). The hitting time tail gives the principal eigenvalue $\gamma_1^2$ without any zero, and a Laplace inversion of the tail recovers $\gamma_1$ to eighteen digits (Proposition 25.3.2, proved under RH, simple zeros and the lower bound $|\xi'(\rho)| \geq e^{-C|\gamma|}$ for the full series, the exponential excess needing only the two term tail; the digits are computed).

3. The Gumbel pieces have unconditional Hadamard products whose atoms are the poles of the archimedean factor, while the atoms $\gamma_k^2$ come from the superposition, which is a sum and not a product (Section 25.4, proved for the pieces and heuristic for the superposition).

4. Pólya's kernel has an explicit string, the Liouville string of Biane, and its

first eigenvalues are computed (Section 25.5).

5. For any candidate kernel with infinitely many zeros the zeros are all real if and only if the Hankel determinants of the moment sequence $\mu_j = \sum_k \widehat{\gamma}_k^{-2j}$ are all positive (Proposition 25.6.1, proved). The first eleven pairs of determinants are positive for the exact kernel, the two term folded kernel and Pólya's kernel (computed). Finite order positivity cannot see an off line pair above a height that grows with the order (heuristic).

6. There are three routes to RH from this material; each stops at a known place, and the folded family gives a reduction of RH to a height rule for all $N$ (Section 25.7, Proposition 25.7.1, proved). The Rouché inequality behind the height rule is certified in ball arithmetic for $N = 1, 2, 3, 4$ and the heights of the table, given the count of zeros of $\Xi$ below the height (Section 25.8); the extension to $N = 5$ is computed.

7. The eigenvalue problem for the shifted Bessel kernels is a rational recurrence, and its eigenvalues are real (proved), but the positivity of the extension and its reading as a string are not shown for the tabulated cases; the cotangent law for fractional shifts, the collision of the de Bruijn and Newman zeros and the masking of the first nonreal pair by finite Hankel orders are treated in Sections 25.8 and 25.9 with the status stated there.

8. Devroye's alternating series gives a sign test for the reciprocal $\Xi$ law, the mean excess function of the hitting time has an expansion with a rate, and a detection rule for the first nonreal pair in the de Bruijn and Newman family is stated and tested (Section 25.10, computed unless stated).

The remaining open problems close the chapter.

## 25.1 The Pólya density as an extreme value law

Write

$$\Phi(u) = \sum_{n\geq 1} \phi_n(u), \qquad \phi_n(u) = \left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right) e^{-\pi n^2 e^{4u}},$$

so that $H(z) = \int_0^\infty \Phi(u)\cos(zu)\,du = \frac{1}{8}\Xi(z/2)$ (see the notation paragraph of the introduction; the identity was checked numerically to $10^{-88}$ at $z = 0$ and to $10^{-60}$ at $z = 150 + 150i$). Let $g_k(x) = e^{-kx}\exp(-e^{-x})/\Gamma(k)$, the density of $-\log G_k$ for a gamma variable $G_k$ of shape $k$. For integer $k$ this is the law of the $k$th largest point of the Gumbel limit process with intensity $e^{-x}dx$, and for real $k$ it is a fractional order statistic in the sense of Stigler [275]. The maximum of $n$ independent exponentials and the Gumbel law are already in the book (Chapter 14, Proposition 14.2.1, and Section 24.7); what follows decomposes the individual terms of the theta series.

THEOREM 25.1.1 (Gumbel decomposition). *With $x = -4u - \log(\pi n^2)$, as densities in $x$,*

$$\phi_n(u)\,du = \pi^{-1/4} n^{-1/2} \Gamma(\tfrac{5}{4}) \left[ \tfrac{5}{8} g_{9/4}(x) - \tfrac{3}{4} g_{5/4}(x) \right] dx.$$

*The signed masses of the terms are $-\frac{1}{8}\pi^{-1/4}\Gamma(\frac{5}{4})n^{-1/2}$, and the formal sum of the masses, regularised by $\zeta$, is*

$$-\tfrac{1}{8}\pi^{-1/4}\Gamma(\tfrac{5}{4})\zeta(\tfrac{1}{2}) = \tfrac{1}{4}\xi(\tfrac{1}{2}) = \int_{\mathbb{R}} \Phi.$$

*With $s = (1+iz)/2$,*

$$\int_{\mathbb{R}} \phi_n(u) e^{izu} du = \tfrac{1}{4} n^{-s} \xi(s)/\zeta(s).$$

*Proof.* Put $y = e^{-x} = \pi n^2 e^{4u}$, so that $e^{9u} = (y/\pi n^2)^{9/4}$, $e^{5u} = (y/\pi n^2)^{5/4}$ and $du = -dx/4$. The coefficients become $2\pi^2 n^4 (\pi n^2)^{-9/4} = 2\pi^{-1/4} n^{-1/2}$ and $3\pi n^2 (\pi n^2)^{-5/4} = 3\pi^{-1/4} n^{-1/2}$, and $\Gamma(\frac{9}{4}) = \frac{5}{4}\Gamma(\frac{5}{4})$ turns $\frac{1}{4}\cdot 2\Gamma(\frac{9}{4})$ into $\frac{5}{8}\Gamma(\frac{5}{4})$. The masses are $\frac{5}{8} - \frac{3}{4} = -\frac{1}{8}$ times the common factor. The $\zeta$ regularised sum is a formal identity (the series $\sum n^{-1/2}$ diverges), and it agrees with $\int_{\mathbb{R}} \Phi = \xi(\frac{1}{2})/4$ because $\Gamma(\frac{5}{4}) = \frac{1}{4}\Gamma(\frac{1}{4})$ and $\xi(\frac{1}{2}) = -\frac{1}{8}\pi^{-1/4}\Gamma(\frac{1}{4})\zeta(\frac{1}{2})$. For the transform, the same substitution gives $(\pi n^2)^{-iz/4}$ times a combination of $\Gamma(w)$ at $w = (9+iz)/4$ and $(5+iz)/4$, which equals $n^{-s}\pi^{-s/2} s(s-1)\Gamma(s/2)/8$. □

The Dirichlet series of $\zeta$ is therefore a sum over translates of one Gumbel kernel along the lattice $\frac{1}{2}\log n$. The tail of the normalised law $p = 4\Phi/\xi(\frac{1}{2})$ is $S(u) \sim (2\pi/\xi(\frac{1}{2}))e^{5u}\exp(-\pi e^{4u})$ (proved, one integration), so $p$ is in the Gumbel domain of attraction of the theorem of Fisher, Tippett and Gnedenko (Proposition 14.10.1). That the maximum of $N$ independent copies converges at the slow rate $1/\log N$ is a heuristic by analogy with the normal law and is not used. Three families of approximations follow.

- *Folded Gumbel.* $\Phi_N^F(u) = \sum_{n \le N} \phi_n(|u|)$. The sharp cutoff of the theta series at $N$ is the omitted set $\{n > N\}$ of Chapter 22 (Section 22.2), and the cosine transform of $\Phi_N^F$ is

$$H_N^F(z) = \tfrac{1}{2}\left[F_N(z) + F_N(-z)\right],$$
$$F_N(z) = \sum_{n \le N} \left[ \tfrac{1}{2} a_n^{2-w_9} \Gamma(w_9, a_n) - \tfrac{3}{4} a_n^{1-w_5} \Gamma(w_5, a_n) \right],$$

  where $a_n = \pi n^2$, $w_j = (j+iz)/4$ and $\Gamma(w,a)$ is the upper incomplete gamma function. For real $z$ this is $\operatorname{Re} F_N(z)$, which is how it is usually written, but off the real axis only the symmetrised form is the transform of $\Phi_N^F$.

- *Two sided Gumbel.* Pólya's kernel of Section 8.7 (equation (8.7.1)),

$$\Phi^P(u) = (4\pi^2 \cosh 9u - 6\pi \cosh 5u) e^{-2\pi \cosh 4u}.$$

  Here $e^{-2\pi \cosh 4u} = \mathbb{P}(V \le u < W)$ for independent variables $V$ and $W$ with $\mathbb{P}(V \le u) = \exp(-\pi e^{-4u})$ and $\mathbb{P}(W > u) = \exp(-\pi e^{4u})$, a Gumbel maximum and a Gumbel minimum, and for real $z$

$$H^P(z) = \operatorname{Re}\left[\pi^2 K_{(9+iz)/4}(2\pi) - \tfrac{3\pi}{2} K_{(5+iz)/4}(2\pi)\right].$$

- *Dyck chain.* With $H_n$ the height of a uniform Dyck path of length $2n$ and

$$Y_n = \sqrt{2/\pi}\, H_n/\sqrt{2n},$$

  the Dirichlet polynomial $F_n(s) = \mathbb{E}[Y_n^s]$ tends to $2\xi(s)$ on $\operatorname{Re} s > 0$, because $\mathbb{E}[Y^s] = 2\xi(s)$ for $Y = \sqrt{2/\pi}$ times the maximum of a Brownian excursion (Biane, Pitman and Yor [49]; this is the clock variable of Chapter 14) and the Dyck height scales to the excursion maximum (the height law of Dyck paths is in de Bruijn, Knuth and Rice [106]).

By [49] the normalised law $p$ is the law of $\frac{1}{2}\log Y$ under the measure with density proportional to $Y^{1/2}$: indeed $\mathbb{E}[Y^{1/2}Y^{iz/2}]/\mathbb{E}[Y^{1/2}] = \xi(s)/\xi(\frac{1}{2})$. The Dyck chain is Markov in the sense of the models of Smith, Tawn and Coles [270], whose subject is the extremes of stationary Markov chains, and its extremes arrive in clusters rather than as independent exceedances; the analogy with the chains of [270] is ours, not theirs. Table 25.1 lists the zeros of the approximations, recomputed for this book. The zero counts are sign changes of the real function $H_N^F(2x)$ (the variable $x = z/2$ of the introduction), with signs certified on a grid of mesh 0.05.

| approximation | real zeros | accuracy |
|---|---|---|
| folded, $N = 2$ | 7 below $x = 70$ | $3.9 \cdot 10^{-5}$ to $\gamma_3$ |
| folded, $N = 3$ | 15 below $x = 100$ | $9.6 \cdot 10^{-6}$ to $\gamma_{11}$ |
| folded, $N = 4$ | 31 below $x = 140$ | $5.8 \cdot 10^{-6}$ to $\gamma_{25}$ |
| two sided (Pólya) | all real | first zero $x = 7.604$ against $\gamma_1 = 14.135$ |
| Dyck chain, $n = 12800$ | zero near $\gamma_1$ off the line | $0.4367 + 14.0457i$ against $0.5 + 14.1347i$ |
| Dyck chain, Richardson in $n^{-1/2}$ | off the line, to $10^{-3}$ | $0.50003 + 14.13473i$ (two steps) |

Table 25.1: Zeros of the extreme value approximations (computed). For the Dyck chain the real parts of the zero near $\gamma_1$ for $n = 800, 1600, 3200, 6400, 12800, 25600$ are $0.2438, 0.3196, 0.3729, 0.4103, 0.4367, 0.4553$, and the extrapolation uses $n = 3200, 6400, 12800$.

The folded family is accurate up to a height $T_N \approx 4(N+1)^2$ in $x$ and then fails. The mechanism is a kink at the origin. Write $R_N = \Phi - \Phi_N^F = \sum_{n>N} \phi_n$ on $u \geq 0$ and $\kappa_N = |R_N'(0)|$, the one sided slope. Its Fourier tail is of order $\kappa_N/z^2$, which overtakes $H(z) \approx |z|^{7/4} e^{-\pi z/8}$ when $x \approx 4(N+1)^2$ (heuristic: a balance of leading orders). The slope is $\kappa_N = 8.27 \cdot 10^{-8}$, $1.39 \cdot 10^{-16}$ and $2.87 \cdot 10^{-28}$ for $N = 2, 3, 4$ (computed); the leading order formula $8\pi^3 (N+1)^6 e^{-\pi(N+1)^2}$ overstates it by 15%, 8% and 5%. The predicted heights are 36, 64 and 100. Measured by the first error above $10^{-2}$ in a tracked zero the breakdown is at $\gamma_5 = 32.9$, $\gamma_{15} = 65.1$ and $\gamma_{30} = 101.3$, and measured by the last real zero tracked it is at $x = 39.5$, 65.0 and 103.7 (computed); with error thresholds from $10^{-3}$ to $10^{-1}$ the heights lie in 30 to 38, 59 to 65 and 96 to 104, so the rule is a statement about the order of magnitude and the leading constant, not a sharp threshold.

For the census of complex zeros below, the exact law of the height $H_n$ of the Dyck chain is used: by the reflection principle the number of Dyck paths of length $2n$ and height at most $h$ is $\sum_{k\in\mathbb{Z}} \left[ \binom{2n}{n+k(h+2)} - \binom{2n}{n-1+k(h+2)} \right]$, out of $\binom{2n}{n}/(n+1)$. At $n = 25600$, with these probabilities as balls at 256 bits, the argument principle on the boundary of $-4 < \operatorname{Re} s < 6$, $0 < \operatorname{Im} s < 60$ counts 13 zeros of the Dirichlet polynomial $F_n$ of the Dyck chain, in the bands $0 < \operatorname{Im} s < 20$, $20 < \operatorname{Im} s < 40$ and $40 < \operatorname{Im} s < 60$ the numbers 1, 5 and 7 that the ordinates $\gamma_k$ have (the number of $\gamma_k$ below 60 is 13). There is no zero with $\frac{1}{2} < \operatorname{Re} s < 6$ and $0 < \operatorname{Im} s < 60$, and the real parts of the 13 zeros run from 0.0907 (the highest, at height 59.126) to 0.4553 (the lowest, at height 14.072). Followed in $n$ from 25600 through 12800, 6400 and 3200 down to 1600, with Newton's method started at the zero found for the next larger $n$, these 13 zeros stay distinct and to the left of the line, and the zero of lowest height lies at distance 0.3082, 0.1544 and 0.0773 from $\frac{1}{2} + i\gamma_1$ for $n = 1600$, 6400 and 25600 (computed, `px_dyck.py`). The count is for $n = 25600$ only: at $n = 1600$ the same contour holds 14 zeros, and Newton's method started at $\frac{1}{2} + i\gamma_k$ does not stay near it for $k = 4, \ldots, 12$ (the starts $k = 7$ and 8 converge to one zero), so the zeros are identified by this following in $n$. This paragraph follows a note of Nicholas Polson merged into the book [232].

The side from which these zeros approach is set by the lattice. By de Bruijn, Knuth and Rice [106], $\mathbb{E}H_n = \sqrt{\pi n} - \frac{3}{2} + o(1)$, while $\sqrt{2n}$ times the mean of the excursion maximum $\sqrt{\pi/2}\, Y$ is $\sqrt{\pi n}$; so to first order $H_n/\sqrt{2n}$ is that maximum moved down by

$$\delta_n = \frac{3}{2\sqrt{2n}}.$$

Heuristically, the reflection formula above makes $\mathbb{P}(H_n \leq h)$ close to $\mathbb{P}(\sqrt{\pi n}\, Y \leq h+2)$, so that $H_n$ behaves like $\sqrt{\pi n}\, Y - 2$ rounded up, and rounding up adds $\frac{1}{2}$ on average (heuristic). For the moments this reads as follows; since $\mathbb{E}Y^{s-1} = 2\xi(s-1)$, it says that $F_n(s)$ is $\mathbb{E}(Y - \sqrt{2/\pi}\, \delta_n)^s$ to first order.

CONJECTURE 25.1.2. *Uniformly on every compact subset of* $\operatorname{Re} s > 0$,

$$F_n(s) = 2\xi(s) - \frac{3s}{\sqrt{\pi n}}\,\xi(s-1) + O\Big(\frac{1}{n}\Big).$$

At $s = 1$ the conjecture says $\mathbb{E}H_n = \sqrt{\pi n} - \frac{3}{2} + O(n^{-1/2})$, which agrees with de Bruijn, Knuth and Rice, and it implies that $F_n \to 2\xi$ locally uniformly on $\operatorname{Re} s > 0$. Status: open; the zeros below agree with its consequence, Proposition 25.1.3 (computed).

PROPOSITION 25.1.3. *Assume Conjecture 25.1.2. Let $\rho$ be a simple zero of $\xi$ and put*

$$c_{\mathrm{D}}(\rho) = \sqrt{\frac{2}{\pi}}\,\frac{\rho\,\xi(\rho-1)}{\xi'(\rho)} = \sqrt{\frac{2}{\pi}}\,\frac{\rho\,\xi(2-\rho)}{\xi'(\rho)}.$$

*There is a constant $C$ such that for all large $n$, $F_n$ has exactly one zero $z_n(\rho)$ in the disc $|s - \rho - \delta_n c_{\mathrm{D}}(\rho)| < C/n$; thus $z_n(\rho) = \rho + \delta_n c_{\mathrm{D}}(\rho) + O(1/n)$. Status: conditional.*

*Proof.* Put $g(s) = s\,\xi(s-1)$. On a closed disc about $\rho$ in $\operatorname{Re} s > 0$, the conjecture and Taylor's formula give $F_n(s) = 2\xi'(\rho)(s-\rho) - \frac{3}{\sqrt{\pi n}}\,g(\rho) + O\big(|s-\rho|^2 + n^{-1/2}|s-\rho| + n^{-1}\big)$. Since $\frac{3g(\rho)}{2\sqrt{\pi n}\,\xi'(\rho)} = \delta_n c_{\mathrm{D}}(\rho)$, on the circle $|s-\rho-\delta_n c_{\mathrm{D}}(\rho)| = C/n$, where $|s-\rho| = O(n^{-1/2})$, this is $2\xi'(\rho)\big(s - \rho - \delta_n c_{\mathrm{D}}(\rho)\big) + O(1/n)$. The linear term has modulus $2|\xi'(\rho)|C/n$, which exceeds the error once $C$ is large, and Rouché's theorem gives exactly one zero inside, as in Proposition 7.2.2. The second form of $c_{\mathrm{D}}$ is the functional equation. □

PROPOSITION 25.1.4. *Let $\rho = \frac{1}{2} + i\gamma$, $\gamma > 0$, be a simple zero on the critical line, with $\theta$ and $Z$ as in Theorem 7.2.3. Then*

$$c_{\mathrm{D}}(\rho) = -\frac{C_{\mathrm{D}}(\gamma)\,e^{-i\Theta(\gamma)}}{Z'(\gamma)}, \qquad C_{\mathrm{D}}(\gamma) = \frac{\sqrt{2}}{\pi}\Big|\frac{3}{2} - i\gamma\Big|\,\Big|\frac{\Gamma(\frac{3}{4} + \frac{i\gamma}{2})}{\Gamma(\frac{1}{4} + \frac{i\gamma}{2})}\Big|\,\big|\zeta(\tfrac{3}{2} + i\gamma)\big| > 0,$$

$$\Theta(\gamma) = \theta(\gamma) + \arctan\tanh\frac{\pi\gamma}{2} + \arg\zeta(\tfrac{3}{2} + i\gamma) - \arctan\frac{3}{2\gamma},$$

*so that $\operatorname{Re} c_{\mathrm{D}}(\rho)$ has the sign of $-\cos\Theta(\gamma)/Z'(\gamma)$. Status: proved.*

*Proof.* Since $\Xi(t) = \xi(\frac{1}{2} + it) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ and $Z(\gamma) = 0$, $\xi'(\rho) = -i\,\Xi'(\gamma)$ with $\Xi'(\gamma) = -\frac{1}{2}(\gamma^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|Z'(\gamma)$, and $\xi(2-\rho) = \xi(\frac{3}{2} - i\gamma) = \overline{\xi(\frac{3}{2} + i\gamma)}$. By the definition of $\theta$, $\pi^{-i\gamma/2}\Gamma(\frac{3}{4} + \frac{i\gamma}{2}) = |\Gamma(\frac{3}{4} + \frac{i\gamma}{2})|e^{i(\theta(\gamma)+\Delta(\gamma))}$ with $\Delta(\gamma) = \arg\Gamma(\frac{3}{4} + \frac{i\gamma}{2}) - \arg\Gamma(\frac{1}{4} + \frac{i\gamma}{2})$, and the reflection formula $\Gamma(\frac{1}{4} + it)\Gamma(\frac{3}{4} - it) = \pi/\sin(\frac{\pi}{4} + i\pi t)$ gives $\Delta(\gamma) = \arg\sin(\frac{\pi}{4} + \frac{i\pi\gamma}{2}) = \arctan\tanh\frac{\pi\gamma}{2}$. Substituting, $\rho(\frac{1}{2} - i\gamma) = \gamma^2 + \frac{1}{4}$ cancels the factor of $\Xi'$, and

$$c_{\mathrm{D}}(\rho) = -i\,\frac{\sqrt{2}}{\pi}\Big(\frac{3}{2} - i\gamma\Big)\Big|\frac{\Gamma(\frac{3}{4} + \frac{i\gamma}{2})}{\Gamma(\frac{1}{4} + \frac{i\gamma}{2})}\Big|\,e^{-i(\theta+\Delta)}\,\frac{\overline{\zeta(\frac{3}{2} + i\gamma)}}{Z'(\gamma)};$$

writing $\frac{3}{2} - i\gamma = |\frac{3}{2} - i\gamma|\, e^{-i(\pi/2 - \arctan(3/2\gamma))}$ gives the statement. □

With $S_{\mathrm{mid}}$ as after Theorem 7.2.3, $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S_{\mathrm{mid}})$ and $Z'(\gamma_n)$ has the sign $(-1)^{n+1}$, so $\operatorname{Re} c_{\mathrm{D}}(\rho_n) < 0$ exactly when $\sin\big(\Theta(\gamma_n) - \theta(\gamma_n) - \pi S_{\mathrm{mid}}\big) > 0$. Here $\Theta - \theta = \frac{\pi}{4} + \pi a_{\mathrm{D}}(\gamma)$, and since $|\arctan\tanh\frac{\pi\gamma}{2} - \frac{\pi}{4}| \le 2e^{-\pi\gamma}$ and, for the branch given by the Euler product, $|\arg\zeta(\frac{3}{2} + i\gamma)| \le \log\zeta(\frac{3}{2}) = 0.9603$, $|a_{\mathrm{D}}(\gamma)| \le \big(0.9603 + \frac{3}{2\gamma} + 2e^{-\pi\gamma}\big)/\pi$, below 0.35 for $\gamma \ge 14$. So $\operatorname{Re} c_{\mathrm{D}}(\rho_n) < 0$ when $-\frac{3}{4} + a_{\mathrm{D}} < S_{\mathrm{mid}} < \frac{1}{4} + a_{\mathrm{D}}$ and $\operatorname{Re} c_{\mathrm{D}}(\rho_n) > 0$ when $\frac{1}{4} + a_{\mathrm{D}} < S_{\mathrm{mid}} < \frac{5}{4} + a_{\mathrm{D}}$, modulo 2 (proved), and under Conjecture 25.1.2 the zero is then approached from the left, respectively from the right (Proposition 25.1.3): a Gram parity, as for the lattice of Section 7.2.1, with the window $(-\frac{3}{4}, \frac{1}{4})$ in place of $(-\frac{1}{2}, \frac{1}{2})$ and an arithmetic correction about three times as large. The first zero with $\operatorname{Re} c_{\mathrm{D}}(\rho_k) > 0$ is $\rho_{64}$, against $\rho_{213}$ there (computed in ball arithmetic, below).

In ball arithmetic, from the first 200 zeros of the stored list, $\xi'(\rho_k) \ne 0$ for every $k \le 200$, $\operatorname{Re} c_{\mathrm{D}}(\rho_k) < 0$ for every $k \le 63$, $c_{\mathrm{D}}(\rho_1) = -6.72018 - 9.54303i$, $c_{\mathrm{D}}(\rho_{64}) = 53.08757 - 266.82638i$ (radius below $10^{-55}$), and among $k \le 200$, $\operatorname{Re} c_{\mathrm{D}}(\rho_k) > 0$ exactly for $k = 64$, 92, 115, 126, 136, 160, 165, 183 and 187 (computed, `lb_c.py`). At $n = 6400$ and 25600, Newton's method started at $\rho_k + \delta_n c_{\mathrm{D}}(\rho_k)$, $k \le 13$, converges to the zeros of the census above, each alone in a square of half side 0.25 by the argument principle, and the distance $|z_n(\rho_k) - \rho_k - \delta_n c_{\mathrm{D}}(\rho_k)|$ falls by a factor between 3.77 and 4.64 from $n = 6400$ to $n = 25600$, for 11 of the 13 zeros between 3.94 and 4.14, as a remainder $O(1/n)$ predicts; for $\rho_1$ at $n = 25600$ the law gives the real part $\frac{1}{2} - 0.04455$ against $\frac{1}{2} - 0.04470$. The zero of $F_n$ near $\rho_{64}$, followed from $n = 10^8$ down to $10^5$, lies to the right of the line at every computed $n$: $\operatorname{Re} z_n(\rho_{64}) - \frac{1}{2}$ is 0.1615, 0.1348, 0.0837, 0.0456, 0.0223, 0.0118 and 0.0061 at $n = 10^5$, $3 \cdot 10^5$, $10^6$, $3 \cdot 10^6$, $10^7$, $3 \cdot 10^7$ and $10^8$, against the first order values $\delta_n \operatorname{Re} c_{\mathrm{D}}(\rho_{64})$ from 0.1781 down to 0.00563, and for $10^6 \le n \le 10^8$ the product $n\,|z_n(\rho_{64}) - \rho_{64} - \delta_n c_{\mathrm{D}}(\rho_{64})|$ lies between $6.33 \cdot 10^4$ and $6.61 \cdot 10^4$. The first order regime needs $\delta_n |c_{\mathrm{D}}(\rho_{64})|$ small: $|c_{\mathrm{D}}(\rho_{64})| = 272.1$, so this is 0.91 at $n = 10^5$ and 0.029 at $n = 10^8$. The neighbours $\rho_{63}$ and $\rho_{65}$ stay to the left, with real parts $\frac{1}{2} - 0.0283$ and $\frac{1}{2} - 0.0227$ at $n = 10^8$ (computed, `lb_zero.py`; Newton's method on ball enclosures of the exact law, with the terms below $10^{-110}$ dropped and not bounded, checked against a second law with another cut; uniqueness by the argument principle, sampled). This part of the section follows a note of Nicholas Polson merged into the book [239].

## 25.2 Smith and Tawn in pictures

The viewpoint of Smith and Tawn enters the chapter in three places. Figure 25.1 shows the second and the third from the data stored for this book; the first has no stored data and is described in words, and the statistical estimates mentioned for it are the source's.

The first is the model of Smith, Tawn and Coles [270]: a first order Markov chain in Gumbel margins whose transitions come from the bivariate logistic extreme value law with $\alpha = \frac{1}{2}$. It is simulated by inverting the conditional distribution

$$F(y \mid x) = \exp\bigl(1/x - (x^{-1/\alpha} + y^{-1/\alpha})^{\alpha}\bigr)\bigl(1 + (x/y)^{1/\alpha}\bigr)^{\alpha-1}$$

in unit Fréchet margins (the formula was rederived from the logistic law and is correct). Exceedances of a high threshold arrive in clusters rather than one at a time. The source reports a runs estimate of the extremal index from the plotted series of about 0.38 and a value of about 0.33 from simulating the tail chain from a very high level, so that the maximum of $N$ steps behaves like $F^{0.33N}$.

The second is the Markov chain behind the Pólya law, a uniform Dyck path rescaled; the left panel of the figure follows the zero of its Dirichlet polynomial $F_n$ near $\frac{1}{2} + i\gamma_1$ for $n = 100$ to 12800 (Table 25.1). The Pólya variable is the law of $\frac{1}{2}\log Y$ under the $Y^{1/2}$ biased measure (above), and the visits within two lattice steps of the maximum form one cluster. The extremes are not a limit over a long series here: the whole law is a cluster functional, which is why the Pólya density is a theta function and not a generalised extreme value law.

The third is the threshold choice problem of Smith and Tawn transferred to the Thorin variable. The right panel of the figure shows its simplest form, the second atom estimated after deflating the first against the threshold $x$, which approaches $\gamma_2$ as the threshold rises (Section 25.3). Prony fits of $K = 6, 8, 10$ exponentials to the tail $\mathbb{P}(T > x)$ on windows starting at a threshold $x_0$ treat the same problem jointly. Raising the threshold removes the bias from the omitted atoms, raising $K$ lets one use a lower threshold, and the two together take $\gamma_1$ to $10^{-39}$ (Section 25.9).

## 25.3 Hitting times and the Thorin variable

Kent's theorem and Krein's converse are in the book (Proposition 24.2.1, Section 8.1.3; Kent [305], Dym and McKean [124], and for generalised diffusions Kotani and Watanabe [206] and Knight [205]). We use them in the following form, which differs from the source in two points. A string with a reflecting end and a Dirichlet end has simple spectrum, so the converse holds for strictly increasing sequences $0 < \lambda_1 < \lambda_2 < \cdots$ with $\sum_k \lambda_k^{-1} < \infty$ only. And the string is not unique: the hitting time law fixes the eigenvalues and not the norming constants (a string with two point masses has three parameters modulo the natural scale and two eigenvalues, so a one parameter family has the same hitting time law). The spectral measure, eigenvalues together with weights, is what determines the string by Krein's algorithm.

Let $\widehat{H}$ be an even real entire function of order less than two with $\widehat{H}(0) > 0$, and put

$$\widehat{L}(s) = \frac{\widehat{H}(0)}{\widehat{H}(2i\sqrt{s})},$$

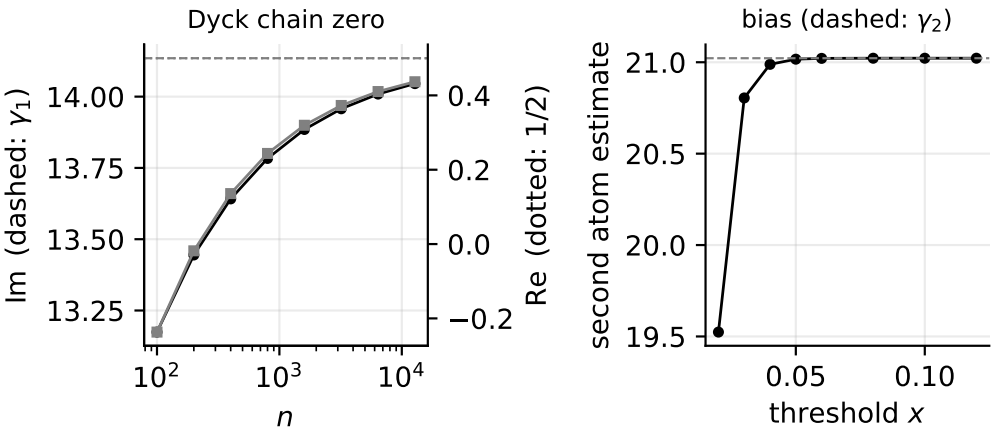



Figure 25.1: Left: real and imaginary part of the zero of the Dirichlet polynomial of the Dyck chain near $\gamma_1$ for $n = 100$ to 12800 (dashed: $\gamma_1$, dotted: the real part $\frac{1}{2}$). Right: estimate of the second atom of the hitting time law after deflating the first, against the threshold $x$ (dashed: $\gamma_2$). Drawn from the stored data of `code/ev_g1_tail.py` by `code/ev_figs.py`; the panel of the source with the logistic Markov chain and its exceedance clusters has no stored data and is not drawn.

which for $\widehat{H} = H$ is $L(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s}) = \prod_k (1 + s/\gamma_k^2)^{-1}$. Because the order is below two, $\widehat{H}(2i\sqrt{s})$ is of order below one in $s$, so it is the canonical product of genus zero over its zeros. Here $\pm\gamma_k$ are the zeros of $\Xi$, so that $\frac{1}{2} + i\gamma_k$ are the nontrivial zeros of $\zeta$: the $\gamma_k$ are the ordinates (imaginary parts of the zeros of $\zeta$), and they are real exactly when the zeros lie on the critical line, in which case $0 < \gamma_1 \le \gamma_2 \le \cdots$.

Corollary 25.3.1. *Write $\pm\gamma_k$ for the zeros of $\Xi$, ordered by real part, so that $\frac{1}{2} + i\gamma_k$ are the nontrivial zeros of $\zeta$ (the $\gamma_k$ are ordinates, real under RH). The following are equivalent.*

1. *All zeros of $\Xi$ are real (RH).*
2. *$L$ is the Laplace transform of a generalised gamma convolution whose Thorin measure $\sum_k \delta_{\gamma_k^2}$ is positive (Polson [421], Proposition 1.2.1).*

*Moreover, all zeros of $\Xi$ are real and simple if and only if there is a Krein string with positive mass whose hitting time $T$ satisfies $\mathbb{E}e^{-sT} = L(s)$. The same statements hold with $\widehat{H}$ and $\widehat{L}$ in place of $H$ and $L$.*

*Proof.* The equivalence of (1) and (2) is Proposition 1.2.1 and Theorem 14.6.1. If the zeros are real and simple, the $\gamma_k^2$ are strictly increasing and Krein's converse

gives a string. If a string exists, its spectrum is real, positive and simple, and by Kent's theorem $\mathbb{E}e^{-sT} = \prod_k (1 + s/\lambda_k)^{-1}$; the zeros of $\widehat{H}(2i\sqrt{s})$ in $s$ are then the $-\lambda_k$, so the zeros of $\widehat{H}$ are real and simple. The same argument covers $\widehat{H}$. □

So a hitting time computation shows that the zeros are real and simple exactly when the string can be exhibited (the book states the string version only with simple zeros, Proposition 24.2.1; simplicity of the zeros of $\Xi$ is not known). Two features of the string are visible without exhibiting it.

Proposition 25.3.2 (Principal eigenvalue from the tail). *If RH holds, the zeros are simple and $|\xi'(\rho)| \geq e^{-C|\gamma|}$ for some constant $C$ and all zeros $\rho = \frac{1}{2} + i\gamma$ (the hypothesis of Theorem 24.1.2), then $\mathbb{P}(T > x) = \sum_k c_k e^{-\gamma_k^2 x}$ with $c_k = \prod_{j \neq k} (1 - \gamma_k^2/\gamma_j^2)^{-1} = 2i\xi(\frac{1}{2})/(\gamma_k \xi'(\frac{1}{2} + i\gamma_k))$. In particular the excess of $T$ over a high threshold is asymptotically exponential with rate $\lambda_1 = \gamma_1^2$, the bottom of the spectrum of the string; this last clause needs only the two term tail of Proposition 24.1.3, which has no hypothesis on $\xi'$.*

This is Proposition 24.1.3 and the hazard rate of Section 8.1.1 in the present notation. The exponential excess follows directly from the expansion; the statement is analogous to the peaks over threshold theorem of Pickands and of Balkema and de Haan [249, 19], which is not used. The estimator is the hazard rate. At a threshold $x$ the ratio of the density to the tail, $f(x)/\mathbb{P}(T > x)$, tends to $\gamma_1^2$, and the Laplace inversion of $L$ (the method of Talbot, at 100 and 130 nodes, agreeing to 20 digits) uses no zero. On the window $x \in [0.18, 0.20]$ the square root of this ratio is $14.1347251417346937\ldots$ at all three of $x = 0.18, 0.19, 0.20$ (the values are $14.13472514173469373756$, $\ldots 3785760$ and $\ldots 3790040$ against $\gamma_1 = 14.13472514173469379045\ldots$), so $\gamma_1$ is recovered to eighteen significant digits (computed). The threshold bias for the later atoms is the point of Smith and Tawn: deflating by the first atom, the second atom is estimated as 21.0167, 21.0212 and 21.02204 from the thresholds $x = 0.05$, 0.06 and 0.1, against $\gamma_2 = 21.02204$ (computed; the source printed values for the third and fourth atoms that were not reproduced and are not repeated). Figure 25.2 shows the error of these estimates of $\gamma_1$.

## 25.4 B splines, Hadamard products and Thorin

Is there a B spline view of the Gumbel law that leads to a Hadamard product and from there to a Thorin representation? The first two steps are in the book. By Rényi's representation the maximum of $n$ independent exponentials is $M_n = \sum_{k \leq n} E_k/k$, a generalised gamma convolution with Thorin measure $\sum_{k \leq n} \delta_k$ and the passage time of a pure death chain; $M_n - \log n$ tends to the Gumbel law $G$, with $\mathbb{E}e^{-sG} = \Gamma(1+s) = \exp\big(-\gamma_E s - \sum_{k \geq 1} [\log(1 + s/k) - s/k]\big)$, a compensated Thorin representation with atoms at the integers (the compensation is needed

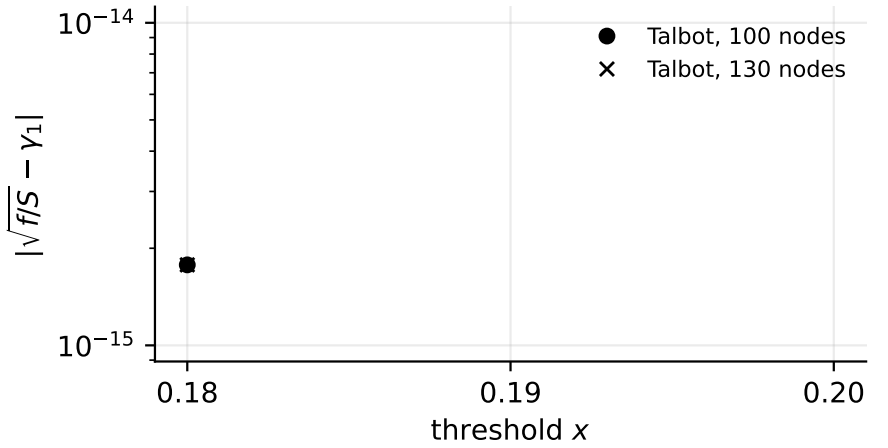



Figure 25.2: Error of the estimate $\sqrt{f(x)/\mathbb{P}(T > x)}$ of $\gamma_1$ from the Laplace inversion of $L$ (Talbot, 100 and 130 nodes) at $x = 0.18, 0.19, 0.20$. Drawn from the stored data of `code/ev_g1_tail.py` by `code/ev_figs.py`; the curve of the tail with the three atom expansion in the source is not drawn, since only these estimates are stored.

because $\sum 1/k$ diverges and is what the centring $\log n$ removes). The density of $M_n$ is, up to the factor $y^{n-1}$ and a Laplace transform, a B spline with the knots $1, \dots, n$, and the same holds for the truncations of the reciprocal $\xi$ law with knots $\gamma_1^2, \dots, \gamma_K^2$ (Proposition 24.7.1 and Section 24.7, which also give the Dirichlet mean form; the density of a sum of exponentials is the divided difference of $e^{-\lambda x}$ in $\lambda$, which is the B spline only after this transform). If RH holds, the truncated laws shifted by the tail mean $r_K = \sum_{k>K} \gamma_k^{-2}$ converge to the reciprocal $\xi$ law, since the difference has mean zero and variance $\sum_{k>K} \gamma_k^{-4} \to 0$; no compensation is needed because $\sum_k \gamma_k^{-2} < \infty$ (the shift is the tail shift of the Kent approximants of Section 8.1.3).

Figure 25.3 shows the supremum distances against $K$. For the reciprocal $\Xi$ law the supremum distance between the shifted truncation and the density obtained by Laplace inversion is 128 and 42 for $K = 1, 2$ (attained at the knot $r_K$, where the truncated density jumps or ends), and 8.1, 0.80 and 0.23 for $K = 5, 20, 40$, against a peak density of 80 (computed; the source printed 114 and 37 for $K = 1, 2$, values on a grid that misses the knot; on a uniform grid of mesh $5 \cdot 10^{-4}$ they are 110 and 36).

What the B spline view does and does not give for the Pólya density is the following.

1. Each Gumbel piece of Theorem 25.1.1 has its own exact chain. The piece $g_k$ is the law of $-\log G_k$, so $\mathbb{E}e^{-sX} = \mathbb{E}G_k^s = \Gamma(k+s)/\Gamma(k)$, and the Weierstrass

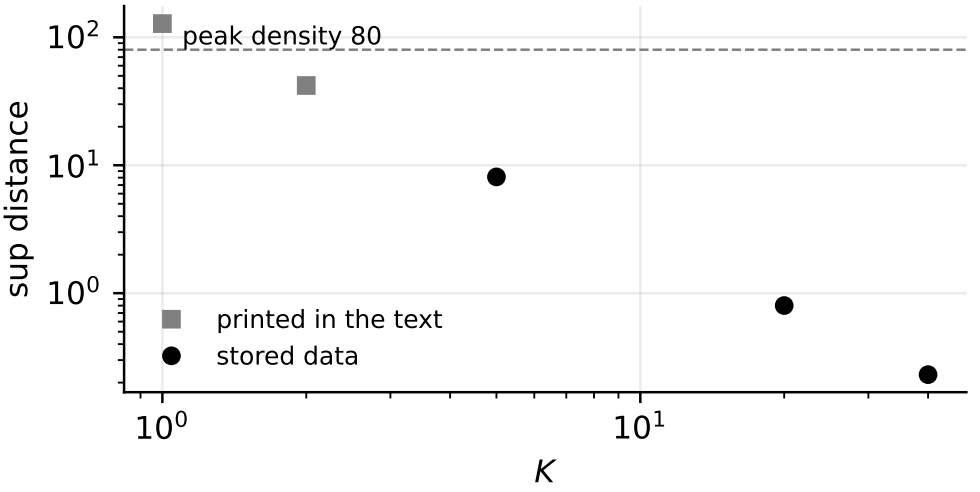



Figure 25.3: Supremum distance between the mean corrected truncation with knots $\gamma_1^2, \ldots, \gamma_K^2$ and the density of the reciprocal $\Xi$ law, against $K$ (dashed: the peak density 80). The values for $K = 5, 20, 40$ are the stored data of `code/ev_g1_tail.py`, those for $K = 1, 2$ are the printed values above; drawn by `code/ev_figs.py`. The density curves of the source, including the Gumbel limit of the maxima of $n$ exponentials, are not drawn.

product gives a compensated Thorin form with atoms at $k, k+1, k+2, \ldots$. For the Pólya pieces these are $\frac{9}{4} + j$ and $\frac{5}{4} + j$ (proved).

2. In $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ this Gumbel structure is the archimedean factor $\Gamma(s/2)$. Its poles are at $s = 0, -2, -4, \ldots$; the pole at 0 is cancelled by the factor $s$ and the others by the trivial zeros $-2, -4, \ldots$ of $\zeta$, so the knots of the Gumbel view correspond to the trivial zeros (proved).

3. The atoms at $\gamma_k^2$ do not come from any product of Gumbel pieces. They come from the theta superposition $\Phi = \sum_n \phi_n$, which translates one Gumbel kernel along the lattice $\frac{1}{2}\log n$ and turns the Gamma factor into $\Gamma(s/2)\zeta(s)$ (Theorem 25.1.1). A superposition is a sum, not a product, and a sum does not preserve Hadamard products or Thorin form (a heuristic remark: it explains why the argument stops, and proves nothing).

So the B spline view leads to a Hadamard product and a Thorin representation in two places, for every Gumbel piece unconditionally and for the reciprocal $\Xi$ law given the zeros. The step that would make it a route to RH, to show that the superposition of Gumbel pieces still has a positive Thorin measure, is the open question of this chapter in another form.

## 25.5 An explicit string: the Liouville operator

Pólya proved that his two sided kernels have only real zeros [441, 447]. The spectral reading of that fact is due to Titchmarsh [497], and its hitting time form is Biane's [48]. Both are in the book. For $a > 0$ the function $g(\tau) = K_\nu(ae^\tau)$ solves $g'' - a^2e^{2\tau}g = \nu^2 g$, the real zeros $z_k$ of $z \mapsto K_{iz}(a)$ are the square roots of the Dirichlet eigenvalues of $-g'' + a^2e^{2\tau}g$ on $(0,\infty)$ with $g(0) = 0$, and the diffusion obtained by the Doob transform of this operator by its positive solution $K_0(ae^\tau)$, started at the entrance boundary at $+\infty$, has hitting time $T_0$ of 0 with

$$\mathbb{E}e^{-sT_0} = \frac{K_0(a)}{K_{\sqrt{s}}(a)} = \prod_k \Big(1 + \frac{s}{z_k^2}\Big)^{-1}$$

(Proposition 8.7.7 and Theorem 8.7.10, proved, with simple spectrum). The product form needs one remark that the source omitted: $K_{\sqrt{s}}(a)$ is of order one half in $s$, because $|K_\nu(a)| \le K_{|\operatorname{Re}\nu|}(a) \le C\,\Gamma(|\nu|+1)(2/a)^{|\nu|}$, so Hadamard's factorisation has genus zero with no exponential factor.

Numerically, a finite difference discretisation of the operator on $(0, 6)$ with $a = 2\pi$ and about 64000 intervals (the source does not state its mesh; the digits are reproduced at that size, and the scheme is of second order) gives $\sqrt{\lambda_k} = 9.768770$, 12.448488, 14.684960, 16.691578, 18.549374, and the zeros of $K_{iz}(2\pi)$ are 9.768770, 12.448488, 14.684960, 16.691579, 18.549375 (computed; the signs of the Bessel values are certified and the zeros are found by bisection, 9.76877008, 12.44848789, 14.68495978, 16.69157894, 18.54937520).

Pólya's kernel involves $K_{(9\pm iz)/4}(2\pi)$ and not $K_{iz}(2\pi)$. He passes from one to the other with the shift lemma: if $G$ is real entire of genus at most one with only real zeros, so is $\frac{1}{2}[G(z+ic) + G(z-ic)]$, which is the transform of $\cosh(cu)$ times the original kernel. The lemma deals with one Bessel term, and the combination of the two terms of $\Phi^P$ is handled in the refinement of Section 16.3 (which also records that the kernel with the second exponential term kept is real rooted). The lemma preserves real zeros, hence by Corollary 25.3.1 the existence of a string when the zeros are simple, but it does not say what the new string is. Writing the transformed string explicitly is Problem 2 of Section 25.8.

## 25.6 The string test from kernel moments

For a general approximation the string is not known, but whether the zeros are real, the first condition for it to exist, can be tested from the kernel alone. Expand

$$\widehat{H}(2i\sqrt{s}) = \int_0^\infty \widehat{\Phi}(u)\cosh(2\sqrt{s}\,u)\,du = \sum_{m\ge 0} \frac{4^m M_{2m}}{(2m)!}s^m,$$

$$M_{2m} = \int_0^\infty u^{2m}\widehat{\Phi}(u)\,du.$$

Taking logarithms gives $\log[\widehat{H}(2i\sqrt{s})/\widehat{H}(0)] = \sum_j (-1)^{j+1}\mu_j s^j/j$ with

$$\mu_j = \sum_k \widehat{\gamma}_k^{-2j},$$

the power sums of the inverse Thorin atoms, equivalently $\kappa_j(T)/(j-1)!$ for the cumulants $\kappa_j(T)$ of the candidate hitting time. Put $w_k = \widehat{\gamma}_k^{-2}$ and $\nu = \sum_k w_k \delta_{w_k}$, so that $\mu_{j+1} = \int y^j \nu(dy)$ whenever the $w_k$ are positive.

PROPOSITION 25.6.1 (Hankel test). *Let $\widehat{H}$ be as in Section 25.3, with infinitely many zeros and $\sum_k |w_k| < \infty$. The zeros of $\widehat{H}$ are all real (and nonzero) if and only if all the Hankel determinants*

$$D_n = \det[\mu_{i+j+1}]_{i,j=0}^{n-1}, \qquad D'_n = \det[\mu_{i+j+2}]_{i,j=0}^{n-1}, \qquad n \geq 1,$$

*are positive.*

*Proof.* If the zeros are real, $\nu$ is a positive measure on $(0,\infty)$ and the determinants are those of the Stieltjes moment problem, hence positive. Conversely, if all are positive, the sequence $(\mu_{j+1})$ is a Stieltjes moment sequence. It grows at most like $\rho^j$ with $\rho = \max_k |w_k|$, so the representing measure $\nu'$ is unique and supported in $[0,\rho]$, and its Stieltjes transform $\int \nu'(dy)/(1-zy)$ is analytic off $[1/\rho,\infty)$ and equals $\sum_{j\geq 0} \mu_{j+1} z^j$ near 0. That series is also $\sum_k w_k/(1-zw_k)$, a meromorphic function (the sum converges locally uniformly off the poles because $\sum |w_k| < \infty$) with poles exactly at the points $1/w_k$ and residues that do not cancel, since terms with equal $w_k$ add with the same sign and distinct $w_k$ give distinct poles. By analytic continuation the poles lie in $[1/\rho,\infty)$, so every $w_k$ is real and positive. □

The continued fraction coefficients built from $D_n$ and $D'_n$ are the masses and lengths of a discrete string, a birth and death chain whose hitting time matches the first $2n$ cumulants (classical, Stieltjes and Krein; the wording of the matching was not checked). So the test is itself a hitting time calculation: one builds the chain and checks that its rates are positive. It is a finite form of the Stieltjes criterion of Theorem 14.6.1, and the moment matching along the Kent approximants is in Section 8.1.4.

The moments $M_{2m}$ were computed by quadrature for the exact kernel, the two term folded kernel and Pólya's two term kernel, then $\mu_j$, then $D_n$ and $D'_n$. At 180 digits all of $D_1,\dots,D_{11}$ and $D'_1,\dots,D'_{11}$ are positive for all three kernels, with the sign of each determinant certified by ball arithmetic from the computed moments (the quadrature error of the moments is not certified, the relative change between two node counts is below $10^{-89}$; computed). For example $D_{11} = 8.99\cdot 10^{-368}$ for the exact kernel, $9.52\cdot 10^{-368}$ for the folded kernel and $1.14\cdot 10^{-350}$ for Pólya's, and $D_6 = 8.1\cdot 10^{-100}$ for the exact kernel. The first moment is $\mu_1 = \sum_k \gamma_k^{-2} = 0.0231049931154190$ for the exact kernel, which equals $\xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$ computed independently by a Cauchy integral; for the two term folded kernel it

is 0.0231049931181204, and for Pólya's kernel 0.0363173599127934. (The source states the precision as 160 digits; the recomputation used 180.)

The positive result for the folded kernel, which does have nonreal zeros above height about $x = 40$ (Table 25.1), shows the limitation of the test. A Hankel matrix of order $n$ sees roughly the first $n$ atoms, and the weights of the atoms decay like $\widehat{\gamma}^{-2j}$, so an off line pair at height $x$ needs an order beyond the number of real atoms below $x$, and precision of the order of $4n \log_{10}(x/\gamma_1)$ digits, before it can make a determinant negative (heuristic; the masking of the first pair is measured in Section 25.9). Finite order positivity certifies a positive string for a truncated moment problem, which is the analogue of verifying RH to a finite height, and it mirrors Li's criterion [336], where finitely many positive coefficients prove nothing about RH.

## 25.7 Is there a path to RH?

Corollary 25.3.1 turns RH with simple zeros into the existence of a string. There are three ways one might try to produce it. In this section and the next the variable of the zeros is written $x = z/2$ when it is real and $w = z/2$ when it is complex, so that the zeros of $H_N^F(2x)$ are the ordinates tracked below; the time of the flow is $t$.

### 25.7.1 Route one: solve the inverse problem for Ξ directly

Find the string whose hitting time has Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ and show its mass is positive. Krein's inverse problem needs the spectral measure, that is the eigenvalues and their weights, and the zeros alone do not determine the string (Proposition 24.2.1 gives the string only up to the weights), so this route is circular unless one has independent access to the mass. A string defines a de Branges space, and de Branges proposed a positivity condition implying RH. Conrey and Li [90] showed that this particular condition fails for $\zeta$. That does not close the route, but it shows that the obvious positivity statements about the string are false.

### 25.7.2 Route two: a convergent family of explicit strings

Real zeros are preserved under locally uniform limits (Hurwitz). Hence RH follows from any sequence $\widehat{H}_m \to H$ locally uniformly in which each $\widehat{H}_m$ has only real zeros, and, for even real entire functions of order less than two with $\widehat{H}_m(0) > 0$, each then has a string if its zeros are simple (Proposition 24.3.1 and Corollary 4.1.2 give the same reduction at the level of hitting times). The candidate families are listed in Section 24.4; they stop short in different ways.

- Pólya's kernels have explicit strings (Section 25.5, Proposition 8.7.7) but do not converge to $\Phi$.

- Hejhal's $m$ term kernels converge to $\Phi$, but only almost all their zeros are real [170].
- The de Bruijn and Newman family $H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du$ has only real zeros for $t \geq \Lambda$, and $\Lambda \geq 0$ (Rodgers and Tao [463]), with $\Lambda \leq 0.22$ (Polymath [456]) and $\Lambda \leq 0.2$ (Platt and Trudgian [419]). Thus the strings exist for $t > \Lambda$, where the zeros are also simple (Section 24.4), and RH is the statement $\Lambda = 0$: that the strings survive the limit $t \downarrow 0$.

### 25.7.3 Route three: the folded family at growing height

The extreme value approximations give a concrete reduction. Here $H_N^F$ is the transform of the folded kernel of the family with $N$ terms, the cosine transform of $\Phi_N^F = \sum_{n\leq N}\phi_n$ on $u > 0$ (Table 25.1), so that $\Phi - \Phi_N^F$ is the omitted set $\{n > N\}$ of Chapter 22 and the case $N = 0$ is the empty sum, which carries no zeros. Proposition 22.1.3, with the integers $n \leq N$ in place of $\mathcal{F}(P)$ and its constant $\frac{1}{2}$ replaced by $\frac{1}{2} - c_E$, where $c_E = \sum_{m>N}(4\pi m^2 - 1)e^{-\pi m^2}$ (Section 22.2), gives $8H_N^F(z)$ at $s = \frac{1}{2} + \frac{iz}{2}$.

Proposition 25.7.1. *Suppose there is $c > 0$ such that for every $N \geq 1$ all zeros of $H_N^F$ in the disc $|z| < 2c(N+1)^2$ are real. Then RH holds.*

*Proof.* $H_N^F \to H$ locally uniformly, since $\Phi - \Phi_N^F$ on $u > 0$ is bounded by $\sum_{n>N}|\phi_n|$, which decays like $e^{-\pi(N+1)^2}$ uniformly with all exponential moments. Let $z_0$ be a zero of $H$ with $|z_0| < 2c(N_0+1)^2$. For $N \geq N_0$ Hurwitz's theorem gives zeros $z_N$ of $H_N^F$ with $z_N \to z_0$, and these lie in the disc, so they are real. Hence $z_0$ is real. □

The hypothesis was checked for $N = 1, 2, 3, 4$ with $c = 3.6$ (Section 25.8, certified in ball arithmetic given the count of zeros of $\Xi$ below the height of each disc), and for $N = 5$ in floating point (Section 25.9, computed). Each $H_N^F$ is a finite sum of incomplete gamma functions, which makes this the most explicit of the three routes, and it is close in spirit to the conjectures of Haglund on incomplete gamma approximants [168]. In string language it asks for a family of strings that agree with the true string up to frequency $4(N+1)^2$. But the proposition is a reformulation, not a reduction in difficulty: for each $N$ the hypothesis implies RH up to height about $c(N+1)^2$, and nothing in the structure of $H_N^F$ is yet known to force it.

### 25.7.4 Verdict

There is no path to RH here in the sense of a proof. There is a precise target, the positivity of the string mass, three ways to approach it, and in each case a known place where the argument stops: the inverse problem needs data that are not available, the $\Lambda = 0$ limit, or a height rule that is equivalent to RH at growing

heights. What the extreme value and hitting time viewpoint adds is computational. The bottom of the spectrum is a tail rate that can be estimated by peaks over threshold to eighteen digits without computing zeros, Hankel positivity gives a finite order test from the kernel moments, which certifies no height (Problem 4), and the folded family turns the height rule into an explicit numerical hypothesis.

## 25.8 Work on the open problems

The five problems of this section and the next are those that the source leaves open; each subsection says what was proved, what was computed and what remains. The numbers are those of the scripts `code/ev_g1_fold.py`, `code/ev_g1_string.py`, `code/ev_dbn.py`, `code/ev_rouche5.py` and `code/ev_prony2.py`.

### 25.8.1 Problem 1: the height rule for fixed $N$

Write $R_N = \Phi - \Phi_N^F = \sum_{n>N} \phi_n$ on $u \geq 0$ and $E_N = H - H_N^F$, so that $E_N(z) = \int_0^\infty R_N(u)\cos(zu)\,du$. The cosine transform of $\phi_n$ at a complex $z$ is $\frac{1}{2}(F_n(z) + F_n(-z))$ with $F_n$ the incomplete gamma sum, and equals $\operatorname{Re} F_n(z)$ only for real $z$.

Lemma 25.8.1 (Kink bound). *For real $x \neq 0$,*

$$E_N(x) = -\frac{R_N'(0)}{x^2} - \frac{1}{x^2}\int_0^\infty R_N''(u)\cos(xu)\,du, \qquad |E_N(x)| \leq \frac{B_N}{x^2},$$

*with $B_N = |R_N'(0)| + \|R_N''\|_{L^1(0,\infty)}$.*

*Proof.* Integrate by parts twice; $R_N$ and $R_N'$ vanish at infinity and the boundary term $R_N(0)\sin(0)$ is zero. □

This is the integration by parts of Theorem 22.2.3 for the cutoff $\{n > N\}$, with $\kappa_N = |R_N'(0)|$ the kink. Numerically $B_N = 1.653 \times 10^{-7}$, $2.783 \times 10^{-16}$ and $5.739 \times 10^{-28}$ for $N = 2, 3, 4$ (and 0.079 for $N = 1$). In every case $R_N'' \geq 0$ on $(0, \infty)$, with the sign decided in ball arithmetic on every cell of a subdivision, so that $\|R_N''\|_{L^1} = |R_N'(0)|$ and $B_N = 2\kappa_N$ exactly (certified, with a rigorous bound for the tail of the series over $n$). The leading order expression $8\pi^3(N+1)^6 e^{-\pi(N+1)^2}$ for $\kappa_N$ overstates it by 15%, 8% and 5% for $N = 2, 3, 4$.

Proposition 25.8.2 (Rouché certificate). *Let $T$ lie strictly between two consecutive ordinates, and suppose $\Xi$ has exactly $2\mathcal{N}(T)$ zeros in $|w| < T$, all real and simple. If $|E_N(2Te^{i\vartheta})| < |H(2Te^{i\vartheta})|$ for $\vartheta \in [0, \pi/2]$, and $H_N^F(2x)$ has $\mathcal{N}(T)$ sign changes on $(0, T)$, then every zero of $H_N^F(2x)$ with $|x| < T$ is real.*

*Proof.* Both functions are even and real on the real axis, so the inequality holds on the whole circle $|z| = 2T$. By Rouché, $H_N^F = H - E_N$ has $2\mathcal{N}(T)$ zeros in the disc, the same number as $H$. The sign changes supply $2\mathcal{N}(T)$ real ones, so there are no others. □

The inequality on the quarter circle was tested in two ways. In floating point it was sampled at 721 points at 250 and 450 bits, computing $E_N$ directly as the sum of the omitted incomplete gamma terms (computed). It was also certified: on each of 3200 to 19600 cells in $\vartheta$ the bound $|E_N(z)| \le \int_0^\infty |R_N(u)| \cosh(u \operatorname{Im} z)\, du$ is evaluated in ball arithmetic (a closed form in incomplete gamma functions with a rigorous tail for $n > N + 5$), $|H|$ is the box value of $\frac{1}{8}\Xi(z/2)$, and the signs of $H_N^F(2x)$ are certified on a grid of step 0.05 (certified). What stays outside ball arithmetic is the count of the zeros of $\Xi$ in the disc, taken from the list of zeros of the book and the verification of RH to this height, which is the hypothesis of the proposition. Table 25.2 gives the results.

| $N$ | $T$ | $\mathcal{N}(T)$ | sampled $\min \lvert H\rvert/\lvert E_N\rvert$ | certified lower bound | $c = T/(N+1)^2$ |
|---|---|---|---|---|---|
| 1 | 14.4 | 1 | 2.45 | 1.0018 | |
| 1 | 17.5 | 1 | 3.65 | 1.0195 | 4.38 |
| 2 | 31.7 | 4 | 151 | 3.49 | |
| 2 | 35.3 | 5 | 36.3 | 2.70 | 3.92 |
| 3 | 51.4 | 10 | $2.83 \times 10^5$ | 2555 | |
| 3 | 57.9 | 12 | 1965 | 64.9 | 3.62 |
| 4 | 86.1 | 23 | $1.22 \times 10^6$ | 35921 | |
| 4 | 97.3 | 28 | 355 | 54.2 | 3.89 |

Table 25.2: Rouché tests for the folded kernels. The sampled minimum is over 721 points of the quarter circle (computed); the lower bound is over the whole quarter circle (certified in ball arithmetic, and weaker than the sampled minimum because the bound is crude). Every row passes both.

In every row the sampled minimum of $|H|/|E_N|$ is attained at $\vartheta = 0$, on the real axis, where Lemma 25.8.1 applies. There the margins $|H(2T)|(2T)^2/B_N$ are 5.77, 262 and 51.4 at the largest heights for $N = 2, 3, 4$; for $N = 1$ they are 0.41 and 0.80, below one, so that for $N = 1$ the kink bound alone does not suffice and only the certificate on the whole circle does. With Proposition 25.8.2 the rows certify that every zero of $H_N^F(2x)$ with $|x| < T$ is real, for $N = 1, 2, 3, 4$, for $c = 3.6$ (given the zero count). The case $N = 0$ is excluded, since $H_0^F = 0$.

The certificates settle the hypothesis of Proposition 25.7.1 only for these four levels, and they also show why the general case is hard. The binding inequality is

$$\tfrac{1}{8}|\Xi(T)| > \frac{B_N}{4T^2}$$

at a point $T$ between consecutive zeros. That is a lower bound for $|\Xi|$ at Gram type points up to height about $4(N+1)^2$, exactly the information Turing's method uses to verify RH to that height [506]. So the height rule for $N$ contains RH verification up to $c(N+1)^2$, and the folded route can run level with numerical verification of RH but not ahead of it (heuristic).

### 25.8.2 Problem 2: explicit strings for shifted Bessel kernels

Pólya's kernels involve $\operatorname{Re} K_{\mu+i\nu}(a)$ with $\mu = 9/4$ and $5/4$. For every $\mu \geq 0$ and $a > 0$, $\nu \mapsto \operatorname{Re} K_{\mu+i\nu}(a)$ is a multiple of the characteristic function, at $\theta = \nu$, of the law of Theorem 13.5.4 with $\ell = 1$, no tilt and one field $\lambda_1 = \mu$, so its zeros are real. For integer shifts the boundary condition of the eigenvalue problem is explicit. The poles of its rational function are written $q_j$ here.

Theorem 25.8.3. *Let $\mu \geq 0$ be an integer and $a > 0$. By the recurrence $K_{\nu+1}(x) = K_{\nu-1}(x) + (2\nu/x)K_\nu(x)$,*

$$\operatorname{Re} K_{\mu+i\nu}(a) = A_\mu(\nu^2)K_{i\nu}(a) + B_\mu(\nu^2)\, aK'_{i\nu}(a)$$

*with real polynomials $A_\mu, B_\mu$. Hence, at a point with $B_\mu(\nu^2) \neq 0$, $\nu$ is a zero exactly when $g(\tau) = K_{i\nu}(ae^\tau)$ solves the Liouville problem $-g'' + a^2e^{2\tau}g = \lambda g$, $\lambda = \nu^2$, with the spectral boundary condition*

$$g'(0) = \beta_\mu(\lambda)\, g(0), \qquad \beta_\mu = -A_\mu/B_\mu.$$

*The first cases are Dirichlet for $\mu = 0$, Neumann ($\beta_1 = 0$) for $\mu = 1$,*

$$\beta_2 = \tfrac{a^2}{2} - \lambda, \qquad \beta_3 = \frac{4(a^2 - 3\lambda)}{a^2 + 8 - 4\lambda}.$$

The theorem is proved (the recurrence; $\beta_2$, $\beta_3$ and every entry of Table 25.3 were rederived independently). The Liouville problem is the one of Proposition 8.7.7, and the shift lemma for Pólya's kernels is in Section 16.3.

Proposition 25.8.4 (Extended boundary problem). *Suppose $\beta_\mu(\lambda) = \alpha - m\lambda - \sum_{j=1}^{J} \sigma_j/(q_j - \lambda)$ with $m \geq 0$, $\sigma_j > 0$ and $q_j$ real. Then the eigenvalue problem of Theorem 25.8.3, away from the poles, is the eigenvalue problem of a self adjoint operator built from the Liouville operator on $(0, \infty)$ with a point mass $m$ at $0$ and $J$ extra coordinates $w_j$ coupled to $g(0)$, with stiffness $q_j$, unit mass and coupling $-\sqrt{\sigma_j}$. Its eigenvalues are real. If in addition the energy form is nonnegative, for example if every $q_j > 0$ and $\alpha \geq \sum_j \sigma_j/q_j$, they are positive, so that the corresponding $\nu$ are real.*

*Proof.* The boundary condition enters the Rayleigh quotient as $\int_0^\infty (g'^2 + a^2e^{2\tau}g^2)\, d\tau + \beta_\mu(\lambda)g(0)^2$. The term $-m\lambda g(0)^2$ moves to the mass side as a point mass. Each term $-\sigma_j g(0)^2/(q_j - \lambda)$ is the Schur complement of an extra coordinate $w_j$: eliminating $w_j = \sqrt{\sigma_j}\, g(0)/(q_j - \lambda)$ from the form $\sum_j [(q_j - \lambda)w_j^2 - 2\sqrt{\sigma_j}\, g(0)w_j]$ leaves $-\sum_j \sigma_j g(0)^2/(q_j - \lambda)$. The resulting energy and mass forms are symmetric and the mass form is positive, so the problem is self adjoint and its eigenvalues are real. The energy form is the sum of the nonnegative integral and of $\alpha g(0)^2 + \sum_j q_j w_j^2 - 2\sum_j \sqrt{\sigma_j}\, g(0)w_j$, which is nonnegative under the stated condition. □

This does not make the extended system a string. A half line with $J$ extra coordinates coupled at 0 does not carry a Dirichlet condition at 0, no hitting time is constructed for it, and Kent's theorem is not available. The reading in which $m > 0$ makes the diffusion sticky at 0 and each pole is an extra state into which the process jumps from 0 and from which it returns is heuristic. What does follow is this: if the zeros $\nu_k$ of $\operatorname{Re} K_{\mu+i\nu}(a)$ are real, positive and simple, then by Krein's converse (Proposition 24.2.1) there is a string whose hitting time has Laplace transform $\prod_k (1 + s/\nu_k^2)^{-1}$, but this string is not identified with the extended system. Table 25.3 records the sign condition. The zeros of $\operatorname{Re} K_{3+i\nu}(2)$ are 1.41977, 3.84551, 5.74106, 7.37072, 8.84981; for $\mu = 4$, $a = 2\pi$ they are 2.72330, 7.25847, 10.37828, 12.82694, 14.96543; for $\mu = 6$, $a = 2\pi$ they are 1.92369, 5.51737, 8.55690, 11.10450, 13.33385 (computed; the source reports that a discretisation of the extended string with 7000 nodes gives the same lowest five eigenvalues to the digits it prints, which was not rerun).

| $\mu$ | $m$ | poles $q_j$ at $a = 2\pi$ | $\sigma_j$ at $a = 2\pi$ | sign condition at $a = 2\pi$ / $a = 2$ |
|---|---|---|---|---|
| 2 | 1 | none | none | yes / yes |
| 3 | 0 | 11.87 | $-3.87$ | no / yes |
| 4 | 1/6 | 7.58 | 1.98 | yes / yes |
| 5 | 0 | 5.02, 59.6 | $-0.165$, 270.6 | no / yes |
| 6 | 1/15 | 3.71, 35.0 | 0.054, 42.2 | yes / yes |
| 7 | 0 | 2.88, 26.0, 195.5 | 0.006, 13.9, 2877 | yes / yes |
| 8 | 1/28 | 2.34, 20.8, 89.1 | 0.005, 5.86, 380 | yes / yes |

Table 25.3: Partial fraction data for $\beta_\mu$ (computed in ball arithmetic, `code/ev_g1_string.py`). The sign condition of Proposition 25.8.4 is that $m \geq 0$ and all $\sigma_j > 0$; it is a hypothesis of the proposition and does not by itself give a string.

When the sign condition fails, as for $\mu = 3$ and 5 at $a = 2\pi$, the zeros are still real by Pólya's shift lemma (Section 16.3), and Krein's converse still supplies a positive string if they are simple; it is simply not this Liouville extension. For Pólya's own orders the recurrence reduces $\mu = 9/4$ and $5/4$ to the base order $1/4$, but $K_{1/4+i\nu}(a)$ is complex for real $\nu$, so the reduction lands on a fractional base string that is not identified here. The problem is therefore reduced to fractional $\mu \in (0, 1)$, which is taken up in Section 25.9.

### 25.8.3 Problem 3: the de Bruijn and Newman strings

Proposition 25.8.5. *For $H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du$, the mean of the candidate hitting time is*

$$\mu_1(t) = \mathbb{E} T_t = 2\mathbb{E}_t[U^2], \qquad \frac{d\mu_1}{dt} = 2 \operatorname{Var}_t(U^2) > 0,$$

*where $U$ has density proportional to $e^{tu^2}\Phi(u)$ on $(0,\infty)$. Equivalently $\mu_1(t) = \sum_k \gamma_k(t)^{-2}$, with $\pm 2\gamma_k(t)$ the zeros of $H_t$ counted with multiplicity.*

*Proof.* $\mu_1$ is the ratio of the first two Taylor coefficients of $H_t(2i\sqrt{s})$, namely $4M_2/(2M_0)$ with $M_{2m}$ the moments of the tilted kernel; differentiate in $t$. □

The moments of the reweighted kernel along the flow are those of Section 8.4.3, and what is added is the monotonicity of the mean. Numerically $\mu_1 = 0.0232010$, 0.0231050, 0.0226358, 0.0190909 at $t = 0.2, 0, -1, -10$, and $d\mu_1/dt = 4.78 \times 10^{-4}$ at $t = 0$ (computed). The mean hitting time increases strictly through $\Lambda$, so it does not see the transition. Nor do low order Hankel determinants: $D_n$ and $D_n'$ are positive for $n \le 5$ on a grid of 13 values of $t$ in $[-10, 0.2]$ (computed), although $H_t$ has nonreal zeros for $t < 0$ (Rodgers and Tao [463]). The zeros that leave the line first are high ones, at Lehmer pairs, and low order moments are dominated by $\gamma_1, \gamma_2, \ldots$. The transition at $\Lambda$ is a high order phenomenon in the string (heuristic).

### 25.8.4 Problem 4: detecting the first nonreal pair

For the two term folded kernel, the argument principle on the boundary of $[36, 150] \times [-8, -0.05]$ in the $w$ plane gives exactly two zeros,

$$w = 43.13891 - 3.28097i, \qquad w = 47.52276 - 6.25251i,$$

a further 17 in $[36, 150]\times[-40, -8]$, and none with real part below 36 and imaginary part above $-40$ (computed, floating point; the winding numbers are 2, 17 and 0). Beyond the breakdown height the zeros of the source peel away into the half plane rather than pairing just off the line (the source's observation, not rerun). The moment Hankel test computed at 120 digits is positive to order 11, and a 240 digit run gives the same signs (computed). If instead one keeps only the seven real atoms and the first pair, $D_6$ is already negative. The remaining off line atoms therefore mask the first pair at order 6 (Table 25.6), while the determinants of the kernel moments are positive to order 11. So a finite order sign is not a local detector. It certifies only that the truncated moment problem has a positive solution, and the order needed to see a given nonreal pair is not determined by the number of real atoms below it (heuristic).

### 25.8.5 Problem 5: higher eigenvalues by joint threshold fits

Replace two point deflation by Prony's method on $2K$ equally spaced values $x_0 + jd$ of the Talbot tail at 80 digits, fitting $K$ exponentials at once; this is the extraction of Section 8.1.2 with the atoms fitted jointly. Table 25.4 shows the errors, as reported by the source. The table was not rerun, and the spacing $d$ of the window is not stated there.

| $K$ | $x_0$ | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ | $\gamma_4$ | $\gamma_5$ | $\gamma_6$ |
|---|---|---|---|---|---|---|---|
| 3 | 0.04 | $4\times10^{-14}$ | $2\times10^{-7}$ | $7\times10^{-4}$ | | | |
| 4 | 0.03 | $3\times10^{-16}$ | $1\times10^{-9}$ | $6\times10^{-6}$ | 0.084 | | |
| 5 | 0.02 | $8\times10^{-16}$ | $3\times10^{-10}$ | $7\times10^{-7}$ | 0.0075 | 0.16 | |
| 6 | 0.015 | $6\times10^{-17}$ | $1\times10^{-11}$ | $2\times10^{-8}$ | $4\times10^{-4}$ | 0.017 | 0.85 |
| 6 | 0.01 | $2\times10^{-10}$ | $1\times10^{-6}$ | $3\times10^{-4}$ | 0.16 | 0.27 | 4.4 |

Table 25.4: Absolute errors of $\gamma_k$ recovered by Prony fits of $K$ exponentials to the tail of the hitting time, window starting at threshold $x_0$ (source values, unchecked).

The pattern is the threshold bias of peaks over threshold fits in exact form. The omitted atoms contribute roughly $e^{-(\gamma_{K+1}^2-\gamma_k^2)x_0}$ to the error in $\gamma_k$, up to conditioning factors (the rule of Section 8.1.2), so raising $K$ allows a lower threshold (heuristic). But lowering the threshold too far, as in the last row, lets atoms beyond $K$ contaminate the fit and loses accuracy everywhere. The optimal pair $(K, x_0)$ trades these two biases against the working precision.

## 25.9 Second round on the open problems

The second round of the source returns to the five problems of Section 25.8 with longer computations. The status of each number is stated where it is printed; nothing in this section is in interval arithmetic.

### 25.9.1 Problem 1: the circle test at $N = 5$

The test of Proposition 25.8.2 was repeated for the five term folded kernel, computing $E_5$ as the sum of the omitted incomplete gamma terms $n = 6, \dots, 14$ (floating point, 30 digits for the circle, 80 digits for the sign changes). Below height 160, $H_5^F(2x)$ has 53 sign changes on a grid of step 0.03 (computed), while $\Xi$ has 58 ordinates there, so the two counts differ by five. The source states that each sign change lies within the grid resolution of an ordinate of $\Xi$; that was not reproduced, since the largest distance from a sign change to the nearest ordinate is 1.04. The circle test, sampled at 721 points of the quarter circle, then gives the rows of Table 25.5, each with the minimum of $|H|/|E_5|$ again on the real axis.

So, on the sampled circles (computed, not interval arithmetic), every zero of $H_5^F(2x)$ with $|x| < 148.7$ is real, giving $c \geq 148.7/36 = 4.13$. The margin falls steadily, roughly an order of magnitude every one or two zeros, and the test first fails at $T = 150.5$, just above the predicted breakdown at $4 \cdot 6^2 = 144$. A failed circle means only that the test stops there, not that a zero has left the line. Across $N = 2, \dots, 5$ the computed constants are $3.92, 3.62, 3.89, 4.13$ (the first three certified in Section 25.8). These are heights at which the test succeeded, not breakdown heights: for $N = 2$ the first nonreal zero is at $w = 43.14 - 3.28i$

| $T$ | $\mathcal{N}(T)$ | sampled $\min_\vartheta \lvert H\rvert/\lvert E_5\rvert$ | result |
|---|---|---|---|
| 128.548 | 42 | $6.01 \times 10^{6}$ | pass |
| 130.333 | 43 | $7.66 \times 10^{5}$ | pass |
| 132.293 | 44 | $3.55 \times 10^{5}$ | pass |
| 134.127 | 45 | $3.59 \times 10^{4}$ | pass |
| 136.436 | 46 | $3.54 \times 10^{4}$ | pass |
| 138.926 | 47 | 1191 | pass |
| 140.430 | 48 | 247 | pass |
| 142.118 | 49 | 157 | pass |
| 144.556 | 50 | 57.1 | pass |
| 146.712 | 51 | 3.11 | pass |
| 148.738 | 52 | 1.36 | pass |
| 150.489 | 53 | 0.052 | fail |

Table 25.5: Rouché tests for $N = 5$ (computed: floating point, sampled on the circle, $E_5$ truncated at $n = 14$).

(Section 25.8), which puts the breakdown at $c \le 43.14/9 = 4.79$. So the height rule $T_N \approx 4(N+1)^2$ is consistent with $c$ between 3.6 and 4.2 at every level tested, and nothing more.

### 25.9.2 Problem 2: fractional shifts and the cotangent law

For noninteger $\mu$ there is no finite recurrence, but each zero $\nu_k$ of $\operatorname{Re} K_{\mu+i\nu}(a)$ still determines the Robin ratio that the Liouville eigenfunction must satisfy there,

$$\beta_k = \frac{g'(0)}{g(0)} = \frac{aK'_{i\nu_k}(a)}{K_{i\nu_k}(a)}, \qquad \lambda_k = \nu_k^2.$$

The problem is whether these values come from an admissible boundary law $\beta(\lambda) = \alpha - m\lambda - \int \sigma(dq)/(q-\lambda)$ with $m \ge 0$ and $\sigma \ge 0$, which is decreasing between its poles.

Proposition 25.9.1 (Cotangent law). *For fixed $a > 0$ and $\mu > 0$ not an integer, as $k \to \infty$,*

$$\beta_k = \cot\Big(\frac{\pi\mu}{2}\Big)\sqrt{\lambda_k}\,\big(1 + O\big((a/2\nu_k)^{2\mu}\big)\big) + O(1).$$

*Sketch.* For $\nu$ large compared with $a$, $K_{\mu+i\nu}(a) = \frac{1}{2}\Gamma(\mu+i\nu)(a/2)^{-\mu-i\nu}\big(1 + O((a/2\nu)^{2\mu})\big)$ up to a relative term of order $a^2/\nu$ in the leading series, which is absorbed in the $O(1)$, the second term of the small argument expansion being smaller by the stated factor. So the zeros satisfy $\arg\Gamma(\mu+i\nu) - \nu\log(a/2) = \frac{\pi}{2} + k\pi$, and Stirling gives $\arg\Gamma(\mu+i\nu) = \arg\Gamma(i\nu) + \frac{\pi\mu}{2} + O(1/\nu)$. For $\mu = 0$ both terms have equal size and $K_{i\nu}(ae^{\tau}) \approx |\Gamma(i\nu)|\cos(\arg\Gamma(i\nu) - \nu\log(a/2) - \nu\tau)$, whose logarithmic derivative at $\tau = 0$ is $\nu\tan(\arg\Gamma(i\nu) - \nu\log(a/2))$. At the zeros the phase is $\frac{\pi}{2} - \frac{\pi\mu}{2}$ modulo $\pi$, giving $\nu\cot(\pi\mu/2)$. $\square$

The proof is a sketch: the phase computation, which gives the leading term, was rederived independently, and the error terms were not, so they are heuristic and the statement is proved only for the leading term. The law matches every integer case of Theorem 25.8.3: $\cot(\pi\mu/2) = 0$ for odd $\mu$, where $\beta$ is bounded, and $\cot(\pi\mu/2) = \infty$ for even $\mu$, where $\beta$ grows linearly in $\lambda$ through the point mass. Numerically, at $a = 2\pi$ the slope of $\beta_k$ against $\sqrt{\lambda_k}$ over the last zeros below $\nu = 45$ is 1.0007 to 1.0008, 0.4095 to 0.4100 and $-0.4178$ to $-0.4172$ for $\mu = \frac{1}{2}, \frac{3}{4}, \frac{5}{4}$, against $\cot(\pi\mu/2) = 1$, 0.4142 and $-0.4142$ (computed; the range is over windows of the last 6 to 12 zeros, because the source does not define the window, and at least 45 digits are needed since $K_{i\nu}(2\pi)$ is of size $10^{-31}$ at $\nu = 45$). For $\mu = \frac{1}{4}$ it is 3.04 to 3.06 against 2.41, consistent with the slow correction $(a/2\nu)^{1/2} \approx 0.28$ at $\nu = 40$.

The sign of $\cot(\pi\mu/2)$ splits the problem in two.

- For $\mu \in (1,2)+2\mathbb{Z}$, including Pólya's $\mu = \frac{5}{4}$, $\beta$ decreases like $-c\sqrt{\lambda}$. Since $\sqrt{\lambda}$ is a Nevanlinna function, this is, heuristically, the growth of an admissible law with a continuous spread of extra masses, $\sigma$ carried by the negative half line and so off the spectrum, rather than finitely many; numerically $\beta_k = 0.37 - 0.417\sqrt{\lambda_k} + 2.43/\sqrt{\lambda_k}$ over the first 24 zeros, with maximal relative error $3.3 \times 10^{-3}$ (computed; the accuracy $5 \times 10^{-4}$ stated by the source was not reproduced by a fit of this form in the maximum norm).

- For $\mu \in (0,1)+2\mathbb{Z}$, including $\mu = \frac{9}{4}$, $\beta$ increases with $\lambda$, so an admissible law needs a pole in every gap between consecutive eigenvalues, that is, coupling to a second string with an interlacing spectrum.

Pólya's two term kernel combines one order of each type. Identifying the continuum extension for $\mu = \frac{5}{4}$ and the interlacing string for $\mu = \frac{9}{4}$ is the refined form of the problem (open).

### 25.9.3 Problem 3: where the de Bruijn and Newman zeros collide

For $t < 0$, $H_t(z) = \mathbb{E}H(z + W)$ with $W$ normal of variance $2|t|$, so $H_t$ can be computed by Gaussian convolution of $H$ on a fine real grid (Section 8.4.1 and Theorem 8.4.20 give the same flow). Counting sign changes of $H_t(2x)$ below $x = 100$, where $\Xi$ has 29 zeros, gives 29 real zeros for $t = -0.25$ and $-0.5$, then 27, 19, 12 and 8 for $t = -1, -2, -4, -8$ (computed, step 0.05, 60 digits). A bisection of the source places the first collision below height 100 at $t \approx -0.881$, near $x \approx 95.5$, for the pair at 94.651 and 95.871, the closest pair below 100 (gap 1.22). The recomputation did not repeat this bisection, so the value is unchecked here; the book's own computation of the collision of this pair, with the same neighbourhood, is in Section 8.4.3. At $t = -4$ a nonreal pair sits near height 33, at $33.1150940 \pm 0.1505390\,i$ (computed). The fourth and fifth zeros, at 30.42

and 32.94 for $t = 0$, have then merged, which is the source's reading and was not checked.

The moment Hankel test sees these collisions, but late. Computed at 130 digits to order 10, all $D_n$ and $D'_n$ are positive at $t = 0$. At $t = -4$ the first negative value is $D'_{10}$, and at $t = -8$ the determinants $D_8, D_9, D_{10}$ and their primed partners are all negative, with $D_1, \dots, D_7$ and their primed partners positive (computed). A nonreal pair of rank about four is therefore detected only at order eight to ten, about twice its rank. The first collision, of rank about 27, is far beyond order ten. In this family the test gives the lower bound $\Lambda > -4$ from kernel moments alone: a negative determinant excludes a positive measure, hence real zeros, at $t = -4$, and nonreal zeros at $t = -4$ give $\Lambda > -4$. This is far weaker than Rodgers and Tao [463], and than the moment only bound of the book's own Hankel test along the flow (Table 8.9), which already gives $t^*_{50} \approx -1.03$ with no failing order up to 50 on $[-1.03, 0]$, so that the order needed to see the collision at $-0.881$ exceeds 50. What the computation adds to that section is the signs at $t = -4$ and $-8$ and the shifted determinants $D'_n$. They show that the Hankel string test does detect a genuine loss of reality when the off line atoms are few, in contrast with the folded kernels of Problem 4.

### 25.9.4 Problem 4: why the first pair is masked

The off line zeros of the two term folded kernel lie on a regular curve. Starting from $43.139 - 3.281i$ and $47.523 - 6.253i$, the source locates 19 zeros $w_j$ in $[36, 150] \times [-40, 0]$, with real parts spaced about 4 apart on average (from the printed end points) and imaginary parts decreasing to $-38.6$ at real part 114.9, finds by the argument principle 16 more with imaginary part in $[-60, -40]$ and 25 more in $[-100, -60]$, and none with real part above 150 and imaginary part above $-40$. Of this, only the count $19 = 2 + 17$ in $[36, 150] \times [-40, 0]$ is recomputed (Section 25.8); the geometry and the further counts are unchecked.

The Hankel determinants then follow from the Andreief (Heine) identity. With atoms $y_i = 1/\gamma_i^2$ and moments $\mu_{j+1} = \sum_i y_i^{j+1}$,

$$D_n = \sum_{|S|=n} \prod_{i \in S} y_i \prod_{i<j \in S} (y_i - y_j)^2,$$

summed over $n$ element sets of atoms. A term containing both members of a conjugate pair $y, \bar{y}$ carries the factor $(y - \bar{y})^2 = -4(\operatorname{Im} y)^2$ and is typically negative. Table 25.6 shows $D_6$ as atoms are added.

The source splits the sum over the 22 atoms of largest modulus by type: all real sets $1.3 \times 10^{-105}$, sets containing a full pair $-3.4 \times 10^{-102}$, and sets containing single members of several pairs $+7.2 \times 10^{-102}$. The first value agrees with the first row of the table; the other two were not recomputed (unchecked), and the selection of the 22 atoms is not defined in the source. If the split is right, the masking is

| atoms used | $D_6$ |
|---|---|
| 7 real atoms only | $1.29 \times 10^{-105}$ |
| plus the first pair | $-1.81 \times 10^{-104}$ |
| plus the first two pairs | $-2.27 \times 10^{-103}$ |
| plus all 19 located pairs | $1.09 \times 10^{-100}$ (unchecked) |
| kernel moments (all zeros) | $8.10 \times 10^{-100}$ |

Table 25.6: The masking of the first nonreal pair in $D_6$ for the two term folded kernel (computed, 250 digits, except the kernel moments row, which is from 120 and 240 digits, and the row marked unchecked, which is the source's value and needs the 19 located zeros).

a cross term effect. The atoms on the curve have moduli $|y_i|$ comparable to the real atoms, because $|\gamma|$ along the curve grows only like the real part, and their mixed products $2\,\mathrm{Re}\prod$ are positive and larger than the negative full pair terms. A nonreal pair is visible to a finite Hankel test only when the other off line atoms are too small or too spread out to compensate, and the test cannot tell which case it is in (heuristic).

### 25.9.5 Problem 5: forty digit eigenvalues

In this subsection $x$ is the hitting time variable. At 140 digits, Prony fits of $K$ exponentials to the survival function of the hitting time on windows $x_0 + 0.003j$, $j < 2K$, give the errors in Table 25.7 (computed; the source used Talbot inversion of degree 160, while the recomputation of this book used the survival series $\sum_k c_k e^{-\lambda_k x}$ over the zeros; the density form of the tail gives larger errors, for example $7 \times 10^{-13}$ instead of $9 \times 10^{-14}$ for $\gamma_1$ at $K = 6$, so the tail used is the survival function). The rows were reproduced at 100 and 130 digits to the one digit printed.

| $K$ | $x_0$ | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ | $\gamma_4$ | $\gamma_5$ | $\gamma_6$ | $\gamma_8$ |
|---|---|---|---|---|---|---|---|---|
| 6 | 0.02 | $9 \times 10^{-14}$ | $1 \times 10^{-9}$ | $3 \times 10^{-7}$ | $6 \times 10^{-4}$ | 0.013 | 0.52 | |
| 8 | 0.02 | $5 \times 10^{-25}$ | $1 \times 10^{-19}$ | $3 \times 10^{-16}$ | $3 \times 10^{-11}$ | $4 \times 10^{-9}$ | $2 \times 10^{-5}$ | 0.11 |
| 10 | 0.015 | $8 \times 10^{-34}$ | $1 \times 10^{-27}$ | $1 \times 10^{-23}$ | $7 \times 10^{-18}$ | $4 \times 10^{-15}$ | $2 \times 10^{-10}$ | $5 \times 10^{-5}$ |
| 10 | 0.02 | $2 \times 10^{-39}$ | $7 \times 10^{-33}$ | $2 \times 10^{-28}$ | $5 \times 10^{-22}$ | $6 \times 10^{-19}$ | $2 \times 10^{-13}$ | $4 \times 10^{-7}$ |

Table 25.7: Absolute errors of $\gamma_k$ from Prony fits to the tail of the reciprocal $\Xi$ hitting time at 140 digits (computed).

With ten atoms the first zero comes out to 39 digits and the eighth to 7, from the Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ alone in the source, whose method differs from the recomputation (survival series over the zeros). For $K = 10$, the only $K$ with two thresholds, the error falls as the threshold rises from 0.015 to 0.02, so the table supports a threshold not below 0.02 at this precision. The trade off between

the bias from omitted atoms and the loss of precision is visible in the columns: each extra atom in the fit buys roughly five to seven more digits in the leading eigenvalues (from $K = 6$ to 8 and from 8 to 10 the error of $\gamma_1$ falls by 11 and 14 digits, so each added pair buys ten to fourteen), and the last atom fitted is always the least accurate because the omitted atoms load onto it. For a Lehmer pair with gap $\delta$ at height $\gamma$ the rates differ by about $2\gamma\delta$, and resolving them requires a window long enough that $e^{-2\gamma\delta x}$ varies appreciably while both terms remain above the working precision, which fixes how many digits a given pair costs (heuristic).

## 25.10 Devroye's series, the mean excess function and detection

This section develops the last part of Nick Polson's chapter [231]: tests of the extreme value picture that read the Laplace transform $L(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, together with a small amount of information about the zeros. Its numbers were recomputed for this book by scripts written from the mathematics alone (the scripts of the source were not used). Every number printed below is one of those recomputations unless it is marked unchecked, and where the recomputation differed from the source the recomputed value is printed. The section has four parts: Devroye's alternating series and its sign pattern, the mean excess function and a theorem of Schoenberg that makes it a criterion, the weak convergence statements behind it, and a detection experiment in the de Bruijn and Newman family. What the book already contains is cited and not restated: the residues $R_k = c_k\gamma_k^2$ and the spectral expansion of the density (Section 8.1.1 and Exercise 8.1), the criterion by Pólya frequency functions (Proposition 8.1.2), the weak closure of the GGC class (Corollary 4.1.2) and the digits rule for Prony fits (Section 8.1.2).

### 25.10.1 Devroye's alternating series and the sign pattern

Devroye's exact samplers for the Jacobi theta laws rest on one device: a density written as a series whose terms alternate in sign and decrease in size is bracketed by its partial sums [110, 115]. The reciprocal $\Xi$ law has such a series, but it is stated here with its hypotheses. Assume RH, assume that the zeros are simple, and assume $|\xi'(\rho)| \geq e^{-C|\gamma|}$ as in Theorem 24.1.2; the zeros are $\rho = \frac{1}{2} + i\gamma_k$ with $\gamma_k$ real, $0 < \gamma_1 < \gamma_2 < \cdots$, and $\pm\gamma_k$ are the zeros of $\Xi$. Then the density of $X = \sum_k E_k/\gamma_k^2$ is

$$f(x) = \sum_{k \geq 1} c_k \gamma_k^2 e^{-\gamma_k^2 x}, \qquad c_k = \prod_{j \neq k} \Big(1 - \frac{\gamma_k^2}{\gamma_j^2}\Big)^{-1} = \frac{2i\,\xi(\frac{1}{2})}{\gamma_k\,\xi'(\frac{1}{2} + i\gamma_k)}.$$

Without the lower bound for $|\xi'(\rho)|$ the book proves only the grouped limit of the residue sums (Theorem 24.1.1), and the partial sums used below are then not

known to converge to $f$. The coefficient $c_k$ is undefined at a multiple zero. The two term tail of $f$ needs none of these hypotheses (Proposition 24.1.3).

PROPOSITION 25.10.1 (Sign pattern). *If the atoms $\gamma_1^2 < \gamma_2^2 < \cdots$ are real and simple, then* $\operatorname{sign} c_k = (-1)^{k-1}$.

*Proof.* Exactly $k-1$ of the factors $1 - \gamma_k^2/\gamma_j^2$ are negative, those with $j < k$. □

REMARK 25.10.2. If a pair of atoms $\lambda_k = \gamma_k^2$ is not real, the two coefficients of the pair are complex conjugates, and for every real atom $\lambda_m$ the pair contributes the factor $|1 - \lambda_m/\lambda_k|^{-2} > 0$ to $c_m$, so the real coefficients still alternate when they are counted among the real atoms: the coefficient of a real atom of rank $r$ among the real atoms has sign $(-1)^{r-1}$. A sign test on the coefficients therefore sees a nonreal pair only if the rank of each atom is known, which is a zero count. The test also reads $\gamma_k$ and $\xi'(\rho_k)$, not $L$ alone.

Devroye's switch point is the $x_k$ beyond which $|c_{k+1}|\gamma_{k+1}^2 e^{-\gamma_{k+1}^2 x} \le |c_k|\gamma_k^2 e^{-\gamma_k^2 x}$, namely

$$x_k = \frac{\log\big(|c_{k+1}|\gamma_{k+1}^2/(|c_k|\gamma_k^2)\big)}{\gamma_{k+1}^2 - \gamma_k^2}.$$

Bracketing at a point $x$ needs the terms to alternate and decrease from the first index on, that is $x \ge \sup_k x_k$.

| law | signs of $c_k$ | $x_1, x_2, x_3$ |
|---|---|---|
| $1/\cosh\sqrt{2s}$ | $+-+-+-+-$ | .111, .0259, .0114 |
| $\sqrt{2s}/\sinh\sqrt{2s}$ | $+-+-+-+-$ | .0936, .0329, .0167 |
| reciprocal $\Xi$ | alternating, first 1700 zeros | .01963, .01533, .01385 |

Table 25.8: Calibration on Devroye's two explicit laws, with atoms $\pi^2(k-\frac12)^2/2$ and $\pi^2k^2/2$, and the reciprocal $\Xi$ law with atoms $\gamma_k^2$. For the two strings $c_k\lambda_k = (-1)^{k-1}\pi(k-\frac12)$ and $c_k = 2(-1)^{k-1}$ in closed form, so that $x_k = \log\frac{2k+1}{2k-1}/(k\pi^2)$ and $x_k = \log\frac{(k+1)^2}{k^2}/\big((2k+1)\pi^2/2\big)$ (computed, and checked against the product at $K = 4000$ factors). For the reciprocal $\Xi$ law the coefficients come from $\xi'(\rho_k)$ at the first 1700 zeros (computed).

For the reciprocal $\Xi$ law the signs alternate for all 1700 coefficients computed (computed): $c_1 = 50.871$, $c_2 = -2664.6$, $c_3 = 3.1420 \times 10^4$, $c_4 = -1.3550 \times 10^6$, and so on. Beyond the first switch point the partial sums bracket the density, which was checked against Talbot inversion at 30 digits.

- At $x = 0.02$, where $f = 80.10420$, the partial sums are 186.93, 16.148, 88.602, 77.170, 80.2026, 80.09549.
- At $x = 0.03$, where $f = 23.433069$, they are 25.352, 23.295, 23.4341, 23.43301, 23.4330693, 23.43306926.

Each sum lies on the opposite side of $f$ from the one before. Over the first 1700 zeros the largest switch point is $x_1 = 0.019627$, and the largest from $k = 4$ on is $0.011656$, at $k = 7$ (computed). This gives an exact Devroye sampler for $X$ on $x \geq 0.01963$, and two sided bounds there, both conditional on RH with simple zeros, on the lower bound for $|\xi'(\rho)|$ above, and on $x_k \leq 0.01963$ for every $k$. The last condition was checked for $k \leq 1699$ only. For large $k$ the switch point depends on the ratios $|\xi'(\rho_k)/\xi'(\rho_{k+1})|$ over the gaps $\gamma_{k+1}^2 - \gamma_k^2$, and no bound for them is known. The sampler covers the part of the law on $x \geq 0.01963$ only. The bounds are computed with floating point arithmetic, not enclosed.

### 25.10.2 Schoenberg's theorem and the mean excess function

The mean excess, or mean residual life, function of $X$ is $e(u) = \mathbb{E}[X - u \mid X > u]$. In the source it is computed without zeros, by one Laplace inversion each for numerator and denominator,

$$\int_u^\infty \mathbb{P}(X > x)\, dx = \mathcal{L}^{-1}\Big[\frac{\mathbb{E}X}{s} - \frac{1 - L(s)}{s^2}\Big](u),$$

with

$$\mathbb{E}X = \frac{\xi''(\frac{1}{2})}{2\xi(\frac{1}{2})} = 0.02310499311541897\ldots$$

The criterion by Pólya frequency functions is Proposition 8.1.2. It rests on Schoenberg's characterisation [474], which is restated here with its nondegeneracy condition and with the Gaussian parameter renamed so that it does not clash with the ordinates $\gamma_k$.

Theorem 25.10.3 (Schoenberg). *An integrable density $f$ is a Pólya frequency function, totally positive of every order, exactly when its bilateral Laplace transform is $1/\Psi(s)$ in a strip around the origin, with $\Psi(s) = Ce^{-\alpha s^2 + \delta s}\prod_k (1 + \delta_k s)e^{-\delta_k s}$, where $\alpha \geq 0$, $\delta$ and $\delta_k$ are real, $\sum \delta_k^2 < \infty$ and $0 < \alpha + \sum \delta_k^2$ (Schoenberg 1951).*

Corollary 25.10.4. *RH holds if and only if $L(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is the Laplace transform of a Pólya frequency density. Under RH that density is log concave, the hazard rate of $X$ is increasing, and the mean excess function decreases monotonically,*

$$e(0) = \sum_k \gamma_k^{-2} = 0.0231050 \;\searrow\; e(\infty) = \gamma_1^{-2} = 0.00500524.$$

*Proof.* The function $1/L(s) = \prod_k (1 + s/\gamma_k^2)$ has the form of Theorem 25.10.3 with $\alpha = 0$ and $\delta_k = \gamma_k^{-2}$, which are real exactly when every $\gamma_k$ is real: an off line zero with $\gamma_k$ not real gives a nonreal $\gamma_k^2$, and a real zero of $\xi$ off the line would lie in $(0, 1)$, where, by the classical fact that $\zeta$ has no zero in $(0, 1)$, there is none. Total positivity of order two is log concavity, which implies an increasing hazard rate,

and an increasing hazard rate implies a decreasing mean residual life (Barlow and Proschan [22]). □

So the mean excess function interpolates between the full Thorin moment $\sum_k \gamma_k^{-2}$ at $u = 0$ and the first atom at $u = \infty$. Its approach to the limit carries the next atom. The book states the analogous hazard limit, with rate $\gamma_2^2 - \gamma_1^2$, in Section 8.1.1 and Exercise 8.3.

PROPOSITION 25.10.5 (Expansion and rate). *Under RH with simple zeros and* $|\xi'(\rho)| \geq e^{-C|\gamma|}$ *as above, and with* $\lambda_k = \gamma_k^2$,

$$e(u) = \frac{\sum_k c_k e^{-\lambda_k u}/\lambda_k}{\sum_k c_k e^{-\lambda_k u}} = \frac{1}{\lambda_1} + A\, e^{-(\lambda_2-\lambda_1)u} + O\big(e^{-\min(\lambda_3-\lambda_1,\, 2(\lambda_2-\lambda_1))u}\big), \qquad A = \frac{c_2}{c_1}\Big(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\Big) > 0.$$

*The excess law of $X$ over $u$ converges to the exponential law with rate $\lambda_1$ in total variation at rate $e^{-(\lambda_2-\lambda_1)u}$. Without simple zeros the two term form follows from the two term tail of Proposition 24.1.3, with $w_k/\lambda_k$ in place of $c_k$ for $k = 1, 2$.*

*Proof.* The first equality is $\int_u^\infty \mathbb{P}(X > x)\,dx / \mathbb{P}(X > u)$ with $\mathbb{P}(X > x) = \sum_k c_k e^{-\lambda_k x}$. Put $\varepsilon = e^{-(\lambda_2-\lambda_1)u}$. The numerator is $\lambda_1^{-1} c_1 e^{-\lambda_1 u}\big(1 + \frac{c_2\lambda_1}{c_1\lambda_2}\varepsilon + O(e^{-(\lambda_3-\lambda_1)u})\big)$ and the denominator is $c_1 e^{-\lambda_1 u}\big(1 + \frac{c_2}{c_1}\varepsilon + O(e^{-(\lambda_3-\lambda_1)u})\big)$. Dividing, the first order term is $\frac{c_2}{c_1}\big(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\big)\varepsilon$ and the next is $O(\varepsilon^2)$. Since $c_2/c_1 < 0$ (Proposition 25.10.1) and $1/\lambda_2 < 1/\lambda_1$, $A > 0$. □

Everything below was recomputed for this book from the residue series over the first 300 zeros at 50 and 70 digits on the grid $u = 0.005, 0.0075, \ldots, 0.2$ (79 points); the source computed it by Talbot inversion from $L$ alone. None of it is enclosed in interval arithmetic.

- Log concavity: $ff'' - f'^2 < 0$ at all 79 points. This is a sampled check. Two of the points, 0.005 and 0.0075, lie below 0.008, where Theorem 24.5.5 starts and where the book's proof is conditional on a saddle point approximation (Proposition 24.5.3).
- Monotonicity: $e(u)$ decreases strictly at every grid step. Beyond $u \approx 0.165$ consecutive values agree with $\gamma_1^{-2} = 0.00500524412359412$ to every printed digit.
- Expansion: Proposition 25.10.5 predicts $A = 0.1436441604$. The observed deviation $e(u) - \gamma_1^{-2}$ against $Ae^{-(\lambda_2-\lambda_1)u}$ is $9.808 \times 10^{-5}$ against $1.006 \times 10^{-4}$ at $u = 0.03$, $2.3625 \times 10^{-7}$ against $2.3637 \times 10^{-7}$ at $u = 0.055$, and $1.3052007 \times 10^{-12}$ against $1.3052008 \times 10^{-12}$ at $u = 0.105$. From $u = 0.13$ on the agreement is to all eight printed digits; the residual at small $u$ is the $\lambda_3$ term.

Read without zeros (as the source does, while the recomputation used the zeros), the expansion gives the first two zeros. An Aitken extrapolation of three values of $e(u)$ estimates $e(\infty) = \gamma_1^{-2}$ and, as $-\log r/h$ for the extrapolated ratio $r$ and the step $h = 0.0075$, the gap $\lambda_2 - \lambda_1$. It gives $\gamma_1 = 14.1347778$, $14.1347266$ and $14.1347252$ from windows starting at $u = 0.03$, $0.04$ and $0.05$, and $\gamma_2 = 20.965$, $21.006$ and $21.019$, against $\gamma_1 = 14.1347251417$ and $\gamma_2 = 21.0220396$. This is weaker than the Prony fits to the tail itself (Table 25.7). The mean excess is most useful as a criterion and as a check, not as a way of computing zeros.

### 25.10.3 Weak convergence

Three weak convergence statements sit behind these diagnostics.

1. Pickands, Balkema and de Haan: the excess law over $u$, rescaled by $e(u)$, converges to a generalised Pareto law [249, 19]. Here the shape is zero and the convergence is exponentially fast (Proposition 25.10.5).

2. Closure: generalised gamma convolutions are closed under weak limits (Bondesson [56]; Corollary 4.1.2), and so are Pólya frequency laws with nondegenerate limits (Karlin [188]). Hence RH is equivalent to the reciprocal $\Xi$ law being a weak limit of hypoexponential laws $\sum_{k \le K} E_k/\lambda_{k,K}$ with real positive rates, a distributional form of the Hurwitz argument. The equivalence is the reduction of Proposition 24.3.1, combined with Corollary 4.1.2.

3. The exponential B splines of Proposition 24.7.1 (Section 25.4) are such a sequence, but they use the zeros. Zero free sequences exist under RH: the Jensen laws of Section 24.6 are GGC laws for every degree $d \le 2 \cdot 10^{17}$, exist for every $d$ exactly when RH holds, and then converge to the law of $X$ (Proposition 24.6.1). Another candidate is a sequence of approximations $\widehat{L}_N(s) = \widehat{H}_N(0)/\widehat{H}_N(2i\sqrt{s})$ with only real zeros that converge to $L$. Pólya's kernels give Pólya frequency laws but do not converge. The folded kernels converge, and their zeros were found to be real up to a height of about $c(N+1)^2$ with computed constants $c$ between 3.6 and 4.4 (certified in ball arithmetic for $N \le 4$ given the count of zeros of $\Xi$ in each disc, computed for $N = 5$; Sections 25.8 and 25.9).

### 25.10.4 Detection in the de Bruijn and Newman family

The family $H_t$ is the one of Chapter 8.4, with the flow time $t$ and the variable of the zeros written $w = z/2$ (real or complex), whose collisions and Hankel test along the flow are in Sections 8.4.2 and 8.4.3. At $t = -4$ the fourth and fifth zeros of $H_t$ have merged, according to the source (not checked here). Evaluated by quadrature against the kernel, $H_{-4}(2w)$ is negative at all 41 points of $[28, 38]$ with step 0.25 (computed; the largest value is $-6.67 \times 10^{-12}$ and the smallest

modulus is at $w = 33.25$), so that it almost touches zero near 33. Root finding locates the pair at

$$w = 33.1150940 \pm 0.1505390\,i, \qquad \gamma^2 = 1096.58679 \pm 9.970228\,i$$

(computed). This explains why a short window diagnostic would be expected to miss it. The pair adds $e^{-1096.6x}\cos(9.97x)$ to the tail, and over a window of length 0.04 the cosine turns by only 0.4 radians. The source reports that a six atom Prony fit on such a window returns spurious atoms, one of them real and a pair, and that the same fit at $t = 0$ returns an equally spurious pair where the true zero is real. This was not recomputed. Short windows cannot separate a slightly nonreal pair from omitted atom bias.

REMARK 25.10.6 (Detection rule, heuristic). To see one radian of oscillation from a nonreal atom $\lambda = a+ib$, a tail window must extend to $x \approx 1/b$. There its term is smaller than the leading term by $e^{-(a-\lambda_1)/b}$, so that the computation needs about $(a - \lambda_1)/(b \ln 10)$ digits beyond the size of the leading term. If the window ends at $x_{\text{end}}$ instead, about $(a - \lambda_1)x_{\text{end}}/\ln 10$ digits are needed. The book states the same kind of rule for the atoms of the real line in Section 8.1.2. This is an order of magnitude estimate and not a theorem.

At $t = -4$ the first count is $(1096.59 - 234.42)/(9.970 \ln 10) \approx 37.6$ digits (computed, with $\gamma_1 = 15.31093766$ taken from Table 25.9, which is unchecked). The window of the source, given below, ends at 0.188, where the pair is smaller than the leading term by about $e^{-162} \approx 10^{-70}$, which is the count for that window.

The source tested the rule with a window $x_j = 0.08 + 0.012j$, $j = 0, \ldots, 9$, which turns the cosine by about one radian, and five atom Prony fits at 120 digits, with $H_{-4}(2i\sqrt{s})$ computed as the Gaussian average $\mathbb{E}H(2i\sqrt{s} + W)$, $W$ normal with variance 8, by 90 point Gauss and Hermite quadrature on $\xi$. Table 25.9 is the outcome as the source reports it. The fits were not run for this book. The only entries independently recomputed are the pair, which is the zero of $H_{-4}$ found directly, and the zeros of the control column, which are $\gamma_1, \ldots, \gamma_5$.

| | $t = 0$ (control) | $t = -4$ |
|---|---|---|
| $\widehat{\gamma}_1$ | 14.13472514 | 15.31093766 |
| $\widehat{\gamma}_2$ | 21.02203964 | 22.5998361 |
| $\widehat{\gamma}_3$ | 25.01085758 | 26.14438156 |
| $\widehat{\gamma}_4$ | 30.42487613 | $33.11509402 - 0.1505390225\,i$ |
| $\widehat{\gamma}_5$ | 32.93506159 | $33.11509402 + 0.1505390225\,i$ |

Table 25.9: Long window threshold fits from the Laplace transform alone, as reported by the source (unchecked: the fits were not recomputed; the pair agrees with the zero of $H_{-4}$ found directly, $33.1150940172 \pm 0.150539022\,i$).

If the table is right, the fit returns for $t = 0$ the atoms $\gamma_1, \ldots, \gamma_5$ and for $t = -4$

a conjugate pair with Thorin atom $1096.58679 \pm 9.970227764\,i$, which agrees with the root found directly to all ten digits printed. A threshold analysis of the tail of the inverse Laplace transform of $1/H_{-4}(2i\sqrt{s})$ (at $t = -4$ this need not be a probability density, so it is not a hitting time tail) would then detect a genuine loss of reality from the transform alone, once the window and the precision follow Remark 25.10.6. Until the fits are recomputed this is a claim of the source, and the rule itself stays heuristic.

The rule makes the limitation of all tail diagnostics quantitative. A pair that has only just left the line has small $\operatorname{Im}\gamma^2 = 2\operatorname{Re}\gamma\operatorname{Im}\gamma$, and the digits required grow like $1/\operatorname{Im}\gamma^2$, so it is the hardest kind to see. This is the Lehmer pair phenomenon in tail form (the flow of a Lehmer pair is in Section 8.4.6), and it is why such tests are verification to a finite height and precision rather than proof.

## 25.11 Remaining open problems

1. Interval arithmetic for the Rouché test at $N = 5$ and for the count of zeros of $\Xi$ in each disc, and the level $N = 6$ near height 196 (Sections 25.8 and 25.9).

2. Identify the continuum extension for $\mu = \frac{5}{4}$ and the interlacing string for $\mu = \frac{9}{4}$, and combine them into a string for Pólya's two term kernel.

3. Give the Hankel order needed to see the de Bruijn and Newman collision at $t \approx -0.881$ near height 95.5. This is answered in part by Table 8.9: no order up to 50 fails on $[-1.03, 0]$, so at the collision time the order needed exceeds 50.

4. Prove that the off line zeros of the folded kernels lie on a curve, with explicit spacing, and decide whether masking persists at all Hankel orders for this family.

5. Turn the Prony scans into an error formula in $(K, x_0, \text{precision})$ and apply it to a Lehmer pair.

6. Find a product structure in the theta superposition of Gumbel pieces that carries the Thorin atoms at $\gamma_k^2$ (Section 25.4).

7. Show that the switch points satisfy $x_k \le x_1$ for every $k$ (Section 25.10.1), from properties of $L$ and a lower bound for $|\xi'(\rho)|$, and use Devroye's bounds for a certified sampler of the reciprocal $\Xi$ law. The alternation itself is proved (Proposition 25.10.1); what is open is the monotonicity of the terms.

8. Higher order total positivity: test $\mathrm{PF}_3$ and beyond for the reciprocal $\Xi$ density from Laplace inversion, since Corollary 25.10.4 makes every order a necessary condition for RH. The ladder of Toeplitz minors of the Pólya density is in Chapter 24.

9. Make the detection rule of Remark 25.10.6 into an algorithm with error control, and estimate its cost for the Lehmer pairs known from verification.

10. One sided approach in the Dyck chain. If $F_n \to 2\xi$ locally uniformly on $\operatorname{Re} s > 0$ and, for every $T > 0$, the Dirichlet polynomial $F_n$ of the Dyck chain (Section 25.1) has no zero in $\frac{1}{2} < \operatorname{Re} s < 1$, $0 < \operatorname{Im} s < T$ for all large $n$, then RH holds: if RH fails, the functional equation and the symmetry $\xi(\bar{s}) = \overline{\xi(s)}$ give a zero $\rho$ of $\xi$ with $\frac{1}{2} < \operatorname{Re}\rho < 1$ and $\operatorname{Im}\rho > 0$, and by Hurwitz's theorem, for $T > \operatorname{Im}\rho$, $F_n$ has zeros in that rectangle for all large $n$. Under Conjecture 25.1.2 the hypothesis holds for $T < \gamma_{64} = 169.9119\ldots$ and fails for every $T > \gamma_{64}$. The zeros of $\xi$ below height 170 are $\rho_1, \ldots, \rho_{64}$, simple and on the line [508]. For $T > \gamma_{64}$, the zero $z_n(\rho_{64})$ of Proposition 25.1.3 tends to $\rho_{64}$ and has $\operatorname{Re} z_n(\rho_{64}) = \frac{1}{2} + \delta_n \operatorname{Re} c_{\mathrm{D}}(\rho_{64}) + O(1/n) > \frac{1}{2}$ for large $n$, since $\operatorname{Re} c_{\mathrm{D}}(\rho_{64}) > 53.08$ (computed in ball arithmetic, Section 25.1). For $T < \gamma_{64}$, $F_n \to 2\xi$ uniformly on the closed rectangle; outside small discs about $\rho_1, \ldots, \rho_{63}$ it carries no zero of $\xi$, hence none of $F_n$ for large $n$, and in the disc about $\rho_k$ the only zero is $z_n(\rho_k)$, to the left of the line for large $n$ because $\operatorname{Re} c_{\mathrm{D}}(\rho_k) < 0$ (computed in ball arithmetic). This is the argument of Corollary 7.2.4 for the Dyck chain: conditionally on Conjecture 25.1.2 the hypothesis fails, and this criterion of one sided approach cannot prove RH along this chain. For $10^5 \le n \le 10^8$ the zero near $\rho_{64}$ lies to the right of the line (computed, Section 25.1). Status: the implication proved, the failure of the hypothesis conditional on Conjecture 25.1.2; open: prove the conjecture, which would also give the locally uniform convergence of the first hypothesis. This item follows notes of Nicholas Polson merged into the book [232, 239].

The scripts of this chapter are listed in `code/README.md`.

# Chapter 26

# Weil positivity as the Thorin condition

Weil's criterion states RH as the positivity of a functional built from the explicit formula. Restricted to Gaussian test functions and their derivatives in the heat time, this functional is the heat trace $W(t)$ of the zeros, and positivity becomes the Thorin condition of the reciprocal $\Xi$ law: RH holds if and only if $W$ is completely monotone. The explicit formula writes $W = \mathcal{A} - \mathcal{P}$ for every $t > 0$, with an archimedean part $\mathcal{A}$ and a prime part $\mathcal{P}$, and both have an extreme value form: $\mathcal{A}$ is $u$ times the Lévy measure of a Gumbel type variable of order $\frac{1}{4}$, and $\mathcal{P}$ is a positive weighted series of Fréchet distribution functions, one for each prime power. This chapter reads the two sides of the explicit formula in that language, maps the three regimes of the Gaussian family, recovers the zeros from the primes by Prony's method, and collects what can be proved and what can only be computed about windows of positivity, certificates from finitely many primes, rigidity of the primes, smoothing of the prime error, energy, Hankel positivity and a one heat time form of RH. It does not prove RH, and it says where each route stops.

The chapter develops a source chapter by Nick Polson [248], together with the computations done for this book. Its numbers were recomputed here by scripts written from the mathematics alone (`code/wt_heat.py`, `code/wt_window.py` and `code/wt_rig.py`, with data in `data/wt_*.json`, listed in `code/README.md`; the scripts of the source were not used). Every number printed below is one of those recomputations unless it is marked unchecked, and where the recomputation differed from the source the recomputed value is printed. An earlier draft of the first half of the source was reviewed first, and its findings are applied here to the final text. Statements of papers that were not read are marked unchecked.

The notation is the book's, with the following changes from the source. The heat time is $t$ (the source writes $x$), and the heat trace of the zeros is the book's $W(t)$ of Section 1.2 (the source writes $\Theta$); the book's $\Pi + \mathcal{A}$ of Table 5.6 is written $\mathcal{A}(t)$ here, and $\mathcal{P}(t)$ is the prime part. The Gaussian kernel is $g_t(u) = (4\pi t)^{-1/2} e^{-u^2/4t}$. Chebyshev's function is $\psi_{\mathrm{C}}(y) = \sum_{n \le y} \Lambda(n)$, and $\psi = \Gamma'/\Gamma$ is the digamma function. The normalised prime error is $D(u) = (\psi_{\mathrm{C}}(e^u) - e^u)e^{-u/2}$. The shifted functional of Section 26.10 is $V(t, a)$. Zeros are indexed as in Sec-

tion 1.2 (one zero of each pair, counted with multiplicity), but the symbols are local and differ from that section: $\rho_j = \beta_j + i\tau_j$ with $\tau_j > 0$, where $\tau_j$ is the real ordinate that Section 1.2 calls $\gamma_k$; $\gamma_j = \tau_j + i(\frac{1}{2} - \beta_j)$ is complex, and $\lambda_j = \gamma_j^2$ is the book's $a_k = -(\rho_k - \frac{1}{2})^2$. So that $\rho_j = \frac{1}{2} + i\gamma_j$ and $\gamma_j$ is real exactly when $\rho_j$ is on the critical line. A sum over positive real $\gamma$ is written only under RH. The first window root is $r_*$ with $\lambda_* = r_*^2$.

Much of the source is already in the book and is cited, not restated: the heat trace, its Frullani and Weil forms and the equivalence of RH with complete monotonicity (Section 1.2, Proposition 1.2.1, Equation (1.2.5)), the digits of Table 5.6 and Table 1.4, the theorems that $W > 0$ and $W' < 0$ for every $t > 0$ (Theorem 5.1.1), that $W$ is completely monotone on $(0, 1.29]$ with no information on the zeros below 0.019 (Theorem 5.1.9) and that $(-1)^k W^{(k)} > 0$ for all $t > 0$ and $k \le 1.16 \times 10^{25}$ (Theorem 5.1.13), the tent version of the sine squared lemma (Theorem 17.6.2), the Gumbel kernels of fractional order (Section 25.1) and the Prony extraction of the atoms (Section 8.1.2). What this chapter adds to the book is the Gumbel and Fréchet readings of $\mathcal{A}$ and $\mathcal{P}$, the archimedean window with $r_*$ the critical point of the Riemann Siegel theta function, the recovery of the zeros from $\Gamma$ and the primes, the prime by prime windows, the rigidity and energy tables, the Hankel form with the first two cumulants computed from the primes and the one heat time theorem.

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it (floating point or high precision, never an interval enclosure unless said), conditional, heuristic or open as stated. The results are these.

1. RH holds if and only if $W$ is completely monotone, if and only if the Thorin measure $\sum_j \delta_{\lambda_j}$ is supported on $(0, \infty)$, if and only if $\mathcal{A} - \mathcal{P}$ is completely monotone (Theorem 26.2.1, proved, with the converse by the pole argument for the resolvent). The Laplace transform of $W$ is meromorphic with a nonzero residue at every $-\lambda_j$, so a nonreal $\lambda_j$ cannot be the pole of a Stieltjes transform.

2. The archimedean density is $u$ times the Lévy density of $-\frac{1}{2}\log G_{1/4}$, and the prime part is a weighted series of Fréchet distribution functions with scales $(\log n)^2/4$ (Section 26.3, proved).

3. $W$ is computed to 110 digits from $\Gamma$ and the primes; Prony's method on 16 such samples gives $\gamma_1, \gamma_2, \gamma_3$ to more than 30 digits and the eighth zero to 4 digits (Table 26.4, computed).

4. The archimedean part $\mathcal{A}$ has a spectral density with exactly one sign change, at $\lambda_* = 39.562 \approx 4\pi^2$. The first sign change $t_n$ of $(-1)^n \mathcal{A}^{(n)}$ satisfies $t_n = n/\lambda_* + \beta + \delta/n + O(n^{-2})$ with explicit $\beta$ and $\delta$ (Propositions 26.8.2 and 26.9.1,

proved; the table is computed). Complete monotonicity of order $n$ of $\mathcal{A}$ holds only on $(0, t_1]$.

5. Each prime power added extends the order one window to about $\log q/2\gamma_1$ (computed, a heuristic balance). A one sided certificate from the primes up to $N_0$ is proved (Proposition 26.9.2); the windows tabulated from it are computed in floating point on a grid, not certified.

6. The variance of $\psi_{\mathrm{C}}$ in short intervals follows the GUE prediction and not the Poisson level (computed, with a heuristic prediction). RH is equivalent to finite discounted energy of $D$ (Proposition 26.9.4, proved); the mean square constant needs simple zeros.

7. The first two cumulants of the reciprocal $\Xi$ law are read off the primes, and RH is equivalent to the positive semidefiniteness of two Hankel families of the normalised cumulants (Section 26.9.5, proved).

8. RH holds if and only if $V(t, a) \geq 0$ for every $a \geq 0$ at one fixed heat time $t$ (Theorem 26.10.1, proved by Landau's theorem for nonnegative functions).

The open problems close the chapter.

## 26.1 Weil's explicit formula and the heat trace

Write the nontrivial zeros as $\rho = \frac{1}{2} + i\gamma$, with the indexing of the introduction. For an even test function $h$, holomorphic in a closed strip wider than $|\operatorname{Im} r| \leq \frac{1}{2}$ and decaying like $O((1 + |\operatorname{Re} r|)^{-2-\eta})$ there, with $g(u) = \frac{1}{2\pi}\int h(r)e^{-iru}\,dr$, the Guinand and Weil explicit formula [163, 512], [288, Theorem 5.12] is

$$\sum_{\rho} h(\gamma_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) - g(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\,\operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\,dr - 2\sum_{n\geq 1}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n), \quad (26.1.1)$$

the sum over all nontrivial zeros with multiplicity. The terms $h(\pm\frac{i}{2})$ come from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, which the factor $s(s-1)$ of $\xi$ removes. Weil's criterion [512, 55] states that RH holds if and only if the right side is nonnegative for every even part of a test function $F(r)\overline{F(\bar{r})}$ in the admissible class. Under RH this is clear, since then $h(\gamma) = |F(\gamma)|^2$ for real $\gamma$. The content is the converse. For test functions of small support the positivity is known: Yoshida proved it for support in a short interval [525], and Connes and Consani showed that the archimedean place controls the sign for small support [87]. These are the results of Chapters 17 and 18, and they use no information on the zeros.

### Gaussian test functions

Take $h_t(r) = e^{-tr^2}$ with $t > 0$. It is entire and decays faster than any power on every strip, so (26.1.1) holds for every $t > 0$, with $g_t(u) = (4\pi t)^{-1/2} e^{-u^2/4t}$. Pairing $\rho$ with $1-\rho$ (the zeros with $\tau > 0$ are closed under $\rho \mapsto 1-\bar\rho$) gives $\sum_\rho h_t(\gamma_\rho) = 2W(t)$ with

$$W(t) = \sum_j e^{-\lambda_j t}, \qquad \operatorname{Re}\lambda_j = \tau_j^2 - (\tfrac12 - \beta_j)^2 \geq \tau_j^2 - \tfrac14.$$

This is the book's unconditional definition of the heat trace (Section 1.2); the zero counting bound $N(T) = O(T\log T)$ gives absolute convergence for every $t > 0$ and after any number of differentiations, and $W$ is real because the nonreal $\gamma_j$ come with their conjugates. Then (26.1.1) becomes $W(t) = \mathcal{A}(t) - \mathcal{P}(t)$, where

$$\mathcal{A}(t) = e^{t/4} - \tfrac12 g_t(0)\log\pi + \frac{1}{4\pi}\int_{\mathbb{R}} e^{-tr^2}\,\operatorname{Re}\psi\big(\tfrac14 + \tfrac{ir}{2}\big)\,dr, \tag{26.1.2}$$

$$\mathcal{P}(t) = \sum_{n\geq 2} \Lambda(n)\, n^{-1/2}\, g_t(\log n). \tag{26.1.3}$$

This is the identity (1.2.5) (with the polar term included in $\mathcal{A}$), and Tables 1.4 and 5.6 give its digits. Table 26.1 adds a few more. At 110 digits the value $\mathcal{A}-\mathcal{P}$ equals the sum over the first 1700 zeros to $10^{-70}$ or better at each $t$ of the table, $t = 0.005, 0.02, 0.05, 0.1, 0.2$ (computed, `code/wt_heat.py`), and the printed digits of the source are reproduced, for example $W(0.02) = 0.018541291731319482254$.

| $t$ | $W(t)$ from the zeros | $\mathcal{A}(t)-\mathcal{P}(t)$ from $\Gamma$ and the primes |
|---|---|---|
| 0.005 | 0.5371919640910408957 | 0.5371919640910408957 |
| 0.02 | 0.018541291731319482254 | 0.018541291731319482254 |
| 0.05 | $4.5878350485735742685 \times 10^{-5}$ | $4.5878350485735742685 \times 10^{-5}$ |
| 0.1 | $2.1047997958223512845 \times 10^{-9}$ | $2.1047997958223512845 \times 10^{-9}$ |
| 0.2 | $4.4301821802236448843 \times 10^{-18}$ | $4.4301821802236448843 \times 10^{-18}$ |

Table 26.1: The explicit formula for the heat trace, agreeing to all printed digits (computed).

## 26.2 Weil positivity is the Thorin condition

Put $H(s) = \xi(\frac12 + \sqrt{s})/\xi(\frac12)$, which is entire in $s$ by the functional equation. By Hadamard's factorisation (order $\frac12$), $H(s) = \prod_j (1 + s/\lambda_j)$ with $\sum_j |\lambda_j|^{-1} < \infty$, and the candidate $L = 1/H$ has the form

$$-\log L(s) = \sum_j \log\Big(1 + \frac{s}{\lambda_j}\Big) = \int_0^\infty (1 - e^{-sx})\,\frac{W(x)}{x}\,dx,$$

by Frullani's integral $\log(1+s/\lambda) = \int_0^\infty (1-e^{-sx})e^{-\lambda x}\,dx/x$ for $\operatorname{Re}\lambda > 0$ (the unconditional statement is [424, Theorem 4.3], quoted in Section 1.2). Under RH every $\lambda_j = \tau_j^2$ is positive, $L(s) = \mathbb{E}e^{-sT}$ for the clock variable $T = \sum_j E_j/\lambda_j$ of Section 1.2 (the reciprocal $\Xi$ law, with $E_j$ independent standard exponentials), its Lévy measure is $W(x)\,dx/x$ and its Thorin measure is $U = \sum_j \delta_{\lambda_j}$, with the multiplicities. Without RH only the analytic identities hold, and $W(x)/x$ is the only possible Lévy density (Theorem 5.1.1 proves $W > 0$ and $W' < 0$ unconditionally).

THEOREM 26.2.1. *The following are equivalent.*

*(i) RH.*

*(ii) Every $\lambda_j$ lies in $(0,\infty)$, that is, the Thorin measure $U = \sum_j \delta_{\lambda_j}$ is supported on $(0,\infty)$ and $T$ is a generalised gamma convolution.*

*(iii) $W$ is completely monotone on $(0,\infty)$: $(-1)^n W^{(n)}(t) \geq 0$ for all $n \geq 0$ and $t > 0$.*

*(iv) $\mathcal{A} - \mathcal{P}$ is completely monotone on $(0,\infty)$, with $\mathcal{A}$ and $\mathcal{P}$ as in* (26.1.2) *and* (26.1.3).

*Proof.* (iii) and (iv) are the same statement, since $W = \mathcal{A} - \mathcal{P}$ for every $t > 0$. Under RH every $\lambda_j = \tau_j^2 > 0$, which gives (ii), and then $W = \int e^{-\lambda t} U(d\lambda)$ is completely monotone (termwise differentiation is justified by the convergence bound above), which gives (iii). For (iii) implies (i) define $S(s) = \sum_j (s+\lambda_j)^{-1}$. The series converges normally away from its poles, and for real $s > \frac{1}{4}$ Fubini applies because $\sum_j (s + \operatorname{Re}\lambda_j)^{-1} < \infty$, so $S(s) = \int_0^\infty e^{-sx} W(x)\,dx$. If $W$ is completely monotone, Bernstein's theorem gives $W(x) = \int_{[0,\infty)} e^{-ux}\mu(du)$ with $\mu \geq 0$, and Tonelli gives $S(s) = \int (s+u)^{-1}\mu(du)$ for $s > \frac{1}{4}$. Finiteness at one such $s$ gives $\int (1+u)^{-1}\mu(du) < \infty$, so the right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. It agrees with the meromorphic series on a real interval, hence on the connected set $\mathbb{C}\setminus(-\infty,0]$ minus the poles of $S$. A pole $-\lambda_j$ outside $(-\infty,0]$ would be removable, but its residue is the positive integer multiplicity of $\lambda_j$. So every $\lambda_j \in [0,\infty)$. Finally $\operatorname{Im}\lambda_j = 2\tau_j(\frac{1}{2} - \beta_j)$ and $\tau_j > 0$, so $\beta_j = \frac{1}{2}$ and $\lambda_j = \tau_j^2 > 0$. This gives (i), and (ii) implies (iii) as above. (ii) implies (i) by the same last step. □

This is Proposition 1.2.1 with the converse spelled out; the uniqueness of exponents of a complex exponential series, which the source asserted, is replaced by the pole argument, which was checked independently (the converse by the resolvent). Statement (iv) is Weil positivity restricted to the cone generated by the Gaussian test functions and their $t$ derivatives, $h(r) = r^{2n}e^{-tr^2}$ for all $n \geq 0$ and $t > 0$ (these are $|r^n e^{-tr^2/2}|^2$ on the real line and equal $(-\partial_t)^n$ of $h_t$). Positivity of $W$ alone is not this statement, and it is proved without RH (Theorem 5.1.1). The prime side converges for every $t > 0$, because $g_t(\log n)$ decays like $e^{-(\log n)^2/4t}$.

Remark 26.2.2 (The Stieltjes form and the parabola). The Laplace transform of $W$ is the meromorphic function $S(s) = \sum_j (\lambda_j + s)^{-1} = H'(s)/H(s)$, a Stieltjes function if and only if RH holds. Its Weil kernel $h_s(r) = (r^2 + s)^{-1} = |r + i\sqrt{s}|^{-2}$ is admissible only when the pole $-i\sqrt{s}$ lies outside the strip, that is $\operatorname{Re}\sqrt{s} > \frac{1}{2}$, which is also where the prime sum $\sum \Lambda(n) n^{-1/2-\sqrt{s}}$ converges. In the $s$ plane this is the outside of the parabola $\operatorname{Re}\sqrt{s} = \frac{1}{2}$. A pole $-\gamma^2$ of $S$ with $\gamma = \gamma_r + i\beta$ lies inside the parabola, because $\operatorname{Re}(-\gamma^2) - \frac{1}{4} + (\operatorname{Im}\gamma^2)^2 = -(1 - 4\beta^2)(\gamma_r^2 + \frac{1}{4}) < 0$ for $|\beta| < \frac{1}{2}$, and for $\beta = 0$ the quantity is $-(\gamma_r^2 + \frac{1}{4}) < 0$ as well. So all the poles of the Stieltjes picture, violations and non violations alike, sit where the Euler product diverges. The Gaussian form of Theorem 26.2.1 avoids this by using test functions that are admissible everywhere.

## 26.3 The archimedean place is a Gumbel Lévy measure

Put $k(u) = e^{-u/2}/(1 - e^{-2u}) = \sum_{m \geq 0} e^{-(2m+\frac{1}{2})u}$. The integral (26.1.2) of the digamma function against the Gaussian can be written in an absolutely convergent form. Start from $\psi(z) = -\gamma_E + \int_0^\infty (e^{-s} - e^{-zs})(1 - e^{-s})^{-1}\, ds$ with $z = \frac{1}{4} + \frac{ir}{2}$ and $s = 2u$, which gives

$$\frac{1}{4\pi} \int e^{-tr^2} \operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big) dr = -\frac{\gamma_E}{2} g_t(0) + \int_0^\infty \frac{g_t(0) e^{-2u} - e^{-u/2} g_t(u)}{1 - e^{-2u}}\, du. \quad (26.3.1)$$

Splitting off $g_t(0) e^{-u/2}$ from the numerator and using $\int_0^\infty (e^{-u/2} - e^{-2u})(1 - e^{-2u})^{-1} du = \frac{1}{2}(\psi(1) - \psi(\frac{1}{4}))$, the archimedean part is

$$\mathcal{A}(t) = e^{t/4} + \tfrac{1}{2}\big[\psi(\tfrac{1}{4}) - \log\pi\big] g_t(0) + \int_0^\infty \big[g_t(0) - g_t(u)\big] k(u)\, du. \quad (26.3.2)$$

The compensation $g_t(0) - g_t(u) = O(u^2)$ is essential. The two pieces $\int g_t(u) k(u)\, du$ and $\int g_t(0) k(u)\, du$ separately diverge, because $k(u) \sim 1/2u$ at the origin, so the archimedean place cannot be set beside the primes as two finite integrals against $g_t$. The identity (26.3.1) was verified numerically to $10^{-71}$ in the source (unchecked here). The prime part is $\mathcal{P}(t) = \int_0^\infty e^{-u/2} g_t(u)\, d\psi_{\mathrm{C}}(e^u)$ with the atom at $u = 0$ left out, and both measures have an infinitely divisible meaning.

- The density $k(u)$ is $u$ times the Lévy density $e^{-u/2}/\big(u(1 - e^{-2u})\big)$ of $-\frac{1}{2}\log G_{1/4}$, where $G_{1/4}$ has the Gamma$(\frac{1}{4})$ law. More generally $-\frac{1}{2}\log G_a$ has Lévy density $e^{-2au}/\big(u(1 - e^{-2u})\big)$ on $u > 0$, with compensation (the density behaves like $1/2u^2$ at the origin), by taking two derivatives of $\log\Gamma(a - i\theta/2) - \log\Gamma(a)$. The variable $-\log G_k$ is the $k$th largest point of a Poisson process with intensity $e^{-v}\, dv$ for an integer $k$ (put $z = e^{-v}$ to get the arrival times of a unit rate process); for a fractional order $a$ it is the log gamma extension of that family, the Gumbel kernel of fractional order of Section 25.1, not an ordinary ranked point. It is a real valued infinitely divisible variable, not a subordinator.

- The prime measure $\Lambda(n)n^{-1/2}\delta_{\log n}$ is $u$ times the atomic measure $\sum_n \frac{\Lambda(n)}{\log n} n^{-\sigma}\delta_{\log n}$ at $\sigma = \frac{1}{2}$. For $\sigma > 1$ this is the Lévy measure $\sum_{p,m\geq 1} p^{-m\sigma}\delta_{m\log p}/m$ of $\log N_\sigma$, with $\mathbb{P}(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, which is compound Poisson with total mass $\log\zeta(\sigma)$. At $\sigma = \frac{1}{2}$ the total mass beyond any point is infinite ($\sum_p p^{-1/2} = \infty$), so the measure is locally finite and arithmetic but it is not a Lévy measure and the law of $N_\sigma$ does not exist. The Gaussian test function is what makes the pairing $\int u\, g_t(u)\, \nu_{1/2}(du) = \mathcal{P}(t)$ finite.

So Weil positivity in Gaussian form compares the compensated archimedean functional (26.3.2) with the prime pairing and asks that the difference be completely monotone in $t$. It is not a difference of two probability Lévy measures.

### Fréchet form of the prime part

As a function of $t$, each prime power enters through

$$\sqrt{4\pi t}\, g_t(\log n) = \exp\Big(-\frac{(\log n)^2}{4t}\Big),$$

the Fréchet distribution function of index one with scale $(\log n)^2/4$, evaluated at $4t/(\log n)^2$ through $\Phi_1(y) = e^{-1/y}$. So

$$\mathcal{P}(t) = \frac{1}{\sqrt{4\pi t}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, \Phi_1\Big(\frac{4t}{(\log n)^2}\Big),$$

a positive weighted series of Fréchet distribution functions whose scales are the squared logarithms of the prime powers. The weights $\Lambda(n)n^{-1/2}$ are not summable ($\sum_{n\leq N}\Lambda(n)n^{-1/2} \sim 2\sqrt{N}$), so this is not a probability mixture, and a finite truncation is one only after division by its total weight. The smallest scale, $(\log 2)^2/4 = 0.120$, belongs to the prime 2, and for $t \downarrow 0$

$$\mathcal{P}(t) = \frac{\log 2}{\sqrt{8\pi t}} \exp\Big[-\frac{(\log 2)^2}{4t}\Big]\Big\{1 + O\Big(\exp\Big[-\frac{(\log 3)^2 - (\log 2)^2}{4t}\Big]\Big)\Big\},$$

proved by isolating $n = 2$ and factoring out the exponential of the next term $n = 3$. The prime part is governed by this one extreme atom as $t \downarrow 0$.

## 26.4 Three regimes

Table 26.2 shows $\mathcal{A}$, $\mathcal{P}$ and $W$ computed at 110 digits (computed, `code/wt_heat.py`; a second run at 130 digits differs by at most $2\times 10^{-110}$ in $W$ at all 23 values of $t$ tried). The columns of $\mathcal{A}$ and $W$ are those of Table 5.6, where $\mathcal{A}$ is the sum of its columns $\Pi$ and $\mathcal{A}$.

| $t$ | $\mathcal{A}(t)$ | $\mathcal{P}(t)$ | $W(t)$ | digits cancelled | share of the prime 2 in $\mathcal{P}$ |
|---|---|---|---|---|---|
| 0.002 | 1.783 | $2.6 \times 10^{-26}$ | 1.783 | 0 | 1.0000 |
| 0.005 | 0.537 | $7.2 \times 10^{-11}$ | 0.537 | 0 | 1.0000 |
| 0.01 | 0.1497 | $8.4 \times 10^{-6}$ | 0.1497 | 0 | 1.0000 |
| 0.02 | 0.02095 | 0.00241 | 0.01854 | 0.05 | 0.9999 |
| 0.05 | 0.05796 | 0.05791 | $4.59 \times 10^{-5}$ | 3.1 | 0.966 |
| 0.1 | 0.16281 | 0.16281 | $2.10 \times 10^{-9}$ | 7.9 | 0.808 |
| 0.2 | 0.30152 | 0.30152 | $4.43 \times 10^{-18}$ | 16.8 | 0.562 |
| 0.3 | 0.39603 | 0.39603 | $9.32 \times 10^{-27}$ | 25.6 | 0.427 |

Table 26.2: Archimedean part, prime part and heat trace (computed). Digits cancelled is $\log_{10}(\mathcal{A}/W)$, which differs from the measure of the caption of Table 5.6.

1. Archimedean dominance, $t \le 0.01$. Here $\mathcal{P}/W \le 6 \times 10^{-5}$ (computed: $5.6 \times 10^{-5}$ at $t = 0.01$), and all of $\mathcal{P}$ comes from the prime 2, through its Fréchet factor $e^{-(\log 2)^2/4t}$. Positivity of $W$ is inherited from the archimedean part, and so are the derivative conditions of the orders up to 6 tested at the grid of $t$ (unchecked as a computation). The book proves more than this: every order $k \ge 1$ for $t \le 0.019$ with no information on the zeros (Theorem 5.1.9(a)), and all orders up to $1.16 \times 10^{25}$ for every $t > 0$ (Theorem 5.1.13). A small $\mathcal{P}/W$ alone controls function values, not derivatives, because the derivatives of $\mathcal{P}$ are not dominated by those of $\mathcal{A}$ uniformly in the order. This regime is the Gaussian counterpart of small support positivity: a Gaussian of variance $2t$ sees essentially nothing beyond $\log 2$.

2. Crossover, $t \approx 0.02$ to 0.05. The prime part becomes comparable to $W$ (at $t = 0.05$ the ratio $\mathcal{P}/W$ is about 1260) but is still almost entirely the prime 2. Positivity here compares one extreme atom with the archimedean density.

3. Cancellation, $t \gtrsim 0.05$. As $t \to \infty$ both $\mathcal{A}$ and $\mathcal{P}$ grow like $e^{t/4}$; on the range of the table $\mathcal{A}$ rises only from 0.02 to 0.40. While $W \approx e^{-\gamma_1^2 t}$ (for large $t$, if $\lambda_1$ is simple and separated), the two parts agree to $\log_{10}(\mathcal{A}/W) \approx \gamma_1^2 t/\ln 10 + \log_{10}\mathcal{A} \approx 87t$ digits ($\gamma_1^2/\ln 10 = 86.767$), and the prime 2's share falls as the bulk of the primes enters. All the information about the zeros lives in this exponentially small difference, and the evaluation of $W$ is very ill conditioned there; this is a statement about conditioning, not a theorem that the information is confined to one interval of $t$.

### 26.4.1 The archimedean positivity window

How far does the archimedean part alone satisfy the Bernstein condition? Combining the two archimedean terms of (26.1.2),

$$\mathcal{A}(t) = e^{t/4} + \int_0^\infty e^{-\lambda t} w(\lambda)\, d\lambda, \qquad w(\lambda) = \frac{\operatorname{Re}\psi(\frac{1}{4} + \frac{i}{2}\sqrt{\lambda}) - \log\pi}{4\pi\sqrt{\lambda}} \tag{26.4.1}$$

(substitute $\lambda = r^2$). So $\mathcal{A}$ is the Laplace transform of a signed measure: an atom of weight one at $\lambda = -\frac{1}{4}$, coming from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, and a density $w$ that is negative for $\lambda < \lambda_* = r_*^2$ and positive above (Lemma 26.8.1). Here $r_*$ solves $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) = \log\pi$. Since $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) - \log\pi = 2\vartheta'(r)$ for the Riemann Siegel theta function $\vartheta$ (Section 5.1), $r_*$ is the critical point of $\vartheta$ (its minimum on $r > 0$). Numerically $r_* = 6.28984$ and $\lambda_* = 39.5620$ (computed, $r_* = 6.2898359888$; $r_* \approx 2\pi$ because $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) \approx \log(r/2)$). Only the digamma piece is a Gumbel Lévy density; the atom and the term in $\log\pi$ are not.

The $n$th derivative weights $w$ by $\lambda^n e^{-\lambda t}$, which peaks at $\lambda = n/t$. So $(-1)^n\mathcal{A}^{(n)}(t) \geq 0$ should hold while $n/t$ stays above $\lambda_*$, that is for $t \lesssim n/\lambda_*$ (heuristic, made precise in Proposition 26.8.2). Table 26.3 gives the first sign change $t_n$ of $(-1)^n\mathcal{A}^{(n)}$, found by bisection at 35 and 45 digits (computed, `code/wt_window.py`; the two runs agree). For $n = 1, \ldots, 6$ it is $0.025332, 0.051040, 0.076135, 0.101334, 0.126561, 0.151805$, within 1 percent of $n/\lambda_*$ with $\lambda_* = 39.5620$.

| order $n$ | first sign change $t_n$ of $(-1)^n\mathcal{A}^{(n)}$ | $n/\lambda_*$ | relative deviation |
|---|---|---|---|
| 1 | 0.025332 | 0.025277 | 0.22% |
| 2 | 0.051040 | 0.050554 | 0.96% |
| 3 | 0.076135 | 0.075830 | 0.40% |
| 4 | 0.101334 | 0.101107 | 0.22% |
| 5 | 0.126561 | 0.126384 | 0.14% |
| 6 | 0.151805 | 0.151661 | 0.095% |

Table 26.3: The first sign change $t_n$ of $(-1)^n\mathcal{A}^{(n)}$ against $n/\lambda_*$, $\lambda_* = r_*^2 = 39.5620$ (computed).

The law $t_n \approx n/\lambda_*$ holds to within 1% for $n \leq 6$. The relative deviation is 0.22% at $n = 1$ and 0.96% at $n = 2$ and decreases from $n = 2$ on; in absolute terms $t_n - n/\lambda_*$ is $0.55, 4.86, 3.05, 2.27, 1.77, 1.44$ in units of $10^{-4}$. The table tests the single inequality $k = n$. Since $t_1 < t_2 < \cdots < t_6$, the conditions of all orders $k \leq n$ hold together only on $(0, t_1]$, so $\mathcal{A}$ is completely monotone of order $n$ only up to $t_1$ and not up to $t_n$. For the full heat trace all six orders are positive at every $t$ tested from 0.002 to 0.2 (unchecked as a computation), as Theorem 5.1.13 proves for all $t$. The binding order is $n = 1$: the archimedean part alone is decreasing only up to $t_1 = 0.02533 \approx 1/(4\pi^2) = 0.025330$, and beyond that point $W$ decreases only because of the prime part, which at $t_1$ is still essentially the prime 2 alone, through $e^{-(\log 2)^2/4t_1} = 0.0087$ (computed, 0.0087253).

This gives a spectral reading under RH of the division of labour. The archimedean place supplies the positive spectral density above $\lambda_* \approx 4\pi^2$ and two defects: negative spectral mass below $\lambda_*$ and the atom at $-\frac{1}{4}$. Subtracting the Laplace representations of $\mathcal{A}$ and $W = \int e^{-\lambda t}U(d\lambda)$ gives

$\mathcal{P}(t) = \int e^{-\lambda t}\,[\delta_{-1/4} + w(\lambda)\,d\lambda - U(d\lambda)]$, so the prime part is not itself a positive Thorin measure; it cancels the defects and leaves the atoms of $U$ at $\lambda_j$, all of which lie above $\lambda_*$ since $\gamma_1^2 = 199.79$ (computed). Inside $t < t_n$ the single condition $(-1)^n \mathcal{A}^{(n)} \geq 0$ holds without the primes, which is a Gaussian analogue of the small support theorems.

## 26.5 Zeros from primes by Prony's method

In the cancellation regime $W(t) = \sum_j e^{-\lambda_j t}$ is a sum of decaying exponentials, so Prony's method applies, and the book uses it for exactly this purpose with the digit law below (Section 8.1.2, Tables 25.4 and 25.7). The difference here is that $W$ comes from (26.1.2) and (26.1.3), from the gamma function and the primes, with no evaluation of $\zeta$ and no zeros. We computed $W$ at 110 digits at $t = 0.02 + 0.006j$, $j = 0, \ldots, 15$, and fitted eight exponentials. The rule that terms with $\log n > \sqrt{4t \cdot 110 \ln 10}$ are below the working precision keeps the prime powers up to 2767 at $t = 0.062$ (403 primes, 438 prime powers) but up to 38 436 at the largest sample $t = 0.11$. Cutting at $n \leq 2767$ at every sample leaves an absolute error $1.2 \times 10^{-62}$ at $t = 0.11$ (against $W = 2.85 \times 10^{-10}$), and the fitted exponents $k \geq 4$ are unchanged by it, while the errors of $\gamma_1, \gamma_2, \gamma_3$ are then $4 \times 10^{-52}$, $5 \times 10^{-43}$ and $2 \times 10^{-34}$ (computed). So the recovery of Table 26.4 is valid from the 403 primes, and the 110 digit evaluation of $W$ itself needs the larger cutoff at the larger $t$.

| $k$ | $\widehat{\gamma}_k$ from primes | $\gamma_k$ | error |
|---|---|---|---|
| 1 | 14.13472514173469379 | 14.13472514173469379 | $1.3 \times 10^{-52}$ |
| 2 | 21.022039638771554993 | 21.022039638771554993 | $1.6 \times 10^{-41}$ |
| 3 | 25.010857580145688763 | 25.010857580145688763 | $2.9 \times 10^{-34}$ |
| 4 | 30.42487612585951321 | 30.42487612585951321 | $7.2 \times 10^{-24}$ |
| 5 | 32.935061587739189691 | 32.935061587739189691 | $3.0 \times 10^{-19}$ |
| 6 | 37.58617815884 | 37.58617815883 | $1.5 \times 10^{-11}$ |
| 7 | 40.9187208 | 40.9187190 | $1.8 \times 10^{-6}$ |
| 8 | 43.3279 | 43.3271 | $7.9 \times 10^{-4}$ |

Table 26.4: Zeros recovered by Prony's method from the heat trace computed from $\Gamma$ and the primes (computed, `code/wt_heat.py`, 110 digits; the source printed errors of $10^{-30}$ for $k \leq 3$, which was the floor of its reference, and the digits shown are those of the source).

That the explicit formula carries the zeros in its prime side is classical [163, 512]. The zeros are the exponential rates of $W$, the omitted prime atoms act as a bias on the fit, and the number of digits needed to see the $k$th zero grows like $(\gamma_k^2 - \gamma_1^2)t/\ln 10$ (heuristic, as in Section 8.1.2); this does not count the cost of evaluating $W$ itself.

## 26.6 The prime point process: first and second order

On the log scale, $u = \log y$, the $\Lambda$ weighted prime powers form a point process with counting function $\psi_{\mathrm{C}}(e^u)$. By the prime number theorem its intensity is $e^u$, which is the intensity $e^{-v}$ of the Gumbel point process with $v = -u$. In this heuristic reading the prime side of the explicit formula is, to first order, the reflected Gumbel process, and the weight $e^{-u/2}$ in (26.1.1) is the square root of that intensity, the normalisation under which a Poisson process with intensity $e^u$ has fluctuations of order one. This is an analogy for the cumulative counts and not an identification of the processes.

RH lives in the second order. By von Koch's theorem [359], RH holds if and only if $\psi_{\mathrm{C}}(y) = y + O(y^{1/2+\epsilon})$ for every $\epsilon > 0$, that is $(\psi_{\mathrm{C}}(y) - y)/y = O(y^{-1/2+\epsilon})$. The error changes sign infinitely often (it is $\Omega_{\pm}(\sqrt{y}\log\log\log y)$ by Littlewood), so this is a growth bound and not an index of regular variation. We compare the primes with the Cramér random model, which keeps $n$ independently with probability $1/\log n$.

- For $10^3 < y \le 10^7$ the normalised fluctuation $|\psi_{\mathrm{C}}(y) - y|/\sqrt{y}$ of the primes is at most 0.777 at integers (attained at $n = 1423$) and 0.803 as a supremum over real $y$ (computed, `code/wt_rig.py`, sieve to $2.1 \times 10^7$). The book quotes the explicit bound $0.94\sqrt{y}$ in Section 9.1.7.

- For the Cramér model the same maximum over 20 realisations lies between 5.7 and 12.6, with median 7.8 (computed); the source's single draw 11.1 is within this range.

The primes are far more rigid than a Poisson process with the same intensity at this scale. The size of $\psi_{\mathrm{C}}(y) - y$ at scale $y$ is set under RH by the sum $\sum 1/|\rho|$ over the low zeros and by Cramér's mean square theorem (Proposition 26.9.4), not by repulsion between pairs of zeros; pair correlation governs the variance in short intervals, treated in Section 26.8.3. Calling the rigidity the prime side counterpart of the repulsion of the zeros is a heuristic.

## 26.7 Does the extreme value approach offer insight?

It offers four things, each of which can be checked.

1. A common language for the two sides of the explicit formula. The archimedean place is $u$ times the Lévy density of $-\frac{1}{2}\log G_{1/4}$, a log gamma variable of fractional order in the family of Section 25.1; the primes are the atomic measure $\sum \Lambda(n) n^{-1/2} \delta_{\log n}$, which is $u$ times the formal Lévy measure of the zeta distribution on the critical line; and Weil positivity in Gaussian form is complete monotonicity of the compensated difference.

2. A map of where positivity is easy and where it is hard. For $t \leq 0.01$ it is inherited from the archimedean part, with the prime 2 a negligible extreme atom. For $t \gtrsim 0.05$ it is a statement about an exponentially small difference between two large quantities. This is consistent with the small support theorems of Yoshida and of Connes and Consani, which use no information on the zeros; with the verified zeros, the orders $n \leq 1.16 \times 10^{25}$ are known at every $t > 0$ (Theorems 5.1.1 and 5.1.13). The boundary of the archimedean part alone is computable: its $n$th derivative condition first fails at $t_n \approx n/\lambda_*$, fixed by where its spectral density changes sign.

3. A computational method. Prony's method on the heat trace computed from the primes recovers the zeros, with the precision cost made explicit (Section 26.5).

4. A heuristic point process reading of RH. To first order the primes on the log scale look like the reflected Gumbel process; RH is a second order statement that their fluctuations stay at square root scale; and numerically they are more rigid than a Poisson process.

What it does not offer is the missing estimate. In the cancellation regime positivity needs control of $\psi_{\mathrm{C}}(y) - y$ at square root scale, uniformly, which is RH in von Koch's form (heuristic), so the extreme value view reformulates the obstruction precisely but does not remove it. The two places where it could still help are the spectrum of complex indices of the second order term of $\psi_{\mathrm{C}}$ (Section 26.8.3) and the rigidity of the primes, which suggests looking for a structural reason, from the Euler product, why the prime process is less variable than Poisson.

## 26.8 Windows, primes and rigidity

### 26.8.1 The archimedean window: sign structure and asymptotics

By (26.4.1), $D_n(t) := (-1)^n \mathcal{A}^{(n)}(t) = (-\frac{1}{4})^n e^{t/4} + I_n(t)$ with $I_n(t) = \int_0^\infty \lambda^n e^{-\lambda t} w(\lambda)\, d\lambda$. Its sign structure is simple.

Lemma 26.8.1. *$w$ has exactly one sign change on $(0, \infty)$, from negative to positive, at $\lambda_* = r_*^2$, $r_* = 6.28984$, and $w'(\lambda_*) > 0$.*

*Proof.* With $z = \frac{1}{4} + \frac{ir}{2}$, $\frac{d}{dr} \operatorname{Re} \psi(z) = -\frac{1}{2} \operatorname{Im} \psi'(z)$ and $\psi'(z) = \sum_{k \geq 0} (z+k)^{-2}$ give $\operatorname{Im} \psi'(z) = -\sum_k (\frac{1}{4} + k)\, r\, |z+k|^{-4} < 0$ for $r > 0$. So $\operatorname{Re} \psi(\frac{1}{4} + \frac{ir}{2})$ increases strictly in $r$, from $\psi(\frac{1}{4}) = -4.227$ to $+\infty$, and crosses $\log \pi$ exactly once, with positive slope (this is the monotonicity of $m = 2\vartheta'$ of Section 5.1). The sign of $w$ is that of the numerator. □

Proposition 26.8.2 (Sign structure). *For every $n \geq 1$, $D_n(t) > 0$ for small $t > 0$. For odd $n$, $D_n$ has exactly one sign change on $(0, \infty)$, at a point $t_n$, and is negative*

*after it. For even $n$, $D_n$ has at most two sign changes, and is positive for large $t$; if it has two, the first is $t_n$ and the second lies where the atom $4^{-n}e^{t/4}$ overtakes the negative part of the density. As $n \to \infty$ a first sign change $t_n$ exists, and $t_n = n/\lambda_* + O(1)$.*

*Proof.* The measure $\lambda^n \mu(d\lambda)$, $\mu = \delta_{-1/4} + w(\lambda)\, d\lambda$, read along increasing $\lambda$, has the sign pattern $(-1)^n$ at $\lambda = -\frac{1}{4}$, then $-$ on $(0, \lambda_*)$ by Lemma 26.8.1, then $+$. That is one sign change for odd $n$ and two for even $n$. The Laplace kernel $e^{-\lambda t}$ is totally positive, so $D_n$ has no more sign changes in $t$ than the measure has in $\lambda$ (Karlin [188, Chapter 5]). As $t \to 0$ the density $w(\lambda) \sim (\log \lambda)/(8\pi\sqrt{\lambda})$ at infinity makes $I_n(t) \to +\infty$, so $D_n > 0$. As $t \to \infty$, $I_n(t) = O(t^{-n-1/2})$ (the density behaves like $c_0 \lambda^{-1/2}$ at zero) while the atom is $e^{t/4}4^{-n}$ in size, so for odd $n$ the atom is negative and $D_n < 0$ for large $t$, which gives exactly one sign change, and for even $n$ the atom is positive and $D_n > 0$ for large $t$, which gives zero or two changes. For the asymptotics, $I_n$ has exactly one zero: write $I_n = B_n - C_n$ with $B_n$ the integral over $(\lambda_*, \infty)$ and $C_n$ the absolute value of the integral over $(0, \lambda_*)$; the ratio $B_n/C_n$ has logarithmic derivative in $t$ equal to minus the difference of two mean values of $\lambda$, the first larger than the second, so it decreases strictly from $+\infty$ to 0. The location of this zero is computed in Proposition 26.9.1: it is $n/\lambda_* + O(1)$, and the atom, of relative size $O(\sqrt{n}b^n)$ with $b < 1$ near $t = n/\lambda_*$, does not move it by more than an exponentially small amount. So for large $n$, $D_n$ changes sign near $n/\lambda_*$, and this is the first change. □

The existence of a first sign change for even $n$ is therefore proved for $n$ large, and computed for $n = 2, 4, 6$. A scan of $D_n$ on $[0.005, 7]$ at 35 digits (computed) finds exactly one sign change for $n = 1$ and $n = 3$, at 0.025332 and 0.076135 (the entries of Table 26.3), and for $n = 2, 4, 6$ a second change at 1.3612, 3.2456 and 5.2669 (computed), as the proposition allows. In Table 26.3 the offset $t_n - n/\lambda_*$ is below $5 \times 10^{-4}$ for $n \le 6$ and decreases from $n = 2$ on, so the $O(1)$ term is small in practice.

### 26.8.2 Prime by prime: each extreme atom buys a window

Let $\mathcal{P}_S$ be the prime part restricted to the prime powers of a finite set of primes $S$ (all powers of its primes included), and let $t_n(S)$ be the first point where $(-1)^n(\mathcal{A} - \mathcal{P}_S)^{(n)}$ fails to be positive. Adding primes in increasing order adds the atoms $\log 2, \log 4, \log 8, \log 3, \ldots$ of the prime point process in groups. Since $\mathcal{A} - \mathcal{P}_S = W + \mathcal{P}_{\mathrm{rest}}$, the two differ by the omitted prime powers. The first omitted one, $q$, contributes about $\Lambda(q) q^{-1/2} g_t(\log q)$, of exponential size $e^{-(\log q)^2/4t}$, while $W$ and its derivatives decay like $e^{-\gamma_1^2 t}$. The two meet at

$$t(S) \approx \frac{\log q}{2\gamma_1}, \qquad q = \text{the smallest prime power not generated by } S. \tag{26.8.1}$$

This is a heuristic balance of exponents. Coefficients, derivative polynomials, the other zeros and the other omitted prime powers contribute corrections, and no uniform bound for them is given, so (26.8.1) is an empirical law for this truncation scheme and not a theorem. Read as an uncertainty principle, a Gaussian of width $\sqrt{2t}$ in $u = \log n$ resolves the first missing atom when its dual width resolves the lowest zero. Table 26.5 gives the error of this law as 3.2 percent for $S$ empty, about 1 percent from $q = 5$ on and 0.17 percent at $q = 101$.

Table 26.5 tests (26.8.1). We computed $t_n(S)$ by an upward scan to the first failure followed by bisection (computed, `code/wt_window.py`, 35 and 45 digits, scanned to 0.8).

| primes $S$ | first omitted $q$ | $\log q/2\gamma_1$ | $t_1(S)$ | $t_2(S)$ | $t_3(S)$ |
|---|---|---|---|---|---|
| none | 2 | 0.0245 | 0.0253 | 0.0510 | 0.0761 |
| 2 | 3 | 0.0389 | 0.0394 | 0.146 | 0.223 |
| 2, 3 | 5 | 0.0569 | 0.0575 | 0.476 | 0.0608 |
| $\le 5$ | 7 | 0.0688 | 0.0696 | $> 0.8$ | 0.0720 |
| $\le 7$ | 11 | 0.0848 | 0.0856 | $> 0.8$ | 0.0872 |
| $\le 13$ | 17 | 0.1002 | 0.1010 | $> 0.8$ | 0.1022 |
| $\le 31$ | 37 | 0.1277 | 0.1289 | $> 0.8$ | 0.1297 |
| $\le 97$ | 101 | 0.1633 | 0.1635 | $> 0.8$ | 0.1641 |

Table 26.5: End of the positivity window of order $n$ when only the primes in $S$ are kept (computed). The entries $> 0.8$ mean no failure up to the end of the scan.

Three facts emerge.

1. The law (26.8.1) holds for the order one window. Its error, as a percentage of $t_1(S)$, is 3.2% for $S$ empty ($q = 2$), between 0.8% and 1.3% for the next six rows (about 1% from $q = 5$ on), and 0.17% at $q = 101$.

2. Among the orders $1, 2, 3$ tested, order 1 always fails first. There is a mechanism for the sign: each omitted term $g_t(\log q)$ is increasing in $t$ for $t < (\log q)^2/2$, since $\partial_t \log g_t(u) = u^2/4t^2 - 1/2t$, so that $-W' \ge -(\mathcal{A} - \mathcal{P}_S)'$ there (Proposition 26.9.2(1)). For the higher derivatives $\partial_t^n g_t(u) = (-1)^n n!\, t^{-n} g_t(u) L_n^{(-1/2)}(u^2/4t)$ is positive once $u^2/4t$ exceeds the largest zero of the Laguerre polynomial $L_n^{(-1/2)}$, a threshold that grows with $n$; for a finite set of orders the omitted primes then subtract from the odd order conditions and add to the even ones on a small common interval. This accounts for the parity effect in the table ($t_2 > 0.8$ once the primes up to 5 are in) but not for a ranking of order 1 against every higher order, and for $S$ empty or $\{2\}$ the table has $t_2 < t_3$, so only order 1 is first. There is no fixed $t > 0$ at which every $t$ derivative of $g_t(u)$, $u > 0$, is nonnegative (otherwise its Taylor series at $t$, of radius $t$, would have a singularity at its

positive real radius). That order 1 binds for all orders is an observation on the computed orders.

3. Keeping the first factor $e^{-(\log q)^2/4t}$ below $e^{-\gamma_1^2 t}$ takes every prime power up to about $e^{2\gamma_1 t}$ in this balance, and each atom $\log q$ extends the order one window to about $\log q/2\gamma_1$. These are heuristic scales for this truncation scheme, not theorems of necessary information (a tail can sometimes be bounded without listing its atoms). The full condition on $(0,\infty)$ would need them all.

### 26.8.3 Rigidity of the prime point process

Goldston and Montgomery [153] proved that, under RH, a strong form of Montgomery's pair correlation conjecture for the zeros is equivalent to an asymptotic for the variance of $\psi_{\mathrm{C}}$ in short intervals. Montgomery and Soundararajan [218] refined the constant under a uniform Hardy and Littlewood prime tuples hypothesis, not under RH alone. The local prediction, the GUE prediction transported to the primes, is

$$V_\delta = \frac{1}{b-a}\int_a^b \frac{[\psi_{\mathrm{C}}((1+\delta)y) - \psi_{\mathrm{C}}(y) - \delta y]^2}{\delta y}\,dy \approx \log\frac{1}{\delta} + 1 - \gamma_E - \log 2\pi,$$

with $[a,b] = [10^6, 10^7]$ here, a normalised mean square (the source's ratio of the variance of the short interval error to $h = \delta y$ with $y$ random is this quantity). A Poisson process with the same intensity, such as the Cramér model, gives instead about $\log y - 1$, averaged over the range 14.37. The difference, $\log y$ against $\log(1/\delta)$, measures the rigidity. Table 26.6 compares them (computed, `code/wt_rig.py`: random points of $(10^6, 10^7)$ and a sieve to $2.1 \times 10^7$, two independent prime samples, standard errors 0.003 to 0.03 and the prime column the first sample; the Cramér column is one realisation with standard errors 0.03 to 0.05).

| $\delta$ | primes | GUE prediction | Cramér model (one realisation) | Poisson $\log y - 1$ |
|---|---|---|---|---|
| $10^{-5}$ | 10.13 | 10.10 | 14.40 | 14.37 |
| $10^{-4}$ | 7.71 | 7.80 | 14.53 | 14.37 |
| $10^{-3}$ | 5.32 | 5.49 | 14.60 | 14.37 |
| $3 \times 10^{-3}$ | 4.04 | 4.39 | 14.81 | 14.37 |
| $10^{-2}$ | 3.22 | 3.19 | 14.76 | 14.37 |
| $10^{-1}$ | 1.10 | 0.89 | 14.55 | 14.37 |

Table 26.6: $V_\delta$ for the primes, the GUE prediction, the Cramér model and the Poisson level (computed). The source printed $10.19, 7.69, 5.34, 4.06, 3.23, 1.10$ for the primes (all within two standard errors of these) and Cramér values 14.19 to 15.22 with a last entry 10.09, which the recomputation does not reproduce.

The primes follow the GUE prediction over four decades of $\delta$: the difference is 0.03, −0.09, −0.17, −0.35, 0.03, 0.21 at the six scales printed, so it is within 0.2 at four of the six and at most 0.35 at $\delta = 3 \times 10^{-3}$ (from the rounded table values). The Cramér model stays near the Poisson level. These finite numerical results are compared with a conjectural variance prediction. They do not establish GUE statistics for the prime point process, since a variance fixes second moments only, and they do not prove an unconditional variance asymptotic. A variance estimate of this kind is a statement under RH in its Goldston and Montgomery form, so by itself it cannot be cited to prove RH; a future unconditional variance bound strong enough could still contribute to a proof, and no impossibility theorem is claimed. That any proof of positivity in the cancellation regime would have to use this property is an opinion, not a result.

### 26.8.4 Second order regular variation

By the explicit formula, with symmetric truncation in the height, $\psi_{\mathrm{C}}(y) - y = -\sum_\rho y^\rho/\rho + O(\log y)$ for $y \geq 2$. Relative to the first order term $y$, the correction is a superposition of terms $y^{\rho-1}$ with complex indices $\rho - 1 = -\frac{1}{2} + \delta + i\gamma$, in the convention $\rho = \frac{1}{2} + \delta + i\gamma$ of Section 5.1.5. Classical second order regular variation (de Haan and Stadtmüller [108]) requires a limit of $(f(ty)/f(t) - y^\alpha)/A(t)$ with $A$ regularly varying, and handles one real second order index. For $f = \psi_{\mathrm{C}}$ no such limit exists: under RH, $(\psi_{\mathrm{C}}(ty) - ty)/\sqrt{t}$ is an almost periodic function of $\log t$, and the second order term is an oscillating sum with a whole spectrum of complex indices. What is true is that $\limsup_{y\to\infty} \log|\psi_{\mathrm{C}}(y) - y|/\log y = \sup \mathrm{Re}\,\rho$ unconditionally, and RH says that this exponent is $\frac{1}{2}$, which is von Koch's criterion. A theory of second order regular variation with complex index spectra would turn RH into the statement that the second order spectrum lies on a vertical line. This is recorded as a reformulation only.

## 26.9 Certificates, smoothing, energy and moments

### 26.9.1 The window law with explicit constants

Carrying Laplace's method in the proof of Proposition 26.8.2 one order further gives the constants of $t_n$. Let $w$ be smooth near its simple zero $\lambda_*$ (Lemma 26.8.1), of polynomial growth at infinity and integrable at zero, and put, uniformly for $c$ near $\lambda_*$,

$$J_n(c) = \frac{(n/c)^{n+1}}{n!} \int_0^\infty \lambda^n e^{-n\lambda/c} w(\lambda)\, d\lambda = \mathbb{E}\, w(cT_n),$$

where $T_n$ has the gamma law of shape $n+1$ and rate $n$. The moments about one are $\mathbb{E}(T_n - 1) = n^{-1}$, $\mathbb{E}(T_n - 1)^2 = n^{-1} + 2n^{-2}$, $\mathbb{E}(T_n - 1)^3 = 5n^{-2} + O(n^{-3})$ and $\mathbb{E}(T_n - 1)^4 = 3n^{-2} + O(n^{-3})$, and a Taylor expansion with gamma concentration

for the remainder gives $J_n(c) = w(c) + A_1(c)/n + A_2(c)/n^2 + O(n^{-3})$ with

$$A_1(c) = c\,w'(c) + \tfrac{1}{2}c^2 w''(c), \qquad A_2(c) = c^2 w''(c) + \tfrac{5}{6}c^3 w'''(c) + \tfrac{1}{8}c^4 w^{(4)}(c).$$

PROPOSITION 26.9.1 (Window law). *The first sign change of $D_n$ satisfies $t_n = n/\lambda_* + \beta + \delta/n + O(n^{-2})$, with*

$$\beta = \frac{1}{\lambda_*}\Big(1 + \frac{\lambda_* w''(\lambda_*)}{2w'(\lambda_*)}\Big), \qquad \delta = \frac{\alpha^2}{\lambda_*^3} - \frac{\eta}{\lambda_*^2},$$

*where $\alpha = -A_1(\lambda_*)/w'(\lambda_*)$ and $\eta = -\big[\tfrac{1}{2}w''(\lambda_*)\alpha^2 + A_1'(\lambda_*)\alpha + A_2(\lambda_*)\big]/w'(\lambda_*)$.*

*Proof.* Put $t = n/c$. Then $D_n(n/c)$ is, up to a positive factor, $J_n(c)$ plus the atom term $(-\frac{1}{4})^n e^{n/4c}$ divided by $n!(c/n)^{n+1}$, which is $O(\sqrt{n}b(c)^n)$ with $b(c) = e^{1+1/4c}/4c$. Since $\operatorname{Re}\psi(\frac{1}{4} + \frac{i}{2}) \le \frac{1}{5} - \gamma_E < \log\pi$ (the value is about $-0.88$) gives $\lambda_* > 1$, one has $b(\lambda_*) < 1$, and near $\lambda_*$ the atom, with its derivatives, is exponentially small. So the root $c_n = n/t_n = \lambda_* + \alpha/n + \eta/n^2 + O(n^{-3})$ is found by comparing powers of $1/n$ in $J_n(c_n) = 0$, which uses $w'(\lambda_*) > 0$ (Lemma 26.8.1) and the implicit function theorem for the remainder. Inverting $t_n = n/c_n$ gives $\beta = -\alpha/\lambda_*^2$ and the stated $\delta$. □

At $\lambda_*$ the book's scripts give $w' = 1.60238 \times 10^{-4}$ and $w'' = -8.1092 \times 10^{-6}$, so $\beta = -2.70691 \times 10^{-5}$ (computed), as in the source. The measured offsets $t_n - n/\lambda_*$ for $n = 2, \dots, 6$ are $4.86, 3.05, 2.27, 1.77, 1.44$ in units of $10^{-4}$, and $n(t_n - n/\lambda_* - \beta)$ is $1.03, 1.00, 1.02, 1.02, 1.03$ in units of $10^{-3}$ (computed: $1.026, 0.995, 1.015, 1.023, 1.027$). The formula above gives $\delta = 1.0365 \times 10^{-3}$ (evaluated from the formula in the review of the source; no script of the book evaluated it). The finite order fit $1.02 \times 10^{-3}$ is an effective coefficient and does not identify the asymptotic $\delta$; it differs from it by 2%, within what a further term would contribute. The case $n = 1$ lies outside the asymptotic regime ($n(t_1 - 1/\lambda_* - \beta) = 0.082 \times 10^{-3}$).

For even $n$ the second sign change of $D_n$ is at $t = 1.3612$, $3.2456$ and $5.2669$ for $n = 2, 4, 6$ (computed). Balancing the positive atom against the negative part of the integral with $w(\lambda) \approx c_0 \lambda^{-1/2}$ near 0, $c_0 = (\psi(\frac{1}{4}) - \log\pi)/4\pi = -0.427505$, gives the equation $4^{-n}e^{t/4} = |c_0|\Gamma(n + \frac{1}{2})t^{-n-1/2}$, with roots 1.98, 3.94 and 5.99 (computed). This captures the growth but overestimates, with relative error falling from 46% to 14% (45.7% and 13.7% at the ends). For large $n$ the second change is $a_* n + O(\log n)$ with $a_*/4$ the solution of $b + 1 + \log b = 0$, that is $a_* = 1.113858$ (unchecked by a script of the book, but $4W(1/e) = 1.11386$): at $t = an$ Laplace's method gives the negative integral as $w(1/a)\, n!\, t^{-n-1}(1 + O(1/n))$, and equating its exponential rate with that of $4^{-n}e^{t/4}$ gives $a/4 + 1 + \log(a/4) = 0$, with saddle $\lambda = 1/a_* = 0.8978$ where $w < 0$ (proved, as an asymptotic for the local balanced root; it does not replace the computed values for small $n$).

### 26.9.2 Certificates from finitely many primes

Proposition 26.9.2 (One sided certificates). *Let $q$ be the smallest prime power omitted from $\mathcal{P}_S$, so that $W = (\mathcal{A} - \mathcal{P}_S) - \mathcal{P}_{\mathrm{rest}}$, where $\mathcal{P}_{\mathrm{rest}}$ collects the omitted prime powers, all at least $q$.*

*(1) For $t < (\log q)^2/2$, $-W'(t) \geq -(\mathcal{A} - \mathcal{P}_S)'(t)$. Hence if $\mathcal{A} - \mathcal{P}_S$ is decreasing on $(0, X]$ with $X < (\log q)^2/2$, so is $W$, with no tail estimate needed.*

*(2) For even orders a tail bound is needed. Let $S$ be all prime powers up to $N_0$ and let $f(y) = y^{-1/2}\partial_t^2 g_t(\log y)$ be nonnegative and decreasing on $(N_0, \infty)$. Then, with $\psi_{\mathrm{C}}(y) < 1.03883\, y$ [465],*

$$W''(t) \geq (\mathcal{A} - \mathcal{P}_{\leq N_0})''(t) - 1.03883\Big(N_0 f(N_0) + \int_{\log N_0}^{\infty} e^{u/2}\partial_t^2 g_t(u)\, du\Big).$$

*The hypotheses on $f$ hold for every $0 < t \leq X$ as soon as $X \leq (\log N_0)^2/[2(5 + \sqrt{10})]$.*

*Proof.* For (1), $\partial_t \log g_t(u) = u^2/4t^2 - 1/2t > 0$ when $t < u^2/2$, so every term of $\mathcal{P}_{\mathrm{rest}}$ has positive $t$ derivative, and the differentiated prime sum converges locally uniformly. For (2), Stieltjes integration by parts gives for $f \geq 0$ decreasing and integrable with $yf(y) \to 0$ that $\sum_{m > N_0} \Lambda(m) f(m) = -\psi_{\mathrm{C}}(N_0) f(N_0) - \int_{N_0}^{\infty} \psi_{\mathrm{C}} f'\, dy \leq 1.03883\big(N_0 f(N_0) + \int_{N_0}^{\infty} f\, dy\big)$, and $\int_{N_0}^{\infty} f(y)\, dy = \int_{\log N_0}^{\infty} e^{u/2}\partial_t^2 g_t(u)\, du$. For the hypotheses put $z = (\log y)^2/4t$ and $Q(z) = z^2 - 3z + \frac{3}{4}$, so that $f(y) = y^{-1/2} g_t(\log y) t^{-2} Q(z)$; the $u$ derivative of $f$ has the sign of $\frac{u}{2t}(Q'(z) - Q(z)) - \frac{1}{2}Q(z)$ with $Q' - Q = -z^2 + 5z - \frac{15}{4}$, and both $f \geq 0$ and $f' \leq 0$ hold when $z \geq (5 + \sqrt{10})/2$. □

We evaluated the lower bound in (2) on the geometric grid $0.01 \cdot 1.08^k$ of $t$, at 30 digits, checking that $f$ is nonnegative and decreasing on a grid of $u$ of step 0.01 (computed, `code/wt_window.py`). The lower bound stays positive up to the points of Table 26.7, and the first failing grid points are 0.0799, 0.1597 and 0.2534, where the bound itself is negative. In Table 26.7 the order two windows are 0.0740, 0.1479 and 0.2346 for the prime powers up to 10, 100 and 1000.

| prime powers kept | order 2 window from the bound | law $\log q/2\gamma_1$ | order 1 window (Table 26.5) |
|---|---|---|---|
| $\leq 10$ | 0.0740 | 0.0848 ($q = 11$) | 0.0856 |
| $\leq 100$ | 0.1479 | 0.1633 ($q = 101$) | 0.1635 |
| $\leq 1000$ | 0.2346 | 0.2447 ($q = 1009$) | |

Table 26.7: Windows for $W'' > 0$ suggested by the bound of Proposition 26.9.2(2) from the primes up to $N_0$ with the Rosser and Schoenfeld tail (computed in floating point on a grid; not certified).

The windows follow the same law and fall short of it by 12.8%, 9.4% and 4.1%, the cost of the explicit tail. The inequalities of the proposition are proved; the table is not a certificate, because floating point positivity at grid points proves neither the signs of the computed quantities nor positivity between the points, and it does not cover $0 < t < 0.01$. A continuous certificate would partition $[\varepsilon, X]$ into intervals $I_i = [a_i, b_i]$, enclose the lower bound $B_N(c_i) \geq L_i$ at a point $c_i \in I_i$ and a bound $|B_N'| \leq M_i$ on $I_i$, and check $L_i > M_i \max(c_i - a_i, b_i - c_i)$; below $\varepsilon$ the book's Theorem 5.1.9(a) covers $(0, 0.019]$ for every order. For every order and every $t$ the theorem is Theorem 5.1.13, which uses the zeros. A related computed remark on the method of that theorem: its bound for part (a) vanishes at $t = 0.0193$ (Section 5.1.3), and a reader of that section found that replacing the proved envelope of the prime terms by the sharp envelope moves the root to about 0.022 (reported as 0.0218 with the supremum over $k \leq 300$ by a reader of that section; no script of the book reproduces it, unchecked), the limit then being set by $E_1$, which vanishes at 0.0261.

### 26.9.3 The heat trace as a Gaussian smoothing of the prime error

Let $D(u) = (\psi_{\mathrm{C}}(e^u) - e^u)e^{-u/2}$ be the normalised prime error on the log scale, so $D(0) = -1$.

Proposition 26.9.3. *For every $t > 0$ (without RH),*

$$W(t) = \Sigma_\infty(t) + \int_0^\infty D(u)\, k_t(u)\, du, \qquad k_t = \partial_u g_t - \tfrac{1}{2} g_t, \tag{26.9.1}$$

*with*

$$\Sigma_\infty(t) = \int_0^\infty g_t(u) e^{-u/2} du - g_t(0)\big(1 + \tfrac{1}{2}\log\pi\big) + \frac{1}{4\pi}\int e^{-tr^2}\,\mathrm{Re}\,\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big) dr.$$

*Proof.* Write $d\psi_{\mathrm{C}}(e^u) = e^u\, du + d(D\, e^{u/2})$ in (26.1.3) and integrate the second part by parts: $\mathcal{P} = \int_0^\infty g_t e^{u/2} du - g_t(0)D(0) - \int_0^\infty D\,(\partial_u g_t - \frac{1}{2} g_t)\, du$, the upper boundary vanishing by Gaussian decay and $\psi_{\mathrm{C}}(y) \ll y$. Since $\int_0^\infty g_t(e^{u/2} + e^{-u/2})\, du = \int_{\mathbb{R}} g_t e^{u/2} du = e^{t/4}$, subtracting $\mathcal{P}$ from (26.1.2) gives the claim. □

The source verified (26.9.1) against the zeros to 18 digits at $t = 0.01, 0.05, 0.1, 0.2$ (unchecked here). The smooth part $\Sigma_\infty$ is of order one and negative: $-3.20$, $-1.77$, $-1.33$, $-0.99$ at those points (unchecked here). So the Gaussian derivative smoothing of $D$ at scale $\sqrt{t}$ must cancel an explicit archimedean quantity to exponential accuracy. By the explicit formula $D(u) \approx -\sum_\rho e^{(\rho - 1/2)u}/\rho$, and the bilateral Gaussian transform $\int_{\mathbb{R}} g_t(u) e^{qu} du = e^{tq^2}$, for $q = \rho - \frac{1}{2} = i\gamma$, is the factor $e^{-t\gamma^2}$ of the heat trace, complex off the critical line; this explains the heat factor, although the half line integral in (26.9.1) also carries a derivative, boundary terms and the archimedean

terms, so it is not a pure Fourier multiplier acting on the zero series. Under RH $W(t) \sim m_{\gamma_1} e^{-\gamma_1^2 t}$ as $t \to \infty$, and an off line zero, with growth $e^{\delta u}$ in $D$, would spoil complete monotonicity. Integrating $W = \mathcal{A} - \mathcal{P}$ term by term is valid on a finite interval $(0, X]$; it must not be read as separate integration over $(0, \infty)$, since $\int_0^\infty g_t(u)\, dt = \infty$ and $\mathcal{A}$ contains $e^{t/4}$, and the convergent full integral comes from the cancellation in $\mathcal{A} - \mathcal{P}$.

### 26.9.4 Variance, energy and RH

Proposition 26.9.4 (Energy form of RH). *RH holds if and only if* $E_\epsilon = \int_0^\infty D(u)^2 e^{-2\epsilon u}\, du < \infty$ *for every* $\epsilon > 0$. *Under RH,*

$$\lim_{U \to \infty} \frac{1}{U} \int_0^U D(u)^2\, du = 2 \sum_{\gamma > 0} \frac{m_\gamma^2}{\frac{1}{4} + \gamma^2},$$

*the sum over distinct positive ordinates, with multiplicities* $m_\gamma$. *This equals* $\sum_\rho |\rho|^{-2} = 2 + \gamma_E - \log 4\pi = 0.0461914179$ *if all zeros are simple, and equals* $2 + \gamma_E - \log 4\pi + 2 \sum_{\gamma > 0} m_\gamma (m_\gamma - 1)/(\frac{1}{4} + \gamma^2)$ *in general.*

*Proof.* For $\operatorname{Re} s > 1$, $\int_0^\infty D(u) e^{-zu} du = \int_1^\infty (\psi_{\mathrm{C}}(y) - y) y^{-s-1} dy = -\frac{\zeta'(s)}{s\zeta(s)} - \frac{1}{s-1}$ with $s = \frac{1}{2} + z$, by the substitution $y = e^u$ and the Dirichlet series of $-\zeta'/\zeta$; the singularity at $s = 1$ cancels, and at a zero $\rho$ of multiplicity $m_\rho$ the residue is $-m_\rho/\rho \neq 0$. If $E_\epsilon < \infty$, then for $\sigma > \epsilon$ the Cauchy and Schwarz inequality gives $\int |D| e^{-\sigma u} du \le E_\epsilon^{1/2} / \sqrt{2(\sigma - \epsilon)}$, and with powers of $u$ the transform is holomorphic on $\operatorname{Re} z > \epsilon$, so no zero has $\operatorname{Re} \rho > \frac{1}{2} + \epsilon$; for every $\epsilon$ this gives RH by the functional equation. Conversely, under RH von Koch's bound $\psi_{\mathrm{C}}(y) - y = O(\sqrt{y} \log^2 y)$ gives $D(u) = O((1+u)^2)$ and $E_\epsilon < \infty$ for every $\epsilon > 0$; neither Paley and Wiener nor a mean square theorem is needed. For the limit, under RH $D(u) = -2 \operatorname{Re} \sum_{\gamma > 0} m_\gamma e^{i\gamma u} / (\frac{1}{2} + i\gamma) + o(1)$ in the $B^2$ sense, and averaging the square of a finite sum over distinct frequencies leaves the squared coefficients, which gives the formula (Cramér [97] for the case of simple zeros; the multiplicity sensitive form is in Akbary, Ng and Shahabi, arXiv 1306.1657, unchecked). The unconditional identity $\sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma_E - \log 4\pi$, with multiplicity, comes from the logarithmic derivative of the Hadamard product at 0 and at 1. □

Table 26.8 checks the energy constant from the primes up to $2 \times 10^7$, integrating $D^2$ exactly on each interval between integers (computed, `code/wt_rig.py`). The last column is the value of the sum under simple zeros, 0.046191 (computed from the identity); finite windows can be consistent with it without proving the limit or the simplicity of the zeros. The mean of $D^2$ is $0.0509, 0.0442, 0.0465, 0.0468$ on the windows $[5, 10], [8, 12], [10, 14], [12, 16.8]$ against 0.0462 for $\sum_\rho |\rho|^{-2}$.

This settles the variance question raised by the rigidity of Section 26.8.3 in three parts.

| window in $u = \log y$ | mean of $D^2$, primes | Cramér model (one realisation) | $\sum_\rho \lvert\rho\rvert^{-2}$ |
|---|---|---|---|
| $[5, 10]$ | 0.0509 | 5.19 | 0.0462 |
| $[8, 12]$ | 0.0442 | 3.43 | 0.0462 |
| $[10, 14]$ | 0.0465 | 2.93 | 0.0462 |
| $[12, 16.8]$ | 0.0468 | 5.14 | 0.0462 |

Table 26.8: Energy of the normalised prime error on the log scale (computed). The source printed 0.0506 for the first window and 7.54, 9.78, 9.14, 7.15 for a Cramér realisation; the Cramér values are heavy tailed and one realisation does not support a growth rate.

1. The short interval variance of Goldston and Montgomery is a statement under RH, so it cannot be used by itself to prove RH.

2. A bound on the energy alone cannot prove $W(t) \geq 0$ at a given $t$ when $\Sigma_\infty(t) < 0$. If $E_\epsilon \leq M^2$, weighted Cauchy and Schwarz gives $\lvert \int D k_t \rvert \leq M K_{t,\epsilon}$ with $K_{t,\epsilon}^2 = \int_0^\infty k_t^2 e^{2\epsilon u} du$, so $W(t) \in [\Sigma_\infty - M K_{t,\epsilon}, \Sigma_\infty + M K_{t,\epsilon}]$. A small enough norm bound could certify a negative sign, and the limitation is exact for norm information: the two functions $D_\pm = \pm M e^{2\epsilon u} k_t / K_{t,\epsilon}$ have the same discounted energy and opposite extreme pairings. The norm depends on $t$ and is not uniformly of order one.

3. The global energy of $D$ is equivalent to RH (Proposition 26.9.4).

In the language of the second order term, RH says that $D$ has finite discounted energy on the log scale, and under RH its energy per unit of $u$ is the sum of Proposition 26.9.4. The Cramér model has finite discounted energy too, almost surely for every $\epsilon > 0$ at once: with $\Psi_C(y) = \sum_{3 \leq n \leq y} B_n \log n$ and $B_n$ Bernoulli with parameter $1/\log n$, one has $\mathbb{E}\Psi_C(y) = y + O(1)$ and $\operatorname{Var} \Psi_C(y) = y \log y - 2y + O(\log y)$, so $\mathbb{E} D_C(u)^2 = O(1 + u)$ and Tonelli gives $\mathbb{E} \int D_C^2 e^{-2\epsilon u} du < \infty$. What it lacks is a bounded mean square per unit of $u$: that grows like $u$ in expectation. That this reflects the Euler product is a heuristic; the energy equivalence for the primes uses the zeta specific transform and is not a characterisation of Euler products.

### 26.9.5 The reciprocal Xi law from primes

Under RH the reciprocal $\Xi$ law $T = \sum_j E_j / \lambda_j$ has Lévy measure $W(x)\, dx/x$ and Thorin measure $U = \sum_j \delta_{\lambda_j}$, so its cumulants are $\kappa_j = \int_0^\infty x^{j-1} W(x)\, dx = (j-1)!\, s_j$ with $s_j = \sum_k \lambda_k^{-j} = \sum_{\gamma > 0} \gamma^{-2j}$. (The $(j-1)!$ appears once: $\int_0^\infty x^{j-1} e^{-\lambda x} dx = (j-1)! \lambda^{-j}$.) The integral identity $\int x^{j-1} W\, dx = (j-1)!\, s_j$ needs no RH: $\operatorname{Re} \lambda_j > 0$ follows from $\lvert \operatorname{Im} \gamma \rvert < \frac{1}{2}$ and the verification of RH to a finite height [419], and Fubini applies since $\sum_\gamma \int x^{j-1} \lvert e^{-\gamma^2 x} \rvert dx =$

$(j-1)!\sum(\operatorname{Re}\gamma^2)^{-j}<\infty$. Each $s_j$ can therefore be computed from $\Gamma$ and the primes alone. For a finite $X$, $\int_0^X g_t(u)\,dt=\sqrt{X/\pi}\,e^{-u^2/4X}-\frac{u}{2}\operatorname{erfc}(u/2\sqrt{X})$ for $u\ge0$, and the archimedean part integrates in closed form against $e^{t/4}$ and $g_t(0)$, leaving one integral against $\operatorname{Re}\psi(\frac14+\frac{ir}{2})$. The source cut at $X=0.25$ and kept the prime powers up to 4000; at 35 digits it printed the following (unchecked here, not recomputed in this book).

| | $\Gamma$ and primes $\le 4000$ | from $\xi$ at $\frac12$ |
|---|---|---|
| $s_1=\sum\gamma^{-2}$ | 0.023104993115418970788932 79 | 0.023104993115418970788933 81 |
| $s_2=\sum\gamma^{-4}$ | 0.000037172599285269686164 61 | 0.000037172599285269686168 87 |

The right column uses $\xi(\frac12+z)=\xi(\frac12)\prod(1+z^2/\gamma^2)$, so $s_1=\xi''(\frac12)/2\xi(\frac12)$ and $s_2=s_1^2-\xi^{(4)}(\frac12)/12\xi(\frac12)$; these are analytic coefficients, and calling them the mean and variance of $T$ is conditional on RH. The columns differ by $1.02\times10^{-24}$ and $2.6\times10^{-25}$, which are the omitted heat tails beyond $X=\frac14$: the first zero alone gives $\int_X^\infty W\ge e^{-\gamma_1^2X}/\gamma_1^2=1.0173\times10^{-24}$, so the cut is not below $10^{-24}$ as the source stated, and for $j=2$ the first zero gives $e^{-\gamma_1^2X}(X/\gamma_1^2+1/\gamma_1^4)=2.594\times10^{-25}$. The relative differences are $4\times10^{-23}$ and $7\times10^{-21}$, so the agreement is to 23 and 20 significant digits, not 24 for both. The omitted prime tail, bounded by the partial summation of Proposition 26.9.2 applied to $G_X(u)=\int_0^X g_t(u)\,dt$ (nonnegative and decreasing in $u$), is about $2\times10^{-31}$ at $N=4000$ (unchecked) and is far smaller.

This is a consistency check and not evidence for RH. What RH adds is that the atoms $\lambda_j^{-1}$ are real and positive. That is a moment problem. Put $m_j=s_{j+1}=\Gamma(j+1)^{-1}\int_0^\infty x^jW\,dx=\kappa_{j+1}/j!$, the normalised cumulants, and $\nu=\sum_j\lambda_j^{-1}\delta_{\lambda_j^{-1}}$, so that $m_j=\int t^j\,\nu(dt)$ and $|m_j|\le SR^j$ with $S=\sum|\lambda_j|^{-1}$ and $R=\sup|\lambda_j|^{-1}$.

Proposition 26.9.5 (Hankel form of RH). *RH holds if and only if the Hankel matrices $H_d=(m_{i+j})_{0\le i,j\le d}$ and $H_d^{(1)}=(m_{i+j+1})_{0\le i,j\le d}$ are positive semidefinite for every $d\ge0$.*

*Proof.* Under RH $\nu$ is a positive measure and the two quadratic forms are $\int|p|^2d\nu$ and $\int t|p|^2d\nu$. Conversely, Stieltjes' theorem gives a positive measure $\mu$ on $[0,\infty)$ with these moments; from $b^{2k}\mu([b,\infty))\le m_{2k}\le SR^{2k}$ it is supported on $[0,R]$. Near $z=0$, $M(z)=\int(1-zt)^{-1}\mu(dt)=\sum_j m_jz^j=\sum_k 1/(\lambda_k-z)$. The integral is holomorphic off $[1/R,\infty)$, while the spectral series is meromorphic, converging normally away from its poles, each of which has residue minus its positive integer multiplicity, so no cancellation occurs. Analytic continuation forces every $\lambda_k$ to be positive, and then $\operatorname{Re}\gamma_k>0$ with $\gamma_k^2>0$ gives $\gamma_k$ real, which is RH. □

Semidefiniteness means all quadratic forms, equivalently all principal minors, and a computation of some leading determinants is not a replacement for it. Each entry is a finite combination of integrals of $\mathcal{A}-\mathcal{P}$ against powers of $t$, convergent at the Gaussian rate in the primes. This is a criterion of the type of the Keiper

and Li criteria and of Theorem 5.1.3, which has Hankel matrices at one heat time; here the entries are moments of the Thorin side. The extreme value structure of $W$ near 0 (Section 26.3) is what makes the integrals converge rapidly.

## 26.10 The sine squared lemma and a single heat time

The Thorin condition of Theorem 26.2.1 asks for every derivative of $W$ at every $t$. The sine squared lemma (1.2.9), in its Lévy and Khintchine form $1-\cos\theta = 2\sin^2(\theta/2)$ (valid for every complex $\theta$), trades that ladder for a one parameter family at a single heat time. The tent version used by Yoshida's window is Theorem 17.6.2.

### 26.10.1 The shifted functional

For $t>0$ and $a\in\mathbb{R}$ put

$$f_t(a)=\sum_j e^{-\lambda_j t}\cos(a\gamma_j), \qquad V(t,a)=W(t)-f_t(a)=\sum_j 2\sin^2\!\big(\tfrac{a\gamma_j}{2}\big)e^{-\lambda_j t},$$

with the indexing of the introduction ($\operatorname{Re}\gamma_j>0$, multiplicities counted). For every complex $z$, $1-\cos az=\frac12(1-e^{iaz})(1-e^{-iaz})$. With $F(z)=(1-e^{iaz})e^{-tz^2/2}/\sqrt{2}$ and $F^*(z)=\overline{F(\bar z)}$, $F(z)F^*(z)=(1-\cos az)e^{-tz^2}$, which is $|F(z)|^2$ on the real axis. So $V$ is the Weil functional of the even part of an admissible test function $FF^*$ (it decays like a Gaussian on every strip). At a nonreal zero the summand need not be nonnegative (at $\gamma=ib$ the weight $2\sin^2(a\gamma/2)$ equals $-2\sinh^2(ab/2)<0$), so termwise nonnegativity holds only under RH. The explicit formula (26.1.1) with $h(r)=e^{-tr^2}\cos(ar)$, for which $g(u)=\frac12\big(g_t(u-a)+g_t(u+a)\big)$, gives

$$f_t(a)=e^{t/4}\cosh\tfrac{a}{2}-\tfrac12 g_t(a)\log\pi+\frac{1}{4\pi}\int_{\mathbb{R}} e^{-tr^2}\cos(ar)\,\operatorname{Re}\psi\big(\tfrac14+\tfrac{ir}{2}\big)\,dr \\ -\frac12\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\big(g_t(\log n-a)+g_t(\log n+a)\big). \quad (26.10.1)$$

At $a=0$ this is $W=\mathcal{A}-\mathcal{P}$. For $a>0$ the shift moves each prime power atom from $\log n$ to $\log n-a$, so the primes near $e^a$ are brought to the centre of the Gaussian, while the pole term grows like $\frac12 e^{t/4}e^{a/2}$ and must be cancelled by them. That cancellation is the prime number theorem near $y=e^a$, read through a Gaussian of width $\sqrt{2t}$ in $\log y$.

Theorem 26.10.1 (One heat time suffices). *Fix $t>0$. RH holds if and only if $V(t,a)\ge0$ for every $a\ge0$.*

*Proof.* Under RH every $\gamma_j$ is real and each term of $V$ is nonnegative. Conversely let $V(t,\cdot)\ge0$ and suppose a zero is off the line, so that some $\gamma_j=\tau_j+i\beta$ has

$\beta \neq 0$; its conjugate occurs as well, so we may take $\beta > 0$. Since $|\operatorname{Im}\gamma| < \frac{1}{2}$, $|\cos(a\gamma)| \le e^{a/2}$ for $a \ge 0$, and $\sum_j |e^{-\gamma_j^2 t}| < \infty$ by the zero counting bound, the Laplace transform

$$L(s) = \int_0^\infty e^{-sa} V(t,a)\, da = \frac{W(t)}{s} - \sum_j e^{-\gamma_j^2 t} \frac{s}{s^2+\gamma_j^2}$$

holds for $\operatorname{Re} s > \frac{1}{2}$. On each compact set avoiding the candidate poles the series converges normally (the summands are $O(e^{-\tau^2 t}/\tau^2)$), so the right side is meromorphic on $\mathbb{C}$, with poles at $s = \pm i\gamma_j$, and a possible pole at $s = 0$. The residue at a pole is $-\frac{1}{2}$ times the multiplicity times $e^{-\gamma^2 t}$, which is nonzero, and distinct ordinates with the same pole coincide, so poles do not cancel. The point $s = -i\gamma_j = \beta - i\tau_j$ is a genuine pole of real part $\beta > 0$, so the abscissa of convergence $\sigma_c$ of the transform is at least $\beta > 0$. Because $V \ge 0$, Landau's theorem says that $L$ is singular at the real point $\sigma_c$: if $L$ were analytic there, choose $s_0 = \sigma_c + d$ with $d > 0$ small, so that the Taylor disk at $s_0$ has radius larger than $d$; its coefficients $(-1)^k L^{(k)}(s_0)/k! = \int a^k e^{-s_0 a} V\, da/k!$ are nonnegative, and Tonelli sums the series at a real point $s_0 - h$ with $h > d$ inside the disk to $\int e^{-(s_0-h)a} V\, da < \infty$, contradicting the definition of $\sigma_c$. The only real candidates for singularities of the right side are $s = 0$ and the points $s = \pm i\gamma_j$ with $\gamma_j$ purely imaginary. The latter cannot occur: $\operatorname{Re}\gamma_j > 0$ by the indexing, which is legitimate because $\xi$ has no real zeros ($\xi(s) > 0$ for real $s$: for $0 < s < 1$ the alternating series gives $\eta(s) > 0$, hence $\zeta(s) < 0$ while $s(s-1) < 0$, for $s > 1$ the factors are positive, for $s < 0$ use the functional equation, and $\xi(0) = \xi(1) = \frac{1}{2}$). And $\sigma_c \ge \beta > 0$ excludes $s = 0$. This contradicts the existence of the real singularity, so every $\gamma_j$ is real. □

Where complete monotonicity used every order $n$ at every $t$, Theorem 26.10.1 uses one $t$ and every shift $a$, and the shift plays the role of the order. By (26.9.1), $f_t(a)$ is the Gaussian smoothing at scale $\sqrt{t}$ of the prime error $D$ near $u = a$, plus explicit archimedean terms, so the theorem is a one sided bound on a smoothed $D$ that is uniform in $a$. That is the quantity between pointwise and energy bounds left open in Section 26.9. It is close in spirit to classical facts about smoothed sums over zeros, and it is presented as a reformulation. The tent version Theorem 17.6.2 is the compactly supported form of the same lemma.

### 26.10.2 Numerical check

The source took $t = 0.02$, the first 120 zeros and the prime powers up to 5000, at 30 digits (unchecked here; an independent calculation reproduced the six values to the printed digits from the first eight ordinates). At $t = 0.02$ the values from the zeros and from $\Gamma$ and the primes agree at every printed digit, for example $V(t, 0.3) = 0.0267570868155$ and $V(t, \log 2) = 0.0357321878430$.

| $a$ | $V(t,a)$ from zeros | $V(t,a)$ from $\Gamma$ and primes, (26.10.1) |
|---|---|---|
| 0.3 | 0.0267570868155 | 0.0267570868155 |
| $\log 2$ | 0.0357321878430 | 0.0357321878430 |
| $\log 3$ | 0.0367065917181 | 0.0367065917181 |
| 1.5 | 0.0313514051449 | 0.0313514051449 |
| $\log 8$ | 0.0264484602913 | 0.0264484602913 |
| 3.0 | 0.0185315145626 | 0.0185315145626 |

The source reported a minimum of $V/W$ of $4.5 \times 10^{-4}$ on a grid of 1600 points in $0 < a \le 8$ (unreproduced; on the grid $a = 8k/1600$ an independent calculation gives about $5.6 \times 10^{-4}$ at $a = 0.89$, and other grids give other values, none a continuous minimum). At this $t$ the lowest zero carries 99.2% of $W$ (0.99198; the source's 99.9% is not possible, because the second zero alone brings the share below 0.99218). The local minima sit near $a \approx 2\pi k/\gamma_1 = 0.4445k$ and are deep for even $k$ (ratio about $4.9 \times 10^{-4}$ at $k = 2$ and $4.2 \times 10^{-4}$ at $k = 4$, unchecked) and shallow for odd $k$ (about 0.016, unchecked), since $2\gamma_2/\gamma_1 = 2.975$ is close to an integer. For smaller $t$ at a fixed positive shift the prime power resonance $a_0 = \log m$ gives a peak, not a dip: $V(t, a_0) = W(t) + \Lambda(m)/(4\sqrt{\pi m t}) + O(1)$ as $t \downarrow 0$.

To see the criterion fail, the source moved $\gamma_1$ off the line, replacing it by the pair $\gamma_1 \pm i\beta$ and keeping the other zeros. The pair contributes $2\,\mathrm{Re}[c(1-\cos(a(\gamma_1+i\beta)))]$ with $c = e^{-t(\gamma_1+i\beta)^2}$, whose leading oscillation at large $a$ is $-e^{-t(\gamma_1^2-\beta^2)}e^{\beta a}\cos(a\gamma_1 + 2t\gamma_1\beta)$. For $\beta = 0.05$ the minimum of $V$ over $0 < a \le 200$ is of the order of $-4\times10^2$ (the source printed $-391$ on a coarse sample; the single shift $a \approx 199.588$ gives $-396.8$ in an independent calculation), and for $\beta = 0.2$ it is about $-4 \times 10^{15}$, both near $a = 200$ (unchecked here). The off line pair grows like $e^{\beta a}$ while oscillating, and the sine squared weights can no longer hold it. This is an experiment on one finite model, not a proof of a general domination.

### 26.10.3 What it costs

The prime side of $V(t,a)$ is governed by the prime powers near $e^a$ in $\log n$. Write $V = V_N - \sum_{m>N} \Lambda(m) m^{-1/2} H_{t,a}(\log m)$ with $H_{t,a}(u) = g_t(u) - \frac{1}{2}[g_t(u-a) + g_t(u+a)] = g_t(u)\big[1 - e^{-a^2/4t}\cosh(au/2t)\big]$ and $V_N$ the same expression with the sum truncated at $N$ and the archimedean part written out. For $a > 0$, $H_{t,a}(u) \le 0$ for $u \ge u_0(t,a) = (2t/a)\,\mathrm{arcosh}(e^{a^2/4t})$, so $\log N \ge u_0$ implies the rigorous one sided bound $V \ge V_N$; for $0 \le a \le A$ the condition $\log N \ge A + \sqrt{2t}$ suffices, by convexity of $g_t$ on $[\sqrt{2t}, \infty)$. At $a = 0$, $V = 0$ exactly, so a positive uniform margin is impossible, and the quotient $V(t,a)/a^2$, whose value at 0 is $-W'(t)/2$, is what a certificate must enclose. This is a proved extension of the certificate argument, with the tail hypotheses to be checked as in Proposition 26.9.2; where the sign condition is unavailable an absolute tail majorant with partial summation is used. The number of primes needed is not simply "up to $e^{a+O(\sqrt{t})}$": to keep a fixed

small absolute tail $\eta$ at large $a$, completing the square gives $\int_L^\infty e^{u/2} g_t(u-a)\,du = \frac{1}{2} e^{a/2+t/4} \operatorname{erfc}\big((L-a-t)/2\sqrt{t}\big)$ and so $L = \log N = a + t + O(\sqrt{t(a+\log(1/\eta))})$ in the bound of that type, and a one sided certificate that drops a helpful signed tail can use a different cutoff. For large $a$ the condition is square root control of $\psi_{\mathrm{C}}$ near $y = e^a$ in a smoothed one sided sense, which is not literally von Koch's pointwise estimate. So the lemma collapses the derivative ladder and turns the Thorin condition into a positive Weil functional at one heat time. The obstruction is moved from large $t$ to large $a$, but it is not removed. The criterion is local in $a$: a failure of RH must show up as a negative value of a single explicit function of one variable.

## 26.11 Is there a path to RH?

The chapter gives four exact forms of RH, each computable from $\Gamma$ and the primes, and one identity.

1. complete monotonicity of $\mathcal{A} - \mathcal{P}$ (Theorem 26.2.1);
2. finite discounted energy of $D$ on the log scale (Proposition 26.9.4);
3. positive semidefiniteness of the two Hankel families of the normalised cumulants of the reciprocal $\Xi$ law (Proposition 26.9.5);
4. nonnegativity of the sine squared functional $V(t,\cdot)$ at a single heat time (Theorem 26.10.1).

The identity (26.9.1) writes $W$ as a Gaussian smoothing of $D$ and is a restatement of the first form; the source listed the nonnegativity of $W$ as a fifth exact form, but $W > 0$ holds unconditionally (Theorem 5.1.1), and positivity of an exponential sum does not force its exponents to be real (the ordinates $1, 2 \pm \frac{i}{4}$ give $e^{-x} + 2e^{-63x/16}\cos x > 0$ for all $x > 0$). Complete monotonicity does give the converse, by the pole argument of Theorem 26.2.1.

The forms split by regime. For small $t$ the position is this. The archimedean part alone is positive to the first sign changes $t_n \approx n/\lambda_*$ (Proposition 26.8.2, proved). This transfers to $W$ at order 1 (Proposition 26.9.2(1), proved). The window obtained from finitely many primes follows the law $\log q/2\gamma_1$ (computed, a heuristic balance, Table 26.5), and Proposition 26.9.2(2) suggests windows at order 2 (computed on a grid, Table 26.7). For every order and every $t$ the theorems are those of Chapter 5, Theorems 5.1.9 and 5.1.13, which use the verified zeros, except part (a) of the first. This is the Gaussian counterpart of the small support results of Yoshida and of Connes and Consani. To prove RH by the prime side one would need the window for every order and every $t$, and the law suggests the cost: reaching $t$ takes prime powers up to about $e^{2\gamma_1 t}$, which is infinite as $t \to \infty$

(heuristic). A finite prime side certificate covers a finite window and no more, whereas the zero side covers all $t$ at orders up to $1.16 \times 10^{25}$.

In the cancellation regime the four forms are heuristically controlled by one estimate, $\psi_{\mathrm{C}}(y) - y$ at square root scale, in forms with different quantifiers: the energy form says it exactly, and a pointwise, a discounted square or a one sided smoothed bound are different statements that the zeta specific transforms relate. The smoothing identity also shows that no bound of norm type alone can prove $W(t) > 0$ at a single $t$ where $\Sigma_\infty(t) < 0$. Here the extreme value view gives a precise map of the obstruction, not a way around it. Two leads remain, both structural. The first is the rigidity of the prime point process, which follows the GUE variance and not the Poisson level; whether the Euler product forces this rigidity is open, and a proof that it does would prove more than RH (heuristic). The second is the Hankel form, which turns RH into the positivity of minors built from the integrals $\int t^j(\mathcal{A} - \mathcal{P})\,dt$. Like the Keiper and Li criteria it is an equivalence and not a reduction, but on the Thorin side its entries converge at the Gaussian rate in the primes and its early minors can be computed to high order. None of this proves RH.

## 26.12 Open problems

Problem 26.1. Make the windows of Table 26.7 rigorous by continuous interval enclosure, with an analytic bound below the first grid point, extend Proposition 26.9.2 to all orders up to some $K$, and push the certified window as far as the primes up to $10^6$ allow, where the law predicts about 0.49.

Problem 26.2. Carry out the shifted certificate of Section 26.10 (the enclosure of $V(t, a)/a^2$ on $0 \le a \le \log N_0$ from the primes up to $N_0$), and find the best heat time $t$ for such certificates.

Problem 26.3. Compute the principal minors of the two Hankel families of Proposition 26.9.5 from the primes to high order, and find which features of the prime point process control their sign. Do the same for Devroye's sign test and the mean excess of the reciprocal $\Xi$ law (Section 25.10).

Problem 26.4. Give a structural reason, from the Euler product, for the GUE rigidity of Table 26.6, or a theory of second order regular variation with a spectrum of complex indices in which RH is the statement that the spectrum lies on a vertical line.

The first open problem of the source, the $1/n$ coefficient $\delta$ of the window law, is answered by Proposition 26.9.1.

## Notes

The chapter follows the source chapter by Nick Polson [248]. Guinand and Weil are credited for the explicit formula, Bombieri for the quadratic functional, Yoshida and Connes and Consani for the small support positivity, Goldston and Montgomery and Montgomery and Soundararajan for the short interval variance, Cramér for the mean square, von Koch for the pointwise criterion, Rosser and Schoenfeld for the bound on $\psi_{\mathrm{C}}$ and de Haan and Stadtmüller for second order regular variation. The converse in Theorem 26.2.1, Theorem 26.10.1 and Proposition 26.9.5 is by the pole argument for a resolvent and by Landau's theorem for nonnegative functions, in the form of the review of the source. The figures of the source are not reproduced; the scripts are `code/wt_heat.py`, `code/wt_window.py` and `code/wt_rig.py`.

# Chapter 27

# The Comtet Majumdar family and the uniform law

Comtet and Majumdar [86] proved, in an appendix on the maximum of a random walker, an identity between an integral of the logarithm of $(1-e^{-k^\mu})/k^\mu$ and the value $\zeta(1/\mu)$ of the Riemann zeta function, and they proved it through a cutoff regularisation. This chapter asks what that identity and that regularisation can do for the programme of the book. The programme is to approximate $\Xi$ by entire functions whose reciprocals are already Laplace transforms of generalised gamma convolutions (GGC), and then to pass to the limit. The answer has two halves. On the arithmetic side the answer is negative: the shifted form of the regularisation of Comtet and Majumdar, and every decreasing damping of the coefficients of the theta series, produces functions that are not in the Laguerre Pólya class, for structural reasons that are proved below. On the Thorin side the same shift, read as a tilt of a Lévy measure, gives a positive object: for the uniform law the Thorin density of the tilted law is strictly positive for every positive tilt, and its Mellin transform is the regularised zeta function of the family. The uniform law is therefore a solved model of the question that the book asks for $\xi$ in Chapter 9, where the corresponding positivity is proved for every basepoint $\alpha \geq 1$.

The chapter develops a source note by Nick Polson [229], together with the computations done for this book. Its numbers were recomputed here by a script written from the mathematics alone (the scripts of the source were not used; the script of this chapter is `code/cg_checks.py`, with its output in `data/cg_checks_*.json` and the logs beside them, listed in `code/README.md`). Every number printed below is one of those recomputations unless it is marked unchecked, and where the recomputation differed from the source the recomputed value is printed. The tables of zeros of the damped kernels (Section 27.5) were not recomputed and are marked unchecked. The notation is the book's, with the following warnings. The kernel $\Phi$ is the kernel of (1.2.3), so that $\Xi(t) = \int_{\mathbb{R}} \Phi(u)\cos(tu)\,du$, and the sum over $n$ inside $\Phi$ carries the terms $\Phi_n$ of Section 27.5. The damping parameter is $\delta$ (elsewhere in the book $\delta$ may denote

an off line displacement, which does not occur here). The Poisson kernel of the book is $P_a$ with the factor $1/\pi$ (Chapter 2); the unnormalised Lorentzian of this chapter is $\ell_\varepsilon(x) = \varepsilon/(\varepsilon^2 + x^2) = \pi P_\varepsilon(x)$. The Thorin density of the uniform law is $\kappa_\delta$; for $\xi$ the book's $D_\alpha$ of Chapter 16 plays the role of $\kappa_\alpha^\xi$. The constant of Comtet and Majumdar is written $C_\mu$ (the letter $\gamma$ is kept for ordinates of zeros), the exponential variables are $e_k$ as in Chapter 2, and the integer part is written $\lfloor t \rfloor$ (the letter $N$ is kept for zero counts). The de Bruijn Newman flow is written with the time $t$ and the kernel $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$ of Section 8.4, not with the parameter $\lambda$ of the source, whose normalisation differs from the literature (Section 27.6).

Several parts of the source are already in the book and are cited, not restated. The Thorin form of the reciprocal $\xi$ function with a basepoint, its Lévy form, the compensated density $v_\alpha - b_\alpha$ and its positivity for every $\alpha \geq 1$ (Proposition 9.1.10, Proposition 9.1.12, Theorem 9.1.18, Remark 9.1.20) replace the whole of the source's section on $\xi$, including its theorem on the threshold $\alpha_1 = 7.7892$, its numerical tables for $\alpha \in (1, \alpha_1)$ and its first open problem, which Proposition 9.1.12 settles. The equivalence of RH with positivity of the uncompensated density for all $\alpha > \frac{1}{2}$ is Corollary 16.12.10 and Corollary 16.12.11 (Chapter 16), and Lagarias's criterion is cited from there. The $2\sin^2$ lemma is (1.2.9) and Lemma 6.2.4, and its Thorin consequence for a Lévy measure is (21.5.8) and Theorem 21.5.8; the de Bruijn Newman theorem is Theorem 7.2.16. The kink mechanism for weighted theta sums is that of Chapter 22 (Lemma 22.1.1, Theorem 22.2.3, Proposition 22.8.1, Proposition 22.8.2). What is not elsewhere in the book is the identity decoded as Spitzer's formula together with the functional equation, the exact expansion of the shifted Comtet Majumdar family with its obstructions, the Gauss weights in the kink theorem with the explicit coefficients, the positivity of the Thorin density of the uniform law for every tilt, its Mellin transform, and the open problems.

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it, conditional, heuristic or open as stated. The results are these.

1. The identity of Comtet and Majumdar holds on the whole critical strip, $0 < \mathrm{Re}\, s < 1$, and is the functional equation of $\zeta$ combined with Spitzer's formula for the maximum of a symmetric stable walk (Proposition 27.1.1, Proposition 27.1.2, proved). The logarithm of the Laplace transform of the uniform law has a regularised signed Lévy representation whose $k$ measure is Lebesgue measure minus counting measure, and $\zeta$ is its Mellin transform (Proposition 27.1.3, proved as improper integrals with a common cutoff).

2. An even entire function of order less than two has a GGC reciprocal exactly when its zeros are real (Lemma 27.2.1), so a family with GGC reciprocals converging to $\Xi$ proves RH (Corollary 27.2.2, proved).

3. The shifted Comtet Majumdar family $\zeta_\delta$ has an exact expansion in $\zeta(s+k)$ (Theorem 27.3.2, proved). Its completion has a pole at $s = 3$ for every $\delta \in (0, 2\pi)$, so its reciprocal is not a GGC transform and not the transform of any positive measure (Proposition 27.3.3, proved; the claim of poles at all odd integers of the source is false and is repaired). Each simple zero moves at an explicit first order rate (Proposition 27.3.3(b), proved), and the first three leave the critical line (computed, Table 27.1). The shifted family is not the literal cutoff of Comtet and Majumdar, and the two differ already in sign (Remark 27.3.4, proved).

4. The uniform law, tilted by $\delta > 0$ in the sense of the $2\sin^2$ lemma, has a strictly positive Thorin density for every $\delta > 0$, found by Poisson summation and convexity of the Cauchy kernel (Theorem 27.4.4, proved; the numerical constant of the proof is computed), and the regularised zeta function is its Mellin transform on $1 < \operatorname{Re} s < 3$ (Theorem 27.4.5, proved).

5. Gauss weights on the terms of the theta series produce, for every $\delta > 0$, a kernel transform with finitely many real zeros and infinitely many non-real zeros (Theorem 27.5.1, proved); symmetrising instead produces poles (Section 27.5).

6. The de Bruijn Newman family is a deformation whose GGC property is Theorem 7.2.16; what is false in the source, a convergence criterion for RH and the normalisation of the thresholds, is repaired (Section 27.6). The remaining problems are open.

## 27.1 The identity decoded

Throughout, $s = 1/\mu$ and

$$g(x) = \frac{1}{e^x - 1} - \frac{1}{x}, \qquad x > 0.$$

Note $g(0^+) = -\frac{1}{2}$, $g(x) \sim -1/x$ as $x \to \infty$, and $g < 0$ on $(0, \infty)$.

Proposition 27.1.1 (Mellin form). *For* $0 < \operatorname{Re} s < 1$,

$$I(s) := \frac{1}{\pi} \int_0^\infty x^{-s} g(x)\, dx = \frac{\Gamma(1-s)\zeta(1-s)}{\pi} = \frac{\zeta(s)}{(2\pi)^s \sin(\pi s/2)}. \tag{27.1.1}$$

*With $s = 1/\mu$ this is the identity of Comtet and Majumdar [86],*

$$\frac{1}{\pi} \int_0^\infty \frac{dk}{k^2} \ln\Big[\frac{1 - e^{-k^\mu}}{k^\mu}\Big] = \frac{\zeta(1/\mu)}{(2\pi)^{1/\mu} \sin(\pi/(2\mu))}, \qquad 1 < \mu \le 2. \tag{27.1.2}$$

*Proof.* The first equality is the classical formula $\Gamma(z)\zeta(z) = \int_0^\infty x^{z-1} g(x)\,dx$ for $0 < \operatorname{Re} z < 1$ [498, Equation 2.7.1] with $z = 1 - s$. The second is the functional equation $\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2)\Gamma(1-s)\zeta(1-s)$. For (27.1.2), substitute $x = k^\mu$ in the left side and integrate by parts once; the boundary terms vanish because $\mu > 1$. □

Comtet and Majumdar state (27.1.2) for real $\mu \in (1, 2]$, that is $s \in [\frac{1}{2}, 1)$. The proposition holds for complex $s$ in the whole strip, in particular on the critical line $s = \frac{1}{2} + it$, which is what makes it relevant to the zeros; for complex $\mu$ only the Mellin form (27.1.1) is the statement proved, the integral in $k$ being formal. The equation numbers of the paper were not checked.

Proposition 27.1.2 (Stable walk reading). *Let $S_n$ be a random walk with symmetric stable steps, $\mathbb{E}e^{ikX} = e^{-|k|^\mu}$, $1 < \mu \le 2$, and $M_n = \max_{0 \le j \le n} S_j$. With $s = 1/\mu$,*

$$\mathbb{E}M_n = \frac{\Gamma(1-s)}{\pi s}\, n^s + \frac{\Gamma(1-s)\zeta(1-s)}{\pi} + O(n^{s-1}).$$

*Hence the constant of Comtet and Majumdar is*

$$C_\mu = I(s) = \frac{\zeta(s)}{(2\pi)^s \sin(\pi s/2)}.$$

*Proof.* Spitzer's identity [271] gives $\mathbb{E}M_n = \sum_{k=1}^n \mathbb{E}S_k^+/k$. By stability $S_k \overset{d}{=} k^s S_1$ and $\mathbb{E}S_1^+ = \frac{1}{2}\mathbb{E}|S_1| = \Gamma(1-s)/\pi$. Hence $\mathbb{E}M_n = \pi^{-1}\Gamma(1-s)\sum_{k \le n} k^{s-1}$, and Euler Maclaurin gives $\sum_{k \le n} k^{s-1} = n^s/s + \zeta(1-s) + \frac{1}{2}n^{s-1} + O(n^{s-2})$. □

So the identity is Spitzer plus Euler Maclaurin plus the functional equation. The map $s \mapsto 1 - s$ is $\mu \mapsto \mu/(\mu - 1)$, the Hölder conjugate index, and the critical line $\operatorname{Re} s = \frac{1}{2}$ is through the self dual point $\mu = 2$, the Gaussian walk. In the normalisation of Proposition 27.1.2, where the Gaussian steps have variance 2, the constant there is $C_2 = \zeta(\frac{1}{2})/\sqrt{\pi} = -0.82392$ (computed); for steps of unit variance it is $\zeta(\frac{1}{2})/\sqrt{2\pi} = -0.58260$ (computed), which is the constant in the paper (their equation 32, not checked). The source printed the second value in the first normalisation. Real indices give real $s \in [\frac{1}{2}, 1)$, where $\zeta < 0$, and this is the content of the paper's Appendix A, which is the statement that there are no zeros on the real segment $[\frac{1}{2}, 1)$. The zeros on the critical line are not zeros of a probabilistic offset constant, since a walk with index $\mu = 2/(1+2it)$ is not defined; the factor relating $\Xi$ to $C_\mu$ is nonzero and the continuation in $s$ is analytic.

### 27.1.1 The uniform law and its logarithm

The integrand of (27.1.2) is the logarithm of $\varphi_U(x) = (1 - e^{-x})/x = \mathbb{E}e^{-xV}$, $V \sim \text{Uniform}(0, 1)$, at $x = k^\mu$.

Proposition 27.1.3. *For $x > 0$,*

$$\ln \frac{1-e^{-x}}{x} = -\lim_{R\to\infty}\Big\{\int_0^R (1-e^{-xt})\frac{dt}{t} - \sum_{1\le n\le R}\frac{1-e^{-xn}}{n}\Big\}, \tag{27.1.3}$$

*that is, with the signed $k$ measure $dt - d\lfloor t\rfloor = d\{t\}$ ($\{t\}$ the fractional part), $\ln\varphi_U(x) = -\int_0^\infty (1-e^{-xt})\,t^{-1}(dt - d\lfloor t\rfloor)$ as an improper integral with a common upper cutoff. For $0 < \operatorname{Re} z < 1$,*

$$\lim_{R\to\infty}\int_0^R t^{-z}\big(d\lfloor t\rfloor - dt\big) = -z\int_0^\infty \{t\}\,t^{-z-1}\,dt = \zeta(z). \tag{27.1.4}$$

*Proof.* The first integral in the braces is $\int_0^{xR}(1-e^{-v})\,dv/v = \log(xR) + \gamma_E + E_1(xR)$, and the sum is $H_{\lfloor R\rfloor} - \sum_{n\le R} e^{-xn}/n$. Since $H_{\lfloor R\rfloor} - \log R \to \gamma_E$, $E_1(xR)\to 0$ and $\sum_{n\ge1} e^{-xn}/n = -\log(1-e^{-x})$, the braces tend to $\log x - \log(1-e^{-x})$, which is (27.1.3). For (27.1.4), Stieltjes integration by parts gives, at a finite cutoff $R$, $\int_0^R t^{-z}(d\lfloor t\rfloor - dt) = -\{R\}R^{-z} - z\int_0^R \{t\}t^{-z-1}\,dt$. The boundary term tends to 0, and the last integral converges absolutely because $\{t\} = t$ below 1 and $\{t\}\le 1$ above. The identification with $\zeta(z)$ is Titchmarsh's formula [498, Equation 2.1.5]. □

The common cutoff matters. Neither the continuous part nor the atomic part is integrable against $t^{-1}$ at infinity, the two parts are mutually singular, and $\int_1^\infty |dt - d\lfloor t\rfloor|/t = \infty$. The difference is therefore not a signed Lévy measure in the sense of Lebesgue integration, and the representation (27.1.3) is a regularised one. After a positive tilt $e^{-\delta t}$ the integrability is restored, and that is what Section 27.4 uses. The uniform law is not infinitely divisible, and the proposition shows how: the measure that would be its Lévy measure is Lebesgue minus counting. In this language (27.1.1) reads $\pi I(s) = \Gamma(1-s)\int t^{s-1}(d\lfloor t\rfloor - dt)$, a signed Thorin type integral.

## 27.2 GGC reciprocals and the Laguerre Pólya class

The Laguerre Pólya class $\mathcal{LP}$ is the class of entire functions that are locally uniform limits of real polynomials with only real zeros. The lemma below is known (Proposition 7.1.15 and Proposition 1.2.1 give it in the form needed there); it is stated and proved here because everything in the chapter is a use of it.

Lemma 27.2.1. *Let $E$ be even, real on $\mathbb{R}$, entire of order less than two, with $E(0) > 0$. Then $w \mapsto E(0)/E(i\sqrt{w})$ is the Laplace transform of a GGC if and only if all zeros of $E$ are real, that is $E \in \mathcal{LP}$. In that case the Thorin measure is $\sum_k \delta_{\tau_k^2}$, where $\pm\tau_k$ are the zeros of $E$.*

*Proof.* An even function of order less than two has the Hadamard product $E(t) = E(0)\prod_k (1 - t^2/\tau_k^2)$ with $\sum_k |\tau_k|^{-2} < \infty$ (order less than two excludes a Gaussian factor $e^{-at^2}$), and $E(i\sqrt{w})/E(0) = \prod_k (1 + w/\tau_k^2)$ is entire in $w$. If every $\tau_k$ is real, the reciprocal is the Laplace transform of $\sum_k e_k/\tau_k^2$, which converges almost surely, a GGC with the stated Thorin measure. Conversely, a GGC transform $\exp\{-bw - \int \ln(1 + w/x)U(dx)\}$ is analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, so its reciprocal has no zeros there, and the zeros $w = -\tau_k^2$ of the product must lie in $(-\infty, 0)$. □

Corollary 27.2.2. *Let $\Xi_\delta$ satisfy the hypotheses of Lemma 27.2.1, let $\Xi_\delta \to \Xi$ locally uniformly as $\delta \to 0$, and let each $\Xi_\delta(0)/\Xi_\delta(i\sqrt{w})$ be a GGC transform. Then RH holds. Conversely, under RH the constant family $\Xi_\delta = \Xi$ qualifies.*

*Proof.* $\mathcal{LP}$ is closed under locally uniform limits (Hurwitz), so $\Xi \in \mathcal{LP}$. Equivalently the transforms converge pointwise on $(0, \infty)$ and the GGC class is closed under weak limits [56]. □

So the question asked of any approximating family is equivalent to RH, and the only realistic aim is a family in which the real zero property is visible for structural reasons. The rest of the chapter tests the families that the identity suggests.

## 27.3 The shifted Comtet Majumdar family

Comtet and Majumdar replace $1/(e^x - 1)$ by $1/(e^x - 1 + \epsilon)$ and $1/x$ by $1/(x + \epsilon)$ (their equations B5 to B10, not checked against the paper). With $1 - \epsilon = e^{-\delta}$ one has $1/(e^x - 1 + \epsilon) = e^\delta/(e^{x+\delta} - 1)$. The family studied in this chapter is the shift $g(x) \mapsto g(x + \delta)$, a variant of their cutoff with the same limit as $\epsilon \to 0$, which differs from it already at the order $\epsilon^{1-s}$ (Remark 27.3.4); every result below is a statement about the shifted family and is not to be attributed to the literal cutoff.

Definition 27.3.1. For $\delta > 0$ set $I_\delta(s) = \frac{1}{\pi}\int_0^\infty x^{-s} g(x + \delta)\, dx$ for $0 < \operatorname{Re} s < 1$ and $\zeta_\delta(s) = (2\pi)^s \sin(\pi s/2)\, I_\delta(s)$, so that $\zeta_0 = \zeta$ by (27.1.1), and $\xi_\delta(s) = \frac{1}{2}s(s - 1)\pi^{-s/2}\Gamma(s/2)\zeta_\delta(s)$, continued meromorphically.

Theorem 27.3.2 (Exact expansion). *For $0 < \delta < 2\pi$,*

$$\pi I_\delta(s) = \Gamma(1-s)F_\delta(s), \qquad F_\delta(s) = \operatorname{Li}_{1-s}(e^{-\delta}) - \Gamma(s)\delta^{-s} = \sum_{k \geq 0} \zeta(1 - s - k)\frac{(-\delta)^k}{k!}, \tag{27.3.1}$$

*and*

$$\begin{aligned}\zeta_\delta(s) &= \sum_{k\ge 0}\Big(\frac{\delta}{2\pi}\Big)^k \frac{(s)_k}{k!}\,\frac{\sin(\pi s/2)}{\sin(\pi(s+k)/2)}\,\zeta(s+k)\\ &= \zeta(s)+\frac{\delta}{2\pi}\,s\tan\frac{\pi s}{2}\,\zeta(s+1)\\ &\quad -\frac{\delta^2}{8\pi^2}\,s(s+1)\zeta(s+2)-\cdots\end{aligned}\tag{27.3.2}$$

*The ratio of sines equals* $(-1)^j$ *for* $k=2j$ *and* $(-1)^j\tan(\pi s/2)$ *for* $k=2j+1$.

*Proof.* Expanding $1/(e^{x+\delta}-1)=\sum_n e^{-n\delta}e^{-nx}$ gives $\int_0^\infty x^{-s}(e^{x+\delta}-1)^{-1}dx = \Gamma(1-s)\,\mathrm{Li}_{1-s}(e^{-\delta})$, and $\int_0^\infty x^{-s}(x+\delta)^{-1}dx=\delta^{-s}\Gamma(s)\Gamma(1-s)$, both absolutely convergent for $0<\operatorname{Re}s<1$. The series for $F_\delta$ is Lindelöf's expansion $\mathrm{Li}_\sigma(e^m)=\Gamma(1-\sigma)(-m)^{\sigma-1}+\sum_k\zeta(\sigma-k)m^k/k!$, valid for $|m|<2\pi$, with $\sigma=1-s$, $m=-\delta$ (the attribution of the expansion was not checked). For (27.3.2) write $A(s)=(2\pi)^s\sin(\pi s/2)/\pi$, so $\zeta(s)=A(s)\Gamma(1-s)\zeta(1-s)$, and substitute $\zeta(1-s-k)=\zeta(s+k)/\{A(s+k)\Gamma(1-s-k)\}$ into each term. Then $\Gamma(1-s)/\Gamma(1-s-k)=(-1)^k(s)_k$ and $A(s)/A(s+k)=(2\pi)^{-k}\sin(\pi s/2)/\sin(\pi(s+k)/2)$. □

The formula (27.3.2) agrees with direct quadrature of $I_\delta$ at $s=0.6$ for $\delta=0.1$ and 1 (relative difference 0 and $1.0\times10^{-21}$) and at $s=0.5+5i$ for the same $\delta$ (relative differences $1.8\times10^{-17}$ and $3.4\times10^{-17}$, limited by the 20 digit quadrature), and the form $F_\delta$ agrees with the Lindelöf sum to $10^{-21}$ (computed, `code/cg_checks.py`, sections `quad` and `poles`).

Proposition 27.3.3 (Obstructions). *Let* $0<\delta<2\pi$.

(a) $\xi_\delta(s)=\frac12 s(s-1)\pi^{(s-1)/2}\Gamma\big(\frac{1-s}{2}\big)F_\delta(s)$ *is meromorphic and not entire. It has a pole at* $s=3$ *for every* $\delta\in(0,2\pi)$ *and poles at* $s=-1,-2,-3,\dots$ *It has a pole at* $s=2m+1$, $m\ge2$, *exactly when* $F_\delta(2m+1)\neq0$, *and* $F_\delta(5)=0$ *at* $\delta=4.7215$ *and* $F_\delta(7)=0$ *at* $\delta=3.0416$ *(computed), so there is no pole at* $s=5$ *and* $s=7$ *for those* $\delta$.

(b) *Each simple zero* $\rho=\frac12+i\gamma$ *of* $\zeta$ *moves to*

$$\rho_\delta=\rho-\frac{\delta}{2\pi}\,\frac{\rho\tan(\pi\rho/2)\,\zeta(\rho+1)}{\zeta'(\rho)}+O(\delta^2),\tag{27.3.3}$$

*and* $\xi_\delta(s)\neq\xi_\delta(1-s)$.

(c) *If* $\xi_\delta(\frac12)\neq0$, *the function* $w\mapsto\xi_\delta(\frac12)/\xi_\delta(\frac12+\sqrt{w})$ *is not the Laplace transform of a GGC, nor of any positive measure.*

*Proof.* (a) From the functional equation, $\pi^{-s/2}\Gamma(s/2)A(s)\Gamma(1-s)=\pi^{(s-1)/2}\Gamma((1-s)/2)$, which gives the formula. $\Gamma((1-s)/2)$ has poles at $s=1,3,5,\dots$, and

the factor $s-1$ removes $s=1$. At $s=2m+1$ a pole remains exactly when $F_\delta(2m+1)\neq 0$. Poisson summation for $1/(e^\delta-1)$ gives, for $n\geq 1$,

$$F_\delta(n+1)=n!\sum_{k\neq 0}(\delta-2\pi ik)^{-n-1}.$$

For $n=2$ this is $F_\delta(3)=4\sum_{k\geq 1}\operatorname{Re}(\delta-2\pi ik)^{-3}$, and $\operatorname{Re}(\delta-2\pi ik)^{-3}=|\delta-2\pi ik|^{-3}\cos 3\theta_k$ with $\theta_k=\arctan(2\pi k/\delta)\in(\pi/4,\pi/2)$ because $\delta<2\pi$, so $3\theta_k\in(3\pi/4,3\pi/2)$ and every term is negative. Hence $F_\delta(3)<0$ and the pole at $s=3$ is present. For larger $m$ the term $k=1$ dominates and its real part changes sign near $\delta=2\pi\tan(\pi/(2m+1))$, so no interval $(0,\delta_0)$ avoids the zeros of $F_\delta(2m+1)$; the values 4.7215 and 3.0416 are zeros of $F_\delta(5)$ and $F_\delta(7)$ (computed at forty digits). The source's argument $F_\delta(2m+1)=-\delta\zeta(-2m-1)+O(\delta^3)$ is not uniform in $m$. At $s=-m$, $m\geq 1$, the factor $\Gamma(s)\delta^{-s}$ has a pole and the other factors are finite and nonzero.

(b) Implicit differentiation of $\zeta_\delta(\rho_\delta)=0$ using the $k=1$ term of (27.3.2). The failure of the functional equation is immediate from (a): a function invariant under $s\mapsto 1-s$ cannot have a pole at $-1$ and be regular at 2.

(c) The pole of $\xi_\delta$ at $s=3$ makes the reciprocal vanish at $w=(5/2)^2=25/4$ (the value $\frac12+\sqrt{w}=3$), whereas the Laplace transform of a nonzero positive measure is strictly positive on $(0,\infty)$. A second reason, for $\delta$ small enough that (b) applies: a zero $\rho_\delta$ with $\operatorname{Re}\rho_\delta>\frac12$ makes the reciprocal have a pole at $w=(\rho_\delta-\frac12)^2\notin(-\infty,0]$, which a GGC transform cannot have. The source's first reason, that the zero set is not symmetric and so the function is not a function of $w$, is not an argument: with the principal root $w\mapsto\xi_\delta(\frac12+\sqrt{w})$ is a function on $\mathbb{C}\setminus(-\infty,0]$, which is all that a GGC transform requires. □

| | $\gamma_1=14.1347$ | $\gamma_2=21.0220$ | $\gamma_3=25.0109$ |
|---|---|---|---|
| slope, real part of (27.3.3) | 1.53709 | 1.95365 | 2.14062 |
| $\delta=0.01$ | 0.015371 | 0.019537 | 0.021406 |
| $\delta=0.05$ | 0.076858 | 0.097694 | 0.107022 |
| $\delta=0.10$ | 0.153723 | 0.195408 | 0.214026 |
| $\delta=0.30$ | 0.461231 | 0.586400 | 0.641850 |
| $\delta=1.00$ | 1.537267 | 1.953476 | 2.137214 |

Table 27.1: Displacement $\operatorname{Re}\rho_\delta-\frac12$ of the first three zeros of the shifted family, by Newton's method from (27.3.2) at 30 and 45 digits (difference below $5\times10^{-25}$; computed). The first row is the real part of the complex slope of (27.3.3), whose imaginary parts $-0.1565$, $+0.0703$, $-0.2961$ show that $\operatorname{Im}\rho_\delta$ moves too. The displacement is linear in $\delta$ only to the digits of the slope for small $\delta$ (at $\delta=1$ the first entry is 1.53727 against 1.53709).

The lesson is that damping applied on the $\zeta(1-s)$ side is carried by the functional equation into the operator series (27.3.2), which mixes $\zeta(s)$ with $\zeta(s+1), \zeta(s+2), \dots$ through the factor $\tan(\pi s/2)$; this is not symmetric and generates poles.

REMARK 27.3.4 (The shift is not the literal cutoff). Write $\epsilon = 1 - e^{-\delta}$. The literal cutoff of Comtet and Majumdar has the integrand $\tilde{g}_\epsilon(x) = 1/(e^x - 1 + \epsilon) - 1/(x+\epsilon) = e^\delta/(e^{x+\delta} - 1) - 1/(x+\epsilon)$, whereas the shifted family has $g(x+\delta) = 1/(e^{x+\delta} - 1) - 1/(x+\delta)$; both the multiplicative factor $e^\delta$ and the second shift differ. In particular $\tilde{g}_\epsilon(0) = \tilde{g}'_\epsilon(0) = 0$ while $g(\delta) < 0$. For $0 < \operatorname{Re} s < 1$ one finds $\pi\tilde{I}_\epsilon(s) = \Gamma(1-s)\big[e^\delta \operatorname{Li}_{1-s}(e^{-\delta}) - \Gamma(s)\epsilon^{-s}\big]$, and with $\delta = \epsilon + \epsilon^2/2 + O(\epsilon^3)$ the two families differ at the order $\epsilon^{1-s}$: $\tilde{I}_\epsilon(s) = I(s) + \Gamma(1-s)\Gamma(s)(1-s/2)\epsilon^{1-s}/\pi + O(\epsilon)$, while the shifted family's correction is analytic and linear in $\delta$. A sign test shows the difference also after continuation. For $x > 0$ one has $e^x - 1 > x$, so $\tilde{g}_\epsilon(x) < 0$, and $\tilde{g}_\epsilon(x) = -x^2/(2\epsilon^2) + O(x^3)$ at 0, so the literal integral converges for $0 < \operatorname{Re} s < 3$, and $\tilde{\zeta}_\epsilon(s) = (2\pi)^s \sin(\pi s/2)\tilde{I}_\epsilon(s) < 0$ for real $1 < s < 2$, whereas Theorem 27.4.5 below gives $\zeta_\delta(s) > 0$ there. The pole expansions, the zero drift and the Thorin Mellin identity of this chapter are proved for the shifted family and cannot be attributed unchanged to the finite $\epsilon$ cutoff. (This corrects the source, which called the cutoff exactly the shift; the sign test and the expansion follow from the formulas above and were not run by a script.)

## 27.4 The uniform law and the $2\sin^2$ lemma

Section 27.3 asked whether the shifted family lies in $\mathcal{LP}$, and the answer was no. This section asks the question that the $2\sin^2$ lemma is built for: given a Lévy measure on the primal side, what is the Thorin measure on the dual side? Asked this way the tilt of the uniform law becomes a positive object, for every $\delta > 0$.

### 27.4.1 The lemma and the Thorin density of a signed Lévy measure

LEMMA 27.4.1 (The $2\sin^2$ lemma). *For $x > 0$ and $w \geq 0$,*

$$\frac{e^{-wx} + wx - 1}{x} = \frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{w^2}{y^2}\Big)\big(1 - \cos xy\big)\,dy. \tag{27.4.1}$$

*In the variables $u = w^2/2$, $z = y^2/2$ this is $e^{-wx} + wx - 1 = \int_0^\infty \log(1+u/z)\,\tau_x(z)\,dz$ with $\tau_x(z) = \frac{x}{\pi\sqrt{2z}}\,2\sin^2\big(x\sqrt{z/2}\big) \geq 0$.*

This is (1.2.9) in the book (see Lemma 6.2.4 and Theorem 21.5.8); the lemma is credited there. A one line proof of (27.4.1) is the following. Both sides vanish at $w = 0$. Differentiating in $w$ and using $\int_0^\infty (w^2 + y^2)^{-1} dy = \pi/2w$ and $\int_0^\infty \cos(xy)(w^2 + y^2)^{-1} dy = \pi e^{-wx}/2w$, the right side gives $\frac{1}{\pi}\int_0^\infty 2w(1 - \cos xy)(w^2 + y^2)^{-1} dy = 1 - e^{-wx}$, the derivative of the left side; the interchange is

justified by writing $\log(1+w^2/y^2)=\int_0^w 2v(v^2+y^2)^{-1}dv$ and applying Tonelli to the nonnegative double integral. The change of variable $z=y^2/2$ gives the second form.

The following statement differs from the corresponding one in Chapter 21 (the identity (21.5.8)) only in its hypothesis, an abstract signed measure $M$ in place of the explicit Lévy measure of $\xi$.

THEOREM 27.4.2 (Thorin density from a signed Lévy measure). *Let $f$ be analytic and zero free on $\operatorname{Re} s>\alpha_0$ and positive on $(\alpha_0,\infty)$. Let $M$ be a signed measure on $(0,\infty)$ with $\int(1\wedge x)e^{-\alpha x}|M|(dx)<\infty$ for every $\alpha>\alpha_0$, such that*

$$\frac{f'}{f}(s)-\frac{f'}{f}(\alpha)=\int_0^\infty \bigl(e^{-\alpha x}-e^{-sx}\bigr)M(dx),\qquad \operatorname{Re} s>\alpha_0,\ \alpha>\alpha_0. \tag{27.4.2}$$

*Then for every $\alpha>\alpha_0$ and $w\ge 0$,*

$$\log\frac{f(\alpha+w)}{f(\alpha)}-w\,\frac{f'}{f}(\alpha)=\int_0^\infty \log\Bigl(1+\frac{w^2}{y^2}\Bigr)\kappa_\alpha(y)\,dy, \tag{27.4.3}$$

$$\kappa_\alpha(y)=\frac{1}{\pi}\int_0^\infty\bigl(1-\cos xy\bigr)e^{-\alpha x}M(dx)=\frac{1}{\pi}\Bigl[\operatorname{Re}\frac{f'}{f}(\alpha+iy)-\frac{f'}{f}(\alpha)\Bigr]. \tag{27.4.4}$$

*If $\kappa_\alpha\ge 0$, then $u\mapsto F_\alpha(u)=\dfrac{f(\alpha)}{f(\alpha+\sqrt{2u})}\exp\bigl\{\sqrt{2u}\,\frac{f'}{f}(\alpha)\bigr\}$ is the Laplace transform of a GGC with Thorin density $\tau_\alpha(z)=\kappa_\alpha(\sqrt{2z})/\sqrt{2z}$.*

*Proof.* Differentiating (27.4.2) in $s$ gives $(f'/f)'(s)=\int xe^{-sx}M(dx)$. The function $\int_0^\infty e^{-\alpha x}(e^{-wx}+wx-1)\,M(dx)/x$ has the same second derivative in $w$ as the left side of (27.4.3), and both vanish together with their first derivatives at $w=0$, so they are equal. Now insert (27.4.1). Fubini is justified by Tonelli applied to $|M|$: by (27.4.1) the double integral of the absolute values is $\int e^{-\alpha x}(e^{-wx}+wx-1)|M|(dx)/x\le\int e^{-\alpha x}\min(w^2x/2,w)\,|M|(dx)<\infty$. The second form of $\kappa_\alpha$ is (27.4.2) at $s=\alpha+iy$, since $\operatorname{Re}(e^{-\alpha x}-e^{-(\alpha+iy)x})=e^{-\alpha x}(1-\cos xy)$. If $\kappa_\alpha\ge 0$ then $\log F_\alpha(u)=-\int\log(1+u/z)\tau_\alpha(z)\,dz$ with a nonnegative density and a finite integral, which is the definition of a GGC transform [56]. □

When $M\ge 0$ the density $\kappa_\alpha$ is automatically nonnegative. The interesting cases are those where $M$ is a signed measure, counting measure minus a smooth density, and $\kappa_\alpha$ is still positive. Formula (27.4.4) gives a test that needs no Lévy measure: $\kappa_\alpha\ge 0$ if and only if the real part of the logarithmic derivative on the vertical line through $\alpha$ is never below its value on the real axis.

COROLLARY 27.4.3 (Zeros on a line give Lorentzians). *Let $a$ be real and let the hypotheses of Theorem 27.4.2 hold. Suppose $f'/f(s)=a+\sum_k(s-\rho_k)^{-1}$ (symmetric summation) with all $\rho_k=\beta_0+it_k$ on one vertical line, the set $\{t_k\}$ symmetric*

*and $0 \notin \{t_k\}$. Then for $\alpha = \beta_0 + \varepsilon$, $\varepsilon > 0$,*

$$\kappa_\alpha(y) = \frac{1}{\pi} \sum_k \big[\ell_\varepsilon(y - t_k) - \ell_\varepsilon(t_k)\big], \tag{27.4.5}$$

*and if $\sum_k t_k^{-2} < \infty$ then $\kappa_\alpha(y)\,dy \to \sum_{t_k>0} \delta_{t_k}(dy)$ vaguely on $(0,\infty)$ as $\varepsilon \downarrow 0$; the right side of (27.4.3) tends to $\sum_{t_k>0} \log(1 + w^2/t_k^2)$.*

*Proof.* $\mathrm{Re}(\alpha + iy - \beta_0 - it)^{-1} = \ell_\varepsilon(y - t)$. As $\varepsilon \downarrow 0$, $\pi^{-1}\ell_\varepsilon(\cdot - t) \to \delta_t$; the constant $\pi^{-1}\sum_k \ell_\varepsilon(t_k) \le \pi^{-1}\varepsilon \sum_k t_k^{-2} \to 0$; and peaks at $t_k < 0$ do not reach $y > 0$. For the last limit the integral against $\log(1+w^2/y^2)$ needs the tail bound $\sum_k O(w^2/t_k^2)$ for the dominated convergence, which the hypothesis $\sum t_k^{-2} < \infty$ supplies. □

### 27.4.2 The tilted uniform law

Take $f = \varphi_U(s) = (1 - e^{-s})/s$. By Proposition 27.1.3 (after the tilt, where the integrals converge absolutely), (27.4.2) holds with $M_U(dx) = dx - d\lfloor x \rfloor$, $f'/f = g$ and $\alpha_0 = 0$, because $g(s) = \sum_{n\ge1} e^{-ns} - 1/s = \int_0^\infty e^{-sx}(d\lfloor x\rfloor - dx)$ and the zeros of $\varphi_U$ are $2\pi i k$, $k \ne 0$, on the line $\mathrm{Re}\, s = 0$. The tilt is $\alpha = \delta$: it replaces $g(x)$ by $g(x + \delta)$ in the Mellin integral, which is the shifted family of Section 27.3.

Theorem 27.4.4 (Thorin density of the tilted uniform law). *For every $\delta > 0$,*

$$\begin{aligned} \pi\kappa_\delta(y) &= \frac{y^2}{\delta(\delta^2 + y^2)} - \frac{1}{2}\coth\frac{\delta}{2}\,\frac{1 - \cos y}{\cosh\delta - \cos y} \\ &= \sum_{m\ge1} \big[\ell_\delta(y - 2\pi m) + \ell_\delta(y + 2\pi m) \\ &\qquad - 2\ell_\delta(2\pi m)\big], \end{aligned} \tag{27.4.6}$$

*and $\kappa_\delta(y) > 0$ for all $y > 0$. Hence $u \mapsto \dfrac{\varphi_U(\delta)}{\varphi_U(\delta + \sqrt{2u})}\, e^{\sqrt{2u}\,g(\delta)}$ is the Laplace transform of a GGC with Thorin density $\tau_\delta(z) = \kappa_\delta(\sqrt{2z})/\sqrt{2z}$. As $\delta \downarrow 0$, $\kappa_\delta(y)\,dy \to \sum_{m\ge1} \delta_{2\pi m}$ vaguely, and the limit of (27.4.3) is*

$$\log\varphi_U(w) + \frac{w}{2} = \log\frac{\sinh(w/2)}{w/2} = \sum_{m\ge1} \log\Big(1 + \frac{w^2}{4\pi^2 m^2}\Big),$$

*so $(w/2)/\sinh(w/2)$ is the Laplace transform in $u = w^2/2$ of $\sum_m e_m/(2\pi^2 m^2)$, a GGC with Thorin measure $\sum_m \delta_{2\pi^2 m^2}$ (a law of Biane, Pitman and Yor type [49]; it is $S_1/4$ in their notation).*

*Proof.* The closed form follows from (27.4.4) with $g(s) = \sum_{n\ge1} e^{-ns} - 1/s$: one has $1/\delta - \mathrm{Re}(\delta + iy)^{-1} = y^2/\{\delta(\delta^2 + y^2)\}$ and $\sum_n e^{-n\delta}(1 - \cos ny) =$

$\frac12\coth(\delta/2)(1-\cos y)/(\cosh\delta-\cos y)$. The Lorentzian form is Corollary 27.4.3, since $g(s)=-\frac12+\sum_{k\neq0}(s-2\pi ik)^{-1}$. For positivity let $P(y)=\sum_{m\in\mathbb{Z}}\ell_\delta(y-2\pi m)=\frac12\sinh\delta/(\cosh\delta-\cos y)$, the periodised Cauchy kernel, so that $\pi\kappa_\delta=[P(y)-\ell_\delta(y)]-[P(0)-\ell_\delta(0)]$.

Case $y\geq\pi$. Here $P(y)\geq P(\pi)=\frac12\tanh(\delta/2)$ and $\ell_\delta(y)\leq\ell_\delta(\pi)$, so

$$\pi\kappa_\delta(y)\geq\tfrac12\tanh\tfrac{\delta}{2}-\tfrac12\coth\tfrac{\delta}{2}+\frac{1}{\delta}-\frac{\delta}{\delta^2+\pi^2}=\frac{\pi^2}{\delta(\delta^2+\pi^2)}-\frac{1}{\sinh\delta}>0,$$

because $\pi^2\sinh\delta>\pi^2\delta+\pi^2\delta^3/6>\delta(\delta^2+\pi^2)$, using $\pi^2/6>1$.

Case $0<y<\pi$ and $\delta\leq\sqrt3\,\pi$. Since $\ell_\delta''(x)=2\delta(3x^2-\delta^2)/(\delta^2+x^2)^3$, $\ell_\delta$ is strictly convex on $[\delta/\sqrt3,\infty)\supset[\pi,\infty)$, and $[2\pi m-y,2\pi m+y]\subset[\pi,\infty)$. So every bracket in (27.4.6) is positive.

Case $0<y<\pi$ and $\delta\geq4.1$. Use $1-\cos y\leq y^2/2$ and $\cosh\delta-\cos y\geq\cosh\delta-1=2\sinh^2(\delta/2)$ in the second term of (27.4.6), and $y^2/\{\delta(\delta^2+y^2)\}\geq y^2/\{\delta(\delta^2+\pi^2)\}$ in the first. Then $\pi\kappa_\delta>0$ provided $R(\delta)=8\tanh(\delta/2)\sinh^2(\delta/2)/\{\delta(\delta^2+\pi^2)\}>1$. The logarithmic derivative of the numerator exceeds $\coth(\delta/2)>1$, while that of the denominator, $1/\delta+2\delta/(\delta^2+\pi^2)$, is at most $2/\pi<1$ for $\delta\geq\pi$. So $R$ is increasing on $[\pi,\infty)$, and $R(4.1)=1.0322$ (computed, 1.032196804). Since $4.1<\sqrt3\,\pi=5.4414$, the three cases cover all $\delta>0$ and $y>0$. The limit statements are Corollary 27.4.3 together with $\varphi_U(s)=e^{-s/2}\sinh(s/2)/(s/2)$. □

The closed form agrees with the Lorentzian sum to fifteen digits at nine points ($\delta=0.1,1,5$, $y=0.5,3,7$), $\kappa_\delta>0$ on a grid of step $1/40$ on $(0,40]$ for ten values of $\delta$ from 0.01 to 20, and $R(4)=0.98037<1$ so the threshold 4.1 is nearly sharp (computed, section `uniform` of `code/cg_checks.py`). So the $2\sin^2$ lemma converts the signed measure $dx-d\lfloor x\rfloor$ of the uniform law, which is not a Lévy measure of an infinitely divisible law, into a strictly positive Thorin density for every $\delta>0$. The mechanism is Poisson summation: the difference between the sum over the integers and the integral is a sum of Lorentzians centred at the dual lattice $2\pi\mathbb{Z}$, and convexity of the Cauchy kernel makes them dominate.

THEOREM 27.4.5 (The shifted zeta as a Thorin Mellin transform). *For $1<\operatorname{Re}s<3$ and every $\delta>0$,*

$$\zeta_\delta(s)=(2\pi)^s\int_0^\infty y^{-s}\kappa_\delta(y)\,dy,\tag{27.4.7}$$

*where $\zeta_\delta$ is the shifted family of Section 27.3, continued analytically (for $\delta<2\pi$ it is the series (27.3.2)). At $\delta=0$ the right side is $(2\pi)^s\sum_m(2\pi m)^{-s}=\zeta(s)$.*

*Proof.* Differentiate (27.4.3) in $w$: $h(w):=g(\delta+w)-g(\delta)=\int_0^\infty 2w(w^2+y^2)^{-1}\kappa_\delta(y)\,dy$. Since $h(w)=O(w)$ at 0 and $h(w)\to-g(\delta)$ at $\infty$, the integral $\int_0^\infty w^{-s}h(w)\,dw$ converges for $1<\operatorname{Re}s<2$. It is the analytic continuation of $\pi I_\delta(s)=\int w^{-s}g(\delta+w)\,dw$ from $0<\operatorname{Re}s<1$: subtracting the value at 0 shifts the Mellin strip. By Tonelli ($\kappa_\delta\geq0$) and $\int_0^\infty w^{1-s}\,2(w^2+y^2)^{-1}dw=\pi y^{-s}/\sin(\pi s/2)$,

we get $I_\delta(s) = \{\sin(\pi s/2)\}^{-1} \int y^{-s}\kappa_\delta(y)\,dy$. Multiply by $(2\pi)^s \sin(\pi s/2)$. The right side of (27.4.7) converges for $1 < \operatorname{Re} s < 3$, since $\kappa_\delta(y) = O(y^2)$ at 0 and $0 \le \kappa_\delta \le 1/(\pi\delta)$, which gives convergence for $\operatorname{Re} s > 1$ and the continuation to that strip. The restriction $\delta < 2\pi$ belongs to the series (27.3.2) only. □

At $s = 2.5$, $\delta = 0.5$ the right side of (27.4.7) is 1.53343255 (computed with $N = 120$ periods of the lattice, error estimate $5 \times 10^{-8}$; $N = 60$ gives 1.5334325) and the series (27.3.2) gives 1.53343255193 (computed). The theorem explains the identity of Comtet and Majumdar: formally their (27.1.2) is the Mellin transform of the Thorin measure $\sum_m \delta_{2\pi m}$ of the uniform law, which diverges in the strip $0 < s < 1$, and the shift smooths this lattice into the positive density $\kappa_\delta$. This is the reading for the shifted family; for the literal cutoff the Mellin transform is negative at real $s \in (1, 2)$ (Remark 27.3.4).

### 27.4.3 The Riemann function, by citation

The same construction applies to $\xi$, and the book has it in full. The Lévy measure of $\xi$ is primes minus a smooth density, $M_\xi = \sum_{n\ge2} \Lambda(n)\delta_{\log n} - e^x\,dx + dx/(e^{2x}-1)$, which is $-\sum_\rho e^{\rho x}$ as a distribution by the explicit formula (Proposition 9.1.10; $\Lambda(n)$ is von Mangoldt's function, which this chapter does not otherwise use). Its Thorin density at the basepoint $\alpha$ is $\pi^{-1}$ times the compensated density $v_\alpha(y) - b_\alpha$ of Chapter 9 (formula (9.1.7)), and Proposition 9.1.12 proves, with no hypothesis on the zeros, that it is positive for every $\alpha \ge 1$ and $y \ne 0$, so the compensated reciprocal is a GGC for every $\alpha \ge 1$; the threshold 7.789, the root 7.789185 of $\frac{1}{8}\zeta(3, \alpha/2 + 1) = (\alpha - 1)^{-3}$ (computed), also appears there. The unconditional GGC property of the uncompensated reciprocal for $\alpha \ge 1$ is Theorem 1.2.2. Positivity of the uncompensated density for every $\alpha > \frac{1}{2}$ is equivalent to RH by Lagarias's criterion (Corollary 16.12.10), and under RH the passage to $\alpha \downarrow \frac{1}{2}$ gives Hadamard's product exactly as Corollary 27.4.3 does for the uniform law (Proposition 9.1.2). Two levels of positivity are distinguished there. Level (1) is positivity of the uncompensated density, which is the GGC property of $\xi(\alpha)/\xi(\alpha + \sqrt{s})$. Level (2) is positivity of the compensated density, the GGC property of the compensated reciprocal $F_\alpha$, which strips the stable component $e^{-b\sqrt{s}}$ of index $\frac{1}{2}$ out of level (1). Level (2) implies level (1) and is logically stronger; no basepoint is known at which (1) holds and (2) fails.

The uniform law is the model in which only level (2) holds. There the uncompensated density is $\pi^{-1} \operatorname{Re} g(\delta + iy) = \pi^{-1}[P(y) - \frac{1}{2} - \ell_\delta(y)]$, which is negative near $y = \pi$ because $P(\pi) = \frac{1}{2}\tanh(\delta/2) < \frac{1}{2}$; the factor $e^{-s/2}$ in $\varphi_U$ is a drift, and compensation is what makes Theorem 27.4.4 work. For $\xi$, which has no such factor, level (1) is the natural statement and level (2) is what Proposition 9.1.12 adds.

Remark 27.4.6 (Zeta of the zeros). Under RH, as $\alpha \downarrow \frac{1}{2}$ the Thorin density

$\kappa_\alpha^\xi$ of $\xi$ ((27.4.4) with $f = \xi$) tends vaguely to $\sum_{\gamma>0}\delta_\gamma$, and the argument of Theorem 27.4.5 suggests $\int_0^\infty y^{-s}\kappa_\alpha^\xi(y)\,dy \to \sum_{\gamma>0}\gamma^{-s}$, the zeta function of the zeros (Voros [360]), for $1 < \operatorname{Re} s < 3$. For the uniform law the same limit is $(2\pi)^{-s}\zeta(s)$ by Theorem 27.4.5. This is a heuristic: the limit interchange needs the bounds $|\kappa_\alpha^\xi(y)| \le C\varepsilon y^2$ near the origin and a tail bound of order $\gamma^{-\sigma} + \varepsilon\gamma^{-2}$ for each Lorentzian against $y^{-\sigma}$, summed with the zero counting estimate, which are sketched and not written out. The range $\operatorname{Re} s > 1$ of the source is wrong: the integral diverges at the origin for real $s \ge 3$ because $\kappa_\alpha^\xi(y) = C_\varepsilon y^2 + O(y^4)$ with $C_\varepsilon > 0$ for small $\varepsilon$.

| | uniform law, shifted | Riemann $\xi$ |
|---|---|---|
| function | $\varphi_U(s) = (1-e^{-s})/s$ | $\xi(s)$ |
| signed $k$ measure $M$ | $dx - d\lfloor x\rfloor$, Lebesgue minus integers | $\sum\Lambda(n)\delta_{\log n} - e^x dx + \frac{dx}{e^{2x}-1}$, primes minus smooth |
| duality | Poisson summation | explicit formula |
| zeros | $2\pi i k$, on $\operatorname{Re} s = 0$ | $\rho$; on $\operatorname{Re} s = \frac12$ under RH |
| tilt | $\alpha = \delta$ (the shift) | $\alpha = \frac12 + \varepsilon$ |
| Thorin density | Lorentzian smoothing of $\sum\delta_{2\pi m}$, positive for all $\delta > 0$ (Theorem 27.4.4); only level (2) | uncompensated and compensated: positive for $\alpha \ge 1$ (Theorem 1.2.2, Proposition 9.1.12); uncompensated for all $\alpha > \frac12$ iff RH |
| limit Thorin measure | $\sum_m \delta_{2\pi m}$ | $\sum_{\gamma>0}\delta_\gamma$ counted with multiplicity, under RH |
| Mellin of Thorin | $(2\pi)^{-s}\zeta(s)$, proved | $\sum_{\gamma>0}\gamma^{-s}$, heuristic |

Table 27.2: The uniform law and the Riemann problem under the $2\sin^2$ lemma. Convergence of the Thorin densities is vague convergence. Without RH the limit at $\alpha = \frac12$ records only the zeros on the critical line (zeros at $\frac12 \pm a + i\gamma$ cancel in the real part), so the limit measure is not a test of RH by itself.

## 27.5 Damping the kernel

The asymmetry of Section 27.3 can be blamed on working with $g$ rather than with the symmetric theta representation. This section applies the same damping, with Gauss weights, to the coefficients in Riemann's kernel. The mechanism is that of Chapter 22 (Lemma 22.1.1, Theorem 22.2.3, Proposition 22.8.1, Proposition 22.8.2, where the weights are the friable ones and the proof of finitely many real zeros and infinitely many nonreal ones is the same); what is not elsewhere in the book is the Gauss weights, the explicit coefficients and the sign pattern. Write

$$\Phi_n(u) = \bigl(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\bigr)\exp\bigl(-\pi n^2 e^{2u}\bigr), \qquad \Phi = 2\sum_{n\ge1}\Phi_n,$$

so that $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(ut)\,du$, $\Phi > 0$ and $\Phi$ is even; the evenness is the theta transformation formula. For $\delta \geq 0$ let $\Phi_\delta = 2\sum_n e^{-\pi n^2\delta}\Phi_n$ on $[0,\infty)$ and $\Xi_\delta(t) = 2\int_0^\infty \Phi_\delta(u)\cos(ut)\,du$. With $x = e^{2u}$ this is the incomplete gamma form of $\Xi$ (Proposition 22.1.3) with weights $e^{-\pi n^2\delta}$, less the constant $\sum_n(1 - e^{-\pi n^2\delta})(4\pi n^2 - 1)e^{-\pi n^2}$ (two integrations by parts, as in Section 22.2), the analogue of the factors $(1-\epsilon)^n$ in the paper of Comtet and Majumdar (their B6, not checked). Since $\Phi_\delta$ decays like $\exp(-\pi e^{2u})$, each $\Xi_\delta$ is even, real on $\mathbb{R}$ and entire of order one, and $\Xi_\delta \to \Xi$ locally uniformly by dominated convergence, and $\Xi_\delta(0) > 0$ because $\Phi_n > 0$ on $[0,\infty)$, so $\Xi_\delta$ satisfies the hypotheses of Lemma 27.2.1. For complex $t$ the transform is $\Xi_\delta(t) = \sum_n e^{-\pi n^2\delta}[B_n(t) + B_n(-t)]$, where $B_n(t) = 2\pi^2 n^4 \Gamma(z_9, \pi n^2)(\pi n^2)^{-z_9} - 3\pi n^2 \Gamma(z_5, \pi n^2)(\pi n^2)^{-z_5}$ with $z_9 = \frac{9}{4} + \frac{it}{2}$, $z_5 = \frac{5}{4} + \frac{it}{2}$ and $\Gamma(\cdot,\cdot)$ the upper incomplete gamma function (this comes from $2\cos(ut) = e^{iut} + e^{-iut}$ before the substitution); on the real axis $B_n(-t) = \overline{B_n(t)}$ and the sum is $2\,\mathrm{Re}\,B_n(t)$, which is the form usually printed, and which is not the analytic continuation off the real axis.

THEOREM 27.5.1. *For every $\delta > 0$, $\Xi_\delta$ has only finitely many real zeros and infinitely many nonreal zeros. Consequently $\Xi_\delta(0)/\Xi_\delta(i\sqrt{w})$ is not the Laplace transform of a GGC.*

*Proof.* A direct computation gives $\Phi_n'(0) = c_n := \pi n^2 e^{-\pi n^2}\bigl(15\pi n^2 - \frac{15}{2} - 4\pi^2 n^4\bigr)$. For $n = 1$ the bracket is $15\pi - 7.5 - 4\pi^2 = 0.1455 > 0$ (computed; the source printed 0.146); for $n \geq 2$ it equals $-n^2(4\pi^2 n^2 - 15\pi) - 7.5 < 0$ because $4\pi^2 n^2 > 15\pi$. Evenness of $\Phi$ gives $\sum_n c_n = \Phi'(0)/2 = 0$. Hence for decreasing weights $w_n = e^{-\pi n^2\delta}$,

$$\tfrac{1}{2}\Phi_\delta'(0) = \sum_n w_n c_n = w_1 c_1 + \sum_{n\geq 2} w_n c_n > w_1 \sum_{n\geq 1} c_n = 0,$$

since $w_n < w_1$ and $c_n < 0$ for $n \geq 2$. Two integrations by parts give

$$\int_0^\infty \Phi_\delta(u)\cos(ut)\,du = -\frac{\Phi_\delta'(0)}{t^2} - \frac{1}{t^2}\int_0^\infty \Phi_\delta''(u)\cos(ut)\,du,$$

and the last integral tends to zero by the Riemann Lebesgue lemma. Thus $t^2\Xi_\delta(t) \to -2\Phi_\delta'(0) < 0$, so $\Xi_\delta < 0$ for all large real $t$ and has finitely many real zeros. If $\Xi_\delta$ had finitely many zeros in all, Hadamard's theorem would give $\Xi_\delta = Pe^{at}$ with $P$ a polynomial; evenness and reality force $a = 0$, and a nonzero polynomial cannot tend to zero along $\mathbb{R}$. So there are infinitely many zeros, all but finitely many nonreal, and Lemma 27.2.1 applies. □

The mechanism is that damping the $n$ sum destroys the evenness of $\Phi$ at $u = 0$, so the even extension $\Phi_\delta(|u|)$ has a kink, and a kink produces a nonoscillating $t^{-2}$ tail which swamps the exponentially small oscillation $|\Xi(t)| \approx t^{7/4}e^{-\pi t/4}$ that

carries the zeros. Heuristically the real zeros stop near the height $T(\delta)$ where these balance, $\frac{\pi}{4}T - \frac{15}{4}\ln T \approx \ln(1/\delta)$; for $\delta = 10^{-3}$ this gives $T \approx 24$ (heuristic), and the source reports that the last real zero is at 24.54 (unchecked, not recomputed).

COROLLARY 27.5.2 (Modular constraints). *Let* $\Phi_w = 2\sum_n w_n\Phi_n$ *with real weights such that* $\sum_n |w_n| n^A e^{-\pi n^2} < \infty$ *for every* $A \geq 0$, *and* $\Xi_w(t) = 2\int_0^\infty \Phi_w(u)\cos(ut)\,du$. *If* $\Xi_w \in \mathcal{LP}$ *and* $\Xi_w$ *is not of the form* $e^{-at^2}$ *times a polynomial, then* $\sum_n w_n \Phi_n^{(2j+1)}(0) = 0$ *for every* $j \geq 0$.

*Proof.* If the first nonzero odd derivative is of order $2j+1$, repeated integration by parts gives $\Xi_w(t) \sim 2(-1)^{j+1}\Phi_w^{(2j+1)}(0)\,t^{-2j-2}$, of eventually constant sign, and the argument of Theorem 27.5.1 applies (an element of $\mathcal{LP}$ with finitely many zeros is $e^{-at^2+bt}P$). □

For $w_n \equiv 1$ these infinitely many linear constraints are the Taylor coefficients at $x = 1$ of Jacobi's identity $\theta(1/x) = \sqrt{x}\,\theta(x)$. The proof of Theorem 27.5.1 shows that the first constraint is violated by every weight sequence with $w_1 \geq w_n \geq 0$ for $n \geq 2$ and at least one strict inequality (under the summability condition of Corollary 27.5.2): subtracting $w_1 \sum c_n = 0$ gives $\sum_{n\geq 2}(w_n - w_1)c_n > 0$. This covers the usual Abel, Gauss and Lerch weights; it is not a statement about arbitrary deformations of the coefficients.

### 27.5.1 Symmetrising does not help

One might restore evenness by defining $\Phi_\delta$ on the whole line through the $n$ sum and taking the Fourier transform over $\mathbb{R}$. Since $\int_{\mathbb{R}} \Phi_n(u)e^{iut}du = \frac{1}{4}s(s-1)\pi^{-s/2}\Gamma(s/2)\,n^{-s}$ with $s = \frac{1}{2} + it$, this gives

$$\Xi_\delta^\natural(t) = \tfrac{1}{2}\big[\xi_\delta^G(\tfrac{1}{2}+it) + \xi_\delta^G(\tfrac{1}{2}-it)\big], \qquad \xi_\delta^G(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\sum_{n\geq 1} n^{-s}e^{-\pi n^2\delta},$$

which is the real part of the Gauss damped completed zeta function on the real axis and its continuation elsewhere. Now $\Phi_\delta(u) \sim -6\pi\big(\sum_n n^2 e^{-\pi n^2\delta}\big)e^{5u/2}$ as $u \to -\infty$, only exponential decay, because the double exponential decay of $\Phi$ at $-\infty$ is again the theta formula. Correspondingly $\xi_\delta^G$ has a pole at $s = -2$ and $\Xi_\delta^\natural$ has poles at $t = \pm\frac{5}{2}i$. The reciprocal $1/\Xi_\delta^\natural(i\sqrt{w})$ then vanishes at $w = 25/4$, exactly as in Proposition 27.3.3(c). The two horns are therefore: damp on one side and get a kink (Theorem 27.5.1); damp symmetrically and get a pole.

### 27.5.2 The numerics of the source

The source tabulates zeros of the damped kernels, from mpmath at 18 to 30 digits with sign changes on a grid and an argument principle count on the box $0 < \operatorname{Re} t < T$, $|\operatorname{Im} t| < 4$. None of it was recomputed for this book except the

three values of $\Phi'_\delta(0)$ below, and the rest is unchecked. For $\delta = 0.05$ and $\delta = 0.01$ the source finds one real zero in $(0, 60)$ (at 14.1002 and 14.1263) and three zeros in the box, against 13 for $\Xi$ itself; for $\delta = 10^{-3}$ a scan to height 110 finds three real zeros (14.1338, 21.0755, 24.5364); the ratio $t^2\Xi_\delta(t)/\{-2\Phi'_\delta(0)\}$ at $t = 100$ is 0.973 for three values of $\delta$; and $\frac{1}{2}\Phi'_\delta(0) = 1.85 \times 10^{-4}$, $1.72 \times 10^{-3}$, $6.34 \times 10^{-3}$ at $\delta = 10^{-3}, 10^{-2}, 5 \times 10^{-2}$, the last three agreeing with a hand recomputation from $c_1 = 0.019749$, $c_2 = -0.019749$, $c_3 = -4.1 \times 10^{-8}$. For the symmetric damping $e^{-\delta \cosh 2u}$ (the kernel $e^{-2\pi \cosh 2u}$ of Pólya's Bessel function is treated in Section 8.7 and Proposition 8.7.16, and the cosh ladder in Chapter 22) the source finds that all zeros below height 45 stay real for $\delta = 0.5$ and 2, eight and six of them, moving up and thinning out as $\delta$ grows (unchecked). If true this distinguishes damping that respects Jacobi's identity from damping that does not, but it is evidence about low zeros only.

## 27.6 The de Bruijn Newman family and what the identity contributes

Corollary 27.5.2 says that an admissible deformation of the kernel must keep the even extension of $\Phi$ smooth at $u = 0$, and multiplying by a smooth even function of $u$ whose derivatives grow at most like $e^{Cu^2}$ does this and keeps the double exponential decay. The classical example is the de Bruijn Newman flow of Section 8.4,

$$H_t(z) = \int_0^\infty e^{tu^2}\, \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du,$$
$$\Phi_{\mathrm{dBN}}(u) = \tfrac{1}{2}\Phi(2u),$$
$$H_0(z) = \tfrac{1}{8}\Xi(z/2).$$

De Bruijn proved that $H_t$ has only real zeros for $t \geq \frac{1}{2}$, the Polymath project for $t \geq 0.22$, Platt and Trudgian for $t \geq 0.2$, and Rodgers and Tao that $\Lambda \geq 0$ (Section 8.4). The source writes the flow with $e^{\lambda u^2}\Phi(u)$ in the variable $z$, where $H_0 = \Xi$; in that normalisation $H_\lambda(z) = 8H_{4\lambda}(2z)$, the time scales by 4, and the literature thresholds become $\lambda \geq \frac{1}{8}$, 0.055 and 0.05. The statement of the source with $\lambda \geq 0.2$ is therefore true (the zeros stay real forward in time) but is not the stated literature result, and the Polymath bound is 0.22, not 0.2. The theorem is Theorem 7.2.16: for $t \geq \Lambda$ the reciprocal $R_t = H_t(0)/H_t(2i\sqrt{s})$ is a GGC transform with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$ where $z_k(t) > 0$ are the zeros of $H_t$, unconditionally for $t \geq 0.2$, and RH holds if and only if $R_t$ is a GGC for arbitrarily small $t > 0$, that is by Rodgers and Tao if and only if $\Lambda = 0$. The source's statement that the transforms converge as $\lambda \downarrow 0$ if and only if RH holds is false in the direction convergence implies RH: $H_t(i\sqrt{w}) > 0$ for every $t$, and dominated convergence gives $H_t(0)/H_t(i\sqrt{w}) \to H_0(0)/H_0(i\sqrt{w})$ for every $w > 0$ without any hypothesis. What is equivalent to RH is the GGC property for arbitrarily small $t$.

The Gaussian multiplier is $\exp(-t\,d^2/dz^2)$ acting on $H_0$, which preserves real zeros forward in time; a decreasing multiplier is the analogue of $t<0$, and Rodgers and Tao prove that every $t<0$ produces nonreal zeros. For the Gaussian family with $t<0$ and for the coefficient damping of Theorem 27.5.1 this is a theorem. That damping is the wrong direction for every decreasing multiplier, whether applied to $n$ or to $u$, is a heuristic and is not proved here (the cosh numerics above, if confirmed, show low zeros staying real), and nothing established in this chapter decides the Laguerre Pólya property of other decreasing multipliers.

What the identity of Comtet and Majumdar does contribute is a probabilistic reading of the signed object that must become positive. By Proposition 27.1.3, $\zeta$ is the Mellin transform of the negative of the signed $k$ measure $dt-d\lfloor t\rfloor$ of the uniform law (equation (27.1.4)). Under RH, $\Xi(0)/\Xi(i\sqrt{w})$ is the GGC transform with Thorin measure the counting measure of the squared ordinates, and the unnormalised $1/\Xi(i\sqrt{w})$ is only its constant multiple with the same Thorin measure. The identity therefore links two counting measures: integers minus Lebesgue on the arithmetic side, zeros on the spectral side. For real $s\in(0,1)$, $g<0$ and $\sin(\pi s/2)>0$ give $\zeta(s)<0$ by (27.1.1), which is the statement that $\zeta$ has no real zeros there.

Problem 27.1. Find the analogue, for the explicit formula, of the convexity and Poisson summation argument of Theorem 27.4.4: a structural reason, not relying on zero locations, why the Lorentzian smoothing of $M_\xi$ against $1-\cos xy$ stays positive in the strip $\frac12<\alpha\le1$. (Positivity for $\alpha\ge1$ is Proposition 9.1.12.)

Problem 27.2. Find a one parameter family of even multipliers $m_\delta(u)\ge1$, $m_\delta\to1$, for which the walk interpretation of Proposition 27.1.2 survives (for instance $m_\delta$ arising from subordinating the stable walk) and for which real zeros of $\int_{\mathbb R} m_\delta\Phi\,e^{iut}du$ can be proved for $\delta$ below the known threshold $t=0.2$ of the flow.

Problem 27.3. The symmetric damping $m_\delta(u)=e^{-\delta\cosh 2u}$ is even and smooth, so every constraint of Corollary 27.5.2 holds for $m_\delta\Phi$, and its improved decay is immediate. Decide whether its transforms lie in $\mathcal{LP}$. It is not the same as weighting $\theta$ by the Jacobi invariant weight $e^{-\delta(x+1/x)/2}$: with $L$ the operator taking $\theta$ to $\Phi$, $L(W\theta)\ne WL\theta$ because the product rule leaves terms in $W'$ and $W''$, so $m_\delta\Phi$ is the right object. For $A,B>0$ one has $\int_{\mathbb R} e^{(a+it)u-Ae^{2u}-Be^{-2u}}\,du=(B/A)^{(a+it)/4}K_{(a+it)/2}(2\sqrt{AB})$ (substitute $x=e^{2u}$ in the integral defining $K$), and each term of $m_\delta\Phi$ has $A=\pi n^2+\delta/2$, $B=\delta/2$ and $a=\frac92$ or $\frac52$, which gives Bessel $K$ sums of Chowla Selberg type (compare Chapter 23); this identity gives no conclusion about zeros. A height of order $\exp(c/\delta)$ at which nonreal zeros appear is only a conjecture suggested by the multiplier $e^{-\delta\cosh 2u}\le1$, with no theorem or numerical certification, and the low lying numerics are not decisive.

Proposition 9.1.12 settles the source's first open problem, compensated posi-

tivity of $\xi$ for $1 < \alpha < 7.789$.

# Chapter 28

# Truncations of the beta lattice and Hermite coefficients

Section 7.2 truncates the lattice of Biane, Pitman and Yor for zeta after $n$ terms. The truncated Mellin transforms are explicit, their zeros move to the zeros of $\xi$ at the first order rate $\kappa(\rho)/n$ (Proposition 7.2.2), the exact form of $\kappa(\rho)$ ties the side of approach to the Riemann Siegel phase and so to Gram's law (Theorem 7.2.3), and a single zero approached from the wrong side, $\rho_{213}$, shows that the one sided conjecture is false (Corollary 7.2.4). Symmetrising by the functional equation removes the mechanism and leaves a statement that is not refuted (Proposition 7.2.6). This chapter carries that programme to the Dirichlet beta function $L(s, \chi_4)$, where the lattice is $\lambda_k = 2\pi^2(k - \frac{1}{2})^2$ and the Mellin transform is the completed $L$ function, and it adds what the zeta section does not have: the orders $m \geq 1$ of cumulant matching, an exact mean corrected truncation as a finite sum of incomplete gamma functions, and a computation of the drift at all 379 zeros below height 500. The second half is a separate subject, the expansion of $\Xi$ in Hermite functions at a small scale, whose coefficients are samples of $\Xi$ at the turning points of the Hermite functions, up to an amplitude factor and an error $O(a^4)$, and have been used to count zeros from the Pólya kernel alone.

The chapter develops two notes of Nick Polson, on the Thorin truncation of $L(s, \chi_4)$ [245] and on progress with open problems about the Hermite coefficients of $\Xi$ [237], together with the computations done for this book (the scripts are `code/tr_*.py`, listed in `code/README.md`). Where the recomputation differed from a note the recomputed value is printed, and a statement that was not recomputed is marked unchecked. Several statements of the notes were restated in the course of the merge: the drift law of the first note holds for a remainder only under a hypothesis on its first unmatched cumulant (Theorem 28.2.2, Proposition 28.2.4), the corollary that no one sided criterion can hold is conditional on one zero with a negative drift (Corollary 28.3.2), and the asymptotic of the second note is proved in an additive form that is uniform in $m$ and fails in the relative form at the zeros of $\Xi$ (Theorem 28.8.1, Proposition 28.8.2).

Notation is the book's, with these warnings. The function $\xi_4(s) = \xi(s, \chi_4)$ is

that of Section 8.5.7, and the letter $\Lambda$, which the first note uses for $\frac{\sqrt{\pi}}{2}\xi_4$, is the de Bruijn Newman constant here. The variable $\omega$ is the exit time of Remark 8.5.23 divided by $2\pi$. The functions $\vartheta_\beta$ and $Z_\beta$ are the Riemann Siegel and Hardy functions of $\chi_4$, written with a subscript so that $\vartheta$ and $Z$ remain those of zeta. The kernel $\Phi$ is that of (1.2.3) (the second note does not define it), $H_t$ is the de Bruijn Newman function of Section 8.4, and $H_n$ with an integer index is the Hermite polynomial of Section 24.6. The letter $a$ is the Hermite scale in the second half and a constant is written $\lambda_* = 2\pi^2$ in the first.

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it, conditional, heuristic or open as stated. A computation in floating point is called computed and never certified; the zero counts of this chapter are of that kind. The results are these.

1. GRH for $\chi_4$ is equivalent to a local zero free condition on the truncated Mellin transforms (Theorem 28.1.1, proved). The truncated zeros move by $\delta_m n^{-(2m+1)}$ for the hard cutoff and the mean correction, and by a multiple of the first unmatched cumulant difference in general (Theorem 28.2.2, Corollary 28.2.3, proved); the formula for $m \geq 2$ with a generalised gamma convolution remainder is false (Proposition 28.2.4, proved), and the drift of the first zero is computed for the three kinds of remainder (Table 28.1).

2. The sign of the real part of the drift is a Gram parity of $\chi_4$ with two corrections (Theorem 28.3.1, proved). Among the 379 zeros below height 500 the mean correction approaches from the left exactly at the five zeros with anomalous Gram parity (computed, Table 28.2). A zero with negative drift excludes a one sided criterion at that order (Corollary 28.3.2, proved), and the premise is computed and not proved.

3. The truncations have transient zeros in the left half strip (Table 28.3, computed for the hard cutoff and the mean correction). The symmetrised truncations satisfy a Hurwitz criterion (Theorem 28.5.1, proved) and keep every simple critical zero on the line locally (Proposition 28.5.2, proved); the heights reached are those of the computed tables, and the mean corrected rows are computed as well.

4. The Hermite coefficients of $\Xi$ at scale $a$ are the Taylor coefficients of $e^{z^2/2}$ times the backward heat flow of $\Xi$ for the time $2a^2$ (Theorem 28.7.1, proved). They sample $\Xi$ at the turning points $a\sqrt{4m+1}$ with an additive error $O(a^4)$ that is uniform in $m$ (Theorem 28.8.1, proved), and the first negative coefficient reads off $\gamma_1$ (Corollary 28.9.1, proved; computed, Table 28.6).

5. The number of sign changes of the coefficients equals the number of zeros of zeta up to height 1000 at scales up to about 5 to 6, with a lag of order one

at the top edge (Remark 28.10.1, computed); what a verified count would prove is Proposition 28.10.2.

The open problems close each half.

## 28.1 The beta lattice and its truncations

Write $\xi_4(s) = \xi(s,\chi_4) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_4)$ for the completed Dirichlet beta function of Section 8.5.7, entire and symmetric about $s=\frac12$, and let $\omega$ be the exit time variable of Remark 8.5.23 divided by $2\pi$. Then

$$\mathbb{E}e^{-\lambda\omega} = \operatorname{sech}\sqrt{\lambda/2} = \prod_{k\ge1}\Big(1+\frac{\lambda}{\lambda_k}\Big)^{-1}, \qquad \lambda_k = \lambda_*\big(k-\tfrac12\big)^2, \quad \lambda_* = 2\pi^2,$$

so that $\omega = \sum_k e_k/\lambda_k$ with independent standard exponential variables $e_k$, and the Thorin measure of $\omega$ is $\sum_k \delta_{\lambda_k}$. By Remark 8.5.23 and the functional equation of $\xi_4$ the Mellin transform is

$$M(z) = \mathbb{E}\,\omega^{-z} = (2\pi)^z\xi_4(2z), \tag{28.1.1}$$

which is $\frac{2}{\sqrt\pi}(2\pi)^z\Lambda(2z)$ in the notation of the source note [245] (its $\Lambda$ is $\frac{\sqrt\pi}{2}\xi_4$; the letter $\Lambda$ is the de Bruijn Newman constant in this book). The nontrivial zeros of $L(s,\chi_4)$ are the points $z=\rho/2$ of the strip $0<\operatorname{Re}z<\frac12$ at which $M$ vanishes, and $g(z)=(2\pi)^{-z}M(z)=\xi_4(2z)$ is symmetric about $\operatorname{Re}z=\frac14$. GRH for $\chi_4$ says that all these zeros lie on $\operatorname{Re}z=\frac14$. This is the lattice of Section 7.2 with the atoms $\lambda_k$ of the Dirichlet beta function in place of the squares $k^2$ of the zeta function.

Split $\omega=\omega_n+T_n$ into the first $n$ atoms and the tail $T_n=\sum_{k>n}e_k/\lambda_k$. With $q_j(n)=\sum_{k>n}\lambda_k^{-j}$ the cumulants of the tail are $\kappa_j(T_n)=(j-1)!\,q_j(n)$, and by the midpoint rule (proved below, Lemma 28.2.1)

$$\kappa_j(T_n) = \frac{(j-1)!}{\lambda_*^j(2j-1)\,n^{2j-1}}\Big(1-\frac{j(2j-1)}{12\,n^2}+O(n^{-4})\Big). \tag{28.1.2}$$

A truncation replaces $T_n$ by a variable $R_n\ge0$ independent of $\omega_n$ and puts $M_n(z)=\mathbb{E}(\omega_n+R_n)^{-z}$. Three choices are used. The first two are definite, the third is a family.

*Hard cutoff* $R_n=0$. With $A_k^{(n)}=\prod_{j\le n,\,j\ne k}\lambda_j/(\lambda_j-\lambda_k)$ the density of $\omega_n$ is $\sum_{k\le n}A_k^{(n)}\lambda_k e^{-\lambda_k w}$, and

$$M_n^{(0)}(z)=\mathbb{E}\omega_n^{-z}=\Gamma(1-z)\sum_{k\le n}A_k^{(n)}\lambda_k^z,$$

an exponential polynomial times $\Gamma(1-z)$. It is holomorphic for $\operatorname{Re}z<n$ (the density behaves like $w^{n-1}$ at 0); the poles of $\Gamma(1-z)$ at $1,\dots,n-1$ cancel and the pole at $n$ does not, so $M_n^{(0)}$ is not entire.

*Mean correction* $R_n = c_n = \mathbb{E}T_n = q_1(n) = \psi'(n+\frac{1}{2})/(2\pi^2)$, a constant. Since $\int_0^\infty (w+c)^{-z}\lambda e^{-\lambda w}dw = \lambda^z e^{\lambda c}\Gamma(1-z,\lambda c)$,

$$M_n^{(1)}(z) = \mathbb{E}(\omega_n + c_n)^{-z} = \sum_{k\le n} A_k^{(n)} \lambda_k^z e^{\lambda_k c_n}\,\Gamma(1-z,\lambda_k c_n),$$

a finite sum of upper incomplete gamma functions with rational weights, entire in $z$. It is of the same incomplete gamma type as the approximants of $\Xi$ of Haglund [168] (Chapter 22, Chapter 25); the weights are those of a Thorin lattice of $L(s,\chi_4)$, not of $\Xi$.

*Matching* $R_n \ge 0$ independent of $\omega_n$ whose first $m$ cumulants equal those of $T_n$. For $m = 0$ nothing is matched, so the hard cutoff is one admissible choice, and the constant $c_n$ is the choice for $m = 1$ with a deterministic remainder. A mean match alone does not force $R_n$ to be constant. The distinction matters for $m \ge 2$ (Section 28.2).

At $n = 40$ and $z = 0.3$ the mean corrected value is $M_{40}^{(1)}(0.3) = 1.70347125$ against $M(0.3) = 1.70347184$, and the hard cutoff at $n = 200$ gives $M_{200}^{(0)}(0.3) = 1.70461$ (computed, `code/tr_drift.py`, by the Laplace integral of the density with no cancellation of the $A_k^{(n)}$; two runs at different precision and step agree to $10^{-34}$; the printed digits are 1.703471255, 1.70347184 and 1.704611458).

Theorem 28.1.1 (Local criterion). *Let $R_n \ge 0$ be independent of $\omega_n$ with $R_n \to 0$ in probability, as for the hard cutoff and for every nonnegative remainder with $\mathbb{E}R_n = c_n$ (Markov's inequality gives $R_n \to 0$ in probability, since $c_n = O(n^{-1})$). Then $M_n \to M$ locally uniformly on $0 < \operatorname{Re} z < \frac{1}{2}$, and GRH for $\chi_4$ holds if and only if every compact subset of $0 < \operatorname{Re} z < \frac{1}{4}$ is free of zeros of $M_n$ for all sufficiently large $n$.*

*Proof.* Let $X_n = \omega_n + R_n$. Coupling the exponential summands gives $\omega_n \uparrow \omega$ almost surely, so $X_n \to \omega$ in probability, and the laws of $X_n$ converge weakly to that of $\omega$. Let $K$ be a compact subset of the strip and choose $b \in (\max_K \operatorname{Re} z, 2)$. For $n \ge 2$, $X_n \ge \omega_2$, so $|X_n^{-z}| \le 1 + \omega_2^{-b}$, which is integrable because the density of $\omega_2$ is $O(w)$ at 0. Weak convergence and this uniform integrability give $\mathbb{E}X_n^{-z} \to \mathbb{E}\omega^{-z}$ for every $z \in K$. The same bound on a slightly larger compact set gives holomorphy and local boundedness, and Vitali's theorem gives local uniform convergence.

If GRH holds, $M$ has no zero on a compact $K \subset \{0 < \operatorname{Re} z < \frac{1}{4}\}$, so $\min_K |M| > 0$ and $M_n \ne 0$ on $K$ for large $n$. Conversely, if $M$ has a zero in $0 < \operatorname{Re} z < \frac{1}{4}$, a small closed disc about it with zero free boundary lies in the half strip, and Rouché's theorem gives zeros of $M_n$ in the disc for every large $n$, contradicting the property ($M$ is not identically zero, being positive on the real segment). A zero in $\frac{1}{4} < \operatorname{Re} z < \frac{1}{2}$ is carried to the left half strip by the functional equation $(2\pi)^{-z}M(z) = (2\pi)^{z-\frac{1}{2}}M(\frac{1}{2}-z)$. □

The theorem converts GRH into a statement about explicit finite objects, but it is an equivalence: all the difficulty is in controlling the zeros of $M_n$ uniformly. It is the analogue of Theorem 7.2.1 for $\chi_4$, and the rest of the section studies how the zeros of $M_n$ move.

## 28.2 The drift of the truncated zeros

For $m = 0$ and zeta the drift of a truncated zero is Proposition 7.2.2. The statement for $\chi_4$ and every order $m$ needs one point that the first version of the source note missed: a remainder that matches the first $m$ cumulants of the tail has a first unmatched cumulant of its own, and the drift depends on the difference of the two. The proofs below are for $r = m+1 \ge 1$ and use the following elementary tail estimate.

Lemma 28.2.1. *For fixed $j \ge 1$, $q_j(n) = \lambda_*^{-j}\{n^{1-2j}/(2j-1) - \frac{j}{12}n^{-2j-1} + O(n^{-2j-3})\}$. In particular*

$$c_n = \frac{1}{\lambda_* n} - \frac{1}{12\lambda_* n^3} + O(n^{-5}),$$
$$v_n = \operatorname{Var} T_n = q_2(n) = \frac{1}{3\lambda_*^2 n^3} - \frac{1}{6\lambda_*^2 n^5} + O(n^{-7}),$$
$$\kappa_3(T_n) = \frac{2}{5\lambda_*^3 n^5} + O(n^{-7}).$$

*Proof.* For $p > 1$, expand $x^{-p}$ about the midpoint of $[k-1, k]$ to fourth order: the quadratic term contributes $p(p+1)(k-\frac12)^{-p-2}/24$ and the remainder sums to $O(n^{-p-3})$. Summing over $k > n$ gives $\sum_{k>n}(k-\frac12)^{-p} = n^{1-p}/(p-1) - (p/24)n^{-p-1} + O(n^{-p-3})$, using the same estimate without the correction for $\sum_{k>n}(k-\frac12)^{-p-2}$. Take $p = 2j$. □

Fix $m \ge 0$ and write $r = m+1$, $p = 2r-1$. Let $R_n \ge 0$ be independent of $\omega_n$ with $\kappa_j(R_n) = \kappa_j(T_n)$ for $1 \le j < r$, and assume, for a constant $C$ independent of $n$ and $j$,

$$\log \mathbb{E}e^{-tR_n} = \sum_{j\ge1} \frac{(-1)^j \kappa_j(R_n)}{j!} t^j \quad (0 \le t < n^2/C), \qquad \frac{|\kappa_j(R_n)|}{j!} \le C^j n^{1-2j}. \tag{28.2.1}$$

This is a sufficient condition and not a consequence of matching finitely many cumulants. It holds for the hard cutoff, the constant remainder and the gamma remainders below, and for the tail $T_n$ itself after increasing $C$, since $\log \mathbb{E}e^{-tT_n} = -\sum_{k>n}\log(1+t/\lambda_k)$ and $q_j(n) \le (2j-1)^{-1}\lambda_*^{-j}(n-\frac12)^{1-2j}$. Put $\Delta_{r,n} = \kappa_r(T_n) - \kappa_r(R_n)$.

THEOREM 28.2.2 (Drift law). *Under these hypotheses, uniformly for $z$ in a compact subset $K$ of $\operatorname{Re} z > 0$,*

$$M_n(z) - M(z) = -\frac{(-1)^r \Delta_{r,n}}{r!}\,(z)_r\, M(z+r) + O_K\big(n^{-p-2} + n^{-2p}\big), \tag{28.2.2}$$

*with $(z)_r = z(z+1)\cdots(z+r-1)$. Let $\rho = \frac12 + i\gamma$ be a simple zero of $\xi_4$ and $z_0 = \rho/2$. For all large $n$ a fixed small disc about $z_0$ contains exactly one zero $z_n$ of $M_n$, it is simple, and*

$$\begin{aligned} z_n - z_0 &= \frac{(-1)^r \Delta_{r,n}}{r!}(z_0)_r \frac{M(z_0+r)}{M'(z_0)} + O\big(n^{-p-2} + n^{-2p}\big), \\ \frac{M(z_0+r)}{M'(z_0)} &= \frac{(2\pi)^r \xi_4(\rho+2r)}{2\,\xi_4'(\rho)}. \end{aligned} \tag{28.2.3}$$

*If $\Delta_{r,n} = d\,n^{-p} + o(n^{-p})$ with $d \neq 0$, then $z_n - z_0 = D_r n^{-p} + o(n^{-p})$ with $D_r = \dfrac{(-1)^r d\,(2\pi)^r (z_0)_r\, \xi_4(\rho+2r)}{2\,r!\,\xi_4'(\rho)} \neq 0$.*

*Proof.* The Laplace transforms satisfy $\mathbb{E}e^{-t(\omega_n+R_n)} = L(t)e^{D_n(t)}$ with $L(t) = \operatorname{sech}\sqrt{t/2}$ and $D_n = \log \mathbb{E}e^{-tR_n} - \log \mathbb{E}e^{-tT_n}$. The first $r-1$ cumulants cancel, so (28.2.1) gives, for $0 \le t \le n^2/(2C)$, $D_n(t) = -\frac{(-1)^r\Delta_{r,n}}{r!}t^r + O(n^{-p-2}t^{r+1})$ with $\Delta_{r,n} = O(n^{-p})$. On $0 \le t \le s_n = A(\log n)^2$ this is $o(1)$, and the exponential gives $e^{D_n} - 1 = -\frac{(-1)^r\Delta_{r,n}}{r!}t^r + O(n^{-p-2}t^{r+1} + n^{-2p}t^{2r})$. For $t \ge s_n$: with $B = \max_K \operatorname{Re} z$, an integer $Q > B$ and $J_n = \lfloor \sqrt{s_n/\lambda_*} \rfloor$, the first $Q$ factors of $\mathbb{E}e^{-t\omega_n}$ are at most $C_Q t^{-Q}$ and each of the factors with $Q < k \le J_n$ is at most $\frac12$, so $\mathbb{E}e^{-t\omega_n} \le C_Q t^{-Q} 2^{-(J_n - Q)}$. Since $R_n \ge 0$, both $L(t)$ and $\mathbb{E}e^{-t(\omega_n+R_n)}$ are bounded by it, and the Mellin integrals over $[s_n,\infty)$ are $O(n^{-N})$ for every $N$ once $A$ is large. The identity $\mathbb{E}X^{-z} = \Gamma(z)^{-1}\int_0^\infty t^{z-1}\mathbb{E}e^{-tX}\,dt$ converges absolutely for $\operatorname{Re} z > 0$ and large $n$, and $\Gamma(z+r)/\Gamma(z) = (z)_r$, so integrating the expansion gives (28.2.2).

By Rouché's theorem a small disc about the simple zero $z_0$ contains exactly one zero $z_n$ of $M_n$, and it is simple. Put $E_n = M_n - M$; then $E_n = O(n^{-p})$ and, by Cauchy's estimate on a slightly larger disc, $E_n' = O(n^{-p})$. Thus $z_n - z_0 = O(n^{-p})$, and Taylor's formula $0 = M'(z_0)(z_n - z_0) + E_n(z_0) + O(n^{-2p})$ gives (28.2.3). The ratio follows from (28.1.1): $M'(z_0) = 2(2\pi)^{z_0}\xi_4'(\rho)$ and $M(z_0+r) = (2\pi)^{z_0+r}\xi_4(\rho+2r)$. Finally $(z_0)_r \neq 0$, $\xi_4'(\rho) \neq 0$, and $\xi_4(\rho+2r) \neq 0$ because $\operatorname{Re}(\rho+2r) > 1$, where the Euler product of $L(s,\chi_4)$ has no zero, so $D_r \neq 0$ when $d \neq 0$. □

COROLLARY 28.2.3 (The two named schemes). *Let $\rho = \frac12 + i\gamma$ be a simple zero of $\xi_4$. Then*

$$\delta_m = \frac{(-1)^{m+1}(z_0)_{m+1}\,\xi_4(\rho+2m+2)}{2(m+1)(2m+1)\,\pi^{m+1}\,\xi_4'(\rho)}, \tag{28.2.4}$$

*and for the hard cutoff* $z_n = z_0 + \delta_0/n + O(n^{-2})$*, for the mean correction* $z_n = z_0 + \delta_1/n^3 + O(n^{-5})$*. For a remainder that satisfies the hypotheses of Theorem 28.2.2 at order* $m$ *and has* $\kappa_{m+1}(R_n) = o(n^{-(2m+1)})$ *the displacement is* $\delta_m n^{-(2m+1)}(1 + o(1))$*.*

*Proof.* If $\kappa_r(R_n) = o(n^{-p})$, then $d = (r-1)!/[\lambda_*^r(2r-1)]$ by Lemma 28.2.1, and $D_r$ of Theorem 28.2.2 is $\delta_m$ of (28.2.4). For the hard cutoff $r = 1$ and $\Delta_{1,n} = c_n$; for the constant $c_n$, $r = 2$, $\Delta_{2,n} = v_n$. The error is $O(n^{-3} + n^{-2})$ for $r = 1$ and $O(n^{-5} + n^{-6})$ for $r = 2$. For the hard cutoff the relative error is $O(n^{-1})$, not $O(n^{-2})$: for $r = 1$ the error term $n^{-2p} = n^{-2}$ of (28.2.2) is the term $\frac{1}{2}c_n^2(z)_2 M(z+2)$ of the expansion of $e^{D_n}$, which is not cancelled by the variance term $v_n = O(n^{-3})$ and has a nonzero coefficient at $z_0$. □

The first zero of $L(s, \chi_4)$ is at $\gamma = 6.0209489$. Table 28.1 compares $\delta_m$ with the measured $n^{2m+1}(z_n - z_0)$ (computed, `code/tr_drift.py`, precision 40 digits with step $h = 1/4$ against 50 digits with step $h = 1/8$, differences below $10^{-26}$). The hard cutoff approaches $\delta_0$ with relative error $(0.27 + 0.51i)/n$, of modulus $0.58/n$, and the mean correction with an error that falls by a factor of 4 at each doubling of $n$, as the orders $O(n^{-1})$ and $O(n^{-2})$ say; the finite $n$ values agree with the predicted constants to two or three digits, not four. For the hard cutoff the real part of $z_n - \frac{1}{4}$ is $9.44 \times 10^{-3}$ at $n = 128$ and $4.71 \times 10^{-3}$ at $n = 256$. For the mean correction it is $5.87 \times 10^{-5}$, $3.10 \times 10^{-4}$, $4.72 \times 10^{-5}$, $6.15 \times 10^{-6}$, $7.76 \times 10^{-7}$, $9.73 \times 10^{-8}$ at $n = 4, 8, 16, 32, 64, 128$. The value at $n = 4$ is small by accident (the law $\operatorname{Re}\delta_1/n^3$ gives $3.2 \times 10^{-3}$ there), and the sequence is not monotone. The hard cutoff would need $n \approx 1.5 \times 10^6$ to reach $8 \times 10^{-7}$, from $\operatorname{Re}\delta_0/(8 \times 10^{-7})$ (computed from the asymptotic law, not from a run).

| $n$ | hard cutoff, $n(z_n - z_0)$ | mean correction, $n^3(z_n - z_0)$ | gamma remainder, $n^5(z_n - z_0)$ |
|---|---|---|---|
| 16 | $1.2424 - 0.5466i$ | $0.1934 - 1.0874i$ | $-1.1423 - 0.4084i$ |
| 32 | $1.2234 - 0.5618i$ | $0.2015 - 1.0853i$ | $-1.1429 - 0.4370i$ |
| 64 | $1.2137 - 0.5692i$ | $0.2035 - 1.0849i$ | $-1.1428 - 0.4443i$ |
| 128 | $1.2089 - 0.5728i$ | $0.2040 - 1.0848i$ | $-1.1428 - 0.4461i$ |
| limit | $\delta_0 = 1.2041 - 0.5763i$ | $\delta_1 = 0.2041 - 1.0848i$ | $\frac{4}{9}\delta_2 = -1.1428 - 0.4467i$ |

Table 28.1: The drift of the first zero of $M_n$ at $\gamma = 6.0209489$ (computed). The last column is the gamma remainder of Proposition 28.2.4, for which $\delta_2 = -2.5713 - 1.0051i$ is not the limit.

### 28.2.1 Why the matching hypothesis cannot be dropped

The source note states the formula (28.2.4) for every $m$ and every remainder that matches $m$ cumulants, for instance a generalised gamma convolution with a few atoms. This is false for $m \geq 2$, because the first unmatched cumulant of $R_n$ enters.

Proposition 28.2.4. *Let $R_n$ have the gamma law with shape $c_n^2/v_n$ and scale $v_n/c_n$, so that $\mathbb{E}R_n = c_n$ and $\operatorname{Var} R_n = v_n$ exactly. It satisfies* (28.2.1)*, and with $r = 3$ the drift law gives $z_n = z_0 + \frac{4}{9}\delta_2\, n^{-5} + O(n^{-7})$. A second remainder $R_n = G_{1,n} + G_{2,n}$, a sum of two independent gamma variables with matched mean and variance and the same natural scale, gives $z_n = z_0 - \frac{1}{9}\delta_2\, n^{-5} + O(n^{-7})$. A bounded remainder $R_n = b_n + hB_n$ with $B_n$ a Bernoulli variable, $h > 0$ small and fixed, matched to the mean and variance, has $z_n - z_0 = cn^{-3} + O(n^{-5})$ with $c \neq 0$.*

*Proof.* For the gamma law $\kappa_3(R_n) = 2v_n^2/c_n$, so $\Delta_{3,n} = 2\big(q_3 - q_2^2/q_1\big) = \frac{8}{45\lambda_*^3 n^5} + O(n^{-7}) = (\frac{4}{9} + O(n^{-2}))\kappa_3(T_n)$ by Lemma 28.2.1, and the difference in brackets is positive for every $n$ by the Cauchy Schwarz inequality applied to $\lambda_k^{-1/2}$ and $\lambda_k^{-3/2}$. Theorem 28.2.2 gives the first claim. For two gamma variables with shapes and scales $\big(\frac{8c_n}{5s_n}, \frac{s_n}{2}\big)$ and $\big(\frac{c_n}{15s_n}, 3s_n\big)$, $s_n = v_n/c_n$, the mean is $c_n$, the variance is $v_n$ and $\kappa_3(R_n) = 4v_n^2/c_n$, so $\Delta_{3,n} = -\frac{2}{45\lambda_*^3 n^5} + O(n^{-7}) = (-\frac{1}{9} + O(n^{-2}))\kappa_3(T_n)$. For the Bernoulli remainder put $p_n = \frac{1}{2}(1 - \sqrt{1 - 4v_n/h^2})$ and $b_n = c_n - hp_n$, so that $R_n \geq 0$ for large $n$ with mean $c_n$ and variance $v_n$ exactly; then $\kappa_3(R_n) = hv_n(1-2p_n)$ is of order $n^{-3}$. Write $M_h(z) = \mathbb{E}(\omega+h)^{-z}$ and $H_h(z) = h^{-2}\big[M_h(z) - M(z) + hzM(z+1)\big] - \frac{1}{2}(z)_2 M(z+2)$. The method of the proof of Theorem 28.2.2 gives $M_n(z) - M(z) = v_n H_h(z) + O_K(n^{-5})$, and Taylor's theorem gives $H_h(z) = -\frac{h}{6}(z)_3 M(z+3) + O(h^2)$, which does not vanish at $z_0$ for small $h$; hence $z_n - z_0 = -H_h(z_0)/(3\lambda_*^2 M'(z_0))\, n^{-3} + O(n^{-5})$ with a nonzero coefficient. □

The computed gamma column of Table 28.1 agrees: the ratio of $n^5(z_n - z_0)$ to $\delta_2$ is 0.4444 at $n = 128$, against $\frac{4}{9} = 0.4444$ (computed). For $m = 2$ the choice of a generalised gamma convolution cannot repair the formula. For every positive measure $U$ the moments $m_j = \int u^{-j} U(du)$ satisfy $m_1 m_3 \geq m_2^2$, so a generalised gamma convolution remainder with the mean and variance of $T_n$ has $\kappa_3(R_n) \geq 2v_n^2/c_n$, which is of exact order $n^{-5}$ and not $o(n^{-5})$ (proved, by the Cauchy Schwarz inequality). The one gamma remainder is the extreme case. So the coefficient $\delta_m$ of (28.2.4) for $m \geq 2$ belongs to a hypothetical remainder with $\kappa_{m+1}(R_n) = o(n^{-(2m+1)})$, and not to the generalised gamma convolutions the note proposes.

## 28.3 The side of approach is a Gram parity

For zeta the exact form of the first order drift is Theorem 7.2.3: the sign of $\operatorname{Re}\kappa(\rho)$ is, up to the phase of $\zeta(\frac{5}{2}+i\gamma)$, the sign of $\sin\theta(\gamma)/Z'(\gamma)$, a Gram parity. The same holds for $\chi_4$ and for every $m$ with one more correction. Put $G(s) = (4/\pi)^{s/2}\Gamma(\frac{s+1}{2})$, so that $\xi_4 = \frac{2}{\sqrt{\pi}} GL$, and let $\vartheta_\beta(x) = \arg G(\frac{1}{2}+ix)$ and $Z_\beta(x) = e^{i\vartheta_\beta(x)} L(\frac{1}{2}+ix, \chi_4)$ be the Riemann Siegel function and the Hardy function of $\chi_4$ (the letters $\vartheta$ and $Z$ are those of zeta in Section 2.2.1, and the subscript is for beta). Then $G(\frac{1}{2} +$

$ix)L(\frac{1}{2}+ix,\chi_4) = |G|\,Z_\beta(x)$ is real, and the Gram points of $\chi_4$ are the solutions of $\vartheta_\beta(x) = j\pi$.

THEOREM 28.3.1 (Phase law). *Let $\rho = \frac{1}{2}+i\gamma$, $\gamma>0$, be a simple zero of $\xi_4$ and let $\delta_m$ be the number* (28.2.4). *Then*

$$\delta_m = \frac{i\,C_m(\gamma)\,e^{i\Theta_m(\gamma)}}{Z'_\beta(\gamma)}, \qquad C_m(\gamma)>0,$$

$$\Theta_m(\gamma) = \vartheta_\beta(\gamma) + \arg L(\rho+2m+2,\chi_4) - \sum_{j=0}^{m}\Big[\arctan\frac{\frac{1}{2}+2j}{\gamma} + \arctan\frac{\frac{3}{2}+2j}{\gamma}\Big],$$

*so that* $\operatorname{sgn}\operatorname{Re}\delta_m = \operatorname{sgn}\big[-\sin\Theta_m(\gamma)\,Z'_\beta(\gamma)\big]$, *zero included. For an actual remainder satisfying the hypotheses of Theorem 28.2.2 and with $\eta_m = \lim_n\big[\kappa_{m+1}(T_n) - \kappa_{m+1}(R_n)\big]/\kappa_{m+1}(T_n)$ the displacement is $\eta_m\delta_m n^{-(2m+1)}(1+o(1))$ when the limit exists and $\eta_m\neq 0$, and the sign of the real part is* $\operatorname{sgn}\big[-\eta_m\sin\Theta_m(\gamma)Z'_\beta(\gamma)\big]$ *when* $\eta_m\sin\Theta_m\neq 0$.

*Proof.* The gamma factor satisfies $G(s+2) = \frac{4}{\pi}\cdot\frac{s+1}{2}G(s)$, so $\xi_4(\rho+2m+2) = \frac{2}{\sqrt{\pi}}(\frac{4}{\pi})^{m+1}\prod_{j=0}^{m}\big(\frac{\rho+1}{2}+j\big)G(\rho)L(\rho+2m+2,\chi_4)$. On the critical line $G(\rho) = |G(\rho)|e^{i\vartheta_\beta(\gamma)}$, and differentiating $G(\frac{1}{2}+ix)L(\frac{1}{2}+ix,\chi_4) = |G|Z_\beta(x)$ at a zero gives $i\,GL'(\rho) = |G|Z'_\beta(\gamma)$, that is $\frac{\sqrt{\pi}}{2}\xi'_4(\rho) = -i|G(\rho)|Z'_\beta(\gamma)$. In (28.2.4) the product $(z_0)_{m+1}\prod_j(\frac{\rho+1}{2}+j) = \prod_{j=0}^{m}(\frac{\rho}{2}+j)(\frac{\rho+1}{2}+j)$ has factors $\frac{1}{4}+j+\frac{i\gamma}{2}$ and $\frac{3}{4}+j+\frac{i\gamma}{2}$ with arguments $\frac{\pi}{2}-\arctan\frac{1/2+2j}{\gamma}$ and $\frac{\pi}{2}-\arctan\frac{3/2+2j}{\gamma}$ for $\gamma>0$. The terms $\frac{\pi}{2}$ total $(m+1)\pi$ and cancel $(-1)^{m+1}$, which leaves $\delta_m = iC_me^{i\Theta_m}/Z'_\beta(\gamma)$ with $C_m = (4/\pi)^{m+1}|\prod|\,|L(\rho+2m+2,\chi_4)|/[2(m+1)(2m+1)\pi^{m+1}]$. Taking real parts, $\operatorname{Re}(ie^{i\Theta}) = -\sin\Theta$. The last statement is Theorem 28.2.2 with $d = \eta_m\lim_n n^{2m+1}\kappa_{m+1}(T_n)$. □

At two consecutive simple zeros $Z'_\beta$ has opposite signs, so the sign of the drift alternates with the parity of the zero index unless $\sin\Theta_m$ changes sign, which is the Gram law of $\chi_4$ read through $\Theta_m$. The two corrections to $\vartheta_\beta$ are the arithmetic phase $\arg L(\rho+2m+2,\chi_4)$ and the Pochhammer phase, which is of exact size $-2(m+1)^2/\gamma + O_m(\gamma^{-3})$ (proved, first order sums). The arithmetic phase is bounded, $|L(\sigma+i\gamma,\chi_4)-1|\le\epsilon_\sigma = \sum_{k\ge1}(2k+1)^{-\sigma}$, but it does not tend to zero at fixed $\sigma>1$: the mean square of $\operatorname{Im}L(\sigma+it,\chi_4)$ over $[0,T]$ tends to $\frac{1}{2}\sum_{n\ge2}\chi_4(n)^2n^{-2\sigma}>0$ (proved, by the orthogonality of distinct frequencies). For zeta the corresponding statement is that $\operatorname{Re}\kappa$ has the sign of $-\cos(\pi S_{\mathrm{mid}})$ up to a shift of the threshold $\frac{1}{2}$ by about 0.11 (Section 7.2).

### 28.3.1 All zeros below height 500

The zeros of $L(s,\chi_4)$ below height 500 were located by sign changes of $Z_\beta$ on two independent grids (steps 0.02 and 0.0137) and refined at two precisions (computed,

`code/tr_zeros.py`, data in `data/tr_zeros.json`). There are 379 zeros, all simple and on the line as far as the computation sees (the minimal gap is 0.1336, against the step 0.02), in agreement with $\vartheta_\beta(500)/\pi = 379.15$ and with $N(500) = 379.1$ from (8.5.8). This is not a certified count: there is no Turing bound, and the certified count in this book stops at height 300, where $N(300) = 203$ (Section 21.6.1). For each zero the numbers $\delta_m$, $0 \le m \le 5$, were computed from (28.2.4), and the sign law of Theorem 28.3.1 holds at all 379 zeros for each $m$ (an identity, so this checks the arithmetic).

Measuring the Gram parity by the index $\lfloor \vartheta_\beta(\gamma)/\pi \rfloor$ minus the zero number, 374 zeros have the offset $-1$ and five do not (two have $-2$ and three have 0). The five exceptions, at $\gamma = 138.7502, 290.2521, 364.4197, 400.8264, 422.5104$, are listed in Table 28.2. For the mean correction ($m = 1$), $\operatorname{Re}\delta_1 < 0$ exactly at these five zeros and at no other. For the hard cutoff ($m = 0$), $\operatorname{Re}\delta_0 < 0$ at the last four, and at 138.750 the arithmetic phase is large enough to turn the sign. For $m \ge 2$ the Pochhammer phase produces further negative real parts at low height: $\operatorname{Re}\delta_m < 0$ at 8, 11, 19 and 29 of the 379 zeros for $m = 2, 3, 4, 5$ (for $m = 2$ at 6.021, 12.988, 138.750, 236.242, 290.252, 364.420, 400.826 and 422.510, and for $m = 5$ at 22 zeros below height 105). These counts concern the formal coefficients $\delta_m$; by Proposition 28.2.4 they are the drift of an actual remainder only after multiplication by $\eta_m$.

| $\gamma$ | zero number | gap to the partner | $\operatorname{Re}\delta_0$ | $\operatorname{Re}\delta_1$ |
|---|---|---|---|---|
| 138.7502 | 78 | 0.2929 | $> 0$ | $< 0$ |
| 290.2521 | 195 | 0.4251 | $< 0$ | $< 0$ |
| 364.4197 | 258 | 0.3840 | $< 0$ | $< 0$ |
| 400.8264 | 291 | 0.1336 | $< 0$ | $< 0$ |
| 422.5104 | 309 | 0.1769 | $< 0$ | $< 0$ |

Table 28.2: The five zeros of $L(s, \chi_4)$ below height 500 with anomalous Gram parity (computed). The partner is the neighbouring zero at the stated gap (the left neighbour for the zero number 291).

Every one of the five lies in a close pair of zeros, and these gaps, 0.293, 0.425, 0.384, 0.134, 0.177, were recomputed. For zeta the analogue is the list of Section 7.2, three zeros among the first 460 approached from the right, which the book relates to $|S_{\mathrm{mid}}| > 0.56$. The five ordinates of Table 28.2 are the evidence for $\chi_4$ at $m = 0$ and $m = 1$.

Corollary 28.3.2 (No one sided criterion). *Fix $m \ge 0$ and a remainder satisfying the hypotheses of Theorem 28.2.2, for which the limit $\eta_m$ exists. If some simple zero $\rho$ of $\xi_4$ on the critical line has $\operatorname{Re}(\eta_m \delta_m) < 0$, then for all large $n$ the function $M_n$ has a zero in $0 < \operatorname{Re} z < \frac{1}{4}$, so no criterion of the form that $M_n$ has no zero in $0 < \operatorname{Re} z < \frac{1}{4}$ for infinitely many $n$ can hold at order $m$ for this remainder.*

*Proof.* By Theorem 28.2.2, $\operatorname{Re} z_n = \frac{1}{4} + n^{-(2m+1)} \operatorname{Re}(\eta_m \delta_m) + o(n^{-(2m+1)})$, which is below $\frac{1}{4}$ for large $n$; and the zero lies in the strip $0 < \operatorname{Re} z < \frac{1}{2}$. □

This is the argument of Corollary 7.2.4, in the other direction. Its premise is computed and not proved: for the mean correction ($m = 1$, $\eta_1 = 1$) the zero $\gamma = 138.7502$ has $\operatorname{Re} \delta_1 < 0$ in floating point, and so do the four others. A zero with $\operatorname{Re} \delta_1 < 0$ proved in ball arithmetic would make the corollary unconditional for $m = 1$, exactly as in Corollary 7.2.4. The source note states more, that infinitely many Gram anomalies of $\chi_4$ would force infinitely many left approaches at every fixed order $m$. This does not follow from the Gram parity alone, because the side is set by $\sin \Theta_m$ and not by $\sin \vartheta_\beta$: at 138.750 the hard cutoff approaches from the right although the zero is anomalous. A precise implication is the following.

Proposition 28.3.3. *Let* $\sigma = 2m + \frac{5}{2}$, $\epsilon_\sigma = \sum_{k \ge 1} (2k+1)^{-\sigma} < 1$ *and* $B_m(\gamma) = \arcsin \epsilon_\sigma + \sum_{j=0}^{m} \big[ \arctan \frac{1/2+2j}{\gamma} + \arctan \frac{3/2+2j}{\gamma} \big]$. *Then* $|\Theta_m(\gamma) - \vartheta_\beta(\gamma)| \le B_m(\gamma)$. *If infinitely many simple critical zeros satisfy* $-\sin \vartheta_\beta(\gamma) Z'_\beta(\gamma) < 0$ *and* $\operatorname{dist}(\vartheta_\beta(\gamma), \pi\mathbb{Z}) > B_m(\gamma)$, *then infinitely many zeros have* $\operatorname{Re} \delta_m < 0$.

*Proof.* The disc of centre 1 and radius $\epsilon_\sigma$ contains $L(\sigma + i\gamma, \chi_4)$, and it is seen from the origin under the angle $\arcsin \epsilon_\sigma$, which bounds $|\arg L|$. The interval between $\vartheta_\beta(\gamma)$ and $\Theta_m(\gamma)$ then contains no multiple of $\pi$, the two sines have the same nonzero sign, and Theorem 28.3.1 applies. □

The hypotheses of Proposition 28.3.3 on the five ordinates, with the margin $B_m(\gamma)$, are not checked here. Whether Gram anomalies with the margin occur infinitely often for $\chi_4$ is open. For zeta Gram's law fails with positive frequency [502], and the analogy with the five failures below 500 is a reason to expect the same for $\chi_4$ (heuristic). The comparison with the de Bruijn Newman constant, that these are the configurations that make $\Lambda \ge 0$, is an analogy, and the theorem of Rodgers and Tao [463] does not prove a truncation obstruction.

## 28.4 Transient zeros

Besides the zeros attached to the zeros of $L$, the truncations have zeros that wander into the left half strip. The counts of Table 28.3 are zeros of $M_n$ in the rectangles $0.02 < \operatorname{Re} z < 0.2499$ (left) and $0.2501 < \operatorname{Re} z < 0.48$ (right) by the argument principle (computed, `code/tr_counts.py`, enclosures of the function values in ball arithmetic but a floating point winding sum, a second run at 1.5 times the precision agrees; not certified).

The zeros of $L(s, \chi_4)$ with $0 < \gamma < 40$, $40 < \gamma < 80$ and $80 < \gamma < 150$ number 14, 23 and 48 (computed, agreeing with (8.5.8)). The source's right counts for the mean correction reach 23 in the band $20 < \operatorname{Im} z < 40$ at $n = 64$ and $n = 128$, and are 14 in the first band at $n = 32$, where the check of this chapter

| truncation | $n$ | $\operatorname{Im} z$ | left | right |
|---|---|---|---|---|
| hard cutoff | 32 | 15 to 60 | 2 | 14 |
| hard cutoff | 64 | 15 to 60 | 2 | 15 |
| mean correction | 32 | 0.5 to 20 | 0 | 13 |
| mean correction | 16 | 20 to 40 | 3 | 3 |
| mean correction | 32 | 20 to 40 | 1 | 20 |
| mean correction | 64 | 20 to 40 | 0 | 23 |
| mean correction | 64 | 40 to 75 | 4 | 37 |
| mean correction | 128 | 20 to 40 | 0 | 23 |

Table 28.3: Zeros of $M_n$ in the left and right half strips. The counts are computed by `code/tr_counts.py` (`data/tr_counts.json`, run on Hopper at two precisions with identical counts). The source note printed 22 for the hard cutoff at $n = 32$, a blank at $n = 64$ and 14 for the mean correction at $n = 32$, $\operatorname{Im} z$ from 0.5 to 20.

reported 13 (the mean correction counts are computed, see the caption). A count of 13 is not explained by a resolution frontier: the zero of $M_{32}^{(1)}$ near $\rho_1/2$ has $\operatorname{Re} z - \frac{1}{4} = 6.15 \times 10^{-6}$ (Table 28.1) and lies in the band $0.2499 < \operatorname{Re} z < 0.2501$ that the rectangles omit. At $n = 64$ the band $40 < \operatorname{Im} z < 75$ holds 37 captured zeros of the 48 in $80 < \gamma < 150$ and four zeros in the left half strip (computed, Table 28.3). The source note gives their ordinates (heights doubled) as 118.82, 126.01, 138.65 and 142.92, which were not recomputed (unchecked). Each is said to lie between the members of a close pair of zeros of $L$, namely $(118.538, 119.453)$, $(125.769, 126.299)$, $(138.457, 138.750)$ and $(142.394, 143.329)$; these eight ordinates were recomputed and are zeros of $L$. The third pair contains the first Gram anomaly. The pairs $(44.618, 45.600)$ and $(67.637, 68.366)$ are also pairs of zeros of $L$ (checked); the note reports that at $n = 32$ the hard cutoff has transient zeros between their members, at $0.2356 + 22.5622i$ and $0.2411 + 34.0550i$, the first gone by $n = 256$ and the second at $\operatorname{Re} z = 0.237$ and moving right at $n = 768$ (unchecked: this was not recomputed, since the hard cutoff at $n = 768$ needs about 1500 digits).

The picture the note draws is that mean correction reduces the transients of the hard cutoff and moves them to the resolution frontier, and that the frontier rises with $n$. In Table 28.3 the left count in the band $20 < \operatorname{Im} z < 40$ is $3, 1, 0, 0$ for $n = 16, 32, 64, 128$ (computed), consistent with a frontier that rises with $n$, while the hard cutoff keeps two left zeros at both $n = 32$ and $n = 64$ in $15 < \operatorname{Im} z < 60$. The rectangles of the count do not contain a neighbourhood of the critical line or of the left boundary. A zero $z_n = \frac{1}{4} + \delta_1 n^{-3} + o(n^{-3})$ with $\operatorname{Re} \delta_1 < 0$ eventually lies in $0.2499 < \operatorname{Re} z < \frac{1}{4}$ and is invisible to the count while remaining to the left of the line; so an empty left rectangle at a fixed height does not exclude such zeros, which are consistent with Theorem 28.1.1 because its compact sets stay away from the line. No uniform frontier is proved, and no statement that the transients occur

only at close pairs.

## 28.5 Symmetrised truncations

The functional equation can be imposed on each truncation. Put $g_n(z) = (2\pi)^{-z}M_n(z) = \mathbb{E}e^{-zV_n}$, with $V_n = \log(2\pi(\omega_n + R_n))$, and

$$F_n(z) = g_n(z) + g_n(\tfrac{1}{2} - z).$$

Then $F_n(\frac{1}{2} - z) = F_n(z)$ and $F_n(\bar{z}) = \overline{F_n(z)}$ on the common domain, so $F_n$ is real on $\operatorname{Re} z = \frac{1}{4}$ and its zeros off the line come in quadruples; this is Proposition 7.2.6 for $\chi_4$. If $p_n$ is the density of $V_n$ and $q_n(v) = e^{-v/4}p_n(v)$, then $F_n(\frac{1}{4} + iy) = \int_{\mathbb{R}}[q_n(v)+q_n(-v)]e^{-iyv}dv$, so $F_n(\frac{1}{4}+iy)/F_n(\frac{1}{4})$ is the characteristic function of the positive even density $[q_n(v) + q_n(-v)]/F_n(\frac{1}{4})$ (proved, by the change of variables; for a remainder without a density use measures).

Theorem 28.5.1 (Hurwitz criterion). *If for infinitely many $n$ all zeros of $F_n$ (for the remainders of Theorem 28.1.1) in $0 < \operatorname{Re} z < \frac{1}{2}$ lie on $\operatorname{Re} z = \frac{1}{4}$, then GRH holds for $\chi_4$.*

*Proof.* By Theorem 28.1.1 and the functional equation $F_n \to 2\xi_4(2z)$ locally uniformly in the strip. Along the subsequence apply Hurwitz's theorem separately to the two open half strips $0 < \operatorname{Re} z < \frac{1}{4}$ and $\frac{1}{4} < \operatorname{Re} z < \frac{1}{2}$: the nonzero limit $2\xi_4(2z)$ has no zero in either. Hurwitz's theorem is not a theorem on a closed line, so this uses the subsequence and does not apply to one fixed approximant. □

Proposition 28.5.2. *Every simple critical zero of $\xi_4(2z)$ is, for all large $n$, the limit of exactly one zero of $F_n$ in a small disc about it, and that zero lies on $\operatorname{Re} z = \frac{1}{4}$.*

*Proof.* Rouché's theorem in a disc centred on the line gives exactly one zero of $F_n$ in it, and the reflection $z \mapsto \frac{1}{2} - \bar{z}$ preserves the disc and the zero set, so the zero lies on the line. This holds even when the corresponding zero of $g_n$ approaches from the left. □

The proposition is local. On a fixed compact region it gives criticality of every zero of $F_n$ for large $n$, but it says nothing about heights that grow with $n$.

For the hard cutoff, counting all zeros in $0.02 < \operatorname{Re} z < 0.48$, $0.5 < \operatorname{Im} z < 60$ against sign changes on the line gives Table 28.4 (computed, `code/tr_counts.py`, the same at 1.5 times the precision). The count 64 equals the number of zeros of $L(s, \chi_4)$ below height 120 found by the same procedure. For $n = 8$ and 16 every zero of $F_n$ in the counted rectangle $0.02 < \operatorname{Re} z < 0.48$, $0.5 < \operatorname{Im} z < 60$ is critical, and the zeros of $F_8$ already approximate those of $L$ (Table 28.5, computed, heights $\gamma = 2\operatorname{Im} z$).

| $n$ | zeros in the strip | on the line | left zeros of $g_n$ |
|---|---|---|---|
| 8 | 64 | 64 | 0 |
| 16 | 64 | 64 | 0 |
| 32 | 64 | 60 | 2 |
| 64 | 64 | 60 | 2 |

Table 28.4: Symmetrised hard cutoff (computed).

| | | | | | | |
|---|---|---|---|---|---|---|
| $F_8$ | 5.9438 | 10.3288 | 12.9300 | 16.5280 | 18.1556 | 21.5568 |
| $L(s,\chi_4)$ | 6.0209 | 10.2443 | 12.9876 | 16.343 | 18.292 | 21.451 |

Table 28.5: The first zeros of $F_8$ and of $L(s,\chi_4)$ (computed).

Remark 28.5.3 (Zeros of $g_n$ to the left). For $n = 32$ and 64 the four off line zeros of $F_n$ and the two zeros of $g_n$ to the left of the line are computed counts. The source note reads this as a mechanism, that a zero of $g_n$ to the left of the line produces an off line pair of $F_n$, in the manner of the Hermite Biehler theorem. That cannot be a general statement. At a zero $z_*$ of $g_n$ one has $F_n(z_*) = g_n(\frac{1}{2} - z_*)$, which need not vanish; and the Hermite Biehler theorem gives real zeros of $E + E^*$ from the absence of zeros of $E$ in a half plane, not the converse. Here is a counterexample with a positive transform. Let $w = z - \frac{1}{4}$, $c = e^{-1/8}$ and $g(z) = (e^w + c)/(e^{-1/4} + c)$, a positive two point Laplace transform with $g(0) = 1$ whose zeros $z = \frac{1}{8} + (2k+1)\pi i$ lie to the left of the line. Its symmetrisation is $F(z) = 2(\cosh w + c)/(e^{-1/4} + c)$, and every solution of $\cosh w = -c$ with $0 < c < 1$ has $\operatorname{Re} w = 0$ (if $w = x + iy$ with $x \neq 0$ then $\sin y = 0$ and $|\cosh w| \geq 1$). So all zeros of $F$ are critical (proved). In the data the off line zeros of $F_{32}$ and $F_{64}$ are four, accompanying two left zeros of $g_n$; that they coincide is an observation of the source, not recomputed.

The modulus inequality $|g_n(\frac{1}{2} - z)| < |g_n(z)|$ on an open set off the line implies $F_n \neq 0$ there, since $|F_n(z)| \geq |g_n(z)| - |g_n(\frac{1}{2} - z)| > 0$ (proved). It is sufficient and not necessary. The source note reports that it fails near the real axis even for $n = 8$ and gives $-0.118$ at $z = 0.01 + 0.5i$ for the minimum of $\log|g_8(z)| - \log|g_8(\frac{1}{2} - z)|$. The recomputed minimum on the grid $x \in [0.01, 0.25]$, $y \in [0.5, 60.5]$ (steps 0.01 and 0.05) is $-0.0282$ at the same point (computed, `data/tr_mod.log`): the inequality fails, but with the smaller value. Its failure does not imply an off line zero. The note adds that $F_8$ has no zeros there because $g_8$ is positive on the real axis; positivity on the real segment gives a zero free neighbourhood $|\operatorname{Im} z| < c/B$ of it only with a bound $|F_8'| \leq B$ and $c = \min F_8$ on the segment, and neither is given.

With the mean correction, below height 40 ($\gamma < 80$) the function $F_{64}$ has all 37 zeros on the critical line and $F_{32}$ has 35 of 37 (computed by `code/tr_counts.py`, `data/tr_counts.json`, at two precisions with identical counts, in agreement with the source). The reading in the note, that the symmetrised mean corrected trun-

cations satisfy the hypothesis below their resolution frontier, rests on two values of $n$ and is an observation.

## 28.6 Where this leaves the route

The relation to Chapter 25 is as for zeta. Theorem 28.1.1 is the form of Proposition 25.7.1 and of Section 25.7 for $\chi_4$, with the Dirichlet beta lattice in place of the folded Gumbel kernels. Theorem 28.2.2 gives the exact rate at which the truncated zeros converge for the hard cutoff and the mean correction, and with the matching hypothesis for every order. Theorem 28.3.1 identifies the side as a Gram parity, and the computation to height 500 shows the sign of the drift of the mean correction to be negative exactly at the five zeros with anomalous Gram parity. What is not proved is that the premise of Corollary 28.3.2 holds (a zero with $\operatorname{Re}\delta_1 < 0$ in ball arithmetic) and that Gram anomalies with a margin recur. The symmetrised truncations avoid the one sided obstruction by Proposition 28.5.2, and Theorem 28.5.1 is a reduction of GRH for $\chi_4$ to a statement for infinitely many $n$ that is not proved or tested beyond the heights of Table 28.4. None of this proves GRH, and nothing in the computation is certified.

### 28.6.1 Open problems on the truncations

Problem 28.1. Prove that Gram anomalies with the margin $B_m(\gamma)$ of Proposition 28.3.3 occur infinitely often for $L(s, \chi_4)$, adapting the arguments known for $\zeta$ [502]. Together with Proposition 28.3.3 this would make Corollary 28.3.2 unconditional at each order $m$ with $\eta_m > 0$.

Problem 28.2. Prove in ball arithmetic that $\operatorname{Re}\delta_1 < 0$ at the zero $\gamma = 138.7502$ of $L(s, \chi_4)$, as is done for zeta in Corollary 7.2.4, and certify the count of 379 zeros below height 500 with a Turing bound.

Problem 28.3. Prove that $F_8$, the symmetrised hard cutoff with eight atoms, has only critical zeros, by combining the phase of $g_8$ at low height with the modulus inequality and a Newton polygon bound at large height. The analogues for zeta are Theorem 7.2.9 and Corollary 7.2.10.

Problem 28.4. Determine the largest $n$ for which the symmetrised mean corrected truncation has only critical zeros in the strip, and decide whether the first failure is related to the first Gram anomaly at 138.75.

Problem 28.5. Find a remainder $R_n$, not a cumulant match, whose drift $\eta\delta_m$ has a sign independent of the Gram parity. By Theorem 28.3.1 this needs a correction whose phase cancels $\vartheta_\beta(\gamma)$, which suggests building $R_n$ from the functional equation and not from the tail.

Problem 28.6. Carry out the analysis of orders $m \geq 1$ for zeta with the lattice of Section 7.2, where $m = 0$ is Theorem 7.2.3, and compare the Gram parity of the corrected phase with the classical failures of Gram's law for zeta (the first near the Gram point $g_{126} \approx 282.455$, where $Z(g_{126}) = -0.0276$).

## 28.7 Hermite coefficients of $\Xi$

The second half of the chapter is about the expansion of $\Xi$ in Hermite functions at a small scale. The source note [237] refers to an earlier chapter of the author for its definitions, which are not in the book, so they are written out here. The kernel is the Pólya kernel $\Phi$ of (1.2.3), even and positive with $\Xi(t) = \int_{\mathbb{R}} \Phi(u) e^{itu} du$; the source note does not define its $\Phi$, and with the kernel of (1.2.3) the numbers of the note that were recomputed agree, except where this chapter says otherwise, whereas the de Bruijn form $\Phi_{\mathrm{dBN}}$ of Section 8.4, with the exponent $e^{4u}$, puts the zeros at $2\gamma$ and is not the one meant. The kernel satisfies $\int |\Phi(u)|(1+|u|)^N e^{Bu^2+C|u|} du < \infty$ for all $N, B, C \geq 0$ (it decays like $\exp(-\pi e^{2|u|})$), and this is all that is used.

### 28.7.1 Definitions

Let $H_n$ be the Hermite polynomial of Section 24.6 (physicists' normalisation, the one that gives $H_{2d}$ in the Hermite identity there), and

$$h_n(x) = \pi^{-1/4} (2^n n!)^{-1/2} H_n(x) e^{-x^2/2},$$

the orthonormal Hermite functions. They satisfy $h_n'' + (2n + 1 - x^2) h_n = 0$, $\int h_n(x) e^{i\xi x} dx = \sqrt{2\pi}\, i^n h_n(\xi)$, and $h_{2m}(0) = (-1)^m \pi^{-1/4} \sqrt{(2m)!}/(2^m m!)$. For a scale $a > 0$ put

$$c_n(a) = \sqrt{a} \int_{\mathbb{R}} \Phi(u) h_n(au)\, du, \qquad t_m = t_m(a) = a\sqrt{4m+1},$$

$$b_m(a) = \sqrt{2\pi}\, (-1)^m c_{2m}(a) = \int_{\mathbb{R}} \Xi(t)\, \frac{h_{2m}(t/a)}{\sqrt{a}}\, dt.$$

The second form of $b_m$ follows from the Fourier relation with $t = ax$. Since $\Phi$ is even, $c_n(a) = 0$ for odd $n$, and $b_m(a)$ is exactly the coefficient of the orthonormal function $h_{2m}(t/a)/\sqrt{a}$ in the expansion of $\Xi$, and the factor $\sqrt{2\pi}(-1)^m$ relates $b_m$ to $c_{2m}$. The number $t_m$ is the turning point of $h_{2m}(t/a)$: the Hermite equation $h_{2m}'' + (4m + 1 - x^2) h_{2m} = 0$ is oscillatory for $|x| < \sqrt{4m+1}$, that is for $|t| < t_m$. Section 24.6 uses the same polynomials $H_{2d}$ in another role: there the Jensen polynomial of degree $d$ is the expectation of a Hermite polynomial of the Brownian variable $V$, whereas here the Hermite functions are integrated against the kernel.

The heat family enters as in Section 8.4. With $H_t$ the de Bruijn Newman function of that section one has $8H_t(2\tau) = \int_{\mathbb{R}} \Phi(v) e^{tv^2/4} \cos(\tau v)\, dv$, so the Gaussian

smoothing

$$G_a(\tau) = \int_{\mathbb{R}} \Phi(u) e^{-a^2u^2/2} \cos(\tau u)\, du = 8H_{-2a^2}(2\tau)$$
$$= \frac{1}{a\sqrt{2\pi}} \int_{\mathbb{R}} \Xi(s) e^{-(\tau-s)^2/(2a^2)} ds$$

is $\Xi$ smoothed by a Gaussian of variance $a^2$ (the last equality because $e^{-a^2u^2/2}$ is the characteristic function of $N(0, a^2)$), and it is the backward heat flow of the kernel for the time $t = -2a^2$.

THEOREM 28.7.1 (Hermite coefficients and the heat flow). *For every $a > 0$,*

$$\sum_{m\ge 0} b_m(a) \frac{z^{2m}}{\sqrt{(2m)!}} = \sqrt{2\pi}\, \pi^{-1/4} \sqrt{a}\, e^{z^2/2}\, G_a\big(\sqrt{2}\, az\big),$$

*and with $\nu_k(a) = \int \Phi(u) e^{-a^2u^2/2} u^{2k} du > 0$,*

$$b_m(a) = \sqrt{2\pi}\, \pi^{-1/4} \sqrt{a}\, \sqrt{(2m)!} \sum_{k=0}^{m} \frac{(-1)^k 2^k a^{2k} \nu_k(a)}{(2k)!\, 2^{m-k} (m-k)!}.$$

*Proof.* The generating function

$$\sum_n h_n(x) w^n/\sqrt{n!} = \pi^{-1/4} e^{-x^2/2 + \sqrt{2}xw - w^2/2}$$

integrated against $\sqrt{a}\, \Phi(u)$ at $x = au$ gives

$$\sum_n c_n(a) w^n/\sqrt{n!} = \pi^{-1/4} \sqrt{a}\, e^{-w^2/2} \int \Phi(u) e^{-a^2u^2/2} e^{\sqrt{2}auw} du.$$

On bounded sets of $w$ the integrand and its derivatives are bounded by a constant times $|\Phi(u)| e^{-a^2u^2/2 + C|u|}$, which is integrable, so the coefficients may be extracted termwise. Put $w = iz$: the odd terms vanish by evenness, $(iz)^{2m} = (-1)^m z^{2m}$ supplies the sign in $b_m$, and $e^{i\sqrt{2}azu}$ integrates to $G_a(\sqrt{2}az)$ by evenness. The moment formula follows by multiplying the series

$$G_a(\sqrt{2}az) = \sum_k (-1)^k 2^k a^{2k} \nu_k z^{2k}/(2k)!$$

by

$$e^{z^2/2} = \sum_j z^{2j}/(2^j j!).$$

Each $\nu_k > 0$ because $\Phi > 0$. □

The moment formula, evaluated in 60 digit arithmetic, agrees with the quadrature value of $b_m$ for $m \le 15$ at $a = 3$ and $a = 1.5$ with relative differences $6.7 \times 10^{-55}$ and $3.1 \times 10^{-59}$, and the two quadratures (steps 1/64 and 1/128) differ by $3.6 \times 10^{-57}$ and $1.5 \times 10^{-60}$ (computed, `code/tr_hermite.py`). Also $b_3(1.5) = 0.320600426659651$ from $\Phi$ and from the integral of $\Xi$ (computed).

Remark 28.7.2. By Rodgers and Tao [463], $\Lambda \ge 0$, so $H_t$ has nonreal zeros for every $t < 0$, and therefore $G_a$ is not real rooted for any $a > 0$, since $t = -2a^2 < 0$ (proved). The factor $e^{z^2/2}$ never vanishes, so $e^{z^2/2} G_a(\sqrt{2}az)$ has exactly the zeros of $G_a(\sqrt{2}az)$. The note says that the coefficients of this product nevertheless recover the real zeros of $\Xi$. That is not a statement about zeros of the product. What is true is that the signs of $b_m(a)$ follow those of $\Xi$ at the points $t_m$ away from the zeros as $a \to 0$ (Theorem 28.8.1).

## 28.8 The turning point asymptotic

The note proposes, as Proposition 1, that $b_m(a) = \sqrt{2a}\, m^{-1/4}\, \Xi(a\sqrt{4m+1})(1 + O(a^2))$ as $a \to 0$ with $t = a\sqrt{4m+1}$ fixed, and derives it by a saddle point argument on the circle $|z| = \sqrt{2m}$ in Theorem 28.7.1. The leading constant is right: the saddles are at $\theta = 0, \pi$ with the exponent $m - 2m\theta^2 + O(m\theta^3)$, the Gaussian in $\theta$ has variance $1/(4m)$, the imaginary shift of the argument of $G_a$ has variance exactly $a^2$, and $\mathbb{E} G_a(t + iaY) = \Xi(t)$ for $Y \sim N(0,1)$ by Fubini, so the average of $G_a$ over the shifts undoes the smoothing; Stirling's formula and the Gaussian integral give $\sqrt{2a}\, m^{-1/4}$. But the error $O(a^2)$ is not derived (only the shift of the sampling point from $2a\sqrt{m}$ to $a\sqrt{4m+1}$ is), the real displacement $-a\sqrt{m}\,\theta^2$ and the cubic phase are not estimated, and the contribution off the saddles is not bounded. The statement as written is false at the zeros of $\Xi$ (Proposition 28.8.2). The theorem below is proved from the Hermite differential equation and replaces it; it states an additive estimate, uniform in $m$, with the exact amplitude.

Put

$$A_m(a) = \sqrt{2\pi a}\,(-1)^m h_{2m}(0) = \sqrt{2}\,\pi^{1/4}\sqrt{a}\,\frac{\sqrt{(2m)!}}{2^m m!} > 0,$$

$$M_k(a_0) = \int_{\mathbb{R}} |\Phi(u)||u|^k e^{a_0^2 u^2/2} du < \infty.$$

Theorem 28.8.1 (Additive asymptotic). *For every $a_0 > 0$, $0 < a \le a_0$ and integer $m \ge 0$,*

$$\Big| \frac{b_m(a)}{A_m(a)} - \Xi(t_m) \Big| \le \frac{M_4(a_0)}{12}\, a^4,$$

$$\frac{b_m(a)}{A_m(a)} = \Xi(t_m) + a^4 Q(t_m) + R_m(a), \qquad |R_m(a)| \le \frac{M_8(a_0)}{672}\, a^8,$$

*with* $t_m = a\sqrt{4m+1}$ *and the even entire function*

$$Q(t) = \frac{\Xi'''(t)}{6t} - \frac{\Xi''(t)}{4t^2} + \frac{\Xi'(t)}{4t^3} \quad (t \neq 0),$$
$$Q(0) = \frac{\Xi^{(4)}(0)}{12} = \frac{1}{12}\int_{\mathbb{R}} \Phi(u) u^4 du.$$

*Proof.* Put $y_m(u) = h_{2m}(au)/h_{2m}(0)$. Then $b_m/A_m = \int \Phi(u) y_m(u) du$, and the Hermite equation gives $y_m'' + (t_m^2 - a^4u^2)y_m = 0$, $y_m(0) = 1$, $y_m'(0) = 0$. For real $t$ and complex $z$ let $y_{z,t}$ be the even solution of $y'' + (t^2 - zu^2)y = 0$ with these initial values. For $u \geq 0$, $y_{z,t} = \cos tu + z\, L_t y_{z,t}$ with $(L_t f)(u) = \int_0^u \frac{\sin t(u-v)}{t} v^2 f(v) dv$ (the kernel is $u - v$ at $t = 0$). Since $|\sin(t(u-v))/t| \leq u - v$, the Volterra series is majorised by the solution of $w = 1 + |z| K w$ with $(Kf)(u) = \int_0^u (u-v) v^2 f(v) dv$. The function $v = e^{\sqrt{|z|}u^2/2}$ satisfies $v'' = (\sqrt{|z|} + |z|u^2)v \geq |z| u^2 v$, so $v \geq 1 + |z| K v$, hence $w \leq v$ and

$$|y_{z,t}(u)| \leq e^{\sqrt{|z|}u^2/2} \quad (u \in \mathbb{R},\ t \in \mathbb{R}).$$

Iterating the integral equation once and twice, and using $K1 = u^4/12$ and $K^2 1 = u^8/672$, gives $|y_{z,t} - \cos tu| \leq \frac{|z|u^4}{12} e^{\sqrt{|z|}u^2/2}$ and $|y_{z,t} - \cos tu - z L_t \cos(t\,\cdot)| \leq \frac{|z|^2 u^8}{672} e^{\sqrt{|z|}u^2/2}$, with constants independent of $t$. Take $z = a^4$, $t = t_m$, $a \leq a_0$ and integrate against $\Phi$: $\int \Phi \cos(t_m u) du = \Xi(t_m)$, and the first inequality follows. The function $f_t = (\frac{u^3}{6t} - \frac{u}{4t^3}) \sin tu + \frac{u^2}{4t^2} \cos tu$ solves $f'' + t^2 f = u^2 \cos tu$ with $f(0) = f'(0) = 0$, so $f_t = L_t \cos(t\,\cdot)$, and differentiating the Fourier integral of $\Xi$ gives $\int \Phi f_t = Q(t)$. At $t = 0$, $f_0 = u^4/12$. $\square$

For the amplitude, Stirling's expansion gives $A_m(a) = \sqrt{2a}\, m^{-1/4}\big(1 - \frac{1}{16m} + \frac{1}{512m^2} + O(m^{-3})\big)$. Hence for $0 < \varepsilon \leq t = t_m \leq T$ (so that $m \to \infty$ uniformly as $a \to 0$)

$$\frac{b_m(a)}{\sqrt{2a}\, m^{-1/4}} = \Big(1 - \frac{1}{16m} + O(m^{-2})\Big)\big(\Xi(t) + O(a^4)\big), \tag{28.8.1}$$

an additive statement, uniform even if the interval contains zeros. At a fixed $t > 0$ with $\Xi(t) \neq 0$ one may divide, and then $b_m(a) = \sqrt{2a}\, m^{-1/4}\, \Xi(t)(1 + O_t(a^2))$, which is the formula of the note; the leading correction is $-1/(16m) = -a^2/(4t^2) + O(a^4)$ and comes from the amplitude and not from the saddle. A uniform relative error holds on a compact set $K \subset (0, \infty)$ with $\inf_K |\Xi| > 0$. With the sampling point as the variable, $n = m + \frac{1}{4} = t^2/(4a^2)$, the amplitude is $A_m = \frac{2a}{\sqrt{t}}\big(1 - \frac{a^4}{8t^4} + O(a^8)\big)$ and

$$b_m(a) = \frac{2a}{\sqrt{t}}\Big\{\Xi(t) + a^4\Big[Q(t) - \frac{\Xi(t)}{8t^4}\Big] + O_{\varepsilon,T}(a^8)\Big\},$$

with no term of order $a^2$ or $a^6$; this form is not uniform near $t = 0$, where Theorem 28.8.1 with the exact $A_m$ is the right one. At $m = 0$ one has $t_0 = a$

and $b_0(a) = A_0(a)\int \Phi(u)e^{-a^2u^2/2}du > 0$, and the formula with $m^{-1/4}$ is undefined; at fixed $m \geq 1$, $b_m/A_m \to \Xi(0) > 0$ as $a \to 0$. Even for $m \geq 1$ the relative formula with the prefactor $\sqrt{2a}$ does not extend to $t = 0$: for $m = 1$, $b_1(a)/(\sqrt{2a}\,\Xi(a\sqrt{5})) \to \pi^{1/4}/\sqrt{2} = 0.9414$ (proved, from $A_1 = \pi^{1/4}\sqrt{a}$).

Proposition 28.8.2. *At every positive real zero $\gamma$ of $\Xi$, of any multiplicity, $b_m(a_m) \neq 0$ for all large $m$, where $a_m = \gamma/\sqrt{4m+1}$. If $Q(\gamma) \neq 0$ then $b_m(a_m) = \frac{2Q(\gamma)}{\sqrt{\gamma}}a_m^5 + O(a_m^9)$. The relative formula of the note therefore fails at the zeros of $\Xi$, where it would give $b_m(a_m) = 0$.*

*Proof.* For fixed real $t$ put $F_t(z) = \int \Phi(u)y_{z,t}(u)du$. By the bound of the proof of Theorem 28.8.1 it is entire in $z$, and $F_t(0) = \Xi(t)$. For $t > 0$ and $z = t^4$ the solution is $y = e^{-t^2u^2/2}$, so $F_t(t^4) = \int \Phi e^{-t^2u^2/2}du > 0$. At $t = \gamma$, $F_\gamma(0) = 0$ but $F_\gamma \not\equiv 0$, so $F_\gamma(z) \neq 0$ for small $z \neq 0$; and $b_m(a_m) = A_m(a_m)F_\gamma(a_m^4)$. If $Q(\gamma) \neq 0$ the Taylor expansion at $z = 0$ gives the leading term, with $A_m(a_m) \sim 2a_m/\sqrt{\gamma}$. □

## 28.9 The first negative coefficient

Let $m^*(a) = \min\{m \geq 0 : b_m(a) < 0\}$, and let $\gamma_1 = 14.134725\ldots$ be the first positive zero of $\Xi$. It is simple (the first $1.5 \times 10^9$ zeros are simple and on the line [508]), so $\Xi > 0$ on $[0, \gamma_1)$ and $\Xi < 0$ just to the right of $\gamma_1$.

Corollary 28.9.1. *For every $0 < \delta < \gamma_1$ and $a$ small, $b_m(a) > 0$ for all $m$ with $t_m \leq \gamma_1 - \delta$, the index $m^*(a)$ exists, and $\lim_{a\to 0} 4a^2m^*(a) = \gamma_1^2$. If $\gamma_a$ is the unique zero near $\gamma_1$ of $t \mapsto \int \Phi(u)y_{a^4,t}(u)du$, then $\gamma_a = \gamma_1 - a^4Q(\gamma_1)/\Xi'(\gamma_1) + O(a^8)$ and $m^*(a) = \lfloor(\gamma_a^2/a^2 - 1)/4\rfloor + 1$ for small $a$.*

*Proof.* Let $c_\delta = \min_{[0,\gamma_1-\delta]}\Xi > 0$. By Theorem 28.8.1, $b_m/A_m \geq c_\delta - Ca^4 > 0$ when $t_m \leq \gamma_1 - \delta$ and $a$ is small, so $m^*(a)$, where it exists, has $t_{m^*} > \gamma_1 - \delta$ and $4a^2m^* = t_{m^*}^2 - a^2 > (\gamma_1 - \delta)^2 - a^2$, which gives $\liminf 4a^2m^* \geq \gamma_1^2$. For the upper bound pick $s > \gamma_1$ with $\Xi(s) < 0$ and $m(a) = \lfloor(s^2/a^2 - 1)/4\rfloor$; then $t_{m(a)} \to s$, the additive error tends to zero, so $b_{m(a)} < 0$ and $\limsup 4a^2m^* \leq s^2$; let $s \downarrow \gamma_1$. For the exact formula, $F(z,t) = \int \Phi y_{z,t}$ is jointly analytic near $(0, \gamma_1)$ (the kernel bound $|\sin(tw)/t| \leq we^{|\operatorname{Im} t|w}$ gives an integrable majorant), $\partial_z F(0,t) = Q(t)$ and $\partial_t F(0,t) = \Xi'(t) \neq 0$, so the analytic implicit function theorem gives a zero $\rho(z) = \gamma_1 - zQ(\gamma_1)/\Xi'(\gamma_1) + O(z^2)$ and $\gamma_a = \rho(a^4)$. For small $a$ the function $F(a^4, \cdot)$ is positive up to $\gamma_a$ and negative just after it, and the grid has spacing $O(a^2)$ there, so the first negative index is the first grid point strictly after $\gamma_a$. □

The hypothesis of the note, an error uniform on $[0, \gamma_1 - \delta]$, gives only the lower bound: the limit needs negativity of $b_m$ just above $\gamma_1$, that is a uniform estimate on an interval that contains a point beyond $\gamma_1$.

Numerically the note states the sharper law $m^*(a) = \lceil(\gamma_1^2/a^2 - 1)/4\rceil$ for $a \leq 1.5$. Table 28.6 (computed, second run at step $h/2$ the same) agrees with it at four

scales. It cannot hold as a statement for every $a$. The threshold inside the rounding is $(\gamma_a^2/a^2-1)/4$, which differs from $(\gamma_1^2/a^2-1)/4$ by $-\gamma_1 Q(\gamma_1)a^2/(2\Xi'(\gamma_1))+O(a^6)$, and $\gamma_a \neq \gamma_1$ for small $a$ (otherwise $F_{\gamma_1}$ would vanish identically, against $F_{\gamma_1}(\gamma_1^4) > 0$), so $\gamma_a - \gamma_1 = c\,a^{4p} + o(a^{4p})$ with $c \neq 0$ and $p \geq 1$. If $c > 0$, at $a = \gamma_1/\sqrt{4m+1}$ the ceiling of the note gives $m$ and the first negative coefficient is $m+1$ for large $m$; if $c < 0$ a slightly smaller $a$ gives the opposite. Such scales exist arbitrarily close to 0, because the displacement of the root is much smaller than the spacing $O(a^2)$ of the grid. At $a = 2$ the formula gives 12.24 and the first negative index is 12, so the ceiling formula, which would give 13, fails there, although $a = 2$ is not in the asymptotic regime; and the reading of $\gamma_1$ to about $10^{-2}$ from $m^*(1/2)$ is only to 0.035 (the value $m^*(\frac{1}{2}) = 200$ places $\gamma_1$ in $(14.116, 14.151]$).

| $a$ | 0.5 | 0.75 | 1 | 1.5 | 2 |
|---|---|---|---|---|---|
| $(\gamma_1^2/a^2-1)/4$ | 199.54 | 88.55 | 49.70 | 21.95 | 12.24 |
| first negative $m$ | 200 | 89 | 50 | 22 | 12 |

Table 28.6: The first negative Hermite coefficient of Ξ (computed).

## 28.10 Counting zeros from Φ alone

### 28.10.1 The ratio table

Table 28.7 gives the ratio $b_m(a)/\big(\sqrt{2a}\,m^{-1/4}\,\Xi(a\sqrt{4m+1})\big)$ at three scales (computed; the source note does not say which $m$, and the indices $m = 9, 36, 81, 144, 182$ at $a = 0.5$, $4, 16, 36, 64, 81$ at $a = 0.75$ and $2, 9, 20, 36, 45$ at $a = 1$ reproduce every printed digit; the sampling points are $t_m = 3.04, 6.02, 9.01, 12.01, 13.50$, $3.09, 6.05, 9.03, 12.02, 13.52$ and $3.00, 6.08, 9.00, 12.04, 13.45$). The last column is the error at $t \approx 9$ divided by $a^2$, which is 0.0033, 0.0036, 0.0041 (the note printed 0.0032 for the first). The table is explained by the amplitude: the printed values are close to $A_m/(\sqrt{2a}\,m^{-1/4}) = 1 - \frac{1}{16m} + \cdots$ (for example $1 - 1/144 = 0.9931$ at $a = 0.5$, $m = 9$), and the term $a^4 Q(t)/\Xi(t)$ of Theorem 28.8.1 is the rest. The coefficients range over four orders of magnitude here. At larger $a$ and greater height the ratio is no longer near 1; the note says it is between 1.5 and 9 at $a = 4$ for $100 < t < 200$ and that no $H_\lambda$ with $\lambda \in [-8, 3]$ fits better, which could not be reproduced (the ratio is not defined near the zeros of Ξ, and it varies in sign over the window).

### 28.10.2 Method and the count

The integrand $\Phi(u)h_{2m}(au)$ is not entire: it is analytic in $|\operatorname{Im} u| < \pi/4$ and the lines $\operatorname{Im} u = \pm\pi/4$ are a natural boundary ((1.2.3) and the text after it), but it decays like $\exp(-\pi e^{2|u|})$ and the trapezoid rule converges geometrically. For a

| $a$ | $t \approx 3$ | $t \approx 6$ | $t \approx 9$ | $t \approx 12$ | $t \approx 13.5$ | error at $t \approx 9$ over $a^2$ |
|---|---|---|---|---|---|---|
| 0.50 | 0.9931 | 0.9983 | 0.9992 | 0.9993 | 0.9986 | 0.0033 |
| 0.75 | 0.9847 | 0.9961 | 0.9980 | 0.9976 | 0.9937 | 0.0036 |
| 1.00 | 0.9699 | 0.9931 | 0.9959 | 0.9938 | 0.9830 | 0.0041 |

Table 28.7: The ratio of $b_m(a)$ to $\sqrt{2a}\, m^{-1/4}\, \Xi(a\sqrt{4m+1})$ (computed).

function analytic in a strip $|\operatorname{Im} u| \le d$ with $M_d = \int (|f(x+id)| + |f(x-id)|)dx < \infty$ the error of the infinite trapezoid rule of step $h$ is at most $M_d/(e^{2\pi d/h} - 1)$ (proved, by shifting the Fourier integral to the edges of the strip and summing two geometric series), with $d < \pi/4$; the constants depend on $a$ and $m$, so the note's statement that the transform of the integrand is negligible beyond $2T$ and that $h = 2\pi/(3T + 100)$ suffices is not proved. The count was recomputed with this step (the note has about 655 nodes at $T = 500$), the Hermite functions by the three term recurrence for all orders at once, and a working precision of the order of $0.34\,T+40$ digits (the coefficients near height $T$ have size $e^{-\pi T/4}$), and the positions of the sign changes were compared with a run at half the step and more digits (computed, `code/tr_hermite.py`; Hopper). The note's column of digits does not follow its own rule and is not printed.

| $T$ | $N(T)$ | coefficients | sign changes of $b_m$ by scale $a$ |
|---|---|---|---|
| 100 | 29 | 625 at $a = 2$ | $a = 2, 3, 4, 5$: 29, 29, 29, 29 |
| 200 | 79 | 625 at $a = 4$ | $a = 4, 5$: 79, 79; $a = 6, 7$: 78, 78; $a = 8$: 77 |
| 500 | 269 | 1736 at $a = 6$ | $a = 6$: 269; 7: 268; 8: 267; 9: 265; 10: 263; 14: 245; 18: 192 |
| 1000 | 649 | 6945 at $a = 6$ | $a = 5, 6$: 648, 648 (edge lag) |
| 1004 | 652 | 7000 at $a = 6$ | $a = 6$: 651 |

Table 28.8: Number of sign changes of $b_m(a)$ for $t_m(a) \le T$ against the zero count $N(T)$ (computed). The coefficient counts of the note (627, 1738, 6947, 7002) were two too large.

Remark 28.10.1 (Observation). For $a$ up to about 5 to 6 the number of sign changes of $(b_m(a))_{t_m \le T}$ equals $N(T)$ at $T = 100, 200, 500$, and at $T = 1000$ it equals $N(T)$ except for the last zero within about one unit of the top of the computed range. In every run above the threshold the count was smaller than $N(T)$.

This is an observation (computed, floating point; the positions agree in the second run at $T = 100$, $T = 200$ and $T = 500$, $a = 6$, and the other rows are single runs), and not a theorem: a universal one sided inequality and an association between sign changes of $b_m$ and zeros of zeta are not proved. The zero counts are those of zeta ($N(T) = 29, 79, 269, 649, 652$, mpmath). The equality is a statement about counts and not about signs: at $T = 200$, $a = 4$ the sign of $b_m$ agrees with

the sign of $\Xi(t_m)$ at 81.8 percent of the indices ($c = 1$; 81.9 percent at $c = 0$), and sampling at $a\sqrt{4m + c}$ for $c \in [0, 3]$ moves this between 79.7 and 81.9 percent (computed). The sign changes are displaced by a few indices relative to the zeros.

### 28.10.3 The edge lag at $T = 1000$

At $T = 1000$ both $a = 5$ and $a = 6$ give 648 sign changes against $N(1000) = 649$. The last zero is $\gamma_{649} = 999.79$, 0.3 below the top of the range. Extending the computation at $a = 6$ to $t \le 1004$ the next sign changes occur between consecutive $t_m$ at $(1000.85, 1000.92)$, $(1002.07, 1002.14)$ and $(1003.36, 1003.44)$, for the zeros 999.79, 1001.35 and 1002.40 (computed), and $\gamma_{652} = 1003.27$ again falls inside the lag at the new edge; the note gives the sign changes at 1000.89, 1002.11, 1003.40, which are inside these intervals. So the sign changes trail the zeros by about one unit at this height, and the count has to be read on $[0, T - \delta]$ with a lag $\delta(a, T)$. The note adds that at low height the turning points sit slightly below the zeros and at $T = 1000$ slightly above, so that $\delta$ changes sign with height; this is an observation that Theorem 28.8.1 does not yet explain.

Proposition 28.10.2 (What a verified count proves). *Let $\widetilde{b}_m/A_m$ be computed values of the normalised coefficients with rigorous errors $e_m$, and let $E_m$ be a rigorous bound for $|b_m/A_m - \Xi(t_m)|$ (for example $M_4(a_0)a^4/12$ from Theorem 28.8.1). If $|\widetilde{b}_m/A_m| > e_m + E_m$ then $\operatorname{sgn}\widetilde{b}_m = \operatorname{sgn}\Xi(t_m)$. Suppose there are $S$ sign changes at pairs of adjacent sampling points below $T$, all with this property. Then $\Xi$ has $S$ zeros in disjoint intervals, and if an independent rigorous method gives $N(T) \le S$, where $N(T)$ is the number of zeros $t$ of $\Xi$ with $0 < \operatorname{Re} t < T$, counted with multiplicity (Turing's method, Section 2.2.1), all those zeros are real and simple.*

*Proof.* The triangle inequality transfers the signs, the intermediate value theorem gives one zero of $\Xi$ in each of the $S$ disjoint intervals between adjacent sampling points, and the upper bound, counted with multiplicity, leaves no room for another zero or a multiple zero. □

A sign change of $b_m$ certifies nothing by itself, because $b_m$ is an integral of $\Xi$ and not a value. The chain of the note, a rigorous version of the asymptotic, a lower bound for $|\Xi|$ at the sampling points and Turing's method, is Proposition 28.10.2 with Theorem 28.8.1 supplying the first step with explicit constants; the sampling values that are zero or uncertain must be enclosed separately. High working precision and agreement after halving the step are diagnostics, not these inequalities, so the counts of Table 28.8 are computed and not verified.

## 28.11 The best scale and the run lengths

Two quantities compete. The grid $t_m(a)$ has spacing $dt_m/dm = 2a^2/t_m$ (exactly $4a^2/(t_{m+1} + t_m)$ between neighbours), so a small $a$ resolves close zeros; the number

of coefficients is $M = T^2/(4a^2) + 1$, so a large $a$ is cheap. The note says that the binding constraint is not resolution, because at the largest successful scale the grid spacing is only 0.12 to 0.17 of the mean gap $2\pi/\log(T/2\pi)$. The recomputed ratios of $2a^2/T$ to the mean gap are 0.22 ($T = 100$, $a = 5$), 0.14 ($T = 200$, $a = 5$), 0.10 ($T = 500$, $a = 6$) and 0.06 ($T = 1000$, $a = 6$) (computed), so the note's range is not reproduced; and the inference does not follow in any case, since two zeros arbitrarily close together can fit between two sampling points whatever the mean gap (the function $(t-c)^2 - \varepsilon^2$ is positive at both ends of a grid interval for small $\varepsilon$). The tables alone do not separate a missed grid crossing from a distortion of the asymptotic; what is observed is that the counts fall short above $a \approx 5$ to 6 and that the usable threshold rises slowly with $T$, from at least 5 at $T = 100$ to between 6 and 7 at $T = 500$. With $a$ near the threshold there are about 6.5 coefficients per zero at $T = 500$ and 10.7 at $T = 1000$ (1736/269 and 6945/649), each costing $O(T)$ nodes at $0.34\,T$ digits. As a verification method this is far slower than Odlyzko and Schönhage [392]; its value is structural, since every quantity is an explicit integral of the positive kernel $\Phi$.

The note also reads the sign pattern through Good and Gaskins [156]: with $\Xi(t) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ for the Hardy function $Z$ and the Riemann Siegel phase $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$, the sign of $b_m$ would rotate by $\theta_m = \vartheta'(t_m)\,dt_m/dm = a^2\log(t_m/2\pi)/t_m$ per step in $m$, an effective complex ratio, with mean sign run $\pi/\theta_m = \pi t/(a^2\log(t/2\pi))$. The rotation law is not derived: the real function $Z$ is not determined by its phase derivative, no recurrence or complex ratio for $b_m$ has been derived, and Theorem 28.8.1 is an asymptotic at bounded height as $a \to 0$ and does not apply at fixed $a = 4$ and increasing height (the attribution to Good and Gaskins is the note's and was not checked). What can be said is an averaged statement.

Proposition 28.11.1 (Average run length). *Let $N(T)$ count the zeros $t$ of $\Xi$ with $0 < \operatorname{Re} t < T$, with multiplicity. Assume the zeros $t$ of $\Xi$ with $T \le \operatorname{Re} t \le T + H$ are real and simple and that the sign changes of $(b_m(a))$ with $t_m \in [T, T+H]$ correspond one to one to them. Then the mean number of coefficient steps per sign change is the number $TH/(2a^2) + O(H^2/a^2 + 1)$ of grid points in the window divided by $N(T+H) - N(T) = \frac{H}{2\pi}\log\frac{T}{2\pi} + O(H^2/T + \log T)$, and for fixed $a$, $H \to \infty$ and $H = o(T)$ the ratio is asymptotic to $\pi T/(a^2\log(T/2\pi))$.*

This is the conditional average that Table 28.9 tests. The predicted values $\pi t/(a^2\log(t/2\pi))$ at the window midpoints $60, 100, 140, 180$ and $a = 4$ are 5.22, 7.10, 8.86, 10.53 (computed, arithmetic). The note's observed values 5.29, 7.19, 8.89, 10.50 were not reproduced, because "mean run" is not defined in the note (edge blocks, which window); with the number of indices in the window divided by the number of sign blocks at $T = 200$, $a = 4$, the observed values are 5.00, 6.94, 8.33, 10.23 (computed). They are consistent with the average and do not support the rotation law.

| window in $t$ ($a = 4$) | 40 to 80 | 80 to 120 | 120 to 160 | 160 to 200 |
|---|---|---|---|---|
| predicted $\pi t/(a^2 \log(t/2\pi))$ | 5.22 | 7.10 | 8.86 | 10.53 |
| observed mean run (indices over sign blocks) | 5.00 | 6.94 | 8.33 | 10.23 |

Table 28.9: Mean sign runs of $b_m(4)$ (computed).

## 28.12 The Thorin shadow

The note concludes that, by the turning point asymptotic, Hermite coefficients of a real even function sample the function at the points $a\sqrt{4m+1}$, so that sign definiteness of the coefficients is asymptotically equivalent to positivity of the function on the real line, and that a Thorin or GGC positivity statement for the reciprocal kernel has a Hermite form that is no stronger and no weaker. This does not follow, and the sentence is not true as stated. Let $f(t) = (\sin t/t)^2$, which is even, entire and nonnegative on the real axis, with $f(t) = \frac{1}{4}\int_{-2}^{2}(2-|u|)e^{itu}du$ and a nonnegative compactly supported kernel. The proof of Theorem 28.8.1 applies to it. At $t = \pi$ one has $f'(\pi) = 0$, $f''(\pi) = 2/\pi^2$, $f'''(\pi) = -12/\pi^3$, so $Q_f(\pi) = -5/(2\pi^4) < 0$, and along $a = \pi/\sqrt{4m+1}$ the normalised coefficient is $a^4 Q_f(\pi) + O(a^8) < 0$ for all large $m$: a nonnegative function can have negative small scale Hermite coefficients at every large $m$ along that sequence (proved). The correct statements are two. For a kernel of the class above, if $\Xi$ (or any function of the class) is strictly positive on a compact interval, then every coefficient with $t_m$ in the interval is positive once $a$ is small (the positive minimum and the additive error, with $A_m > 0$). Conversely, if all these coefficients are nonnegative for every small $a$, the function is nonnegative on the interval (choose grid points converging to any $t$ and pass to the limit in the normalised formula; at $t = 0$ use $m = 0$). Neither characterises the Thorin or GGC property of a reciprocal kernel. The Hermite coefficients add computability, and no result that says more is known here.

## 28.13 Notes and open problems

The two halves of this chapter develop two notes of Nick Polson [245, 237]; the definitions of the second were supplied by the book, the drift law of the first was restated with the matching hypothesis, and the saddle point statement of the second was replaced by Theorem 28.8.1, whose proof is the Volterra argument above. The scripts are `code/tr_zeros.py` (the zeros of $L(s, \chi_4)$ below height 500, the Gram offsets and the numbers $\delta_m$), `code/tr_drift.py`, `code/tr_counts.py`, `code/tr_hermite.py`, and the logs in `data/tr_*.json` and `data/tr_*.log`; they are listed in `code/README.md`. Everything computed here is numerical, and nothing is certified in ball arithmetic. The Lehmer pair near $t = 7005$ that the note mentions consists of the zeros 6709 and 6710 at 7005.062866 and 7005.100565, a

gap of 0.0377 (computed).

Problem 28.7. Evaluate the constants of Theorem 28.8.1 explicitly (the moments $M_4(a_0)$ and $M_8(a_0)$ and a bound for $Q$) and use Proposition 28.10.2 with Turing's method to count the zeros of $\Xi$ up to a given height from the coefficients alone, including the edge lag $\delta(a, T)$.

Problem 28.8. Explain the distortion of the counts at large $a$, and find a modified coefficient sequence, perhaps with $e^{z^2/2}$ replaced by another multiplier in Theorem 28.7.1, that keeps the count exact at larger $a$ and so cuts the number $M = T^2/(4a^2)$ of coefficients.

Problem 28.9. Push the count to $T = 2000$ and to the Lehmer pair near $t = 7005$, where a sign change count is the natural place for a close pair to hide.

Problem 28.10. Find a Hermite statement that is equivalent to a Thorin or GGC positivity property of the reciprocal kernel, or show that none exists, given that sign definiteness of the Hermite coefficients is not equivalent to positivity of the function (the example $(\sin t/t)^2$).

# Chapter 29

# Completeness of the secant dilates and the Möbius problem

Chapter 6 writes the exponential as a Möbius weighted sum of hyperbolic secants (Proposition 6.1.3) and records the question of Lai and Vere-Jones [322] on the completeness of the dilates $\operatorname{sech}(\lambda x)$ as open (Problem 7.10 asks for its relation to the Nyman Beurling criterion). This chapter answers the question in weighted $L^2$ spaces on the half line and says exactly where arithmetic enters. The answer has two halves. In one weight the integer dilates $\operatorname{sech}(nx)$ are complete unconditionally, by Carleman's formula, whatever the zeros of $L(s,\chi_4)$ do. In two weights at once completeness is equivalent to a zero free half plane of $L(s,\chi_4)$, with the generalised Riemann hypothesis at the threshold. The second half is a Nyman Beurling type criterion for the Dirichlet beta function, with an analogue for $\zeta$.

The chapter develops two source notes by Nick Polson, one on the secant dilates, Lerch's theorem and the Möbius function (Sections 29.1 to 29.8) and one that works its four open problems (Sections 29.10 to 29.16), together with the computations done for this book (`code/lv_sech_tables.py`, `code/lv_gram.py` and their outputs `data/lv_sech_tables.json`, `data/lv_gram.json`, listed in `code/README.md`; the scripts of the sources were not used). Every number printed is one of these recomputations, and where the source differed the recomputed value is printed. The completeness question of Lai and Vere-Jones is posed in [322] as recorded in Chapter 6; the paper itself was not consulted for this chapter, and no statement here is attributed to it beyond the question and the identity of Proposition 6.1.3.

Notation. The book's Re, Im, sech, $\mathbb{R}$, $\mathbb{C}$ and the cut plane $\mathbb{C}\setminus(-\infty,0]$ are used. For $c>0$ the weighted space is $\mathcal{W}_c = L^2((0,\infty),x^{2c-1}dx)$; $\beta(s)=L(s,\chi_4)=1-3^{-s}+5^{-s}-\cdots$ is the Dirichlet beta function, a symbol local to this chapter (the real part of a zero is always written $\operatorname{Re}\rho$); $\mu$ is the Möbius function, $M_\chi(x)=\sum_{n\le x}\mu(n)\chi_4(n)$ and $M(x)=\sum_{n\le x}\mu(n)$. The Dirichlet polynomial $\mathcal{A}_N(s)=\sum_{n\le N}\mu(n)\chi_4(n)n^{-s}$ is written with a script letter because $A_N$ is the coefficient of Theorem 6.1.5. The completed beta function is $\xi(s,\chi_4)$ of Section 8.5.7. The probability measures of one pair of zeros (Section 29.14) are written $Q_{a,\gamma}$. Status

words follow the rest of the book: proved, computed, conditional, heuristic, open. Results that rest on the quasi Riemann hypothesis [398, 399, 352] are conditional on it, and on one further point: the preprint is announced for Dirichlet $L$ functions, and whether its theorem covers the character modulo 4 has not been checked for this book.

The results are these.

1. The integer dilates $\{\operatorname{sech}(nx)\}$ are complete in $L^1$ with the weight of Lerch's theorem, and in every $\mathcal{W}_c$, $c > 0$, unconditionally; more generally $\{\operatorname{sech}(\lambda x)\}_{\lambda\in\Lambda}$, $\Lambda \subset [1,\infty)$, is complete in $\mathcal{W}_c$ whenever $\sum_\Lambda 1/\lambda = \infty$ (Theorems 29.1.4 and 29.3.2, proved). The proof of the second statement replaces Lerch by Carleman's formula and uses no arithmetic.

2. Completeness in two weights $\mathcal{W}_a \cap \mathcal{W}_b$ for all $\theta < a < b < 1$ is equivalent to $L(s,\chi_4) \neq 0$ for $\operatorname{Re} s > \theta$, and to convergence of the Möbius secant series in every $\mathcal{W}_c$, $\theta < c < 1$ (Theorem 29.5.2, proved); completeness in two weights straddling a strip forces that strip to be zero free (Proposition 29.5.1, proved). The zeta kernel $1/(e^x-1)-1/x$ has the same two statements, with $\zeta$ and RH (Theorem 29.6.1, proved). The consequences of the quasi Riemann hypothesis are in Section 29.7 (conditional).

3. Under a polynomial bound for $1/\beta$ on the line $\operatorname{Re} s = c$, which holds for $c \geq 1$, conditionally on a zero free half plane for $c$ to its right, and for $c > 0$, $c \neq \frac{1}{2}$ under GRH, the condition $\sum_\Lambda 1/\lambda = \infty$ is also necessary (Theorem 29.10.2, proved); the line $c = \frac{1}{2}$ stays open. The $L^1$ criterion of Salem for the integer dilates, and its beta analogue, are Theorem 29.11.1 (proved).

4. The optimal distance of $e^{-x}$ from the span of $\operatorname{sech}(nx)$, $n \leq N$, is given by a Gram system (computed, Table 29.2); the optimal coefficients are huge, and the explanation by the frequencies $\log n$ is heuristic (Section 29.12). The residual identity of Proposition 29.12.1, with its factor 2, is proved.

5. For the Meixner family of order $t$ the Mellin transform is $M_t$ of Theorem 29.13.1, single weight completeness holds throughout, and for $t = 2$ the Möbius route is unconditional (Theorem 29.13.3, proved).

6. For $\xi(s,\chi_4)$ the equivalence of the zero free half plane with a generalised gamma convolution form of $\xi(\theta,\chi_4)/\xi(\theta+\sqrt{s},\chi_4)$ and the explicit Thorin measure are Theorem 29.14.2 (proved; for $\xi$ this is Theorem 1.2.2); the link with the two weight criterion is new (Section 29.8).

The open problems close the chapter.

## 29.1 The secant family and the identity

On the whole line no symmetric scale family can be complete, because odd statistics have mean zero, so the dilates are folded to $(0,\infty)$. The symmetric Meixner density of order $t>0$ is proportional to $|\Gamma(\frac{t}{2}+\frac{ix}{2})|^2$ (Section 6.2.3), and $t=1$ gives the hyperbolic secant density because $|\Gamma(\frac12+iy)|^2=\pi/\cosh(\pi y)$. Folded to $x>0$ it is $p_\lambda(x)=\frac{2\lambda}{\pi}\operatorname{sech}(\lambda x)$, and a family $\{p_\lambda:\lambda\in\Lambda\}$ is complete when $\mathbb{E}_\lambda g(X_\lambda)=0$ for all $\lambda\in\Lambda$ forces $g=0$ almost everywhere. The smallest natural parameter set is $\Lambda=\mathbb{N}$, so completeness says that the dilates $\operatorname{sech}(nx)$ separate functions; this is the integer case of the question of Lai and Vere-Jones [322] as recorded in Chapter 6.

Lemma 29.1.1. *For $y>0$, $\operatorname{sech} y = 2\sum_{m\ge1}\chi_4(m)e^{-my}$, and $\sum_{n\ge1}\operatorname{sech}(ny)\le 2/(e^y-1)$.*

*Proof.* The first identity is the geometric series of Proposition 6.1.3. For the bound use $\operatorname{sech}(ny)\le 2e^{-ny}$. $\square$

Lemma 29.1.2. *For $\operatorname{Re} s>0$ and $\lambda>0$, $\int_0^\infty x^{s-1}\operatorname{sech}(\lambda x)\,dx = 2\lambda^{-s}\Gamma(s)\beta(s)$.*

*Proof.* For $\lambda=1$ integrate Lemma 29.1.1 termwise for $\operatorname{Re} s>1$, where $\sum_m\int_0^\infty x^{\sigma-1}e^{-mx}dx=\Gamma(\sigma)\zeta(\sigma)<\infty$, and continue analytically; the integral is holomorphic for $\operatorname{Re} s>0$ and $\beta$ is entire. The substitution $y=\lambda x$ gives general $\lambda$. $\square$

By Proposition 6.1.3, $e^{-x}=\frac12\sum_{n\ge1}\mu(n)\chi_4(n)\operatorname{sech}(nx)$ with absolute convergence for $x>0$, and replacing $x$ by $kx$,

$$e^{-kx}=\frac12\sum_{n\ge1}\mu(n)\chi_4(n)\operatorname{sech}(nkx),\qquad k\in\mathbb{N}, \tag{29.1.1}$$

and every function on the right is again one of the dilates. In Mellin language the identity is the factorisation $\Gamma(s)=\frac12\cdot 2\Gamma(s)\beta(s)\cdot\beta(s)^{-1}$ with $\beta(s)^{-1}=\sum\mu(n)\chi_4(n)n^{-s}$.

Theorem 29.1.3 (Lerch). *Let $\int_0^\infty|h(x)|e^{-x}dx<\infty$. If $\int_0^\infty e^{-kx}h(x)\,dx=0$ for all $k\in\mathbb{N}$, then $h=0$ almost everywhere.*

*Proof.* With $u=e^{-x}$ and $q(u)=h(-\log u)$ we have $q\in L^1(0,1)$ and $\int_0^1 u^j q(u)\,du=0$ for $j\ge0$. The measure $q\,du$ annihilates the polynomials, hence $C[0,1]$ by Weierstrass, hence $q=0$. $\square$

Theorem 29.1.4 (completeness in $L^1$). *Let $g$ satisfy*

$$\int_0^1\frac{|g(x)|}{x}\,dx+\int_1^\infty|g(x)|e^{-x}dx<\infty. \tag{29.1.2}$$

*If $\int_0^\infty g(x)\operatorname{sech}(nx)\,dx = 0$ for all $n \in \mathbb{N}$, then $g = 0$ almost everywhere. The same holds with $\mathbb{N}$ replaced by a set $\Lambda \subset (0,\infty)$ containing an interval, provided the integrals $\int g\operatorname{sech}(\lambda x)\,dx$, $\lambda \in \Lambda$, converge absolutely and vanish.*

*Proof.* By Lemma 29.1.1, $\sum_n |\operatorname{sech}(nkx)| \le 2/(e^{kx}-1)$, which is $O(1/x)$ at the origin and $O(e^{-x})$ at infinity, so (29.1.2) justifies integrating (29.1.1) termwise against $g$. Every term vanishes, hence $\int g e^{-kx}dx = 0$ for all $k$, and Theorem 29.1.3 applies. For the interval clause, let $I \subset \Lambda$ be an interval and $\lambda_0$ a point of its interior. Absolute convergence at $\lambda_0$ gives $\int_1^\infty |g|e^{-\lambda_0 x}dx < \infty$, because $\operatorname{sech}(\lambda_0 x) \ge e^{-\lambda_0 x}$. The function $F(z) = \int_0^\infty g(x)\operatorname{sech}(zx)\,dx$ is then holomorphic on $\operatorname{Re} z > \lambda_0$ (on compact subsets the large $x$ part is dominated by $C|g(x)|e^{-(\lambda_0+\delta)x}$ and the small $x$ part by $C|g(x)|$), and it vanishes on a subinterval of $I$ to the right of $\lambda_0$, hence on the half plane. Take an integer $K > \max(1,\lambda_0)$. The integrals $\int g(x)\operatorname{sech}(nkx)\,dx$ vanish for all integers $n \ge 1$ and $k \ge K$, so (29.1.1) (again integrated termwise, the majorant being $|g|\,2/(e^{kx}-1)$ with $k \ge K$) gives $\int g(x)e^{-kx}dx = 0$ for $k \ge K$. Apply Theorem 29.1.3 to $h(x) = g(x)e^{-(K-1)x}$, whose moments of orders $1, 2, \dots$ are these integrals. $\square$

The proof of the first part uses only $\sum_{d\mid N}\mu(d) = [N = 1]$, the complete multiplicativity of $\chi_4$ and $|\mu| \le 1$. The condition $\int_0^1 |g|/x < \infty$ is the $L^1$ shadow of the half plane $\operatorname{Re} s > 1$ in which the Möbius series converges absolutely. The theorem is proved here in the form stated; it is not claimed to be the form in which the question was posed in [322].

## 29.2 Weighted spaces and the Mellin line

Let $\mathcal{W}_c = L^2((0,\infty), x^{2c-1}dx)$. The change of variable $x = e^u$ and Plancherel give

$$\langle f, g\rangle_c = \int_0^\infty f\bar{g}\,x^{2c-1}dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(c+it)\overline{G(c+it)}\,dt, \qquad F(s) = \int_0^\infty f(x)x^{s-1}dx. \tag{29.2.1}$$

So $\mathcal{W}_c$ is the space of functions whose Mellin transform lives on the line $\operatorname{Re} s = c$. For $c > 0$ both $e^{-x}$ and every $\operatorname{sech}(\lambda x)$ lie in $\mathcal{W}_c$, and dilation acts by $\|f(k\,\cdot)\|_c = k^{-c}\|f\|_c$.

Lemma 29.2.1 (reduction). *For $c > 0$, or for $\mathcal{W}_a \cap \mathcal{W}_b$ with $0 < a < b$ and norm $\|f\|_a + \|f\|_b$, the exponentials $\{e^{-kx}\}_{k\in\mathbb{N}}$ are complete. Hence a system whose linear span is invariant under the dilations $f \mapsto f(k\,\cdot)$, $k \in \mathbb{N}$, is complete as soon as its closed span contains $e^{-x}$.*

*Proof.* A functional on $\mathcal{W}_c$ is $f \mapsto \int fh\,dx$ with $h = \bar{g}x^{2c-1}$, $g \in \mathcal{W}_c$, and $\int |h|e^{-x} \le \|g\|_c\|e^{-x}\|_c$; apply Theorem 29.1.3. On the intersection the dual is

the sum of the two duals (Hahn Banach) and the same estimate applies to each piece. If $f_j \to e^{-x}$ with $f_j$ in the span, then $f_j(k\,\cdot) \to e^{-k\,\cdot}$ in the same space and $f_j(k\,\cdot)$ is again in the span. □

Write $S_N(x) = \frac{1}{2}\sum_{n\le N} \mu(n)\chi_4(n)\operatorname{sech}(nx)$. By Lemma 29.1.2 and (29.2.1),

$$\|S_N - e^{-x}\|_c^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\Gamma(c+it)|^2 \left|\beta(c+it)\mathcal{A}_N(c+it) - 1\right|^2 dt, \tag{29.2.2}$$

so convergence of the Möbius secant series in $\mathcal{W}_c$ is the statement that $\mathcal{A}_N \to 1/\beta$ on the line $\operatorname{Re} s = c$ in a norm weighted by $|\Gamma(c+it)|^2 \asymp |t|^{2c-1}e^{-\pi|t|}$.

THEOREM 29.2.2 (the classical range). *For $c \ge 1$, $S_N \to e^{-x}$ in $\mathcal{W}_c$.*

*Proof.* For $c > 1$ use $|\mathcal{A}_N| \le \zeta(c)$, $|\beta| \le \zeta(c)$ and dominated convergence in (29.2.2). For $c = 1$ the prime number theorem for $\mu\chi_4$ in the form $M_\chi(x) \ll x\exp(-a\sqrt{\log x})$ and partial summation,

$$\mathcal{A}_N(s) = \frac{M_\chi(N)}{N^s} + s\int_1^N \frac{M_\chi(u)}{u^{s+1}}du, \tag{29.2.3}$$

give $|\mathcal{A}_N(1+it) - 1/\beta(1+it)| \le C(1+|t|)\varepsilon_N$ with $\varepsilon_N \to 0$ (the integral converges on $\operatorname{Re} s = 1$ by the same bound for $M_\chi$); with $|\beta(1+it)| \ll \log(2+|t|)$, (29.2.2) tends to zero. □

## 29.3 Completeness in one weight is unconditional

The Möbius series is one particular approximation of $e^{-x}$. Completeness asks only that some approximation exists, and in a single weight this needs no arithmetic. The tool is Carleman's formula for a half plane (Titchmarsh [355]; Boas [27]): if $f$ is analytic in a neighbourhood of $\operatorname{Re} w \ge 0$, $f \not\equiv 0$, with zeros $r_k e^{i\theta_k}$, then for $R > 1$

$$\sum_{1<r_k<R}\Big(\frac{1}{r_k} - \frac{r_k}{R^2}\Big)\cos\theta_k = \frac{1}{\pi R}\int_{-\pi/2}^{\pi/2} \log|f(Re^{i\theta})|\cos\theta\, d\theta + \frac{1}{2\pi}\int_1^R \Big(\frac{1}{y^2} - \frac{1}{R^2}\Big)\log|f(iy)f(-iy)|\, dy + O_f(1). \tag{29.3.1}$$

The form of the formula was checked against the standard statement; the section numbers of the two books are not claimed.

LEMMA 29.3.1 (uniqueness on a divergent set). *Let $\Phi$ be analytic on $\operatorname{Re} w > 0$ with $|\Phi(w)| \le C(|w|/\operatorname{Re} w)^B$ for some $B \ge 0$. If $\Phi(\lambda) = 0$ for every $\lambda$ in a set $\Lambda \subset [1,\infty)$ with $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$, then $\Phi \equiv 0$.*

*Proof.* If $\Lambda$ has a finite accumulation point, $\Phi \equiv 0$ by the identity theorem, so let the points of $\Lambda$ be isolated. Fix $\varepsilon \in (0,1)$ and apply (29.3.1) to $f(w) = \Phi(w+\varepsilon)$, which is analytic on $\operatorname{Re} w > -\varepsilon$, assuming $f \not\equiv 0$. Its zeros include $\lambda - \varepsilon$, $\lambda \in \Lambda$, on the positive axis (the finitely many with $\lambda - \varepsilon \le 1$ lie outside the range of the sum and cost $O(1)$). The other zeros of $f$ contribute nonnegative terms ($\cos\theta_k \ge 0$, $r_k < R$). Since $\frac{1}{r} - \frac{r}{R^2} \ge \frac{3}{4}\cdot\frac{1}{r}$ for $r < R/2$, the left side is at least $\frac{3}{4}\sum_{\lambda<R/2} 1/\lambda - O(1) \to \infty$. On the right, for $w = Re^{i\theta}$ we have $|w+\varepsilon|/\operatorname{Re}(w+\varepsilon) \le 2/\cos\theta$, so $\log|f| \le C' + B\log(1/\cos\theta)$ and the first integral is at most $O(1/R)$, because $\int \log(1/\cos\theta)\cos\theta\, d\theta < \infty$. On the imaginary axis $|f(\pm iy)| \le C(1+|y|/\varepsilon)^B$, so the second integral is at most $C_\varepsilon \int_1^\infty \log(2+y)\, y^{-2} dy < \infty$, the weight $\frac{1}{y^2} - \frac{1}{R^2}$ being nonnegative. The right side is bounded above, a contradiction. Hence $f \equiv 0$. $\square$

THEOREM 29.3.2 (completeness in one weight). *Let $c > 0$ and let $\Lambda \subset [1,\infty)$ satisfy $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$. Then $\{\operatorname{sech}(\lambda x)\}_{\lambda\in\Lambda}$ is complete in $\mathcal{W}_c$. In particular the integer dilates are complete in every $\mathcal{W}_c$, and so are the prime dilates $\{\operatorname{sech}(px)\}_{p \text{ prime}}$.*

*Proof.* Let $g \in \mathcal{W}_c$ be orthogonal to every $\operatorname{sech}(\lambda x)$, and let $G$ be its Mellin transform on $\operatorname{Re} s = c$. By Lemma 29.1.2 and (29.2.1),

$$\int_{-\infty}^{\infty} H(t)\lambda^{-it} dt = 0 \quad (\lambda \in \Lambda), \qquad H(t) = \Gamma(c+it)\beta(c+it)\overline{G(c+it)}.$$

Stirling and the convexity bound give $|\Gamma(c+it)\beta(c+it)| \le C(1+|t|)^A e^{-\pi|t|/2}$, and $G \in L^2$, so $H \in L^1$. For $w = |w|e^{i\varphi}$ with $|\varphi| < \pi/2$ define

$$\Phi(w) = \int_{-\infty}^{\infty} H(t)\, w^{-it}\, dt, \qquad |w^{-it}| = e^{t\varphi}.$$

By Cauchy Schwarz,

$$|\Phi(w)| \le C\|G\|_2 \Big( \int (1+|t|)^{2A} e^{-(\pi - 2|\varphi|)|t|} dt \Big)^{1/2} \le C' \Big( \frac{\pi}{2} - |\varphi| \Big)^{-A-1/2},$$

and since $\frac{\pi}{2} - |\varphi| \ge \cos\varphi = \operatorname{Re} w/|w|$, $\Phi$ satisfies the growth bound of Lemma 29.3.1 with $B = A + \frac{1}{2}$. It is analytic and vanishes on $\Lambda$, so $\Phi \equiv 0$. On the positive axis $\Phi(e^\xi) = \int H(t) e^{-i\xi t} dt$ is the Fourier transform of $H \in L^1$, so $H = 0$ almost everywhere. The zeros of $\beta$ on the line $\operatorname{Re} s = c$ form a discrete set, hence $G = 0$ almost everywhere and $g = 0$. $\square$

REMARK 29.3.3 (the threshold, heuristic). If the Mellin transform of a kernel is bounded by $e^{-\alpha|t|}$ times a power, the function $\Phi$ above is analytic at least in the sector $|\arg w| < \alpha$. The map $w \mapsto w^{\pi/(2\alpha)}$ sends the sector to a half plane and $\lambda$ to $\lambda^{\pi/(2\alpha)}$, so the Carleman sum becomes $\sum \lambda^{-\pi/(2\alpha)}$; for $\Lambda = \mathbb{N}$ this diverges if and only if $\alpha \ge \pi/2$. The gamma factor of sech decays at precisely $\alpha = \pi/2$. This is the threshold of the method: failure of completeness for $\alpha < \pi/2$ is not shown, and the sharpness question is Open Problem 29.1.

REMARK 29.3.4 (Müntz and Szász). For the kernel $e^{-x}$ itself ($\beta \equiv 1$) the proof of Theorem 29.3.2 gives the sufficiency half of the Müntz and Szász theorem: under $u = e^{-x}$, completeness of $\{e^{-\lambda x}\}$ in $\mathcal{W}_{1/2}$ is completeness of $\{u^{\lambda - 1/2}\}$ in $L^2(0,1)$, and divergence of $\sum 1/\lambda$ is the classical sufficient condition. Lerch's theorem is the case $\Lambda = \mathbb{N}$.

Three consequences are worth stating plainly. First, taking $c = \frac{1}{2}$, the integer dilates of sech are complete in $L^2(0,\infty)$, even though $\beta$ has infinitely many zeros on the very line $\operatorname{Re} s = \frac{1}{2}$ on which the Mellin transform lives. Second, no property of $\mu$ is needed for completeness in one weight. Third, the same proof applies to any kernel whose Mellin transform on the line is $\Gamma(s)$ times a polynomially bounded function vanishing only on a null set of the line, in particular to the zeta kernel of Section 29.6. Theorem 29.3.2 therefore settles the one weight form of the completeness question recorded in Chapter 6, unconditionally; only the two weight form is equivalent to a zero free half plane (Theorem 29.5.2).

## 29.4 The Möbius route in the strip $\frac{1}{2} < c < 1$

Where arithmetic does enter is in the explicit approximation by the Möbius series.

LEMMA 29.4.1 (partial summation). *Suppose $M_\chi(x) = O_\varepsilon(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$, with $\frac{1}{2} \le \theta < 1$. Then for $\sigma = \operatorname{Re} s > \theta$, $t = \operatorname{Im} s$ and $0 < \varepsilon < \sigma - \theta$, with constants depending on $\sigma$ and $\varepsilon$,*

$$|\mathcal{A}_N(s)| \le C(1+|t|), \qquad |\mathcal{A}_N(s) - 1/\beta(s)| \le C(1+|t|)N^{\theta+\varepsilon-\sigma}.$$

*Proof.* In (29.2.3) the integral converges absolutely for $\sigma > \theta$ to an analytic function equal to $1/(s\beta(s))$ for $\sigma > 1$, hence throughout; the boundary term and the tail $|s| \int_N^\infty u^{\theta+\varepsilon-\sigma-1} du$ give the rate. □

THEOREM 29.4.2 (the strip). *Under the hypothesis of Lemma 29.4.1, for every $c \in (\theta, 1)$,*

$$\|S_N - e^{-x}\|_c = O_\varepsilon(N^{\theta-c+\varepsilon}).$$

*Under the generalised Riemann hypothesis for $\beta$ this holds for every $c > \frac{1}{2}$.*

*Proof.* Insert Lemma 29.4.1 in (29.2.2). The integral $\int |\Gamma\beta|^2 (1+|t|)^2 dt$ on the line $\operatorname{Re} s = c$ is finite by Stirling and convexity. □

PROPOSITION 29.4.3 (classical equivalences). *For $\frac{1}{2} \le \theta < 1$ the following are equivalent: (a) $M_\chi(x) = O(x^{\theta+\varepsilon})$ for all $\varepsilon > 0$; (b) $\sum \mu(n)\chi_4(n)n^{-s}$ converges for $\operatorname{Re} s > \theta$; (c) $\beta(s) \neq 0$ for $\operatorname{Re} s > \theta$.*

*Proof.* (a) implies (b) by partial summation. (b) implies (c): the sum is analytic on $\operatorname{Re} s > \theta$ and equals $1/\beta$ for $\operatorname{Re} s > 1$, so $1/\beta$ continues analytically there and $\beta$ has no zeros. (c) implies (a) is Littlewood's theorem for $L(s, \chi_4)$: a zero free half

plane gives $1/\beta(\sigma+it) \ll |t|^{\varepsilon}$ for $\sigma > \theta$ by the Borel and Carathéodory argument, and Perron's formula then bounds $M_\chi$; compare Titchmarsh [498, Section 14.25] for $\zeta$, whose proof carries over to the character. □

## 29.5 Two weights: where the zeros are felt

The one weight proof breaks down for an intersection of two weights for a structural reason. If $f \in \mathcal{W}_a \cap \mathcal{W}_b$ and $a < \sigma < b$, splitting at $x = 1$ and using Cauchy Schwarz with weight $a$ on $(0,1)$ and weight $b$ on $(1,\infty)$ gives

$$\Big|\int_0^\infty f(x)x^{s-1}dx\Big| \le \frac{\|f\|_a}{\sqrt{2(\sigma-a)}} + \frac{\|f\|_b}{\sqrt{2(b-\sigma)}}. \tag{29.5.1}$$

So evaluation of the Mellin transform at any point of the open strip is continuous, and zeros of $\beta$ in the strip are inherited by limits. In a single weight there is no strip and evaluation off the line is not continuous.

Proposition 29.5.1 (necessity). *Let $0 < a < b$. If $e^{-x}$ is a limit in $\mathcal{W}_a \cap \mathcal{W}_b$ of finite combinations of the* $\operatorname{sech}(nx)$*, then $\beta(s) \neq 0$ for $a < \operatorname{Re} s < b$.*

*Proof.* Each approximant has Mellin transform $2\Gamma(s)\beta(s)D(s)$ with $D$ a Dirichlet polynomial. By (29.5.1) these converge pointwise in the strip to $\Gamma(s)$, which is nonzero; a zero $\rho$ of $\beta$ in the strip would force $\Gamma(\rho) = 0$. □

Theorem 29.5.2 (two weight criterion). *For $\frac{1}{2} \le \theta < 1$ the following are equivalent:*

*(i) $L(s,\chi_4) \neq 0$ for $\operatorname{Re} s > \theta$;*

*(ii) $\{\operatorname{sech}(nx)\}_{n\in\mathbb{N}}$ is complete in $\mathcal{W}_a \cap \mathcal{W}_b$ for all $\theta < a < b < 1$;*

*(iii) $S_N \to e^{-x}$ in $\mathcal{W}_c$ for every $c \in (\theta, 1)$.*

*With $\theta = \frac{1}{2}$, (i) is the generalised Riemann hypothesis for $L(s,\chi_4)$.*

*Proof.* (i) gives (a) of Proposition 29.4.3, then Theorem 29.4.2 gives (iii). Convergence in each $\mathcal{W}_c$, $c \in [a,b]$, gives convergence in $\mathcal{W}_a \cap \mathcal{W}_b$, and Lemma 29.2.1 gives (ii). If (ii) holds and $\beta(\rho) = 0$ with $\theta < \operatorname{Re}\rho < 1$, choose $a < \operatorname{Re}\rho < b$ in $(\theta, 1)$ and apply Proposition 29.5.1; there are no zeros with $\operatorname{Re}\rho \ge 1$. Thus (ii) implies (i). □

The picture is now as follows. Completeness in one weight is free (Theorem 29.3.2). Completeness in two weights straddling a strip forces that strip to be zero free (Proposition 29.5.1), and completeness for all pairs $\theta < a < b < 1$ is equivalent to a zero free half plane (Theorem 29.5.2); whether a zero free strip $a < \operatorname{Re} s < b$ alone gives completeness in $\mathcal{W}_a \cap \mathcal{W}_b$ is not shown. At a single

endpoint $c < 1$, convergence of $S_N$ in $\mathcal{W}_c$ does imply zero freeness strictly to the right of $c$, by combining it with the convergence at any $b > 1$ and applying Proposition 29.5.1 in $(c, b)$, but a bare zero free half plane $\operatorname{Re} s > c$ does not by this proof give convergence on the line $c$ itself. The properties of $\mu$ that matter form a ladder.

(P0) Nothing: completeness in one weight, every $c > 0$.

(P1) Inversion $\sum_{d|N} \mu(d) = [N = 1]$: the pointwise identity of Proposition 6.1.3.

(P2) $|\mu| \le 1$: the $L^1$ theorem and convergence in $\mathcal{W}_c$, $c > 1$.

(P3) The prime number theorem for $\mu\chi_4$: convergence at $c = 1$.

(P4) Power cancellation $M_\chi(x) = O(x^{\theta+\varepsilon})$: convergence and two weight completeness for $c \in (\theta, 1)$, equivalent to a zero free half plane for $\beta$. Square root cancellation is GRH for $\beta$.

The arithmetic function is the twisted $\mu\chi_4$, not $\mu$: the alternating signs of the secant produce $\beta$, not $\zeta$. The barrier $c = \frac{1}{2}$ is genuine: $\beta$ vanishes at $\frac{1}{2} + 6.0209489\ldots i$ (computed, 6.0209489046976, `data/lv_sech_tables.json`), so two weights straddling $\frac{1}{2}$ can never work, and $M_\chi(x) \neq o(x^{1/2})$, since $M_\chi(x) = o(x^{1/2})$ would give $(\sigma - \frac{1}{2})/|\beta(\sigma + i\gamma)| \to 0$ as $\sigma \downarrow \frac{1}{2}$ at a zero, which is impossible.

## 29.6 The zeta version

To involve $\mu$ itself, use the Bose kernel with its pole removed,

$$\varphi(x) = \frac{1}{e^x - 1} - \frac{1}{x}, \qquad \int_0^\infty x^{s-1}\varphi(x)\,dx = \Gamma(s)\zeta(s) \quad (0 < \operatorname{Re} s < 1). \tag{29.6.1}$$

Since $\varphi(x) \to -\frac{1}{2}$ at the origin and $\varphi(x) \sim -1/x$ at infinity, $\varphi \in \mathcal{W}_c$ exactly for $0 < c < 1$. The cosecant $\operatorname{csch} x$, with transform $2\Gamma(s)(1 - 2^{-s})\zeta(s)$, is unsuitable because the pole of $\zeta$ removes it from $\mathcal{W}_c$ for $c \le 1$. Write $T_N(x) = \sum_{n \le N} \mu(n)\varphi(nx)$.

Theorem 29.6.1. *(1) For every $c \in (0, 1)$ the dilates $\{\varphi(nx)\}_{n\in\mathbb{N}}$ are complete in $\mathcal{W}_c$, unconditionally. (2) Let $\frac{1}{2} \le \theta < 1$ and suppose $M(x) = O_\varepsilon(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$. Then $T_N \to e^{-x}$ in $\mathcal{W}_c$ with $\|T_N - e^{-x}\|_c = O_\varepsilon(N^{\theta-c+\varepsilon})$ for $\theta < c < 1$ and $0 < \varepsilon < c - \theta$. (3) For $\frac{1}{2} \le \theta < 1$, completeness of $\{\varphi(nx)\}$ in $\mathcal{W}_a \cap \mathcal{W}_b$ for all $\theta < a < b < 1$ is equivalent to $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$ (the pole at 1 being allowed); with $\theta = \frac{1}{2}$ it is equivalent to the Riemann hypothesis.*

*Proof.* The proofs of Theorems 29.3.2, 29.4.2, 29.5.2 carry over with $(\beta, \mu\chi_4, \operatorname{sech})$ replaced by $(\zeta, \mu, \varphi)$, with the following changes. The Mellin transform of $\varphi(n\,\cdot)$ is $n^{-s}\Gamma(s)\zeta(s)$ on $0 < \operatorname{Re} s < 1$, with no factor 2, and $|\zeta(c + it)| \ll (1 + |t|)^{1/2}$ there,

so $|\Gamma\zeta| \le C(1+|t|)^A e^{-\pi|t|/2}$ and the proof of part (1) is that of Theorem 29.3.2; $\zeta$ has discrete zeros on the line. For part (2) the Mellin transform of $T_N - e^{-x}$ on the line $\operatorname{Re} s = c$ is $\Gamma(s)\big(\zeta(s)\mathcal{A}_N(s) - 1\big)$ with $\mathcal{A}_N = \sum_{n\le N} \mu(n) n^{-s}$, and Lemma 29.4.1 holds for $\mathcal{A}_N$ and $1/\zeta$ with $M$ in place of $M_\chi$ ($1/\zeta$ is analytic on $\operatorname{Re} s > \theta$ because $\zeta$ has no zeros there, and $\zeta$ is analytic on $0 < \operatorname{Re} s < 1$). For part (3) Proposition 29.5.1 holds with $\zeta$ because the approximants have Mellin transform $\Gamma\zeta D$ and converge pointwise in the strip $a < \operatorname{Re} s < b \subset (0,1)$ to $\Gamma$; zeros with $\operatorname{Re}\rho \ge 1$ do not exist; and the classical equivalence of Proposition 29.4.3 for $\zeta$ is Littlewood's theorem. □

The contrast with the secant is that $\varphi$ has no unconditional Möbius range at all: its Mellin line must lie inside $0 < c < 1$. Part (3) is a Nyman and Beurling type criterion on the half line, and part (1) explains why it needs two weights: in the classical Nyman and Beurling problem the depth comes from restricting to $(0,1)$, which makes Mellin transforms analytic in a half plane, while here the second weight plays that role.

## 29.7 What the quasi Riemann hypothesis gives

In October 2026 OpenAI released a preprint [398] (dated 30 September) announcing a zero free half plane $\operatorname{Re} s > \frac{7}{8}$ for Dirichlet $L$ functions, with a formalisation in Lean, and a companion [399] with the weaker half plane $\operatorname{Re} s > \frac{11}{12}$; Liu [352] claims a small improvement of the exponent $\frac{7}{8}$. A preprint attributed to Thorner, extending a zero free half plane $\operatorname{Re} s > \frac{31}{32}$ to the Dedekind zeta function of every number field, is cited here from the source as unchecked [354]. None of these has been refereed, and none has been checked for this book. Every statement below that uses them is conditional on them, and those about $\beta$ on one further point: the $\frac{7}{8}$ theorem is announced for Dirichlet $L$ functions, and that it covers $L(s,\chi_4)$ is assumed here and is a question for the authors. If these results stand, the functional equation puts the nontrivial zeros of $\zeta$ and of $\beta$ in $\frac{1}{8} \le \operatorname{Re}\rho \le \frac{7}{8}$.

| input | secant, $\mu\chi_4$ | zeta kernel, $\mu$ |
|---|---|---|
| none | one weight, all $c > 0$ | one weight, $0 < c < 1$ |
| PNT | Möbius series, $c \ge 1$ (rate subexponential in $\sqrt{\log N}$) | nothing new |
| Dedekind QRH ($\frac{31}{32}$), unchecked | $c > \frac{31}{32}$ | $\frac{31}{32} < c < 1$ |
| QRH for Dirichlet $L$ ($\frac{7}{8}$), conditional | $\frac{7}{8} < c$ | $\frac{7}{8} < c < 1$ |
| GRH | $c > \frac{1}{2}$ | $\frac{1}{2} < c < 1$ |

For a zero free half plane $\operatorname{Re} s > \theta$ the entries mean: the Möbius series converges in $\mathcal{W}_c$ at the rate $N^{\theta - c + \varepsilon}$ for every $c > \theta$ (secant) or $\theta < c < 1$ (zeta kernel), and two weight completeness holds on that range. For the secant kernel and $c > 1$

the elementary absolute convergence argument gives $O_c(N^{1-c})$, and at $c = 1$ the prime number theorem gives $O((1 + \sqrt{\log N})e^{-a\sqrt{\log N}})$; neither is a power law with $\theta = 1$, so the PNT row is not an instance of the rate $N^{\theta-c+\varepsilon}$. The input $\frac{7}{8}$ pushes the Möbius route below $c = 1$, from $c \geq 1$ down to $c > \frac{7}{8}$, for both kernels at once if the theorem covers $\beta$ as well as $\zeta$. The stated half plane does not supply convergence in the remaining range $\frac{1}{2} < c \leq \frac{7}{8}$; GRH is sufficient there, whereas a weaker zero free bound already gives convergence above its own boundary. For the one weight question QRH is irrelevant, since that question was settled unconditionally.

## 29.8 Relation to the Thorin programme

Three remarks connect this part to the rest of the book. The second and third are analogies (heuristic), not consequences of the theorems above.

First, the secant is a Thorin object, in the form of Lemma 6.1.4: $\operatorname{sech}\sqrt{2s} = \prod_{k\geq 1}(1 + s/\lambda_k)^{-1}$ with $\lambda_k = (2k-1)^2\pi^2/8$ is the Laplace transform of the exit time of Brownian motion from $(-1, 1)$, a generalised gamma convolution with discrete Thorin measure at the $\lambda_k$ (Table 1.8). Dilation of sech corresponds to a time scaling of that law after the substitution $x = \sqrt{2s}$.

Second, the book's Thorin type expansion of $1/\zeta(\frac{1}{2} + \sqrt{s})$ (Theorem 6.1.5) applies Proposition 6.1.3 pointwise at $x = \sqrt{s}\log n$ to the Dirichlet series of $1/\zeta$. Theorems 29.5.2 and 29.6.1 are statements in $L^2$ in the variable $x$ about approximating $e^{-x}$ in weighted spaces; they are not statements about that series in $s$, and the kernel $\varphi$ is not used in it. The parallel is this: the convergence of the construction needs the same zero free region as the Möbius series, and no new zero free region comes out of it.

Third, the dichotomy between one and two weights is a pattern for the programme. Statements made on a single Mellin line (completeness, $L^2$ approximation) are cheap, because point evaluations off the line are not controlled, while statements uniform on a strip carry analyticity into the strip and force zero freeness there (Proposition 29.5.1). A GGC Laplace transform is analytic and zero free in a half plane, so a Thorin representation of $1/\xi$ is of the second kind; Section 29.14 makes this precise for $\xi(s, \chi_4)$.

## 29.9 Numerical check

The weighted errors $\|S_N - e^{-x}\|_c$ and $\|T_N - e^{-x}\|_c$ were computed by quadrature on logarithmic grids (`code/lv_sech_tables.py`, output `data/lv_sech_tables.json`); the secant table was computed on two grid steps and agrees to $10^{-15}$ relative, the zeta kernel table to $5{\cdot}10^{-8}$, the part of the zeta integrals beyond $x = 60$ being added exactly ($T_N - e^{-x} = -m_N/x$ there, $m_N = \sum_{n\leq N}\mu(n)/n$). The pointwise identity of Proposition 6.1.3 is proved, and the check of it at

$x = 0.05, 0.3, 1, 3$ in the source is not reproduced here.

| | secant, $\|S_N - e^{-x}\|_c$ | | | |
|---|---|---|---|---|
| $N$ | $c = 0.60$ | $c = 0.75$ | $c = 1.00$ | $c = 1.25$ |
| 10 | $7.3 \times 10^{-3}$ | $5.1 \times 10^{-3}$ | $2.9 \times 10^{-3}$ | $1.7 \times 10^{-3}$ |
| $10^2$ | $3.9 \times 10^{-2}$ | $1.9 \times 10^{-2}$ | $6.3 \times 10^{-3}$ | $2.2 \times 10^{-3}$ |
| $10^3$ | $3.5 \times 10^{-2}$ | $1.2 \times 10^{-2}$ | $2.2 \times 10^{-3}$ | $4.2 \times 10^{-4}$ |
| $10^4$ | $5.3 \times 10^{-2}$ | $1.3 \times 10^{-2}$ | $1.2 \times 10^{-3}$ | $1.3 \times 10^{-4}$ |
| | zeta kernel, $\|T_N - e^{-x}\|_c$ | | | |
| $N$ | $c = 0.75$ | $c = 0.875$ | $c = 0.95$ | |
| 10 | $2.1 \times 10^{-1}$ | $2.3 \times 10^{-1}$ | $3.1 \times 10^{-1}$ | |
| $10^2$ | $1.2 \times 10^{-1}$ | $1.0 \times 10^{-1}$ | $1.2 \times 10^{-1}$ | |
| $10^3$ | $3.0 \times 10^{-2}$ | $1.9 \times 10^{-2}$ | $1.9 \times 10^{-2}$ | |
| $10^4$ | $2.6 \times 10^{-2}$ | $1.2 \times 10^{-2}$ | $1.0 \times 10^{-2}$ | |

Table 29.1: Weighted errors of the Möbius approximants (computed). The column $c = 0.95$ of the source was $2.9 \times 10^{-1}$, $1.1 \times 10^{-1}$, $1.8 \times 10^{-2}$, $9.8 \times 10^{-3}$, a few percent low.

For $c \geq 1$ Theorem 29.2.2 gives convergence but no rate, and at $c = 1$ the computed errors rise from $N = 10$ to $N = 100$. Inside the strip the GRH rates $N^{1/2-c}$ are slow ($N^{-0.25}$ at $c = 0.75$, $N^{-0.1}$ at $c = 0.6$) and are masked at this range by the oscillation of $M_\chi(N)$, whose values at $N = 10, 10^2, 10^3, 10^4$ are $2, 3, 6, -24$; the error is concentrated at $x \lesssim 1/N$, where $S_N(x) \approx \frac{1}{2} M_\chi(N)$ (heuristic, a Taylor approximation). The finite computations cannot prove an asymptotic rate or a zero free half plane. The largest value of $|M_\chi(n)|/\sqrt{n}$ for $n \leq 10^4$ is $2/\sqrt{3} = 1.1547$, attained at $n = 3$, so it says nothing about large $n$. The zeta kernel series converges only conditionally pointwise, since $\varphi(nx) \approx -1/(nx)$ and $\sum \mu(n)/n = 0$ is the prime number theorem; its weighted errors nevertheless decrease steadily.

## 29.10 Necessity of the divergence condition

The sufficiency proof of Theorem 29.3.2 used Carleman's formula and nothing about $\beta$. The necessity proof below divides by $\beta$ on the line, so it needs a hypothesis. Write $K_c(t) = \Gamma(c+it)\beta(c+it)$, which is half the Mellin transform of sech on the line $\operatorname{Re} s = c$, and let $c > 0$ (the spaces $\mathcal{W}_c$ contain the dilates only for $c > 0$).

($\mathrm{N}_c$) $\quad 1/\beta(c + it) \ll (1 + |t|)^A$ for some $A \geq 0$.

Lemma 29.10.1. *Let $c > 0$. Then ($\mathrm{N}_c$) holds unconditionally for $c \geq 1$. If $\beta$ has no zeros in $\operatorname{Re} s > \theta$, it holds for every $c > \theta$ and, by the functional equation, for every $c \in (0, 1 - \theta)$. In particular it holds for all $c > 0$ outside $[\frac{1}{8}, \frac{7}{8}]$ given a zero free half plane $\operatorname{Re} s > \frac{7}{8}$ for $L(s, \chi_4)$ (conditional on the quasi Riemann hypothesis of Section 29.7), and for all $c > 0$, $c \neq \frac{1}{2}$, under GRH for $\beta$.*

*Proof.* For $c > 1$, $|1/\beta(c+it)| \le \zeta(c)$; for $c = 1$, $1/\beta(1+it) \ll \log(2+|t|)$ as for $\zeta$. Given a zero free half plane $\operatorname{Re} s > \theta$, the Borel and Carathéodory argument of Titchmarsh [498, Theorem 14.2], which uses only the zero free region, gives $\log \beta(\sigma+it) \ll (\log|t|)^B$ with $B \le 1$ for $\sigma > \theta$, hence $(\mathrm{N}_c)$. For $c < 1-\theta$ put $s = 1-c-it$, so that $\operatorname{Re} s > \theta$ and the functional equation $\beta(1-s) = (2/\pi)^s \sin(\pi s/2)\Gamma(s)\beta(s)$ gives $\beta(c+it) = (2/\pi)^s \sin(\pi s/2)\Gamma(s)\beta(s)$. Since $|\sin(\pi s/2)\Gamma(s)| \asymp |t|^{1/2-c}$ for large $|t|$ on the line $\operatorname{Re} s = 1-c \in (\theta, 1)$, which is not an even integer, $1/\beta(c+it)$ is bounded by $|t|^{c-1/2}$ times a polynomial bound for $1/\beta(s)$. □

THEOREM 29.10.2 (necessity). *Let $c > 0$ and assume $(\mathrm{N}_c)$. If $\Lambda \subset [1,\infty)$ and $\sum_\Lambda 1/\lambda < \infty$, then $\{\operatorname{sech}(\lambda x)\}_{\lambda\in\Lambda}$ is not complete in $\mathcal{W}_c$. So under $(\mathrm{N}_c)$ the dilates are complete in $\mathcal{W}_c$ if and only if $\sum_\Lambda 1/\lambda = \infty$.*

*Proof.* Step 1, a Blaschke product. Put $B(w) = \prod_{\lambda\in\Lambda} \frac{1-w/\lambda}{1+w/\lambda}$. Since $\log\frac{1-w/\lambda}{1+w/\lambda} = -2w/\lambda + O(|w|^2/\lambda^2)$ and $\sum 1/\lambda < \infty$, the product converges uniformly on compact subsets of $\mathbb{C}\setminus(-\Lambda)$ to an analytic function with zeros exactly at $\Lambda$ and poles at $-\Lambda$. Each factor has modulus at most 1 on $\operatorname{Re} w \ge 0$ and modulus 1 on the imaginary axis, so $|B| \le 1$ on $\operatorname{Re} w \ge 0$. On the imaginary axis

$$\Big|\frac{d^j}{dw^j}\log B(iy)\Big| = (j-1)!\Big|\sum_\lambda\Big[\frac{1}{(\lambda-iy)^j} + \frac{(-1)^{j-1}}{(\lambda+iy)^j}\Big]\Big| \le 2(j-1)!\sum_\lambda \frac{1}{\lambda},$$

because $|\lambda \pm iy| \ge \lambda \ge 1$. Hence every derivative $B^{(j)}$ is bounded on the imaginary axis.

Step 2, the strip. Let $S = \{z : |\operatorname{Im} z| < \pi/2\}$ and $w = e^z$, which maps $S$ onto the right half plane and $\partial S$ onto the imaginary axis. Put $\widetilde{\Phi}(z) = B(e^z)e^{-z^2}$. It is analytic on $|\operatorname{Im} z| < \pi$ (the poles of $B(e^z)$ are at $z = \log\lambda + (2j+1)i\pi$), vanishes at $z = \log\lambda$, $\lambda\in\Lambda$, and $|\widetilde{\Phi}(\xi+i\eta)| \le e^{\pi^2/4-\xi^2}$ on $\overline{S}$. On the boundary lines $z = \xi \pm i\pi/2$, $\frac{d^k}{d\xi^k}[B(e^z)] = \sum_{j=1}^k S(k,j)e^{jz}B^{(j)}(e^z)$ with Stirling numbers $S(k,j)$, which by Step 1 is $O_k(\sum_{j\le k} e^{j\xi}) = O_k(e^{k\max(\xi,0)})$, while the derivatives of $e^{-z^2}$ are polynomials times $e^{-z^2}$. Every $\xi$ derivative of $\widetilde{\Phi}$ on either boundary line is therefore in $L^2(d\xi)$.

Step 3, the inverse transform. Put $H(t) = \frac{1}{2\pi}\int \widetilde{\Phi}(\xi)e^{it\xi}d\xi$. Shifting the contour to $\operatorname{Im} z = \eta$, allowed by the Gaussian decay, gives $H(t)e^{t\eta} = \frac{1}{2\pi}\int \widetilde{\Phi}(\xi+i\eta)e^{it\xi}d\xi$ for $|\eta| \le \pi/2$, so $H(t)e^{\pm\pi t/2}$ is the inverse Fourier transform of the boundary values. Since these have all derivatives in $L^2$, Plancherel gives $H(t)e^{\pi|t|/2}(1+|t|)^{A'} \in L^2(\mathbb{R})$ for every $A' \ge 0$, in particular $H(t)e^{\pi|t|/2}(1+|t|)^{A'} \in L^1$, so inversion holds on the closed strip: $\int H(t)e^{-itz}dt = \widetilde{\Phi}(z)$. Hence $\int H(t)\lambda^{-it}dt = \widetilde{\Phi}(\log\lambda) = 0$ for $\lambda\in\Lambda$, and $H \not\equiv 0$ because $\widetilde{\Phi} \not\equiv 0$.

Step 4, division by the kernel. Define $G$ on $\operatorname{Re} s = c$ by $\overline{G(c+it)} = H(t)/K_c(t)$. By Stirling, $|1/\Gamma(c+it)| \asymp e^{\pi|t|/2}|t|^{1/2-c}$ for large $|t|$, and by $(\mathrm{N}_c)$, $|G(c+it)| \le$

$C|H(t)|e^{\pi|t|/2}(1+|t|)^{A+1/2-c} \in L^2$. Let $g \in \mathcal{W}_c$ have Mellin transform $G$; $g \neq 0$. By (29.2.1) and Lemma 29.1.2,

$$\langle \operatorname{sech}(\lambda\,\cdot), g\rangle_c = \frac{\lambda^{-c}}{\pi}\int \lambda^{-it}K_c(t)\overline{G(c+it)}\,dt = \frac{\lambda^{-c}}{\pi}\int H(t)\lambda^{-it}dt = 0$$

for every $\lambda \in \Lambda$. □

Corollary 29.10.3. *For $c \geq 1$, for $c > \frac{7}{8}$ given the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ for $L(s,\chi_4)$ (conditional), and for every $c > 0$, $c \neq \frac{1}{2}$, under GRH for $\beta$, $\{\operatorname{sech}(\lambda x)\}_{\lambda\in\Lambda}$, $\Lambda \subset [1,\infty)$, is complete in $\mathcal{W}_c$ if and only if $\sum_\Lambda 1/\lambda = \infty$.*

Remark 29.10.4 (the barrier $c = \frac{1}{2}$). At $c = \frac{1}{2}$ the line carries the zeros $\frac{1}{2} + i\gamma$ of $\beta$ and $(\mathrm{N}_{1/2})$ fails. The construction survives if $H$ can be made to vanish at each $\gamma$ to the order of the zero, with $H/\beta$ still square integrable. Vanishing of $H$ at a point $\gamma$ is the linear condition $\int \widetilde{\Phi}(\xi)e^{i\gamma\xi}d\xi = 0$, a moment condition and not a zero of $\widetilde{\Phi}$, so the Blaschke product does not produce it; finitely many such conditions could be imposed, but this does not handle all the zeros or the square integrability after division. Whether the zeros of $\beta$ on the critical line can rescue completeness for a set with $\sum 1/\lambda < \infty$ is open (Problem 29.1). Numerically, for $\Lambda = \{n^2\}$ the distance from $e^{-x}$ to the span decreases only slowly at $c = \frac{1}{2}$, as it does at $c = \frac{3}{2}$ where Theorem 29.10.2 applies, and at $N = 24$ it is more than an order of magnitude above the distance for the integers (Table 29.4). This is consistent with a positive limit and so with the critical line not being special for this question; it is not decisive, since incompleteness of a span does not put the particular target $e^{-x}$ outside it.

Remark 29.10.5. The argument is the half plane Blaschke construction of Müntz and Szász, transported to the strip. What the strip adds is the Gaussian factor $e^{-z^2}$, which supplies the Sobolev regularity of the boundary values needed to absorb the polynomial part of $1/K_c$, and what the kernel adds is the division by $\beta$, which is where $(\mathrm{N}_c)$ enters. The same proof gives necessity for the zeta kernel $\varphi$ of (29.6.1), $0 < c < 1$, under the corresponding hypothesis $1/\zeta(c+it) \ll (1+|t|)^A$.

## 29.11 Wiener, Kac and Salem: the $L^1$ picture

The completeness problems of this chapter have an older $L^1$ form, treated in Section 10.1.6 and Section 10.1.7: Wiener's theorem [515] in its homogeneous form [294], and Salem's criterion [468] for the Fermi kernel $1/(e^x+1)$, whose Mellin transform is $\Gamma(s)\eta(s)$ (Proposition 10.1.12, Corollary 10.1.13). In dilation form, for $k$ on $(0,\infty)$ with $\int_0^\infty |k(x)|x^{\sigma-1}dx < \infty$, the span of $\{x^{\sigma-1}k(\lambda x) : \lambda > 0\}$ is dense in $L^1(0,\infty)$ if and only if the Mellin transform of $k$ has no zero on the line $\operatorname{Re} s = \sigma$, and the bounded annihilators that exist when it vanishes at $\sigma + i\gamma$ are the characters $x^{i\gamma}$ (under $x = e^u$ translates become dilates). The beta analogue

for the secant kernel is

$$\overline{\operatorname{span}}\{x^{\sigma-1}\operatorname{sech}(\lambda x) : \lambda > 0\} = L^1(0,\infty) \iff \beta(\sigma+it) \neq 0 \ \text{ for all } t \in \mathbb{R}, \tag{29.11.1}$$

for $\sigma > 0$. This holds for $\sigma \geq 1$, fails for $\sigma = \frac{1}{2}$ (the zero $\frac{1}{2} + 6.02\ldots i$), and for $\frac{1}{2} < \sigma < 1$ it holds for every such $\sigma$ if and only if GRH holds for $\beta$; the mirror lines $0 < \sigma < \frac{1}{2}$ carry the same zero free statements by the functional equation. At $\sigma = 1$ Salem's own kernel is incomplete, because $1 - 2^{1-s}$ vanishes at $1 + 2\pi i k/\log 2$ for nonzero integers $k$ (the zero at $k = 0$ cancels the pole of $\zeta$, leaving $\eta(1) = \log 2 \neq 0$); the secant kernel has no such defect.

Salem used every positive dilation. The Carleman argument removes that.

THEOREM 29.11.1 (Salem's criterion on a divergent set). *Let $\sigma > 0$, let $k$ be $1/(e^x+1)$ or $\operatorname{sech} x$, and let $\Lambda \subset [1,\infty)$ satisfy $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$. Then $\{x^{\sigma-1}k(\lambda x)\}_{\lambda\in\Lambda}$ is complete in $L^1(0,\infty)$ if and only if the Mellin transform of $k$ has no zero on the line $\operatorname{Re} s = \sigma$. In particular, for $\frac{1}{2} < \sigma < 1$, the integer dilates of the Fermi kernel are complete in $L^1$ if and only if $\zeta$ has no zero on $\operatorname{Re} s = \sigma$, and the integer dilates of* sech *if and only if $\beta$ has none.*

*Proof.* If the transform vanishes at $\sigma + i\gamma$ then $x^{i\gamma}$ is a bounded annihilator of every dilate. Conversely let $\varphi \in L^\infty$ annihilate $x^{\sigma-1}k(\lambda x)$ for $\lambda \in \Lambda$ and put $\Phi(w) = \int_0^\infty x^{\sigma-1}\varphi(x)k(wx)\,dx$, $\operatorname{Re} w > 0$. Both kernels extend analytically to $\operatorname{Re} w > 0$, since $e^{wx}+1$ and $\cosh(wx)$ vanish only for $wx \in i\mathbb{R}$. For $w = re^{i\psi}$, $|\psi| < \pi/2$, split the integral at $x = 1/r$. On $x \leq 1/r$ the kernel is bounded by an absolute constant $C_0$, since $|wx| \leq 1$ keeps $wx$ away from the zeros. On $x > 1/r$, $|e^{wx}+1| \geq e^{rx\cos\psi} - 1$ and $|\cosh(wx)| \geq \sinh(rx\cos\psi) \geq \frac{1}{2}(e^{rx\cos\psi}-1)$, so after $y = rx\cos\psi$ and with a further absolute constant,

$$|\Phi(w)| \leq \|\varphi\|_\infty \Big[\frac{C_0}{\sigma} r^{-\sigma} + C_1 (r\cos\psi)^{-\sigma} \int_{\cos\psi}^\infty \frac{y^{\sigma-1}\,dy}{e^y - 1}\Big]$$
$$\leq C_\sigma \|\varphi\|_\infty\, r^{-\sigma} (\cos\psi)^{-\sigma-1} \log\frac{2}{\cos\psi}$$

for $r \geq 1$. So $\Phi(w+1)$ is analytic in $\operatorname{Re} w > 0$, and since $|w+1|/\operatorname{Re}(w+1) \leq |w|/\operatorname{Re} w$ it is bounded by $C(|w|/\operatorname{Re} w)^{\sigma+2}$. If $\Lambda \cap [1,2)$ is infinite, $\Phi$ vanishes on a set with a limit point in $[1,2]$ and $\Phi \equiv 0$ by the identity theorem; otherwise $\Phi(w+1)$ vanishes at $w = \lambda - 1$, $\lambda \in \Lambda$, $\lambda \geq 2$, a set in $[1,\infty)$ with divergent reciprocal sum. Lemma 29.3.1 gives $\Phi \equiv 0$. In particular $\varphi$ annihilates $x^{\sigma-1}k(\lambda x)$ for every $\lambda > 0$, and the dilation form of Wiener's theorem gives $\varphi = 0$. □

Three completeness statements now stand side by side for the same dilates, each governed by which point evaluations of the Mellin transform are continuous on the space.

| space | obstruction | annihilators |
|---|---|---|
| $L^1$, Salem | zeros on $\operatorname{Re} s = \sigma$ | $x^{i\gamma}$ |
| $\mathcal{W}_c$ | none | none |
| $\mathcal{W}_a \cap \mathcal{W}_b$ | zeros in $a < \operatorname{Re} s < b$ | $f \mapsto \widehat{f}(\rho)$ |

In $L^1$ the Mellin transform is continuous on the line and the dual contains the characters, so a zero on the line is seen. In $L^2$ with one weight the transform is only defined almost everywhere on the line, the characters are not square integrable, and nothing off the line is visible: Theorem 29.3.2 holds even at $c = \frac{1}{2}$, where Salem's family is unconditionally incomplete. With two weights the bound (29.5.1) makes evaluation continuous throughout the strip, and the annihilator attached to a zero $\rho$ is the functional $f \mapsto \widehat{f}(\rho)$. Salem's criterion and Theorem 29.5.2 are thus the boundary and interior versions of one statement, the first detecting zeros on a line and the second zeros in a strip. The quasi Riemann hypothesis is a statement of the second kind: if it covers $L(s, \chi_4)$, it makes the two weight criterion hold on $(\frac{7}{8}, 1)$ (conditional) but, with only this half plane, does not resolve zero freeness on individual lines in $\frac{1}{8} \le \operatorname{Re} s \le \frac{7}{8}$.

Remark 29.11.2. Nyman and Beurling's criterion is a fourth regime: the dilates are restricted to $(0, 1)$, which makes the Mellin transforms of the space analytic in a half plane, so every zero in $\operatorname{Re} s > \frac{1}{2}$ is seen in a single $L^2$ norm, by the bound $|\int_0^1 f(x) x^{s-1} dx| \le \|f\|_2 / (2 \operatorname{Re} s - 1)^{1/2}$ for $\operatorname{Re} s > \frac{1}{2}$. The space $\mathcal{W}_a \cap \mathcal{W}_b$ is the half line substitute for that restriction. Together with Section 29.5 this answers the first half of Problem 7.10, in the variable $x$ on $\mathcal{W}_c$; the $L^2$ version in $s$ on the boundary $\operatorname{Re} \sqrt{s} = \frac{1}{2}$ is not addressed.

## 29.12 Explicit approximants in one weight

### 29.12.1 The Gram system

Let $d_N(c)$ be the distance in $\mathcal{W}_c$ from $e^{-x}$ to the span of $\operatorname{sech}(nx)$, $n \le N$. With $G_{nm} = \langle \operatorname{sech}(n\,\cdot), \operatorname{sech}(m\,\cdot) \rangle_c$ and $b_n = \langle e^{-x}, \operatorname{sech}(n\,\cdot) \rangle_c$,

$$d_N(c)^2 = \|e^{-x}\|_c^2 - b^{\mathsf{T}} G^{-1} b, \qquad \|e^{-x}\|_c^2 = \frac{\Gamma(2c)}{2^{2c}}, \tag{29.12.1}$$

and the optimal coefficients are $a = G^{-1} b$ (the matrix $G$ is positive definite, since a nonzero finite combination of distinct dilates cannot vanish). The data have closed forms. Expanding $\operatorname{sech}(nx) = 2 \sum_m \chi_4(m) e^{-mnx}$,

$$b_n = 2\Gamma(2c) \sum_{m \ge 1} \frac{\chi_4(m)}{(1+nm)^{2c}}, \qquad G_{nm} = \frac{2(nm)^{-c}}{\pi} \int_{-\infty}^{\infty} |K_c(t)|^2 \cos\bigl(t \log \tfrac{m}{n}\bigr) dt,$$

the first an alternating series convergent for every $c > 0$, the second a rapidly convergent integral because $|K_c(t)|^2 \asymp |t|^{2c-1} e^{-\pi|t|} |\beta(c + it)|^2$.

### 29.12.2 What the numbers say

Table 29.2 gives $d_N(c)$, the largest optimal coefficient, and the error of the Möbius approximant $S_N$ for comparison, recomputed in multiple precision (`code/lv_gram.py`, `data/lv_gram.json`; two precisions and step sizes agree to $3 \times 10^{-28}$ relative in every $d_N$ and $\max|a_n|$; the source used 40 digits, which is not enough at $N = 32$).

| $c$ | $N$ | $d_N(c)$ | $\max_n \lvert a_n\rvert$ | $\lVert S_N - e^{-x}\rVert_c$ |
|---|---|---|---|---|
| 0.60 | 4 | $8.1 \times 10^{-3}$ | 0.55 | $4.8 \times 10^{-2}$ |
| | 8 | $9.7 \times 10^{-4}$ | 72 | $7.3 \times 10^{-3}$ |
| | 16 | $1.6 \times 10^{-4}$ | $3.3 \times 10^{6}$ | $7.8 \times 10^{-2}$ |
| | 32 | $9.5 \times 10^{-5}$ | $7.3 \times 10^{17}$ | $6.2 \times 10^{-2}$ |
| 0.75 | 4 | $5.8 \times 10^{-3}$ | 0.48 | $4.0 \times 10^{-2}$ |
| | 8 | $5.7 \times 10^{-4}$ | 54 | $5.1 \times 10^{-3}$ |
| | 16 | $1.0 \times 10^{-4}$ | $1.2 \times 10^{6}$ | $4.7 \times 10^{-2}$ |
| | 32 | $6.5 \times 10^{-5}$ | $7.2 \times 10^{16}$ | $3.5 \times 10^{-2}$ |
| 1.00 | 4 | $3.7 \times 10^{-3}$ | 0.49 | $3.0 \times 10^{-2}$ |
| | 8 | $2.8 \times 10^{-4}$ | 36 | $2.9 \times 10^{-3}$ |
| | 16 | $5.5 \times 10^{-5}$ | $4.0 \times 10^{5}$ | $2.1 \times 10^{-2}$ |
| | 32 | $2.8 \times 10^{-5}$ | $5.4 \times 10^{17}$ | $1.4 \times 10^{-2}$ |

Table 29.2: Optimal distance, size of the optimal coefficients and the Möbius error (computed). The source printed $2.4 \times 10^{15}$, $1.5 \times 10^{16}$, $2.7 \times 10^{16}$ at $N = 32$, off by a factor of 5 to 300.

The optimal distance is about one order of magnitude below the Möbius error at $N = 4$ and 8 (ratios 0.10 to 0.17), and two to three orders below it at $N = 16$ and 32 (ratios $1.5 \times 10^{-3}$ to $2.6 \times 10^{-3}$). The optimal coefficients explode, between $10^{16}$ and $10^{18}$ at $N = 32$: the problem is ill conditioned, and beyond $N \approx 16$ the optimal approximant is numerically useless in ordinary precision. The Möbius approximant, whose coefficients are $0, \pm\frac{1}{2}$, is stable but converges slowly and not monotonically, its error tracking the oscillation of $M_\chi(N)$. This is the same trade off as in the Nyman and Beurling problem, where Báez-Duarte [16] observed that the natural Möbius approximants are far from optimal.

### 29.12.3 Why the coefficients explode

The explanation is heuristic. On the line $\operatorname{Re} s = c$ the approximation problem is to make $2\beta D_N - 1$ small in $L^2(|\Gamma(c+it)|^2 dt)$, where $D_N(c+it) = \sum_{n \le N} a_n n^{-c-it}$. The weight $e^{-\pi|t|}$ confines the problem to a segment $|t| \le T$. On a segment the exponentials $e^{-it \log n}$, $n \ge 1$, are complete in $L^2(-T, T)$ for every fixed $T$: if $h$ annihilates them, $E(z) = \int_{-T}^{T} h(t) e^{-itz} dt$ is entire of exponential type at most $T$, so has $O(r)$ zeros in $|z| \le r$, while $E$ vanishes at the $e^r$ points $\log n \le r$ with $n \le e^r$, so $E = 0$. Near linear dependence is quantitative: $\lVert e^{-it \log(n+1)} - e^{-it \log n} \rVert^2_{L^2(-T,T)} \le$

$\frac{2T^3}{3}\log(1+\frac{1}{n})^2$, so the finite Gram matrices have no uniform lower spectral bound. This proves ill conditioning of the restricted systems; it does not determine the size of the optimal coefficient vector for this target, and the effective cutoff of the segment depends on the coefficient growth through $\log\big((1+\|a\|_{N,c})/\varepsilon\big)$, $\|a\|_{N,c}=\sum_{n\le N}|a_n|n^{-c}$. The qualitative picture, completeness together with increasingly close frequencies, is what Table 29.2 shows.

PROPOSITION 29.12.1 (what the weight sees). *Let $c>0$, $T\ge 1$ and $a_n$ be arbitrary coefficients, with $D_N(s)=\sum_{n\le N}a_n n^{-s}$ and $\|a\|_{N,c}=\sum_{n\le N}|a_n|n^{-c}$. Then $\|e^{-x}-\sum_{n\le N}a_n\operatorname{sech}(nx)\|_c^2=\frac{1}{2\pi}\int_{\mathbb{R}}|\Gamma(c+it)|^2|1-2\beta(c+it)D_N(c+it)|^2dt$, and*

$$\Big\|e^{-x}-\sum_{n\le N}a_n\operatorname{sech}(nx)\Big\|_c^2\le\frac{1}{2\pi}\int_{|t|\le T}|\Gamma(c+it)|^2\big|1-2\beta D_N\big|^2dt + C_c\,e^{-\pi T}T^{2c}(1+\|a\|_{N,c})^2.$$

*Where $\beta\neq 0$ the integrand of the first term is $|K_c(t)|^2|2D_N-1/\beta|^2$.*

*Proof.* The identity is (29.2.1) with Lemma 29.1.2; the Mellin transform of $\operatorname{sech}(nx)$ carries the factor 2, which is why $2D_N-1/\beta$ and not $D_N-1/\beta$ appears (for $a_n=\mu(n)\chi_4(n)/2$ the first term is the error of $S_N$, which tends to 0, and the source's form would give the error of an approximant with twice the coefficients). For the tail $|t|>T$ use $|D_N|\le\|a\|_{N,c}$, $|\Gamma(c+it)|^2\ll_c|t|^{2c-1}e^{-\pi|t|}$ and $|\beta(c+it)|\ll_c(1+|t|)^{1/2}$ (functional equation, absolute convergence to the right of 1 and interpolation), so the tail is at most $C(1+\|a\|_{N,c})^2\int_T^\infty t^{2c}e^{-\pi t}dt\ll_c e^{-\pi T}T^{2c}(1+\|a\|_{N,c})^2$ for $T\ge 1$. The first form holds for every $T>0$ if $T^{2c}$ is replaced by $(1+T)^{2c}$ (take all $a_n=0$ and $T\downarrow 0$ to see that some such change is needed). □

A computed check of the identity at $N=8$, $c=0.75$, $a_n=\mu(n)\chi_4(n)/2$: the $x$ space error squared is $2.5846230428\times10^{-5}$ and so is $\frac{1}{2\pi}\int|K_c|^2|2D_N-1/\beta|^2dt$, while the form with $D_N-1/\beta$ gives $7.8\times10^{-2}$.

So an upper bound for $d_N(c)$ follows from approximating $1/\beta$ on a growing segment $|t|\le T_N$ by Dirichlet polynomials with coefficients of size $\|a\|_{N,c}=e^{o(T_N)}$ and a vanishing segment residual (a sufficient route, not an equivalence); the obstruction to an explicit rate is that the coefficient growth is not controlled.

REMARK 29.12.2 (what is known conditionally). If $\beta$ has no zeros in $\operatorname{Re}s>\theta$, then $d_N(c)\le\|S_N-e^{-x}\|_c=O_\varepsilon(N^{\theta-c+\varepsilon})$ for $c\in(\theta,1)$ by Theorem 29.4.2, so $d_N(c)$ decays polynomially; given the quasi Riemann hypothesis for $L(s,\chi_4)$ (conditional) this is a polynomial rate for $c>\frac{7}{8}$. The question of a lower bound is open. In the Nyman and Beurling setting Burnol [39] proves a zero dependent asymptotic lower bound of order $1/\log N$ for the squared distance $\delta_N^2$, of the form $\liminf\delta_N^2\log N\ge\sum_{\operatorname{Re}\rho=1/2}m_\rho^2/|\rho|^2$; it comes from the continuity of point evaluation at the zeros, which the single weight $\mathcal{W}_c$ lacks, and it does not transfer

to this Hilbert space. So either there is no polynomial lower bound in one weight, or it has a different source. We leave this as the refined form of Problem 29.2.

## 29.13 The Meixner family

For $t > 0$ let $f_t(x) = |\Gamma(\frac{t}{2} + \frac{ix}{2})|^2$, $x > 0$, the symmetric Meixner density of order $t$ up to normalisation (Section 6.2.3); $f_1(x) = \pi \operatorname{sech}(\pi x/2)$ and $f_2(x) = (\pi x/2)/\sinh(\pi x/2)$.

Theorem 29.13.1 (Mellin transform of the Meixner density). *For $0 < \operatorname{Re} s < 1$,*

$$M_t(s) = \int_0^\infty x^{s-1} f_t(x)\, dx = \frac{2^s \pi\, \Gamma(t)}{\sin(\pi s/2)}\, Z_t(1-s), \tag{29.13.1}$$

*where $Z_t$ is entire, given by*

$$\begin{aligned} Z_t(w) &= \frac{1}{\Gamma(w)} \int_0^\infty u^{w-1} (2\cosh(u/2))^{-t}\, du \\ &= \sum_{k\ge 0} (-1)^k \binom{t+k-1}{k} \Big(k + \frac{t}{2}\Big)^{-w} \qquad (\operatorname{Re} w > t-1). \end{aligned}$$

*For $t = 1, 2, 3$,*

$$Z_1(w) = 2^w \beta(w), \qquad Z_2(w) = \eta(w-1) = (1 - 2^{2-w})\zeta(w-1),$$
$$Z_3(w) = 2^{w-3}\big(\beta(w) - \beta(w-2)\big).$$

*For integer $t$, $Z_t$ is a linear combination of $\eta(w-j)$, $j$ odd, when $t$ is even, and of $2^{w-2i}\beta(w-2i)$ when $t$ is odd.*

*Proof.* The beta integral with $r = e^u$ gives the Fourier representation

$$|\Gamma(a+iy)|^2 = \Gamma(2a) \int_{\mathbb{R}} e^{iyu} (2\cosh(u/2))^{-2a} du,$$

hence with $a = \frac{t}{2}$, $y = \frac{x}{2}$,

$$f_t(x) = 2\Gamma(t) \int_0^\infty \cos(xu/2)(2\cosh(u/2))^{-t} du.$$

Insert an Abel regulator $e^{-\epsilon x}$ in the Mellin integral, so that Fubini applies: the inner integral is

$$\tfrac{1}{2}\Gamma(s)[(\epsilon - iu/2)^{-s} + (\epsilon + iu/2)^{-s}],$$

bounded by $C_s u^{-\operatorname{Re} s}$ uniformly in $\epsilon > 0$ for $0 < \operatorname{Re} s < 1$, which is integrable against $(2\cosh(u/2))^{-t}$; dominated convergence as $\epsilon \downarrow 0$ gives

$$\begin{aligned} M_t(s) &= 2\Gamma(t)\, 2^s \Gamma(s) \cos(\pi s/2) \int_0^\infty u^{-s} (2\cosh(u/2))^{-t} du \\ &= 2\Gamma(t) 2^s \Gamma(s) \cos(\pi s/2) \Gamma(1-s) Z_t(1-s), \end{aligned}$$

and

$$\Gamma(s)\Gamma(1-s)\cos(\pi s/2) = \pi/(2\sin(\pi s/2))$$

gives (29.13.1). At $u=0$, $(2\cosh(u/2))^{-t}$ has an even analytic expansion $\sum b_j u^{2j}$; subtracting finitely many terms on $(0,1)$ continues $\int_0^\infty u^{w-1}(\cdots)du$ meromorphically with simple poles at $w=-2j$ only, and division by $\Gamma(w)$ cancels them, so $Z_t$ is entire (and vanishes at $-1,-3,\dots$). Absolute termwise integration of $(2\cosh(u/2))^{-t} = \sum_k \binom{-t}{k} e^{-(k+t/2)u}$ gives the series for $\operatorname{Re} w > t$, and analytic continuation gives it for $\operatorname{Re} w > t-1$, where the alternating series converges. For $t=1$, $(k+\frac12)^{-w} = 2^w(2k+1)^{-w}$. For $t=2$, $\binom{k+1}{k}(k+1)^{-w} = (k+1)^{1-w}$. For $t=3$,

$$\binom{k+2}{2} = \tfrac12\big((k+\tfrac32)^2 - \tfrac14\big) \quad\text{and}\quad \sum_k (-1)^k (k+\tfrac32)^{-u} = 2^u(1-\beta(u)),$$

the constant terms cancelling. For integer $t$, $\binom{k+t-1}{t-1}$ is a polynomial in $m = k+\frac{t}{2}$, odd of degree $t-1$ for even $t$ and even for odd $t$, vanishing at

$$m = 1,\dots,\tfrac{t}{2}-1, \quad \text{respectively} \quad m = \tfrac12,\dots,\tfrac{t-2}{2},$$

so the sums may be started at $m=1$, respectively $m=\frac12$, and become shifted $\eta$ or $\beta$ series. □

The formulas for $t=2,3$ are elementary and were not recomputed numerically; the statement for general $t$ is proved above and the explicit combinations are stated for integer $t$ only.

Corollary 29.13.2 (one weight completeness throughout the family). *For every $t>0$ and every $c>0$, $\{f_t(\lambda x)\}_{\lambda\in\Lambda}$ with $\Lambda\subset[1,\infty)$ is complete in $\mathcal{W}_c$ whenever $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$; in particular the integer dilates are complete.*

*Proof.* $f_t$ is bounded at 0 and $f_t(x) \sim 2\pi(x/2)^{t-1}e^{-\pi x/2}$, so $f_t \in \mathcal{W}_c$ for all $c>0$. For $c\ge 1$ the bounds on $M_t$ below refer to its continuation from $0<\operatorname{Re} s<1$, and at even $c$ the pole of $1/\sin(\pi s/2)$ cancels a zero of $Z_t$. On $\operatorname{Re} s = c$, $|1/\sin(\pi s/2)| \asymp e^{-\pi|\tau|/2}$, and $Z_t(1-c-i\tau)$ is polynomially bounded: with $a_w(k) = \frac{\Gamma(k+t)}{\Gamma(t)\Gamma(k+1)}(k+\frac{t}{2})^{-w}$ and $\Delta a(k) = a(k)-a(k+1)$, alternating summation gives $Z_t(w) = \sum_{j<r} 2^{-j-1}\Delta^j a_w(0) + 2^{-r}\sum_{k\ge0}(-1)^k\Delta^r a_w(k)$, and on a fixed line $|\Delta^r a_w(k)| \le C(1+|\operatorname{Im} w|)^r(1+k)^{t-1-\operatorname{Re} w-r}$ by Stirling and Cauchy's estimate, so for $r > t-\operatorname{Re} w$ the last series converges absolutely with a polynomial bound in $|\operatorname{Im} w|$. The function $Z_t$ is not identically zero ($M_t(c)>0$), so $M_t(c+i\tau)\neq 0$ for almost every $\tau$, and $|M_t(c+i\tau)| \le C(1+|\tau|)^A e^{-\pi|\tau|/2}$. These are the only properties of the kernel used in the proof of Theorem 29.3.2. □

The hypothesis $\Lambda\subset[1,\infty)$ cannot be dropped: for $\Lambda = \{2^{-n} : n\ge 1\}$, whose reciprocal sum diverges, take a smooth $h\neq 0$ supported in a short interval

avoiding the zeros of $M_t(c+i\tau)$ together with its translate by $P = 2\pi/\log 2$, put $H(\tau) = h(\tau) - h(\tau - P)$ and $\overline{G} = H/M_t$; then $\int H(\tau)\lambda^{-i\tau}d\tau = 0$ for every $\lambda = 2^{-n}$, and $G$ is a nonzero orthogonal vector. The threshold of Remark 29.3.3 is universal in the Meixner family: $f_t$ is analytic in the open right half plane, with poles on the imaginary axis at $\pm i(t+2k)$, and the rate $\pi/2$ is the half opening of that sector. What changes with $t$ is the arithmetic.

THEOREM 29.13.3 (order two is free). *Let $c_n = \mu(n)$ for odd $n$ and $c_n = 0$ for even $n$. Then for every $x > 0$, with absolute convergence,*

$$e^{-x} = \frac{1}{2}\sum_{n\geq 1} c_n \operatorname{csch}(nx), \qquad xe^{-x} = \frac{1}{2}\sum_{n\geq 1} \frac{c_n}{n}\,(nx)\operatorname{csch}(nx), \tag{29.13.2}$$

*and $\sum_n (c_n/n)n^{-s} = 1/\big((1-2^{-s-1})\zeta(s+1)\big)$ converges absolutely for $\operatorname{Re} s > 0$. Consequently the series for $xe^{-x}$ converges absolutely in every $\mathcal{W}_c$, $c > 0$, and $\{(nx)\operatorname{csch}(nx)\}_{n\in\mathbb{N}}$ is complete in every $\mathcal{W}_c$ and in every $\mathcal{W}_a \cap \mathcal{W}_b$, $0 < a < b$, unconditionally.*

*Proof.* (The source writes $c_n = \sum_{2^j|n} \mu(n/2^j)$; for $n = 2^k m$ with $m$ odd and $k \geq 1$ the two nonzero terms $\mu(m)$ and $\mu(2m) = -\mu(m)$ cancel, which gives the form above.) Since $\operatorname{csch}(nx) = 2\sum_{m \text{ odd}} e^{-mnx}$, absolute convergence at fixed $x > 0$ permits grouping by $mn = N$: $\frac{1}{2}\sum_n c_n \operatorname{csch}(nx) = \sum_{N \text{ odd}} e^{-Nx}\sum_{n|N}\mu(n) = e^{-x}$. Multiply by $x$ for the second identity. The Mellin transform of $(nx)\operatorname{csch}(nx)$ is $2n^{-s}\Gamma(s+1)(1-2^{-s-1})\zeta(s+1)$ for $\operatorname{Re} s > 0$. Put $h(x) = x\operatorname{csch} x \in \mathcal{W}_c$; then $\sum |c_n/n|\,\|h(n\cdot)\|_c \leq \|h\|_c \sum n^{-c-1} < \infty$, so the series converges absolutely in each $\mathcal{W}_c$, and simultaneously in $\mathcal{W}_a \cap \mathcal{W}_b$, to its pointwise value $xe^{-x}$. The span is invariant under integer dilations, so its closure contains every $nxe^{-nx}$. These are complete: if $g$ is orthogonal to them in $\mathcal{W}_a \cap \mathcal{W}_b$ (or $\mathcal{W}_c$) and $w(x) = x^{2a-1} + x^{2b-1}$ (respectively $x^{2c-1}$), the finite complex measure $d\nu = xe^{-x}\bar{g}\,w\,dx$ has $\int e^{-(n-1)x}d\nu = 0$ for $n \geq 1$; pushing forward by $y = e^{-x}$ and using the density of polynomials in $C[0,1]$ gives $\nu = 0$, hence $g = 0$. □

The function $M_2(s) = 2(2/\pi)^s\Gamma(s+1)(1-2^{-s-1})\zeta(s+1)$ has no zeros in $\operatorname{Re} s > 0$; the zeros of $Z_2(1-s) = \eta(-s)$ at the positive even integers cancel the apparent poles of $1/\sin(\pi s/2)$, and the nontrivial zeros of $\zeta$ mapped by $s \mapsto -s$ lie to the left of the admissible half plane (the critical line of $\eta(-s)$ is $\operatorname{Re} s = -\frac{1}{2}$). So there is nothing for the two weight criterion to detect, and Theorem 29.13.3 is the Meixner analogue of the classical range $c \geq 1$ for the secant, now covering every weight. For integer $t \geq 3$ the zero set of $Z_t(1-s)$ in the strip is that of a combination such as $\beta(1-s) - \beta(-1-s)$ or $\eta(-s) - \eta(-2-s)$; it is not governed by a single $L$ function, and for noninteger $t$ the series $Z_t$ remains the object to study.

## 29.14 The Thorin form of the two weight criterion

Let $\Xi$ denote $\xi$ or $\xi(\cdot,\chi_4)$, with $\xi(s,\chi_4) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})\beta(s)$ as in Section 8.5.7. Both are entire of order one, real and positive on the real axis, satisfy $\Xi(s) = \Xi(1-s)$, and have the Hadamard product $\Xi(s) = \Xi(0)\prod_\rho(1-s/\rho)$ over the nontrivial zeros, with $\rho$ and $\bar\rho$ paired (Davenport [104]); grouping the zeros into reflected quartets makes the product invariant under $s \mapsto 1-s$, so the functional equation forces the exponential factor of Hadamard's theorem to be trivial. For $\theta \geq \frac12$ put

$$F_\theta(s) = \frac{\Xi(\theta)}{\Xi(\theta+\sqrt{s})}, \qquad s \in \mathbb{C}\setminus(-\infty,0], \tag{29.14.1}$$

with the principal square root. A probability law on $[0,\infty)$ is a generalised gamma convolution (GGC) if its Laplace transform is $\exp(-as - \int_0^\infty \log(1+s/u)U(du))$ with $a \geq 0$ and a Thorin measure $U$, a positive Radon measure on $(0,\infty)$ with $\int_{(0,1)} |\log u| U(du) < \infty$ and $\int_1^\infty u^{-1}U(du) < \infty$ (local finiteness is part of the definition: the two integrals alone allow infinite mass accumulating at 1); the Thorin measure is unique [56].

For $\Xi = \xi$ and $\theta \geq 1$, and more generally when $\theta > \operatorname{Re}\rho$ for every zero, the GGC statement and the Thorin density are Theorem 1.2.2 of Chapter 2 (with the density (1.2.6) and its Poisson structure, Proposition 1.2.3); the strip form is Theorem 10.1.2 and Theorem 10.1.3, and Chapter 9 computes the Thorin measure at the basepoint one. What is new here is the case of $\xi(s,\chi_4)$, the description of one conjugate pair of zeros as the law of $|\gamma + aC|$, and the link with the two weight criterion of Theorem 29.5.2.

Lemma 29.14.1 (one pair of zeros). *Let $a \geq 0$ and $\gamma > 0$. For $s \in \mathbb{C}\setminus(-\infty,0]$,*

$$\log\frac{(a+\sqrt{s})^2+\gamma^2}{a^2+\gamma^2} = \int_0^\infty \log\Big(1+\frac{s}{u}\Big)Q_{a,\gamma}(du),$$

*where $Q_{0,\gamma} = \delta_{\gamma^2}$ and, for $a > 0$, $Q_{a,\gamma}$ is the probability measure on $(0,\infty)$ with density $q_{a,\gamma}(u) = \frac{1}{2\pi\sqrt{u}}\big[\frac{a}{a^2+(\sqrt{u}-\gamma)^2} + \frac{a}{a^2+(\sqrt{u}+\gamma)^2}\big]$. In the variable $v = \sqrt{u}$, $Q_{a,\gamma}$ is the law of $|\gamma + aC|$ with $C$ standard Cauchy.*

*Proof.* Both sides vanish at $s = 0$, so it suffices to compare $s$ derivatives for $s > 0$ and continue analytically (the positivity of the measure and the bounds below justify differentiation under the integral). The left side has derivative $(a+\sqrt{s})/(\sqrt{s}((a+\sqrt{s})^2+\gamma^2))$. With $u = v^2$ the right side has derivative $\frac{1}{\pi}\int_{\mathbb{R}} \frac{1}{v^2+s}\frac{a}{a^2+(v-\gamma)^2}dv$, the Poisson integral of $v \mapsto 1/(v^2+s)$ at the point $\gamma + ia$ of the upper half plane. Since $1/(v^2+s) = \operatorname{Re}[1/(\sqrt{s}(\sqrt{s}-iv))]$ and $z \mapsto 1/(\sqrt{s}(\sqrt{s}-iz))$ is bounded and analytic in $\operatorname{Im} z > 0$, the Poisson integral equals $\operatorname{Re}[1/(\sqrt{s}(\sqrt{s}+a-i\gamma))]$, which is the left derivative. The mass is 1 because the two Cauchy densities integrate over $v > 0$ to the integral of one Cauchy density over $\mathbb{R}$; the case $a = 0$ is the limit. The condition $a \geq 0$ is indispensable. $\square$

THEOREM 29.14.2 (Thorin form). *Let $\theta \geq \frac{1}{2}$. The following are equivalent.*

*(i)* $\Xi$ *has no zeros in* $\operatorname{Re} s > \theta$.

*(ii)* $F_\theta$ *is the Laplace transform of a generalised gamma convolution.*

*(iii)* $F_\theta$ *is analytic in* $\mathbb{C} \setminus (-\infty, 0]$.

*When they hold, the Thorin measure of* $F_\theta$ *is*

$$U_\theta = \sum_{\rho,\ \operatorname{Im}\rho>0} m_\rho \, Q_{\theta-\operatorname{Re}\rho,\ \operatorname{Im}\rho}, \tag{29.14.2}$$

*over the distinct zeros in the upper half plane with multiplicity* $m_\rho$*, and* $F_\theta(s) = \exp(-\int_0^\infty \log(1+s/u)U_\theta(du))$ *with no linear term. For* $\Xi = \xi(\cdot, \chi_4)$*, (i) is equivalent to two weight completeness of the secant dilates on* $(\theta, 1)$ *(Theorem 29.5.2, for* $\theta < 1$*); for* $\Xi = \xi$*, to two weight completeness of the dilates of* $\varphi$ *(Theorem 29.6.1).*

*Proof.* (ii) implies (iii) because a GGC Laplace transform is analytic in $\mathbb{C}\setminus(-\infty, 0]$. (iii) implies (i): a zero $\rho$ with $\operatorname{Re}\rho > \theta$ makes $F_\theta$ have a pole at $s = (\rho - \theta)^2$, which is not on $(-\infty, 0]$ since $\operatorname{Re}(\rho-\theta) > 0$ (and $\Xi(\theta) > 0$, the zero is nonreal and the square root map is locally invertible there). (i) implies (ii): by the Hadamard product and the pairing, for $s > 0$,

$$\log \frac{\Xi(\theta+\sqrt{s})}{\Xi(\theta)} = \sum_{\operatorname{Im}\rho>0} m_\rho \log \frac{(a_\rho+\sqrt{s})^2+\gamma_\rho^2}{a_\rho^2+\gamma_\rho^2},$$
$$a_\rho = \theta - \operatorname{Re}\rho \in [0, \theta],\ \gamma_\rho = \operatorname{Im}\rho,$$

with $a_\rho \geq 0$ by (i); the summands are nonnegative and $O_{\theta,s}(\gamma_\rho^{-2})$, so the series converges normally. Lemma 29.14.1 turns each term into $\int \log(1+s/u)Q_{a_\rho,\gamma_\rho}(du)$, and by Tonelli the sum of the positive measures is a Thorin measure: for large $\gamma$, $0 \leq a \leq \theta$ and $0 < u < 1$, $q_{a,\gamma}(u) \leq C_\theta \gamma^{-2} u^{-1/2}$, which makes the small $u$ integral summable (finitely many low zeros cost nothing), and $\int_1^\infty u^{-1} Q_{a,\gamma}(du)$ is the integral of $v^{-2}$ against the folded Cauchy law over $v \geq 1$, which is $\ll_\theta \gamma^{-2}$ (on $|v-\gamma| < \gamma/2$ use $v^{-2} \leq 4\gamma^{-2}$ and mass at most one, elsewhere the density is at most $C_\theta\gamma^{-2}$ and $\int_1^\infty v^{-2}dv < \infty$; for $a = 0$ the atom at $\gamma^2$ gives the same). Hence $U_\theta$ is a locally finite Thorin measure, $F_\theta(s) = \exp(-\int \log(1+s/u)U_\theta(du))$ for $s > 0$, and both sides continue to $\mathbb{C} \setminus (-\infty, 0]$. The absence of a linear term is the absence of an exponential factor in the Hadamard product. The completeness statements are Theorems 29.5.2 and 29.6.1. □

COROLLARY 29.14.3 (Thorin representations above the zero free abscissa). *For every* $\theta \geq 1$*,* $\xi(\theta)/\xi(\theta+\sqrt{s})$ *and* $\xi(\theta, \chi_4)/\xi(\theta+\sqrt{s}, \chi_4)$ *are Laplace transforms of GGCs with the absolutely continuous Thorin measures* (29.14.2)*, since the zeros have* $\operatorname{Re}\rho < 1$*. Conditionally on the zero free half plane* $\operatorname{Re} s > \frac{7}{8}$ *of the quasi*

*Riemann hypothesis (for $\beta$, on the theorem covering $L(s,\chi_4)$), the GGC representation holds for every $\theta \geq \frac{7}{8}$, and the Thorin measure is absolutely continuous for $\theta > \frac{7}{8}$, and at $\theta = \frac{7}{8}$ if no zero lies on $\operatorname{Re} s = \frac{7}{8}$; a zero on that line contributes an atom $m_\rho \delta_{\gamma_\rho^2}$. For $\theta = \frac{1}{2}$ the statement for $\xi$ is the Riemann hypothesis and the Thorin measure is then $\sum_{\gamma>0} m_\gamma \delta_{\gamma^2}$.*

The last sentence is the reciprocal Xi representation around which the book is organised; Theorem 29.14.2 embeds it in a one parameter family and says what the family looks like above the zero free abscissa.

THEOREM 29.14.4 (the Cauchy semigroup of Thorin measures; for $\xi$ the Cauchy flow of Proposition 1.2.3 and [424, Chapter 8]). *Suppose $\Xi$ has no zeros in $\operatorname{Re} s > \theta_0$. For $\theta \geq \theta_0$ let $\widetilde{U}_\theta = \frac{1}{2}\sum_\rho m_\rho \operatorname{Cauchy}(\operatorname{Im}\rho, \theta - \operatorname{Re}\rho)$ be the symmetric measure on $\mathbb{R}$ (in the variable $v$), summed over the distinct zeros of both half planes, with $\operatorname{Cauchy}(\gamma, 0) = \delta_\gamma$. Then $U_\theta$ is the image of $\widetilde{U}_\theta$ under $v \mapsto v^2$, and for $\theta_0 \leq \theta < \theta'$,*

$$\widetilde{U}_{\theta'} = \widetilde{U}_\theta * \operatorname{Cauchy}(0, \theta' - \theta).$$

*In particular, if all zeros lie on $\operatorname{Re} s = \theta_0$ (which for the completed functions forces $\theta_0 = \frac{1}{2}$), then $\widetilde{U}_\theta$ is the Poisson smoothing at width $\theta - \theta_0$ of $\frac{1}{2}\sum_\rho m_\rho \delta_{\operatorname{Im}\rho}$, and $\widetilde{U}_\theta \to \frac{1}{2}\sum_\rho m_\rho \delta_{\operatorname{Im}\rho}$ vaguely as $\theta \downarrow \theta_0$.*

*Proof.* Each of $\operatorname{Cauchy}(\gamma, a)$ and $\operatorname{Cauchy}(-\gamma, a)$ is carried by $v \mapsto v^2$ to $Q_{a,\gamma}$ (Lemma 29.14.1), so the factor $\frac{1}{2}$ makes the image equal to (29.14.2); without it the image of the sum over all zeros is $2U_\theta$. The Cauchy laws form a convolution semigroup in the scale, $\operatorname{Cauchy}(\gamma, a) * \operatorname{Cauchy}(0, b) = \operatorname{Cauchy}(\gamma, a + b)$, and $\theta' - \operatorname{Re}\rho = (\theta - \operatorname{Re}\rho) + (\theta' - \theta)$; Tonelli permits summing the positive measures, which are locally finite by the tail estimate of Theorem 29.14.2. For vague convergence, a compactly supported continuous test function sees each Cauchy term converge to its atom, and the contribution of large $|\gamma|$ is bounded by $C\sum \gamma^{-2}$ uniformly for widths in a bounded interval. The measures have infinite total mass, so weak convergence is not the notion. □

Three remarks on what this gives the programme.

First, a statement that is checkable. For $\theta \geq 1$ the Thorin measure of $\xi(\theta)/\xi(\theta + \sqrt{s})$ is the Cauchy smoothing at width $\theta - \frac{1}{2}$ of the zero measure, plus a correction from any zeros off the critical line. For $\theta = 1$ and the first 1000 zeros, taken on the critical line, the sum $\sum_\gamma \int \log(1 + s/u) Q_{1/2,\gamma}(du)$ plus a smooth tail from the zero density reproduces $\log \xi(1 + \sqrt{s})/\xi(1)$ (Table 29.3, computed, `data/lv_sech_tables.json`). The tail is a model and not a bound, so the totals are a computation and not a certified digit guarantee. A Thorin measure obtained by any other computational route must agree with this one; the comparison at level $\theta \geq 1$ is a test of the computational machinery with no hypothesis attached.

Second, the right metric. Above the zero free abscissa the Thorin measure is absolutely continuous and the RH measure $\sum \delta_{\gamma^2}$ is atomic, and both have

| $s$ | $\log \xi(1+\sqrt{s})/\xi(1)$ | zero sum | tail (model) | total |
|---|---|---|---|---|
| 1 | 0.0461176 | 0.044678 | 0.00144 | 0.046118 |
| 4 | 0.137917 | 0.133599 | 0.00432 | 0.137917 |
| 16 | 0.454853 | 0.440461 | 0.0144 | 0.454854 |

Table 29.3: The Thorin check at $\theta = 1$ (computed). The first column is the closed forms $\log(\pi/3)$, $\log(3\zeta(3)/\pi)$, $\log(15\zeta(5)/\pi^2)$ evaluated; the source's digits differed in the last place.

infinite mass, so their total variation distance is infinite and not 2; the meaningful comparison is vague convergence (Theorem 29.14.4), or a metric after a weight with finite integral. The claim that a zero to the right of $\theta$ would show up as a negative bump of mass $-1$ at $\gamma$, obtained by putting $a < 0$ in $Q_{a,\gamma}$, is not true as a characterisation and is not used: formula (29.14.2) was proved only for $a \geq 0$. For $a = -h < 0$ one has $L_{-h}(s) = -L_h(s) + \log \frac{(s+h^2+\gamma^2)^2 - 4h^2 s}{(h^2+\gamma^2)^2}$, where $L_a(s) = \log \frac{(a+\sqrt{s})^2+\gamma^2}{a^2+\gamma^2}$; a negative Cauchy density accounts for $-L_h$ only, and the last term has its zeros at $(h \pm i\gamma)^2$, off the cut, which are precisely the singularities that obstruct the GGC representation. At $\theta = \frac{1}{2}$ an off critical reflected pair with widths $-h$ and $+h$ has formally cancelling densities, while $F_{1/2}$ still has off cut poles; so positivity of the formal signed density does not characterise RH (compare Remark 1.2.4, where a positive density with a zero to its right is exhibited). What is proved is Theorem 29.14.2: RH is equivalent to analyticity of $F_{1/2}$ on $\mathbb{C} \setminus (-\infty, 0]$, or to its GGC property, and above a zero free boundary the positive Cauchy mixture is the genuine Thorin measure. No theorem here identifies the first failure of the GGC property with the appearance of a negative part.

Third, the flow. Above the zero free abscissa $\theta \mapsto \widetilde{U}_\theta$ is the Poisson extension of the already smoothed boundary measure $\widetilde{U}_{\theta_0}$, a Cauchy semigroup running forward, and it is the Poisson extension of an atomic zero measure on one line only when all real parts agree. The two weight criterion of Theorem 29.5.2 is, through Theorem 29.14.2, the statement that $F_\theta$ is a GGC at every level above $\theta$. Forward Cauchy convolution is a positive operation; the inverse is not automatically one, so running the semigroup downward from a computed $\widetilde{U}_1$ is an inversion problem, not a theorem (Problem 29.5).

## 29.15 Numerical checks of the sharpness experiment

The sharpness experiment compares $\Lambda = \{n^2\}$ with $\Lambda = \{n\}$ at $c = \frac{3}{2}$ and at $c = \frac{1}{2}$ (computed with the Gram system of Section 29.12, `code/lv_gram.py`).

For the squares the distance flattens, consistent with Theorem 29.10.2 at $c = \frac{3}{2}$ (the theorem gives a nonzero orthogonal vector, not that $e^{-x}$ lies outside the closed span), and it behaves similarly at $c = \frac{1}{2}$; for the integers it keeps falling. The ratio

| | $c = 3/2$ | | $c = 1/2$ | |
|---|---|---|---|---|
| $N$ | squares | integers | squares | integers |
| 4 | $9.9 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.2 \times 10^{-2}$ | $1.0 \times 10^{-2}$ |
| 8 | $6.5 \times 10^{-3}$ | $1.3 \times 10^{-4}$ | $7.2 \times 10^{-3}$ | $1.4 \times 10^{-3}$ |
| 16 | $5.2 \times 10^{-3}$ | $1.6 \times 10^{-5}$ | $4.9 \times 10^{-3}$ | $2.4 \times 10^{-4}$ |
| 24 | $4.8 \times 10^{-3}$ | $9.3 \times 10^{-6}$ | $4.2 \times 10^{-3}$ | $1.2 \times 10^{-4}$ |

Table 29.4: Distance from $e^{-x}$ to the span of the first $N$ dilates, squares against integers (computed).

of squares to integers is 5, 49, 322, 519 at $c = \frac{3}{2}$ and 1.1, 5.0, 20, 35 at $c = \frac{1}{2}$, so at $c = \frac{1}{2}$ it exceeds one order of magnitude from $N = 16$ on and does not reach two.

## 29.16 Open problems

Problem 29.1 (sharpness at $c = \frac{1}{2}$). Does every $\Lambda \subset [1, \infty)$ with $\sum 1/\lambda < \infty$ admit a nonzero annihilator of $\{\operatorname{sech}(\lambda x)\}$ in $\mathcal{W}_{1/2}$, that is, can the zeros of $\beta$ on the critical line be used to build, from a convergent set, functions that the Blaschke construction cannot see? A separate target question is whether $e^{-x}$ has positive distance from the closed span of $\{\operatorname{sech}(n^2 x)\}$ in $\mathcal{W}_{1/2}$. The numerical evidence for the squares (Table 29.4) is consistent with a positive limit and proves nothing.

Problem 29.2 (rate and lower bound for $d_N(c)$). Find an unconditional rate for $d_N(c)$, $\frac{1}{2} < c < 1$, or a lower bound. By Proposition 29.12.1 a sufficient route to an upper bound is approximating $1/\beta$ on the segment $|t| \le T_N$ by Dirichlet polynomials with coefficients at most $e^{o(T_N)}$.

Problem 29.3 (the strip converse). Does a zero free strip $a < \operatorname{Re} s < b$ alone give completeness of $\{\operatorname{sech}(nx)\}$ in $\mathcal{W}_a \cap \mathcal{W}_b$? Proposition 29.5.1 is the converse direction, and Theorem 29.5.2 needs the whole half plane.

Problem 29.4 (Meixner order $t \ge 3$). Describe the zero sets of $Z_t(1 - s)$ in the strip for integer $t \ge 3$, beginning with $\beta(1 - s) - \beta(-1 - s)$. Are there infinitely many zeros in $0 < \operatorname{Re} s < 1$, and is there a two weight criterion with an arithmetic meaning?

Problem 29.5 (below the zero free abscissa). Is there a direct characterisation, not passing through the zeros, of which positive measures $\widetilde{U}_1$ arise at level $\theta = 1$ from an $L$ function? Given $\widetilde{U}_1$, does the level one Thorin measure together with an explicit analytic reconstruction determine the first abscissa at which $F_\theta$ develops an off cut pole or loses the GGC property? The level one function $\Xi(1 + \sqrt{s})$ is available from the absolutely convergent Euler product, but recovering $\widetilde{U}_1$ from it is an ill posed inversion.

# Chapter 30

# The Thorin measure at level one and the backward Cauchy flow

Chapter 29 shows that for $\theta \geq 1$ the function $\xi(\theta)/\xi(\theta+\sqrt{s})$ is, unconditionally, the Laplace transform of a generalised gamma convolution, that its Thorin measure is the Cauchy smoothing of the zero measure at width $\theta - \operatorname{Re}\rho$, and that the family $\theta \mapsto U_\theta$ is a Cauchy semigroup above the zero free abscissa (Theorems 29.14.2 and 29.14.4). It asks whether the semigroup can be run backward from a level one measure computed without the zeros (Problem 29.5). This chapter does three things with that question.

First, it computes the level one measure in closed form. Its density is $\pi^{-1}\operatorname{Re}(\xi'/\xi)(1+iv)$, a smooth gamma part minus the prime sum $\pi^{-1}\sum_{n\geq 2}\Lambda(n)n^{-1}\cos(v\log n)$, read as a compensated limit because the series itself diverges. The Fourier transform of the measure is, apart from the transform of the smooth part and a constant 1 from the pole of $\zeta$, a sum of point masses $-\Lambda(n)/n$ at $\omega = \pm\log n$: the level one measure carries the primes in its spectrum and the zeros as the centres of its Cauchy components, and the identity between the two descriptions is a Weil type explicit formula for one analytic test function evaluated at the zeros.

Second, it studies the backward flow. On the Fourier side the semigroup multiplies by $e^{-(\theta-\theta')|\omega|}$, so going from level 1 to level $\theta'$ multiplies by $e^{(1-\theta')|\omega|}$. The band limited backward flow converges to the level $\theta'$ measure if and only if $\zeta$ has no zeros to the right of $\theta'$; a zero to the right makes the limit fail to be a measure $\mu$ with $\int |\mu|(dv)/(1+v^2) < \infty$. Numerically, from samples of the level one density on the line $\operatorname{Re} s = 1$ only, the flow places a peak within 0.035 of each of the 109 zeros tested, in $100 < v < 300$.

Third, it treats the backward flow as a positive linear inverse problem of Vardi and Lee type with the Cauchy kernel as the blur. The EM algorithm of Shepp and Vardi, an information geometric analogue of von Neumann's alternating projection (Csiszár and Tusnády), gives a positivity preserving backward flow with a diagnostic, the total Kullback Leibler residual; its data augmentation Gibbs

sampler is its Monte Carlo version; and generative Bayesian computation amortises a posterior under a GUE prior, trained on simulations. The EM inversion locates the 109 zeros without ringing, and a redesigned injection test separates an injected off line quartet from a control; the GBC implementation, rerun here with a corrected forward model, counts the zeros in unit bins almost exactly.

The chapter ends by asking whether any of this is a path to the Riemann hypothesis. It is not. It is a dictionary in which RH reads as the statement that the backward flow runs all the way to $\theta = \frac{1}{2}$, and in which the exponential amplification $e^{(1-\theta')|\omega|}$ is the classical error term $\psi(x) - x = O(x^{\theta'+\varepsilon})$ in Fourier clothing.

The chapter develops a source chapter by Nick Polson [240], together with the computations done for this book (the scripts `code/lo_common.py`, `code/lo_flow.py`, `code/lo_em.py` and `code/lo_gbc.py`, their outputs `data/lo_*.json` and logs, listed in `code/README.md`; the scripts of the source were not used, and the sampler of $u_1$ and the EM iteration are those of Chapter 31, `code/cx_em.py`). Every number printed below is one of these recomputations unless it is marked as the author's or the review's, and where the recomputation differed from the source the recomputed value is printed. Claims of the source that could not be reproduced are dropped or restated as such in the text.

Notation. The book's Re, $\mathbb{R}$ and the cut plane $\mathbb{C} \setminus (-\infty, 0]$ are used; $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, a zero is $\rho = \sigma_\rho + i\gamma_\rho$, and $P_c(v) = (c/\pi)/(c^2+v^2)$ is the Cauchy kernel of width $c > 0$, with $P_0 = \delta_0$ (the Poisson kernel of Section 1.2, with the same index). The Thorin measure of this chapter is written $\mathfrak{U}_\theta$ (Section 30.1). It carries no factor $\frac{1}{2}$ and is therefore twice the symmetric measure $\widetilde{U}_\theta$ of Theorem 29.14.4; its density is the density $\rho_{\theta-1/2}$ of Chapter 19 (Theorem 19.2.2). The digamma function is written $\Gamma'/\Gamma$, because $\psi(x) = \sum_{n \le x} \Lambda(n)$ is Chebyshev's function here; the grid spacing of the numerics is $h$ and the test function of the explicit formula is $\varphi$, the Fourier transform being $\widehat{f}(\omega) = \int e^{-i\omega v} f(v)\, dv$.

Status words follow the rest of the book. A statement is proved when a proof or a pointer is given, computed when a script of the book produced it, conditional, heuristic or open as stated; nothing in this chapter is certified in ball arithmetic except the sampler of $u_1$, which is evaluated with python-flint balls at 128 bits and checked at 192. The results are these.

1. The level one Thorin measure of $\xi$ has the explicit real analytic density (30.1.3), for every real $v$ including $v = 0$, where the prime sum is a compensated limit with the value $u_1(0) = (2 + \gamma_E - \log 4\pi)/(2\pi)$ (Theorem 30.1.1, proved; for $v \neq 0$ Propositions 9.1.2 and 9.1.3). Its positivity is automatic from the Euler product; the nonvanishing of $\zeta$ on $\operatorname{Re} s = 1$ is equivalent to the absence of atoms (Section 30.6, proved).

2. The Fourier transform of the measure is the sum of prime atoms, the trans-

form of the smooth part and the constant 1, and the identity of the two descriptions is (30.2.2), an explicit formula for one analytic test function (Proposition 30.2.1, proved). The spectral peaks lie at $\log n$ to $10^{-4}$ (Table 30.1, computed).

3. The band limited backward flow converges at level $\theta'$ if and only if $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$, and RH is the convergence at every $\theta' > \frac{1}{2}$ (Theorem 30.3.1, proved). The flow recovers 109 zeros to 0.035 at $\theta' = \frac{3}{4}$ and to 0.027 at $\theta' = \frac{1}{2}$; the sharp cutoff rings and the Fejér window does not (Section 30.3.1, computed).

4. The existence of a positive solution of the deconvolution $P_{1-\theta'} * x = u_1$ is the zero free half plane $\operatorname{Re} s > \theta'$, and the total Kullback Leibler residual is bounded away from zero as the grid is refined exactly when a zero lies to the right of $\theta'$ (Proposition 30.4.1, proved, with a counterexample to the residual per unit length). The EM runs and a quartet injection test with controls are computed (Section 30.4.4).

5. The data augmentation sampler needs an exposure and pseudo counts; it is a method and carries no result. Generative Bayesian computation under a GUE prior counts the zeros in unit bins with 98 per cent exact bins on the 200 test bins, for a corrected forward model (Section 30.5, computed; the formulation is heuristic).

6. None of this is a path to RH (Section 30.6); four open problems close the chapter (Section 30.7).

## 30.1 The level one measure

Throughout, $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, zeros are $\rho = \sigma_\rho + i\gamma_\rho$ counted with multiplicity, and $\theta_0 = \sup \sigma_\rho \le 1$. For every $\theta \ge \theta_0$, that is for every $\theta$ such that $\zeta$ has no zeros in $\operatorname{Re} s > \theta$ (in particular for every $\theta \ge 1$, unconditionally), let

$$\mathfrak{U}_\theta = \sum_\rho \operatorname{Cauchy}(\gamma_\rho, \theta - \sigma_\rho), \qquad \operatorname{Cauchy}(\gamma, 0) = \delta_\gamma, \tag{30.1.1}$$

the sum over the zeros of both half planes, a positive measure on $\mathbb{R}$ in the variable $v$ with $\int \mathfrak{U}_\theta(dv)/(1+v^2) < \infty$. It is twice the symmetric measure $\widetilde{U}_\theta$ of Theorem 29.14.4, and the image of its restriction to $v > 0$ under $v \mapsto v^2$ is the Thorin measure $U_\theta$, each conjugate pair of zeros contributing one term.

Theorem 30.1.1 (density of the Thorin measure; at $\theta = 1$, $v \ne 0$, Propositions 9.1.2 and 9.1.3). *If $\theta > \sigma_\rho$ for every $\rho$ (for instance if $\theta > \theta_0$, and for every $\theta \ge 1$ because $\zeta(1+it) \ne 0$), the measure $\mathfrak{U}_\theta$ is absolutely continuous with real*

*analytic density*

$$u_\theta(v) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\theta+iv) = \frac{1}{\pi}\,\partial_\sigma \log|\xi(\sigma+iv)|\Big|_{\sigma=\theta}, \tag{30.1.2}$$

*and* $\mathfrak{U}_\theta((0,V]) = \frac{1}{\pi}\arg\xi(\theta+iV)$ *for the branch with* $\arg\xi(\theta)=0$. *At* $\theta=1$,

$$\begin{aligned} u_1(v) = \frac{1}{\pi}\Big[&\frac{1}{1+v^2} - \frac{\log\pi}{2} + \frac{1}{2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac{1+iv}{2}\Big) \\ &- \lim_{X\to\infty}\Big(\sum_{n\le X}\frac{\Lambda(n)}{n}\cos(v\log n) - \frac{\sin(v\log X)}{v}\Big)\Big] \end{aligned} \tag{30.1.3}$$

*for every real* $v$, *where* $\sin(v\log X)/v$ *means* $\log X$ *at* $v=0$. *The limit exists, locally uniformly in* $v$, *and equals* $\operatorname{Re}C(1+iv)$ *with* $C(s) = -\zeta'(s)/\zeta(s) - 1/(s-1)$: *it is* $-\operatorname{Re}(\zeta'/\zeta)(1+iv)$ *for* $v\neq 0$ *and minus Euler's constant at* $v=0$, *while the cosine series itself has no limit for any real* $v$ *(compare Proposition 9.1.3). For* $\theta>1$,

$$\begin{aligned} u_\theta(v) = \frac{1}{\pi}\Big[&\frac{\theta}{\theta^2+v^2} + \frac{\theta-1}{(\theta-1)^2+v^2} - \frac{\log\pi}{2} \\ &+ \frac{1}{2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac{\theta+iv}{2}\Big) - \sum_{n\ge 2}\frac{\Lambda(n)}{n^\theta}\cos(v\log n)\Big], \end{aligned}$$

*the prime sum converging absolutely.*

*Proof.* Each term of (30.1.1) is a probability measure, and for $0\le c\le\theta$

$$\int \frac{P_c(v-\gamma)}{1+v^2}\,dv = \pi P_{1+c}(\gamma) = \frac{1+c}{(1+c)^2+\gamma^2} \le \frac{1+\theta}{1+\gamma^2};$$

since $\sum_\rho (1+\gamma_\rho^2)^{-1} < \infty$, the sum converges to a positive measure with $\int \mathfrak{U}_\theta(dv)/(1+v^2) < \infty$. If every width $\theta-\sigma_\rho$ is positive, fix $R>0$. The finitely many zeros with $|\gamma_\rho|\le R+2\theta+2$ give Cauchy densities of positive width, which are real analytic; for the others the densities, continued to $|\operatorname{Re} z|\le R$, $|\operatorname{Im} z|\le\frac12$, are holomorphic and bounded by $2\theta/(\pi(|\gamma_\rho|-R)^2)$, so their sum is holomorphic there. Hence $\mathfrak{U}_\theta$ has a real analytic density.

By the Hadamard product with conjugate pairing, $(\xi'/\xi)(s) = \sum_\rho 1/(s-\rho)$, so

$$\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\theta+iv) = \frac{1}{\pi}\sum_\rho \frac{\theta-\sigma_\rho}{(\theta-\sigma_\rho)^2+(v-\gamma_\rho)^2},$$

which is the density of (30.1.1). The second form in (30.1.2) is the Cauchy Riemann equation for $\log\xi$, and integrating $\partial_v \arg\xi = \partial_\sigma\log|\xi|$ gives the mass formula. For (30.1.3), $\xi'/\xi = 1/s + 1/(s-1) - \frac12\log\pi + \frac12\Gamma'/\Gamma(s/2) + \zeta'/\zeta$, so $u_1 = g_1 - \pi^{-1}\operatorname{Re}C(1+iv)$, where $g_1$ collects the first three terms of (30.1.3)

and $C(s) = -\zeta'(s)/\zeta(s) - 1/(s-1)$ is analytic on a neighbourhood of $\operatorname{Re} s \geq 1$, because $\zeta(1+it) \neq 0$ and the pole of $\zeta$ cancels; at $v = 0$ this is the value of the continuous function $u_1$. Write $\sum_{n \leq t} \Lambda(n) = t + R(t)$ with $R(t) = O(te^{-c\sqrt{\log t}})$. Partial summation gives, for $\operatorname{Re} s \geq 1$,

$$\sum_{n \leq X} \frac{\Lambda(n)}{n^s} - \int_1^X \frac{dt}{t^s} = \frac{R(X)}{X^s} - R(1) + s \int_1^X \frac{R(t)}{t^{s+1}}\, dt,$$

and as $X \to \infty$ the right side converges, uniformly for $s$ in compact subsets of $\operatorname{Re} s \geq 1$, because $\int^\infty e^{-c\sqrt{\log t}}\, dt/t < \infty$. For $\operatorname{Re} s > 1$ the left side tends to $C(s)$, so by continuity the limit is $C(s)$ on $\operatorname{Re} s = 1$ too. At $s = 1 + iv$ the real part of $\int_1^X t^{-s}\, dt$ is $\sin(v \log X)/v$, and $\log X$ at $v = 0$, which has no limit as $X \to \infty$; this gives the compensated limit in (30.1.3) and the divergence of the cosine series. For $\theta > 1$ the Dirichlet series converges absolutely, and $\operatorname{Re} 1/(s-1) = (\theta - 1)/((\theta-1)^2 + v^2)$ at $s = \theta + iv$. □

At $v = 0$ the compensated limit is $\lim(\sum_{n \leq X} \Lambda(n)/n - \log X) = -\gamma_E$, and (30.1.3) gives $u_1(0) = (1 + \gamma_E/2 - \log 2 - \frac{1}{2}\log \pi)/\pi = (2 + \gamma_E - \log 4\pi)/(2\pi) = 0.00735159249234$, the value of the continuous function $u_1$ at the origin (computed, to 30 digits, `code/lo_common.py`; the values $u_1(10^{-3}) - u_1(0) = 3.5 \times 10^{-11}$ and $u_1(0.05) - u_1(0) = 8.8 \times 10^{-8}$ confirm it). It is also $(2\pi)^{-1} \sum_\rho 1/(\rho(1-\rho))$.

Formula (30.1.3) is the answer to the first question of this chapter. The level one Thorin measure of the reciprocal xi function is an explicit function of the primes, with no reference to zeros, and Theorem 29.14.2 says it is a positive measure. Positivity of (30.1.3) is itself a statement about primes: it says that the limit in (30.1.3) is at most $1/(1+v^2) - \frac{1}{2}\log \pi + \frac{1}{2} \operatorname{Re} \Gamma'/\Gamma((1+iv)/2)$, the right side being $\frac{1}{2}\log(v/2\pi) + O(1/v^2)$. This inequality follows from the Euler product (Proposition 9.1.2). Numerically, the mass of $u_1$ on $[0, 400]$ is 201.67 and there are 202 zeros below height 400 (computed: Simpson on step 0.05 gives 201.66682, and the formula below over the ordinates of `data/zeros_1700.txt` plus the smooth tail gives 201.66681), the difference coming from the Cauchy smoothing of the zero count,

$$\mathfrak{U}_1((0, V]) = \frac{1}{\pi} \sum_{\gamma_\rho > 0} \Big[ \arctan \frac{V - \gamma_\rho}{1 - \sigma_\rho} + \arctan \frac{V + \gamma_\rho}{1 - \sigma_\rho} \Big].$$

## 30.2 The Fourier side: primes

Write $\widehat{\mu}(\omega) = \int e^{-i\omega v} \mu(dv)$ for the Fourier transform of a tempered measure.

Proposition 30.2.1 (spectrum of the level one measure). *For $\theta \geq 1$ let $g_\theta(v) = \frac{1}{\pi}\big[\theta/(\theta^2 + v^2) - \frac{1}{2}\log \pi + \frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma}((\theta + iv)/2)\big]$. As tempered distributions,*

$$\widehat{\mathfrak{U}_1}(\omega) = \widehat{g_1}(\omega) + 1 - \sum_{n \geq 2} \frac{\Lambda(n)}{n} \big(\delta_{\log n} + \delta_{-\log n}\big)(\omega), \tag{30.2.1}$$

*the constant* $1$ *being the transform of the atom* $\delta_0$ *that the pole of* $\zeta$ *at* $s=1$ *contributes in the limit* $\theta\downarrow 1$*; on the zero side* $\widehat{\mathfrak{U}_1}(\omega)=\sum_\rho e^{-(1-\sigma_\rho)|\omega|}e^{-i\omega\gamma_\rho}$*, the limit of the sums over* $|\gamma_\rho|<T$*. Equivalently, for every* $\varphi\in\mathcal{S}(\mathbb{R})$*,*

$$\sum_\rho \big(P_{1-\sigma_\rho}*\varphi\big)(\gamma_\rho)=\int g_1\varphi\,dv+\varphi(0)-\frac{1}{2\pi}\sum_{n\ge 2}\frac{\Lambda(n)}{n}\big(\widehat{\varphi}(\log n)+\widehat{\varphi}(-\log n)\big), \tag{30.2.2}$$

*with* $P_c$ *the Cauchy kernel of width* $c$*. For* $\theta\ge 1$ *the semigroup* $\mathfrak{U}_\theta=P_{\theta-1}*\mathfrak{U}_1$ *reads, since* $g_\theta=P_{\theta-1}*g_1$*,* $\widehat{\mathfrak{U}_\theta}(\omega)=\widehat{P_{\theta-1}*\mathfrak{U}_1}(\omega)=e^{-(\theta-1)|\omega|}\widehat{\mathfrak{U}_1}(\omega)=\widehat{g_\theta}(\omega)+e^{-(\theta-1)|\omega|}-\sum_{n\ge2}\Lambda(n)n^{-\theta}(\delta_{\log n}+\delta_{-\log n})(\omega)$*, which on the prime side is the replacement of* $\Lambda(n)/n$ *by* $\Lambda(n)n^{-\theta}$*.*

*Proof.* For $1<\theta\le 2$ the formula of Theorem 30.1.1 reads

$$u_\theta=g_\theta+P_{\theta-1}-\pi^{-1}\sum_n\Lambda(n)n^{-\theta}\cos(v\log n),$$

absolutely convergent. Integrating against $\varphi\in\mathcal{S}(\mathbb{R})$ and using

$$\int\varphi(v)\cos(v\log n)\,dv=\tfrac12\big(\widehat{\varphi}(\log n)+\widehat{\varphi}(-\log n)\big),$$

we get

$$\sum_\rho(P_{\theta-\sigma_\rho}*\varphi)(\gamma_\rho)=\int g_\theta\varphi\,dv+(P_{\theta-1}*\varphi)(0)-\frac{1}{2\pi}\sum_{n\ge2}\frac{\Lambda(n)}{n^\theta}\big(\widehat{\varphi}(\log n)+\widehat{\varphi}(-\log n)\big).$$

Let $\theta\downarrow1$. On the left, splitting the integral at $|v-\gamma|=|\gamma|/2$ gives $|(P_c*\varphi)(\gamma)|\le C_\varphi/(1+\gamma^2)$ uniformly for $0\le c\le2$; each term is continuous in $c$ and $\sum_\rho(1+\gamma_\rho^2)^{-1}<\infty$, so dominated convergence over the zeros applies. On the right, $g_\theta\to g_1$ pointwise with $|g_\theta(v)|\le C\log(2+|v|)$ for $1\le\theta\le2$, $(P_{\theta-1}*\varphi)(0)\to\varphi(0)$, and $\sum_n\Lambda(n)n^{-1}|\widehat{\varphi}(\pm\log n)|<\infty$ because $\widehat{\varphi}(\omega)=O((1+|\omega|)^{-3})$ and $\sum_{n\le x}\Lambda(n)/n=\log x+O(1)$. This proves (30.2.2). Its left side is $\int\varphi\,d\mathfrak{U}_1$, and $\int f\varphi\,dv=\frac{1}{2\pi}\int\widehat{f}(\omega)\widehat{\varphi}(-\omega)\,d\omega$, so (30.2.2) for all $\varphi\in\mathcal{S}(\mathbb{R})$ is (30.2.1). The zero side is the transform of (30.1.1) term by term, $\widehat{\mathrm{Cauchy}(\gamma,c)}(\omega)=e^{-c|\omega|}e^{-i\omega\gamma}$, the sums over $|\gamma_\rho|<T$ converging in $\mathcal{S}'$ by the same bound. For $\theta>1$ the computation without the limit gives the last formula, and $\mathfrak{U}_\theta=P_{\theta-1}*\mathfrak{U}_1$ holds term by term because $P_{\theta-1}*P_c=P_{\theta-1+c}$. □

The identity (30.2.2) is an explicit formula of Weil's type [512]. For a conjugate pair of zeros $\rho,\bar\rho$, with

$$H_\varphi(z)=\frac{1}{2\pi}\int_0^\infty\big(\widehat{\varphi}(t)+\widehat{\varphi}(-t)\big)e^{(z-1)t}\,dt,$$

one has $(P_{1-\sigma} * \varphi)(\gamma) + (P_{1-\sigma} * \varphi)(-\gamma) = H_\varphi(\rho) + H_\varphi(\bar{\rho})$, so the left side of (30.2.2) is $\sum_\rho H_\varphi(\rho)$, summed over conjugate pairs: one test function evaluated at the zeros. Under RH it is the classical explicit formula tested against $P_{1/2} * \varphi$ (compare Theorem 20.1.1). With Theorem 20.1.1, it identifies the Thorin measure of this book with the object that the explicit formula has always been about. As a check, for $\varphi(v) = e^{-v^2/2}$ the two sides of (30.2.2) agree to thirteen digits in the form $\int (u_1 - g_1)\varphi \, dv = \varphi(0) - (2\pi)^{-1} \sum_n \Lambda(n) n^{-1} (\widehat{\varphi}(\log n) + \widehat{\varphi}(-\log n))$, both equal to 0.4267700651131 (computed in the review of the source with 120 point Gauss and Legendre quadrature and 20 digit arithmetic, not by a script of the book).

The spectral peaks of $u_1$ sit at $\log 2, \log 3, \log 4, \log 5, \log 7, \ldots$ with heights roughly proportional to $\Lambda(n)/n$. Table 30.1 shows them recovered from a Hann windowed Fourier transform of $u_1$ sampled on $[-400, 400]$ at spacing 0.05 (by symmetry), padded to a frequency step of $3 \times 10^{-5}$ (`code/lo_flow.py`, `data/lo_flow.json`).

| $n$ | peak $\omega$ | $e^\omega$ | relative height | $\Lambda(n)/n$ relative to $n = 3$ |
|---|---|---|---|---|
| 2 | 0.6931 | 2.000 | 0.950 | 0.946 |
| 3 | 1.0986 | 3.000 | 1.000 | 1.000 |
| 4 | 1.3863 | 4.000 | 0.449 | 0.473 |
| 5 | 1.6095 | 5.000 | 0.872 | 0.879 |
| 7 | 1.9459 | 7.000 | 0.747 | 0.759 |
| 8 | 2.0794 | 8.000 | 0.202 | 0.237 |
| 9 | 2.1972 | 9.000 | 0.303 | 0.333 |
| 11 | 2.3979 | 11.000 | 0.576 | 0.595 |
| 13 | 2.5650 | 13.000 | 0.517 | 0.539 |
| 17 | 2.8332 | 17.000 | 0.430 | 0.455 |
| 19 | 2.9444 | 19.000 | 0.397 | 0.423 |
| 23 | 3.1355 | 23.000 | 0.343 | 0.372 |
| 29 | 3.3673 | 29.000 | 0.286 | 0.317 |
| 31 | 3.4340 | 31.001 | 0.271 | 0.302 |

Table 30.1: Spectral peaks of the level one Thorin measure (computed). Every peak lies within $10^{-4}$ of the logarithm of a prime or prime power. The heights, relative to that of $n = 3$, depend on the window and follow the relative value of $\Lambda(n)/n$ to within 0.04.

## 30.3 The backward flow

Fix $\frac{1}{2} \le \theta' < 1$, put $a = 1 - \theta'$, and let $\Delta_\Omega(\omega) = (1 - |\omega|/\Omega)_+$ be the Fejér window. Let $K_{\theta',\Omega}$ be the inverse Fourier transform of $\Delta_\Omega(\omega) e^{a|\omega|}$; this multiplier is even, continuous, compactly supported and has a derivative of bounded variation, so $K_{\theta',\Omega}$ is smooth and $|K_{\theta',\Omega}(v)| \le C_{\Omega,a}/(1 + v^2)$. The band limited backward flow

from level one to level $\theta'$ is

$$B_{\theta',\Omega}\mathfrak{U}_1 = \mathfrak{U}_1 * K_{\theta',\Omega}, \tag{30.3.1}$$

a smooth function of growth $O(\log(2+|v|))$, since $\mathfrak{U}_1([t,t+1]) = O(\log(2+|t|))$; formally its transform is $\Delta_\Omega(\omega)e^{a|\omega|}\widehat{\mathfrak{U}_1}(\omega)$. Let $\Phi_\Omega(v) = (1-\cos\Omega v)/(\pi\Omega v^2)$ be the Fejér kernel, the inverse transform of $\Delta_\Omega$.

THEOREM 30.3.1 (dichotomy for the backward flow). *Let $\frac{1}{2} \le \theta' < 1$ and $a = 1-\theta'$.*

*(i) If $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$, then $B_{\theta',\Omega}\mathfrak{U}_1 = \mathfrak{U}_{\theta'} * \Phi_\Omega$, with $\mathfrak{U}_{\theta'}$ as in* (30.1.1)*, which has atoms at the zeros with $\sigma_\rho = \theta'$. As $\Omega \to \infty$, $B_{\theta',\Omega}\mathfrak{U}_1 \to \mathfrak{U}_{\theta'}$ vaguely, and $\int B_{\theta',\Omega}\mathfrak{U}_1(v)\,dv/(v^2+s) \to \int \mathfrak{U}_{\theta'}(dv)/(v^2+s)$ for every $s>0$. If $\theta' > \sigma_\rho$ for every $\rho$, the convergence is locally uniform as densities.*

*(ii) If $\zeta$ has a zero $\rho$ with $\sigma_\rho > \theta'$, there is no signed measure $\mu$ with $\int |\mu|(dv)/(1+v^2) < \infty$ such that $\int B_{\theta',\Omega}\mathfrak{U}_1(v)\,dv/(v^2+s) \to \int \mu(dv)/(v^2+s)$ as $\Omega \to \infty$ for all sufficiently large $s$.*

*Proof.* Write $f_s(v) = 1/(v^2+s)$, so that $f_s = (\pi/\sqrt{s})P_{\sqrt{s}}$ and $\widehat{f_s}(\omega) = (\pi/\sqrt{s})e^{-\sqrt{s}|\omega|}$. The measures below are positive or satisfy $\int |\mu|/(1+v^2) < \infty$, and the kernels are $O(1/(1+v^2))$, so every use of Fubini's theorem is absolute.

(i) Every zero has $\sigma_\rho \le \theta'$, so $\mathfrak{U}_{\theta'}$ is defined and, term by term, $\mathfrak{U}_1 = \mathfrak{U}_{\theta'} * P_a$, because $P_c * P_a = P_{c+a}$ for $c \ge 0$ (Theorem 29.14.4). The integrable function $P_a * K_{\theta',\Omega}$ has transform $e^{-a|\omega|}\Delta_\Omega(\omega)e^{a|\omega|} = \Delta_\Omega(\omega)$, so $P_a * K_{\theta',\Omega} = \Phi_\Omega$ and $B_{\theta',\Omega}\mathfrak{U}_1 = \mathfrak{U}_{\theta'} * P_a * K_{\theta',\Omega} = \mathfrak{U}_{\theta'} * \Phi_\Omega$. The kernel $\Phi_\Omega$ is nonnegative, has integral one and satisfies $\Phi_\Omega(u) \le 2/(\pi\Omega u^2)$. For $f$ continuous with support in $[-R,R]$, $\Phi_\Omega * f \to f$ uniformly and $|\Phi_\Omega * f(v)| \le 8\|f\|_1/(\pi\Omega v^2)$ for $|v| \ge 2R$, so $|\Phi_\Omega * f - f| \le \eta_\Omega/(1+v^2)$ with $\eta_\Omega \to 0$, and $\int f\,d(\mathfrak{U}_{\theta'} * \Phi_\Omega) = \int (\Phi_\Omega * f)\,d\mathfrak{U}_{\theta'} \to \int f\,d\mathfrak{U}_{\theta'}$. For $f = f_s$, $\Phi_\Omega * f_s \to f_s$ pointwise and $\Phi_\Omega * f_s \le C_s/(1+v^2)$ for $\Omega \ge 1$, and dominated convergence gives the second statement. If $\theta' > \sigma_\rho$ for every $\rho$, $\mathfrak{U}_{\theta'}$ has a continuous density $u_{\theta'}$ (Theorem 30.1.1). For $|v| \le R$ and $0 < \delta \le 1$, $|\Phi_\Omega * u_{\theta'}(v) - u_{\theta'}(v)|$ is at most the largest change of $u_{\theta'}$ over distances $\delta$ on $[-R-1, R+1]$, plus $(2/(\pi\Omega)) \int_{|t|\ge\delta} u_{\theta'}(v-t)t^{-2}\,dt + (4/(\pi\Omega\delta)) \max_{[-R,R]} u_{\theta'}$, and the last two terms are $O_{R,\delta}(1/\Omega)$.

(ii) Let $s > a^2$ and $b = \sqrt{s} - a > 0$. The integrable function $K_{\theta',\Omega} * f_s$ has transform $(\pi/\sqrt{s})\Delta_\Omega(\omega)e^{-b|\omega|}$, so $K_{\theta',\Omega} * f_s = (\pi/\sqrt{s})\Phi_\Omega * P_b = (b/\sqrt{s})\Phi_\Omega * f_{b^2}$ and

$$\int B_{\theta',\Omega}\mathfrak{U}_1(v)f_s(v)\,dv = \int (K_{\theta',\Omega} * f_s)\,d\mathfrak{U}_1 = \frac{b}{\sqrt{s}}\int (\Phi_\Omega * f_{b^2})\,d\mathfrak{U}_1$$
$$\longrightarrow \frac{b}{\sqrt{s}}\int \frac{\mathfrak{U}_1(dv)}{v^2+b^2}$$

as $\Omega \to \infty$, by the domination used in (i). Since $\int P_c(v-\gamma) f_{b^2}(v)\,dv = (\pi/b)P_{b+c}(\gamma) = b^{-1}\operatorname{Re}(b+c-i\gamma)^{-1}$, summing over the zeros with $c = 1-\sigma_\rho$ gives

$$\int \frac{\mathfrak{U}_1(dv)}{v^2+b^2} = \frac{1}{b}\sum_\rho \operatorname{Re}\frac{1}{1+b-\rho} = \frac{1}{b}\,\frac{\xi'}{\xi}(1+b),$$

so the limit is $(\xi'/\xi)(\theta'+\sqrt{s})/\sqrt{s}$. If a measure $\mu$ as in (ii) existed, then for all large $s>0$

$$G(s) := \int \frac{\mu(dv)}{v^2+s} = \frac{1}{\sqrt{s}}\,\frac{\xi'}{\xi}(\theta'+\sqrt{s}).$$

On a compact subset of $\mathbb{C}\setminus(-\infty,0]$ one has $|v^2+s| \ge c(1+v^2)$, so $G$ is holomorphic on $\mathbb{C}\setminus(-\infty,0]$. The right side, with the principal square root, is meromorphic on $\mathbb{C}\setminus(-\infty,0]$, and the identity theorem on the connected set $\mathbb{C}\setminus(-\infty,0]$ minus its poles makes $G$ equal to it there. Squaring maps $\operatorname{Re} w > 0$ onto $\mathbb{C}\setminus(-\infty,0]$, so $s_0 = (\rho-\theta')^2 \in \mathbb{C}\setminus(-\infty,0]$ with $\sqrt{s_0} = \rho-\theta'$, and near $s_0$ the right side is $2m_\rho/(s-s_0)+O(1)$, where $m_\rho \ge 1$ is the multiplicity of $\rho$, while $G$ is bounded near $s_0$. This contradiction proves (ii). □

The theorem is the Thorin criterion of Chapter 29 in dynamical form: RH is the statement that the backward flow from the explicit level one measure converges at every level $\theta' > \frac12$. Part (ii) is proved by the same pole argument as in Theorem 29.14.2. The flow does not escape the equivalence; it relocates the difficulty into the multiplier $e^{(1-\theta')|\omega|}$, which amplifies whatever is not known about $\widehat{\mathfrak{U}_1}$ at high frequency. On the prime side, high frequency means large $n$: the content of $\widehat{\mathfrak{U}_1}$ on $|\omega| \le \Omega$ is $\sum_{n\le e^\Omega}\Lambda(n)/n$, and the backward flow to level $\theta'$ heuristically asks for $\sum_{n\le x}\Lambda(n)n^{-\theta'}$ to behave, which is the statement $\psi(x)-x = O(x^{\theta'+\varepsilon})$ for every $\varepsilon>0$ (the classical equivalence with the zero free half plane $\operatorname{Re} s > \theta'$).

### 30.3.1 Numerical backward flow

The run is `code/lo_flow.py` (`data/lo_flow.json`). We sampled $u_1$ on $[0,400]$ at spacing 0.05 from (30.1.2) with $\theta=1$, evaluating $\zeta$ and $\zeta'$ on the line $\operatorname{Re} s = 1$ only (python-flint, 128 bits), with the analytic value $u_1(0)$ at $v=0$, extended by symmetry, applied a Hann window and the multiplier of (30.3.1) with a sharp cutoff at $\Omega$, not the Fejér window of (30.3.1), inverted and divided by the window. A peak is a local maximum of the output in $100 < v < 300$ of height at least a quarter of the maximum, and the peaks are compared with the 109 zeros of $\zeta$ in that range ($N(300)-N(100) = 138-29$). The Fejér variant multiplies the sharp multiplier by $1-|\omega|/\Omega$.

At $\theta' = 0.75$ the flow produces a positive function with 109 peaks, each within 0.035 of a zero, as Theorem 30.3.1(i) predicts if $\zeta$ has no zeros in $\operatorname{Re} s > 3/4$: the level $\frac34$ measure is then a smooth positive density whose peaks lie near the zeros. At $\theta' = 0.5$ the peaks lie within 0.027 of the zeros, but the function oscillates below

| $\theta'$ | $\Omega$ | cutoff | peaks | zeros | max error | min value |
|---|---|---|---|---|---|---|
| 0.75 | 40 | sharp | 109 | 109 | 0.035 | +0.075 |
| 0.50 | 40 | sharp | 109 | 109 | 0.027 | −3.2 |
| 0.50 | 60 | sharp | 213 | 109 | 0.042 | −16.5 |
| 0.50 | 60 | Fejér | 109 | 109 | 0.025 | −0.020 |

Table 30.2: The Fourier backward flow from samples of $u_1$ on $\operatorname{Re} s = 1$ (computed). The error is the largest distance between a zero in $100 < v < 300$ and the nearest peak. At $\Omega = 60$ the sharp cutoff has 213 peaks of height at least a quarter of the maximum, 109 of them at the zeros, and the maximum is +26.6.

zero between them. This comes from the sharp cutoff: its kernel $\sin(\Omega v)/(\pi v)$ has negative lobes, the first of depth about $0.22\,\Omega/\pi \approx 2.8$ at $\Omega = 40$, whereas under RH the Fejér approximant $\mathfrak{U}_{1/2} * \Phi_\Omega$ of Theorem 30.3.1(i) is nonnegative, because $\Phi_\Omega \geq 0$. Theorem 30.3.1 does not cover the sharp cutoff. The Fejér run at $\Omega = 60$ is clean: 109 peaks, error 0.025 and minimum −0.020 (the small negative values come from the Hann window and its division); at $\Omega = 40$ its minimum is −0.011.

The cutoff is limited by the accuracy of the samples: at level $\frac{1}{2}$ the multiplier at the cutoff is $e^{\Omega/2}$, about $10^{13}$ at $\Omega = 60$, so an error of $10^{-7}$ in the single sample at $v = 0$ already produces oscillations of order $10^2$ in $100 < v < 300$. This is what the source printed: its $\Omega = 60$ row, with 1022 peaks, error 1.9 and minimum −430, has its minimum reproduced only if the sample at $v = 0$, the pole of $\zeta$, is replaced by the sample at $v = 0.05$ (an error of $8.8 \times 10^{-8}$; minimum −430.2, maximum +430.9; the peak count and the error are not reproduced, the source giving no peak rule), and the source read it as amplified rounding error. With the analytic $u_1(0)$ the output of the sharp cutoff at $\Omega = 60$ is not noise but the ringing of the cutoff: each atom gives a sinc of height $\Omega/\pi \approx 19$ and the sidelobes of 213 peaks follow; double precision rounding alone would give oscillations of order $3 \times 10^{-4}$. The line $\operatorname{Re} s = 1$ determines the zeros, and the 109 zeros tested are read off it to 0.025 to 0.035 (two decimals, not the three decimals and the first two hundred zeros of the source), but at level $\frac{1}{2}$ the precision required of the level one data grows like $e^{\Omega/2}$ with the cutoff.

## 30.4 The backward flow as a Bayesian inverse problem

### 30.4.1 Positive linear inverse problem

Discretise. Let $y_i = u_1(v_i)$ on a grid $v_i$ of spacing $h$ covering the data window, let $x_j \geq 0$ be the unknown densities of $\mathfrak{U}_{\theta'}$ on a grid $w_j$ of the same spacing covering a larger support (cell masses $hx_j$), and let

$$y = Ax, \qquad A_{ij} = hP_{1-\theta'}(v_i - w_j). \tag{30.4.1}$$

Every entry of $A$ is positive, and a column centred well inside the data window has $\sum_i A_{ij} \approx 1$. This is a positive linear inverse problem in the sense of Vardi and Lee [356]: the same structure as emission tomography, image deblurring and the portfolio problem of their title, with the Cauchy kernel as the blur. Two features distinguish it from the usual instances. The data are exact, so the problem is one of existence rather than estimation. And the existence question is the zero free half plane $\operatorname{Re} s > \theta'$, the Riemann hypothesis when $\theta' = \frac{1}{2}$: by Proposition 30.4.1(a), a positive solution of the continuous problem exists if and only if $\zeta$ has no zeros to the right of $\theta'$.

### 30.4.2 Likelihood, prior, posterior

The data $y_i$ are exact real numbers, so a Poisson model is a device, but a useful one. Read $hy_i$ as the expected count in cell $i$ of a Poisson process whose intensity is the Cauchy blur of an unknown positive measure: the complete data are counts $z_{ij}$ of mass originating in cell $j$ and observed in cell $i$, Poisson with means $hA_{ij}x_j$, and the observed counts are $\sum_j z_{ij}$. Replacing these counts by the data $hy_i$, the log likelihood of $x$ is, up to constants and the factor $h$,

$$\ell(x) = \sum_i \big( y_i \log (Ax)_i - (Ax)_i \big),$$

concave in $x$, and maximising it over $x \geq 0$ minimises the generalised Kullback Leibler divergence $\mathrm{KL}(y \,\|\, Ax) = \sum_i [y_i \log (y_i/(Ax)_i) - y_i + (Ax)_i]$, which is defined for real data. The EM algorithm with the $z_{ij}$ as missing data is the Shepp and Vardi iteration [268]

$$x_j^{(k+1)} = \frac{x_j^{(k)}}{\sum_i A_{ij}} \sum_i A_{ij} \frac{y_i}{(Ax^{(k)})_i}, \tag{30.4.2}$$

which is also the Richardson and Lucy deblurring algorithm [263, 212]. It preserves positivity at every step, increases the likelihood monotonically, and, from a strictly positive start, converges to a maximum likelihood solution [362]. Csiszár and Tusnády [102] identified it as alternating $I$ projection between the set of blurred positive measures and the set of distributions consistent with the data, an information theoretic analogue of von Neumann's alternating projection theorem for subspaces of a Hilbert space.

A Bayesian completes the model with a prior on $x$, the conjugate choice being a product of gamma laws. A sampler needs integer counts, which exact data do not supply: fix an exposure $\tau > 0$ and let $N_i$ be the integer nearest to $\tau h y_i$, modelled as $N_i = \sum_j Z_{ij}$ with independent $Z_{ij}$, Poisson with means $\tau h A_{ij} x_j$. Given $x$ and $N$, the vector $(Z_{ij})_j$ is multinomial with total $N_i$ and probabilities $A_{ij}x_j/(Ax)_i$; given $Z$, with independent $\mathrm{Gamma}(\alpha_j, \beta_j)$ priors, $x_j$ is gamma with shape $\alpha_j + \sum_i Z_{ij}$ and rate $\beta_j + \tau h \sum_i A_{ij}$. Alternating the two steps is a data

augmentation Gibbs sampler, the Monte Carlo counterpart of (30.4.2) in which the E step is replaced by simulation: the deterministic EM fixed point becomes a posterior over positive measures, and the uncertainty that EM hides becomes the spread of the posterior. That spread depends on the exposure $\tau$, which exact data do not determine. The sampler is a method; no run of it is reported here (Problem 30.1).

The generative Bayesian computation of the next section amortises the posterior of a different model. Instead of running a Gibbs chain for each data set, one simulates pairs from a prior on configurations of zeros and the forward map, trains a quantile network to represent conditional quantiles of chosen functionals given the data, and evaluates it at the observed data. The three methods share the forward map but not the prior or the unknown: EM gives the posterior mode under a flat prior on the grid masses, data augmentation samples the posterior under a gamma prior on the same masses, and GBC regresses functionals of a GUE configuration of zeros on simulated data.

### 30.4.3 What positivity buys

The Fourier backward flow of Section 30.3 ignores positivity, which is why the sharp cutoff rings below zero at $\theta' = \frac{1}{2}$ and why its error is amplified by $e^{(1-\theta')\Omega}$. The EM inversion enforces positivity and uses the whole frequency content of the data, so it is the natural positivity preserving backward flow, and its total residual is the diagnostic of Proposition 30.4.1.

Proposition 30.4.1 (the EM residual as a diagnostic). *Let $\theta' \in [\frac{1}{2}, 1)$ and $a = 1 - \theta'$.*

*(a) There is a positive measure $x$ on $\mathbb{R}$ with $\int x(dv)/(1+v^2) < \infty$ and $P_a * x = u_1$ if and only if $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$. It is then unique and equals $\mathfrak{U}_{\theta'}$.*

*(b) Let $h_k \to 0$ and $L_k \to \infty$ with $h_k^2 L_k \log L_k \to 0$. For positive measures $x = \sum_j x_j \delta_{jh_k}$ on the grid $h_k\mathbb{Z}$ with $\sum_j x_j/(1 + j^2 h_k^2) < \infty$, let $m_i(x) = (P_a * x)(v_i)$ at the data points $v_i \in h_k\mathbb{Z} \cap [-L_k, L_k]$, let $y_i = u_1(v_i)$, and let*

$$D_k = \inf_x \, h_k \sum_i \Big( y_i \log \frac{y_i}{m_i(x)} - y_i + m_i(x) \Big)$$

*be the total, not normalised, Kullback Leibler residual. Then $D_k \to 0$ if $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$, and $\liminf_k D_k > 0$ if it has one.*

*Proof.* (a) is Theorem 19.2.2(a) with $\varepsilon_0 = \frac{1}{2}$ and $\varepsilon_1 = \theta' - \frac{1}{2}$.

Write $d(y, m) = y \log(y/m) - y + m$; then $(\sqrt{y} - \sqrt{m})^2 \le d(y, m) \le (y-m)^2/m$ for $y, m > 0$. Also $|\partial_t \log P_a(t)| = 2|t|/(a^2 + t^2) \le 1/a$, so $P_a(t')$ lies between $P_a(t) e^{-|t-t'|/a}$ and $P_a(t) e^{|t-t'|/a}$.

(b), no zeros in $\operatorname{Re} s > \theta'$. Let $x_j = \mathfrak{U}_{\theta'}([jh_k - h_k/2, jh_k + h_k/2))$, which satisfies the summability condition because $\mathfrak{U}_{\theta'}$ does. By (a), $y_i = \int P_a(v_i - w)\, \mathfrak{U}_{\theta'}(dw)$,

and moving each $w$ to the centre of its cell changes $P_a(v_i - w)$ by a factor in $[e^{-h_k/2a}, e^{h_k/2a}]$, so $|m_i(x) - y_i| \le (e^{h_k/2a} - 1)y_i$ and $d(y_i, m_i(x)) \le Ch_k^2 y_i$ for $h_k \le a$. As $u_1 = P_a * \mathfrak{U}_{\theta'}$ gives $|u_1'| \le u_1/a$, we have $h_k \sum_i y_i \le e^{h_k/2a} \int_{-L_k-1}^{L_k+1} u_1\,dv$, which is $O(L_k \log L_k)$ by the Riemann von Mangoldt formula: the zeros with $|\gamma_\rho| \le 2L_k + 2$ contribute at most their number, and the others $O(\log L_k)$. Hence $D_k = O(h_k^2 L_k \log L_k) \to 0$. Atoms of $\mathfrak{U}_{\theta'}$, at zeros with $\sigma_\rho = \theta'$, are allowed.

(b), a zero in $\operatorname{Re} s > \theta'$. Suppose that $D_k \to 0$ along a subsequence, and choose grid measures $x_k$ with total residual at most $D_k + 1/k$. The function $F_k = P_a * x_k$ is smooth and positive with $|F_k'| \le F_k/a$, and $h_k \sum_{|v_i| \le R} (\sqrt{y_i} - \sqrt{F_k(v_i)})^2 \to 0$ for every $R > 0$. Since $F_k \ge e^{-1/a} F_k(0)$ on $[-1, 1]$, the case $R = 1$ shows that $F_k(0) = (a/\pi) \int x_k(dw)/(a^2 + w^2)$ is bounded. So the measures $\nu_k(dw) = x_k(dw)/(a^2 + w^2)$ have bounded mass, and along a further subsequence they converge weakly on the compactification $\mathbb{R} \cup \{\infty\}$ to a finite measure $\nu$. Now $F_k(v) = \int \kappa_v\,d\nu_k$ with $\kappa_v(w) = (a/\pi)(a^2 + w^2)/(a^2 + (v - w)^2)$, which is jointly continuous on $[-R, R] \times (\mathbb{R} \cup \{\infty\})$ with $\kappa_v(\infty) = a/\pi$. Hence $F_k \to F$ uniformly on $[-R, R]$, where $F = P_a * x + c$ with $x = (a^2 + w^2)\nu$ on $\mathbb{R}$ and $c = (a/\pi)\nu(\{\infty\}) \ge 0$, and Riemann sums give $\int_{-R}^{R} (\sqrt{u_1} - \sqrt{F})^2\,dv = 0$. As $R$ is arbitrary and $P_a * (c\,dv) = c$, the positive measure $x + c\,dv$ has $\int (x + c\,dv)/(1 + v^2) < \infty$ and $P_a * (x + c\,dv) = u_1$, which (a) excludes. So $\liminf_k D_k > 0$. □

The total residual cannot be replaced by the residual per unit length of the data window: when only finitely many zeros lie to the right of $\theta'$, the latter tends to zero, since an atom of mass $m_\rho$ at each of their ordinates leaves the defect $\sum_\rho (P_{1-\sigma_\rho} - P_a)(v - \gamma_\rho)$, whose total residual is finite. The proof of (b) also shows that the residual restricted to some fixed window $[-R, R]$ stays bounded away from zero, and the same argument works for every $R > 0$ since $u_1$ and $P_a * x + c$ are real analytic, but the proposition gives no rate. The runs below, with a fixed data window and a finite unknown support, illustrate it; no finite computation certifies either alternative.

### 30.4.4 Numerical EM inversion

The runs are `code/lo_em.py` (`data/lo_em_table.json`, `data/lo_em_inject.json`), with the EM iteration (30.4.2) of `code/cx_em.py` (the convolution by FFT agrees with the dense product to $2 \times 10^{-15}$). Data are $u_1$ on $[100, 300]$, the unknown is supported on $[80, 320]$, spacing 0.05, flat start with the total mass matched. The residual printed is the plain sum $\sum_i [y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i]$ over the 4001 data points, that is $\mathrm{KL}(y \,\|\, Ax)$ of Proposition 30.4.1 divided by the spacing $h$, not per point and not per unit length; it changes by 1.6 per cent between spacings 0.05 and 0.1, so two digits at most are meaningful. A peak is a local maximum of prominence at least 1 per cent of the maximum, and $\min x$ is the smallest $x_j$.

| $\theta'$ | steps | KL residual | peaks | max error | $\min x$ |
|---|---|---|---|---|---|
| 0.75 | 300 | $2.4 \times 10^{-5}$ | 109 of 109 | 0.040 | $7.7 \times 10^{-2}$ |
| 0.50 | 300 | $4.9 \times 10^{-2}$ | 109 of 109 | 0.085 | $2.7 \times 10^{-5}$ |
| 0.50 | 3000 | $4.0 \times 10^{-3}$ | 109 of 109 | 0.030 | $9.9 \times 10^{-12}$ |

Table 30.3: EM inversion of $u_1$ on $[100, 300]$ (computed). The source printed $4.1 \times 10^{-3}$ in the last row; the recomputation gives $4.05 \times 10^{-3}$, and the grid sensitivity of the residual is of that size.

At $\theta' = 0.75$ the iteration converges quickly to a strictly positive density with one peak per zero; at $\theta' = 0.5$ the target is atomic, the iteration sharpens slowly, as EM does when the truth lies on the boundary of the positive cone, and after 3000 steps the mass has concentrated at the zeros to 0.03 with the density between them driven to $10^{-11}$. There is no ringing.

With an off line pair injected at $\sigma = 0.75$ and $\theta' = 0.75$, the source reports a residual after 3000 steps of $1.6 \times 10^{-5}$ against $2.3 \times 10^{-7}$ for the real $u_1$, a factor of about 70; the recomputation gives $1.62 \times 10^{-5}$ and $2.27 \times 10^{-7}$ (ratio 71, and 53 on the data window $[150, 250]$) for a quartet at $\sigma = \theta'$ that keeps the total mass, and the source's local misfit near the injected zero could not be reproduced, because its definition is not given. This zero lies on the level, not to its right, so a positive solution still exists: at level one the pair adds $P_{1/4}(v - \gamma) + P_{3/4}(v - \gamma)$, which is $P_{1/4} * (\delta_\gamma + P_{1/2}(\cdot - \gamma))$. The larger residual therefore measures the slow concentration of EM onto an atom, as in the $\theta' = 0.5$ rows, and does not illustrate Proposition 30.4.1(b), which needs $\sigma > \theta'$.

The test was therefore redesigned. The two consecutive line zeros nearest the centre of the window ($\gamma = 198.0153$ and $201.2648$, midpoint $199.6400$) are removed and replaced by the quartet $\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho$ with real part $\sigma$ at the midpoint (widths $1 - \sigma$ and $\sigma$ at the midpoint and at its mirror image, total mass unchanged to $10^{-4}$). The control puts two line zeros at the midpoint. Table 30.4 gives the total residual at 10000 steps and the ratio of quartet to control at 300, 1000, 3000 and 10000 steps, on the data windows $[100, 300]$ (unknown $[80, 320]$) and $[150, 250]$ (unknown $[130, 270]$).

| $\theta'$, $\sigma$ | window | real $u_1$ | control | quartet | ratio at $300, \dots, 10^4$ steps |
|---|---|---|---|---|---|
| 0.50, 0.75 | $[100, 300]$ | $1.09 \times 10^{-3}$ | $1.08 \times 10^{-3}$ | $1.85 \times 10^{-2}$ | 1.4, 2.4, 5.4, 17 |
| 0.50, 0.75 | $[150, 250]$ | $5.0 \times 10^{-4}$ | $5.0 \times 10^{-4}$ | $1.78 \times 10^{-2}$ | 1.8, 3.8, 10, 36 |
| 0.75, 0.85 | $[100, 300]$ | $3.2 \times 10^{-8}$ | $3.3 \times 10^{-8}$ | $7.03 \times 10^{-3}$ | 366, 4000, 31000, 220000 |
| 0.75, 0.85 | $[150, 250]$ | $5.9 \times 10^{-8}$ | $5.9 \times 10^{-8}$ | $7.02 \times 10^{-3}$ | 627, 4500, 23000, 120000 |
| 0.75, 0.75 | $[100, 300]$ | $3.2 \times 10^{-8}$ | $3.3 \times 10^{-8}$ | $4.4 \times 10^{-6}$ | 9.0, 30, 72, 136 |

Table 30.4: Quartet injection test with controls (computed, `code/lo_em.py inject`); total residual after 10000 EM steps. The last row is the boundary case $\sigma = \theta'$.

The redesigned test separates the situations. At $\theta' = 0.75$ and $\sigma = 0.85 > \theta'$ the quartet residual stops at $7.03 \times 10^{-3}$, the same for both windows to 0.3 per cent: a floor, with no positive solution, as Proposition 30.4.1(b) predicts for a zero to the right of $\theta'$, while the control and the real data decay below $10^{-7}$. At $\theta' = 0.5$ and $\sigma = 0.75$ the control falls like $1/k$ ($4.9 \times 10^{-2}$, $1.3 \times 10^{-2}$, $4.0 \times 10^{-3}$, $1.1 \times 10^{-3}$ at 300, 1000, 3000, 10000 steps) while the quartet leaves $6.9 \times 10^{-2}$, $3.1 \times 10^{-2}$, $2.2 \times 10^{-2}$, $1.85 \times 10^{-2}$; the floor is not reached within 10000 steps and the ratio keeps growing, so this row shows slower decay than the control and not a plateau. The boundary case $\sigma = \theta'$ keeps decaying, like $1/k$, but more slowly than the control (ratio 9.0 at 300 steps, 136 at 10000), because a positive solution exists and EM is slow onto an atom. The plain sum is $1/h$ times the total residual of the proposition, a fixed factor at fixed spacing; a finite run certifies neither alternative.

## 30.5 Generative Bayesian computation for the backward flow

The backward flow is a deconvolution, and a deconvolution is a Bayesian inverse problem once one says what is believed about the unknown. The unknown is the configuration $Z = \{\gamma\}$ of zeros in a window; the forward map is the Cauchy semigroup,

$$Z \mapsto \sum_{\gamma \in Z} \mathrm{Cauchy}(\gamma, \tfrac{1}{2})$$

on the window, plus the contribution of the zeros outside the window, and nothing else: by Theorem 30.1.1 the zero sum is the whole of $u_1$, and the smooth part $g_1$ belongs to the prime side decomposition (30.1.3). The data are the values of $u_1$ on a grid, exact to rounding. A prior for $Z$ is supplied by Montgomery's pair correlation [374] and Odlyzko's computations [391]: locally the zeros look like the eigenvalues of a large GUE matrix, unfolded to density $\frac{1}{2\pi} \log \frac{v}{2\pi}$.

Generative Bayesian computation (Polson and Sokolov [255, 256]) handles such a problem without likelihoods or Markov chains. One simulates $(Z_i, y_i)$ from the prior and the forward map, trains a deep quantile network $q_\tau(y)$ to predict functionals of $Z$ from $y$ at quantile levels $\tau$, and evaluates the trained network at the observed $y$. The network is meant as an amortised posterior: by the noise outsourcing theorem [187] any conditional law is the push forward of a uniform variable through a measurable map, and the quantile network learns that map for the functionals of interest; in practice it estimates marginal conditional quantiles of those functionals, not their joint law. Here the functionals are the counts of zeros in eight unit bins at the centre of a window of length 16, which determine the zero positions to within the bin width and are what a deconvolution has to get right. The formulation is heuristic; what follows is a computation.

### 30.5.1 Implementation

The run is `code/lo_gbc.py` (`data/lo_gbc.json`), with two seeds, in 170 seconds on 8 cores. Training set: 24,000 windows. For each window a density $d$ is drawn uniformly from $[0.45, 0.8]$, which covers the heights $150 \le v \le 350$, where the zero density $(2\pi)^{-1}\log(v/2\pi)$ runs from 0.50 to 0.64 (and in fact the heights 106 to 958); a GUE matrix $H = (A + A^*)/2$ of size $M = 400$, with $A = X + iY$ and $X, Y$ independent standard normal matrices, so that the off diagonal entries have variance 1 and the spectral edge is $2\sqrt{M} = 40$; the eigenvalues $x$ are unfolded to unit spacing by the semicircle law, $e = MF(x/(2\sqrt{M}))$ with $F$ the semicircle distribution function, and rescaled to density $d$ (the measured number of zeros per unit length divided by $d$ is 0.998 over the 24,000 windows); the forward map is evaluated on the grid of spacing 0.1 as the sum of $P_{1/2}(t - z)$ over the simulated zeros within $150/d$ of the centre, plus the Cauchy smoothing of the mean density $d$ for the zeros beyond; the bin counts are the targets. Dividing the eigenvalues by $\sqrt{2M}$ instead of $2\sqrt{M}$ would give the density $d/\sqrt{2}$, which does not cover the real density above $v = 220$. Network: two hidden layers of 256 rectified units, 24 outputs (eight bins at $\tau = 0.1, 0.5, 0.9$), pinball loss, Adam, 25 epochs, plain numpy. Test set: windows of the real $u_1$ centred at $v = 150, 158, \ldots, 342$ (25 windows, 200 bins), with the true counts from the zeros. The forward map with the real zeros within $150/d$, $d = (2\pi)^{-1}\log(c/2\pi)$ at the window centre $c$, and the mean density beyond differs from $u_1$ by at most $4.4 \times 10^{-4}$ on these windows (mean $1.9 \times 10^{-4}$, against $\max u_1 = 1.07$): the modelling error of the outside zeros, the mirror zeros and the density gradient.

### 30.5.2 Results

The two seeds are given as seed 1 and seed 2. The pinball loss at epoch 25 is 0.0157 and 0.0166 on the training simulations, and 0.0161 and 0.0175 on held out simulations. On the 200 real bins the rounded posterior median equals the true count in 196 and 197 bins (98.0 and 98.5 per cent) and is within one of it in all 200; the errors are 3 undercounts and 1 overcount, and 3 undercounts, and about half of them fall in the five bins that hold two or more zeros, where two Cauchy bumps of width $\frac{1}{2}$ merge into one. On held out simulations the quantile levels are calibrated to within about 0.05 for integer counts: $P(N \le q_{0.1}) = 0.086$ and 0.095, $P(N \le q_{0.5}) = 0.486$ and 0.449, $P(N \le q_{0.9}) = 0.892$ and 0.891, and the interval $[q_{0.1}, q_{0.9}]$ covers 0.806 and 0.796 (nominal 0.8); on the real windows it covers 0.805 and 0.775. The first run of the source reported 68 per cent exact bins, 99 per cent within one, and uncalibrated quantiles, with training inputs and test inputs that did not match (the smooth part $g_1$ was counted twice); with the corrected model the network counts almost exactly, and the earlier numbers are not those of this model.

A misfit statistic, the root mean square distance over $|t| \le 4$ between $y$ and

the forward map of the reconstruction that places the rounded median counts at the bin centres, is 0.173 (standard deviation 0.031) on the real windows and 0.331 (0.154) and 0.307 (0.132) on the same windows with an injected quartet at $\sigma = 0.75$ (total mass kept), for the two seeds; with two line zeros placed at the midpoint instead it is 0.190 and 0.186. The injected misfit exceeds the real one in 20 and 21 of the 25 windows. The source's statistic, which did not separate real from injected windows (0.31 against 0.34), is not defined in the source and was not reproduced; the statistic here is a plug in misfit and not a posterior predictive check proper.

This is a first implementation that counts the zeros in unit bins and does not locate them. The EM inversion of Section 30.4.4, which uses the same positivity and no prior, locates the zeros to 0.03, while GBC gives bin counts only, which throw away the fine structure of the bumps, and the two are not compared on one metric. A GBC inversion that targets the EM solution itself, or the positions of the zeros as continuous outputs, with the EM reconstruction supplied as a feature, is the right next step and is left as an open problem.

### 30.5.3 What the Bayesian view adds

The GUE prior encodes the critical line: every simulated configuration has all its zeros at real part $\frac{1}{2}$, so every simulated $y$ is a sum of Cauchy bumps of width exactly $\frac{1}{2}$. A zero at $\sigma \neq \frac{1}{2}$ produces, with its mirror, bumps of widths $1-\sigma$ and $\sigma$ in place of the two bumps of width $\frac{1}{2}$ of two zeros on the line. Such a window is atypical under the prior, and a misfit statistic, the distance between the data and the forward map of the posterior median configuration, is meant to flag it; in the run above it separates the injected windows from the real ones on average, and a posterior predictive check proper would simulate from the posterior rather than plug in its median. The method can do two things that the Fourier deconvolution does not: regularise the ill posed inversion with a prior rather than with a frequency cutoff, and report uncertainty and misfit, to the extent that its quantiles are calibrated. What it cannot do is establish the absence of off line zeros. The prior assumes them away, and the misfit statistic can only flag a window where the data disagree with that assumption, at the resolution the window allows.

In the language of the book on Hilbert's sixth problem [257], the forward map is the Poisson semigroup, the subordination of the heat semigroup by the stable one half law, and Cauchy$(\gamma, a)$ is the law of $\gamma + B(T_a)$ for a Brownian motion $B$ and the hitting time $T_a$ of level $a$ by an independent Brownian motion. Under RH the level $\theta$ Thorin measure is thus the zero measure run through Brownian motion to the hitting time of level $\theta - \frac{1}{2}$, and the backward flow is the unstopping of that Brownian motion; without RH the zeros carry widths $\theta - \sigma_\rho$ and the unconditional version is Section 9.2.3. That book reads this in thermodynamic terms; for the infinite measures here entropy is not defined, and the reading is heuristic, not an

independent handle on RH.

## 30.6 Is this a path to RH?

No, and it is worth being precise about why, because the chapter produces several objects that look like handles.

The level one measure is explicit and unconditional. Its positivity, which is what the Thorin theory gives for free, is automatic: by the Euler product every zero has $\sigma_\rho \le 1$, so every term of (30.1.1) at $\theta = 1$ is a positive measure. The nonvanishing of $\zeta$ on the line $\operatorname{Re} s = 1$ is equivalent to something else, the absence of atoms: $\mathfrak{U}_1$ has no atoms, and then has a continuous density, if and only if $\zeta(1 + it) \neq 0$ for every real $t$. Indeed a zero $1 + i\gamma_0$ would contribute the atom $\mathrm{Cauchy}(\gamma_0, 0) = \delta_{\gamma_0}$, which the other terms, all positive, cannot cancel, and if every width $1 - \sigma_\rho$ is positive the proof of Theorem 30.1.1 gives a continuous density. Everything to the left of the line enters only through the backward flow.

The backward flow is a deconvolution by $e^{(1-\theta')|\omega|}$. Its convergence at level $\theta'$ is equivalent, by Theorem 30.3.1, to the zero free half plane $\operatorname{Re} s > \theta'$, and the proof of that equivalence is the pole argument of Chapter 29. On the prime side the flow is the passage from the weights $\Lambda(n)/n$ to $\Lambda(n)n^{-\theta'}$, and its convergence is equivalent to the classical bound $\psi(x) - x = O(x^{\theta'+\varepsilon})$ for every $\varepsilon > 0$. The dictionary is exact and nothing is lost, but nothing is gained either: the error term in the prime number theorem has become the growth of a Fourier multiplier.

The Bayesian inversion, whether by EM, by data augmentation, or by generative Bayesian computation, is a numerical method for the inverse problem. Its residual is a diagnostic equivalent to the zero free half plane (Proposition 30.4.1), but equivalent is the operative word: the total residual stays bounded away from zero as the discretisation is refined exactly when there is a zero to the right of $\theta'$, and showing that it tends to zero at every $\theta' > \frac{1}{2}$ is RH. The GUE prior of the GBC version assumes the critical line outright. These are tools for computation and verification, not for proof.

What the chapter does contribute is narrower: the prime formula of Propositions 9.1.2 and 9.1.3 extended to every level $\theta \ge 1$ and to $v = 0$; the explicit formula of Theorem 20.1.1 read at level one, for one analytic test function; a dichotomy theorem for the backward flow; the total residual theorem for the discretised inversion; and a computation in which 109 zeros are read off the line $\operatorname{Re} s = 1$ by deconvolution to 0.03, with a first GBC implementation that recovers bin counts. The question that remains is Problem 29.5, now sharpened: find a property of the explicit measure (30.1.3), visible from the primes, that is preserved by the backward Poisson flow and forces it to remain a measure. Positivity is not such a property, since it is not preserved backward. We do not know one.

## 30.7 Open problems

Problem 30.1. Implement the data augmentation sampler of Section 30.4 for the real $u_1$ and compare its posterior spread at $\theta' = 0.75$ and $\theta' = 0.5$ with the EM point estimate; then build the GBC inversion with the EM reconstruction as an input feature and zero positions as continuous outputs, and calibrate its quantiles on held out simulations.

Problem 30.2. Characterise, without reference to zeros, the functions of the form (30.1.3): which densities $\frac{1}{\pi}[g(v) - \lim_X(\sum_{n\le X} a_n \cos(v\log n) - \sin(v\log X)/v)]$ with $a_n \ge 0$ supported on prime powers are Thorin densities at level one of a function $\Xi(1)/\Xi(1+\sqrt{s})$ with $\Xi$ in the Selberg class? The Thorin conditions constrain the growth and the integrability near $u = 0$ but say nothing about the arithmetic of the $a_n$. (The wording is the author's; the Selberg class needs the completed function, the gamma factors, the conductor and the poles to be specified, and the sign of the coefficients of $-L'/L$.)

Problem 30.3. Quantify the backward flow. For $\theta' \in (\theta_0, 1)$ give the rate of locally uniform convergence of $B_{\theta',\Omega}\mathfrak{U}_1$ to $\mathfrak{U}_{\theta'}$ in terms of $\Omega$ and $\theta' - \theta_0$, and give the matching lower bound in case (ii): with $\sigma_{\max} = \sup \sigma_\rho > \theta'$ attained by a zero, show that $\sup_{|v|\le V} |B_{\theta',\Omega}\mathfrak{U}_1(v)|$ grows like $e^{(\sigma_{\max}-\theta')\Omega}$ up to polynomial factors. For one zero with $d = \sigma - \theta' > 0$ the Fejér inverse at distance $t$ from its ordinate is $(\pi\Omega)^{-1}\operatorname{Re}[(e^{z\Omega} - 1 - z\Omega)/z^2]$ with $z = d + it$, which at $t = 0$ is asymptotic to $e^{d\Omega}/(\pi d^2\Omega)$; with infinitely many zeros the supremum need not be attained and the cancellations among them are part of the problem.

Problem 30.4. Replace the GUE prior by a prior that allows a fraction of zeros off the line, for instance pairs at $\frac{1}{2} \pm \delta + i\gamma$ with $\delta$ drawn from a prior on $[0, \frac{1}{2})$, and train the quantile network to report the posterior of $\delta$ for each detected zero. The posterior of $\delta$ at the real $u_1$ is then a window by window numerical diagnostic at the resolution the data allow; being conditional on the prior, it is not a certificate.

# Chapter 31

# Complexity and acceleration of the backward flow

Chapter 30 turned the backward Cauchy flow into a positive linear inverse problem and solved it by the EM algorithm of Vardi and Lee. This chapter asks three questions about that computation. What does it cost, and in particular what does log concave sampling in the sense of Applegate, Kannan and Polson say about it? Can the EM iteration be accelerated, by Nesterov momentum or otherwise, and what does the accelerated iteration compute? And does the representation of the Cauchy kernel as a scale mixture of normals, the oldest tool in the Bayesian robustness literature, help?

The chapter begins by stating the deconvolution problem precisely, including the matrix that the computations invert. It goes on to pictures of the Thorin measures at levels 1, 7/8, 3/4 and 1/2, a theorem on what their positivity means, an integer valued mass deficit that decides positivity up to any finite height, a level mass function whose jumps sit at the real parts of off line zeros, a certification in ball arithmetic of these statements up to height 1419.9, past the thousandth zero, and a frank account of the limits of the computation. One theorem closes the first part, in three forms: the level one Thorin density has its global minimum at the origin, unconditionally (Proposition 9.1.11); given the quasi Riemann hypothesis, announced and not refereed, the same holds at every level in $[7/8, 1]$; and in general, for every $\theta \in (\frac{1}{2}, 1]$, the zero free half plane $\operatorname{Re} s > \theta$ is equivalent to the global minimum of $u_{\theta'}$ at the origin for all $\theta' \in [\theta, 1]$, with the Riemann hypothesis as the case $\theta \downarrow \frac{1}{2}$ (Theorem 9.1.18). An a posteriori bound certifies the window masses and centroids of the backward flow from its residual, in ball arithmetic. A smoothed prime sum evaluates $u_1$ at any height to a fixed accuracy in time polynomial in the logarithm of the height, given that the zeros near that height lie on the critical line (verified below $3 \times 10^{12}$; at every height conditionally on the announced half plane $\operatorname{Re} s > 7/8$), and a nonuniform FFT gives a million heights in a second (the author's timing, not rerun); the same explicit formula, read backwards, is the heuristic reason why $\mathfrak{U}_{1/2}$ (equal to $2\mathfrak{U}_{1/2}$ of Theorem 29.14.4, see the notation below) cannot be computed efficiently by deconvolution, and how

the conjecture of Hilbert and Pólya reads in these terms.

Then the answers. The cost splits into three layers. Evaluating the level one measure needs $\zeta$ on the line $\operatorname{Re} s = 1$, for which the best algorithms are polynomial in the height $T$ and not in $\log T$; this is the bottleneck and the Thorin formulation does nothing to it. The inversion, given the data, is a convex minimisation, and its Bayesian version is a log concave sampling problem, to which the theorem of Applegate, Kannan and Polson applies with a ratio of radii that is not bounded here; the ill posedness appears as a precision requirement that is linear in the resolution for the linear inverse, hence polynomial in the bits requested. Verification is Turing's method and costs $T^{1+\varepsilon}$. The EM iteration is a majorise minimise scheme in the Kullback Leibler geometry; Nesterov momentum applied in the logarithmic domain keeps positivity, converges an order of magnitude faster at $\theta' = 1/2$ in the runs, and in every run reached the same object, the reverse information projection of the data onto the cone of Cauchy blurred positive measures, which is the Thorin measure at level $\theta'$ when it exists (Proposition 31.3.4). The scale mixture of normals turns the Cauchy blur into a heat flow at a random time with the first passage law of Brownian motion, gives a conditionally Gaussian data augmentation sampler with explicit inverse gamma updates, and identifies the Thorin measure at level $\theta$ with the subordinated heat trace of the zeros; it changes nothing in the complexity classification, and nothing in the arithmetic.

The chapter develops a source chapter on complexity and acceleration of the backward flow by Nick Polson, together with the computations done for this book. Its numbers were recomputed or checked here by scripts written from the mathematics alone (the scripts of the source were not used): `code/cx_em.py` for the EM tables of Section 31.4, `code/cx_u_certify.py` for Theorem 31.9.6, `code/cx_min.py` for the global minimum certificates and `code/cx_slope_X.py` for Proposition 31.11.4, with outputs `data/cx_*.json`, listed in `code/README.md`. Every number printed is one of those recomputations unless it is marked as the author's, and where a recomputation differed from the source the recomputed value is printed. The a posteriori certificate (Theorem 31.11.8) was computed by the author and was not rerun for this book. Several parts of the source are in the book already and are cited, not restated: the semigroup of Cauchy convolutions and the Thorin measure (Theorem 29.14.4, Theorem 29.14.2), the existence criterion for the deconvolution (Theorem 19.2.2), the density, the backward multiplier and the EM set up of Chapter 30, the global minimum of the level one density and its equivalent forms (Proposition 9.1.11, Theorem 9.1.18) and the zero counts (HSW22 [176], in the form used in Chapter 9). What this chapter adds is the theory of EM for the backward flow with its repaired proofs, the acceleration experiments, the analysis of the von Neumann and Ulam estimator, the mass deficit identity, and the certificates of the positivity of $u_{3/4}$ and $u_{7/8}$ on $[0, 1420]$.

The notation is the book's, with the following warnings. A level $\theta$ is the real part of a point $s = \theta + iv$, and $u_\theta(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\theta + iv)$ is the density of

Chapters 29 to 33; in the notation of Chapter 19 it is $\rho_{\theta-1/2}(v)$, the ordinate being $v$ here. The Cauchy density is $P_a(x) = (a/\pi)/(a^2 + x^2)$. The symmetric Thorin measure $\mathfrak{U}_\theta$ of equation (30.1.1) has density $\sum_\rho P_{\theta-\sigma_\rho}(v - \gamma_\rho)$ with no factor $\frac{1}{2}$, so $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ with the $\widetilde{U}_\theta$ of Theorem 29.14.4. The source's Chapters 2 and 3 are Chapters 29 and 30 of the book. The word certified is used only for a computation in ball arithmetic whose sentence says what the certificate covers. The announced zero free half plane $\operatorname{Re} s > 7/8$ (OpenAI [398, 399], with a formalisation in Lean, and Liu [352]; announced in October 2026 and not yet refereed) is never used as a theorem: a result that uses it is stated as conditional on it.

Status words follow the rest of the book: proved, computed, conditional, heuristic or open. The results are these.

1. The Thorin measure at level $\theta'$ is the solution of a positive deconvolution; the matrix $A$ of its discretisation, with cell densities as unknowns, has the properties (a) to (e) of Section 31.1 (proved), including the kernel of dimension exactly $2(N - M)$ when the unknown window is larger. The level one data must be supplied to absolute precision $e^{-(1-\theta')\Omega}$ for the linear inverse (Proposition 31.2.2, proved). Log concavity of the Poisson posterior is proved (Theorem 31.2.1); the sampling clause is a cited theorem and is conditional on a bound for the ratio of radii that is not proved here.

2. EM is a majorise minimise scheme with the rate $D(x^*\|x^{(0)})/k$ in the consistent case (Theorem 31.3.1, proved, with the repaired step) and local linear convergence in the interior (Theorem 31.3.2, proved; its injectivity hypothesis fails in the runs). What EM computes: under three growth conditions its limits converge vaguely to $\mathfrak{U}_{\theta'}$ when $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$, and the residual stays bounded away from zero when it has one (Proposition 31.3.4, proved).

3. Log domain Nesterov momentum with restart and SQUAREM are an order of magnitude faster than EM at $\theta' = 1/2$ on a window; on the full range (202 zeros below height 400) and on the first thousand zeros (994 of 995 zeros below 1415 found as peaks) they locate the zeros to about $10^{-2}$ (Section 31.4, computed; the rates $O(1/k)$ and $O(1/k^2)$ and the identification of the limit are observed, not proved).

4. The von Neumann and Ulam estimator for the backward flow has infinite variance whenever $h < 0.5221\,\pi(1 - \theta')$ (Proposition 31.5.1, proved); the positive reverse walk is the EM step in expectation. The Cauchy kernel is a first passage mixture of Gaussians, giving the subordinated heat trace of the zeros and an inverse gamma data augmentation sampler (Proposition 31.6.1, proved; the sampler comparison is computed).

5. The level 7/8 Thorin measure is computed by three routes (computed); its identification as a Thorin measure is conditional on the announced half plane.

The positivity theorem (Theorem 31.9.1), the mass deficit identity (Proposition 31.9.2) and the level mass function (Corollary 31.9.3) are proved; the deficit tables, the $U_{3/4}$ and $U_{1/2}$ tables and Turing's check in Thorin bookkeeping are computed. Decidability by level is proved for finite heights (Section 31.10).

6. Certified positivity (Theorem 31.9.6, computed in ball arithmetic at two precisions, 96 and 160 bits): the densities $u_{3/4}$ and $u_{7/8}$ are positive on $[0, 1420]$, every certified lower bound exceeding $4 \times 10^{-5}$ in the recomputation; the argument change of $\xi$ gives exactly 1000 zeros below height 1419.9, all on the critical line and simple, which is known (Platt and Trudgian [419]); the positivity of $u_{3/4}$ and $u_{7/8}$ on this range is what the certificate adds.

7. The global minimum of the Thorin density is at the origin: at level 1 this is Proposition 9.1.11 (the proof of Theorem 31.10.2(i) is an independent computed one); at level 7/8 it holds conditionally on the announced half plane and is computed, with least certified bound on $[13.5, 1420]$ equal to 0.0360 at $\theta = 1$ and 0.0272 at $\theta = 7/8$, both at $v = 17.49$ (Theorem 31.10.2); the Riemann hypothesis form is Theorem 9.1.18(vi) (Theorem 31.10.3). The uniform law and the lower bound for the excess, $W \geq w^*$ with its brackets (Theorems 31.11.1 and 31.11.3), are computed, and conditional on the announced half plane at level 7/8.

8. The slope of the level one density has certified infima near 0.1758 to 0.1793 (Proposition 31.11.4, computed), and under the Riemann hypothesis the limiting bracket $X$ has its minimum in the first gap at $v^* = 17.4979 \pm 0.0001$, with $0.1838985 \leq X(v^*) \leq 0.1838988$ (computed, conditional on RH); that this is its infimum over $13.5 \leq |v| \leq 1418.9$ is the author's computation, not rerun.

9. The a posteriori bound for the window masses and centroids of the backward flow, from its residual, is proved (Proposition 31.11.5); the certificate of Theorem 31.11.8 is computed by the author and not rerun here, and is conditional on the absence of zeros to the right of $\theta'$ (the Riemann hypothesis at $\theta' = \frac{1}{2}$); it certifies Gaussian weighted masses and centroids, and single zeros only through Corollary 31.11.9 (proved).

10. The smoothed prime sum for $u_1$ has an exact identity (proved) and an error bound proved under the local hypothesis on the zeros, with the pole term that the source had omitted (Theorem 31.11.10); conditionally on the announced half plane it holds at every height with $X$ of order $(\log v/\varepsilon)^8$; the nonuniform FFT experiment is computed (by the author, not rerun).

11. The explicit formula read backwards, and the efficiency of the backward flow (Section 31.11.10), and the reading of the Hilbert and Pólya conjecture as

a resolvent trace (Section 31.11.11), are heuristic. The open problems close the chapter.

## 31.1 The deconvolution problem, precisely

### 31.1.1 Why the Thorin measure is a deconvolution

Fix $\frac{1}{2} \le \theta' < \theta$ and write $P_a(x) = (a/\pi)/(a^2 + x^2)$ for the density of the Cauchy law of scale $a$. Theorem 29.14.4 says that if $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$ then

$$\mathfrak{U}_\theta = P_{\theta-\theta'} * \mathfrak{U}_{\theta'}. \tag{31.1.1}$$

The left side is known at $\theta = 1$ without any information about zeros: its density is $u_1(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + iv)$ (Theorem 30.1.1). The right side contains the unknown $\mathfrak{U}_{\theta'}$. So the Thorin measure at level $\theta'$ is, by the semigroup property, the solution $\mu$ of the convolution equation

$$P_{1-\theta'} * \mu = \mathfrak{U}_1, \qquad \mu \ge 0, \tag{31.1.2}$$

and finding it is a deconvolution. Three facts fix the nature of the problem.

Uniqueness. The Fourier transform of $P_a$ is $e^{-a|\omega|}$, which never vanishes, so $\mu \mapsto P_a * \mu$ is injective on signed measures $\mu$ with $\int (1 + v^2)^{-1} d|\mu| < \infty$ (Theorem 19.2.2): (31.1.2) has at most one solution.

Existence. A positive solution of (31.1.2) exists if and only if $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$, and it is then $\mathfrak{U}_{\theta'}$ (Theorem 19.2.2). The deconvolution is therefore not an estimation problem but an existence problem: existence at $\theta'$ is the half plane $\operatorname{Re} s > \theta'$, and at all $\theta' > \frac{1}{2}$ together it is the Riemann hypothesis.

Ill posedness. The inverse of $\mu \mapsto P_a * \mu$ has Fourier symbol $e^{a|\omega|}$: the problem is ill posed in Hadamard's sense, with exponential rather than polynomial growth of the inverse. A perturbation of $\mathfrak{U}_1$ of size $\delta$ at frequency $\omega$ becomes $\delta e^{(1-\theta')|\omega|}$ in $\mu$.

Two inversions should be distinguished. The Thorin measure of $F_\theta(s) = \xi(\theta)/\xi(\theta + \sqrt{s})$ at its own level is obtained from $F_\theta$ by Stieltjes inversion, $u_\theta(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\theta + iv)$, which is the boundary value of $-(\log F_\theta)'$ on the cut and is immediate once $\xi'/\xi$ is available at level $\theta$. The passage between levels is (31.1.2), and its difficulty is the difficulty of continuing $\xi'/\xi$ from the line $\operatorname{Re} s = 1$ to the line $\operatorname{Re} s = \theta'$: the Cauchy kernel is the Poisson kernel of the half plane $\operatorname{Re} s > \theta'$, so (31.1.1) is the Poisson integral formula for the harmonic function $\partial_\sigma \log|\xi|$, and deconvolution is harmonic continuation backward, into the region where the function may fail to be harmonic because of a zero.

### 31.1.2 The matrix $A$

Discretise (31.1.2) with a common spacing $h$. The data grid is $v_i = ih$, $|i| \le M$, covering $[-V, V]$ with $V = Mh$, and the unknown grid is $w_j = jh$, $|j| \le N$,

covering $[-W, W]$ with $W = Nh > V$. The unknown is the vector of cell densities $x_j = h^{-1}\mu([w_j - \frac{h}{2}, w_j + \frac{h}{2}))$, so that $\mu \approx \sum_j x_j h\, \delta_{w_j}$, and the data vector is $y_i = u_1(v_i)$. Then

$$A_{ij} = h\, P_{1-\theta'}(v_i - w_j) = \frac{h(1-\theta')/\pi}{(1-\theta')^2 + (i-j)^2 h^2}, \tag{31.1.3}$$

so $A \in \mathbb{R}^{(2M+1)\times(2N+1)}$, and $(Ax)_i = \sum_j x_j h P_{1-\theta'}(v_i - w_j)$ is the midpoint rule for the density of $P_{1-\theta'} * \mu$ at $v_i$. (Chapter 30 writes the same products with the cell masses $hx_j$ as unknowns and the matrix $P_{1-\theta'}(v_i - w_j)$; masses together with the factor $h$ in $A$ would count it twice. The residuals of both chapters include the factor $h$.) The properties of $A$ that the chapter uses are the following.

(a) Every entry is positive, so $A$ maps the orthant into the orthant and $y = Ax$ is a positive linear inverse problem in the sense of Vardi and Lee [356].

(b) $A_{ij}$ depends only on $i - j$: $A$ is the restriction to $|i| \le M$, $|j| \le N$ of the symmetric Toeplitz operator with kernel $k_m = hP_{1-\theta'}(mh)$, $m \in \mathbb{Z}$. Hence $Ax$ and $A^{\mathsf{T}} r$ are both convolutions with $k$, restricted to the appropriate index range, and can be computed by the fast Fourier transform after circulant embedding in length $L \ge 2(2N+1)$. The dense form has $(2M+1)(2N+1)$ entries, $4001 \times 4801$ for the window runs below; the FFT form costs $O(L \log L)$ per product, which is what makes the full range feasible.

(c) The column sums $c_j = \sum_{|i|\le M} A_{ij} = h \sum_{|i|\le M} P_{1-\theta'}((i-j)h)$ equal 1 up to the discretisation error and the Cauchy tail outside the data window: for $|w_j| < V$,

$$\begin{aligned} c_j = 1 + O\big(e^{-2\pi(1-\theta')/h}\big) - \frac{1-\theta'}{\pi}\Big(\frac{1}{V - w_j} + \frac{1}{V + w_j}\Big) \\ + O\big(h(V - |w_j|)^{-2} + (V - |w_j|)^{-3}\big), \end{aligned}$$

because $h \sum_{m\in\mathbb{Z}} P_a(mh) = \coth(\pi a/h)$ and each tail is $\pi^{-1} \arctan\big(a/(V \mp w_j + \frac{h}{2})\big)$ up to the midpoint error. They are what EM divides by, and the division is exactly what accounts for mass that leaves the data window. This is also why $W > V$: zeros just outside $[-V, V]$ contribute Cauchy tails to the data inside, and the unknown grid must have room for them.

(d) The symbol of the bi infinite Toeplitz operator is, by Poisson summation,

$$\begin{aligned} \phi(\xi) &= \sum_{m\in\mathbb{Z}} k_m e^{im\xi} = \sum_{\ell\in\mathbb{Z}} e^{-(1-\theta')|\xi + 2\pi\ell|/h} \\ &= \frac{\cosh\big(r(\pi - |\xi|)\big)}{\sinh(\pi r)}, \qquad |\xi| \le \pi, \quad r = \frac{1-\theta'}{h}, \end{aligned}$$

so the nonzero singular values of $A$ lie between $1/\sinh(\pi r)$ and $\coth(\pi r)$, and the condition number of the bi infinite operator is $\cosh(\pi r)$, of order $e^{(1-\theta')\pi/h}$, which is $e^{78}$ for $h = 0.02$ and $\theta' = 1/2$. The frequency $\omega = \xi/h$ is the one of the multiplier $e^{(1-\theta')|\omega|}$ of Chapter 30 (equation (30.3.1)), and the singular value at frequency $\omega$ is $e^{-(1-\theta')|\omega|}$, the Fourier symbol of the Cauchy kernel.

(e) If $N \le M$, $A$ is injective. If $N > M$, $A$ has rank $2M + 1$ and a kernel of dimension exactly $2(N - M)$, for every $h$. Every entry of $A$ is positive, so the kernel contains no nonzero $x \ge 0$: its vectors are signed, and when $2(N - M) \le M + 1$, that is when the unknown window exceeds the data window by at most about half its length, none of them vanishes on the data window. EM updates every coordinate. The minimisers of $\mathrm{KL}(y\|Ax)$ over $x \ge 0$ share one image $z^* = Ax$ and form the polytope $\{x \ge 0 : Ax = z^*\}$; the EM limit is a point of it that may depend on the start. For fixed windows the continuous operator $\mu \mapsto (P_{1-\theta'} * \mu)|_{[-V,V]}$ on measures carried by $[-W, W]$ is injective, because $P_{1-\theta'} * \mu$ is real analytic, but its inverse is unbounded. So the uniqueness clause of Theorem 31.3.1 and the hypothesis of Theorem 31.3.2 fail in the runs below, where $N > M$, while the bound on $f$ in Theorem 31.3.1 holds.

*Proof of (c) to (e).* For (c) and (d), Poisson summation gives $\sum_m hP_a(mh)e^{im\xi} = \sum_\ell e^{-a|\xi+2\pi\ell|/h}$, which sums to the displayed hyperbolic quotient, and at $\xi = 0$ to $\coth(\pi r)$. For (e), with $r = (1-\theta')/h$ every finite section $B$ of the Toeplitz operator satisfies $z^{\mathsf{T}}Bz = \frac{1}{2\pi}\int_{-\pi}^{\pi} \phi(\xi)\left|\sum_j z_j e^{ij\xi}\right|^2 d\xi \ge \phi(\pi)|z|^2$, and $\phi(\pi) = 1/\sinh(\pi r) > 0$. If $N \le M$ the rows $|i| \le N$ of $A$ form such a section; if $N > M$ the central $(2M + 1)$ block is one, so $A$ has full row rank and nullity $(2N + 1) - (2M + 1)$. When $2(N - M) \le M + 1$, a kernel vector carried by $|j| > M$ gives $\sum_j z_j(a^2 + (v - w_j)^2)^{-1} = 0$ at the $2M + 1$ points $v_i$; this rational function has distinct poles $w_j \pm ia$ and a numerator of degree at most $4(N - M) - 2 < 2M + 1$, so all $z_j = 0$. The function $\mathrm{KL}(y\|z)$ is strictly convex in $z$ for $y > 0$. For $\mu$ carried by $[-W, W]$, $P_a * \mu$ extends holomorphically to $|\operatorname{Im} v| < a$, so vanishing on $[-V, V]$ forces $P_a * \mu = 0$ and $\mu = 0$. □

### 31.1.3 Three inverses of the same matrix

The three computations of Chapter 30 are three inverses of (31.1.3).

The Fourier backward flow is the truncated singular value inverse. For the bi infinite operator the singular vectors are the Fourier modes and the singular values are $e^{-(1-\theta')|\omega|}$, so the pseudo inverse with all singular values below $e^{-(1-\theta')\Omega}$ discarded is the multiplier $e^{(1-\theta')|\omega|}\mathbf{1}_{|\omega|\le\Omega}$, which is $B_{\theta',\Omega}$ of equation (30.3.1) with a sharp window. It is linear, ignores positivity, and rings.

EM is the maximum likelihood inverse under the Poisson model: the minimiser of $\mathrm{KL}(y\|Ax)$ over $x \ge 0$. It uses every singular value, enforces positivity, and in

the consistent case converges to a minimiser, unique when $A$ is injective, at the rate of Theorem 31.3.1.

The Bayesian inverse is the posterior over $x \geq 0$ under the Poisson likelihood and a gamma prior, sampled by data augmentation or by the random walk of Theorem 31.2.1; its mode is the EM solution when the prior is flat.

## 31.2 Three layers of cost

Fix a height $T$, a level $\theta' \in [\tfrac12, 1)$, a grid spacing $h$, and an accuracy $\varepsilon$.

### 31.2.1 The data

The level one density $u_1(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1+iv)$ requires $\zeta(1+iv)$ and $\zeta'(1+iv)$. The Riemann Siegel formula costs $v^{1/2+\varepsilon}$ per evaluation; Hiary's algorithms cost $v^{1/3+\varepsilon}$ [171] and $v^{4/13+\varepsilon}$ (the exponents are the source's, unchecked); the Odlyzko and Schönhage method [392] evaluates $\zeta$ at all heights up to $T$ on a grid of spacing $T^{-1/2}$ in time $T^{1+\varepsilon}$, which is $T^{\varepsilon}$ per point. All of these are polynomial in $T$ and exponential in the bit length of $T$. No algorithm polynomial in $\log T$ is known for $\zeta$ at a single point of a vertical line. So the data layer costs $T^{1+\varepsilon}$ for the measure up to height $T$, independently of everything that follows.

### 31.2.2 The inversion

The discretised problem of Chapter 30 is $y = Ax$ with $A_{ij} = hP_{1-\theta'}(v_i - w_j)$, $n$ unknowns $x_j \geq 0$, and the Poisson log likelihood

$$\ell(x) = \sum_{i} \big( y_i \log (Ax)_i - (Ax)_i \big). \tag{31.2.1}$$

Theorem 31.2.1 (log concavity). *The function $\ell$ is concave on the open orthant $x > 0$. With independent gamma priors $x_j \sim \operatorname{Gamma}(\alpha_j, \beta_j)$, $\alpha_j \geq 1$, the posterior density $\pi(x) \propto e^{\ell(x)} \prod_j x_j^{\alpha_j - 1} e^{-\beta_j x_j}$ is log concave on the orthant. Consequently, for every $\varepsilon > 0$, a sample from a distribution within total variation $\varepsilon$ of $\pi$ can be drawn by a random walk in time polynomial in $n$, $\log(1/\varepsilon)$ and the ratio of the radii of a ball containing most of the mass of $\pi$ and a ball on which $\pi$ is bounded below, given an oracle for $\ell$.*

*Proof.* Each $(Ax)_i$ is a positive linear form, log is concave and increasing, and $y_i \geq 0$, so $y_i \log (Ax)_i$ is concave; $-(Ax)_i$ is linear. The log prior $(\alpha_j - 1)\log x_j - \beta_j x_j$ is concave for $\alpha_j \geq 1$. The orthant is convex. The sampling statement is the random polynomial time theorem for log concave distributions of Applegate, Kannan and Polson [5] and Applegate and Kannan [4], in the form given by Lovász and Vempala [213]; the condition on radii is their roundness condition, and it is not verified here. $\square$

So sampling the posterior to total variation $\varepsilon$ takes time polynomial in $n$, $\log(1/\varepsilon)$ and the ratio of radii, given an oracle for $\ell$; that ratio is not bounded here. The EM limit minimises a convex function and is computed to accuracy $\varepsilon$ in function value in deterministic polynomial time. We expect that the ill posedness of the deconvolution does not appear in the running time; it appears in the precision of the data that the computation must be given.

Proposition 31.2.2 (precision). *Let $\Omega$ be the largest frequency to be resolved in the level $\theta'$ measure. Then, for the linear inverse, the level one data must be supplied to absolute precision, on the scale of $u_1$, $e^{-(1-\theta')\Omega}$, that is, to $(1-\theta')\Omega/\log 2$ binary digits, and this is sharp for the linear inverse: an error of size $\delta$ at frequency $\omega$ in $\widehat{u_1}$ produces an error of size $\delta e^{(1-\theta')|\omega|}$ in the inverted measure.*

*Proof.* The backward multiplier is $e^{(1-\theta')|\omega|}$ (equation (30.3.1)). □

The number of bits grows linearly in $\Omega$, so resolving $\Omega$ is polynomial in $\Omega$ in bit complexity. Resolving the zeros near height $T$ needs $\Omega$ of order $\log T$, their mean spacing being $2\pi/\log T$ (close pairs need more), so the precision needed to read the zeros off the line $\operatorname{Re} s = 1$ at level $\theta'$ is $(1-\theta')\log_2 T$ bits plus a constant, which is polynomial in $\log T$. The precision layer is cheap. It is the evaluation of $\zeta$ to that precision at height $T$ which is not.

### 31.2.3 Verification

Certifying that no zero lies to the right of $\theta'$ below height $T$ is Turing's method [506], an application of the argument principle to $\xi$ on a rectangle, and costs $T^{1+\varepsilon}$. The EM residual of Proposition 31.3.4(b) is a diagnostic for the same statement; heuristically, a residual computed to precision $\varepsilon$ on a window resolved to frequency $\Omega$ detects a zero to the right of $\theta'$ only beyond a distance set by $\varepsilon$ and $\Omega$ (no bound is proved here), and obtaining that precision costs the data layer.

Remark 31.2.3. Deciding the Riemann hypothesis is not known to lie in any complexity class, and the three layers above do not place it in one. What they give is conditional: given $\zeta$ on the line $\operatorname{Re} s = 1$ to the required precision, the EM limit at a level above the zero free abscissa is a convex minimisation, and the posterior over it can be sampled in the time of Theorem 31.2.1, whose ratio of radii is not bounded here.

## 31.3 EM as majorise minimise

Write $f(x) = -\ell(x) = \sum_i \big((Ax)_i - y_i \log (Ax)_i\big)$, convex on the orthant, and $c_j = \sum_i A_{ij}$. The EM iteration of Chapter 30 is

$$x_j^{+} = x_j \, \frac{1}{c_j} \sum_i A_{ij} \frac{y_i}{(Ax)_i} = x_j \Big(1 - \frac{\partial_j f(x)}{c_j}\Big). \tag{31.3.1}$$

The bound of Theorem 31.3.1 below and the Jacobian of Theorem 31.3.2 are classical for EM with a Poisson likelihood [356, 102]; the proofs are given to fix the constants and the hypotheses. Three readings of (31.3.1) are useful.

It is an MM algorithm. Jensen's inequality for $-\log(Ax)_i$, with weights proportional to $A_{ij}x_j^{(k)}$, majorises $f$ at $x^{(k)}$ by a separable surrogate, and (31.3.1) is the exact minimiser of the surrogate. Hence $f(x^{(k+1)}) \le f(x^{(k)})$, with equality only at a fixed point, and every fixed point with $x > 0$ is a minimiser of the convex $f$.

It is related to mirror descent. With the entropy mirror map, the mirror descent step of size $\eta$ is

$$x_j^+ = x_j \exp(-\eta\, \partial_j f(x));$$

(31.3.1) is the first order expansion of this with $\eta = 1/c_j$. In the language of relative smoothness [24], $f$ is $\|y\|_1$ smooth relative to the Burg entropy $-\sum_j \log x_j$ on the orthant (as the source states it, with the constant unchecked), which places EM near the Bregman proximal methods with their sublinear guarantees; EM is not literally the Bregman step for the Burg entropy. A direct argument gives the rate with an explicit constant.

THEOREM 31.3.1 (rate of EM in the consistent case). *Normalise the columns so that $c_j = 1$, and suppose the problem is consistent: $y = Ax^*$ for some $x^* \ge 0$. Then for every $x^{(0)} > 0$ and $k \ge 1$ the EM iterates satisfy*

$$f(x^{(k)}) - f(x^*) \le \frac{D(x^* \,\|\, x^{(0)})}{k}, \quad D(x^*\|x) = \sum_j \Big( x_j^* \log \frac{x_j^*}{x_j} - x_j^* + x_j \Big).$$

*If moreover $y > 0$ and $A$ is injective, the minimiser is unique and $x^{(k)} \to x^*$.*

*Proof.* For every $x > 0$, $\sum_i (Ax)_i = \sum_j x_j$ because $c_j = 1$, and $Ax^* = y$ gives $f(x^*) = \sum_i y_i - \sum_i y_i \log y_i$, so

$$\begin{aligned} f(x) - f(x^*) &= \sum_i \Big( y_i \log \frac{y_i}{(Ax)_i} - y_i + (Ax)_i \Big) \\ &= \sum_i y_i \log \frac{y_i}{(Ax)_i} + \sum_j x_j - \sum_i y_i \;\ge 0. \end{aligned}$$

Summing (31.3.1) over $j$ with $c_j = 1$ gives $\sum_j x_j^+ = \sum_i y_i$. Now

$$\begin{aligned} D(x^*\|x^+) - D(x^*\|x) &= \sum_j x_j^* \log \frac{x_j}{x_j^+} + \sum_j x_j^+ - \sum_j x_j \\ &\le -\sum_j x_j^* \sum_i A_{ij} \log \frac{y_i}{(Ax)_i} + \sum_i y_i - \sum_j x_j \end{aligned}$$

by Jensen's inequality with the weights $A_{ij}$, $\sum_i A_{ij} = 1$, applied to $x_j^+/x_j = \sum_i A_{ij} y_i/(Ax)_i$. The first term on the right is $-\sum_i (Ax^*)_i \log(y_i/(Ax)_i) = -\sum_i y_i \log(y_i/(Ax)_i)$, so the right side is $-(f(x) - f(x^*))$. Thus $D(x^*\|x^{(j+1)}) \le D(x^*\|x^{(j)}) - (f(x^{(j)}) - f(x^*))$ for every $j \ge 0$. Summing over $j < k$ and using that $f(x^{(j)})$ is nonincreasing, $k\,(f(x^{(k)}) - f(x^*)) \le \sum_{j<k} (f(x^{(j)}) - f(x^*)) \le D(x^*\|x^{(0)})$. If $y > 0$ and $A$ is injective, $f$ is strictly convex, since $t \mapsto -y_i \log t$ is strictly convex and $x \mapsto Ax$ is one to one, so the minimiser is unique; the iterates are bounded by the mass identity, and since $f(x^{(k)}) \to f(x^*)$ every limit point is a minimiser, hence equals $x^*$. □

The constant explains the slow convergence at the boundary. When the target is atomic, $x^*$ puts mass about one on a few cells of width $h$ around each zero, and $D(x^*\|x^{(0)})$ for a flat start is of the order $N \log(1/(hd))$ with $N$ the number of zeros and $d$ their density (a heuristic estimate): the bound degrades like the entropy of the target relative to the start. In the interior, where $x^*$ is a smooth positive density, the convergence is linear, and the rate can be computed.

THEOREM 31.3.2 (local linear convergence in the interior). *In the setting of Theorem 31.3.1, with $c_j = 1$, suppose $x^* > 0$ componentwise and $A$ injective. Let $B = \operatorname{diag}(y)^{-1/2} A \operatorname{diag}(x^*)^{1/2}$ and $S = B^{\mathsf{T}} B$. Then the eigenvalues of $S$ lie in $(0, 1]$, with 1 attained, and the Jacobian of the EM map at $x^*$ is similar to $I - S$, so its spectrum lies in $[0, 1)$ and*

$$\|x^{(k)} - x^*\| = O\big((1 - \lambda_{\min}(S))^k\big)$$

*for $x^{(0)}$ near $x^*$. The rate $1 - \lambda_{\min}(S)$ is that of the linearised iteration along the eigenvector $\operatorname{diag}(x^*)^{1/2} s$ of the Jacobian, $s$ an eigenvector of $S$ for $\lambda_{\min}$, and $\lambda_{\min}(S) \ge \min_j x_j^* \cdot \sigma_{\min}(A)^2 / \max_i y_i$.*

*Proof.* Write $M(x)_j = x_j \sum_i A_{ij} y_i/(Ax)_i$ for the EM map. Differentiating,

$$\frac{\partial M_j}{\partial x_k} = \delta_{jk} \sum_i A_{ij} \frac{y_i}{(Ax)_i} - x_j \sum_i A_{ij} A_{ik} \frac{y_i}{(Ax)_i^2}.$$

At $x^*$ the first sum is $c_j = 1$ because $Ax^* = y$, and the second is $(\operatorname{diag}(x^*) A^{\mathsf{T}} \operatorname{diag}(y)^{-1} A)_{jk}$. So $J = I - \operatorname{diag}(x^*) A^{\mathsf{T}} \operatorname{diag}(y)^{-1} A = I - \operatorname{diag}(x^*)^{1/2}\, S \operatorname{diag}(x^*)^{-1/2}$, which is similar to $I - S$. $S$ is symmetric positive semidefinite and positive definite when $A$ is injective and $x^* > 0$. For the upper bound on its spectrum, $BB^{\mathsf{T}} = \operatorname{diag}(y)^{-1/2} A \operatorname{diag}(x^*) A^{\mathsf{T}} \operatorname{diag}(y)^{-1/2}$ has the same nonzero eigenvalues as $S$, and $P = A \operatorname{diag}(x^*) A^{\mathsf{T}}$ is a positive matrix with row sums $(P\mathbf{1})_i = \sum_j A_{ij} x_j^* c_j = y_i$. Hence $\operatorname{diag}(y)^{-1} P$ is a positive stochastic matrix, with Perron root 1 and all eigenvalues of modulus at most 1, and $BB^{\mathsf{T}}$ is similar to it. So the spectrum of $S$ lies in $(0, 1]$, that of $J$ in $[0, 1)$, and the local linear rate follows from the contraction of the linearisation. The lower bound on $\lambda_{\min}(S)$ is $\min_j x_j^* \, \sigma_{\min}(A)^2 / \max_i y_i$ from the definition of $B$. □

Two consequences, for the linearised iteration. First, the spectrum of $J$ is nonnegative, so each linearised error mode decays monotonically and EM does not oscillate near $x^*$. Second, the rate is bounded through $\sigma_{\min}(A)^2$, which by the symbol computation of Section 31.1 (d) is at least $1/\sinh^2(\pi r)$, of order $e^{-2(1-\theta')\pi/h}$, for the frequencies the grid can represent: the linear regime is real but its rate constant can be exponentially close to one for the highest modes, which is the ill posedness again. At the boundary level $\theta' = 1/2$ the hypothesis $x^* > 0$ fails between the zeros, and there the inactive components converge sublinearly, at the rate of Theorem 31.3.1 (an observation); the full range runs, where $A$ has a kernel and the theorem does not apply, behave in the same way.

Remark 31.3.3 (a limit on acceleration). Dragomir, Taylor, d'Aspremont and Bolte [122] showed that for problems that are merely smooth relative to a Bregman distance, no Bregman first order method can improve the $O(1/k)$ rate in general. Any acceleration of EM for the backward flow therefore exploits the particular structure of the problem rather than a general theorem, and we give none; the $O(1/k^2)$ behaviour reported below is observed, not proved.

It is an information projection. The minimisers of $f$ are the minimisers of $\mathrm{KL}(y \,\|\, Ax)$, so the EM limit is the reverse $I$ projection of the data onto the convex cone $\{Ax : x \geq 0\}$, and the iteration is the alternating minimisation of Csiszár and Tusnády [102] between that cone and the data, the information geometric form of von Neumann's alternating projections.

Proposition 31.3.4 (what EM computes). *Let $\frac{1}{2} \leq \theta' < 1$ and $a = 1 - \theta'$. For a grid of spacing $h$ with data window $[-V, V]$ and unknown window $[-W, W]$, $W > V$, let*

$$R_h(x) = h \sum_{|i| \leq M} \Big( y_i \log \frac{y_i}{(Ax)_i} - y_i + (Ax)_i \Big), \qquad R_h^* = \min_{x \geq 0} R_h(x),$$

*the residual quoted in the tables below (it is not divided by the length of the data window), let $x_h$ be any minimiser, for instance the limit of EM from a positive start, and let $\mu_h = \sum_j x_{h,j} h \, \delta_{w_j}$. Consider a sequence of grids with $h \to 0$ and $V \to \infty$.*

*(a) If $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$ and*

$$\frac{(\log W)^2}{W - V} \to 0, \qquad \frac{(W - V)^2 \log V}{V^2} \to 0, \qquad h^2 V \log^2 V \to 0,$$

*then $R_h^* \to 0$ and $\mu_h \to \mathfrak{U}_{\theta'}$ vaguely.*

*(b) If $\zeta$ has a zero with real part greater than $\theta'$, then $\liminf R_h^* > 0$, whatever the choice of $W > V$.*

*Proof.* Constants $C$ depend on $\theta'$ only. For $y, z > 0$,

$$(\sqrt{y} - \sqrt{z})^2 \;\le\; y \log \frac{y}{z} - y + z \;\le\; \frac{(y-z)^2}{z}, \tag{31.3.2}$$

which with $t = y/z$ reduces to $s \log s - s + 1 \ge 0$, $s = \sqrt{t}$, on the left and to $t - 1 - \log t \ge 0$ on the right. Since $|P_a'(x)| = \frac{2|x|}{a^2+x^2} P_a(x) \le P_a(x)/a$, every positive measure $\nu$ with $g = P_a * \nu$ finite satisfies

$$g(v) \le e^{|v-u|/a} g(u) \qquad (u, v \in \mathbb{R}), \tag{31.3.3}$$

and so does $P_a$ itself. We use $u_1(0) \le u_1(v) \le C \log(|v| + 3)$ (Proposition 9.1.11 and $\zeta'/\zeta(1+iv) = O(\log|v|)$), and that a measure $\sum_\rho P_{a_\rho}(\cdot - \gamma_\rho)$ with $0 \le a_\rho \le \frac12$ gives mass at most $C \log(|t| + 3)$ to $[t, t+1]$, by the zero count.

*Step 1 (comparison point; for (a) only).* Put $x_j^* = h^{-1} \mathfrak{U}_{\theta'}([w_j - \frac{h}{2}, w_j + \frac{h}{2}))$ and $z^* = Ax^*$. Since $y_i = \int P_a(v_i - w)\, d\mathfrak{U}_{\theta'}(w)$ and $|P_a(x+s) - P_a(x)| \le \frac{|s|}{a} e^{|s|/a} P_a(x)$ by (31.3.3), for $h \le a$

$$|z_i^* - y_i + \tau_i| \le \frac{h}{a} y_i, \qquad \tau_i = \int_{|w| \ge W + h/2} P_a(v_i - w)\, d\mathfrak{U}_{\theta'}(w) \le \frac{C \log W}{W - |v_i|}.$$

So $z_i^* \ge u_1(0)/2$ for all $i$ once $h$ is small and $W - V$ is large, and by the right side of (31.3.2)

$$R_h^* \le R_h(x^*) \le \frac{4}{u_1(0)}\, h \sum_i \Big( \frac{h^2 y_i^2}{a^2} + \tau_i^2 \Big) \le C \Big( h^2 V \log^2 V + \frac{(\log W)^2}{W - V} \Big) \to 0.$$

*Step 2 (local bounds; needs only $R_h^* \to 0$).* Write $z = Ax_h$ and $g_h = P_a * \mu_h$, so that $z_i = g_h(v_i)$ exactly. By the left side of (31.3.2), $h \sum_i (\sqrt{y_i} - \sqrt{z_i})^2 \le R_h^*$; with $z_i \le 2y_i + 2(\sqrt{z_i} - \sqrt{y_i})^2$ and Cauchy and Schwarz, for every interval $J \subset [-\varrho, \varrho]$, $\varrho \le V$,

$$h \sum_{v_i \in J} z_i \le C(|J| + h) \log(\varrho + 3) + 2R_h^*,$$

$$h \sum_{|v_i| \le \varrho} |z_i - y_i| \le (R_h^*)^{1/2} \Big( h \sum_{|v_i| \le \varrho} (\sqrt{z_i} + \sqrt{y_i})^2 \Big)^{1/2} \to 0.$$

By (31.3.3), $\int_J g_h \le e^{h/(2a)} h \sum_{v_i \in J + [-h,h]} z_i$, and with the smoothness of $u_1$ this gives $g_h \to u_1$ in $L^1_{\mathrm{loc}}(\mathbb{R})$. If $K + [-2,2] \subset [-V, V]$, then $\mu_h(K)\, P_a([-1,1]) \le \int_{K+[-1,1]} g_h$ with $P_a([-1,1]) = \frac{2}{\pi} \arctan \frac{1}{a} > \frac{7}{10}$, so

$$\mu_h([t, t+1]) \le C \log(|t| + 3) + 3R_h^* \qquad (|t| \le V - 3).$$

Hence $\{\mu_h\}$ is vaguely precompact, every vague limit $\mu$ satisfies $\mu([t, t+1]) \le C \log(|t| + 3)$, so $\int (1 + w^2)^{-1} d\mu < \infty$, and $g_h(0) \le e^{1/a}\, \frac12 \int_{-1}^1 g_h \le C$.

*Step 3 (the limit equation).* Let $\mu_h \to \mu$ vaguely along a subsequence, fix $\varrho_0$ and take $R \ge 2\varrho_0$ with $\mu(\{\pm R\}) = 0$. For $|v| \le \varrho_0$ split $g_h(v)$ over $|w| \le R$, $R < |w| \le V-3$ and $V-3 < |w| \le W$. The first part tends to $\int_{|w|\le R} P_a(v-w)\,d\mu(w)$ uniformly in $|v| \le \varrho_0$. By Step 2 the second is at most $\sum_{k\ge 0} \mu_h(2^k R < |w| \le 2^{k+1}R)\,\frac{4a}{\pi(2^k R)^2} \le C\frac{\log R}{R}$, uniformly in $h$. In the third, $\left|\frac{a^2+w^2}{a^2+(v-w)^2} - 1\right| \le C\varrho_0/V$, so it equals $\kappa_h(1+O(\varrho_0/V))$ with $\kappa_h = \int_{V-3<|w|\le W} P_a(w)\,d\mu_h(w) \le g_h(0) \le C$. Along a further subsequence $\kappa_h \to \kappa \ge 0$; letting $h \to 0$ and then $R \to \infty$, $g_h \to P_a * \mu + \kappa$ uniformly on compact sets, and with Step 2

$$P_a * \mu + \kappa = u_1 \ \text{ on } \mathbb{R}, \qquad \mu \ge 0, \quad \kappa \ge 0. \tag{31.3.4}$$

*Step 4 (no constant; for (a) only).* For $v \in [V-5, V-4]$ and $V-3 < w \le W$, $P_a(v-w) \ge \frac{a/\pi}{a^2+(W-V+5)^2}$; integrating over $v \in [V-5, V-4]$ and using Step 2, $\mu_h((V-3, W]) \le C(W-V+5)^2 \log V$, and likewise on the negative side. Hence

$$\kappa_h \le \frac{a/\pi}{(V-3)^2}\,\mu_h(V-3 < |w| \le W) \le C\,\frac{(W-V+5)^2 \log V}{V^2} \to 0,$$

and $\kappa = 0$.

*Proof of (a).* By Step 1, $R_h^* \to 0$, so Steps 2 to 4 apply, and every subsequential vague limit satisfies $P_a * \mu = u_1 = P_a * \mathfrak{U}_{\theta'}$. For the signed measure $\nu = \mu - \mathfrak{U}_{\theta'}$, which has $\int (1+w^2)^{-1} d|\nu| < \infty$, the Poisson extension $F(t,v) = (P_t * \nu)(v)$ is harmonic in $t > 0$ and vanishes at $t = a$, hence for $t \ge a$ because $P_t = P_{t-a} * P_a$, hence for all $t > 0$ by real analyticity; so $\nu = \lim_{t\downarrow 0} F(t,\cdot) = 0$. Every subsequence of $(\mu_h)$ thus has a further subsequence converging to $\mathfrak{U}_{\theta'}$, and $\mu_h \to \mathfrak{U}_{\theta'}$ vaguely.

*Proof of (b).* Suppose $R_h^* \to 0$ along a subsequence. Steps 2 and 3 use neither the hypothesis nor the growth conditions of (a), so (31.3.4) holds for some $\mu \ge 0$ and $\kappa \ge 0$. For $\sigma > \theta'$ put $H(\sigma + iv) = \pi\big((P_{\sigma-\theta'} * \mu)(v) + \kappa\big)$, a harmonic function on $\operatorname{Re} s > \theta'$. For $\sigma \ge 1$, $P_{\sigma-\theta'} * \mu = P_{\sigma-1} * (u_1 - \kappa)$ and $\pi P_{\sigma-1} * u_1(v) = \operatorname{Re}(\xi'/\xi)(\sigma+iv)$ (Theorem 29.14.4 at levels at least 1), so $H = \operatorname{Re}\,\xi'/\xi$ on $\operatorname{Re} s \ge 1$. Both are harmonic on the connected open set $\{\operatorname{Re} s > \theta'\}$ minus the zeros of $\xi$, so they agree there. Near a zero $\rho_0$ of multiplicity $m$ with $\operatorname{Re}\rho_0 > \theta'$, $\operatorname{Re}(\xi'/\xi)(s) = \operatorname{Re}\frac{m}{s-\rho_0} + O(1)$ is unbounded while $H$ is continuous at $\rho_0$, a contradiction. Hence $\liminf R_h^* > 0$. □

Remark 31.3.5. (1) Part (b) is qualitative: it gives no lower bound for $\liminf R_h^*$ in terms of the distance of the zero from the line $\operatorname{Re} s = \theta'$, which is why the residual is a diagnostic and not a certificate. (2) The residual must not be divided by the length $2V$ of the data window: a defect confined to bounded heights costs a bounded total residual, which the division sends to 0. For instance, with $0 < b < a$, $0 < \epsilon < \pi b$ and data $y = 1 - \epsilon P_b > 0$, the measure $dv$ has bounded total residual, so the normalised residual tends to 0, yet no $\mu \ge 0$, $\kappa \ge 0$ satisfy $P_a * \mu + \kappa = y$: away from $\omega = 0$ the Fourier transform of $\mu$ would be $-\epsilon e^{(a-b)|\omega|}$, which is not

tempered. (3) Without the second growth condition in (a), the argument does not exclude a constant $\kappa > 0$ in (31.3.4), left by mass beyond the data window; then $\mu = \mathfrak{U}_{\theta'} - \kappa\, dv$, a positive measure whenever $\kappa \le \inf u_{\theta'}$, so uniqueness of the continuous equation alone does not identify the limit.

The second case is worth a sentence. Where the exact backward flow fails to be a measure, EM does not diverge; it returns a positive measure whose residual stays bounded away from zero as the grid is refined, Proposition 31.3.4(b). This is the behaviour one wants from a diagnostic.

## 31.4 Acceleration

EM is slow when the solution lies on the boundary of the cone, which is exactly the case at $\theta' = 1/2$ where the target is atomic: 3000 steps gave a residual of $4 \times 10^{-3}$ in Chapter 30. Two accelerations were tested. Everything in this section was recomputed for this book by the script `code/cx_em.py` (Cauchy products as FFT convolutions, data $u_1$ from $\xi'/\xi$ in ball arithmetic of python-flint at 128 bits), with outputs `data/cx_em_*.json`; the author's runs were not available, and where the recomputation differs from the source the recomputed value is printed. The conventions that the numbers depend on are these. The KL residual is $R = h\sum_i \big(y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i\big)$, the integral over the data grid, as in Proposition 31.3.4. The start is flat, $x_j = \sum_i y_i / \sum_{i,j} A_{ij}$. A peak is a local maximum of $x$ of prominence at least 0.05, found by `scipy.signal.find_peaks`, and its position is the grid point unless the text says that it was refined by a parabola. The basin mass of a zero is $h\sum x_j$ between the midpoints of consecutive zeros (the first basin starts at $v = 0$, the last ends at $v = 400$). Steps are EM evaluations. Timings are of the order of magnitude only, since the hardware and the evaluator differ from the source's.

### 31.4.1 Nesterov momentum in the logarithmic domain

Momentum in the variable $x$ would leave the orthant. In the mirror geometry the natural momentum is geometric: with $z = \log x$,

$$z_{\mathrm{ext}}^{(k)} = z^{(k)} + \beta_k\big(z^{(k)} - z^{(k-1)}\big), \qquad x^{(k+1)} = \mathrm{EM}\big(\exp z_{\mathrm{ext}}^{(k)}\big), \tag{31.4.1}$$

with $\beta_k = (t_k - 1)/t_{k+1}$, $t_{k+1} = (1 + \sqrt{1 + 4t_k^2})/2$, the Nesterov sequence [222], and adaptive restart [224]: if $f$ increases, take a plain EM step and reset $t$ to 1. Positivity is automatic. This is related to accelerated mirror descent in the sense of Krichene, Bayen and Bartlett [189], with the EM surrogate as the proximal step.

### 31.4.2 SQUAREM

The SQUAREM of Varadhan and Roland [361] takes two EM steps, forms the differences $r = x_1 - x_0$ and $q = (x_2 - x_1) - r$, and extrapolates $x_0 - 2\alpha r + \alpha^2 q$ with $\alpha = -\|r\|/\|q\|$, followed by one more EM step. It is a quasi Newton extrapolation of the EM map and costs three EM evaluations per iteration. It needs two safeguards, and the step counts depend on them: here the extrapolated point is clipped below at $10^{-30}$ to keep positivity, and the iteration falls back to $x_2$ if $f$ at the stabilised point is not smaller than at $x_2$.

### 31.4.3 Results on a window

Steps are counted in EM evaluations; levels are values of the KL residual; data on $[100, 300]$ at spacing 0.05, unknown on $[80, 320]$, dense matrix.

| $\theta'$ | method | steps to $10^{-2}$ | to $4 \cdot 10^{-3}$ | to $10^{-3}$ | to $3 \cdot 10^{-4}$ |
|---|---|---|---|---|---|
| 0.75 | EM | 25 | 36 | 64 | 104 |
| 0.75 | Nesterov | 11 | 14 | 20 | 28 |
| 0.75 | SQUAREM | 9 | 15 | 21 | 27 |
| 0.50 | EM | 1305 | never | never | never |
| 0.50 | Nesterov | 85 | 131 | 250 | 775 |
| 0.50 | SQUAREM | 114 | 249 | 993 | never |

("Never" means within 3000 steps.) After 3000 EM evaluations at $\theta' = 1/2$ the residuals are $4.05 \times 10^{-3}$ for EM, $5.7 \times 10^{-5}$ for Nesterov and $8.2 \times 10^{-4}$ for SQUAREM. All three recover the 109 zeros in $(100, 300)$ to 0.03 at worst (maximum distance 0.022, 0.010 and 0.014 at $\theta' = 1/2$ with peaks refined by a parabola, 0.017, 0.027 and 0.020 at $\theta' = 3/4$; at $\theta' = 1/2$ Nesterov and SQUAREM also show 115 and 113 peaks, the extra ones small) and keep every $x_j$ positive, the smallest being of order $10^{-52}$.

### 31.4.4 Results on the full range by FFT

The matrix $A$ is Toeplitz, so $Ax$ and $A^{\mathsf{T}} r$ are convolutions with the Cauchy kernel and cost $O(n \log n)$ by the fast Fourier transform. This allows the whole range. We took the data $u_1$ on $[-400, 400]$ by symmetry at spacing 0.02, the unknown on $[-440, 440]$ (44,001 cells), a flat start, and compared the 202 zeros below height 400 with the peaks of the result. The basin mass is the mass of $x$ between the midpoints of consecutive zeros, which should be one.

| $\theta'$ | method | steps | KL | peaks | err (max/mean) | mass |
|---|---|---|---|---|---|---|
| 0.75 | EM | 400 | $4.4 \cdot 10^{-5}$ | 202 | 0.045 / 0.0067 | $0.999 \pm 0.043$ |
| 0.75 | Nest. | 400 | $6.9 \cdot 10^{-8}$ | 202 | 0.045 / 0.0063 | $0.999 \pm 0.042$ |
| 0.50 | EM | 3000 | $1.6 \cdot 10^{-2}$ | 201 | $2.07^{\dagger}$ / 0.013 | $1.001 \pm 0.038$ |
| 0.50 | Nest. | 3000 | $1.8 \cdot 10^{-4}$ | 205 | 0.010 / 0.0055 | $1.001 \pm 0.031$ |

The source printed $6.5\cdot10^{-5}$ and $2.6\cdot10^{-5}$ at $\theta'=3/4$ and $1.5\cdot10^{-2}$ and $1.2\cdot10^{-4}$ at $\theta'=1/2$; the Nesterov value at $3/4$ differs by a factor of 400 and is the least stable entry (it depends on the restart details). Errors are for grid point peaks, except † (peaks refined by a parabola, one zero without a peak: with 202 grid point peaks the maximum is 0.029 and the value depends on the peak rule); the 205 peaks of the last row include three small spurious ones; the mass spreads are dominated by the two edge basins (first 1.054, last 0.517 at $\theta'=3/4$) and are 0.026 in the interior at $3/4$ and 0.006 for accelerated EM at $1/2$. A run takes of the order of a minute on one cluster core (57 s for EM and 82 s for Nesterov at $h=0.02$). At $\theta'=1/2$ plain EM has not separated one close pair after 3000 steps, which accounts for the large maximum error; accelerated EM separates every pair, places all 202 zeros to 0.010 at worst and 0.0055 on average, and assigns each a basin mass of one with standard deviation 3.1 per cent, the largest deviations being the first zero, whose basin starts at $v=0$, and the last, whose basin ends at the edge. So the first two hundred zeros of $\zeta$, with their multiplicity one (Theorem 31.9.6), are read off the line $\operatorname{Re} s=1$ by a positivity preserving iteration in a minute.

The rates can be checked against Theorem 31.3.1. At $\theta'=1/2$, with $R^*$ the least residual observed, the product $k\,(R(x^{(k)})-R^*)$ for plain EM is 54.7, 51.9, 49.5 and 48.1 at $k=500$, 1000, 2000, 3000 (the source printed 52.9, 49.4, 46.1, 44.0): bounded and slowly decreasing, the $O(1/k)$ regime of the theorem; on a grid of spacing 0.04, 0.02 and 0.01 the first three values agree to four digits, so the constant does not depend on the grid. For accelerated EM the product $k^2(R(x^{(k)})-R^*)$ is 518, 355 and 416 at $k=500$, 1000, 2000 (the source: 411, 295, 313): bounded, the empirical $O(1/k^2)$, with a sawtooth caused by the restarts. At $\theta'=0.75$ the product $k\,(R(x^{(k)})-R^*)$ for EM falls from 0.32 at $k=50$ to 0.018 at $k=400$, faster than $1/k$: the interior linear regime.

### 31.4.5 The first thousand zeros

The same code run to height 1450, with $h=0.025$, the unknown on $[-1490,1490]$ (119,201 cells) and 4000 accelerated steps, gives the following (reproduced by the recomputation). Below height 1415 there are 995 zeros; the inversion finds 994 peaks. The one failure is the closest pair in the range, 1329.0435 and 1329.2050, gap 0.16, which the iteration has not yet separated and reports as a single peak at 1329.125 carrying mass two (2.0012); the next closest pair, gap 0.22 at 1054.78 and 1055.00, is resolved. Over the 995 zeros the distance from each zero to the nearest grid point peak has mean 0.0065, median 0.0063, 99th percentile 0.015 (with the peaks refined by a parabola, 0.0015, 0.0011 and 0.0095), and exceeds 0.03 only for the two members of the merged pair. The basin masses are $1.0007\pm0.0053$, every zero simple. The residual after 4000 steps is $3.8\times10^{-4}$; more steps are expected to separate the last pair, as the window runs at $\theta'=1/2$ showed for gaps of this size.

The timing is the point of Section 31.2 made concrete, but only in order of

magnitude. The source reports 1375 seconds for $u_1$ on $[0, 1450]$ at spacing 0.025 (58,000 points) against 151 seconds for the inversion. With the evaluator of this book (python-flint, 128 bits, eight processes) the data took about ten seconds and the inversion a few hundred seconds, so which layer dominates at height 1450 depends on the evaluator; what does not depend on it is that the cost of the data layer grows with $T$ like $T^{1+\varepsilon}$ and the inversion only like $n \log n$.

### 31.4.6 What the accelerated iteration computes

Restart falls back to EM, so accepted steps never increase $f$; if the accelerated iterates converge to a positive limit, the limit is an EM fixed point with positive coordinates, hence a minimiser of $f$, and in every run the accelerated iteration reached the same object as EM, the reverse $I$ projection of Proposition 31.3.4, hence $\mathfrak{U}_{\theta'}$ when it exists. Convergence and the observed rate are not proved here. What changes is the path. Su, Boyd and Candès [266] showed that Nesterov's sequence is a discretisation of the damped second order equation $\ddot{z} + (3/t)\dot{z} + \nabla f = 0$, a Euclidean analogue of the accelerated flow here. With noisy data, where the EM iteration count acts as a regularisation parameter in the manner of Landweber iteration, acceleration reaches the overfitting regime correspondingly sooner, and the restart rule becomes the stopping rule.

## 31.5 Von Neumann's Monte Carlo inversion

Before Markov chain Monte Carlo there was von Neumann's Monte Carlo, invented with Ulam at Los Alamos to invert matrices by random walks and written up by Forsythe and Leibler [139]. It applies to the backward flow directly, and seeing why it fails there and why the Bayesian inversion succeeds is instructive.

### 31.5.1 The von Neumann and Ulam estimator

To solve $x = Hx + b$ when the Neumann series $x = \sum_{k \ge 0} H^k b$ converges, choose a transition matrix $p_{jk} > 0$ wherever $H_{jk} \neq 0$ and a stopping probability $q_j > 0$. Start a walk at $j_0 = j$, move $j_0 \to j_1 \to \cdots$ according to $p$, stop after each step with probability $q$ (the walk is at state $j_\tau$ when it stops), and accumulate the weights $w_k = \prod_{l<k} H_{j_l j_{l+1}} / (p_{j_l j_{l+1}} (1 - q_{j_l}))$. The estimator

$$\widehat{x}_j = w_\tau \, \frac{b_{j_\tau}}{q_{j_\tau}}$$

has expectation $\sum_{k \ge 0} (H^k b)_j$ term by term, because the probability of a path of length $k$ that stops at its end is $\prod_{l<k} (1 - q_{j_l}) p_{j_l j_{l+1}} \cdot q_{j_k}$; this is a statement about an absolutely convergent series, which needs $\rho(|H|) < 1$. Its second moment from the state $j$ is $\sum_{m \ge 0} (Q^m g)_j$ with $Q_{jk} = H_{jk}^2 / ((1 - q_j) p_{jk})$ and $g_j = b_j^2 / q_j$, so it is finite when $\rho(Q) < 1$; and by the Cauchy and Schwarz step in the proof of

Proposition 31.5.1 it is infinite for every admissible $p, q$ when $b > 0$ and $\rho(|H|) > 1$. That $\rho(|H|) < 1$ is also sufficient for suitable $p, q$ is a standard fact; for row sums of $|H|$ below one it is shown at the end of the proof of Proposition 31.5.1.

### 31.5.2 Applied to the backward flow

The deconvolution $y = Ax$ is the fixed point equation $x = (I - A)x + y$ with $A$ square ($M = N$), so $H = I - A$, $b = y$, and the Neumann series is the Richardson iteration $x^{(k+1)} = x^{(k)} + (y - Ax^{(k)})$ with unit step, the linear cousin of EM. Since $A$ is the Cauchy convolution, a step of the walk under $p \propto |H|$ is a Cauchy jump, and the walk is a Cauchy random walk with alternating signs in the weights: the von Neumann and Ulam estimator for the backward flow is a signed Cauchy random walk started at $v$ and evaluating $y$ at its positions. Its variance is not finite.

Proposition 31.5.1 (infinite variance). *Let $M = N$, so that $A$ is the square section of order $2N + 1$ of the Toeplitz matrix* (31.1.3) *and $x = (I - A)x + y$ is defined, and put $t = h/(\pi(1 - \theta'))$. If $t < t_0 = 0.52219\ldots$, the root in $(0, 1)$ of $\coth(1/t) = 2t$, in particular if $h < \pi(1 - \theta')/2$, then the bi infinite operator $|I - A|$ has spectral radius $1 - 2t + \coth(1/t) > 1$, and so do its sections for all $N \geq N_0(h, \theta')$. For such a section consider the von Neumann and Ulam estimator of $\sum_{k\geq 0}(I-A)^k y$ which, at state $j$, stops with probability $q_j \in (0, 1)$ and otherwise moves to $k$ with probability $(1-q_j)p_{jk}$, $p_{jk} > 0$ wherever $(I-A)_{jk} \neq 0$, and returns the product of the weights $(I - A)_{jk}/((1 - q_j)p_{jk})$ along the path times $y_j/q_j$ at the stopping state $j$. Its second moment is infinite from every starting state. If $t_0 < t \leq 1$, the spectral radius of $|I - A|$ is below one for the bi infinite operator and all its sections, and the conclusion fails.*

*Proof.* For $t \leq 1$ the diagonal entry is $A_{jj} = t$, so $|I-A| = (1-2t)I+A$ entrywise, with symbol $1-2t+\phi(\xi)$ and $\phi(\xi) = \cosh((\pi-|\xi|)/(\pi t))/\sinh(1/t)$, the symbol of (31.1.3). For a Laurent operator with nonnegative summable entries the spectral radius on $\ell^2(\mathbb{Z})$ is the maximum of the symbol, here its value $1-2t+\coth(1/t)$ at $\xi = 0$, which is also the row sum. The function $\coth(1/t) - 2t$ decreases on $(0, 1]$, its derivative being $(x/\sinh x)^2 - 2 < 0$ with $x = 1/t$; it is positive near 0 and equals $\coth 1 - 2 < 0$ at $t = 1$, so the spectral radius exceeds one exactly when $t < t_0$. For a section, a symmetric nonnegative matrix, the spectral radius is the largest Rayleigh quotient; the normalised constant vector has quotient tending to the row sum as $N \to \infty$, which gives $N_0$. If $t_0 < t \leq 1$, every row sum of every section is at most $1 - 2t + \coth(1/t) < 1$. With $r_j = \sum_k |H_{jk}|$ the row sums of $H = I - A$, the choice $q_j = 1 - r_j$ and $p_{jk} = |H_{jk}|/r_j$ gives $(1 - q_j)p_{jk} = |H_{jk}|$ and $Q_{jk} = |H_{jk}|$, so $\rho(Q) \leq \max_j r_j < 1$ and the second moment $\sum_m (Q^m g)_j$ is finite: for this $p, q$ the conclusion fails.

Let $H = I - A$ be a section with $\lambda = \rho(|H|) > 1$ and Perron vector $w > 0$ (all entries of $|H|$ are positive). With $Q_{jk} = H_{jk}^2/((1-q_j)p_{jk})$, summing over the paths

gives the second moment $\sum_{m\ge0}(Q^m g)_{j_0}$ from the state $j_0$, where $g_j = y_j^2/q_j > 0$ since $y = u_1 > 0$. By Cauchy and Schwarz,

$$\begin{aligned}\lambda w_j = \sum_k |H_{jk}| w_k &\le \Big(\sum_k Q_{jk} w_k^2\Big)^{1/2} \Big(\sum_k (1-q_j)p_{jk}\Big)^{1/2} \\ &\le \big(Qw^{(2)}\big)_j^{1/2}, \qquad w^{(2)} = (w_k^2)_k,\end{aligned}$$

so $Qw^{(2)} \ge \lambda^2 w^{(2)}$ and $Q^m w^{(2)} \ge \lambda^{2m} w^{(2)}$. As $g \ge c\, w^{(2)}$ for some $c > 0$, the series diverges. (The same argument without squares shows that the estimator is not even integrable: $E|\widehat{x}_j| = \sum_k (|H|^k y)_j = \infty$.) □

All the grids of the runs of this chapter are in the first case ($t = 0.0127$ for $h = 0.02$ and $\theta' = \frac{1}{2}$). So the oldest Monte Carlo inversion exists for the backward flow on paper and is useless in practice, and the reason is sign: the matrix $I - A$ has entries of both signs, the walk carries alternating weights, and their cancellation is what the Neumann series relies on and what a sample cannot reproduce. This is the usual fate of von Neumann and Ulam on ill conditioned systems, and the Cauchy deconvolution, with condition number of order $e^{(1-\theta')\pi/h}$, is as ill conditioned as they come.

### 31.5.3 The positive walk

The Bayesian inversion is also a Monte Carlo of a random walk, but a positive one. Under the current $x$, read $A$ as a Markov kernel from sources $w_j$ to detectors $v_i$: a unit of mass at $w_j$ lands at $v_i$ with probability $A_{ij}/c_j$, a Cauchy jump. Bayes' rule reverses it: a unit of mass observed at $v_i$ came from $w_j$ with probability

$$\pi_{ij}(x) = \frac{A_{ij} x_j}{(Ax)_i},$$

a reverse Cauchy jump weighted by the current estimate of the sources. The EM step is the expectation of this reverse walk,

$$x_j^+ = \frac{1}{c_j} \sum_i y_i\, \pi_{ij}(x)\,:$$

send every unit of observed mass one step backward in expectation. The data augmentation sampler of Chapter 30 replaces the expectation by a sample: draw the source of each unit of observed mass from $\pi_{ij}(x)$ (the $y_i$ are real densities, not integer counts, so the units are pseudo counts and the sampler is a computational device and not an exact likelihood), then draw $x$ from its gamma conditional given the counts. The weights $\pi_{ij}$ are probabilities, so the sampled reverse step has the variance of a multinomial and no more. This is a Monte Carlo inversion of a different kind: not the Neumann series sampled by a signed walk, which diverges,

but the EM fixed point sampled by a positive reverse walk, which is a Gibbs sampler. The nonlinearity of the EM map, $x$ appearing in $\pi_{ij}(x)$, is exactly what removes the signs; it is positivity paying for itself once more.

Remark 31.5.2. Two other ideas of von Neumann meet in the backward flow: his alternating projection theorem is, through Csiszár and Tusnády, the structure of the EM iteration itself (a reading, not used as a theorem), and the signed random walk shows what goes wrong without positivity.

## 31.6 The Cauchy kernel as a scale mixture of normals

The Cauchy law is a scale mixture of normals, a fact used for robust regression since Andrews and Mallows [6] and in Bayesian computation since Carlin and Polson [95]. Here the mixture has a probabilistic meaning that ties the chapter to the hitting time chapters.

Proposition 31.6.1 (subordination). *Let $B$ be a standard Brownian motion and $T_a$ the first passage time of an independent Brownian motion to level $a>0$, with density $\ell_a(t)=a(2\pi)^{-1/2}t^{-3/2}e^{-a^2/(2t)}$. Then $\gamma+B(T_a)$ has the Cauchy law of centre $\gamma$ and scale $a$, that is*

$$\frac{a/\pi}{a^2+(v-\gamma)^2}=\int_0^\infty \frac{e^{-(v-\gamma)^2/(2t)}}{\sqrt{2\pi t}}\,\ell_a(t)\,dt, \tag{31.6.1}$$

*and on the Fourier side $e^{-a|\omega|}=\int_0^\infty e^{-t\omega^2/2}\ell_a(t)\,dt$. Consequently, if $\zeta$ has no zeros in $\operatorname{Re} s>\theta_0$ and $\theta>\theta_0$,*

$$\mathfrak{U}_\theta=\sum_\rho \int_0^\infty \big(e^{t\Delta/2}\delta_{\gamma_\rho}\big)\,\ell_{\theta-\sigma_\rho}(t)\,dt, \tag{31.6.2}$$

*the Thorin measure at level $\theta$ is the sum over the zeros of the heat flow of $\delta_{\gamma_\rho}$ run to the random time $T_{\theta-\sigma_\rho}$, and under RH it is the heat flow of $\sum_\rho \delta_{\gamma_\rho}$ to the single random time $T_{\theta-1/2}$.*

*Proof.* The Fourier identity is the Laplace transform of the first passage law, $Ee^{-\lambda T_a}=e^{-a\sqrt{2\lambda}}$ at $\lambda=\omega^2/2$; the density identity is its inverse transform; (31.6.2) is equation (30.1.1) (the sum of Cauchy terms $P_{\theta-\sigma_\rho}(\cdot-\gamma_\rho)$, a sum of positive measures) with each Cauchy term written as the mixture. □

Formula (31.6.2) connects the Thorin measure to the heat trace of the zeros, the sum of $e^{-t\gamma^2/2}$ over the zeros (compare Section 9.2.3): under RH the Fourier transform of $\mathfrak{U}_\theta$ is

$$\int_0^\infty \Big(\sum_\gamma e^{-i\omega\gamma}\Big)e^{-t\omega^2/2}\,\ell_{\theta-1/2}(t)\,dt,$$

the Fourier transform of the zero measure multiplied by the subordinated Gaussian. It also explains, in one line, why the backward Cauchy flow is less ill posed than a backward heat flow: the forward multiplier $e^{-(1-\theta')|\omega|} = \int_0^\infty e^{-t\omega^2/2}\ell_{1-\theta'}(t)\,dt$ is an average of heat multipliers, and for large $|\omega|$ the average is carried by small $t$, where $\ell_a(t)$ behaves like $e^{-a^2/(2t)}$, so the Gaussian decay averages to the exponential $e^{-a\sqrt{2\lambda}}$ at $\lambda = \omega^2/2$. The backward multiplier $e^{(1-\theta')|\omega|}$ is the reciprocal of this average, not the average of the reciprocals $e^{t\omega^2/2}$, whose integral against $\ell_a$ diverges for every $\omega \neq 0$ because $\ell_a(t) \sim a(2\pi)^{-1/2}t^{-3/2}$ as $t \to \infty$; it therefore grows exponentially and not like a Gaussian.

### 31.6.1 Data augmentation with latent times

The mixture gives the Bayesian inversion a conditionally Gaussian structure. Augment the complete data of Chapter 30: each unit of mass observed at $v_i$ (a pseudo count, as above) carries one latent time $t_i$ with prior $\ell_a$, $a = 1 - \theta'$, and given $t_i$ and its source $w_j$ its displacement is $N(0, t_i)$. Then

$$\begin{aligned} p(t_i \mid v_i, w_j) &\propto t_i^{-1/2} e^{-(v_i - w_j)^2/(2t_i)}\, t_i^{-3/2} e^{-a^2/(2t_i)} \\ &= t_i^{-2} \exp\Big(-\frac{a^2 + (v_i - w_j)^2}{2t_i}\Big), \end{aligned}$$

an inverse gamma law with shape 1 and scale $\frac{1}{2}(a^2 + (v_i - w_j)^2)$, explicit and cheap to sample. Given the times, the allocation of the observed mass at $v_i$ to the sources is multinomial with Gaussian weights

$$x_j \exp\big(-(v_i - w_j)^2/(2t_i)\big),$$

normalised over $j$, and given the allocations the $x_j$ are gamma as before. The three block Gibbs sampler alternates these steps; the allocation step is dense, since each observation is allocated among all the sources, so a sweep costs of the order of $(2M + 1)(2N + 1)$ operations. This is the Carlin and Polson augmentation [95] applied to the forward map rather than to an error law, and it is the Monte Carlo inversion of Chapter 30 with one extra block.

### 31.6.2 Does the mixture help?

For the complexity classification, no. The marginal posterior in $x$ is the log concave law of Theorem 31.2.1 with or without the latent times, and the augmented posterior in $(x, t)$ is not jointly log concave, the inverse gamma factors being log convex in $t$ on part of their range; the random walk guarantee applies to the marginal, which is what one samples. For the data layer, no: the latent times do not touch the evaluation of $\zeta$. For the sampler, yes in the usual way: the conditionally Gaussian blocks mix well, and each conditional is a standard law. For

understanding, yes: (31.6.2) says that the Thorin semigroup is Brownian motion run to a first passage time, that the levels $\theta$ index the level of the barrier, and that RH is the statement that one barrier, at distance $\theta - \frac{1}{2}$, serves every zero. The backward flow is the un stopping of that Brownian motion, and its ill posedness is the usual one for the backward heat equation, tamed by the first passage mixture to an exponential rather than Gaussian amplification.

## 31.7 What is being computed

Collecting the chapter: given $\zeta$ on the line $\operatorname{Re} s = 1$ to an absolute precision on the scale of $u_1$, which for the linear inverse costs $(1 - \theta') \log_2 T$ bits near height $T$ (Proposition 31.2.2), which is the expensive part and the only part that grows faster than the output, the EM iteration and its accelerated forms compute the information projection of the level one Thorin measure onto the cone of Cauchy blurred positive measures at level $\theta'$. When $\zeta$ has no zeros to the right of $\theta'$ the limit is the Thorin measure $U_{\theta'}$ of $\xi(\theta')/\xi(\theta' + \sqrt{s})$ (Proposition 31.3.4(a), under its conditions on the windows). Under the Riemann hypothesis this is the Brownian image of the zero measure stopped at the first passage of level $\theta' - \frac{1}{2}$, and the zeros are its peaks. When there is a zero to the right of $\theta'$, the residual of the grid problems stays bounded away from zero as the grid is refined (Proposition 31.3.4(b)); the residual is a diagnostic, not a certificate. The posterior over the grid problem is log concave (Theorem 31.2.1). None of this reaches the arithmetic, which lives in the evaluation of $\zeta$ and costs polynomial time in $T$ by every known method.

The numbers of the rest of this part are of two kinds. Those of Theorem 31.9.6, of Theorem 31.10.2 and of the EM runs of Section 31.9.6 were recomputed for this book by scripts written from the mathematics (`code/cx_u_certify.py`, `code/cx_min.py`, `code/cx_slope_X.py`, `code/cx_em.py`, with the data files `data/cx_*.json`), and where the recomputation differed from the source the recomputed value is printed. The other numbers (the three routes of Section 31.8, the mass deficit and level mass tables, the tables of $U_{3/4}$, the timings) are the author's, were not rerun, and are marked as such where it matters. Status words are proved, computed, conditional, heuristic and open. The author's figures are not reproduced; the text says in words what they show.

## 31.8 Thorin measures at level 7/8

### 31.8.1 Thorin measures, not primes

What the backward flow computes well is Thorin measures. At every level $\theta \geq 1$ the Thorin measure of $\xi(\theta)/\xi(\theta + \sqrt{s})$ has the explicit density

$$\frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\theta + iv),$$

with the prime sum $-\sum \Lambda(n)n^{-\theta}\cos(v\log n)$ inside it (the sign is that of $\zeta'/\zeta = -\sum \Lambda(n)n^{-s}$; the series converges absolutely for $\theta > 1$ and at $\theta = 1$ stands for the value of $\zeta'/\zeta$ on the line), computable from $\zeta'/\zeta$ on one line with no zeros and no inversion (Chapter 30 and Theorem 29.14.2). Below level one, at every $\theta'$ above the zero free abscissa, the Thorin measure is the EM limit from the level one data, positivity preserving, and at $\theta' = \frac{1}{2}$ it is resolved down to the atoms, where it is the zero measure itself; the thousand zero run of Section 31.4.5 is a computation of the level one half Thorin measure of the reciprocal $\xi$ function from the Euler product side. And the target for the Thorin computations is fixed: under the Riemann hypothesis the measure at level $\theta > \frac{1}{2}$ is the Cauchy smoothing of the zero measure at width $\theta - \frac{1}{2}$, so a computed Thorin measure is to be compared with that smoothed density, not with the atomic measure in total variation, where the two measures are mutually singular whatever the truth.

What the flow does not compute is primes. The primes are the spectrum of the level one measure: the Fourier transform of the level one density carries point masses at $\pm\log n$ of weight proportional to $-\Lambda(n)/n$, and they are put in through the Euler product when $u_1$ is evaluated; reading them off the Fourier transform, as the level one chapter does (Chapter 30), recovers what was supplied. The deconvolution runs from the level one measure towards the zeros, and the primes appear in it only as the frequencies of the data. The reverse passage, from zeros to primes, is the classical computation of $\psi(x)$ by the explicit formula, which nothing in this chapter changes.

### 31.8.2 The level 7/8 measure

The quasi Riemann hypothesis (Section 29.7) is the zero free half plane $\operatorname{Re} s > 7/8$, announced in October 2026 by OpenAI, Liu and Thorner with a formalisation in Lean and not yet refereed [398, 399, 352]. It makes $\theta = 7/8$ the lowest level at which the Thorin representation is currently claimed; every statement below that rests on it is conditional on it, as in Chapter 29:

$$\frac{\xi(7/8)}{\xi(7/8+\sqrt{s})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{u}\Big)U_{7/8}(du)\Big), \tag{31.8.1}$$

with $U_{7/8}$ a positive Thorin measure (Theorem 29.14.2). Here is how to compute it, three ways, with the agreement between them as the check. Throughout, $u(v)$ denotes the density of the symmetric measure $\mathfrak{U}_{7/8}$ in the variable $v$; the Thorin measure $U_{7/8}$ in the variable $u = v^2$ is the image of the restriction of $\mathfrak{U}_{7/8}$ to $v > 0$, so that $\int g(u)U_{7/8}(du) = \int_0^\infty g(v^2)u(v)\,dv$, and its density is $u(\sqrt{u})/(2\sqrt{u})$.

Route (a), direct. By the explicit density of the level one chapter, $u(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac{7}{8}+iv)$, with

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{\log\pi}{2} + \frac{1}{2}\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s),$$

evaluated at $s = \frac{7}{8} + iv$. This needs $\zeta$ and $\zeta'$ inside the critical strip, which any multiprecision library supplies by analytic continuation; on $[0, 400]$ at spacing 0.05 it took 100 seconds in the author's run. The formula defines a signed density unconditionally; the announced half plane is what makes it the Thorin measure of (31.8.1). Its minimum on $[0, 400]$ is $0.0055 > 0$ and its mass on $[0, 400]$ is 201.63 against 202 zeros, the deficit being the Cauchy tails of width 3/8 at the ends.

Route (b), deconvolution. Run the accelerated EM of Section 31.4 on the level one data with the Cauchy kernel of width $1 - \frac{7}{8} = \frac{1}{8}$. The condition number $e^{\pi/(8h)}$ is $e^{7.9}$ at $h = 0.05$, mild, and 400 steps reach a residual of $2 \times 10^{-10}$ (the author's run). This route uses $\zeta$ on the line $\operatorname{Re} s = 1$ only, never inside the strip. Away from the edges of the data window it agrees with route (a) to $6 \times 10^{-4}$ absolutely and 0.33 per cent relatively.

Route (c), zeros. Under the Riemann hypothesis, which is verified far beyond height 400 [419],

$$u(v) = \sum_{\rho} \frac{(3/8)/\pi}{(3/8)^2 + (v - \gamma_\rho)^2},$$

summed over all zeros with both signs of $\gamma$. With the first 1000 zeros and a tail from the zero density $\frac{1}{2\pi} \log \frac{v}{2\pi}$ above the last one, this agrees with route (a) to $9 \times 10^{-5}$ absolutely and 1.6 per cent relatively, the relative discrepancy being the crude tail at the top of the range.

Finally the identity (31.8.1) itself, with route (a) on the right. Minus the logarithm of the left side is $\log \xi(\frac{7}{8} + \sqrt{s}) - \log \xi(\frac{7}{8})$ and the right side is $\int_0^\infty \log(1 + s/v^2)\, u(v)\, dv$. At $s = 1$ the left side is 0.040368, the integral over $[0, 400]$ is 0.038361 and the tail from the zero density 0.002051, total 0.040412; at $s = 4$ the three numbers are 0.126495, 0.118335, 0.008202, total 0.126537. The agreement is to the accuracy of the tail.

### 31.8.3 What this gives the programme

Route (b) is the one to notice. It produces the Thorin measure at level 7/8, a positive measure whose existence follows from the announced half plane, from data on the line $\operatorname{Re} s = 1$ where the Euler product is in force, by a positivity preserving iteration with no reference to zeros and no analytic continuation of $\zeta$ into the strip. The same computation at any $\theta'$ between 1/2 and 7/8 is defined and runs; what the half plane adds at 7/8 is the guarantee that its limit is a Thorin measure rather than an information projection with a residual (Proposition 31.3.4). Running it at $\theta' < 7/8$ and watching the residual is therefore the computational form of the question whether the zero free half plane extends; the residual is a diagnostic at the resolution the data allow, not a certificate, because Proposition 31.3.4(b) is qualitative.

## 31.9 Pictures of the Thorin measures and the decision of positivity

### 31.9.1 Pictures

The author's figures are not reproduced here; this is what they show. The Thorin densities $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$ on $[0, 60]$ at the five levels $\theta = 0.55, 0.6, 0.75, 7/8, 1$ are each computed directly on their own line with no use of the zeros. It is the Cauchy semigroup (Proposition 1.2.3) seen from the side: as $\theta$ decreases to $\frac{1}{2}$ the peaks at the zeros sharpen like $1/(\pi(\theta - \frac{1}{2}))$ and the troughs between them deepen, and at $\theta = \frac{1}{2}$ the picture becomes the zero measure.

The level $7/8$ measure on $[0, 400]$, in the variable $v$, oscillates about $(2\pi)^{-1}\log(v/2\pi)$, the density of the zeros, as it must, since the average of a sum of Cauchy bumps is the density of their centres. In the variable $u = v^2$ the Thorin measure $U_{7/8}$ has density $g(u) = u_{7/8}(\sqrt{u})/(2\sqrt{u})$, and its two asymptotes make the Thorin conditions visible: $g(u) \sim u_{7/8}(0)/(2\sqrt{u})$ as $u \to 0$, so $\int_0^1 |\log u|\, U_{7/8}(du) < \infty$, and $g(u) \approx \log(\sqrt{u}/2\pi)/(4\pi\sqrt{u})$ on average as $u \to \infty$, so $\int_1^\infty u^{-1} U_{7/8}(du) < \infty$. The total mass is infinite, consistent with $F_{7/8}(s) = \xi(7/8)/\xi(7/8 + \sqrt{s})$ decaying faster than any power of $s$, in fact like $\exp(-\frac{1}{2}\sqrt{s}\log\sqrt{s})$.

The three routes of Section 31.8 are compared in a third figure: the EM deconvolution from level one and the sum over zeros, each against the direct evaluation. Both differences are below $10^{-3}$ on the whole range and smooth, the first coming from the edge of the data window and the second from the crude tail above the thousandth zero.

### 31.9.2 What positivity means

Theorem 31.9.1 (positivity). *For $\frac{1}{2} \le \theta < 1$ the following are equivalent.*

*(i) $U_\theta$ is a positive Thorin measure, that is, $F_\theta(s) = \xi(\theta)/\xi(\theta + \sqrt{s})$ is the Laplace transform of a generalised gamma convolution.*

*(ii) $\zeta$ has no zeros in $\operatorname{Re} s > \theta$.*

*(iii) $\operatorname{Re}(\xi'/\xi)(s) > 0$ for $\operatorname{Re} s > \theta$; that is, $u_\sigma(v) > 0$ for every $\sigma > \theta$ and every $v$.*

*For $\theta = \frac{1}{2}$ the equivalence of (ii) and (iii) is Lagarias's theorem [320].*

*Proof.* (i) and (ii) are equivalent by Theorem 29.14.2. (ii) implies (iii) term by term, since $\operatorname{Re}(\xi'/\xi)(s) = \sum_\rho (\sigma - \sigma_\rho)/|s - \rho|^2$, the sum taken over the zeros in pairs $\rho$, $1 - \bar{\rho}$, and every $\sigma - \sigma_\rho$ is positive. For (iii) implies (ii), let $\rho_0 = \sigma_0 + i\gamma_0$ be a zero of multiplicity $m$ with $\sigma_0 > \theta$. Near $\rho_0$, $\xi'/\xi(s) = m/(s - \rho_0) + h(s)$ with $h$ analytic, so $\operatorname{Re}(\xi'/\xi)(\sigma_0 - \varepsilon + i\gamma_0) = -m/\varepsilon + O(1)$, which is negative for small $\varepsilon$, and $\sigma_0 - \varepsilon > \theta$. □

Positivity on the single line $\operatorname{Re} s = \theta$, $u_\theta \geq 0$, is necessary for (ii) but not sufficient. An off line zero $\rho_0$ puts a negative bump of height $1/(\pi(\sigma_0 - \theta))$ at $v = \gamma_0$, and its mirror $1 - \bar{\rho}_0$ a positive bump of height $1/(\pi(\theta + \sigma_0 - 1))$. The remaining zeros add a background which, once $\theta - \frac{1}{2}$ exceeds the mean spacing of the zeros near $\gamma_0$, is about the zero density

$$d(\gamma_0) = \frac{1}{2\pi} \log \frac{\gamma_0}{2\pi}.$$

Heuristically, since the background grows with the height, a zero high enough up is invisible to pointwise positivity on a line to its left; the book has the corresponding question as open (Remark 1.2.4). At height 60 the opposite holds, as the last column of Table 31.1 shows.

### 31.9.3 The mass deficit

The invariant that does not mask is the mass.

Proposition 31.9.2 (mass deficit). *Let $\frac{1}{2} < \theta < 1$ and $T > 0$, and suppose no zero lies on the segment $\operatorname{Re} s = \theta$, $0 \leq \operatorname{Im} s \leq T$, nor on the segment $\operatorname{Im} s = T$, $\theta \leq \operatorname{Re} s \leq 1$. Let $M_\sigma(T) = \int_0^T u_\sigma(v)\, dv$ and let $N(\theta, T)$ be the number of zeros with $\operatorname{Re} \rho > \theta$ and $0 < \operatorname{Im} \rho < T$. Then*

$$2N(\theta, T) = M_1(T) - M_\theta(T) + \frac{1}{\pi}\Delta_{\mathrm{top}}(\theta, T), \tag{31.9.1}$$

*where $\Delta_{\mathrm{top}}(\theta, T) = -\int_\theta^1 \operatorname{Im}(\xi'/\xi)(\sigma + iT)\, d\sigma$ is the change of $\arg \xi$ along the top side.*

*Proof.* Apply the argument principle to $\xi$ on the rectangle with corners $\theta$, 1, $1+iT$, $\theta + iT$, traversed counterclockwise. There are no zeros on $\operatorname{Re} s = 1$, and $\xi$ is real and positive on $[\theta, 1]$, so the bottom side contributes nothing. By the Cauchy Riemann equations $\partial_v \arg \xi(\sigma + iv) = \operatorname{Re}(\xi'/\xi)(\sigma + iv) = \pi u_\sigma(v)$, so the right side contributes $\pi M_1(T)$ and the left side, traversed downward, $-\pi M_\theta(T)$. The top side contributes $\Delta_{\mathrm{top}}$, since $\partial_\sigma \arg \xi = \operatorname{Im}(\xi'/\xi)$. The total is $2\pi N(\theta, T)$. □

So the Thorin measure at level $\theta$ carries exactly two units less mass up to height $T$ than the measure at level one, for each zero to its right, after the explicit correction along the top. The deficit is an integer, so computing it to an accuracy better than one half decides it. The identity is the argument principle, which is what Backlund and Turing use to count zeros; the Thorin picture supplies the bookkeeping, and a semigroup reading of each term, but not a cheaper computation. As $\theta$ decreases to $\frac{1}{2}$ with no zeros off the line below $T$, $M_\theta(T)$ tends to the number of zeros on the critical line below $T$, and (31.9.1) becomes the classical formula $N(T) = M_1(T) + \pi^{-1}\Delta_{\mathrm{top}}(\frac{1}{2}, T)$, which is Backlund's count and holds

with $N(T)$ the number of all zeros below $T$, with no hypothesis, when $T$ is not an ordinate.

Table 31.1 evaluates (31.9.1) at $T = 100$ for the real $\xi$, and for a synthetic $\tilde{\xi}$ in which the consecutive zeros $\frac{1}{2} + 59.347i$ and $\frac{1}{2} + 60.832i$ and their conjugates are replaced by the off line quadruple $0.85 \pm 60.089i$, $0.15 \pm 60.089i$. The factor that does this is a ratio of products of two quadratics, with the exact zeros in the denominator, symmetric under $s \mapsto 1 - s$ and positive on the real axis, so $\tilde{\xi}$ is entire of order one with the functional equation of $\xi$. The arguments were tracked by continuous variation on a grid of spacing 0.02 in $v$ and 0.0005 in $\sigma$. The table is the author's computation and was not rerun.

| $\theta$ | $M_\theta(100)$ | $\Delta_{\mathrm{top}}/\pi$ | $2N$ for $\xi$ | $2N$ for $\tilde{\xi}$ | $\min \tilde{u}_\theta$ |
|---|---|---|---|---|---|
| 0.55 | 29.0109 | −0.1002 | 0.0000 | 2.0000 | −0.26 |
| 0.60 | 29.0218 | −0.0893 | 0.0000 | 2.0000 | −0.55 |
| 0.65 | 29.0328 | −0.0784 | 0.0000 | 2.0000 | −0.94 |
| 0.70 | 29.0438 | −0.0674 | 0.0000 | 2.0000 | −1.51 |
| 0.75 | 29.0548 | −0.0563 | 0.0000 | 2.0000 | −2.57 |
| 0.80 | 29.0659 | −0.0452 | 0.0000 | 2.0000 | −5.38 |
| 0.875 | 29.0827 | −0.0284 | 0.0000 | 0.0000 | 0.006 |
| 0.90 | 29.0884 | −0.0228 | 0.0000 | 0.0000 | 0.006 |
| 0.95 | 29.0997 | −0.0114 | 0.0000 | 0.0000 | 0.007 |
| 1.00 | 29.1112 | 0 | 0.0000 | 0.0000 | 0.007 |

Table 31.1: The mass deficit at $T = 100$, where $M_1(100) = 29.1112$ and $N(100) = 29$ (the author's computation, not rerun).

For the real $\xi$ the deficit vanishes to four decimals at every level of the table, and $M_\theta(100)$ decreases almost linearly, with slope 0.223 per unit of $\theta$, towards 29.000 as $\theta \to \frac{1}{2}$: the smoothed zero count becomes the zero count. For $\tilde{\xi}$ the deficit is 2 exactly when $\theta < 0.85$ (the injected zero is the only one to the right of such a $\theta$) and 0 above, and the table shows it at 2.0000 for $\theta \le 0.80$ and 0.0000 for $\theta \ge 0.875$; at this height the synthetic density is also visibly negative near 60 at every level of the table below 0.85.

Corollary 31.9.3 (the level mass function). *Let $T > 0$ not be the ordinate of a zero. For $\theta \in (\frac{1}{2}, 1]$ not the real part of a zero below height $T$,*

$$M_\theta(T) = M_1(T) + \frac{1}{\pi}\Delta_{\mathrm{top}}(\theta, T) - 2N(\theta, T).$$

*The first two terms form a real analytic function of $\theta$, since $\xi$ has no zero on the line* $\operatorname{Im} s = T$*; its derivative is*

$$\frac{1}{\pi} \operatorname{Im} \frac{\xi'}{\xi}(\theta + iT).$$

*The last term is a step function which drops by* $2m$ *as* $\theta$ *decreases through* $\operatorname{Re}\rho$*, for each zero* $\rho$ *of multiplicity* $m$ *with* $\operatorname{Re}\rho > \frac{1}{2}$ *and* $0 < \operatorname{Im}\rho < T$*. Hence* $\theta \mapsto M_\theta(T)$ *extends continuously to* $(\frac{1}{2}, 1]$ *if and only if every zero below height* $T$ *lies on the critical line, and its jumps occur exactly at the real parts of the off line zeros, of size twice the total multiplicity of the zeros with that real part and ordinate below* $T$*.*

So the level mass function reads off real parts: run $\theta$ down from 1, and a zero off the line shows itself by a jump of exactly 2 at its real part. At $T = 100$, for $\xi$ the masses on the ten vertical lines of the table agree with the smooth part, computed from one horizontal sweep, to $2.3 \times 10^{-5}$, and the smooth part has slope 0.223 per unit of $\theta$; for $\tilde{\xi}$ they lie exactly 2.0000 below it for $\theta < 0.85$ and on it above (the author's computation, not rerun). The corollary describes what the Thorin masses record. It does not locate real parts more cheaply than the argument principle, since the jumps can only be seen by computing along vertical lines.

### 31.9.4 Positivity of $U_{7/8}$

Given the zero free half plane $\operatorname{Re} s > 7/8$ (announced, not refereed, as above), Theorem 31.9.1 makes $U_{7/8}$ positive: every term of $u_{7/8} = \sum_\rho P_{7/8-\sigma_\rho}(v - \gamma_\rho)$, with $P_a$ the Cauchy density of width $a$ and $P_0 = \delta_0$, is nonnegative, a zero on the line $\operatorname{Re} s = 7/8$, if there were one, contributing an atom. The computation displays the margin. On $[0, 400]$ the minimum of $u_{7/8}$ is 0.00551, attained at $v = 0$, where $u_{7/8}(0) = \pi^{-1}(\xi'/\xi)(7/8) = 0.005514663528$; this value is positive unconditionally, because $\xi(\frac{1}{2} + t) = \int \Phi(x)\cosh(tx)\,dx$ with Riemann's kernel $\Phi$ positive, so $\xi$ increases on $(\frac{1}{2}, \infty)$. Away from the origin the minimum on $[15, 400]$ is 0.0272, at $v = 17.5$ between the first two zeros; with the second difference estimate $\max|u''| \approx 11.8$ on the grid of spacing 0.05, the grid minimum less $h^2/8 \cdot \max|u''|$ is 0.0235. Theorem 31.9.6 certifies the positivity of $u_{7/8}$ in ball arithmetic on $[0, 1420]$, and Theorem 31.10.2 certifies the lower bound 0.0272 for $u_{7/8}$ on $[13.5, 1420]$, the least certified bound lying at $v = 17.49$. The mass deficit at $\theta = 7/8$, $T = 100$, is zero.

### 31.9.5 The measure $U_{3/4}$ below level $7/8$

Level $3/4$ lies below $7/8$, so the positivity of $U_{3/4}$ at every height is not available from the announced half plane: by Theorem 31.9.1 it is equivalent to the zero free half plane $\operatorname{Re} s > 3/4$, which is open. We computed it as at $7/8$, with the direct route on a finer grid, spacing 0.02 and 20,001 points, because the peaks now have width $1/4$. The table is the author's computation and was not rerun, apart from the value at the origin.

| quantity | value |
|---|---|
| $\min u_{3/4}$ on $[0,400]$ | 0.00368 at $v=0$ |
| $\min u_{3/4}$ on $[15,400]$ | 0.0182 at $v=17.5$ |
| $\max\lvert u''\rvert$; margin | 40.5; 0.0162 |
| mass on $[0,400]$ | 201.60 against $N(400)=202$ |
| zeros plus tail minus direct | $5.7\times10^{-5}$ |
| EM: residual; relative error | $1.5\times10^{-9}$; 1.1 per cent |
| EM density: minimum | $4.0\times10^{-3}$ |
| $2N(\frac{3}{4},400.9)$ | $202.1778-202.0891-0.0887=0.0000$ |

The EM run used the Cauchy kernel of width $1/4$, condition number $e^{\pi/(4h)}=e^{15.7}$ at $h=0.05$, and 1500 accelerated steps; the height $T=400.9$ for the deficit lies between the zeros 399.985 and 401.839. The minimum at the origin equals $\pi^{-1}(\xi'/\xi)(3/4)=0.003676904191$, and the margin is the grid minimum on $[15,400]$ less $\frac{h^2}{8}\max\lvert u''\rvert$. So $u_{3/4}$ is positive pointwise on $[0,400]$, by a margin of 0.0037 at the origin and 0.016 elsewhere, and its mass deficit to height 400.9 is zero: no zero lies to the right of $3/4$ below that height. That is decided at this height, certified on $[0,1420]$ by Theorem 31.9.6, and was of course known from the verification of the Riemann hypothesis far higher [419]. Above the heights any computation reaches, the positivity of $U_{3/4}$ is the open zero free half plane $\operatorname{Re}s>3/4$. In the author's comparison figure the measure $U_{3/4}$ against $U_{7/8}$ shows the peaks at the zeros higher and the troughs deeper, the Cauchy semigroup run one eighth further towards the zeros.

### 31.9.6 All the way down: $U_{1/2}$

At $\theta=\frac{1}{2}$ the densities collapse. Since $\xi(\frac{1}{2}+iv)=\Xi(v)$ is real, $\operatorname{Re}(\xi'/\xi)(\frac{1}{2}+iv)=0$ away from the zeros, and the measure is purely atomic: $\pi\,\mathfrak{U}_{1/2}=\partial_v\arg\xi(\frac{1}{2}+iv)$, which jumps by $\pi m$ at a zero of multiplicity $m$ on the line. Under the Riemann hypothesis $\mathfrak{U}_{1/2}=\sum_\rho\delta_{\gamma_\rho}$ and $U_{1/2}=\sum_{\gamma>0}m_\gamma\delta_{\gamma^2}$, the Thorin measure of the reciprocal $\xi$ function around which the book is organised. It can be computed three ways, and the author's last figure shows two of them: the cumulative masses $M_\theta(v)$ at $\theta=1,0.75,0.55$ converging to the zero staircase $N(v)$, which is the cumulative mass at level $\frac{1}{2}$, and the EM backward flow at $\theta'=\frac{1}{2}$ from level one data on a logarithmic scale, with the sign changes of $Z$ marked.

Directly at level $\frac{1}{2}$, the atoms of odd multiplicity are the sign changes of Hardy's function $Z(v)=e^{i\vartheta(v)}\zeta(\frac{1}{2}+iv)$, which is real with the sign of $\Xi(v)$ up to a fixed sign. A scan of $[1,400.5]$ at spacing 0.02, 19,976 evaluations of $Z$ in 122 seconds (the author's run), followed by root finding, gives 202 atoms below height 400, agreeing with the tabulated zeros to machine precision. This is the classical computation of zeros and is included for comparison.

By the backward flow at $\theta'=\frac{1}{2}$ from level one (Section 31.4), on the full range $[-400,400]$ at $h=0.02$ with the logarithmic domain Nesterov scheme and

3000 steps, the recomputation finds the peaks of the EM density (three extra small peaks at prominence 0.05) within 0.010 of the zeros, on average 0.0055 (peaks located at grid points), and the basin masses $1.0013 \pm 0.0313$ (the spread is dominated by the two basins at the edges of the window, and is 0.0061 for the interior ones); a plain EM run of the same length has a residual two orders of magnitude larger and one zero without a peak. The author's centroid statements (the basin centroids lie on average 0.0014 and at most 0.081 from the zeros, the density between the atoms falls to $10^{-12}$) were not rerun. To height 1415 (the run of Section 31.4.5) the basin masses are $1.0007 \pm 0.0053$ in the recomputation, with the one close pair, 1329.0435 and 1329.2050 (gap 0.16), that the flow does not resolve: it gives one peak, at 1329.125, with the mass 2.001 of the two basins.

As the limit of the cumulative masses, $M_\theta(v)$ approaches the staircase $N(v)$ as $\theta$ decreases: $M_1(100) = 29.1112$, $M_{0.75}(100) = 29.0548$, $M_{0.55}(100) = 29.0109$, and $N(100) = 29$.

Is $U_{1/2}$ positive? That is the Riemann hypothesis. If a zero leaves the line, $F_{1/2}$ acquires a pole in the cut plane and no Thorin measure exists at level $\frac{1}{2}$ (Theorem 29.14.2). To certify positivity up to a height $T$ one needs more than the sign changes, which count the zeros of odd multiplicity on the line; one needs the total count $N(T)$ independently, and Backlund's count above supplies it from level one data:

$$N(T) = M_1(T) + \frac{1}{\pi}\Delta_{\mathrm{top}}(\tfrac{1}{2}, T).$$

At $T = 400.9$ this gives $202.1778 - 0.1778 = 202.0000$, and there are 202 sign changes of $Z$ below 400.9. Every zero below height 400.9 is therefore on the line and simple, which Theorem 31.9.6 certifies to height 1419.9, and $\mathfrak{U}_{1/2}$ restricted to $|v| < 400.9$ is the sum of unit atoms at the ordinates. This is Turing's verification of the Riemann hypothesis to a given height [506], written in Thorin bookkeeping: the level one mass counts the zeros, the level one half atoms locate them on the line, and the two counts agree. Platt and Trudgian have carried the verification to height $3 \times 10^{12}$ [419]; no computation can carry it to every height.

### 31.9.7 Can positivity be decided?

By level.

For $\theta \geq 1$ positivity is an unconditional theorem, and for $\theta \geq 7/8$ it holds conditionally on the announced half plane. There is nothing to decide; the computation displays it.

For $\frac{1}{2} < \theta < 7/8$ and a finite height $T$, positivity up to height $T$, meaning $N(\theta, T) = 0$, is decidable: compute the deficit (31.9.1) to accuracy better than one half in interval arithmetic, provided no zero lies on the boundary of the rectangle, which for $T$ within the range where the Riemann hypothesis has been verified is known. For $\theta = \frac{1}{2}$ the same decision is made by Backlund's count together with the sign changes. Theorem 31.9.6 does this in ball arithmetic at $\theta = 3/4$ and

$T = 1419.9$, in about ten minutes on one core in the author's run and in half a minute on a cluster node in ours.

For $\frac{1}{2} \le \theta < 7/8$ and all heights, positivity is the zero free half plane $\operatorname{Re} s > \theta$, and for $\theta = \frac{1}{2}$ the Riemann hypothesis. It is refutable by a finite computation if false: an offending zero lies in some finite rectangle, over which the deficit is a positive even integer. It is not confirmable by checking finitely many heights without an argument for the rest, if true, since every computation sees a finite height. That is why the half plane at $7/8$ is claimed by a proof and not by a computation, and why no computation in this book, however accurate, settles the positivity of $U_\theta$ below $7/8$.

### 31.9.8 Computational details

The author's computations used Python with `mpmath` [376] at 15 significant digits for $\zeta$, $\zeta'$, $\log\Gamma$, $\psi$ and Hardy's $Z$, `numpy` for the fast Fourier transforms and the EM iterations, and `matplotlib` for the figures. The first 1000 zeros were taken from `mpmath.zetazero`. The recomputations of this book used Arb through `python-flint` [290, 259] at 96 to 160 bits.

Densities. $u_\theta(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\theta + iv)$ was evaluated from

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2}\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s).$$

The grid spacing was chosen so that each Cauchy peak of width $\theta - \frac{1}{2}$ is covered by at least ten points: 0.05 at $\theta = 7/8$ and 1, 0.02 at $\theta = 3/4$, 0.01 at $\theta = 0.55$ and 0.6.

Arguments. The masses $M_\theta(T)$ and the top correction were obtained by continuous variation of $\operatorname{Im} \log \xi$, computed as $\operatorname{Im}[\log s + \log(s-1) - \frac{s}{2}\log\pi + \log\Gamma(s/2) + \log\zeta(s)]$ and unwrapped along the path. The unwrapping is valid when the phase increment per step is below $\pi$; since $\partial_v \arg \xi = \pi u_\theta$ and $\pi u_\theta \le 1/(\theta - \frac{1}{2})$ plus a background of order $\log v$, a step of 0.02 gives increments below 0.5 radians at $\theta = 0.55$. The horizontal sweeps used step 0.0005 in $\sigma$. The integer deficits came out within $10^{-4}$ of integers in every case.

Deconvolution. The unknowns are the cell densities $x_j$ on the unknown grid, so that $A_{ij} = hP_{1-\theta'}(v_i - w_j)$ and the measure is $\sum_j x_j h\,\delta_{w_j}$ (equation (31.1.3)). The data grid was symmetric, $[-400, 400]$ by evenness of $u_1$, the unknown grid $[-440, 440]$, the Toeplitz products by FFT after circulant embedding, the start flat, $x_j = \sum_i y_i / \sum_{i,j} A_{ij}$, and the acceleration the logarithmic domain Nesterov scheme of Section 31.4 with restart. The quoted residuals are the Kullback Leibler values $R = h\sum_i \big(y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i\big)$, with the factor $h$ (without it every value is $1/h$ times larger); the steps counted are evaluations of the EM map. A peak of an EM density is a local maximum of prominence at least 0.05, located at its grid point unless said to be refined; the basin of a zero runs between the

midpoints to its neighbours (the first from $v = 0$, the last to $v = 400$) and its mass is $h \sum x_j$ over the basin.

The author's timings on one core, with `mpmath` for the data and only of the order of magnitude (the recomputation with Arb at 128 bits evaluated $u_1$ at 58,001 points in seconds on eight processes of a cluster, so that the statement that the data layer dominates the cost holds for `mpmath` and not for Arb):

| computation | seconds |
|---|---|
| $u_1$ on $[0, 400]$, 8001 points | 103 |
| $u_{7/8}$ on $[0, 400]$, 8001 points | 100 |
| $u_{3/4}$ on $[0, 400]$, 20,001 points | 239 |
| $u_\theta$ on $[0, 60]$ at five levels | 60 |
| arguments on ten vertical lines to $T = 100$ and one sweep | 148 |
| arguments to $T = 400.9$ at $\theta = 3/4$ and 1, two sweeps | 207 |
| EM at width 1/8, 400 steps, 17,601 unknowns | 3 |
| EM at width 1/4, 1500 steps | 11 |
| sign changes of $Z$ on $[1, 400.5]$ | 122 |
| $u_1$ on $[0, 1450]$, 58,001 points; EM to height 1450 | 1375; 151 |

The positivity of $u_{3/4}$ and $u_{7/8}$ on $[0, 1420]$ and the zero counts above are certified in ball arithmetic in Section 31.9.9; the other numbers above are floating point.

### 31.9.9 Certification in ball arithmetic

Every positivity statement and integer count above was first obtained in floating point. This subsection turns them into computer assisted statements by repeating the computations in ball arithmetic, using the Arb library [290]. In ball arithmetic each quantity is an interval guaranteed to contain the true value, including all rounding and truncation errors in $\zeta$, $\Gamma$ and their derivatives. Two lemmas reduce the infinitely many points of a line or a segment to finitely many ball evaluations.

Lemma 31.9.4 (cells). *Let $L = \xi'/\xi$, $u(v) = \pi^{-1} \operatorname{Re} L(\theta + iv)$, and let $\xi$ have no zero in the closed cell $\{\theta + iv : |v - m| \le r\}$. Then for $|t| \le r$*

$$u(m + t) \;\ge\; u(m) - r\,|u'(m)| - \frac{r^2}{2\pi} \sup_{|v-m|\le r} |L''(\theta + iv)|,$$

*where $u'(m) = -\pi^{-1} \operatorname{Im} L'(\theta + im)$.*

*Proof.* Taylor's theorem with the Lagrange remainder, using $\partial_v L(\theta + iv) = iL'$ and $\partial_v^2 L(\theta + iv) = -L''$, so that $|u''| \le |L''|/\pi$. □

In practice $u(m)$ and $u'(m)$ are evaluated at the exact midpoint, where the balls are tight, and the supremum is bounded by evaluating the power series of

$L$ on the whole cell as a ball, which overestimates but is multiplied by $r^2/2$. A cell whose bound is not certifiably positive, or on which the ball enclosure of $\xi$ contains 0, is bisected.

Lemma 31.9.5 (segment). *Let $[a,b]$ be a segment of length $d$ with*

$$d \sup_{s\in[a,b]} |\xi'(s)| < |\xi(a)|.$$

*Then $\xi$ has no zero on the segment, and the continuous change of $\arg\xi$ along it is the principal value $\operatorname{Arg}(\xi(b)/\xi(a))$, which lies in $(-\pi/2,\pi/2)$.*

*Proof.* For $s\in[a,b]$, $|\xi(s)-\xi(a)| \le d\sup|\xi'| < |\xi(a)|$, so $\xi(s)/\xi(a)$ lies in the open disk of radius less than one about 1, which lies in the half plane $\operatorname{Re} w > 0$. The continuous argument of $\xi(s)/\xi(a)$ therefore stays in $(-\pi/2,\pi/2)$ along the segment and ends at the principal value. □

The cells have exact dyadic endpoints, the segments exact dyadic endpoints except for the corner $T^* = 14199/10$, which is one ball shared by all paths, so that the contours close and every enclosure contains the value at the exact corner. A version that accumulated endpoints in floating point produced an enclosure of the deficit that excluded its true value 0 by $10^{-17}$, which is how the closure requirement was noticed.

Theorem 31.9.6 (certified). *Let $T^* = 1419.9$. Then:*

*(i) $u_{3/4}(v) > 0$ and $u_{7/8}(v) > 0$ for $0 \le v \le 1420$;*

*(ii) $N(T^*) = 1000$;*

*(iii) $N(\frac34, T^*) = 0$, with $M_{3/4}(T^*) = 1000.01952570266\ldots$ to within $10^{-20}$;*

*(iv) $\Xi$ has 1000 certified sign changes in $(0,T^*)$.*

*Consequently every zero of $\zeta$ with $0 < \operatorname{Im}\rho < T^*$ lies on the critical line and is simple; the measure $\pi^{-1}\partial_v \arg\xi(\frac12+iv)$ of Section 31.9.6 restricted to $0 \le v < T^*$, which under the Riemann hypothesis is the Thorin measure $\mathfrak{U}_{1/2}$, is the sum of 1000 unit atoms at the ordinates; and $u_{3/4}$ and $u_{7/8}$ have positive densities there, by (i).*

*Proof.* Computed, computer assisted, in the ball arithmetic of Arb at 96 bits and again at 160 bits (`code/cx_u_certify.py`, `data/cx_u_certify_96.json` and `data/cx_u_certify_160.json`); two standard facts are used and not certified, that $\xi > 0$ on $(\frac12,\infty)$ and Backlund's relation $\pi N(T) = \Delta\arg\xi$ along $2 \to 2+iT \to \frac12+iT$. For (i), Lemma 31.9.4 on the dyadic cells of width $2^{-7}$ over $[0,1420]$, bisected once where necessary, gives 182,409 cells for $\theta = 3/4$ and 181,781 for $\theta = 7/8$ at 96 bits (182,512 and 181,783 at 160 bits; the counts belong to the method and the

grid, not to $u$); every certified lower bound is positive, and in both runs and at both levels it is at least $4 \times 10^{-5}$. The least bound of the certificate depends on the grid and the precision and is not the minimum of $u$, which for $u_{3/4}$ and $u_{7/8}$ on $[0, 1420]$ is attained near 0, where $u_{3/4}(0) = \pi^{-1}(\xi'/\xi)(3/4) = 0.003676904191347$ and $u_{7/8}(0) = 0.005514663527790$ (balls of radius below $10^{-17}$). For (ii), Lemma 31.9.5 along the path $2 \to 2 + iT^* \to \frac{1}{2} + iT^*$, which gives $\pi N(T^*)$ as the change of argument because $\xi$ is real and positive on $[\frac{1}{2}, 2]$ and real on the critical line: the enclosure contains 1000 and has width below $10^{-20}$ (11,470 segments on $\operatorname{Re} s = 2$ and 96 pieces of length $1/64$ on the top side at 96 bits). For (iii), Lemma 31.9.5 around the rectangle $[\frac{3}{4}, 2] \times [0, T^*]$, using Proposition 31.9.2 with the right side moved from $\operatorname{Re} s = 1$ to $\operatorname{Re} s = 2$, which changes nothing since there are no zeros with $\operatorname{Re} s \geq 1$ and keeps the balls away from the pole of $\zeta$: the enclosure of $2N(\frac{3}{4}, T^*)$ has radius below $10^{-20}$ about 0 and contains the single integer 0 (32,387 segments on $\operatorname{Re} s = \frac{3}{4}$), and $M_{3/4}(T^*) = 1000.0195257026609355$ with the same twenty digits at both precisions. For (iv), the signs of $\Xi(v) = \xi(\frac{1}{2} + iv)$ at the 70,996 points $v = k/50$, $k = 0, \ldots, 70995$, are certified, none undetermined, with $\Xi(0) > 0$, and change 1000 times. Since the sign changes give at least 1000 distinct zeros of odd multiplicity on the line and (ii) gives exactly 1000 zeros in all, each zero is on the line and simple. The statement about $\pi^{-1}\partial_v \arg \xi$ is (ii) with (iv); the positivity statement is (i); (iii) is implied by (ii) and (iv) and is an independent check.

□

| certificate | enclosure at 96 bits (160 bits) | size |
|---|---|---|
| (i) $u_{3/4} > 0$ on $[0, 1420]$ | least bound at least $4 \times 10^{-5}$ | 182,409 cells |
| (i) $u_{7/8} > 0$ on $[0, 1420]$ | least bound at least $4 \times 10^{-5}$ | 181,781 cells |
| (ii) $N(1419.9)$ | $1000 \pm 4 \times 10^{-22}$ ($\pm 2 \times 10^{-41}$) | 11,470 + 96 segments |
| (iii) $2N(\frac{3}{4}, 1419.9)$ | $0 \pm 1.4 \times 10^{-21}$ ($\pm 7 \times 10^{-41}$) | 32,387 segments |
| (iv) signs of $\Xi$ | 1000 changes, none undetermined | 70,996 points |

Table 31.2: The certificates of Theorem 31.9.6, as recomputed. The whole run takes 29 seconds at 96 bits and 46 at 160 bits on 24 cores of a cluster; the author's run took about ten minutes on one core.

The author's run to height 400.9 certifies $N(400.9) = 202$, an enclosure of radius below $10^{-16}$ for $2N(\frac{3}{4}, 400.9)$, 202 sign changes, and both positivity statements on $[0, 400]$, in under two minutes (not rerun).

Parts (ii) to (iv) are known: the Riemann hypothesis has been verified to height $3 \times 10^{12}$ [419], by Turing's method written here in Thorin masses. What the theorem adds is part (i), the positivity of the Thorin densities at levels 3/4 and 7/8 on $[0, 1420]$ as a computer assisted statement, and the integer count that goes with it; it is fast enough to run on a laptop.

## 31.10 The limits of the computation

Can the computations of this chapter be pushed past what is known? For $\zeta$, no, and the reasons are worth stating precisely.

Height. Theorem 31.9.6 reaches height 1419.9, the first thousand zeros, in minutes. The record is that of Platt and Trudgian: the Riemann hypothesis verified rigorously to height $3\times10^{12}$ [419], about $1.2\times10^{13}$ zeros. That computation combines Turing's method with the fast multi evaluation of Odlyzko and Schönhage [392], so it pays a few operations per zero rather than one full evaluation of $\zeta$ per point. The mass deficit of Proposition 31.9.2 is the argument principle, which is Turing's method written in Thorin masses, so its known cost is that of the zero verification, of order $T^{1+\varepsilon}$ with the fast methods. Matching the record would mean rebuilding that machinery and buying comparable computing time.

All heights. Below $7/8$, positivity of $U_\theta$ at every height is the zero free half plane $\operatorname{Re} s>\theta$ (Theorem 31.9.1). As Section 31.9.7 explains, a computation can refute that statement if it is false and cannot confirm it if it is true by checking finitely many heights without an argument for the rest, because every computation stops at a finite height $H$. Above $H$ an argument is needed, and that argument is the open problem.

Combining the two gains nothing either. Below $3\times10^{12}-\frac12$ every zero is on the line, so $U_\theta$ restricted to that range is known to be positive for every $\theta>\frac12$ (Corollary 11.4.7). Above it, the announced half plane gives positivity for $\theta\ge7/8$, conditionally, and nothing below. Every computation in these chapters lies inside what is already known. The frontier that moves the half plane is analytic: zero density estimates such as those of Guth and Maynard [166], and the announced, not yet refereed, half plane $\operatorname{Re} s>7/8$.

What computation can still add is narrower. The rest of this section states two results about the Thorin densities themselves, one unconditional and one conditional on the announced half plane; the unconditional one is already in the book (Proposition 9.1.11), and the second form of the other is Theorem 9.1.18(vi). They rest on a lemma.

Lemma 31.10.1 (pairing). *Let $\nu=\sum_\rho\delta_{\gamma_\rho}$ be the counting measure of the ordinates of the nontrivial zeros, with multiplicity and both signs, and $P_a$ the Cauchy density of width $a$, with $P_0=\delta_0$. If $\zeta$ has no zeros in $\operatorname{Re} s>\theta_0$, where $\frac12\le\theta_0\le1$, then for every $\theta\ge\theta_0$, as measures,*

$$\mathfrak{U}_\theta\ \ge\ \tfrac12\,P_{2\theta-1}*\nu.$$

*In particular the bound holds unconditionally for $\theta\ge1$, and for $\theta\ge7/8$ it holds conditionally on the announced half plane. Under the Riemann hypothesis $\mathfrak{U}_\theta=P_{\theta-1/2}*\nu$ exactly.*

*Proof.* Group the zeros into orbits under $\rho\mapsto1-\bar\rho$, which preserves the ordinate. An orbit of two zeros $\sigma+i\gamma$, $1-\sigma+i\gamma$ contributes $P_a+P_b$ at the point $\gamma$, with widths

$a = \theta - \sigma$ and $b = \theta - 1 + \sigma$, both nonnegative because $\theta \ge \theta_0 \ge \max(\sigma, 1-\sigma)$, and $a+b = 2\theta - 1$. For fixed $x \ne 0$ write $\pi P_a(x) = a\,h(a)$ with $h(a) = 1/(a^2+x^2)$, which is decreasing in $a \ge 0$. Then

$$a\,h(a) + b\,h(b) \;\ge\; a\,h(a+b) + b\,h(a+b) = (a+b)\,h(a+b),$$

that is $P_a(x) + P_b(x) \ge P_{a+b}(x) = \frac{1}{2} \cdot 2P_{2\theta-1}(x)$, which is the claim for the two zeros of the orbit (the two zeros count twice in $\nu$); at $x = 0$ the inequality holds as measures because a width zero term is an atom. An orbit of one zero, on the critical line, contributes $P_{\theta-1/2}$, and $P_{\theta-1/2}(x) \ge \frac{1}{2}P_{2\theta-1}(x)$ by the same monotonicity with $a = \theta - \frac{1}{2}$, $a+b = 2a$. Summing over orbits gives the lemma. □

The lemma says that the bound survives when off line pairs move to the edges of the strip allowed, one zero of each pair becoming an atom; even then the density keeps at least half of the smoothing it has under the Riemann hypothesis, at twice the width. Combined with the cell certificate and an explicit zero count, it proves the following.

THEOREM 31.10.2 (global minimum). *(i) Unconditionally, for every real $v \ne 0$,*

$$u_1(v) \;>\; u_1(0) = \frac{1}{\pi}\Big(1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi\Big) = 0.0073515924923\ldots,$$

*where $\gamma_E$ is Euler's constant; this is Proposition 9.1.11. (ii) If $\zeta$ has no zeros in $\operatorname{Re} s > 7/8$ (the announced half plane), then, as measures, $\mathfrak{U}_{7/8} \ge u_{7/8}(0)\,dv$ with $u_{7/8}(0) = \pi^{-1}(\xi'/\xi)(7/8) = 0.0055146635278\ldots$, and $u_{7/8}(v) > u_{7/8}(0)$ wherever $v \ne 0$ is not the ordinate of a zero on the line $\operatorname{Re} s = 7/8$.*

*Proof.* Part (i) is Proposition 9.1.11; the argument below is the same in three ranges and gives (ii) as well. It is computed, computer assisted (`code/cx_min.py`, `data/cx_c2_*.json`). Write $\theta$ for 1 or 7/8; both functions are even. Three ranges.

$|v| \le 13.5$. Here $u_\theta = \sum_\rho P_{\theta-\sigma_\rho}(v - \gamma_\rho)$ with every width in $[0, 1)$ for $\theta = 1$, and in $[0, \frac{3}{4}]$ for $\theta = 7/8$ under the hypothesis. The Cauchy density $P_a(x)$ is convex for $|x| \ge a/\sqrt{3}$, and every zero has $|\gamma_\rho| \ge 14.1347$ (Theorem 31.9.6), so $|v - \gamma_\rho| \ge 0.63 > 1/\sqrt{3}$. Hence $u_\theta$ is a convergent sum of convex functions, strictly convex and even on $[-13.5, 13.5]$, with its unique minimum at 0. The value $u_1(0) = \pi^{-1}(\xi'/\xi)(1) = \pi^{-1}\sum_\rho 1/\rho$ is $\pi^{-1}$ times the classical constant $1 + \gamma_E/2 - \frac{1}{2}\log 4\pi$.

$13.5 \le |v| \le 1420$. Lemma 31.9.4 with the threshold $u_\theta(0)$ in place of 0: on 175,813 cells of radius 0.004 for each of $\theta = 1$ and $\theta = 7/8$, with no bisection, every certified lower bound exceeds the upper end of the ball for $u_\theta(0)$, the smallest being 0.0360 at $v = 17.49$ for $\theta = 1$ and 0.0272 at $v = 17.49$ for $\theta = 7/8$ (the margins over $u_\theta(0)$ are 0.0287 and 0.0217). The computation evaluates $\xi'/\xi$ on the line and uses no hypothesis.

$|v| \ge 1420$. By Lemma 31.10.1, unconditional at $\theta = 1$ and conditional at $7/8$, $u_\theta(v) \ge \frac12 \sum_\gamma P_{2\theta-1}(v-\gamma) \ge \frac12\, n(v)\, P_{2\theta-1}(8)$, where $n(v)$ is the number of ordinates in $(v-8, v+8]$ (positive ordinates, with multiplicity). Hasanalizade, Shen and Wong [176] (the form used in Proposition 9.1.11) give $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78| \le E(T)$ with $E(T) = 2.5167 + 1/(50T)$ for $e \le T \le X_0 = 30\,610\,046\,000$ and $E(T) = 0.1095\log T + 0.2042\log\log T + 3.0305 + 1/(50T)$ for $T > X_0$. Hence $n(v) \ge D(v) - E(v+8) - E(v-8)$ with $D(v) = M(v+8) - M(v-8)$ and $M(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. The right side is $8.7678\ldots$ at $v = 1418.9$, so more than $8.767$, and it increases with $v$ as long as $v + 8 \le X_0$, because $M'(T) = \log(T/2\pi)/(2\pi)$ increases, so $D$ increases, and $E$ decreases; for larger $v$ it is at least 44 at $X_0 - 8$ and increases (a numerical check of the derivative, in the recomputation). So $n(v) \ge 9$ for $v \ge 1420$, since $n$ is an integer, and $u_\theta(v) \ge 4.5\, P_{2\theta-1}(8)$, which is $0.0220$ for $\theta = 1$ and $0.0166$ for $\theta = 7/8$, more than twice $u_\theta(0)$ in both cases. The conclusion would survive a loss of five ordinates at $\theta = 1$ and of six at $\theta = 7/8$. $\square$

The same argument, with the Riemann hypothesis in place of the zero free half plane, covers every level at once, and the conclusion turns out to be equivalent to the hypothesis.

THEOREM 31.10.3 (a Thorin form of the Riemann hypothesis). *The Riemann hypothesis holds if and only if, for every $\theta \in (\frac12, 1]$, the Thorin density $u_\theta$ attains its global minimum at $v = 0$; that is,*

$$\operatorname{Re}\frac{\xi'}{\xi}(\theta + it) \;\ge\; \frac{\xi'}{\xi}(\theta) \qquad \text{for all real } t \text{ and } \tfrac12 < \theta \le 1.$$

*Under the hypothesis the minimum is attained only at $t = 0$.*

*Proof.* This is the equivalence of Theorem 9.1.18(vi), $v_\alpha(\theta) > b_\alpha$ for $\alpha \in (\frac12, 1)$, $\theta \ne 0$, in the form with the weak inequality; the proof here uses the cell certificate of Theorem 31.10.2 and an explicit zero count, and the forward direction is computer assisted. Suppose the hypothesis, and let $a = \theta - \frac12 \in (0, \frac12]$. Then $u_\theta = \sum_\rho P_a(v - \gamma_\rho)$, so

$$\frac{u_\theta(v)}{a} = \frac{1}{\pi}\sum_\rho \frac{1}{a^2 + (v-\gamma_\rho)^2} \;\ge\; \frac{1}{\pi}\sum_\rho \frac{1}{\frac14 + (v-\gamma_\rho)^2} = 2u_1(v),$$

$$\frac{u_\theta(0)}{a} \;\le\; \frac{1}{\pi}\sum_\rho \frac{1}{\gamma_\rho^2}.$$

Since $\gamma_\rho^2 \ge \gamma_1^2$ for every zero, the last sum is at most

$$\Big(1 + \frac{1}{4\gamma_1^2}\Big)\frac{1}{\pi}\sum_\rho \frac{1}{\frac14 + \gamma_\rho^2} = \Big(1 + \frac{1}{4\gamma_1^2}\Big)\frac{2 + \gamma_E - \log 4\pi}{\pi} \le 0.01473.$$

For $13.5 \le |v| \le 1420$, Theorem 31.10.2 gives $2u_1(v) \ge 0.0720$ (twice the certified 0.0360). Beyond 1420, under the hypothesis, $u_1(v) = \sum_\gamma P_{1/2}(v-\gamma) \ge 9P_{1/2}(8)$ by the count of that proof, so $2u_1(v) \ge 18P_{1/2}(8) \ge 0.0445$. Both exceed 0.01473, so $u_\theta(v) > u_\theta(0)$ there. For $|v| \le 13.5$ the convexity argument applies with every width equal to $a \le \frac12$. Conversely, if a zero $\rho_0 = \sigma_0 + i\gamma_0$ has $\sigma_0 > \frac12$, take $\theta = \sigma_0 - \varepsilon$; then $\operatorname{Re}(\xi'/\xi)(\theta + i\gamma_0) = -m/\varepsilon + O(1) < 0$, while $(\xi'/\xi)(\theta) > 0$ unconditionally, because $\xi(\frac12 + x) = \int \Phi(y)\cosh(xy)\,dy$ with Riemann's kernel $\Phi$ positive increases for $x > 0$. So the minimum is not at 0. □

Theorem 31.11.1 below extends part (ii) of Theorem 31.10.2 to every level in $[7/8, 1]$. Theorem 31.10.3 sits beside the reformulations of Lagarias [320], $\operatorname{Re}(\xi'/\xi) > 0$ on $\operatorname{Re} s > \frac12$, and of Sondow and Dumitrescu [482], that $|\xi(\sigma + it)|$ increases in $\sigma$ for $\sigma > \frac12$. It strengthens both under the hypothesis: the rate $\partial_\sigma \log|\xi(\sigma + it)|$ is not merely positive but smallest on the real axis, at every level between $\frac12$ and 1. Its forward direction is a computer assisted proof, half a minute of ball arithmetic and an explicit zero count, and like everything else in this chapter it is a reformulation, not progress towards the hypothesis.

Two other directions are open to computation. The certificate scripts transfer to any completed $L$ function with a ball arithmetic evaluator; for Dirichlet $L$ functions, including the Dirichlet beta function, and for higher degree $L$ functions, such as that of Ramanujan's $\Delta$ or the Dedekind zeta functions of number fields, whether they would add anything to the published verifications is not examined here. And the Thorin measures at levels 7/8 and 3/4, the generalised gamma convolutions they define, and their margins are tables of numbers; they are not theorems about zeros.

## 31.11 Open problems worked

Notation of this section, as in the rest of the chapter: $\theta$ is the level and $v$ the height, $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$, $P_a(x) = a/(\pi(a^2 + x^2))$, and a zero is $\rho = \sigma_\rho + i\gamma_\rho$ with $\sigma_\rho$ and $\gamma_\rho$ real (so $\gamma_\rho$ is the height of the zero, which is the book's ordinate of $\rho$ only under the Riemann hypothesis). The symmetric Thorin measure $\mathfrak{U}_\theta$ in the variable $v$ has density $\sum_\rho P_{\theta-\sigma_\rho}(v-\gamma_\rho)$ with no factor $\frac12$, so it is twice the measure $\widetilde{U}_\theta$ of Theorem 29.14.4; it exists as a positive measure when $\zeta$ has no zeros in $\operatorname{Re} s > \theta$, and then $P_a * \mathfrak{U}_\theta = \mathfrak{U}_{\theta+a}$, in particular $P_{1-\theta} * \mathfrak{U}_\theta = u_1\,dv$ (Theorem 29.14.4). The zero free half plane $\operatorname{Re} s > \frac78$ is announced, formalised in Lean and not refereed [398, 399, 352], as described in Section 29.7: every statement below that uses it is conditional on it.

Three open problems were posed for this chapter: the rate of EM and accelerated EM at the boundary, the value of the latent Brownian times in the sampler, and a fast algorithm for $u_1$. The first is answered by experiment (observed, not proved), the second by a comparison inside the scheme together with

an experiment, and the third at fixed accuracy. The global minimum problem of Section 31.10 is solved there, by Theorems 31.10.2 and 31.10.3; its extensions, to every level in $[\frac{7}{8}, 1]$ and then to every level under the matching hypothesis, follow, with the slope of the excess. The a posteriori bound for the backward flow is worked up from a duality in floating point to a certificate in ball arithmetic, with Gaussian weighted centroids as well as masses, at the levels $\frac{1}{2}$ and $\frac{3}{4}$. Two short subsections at the end say in what sense $\mathfrak{U}_{1/2}$ is and is not computed efficiently, and what a von Neumann representation of it would be.

### 31.11.1 The rate at the boundary

The question was whether the factor of about 15 by which acceleration beat EM at $\theta' = \frac{1}{2}$ is generic, and how the constants depend on the grid. What the runs show is a constant that does not depend on the grid, an EM exponent close to 1 for the first few thousand steps that then rises, and an accelerated scheme whose restarts make the objective a sawtooth with no single exponent. All of this is observed, not proved. The runs are those of `code/cx_em.py` (jobs gridind, expo and restart; data in `data/cx_em_gridind_*.json`, `data/cx_em_expo_*.json`, `data/cx_em_restart.json`), with the flat start $x_j = \sum_i y_i / \sum_{i,j} A_{ij}$, the residual $R = h \sum_i [y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i]$ with the factor $h$, and steps counted as evaluations of the EM map.

Grid independence. The full range problem of the acceleration section at $\theta' = \frac{1}{2}$ was run for 3000 steps at $h = 0.04$, 0.02 and 0.01. With $R^*$ the smallest residual reached by either method, $k(R_k - R^*)$ for EM is 54.7, 51.9, 49.5 and 48.1 at $k = 500$, 1000, 2000 and 3000, the same on the three grids to four digits, and the number of EM steps divided by the number of accelerated steps needed to reach EM's residual at 3000 steps is 23.1 (3000/130) on all three. The iteration is read as a discretisation of the continuous Richardson and Lucy iteration for measures, whose iterates over thousands of steps stay resolved by the grid. The bound $D(x^* \| x^{(0)})$ of Theorem 31.3.1, which grows like $\log(1/h)$ at an atomic target and is infinite in the continuum, is therefore not the active constraint on these grids, and the constant of order 50 appears to be a property of the continuous problem.

The exponent. A reference value $R^*$ taken from a finite run biases every exponent estimated from $R_k - R^*$, so we use the three point exponent

$$p(k) = \log_2 \frac{R_k - R_{2k}}{R_{2k} - R_{4k}},$$

which equals $p$ exactly when $R_k - R^* = Ck^{-p}$ and involves no reference value. For EM it is 1.080, 1.067, 1.062, 1.054 at $k = 500$, 1000, 2000, 4000 at $h = 0.08$, equal to about $10^{-3}$ on the grids 0.04 and 0.02; it decreases slowly and stays close to 1. After about 4000 steps it rises, to 1.10 at 8000, 1.22 at 16,000, 1.53 at 32,000 and 1.84 at 50,000 in a run of 200,000 steps at $h = 0.08$. The residual itself is $9.0 \times 10^{-4}$, $5.8 \times 10^{-4}$ and $5.0 \times 10^{-4}$ at $5 \times 10^4$, $10^5$ and $2 \times 10^5$ steps. A

geometric rate $R_k - R^* \approx Ce^{-k/K}$ is not claimed: no estimator of $K$ is fixed, and the estimate from three equally spaced differences gives $K \approx 32{,}000$ at $k = 5 \times 10^4$ only. The reading consistent with the theory of EM on a cone is a sublinear phase, at the rate $1/k$ that multiplicative updates have at a boundary optimum, while the mass on cells away from the atoms is driven to zero, followed by the local linear convergence of Theorem 31.3.2 on the identified support, at a rate set by the curvature at the atoms. We state this as an interpretation. What is observed is that the constant of order 50 describes the first few thousand steps and that the upturn after 4000 steps appears in the longest run; no exponent is claimed to be generic.

Acceleration. For the logarithmic domain scheme with function restart the three point exponent reads 2.8, 0.6, 1.2, 1.2, 1.3 at $k = 500$, 1000, 2000, 4000, 5000, a sawtooth whose single values depend on the restart instants; its running value at $k = 2000$, 3000, 5000 is 1.2, 1.4, 1.3. The speedup over EM, the ratio of step counts to a common residual at $h = 0.04$, is 13.5, 18.9, 29.6, 41.7 and 39.1 at the residuals EM reaches after 1000, 2000, 5000, 10,000 and 20,000 steps: it grows and saturates near 40. Five restart rules were compared at $h = 0.04$: function restart, no restart, a gradient restart (here the step against the momentum in the logarithmic variables, an adaptation of the rule of O'Donoghue and Candès [224]), a fixed restart every 500 steps, and a momentum capped at 0.9. They reach EM's residual at 20,000 steps in 512, 344, 344, 344 and 1991 steps, with final values 2.0, 1.7, 1.5, 2.8 and $18 \times 10^{-5}$ after 20,000 steps. The capped momentum, a heavy ball iteration, is about four times slower than function restart. The exponent without restart is erratic in the rerun (between 0.2 and 0.3 at $k = 4000$ and 5000, not monotone earlier), so no exponent is claimed for it. So no restart rule is uniformly best in these runs: function restart is the safe one, since it never increases $f$, and here it costs about one and a half times the steps of the others. There is no generic speedup factor, and the accelerated scheme is the one to run.

### 31.11.2 The latent times do not help the sampler

The question was whether the three block sampler of the section on the Cauchy kernel as a scale mixture, which carries a Brownian time for every unit of observed mass (a pseudo count), mixes better than the two block sampler near $\theta' = \frac{1}{2}$. In the comparison below it does not, and for this scan there is a general reason. The two block sampler is the three block sampler with the latent times integrated out analytically, since the Cauchy kernel is the time mixture of Gaussians. Let $Z$ be the allocations, $X$ the masses and $T$ the times, with $X$ and $T$ independent given $Z$. The allocation marginal of the three block scan is data augmentation through $(X, T)$, that of the collapsed scan is data augmentation through $X$, and for centred $g(Z)$ their reversible kernels satisfy, by conditional Jensen,

$$\langle g, K_3 g \rangle = \mathbb{E}\big[\mathbb{E}(g \mid X, T)^2\big] \;\geq\; \mathbb{E}\big[\mathbb{E}(g \mid X)^2\big] = \langle g, K_2 g \rangle,$$

so the additional augmentation cannot improve the $L^2$ spectral contraction of the allocation marginal. This is the comparison of Liu, Wong and Kong for grouping and collapsing [211], in the scope in which it is proved here: it orders neither wall clock efficiency nor every finite run autocorrelation estimate for other scans and observables.

We ran both samplers on real data (the author's runs, not repeated in this book). The level one density on the window $100 \le v \le 120$, sampled at spacing $h = 0.05$, was turned into Poisson pseudo counts with means

$$200\, h\, u_1(v_i).$$

The unknown was on $[98, 122]$, the prior Gamma$(1, 0.01)$ per cell, the start the EM solution, and each run had 500 burn in and 3000 recorded sweeps. Integrated autocorrelation times were estimated by the initial positive sequence of Geyer [148], for the mass within 0.5 of a zero, the mass in the trough between two zeros, and the log likelihood.

| $\theta'$ | sampler | near a zero | trough | log likelihood | seconds |
|---|---|---|---|---|---|
| 0.75 | two block | 4.3 | 7.1 | 1.8 | 22 |
| 0.75 | three block | 4.9 | 9.2 | 2.9 | 65 |
| 0.55 | two block | 5.4 | 7.8 | 2.5 | 23 |
| 0.55 | three block | 6.4 | 10.4 | 2.5 | 59 |

The latent times lengthen the autocorrelation of the two masses by 14 to 19 per cent (near a zero) and 30 to 33 per cent (trough), and that of the log likelihood by 61 per cent at $\theta' = 0.75$ and not at all at $\theta' = 0.55$; they multiply the cost per sweep by 2.6 to 3, at both levels, including the level near $\frac{1}{2}$ where the conjecture expected them to help. The posterior means agree, 0.80 for the mass near the zero in both samplers. The scale mixture of normals keeps its value as an interpretation, Brownian motion stopped at a first passage time, and as a source of proposals, but not as a device for speed.

### 31.11.3 The global minimum at every level between 7/8 and 1

The second open problem asked for Theorem 31.10.2(ii) at every level $\theta \in [\frac{7}{8}, 1]$ rather than at $\frac{7}{8}$ alone. The difficulty was the middle range, where a certificate uniform in $\theta$ seemed to require a two dimensional search. A monotonicity argument removes the second dimension. The theorem is conditional on the announced half plane.

Theorem 31.11.1 (global minimum, uniform in the level; conditional). *Suppose $\zeta$ has no zeros in $\operatorname{Re} s > \frac{7}{8}$. Then for every $\theta \in [\frac{7}{8}, 1]$ and every real $v \neq 0$ that is not the height of a zero on $\operatorname{Re} s = \theta$,*

$$u_\theta(v) \;>\; u_\theta(0) = \frac{1}{\pi} \frac{\xi'}{\xi}(\theta),$$

*and the singular part of $\mathfrak{U}_\theta$ consists of positive atoms, which occur only if $\theta = \frac{7}{8}$ and a zero lies on $\operatorname{Re} s = \frac{7}{8}$; so $\mathfrak{U}_\theta - u_\theta(0)\,dv$ is a positive measure. Unconditionally, $\theta \mapsto u_\theta(0)$ is increasing on $[\frac{1}{2}, \infty)$.*

*Proof.* Write $g(\theta, v) = u_\theta(v) - u_\theta(0)$ and $a_\rho = \theta - \sigma_\rho \in [\theta - \frac{7}{8}, \theta - \frac{1}{8}] \subset [0, \frac{7}{8}]$, which uses the hypothesis and the functional equation. Monotonicity of $u_\theta(0)$: $\xi(\frac{1}{2} + x)$ is a positive multiple of $\int_0^\infty \Phi(y)\cosh(xy)\,dy$ with the positive density $\Phi$ of (1.2.3), and the logarithm of the Laplace transform of a positive measure is strictly convex (Hölder), so $(\xi'/\xi)(\frac{1}{2} + x)$ is strictly increasing in $x$. In particular $u_\theta(0) \le u_1(0) = 0.00736$ on the range.

$|v| \le 13.5$: every term $P_{a_\rho}(v - \gamma_\rho)$ with $a_\rho > 0$ is convex, since $|v - \gamma_\rho| \ge 0.63 > \frac{7}{8}/\sqrt{3}$; the sum is strictly convex and even, with its minimum at 0. This is the argument of Theorem 31.10.2, valid for all $\theta \le 1$ at once.

$|v| \ge 1418.9$: Lemma 31.10.1 and the zero count of [176] give at least seven ordinates in $(v - 8, v + 8]$ (the explicit bound is 8.76786 at $v = 1418.9$, so there are at least nine, and it increases with $v$, the derivative of its main term $(2\pi)^{-1}\log\frac{v+8}{v-8} \ge 8/(\pi v)$ dominating the derivatives of the error terms, which are $O(1/v)$; this was checked numerically in the computation of Section 31.11.3, `data/cx_c2_ords.json`). Seven suffice: $u_\theta(v) \ge \frac{7}{2}P_{2\theta-1}(8) \ge \frac{7}{2}P_{3/4}(8) = 0.01294 > u_1(0) \ge u_\theta(0)$, using that $P_c(8)$ increases in $c$ on $[\frac{3}{4}, 1]$.

$13.5 \le |v| \le 1418.9$: by Theorem 31.9.6 every zero with $|\gamma| < 1419.9$ lies on the critical line and is one of the 1000 certified ordinates (and their negatives), so its width is exactly $a = \theta - \frac{1}{2} \le \frac{1}{2}$, and the zeros with $|\gamma| \ge 1419.9$ are at distance at least 1 from $v$. Two cases.

If some certified zero has $|v - \gamma| \le 1.05$, then $u_\theta(v) \ge P_{\theta-1/2}(1.05) \ge P_{3/8}(1.05) = 0.0960 > u_\theta(0)$, since $P_c(1.05)$ increases in $c$ on $[\frac{3}{8}, \frac{1}{2}]$.

If every certified zero has $|v - \gamma| \ge 0.95$, differentiate in $\theta$:

$$\partial_\theta u_\theta(v) = \frac{1}{\pi}\sum_\rho f\big(|v - \gamma_\rho|, a_\rho\big), \qquad f(d, a) = \frac{d^2 - a^2}{(d^2 + a^2)^2}.$$

For the far zeros, $d \ge 1 > \frac{7}{8} \ge a$, so their terms are positive. For the certified zeros, $a \le \frac{1}{2}$, and $f$ is decreasing in $a$ for $d > a/\sqrt{3}$ and decreasing in $d$ for $d > \sqrt{3}\,a$, both satisfied when $d \ge 0.95$ and $a \le \frac{1}{2}$. Hence

$$\partial_\theta u_\theta(v) \;\ge\; \frac{1}{\pi}\sum_{|\gamma|<1419.9} f\big(|v - \gamma|, \tfrac{1}{2}\big) =: S(v),$$

a function of the certified ordinates alone, independent of $\theta$. On the other side, $\partial_\theta u_\theta(0) = \pi^{-1}\sum_\rho(\gamma_\rho^2 - a_\rho^2)/(a_\rho^2 + \gamma_\rho^2)^2 \le \pi^{-1}\sum_\rho \gamma_\rho^{-2}$. The sum over the certified ordinates, both signs, is at most 0.044804, and the sum over $|\gamma| \ge 1419.9$ is at most 0.001442 (the computed value is 0.0014417) by partial summation against

the bound of [176] for $N(T)$, so $\partial_\theta u_\theta(0) \le c^* = 0.0147225$ for every $\theta$. The ball arithmetic computation described below certifies $S(v) > c^*$ on every cell of the second case. Therefore $\partial_\theta g \ge 0$ on $[\frac{7}{8}, 1]$, and $g(\theta, v) \ge g(\frac{7}{8}, v) > 0$ by Theorem 31.10.2(ii).

Atoms: a zero contributes an atom to $\mathfrak{U}_\theta$ only if its width $a_\rho = 0$, that is $\sigma_\rho = \theta$, and $\sigma_\rho \le \frac{7}{8} \le \theta$ forces $\theta = \frac{7}{8}$. □

The computation (`code/cx_min.py`, part S; data `data/cx_c2_S.json`) partitions $[13.5, 1418.9]$ into cells of radius 0.1, bisected where neither case applies to the whole cell. The ordinates enter as the certified sign change enclosures. A cell is of the first kind if for one certified ordinate the largest distance between a point of the cell and a point of its enclosure is at most 1.05, and of the second kind if the smallest distance between the cell and every enclosure is at least 0.95; then $S$ is bounded below by evaluating each $f(\cdot, \frac{1}{2})$ at the largest such distance, since $f$ decreases in $d$. On the grid of the text (7,245 cells in all) 6,733 cells are of the first kind and 512 of the second, every cell is certified, and the least certified value of $S$ on a cell of the second kind is 0.066 against $c^* = 0.0147225$. The run takes seconds on a cluster node. The two dimensional search that the open problem anticipated was never needed: the level enters only through the two monotonicities, in $a$ for the near zeros and in sign for the far ones.

Corollary 31.11.2. *Given the zero free half plane* $\operatorname{Re} s > \frac{7}{8}$*, for every* $\theta \in [\frac{7}{8}, 1]$ *the Thorin measure* $\mathfrak{U}_\theta$ *dominates the multiple* $u_\theta(0)\,dv$ *of Lebesgue measure, with* $u_\theta(0)$ *increasing from* 0.0055147 *at* $\theta = \frac{7}{8}$ *to* 0.0073516 *at* $\theta = 1$.

### 31.11.4 Below 7/8: a level by level theorem

The problem left open after Theorem 31.11.1 was its extension to levels below $\frac{7}{8}$ under the corresponding hypothesis. Working it produced a cleaner statement than either of the theorems it extends, and a cleaner proof. The case $\theta = 1$ is Proposition 9.1.11 and the case $\theta \downarrow \frac{1}{2}$ is Theorem 9.1.18(vi), both already in the book.

Theorem 31.11.3 (level by level). *Let* $\frac{1}{2} < \theta \le 1$. *The following are equivalent.*

*(i)* $\zeta$ *has no zeros in* $\operatorname{Re} s > \theta$.

*(ii) For every* $\theta' \in [\theta, 1]$ *the function* $u_{\theta'}$ *attains its global minimum at the origin:* $u_{\theta'}(v) \ge u_{\theta'}(0)$ *for all real* $v$ *that are not heights of zeros on* $\operatorname{Re} s = \theta'$.

*When they hold,* $\mathfrak{U}_{\theta'} \ge u_{\theta'}(0)\,dv$ *as measures for* $\theta' \in [\theta, 1]$*, and for every* $\theta' \in [\theta, 1]$*, with* $c = 2\theta' - 1$ *and the sum over all zeros,*

$$u_{\theta'}(v) - u_{\theta'}(0) \;\ge\; \frac{c}{2\pi}\Big(\sum_\rho \frac{1}{c^2 + (v - \gamma_\rho)^2} - \pi w^*\Big), \tag{31.11.1}$$

*with* $w^* = 0.0294450$*, and the bracket is at least* $\pi(0.06174 - w^*) > 0.1014$ *for* $13.5 \le |v| \le 1418.9$ *and at least* $7/65 - \pi w^* > 0.0151$ *for* $|v| \ge 1418.9$*.*

*Proof.* (ii) implies (i) exactly as in Theorem 31.10.3: a zero $\rho_0$ of multiplicity $m$ with $\sigma_0 > \theta$ makes $u_{\theta'}(\gamma_0) = -m/\varepsilon + O(1)$ at the level $\theta' = \sigma_0 - \varepsilon \in [\theta, 1]$, which is negative, while $u_{\theta'}(0) = \pi^{-1}(\xi'/\xi)(\theta') > 0$.

(i) implies (ii). Fix $\theta' \in [\theta, 1]$, write $c = 2\theta' - 1 \in (0, 1]$, and let $a_\rho = \theta' - \sigma_\rho \ge 0$. Two bounds, one on each side.

At the origin, grouping the zeros into orbits under $\rho \mapsto 1 - \bar{\rho}$, the widths in an orbit sum to $c$ (a pair) or equal $c/2$ (a zero on the line), so

$$u_{\theta'}(0) = \frac{1}{\pi} \sum_{\rho} \frac{a_\rho}{a_\rho^2 + \gamma_\rho^2} \;\le\; \frac{1}{\pi} \sum_{\text{orbits}} \frac{c}{\gamma^2} \;\le\; \frac{c}{\pi} \sum_{\rho} \frac{1}{\gamma_\rho^2} \;\le\; c\, c^*,$$

with $c^* = 0.0147225$, the certified value of Theorem 31.11.1: the certified ordinates below 1419.9 contribute at most 0.044804 to $\sum_\rho \gamma_\rho^{-2}$ and the rest at most 0.001442 by [176]. (An orbit contributes at most $\frac{c}{2} \sum \gamma^{-2}$ over its zeros, which would halve the bound; the weaker form is what the printed constants use.)

Away from the origin, Lemma 31.10.1 gives $u_{\theta'}(v) \ge \frac{1}{2} \sum_\rho P_c(v - \gamma_\rho) = \frac{c}{2\pi} \sum_\rho \frac{1}{c^2 + (v - \gamma_\rho)^2} \ge \frac{c}{2\pi} \sum_\rho \frac{1}{1 + (v - \gamma_\rho)^2}$, since $c \le 1$. So (31.11.1) holds with $w^* = 2c^*$, and (ii) at $v$ follows from

$$W(v) = \frac{1}{\pi} \sum_{\rho} \frac{1}{1 + (v - \gamma_\rho)^2} \;\ge\; 2c^* = w^*, \tag{31.11.2}$$

a condition on the ordinates alone, with no $\theta'$ in it. Dropping the zeros above height 1419.9 and those of negative height only lowers $W$. Ball arithmetic certifies (31.11.2) on $[13.5, 1418.9]$ with the 1000 certified positive ordinates, 7027 cells of radius 0.1, each term evaluated at the farthest point of the cell; the smallest certified cell bound is $W \ge 0.06174$, in the cell about $v = 17.4$, between the first two zeros (`code/cx_min.py`, part W; `data/cx_c2_W.json`), and the bracket is at least $\pi(0.06174 - w^*) = 0.10146$. With the negative ordinates included the same cells give 0.0653. For $|v| \ge 1418.9$ the seven ordinates within distance 8 give $W \ge 7/(65\pi) > 0.03427 > w^*$ and the bracket is at least $7/65 - \pi w^* = 0.01518$ (with the nine ordinates of the count of [176] it is $9/65 - \pi w^* = 0.0459$). For $|v| \le 13.5$ every term $P_{a_\rho}(v - \gamma_\rho)$ is convex, because $a_\rho \le \theta' - (1 - \theta) \le 1$ and $|v - \gamma_\rho| \ge 0.63 > 1/\sqrt{3}$, so $u_{\theta'}$ is strictly convex and even there and its minimum is at 0. The excess (31.11.1) is the difference of the two bounds. The positivity of the atoms gives the measure statement. □

Theorem 31.11.3 contains the earlier results as special cases. With $\theta = 1$ it gives the weak form of Theorem 31.10.2(i), with $\theta = \frac{7}{8}$ it is Theorem 31.11.1, and letting $\theta \downarrow \frac{1}{2}$ it is Theorem 31.10.3. Its proof is shorter than any of theirs because

the comparison is made directly between the pairing lower bound and the orbit upper bound, both of which scale linearly in $c$, so that the level drops out and a single certified function of the ordinates, $W$, decides every level at once. The monotonicity in $\theta$ that Theorem 31.11.1 needed is not used.

The quantitative content answers the question posed: as $\theta' \downarrow \frac{1}{2}$ the excess of the Thorin density over its value at the origin has a lower bound linear in $c = 2\theta' - 1$, with an explicit slope, at least $0.1014/(2\pi) = 0.0161$ per unit of $c$ on the certified range and $0.0151/(2\pi) = 0.0024$ beyond it. The slope is not sharp; the pairing lemma halves the density, and $W$ was bounded with $c = 1$ in place of $c$. The order is right: under the Riemann hypothesis, at the midpoint between two consecutive zeros with gap $G$, $u_{\theta'}(v) = \sum P_{c/2}(v-\gamma) \approx \frac{c}{2\pi} \cdot \frac{8}{G^2}$ for small $c$, and $u_{\theta'}(0) \approx c\,c^*/2$, so the excess is of order $c$ at fixed $v$ (it is of order $1/c$ at a zero). The slope itself is pinned down in the next subsection.

### 31.11.5 The slope of the excess

The losses in (31.11.1) are avoidable for the zeros whose position is known. By Theorem 31.9.6 every zero with $|\gamma| < 1419.9$ lies on the critical line, so at level $\theta'$ its width is exactly $c/2$ and its contribution to $u_{\theta'}$ is exactly $P_{c/2}(v-\gamma)$, with no pairing. Only the zeros above height 1419.9, whose real parts are unknown, need Lemma 31.10.1 and the orbit bound. Writing $V_c(v) = \sum_{|\gamma|<1419.9} \big(\frac{c^2}{4} + (v-\gamma)^2\big)^{-1}$, both signs of $\gamma$ included, the two sides become

$$u_{\theta'}(v) \;\geq\; \frac{c}{2\pi}\Big(V_c(v) + \sum_{|\gamma|\geq 1419.9} \frac{1}{c^2 + (v-\gamma)^2}\Big),$$

$$u_{\theta'}(0) \;\leq\; \frac{c}{2\pi}\Big(V_c(0) + 2 \sum_{|\gamma|\geq 1419.9} \frac{1}{\gamma^2}\Big),$$

and dropping the positive far sum on the left,

$$u_{\theta'}(v) - u_{\theta'}(0) \;\geq\; \frac{c}{2\pi}\, B_c(v), \qquad B_c(v) = V_c(v) - V_c(0) - 2 \sum_{|\gamma|\geq 1419.9} \frac{1}{\gamma^2}, \tag{31.11.3}$$

where the last sum is at most 0.001442 by [176], so that $B_c(v) \geq V_c(v) - V_c(0) - 0.0029$ (the far sum in the display below is written with $c^2$ in place of $\frac{c^2}{4}$, which is the weaker bound). On $13.5 \leq |v| \leq 1418.9$ the function $B_c$ decreases in $c$ (its derivative in $c$ is $\frac{c}{2}\sum_\gamma \big[(\frac{c^2}{4} + \gamma^2)^{-2} - (\frac{c^2}{4} + (v-\gamma)^2)^{-2}\big]$, in which the first sum is below $\sum \gamma^{-4} < 7.5 \times 10^{-5}$ while the nearest certified ordinate, within 3.45 of $v$, alone gives more than $(\frac{1}{4} + 3.45^2)^{-2} > 6.7 \times 10^{-3}$ to the second; single terms need not decrease), so its value at $c = 0$, which involves only $\sum (v-\gamma)^{-2}$ over the certified ordinates, is a lower bound for the slope as $\theta' \downarrow \frac{1}{2}$.

Proposition 31.11.4 (the slope). *Under the hypothesis of Theorem 31.11.3, for every $\theta' \in [\theta, 1]$ and $c = 2\theta' - 1$,*

$$\inf_{13.5 \le |v| \le 1418.9} B_c(v) \;\ge\; \begin{cases} 0.17575 & c = 1, \\ 0.17838 & c = \tfrac{1}{2}, \\ 0.17904 & c = \tfrac{1}{4}, \\ 0.17927 & c \downarrow 0, \end{cases}$$

*each infimum being approached at $v = 17.49$ to $17.50$, between the first two zeros; and, for $|v| \ge 1418.9$,*

$$B_c(v) + \sum_{|\gamma| \ge 1419.9} \frac{1}{c^2 + (v-\gamma)^2} \;\ge\; \frac{7}{c^2+64} - V_c(0) - 0.0029 \;\ge\; 0.0599,$$

*while $B_c(v)$ itself tends to $-V_c(0) - 2\sum_{|\gamma|\ge 1419.9} \gamma^{-2} < 0$ as $|v| \to \infty$. Under the Riemann hypothesis the limiting bracket*

$$X(v) = \lim_{c \downarrow 0} \frac{2\pi}{c}\big(u_{\theta'}(v) - u_{\theta'}(0)\big) = \sum_{\rho} \Big( \frac{1}{(v-\gamma_\rho)^2} - \frac{1}{\gamma_\rho^2} \Big)$$

*has, in the first gap between the zeros $14.1347$ and $21.0220$, a unique minimiser $v^* \in [17.4978, 17.4980]$ (below the midpoint $17.5784$ of the gap), and*

$$0.1838985448 \;\le\; X(v^*) \;\le\; 0.1838987563;$$

*on the rest of $13.5 \le |v| \le 1418.9$, $V_0(v) - V_0(0) \ge 0.184074$ (the author's computation, not rerun).*

*Proof.* First clause, ball arithmetic as for (31.11.2): the interval $[13.5, 1418.9]$ is cut into cells of radius $0.1$, $V_c$ is bounded below on a cell by evaluating each term at the farthest point of the cell from the enclosure of $\gamma$, $V_c(0)$ is bounded above using the lower endpoints of the enclosures, and the cells whose lower bound falls below the best upper bound plus $0.005$ are bisected. The printed values are valid lower bounds; a rerun with bisection to radius $2^{-14}$ and the enclosures of width $0.02$ gives $0.175808$, $0.178430$, $0.179098$, $0.179322$ at $v = 17.493$ to $17.498$ (`code/cx_min.py`, part B; `data/cx_c2_B.json`). Far range: seven ordinates lie within distance $8$ of $v$; each contributes at least $(c^2+64)^{-1}$, through $V_c(v)$ if it is certified and through the far sum of the first display otherwise. The minimum over $c \in [0,1]$ of $7/(c^2+64) - V_c(0) - 0.0029$ is $0.0600$ (at $c = 1$, using $V_1(0) \le 0.044785$), so $0.0599$ holds.

The last statement. Under the Riemann hypothesis $X(v) = [V(v) - V(0)] + F(v)$ with $V = V_0$ over the certified ordinates and $F(v) = \sum_{\gamma \ge 1419.9} f_v(\gamma)$, where

$$f_v(\gamma) = \frac{1}{(\gamma - v)^2} + \frac{1}{(\gamma+v)^2} - \frac{2}{\gamma^2} = \frac{2v^2(3\gamma^2 - v^2)}{\gamma^2(\gamma^2 - v^2)^2} \;\ge\; 0 \qquad (\gamma > v),$$

so the uncertified zeros only raise $X$, and $X \ge V - V(0)$ on the whole range. Under the Riemann hypothesis $X''(v) = 6\sum_\rho (v-\gamma_\rho)^{-4} > 0$ between consecutive ordinates, so $X$ is strictly convex on $(14.1347, 21.0220)$ and has a unique minimiser $v^*$ there; on $[13.5, 14.1347)$ it is increasing, being even and convex on $(-\gamma_1, \gamma_1)$, and $X(13.5) > 2$, the nearest ordinate alone contributing 2.48 while $\sum_\rho \gamma_\rho^{-2} = 0.0462$. For $0 < v < \gamma$, $\partial_v f_v(\gamma) = 4v(3\gamma^2 + v^2)/(\gamma^2 - v^2)^3 > 0$, so $F' \ge 0$. With the 1000 certified ordinates (python-flint, 160 and 320 bits, `code/cx_slope_X.py`, `data/cx_slope_X.json`) the derivative $Q'$ of $Q = V - V(0)$ is enclosed by $Q'(17.4978) \in [-1.0391209, -1.0367081] \times 10^{-5}$ and $Q'(17.4980) \in [7.2326504, 7.2567776] \times 10^{-6}$, including a rigorous bound $0 \le F' \le 2.41 \times 10^{-8}$ for the tail (partial summation against [176]). So $X'$ changes sign exactly once in $[17.4978, 17.4980]$. For the value, tangent lines of $Q$ at the two endpoints give the lower bound, and $Q(17.4979)$ plus the tail bound $2.11 \times 10^{-7}$ the upper. The statement for the rest of the range is the cell computation of the first clause for $V - V(0)$ with the enclosures of the author, not rerun: the rerun of the first clause gives $V - V(0) \ge B_0 \ge 0.17927$ only. □

The certified values for the four levels differ by two per cent, so the slope of the excess is essentially independent of the level: in the notation of (31.11.1), the bracket is at least 0.1757 rather than 0.1014 on the middle range and at least 0.0599 rather than 0.0151 beyond it, and the slope per unit $c$ is at least $0.1757/(2\pi) = 0.0279$ against the earlier 0.0161. The computation also answers the two questions of the open problem. By the author's computation (the last clause of Proposition 31.11.4, not rerun) the infimum over $13.5 \le v \le 1418.9$ is attained between the first two zeros, at $v^* = 17.4979$, not at their midpoint 17.578 but pushed below the midpoint by the zeros above; and its value as $\theta' \downarrow \frac{1}{2}$ is 0.18390 to five digits under the Riemann hypothesis, the uncertified zeros contributing less than $3 \times 10^{-7}$ because their paired terms $f_v(\gamma)$ are of order $v^2/\gamma^4$ rather than $1/\gamma^2$. The two neighbours alone give $8/G^2 = 0.1687$ with $G = 6.887$ at the midpoint of the gap (0.1689 at $v^*$), the other 1998 certified ordinates of both signs add 0.060, and $V(0)$ takes away 0.0448. The unconditional certificate (31.11.3) is weaker than this by the orbit bound at the origin, $2\sum_{\text{far}} \gamma^{-2} \le 0.0029$, which is the price of not assuming where the far zeros lie.

### 31.11.6 A posteriori bounds for the backward flow

The fourth open problem asked for a bound on the distance between the EM output and the Thorin measure in terms of the computed residual, so that quantities read from the output become certified. The bound is a duality.

Proposition 31.11.5 (a posteriori window bound). *Let $\theta' \in [\frac{1}{2}, 1)$, $a = 1 - \theta'$, and suppose $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$, so that $\mathfrak{U}_{\theta'}$ exists and $P_a * \mathfrak{U}_{\theta'} = u_1\, dv$. Let $\mu$ be a finite positive measure, for instance $\mu = \sum_j x_j h\, \delta_{w_j}$ built from an EM iterate,*

*and let* $r = P_a * \mu - u_1$ *on* $\mathbb{R}$. *For a test function* $\varphi$ *with* $\int e^{a|\omega|}|\widehat{\varphi}(\omega)|\,d\omega < \infty$ *put* $\psi = \mathcal{F}^{-1}[e^{a|\omega|}\widehat{\varphi}]$, *and assume that* $\psi \cdot (P_a * \mu + u_1)$ *is integrable (this holds for the Gaussian windows below, by the tails of Lemma 31.11.6). Then*

$$\Big|\int \varphi\,d\mu - \int \varphi\,d\mathfrak{U}_{\theta'}\Big| \;\le\; \int_{\mathbb{R}} |r(v)|\,|\psi(v)|\,dv.$$

*Proof.* $\widehat{P_a} = e^{-a|\omega|}$, so $P_a * \psi = \varphi$. By Fubini, justified by the integrability hypothesis applied to the absolute integrand, $\int \varphi\,d\mu = \int \psi\,(P_a * \mu)\,dv$ and $\int \varphi\,d\mathfrak{U}_{\theta'} = \int \psi\,u_1\,dv$. Subtract. □

Three remarks fix how the bound is used. The residual is needed on all of $\mathbb{R}$: inside the data window it is the EM residual, between the data window and the edge of the grid carrying $\mu$ it is computable from $u_1$, and beyond that grid $r = P_a * \mu - u_1$ with $P_a * \mu(v) \le a\|\mu\|/(\pi(a^2 + (|v| - W)^2))$ for $\operatorname{supp}\mu \subset [-W, W]$ and $|v| > W$, against the decay of $\psi$, which is $|\psi(v)| \sim a\sigma\sqrt{2\pi}/(\pi(v-c)^2)$ for the Gaussian window $\varphi = e^{-(v-c)^2/2\sigma^2}$, from the kink of $e^{a|\omega|}$ at the origin. The data must be known to certified accuracy on the computation grid: in the author's experience, interpolating $u_1$ from a coarser grid introduces errors of $10^{-3}$, larger than the EM residual, and the bound fails. And the tail must be measured from the window centre, not from the origin. Centroid certificates come from the odd test function $\varphi_1(v) = (v-c)\varphi(v)$: with certified mass $M \pm b_0$ and certified first moment $m_1 \pm b_1$, the $\varphi$ weighted centroid of $\mathfrak{U}_{\theta'}$ lies within $(|m_1| + b_1)/(M - b_0)$ of $c$. This is a statement about a weighted centroid; at $\theta' = \frac{1}{2}$, under the Riemann hypothesis, an interval for a single zero needs in addition a bound on the contribution of the other zeros, Corollary 31.11.9 below.

In floating point, with exact data at spacing 0.02 on $[-400, 400]$, the unknown on $[-440, 440]$ and 4000 accelerated steps, the author found that the bound holds for every one of the 198 windows centred at the zeros in $(5, 395)$, for masses and for moments, at $\theta' = \frac{1}{2}$ and $\theta' = \frac{3}{4}$ and for $\sigma = 0.5$ and 0.3; the figures are superseded by the certified ones of the next subsection, which they agree with once the tails are separated. These computations were not repeated in this book.

### 31.11.7 The certificate

Two things stood between the last subsection and a certificate in the sense of Section 31.9.9: the dual function $\psi$ came out of a floating point transform, and the bound $\int |r||\psi|$ was a Riemann sum standing in for an integral. Both are removed by changing the test function slightly, from the Gaussian $\varphi$ to the function that the grid actually sees.

Lemma 31.11.6 (the dual function in closed form). *For* $\varphi(v) = e^{-(v-c)^2/2\sigma^2}$ *and* $a > 0$, *with* $q = a + i(v - c)$ *and*

$$E(q) = e^{q^2/2\sigma^2}\operatorname{erfc}\Big(-\frac{q}{\sigma\sqrt{2}}\Big),$$

*the dual functions of $\varphi$ and of $\varphi_1(v) = (v-c)\varphi(v)$ are*

$$\psi(v) = \operatorname{Re} E(q), \qquad \psi_1(v) = \operatorname{Im}\big(q\,E(q)\big) = -\sigma^2\psi'(v).$$

*Both extend to entire functions of $v$: with $q_\pm = a \pm i(z-c)$, $\Psi(z) = \frac12[E(q_+) + E(q_-)]$ and $\Psi_1(z) = \frac{1}{2i}[q_+E(q_+) - q_-E(q_-)]$. For every $x = v - c \neq 0$,*

$$|\psi(v)| \le \frac{\sigma\sqrt{2\pi}}{\pi}\Big(\frac{a}{x^2} + \frac{B_4}{x^4}\Big), \qquad |\psi_1(v)| \le \frac{\sigma^3\sqrt{2\pi}}{\pi}\Big(\frac{2a}{|x|^3} + \frac{B_5}{|x|^5}\Big),$$

*where, with $g(\omega) = e^{a\omega - \sigma^2\omega^2/2}$ and $G(\omega) = \omega g(\omega)$, $B_4 = |a^3 - 3\sigma^2 a| + \int_0^\infty |g^{(4)}|$ and $B_5 = 4|a^3 - 3\sigma^2 a| + \int_0^\infty |G^{(5)}|$.*

*Proof.* By definition,

$$\psi(v) = \frac{1}{2\pi}\int e^{a|\omega|}\widehat{\varphi}(\omega)e^{i\omega v}d\omega = \frac{\sigma\sqrt{2\pi}}{\pi}\int_0^\infty g(\omega)\cos(\omega x)\,d\omega,$$
$$\int_0^\infty e^{-p\omega^2+q\omega}d\omega = \tfrac12\sqrt{\pi/p}\,e^{q^2/4p}\operatorname{erfc}\big(-q/2\sqrt{p}\big),$$

and the second, classical, integral with $p = \sigma^2/2$ and $q = a + ix$ gives the first formula; at $a = 0$ it reduces to $\varphi$, since erfc of an imaginary argument has real part 1. Since $\overline{E(q)} = E(\bar q)$, the real part is $\frac12[E(q_+) + E(q_-)]$ on the real axis, which is entire in $z$. Differentiating, $E'(q) = (q/\sigma^2)E(q) + \sqrt{2/\pi}/\sigma$, so $\psi'(v) = \operatorname{Re}(iE'(q)) = -\sigma^{-2}\operatorname{Im}(qE(q))$; and $\widehat{\varphi_1} = -i\sigma^2\omega\widehat{\varphi}$ gives $\psi_1 = -\sigma^2\psi'$. The tails come from integrating by parts:

$$\int_0^\infty g\cos(\omega x)\,d\omega = -\frac{g'(0)}{x^2} + \frac{g'''(0)}{x^4} + \frac{1}{x^4}\int_0^\infty g^{(4)}\cos(\omega x)\,d\omega,$$
$$\int_0^\infty G\sin(\omega x)\,d\omega = -\frac{G''(0)}{x^3} + \frac{G''''(0)}{x^5} + \frac{1}{x^5}\int_0^\infty G^{(5)}\cos(\omega x)\,d\omega,$$

with $g'(0) = a$, $g'''(0) = a^3 - 3\sigma^2 a$, $G''(0) = 2a$ and $G''''(0) = 4(a^3 - 3\sigma^2 a)$; the second integral is the one for $\psi_1 = \frac{\sigma^3\sqrt{2\pi}}{\pi}\int_0^\infty G\sin(\omega x)\,d\omega$. □

The integrals $\int|g^{(4)}|$ and $\int|G^{(5)}|$ are enclosed by writing $g^{(k)} = H_k g$ with the polynomials $H_{k+1} = H_k' + (a - \sigma^2\omega)H_k$, replacing the coefficients of $H_k$ by their absolute values, and integrating in balls. The author reports $B_4 = 38.0$ and $B_5 = 550.6$ at $a = \frac12$, $\sigma = 0.3$ (not rerun); the constants are generous, but they multiply $x^{-4}$ and $x^{-5}$ and matter only where the bound is used, at distances above 440.

Now fix the grid $v_i = ih$, $|v_i| \le 880$, and let $\psi_h = \sum_i \psi(v_i)h\,\delta_{v_i}$ and $\varphi_h = P_a * \psi_h$. For any finite positive $\mu$ with $P_a * \mu$ finite at the grid points, and $U = \mathfrak{U}_{\theta'}$,

$$\int \varphi_h\,d\mu - \int \varphi_h\,d\mathfrak{U}_{\theta'} = \sum_i \psi(v_i)\,h\,r_i, \qquad r_i = (P_a * \mu)(v_i) - u_1(v_i), \quad (31.11.4)$$

exactly, by Fubini (the sum defining $\psi_h$ is finite, so the interchange is absolute) and $P_a * \mathfrak{U}_{\theta'} = u_1$. There is no integral left: the right side is a finite sum of balls once $r_i$ and $\psi(v_i)$ are balls. For $\mu = \sum_j x_j h \delta_{w_j}$ on the grid, with $x_j$ the cell densities, $(P_a * \mu)(v_i) = \sum_j x_j h P_a(v_i - w_j)$ is a polynomial product, which Arb [290] multiplies with rigorous error bounds in a fraction of a second; the $x_j$ are the double precision EM output, taken as exact rationals. The identity does not depend on EM having converged, or on $\mu$ having been produced by EM. What remains is to compare $\varphi_h$ with $\varphi$:

$$\int \varphi \, d\mu - \int \varphi \, d\mathfrak{U}_{\theta'} = \sum_i \psi(v_i) h r_i + \int (\varphi - \varphi_h) \, d\mu - \int (\varphi - \varphi_h) \, d\mathfrak{U}_{\theta'}.$$

Write $\varphi_h^{\infty}$ for the same sum over the whole lattice $h\mathbb{Z}$. Then $\varphi - \varphi_h = (\varphi - \varphi_h^{\infty}) + \sum_{|v_i|>880} \psi(v_i) h P_a(\cdot - v_i)$, the first term the trapezoid error of an analytic integrand, the second a truncation controlled by the tails of Lemma 31.11.6.

Lemma 31.11.7 (trapezoid error). *Let $\Psi$ be the entire extension of $\psi$, $f_w(v) = \Psi(v) P_a(w - v)$, and assume $h \le \pi\sigma^2/(2a)$. For every real $w$,*

$$|\varphi(w) - \varphi_h^{\infty}(w)| \;\le\; \min\Big( \frac{4 S_\Psi}{e^{\pi a/h} - 1}, \; \frac{E_{\mathrm{dec}}}{(w-c)^2} \Big),$$

*where $S_\Psi = 2e^{9a^2/8\sigma^2} + 1$ bounds $|\Psi|$ on the strip $|\operatorname{Im} v| \le a/2$, and*

$$E_{\mathrm{dec}} = \frac{\sigma\sqrt{2\pi}}{2\pi} \, (4a + J) \, \frac{2e^{-2\pi a/h}}{1 - e^{-2\pi a/h}},$$
$$J = \int_{\mathbb{R}} \big( (2a + \sigma^2|\omega|)^2 + \sigma^2 \big) e^{2a|\omega| - \sigma^2\omega^2/2} \, d\omega;$$

*at $w = c$ the second term of the minimum is $+\infty$. The same holds for $\psi_1$ and $\varphi_1$ with $b = \frac{3a}{2}$, $S_1 = \sigma^3\sqrt{2\pi}\,\pi^{-1} \int_0^\infty t e^{bt - \sigma^2 t^2/2} dt$ in place of $S_\Psi$ with numerator $4S_1$, $C_1 = \sup_{t \ge 0} t e^{2at - \sigma^2 t^2/2}$, $J_1 = \int_{\mathbb{R}} \big[ 2(2a + \sigma^2|\omega|) + |\omega| \big( (2a + \sigma^2|\omega|)^2 + \sigma^2 \big) \big] e^{2a|\omega| - \sigma^2\omega^2/2} d\omega$, and*

$$E_{\mathrm{dec},1} = \frac{\sigma^3\sqrt{2\pi}}{2\pi} \, (2aC_1 + J_1) \, \frac{2e^{-2\pi a/h}}{1 - e^{-2\pi a/h}}.$$

*Proof.* The uniform bound is the theorem of Trefethen and Weideman [500] for the trapezoid rule on $\mathbb{R}$: if $f$ is analytic in $|\operatorname{Im} v| < d$, tends to zero there, and $\int |f(v + ib)| dv \le M$ for $|b| < d$, then

$$\Big| h \sum_k f(kh) - \int f \Big| \le \frac{2M}{e^{2\pi d/h} - 1}.$$

Take $d = a/2$, inside the poles of $P_a(w - \cdot)$ at $w \pm ia$. On the strip,

$$|a^2 + (t - ib)^2| = |t - i(b + a)| \, |t - i(b - a)| \ge t^2 + \tfrac{a^2}{4},$$

so $\int |P_a(w-v-ib)|\,dv \le (a/\pi)(2\pi/a) = 2$ and $M \le 2\sup|\Psi|$. For

$$\Psi(v+ib) = \tfrac{1}{2}\big[E(a-b+i(v-c)) + E(a+b-i(v-c))\big]$$

write $E(q) = 2e^{z^2} - w(iz)$ with $z = q/\sigma\sqrt{2}$ and $w$ the Faddeeva function, which satisfies $|w(\zeta)| \le 1$ for $\operatorname{Im}\zeta \ge 0$, that is for $\operatorname{Re} q > 0$; and $|e^{z^2}| = e^{((\operatorname{Re} q)^2 - (\operatorname{Im} q)^2)/2\sigma^2} \le e^{9a^2/8\sigma^2}$. Hence $S_\Psi$. The decaying bound is Poisson summation: $\varphi_h^{\infty}(w) - \varphi(w) = \sum_{k\neq 0} \widehat{f_w}(2\pi k/h)$ and

$$\widehat{f_w}(\omega_0) = \frac{e^{-i\omega_0 w}}{2\pi}\int \widetilde{G}(\omega) e^{i\omega(w-c)} d\omega,$$
$$\widetilde{G}(\omega) = \sigma\sqrt{2\pi}\, e^{a|\omega| - a|\omega-\omega_0| - \sigma^2\omega^2/2},$$

which is smooth except for kinks at 0 and $\omega_0$. Two integrations by parts give $(w-c)^{-2}$ times the sum of the jumps of $\widetilde{G}'$, each at most $2a\sigma\sqrt{2\pi}e^{-a|\omega_0|}$ since $e^{2a|\omega_0| - \sigma^2\omega_0^2/2} \le 1$ for $|\omega_0| \ge 4a/\sigma^2$, which holds for every sampling frequency $|\omega_0| \ge 2\pi/h$ exactly when $h \le \pi\sigma^2/(2a)$; plus $\int|\widetilde{G}''| \le \sigma\sqrt{2\pi}e^{-a|\omega_0|}J$, using $e^{a|\omega| - a|\omega-\omega_0|} \le e^{-a|\omega_0| + 2a|\omega|}$. Summing the geometric series over $k$ gives $E_{\mathrm{dec}}$. For $\psi_1$ the transform carries the factor $-i\sigma^2\omega$, the strip bound is the Fourier integral of $\Psi_1$ on the strip, and the second derivative of $\omega\,\widetilde{G}$ is bounded in the same way, which gives the constants $S_1$, $C_1$, $J_1$ (derived in the review of the source, not rerun). $\square$

With $h = 0.02$ the condition $h \le \pi\sigma^2/(2a)$ holds for all four parameter pairs of the table, $e^{\pi a/h}$ is $e^{78.5}$ at $a = \frac{1}{2}$ and $e^{39.3}$ at $a = \frac{1}{4}$, and the constants are below $10^{-30}$ and $10^{-15}$ respectively; they are carried through the computation, and they are the reason the grid spacing was never allowed to grow. The last ingredient is a bound on the mass of $\mathfrak{U}_{\theta'}$, which is unknown but dominated through the forward equation: for $L > 0$,

$$\mathfrak{U}_{\theta'}([-W,W]) \;\le\; \frac{2M_1(W+L)}{1 - 2P_a((L,\infty))},$$
$$M_1(T) = \int_0^T u_1(v)\,dv = \pi^{-1}\operatorname{Im}\log\xi(1+iT),$$

since $P_a * \mathfrak{U}_{\theta'} \ge P_a * (\mathfrak{U}_{\theta'}|_{[-W,W]})$ and the latter integrates to at least $P_a([-L,L])\,\mathfrak{U}_{\theta'}([-W,W])$ over $[-W-L, W+L]$. The function $M_1$ is explicit up to $\arg\zeta(1+iT)$ (continuous branches of the logarithms, continued horizontally from $\sigma = +\infty$):

$$\pi M_1(T) = \arctan T + \tfrac{\pi}{2} - \tfrac{T}{2}\log\pi + \operatorname{Im}\log\Gamma\big(\tfrac{1+iT}{2}\big) + \arg\zeta(1+iT),$$

and $|\arg\zeta(1+iT)| \le 24.303 + \log\log T + 0.498$ for $T \ge 13$, by integrating the explicit bound $|\zeta'/\zeta(\sigma+it)| \le 24.303\log t$ on $\sigma \ge 1$ from 1 to $1+1/\log T$, the bound

$-\zeta'/\zeta(\sigma) \le 1/(\sigma-1)$ from there to 2, and $\log\zeta(2) = 0.4977$ beyond. The explicit bound is Leong's [208]; the constant 24.303 is the author's, and the corollary of that paper is reported (unchecked) to give $18.062\log t$ for $\sigma \ge 1$, $t \ge 5$, so 24.303 is conservative. So $M_1$ is a ball, and the mass of $\mathfrak{U}_{\theta'}$ on $[-W, W]$ is bounded by an increasing explicit function $\overline{U}(W)$, with $\overline{U}(880) = 1286.4$ at $a = \frac12$ (the author's value, not rerun).

THEOREM 31.11.8 (certified windows; conditional, computed by the author). *Let $\theta' \in \{\frac12, \frac34\}$, $a = 1-\theta'$, $\sigma \in \{0.3, 0.5\}$, suppose $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$ (at $\theta' = \frac12$ this is the Riemann hypothesis; at $\theta' = \frac34$ it is Theorem 31.9.6 up to height 1419.9 and an assumption beyond), and let $\mu$ be the accelerated EM output after 4000 steps on the grid of spacing $h = 0.02$ on $[-440, 440]$. For each of the 198 windows $\varphi$ of width $\sigma$ centred at the grid point $c$ nearest a zero in $(5, 395)$, with $\varphi_1 = (v-c)\varphi$,*

$$\Big|\int \varphi\, d\mu - \int \varphi\, d\mathfrak{U}_{\theta'}\Big| \le B_0(c), \qquad \Big|\int \varphi_1\, d\mu - \int \varphi_1\, d\mathfrak{U}_{\theta'}\Big| \le B_1(c),$$

$$B_0(c) = h \sum_{|v_i|\le 880} |\psi(v_i)||r_i| + \|\mu\|\,\varepsilon_\mu + \overline{U}(880)\,\varepsilon_U + T(c),$$

$$T(c) = \int_{|w|>880} D(|w| - |c|)\, d\mathfrak{U}_{\theta'}(w),$$

*and $B_1$ likewise with $\psi_1$, where $\varepsilon_\mu, \varepsilon_U$ are the uniform bounds on $|\varphi - \varphi_h|$ over the support of $\mu$ and over $[-880, 880]$ given by Lemma 31.11.7 and the tails of Lemma 31.11.6, $D$ is the explicit decreasing majorant*

$$D(d) = \frac{E_{\mathrm{dec}}}{d^2} + \frac{K_{\mathrm{far}}\, a}{\pi(1-\lambda)^2 d^2} + \Big(1 + \frac{h}{\pi a}\Big)\bar{k}(\lambda d), \qquad \lambda = 0.9,$$

*with $\bar{k}$ the tail bound of Lemma 31.11.6 and $K_{\mathrm{far}}$ a bound on $\sum_{|v_i|>880} |\psi(v_i)|h$, and the last integral is bounded by Abel summation against $\overline{U}$ over blocks $W_k = 880 \cdot 1.005^k$ up to $10^7$ and by $\overline{U}(w) \le w\log w$ beyond. Every quantity is a ball. The 792 pairs of bounds (mass and moment, for 198 windows, two levels and two widths) were compared with floating point reference values and agree. The certified figures are in the table below.*

The finite computations of this theorem are the author's and were not rerun in this book (the script of the source is `certify_apost.py`); the proof of the bound given those computations is the identity (31.11.4), Lemma 31.11.7 and the mass bound. The majorant $D$ is valid after Lemma 31.11.7: split the omitted sample points into $|v_i - c| \le \lambda d$, where $P_a(w - v_i) \le a/[\pi(1-\lambda)^2 d^2]$, and the complement, where $|\psi(v_i)| \le \bar{k}(\lambda d)$ and $h\sum_i P_a(w - v_i) \le 1 + h/(\pi a)$. The Abel summation is an upper sum: if $D$ is decreasing, integration by parts gives $\int_{|w|>V} D(|w| - |c|)\, d\mathfrak{U} \le \int_V^\infty [-D'(R - |c|)]\, \overline{U}(R)\, dR$, and on a block $[W_k, W_{k+1}]$ the integral is at most $\overline{U}(W_{k+1})[D(W_k - |c|) - D(W_{k+1} - |c|)]$.

| $\theta'$ | $\sigma$ | mass bound $B_0$ | | | moment $B_1$ | | centroid bound | |
|---|---|---|---|---|---|---|---|---|
| | | max | mean | far tail | max | mean | median | max |
| $1/2$ | 0.5 | $2.2\times10^{-2}$ | $5.4\times10^{-3}$ | $1.7\times10^{-3}$ | $1.8\times10^{-3}$ | $1.1\times10^{-3}$ | 0.027 | 0.20 |
| $1/2$ | 0.3 | $1.4\times10^{-2}$ | $5.1\times10^{-3}$ | $1.1\times10^{-3}$ | $2.0\times10^{-3}$ | $1.2\times10^{-3}$ | 0.0072 | 0.11 |
| $3/4$ | 0.5 | $5.7\times10^{-3}$ | $1.7\times10^{-3}$ | $8.3\times10^{-4}$ | $2.2\times10^{-4}$ | $1.1\times10^{-4}$ | 0.042 | 0.14 |
| $3/4$ | 0.3 | $3.4\times10^{-3}$ | $1.0\times10^{-3}$ | $4.9\times10^{-4}$ | $3.6\times10^{-5}$ | $1.2\times10^{-5}$ | 0.0068 | 0.071 |

The far tail column is the mean of the last term of $B_0$, the price of not knowing $\mathfrak{U}_{\theta'}$ beyond 880 except through its mass bound; it is a fifth to a half of the mean bound, and would shrink if the grid were extended, since $\overline{U}$ grows only logarithmically faster than the width. The main term, the finite sum in (31.11.4), has mean $3.6\times10^{-3}$ at $\theta'=\frac12$, $\sigma=0.3$; the floating point bound of the previous subsection, with its own tail estimate, was $4.5\times10^{-3}$, and the two agree once the tails are separated. The certificate bounds Gaussian weighted masses and centroids, and nothing finer. At $\theta'=\frac12$ every window has certified mass at least 0.9929, so, since an interval of length one below height 400 holds at most $n(400)<9.34$ ordinates (the count of Theorem 31.11.10), each window has a zero within 0.73 of its centre at $\sigma=0.3$ and within 1.22 at $\sigma=0.5$: if no zero lay within $r$ of the centre, the Gaussian mass would be at most $2n\sum_{k\ge0}e^{-(r+k)^2/(2\sigma^2)}$, which is 0.967 and 0.952 at these radii, below 0.9929 (the critical radii are 0.7267 and 1.2115). The certified centroid lies within 0.05 of the window centre for 192 of the 198 windows at $\sigma=0.3$ (as reported by the author), the six exceptions being the close pairs, where a neighbouring atom at distance $d$ moves the centroid by an amount of the order of $d\,e^{-d^2/2\sigma^2}$. At $\theta'=\frac34$ the target is the Thorin density $\sum_\rho P_{3/4-\sigma_\rho}(v-\gamma_\rho)$, which is $\sum P_{1/4}(v-\gamma)$ when the zeros are on the line, the window mass is about 0.7, the bounds are five times tighter because the multiplier $e^{|\omega|/4}$ is milder, and the certified quantities are masses and centroids of the Thorin measure at level $\frac34$ itself. The computation takes thirty seven minutes on two cores, almost all of it in the 176,002 evaluations of erfc at complex arguments that build the tables of $\psi$ and $\psi_1$ along the grid; the convolutions, the window sums and the Abel summations are polynomial products and short loops. The serialisation of the EM vector and of the data balls must be lossless for the bound to be a certificate; this was not repeated here.

Corollary 31.11.9 (from centroids to zeros). *Assume the Riemann hypothesis, so that* $\mathfrak{U}_{1/2}=\sum_\rho\delta_{\gamma_\rho}$. *Let* $\varphi(v)=e^{-(v-c)^2/(2\sigma^2)}$, *let* $\int\varphi\,d\mathfrak{U}_{1/2}\in[\widehat{M}-B_0,\widehat{M}+B_0]$ *with* $\widehat{M}>B_0$ *and* $\int(v-c)\varphi\,d\mathfrak{U}_{1/2}\in[\widehat{m}-B_1,\widehat{m}+B_1]$ *be certified, let* $\gamma_*$ *be an ordinate with* $|\gamma_*-c|\le R$, *and let* $\Theta\ge\sum_{\gamma\ne\gamma_*}(|\gamma-c|+R)\varphi(\gamma)$, *the sum over the other ordinates with multiplicity. Then*

$$|\gamma_*-c|\ \le\ \frac{|\widehat{m}|+B_1+\Theta}{\widehat{M}-B_0}.$$

*Proof.* With $M=\int\varphi\,d\mathfrak{U}_{1/2}$, $\int(v-c)\varphi\,d\mathfrak{U}_{1/2}=(\gamma_*-c)M+\sum_{\gamma\ne\gamma_*}(\gamma-\gamma_*)\varphi(\gamma)$ and

$|\gamma - \gamma_*| \le |\gamma - c| + R$. Hence $|\gamma_* - c|\, M \le |\widehat{m}| + B_1 + \Theta$, and $M \ge \widehat{M} - B_0 > 0$. $\square$

Three remarks. First, the windows are centred at grid points, within 0.01 of the zeros, so that $\psi(v_i - c)$ is a shift of one table; the centroid certificate is unaffected, since it reports the offset of the weighted centre from $c$. Below height 1419.9 the ordinates are already certified (Theorem 31.9.6), so $\Theta$ is computable but Corollary 31.11.9 locates nothing new there: the content of the certificate is the window masses and centroids, and at $\theta' = \frac{3}{4}$ the Thorin density itself. Second, the hypothesis that $\zeta$ has no zeros in $\operatorname{Re} s > \theta'$ is used only to know that $\mathfrak{U}_{\theta'}$ exists and satisfies $P_a * \mathfrak{U}_{\theta'} = u_1$; at $\theta' = \frac{1}{2}$ the hypothesis is the Riemann hypothesis. Third, the bound is a statement about $\mathfrak{U}_{\theta'}$, not about zeros; at $\theta' = \frac{1}{2}$, under the Riemann hypothesis, it becomes a statement about Gaussian weighted sums over the zeros, and about single zeros only through Corollary 31.11.9.

### 31.11.8 An iteration adapted to the certificate

The certificate's main term is $h \sum_i |\psi_c(v_i)||r_i|$, a weighted $\ell^1$ norm of the residual, and the question was whether an iteration aimed at it does better than EM, which minimises the Kullback and Leibler divergence. We compared, at $\theta' = \frac{1}{2}$ on the grid of spacing 0.02, with $\sigma = 0.3$ and the weights $w_i = \sum_c |\psi_c(v_i)|$ summed over the 198 windows: accelerated EM; FISTA with nonnegativity on the weighted least squares $\frac{1}{2}\sum_i w_i (Ax - y)_i^2$, which bounds the sum over the windows of the weighted $\ell^1$ norms by Cauchy and Schwarz, $h \sum_i w_i |r_i| \le h (\sum_i w_i)^{1/2} (\sum_i w_i r_i^2)^{1/2}$ (an aggregate bound, not a bound for each window); FISTA on the plain least squares; and accelerated EM followed by a weighted FISTA polish. The figure of merit is the mean over the windows of the main term. The experiment is the author's and was not repeated in this book.

| iteration (steps) | main term | KL |
|---|---|---|
| accelerated EM (4000) | $5.32 \times 10^{-5}$ | $1.2 \times 10^{-4}$ |
| weighted FISTA (4000) | $7.38 \times 10^{-5}$ | 0.95 |
| plain FISTA (4000) | $5.26 \times 10^{-5}$ | $1.4 \times 10^{-4}$ |
| EM (4000), weighted FISTA polish (1000) | $4.12 \times 10^{-5}$ | $4.3 \times 10^{-4}$ |
| accelerated EM (5000) | $4.73 \times 10^{-5}$ | $9.8 \times 10^{-5}$ |

The weighted objective from a cold start is worse on its own figure of merit, and much worse in divergence: the weights concentrate on the peaks of $u_1$, where the deconvolution is most ill conditioned, and the iteration spends its steps there. Plain least squares and EM are indistinguishable. The polish gains thirteen per cent over the same number of EM steps. The reading is that the main term is dominated by the residual within a unit of each zero, where every objective fits equally badly, the limit being the conditioning of the problem rather than the choice of loss. So in this experiment the certificate does not reward an iteration

of its own, and the route to a tighter certificate is more steps of the accelerated scheme, or a finer grid near the zeros; one experiment does not show that no other objective helps.

### 31.11.9 A fast algorithm for $u_1$ at fixed accuracy

The third open problem asked for an algorithm for $u_1(v)$ polynomial in $\log v$. Taken literally, with the accuracy $\varepsilon$ also entering polynomially in $\log(1/\varepsilon)$, it would follow from such an algorithm for $\zeta$ and $\zeta'$ on $\operatorname{Re} s = 1$, and is open. With the accuracy fixed it has a positive answer, which comes from the explicit formula: given that the zeros near the height $v$ lie on the critical line (verified below $3\times10^{12}$ [419]; at every height conditionally on the announced half plane $\operatorname{Re} s > \frac{7}{8}$, with a larger exponent), the primes below $X$ determine $u_1$ at that height to accuracy of order $X^{-1/2}\log v$, so the cost of one value is $O(X)$ whatever the height, and a nonuniform fast Fourier transform gives a whole grid of heights for the price of one.

Theorem 31.11.10 (smoothed prime sum). *Let $v \ge 2$, $X \ge 2$, $s = 1 + iv$, $L = \zeta'/\zeta$, and*

$$u_1^X(v) = \frac{1}{\pi}\Big[\frac{1}{1+v^2} - \frac{\log\pi}{2} + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+iv}{2}\Big) - \sum_{n\ge2}\frac{\Lambda(n)}{n}\,e^{-n/X}\cos(v\log n)\Big].$$

*Then, with $R_X(s) = (2\pi i)^{-1}\int_{\operatorname{Re} w=-3/2}(-L(s+w))\Gamma(w)X^w\,dw$,*

$$\pi\big(u_1(v) - u_1^X(v)\big) = -\operatorname{Re}\sum_\rho \Gamma(\rho-s)X^{\rho-s} + \operatorname{Re}\,\Gamma(1-s)X^{1-s} + \operatorname{Re}\big[X^{-1}L(s-1)\big] + \operatorname{Re} R_X(s),$$

*with $|R_X(s)| \le 2.5X^{-3/2}\log(v+2)$, and if every zero with $|\gamma - v| \le Y$, $Y \ge 1$, lies on the critical line,*

$$|u_1(v) - u_1^X(v)| \le \frac{1}{\pi}\Big[c_\Gamma\, n(v)\,X^{-1/2} + 2.6\,c'_\Gamma\, n(v+Y+1)\,e^{-\pi Y/2} + \sqrt{\frac{\pi}{v\sinh\pi v}} + \frac{27\log v}{X} + \frac{2.5\log(v+2)}{X^{3/2}}\Big],$$

*where $n(t) = 0.39\log t + 7$ bounds the number of ordinates in any interval of length one below height $t$, by [176, (1.7), (2.7) and Corollary 1.4]; $c_\Gamma = \sup_{v\ge2}\sum_{j\in\mathbb{Z}}\sup_{j\le y<j+1}|\Gamma(-\frac12+iy)|\,\frac{n(v+|j|+1)}{n(v)} \le 8.34$ (without the factor $n(v+|j|+1)/n(v)$ the sum is 8.14376); and $c'_\Gamma = \sup\{|\Gamma(x+iy)|e^{\pi|y|/2} : -1 < x < 0,\ |y| \ge 1\} \le 2.51$.*

*Proof.* $\sum_n \Lambda(n) n^{-s} e^{-n/X} = (2\pi i)^{-1} \int_{(2)} (-L)(s+w)\Gamma(w) X^w \, dw$ by Mellin inversion of $e^{-y}$. Shift the contour to $\operatorname{Re} w = -\frac{3}{2}$. This crosses the pole of $\Gamma$ at $w=0$, with residue $-L(s)$, the zeros $w = \rho - s$ (real part in $(-1,0)$), with residues $-\Gamma(\rho-s)X^{\rho-s}$ times the multiplicity, the pole $w = 1-s$ of $L$, with residue $\Gamma(1-s)X^{1-s}$, and the pole $w=-1$ of $\Gamma$, with residue $X^{-1}L(s-1)$; it crosses none of the trivial zeros ($\operatorname{Re} w = -2k-1 \le -3$) or of the later poles of $\Gamma$. The horizontal segments vanish along heights that avoid the zeros, by the exponential decay of $\Gamma$. Taking real parts and using $\pi u_1(v) = \operatorname{Re}[1/s - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2)] + \operatorname{Re} L(s)$ gives the identity. On $\operatorname{Re} w = -\frac{3}{2}$, $\operatorname{Re}(s+w) = -\frac{1}{2}$; by the functional equation $L(z) = \chi'/\chi(z) - L(1-z)$ with $\operatorname{Re}(1-z) = \frac{3}{2}$, where the Dirichlet series converges absolutely, so $|L(1-z)| \le -\zeta'/\zeta(\frac{3}{2}) < 1.5053$. Here $\chi'/\chi(z) = \log 2\pi + \frac{\pi}{2}\cot\frac{\pi z}{2} - \psi(1-z)$ with $|\cot\frac{\pi z}{2}| = 1$ on $\operatorname{Re} z = -\frac{1}{2}$, and Binet's formula $\psi(a) = \log a - \frac{1}{2a} - 2\int_0^\infty \frac{t\,dt}{(t^2+a^2)(e^{2\pi t}-1)}$, with $|t^2+a^2| \ge \frac{9}{4}$ for $\operatorname{Re} a = \frac{3}{2}$, gives $|\psi(\frac{3}{2}+i\tau)| \le \log(|\tau|+2) + \frac{\pi}{2} + \frac{1}{3} + \frac{1}{27}$. For $w = -\frac{3}{2} + iy$, as $\log(|v+y|+2) \le \log(v+2) + \log(1+|y|)$, this gives $|L(s+w)| \le \log(v+2) + \log(1+|y|) + 6.856$, and with $\int_{\mathbb{R}} |\Gamma(-\frac{3}{2}+iy)|\,dy < 2.527$ and $\int_{\mathbb{R}} |\Gamma(-\frac{3}{2}+iy)|\log(1+|y|)\,dy < 0.798$ (computed), $|R_X(s)| \le X^{-3/2}\big(0.41\log(v+2) + 2.89\big) \le 2.5 X^{-3/2}\log(v+2)$, since $\log(v+2) \ge \log 4$.

For the bound: the pole term is $\operatorname{Re}\Gamma(-iv)X^{-iv}$, of modulus at most $|\Gamma(-iv)| = (\pi/(v\sinh\pi v))^{1/2}$ since $|X^{-iv}| = 1$. For $v \ge 5$, $|L(iv)| \le 27\log v$ by the functional equation, $|\chi'/\chi(iv)| \le \log v$ and the explicit bound of [208] for $|L(1-iv)|$ on $\operatorname{Re} s = 1$; for $2 \le v \le 5$ it is a direct evaluation (the maximum of $|L(iv)|/\log v$ on $[2,60]$ is 1.014). For the zeros on the line, $|\Gamma(-\frac{1}{2}+iy)|^2 = \pi/(\cosh(\pi y)(\frac{1}{4}+y^2))$ and $|X^{\rho-s}| = X^{-1/2}$; the zeros are counted in the unit intervals $\gamma - v \in [j, j+1)$, each holding at most $n(v+|j|+1)$ ordinates, and the resulting sum is $c_\Gamma n(v) X^{-1/2}$. A zero at distance more than $Y$ contributes at most $c'_\Gamma e^{-\pi|\gamma-v|/2}$ ($|X^{\rho-s}| \le 1$ and $|\Gamma(\sigma_\rho - 1 + i(\gamma-v))| \le c'_\Gamma e^{-\pi|\gamma-v|/2}$), and $\sum_{k\ge 0} 2e^{-\pi k/2} n(v+Y+k+1) \le 2.6\, n(v+Y+1)$. The constants 8.3388, 8.14376 and 2.509 were recomputed in the reconciliation of the reviews, as was the count $n(t)$ against [176]. □

The pole term cannot be dropped. At $v=2$ every zero has $|\gamma - 2| \ge 12$ and lies on the critical line below $3\times 10^{12}$; taking $Y = 10^{12}$, the bound without the pole term tends to 0 as $X \to \infty$. But $\operatorname{Re}\Gamma(-2i)X^{-2i} = |\Gamma(-2i)|\cos(\arg\Gamma(-2i) - 2\log X)$ equals $|\Gamma(-2i)| = 0.0766$ at $X_k = \exp((\arg\Gamma(-2i) + 2\pi k)/2)$, and the other terms tend to 0, so $\limsup_{X\to\infty} |u_1(2) - u_1^X(2)| \ge 0.0766/\pi = 0.0244$. The pole term is exponentially small at large heights, which is why the experiments below are not affected.

The hypothesis on the zeros within $Y$ of $v$ is verified for $v + Y \le 3\times 10^{12}$ with $Y$ of a few dozen [419], and $e^{-\pi Y/2}$ is then negligible. Beyond, under the zero free region alone, $|X^{\rho-s}| \le X^{-c/\log v}$ and the first term decays only when $\log X \gg \log v$. Conditionally on the announced half plane $\operatorname{Re} s > \frac{7}{8}$ [398, 399, 352], every zero has $\frac{1}{8} \le \beta \le \frac{7}{8}$, so $|X^{\rho-s}| \le X^{-1/8}$ and $\operatorname{Re}(\rho - s)$ stays in $[-\frac{7}{8}, -\frac{1}{8}]$,

away from the poles of $\Gamma$; the same count gives $|u_1(v) - u_1^X(v) - \pi^{-1}\operatorname{Re}\Gamma(1-s)X^{1-s}| \le C\log(v+2)(X^{-1/8}+X^{-1})$ at every height, so that a fixed accuracy $\varepsilon$ costs $X$ of order $(\log v/\varepsilon)^8$. The number of terms is $O(X)$, since the sum can be cut at $n = 30X$ with a dropped tail of about $e^{-30}\log X$ (for a tolerance $\delta$ one cuts at about $X\log(1/\delta)$), and the cost of one value is $O(X)$ after a sieve of $O(X\log\log X)$, independent of $v$ in the count of arithmetic operations; the bit cost adds the evaluation of the phases $v\log n$, which needs $\log n$ to about $\log(v/\varepsilon)$ bits. For accuracy $\varepsilon$ take $X = (c_\Gamma n(v)/\varepsilon)^2$: the cost is $O(\varepsilon^{-2}\log^2 v)$ operations, polynomial in $\log v$ and in $1/\varepsilon$.

On a grid $v_k = kh$ the sum is

$$\operatorname{Re}\sum_n a_n e^{ik(h\log n)}, \qquad a_n = \frac{\Lambda(n)}{n}\,e^{-n/X},$$

a type one nonuniform discrete Fourier transform with the frequencies $h\log n$, and a nonuniform FFT evaluates it at $K$ heights with $M$ retained prime powers in $O(K\log K + M\log(1/\delta))$ operations for a transform accuracy $\delta$ [138], to which the coefficients, the sieve and the smooth gamma factor part are to be added. The experiment is the author's (script `fast_u1.py` of the source) and was not rerun in this book. With $X = 10^6$ and 1.86 million prime powers below $3\times10^7$, the error $|u_1 - u_1^X|$ at the heights $10^3$, $10^4$, $10^5$, $10^6$ and $10^7$ is reported as $9\times10^{-4}$, $4\times10^{-4}$, $8\times10^{-4}$, $1.1\times10^{-3}$ and $1.4\times10^{-3}$, against Arb [290]; averaged over fifty one heights near $10^6$ the error is $0.05\log v\,X^{-1/2}$ for $X = 10^4$, $10^5$ and $10^6$, which is the law of the theorem with a constant about twenty times smaller than the leading coefficient of the proved bound ($c_\Gamma\cdot 0.39/\pi = 1.0$ per $\log v$), the zero sum being oscillatory. The nonuniform FFT (`finufft`) is reported to give $u_1^X$ at 1,048,576 heights at spacing 0.02, up to height 20,972, in 0.4 seconds for the prime sums alone (the smooth part is not in the timing and the thread count is not fixed), agreeing with the direct sums to $10^{-8}$; Arb needs about two milliseconds per height, two thousand seconds for the same grid, but gives twenty digits.

So the practical picture of the first layer of cost, the data, is this. Twenty digits at one height cost of order $v^{1/2}$ by the formula of Riemann and Siegel [127, 478] and of order $v^{1/3}$ by Hiary [171]; three digits at a million heights cost a second by the primes; the deconvolution needs four or five digits on a grid, which by the observed law above the smoothed sum delivers with $X = 10^8$ to $10^{10}$ (the proved bound asks for more), at the edge of a laptop, and Arb delivers in minutes up to the heights of this chapter. What stays open is the question in its sharp form, $\log(1/\varepsilon)$ rather than $1/\varepsilon$, which would follow from such an algorithm for $\zeta$ and $\zeta'$ on the line.

### 31.11.10 Is $\mathfrak{U}_{1/2}$ computed efficiently?

The Cauchy deconvolution at the bottom level is the equation

$$P_{1/2} * \mathfrak{U}_{1/2} = u_1, \qquad \mathfrak{U}_{1/2} = \sum_\rho \delta_{\gamma_\rho} \text{ under the Riemann hypothesis,}$$

that is, $u_1(v) = \sum_\rho P_{1/2}(v - \gamma_\rho)$: the level one density is the Poisson integral of the counting measure of the zeros at distance $\frac{1}{2}$ from the line. To compute $\mathfrak{U}_{1/2}$ is to undo that Poisson integral. In Fourier variables the equation reads $e^{-|\omega|/2}\,\widehat{\mathfrak{U}_{1/2}}(\omega) = \widehat{u_1}(\omega)$, and by the prime sum of Theorem 31.11.10 the spectrum $\widehat{u_1}$ consists of atoms of weight $-\Lambda(n)/n$ at $\omega = \pm\log n$ and a smooth part from the Gamma factor (the prime part of $u_1$ is $-\pi^{-1}\sum \Lambda(n)n^{-1}\cos(v\log n)$ and $\widehat{\cos} = \pi(\delta_{\log n}+\delta_{-\log n})$). So the deconvolution multiplies the atom at $\log n$ by $e^{(\log n)/2} = \sqrt{n}$:

$$\widehat{\mathfrak{U}_{1/2}}: \quad -\frac{\Lambda(n)}{n} \;\longmapsto\; -\frac{\Lambda(n)}{\sqrt{n}} \qquad \text{at } \omega = \pm\log n,$$

which is the Fourier side of Weil's explicit formula,

$$\sum_\rho \widehat{g}(\gamma_\rho) = -\sum_n \frac{\Lambda(n)}{\sqrt{n}}\big(g(\log n) + g(-\log n)\big) + \cdots .$$

The backward Cauchy flow from level 1 to level $\frac{1}{2}$ is the explicit formula, and the multiplier $e^{|\omega|/2}$ is the factor $\sqrt{n}$ that turns the series $\sum \Lambda(n)/n^{1+iv}$, convergent on $\operatorname{Re} s = 1$ only in a smoothed sense, into the series $\sum \Lambda(n)/n^{1/2+iv}$, which has no meaning except in the sense of distributions.

This is the heuristic answer to the question of efficiency. To resolve $\mathfrak{U}_{1/2}$ at scale $\delta$ one needs its spectrum to frequency $\Omega \approx \pi/\delta$, hence the primes to $e^{\Omega}$, exponentially many; equivalently, the data $u_1$ must be known to accuracy $e^{-\Omega/2}$, since that is the factor by which errors at frequency $\Omega$ are amplified. The problem is exponentially ill posed in the resolution, and no linear band limited inversion of $u_1$ escapes this; EM shows no sign of escaping it. This is a statement about worst case stability, not an operation count lower bound for every method: for a single unit atom with blur $y(v) = P_a(v-\gamma)$, $\gamma = a\,[1/y(-t) - 1/y(t)]/(4\pi t)$ for any $t \neq 0$, so extra structure changes the inverse problem. The backward flow with 4000 accelerated steps on a grid of spacing 0.02 certifies Gaussian weighted centroids with an uncertainty of about $10^{-3}$ (the moment bound $B_1$ of Theorem 31.11.8, divided by the mass), not the atoms (Corollary 31.11.9), and further digits cost further steps, at the observed rates of Section 31.11.1, or a finer grid with data accurate to about $e^{-\pi/2h}$. In the complexity class sense the computation is cheap per step: an FFT step costs $O(T\log T)$ on a grid to height $T$, and the number of steps for a fixed resolution was observed to be grid independent; no bound on the number of steps uniform in $T$, the least gap or the accuracy is proved here,

so the total $T^{1+\varepsilon}$ of the method of Odlyzko and Schönhage [225] for all zeros to height $T$ is not claimed for the deconvolution. In the sense that matters, it is not efficient: direct zero finding gives $\gamma$ to far higher accuracy than the backward flow in less time, because it uses the functional equation, which carries the information of the spectrum beyond any $\Omega$ in closed form, where the deconvolution has only the data below $\Omega$. The efficient algorithm for $\mathfrak{U}_{1/2}$ is Turing's [506], not EM; what the deconvolution contributes is the certificate, a bound on the measure from the residual alone, and the levels above $\frac{1}{2}$, where the Thorin measure is a density and no zero finder produces it.

### 31.11.11 A von Neumann representation of $\mathfrak{U}_{1/2}$

Two things could be meant, and both have an answer. In the sense of the von Neumann and Ulam estimator of the earlier section, $\mathfrak{U}_{1/2} = (I - (I - A))^{-1}u_1$ with $A = P_{1/2}*$ has the Neumann series $\sum_{k\geq 0}(I - A)^k u_1$, which converges at every frequency, since the symbol $1 - e^{-|\omega|/2}$ lies in $[0, 1)$, but to the unbounded operator with symbol $e^{|\omega|/2}$. Its partial sums are the Richardson iterates with unit step: the sum to $K$ terms applied to $u_1$ has symbol $[1 - (1 - e^{-|\omega|/2})^{K+1}]e^{|\omega|/2}$, so relative to the target the effective filter is $1 - (1 - e^{-|\omega|/2})^{K+1}$, which is near 1 for $|\omega| \leq 2\log K$ and near 0 beyond. The truncated Neumann series is therefore a soft band limited inverse with resolution of order $\pi/(2\log K)$, and the resolution 0.1 costs $K = e^{16}$ terms. The random walk that samples this series has kernel $|I - A| = \delta + P_{1/2}$, of total mass 2, and the estimator with random termination has infinite second moment (Proposition 31.5.1 for the grid version below the threshold $h < 0.52\pi(1 - \theta')$); finitely truncated walks have finite second moments. That is the von Neumann representation in the Monte Carlo sense, and it is exact, convergent and useless, for the reason of the previous subsection.

In the other sense, von Neumann's spectral theorem, the question is whether $\mathfrak{U}_{1/2}$ is the spectral measure of a natural self adjoint operator, and that is the conjecture of Hilbert and Pólya. When $\mathfrak{U}_{1/2}$ is a positive measure, that is under the Riemann hypothesis, a self adjoint operator with spectral measure $\mathfrak{U}_{1/2}$ exists trivially (a diagonal one, with the heights $\gamma_\rho$ as eigenvalues), so existence alone is not the conjecture. What is unconditional is the shape of the answer at the levels above. If $H$ is self adjoint with spectral measure $\mu_H = \sum \delta_\gamma$, then for $a > 0$ the resolvent is not trace class, since $N(T) \asymp T\log T$, but its imaginary part is, and

$$\frac{1}{\pi}\operatorname{Tr}\operatorname{Im}(H - v - ia)^{-1} = \sum_\gamma \frac{a/\pi}{(\gamma - v)^2 + a^2} = (P_a * \mu_H)(v),$$

so the Thorin density at level $\theta = \frac{1}{2} + a$ is the trace of the imaginary part of the resolvent of $H$ at height $a$ above the real axis, and the backward Cauchy flow $\theta \downarrow \frac{1}{2}$ is the Stieltjes inversion formula, the boundary value of the resolvent that recovers the spectral measure. The chapter has computed, unconditionally, the

function that would be that trace:

$$u_\theta(v) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\theta + iv) = \frac{1}{\pi}\sum_\rho \operatorname{Re}\frac{1}{\theta + iv - \rho} = \frac{1}{\pi}\sum_\rho \frac{a_\rho}{a_\rho^2 + (v - \gamma_\rho)^2},$$

with $a_\rho = \theta - \sigma_\rho$, which is the trace of the imaginary part of the resolvent of the operator $H$ with spectrum $\{\gamma_\rho\}$ at height $a$ exactly when all $a_\rho$ equal $a$, that is when all zeros are on the line. So the densities $u_\theta$, $\theta > \frac{1}{2}$, are the traces $\pi^{-1}\operatorname{Tr}\operatorname{Im}(H - v - i(\theta - \frac{1}{2}))^{-1}$ of the one operator $H$ with spectrum $\{\gamma_\rho\}$ if and only if the Riemann hypothesis holds. (If a zero has $\sigma_0 > \frac{1}{2}$, then at $v = \gamma_0$ the density $u_\theta$ is unbounded as $\theta \downarrow \sigma_0$, while the trace of Im of the resolvent is finite for every $\theta > \frac{1}{2}$.) The conjecture of Hilbert and Pólya asks for a natural $H$. What the Thorin measures at levels $\theta > \frac{1}{2}$ provide is the resolvent trace without the operator: $\xi'/\xi$ plays the part of the trace of the resolvent at $z = v + i(\theta - \frac{1}{2})$, and the certificate of Theorem 31.11.8 is a bound on the boundary measure obtained from the resolvent at height $\frac{1}{2}$, which is the content of the Stieltjes inversion made quantitative. Nothing in this produces $H$; it says what $H$ would have to do.

## 31.12 Open problems

Problem 31.1. Prove the two phase picture of Section 31.11.1 for the continuous Cauchy deconvolution at an atomic target: a sublinear phase with $k(R_k - R^*)$ near a constant of order 50 (observed: 54.7, 51.9, 49.5, 48.1 at $k = 500$, 1000, 2000, 3000), lasting of the order of $10^4$ steps on this data, followed by faster convergence on the support of the atoms with a grid independent rate; and compute that rate from the curvature of the divergence at the atoms. For the accelerated scheme with restart, explain the observed three point exponents, which oscillate between 0.6 and 2.8 with a running value near 1.3.

Problem 31.2. Proposition 31.11.4 gives the limiting slope under the Riemann hypothesis to six digits, 0.183899 at $v^* = 17.4979$. Is there a closed form? By the Hadamard product, under the Riemann hypothesis, $\log|\xi(\frac{1}{2} + iv)| = \log\xi(\frac{1}{2}) + \sum_{\gamma>0}\log|1 - v^2/\gamma^2|$, so with $\mathcal{C}(v) = \partial_v^2 \log|\xi(\frac{1}{2} + iv)|$ the bracket is $X(v) = \mathcal{C}(0) - \mathcal{C}(v)$: the slope of the excess is the drop of the second logarithmic derivative of $\xi$ along the critical line from the origin to $v$, and $v^*$ is the zero of $\mathcal{C}'$ between the first two zeros. Whether $\mathcal{C}(v^*)$ has an expression in known constants is not clear.

Problem 31.3. Theorem 31.11.10 evaluates $u_1$ to accuracy $\varepsilon$ in time polynomial in $\log v$ and in $1/\varepsilon$. Is there an algorithm polynomial in $\log v$ and $\log(1/\varepsilon)$? Such an algorithm for $\zeta$ and $\zeta'$ on $\operatorname{Re} s = 1$ would give one; the converse is not known, and the question is open. The smoothed sum cannot be the route, since the zeros themselves contribute $X^{-1/2}$.

Problem 31.4. The certificate's main term is set by the residual within a unit of each zero, which no objective reduces faster than the conditioning allows (as observed in Section 31.11.8). Quantify this: for a single atom at $\gamma$ blurred by $P_a$ and observed exactly, what is the best achievable $h \sum_i |\psi_\gamma(v_i)||r_i|$ after $k$ steps of any first order method on the cone, as a function of $k$, $h$ and $a$? The answer would say how far Theorem 31.11.8 can be pushed at a given cost.

# Chapter 32

# The explicit formula as a deconvolution

Chapters 30 and 31 write the Thorin measure at level $\theta$ as the Cauchy smoothing of the zeros at width $\theta - \frac{1}{2}$ (under the Riemann hypothesis; in general the widths are $\theta - \sigma_\rho$), turn the passage from level 1 to level $\theta$ into a deconvolution, and solve it by EM to an accuracy of about $10^{-3}$ in the positions of the zeros, with a certificate. This chapter looks at the bottom of that flow. At level $\frac{1}{2}$, under the Riemann hypothesis, the Thorin measure is the counting measure of the zeros, the data are the level one density $u_1(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1+iv)$, and the deconvolution, written in Fourier variables, is Weil's explicit formula: the multiplier $e^{|\omega|/2}$ that undoes the Cauchy kernel is the factor $\sqrt{n}$ that turns the weights $\Lambda(n)/n$ of the level one spectrum into the weights $\Lambda(n)/\sqrt{n}$ of the formula. That identification says how ill posed the problem is. For a general measure the resolution $\delta$ costs, heuristically, the primes to $e^{\pi/\delta}$, or data accurate to $e^{-\pi/2\delta}$: exponential. For the measure at hand, which is a finite sum of atoms on every window, it costs much less, because the Fourier transform of a sum of Lorentzians is a sum of damped exponentials, and a sum of damped exponentials is recovered from its samples by Prony's method with an error that, for a fixed configuration of atoms, is proportional to the data error. The matrix pencil applied to the tapered Fourier transform of $u_1$ on a window of width 60 returns, in the computation reported by the author, the ordinates and real parts of the central zeros to within $8 \times 10^{-10}$, from level one data alone, in seconds; a synthetic pair of zeros placed off the line at $0.85 \pm 60.089i$ is returned with real parts 0.85 and 0.15 to seven digits. So sparse methods solve the deconvolution many orders of magnitude better than EM on the same data, and the reason is structural: EM solves the problem on a dense grid, where it is exponentially ill posed, and the matrix pencil solves it in the model where it is polynomially ill posed. A posterior concentration theorem makes the recovery Bayesian, and the functional equation makes it singular: off line zeros come in pairs at $\frac{1}{2} \pm \delta$, the data depend on $\delta$ only through $\delta^2$, and the real part of a pair is resolved at the rate $\tau^{1/2}$ when the data are accurate to $\tau$, with Bayes factors for the line against the pair that grow only like $\tau^{-1/2}$. None of this is a path to the hypothesis: the method reads real parts, it does not constrain

them.

The chapter develops a source chapter on the explicit formula, sparse recovery and Bayesian detection of the zeros by Nick Polson [230], together with the computations done for this book. Its numbers were recomputed or checked here by scripts written from the mathematics alone (the scripts of the source were not used; the scripts of this chapter are `code/es_*.py` and `code/ex_zeros3e12.py`, with outputs `data/es_*.json` and `data/ex_zeros3e12.json`, listed in `code/README.md`). Every number printed below is one of those recomputations unless it is marked as the author's: a figure marked computed by the author's scripts was not rerun for this book, and where a recomputation differed from the source the recomputed value is printed. Several parts of the source are in the book already and are cited, not restated: the Lorentzian representation of $u_1$ and the Hadamard product (Proposition 1.2.3), the criterion that the deconvolution has a positive solution (Theorem 19.2.2), the explicit formula for even test functions read as the deconvolved functional (Theorem 20.1.1, Section 20.1.1), the EM, grid and certificate methods (Chapters 30 and 31) and the survey of Cauchy deconvolution (Chapter 33). What is new here is the form of Theorem 20.1.1 for all compactly supported test functions, with its pole term, the taper theorem and its use with the matrix pencil, the computations, and the posterior theory with its singular case.

The notation is the book's, with the following warnings. A level $\theta$ is the real part of a point $s = \theta + iv$, and $u_\theta(v) = \pi^{-1}\,\mathrm{Re}(\xi'/\xi)(\theta + iv)$ is the density of Chapters 29 to 33; in the notation of Chapter 19 (equation (1.2.7)) it is $\rho_{\theta-1/2}(v)$, the ordinate being $v$ here. For a zero $\rho = \sigma_\rho + i\gamma_\rho$ the width is $a_\rho = 1 - \sigma_\rho \in (0,1)$ at level 1. The Cauchy density is $P_a(x) = (a/\pi)/(a^2+x^2)$, $\mathfrak{U}_\theta = \sum_\rho \mathrm{Cauchy}(\gamma_\rho, \theta - \sigma_\rho)$ is the Thorin measure with density $\sum_\rho P_{\theta-\sigma_\rho}(v - \gamma_\rho)$ and no factor $\frac{1}{2}$, as in Chapter 31, so that $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ with $\widetilde{U}_\theta$ of Theorem 29.14.4, $\gamma_E$ is Euler's constant and $\Lambda$ the von Mangoldt function. The source's Chapters 3 and 4 are Chapters 30 and 31 of the book. In the Bayesian sections $\tau$ is the standard deviation of each real coordinate of the noise. The word certified is used only for a computation in ball arithmetic whose sentence says what the certificate covers: the sign changes of Hardy's function $Z$ in Sections 32.8.1, 32.8.6 and 32.8.7. Everything else done in floating point is computed.

Status words follow the rest of the book: proved, computed, conditional, heuristic or open. The results are these.

1. The explicit formula of Guinand and Weil is the Fourier side of the deconvolution from level 1 to level $\frac{1}{2}$: for every test function in $C_c^\infty(\mathbb{R})$, unconditionally, the zero side $\sum_\rho \int g\, e^{(\sigma_\rho - 1/2)|\omega|} e^{-i\omega\gamma_\rho} d\omega$ equals the prime side with the pole term and the finite part of the Gamma factor; for even $g$ this is Theorem 20.1.1, and under the Riemann hypothesis it reads $\sum_\rho e^{-i\omega\gamma_\rho} = e^{|\omega|/2}(1 + \widehat{G}) - \sum \Lambda(n) n^{-1/2}(\delta_{\log n} + \delta_{-\log n})$ (Theorem 32.1.1, proved; the reading is conditional). A Gaussian perturbation of the measure

that data of accuracy $\varepsilon$ cannot see, of total variation 1, exists as soon as $0.72\,e^{-\Omega/2} \le 2\varepsilon$ (Proposition 32.1.2, proved); the consequence that resolution $\delta$ needs the primes to $e^{\pi/\delta}$ is a heuristic remark (Remark 32.1.3).

2. The tapered Fourier transform of $u_1$ on a window is a sum of damped exponentials, one per zero, with the pole $e^{\Delta(-a_\rho+i(\gamma_\rho-c))}$ carrying both coordinates of the zero, plus a remainder of order $e^{-\ell^2\omega^2/2}$ (Theorem 32.2.1, proved, with no Gamma terms). The matrix pencil recovers the poles of an exact sum (Lemma 32.2.2, proved). Its error analysis is heuristic. At height 200 the eleven zeros within 10 of the centre are placed to within $8\times10^{-10}$ in both coordinates, and a synthetic pair off the line at 60.089 is read to seven digits (Section 32.3, computed by the author's scripts, not rerun).

3. Posterior concentration on a window holds at the rate $\tau$ for zeros with separated ordinates, under explicit hypotheses (Theorem 32.4.1, proved); the hypotheses do not hold at the settings of the computations (Remark 32.4.2, computed), and concentration cannot prove the hypothesis, since a finite computation settles finitely many zeros.

4. The functional equation makes the model singular: with one amplitude common to the two members, a pair at $\frac12 \pm \delta$ enters the data through $\delta^2$ (Lemma 32.5.1, proved). The posterior of $\delta^2$ concentrates at the rate $\tau$, so the real part is resolved at the rate $\tau^{1/2}$ when the line is the truth; no method, neural networks included, separates $\delta = 0$ from $\delta = c\,(\tau/\|t_\perp\|)^{1/2}$, $t_\perp$ the projected tangent of the pair, with error probabilities summing to less than $1 - c^2/2 - O(\tau)$; and the matched filter and the likelihood ratio test attain the scale $\tau^{1/2}$ (Theorem 32.5.2, proved for data simulated from the model in a window without aliasing, not for the runs on the zeros of $\zeta$).

5. Bayes factors with the right alternative, the pair and not a single zero: under the line the Bayes factor of the line against the pair is of order $\tau^{-1/2}$ in probability, under a fixed pair it is exponentially small, and against a single zero alternative it is of order $\tau^{-1}$ (Theorem 32.5.4, proved in the common amplitude model). For the zero at 201.264752, at an assumed noise level, the computed $\log_{10}$ of the Bayes factor is 6.1 against the pair and 11.8 against the single zero alternative: the functional equation takes roughly the square root of the evidence (computed on an assumed design, that of the detection experiment, an illustration).

6. An adaptive prior, the sine kernel process on the distinct ordinates at the Riemann and von Mangoldt density and a spike and slab on the pairs for the real parts, is a heuristic choice resting on a conjecture about the ordinates; its properties used here are proved (Remarks 32.5.3 and 32.6.1). A detection experiment compares the pencil and a plug in matched filter with the benchmark of Theorem 32.5.2(iii) (computed).

7. The section on the open problems of the source works them: the certification of the pencil's output by sign changes and a winding number (logic proved, eleven zeros near height 200 computed), close pairs, the levels $\frac{3}{4}$ and 0.6 and the signed level $\theta$ transform (Proposition 32.8.1, proved), the level as a cost parameter, and many windows (173 zeros on $[100, 400]$ from 31 windows, computed by the author).

8. A window at height $c = 3 \times 10^{12} + 100$ lies inside the range where Platt and Trudgian [419] verify the hypothesis, $0 < \gamma \le 3\,000\,175\,332\,800$, so the run is a demonstration at a height near the limit of the verification: the data are values of $\xi'/\xi$ at $\operatorname{Re} s = 0.6$ computed in ball arithmetic, no primes are used and no zero unknown before is found. The pencil returns 52 components; an independent scan of Hardy's function finds exactly 52 certified sign changes there, and the one by one comparison with the pencil's ordinates has not been made. The selection of components in the source's runs presumes the real part $\frac{1}{2}$ (Section 32.8.6, computed).

9. A window at $c = 10^{13} + 100$, above the verified height by a factor 3.3, is run with the book's own pencil, whose selection uses the width of each pole: 53 components, real parts within $5 \times 10^{-9}$ of $\frac{1}{2}$ (computed, an uncertified estimate), every ordinate within $10^{-7}$ of one of 53 certified sign changes of $Z$ and no certified sign change without a pencil ordinate. The count of zeros in the window and the real parts are not certified, and nothing is proved about any zero outside the 53 (Section 32.8.7).

The open problems close the chapter.

## 32.1 The deconvolution at level one half

### 32.1.1 The forward equation

Write $s = 1+iv$ and, for a zero $\rho = \sigma_\rho + i\gamma_\rho$, $a_\rho = 1-\sigma_\rho \in (0,1)$. Proposition 1.2.3 and the Hadamard product give, with the zeros paired as $\rho, \bar\rho$ so that the sum converges,

$$u_1(v) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(1+iv) = \frac{1}{\pi} \sum_\rho \frac{a_\rho}{a_\rho^2 + (v-\gamma_\rho)^2} = \sum_\rho P_{a_\rho}(v-\gamma_\rho), \qquad (32.1.1)$$

a sum of Cauchy densities $P_a(x) = (a/\pi)/(a^2+x^2)$ of widths $a_\rho$ centred at the ordinates. Under the Riemann hypothesis every width is $\frac{1}{2}$ and

$$u_1 = P_{1/2} * \mathfrak{U}_{1/2}, \qquad \mathfrak{U}_{1/2} = \sum_\rho \delta_{\gamma_\rho} : \qquad (32.1.2)$$

the level one density is the Poisson integral of the zero counting measure at distance $\frac{1}{2}$ from the line. In general (32.1.1) is the Poisson integral of the positive measure

$\sum_\rho \delta_{(\gamma_\rho, a_\rho)}$, each atom smoothed at its own distance $a_\rho$, and Theorem 19.2.2 shows that a positive solution $\mu$ of $P_a * \mu = u_1$ exists for a given $a \in (0, \frac12]$ if and only if no zero has $a_\rho < a$.

### 32.1.2 The Fourier side

With $\widehat{f}(\omega) = \int f(v) e^{-i\omega v} dv$, $\widehat{P_a}(\omega) = e^{-a|\omega|}$, so termwise

$$\widehat{u_1}(\omega) = \sum_\rho e^{-a_\rho|\omega|} e^{-i\omega\gamma_\rho}, \tag{32.1.3}$$

and under the Riemann hypothesis $\widehat{u_1} = e^{-|\omega|/2}\,\widehat{\mathfrak{U}_{1/2}}$ with $\widehat{\mathfrak{U}_{1/2}}(\omega) = \sum_\rho e^{-i\omega\gamma_\rho}$, a tempered distribution. On the other hand, from the Dirichlet series, as tempered distributions,

$$\widehat{u_1} = 1 + \widehat{G} - \sum_{n\ge2} \frac{\Lambda(n)}{n}\big(\delta_{\log n} + \delta_{-\log n}\big), \tag{32.1.4}$$

where the 1 is the transform of $\delta_0$, the boundary value on $\operatorname{Re} s = 1$ of $\pi^{-1}\operatorname{Re}(s-1)^{-1}$, which the prime series does not contain, and $G(v) = \pi^{-1}[1/(1+v^2) - \frac12\log\pi + \frac12 \operatorname{Re}\psi(\frac{1+iv}{2})]$ is the Gamma part, whose transform is the distribution

$$\langle \widehat{G}, g\rangle = (1 - \log\pi - \gamma_E)\, g(0) + \sum_{k\ge1}\Big[\frac{g(0)}{k+1} - \int_{\mathbb{R}} g(\omega)\, e^{-(2k+1)|\omega|}\, d\omega\Big]. \tag{32.1.5}$$

Away from the origin $\widehat{G}(\omega) = -e^{-3|\omega|}/(1-e^{-2|\omega|}) = -1/(2|\omega|) + 1 + O(|\omega|)$, the transform of the growth $(2\pi)^{-1}\log|v|$ of $G$: a singularity that is not integrable, and (32.1.5) is its finite part regularisation. Multiplying (32.1.4) by $e^{|\omega|/2}$ multiplies the atom at $\pm\log n$ by $\sqrt{n}$.

Theorem 32.1.1 (the explicit formula is the deconvolution). *For every $g \in C_c^\infty(\mathbb{R})$, with the sum over $\rho$ absolutely convergent,*

$$\begin{aligned}\sum_\rho \int_{\mathbb{R}} g(\omega)\, e^{(\sigma_\rho - \frac12)|\omega|}\, e^{-i\omega\gamma_\rho}\, d\omega &= \int_{\mathbb{R}} g(\omega)\, e^{|\omega|/2}\, d\omega \\ &\quad - \sum_{n\ge2} \frac{\Lambda(n)}{\sqrt{n}}\big(g(\log n) + g(-\log n)\big) + (1 - \log\pi - \gamma_E)\, g(0) \\ &\quad + \sum_{k\ge1}\Big[\frac{g(0)}{k+1} - \int_{\mathbb{R}} g(\omega)\, e^{-(2k+\frac12)|\omega|}\, d\omega\Big]. \end{aligned}\tag{32.1.6}$$

*For even $g$ this is Theorem 20.1.1, the explicit formula of Guinand [163] and Weil [512]; for odd $g$ both sides vanish. Under the Riemann hypothesis the left side is $\sum_\rho \widehat{g}(\gamma_\rho)$, and* (32.1.6) *says that on $C_c^\infty(\mathbb{R})$*

$$\sum_\rho e^{-i\omega\gamma_\rho} = e^{|\omega|/2}\big(1 + \widehat{G}(\omega)\big) - \sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\big(\delta_{\log n} + \delta_{-\log n}\big),$$

*where $e^{|\omega|/2}\widehat{G}$ is the functional $g \mapsto \langle \widehat{G}, e^{|\cdot|/2} g\rangle$ given by* (32.1.5)*. The left side is tempered; the right side is tempered only as a whole, since the prime measure has mass about $2e^{X/2}$ on $[0, X]$.*

*Proof.* (a) For $\sigma > 1$, $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \zeta'/\zeta(s)$. Let $\sigma \downarrow 1$ with $s = \sigma + iv$. The left side tends to $\pi u_1$ in $\mathcal{S}'(\mathbb{R})$, since $\xi'/\xi$ is analytic across $\operatorname{Re} s = 1$ and grows like $\log|v|$. On the right, $\operatorname{Re} 1/s \to 1/(1+v^2)$, $\frac{1}{2}\operatorname{Re}\psi(s/2) \to \frac{1}{2}\operatorname{Re}\psi(\frac{1+iv}{2})$, $\pi^{-1}\operatorname{Re}(s-1)^{-1} = P_{\sigma-1}(v) \to \delta_0$, and $\operatorname{Re}\zeta'/\zeta(s) = -\sum_n \Lambda(n) n^{-\sigma}\cos(v\log n)$, whose transform $-\pi\sum_n \Lambda(n)n^{-\sigma}(\delta_{\log n} + \delta_{-\log n})$ converges in $\mathcal{S}'$ as $\sigma \downarrow 1$ by dominated convergence, the measure $\sum_n \Lambda(n) n^{-1}\delta_{\log n}$ having mass $O(X)$ on $[0, X]$. This gives (32.1.4). The digamma series gives $G(v) = (1 - \log\pi - \gamma_E)/2\pi + \sum_{k\ge 1}[1/(2\pi(k+1)) - P_{2k+1}(v)]$, the $1/(1+v^2)$ cancelling the term $k = 0$, and transforming termwise gives (32.1.5); the series converges because $\int g(\omega)e^{-b|\omega|}d\omega = 2g(0)/b + O(b^{-2})$ and $1/(k+1) - 2/(2k+1) = O(k^{-2})$.

(b) By (32.1.1) the partial sums of $\sum_\rho P_{a_\rho}(v - \gamma_\rho)$ are positive and increase to $u_1 = O(\log|v|)$, so the series converges in $\mathcal{S}'$ and $\widehat{u_1} = \sum_\rho e^{-a_\rho|\omega|}e^{-i\omega\gamma_\rho}$.

(c) Let $g \in C_c^\infty(\mathbb{R})$ and $\varphi = e^{|\omega|/2}g$, which is Lipschitz with compact support and has $\varphi'$ of bounded variation. Let $\varphi_\epsilon = \varphi * \eta_\epsilon$ with a mollifier $\eta_\epsilon$: then $\varphi_\epsilon \in C_c^\infty$ with supports in a fixed compact set, $\varphi_\epsilon \to \varphi$ uniformly, and $\|\varphi_\epsilon'\|_\infty$ and the total variation of $\varphi_\epsilon'$ are bounded uniformly in $\epsilon$. Pair (a) and (b) with $\varphi_\epsilon$. On the zero side the term of $\rho$ is the transform of $e^{-a_\rho|\omega|}\varphi_\epsilon$ at $\gamma_\rho$; two integrations by parts, the second against the measure $d\big(e^{-a_\rho|\omega|}\varphi_\epsilon\big)'$ of bounded mass, bound it by $C(1+\gamma_\rho^2)^{-1}$ uniformly in $\epsilon$ and $\rho$, and $\sum_\rho(1+\gamma_\rho^2)^{-1} < \infty$, so by dominated convergence the zero side tends to $\sum_\rho \int g(\omega)e^{(1/2-a_\rho)|\omega|}e^{-i\omega\gamma_\rho}d\omega$, the left side of (32.1.6), since $\frac{1}{2} - a_\rho = \sigma_\rho - \frac{1}{2}$. On the other side the prime sum is finite on the support, $\int\varphi_\epsilon \to \int g\, e^{|\omega|/2}$, and $\langle\widehat{G}, \varphi_\epsilon\rangle \to \langle\widehat{G}, \varphi\rangle$, because each term of (32.1.5) converges and $|\varphi_\epsilon(\omega) - \varphi_\epsilon(0)| \le C|\omega|$ bounds the $k$-th term by $C/k^2$ uniformly. This is (32.1.6). Under the Riemann hypothesis $a_\rho = \frac{1}{2}$, and $N(T) = O(T\log T)$ makes $\sum_\rho e^{-i\omega\gamma_\rho}$ tempered. □

The theorem is classical, and that is the point, made already in Section 20.1.1: the backward Cauchy flow from level 1 to level $\frac{1}{2}$, the object that Chapters 30 and 31 computed by EM, is the explicit formula, and the statement of Theorem 19.2.2 that the Thorin measure at level $\theta$ is the solution of $P_{1-\theta} * \mu = u_1$ is, at $\theta = \frac{1}{2}$ and under the Riemann hypothesis, the statement that the zeros are the Fourier transform of the primes. The multiplier $e^{|\omega|/2}$ of the deconvolution, whose unboundedness made the problem ill posed, is the passage from $\Lambda(n)/n$ to $\Lambda(n)/\sqrt{n}$, from a tempered measure, of mass about $X$ on $[0, X]$, to one of mass about $2e^{X/2}$, which pairs only with test functions of compact support. Everything that was said about the conditioning of the grid matrix in Chapter 31 is a statement about this passage.

### 32.1.3 What the ill posedness costs

Proposition 32.1.2 (an unresolvable perturbation). *Let* $\Omega > 0$, $L \geq 1$, $\chi_L(v) = (2\pi L^2)^{-1/2} e^{-v^2/2L^2}$, $N = \int_{\mathbb{R}} |\cos(\Omega v)|\, \chi_L(v)\, dv$ *and* $\nu = N^{-1} \cos(\Omega v)\, \chi_L(v)\, dv$. *Then* $\|\nu\|_{TV} = 1$, $\widehat{\nu}$ *is concentrated within* $O(1/L)$ *of* $\pm\Omega$, *and*

$$\|P_{1/2} * \nu\|_\infty \leq \sqrt{\pi/8}\; L^{-1} e^{1/8L^2}\, e^{-\Omega/2} \leq 0.72\, e^{-\Omega/2}.$$

*Consequently, if* $0.72\, e^{-\Omega/2} \leq 2\varepsilon$ *and* $\mu$ *is a positive measure with* $\mu \geq N^{-1}\chi_L\, dv$, *then* $\mu$ *and* $\mu + \nu$ *are positive measures at total variation distance* 1 *whose data* $P_{1/2} * \mu$ *and* $P_{1/2} * (\mu + \nu)$ *lie within* $\varepsilon$ *in* $L^\infty$ *of one function: data of accuracy* $\varepsilon$ *do not distinguish them.*

*Proof.* $\widehat{\chi_L}(\omega) = e^{-L^2\omega^2/2}$, so $\widehat{\nu}(\omega) = (2N)^{-1}\big(e^{-L^2(\omega-\Omega)^2/2} + e^{-L^2(\omega+\Omega)^2/2}\big)$. By Fourier inversion, the evenness of $e^{-|\omega|/2}$ and $e^{-|\omega|/2} \leq e^{-\omega/2}$,

$$\begin{aligned} \|P_{1/2} * \nu\|_\infty &\leq \frac{1}{2\pi} \int_{\mathbb{R}} e^{-|\omega|/2} |\widehat{\nu}(\omega)|\, d\omega \\ &\leq \frac{1}{2\pi N} \int_{\mathbb{R}} e^{-\omega/2} e^{-L^2(\omega-\Omega)^2/2}\, d\omega = \frac{e^{1/8L^2}}{\sqrt{2\pi}\, L\, N}\, e^{-\Omega/2}. \end{aligned}$$

From $|\cos x| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{k\geq 1} (-1)^{k+1} \frac{\cos 2kx}{4k^2-1}$ and $\int \cos(2k\Omega v)\chi_L(v)\, dv = e^{-2k^2L^2\Omega^2}$,

$$N = \frac{2}{\pi} + \frac{4}{\pi} \sum_{k\geq 1} (-1)^{k+1} \frac{e^{-2k^2L^2\Omega^2}}{4k^2 - 1} \geq \frac{2}{\pi},$$

the series being alternating with decreasing terms. With $N \geq 2/\pi$ this is the bound, and $\sqrt{\pi/8}\, e^{1/8} = 0.7101$. For the last statement, $\nu \geq -N^{-1}\chi_L$, so $\mu + \nu \geq 0$, and the midpoint of the two data lies within $\frac{1}{2}\|P_{1/2} * \nu\|_\infty \leq \varepsilon$ of each. □

Remark 32.1.3 (heuristic). Identifying resolution at scale $\delta$ with frequency $\Omega = \pi/\delta$, Proposition 32.1.2 suggests that a general density cannot be resolved at scale $\delta$ from data less accurate than about $\frac{1}{2}e^{-\pi/2\delta}$, and, since in (32.1.6) a test function supported in $[-\Omega, \Omega]$ involves $\Lambda(n)$ only for $n \leq e^{\Omega}$, that the primes to $e^{\pi/\delta}$ are the matching arithmetic input. Neither is a theorem: no loss function is fixed, no method is excluded, and both fail for measures with few atoms, as the computations below show.

For $\delta = 0.1$ the heuristic gives $\varepsilon \approx 10^{-7}$ and primes to $e^{31} \approx 10^{13}$; for $\delta = 0.01$ it is $3 \times 10^{-69}$ and $10^{136}$. The EM computation of Chapter 31 reached $\delta \approx 10^{-3}$ in the positions of well separated atoms from data accurate to $10^{-20}$, which is better than the heuristic allows for a general measure, and the reason is that the target is not a general measure. Proposition 32.1.2 is about the worst perturbation; a positive measure that is a sum of a few atoms per unit length has far fewer degrees of freedom than a density, and the question is whether a method can use that.

## 32.2 Sparse recovery

### 32.2.1 Positivity and sparsity

Two facts from the theory of super resolution apply directly. First, Candès and Fernandez Granda [72] proved that a measure $\mu = \sum_k c_k \delta_{t_k}$ on $[0,1]$ is recovered exactly from its Fourier coefficients $\widehat{\mu}(\omega)$, $|\omega| \le \Omega$, by minimising the total variation subject to the data, provided the atoms are separated by at least $2 \cdot 2\pi/\Omega$; with noise $\varepsilon$ the recovery error is $O(\varepsilon)$ times a super resolution factor, polynomial in $\Omega\delta$ rather than exponential [71]. Second, for positive measures the separation is not needed at all: de Castro and Gamboa [107], and Schiebinger, Robeva and Recht [267], showed that a positive measure with $k$ atoms is the unique positive measure with its moments of orders 0 to $2k$, and is recovered by a linear program. Our measure is positive, and on a window of length $W$ at height $v$ it has about $W \log(v/2\pi)/2\pi$ atoms separated on average by $2\pi/\log(v/2\pi)$, with occasional pairs much closer. So the problem $P_{1/2} * \mu = u_1$, $\mu \ge 0$, restricted to a window and to the frequencies the data support, is of the kind these theorems treat.

The gridded convex methods of Chapter 31 already respect positivity, and the sparse vertex solutions of nonnegative least squares are a step toward sparsity; but on a grid of spacing $h$ the model has $W/h$ degrees of freedom, the unknown is a density, and the conditioning is that of Proposition 32.1.2. The gain from sparsity comes only when the model is the set of sums of atoms, and for that the natural tool is on the Fourier side.

### 32.2.2 Lorentzians become exponentials

Theorem 32.2.1 (tapered transform). *Let $w(v) = e^{-(v-c)^2/2\ell^2}$, $\ell > 0$, and*

$$F(\omega) = \int_{\mathbb{R}} u_1(v)\, w(v)\, e^{i\omega(v-c)}\, dv.$$

*Then for $\omega\ell^2 > 1$*

$$F(\omega) = \sum_{\rho} w(\gamma_\rho + i a_\rho)\, e^{-a_\rho \omega} e^{i\omega(\gamma_\rho - c)} + R(\omega), \tag{32.2.1}$$

*where $w(x+iy) = w(x) e^{y^2/2\ell^2} e^{-iy(x-c)/\ell^2}$ is the entire extension of the taper, the series converges absolutely, and, with $Y = \omega\ell^2$,*

$$|R(\omega)| \le \frac{1}{\pi}\, e^{-\ell^2\omega^2/2} \sum_{\rho} a_\rho \int_{\mathbb{R}} \frac{e^{-(x-c)^2/2\ell^2}\, dx}{(x-\gamma_\rho)^2 + (Y - a_\rho)^2},$$

$$\int_{\mathbb{R}} \frac{e^{-(x-c)^2/2\ell^2}\, dx}{(x-\gamma)^2 + B^2} \le \min\Big(\frac{\pi}{B}, \frac{\sqrt{2\pi}\,\ell}{B^2}\Big),$$

*so that, heuristically, $R$ is of the order $e^{-\ell^2\omega^2/2}$ times the number of zeros per unit length at height $c$, divided by $\omega\ell$. In particular, for $\omega_j = \omega_0 + j\Delta$ with $\omega_0\ell^2 > 1$,*

$$F(\omega_j) = \sum_\rho A_\rho\, z_\rho^j + R(\omega_j), \qquad |R(\omega_j)| \le C_0\, e^{-\ell^2\omega_j^2/2},$$
$$A_\rho = w(\gamma_\rho + ia_\rho) e^{-a_\rho\omega_0} e^{i\omega_0(\gamma_\rho - c)}, \qquad z_\rho = e^{\Delta(-a_\rho + i(\gamma_\rho - c))},$$

*where $C_0$ is the bound on $|R(\omega_0)|$ divided by $e^{-\ell^2\omega_0^2/2}$. The Gamma factor does not enter:* (32.1.1) *is all of $u_1$, and the Lorentzians $P_{2k+1}$ centred at $0$, which the trivial zeros $-2k$ contribute to $\operatorname{Re}\zeta'/\zeta$, are cancelled in $\xi'/\xi$ by the poles of $\Gamma(s/2)$.*

*Proof.* By (32.1.1), $u_1 = \sum_\rho P_{a_\rho}(\cdot - \gamma_\rho)$ exactly, with positive terms, and $\int u_1 w < \infty$; so $F(\omega) = \sum_\rho \int P_{a_\rho}(v - \gamma_\rho)\phi(v)\,dv$ with $\phi(v) = w(v)e^{i\omega(v-c)}$, which is entire with

$$|\phi(x + iy)| = e^{-(x-c)^2/2\ell^2}\, e^{y^2/2\ell^2 - \omega y}.$$

Write $P_a(v-\gamma) = (2\pi i)^{-1}[(v-\gamma-ia)^{-1} - (v-\gamma+ia)^{-1}]$ and move the integral over $\mathbb{R}$ to the line $\operatorname{Im} v = Y$, $Y = \omega\ell^2$, where $|\phi|$ is smallest. Since $0 < a < 1 < Y$, the first term picks up the residue at $v = \gamma + ia$, the second has no pole in between, the vertical sides contribute nothing by the Gaussian decay in $x$, and

$$\int_{\mathbb{R}} P_a(v-\gamma)\phi(v)\,dv = \phi(\gamma + ia) + \frac{1}{2\pi i}\int_{\operatorname{Im} v = Y} \frac{2ia\,\phi(v)}{(v-\gamma)^2 + a^2}\,dv.$$

On the line, $|\phi| = e^{-(x-c)^2/2\ell^2} e^{-\ell^2\omega^2/2}$ and $|(v-\gamma)^2 + a^2| = |v-\gamma-ia|\,|v-\gamma+ia| \ge (x-\gamma)^2 + (Y-a)^2$, which gives the term of the bound attached to $\rho$; the two bounds for the integral follow from $\int ((x-\gamma)^2 + B^2)^{-1}dx = \pi/B$ and $\int e^{-(x-c)^2/2\ell^2}dx = \sqrt{2\pi}\,\ell$. The sums over $\rho$ converge: $|w(\gamma+ia)| \le e^{1/2\ell^2} e^{-(\gamma-c)^2/2\ell^2}$, the integrals are $O((\gamma-c)^{-2})$ uniformly because $Y - a \ge Y - 1 > 0$, and $N(T) = O(T\log T)$. Finally $\phi(\gamma+ia) = w(\gamma+ia)e^{i\omega(\gamma-c)}e^{-a\omega}$, and the bound at $\omega_j$ holds because the bound on $|R(\omega)|$ divided by $e^{-\ell^2\omega^2/2}$ decreases in $\omega$. □

A hypothesis of the form $\omega\ell^2 > a_\rho$ is needed, and $\omega\ell^2 > 1$ suffices: for $\omega\ell^2 \le a_\rho$ the line $\operatorname{Im} v = \omega\ell^2$ does not reach the pole, and the residue is not picked up. The settings of the computations below ($\omega_0\ell^2 = 500$ at height 200, 20 at height $3\times10^{12}$) satisfy it.

The content of the theorem is in the last display. The samples of the tapered transform are a finite sum of damped complex exponentials in $j$, one per zero that the taper sees, plus a remainder, with the pole $z_\rho$ carrying both coordinates of the zero: its modulus gives $a_\rho = 1 - \sigma_\rho$, its argument gives $\gamma_\rho - c$ modulo $2\pi/\Delta$. Nothing is assumed about the real parts in the statement; equation (32.2.1) is (32.1.1) and nothing else.

### 32.2.3 Prony and the matrix pencil

Recovering the poles of a sum of exponentials from $J$ equally spaced samples is Prony's problem, and the matrix pencil of Hua and Sarkar [183] solves it stably. Form the Hankel matrix $H_{ik} = F(\omega_{i+k})$, $0 \le i < J-L$, $0 \le k < L$, with $L \approx J/2$, and its two submatrices $H_0$, $H_1$ obtained by deleting the last and the first row. If the samples are an exact sum of $M$ exponentials with distinct nonzero poles and nonzero amplitudes, and $M \le \min(L, J-L-1)$, then $H_0$ and $H_1$ have rank $M$ and the values of $z$ at which $H_1 - zH_0$ drops rank are the poles $z_\rho$ (Lemma 32.2.2). With noise, one takes the rank $M$ truncated singular value decomposition $H_0 \approx U_M \Sigma_M V_M^*$, chooses $M$ by the gap in the singular values, and computes the eigenvalues of $U_M^* H_1 V_M \Sigma_M^{-1}$. The amplitudes $A_\rho$ follow by least squares against the Vandermonde matrix of the recovered poles, and a component is accepted as a zero when the modulus of its amplitude matches the prediction $|w(\gamma_\rho + i a_\rho)| e^{-a_\rho \omega_0}$ of Theorem 32.2.1, a diagnostic and not a certificate (in the runs at levels $\theta < 1$ below, the prediction is evaluated at the width $\theta - \tfrac{1}{2}$ of a zero on the line); spurious components, which absorb the zeros at the edge of the taper and the wrapped poles of zeros farther away, have amplitudes orders of magnitude off the profile.

Lemma 32.2.2 (exact matrix pencil). *Let $f_j = \sum_{m=1}^{M} A_m z_m^j$, $0 \le j < J$, with distinct nonzero $z_m$, nonzero $A_m$ and $M \le \min(L, J-L-1)$, and let $H_0 = (f_{i+k})$ and $H_1 = (f_{i+k+1})$, $0 \le i \le J-L-2$, $0 \le k < L$. Then* $\operatorname{rank} H_0 = \operatorname{rank} H_1 = M$, $\operatorname{rank}(H_1 - zH_0) < M$ *exactly when $z \in \{z_1, \dots, z_M\}$, and if $H_0 = U_M \Sigma_M V_M^*$ is a reduced singular value decomposition then $U_M^* H_1 V_M \Sigma_M^{-1}$ is similar to* $\operatorname{diag}(z_1, \dots, z_M)$.

*Proof.* With $V = (z_m^i)$ of size $(J-L-1) \times M$, $W = (z_m^k)$ of size $L \times M$, $D = \operatorname{diag}(A_m)$ and $Z = \operatorname{diag}(z_m)$, we have $H_0 = VDW^T$ and $H_1 = VDZW^T$. $V$ and $W$ are Vandermonde with distinct nodes and at least $M$ rows, so both have rank $M$; hence $\operatorname{rank} H_0 = \operatorname{rank} H_1 = M$ and $\operatorname{rank}(H_1 - zH_0) = \operatorname{rank}(Z - zI) = M - \#\{m : z_m = z\}$. The columns of $U_M$ span the column space of $V$, so $P = U_M^* V$ is invertible; from $H_0 V_M = U_M \Sigma_M$ we get $PD(W^T V_M) = \Sigma_M$, so $W^T V_M = D^{-1} P^{-1} \Sigma_M$ and $U_M^* H_1 V_M \Sigma_M^{-1} = PDZD^{-1}P^{-1} = PZP^{-1}$. $\square$

Heuristically, the error analysis is the standard one for the pencil: if the samples carry an error $\eta$ and the exponentials are separated, the poles move by $O(\eta / |A_{\min}|)$ times a condition number polynomial in $J$ and in the separation, not exponential in the bandwidth. The bandwidth enters through $|w(\gamma_\rho + i a_\rho)| e^{-a_\rho \omega_{J-1}}$: at the largest frequency used, the signal of a zero is $e^{-\Omega/2}$ times its amplitude, and the data error at that frequency is $\eta$, so the effective noise is $\eta e^{\Omega/2}$. This is the same $e^{\Omega/2}$ as in Proposition 32.1.2, but it now multiplies the data error, not the resolution: with data exact to $10^{-20}$ and $\Omega = 45$ the effective

noise is $10^{-10}$, and with double precision transforms, $\eta = 10^{-16}$, it is $10^{-6}$ at the top frequency and far smaller at the frequencies that carry most of the signal.

## 32.3 The computation

The numbers of this section are the author's, computed by the scripts of the source and not rerun for this book. The data are $u_1$ on $[150, 250]$ at spacing 0.02, enclosed in balls of radius below $5 \times 10^{-21}$ by the script of Chapter 31; the taper is centred at $c = 200$ with $\ell = 10$, so that the zeros within 15 of the centre have taper weight at least 0.3 and those beyond 30 weight below 0.012. The transform $F(\omega_j)$ is computed as a Riemann sum over the data in ball arithmetic, at $\omega_j = 5 + 0.1j$, $j < 400$, so $\Omega = 45$, and converted to double precision. The balls enclose the finite sum, not $F$: the aliasing error of the sum is of the order $e^{-a(2\pi/h - \Omega)} = e^{-135}$ with $a = \frac{1}{2}$ and $h = 0.02$, but the truncation of the data at $c \pm 5\ell$, where the taper is $e^{-12.5} = 3.7 \times 10^{-6}$, is not bounded. The Hankel matrix is $200 \times 200$; the singular values fall from 1 to $10^{-9}$ over the first sixty and the model order is taken there. At $\Delta = 0.1$ the ordinates are determined modulo $2\pi/\Delta = 62.8$, so zeros farther than 31.4 from the centre appear wrapped into the window, with taper weights below $e^{-4.9}$; the amplitude test sets them aside, and only ordinates within 25 of the centre are reported. The whole computation takes two minutes, almost all in the ball arithmetic transform; in double precision it takes a second.

| | zeros | position error | | error in $\sigma_\rho$ | |
|---|---|---|---|---|---|
| $\lvert\gamma - c\rvert \le$ | | max | median | max | median |
| 10 | 11 | $4.5 \times 10^{-10}$ | $1.8 \times 10^{-10}$ | $7.9 \times 10^{-10}$ | $2.3 \times 10^{-10}$ |
| 15 | 17 | $5.0 \times 10^{-7}$ | $2.5 \times 10^{-10}$ | $8.4 \times 10^{-7}$ | $3.8 \times 10^{-10}$ |
| 20 | 21 | $5.0 \times 10^{-7}$ | $4.5 \times 10^{-10}$ | $8.4 \times 10^{-7}$ | $7.9 \times 10^{-10}$ |
| 25 | 28 | $4.1 \times 10^{-5}$ | $2.7 \times 10^{-9}$ | $4.0 \times 10^{-5}$ | $1.1 \times 10^{-8}$ |

Table 32.1: The matrix pencil at height 200, level one data, taper $\ell = 10$: errors in the position and in the real part of the recovered zeros, by distance from the centre (computed by the author's scripts, not rerun).

The eleven zeros within 10 of the centre are placed to within $8 \times 10^{-10}$ in both coordinates; the error grows toward the edge of the taper as the amplitudes fall, as the heuristic error analysis predicts. For comparison, the backward flow of Chapter 31 on the same data, with 4000 accelerated EM steps and the certificate, placed the centroids to about $10^{-3}$. The gain is seven orders of magnitude at the centre, from the same $u_1$. The modulus of the tapered transform $F(\omega)$ follows the line $e^{-\omega/2}$, so that the signal falls by $e^{-20}$ over the frequencies used (the author's figure is not reproduced).

The real parts are the new element. Every recovered pole has $|z_\rho| = e^{-\Delta a_\rho}$ with $a_\rho = 0.5000000$ to the digits shown; the fit does not constrain the real parts, and

would have reported otherwise. To make the point we added to $u_1$ the Lorentzians of a fictitious pair of zeros at $0.85 \pm 60.089i$ and $0.15 \pm 60.089i$, which is what a zero off the line and its reflection would contribute, and ran the pencil on a window centred at 60. It returned two components at $\gamma = 60.08900000$ and $60.08899905$ with $a = 0.15000000$ and $0.84999991$, beside the genuine zeros $59.347044$ and $60.831779$ with $a = 0.500000$. The pencil reads the real parts of the zeros off the level one data, to seven digits, without any assumption.

## 32.4 Posterior concentration, and what it cannot do

One may ask for a posterior concentration result for the recovery of $\mathfrak{U}_{1/2}$, and hence for a proof of the Riemann hypothesis. The first can be done under hypotheses; the second cannot, and the theorem itself shows why.

Fix a window, centre $c$ and taper $\ell$, a sampling step $\Delta$, the samples $\omega_j = \omega_0 + j\Delta$, $0 \le j < J$, with $\omega_0 \ell^2 > 1$, and a radius $R$ with $2R\Delta < 2\pi$, so that $\gamma \mapsto e^{i\Delta(\gamma - c)}$ is one to one on $[c-R, c+R]$. By Theorem 32.2.1 the data vector $F = (F(\omega_j))_j$ is

$$F = \Phi(\vartheta^0) + b, \qquad \Phi(\vartheta)_j = \sum_{m=1}^{M} A_m \, z_m^j, \qquad z_m = e^{\Delta(-a_m + i(\gamma_m - c))},$$

where $\vartheta^0$ collects the zeros with $|\gamma_m - c| \le R$ and $b$ is the deterministic bias from the zeros beyond $R$ and the remainder of Theorem 32.2.1; its size is about $2n(c)(\ell^2/R)e^{-R^2/2\ell^2}$ per sample, with $n(c)$ the density of zeros at height $c$. To make a Bayesian statement we model the observation as $Y = F + \tau\xi$, where $\xi \in \mathbb{C}^J$ has independent standard normal real and imaginary parts, and put a prior on $\vartheta$.

Theorem 32.4.1 (posterior concentration on a window). *Fix $s > 0$, $0 < a_- < a_+ < 1$, $0 < \alpha_- < A_+$ and an integer $M_{\max}$ with $J \ge 2M_{\max}$. For $0 \le M \le M_{\max}$ let $K_M$ be the compact set of $\vartheta = (\gamma_m, a_m, A_m)_{m \le M}$ with $c - R \le \gamma_1 < \cdots < \gamma_M \le c + R$, $\gamma_{m+1} - \gamma_m \ge s$, $a_m \in [a_-, a_+]$ and $\alpha_- \le |A_m| \le A_+$. Let the prior be $\Pi = \sum_M p_M \Pi_M$, where $\Pi_M$ has a density on $K_M$ with values between $\pi_- > 0$ and $\pi_+$. Let $\vartheta^0$ be an interior point of $K_{M^0}$ with $p_{M^0} > 0$, and let $Y = \Phi(\vartheta^0) + b + \tau\xi$ with $\|b\|_2 \le B\tau$. Put $d(\vartheta, \vartheta^0) = \max_m \big(|\gamma_m - \gamma_m^0| + |a_m - a_m^0| + |A_m - A_m^0|\big)$ if $M = M^0$ and $d = \infty$ otherwise. Then there are $\tau_0, r_0, K, \kappa_0 > 0$, depending on $\vartheta^0$, the sets $K_M$, the prior, $J$, $\Delta$ and $B$, such that for $\tau \le \tau_0$ and $K\tau \le \varepsilon \le r_0$, with probability at least $1 - e^{-\kappa_0 \varepsilon^2/\tau^2}$,*

$$\Pi\big(d(\vartheta, \vartheta^0) \le \varepsilon \,\big|\, Y\big) \;\ge\; 1 - e^{-\kappa_0 \varepsilon^2/\tau^2}.$$

*In particular, for fixed $\varepsilon_1 > 0$ and data with $\|b\|_2 \le B\tau$, as $\tau \to 0$ the posterior probability that every component has $|\sigma_m - \frac{1}{2}| \le \varepsilon_1$ tends to 1 in probability if every zero of $\vartheta^0$ has $|\sigma_m^0 - \frac{1}{2}| < \varepsilon_1$, and to 0 if one has $|\sigma_m^0 - \frac{1}{2}| > \varepsilon_1$.*

*Proof. Identifiability.* On each $K_M$ the poles $z_m$ are distinct and nonzero, and $(\gamma, a) \mapsto z$ is one to one on $[c-R, c+R] \times (0,1)$ because $2R\Delta < 2\pi$. If $\Phi(\vartheta) = \Phi(\vartheta^0)$ with $\vartheta \in K_M$, the difference is a combination of at most $M + M^0 \le J$ distinct exponentials vanishing at $J$ consecutive indices; the Vandermonde matrix of the distinct poles has full column rank, so after merging equal poles every coefficient vanishes, and since all amplitudes are nonzero each pole of $\vartheta$ is a pole of $\vartheta^0$ with the same amplitude and conversely. Hence $M = M^0$ and, the labels being fixed by the order of the $\gamma_m$, $\vartheta = \vartheta^0$. By continuity and compactness $g = \min_{M \neq M^0} \min_{K_M} \|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 > 0$. In the $4M^0$ real coordinates the derivative of $\Phi$ at $\vartheta^0$ is injective: with respect to $(z_m, A_m)$ it is the $J \times 2M^0$ matrix with columns $(z_m^0)^j$ and $j A_m^0 (z_m^0)^{j-1}$, whose first $2M^0$ rows have determinant $\pm \prod_m A_m^0 \prod_{m<n} (z_n^0 - z_m^0)^4 \neq 0$, and $dz = z\Delta(-da + i\, d\gamma)$ with $z \neq 0$. So $\|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 \ge \lambda_0 d$ for $d \le r_1$, and on the compact set $\{d \ge r_1\} \cap K_{M^0}$ the left side has a positive minimum; hence $\|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 \ge \lambda d(\vartheta, \vartheta^0)$ on $K_{M^0}$ for some $\lambda > 0$, and $\|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 \le L_\Phi d$ on the $d$-ball of radius $r_1$ about $\vartheta^0$, which lies in $K_{M^0}$ if $r_1$ is small.

*Noise.* Put $u = \Phi(\vartheta) - \Phi(\vartheta^0)$ and $e = b + \tau\xi$. The log likelihood ratio of $\vartheta$ to $\vartheta^0$ is $\big(2\,\mathrm{Re}\langle u, e\rangle - \|u\|^2\big)/2\tau^2$. On the event $E_t = \{\|\xi\|_2 \le \sqrt{2J} + \sqrt{2t}\}$, of probability at least $1 - e^{-t}$ by Gaussian concentration, $\|e\| \le q\tau$ with $q = B + \sqrt{2J} + \sqrt{2t}$, so the ratio lies between $e^{-\|u\|^2/2\tau^2 - q\|u\|/\tau}$ and $e^{-\|u\|^2/4\tau^2 + q^2}$, by $-x^2/2 + qx \le -x^2/4 + q^2$.

*Denominator.* On $\{M = M^0,\ d \le \tau\}$, $\tau \le r_1$, we have $\|u\| \le L_\Phi \tau$, and this set has prior mass at least $p_{M^0} \pi_- v_0 \tau^D$, $D = 4M^0$, where $v_0 \tau^D$ is the volume of the $d$-ball of radius $\tau$. So the integral of the ratio against $\Pi$ is at least $p_{M^0} \pi_- v_0 \tau^D e^{-L_\Phi^2/2 - L_\Phi q}$.

*Numerator.* Since $e^{-\lambda^2 d^2/4\tau^2} \le e^{-\lambda^2\varepsilon^2/8\tau^2} e^{-\lambda^2 d^2/8\tau^2}$ for $d > \varepsilon$, the integral of the ratio over $\{M = M^0,\ d > \varepsilon\}$ is at most

$$p_{M^0} \pi_+ e^{q^2} e^{-\lambda^2\varepsilon^2/8\tau^2} \int_{\mathbb{R}^D} e^{-\lambda^2 d(\vartheta,\vartheta^0)^2/8\tau^2}\, d\vartheta = p_{M^0} \pi_+ C_D \Big(\frac{\tau}{\lambda}\Big)^D e^{q^2} e^{-\lambda^2\varepsilon^2/8\tau^2},$$

and the integral over $\{M \neq M^0\}$ is at most $e^{q^2} e^{-g^2/4\tau^2}$.

*Conclusion.* The powers $\tau^D$ cancel:

$$\Pi(d > \varepsilon \mid Y) \le C_1 e^{q^2 + L_\Phi q} \big( e^{-\lambda^2\varepsilon^2/8\tau^2} + \tau^{-D} e^{-g^2/4\tau^2} \big).$$

For $\varepsilon \le r_0 := g/\lambda$ and $\tau \le \tau_0$, with $\tau_0 \le r_1$ so small that $\tau^{-D} e^{-g^2/8\tau^2} \le 1$, the second term is at most the first. Since $q^2 + L_\Phi q \le 2q^2 + L_\Phi^2 \le C_2 + 12t$, on $E_t$ we get $\Pi(d > \varepsilon \mid Y) \le C_3 e^{12t} e^{-\lambda^2\varepsilon^2/8\tau^2}$. Take $t = \lambda^2\varepsilon^2/192\tau^2$: then $\Pi(d > \varepsilon \mid Y) \le C_3 e^{-\lambda^2\varepsilon^2/16\tau^2} \le e^{-\lambda^2\varepsilon^2/32\tau^2}$ once $\varepsilon \ge K\tau$ with $K^2\lambda^2 \ge 32 \log \max(C_3, 1)$, on an event of probability at least $1 - e^{-\lambda^2\varepsilon^2/192\tau^2}$. This is the claim with $\kappa_0 = \lambda^2/192$. For the last statement, $|\sigma_m - \sigma_m^0| = |a_m - a_m^0| \le d$; apply the bound with the fixed radius $\varepsilon = \varepsilon_1 - \max_m |\sigma_m^0 - \tfrac12|$ in the first case and $\varepsilon = \tfrac12(|\sigma_m^0 - \tfrac12| - \varepsilon_1)$ in the second, reduced to $r_0$ if necessary. $\square$

REMARK 32.4.2. The theorem needs the bias below the noise, $\|b\|_2 \le B\tau$ with $B$ of order one, and no aliasing, $R < \pi/\Delta$. At the settings of the computations of this chapter these are incompatible. With $\ell = 10$, $\Delta = 0.1$ and $\tau = 10^{-16}$ at height 200, the bias falls below $\tau$ only for $R \ge 86$, while $\pi/\Delta = 31.4$; with $\ell = 2$ and $\Delta = 0.2$ at height $3 \times 10^{12}$ it needs $R \ge 17.4$, while $\pi/\Delta = 15.7$. With $R$ just below $\pi/\Delta$ the bias is about $2 \times 10^{-3}$ per sample at height 200, and the theorem then holds only with $B$ of the order of $\|b\|_2/\tau$, that is at radii of the order of the bias divided by $\lambda$, which is no information. The theorem describes a computation with $\Delta < \pi/86 = 0.036$ at $\ell = 10$, about 1100 samples on $[5, 45]$, or one in which the amplitudes are tied to $(\gamma_m, a_m)$ through the taper, which removes the aliasing; that variant is not proved here.

The separation $s > 0$ excludes two components at one ordinate, and so the pairs $\frac{1}{2} \pm \delta$ that the functional equation imposes on zeros off the line: such a configuration is not an interior point of any $K_M$, and the rate $\tau$ of the theorem does not apply to it. Near the line the two poles of such a pair merge; Section 32.5 treats that singular case.

Under its hypotheses the theorem is a useful statement: the posterior for the zeros in a window, including their real parts, contracts to the truth at the rate of the data accuracy. It is the Bayesian form of Theorem 32.2.1. Its hypotheses do not hold at the settings of the computations above (Remark 32.4.2): the errors of $10^{-10}$ to $10^{-7}$ that the matrix pencil reaches from data with $\tau = 10^{-16}$ are an observation, not an instance of the theorem. Its constants, however, depend on the separation of the poles, and Section 32.5 shows that this dependence is not a technicality: the functional equation forces off line zeros to come in pairs whose poles merge as they approach the line, and near the line the rate for the real part of a pair is $\tau^{1/2}$, not $\tau$ (Theorem 32.5.2).

The theorem concentrates the posterior at $\vartheta^0$, whatever $\vartheta^0$ is. It says that the data decide, in the limit of exact data, the real parts of finitely many zeros to any finite precision; it does not say what they decide. If the Riemann hypothesis is true, the posterior probability that the zeros in the window lie within $\varepsilon$ of the line tends to one; if it is false for some zero in the window, the same probability tends to zero; and the theorem is indifferent between the two cases, because posterior consistency is a statement about the mechanism, not about the parameter. To prove the hypothesis from it one would need, first, $\varepsilon = 0$, which a posterior at a positive noise level does not give, and second, all windows at once, which no finite computation covers. The second is the obstacle that the verification of the hypothesis to height $3 \times 10^{12}$ meets: a finite computation settles finitely many zeros, and the hypothesis is a statement about all of them. The first is particular to the posterior: sign changes of $Z$ together with a count of zeros prove $\sigma = \frac{1}{2}$ exactly for every zero they cover, while a posterior at a positive noise level gives the real parts only to finite precision. Nor is there a loophole in the limit $\tau \to 0$: the data $u_1$ are not random, the noise is a modelling device, and the limit of the posterior

is the point mass at the true zeros, which is where the problem started. We record this because posterior concentration is exactly the right description of what the pencil computation does, but the step from concentration to a theorem about the parameter is the step that no statistical argument can make: consistency means the posterior ends at the truth, and the truth here is the Riemann hypothesis or its negation, undetermined by the theorem.

What the theorem does add to the programme is a precise sense in which $\mathfrak{U}_{1/2}$ is recoverable: under its hypotheses the atomic model is identifiable from the level one data on a window without aliasing, the posterior contracts at the rate of the data accuracy, and the real parts are part of what is recovered. That is the sparse answer to the deconvolution, and it is as far as the data go.

## 32.5 The functional equation makes the problem singular

### 32.5.1 Pairs, not zeros

Theorem 32.4.1 treats each zero as a free parameter $(\gamma, a)$. The zeros are not free: $\xi(s) = \xi(1-s)$, so a zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\delta > 0$ comes with its partner $1 - \bar{\rho} = \frac{1}{2} - \delta + i\gamma$ at the same ordinate. At level $\theta$ the pair has widths $a_0 \mp \delta$ with $a_0 = \theta - \frac{1}{2}$, and its two poles $e^{\Delta(-(a_0 \mp \delta) + i(\gamma - c))}$ merge as $\delta \to 0$ into one pole of doubled amplitude, the contribution of a double zero on the line. The pair enters the samples with one amplitude common to its two members, so that the contribution of a pair is $A\,[f(a_0 - \delta) + f(a_0 + \delta)]$ with one free complex amplitude $A$, and $\delta = 0$ is a zero on the line of free amplitude $2A$. The right alternative to "this zero is on the line" is therefore not "this zero has width $a_0 + \varepsilon$", which the functional equation forbids, but "this zero is a pair at $\frac{1}{2} \pm \delta$". The two alternatives behave very differently.

Lemma 32.5.1 (evenness). *In the notation of Theorem 32.2.1 at level $\theta$, with $0 \le \delta < a_0$, write the contribution of one zero of width $a$ at ordinate $\gamma$ to the samples as $f(a)_j = \exp(-q^2/2\ell^2 + i\omega_j q)$ with $q = (\gamma - c) + ia$. The contribution of the pair at $\frac{1}{2} \pm \delta$ is $f(a_0 - \delta) + f(a_0 + \delta)$, an even entire function of $\delta$, hence an entire function $g(\eta)$ of $\eta = \delta^2$, with*

$$g(\eta) = 2f(a_0) + \eta\, t + O(\eta^2),$$
$$t_j = f''(a_0)_j = f(a_0)_j \Big[ \Big( \omega_j + \frac{i q_0}{\ell^2} \Big)^2 + \frac{1}{\ell^2} \Big], \quad q_0 = (\gamma - c) + i a_0.$$

*Proof.* Swapping the two members of the pair changes $\delta$ to $-\delta$ and leaves the sum unchanged, so the sum is even; it is entire in $\delta$ because $f$ is entire in $a$. An even entire function is an entire function of $\delta^2$. The coefficient of $\delta^2$ is $f''(a_0)$, and $\partial_a \log f = -(\omega_j + iq/\ell^2)$, $\partial_a^2 \log f = 1/\ell^2$, which gives $t$. □

With the amplitudes of Theorem 32.2.1, the limit $\delta \to 0$ of a pair is a double zero, $2f(a_0)$, and a simple zero, $f(a_0)$, is not nested in the pair model. The singular

analysis therefore compares a double zero with a pair, or, in the common amplitude model, a zero of free amplitude with a pair of one free common amplitude; for a simple zero the amplitude is $A^0 = \frac{1}{2}$ and the tangent is $t = f''(a_0)/2$. Freeing the amplitudes of the two members independently discards the multiplicity: the contribution of the pair is then not even in $\delta$, and the joint Jacobian is rank deficient at $\delta = 0$. We use the common amplitude model throughout this section, and the statements about "a zero" below are statements in the free amplitude model.

So the data depend on the displacement $\delta$ only through $\delta^2$: the Fisher information for $\delta$ vanishes at $\delta = 0$, and the model is singular there, in the way a mixture with two merging components is singular [80, 223]. In the parameter $\eta \geq 0$ it is regular, one sided at $\eta = 0$, when $t_\perp \neq 0$ (Theorem 32.5.2).

THEOREM 32.5.2 (the rate for the real part). *Fix a window, a level $\theta > \frac{1}{2}$ with $a_0 = \theta - \frac{1}{2}$, the samples $\omega_j = \omega_0 + j\Delta$, $0 \leq j < J$, and $f$ as in Lemma 32.5.1. Let one ordinate $\gamma$ carry either a zero on the line or a pair at $\frac{1}{2} \pm \delta$, $0 \leq \delta \leq \delta_+ < a_0$, entering the samples as $A\, g(\eta;\gamma)$, where $g(\eta;\gamma) = f(a_0 - \sqrt{\eta}) + f(a_0 + \sqrt{\eta})$, $\eta = \delta^2$, and $A$ is one complex amplitude common to the two members; let the other $M$ zeros of the window enter as $\sum_{m \leq M} A_m z_m^j$ as in Theorem 32.4.1, labelled by increasing ordinate. Write $\vartheta' = (\gamma, A, (\gamma_m, a_m, A_m)_{m \leq M})$, of real dimension $D' = 3 + 4M$, and $\Phi(\eta, \vartheta') \in \mathbb{C}^J = \mathbb{R}^{2J}$ for the noiseless samples, and assume:*

*(H1) $Y = \Phi(\eta^0, \vartheta'^0) + b + \tau\xi$, where $\xi \in \mathbb{C}^J$ has independent standard normal real and imaginary parts, as in Theorem 32.4.1, so that $\tau$ is the standard deviation of each real coordinate, and $b \in \mathbb{C}^J$ is deterministic with $\|b\|_2 \leq B\tau$;*

*(H2) the parameter set is $\Theta = [0, \eta_+] \times K'$ with $\eta_+ = \delta_+^2$ and $K'$ compact, $\eta^0 \in [0, \eta_+)$, $\vartheta'^0$ is interior to $K'$, and $\Phi$ is injective on $\Theta$;*

*(H3) the real $2J \times D'$ Jacobian $T'$ of $\vartheta' \mapsto \Phi(\eta^0, \vartheta')$ at $\vartheta'^0$ has rank $D'$; with $t = \partial_\eta \Phi(\eta^0, \vartheta'^0)$, which is $A^0 f''(a_0)$ when $\eta^0 = 0$, $P'$ the orthogonal projection of $\mathbb{R}^{2J}$ onto the range of $T'$ and $t_\perp = (I - P')t$, we have $\|t_\perp\| > 0$;*

*(H4) the prior puts mass $1 - p \in [0, 1)$ at $\eta = 0$ and has density $p\,\pi_\eta(\eta)$ on $(0, \eta_+]$, $\pi_\eta(\eta) = \pi(\sqrt{\eta})/2\sqrt{\eta}$ with $\pi$ continuous and positive on $[0, \delta_+]$, and, independently, $\vartheta'$ has a density $p'$ bounded on $K'$, continuous and positive at $\vartheta'^0$.*

*Let $r_\tau = \tau/\|t_\perp\|$ and let $\tau \to 0$ with $B$ and $\vartheta'^0$ fixed, and with $\eta^0$ fixed in (i) and (ii). Then:*

*(i) for every $M_\tau \to \infty$, $\Pi(|\eta - \eta^0| > M_\tau r_\tau \mid Y) \to 0$ in probability; hence $\Pi(\delta > (M_\tau r_\tau)^{1/2} \mid Y) \to 0$ if $\delta^0 = 0$, and $\Pi(|\delta - \delta^0| > M_\tau r_\tau/\delta^0 \mid Y) \to 0$ if $\delta^0 > 0$;*

*(ii)* *let* $\eta^0 = 0$ *and, for* $0 < \delta_1 \le \delta_+$, *let* $D(\delta_1) = \min_{\vartheta' \in K'} \|\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta')\|$, *attained at* $\vartheta'^*$; *then* $D(\delta_1) = \delta_1^2 \|t_\perp\| + O(\delta_1^4)$, *the Kullback Leibler divergence of the law of* $Y$ *under* $(\delta_1^2, \vartheta'^0)$ *from the law under* $(0, \vartheta'^*)$ *is* $D(\delta_1)^2/2\tau^2$, *and for* $\delta_1 = c\, r_\tau^{1/2}$ *the two error probabilities of any test of* $\delta = 0$ *against* $\delta = \delta_1$, *whatever the method, sum to at least* $1 - c^2/2 - O(\tau)$;

*(iii)* *let* $\eta^0 = v\, r_\tau$ *with* $v \ge 0$ *fixed* ($v = 0$ *is the null), let* $\|b\|_2 = o(\tau)$, *take* $t_\perp$ *at* $\eta = 0$, *and let* $Z = \langle t_\perp, r\rangle / (\tau \|t_\perp\|)$ *with* $r$ *the residual of the least squares fit of* $\Phi(0, \cdot)$ *over* $K'$; *then* $Z = v + \zeta_\perp + o_P(1)$ *with* $\zeta_\perp$ *standard Gaussian, so the one sided matched filter* $Z > z_\alpha$ *has asymptotic level* $\alpha$ *and asymptotic power* $1 - N(z_\alpha - v)$, $N$ *the standard normal distribution function, which is* $\frac{1}{2}$ *at* $\delta^* = (z_\alpha r_\tau)^{1/2}$; *and twice the logarithm of the generalised likelihood ratio of* $\eta \ge 0$ *against* $\eta = 0$ *is* $\max(Z, 0)^2 + o_P(1)$, *so that for* $\alpha < \frac{1}{2}$ *the generalised likelihood ratio test at level* $\alpha$ *is asymptotically the matched filter.*

Remark 32.5.3. (H2) and (H3) hold, and $\|t_\perp\| > 0$ is automatic when $\eta^0 = 0$, if $J \ge 2M + 4$, $2R\Delta < 2\pi$ as in Theorem 32.4.1, the amplitudes in $K'$, $A$ included, are bounded away from 0, the ordinate $\gamma$ ranges over an interval $I$ of length less than $2\pi/\Delta$, and the poles $z_m$ keep a distance at least $s > 0$ from one another and from the set $S = \{e^{\Delta(-a+i(\gamma'-c))} : |a - a_0| \le \delta_+,\ \gamma' \in I\}$ that contains the poles of the pair. Indeed $f(a)_j = f(a)_0\, z(a)^j$ with $z(a) = e^{\Delta(-a+i(\gamma-c))}$ and $f(a)_0 \ne 0$, so two parameters with the same samples give two exponential sums whose difference has at most $2M+4 \le J$ distinct poles and vanishes for $0 \le j < J$; the Vandermonde matrix of these poles has full column rank, so the two sums have the same poles with the same coefficients. The poles in $S$ are those of the pair: one pole means $\eta = 0$, two poles of moduli $e^{-\Delta(a_0 \mp \sqrt{\eta})}$ give $\eta > 0$, their common argument gives $\gamma$, and the coefficients divided by $f(a_0 \mp \sqrt{\eta})_0$ give $A$; the remaining poles give the $(\gamma_m, a_m, A_m)$, the ordinates by $2R\Delta < 2\pi$ and the labels by their order. For (H3) at $\eta^0 = 0$, the real columns of $T'$ lie in the complex span of $(z_m^j)_j$, $(jz_m^j)_j$, $m \le M$, and $(z_0^j)_j$, $(jz_0^j)_j$, $z_0 = z(a_0)$: four real columns per zero, $z_m^j$, $iz_m^j$, $iA_m \Delta j z_m^j$, $-A_m \Delta j z_m^j$, and three for the pair, independent over $\mathbb{R}$ because the confluent Vandermonde matrix of these $2M + 2$ complex columns has full rank for $J \ge 2M + 2$. Finally $t = A^0 f''(a_0)$ and $\partial_a^2 z(a)^j = \Delta^2 j^2 z(a)^j$, so $t$ has the component $A^0 \Delta^2 f(a_0)_0\, (j^2 z_0^j)_j$ with a nonzero coefficient, and the confluent Vandermonde matrix with the further column $(j^2 z_0^j)_j$ has full rank for $J \ge 2M+3$; so $t$ is not in the complex span of the columns of $T'$, a fortiori not in their real span, and $t_\perp \ne 0$. With a prior on $M$ as in Theorem 32.4.1 (amplitudes bounded below, $M \le M_{\max}$) and $J \ge 2M_{\max} + 4$, Step 1 of the proof extends across $M$ by the Vandermonde argument of that theorem, and (i) holds unchanged. Like Theorem 32.4.1, the theorem needs the bias below the noise and no aliasing, so by Remark 32.4.2 it does not cover the runs on the zeros of $\zeta$ in this chapter; it covers data simulated from the model with $b = 0$ in a window of radius below $\pi/\Delta$.

*Proof.* Write $\vartheta = (\eta, \vartheta')$, $\vartheta^0 = (\eta^0, \vartheta'^0)$, $u(\vartheta) = \Phi(\vartheta) - \Phi(\vartheta^0)$ and $n = b + \tau\xi$, so that the log likelihood ratio of $\vartheta$ to $\vartheta^0$ is $\ell(\vartheta) = (2\langle u, n\rangle - \|u\|^2)/2\tau^2$, with $\langle\cdot,\cdot\rangle$ the real inner product of $\mathbb{R}^{2J}$. By Lemma 32.5.1, $\Phi$ is real analytic on a neighbourhood of $\Theta$, including $\eta < 0$, so there are $L$ and $C_2$ with $\|u(\vartheta)\| \le L|\vartheta - \vartheta^0|$, $\|J_0\| \le L$ and $\|u(\vartheta) - J_0(\vartheta - \vartheta^0)\| \le C_2|\vartheta - \vartheta^0|^2$ on $\Theta$, where $J_0 = [t, T']$.

*Step 1 (separation).* By (H3), $t$ is not in the range of $T'$, so $J_0$ has full column rank and $\lambda = \sigma_{\min}(J_0) > 0$; hence $\|u(\vartheta)\| \ge \lambda|\vartheta - \vartheta^0|/2$ for $|\vartheta - \vartheta^0| \le r_1 = \lambda/2C_2$. By (H2), continuity and compactness, $g_1 = \min\{\|u(\vartheta)\| : \vartheta \in \Theta,\ |\vartheta - \vartheta^0| \ge r_1\} > 0$. With $\lambda_1 = \min(\lambda/2, g_1/r_1)$, $\|u(\vartheta)\| \ge \lambda_1 \min(|\vartheta - \vartheta^0|, r_1)$ on $\Theta$; note that $\lambda_1 \le \lambda/2 \le \sigma_{\min}(T')/2$.

*Step 2 (localisation).* Let $E_\tau = \{\|\xi\| \le \sqrt{2J} + 2(\log 1/\tau)^{1/2}\}$, of probability at least $1 - \tau^2$ by Gaussian concentration. On $E_\tau$, $\|n\| \le \tau\Lambda_\tau$ with $\Lambda_\tau = B + \sqrt{2J} + 2(\log 1/\tau)^{1/2}$, hence $\ell \le (2\|u\|\tau\Lambda_\tau - \|u\|^2)/2\tau^2$ everywhere and $\ell \ge -L\Lambda_\tau - L^2/2$ on $B_\tau = \{\vartheta \in \Theta : |\vartheta - \vartheta^0| \le \tau\}$. By (H4) the prior has a density bounded below by a positive constant on $\Theta$ near $\vartheta^0$, and $B_\tau$ contains at least half a ball, so $\Pi(B_\tau) \ge c_0\tau^{1+D'}$ and $\int e^\ell\, d\Pi \ge c_0\tau^{1+D'}e^{-L\Lambda_\tau - L^2/2}$. Let $\varepsilon_\tau = K\tau(\log 1/\tau)^{1/2}$ with $\lambda_1 K > \max(8, 2(1 + D')^{1/2})$. For small $\tau$, $\varepsilon_\tau \le r_1$ and $\lambda_1\varepsilon_\tau \ge 4\tau\Lambda_\tau$; as $2x\tau\Lambda_\tau - x^2 \le -x^2/2$ for $x \ge 4\tau\Lambda_\tau$, $\ell \le -\lambda_1^2\varepsilon_\tau^2/4\tau^2$ on $\{|\vartheta - \vartheta^0| > \varepsilon_\tau\}$. Hence, on $E_\tau$,

$$\Pi\big(|\vartheta - \vartheta^0| > \varepsilon_\tau \bigm| Y\big) \le c_0^{-1} \exp\Big(-\Big(\frac{\lambda_1^2K^2}{4} - 1 - D'\Big)\log\frac{1}{\tau} + L\Lambda_\tau + \frac{L^2}{2}\Big) \to 0.$$

The radius $\varepsilon_\tau$ only localises the posterior; the rate comes from Steps 3 and 4.

*Step 3 (linearisation).* On $U_\tau = \{\vartheta \in \Theta : |\vartheta - \vartheta^0| \le \varepsilon_\tau\}$ write $\vartheta - \vartheta^0 = (h_\eta, h')$ and $k = h' + h_\eta w$ with $w = (T'^{\mathsf T}T')^{-1}T'^{\mathsf T}t$, so that $J_0(\vartheta - \vartheta^0) = h_\eta t_\perp + T'k$, the two terms being orthogonal. Then $\ell = \ell_1(h_\eta) + \ell_2(k) + \rho$ with

$$\ell_1(x) = \frac{2x\langle t_\perp, n\rangle - x^2\|t_\perp\|^2}{2\tau^2}, \qquad \ell_2(k) = \frac{2\langle T'k, n\rangle - \|T'k\|^2}{2\tau^2},$$

and, on $E_\tau$, $|\rho| \le (2C_2\varepsilon_\tau^2\tau\Lambda_\tau + 2LC_2\varepsilon_\tau^3 + C_2^2\varepsilon_\tau^4)/2\tau^2 = O(\tau(\log 1/\tau)^{3/2})$ uniformly on $U_\tau$.

*Step 4 (the marginal of $\eta$).* Let $W = \langle t_\perp, n\rangle/(\tau\|t_\perp\|) = \zeta_\perp + \langle t_\perp, b\rangle/(\tau\|t_\perp\|)$, where $\zeta_\perp = \langle t_\perp, \xi\rangle/\|t_\perp\|$ is standard Gaussian and the second term is at most $B$ in modulus; so $W = O_P(1)$ and $\ell_1(vr_\tau) = Wv - v^2/2$. Let $I_2 = \int_{\mathbb{R}^{D'}} e^{\ell_2(k)}\, dk$, a Gaussian integral centred at $\hat k = (T'^{\mathsf T}T')^{-1}T'^{\mathsf T}n$ with covariance $\tau^2(T'^{\mathsf T}T')^{-1}$; on $E_\tau$, $|\hat k| \le \tau\Lambda_\tau/\sigma_{\min}(T') \le \varepsilon_\tau/4$ for small $\tau$, because $\sigma_{\min}(T')K \ge 2\lambda_1 K > 16$ and $\Lambda_\tau \le 4(\log 1/\tau)^{1/2}$ for small $\tau$. Put $S_\tau = U_\tau \cap \{|h_\eta| > M_\tau r_\tau\}$ and $S'_\tau = U_\tau \cap \{|h_\eta| \le r_\tau\}$. The change of variables $(h_\eta, h') \mapsto (h_\eta, k)$ has Jacobian 1. Bounding each $k$ integral by $I_2$,

$$\int_{S_\tau} e^\ell\, d\Pi \le e^{o(1)} \sup_{K'} p'\ I_2 \int_{|x| > M_\tau r_\tau} e^{\ell_1(x)}\, p\, \pi_\eta(\eta^0 + x)\, dx.$$

For $|h_\eta| \le r_\tau$ the $k$ section of $U_\tau$ contains the ball of radius $\varepsilon_\tau/2$ about $\hat{k}$, since there $|h'| \le |\hat{k}| + \varepsilon_\tau/2 + r_\tau|w| \le (\frac{3}{4} + o(1))\varepsilon_\tau$, and the Gaussian gives this ball the mass $(1 - o(1))I_2$, its radius being large against $\tau/\sigma_{\min}(T')$; so

$$\int_{S'_\tau} e^{\ell}\, d\Pi \ge e^{-o(1)}(1 - o(1)) \inf_{U_\tau} p'\ I_2 \int_{|x| \le r_\tau} e^{\ell_1(x)}\, p\, \pi_\eta(\eta^0 + x)\, dx.$$

If $\eta^0 > 0$, then $\pi_\eta(\eta^0 + x) = \pi_\eta(\eta^0)(1 + o(1))$ for $|x| \le \varepsilon_\tau$, and with $x = v r_\tau$ the ratio of the right sides is at most $(1 + o(1)) \int_{|v| > M_\tau} e^{Wv - v^2/2} dv \big/ \int_{|v| \le 1} e^{Wv - v^2/2} dv$, which tends to 0 in probability because $W = O_P(1)$; the atom at $\eta = 0$ lies outside $U_\tau$ and is covered by Step 2. If $\eta^0 = 0$, then $x = \eta \ge 0$, $\pi_\eta(\eta) = \pi(0)(1 + o(1))/2\sqrt{\eta}$ on $(0, \varepsilon_\tau]$, the factor $\pi(0) r_\tau^{1/2}/2$ cancels, and the ratio is at most $(1 + o(1)) \int_{M_\tau}^{\infty} v^{-1/2} e^{Wv - v^2/2} dv \big/ \int_0^1 v^{-1/2} e^{Wv - v^2/2} dv \to 0$; the atom at $\eta = 0$ lies in $S'_\tau$ and only increases the denominator. As $\Pi(|\eta - \eta^0| > M_\tau r_\tau \mid Y) \le \Pi(U_\tau^c \mid Y) + \Pi(S_\tau \mid Y)/\Pi(S'_\tau \mid Y)$, this proves (i) for $\eta$; the statements for $\delta$ follow from $|\sqrt{\eta} - \sqrt{\eta^0}| \le |\eta - \eta^0|^{1/2}$ and $|\sqrt{\eta} - \sqrt{\eta^0}| \le |\eta - \eta^0|/\sqrt{\eta^0}$.

*(ii).* The two laws are Gaussian with covariance $\tau^2 I_{2J}$ and means differing by $\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta'^*)$, $b$ cancelling, so the divergence is $D(\delta_1)^2/2\tau^2$. The choice $\vartheta' = \vartheta'^0 + \delta_1^2 w$, in $K'$ for small $\delta_1$, gives $\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta') = \delta_1^2 t_\perp + O(\delta_1^4)$, so $D(\delta_1) \le \delta_1^2 \|t_\perp\| + O(\delta_1^4)$. Conversely, let $r_2 = \min(r_1, \sigma_{\min}(T')/2C_2)$. If $|\vartheta' - \vartheta'^0| \ge r_2$, Step 1 and the triangle inequality give $\|\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta')\| \ge \lambda_1 r_2 - L\delta_1^2 > \delta_1^2 \|t_\perp\| + O(\delta_1^4)$ for small $\delta_1$. If $h' = \vartheta' - \vartheta'^0$ has $|h'| < r_2$, the difference is $\delta_1^2 t - T'h' + e$ with $\|e\| \le C_2(\delta_1^4 + |h'|^2)$; its norm is at least $\sigma_{\min}(T')|h'|/2 - \delta_1^2\|t\| - C_2\delta_1^4$, which exceeds the upper bound for $D(\delta_1)$ unless $|h'| = O(\delta_1^2)$, and at least $\|(I - P')(\delta_1^2 t - T'h')\| - \|e\| = \delta_1^2\|t_\perp\| - C_2(\delta_1^4 + |h'|^2)$. So $D(\delta_1) = \delta_1^2\|t_\perp\| + O(\delta_1^4)$. By Le Cam's two point lemma and Pinsker's inequality, the two error probabilities of any test sum to at least $1 - \mathrm{TV} \ge 1 - (\mathrm{KL}/2)^{1/2} = 1 - D(\delta_1)/2\tau$, which is $1 - c^2/2 - O(\tau)$ at $\delta_1 = c\, r_\tau^{1/2}$.

*(iii).* Let $\hat{\vartheta}'$ minimise $\|Y - \Phi(0, \vartheta')\|$ over $K'$. Since $\|Y - \Phi(0, \vartheta'^0)\| \le \|n\| + L\eta^0 = O_P(\tau)$, also $\|\Phi(0, \hat{\vartheta}') - \Phi(0, \vartheta'^0)\| = O_P(\tau)$, and Step 1 with the base point $(0, \vartheta'^0)$ gives $|\hat{\vartheta}' - \vartheta'^0| = O_P(\tau)$; so with probability tending to one $\hat{\vartheta}'$ is interior and satisfies the normal equations. Expanding, $Y - \Phi(0, \hat{\vartheta}') = n + \eta^0 t - T'(\hat{\vartheta}' - \vartheta'^0) + O_P(\tau^2)$ and $T'^{\mathsf{T}}(Y - \Phi(0, \hat{\vartheta}')) = O_P(\tau^2)$, so $r = (I - P')(n + \eta^0 t) + O_P(\tau^2)$ and $Z = \zeta_\perp + \langle t_\perp, b\rangle/(\tau\|t_\perp\|) + v + O_P(\tau)$, the middle term being at most $\|b\|_2/\tau \to 0$. The level and the power follow, and the power is $\frac{1}{2}$ when $v = z_\alpha$, that is $\eta^0 = z_\alpha r_\tau$. The least squares fit over $\eta \ge 0$ and $\vartheta'$ is consistent in the same way, and in the linearised model the minimum of the residual sum of squares is $\|r\|^2 - \max(\langle t_\perp, r\rangle, 0)^2/\|t_\perp\|^2 + O_P(\tau^3)$; so twice the log likelihood ratio is $\max(Z, 0)^2 + o_P(1)$, and for $\alpha < \frac{1}{2}$ the events $\max(Z, 0)^2 > z_\alpha^2$ and $Z > z_\alpha$ coincide. □

Part (i) is proved, with the conventions of Theorem 32.4.1: the noise, a bias bound $\|b\|_2 \le B\tau$, the separation and no aliasing conditions, and a rate with no

logarithmic factor; the second rate of (i), $\tau/(\delta_0\|t_\perp\|)$, is claimed only for fixed $\delta_0 > 0$, since under the spike of Theorem 32.5.4 it fails in the intermediate range. Part (ii) is proved and shows that no method does better than the rate $\tau^{1/2}$ for the real part, and (iii) is proved for the matched filter.

The scale is visible in the data. We took the window about 60 at level 0.6 with taper $\ell = 3$ and the samples $\omega_j = 2 + 0.2j$, $j < 100$ (the design of the detection experiment below, assumed because the source states none for this window), with a pair at $\frac{1}{2} \pm \delta$ added at 60.089 to the zeros, the pair fitted with one free common amplitude, and the noise $\tau = 1.7 \times 10^{-11}$ of the source, an estimate from a residual, taken as the standard deviation per real coordinate. In this design $\|t_\perp\| = 323.8$ for the free amplitude and 406.5 for an amplitude fixed by the model, the matched filter has $\delta^* = (1.645\,\tau/\|t_\perp\|)^{1/2} = 2.9 \times 10^{-7}$, and the Bayes factor, which pays an Occam factor, crosses one at about $3 \times 10^{-7}$ (`code/es_bf.py`, `data/es_bf_src.json`, computed, one noise draw per $\delta$: a pair at $10^{-6}$ is already detected, with a profile likelihood gain of 380 in $2\log$). The data accuracy of the source is $10^{-11}$, and in the design above a real part is resolved to a few times $10^{-7}$. The comparison with Theorem 32.5.2 is with its scale only: the error of the run is deterministic, one noise draw, and Theorem 32.5.2 does not cover the reported runs (Remark 32.5.3).

### 32.5.2 Bayes factors with the right alternative

Theorem 32.5.4 (Bayes factors). *In the setting of Theorem 32.5.2, put the spike and slab prior on the pair: $\delta = 0$ with probability $1 - p$, and otherwise $\delta$ with a density $\pi$ continuous and positive on $[0, \delta_+]$. Let $s = \|t_\perp\|/\tau$, $G(Z) = \int_0^\infty u^{-1/2} e^{Zu - u^2/2}\,du$, and $Z$ the statistic of Theorem 32.5.2(iii).*

*(a) If $\eta^0 = 0$, or $\eta^0 = v\,r_\tau$ with $v$ fixed, and $\|b\|_2 = o(\tau)$, and $\vartheta'$ has the same prior under the line and under the pair, the Bayes factor for the line against the pair is*

$$\mathrm{BF}_{01} = \frac{2\,s^{1/2}}{\pi(0)\,G(Z)}\,\big(1 + o_P(1)\big);$$

*under the null $Z$ is asymptotically standard Gaussian and $\mathrm{BF}_{01} \asymp \tau^{-1/2}$ in probability.*

*(b) Under a pair with fixed $\delta \in (0, \delta_+)$, if $p'$ is bounded below on $K'$ and every point of $K'$ has prior mass at least $c\tau^{D'}$ within distance $\tau$, then $\log \mathrm{BF}_{01} = -D(\delta)^2(1 + o_P(1))/2\tau^2$, with $D(\delta) = \delta^2\|t_\perp\| + O(\delta^4)$ as in Theorem 32.5.2(ii).*

*(c) Against the alternative of a single zero of width $a_0 + \varepsilon$, with $\varepsilon$ having a density positive and continuous at 0 and $\|(I - P')\partial_\varepsilon \Phi\| > 0$, the same data give $\mathrm{BF}_{01} \asymp \tau^{-1}$ in probability under the null.*

*Proof.* (a) $\mathrm{BF}_{10} = m_1/m_0$ with $m_1 = \iint_{\eta>0} e^{\ell(\eta,\vartheta')}\pi_\eta(\eta)\,p'(\vartheta')\,d\eta\,d\vartheta'$ and $m_0 = \int e^{\ell(0,\vartheta')}p'(\vartheta')\,d\vartheta'$, $\ell$ the log likelihood ratio to $(0,\vartheta'^0)$. The offset $\Phi(\eta^0,\vartheta'^0) - \Phi(0,\vartheta'^0) = \eta^0 t + O(\tau^2)$ is $O(\tau)$ and can be carried in $b$ in Steps 2 and 3 of the proof of Theorem 32.5.2, which then apply to $m_0$ and to $m_1$: the parts outside $U_\tau$ are negligible, and on $U_\tau$ the integrand factorises as in Step 3 with $W$ replaced by $W_0 = \langle t_\perp, n+\eta^0 t\rangle/(\tau\|t_\perp\|) = Z + o_P(1)$, by the proof of (iii). The $k$ integrals give $p'(\vartheta'^0)I_2(1+o_P(1))$ in both (in $m_1$ it suffices, as in Step 4, to control the $k$ sections for $\eta \le M_\tau r_\tau$, the rest being negligible), and, with $\eta = u\,r_\tau$ and $\varepsilon_\tau/r_\tau \to \infty$,

$$\int_0^{\varepsilon_\tau} e^{W_0\eta/r_\tau - \eta^2/2r_\tau^2}\,\frac{\pi(\sqrt{\eta})}{2\sqrt{\eta}}\,d\eta = \frac{\pi(0)}{2}\,r_\tau^{1/2}\,G(W_0)\,(1+o(1)).$$

Since $r_\tau^{1/2} = s^{-1/2}$ and $G$ is continuous and positive, $\mathrm{BF}_{10} = \frac{1}{2}\pi(0)s^{-1/2}G(Z)(1+o_P(1))$. Under the null $Z = \zeta_\perp + o_P(1)$, so $G(Z)$ is bounded above and away from 0 in probability.

(b) Take $\ell$ relative to the truth $(\delta^2,\vartheta'^0)$ and work on $E_\tau$. For the slab, $\ell \le \sup_x(2x\tau\Lambda_\tau - x^2)/2\tau^2 = \Lambda_\tau^2/2$ and Step 2 bounds $m_1$ below, so $|\log m_1| = O(\log 1/\tau)$. For the spike, $\|u(0,\vartheta')\| \ge D(\delta)$ for every $\vartheta'$, so $\ell(0,\vartheta') \le -(D^2 - 2D\tau\Lambda_\tau)/2\tau^2$ once $D \ge \tau\Lambda_\tau$; and within distance $\tau$ of the minimiser $\vartheta'^*$, $\|u(0,\vartheta')\| \le D + L\tau$ and $\ell(0,\vartheta') \ge -(D+L\tau)^2/2\tau^2 - (D+L\tau)\Lambda_\tau/\tau$, on a set of prior mass at least $c\tau^{D'}$. Hence $\log m_0 = -D^2/2\tau^2 + O(\Lambda_\tau/\tau) + O(\log 1/\tau)$, and $\log \mathrm{BF}_{01} = \log m_0 - \log m_1 = -D^2(1+o(1))/2\tau^2$.

(c) As in (a), with the regular coordinate $\varepsilon$ in place of $\eta$: the $\varepsilon$ integral is $\pi_\varepsilon(0)\,\tau\,\|(I-P')\partial_\varepsilon\Phi\|^{-1}\int e^{Z_\varepsilon v - v^2/2}\,dv\,(1+o_P(1))$, over the half line or the line allowed by the sign of $\varepsilon$, which is of order $\tau$; so $\mathrm{BF}_{10} \asymp \tau$. □

The formula needs the statistic $Z$ and the scale $s$, so that its use on data is a computation, and the one for the real zero at 201.264752 is an illustration under assumptions. The source does not give the noise level of that window. With $\pi$ uniform on $(0,\frac{1}{2})$, the design of the detection experiment below (level 0.6, $\ell = 3$, $\omega_j = 2 + 0.2j$, an assumption) and $\tau = 1.7 \times 10^{-11}$ assumed, the simple zero, of amplitude $A^0 = \frac{1}{2}$, has $\|t_\perp\| = 148.6$ and $s = \|t_\perp\|/\tau = 8.74 \times 10^{12}$; the formula of Theorem 32.5.4 gives $\log_{10}\mathrm{BF}_{01} = 6.14$ at $Z = 0$ and 5.51 at $Z = 1.7$, and the integral computed exactly gives a median 6.10 over 40 noise draws, with range 5.55 to 6.48. The single zero alternative gives 11.83 at $Z_1 = 0$ and 10.92 at $Z_1 = 1.7$, and 9.66 to 11.78 for the other ten central zeros at $Z_1 = 0$, the spread arising from the norm of the projected tangent, which varies from 0.2 to 29 over the central zeros (computed, `code/es_bf.py`, `data/es_bf_src.json`). So, under these assumptions, one window of data at this accuracy is worth a factor of about $10^6$ per zero in favour of the line against the pair, against about $10^{12}$ against the single zero alternative: the functional equation takes the square root of the evidence (computed in the free amplitude model; heuristic beyond it).

## 32.6 An adaptive prior, and a ReLU network

### 32.6.1 Where to look: the ordinates

The prior of Theorem 32.4.1 put the ordinates uniformly on the window. A prior that regularises where the zeros are has two pieces of information to use. The first is unconditional: the zeros have density $d(t) = \log(t/2\pi)/2\pi$ at height $t$, with an explicit error, so the number in a window of fixed length $W$ is $Wd(c) + O(\log c)$ (proved, Riemann and von Mangoldt). The second is conjectural and enters only as a prior: conjecturally (the pair correlation theorem of Montgomery [374], under RH and for restricted support, and the computations of Odlyzko [391]) the local statistics of the ordinates are those of the sine kernel determinantal process. So take the prior on the ordinates in a window to be the determinantal point process with kernel

$$K(x,y) = \frac{\sin \pi d(c)(x-y)}{\pi(x-y)},$$

which adapts to the height through $d(c)$ and repels: its Janossy density vanishes like $\prod_{i<j}(x_i - x_j)^2$ at coincidences. Three consequences. At a configuration of distinct ordinates the rates of Theorems 32.4.1 and 32.5.2 are unchanged, the prior density being continuous and positive there, since the prior mass of a ball of radius $\tau$ about it is still of order $\tau^{\dim}$; only the constants move. The posterior on the number $M$ of zeros in the window is pulled toward $Wd(c)$, a prior tendency and the Bayesian form of the count that Section 32.8.6 could not certify at $3 \times 10^{12}$ (heuristic). And spurious components that sit on top of genuine ones are penalised, a priori and by the two point correlation, by the factor $(\pi d\,|x_i - x_j|)^2/3$, which is $1 \times 10^{-4}$ at the density $d = 0.569$ of height 224 and a separation 0.0097, and which may help the many window scheme remove them (heuristic). The repulsion is a conjecture about the zeros and enters only as a prior. A determinantal process has almost surely no two points at one ordinate, so the process is put on the distinct ordinates, at the intensity $d(c)$, a pair at $\frac{1}{2} \pm \delta$ counting as one point marked by $\delta$; put on the zeros themselves it would give prior probability zero to every off line pair, and so decide the real parts before the data. With this convention the ordinate prior is the same under the line and under the pair, and to leading order as $\tau \to 0$ it leaves the Bayes factor of Theorem 32.5.4 unchanged (Remark 32.6.1); at finite $\tau$ it can still move the posterior of the real parts, as any prior on parameters that the likelihood couples with them can.

Remark 32.6.1. For a determinantal process with continuous kernel $K$ and intensity $d$ the second correlation function is $\rho_2(x,y) = d^2 - K(x,y)^2 \le d^2$, so the expected number of ordered pairs of distinct points of a window of length $W$ at distance less than $\epsilon$ is at most $2\epsilon W d^2$, which tends to 0 with $\epsilon$: almost surely no two points coincide, while a pair at $\frac{1}{2} \pm \delta$, and a double zero, put two zeros at one ordinate. On the distinct ordinates the prior density $p'$ of $\vartheta'$ in Theorem 32.5.2 (the

Janossy density of the ordinates times the priors of the amplitudes and widths) does not depend on $\eta$, and it is continuous and positive at distinct ordinates; in the proof of Theorem 32.5.4(a) the integrals over $\vartheta'$ give $p'(\vartheta'^0)I_2(1+o_P(1))$ in both marginal likelihoods, so $p'$ cancels from $\mathrm{BF}_{01}$ to leading order. At finite noise it does not cancel from the posterior of a real part: if $Y_1=\gamma+a+\sigma\epsilon_1$ and $Y_2=\gamma+\sigma\epsilon_2$, with $\epsilon_1,\epsilon_2$ independent standard normal and independent priors $\gamma\sim N(m,v_\gamma)$, $a\sim N(0,v_a)$, then

$$E[a\mid Y_1,Y_2]=\frac{v_a(Y_1-m_2)}{v_a+v_2+\sigma^2},\qquad m_2=\frac{\sigma^2 m+v_\gamma Y_2}{v_\gamma+\sigma^2},\quad v_2=\frac{v_\gamma\sigma^2}{v_\gamma+\sigma^2},$$

which depends on the prior mean $m$ of the ordinate whenever $\sigma>0$ (given $Y_2$, $\gamma$ is $N(m_2,v_2)$, and then $Y_1$ given $a$ is $N(m_2+a,v_2+\sigma^2)$).

### 32.6.2 What to believe: the real parts

For the real parts the adaptive prior is the spike and slab on pairs of Theorem 32.5.4: the functional equation supplies the structure, the spike at $\delta=0$ supplies the shrinkage, and the slab keeps the alternative honest. This is the prior under which a computation like that of Section 32.8.6 should be read, and Theorem 32.5.4 says what it can deliver: under the line, Bayes factors of order $\tau^{-1/2}$ in probability per pair, the other pairs held on the line, in the free amplitude model (the posterior odds are $(1-p)\,\mathrm{BF}_{01}/p$), and a resolution of $(\tau/\|t_\perp\|)^{1/2}$ in the real part.

### 32.6.3 A ReLU network

The question was whether a ReLU network can serve as the adaptive prior or regulariser. Polson and Ročková [252] proved that deep ReLU networks with a spike and slab prior on their weights have posteriors that contract at the near minimax rate for Hölder functions, adapting to the unknown smoothness. That theorem is about nonparametric regression and gives no rate for the window model, and what it suggests here is heuristic; part (ii) of Theorem 32.5.2 bounds every method, networks included. Where a network can help is in amortisation: it learns a classifier of the line against the pair, which at the population optimum is the posterior probability of the pair under the training law, and costs one pencil fit and one forward pass per window. In the author's experiment the training law is a recentred circular unitary ensemble of 40 points truncated to the window, with a log uniform slab on $[10^{-7.5},0.09]$ for the displacement of the pair, which is neither the sine kernel process nor the spike and slab of Theorem 32.5.4 (this is read from the author's script). That experiment, with two ReLU detectors (a network with two hidden layers of 256 units on the raw transform and the same network on the residual of the fit on the line), was run by the author and is unchecked: its two rows were not reproduced here and are not printed.

The pencil and a plug in matched filter were rerun for this book on the window model of Theorem 32.2.1 at level 0.6 with taper $\ell = 3$ and 100 samples $\omega_j = 2+0.2j$ (the sampling that the author's script appears to use; the chapter does not give it), ordinates drawn from the circular unitary ensemble scaled to the density at height 200, and the zero nearest the centre either a double zero on the line or a pair at $\frac{1}{2} \pm \delta$; the noise has complex root mean square $\tau$, that is standard deviation $\tau/\sqrt{2}$ per real coordinate. Two detectors, each calibrated to five per cent false alarms on 1200 null windows: the pencil statistic $\max|\hat{a} - a_0|$ over the components within 0.5 of the centre, and a plug in version of the matched filter of Theorem 32.5.2(iii) in which the tangent $t$ is that of the central zero at the pencil's ordinate and $t_\perp$ is orthogonal to the tangent columns (amplitude and pole) of every pencil component (this is our reading of "built from the pencil's components"; the source does not define it, and no efficiency claim is made for any plug in filter). The quantity $\delta^*$ is the smallest displacement detected with power one half.

| | detectable $\delta^*$ at noise $\tau =$ | | | | |
|---|---|---|---|---|---|
| detector | $10^{-10}$ | $10^{-8}$ | $10^{-6}$ | $10^{-4}$ | $10^{-2}$ |
| benchmark $0.0537\,\tau^{1/2}$ | $5.4 \times 10^{-7}$ | $5.4 \times 10^{-6}$ | $5.4 \times 10^{-5}$ | $5.4 \times 10^{-4}$ | $5.4 \times 10^{-3}$ |
| pencil | $8.5 \times 10^{-7}$ | $7.0 \times 10^{-6}$ | $6.1 \times 10^{-5}$ | $9.1 \times 10^{-4}$ | $7.2 \times 10^{-3}$ |
| plug in filter | $1.0 \times 10^{-6}$ | $1.3 \times 10^{-5}$ | $1.3 \times 10^{-4}$ | $1.0 \times 10^{-3}$ | $1.2 \times 10^{-2}$ |

The benchmark is Theorem 32.5.2(iii) at the half power point, $\delta^* = (1.645\,\tau'/\|t_\perp\|)^{1/2}$, with the median 403 of $\|t_\perp\|$ over 300 windows (10th and 90th percentiles 382 and 407) and $\tau' = \tau/\sqrt{2}$, the standard deviation per real coordinate; with $\tau$ itself as the standard deviation per real coordinate the constant is 0.0639 and the row is $6.4 \times 10^{-7}$ to $6.4 \times 10^{-3}$. It is the half power point at the median, a benchmark and not a bound (a law of $\|t_\perp\|$ with two values can have a smaller average power at it). The pencil and plug in rows are the means of two seeds (150 windows per $\delta$, 16 values of $\delta$; `code/es_detect.py`, `code/es_detect_all2.py`, `data/es_detect_src.log`, computed).

Two findings. First, the law $\delta^* \propto \tau^{1/2}$ of Theorem 32.5.2 holds over eight decades of noise for the pencil, with $\delta^*/\tau^{1/2}$ between 0.057 and 0.093, within a factor 1.7 of the benchmark; a data accuracy of $10^{-10}$ resolves a real part to about $10^{-6}$ in this design. Second, the plug in filter is 1.1 to 2.1 times worse than the pencil in our implementation. The source's rows were not reproduced; they depend on the pencil's order rule and on the definition of the plug in filter, which the source does not give.

So the answer to the question has two parts. A ReLU network is not a regulariser that changes what can be found: the bound in part (ii) of Theorem 32.5.2 holds for it, and a network trained on residuals is a learned test that cannot beat it. Whether such a learned test is more robust than the pencil is a matter for the author's experiment, which is unchecked. The regularisation that matters is in the prior: the sine kernel for where the zeros are, and the spike and slab on pairs for

what their real parts may be. None of this reaches the hypothesis. Each window at accuracy $\tau$ gives, under the line, Bayes factors of order $\tau^{-1/2}$ in probability per pair, the other pairs held on the line, in the free amplitude model, and a resolution of order $\tau^{1/2}$ in the real part, finitely many zeros at finite precision, as before.

## 32.7 What this does and does not do

Sparse methods solve the deconvolution. The problem $P_{1/2} * \mu = u_1$ is exponentially ill posed for a general $\mu$ and polynomially ill posed for a sum of atoms, and the matrix pencil, which works in the second model, recovers the atoms to ten digits where EM, which works in the first, reaches three. The structural reason is Theorem 32.2.1: the explicit formula, read as a deconvolution and tapered, is a sum of damped exponentials, Prony's problem, and the level one data determine the poles. Two further things follow. The pencil does not need to assume a width: it returns $a_\rho$ and so places each zero in the strip, which the backward flow at a fixed level cannot do, since that flow presupposes a width. And the certificate question of Chapter 31 changes character: a certified version of the pencil would need interval versions of the singular value decomposition and the eigenvalue problem, which exist, and a bound linking the matrix model to $\xi$ (the remainder of Theorem 32.2.1 and the far zeros), which does not, and would then give certified zeros from the line $\operatorname{Re} s = 1$; we have not done it, and it is the first open problem below.

What it does not do is also clear. The data $u_1$ to twenty digits are $\operatorname{Re}(\xi'/\xi)$ on $\operatorname{Re} s = 1$ to twenty digits, which is computed from $\zeta$ by the same algorithms that find zeros directly; the pencil is a different route from the same information, not a cheaper one. It reads the real parts; it does not constrain them. The one thing it offers the Thorin programme is a sharper statement of what the programme computes: the backward flow at level $\theta$ is the resolvent trace $\pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$, whose Fourier transform is $\sum_\rho \operatorname{sgn}(\theta - \sigma_\rho) e^{-|\theta-\sigma_\rho||\omega|} e^{-i\omega\gamma_\rho}$ (Proposition 32.8.1), and the Riemann hypothesis is the statement that every damping rate $|\theta - \sigma_\rho|$ at level $\theta$ is $\theta - \frac{1}{2}$, which is what the pencil measures and what no measurement proves.

A Bayesian reader will recognise the two models as two priors on the same unknown; we state the comparison as an analogy, since no posterior was computed here. The grid prior of Chapter 31, a product of gamma laws on the cells, is a prior on densities, and the EM solution is a point estimate under that model, with its $10^{-3}$. A spike prior, a point process of atoms with amplitudes bounded away from zero, is a prior on sums of atoms; in the setting of Theorem 32.4.1 its posterior concentrates at the true atoms at the rate of the noise, and the matrix pencil, which is exact up to the remainder of Theorem 32.2.1 when the data come from finitely many atoms, reaches the same atoms without computing a posterior. The deconvolution is the same; what changed is the model, and in an exponentially ill

posed problem the model is everything.

## 32.8 Questions taken up

This section takes up the questions left open by the first sections: how to certify the pencil's output, how close pairs behave, what the levels above one half add, which level to use, how to combine windows, and whether the method works at a large height. The tables of this section are the authors' computations at heights near 200, not rerun for this book, except where a sentence says otherwise.

### 32.8.1 Certifying the pencil

The pencil proposes; ball arithmetic disposes. For each of the eleven zeros within 10 of the centre $c = 200$, with the pencil's value $\gamma$, two checks were made in Arb at 80 bits. First, $\xi(\frac{1}{2} + it)$ changes sign on $[\gamma - 2 \times 10^{-9}, \gamma + 2 \times 10^{-9}]$, which certifies a zero on the line within $2 \times 10^{-9}$ of $\gamma$. Second, the winding number of $\xi$ around the circle of radius $10^{-3}$ about $\frac{1}{2} + i\gamma$, computed as $(2\pi i)^{-1} \oint \xi'/\xi$ by Arb's rigorous integrator, is the ball $1.0000 \pm 10^{-9}$, hence exactly 1: there is one zero in the disc, it is simple, and it is the one on the line. All eleven pass. This is not the same as Rump's verified eigenvalue bounds [265] for the pencil itself, which would enclose the eigenvalues of the matrix model and would still need a bound linking that model to $\xi$; it is what one wants in practice: the pencil locates, the argument principle certifies, and together they give certified zeros from level one data in the time the pencil takes plus a few seconds per zero. For a zero off the line the same winding number around a circle of radius less than $|\sigma - \frac{1}{2}|$ about the pencil's $\sigma + i\gamma$ would certify it with the same rigour, which is more than a sign change can do.

### 32.8.2 Close pairs

The pencil's resolution of a close pair was tested on synthetic data built from Theorem 32.2.1: the Lorentzians of the true zeros within 15 of the centre, with those within 1.5 of the centre removed and a pair at $c \pm d/2$ put in their place, for $d$ from 1 down to 0.01, with Gaussian noise $\eta = 10^{-16}$ or $10^{-12}$ on the samples and bandwidths $\Omega = 25$, 45, 65. The error in the recovered separation is below $10^{-12}$ at $\eta = 10^{-16}$ and below $5 \times 10^{-9}$ at $\eta = 10^{-12}$, for every $d$ down to 0.01 and every $\Omega$: the loss from closeness is a factor of a few hundred at most over two decades of separation, a polynomial loss consistent with the bounds of Donoho and of Li and Liao [121, 210], not the exponential one of Proposition 32.1.2. The closest real pair below height 1420 reported by the authors, at 1329.043518 and 1329.205019 with gap 0.1615 (not checked here), is recovered from the real data to about $2 \times 10^{-8}$ at level 1 and to about $3 \times 10^{-11}$ at level 0.6. Down to a separation of 0.01, close pairs are not the limitation of the method.

### 32.8.3 Levels above one half

The pencil was run on $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$ for $\theta = 1$, $\frac{3}{4}$ and 0.6 on the same window. The zeros enter with a sign, and the sign matters for a zero to the right of $\theta$.

Proposition 32.8.1 (levels below the zeros). *Let $\frac{1}{2} < \theta \le 1$ with $\theta \neq \sigma_\rho$ for every $\rho$, and $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$. Then*

$$u_\theta(v) = \sum_\rho \operatorname{sgn}(\theta - \sigma_\rho)\, P_{|\theta-\sigma_\rho|}(v - \gamma_\rho),$$

$$\widehat{u_\theta}(\omega) = \sum_\rho \operatorname{sgn}(\theta - \sigma_\rho)\, e^{-|\theta-\sigma_\rho||\omega|} e^{-i\omega\gamma_\rho},$$

*and for $\omega\ell^2 > 1$ the tapered transform $F_\theta(\omega) = \int u_\theta(v)w(v)e^{i\omega(v-c)}dv$ is*

$$F_\theta(\omega) = \sum_\rho \operatorname{sgn}(\theta - \sigma_\rho)\, w(\gamma_\rho + i|\theta - \sigma_\rho|)\, e^{-|\theta-\sigma_\rho|\omega} e^{i\omega(\gamma_\rho - c)} + R_\theta(\omega),$$

*with $R_\theta$ bounded as in Theorem 32.2.1 with $a_\rho$ replaced by $|\theta - \sigma_\rho|$. A zero with $\sigma_\rho > \theta$ therefore gives the pole $e^{\Delta(-(\sigma_\rho-\theta)+i(\gamma_\rho-c))}$, inside the unit circle on the ring of a zero at $2\theta - \sigma_\rho$, with an amplitude of sign opposite to the taper profile, while its partner $1 - \sigma_\rho < \theta$ gives a pole at the same ordinate with the profile's sign. The Riemann hypothesis holds if and only if every damping rate $|\theta - \sigma_\rho|$ equals $\theta - \frac{1}{2}$.*

*Proof.* With the zeros paired, $\xi'/\xi(s) = \sum_\rho (s - \rho)^{-1}$, and $\operatorname{Re}(\theta + iv - \rho)^{-1} = (\theta - \sigma_\rho)/((\theta - \sigma_\rho)^2 + (v - \gamma_\rho)^2) = \pi \operatorname{sgn}(\theta - \sigma_\rho) P_{|\theta-\sigma_\rho|}(v - \gamma_\rho)$; the series converges absolutely, its terms being $O(\gamma_\rho^{-2})$. Transform termwise with $\widehat{P_b}(\omega) = e^{-b|\omega|}$. For $F_\theta$ apply the proof of Theorem 32.2.1 to each $P_{|\theta-\sigma_\rho|}$, whose width lies in $(0, 1)$, below $\omega\ell^2$. A pole of modulus $e^{-\Delta b}$ is read as the width $b$, that is as the real part $\theta - b$, which for $b = \sigma_\rho - \theta$ is $2\theta - \sigma_\rho$. For the last statement: if $|\theta - \sigma_\rho| = \theta - \frac{1}{2}$ for every $\rho$, each $\sigma_\rho$ is $\frac{1}{2}$ or $2\theta - \frac{1}{2}$; but if $\sigma_\rho = 2\theta - \frac{1}{2}$ then $1 - \sigma_\rho = \frac{3}{2} - 2\theta$ is also the real part of a zero, with rate $3(\theta - \frac{1}{2}) \neq \theta - \frac{1}{2}$. □

For $\sigma_\rho < \theta$ the poles are $e^{\Delta(-(\theta-\sigma_\rho)+i(\gamma_\rho-c))}$ (equation (32.1.1) with $a_\rho = \theta - \sigma_\rho$ and Proposition 32.8.1). Three findings, from the authors' run.

| $\theta$ | position error, max | median | error in $\sigma_\rho$, max | median |
|---|---|---|---|---|
| 1 | $4.5 \times 10^{-10}$ | $1.8 \times 10^{-10}$ | $7.9 \times 10^{-10}$ | $2.3 \times 10^{-10}$ |
| 3/4 | $3.3 \times 10^{-11}$ | $1.5 \times 10^{-11}$ | $5.3 \times 10^{-11}$ | $1.4 \times 10^{-11}$ |
| 0.6 | $2.4 \times 10^{-11}$ | $8.0 \times 10^{-12}$ | $2.3 \times 10^{-11}$ | $7.2 \times 10^{-12}$ |

First, the real parts agree across levels: $\sigma_\rho = \theta - a_\rho$ is 0.50000000 for every one of the eleven central zeros at all three levels, to eight digits, a consistency check that

the computation of $u_\theta$ passes. Second, lower levels are better conditioned, by an order of magnitude from 1 to $\frac{3}{4}$ and a little more to 0.6: the damping $e^{-(\theta-1/2)\omega}$ is milder, so the signal at the top of the band is $e^{-4.5}$ at $\theta = 0.6$ against $e^{-22.5}$ at $\theta = 1$, and the pencil's noise amplification falls accordingly. Third, and this is the point of the exercise, nothing in the method needs the level to be above the zeros: $u_\theta$ is defined for every $\theta$ that is not the real part of a zero, and by Proposition 32.8.1 the Lorentzians of zeros to the right of $\theta$ have negative width and appear as poles inside the unit circle, of modulus $e^{-\Delta(\sigma_\rho-\theta)}$, with amplitudes of the opposite sign to the taper profile, so that the pencil reads them at $2\theta - \sigma_\rho$; the sign of the amplitude is the test, a test that the amplitude filter used in these runs, which compares moduli, does not perform. The pencil at level $\theta$ could therefore be a test for zeros in $\operatorname{Re} s > \theta$ on a window through the sign of the amplitude (not implemented in the runs reported here, open), with a certificate available by the winding number of Section 32.8.1. The recovered poles at the three levels lie on three concentric rings, one per level, all at $\sigma_\rho = \frac{1}{2}$: the argument of a pole is the ordinate and its modulus is $e^{-\Delta(\theta-\sigma_\rho)}$. The selection of components in these runs compares the amplitude with the taper profile at the width $\theta - \frac{1}{2}$ of a zero on the line (two independent readings of the authors' scripts agree), so that these runs presume the real part $\frac{1}{2}$ in the selection step.

### 32.8.4 The level as a cost parameter

Which level should one use? In a separate run, the pencil was run at $\theta = 1$, $\frac{3}{4}$, 0.6, 0.55, 0.52 and 0.51 on the window about 200, with the grid spacing reduced as the level drops so that the leading aliasing error $e^{-(\theta-1/2)(2\pi/h-\Omega)}$ of the transform stays below $3 \times 10^{-12}$: $h = 0.02$ down to 0.6, then 0.01, 0.004 and 0.002.

| $\theta$ | $h$ | data points | position error, median | error in $\sigma_\rho$, median |
|---|---|---|---|---|
| 1 | 0.02 | 5001 | $1.8 \times 10^{-10}$ | $2.3 \times 10^{-10}$ |
| 3/4 | 0.02 | 5001 | $1.5 \times 10^{-11}$ | $1.3 \times 10^{-11}$ |
| 0.6 | 0.02 | 5001 | $8.0 \times 10^{-12}$ | $7.2 \times 10^{-12}$ |
| 0.55 | 0.01 | 10001 | $3.7 \times 10^{-12}$ | $4.2 \times 10^{-12}$ |
| 0.52 | 0.004 | 25001 | $2.8 \times 10^{-12}$ | $3.9 \times 10^{-12}$ |
| 0.51 | 0.002 | 50001 | $9.9 \times 10^{-13}$ | $1.4 \times 10^{-12}$ |

The error falls by a factor twenty from level 1 to level 0.6 and then only slowly, by another factor eight down to 0.51, while the number of data points, and with it the cost of the data, grows like $1/(\theta - \frac{1}{2})$. The gain presumably saturates because the pencil's error is set by the lower frequencies, where the signal is large at every level, and not by the top of the band where the damping $e^{-(\theta-1/2)\Omega}$ differs most (heuristic). So the answer to the problem is that the level is a weak parameter: anything in $[0.55, 0.75]$ is within a factor of twenty of the best, and $\theta = 0.6$ with the grid 0.02 is a good default, costing the same data as level 1 and giving twenty

times the accuracy. There is no level below which the data become as expensive as finding zeros directly, since Arb evaluates $\zeta$ at a point of the strip at about the cost of a point on the line at the same height; what grows is the number of points, and only linearly.

### 32.8.5 Many windows

A range of heights is covered by combining windows, and the simplest rule works: tapers of width $\ell = 10$ centred every 10 units, each zero taken from the window whose centre is nearest to it, with half a unit of overlap. On $[100, 400]$ at level 0.6 this is 31 windows, and all 173 zeros in the range are recovered (the count agrees with $N(400) - N(100) = 173$), with median position error $2.4 \times 10^{-12}$, maximum $2.6 \times 10^{-11}$, and similar figures for the real parts; the pencils take four seconds, the data on $[40, 460]$ at spacing 0.02 seven. Seven spurious components survive the amplitude filter, all at the edges of windows and all with amplitude at the lower limit of the filter; a stricter filter, or the agreement of two overlapping windows, should remove them. The cost to height $T$ is about $T/h$ evaluations of $\xi'/\xi$ off the line, fifty per unit height at $h = 0.02$, each about as expensive as an evaluation of $\zeta$ at that height, plus $T/10$ pencils whose order grows like $\log T$ with the density of zeros; a sign change scan needs a few evaluations of $Z$ per zero, on the line. Both counts grow like $T$ up to logarithms and each evaluation costs about the same, so the constants decide, and they favour the direct methods by a wide margin; the pencil's advantage is the real parts, which the sign changes never see.

### 32.8.6 A window near the limit of the verified range

We ran the pencil at a large height, near the limit of the rigorous verification: Platt and Trudgian [419] verify the hypothesis for $0 < \gamma \le 3\,000\,175\,332\,800$, and we took the window centred at $c = 3 \times 10^{12} + 100$, inside that range, at level $\theta = 0.6$ with taper $\ell = 2$, chosen because the zeros there are 4.3 per unit height and a window of $\pm 6$ holds about fifty of them. No prime sum is used: the data are values of $\xi'/\xi$ at $\operatorname{Re} s = 0.6$. They are $u_{0.6}$ at 1001 points at spacing 0.02, each from the first two Taylor coefficients of $\xi$ at $0.6 + iv$ in Arb at 160 bits, 3.8 seconds a point on our machine, since the Riemann and Siegel sum has $7 \times 10^5$ terms there and the Gamma factor costs 46 bits of the working precision at this height; the 1001 points took 32 minutes on two cores. The pencil used $\omega \in [5, 85]$ at spacing 0.2, a Hankel matrix of order 200, and model order 90 by the singular value gap. This is the authors' computation (scripts `pencil_3e12.py` and `pencil_3e12_data.py` of the source, which this book did not run), computed and not rerun.

It returned 52 components with amplitudes on the taper's profile within 6 of the centre, at mean spacing 0.2325 between the extreme ordinates, the density $\log(c/2\pi)/2\pi = 4.28$ per unit predicting 51 (the smooth count is 51.36). Every

one of them has

$$\sigma_\rho = 0.5000000 \pm 1.4 \times 10^{-8},$$

the real part read from the modulus of the pole, which the fit does not constrain. The components are selected by comparing the moduli of their amplitudes with the taper profile computed at the width $\theta - \frac{1}{2}$ of a zero on the line, so a zero far from the line would have been discarded by the selection; that the selection discarded no zero on the line is shown by the count of sign changes below. The ordinates are then certified on the line by sign changes of Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, computed in balls with $\vartheta$ from $\log\Gamma$ at 160 bits, on the intervals $[\gamma - 10^{-7}, \gamma + 10^{-7}]$: all 52 of the 52 intervals contain a sign change, which places a zero on the line within $10^{-7}$ of each pencil ordinate (the authors' certificate, not rerun). An independent scan of Hardy's function in ball arithmetic at 160 bits (`code/ex_zeros3e12.py`, `data/ex_zeros3e12.json`) finds exactly 52 certified sign changes in $[c - 6, c + 6]$, at ordinates from $c - 5.91$ to $c + 5.95$, with brackets of width below $10^{-10}$; the comparison with the pencil's ordinates, one by one, has not been made (the pencil's list is not available), and a sign scan cannot see two zeros inside one step, so the count 52 is not a proof of completeness. The winding numbers, which closed the certificate for the eleven zeros near height 200, did not complete here. Arb's rigorous integrator, fed $\xi'/\xi$ on a circle of radius 0.03 about the pencil's zero nearest the centre, ran for over an hour at 3.8 seconds an evaluation without converging, and the alternative of enclosing $\xi$ on arcs of the circle by ball evaluation failed because Arb's enclosure of $\zeta$ at a ball argument of radius $10^{-4}$ at this height already contains zero: the derivative bounds it uses are not tuned to the Riemann and Siegel range. So what is certified at $3 \times 10^{12}$ is the existence of a zero on the line within $10^{-7}$ of each of the 52 ordinates the pencil read, and what is not certified is that these are all the zeros in the window, which would follow from a winding number or from Turing's method [506] with the explicit bounds for $S(T)$ [501], and which the pencil's amplitude profile supports without proving. A rigorous argument principle at large height needs certified nonzero values of $\zeta$ on a mesh of the circle together with an explicit bound for $|\zeta'|$ on a small disc, available from the explicit convexity bounds, and is the natural next step.

What this is and is not. It is a demonstration that the pencil runs at a height near the limit of the rigorous verification, recovers the zeros of a window from data on the line $\operatorname{Re} s = 0.6$, and reports their real parts, which the fit does not constrain, although the selection of components presumes the real part $\frac{1}{2}$. It is not a finding about the zeros, and the window lies inside the verified range. Zeros at and far above this height have been computed before: Gourdon's unrigorous computation of the first $10^{13}$ zeros [157] reaches about $2.4 \times 10^{12}$, and Odlyzko's computations near the $10^{20}$th and higher zeros [393, 395] located long runs of zeros on the line by the same sign change test used here (these heights and attributions are the source's and were not checked here). Nor is reading a real part a stronger

test than counting: an off line pair in a window shows up in Turing's method as two missing sign changes against the count from $N(T)$, and the pencil adds nothing to that test. Two of its outputs are also weaker than they look. The real parts $0.5000000 \pm 1.4 \times 10^{-8}$ come from a floating point pencil and are an estimate, not a certificate; what is certified is a sign change of $Z$ near each ordinate, which is the standard verification. And since the count in the window is not certified, an off line pair the pencil missed is not excluded by anything computed here. What the run does establish is narrower: the deconvolution route of this chapter, from data on $\operatorname{Re} s = 0.6$ to positions and real parts, works at height $3 \times 10^{12}$, with real parts within $1.4 \times 10^{-8}$ of $\frac{1}{2}$, at a cost of half an hour per window, and its output agrees with the authors' sign change certificate for every zero it reports.

### 32.8.7 Above the verified range

The verified range of Platt and Trudgian ends at $3\,000\,175\,332\,800$, and the window centred at $c = 10^{13} + 100$ lies above it, by a factor 3.33 in height. We repeated the computation there with scripts written for this book from the mathematics (the scripts of the source were not used): `code/es_pencil.py` generates the data and runs the pencil, and `code/es_zeros1e13.py` is the independent scan of Hardy's function; the results are in `data/es_pencil1e13.json`, `data/es_zeros1e13.json` and, for the data, `data/es_u06_1e13.json`. The setting is that of the previous subsection: level 0.6, taper $\ell = 2$, 1001 values $u_{0.6}(v_k)$ at $v_k = c - 10 + 0.02k$, each a ball at 160 bits from the first two Taylor coefficients of $\zeta$ at $0.6 + iv_k$ (the largest radius is $1.6 \times 10^{-27}$, and re-evaluating the first point of every task at 200 bits changes it by at most $1.7 \times 10^{-47}$), at 2.6 seconds a point on one core, 2634 core seconds in all; the pencil uses $\omega \in [5, 85]$ at spacing 0.2 (401 samples) and a Hankel matrix of order 200 in double precision, in 0.7 seconds. The model order is 132, the number of singular values above ten times the plateau of the spectrum: the spectrum decays gradually, a gap rule is not usable, and a smaller order loses zeros.

The pencil returns 53 components with amplitude on the taper's profile within 6 of the centre, at ordinates from $c-5.7$ to $c+5.8$, the smooth count of the window being 53.66; the ratio of the amplitude to the profile is 1 to within $1.6 \times 10^{-7}$ for all 53. The real part of each pole is read from its modulus, and the filter that selects the components compares the amplitude with the taper profile computed at the pole's own width, so that no real part is assumed (this is the difference from the runs of the source, whose selection uses the width $\theta - \frac{1}{2}$). The largest $|\sigma_\rho - \frac{1}{2}|$ over the 53 poles is $4.7 \times 10^{-9}$ (median $7.1 \times 10^{-11}$), an estimate from a double precision pencil and not a certificate.

The scan finds 53 certified sign changes of Hardy's function in $[c - 6, c + 6]$: each step of 0.02 has opposite signs of $Z$ at its ends in ball arithmetic at 160 bits, and the bisection brackets of width $10^{-10}$ were certified at both ends again at 192 bits. Every pencil ordinate lies within $10^{-7}$ of the midpoint of a certified bracket,

and every certified bracket has a pencil ordinate: all 53 are matched, no pencil component and no sign change is left over, and the largest ordinate difference is $3.9\times10^{-9}$ (median $5.7\times10^{-11}$). The result is stable: with the model order changed by $\pm5$, 53 components remain, the ordinates move by at most $3.1\times10^{-9}$ and the real parts by at most $2.1\times10^{-9}$. Against the scan, the largest ordinate difference over the model order is $1.1\times10^{-3}$ at order 70, $1.9\times10^{-4}$ at 80, $4.8\times10^{-6}$ at 90, $5.4\times10^{-9}$ at 95 and $4.9\times10^{-9}$ at 100, and below order 65 the pencil finds only between 8 and 49 of the 53 zeros. The same pencil was validated on known zeros, at $c=200$ (seven zeros) and $c=1000$ (ten zeros): all are matched, the largest ordinate difference is $7.1\times10^{-15}$ and $1.4\times10^{-14}$, and the largest $|\sigma-\frac12|$ is $1.7\times10^{-14}$ and $2.3\times10^{-14}$.

The status is this. Computed: the 53 components, their ordinates and real parts. Certified: 53 zeros of $\zeta$ lie on the critical line in $[c-6,c+6]$, each located to $10^{-10}$ (this is a certified computation above the height verified in the literature, but it certifies only these 53 zeros on the line). Not certified: the count (a sign scan cannot see two zeros inside one step of 0.02, the smallest gap between consecutive zeros found is 0.0326, so a pair closer than the step is not excluded, and no Turing method or winding number was run; the pencil independently returns the same 53 and no further pole on the amplitude profile, which supports the count without proving it), and the real parts (the $4.7\times10^{-9}$ is an estimate; it says that no component of the window is off the line by more than the pencil's error, and it is not a proof of the Riemann hypothesis in the window: a zero outside the window, or hidden in a component of the profile, is not excluded). Nothing is certified off the line. The data enter the pencil as midpoints in double precision, an error of about $10^{-16}$, and the accuracy falls at the edge of the taper: the further 14 poles kept at 6 to 7.5 from the centre, outside the window and not compared with the scan, have $|\sigma-\frac12|$ up to $2.6\times10^{-8}$. No priority is claimed: unrigorous computations of zeros at and far above this height exist [157, 393, 395], and the sign change scan is the standard method; what this subsection records is that the deconvolution route of this chapter, from data on $\operatorname{Re}s=0.6$, reproduces the certified zeros of a window of this height with real parts read within $5\times10^{-9}$ of $\frac12$.

## 32.9 Open problems

Problem 32.1. Rigorous pencil. The certification of Section 32.8.1 goes through the argument principle after the fact, and a verified pencil in the sense of Rump would need, besides verified linear algebra on ball data, a perturbation bound for the pencil under the model error of Theorem 32.2.1, the far zeros and the remainder of Theorem 32.2.1, which make the Hankel matrix of exact rank only in the limit. State and prove that bound, and compare the width of the resulting enclosures with the winding number route. Status: open.

Problem 32.2. Spurious components. At the edges of overlapping windows a few components pass the amplitude filter without being zeros. Characterise them: they should be the pencil's representation of the taper's tail of the next zeros outside the window, with poles close to the genuine ones but amplitudes off the profile. A rule based on the agreement of two windows, or on the phase of the amplitude, which Theorem 32.2.1 predicts as $e^{i(\omega_0 - a/\ell^2)(\gamma - c)}$, should remove all of them; under explicit hypotheses on the separation and the amplitudes, prove that it does. Status: open.

Problem 32.3. Higher and wider. The runs near and above the verified range covered one window each, of fifty two and fifty three zeros, with data costing half an hour on two cores at $3 \times 10^{12}$ and 2.6 seconds a point at $10^{13}$. The cost per point is the Riemann and Siegel sum, $t^{1/2}$ terms, and Hiary's method [171] would bring it to $t^{1/3+o(1)}$; the number of points per zero is fixed by the taper and the spacing. Make the argument principle rigorous at large height, with an explicit bound for $|\zeta'|$ on a small disc from the explicit convexity bounds, so that the count in the window is certified and not only the existence on the line. Estimate the height at which a window of the pencil, with its full certificate, is feasible on a laptop in a day, and run it; and determine whether the many window scheme at a fixed height, say $10^3$ zeros above $3\,000\,175\,332\,800$, is competitive with Turing's method for that range when the real parts are wanted as output and not only tested. Status: open.

Problem 32.4. The adaptive prior in full. Place the sine kernel process on marked distinct ordinates, a pair counting as one point marked by its displacement, and implement the posterior on the number, the positions and the marks of the zeros in a window, with the spike and slab on the marks, by reversible jump or by a determinantal birth and death sampler started at the pencil; measure on the many window run whether it removes the spurious components. State the count with its multiplicity (a double zero counts twice), and report the posterior probability of each count, conditional on the model, separately from the argument principle certificate, which it does not replace. Status: open.

Problem 32.5. The constant in the rate. Theorem 32.5.2 gives $\delta^* = (z_\alpha \tau / \|t_\perp\|)^{1/2}$, with $\tau$ the standard deviation per real coordinate, and $\|t_\perp\|$ depends on the taper, the bandwidth and the level. Find the design that maximises $\|t_\perp\|/\tau$, the information and not $\|t_\perp\|$ alone, for a given cost, and determine how it scales with the height, through the gaps to the neighbouring zeros, which enter $t_\perp$ through the projection, and not through the mean density alone. Status: open.

# Chapter 33

# Cauchy deconvolution

Every computation in Chapters 30 to 32 inverts the same operator: convolution with the Cauchy density $P_a(x) = (a/\pi)/(a^2 + x^2)$. The Thorin measure at level $\theta'$ is the solution, when a positive one exists, of $P_{1-\theta'} * \mu = u_1$; the explicit formula is that equation at $\theta' = \frac{1}{2}$, in measures under RH (Theorem 19.2.2(b)) and in distributions unconditionally (Theorem 20.1.1); the matrix pencil solves it for atoms. This chapter collects what is known about the operator in its own right, proves the facts the book uses, and places the book's results in the literature.

The facts fit in a few sentences. Convolution with $P_a$ is harmonic extension to height $a$ in the upper half plane, so Cauchy deconvolution is the Cauchy problem for Laplace's equation, Hadamard's original example of an ill posed problem [167]. It is unique when it is solvable. The forward map preserves positivity and the inverse does not: positive data can come from a signed source, and whether a positive source exists is a statement about positive harmonic functions, which Herglotz and Riesz characterised. Positivity of the deconvolution is monotone in the width, so every positive datum has a critical width beyond which no positive source exists; for the Thorin family that width is 1 minus the supremum of the real parts of the zeros. Stability is Hölder between levels, with the exponent given by the three lines theorem, and logarithmic at the boundary, and both rates are attained. In statistics Cauchy noise is supersmooth of order one, and the minimax rates of density deconvolution are logarithmic; Bayesian posteriors contract at logarithmic rates in the cases treated (Section 33.4.2). All of this changes when the source is a sum of atoms: the problem becomes Prony's, and the error is Lipschitz in the data error, with a constant that grows polynomially as atoms approach, except near merging atoms, where it is the square root.

The chapter develops a survey chapter on Cauchy deconvolution by Nick Polson (the last of the source chapters, of which Chapters 29 to 32 are the earlier ones), together with the checks made for this book. The author's figure is not reproduced: its two panels are described in words and in Table 33.1. What is new in the book is the generality of the statements (a datum that is any nonnegative function, a source that is any measure of the class $\int (1 + x^2)^{-1} d|\mu| < \infty$), the

example of Section 33.2, the three lines bound with its sharpness, the bound for finite sources (Proposition 33.3.2), the positive two level bound (Corollary 33.3.3), the logarithmic bound and the survey of the statistical, algorithmic and physical literature. The specialisation to the Thorin family, the uniqueness argument and the unboundedness of the deconvolution are in Chapters 19 and 20 and are cited.

The notation is the book's, with the following warnings. A level $\theta$ is the real part of a point $s = \theta + iv$, and $u_\theta(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\theta + iv)$ is the density of Chapters 30 to 32; in the notation of Chapter 19, which extends equation (1.2.7), it is $\rho_{\theta-1/2}(v)$, the ordinate being $v$ here. The measure $\mathrm{Cauchy}(\gamma, a)$ of Chapter 29 is $P_a(\cdot - \gamma)\,dx$. The Thorin measure of Chapters 30 to 32 is $\mathfrak{U}_\theta = \sum_\rho \mathrm{Cauchy}(\gamma_\rho, \theta - \sigma_\rho)$ with density $u_\theta$, and $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ with $\widetilde{U}_\theta$ the measure of Theorem 29.14.4, which has the factor $\frac{1}{2}$. The harmonic extension of a measure is written $h$, and not $U$ as in the source. The largest real part of a zero is $\sigma_{\max} = \sup \operatorname{Re} \rho$ (the source's $\Theta$). Widths are $a, c, d, w \geq 0$, except that $b$ may be negative with $b > -w$ (Theorem 33.3.1, Proposition 33.3.2), the example of Section 33.2 has its own letters $q$, $r$ and $m$, and $\eta$ is the linear coefficient of Theorem 33.2.3. Throughout, a measure $\mu$ is called admissible when $\int (1 + x^2)^{-1} d|\mu|(x) < \infty$, the zero counting measure $\sum_\gamma \delta_\gamma$ being the example that matters.

Status words follow the rest of the book: proved, computed, conditional, heuristic or open. Statements taken from the literature are marked cited and carry the reference; the notes of the last section name the entry that the citation verifier has not confirmed and the statements not checked against the papers. The results are these.

1. Uniqueness of Cauchy deconvolution, the description of the range of $P_a*$ on $L^2$ and the failure of the naïve description of it (Propositions 33.1.1 and 33.1.2, proved).

2. A positive datum with a signed source, with the closed form of its critical width (Example 33.2.1, proved; the numbers are computed), the critical width of a general positive datum, attained and possibly infinite (Proposition 33.2.2, proved), $c^*(u_1) = 1 - \sigma_{\max}$ (Theorem 19.2.2(a), cited), and the Herglotz characterisation of the data that have a positive source (Theorem 33.2.3, proved), with Lagarias's criterion as its case $c = \frac{1}{2}$ (cited).

3. Hölder stability between levels, sharp (Theorem 33.3.1, proved); the same for finite signed sources with the sharp exponent $(b + w)/(a + w)$ (Proposition 33.3.2, proved); a two level bound for positive sources on the line (Corollary 33.3.3, proved); the logarithmic bound at the boundary, sharp in order and in the constant (Theorem 33.3.4, proved). The explanation of why $\widetilde{U}_{3/4}$ is recovered better than $\widetilde{U}_{1/2}$ is heuristic, and a test of the exponent on Lorentzians is computed (Table 33.1).

4. A survey of density deconvolution, Bayesian inverse problems and positiv-

ity preserving algorithms (cited), and a polynomial upper rate for sources analytic in a strip (Proposition 33.4.1, proved).

5. Atomic sources and merging atoms, and the analytic continuation problem of many body physics (cited, with the proofs of Chapters 31 and 32), and three open problems.

## 33.1 The operator

### 33.1.1 Semigroup and harmonic extension

With $\widehat{f}(\omega) = \int f(x)e^{-i\omega x}dx$, the transform of $P_a$ is $e^{-a|\omega|}$, so

$$P_a * P_b = P_{a+b}, \qquad P_a = e^{-a|D|}, \quad |D| = (-\partial_x^2)^{1/2}:$$

the Cauchy densities form the Poisson semigroup, the symmetric stable semigroup of index one (Theorem 1.2.2 and Proposition 1.2.3 (Section 1.2) give the same facts for the clock of $\xi$; Theorem 29.14.4 gives it for the Thorin measures). For an admissible measure $\mu$ the function

$$h(x,y) = (P_y * \mu)(x), \qquad y > 0,$$

is harmonic in the upper half plane, being the real part of $\pi^{-1}i\int\big((z-t)^{-1} + t(1+t^2)^{-1}\big)d\mu(t)$ at $z = x + iy$. The integral converges absolutely because $(z - t)^{-1} + t(1 + t^2)^{-1} = (1 + tz)/((z - t)(1 + t^2)) = O_z(t^{-2})$, and the added term is real, so after multiplication by $i$ it changes only the imaginary part. (Without the added term the integral diverges for Lebesgue measure and for the zero counting measure, since $\sum_\gamma 1/|\gamma| = \infty$.) The boundary values of $h$ are $\mu$ in the vague sense. So $P_a * \mu$ is the restriction of the harmonic extension of $\mu$ to the line $y = a$, and the equation

$$P_a * \mu = u \tag{33.1.1}$$

asks for the boundary values of a harmonic function known on a parallel line. A second description, used in Chapter 31 (Proposition 31.6.1) and in Section 9.2.3, is probabilistic: $P_a$ is the law of planar Brownian motion started at height $a$ when it first hits the real axis, equivalently a Gaussian with a random variance distributed as the first passage time of one dimensional Brownian motion to level $a$.

### 33.1.2 Uniqueness and the range

Proposition 33.1.1 (uniqueness). *If $\mu_1, \mu_2$ are admissible measures and $P_a*\mu_1 = P_a * \mu_2$, then $\mu_1 = \mu_2$.*

*Proof.* Let $\nu = \mu_1 - \mu_2$ and $g(x,y) = (P_y * \nu)(x)$, harmonic on $y > 0$. By the semigroup property, $g(\cdot, y) = P_{y-a} * g(\cdot, a) = 0$ for $y > a$. The semigroup

property applies to the signed measure $\nu$: $|g(\cdot,a)| \le P_a * |\nu|$ and $P_{y-a} * (P_a * |\nu|) = P_y * |\nu| < \infty$ by Tonelli's theorem, so Fubini's theorem gives $P_{y-a} * g(\cdot,a) = g(\cdot,y)$ with absolute convergence. So $g$ vanishes on an open set and hence on the whole half plane. An admissible measure is the vague boundary value of its Poisson integral, so $\nu = 0$. □

This is the uniqueness part of Theorem 19.2.2, with the same argument, stated for a general admissible measure.

The inverse is formally multiplication by $e^{a|\omega|}$, which is not the Fourier transform of any tempered distribution, so the inverse is not a convolution with a tempered distribution; it is defined only on the range of $P_a*$. If $u = P_a * \mu$, the Poisson extension of $u$ continues harmonically to the depth $a$ below the line, as $P_{y+a} * \mu$. The converse fails without conditions on the boundary values at depth $a$: $u(x) = 2 + \cos(x^2)$ is positive and bounded, and its Poisson extension continues to an entire harmonic function, but $u \ne P_a * \mu$ for every $a > 0$ and every admissible $\mu$. For positive sources Theorem 33.2.3 characterises the range exactly. On $L^2(\mathbb{R})$ the range is $\{u : \int e^{2a|\omega|} |\widehat{u}(\omega)|^2 d\omega < \infty\}$, and the inverse is unbounded in the $L^2$ norm and in every Sobolev norm. This is the source of everything that follows. Chapter 19 makes the same remark for the Thorin density (the paragraph after Theorem 19.2.2). The statements are proved as follows.

PROPOSITION 33.1.2 (the range). *(a) For $f \in L^2(\mathbb{R})$, $\int e^{2a|\omega|} |\widehat{P_a * f}|^2 d\omega = \int |\widehat{f}|^2 d\omega$, and every $u$ with $\int e^{2a|\omega|} |\widehat{u}|^2 < \infty$ is $P_a * f$ for an $f \in L^2$. (b) For all real $s, t$, $\sup \|f\|_{H^t} / \|P_a * f\|_{H^s} = \infty$. (c) The function $u = 2 + \cos(x^2)$ is not $P_a * \mu$ for any $a > 0$ and any admissible $\mu$, although its Poisson extension continues to an entire harmonic function.*

*Proof.* (a) $\widehat{P_a * f} = e^{-a|\omega|} \widehat{f}$, and conversely $\widehat{f} = e^{a|\omega|} \widehat{u}$ defines $f \in L^2$ with $P_a * f = u$. (b) Let $f_N$ be real with $\widehat{f_N}$ the indicator of $[N, N+1] \cup [-N-1, -N]$. Then $\|P_a * f_N\|_{H^s} \le e^{-aN} \|f_N\|_{H^s}$, and $\|f_N\|_{H^t} / \|f_N\|_{H^s}$ lies between two powers of $N$, so $\|f_N\|_{H^t} / \|P_a * f_N\|_{H^s} \ge c N^{-|t-s|} e^{aN} \to \infty$. (c) The Poisson extension $h = P_y * u$ is continuous on $y \ge 0$, and $g(x,y) = 2 + \operatorname{Re} \cos((x+iy)^2)$ is an entire harmonic function with $g(x,0) = u(x)$. So $h - g$ is harmonic on $y > 0$, continuous on $y \ge 0$ and zero on $y = 0$; by the Schwarz reflection principle it extends to an entire harmonic function, and so does $h = g + (h - g)$. Suppose $u = P_a * \mu$ with $\mu$ admissible. Then $u''' = (\partial_x^3 P_a) * \mu$ and $|\partial_x^3 P_a(x-t)| \le C_a (1 + (x-t)^2)^{-1} \le 2C_a (1+x^2)(1+t^2)^{-1}$, because $1 + t^2 \le 2(1+x^2)(1+(x-t)^2)$. So $|u'''(x)| \le C(1+x^2)$. But $u'''(x) = 8x^3 \sin(x^2) - 12x \cos(x^2)$ is not $O(1+x^2)$. □

## 33.2 Positivity

### 33.2.1 The forward map preserves it, the inverse does not

$P_a$ is a probability density, so $\mu \geq 0$ implies $P_a * \mu \geq 0$, and the semigroup property carries positivity up the levels: if $P_c * \mu \geq 0$ for some $c$ then $P_{c'} * \mu \geq 0$ for every $c' \geq c$. The converse fails, and the failure is quantitative.

Example 33.2.1 (positive data, signed source). Let $d > 0$, $0 < q < 1$ and $\mu = \delta_0 - q\delta_d$. Then $P_a * \mu = P_a - qP_a(\cdot - d) \geq 0$ on $\mathbb{R}$ if and only if

$$q \;\leq\; \frac{r-d}{r+d}, \qquad r = \sqrt{d^2 + 4a^2}.$$

*Proof.* Positivity is $q \leq \inf_x f(x)$ with $f(x) = (a^2 + (x-d)^2)/(a^2 + x^2)$. Setting $f' = 0$ gives $x^2 - dx - a^2 = 0$, and at the root $x^* = (d+r)/2$, using $x^{*2} = dx^* + a^2$, the numerator is $r(r-d)/2$ and the denominator $r(r+d)/2$. The other root is a maximum, and $f \to 1$ at infinity. □

By Proposition 33.1.1 the signed measure is the only source of these data at the width $a$, so a positive datum can have no positive source at the width $a$ (at width 0 the datum is its own source). The example also shows the structure that the Thorin family has. Deconvolving $u = P_a * \mu$ only part of the way, by $P_c$ with $c < a$, gives $P_{a-c} * \mu = P_{a-c} - qP_{a-c}(\cdot - d)$, which is positive if and only if $q \leq (r_c - d)/(r_c + d)$ with $r_c = \sqrt{d^2 + 4(a-c)^2}$; that is, if and only if

$$c \;\leq\; c^* = a - \tfrac{1}{2}\sqrt{m^2 - d^2}, \qquad m = \frac{1+q}{1-q}\, d,$$

and $c^* \geq 0$ if and only if the datum is positive. Positivity holds up to a critical width and is lost beyond it. For $q = 9 - 4\sqrt{5} \approx 0.0557$, $a = \frac{1}{2}$ and $d = 1$ one has $m = \sqrt{5}/2$ and $c^* = \frac{1}{4}$ exactly: the datum is positive (the bound of the example is $3 - 2\sqrt{2} \approx 0.1716$), its partial deconvolutions are positive for $c < \frac{1}{4}$, touch zero at $c = \frac{1}{4}$ and are negative beyond. (Computed: with $q = 0.0557$ the critical width is 0.25007, so the value $\frac{1}{4}$ needs $q = 9 - 4\sqrt{5}$.)

Proposition 33.2.2 (the critical width). *For $u \geq 0$ with $\int u(x)(1+x^2)^{-1}dx < \infty$ let $c^*(u) = \sup\{c \geq 0 :\ u = P_c * \nu$ for some measure $\nu \geq 0\} \in [0, \infty]$, where $P_0 * \nu = \nu$, so that $c = 0$ always qualifies. Then a positive source at width $c$ exists for every finite $c \leq c^*(u)$ and for no $c > c^*(u)$; in particular the supremum is attained when it is finite.*

*Proof.* If $u = P_c * \nu$ with $\nu \geq 0$ and $0 \leq c' < c$, then $u = P_{c'} * (P_{c-c'} * \nu)$ with $P_{c-c'} * \nu \geq 0$, by Tonelli and the semigroup property, so the admissible widths form an interval containing 0. Let $0 < c^*(u) < \infty$ and let $0 < c_n \uparrow c^*(u)$ be admissible. Then $u = P_{c_1} * \nu_1$ is continuous, and by Theorem 33.2.3 the harmonic extension

$h$ of $u$ extends to a nonnegative harmonic function $V_n$ on $y > -c_n$. Since $V_n$ and $V_m$ agree with $h$ on $y > 0$, they agree on the connected set $y > -\min(c_n, c_m)$, so together they define a nonnegative harmonic function on $y > -c^*(u)$ that extends $h$. Theorem 33.2.3 with $c = c^*(u)$ gives $u = P_{c^*(u)} * \nu$ with $\nu \geq 0$. □

The critical width is not capped at the width of any given deconvolution, so it does not presuppose that a source exists at that width. For the Thorin family it is the following. Let $u_1 = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + iv)$, the density of level one. By Theorem 19.2.2(a), with $\varepsilon_0 = \frac{1}{2}$ and $\varepsilon_1 = \frac{1}{2} - c$, a positive source at width $c \in (0, \frac{1}{2}]$ exists if and only if $\zeta$ has no zeros in $\operatorname{Re} s > 1 - c$, zeros on $\operatorname{Re} s = 1 - c$ contributing atoms; the source is then $u_{1-c}\, dv$ (the Thorin measure $\mathfrak{U}_{1-c}$), which is a function when $c < 1 - \sigma_{\max}$ and has atoms when $c = 1 - \sigma_{\max}$ and some zero has real part $\sigma_{\max}$. For $c > \frac{1}{2}$ there is no source at all, since its Poisson integral would be harmonic on $y > -c$ and equal to $\pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + y + iv)$ off the zeros, which has poles at the zeros on the critical line (compare Theorems 29.14.2, 29.14.4 and 30.3.1). So

$$c^*(u_1) = 1 - \sigma_{\max}, \qquad \sigma_{\max} = \sup\{\operatorname{Re} \rho\},$$

and the Riemann hypothesis is the statement $c^*(u_1) = \frac{1}{2}$: the critical width of the level one density is as large as the functional equation allows. This is Theorem 19.2.2, where the sentence after the proof calls $\sigma_{\max} - \frac{1}{2}$ the depth to which the Thorin density can be deconvolved into a positive measure; Remark 1.2.4 says why positivity of the density on the single line is weaker.

### 33.2.2 When does a positive source exist

Theorem 33.2.3 (existence of a positive source). *Let $u \geq 0$ be continuous with $\int u(x)(1 + x^2)^{-1} dx < \infty$, and let $h(x, y) = (P_y * u)(x)$ be its harmonic extension, $y > 0$. For $c > 0$ the following are equivalent.*

*(i) $u = P_c * \nu$ for a measure $\nu \geq 0$.*

*(ii) $h$ extends to a nonnegative harmonic function on the half plane $y > -c$.*

*Proof.* If (i) holds, $V(x, y) = (P_{y+c} * \nu)(x)$ is harmonic and nonnegative on $y > -c$ and equals $h$ for $y \geq 0$ by the semigroup property. Conversely, a nonnegative harmonic function $V$ on $y > -c$ is, by the Herglotz and Riesz representation for the half plane (see [2]), of the form $V(x, y) = \eta(y + c) + (P_{y+c} * \nu)(x)$ with $\eta \geq 0$, $\nu \geq 0$ and $\int (1 + t^2)^{-1} d\nu < \infty$. In fact $\eta = 0$. At $y = 0$ the representation says $u = \eta c + P_c * \nu$, since $V = h$ on $y \geq 0$ and $u$ is continuous, so for $y > 0$ we have both $V(\cdot, y) = \eta(y + c) + P_{y+c} * \nu$ and $h(\cdot, y) = P_y * u = \eta c + P_{y+c} * \nu$, whence $\eta y = 0$. Thus $u = P_c * \nu$. □

For the level one density the harmonic extension is explicit. Smoothing raises the level, $P_y * u_1 = u_{1+y}$, and the formula continues below the line: $u_\theta$ is harmonic in $(\theta, v)$ wherever $\xi$ has no zeros. So condition (ii) for $u_1$ at width $c$ says that $\xi$ has no zeros on $\operatorname{Re} s > 1 - c$ and $\operatorname{Re}(\xi'/\xi)(s) \geq 0$ there, and at $c = \frac{1}{2}$ it is Lagarias's criterion [320]: the Riemann hypothesis is equivalent to $\operatorname{Re}(\xi'/\xi)(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$ (the conditions $\geq 0$ and $> 0$ agree by the minimum principle). In the language of analytic continuation, (ii) says that $h$ is the real part of an analytic function of positive real part on $\operatorname{Im} z > -c$, a Pick function after a rotation.

## 33.3 Stability

### 33.3.1 Hölder between levels

Theorem 33.3.1 (three lines). *Let $b_0 < b < a$ and let $\mu$ be a signed measure with $P_{b_0} * \mu \in L^2$, where for $b_0 < 0$ this means $\mu = P_{|b_0|} * \lambda$ with $\lambda \in L^2$, and $P_{b_0} * \mu$ stands for $\lambda$. Then*

$$\|P_b * \mu\|_2 \;\leq\; \|P_a * \mu\|_2^{\frac{b-b_0}{a-b_0}} \; \|P_{b_0} * \mu\|_2^{\frac{a-b}{a-b_0}}.$$

*In particular, if two sources agree in their data at width $a$ to within $\varepsilon$ in $L^2$ and both are bounded by $M$ at width $b_0$, they agree at width $b$ to within $\varepsilon^\kappa (2M)^{1-\kappa}$, $\kappa = (b - b_0)/(a - b_0)$. The exponent is attained in rate.*

*Proof.* Let $F$ be the Fourier transform of $P_{b_0} * \mu \in L^2$ (of $\lambda$ when $b_0 < 0$). For $y > b_0$, $P_y * \mu = P_{y-b_0} * (P_{b_0} * \mu)$, so $\|P_y * \mu\|_2^2 = (2\pi)^{-1} \int e^{-2(y-b_0)|\omega|} |F|^2 d\omega$, and with $\kappa = (b - b_0)/(a - b_0)$,

$$e^{-2(b-b_0)|\omega|}|F|^2 = \big(e^{-2(a-b_0)|\omega|}|F|^2\big)^\kappa \big(|F|^2\big)^{1-\kappa},$$

so Hölder's inequality with exponents $1/\kappa$ and $1/(1-\kappa)$ gives the bound. Equivalently, $y \mapsto \log \|P_y * \mu\|_2$ is convex, the three lines theorem for the harmonic extension. For sharpness take $F$ concentrated near a frequency $\Omega$ with $e^{-(a-b_0)\Omega} = \varepsilon/M$: both sides are then equal up to constants. □

The exponent depends on how far below the target level an a priori bound is available. For the Thorin family under the Riemann hypothesis the data are at level 1, which is width $a = \frac{1}{2}$ above the zero counting measure, and the level $\theta'$ measure is at width $b = \theta' - \frac{1}{2}$. The zero measure is atomic, so Theorem 33.3.1 admits no level below it ($b_0 < 0$ would require $\sum_\gamma \delta_\gamma = P_{|b_0|} * \lambda$ with $\lambda \in L^2$, a smooth function), and it has infinite mass. For the zeros in a window, a finite measure, Proposition 33.3.2 below with $w = 0$, $a = \frac{1}{2}$ and $b = \theta' - \frac{1}{2}$ gives the exponent $2\theta' - 1$: about $\frac{1}{2}$ at $\theta' = \frac{3}{4}$, falling to 0 at $\theta' = \frac{1}{2}$. The application to the Thorin family is heuristic, since a window does not commute with $P_y$ and the zeros outside it carry infinite mass. This explains, heuristically, why $\widetilde{U}_{3/4}$ was

recovered so much better than $\widetilde{U}_{1/2}$ in Chapters 30 and 31. Table 33.1 shows the exponent $(b+w)/(a+w)$ of Proposition 33.3.2 on a test source of Lorentzians of width $w$, to within 0.01.

PROPOSITION 33.3.2 (finite sources). *Let $\nu$ be a finite signed measure of total variation at most $B$, let $w \ge 0$ and $h = P_w * \nu$ (with $P_0 * \nu = \nu$, and $P_b * h$ meaning $P_{b+w} * \nu$ when $b < 0$), and let $-w < b < a$. Then*

$$\|P_b * h\|_2 \le \sqrt{2}\, D^{1-\kappa}\, \|P_a * h\|_2^{\kappa}, \qquad \kappa = \frac{b+w}{a+w}, \quad D = \frac{B}{\sqrt{2\pi(b+w)}},$$

*and no larger exponent holds uniformly over this class.*

*Proof.* The transform of $h$ satisfies $|\widehat{h}(\omega)| \le B e^{-w|\omega|}$. Let $\varepsilon = \|P_a * h\|_2$ and $\Omega \ge 0$. For $|\omega| \le \Omega$, $e^{-2b|\omega|} = e^{2(a-b)|\omega|} e^{-2a|\omega|} \le e^{2(a-b)\Omega} e^{-2a|\omega|}$, so by Plancherel

$$\begin{aligned} \|P_b * h\|_2^2 &= \frac{1}{2\pi} \int e^{-2b|\omega|} |\widehat{h}(\omega)|^2 d\omega \\ &\le e^{2(a-b)\Omega} \varepsilon^2 + \frac{B^2}{2\pi} \int_{|\omega|>\Omega} e^{-2(b+w)|\omega|} d\omega \\ &= e^{2(a-b)\Omega} \varepsilon^2 + D^2 e^{-2(b+w)\Omega}. \end{aligned}$$

If $\varepsilon < D$, take $e^{(a+w)\Omega} = D/\varepsilon$: both terms equal $D^{2(1-\kappa)} \varepsilon^{2\kappa}$. If $\varepsilon \ge D$, take $\Omega = 0$: then $\|P_b * h\|_2 \le D \le D^{1-\kappa} \varepsilon^{\kappa}$. For sharpness let $\phi(x) = (2\pi)^{-1} \big(\sin(x/2)/(x/2)\big)^2$, so that $\phi \ge 0$, $\int \phi = 1$ and $\widehat{\phi}(\omega) = (1 - |\omega|)_+$, and let $\nu_\Omega = B\phi(x) \cos(\Omega x)\, dx$ with $\Omega \ge 2$, of total variation at most $B$, and $h_\Omega = P_w * \nu_\Omega$. Then $\widehat{\nu_\Omega} = \frac{B}{2}\big(\widehat{\phi}(\cdot - \Omega) + \widehat{\phi}(\cdot + \Omega)\big)$ has two disjoint bumps, and for every $y > -w$, $\|P_y * h_\Omega\|_2 = B e^{-(y+w)\Omega} c_\Omega(y)$ with $(6\pi)^{-1/2} e^{-(y+w)} \le c_\Omega(y) \le (6\pi)^{-1/2} e^{y+w}$. So $\|P_b * h_\Omega\|_2 / \|P_a * h_\Omega\|_2^{\kappa}$ lies between two positive multiples of $B^{1-\kappa}$, independent of $\Omega$, while $\|P_a * h_\Omega\|_2 \to 0$ as $\Omega \to \infty$; this excludes every exponent above $\kappa$. □

With $w = 0$ this is the bound for a finite signed measure at the data width $a$, and with $b = 0$ and $w > 0$ it gives the exponent $w/(a+w)$ for recovering $h = P_w * \nu$ from the data at width $a$: the improvement comes from the analytic smoothing and the mass bound, not from positivity, and if $w \to 0$ or the mass bound is removed no uniform positive exponent follows. It is stated here for a finite mass; in Theorem 33.3.1 the mass is replaced by an $L^2$ bound at a level below the target.

If the datum and both sources are nonnegative, the mass in a window is controlled by the data alone, and the three lines theorem needs no a priori bound.

COROLLARY 33.3.3 (positive sources). *If $\mu_1, \mu_2 \ge 0$, $u_j = P_a * \mu_j \in L^2$ and $\|u_1 - u_2\|_2 \le \varepsilon$, then for $0 < d < b < a$*

$$\|P_b * (\mu_1 - \mu_2)\|_2 \le \varepsilon^{\kappa} \big((a/d)(\|u_1\|_2 + \|u_2\|_2)\big)^{1-\kappa}, \qquad \kappa = \frac{b-d}{a-d}.$$

Table 33.1: Stability between levels, computed. The source is a sum of 40 Lorentzians of width $w = 0.05$ on a period 400, the data are at width $a = \frac{1}{2}$ with white noise of size $\varepsilon$ at every frequency, and $P_b * \mu$ is recovered by an oracle spectral cutoff, with $\varepsilon$ from $10^{-12}$ to $10^{-3}$. The fitted slope is that of log of the error in $P_b * \mu$ against $\log \varepsilon$; the predicted slope is the exponent $(b+w)/(a+w)$ of Proposition 33.3.2. Nineteen values of $\varepsilon$ (the error does not depend on the noise draw); script `code/cd_slopes.py`, data `data/cd_slopes.json`. All four fitted slopes are within 0.004 of the prediction. Status: computed, illustration.

| level $b$ | $\frac{3}{8}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | 0 |
|---|---|---|---|---|
| predicted $(b+w)/(a+w)$ | 0.7727 | 0.5455 | 0.3182 | 0.0909 |
| fitted slope | 0.769 | 0.545 | 0.318 | 0.091 |

*Proof.* For $0 < d < a$, $P_d/P_a = (d/a)(a^2+x^2)/(d^2+x^2) \le a/d$ pointwise, so $P_d * \mu_j \le (a/d) P_a * \mu_j$ pointwise, and $\|P_d * (\mu_1 - \mu_2)\|_2 \le (a/d)(\|u_1\|_2 + \|u_2\|_2)$. Theorem 33.3.1 applies with $b_0 = d$. □

This is the whole line version, with data in $L^2$ on all of $\mathbb{R}$; for data on a window the question stays open (Problem 33.3).

### 33.3.2 Logarithmic at the boundary

Theorem 33.3.4 (the boundary). *Suppose $\|\mu\|_{H^s} \le M$ for some $s > 0$ and $\|P_a * \mu\|_2 \le \varepsilon$, with $\varepsilon/M$ small enough, depending on $a$ and $s$. Then*

$$\|\mu\|_2 \;\le\; C_s\, M \Big( \frac{2a}{\log(M/\varepsilon)} \Big)^s ,$$

*and no bound of smaller order holds uniformly over the class: the modulus of continuity of the inverse is of this order.*

*Proof.* Split the frequencies at $\Omega$. Below, $|\widehat{\mu}| \le e^{a\Omega} |\widehat{P_a * \mu}|$, contributing at most $e^{a\Omega}\varepsilon$ to the norm; above, the $H^s$ bound gives at most $M\Omega^{-s}$. With $\Omega = (2a)^{-1} \log(M/\varepsilon)$ the first term is $(M\varepsilon)^{1/2}$, of smaller order, and the second is the bound. For the lower bound, a source with transform concentrated near the frequency $\Omega = a^{-1}\log(M/\varepsilon)$ and $H^s$ norm $M$ has $L^2$ norm of order $M\Omega^{-s}$ and data of size at most $\varepsilon$ (of size $o(\varepsilon)$ in fact, $\varepsilon\Omega^{-s}$). □

The cutoff at $2a$ is a convenient one. The sharp modulus $\sup\{\|\mu\|_2 : \|\mu\|_{H^s} \le M,\ \|P_a * \mu\|_2 \le \varepsilon\}$ equals $(1 + o(1))\, M\big(a/\log(M/\varepsilon)\big)^s$ as $\varepsilon/M \to 0$: in the upper bound take $\Omega = a^{-1}\big(L - (s+1)\log L\big)$ with $L = \log(M/\varepsilon)$, which makes the low frequency term $ML^{-s-1}$ and the high frequency term $(1+o(1))M(a/L)^s$; the lower bound is the one given.

So at the boundary the inverse loses all but a logarithm. The two theorems together describe the backward Cauchy flow of the Thorin programme: heuristically, Hölder from level 1 to any level above $\frac{1}{2}$ for the zeros in a window, with the exponent $2\theta' - 1$ of Proposition 33.3.2, and logarithmic at $\frac{1}{2}$ for a source in a Sobolev ball (Theorem 33.3.4), which the zero measure is not.

## 33.4 Statistics

### 33.4.1 Supersmooth noise

In density deconvolution one observes $Y_i = X_i + E_i$, $i \le n$, with $E_i$ of known law and wants the density $f$ of $X_i$. The difficulty is set by the decay of the characteristic function $\phi_E$ of the noise. Cauchy noise has $\phi_E(\omega) = e^{-a|\omega|}$ and is supersmooth of order one in the classification of Fan [132], who showed that for supersmooth noise the minimax rate is logarithmic in $n$ (cited). Carroll and Hall [75] found this for Gaussian noise before Fan. Stefanski and Carroll [273] introduced the deconvoluting kernel estimators, which attain these rates [75, 132]; the estimator has a Fourier cutoff at a frequency of order $\log n$. Heuristically, the logarithm is the statistical form of Theorem 33.3.4 with $\varepsilon = n^{-1/2}$.

The rates improve when the target is smooth in the strong sense. Butucea and Tsybakov [40, 41] constructed sharp asymptotic minimax estimators when the target is itself supersmooth, with rates much faster than the logarithmic ones (cited). The case of Cauchy noise and a target analytic in a strip is simpler and is a bound of the form of Proposition 33.3.2, stated here without attribution.

Proposition 33.4.1 (a strip). *Let $E_i$ be Cauchy noise of width $a$ and let $f$ satisfy $\int e^{2w|\omega|}|\widehat{f}(\omega)|^2 d\omega \le 2\pi M^2$ with $w > 0$. The spectral cutoff estimator $\widehat{f}_n(x) = (2\pi)^{-1}\int_{|\omega|\le\Omega} e^{a|\omega|}\widehat{\phi}_n(\omega)e^{i\omega x}d\omega$, where $\widehat{\phi}_n(\omega) = n^{-1}\sum_{j\le n} e^{-i\omega Y_j}$ is the empirical transform of $Y_1,\dots,Y_n$ in the convention of Section 33.1, with $\Omega = \log n/(2(a+w))$, has root mean integrated squared error $O(n^{-w/(2(a+w))})$. This is an upper bound.*

*Proof.* The transform of the law of $Y$ is $\widehat{f}(\omega)e^{-a|\omega|}$ and $\mathbb{E}|\widehat{\phi}_n(\omega) - \widehat{f}(\omega)e^{-a|\omega|}|^2 \le 1/n$. By Plancherel, the integrated variance is at most $(2\pi n)^{-1}\int_{|\omega|\le\Omega} e^{2a|\omega|}d\omega \le e^{2a\Omega}/(2\pi a n)$, and the integrated squared bias is $(2\pi)^{-1}\int_{|\omega|>\Omega}|\widehat{f}|^2 \le M^2 e^{-2w\Omega}$. With $e^{2(a+w)\Omega} = n$ both are of order $n^{-w/(a+w)}$. □

A source of Lorentzians of width $w$ (the test source of Table 33.1) is analytic in every strip of width less than $w$, and the exponent $(b+w)/(a+w)$ of Proposition 33.3.2 is positive even at $b = 0$. Meister's book [215] collects the theory.

### 33.4.2 Bayesian posteriors

For linear inverse problems with Gaussian priors, Knapik, van der Vaart and van Zanten [201] computed the posterior contraction rates in terms of the decay of the singular values, for polynomially decaying singular values only, the mildly ill posed case (cited). Cauchy deconvolution is severely ill posed: on a periodic domain its singular values are $e^{-a|k|}$ (convolution with $P_a$ on $L^2(\mathbb{R})$ is not compact, being bounded below on the functions with spectrum in $[-1,1]$, so a periodic or windowed setting is needed). The severely ill posed case was treated by the same authors for the recovery of the initial condition of the heat equation [202], the Gaussian analogue of the backward Cauchy flow. For mixtures, where the unknown is a mixing measure seen through a known kernel, Nguyen [223] gave rates in Wasserstein distance for mixing measures with compact support: polynomial for ordinary smooth kernels and logarithmic for supersmooth ones, of order $(\log n)^{-1/\beta}$ for a kernel of supersmooth order $\beta$, $(\log n)^{-1/2}$ for the Gaussian kernel; for the Cauchy kernel, $\beta = 1$, the hypotheses of the paper are not checked here. Gao and van der Vaart [145] treated Dirichlet process mixtures of Laplace kernels (the noise is Laplace, not Cauchy). No posterior theorem for Cauchy deconvolution is cited. By analogy the Bayesian picture is the deterministic one with $\varepsilon$ replaced by the noise level, and this is a heuristic: prior mass, testing and support conditions enter.

### 33.4.3 Positivity in the algorithms

Which methods respect positivity is a practical question with a clean answer. Spectral cutoff, Tikhonov regularisation and Landweber iteration are linear and do not: they return signed estimates even when a positive source exists. The positivity preserving methods are nonlinear: nonnegative least squares (Section 8.6.1), maximum entropy, and the EM algorithm of Richardson and Lucy [263, 212], studied for positive linear inverse problems by Vardi and Lee [356], whose multiplicative update keeps every iterate positive from a positive start (a zero coordinate stays zero) and on a finite grid converges to the Kullback Leibler projection of the data onto the cone of blurred positive measures. Chapters 30 and 31 used EM for this reason, and its accelerated form in the logarithmic domain (Section 31.4) for the same reason. When no positive source exists, which by Proposition 33.2.2 is the case beyond the critical width, these methods still return positive estimates, each its own projection (least squares for nonnegative least squares, Kullback Leibler for EM), and the residual, compared with the data error, is what detects the failure (Proposition 30.4.1): positivity of the method is not evidence of positivity of the source.

## 33.5 Atomic sources

### 33.5.1 From logarithmic to polynomial

The rates above are for sources in smoothness classes. A source that is a finite sum of atoms, $\mu = \sum_k c_k \delta_{t_k}$, is not in any of the smoothness classes, and for it the problem is different in kind. The data are a sum of Lorentzians, $\sum_k c_k P_a(x - t_k)$, whose transform is the sum of exponentials $\sum_k c_k e^{-a|\omega| - i\omega t_k}$, and after a taper, as in Theorem 32.2.1, the samples on an arithmetic progression of frequencies are a sum of damped complex exponentials. Recovering their poles is Prony's problem, solved stably by Hankel matrix methods: the matrix pencil of Hua and Sarkar [183], and the Hankel singular value decomposition used in nuclear magnetic resonance spectroscopy since Barkhuijsen, de Beer and van Ormondt [21], where Lorentzian line shapes and their damped exponential transforms are the daily object. For a fixed number of separated atoms with amplitudes bounded below and no aliasing, the error is Lipschitz in the data error, with a constant that grows polynomially as atoms approach each other, except near merging atoms, where it is the square root; this is super resolution, studied by Donoho [121], Candès and Fernandez Granda [72], and, for positive measures without a separation condition, Schiebinger, Robeva and Recht [267], with sharp constants for close atoms by Li and Liao [210] (cited). Chapter 32 gives the comparison of EM on a grid with the pencil on the zeros of $\zeta$, with its numbers and the noise level of the data; they are not repeated here.

### 33.5.2 Merging atoms

The polynomial rate has one exception that matters for $\zeta$. If two atoms approach each other symmetrically, the data depend on their separation only through its square, the Fisher information for the separation vanishes, and the rate for the separation is the square root of the data error; this is the case of two atoms merging symmetrically (with more merging atoms the exponent can be smaller). It is the singular rate of finite mixtures (Chen [80]; Nguyen [223]) in its simplest case, and Chapter 32 met it in the form forced by the functional equation: a zero off the line comes with its partner at the same ordinate, their widths at level $\theta$ are $\theta - \frac{1}{2} \mp \delta$, the two poles merge as $\delta \to 0$, and a data error $\tau$ determines $\delta$ only to order $\tau^{1/2}$ (Lemma 32.5.1 and Theorem 32.5.2).

### 33.5.3 Analytic continuation in physics

A related problem is solved daily in many body physics, where a spectral function $A(\omega) \geq 0$ is to be recovered from its Cauchy transform $G(z) = \int A(\omega)(z - \omega)^{-1} d\omega$ known at points away from the real axis, the Matsubara frequencies. Its imaginary part at height $y$ is $-\pi(P_y * A)$, but the observation operator differs from the one above: the data sample the Poisson integral at $x = 0$ and several heights, not along

a line, so uniqueness and stability do not transfer (finite data are not unique). The problem shares these features: it is severely ill posed, unconstrained methods produce negative spectral weight, and the standard tools are maximum entropy (Jarrell and Gubernatis [185]) and, more recently, Nevanlinna interpolation (Fei, Yeh and Gull [135]), which builds positivity in by parametrising the Pick functions of Theorem 33.2.3 directly. Section 31.11.11 gives the identity that links the two: the Thorin density is $u_\theta(v) = \pi^{-1}\operatorname{Tr}\big[a((H-v)^2+a^2)^{-1}\big]$ with $a = \theta - \frac{1}{2}$, for a hypothetical self adjoint $H$ whose spectrum is the set of ordinates (so under RH); the resolvent of $H$ itself is not trace class. The analytic continuation of physics and the backward flow of the Thorin programme are related problems.

## 33.6 What is known and what is not

Known: uniqueness; the failure of positivity under the inverse and its characterisation by positive harmonic extension; Hölder stability between levels with the three lines exponent and logarithmic stability at the boundary, both sharp; logarithmic minimax rates for smooth sources and polynomial upper rates for sources analytic in a strip; Lipschitz error for separated atomic sources, with the square root rate for merging atoms; and a range of positivity preserving algorithms whose limits are projections on a finite grid.

Two features of this book's use of the operator are the following. The first is the systematic use of the whole family of partial deconvolutions, the Thorin family $\mathfrak{U}_\theta = P_{\theta-\theta'} * \mathfrak{U}_{\theta'}$, with positivity at each level, and its critical width $c^*(u_1) = 1 - \sigma_{\max}$ (Theorem 19.2.2(a)). The second is the identification of the full deconvolution of the level one density with Weil's explicit formula (Theorem 20.1.1 and Chapter 32), which is classical in substance.

Problem 33.1. Estimating the critical width. Given noisy data $u + \text{noise}$ with $u \geq 0$ and $\int u(x)(1+x^2)^{-1}dx < \infty$, estimate $c^*(u)$ of Proposition 33.2.2. For atomic sources with positive weights it is the smallest width among the poles, estimable by the pencil on a class with bounded order, weights bounded below, separation and no aliasing; without weights bounded below there is no uniform rate, since $c^*$ is discontinuous ($(1-\delta)P_A + \delta P_r$ has $c^* = r < A$ for every $\delta > 0$). For general sources it is a boundary of a convex cone in an infinite dimensional space. Status: open.

Problem 33.2. Positivity and rates. Positivity does not improve the logarithmic minimax rate over smoothness classes of densities in the worst case (the lower bounds of Fan [132] are over such classes), and sparsity improves it to a polynomial one. Imposing a prescribed number of sign changes of a real Fourier transform does not help: two positive densities whose transforms are $e^{-|\omega|}$ and that function plus two nonnegative bumps near $\pm\Omega$ both have no sign change, and keep the logarithmic obstruction. Positive sources of the form $P_w * \nu$ with $\nu \geq 0$ of mass at

most $B$ have the polynomial exponent $w/(a+w)$ of Proposition 33.3.2 at $b=0$, but that comes from the smoothing and the mass bound, not from positivity, and the zero counting measure has infinite mass. Open: is the Thorin family in a class on which positivity alone gives a polynomial rate?

Problem 33.3. Three lines with positivity on a window. Theorem 33.3.1 needs an a priori bound at a level below the target, and Corollary 33.3.3 removes it for data in $L^2$ on the whole line. For data known only on a window, the mass in a window of a positive source is controlled by the data alone, as in the a posteriori bound of Proposition 31.11.5. Does positivity give a Hölder bound between levels from data on a window, without any a priori bound below, with an exponent determined by the data? Status: open.

## Notes

The chapter develops the survey chapter on Cauchy deconvolution by Nick Polson, with the corrections and additions of the book's reviews. The bibliography entries that are new for this chapter have been checked against CrossRef or OpenAlex, except [167], and the statements about the contents of four of the papers (Knapik et al. 2011 and 2013, Butucea and Tsybakov, Fan, Gao and van der Vaart) are those of a web check made for this chapter and about the others those of the reviews, not a reading of the papers here (in particular the supersmooth order $\beta$ of Nguyen [223], and the form of the rates of [202], which is not stated).

# Chapter 34

# Cauchy deconvolution from $\mathfrak{U}_1$ to $\mathfrak{U}_{1/2}$

Under the Riemann hypothesis the level one density of Chapters 30 to 33 is the Cauchy smoothing of the zero counting measure, $u_1 = P_{1/2} * \mathfrak{U}_{1/2}$ (equation (32.1.2)), and the Thorin measure $\mathfrak{U}_\theta$ at every level $\theta \in (\frac{1}{2}, 1)$ is the same smoothing at scale $\theta - \frac{1}{2}$. Can the deconvolution from $\mathfrak{U}_1$, which the Euler product makes accessible, to $\mathfrak{U}_{1/2}$, which is the object of the Riemann hypothesis, be carried out, and can the particle filtering methods of the book on Hilbert's sixth problem [257], accelerated by generative Bayesian computation (GBC) [255], make progress on it? This chapter answers in three layers. The identities and criteria that relate the two measures are recalled with their labels (Section 34.1). A computational method follows: a particle filter whose time is the level, running the Cauchy flow backwards from $\theta = 1$ towards $\theta = \frac{1}{2}$ in a window of heights, and a GBC screen that replaces most of its work by one pass of a network. And the gap between local computation and the Riemann hypothesis, which is a statement for all heights, is recorded together with what a proof along these lines would still need (Section 34.10).

The chapter develops a note by Nick Polson [228], with the computations done for this book. Several parts of the note are already in the book, in the same or in a stronger form, and are cited by label in Section 34.1 and not restated: the density and the Hadamard sum, the transport identity, the deconvolution criterion (Theorem 19.2.2, Proposition 33.2.2) and Thorin positivity as a zero free half plane (Theorem 31.9.1, Lagarias [320], Sondow and Dumitrescu [482]). The numbers of the chapter are of two kinds. The zeros and counts of the Davenport and Heilbronn function, the contour moments, the values of the density and the negative masses, the information numbers, the linear programs and polynomial tables and the nested sampling scores were recomputed for this book by scripts written from the mathematics (`code/cf_dhzeros.py`, `cf_moments.py`, `cf_xi.py`, `cf_info.py`, `cf_lp.py`, `cf_arith.py` and `cf_ns.py`, with the data files `data/cf_*.json`), and where the recomputation differed from the note the recomputed value is printed. The runs of the particle filter, of the GBC networks, of the concentration experiments and of the nested sampler are runs of the note's own code, which was

not run for the book: they are printed as the note reports them and are marked computed by the author.

Status words follow the rest of the book: proved, computed, conditional, heuristic or open. Statements at the level $\frac{7}{8}$ use the quasi Riemann hypothesis QRH($\frac{7}{8}$), announced by OpenAI [398] and formalised in Lean, and not refereed (Section 29.7); every one of them is conditional on it. Nothing in this chapter proves the Riemann hypothesis.

1. The notation and the cited criteria (Section 34.1): the right limit at a level through a zero (Remark 34.1.1), the transport identity (Proposition 34.1.2, proved), and local positivity of the density from any zero free region that narrows with the height, in particular the regions of de la Vallée Poussin and of Vinogradov and Korobov (Corollary 34.1.3, proved). The formula of Balazard, Saias and Yor is proved at every level in Chapter 35.

2. Two test functions (Section 34.2): $\zeta$ and the Davenport and Heilbronn function, the control with zeros off the line, whose four zeros off the line below height 200 are computed and certified (Proposition 34.2.1, proved; the zeros computed and certified in ball arithmetic).

3. A particle filter along the Cauchy flow (Section 34.3): the model and the sampler, the detectability of a symmetric pair (Proposition 34.3.1, proved), and the information carried by each level, which is first order for the filter as run and second order for a pair (heuristic).

4. GBC (Section 34.4): the contour moments, which determine the zeros in a rectangle exactly (Proposition 34.4.1, proved; Table 34.4 computed), the learned screen and its calibration (computed by the author), and its use to initialise the filter.

5. Concentration of the particle approximation and the window prediction of $u_{3/4}$ (Sections 34.5 and 34.6): a concentration inequality for the idealised system (Theorem 34.5.1, proved), which does not cover the samplers run, and the prediction from a fit to two levels (computed by the author, Proposition 34.6.1, proved).

6. Negative mass and arithmetic inequalities (Sections 34.7 and 34.8): the unconditional bound on the negative part of $\mathfrak{U}_\theta$ in terms of zero density (Theorem 34.7.1, proved), the converse results, the zero free regions that follow from positivity of the density at the heights 0, $t$ and $2t$ (Theorem 34.8.1, proved), the barrier that the Davenport and Heilbronn function sets (Proposition 34.8.3, proved), and the comparison of what multilevel schemes can give: a computed estimate of about $1/(11.3 \log \gamma)$, a proved cap, and the constants that pair each zero with its reflection (Propositions 34.8.7 and 34.8.10).

7. Nested sampling over heights and the open questions (Sections 34.9 and 34.10): the normalised margin of $\zeta$ and the zeros of the Davenport and Heilbronn function near height 1000 (computed), and what would close the gap.

## 34.1 What the book already proves

### 34.1.1 Notation

The notation is the book's. The completed function is $\xi$ of Section 30.1, the zeros are $\rho = \sigma_\rho + i\gamma_\rho$ with multiplicity, a level is $\theta$ (the real part of $s = \theta + iv$), and

$$u_\theta(v) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\theta + iv) = \frac{1}{\pi}\,\partial_\sigma \log|\xi(\sigma + iv)|\Big|_{\sigma=\theta}$$

is the density of Theorem 30.1.1. The measure $\mathfrak{U}_\theta = \sum_\rho \operatorname{Cauchy}(\gamma_\rho, \theta - \sigma_\rho)$ of equation (30.1.1) lives on the line, in the variable $v$, and carries no factor $\frac12$: $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ with the $\widetilde{U}_\theta$ of Theorem 29.14.4, and the Thorin measure $U_\theta$ of Section 29.14, in the variable $x = v^2$, is the image of the restriction of $\mathfrak{U}_\theta$ to $v > 0$. The symbol $U_\theta$ is used only for it. The Cauchy density is $P_a(x) = (a/\pi)/(a^2 + x^2)$ for $a > 0$, with $P_a * P_b = P_{a+b}$ and $\widehat{P_a}(\omega) = e^{-a|\omega|}$, and we write $Q_a(x) = (a/\pi)/(a^2 + x^2)$ for every real $a \neq 0$, so that $Q_a = -P_{|a|}$ is a Cauchy density with negative mass when $a < 0$, and $Q_0(x) = 0$ for $x \neq 0$. For $\theta$ not equal to the real part of a zero at the height in question the Hadamard sum (equation (32.1.1) at $\theta = 1$, the proof of Theorem 31.9.1 at every level) reads

$$u_\theta(v) = \sum_\rho Q_{\theta-\sigma_\rho}(v - \gamma_\rho), \tag{34.1.1}$$

absolutely convergent because $\sum_\rho |\gamma_\rho|^{-2} < \infty$. The function $(\sigma, v) \mapsto \pi^{-1}\operatorname{Re}(\xi'/\xi)(\sigma + iv)$ is harmonic off the zeros, as the real part of an analytic function, and positivity of $u_\theta$ at one level is the statement $\partial_\sigma \log|\xi| \geq 0$ there; positivity at every level $\theta > \sigma_0$ is the monotone increase of $|\xi(\sigma + iv)|$ in $\sigma > \sigma_0$ studied by Sondow and Dumitrescu [482] and by Matiyasevich, Saidak and Zvengrowski [219].

Remark 34.1.1 (a level through a zero). At a level $\theta = \sigma_\rho$ the term $Q_{\theta-\sigma_\rho}(v - \gamma_\rho)$ of (34.1.1) has no meaning at $v = \gamma_\rho$. For every $\theta > \frac12$ we take $\mathfrak{U}_\theta$ to be the vague limit of $u_{\theta'}(v)\,dv$ as $\theta' \downarrow \theta$. Each term converges vaguely, and the far terms are dominated by $|\gamma_\rho|^{-2}$ uniformly for $\theta'$ in a compact set, so the limit exists; it is the measure of equation (30.1.1) with an atom at the ordinate of every zero with $\sigma_\rho = \theta$ and a negative Cauchy term $Q_{\theta-\sigma_\rho}$ for every zero with $\sigma_\rho > \theta$. At $\theta = \frac12$ it is $\mathfrak{U}_{1/2} = \sum_{\sigma_\rho=1/2} \delta_{\gamma_\rho}$ plus the contribution of the zeros off the line, which vanishes (Proposition 34.2.1).

### 34.1.2 The transport identity and the criteria

Proposition 34.1.2 (Cauchy transport). *For $\theta > \frac{1}{2}$ and $\theta + iv$ not a zero,*

$$u_\theta = P_{\theta-1/2} * \mathfrak{U}_{1/2} + R_\theta, \qquad R_\theta(v) = \sum_{\sigma_\rho > 1/2} \big[Q_{\theta-\sigma_\rho} + Q_{\theta-1+\sigma_\rho}\big](v - \gamma_\rho), \quad (34.1.2)$$

*the sum over the zeros off the critical line, each taken together with its reflection $1 - \bar\rho$, which has the same ordinate. In particular $u_1 = P_{1/2} * \mathfrak{U}_{1/2} + R_1$: the density at level one is the Cauchy smoothing of $\mathfrak{U}_{1/2}$ at scale $\frac{1}{2}$ plus one Cauchy pair $P_{1-\sigma_\rho} + P_{\sigma_\rho}$ for each such zero. The pair is positive at $\theta = 1$ and is a signed term only for $\theta < \sigma_\rho$.*

*Proof.* Split the sum (34.1.1) into the zeros on the line, each contributing $P_{\theta-1/2}(\cdot - \gamma_\rho) = P_{\theta-1/2} * \delta_{\gamma_\rho}$, and the pairs $\rho$, $1 - \bar\rho$ off it. □

This is Theorem 29.14.4 above the abscissa of the last zero, and equation (32.1.2) under the Riemann hypothesis, where $R_\theta = 0$; what is stated here is the identity of densities at every level, the second term being the defect. In Fourier terms a zero on the line contributes $e^{-(\theta-1/2)|\omega|}e^{i\gamma\omega}$, while a pair at distance $\delta = \sigma_\rho - \frac{1}{2}$ contributes $2\cosh(\delta\omega)\, e^{-(\theta-1/2)|\omega|}e^{i\gamma\omega}$ for $\theta > \sigma_\rho$. Deconvolving by $P_{1/2}$ from $\theta = 1$ multiplies by $e^{|\omega|/2}$, and the result $2\cosh(\delta\omega)$ is not the Fourier transform of any tempered distribution (Chapter 33). This is the whole problem in one line.

Three statements are known and are not reproved. First, the deconvolution criterion: the Riemann hypothesis holds if and only if $u_1 = P_{1/2} * \nu$ for a positive measure $\nu$ with $\int (1+v^2)^{-1}\nu(dv) < \infty$, and then $\nu = \mathfrak{U}_{1/2}$ (Theorem 19.2.2(b); the critical width of a positive datum is Proposition 33.2.2, with the Herglotz representation of Theorem 33.2.3, in the context of Chapter 33). Second, Thorin positivity is a zero free half plane: for $\frac{1}{2} \le \sigma_0 < 1$ the measures $\mathfrak{U}_\tau$ are positive for every $\tau > \sigma_0$ if and only if $\xi$ has no zero with $\sigma_\rho > \sigma_0$, and for $\sigma_0 = \frac{1}{2}$ this is the theorem of Lagarias [320] in the form of Sondow and Dumitrescu [482] (Theorem 31.9.1); positivity at the single level $\sigma_0$ is necessary and is not known to be sufficient (Remark 1.2.4). Third, at levels $\theta \ge 1$ the density is positive unconditionally, $u_\theta = \sum_\rho P_{\theta-\sigma_\rho}(\cdot - \gamma_\rho) > 0$ (Theorem 30.1.1, Corollary 29.14.3). The pairing that underlies every statement of this chapter about a fixed height is that of Proposition 3.3.1(ii), used in Corollary 11.4.7(1) and Lemma 31.10.1.

### 34.1.3 Local positivity from a zero free region

A zero free region that narrows with the height is not a half plane, so Theorem 31.9.1 does not apply to it directly. The pairing gives the local statement.

Corollary 34.1.3 (local positivity from a zero free region). *Let $\eta : [0,\infty) \to (0, \frac{1}{2}]$ be nonincreasing and suppose that every zero satisfies $\sigma_\rho \le 1 - \eta(|\gamma_\rho|)$. Then $u_\theta(v) > 0$ whenever $\theta > \frac{1}{2}$, $\theta \ge 1 - \eta(|v| + \frac{1}{2})$ and $\xi(\theta + iv) \ne 0$.*

*Proof.* Group the zeros in the Hadamard sum (32.1.1) into orbits of $\rho \mapsto 1 - \bar\rho$. As in the proof of Corollary 11.4.7(1) (Proposition 3.3.1(ii)), a zero on the critical line contributes $P_{\theta-1/2}(v - \gamma_\rho) > 0$, an orbit with $\sigma_\rho \le \theta$ contributes a nonnegative amount, and an orbit with $\sigma_\rho > \theta$ is negative only where $|v - \gamma_\rho| < \sqrt{(\sigma_\rho - \theta)(\sigma_\rho + \theta - 1)} < \frac{1}{2}$. A zero with $\sigma_\rho > \theta$ and $|v - \gamma_\rho| < \frac{1}{2}$ would have $|\gamma_\rho| < |v| + \frac{1}{2}$, hence $\sigma_\rho \le 1 - \eta(|\gamma_\rho|) \le 1 - \eta(|v| + \frac{1}{2}) \le \theta$, a contradiction. So every orbit contributes a nonnegative amount, and the zero $\frac{1}{2} + i\gamma_1$ a positive one. □

With the region of de la Vallée Poussin ($\eta(T)$ a constant over $\log T$, Proposition 9.1.4) or of Vinogradov and Korobov ($\eta(T) = c(\log T)^{-2/3}(\log\log T)^{-1/3}$ [509]), taken as $\frac{1}{2}$ at small heights, this is positivity of $u_\theta$ at height $v$ for $\theta$ within that width of the line $\operatorname{Re} s = 1$, the width taken at height $|v| + \frac{1}{2}$. It is local in $v$, as it must be: the width shrinks with height, and positivity at the single level $\theta$ does not by itself give a zero free half plane. Positivity of $\mathfrak{U}_\tau$ for every $\tau > \sigma$ is QRH($\sigma$) (Theorem 31.9.1), while positivity at the single level $\sigma$ is necessary and not known to be sufficient (Remark 1.2.4).

### 34.1.4 Two cited identities

Two further identities of the note are in the book or in the next chapter. The logarithmic version of the transport, in which the defect of $\log|h|$ becomes a Green potential of the zeros off the line, is the formula of Balazard, Saias and Yor [38]: Chapter 35 proves the formula of Balazard, Saias and Yor at every level $\theta$ (Theorem 35.2.1); here it enters only through Proposition 34.8.3. And the Cauchy density is the harmonic measure of a half plane: Cauchy$(\gamma, a)$ is the law of $\gamma + B(T_a)$ for a Brownian motion $B$ and the hitting time $T_a$ of level $a$ by an independent Brownian motion (the closing remark of Section 30.5, Section 9.2.3), so that under the Riemann hypothesis $u_\theta(v)$ is the summed hitting density of the critical zeros for planar Brownian motion started at $\theta + iv$. The forward flow is a Markov semigroup; the backward flow has no positive kernel, it is an inverse problem, and the natural particle method for it is a Bayesian filter and not a forward simulation.

## 34.2 Two test functions

Proposition 34.2.1 (the two ends). *(a) (cited) For $\theta \ge 1$, unconditionally, $u_\theta(v) = \sum_\rho P_{\theta-\sigma_\rho}(v - \gamma_\rho) > 0$; in particular $\mathfrak{U}_1 = \sum_\rho P_{1-\sigma_\rho}(\cdot - \gamma_\rho)$ is a positive mixture of Cauchy laws centred at the ordinates, with scales in $(0,1)$. This is Theorem 30.1.1 and Corollary 29.14.3, and it holds for $\zeta$ because $\zeta(1 + it) \ne 0$.*

(b) *(proved) As $\theta \downarrow \frac{1}{2}$, in the sense of distributions, $u_\theta \to \mathfrak{U}_{1/2} = \sum_{\sigma_\rho = 1/2} \delta_{\gamma_\rho}$. Zeros off the line are invisible in $\mathfrak{U}_{1/2}$: a pair $\sigma_\rho$, $1-\sigma_\rho$ contributes $Q_{\theta-\sigma_\rho} + Q_{\theta-1+\sigma_\rho} \to Q_{1/2-\sigma_\rho} + Q_{\sigma_\rho - 1/2} = 0$.*

*Proof.* (b) For a zero on the line $P_{\theta-1/2} \to \delta_0$. For a pair, $\theta - \sigma_\rho$ and $\theta - 1 + \sigma_\rho$ tend to $\frac{1}{2} - \sigma_\rho \neq 0$ and $\sigma_\rho - \frac{1}{2} \neq 0$, and $a \mapsto Q_a$ is continuous from $\mathbb{R} \setminus \{0\}$ to $L^1(\mathbb{R})$, so the pair terms converge to $-P_{\sigma_\rho - 1/2} + P_{\sigma_\rho - 1/2} = 0$. The far terms are bounded by $|\gamma_\rho|^{-2}$ uniformly for $\theta \in [\frac{1}{2}, 1]$, which gives dominated convergence on compact sets. □

In the variable $x = v^2$ this is the statement that $U_{1/2} = \sum_\gamma \delta_{\gamma^2}$ is the discrete measure of the Riemann hypothesis programme.

All computations of this chapter use $\zeta$ and the Davenport and Heilbronn function, which has no Euler product and has zeros off the critical line. With $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1) = 0.28407904384\ldots$ it is [111, 498]

$$f(s) = 5^{-s}\Big[\zeta(s, \tfrac{1}{5}) + \kappa\,\zeta(s, \tfrac{2}{5}) - \kappa\,\zeta(s, \tfrac{3}{5}) - \zeta(s, \tfrac{4}{5})\Big],$$

and the completed function $\Lambda(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$ is entire and satisfies $\Lambda(s) = \Lambda(1-s)$; the identity is that of Davenport and Heilbronn [111] (see [498]), and as a diagnostic the relative difference $|\Lambda(s) - \Lambda(1-s)|/|\Lambda(s)|$ is at most $1.1 \times 10^{-87}$ at four points in ball arithmetic. The function is treated in Section 24.17.2; here it is the control. The zeros of $f$ off the line below height 200 were found by Spira [484]; refined by Newton's method in ball arithmetic (`code/cf_dhzeros.py`, `data/cf_dhzeros.json`) they are

$$0.8085171825 + 85.6993484854\,i, \qquad 0.6508300806 + 114.1633427308\,i,$$
$$0.5743560505 + 166.4793059132\,i, \qquad 0.7242576946 + 176.7024612429\,i,$$

together with their reflections in the critical line, each certified to be the only zero in a square of half width $10^{-16}$ about the centre, so that all ten printed decimals are correct. These are all the zeros off the line below height 200 (computed, certified in ball arithmetic): $\Lambda$ has 130 zeros with $0 < \gamma < 200$, counted by the argument principle on a rectangle, and 122 sign changes of $\Lambda(\frac{1}{2} + it)$ on the segment, with certified signs on a grid of step $\frac{1}{50}$; the difference $8 = 2 \cdot 4$ is the number of off line zeros and their reflections, and equality means that no zero on the line is missed or multiple and that no zero off the line other than these four and their reflections exists.

For $\Lambda$ the density $u_\theta = \pi^{-1}\operatorname{Re}\Lambda'/\Lambda(\theta + iv)$ plays the role of $u_\theta$ for $\xi$. The Hadamard sum (34.1.1), the transport identity of Proposition 34.1.2 and the deconvolution criterion with the base line $\operatorname{Re} s = 1$ replaced by a line to the right of all zeros apply to it ($\operatorname{Re} s = 2$ works, since $|f(s) - 1| \le \zeta(2) - 1 < 1$ there, as $|\kappa| < 1$); Proposition 34.2.1(a) does not apply, since $f$ has infinitely many zeros

in $\operatorname{Re} s > 1$ [111]. Proposition 34.8.3 makes this precise. The principle of such a test is that of Section 24.17.2: a method that detects zeros off the line of $\xi$ must detect them for $\Lambda$.

For $\zeta$ the densities $u_\theta$ on the window $[60, 120]$ stay positive at every level computed (the zeros of $\zeta$ at these heights are on the line [419]) and sharpen into spikes at the ordinates as $\theta$ decreases, which is the transport identity with $R_\theta = 0$. For $\Lambda$ the densities at $\theta = 1$ and $\theta = 0.9$ are positive on this window (computed; the zeros of $f$ in the window have $\sigma_\rho \le 0.8085$). They turn negative near $v = 85.70$ once $\theta < 0.8085$ and near $v = 114.16$ once $\theta < 0.6508$: at the ordinates the minima of $u_\theta$ over $[60, 120]$ are $-36.8$ at $\theta = 0.8$ and $-4.96$ at $\theta = 0.6$ (computed, `code/cf_xi.py`; the grid values $-36.6$ and $-4.6$ of the note are grid minima). This is the signed defect of Proposition 34.1.2, seen directly.

## 34.3 A particle filter along the Cauchy flow

### 34.3.1 The state space model

Fix a window of heights. The parameter is the local configuration $z = \{(\gamma_j, \sigma_j)\}_{j=1}^{K}$ of the zeros with ordinates in a slightly larger window, with $K$ fixed by the argument principle (a contour integral of $\xi'/\xi$, computed once). The observation at level $\theta_k$ is the vector $y_k = (u_{\theta_k}(v_i))_i$ on a grid, computed to high precision. Equation (34.1.1) gives the observation equation

$$y_k(v) = \sum_{j=1}^{K} Q_{\theta_k - \sigma_j}(v - \gamma_j) + b_k(v) + \varepsilon_k(v),$$

where $b_k$ is the contribution of the zeros outside the window. For a distant zero the pair terms equal $2(\theta - \tfrac12)/(\pi(v-\gamma)^2)$ to leading order, so we model $b_k(v) = (\theta_k - \tfrac12)(c_0 + c_1 w + c_2 w^2 + c_3 w^3) + (\theta_k - \tfrac12)^3 c_4$ with $w$ the centred height. The coefficients $c$ enter linearly and are integrated out in closed form under a Gaussian prior, by the Woodbury identity, so the particles carry only $(\gamma_j, \sigma_j)$. The error $\varepsilon$ is a tolerance, not noise: the data are deterministic, and $\varepsilon = 0.01$ absorbs the background approximation. Every statement below about posteriors, information and coverage refers to this Gaussian tolerance likelihood.

The levels $1 = \theta_1 > \theta_2 > \cdots > \theta_n$ play the role of time. With the background coefficients shared by the levels and integrated out jointly, the filtering distributions are

$$\pi_k(z) \;\propto\; \pi_0(z)\, p(y_1, \dots, y_k \mid z),$$

the Feynman Kac flow of [257] with potentials $G_k = p(y_k \mid y_1, \dots, y_{k-1}, z)$ and Markov moves invariant for $\pi_k$. The prior is uniform: ordinates uniform in the window and every $\sigma_j$ uniform on $(0, 1)$, so that the functional equation is not imposed on the particles (Section 34.3.4).

### 34.3.2 Algorithm

1. Draw $N$ particles from the prior.
2. For $k = 1, \dots, n$: move from $\pi_{k-1}$ to $\pi_k$ through the tempered targets $\pi_{k-1}^{1-\lambda}\pi_k^{\lambda}$, choosing each increment of $\lambda$ by bisection so that the effective sample size of the incremental weights is $N/2$; resample systematically; rejuvenate by componentwise random walk Metropolis moves with scales set from the current particle cloud.
3. Report the posterior of every $\sigma_j$, and of $\delta^* = \max_j |\sigma_j - \frac{1}{2}|$ over the ordinates in the interior of the window.

This is the sequential Monte Carlo sampler of Chopin [81] and Del Moral, Doucet and Jasra [105], with the Cauchy level as time; the book's account of sequential Monte Carlo for the explicit formula is Section 32.6, and the single level Bayesian inversion is Section 30.4, with its path of Section 30.6.

### 34.3.3 Results for $\zeta$

The runs of this section and the next are the note's runs (computed by the author, not recomputed for the book). We used $N = 1000$ particles, the window $[78, 94]$ with 401 grid points, $K = 10$ zeros in $[74, 98]$, and levels $\theta \in \{1, 0.9, 0.8, 0.7, 0.65, 0.6\}$. Table 34.1 gives the posterior for the zeros in the data window and Table 34.2 the posterior of $\delta^*$ as levels are added. The ordinates are recovered to within $10^{-4}$ and every $\sigma_j$ is pinned at $\frac{1}{2}$ to within $4 \times 10^{-4}$. The bound on $\delta^*$ falls by a factor of fourteen between $\theta = 1$ and $\theta = 0.6$. Each $\sigma_j$ is a free parameter, so the information is first order, $\int (\partial_a P_a)^2 \, dx = 1/(4\pi a^3)$, and $(\sum_l a_l^{-3}/a_1^{-3})^{1/2} = 13.6$ for these levels (Section 34.3.4); Proposition 34.3.1, which concerns a symmetric pair, would give 7.8. Zeros just outside the data window are poorly identified, as they should be; they act as nuisance parameters.

| posterior median $\gamma$ | posterior median $\sigma_j$ | 95% interval for $\sigma_j$ |
|---|---|---|
| 79.3373 | 0.49999 | [0.49964, 0.50036] |
| 82.9104 | 0.50000 | [0.49966, 0.50035] |
| 84.7355 | 0.50002 | [0.49967, 0.50038] |
| 87.4253 | 0.49999 | [0.49964, 0.50034] |
| 88.8091 | 0.49999 | [0.49961, 0.50036] |
| 92.4919 | 0.50001 | [0.49967, 0.50037] |

Table 34.1: Particle filter posterior for the zeros of $\zeta$ in the data window $[78, 94]$ after all levels down to $\theta = 0.6$ (computed by the author). The true ordinates are 79.3374, 82.9104, 84.7355, 87.4253, 88.8091 and 92.4919.

| levels absorbed down to $\theta =$ | 1 | 0.9 | 0.8 | 0.7 | 0.65 | 0.6 |
|---|---|---|---|---|---|---|
| median of $\delta^*$ ($\times 10^{-3}$) | 3.65 | 2.15 | 1.31 | 0.73 | 0.46 | 0.26 |
| 95% quantile of $\delta^*$ ($\times 10^{-3}$) | 6.43 | 3.69 | 2.33 | 1.34 | 0.85 | 0.46 |
| likelihood evaluations ($\times 10^{6}$) | 1.86 | 2.46 | 2.95 | 3.43 | 3.79 | 4.15 |

Table 34.2: The backward Cauchy flow as a filter: the posterior of $\delta^* = \max|\sigma_j - \frac{1}{2}|$ over the zeros of $\zeta$ in $[80, 92]$ as levels are added, with cumulative cost (computed by the author). The bounds measure the displacement of one unpaired zero from the line, which the functional equation forbids.

For a pair $\frac{1}{2} \pm \delta$ the same data bound $\delta$ only at about $4.4 \times 10^{-3}$ (one sided 95%, second order Gaussian approximation, six levels; 0.0345 after level one alone), ten times weaker than the bound on an unpaired displacement (computed, `code/cf_info.py`). A rerun with the pair parametrisation of Section 32.5 (Lemma 32.5.1, Theorem 32.5.2), each ordinate on the line or a pair $\frac{1}{2} \pm \delta$ with $\eta = \delta^2 \geq 0$, has not been done.

The cost is the obstacle. The run used $4.15 \times 10^6$ likelihood evaluations for a window of length 16 containing six zeros. For the Davenport and Heilbronn window, with seventeen zeros and a uniform prior, the tempering stalled at $\lambda \approx 0.003$ after the first several hundred thousand evaluations: the posterior is so concentrated relative to the prior that the bridge needs thousands of steps. Optimising this is the job of GBC (Section 34.4).

### 34.3.4 How much information each level carries

Proposition 34.3.1 (detectability). *Let $a = \theta - \frac{1}{2} > \delta > 0$. At the centre $v = \gamma$, a pair of zeros at $\frac{1}{2} \pm \delta + i\gamma$ exceeds a double zero on the line at $\frac{1}{2} + i\gamma$ by*

$$\Delta(a, \delta) = \frac{1}{\pi}\Big(\frac{1}{a - \delta} + \frac{1}{a + \delta}\Big) - \frac{2}{\pi a} = \frac{2\delta^2}{\pi a(a^2 - \delta^2)}.$$

*In the parameter $\eta = \delta^2$ the pair differs from the double zero by $\eta\, \partial_a^2 P_a + O(\eta^2)$, and*

$$\int_{\mathbb{R}} \big(\partial_a^2 P_a(x)\big)^2 dx = \frac{3}{4\pi a^5}.$$

*Proof.* Direct computation, with $\partial_a^2 P_a(x) = 2a(a^2 - 3x^2)/\big(\pi(a^2 + x^2)^3\big)$ and the scaling $P_a(x) = a^{-1} P_1(x/a)$. The integral was also checked by quadrature at 40 digits (`code/cf_info.py`). □

Two consequences concern a symmetric pair, the situation of Section 32.5. First, the information about $\delta$ is second order: the Fisher information at $\delta = 0$ vanishes, and one works with $\eta = \delta^2$. Second, the information grows like $a^{-5}$ as

the level approaches the line. With observation error $\varepsilon$ on a grid of spacing $\Delta v$, the level $a$ resolves $\delta$ once roughly $\delta^4 \gtrsim (4\pi/3)\,\varepsilon^2 \Delta v\, a^5$ (heuristic, for the pair model). Coarse levels are well posed and uninformative; fine levels are informative and ill posed, which is the structure that sequential Monte Carlo handles well: the level is the filter's time, and each new level sharpens the likelihood of a pair in $\eta$ by a factor of order $((\theta_{k-1} - \frac{1}{2})/(\theta_k - \frac{1}{2}))^5$. The filter of this section does not impose the pairing, so its information is first order, as follows.

### 34.3.5 Information rates

For a single zero with free real part $\sigma$ (no pairing imposed), $\partial_\sigma Q_{\theta-\sigma} = -\partial_a P_a$ and $\int (\partial_a P_a)^2 dx = 1/(4\pi a^3)$. With error $\varepsilon$ on a grid of spacing $\Delta v$, the Fisher information after levels $a_1, \dots, a_k$, with $a_l = \theta_l - \frac{1}{2}$, is about $(4\pi\varepsilon^2\Delta v)^{-1} \sum_{l \le k} a_l^{-3}$, so the Bernstein and von Mises posterior standard deviation of the Gaussian tolerance model is

$$\mathrm{sd}(\sigma_j) \approx \varepsilon \sqrt{4\pi\Delta v} \Big( \sum_{l \le k} a_l^{-3} \Big)^{-1/2} \qquad \text{(heuristic)}.$$

Table 34.3 compares this with the particle filter of this section. The agreement is within 8% at the first level and exact at the last, so the filter matches the first order Fisher information of the model with free real parts; a deterministic tolerance likelihood gives no frequentist coverage, and the efficient information for the full multizero model with unknown background is not that of an isolated zero, so no more is claimed. The ratios of the median bound of Table 34.2 to its value at level one (observed 1.70 to 1.74, 2.76 to 2.79, 4.80 to 5.00, 7.56 to 7.93, 13.98 to 14.04) follow the first order law, $(\sum_l a_l^{-3}/a_1^{-3})^{1/2} = 1.72$, 2.75, 4.82, 7.76, 13.61 for the levels 0.9 to 0.6, and not the law of a pair, $(\sum_l a_l^{-5}/a_1^{-5})^{1/4} = 1.42$, 2.03, 3.27, 4.79, 7.77 (computed, `code/cf_info.py`).

| levels absorbed down to $\theta =$ | 1 | 0.9 | 0.8 | 0.7 | 0.65 | 0.6 |
|---|---|---|---|---|---|---|
| predicted sd of $\sigma_j$ ($\times 10^{-4}$) | 25.1 | 14.6 | 9.1 | 5.2 | 3.2 | 1.84 |
| particle filter, mean over six zeros | 27.1 | 15.3 | 9.4 | 5.3 | 3.3 | 1.84 |

Table 34.3: Posterior standard deviation of $\sigma_j$ for the six zeros of $\zeta$ in $[78, 94]$, against the first order Fisher information formula ($\varepsilon = 0.01$, $\Delta v = 0.04$). The predicted row is computed (`code/cf_info.py`); the filter row is the author's run.

When the functional equation is imposed, a zero on the line is compared with a symmetric pair and the information is second order (Proposition 34.3.1): the relevant quantity is $\sum_l a_l^{-5}$ and the resolvable $\delta$ scales like $a_{\min}^{5/4}$ (heuristic). Both formulas say the same thing: information comes from the levels nearest the line. The levels $\theta \ge 0.9$ supply 1.6% of $\sum a^{-3}$ and 0.1% of $\sum a^{-5}$ in this schedule.

Problem 34.1. What is the best possible resolution of the pair parameter $\eta = \delta^2$ from data at levels $\theta \geq \theta_{\min}$, and does the particle filter with the pair parametrisation of Section 32.5 reach it? (open)

## 34.4 Optimising the filter with generative Bayesian computation

### 34.4.1 Contour moments

Before learning a summary, we recall what exact contour data give.

Proposition 34.4.1 (contour moments). *Let $R = [\theta_0, \theta_1] \times [T_1, T_2]$ with $\xi \neq 0$ on $\partial R$, let $c \in \mathbb{C}$ and*

$$M_k(R) = \frac{1}{2\pi i} \oint_{\partial R} (s - c)^k \, \frac{\xi'}{\xi}(s) \, ds \qquad (k \geq 0).$$

*Then $M_k(R) = \sum_{\rho \in R} (\rho - c)^k$, with multiplicity. In particular $K = M_0(R)$ is the number of zeros in $R$, and $M_1(R), \dots, M_K(R)$ determine them: with $e_0 = 1$ and $j e_j = \sum_{r=1}^{j} (-1)^{r-1} e_{j-r} M_r(R)$ for $1 \leq j \leq K$, the numbers $\rho - c$, $\rho \in R$, are the roots of $z^K - e_1 z^{K-1} + \cdots + (-1)^K e_K$.*

*Proof.* The function $(s-c)^k \xi'(s)/\xi(s)$ is meromorphic inside $R$, with a simple pole at each zero $\rho$ of multiplicity $m$, of residue $m(\rho - c)^k$; the residue theorem gives the first claim. The power sums $M_1, \dots, M_K$ of the $K$ numbers $\rho - c$ determine their elementary symmetric functions $e_1, \dots, e_K$ by Newton's identities, which is the recursion displayed, and these are the coefficients of the monic polynomial with those roots. □

Remark 34.4.2. This is the method of Delves and Lyness [117]. The moments use $\xi'/\xi$ itself on the whole of $\partial R$, horizontal edges included, at abscissae at least $\theta_0$; with exact contour integrals they give the zeros in $R$ exactly. They are not functions of the grid values used by the filter, so they are not a sufficient statistic of its data. The real part of an analytic function on the two vertical edges does not determine it: with $a = \theta_0$, $b = \theta_1$ and $T_1 < t_0 < T_2$, the entire function $g(s) = \cos\big(\pi(s - a - it_0)/(b - a)\big)$ is real and nonzero on both vertical lines, $g'/g$ is purely imaginary there, and $g$ vanishes at $\frac{a+b}{2} + it_0 \in R$; so $\xi$ and $\xi g$ have the same density on both vertical edges and different zeros in $R$. Within the class of functions like $\xi$, of order one with zeros in the strip, the density on one segment of the line $\operatorname{Re} s = 1$ does determine $\xi'/\xi$ (Proposition 34.6.1(a)), but only through analytic continuation, which no finite grid of values carries.

So in the noise free limit the deconvolution in a window is computed by a handful of contour integrals, to twelve digits for the zero of the last row. What makes the problem hard is not the window; it is that there are infinitely many

| function | rectangle | $M_0$ | recovered zeros right of $\theta_0$ |
|---|---|---|---|
| $\zeta$ | $[0.6,3]\times[80,92]$ | 0 (radius $3\times10^{-48}$) | none |
| $\zeta$ | $[0.55,3]\times[60,120]$ | 0 (radius $7\times10^{-46}$) | none |
| Davenport and Heilbronn | $[0.6,3]\times[80,92]$ | 1 (radius $3\times10^{-48}$) | $0.808517182457 + 85.699348485378\,i$ |

Table 34.4: Contour moments, computed in ball arithmetic at 128 and 192 bits (`code/cf_moments.py`, `data/cf_moments.json`); the two precisions agree, and the integrals are enclosed, so that each $M_0$ is an enclosure containing one integer. The note printed $7\times10^{-22}$ and 0.006 for the first two entries, floating point quadrature residuals (the second from zeros at distance 0.05 from the edge); the exact values are 0. The zero of the last row is the Newton root with $K=1$ and $M_1$ as an interval.

windows. The role of GBC is to make each window nearly free, with its uncertainty quantified, when the data are a grid of noisy values rather than exact contour integrals.

### 34.4.2 GBC: learned summary and quantile network

Following Polson and Sokolov [255], and the semi automatic summaries of Fearnhead and Prangle [141], we simulate from the joint law of parameters and data, learn the conditional mean as a summary, and then learn the posterior quantile function given the summary (the book's account is Section 30.5).

*Simulator.* A window of half width 4 around a centre $T$; ordinates from a rigid point process (Gamma(4) spacings, density uniform on $[0.35,0.8]$ per unit, covering the Davenport and Heilbronn function and $\zeta$ at heights up to a few hundred; the density of $\zeta$ is 0.35 at height 56.8 and lower below, so the windows of $\zeta$ centred at 30 to 55 lie outside the training range); with probability $\frac12$ one zero in the central interval $|\gamma-T|\le3$ is replaced by a symmetric pair at distance $\delta$, mostly uniform on $[0.02,0.45]$; with probability 0.2 a decoy pair outside the central interval; a random quadratic background proportional to $\theta-\frac12$; observations $u_\theta$ on 161 points at $\theta\in\{1,0.85,0.7,0.6\}$ with error 0.003. The target is $\delta^*=\max\{|\sigma_\rho-\frac12|:|\gamma_\rho-T|\le3\}$, equal to 0 when the central zeros lie on the line.

*Stage one: summary.* A network $S$ (two hidden layers, 256 and 128 units; the code's input has 652 entries for the 644 grid values, and the note does not say what the other eight are) is trained by squared error to $S(y)\approx\mathbb{E}[(\delta^*,\,\mathbf{1}\{\delta^*>0\},\,\gamma^*)\mid y]$, with $\gamma^*$ the ordinate of the displaced zero. In the code the location loss is masked

by the indicator that a displaced pair is present, so the third output estimates the location given that a pair is present, $\mathbb{E}[A\gamma^* \mid y]/\mathbb{E}[A \mid y]$ with $A = \mathbf{1}\{\delta^* > 0\}$, and the second network receives the three outputs together with eight hand built minima of the data, an eleven dimensional conditioning vector. By the argument of Fearnhead and Prangle the conditional mean is the optimal summary for estimating these targets under quadratic loss, when the network equals it.

*Stage two: posterior by noise outsourcing.* A second network $Q(\tau, S(y))$ is trained by the pinball loss $\rho_\tau(\delta^* - Q)$ with $\tau \sim U(0,1)$ as an input. By the noise outsourcing theorem [187], $\delta^* \overset{d}{=} Q(U, S(y))$ with $U$ uniform and independent, for the exact conditional quantile function given $S(y)$, so posterior draws cost one forward pass each; the network approximates that function, and the posterior given $S(y)$ is the posterior given $y$ only if $S$ is sufficient (as in Section 30.5). The network is not constrained to be monotone in $\tau$.

*Training and calibration.* With 120,000 training and 20,000 test simulations, training took 111 seconds on two cores. The root mean square error of the first output of $S$ for $\delta^*$ is 0.016. Calling a window flagged when the posterior median exceeds 0.03 gives a true positive rate of 0.978 and a false positive rate of 0.002; detection is essentially perfect for $\delta^* \geq 0.05$ and falls to 0.67 for $\delta^* \in [0.02, 0.05]$ (computed by the author). Coverage of the posterior quantiles is in Table 34.5.

| nominal $\tau$ | 0.05 | 0.25 | 0.50 | 0.75 | 0.95 |
|---|---|---|---|---|---|
| empirical | 0.049 | 0.182 | 0.467 | 0.725 | 0.917 |

Table 34.5: Coverage of the GBC posterior quantiles of $\delta^*$ on held out simulations (computed by the author). The middle quantiles undercover, an approximation and calibration error of the quantile network; at an atom exact quantiles overcover. The upper tail is slightly short and is the place to add conformal calibration.

### 34.4.3 Screening real windows

We applied the trained networks to 35 windows centred at $T = 30, 35, \ldots, 200$ for each function. Each window costs 644 evaluations of $\xi'/\xi$ (of $\Lambda'/\Lambda$ for the Davenport and Heilbronn function) and then 6.7 milliseconds of network time. Table 34.6 lists the flagged windows.

### 34.4.4 GBC inside the particle filter

The GBC posterior is fast but approximate; the particle filter targets the model posterior but is slow. We combine them in the standard way, using the learned posterior as the starting distribution of the particle system.

| window centre | GBC location | true zero | GBC median $\delta^*$ | true $\delta$ |
|---|---|---|---|---|
| 85 | 85.68 | $0.8085 + 85.6993\,i$ | 0.314 | 0.3085 |
| 115 | 114.08 | $0.6508 + 114.1633\,i$ | 0.143 | 0.1508 |
| 165 | 166.47 | $0.5744 + 166.4793\,i$ | 0.086 | 0.0744 |
| 175 | 176.72 | $0.7243 + 176.7025\,i$ | 0.211 | 0.2243 |
| 180 | 176.58 | same zero, at the edge | 0.361 | 0.2243 |

Table 34.6: Davenport and Heilbronn windows flagged by GBC (computed by the author; the zeros are those of Section 34.2). The four zeros off the line below height 200 are flagged, and located to within 0.1 in the four windows that target them. The window at 180 sees the zero at 176.70 near its edge, outside its target interval $[177, 183]$, and overstates $\delta$: a boundary false alarm, a boundary effect of the simulator. No other window of either function is flagged; for all 35 windows of $\zeta$ the posterior median is 0 and the 95% quantile is at most 0.007.

1. Screen every window with GBC.

2. In a flagged window, build a proposal $q$: ordinates near the peaks of $u_{\theta_n}$ at the lowest level for zeros on the line, a pair near each negative dip with $\delta$ drawn from the GBC posterior by noise outsourcing, and diffuse components for the rest.

3. Run the tempered sampler from $q$ to the posterior, through the bridge $q^{1-\lambda}(\pi_0 L)^{\lambda}$, whose potentials are powers of $\pi_0 L/q$ and so involve $\log \pi_0 - \log q$, with the same resample and move steps as in Section 34.3.

In practice step 3 works far better when the proposal is first moved to the posterior mode by a Gauss Newton fit started from the GBC output, with $q$ the Gaussian with four times the inverse Gauss Newton Hessian as covariance. The results:

For $\zeta$ the four zeros inside $[80, 92]$ are recovered as before, with 95% intervals for $\sigma_j$ of half width $4 \times 10^{-4}$ around $\frac{1}{2}$. For the Davenport and Heilbronn function the filter finds the pair at $\gamma = 85.6993$ with $\sigma = 0.80851$ (interval $[0.80851, 0.80852]$) and $\sigma = 0.18994$ (interval $[0.18867, 0.19084]$). The first agrees with the true 0.808517 to $10^{-5}$. The second misses the true 0.191483 by 0.0015, and the intervals for several zeros on the line exclude $\frac{1}{2}$ by up to $6 \times 10^{-4}$. Both are symptoms of misspecification: with 12 zeros in a window of length 18 the cubic background is too crude for a tolerance of $\varepsilon = 0.01$, and the posterior is overconfident. A larger zero window, or a background built from the Riemann von Mangoldt density, should help; the last paragraph of the next subsection confirms the first for these data.

| sampler | window | steps | evaluations | seconds |
|---|---|---|---|---|
| plain filter, uniform prior, $\zeta$ | $[78, 94]$ | many | $4.15 \times 10^6$ | minutes |
| GBC, Gauss Newton, filter, $\zeta$ | $[80, 92]$ | 7 | $3.6 \times 10^4$ | 3.6 |
| plain filter, uniform prior, Davenport and Heilbronn | $[78, 94]$ | stalled | n/a | stopped |
| GBC, Gauss Newton, filter, Davenport and Heilbronn | $[80, 92]$ | 1020 | $5.1 \times 10^6$ | 645 |

Table 34.7: Costs of reported runs (computed by the author). The two runs for $\zeta$ differ in window ($[78, 94]$ with six zeros against $[80, 92]$ with four), in zeros and in data, so their ratio of about 115 in evaluations is a comparison of two different runs and not a matched speedup. For Davenport and Heilbronn the GBC initialised filter succeeds where the plain filter stalls (at tempering exponent $\lambda \approx 0.003$), but the posterior is strongly non Gaussian because the level $\theta = 0.8$ lies within 0.0085 of $\sigma = 0.8085$, so the bridge from the Gaussian proposal still needs many steps.

### 34.4.5 Three computational questions

*Sine kernel statistics in the simulator.* We replaced the rigid Gamma(4) spacing of the GBC simulator by eigenphases of Haar unitary matrices (the circular unitary ensemble), rescaled to the local density, whose local statistics approximate the sine kernel correlations conjectured for the zeros of $\zeta$ (the sine process is the limit as the size grows, and the matrix size is not stated in the note, heuristic); the empirical spacing variance is 0.16 against 0.25 before. Retraining both networks took 285 seconds. On held out simulations of the new simulator the 95% quantile has coverage 0.952, against 0.917 for the rigid networks on theirs. On the real windows the screen again flags exactly the four Davenport and Heilbronn zeros, plus the edge window at 180, and nothing for $\zeta$ (largest posterior median 0.001). The estimates of $\delta$ improve about threefold (Table 34.8, computed by the author).

| zero | $0.8085 + 85.70\,i$ | $0.6508 + 114.16\,i$ | $0.5744 + 166.48\,i$ | $0.7243 + 176.70\,i$ |
|---|---|---|---|---|
| true $\delta$ | 0.3085 | 0.1508 | 0.0744 | 0.2243 |
| rigid simulator | 0.314 | 0.143 | 0.086 | 0.211 |
| sine kernel simulator | 0.312 | 0.149 | 0.079 | 0.228 |

Table 34.8: GBC posterior medians of $\delta$. Mean absolute error 0.0096 with the rigid simulator and 0.0034 with the sine kernel simulator.

*Conformal calibration.* For the rigid networks, split conformal calibration on 10,000 fresh simulations shifts the 95% upper bound by an order statistic of the calibration residuals, 0.0015, and raises its coverage on 20,000 test simulations from 0.915 to 0.948. Conditional on an off line pair being present, coverage rises

only from 0.885 to 0.906: marginal conformal calibration does not repair conditional coverage. A Mondrian version partitions by a label that must be observed at prediction time; the indicator $\mathbf{1}\{\delta^* > 0\}$ is not, so the partition would have to use the network's own flag (open). For the sine kernel networks the shift is $-0.0002$ and no correction is needed.

*Removing the misspecification.* Section 34.4.4 reported that the filter missed the partner zero, with $\sigma = 0.19148$ at height 85.6993, by 0.0015 and was overconfident on the zeros on the line. We attributed this to too narrow a zero window. Widening it to $[74, 98]$ with $K = 17$ zeros, counted by the argument principle, and data on $[78, 94]$ at the five levels other than $\theta = 0.8$, the Gauss Newton fit started from the GBC output converged in 82 evaluations and 4.7 seconds, with essentially zero residual. With Laplace standard errors it gives

$$\sigma = 0.80852 \pm 0.00015, \qquad 1 - \sigma = 0.19149 \pm 0.00129, \qquad \gamma = 85.6993 \pm 0.0001,$$

against the true 0.808517, 0.191483 and 85.699348. Every zero on the line in the data window comes out as $0.50000 \pm 0.00019$. The wider window fit removes the discrepancy of the narrower fit for these data. The six components outside the data window are unidentified and drift to the boundary of the prior box, as nuisance parameters should. They are also why the full tempered sampler on this wider window did not finish in forty minutes: the posterior in those coordinates is flat and truncated, far from the Gaussian proposal. Integrating the unobserved zeros into the background, through the Riemann von Mangoldt density, rather than carrying them as particles, is a remedy that has not been tried.

## 34.5 Concentration of the particle approximation

We now prove a non asymptotic concentration inequality for the particle approximation, in the style of Del Moral [109]. It applies to the particle system defined below: multinomial resampling, with potentials $G_k$ and mutation kernels $M_k$ fixed in advance. A tempering bridge is also a Feynman Kac flow, so the plain filter of Section 34.3 with a fixed schedule and fixed kernels is covered, and so is the GBC initialised filter when in addition $\log \pi_0 - \log q$ has finite oscillation $B$, since its potentials are powers of $\pi_0 L / q$ (in Corollary 34.5.4 replace $\Omega_\varsigma$ by $\Omega_\varsigma + B$). The samplers run in Sections 34.3 and 34.4 resample systematically, choose their tempering increments and move scales from the particles, and move by random walk Metropolis kernels; the theorem does not cover them (Lemma 34.5.3), and the $N^{-1/2}$ behaviour in Table 34.9 is computed. The statement is not the posterior concentration of Theorem 32.4.1, which describes how the posterior contracts as the noise level tends to zero; here the posterior is fixed and the question is the Monte Carlo error of the sampler. Throughout, the posterior is that of the Gaussian tolerance likelihood of Section 34.3, in which $\varepsilon$ is a tolerance and not a noise

level, and every statement about it refers to that likelihood. Theorem 34.5.1, Remark 34.5.2, Lemma 34.5.3, the stability bound and Corollary 34.5.4 are proved, and the two remarks that close the section are heuristic; the symbols $\epsilon$ (a minorisation constant) and $\varepsilon$ (the tolerance) are different.

### 34.5.1 Setting

Let $\eta_0$ be a probability measure on a measurable space $E$, let $G_0, \dots, G_{n-1}$ be positive bounded potentials with $\sup G_k / \inf G_k < \infty$, and let $M_1, \dots, M_n$ be Markov kernels. Write $\Psi_G(\mu)(dx) = G(x)\mu(dx)/\mu(G)$ and

$$\eta_{k+1} = \Phi_{k+1}(\eta_k), \qquad \Phi_{k+1}(\mu) = \Psi_{G_k}(\mu) M_{k+1}.$$

In the filter, $G_k$ is the incremental likelihood (or its tempered power) and $M_{k+1}$ the Metropolis kernel that leaves $\eta_{k+1}$ invariant. The particle system draws $X_0^1, \dots, X_0^N$ independently from $\eta_0$ and, given generation $k$, draws $X_{k+1}^1, \dots, X_{k+1}^N$ independently from $\Phi_{k+1}(\eta_k^N)$, where $\eta_k^N = N^{-1}\sum_i \delta_{X_k^i}$. This is multinomial resampling followed by mutation.

Put $\mathcal{Q}_{k+1}(x, dy) = G_k(x) M_{k+1}(x, dy)$, $\mathcal{Q}_{k,n} = \mathcal{Q}_{k+1} \cdots \mathcal{Q}_n$ (the identity when $k = n$),

$$G_{k,n} = \mathcal{Q}_{k,n}(1), \qquad \mathcal{P}_{k,n}(f) = \frac{\mathcal{Q}_{k,n}(f)}{\mathcal{Q}_{k,n}(1)}, \qquad \Phi_{k,n}(\mu)(f) = \frac{\mu\mathcal{Q}_{k,n}(f)}{\mu\mathcal{Q}_{k,n}(1)},$$

so that $\eta_n = \Phi_{k,n}(\eta_k)$. Define the oscillation ratio and the Dobrushin coefficient

$$g_{k,n} = \sup_{x,y} \frac{G_{k,n}(x)}{G_{k,n}(y)}, \qquad \varkappa_{k,n} = \sup_{x,y} \big\| \mathcal{P}_{k,n}(x,\cdot) - \mathcal{P}_{k,n}(y,\cdot) \big\|_{\mathrm{TV}},$$

$$S_n = \sum_{k=0}^{n} g_{k,n}\, \varkappa_{k,n}.$$

Theorem 34.5.1 (concentration). *For every measurable $f$ with $\operatorname{osc}(f) \le 1$, every $N$ and every $x > 0$,*

$$\mathbb{P}\big( |\eta_n^N(f) - \eta_n(f)| \ge x \big) \;\le\; 2(n+1) \exp\Big( -\frac{N x^2}{2 S_n^2} \Big),$$

$$\mathbb{E}\big[ (\eta_n^N(f) - \eta_n(f))^2 \big]^{1/2} \le \frac{S_n}{\sqrt{N}}.$$

*Proof.* Since $\Phi_{k,n} \circ \Phi_k = \Phi_{k-1,n}$, with the convention $\Phi_0(\eta_{-1}^N) = \eta_0$ the error telescopes:

$$\eta_n^N(f) - \eta_n(f) = \sum_{k=0}^{n} \Big[ \Phi_{k,n}(\eta_k^N)(f) - \Phi_{k,n}\big(\Phi_k(\eta_{k-1}^N)\big)(f) \Big].$$

For probability measures $\mu, \nu$, using $\nu\mathcal{Q}_{k,n}\big(f - \Phi_{k,n}(\nu)f\big) = 0$,

$$\Phi_{k,n}(\mu)(f) - \Phi_{k,n}(\nu)(f) = \frac{(\mu-\nu)\big(G_{k,n}\,h\big)}{\mu(G_{k,n})}, \qquad h = \mathcal{P}_{k,n}f - \Phi_{k,n}(\nu)(f).$$

Because $\Phi_{k,n}(\nu)(f)$ is a weighted average of the values of $\mathcal{P}_{k,n}f$, we have $|h| \le \operatorname{osc}(\mathcal{P}_{k,n}f) \le \varkappa_{k,n}$. Write $G_{k,n}h/\mu(G_{k,n}) = \tilde{h}\cdot\|G_{k,n}\|_\infty/\mu(G_{k,n})$ with $\tilde{h} = G_{k,n}h/\|G_{k,n}\|_\infty$, so that $|\tilde{h}| \le \varkappa_{k,n}$ and $\|G_{k,n}\|_\infty/\mu(G_{k,n}) \le g_{k,n}$. Take $\mu = \eta_k^N$ and $\nu = \Phi_k(\eta_{k-1}^N)$. Then $\tilde{h} = \tilde{h}_k$ is measurable with respect to generation $k-1$, and conditionally on that generation $\eta_k^N(\tilde{h}_k)$ is a mean of $N$ independent variables with values in an interval of length $2\varkappa_{k,n}$ and mean $\nu(\tilde{h}_k)$. Hoeffding's inequality gives

$$\mathbb{P}\big(|Z_k| \ge u\big) \le 2\exp\big(-Nu^2/(2\varkappa_{k,n}^2)\big), \qquad Z_k = (\eta_k^N - \nu)(\tilde{h}_k),$$

and the $k$th term is at most $g_{k,n}|Z_k|$ in absolute value. Allocate $x$ in proportion to $g_{k,n}\varkappa_{k,n}$: if $\sum_k g_{k,n}|Z_k| \ge x$ then $|Z_k| \ge x\varkappa_{k,n}/S_n$ for some $k$, and the union bound gives the tail estimate. For the second moment, Minkowski's inequality and $\mathbb{E}[Z_k^2] \le \varkappa_{k,n}^2/N$ give $\|\sum_k g_{k,n}|Z_k|\|_2 \le S_n/\sqrt{N}$. □

Remark 34.5.2 (a tail bound without the factor $n+1$). Under the hypotheses of Theorem 34.5.1, for every $f$ with $\operatorname{osc}(f) \le 1$, every $N$ and every $x > 0$,

$$\mathbb{P}\big(|\eta_n^N(f) - \eta_n(f)| \ge x\big) \le 2\exp\Big(-\frac{Nx^2}{2S_n^2}\Big).$$

*Proof.* Terms with $\varkappa_{k,n} = 0$ vanish; for the others put $w_k = g_{k,n}\varkappa_{k,n}/S_n$ and $V_k = |Z_k|/\varkappa_{k,n}$, with $Z_k$ as in the proof of Theorem 34.5.1. Conditionally on generation $k-1$ (on nothing for $k=0$), $Z_k/\varkappa_{k,n}$ is a mean of $N$ independent centred variables with values in intervals of length at most 2, so Hoeffding's lemma gives $\mathbb{E}\big[e^{\pm\lambda Z_k/\varkappa_{k,n}} \mid \text{generation } k-1\big] \le e^{\lambda^2/(2N)}$, hence $\mathbb{E}\,e^{\lambda V_k} \le 2e^{\lambda^2/(2N)}$ for $\lambda \ge 0$. The proof of Theorem 34.5.1 gives $|\eta_n^N(f) - \eta_n(f)|/S_n \le \sum_k w_k V_k$ with $\sum_k w_k = 1$, so by convexity of the exponential

$$\mathbb{E}\exp\Big(\lambda\,\frac{|\eta_n^N(f) - \eta_n(f)|}{S_n}\Big) \le \sum_k w_k\,\mathbb{E}\,e^{\lambda V_k} \le 2e^{\lambda^2/(2N)}.$$

Markov's inequality with $\lambda = Nx/S_n$ gives the bound. □

### 34.5.2 Stability gives bounds uniform in the number of steps

$S_n$ can grow with $n$. It does not when the mutation kernels mix. Suppose, as in Del Moral [109], that

$$M_k(x,\cdot) \ge \epsilon\, M_k(y,\cdot) \text{ for all } x, y, \qquad \sup G_k \le e^c \inf G_k \text{ for all } k.$$

Then $g_{k,n} \le e^c/\epsilon$, because $G_{k,n} = G_k\, M_{k+1}(G_{k+1,n})$ and the minorisation bounds the ratio of the second factor by $1/\epsilon$. The one step kernels of $\mathcal{P}_{k,n}$, namely

$$R_j(x, dy) = \frac{M_j(x, dy)\, G_{j,n}(y)}{M_j(G_{j,n})(x)},$$

satisfy $R_j(x,\cdot) \ge \epsilon^2 R_j(y,\cdot)$ (the ratio of the densities of $R_j(x,\cdot)$ and $R_j(y,\cdot)$ is $\big[M_j(x,dz)/M_j(y,dz)\big] \cdot \big[M_j(G_{j,n})(y)/M_j(G_{j,n})(x)\big]$, and each factor is at least $\epsilon$), so $\varkappa_{k,n} \le (1-\epsilon^2)^{n-k}$, since $\mathcal{P}_{k,n} = R_{k+1} \cdots R_n$. Hence $S_n \le 1 + e^c(1-\epsilon^2)\epsilon^{-3} \le e^c\epsilon^{-3}$ for every $n$.

The hypothesis is strong, and the kernels of the filter do not satisfy it.

Lemma 34.5.3 (Metropolis kernels are not ratio minorised). *Let $M$ be a Markov kernel on $E$, and suppose there are points $x, y$ and a measurable set $B$ with $M(y,B) > 0 = M(x,B)$. Then no $\epsilon > 0$ satisfies $M(x',\cdot) \ge \epsilon M(y',\cdot)$ for all $x', y'$. This is the case for a random walk Metropolis kernel whose proposals have a density, at any $y$ with positive rejection probability ($B = \{y\}$, $x \ne y$), for its powers, and for componentwise Metropolis sweeps ($B = \{z : z_1 = y_1\}$, $x_1 \ne y_1$, when rejection of the first coordinate has positive probability at $y$).*

*Proof.* $M(x,B) = 0 < \epsilon M(y,B)$. From $x$ the chain reaches the point $y$, or the hyperplane $z_1 = y_1$, only by accepting a proposal that lands there, an event of probability zero, in one step and in $m$ steps; from $y$ a rejection keeps it in $B$ with positive probability. □

If $M_k$ has a density $m_k(x,z)$ with respect to a common reference measure with $\alpha_k h_k(z) \le m_k(x,z) \le \alpha_k' h_k(z)$ for all $x, z$, then $M_k(x,\cdot) \ge (\alpha_k/\alpha_k')M_k(y,\cdot)$; exact Gibbs sweeps are an example when the target density is bounded above and below on $\Theta_\varsigma$ (defined below). With $S_n \le e^c\epsilon^{-3}$, Remark 34.5.2 gives $\mathbb{P}(|\eta_n^N(f) - \eta_n(f)| \ge x) \le 2\exp(-\frac{1}{2}Nx^2\epsilon^6 e^{-2c})$ for every $n$. The number of steps does not enter the bound.

### 34.5.3 The Cauchy flow

Corollary 34.5.4 (concentration along the Cauchy flow). *Fix $\varsigma > 0$ and restrict the parameter space to $\Theta_\varsigma = \{z : |\theta_l - \sigma_j| \ge \varsigma$ for all levels $\theta_l$ and zeros $j\}$, with ordinates in the window. Then the total log likelihood has finite oscillation*

$$\operatorname{osc}(\ell) \le \Omega_\varsigma = \frac{1}{2\varepsilon^2} \sum_{l,i} \Big( |y_l(v_i)| + \frac{K}{\pi\varsigma} \Big)^2,$$

*and a tempering schedule with $n = \max(1, \lceil \Omega_\varsigma / c \rceil)$ equal increments has potentials with $\sup G / \inf G \le e^c$. If the mutation kernels satisfy the ratio minorisation with constant $\epsilon$ (by Lemma 34.5.3 this needs kernels with densities, as in the*

*condition above), then for every posterior functional with oscillation at most one, in particular for the posterior probability that* $\max_j |\sigma_j - \frac{1}{2}|$ *exceeds any threshold,*

$$\mathbb{P}\big(|\eta_n^N(f) - \eta_n(f)| \geq x\big) \leq 2\exp\Big(-\tfrac{1}{2}Nx^2\epsilon^6 e^{-2c}\Big) \qquad \text{for every } N \text{ and } x > 0.$$

*Without the minorisation, Theorem 34.5.1 holds with its finite* $S_n$*, which is not uniform in* $n$*.*

*Proof.* On $\Theta_\varsigma$ each term $|Q_{\theta_l-\sigma_j}(v)|$ is at most $1/(\pi\varsigma)$. The marginal likelihood after integrating out the background is $\exp(-\frac{1}{2}r^\top\Sigma^{-1}r)$ with $\Sigma \succeq \varepsilon^2 I$, so its logarithm lies between $-\|r\|^2/(2\varepsilon^2)$ and 0, and $\|r\|^2$ is at most the sum in the display, since each entry of $r$ is at most $|y_l(v_i)| + K/(\pi\varsigma)$ in absolute value. A potential $L^{\Delta\lambda}$ then has $\sup / \inf \leq e^{\Delta\lambda\,\Omega_\varsigma} \leq e^c$ for $\Delta\lambda = 1/n \leq c/\Omega_\varsigma$. The rest is Theorem 34.5.1, Remark 34.5.2 and the stability bounds. □

Two remarks keep this honest. The restriction to $\Theta_\varsigma$ is needed: when a candidate zero approaches a grid point $(\theta_l, v_i)$, that is, when $\sigma_j \to \theta_l$ with $\gamma_j \to v_i$, the term $Q_{\theta_l-\sigma_j}$ blows up, the likelihood tends to zero, and $\inf G = 0$. This is the same singularity that makes the deconvolution informative, seen from the sampler's side. Under QRH($\theta$) the truncation is automatic for levels above $\theta + \varsigma$, provided the prior is restricted to $1 - \theta \leq \sigma_j \leq \theta$; the sampler's prior is uniform on $(0, 1)$. Second, the constants are explicit but pessimistic: $\Omega_\varsigma$ is of order $10^6$ for the grids of the source note (not recomputed here), and the minorisation constant, when it exists, enters as $\epsilon^6$. The theorem guarantees the $N^{-1/2}$ rate and exponential tails with a computable constant for the system it covers; it does not predict the constant we observe.

### 34.5.4 The rate in practice

To see the $N^{-1/2}$ rate, the source note reran the GBC initialised filter for $\zeta$ twelve times at each of $N \in \{250, 500, 1000, 2000\}$ and measured the spread across runs of three posterior functionals for the zero at 84.7355: the posterior mean of $\sigma$, the posterior mean of $\gamma$, and the posterior probability that $|\sigma - \frac{1}{2}| > 2 \times 10^{-4}$, which has oscillation one. These runs were not repeated for this book (they need the code, data and seeds of the source); Table 34.9 is computed by the source, and the sampler is the systematic and adaptive one that Theorem 34.5.1 does not cover.

For the probability functional, Theorem 34.5.1 gives root mean square error at most $S_n/\sqrt{N}$. The observed 0.038 at $N = 2000$ corresponds to an effective $S_n \approx 1.7$, far below what the worst case constants of Corollary 34.5.4 allow. This is a lower estimate, since the theorem bounds the root mean square error, bias included, while the spread across runs measures less. That the adaptive rule, which keeps the effective sample size at $N/2$, is what controls the constant in practice is a heuristic, and it is the adaptive version that remains to be analysed.

| $N$ | 250 | 500 | 1000 | 2000 |
|---|---|---|---|---|
| sd of posterior mean of $\sigma$ ($\times 10^{-5}$) | 6.5 | 5.5 | 3.2 | 2.8 |
| sd of posterior mean of $\gamma$ ($\times 10^{-5}$) | 6.9 | 6.7 | 2.5 | 2.3 |
| sd of posterior probability | 0.104 | 0.062 | 0.048 | 0.038 |

Table 34.9: Monte Carlo spread over twelve runs, computed in the source note. Least squares log log slopes of the printed values are $-0.44$, $-0.62$ and $-0.47$, against $-1/2$ from Theorem 34.5.1; with twelve runs per cell each standard deviation has a relative error of about 20%, so the slopes carry about $\pm 0.15$.

### 34.5.5 Two open problems

Remark 34.5.5 (the continuous level limit; heuristic, open). Let $\eta_\lambda \propto q^{1-\lambda}(\pi_0 L)^\lambda$ be the tempered flow, with $V = \log(\pi_0 L) - \log q$. Differentiating the density gives, exactly,

$$\frac{d}{d\lambda}\eta_\lambda(f) = \mathrm{Cov}_{\eta_\lambda}(V, f),$$

and Metropolis moves that leave $\eta_\lambda$ invariant do not change $\eta_\lambda(f)$. The particle version is a Moran system: each particle moves by the generator $\mathcal{L}_\lambda$ of the moves and, at rate $(V(X^j) - V(X^i))_+/N$, particle $i$ is replaced by a copy of particle $j$; its mean drift for the empirical measure is $\mathrm{Cov}_{\eta_\lambda^N}(V, f)$. As the level spacing tends to zero one expects convergence at rate $N^{-1/2}$ and a Gaussian limit for the fluctuations, by the martingale problem method (Ethier and Kurtz [128], as used by Del Moral and Miclo [118]), with an asymptotic variance that in discrete time is $\sum_k \eta_k\big[(G_{k,n}/\eta_k G_{k,n})^2(\mathcal{P}_{k,n}f - \eta_n f)^2\big]$ (the formula of the source, quoted from the literature and unchecked here, see [109]). This is heuristic: no scaling of the likelihood increments, the tolerance $\varepsilon$, the mutation times and the resampling rates has been fixed, and giving every new level the same tolerance $\varepsilon$ multiplies in more and more information, so that the log likelihood of a mesh $h$ is of order $1/h$ and need not converge to a finite potential. On a finite state space with bounded potentials and mutation generators the limit is the corresponding ordinary differential equation and the Moran approximation has error $O(N^{-1/2})$; for the Cauchy flow $V$ is unbounded below near the levels, as in Corollary 34.5.4, and on $\Theta_\varsigma$ it is bounded. A continuous time limit theorem for the singular potentials, with a Lyapunov function that controls how often particles approach the singular set $\{\sigma_j = \theta_l\}$, and the corresponding result for the adaptive schedule (Beskos, Jasra, Kantas and Thiery [25]), are open problems.

Remark 34.5.6 (concentration under adaptive tempering; heuristic, open). Theorem 34.5.1 assumes a fixed schedule. The filter of Section 34.4 chooses each tempering increment so that the effective sample size of the incremental weights is $N/2$, which says that the empirical relative second moment $\eta_k^N(G_k^2)/\eta_k^N(G_k)^2$

equals 2. The proof of Theorem 34.5.1 uses the oscillation ratio $g_{k,n}$ only to bound $\|G_{k,n}\|_\infty/\eta_k^N(G_{k,n})$. A natural idea is to replace that sup norm bound by a second moment bound, so that by Cauchy Schwarz each term of the telescoping sum is controlled by $\eta_k(G_{k,n}^2)/\eta_k(G_{k,n})^2$, the quantity that the adaptive rule keeps near 2 for one step. The non asymptotic variance theorem of Cérou, Del Moral and Guyader [77] is reported to give the resulting bound for a fixed schedule with such relative moments, and Beskos, Jasra, Kantas and Thiery [25] show that the adaptive schedule converges to its deterministic limit (both statements are the source's and the papers were not read for this book). Combining the two, with the singular potentials of the Cauchy flow kept away from the levels as in Corollary 34.5.4, would give a version of Theorem 34.5.1 whose constant is set by the effective sample size threshold rather than by $\sup G/\inf G$. This sketch is heuristic: control of the one step relative second moment says nothing about $\eta_k(G_{k,n}^2)/\eta_k(G_{k,n})^2$ for $k < n-1$, because $G_{k,n}$ involves all later potentials, so a multistep moment bound does not follow. That version, the one that would match the constants observed, is open.

## 34.6 From $\mathfrak{U}_1$ to $\mathfrak{U}_{3/4}$

The level $\theta = \frac{3}{4}$ sits halfway between the two ends of the deconvolution, and it is the level of QRH(3/4). There are two ways to obtain $\mathfrak{U}_{3/4}$.

*Directly.* $\mathfrak{U}_{3/4}\big((0,V]\big) = \int_0^V u_{3/4}(v)\,dv$ with $u_{3/4}(v) = \pi^{-1}\operatorname{Re}\xi'/\xi(\frac{3}{4}+iv)$, evaluated to high precision. This is a forward computation and needs no filter. On $[78, 94]$, on a grid of spacing 0.02, we find the following (computed, `code/cf_xi.py`, `data/cf_xi.json`).

- For $\zeta$, $u_{3/4}$ is positive on the grid, with minimum 0.0734 at $v = 90.68$ between the zeros 88.81 and 92.49, and $\mathfrak{U}_{3/4}\big((78,94]\big) = 6.162$, close to the six zeros in the window spread by Cauchy kernels of scale 1/4. Positivity for every $\theta > \frac{1}{2}$ at these heights is Corollary 11.4.7, not a consequence of the grid values.

- For Davenport and Heilbronn, $u_{3/4}$ has a negative spike of depth $-4.80$ at $v = 85.70$, and the negative part of $\mathfrak{U}_{3/4}$ on the window has mass 0.578. Its total increment is 8.763. The spike comes from the zero $0.8085 + 85.699i$, whose real part exceeds $\frac{3}{4}$: it contributes the negative Cauchy term $Q_{3/4-0.8085}$ of scale 0.0585, partly cancelled by the positive terms (Theorem 34.7.1).

*By the backward flow.* The test of the method is to predict $u_{3/4}$ from a fit to the two levels $\theta \in \{1, 0.9\}$: fit the filter to these levels and push each posterior particle

forward to $\theta = \frac{3}{4}$ through the Hadamard sum (32.1.1),

$$u_{3/4}(v \mid z) = \sum_j Q_{3/4-\sigma_j}(v - \gamma_j) + b_{3/4}(v \mid z),$$

with the background coefficients refitted given $z$ (to their conditional mean, so that the band below is the band of the conditional mean prediction and omits the uncertainty of those coefficients). This is the Cauchy deconvolution in action: the data are on lines to the right, and the prediction is to the left. Since $Q_{3/4-\sigma}$ changes sign when $\sigma$ crosses $\frac{3}{4}$, the prediction is only as good as the posterior for each $\sigma_j$ relative to $\frac{3}{4}$, which is the detectability problem of Proposition 34.3.1 in another form.

The results of the source note, with $N = 1000$ particles and the GBC initialised filter of Section 34.4 (its initial draws come from the GBC screen, which uses the levels 1, 0.85, 0.7 and 0.6), were not repeated for this book (they need the code, data and seeds of the source; the figure of the source is not reproduced). They are computed, as follows.

- For $\zeta$, the filter needed 7 tempering steps and $3.6 \times 10^4$ likelihood evaluations (one second). The predicted density of $\mathfrak{U}_{3/4}$ on $[80, 92]$ has median absolute error 0.0012 and maximum error 0.008 against the direct computation, and the 95% band covers the truth at every grid point.

- For Davenport and Heilbronn, again 7 steps and two seconds. From the fit to the levels 1 and 0.9 the filter places the pair at $\gamma = 85.699$ with real parts $\sigma = 0.8082$ (interval $[0.8080, 0.8084]$, which excludes the true value 0.80852, so the posterior is overconfident, as in Section 34.4) and $\sigma = 0.1987$, against the true 0.8085 and 0.1915. It then predicts the negative spike of $\mathfrak{U}_{3/4}$: median absolute error 0.0019 and maximum error 0.042 against a spike of depth 4.8, with the band covering 80% of grid points. The misses are at the bottom of the spike, where a bias of $3 \times 10^{-4}$ in $\sigma$ is amplified by a factor of order $1/(\pi(\sigma - \frac{3}{4})^2) \approx 90$.

So, within the window model of Section 34.3 (the zeros in the window and a cubic background), the filter predicts $u_{3/4}$ on the window from the levels 1 and 0.9, to about two digits at the bottom of the spike and to three for the median error, including the failure of positivity caused by a zero off the line between the data and the target. The run was initialised from GBC draws computed from the levels 1, 0.85, 0.7 and 0.6, so it is not a computation from $\theta \geq 0.9$ alone; a clean rerun with a screen trained on the levels $\{1, 0.9\}$ only has not been done (a question for the authors). Exact data on a window determine $u_{3/4}$ only through analytic continuation (Proposition 34.6.1), which is unstable. Positivity of $\mathfrak{U}_\tau$ at all heights for every $\tau > \frac{3}{4}$ is QRH(3/4) (Theorem 31.9.1), and no finite computation gives that.

PROPOSITION 34.6.1 (what one level determines). *(a) Let $F_1, F_2$ be real entire functions of order at most one with all zeros in $0 < \operatorname{Re} s < 1$, and let $u_\theta^{(j)}(v) = \pi^{-1}\operatorname{Re}(F_j'/F_j)(\theta+iv)$. If $u_1^{(1)} = u_1^{(2)}$ on an interval, then $F_1'/F_1 = F_2'/F_2$; in particular $u_\theta^{(1)} = u_\theta^{(2)}$ for every $\theta$.*

*(b) The order hypothesis cannot be dropped: $F = \xi e^G$ with $G(s) = -(10\pi)^{-1}\cos(10\pi(s-1))$ is real entire with the zeros of $\xi$, its densities at the levels 1 and 0.9 equal $u_1$ and $u_{0.9}$ on the whole line, and its density at $\frac34$ is $u_{3/4}(v) - \pi^{-1}\cosh(10\pi v)$.*

*(c) If no zero has $\sigma_\rho \ge \frac34$ (in particular under RH), then $u_1 = P_{1/4} * u_{3/4}$: recovering $u_{3/4}$ from $u_1$ inverts a Poisson smoothing, whose Fourier multiplier is $e^{-|\omega|/4}$, so an error at frequency $\omega$ grows by the factor $e^{|\omega|/4}$.*

*Proof.* (a) Neither $F_j$ vanishes on $\operatorname{Re} s = 1$, so $v \mapsto u_1^{(j)}(v)$ is real analytic and the two densities agree on $\mathbb{R}$. By Hadamard's factorisation, $\pi u_\theta^{(j)}(v) = c_j + \sum_\rho (\theta - \sigma_\rho)/((\theta-\sigma_\rho)^2 + (v-\gamma_\rho)^2)$, with a real constant $c_j$ and an absolutely convergent sum over the zeros $\rho = \sigma_\rho + i\gamma_\rho$ of $F_j$, because $\sum_\rho |\rho|^{-2} < \infty$. For $\theta > 1$ every term of the sum is positive and $P_{\theta-1} * P_{1-\sigma_\rho}(\cdot - \gamma_\rho) = P_{\theta-\sigma_\rho}(\cdot - \gamma_\rho)$, so by Tonelli $u_\theta^{(j)} = P_{\theta-1} * u_1^{(j)}$. Hence $\operatorname{Re}(F_1'/F_1 - F_2'/F_2) = 0$ on $\operatorname{Re} s > 1$, so $F_1'/F_1 - F_2'/F_2$ is an imaginary constant there, which is 0 because both functions are real on the real axis; by the identity theorem they agree on $\mathbb{C}$.

(b) $G'(s) = g(s) = \sin(10\pi(s-1))$ and $F'/F = \xi'/\xi + g$. On $\operatorname{Re} s = 1$, $g(1+iv) = i\sinh(10\pi v)$; on $\operatorname{Re} s = 0.9$, $g(0.9+iv) = \sin(-\pi + 10\pi i v) = -i\sinh(10\pi v)$; both are purely imaginary. On $\operatorname{Re} s = \frac34$, $g(\frac34 + iv) = \sin(-\frac{5\pi}{2} + 10\pi i v) = -\cosh(10\pi v)$.

(c) By the Hadamard sum (32.1.1), every zero then contributes $P_{3/4-\sigma_\rho}(\cdot - \gamma_\rho)$ to $u_{3/4}$, with $\frac34 - \sigma_\rho > 0$, and $P_{1/4} * P_{3/4-\sigma_\rho} = P_{1-\sigma_\rho}$; sum by Tonelli. Finally $\int e^{i\omega x} P_a(x)\,dx = e^{-a|\omega|}$. □

## 34.7 Negative mass and converse results

Throughout this section $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta+iv)$ is the density of $\mathfrak{U}_\theta$, real parts of zeros are $\sigma_\rho$ and levels are $\theta$, as in Section 34.1. By the Hadamard sum (32.1.1) taken at level $\theta$,

$$u_\theta(v) = \sum_\rho P_{\theta-\sigma_\rho}(v-\gamma_\rho), \qquad P_a(x) = \frac{a}{\pi(a^2+x^2)} \quad (a \in \mathbb{R}),$$

the sum over the zeros of both half planes, paired as $\rho, \bar\rho$ so that it converges absolutely, with $P_0 = \delta_0$. For $\theta \ge \sup \sigma_\rho$ every term is nonnegative, and this is the measure of (30.1.1). For smaller $\theta$ the terms with $\sigma_\rho > \theta$ are negative, and the question of this section is how much they can cost. The results are of three

kinds. Zero density bounds the negative mass (Theorem 34.7.1, proved), and a computed example for the Davenport and Heilbronn function shows the bound at work (Table 34.10, computed). A converse, in which negative mass forces zeros, is proved for separated zeros and in a weighted form for all zeros (Proposition 34.7.2, Corollary 34.7.3, Theorem 34.7.4). Under QRH($\theta_0$) the density has a margin (Proposition 34.7.5, conditional on QRH($\theta_0$)).

### 34.7.1 How negative the density can be

Problem 34.2. How negative can $\mathfrak{U}_\theta$ be, for $\frac12 < \theta < 1$, without assuming RH?

Write $N(\theta, T)$ for the number of zeros with $\sigma_\rho > \theta$ and $0 < \gamma_\rho \le T$, counted with multiplicity, and $u_\theta^- = \max(-u_\theta, 0)$.

Theorem 34.7.1 (zero density controls the negative Thorin mass). *Let $\frac12 < \theta < 1$ and $T \ge 2$.*

*(a)* $\displaystyle\int_0^T u_\theta^-(v)\,dv \le N\big(\theta, T + \tfrac12\big).$

*(b) If $T$ is not an ordinate and no zero with $0 < \gamma_\rho < T$ has real part $\theta$, then*

$$\mathfrak{U}_\theta\big((0,T]\big) = \int_0^T u_\theta(v)\,dv = \frac{T}{2\pi}\log\frac{T}{2\pi e} - 2N(\theta,T) + O(\log T).$$

*(c) Consequently, for $T$ as in (b), Selberg's density theorem $N(\theta,T) \ll T^{1-(\theta-\frac12)/4}\log T$ [477] shows that the negative part of $\mathfrak{U}_\theta$ on $(0,T]$ is a fraction $O\big(N(\theta,T+\frac12)/(T\log T)\big) = O\big(T^{-(\theta-\frac12)/4}\big)$ of its mass.*

*Proof.* (a) Group the zeros in the Hadamard sum into orbits of $\rho \mapsto 1-\bar\rho$, which preserves ordinates. A zero on the critical line contributes $P_{\theta-1/2}(v-\gamma_\rho) > 0$, and an orbit with $\frac12 < \sigma_\rho \le \theta$ contributes $Q_{\theta+\sigma_\rho-1}(v-\gamma_\rho) + Q_{\theta-\sigma_\rho}(v-\gamma_\rho) \ge 0$, with $Q_0 = 0$ off the ordinate. An orbit with $\sigma_\rho > \theta$ contributes $P_a(v-\gamma_\rho) - P_b(v-\gamma_\rho)$ with $a = \theta + \sigma_\rho - 1 > b = \sigma_\rho - \theta > 0$, and

$$\pi\big(P_a(x) - P_b(x)\big) = \frac{(a-b)(x^2-ab)}{(a^2+x^2)(b^2+x^2)},$$

so its negative part is at most $P_b(x)$ for $x^2 < ab$ and vanishes for $x^2 \ge ab$, where $ab < \big(\frac{a+b}{2}\big)^2 = (\sigma_\rho - \frac12)^2 < \frac14$ (as in the proof of Proposition 3.3.1(ii)). Since the negative part of a sum is at most the sum of the negative parts, $u_\theta^-(v) \le \sum_{\sigma_\rho>\theta} P_{\sigma_\rho-\theta}(v-\gamma_\rho)\,\mathbf{1}\{|v-\gamma_\rho| < \frac12\}$, the sum over zeros of both signs of $\gamma_\rho$. On $[0,T]$ only zeros with $-\frac12 < \gamma_\rho < T + \frac12$ contribute, each at most $\int P_b = 1$, and no zero has $|\gamma_\rho| < \frac12$; this gives (a).

(b) Apply the argument principle to $\xi$ on the rectangle with vertices $\theta$, $2$, $2+iT$, $\theta+iT$, positively oriented. It contains the $N(\theta,T)$ zeros with $\sigma_\rho > \theta$ and

$0 < \gamma_\rho < T$, and $\xi$ is real and positive on $[\theta, 2]$. Writing $\Delta_I$ for the continuous change of $\arg \xi$ along a segment $I$,

$$\Delta_{[\theta,\theta+iT]} = \Delta_{[2,2+iT]} + \Delta_{[2+iT,\theta+iT]} - 2\pi N(\theta, T).$$

The left side is $\pi \int_0^T u_\theta$ (Theorem 30.1.1). As in the proof of the Riemann von Mangoldt formula, $\Delta_{[2,2+iT]} + \Delta_{[2+iT,\frac{1}{2}+iT]} = \pi N(T)$, and the change along the horizontal segment from $\frac{1}{2} + iT$ to $\theta + iT$ is $O(\log T)$, by Stirling's formula for the Gamma factor and, for $\zeta$, by the bound that gives $S(T) = O(\log T)$. With $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T)$ this is (b); it is Proposition 31.9.2 in the present notation.

(c) By (b) and $N(\theta, T) = o(T \log T)$, $\mathfrak{U}_\theta((0, T]) = \frac{T}{2\pi} \log T \, (1 + o(1))$; divide (a) by it and apply Selberg's bound to $N(\theta, T + \frac{1}{2})$. □

The display in (b) without the term $-2N(\theta, T)$ holds under RH and is equivalent to $N(\theta, T) = O(\log T)$, which is open. Theorem 11.4.2 gives better exponents for statements of this kind (Ingham below $\frac{7}{10}$, Guth and Maynard above), so (c) is a consequence of it; what (a) adds is the constant, with no tail term. Ingham's bound $N(\theta, T) \ll T^{3(1-\theta)/(2-\theta)} \log^5 T$ is better than Selberg's near $\theta = 1$, the density hypothesis $N(\theta, T) \ll T^{2(1-\theta)+\epsilon}$ would make the fraction $O(T^{1-2\theta+\epsilon})$, and RH makes it zero. Define the Thorin defect exponent, with the convention $\log 0 = -\infty$,

$$\operatorname{def}(\theta) = \limsup_{T \to \infty} \frac{\log \int_0^T u_\theta^-(v)\, dv}{\log T}.$$

By Theorem 34.7.1(a), $\operatorname{def}(\theta)$ is at most the density exponent $\limsup \log N(\theta, T)/\log T$ (proved). It equals $-\infty$ under RH.

For the Davenport and Heilbronn function $f$ (Section 34.2) the bound can be seen at work. Its density $u_\theta$ is the same expression with $\Lambda_{\mathrm{DH}}$ of Proposition 34.8.3 in place of $\xi$, and the pairing argument of (a) applies to its zeros with $\frac{1}{2} < \sigma_\rho < 1$. Table 34.10 gives the negative mass of $u_\theta$ on intervals of length 3 centred at the two lowest zeros off the line, $0.8085\ldots + 85.6993\ldots i$ and $0.6508\ldots + 114.1633\ldots i$ (certified, Section 34.2). Within each interval only the zero at its centre can make $u_\theta$ negative, by the pairing, and it contributes at most one when $\sigma_\rho > \theta$ and nothing when $\sigma_\rho \le \theta$. The mass is largest just below $\sigma_\rho$ and decays as the level moves further left, because the negative Cauchy term spreads out and is absorbed by the positive background of the other zeros.

Problem 34.3 (converse). Does $\operatorname{def}(\theta) \le A(1-\theta)$ for all $\theta$ imply $N(\theta, T) \ll T^{A(1-\theta)+\epsilon}$? A zero with $\sigma_\rho > \theta$ forces negative mass at levels just below $\sigma_\rho$ (Theorem 31.9.1), but nearby zeros with $\sigma_\rho$ just below the level contribute large positive terms, and the cancellation is not controlled. A converse would make the negative Thorin mass a proxy for zero density. This is open in general; the next subsection settles the case of separated zeros, and bounds the general case from below.

| level $\theta$ | 0.9 | 0.79 | 0.7 | 0.6 |
|---|---|---|---|---|
| negative mass near 85.70 ($\sigma_\rho = 0.8085$) | 0 | 0.763 | 0.424 | 0.195 |
| bound for this interval | 0 | 1 | 1 | 1 |
| negative mass near 114.16 ($\sigma_\rho = 0.6508$) | 0 | 0 | 0 | 0.454 |
| bound for this interval | 0 | 0 | 0 | 1 |

Table 34.10: Negative mass of $u_\theta$ for the Davenport and Heilbronn function on intervals of length 3 centred at its two lowest zeros off the line (computed, `code/cf_xi.py`, `data/cf_xi.json`). The bound is the number of zeros with $\sigma_\rho > \theta$ at the centre.

### 34.7.2 A partial converse

For a zero $\rho$, of multiplicity $k(\rho)$, let $d(\rho)$ be the distance from $\rho$ to the nearest other zero, and

$$\mathrm{cr}(\rho) = \sum_{\rho' \neq \rho,\ |\gamma_{\rho'} - \gamma_\rho| \le 2} \frac{k(\rho')}{|\rho' - \rho|},$$

the sum over distinct zeros, the reflected zero $1 - \bar\rho$ included. Multiplicity enters as a weight, and $d(\rho)$ is the distance to a distinct zero location.

Proposition 34.7.2 (a separated zero forces negative Thorin mass). *Let $\rho_0 = \sigma_0 + i\gamma_0$ be a zero of multiplicity $k$ with $\sigma_0 > \frac12$ and $\gamma_0 \ge 3$. There is an absolute constant $C_1 \ge 1$ such that if*

$$r \le \min\Big(1,\ \sigma_0 - \tfrac12,\ \tfrac12 d(\rho_0),\ \frac{k}{\pi\left(\mathrm{cr}(\rho_0) + C_1 \log \gamma_0\right)}\Big),$$

*then*

$$\int_{\sigma_0 - r}^{\sigma_0} \int_{\gamma_0 - r}^{\gamma_0 + r} u_\theta^-(v)\, dv\, d\theta \ \ge\ \frac{k\, r}{\pi}.$$

*Proof.* Let $D^-$ be the half disc $\{|s - \rho_0| < r,\ \mathrm{Re}\, s < \sigma_0\}$, which lies in $\{\mathrm{Re}\, s > \frac12\}$ and in the rectangle of the claim. Since $u_\theta^- \ge -u_\theta$, the integral is at least $-\iint_{D^-} u\, dA$, with $u(\theta + iv) = u_\theta(v)$. Split $u$ by the Hadamard sum. In polar coordinates $s - \rho_0 = ue^{i\phi}$ with $\phi \in (\frac{\pi}{2}, \frac{3\pi}{2})$ the own term is

$$\frac{k}{\pi} \iint_{D^-} \mathrm{Re}\, \frac{1}{s - \rho_0}\, dA = \frac{k}{\pi} \int_0^r \int_{\pi/2}^{3\pi/2} \cos\phi\, d\phi\, du = -\frac{2kr}{\pi}.$$

A zero $\rho'$ with $|\gamma_{\rho'} - \gamma_0| \le 2$ is at distance at least $2r$ from $\rho_0$, so $|s - \rho'| \ge |\rho' - \rho_0|/2$ on $D^-$ and its term is at most $\pi^{-1} k(\rho') \big(\frac{\pi r^2}{2}\big)\big(\frac{2}{|\rho' - \rho_0|}\big)$ in absolute value; these sum to at most $r^2 \mathrm{cr}(\rho_0)$. For $|\gamma_{\rho'} - \gamma_0| > 2$ we have $|\theta - \sigma_{\rho'}| \le 1$ and $|v - \gamma_{\rho'}| \ge$

$|\gamma_{\rho'}-\gamma_0|/2$, so $|\mathrm{Re}(s-\rho')^{-1}|\le 4/(\gamma_{\rho'}-\gamma_0)^2$, and $\sum_{|\gamma'-\gamma_0|>2}(\gamma'-\gamma_0)^{-2}\ll\log\gamma_0$; these contribute at most $\frac12 C_1 r^2\log\gamma_0$. Hence

$$-\iint_{D^-} u\,dA \;\ge\; \frac{2kr}{\pi}-r^2\big(\mathrm{cr}(\rho_0)+\tfrac12 C_1\log\gamma_0\big) \;\ge\; \frac{kr}{\pi}$$

by the choice of $r$. □

Corollary 34.7.3 (two sided control for separated zeros). *Let $\frac12<\theta_0<1$, $T\ge 3$ and $E(\theta_0,T)=\int_{\theta_0}^1\int_0^{T+1}u_\theta^-(v)\,dv\,d\theta$. Let $N_{\mathrm{sep}}(\theta_0,T;c,\delta)$ count the zeros with $\sigma_\rho>\theta_0$, $3\le\gamma_\rho\le T$, $\mathrm{cr}(\rho)\le c$ and distance at least $2\delta$ from every other zero. With $r=\min\big(\delta,\ 1/(\pi(c+C_1\log T))\big)$,*

$$\begin{aligned}\frac{r}{\pi}\,N_{\mathrm{sep}}(\theta_0+r,T;c,\delta) \;&\le\; E(\theta_0,T) \;\le\; \int_{\theta_0}^1 N\big(\theta,T+\tfrac32\big)\,d\theta\\ &=\sum_{\sigma_\rho>\theta_0,\ 0<\gamma_\rho\le T+\frac32}(\sigma_\rho-\theta_0).\end{aligned}$$

*Proof.* The half discs of distinct separated zeros are disjoint, and for $\sigma_\rho>\theta_0+r$ they lie in $\{\theta_0<\theta<1,\ 0<v<T+1\}$, so the lower bound adds Proposition 34.7.2 over them (with $r\le\sigma_\rho-\frac12$ because $\theta_0>\frac12$, and $r<1$). The upper bound integrates Theorem 34.7.1(a), applied with $T+1$ in place of $T$, over $\theta$. □

The lower bound counts only separated zeros and the upper bound counts every zero, weighted by its distance to the level $\theta_0$, so the two bounds meet only when the zeros to the right of $\theta_0$ are separated at scale $\delta\gg 1/\log T$ with crowding $c=O(\log T)$; then $r$ is of order $1/\log T$ and the two sides agree up to powers of $\log T$. Clusters are the open case: if many zeros off the line crowd together, their positive and negative Cauchy terms can cancel, and the negative mass can undercount them. Theorem 34.7.4 quantifies this instead of excluding it.

Theorem 34.7.4 (every zero off the line forces negative Thorin mass). *Let $\theta_0>\frac12$ and, for a zero $\rho$ with $\sigma_\rho>\theta_0$,*

$$r(\rho)=\min\Big(1,\ \sigma_\rho-\theta_0,\ \tfrac12 d(\rho),\ \frac{k(\rho)}{\pi\,\big(\mathrm{cr}(\rho)+C_1\log\gamma_\rho\big)}\Big),$$

*with $C_1$ the constant of Proposition 34.7.2. Then for $T\ge 3$, with $E$ as in Corollary 34.7.3,*

$$E(\theta_0,T)\;\ge\;\frac{1}{\pi}\sum_{\sigma_\rho>\theta_0,\ 3\le\gamma_\rho\le T}k(\rho)\,r(\rho),$$

*the sum running over distinct zeros. Moreover, for an absolute $c>0$,*

$$r(\rho)\ge c\min\big(\sigma_\rho-\theta_0,\ \min(d(\rho),1)/\log\gamma_\rho\big),$$

*and so*

$$E(\theta_0,T)\ \gg\ \frac{1}{\log T}\sum_{\sigma_\rho>\theta_0,\ 3\le\gamma_\rho\le T}\min\big((\sigma_\rho-\theta_0)\log T,\ d(\rho),\ 1\big).$$

*Proof.* The proof of Proposition 34.7.2 applies to each zero with its own radius $r(\rho)$, and the half discs $D^-(\rho)$ are pairwise disjoint because $r(\rho)\le d(\rho)/2$. They lie in $\{\theta_0<\theta<1,\ 0<v<T+1\}$, so their contributions add. For the second statement, at most $C_2\log\gamma_\rho$ zeros, counted with multiplicity, satisfy $|\gamma'-\gamma_\rho|\le 2$, each at distance at least $d(\rho)$, so $\mathrm{cr}(\rho)\le C_2\log\gamma_\rho/d(\rho)$. Replacing $d(\rho)$ by $\min(d(\rho),1)$ in this bound gives $k/(\pi(\mathrm{cr}+C_1\log\gamma_\rho))\ge\min(d(\rho),1)/(\pi(C_2+C_1)\log\gamma_\rho)$, and, enlarging $C_1$ and $C_2$ if necessary, $\frac12 d(\rho)$ is larger than this for $\gamma_\rho\ge 3$. The last display follows from $k(\rho)\ge 1$ and $\log\gamma_\rho\le\log T$. □

So every zero off the line pays, in negative Thorin mass integrated over levels and heights, at least its isolation distance divided by $\log T$, or its distance from the level $\theta_0$ if that is smaller; a cluster of $M$ zeros at mutual distances at least $d\le 1$, each at least $d/\log T$ to the right of $\theta_0$, pays at least about $Md/\log T$. Suppose, as a hypothesis (conditional, and heuristic since no zero off the line is known), that every zero with $\sigma_\rho>\theta_0$ has $d(\rho)\ge c'/\log\gamma_\rho$, which is the typical isolation for zeros with the sine kernel statistics of the zeros of $\zeta$. Then Theorem 34.7.4 and Corollary 34.7.3 give, for every $\epsilon>0$,

$$\frac{N(\theta_0+\epsilon,T)}{(\log T)^2}\ \ll_\epsilon\ E(\theta_0,T)\ \ll\ \int_{\theta_0}^{1}N\big(\theta,T+\tfrac32\big)\,d\theta,$$

a two sided equivalence between negative Thorin mass and zero density up to powers of $\log T$ (conditional on that hypothesis). The step from isolation of a typical zero to every zero is not proved.

### 34.7.3 A check on the Davenport and Heilbronn function

Table 34.11 illustrates the mechanism of Proposition 34.7.2 on the Davenport and Heilbronn function, by integrating $u_\theta^-$ numerically over the left half disc of radius $r$ around each zero off the line. In every row the negative mass lies between $r/\pi$ and the own term $2r/\pi$; the gap measures the cancellation from the other zeros. The table illustrates the mechanism and does not check the proposition: the lower bound $r/\pi$ is proved only for $r$ below the explicit but unspecified threshold of the proposition, with the constant $C_1$ not given.

### 34.7.4 Positivity at 7/8 and what it gives

Problem 34.4. What does QRH($\theta_0$), the absence of zeros with $\sigma_\rho>\theta_0$, give in this framework, for instance at $\theta_0=\frac78$?

| zero | $r$ | negative mass on $D^-$ | $r/\pi$ | $2r/\pi$ |
|---|---|---|---|---|
| $0.8085 + 85.6993\,i$ | 0.05 | 0.0296 | 0.0159 | 0.0318 |
| $0.8085 + 85.6993\,i$ | 0.10 | 0.0545 | 0.0318 | 0.0637 |
| $0.8085 + 85.6993\,i$ | 0.14 | 0.0710 | 0.0446 | 0.0891 |
| $0.6508 + 114.1633\,i$ | 0.05 | 0.0275 | 0.0159 | 0.0318 |
| $0.6508 + 114.1633\,i$ | 0.10 | 0.0457 | 0.0318 | 0.0637 |
| $0.6508 + 114.1633\,i$ | 0.14 | 0.0530 | 0.0446 | 0.0891 |

Table 34.11: Negative mass of $u_\theta$ on the left half disc around the zeros of the Davenport and Heilbronn function off the line, on a $40 \times 81$ grid, as reported in the source note and not recomputed here. In every row it lies between $r/\pi$ and $2r/\pi$.

A zero free half plane $\operatorname{Re} s > \frac{7}{8}$ is announced for Dirichlet $L$ functions in a preprint of OpenAI with a Lean formalisation, with a small improvement claimed by Liu and a related preprint attributed to Thorner (unchecked), none of them refereed (Section 29.7); every statement below that uses $\mathrm{QRH}(\frac{7}{8})$ is conditional on those announcements. By Theorem 31.9.1, $\mathrm{QRH}(\theta_0)$ is equivalent to positivity of $\mathfrak{U}_\tau$ for every $\tau > \theta_0$, while positivity at the single level $\theta_0$ is necessary and not known to be sufficient (Remark 1.2.4). The Balazard, Saias and Yor criterion for $\mathrm{QRH}(\theta_0)$ as a vanishing defect of Cauchy averages of $\log|\zeta|$, with its computations at $\theta_0 = \frac{7}{8}$, is in Theorem 35.2.1 (Chapter 35). Here is the margin.

Proposition 34.7.5 (positivity margin, conditional on $\mathrm{QRH}(\theta_0)$). *Assume* $\mathrm{QRH}(\theta_0)$ *with* $\frac{1}{2} \le \theta_0 < 1$. *There are constants* $c > 0$ *and* $v_0$ *such that* $u_\theta(v) \ge c\,(\theta - \theta_0)\log v$ *for* $\theta_0 < \theta \le 1$ *and* $v \ge v_0$.

*Proof.* Every zero has $1 - \theta_0 \le \sigma_\rho \le \theta_0$, so every term $P_{\theta-\sigma_\rho}(v - \gamma_\rho)$ of the Hadamard sum has width $\theta - \sigma_\rho \in [\theta - \theta_0, 1)$ and is positive. For $|v - \gamma_\rho| \le 1$ it is at least $(\theta - \theta_0)/(2\pi)$, because $a^2 + x^2 \le 2$. By the Riemann von Mangoldt formula, the number of zeros with $|\gamma_\rho - v| \le 1$ is $\pi^{-1}\log(v/2\pi) + O(1) + S(v+1) - S(v-1)$, and the explicit bound $|S(T)| \le 0.1095\log T + 0.2042\log\log T + 3.0305$ of Hasanalizade, Shen and Wong [176] (as used in Proposition 9.1.11) leaves at least $\big(\pi^{-1} - 0.219 - o(1)\big)\log v \ge 0.09\log v$ of them for large $v$, since $\pi^{-1} = 0.3183\ldots$. Summing gives the claim with $c = 0.09/(2\pi)$. $\square$

The particle filter can start at the level $\theta = \frac{7}{8}$ without any hypothesis, since the levels at or above 0.9 carry only 1.6% of $\sum a_l^{-3}$ and 0.1% of $\sum a_l^{-5}$ in the schedule of Section 34.3 (computed). Conditionally on $\mathrm{QRH}(\frac{7}{8})$ the prior can also be truncated to real parts in $[\frac{1}{8}, \frac{7}{8}]$. Together this helps only a little, because neither touches the informative levels between $\frac{1}{2}$ and $\frac{7}{8}$.

### 34.7.5 Certificates

Problem 34.5. Turn the screen of Section 34.4 into a proof in each window.

The posterior of the screen is calibrated on simulations, so it is evidence and not proof. A certificate in a window needs the contour moment $M_0$ of Proposition 34.4.1 computed in interval arithmetic and shown to be less than $\frac{1}{2}$; this is the argument principle, as in Section 31.10 (Theorem 31.9.6 and Section 31.9.7 treat certified counts). A certificate is valid whatever its cost; the screen would move the verified height $3 \times 10^{12}$ [419] only if it were cheaper per zero than the existing methods, which we do not expect. The role of the screen is to choose where to spend the expensive rigorous integrations, and how close to the line the rectangle can be placed.

## 34.8 Zero free regions from positivity

Section 34.7 used the Hadamard sum, in which every zero contributes a Cauchy term. Here the Euler product enters, as an upper bound on a positive combination of values of $u_\theta$ at several heights and, in the multilevel form, at several levels. We characterise the admissible combinations, optimise the zero free constant they give, and prove that no member of the class gives more than a region of width $c/\log\gamma$ with $c < 1$ (Proposition 34.8.7, proved). The linear programming values, the Stechkin constant and the optimised constants are computed. Where the book already proves a statement (the de la Vallée Poussin region is Proposition 9.1.4) it is cited, and what is new is the general form of the inequality.

### 34.8.1 The Euler product as an inequality between Cauchy data

From $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, for $s = \theta + iv$,

$$\begin{aligned} \pi u_\theta(v) &= g_\theta(v) - \Pi_\theta(v), \\ g_\theta(v) &= \operatorname{Re}\Big[\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi\big(\tfrac{s}{2}\big)\Big], \\ \Pi_\theta(v) &= -\operatorname{Re}\frac{\zeta'}{\zeta}(\theta + iv). \end{aligned} \tag{34.8.1}$$

The gamma part $g_\theta$ is explicit and carries no arithmetic. The arithmetic is in $\Pi_\theta(v) = \sum_{n\ge 2}\Lambda(n)n^{-\theta}\cos(v\log n)$, which for $\theta > 1$ is a cosine series with non-negative coefficients ($\Lambda$ is the von Mangoldt function). That is the whole content of the Euler product used below.

Theorem 34.8.1 (de la Vallée Poussin in Thorin form, general version; the first display is Proposition 9.1.4(i) in the present notation). *For $\theta > 1$ and every real $v$,*

$$3u_\theta(0) + 4u_\theta(v) + u_\theta(2v) \;\le\; \frac{1}{\pi}\big[3g_\theta(0) + 4g_\theta(v) + g_\theta(2v)\big].$$

*More generally, for every cosine polynomial* $p(\omega)=\sum_{k=0}^{K}a_k\cos k\omega$ *with* $p\ge 0$ *and* $a_k\ge 0$, $\sum_k a_k u_\theta(kv)\le \pi^{-1}\sum_k a_k g_\theta(kv)$.

*Proof.* $\sum_k a_k\Pi_\theta(kv)=\sum_n \Lambda(n)n^{-\theta}p(v\log n)\ge 0$, and $\Pi_\theta=g_\theta-\pi u_\theta$ by (34.8.1). The first display is $p=3+4\cos\omega+\cos 2\omega=2(1+\cos\omega)^2$. □

Corollary 34.8.2 (zero free region). *Let* $c_*=(14-8\sqrt{3})/5=0.028719\ldots$ *For every* $c<c_*$ *there is* $\gamma_c$ *such that every zero with* $\gamma_\rho\ge\gamma_c$ *has* $1-\sigma_\rho\ge c/\log\gamma_\rho$.

*Proof.* Take $v=\gamma_0$ and $\theta=1+a/\log\gamma_0$. For $\theta>1$ every term of the Hadamard sum is positive, so $\pi u_\theta(\gamma_0)\ge 1/(\theta-\sigma_0)$, and $u_\theta(0), u_\theta(2\gamma_0)\ge 0$. Since $g_\theta(0)=1/(\theta-1)+O(1)$ and $g_\theta(v)=\frac12\log v+O(1)$, Theorem 34.8.1 gives $4/(\theta-\sigma_0)\le(3/a+\frac52+o(1))\log\gamma_0$. Rearranging, $(1-\sigma_0)\log\gamma_0\ge 4a/(3+\frac52 a)-a-o(1)$. The maximum over $a>0$ is attained at $a=(4\sqrt{3}-6)/5=0.18564\ldots$, where $3+\frac52 a=2\sqrt{3}$, and equals $a(2/\sqrt{3}-1)=c_*$. The $o(1)$ is why only $c<c_*$ follows. □

The choice $a=\frac14$ gives the constant $\frac{3}{116}$ of Proposition 9.1.4. The constant $c_*$ is the crude one: sharper cosine polynomials and smoothing give 1/5.69693 (Kadiri [186]), 1/5.573412 (Mossinghoff and Trudgian [220]) and 1/5.558691 (Mossinghoff, Trudgian and Yang [221], for $|t|\ge 2$), constants as quoted in those papers and not checked here, and exponential sum estimates give the Vinogradov and Korobov shape [509]. None of this reaches a fixed $\theta<1$.

With the region of de la Vallée Poussin ($\eta(T)$ a constant over $\log T$, Proposition 9.1.4) or of Vinogradov and Korobov ($\eta(T)=c(\log T)^{-2/3}(\log\log T)^{-1/3}$), taken as $\frac12$ at small heights, Corollary 34.1.3 is positivity of $u_\theta$ at height $v$ for $\theta$ within that width of the line $\operatorname{Re}s=1$, the width taken at height $|v|+\frac12$. It is local in $v$, as it must be: the width shrinks with height. Positivity of $\mathfrak{U}_\tau$ for every $\tau>\theta$ is QRH($\theta$) (Theorem 31.9.1), while positivity at the single level $\theta$ is necessary and not known to be sufficient (Remark 1.2.4).

### 34.8.2 The Davenport and Heilbronn barrier

The Euler product is missing for the function $f$ of Davenport and Heilbronn (Section 34.2), and the inequality of Theorem 34.8.1 is what fails. Everything else of the framework survives.

Proposition 34.8.3 (the Davenport and Heilbronn barrier). *Let* $\Lambda_{\mathrm{DH}}(s)=(5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$, *which is entire of order one, real on the real axis and satisfies* $\Lambda_{\mathrm{DH}}(s)=\Lambda_{\mathrm{DH}}(1-s)$. *The Hadamard sum, the identity* (34.1.2), *Theorem 31.9.1 and Proposition 34.3.1 hold for* $\Lambda_{\mathrm{DH}}$ *with the same proofs; so does Theorem 35.2.1 with* $h=f$ *and the sum over all zeros of* $f$ *to the right of* $\theta$; *and so does Theorem 19.2.2 with the line* $\operatorname{Re}s=1$ *replaced by* $\operatorname{Re}s=2$, *to the right of all zeros since* $|f(s)-1|\le\zeta(2)-1<1$ *there. The function* $f$ *has infinitely many zeros with* $\sigma_\rho>1$ *[111], so* $\Lambda_{\mathrm{DH}}$ *violates* QRH($\theta$) *for every* $\theta<1$;

*its lowest zeros off the line,* $0.8085\ldots+85.6993\ldots i$ *and* $0.6508\ldots+114.1633\ldots i$ *[484], already violate* $\mathrm{QRH}(\frac{3}{4})$ *and* $\mathrm{QRH}(\frac{5}{8})$.

The step from (ii) to (i) in Theorem 19.2.2 uses that $\xi'/\xi$ is analytic in $\operatorname{Re} s > 1$, which fails for $\Lambda_{\mathrm{DH}}$; positivity on the line $\operatorname{Re} s = 1$ (Theorem 30.1.1) and the constant of Proposition 34.7.5 are specific to $\zeta$. So these properties do not prove $\mathrm{QRH}(\theta)$ for any $\theta < 1$: some further property is needed, the Euler product being the natural one. Where it enters is (34.8.1): for $\theta > 1$ the prime sum has nonnegative cosine coefficients, so a nonnegative cosine polynomial applied to it bounds a combination of values of $u_\theta$ at the heights $0, v, 2v$ from above by explicit gamma terms. A zero close to $1 + iv$ would make $u_\theta(v)$ too large, so the bound pushes zeros away from the line $\operatorname{Re} s = 1$. For $f$ the corresponding combination has no sign: the source note reports that $3A_\theta(0) + 4A_\theta(v) + A_\theta(2v)$, with $A_\theta = -\operatorname{Re} f'/f$, is positive for $\zeta$ on $1 \le v \le 200$ (minima 142.2, 23.3 and 5.0 at $\theta = 1.02$, 1.1 and 1.3) and negative for $f$ on more than half of that interval at every level, with its minimum near $v = 85.72$, the height of the zero at 0.8085 (values $-17.5$, $-12.2$, $-6.3$); the same note reports that the optimised curvature kernel of Table 34.12 (level $x = 0.382$, degree 4 polynomial), on 700 heights in $[10, 200]$, is positive for $\zeta$ (minimum 10.84) and negative for $f$ on 55% of the heights (minimum $-5.97$ at $v = 85.84$). These values are the source's, not recomputed here, and the source's figure is not reproduced.

### 34.8.3 Multilevel inequalities

Fix $v \ge 2$, write $L = \log v$, and parametrise levels to the right of the line $\operatorname{Re} s = 1$ by $\theta = 1 + x/L$ with $x > 0$. A level kernel is a finite signed measure $w$ with compact support in $(0, \infty)$; it acts on the density by $w[u](y) = \int u_{1+x/L}(y)\, w(dx)$, and similarly on $g$ and $\Pi$. Its Laplace transform is $\phi_w(\lambda) = \int e^{-x\lambda}\, w(dx)$.

Theorem 34.8.4 (multilevel inequality). *Let* $p(\omega) = \sum_{k=0}^{K} a_k \cos k\omega \ge 0$ *with* $a_k \ge 0$*, and let* $w$ *be a level kernel with* $\phi_w(\lambda) \ge 0$ *for* $\lambda > 0$*. Then for every* $v \ge 2$*,*

$$\sum_{k=0}^{K} a_k\, w[u](kv) \;\le\; \frac{1}{\pi} \sum_{k=0}^{K} a_k\, w[g](kv).$$

*Proof.* By (34.8.1), $\pi\, w[u] - w[g] = -w[\Pi]$, and, the support of $w$ being compact in $(0, \infty)$ so that the series and the integral may be interchanged,

$$\sum_{k} a_k\, w[\Pi](kv) = \sum_{n \ge 2} \Lambda(n)\, n^{-1}\, p(v \log n)\, \phi_w(\log n / L) \;\ge 0. \qquad \square$$

The single level case $w = \delta_x$ is Theorem 34.8.1, since then $\phi_w(\lambda) = e^{-x\lambda}$ (and $w$ is a point mass, a measure with compact support). The second ingredient controls the zeros.

LEMMA 34.8.5 (admissible kernels). *Let $w$ be a level kernel supported in $[x_0, x_1] \subset (0,\infty)$, and let $F_w(z) = \int w(dx)/(z+x)$. The following are equivalent.*

*(i)* $\int w(dx)\, P_{x+y}(\tau) \geq 0$ *for all $y \geq 0$ and all real $\tau$; that is, every zero with $\sigma_\rho \leq 1$, seen through the kernel, contributes a nonnegative amount (widths and distances in units of $1/L$).*

*(ii)* $\operatorname{Re} F_w(i\tau) = \int w(dx)\, x/(x^2+\tau^2) \geq 0$ *for all real $\tau$.*

*(iii)* $\int_0^\infty \cos(\tau\lambda)\, \phi_w(\lambda)\, d\lambda \geq 0$ *for all $\tau$; that is, the even extension of $\phi_w$ is positive definite.*

*When they hold, $|\phi_w(\lambda)| \leq \phi_w(0)$ for all $\lambda$.*

*Proof.* $\pi \int w(dx) P_{x+y}(\tau) = \operatorname{Re} F_w(y+i\tau)$ is harmonic in $\operatorname{Re} z > -x_0$, tends to 0 at infinity, and equals the left side of (ii) on the imaginary axis; the minimum principle on the right half plane gives (ii)⇒(i), and (i)⇒(ii) is the case $y = 0$. Since $F_w(z) = \int_0^\infty e^{-z\lambda} \phi_w(\lambda)\, d\lambda$, (ii) and (iii) are the same statement. The last claim is Bochner's theorem. □

Call $(p, w)$ admissible if $p$ is as in Theorem 34.8.4 and $w$ satisfies $\phi_w \geq 0$ and the conditions of Lemma 34.8.5. For two levels, $w = \delta_{x_1} - \kappa\, \delta_{x_2}$ with $x_1 < x_2$, admissibility is exactly $\kappa \leq x_1/x_2$ (the cosine transform of $w$ is $x_1/(x_1^2+\tau^2) - \kappa x_2/(x_2^2+\tau^2)$, which is nonnegative for all $\tau$ exactly then). Stechkin's device, which pairs each zero with its reflection, uses a two level kernel with $\kappa$ above this bound, so it lies outside the class (Proposition 34.8.10).

COROLLARY 34.8.6 (zero free constant of an admissible pair). *Let $(p, w)$ be admissible, with $w$ of compact support in $(0,\infty)$ and $\phi_w(0) > 0$; put $I = \int_0^\infty \phi_w$, $J(y) = \int_0^\infty e^{-y\lambda} \phi_w(\lambda)\, d\lambda$ and $A = a_1 + \cdots + a_K$. If $(a_1 - a_0) I > \frac{A}{2} \phi_w(0)$, the equation $a_1 J(y) = a_0 I + \frac{A}{2}\phi_w(0)$ has a unique root $y^*(p, w) > 0$, and every zero with $\gamma_\rho$ large has $(1-\sigma_\rho)\log\gamma_\rho \geq y^*(p,w) - o(1)$. Otherwise put $y^*(p,w) = 0$: the scheme gives no positive constant.*

*Proof.* $J$ is continuous and strictly decreasing on $[0,\infty)$, since $\phi_w \geq 0$ is positive near 0, with $J(0) = I$ and $J(y) \to 0$; the hypothesis puts the right side strictly between $a_1 J(0)$ and 0, which gives a unique root. Apply Theorem 34.8.4 at $v = \gamma_0$. By Lemma 34.8.5 every zero contributes a nonnegative amount to the left side, so it is at least $a_1$ times the contribution of $\rho_0$ at height $\gamma_0$, which is $(L/\pi)J(y)$ with $L = \log\gamma_0$ and $y = (1-\sigma_0)L$. On the right, $g_{1+x/L}(0) = L/x + O(1)$ and $g_{1+x/L}(k\gamma_0) = \frac{1}{2}L + O(1)$ for $k \geq 1$, uniformly on the support of $w$, and $\int w(dx)/x = I$. Dividing by $L/\pi$ gives $a_1 J(y) \leq a_0 I + \frac{A}{2}\phi_w(0) + o(1)$. For $\epsilon > 0$, $a_1 J(y^* - \epsilon)$ exceeds the right side by a fixed amount, so $y < y^* - \epsilon$ fails for $\gamma_0$ large. □

### 34.8.4 Optimising the constant

We computed $y^*$ numerically (`code/cf_arith.py`, `data/cf_arith.json`), over cosine polynomials $p = |\sum_j b_j e^{ij\omega}|^2$ with $b_j \geq 0$, which makes every $a_k \geq 0$, and three families of kernels. The first is a single level, for which $y^* = 2(\sqrt{a_1} - \sqrt{a_0})^2/A$ at the best level. The second is two levels with $\kappa = x_1/x_2$, where the optimum pushes $x_2 \to x_1$. In that limit the kernel becomes

$$w[u] \;\longrightarrow\; u_\theta - (\theta - 1)\,\partial_\theta u_\theta = u_\theta + \frac{\theta - 1}{\pi}\,\partial_v^2 \log|\xi(\theta + iv)|,$$

the density corrected by the curvature of $\log|\xi|$ in $v$, using harmonicity, after division by $1 - x_1/x_2$ (the derivative kernels are limits of finite measure kernels and not covered by Theorem 34.8.4 itself). Its Laplace transform is $\phi(\lambda) = e^{-x\lambda}(1 + x\lambda)$. The third family is a single level with derivatives, $\phi(\lambda) = e^{-x\lambda} q(\lambda)$ with $q$ a polynomial, constrained by $q \geq 0$ and by $\sum_j q_j\, j!\, \mathrm{Re}(x + i\tau)^{-j-1} \geq 0$, both imposed on finite grids.

The curvature correction (two levels) improves every polynomial by about 30%. Adding derivatives of the level kernel gives a further 13% at degree 4 and a little more at degree 5. All of this sits below the cap of the next proposition, which is not sharp, and the computation of the last subsection puts the best of these kernels within about 5% of the linear programming values (computed).

### 34.8.5 The $1/\log\gamma$ barrier

Proposition 34.8.7 (no admissible pair beats $1/\log\gamma$). *Let $(p, w)$ be admissible with $a_1 > a_0$, normalised by $\phi_w(0) = 1$, and let $y > 0$ satisfy $a_1 \int_0^\infty e^{-y\lambda}\phi_w(\lambda)\,d\lambda \geq a_0 \int_0^\infty \phi_w(\lambda)\,d\lambda + \frac{A}{2}$. Then*

$$y \;\leq\; \frac{2}{A}\Big(a_1 - a_0 - a_0 \log\frac{a_1}{a_0}\Big) \;<\; \frac{2(a_1 - a_0)}{A} \;\leq\; 1.$$

*In particular $y^*(p, w)$ of Corollary 34.8.6 obeys these bounds; the first is* 0.0548 *for $3 + 4\cos\omega + \cos 2\omega$ and* 0.1051 *for the degree* 4 *polynomial of Table 34.12.*

*Proof.* A nonzero nonnegative cosine polynomial has $a_0 > 0$. By Lemma 34.8.5, $0 \leq \phi_w \leq \phi_w(0) = 1$. The function $h(\lambda) = a_1 e^{-y\lambda} - a_0$ is positive exactly for $\lambda < \lambda_* = y^{-1}\log(a_1/a_0)$, so

$$\frac{A}{2} \leq \int_0^\infty h\,\phi_w \leq \int_0^{\lambda_*} h = \frac{a_1 - a_0 - a_0\log(a_1/a_0)}{y},$$

which is the first inequality. The second holds because $\log(a_1/a_0) > 0$. Finally $a_1 \leq A$, and $a_1 = \pi^{-1}\int_0^{2\pi} p(\omega)\cos\omega\,d\omega \leq \pi^{-1}\int_0^{2\pi} p = 2a_0$, so $2(a_1 - a_0)/A \leq 2(1 - a_0/a_1) \leq 1$. □

| kernel | cosine polynomial | $c = y^*$ | $1/c$ |
|---|---|---|---|
| single level | $3 + 4\cos\omega + \cos 2\omega$ | 0.0287 | 34.8 |
| single level | best of degree 2 | 0.0376 | 26.6 |
| single level | best of degree 3 | 0.0542 | 18.5 |
| single level | best of degree 4 to 8 | 0.0573 | 17.4 |
| two levels, curvature limit | $3 + 4\cos\omega + \cos 2\omega$ | 0.0377 | 26.5 |
| two levels, curvature limit | best of degree 3 | 0.0704 | 14.2 |
| two levels, curvature limit | best of degree 4 | 0.0742 | 13.5 |
| level with derivatives, $\deg q = 2$ | best of degree 4 | 0.0800 | 12.5 |
| level with derivatives, $\deg q = 3$ | best of degree 4 | 0.0826 | 12.1 |
| level with derivatives, $\deg q = 4$ | best of degree 4 | 0.0841 | 11.9 |
| level with derivatives, $\deg q = 5$ | best of degree 4 | 0.0849 | 11.8 |
| cap of Proposition 34.8.7 | best of degree 4 | $\le 0.1051$ | $\ge 9.5$ |
| one far level paired (Proposition 34.8.10), outside the class | best of degree 4 | 0.10367 | 9.646 |

Table 34.12: Zero free constants $1 - \sigma_\rho \ge c/\log\gamma_\rho$ from admissible pairs, in the scheme of Corollary 34.8.6 (computed). The optimal polynomial of degree 4 has $a \approx (1, 1.705, 1.044, 0.425, 0.089)$, and the single level rows to degree 8 are over squares with $b_j \ge 0$ (with only $p \ge 0$ and $a_k \ge 0$ the value grows by about 1% at degree 8); the best level for the derivative family at degree 4 is $x = 0.89$. The rows with derivatives run to $\deg q = 5$ and are optima on discrete constraint grids (computed); at $\deg q = 4$ and 5 the kernels are not certified admissible (at $\deg q = 5$ a dense audit of positive definiteness gives $-1.6 \times 10^{-6}$). The cap of the source note, 0.432 for the degree 4 polynomial, is replaced by the sharper cap of Proposition 34.8.7. The last row is outside the class of Lemma 34.8.5.

Within this class the scheme of Corollary 34.8.6 gives zero free regions $1-\sigma_\rho \ge c/\log\gamma_\rho$ with $c < 1$, never a region of fixed width. The proof uses $\phi_w \le \phi_w(0)$, which kernels that pair each zero with its reflection violate (Proposition 34.8.10), so it says nothing about them. The mechanism is that the term $\frac{A}{2}\phi_w(0)\log v$ is the Cauchy smoothing of the Riemann von Mangoldt density of the zeros, entering through the gamma factor $g$, while positive definiteness forces the weight of the kernel on the zero itself to be at most $\phi_w(0)/y$, so that the zero cannot outweigh the density once $y$ is of order one.

Remark 34.8.8 (what goes beyond; heuristic). The Vinogradov and Korobov region beats $1/\log v$ by replacing the gamma bound $g_\theta(v) \approx \frac{1}{2}\log v$ with an upper

bound for $\log|\zeta|$ near $\theta = 1$ from exponential sums. Through Jensen's formula this replaces the density term $\frac{1}{2}\log v$ by a quantity of order $(\log v)^{2/3}(\log\log v)^{1/3}$ in a disc of radius about $(\log v)^{-2/3}$ around $1+iv$; in the Thorin language that is a local upper bound on the mass of $\mathfrak{U}_\theta$ near height $v$, sharper than the density. A fixed zero free strip would need a uniform bound of the type $\log|\zeta(\theta+iv)| = o(\log v)$ for $\theta$ in a fixed neighbourhood of 1, together with an inequality that converts it into exclusion of zeros. Even the Lindelöf hypothesis is not known to suffice. The suggestion that the method alone yields only regions of width $(\log v)^{-2/3}$ up to logarithms is a heuristic and unchecked.

### 34.8.6 The computed ceiling, and Stechkin's pairing

The kernels of Table 34.12 reach 0.085 while Proposition 34.8.7 caps them at 0.105. To see where the truth lies we computed the best constant attainable in the scheme of Corollary 34.8.6 by any smoothing in a larger formal class: every $\phi$ with $0 \le \phi \le 1$ and $\phi(0) = 1$ whose even extension is positive definite, not only Laplace transforms of level kernels. This is the class in which the proof of Proposition 34.8.7 still runs; whether each such $\phi$ comes from a level kernel is not claimed. For a fixed cosine polynomial and a fixed $y$, whether some $\phi$ certifies $y$ is a linear program:

$$\max_{\phi} \Big\{ a_1 \int_0^{\lambda_{\max}} e^{-y\lambda}\phi - a_0 \int_0^{\lambda_{\max}} \phi \Big\} \quad \text{subject to}$$

$$0 \le \phi \le 1, \qquad \phi(0) = 1, \qquad \int_0^{\lambda_{\max}} \cos(\tau\lambda)\phi(\lambda)\,d\lambda \ge 0 \text{ for all } \tau,$$

and $y$ is certified exactly when the maximum is at least $A/2$. For a fixed $\phi$, optimising the polynomial (with $p \ge 0$ imposed on a grid of angles and $a_k \ge 0$) is again a linear program. We alternate the two and bisect on $y$; the joint problem is not linear, so this is a search and not a global optimisation. We discretise with $\lambda_{\max} = 12$ and 300 points in $\lambda$ and impose positive definiteness at finitely many frequencies $\tau$.

Table 34.13 reports what the discretised linear program finds in the scheme of Corollary 34.8.6, in which every zero contributes nonnegatively (Lemma 34.8.5(i)): about $1/(11.3\log\gamma)$ for polynomials of degree at most 8. The values are computed, not certified: positive definiteness is imposed at finitely many frequencies, the support is truncated at $\lambda_{\max}$, and the polynomial and the smoothing are optimised alternately. For $3 + 4\cos\omega + \cos 2\omega$, with $\lambda_{\max} = 12$ and 300 points, 460 equally spaced frequencies on $[0, 400]$ give 0.0460 and a grid concentrated at low frequencies gives 0.0450, while Proposition 34.8.7 proves the bound 0.0548. On grids that resolve the origin the best level kernel, 0.0849, is about 4% below the computed value 0.0881 for the whole class at degree at most 8; raising the degree of the polynomial beyond 4 adds about 1%.

| problem | ceiling $c$ | $1/c$ |
|---|---|---|
| $p = 3 + 4\cos\omega + \cos 2\omega$, smoothing optimised; $\lambda_{\max} = 12$, 300 points, 460 uniform frequencies to 400 | 0.0460 | 21.7 |
| same, 699 frequencies concentrated below 25 | 0.0450 | 22.2 |
| same, $\lambda_{\max} = 20$, 600 points, frequencies concentrated below 25 | 0.0451 | 22.2 |
| degree 4 polynomial of Table 34.12, smoothing optimised | 0.0874 | 11.4 |
| polynomial of degree $\le 4$ and smoothing both optimised | 0.0874 | 11.4 |
| polynomial of degree $\le 8$ and smoothing both optimised | 0.0881 | 11.35 |
| best level kernel (Table 34.12) | 0.0849 | 11.8 |
| cap of Proposition 34.8.7, degree 4 | 0.1051 | 9.5 |
| one far level paired (Proposition 34.8.10), degree 4, outside the class | 0.10367 | 9.646 |

Table 34.13: Ceiling of the scheme of Corollary 34.8.6 over nonnegative positive definite smoothings, by discretised linear programming (computed, `code/cf_lp.py`, `data/cf_lp.json`; not certified). Refining the frequency grid near the origin moves the value for $3 + 4\cos\omega + \cos 2\omega$ from 0.0460 to 0.0450; on the stated uniform grid, changing the number of points or $\lambda_{\max}$ moves the third digit. The rows with the optimised polynomial are lower estimates, since the alternating searches end at local optima (the degree 4 gain over the fixed polynomial is below $10^{-4}$).

Pairing each zero with its reflection $1 - \bar\rho$ leaves the class. The kernel of the pairing, $\delta_x - \kappa\delta_{x_1}$ with $x_1 = (\theta_1 - 1)L$ of order $L$, has the cosine transform $x/(x^2 + \tau^2) - \kappa x_1/(x_1^2 + \tau^2)$, which is negative for large $\tau$ because $x < \kappa x_1$, so it is not admissible and there is no contradiction with the linear program. The following lemma is the pair inequality behind Stechkin's device [272], proved here by direct algebra.

LEMMA 34.8.9 (Stechkin's pair inequality). *Let $\theta > 1$, $\theta_1 = \frac{1}{2}\big(1 + \sqrt{1 + 4\theta^2}\big)$ and $\kappa = (2\theta - 1)/(2\theta_1 - 1)$. For $0 \le b \le 1$ and real $\tau$ put*

$$F_\theta(b, \tau) = \operatorname{Re}\Big[\frac{1}{\theta - b + i\tau} + \frac{1}{\theta - 1 + b + i\tau}\Big].$$

*Then $F_\theta(b, \tau) \ge \kappa\, F_{\theta_1}(b, \tau)$.*

*Proof.* Put $c = \theta - \frac{1}{2}$, $c_1 = \theta_1 - \frac{1}{2}$, $e = (b - \frac{1}{2})^2 \le \frac{1}{4}$, $q = \tau^2$ and $K = 4eq$. With

$p=\theta-b$ and $p'=\theta-1+b$, both positive,

$$F_\theta=\frac{p}{p^2+q}+\frac{p'}{p'^2+q}=\frac{(p+p')(pp'+q)}{(p^2+q)(p'^2+q)}=\frac{2cX}{X^2+K},\qquad X=c^2-e+q,$$

because $p+p'=2c$, $pp'=c^2-e$ and $(p^2+q)(p'^2+q)=(c^2-e+q)^2+4eq$. In the same way $F_{\theta_1}=2c_1X_1/(X_1^2+K)$ with $X_1=X+D$ and $D=c_1^2-c^2=\theta>0$. Since $\kappa=c/c_1$,

$$F_\theta-\kappa F_{\theta_1}=2c\Big[\frac{X}{X^2+K}-\frac{X+D}{(X+D)^2+K}\Big]=\frac{2cD\,\big(X(X+D)-K\big)}{(X^2+K)\big((X+D)^2+K\big)},$$

and

$$X(X+D)-K=q^2+(c^2+c_1^2-6e)\,q+(c^2-e)(c_1^2-e).$$

Here $c^2\ge\frac14\ge e$, $c_1^2>c^2$ and $c^2+c_1^2=2\theta^2-\theta+\frac12\ge\frac32\ge 6e$, so every coefficient is nonnegative. □

Proposition 34.8.10 (one far level and no smoothing). *Let* $p(\omega)=\sum_{k=0}^K a_k\cos k\omega\ge 0$ *with* $a_k\ge 0$ *and* $a_1>a_0$, *let* $A=a_1+\cdots+a_K$ *and* $\kappa_*=1/\sqrt5$. *Every zero* $\rho_0=\sigma_0+i\gamma_0$ *with* $\gamma_0$ *large has*

$$(1-\sigma_0)\log\gamma_0\ \ge\ \frac{2(\sqrt{a_1}-\sqrt{a_0})^2}{(1-\kappa_*)A}-o(1).$$

*For* $p=3+4\cos\omega+\cos2\omega$ *the constant is* 0.05195*; for the degree* 4 *polynomial of Table 34.12,* $a\approx(1,1.705,1.044,0.425,0.089)$, *it is* $0.10367=1/9.646$ *(computed values; the proposition is proved).*

*Proof.* Let $L=\log\gamma_0$, $\theta=1+x/L$ with $x>0$, and $\theta_1$, $\kappa$ as in Lemma 34.8.9, so that $\kappa=\kappa_*+O(1/L)$. For $\theta'>1$ write $\pi u_{\theta'}=g_{\theta'}-\Pi_{\theta'}$ as in (34.8.1). Since $n^{-\theta}-\kappa n^{-\theta_1}>0$ for $n\ge2$,

$$\sum_k a_k\big[\Pi_\theta(k\gamma_0)-\kappa\Pi_{\theta_1}(k\gamma_0)\big]=\sum_{n\ge2}\Lambda(n)\big(n^{-\theta}-\kappa n^{-\theta_1}\big)p(\gamma_0\log n)\ge0,$$

so $\sum_k a_k[u_\theta-\kappa u_{\theta_1}](k\gamma_0)\le\pi^{-1}\sum_k a_k[g_\theta-\kappa g_{\theta_1}](k\gamma_0)$. By the Hadamard sum, grouping the zeros into orbits of $\rho\mapsto1-\bar\rho$, which preserves ordinates, $\pi[u_\theta-\kappa u_{\theta_1}](y)$ is a sum over orbits of $F_\theta(\sigma_\rho,y-\gamma_\rho)-\kappa F_{\theta_1}(\sigma_\rho,y-\gamma_\rho)$ (half of it for a zero on the critical line), each nonnegative by Lemma 34.8.9. Keep only the orbit of $\rho_0$ in the term $k=1$; at $y=\gamma_0$ it contributes at least $1/(\theta-\sigma_0)-O(1)$. On the right, $g_\theta(0)-\kappa g_{\theta_1}(0)=L/x+O(1)$, from the pole of $\zeta$, and $g_\theta(k\gamma_0)-\kappa g_{\theta_1}(k\gamma_0)=(1-\kappa)\frac{L}{2}+O(1)$ for $k\ge1$, by Stirling's formula. With $y=(1-\sigma_0)L$, so that $\theta-\sigma_0=(x+y)/L$, division by $L/\pi$ gives

$$\frac{a_1}{x+y}\le\frac{a_0}{x}+\frac{(1-\kappa_*)A}{2}+o(1),$$

and the choice $x=2(\sqrt{a_0a_1}-a_0)/((1-\kappa_*)A)$ gives the bound. □

The Stechkin constant over polynomials is the single level optimum divided by $1-1/\sqrt{5}$: the best polynomials of degree 2, 3 and 4 give $0.06809 = 1/14.687$, $0.09800 = 1/10.204$ and $0.10367 = 1/9.646$ (computed). The last is the constant 9.646 of Rosser and Schoenfeld [264], an attribution recalled from the literature and not verified here. The constants $1/(5.573412\log\gamma)$ [220] and $1/(5.558691\log\gamma)$ [221] combine the pairing with smoothing and a finer accounting. Changing bounded error terms cannot change an asymptotic constant, and the gap between the computed ceiling of the class and the constants that use pairing is open.

## 34.9 Nested sampling over heights

Every computation of the earlier sections lives below height 200. Since $\xi'/\xi$ costs about 0.07 seconds even at height $10^7$ (Riemann and Siegel in mpmath, a timing reported by the source), the question is how to spend evaluations at large heights. Nested sampling [269] concentrates evaluations on the most suspicious heights, and under idealised assumptions (Subsection 34.9.1) it estimates the prior volume of the set of near failures of Thorin positivity. The runs below are computed. Their heights and scores are recomputed in ball arithmetic (`code/cf_ns.py`, `data/cf_ns.json`); the run parameters, the volumes $\log X$ and the run times are the source's, and they need its code and seeds.

### 34.9.1 The score and the algorithm

At height $t$ define the normalised Thorin density

$$s(t) = \frac{u_{\theta_0}(t)}{\varrho(t)}, \qquad \theta_0 = 0.6, \qquad \varrho(t) = \frac{1}{2\pi}\log\frac{qt}{2\pi},$$

with $q = 1$ for $\zeta$ and $q = 5$ for the Davenport and Heilbronn function, so that $s$ has mean about one. A negative value of $s(t)$ forces a zero $\rho$ with $\sigma_\rho > \theta_0$ and $|t-\gamma_\rho| < \sqrt{(\sigma_\rho-\theta_0)(\sigma_\rho+\theta_0-1)} < \frac{1}{2}$, by the pairing in the proof of Proposition 3.3.1(ii). A zero with $\sigma_\rho > \theta_0$ makes $u_\theta$ negative near its height at levels $\theta$ just below $\sigma_\rho$ (Theorem 31.9.1); at the fixed level $\theta_0$ it need not (Lemma 11.4.9, Remark 1.2.4), and Theorem 34.7.4, which integrates over levels, does not bound $s$. For $\zeta$ the positivity of $s$ below height $3\times10^{12}-\frac{1}{2}$ is Corollary 11.4.7(1) (with the verification of Platt and Trudgian [419]); the computation measures the margin. For $\zeta$ the lower tail of $s$ is governed by the largest gaps between consecutive zeros. In unfolded coordinates $x = \varrho t$, and keeping only the zeros near $t$, $s \approx \sum_j P_b(x-x_j)$ with $b = (\theta_0-\frac{1}{2})\varrho(t)$.

Take the prior $t \sim U[T_1, T_2]$ and the likelihood $-s(t)$. Nested sampling keeps $N$ live heights. At iteration $k$ it removes the live height with the largest $s$, records the threshold $\ell_k$, and replaces it by a copy of another live height moved by $M$ Metropolis steps for the uniform law on $\{s < \ell_k\}$. Each step is a local Gaussian move of

a quarter of the mean spacing with probability 0.9, or a global uniform proposal with probability 0.1. Under exact independent replacements and a continuous score distribution, the prior volume $X_k = P(s(t) < \ell_k)$ satisfies that $-\log X_k$ is Gamma distributed with shape $k$ and rate $N$, so $\log X_k = -k/N \pm \sqrt{k}/N$ in the sense of mean and standard deviation [269, 82], and the error of the log volume is about $\sqrt{\log(1/X)/N}$, which does not vanish for rare sets. The implemented clones are not independent, so these statements are the ideal model and not an error certificate. The method is related to adaptive multilevel splitting [76]; the concentration theorem of Section 34.5 is not shown to cover it, since its thresholds are random and its potentials vanish.

### 34.9.2 Results

*Davenport and Heilbronn* On $t \in [1000, 1100]$, with $N = 50$ live heights, $M = 4$ moves and 300 iterations (1250 evaluations of $\Lambda'/\Lambda$, twelve minutes), no point of the initial uniform sample had $s < 0$. Nested sampling crossed $s = 0$ at iteration 239, so the set of heights with $s < 0$ has estimated prior volume $X = e^{-4.78 \pm 0.31} \approx 0.008$, about 0.8 units of height. Its negative points form two clusters, near 1012.0 and 1018.8. Newton's method started there gives two zeros of $f$ off the line, computed in ball arithmetic and certified by `code/cf_dhzeros.py` (one zero in a square of half width $10^{-18}$ around each centre, at 256 and 384 bits),

$$\rho = 0.691404808984 + 1012.019981489088\, i,$$
$$\rho = 0.698097971267 + 1018.795762313197\, i,$$

together with their reflections. Both zeros are in the table of Balanzario and Sánchez-Ortiz [28], and zeros off the line of this function are classical [111, 484, 28]. The argument principle in interval arithmetic, with $f \neq 0$ for $\operatorname{Re} s \geq 2$ since $|f(s) - 1| \leq \zeta(2) - 1 < 1$ there, gives 0 zeros in $[0.6, 2] \times [1000, 1010]$ and 2 in $[0.6, 2] \times [1010, 1020]$ (computed, certified). So nested sampling found every zero with $\beta > 0.6$ in $[1000, 1020]$, and the two it found are all of them there. The rest of the band $[1020, 1100]$ was not counted.

*$\zeta$ at heights $10^6$ to $10^{10}$* The source ran nested sampling on bands of length $10^4$ starting at $10^6$, $10^7$, $10^9$ and $10^{10}$. Each band contains between 19,000 and 34,000 zeros.

For the most suspicious heights of the bands at $10^6$ and $10^7$ the source reports argument principle counts of zeros with $\beta > 0.6$ in $[0.6, 2] \times [t - \frac{1}{2}, t + \frac{1}{2}]$ equal to 0 (floating point quadrature, not recomputed here), as they must be, since these heights lie inside the range verified by Platt and Trudgian [419]. The minima sit in gaps of normalised length 2.44 (at 1009611.6) and 2.62 (at 10005411.2). At $10^9$ and $10^{10}$ the runs are shorter and the live points collapse into a single gap, so those minima (0.252 and 0.292) are upper bounds on the band minimum. The

| band | $N, M, K$ | evals | $\log X$ | min $s$ | height, normalised gap |
|---|---|---|---|---|---|
| $10^6$ | 100, 8, 1200 | 9700 | $-12$ | 0.175 | 1009611.61, gap 2.44 |
| $10^7$ | 100, 8, 1200 | 9700 | $-12$ | 0.182 | 10005411.18, gap 2.62 |
| $10^9$ | 100, 8, 1000 | 8100 | $-10$ | 0.252 | 1000004287.61, gap 2.54 |
| $10^{10}$ | 50, 4, 400 | 1650 | $-8$ | 0.292 | 10000009878.55, gap 2.38 |

Table 34.14: Nested sampling for $\zeta$ on bands of length $10^4$, with $N$ live heights, $M$ moves per replacement and $K$ iterations (computed). In every band the score stays positive. The column min $s$ is the smallest value found by the run, not the band minimum. The scores at the printed heights and the normalised gaps, measured between consecutive zeros found as sign changes of Hardy's $Z$ function, are recomputed with enclosure radius below $6 \times 10^{-12}$ (`code/cf_ns.py`); the run parameters and $\log X$ are the source's. The smallest values sit in the largest normalised gaps.

source reports that the sine kernel predicts the $10^{-4}$ quantile of $s$ at 0.231 for $10^9$ and the $10^{-3}$ quantile at 0.287 for $10^{10}$, at the matching $b = 0.30$ and 0.34 (not recomputed).

*Checking the volume estimates, and a failure mode* A plain Monte Carlo sample of 5000 uniform heights in the band at $10^6$ gives $P(s < 0.399) = 0.130$, $P(s < 0.235) = 0.0104$ and $P(s < 0.2005) = 0.0018$. Nested sampling put these levels at $X = 0.10$, 0.010 and 0.0010. The two agree to within the stated error down to $X \approx 10^{-2}$ in this one comparison, and nested sampling underestimates by a factor of about two at $10^{-3}$. The Monte Carlo sample also found a point with $s = 0.142$, in a gap of normalised length 2.87 at height 1004606.8 (both values recomputed: $s = 0.14185$, gap 2.865), deeper than anything nested sampling reached. The reason is structural. The set $\{s < \ell\}$ is a union of thousands of short intervals, one inside each large gap, and local moves cannot travel between them; once global proposals are rarely accepted, the live points collapse into a few intervals. No reliability threshold in the volume is established: the estimates agree down to about $10^{-2}$ in the comparison above and disagree beyond it, and beyond that nested sampling serves as a search heuristic, not as an exhaustive search. The remedy is to propose new heights at the large gaps directly, located cheaply by sign changes of Hardy's $Z$ function. That is a gap guided form of nested sampling, which the source leaves for a next round.

*Comparison with the sine kernel* The source compares the tails with the prediction of the sine kernel process, modelled by circular unitary matrices of size 600 at the matching value of $b$, a finite unitary ensemble and not the infinite sine process; the uncertainty of the extreme tail, which comes from few matrices, is not estimated. At volumes $10^{-1}$ and $10^{-2}$ the band at $10^7$ gives 0.394 against 0.410 and

0.258 against 0.259, and the band at $10^6$ agrees to about 10% (not recomputed). Deeper in the tail the zeros of $\zeta$ give fewer small values of $s$ than the sine kernel model, both in nested sampling and in plain Monte Carlo (0.0018 against about 0.004 at $s < 0.2005$). This is the direction of the finite height deficit of large gaps, which Bogomolny, Bohigas, Leboeuf and Monastra describe by a unitary ensemble of effective size $\log(t/2\pi)/\sqrt{12\Lambda_0}$ with $\Lambda_0 = 1.57314\ldots$ [52]. The coefficient is 0.2302 (not $0.2887 = 1/\sqrt{12}$), and the sizes are 2.76 at $10^6$, 3.29 at $10^7$, 4.35 at $10^9$ and 4.88 at $10^{10}$ (computed, `code/cf_info.py`). The comparison of these sizes with the finite height data is a hypothesis (heuristic) and has not been checked quantitatively.

### 34.9.3 What this buys

Nested sampling does not verify RH at new heights. Exhaustive verification by Turing's method already reaches $3 \times 10^{12}$ [419], and a randomised search cannot replace it. What it gives is a cheap, scale free way to measure how close $\zeta$ comes to violating the positivity of $u_{\theta_0}$ at a given height. The smallest margins found are between 0.14 and 0.29 of the mean density at heights $10^6$ to $10^{10}$, always in the largest gaps between consecutive zeros. Their lower tail was compared with the sine kernel model and agreed down to volumes of about $10^{-2}$ in the comparison above, with fewer extreme gaps than the model beyond that (computed, heuristic as a statement about the tail). It also gives a heuristic search, which found two zeros off the line of the Davenport and Heilbronn function in $[1010, 1020]$. Lifting it to a certificate needs the gap guided proposals above and an exhaustive count, which brings it back to Turing's method; the certificates that the book has are in Section 31.9.7, Theorem 31.9.6, Theorem 31.10.2 and Section 31.10.

## 34.10 What would close the gap

The proved statements of this chapter are Theorems 34.7.1, 34.5.1, 34.8.1, 34.8.4 and 34.7.4, Corollaries 34.1.3, 34.8.2, 34.7.3 and 34.8.6, Lemmas 34.8.5 and 34.8.9, and Propositions 34.2.1, 34.3.1, 34.4.1, 34.6.1, 34.7.2, 34.8.7, 34.8.10 and 34.8.3; Proposition 34.7.5 is conditional on QRH($\theta$), and the formula of Balazard, Saias and Yor is Theorem 35.2.1. Together with the results recalled in Section 34.1 they place the reciprocal Xi programme in a precise position.

1. $\mathfrak{U}_1$ is unconditionally a positive Cauchy mixture, and $\mathfrak{U}_{1/2}$ is the counting measure of the critical zeros. The passage between them is one step of the Cauchy semigroup, and RH is exactly the statement that this step can be inverted within positive measures (Theorem 19.2.2).

2. Positivity of $\mathfrak{U}_\tau$ for every $\tau > \sigma$ is QRH($\sigma$), a zero free half plane (Theorem 31.9.1), while positivity at the single level $\sigma$ is necessary and not known

to be sufficient (Remark 1.2.4). There is no slack: the Thorin route to RH is the zero free region problem in another language.

3. Unconditionally, $\mathfrak{U}_\theta$ is positive up to a fraction $O(T^{-(\theta-1/2)/4})$ of its mass on $(0,T]$ (Theorem 34.7.1, proved with Selberg's density theorem). This is the unconditional content of the Thorin picture inside the strip.

4. The Davenport and Heilbronn function satisfies the Hadamard sum, the identity (34.1.2), Theorem 31.9.1, Proposition 34.3.1, Theorem 35.2.1 and Theorem 19.2.2 (with the line $\operatorname{Re} s = 2$), and it has zeros off the line (Proposition 34.8.3). Its zeros with $\beta > 1$ show that these properties do not prove $\mathrm{QRH}(\theta)$ for any $\theta < 1$: some further property is needed, the Euler product being the natural one. It enters as an upper bound on positive combinations of the Thorin density at several heights (Theorem 34.8.1), which $\zeta$ satisfies and the Davenport and Heilbronn function violates, most strongly at the height of its zero off the line. With the Cauchy structure this yields positivity within $c/\log t$ of $\theta = 1$ (Corollary 34.8.2, with the local form of Corollary 34.1.3). Combining levels as well as heights improves the best single level constant by about 48% (computed, Table 34.12), but in the class in which every zero contributes nonnegatively no inequality gives more than $1 - \beta \geq c/\log\gamma$ with a constant $c < 1$ (Proposition 34.8.7): the density of the zeros, entering through the gamma factor, caps it. The ceiling of that class, computed by a discretised linear program and not certified, is about $1/(11.3\log\gamma)$ (Table 34.13). Pairing each zero with its reflection leaves the class (Proposition 34.8.10), so the barrier says nothing about it.

5. Every zero off the line forces negative Thorin mass at least its isolation distance over $\log T$ (Theorem 34.7.4); for separated zeros the negative mass and the zero count control each other up to a factor $\log T$ when the isolation is at least of order $1/\log T$ and the counts are of order $\log T$ or smaller (Corollary 34.7.3).

The computational statements are local, including the nested sampling of Section 34.9, which samples heights up to $10^{10}$ and certifies only the windows that are counted with interval arithmetic. The particle filter recovers the zeros of $\zeta$ in a window to four decimals and bounds their distance from the line by $5 \times 10^{-4}$ at the 95% level under the Gaussian tolerance model, and it matches the first order Fisher information of the model with free real parts (heuristic; Section 34.3). The GBC screen scans a window in milliseconds and found all four zeros off the line of the Davenport and Heilbronn function below height 200, with one boundary false alarm (the window centred at 180, Table 34.6) and none for $\zeta$. The contour moments recover a zero off the line to ten digits, computed in ball arithmetic (Table 34.4). None of this extends past the verified height $3 \times 10^{12}$, and none of it is a proof.

A proof along these lines needs one of two things. The first is an arithmetic input, coming from the Euler product, that controls $u_\theta$ uniformly in $t$ for some $\theta < 1$ beyond the Vinogradov and Korobov boundary. Section 34.8 shows that positivity inequalities of de la Vallée Poussin type, in which every zero contributes nonnegatively, cannot supply it: they are capped at $c/\log t$ with $c < 1$, and in the computed basic form at about $1/(11.3 \log t)$. The input could instead be a growth bound for $\zeta$ near $\theta = 1$, as in Vinogradov and Korobov, but uniform (heuristic). The second is a density estimate strong enough to force the negative Thorin mass to vanish. Theorem 34.7.4 shows this direction is not lost to clustering: every zero off the line pays negative mass in proportion to its isolation. But the density estimate needed is itself equivalent to the absence of such zeros. Both routes are open, and both are hard in the classical sense.

# Chapter 35

# Balazard, Saias and Yor meet Thorin

Balazard, Saias and Yor [38] proved in 1999 that the Riemann Hypothesis is equivalent to a single Cauchy average of $\log|\zeta|$ on the critical line being zero. The Thorin programme of this book studies the measures $\mathfrak{U}_\sigma$ with density $m_\sigma = \pi^{-1}\operatorname{Re}\xi'/\xi(\sigma+it)$ and their images $U_\sigma$, the Thorin measures of $\xi(\sigma)/\xi(\sigma+\sqrt{s})$ (equation (30.1.1) and Chapter 30). This chapter shows that the two are the same harmonic analysis, one derivative in $\sigma$ and one archimedean term apart, and then solves the resulting Cauchy averages numerically by quadrature, by plain Monte Carlo and by a vertical likelihood sampler after Polson and Scott (Section 35.6; computed in code/bsy_mc.py, where the plain sampler has the smaller standard error). Chapter 34 is the companion: it cites Theorem 35.2.1 for the criterion, and its particle filter and nested sampling over heights supply the scans of Section 35.6.

The chapter develops a note by Nick Polson and Vadim Sokolov [227], with the computations done for this book. Several parts of the note are already in the book, in the same or in a stronger form, and are cited by label in Section 35.1 and not restated: the density of $\mathfrak{U}_\sigma$ and its Hadamard sum, the Cauchy semigroup, the Poisson structure of the Thorin measures and the deconvolution criterion. The numbers of the chapter are of two kinds. The zeros and counts behind Theorem 35.4.3 were certified for this book in ball arithmetic (`code/cf_dhzeros.py`, data file `data/cf_dhzeros.json`). The other tables were recomputed in floating point by scripts written from the mathematics (`code/bsy_zdh.py`, the shared evaluator, and the files `code/bsy_*.py` with their data files `data/bsy_*.json`, listed in `code/README.md`); where the recomputation differed from the note, the recomputed value is printed. Nothing here is certified except where the sentence says so.

Status words follow the rest of the book: proved, computed, conditional, heuristic or open. The statements at the level $\frac{7}{8}$ use the quasi Riemann hypothesis QRH($\frac{7}{8}$), announced by OpenAI [398] and formalised in Lean, and not refereed (Section 29.7); every value at the level $\frac{7}{8}$ is conditional on it. The notation is that of Chapters 30 to 34: $\mathfrak{U}_\sigma$ is the measure on $\mathbb{R}$ without the factor $\frac{1}{2}$, $P_a(x) = a/(\pi(a^2+x^2))$ is the Cauchy density of scale $a$ and, for $a<0$, the same

formula, so that $P_{-a} = -P_a$ as in Proposition 1.2.3 (Chapter 34 writes $Q_a$ for this signed kernel), and QRH($\theta$) is the statement that $\zeta$ has no zero with $\beta > \theta$.

The chapter contains:

1. a dictionary between the logarithmic level (Balazard, Saias and Yor) and the Thorin level: the Poisson integral becomes a Cauchy smoothing of $\mathfrak{U}_{1/2}$, and the Green potential of each zero off the line becomes a signed Cauchy defect, with an archimedean term that the logarithmic level does not see (Section 35.3);
2. the criterion at every level $\theta \in [\frac{1}{2}, 1)$ and every height, as the Nevanlinna formula of the half plane for $\zeta(s)(s-1)/(s-2\theta+1)$, which characterises QRH($\theta$); the case $\theta = \frac{1}{2}$, $t_0 = 0$ is the theorem of Balazard, Saias and Yor (Section 35.2);
3. the reading of the $2\sin^2$ lemma (equation (1.2.9)): the Lévy density of the reciprocal Xi function is positive with no hypothesis, and RH is equivalent to that reciprocal being an extended generalised gamma convolution (EGGC) with the Thorin measure $\mathfrak{U}_{1/2}$ (Section 35.4);
4. a probabilistic reading of the criterion by Brownian motion and the Riesz decomposition, in the spirit of Yor (Section 35.5), and its complex form: under QRH($\theta$) the boundary values of $\log h_\theta$ are the stopped value of a time changed complex Brownian motion, so $\log|\zeta|$ and $\arg h_\theta$ have equal variances and zero covariance under harmonic measure at every height, and without the hypothesis the defects of these moments are integrals along horizontal cuts from the zeros to the line (Proposition 35.5.4);
5. a proof that the $2\sin^2$ Lévy density of the Davenport and Heilbronn function is positive although it has zeros off the line, the three finite zero computations behind it being certified in ball arithmetic, so infinite divisibility of the reciprocal does not force the zeros onto the line (Section 35.4);
6. a vertical (layer cake) form of the criterion, in which RH says that the upper and lower tails of $\log|\zeta|$ under harmonic measure balance, with numerical solutions by quadrature, by plain and vertical Monte Carlo, and by the weighted Monte Carlo of Yakowitz, Krimmel and Szidarovszky with an exact Dirichlet polynomial control; an exact local form of the criterion on a half disc, for which the sorted trapezoid rule has mean square error $O(N^{-4})$, and $O(N^{-5})$ after a change of variable (proved); and the truncated whole line average, whose tail is $O(T^{-2}\log T)$ under RH [26] (computed in Table 35.6; Section 35.6);
7. a smoothed Riesz identity in the plane, in which the Cauchy kernel is replaced by a compactly supported bump. The integrand is then smooth away from the zeros, the Voronoi and Delaunay forms of the weighted rule have root mean square error $O(N^{-3/2})$ at independent uniform nodes (proved,

Theorem 35.7.3, which gives $O(N^{-1/2-2/d})$ in dimension $d$; computed at jittered nodes), a shifted grid reaches machine precision in a box free of zeros, and deflation handles a zero in the box; the same count is a linear functional of the horizontal logarithmic derivative alone (Section 35.7).

Assuming RH, Bui, Lester and Milinovich [26] bound the tail of the truncated Balazard, Saias and Yor integral and disprove a conjecture of Borwein, Bradley and Crandall [23] on its rate of decay; Section 35.6.6 uses their results. Nothing here proves RH. Except for the positivity of $\mathfrak{U}_\sigma$ for $\sigma \ge 1$ and Theorem 35.5.2, every identity of Sections 35.1 to 35.5 holds, with the changes listed in Section 35.8, for the Davenport and Heilbronn function, which has zeros off the line; Section 35.8 records what that implies.

## 35.1 Thorin measures along vertical lines

Write $\rho = \beta + i\gamma$ for the zeros of $\zeta$ (the zeros of $\xi$), counted with multiplicity. For $\sigma > \frac12$ that is not the real part of a zero let $\mathfrak{U}_\sigma$ be the measure on $\mathbb{R}$ with density $m_\sigma(t) = \pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it)$ in the variable $t$. For $\sigma \ge \theta_0 = \sup \beta$ this is the positive measure of equation (30.1.1), with $m_\sigma = u_\sigma$ of Theorem 30.1.1; for $\frac12 < \sigma < \theta_0$ it is a signed measure. At a level $\sigma = \beta$ through a zero the density has a simple pole, and $\mathfrak{U}_\sigma$ is the right limit as in Remark 34.1.1. The Thorin measure $U_\sigma$ is the image of the restriction of $\mathfrak{U}_\sigma$ to $t > 0$ under $t \mapsto t^2$, so that $\pi U_\sigma\big((0, t^2]\big) = \arg \xi(\sigma + it)$ with the argument followed continuously up the vertical line. When $\xi(\sigma)/\xi(\sigma+\sqrt{s})$ is a generalised gamma convolution (GGC) this is its Thorin measure; for $\sigma > 1$ it always is (Theorem 1.2.2, Corollary 29.14.3).

Lemma 35.1.1 (Cauchy and Riemann). *The density of $\mathfrak{U}_\sigma$ in the variable $t$ is*

$$m_\sigma(t) = \frac{1}{\pi}\,\partial_t \arg \xi(\sigma + it) = \frac{1}{\pi}\,\partial_\sigma \log |\xi(\sigma + it)| = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it),$$

*a harmonic function of $(\sigma, t)$ away from the zeros.*

This is the Cauchy and Riemann equations for $\log \xi$, as in the proof of Theorem 30.1.1 and Proposition 3.4.1. Hadamard's product gives, absolutely convergently (Proposition 1.2.3),

$$m_\sigma(t) = \sum_\rho P_{\sigma-\beta}(t - \gamma), \tag{35.1.1}$$

the sum over all zeros, with the signed $P_a$ of Proposition 1.2.3. Hence $\mathfrak{U}_1 = \sum_\rho P_{1-\beta}(\cdot - \gamma)$ is unconditionally a positive Cauchy mixture, and as $\sigma \downarrow \frac12$ the measure tends vaguely (Theorem 29.14.4; Proposition 34.2.1) to $\mathfrak{U}_{1/2} = \sum_{\beta=1/2} \delta_\gamma$, the counting measure of the critical zeros. Pairs off the line cancel in this limit, since $P_{1/2-\beta} + P_{\beta-1/2} = 0$.

## 35.2 The Balazard, Saias and Yor criterion at every level and height

Fix $\theta \in [\frac{1}{2}, 1)$ and a height $t_0$. Put

$$h_\theta(s) = \zeta(s)\,\frac{s-1}{s-2\theta+1},$$

which is analytic in $\{\operatorname{Re} s > \theta\}$ and satisfies $|h_\theta| = |\zeta|$ on the line $\operatorname{Re} s = \theta$, because $2\theta - 1$ is the reflection of 1 in that line. For a zero $\rho$ with $\beta > \theta$ let

$$G_\theta(s,\rho) = \log\left|\frac{s-2\theta+\bar\rho}{s-\rho}\right| \ge 0$$

be the Green function of the half plane $\{\operatorname{Re} s > \theta\}$ with pole at $\rho$.

Theorem 35.2.1 (Balazard, Saias and Yor at level $\theta$). *Let $\frac{1}{2} \le \theta < 1$, $H = \{\operatorname{Re} s > \theta\}$ and $h_\theta(s) = \zeta(s)\,(s-1)/(s-2\theta+1)$. For every $s = \sigma + it \in H$ with $h_\theta(s) \ne 0$,*

$$\int_{-\infty}^{\infty} \log|\zeta(\theta+i\tau)|\,P_{\sigma-\theta}(t-\tau)\,d\tau - \log|h_\theta(s)| = \sum_{\beta>\theta} G_\theta(s,\rho) \ge 0, \qquad (35.2.1)$$

*the integral absolutely convergent and the sum, over the zeros with $\beta > \theta$ counted with multiplicity, convergent. In particular*

$$D_\theta(t_0) := \int_{-\infty}^{\infty} \log|\zeta(\theta+i\tau)|\,P_{1-\theta}(t_0-\tau)\,d\tau - \log|h_\theta(1+it_0)| = \sum_{\beta>\theta} G_\theta(1+it_0,\rho),$$

*and* QRH($\theta$) *holds if and only if $D_\theta(t_0) = 0$ for one (equivalently every) $t_0$. At $\theta = \frac{1}{2}$, $t_0 = 0$ this is the theorem of Balazard, Saias and Yor [38]:*

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\log|\zeta(\frac{1}{2}+it)|}{\frac{1}{4}+t^2}\,dt = \sum_{\beta>1/2} \log\left|\frac{\rho}{1-\rho}\right|.$$

*Proof.* Write $F = h_\theta$. Since $2\theta - 1 < \theta$, $F$ is analytic on a neighbourhood of $\overline{H}$ (the pole of $\zeta$ at 1 is cancelled by $s-1$); its zeros in $H$ are those of $\zeta$ with $\beta > \theta$, with the same multiplicities, and $|F| = |\zeta|$ on $\partial H$, because $2\theta - 1$ is the reflection of 1 in $\partial H$. Every $s \in \overline{H}$ is at least as close to 1 as to $2\theta - 1$, so $|F(s)| \le |\zeta(s)|$ for $|s-1| \ge 1$, while $F$ is bounded on $\overline{H} \cap \{|s-1| \le 1\}$. The convexity bound for $\zeta$ in $\theta \le \sigma \le 2$ [498, Chapter V] and $|\zeta(s)| \le \zeta(2)$ for $\sigma \ge 2$ give a constant $A$ with $\log|F(s)| \le A\log(2+|s|)$ on $\overline{H}$. Hence $F$ is of exponential type in $\overline{H}$ and $\int \log^+|F(\theta+i\tau)|\,(1+\tau^2)^{-1}d\tau < \infty$, and Nevanlinna's formula for a half plane [209] gives, for $s \in H$ not a zero,

$$\log|F(s)| = \int_{-\infty}^{\infty} \log|F(\theta+i\tau)|\,P_{\sigma-\theta}(t-\tau)\,d\tau - \sum_{\beta>\theta} G_\theta(s,\rho) + \kappa\,(\sigma-\theta),$$

$$\kappa = \limsup_{r\to\infty} \frac{\log|F(\theta+r)|}{r},$$

with the integral absolutely convergent and the sum convergent. Since $F(\theta+r)\to 1$ as $r\to\infty$, $\kappa=0$, which is (35.2.1). Every term of the sum is positive, because $|s-2\theta+\bar\rho|^2-|s-\rho|^2=4(\sigma-\theta)(\beta-\theta)>0$. At $s=1+it_0$, which is never a zero, this is the formula for $D_\theta(t_0)$; the sum is empty exactly under $\mathrm{QRH}(\theta)$, and otherwise positive at every $t_0$. The term $\kappa(\sigma-\theta)$ cannot be dismissed by finite order: $F(s)=e^{-c(s-\theta)}$ with $c>0$ has order one, no zeros and $|F|=1$ on $\partial H$, but $\log|F(1+it_0)|=-c(1-\theta)$; for it $\kappa=-c$. □

The theorem is the Riesz decomposition of the subharmonic function $\log|h_\theta|$ on the half plane; the formula of the half plane is cited from [209] without a page. Two consequences are used below. The first is the case $\theta=\frac12$ at a general point $\sigma+it$,

$$\log|h_{1/2}(\sigma+it)|=\big(P_{\sigma-1/2}*\log|\zeta(\tfrac12+i\,\cdot)|\big)(t)-\sum_{\beta>1/2}G_{1/2}(\sigma+it,\rho),\quad(35.2.2)$$

The second is the form at $\theta=\frac78$, $t_0=0$, where $h_{7/8}(1)=\frac{1}{2-2\theta}=4$: the Cauchy average of $\log|\zeta(\frac78+i\tau)|$ at scale $\frac18$ equals $\log 4$ if and only if $\mathrm{QRH}(\frac78)$ holds. That equivalence is unconditional; that the average does equal $\log 4$ is conditional on $\mathrm{QRH}(\frac78)$, announced and not refereed (Section 29.7). Table 35.3 lists the defects computed at the levels $\frac12$, $\frac34$ and $\frac78$.

Remark 35.2.2 (local versus global). A zero at height $\gamma$ with $\beta=\theta+\delta$ contributes about $2\delta(1-\theta)/\gamma^2$ to $D_\theta(0)$. That is below $10^{-25}$ beyond the verified height $3\times10^{12}$ [419], so the global criterion is numerically blind: summed over all possible zeros above that height, $D_\theta(0)$ is below $7.4\times10^{-13}$ at $\theta=\frac12$, $1.85\times10^{-13}$ at $\theta=\frac34$ and $4.7\times10^{-14}$ at $\theta=\frac78$ (computed from the zero counting bound of Hasanalizade, Shen and Wong [176], `code/bsy_tailbound.py`; the last bound is $4.62\times10^{-14}$ to three digits). Centred at $t_0=\gamma$, the same zero contributes $\log\frac{1-\theta+\delta}{1-\theta-\delta}\approx\frac{2\delta}{1-\theta}$, which is of order $\delta$ itself. For the Davenport and Heilbronn function at $\theta=\frac34$ the zero $0.8085\ldots+85.6993\ldots i$ gives $\log\frac{0.308517}{0.191483}=0.4770$ at its own height (Table 35.2 has the computed value 0.4770) and about $4\times10^{-6}$ seen from $t_0=0$. The criterion should be applied with the Cauchy kernel centred where the zeros are. Computing $D_{7/8}(t_0)$ on a grid of heights is a practical local test of $\mathrm{QRH}(\frac78)$ at those heights, and the generative screen of Section 34.4 tells where to centre it; it is not a proof, because it is local.

## 35.3 The dictionary: one derivative in $\sigma$

Equation (35.2.1) with $\theta=\frac12$ at a general point of the half plane is the identity (35.2.2). Differentiating it in $\sigma$ gives the Thorin statement, in two steps: the transport of $\mathfrak{U}_{1/2}$ to $\mathfrak{U}_\sigma$, and the archimedean term.

THEOREM 35.3.1 (transport). *For $\sigma > \frac12$,*

$$m_\sigma = P_{\sigma-1/2} * \mathfrak{U}_{1/2} + \sum_{\beta>1/2} \big[P_{\sigma-\beta} + P_{\sigma-1+\beta}\big](\cdot - \gamma).$$

*Proof.* The formula is (35.1.1) split into zeros on the line and zeros off it, a zero $\rho$ with $\beta > \frac12$ being paired with $1-\bar\rho$ at the same height $\gamma$. □

REMARK 35.3.2 (zero by zero). A zero on the line enters (35.2.2) through the logarithmic singularity of the boundary values, whose Poisson extension has $\sigma$ derivative $\pi P_{\sigma-1/2}(\cdot-\gamma)$. A pair $\rho, 1-\bar\rho$ off the line enters through two terms. The Poisson extension of $\log|(\cdot-\rho)(\cdot-1+\bar\rho)|$ from the line is $2\log|s-1+\bar\rho|$, with $\sigma$ derivative $2\pi P_{\sigma-1+\beta}$. The Green term $-G_{1/2}(s,\rho)$ has $\sigma$ derivative $\pi(P_{\sigma-\beta} - P_{\sigma-1+\beta})$. The sum is $\pi(P_{\sigma-\beta}+P_{\sigma-1+\beta})$. The three derivatives are direct: $\partial_\sigma \log|\sigma+it-c| = \operatorname{Re}(s-c)^{-1} = \pi P_{\sigma-\operatorname{Re}c}(t-\operatorname{Im}c)$, and $2\log|s-1+\bar\rho|$ is harmonic on $\{\operatorname{Re}s > \frac12\}$ (its singularity $1-\bar\rho$ lies to the left of the line) with the right boundary values.

Differentiating (35.2.2) in $\sigma$ does not give $m_\sigma$ directly, because $\pi m_\sigma = \partial_\sigma \log|\xi|$ while (35.2.2) concerns $\log|h_{1/2}|$, and $\xi = C\,h_{1/2}$ with

$$C(s) = \frac{\xi(s)}{h_{1/2}(s)} = \tfrac12 s^2 \pi^{-s/2}\Gamma(\tfrac{s}{2}),$$

analytic and zero free on $\{\operatorname{Re}s > 0\}$. With $a_\sigma(t) = \pi^{-1}\operatorname{Re}(C'/C)(\sigma+it) = \pi^{-1}\operatorname{Re}\big[\frac{2}{s} - \frac12\log\pi + \frac12\psi(\frac{s}{2})\big]$, for $\sigma > \frac12$,

$$m_\sigma(t) = \frac{1}{\pi}\,\partial_\sigma\big(P_{\sigma-1/2} * \log|\zeta(\tfrac12 + i\,\cdot)|\big)(t) + a_\sigma(t) + \sum_{\beta>1/2}\big[P_{\sigma-\beta} - P_{\sigma-1+\beta}\big](t-\gamma), \tag{35.3.1}$$

equivalently, by Theorem 35.3.1, $\frac{1}{\pi}\partial_\sigma(P_{\sigma-1/2} * \log|\zeta(\frac12+i\cdot)|) = P_{\sigma-1/2} * \mathfrak{U}_{1/2} + \sum_{\beta>1/2} 2P_{\sigma-1+\beta}(\cdot-\gamma) - a_\sigma$. The term $a_\sigma$ cannot be absorbed into the Poisson integral: $\log|C(\frac12+it)| = -\frac{\pi}{4}|t| + O(\log|t|)$, whose Cauchy average diverges, which is why the criterion is stated for $\zeta$ and not for $\xi$.

*Proof.* On $\{\operatorname{Re}s > \frac12\}$, $\log|\xi| = \log|h_{1/2}| + \log|C|$, so $\pi m_\sigma = \partial_\sigma \log|h_{1/2}| + \pi a_\sigma$ by Lemma 35.1.1. Differentiate (35.2.2): the Poisson integral is harmonic, and with $\partial_\sigma \log|s-c| = \operatorname{Re}(s-c)^{-1} = \pi P_{\sigma-\operatorname{Re}c}(t-\operatorname{Im}c)$ (signed $P$),

$$\partial_\sigma G_{1/2}(s,\rho) = \partial_\sigma \log|s-1+\bar\rho| - \partial_\sigma\log|s-\rho| = \pi\big[P_{\sigma-1+\beta} - P_{\sigma-\beta}\big](t-\gamma),$$

the differentiated series converging locally uniformly off the zeros. This gives (35.3.1); the second form follows by substituting $m_\sigma = P_{\sigma-1/2} * \mathfrak{U}_{1/2} + \sum_{\beta>1/2}[P_{\sigma-\beta} + P_{\sigma-1+\beta}](\cdot-\gamma)$. The asymptotics of $\log|C|$ are Stirling's formula. □

| logarithmic level | Thorin level |
|---|---|
| $\log\lvert\xi\rvert$ | $\pi\, m_\sigma = \partial_\sigma \log\lvert\xi\rvert$ |
| Poisson integral of $\log\lvert\zeta(\frac12 + i\cdot)\rvert$ | $P_{\sigma-1/2} * \mathfrak{U}_{1/2}$ plus $2P_{\sigma-1+\beta}$ per zero with $\beta > \frac12$, minus $a_\sigma$ |
| minus the Green potential, $-G(\cdot,\rho) \le 0$ | signed defect $P_{\sigma-\beta} - P_{\sigma-1+\beta}$ |
| average equals $\log\lvert h(1)\rvert$ | $\mathfrak{U}_\sigma \ge 0$ for $\sigma > \frac12$ |
| $D_\theta \equiv 0$ | $\mathfrak{U}_\sigma \ge 0$ for $\sigma > \theta$ |
| $D$ of order $\delta$ at $t_0 = \gamma$ | negative spike of $m_\sigma$ at $\gamma$ |

Table 35.1: The dictionary between the two levels.

Corollary 35.3.3 (equivalent forms of RH). *The following are equivalent: (i) RH; (ii) $D_{1/2}(t_0) = 0$ for some $t_0$; (iii) $\mathfrak{U}_\sigma$ is positive for every $\sigma \in (\frac12, 1)$; (iv) $\mathfrak{U}_1 = P_{1/2} * \nu$ for a positive measure $\nu$ on $\mathbb{R}$ with $\int (1+t^2)^{-1}\nu(dt) < \infty$, and then $\nu = \mathfrak{U}_{1/2}$. More generally, for $\theta \in [\frac12, 1)$, QRH($\theta$) holds if and only if $D_\theta \equiv 0$, if and only if $\mathfrak{U}_\sigma \ge 0$ for all $\sigma > \theta$.*

(i)⇔(ii) is Theorem 35.2.1. Statement (iii) and its form for QRH($\theta$) are Lagarias's argument (Remark 1.2.4); (iv) is Theorem 19.2.2(a).

*Proof.* Let $\rho_0$ be a zero with $\beta_0 > \theta$. By (35.1.1), $m_\sigma(\gamma_0) \to -\infty$ as $\sigma \uparrow \beta_0$, because the term of $\rho_0$ is $P_{\sigma-\beta_0}(0) = 1/(\pi(\sigma-\beta_0))$ and the other terms stay bounded near $\gamma_0$; so $m_\sigma$ is negative on a neighbourhood of $\gamma_0$ for $\sigma \in (\theta, \beta_0)$ close to $\beta_0$, and $\mathfrak{U}_\sigma$ is not positive. This gives (iii)⇒(i), and its form for QRH($\theta$). Conversely, under QRH($\theta$) every term of (35.1.1) with $\sigma > \theta$ is positive. For (iv), if RH holds, Theorem 35.3.1 at $\sigma = 1$ gives $\nu = \mathfrak{U}_{1/2}$. Conversely a positive $\nu$ makes $P_{\sigma-1/2} * \nu$ a positive harmonic function on $\{\sigma > \frac12\}$ equal to $m_\sigma$ for $\sigma \ge 1$ (the two agree at $\sigma = 1$ and are harmonic with the growth of Poisson integrals; this is the argument of Theorem 19.2.2(a)). By unique continuation they agree off the zeros, which is impossible near a zero with $\beta > \frac12$, where $m_\sigma$ is unbounded of both signs. So RH holds, and then $P_{\sigma-1/2} * \nu = P_{\sigma-1/2} * \mathfrak{U}_{1/2}$ gives $\nu = \mathfrak{U}_{1/2}$. □

## 35.4 The 2 sin² lemma, the extended gamma convolution and the Davenport and Heilbronn trace

The $2\sin^2$ lemma (equation (1.2.9)) is the Lévy and Khintchine representation written with $1 - \cos(xu) = 2\sin^2(xu/2)$. Applied to the factors of the reciprocal Xi function along the real direction, with Frullani's integral $\log(1 + x^2/c^2) = \int (1 - \cos xu)\, e^{-c|u|}\, du/|u|$ for $\operatorname{Re} c > 0$, it gives for real $x$

$$\log \frac{\xi(\frac12 + x)}{\xi(\frac12)} = \int_{-\infty}^{\infty} 2\sin^2\Big(\frac{xu}{2}\Big)\, \frac{K(u)}{|u|}\, du, \qquad K(u) = \sum_{\gamma>0} e^{-\gamma|u|} \cos\big((\beta - \tfrac12)u\big), \tag{35.4.1}$$

the sum over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$, with multiplicity. Under RH, $K(u) = \sum_{\gamma>0} e^{-\gamma|u|}$ is the Poisson trace $K_{1/2}$ of Section 10.1.4, and the left side is $\sum_{\gamma>0} \log(1 + x^2/\gamma^2)$.

Proposition 35.4.1 (the Lévy density is positive with no hypothesis). *$K(u) > 0$ for every $u \neq 0$. So $x \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2} + x)$ is, unconditionally, the characteristic function of an infinitely divisible law with Lévy density $K(u)/|u|$ (Theorem 5.1.1). Under RH it is moreover an extended generalised gamma convolution (EGGC) $\exp\big(-\int_{(0,\infty)} \log(1 + x^2/y^2)\, U(dy)\big)$ with $U = \sum_{\gamma>0} \delta_\gamma$, the restriction of $\mathfrak{U}_{1/2}$ to $(0, \infty)$.*

*Proof.* Write $\delta = \beta - \frac{1}{2}$, so $|\delta| < \frac{1}{2}$. For $0 < |u| \leq \pi$, $|\delta u| < \frac{\pi}{2}$ and every term of $K(u)$ is positive. Let $|u| > \pi$. The zeros with $0 < \gamma < 100$ lie on the line [419], the lowest at $\gamma_1 = 14.1347\ldots$, and a zero off the line contributes at least $-e^{-\gamma|u|}$, so

$$e^{\gamma_1|u|} K(u) \geq 1 - \sum_{\gamma \geq 100} e^{-(\gamma-\gamma_1)\pi}.$$

Pairing $\rho$ with $1 - \bar{\rho}$, $\operatorname{Re}\frac{1}{\rho} + \operatorname{Re}\frac{1}{1-\bar{\rho}} = \frac{\beta}{|\rho|^2} + \frac{1-\beta}{|1-\rho|^2} \geq \frac{1}{1+\gamma^2}$, hence $\sum_\rho (1+\gamma^2)^{-1} \leq 2\sum_\rho \operatorname{Re}\frac{1}{\rho} = 2 + \gamma_E - \log 4\pi < 0.05$ [104, Chapter 12]. Since $(1+\gamma^2)e^{-\pi\gamma}$ decreases for $\gamma \geq 1$, the last sum is at most $0.05\,(1 + 10^4)\, e^{-\pi(100-\gamma_1)} < 10^{-113}$, and $K(u) > 0$. The display follows from the Hadamard product of $\xi(\frac{1}{2} + x)$ over pairs and quadruples of zeros and Frullani's integral with $c = \gamma - i\delta$; positivity of $K$ and the Lévy and Khintchine formula give infinite divisibility. Under RH, $\exp(-\sum_{\gamma>0} \log(1 + x^2/\gamma^2)) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + x)$. □

So positivity of the Lévy density carries no information on RH: it is proved with no hypothesis (the numbers are $2+\gamma_E-\log 4\pi = 0.046191$ and $10001\, e^{-\pi(100-\gamma_1)} = 7.0 \times 10^{-114}$, computed). The book has a stronger form: with no hypothesis the law is self decomposable and $K$ decreases on $(0, \infty)$ (Section 1.3.2, whose $k$ is this $K$ because $\sqrt{a_j} = \gamma - i(\beta - \frac{1}{2})$; Theorem 5.1.1); the proof above is direct. The EGGC property is what sees RH.

Proposition 35.4.2 (RH as an EGGC statement). *RH holds if and only if $\xi(\frac{1}{2})/\xi(\frac{1}{2} + x)$ is the characteristic function of an EGGC. When it is, its Thorin measure is the restriction of $\mathfrak{U}_{1/2}$ to $(0, \infty)$, the measure that the Cauchy deconvolution of Corollary 35.3.3(iv) recovers from $\mathfrak{U}_1$.*

*Proof.* An EGGC characteristic function $\exp\big(-\int \log(1+x^2/y^2)\, U(dy)\big)$ is analytic off the imaginary axis; a drift or a Gaussian part would change the exponent by an entire function and would not affect this. A zero $\frac{1}{2}+\delta+i\gamma$ with $\delta \neq 0$ produces a pole of $1/\xi(\frac{1}{2}+x)$ at $x = \delta + i\gamma$, off that axis. Conversely under RH the product formula above exhibits the EGGC. □

This is Theorem 10.1.2 at the centre (see also Section 1.3.2).

For a function with zeros off the line the same computation goes through. Let $\Lambda$ be entire, with $\Lambda(\frac12+x)$ even, real on the real axis, of order one, with $\Lambda(\frac12)\neq 0$ and no real zeros, whose zeros are exactly, with multiplicity, the pairs $\pm i\gamma'$ on the imaginary axis and the quadruples $\pm\delta\pm i\gamma$ with $\gamma>0$ and $\delta\neq 0$ bounded. Frullani's integral with $c=\gamma-i\delta$, $\operatorname{Re}c>0$, gives

$$\log\frac{\Lambda(\frac12+x)}{\Lambda(\frac12)}=\int(1-\cos xu)\,\frac{K_\Lambda(u)}{|u|}\,du,$$
$$K_\Lambda(u)=\sum_{\gamma'>0}e^{-\gamma'|u|}+2\sum_{\text{quadruples}}e^{-\gamma|u|}\cos(\delta u),$$

and by uniqueness of the Lévy and Khintchine representation, $\Lambda(\frac12)/\Lambda(\frac12+x)$ is an infinitely divisible characteristic function exactly when $K_\Lambda\geq 0$. For the converse, suppose $\Lambda(\frac12)/\Lambda(\frac12+x)$ is the characteristic function of an infinitely divisible law, with Gaussian coefficient $c\geq 0$ and Lévy measure $\nu$. The function is analytic near 0, so the law has a finite second moment and $\int u^2\nu(du)<\infty$. Differentiating both exponents twice shows that the finite measures $c\,\delta_0+u^2\nu(du)$ and $|u|K_\Lambda(u)\,du$ (finite, since $\int|u||K_\Lambda(u)|\,du\leq 2\sum\gamma^{-2}<\infty$) have the same Fourier transform; so $c=0$, $\nu(du)=K_\Lambda(u)\,du/|u|$ and $K_\Lambda\geq 0$, everywhere since $K_\Lambda$ is continuous on $u\neq 0$. Off line zeros make terms of $K_\Lambda$ change sign, but they need not make $K_\Lambda$ negative (Remark 22.9.2; for the function of Exercise 1.7, $K_\Lambda(u)\geq(e^{|u|}-1)^{-1}-2e^{-10.5|u|}>0$, and by the argument of Proposition 35.4.2 its reciprocal is not an EGGC).

THEOREM 35.4.3 (infinite divisibility without the EGGC property). *Let $f$ be the Davenport and Heilbronn function, $\Lambda(s)=(5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$, so that $\Lambda(s)=\Lambda(1-s)$, and let $K_{\mathrm{DH}}$ be its trace in the display above. Then $K_{\mathrm{DH}}(u)>0$ for every $u\neq 0$, and $\Lambda(\frac12)/\Lambda(\frac12+x)$ is the characteristic function of an infinitely divisible law. That law is not an EGGC, because $\Lambda$ has a zero off the line, near $0.8085+85.6993\,i$ [484, 28]. The proof rests on three finite computations, certified in ball arithmetic: (a) $\Lambda$ has a zero $\frac12+ig$ with $5.09<g<5.10$; (b) $\Lambda$ has no zero $\beta+i\gamma$ with $\beta\neq\frac12$ and $0<\gamma<85.69$; (c) $\Lambda$ has exactly one zero in a square of half width $10^{-16}$ about $0.8085171824566374+85.6993484853775922\,i$.*

*Proof.* Real zeros. With $\kappa=(\sqrt{10-2\sqrt5}-2)/(\sqrt5-1)\in(0,1)$ the coefficients of $f$ are $1,\kappa,-\kappa,-1,0$ modulo 5, and for real $\sigma>0$

$$f(\sigma)=\sum_{k\geq 0}\Big[(5k+1)^{-\sigma}-(5k+4)^{-\sigma}+\kappa\big((5k+2)^{-\sigma}-(5k+3)^{-\sigma}\big)\Big]>0,$$

each bracket being positive and $O(k^{-\sigma-1})$; the grouped series is analytic in $\sigma>0$ and equals $f$ for $\sigma>1$. So $\Lambda>0$ on $[\frac12,\infty)$ and, by the functional equation, on $\mathbb{R}$.

A strip. For $\sigma \geq 2$, $|f(s)-1| \leq \zeta(2)-1 < 1$, so every zero of $\Lambda$ has $|\beta - \frac{1}{2}| < \frac{3}{2}$.

A weighted bound. Write the zeros of $F(x) = \Lambda(\frac{1}{2}+x)$ with positive imaginary part (the upper zeros) as $\delta + i\gamma$, and $c = \gamma - i\delta$, so that $F(x)/F(0) = \prod(1+x^2/c^2)$ over the upper zeros with multiplicity. At $x = \frac{5}{2}$, $\frac{\Lambda'}{\Lambda}(3) = \sum 5\ \mathrm{Re}(x^2+c^2)^{-1}$, and with $A = \gamma^2 + x^2 - \delta^2 \geq \gamma^2$, $\mathrm{Re}(x^2+c^2)^{-1} = (A + 4\gamma^2\delta^2/A)^{-1} \geq (\gamma^2 + \frac{61}{4})^{-1}$. Since $\frac{\Lambda'}{\Lambda}(3) = \frac{1}{2}\log\frac{5}{\pi} + \frac{1}{2}\psi(2) + \frac{f'}{f}(3)$, $|f(3)-1| \leq \zeta(3)-1 < \frac{1}{4}$ and $|f'(3)| \leq -\zeta'(3) < \frac{1}{4}$, we get $\frac{\Lambda'}{\Lambda}(3) < \frac{1}{4} + \frac{1}{2} + \frac{1}{3} < 2$, so

$$\sum_{\text{upper zeros}} \Big(\gamma^2 + \frac{61}{4}\Big)^{-1} < \frac{2}{5}.$$

Positivity. $K_{\mathrm{DH}}(u) = \sum_{\text{upper zeros}} e^{-\gamma|u|}\cos(\delta u)$. For $0 < u \leq \pi/3$ every term is positive, since $|\delta u| < \frac{\pi}{2}$. For $u \geq \pi/3$ keep the term of the zero in (a), drop the other terms of zeros on the line, and bound each term of a zero off the line below by $-e^{-\gamma u}$. By (b), the weighted bound and the decrease of $(t^2 + \frac{61}{4})e^{-(t-5.1)\pi/3}$ for $t \geq 85.69$,

$$\begin{aligned} e^{gu}K_{\mathrm{DH}}(u) &\geq 1 - \sum_{\text{off line, upper}} e^{-(\gamma-5.1)\pi/3} \\ &\geq 1 - \frac{2}{5}\Big(85.69^2 + \frac{61}{4}\Big)e^{-(85.69-5.1)\pi/3} > 1 - 10^{-33}. \end{aligned}$$

So $K_{\mathrm{DH}} > 0$ on $(0,\infty)$, and the representation above gives infinite divisibility. By (c) $\Lambda$ has a zero off the line, so $1/\Lambda(\frac{1}{2}+x)$ has a pole off the imaginary axis and the law is not an EGGC (proof of Proposition 35.4.2).

The computations. The script `code/cf_dhzeros.py` (data `data/cf_dhzeros.json`) works in Arb balls at two precisions, with every count and every zero agreeing at both. (a) $\mathrm{Re}\,\Lambda(\frac{1}{2}+5.09i) = 3.564\times10^{-4}$ and $\mathrm{Re}\,\Lambda(\frac{1}{2}+5.10i) = -4.974\times10^{-4}$, each with a certified sign, so the real function $\Lambda(\frac{1}{2}+it)$ changes sign on $[5.09, 5.10]$; the zero is $\gamma_1 = 5.09416\ldots$. (b) The argument principle on the rectangle $[-2,3]\times[-85.69, 85.69]$ gives 86 zeros, so 43 in the upper half; the certified sign changes of $\Lambda(\frac{1}{2}+it)$ on $(0, 85.69)$ number 43, $\Lambda$ is certified nonzero on the real axis, and so all 43 upper zeros are on the line and simple, and none lies off it. (c) The square contains exactly one zero, certified by the argument principle. The zero lies at distance 0.3085 to the right of the line. These certified finite computations make the theorem proved. The floating point zero list to height 300 (`code/bsy_levy.py`: 204 sign changes of $\Lambda(\frac{1}{2}+it)$ and 5 quadruples, at heights 85.6993, 114.1633, 166.4793, 176.7025 and 240.4047, and $204 + 2\cdot 5 = 214$ by the argument principle) is computed evidence only; the proof uses just the part below 85.69.

□

Remark 35.4.4. If a narrower strip is wanted, $\sigma_1 = 1.4$ gives $|\beta - \frac{1}{2}| < 0.9$ and the threshold $\pi/1.8 = 1.7453$ for the first range of $u$ (the sum $\sum_{n\geq 2}|a_n|n^{-\sigma}$ equals

$0.98468 < 1$ at $\sigma = 1.4$ and $1.00428 > 1$ at 1.3938; the root of the equation sum $= 1$ lies in $[1.39492, 1.39531]$, `code/bsy_strip.py`). The theorem does not use it.

Figure 35.1 shows $K_{\mathrm{DH}}$ computed two ways. One is the sum over the zeros up to height 300. The other is the explicit formula, which uses no zeros:

$$K_{\mathrm{DH}}(u) = \frac{1}{\pi}\,\mathrm{Re}\big[e^{i\alpha u}A(u)\big] + \frac{1}{\pi}\int_0^{\alpha} \frac{\Lambda'}{\Lambda}\Big(\tfrac{1}{2}+v\Big)\sin(vu)\,dv,$$

$$A(u) = \int_0^{\infty} \frac{\Lambda'}{\Lambda}(c+it)\,e^{-tu}\,dt,$$

with $c = 3$, $\alpha = c - \frac{1}{2}$, and $f'/f$ expanded as a Dirichlet series. It follows from the residue theorem on the rectangles $1 - c \le \mathrm{Re}\,s \le c$, $0 \le \mathrm{Im}\,s \le T$, with $T \to \infty$ through heights away from the zeros, the functional equation and the reality of $\Lambda'/\Lambda$ on the real axis. The two computations agree to a relative error of $4 \times 10^{-13}$ for $0.1 \le u \le 0.5$ and $10^{-9}$ for $0.1 \le u \le 2$, at the sampled points (computed, floating point, `code/bsy_levy.py`), and the off line terms of the zeros below height 300 stay below 0.02 percent of $K_{\mathrm{DH}}$ there; the share reaches 4 percent only at $u = 0.01$, outside the sampled range and where the zero sum cut at height 300 has not converged. So the $2\sin^2$ Lévy density can stay positive with zeros off the line. RH is the EGGC property, and infinite divisibility alone does not see it.

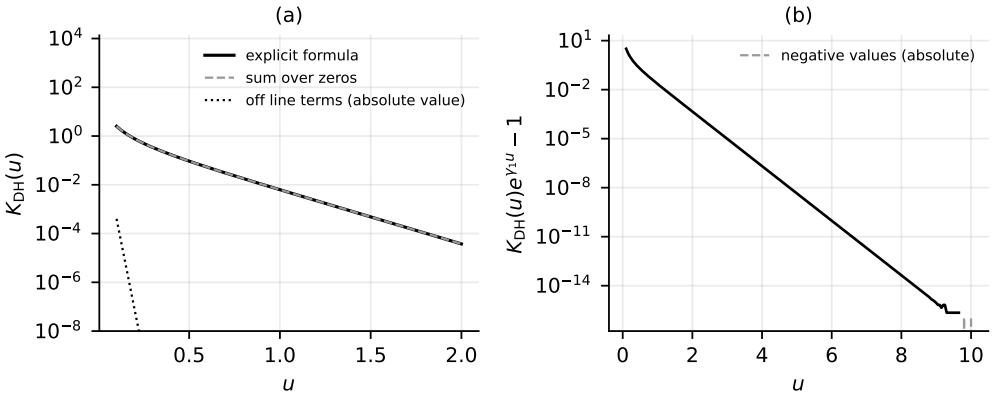



Figure 35.1: The Davenport and Heilbronn Lévy density. Left: $K_{\mathrm{DH}}$ by the explicit formula and by the sum over zeros, with the off line terms in absolute value. Right: $K_{\mathrm{DH}}(u)e^{\gamma_1 u} - 1$ on a logarithmic scale; the quantity $K_{\mathrm{DH}}(u)e^{\gamma_1 u}$ tends to 1, from above for large $u$ since the first zero is simple and the second lies on the line. Beyond $u = 9$ the right panel shows rounding noise.

Positivity of the trace is not automatic for Dirichlet series with a functional equation, as the following remark shows.

Remark 35.4.5 (a negative trace). Let $B(s) = L(s,\chi_4)$, $\Lambda_B(s) = (4/\pi)^{s/2}\Gamma(\frac{s+1}{2})B(s)$, let $p < q < 2p$ be odd primes, and put $a = \frac12\log(pq)$, $b = \frac12\log(p/q)$ and $P(s) = 1 + 2\sqrt{p}\,p^{-s} + 2\sqrt{q}\,q^{-s} + \sqrt{pq}\,(pq)^{-s}$. Then

$$\Big(\frac{4pq}{\pi}\Big)^{s/2}\Gamma\Big(\frac{s+1}{2}\Big)P(s)B(s) = 2(pq)^{1/4}\Lambda_B(s)\big[\cosh(a(s-\tfrac12)) + 2\cosh(b(s-\tfrac12))\big],$$

which is entire of order one, real and positive on $\mathbb{R}$, and invariant under $s \mapsto 1-s$. With $B_p = B\,(1-\chi_4(p)p^{-s})$, the $L$ function of the character modulo $4p$ induced by $\chi_4$, $p^{-s}B = \chi_4(p)(B - B_p)$, and similarly for $q$ and $pq$, so $PB$ is a real linear combination of $L$ functions of characters modulo 4, $4p$, $4q$ and $4pq$. With $z = a(s-\frac12)$ and $\epsilon = b/a$, at a zero of $H_\epsilon(z) = \cosh z + 2\cosh(\epsilon z)$ we have $|\sinh x| \le 2\cosh(x/2)$, so $|x| < 3$; by Rouché's theorem against $\cosh z + 2$, for $|\epsilon|$ small $H_\epsilon$ has exactly two zeros with $0 \le \operatorname{Im} z \le 2\pi$, near $\pm\operatorname{arcosh} 2 + i\pi$. For $p$ large they give the lowest zeros of the completed function above the real axis, a pair $\frac12 \pm \delta_0 + i\gamma_0$ with $\delta_0 > 0$ and $\gamma_0$ below the first zero of $\Lambda_B$; hence the trace satisfies $K(u) = 2e^{-\gamma_0 u}\cos(\delta_0 u) + O(e^{-(\gamma_0+\eta)u})$ for some $\eta > 0$, and $K((2k+1)\pi/\delta_0) < 0$ for all large $k$. So the reciprocal of this completed function is not infinitely divisible. For $p = 101$, $q = 103$: $\delta_0 = 0.28475$, $\gamma_0 = 0.67928$ and $K(\pi/\delta_0) = -1.11 \times 10^{-3}$ (computed in floating point from the zeros of $H_\epsilon$ with $m \le 5$ and the first five zeros of $L(s,\chi_4)$).

The primitive version of the question stays open (Open problem 35.6).

## 35.5 Brownian motion and a probabilistic reading of the criterion

The Cauchy kernel $P_{1-\theta}(t_0 - \cdot)$ is the law of the height at which planar Brownian motion $B$ started at $1 + it_0$ first hits the line $\operatorname{Re} s = \theta$. As in Section 9.2.3, the Cauchy process is Brownian motion run along the first passage times of an independent Brownian motion, indexed by the level.

Proposition 35.5.1 (the criterion as optional stopping). *Let $B$ be a planar Brownian motion started at $s_0 = 1 + it_0$, $\tau$ its hitting time of the line $\{\operatorname{Re} s = \theta\}$ and $H = \{\operatorname{Re} s > \theta\}$. Then $\log|h_\theta(B_\tau)|$ is integrable and*

$$\mathbb{E}\big[\log|h_\theta(B_\tau)|\big] = \log|h_\theta(s_0)| + \sum_{\beta>\theta} G_\theta(s_0,\rho).$$

*The process $\log|h_\theta(B_{t\wedge\tau})|$ is a local martingale and a submartingale; it is a uniformly integrable martingale if and only if $\zeta$ has no zero with $\beta > \theta$. Each zero contributes its Riesz mass $2\pi$, times $\frac12$, times the value at $\rho$ of $\pi^{-1}G_\theta(s_0,\cdot)$, the density of the expected occupation measure of $B$ killed at $\tau$. Planar Brownian motion almost surely never visits a given point, so no local time at the zeros is involved.*

*Proof.* The law of $B_\tau$ is $P_{1-\theta}(t_0-\tau)\,d\tau$ on the line (Section 9.2.3) and $|h_\theta| = |\zeta|$ there, so integrability and the identity are Theorem 35.2.1. The function $\log|h_\theta|$ is harmonic on $H$ off the zeros, which $B$ almost surely never visits; stopping at the exit times of the compact sets $\{|s| \le n,\ \operatorname{Re} s \ge \theta + \frac{1}{n},\ |s-\rho| \ge \frac{1}{n} \text{ for all } \rho\}$ shows that $\log|h_\theta(B_{t\wedge\tau})|$ is a local martingale. By (35.2.1), $\log|h_\theta| = v - \sum_{\beta>\theta} G_\theta(\cdot,\rho)$ on $H$, where $v(z) = \mathbb{E}_z \log|\zeta(B_\tau)|$; by the strong Markov property $v(B_{t\wedge\tau}) = \mathbb{E}[\log|\zeta(B_\tau)| \mid \mathcal{F}_{t\wedge\tau}]$ is a uniformly integrable martingale. Each $G_\theta(B_{t\wedge\tau},\rho)$ is a positive local martingale, hence a supermartingale, and so is their sum, by monotone convergence and $\sum G_\theta(s_0,\rho) < \infty$. So $\log|h_\theta(B_{t\wedge\tau})|$ is a submartingale. If there is a zero with $\beta > \theta$, its terminal value has expectation strictly above its initial value, so it is not a uniformly integrable martingale; if there is none it equals $v(B_{t\wedge\tau})$. Finally $\pi^{-1}G_\theta(\cdot,z)$ is the Green function of $H$ for the generator $\frac{1}{2}\Delta$, since $\frac{1}{2}\Delta\,\pi^{-1}G_\theta(\cdot,z) = -\delta_z$ and $G_\theta(\cdot,z) = 0$ on $\partial H$, which gives the occupation statement. □

So RH says that $\log|h_{1/2}|$, whose boundary values are $\log|\zeta|$, is a fair game for Brownian motion stopped on the critical line, and each zero off the line is a source of drift. Under RH, $m_\sigma(t)$ is the hitting density of the critical line for Brownian motion from $\sigma + it$, summed over the critical zeros, and a zero off the line is the one thing that can make that density negative.

### 35.5.1 The complex logarithm: a conformal martingale

Proposition 35.5.1 uses only the real part of $\log h_\theta$. Under the quasi Riemann hypothesis the whole complex logarithm is analytic on the half plane, and conformal invariance then fixes all the analytic moments of $\log h_\theta$ on the line under harmonic measure; with the energy identity below they give its second order structure.

Theorem 35.5.2 (moments of $\log h_\theta$ under harmonic measure). *Assume* QRH($\theta$), $\frac{1}{2} \le \theta < 1$. *Let $L$ be the analytic logarithm of $h_\theta$ on $H = \{\operatorname{Re} s > \theta\}$ with $L(\sigma+it) \to 0$ as $\sigma \to +\infty$. For almost every $\tau$ the limit $Z(\tau) = X(\tau) + iY(\tau) = \lim_{\sigma\downarrow\theta} L(\sigma+i\tau)$ exists, $X(\tau) = \log|\zeta(\theta+i\tau)|$, and $Y(\tau) = \arg h_\theta(\theta+i\tau)$ continued along the horizontal path from $+\infty$. Let $\tau$ have law $P_{1-\theta}(t_0-\cdot)$ and $z_0 = x_0+iy_0 = L(1+it_0)$. Then $\mathbb{E}|Z|^p < \infty$ for every $p < \infty$, $\mathbb{E}[Z^k] = z_0^k$ for every $k \ge 1$, and in particular $\mathbb{E}X = x_0$, $\mathbb{E}Y = y_0$, $\operatorname{Var} X = \operatorname{Var} Y$, $\operatorname{Cov}(X,Y) = 0$, with*

$$\operatorname{Var} X = \operatorname{Var} Y = \frac{1}{\pi}\iint_H G_\theta(1+it_0,w)\left|\frac{h_\theta'}{h_\theta}(w)\right|^2 dA(w).$$

*Proof.* Under QRH($\theta$), $h_\theta$ is analytic and zero free on $H$ ($h_\theta(1) = 1/(2-2\theta)$), so $L$ exists. By Theorem 35.2.1 with an empty sum, $\operatorname{Re} L = \log|h_\theta|$ is the Poisson integral of $X$, and $X$ has every moment under $P_{1-\theta}(t_0-\cdot)$ by Lemma 35.6.1. The map $w = (s-1-it_0)/(s-2\theta+1-it_0)$ sends $H$ onto the unit disc, $1+it_0$ to 0, and the harmonic measure to normalised arc length; put $F = L \circ s(w)$. Then

$\operatorname{Re} F$ is the Poisson integral of a function in $L^p$ of the circle for every $p < \infty$, so by M. Riesz's theorem on conjugate functions [123, Theorem 4.1] $F \in H^p$ for every $p < \infty$. Hence $F$ has nontangential limits $F^*$ almost everywhere, with $F^* = Z$: at boundary points where $h_\theta \neq 0$, $L$ is continuous up to the line and the horizontal convention gives the same value, and the boundary zeros form a null set. For $k \geq 1$, $F^k \in H^1$, so $\mathbb{E}[Z^k] = \int (F^*)^k \frac{d\phi}{2\pi} = F(0)^k = z_0^k$. With $k = 1, 2$, $\mathbb{E}[(Z - z_0)^2] = 0$, that is $\mathbb{E}(X - x_0)^2 = \mathbb{E}(Y - y_0)^2$ and $\mathbb{E}(X - x_0)(Y - y_0) = 0$. Finally, by the Littlewood and Paley identity, $\mathbb{E}|Z - z_0|^2 = \|F - F(0)\|_{H^2}^2 = \frac{2}{\pi} \iint_{\mathbb{D}} |F'(w)|^2 \log \frac{1}{|w|}\, dA(w)$, and since $\log(1/|w(s)|) = G_\theta(1 + it_0, s)$ and $|F'(w)|^2 dA(w) = |L'(s)|^2 dA(s)$, this is $\frac{2}{\pi} \iint_H G_\theta(1 + it_0, s)|L'(s)|^2 dA(s)$, half of it for each coordinate. □

Remark 35.5.3 (the Brownian reading). For planar Brownian motion $B$ from $1 + it_0$, stopped at its hitting time $T$ of the line, $L(B_{t\wedge T}) = z_0 + W_{A_{t\wedge T}}$ with $A_t = \int_0^t |L'(B_r)|^2\, dr$ and a complex Brownian motion $W$ (Lévy's theorem; the clock is adapted, not independent of $W$). The bound $|F|^p \leq P[|F^*|^p]$ of the proof is the harmonic majorant that makes the stopped powers uniformly integrable, so $\mathbb{E}[Z^k] = z_0^k$ is optional stopping. A zero of multiplicity $q$ to the right of the line removes this argument and makes $Y$ jump upwards by $2\pi q$ at its height; a zero on the line itself gives a jump $\pi q$ already under QRH($\theta$).

The coordinate $Y$ is $\arg h_\theta$ continued along the horizontal path, not the raw argument of $\zeta$. The raw argument differs from it by an explicit term, of the order of $2 \arctan \frac{1-\theta}{t}$, and the identities of the theorem fail for it: at $\theta = \frac{1}{2}$, $t_0 = 0$ under RH the variance of $\arg \zeta$ exceeds that of $\log|\zeta|$ by more than $\pi^2/3$. Indeed, with $w = (s-1)/s$ the raw argument is $Y - \phi$, where $\phi = \operatorname{Arg} w(\frac{1}{2} + i\tau)$ is uniform on $(-\pi, \pi)$, so $\operatorname{Var} \phi = \pi^2/3$. Writing $F(w) = \sum_{n\geq 1} a_n w^n$ as in the proof of Theorem 35.5.2 and using $\mathbb{E}[\phi \sin n\phi] = (-1)^{n+1}/n$ gives $\operatorname{Cov}(Y, \phi) = -\int_0^1 F(-r)\, dr/r$, and $F(-r) = \log h_{1/2}(\frac{1}{1+r}) > 0$ because $h_{1/2}(s) = 1 + (1 - s)\int_1^\infty \{x\} x^{-s-1}\, dx > 1$ for $\frac{1}{2} < s < 1$. Hence $\operatorname{Var}(Y - \phi) = \operatorname{Var} X + \frac{\pi^2}{3} - 2\operatorname{Cov}(Y, \phi) > \operatorname{Var} X + \frac{\pi^2}{3}$.

The moments do not determine the joint law of $(X, Y)$ by themselves; the law is the image of the Cauchy law under $L$. What the theorem adds is the moment identities and the variance formula. Selberg's central limit theorem says that for $t$ uniform on $[T, 2T]$, $\log|\zeta(\frac{1}{2} + it)|$ and $\arg \zeta(\frac{1}{2} + it)$ are asymptotically independent normals with the same variance $\frac{1}{2} \log\log T$. Theorem 35.5.2 is an exact finite height counterpart: under QRH($\theta$), at every height, the two coordinates have equal variances and zero covariance under harmonic measure. A zero to the right of the line removes this proof: the logarithm then has a branch point in the half plane, and $Y$ jumps by $2\pi q$ at the height of a zero of multiplicity $q$.

Table 35.2 tests Theorem 35.5.2. For $\zeta$ the variances of $\log|\zeta|$ and $\arg h_\theta$ agree and the covariance vanishes to $2.1 \times 10^{-5}$ in every row, the size of the

| | $\theta$ | $t_0$ | $\mathbb{E}X - x_0$ | $\mathbb{E}Y - y_0$ | $\operatorname{Var} X$ | $\operatorname{Var} Y$ | $\operatorname{Cov}(X,Y)$ |
|---|---|---|---|---|---|---|---|
| $\zeta$ | $\frac{1}{2}$ | 0 | 0.0000 | 0.0000 | 0.1415 | 0.1415 | 0.0000 |
| $\zeta$ | $\frac{1}{2}$ | 85.7 | 0.0000 | 0.0000 | 0.7056 | 0.7056 | 0.0000 |
| $\zeta$ | $\frac{1}{2}$ | 1000 | 0.0000 | 0.0000 | 1.0539 | 1.0539 | 0.0000 |
| $\zeta$ | $\frac{1}{2}$ | 5000 | 0.0000 | 0.0000 | 1.1298 | 1.1298 | 0.0000 |
| $\zeta$ | $\frac{3}{4}$ | 85.7 | 0.0000 | 0.0000 | 0.1341 | 0.1341 | 0.0000 |
| DH | $\frac{1}{2}$ | 85.7 | +1.4407 | −0.0016 | 0.6410 | 4.4414 | −0.0727 |
| DH | $\frac{3}{4}$ | 85.7 | +0.4770 | −0.0003 | 1.0124 | 3.2682 | −0.0162 |

Table 35.2: Moments of $Z = \log h_\theta(\theta + i\tau)$ under $P_{1-\theta}(t_0 - \cdot)$, computed in floating point (`code/bsy_cm.py`, shared evaluator `code/bsy_zdh.py`): Poisson weighted panels between the zeros with tanh sinh nodes, midpoint cells beyond distance 1000 out to $|\tau - t_0| \le 13000$, and the mass beyond the window ($2.45 \times 10^{-5}$ at $\theta = \frac{1}{2}$, $1.22 \times 10^{-5}$ at $\theta = \frac{3}{4}$) estimated from the sample moments of the outer window. Entries are from the finer of two meshes; the larger of the difference between the meshes and half the tail correction is at most $2.1 \times 10^{-5}$ in every entry (the row at height 5000 is in `data/bsy_cm_5000.json`). For $\zeta$ the equal variance and zero covariance identities of Theorem 35.5.2 hold to that accuracy; for Davenport and Heilbronn (DH), with $h = f$, they fail at both levels.

differences between the two meshes. For Davenport and Heilbronn the variance of the argument is 6.9 times that of the modulus at $\theta = \frac{1}{2}$ and 3.2 times at $\theta = \frac{3}{4}$, as the jump of $Y$ at the height of the zero suggests (an observation, not a consequence of the jump), and the first moment shows the defect of Theorem 35.2.1. At the three positive heights shown the variance at $\theta = \frac{1}{2}$ is close to $\frac{1}{2}\log\log t_0$ (0.71 against 0.75 at height 85.7, 1.05 against 0.97 at height 1000, 1.13 against 1.07 at height 5000), the scale of Selberg's theorem, which concerns $t$ uniform on $[T, 2T]$ and implies nothing for a window of fixed width at a given centre.

### 35.5.2 The defect of the complex moments

Without QRH($\theta$) the moments of $Z$ have explicit defects. The contour argument of Bui, Lester and Milinovich [26] moves the line of integration across horizontal cuts from the zeros for the logarithm itself; the extension to every polynomial moment is proved here.

Proposition 35.5.4 (complex moments through the cuts). *Let $\frac{1}{2} \le \theta < 1$, $z = 1 + it_0$, $z^* = 2\theta - 1 + it_0$ and $K(s) = \frac{1}{s - z^*} - \frac{1}{s - z}$. For each ordinate $\gamma$ of a zero with $\beta > \theta$ let $b_\gamma$ be the largest real part of the zeros with that ordinate, $C_\gamma = [\theta + i\gamma,\, b_\gamma + i\gamma]$, and for $x + i\gamma \in C_\gamma$ let $M_\gamma(x)$ be the number of zeros $\beta' + i\gamma$ with $\beta' > x$, counted with multiplicity. Let $L$ be the logarithm of $h_\theta$ on $\{\operatorname{Re} s > \theta\}$ minus the cuts, continued along horizontal paths from $+\infty$, where it tends to 0; let $z_0 = L(z)$ and let $L_-$ be the boundary value from below on $C_\gamma$, so that the value above minus the value below is $2\pi i M_\gamma$. Let $Z(\tau) = \lim_{\sigma \downarrow \theta} L(\sigma + i\tau)$, which exists*

*for almost every $\tau$, and $\tau \sim P_{1-\theta}(t_0 - \cdot)$. Then for every polynomial $F$*

$$\mathbb{E}F(Z) - F(z_0) = \sum_\gamma \frac{1}{2\pi i} \int_{C_\gamma} \big[F(L_- + 2\pi i M_\gamma) - F(L_-)\big] K(s)\, ds,$$

*each integral taken from left to right and the sum absolutely convergent. In particular*

$$D := \mathbb{E}Z - z_0 = \sum_{\beta>\theta} \Big[\log \frac{\rho - z^*}{\rho - z} - \log \frac{\theta + i\gamma - z^*}{\theta + i\gamma - z}\Big],$$

*the sum over the zeros with multiplicity and the logarithms continued along the cut, and*

$$\mathbb{E}Z^2 - z_0^2 = 2 \sum_\gamma \int_{C_\gamma} M_\gamma \,(L_- + \pi i M_\gamma)\, K(s)\, ds.$$

*The real part of the formula for $D$ is Theorem 35.2.1 at $1 + it_0$. The second order balance $\operatorname{Var} X = \operatorname{Var} Y$, $\operatorname{Cov}(X, Y) = 0$ holds exactly when $2\sum_\gamma \int_{C_\gamma} M_\gamma\,(L_- + \pi i M_\gamma - z_0)\, K\, ds = D^2$. When the zeros with $\beta > \theta$ are simple with distinct ordinates, $M_\gamma = 1$ on every cut and $L_- + \pi i$ is the mean of the two boundary values.*

*Proof.* On the line $K(\theta + i\tau) = 2a/(a^2 + (\tau - t_0)^2)$ with $a = 1 - \theta$, so $K(s)\, ds/(2\pi i) = P_{1-\theta}(t_0 - \tau)\, d\tau$ there and $\mathbb{E}F(Z) = \frac{1}{2\pi i} \int F(L) K\, ds$ along the line upwards. Take the rectangle $\theta \le \operatorname{Re} s \le X$, $|\operatorname{Im} s - t_0| \le T$, with $t_0 \pm T$ not ordinates of zeros; remove thin strips along the cuts, small discs about their right ends and about the zeros on them, and small half discs about the zeros on the line, and apply the residue theorem to $F(L)K$ on the rest. Its only pole there is $z$, with residue $-F(z_0)$. With the positive orientation the line is traversed downwards and the upper bank of each cut from left to right, so, as the strips, discs and half discs shrink, $\frac{1}{2\pi i} \int F(L) K\, ds$ along the line upwards equals $F(z_0)$ plus the cut terms of the statement plus $\frac{1}{2\pi i}$ times the integrals over the other three sides; the discs and half discs contribute $O(r(1 + |\log r|)^{\deg F}) \to 0$. In a fixed vertical strip $L(s) = A(s) + \sum m_\rho \log(s - \rho)$, the sum over the $O(\log(|t| + 2))$ zeros with $|\gamma - t| \le 1$, where $A = O(\log(|t| + 2))$ (remove these factors in a disc and apply the Borel and Carathéodory lemma to the rest); so the integrals of $|L|^p$ over horizontal segments and over vertical segments of unit length are $O_p(\log^{c_p}(|t| + 2))$. As $K = O(t^{-2})$ in the strip, the horizontal sides contribute $o(1)$ as $T \to \infty$, and the sum over the cuts converges absolutely, since there are $O(T \log T)$ zeros up to height $T$ and every cut is shorter than $\frac{1}{2}$. On the right side $L(s) = O(2^{-\operatorname{Re} s}) + O(|s|^{-1})$, which contributes $o(1)$ as $X \to \infty$. For $F(w) = w$ the jump is $2\pi i M_\gamma$, and $\int_{C_\gamma} M_\gamma K\, ds$ is the sum over the zeros with ordinate $\gamma$ of $\int_{\theta + i\gamma}^{\rho} K\, ds$; for $F(w) = w^2$ the jump is $4\pi i M_\gamma (L_- + \pi i M_\gamma)$. Finally $\mathbb{E}Z^2 - (\mathbb{E}Z)^2 = (\mathbb{E}Z^2 - z_0^2) - 2z_0 D - D^2$. □

The same holds for the Davenport and Heilbronn function with $h = f$, provided no zero has $\gamma = t_0$ and $\beta \geq 1$, so that $z$ lies off the cuts and $f(z) \neq 0$; the proof uses the local factorisation and the Dirichlet series bounds of $f$ in place of those of $\zeta$.

We checked Proposition 35.5.4 on the Davenport and Heilbronn function at $t_0 = 85.7$, with the zeros with $\beta > \theta$ below height 700 and their conjugates (computed, floating point; `code/bsy_cuts.py`, `data/bsy_cuts.json`): $L$ is continued along each cut, the cut integrals use Gauss and Legendre nodes, and the jump across every cut is $2\pi i$ to 7 digits. At $\theta = \frac{1}{2}$ the cut formula gives $\mathbb{E}Z - z_0 = 1.4407 - 0.0016\,i$ and the balance defect $\operatorname{Var} X - \operatorname{Var} Y + 2i \operatorname{Cov}(X, Y) = -3.8003 - 0.1455\,i$, against $1.4407 - 0.0016\,i$ and $-3.8004 - 0.1455\,i$ from the unrounded entries of Table 35.2; at $\theta = \frac{3}{4}$ it gives $0.4770 - 0.0003\,i$ and $-2.2558 - 0.0323\,i$, against $0.4770 - 0.0003\,i$ and $-2.2558 - 0.0323\,i$.

For one simple zero with $\beta > \theta$ the balance condition reads exactly $\bar{L} - z_0 = \frac{1}{2}D$ when $D \neq 0$, where $\bar{L} = D^{-1} \int_C (L_- + \pi i) K \, ds$ is a mean of $L_- + \pi i$ along its cut with the complex weight $K/D$. For a zero far from $t_0$ the weight is nearly constant along the short cut, so $\bar{L}$ nearly does not depend on $t_0$, while $D \to 0$; balance at every such $t_0$ would make $z_0 = \log h_\theta(1 + it_0)$ nearly constant, which it is not. This suggests that the balance fails for most $t_0$ when a zero lies to the right of the line (heuristic: the variation of $\bar{L}$ with $t_0$ has not been bounded). With several cuts the contributions can cancel, and no proof is known (Open problem 35.5).

## 35.6 The Monte Carlo solution

### 35.6.1 The vertical representation

Write $f(\tau) = \log|\zeta(\theta + i\tau)|$ (or $\log|h|$ for another function) and $P = P_{1-\theta}(t_0 - \cdot)$. The layer cake formula gives

$$\mathbb{E}_P[f] = \int_0^\infty P(f > \lambda)\, d\lambda - \int_0^\infty P(f < -\lambda)\, d\lambda =: A^+ - A^-.$$

In this vertical form Theorem 35.2.1 says that QRH($\theta$) holds exactly when the upper and lower tail areas of $\log|\zeta|$ under harmonic measure differ by $\log|h_\theta(1 + it_0)|$, for one (equivalently every) $t_0$. The lower tail $P(f < -\lambda)$ is the harmonic measure of the neighbourhoods $\{|\zeta| < e^{-\lambda}\}$ of the zeros on the line, which decays like $e^{-\lambda}$ when $\theta = \frac{1}{2}$ and the zeros are simple (heuristic on the whole line).

Lemma 35.6.1 (finite moments). *For $\frac{1}{2} \leq \theta < 1$, every $t_0$ and every $p < \infty$, $\mathbb{E}_P[|f|^p] < \infty$. Hence plain Monte Carlo with exact Cauchy draws is unbiased, needs no truncation of the heavy tail, and has root mean square error $(\operatorname{Var}_P f/n)^{1/2}$.*

*Proof.* The density of $P$ is $\ll_{t_0} (1 + t^2)^{-1}$, $|\zeta(\theta - it)| = |\zeta(\theta + it)|$, and $f$ has only logarithmic singularities on bounded intervals, so $\int_{-2}^{2} |f|^p \, dP < \infty$, and it suffices

to show $\int_T^{T+1} |f(t)|^p\,dt \ll_p \log^p T$ for $T \geq 2$, since $\sum_T T^{-2}\log^p T < \infty$. By the convexity bound, $\log^+ |\zeta(\theta+it)| \ll \log T$ on $[T, T+1]$. By the local formula $\log|\zeta(s)| = \sum_{|\gamma-t|\leq 1} \log|s-\rho| + O(\log t)$, uniform in $-1 \leq \sigma \leq 2$ [498, Chapter IX], and $|\theta + it - \rho| \geq |t-\gamma|$,

$$\log^- |\zeta(\theta+it)| \leq \sum_{|\gamma-t|\leq 1} \log^- |t-\gamma| + O(\log T).$$

The number $n_T$ of zeros with $|\gamma - T| \leq 2$ is $O(\log T)$. By Hölder's inequality for the sum over $n_T$ terms,

$$\int_T^{T+1} \Big( \sum_{|\gamma-t|\leq 1} \log^- |t-\gamma| \Big)^p\,dt \leq n_T^{p-1} \sum_{|\gamma-T|\leq 2} \int_{\mathbb{R}} \big(\log^- |x|\big)^p\,dx \ll_p \log^p T,$$

since $\int_{\mathbb{R}} (\log^- |x|)^p\,dx = 2\,\Gamma(p+1)$. □

### 35.6.2 Three estimators

*Quadrature* Adaptive Gauss quadrature on $|\tau - t_0| \leq W$, with the zeros on the line used as breakpoints so that the logarithmic singularities are endpoint singularities. The neglected Cauchy mass is $\frac{2}{\pi}\arctan\frac{1-\theta}{W} \approx \frac{2(1-\theta)}{\pi W}$; the neglected part of the integral needs its own bound (Cauchy and Schwarz with Lemma 35.6.1).

*Plain Monte Carlo* Draw $\tau_i = t_0 + (1-\theta)\tan\big(\pi(u_i - \frac{1}{2})\big)$ with $u_i$ uniform, and average $f(\tau_i)$. Lemma 35.6.1 makes this unbiased with a finite standard error.

*Vertical likelihood Monte Carlo* Following Polson and Scott [358], sample instead from a law that reweights the levels of $f$,

$$P_w(d\tau) \;\propto\; P(d\tau)\, w\big(f(\tau)\big),$$

and recover $\mathbb{E}_P[f]$ and the vertical curves $P(f > \lambda)$, $P(f < -\lambda)$ by self normalised weights $1/w$. We use $w(f) = |f - c| + \epsilon$, with $c$ a pilot estimate, a regularised form of the weight $|f - \mathbb{E}_P f|$, which is optimal for independent draws (Remark 35.6.2). We sample $P_w$ by a Metropolis chain mixing Cauchy independence proposals with local moves. The point of the weighting is to visit the deep levels $\{f < -\lambda\}$ near the zeros and the upper tail more often than plain sampling does (computed, code/bsy_mc.py; Table 35.3 and below).

Remark 35.6.2 (the optimal weight). Let $\mu = \mathbb{E}_P f$, $w > 0$, $Z = \mathbb{E}_P w$ and $q = Pw/Z$. For independent draws $\tau_i \sim q$ the self normalised estimator $\hat\mu = \sum_i f(\tau_i)/w(\tau_i) \big/ \sum_i 1/w(\tau_i)$ satisfies $n\,\mathrm{Var}(\hat\mu) \to Z\,\mathbb{E}_P[(f-\mu)^2/w]$, since $\hat\mu - \mu = Z\,n^{-1}\sum_i (f-\mu)(\tau_i)/w(\tau_i) + o_P(n^{-1/2})$ and $\mathbb{E}_q[(f-\mu)^2/w^2] = Z^{-1}\mathbb{E}_P[(f-\mu)^2/w]$. By the Cauchy and Schwarz inequality the limit is at least $(\mathbb{E}_P|f-\mu|)^2$, with

equality exactly for $w \propto |f-\mu|$, and $(\mathbb{E}_P|f-\mu|)^2 \le \mathrm{Var}_P f$, with equality only if $|f-\mu|$ is constant. So for independent draws the weight $|f-\mu|$ strictly beats plain Monte Carlo for these integrands, and $|f-c|+\epsilon$ with a pilot $c$ approximates it; this is the standard optimal proposal of self normalised importance sampling. For the integrands of Table 35.3 the factor $\mathrm{Var}_P f/(\mathbb{E}_P|f-\mu|)^2$ ranges over the nine rows from 1.35 (DH, $\theta=\frac34$, $t_0=85.7$) to 2.46 ($\zeta$, $\theta=\frac78$, $t_0=85.7$; the nine values are 1.35, 1.41, 1.43, 1.48, 1.55, 1.86, 2.02, 2.13 and 2.46), with an error of up to 0.03 (computed by quadrature, `code/bsy_isbound.py`), so for independent draws no weight gains more than that over plain Monte Carlo. For a Markov chain the limit is $Z^2\big[\mathrm{Var}_q Y_0 + 2\sum_{j\ge1}\mathrm{Cov}_q(Y_0,Y_j)\big]$ with $Y=(f-\mu)/w$, when a central limit theorem holds; negative autocorrelation can bring it below the value for independent draws, so the factor above does not bound Markov samplers, and the optimum changes.

### 35.6.3 Results

Table 35.3 reports the local defect $D_\theta(t_0)$ for $\zeta$ and for the Davenport and Heilbronn function (with $h=f$, which is entire), whose zero $0.808517182457 + 85.699348485378\,i$ lies off the line (certified in Section 34.2; see also [484, 28]). The prediction for $\zeta$ is zero, under RH for $\theta=\frac12$ and under QRH($\theta$) for the other levels. For Davenport and Heilbronn it is the sum of the Green potentials of its zeros with $\beta>\theta$, at least that of this zero (the other zeros below height 200 add about $2\times10^{-4}$ at $\theta=\frac12$ and less than $10^{-5}$ at $\theta=\frac34$): $\log(0.8085/0.1915)\approx1.440$ at $\theta=\frac12$ and $\log(0.3085/0.1915)\approx0.477$ at $\theta=\frac34$. The table prints the sum over the zeros below height 700.

The windowed values of the quadrature agree with the predictions to the size of the neglected tail, which is larger than the printed values suggest. For $\zeta$ at $(\frac12,0)$ the Cauchy and Schwarz bound for the neglected part, over blocks of unit length, is $5.3\times10^{-3}$ (computed up to height 1000) plus $1.1\times10^{-3}$ (extrapolated, heuristic) at $W=60$, and $1.4\times10^{-3}$ plus $1.1\times10^{-3}$ at $W=200$; at $(\frac78,0)$ it is $6.3\times10^{-4}$ plus $1.7\times10^{-4}$ and $1.6\times10^{-4}$ plus $1.7\times10^{-4}$; for Davenport and Heilbronn at $(\frac34,85.7)$ it is $1.0\times10^{-3}$ plus $6.2\times10^{-4}$ and $2.0\times10^{-4}$ plus $6.2\times10^{-4}$. The two Monte Carlo columns agree with the quadrature values within 2.4 standard errors (plain Monte Carlo for $\zeta$ at $(\frac12,85.7)$ is 2.4 off, the other seventeen entries are within 2), and at height 100 neither sampler resolves the defect $1.49\times10^{-4}$, which is far below their standard errors 0.005 and 0.013. The local character of the criterion is visible in the last rows: centred at the zero the defect of Davenport and Heilbronn at $\theta=\frac34$ is the order one number 0.477, and seen from height 100 the same zero gives $1.4\times10^{-4}$.

At $\theta=\frac78$ and $t_0=0$ the criterion of Theorem 35.2.1 reads

$$\frac{1}{\pi}\int_{-\infty}^{\infty}\log\left|\zeta\left(\tfrac78+i\tau\right)\right|\frac{1/8}{1/64+\tau^2}\,d\tau=\log 4+D_{7/8}(0),$$

| $(\theta, t_0)$ | prediction | quad. $W = 60$ | quad. $W = 200$ | plain MC | vertical MC |
|---|---|---|---|---|---|
| $\zeta$ $(\frac{1}{2}, 0)$ | 0 | $-4.1 \times 10^{-5}$ | $5.0 \times 10^{-6}$ | $-0.0040 \pm 0.0026$ | $-0.0021 \pm 0.0055$ |
| $\zeta$ $(\frac{7}{8}, 0)$ | 0 | $-7.9 \times 10^{-6}$ | $8.7 \times 10^{-7}$ | $-0.0032 \pm 0.0049$ | $-0.0173 \pm 0.0132$ |
| $\zeta$ $(\frac{1}{2}, 85.7)$ | 0 | $1.79 \times 10^{-4}$ | $-1.5 \times 10^{-6}$ | $-0.0146 \pm 0.0061$ | $-0.0064 \pm 0.0141$ |
| $\zeta$ $(\frac{3}{4}, 85.7)$ | 0 | $5.8 \times 10^{-5}$ | $-7.9 \times 10^{-7}$ | $-0.0022 \pm 0.0026$ | $-0.0074 \pm 0.0055$ |
| $\zeta$ $(\frac{7}{8}, 85.7)$ | 0 | $2.2 \times 10^{-5}$ | $-3.8 \times 10^{-7}$ | $-0.0011 \pm 0.0016$ | $0.0037 \pm 0.0032$ |
| DH $(\frac{1}{2}, 85.7)$ | 1.4407 | 1.4405 | 1.4406 | $1.4374 \pm 0.0133$ | $1.4727 \pm 0.0229$ |
| DH $(\frac{3}{4}, 85.7)$ | 0.4770 | 0.47697 | 0.47698 | $0.4759 \pm 0.0159$ | $0.4740 \pm 0.0312$ |
| DH $(\frac{3}{4}, 100)$ | $1.49 \times 10^{-4}$ | $1.16 \times 10^{-4}$ | $1.47 \times 10^{-4}$ | $0.0099 \pm 0.0054$ | $0.0139 \pm 0.0130$ |
| DH $(\frac{7}{8}, 85.7)$ | 0 | $-3.8 \times 10^{-6}$ | $-3.6 \times 10^{-7}$ | $-0.0160 \pm 0.0141$ | $-0.0311 \pm 0.0260$ |

Table 35.3: Local defects $D_\theta(t_0)$ (computed, floating point). Davenport and Heilbronn is abbreviated DH. The quadrature columns are the integral over $|\tau - t_0| \leq W$ minus $\log|h_\theta(1 + it_0)|$, with the zeros as breakpoints (code/bsy_defects.py): they are realised truncation errors of the window, not accuracies, and change with $W$. Outside the window the Cauchy mass is 0.0053 at $\theta = \frac{1}{2}$, 0.0027 at $\theta = \frac{3}{4}$ and 0.0013 at $\theta = \frac{7}{8}$ for $W = 60$. Monte Carlo (code/bsy_mc.py, computed; the seeds are stored in data/bsy_mc.json) uses $n = 20{,}000$ evaluations for $\zeta$ and 4,000 for DH. Plain Monte Carlo uses exact Cauchy draws, with standard error $\mathrm{sd}/\sqrt{n}$. The vertical sampler takes a pilot mean $c$ from $n/20$ draws, $\epsilon = 0.05$ times the standard deviation of the pilot, an even mixture of Cauchy independence proposals and Gaussian steps of scale $(1 - \theta)/2$, and a burn in of $n/20$ steps, with batch means standard errors (50 batches). The Davenport and Heilbronn prediction is the sum of the Green potentials of the zeros found below height 700; for $\theta = \frac{7}{8}$ none has $\beta > \frac{7}{8}$, but $f$ has zeros with $\beta > 1$ [111], so the exact defect is positive.

with $D_{7/8}(0) = 0$ under QRH($\frac{7}{8}$); the quadrature gives $\log 4$ to within the window error of the table. With no hypothesis, $D_\theta(0)$ is the sum of the Green potentials of the zeros with $\beta > \theta$, each at most $2(1 - \theta)(\beta - \theta)/\gamma^2$, and by the verification of RH to height $3 \times 10^{12}$ [419] and the explicit zero counting bound of [176] the sum over all possible zeros above it is below $7.4 \times 10^{-13}$ at $\theta = \frac{1}{2}$, $1.85 \times 10^{-13}$ at $\theta = \frac{3}{4}$ and $4.7 \times 10^{-14}$ at $\theta = \frac{7}{8}$ (computed, code/bsy_tailbound.py). The global criterion is therefore numerically blind to any zero above the verified range, and is useful only locally: computing $D_{7/8}(t_0)$ on a grid of heights is a practical test of QRH($\frac{7}{8}$) near each height, and not a proof.

All three estimators agree with the predictions. Quadrature gives the smallest realised errors in one dimension, with the caveat above. Plain Monte Carlo has the smaller standard error of the two sampling methods; the vertical sampler pays for its Markov dependence and its importance weights. What the vertical sampler does buy is more visits to the deep levels: at $\theta = \frac{1}{2}$ it visits $\{f < -2\}$ more often than plain sampling (computed: 315 visits against 25 for $\zeta$ at $(\frac{1}{2}, 0)$, 2952 against 679 at $(\frac{1}{2}, 85.7)$, 76 against 43 for Davenport and Heilbronn), without reaching a deeper minimum of $f$. So it samples the lower tail $A^-$, the harmonic measure of the neighbourhoods of the zeros, more densely, at the price of a larger standard

error.

Figure 35.2 shows the vertical picture. For $\zeta$ the running balance $A^+(\Lambda) - A^-(\Lambda) - \log|h(1+it_0)|$ settles at zero once the level cutoff passes about 4. For Davenport and Heilbronn it settles at about 1.44, near the Green potential 1.44 of the zero off the line, read as an imbalance between the upper and lower tails.

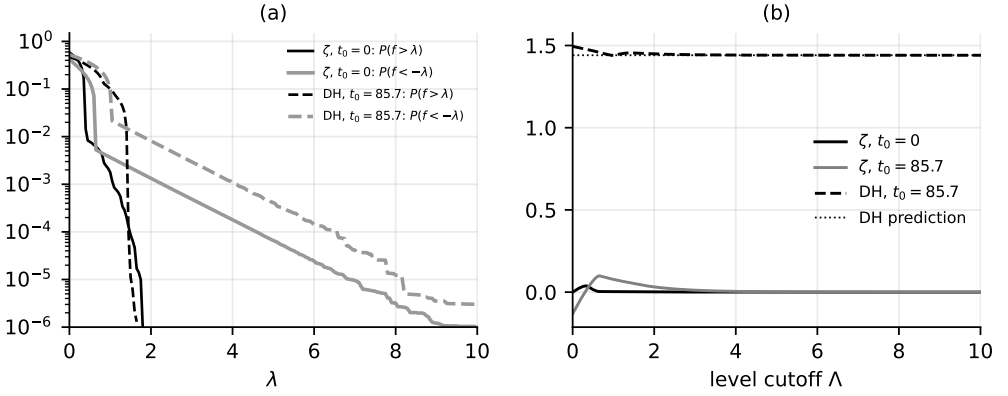



Figure 35.2: Vertical curves at $\theta = \frac{1}{2}$ (computed, floating point; `code/bsy_vertical.py`, `data/bsy_vertical.json`, figure by `code/bsy_figs.py`). Left: $P(f > \lambda)$ (black) and $P(f < -\lambda)$ (grey) under the Cauchy law centred at $t_0$, for $\zeta$ at $t_0 = 0$ (solid) and Davenport and Heilbronn at $t_0 = 85.7$ (dashed), $\lambda \le 10$, logarithmic vertical axis. Right: the running balance $\mathbb{E}[\mathrm{clip}(f, -L, L)] - \log|h(1+it_0)|$, which converges to the defect $D_{1/2}(t_0)$, for $\zeta$ at $t_0 = 0$ and 85.7 and for Davenport and Heilbronn at $t_0 = 85.7$, with the prediction 1.4407 dotted; window $|\tau - t_0| \le 200$, neglected Cauchy mass 0.0016.

### 35.6.4 Weighted Monte Carlo of Yakowitz, Krimmel and Szidarovszky

The map $\tau(u) = t_0 + a\tan\big(\pi(u - \frac{1}{2})\big)$, with $a = 1 - \theta$, writes every Cauchy average as an integral over the unit interval, $\mathbb{E}_P[f] = \int_0^1 g(u)\,du$ with $g = f \circ \tau$. Yakowitz, Krimmel and Szidarovszky [523] proposed sorting uniform nodes $u_{(1)} < \cdots < u_{(N)}$ and applying the trapezoid rule:

$$T_N = \sum_{i=0}^{N} \big(u_{(i+1)} - u_{(i)}\big)\,\frac{g(u_{(i)}) + g(u_{(i+1)})}{2}, \qquad u_{(0)} = 0,\ u_{(N+1)} = 1.$$

We set $g(0) = g(1) = 0$, a convention: $g$ has no limit at the ends, and 0 is the mean of $\log|\zeta|$ on the line $\mathrm{Re}\, s = \theta$ for $\theta > \frac{1}{2}$, and for $\theta = \frac{1}{2}$ under RH. Their theorem: if $g \in C^2[0,1]$ then $\mathbb{E}(T_N - \mathbb{E}_P f)^2 = O(N^{-4})$, against $O(N^{-1})$ for the sample mean.

Lemma 35.6.8 below gives the leading bias. The same weights define a rule for any node set, although the theorem does not transfer to dependent nodes, so the states visited by the vertical likelihood chain can be used as nodes, the spacings replacing the importance weights $1/w$.

The spacings remove the variance that the sample mean spends on the smooth part of $g$, and the Euler product supplies a second, exact reduction.

Lemma 35.6.3 (exact Cauchy means). *For any Dirichlet polynomial* $D(s) = \sum_{n \le M} b_n n^{-s}$ *and* $a = 1 - \theta$,

$$\mathbb{E}_P\big[\operatorname{Re} D(\theta + i\tau)\big] = \operatorname{Re} D(1 + it_0),$$
$$\mathbb{E}_P\Big[\log\Big|\frac{\theta + 1 + i\tau}{\theta - 1 + i\tau}\Big|\Big] = \log\Big|\frac{2 + it_0}{2 - 2\theta + it_0}\Big|.$$

*Proof.* The characteristic function of the Cauchy law gives $\mathbb{E}_P[n^{-i\tau}] = n^{-it_0} e^{-a \log n}$, so each term $n^{-\theta - i\tau}$ averages to $n^{-1-it_0}$. For the second, $|\theta - 1 + i\tau| = |\theta - (2\theta - 1) + i\tau|$ on the line, and $\log|s + 1| - \log|s - 2\theta + 1|$ is harmonic on $\{\operatorname{Re} s > \theta\}$; its Poisson integral at $1 + it_0$ is its value there. □

With $D_M$ the Selberg smoothed truncation of $\log \zeta$, $b_n = \frac{\Lambda(n)}{\log n}\big(1 - \frac{\log n}{\log M}\big)$, we apply the trapezoid rule to $g - \operatorname{Re} D_M \circ \tau - \ell \circ \tau$, where $\ell$ is the pole term of Lemma 35.6.3, and add back the exact means. Table 35.4 and Figure 35.3 report root mean square errors over 400 replications, at $t_0 = 0$, against the values 0 and $\log 4$ that the criterion gives under RH and QRH($\frac{7}{8}$); with no hypothesis they hold to within $7.4 \times 10^{-13}$ and $4.7 \times 10^{-14}$ (above).

| $\theta$ | $N$ | plain | YKS | YKS with control | gain |
|---|---|---|---|---|---|
| $\frac{1}{2}$ | 500 | $1.7 \times 10^{-2}$ | $9.6 \times 10^{-3}$ | $5.3 \times 10^{-3}$ | 3.2 |
| $\frac{1}{2}$ | 2000 | $8.4 \times 10^{-3}$ | $4.0 \times 10^{-3}$ | $2.6 \times 10^{-3}$ | 3.3 |
| $\frac{1}{2}$ | 8000 | $4.2 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | 3.7 |
| $\frac{1}{2}$ | 16000 | $3.1 \times 10^{-3}$ | $9.7 \times 10^{-4}$ | $7.2 \times 10^{-4}$ | 4.3 |
| $\frac{7}{8}$ | 500 | $3.0 \times 10^{-2}$ | $2.3 \times 10^{-3}$ | $4.0 \times 10^{-4}$ | 74 |
| $\frac{7}{8}$ | 2000 | $1.5 \times 10^{-2}$ | $1.3 \times 10^{-3}$ | $2.3 \times 10^{-4}$ | 65 |
| $\frac{7}{8}$ | 8000 | $7.7 \times 10^{-3}$ | $4.8 \times 10^{-4}$ | $1.0 \times 10^{-4}$ | 75 |
| $\frac{7}{8}$ | 16000 | $5.1 \times 10^{-3}$ | $2.9 \times 10^{-4}$ | $6.9 \times 10^{-5}$ | 75 |

Table 35.4: Root mean square error of the Cauchy average of $\log|\zeta(\theta + i\tau)|$, $t_0 = 0$, over 400 replications (computed, floating point; code/bsy_yks.py, data/bsy_yks_r400.json; the sampling error of each entry is about 3.5% and larger for the heavy tailed plain estimator). The control uses $M = 100$ at $\theta = \frac{1}{2}$ and $M = 1000$ at $\theta = \frac{7}{8}$. Gain is the ratio of plain to YKS with control. Fitted slopes over $500 \le N \le 16{,}000$ at $\theta = \frac{1}{2}$ and $\frac{7}{8}$: plain $-0.49$ and $-0.50$, YKS $-0.66$ and $-0.60$, YKS with control $-0.58$ and $-0.52$.

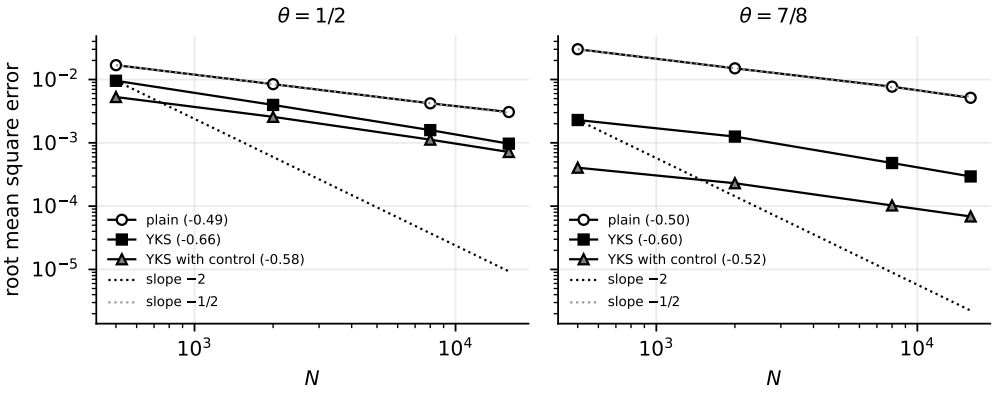



Figure 35.3: Root mean square error against $N$ (from the 400 replications of Table 35.4; code/bsy_yks.py, data/bsy_yks_r400.json), drawn for $\theta = \frac{1}{2}$ and $\frac{7}{8}$ for plain sampling, YKS and YKS with control, with the fitted slopes in the legend. The dotted lines have slope $-2$ (anchored at the first YKS point), the rate of the theorem of Yakowitz, Krimmel and Szidarovszky for integrands in $C^2[0,1]$, and slope $-\frac{1}{2}$ (anchored at the first plain point).

The gain is large at $\theta = \frac{7}{8}$. There the trapezoid weights alone reduce the error by a factor between 12 and 18, and with the control by a factor between 65 and 75, equivalent to between 4,000 and 6,000 times as many plain samples. At $\theta = \frac{1}{2}$ the factor is between 3 and 4.3. With coefficient one the control alone makes the sample mean worse: over 400 replications its root mean square error is 3.3 times that of plain sampling at $\theta = \frac{1}{2}$ and 1.04 times at $\theta = \frac{7}{8}$ (computed, code/bsy_yks.py, data/bsy_yks_r400.json); it helps only once the trapezoid weights have removed the variance of the smooth part.

The rate, however, is not $N^{-2}$. The fitted slopes over $500 \le N \le 16{,}000$ are between $-0.49$ and $-0.66$ for every method. The hypothesis $g \in C^2[0,1]$ fails, and indeed $g \notin C^1[0,1]$:

Proposition 35.6.4 (why the $N^{-4}$ bound does not apply). *Let $\frac{1}{2} \le \theta < 1$, $f(\tau) = \log|\zeta(\theta + i\tau)|$ and $g = f \circ \tau$. If $\theta = \frac{1}{2}$, $g$ is unbounded near $\tau^{-1}(\gamma)$ for every ordinate $\gamma$ of a zero on the line. If $\frac{1}{2} < \theta < 1$, $g'$ is unbounded near $u = 0$ and near $u = 1$. In both cases $g \notin C^1[0,1]$.*

*Proof.* For $\theta = \frac{1}{2}$, $f(\tau) \to -\infty$ as $\tau \to \gamma$. Let $\frac{1}{2} < \theta < 1$ and suppose $|g'| \le C$ on $(1 - \eta, 1)$. Then $g(u)$ has a limit as $u \to 1$, so $|\zeta(\theta + i\tau)| \to e^L$ for some $L$ as $\tau \to \infty$. The mean value theorems for $\sigma > \frac{1}{2}$ [498, Chapter VII] give $T^{-1}\int_1^T |\zeta(\theta + it)|^2 dt \to \zeta(2\theta)$ and $T^{-1}\int_1^T |\zeta(\theta + it)|^4 dt \to \zeta(2\theta)^4/\zeta(4\theta)$, so $e^{2L} = \zeta(2\theta)$ and $e^{4L} = \zeta(2\theta)^4/\zeta(4\theta)$, whence $\zeta(2\theta)^2 = \zeta(4\theta)$. This is false, since $\zeta(4\theta) =$

$\sum_{n=m^2} n^{-2\theta} < \sum_n d(n)\, n^{-2\theta} = \zeta(2\theta)^2$. The end $u = 0$ is the same, because $|\zeta(\theta - i\tau)| = |\zeta(\theta + i\tau)|$. □

Remark 35.6.5 (a heuristic rate). For a single component $f(\tau) = \cos(k\tau)$ the following count suggests a mean square error of order $\sqrt{ak}\, N^{-3/2}$ for the sorted trapezoid rule, that is a root mean square error of order $N^{-3/4}$ (heuristic). Near $u = 1$, $\tau - t_0 \approx a/(\pi(1-u))$, and $\tau'(u) = \pi\big(a^2 + (\tau - t_0)^2\big)/a$. A cell of width $h \approx 1/N$ in $u$ has width $\Delta \approx h\pi(\tau - t_0)^2/a$ in $\tau$ for large $\tau$. Cells with $k\Delta \lesssim 1$ contribute variance of order $h^2(k\Delta)^4$; cells beyond $T_N \asymp (aN/k)^{1/2}$, where $k\Delta \gtrsim 1$, behave like independent samples and contribute of order $h^2$ each. There are about $Na/(\pi\tau^2)$ cells per unit of $\tau$. The two sums are each of order $\sqrt{ak}\, N^{-3/2}$.

A test with $f(\tau) = \cos(\tau \log 2)$ gives the slope $-0.74$ for the trapezoid rule at both levels over $500 \le N \le 128{,}000$ (and $-0.50$ for plain sampling), close to the predicted $-\frac{3}{4}$. The Selberg smoothed polynomials $\operatorname{Re} D_{1000}$ give $-0.67$ at $\theta = \frac{7}{8}$ and $-0.71$ at $\theta = \frac{1}{2}$. The true integrands give $-0.60$ and $-0.66$ for the trapezoid rule alone, and only the plain slope is near $-\frac{1}{2}$. The likely cause (heuristic) is the features that the polynomial lacks: the logarithmic singularity at each zero on the line, and the dips of width $\theta - \frac{1}{2}$ at the heights of the zeros, whose number grows like $\log \tau$. Each of these adds error at the scale of one cell.

The $N^{-4}$ rate does not transfer to the vertical scale either, for two reasons. First, the vertical representation needs the masses $X(\lambda) = P(f < -\lambda)$ at the nodes. In nested sampling and in the vertical likelihood chain these are estimated, with a Monte Carlo error (of order $N^{-1/2}$ for independent draws) that a better quadrature rule alone does not remove (heuristic). Second, even computed exactly, the quantile function of $f$ under $P$ has a logarithmic singularity at 0 when $\theta = \frac{1}{2}$, because $P(f < -\lambda) \asymp e^{-\lambda}$ (heuristic, as above), and a flat point at every nondegenerate critical value of $f$, where the law of $f$ has an inverse square root singularity, one for each gap between consecutive zeros. Whether the vertical chain's visited states, which concentrate nodes at the deep levels and leave wider gaps elsewhere, work as nodes of the trapezoid rule is not tested here; its error is governed by the widest gaps where $g$ oscillates (heuristic).

In one dimension, quadrature with the zeros as breakpoints remains the method of choice. The case for the Yakowitz weights with the Dirichlet control is in averages that quadrature cannot reach cheaply, such as the double integrals over level and height, where the trapezoid rule generalises to Voronoi and Delaunay weights. Section 35.7 carries this out. On the line itself, Section 35.6.5 gives an exact local identity at the rates of the theory, and Section 35.6.6 a truncation of the whole line average whose tail is bounded under RH.

### 35.6.5 An exact identity on a half disc

Proposition 35.6.4 concerns the Cauchy map. Whether another parametrisation of the whole line gives an integrand smooth enough for the rates of Yakowitz,

Krimmel and Szidarovszky is not settled here: for a positive integrand the distribution function of $fP$ makes the transformed integrand constant, but it needs the integral itself. A change of domain does work. Cut the line at the heights $t_0 \pm R$ and close the contour by a semicircle in the half plane, where $\log|h|$ is harmonic away from the zeros; the Cauchy weight becomes the harmonic measure of a half disc, and the identity stays exact.

THEOREM 35.6.6 (Balazard, Saias and Yor on a half disc). *Let $c = \theta + it_0$, $R > 0$, $H = \{|s-c| < R,\ \operatorname{Re} s > \theta\}$, $z \in H$, and let $\omega_z$ be the harmonic measure of $\partial H$ seen from $z$. Let $h \not\equiv 0$ be analytic on a neighbourhood of $\overline{H}$, with no zero on the closed arc $\{|s-c| = R,\ \operatorname{Re} s \geq \theta\}$, with zeros $\rho_1, \dots, \rho_n$ of multiplicities $m_1, \dots, m_n$ on the open diameter, and with $h(z) \neq 0$. Then*

$$\int_{\partial H} \Big( \log|h| - \sum_j m_j \log|s-\rho_j| \Big)\, d\omega_z = \log|h(z)| - \sum_j m_j \log|z-\rho_j| + \sum_{\rho \in H} G_H(z,\rho),$$

*the last sum over the zeros of $h$ in $H$ with multiplicity, and the integrand on the left is real analytic on each of the two closed pieces of $\partial H$. In the coordinate $w = (s-c)/R$, with $w_z = (z-c)/R$, $w^* = -\overline{w_z}$ the reflection of $w_z$ in the diameter and $G_D(a,w) = \log|(1-\bar{a}w)/(w-a)|$,*

$$G_H(z,\cdot) = G_D(w_z,\cdot) - G_D(w^*,\cdot),$$

$$\frac{d\omega_z}{d\phi} = \frac{1-|w_z|^2}{2\pi|e^{i\phi}-w_z|^2} - \frac{1-|w^*|^2}{2\pi|e^{i\phi}-w^*|^2}$$

*on the arc $w = e^{i\phi}$, $|\phi| \leq \frac{\pi}{2}$, and on the diameter $w = iy$, $|y| \leq 1$, the density with respect to $dy$ is $\frac{1}{2\pi}\partial_x\big[G_D(w_z, x+iy) - G_D(w^*, x+iy)\big]$ at $x = 0$. Both densities are real analytic up to the corners $w = \pm i$, where they vanish to first order. In particular $h$ has no zero in $H$ if and only if the identity holds without the Green sum.*

*Proof.* Put $q(s) = h(s)\prod_j (s-\rho_j)^{-m_j}$. Then $q$ is analytic on a neighbourhood of $\overline{H}$ with no zero on $\partial H$, so $\log|q| = \log|h| - \sum_j m_j \log|s-\rho_j|$ is real analytic on a neighbourhood of $\partial H$. Green's identity on $H$ with small discs about the zeros of $q$ removed (on the circle of radius $r$ about a zero of multiplicity $m$ the normal derivative of $\log|q|$ integrates to $2\pi m + O(r)$) gives the Riesz decomposition $\log|q(z)| = \int_{\partial H} \log|q|\, d\omega_z - \sum_{\rho\in H} G_H(z,\rho)$, which is the identity. The function $G_D(w_z,\cdot) - G_D(w^*,\cdot)$ vanishes on the unit circle and, because $w^*$ is the reflection of $w_z$ in the imaginary axis, on the diameter, and its only singularity in the half disc is the logarithmic pole at $w_z$; so it is the Green function of $H$. The harmonic measure density is its inner normal derivative divided by $2\pi$, which gives the two

formulas. They are rational in the boundary point, and at the right angled corners $w = \pm i$ each has a simple zero, as the reflection principle predicts ($r^{\pi/(\pi/2)-1} = r$ at distance $r$ from a corner). Since $G_H > 0$ on $H$, the Green sum vanishes exactly when $H$ contains no zero. □

REMARK 35.6.7 (the half plane as a limit). Let $h = h_\theta$ and $z = 1 + it_0$. Each $\log|s - \rho_j|$ is the Poisson integral of its boundary values (it is the real part of $\log(s-\rho_j)$, which lies in $H^p$ of the half plane for every $p < \infty$), so the identity also holds with the subtracted terms removed on both sides. As $R \to \infty$ the density on the diameter increases to the Cauchy density $P_{1-\theta}(t_0 - \cdot)$, since harmonic measure on the common boundary increases with the domain, and the Green sum increases to the sum of Theorem 35.2.1. On the arc the density is at most $CR^{-1}\cos\phi$, so the part of the arc in $\theta \le \operatorname{Re} s \le 2$ has harmonic measure $O(R^{-3})$; along radii $R_k \to \infty$ with $|t_0 \pm R_k - \gamma| \ge c/\log R_k$ for every zero, $|\log|h_\theta|| = O(\log^2 R_k)$ there, while $|\log|h_\theta(s)|| = O(2^{-\operatorname{Re} s} + R_k^{-1})$ on the rest of the arc. So the arc term tends to 0, and with Lemma 35.6.1 and dominated convergence on the diameter Theorem 35.6.6 becomes Theorem 35.2.1 at $1 + it_0$. That the arc carries vanishing harmonic measure would not suffice by itself: for $h(s) = e^{s-\theta}$ the arc contributes $1 - \theta$ for every $R$.

Wunderlich [363] gives other finite radius forms of the criterion, by Jensen's formula on discs with the trivial zeros removed explicitly, and a relative entropy form: if $X$ has the density $6/(\pi(1+t^2)(4+t^2))$, which is twice the Cauchy$(0,1)$ density minus the Cauchy$(0,2)$ density, $\Xi(t) = \xi(\frac12 + it)$ is Riemann's Xi function, $\mathbb{E}\log|\Xi(X)|$ is finite, $W = |\Xi(X)/\Xi(0)|^2$, and $\nu$ is the law with density $W/\mathbb{E}W$ with respect to the law $\lambda$ of $X$, then $D_{\mathrm{KL}}(\lambda\|\nu) = \log\mathbb{E}W - \mathbb{E}\log W$, with no hypothesis; in his formulation RH is the saturation of a separate bound. The half disc sits on the line and is local in height, which is what the quadrature rates below use.

LEMMA 35.6.8 (the sorted trapezoid rule). *Let $g \in C^4[a,b]$ and let $T_N$ be the sorted trapezoid rule at $N$ independent uniform nodes on $[a,b]$ with the end points added, as in Section 35.6 (there $g$ vanishes at the end points, here its end values are used). Then*

$$\mathbb{E}T_N - \int_a^b g = \frac{(b-a)^2\big(g'(b) - g'(a)\big)}{2(N+1)(N+2)} + O(N^{-3}), \qquad \operatorname{Var} T_N = O(N^{-5}).$$

*So the mean square error is $O(N^{-4})$, and $O(N^{-5})$ when $g'(a) = g'(b)$.*

*Proof.* Take $[a,b] = [0,1]$ and write $T_N - \int_0^1 g = \sum_{i=0}^{N} e(u_{(i)}, u_{(i+1)})$, $u_{(0)} = 0$, $u_{(N+1)} = 1$, where $e(x, x+h) = \frac{h}{2}(g(x)+g(x+h)) - \int_x^{x+h} g = \frac{h^3}{12}g''(x+\frac{h}{2}) + O(h^5)$ and $|e(x,x+h)| \le \frac{h^3}{12}\|g''\|_\infty$. The inner intervals form a point process in (left end, length) of intensity $N(N-1)(1-h)^{N-2}$ on $\{x > 0,\ h > 0,\ x+h < 1\}$, and the two

outer ones have lengths of density $N(1-h)^{N-1}$. Since $\int_0^{1-h} g''(x+\frac{h}{2})\,dx = g'(1-\frac{h}{2}) - g'(\frac{h}{2}) = g'(1) - g'(0) + O(h)$ and $N(N-1)\int_0^1 (1-h)^{N-2}h^3\,dh = \frac{6}{(N+1)(N+2)}$, the inner intervals give the main term with error $O(N^2\int_0^1 (1-h)^{N-2}h^4 dh) = O(N^{-3})$; the outer intervals give $O(N\int_0^1 (1-h)^{N-1}h^3 dh) = O(N^{-3})$, and the $O(h^5)$ terms give $O(\mathbb{E}\sum_i h_i^5) = O(N^{-4})$. Replacing one node by an independent copy merges its two adjacent intervals and splits the interval containing the copy, so at most six intervals change, each a spacing, a size biased spacing or a sum of two spacings, with sixth moment $O(N^{-6})$. Hence $\mathbb{E}(T_N - T'_N)^2 = O(N^{-6})$, and the Efron and Stein inequality gives $\operatorname{Var} T_N \le \frac{N}{2}\mathbb{E}(T_N - T'_N)^2 = O(N^{-5})$. □

Each piece of the boundary integral in Theorem 35.6.6 is a smooth integrand on a closed interval that vanishes at the ends. By Lemma 35.6.8 the sorted trapezoid rule with $N/2$ nodes on each piece has mean square error $O(N^{-4})$; its bias $\frac{2(b-a)^2(g'(b)-g'(a))}{(N+2)(N+4)} + O(N^{-3})$ vanishes to that order only if the end slopes agree. The change of variable $y = \Phi(v) = v + \sin(\pi v)/\pi$ on each piece, written on $[-1,1]$, makes the transformed integrand $g(\Phi(v))\Phi'(v)$ vanish to at least fifth order at the ends, because $\Phi'(\pm(1-r)) = \frac{\pi^2}{2}r^2 + O(r^4)$ and $1 - |\Phi(\pm(1-r))| = \frac{\pi^2}{6}r^3 + O(r^5)$; both end slopes are then zero, and by the same lemma the mean square error is $O(N^{-5})$. These are upper bounds (proved).

Table 35.5 and Figure 35.4 test this with $R = 1$, $t_0 = 85.7$ and $z = c + \frac{1}{2}R$, over 200 replications with $N/2$ nodes on each piece (computed, floating point; `code/bsy_halfdisk.py`, `data/bsy_halfdisk.json`), in three cases: $\zeta$ at $\theta = \frac{3}{4}$, where nothing is subtracted; $\zeta$ at $\theta = \frac{1}{2}$, where the zero $\frac{1}{2} + 84.7354\ldots i$ on the diameter is subtracted; and the Davenport and Heilbronn function at $\theta = \frac{1}{2}$, with its zero $0.8085\ldots + 85.6993\ldots i$ in the half disc, so that the right side includes $G_H(z,\rho) = 1.12940$; no zero of it lies on the diameter. Gauss and Legendre quadrature of the boundary integral agrees with the right side to $5\times10^{-15}$ (largest difference $4.2\times10^{-15}$). The sorted trapezoid rule falls with slopes between $-2.07$ and $-2.28$, within the $N^{-4}$ mean square bound, its error mostly bias (the bias term of Lemma 35.6.8 is within 8 percent of the measured mean error for $N \ge 4096$); with the change of variable the slopes are between $-2.49$ and $-2.51$, consistent with the $N^{-5}$ bound; plain Monte Carlo falls as $N^{-1/2}$. At $N = 32{,}768$ the weighted rules are between $1.5\times10^5$ and $1.5\times10^6$ times more accurate than plain sampling. The change of variable has the steeper slope, but at these $N$ it is not the more accurate rule for $\zeta$ and DH at $\theta = \frac{1}{2}$ (whether the two rules cross beyond $N = 32{,}768$ was not computed).

So the mean square rates $N^{-4}$ and $N^{-5}$ of one dimensional weighted Monte Carlo are reached on the line by an exact local form of the criterion, which still detects zeros off the line through the Green function of the half disc. For the Cauchy average over the whole line no such rate is proved (Section 35.6.6 and Open problem 35.3).

| case | target | plain | sorted trapezoid | with $\Phi$ |
|---|---|---|---|---|
| $\zeta$, $\theta = \frac{3}{4}$ | 0.028322 | $1.2 \times 10^{-3}$ $(-0.49)$ | $3.4 \times 10^{-9}$ $(-2.07)$ | $2.7 \times 10^{-9}$ $(-2.49)$ |
| $\zeta$, $\theta = \frac{1}{2}$ | 0.048774 | $2.7 \times 10^{-3}$ $(-0.52)$ | $1.9 \times 10^{-9}$ $(-2.28)$ | $5.1 \times 10^{-9}$ $(-2.49)$ |
| DH, $\theta = \frac{1}{2}$ | $-0.365254$ | $2.1 \times 10^{-3}$ $(-0.50)$ | $7.8 \times 10^{-9}$ $(-2.16)$ | $1.4 \times 10^{-8}$ $(-2.51)$ |

Table 35.5: Root mean square error at $N = 32{,}768$ for the half disc identity of Theorem 35.6.6, $R = 1$, $t_0 = 85.7$, 200 replications, with the slopes fitted for $N \geq 1024$ in brackets (computed, floating point; `code/bsy_halfdisk.py`, `data/bsy_halfdisk.json`). The target is the right side of the identity; for the Davenport and Heilbronn function (DH) it includes the Green term $G_H(z, \rho)$.

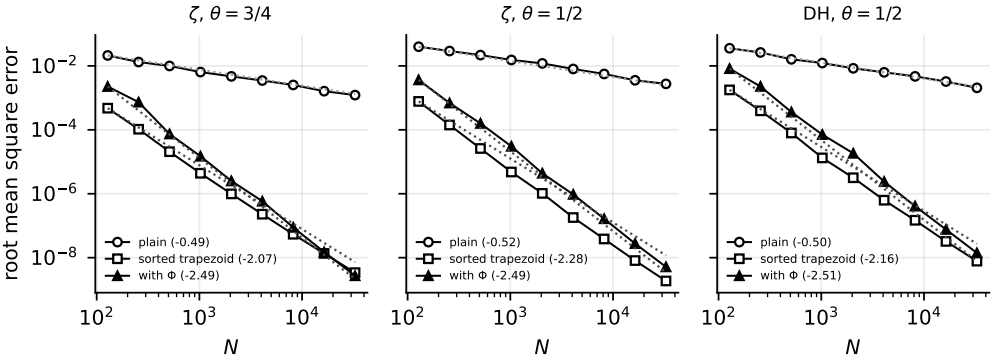



Figure 35.4: Root mean square error against $N$ for the half disc identity (data of Table 35.5; figure by `code/bsy_figs.py`). The dotted lines have slopes $-\frac{1}{2}$, $-2$ and $-\frac{5}{2}$.

### 35.6.6 Truncation on the whole line

Proposition 35.6.4 concerns the integrand after the Cauchy map. On the line itself the Cauchy tail beyond a height can instead be bounded directly. Assume RH. Then

$$\int_T^t \log|\zeta(\tfrac{1}{2}+iu)|\,du = O\Big(\frac{\log t}{(\log\log t)^2}\Big) \qquad (3 \leq T \leq t),$$

the estimate behind Theorem 1.1 of Bui, Lester and Milinovich [26]. Let $A(t)$ be the left side with $T$ replaced by $t_0 + T$ and $w = P_{1/2}(t_0 - \cdot)$. Integration by parts gives $\int_{t_0+T}^{\infty} \log|\zeta(\frac{1}{2}+i\tau)|\,w(\tau)\,d\tau = -\int_{t_0+T}^{\infty} A(\tau)w'(\tau)\,d\tau$, since $A(t)w(t) \to 0$, and $w'(\tau) = O_{t_0}(\tau^{-3})$. With the same bound below $t_0 - T$,

$$\int_{|\tau-t_0|>T} \log|\zeta(\tfrac{1}{2}+i\tau)|\,P_{1/2}(t_0-\tau)\,d\tau = O_{t_0}\Big(\frac{\log T}{T^2(\log\log T)^2}\Big)$$

(conditional on RH). At $t_0 = 0$ this is [26, Theorem 1.1], where it is also shown that the tail is not $O(T^{-2})$. At a level $\theta \in (\frac{1}{2}, 1)$, under RH, the integral of $\log\zeta$

around the rectangle with vertices $\theta + iT$, $\theta + it$ and $+\infty$ bounds the primitive of $\log|\zeta(\theta+iu)|$ by $O_\theta(\log t)$, and the tail by $O_\theta(T^{-2}\log T)$.

For the rule place $N = 2T/h$ nodes $\tau_j = t_0 - T + h(j+U)$, $0 \le j < N$, with one shift $U$ uniform on $[0,1)$. At $\theta = \frac{3}{4}$ nothing is deflated. At $\theta = \frac{1}{2}$ deflate each zero $\rho_k = \frac{1}{2} + i\gamma_k$ with $|\gamma_k - t_0| \le T + 1$ by

$$u_k(s) = \log\Big|\frac{(s-\rho_k)(s-\rho_k+2)}{(s-\rho_k+1)^2}\Big|,$$

which is harmonic on $\operatorname{Re} s > \frac{1}{2}$ and $O(|s|^{-2})$; on the line it is $\frac{1}{2}\log\frac{x^2(4+x^2)}{(1+x^2)^2}$ with $x = \tau - \gamma_k$, about $x^{-2}$ for large $|x|$, and its integral over $\mathbb{R}$ is 0. The estimate is $h\sum_j\big[(f - \sum_k u_k)P\big](\tau_j) + \sum_k \int_{|\tau-t_0|\le T} u_k P\,d\tau$, the window means of the $u_k$ computed by quadrature. The shifted grid is unbiased for the integral over the window, so the mean error over $U$ is exactly minus the Cauchy tail beyond $T$, and one shift adds a discretisation error of mean zero. Adding back instead the whole line means $u_k(1+it_0)$ would add the tails $\sum_k \int_{|\tau-t_0|>T} u_k P\,d\tau$, about the value of $P$ at the edge times the sum of the reciprocal distances of the zeros to the edges; for this variant at $\theta = \frac{1}{2}$ the measured $T^2 \cdot |\text{error}|$ grows from 0.15 at $T = 10$ to 1.24 at $T = 640$, more than 97 percent of the error at $T = 640$ being the deflation tails (computed), so it is not the tail of $\log|\zeta|$.

| | $\theta = \frac{3}{4}$, $h = 0.02$ | | | $\theta = \frac{1}{2}$, $h = 0.01$, zeros deflated | | | |
|---|---|---|---|---|---|---|---|
| $T$ | error | spread | $T^2 \cdot \lvert\text{error}\rvert$ | error | spread | $T^2 \cdot \lvert\text{error}\rvert$ | zeros |
| 10 | $-9.47 \times 10^{-4}$ | $5.5 \times 10^{-8}$ | 0.095 | $-2.09 \times 10^{-3}$ | $8.1 \times 10^{-7}$ | 0.209 | 10 |
| 40 | $-3.40 \times 10^{-5}$ | $2.2 \times 10^{-8}$ | 0.054 | $-5.37 \times 10^{-5}$ | $1.3 \times 10^{-8}$ | 0.086 | 33 |
| 160 | $+4.36 \times 10^{-6}$ | $9.8 \times 10^{-9}$ | 0.112 | $+1.05 \times 10^{-5}$ | $9.4 \times 10^{-9}$ | 0.268 | 123 |
| 640 | $+6.47 \times 10^{-8}$ | $8.2 \times 10^{-10}$ | 0.027 | $+7.84 \times 10^{-8}$ | $5.5 \times 10^{-10}$ | 0.032 | 742 |
| 1280 | $-1.36 \times 10^{-7}$ | $1.8 \times 10^{-10}$ | 0.223 | | | | |

Table 35.6: Truncated Cauchy average of $\log|\zeta(\theta+i\tau)|$ over $|\tau - t_0| \le T$ at $t_0 = 85.7$, minus $\log|h_\theta(1+it_0)|$; as in Remark 35.2.2, the zeros above $3 \times 10^{12}$ move the whole line average away from $\log|h_\theta(1+it_0)|$ by less than $10^{-12}$, with no hypothesis. Mean error and spread over 10 random shifts of the grid (computed, floating point; `code/bsy_trunc.py`, `data/bsy_trunc.json`). At $\theta = \frac{1}{2}$ the zeros with $|\gamma - t_0| \le T + 1$ (the zero range convention, counted in the last column) are deflated and their window means added back (the unbiased rule). The shifted grid is unbiased for the window integral, so the mean error is minus the Cauchy tail beyond $T$, up to the spread. In this range $T^2 \cdot |\text{error}|$ stays below 0.27; the bounds of [26] allow slow growth.

Table 35.6 shows the behaviour at $t_0 = 85.7$. In this range the measured error is the tail, which under RH is $O(T^{-2}\log T)$, that is $O(N^{-2}\log N)$ at a fixed step, and the spread over the shifts is at most $8.1 \times 10^{-7}$ (at $T = 10$, $\theta = \frac{1}{2}$) and below $1.0 \times 10^{-8}$ for $T \ge 160$. The error changes sign with $T$ (between $T = 40$

and 160 and again, at $\theta = \frac{3}{4}$, between 640 and 1280), a bounded oscillation of $T^2 \cdot |\text{error}|$ consistent with $T^{-2}$ up to logarithms in this range (computed). This is computed, not a theorem about the rule: at a fixed step the discretisation error of one shift need not tend to 0 as $T$ grows (besides an endpoint term of order $h$ times the integrand at the edges of the window, it contains an aliasing term, of order $e^{-2\pi b/h}$ for an integrand analytic in a strip of width $b$, which in general does not decay as $T$ grows), so a rate in $N$ needs bounds on the deflated integrand, or a step that shrinks with $N$ (Open problem 35.3). The tail is bounded under RH, not corrected; only the half disc of Theorem 35.6.6 gives an exact identity at these rates.

With no hypothesis, Bui, Lester and Milinovich [26, Theorem 1.2] prove that for $T \geq 3$

$$I(T) = \int_{-T}^{T} \frac{\log|\zeta(\frac{1}{2}+it)|}{\frac{1}{4}+t^2}\,dt = 2\pi \sum_{\substack{|\gamma|\leq T\\ \beta>1/2}} \log\Big|\frac{\rho}{1-\rho}\Big| + O\Big(\frac{\log T}{T^2}\Big):$$

up to that error the truncated integral is the sum over the zeros below height $T$, the global counterpart of centring the kernel at the height of interest. With Theorem 35.2.1 the tail is $I(\infty) - I(T) = 2\pi \sum_{|\gamma|>T,\,\beta>1/2} \log|\rho/(1-\rho)| + O(\log T/T^2)$, which is $O(\log T/T)$ with no hypothesis, since each term lies between 0 and $(2\beta-1)/(2|1-\rho|^2) \leq 1/(2\gamma^2)$ and there are $O(T\log T)$ zeros up to height $T$. Under RH their Omega theorem disproves the conjecture $I(T) = O(T^{-2})$ of Borwein, Bradley and Crandall [23], as stated in [26].

### 35.6.7 Scanning heights

The local criterion invites a scan over $t_0$. Chapter 34 provides two tools for it, the particle filter along the Cauchy flow (Section 34.3) and nested sampling over heights (Section 34.9), with the results of Tables 34.6 and 34.14. Once a suspicious height is found, the defect $D_\theta(t_0)$ of Theorem 35.2.1 gives a check, in floating point: centred at a zero at distance $\delta$ to the right of the line $\operatorname{Re} s = \theta$ it is at least $\log\frac{1-\theta+\delta}{1-\theta-\delta} \approx \frac{2\delta}{1-\theta}$.

## 35.7 Two dimensions: the smoothed Riesz identity

The one dimensional rule stalls between $N^{-1/2}$ and $N^{-2/3}$ because the Cauchy kernel has a heavy tail and $\log|\zeta|$ never stops oscillating. Both come from the boundary formulation: Theorem 35.2.1 tests $\log|h_\theta|$ against the Green function of a half plane, whose normal derivative on the line is $2\pi$ times the Cauchy density. Testing against a smooth bump of compact support instead gives an identity with no tail at all, at the price of one more dimension.

THEOREM 35.7.1 (smoothed Riesz identity). *Let $f$ be meromorphic on a neighbourhood of a compact set $K \subset \mathbb{C}$, not identically zero on any component of the neighbourhood, and let $\varphi \in C_c^\infty$ with support in $K$. Then*

$$I(\varphi) := \iint \log|f(\sigma+it)|\,\Delta\varphi(\sigma,t)\,d\sigma\,dt = 2\pi\Big(\sum_{f(\rho)=0}\varphi(\rho) - \sum_{f(p)=\infty}\varphi(p)\Big),$$

*with zeros and poles counted with multiplicity. The first identity is the classical distributional identity $\Delta\log|f| = 2\pi\big(\sum\delta_\rho - \sum\delta_p\big)$ paired with the test function $\varphi$. Equivalently, for $f=\xi$, $-\iint\big(\pi\,m_\sigma(t)\,\partial_\sigma\varphi + \partial_t\log|\xi|\,\partial_t\varphi\big)\,d\sigma\,dt = 2\pi\sum\varphi(\rho)$, so the Thorin density $m_\sigma$ is the $\sigma$ component of the field whose divergence counts the zeros.*

*Proof.* In the sense of distributions $\Delta\log|f| = 2\pi\big(\sum\delta_\rho - \sum\delta_p\big)$, because $\log|f| = \log|g| + \sum\log|s-\rho| - \sum\log|s-p|$ with $g$ analytic and zero free near $K$, $\log|g|$ harmonic, $\Delta\log|s-\rho| = 2\pi\delta_\rho$ and $\Delta\log|s-p|^{-1} = -2\pi\delta_p$ at a pole. Pair with $\varphi$ and integrate by parts twice. The second form is one integration by parts, with $\partial_\sigma\log|\xi| = \pi m_\sigma$ (equation (30.1.1) and Theorem 30.1.1); $\nabla\log|\xi|$ is locally integrable since its singularities are of order $1/|s-\rho|$. □

Theorem 35.2.1 corresponds formally to the case where $\varphi$ is the Green function of the half plane $\{\operatorname{Re} s > \theta\}$ with pole at $1+it_0$. That choice is singular at the pole and not compactly supported, and integrating out $\sigma$ leaves the Cauchy average on the boundary line. With a bump in its place, the two facts that blocked the one dimensional rule disappear. If $f$ has no zeros in the support, the integrand $g = \log|f|\,\Delta\varphi$ is in $C_c^\infty$, and $I(\varphi) = 0$ exactly. A zero $\rho$ in the support moves $I$ by $2\pi\varphi(\rho)$, which is of order one unless $\rho$ is near the edge of the support. With a partition of unity $\sum_k\varphi_k = 1$ on a region, $\sum_k I(\varphi_k) = 2\pi$ times the number of zeros in the region, plus $2\pi\sum_k\varphi_k(\rho)$ for each zero $\rho$ where $0 < \sum_k\varphi_k < 1$, so overlapping bumps leave no blind spots. This is a smoothed form of the argument principle. The numerical point is that the integrand is smooth and compactly supported when no zero lies in the support (after deflation otherwise), which is the setting in which weighted Monte Carlo pays.

### 35.7.1 WEIGHTED MONTE CARLO IN THE PLANE

Map the box to $[-1,1]^2$ and take $\varphi(x,y) = b(x)b(y)$ with $b(x) = e^{-1/(1-x^2)}$ on $|x|<1$. The rules we compare all use $N$ nodes:

1. plain Monte Carlo, the sample mean at uniform nodes;
2. Voronoi weights: each uniform node is weighted by the area of its Voronoi cell, the two dimensional form of weighting by spacings;

3. Delaunay weights: piecewise linear interpolation on the Delaunay triangulation of the nodes and the four corners, the two dimensional trapezoid rule; at uniform nodes and at jittered nodes, one uniform point in each cell of a $\sqrt{N} \times \sqrt{N}$ grid;
4. the randomly shifted product trapezoid rule on a $\sqrt{N} \times \sqrt{N}$ grid;
5. deflation: replace $\log|f|$ by $\log|f(s)/(s-\hat{\rho})|$ for an estimate $\hat{\rho}$ of a zero, and add back $2\pi\varphi(\hat{\rho})$, which is exact by Theorem 35.7.1 for meromorphic $f$.

All of these are unbiased or asymptotically unbiased, and the spread over replications estimates their random error, not their bias.

PROPOSITION 35.7.2 (rates). *After an affine change of variables the box is the torus $[0,1)^2$; the shifted product trapezoid rule with $N = m^2$ nodes is $Q(\delta) = m^{-2}\sum_{j\in\{0,\dots,m-1\}^2} g(j/m+\delta)$, with $\delta$ uniform on $[0,1/m)^2$. (i) If $g$ has a smooth periodic extension (for example the integrand $\log|f|\,\Delta\varphi$ with the flat bump of the text and no zero in the box), then $Q(\delta) - \int g = O(N^{-k})$ for every $k$, uniformly in $\delta$, and $\mathbb{E}_\delta Q = \int g$. (ii) If $g = \psi \log|s-\rho|$ with $\psi \in C_c^\infty((0,1)^2)$ and $\rho \in (0,1)^2$, then $\mathbb{E}_\delta Q = \int g$ and $\left(\mathbb{E}_\delta|Q - \int g|^2\right)^{1/2} = O(N^{-1})$. For a fixed shift whose nodes all lie at distance at least $c/m$ from $\rho$, $|Q - \int g| = O_c(N^{-1}\log N)$. If $\psi(\rho) \neq 0$, no bound uniform in the shift holds. (iii) On the box, for $g$ as in Theorem 35.7.3 (for example the integrand of the text when $f$ has no zero in the closed box), Voronoi weights and Delaunay weights at independent uniform nodes have root mean square error $O(N^{-3/2})$, against $N^{-1/2}$ for the sample mean (Theorem 35.7.3); at jittered nodes the same rate is computed, not proved.*

*Proof.* For $g$ square integrable on the torus, Poisson summation gives, in $L^2(d\delta)$, $Q(\delta) - \int g = \sum_{\ell\neq 0} \hat{g}(m\ell)\, e^{2\pi i m\langle \ell,\delta\rangle}$ with $\hat{g}(k) = \int g(x) e^{-2\pi i\langle k,x\rangle} dx$. These exponentials are orthonormal for $\delta$ uniform on $[0,1/m)^2$, so $\mathbb{E}_\delta Q = \int g$ and $\mathbb{E}_\delta|Q - \int g|^2 = \sum_{\ell\neq 0} |\hat{g}(m\ell)|^2$. (i) $\hat{g}$ decays faster than every power, and the series converges absolutely, uniformly in $\delta$. (ii) In the sense of distributions $\Delta g = 2\pi\psi(\rho)\delta_\rho + q$ with $q = 2\nabla\psi\cdot\nabla\log|s-\rho| + \Delta\psi\,\log|s-\rho| \in L^1$, so $4\pi^2|k|^2|\hat{g}(k)| \le 2\pi|\psi(\rho)| + \|q\|_1$ and $\mathbb{E}_\delta|Q - \int g|^2 \le Cm^{-4}\sum_{\ell\neq 0}|\ell|^{-4} = O(N^{-2})$. For the separated bound put $h = 1/m$: on a cell at distance $r \ge 2h$ from $\rho$ the trapezoid error is $O(h^4 r^{-2})$, and the $O(r\,dr/h^2)$ cells at distance about $r$ give $O(h^2\log(1/h))$ in total; the $O(1)$ cells within $2h$ of $\rho$ contribute $O(h^2\log(1/h))$ to $\int g$ and, by the separation, $O(h^2\log(1/(ch)))$ to $Q$. If $\psi(\rho) \neq 0$, a node at distance $\eta$ from $\rho$ contributes $m^{-2}\psi(\rho)\log\eta + O(m^{-2})$, which is unbounded as $\eta \to 0$, so no bound uniform in $\delta$ holds. (iii) is Theorem 35.7.3 with $d = 2$. □

### 35.7.2 A proof of the $N^{-3/2}$ rate

The rate in Proposition 35.7.2(iii) can be proved at independent uniform nodes, in every dimension, for the Voronoi rule and for the Delaunay rule with the corners

of the box added. Write $x_{\mathrm{nn}}(x)$ for the node nearest to $x$, $d(x) = |x - x_{\mathrm{nn}}(x)|$ and $v(x) = x_{\mathrm{nn}}(x) - x$. Then $\frac{1}{2}d(x)^2 = \frac{1}{2}|x|^2 - \max_i\big(x \cdot X_i - \frac{1}{2}|X_i|^2\big)$ is Lipschitz with gradient $-v(x)$ almost everywhere, so the displacement to the nearest node is a gradient field, and the first order error of the Voronoi rule integrates by parts into a second order quantity.

THEOREM 35.7.3 (weighted Monte Carlo for integrands flat at the boundary). *Let $Q = [-1,1]^d$, let $X_1, \dots, X_N$ be independent and uniform on $Q$, and let $g \in C^3(Q)$ satisfy (a) $g = 0$ within distance $\delta > 0$ of $\partial Q$, or (b) $g \in C^\infty(Q)$ with every derivative vanishing on $\partial Q$. Let $V_N = \sum_i |V_i \cap Q|\, g(X_i)$ be the Voronoi rule, with the cells clipped to $Q$, and $D_N = \int_Q I_N g$ the Delaunay rule, where $I_N g$ interpolates $g$ linearly on the Delaunay triangulation of the nodes and the $2^d$ corners of $Q$. Then*

$$V_N - \int_Q g = \frac{1}{2}\int_Q d(x)^2\, \Delta g(x)\, dx + \frac{1}{2}\int_Q v^{\mathsf{T}} D^2 g(x)\, v\, dx + \int_Q r(x)\, dx,$$
$$|r(x)| \le \tfrac{1}{6}\|D^3 g\|_\infty |v|^3,$$

*and for $A_N = V_N$ and for $A_N = D_N$*

$$\Big|\mathbb{E}A_N - \int_Q g\Big| = \begin{cases} O(Ne^{-cN\delta^d}) & \text{in case (a),} \\ O(N^{-k}) \text{ for every } k & \text{in case (b),} \end{cases} \qquad \operatorname{Var} A_N = O(N^{-1-4/d}).$$

*So the root mean square error is $O(N^{-1/2-2/d})$ in every dimension: $N^{-5/2}$ on the line and $N^{-3/2}$ in the plane. The constants depend on $g$, $d$ and, in case (a), $\delta$. Case (b) includes $g = \log|f|\,\Delta\varphi$ for the bump $\varphi$ of Section 35.7, mapped to $Q$, when $f$ is analytic and zero free on a neighbourhood of the closed box.*

*Proof.* Identity. The Voronoi rule is $\int_Q g(x_{\mathrm{nn}}(x))\, dx$, so its error is $\int_Q [g(x+v) - g(x)]\, dx$; expand to third order. The first order term is $\int_Q \nabla g \cdot v = -\int_Q \nabla g \cdot \nabla(\frac{1}{2}d^2) = \frac{1}{2}\int_Q d^2 \Delta g$, with no boundary term because $\nabla g = 0$ on $\partial Q$ in both cases.

Bias of the Voronoi rule. By exchangeability $\mathbb{E}V_N = \int_Q g(y)\, w_N(y)\, dy$ with

$$w_N(y) = \frac{N}{|Q|}\int_Q \Big(1 - \frac{|B(x, |x-y|) \cap Q|}{|Q|}\Big)^{N-1} dx,$$

which is $N/|Q|$ times the expected volume of the cell of a node placed at $y$. Let $r = \frac{1}{2}\operatorname{dist}(y, \partial Q)$ and $\kappa_d = |B(0,1)|$. For $|x-y| \le r$ the ball $B(x, |x-y|)$ lies in $Q$ and the integrand is $(1 - \kappa_d |x-y|^d/|Q|)^{N-1}$. For $x \in Q$ and $\rho \le 2$, $|B(x,\rho) \cap Q| \ge 4^{-d}\kappa_d \rho^d$ (for $\rho \le 1$ the ball contains its part in the orthant at $x$ that points to the centre of $Q$, which lies in $Q$, and for $1 < \rho \le 2$ use the radius 1), so for $|x-y| > r$ the integrand is at most $e^{-c_1 N r^d}$. Since $\frac{N}{|Q|}\int_{\mathbb{R}^d}(1 - \kappa_d|u|^d/|Q|)_+^{N-1} du = N\int_0^1 (1-t)^{N-1} dt = 1$ and the two integrands agree for $|x-y| \le r$, $|w_N(y) - 1| \le$

$CNe^{-c_1Nr^d}$. Hence $|\mathbb{E}V_N - \int_Q g| \le CN \int_Q |g(y)|\, e^{-c_1N(\mathrm{dist}(y,\partial Q)/2)^d}\, dy$. In case (a) this is $O(Ne^{-c_1N(\delta/2)^d})$. In case (b) Taylor's formula at the nearest boundary point gives $|g(y)| \le C_m \,\mathrm{dist}(y,\partial Q)^m$ for every $m$, and the bound is $O(N^{1-(m+1)/d})$ for every $m$.

Variance of the Voronoi rule. Let $R$ be the largest distance from a node to the points of its clipped cell. If $R > \rho$ for a node $y$, the convex cell contains a point $x$ with $|x-y| = \rho$, and $B(x,\rho)$ holds no other node; one of a fixed set of $C_d$ points $x'$ on the sphere $|x'-y| = \rho$ has $|x-x'| \le \frac{\rho}{4}$, and then $B(x,\frac{\rho}{2}) \subset B(x',\frac{3}{4}\rho) \subset B(x,\rho)$, so $B(x',\frac{3}{4}\rho)$ holds no other node and, as $x \in Q$, meets $Q$ in volume at least $4^{-d}\kappa_d(\rho/2)^d$. So $P(R > \rho \mid y) \le C_d e^{-cN\rho^d}$ for $\rho \le 4$, uniformly in $y$, and $\mathbb{E}R^p = O(N^{-p/d})$ for every $p$. The node nearest to $y$ is at distance at most $2R$, because the midpoint lies in the cell of $y$; so removing $y$ moves the nearest node of each point of its cell to distance at most $3R$. Write $V_N - \int_Q g = \int_Q \frac{1}{2}d^2\Delta g\, dx + \int_Q \mathcal{R}(x,v(x))\, dx$ with $\mathcal{R}(x,v) = g(x+v) - g(x) - \nabla g(x)\cdot v$, $|\mathcal{R}| \le \frac{1}{2}\|D^2g\|_\infty |v|^2$. Replacing $X_1$ by an independent copy $X_1'$ changes the integrands only on the old cell of $X_1$ and the new cell of $X_1'$, where $d$ and $|v|$ are at most $3\max(R,R')$ before and after, $R$ and $R'$ being the two cell radii. So $|V_N - V_N'| \le C(R^{d+2} + R'^{d+2})$, $\mathbb{E}(V_N - V_N')^2 = O(N^{-2-4/d})$, and the Efron and Stein inequality gives $\mathrm{Var}\, V_N \le \frac{N}{2}\mathbb{E}(V_N - V_N')^2 = O(N^{-1-4/d})$.

The Delaunay rule. Linear interpolation has no first order error, since $\sum_j \lambda_j(x)(y_j - x) = 0$ for the barycentric coordinates of $x$ in its simplex $S$, and $|I_Ng - g| \le \frac{1}{2}\|D^2g\|_\infty \,\mathrm{diam}(S)^2$ on $S$, whatever the shape of $S$. In case (a) identify opposite faces of $Q$ to make a flat torus, on which $g$ extends smoothly. There the Delaunay weights $w_i = \int \lambda_i$ are nonnegative, sum to $|Q|$ and move with translations of the configuration, so $\mathbb{E}(w_i \mid X_i = x) = |Q|/N$ and the torus rule is exactly unbiased. Fix $\eta$ small compared with $\delta$. Outside an event of probability $O(e^{-cN\eta^d})$ every ball of radius $\eta$ in a fixed finite net inside $Q$ contains a node. On that event the empty circumball of a simplex that meets the support of $g$ has radius $O(\eta)$, because it contains a ball inside $Q$ of radius comparable to the smaller of its radius and $\delta$; so the simplices that carry $g$, and the weights of the nodes in the support, are the same for the box with its corners and for the torus. Both rules are bounded by $|Q|\,\|g\|_\infty$, so the box rule has bias $O(e^{-cN\delta^d})$. Inserting a node changes the triangulation only in the union of the simplices whose circumballs contain it; on the same event the part of it that carries $g$ lies within a radius $R$ of the node with $P(R > \rho) \le Ce^{-cN\rho^d}$, by the net argument above, and by the value bound the rule changes by $O(R^{d+2})$; the Efron and Stein inequality gives $\mathrm{Var}\, D_N = O(N^{-1-4/d})$ as before. In case (b) write $g = g\chi + g(1-\chi)$ with a cutoff $\chi$ equal to 1 at distance at least $2\delta_N$ from $\partial Q$ and to 0 within $\delta_N$, $\delta_N = N^{-1/(2d)}$, whose derivatives of order $j$ are $O(\delta_N^{-j})$. The second part changes both the rule and the integral by at most $|Q| \sup|g(1-\chi)| = O(N^{-k})$ for every $k$, since the weights are nonnegative and

sum to $|Q|$. The first part satisfies (a) with $\delta = \delta_N$, so its bias is $O(e^{-cN^{1/2}})$, and its second derivatives are bounded uniformly in $N$, because the derivatives of $g$ within $2\delta_N$ of $\partial Q$ are $O(\delta_N^m)$ for every $m$; so its variance is $O(N^{-1-4/d})$. □

In one dimension Lemma 35.6.8 gives the rate of Yakowitz, Krimmel and Szidarovszky its exact form: their mean square $N^{-4}$ is the squared bias $\frac{(b-a)^4(g'(b)-g'(a))^2}{4N^4}$ to leading order, and when $g'(a) = g'(b)$ the mean square error is $O(N^{-5})$ (for $g \in C^4$ by the lemma, and for $g$ as in Theorem 35.7.3 with $d = 1$ by that theorem). The integrand of Section 35.7 satisfies (b) when $f$ has no zero in the closed box, so for $\zeta$ in Figure 35.6 the Voronoi and Delaunay rules at independent uniform nodes have root mean square error $O(N^{-3/2})$ (proved; the fitted slopes are $-1.43$ and $-1.40$), and at jittered nodes the slope $-1.54$ is computed. With a zero in the box the integrand is not $C^3$, and the slopes between $-1.0$ and $-1.3$ are outside the theorem.

Remark 35.7.4. If $g$ is smooth on $Q$ but does not vanish near $\partial Q$, the integration by parts leaves the boundary term $-\frac{1}{2}\oint_{\partial Q} d^2\, \partial_n g$, and $\int_Q \Delta g = \oint_{\partial Q} \partial_n g$ need not vanish, so the bias of these rules is in general of order $N^{-2/d}$ (affine $g$ is an exception); the exponential rows of Table 35.7 show it (computed).

| integrand and rule | $N = 1000$ | largest $N$ | slope | bound |
|---|---|---|---|---|
| bump, $d = 1$, Voronoi | $1.2 \times 10^{-6}$ | $3.5 \times 10^{-14}$ (1,024,000) | $-2.51$ | $-\frac{5}{2}$ |
| $e^x$, $d = 1$, trapezoid | $4.8 \times 10^{-6}$ | $4.5 \times 10^{-12}$ (1,024,000) | $-2.00$ | $-2$ |
| bump, $d = 2$, Voronoi | $1.4 \times 10^{-4}$ | $5.1 \times 10^{-9}$ (1,024,000) | $-1.46$ | $-\frac{3}{2}$ |
| bump, $d = 2$, Delaunay | $1.1 \times 10^{-4}$ | $4.2 \times 10^{-9}$ (1,024,000) | $-1.42$ | $-\frac{3}{2}$ |
| $e^{x+y/2}$, $d = 2$, Delaunay | $1.0 \times 10^{-2}$ | $1.2 \times 10^{-5}$ (1,024,000) | $-0.98$ | $-1$ |

Table 35.7: Root mean square error of the weighted rules at independent uniform nodes on $Q = [-1, 1]^d$, 200 replications in one dimension and 30 in two (about 5 percent noise in each root mean square error in one dimension and about 13 percent in two), the largest $N$ in brackets, slopes fitted for $N \geq 4000$ (computed, floating point; `code/bsy_wmc.py`, `data/bsy_wmc.json`). The bump $\prod_i b(x_i/0.9)$ vanishes near $\partial Q$ (case (a) of Theorem 35.7.3); the exponentials do not, and their error is the bias of Remark 35.7.4. The slopes are consistent with the last column, the exponent of Theorem 35.7.3, of Lemma 35.6.8 for the trapezoid row and of the Remark for the last row.

### 35.7.3 Results

The box is $0.6 \leq \sigma \leq 1.4$, $83.7 \leq t \leq 87.7$, centred on the Davenport and Heilbronn zero at height 85.70 (Figure 35.5). For $\zeta$ there are no zeros in the box, so $I(\varphi) = 0$. For Davenport and Heilbronn the box contains $\rho = 0.808517182457 +$

$85.699348485378\,i$, so $I(\varphi) = 2\pi\varphi(\rho) = 0.63165631$. Both functions are evaluated by a vectorised Euler and Maclaurin formula, accurate to about $10^{-12}$, so a million evaluations take seconds. Table 35.8 and Figure 35.6 give root mean square errors over 20 replications (computed, floating point; code/bsy_riesz2d.py, data/bsy_riesz2d.json; the relative standard error of an entry is 16%).

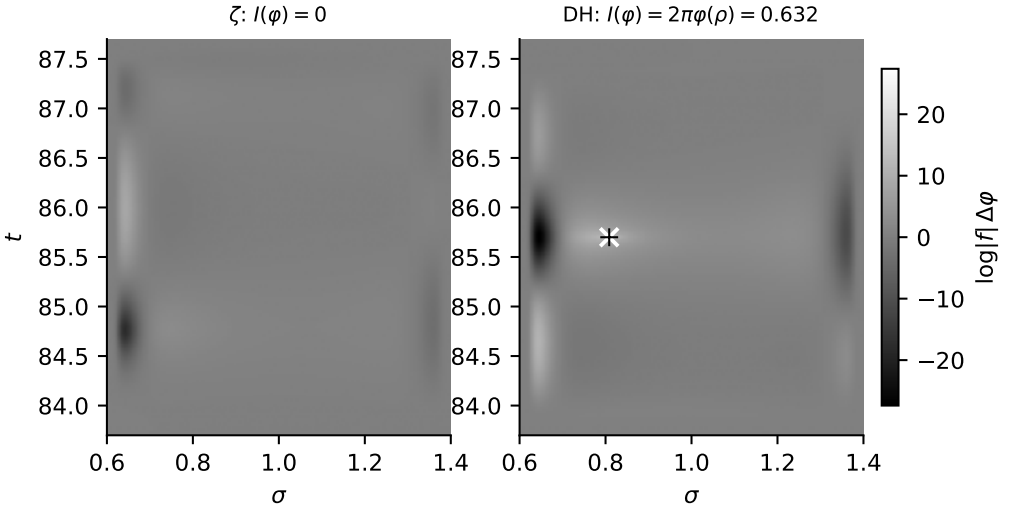



Figure 35.5: The integrand $\log|f|\,\Delta\varphi$ on the box ($121\times121$ points, one grey scale for both functions; code/bsy_riesz2d.py). The cross marks the Davenport and Heilbronn zero. The large values are at the left and right edges of the box, where $\Delta\varphi$ is largest and $\log|f|$ is large; the picture does not show the zero, which is why the two look alike. Their integrals are 0 and 0.632.

| | $N$ | plain | Voronoi | Delaunay | jittered | grid | deflated |
|---|---|---|---|---|---|---|---|
| $\zeta$ | 1024 | $1.2\times10^{-1}$ | $4.3\times10^{-2}$ | $4.5\times10^{-2}$ | $2.0\times10^{-2}$ | $3.0\times10^{-3}$ | |
| $\zeta$ | 16384 | $3.9\times10^{-2}$ | $1.7\times10^{-3}$ | $1.5\times10^{-3}$ | $4.4\times10^{-4}$ | $1.9\times10^{-6}$ | |
| $\zeta$ | 262144 | $9.0\times10^{-3}$ | $3.2\times10^{-5}$ | $2.4\times10^{-5}$ | $5.6\times10^{-6}$ | $2.2\times10^{-14}$ | |
| DH | 1024 | $2.3\times10^{-1}$ | $5.8\times10^{-2}$ | $5.3\times10^{-2}$ | $3.3\times10^{-2}$ | $1.5\times10^{-2}$ | $2.7\times10^{-2}$ |
| DH | 16384 | $6.7\times10^{-2}$ | $3.0\times10^{-3}$ | $2.5\times10^{-3}$ | $1.0\times10^{-3}$ | $6.9\times10^{-4}$ | $1.5\times10^{-5}$ |
| DH | 262144 | $1.2\times10^{-2}$ | $9.5\times10^{-5}$ | $7.4\times10^{-5}$ | $6.1\times10^{-5}$ | $4.8\times10^{-5}$ | $3.5\times10^{-6}$ |

Table 35.8: Root mean square error of $I(\varphi)$ (computed, floating point, 20 replicates; code/bsy_riesz2d.py, data/bsy_riesz2d.json; the relative standard error of an entry is 16%). Jittered is Delaunay on jittered nodes; deflated grid uses an estimate of the zero that is off by $1.4\times10^{-4}$, along $(1+i)/\sqrt{2}$.

The fitted slopes for $N \geq 4096$ are consistent with Proposition 35.7.2. For $\zeta$, with no zero in the box, they are $-0.51$ for plain Monte Carlo, $-1.43$ for Voronoi weights, $-1.40$ for Delaunay weights and $-1.54$ for Delaunay at jittered nodes (the

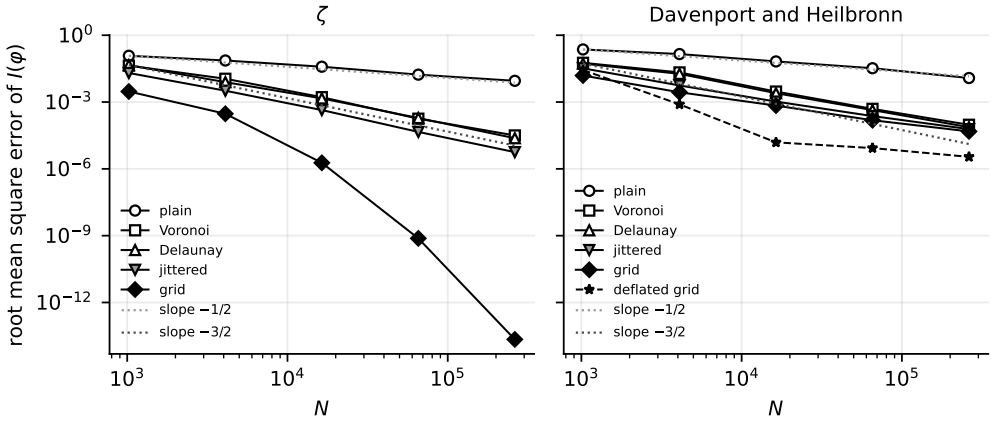



Figure 35.6: Root mean square error of $I(\varphi)$ against $N$ in two dimensions, 20 replicates, for $\zeta$ (left) and Davenport and Heilbronn (right), with the plain, Voronoi, Delaunay, jittered, grid and (Davenport and Heilbronn) deflated grid rules, the deflating zero being off by $1.4 \times 10^{-4}$. The dotted lines have slopes $-\frac{1}{2}$ and $-\frac{3}{2}$ (code/bsy_riesz2d.py, data/bsy_riesz2d.json, figure by code/bsy_figs.py).

observed slopes lie between $-1.4$ and $-1.55$, so $N^{-3/2}$ is an upper estimate). The shifted grid falls faster than any power, to $2.2 \times 10^{-14}$ at $N = 262{,}144$, which is at the accuracy of the function values. Two dimensions thus give what one dimension could not. The rate of the spacings rule is $N^{-3/2}$ (proved at independent uniform nodes, Theorem 35.7.3; computed at jittered nodes), a factor of $N$ better than plain sampling, and the grid is limited only by arithmetic. This answers, in two dimensions, the question raised at the start of Section 35.6.5: the change that makes the integrand smooth is not a change of variable on the line but a move off it.

With a zero in the box the logarithmic singularity takes over at large $N$. All the weighted rules slow to slopes between $-1.0$ and $-1.3$; for the grid this is the rate of Proposition 35.7.2(ii). Deflation with the exact zero restores the smooth rate. With an estimate at distance $\varepsilon$ the residual keeps two logarithmic singularities, and Proposition 35.7.5 gives two regimes: while the grid does not separate them ($\varepsilon\sqrt{N} \lesssim 1$) the error is at most of order $\varepsilon N^{-1/2}$ up to a logarithm, so at fixed $N$ it is proportional to $\varepsilon$ (about $0.03\,\varepsilon$ at $N = 262{,}144$ for $\varepsilon \le 10^{-4}$ in Table 35.9, and the ratio is 0.014 at $N = 2^{20}$, so it is the bound $\varepsilon N^{-1/2}$ that holds and not a fixed constant, where $\hat{\rho} = \rho + \varepsilon(1+i)/\sqrt{2}$); once $\varepsilon\sqrt{N} \gtrsim 1$ it falls like $N^{-1}$. There is no floor. With the zero refined by Newton's method the error is of order $5 \times 10^{-14}$ at $N = 262{,}144$ (Table 35.9, $\varepsilon = 0$). Deflated Delaunay at jittered nodes keeps the slope $-1.52$. The singularity is integrable and mild, and uniform nodes already reach $10^{-4}$ at $N = 262{,}144$ without it being treated (heuristic as a reason; we did

not test concentrating nodes near the zero).

Proposition 35.7.5 (approximate deflation). *Let $\psi \in C_c^\infty((0,1)^2)$, $\rho, \hat\rho \in (0,1)^2$ with $0 < \varepsilon = |\hat\rho - \rho| \le \frac12$, and $r = \psi\,(\log|s-\rho| - \log|s-\hat\rho|)$. The shifted product trapezoid rule of Proposition 35.7.2 with $N$ nodes is unbiased for $\int r$, and its root mean square error over the shift is at most*

$$C\Big(\varepsilon N^{-1/2}\big(1+\log^+\tfrac{1}{\varepsilon\sqrt{N}}\big)^{1/2} + \varepsilon\log\tfrac{2}{\varepsilon}\,N^{-1}\Big) \quad \text{if } \varepsilon\sqrt{N}\le 1,$$
$$C\,N^{-1} \quad \text{if } \varepsilon\sqrt{N}\ge 1,$$

*with $C$ depending only on $\psi$. So for every fixed $\varepsilon$ the error tends to zero: approximate deflation leaves no floor.*

*Proof.* As in Proposition 35.7.2, the rule is unbiased and $\mathbb{E}_\delta|Q - \int r|^2 = \sum_{\ell\neq 0}|\hat r(m\ell)|^2$, $m = \sqrt{N}$. In the sense of distributions $\Delta r = 2\pi\big(\psi(\rho)\delta_\rho - \psi(\hat\rho)\delta_{\hat\rho}\big) + q_\varepsilon$, where $q_\varepsilon = 2\nabla\psi\cdot\nabla\ell_\varepsilon + \Delta\psi\,\ell_\varepsilon$ and $\ell_\varepsilon = \log|s-\rho| - \log|s-\hat\rho|$. Since $|\nabla\ell_\varepsilon| = \varepsilon/(|s-\rho||s-\hat\rho|)$, and $|\ell_\varepsilon| \le 2\varepsilon/|s-\rho|$ for $|s-\rho|\ge 2\varepsilon$, $\|q_\varepsilon\|_1 \le C\varepsilon\log(2/\varepsilon)$. With $|\psi(\rho)e^{-2\pi i\langle k,\rho\rangle} - \psi(\hat\rho)e^{-2\pi i\langle k,\hat\rho\rangle}| \le |\psi(\rho)|\min(2, 2\pi|k|\varepsilon) + \|\nabla\psi\|_\infty\varepsilon$,

$$|\hat r(k)| \le C\,\frac{\min(1,|k|\varepsilon) + \varepsilon\log(2/\varepsilon)}{|k|^2}.$$

If $m\varepsilon \ge 1$ the sum is $O(m^{-4})$. If $m\varepsilon < 1$, split at $|\ell| = 1/(m\varepsilon)$: $\sum_{1\le|\ell|\le 1/(m\varepsilon)}\varepsilon^2 m^{-2}|\ell|^{-2} \le C\varepsilon^2 m^{-2}(1+\log\frac{1}{m\varepsilon})$ and $\sum_{|\ell|>1/(m\varepsilon)} m^{-4}|\ell|^{-4} \le C\varepsilon^2 m^{-2}$, while the second part of the bound on $\hat r$ contributes $C\varepsilon^2\log^2(2/\varepsilon)\,m^{-4}$. Take square roots. □

What this does and does not give for RH is the same as before. With the partition of unity, the smoothed identity is an argument principle count with weights that are $C^\infty$, so a box free of zeros is indicated, in floating point and without a certificate, to near machine precision with a few hundred thousand evaluations. Turing's method already confirms that, rigorously and far higher up. The gain is on the Monte Carlo side: the Balazard, Saias and Yor average in one dimension is a hard integral, while its localised Riesz form in two dimensions is smooth away from the zeros. The weights of Yakowitz, Krimmel and Szidarovszky, in their Voronoi and Delaunay forms, reach $N^{-3/2}$, proved at independent uniform nodes (Theorem 35.7.3) and computed at jittered nodes. Whether this extends to node sets produced by the particle filter or by nested sampling, which are dependent, is open (Open problem 35.4).

### 35.7.4 Counting with the horizontal logarithmic derivative

Theorem 35.7.6 (zeros from the horizontal logarithmic derivative). *Let $f \not\equiv 0$ be meromorphic on a connected neighbourhood of $\{\operatorname{Re} s \ge \sigma_0\}$ and equal, for $\operatorname{Re} s$*

| $\varepsilon$ | $N = 1024$ | 4096 | 16,384 | 65,536 | 262,144 | 1,048,576 |
|---|---|---|---|---|---|---|
| 0 | $3.2 \times 10^{-2}$ | $1.0 \times 10^{-3}$ | $8.1 \times 10^{-6}$ | $4.0 \times 10^{-9}$ | $5.1 \times 10^{-14}$ | $1.2 \times 10^{-15}$ |
| $10^{-8}$ | $3.2 \times 10^{-2}$ | $1.0 \times 10^{-3}$ | $8.1 \times 10^{-6}$ | $4.1 \times 10^{-9}$ | $3.1 \times 10^{-10}$ | $1.4 \times 10^{-10}$ |
| $10^{-6}$ | $3.2 \times 10^{-2}$ | $1.0 \times 10^{-3}$ | $8.0 \times 10^{-6}$ | $5.8 \times 10^{-8}$ | $3.1 \times 10^{-8}$ | $1.4 \times 10^{-8}$ |
| $10^{-4}$ | $3.2 \times 10^{-2}$ | $1.0 \times 10^{-3}$ | $1.1 \times 10^{-5}$ | $5.7 \times 10^{-6}$ | $3.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ |
| $10^{-3}$ | $3.3 \times 10^{-2}$ | $9.3 \times 10^{-4}$ | $1.5 \times 10^{-4}$ | $4.4 \times 10^{-5}$ | $2.1 \times 10^{-5}$ | $1.0 \times 10^{-5}$ |
| $10^{-2}$ | $3.4 \times 10^{-2}$ | $2.0 \times 10^{-3}$ | $9.6 \times 10^{-4}$ | $2.9 \times 10^{-4}$ | $3.0 \times 10^{-5}$ | $1.5 \times 10^{-5}$ |
| none | $1.7 \times 10^{-2}$ | $2.9 \times 10^{-3}$ | $8.2 \times 10^{-4}$ | $1.8 \times 10^{-4}$ | $4.2 \times 10^{-5}$ | $7.2 \times 10^{-6}$ |

Table 35.9: Deflated shifted grid for Davenport and Heilbronn, root mean square error over 10 shifts, as a function of the error $\varepsilon$ in the deflating zero (computed, floating point; code/bsy_defl_dh.py, data/bsy_defl_dh.json; the relative standard error of an entry is 22%). The last row is the grid without deflation. A synthetic test of the two regimes of Proposition 35.7.5 (code/bsy_defl.py) shows the error falling to $N = 2^{22}$ for every $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, with slope near $-0.6$ while $\varepsilon\sqrt{N} \le 1$ and near $-1.1$ afterwards; the constant depends on the bump and the box: at $\varepsilon = 10^{-4}$, $N = 262{,}144$ the ratio of the error to $\varepsilon$ is 0.030 here and 0.0050 in the synthetic test, a factor 6.

*large, to an absolutely convergent Dirichlet series $1 + \sum_{n\ge 2} a_n n^{-s}$, such as $\zeta$ or the Davenport and Heilbronn function. Let $d_f(u,t) = \pi^{-1}\operatorname{Re}(f'/f)(u+it)$, let $\varphi \in C_c^\infty(\{\operatorname{Re} s > \sigma_0\})$, and put $\Phi(u,t) = \int_{-\infty}^{u} \varphi(\sigma,t)\,d\sigma$ and $K_\varphi = \partial_u\varphi + \partial_t^2\Phi$. Then*

$$\sum_{f(\rho)=0} \varphi(\rho) - \sum_{f(p)=\infty} \varphi(p) = -\frac{1}{2}\int_{-\infty}^{\infty}\int_{\sigma_0}^{\infty} d_f(u,t)\,K_\varphi(u,t)\,du\,dt,$$

*zeros and poles counted with multiplicity, the double integral absolutely convergent. $K_\varphi$ is smooth; to the right of the support of $\varphi$ it equals $\partial_t^2\Phi(\infty,t)$ and does not depend on $u$.*

*Proof.* Theorem 35.7.1 for meromorphic $f$ gives $2\pi J = \iint \log|f|\,\Delta\varphi$, with $J$ the left side. Choose $b$ with $\sum |a_n| \log n\, n^{-b} < \infty$. For $u \ge b+1$ large enough that $|f-1| \le \frac12$, $|f'/f| \le C n_0^{-u}$ uniformly in $t$, where $n_0$ is the least $n \ge 2$ with $a_n \ne 0$ (if there is none, $f \equiv 1$ and both sides vanish). For almost every $t$ the horizontal ray meets no zero or pole, and since $f(u+it) \to 1$, $\log|f(\sigma+it)| = -\pi\int_\sigma^\infty d_f(u,t)\,du$. Substitute and exchange the integrals: $d_f$ is locally integrable (near a zero or pole $a$ of order $q$ it is $q\,\operatorname{Re}[(s-a)^{-1}]/\pi + O(1)$), it decays exponentially in $u$ uniformly in $t$, and $K_\varphi$ is bounded and vanishes for $t$ outside the projection of the support. Finally $\int_{-\infty}^{u}\Delta\varphi(\sigma,t)\,d\sigma = \partial_u\varphi(u,t) + \partial_t^2\Phi(u,t)$. □

Remark 35.7.7. (i) For $f = \zeta$, $d_\zeta = m_\sigma - \pi^{-1}\operatorname{Re}\big[\frac{1}{s} + \frac{1}{s-1} - \frac12\log\pi + \frac12\psi(\frac{s}{2})\big]$, where $m_\sigma$ is the Thorin density of Section 35.1; so the Thorin density gives

the count after this known term is subtracted, and $m_\sigma$ itself cannot be substituted ($\log|\xi(u+it)|$ does not tend to 0). (ii) Local form: for every $\varphi \in C_c^\infty$, $\iint m_\sigma(t)\,\Delta\varphi(\sigma,t)\,d\sigma\,dt = -2\sum_\rho \partial_\sigma\varphi(\rho)$, which uses $m_\sigma$ only on the support of $\varphi$; the weights $-2\partial_\sigma\varphi$ integrate to zero along every horizontal line, so a region is counted with signed weights in its collar (for $\varphi = -\frac{1}{2}(\sigma-c)\chi$ with $\chi = 1$ near a region $R$, zeros in $R$ have weight one and zeros where $0 < \chi < 1$ have weight $\chi + (\operatorname{Re}\rho - c)\partial_\sigma\chi$). (iii) Cutoff: if $U$ lies to the right of the support and $A(t) = \Phi(\infty,t)$, then $J + \frac{1}{2}\int\int_{\sigma_0}^{U} d_f K_\varphi\,du\,dt = \frac{1}{2\pi}\int A''(t)\log|f(U+it)|\,dt$. (iv) Deflation by $R(s) = (s-\hat\rho+1)/(s-\hat\rho)$: $J_f = J_{fR} + \varphi(\hat\rho) - \varphi(\hat\rho-1)$.

For (ii): $m_\sigma = \pi^{-1}\partial_\sigma \log|\xi|$ is locally integrable and $\Delta m_\sigma = \pi^{-1}\partial_\sigma\Delta\log|\xi| = 2\sum_\rho \partial_\sigma\delta_\rho$ in the sense of distributions. For (iii): $K_\varphi = A''$ for $u \geq U$ and the integral of $d_f$ from $U$ to $\infty$ is $-\pi^{-1}\log|f(U+it)|$. For (iv): add the divisors of $f$ and $R$; the formula of the theorem holds for $fR$, since its proof uses only $\log|fR(u+it)| \to 0$ and the absolute convergence of the double integral, and $|R'/R| = |(s-\hat\rho)(s-\hat\rho+1)|^{-1} = O(u^{-2})$ uniformly in $t$.

So the zero count is a linear functional of the horizontal component $d_f$ alone, on the whole half strip to the right of the support (the integrand $d_f K_\varphi$ is not smooth at the zeros); the second component of the field in Theorem 35.7.1 is not needed. On the box of Section 35.7, with Gauss and Legendre nodes in $u$ and the periodic trapezoid rule in $t$, the functional with the tail of Remark (iii) added returns 0 for $\zeta$ to rounding error, below $10^{-14}$ (whose pole at 1 lies outside the box). For Davenport and Heilbronn it converges slowly and irregularly to $\varphi(\rho) = 0.1005312239$, with errors between $10^{-5}$ and $10^{-4}$ at 128 to 1024 nodes in $t$ ($-8.6\times10^{-6}$ at 512, a coincidence of the grid), since the integrand has a singularity of order $1/|s-\rho|$. Deflated by $R(s) = (s-\hat\rho+1)/(s-\hat\rho)$ with $\hat\rho = \rho$, it agrees with $\varphi(\rho)$ to $4\times10^{-16}$. The factor $R$ tends to 1 only like $1/u$, so its tail beyond the last node, $-\pi^{-1}\log|R(u_{\max}+it)|$, is added exactly; the zero of $R$ at $\hat\rho - 1$ lies outside the support, so only $\varphi(\hat\rho)$ is added back. With an error $\varepsilon$ in the deflating zero the result moves by $8\times10^{-10}$ at $\varepsilon = 10^{-6}$ and $10^{-7}$ at $\varepsilon = 10^{-4}$ (512 nodes in $t$). These are floating point values with $f'/f$ evaluated by the Euler and Maclaurin formula (code/bsy_count.py, computed); with the tail of $R$ alone, and not the tail of $f$, the result needs $u_{\max}$ near 40 for an error of $10^{-14}$.

## 35.8 Limits

Every statement of Sections 35.1 to 35.5 uses only Hadamard's product, the functional equation and order one growth, with two exceptions: the positivity of $\mathfrak{U}_\sigma$ for $\sigma \geq 1$ (Section 35.1 and Corollary 35.3.3(iv)) uses $\beta < 1$, the non-vanishing of $\zeta$ on $\operatorname{Re} s = 1$, which comes from the Euler product; and Theorem 35.5.2 assumes QRH($\theta$). The other statements hold for the completed Davenport and Heilbronn function [111], which has zeros off the line [484, 28], with three changes: $h_\theta = f$, since $f$ is entire; the archimedean term in (35.3.1) becomes

$\pi^{-1}\operatorname{Re}[\frac{1}{2}\log\frac{5}{\pi}+\frac{1}{2}\psi(\frac{s+1}{2})]$; and every sum over zeros with $\beta>\theta$ includes the zeros with $\beta\ge 1$. So the dictionary, the EGGC form and the Brownian reading are reformulations; none of them can prove RH.

The arithmetic enters only through the Euler product. In the Thorin language it appears as an upper bound on positive combinations of $m_\sigma$ at several heights, for example $3m_\sigma(0)+4m_\sigma(t)+m_\sigma(2t)$ for $\sigma>1$ (Proposition 9.1.4), which the Davenport and Heilbronn function, having no Euler product, need not satisfy. Such inequalities give zero free regions of width $c/\log t$.

Remark 35.8.1 (the explicit formula at a Cauchy smoothed level). Let $F\ge 0$ have a Fourier transform supported in $[-L,L]$, Lipschitz, with derivative of bounded variation, as for the class $\mathcal{B}$ of Section 20.1.1; let $F_c=F(\cdot-c)$, and let $\frac{1}{2}<\sigma<1$ not be the real part of a zero (at such a level take right limits in $\sigma$). The explicit formula of Theorem 20.1.1, applied to the even part of $P_{\sigma-1/2}*F_c$, has on its prime side only the prime powers $n\le e^L$, besides the archimedean terms. On its zero side a zero with $\beta<\sigma$ contributes $\int F_c\,P_{\sigma-\beta}(\cdot-\gamma)$, its term in $\int F_c\,d\mathfrak{U}_\sigma$ by (35.1.1); so if no zero has $\beta>\sigma$ the formula gives $\int F_c\,d\mathfrak{U}_\sigma\ge 0$, which is the content of Theorem 20.1.2 for $F=|\hat{h}|^2$. A zero with $\beta>\sigma$ contributes instead the continuation in $\beta$ of that term, which exceeds its signed term $-\int F_c\,P_{\beta-\sigma}(\cdot-\gamma)$ in $\int F_c\,d\mathfrak{U}_\sigma$ by $F_c(\gamma+i(\beta-\sigma))+F_c(\gamma-i(\beta-\sigma))$, by Fourier inversion at complex arguments as in the proof of Theorem 20.1.1; these corrections need not be positive. So the primes compute the continued Weil functional, which equals the Thorin integral $\int F_c\,d\mathfrak{U}_\sigma$ in general only when no zero lies to the right of $\sigma$; positivity of $\mathfrak{U}_\sigma$ is not a finite computation with the primes.

The Monte Carlo side shows only what it computes, in floating point. It can show that the defect is zero to a given accuracy in a given window, but a zero at height $\gamma$ moves the global average by at most $2(1-\theta)(\beta-\theta)/\gamma^2$.

## 35.9 Open problems

Problem 35.1. Find an implementable sampler for $P_w$ with $w$ close to $|f-\mathbb{E}_P f|$, or another weight, whose variance per unit of computing cost (pilot, burn in, autocorrelation and evaluations of $f$ included) is below that of plain Monte Carlo for the defect $D_\theta(t_0)$, or show that the Metropolis chains of Section 35.6 cannot reach it. For independent draws no weight gains more than the factor $\operatorname{Var}_P f/(\mathbb{E}_P|f-\mathbb{E}_P f|)^2$ of Remark 35.6.2, which is between 1.35 and 2.46 for the integrands of Table 35.3 (computed; `code/bsy_isbound.py`); a Markov chain is not bound by it.

Problem 35.2. Section 35.7 replaces the negative Thorin mass $\iint m_\sigma(t)^-\,d\sigma\,dt$, a nonlinear functional with a kink along the curve $m_\sigma=0$, by the linear functional $I(\varphi)$, and Theorem 35.7.6 writes the count through the horizontal logarithmic derivative alone. Near an undeflated zero $m_\sigma^-$ is integrable but not square integrable, so independent uniform nodes give infinite variance there. Determine the

root mean square rate of the Voronoi and Delaunay rules for $\iint_R m_\sigma^-$ on a bounded region $R$ free of zeros, or after exact deflation of the zeros in $R$. Theorem 35.7.3 does not apply to this nonsmooth functional.

PROBLEM 35.3. On a half disc the sorted trapezoid rule reaches the mean square rates $N^{-4}$ and $N^{-5}$ for an exact local form of the criterion (Theorem 35.6.6 and Lemma 35.6.8). For the whole line: prove a rate in $N$ for the truncated rule of Section 35.6.6 under RH, with a step that shrinks with $N$ and with the discretisation and deflation errors bounded; find a correction for the tail beyond $T$ that is finite, efficiently computable and certified, as the half disc gives for its arc. Are the empirical slopes $-0.60$ and $-0.66$ of the sorted trapezoid rule for $\zeta$ (Table 35.4) a preasymptotic effect of the zeros, or the true rate?

PROBLEM 35.4. Theorem 35.7.3 proves the rate $N^{-1/2-2/d}$ at independent uniform nodes. Extend it to dependent node sets, such as those of the particle filter along the Cauchy flow (Section 34.3) or of nested sampling, under explicit coverage and mixing hypotheses on the nodes (some hypothesis is needed: nodes that all coincide give one evaluation); for deterministic quasi uniform meshes the rate can be $N^{-1}$. For a box containing an undeflated zero, where plain Monte Carlo keeps the rate $N^{-1/2}$, is the root mean square error of the Voronoi and Delaunay rules at independent uniform nodes of order $N^{-1}$ up to logarithms, as the slopes between $-1.0$ and $-1.3$ of Section 35.7 suggest?

PROBLEM 35.5. Fix $\theta \in [\frac12, 1)$ and let $Z_{t_0}$ be the boundary value of the horizontal branch of $\log h_\theta$ under $P_{1-\theta}(t_0 - \cdot)$. If $\mathbb{E}[Z_{t_0}^2] = (\mathbb{E} Z_{t_0})^2$ for every $t_0$, does $\mathrm{QRH}(\theta)$ follow? Proposition 35.5.4 reduces it to the identity $2\sum_\gamma \int_{C_\gamma} M_\gamma (L_- + \pi i M_\gamma - z_0) K \, ds = D^2$ for every $t_0$, with $K$ the cut kernel there. Under RH, is $\mathrm{Var} \log|\zeta(\frac12 + i\tau)| = \frac12 \log\log t_0 + O(1)$ under $P_{1/2}(t_0 - \cdot)$, for every $t_0$ or for $t_0$ outside a set of density zero; and does $(Z_{t_0} - \mathbb{E} Z_{t_0})/(\frac12 \log\log t_0)^{1/2}$ converge in law to the standard normal law of $\mathbb{R}^2$ when $t_0$ is uniform on $[T, 2T]$?

PROBLEM 35.6. Is there a linear combination of $L(s, \chi)$ over primitive characters $\chi$ of one modulus, with a functional equation and zeros off the line, whose trace $K$ is negative somewhere? With imprimitive characters allowed, Remark 35.4.5 gives one. If the lowest zero $\frac12 + i\gamma_1$ of such a function is on the line, the argument of the proof of Theorem 35.4.3, with $\pi/3$ replaced by $\pi/(2\delta_{\max})$, where $\delta_{\max}$ is the largest $|\beta - \frac12|$, gives a positive trace whenever $\sum e^{-(\gamma-\gamma_1)\pi/(2\delta_{\max})} < 1$, the sum over the upper zeros off the line with multiplicity. So a negative trace needs the opposite inequality, and if there are finitely many, $M$, upper zeros off the line, one of them lies within $\frac{2\delta_{\max}}{\pi} \log M$ of $\gamma_1$.

# Appendix A

# Code and reproducibility

The scripts, the data files they write and the reruns made for the book are not printed here: they are released with the PDF, in the book's repository, `https://github.com/VadimSokolov/computing_rh`, beside the LaTeX source of the book. Most scripts are in the folder `code/` and the data files they write in `data/`, the reruns and reconstructions made for the book in `repro/`, and the scripts of some chapters in other folders. The file `code/README.md` lists every script with its purpose, the data files it reads and writes, the results of the book that it produces and its usage, and describes the reruns of September 2026, the computations that have no script in the archive and the gaps that remain.

The scripts run with Python 3.11 and the libraries of `requirements.txt`: `python-flint` 0.9.0, the Python interface to FLINT and Arb [290], for ball arithmetic and the certificates, and `mpmath` [376] for reference values. The long computations ran on the Hopper cluster of George Mason University. The scripts of the later chapters keep a log of each run beside the data file, those of Chapters 17 to 21 take a `--quick` option that runs a reduced case in seconds, and where a chapter printed a value that was recomputed for the book, the data file or the log records the value computed beside it and says whether the two agree. Where the book says that a statement is proved in ball arithmetic, the script named beside it is the certificate, and its output records the enclosures.

## A.1 The Pólya evaluators

The evaluator of Section 2.1 on the real $u$ axis is `code/polya_flint.py`. In ball arithmetic it computes the node values $\Phi_j = \Phi(jh)$, $0 \le j \le J$, from the theta series (2.1.1), and from them $P(E) + P(E^{-1})$ and $Q(E) - Q(E^{-1})$ with $E = e^{sh}$ and the polynomials $P$ and $Q$ of Section 2.1.1, the trapezoid approximations to $\xi(\frac{1}{2} + s)$ and $\xi'(\frac{1}{2} + s)$. The defaults are 240 bits, $h = 1/200$ and $|u| \le 3$, so that $J = 600$. Most runs use 420 bits, the parameters of Section 2.1.1; the Keiper Li coefficients of Section 8.3 use 600 bits, and the Selberg computation of Section 3.5 uses 300.

The tilted evaluator `code/tilted.py` implements Proposition 2.1.1. With the nodes $u_j = x_j + i\eta$, $x_j = jh$, $|j| \le J$, where $J = \lceil x_{\max}/h \rceil$ for the cut $|x| \le x_{\max}$, it returns the trapezoid sums of the two integrals of the proposition. The production scripts evaluate each exponential directly, not by Horner's rule, whose ball radii grow as described in Section 2.1.5. The Lehmer pair and its de Bruijn Newman flow use $\delta = 0.01$, 256 bits, $h = 6 \times 10^{-4}$ and $|x| \le 3.9$; the count at height 1000 in Section 2.2.3.2 uses $\delta = 0.1$, 200 bits, $h = 0.004$ and $|x| \le 4.2$; the Laplace transform of the clock for Sections 8.1 and 8.2 switches to the tilted evaluator with $\delta = 0.1$, 320 bits, $h = 0.003$ and $|x| \le 4.5$ when $|\operatorname{Im} s| \ge 120$; and the runs of Table 2.2 take $\delta$, the precision, $h$, the cut and the heights from the command line, which `code/README.md` records.

In both evaluators the radii of the returned balls bound the rounding errors of the finite sums, and nothing else. Three errors lie outside them: the discretisation error of the trapezoid rule, bounded in Section 2.1.1; the cut of the axis at $|u| \le 3$, where $\Phi < 10^{-500}$, or at $|x| \le x_{\max}$ on the tilted line; and the truncation of the theta series, whose loop stops at the first term with midpoint below $2^{-b-20}$ at working precision $b$ and does not add the tail to the radius. The stopping test compares double precision numbers, so for $b \ge 1055$ its threshold underflows to zero and the test is never met. The untilted route needs more than 1054 bits for thirty digits above height about 840 (Table 2.3), and there the test has to be made in ball arithmetic. The runs in this book use at most 600 bits and are not affected.

# Appendix B

# A short history of computations on the zeros

Every verification in Table B.1 from Backlund's on rests on an evaluator of $\zeta$ on or near the critical line and a certificate that no zero has been missed (Section 2.2). The window on which Chapters 3 to 5 measure distances, the 79 zeros below height 200, is the range that Backlund verified by hand in 1914 [14].

Where Turing bounds an average of $S$ to certify a count (Section 2.2.1), the zero density identity of Section 3.4 is exact, and its value is an integer at every resolution; at the resolution $\frac{1}{2}$, the basepoint one of Section 9.1, it is Backlund's method with the vertical side moved from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = 1$, and it gives the count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$ with no hypothesis, which Section 2.2.3.2 computes at height 1000. Lehmer's close pair near height 7005 [333] is computed from the Pólya kernel in Section 2.2.3.1 and followed under the de Bruijn Newman flow in Section 8.4. The Odlyzko Schönhage algorithm, with Odlyzko's zeros as the reference measure, is one of the evaluators that Section 2.2.2 proposes for the Thorin comparison at large height.

| year | authors | result |
|---|---|---|
| 1859 | Riemann | the first zeros, in unpublished notes later found by Siegel [478] |
| 1903 | Gram | the first 15 zeros; Gram points [158] |
| 1914 | Backlund | all 79 zeros below height 200 [14, 15] |
| 1925 | Hutchinson | all 138 zeros below height 300 [278] |
| 1936 | Titchmarsh | the first 1041 zeros, all below height 1468, by the Riemann Siegel formula, computed with the help of Comrie [496] |
| 1953 | Turing | all 1104 zeros below height 1540, on the Manchester computer; Turing's method [506] |
| 1956 | Lehmer | the first 25000 zeros; the close pair near height 7005 [333, 334] |
| 1969 | Rosser, Yohe, Schoenfeld | the first $3.5 \times 10^6$ zeros; Rosser's rule [466] |
| 1986 | van de Lune, te Riele, Winter | the first $1.5 \times 10^9$ zeros [508] |
| 1987 | Odlyzko | $10^5$ zeros near the $10^{12}$th zero, and the statistics of their spacings [391] |
| 1988 | Odlyzko, Schönhage | fast evaluation of $\zeta$ at many points [392] |
| 2004 | Gourdon | the first $10^{13}$ zeros, not interval certified [157] |
| 2021 | Platt, Trudgian | RH verified rigorously below height $3 \times 10^{12}$ [419] |

Table B.1: Landmarks in the computation of the zeros of $\zeta$. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [127, Chapters 6 to 8].

# Appendix C

# Notation

The tables list the notation used in more than one section; the first five cover the book and record the second meanings that some letters carry in particular chapters and sections. Sums and products over $k$ run over the pairs $\{\rho, 1-\rho\}$.

| symbol | meaning | defined in |
|---|---|---|
| $\xi$, $\Xi$ | the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ | Section 1.2 |
| $\rho$, $\gamma_k$ | a zero $\rho = \beta + i\gamma$ of $\xi$, that is a nontrivial zero of $\zeta$; the ordinates $0 < \gamma_1 \le \gamma_2 \le \cdots$ of the zeros with $\gamma > 0$ | Section 1.2 |
| $a_k$ | $-(\rho_k - \frac{1}{2})^2$, one for each pair $\{\rho, 1-\rho\}$, equal to $\gamma_k^2$ on the line (in Section 8.3, the Taylor coefficients of $\log \xi(1+w)$) | (1.2.1) |
| $\Phi$, $p$, $X$ | the Pólya kernel in Titchmarsh's normalisation; the Pólya density $p = \Phi/\Xi(0)$; a random variable with density $p$ (in Section 6.2, $p$ is the Meixner density; in Section 4.3, $X$ is the clock variable; in Sections 9.1 and 11.3, $X$ is the prime cutoff) | (1.2.3), (1.2.4) |
| $S_2$, $Y$, $J$ | $S_2 = \sum_{k\ge1} \frac{2}{\pi^2 k^2}(e_k + e_k')$ with independent standard exponential variables, the sum of two independent copies of the first passage time $J$ of the level 1 by a three dimensional Bessel process started at 0; $Y = \frac{1}{2}\log(\pi S_2/2)$, with $\mathbb{E}e^{sY} = 2\xi(s)$; $X$ is $Y$ tilted by $e^{Y/2}$ | Section 1.2, Section 9.1.1 |
| $u$, $z$, $\theta$ | the variable $u = s^2$ of the clock (also the variable of $\Phi$); the variable of Thorin measures; the chart $\theta = \sqrt{z}$, which is also the ordinate (the Kent approximants of Section 8.1 write $w$ for $u$) | Section 1.1 |
| $G(u)$ | $\xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$; the clock is $1/G$ (in Section 7.1.3, $G$ is a Laguerre Pólya function) | Section 1.1 |

| symbol | meaning | defined in |
|---|---|---|
| $\mathcal{G}(u,t)$ | the Gaussian smoothing $\sum e^{-2t(\tilde{\gamma}-u)^2}$ of the zeros at resolution $1/(2\sqrt{t})$, with $\tilde{\gamma}=\gamma-i\delta$ for a zero $\frac{1}{2}+\delta+i\gamma$ with $\gamma>0$ | Section 5.1.5 |
| $T$, $e_\gamma$ | the clock variable, with $\mathbb{E}e^{-uT}=1/G(u)$, which exists with no hypothesis and is the centre clock of Chapters 9 to 11; under RH, $T=\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables $e_\gamma$ ($T$ is also a height, as in $N(T)$) | Section 1.2, Proposition 1.2.1 |
| $U$, $U_\alpha$ | the Thorin measure of a GGC $H$, with $\mathbb{E}e^{-uH}=\exp\{-du-\int\log(1+u/z)\,U(dz)\}$ and drift $d$; the signed measure $U_\alpha(dz)=v_\alpha(\sqrt{z})\,dz/(2\pi\sqrt{z})$ at a basepoint $\alpha$; for the clock, the Thorin functional $\sum_k \delta_{a_k}$ | Section 1.2, (1.2.6), Section 5.3.1 |
| $k$ | the canonical function of an infinitely divisible law on $[0,\infty)$ with Lévy density $k(t)/t$ ($k(y)$ in Sections 5.1 and 6.2) | Section 1.3 |

| symbol | meaning | defined in |
|---|---|---|
| $U_{\mathrm{RH}}$, $U_{\mathrm{line}}$, $U^{\pm}_{\mathrm{RH}}$ | $\sum_{\gamma>0}\delta_{\gamma^2}$; the same sum over the zeros on the line; $\sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$ | (1.1.1), Proposition 1.2.3 |
| $\mathcal{W}$, $T_{\mathcal{W}}$, $U_{\mathcal{W}}$, $\mathsf{U}$ | the principal branch of Lambert's function, $\mathcal{W}(u)e^{\mathcal{W}(u)}=u$ (not the heat trace $W$); the Lambert law, with Laplace transform $\mathcal{W}(u)/u=e^{-\mathcal{W}(u)}$; its Thorin measure; a variable uniform on $(0,1)$ (not the Thorin functional $U$) | Sections 11.5.1, 11.5.3 |
| $\varphi_\alpha$, $T_\alpha$ | $\xi(\alpha)/\xi(\alpha+\sqrt{u})$, the clock at a basepoint $\alpha\geq\frac{1}{2}$, with $\varphi_{1/2}=1/G$; the variable with $\mathbb{E}e^{-uT_\alpha}=\varphi_\alpha(u)$, which exists for every $\alpha>\frac{1}{2}$ and is a GGC if and only if no zero has $\beta>\alpha$, in particular for every $\alpha\geq 1$; $T_{1/2}=T$ | Theorems 1.2.2, 9.2.5 |
| $\widetilde{T}$, $\widetilde{T}_\alpha$, $\sigma^{(\alpha)}$ | the centre clock tilted by $e^{-T/4}$, and by $e^{-(\alpha-\frac{1}{2})^2T}$, so that $\widetilde{T}=\widetilde{T}_1$; the subordinator with Laplace exponent $u+2(\alpha-\frac{1}{2})\sqrt{u}$, with $T_\alpha\stackrel{d}{=}\sigma^{(\alpha)}_{\widetilde{T}_\alpha}$ | Theorems 9.2.2, 9.2.5 |
| $H_a$, $H_a^{(c)}$ | the first passage time of $B_t+ct$ to the level $a\geq 0$, for a standard Brownian motion $B$, and $H_a=H_a^{(0)}$; unrelated to $H_t$ | Section 9.2 |
| $Y_\alpha$, $Z$ | the normal scale mixture $Y_\alpha=\sqrt{2T_\alpha}\,Z$, with $Z$ standard normal and independent of $T_\alpha$, whose characteristic function is $\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$ (in Section 11.1, $Z$ is the subordinator with $T\stackrel{d}{=}\int_0^\infty e^{-t}\,dZ_t$) | Proposition 10.1.1 |

| symbol | meaning | defined in |
|---|---|---|
| $W(t)$, $c_k(t)$ | the heat trace $\sum_k e^{-a_k t}$, also written $\int e^{-tz}U(dz)$; the signed derivatives $(-1)^k W^{(k)}(t)$ (in Chapters 6 and 7 the time is $x$, or $y$ where $x$ is a Lévy variable, as in the proof of Lemma 6.2.5 and from Proposition 6.2.16 to the end of Section 6.2; the numbers $c_k(u)$ of Section 4.1 are local to it) | Sections 1.2, 5.1 |
| $m_n$, $\mathcal{H}_N$, $\mathcal{H}'_N$, $\iota_N$, $\iota'_N$ | the Stieltjes moments $\sum_k a_k^{-n-1}$, with $m_0 = \mathbb{E}T$; the Hankel matrices $[m_{i+j}]_{i,j<N}$ and $[m_{i+j+1}]_{i,j<N}$; the numbers of their negative eigenvalues, which count the zeros off the line (Section 8.4 uses the same symbols for the moments $m_n(t)$ of $H_t$) | Section 5.1, Theorem 5.1.7 |
| $W^{[j]}$, $\beta_j$, $B_j$ | the ladder functions $(-t\frac{d}{dt})^j W$, with $W^{[0]} = W$ (along the de Bruijn Newman flow $W^{[j]}(x;t)$, with $x$ the time of the heat trace and $t$ that of the flow); the polynomials with $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)e^{-at}$; the Touchard polynomials, $\beta_j(u) = (-1)^j B_j(-u)$ | Section 11.5.4 |
| $L_n$, $L_\infty$ | Urbanik's classes of multiply self decomposable laws, with $L_0$ the self decomposable laws; unrelated to the Post Widder functionals $L_n[W]$ of Section 7.2.2 | Section 11.5.4.1 |
| $Z^{[j]}$ | the $j$th driving process of the centre clock, $Z^{[1]} = Z$, with Laplace exponent $(u\frac{d}{du})^j \log G$ | Lemma 11.5.7 |
| $g$, $g_k$, $H$, $\lambda^*$ | the candidate generating exponent $\Psi_Z \circ H$ and its Taylor coefficients, with $H$ the inverse of $\log G$; the branch point $\log\lvert G(u^*)\rvert + i\pi$ of $H$ | Section 11.5.4.4 |
| $\Phi_t$, $\widehat{g}$, $R_g$ | the damped Pólya kernel $e^{tv^2/4}\Phi(v)$; the cosine transform $\int g(v)\cos(\theta v)\,dv$ of an even kernel; $\widehat{g}(0)/\widehat{g}(i\sqrt{s})$, with $R_{\Phi_t} = R_t$ | Section 11.5.5.1 |
| $X_t$, $p_t$, $Q_t$, $\varrho_t$, $h_t$ | the variable with density $p_t = \Phi_t/\Xi_t(0)$, $X_0 = X$; the radius of Khinchine's mixture, $X_t = UQ_t$; the radius of the arcsine mixture, $X_t = \varrho_t \cos\Theta$, and its density | Section 11.5.5.2 |
| $\mathcal{B}$, $f$, $\mathcal{B}_g$, $\mathcal{B}_t$ | $\frac{\pi}{2}S_2$, with $\mathbb{E}\mathcal{B}^{s/2} = 2\xi(s)$; its density $b^{-5/4}\Phi(\frac{1}{2}\log b)$; the variable with density proportional to $b^{-5/4}g(\frac{1}{2}\log b)$; $\mathcal{B}_{\Phi_t}$ | Section 11.5.5.3 |
| $\ell$, $\kappa_k(\lambda)$, $c_k(\lambda)$, $\sigma_\lambda$ | a Laplace transform $\mathbb{E}e^{-\lambda\mathcal{V}}$; the cumulants $(-1)^k(\log\ell)^{(k)}(\lambda)$; $\kappa_{k+1}(\lambda)/k!$, the moments of $\sigma_\lambda$ for a GGC (not the $c_k(t)$ of the heat trace) | (11.5.9) |
| $\mathcal{K}_n(\lambda)$, $\mathcal{K}'_n(\lambda)$ | the Hankel matrices $[c_{i+j}(\lambda)]_{i,j<n}$ and $[c_{i+j+1}(\lambda)]_{i,j<n}$, positive semidefinite for a GGC | Section 11.5.5.3 |

| symbol | meaning | defined in |
|---|---|---|
| $\ell_t$, $\dot{\ell}$, $\mathcal{J}$ | $\mathbb{E}e^{-\lambda\mathcal{B}_t}$; its derivative in $t$ at $t=0$, up to a term in $\ell_0$; the first order change of the Thorin measure in a gap, in the proof of Proposition 11.5.15 | Section 11.5.5.3 |
| $\vartheta_a$, $\xi(s,a)$, $K_a$, $\mathcal{B}_a$ | the theta series $\sum n^p a_n e^{-\pi n^2 x/q}$ of a periodic sequence; its completed Dirichlet series; the kernel $2e^{(p+1/2)v}\vartheta_a(e^{2v})$; $\mathcal{B}_{K_a}$ (for $a=\chi_{-4}$, $1/\mathcal{B}_a$ is the variable $Z$ of Remark 8.5.23) | Section 11.5.5.4 |
| $\chi_d$, $S_a$, $\Sigma_a$ | the real primitive character of discriminant $d$ ($\chi_{-4}=\chi_4$); the positivity sum of Lemma 11.5.17; the numerator of the reflected transform (11.5.10) | Section 11.5.5.4 |
| $f_{\mathrm{DH}}$, $\varkappa$, $f_r$ | the function of Davenport and Heilbronn; its coefficient 0.28407904…; the self dual family modulo 7 | Section 11.5.5.4 |
| $k_n$ | the inverse Laplace transform of Li's $\varphi_n$, $\lambda_n=\int k_n W$ | Problem 11.4 |

| symbol | meaning | defined in |
|---|---|---|
| $\varepsilon$ | the resolution; the basepoint is $\frac{1}{2}+\varepsilon$ | Section 1.1 |
| $v_\alpha(r)$ | $\mathrm{Re}(\xi'/\xi)(\alpha+ir)$, which is $\pi$ times the density of $U_\alpha$ in the chart $\theta=\sqrt{z}$ (Chapters 9 to 11 write $v_\alpha(\theta)$) | Section 1.2 |
| $b_\alpha$, $\Pi_\alpha$ | $b_\alpha=\frac{\xi'}{\xi}(\alpha)=v_\alpha(0)$, the drift of the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)}=b_\alpha s+\int_0^\infty(e^{-sx}-1+sx)\,\Pi_\alpha(dx)$ (for $\alpha\geq 1$, and for $\frac{1}{2}<\alpha<1$ when $\Theta<\alpha$), with $b_1=\sum_\rho\rho^{-1}=\lambda_1$; its signed Lévy measure, with atoms $\Lambda(n)/(n^\alpha\log n)$ at $\log n$ | Proposition 9.1.10, Section 9.1.5 |
| $A$, $A_\alpha$, $Q$, $\nu_1$ | the gamma piece $A(\theta)=\frac{1}{2}[\mathrm{Re}\,\psi(\frac{3+i\theta}{2})-\psi(\frac{3}{2})]$ and the compensated prime piece $Q(\theta)=\lim_{X\to\infty}\int_0^{\log X}2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ of $v_1=b_1+A+Q$; the gamma piece $A_\alpha$ at $\alpha$; the prime measure $\nu_1=\sum_n\Lambda(n)n^{-1}\delta_{\log n}-dx$ on $(0,\infty)$, compensated by Lebesgue measure and unrelated to $\nu(\sigma,T)$ (Section 11.3 uses $A$, $A_\alpha$ and $Q$ for $L(s,\chi_{12})$) | (9.1.6), Proposition 9.1.3 |
| $S$, $T_1^\circ$; $S_\alpha$, $T_\alpha^\circ$ | the positive stable variable of index one half with $\mathbb{E}e^{-uS}=e^{-b_1\sqrt{u}}$; the GGC with no drift, independent of $S$, whose Thorin density in the chart $\theta=\sqrt{z}$ is $(A+Q)/\pi$, so that $T_1\overset{d}{=}S+T_1^\circ$ (unrelated to $S(t)$ and $S_2$); the same at a basepoint $\alpha\geq 1$, with $b_\alpha$ and $(v_\alpha-b_\alpha)/\pi$, so that $T_\alpha\overset{d}{=}S_\alpha+T_\alpha^\circ$ | Propositions 9.1.11, 9.1.12 |

| symbol | meaning | defined in |
|---|---|---|
| $V_\alpha$, $V$ | the variable with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, whose density $e^{(\alpha-\frac12)y}p(y)\,\xi(\frac12)/\xi(\alpha)$ is a tilt of the Pólya density, with no hypothesis; $V_{1/2} = X$ and $V = V_1$ | Section 1.2, Section 9.1.1 |
| $\rho_\varepsilon(\theta)$ | the Thorin density at basepoint $\frac12+\varepsilon$ in the chart $\theta = \sqrt{z}$, $\pi^{-1}v_{1/2+\varepsilon}(\theta)$ | (1.2.7) |
| $\mu_\varepsilon$ | the signed measure $U_{1/2+\varepsilon}$, with density $\rho_\varepsilon(\theta)\,d\theta$ in the chart $\theta$; the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac12+\varepsilon > \operatorname{Re}\rho$ for every zero, and called the Thorin measure at resolution $\varepsilon$ in general ($\mu(n)$ is the Möbius function) | Section 1.2 |

| symbol | meaning | defined in |
|---|---|---|
| $P_a$ | the Poisson (Cauchy) kernel $a/(\pi(a^2+x^2))$, with $P_{-a} = -P_a$ | Section 1.2 |
| $d_{\mathrm{match}}(\varepsilon)$ | $\lVert\mu_\varepsilon - P_\varepsilon * U^{\pm}_{\mathrm{RH}}\rVert_{\mathrm{TV}}$, the distance to the RH measure smoothed at the same resolution, which vanishes under RH | Section 3.2 |
| $\tilde{d}_{\mathrm{match}}(\varepsilon)$ | $\lVert\mu_\varepsilon - P_\varepsilon * \widetilde{U}\rVert_{\mathrm{TV}}$, the matched distance computed against the model $\widetilde{U}$ of the zero list (the first 1700 ordinates and the density $\pi^{-1}\vartheta'$ beyond them, each with its reflection) | Section 3.2 |
| $\kappa_{\mathrm{cell}}$ | the cell constant $\pi^{-1}\sum_k \lvert\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\rvert$ of the 79 cells $(b_{k-1}, b_k]$ below height 200, so that $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$ | Section 3.2 |
| $M_\varepsilon(\theta)$ | the cumulative mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$, with the branch of the argument that vanishes at $\theta = 0$ (the Mertens function $M(t) = \sum_{d\le t}\mu(d)$ in Section 10.3) | Section 2.1.2, Section 3.4 |
| $\tau_\varepsilon(T)$ | $\pi^{-1}$ times the change of $\arg\xi$ along the segment from $\frac12+\varepsilon+iT$ to $\frac12+iT$, so that $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac12+\varepsilon, T)$ ($\tau(n)$ is Ramanujan's function in Section 8.5) | Proposition 3.4.2 |
| $N(T)$, $\nu(\sigma,T)$ | the number of zeros with $0 < \operatorname{Im}\rho \le T$; the number of those with $\operatorname{Re}\rho > \sigma$ (Sections 3.4 and 11.4 write $N(\sigma,T)$ for the number with $\operatorname{Re}\rho \ge \sigma$) | Section 1.1 |
| $S(t)$, $S_\varepsilon(t)$ | $\pi^{-1}\arg\zeta(\frac12+it)$, with the argument continued along $2 \to 2+it \to \frac12+it$; $\pi^{-1}\arg\zeta(\frac12+\varepsilon+it)$, continued along the line $\operatorname{Re}s = \frac12+\varepsilon$ from $S_\varepsilon(0+) = -1$ if $0 < \varepsilon < \frac12$ and from 0 if $\varepsilon > \frac12$ | Sections 2.2.1, 3.4.1 |
| $\vartheta(t)$ | the Riemann Siegel theta function $\operatorname{Im}\log\Gamma(\frac14+\frac{it}{2}) - \frac{t}{2}\log\pi$ | Section 2.2 |

| symbol | meaning | defined in |
|---|---|---|
| $Z(t)$ | Hardy's function $e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$, real for real $t$ | Section 2.2 |
| $\Theta$ | a height, the top of a range of ordinates, with $\Theta = 200$ for the window of Chapters 2 to 5; in Chapters 6 and 7 and Sections 9.1 and 10.2, $\sup \operatorname{Re} \rho$ | Section 2.1.2, Section 6.1 |
| $T_0$ | the height $3 \times 10^{12}$ below which RH has been verified [419] | Section 1.1 |
| $\lambda_n$ | the Keiper Li coefficients $\sum_\rho [1-(1-1/\rho)^n]$ | Section 8.3 |
| $H_t$, $\Lambda$, $\Phi_{\mathrm{dBN}}$ | the de Bruijn Newman flow, normalised by $H_0(z) = \frac{1}{8}\Xi(z/2)$; the de Bruijn Newman constant; the kernel $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$ of the flow ($\Lambda(n)$ is von Mangoldt's function) | Section 8.4 |
| $P(\theta)$, $\nu_n$, $A_P$, $K^P$ | Pólya's Bessel function $P(\theta) = K_{i\theta/2}(2\pi)$; the zeros of $\nu \mapsto K_{i\nu}(2\pi)$, so that $P$ vanishes at $\theta = \pm 2\nu_n$; the operator $-\frac{d^2}{dx^2} + e^{2x}$ on $[\log 2\pi, \infty)$ with a Dirichlet condition, whose eigenvalues are the $\nu_n^2$; the trace $K^P(y) = \sum_n e^{-2\nu_n y}$ | Section 8.7 |

| symbol | meaning | defined in |
|---|---|---|
| $\delta$ | the displacement of an off line zero $\frac{1}{2} \pm \delta + i\gamma$, written $d$ in Sections 5.2, 5.3, 5.4, 8.3, 8.4 and 12.2; in Chapter 2 and where its parameters are quoted, the tilt margin, with $0 < \delta < \pi/4$; in Section 4.3, the angle of a zero | Propositions 1.2.3, 2.1.1 |
| $\eta$ | the tilted height $\eta = \pi/4 - \delta$ (Dedekind's eta function in Sections 8.5, 4.3 and 11.3; Dirichlet's eta function $\eta(s) = (1-2^{1-s})\zeta(s)$ in Section 10.1 and, as $\eta(1+v) = (1-2^{-v})\zeta(1+v)$, in Chapters 6 and 7) | Proposition 2.1.1 |
| $K_\eta(\varepsilon)$ | the kernel constant $\int_{\mathbb{R}} \lvert\Phi(x \pm i\eta)\rvert\, e^{\varepsilon\lvert x\rvert}\, dx$ of the trapezoid bound, with $K_\eta = K_\eta(0)$ | Section 2.1.1 |
| $\psi$ | the digamma function $\Gamma'/\Gamma$ (Chebyshev's function $\sum_{n \le y} \Lambda(n)$ at real $y \ge 1$ in Sections 7.1, 9.1, 10.2 and 11.4, and $\psi(y, \chi_{12})$ in Section 11.3) | Section 1.2 |
| $\gamma_E$ | Euler's constant | Section 5.1 |
| $\mathcal{N}$ | the standard normal distribution function, with $\overline{\mathcal{N}} = 1 - \mathcal{N}$ | Section 8.2 |
| $\xi(s, \chi_{12})$, $K$ | the completed $L$ function $(12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12}) = \int_0^\infty \eta(ix) x^{s/2}\, dx/x$; its kernel $K(u) = 2e^{u/2}\eta(ie^{2u})$, with $\xi(\frac{1}{2}+s, \chi_{12}) = \int_{\mathbb{R}} K(u) e^{su}\, du$ (a variable $Y$ with density $K/\xi(\frac{1}{2}, \chi_{12})$ plays the role of $X$) | Section 8.5.1 |

| symbol | meaning | defined in |
|---|---|---|
| $\xi(s,\chi_4)$, $K_4$, $Y_4$ | the completed Dirichlet beta function $(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_4)$, equal to $\int_0^\infty \eta(ix)^3 x^{(s+1)/2}\,dx/x$; its kernel $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$, with $\xi(\frac{1}{2}+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}\,du$; the variable with density $K_4/\xi(\frac{1}{2},\chi_4)$ | Section 8.5.7.1 |
| $R(x)$, $R_h(x)$ | Riesz's function $x\sum_{n\geq 1}\mu(n)n^{-2}e^{-x/n^2}$; its twist $x\sum_{n\geq 1}\mu(n)n^{-2}(x/n^2)^{ih}e^{-x/n^2}$, with $R_0 = R$ | Sections 10.1.8, 10.2.1 |
| $\xi_K(s)$, $C_K$ | the completed Dedekind zeta function $s(s-1)\lvert d_K\rvert^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$ of a number field $K$, with $\xi_{\mathbb{Q}} = 2\xi$; the constant of $\sum_{N(\mathfrak{a})\leq x}\mu(\mathfrak{a})^2/N(\mathfrak{a})$, with $C = C_{\mathbb{Q}} = 1.7171377$ | Sections 10.3.1, 10.3.3 |
| $\lVert\cdot\rVert_{\mathrm{TV}}$ | the total variation $\lvert\cdot\rvert(E)$ of a signed measure on the window $E$ | Section 1.1 |
| GGC, HCM, ID, SD, BO, EGGC | generalised gamma convolution; hyperbolically completely monotone; infinitely divisible; self decomposable; the Bondesson class; the extended class of generalised gamma convolutions, the weak limits of convolutions of gamma laws and their reflections | Section 1.3, Section 1.3.2 |

| symbol | meaning | defined in |
|---|---|---|

Sections 4.2 and 4.3 use the following notation.

| symbol | meaning | defined in |
|---|---|---|
| $s$, $\varphi(s)$ | the variable $u$ of the clock; the clock $\varphi_{1/2}(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ | Section 4.2 |
| $W$ | the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace | Section 4.2 |
| $T_\alpha$ | Bosch's variable $(Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with independent positive $\alpha$ stable $Z_\alpha, \tilde{Z}_\alpha$; here $\alpha$ is a stability index, not a basepoint | Section 4.2 |
| $X$, $p$, $\varepsilon_h$, $\delta$, $\theta$, $\vartheta$ | under RH, the clock variable $T$; an exponent, not the Pólya density; an elasticity, not a resolution; the angle of a zero; angles, not the ordinate and the Riemann Siegel function | Section 4.3 |
| $\psi$, $S_{\mathrm{on}}$, $R_{\mathrm{off}}$ | $-\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$, the function of the Stieltjes form; its sums over the real and over the nonreal $\gamma$ (in Section 11.4.7, the parts of $G'/G$ in the variable $u$) | Theorems 4.2.4, 4.3.9 |
| $D_\gamma$, $\mathcal{E}_{T_0}$, $\theta_0$ | for a nonreal zero $\gamma = a + ib$, the closed disc $\lvert s - (b^2 - a^2)\rvert \le 2ab$ with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$; the parabolic sliver $\{x + iy :\ x < -T_0^2 + T_0 + \frac{1}{4},\ y^2 < \frac{1}{2} - x\}$ that contains every $D_\gamma$; its angle at the tip, seen from the origin | Theorem 4.3.9 |

The numerical checks behind Section 4.3 come from one script in the archive (`code/README.md`) and, for Theorem 4.3.9, Remark 4.3.10 and Figure 4.2, from `pakes_disc.py`.

Chapters 6 and 7 use the following notation.

| symbol | meaning | defined in |
|---|---|---|
| $s$, $t$, $x$ | the variable $u$ of the clock; the variable $z$ of Thorin measures; the time of the heat trace $W(x)$ | Section 6.1 |
| $\tau$, $z$ | the ordinate; the argument of $\zeta$ and $\xi$ | Section 6.1 |
| $G$, $R$, $\varphi$ | the gamma factor $\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$; the reciprocal Xi function $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, the clock $1/G$ of Section 1.2; the normalised reciprocal zeta function $\varphi(s) = \zeta(\frac{1}{2})/\{(1-4s)\zeta(\frac{1}{2} + \sqrt{s})\}$, with $\varphi = RG$ | (6.1.4) |

| symbol | meaning | defined in |
|---|---|---|
| $R_\sigma$, $\varphi_\sigma$, $R_t$ | $\xi(\sigma)/\xi(\sigma+\sqrt{s})$, which the rest of the book writes $\varphi_\sigma$; the normalised reciprocal zeta function $\zeta(\sigma)/\{(1-s/(1-\sigma)^2)\zeta(\sigma+\sqrt{s})\}$ at $\sigma$, with $\varphi_{1/2}=\varphi$, not the $\varphi_\sigma$ of the rest of the book; $H_t(0)/H_t(2i\sqrt{s})$, with $R_0=R$ (also in Section 8.4, which writes $u$ for $s$) | Sections 6.1, 7.2 |
| $\theta$, $c$, $c_n$ | the Riemann Siegel theta function ($\vartheta$ elsewhere); $c=-\theta'(0)=\zeta'(\frac12)/\zeta(\frac12)=2.6860917$ (also quoted in Sections 5.1, 8.5 and in `code/README.md`); $c_n=2n+\frac12$ | Section 6.1, (6.1.6) |
| $W_0(x)$, $w(x)$, $M(t)$ | the archimedean part $\frac{1}{\pi}\int_0^\infty e^{-x\tau^2}\theta'(\tau)\,d\tau$ of $W(x)$; the canonical function $w=W-W_0$ of $\varphi$; $N(\sqrt{t})-\theta(\sqrt{t})/\pi=1+S(\sqrt{t})$ | Section 6.1 |
| $X_\varphi$, $X_R$ | random variables with Laplace transforms $\varphi$ and $R$, so that $X_R$ has the law of the clock variable $T$ | Section 6.2 |
| $\kappa$, $Y_\kappa$ | a Poisson width; the variable with $\mathbb{E}e^{-sY_\kappa}=\kappa/(\kappa+\sqrt{s})$, whose Thorin density is a Poisson kernel of width $\kappa$ in $\sqrt{t}$ | Section 6.2 |
| $\eta(1+v)$, $\chi_4$, $\mathcal{S}_n$ | Dirichlet's eta function; the nontrivial character modulo 4; the truncated lattice variables $\mathcal{S}_n=\sum_{k\le n}k^{-2}(e_k+e'_k)$, with $\mathcal{S}=\frac{\pi^2}{2}S_2$ and tails $\mathcal{T}_n=\mathcal{S}-\mathcal{S}_n$ (also in Section 11.1) | Sections 6.1, 7.2 |
| $B_N$, $h_\lambda$ | the weights $B_N=\sum_{n^m=N}d_m a_n$ of the Vere-Jones representation, with $a_n=\Lambda(n)/(\sqrt{n}\log n)$ and $\sum_n d_n n^{-v}=1/\eta(1+v)$, which vanish unless $N$ is a prime power $p^n$ and are then sums of necklace polynomials at $p^{-1/2}$; the heat trace of the sawtooth measure $U_\lambda$, a Jacobi theta function | Section 6.1, Theorem 7.1.8 |

# Bibliography


[1] J. Abate and P. P. Valkó. Multi-precision Laplace transform inversion. *International Journal for Numerical Methods in Engineering* 60 (2004), 979–993.

[2] S. Axler, P. Bourdon and W. Ramey. *Harmonic Function Theory*, second edition. Springer, 2001.

[3] L. Alpöge and R. Furman. More than two thirds of the zeta zeros are simple and on the critical line. arXiv:2608.13637, 2026.

[4] D. Applegate and R. Kannan. Sampling and integration of near log-concave functions. In *Proceedings of the 23rd Annual ACM Symposium on Theory of Computing*, 1991, 156–163.

[5] D. Applegate, R. Kannan and N. G. Polson. Random polynomial time algorithms for sampling from joint distributions. Technical report, Department of Statistics, Carnegie Mellon University, 1990.

[6] D. F. Andrews and C. L. Mallows. Scale mixtures of normal distributions. *Journal of the Royal Statistical Society Series B (Methodological)* 36 (1974), 99–102.

[7] S. V. Anulova. On Markov stopping times with a given distribution for a Wiener process. *Theory of Probability and its Applications* 25 (1980), 362–366.

[8] J. Arias de Reyna. High precision computation of Riemann's zeta function by the Riemann-Siegel formula, I. *Mathematics of Computation* 80 (2011), 995–1009.

[9] O. Arizmendi, T. Hasebe and N. Sakuma. On the law of free subordinators. *ALEA Latin American Journal of Probability and Mathematical Statistics* 10 (2013), 271–291.

[10] O. Arizmendi and D. Perales. Cumulants for finite free convolution. *Journal of Combinatorial Theory, Series A* 155 (2018), 244–266.

[11] J. V. Armitage. Zeta functions with a zero at $s = \frac{1}{2}$. *Inventiones Mathematicae* 15 (1971), 199–205.

[12] R. Arratia, L. Goldstein and F. Kochman. Size bias for one and all. *Probability Surveys* 16 (2019), 1–61.

[13] M. Aissen, I. J. Schoenberg and A. M. Whitney. On the generating functions of totally positive sequences I. *Journal d'Analyse Mathématique* 2 (1952), 93–103.

[14] R. J. Backlund. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences* 158 (1914), 1979–1981.

[15] R. J. Backlund. Über die Nullstellen der Riemannschen Zetafunktion. *Acta Mathematica* 41 (1918), 345–375.

[16] L. Báez-Duarte. A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis. *Atti della Accademia Nazionale dei Lincei, Rendiconti Lincei, Matematica e Applicazioni* 14 (2003), 5–11.

[17] L. Báez-Duarte. On Maslanka's representation for the Riemann zeta function. *In-*

*ternational Journal of Mathematics and Mathematical Sciences* (2010), Article ID 714147. doi:10.1155/2010/714147, arXiv:math/0307214.

[18] R. C. Baker and J. Pintz. The distribution of square-free numbers. *Acta Arithmetica* 46 (1985), 73–79.

[19] A. A. Balkema and L. de Haan. Residual life time at great age. *Annals of Probability* 2 (1974), 792–804.

[20] W. D. Banks. On the OpenAI proof that Landau Siegel zeros do not exist. arXiv:2610.15483, 2026.

[21] H. Barkhuijsen, R. de Beer and D. van Ormondt. Improved algorithm for noniterative time domain model fitting to exponentially damped magnetic resonance signals. *Journal of Magnetic Resonance* 73 (1987), 553–557.

[22] R. E. Barlow and F. Proschan. *Statistical Theory of Reliability and Life Testing.* Holt, Rinehart and Winston, 1975.

[23] J. M. Borwein, D. M. Bradley and R. E. Crandall. Computational strategies for the Riemann zeta function. *Journal of Computational and Applied Mathematics* 121 (2000), no. 1–2, 247–296. doi:10.1016/s0377-0427(00)00336-8.

[24] H. H. Bauschke, J. Bolte and M. Teboulle. A descent lemma beyond Lipschitz gradient continuity: first-order methods revisited and applications. *Mathematics of Operations Research* 42 (2017), 330–348.

[25] A. Beskos, A. Jasra, N. Kantas and A. Thiery. On the convergence of adaptive sequential Monte Carlo methods. *Annals of Applied Probability* 26 (2016), 1111–1146.

[26] H. M. Bui, S. J. Lester and M. B. Milinovich. On Balazard, Saias, and Yor's equivalence to the Riemann Hypothesis. *Journal of Mathematical Analysis and Applications* 409 (2014), no. 1, 244–253. doi:10.1016/j.jmaa.2013.06.063, arXiv:1306.0856.

[27] R. P. Boas. *Entire Functions.* Academic Press, 1954.

[28] E. Balanzario and J. Sánchez-Ortiz. Zeros of the Davenport–Heilbronn counterexample. *Mathematics of Computation* 76 (2007), 2045–2049.

[29] Á. Baricz and F. Štampach. The Hurwitz-type theorem for the regular Coulomb wave function via Hankel determinants. *Linear Algebra and its Applications* 548 (2018), 259–272.

[30] O. E. Barndorff-Nielsen and D. R. Cox. Edgeworth and saddle-point approximations with statistical applications. *Journal of the Royal Statistical Society B* 41 (1979), 279 ff.

[31] O. E. Barndorff-Nielsen, J. Kent and M. Sørensen. Normal variance-mean mixtures and $z$ distributions. *International Statistical Review* 50 (1982), 145 ff.

[32] O. E. Barndorff-Nielsen, M. Maejima and K. Sato. Some classes of multivariate infinitely divisible distributions admitting stochastic integral representations. *Bernoulli* 12 (2006), 1–33.

[33] O. E. Barndorff-Nielsen and S. Thorbjørnsen. Self-decomposability and Lévy processes in free probability. *Bernoulli* 8 (2002), 323–366.

[34] O. E. Barndorff-Nielsen and S. Thorbjørnsen. A connection between free and classical infinite divisibility. *Infinite Dimensional Analysis, Quantum Probability and Related Topics* 7 (2004), 573–590.

[35] T. Barnet-Lamb, D. Geraghty, M. Harris and R. Taylor. A family of Calabi–Yau varieties and potential automorphy II. *Publications of the Research Institute for Mathematical Sciences* 47 (2011), 29–98.

[36] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer. Counting zeros of

Dirichlet $L$-functions. *Mathematics of Computation* 90 (2021), 1455–1482.
[37] H. Bercovici and V. Pata. Stable laws and domains of attraction in free probability theory (with an appendix by P. Biane). *Annals of Mathematics* 149 (1999), 1023–1060.
[38] M. Balazard, E. Saias and M. Yor. Notes sur la fonction $\zeta$ de Riemann, 2. *Advances in Mathematics* 143 (1999), 284–287.
[39] J.-F. Burnol. A lower bound in an approximation problem involving the zeros of the Riemann zeta function. *Advances in Mathematics* 170 (2002), 56–70.
[40] C. Butucea and A. B. Tsybakov. Sharp optimality in density deconvolution with dominating bias, I. *Theory of Probability and Its Applications* 52 (2008).
[41] C. Butucea and A. B. Tsybakov. Sharp optimality in density deconvolution with dominating bias, II. *Theory of Probability and Its Applications* 52 (2008), 237–249.
[42] H. Bercovici and D. Voiculescu. Free convolution of measures with unbounded support. *Indiana University Mathematics Journal* 42 (1993), 733–773.
[43] C. Berg. Correction to a paper by A. G. Pakes. *Journal of the Australian Mathematical Society* 76 (2004), 67–74.
[44] C. Berg and G. Forst. Multiply self-decomposable probability measures on $\mathbb{R}_+$ and $\mathbb{Z}_+$. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 62 (1983), 147–163.
[45] M. V. Berry and J. P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41 (1999), 236–266.
[46] A. Beurling. A closure problem related to the Riemann zeta-function. *Proceedings of the National Academy of Sciences* 41 (1955), 312–314.
[47] P. Biane. On the free convolution with a semi-circular distribution. *Indiana University Mathematics Journal* 46 (1997), 705–718.
[48] P. Biane. Matrix valued Brownian motion and a paper by Pólya. In *Séminaire de Probabilités XLII*, Lecture Notes in Mathematics, Springer, 2009, 171–185.
[49] P. Biane, J. Pitman and M. Yor. Probability laws related to the Jacobi theta and Riemann zeta functions, and Brownian excursions. *Bulletin of the American Mathematical Society* 38 (2001), 435–465.
[50] N. H. Bingham, C. M. Goldie and J. L. Teugels. *Regular Variation*. Encyclopedia of Mathematics and its Applications 27, Cambridge University Press, 1987.
[51] L. I. Bluestein. A linear filtering approach to the computation of discrete Fourier transform. *IEEE Transactions on Audio and Electroacoustics* 18 (1970), 451–455.
[52] E. Bogomolny, O. Bohigas, P. Leboeuf and A. G. Monastra. On the spacing distribution of the Riemann zeros: corrections to the asymptotic result. *Journal of Physics A: Mathematical and General* 39 (2006), 10743–10754.
[53] H. Bohr and E. Landau. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences* 158 (1914), 106–110.
[54] E. Bombieri and J. C. Lagarias. Complements to Li's criterion for the Riemann hypothesis. *Journal of Number Theory* 77 (1999), 274–287.
[55] E. Bombieri. Remarks on Weil's quadratic functional in the theory of prime numbers, I. *Rendiconti Lincei. Matematica e Applicazioni* (9) 11 (2000), 183–233.
[56] L. Bondesson. *Generalized Gamma Convolutions and Related Classes of Distributions and Densities*. Lecture Notes in Statistics 76, Springer, 1992.
[57] L. Bondesson. A class of probability distributions that is closed with respect to addition as well as multiplication of independent random variables. *Journal of Theoretical Probability* 28 (2015), 1063–1081.

[58] P. Bosch. HCM property and the half-Cauchy distribution. *Probability and Mathematical Statistics* 35 (2015), 191–200.
[59] P. Bosch and T. Simon. A proof of Bondesson's conjecture on stable densities. *Arkiv för Matematik* 54 (2016), 31–38.
[60] P. Bourgade, T. Fujita and M. Yor. Euler's formulae for $\zeta(2n)$ and products of Cauchy variables. *Electronic Communications in Probability* 12 (2007), 73–80.
[61] P. Bourgade, C. P. Hughes, A. Nikeghbali and M. Yor. The characteristic polynomial of a random unitary matrix: a probabilistic approach. *Duke Mathematical Journal* 145 (2008), 45–69.
[62] J. Bourgain. On large values estimates for Dirichlet polynomials and the density hypothesis for the Riemann zeta function. *International Mathematics Research Notices* 2000 (2000), 133–146.
[63] C. J. Bouwkamp. A conjectured definite integral (A. H. Nuttall), solution. *SIAM Review* 28 (1986), 568–569.
[64] R. P. Brent. On the zeros of the Riemann zeta function in the critical strip. *Mathematics of Computation* 33 (1979), 1361–1372.
[65] S. Broadbent, H. Kadiri, A. Lumley, N. Ng and K. Wilk. Sharper bounds for the Chebyshev function $\theta(x)$. *Mathematics of Computation* 90 (2021), 2281–2315.
[66] M. Broadie, P. Glasserman and S. Kou. A continuity correction for discrete barrier options. *Mathematical Finance* 7 (1997), 325–349.
[67] D. Bump, K.-K. Choi, P. Kurlberg and J. Vaaler. A local Riemann hypothesis, I. *Mathematische Zeitschrift* 233 (2000), 1–18.
[68] J. Büthe. An analytic method for bounding $\psi(x)$. *Mathematics of Computation* 87 (2018), 1991–2009.
[69] N. Calkin and H. S. Wilf. Recounting the rationals. *The American Mathematical Monthly* 107 (2000), 360–363.
[70] B. Candelpergher. A new expansion of the Riemann zeta function. arXiv:2512.11405, 2025.
[71] E. J. Candès and C. Fernandez-Granda. Super-resolution from noisy data. *Journal of Fourier Analysis and Applications* 19 (2013), 1229–1254.
[72] E. J. Candès and C. Fernandez-Granda. Towards a mathematical theory of super-resolution. *Communications on Pure and Applied Mathematics* 67 (2014), 906–956.
[73] D. A. Cardon. Sums of exponential functions having only real zeros. *Manuscripta Mathematica* 113 (2004), 307–317.
[74] D. A. Cardon. Fourier transforms having only real zeros. *Proceedings of the American Mathematical Society* 133 (2005), 1349–1356.
[75] R. J. Carroll and P. Hall. Optimal rates of convergence for deconvolving a density. *Journal of the American Statistical Association* 83 (1988), 1184–1186.
[76] F. Cérou, P. Del Moral, T. Furon and A. Guyader. Sequential Monte Carlo for rare event estimation. *Statistics and Computing* 22 (2012), 795–808.
[77] F. Cérou, P. Del Moral and A. Guyader. A nonasymptotic theorem for unnormalized Feynman Kac particle models. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 47 (2011), 629–649.
[78] M. A. Chaudhry and A. Qadir. Extension of Hardy's class for Ramanujan's interpolation formula and master theorem with applications. *Journal of Inequalities and Applications* 2012 (2012), paper 52.
[79] X. Chen, L. Cheng, J. Chadam and D. Saunders. Existence and uniqueness of solutions to the inverse boundary crossing problem for diffusions. *Annals of Applied*

*Probability* 21 (2011), 1663–1693.
[80] J. Chen. Optimal rate of convergence for finite mixture models. *Annals of Statistics* 23 (1995), 221–233.
[81] N. Chopin. A sequential particle filter method for static models. *Biometrika* 89 (2002), 539–551.
[82] N. Chopin and C. P. Robert. Properties of nested sampling. *Biometrika* 97 (2010), 741–755.
[83] K. L. Chung. Excursions in Brownian motion. *Arkiv för Matematik* 14 (1976), 155–177.
[84] Z. Ciesielski and S. J. Taylor. First passage times and sojourn times for Brownian motion in space and the exact Hausdorff measure of the sample path. *Transactions of the American Mathematical Society* 103 (1962), 434–450.
[85] M. W. Coffey and G. Csordas. On the log-concavity of a Jacobi theta function. *Mathematics of Computation* 82 (2013), 2265–2272.
[86] A. Comtet and S. N. Majumdar. Precise asymptotics for a random walker's maximum. *Journal of Statistical Mechanics* (2005), P06013; arXiv cond-mat/0506195.
[87] A. Connes and C. Consani. Weil positivity and trace formula, the archimedean place. *Selecta Mathematica (N.S.)* 27 (2021), Art. 77, arXiv:2006.13771.
[88] A. Connes, C. Consani and H. Moscovici. Zeta spectral triples. arXiv:2511.22755, 2025.
[89] J. B. Conrey. Zeros of derivatives of Riemann's $\xi$-function on the critical line. *Journal of Number Theory* 16 (1983), 49–74.
[90] J. B. Conrey and X.-J. Li. A note on some positivity conditions related to zeta and $L$-functions. *International Mathematics Research Notices* 2000 (2000), 929–940.
[91] J. B. Conrey, D. W. Farmer, J. P. Keating, M. O. Rubinstein and N. C. Snaith. Integral moments of $L$-functions. *Proceedings of the London Mathematical Society* 91 (2005), 33–104.
[92] P. Contucci, P. Kleban and A. Knauf. A fully magnetizing phase transition. *Journal of Statistical Physics* 97 (1999), 523–539.
[93] P. Contucci and A. Knauf. The phase transition of the number-theoretical spin chain. *Forum Mathematicum* 9 (1997), 547–568.
[94] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey and D. E. Knuth. On the Lambert $W$ function. *Advances in Computational Mathematics* 5 (1996), 329–359.
[95] B. P. Carlin and N. G. Polson. Inference for nonconjugate Bayesian models using the Gibbs sampler. *Canadian Journal of Statistics* 19 (1991), 399–405.
[96] H. Cramér. Studien über die Nullstellen der Riemannschen Zetafunktion. *Mathematische Zeitschrift* 4 (1919), 104–130.
[97] H. Cramér. Ein Mittelwertsatz in der Primzahltheorie. *Mathematische Zeitschrift* 12 (1922), 147–153.
[98] T. Craven, G. Csordas and W. Smith. The zeros of derivatives of entire functions and the Pólya-Wiman conjecture. *Annals of Mathematics* 125 (1987), 405–431.
[99] G. Csordas, T. S. Norfolk and R. S. Varga. The Riemann hypothesis and the Turán inequalities. *Transactions of the American Mathematical Society* 296 (1986), 521–541.
[100] G. Csordas, A. M. Odlyzko, W. Smith and R. S. Varga. A new Lehmer pair of zeros and a new lower bound for the de Bruijn-Newman constant $\Lambda$. *Electronic Transactions on Numerical Analysis* 1 (1993), 104–111.
[101] G. Csordas, W. Smith and R. S. Varga. Lehmer pairs of zeros, the de Bruijn-

Newman constant $\Lambda$, and the Riemann hypothesis. *Constructive Approximation* 10 (1994), 107–129.

[102] I. Csiszár and G. Tusnády. Information geometry and alternating minimization procedures. *Statistics and Decisions*, Supplement Issue 1 (1984), 205–237.

[103] H. B. Curry and I. J. Schoenberg. On Pólya frequency functions IV: the fundamental spline functions and their limits. *Journal d'Analyse Mathématique* 17 (1966), 71–107.

[104] H. Davenport. *Multiplicative Number Theory*, third edition, revised by H. L. Montgomery. Graduate Texts in Mathematics 74, Springer, 2000.

[105] P. Del Moral, A. Doucet and A. Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society, Series B* 68 (2006), 411–436.

[106] N. G. de Bruijn, D. E. Knuth and S. O. Rice. The average height of planted plane trees. In *Graph Theory and Computing*, 15–22. 1972.

[107] Y. de Castro and F. Gamboa. Exact reconstruction using Beurling minimal extrapolation. *Journal of Mathematical Analysis and Applications* 395 (2012), 336–354.

[108] L. de Haan and U. Stadtmüller. Generalized regular variation of second order. *Journal of the Australian Mathematical Society* 61 (1996), 381–395.

[109] P. Del Moral. *Feynman Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, 2004.

[110] L. Devroye. On exact simulation algorithms for some distributions related to Jacobi theta functions. *Statistics and Probability Letters* 79 (2009), 2251–2259.

[111] H. Davenport and H. Heilbronn. On the zeros of certain Dirichlet series I, II. *Journal of the London Mathematical Society* 11 (1936), 181–185 and 307–312.

[112] N. G. de Bruijn. The roots of trigonometric integrals. *Duke Mathematical Journal* 17 (1950), 197–226.

[113] H. Delange. Une remarque sur la dérivée logarithmique de la fonction zêta de Riemann. *Colloquium Mathematicum* 53 (1987), 333–335.

[114] P. Deligne. La conjecture de Weil. I. *Publications Mathématiques de l'IHÉS* 43 (1974), 273–307.

[115] L. Devroye. *Non-Uniform Random Variate Generation*. Springer, 1986.

[116] D. K. Dimitrov and F. R. Lucas. Higher order Turán inequalities for the Riemann $\xi$ function. *Proceedings of the American Mathematical Society* 139 (2011), 1013–1022.

[117] L. M. Delves and J. N. Lyness. A numerical method for locating the zeros of an analytic function. *Mathematics of Computation* 21 (1967), 543–560.

[118] P. Del Moral and L. Miclo. Branching and interacting particle systems approximations of Feynman Kac formulae with applications to nonlinear filtering. *Séminaire de Probabilités XXXIV*, Lecture Notes in Mathematics 1729, 2000, 1–145.

[119] A. Dobner. A proof of Newman's conjecture for the extended Selberg class. *Acta Arithmetica* 201 (2021), 29–62.

[120] W. F. Donoghue, Jr. *Monotone Matrix Functions and Analytic Continuation*. Grundlehren der mathematischen Wissenschaften 207, Springer, 1974.

[121] D. L. Donoho. Superresolution via sparsity constraints. *SIAM Journal on Mathematical Analysis* 23 (1992), 1309–1331.

[122] R.-A. Dragomir, A. B. Taylor, A. d'Aspremont and J. Bolte. Optimal complexity and certification of Bregman first-order methods. *Mathematical Programming* 194 (2022), 41–83.

[123] P. L. Duren. *Theory of $H^p$ Spaces*. Academic Press, 1970.

[124] H. Dym and H. P. McKean. *Gaussian Processes, Function Theory, and the Inverse*

*Spectral Problem*. Probability and Mathematical Statistics 31, Academic Press, 1976.
[125] F. J. Dyson. A Brownian-motion model for the eigenvalues of a random matrix. *Journal of Mathematical Physics* 3 (1962), 1191–1198.
[126] A. Edrei. On the generating functions of totally positive sequences II. *Journal d'Analyse Mathématique* 2 (1953), 104–109.
[127] H. M. Edwards. *Riemann's Zeta Function*. Academic Press, 1974.
[128] S. N. Ethier and T. G. Kurtz. *Markov Processes: Characterization and Convergence*. Wiley, 1986.
[129] E. Ekström and S. Janson. The inverse first-passage problem and optimal stopping. *Annals of Applied Probability* 26 (2016), 3154–3177.
[130] R. S. Ellis, J. L. Monroe and C. M. Newman. The GHS and other correlation inequalities for a class of even ferromagnets. *Communications in Mathematical Physics* 46 (1976), 167–182.
[131] P. Embrechts, C. M. Goldie and N. Veraverbeke. Subexponentiality and infinite divisibility. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 49 (1979), 335–347.
[132] J. Fan. On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19 (1991), 1257–1272.
[133] D. W. Farmer. Jensen polynomials are not a plausible route to proving the Riemann Hypothesis. *Advances in Mathematics* 411 (2022), 108781; arXiv:2008.07206.
[134] D. W. Farmer and H. Ki. Landau–Siegel zeros and zeros of the derivative of the Riemann zeta function. *Advances in Mathematics* 230 (2012), 2048–2064.
[135] J. Fei, C.-N. Yeh and E. Gull. Nevanlinna analytical continuation. *Physical Review Letters* 126 (2021), 056402.
[136] W. Feller. *An Introduction to Probability Theory and Its Applications*, volume II, second edition. Wiley, 1971.
[137] J. Fiala, P. Kleban and A. Özlük. The phase transition in statistical models defined on Farey fractions. *Journal of Statistical Physics* 110 (2003), 73–86.
[138] A. H. Barnett, J. Magland and L. af Klinteberg. A parallel nonuniform fast Fourier transform library based on an "exponential of semicircle" kernel. *SIAM Journal on Scientific Computing* 41 (2019), C479–C504.
[139] G. E. Forsythe and R. A. Leibler. Matrix inversion by a Monte Carlo method. *Mathematical Tables and Other Aids to Computation* 4 (1950), 127–129.
[140] S. Fourati. On some analytical properties of stable densities. arXiv:1604.07705, 2016.
[141] P. Fearnhead and D. Prangle. Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation. *Journal of the Royal Statistical Society, Series B* 74 (2012), 419–474.
[142] S. A. Fulling. Nonuniqueness of canonical field quantization in Riemannian spacetime. *Physical Review D* 7 (1973), 2850–2862.
[143] W. Gabcke. *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*. Dissertation, Georg-August-Universität Göttingen, 1979.
[144] P. X. Gallagher. Primes and powers of 2. *Inventiones Mathematicae* 29 (1975), 125–142.
[145] F. Gao and A. W. van der Vaart. Posterior contraction rates for deconvolution of Dirichlet Laplace mixtures. *Electronic Journal of Statistics* 10 (2016), 608–627.
[146] G. Gasper. Using integrals of squares of certain real-valued special functions to

prove that the Pólya $\Xi^*(z)$ function, the functions $K_{iz}(a)$, $a > 0$, and some other entire functions have only real zeros. arXiv:0801.2996, 2008.

[147] I. M. Gel'fand and N. Ya. Vilenkin. *Generalized Functions, Volume 4: Applications of Harmonic Analysis*. Academic Press, 1964.

[148] C. J. Geyer. Practical Markov chain Monte Carlo. *Statistical Science* 7 (1992), 473–483.

[149] S. M. Gonek, C. P. Hughes and J. P. Keating. A hybrid Euler-Hadamard product for the Riemann zeta function. *Duke Mathematical Journal* 136 (2007), no. 3, 507–549. doi:10.1215/s0012-7094-07-13634-2, arXiv:math/0511182.

[150] B. V. Gnedenko. Sur la distribution limite du terme maximum d'une série aléatoire. *Annals of Mathematics* 44 (1943), 423–453.

[151] B. V. Gnedenko and A. N. Kolmogorov. *Limit Distributions for Sums of Independent Random Variables*. Translated by K. L. Chung. Addison-Wesley, Cambridge, MA, 1954; revised edition 1968.

[152] C. Goldie. A class of infinitely divisible random variables. *Proceedings of the Cambridge Philosophical Society* 63 (1967), 1141–1143.

[153] D. A. Goldston and H. L. Montgomery. Pair correlation of zeros and primes in short intervals. In *Analytic Number Theory and Diophantine Problems*, Birkhäuser, 1987, 183–203.

[154] S. M. Gonek and H. Ki. Pair correlation of zeros of the real and imaginary parts of the Riemann zeta function. *Journal of Number Theory* 186 (2018), 35–61.

[155] S. M. Gonek and H. L. Montgomery. Zeros of a family of approximations of the Riemann zeta-function. *International Mathematics Research Notices* 2013 (2013), 4712–4733.

[156] I. J. Good and R. A. Gaskins. Density estimation and bump hunting by the penalized likelihood method. *Journal of the American Statistical Association* 75 (1980), 42.

[157] X. Gourdon. The $10^{13}$ first zeros of the Riemann Zeta function, and zeros computation at very large height. Manuscript, 2004, `http://numbers.computation.free.fr/Constants/Miscellaneous/zetazeros1e13-1e24.pdf`, accessed 27 September 2026.

[158] J.-P. Gram. Note sur les zéros de la fonction $\zeta(s)$ de Riemann. *Acta Mathematica* 27 (1903), 289–304.

[159] M. Griffin, K. Ono, L. Rolen and D. Zagier. Jensen polynomials for the Riemann zeta function and other sequences. *Proceedings of the National Academy of Sciences* 116 (2019), 11103–11110.

[160] R. B. Griffiths, C. A. Hurst and S. Sherman. Concavity of magnetization of an Ising ferromagnet in a positive external field. *Journal of Mathematical Physics* 11 (1970), 790–795.

[161] B. Grigelionis. Processes of Meixner type. *Lithuanian Mathematical Journal* 39 (1999), 33–41.

[162] J. Grommer. Ganze transzendente Funktionen mit lauter reellen Nullstellen. *Journal für die reine und angewandte Mathematik* 144 (1914), 114–166.

[163] A. P. Guinand. A summation formula in the theory of prime numbers. *Proceedings of the London Mathematical Society* 50 (1948), 107–119.

[164] J. Gunns and C. Hughes. The effect of repeated differentiation on $L$-functions. *Journal of Number Theory* 194 (2019), 30–43.

[165] A. Gut. Some remarks on the Riemann zeta distribution. *Revue Roumaine de*

*Mathématiques Pures et Appliquées* 51 (2006).
[166] L. Guth and J. Maynard. New large value estimates for Dirichlet polynomials. *Annals of Mathematics* 203 (2026), 623–675.
[167] J. Hadamard. *Lectures on Cauchy's Problem in Linear Partial Differential Equations*. Yale University Press, 1923.
[168] J. Haglund. Some conjectures on the zeros of approximates to the Riemann Ξ-function and incomplete gamma functions. arXiv:0910.5228, 2009.
[169] G. H. Hardy. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences Paris* 158 (1914), 1012–1014.
[170] D. A. Hejhal. On a result of G. Pólya concerning the Riemann $\xi$-function. *Journal d'Analyse Mathématique* 55 (1990), 59–95.
[171] G. A. Hiary. Fast methods to compute the Riemann zeta function. *Annals of Mathematics* 174 (2011).
[172] G. A. Hiary. An explicit van der Corput estimate for $\zeta(1/2 + it)$. *Indagationes Mathematicae* 27 (2016), 524–533.
[173] G. H. Hardy and J. E. Littlewood. Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes. *Acta Mathematica* 41 (1916), 119–196.
[174] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*, sixth edition, revised by D. R. Heath-Brown and J. H. Silverman. Oxford University Press, 2008.
[175] P. Hartman and G. S. Watson. "Normal" distribution functions on spheres and the modified Bessel functions. *Annals of Probability* 2 (1974), 593–607.
[176] E. Hasanalizade, Q. Shen and P.-J. Wong. Counting zeros of the Riemann zeta function. *Journal of Number Theory* 235 (2022), 219–241.
[177] T. Hasebe and S. Thorbjørnsen. Unimodality of the freely selfdecomposable probability laws. *Journal of Theoretical Probability* 29 (2016), 922–940.
[178] C. B. Haselgrove. A disproof of a conjecture of Pólya. *Mathematika* 5 (1958), 141–145.
[179] D. R. Heath-Brown and J. C. Puchta. Integers represented as a sum of primes and powers of two. *Asian Journal of Mathematics* 6 (2002), 535–566.
[180] E. Hecke. Über die Bestimmung Dirichletscher Reihen durch ihre Funktionalgleichung. *Mathematische Annalen* 112 (1936), 664–699.
[181] H. Hedenmalm, P. Lindqvist and K. Seip. A Hilbert space of Dirichlet series and systems of dilated functions in $L^2(0,1)$. *Duke Mathematical Journal* 86 (1997), 1–37.
[182] C. Hermite. Extrait d'une lettre de Mr. Ch. Hermite de Paris à Mr. Borchardt de Berlin sur le nombre des racines d'une équation algébrique comprises entre des limites données. *Journal für die reine und angewandte Mathematik* 52 (1856), 39–51.
[183] Y. Hua and T. K. Sarkar. Matrix pencil method for estimating parameters of exponentially damped undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 38 (1990), 814–824.
[184] I. A. Ibragimov. On the composition of unimodal distributions. *Theory of Probability and its Applications* 1 (1956), 255–260.
[185] M. Jarrell and J. E. Gubernatis. Bayesian inference and the analytic continuation of imaginary-time quantum Monte Carlo data. *Physics Reports* 269 (1996), 133–195.
[186] H. Kadiri. Une région explicite sans zéros pour la fonction $\zeta$ de Riemann. *Acta*

*Arithmetica* 117 (2005), 303–339.
[187] O. Kallenberg. *Foundations of Modern Probability*, 2nd ed. Springer, 2002.
[188] S. Karlin. *Total Positivity*, Volume I. Stanford University Press, 1968.
[189] W. Krichene, A. Bayen and P. L. Bartlett. Accelerated mirror descent in continuous and discrete time. *Advances in Neural Information Processing Systems* 28 (2015).
[190] D. P. Kennedy. The distribution of the maximum Brownian excursion. *Journal of Applied Probability* 13 (1976), 371–376.
[191] H. Ki. All but finitely many nontrivial zeros of the approximations of the Epstein zeta function are simple and on the critical line. *Proceedings of the London Mathematical Society* 90 (2005), 321–344.
[192] H. Ki. On the nontrivial zeros of modified Epstein zeta functions. *Comptes Rendus Mathématique. Académie des Sciences Paris* 342 (2006), 79–81.
[193] H. Ki. The Riemann $\Xi$ function under repeated differentiation. *Journal of Number Theory* 120 (2006), 120–131.
[194] H. Ki. On the zeros of approximations of the Ramanujan $\Xi$ function. *The Ramanujan Journal* 17 (2008), 123–143.
[195] H. Ki. The zeros of the derivative of the Riemann zeta function near the critical line. *International Mathematics Research Notices* (2008), rnn064.
[196] H. Ki. On the zeros of Weng's zeta functions. *International Mathematics Research Notices* (2010), 2367–2393.
[197] H. Ki and Y. O. Kim. A generalization of Newman's result on the zeros of Fourier transforms. *Computational Methods and Function Theory* 2 (2002), 449–467.
[198] H. Ki and Y. O. Kim. De Bruijn's question on the zeros of Fourier transforms. *Journal d'Analyse Mathématique* 91 (2003), 369–387.
[199] H. Ki, Y. Komori and M. Suzuki. On the zeros of Weng zeta functions for Chevalley groups. *Manuscripta Mathematica* 148 (2015), 119–176.
[200] P. Kleban and A. E. Özlük. A Farey fraction spin chain. *Communications in Mathematical Physics* 203 (1999), 635–647.
[201] B. T. Knapik, A. W. van der Vaart and J. H. van Zanten. Bayesian inverse problems with Gaussian priors. *Annals of Statistics* 39 (2011), 2626–2657.
[202] B. T. Knapik, A. W. van der Vaart and J. H. van Zanten. Bayesian recovery of the initial condition for the heat equation. *Communications in Statistics, Theory and Methods* 42 (2013), 1294–1313.
[203] A. Knauf. On a ferromagnetic spin chain. *Communications in Mathematical Physics* 153 (1993), 77–115.
[204] A. Knauf. The number-theoretical spin chain and the Riemann zeroes. *Communications in Mathematical Physics* 196 (1998), 703–731.
[205] F. B. Knight. Characterization of the Lévy measures of inverse local times of gap diffusion. In *Seminar on Stochastic Processes*, Birkhäuser, 53–78. 1981.
[206] S. Kotani and S. Watanabe. Krein's spectral theory of strings and generalized diffusion processes. *Lecture Notes in Mathematics* 923 (1982), 235–259.
[207] J. C. Lagarias. Hilbert spaces of entire functions and Dirichlet $L$-functions. In *Frontiers in Number Theory, Physics, and Geometry I*, Springer, 2006, 365–377.
[208] N. Leong. Explicit estimates for the logarithmic derivative and the reciprocal of the Riemann zeta function. Preprint arXiv:2405.04869, 2024.
[209] B. Ya. Levin. *Lectures on Entire Functions*. American Mathematical Society, 1996.
[210] W. Li and W. Liao. Stable super resolution limit and smallest singular value of restricted Fourier matrices. *Applied and Computational Harmonic Analysis* 51 (2021),

118–156.

[211] J. S. Liu, W. H. Wong and A. Kong. Covariance structure of the Gibbs sampler with applications to the comparisons of estimators and augmentation schemes. *Biometrika* 81 (1994), 27–40.

[212] L. B. Lucy. An iterative technique for the rectification of observed distributions. *Astronomical Journal* 79 (1974), 745–754.

[213] L. Lovász and S. Vempala. The geometry of logconcave functions and sampling algorithms. *Random Structures and Algorithms* 30 (2007), 307–358.

[214] K. Maślanka. Hypergeometric-like representation of the zeta-function of Riemann. arXiv:math-ph/0105007, 2001.

[215] A. Meister. *Deconvolution Problems in Nonparametric Statistics*. Springer, 2009.

[216] O. Miyatake. Note on Riemann's $\xi$ function. *J. Inst. Polytech. Osaka City Univ. Ser. A* 1 (1951), no. 2, 61–70.

[217] O. Miyatake. On the distribution of zero points of a function which is related to Riemann's $\xi$ function. *J. Inst. Polytech. Osaka City Univ. Ser. A* 1 (1951), no. 2, 39–59.

[218] H. L. Montgomery and K. Soundararajan. Primes in short intervals. *Communications in Mathematical Physics* 252 (2004), 589–617.

[219] Yu. Matiyasevich, F. Saidak and P. Zvengrowski. Horizontal monotonicity of the modulus of the zeta function, Dirichlet $L$-functions, and related functions. *Acta Arithmetica* 166 (2014), 189–200.

[220] M. J. Mossinghoff and T. S. Trudgian. Nonnegative trigonometric polynomials and a zero free region for the Riemann zeta function. *Journal of Number Theory* 157 (2015), 329–349.

[221] M. J. Mossinghoff, T. S. Trudgian and A. Yang. Explicit zero free regions for the Riemann zeta function. *Research in Number Theory* 10 (2024). doi:10.1007/s40993-023-00498-y. arXiv:2212.06867.

[222] Y. Nesterov. A method of solving a convex programming problem with convergence rate $O(1/k^2)$. *Soviet Mathematics Doklady* 27 (1983), 372–376.

[223] X. Nguyen. Convergence of latent mixing measures in finite and infinite mixture models. *Annals of Statistics* 41 (2013), 370–400.

[224] B. O'Donoghue and E. J. Candès. Adaptive restart for accelerated gradient schemes. *Foundations of Computational Mathematics* 15 (2015), 715–732.

[225] A. M. Odlyzko and A. Schönhage. Fast algorithms for multiple evaluations of the Riemann zeta function. *Transactions of the American Mathematical Society* 309 (1988), 797–809.

[226] N. G. Polson. Bessel spin ferromagnets: Lee Yang level repulsion, the GHS inequality, and a moment criterion for the Riemann Hypothesis. Manuscript, October 2026.

[227] N. G. Polson and V. Sokolov. Balazard, Saias and Yor meet Thorin: Cauchy averages and their Monte Carlo solution. Manuscript, October 2026.

[228] N. G. Polson. Cauchy deconvolution from $U_1$ to $U_{1/2}$: particle filtering and generative Bayesian computation along the Cauchy flow of the Riemann xi function. Manuscript, October 2026.

[229] N. G. Polson. The Comtet and Majumdar identity, the $2\sin^2$ lemma, and approximating families for the reciprocal Xi function. Manuscript, October 2026.

[230] N. G. Polson. The explicit formula as a deconvolution. Manuscript, October 2026.

[231] N. G. Polson. Extreme values, hitting times and a path to RH. Manuscript, October 2026.

[232] N. G. Polson. The excursion maximum behind the pole cancellation: work on the open problems. Manuscript, October 2026.
[233] N. G. Polson. Ferromagnets and the Riemann zeros. Manuscript, October 2026.
[234] N. G. Polson. A Gumbel mixture path to the Riemann Hypothesis. Manuscript, October 2026.
[235] N. G. Polson. The Gamma skeleton, heat flow, and the arithmetic displacement: work on the open problems of the Gumbel mixture path. Manuscript, October 2026.
[236] N. G. Polson. The natural EGGC family for the reciprocal Xi function: heat flow, the $2\sin^2$ lemma, $z$ distributions and a class on $(0, \infty)$. Manuscript, October 2026.
[237] N. G. Polson. Progress on the open problems. Manuscript, October 2026.
[238] N. G. Polson. Hitting times and the reciprocal Xi function. Manuscript, October 2026.
[239] N. G. Polson. How close can a finite approximant get? The lattice bias law for Dyck zeros and the universal Lehmer obstruction. Manuscript, October 2026.
[240] N. G. Polson. The Thorin measure at level one and the backward Cauchy flow. Manuscript, October 2026.
[241] N. G. Polson. The Lee Yang zeta function of a hard wall ferromagnet: Riemann's zeta as the constant weight case, and the rigidity of the critical line. Manuscript, October 2026.
[242] N. G. Polson. A path to the Riemann Hypothesis: ferromagnetic skeleton, Coulomb gas, and the one missing step. Manuscript, October 2026.
[243] N. G. Polson. The Xi Pólya Gamma law III: self decomposability without RH, Li's coefficients as cumulants, the motion of zeros, and Fejér mixtures. Manuscript, October 2026.
[244] N. G. Polson. Pole cancellation: why no component of Riemann's mixture is ferromagnetic, and what a ferromagnetic solution must therefore be. Manuscript, October 2026.
[245] N. G. Polson. Thorin truncation and the zeros of $L(s, \chi_4)$. Manuscript, October 2026.
[246] N. G. Polson. Two uniform spins: the exact ferromagnet behind $\xi$, Riemann's contour proof as a Lee Yang trace, and the transform RH asks about. Manuscript, October 2026.
[247] N. G. Polson. Williams, ferromagnetic random variables and the Riemann Xi function: a ferromagnetic family and its reciprocal. Manuscript, October 2026.
[248] N. G. Polson. Weil positivity as the Thorin condition. Manuscript, October 2026.
[249] J. Pickands. Statistical inference using extreme order statistics. *Annals of Statistics* 3 (1975), 119–131.
[250] N. G. Polson. The Euler product of $1/\zeta$ as a Lévy measure, the $2\sin^2$ lemma, and the reciprocal Xi function as a generalized gamma convolution. Note, October 2026.
[251] N. G. Polson. A residue expansion of the reciprocal Xi function and generalized gamma convolution methods. Note, October 2026.
[252] N. G. Polson and V. Ročková. Posterior concentration for sparse deep learning. *Advances in Neural Information Processing Systems* 31 (2018).
[253] N. G. Polson. The reciprocal Xi function as the Laplace transform of an EGGC: mixing laws, the Euler product, Selberg and Barnes. Note, October 2026.
[254] N. G. Polson and J. G. Scott. Local shrinkage rules, Lévy processes and regularized regression. *Journal of the Royal Statistical Society, Series B* 74 (2012), 287–311.
[255] N. G. Polson and V. Sokolov. Generative AI for Bayesian computation. *Entropy* 27

(2025), 683. arXiv:2305.14972.
[256] N. G. Polson and V. Sokolov. *Generative Bayesian Computation: Quantile Neural Networks for Inference and Surrogates*. Chapman and Hall/CRC, 2026.
[257] N. G. Polson and V. Sokolov. *Hilbert's Sixth Problem*. KDP, 2026.
[258] N. G. Polson, J. G. Scott and J. Windle. Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables. *Journal of the American Statistical Association* 108 (2013), 1339–1349.
[259] The python-flint developers. *python-flint: Python bindings for FLINT and Arb*. Software, `https://github.com/flintlib/python-flint`.
[260] H. Rademacher. On the Phragmén–Lindelöf theorem and some applications. *Mathematische Zeitschrift* 72 (1959), 192–204.
[261] A. Rényi. On the theory of order statistics. *Acta Mathematica Academiae Scientiarum Hungaricae* 4 (1953), 191–231.
[262] G. O. Roberts, A. Gelman and W. R. Gilks. Weak convergence and optimal scaling of random walk Metropolis algorithms. *Annals of Applied Probability* 7 (1997), 110–120.
[263] W. H. Richardson. Bayesian based iterative method of image restoration. *Journal of the Optical Society of America* 62 (1972), 55–59.
[264] J. B. Rosser and L. Schoenfeld. Sharper bounds for the Chebyshev functions $\theta(x)$ and $\psi(x)$. *Mathematics of Computation* 29 (1975), 243–269.
[265] S. M. Rump. Verification methods: rigorous results using floating point arithmetic. *Acta Numerica* 19 (2010), 287–449.
[266] W. Su, S. Boyd and E. J. Candès. A differential equation for modeling Nesterov's accelerated gradient method: theory and insights. *Journal of Machine Learning Research* 17 (2016), 1–43.
[267] G. Schiebinger, E. Robeva and B. Recht. Superresolution without separation. *Information and Inference* 7 (2018), 1–30.
[268] L. A. Shepp and Y. Vardi. Maximum likelihood reconstruction for emission tomography. *IEEE Transactions on Medical Imaging* 1 (1982), 113–122.
[269] J. Skilling. Nested sampling for general Bayesian computation. *Bayesian Analysis* 1 (2006), 833–859.
[270] R. L. Smith, J. A. Tawn and S. G. Coles. Markov chain models for threshold exceedances. *Biometrika* 84 (1997), 249–268.
[271] F. Spitzer. A combinatorial lemma and its application to probability theory. *Transactions of the American Mathematical Society* 82 (1956), 323–339.
[272] S. B. Stechkin. Zeros of the Riemann zeta function. *Matematicheskie Zametki* 8 (1970), 419–429.
[273] L. A. Stefanski and R. J. Carroll. Deconvolving kernel density estimators. *Statistics* 21 (1990), 169–184.
[274] M. Stern. Ueber eine zahlentheoretische Funktion. *Journal für die reine und angewandte Mathematik* 55 (1858), 193–220.
[275] S. M. Stigler. Fractional order statistics, with applications. *Journal of the American Statistical Association* 72 (1977), 544–550.
[276] G. 't Hooft. On the quantum structure of a black hole. *Nuclear Physics B* 256 (1985), 727–745.
[277] G. Hurst. Computations of the Mertens function and improved bounds on the Mertens conjecture. *Mathematics of Computation* 87 (2018), 1013–1028.
[278] J. I. Hutchinson. On the roots of the Riemann zeta function. *Transactions of the*

*American Mathematical Society* 27 (1925), 49–60.
[279] M. N. Huxley. On the difference between consecutive primes. *Inventiones Mathematicae* 15 (1972), 164–170.
[280] Y. Ihara. On the Euler–Kronecker constants of global fields and primes with small norms. In *Algebraic Geometry and Number Theory*, V. Ginzburg, ed., Progress in Mathematics 253, Birkhäuser, 2006, 407–451.
[281] A. M. Iksanov, Z. J. Jurek and B. M. Schreiber. A new factorization property of the selfdecomposable probability measures. *Annals of Probability* 32 (2004), 1356–1369.
[282] A. E. Ingham. Mean-value theorems in the theory of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (2) 27 (1928), 273–300.
[283] A. E. Ingham. *The Distribution of Prime Numbers*. Cambridge Tracts in Mathematics and Mathematical Physics 30, Cambridge University Press, 1932.
[284] A. E. Ingham. On the estimation of $N(\sigma, T)$. *Quarterly Journal of Mathematics* 11 (1940), 291–292.
[285] A. E. Ingham. On two conjectures in the theory of numbers. *American Journal of Mathematics* 64 (1942), 313–319.
[286] H. Inoue. Expansion of Dirichlet $L$-function on the critical line in Meixner-Pollaczek polynomials. arXiv:1412.1220, 2014.
[287] K. Itô and H. P. McKean. *Diffusion Processes and Their Sample Paths*. Springer, 1965.
[288] H. Iwaniec and E. Kowalski. *Analytic Number Theory*. American Mathematical Society Colloquium Publications 53, American Mathematical Society, 2004.
[289] L. F. James, B. Roynette and M. Yor. Generalized gamma convolutions, Dirichlet means, Thorin measures, with explicit examples. *Probability Surveys* 5 (2008), 346–415.
[290] F. Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers* 66 (2017), 1281–1292.
[291] D. R. Johnston and A. Yang. Some explicit estimates for the error term in the prime number theorem. *Journal of Mathematical Analysis and Applications* 527 (2023), 127460.
[292] L. Jones and N. O'Connell. Weyl chambers, symmetric spaces and number variance saturation. *ALEA Latin American Journal of Probability and Mathematical Statistics* 2 (2006), 91–118.
[293] Z. J. Jurek and W. Vervaat. An integral representation for selfdecomposable Banach space valued random variables. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 62 (1983), 247–262.
[294] M. Kac. A remark on Wiener's Tauberian theorem. *Proceedings of the American Mathematical Society* 16 (1965), 1155–1157.
[295] G. A. Kalugin and D. J. Jeffrey. Unimodal sequences show Lambert $W$ is Bernstein. arXiv:1011.5940, 2010.
[296] G. A. Kalugin, D. J. Jeffrey and R. M. Corless. Stieltjes, Poisson and other integral representations for functions of Lambert $W$. Preprint, arXiv:1103.5640, 2011; published as [297, 298].
[297] G. A. Kalugin, D. J. Jeffrey, R. M. Corless and P. B. Borwein. Stieltjes and other integral representations for functions of Lambert $W$. *Integral Transforms and Special Functions* 23 (2012), 581–593.
[298] G. A. Kalugin, D. J. Jeffrey and R. M. Corless. Bernstein, Pick, Poisson and related integral expressions for Lambert $W$. *Integral Transforms and Special Functions* 23

(2012), 817–829.
[299] Y. Kasahara. Tauberian theorems of exponential type. *Journal of Mathematics of Kyoto University* 18 (1978), 209–219.
[300] N. M. Katz and P. Sarnak. Zeroes of zeta functions and symmetry. *Bulletin of the American Mathematical Society* 36 (1999), 1–26.
[301] J. P. Keating and N. C. Snaith. Random matrix theory and $\zeta(1/2 + it)$. *Communications in Mathematical Physics* 214 (2000), 57–89.
[302] B. Kerr. A simpler proof that there are no Landau Siegel zeros. arXiv:2610.21099, 2026.
[303] J. B. Keiper. Power series expansions of Riemann's $\xi$ function. *Mathematics of Computation* 58 (1992), 765–773.
[304] J. T. Kent. Some probabilistic properties of Bessel functions. *Annals of Probability* 6 (1978), 760–770.
[305] J. T. Kent. Eigenvalue expansions for diffusion hitting times. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 52 (1980), 309–319.
[306] J. T. Kent. The spectral decomposition of a diffusion hitting time. *Annals of Probability* 10 (1982), 207–219.
[307] D. Kershaw. Some extensions of W. Gautschi's inequalities for the gamma function. *Mathematics of Computation* 41 (1983), 607–611.
[308] A. Ya. Khinchin. *Limit Laws for Sums of Independent Random Variables* (in Russian). ONTI, Moscow, 1938.
[309] H. Ki and Y. O. Kim. On the number of nonreal zeros of real entire functions and the Fourier-Pólya conjecture. *Duke Mathematical Journal* 104 (2000), 45–73.
[310] H. Ki, Y.-O. Kim and J. Lee. On the de Bruijn–Newman constant. *Advances in Mathematics* 222 (2009), 281–306.
[311] Y. O. Kim. A proof of the Pólya-Wiman conjecture. *Proceedings of the American Mathematical Society* 109 (1990), 1045–1052.
[312] R. Koekoek, P. A. Lesky and R. F. Swarttouw. *Hypergeometric Orthogonal Polynomials and Their q-Analogues*. Springer, 2010.
[313] T. Konstantopoulos, P. Patie and R. Sarkar. A new class of solutions to the van Dantzig problem, the Lee–Yang property, and the Riemann hypothesis. *Annales de l'Institut Fourier* 74 (2024), 377–421.
[314] J. Korevaar. Distribution proof of Wiener's Tauberian theorem. *Proceedings of the American Mathematical Society* 16 (1965), 353–355.
[315] N. M. Korobov. Estimates of trigonometric sums and their applications. *Uspekhi Matematicheskikh Nauk* 13 (1958), no. 4, 185–192.
[316] G. K. Kristiansen. A proof of Steutel's conjecture. *Annals of Probability* 22 (1994), 442–452.
[317] A. B. J. Kuijlaars and W. Van Assche. The asymptotic zero distribution of orthogonal polynomials with varying recurrence coefficients. *Journal of Approximation Theory* 99 (1999), 167–197.
[318] A. Kuznetsov. Integral representations for the Dirichlet $L$-functions and their expansions in Meixner–Pollaczek polynomials and rising factorials. *Integral Transforms and Special Functions* 18 (2007), 827–835.
[319] A. Kuznetsov. Expansion of the Riemann $\Xi$ function in Meixner–Pollaczek polynomials. *Canadian Mathematical Bulletin* 51 (2008), 561–569.
[320] J. C. Lagarias. On a positivity property of the Riemann $\xi$-function. *Acta Arithmetica* 89 (1999), 217–234.

[321] J. C. Lagarias. Zero spacing distributions for differenced $L$-functions. *Acta Arithmetica* 120 (2005), 159–184.
[322] C. D. Lai and D. Vere-Jones. Odd man out: the Meixner hypergeometric distribution. *Australian Journal of Statistics* 21 (1979), 256–265.
[323] J. Lamperti. An occupation time theorem for a class of stochastic processes. *Transactions of the American Mathematical Society* 88 (1958), 380–387.
[324] Y. Lamzouri. A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line. arXiv:2609.02882, 2026.
[325] E. Landau. Über die Nullstellen der Zetafunktion. *Mathematische Annalen* 71 (1912), 548–564.
[326] H. J. Landau and H. O. Pollak. Prolate spheroidal wave functions, Fourier analysis and uncertainty, III: the dimension of the space of essentially time and band limited signals. *Bell System Technical Journal* 41 (1962), 1295–1336.
[327] H. J. Landau and H. Widom. Eigenvalue distribution of time and frequency limiting. *Journal of Mathematical Analysis and Applications* 77 (1980), 469–481.
[328] S. Lang. *Algebraic Number Theory*, second edition. Graduate Texts in Mathematics 110, Springer, 1994.
[329] C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. Prentice Hall, 1974.
[330] I. Lazarević. Certaines inégalités où interviennent les fonctions hyperboliques. *Univerzitet u Beogradu, Publikacije Elektrotehničkog Fakulteta, Serija Matematika i Fizika* 159–170 (1966), 41–48.
[331] T. D. Lee and C. N. Yang. Statistical theory of equations of state and phase transitions. II. Lattice gas and Ising model. *Physical Review* 87 (1952), 410–419.
[332] R. S. Lehman. On the distribution of zeros of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (3) 20 (1970), 303–320.
[333] D. H. Lehmer. On the roots of the Riemann zeta-function. *Acta Mathematica* 95 (1956), 291–298.
[334] D. H. Lehmer. Extended computation of the Riemann zeta-function. *Mathematika* 3 (1956), 102–108.
[335] B. Ja. Levin. *Distribution of Zeros of Entire Functions*. Translations of Mathematical Monographs 5, American Mathematical Society, 1964.
[336] X.-J. Li. The positivity of a sequence of numbers and the Riemann hypothesis. *Journal of Number Theory* 65 (1997), 325–333.
[337] H. Z. Li. The number of powers of 2 in a representation of large even integers by sums of such powers and of two primes. *Acta Arithmetica* 92 (2000), 229–237.
[338] H. Z. Li. The number of powers of 2 in a representation of large even integers by sums of such powers and of two primes, II. *Acta Arithmetica* 96 (2001), 369–379.
[339] E. H. Lieb and A. D. Sokal. A general Lee-Yang theorem for one-component and multicomponent ferromagnets. *Communications in Mathematical Physics* 80 (1981), 153–179.
[340] G. D. Lin and C. Y. Hu. The Riemann zeta distribution. *Bernoulli* 7 (2001), 817–828.
[341] A. Lindner, L. Pan and K. Sato. On quasi-infinitely divisible distributions. *Transactions of the American Mathematical Society* 370 (2018), 8483–8520.
[342] Yu. V. Linnik. Prime numbers and powers of two. *Trudy Matematicheskogo Instituta imeni V. A. Steklova* 38 (1951), 152–169.
[343] Yu. V. Linnik. Addition of prime numbers with powers of one and the same number.

*Matematicheskii Sbornik* 32 (1953), 3–60.

[344] Yu. V. Linnik. *Decomposition of Probability Distributions*. Oliver and Boyd, 1964.

[345] J. E. Littlewood. On the zeros of the Riemann zeta-function. *Proceedings of the Cambridge Philosophical Society* 22 (1924), 295–318.

[346] J. Liu, M. C. Liu and T. Wang. The number of powers of 2 in a representation of large even integers, I. *Science in China Series A* 41 (1998), 386–398.

[347] J. Liu, M. C. Liu and T. Wang. On the almost Goldbach problem of Linnik. *Journal de Théorie des Nombres de Bordeaux* 11 (1999), 133–147.

[348] The LMFDB Collaboration. *The L-functions and modular forms database*, `https://www.lmfdb.org`, accessed 27 September 2026.

[349] K. Löwner. Über monotone Matrixfunktionen. *Mathematische Zeitschrift* 38 (1934), 177–216.

[350] R. Lugannani and S. Rice. Saddle point approximation for the distribution of the sum of independent random variables. *Advances in Applied Probability* 12 (1980), no. 2, 475–490. doi:10.2307/1426607.

[351] D. Ostrovsky. Theory of Barnes beta distributions. *Electronic Communications in Probability* 18 (2013), paper no. 59, 1–16. doi:10.1214/ECP.v18-2445.

[352] B. Liu. Improved zero-free half-planes for the quasi-Riemann hypothesis. arXiv:2610.12234 [math.NT], submitted 8 October 2026, v2 12 October 2026. Preprint, not refereed.

[353] B. Liu. Zero-free half-planes for Hecke $L$-functions over number fields. arXiv:2610.26835, submitted 19 October 2026. Preprint, not refereed.

[354] J. Thorner. A quasi Riemann hypothesis for Dedekind zeta functions. Preprint 2610.00200, 2026, proofs attributed to AxiomProver. Unchecked, not refereed.

[355] E. C. Titchmarsh. *The Theory of Functions*, second edition. Oxford University Press, 1939.

[356] Y. Vardi and D. Lee. From image deblurring to optimal investments: maximum likelihood solutions for positive linear inverse problems. *Journal of the Royal Statistical Society, Series B* 55 (1993), 569–598.

[357] W. Vervaat. A relation between Brownian bridge and Brownian excursion. *The Annals of Probability* 7 (1979), 143–149.

[358] N. G. Polson and J. G. Scott. Vertical likelihood Monte Carlo. arXiv:1409.3601.

[359] H. von Koch. Sur la distribution des nombres premiers. *Acta Mathematica* 24 (1901), 159–182.

[360] A. Voros. *Zeta Functions over Zeros of Zeta Functions*. Lecture Notes of the Unione Matematica Italiana 8, Springer, 2010.

[361] R. Varadhan and C. Roland. Simple and globally convergent methods for accelerating the convergence of any EM algorithm. *Scandinavian Journal of Statistics* 35 (2008), 335–353.

[362] Y. Vardi, L. A. Shepp and L. Kaufman. A statistical model for positron emission tomography. *Journal of the American Statistical Association* 80 (1985), 8–20.

[363] H. Wunderlich. Finite-radius Jensen and relative-entropy formulations of the Balazard-Saias-Yor criterion. arXiv:2610.10584, 2026.

[364] Y. Xu and H. Sang. Amplified moments and a refinement of the quasi-Riemann hypothesis. arXiv:2610.15291, October 2026. Preprint, not refereed.

[365] T. Xylouris. On the least prime in an arithmetic progression and estimates for the zeros of Dirichlet $L$-functions. *Acta Arithmetica* 150 (2011), no. 1, 65–91. doi:10.4064/aa150-1-4.

[366] E. Lukacs. Contributions to a problem of D. van Dantzig. *Theory of Probability and its Applications* 13 (1968), 116–127.
[367] S. Mandelbrojt. *Séries de Dirichlet. Principes et méthodes*. Gauthier-Villars, 1969.
[368] V. A. Marčenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR, Sbornik* 1 (1967), 457–483.
[369] A. W. Marcus, D. A. Spielman and N. Srivastava. Interlacing families II: mixed characteristic polynomials and the Kadison Singer problem. *Annals of Mathematics* 182 (2015), 327–350.
[370] A. W. Marcus, D. A. Spielman and N. Srivastava. Finite free convolutions of polynomials. *Probability Theory and Related Fields* 182 (2022), 807–848.
[371] S. Mase. Decomposition of infinitely divisible characteristic functions with absolutely continuous Poisson spectral measure. *Annals of the Institute of Statistical Mathematics* 27 (1975), 289–298.
[372] Math StackExchange question 2063166. Dirichlet series associated to squared Möbius. December 2016, `https://math.stackexchange.com/q/2063166`, accessed 9 October 2026.
[373] N. Metropolis and G.-C. Rota. Witt vectors and the algebra of necklaces. *Advances in Mathematics* 50 (1983), 95–125.
[374] H. L. Montgomery. The pair correlation of zeros of the zeta function. In *Analytic Number Theory*, H. G. Diamond, ed., Proceedings of Symposia in Pure Mathematics 24, American Mathematical Society, 1973, 181–193.
[375] H. L. Montgomery. Zeros of approximations to the zeta function. In *Studies in Pure Mathematics: To the Memory of Paul Turán*, Birkhäuser, Basel, 1983, 497–506.
[376] The mpmath development team. *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.3.0, 2023.
[377] T. Nakamura. A complete Riemann zeta distribution and the Riemann hypothesis. *Bernoulli* 21 (2015), 604–617.
[378] T. Nakamura. Zeta distributions generated by Dirichlet series and their (quasi) infinite divisibility. arXiv:2209.13257, 2022.
[379] C. M. Newman. Zeros of the partition function for generalized Ising systems. *Communications on Pure and Applied Mathematics* 27 (1974), 143–159.
[380] C. M. Newman. Inequalities for Ising models and field theories which obey the Lee-Yang theorem. *Communications in Mathematical Physics* 41 (1975), 1–9.
[381] C. M. Newman. Fourier transforms with only real zeros. *Proceedings of the American Mathematical Society* 61 (1976), 245–251.
[382] C. M. Newman. The GHS inequality and the Riemann hypothesis. *Constructive Approximation* 7 (1991), 389–399.
[383] C. M. Newman and W. Wu. Constants of de Bruijn–Newman type in analytic number theory and statistical physics. *Bulletin of the American Mathematical Society* 57 (2020), 595–614.
[384] C. M. Newman and W. Wu. Lee–Yang property and Gaussian multiplicative chaos. *Communications in Mathematical Physics* 369 (2019), 153–170.
[385] T. Newton. Kernel-concavity obstructions and Fourier-side Laguerre bounds for the Riemann $\Xi$ programme. Zenodo, 2026, `https://doi.org/10.5281/zenodo.23461586`, accessed 20 October 2026.
[386] A. Nica and R. Speicher. *Lectures on the Combinatorics of Free Probability*. London Mathematical Society Lecture Note Series 335, Cambridge University Press, 2006.
[387] A. Nikeghbali and M. Yor. The Barnes $G$ function and its relations with sums and

products of generalized gamma convolution variables. *Electronic Communications in Probability* 14 (2009), 396–411.

[388] A. A. Novikov. Martingales, Tauberian theorem, and strategies of gambling. *Theory of Probability and its Applications* 41 (1997), 716–729.

[389] A. H. Nuttall. A conjectured definite integral, problem. *SIAM Review* 27 (1985), 573.

[390] B. Nyman. *On the One-Dimensional Translation Group and Semi-Group in Certain Function Spaces*. Thesis, University of Uppsala, 1950.

[391] A. M. Odlyzko. On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation* 48 (1987), 273–308.

[392] A. M. Odlyzko and A. Schönhage. Fast algorithms for multiple evaluations of the Riemann zeta function. *Transactions of the American Mathematical Society* 309 (1988), 797–809.

[393] A. M. Odlyzko. *The $10^{20}$-th zero of the Riemann zeta function and 175 million of its neighbors*. Unpublished manuscript, 1992, `https://www-users.cse.umn.edu/~odlyzko/unpublished/zeta.10to20.1992.pdf`, accessed 27 September 2026.

[394] A. M. Odlyzko. An improved bound for the de Bruijn–Newman constant. *Numerical Algorithms* 25 (2000), 293–303.

[395] A. M. Odlyzko. The $10^{22}$-nd zero of the Riemann zeta function. In *Dynamical, Spectral, and Arithmetic Zeta Functions*, M. L. Lapidus and M. van Frankenhuysen, eds., Contemporary Mathematics 290, American Mathematical Society, 2001, 139–144.

[396] A. M. Odlyzko. *Tables of zeros of the Riemann zeta function*. Online, `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`, accessed 27 September 2026.

[397] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain, eds. *NIST Digital Library of Mathematical Functions*. `https://dlmf.nist.gov/`, Release 1.2.8 of 2026-09-15.

[398] OpenAI. The quasi-Riemann hypothesis: a zero-free half-plane $\mathrm{Re}(s) > 7/8$. Preprint dated 30 September 2026, released 6 October 2026 with a formalisation in Lean, `https://github.com/openai/math` (family 003), accessed 7 October 2026.

[399] OpenAI. The quasi-Riemann hypothesis. Preprint dated 5 October 2026, released 6 October 2026, `https://github.com/openai/math` (family 003), accessed 7 October 2026.

[400] A. G. Pakes. Length biasing and laws equivalent to the log-normal. *Journal of Mathematical Analysis and Applications* 197 (1996), 825–854.

[401] A. G. Pakes. Remarks on converse Carleman and Krein criteria for the classical moment problem. *Journal of the Australian Mathematical Society* 71 (2001), 81–104.

[402] A. G. Pakes. Structure of Stieltjes classes of moment-equivalent probability laws. *Journal of Mathematical Analysis and Applications* 326 (2007), 1268–1290.

[403] A. G. Pakes. Lambert's $W$, infinite divisibility and Poisson mixtures. *Journal of Mathematical Analysis and Applications* 378 (2011), 480–492.

[404] A. G. Pakes. On the preservation of infinite divisibility under length-biasing. *Journal of Probability* 2014 (2014), 703697.

[405] A. G. Pakes. The Lambert $W$ function, Nuttall's integral, and the Lambert law. *Statistics and Probability Letters* 139 (2018), 53–60.

[406] A. G. Pakes. Self-decomposable laws from continuous branching processes. *Journal*

*of Theoretical Probability* 33 (2020), 361–395.
[407] A. G. Pakes. Self-decomposability and the Lambert law. *Axioms* 15 (2026), 574.
[408] A. G. Pakes, W.-L. Hung and J.-W. Wu. Criteria for the unique determination of probability distributions by moments. *Australian and New Zealand Journal of Statistics* 43 (2001), 101–111.
[409] A. G. Pakes and R. Khattree. Length-biasing, characterizations of laws and the moment problem. *Australian Journal of Statistics* 34 (1992), 307–322.
[410] A. G. Pakes, T. Sapatinas and E. B. Fosam. Characterizations, length-biasing, and infinite divisibility. *Statistical Papers* 37 (1996), 53–69.
[411] V. Pérez-Abreu and N. Sakuma. Free generalized gamma convolutions. *Electronic Communications in Probability* 13 (2008), 526–539.
[412] G. Peskir. On integral equations arising in the first-passage problem for Brownian motion. *Journal of Integral Equations and Applications* 14 (2002), 397–423.
[413] J. Pintz and I. Z. Ruzsa. On Linnik's approximation to Goldbach's problem, I. *Acta Arithmetica* 109 (2003), 169–194.
[414] J. Pintz and I. Z. Ruzsa. On Linnik's approximation to Goldbach's problem, II. *Acta Mathematica Hungarica* 161 (2020), 569–582.
[415] J. Pitman and M. Yor. Bessel processes and infinitely divisible laws. In *Stochastic Integrals*, Lecture Notes in Mathematics 851, Springer, 1981, 285–370.
[416] J. Pitman and M. Yor. Infinitely divisible laws associated with hyperbolic functions. *Canadian Journal of Mathematics* 55 (2003), 292–330.
[417] D. J. Platt. Numerical computations concerning the GRH. *Mathematics of Computation* 85 (2016), 3009–3027.
[418] D. J. Platt and T. S. Trudgian. Linnik's approximation to Goldbach's conjecture, and other problems. *Journal of Number Theory* 153 (2015), 54–62.
[419] D. J. Platt and T. S. Trudgian. The Riemann hypothesis is true up to $3 \cdot 10^{12}$. *Bulletin of the London Mathematical Society* 53 (2021), 792–797.
[420] N. G. Polson, J. G. Scott and J. Windle. The Bayesian bridge. *Journal of the Royal Statistical Society, Series B (Statistical Methodology)* 76 (2014), 713; sections and theorems are cited by the numbering of arXiv:1109.2279v2.
[421] N. G. Polson. On Hilbert's 8th problem. arXiv:1708.02653v24, 2026.
[422] N. G. Polson. On Hilbert's 8th problem. arXiv:1708.02653v5, 2017.
[423] N. G. Polson. Riemann, Thorin, van Dantzig pairs, Wald couples and Hadamard factorisation. arXiv:1804.10043, 2018.
[424] N. G. Polson. *On Hilbert's 8th Problem: the reciprocal Xi clock, Thorin measures, van Dantzig pairs, zero free approximations, hitting times and the reciprocal zeta function*, first edition. Independently published, 2026.
[425] N. G. Polson. The Dirichlet beta function in Thorin form. Manuscript, October 2026.
[426] N. G. Polson. Biane on Pólya: hitting times, Thorin measures and a Wick rotation. Manuscript, October 2026.
[427] N. G. Polson. Thorin measures of the Dedekind eta function. Manuscript, September 2026.
[428] N. G. Polson. Maxima of exponentials, incomplete gamma continuations, and the Gamma(2) structure of $L$ functions. Manuscript, October 2026.
[429] N. G. Polson. The Pólya density as a maximum order statistic and the Thorin measure of the reciprocal Xi function. Manuscript, October 2026.
[430] N. G. Polson. Thorin type representations of the reciprocal zeta function on the

critical line. Manuscript, September 2026, now Part IX of the first edition of *On Hilbert's 8th Problem* (2026).

[431] N. G. Polson. The squarefree Dirichlet series $\zeta(s)/\zeta(2s)$: a summary of Math StackExchange 2063166, Pólya densities, Gamma(2) smoothing and Thorin measures of $\Xi$. Manuscript, October 2026.

[432] N. G. Polson. Gamma, reciprocal Gamma, $\eta^3$ and $\Delta$: Thorin forms and the Riemann $\Xi$ function. Manuscript, October 2026.

[433] N. G. Polson. Wigner's semicircle, free probability and the Thorin framework for the reciprocal Xi function. Manuscript, version 7, October 2026.

[434] N. G. Polson. Pakes, the Lambert law and the Thorin condition for the Riemann Xi function. Manuscript, October 2026.

[435] N. G. Polson. The Xi Pólya Gamma law VIII: why the exponent is $\cos(2\pi/9)$: pole cancellation, a Meixner convolution, and the structure of the Pólya mixture. Manuscript, October 2026.

[436] N. G. Polson. The Xi Pólya Gamma law X: interlacing families: the cosh family $\operatorname{Re}\xi(\sigma + it)$, the variance shrink theorem, and the zero free region as the extent of an interlacing family. Manuscript, October 2026.

[437] N. G. Polson and V. Sokolov. Computing the Thorin measure of the reciprocal Xi function: a Pólya route and a total variation comparison with the Riemann atoms. Manuscript, September 2026.

[438] N. G. Polson and V. Sokolov. Four attacks on the computational path: a working report. Manuscript, 7 October 2026.

[439] N. G. Polson and V. Sokolov. *On Langlands Conjectures: Probabilistic Methods*, Chapter 21. Draft, 2026.

[440] G. Pólya. Über Annäherung durch Polynome mit lauter reellen Wurzeln. *Rendiconti del Circolo Matematico di Palermo* 36 (1913), 279–295.

[441] G. Pólya. Algebraische Untersuchungen über ganze Funktionen vom Geschlechte Null und Eins. *Journal für die reine und angewandte Mathematik* 145 (1915), 224–249.

[442] G. Pólya. Über die Nullstellen gewisser ganzer Funktionen. *Mathematische Zeitschrift* 2 (1918), 352–383.

[443] G. Pólya. Verschiedene Bemerkungen zur Zahlentheorie. *Jahresbericht der Deutschen Mathematiker-Vereinigung* 28 (1919), 31–40.

[444] G. Pólya. Über die Nullstellen sukzessiver Derivierten. *Mathematische Zeitschrift* 12 (1922), 36–60.

[445] G. Pólya. Über die Existenz unendlich vieler singulärer Punkte auf der Konvergenzgeraden gewisser Dirichletscher Reihen. *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse* (1923), 45–50.

[446] G. Pólya. On the zeros of an integral function represented by Fourier's integral. *Messenger of Mathematics* 52 (1923), 185–188.

[447] G. Pólya. Bemerkung über die Integraldarstellung der Riemannschen $\xi$-Funktion. *Acta Mathematica* 48 (1926), 305–317.

[448] G. Pólya. On the zeros of certain trigonometric integrals. *Journal of the London Mathematical Society* 1 (1926), 98–99.

[449] G. Pólya. Über trigonometrische Integrale mit nur reellen Nullstellen. *Journal für die reine und angewandte Mathematik* 158 (1927), 6–18.

[450] G. Pólya. Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen. *Det Kongelige Danske Videnskabernes Selskab, Mathematisk-fysiske*

*Meddelelser* 7 (1927), no. 17, 3–33.
[451] G. Pólya. Untersuchungen über Lücken und Singularitäten von Potenzreihen. *Mathematische Zeitschrift* 29 (1929), 549–640.
[452] G. Pólya. Some problems connected with Fourier's work on transcendental equations. *The Quarterly Journal of Mathematics* (Oxford) 1 (1930), 21–34.
[453] G. Pólya. On the zeros of the derivatives of a function and its analytic character. *Bulletin of the American Mathematical Society* 49 (1943), 178–191.
[454] G. Pólya. *Collected Papers, Vol. II: Location of Zeros*, edited by R. P. Boas. Mathematicians of Our Time 8, MIT Press, 1974.
[455] G. Pólya and J. Schur. Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen. *Journal für die reine und angewandte Mathematik* 144 (1914), 89–113.
[456] D. H. J. Polymath. Effective approximation of heat flow evolution of the Riemann $\xi$ function, and a new upper bound for the de Bruijn–Newman constant. *Research in the Mathematical Sciences* 6 (2019), paper 31.
[457] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler. More than five-twelfths of the zeros of $\zeta$ are on the critical line. *Research in the Mathematical Sciences* 7 (2020), article 2.
[458] R. Remmert. *Classical Topics in Complex Function Theory*, translated by L. Kay. Graduate Texts in Mathematics 172, Springer, 1998.
[459] M. Riesz. Sur l'hypothèse de Riemann. *Acta Mathematica* 40 (1916), 185–190.
[460] H. Robbins. A remark on Stirling's formula. *American Mathematical Monthly* 62 (1955), 26–29.
[461] G. O. Roberts. Asymptotic approximations for Brownian motion boundary hitting times. *Annals of Probability* 19 (1991), 1689–1731.
[462] G. O. Roberts and C. F. Shortland. The hazard rate tangent approximation for boundary hitting times. *Annals of Applied Probability* 5 (1995), 446–460.
[463] B. Rodgers and T. Tao. The de Bruijn–Newman constant is non-negative. *Forum of Mathematics, Pi* 8 (2020), e6.
[464] D. Romik. Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases. *Acta Arithmetica* 200 (2021), 259–329.
[465] J. B. Rosser and L. Schoenfeld. Approximate formulas for some functions of prime numbers. *Illinois Journal of Mathematics* 6 (1962), 64–94.
[466] J. B. Rosser, J. M. Yohe and L. Schoenfeld. Rigorous computation and the zeros of the Riemann zeta-function. In *Information Processing 68: Proceedings of IFIP Congress 1968*, volume 1, A. J. H. Morrell, ed., North-Holland, 1969, 70–76.
[467] R. Rumely. Numerical computations concerning the ERH. *Mathematics of Computation* 61 (1993), 415–440.
[468] R. Salem. Sur une proposition équivalente à l'hypothèse de Riemann. *Comptes Rendus de l'Académie des Sciences* 236 (1953), 1127–1128.
[469] P. Salminen and P. Vallois. On maximum increase and decrease of Brownian motion. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 43 (2007), 655–676.
[470] K. Sato. Class $L$ of multivariate distributions and its subclasses. *Journal of Multivariate Analysis* 10 (1980), 207–232.
[471] K. Sato. *Lévy Processes and Infinitely Divisible Distributions*. Cambridge Studies in Advanced Mathematics 68, Cambridge University Press, 1999.

[472] A. Saumard and J. A. Wellner. Log-concavity and strong log-concavity: a review. *Statistics Surveys* 8 (2014), 45–114.
[473] R. L. Schilling, R. Song and Z. Vondraček. *Bernstein Functions: Theory and Applications*, second edition. Walter de Gruyter, 2012.
[474] I. J. Schoenberg. On Pólya frequency functions. I. The totally positive functions and their Laplace transforms. *Journal d'Analyse Mathématique* 1 (1951), 331–374.
[475] W. Schoutens and J. L. Teugels. Lévy processes, polynomials and martingales. *Communications in Statistics. Stochastic Models* 14 (1998), 335–349.
[476] K. Seip. *Interpolation and Sampling in Spaces of Analytic Functions*. University Lecture Series 33, American Mathematical Society, 2004.
[477] A. Selberg. Contributions to the theory of the Riemann zeta-function. *Archiv for Mathematik og Naturvidenskab* 48 (1946), 89–155.
[478] C. L. Siegel. Über Riemanns Nachlaß zur analytischen Zahlentheorie. *Quellen und Studien zur Geschichte der Mathematik, Astronomie und Physik, Abteilung B: Studien* 2 (1932), 45–80.
[479] G. Sierra. The Riemann zeros as energy levels of a Dirac fermion in a potential built from the prime numbers in Rindler spacetime. *Journal of Physics A: Mathematical and Theoretical* 47 (2014), 325204.
[480] B. Simon and R. B. Griffiths. The $(\phi^4)_2$ field theory as a classical Ising model. *Communications in Mathematical Physics* 33 (1973), 145–164.
[481] D. Slepian and H. O. Pollak. Prolate spheroidal wave functions, Fourier analysis and uncertainty, I. *Bell System Technical Journal* 40 (1961), 43–63.
[482] J. Sondow and C. Dumitrescu. A monotonicity property of Riemann's xi function and a reformulation of the Riemann hypothesis. *Periodica Mathematica Hungarica* 60 (2010), 37–40.
[483] R. Spira. Zeros of approximate functional approximations. *Mathematics of Computation* 21 (1967), 41–48.
[484] R. Spira. Some zeros of the Titchmarsh counterexample. *Mathematics of Computation* 63 (1994), 747–748.
[485] L. A. Stefanski. A normal scale mixture representation of the logistic distribution. *Statistics & Probability Letters* 11 (1991), 69–70.
[486] F. W. Steutel. Note on the infinite divisibility of exponential mixtures. *Annals of Mathematical Statistics* 38 (1967), 1303–1305.
[487] F. W. Steutel. Some recent results in infinite divisibility. *Stochastic Processes and their Applications* 1 (1973), 125–143.
[488] F. W. Steutel and J. G. F. Thiemann. On the independence of integer and fractional parts. *Statistica Neerlandica* 43 (1989), 53–59.
[489] F. W. Steutel and K. van Harn. *Infinite Divisibility of Probability Distributions on the Real Line*. Marcel Dekker, 2004.
[490] A. Talbot. The accurate numerical inversion of Laplace transforms. *Journal of the Institute of Mathematics and its Applications* 23 (1979), 97–120.
[491] M. Tanaka. A numerical investigation on cumulative sum of the Liouville function. *Tokyo Journal of Mathematics* 3 (1980), 187–189.
[492] J. T. Tate. Fourier analysis in number fields and Hecke's zeta-functions. In *Algebraic Number Theory*, J. W. S. Cassels and A. Fröhlich, eds., Academic Press, 1967, 305–347.
[493] O. Thorin. On the infinite divisibility of the Pareto distribution. *Scandinavian Actuarial Journal* (1977), 31–40.

[494] O. Thorin. On the infinite divisibility of the lognormal distribution. *Scandinavian Actuarial Journal* (1977), 121–148.
[495] O. Thorin. An extension of the notion of a generalized Γ-convolution. *Scandinavian Actuarial Journal* (1978), 141–149.
[496] E. C. Titchmarsh. The zeros of the Riemann zeta-function. *Proceedings of the Royal Society of London, Series A* 157 (1936), 261–263.
[497] E. C. Titchmarsh. *Eigenfunction Expansions Associated with Second-Order Differential Equations*. Clarendon Press, Oxford, 1946.
[498] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*, second edition, revised by D. R. Heath-Brown. Oxford University Press, 1986.
[499] P. Turán. On some approximative Dirichlet polynomials in the theory of the zeta function of Riemann. *Danske Vid. Selsk. Mat.-Fys. Medd.* 24 (1948).
[500] L. N. Trefethen and J. A. C. Weideman. The exponentially convergent trapezoidal rule. *SIAM Review* 56 (2014), 385–458.
[501] T. S. Trudgian. Improvements to Turing's method. *Mathematics of Computation* 80 (2011), 2259–2279.
[502] T. S. Trudgian. On the success and failure of Gram's Law and the Rosser Rule. *Acta Arithmetica* 148 (2011), 225–256.
[503] T. S. Trudgian. An improved upper bound for the argument of the Riemann zeta-function on the critical line II. *Journal of Number Theory* 134 (2014), 280–292.
[504] K.-M. Tsang. Some Ω-theorems for the Riemann zeta-function. *Acta Arithmetica* 46 (1986), 369–395.
[505] N. Tschebotareff. Über die Realität von Nullstellen ganzer transzendenter Funktionen. *Mathematische Annalen* 99 (1928), 660–686.
[506] A. M. Turing. Some calculations of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (3) 3 (1953), 99–117.
[507] K. Urbanik. Limit laws for sequences of normed sums satisfying some stability conditions. In *Multivariate Analysis III*, P. R. Krishnaiah, ed., Academic Press, 1973, 225–237.
[508] J. van de Lune, H. J. J. te Riele and D. T. Winter. On the zeros of the Riemann zeta function in the critical strip. IV. *Mathematics of Computation* 46 (1986), 667–681.
[509] I. M. Vinogradov. A new estimate of the function $\zeta(1+it)$. *Izvestiya Akademii Nauk SSSR, Seriya Matematicheskaya* 22 (1958), 161–164.
[510] D. Voiculescu. Addition of certain non-commuting random variables. *Journal of Functional Analysis* 66 (1986), 323–346.
[511] T. Wang. On Linnik's almost Goldbach theorem. *Science in China Series A* 42 (1999), 1155–1172.
[512] A. Weil. Sur les "formules explicites" de la théorie des nombres premiers. *Communications du Séminaire Mathématique de l'Université de Lund*, tome supplémentaire (1952), 252–265.
[513] G. C. Wick. Properties of Bethe-Salpeter wave functions. *Physical Review* 96 (1954), 1124–1134.
[514] D. V. Widder. *The Laplace Transform*. Princeton Mathematical Series 6, Princeton University Press, 1941.
[515] N. Wiener. Tauberian theorems. *Annals of Mathematics* 33 (1932), 1–100.
[516] N. Wiener. *The Fourier Integral and Certain of Its Applications*. Cambridge University Press, 1933.
[517] E. P. Wigner. On the distribution of the roots of certain symmetric matrices. *Annals*

*of Mathematics* 67 (1958), 325–327.

[518] D. Williams. Brownian motion and the Riemann zeta-function. In *Disorder in Physical Systems*, G. R. Grimmett and D. J. A. Welsh, eds., Oxford University Press, 1990, 361–372.

[519] R. E. Williamson. Multiply monotone functions and their Laplace transforms. *Duke Mathematical Journal* 23 (1956), 189–207.

[520] S. J. Wolfe. On a continuous analogue of the stochastic difference equation $X_n = \rho X_{n-1} + B_n$. *Stochastic Processes and their Applications* 12 (1982), 301–312.

[521] M. Yamazato. Unimodality of infinitely divisible distribution functions of class $L$. *Annals of Probability* 6 (1978), 523–531.

[522] A. C. Yao and D. E. Knuth. Analysis of the subtractive algorithm for greatest common divisors. *Proceedings of the National Academy of Sciences of the United States of America* 72 (1975), 4720–4722.

[523] S. Yakowitz, J. E. Krimmel and F. Szidarovszky. Weighted Monte Carlo integration. *SIAM Journal on Numerical Analysis* 15 (1978), 1289–1300.

[524] M. Yor. Loi de l'indice du lacet brownien, et distribution de Hartman-Watson. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 53 (1980), 71–95.

[525] H. Yoshida. On Hermitian forms attached to zeta functions. In *Zeta Functions in Geometry (Tokyo, 1990)*, N. Kurokawa and T. Sunada, eds., Advanced Studies in Pure Mathematics 21, Kinokuniya, 1992, 281–325.

[526] X. Zhu. Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law. arXiv:2608.24827v2, 2 September 2026.